# Infrastructure Operations

Use **VCF Operations** as a VI administrator to optimize and monitor compute, storage, and network of your VCF platform. These operational capabilities include performance optimization, visibility and observability of infrastructure and operations, global configuration management, monitoring, troubleshooting, log analytics, end-to-end network monitoring, energy efficiency monitoring, and other.

**In This Chapter**


- Integrating Data Sources with VCF Operations
- Configuring and Managing Policies
- Configuring Alerts and Using Actions
- Managing Inventory
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- Extending Monitoring Capabilities


**Getting to Know VCF**

To apply this documentation, you must be acquainted with the VCF Product Overview, VCF Design, and VCF Release Notes.

**SDKs, APIs, and CLI for VCF Administration**

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See Administration SDKs, APIs, and CLI.

## Integrating Data Sources with VCF Operations

You can extend the monitoring capabilities of VCF Operations by installing and configuring integrations in VCF Operations to connect to and analyze data from external data sources in your environment. Once connected, you can use VCF Operations to monitor and manage objects in your environment. These integrations are also referenced as management packs.

An integration can be a connection to a data source, or include predefined dashboards, widgets, alerts, and views.

Integrations include accounts, dashboards, reports, alerts, and other content. The accounts comprise of adapters, using which VCF Operations manages communication and integration with other products, applications, and functions. When an integration is installed and the accounts are configured, you can use the VCF Operations analytics and alerting tools to manage the objects in your environment.

VMware integrations include native, non-native, and external management packs.

For a new deployment of VCF Operations, the status of integrations is as follows:

**Table 1174: Integration Activation Status**

| Integration Name | Activated by Default? | Can be Deactivated? |
|---|---|---|
| vCenter | Yes | No |
| VMware Cloud Foundation | Yes | Yes |
| vSphere Supervisor | Yes | No |
| VCF for vSphere Replication | Yes | Yes |
| vSAN | Yes | No |
| Service Discovery | Yes | No |
| VCF Automation for VM Apps Organization | No | Yes |
| VCF Automation for All Apps Organization | No | Yes |
| OS and Application Monitoring | Yes | No |
| VCF for Networks | No | Yes |
| NSX | Yes | No |
| Ping | No | Yes |
| PCI Compliance | No | Yes |
| ISO Compliance | No | Yes |
| HIPAA Compliance | No | Yes |
| FISMA Compliance | No | Yes |
| DISA Compliance | No | Yes |
| CIS Compliance | No | Yes |

The list of activated and available integrations, accounts, and the full repository of integrations are all available from a central **Integrations** page in VCF Operations . This page can be accessed from the left menu by clicking **Operate** > **Administration** > **Integrations** > **Repository**.

### Non-Native Management Packs and Management Pack Builder

If a technology has no native management pack, you can install a non-native pack or can build your own using VCF Operations Management Pack Builder.

With the Management Pack Builder, you can create custom management packs for use in VCF Operations. This is a no-code solution for bringing in data from an external API and either creating new resources or extending your VMware and third party resources with new data, relationships, and events. For more information, see Management Pack Builder.

### Upgrade Considerations

The native integrations in VCF Operations are reinstalled if VCF Operations is upgraded.

## The Integrations Page in VCF Operations

You can activate integrations, install integrations, add, import, or export accounts from the **Integrations** page in VCF Operations. Integrations are also referenced as Management Packs in VCF Operations.

### How Integrations Work

Each account contains adapters using which VCF Operations integrates with other products, applications, and monitors their function and health. Integrations can include dashboards, reports, alerts, and accounts.

### Where You Find Integrations

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Integrations**. By default, the **Accounts** tab opens. Use this tab to add, import, or export accounts. For more information, see The Accounts Tab.

Click the **Repository** tab to view the installed and available integrations. You can also install integrations that are delivered as PAK files. For more information, see The Repository Tab.

Click the **Solutions Catalog** tab to add and manage VCF Operations management packs from VCF Solutions Catalog. For more information see, The Solutions Catalog Tab.

> **Note:**
> For more information on pre-activated integrations, see Integrating Data Sources with VCF Operations.

### Collection Notifications

The **Notifications** bell icon on the top menu provides quick access to status and critical notifications that are related to data collections. The icon indicates whether notifications exist, and whether any of them are critical.

The list displays notifications about the data collections that are in progress, and indicates whether any of them have critical issues. The list groups the data collection notifications that are in progress into a single entry at the bottom of the list. To view the details about a collection, expand the notification.

Each notification displays the status of the last or current data collection, the associated adapter instance, and the time since the collection completed or an issue was identified. You can click a notification to open the **Integrations** page, where you can see further details, and manage adapter instances.

If problems occur with the data collections, VCF Operations identifies those problems during each 5-minute collection cycle.

Go to **Collection Status** to monitor the data collection, for more information, see Monitoring Data Collection .

## The Accounts Tab

You can view and add accounts from the **Accounts** tab on the Integration page in VCF Operations. You can add accounts for integrations or you can add accounts after activating integrations from the **Repository** tab.

### Where You Find the Accounts tab

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Integrations**. The **Accounts** tab opens by default. The accounts tab lists all the accounts that are configured for the different integrations available in VCF Operations. You can filter accounts to view accounts specific to an integration type.

### What You Can Do in the Accounts tab

You can add and configure accounts that are associated with integrations that are pre-activated or those that you activate in VCF Operations . After you have configured the account, VCF Operations can collect data from or can send data to the target system. You can access the accounts page at any time to modify your adapter configurations.

**Table 1175: Accounts Page Options**

| Option | Description |
|---|---|
| Add | Click **Add** to configure new accounts for integrations. Click **Yes** in the **Installation Required** wizard if the integration is not installed. For more information on integrations, see Integrating Data Sources with VCF Operations.<br><br>**Note:**<br>You can also activate the integration from the Repository page and then add an account from the Accounts tab. For more information on activating available integrations from the Repository tab, see The Repository Tab. |
| Horizontal Ellipses | As a VCF Operations admin, you can back up the adapter configurations before upgrading. Click the horizontal ellipses and then click **Export** to export all the adapter configurations. To import them into a different VCF Operations instance, click **Import**. For more information, see Exporting and Importing Accounts. |
| Collection Status | View the overview of the data that is being collected at the cluster level in VCF Operations. |
| Credentials | You can configure or reconfigure credentials that you use to activate an adapter instance in VCF Operations. For more information, see Configuring Credentials in Integrations. |

After configuring accounts, you can view the account details and can edit them in the **Accounts** page.

**Table 1176: Accounts Grid Options**

| | |
|---|---|
| Vertical Ellipses | To view specific account details, expand the account type. Click the **>** icon on the left of the account and then click the vertical ellipses to change the account configurations.<br><br>• **Edit**: Allows you to edit the integrated adapter instance.<br>• **Delete**: Removes the adapter instance and clears the objects associated with the instance from the system, including historical data and role assignments.<br><br>**Note:**<br><br>When you delete an account, you can choose to delete any related objects by selecting the checkbox, **Delete related objects**. If you do not wish to delete the related objects immediately, leave the check box unselected. The related objects are kept in the inventory for the duration of the retention period specified in the **Global Settings** page. If you recover an adapter instance before the end of the retention period, the related objects are unmarked and not deleted.<br>• **Delete All**: Available for adapter instances with multiple sub-adapter configurations (such as the vCenter adapter). It removes the adapter instance along with its child adapter instances.<br><br>**Note:**<br><br>When you delete an account, you can choose to delete any related objects by selecting the checkbox, **Delete related objects**. If you do not wish to delete the related objects immediately, leave the check box unselected. The related objects are kept in the inventory for the duration of the retention period specified in the **Global Settings** page. If you recover an adapter instance before the end of the retention period, the related objects are unmarked and not deleted.<br>• **Start/ Stop Collecting**: Starts or Stops the data collection process.<br>• **Start/ Stop Collecting All**: Available for adapter instances with multiple sub-adapter configurations (such as the vCenter adapter). It starts or stops the data collection process of the adapter instance along with its child adapter instances.<br>• **Go to details**: Open the object in the object browser. |
| Name | Name that the vendor or manufacturer gave to the solution. |

| | |
|---|---|
| Status | Indicates the status of the solution and whether the adapter is collecting any data.<br>• Not Started: The adapter is not ready for data collection.<br>• Not Applicable: Data monitoring is not supported for the license that is applied to the adapter.<br>• Collecting: The adapter is ready for data collection.<br>• Warning: There is an issue that needs attention. |
| Description | Indicates what the solution monitors or what data source its adapter connects to. |
| Managed by VCF Operations | This column applies only to vCenter accounts including the ones under VCF domains. It displays the IP of the VCF Operations managing the adapter instance or "current instance" if the adapter instance is managed by the current VCF Operations instance. |
| Collector | Shows which VCF Operations node or VCF Operations collector appliance is collecting the data for the adapter. |
| Version | This column applies only to vCenter and VMware Cloud Foundation accounts. It indicates the version of the adapter instance. |

**Adding Accounts**

You can add and configure accounts associated with integrations that are provided with or that you add to VCF Operations. After you have configured the account, VCF Operations can communicate with the target system. You can access the **Accounts** tab in the **Integrations** page at any time to modify your adapter configurations.

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Integrations**. In the Accounts tab, click **Add** and select the integration you want to manage. Configure cloud proxies and activate the integration before adding and configuring accounts. For more information, see Add a Cloud Proxy and The Repository Tab.

## Exporting and Importing Accounts

As a VCF Operations admin, you can back up the adapter configurations before upgrading, export all the adapter configurations, and import them into a different VCF Operations instance.

> **Note:**
>
> Users with "Export" permission can export and users with "Import" permission can import the adapter configurations. For more information, see Access Control: Roles Tab.

1. Export the adapter configuration.

   > **Note:**
   >
   > You cannot export default VMware Cloud Foundation and vCenter accounts that were configured in VCF Operations using the VCF Installer.

   a) In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Integrations**.
   b) In the Accounts tab, select the adapter configuration(s) that you want to export, click the horizontal ellipses, and select **Export**.
   c) Set up a new password to export data. The password should be at least 14 characters long and must include at least one numerical character, upper and lower case character, and special character.
   d) Click **Export**.

   The adapter configurations are exported in .zip format. A password is used to encrypt the data. Use the same password while importing this file.

2. Import the adapter configuration.

   > **Note:**
   > Before importing the content, ensure that you have exported the adapter configurations.

a) In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Integrations**.
b) Click the horizontal ellipses and select **Import**.
c) Click **Browse** to select the `.zip` file and enter the password that you had set while exporting the content.
d) If there is a conflict while importing the adapter configurations, you can either overwrite the existing adapter configurations or skip the import, which is the default option.
e) Click **Import** to import adapter configurations to the destination setup.

## Configuring Credentials in Integrations

Credentials are the user accounts that VCF Operations uses to activate one or more integrations and associated adapters, and to establish communication with the target data sources. The credentials are supplied when you configure each adapter. You can create new credentials and modify the credential settings outside the adapter configuration or during adapter configuration to accommodate changes to your environment.

If you are modifying credentials to accommodate changes based on your password policy, you must validate your connection so that the adapters configured with these credentials begin using the new user name and password to communicate between VCF Operations and the target system.

To edit credential settings, from the left menu, click **Administration** > **Integrations**. In the Accounts tab, click **Credentials** .

By default, the Credential Ownership Enforcement option is activated. When this option is activated, you can view, add, modify, or delete only the credentials that you created or that were assigned to you. You can view and use unassigned credentials only if you have the required permissions. You can also deactivate the **Credential Ownership Enforcement** option from **Global Settings** page to be able to modify credentials created and owned by others. For more information, see List of Global Settings.

### Where You Find Credentials

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Integrations**. In the **Accounts** tab, click **Credentials**.

**Table 1177: Credentials Options**

| Option | Description |
|---|---|
| Toolbar options | Manages the selected credential.<br>• **Add**: Adds new credentials for an adapter type that you can later apply when configuring an adapter. For more information, see Adding Credentials. |
| Edit/Delete credentials | • Click the **Vertical Ellipses** to perform any one of the following actions on existing credentials:<br>– **Edit**: Modifies the selected credentials, usually when the user name and password require a change. The change is applied to the current adapter credentials after you validate the connection and the data source continues to communicate with VCF Operations.<br>– **Delete**: Removes the selected credentials from VCF Operations. Commonly used to delete misconfigured credentials. You cannot delete a credential in active use. To remove it, first reconfigure or delete the adapter, then delete the credential. |

### Adding Credentials

To configure or reconfigure credentials that you use to activate an adapter instance, you must provide the collection configuration settings, for the target system. You can also modify the connection settings for an existing credential instance.

### Where You add Credentials

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Integrations**. In the **Accounts** tab, click **Credentials**, and then click **Add**. The **Manage Credentials** dialog box opens.

### Manage Credentials Options

The **Manage Credentials** dialog box is used to add new or modify existing adapter credentials. The dialog box varies depending on the type of adapter and whether you are adding or editing. The following options describe the basic options. Depending on the adapter instance, the options other than the basic ones vary.

**Table 1178: Manage Credential Add or Edit Options**

| Option | Description |
|---|---|
| Adapter Type | Adapter type for which you are configuring the credentials. |
| Credential Kind | Credentials associated with the adapter. Some adapters support multiple types of credentials. For example, one type can define a user name and password, and another can define a pass code and key phrase. The combination of adapter and credential type affects the additional configuration options. |
| Credential Name | Descriptive user-defined name that you provide to manage the credentials. Not the account user name. |
| User Name | User account credentials that are used in the adapter configuration to connect VCF Operations to the target system. |
| Password | Password for the provided credentials. |

## The Repository Tab

The repository tab lists all the installed and available integrations in VCF Operations. Integrations are also referenced as Management Packs and Solutions.

### Where You Find the Repository Tab

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Integrations**, and then click the **Repository** tab. The page displays installed and available integrations as clickable cards.

### What Can You Do in the Repository Tab

In the **Repository** tab, when you click an integration card, the details page is displayed.

The management pack details page consists of the following three tabs:

- Overview: Describes the management pack and its purpose.
- Metrics: Displays a list of metrics and properties for the various object types as defined by the adapter.

  **Note:**

  This list of metrics and properties can display additional metrics and properties after account creation since other sources can also publish them.

  The actual list of published metrics and properties depends from policy customization and monitored environment.
- Content: Displays the content types and the data defined by that management pack.

**Table 1179: Management Pack Details Page Options**

| Options | Descriptions |
|---|---|
| Name | Name of the management pack, the name of the vendor or manufacturer who created the management pack, and version number. |
| Activate | Installs the management pack. You can configure cloud management packs after activation from the **Repository** or **Accounts** tab of the **Integrations** page.<br>The activation starts only if all the cluster's nodes are accessible.<br>**Note:**<br>Some of the pre-installed management packs are activated by default. You can configure them from the **Management Pack Details** page or the **Accounts** tab of the **Integrations** page |
| Add Account | The add account button is available only after the management pack is activated. For more information on the accounts which are activated by default, see, Integrating Data Sources with VCF Operations. |
| **Horizontal Ellipses** > **Reset Default Content** | This option is only available for the vCenter management pack.<br>After you update your instance of VCF Operations and select the Reset Default Content option, the existing default alert definitions and symptom definitions are overwritten with the newer versions from the update. This includes your existing compliance alert definitions.<br>**Note:**<br>Clone any customized content before you proceed, because a reset overwrites user modifications. |
| **Horizontal Ellipses** > **Deactivate** | You can uninstall a particular integration with its associated data, metadata, and the out of the box content. Select **I understand the risk and agree** to uninstall an integration. |
| **Horizontal Ellipses** > **Upgrade** | Acter you activate a management pack, if an upgrade is available, you can upgrade it to its latest version. To do so, select the select the management pack, click the horizontal ellipsis, and then click **Upgrade**. |

## Adding a Management Pack

Management packs are delivered as PAK files that you upload, license, and install.

### How Management Packs Work

When you add a management pack, you configure adapters that manage the communication and integration between VCF Operations and other products, applications, and functionality.

### Where You Add Management Packs

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Integrations**. Click the **Repository** tab and then click **Add** to install other management packs.

### Add Solutions Wizard Options

The wizard includes three pages where you locate and upload a PAK file, accept the EULA and install, and review the installation.

Before you install the PAK file, or upgrade your VCF Operations instance, clone any customized content to preserve it. Customized content can include alert definitions, symptom definitions, recommendations, and views. While upgrading to the latest version, you can select the **Install the PAK file even if it is already installed** and the **Reset Default Content** options.

**Table 1180: Wizard Options**

| Option | Description |
|---|---|
| Page 1 | |
| Select a Management pack to install | Navigate to your copy of a management pack PAK file. |
| Download Management Pack | When you click **Get** in the **Repository** page, the management pack is automatically downloaded. You can view the Name, Description, and Version of the management pack that is installed. |
| Upload | To stage the management pack for installation, copy the PAK file to VCF Operations. |
| Install the PAK file even if it is already installed | If the PAK file was already uploaded, reload the PAK file using the current file, but leave user customization in place. Do not overwrite or update the solution alerts, symptoms, recommendations, and policies. |
| The PAK file is unsigned | A warning appears if the PAK file is not signed with a digital signature that VMware provides. The digital signature indicates the original developer or publisher and provides the authenticity of the management pack. If installing a PAK file from an untrusted source is a concern, confirm that the management pack is signed and certified with the management pack distributor before proceeding with the installation. |
| Page 2 | |
| I accept the terms of the agreement | Read and agree to the end-user license agreement.<br>**Note:**<br>Click **Next** to install the management pack. The installation starts only if the VCF Operations cluster is online and all the nodes in the cluster are accessible. |
| Page 3 | |
| Installation Details | Review the installation progress, including the VCF Operations nodes where the adapter was installed. |

## The Solutions Catalog Tab

You can add, manage, create, and download management packs in the Solutions Catalog tab in VCF Operations.

### About Solutions Catalog

You can use the Solutions Catalog to discover and deploy curated, trusted, and certified solutions in your VCF environment. From the Solutions Catalog tab, you can:

- View and download available management packs from VCF Solutions Catalog.
- Upload management packs .zip files in VCF Operations.
- Install, upgrade, or delete downloaded or uploaded management packs.
- Create custom management packs in VCF Operations.

If your VCF Operations instance is able to connect to VCF Solutions Catalog (requires outbound Internet connectivity from VCF Operations), you can view, download, and install available management packs in the Solutions Catalog tab. For more information, see Adding Management Packs from Solutions Catalog in Connected Mode.

If your VCF Operations instance cannot connect to VCF Solutions Catalog, you can manually upload management pack .zip files in the Solutions Catalog tab. For more information, see Adding Management Packs from Solutions Catalog in Disconnected Mode.

If you want to create a custom management pack in VCF Operations, click **Create Management Pack** on the Solutions Catalog tab to create one. Once done, you can install it in VCF Operations. For more information, see Management Pack Builder.

**Migrating Management Packs**

When you upgrade to VCF 9.1 from a prior version of VCF Operations, the upgrade migrates the installed management packs.

- Migration Status**:**
  - Only installed management packs are migrated.
  - Downloaded but not installed management packs are removed. They must be re-downloaded and installed after the upgrade.
- Post-Migration Limitation**:**
  - You cannot view the **Content** and **Resource & Support** tabs for migrated management packs.
- Disconnected Mode:
  - If VCF Operations is upgraded in disconnected mode, the icons for the migrated management packs will not be displayed. To resolve this, you must switch to connected mode and either manually sync the management pack inventories or wait for the scheduled sync which runs every 12 hours.

## Adding Management Packs from Solutions Catalog in Connected Mode

If your VCF Operations instance is in connected mode, you can view and download available management packs in the Solution Catalog tab in VCF Operations.

If your VCF Operations instance can successfully sync with VCF Solutions Catalog, all available management packs that are associated with VCF Operations are displayed in the **Available** section of the Solutions Catalog tab.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, select **Administration** > **Integrations**, and then click the **Solutions Catalog** tab.
   a) (Optional) Select a management pack and click **View Details** and click the **Overview**, **Resource & Support**, and **Tasks** tabs to view the management pack details.

      **Note:**

      All actions performed on a management pack are displayed in the **Tasks** tab. You can review execution details for any task, and access the corresponding error logs in the event of a failure.

   On the **Solutions Catalog** tab, the **Available** section is displayed by default. All the management packs that are compatible with the current version of VCF Operations are displayed.
2. Use the version selector to view the available versions and then select the version you want.

   **Note:**

   Only compatible versions of the management pack are displayed in the version selector.
3. Click **Download**, accept the **EULA Agreement**, and then click **Download** again.

The management pack is downloaded in VCF Operations. All downloaded management packs are listed in the **Downloaded** section of the Solutions Catalog tab.

After you download a management pack in VCF Operations, you need to install it before you can use it. For more information, see Installing and Uninstalling a Management Pack from Solutions Catalog.

## Adding Management Packs from Solutions Catalog in Disconnected Mode

If your VCF Operations instance is in disconnected mode, you can download the management pack from VCF Solutions Catalog and then upload it manually in the Solution Catalog tab in VCF Operations.

If your VCF Operations instance is unable to connect to VCF Solutions Catalog, no management pack will be displayed in the **Available** section of the **Solutions Catalog** tab. In such cases, you can manually download management packs and then upload them in VCF Operations. Optionally you can also upload a management pack that is not available in the Solutions Catalog.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, select **Administration** > **Integrations**, and then click the **Solutions Catalog** tab.
2. On the **Solutions Catalog** tab, click the VCF Solutions Catalog link to open the VCF Solutions Catalog webpage.
3. Manually download the desired management pack for VCF Operations. For more information, see VCF Solutions Catalog Overview.
   a) Log in to VCF Solutions Catalog webpage and then click **Download for VCF-OPS** to download the management pack .zip file.
   b) Enter the **Key**.

      The Key is the asset key that is used when you download the management pack PAK file from VCF Solutions Catalog.

      The management pack .zip file is downloaded.
4. Unzip the management pack .zip file, return to the **Solutions Catalog** tab, and then click **Upload**.
5. Browse and select the management pack .zip file and then enter the **Key**.

   You must enter the same key that you entered at the time of download.

The management pack is uploaded in VCF Operations. All uploaded management packs are listed in the **Downloaded** section of the Solutions Catalog tab.

After you upload a management pack in VCF Operations, you must install it. For more information, see Installing and Uninstalling a Management Pack from Solutions Catalog.

### Installing and Uninstalling a Management Pack from Solutions Catalog

After downloading or uploading a management pack in VCF Operations, you must install it.

- Ensure you have downloaded or uploaded the management pack you want to install. For more information, see Adding Management Packs from Solutions Catalog in Connected Mode and Adding Management Packs from Solutions Catalog in Disconnected Mode.

To use a management pack, you must install it.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, select **Administration** > **Integrations**, and then click the **Solutions Catalog** tab.
2. Click **Downloaded** to open the **Downloaded** section to see the list of management packs that have been downloaded from the solutions catalog.
3. Locate the management pack you want to install and click **View Details**.
   a) (Optional) Click the **Overview**, **Metrics**, **Content**, **Resource & Support**, and **Tasks** tabs to review the management pack details.

      **Note:**

      Management packs that were downloaded but not installed before an upgrade are not migrated. If a previously downloaded management pack does not appear on this page after upgrading to VCF 9.1, you must download it again.
4. 
5. Click **Install**, verify the **Component Confirmation**, and then click **Install** again.

   The management pack is installed.

6. Optional: Click **Sync Installed Solution** to sync the management pack inventories on demand. By default, the inventory in the VCF Operations **Repository** tab is synced with the VCF Solutions Catalog page every 12 hours.
7. Optional: Click **Check for Updates** to check for upgrades to previously downloaded management packs.

   If an update is available for any management pack, the **Upgrade** button will become active, allowing you to download the newer version. Upgrading a management pack will overwrite the old version's content with the new version's content.
8. Optional: You can uninstall or delete a management pack that is no longer required. Uninstalling or deleting a management pack, automatically deletes its associated accounts.
   - Uninstall: Removes an installed management pack from the environment and moves it back to the **Available** section of the Solutions Catalog tab, where it can be installed again. Locate the management pack you want to uninstall and click **Uninstall**. Confirm that you want to uninstall the management pack, and then click **Uninstall** again. The management pack is uninstalled.
   - Delete: Permanently removes a management pack from the Solutions Catalog. Before you can delete a management pack, you must first uninstall it. Locate the management pack you want to delete and click **Delete**. Confirm that you want to delete the management pack, and then click **Delete** again. The management pack is deleted.

All installed management packs are displayed on the **Administration** > **Integrations** > **Repository** tab. After installation, you must add accounts for each management pack. Each account contains adapters using which VCF Operations integrates with the management pack and monitors their function and health. For more details, see The Repository Tab.

If you delete a management pack from the respository page, only the management pack icon is removed from the solutions catalog page. To remove the management pack completely from the solutions catalog page, either manually sync the management pack inventories or wait for the scheduled sync which runs every 12 hours.

## Management Pack Builder

The Management Pack Builder enables the creation of custom management packs for use in VCF Operations. This is a no-code solution for bringing in data from an external API and either creating new resources or extending VCF Operations and third party objects with new data, relationships, and events.

### Backwards Compatibility

In VCF Operations, there is no separate installation required. Previously, you needed to deploy a separate OVA virtual appliance, which involved extra setup steps. Now, it is pre-integrated into the VCF Operations platform, reducing the initial installation time and eliminating compatibility issues.

The Management Pack Builder retains backward compatibility. Existing designs can be imported directly, preserving previous work while leveraging new efficiencies and improved performance. This ensures a smooth transition for teams who previously relied on the standalone Management Pack Builder.

### Where you find the Management Pack Builder

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Developer Center** > **Management Pack Builder**.

### What Can You Do in the Management Pack Builder page

You can use the **Management Pack Builder** page to design a new management pack, import or export a design, edit an existing design, or delete an existing design. After you design a management pack, you can install it.

**Table 1181: Accounts Page Options**

| Option | Description |
|---|---|
| Create New | To create a new management pack, you must perform the following steps:<br>• Connect: Connect to a data source to collect data.<br>• Transform: Use the data from the connected source to define the VCF Operations objects.<br>• Verify: Review and validate the management pack design.<br>• Install: Install the management pack.<br>To create a new management pack, click **Create New**, enter the **Name** and **Description**, and select the **Type**. You cannot change the management pack type after you create it.<br>**HTTP API**: Use the HTTP API type to create a management pack that collects data from an HTTP API to create objects, relationships, and events in VCF Operations. To design a management pack using HTTP, see Designing a Management Pack using HTTP.<br>**Prometheus**: Use the Prometheus type to create a management pack that collects data from Prometheus which includes objects and relationships. To design a management pack using Prometheus, see Designing a Management Pack using Prometheus. |
| Import Design | If you do not have any designs, you can import a JSON design file in VCF Operations. For more information, see Importing or Exporting a Design. |

After you create or import a design, you can view, export, edit, or delete it in the **Management Pack Builder** page.

**Table 1182: Designs Grid Options**

| Option | Description |
|---|---|
| Design | Displays the name of the design. |
| Type | Displays the design type: HTTP API or Prometheus. |
| Design Status | Indicates the status of the design.<br>• Valid: The design contains all components required to create a management pack. It has not been tested using the "Verify" step.<br>• Invalid: The design has an issue that needs to be resolved before it can be tested and installed.<br>• Verified: The design contains all components required to create a management pack and the "Verify" step has run successfully.<br>• Incomplete: The design does not contain all the components needed to create a management pack. |
| Install Status | Indicates installation status of the design.<br>• Installed<br>• Uninstalled<br>• Draft<br>• Out of date |
| Last Edited | Indicates when the design was last modified. |
| Edit | Click a design to edit it. |
| Export | Export a design as a JSON file. For more information, see Importing or Exporting a Design. |
| Delete | Delete the design. |

## Before you Start your Design

Before starting, you should make sure you have access to the reference environment including hostname, user information, and API documentation. It is important to think about what you want to include in it. The main items to understand are objects, requests, and relationships.

**Things to Consider for designing a Management Pack using HTTP**

- Objects: Objects are the core item that we will be adding or updating in VCF Operations . You should think of them as the nouns of your technology. An example is a network switch. Your nouns for a switch could be the switch, ports, CPUs, and fans. These would all become objects in your design. For every object, you will provide the metrics and properties by adding one or more requests. VCF Operations manages metrics by the objects and has limits based on number of objects and metrics. Please review the sizing guide for the appropriate sizing of VCF Operations when designing your objects.

  You can also extend existing objects in your VCF Operations environment such as VMs, Host Systems, or third-party resources. By extending those objects, you can add metrics or properties to the existing object allowing custom information to be used as if it was part of the original resource.

  **Note:**

  VCF Operations manages metrics by objects and imposes limits based on the number of objects.

- Requests: These are the API requests used to add metrics and properties. To understand what is available on these requests, reference the API documentation. In our switch example, the requests to create port objects could include an interface endpoint that lists all the ports and a stats endpoint that show the performance of the port.
- Relationships: Relationships associate objects to each other in a parent to child relationship. This creates a topology for your objects allowing VCF Operations to understand the connectivity between your objects. In our switch example, the switch would be the parent to the ports, CPUs, and fans. You can also add relationships to VCF Operations objects like hosts, VMs, or third-party resources like data arrays.
- Events: Events represent a message reported by your endpoint pertaining to a VCF Operations object with an associated severity. Events can be used to impact the health of an object using symptoms. These symptoms also can be used to create alerts.

**Things to Consider for designing a Management Pack using Prometheus**

- Objects: Objects are the core item that we will be adding or updating in VCF Operations. You should think of them as the nouns of your technology. In a Prometheus design, objects are associated with a specific Prometheus job. For every object type, you will select the Prometheus metrics you would like to include and the labels on those metrics which will be mapped to identifiers and properties on the object. VCF Operations manages metrics by the objects and has limits based on number of objects and metrics. Please review the sizing guide for the appropriate sizing of VCF Operations when designing your objects.

  You can also extend existing objects in your environment such as VMs, Host Systems, or third-party resources. By extending those objects, you can add metrics and properties to the existing object, allowing custom information to be used as if it was part of the original resource.

  **Note:**

  VCF Operations manages metrics by objects and imposes limits based on the number of objects.

- Relationships: Relationships associate objects to each other in a parent to child relationship. This creates a topology for your objects allowing VCF Operations to understand the connectivity between your objects. You can also add relationships to objects like hosts, VMs, or third-party resources.

**Creating A Regular Expression**

A regular expression is a sequence of characters that define a search pattern, mainly for use in string matching. Management Pack Builder supports Java-style regular expressions. When using a **Regular Expression** in Management Pack Builder you can enter in the expression in the **Regex** field. When the test button is pressed the **Sample Value** will be run through the expression and the result will be displayed in the **Sample Output**.

Regular expressions can be used in the following contexts:

- When defining fields that should match between objects when creating a relationship.
- When defining fields that should match between an event and the object it could be assigned to when creating an event.

**Variables**
Custom authentication allows the user to add custom variables to be used in the advanced section of the source creation. This is generally used with permanent session tokens, or any variable sent with every request.

## Designing a Management Pack using HTTP

VCF Operations offers a wizard that makes it easy for you to design a management pack based on an HTTP API.

Building a new HTTP API-based management pack consists of the following steps. Once you start your design, you can save your work and pick up from where you left off. You can edit an existing design at any step.

- Define a Data Source: In this initial step, you define the data source from which the management pack will collect information. This involves specifying the external API or system that the management pack will integrate with, setting up the necessary connection parameters, and ensuring that the data source is accessible for data retrieval.
- Set up API Requests: Here, you configure the specific API requests or queries that will be used to fetch data from the defined source. This includes detailing the endpoints, request methods, parameters, and any necessary authentication details to ensure successful data extraction.
- Define Objects: In this step, you define the objects that represent the data entities retrieved from the source. This involves mapping the data to meaningful resource types within VCF Operations, specifying their properties, and determining how they will be represented in the monitoring environment.
- Specify Relationships (Optional): This step focuses on establishing relationships between the defined objects. By specifying how these objects interrelate, you can create a structured representation of the data, which aids in understanding dependencies and interactions within the monitored environment.
- Define Events (Optional): Here, you create an event from an API by defining the message, severity, and assoicated objects. You can configure symptoms and alerts for significant events in VCF Operations after designing the management pack.
- Verify your Design: In this step, you validate the configurations made in the previous steps. This involves testing the data collection, ensuring that objects and relationships are correctly defined, and confirming that events trigger as expected. It’s a crucial step to ensure the management pack functions as intended before deployment.
- Install: This is the final step. After you have reviewed and verified the design, you can install the management pack.

**Define a Source**
Start by identifying the source system or API from which you intend to pull data. This step includes specifying host name, connection settings, and authentication protocols.

**Sub-sections in Define Source**

- Data Source Connections
- Get Session (visible if you turn on session authentication)
- Global Settings
- Release Session (visible if you turn on session authentication)
- Test Connection

**Data Source Connections**

- Collector: The collector node or cloud proxy from which the Management Pack Builder adapter instance will make API requests during the design process.
- Host Name:The target you're pulling data from.
- Port:The port to use.
- TLS Configuration: Select from one of the following options:
  - Verify: Use TLS and verify the target's certificate
  - No Verify: Use TLS but don't verify the target's certificate
  - No TLS: Do not use TLS at all
- Base Path: Path to be added to all API calls defined in this design. You can leave it blank and use the full path for all requests if there is no shared prefix.

- Preview API Path: A preview of your full path.

**Authentication Details**

Configure the authentication details for your API requests.

Choose between basic authentication, custom authentication, and no authentication. The configuration here is used for every subsequent call, so you only need to configure authentication once. If you need to create a session token, before making connection requests, turn on **Use Session Authentication**.

The Management Pack Builder can use three different authentication source types:

- Basic: Specify a username and password. The header is automatically created.
  - Username: The user on reference system that is authorized to query the intended API endpoint.
  - Password: User's password for authentication.
- Custom**:** Add authentication fields and headers.
  - Use custom authentication when you need to add custom variables to be used in the advanced section of the source creation.
  - This is generally used with permanent session tokens or any variable sent with every request if session authentication is not used.
  - It is generally used for login credentials that will be used in POST a post body to get the session token when using session authenication.
- No Authentication**:** Use for public and non-secured APIs.

You can next choose to turn on **Session Authentication**. Session authentication is a method in which a session is created on login and stores information from the response body and/or headers as defined by the user. Common fields include a token or a session ID. When running a collection, the session fields are retrieved once, then used on all subsequent requests. If a release session request is defined, the session is released at the end of the collection.

Once you are done, click **NEXT**.

**Get Session**

Define the **HTTP Method** and **API Path** when using session authentication.

- HTTP Method: Based on the API requirements this will be either a GET, POST, PUT, PATCH.
- API Path: This is the path to the session creation resource as defined by the API documentation.

Next, set up additional request settings, such as headers, query parameters, and body

- HTTP Headers: This is used to assign headers to the request that is used to retrieve the session. This only needs to be used if the API requires custom values in the header. If using basic credentials, the Authorization header is automatically added and thus does not need further configuration.
- Query Parameters: This section allows the user to add custom query prompts to the end of the URL request. This is optional and only necessary if the API documentation calls for it.
- Body: This allows you to create a body to be sent to the session creation request. Some APIs require the body to contain information like user and password. This is optional and only needs to be used if the API documentation indicates it.

Click **Test** to see the test response. VCF Operations displays the **Body**, **Headers** and **Log**.

Once you are done, click **NEXT**.

**Global Settings**

Define the headers which are used to pass data with every call in the request header. Session data should be put here so that every call includes the session ID or token information.

Once you are done, click **NEXT**.

### Release Session

Defining a **Release Session Request** is recommended when using session authentication. This request will log out the session, reduce session leaks, and improve security.

- HTTP Method: Based on the API requirements this will be either a GET, POST, PUT, PATCH, DELETE.
- API Path: This is the path to the API endpoint you want to use to release the session.

Once you are done, click **NEXT**.

### Test Session

In this step, you to define the **HTTP Method** being used (GET, POST, PUT, PATCH), the API path, and see a preview of the fully qualified path. Then, you click **TEST CONNECTION**.

- HTTP Method: Based on the API requirements this will be either a GET, POST, PUT, PATCH.
- API Path: This is the path to test that your defined authentication works for your API.

The test connection confirms that all the source data is set up correctly. If using basic authentication, this will send one request including the basic authentication header. If using the session, this will make up to three requests to create the session, make a call to the test endpoint, then make a call to delete the session.

The response section will show the response to the test connection endpoint. You can see the body and headers of the response as well as the logs generated by the request. You can download the test response as a JSON file.

Once you are done, click **SAVE SOURCE**.

**Add Requests**
This is the second step of the wizard. You can add all the API calls used by the design to provide data to the objects and events. You will need to reference your API documentation to determine the endpoint and settings necessary for this section.

### Add API Request and API Pagination

In the **Add API Request** section, define the endpoint to be used for the request. Note: the source hostname, port, and base path will be prepended to this path for the request URL.

- HTTP Method: This is the method used for the request. It will be either a GET, PUT, PATCH, or POST.
- API Path: Endpoint that needs to be call to retrieve the data.
- Name: Name for this request used in future settings in the design.

Advanced (optional)

In this section you can add specific data to the requests body, headers, and query parameters, as defined by the API documentation.

Once you are done, click **SAVE REQUEST**.

You can also add paging to your API call in the **Add API Pagination** section. You must make the request prior to configuring paging. Management Pack Builder supports two main methods of pagination: offset and pages. Reference your API documentation to determine whether paging is necessary and, if so, which method to use.

- Paging Parameter: This is the parameter name to use to indicate which page we are requesting.
- Starting Offset: This is the number that represents the beginning of the API's results. Often, "Pages" start at 1 and "Offset" starts at 0. Confirm the starting number in the API documentation.
- Limit Parameter: This is the parameter name for the number of records per page.
- Limit Value: The number of records returned per page.
- Data List Returned from Paged Request: List of data being returned. When the count of this is less than the limit value, it means that the end of the data has been reached and no more paged requests will be made.

### Chain From Other API Calls

To add a chained request, return to the wizard, go to the **Requests** step, and select the parent request from requests list, then select **Add Chained**.

- In the **Add Chaining** section, select a list from the parent request's response which contains the attribute(s) to be used in the child request.
- System Attributes: For each attribute that you would like to use in the child request, select **Add Field**. Select the **Attribute**. If desired, update the **Parameter**. The **Parameter** will be used to define the usage key, so make sure that it is meaningful in the context of the design. If the parent request had a value for the selected attribute, it will populate by default. Regardless, you can manually set the value here. This is the value that will be substituted for the usage key later when testing the child request, so make sure that it is a valid value which will cause the child request to succeed.
- In the **Add API Request** section, define the child request like any other request, but now you can use the attribute(s) defined in the previous step. You can insert the usage key in the path, headers, query parameters, and/or body of the child request. To insert the attribute usage key, click the clipboard icon on the right side of the field, then select the attribute you want to insert. It will be inserted where the cursor is currently located in the field. When the request is run via **MAKE REQUEST**, all usage keys will be replaced with the sample values defined in the previous step.

Once you are done, click **SAVE REQUEST**.

**Add Objects**

In the third step, you define the objects that will be part of your custom management pack. You can define new objects or extend existing objects from VCF Operations with new metrics and properties.

**Sub-Sections in Add New and Add Existing Objects**

- Object Type
- Attributes from API
- Properties and Metrics
- Name and Identifiers (Section appears only when adding new objects.)
- Object Bindings (Section appears when there is an existing object or when the attribue selections require it)

**Object Type**

- Add New Object Type: The object type is the user-facing label that will be displayed for this type of object in VCF Operations when this management pack is installed. Example object types are Switch, Port, etc. You can also choose the icon that will be used in VCF Operations. Once you are done, click **NEXT**.
- Add Existing Object: To extend an existing object, choose a resource type to extend from the list of resource types on your VCF Operations instance. You can filter by the resource type or page through the list of available options.

**Attributes from API**

From the list of **Requests**, you select one or more requests to pull into this object. You will then select all the data entries you want to add to your object from the list. You can search the list of attributes for a specific attribute or scroll through the list.

Once you are done, click **NEXT**.

**Properties and Metrics**

Here you will see the metric and property definitions you have attached to the object.

- Label: The name of the metric or property visible in VCF Operations.
- Datatype: Defines how the data is rendered. If you select decimal or one of the counter options and the data cannot be converted to a number, the data will be discarded.
  - String: Returns the value of the attribute as a string. This datatype is allowed only on properties, and not metrics.
  - Decimal: Returns the value of the attribute as a number.
  - Counter Delta: Calculates the change in the value between collections.
  - Counter Rate: Calculates the change in value between collections over time.
- Unit: If applicable, this is where you can define the unit for the data point. This will allow auto-scaling in dashboards and reports.
- Property: This drop-down list indicates whether the data is a property.

- KPI: This toggle indicates whether the data will be treated as a KPI. KPIs are key metrics that are used in dashboards and resource detail pages. They should be metrics that are important to the status, performance, and health of the object.
- Expanded row view:
  - Request: Shows the call and data path from which this metric is pulled.
  - List: List in the request response from which the data is pulled.
  - Attribute: Name of the attribute in the response being pulled.
  - Example: Shows the value of the first returned data point from the request.

  Once you are done, click **NEXT**.

**Name and Identifiers**

Define the identity of the object in this step.

- Object Instance Name: This is the property from the list above that is used as the object's name in VCF Operations.
- Object Identifiers: Define the properties using the smallest set of properties that will accurately and uniquely identify each object.

Once you are done, click **SAVE OBJECT**. If you see Attribute Bindings, proceed to the next step.

**Attribute Bindings**

When combining data from multiple JSON or XML arrays into a single new object, it's necessary to establish a base array and select a corresponding attribute from each of the other arrays for mapping. You will only see this section if it is required.

When extending an existing object, all arrays must link to a property or identifier on the existing object in VCF Operations.

- API Attribute: The data from the selected JSON or XML list in the API response that you match to an existing property or identifier.
- Object Property: A matching property or identifier on the object. For a new object, it is selected from the base list. For an extended object, it is selected from the identifiers and properties of the existing object.

### Best Practices For Metrics and Properties

Follow these best practices when you create objects from new metrics and properties.

**Best Practices**

- Objects created from new metrics and properties should be represent stable objects, not transient objects.
- Metrics should represent measurable performance characteristics. Example: CPU Utilization %, Network Throughput, etc. Properties should represent relatively stable configuration values. Example: version, power status, etc.
- Properties should have a small number of possible values.
- Property values should not change frequently from collection to collection.
- Opt for defining metrics using reasonable unit sizes and prefix multipliers to avoid publishing excessively large values.
- The depth of the relationships of the objects should not be too deep (for example 100).
- Avoid defining artificial relationships that do not exist in the end-point.

### Define Object Relationships

Relationships are a parent to child connection that is used in VCF Operations to define the interaction between interconnected resources. Parent to child relationships are the only supported type of relationships with the Management Pack Builder.

**Parent and Child object**

Define the objects used to create the relationships. Parent objects are higher in the topology.

**Matching Properties**

Relationships in Management Pack Builder are created by matching all resources with matching properties. For example, a Switch has a **switch ID** property and the Port has a **parent switch ID** property. The parent object properties and the properties of any relevant child object(s) will always match, e.g. **switch ID** with **parent switch ID**. With multiple matching sets, all sets of properties must match for the relationship to be created. This can be either case-sensitive or insensitive and use basic or advanced matching.

Once you are done, click **SAVE RELATIONSHIP**.

**Add New Events**

Map events from the target system into the VCF Operations event model. This includes defining event triggers and mapping severity levels, and linking them to VCF Operations objects.

**Sub-sections in this step**

- Add Events from API
- Severity
- Related Objects

**Add Events from API**

- Event Name: The user-friendly name of the event. This will be prefixed on the event message in VCF Operations.
- API Request: This is the request endpoint setup in the **Requests** section to use to create events.
- Event List: The location of events in the API response.
- Event Message: This is the event message that will show in VCF Operations. Often in APIs this would be the event description. You can create the message using one or more fields from the response and type any static information as well. You can also edit the value of the fields using regular expressions. See the section on regular expressions for further help on this.
  Once you are done, click **NEXT**.

**Severity**

Events in VCF Operations are ranked with a severity from Info to Immediate. Please reference your API documentation for a list of valid severities they support.

- Event Severity: Define API response fields and static text which form the event severity definition. You can also adjust the value of the fields using regular expression.
- Map Severities: Add mappings from the VCF event severity (Event Severity) to the corresponding value expected from the API (API Response Severity). For example, if the API returns "critical," you would map to the VCF "Critical" severity. Sometimes an API will have a severity that does not directly correspond to a VCF severity. In that case, the mapping is up to your discretion. For example, if any API returns an event severity of Severe, you must map that to the VCF severity "Critical," or "Immediate."
- Set all unmatched API response Severities to: Specify a default severity. If an event severity is returned from the API that does not match one of the mappings you created, the associated event will be set to the default severity specified here.

Once you are done, click **NEXT**.

**Related Objects**

Relate events to objects by matching. Select "All" if you want to related events to more than one object in the case that multiple related objects match. Select "First" if you only want the first related object to match.

The order defined in this section is the order in which the management pack will evaluate matches.

Add related object: For each related object, first select the object type (**Select object**) to relate to. By default, you can choose any object types included in the current design. If you select "All objects", you can also choose from existing object types in VCF Operations. Next, create a mapping from the object properties to the API response. You can use any combination of object properties, API response fields, static text, and regular expressions to make the mapping work. In

the top-right, you can choose whether the matching logic should be case-sensitive. If you define multiple related objects and selected "Relate events to objects by matching: First", you can use the arrows on the left to change the order in which the mappings will be evaluated.

Once you are done, click **SAVE EVENT**.

#### Verify your Design

Before final deployment, you can test your design to ensure that the management pack collects the correct data and accurately represents the target system within VCF Operations.

#### Perform Collection

Click **PERFORM COLLECTION** to verify your design.

VCF Operations runs the collection and displays the following information:

- Summary: Displays the **Collection Status**, **Collection Time**, **Objects Collected**, **Metrics Collected**, **Properties Collected**, **Relationships Found**, and **Events Collected**.
- Log: Displays the last 1000 lines. You can download the log to view all the lines. You can change the logging level to see more detailed logs.
- Objects: Displays the details for the objects.
- Events: Displays the events for the objects.

Once you are done, go to the final step, **INSTALL**.

#### Install

Once testing is complete, click **REVIEW CONFIGURATION** or directly click **INSTALL**. VCF Operations installs the management pack, making it available for live monitoring.

### Designing a Management Pack using Prometheus

VCF Operations offers a wizard that makes it easy for you to design a management pack from Prometheus data.

Building a Prometheus-based management pack consists of the following steps. Once you start your design, you can save your work and pick up from where you left off. You can edit an existing design at any step.

- Define a Data Source: In this initial step, you define the Prometheus data source from which the management pack will collect information. You must specify the Prometheus instance that the management pack will integrate with, setting up the authentication, and ensuring that Prometheus is accessible for data retrieval. For more information, see Define a Source.
- Define Objects: In this step, you map metrics that are retrieved from Prometheus to meaningful objects and metrics within VCF Operations. For more information, see Add Objects.
- Specify Relationships: This step focuses on establishing relationships between the defined objects. By specifying how these objects interrelate, you can create a structured representation of the data, which aids in understanding dependencies and interactions within the monitored environment. For more information, see Define Object Relationships.
- Verify your Design: In this step, you validate the configurations that were made in the previous steps. It also involves testing the data collection and ensuring that objects and relationships are correctly defined. It’s a crucial step to ensure the management pack functions as intended before deployment. For more information, see Define Object Relationships.
- Install: As the final step, after you have reviewed and verified the design, you can install the management pack. For more information, see Install.

#### Define a Source

Start by identifying the Prometheus instance from which you intend to pull data. This step includes specifying the IP address or FQDN, API port, TLS configuration, and authentication settings.

**Data Source Connections**

- Collector: The collector node or cloud proxy from which the Management Pack Builder adapter instance will make requests to Prometheus during the design process.
- Prometheus Host: the IP address, hostname, or FQDN for the Prometheus instance that will be used during the design process.
- Port: The port hosting the Prometheus APIs. The default is 9090.
- TLS Configuration: Select from one of the following options:
  - Verify: Use TLS and verify the target's certificate
  - No Verify: Use TLS but do not verify the target's certificate
  - No TLS: Do not use TLS at all
- Authentication Type: Select from one of the following options:
  - Basic: Specify a username and password
  - No Authentication: No credentials are required

Once the form is complete, click **Test Connection**. The test connection confirms that the Prometheus instance is accessible with the provided configuration information and the API is returning the expected data.

The **Test Response** section shows you the list of jobs available on the Prometheus instance. In Prometheus, each type of monitored technology typically corresponds to one job. For example, all Clickhouse-related metrics are defined with the “clickhouse” job. By default, Prometheus defines its own self-monitoring metrics under the “prometheus” job.

The **Test Response** section also shows a **Log** tab, where you can review the test connection logs. If you are seeing a failed test connection, review the logs to determine the root cause of the failure.

Once you are done, click **Save Source**.

### Add Objects

Next, select Prometheus metrics that you want to convert into VCF Operations metrics and objects. Prometheus metrics have labels which can be used to uniquely identify objects. Prometheus metrics can be filtered and processed during their conversion to VCF Operations metrics.

**Add New Object Type**

1. Object Type:
   - Object Type: The type which will be assigned and visible in VCF Operations.
   - Prometheus Job: In Prometheus, each type of monitored technology typically corresponds to one job. Select which type of technology will be associated with this object type.
   - Object Icon: Select an icon to represent this object type. By default, all icons are shown. Select a category to filter the icons shown. After a management pack is installed, custom icons can optionally be uploaded in **Administration** > **Control Panel** > **Icons**.
2. Prometheus Data: Prometheus metrics related to the job selected previously are shown. Select the metrics that should comprise the new VCF Operations object type. Use the search bar to filter on metric name and label.

   **Note:**

   When defining object identifiers, you will only be able to select labels that are shared across all the selected metrics. All Prometheus metrics have the “instance” label by default.
3. Identifiers and Name: Select one or more labels to uniquely identify the objects in VCF Operations. You can optionally update the text that will be shown in the VCF Operations UI.

   After the identifying labels are defined, select one of them to be used as the object name.
4. Metrics Mapping: Each Prometheus metric selected previously is shown. A Prometheus metric can be mapped to one or more VCF Operations metrics on the new object type.
   - Metric Label: Optionally update the metric label to be shown in VCF Operations UI.
   - Function: The selection for function varies depending on the type of Prometheus metric:
     - Gauge metrics always use the **Current Total** function. This means the VCF Operations metric will report the sum of the latest values returned.
     - Counter and summary metrics can use a **Delta** or **Rate** function.

        - Delta: The VCF Operations metric will report the difference between the current value and previous value of the Prometheus metric.
        - Rate: The VCF Operations metric will report the difference between the current value and previous value of the Prometheus metric, divided by the time elapsed between collections.
    - Units: Specify the unit for this metric.
    - Filters: Optionally define one or more filters to include only a subset of the Prometheus metric's time series in the VCF Operations metric calculation. For example, you may want to split an HTTP Response metric into multiple VCF Operations metrics by filtering on response codes (2xx -> Successes, 4xx -> Failures, etc.).
        - =: Include time series where the specified label matches the provided text exactly.
        - !=: Include time series where the specified label does not match the provided text exactly.
        - =~: Include time series where the specified label matches the provided regex. Supports the same RE2 regex syntax that Prometheus supports.
        - !~: Include time series where the specified label does not match the provided regex. Supports the same RE2 regex syntax that Prometheus supports.
5. KPI: Indicates whether the data will be treated as a key performance indicator. KPIs are key metrics that are used in dashboards and resource detail pages. They should be metrics that are important to the status, performance, and health of the object.

**Extend Existing Object Type**

To extend an object type that is defined by another management pack already installed in VCF Operations with Prometheus data, select **Extend Existing**.

1. Object Type:
    - Select Existing Object Type: Select the type of object that will be extended with additional metrics. Use the search bar in the top-right to filter on **Object Type** and **Adapter**.
    - Prometheus Job: In Prometheus, each type of monitored technology typically corresponds to one job. Select which type of technology will be associated with this object type.
2. Prometheus Data: Prometheus metrics related to the job selected previously are shown. Select the metrics that should be added to the existing VCF Operations object type. Use the search bar to filter on metric name and label.

    **Note:**

    When defining object identifiers, you will only be able to select labels that are shared across all the selected metrics. All Prometheus metrics have the "instance" label by default.
3. Binding Labels: Select one or more binding labels from the selected Prometheus metrics to link to the existing objects in VCF Operations.
4. Object Binding: Use a combination of one or more binding labels and static text to define a matching pattern between the Prometheus metric labels and the existing object type's properties and identifiers. You may optionally apply a regex to any label, property, or identifier to modify the text that will be used for matching.
5. Metrics Mapping: Each Prometheus metric selected previously is shown. A Prometheus metric can be mapped to one or more VCF Operations metrics on the new object type.
    - Metric Label: Optionally update the metric label to be shown in VCF Operations UI.
    - Function: The selection for function varies depending on the type of Prometheus metric:
        - Gauge metrics always use the **Current Total** function. This means the VCF Operations metric will report the sum of the latest values returned.
        - Counter and summary metrics can use a **Delta** or **Rate** function.
            - Delta: The VCF Operations metric will report the difference between the current value and previous value of the Prometheus metric.
            - Rate: The VCF Operations metric will report the difference between the current value and previous value of the Prometheus metric, divided by the time elapsed between collections.
    - Units: Specify the unit for this metric.
    - Filters: Optionally define one or more filters to include only a subset of the Prometheus metric's time series in the VCF Operations metric calculation. For example, you may want to split an HTTP Response metric into multiple VCF Operations metrics by filtering on response codes (2xx -> Successes, 4xx -> Failures, etc.).
        - =: Include time series where the specified label matches the provided text exactly.

- `!=`: Include time series where the specified label does not match the provided text exactly.
- `=~`: Include time series where the specified label matches the provided regex. Supports the same RE2 regex syntax that Prometheus supports.
- `!~`: Include time series where the specified label does not match the provided regex. Supports the same RE2 regex syntax that Prometheus supports.

6. KPI: Indicates whether the data will be treated as a key performance indicator. KPIs are key metrics that are used in dashboards and resource detail pages. They should be metrics that are important to the status, performance, and health of the object.

#### What To Do Next

- Define Object Relationships: For more information, see Define Object Relationships.
- Verify your Design: For more information, see Verify your Design.
- Install: For more information, see Install.

## Importing or Exporting a Design

A JSON file is created when exporting a design. A JSON file can be imported into VCF Operations. The file is verified as a valid design file and the author and description is shown. Only valid designs can be exported.

#### Importing a Design

To import a design, click the **Import** button in the Management Pack Builder page. Browse to the location of the JSON file in your system. Only one design can be imported at a time.

VCF Operations verifies the validity and displays the name and description of the design. Ensure that another design of the same name does not exist. The design that you are importing must have a unique name.

After you import the design, it is displayed in the Management Pack Builder page. Verify the information in the **Design Status** column. You must provide the reference environment to test the design against. Click the imported design, and in the **Source** step of the wizard, specify the required fields.

#### Exporting a Design

Before you export a design, ensure that it does not contain private information, such as IP addresses, credentials, or IDs, outside of end-user configurable fields. Before sharing or storing your exported design in an unsecured location, review the export file to ensure it does not include any private information.

A zip file containing the JSON file is exported. You must extract the JSON file from the zip file before you can import it.

## Troubleshooting a Design

1. Request Log: Every time you make a test request or perform a test collection in Management Pack Builder, there are tabs which show the body, headers, and logs. In the case of an error response code, review the logs and headers for information regarding the cause of the error.
2. Third-party API documentation: ensure that the request you defined complies with the requirements described in the third-party API documentation.
3. Management Pack Builder Adapter logs: located under `$VCOPS_BASE/user/log/adapters/MPBAdapter` on the node or cloud proxy selected under **Source**. Review these logs for any additional errors or warnings.
4. If you suspect an error with the Management Pack Builder tool itself, you can open a support case, or review the server logs at `/usr/lib/vmware-vcops/user/log/integration-designer-controller.log`. You can also review the Suite API logs at `/usr/lib/vmware-vcops/user/log/suite-api for additional errors.`

   **Note:**

   VMware Support is limited to the functionality and performance of the Management Pack Builder application. Support does not include troubleshooting or designing customer-specific management packs or custom API

integrations. For questions about specific management pack designs, please post in the Management Pack Builder Community Page.

## Monitoring Multiple Cloud Accounts in VCF Operations

You can now monitor all your VMware cloud operations from the Overview page in VCF Operations. This overview page provides a comprehensive summary of your VMware cloud accounts which includes their geo-location, inventory, appliance health and management, and workload operations. You can also view the alerts and the cost for your entire VMware Cloud Infrastructure or individual VMware clouds.

### Where You Find the VCF Overview Tab

The overview page provides a consolidated view of all VMware cloud accounts on the **All VMware Clouds** page. The different tiles provide insights into the location, health, cost, and inventory of your cloud accounts. When you first login, only vCenter and VCF cards are visible.

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Home**. By default, the **VCF Overview** tab opens.

> **Note:**
>
> No data displays for cloud accounts that do not have a configured account. To view data for the other VMware clouds or public clouds, you must configure cloud accounts from the **Administration** > **Integrations** page. For more information, see The Accounts Tab.

### About the VCF Overview Tab

**Monitoring Accounts** tile: View the total number of configured accounts under each account type (VCF, vCenter, Domains) that are collecting data.

**Inventory** tile: View the summarized resources count of the collecting accounts, for example, Datacenter, Cluster, Host, VM, Datastore, vSphere Distributed Switch, vSphere Distributed Port Group) for the whole cloud infrastructure.

**Findings** tile: View the diagnostic findings of your VMware Clouds to track the health of your cloud accounts and manage their health.

**Certificates** tile: Certificates information is available only for VCF license entitled vCenter instances. For more information, see Licensing Overview.

**Overall Capacity| Capacity Remaining| Time Remaining**: View the overall capacity, capacity remaining, and the time remaining before the capacity runs out. If no data displays as part of the capacity and time remaining sections, check your account configuration, and make sure they are collecting data.

**Cost**: View the total cost of ownership, potential savings, and realized savings. The amount is calculated from the first of the month to the current date. This value resets at the beginning of every month.

**Alerts** tile: View the total number of alerts need attention. Alerts of all cloud accounts are consolidated and represented graphically dating back a week from the current date dynamically.

**Green Score** tile: View the status of your VMware Cloud that includes its green score, power consumption, carbon footprint, and overall environmental impact. To view the Green Score information in more detail, click **View Green Score**. For more information, see Configuring Green Score.

**Compliance** tile: View the compliance scores and misconfigurations of your cloud accounts.

### VMware Cloud - Example: vCenter Account

The values displayed in the individual VMware cloud accounts like vCenter are based on their total number of accounts.

> **Note:**
>
> To view the geographical location of the accounts configured for vCenter and VCF, you must assign them to a physical data center. For more information, see Adding Physical Data Centers in VCF Operations.

**Monitoring Accounts** tile: View the total number of vCenter instances that are collecting data.

**Inventory** tile: View the object types of the data collecting vCenter instances, for example, Datacenter, Cluster, Host, VM, Datastore, vSphere Distributed Switch, vSphere Distributed Port Group. Unspecified object types are not reflected in the tile.

**Top Objects Growth Trend** tile: Displays the growth trend of the top objects dating back a month from the current date dynamically.

# VCF Account in VCF Operations

In VCF a VCF domain is a policy-based resource construct with specific availability and performance attributes. It combines compute (vSphere), storage (vSAN), networking (NSX), and VCF Operations fleet management to form a single consumable entity that creates logical resource pools across compute, storage, and networking.

There are two types of VCF domains - the management domain and the workload domain. For more information, see Overview.

Configure the VCF account in VCF Operations to monitor these constructs of VCF.

## Configuring a Standalone VCF Account in VCF Operations

You must configure a VCF account to monitor a VCF Instance.

- Configure a physical data center in VCF Operations. For more information, see Adding Physical Data Centers in VCF Operations

After configuration, all VCF domains within the instance are automatically discovered and configured along with their connection details and credentials of underlying adapters for vCenter, vSAN, and NSX.

> **Note:**
>
> All instances in the VCF Fleet are automatically added to VCF Operations and do not require manual creation of VCF accounts.

1. From the left menu, click **Administration** > **Integrations**.
2. On the **Accounts** tab, click **Add**.
3. On the **Accounts Types** page, click **VMware Cloud Foundation**.
4. Enter a Name and Description for the account.
5. Optional: Select the **Physical Data Center** you want to associate with the VCF account.
6. Enter the FQDN for the **SDDC Manager** you are trying to connect.
7. To add credentials for the VCF Instance, click the **Add** icon, and enter the required credentials.

   The credentials must be associated with the admin role in VCF.

   > **Note:**
   > If you are using an AD or domain integrated account for authenticating your VCF Instance, ensure you enter the username as username@domain or domain\username.
8. Determine which VCF Operations cloud proxy/ group is used to manage the account. From the **Cloud Proxy/ Group** drop-down, select the cloud proxy that you deployed as the collector for this VCF account.

   If you have multiple cloud proxies/ groups in your environment, you can distribute the workload to optimize performance. Select the cloud proxy/ group to manage the adapter processes for this instance. The **SDDC Manager** FQDN must be reachable from selected cloud proxy/ group.
9. Click **Validate Connection** to validate the connection.

   The Review and Accept Certificate wizard appears. Click **Accept** to continue the validation.

10. Optional: In the **Log Operations** section, activate or deactivate the collection of SDDC Manager logs. After you enable this setting, you can use the Log Management features and capabilities to analyze SDDC Manager logs.

    **Note:**

    This is supported for SDDC Manager 9.1.1 and higher only.

    a) This setting is activated by default. Activating this setting overrides any log configuration that is done on the SDDC Manager.
    b) By default, logs are collected directly into the Logs cluster. To collect logs into a unified cloud proxy instead, you must set one up. Note that the default unified cloud proxy cannot be used for this collection type. An advanced unified cloud proxy must be configured instead.
    c) To add filters and other settings, see Managing Collection from Log Sources.

11. Optional: Click **Advanced Settings**.

    - **Enable Domain Monitoring on Creation**: Change this setting to **true** to enable data collection and monitoring of newly discovered and configured domains.

      **Note:**

      By default, this setting is set to **false**. If you configure a VCF account with the default setting, data collection and monitoring of domains is disabled. If you edit the VCF account and change this setting to **true**, data collection and monitoring is enabled but it is done only for the newly discovered domains.
    - **VCF Configuration Limit File Name**: Enter the name of the file that contains the VCF configuration max soft and hard limits and their configured values in VCF Operations.

12. Click **Add** and select an add option for this VCF instance.

    **Note:**

    License and vSphere Client plug-in management is supported only for VCF accounts of versions 9.0 or later. If your VCF account is 9.0 or later and is not managed by any VCF Operations instance, the current VCF Operations instance takes up the license and vSphere Client plug-in management. If you do not have the required privileges for license and vSphere Client plug-in management, VCF account will be added only for monitoring purposes. For more information, see Use and Configure Your vSphere Client.

    After you integrate a VCF account with a VCF Operations instance for license management, it takes up to 15 minutes before the VCF account appears in the **License** tab of VCF Operations.

    If your VCF account is a version before 9.0, you can only activate vSphere Client plug-in management. If your VCF account is a version before 9.0, and is not managed by any VCF Operations instance, the current VCF Operations instance takes up the vSphere Client plug-in management. If you do not have the required privileges for vSphere Client plug-in management, VCF account will be added only for monitoring purposes.

    - **Add all associated vCenter systems only for monitoring and fleet management**: Use this option if you want multiple VCF Operations instances to monitor and provide fleet management to vCenter systems of the VCF account you are adding.
    - **Add all associated vCenter systems for monitoring, fleet management, and license and vSphere Client plug-in management by replacing the current integration**: Use this option to replace existing VCF Operations integration with a newer integration. For versions 9.0 or later, you can use this option if another VCF Operations provides license and vSphere Client plug-in management for vCenter you are adding and you want to replace it with the current VCF Operations. After you remove the current integration, you must assign new licenses for this vCenter. For more information, see Assign a Primary License to a vCenter Instance.

      For versions before 9.0, license management is not applicable. You can use this option if another VCF Operations provides vSphere Client plug-in management for the management domain vCenter for the VCF account you are adding. All current integrations of all associated vCenter systems get replaced.

      **Note:**

      If the management domain vCenter does not have a license and vSphere Client plug-in management, the current VCF Operations starts to manage it automatically.

For VCF 9.0 and later, the management domain vCenter, vSAN, and NSX are created using system-managed credentials. Workload domains are configured after one collection cycle.

If Log Collection and Network and Flow Collection are activated for vCenter and NSX, separate system-managed credentials are created for both.

The VCF account, with the configured domains, is added to the **Accounts** page. The newly added VCF account does not start monitoring the data automatically and you must manually initiate the data collection.

13. On the **Accounts** tab, locate the VCF account, click the vertical ellipsis, and then click **Start Collecting All**.
14. Optional: You can use your VCF account to monitor and manage license and vSphere client plug-in.
    - To bulk activate license and vSphere Client plug-in management for all your VCF accounts of versions 9.0 or later, select the VCF account, and click **Activate Management**.
    - To activate license and vSphere Client plug-in management for a particular VCF account, select the VCF account, click the vertical ellipsis, click **Edit**, and then click **Manage Integrations**. You must have the required privileges to manage integration.

The VCF account is added and starts collecting data in VCF Operations.

You can view the operations of your VCF accounts and their domains from the VCF Health page. For more information, see What is VCF Health in Diagnostics.

If you are using **SDDC Manager** 9.0 or later, you can also monitor your VCF account and collect data seamlessly using system-managed credentials. After you create a VCF account in VCF Operations, you can edit the VCF account and activate **System Managed Credentials.** Activating system-managed credentials for your VCF account does not affect the underlying vCenter and NSX accounts. Newly created vCenter and NSX accounts are automatically configured using system-managed credentials if the corresponding vCenter and NSX endpoints are 9.0. For password rotation, click **Rotate** to rotate your system-managed credential password. You must rotate system-managed credentials for individual endpoints. For more information, see Account Management Design.

Optionally, you can also configure the vCenter, vSAN, NSX, and Service Discovery instances in the VCF account.

- To configure a vCenter account, see Configuring a vCenter Account in VCF Operations.
- To configure a vSAN account, see Configuring a vSAN Account.
- To configure the NSX account, see Configuring the NSX Account in VCF Operations.
- To configure Service Discovery, see Configure Service and Application Discovery.

# vCenter Account in VCF Operations

The vCenter account connects VCF Operations to vCenter instances. You collect data and metrics from those vCenter instances, monitor them, and run actions on them.

VCF Operations evaluates the data in your environment, identifying trends in object behavior, predicting possible problems and future capacity for objects in your system based on those trends, and alerting you when an object exhibits defined symptoms.

## Configuring the vCenter Account

You can configure a vCenter account in VCF Operations to connect your VCF Operations environment to your vCenter instances.

## How Adapter Credentials Work

The vCenter credentials that you use to connect VCF Operations to a vCenter instance, determines what objects VCF Operations monitors. Ensure that you configure adapters and users correctly, and to avoid some of the following issues.

- If you configure the adapter to connect to a vCenter instance with credentials that have permission to access only one of your three ESX hosts, every user who logs in to VCF Operations sees only the one ESX host, even when an individual user has privileges on all three of the ESX hosts in the vCenter.

- If the provided credentials have limited access to objects in the vCenter, even VCF Operations administrative users can run actions only on the objects for which the vCenter credentials have permissions.
- If the provided credentials have access to all the objects in the vCenter, any VCF Operations user who runs actions is using the VCF Operations credentials.

**Controlling User Access to Actions**

Use the vCenter adapter to run actions on vCenter from VCF Operations. If you choose to run actions, you must control user access to the objects in your vCenter environment. You control user access for local users based on how you configure user privileges in VCF Operations. If users log in using their vCenter account, then the way their account is configured in vCenter determines their privileges.

For example, you might have a vCenter user with a read-only role in vCenter. If you give this user the VCF Operations*Power User* role in vCenter rather than a more restrictive role, the user can run actions on objects because the adapter is configured with credentials that has privileges to change objects. To avoid this type of unexpected result, configure local VCF Operations users and vCenter users with the privileges you want them to have in your environment.

To configure a vCenter account, see Configuring a vCenter Account in VCF Operations.

## Configuring a vCenter Account in VCF Operations

To manage your vCenter instances in VCF Operations, you must configure an account for each vCenter instance. The account requires the credentials that are used for communication with the target vCenter.

- Verify that you know the vCenter credentials that have sufficient privileges to connect and collect data, see Privileges Required for Configuring a vCenter Account. If the provided credentials have limited access to objects in vCenter, all users, regardless of their vCenter privileges see only the objects that the provided credentials can access. At a minimum, the user account must have Read privileges and the Read privileges must be assigned at the datacenter or vCenter level.
- You must install the Log management appliance for VCF to enable log collection and analysis in VCF Operations. See *Deploy, Converge, Upgrade* for more information on deployment and upgrade. For more information on log analysis in VCF Operations, see Configuring and Analyzing Logs.

  **Note:**

  Any account credentials that you add are shared with other account administrators and VCF Operations collector hosts. Other administrators might use these credentials to configure a new account or to move an account to a new host.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Integrations**.
2. On the **Accounts** tab, click **Add**.
3. On the Accounts Type page, click **vCenter**.
4. Enter a display name and description for the account.

   **Note:**

   Enter the name for the vCenter instance as you want it to appear in VCF Operations. A common practice is to include the IP address so that you can readily identify and differentiate between instances.
5. Optional: Select the **Physical Data Center** you want to associate with the vCenter account.

   If no physical data center is created or if you want to create a new physical data center for your account, you can add a new physical data center. For more information, see Adding Physical Data Centers in VCF Operations.
6. In the vCenter text box, enter the FQDN or IP address of the vCenter instance to which you are connecting.

   **Note:**

   While configuring VCF Single Sign-On, enter the of the FQDN of the vCenter instance to which you are connecting. vCenter instances configured using IP address are not included in VCF accounts. If you want the vCenter instance to be part of the VCF account, enter the FQDN.

7. To add credentials for the vCenter instance, click the **Add** icon, and enter the required credentials.

   The vCenter credential must have `Performance > Modify intervals` permission activated in the target vCenter to collect VM guest metrics. Optionally, you can use alternate user credentials for actions. Enter an **Action User Name** and **Actions Password**. If you do not enter an **Action User Name** and **Actions Password**, the default user specified is considered for actions.

   **Note:**

   Credentials are stored in VCF Operations and can be used for one or more instances of the vCenter.

   **Note:**

   To monitor application services and operating systems, it is recommended that you enter action credentials with guest operations privileges. For more information, see Privileges Required for Configuring a vCenter Account.

8. Determine which VCF Operations cloud proxy/ group is used to manage the account. From the **Cloud Proxy/ Group** drop-down, select the cloud proxy you deployed as the collector for this vCenter account.

   **Note:**

   The vCenter FQDN or IP address must be reachable from the selected cloud proxy/ group. If you have multiple cloud proxies/ groups in your environment, you can distribute the workload to optimize performance. Select the cloud proxy/ group to manage the adapter processes for this instance.

9. Click **Validate Connection** to validate the connection with your vCenter instance.

10. In the **Review and Accept Certificate** dialog box, review the certificate information.

    - If the certificate presented in the dialog box matches the certificate for your target vCenter instance, click **OK**.
    - If you do not recognize the certificate as valid, click **Cancel**. The validation test fails and the connection to vCenter instance is incomplete. To complete the adapter configuration, you must provide a valid vCenter URL and verify that the certificate on the vCenter is valid.

11. The account is configured to run actions on objects in vCenter from VCF Operations. If you do not want to run actions, deselect **Activate** for Operational Actions.

12. Optional: To collect logs, click **Activate Log Collection**. For more information, see Configuring and Analyzing Logs.

    a) This setting is activated by default.

    a) By default, logs are collected directly into the Logs cluster. To collect logs into a unified cloud proxy instead, you must set one up. Note that the default unified cloud proxy cannot be used for this collection type. An advanced unified cloud proxy must be configured instead.

    **Note:**

    When you export vCenter information from one VCF Operations instance and import it to another, the log collection information is not transferred.

13. Optional: To collect network and flow, click **Activate Network and Flow**. For more information, see Add VMware vCenter.

14. To modify the advanced options regarding collectors, object discovery, or change events, expand the **Advanced Settings**.

    For information about these advanced settings, see Configuring Advanced Settings for a vCenter Account.

15. Click **Add** to select an additional option for this vCenter instance.

    **Note:**

    License and vSphere Client plug-in management is supported only for vCenter accounts of version 9.0 or later. If your vCenter is 9.0 or later and is not managed by any VCF Operations instance, the current VCF Operations instance takes up the license and vSphere Client plug-in management. If you do not have the required privileges for license and vSphere Client plug-in management, the vCenter account will be added only for monitoring purposes. For more information, see Use and Configure Your vSphere Client.

After you integrate a vCenter account with a VCF Operations instance for license management, it takes up to 15 minutes before the vCenter account appears in the **License** tab of VCF Operations.

If you have an earlier version of vCenter accounts, you can only activate vSphere Client plug-in management. If your vCenter is pre 9.0, and is not managed by any VCF Operations instance, the current VCF Operations instance takes up the vSphere Client plug-in management. If you do not have the required privileges for vSphere Client plug-in management, vCenter account will be added only for monitoring purposes.

- **Add vCenter only for monitoring**: Use this option if you want to proceed with the existing VCF Operations integration and use the new vCenter account for monitoring purposes only.
- **Add vCenter for monitoring and license and vSphere Client plug-in management by replacing the current integration**: Use this option to replace existing VCF Operations integration with a newer integration.

  For versions 9.0 or later, you can use this option if another VCF Operations provides license and vSphere Client plug-in management for vCenter you are adding and you want to replace it with the current VCF Operations. After you remove the current integration, you must assign new license to this vCenter instance. For more information, see Assign a Primary License to a vCenter Instance.

  For versions prior to 9.0, license management is not applicable. You can use this option if another VCF Operations provides vSphere Client plug-in management for vCenter you are adding and you want to replace it with the current VCF Operations.

Newly created vCenter accounts do not start monitoring the data automatically and you must manually initiate the data collection.

16. On the Accounts tab, locate the vCenter account, click the vertical ellipsis, and then click **Start Collecting**.
17. Optional: You can use your vCenter account to monitor and manage license and vSphere client plug-in.
    - To bulk activate license and vSphere Client plug-in management for all your vCenter accounts of versions 9.0 or later, select the vCenter account, and click **Activate Management**.
    - To activate license and vSphere Client plug-in management for a particular vCenter account, select the vCenter account, click the vertical ellipsis, click **Edit**, and then click **Manage Integrations**. You must have the required privileges to manage integration. For more information, see Privileges Required for Configuring a vCenter Account.

The vCenter account is added to the list and VCF Operations begins collecting metrics, properties, and events from the vCenter instance. Depending on the number of managed objects, the initial collection can take more than one collection cycle. A standard collection cycle begins every 5 minutes.

For information about the network port that VCF Operations uses to communicate with a vCenter instance and VCF Operations components, see VMware Ports and Protocols.

You can activate vSAN configuration for your account. For more information, see Configuring a vSAN Account.

You can use the vCenter for service discovery, see Configure Service and Application Discovery.

### Privileges Required for Configuring a vCenter Account

To configure your vCenter adapter instance in VCF Operations, you need sufficient privileges to monitor and collect data and to perform vCenter actions. You can configure these permissions as a single role in vCenter to be used by a single service account or configure them as two independent roles for two separate service accounts.

The vCenter account requires credentials with the following privileges configured in vCenter to be able to monitor and collect data and perform necessary actions in the vCenter instance.

**Note:**

The vCenter role is created with a **Read Only** role with three system-defined privileges: **System.Anonymous**, **System.View**, and **System.Read**. For more information, see Using vCenter Roles to Assign Privileges.

**Table 1183: Privileges for Configuring a vCenter Adapter: Monitoring and Data Collection**

| Task | Privilege |
|---|---|
| Property Collection | **System > Anonymous**: This privilege is added automatically when you create a user account and it is not visible in vSphere.<br>**System> View** |
| Performance Metrics Collection | **Performance > Modify intervals**<br>**System > Read**: This privilege is added automatically when you create a user account and it is not visible in vSphere. |
| Objects Discovery<br>Events Collection | **Storage views > View**<br>**Datastore > Browse datastore**<br>**System > View**: This privilege is added automatically when you create a user account and it is not visible in vSphere. |
| Orphaned Disk | **Datastore > Browse datastore** |
| Service Discovery | For credential-based service discovery<br>**Virtual machine > Guest operations > Guest operation alias modification**<br>**Virtual machine > Guest operations > Guest operation alias query**<br>**Virtual machine > Guest operations > Guest operation modifications**<br>**Virtual machine > Guest operations > Guest operation program execution**<br>**Virtual machine > Guest operations > Guest operation queries** |
| | For credential-less service discovery<br>**Virtual machine > Service configuration > Manage service configurations**<br>**Virtual machine > Service configuration > Modify service configuration**<br>**Virtual machine > Service configuration > Query service configurations**<br>**Virtual machine > Service configuration > Read service configuration**<br>For more information, see Configure Service and Application Discovery. |
| Collect Tags | **Global > Global tag**<br>**Global > Health**<br>**Global > Manage custom attributes**: This privilege is required only if the tags are associated with custom attributes.<br>**Global > System tag**<br>**Global > Set custom attribute**<br>**GlobalPermissions access** |
| Authorization Manager | **System> View** |
| Provisioning Checker | **System> View** |
| External Statistics Manager | **ExternalStatsProvider.Register**<br>**ExternalStatsProvider.Unregister**<br>**System.View**<br>**ExternalStatsProvider.Update** |
| Storage Resource Manager | **System.View** |
| Configuration Management | **Administrators Group**<br>User must be part of the Administrators Group. |

| Task | Privilege |
| --- | --- |
| Optimize Container<br>Schedule Optimize Container<br>Automate Optimize Container | • **AutoDeploy -> Rule -> Create**<br>• **AutoDeploy -> Rule -> Delete**<br>• **AutoDeploy -> Rule -> Edit**<br>• **AutoDeploy -> RuleSet -> Activate**<br>• **AutoDeploy -> RuleSet -> Edit**<br>• **Datastore -> Allocate space**<br>• **Global -> Global tag**<br>• **Global -> System tag**<br>• **Host -> Inventory -> Manage Cluster Lifecyle**<br>• **Host -> Inventory -> Modify cluster**<br>• **Resource -> Assign virtual machine to resource pool**<br>• **Resource -> Migrate powered off virtual machine**<br>• **Resource -> Migrate powered on virtual machine**<br>• **Resource -> Query vMotion**<br>• **Storage views -> Configure service**<br>• **Storage views -> View**<br>• **Virtual machine -> Edit inventory > Move**<br>Privilege required for vCenter version 7.x:<br>• **Profile-driven storage -> Profile-driven storage update**<br>• **Profile-driven storage -> Profile-driven storage view**<br>Privilege required for vCenter version 8.x:<br>• **VM storage policies -> Apply VM storage policies**<br>• **VM storage policies -> Update VM storage policies**<br>• **VM storage policies -> VM storage policies edit permissions**<br>• **VM storage policies -> VM storage policies view permissions**<br>• **VM storage policies -> View VM storage policies** |
| Getting storage policies for VMs and datastores and checking VM compatiblity with datastore | **System > Anonymous**<br>**System> View** |
| Provide data to vSphere Predictive DRS | **External stats provider > Update**<br>**External stats provider > Register**<br>**External stats provider > Unregister**<br>**vSphere Stats Privileges > Collect Stats Data**<br>**vSphere Stats Privileges > Modify Stats Configuration**<br>**vSphere Stats Privileges > Query Stats Data** |
| DRS rule management | **Host.Inventory.EditCluster** |
| vSphere Client plug-in | **Extension > Register extension**<br>**Extension > Unregister extension**<br>**Extension > Update extension** |
| Monitoring Health of vCenter Services using VMware Infrastructure Health | User must be part of the **SystemConfiguration.Administrator** group or have **Administrato r** permission. |
| Session Manager: Acquire and manage vCenter sessions | **System.Anonymous**<br>**System.View**<br>**Sessions.TerminateSession** |
| Add License to vCenter | Membership in the**LicenseService.Administrators** Single Sign-On group.<br>**Global > Licenses**<br>**Certificate Management > Create/ Delete** (below Admins Priv) |
| Register/unregister VCF Operations in vCenter | **System> View** |
| Authentication on VCF Operations using vCenter User and apply actions | **privilege.Global.com.vmware.label > vRealize Operations Read Only Role**<br>**privilege.Global.com.vmware.label > vRealize Operations Power User Role** |

| Task | Privilege |
|---|---|
| Discovering WCP related objects and collecting metrics and properties | **Administrator**<br>**Note:**<br>Users with Non-Administrator or custom role must be added to the **ServiceProviderUser** group. The ServiceProviderUsers is a group in the vCenter Single Sign-On Domain.<br>**System.Read** |

**Table 1184: Privileges for Configuring a vCenter Adapter: Performing vCenter Actions**

| Actions | Privilege |
|---|---|
| vCenter Linking | **VCenter Linking> Administer** |
| **Create/Update/Delete** VcTrusts/VcIdentity | **VcTrusts/VcIdentity> Create/Update/Delete** (Admins privileges)<br>**VcTrusts/VcIdentity> Create/Update/Delete** (below Admins privileges) |
| Modify cluster | **Host.Inventory.EditCluster** |
| Execute Custom Script | **Localization: Guest operations > Guest operation program execution**<br>**Virtual machine > Guest operations query**<br>**Localization: Guest operations -> Guest operation queries**<br>**Note:**<br>• The ESX instance that hosts the VMs where the custom script should be run, must have HTTPS access to port 443 from the collector node on which the vCenter adapter instance is configured.<br>• The user provided in the script must have read and write privileges to the temp directory (execute privilege is also required for this directory in Linux systems). For Windows systems, the path can be taken from the environment variable *TEMP*. For Linux systems, it is `/tmp` and/or `/var/tmp`. |
| Power Off VM | **Virtual machine > Interaction > Power off** |
| Power On VM | **Virtual machine > Interaction > Power on** |
| Set Memory for VM | **Virtual machine > Change Configuration > Change Memory** |
| Set CPU Count for VM | **Virtual machine > Change Configuration > Change CPU count** |
| Set CPU Resources for VM<br>Set Memory Resources for VM | **Virtual machine > Change Configuration > Change resource** |
| Create Snapshot for VM | **Virtual machine > Snapshot management > Create snapshot** |
| Delete Unused Snapshot for VM<br>Delete Unused Snapshots For VM Express | **Virtual machine > Snapshot management > Remove snapshot** |
| Delete Unused Snapshots for Datastore<br>Delete Unused Snapshots For Datastore Express | **Virtual machine > Snapshot management > Remove snapshot** |
| Delete Powered Off VM | **Virtual machine > Edit Inventory > Remove** |
| Delete Idle VM | **Virtual machine > Edit Inventory > Remove** |
| Shut Down Guest OS for VM | **Virtual machine > Interaction > Power off** |

| Actions | Privilege |
| --- | --- |
| Move VM | • **Resource > Assign virtual machine to resource pool**<br>• **Resource > Migrate powered off virtual machine**<br>• **Resource > Migrate powered on virtual machine**<br>• **Datastore > Allocate space**<br>• **Virtual machine -> Edit Inventory > Move**<br>**Note:**<br>Combining these four permissions allows the service account to perform Storage vMotion and regular vMotion of an object therefore allowing VCF Operations to perform the given operations. |
| Set DRS Automation | **Host > Inventory > Modify cluster** |
| Provide data to vSphere Predictive DRS | **External stats provider > Update**<br>**External stats provider > Register**<br>**External stats provider > Unregister** |
| Reboot Guest OS for VM | **Virtual machine > Interaction > Reset** |
| Events, tasks, and alarms collection | **System > View**<br>**Note:**<br>**System > View** is a system-defined privilege. When you add a custom role and do not assign any privileges to it, the role is created as a Read Only role with three system-defined privileges:<br>• **System > Anonymous**<br>• **System > View**<br>• **System > Read**<br>**Note:**<br>You must configure the permission on the top-level folder within the vCenter Server inventory, and verify that the **Propagate to children** check box is selected. |
| Syslog configuration on ESX hosts | • **Host > Configuration> Change settings**<br>• **Host > Configuration> Network configuration**<br>• **Host > Configuration> Advanced Settings**<br>• **Host > Configuration> Security profile and firewall**<br>• **Certificates > Manage certificates** - required only if ESXi syslog collection is configured via SSL (Secure collection)<br>**Note:**<br>You must configure the permission on the top-level folder within the vCenter Server inventory, and verify that the **Propagate to children** check box is selected. |

**Table 1185: Privileges for collecting compliance-related properties**

| Task | Privilege/Role |
|---|---|
| Collecting the following properties on the vCenter adapter instance object:<br>• Appliance\|Maximum Number of Failed Login Attempts<br>• Appliance\|Unlock time<br>• Appliance\|Time Interval Between Failures<br>• Appliance\|Is Root Account Enabled<br>• Appliance\|Root Account Expiration Date<br>• Appliance\|Console Cli Enabled<br>• Appliance\|Has Recovery Backup Jobs<br>• Appliance\|Has Networking Firewall Inbound Rules<br>• Appliance\|Has Logging Forwarding Configs<br>• Appliance\|Time Sync Mode | **Administrator** |

For more information about tasks and privileges, see Defined Privileges.

## Configure User Access for Actions

To ensure that users can run actions in VCF Operations, you must configure user access to the actions.

You can create multiple roles, each granting access to specific actions. Administrators and super users already have full permissions. To simplify management, assign roles to user groups instead of configuring individual users.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Access Control** tile.
   a) Click the **Roles** tab.
   b) Click the **Add** icon, and enter a name and description for the role.
   c) To **Assign Permissions**, select the permissions. You can select a whole permission group or individual permissions within a group.
   d) Select one or more permissions, and then click **SAVE**.
2. To create a user group, return to the **Access Control** page.
   a) Click the **User Groups** tab, and click the **Add** icon.
   b) Enter a name and description for the group.
   c) To **Assign Roles and Scopes**, select a role for the user group and assign a scope for each role. You can add one or more roles and assign scope for each role.
   d) To **Assign Users**, select the users you want to assign to the user group.
   e) Click **SAVE**.

Test the users that you assigned to the group. Log out, and log back in as one of the users. Verify that this user can run the expected actions on the selected adapter.

## Configuring Advanced Settings for a vCenter Account

To begin monitoring your environment with VCF Operations, you configure the vCenter account. The solution includes the vCenter account that collects data from the target vCenter instances. For more information, see Configuring a vCenter Account in VCF Operations.

### Where You Find the Solution- vCenter

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Integrations**. On the **Accounts** tab, click **ADD**, and then select **vCenter**.

## Advanced Settings Options for Configuring a vCenter Account

Configure and modify the vCenter account on the Account Information page.

| Option | Description |
|---|---|
| Advanced Settings | Provides options related to designating specific collectors to manage this account, managing object discovery and change events. |
| Auto Discovery | Determines whether new objects added to the monitored system are discovered and added to VCF Operations after the initial configuration of the account.<br>• If the value is true, VCF Operations collects the information about any new objects that are added to the monitored system after the initial configuration. For example, if you add more ESX hosts and virtual machines, these objects are added during the next collections cycle. This is the default value.<br>• If the value is false, VCF Operations monitors only those objects that are present on the target system when you configure the account. |
| Process Change Events | Determines whether the account uses an event collector to collect and process the events generated in the vCenter instance.<br>• If the value is true, the event collector collects and publishes events from vCenter. This is the default value.<br>• If the value is false, the event collector does not collect and publish events. |
| Process VC Tasks | Determines whether the account uses a task collector to collect and process tasks that are created in the vCenter instance.<br>• If the value is true, the task collector collects and publishes tasks from vCenter instance. This is the default value.<br>• If the value is false, the event collector does not collect and publish tasks. |
| Activate Collecting vSphere Distributed Switch<br>Activate Collecting Virtual Machine Folder<br>Activate Collecting vSphere Distributed Port Group | When set to false, reduces the collected data set by omitting collection of the associated category. |
| Maximum Number Of Virtual Machines Collected | Reduces the collected data set by limiting the number of virtual machine collections.<br>To omit data on virtual machines and have VCF Operations collect only host data, set the value to zero. |
| Provide data to vSphere Predictive DRS | vSphere Predictive DRS proactively load balances vCenter cluster to accommodate predictable patterns in the cluster workload.<br>VCF Operations monitors virtual machines running in vCenter, analyzes longer-term historical data, and provides forecast data about predictable patterns of resource usage to Predictive DRS. Based on these predictable patterns, Predictive DRS moves to balance resource usage among virtual machines.<br>Predictive DRS must also be activated for the vSphere Clusters managed by vCenter instances monitored by VCF Operations. Refer to the vSphere Resource Management for details on activating Predictive DRS on a per vSphere cluster basis.<br>When set to true, designates VCF Operations as a predictive data provider, and sends predicative data to vCenter . You can only register a single active Predictive DRS data provider with vCenter at a time. |
| Activate Actions | Activating this option helps in triggering the actions that are related to vCenter. |
| Cloud Type | Identifies the type of vCenter that is used in VCF Operations. |
| **vCenter** ID | A globally unique identifier associated with the vCenter instance. |
| Deactivate collecting Guest File Systems with names containing | Provide a comma-separated list of strings. If these strings are found in any guest files system mount point name, that guest file system will not be collected. |
| Enable vSphere Supervisor Collection | Enables the collection of the auto discovered vSphere Supervisor cluster. |

| Option | Description |
|---|---|
| Activate vCenter -initiated metric collection | If the value is true, it enables the collection of Host and VM level GPU metrics.<br>Change the value to false to stop metric collection. It may take up to 10 minutes to completely disable metric collection. |
| vCenter- initiated metric collection sample interval | The sample interval in seconds between vCenter initiated metric collection. To change the sample interval, you must first disable vCenter initiated metric collection. Update the collection sample interval before restarting the metric collection. |

You can find the vSphere Hardening Guides at VMware Security Hardening Guides.

Click **Save** to finish configuration of the solution.

# vSAN Account in VCF Operations

You can make vSAN operational in a production environment by using dashboards to evaluate, manage, and optimize the performance of vSAN objects and vSAN-activated objects in your vCenter system.

vSAN extends the following features:

- Discovers vSAN disk groups in a vSAN datastore.
- Identifies the vSAN-activated cluster compute resource, host system, and datastore objects in a vCenter system.
- Automatically adds related vCenter components that are in the monitoring state.
- Support for vSAN datastores in workload optimization with cross-cluster rebalance actions.
  - You can move VMs from one vSAN datastore to another vSAN datastore.
  - You can optimize the container if all the vSAN clusters are not in resync state.
  - VMs with different storage policies for each disk or VMs with different types of storage for each disk will not be moved.
  - You can generate a re-balance plan only if sufficient disk space is available at the destination vSAN datastore (The vSAN datastore slack space is also considered).
  - The storage policy assigned to the VM is considered during the workload optimization (Compatibility check is performed against the storage policy).
  - VM migration from vSAN datastore to vSAN stretched clusters is not supported.

## Configuring a vSAN Account

You can manage and monitor the health and performance of your vSAN clusters and remediate alerts in VCF Operations.

- You must have a vCenter Adapter instance configured and monitoring the same vCenter instance that is used to monitor the vSAN and Storage Devices.

  **Note:**

  Only vCenter systems that are configured for both vCenter adapter and vSAN adapter appear in the inventory tree under the vSAN and storage devices. Verify that vCenter that you use to configure the vSAN adapter instance is also configured as a vCenter adapter instance for the VMware vSphere solution. If not, add a vCenter adapter instance for that vCenter. For more information, see Configuring a vCenter Account in VCF Operations.

- You must open port 5989 between the host and any VCF Operations node on which the vSAN adapter resides. This is applicable when the vSAN version in vSphere is 6.6 or lower.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Integrations**.
2. On the **Accounts** tab, select the vCenter instance.

   You must configure the vCenter instance before you configure vSAN on it. To configure the vCenter instance, see Configuring a vCenter Account in VCF Operations.

3. Click the vSAN tab and click the **vSAN configuration** toggle switch to activate vSAN.

   **Note:**

   Once the vSAN adapter instance is activated and saved, the activated vSAN configuration option is not visible.

4. The credentials provided for the vCenter instance are also used for vSAN adapter instance. If you do not want to use these credentials, you can click **Use alternate credentials** option.
   a) Click the plus sign next to the Credential field and enter the details in the **Manage Credentials** dialog box.
   b) Enter the credential name, vCenter user name, and password and click **OK**.
5. Choose **Enable SMART data collection**, to activate SMART data collection for physical disk devices.
6. Click **Validate Connection** to validate the connection with your vCenter instance.
7. Accept the vCenter security certificate.
8. Click **Add** to save the configurations.

   Newly created vCenter account with vSAN activated do not start monitoring the data automatically and you must manually initiate the data collection.
9. On the Accounts tab, locate the vCenter account, click the vertical ellipsis and then click **Start Collecting All**.

The vSAN account starts collecting data. The newly connected vSAN account information will appear on all vSAN dashboards after five minutes. You can get an overview of the vSAN health and performance in the Storage Operations page. For more information, see Monitoring Storage Operations. To view vSAN performance for each cluster individually and to drill down info-specific performance criteria, use the vSAN Performance dashboard.

To verify that the adapter is configured and collecting data from vSAN objects, wait a few collection cycles, then view application-related data.

- Inventory. Verify that all the objects related to the vSAN instance are listed. Objects should be in the collecting state and receiving data.
- Dashboards. Verify that vSAN Capacity Overview, Migrate to vSAN, vSAN Operations Overview, and Troubleshoot vSAN, are added to the default dashboards.
- Under **Inventory** > **vSAN and Storage Devices**, verify that the vSAN hierarchy includes the following related vCenter system objects:
  – vSAN World
  – Cache Disk
  – Capacity Disk
  – vSAN-activated vCenter clusters
  – vSAN Fault Domains (optional)
  – vSAN-activated Hosts
  – vSAN Data stores
  – vSAN Disk Groups
  – vSAN Data store related VMs
  – vSAN Witness Hosts (optional)

## Verify that the Adapter Instance is Connected and Collecting Data

You configured an adapter instance of vSAN with credentials for a vCenter. Now you want to verify that your adapter instance can retrieve information from vSAN objects in your environment.

**Table 1186: Object Types that vSAN Discovers**

| Object Type | Description |
|---|---|
| vSAN Adapter Instance | The vSAN instance. |
| vSAN Cluster | vSAN clusters in your data center. |

| Object Type | Description |
|---|---|
| vSAN Datastore | vSAN datastores in your data center. |
| vSAN Disk Group | A collection of SSDs and magnetic disks used by vSAN. |
| vSAN Fault Domain | A tag for a fault domain in your data center. |
| vSAN Host | vSAN hosts in your data center. |
| vSAN Witness Host | A tag for a witness host of a stretched cluster, if the stretched cluster feature is activated on the vSAN cluster. |
| vSAN World | A vSAN World is a group parent resource for all vSAN adapter instances. vSAN World displays aggregated data of all adapter instances and a single root object of the entire vSAN hierarchy. |
| Cache Disk | A local physical device on a host used for storing VM files in vSAN. |
| Capacity Disk | A local physical device on a host used for read or write caching in vSAN |

The vSAN adapter also monitors the following objects discovered by the VMware vSphere adapter.

- Cluster Compute Resources
- Host System
- Datastore

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory** > **Adapter Instances**, and then click **vSAN Adapter Instance**.
2. Select the adapter instance name to display the list of objects discovered by your adapter instance.
3. Slide the display bar to the right to view the object status.

| Object Status | Description |
|---|---|
| Collection State | If green, the object is connected. |
| Collection Status | If green, the adapter is retrieving data from the object. |

4. Deselect the adapter instance name and expand the **Object Types** tag.
   Each Object Type name appears with the number of objects of that type in your environment.

If objects are missing or not transmitting data, check to confirm that the object is connected. Then check for related alerts.

To ensure that the vSAN adapter can collect all performance data, the vSAN performance service must be activated in vSphere. For instructions on how to activate the service, see Turn on vSAN Performance Service in KB 326483.

To collect capacity metrics for Cache Disks, ensure that the Performance Service is enabled for the vSAN cluster. For more information, see Broadcom knowledge base article 405036.

If the vSAN performance service is deactivated or experiencing issues, an alert is triggered for the vSAN adapter instance and the following errors appear in the adapter logs.

```
ERROR com.vmware.adapter3.vsan.metricloader.VsanDiskgroupMetricLoader.collectMetrics
   - Failed to collect performance metrics for Disk Group
com.vmware.adapter3.vsan.metricloader.VsanDiskgroupMetricLoader.collectMetrics
   - vSAN Performance Service might be turned OFF.
com.vmware.adapter3.vsan.metricloader.VsanDiskgroupMetricLoader.collectMetrics
   - (vim.fault.NotFound)
   {
   faultCause = null,
   faultMessage = (vmodl.LocalizableMessage)
      [
         com.vmware.vim.binding.impl.vmodl.LocalizableMessageImpl@98e1294
      ]
   }
```

## vSAN Log Analytics Enhancements

When VCF Operations is integrated with Log management, you can view and troubleshoot Log management object issues within VCF Operations. Earlier you could troubleshoot issues related only to vCenter objects, but now you can troubleshoot issues related to vSAN also.

The enhancements to vSAN log analytics include use of specific queries to retrieve log information for the following vSAN objects:

- vSAN Cluster
- Witness Host
- Disk Group
- Cache Disk
- Capacity Disk

### Where You Find vSAN Object Logs

Navigate to the vSAN Object Details page, and click the **Logs** tab.

> **Note:**
>
> If you are not logged in to Log management, then VCF Operations prompts you to log in to Log management with your login credentials.

VCF Operations uses special queries for each object type. Using the special queries for vSAN objects, you can perform the following actions:

- View interactive analytics for the selected vSAN object.
- Retrieve log details for the vSAN object.
- Analyze and troubleshoot issues related to the vSAN object.

## Monitoring Storage Operations

The Storage Operations page is a centralized storage dashboard for vSAN and non-vSAN storage types in VCF Operations.

To get an overview of your datastore inventory, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Overview** > **Storage**.

Use the storage operations page to view storage-related alerts and view the usage and distribution of your vSAN and non-vSAN datastores. You can filter vSAN and non-vSAN datastores for a more detailed overview.

> **Note:**
>
> To view vSAN related data on the storage operations page, you must configure a vSAN enabled vCenter account and ensure that the vCenter instance is collecting data in VCF Operations. For more information, see Configuring a vCenter Account in VCF Operations.

You can view the vSAN Cluster Health Score and the different types of vSAN clusters available in VCF Operations. Additionally you can hover on any cluster pin to view the basic performance metrics like IOPS, Latency, and Throughput for that particular vSAN cluster. The storage operations page also displays the **vSAN Clusters** table that displays the vSAN cluster information.

### vSAN Cluster Performance Widget

The vSAN cluster performance widget provides an overview of the performance of the vSAN clusters in relation to one another. To view performance diagnostics of all the vSAN clusters, click **View Diagnostics**. The **Performance Insights** tab opens and displays two types of performance insights:

- **Proactive insights**: Proactive insights report any actively ongoing critical vSAN cluster issues that impact VM latency and require immediate resolution. To activate proactive insights for your clusters, ensure that VCF Operations for Real Time Data Service is activated. For more information, see Deploying VCF Operations for Real Time Data Service.

You can deactivate proative insights or conditionally deactivate proactive insights for some clusters. For more information, see KB article 420866.

- **On demand insights**: On demand insights only appears after you manually run diagnostics for a target vSAN cluster. To run on demand diagnostics, click **Run New Diagnostics**, select a **Diagnostic goal**, select the target vSAN cluster, and then set the desired time range.
  - **Troubleshooting**: Run performance diagnostics for a cluster already in use with a running workload. This helps identify and troubleshoot performance issues impacting the workload.
  - **Benchmarking and Optimizing**: Run performance diagnostics for a new cluster before adding any workload. This enables diagnostics with a **Benchmark goal**.

To view detailed information about a performance insight, click the expand icon. The details pane shows the cluster information, diagnostic results, and suggested remediation steps.

After resolving an issue reported by a proactive insight, it is removed from the **Performance Insights** table when the next proactive insight runs. If you resolve an issue reported by on demand diagnostics, you must run new diagnostics to confirm that the issue is resolved.

**Note:**

If you run on-demand diagnostics on a HCIBench and the recommendation is to tune IO parameters to get closer to the goal, re-run HCIBench run with the recommended IO parameters. Once done, you can re-run on-demand diagnostics on the new HCIBench to confirm that the issue is resolved.

# NSX Account in VCF Operations

The NSX adapter allows you to retrieve alerts and findings from NSX to VCF Operations.

The NSX adapter supports adapter configuration using VIDM for NSX versions 3.0 and above. The roles and permissions associated with the VIDM users collecting the NSX adapter data are:

| Roles | Permissions |
|---|---|
| Enterprise Admin | Collect all data. |
| Network engineer | • Collect all the NSX resources except the Load Balancer and collect limited routers data.<br>Router data collected:<br>– Tier 0 router connected to logical switch.<br>– Tier 1 router created from vCloud Director. |
| • Security Engineer<br>• Security Operator<br>• Auditor | Collect all data except the load balancer. |
| • LB Admin<br>• LB Auditor<br>• Netxpartner Admin | Cannot collect any data. |

## Configuring the NSX Account in VCF Operations

You can configure your NSX account in VCF Operations.

To view the roles and permissions that are associated with the VIDM users collecting the NSX adapter, see NSX Account in VCF Operations

- To use Principal Identities authentication in VCF Operations you must have created a Principal Identity user in NSX.
- Ensure that you have the client certificate and the key to authenticate the Principal Identity users in VCF Operations.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Integrations**.
2. On the **Accounts** tab, click **Add**, and on the **Accounts Types** page, select **NSX**.
3. Configure the adapter instance.

| Option | Action |
|---|---|
| Name | Enter the name for the NSX instance as you want it to appear in VCF Operations. |
| Description | Enter any additional information that helps you manage your instances. |

4. Configure the connection.

| Option | Action |
|---|---|
| Virtual IP/ NSX | Enter the FQDN, the IP address, or the Virtual IP of the NSX. Both IPv4 address and IPv6 address formats are supported. |
| Credential | Select the credential that you want to use to sign on to the environment from the drop-down menu. To add new credentials to access the NSX environment, click the plus sign.<br>1. From the **Manage Credential** dialog, you can click the **Credential Kind** drop-down and select **NSX Client Certificate Crentials** or **NSX Credentials**.<br>The NSX Client Certificate allows you to use Principal Identities user or certificate-based client for authentication and the NSX Credential allows you to use local administrator or VMware Identity Manager for authentication.<br>2. Enter the credential details based on your selection.<br>For NSX Credentials, enter the following details.<br>• Credential Name - The name by which you are identifying the configured credentials.<br>• User Name - The user name of the NSX instance.<br>• Password - The password of the NSX instance.<br>For NSX Client Certificate Credentials, enter the following details.<br>• Credential Name - The name by which you are identifying the configured credentials.<br>• Client Certificate Data - Enter the value of the client certificate data associated with the principal user.<br>• Client Key Data - Enter the value of the client key data associated with the principal user. |

| Option | Action |
|---|---|
| Cloud Proxy/Group | Determine which VCF Operations cloud proxy/ group is used to manage the account. If you have multiple cloud proxies/ groups in your environment, and you want to distribute the workload to optimize performance, select the cloud proxy/ group to manage the adapter processes for this instance.<br>**Review and Accept Untrusted Certificates**<br>When you upgrade VCF Operations, the NSX adapter moves to a warning state and the data collection stops. This happens only when you have an adapter that presents a self-signed certificate, or a certificate signed by an untrusted Certification Authority.<br>To continue the adapter configuration, you must Validate the Connection, where you are prompted to review and accept the certificate.<br>**Note:**<br>If you have a multi-node cluster configuration, then you are prompted to review and accept the certificate for each node. |
| Log Operations | |
| Collect Logs | Select the **Activate Log Collection** checkbox to enable log collection. To control log volume, and other configuration details, see Configuring and Analyzing Logs. |
| Network Operations | |
| Network & Flow | Select the option to activate the network and flow.<br>Install the operations-network appliance from **Fleet Management** > **Lifecycle** to activate this option |
| Advanced Settings | |
| Exclude Resources from Monitoring | Select specific NSX resources that you want to exclude from monitoring in VCF Operations. Once the exclusion is applied, the monitoring for the specified resources is disabled.<br>**Note:**<br>Earlier the Management Pack for NSX would collect all the data and monitor all the resource kinds associated with the Management Pack, but now it allows you to exclude specific resources from monitoring. |
| Exclude Management Services from Monitoring | Select specific NSX management services that you want to exclude from monitoring in VCF Operations. Once the exclusion is applied, the monitoring for the specified services is disabled.<br>**Note:**<br>Earlier the Management Pack for NSX would collect all the data and monitor all the services associated with the Management Pack, but now it allows you to exclude specific services from monitoring. |
| Exclude Logical Router Services from Monitoring | Select the specific router service you want to exclude from monitoring in VCF Operations. Once the exclusion is applied, the monitoring for the specified services is disabled. |
| Auto Discovery | Set the **Auto Discovery** to True or False.<br>• **Auto Discovery - True** - Activates auto-discovery of the new objects added to the monitored system. By default, this field is set to True.<br>• **Auto Discovery - False** - Deactivates auto discovery, you must manually discover the objects from the system which you want to monitor. |

| Option | Action |
| --- | --- |
| Import Alerts from NSX | You can directly import alerts from NSX into VCF Operations by setting the **Import Alerts from NSX** field to **True**.<br>Depending on the number of alerts and the performance of the system, importing alerts into VCF Operations may take some time.<br>By default, this field is set to **False**. |

5. Click **Add** to save the configurations.

   The adapter instance is added to the list. Newly created NSX accounts do not start monitoring the data automatically and you must manually initiate the data collection.

6. On the **Accounts** tab, locate the NSX account, click the vertical ellipsis, and then click **Start Collecting All**.

Verify that the adapter is configured and is collecting data.

### Support for Principal Identities Authentication for NSX

VCF Operations supports authentication of Principal Identities (PI) using NSX . The Principal Identities (PI) are unique users in NSX who can create an object and ensure that the object can only be modified or deleted by the same identity. The authentication of principal identities is only supported through a client certificate. The principal identities authentication is local to NSX and it is possible to assign a predefined Role-based access control (RBAC) role to the principal identity.

Principal Identities are used by third-party applications or cloud management platforms such as Open stack, and Pivotal Container Services (PKS) to ensure that an administrator does not modify the NSX configuration which can generate a mismatch between their view of the NSX environment and the actual configuration.

## VCF Automation for All Apps Organization Account in VCF Operations

The VCF Automation for All Apps Organization management pack expands the operational capabilities available to users, helping partners effectively meter, price, and bill their cloud services using VMware's chargeback technology. It enables flexible segmentation of virtual environments to meet tenant requirements and supports various billing strategies. By offering accurate infrastructure monitoring and customizable pricing models, the solution ensures tenants have a transparent view of their resource consumption, making billing verification simple. Both service providers and tenants benefit from comprehensive reports, real-time alerts, and intuitive dashboards that display usage patterns, resource allocation, and pricing information.

#### Overview of the VCF Automation for All Apps Organization Management Pack

The Management Pack for VCF Automation for version 8.x enabled integration with VMware Aria Automation 8.x environments, providing cloud administrators with monitoring, alerting, reporting, metering, and pricing capabilities. With the strategic consolidation of VMware Cloud Director and VMware Aria Automation into VCF Automation 9.0, existing organizations will persist as VM Apps organizations, while newly created organizations will be categorized as All Apps organizations.

With the release of VCF, VCF Automation, VMware Cloud Director, and Supervisor functionality are integrated into a single product called VCF Automation 9.0. This consolidation empowers customers to effectively manage IT catalogs, multi-tenancy, and modern application development for both enterprises and service providers. When you upgrade your existing vRealize Automation 8.x environment to the latest VCF Automation 9.0 version, your existing organizations will remain unchanged and will be classified VM Apps organizations. Any new tenant created after the upgrade is included in the All Apps organization.

To support these enhancements, you must deploy VCF Automation for All Apps Organization in clean deployments where you are deploying an instance of VCF newly and you do not have VMware Aria Automation 8.x management pack deployed. One adapter can collect information for all the organizations and resources. For more information on Chargeback, see the topic, Performing VCF Automation Based Multitenancy in VCF Operations.

If you have a VMware vSphere Foundation license, the VCF Automation for All Apps Organization integration is not available.

## Considerations for Deploying the Management Pack for VCF Automation for All Apps Organization

The integration and upgrade is based on whether the VMware Aria Automation 8.x management pack was previously installed and whether VCF 9.0 is installed as a clean deployment or an upgraded environment.

**Deploying the VCF Automation 9.0 Management Pack in VCF 9.0**

- VCF 9.0 must be a clean deployment and you should not have VMware Aria Automation 8.x management pack configured.
- **VCF Automation for All Apps Organization** adapter does not support upgraded installations.

If you have a VCF license, the VCF Automation for All Apps Organization adapter is automatically created after VCF Automation is installed. You can deploy VCF Automation in the following two ways:

- Using VCF Operations. For more information, see Deploying VCF Automation as a Day-N Operation.
- Using the VCF Installer. For more information, see Deploy Components by Using VCF Installer to Complete the Converging to VCF Process.

### VCF Automation for All Apps Organization Metrics and Content

To see a list of metrics and content for the VCF Automation for All Apps Organization management pack, go to the VCF Automation for All Apps Organization tile by clicking the **Repository tab** under **Administration** > **Integrations**.

# VCF Automation for VM Apps Organization Account in VCF Operations

VCF Automation for VM Apps Organization management pack enhances the VCF Operations platform by providing cloud-aware visibility into infrastructure operations. This integration allows administrators to monitor the health, performance, and capacity risks of connected cloud environments. Through this functionality, users can import and synchronize cloud accounts into the VCF Operations interface, gaining better insight into cloud zone performance. It also supports efficient workload placement for VMs within managed clusters and offers tools to diagnose and resolve issues with vSphere endpoints directly from the operations dashboard.

You can use VCF Automation for VM Apps Organization to perform some of the following key tasks:

- Gain visibility into the performance and health of cloud zones integrated with VCF Operations.
- Import and synchronize existing cloud accounts from VCF Automation to VCF Operations.
- Manage the workload placement of VMs that are part of the clusters managed by VCF Automation.
- Integrate and troubleshoot vSphere endpoint issues associated with VCF Automation using the VCF Operations dashboard.

  **Note:**

  In this release only vSphere endpoints are supported.

## Configuring the Management Pack for VCF Automation for VM Apps Organization

If you have a VCF license, you can deploy VCF Automation for VM Apps Organization from the VCF Operations repository. This adapter works with a greenfield deployment of VCF 9.0 when you have already installed and configured the VMware Aria Automation 8.x Management Pack.

You must know the following details from the steps in the VM Apps Organization Portal:

- Organization ID
- Credential name and the Refresh Token

For more information, see the topic, Create a Service Account and Get a Refresh Token in VCF Automation for VM Apps.

If VMware Aria Automation 8.x Management Pack was already configured in Aria Operations, and you have upgraded to VCF 9.0, the management pack functions without requiring changes. This is provided that the VMware Aria Automation 8.x APIs remain compatible with VCF Automation 9.0 issued tokens and Cloud Services Platform APIs.

If you are deploying a new VCF 9.0 instance, and you previously had VMware Aria Automation 8.x Management Pack configured, then you must activate theVCF Automation for VM Apps Organization adapter.

1. From the left menu, click **Administration** > **Integrations**.
2. On the **Repository** tab, find the **VCF Automation for VM Apps Organization** tile.
3. Click the **Activate** button. Click **Yes** in the confirmation dialog box.
4. Fill the FQDN of the VCF Automation server. Enter the organization ID from the VM Apps Organization Portal in the **VCF Automation Org ID** field.
5. Set **Auto Discovery** to true.
6. Click the **+** button next in the **Credential** field. Select **Tenant Scoped Service Account** from the drop-down, and enter the Credential Name and the Refresh Token.
7. From the **Cloud Proxy/ Groups** drop-down menu, select the cloud proxy group.
8. Click **Advanced Settings**. From the **User Count** drop-down menu, select the number of user resources to be imported from VCF Automation. The User Count options are 20, 100, 200, 300, 400, and All Users.
9. Click **OK**.
10. Click the **Add** button to add the adapter instance to VCF Operations.

## Advanced Workload Placement for Allocation Model

VCF Operations supports allocation aware advanced workload placement between VCF Automation for VM Apps Organization tenant and VCF Operations for initial placement of virtual machines. The placement recommendation is based on allocation settings that are defined in the VCF Operations policy. The advanced workload placement awareness for allocation works with the existing demand model.

### Need for Allocation Model Awareness

The advanced workload placement based on demand model, depends on the actual demand for resources in a cluster and datastore. If the allocation is based only on the utilization of resources, then it might result in over allocation or over provisioning of resources in clusters. To avoid this, VCF Operations provides allocation model awareness. Allocation model awareness addresses issues related to over allocation or over provisioning by providing you the option of setting the appropriate overcommit ratios in the VCF Operations policy.

The advanced workload placement feature uses the demand model by default and cannot be turned off. To activate the allocation model in addition to the demand model, in VCF Operations configure the appropriate overcommit ratios in policy settings for the preferred clusters, datastores, and datastore clusters.

### Prerequisites

- VCF Operations is configured as an endpoint in VCF Automation for VM Apps Organization tenant. This happens automatically in case of VCF Operations.
- Advanced placement policy in the cloud zone must be activated.
- vCenter Cloud account instance on which the initial provision is done should be same acrossVCF Operations and VCF Automation for VM Apps Organization tenant.
- Overcommit ratios in VCF Operations policy must be configured for the following:
  - For CPU and Memory overcommit ratio - Cluster Compute Resource.
  - For Disk overcommit ratio - Datastore and Datastore Cluster.

### How to Activate the Advanced Placement Policy

To activate the advanced placement policy, in VCF Automation Cloud Assembly Service, navigate to **Infrastructure** > **Cloud Zone** > **Summary** tab and set the placement policy to **ADVANCED**.

**Note:**

If VCF Operations returns no recommendations, then under the placement policy, you can specify if you want VCF Automation for VM Apps Organization tenant to fall back to its default placement using the toggle option.

**How to Activate Allocation Awareness in Advanced Workload Placement Feature**

To activate Allocation awareness in VCF Operations perform the following actions.

1. From the left menu, click **Infrastructure Operations** > **Configurations**, and then click the **Policy Definition** tile.
2. Select the Active policy which is assigned to the Cluster Compute resource under Cloud Zones and make the required changes.
3. Click **Edit Policy** and navigate to the **Capacity** tile.
4. Select the **Object Type** as Cluster Compute Resource and activate the Allocation Model.
5. Set the overcommit ratio as per your requirement and click **Save**.
6. Repeat Step 3 to Step 5 for datastore cluster, vSAN datastores and local datastores.

After the configuration is completed, VCF Automation makes a provisioning request to VCF Operations , and the advanced workload placement engine calculates the recommendation and shares it with VCF Automation for VM Apps Organization tenant. To know the changes after the configuration, view the Allocation metrics and Demand metrics for the cluster.

**Note:**
The allocation model awareness enhancement is limited to advanced workload placement feature. This capability is not extended to workload optimization feature.

## Object Types

VCF Automation brings in cloud accounts and their relationships from VCF Automation for VM Apps Organization tenant into VCF Operations for operational analysis. You can use the following items in the virtual infrastructure as object types in VCF Operations.

- Cloud Zone
- Blueprint
- Project
- Deployment
- Cloud Account
- User
- Organization
- Cloud Automation Services World

## Workload Placement

In VCF Operations , you can configure VCF Automation for VM Apps Organization tenant instances to work with VCF Operations instances. Using VCF Operations you can monitor the placement of existing workloads and optimize the resource usage.

- Verify that the user has privileges of Organizational Owner and Cloud Assembly Administrator set in VCF Automation for VM Apps Organization tenant.
- You must know the vCenter credentials and have the necessary permissions to connect and collect data.
- Verify that VCF Automation for VM Apps Organization tenant is activated from **Administration** > **Integrations** in VCF Operations. See, Configuring the Management Pack for VCF Automation for VM Apps Organization.
- VCF Operations must have the same vCenter Cloud Account configured to match with VCF Automation for VM Apps Organization tenant.
- Ensure that integration is activated for VCF Operations and VCF Automation for VM Apps Organization tenant.

1. Click the Workload Placement tab under **Capacity** > **Optimize**.
2. Click the **View** filter drop-down menu and select the **VRA Managed** objects.

   All the Cloud Zones related to the vCenter are displayed in VCF Operations.
3. Click the **Cloud Zone** you want to optimize.
4. Based on the operational intent, click **Optimize Now**.

   The system creates an optimization plan, which depicts BEFORE and (projected) AFTER workload statistics for the optimization action.
5. If you are satisfied with the projected results of the optimization action, click **NEXT**.
6. Review the optimization moves, then click **BEGIN ACTION**.

   In the scope of VCF Automation for VM Apps Organization tenant integration, VCF Operations sends a move migration request directly to VCF Automation for VM Apps Organization tenant. In the earlier versions, the migration request was sent to the vCenter.

To verify that the optimization action is complete, select **Administration** from the left menu, and click **Recent Tasks** . In the **Recent Tasks** page, use the Status function on the menu bar to locate your action by its status. You can also search using a range of filters. For example, first filter on Starting Time and scroll to the time when you began the action, then select the Object Name filter. Finally, enter the name of one of the VMs in the rebalance plan.

## Pricing for VCF Automation for VM Apps Organization tenant Components in VCF Operations

After you integrate VCF Automation for VM Apps Organization tenant private cloud adapter instances with VCF Operations you can calculate the cost of deployments, projects, and virtual machines of the selected cloud adapter. Pricing provides an overview of the costs related to the cloud environment, cloud resources, and the costs associated with the project.

### How the Pricing Works in VCF Automation for VM Apps Organization Tenant

- VCF Operations understands the constructs defined in VCF Automation for VM Apps Organization tenant and calculates the CPU, RAM, Storage and Additional prices for Projects, Deployments, and virtual machines.
- A single project can have multiple deployments and a single deployment can have multiple virtual machines associated with the deployment.
- Pricing for multiple virtual machines associated with the deployment is the sum of all the resources associated with individual virtual machines.
- If a single project has multiple deployments, then the project pricing is equal to the sum of individual deployments. The deployment can have multiple virtual machines and resources associated with it.
- On day one, the pricing is equal to the cost of resources defined in VCF Operations.
- On day two, the price is calculated using the following formula.
  - Cost of resources for the present day – Cost of resources for the previous day.
- If in case the pricing does not happen as per the definition, then the partial price is set to true, and the pricing is calculated based on the previous days price.
- In VCF Operations, the following new dashboards are included to view the pricing details for the VCF Automation for VM Apps Organization tenant instances.
  - Cloud Automation Environment Overview
  - Cloud Automation Project Cost Overview
  - Cloud Automation Resource Consumption Overview
  - Cloud Automation Top-N Dashboard

### Data Collection Enhancements in VCF Automation for VM Apps Organization Tenant for Pricing in VCF Operations

The following enhancements have been made for the data collection process from VCF Automation for VM Apps Organization tenant for pricing purposes.

- Collect cloud zones with relation to clusters and resources pools from VCF Automation for VM Apps Organization tenant to VCF Operations.
- Collect Projects from VCF Automation for VM Apps Organization tenant with relation to deployments.
- Include project, cloud zone, and blueprint as properties in virtual machines that are deployed in VCF Automation for VM Apps Organization tenant.

#### Upfront Price Support for VCF Automation for VM Apps Organization Tenant Private Cloud Components

VCF Operations supports upfront pricing for VCF Automation for VM Apps Organization in the following ways:

- VCF Operations uses rate cards to provide upfront cost estimates of catalog items just before deployment.
- VCF Automation for VM Apps Organization tenant retrieves the deployment cost and estimated cost from VCF Operations.
- VCF Automation for VM Apps Organization tenant user interface allows you to customize the pricing policies and assign them to the projects or cloud zones.
- If VCF Automation for VM Apps Organization tenant does not specify the pricing policy, then the price is calculated using the VCF Operations cost calculation policy.
- If a custom pricing policy is set for a price calculation, then the deployment and upfront catalog price computation is done as per the custom policy.

#### Upfront Price Support for VMware Cloud on AWS Resources

VCF Operations supports upfront pricing for VMware Cloud on AWS resources in the following ways:

- VCF Operations supports upfront pricing for VMware Cloud on AWS only if rate-based pricing is configured in VCF Automation for VM Apps Organization tenant for VMware Cloud on AWS resources.
- VCF Operations does not support cost-based computation for VMware Cloud on AWS resources.

## Managing Public Cloud Endpoints with VCF Automation for VM Apps Organization Tenant Integration

With the VCF Operations management pack, you can monitor deployments made to public cloud endpoints such as Amazon Web Services (AWS), Google Cloud Platform (GPC), and Microsoft Azure. You can monitor the performance, health, utilization, and availability attributes of deployments made to public cloud endpoints.

This integration supports the VCF Operations management packs for public clouds – MP for AWS, GCP, and Azure. Hence, the public cloud management packs are a prerequisite for this enhancement.

VCF Operations will display the VCF Automation for VM Apps Organization tenant deployed by public cloud resources as long as the same resources are also monitored by VCF Operations for the respective public cloud management packs.

This enhancement also shows all the cloud accounts and cloud zones that are part of public cloud end points.

> **Note:**
>
> To monitor the VCF Automation for VM Apps Organization tenant resources deployed to AWS, GCP, and Azure, you must ensure that the cloud accounts are configured in both VCF Automation for VM Apps Organization tenant and VCF Operations.

#### Import Account Functionality

Users may import AWS, Azure, and vCenter accounts automatically through the **Import Account** option. However, for the GCP account you should manually add the cloud account through **Add Account** option. The credential used to configure the adapter must match the one used in VCF Automation for VM Apps Organization tenant. The traversal specification of VCF Automation for VM Apps Organization tenant has been enhanced to include details about AWS, Azure, and GCP accounts.

## Cloud Zones in VCF Operations

Cloud zones help you to group a set of compute resources and assign capability tags to the zone. The cloud zone is based on accounts/regions, so you must have at least one cloud account configured before you can create a cloud zone. Cloud zones define where and how blueprints configure deployments. You can have one or many cloud zones assigned to each project based on priority and limits.

### How Cloud Zones Work

After you integrate VCF Automation for VM Apps Organization tenant with VCF Operations, you can retrieve cloud zones into VCF Operations. The **Cloud Zones** option is hidden from the user until the integration with VCF Automation is activated from the integration page under **Administration** > **Integrations**.

The Cloud Zones option is activated in VCF Operations , only if the following conditions are met.

- VCF Automation for VM Apps Organization tenant management pack is integrated successfully in VCF Operations.
- VCF Automation for VM Apps Organization tenant objects are discovered in VCF Operations.
- VCF Automation for VM Apps Organization tenant accounts and Aria Operations vCenter Cloud Accounts are synchronized.

All the Cloud Zone objects which are existing in VCF Automation for VM Apps Organization tenant environment, are discovered inVCF Operations. Cloud zones, whose dependent clusters are not discovered in VCF Operations, are not represented in Capacity Overview, Reclaim, and Workload Optimization pages.

# Ping Account in VCF Operations

In VCF Operations, you can configure the ping functionality to verify the availability of end points that exist in your VMware Cloud Foundation platform. The ping functionality is configured at the adapter instance for IP addresses, group of IP addresses, and FQDN's.

Activate the Ping adapter before you configure it.

- If you have multiple adapter instances running on different collectors and both are pinging the same address, you can still get statistics from both the adapter instances for the same IP.
- The FQDN names are checked for validity, the FQDN validation relies on RFC1034 and RFC1123, and only top level domains of the internet are validated. The `.local` domain is not supported as it does not fall into the list of top-level domains in the Domain Name System (DNS) of the Internet.

1. From the left menu, click **Administration** > **Integrations**.
2. In the Accounts tab, click **Add**.
3. Click **Ping**.
4. If the management pack has not been installed previously, Click **Yes** in the dialog box to install the management pack.
5. Configure the Ping adapter instance.

| Option | Description |
|---|---|
| **Name** | Enter a name for the adapter instance. |
| **Description** | Enter the description of the adapter instance. |
| **Unique Name** | Specify the name for the adapter instance. You can use the name to view the metrics published for the adapter instance. |
| **Address List** | Specify the IP address, IP address range, and the FQDN which must be pinged. |
| **Configuration Filename** | Specify the name of the configuration file. The configuration file contains the IP addresses, CIDR information, and FQDN details as a comma-separated file. |
| **Collector / Group** | Select the collector from which this adapter instance must run. |
| **Validate Connection** | Click to check whether the connection is successful or not. |
| **Advanced Settings** | To configure the advanced settings, click the drop-down menu. |

| Option | Description |
|---|---|
| **Wait Interval Time (second)** | Specify the time interval in seconds to wait before running the next batch. Range: 0-300 seconds. |
| **Batch Size** | Specify the number of request packets to send to each target. Range: 20-100. |
| **Interval (millisecond)** | Specify the time the fping waits between successive packets to an individual targets. Greater or equal to 2000 milli seconds. |
| **DNS Name Resolve Interval (minute)** | Specify the time at which you must resolve the DNS name for the next cycle. Minimum value is 15 minutes. |
| **Packet Size (byte)** | Specify the byte size of the packet when you ping. Range: 56-65536 bytes. |
| **Don't Fragment** | Select **False** to fragment the packet and **True** to not fragment the packet. |
| **Generate FQDN Child IPs** | Select **True** to create IP objects by resolved names and add as child of FQDN. |

6. Click **Add**.

   The adapter instance is added to the list. Newly created ping accounts do not start monitoring the data automatically and you must manually initiate the data collection.

After you configure the Ping adapter instance, you can view the adapter details from **Administration** > **Integrations** > **Repository**.

# VCF Operations for networks Account in VCF Operations

The VCF Operations for networks adapter activates integration of VCF Operations with VCF Operations for networks. VCF Operations for networks provides network visibility and analytics to minimize risk during application migration, optimize network performance, manage and scale NSX, and Kubernetes deployments.

- Cloud Integration can be done, when VCF Operations for networks and VCF Operations services are present in the same organization.
- For the cloud integration to be possible, you must have VCF Operations for networks and VCF Operations services in the same geographical location.

This adapter gets problem events from VCF Operations for networks and publishes the alerts in VCF Operations. Alerts are mapped correctly to the common objects between VCF Operations for networks and VCF Operations. Common objects supported in this adapter are vCenter and NSX For the common objects, VCF Operations supports launch-in-context to VCF Operations for networks. This allows the user to perform deep network troubleshooting with the VCF Operations for networks as the context.

The VCF Operations for networks adapter only supports VCF Operations for networks versions 5.2 and above. The VCF Operations for networks adapter can be installed and configured with the on-prem versions of VCF Operations. The VCF Operations for networks adapter does not support cross platform configuration, it should be on-prem VCF Operations to on-prem VCF Operations for networks.

## Configuring VCF Operations for networks

Configure an instance of the VCF Operations for networks in VCF Operations.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Integrations**.
2. From the **Accounts** tab in the **Integrations** page, click **Add Account** and then select the **VCF for Networks** card. You can also activate the management pack from the **Repository** tab, where you will find the card in the **Available Integrations** section. After the management pack is activated, click **Add Account**.
3. Configure the adapter instance.

| Option | Description |
|---|---|
| **Name** | Enter a name for the adapter. |

| Option | Description |
|---|---|
| **Description** | Enter a description. |
| **FQDN/IP** | The FQDN or the IP address of VCF Operations for networks.<br><br>**Note:**<br><br>It is recommended to use the IP address during the VCF Operations for networks instance creation.<br><br>If a VCF Operations for networks instance was configured in VCF Operations 8.18.x using an FQDN instead of an IP address, an automated migration occurs during the upgrade to VCF Operations 9.1.<br><br>• A new adapter instance is automatically created in the **Integrations** > **Accounts** page using the IP address of the server.<br>• The original FQDN-based instance is placed into a stopped state.<br>• All existing objects, data, and relationships are automatically migrated to the new IP-based instance as child objects, ensuring no data loss.<br>• The stopped FQDN-based instance is no longer required and can be safely removed from VCF Operations. |
| **Credential** | Select and add the credential you want to use to sign on to the environment from the drop-down menu. To add new credentials to access the environment of this management pack, click the plus sign.<br><br>• **Credential Kind**. Select and configure the Credential Type. You can select either the Local, LDAP, or VIDM<br><br>**Note:**<br><br>This Management pack supports only the Local, LDAP, and vIDM users that are added in the **User Management** settings of VCF Operations for networks.<br><br>– Local - Network Credentials. Enter the credential name, user name of the local user configured in VCF Operations for networks, and password.<br>– LDAP - Network Credentials. Enter the credential name, LDAP domain configured in VCF Operations for networks, LDAP user name, and LDAP password.<br>– vIDM - Network Insight Credentials. Enter the credential name, vIDM FQDN/IP integrated with VCF Operations for networks, vIDM user name, and vIDM password for that vIDM user.<br>**Credential Name**. |
| **Cloud Proxy/ Group** | Select the required cloud proxy group . |
| **Validate Connection** | Test Connection should be successful. |

4. The VCF Operations for networks instance collects events based on common data sources between VCF Operations and VCF Operations for networks. When you deactivate the **Import problem events as based on common data sources** option, all the events are imported into the VCF Operations.
5. You can collect user-defined events of VCF Operations for networks as notifications in VCF Operations. To do so, activate the **Import User defined events as Notifications**.
6. Select the severity of the problem events you want to import. By default, all the problem events with moderate and critical severities are imported.
7. Click **Add**.

   The VCF Operations for networks instance is added to the list. Newly created VCF Operations for networks accounts do not start monitoring the data automatically and you must manually initiate the data collection.

## Viewing Dropped Flow Alert for Check Point and Palo Alto

You can view dropped flow alerts for Check Point and Palo Alto devices in VCF Operations.

- Ensure VCF Operations for networks is installed in VCF Operations.
- Ensure log management is configured as a syslog server in Check Point and Palo Alto devices.
- Ensure ports 514 and 1514 are open so that they can communicate with log management.

Before VCF 9.1, the Check Point and Palo Alto dropped flow alerts were raised in VCF Operations for networks. Starting with VCF 9.1, you do not need to add the 9.1 version of the log management data source in VCF Operations for networks to receive the Check Point and Palo Alto dropped flow alerts. These alerts are raised and can be viewed in VCF Operations.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configuration**.
   - To view the alert definitions, click **Alert Definitions**, and filter the alerts with the object type set to **Log Fallback Object**. For more information, see Alert Definitions in VCF Operations.
   - To view the symptom definitions, click **Symptom Defitions**, and then click **Logs**. For more information, see Symptom Definitions in VCF Operations.

   The following dropped flow alerts are available:

   - Dropped flows are detected on a Check Point device through VCF Operations for logs.
   - Dropped flows are detected on a Palo Alto device through VCF Operations for logs.
2. To view raised alerts, in the left navigation pane, click **Alerts**, group the alerts by object type, and expand the **Log Fallback Object**.

   **Note:**

   When log management reports a dropped flow received from a Check Point or a Palo Alto device, VCF Operations will raise an alert.
3. Click on an alert to view alert details and check for symptoms.

   For example, the alert for the Check Point device displays the following symptom message: “The Warning symptom 'Dropped flows are detected on Check Point device through log management has been observed on Log Fallback Object." For more information, see Understanding Alerts in VCF Operations.
4. In the **Recommendations** section of the alert, click **analyze logs** to open the log analysis page with the query that is required to filter the dropped flow log messages.
5. In the **Stream** section, expand the log entry to view the required field information for the dropped flow. For more information, see Configuring and Analyzing Logs.

The dropped flow alerts for the Check Point or Palo Alto device are raised in VCF Operations, and the associated log messages are available for analysis in log management.

# vSphere Supervisor Monitoring

The management pack that is introduced in VCF 9.0 helps you monitor the resource utilization and performance of vSphere Supervisor and VKS clusters and resources by using VCF Operations. This is an internal, native management pack that is activated by default. Data collection is enabled through the vCenter integration.

## What is vSphere Supervisor Monitoring

VCF 9.0 introduces an integrated monitoring solution in the form of the vSphere Supervisor management pack to track Kubernetes resource allocation, availability, and performance.

vSphere Supervisor management pack is a native, but internal management pack which is automatically installed when you configure the vCenter integration. This pack enables automated resource detection and monitoring specifically for vSphere Supervisor and vSphere Kubernetes Service VKS clusters. When you enable a setting in the vCenter integration, VCF Operations automatically detects and creates vSphere Supervisor adapter instances for streamlined monitoring. For more details, see Enabling Monitoring for vSphere Supervisor.

Previously, the Management Pack for Kubernetes monitored Kubernetes clusters using a pull-based model. The vSphere Supervisor management pack monitors vSphere Supervisor and VKS clusters using push based metric collection. Unlike the traditional pull-based approach, the vSphere Supervisor adapter uses Telegraf and the data is pushed directly to a Cloud Proxy. For more information on Telegraf, see, Monitoring Application Services using Product-Managed Telegraf. The Cloud Proxy acts as an intermediary to store and forward data, ensuring reliability even if temporary network disruptions occur. To see a list of metrics collected, go to the topic, vSphere Supervisor Metrics.

### How vSphere Supervisor Monitoring Works

The Application Monitoring adapter is responsible for collecting metrics and organizing the resource inventory. For every Cloud Proxy which is configured in VCF Operations, an Application Monitoring adapter is automatically created for it.

Telegraf collects metrics from different Kubernetes objects inside the vSphere Supervisor and VKS clusters and forwards them to the Cloud Proxy. Monitoring of Kubernetes objects such as vSphere Pods and containers is supported. vSphere Pods are only supported with Supervisors that are configured with NSX as the networking stack. For vSphere Supervisor, monitoring of these objects is enabled by default. For VKS clusters, you must enable it.

During the first collection cycle, certificates are created, and the endpoint details are sent to the vSphere Supervisor. The vSphere Supervisor adapter ensures that metrics are directed to the appropriate Cloud Proxy.

Once the Application Monitoring adapter receives metrics, it organizes and stores the inventory. If needed, you can switch to any other Cloud Proxy, and the vSphere Supervisor adapter will automatically update the endpoint details.

## Key Features of the vSphere Supervisor Management Pack

The vSphere Supervisor management pack provides enhanced observability and lifecycle management for VKS.

| | |
|---|---|
| **Push-Based Metrics Collection** | The vSphere Supervisor adapter introduces a push-based model that leverages Telegraf, enabling improved efficiency in metric collection. The metrics are pushed into a Cloud Proxy. This ensures faster and more reliable metric collection. |
| **Seamless Integration** | Enabling the vSphere Supervisor management pack is simple, and allows for automatic detection and configuration of Kubernetes clusters within vCenter instances. |
| **Support for vSphere Supervisor and VKS Clusters** | • vSphere Supervisor **:** Serves as a platform for deploying VKS clusters and ensure overall system stability.<br>• VKS clusters**:** Streamline Kubernetes clusters that host application workloads and run user-deployed applications.<br>The vSphere Supervisor management pack can monitor both VKS clusters and vSphere Supervisor, ensuring quick identification of performance issues. |
| **Automated Cluster Discovery** | When enabled, the vSphere Supervisor adapter identifies all the vSphere Supervisor and VKS clusters in a vCenter environment and creates the necessary Application Monitoring adapter instances for monitoring. |

| | |
|---|---|
| **Enhanced Observability with Dashboards and Metrics** | The vSphere Supervisor adapter provides details about the vSphere Supervisor and VKS cluster infrastructure, and applications running in the VKS clusters. |

## Steps to Monitor vSphere Supervisor and Resources

Use VCF Operations to enable the vSphere Supervisor management pack, and monitor and troubleshoot vSphere Supervisor and VKS clusters.

### Enabling Monitoring for vSphere Supervisor

You enable or disable vSphere Supervisor monitoring when configuring the vCenter cloud account integration in VCF Operations. The vSphere Supervisor internal management pack is enabled or disabled based on this setting. One vCenter cloud account can have multiple Supervisors enabled. When the setting is enabled, an adapter instance for each supervisor is automatically created and monitored. When it is disabled, the adapter instance is still available, but metrics are not collected.

- Ensure that you are logged in to VCF Operations with the required permissions to add or edit a vCenter cloud account.

1. From the left menu, click **Administration** > **Integrations**.
2. On the **Accounts** tab, click **Add**. On the Accounts Type page, click **vCenter**. For more information about configuring a vCenter cloud account, see the topic, Configure a Cloud Account.
3. Alternatively, select an existing vCenter account and click **Edit** from the vertical ellipsis.
4. Expand **Advanced Settings**.
5. Verify that the value of **Enable vSphere Supervisor Collection** is set to true. This is the default setting.
6. Once you have configured the rest of the settings, click **ADD**.

After enabling vSphere Supervisor monitoring the following settings are automatically configured:

- vSphere Supervisor sends data to the Cloud Proxy VM running the Application Monitoring adapter.
- The Telegraf package defines which metrics are collected and how often they are sent.

To verify that the adapter is collecting data, go to **Inventory (Detailed View)** > **Integrations** > **vSphere Supervisor** and check the **Summary** tab in the **Supervisor World** object.

To see the vSphere Supervisor adapters which are configured and collecting metrics, go to **Infrastructure Operations** > **Dashboards and Reports** > **Inventory** and look for the **vSphere Supervisor** dashboards.

### Enabling Monitoring for VKS Clusters

VCF 9.1 leverages the VKS add-on management system to install Telegraf and Prometheus in VKS clusters to enable automated monitoring of VKS Clusters. For more information on the VKS add-on management system, see Installing Standard Packages on VKS Clusters.

#### Monitoring Prerequisites for VKS Clusters

Before you can enable monitoring for VKS clusters, ensure that the following prerequisites are met in your VCF environment.

- Work with the VCF Operations administrator to ensure that vCenter and its associated Supervisor are set up for monitoring in VCF Operations . See Steps to Monitor vSphere Supervisor and Resources.
- Ensure that the Metrics Aggregator VCF Service is installed in the Supervisor. This service is installed by default with VCF Automation. You can verify that the service is installed correctly using the Provider Console in VCF Automation. See Using Services with VCF Automation.
- Supervisor must have the addon repository installed that supports the automated installation of Telegraf and Prometheus addons for monitoring purposes. Ensure the required Telegraf and Prometheus AddonRepository, Addon, AddonConfigDefinition CR's are available in Supervisor. See Managing Addons in VKS Clusters for more information.

After the prerequisites are met, any new VKS cluster is automatically monitored. If you want to disabled monitoring, see Disable Automated Monitoring on VKS Clusters using VCF Automation or Disable Monitoring on VKS Clusters using VCF CLI.

#### Verify Authentication and Security

Supervisor generates the certificates for each Telegraf instance in a VKS Cluster. These certificates are rotated by the cert-manager running on the Supervisor and automatically updated in the VKS Cluster. Metrics Aggregator terminates the TLS connection from the VKS Cluster, thus securing the communication between the Supervisor and VKS clusters. Metrics Aggregator uses the certificates provided by Cloud Proxy for communication between Metrics Aggregator and Cloud Proxy. Cloud Proxy terminates the TLS Connection thereby securing communication between the Supervisor and Cloud Proxy.

For more information on monitoring VKS Clusters, see Monitoring VKS Clusters Using VCF Operations.

### Disabling Monitoring for the Supervisor

You can disable monitoring for Supervisors from VCF Operations. After you disable monitoring, the objects are not displayed in the Object Relationship chart in the Inventory when you browse to the vSphere Supervisor object. The endpoint is removed from the Telegraf configuration of the Supervisor.

1. Go to the **Inventory Management** tile in the **Inventory** > **Configurations**.
2. In the search tab, enter the name of the Supervisor for which you want to disable monitoring. This is a quick way to find the object. Alternately, you can browse the list of objects and locate the objects.
3. Click the edit icon to edit the object. The **Edit Object** dialog box opens.
4. From the drop-down, select **No** for the **Enable Monitoring** setting under the **Advanced Settings**. The default value is **Yes**.
5. Click **OK**.

### Viewing Supervisor Metrics

This table lists the metrics which the Supervisor management pack collects. Go to the **Metrics** tab in the **vSphere Supervisor** object page by navigating to **Administration** > **Integrations** > **Repository** to view their details.

#### Supervisor Metrics

A list of the Supervisor metrics available in VCF Operations is in the following table:

| Object | Element | Metric | Type | Unit |
|---|---|---|---|---|
| Container | CPU | Usage | Metric | Millicores |
| Container | CPU | Throttled | Metric | % |
| Container | CPU | Usage | Metric | % |
| Container | CPU | Requests | Property | Millicores |
| Container | CPU | Limits | Property | Millicores |
| Container | Memory | Usage | Metric | MB |
| Container | Memory | Usage | Metric | % |
| Container | Memory | Limits | Property | MB |
| Container | Memory | Fail Count | Metric | |
| Container | Memory | Requests | Property | MB |
| Container | Summary | Restart Count | Property | Count |
| Container | Performance | Overall Performance | Metric | |
| Container | Images | Image Name | Property | |
| Container | Container readiness | Container Ready | Property | |

| | | | | |
|---|---|---|---|---|
| Pod | CPU | Usage | Metric | Millicores |
| Pod | CPU | Worst Throttled Container | Metric | % |
| Pod | CPU | Requests | Property | Millicores |
| Pod | CPU | Limits | Property | Millicores |
| Pod | Memory | Usage | Metric | MB |
| Pod | Memory | Limits | Property | MB |
| Pod | Memory | Requests | Property | MB |
| Pod | Memory | Total Out of Memory Events | Metric | Count |
| Pod | Network | Receive Packets Dropped | Metric | % |
| Pod | Network | Transmit Packets Dropped | Metric | % |
| Pod | Network | Receive Errors Total | Metric | % |
| Pod | Network | Transmit Errors Total | Metric | % |
| Pod | Summary | Status | Property | |
| Pod | Summary | Total Restarts | Computed Metric | |
| Deployment | CPU | Usage | Metric | Millicores |
| Deployment | CPU | Top 95% CPU Throttling Workload's | Metric | % |
| Deployment | CPU | Requests | Property | Millicores |
| Deployment | CPU | Limits | Property | Millicores |
| Deployment | Memory | Usage | Metric | MB |
| Deployment | Memory | Limits | Property | MB |
| Deployment | Memory | Requests | Property | MB |
| Deployment | Memory | Top 95% Out Of Memory Events | Metric | Count |
| Deployment | Network | Top 95% Network Receive Packets Drop | Metric | % |
| Deployment | Network | Top 95% Network Transmit Packets Drop | Metric | % |
| Deployment | Network | Top 95% Network Receive Errors Total | Metric | % |
| Deployment | Network | Top 95% Network Transmit Errors Total | Metric | % |
| Deployment | Status | Replicas | Metric | |
| Deployment | Summary | No of Containers | Metric | |
| Deployment | Summary | No of Pods | Metric | |
| Deployment | Summary | Pods Not Running | Metric | |
| Deployment | Summary | Status | Metric | |
| Deployment | Status | Ready Replica's | Metric | |
| Deployment | Performance | Overall Performance | Metric | % |
| Deployment | Performance | Network Override | Metric | % |
| Node | CPU | Usage | Metric | Millicores |
| Node | CPU | Top 95% CPU Throttling Workload's | Metric | % |
| Node | CPU | Requests | Property | Millicores |
| Node | CPU | Limits | Property | Millicores |

| | | | | |
|---|---|---|---|---|
| Node | Memory | Usage | Metric | MB |
| Node | Memory | Limits | Property | MB |
| Node | Memory | Requests | Property | MB |
| Node | Network | Top 95% Network Receive Packets Drop | Metric | % |
| Node | Network | Top 95% Network Transmit Packets Drop | Metric | % |
| Node | Network | Top 95% Network Receive Errors Total | Metric | % |
| Node | Network | Top 95% Network Transmit Errors Total | Metric | % |
| Node | API Server Metrics | Request Rate | Metric | |
| Node | API Server Metrics | Request Error Rate | Metric | |
| Node | Controller Manager Metrics | Work Queue Depth | Metric | |
| Node | ETCD Metrics | Is Leader | Metric | |
| Node | ETCD Metrics | Leader changes | Metric | |
| Node | ETCD Metrics | Proposals Applied | Metric | |
| Node | ETCD Metrics | Proposals Committed | Metric | |
| Node | ETCD Metrics | Proposals Pending | Metric | |
| Node | ETCD Metrics | Server Size | Metric | bytes |
| Node | ETCD Metrics | Server Used | Metric | bytes |
| Node | API Server | API Server Up Status | Property | |
| Node | Mvcc DB | Mvcc DB Total Size | Metric | |
| Node | Mvcc DB | Mvcc DB Size In Use | Metric | |
| Node | ETCD Metrics | Etcd Disk Space Usage | Metric | |
| Node | ETCD Metrics | Etcd Server Has Leader | Metric | |
| Node | ETCD Metrics | Top 95% Etcd Disk WAL Fsync Duration | Metric | |
| Node | ETCD Metrics | Top 95% Etcd Disk Backend Commit Duration | Metric | |
| Node | ETCD Metrics | Etcd Network Peer Sent Failures | Metric | |
| Node | Memory | Total Used | Metric | GB |
| Node | CPU | Allocatable CPU | Property | Core |
| Node | Summary | Capacity CPU | Property | Core |
| Node | Summary | Allocatable Memory | Property | GB |
| Node | Summary | Capacity Memory | Property | GB |
| Node | Summary | External IP | Property | |
| Node | Summary | Internal IP | Property | |
| Node | Summary | Number of pods | Computed Metric | |
| Node | Summary | Node Condition - Memory | Property | |
| Node | Summary | Node Condition - Disk | Property | |
| Node | Summary | Node Condition - PID | Property | |
| Node | Summary | Container run-time version | Property | |
| Node | Summary | Kernel version | Property | |

| | | | | |
|---|---|---|---|---|
| Node | Summary | Kube proxy version | Property | |
| Node | Summary | Kubelet Version | Property | |
| Node | Summary | OS Image | Property | |
| Node | Summary | MOID | Property | |
| Node | Summary | ETCD Status | Property | |
| Node | Summary | Controller Manager Status | Property | |
| Node | Summary | API Server Status | Property | |
| Node | Summary | Scheduler Status | Property | |
| Node | Summary | Kubelet Status | Property | |
| Node | Summary | Role | Property | |
| Node | Summary | Status | Property | |
| Namespace | CPU | Usage | Metric | Millicores |
| Namespace | CPU | Top 95% CPU Throttling Workload's | Metric | % |
| Namespace | CPU | Requests | Property | Millicores |
| Namespace | CPU | Limits | Property | Millicores |
| Namespace | Memory | Usage | Metric | MB |
| Namespace | Memory | Limits | Property | MB |
| Namespace | Memory | Requests | Property | MB |
| Namespace | Memory | Top 95% Out Of Memory Events | Metric | Count |
| Namespace | Network | Top 95% Network Receive Packets Drop | Metric | % |
| Namespace | Network | Top 95% Network Transmit Packets Drop | Metric | % |
| Namespace | Network | Top 95% Network Receive Errors Total | Metric | % |
| Namespace | Network | Top 95% Network Transmit Errors Total | Metric | % |
| Namespace | Summary | No of Pods | Metric | Count |
| Namespace | Summary | No of Containers | Metric | Count |
| Namespace | Summary | No of Namespaces | Metric | Count |
| Namespace | Summary | No of Deployments | Metric | Count |
| Namespace | Summary | No of Statefulsets | Metric | Count |
| Namespace | Summary | No of Daemonsets | Metric | Count |
| Namespace | Summary | Number of Pods Not running | Metric | |
| Cluster | Summary | Number of Pods Not running | Metric | |
| Cluster | CPU | Usage | Metric | Cores |
| Cluster | CPU | Top 95% CPU Throttling Workload's | Metric | % |
| Cluster | CPU | Requests | Metric | Cores |
| Cluster | CPU | Limits | Metric | Cores |
| Cluster | Memory | Usage | Metric | MB |
| Cluster | Memory | Limits | Property | GB |
| Cluster | Memory | Requests | Property | GB |

| | | | | |
|---|---|---|---|---|
| Cluster | Memory | Top 95% Out Of Memory Events | Metric | Count |
| Cluster | Network | Top 95% Network Receive Packets Drop | Metric | % |
| Cluster | Network | Top 95% Network Transmit Packets Drop | Metric | % |
| Cluster | Network | Top 95% Network Receive Errors Total | Metric | % |
| Cluster | Network | Top 95% Network Transmit Errors Total | Metric | % |
| Cluster | Summary | No of Nodes | Metric | Count |
| Cluster | Summary | No of Pods | Metric | Count |
| Cluster | Summary | No of Containers | Metric | Count |
| Cluster | Summary | No of Namespaces | Metric | Count |
| Cluster | Summary | No of Deployments | Metric | Count |
| Cluster | Summary | No of Statefulsets | Metric | Count |
| Cluster | Summary | No of Daemonsets | Metric | Count |
| Cluster | Summary | No of Pods not Running | Metric | Count |
| Cluster | Summary | CPU Allocatable | Computed Metric | Cores |
| Cluster | Summary | CPU Capacity | Computed Metric | Cores |
| Cluster | Summary | Memory Allocatable | Computed Metric | GB |
| Cluster | Summary | Memory Capacity | Computed Metric | GB |
| Cluster | Summary | Node(s) Down | Computed Metric | |
| Cluster | Summary | Kubelet(s) Down | Computed Metric | |
| VKS Cluster | Summary | Available | Metric | |
| VKS Cluster | Summary | Addons Available | Metric | |
| VKS Cluster | Summary | Total Capacity | Metric | GB |
| VKS Cluster | Summary | Total Usage | Metric | GB |
| VKS Cluster | Summary | Total Available | Metric | GB |
| VKS Cluster | Summary | Usage | Metric | % |
| VKS Cluster | Summary | Number of PVCs with utilization greater than 80 | Metric | |
| IaaS Controllers | Summary | <Controller_Name> Workqueue Depth | Metric | Percent |
| CSI Controller | Summary | Top 95% vSphere Request Ops | Metric | Percent |
| CSI Controller | Summary | Top 95% CNS Volume Ops | Metric | |
| Supervisor service | Summary | Status | Property | |
| Supervisor service | Summary | Version | Property | |
| Supervisor service | Summary | Service Id | Property | |
| Supervisor service | Summary | Namespace | Property | |
| Persistent Volume Claim (Supervisor only) | Summary | Storage Class | Property | |
| Persistent Volume Claim (Supervisor only) | Summary | Volume Mode | Property | |
| Persistent Volume Claim (Supervisor only) | Summary | Status | Property | |

| | | | | |
|---|---|---|---|---|
| Persistent Volume Claim (Supervisor only) | Summary | Storage Request | Property | GB |
| Persistent Volume Claim (Supervisor only) | Summary | Persistent Volume Capacity | Property | GB |
| Persistent Volume Claim (Supervisor only) | Summary | Persistent Volume Name | Property | |

## Viewing vSphere Supervisor Dashboards

When the vSphere Supervisor management pack is enabled, you can view the metrics and resource information in VCF Operations using the three types of dashboards.

### Where You Find the vSphere Supervisor Dashboards

Go to **Infrastructure Operations** > **Dashboards and Reports** > **Inventory**. Expand the **All** folder and look for the **vSphere Supervisor** dashboards.

### Types of vSphere Supervisor Dashboards

- Kubernetes Inventory
  - Infrastructure details in your vSphere Supervisor, via nodes.
  - Pod and container status via namespace.
  - VKS Clusters lifecycle metrics.
- Kubernetes Performance
  - CPU and memory usage.
  - vSphere Supervisor and VKS Clusters performance.
- vSphere Supervisor
  - Monitor the critical components of the vSphere Supervisor.

### Using the vSphere Supervisor Dashboards

Together, these dashboards offers the following capabilities:

- High-Level Cluster Overview
  - The dashboard offers a summary of cluster health, showing key metrics such as CPU usage, memory consumption, and node availability.
  - You can quickly spot clusters with performance bottlenecks before diving into detailed analysis.
- Drill-Down Approach for Root Cause Analysis
  - If an issue is detected at the cluster level, you can navigate into child objects such as namespaces, nodes, and pods to pinpoint the source.
  - Example: If CPU throttling is detected at the cluster level, users can filter to see which namespaces or pods are consuming excessive CPU.
- Real-Time Monitoring with Push-Based Data Collection
  - Unlike traditional pull-based methods, metrics are pushed to the Cloud Proxy in real time, allowing for faster detection of performance spikes.
  - Users can correlate metrics over time, helping them understand trends that may lead to potential failures.
- Supervisor and Guest Cluster Insights
  - The Supervisor Dashboard monitors critical lifecycle components. If a Supervisor goes down, it may impact all associated VKS Clusters.
  - You can see which clusters are affected and take corrective action immediately.
- Debugging Pods and Containers
  - The dashboard provides alerts on failed pods, frequent container restarts, and out-of-memory (OOM) errors.
  - You can filter by node or namespace to isolate problematic workloads.

### Mapping Supervisor and VKS Cluster Nodes to VM and Node objects

All Kubernetes Nodes of the Supervisor or VKS clusters are represented in VCF Operations as two different objects.

1. As a Kubernetes Node that is created based on metrics that are collected through Telegraf.
2. As a vCenter VM or ESX host created based on inventory and metrics that are collected from vAPI and the ESX host.

These objects will be mapped to each other so that the correlation is clear.

Tthe following mappings are valid for the Supervisor:

- Supervisor Control Plane VM -> Node (control plane)
- ESX host - Node(worker)
- ESX host - Node(worker)

Following are example mappings for VKS clusters:

- VM -> Node (control plane)
- VM -> Node (worker)
- VM -> Node (worker)

# Configuring and Managing Policies

To create a policy, you can inherit the settings from an existing policy, and you can modify the settings in existing policies if you have adequate permissions. After you create a policy, or edit an existing policy, you can apply the policy to one or more groups of objects.

## Policies

A policy is a set of rules that you define for VCF Operations to use to analyze and display information about the objects in your environment. You can create, modify, and administer policies to determine how VCF Operations displays data in dashboards, views, and reports.

### How Policies Relate to Your Environment

VCF Operations policies support the operational decisions established for your IT infrastructure and business units. With policies, you control what data VCF Operations collects and reports on for specific objects in your environment. Each policy can inherit settings from other policies, and you can customize and override various analysis settings, alert definitions, and symptom definitions for specific object types, to support the service Level agreements and business priorities established for your environment.

When you manage policies, you must understand the operational priorities for your environment, and the tolerances for alerts and symptoms to meet the requirements for your business critical applications. Then, you can configure the policies so that you apply the correct policy and threshold settings for your production and test environments.

Policies define the settings that VCF Operations applies to your objects when it collects data from your environment. VCF Operations applies policies to newly discovered objects, such as the objects in an object group. For example, you have an existing VMware adapter instance, and you apply a specific policy to the group named World. When a user adds a new virtual machine to the vCenter instance, the VMware adapter reports the virtual machine object to VCF Operations. The VMware adapter applies the same policy to that object, because it is a member of the World object group.

To implement capacity policy settings, you must understand the requirements and tolerances for your environment, such as CPU use. Then, you can configure your object groups and policies according to your environment.

- For a production environment policy, a good practice is to configure higher performance settings, and to account for peak use times.
- For a test environment policy, a good practice is to configure higher utilization settings.

VCF Operations applies the policies in the priority order, as they appear in the priority column. When you establish the priority for your policies, VCF Operations applies the configured settings in the policies according to the policy rank order to analyze and report on your objects. You must have privileges to access specific features in the VCF Operations user interface. The roles associated with your user account determine the features you can access and the actions you can perform. To set and change the priority of any active policy complete the following steps:

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile.
   In the **Policy Definiton** page, click the horizontal ellipsis, and click **Reorder Policies**.

   **Note:**
   The Reorder Policies option is activated only if there is more than one active policy.
2. In the Reorder Policies window, select the policy and drag it up or down to change the priority.
3. Click **ok** to save the changes made to the priority.

The priority for the Default Policy is always designated with the letter D, and the other active policies are prioritized with numbers 1, 2, and so on. Policy with priority 1 indicates the highest priority. When you assign an object to be a member of multiple object groups, and you assign a different policy to each object group, VCF Operations associates the highest ranking policy with that object.

When multiple policies apply to an object in VCF Operations, the policy with the highest priority is enforced on the object. The only exception is direct assignment: a lower-priority policy is enforced if it is assigned directly to the object.

**Table 1187: Configurable Policy Rule Elements**

| Policy Rule Elements | Thresholds, Settings, Definitions |
|---|---|
| Workload | Configure symptom thresholds for Workload. |
| Time Remaining | Configure thresholds for the Time Remaining. |
| Capacity Remaining | Configure thresholds for the Capacity Remaining. |
| Maintenance Schedule | Sets a time to perform maintenance tasks. |
| Attributes | An attribute is a collectible data component. You can activate or deactivate metric, property, and super metric attributes for collection, and set attributes as key performance indicators (KPIs). A KPI is the designation of an attribute that indicates that the attribute is important in your own environment. |
| Alert Definitions | Activate or deactivate combinations of symptoms and recommendations to identify a condition that classifies as a problem. |
| Symptom Definitions | Activate or deactivate test conditions on properties, metrics, or events. |

### How Upgrades Affect Your Policies

After you upgrade VCF Operations from a previous version, you might find newly added or updated default settings of policies such as, new alerts and symptoms. Hence, you must analyze the settings and modify these settings to optimize them for your current environment. If you apply the policies used with a previous version of VCF Operations, the manually modified policy settings remain unaltered.

## Policy Decisions and Objectives

Implementing policy decisions in VCF Operations is typically the responsibility of the Infrastructure Administrator or the Virtual Infrastructure Administrator, but users who have privileges can also create and modify policies.

You must be aware of the policies established to analyze and monitor the resources in your IT infrastructure.

- If you are a Network Operations engineer, you must understand how policies affect the data that VCF Operations reports on objects, and which policies assigned to objects report alerts and issues.
- If you are the person whose role is to recommend an initial setup for policies, you typically edit and configure the policies in VCF Operations.
- If your primary role is to assess problems that occur in your environment, but you do not have the responsibility to change the policies, you must still understand how the policies applied to objects affect the data that appears in VCF Operations. For example, you might need to know which policies apply to objects that are associated with particular alerts.
- If you are a typical application user who receives reports from VCF Operations, you must have a high-level understanding of the operational policies so that you can understand the reported data values.

# Policy Definitions

The **Policy Definition** page displays the base settings, default policy, and other best practice policies that VCF Operations includes. You can use the **Policy Definition** page to create your own policies. The **Policy Definition** page includes all the configurable settings for the policy elements, such as workload, capacity and time remaining, and so on.

## How Policy Definitions Work

Use the options in **Policy Definition** page to create your own policy from an existing policy, or to override the settings from an existing policy so that you can apply the new settings to groups of objects. You can also import or export a policy and reorder the policies.

Select a policy to display its details in the right pane. The right pane displays a high-level overview of all the details and options for that policy where these details are categorized in tabs. Expand each category to view all the related details.

When you add or edit a policy, you access the policy workspace where you select the base policies and override the settings for metrics and properties, alerts and symptoms, capacity, compliance, workload automation, and groups and objects. In this workspace, you can also apply the policy to objects and object groups. To update the policy associated with an object or object group, the role assigned to your user account must have the Manage Association permission activated for policy management.

## Where You Manage the Policy Definitions

To manage policy definitions, In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile. A list of policies available to use for your environment are displayed.

**Table 1188: Policy Definition Options**

| Option | Description |
|---|---|
| ADD | Create a policy from an existing policy. |
| Horizontal Ellipsis > <Option> | • Edit. Customize the policy so that you can override settings for VCF Operations to analyze and report data about the associated objects.<br>• Delete. Remove a policy from the list.<br>• Set Default Policy. You can set any policy to be the default policy, which applies the settings in that policy to all objects that do not have a policy applied. When you set a policy to be the default policy, the priority is set to D, which gives that policy the highest priority.<br>• Export. Downloads the policy.<br>• Import. Allows you to import policies. To import:<br>– Click the **Import** option from the horizontal ellipsis.<br>– Click **Browse** and select the file to import.<br>– Select if you want to Overwrite or Skip the file in case of a conflict.<br>– Click **Import** to import the policy, and click **Done**.<br>**Note:**<br>To import or export a policy, the role assigned to your user account must have the Import or Export permissions activated for policy management.<br>• Reorder Policies. Change the priority of the active policies. |
| Filters | Limits the list based on the text you type.<br>You can also filter by:<br>• Name<br>• Description<br>• Modified By |

| Option | Description |
|---|---|
| Policies data grid | VCF Operations displays the high-level details for the policies.<br>• Name. Name of the policy as it appears in the Add or Edit Policy workspace, and in areas where the policy applies to objects, such as in Custom Groups.<br>• Status: Indicates whether the policy is active or inactive.<br>• Priority: Indicates the priority of the policy, if set.<br>• Description. Meaningful description of the policy, such as which policy is inherited, and any specific information users need to understand the relationship of the policy to one or more groups of objects.<br>• Last Modified. Date and time that the policy was last modified.<br>• Modified by: Displays the user who last modified the policy.<br>To view the data grid with the list of policies, click **Policy Definition** at the top of the page. |
| Policies data grid > Right Pane | The right pane displays the name and description of the policy from which the settings are inherited, the policy priority, and the option to edit the policy. From the right pane, you can view the complete group of settings that include both customized settings and the settings inherited from the base policies selected when the policy was created.<br>**Note:**<br>The right pane is only displayed after you select a policy from the grid view.<br>• Metrics and Properties: Displays all the attribute types included in the policy. Attribute type includes, metrics properties, and super metrics.<br>• Alerts and Symptoms: Displays all the alert and symptom definitions included in the policy. The Alert Definitions tabs display an overview of the alert definition, criticality, symptom, and state. The Symptoms Definitions tab displays an overview of the symptom name, criticality, and the metric name.<br>• Capacity: Displays an overview of all the thresholds of the objects included in the policy.<br>• Maintenance Schedule: Displays maintenance schedule elements that are defined locally in the policy. You can view and override the policy Maintenance Schedule analysis setting.<br>• Compliance: Displays the compliance thresholds inherited from the base policy or set while creating the policy.<br>• Workload Automation: Displays the details of the workload optimized in your environment per your definition.<br>• VC Pricing: Displays vCenter pricing details. You can add and assign new pricing cards to a vCenter instance and clusters in VCF Operations.<br>• Configuration Management Templates: Displays the vCenter or cluster configuration management templates. If you have not assigned your vCenter or cluster configuration templates to policies at the time of template creation, you can do so after the template creation process.<br>• Groups and Objects: Displays the object or object groups associated with the selected policy and the names of the objects in your environment, their object types, and associated adapters. When a parent group exists for an object, it is shown here. |

## Operational Policies

Determine how to have VCF Operations monitor your objects, and how to notify you about problems that occur with those objects.

VCF Operations Administrators assign policies to objects or object groups and applications to support Service Level Agreements (SLAs) and business priorities. When you use policies with objects or object groups, you ensure that the rules defined in the policies are quickly put into effect for the objects in your environment.

With policies, you can:

- Activate and deactivate alerts.
- Control data collections by persisting or not persisting metrics on the objects in your environment.
- Configure the product analytics and thresholds.
- Monitor objects and applications at different service levels.

- Prioritize policies so that the most important rules override the defaults.
- Understand the rules that affect the analytics.
- Understand which policies apply to objects or object groups.

VCF Operations includes a library of built-in active policies that are already defined for your use. VCF Operations applies these policies in priority order.

When multiple policies apply to an object in VCF Operations, the policy with the highest priority is enforced on the object. The only exception is direct assignment: a lower-priority policy is enforced if it is assigned directly to the object.

When you apply a policy to an object or an object group, VCF Operations collects data from the objects based on the thresholds, metrics, super metrics, attributes, properties, alert definitions, and problem definitions that are activated in the policy.

The following examples of policies might exist for a typical IT environment.

- Maintenance: Optimized for ongoing monitoring, with no thresholds or alerts.
- Critical Production: Production environment ready, optimized for performance with sensitive alerting.
- Important Production: Production environment ready, optimized for performance with medium alerting.
- Batch Workloads: Optimized to process jobs.
- Test, Staging, and QA: Less critical settings, fewer alerts.
- Development: Less critical settings, no alerts.
- Low Priority: Ensures efficient use of resources.
- Default Policy: Default system settings.

## Types of Policies

There are three types of policies such as default policies, custom policies, and policies that are offered with VCF Operations.

### Custom Policies

You can customize the default policy and base policies included with VCF Operations for your own environment. You can then apply your custom policy to an individual object or groups of objects, such as the objects in a cluster, or virtual machines and hosts, or to a group that you create to include unique objects and specific criteria.

You must be familiar with the policies so that you can understand the data that appears in the user interface, because policies drive the results that appear in the VCF Operations dashboards, views, and reports.

To determine how to customize operational policies and apply them to your environment, you must plan ahead. For example:

- Must you track CPU allocation? If you overallocate CPU, what percentage must you apply to your production and test objects?
- Will you overallocate memory or storage? If you use High Availability, what buffers must you use?
- How do you classify your logically defined workloads, such as production clusters, test or development clusters, and clusters used for batch workloads? Or, do you include all clusters in a single workload?
- How do you capture peak use times or spikes in system activity? In some cases, you might need to reduce alerts so that they are meaningful when you apply policies.

When you have privileges applied to your user account through the roles assigned, you can create and modify policies, and apply them to objects. For example:

- Create a policy from an existing base policy, inherit the base policy settings, then override specific settings to analyze and monitor your objects.
- Use policies to analyze and monitor vCenter objects and non- vCenter objects.
- Set custom thresholds for capacity settings on all object types to have VCF Operations report on workload, and so on.
- Activate specific attributes for collection, including metrics, properties, and super metrics.
- Activate or deactivate alert definitions and symptom definitions in your custom policy settings.
- Apply the custom policy to an individual object or groups of objects.

When you use an existing policy to create a custom policy, you override the policy settings to meet your own needs. You set the allocation and demand, the overcommit ratios for CPU and memory, and the thresholds for capacity risk and buffers. To allocate and configure what your environment is actually using, you use the allocation model and the demand model together. Depending on the type of environment you monitor, such as a production environment versus a test or development environment, whether you over allocate at all and by how much depends on the workloads and environment to which the policy applies. You might be more conservative with the level of allocation in your test environment and less conservative in your production environment.

When you establish the priority for your policies, VCF Operations applies the configured settings in the policies according to the policy rank order to analyze and report on your objects. When you assign an object to be a member of multiple object groups, and you assign a different policy to each object group, VCF Operations associates the highest ranking policy with that object.

Your policies are unique to your environment. Because policies direct VCF Operations to monitor the objects in your environment, they are read-only and do not alter the state of your objects. For this reason, you can override the policy settings to fine-tune them until VCF Operations displays the results that are meaningful and that affect for your environment. For example, you can adjust the capacity buffer settings in your policy, and then view the data that appears in the dashboards to see the effect of the policy settings.

## Default Policy in VCF Operations

The default policy is a set of rules that applies to most of your objects.

The Default policy is marked with the letter D in the Priority column and can apply to any number of objects.

All the Default policies appear in the Default Policy group in the policies library, even if that policy is not associated with an object group. When an object group does not have a policy applied, VCF Operations associates the Default policy with that group.

A policy can inherit the Default policy settings, and those settings can apply to various objects under several conditions.

The policy that is set to Default always takes the lowest priority. If you attempt to set two policies as the Default policy, the first policy that you set to Default is initially set to the lowest priority. When you set the second policy to Default, that policy then takes the lowest priority, and the earlier policy that you set to Default is set to the second lowest priority.

You can use the Default policy as the base policy to create your own custom policy. You modify the default policy settings to create a policy that meets your analysis and monitoring needs. When you start with the Default policy, your new policy inherits all the settings from the Default base policy. You can then customize your new policy and override these settings.

The data adapters and solutions installed in VCF Operations provide a collective group of base settings that apply to all objects. In the policy navigation tree in the policies library, these settings are called Base Settings. The Default policy inherits all the base settings by default.

## Policies Provided with VCF Operations

VCF Operations includes sets of policies that you can use to monitor your environment, or as the starting point to create your own policies.

Verify that you are familiar with the policies provided with VCF Operations so that you can use them in your own environment, and to include settings in new policies that you create.

### Where You Find the Policies Provided with VCF Operations Policies

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile to see the policies provided with VCF Operations.

### Policies That VCF Operations Includes

All policies exist under the Base Settings, because the data adapters and solutions installed in your VCF Operations instance provide a collective group of base settings that apply to all objects. In the policies library, these settings are called Base Settings.

The Base Settings policy is the umbrella policy for all other policies, and appears at the top of the policy list in the policies library. All the other policies reside under the Base Settings, because the data adapters and solutions installed in your VCF Operations instance provide a collective group of base settings that apply to all objects.

The configuration based policy set includes policies provided with VCF Operations that you use for specific settings on objects to report on your objects. This set includes several types of policies:

- Efficiency alerts policies for infrastructure objects and virtual machines
- Health alerts policies for infrastructure objects
- Overcommit policies for CPU and Memory
- Risk alerts policies for infrastructure objects and virtual machines

The Default Policy includes a set of rules that applies to most of your objects.

## Using the Policy Workspace to Create and Modify Operational Policies

You can use the workflow in the policy workspace to create local policies quickly, and update the settings in existing policies. Select a base policy to use as the source for your local policy settings, and modify the thresholds and settings used for analysis and collection of data from objects or object groups in your environment. A policy that has no local settings defined inherits the settings from its base policy to apply to the associated objects or object groups.

Verify that objects or object groups exist for VCF Operations to analyze and collect data, and if they do not exist, create them. See Managing Custom Object Groups in VCF Operations.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile.
2. Click **Add** to add a policy or you can select a policy and click **Edit Policy** to edit an existing policy.

   You can add and edit policies and remove certain policies. You can use the Base Settings policy or the Default Policy as the root policy for the settings in other policies that you create. You can set any policy to be the default policy.
3. In the Create Policies workspace, assign a name to the policy, and enter the description.

   Give the policy a meaningful name and description so that all users know the purpose of the policy.
4. From the **Inherit From** drop-down, select one or more policies to use as a baseline to define the settings for your new local policy.

   You can use any of the policies provided with VCF Operations as a baseline source for your new policy settings.
5. Click **Create Policy**.

   The Create Policies workspace provides the options to customize your policy.
6. Click **Metrics and Properties**. In this workspace, select the metric, property, or super metric attributes to include in your policy.

   VCF Operations collects data from the objects in your environment based on the metric, property, or super metric attributes that you include in the policy.

   a) Click **Save** and return to the create policies workspace.
7. Click **Alerts and Symptoms**. In this workspace, select the alert definitions and symptom definitions, and activate or deactivate them as required for your policy.

   VCF Operations identifies problems on objects in your environment and triggers alerts when conditions occur that qualify as problems.

a) Click **Save** and return to the create policies workspace.

8. Click **Capacity**. In this workspace, select and override the situational settings such as committed projects to calculate capacity, time remaining, and other detailed settings.
   a) Click **Save** and return to the create policies workspace.
9. Click **Maintenance Schedule**. In this workspace, select and override the policy Maintenance Schedule analysis setting.
   a) Click **Save** and return to the create policies workspace.
10. Click **Compliance**. In this workspace, set the compliance threshold required for your policy.
    a) Click **Save** and return to the create policies workspace.
11. Click **Workload Automation**. In this workspace, select the optimization settings required for your policy.

    Click the lock icon to unlock and configure the workload automation options specific for your policy. When you click the lock icon to lock the option, your policy inherits the parent policy settings.
    a) Click **Save** and return to the create policies workspace.
12. Click **VC Pricing**. In this workspace you can add and assign new pricing cards to vCenter and clusters in VCF Operations.

    Click the lock icon to unlock and configure the VC Pricing options specific for your policy. When you click the lock icon to lock the option, your policy inherits the parent policy settings.

    a) Click **Save** and return to the create policies workspace.
13. Click **Configuration Management Templates**. In this workspace, if you have not assigned your vCenter or cluster configuration templates to policies at the time of template creation, you can do so after the template creation process.
    a) Click **Save** and return to the create policies workspace.
14. Click **Groups and Objects**. In this workspace, select one or more groups and objects to which the policy applies.

    VCF Operations monitors the objects according to the settings in the policy that is applied to the object or the object group, triggers alerts when thresholds are violated, and reports the results in the dashboards, views, and reports. If you do not assign a policy to one or more objects or object groups, VCF Operations does not assign the settings in that policy to any objects, and the policy is not active. For an object or an object group that dos not have a policy assigned, VCF Operations associates the object group with the Default Policy.

    Filter the object types, and modify the settings for those object types so that VCF Operations collects and displays the data that you expect in the dashboards and views.
    a) Click **Save** and return to the create policies workspace.

After VCF Operations analyzes and collects data from the objects in your environment, review the data in the dashboards and views. If the data is not what you expected, edit your local policy to customize and override the settings until the dashboards display the data that you need.

## Policy Workspace in VCF Operations

The policy workspace allows you to quickly create and modify policies. To create a policy, you can inherit the settings from an existing policy, and you can modify the settings in existing policies if you have adequate permissions. After you create a policy, or edit an existing policy, you can apply the policy to one or more objects or object groups.

### How the Policy Workspace Works

Every policy includes a set of packages, and uses the defined problems, symptoms, metrics, and properties in those packages to apply to specific objects or object groups in your environment. You can view the details for the settings inherited from the base policy, and display specific settings for certain object types. You can override the settings of other policies, and include additional policy settings to apply to the object types.

Use the **Add** and **Edit** options to create policies and edit existing policies.

### Where You Create and Modify a Policy

To create and modify policies, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile. Click **Add** to add a policy or select the required policy, and then in the right pane, click **Edit Policy** to edit the policy. The policy workspace is where you select the base policies, and customize and override the settings for analysis, metrics, properties, alert definitions, and symptom definitions. In this workspace, you can apply the policy to objects or object groups.

To remove a policy from the list, select the policy, click the horizontal ellipse, and select **Delete**.

### Policy Workspace Options

The policy workspace includes a step-by-step workflow to create and edit a policy, and apply the policy to custom object groups.

## Getting Started Details

When you create a policy, you must give the policy a meaningful name and description so that users know the purpose of the policy.

### Where You Assign the Policy Name and Description

To add a name and description to a policy, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile. Click **Add** to add a policy or select the required policy, and then in the right pane, click **Edit Policy** to edit a policy. The name and description appear in the Create or Edit policy workspace. Click the pencil icon to edit the field

**Table 1189: Name and Description Options in the Create or Edit Policy Workspace**

| Option | Description |
|---|---|
| Name | Name of the policy as it appears in the Create or Edit Policy screens, and in areas where the policy applies to objects, such as Custom Groups. |
| Description | Meaningful description of the policy. For example, use the description to indicate which policy is inherited, and any specific information that users must understand the relationship of the policy to one or more groups of objects. |
| Inherit From | The base policy that is used as a starting point. All settings from the base policy will be inherited as default settings in your new policy. You can override these settings to customize the new policy.<br>Select a base policy to inherit the policy settings as a starting point for your new policy.<br>You cannot edit this setting after a policy is created. |

## Select the Inherited Policy Details

You can use any of the policies provided with VCF Operations as a baseline source for your policy settings when you create a policy.

In the policy content area, you can perform the following actions:

- View the packages and elements for the inherited policy and additional policies that you selected to override the settings.
- Compare the differences in settings highlighted between these policies.
- Display object types.

To create a policy, select a base policy to inherit your new custom policy inherits settings. To override some of the settings in the base policy according to the requirements for the service level agreement for your environment, you can select and apply a separate policy for a management pack solution. The override policy includes specific settings defined for the types of objects to override, either manually or that an adapter provides when it is integrated with VCF Operations. The settings in the override policy overwrite the settings in the base policy that you selected.

When you select and apply a policy to use to overwrite the settings that your policy inherits from the base policy, the policy that you select appears in the policy settings cards.

Click each card to display the inherited policy configuration, and your policy, and displays a preview of the selected policy settings. When you select one of the policy cards, you can view the number of activated and deactivated alert definitions, symptom definitions, metrics and properties, and the number of activated and deactivated changes.

When you select the Groups and Objects card, you select the objects to view so that you can see which policy elements apply to the object type. For example, when you select the StorageArray object type, the workspace displays the local packages for the policy and the object group types with the number of policy elements in each group.

You can preview the policy settings for all object types, only the object types that have settings changed locally, or settings for new object types that you add to the list, such as Storage Array storage devices.

#### Where You Select and Override Base Policies Settings

To select a base policy to use as a starting point for your own policy, and to select a policy to override one or more settings that your policy inherits from the base policy, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile. Click **Add** to add a policy. In the Create policies workspace, add a name and description for the policy and from the **Inherit From** drop-down, select the base policy. The policy configuration, objects, and preview appear in cards below this drop-down.

## Capacity Details

You can filter the object types, and modify the settings for those object types so that VCF Operations applies these settings. The data that you expect then appears in the dashboards and views.

#### How the Capacity Workspace Works

When you turn on and configure the Capacity settings for a policy, you can override the settings for the policy elements that VCF Operations uses to trigger alerts and display data. These types of settings include symptom thresholds based on alerts, situational settings such as committed projects to calculate capacity and time remaining, and other detailed settings.

Policies focus on objects and object groups. When you configure policy settings for your local policy, you must consider the object type and the results that you expect to see in the dashboards and views. If you do not change these settings, your local policy retains the settings that your policy inherited from the base policy that you selected.

#### Where You Set the Policy Capacity Settings

To set the capacity settings for your policy, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile. Click **Add** to add a policy or select the required policy.

In the right pane, click **Edit Policy** to edit a policy. In the <policy name> [Edit] workspace, click the **Capacity** card. The capacity settings for host systems, virtual machines, and other object types that you select appears in the workspace.

You can also edit the capacity settings while working on the objects under the **Inventory** > **Capacity** tab. In the **Capacity** tab under **Inventory**, click the **Foundation Policy** drop-down and select **Edit Capacity Setting**.

**Table 1190: Capacity Settings in the Create or Edit Policy Workspace**

| Option | Description |
|---|---|
| Risk Level Configurations | You can set the risk level for the time that is remaining when the forecasted total need of a metric reaches usable capacity. Click the lock icon to override the settings and change the thresholds for your policy.<br>The following are the risk level settings. Use the slider below the graphical display to change the risk level. You can move the slider between Aggressive and Conservative.<br>• Conservative. Use this option for production and mission-critical workloads. This tells the capacity engine to create projections using all peak values (regular or occasional) that have been identified in the historical demand. Use this option to include the upper range of the data. The projection will be based on the high utilization points. Select Conservative for VMs with utilization spikes.<br>• Aggressive. Use this option for non-critical workloads. |
| Business Hours Schedule | Configure business hours as per your time zone, for calculation of capacity analysis and projections. VCF Operations considers the business hours for all objects using the current policy. During non-business hours, VMs could be running other data center activities such as OS upgrades, virus scans, etc after working hours, and hence may not appear to be idle. When you mark business hours schedules, VCF Operations can analyze after hours metrics for inventory, compliance, troubleshooting and other purposes. The reclamation and right sizing analysis and recommendations are based on the business hours and ignore spikes after business hours. Since the business hours schedule are based on policies, different objects can have different business hours. The capacity charts will be based on business hours.<br>**Note:**<br>You can set business hours schedule for VMs and clusters only.<br>**Note:**<br>After you specify business hours, the capacity forecast for the object will be based on the business hours and not 24 hours.<br>Click the lock icon on the left of each element to override the settings and change the thresholds for your policy. |
| Filters | Select the object type by which you want to filter. You can filter by Object Types, Local Changes, and Unsaved Changes. |
| Capacity Settings | Select an object to view the policy elements and settings for the object type so that you can have VCF Operations analyze the object type.<br>You can view and modify the settings for the following policy elements:<br>• Storage consideration for capacity calculation.<br>**Note:**<br>This option is only available for the Cluster Compute Resource.<br>• Allocation Model<br>• Custom Profile<br>• Capacity Buffer<br>Click the lock icon on the left of each element to override the settings and change the thresholds for your policy. |

| Option | Description |
|---|---|
| Criticality Thresholds and Metrics | There are two tabs in this settings.<br>Click the lock icon on the left of each element to override the settings and change the thresholds for your policy.<br>**Criticality Thresholds Tab**: You can view and modify the threshold settings for the following policy elements:<br>• Time Remaining<br>• Capacity Remaining<br>• Workload<br>**Custom Metrics Tab**: In the custom metrics tab, you can configure VCF Operations to use custom metrics in all the capacity calculations. The metrics that you configure in this tab replaces the default metrics that the VCF Operations capacity engine uses. When defining the custom metrics, you can select the metrics shipped with VCF Operations, or select super metrics. Only metrics which have the same unit as the internal metric used by the capacity engine, or a metric which has no unit, are displayed.<br>**Note:**<br>Enabling custom metrics in the capacity calculations is an advanced configuration. Custom metrics alter the way VCF Operations calculates capacity across your environment. Use this setting only when needed.<br>You can view and modify the custom metrics settings for all non-allocation capacity models. For example, for a data center, you can set custom metrics settings for Total Capacity and Utilization for Memory, CPU and Demand. When you click the **Edit** icon beside the total Capacity and Utililization settings, a list of available metrics opens in the right pane. Double click a default metric or super metric from the list to select it. Click **RESET TO DEFAULT** to revert your changes. Changes that you make take effect after the next collection cycle. |

Click **Save** to save the changes.

The local changes made will appear under **Policy Definition** > **Default Policy** > **Capacity** section. You can also view the preview of changes in the Capacity card.

### Policy Workload Element

Workload is a measurement of the demand for resources on an object. You can turn on and configure the settings for the Workload element for the object types in your policy.

#### How the Workload Element Works

The Workload element determines how VCF Operations reports on the resources that the selected object group uses. The resources available to the object group depend on the amount of configured and usable resources.

- A specific amount of physical memory is a configured resource for a host system, and a specific number of CPUs is a configured resource for a virtual machine.
- The usable resource for an object or an object group is a subset of, or equal to, the configured amount.
- The configured and usable amount of a resource can vary depending on the type of resource and the amount of virtualization overhead required, such as the memory that an ESX host machine requires to run the host system. When accounting for overhead, the resources required for overhead are not considered to be usable, because of the reservations required for virtual machines or for the high availability buffer.

### Where You Override the Policy Workload Element

To view and override the policy workload capacity setting, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile. Click **Add** to add a policy or select the required policy. In the right pane, click **Edit Policy** to edit a policy. In the <policy name> [Edit] workspace, click the **Capacity** card. The workload settings for the object type that you have selected appear in the workspace.

View the Workload policy element, and configure the settings for your policy.

If you do not configure the policy element, your policy inherits the settings from the selected base policy.

**Table 1191: Policy Workload Element Settings in the Create or Edit Policies Workspace**

| Option | Description |
|---|---|
| Lock icon | Allows you to override the policy element settings so that you can customize the policy to monitor the objects in your environment. |
| Workload | Allows you to set the number of collection cycles it takes to trigger or clear an alert. |

## Policy Time Remaining Element

The Time remaining element is a measure of the amount of time left before your objects run out of capacity.

### How the Time Remaining Element Works

The Time Remaining element determines how VCF Operations reports on the available time until capacity runs out for a specific object type group.

- The time remaining indicates the amount of time that remains before the object group consumes the capacity available. VCF Operations calculates the time remaining as the number of days remaining until all the capacity is consumed.
- To keep the Time Remaining more than the critical threshold setting or to keep it green, your objects must have more days of capacity available.

### Where You Override the Policy Time Remaining Element

To view and override the policy Time Remaining capacity setting, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile. Click **Add** to add a policy or select the required policy. In the right pane, click **Edit Policy** to edit a policy. In the <policy name> [Edit] workspace, click the **Capacity** card. The time remaining settings for the object type that you have selected appear in the workspace.

View the Time Remaining policy element and configure the settings for your policy.

If you do not configure the policy element, your policy inherits the settings from the selected base policy.

**Table 1192: Policy Time Remaining Element Settings in the Create or Edit Policies Workspace**

| Option | Description |
|---|---|
| Lock icon | Allows you to override the policy element settings so that you can customize the policy to monitor the objects in your environment. |
| Time Remaining | Allows you to set the number of days until capacity is projected to run out based on your current consumption trend. |

## Policy Capacity Remaining Element

Capacity is a measurement of the amount of memory, CPU, and disk space for an object. You can turn on and configure the settings for the Capacity Remaining element for the object types in your policy.

### How the Capacity Remaining Element Works

The Capacity Remaining element determines how reports on the available capacity until resources run out for a specific object type group.

- The capacity remaining indicates the capability of your environment to accommodate workload.
- Usable capacity is a measurement of the percentage of capacity available, minus the capacity affected when you use high availability.

### Where You Override the Policy Capacity Remaining Element

To view and override the policy Capacity Remaining analysis setting, in the VCF Operations s navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile. Click **Add** to add a policy or select the required policy. In the right pane, click **Edit Policy** to edit a policy. In the <policy name> [Edit] workspace, click the **Capacity** card. The capacity remaining settings for the object type that you have selected appears in the workspace.

View the Capacity Remaining policy element and configure the settings for your policy.

If you do not configure the policy element, your policy inherits the settings from the selected base policy.

**Table 1193: Policy Capacity Remaining Element Settings in the Create or Edit Policies Workspace**

| Option | Description |
|---|---|
| Lock icon | Allows you to override the policy element settings so that you can customize the policy to monitor the objects in your environment. |
| Capacity Remaining | Allows you to set the percentage at which the capacity remaining alerts must be triggered. |

## Policy Allocation Model Element

Allocation model defines how much CPU, memory, or disk space is allocated to objects in a datastore, cluster or datastore cluster. In the policy, you can turn on the Allocation Model element and configure the resource allocation for the objects.

### How the Allocation Model Element Works

The Allocation Model element determines how calculates capacity when you allocate a specific amount of CPU, memory, and disk space resource to datastores, clusters or datastore clusters. You can specify the allocation ratio for either one, or all of the resource containers of the cluster. Unlike the demand model, the allocation model is used for capacity calculations only when you turn it on in the policy.

The allocation model element also affects the reclaimable resources for memory and storage in Reclaim page. When you turn on the Allocation Model element in the policy, the tabular representation of the VMs and snapshots in the selected data center from which resources can be reclaimed displays reclaimable memory and disk space based on the overcommit values.

### Where You Override the Allocation Model Element

To view and override the policy workload analysis setting, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile.

Click **Add** to add a policy or select the required policy, and then in the right pane, click **Edit Policy** to edit a policy. In the <policy name> [Edit] workspace, click the **Capacity** card.

The allocation model settings for the object type that you selected appear in the workspace.

Click the unlock icon next to Allocation Model to set the overcommit ratios.

| Option | Description |
|---|---|
| Set overcommit ratio, to enable Allocation Model | Allows you to set the overcommit ratio for CPU, memory, or disk space. Select the check box next to the resource container you want to edit and change the overcommit ratio value. |

**Policy Custom Profile Element**
The custom profile element lets you apply a custom profile which shows how many more of a specified object can fit in your environment depending on the available capacity and object configuration.

**Where You Define the Custom Profiles**

To define a custom profile, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Custom Profiles ton** tile. Click **Add** to define a new custom profile.

**Where You Select the Custom Profile Element**

To view and override the policy Custom Profile analysis setting, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile. Click **Add** to add a policy or select the required policy. In the right pane, click **Edit Policy** to edit a policy. In the <policy name> [Edit] workspace, click the **Capacity** card. The custom profile element for the object types such as datastores, clusters and datastore clusters that you selected appear in the workspace. Click the lock icon to unlock the section and make changes.

**Policy Capacity Buffer Element**
The capacity buffer element lets you add buffer for capacity and cost calculation. For vCenter objects, you can add buffer to CPU, Memory, and Disk Space for the Demand and Allocation models. You can add capacity buffer to datastores, clusters and datastore clusters. The values that you define here affect the cluster cost calculation. The time remaining, capacity remaining, and recommended values are calculated based on the buffer. For WLP, capacity buffer is first considered and then the headroom that you have defined is considered.

**Where You Define the Capacity Buffer**
To view and override the policy Capacity Buffer analysis setting, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile. Click **Add** to add a policy or select the required policy. In the right pane, click **Edit Policy** to edit a policy. In the <policy name> [Edit] workspace, click the **Capacity** card. The Capacity Buffer for the object type that you selected appears in the workspace. Click the lock icon to unlock the section and make changes.

**How the Capacity Buffer Element Works**

The Capacity Buffer element determines how much extra headroom you have and ensures that you have extra space for growth inside the cluster when required. The value of the usable capacity reduces by the buffer amount that you specify here. The default buffer value is zero. If you are upgrading from a previous version of VCF Operations, the buffer values are carried forward to the new version.

The capacity buffer value that you specify for the Allocation model is considered only if you have activated allocation model in the policy.

Starting from version 8.6, capacity buffer is depreciated from cluster compute resources. The overcommit ratio setting (from the allocation model) and buffer settings, if set for the datastore object, takes precedence for the disk space related to datastore cluster and cluster objects. If these settings are not set, then, from a cost calculation perspective, the settings of datastore cluster and cluster (if the settings are missing for the datastore cluster as well), are used. The allocation and buffer settings made on the cluster does not impact the underlying datastores (as they do not inherit these settings), and the same works vice-versa, settings made for datastores are not propagated to the cluster.

The following tables display the capacity buffer that you can define based on the vCenter Adapter object types:

| Object Type | Valid Models for Capacity Buffer |
|---|---|
| CPU | Demand<br>Allocation |
| Memory | Demand<br>Allocation |
| Disk Space | Demand<br>Allocation |

## Maintenance Schedule Details

You can set a time to perform maintenance tasks for each policy.

### Where You Override the Policy Maintenance Schedule Element

To view and override the policy Maintenance Schedule analysis setting, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile. Click **Add** to add a policy or select the required policy. In the right pane, click **Edit Policy** to edit a policy. In the Create or Edit policies workspace, click **Maintenance Schedule**.

If you do not configure the policy element, your policy inherits the settings from the selected base policy.

**Table 1194: Policy Maintenance Schedule Element Settings in the Create or Edit Policies Workspace**

| Option | Description |
|---|---|
| Select Object Type | Select the object type. |
| Filters | You can filter by Local Changes and Unsaved Changes. Select **Yes** or **No** from the drop-down and click **Apply** to apply the filters. |
| Lock icon | Allows you to override the policy element settings so that you can customize the policy to monitor the objects in your environment. |
| Maintenance Schedule | Sets a time to perform maintenance tasks. During maintenance, VCF Operations does not calculate analytics. |

## Compliance Details

Compliance is a measurement that ensures that the objects in your environment meet industrial, governmental, regulatory, or internal standards. You can unlock and configure the settings for the compliance for the object types in your policy.

### Where You Override the Policy Compliance

To view and override the policy compliance setting, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile. Click **Add** to add a policy or select the required policy. In the right pane, click **Edit Policy** to edit a policy. In the Create or Edit policy workspace, click **Compliance**

View the compliance thresholds and configure the settings for your policy.

If you do not configure the policy element, your policy inherits the settings from the selected base policy.

**Table 1195: Compliance Settings in the Create or Edit Policies Workspace**

| Option | Description |
|---|---|
| Lock icon | Allows you to override the policy element settings so that you can customize the policy to monitor the objects in your environment. |
| Compliance | Allows you to set the compliance score threshold based on the number of violations against those standards. |

## Workload Automation Details

You can set the workload automation options for your policy, so that VCF Operations can optimize the workload in your environment as per your definition.

### How the Workload Automation Workspace Works

You click the lock icon to unlock and configure the workload automation options specific for your policy. When you click the lock icon to lock the option, your policy inherits the parent policy settings.

### Where You Set the Policy Workload Automation

Access this screen through the Policies pages:

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile.
2. Select a policy that you want to modify. Ideally, this should be an active policy. Or, click the **ADD** button to add a new policy.
3. Select the **Workload Automation** card to review the changes, or click **EDIT POLICY** to make changes.

**Table 1196: Workload Automation in the Create or Edit Policies Workspace**

| Option | Description |
|---|---|
| Workload Optimization | Select a goal for workload optimization.<br>Select **Balance** when workload performance is your first goal. This approach proactively moves workloads so that the resource utilization is balanced, leading to maximum headroom for all resources.<br>Select **Moderate** when you want to minimize the workload contention.<br>Select **Consolidate** to proactively minimize the number of clusters used by workloads. You might be able to repurpose resources that are freed up. This approach is good for cost optimization, while making sure that performance goals are met. This approach might reduce licensing and power costs. |
| Cluster Headroom | Headroom establishes a required capacity buffer, for example, 20 percent. It provides you with an extra level of control and ensures that you have extra space for growth inside the cluster when required. Defining a large headroom setting limits the systems opportunities for optimization.<br>**Note:**<br>vSphere HA overhead is already included in useable capacity and this setting does not impact the HA overhead. |

| Option | Description |
|---|---|
| Change Datastore | Click the lock icon to select one of the following options:<br>• Do not allow Storage vMotion.<br>• Allow Storage vMotion. This is selected by default.<br>Using this option, you can select what type of virtual machines VCF Operations moves first to address workload.<br>• Storage Stress for Eviction<br>Select the checkbox to include the storage stress in the eviction criteria. In previous versions, cluster stress levels were determined solely by CPU and memory usage, meaning eviction decisions were not triggered by storage constraints. Starting with version 9.1, storage utilization contributes to the overall cluster stress calculation. Eviction decisions can be initiated when a cluster's storage capacity is constrained. |
| Target Network Policy Setting for WLP | Click the lock icon to select the following option:<br>• Generate a Target Network mapping<br>When you select this checkbox, the Workload Placement algorithm in will automatically choose compatible target network, while making the decision to move the VM for the optimization. For choosing compatible network WLP engine will consider the segment path and logical switch UUID of the Distributed Port Group.<br>Workload Optimization Across networks is supported when the optimization candidate clusters are assigned with different port groups (configured with NSX). These port groups configured via NSX have same segmentID and Logical Switch UUID. To enable this ability, check the respective setting in Workload Automation policy settings.<br>**Note:**<br>Segment ID and logical switch UUID properties are published on the VC port groups by NSX. So Worload Placement cannot provide a target network if it is not a NSX configuration and those properties are missing.<br>This setting is not selected by default. |

## Configuring vCenter Pricing Details

You can add and assign new pricing cards to vCenter and Clusters in VCF Operations. The pricing card can be cost-based or rate-based, you can customize the cost-based pricing card and rate-based pricing card as per your requirement. After configuring the pricing card, you can assign it to one more vCenter or Clusters based on your pricing strategy.

If you want to copy the vCenter pricing settings from the policy currently being edited to another policy, click **Copy local changes to other policy** and select the policy to which you want to copy the settings. The copied pricing configuration will override any existing local pricing configuration, in the target policy.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile.
2. Select the required policy or click **Add** to add a new policy.
3. In the right pane, click **Edit Policy**.
4. In the <policy name> [Edit] workspace, click the **VC Pricing** card.
5. Click the Lock icon to override parent policy settings.
6. Select if you want to activate or deactivate the pricing engine.
7. Configure **Basic Charges**: Click the Lock icon to edit the parent policy settings. Pricing can be performed either on a cost basis or independent of it by specifying rate cards. The factor entered here is multiplied by the cost calculated as a derivative of cost drivers.
    1. Based on Cost/Based on Rate: Select if you want to pricing card to be cost-based or rate-based.
       The following options appear if you select the **Based on Cost** option:
        - CPU Cost: Enter a valid CPU cost factor.
        - Memory Cost: Enter a valid memory cost factor.
        - Storage Cost: Enter a valid storage cost factor.

- Additional Cost: Enter a valid additional cost factor.

The following options appear if you select the **Based on Rate** option:

- CPU Rate: Enter the CPU Rate per vCPU, the charging period, and how to charge for the resources.
- Memory Rate: Enter the memory rate per GB, the charging period, and how to charge for the resources.
- Storage Rate: Enter the storage rate per GB, the charging period, and how to charge for the resources.

8. Configure **Guest OS Rate**: Click **Guest OS Rate** in the left pane and then click the Lock icon to edit the parent policy settings. These are additional charges that have to be included based on the operating system running on the virtual machine. The name of the operating system should match exactly as discovered by VMware Tools.
   1. Click **Create Guest OS Rate** and enter the following details:
      – Guest OS Name: Enter a guest OS name.
      – Charge Period: The Charge Period indicates the frequency of charging.
      – Base Rate: Enter a base rate.

      The guest OS rates that you add appear in the table below. To edit or delete the entries, click the vertical ellipses and select the desired option.
9. **Tags**: Click **Tags** in the left pane and then click the Lock icon to edit the parent policy settings. Tag-based charges can be used to charge for value-added services such as antivirus database disaster recovery and other applications. These applications are to be represented as vCenter tags on the VMs for these charges to work.
   1. Recurring Charges: Recurring charges represent repeating charges such as monthly license fees for antivirus software. Click **Add Recurring Tag** and enter the following details:
      – Tag Category: Enter a tag key.
      – Tag Value: Enter a tag value.
      – Base Rate: Enter a base rate.
      – Charge Period: The Charge Period indicates the frequency of charging.
      – Charge Based on Power State: This decides whether the charge should be applied based on the power state of the VM.

      The tags that you add appear in the table below. To edit or delete the entries, click the vertical ellipses and select the desired option.
   2. One Time Tag: Tag-based one-time charges can be used to represent incidental charges such as charges for addressing a support ticket or charges for applying an operating systems patch. Click **Add One Time Tag** and enter the following details:
      – Tag Category: Enter a tag key.
      – Tag Value: Enter a tag value.
      – Base Rate: Enter a base rate.

      The tags that you add appear in the table below. To edit or delete the entries, click the vertical ellipses and select the desired option.
   3. Rate Factor Tag: Rate factors are multiplication factors applied to already calculated charges. For example, to add a 50% premium on storage, set a rate factor of 1.5 to storage charge. Click **Rate Factor Tag** and enter the following details:
      – Tag Category: Enter a tag key.
      – Tag Value: Enter a tag value.
      – Charge Applies To: Select what the charge applies to.
      – Rate Factor: Enter a valid number. For example, if you want to increase the price of CPU which has a tag 'Tag1-Value1' by 20% then select `CPU Charge` from the **Charge Applies To** drop-down list and enter `1.2` in **Rate Factor**.

      The tags that you add appear in the table below. To edit or delete the entries, click the vertical ellipses and select the desired option.
10. Configure **Overall Charges**: Click **Overall Charges** in the left pane and then click the Lock icon to edit the parent policy settings. Overall Charges are flat charges that are applied to VMs that match this policy.
    1. VM Setup Charges: Enter a valid setup fee. This is to charge for the setup of the VMs.
    2. Recurring Charges: Enter a valid number.

3. Charge Period: The Charge Period indicates the frequency of charging.

You can assign policies to the required Organization/Organization VDC under **Operate** > **Administration** > **Configurations**, and then click the **Policy Assignment** tile. For details, see Assigning Policies.

## Metrics and Properties Details

You can select the attribute type to include in your policy so that VCF Operations can collect data from the objects in your environment. Attribute types include metrics, properties, and super metrics. You activate or deactivate each metric, and determine whether to inherit the metrics from base policies that you selected in the workspace.

### How the Collect Metrics and Properties Workspace Works

When you create or customize a policy, you can override the base policy settings to have VCF Operations collect the data that you intend to use to generate alerts, and report the results in the dashboards.

To define the metric and super metric symptoms, metric event symptoms, and property symptoms, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Symptom Definitions** tile.

### Where You Override the Policy Attributes

To override the attributes and properties settings for your policy, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile. Click **Add** to add a policy or select the required policy. In the right pane, click **Edit Policy** to edit a policy. In the Create or Edit policy workspace, click **Metrics and Properties**. The attributes and properties settings for the selected object types appear in the workspace.

You can also edit the metrics and properties while working on the objects under the **Inventory** > **Metrics** tab. In the **Metrics** tab under **Inventory**, click the **Foundation Policy** drop-down and select **Edit Metrics Collection**.

**Table 1197: Metrics and Properties Options**

| Option | Description |
|---|---|
| Actions | Select one or more attributes and select activate, deactivate, or inherit to change the state and KPI for this policy. |
| Filter options | Deselect the options in the **Attribute Type**, **State**, **KPI**, and **DT** drop-down menus, to narrow the list of attributes.<br>• <br>Activated. Indicates that an attribute will be calculated.<br>• <br>Activated (Force). Indicates state change due to a dependency.<br>• <br>Deactivated. Indicates that an attribute will not be calculated.<br>• <br>Inherited. Indicates that the state of this attribute is inherited from the base policy and will be calculated.<br>• <br>Inherited. Indicates that the state of this attribute is inherited from the base policy and will not be calculated.<br>The KPI determines whether the metric, property, or super metric attribute is considered to be a key performance indicator (KPI) when VCF Operations reports the collected data in the dashboards. Filter the KPI states to display attributes with KPI activated, deactivated, or inherited for the policy. |
| Object Type | Filters the attributes list by object type. |
| Page Size | The number of attributes to list per page. |

| Option | Description |
|---|---|
| Attributes data grid | Display the attributes for a specific object type.<br>• Name. Identifies the name of the metric or property for the selected object type.<br>• Type. Distinguishes the type of attribute to be either a metric, property, or super metric.<br>• Adapter Type. Identifies the adapter used based on the object type selected, such as Storage Devices.<br>• Object Type. Identifies the type of object in your environment, such as StorageArray.<br>• State. Indicates whether the metric, property, or super metric is inherited from the base policy.<br>• KPI. Indicates whether the key performance indicator is inherited from the base policy. If a violation against a KPI occurs, VCF Operations generates an alert.<br>• DT. Indicates whether the dynamic threshold (DT) is inherited from the base policy. |

## Alert and Symptom Details

You can activate or deactivate alert and symptom definitions to have VCF Operations identify problems on objects in your environment and trigger alerts when conditions occur that qualify as problems. You can automate alerts.

### How the Alert and Symptom Definitions Workspace Works

VCF Operations collects data for objects and compares the collected data to the alert definitions and symptom definitions defined for that object type. Alert definitions include associated symptom definitions, which identify conditions on attributes, properties, metrics, and events.

You can configure your local policy to inherit alert definitions from the base policies that you select, or you can override the alert definitions and symptom definitions for your local policy.

Before you add or override the alert definitions and symptom definitions for a policy, familiarize yourself on the available alerts and symptoms.

- To view the available alert definitions, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Alert Definitions** tile.
- To view the available symptom definitions, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Symptom Definitions** tile. Symptom definitions are available for metrics, properties, messages, faults, smart early warnings, and external events.

A summary of the number of problem and symptoms that are activated and deactivated, and the difference in changes of the problem and symptoms as compared to the base policy, appear in the Analysis Settings pane of the policies workspace.

### Where You Override the Alert Definitions and Symptom Definitions

To override the alert definitions and symptom definitions for your policy, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile. Click **Add** to add a policy or select the required policy. In the right pane, click **Edit Policy** to edit a policy. In the Create or Edit policies workspace, click **Alerts and Symptoms**. The definitions appear in the workspace.

### Policy Alert Definitions and Symptom Definitions

You can override the alert definitions and symptom definitions for each policy.

#### Policy Alert Definitions

Each policy includes alert definitions. Each alert uses a combination of symptoms and recommendations to identify a condition that classifies as a problem, such as failures or high stress. You can activate or deactivate the alert definitions in your policy, and you can set actions to be automated when an alert triggers.

### How the Policy Alert Definitions Work

VCF Operations uses problems to trigger alerts. A problem manifests when a set of symptoms exists for an object, and requires you to take action on the problem. Alerts indicate problems in your environment. VCF Operations generates

alerts when the collected data for an object is compared to alert definitions for that object type and the defined symptoms are true. When an alert occurs, VCF Operations presents the triggering symptoms for you to take action.

Some of the alert definitions include predefined symptoms. When you include symptoms in an alert definition, and activate the alert, an alert is generated when the symptoms are true.

The Alert Definitions pane displays the name of the alert, the number of symptoms defined, the adapter, object types such as host or cluster, and whether the alert is activated as indicated by **Local**, deactivated as indicated by **not Local**, or inherited. Alerts are inherited with a green checkmark by default, which means that they are activated.

You can automate an alert definition in a policy when the highest priority recommendation for the alert has an associated action.

To view a specific set of alerts, you can select the badge type, criticality type, and the state of the alert to filter the view. For example, you can set the policy to send fault alerts for virtual machines.

### Where You Modify the Policy Alert Definitions

To modify the alerts associated with policies, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile. Click **Add** to add a policy or select the required policy. In the right pane, click **Edit Policy** to edit a policy. In the Create or Edit policies workspace, click **Alerts and Symptoms**. The alert definitions and symptom definitions for the selected object types appear in the workspace.

**Table 1198: Alert Definitions in the Create or Edit Policies Workspace**

| Option | Description |
|---|---|
| Object Type | Filters the alert definitions list by object type. |
| Filters | Limits the list based on the text you type.<br>You can also filter by:<br>• Name<br>• Criticality<br>• Impact<br>• State<br>• Automate<br>• Local Changes<br>• Unsaved Changes<br>Impact indicates the health, risk, and efficiency badges to which the alerts apply.<br>Criticality indicates the information, critical, immediate, warning, or automatic criticality types to which the alert definition applies.<br>Automate indicates the actions that are activated for automation when an alert triggers, or actions that are deactivated or inherited. Actions that are activated for automation might appear as inherited with a green checkmark, because policies can inherit settings from each other. For example, if the Automate setting in the base policy is set to **Local** with a green checkmark, other policies that inherit this setting will display the setting as inherited with a green checkmark.? |
| Actions | Select one or more alert definitions and select activate, deactivate, or inherit to change the state for this policy. |
| Page Size | The number of alert definitions to list per page. |

| Option | Description |
|---|---|
| Alert Definitions data grid | Displays information about the alert definitions for the object types. The full name for Alert definition and the criticality icon appear in a tooltip when you hover the mouse over the Alert Definition name.<br>• Alert Definition. Meaningful name for the alert definition.<br>• State. Alert definition state, either activated, deactivated, or inherited from the base policy.<br>• Automate. When the action is set to Local, the action is activated for automation when an alert triggers. Actions that are activated for automation might appear as inherited with a green checkmark, because policies can inherit settings from each other. For example, if the Automate setting in the base policy is set to **Local** with a green checkmark, other policies that inherit this setting will display the setting as inherited with a green checkmark.?<br>• Symptom. Number of symptoms defined for the alert.<br>• Criticality. Indicates the criticality of the alert.<br>• Actionable Recommendations. Only recommendations with actions in the first priority, as they are the only ones you can automate.<br>• Adapter. Data source type for which the alert is defined.<br>• Object Type. Type of object to which the alert applies. |

If you do not configure the package, the policy inherits the settings from the selected base policy.

### Activating the Deactivated Alerts

Several out-of-the-box alerts have been deactivated to enhance your alert experience and reduce alert noise in your environment. The alerts that are triggered for these deactivated alerts are auto-cancelled and as a result, you may experience a dip in the number of alerts triggered. However, you can still activate these alerts in specific policies.

The reason for deactivating these alerts is that there could be an overwhelming number of alerts when alerts are turned on for all objects, making it difficult to identify the ones that need immediate attention. It is recommended to exercise caution while activating the deactivated alerts for applicable policies.

Read the KB article, KB 91410 to know the list of deactivated alerts.

Perform the following steps to activate the deactivated alerts:

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile.
2. Select the required policy and in the right pane, click **Edit Policy**, and then select the **Alerts and Symptoms** tile.
3. Go to Filters and enter the name of the deactivated alert, and click **Apply**. You can refer to the KB article, KB 91410 for the list of deactivated alerts.
4. Select **Activated** from the **State** drop-down list or click **Actions** > **State** > **Activated**.
5. Click **Save**.

   **Note:**
   You can also activate all the deactivated alerts at once. To do this, filter the alerts by **Deactivated** State, click on the **Select All** option, and from the **Actions** drop-down list, click **State** > **Activated**.

   **Note:**
   You can also activate the deactivated alerts by creating a separate policy, adding custom groups in that policy, and activating the alert definitions in the required policy. By doing this, the deactivated alerts are activated in the user-defined policy and will apply to the objects in the custom group. For more details on custom groups, see Managing Custom Object Groups in VCF Operations.

The deactivated alerts are now active.

**Policy Symptom Definitions**

Each policy includes a package of symptom definitions. Each symptom represents a distinct test condition on a property, metric, or event. You can activate or deactivate the symptom definitions in your policy.

## How the Policy Symptom Definitions Work

VCF Operations uses symptoms that are activated to generate alerts. When the symptoms used in an alert definition are true, and the alert is activated, an alert is generated.

When a symptom exists for an object, the problem exists and requires that you take action to solve it. When an alert occurs, VCF Operations presents the triggering symptoms, so that you can evaluate the object in your environment, and with recommendations for how to resolve the alert.

To assess objects for symptoms, you can include symptoms packages in your policy for metrics and super metrics, properties, message events, and faults. You can activate or deactivate the symptoms to determine the criteria that the policy uses to assess and evaluate the data collected from the objects to which the policy applies. You can also override the threshold, criticality, wait cycles, and cancel cycles.

The Symptoms pane displays the name of the symptom, the associated management pack adapter, object type, metric or property type, a definition of the trigger such as for CPU usage, the state of the symptom, and the trigger condition. To view a specific set of symptoms in the package, you can select the adapter type, object type, metric or property type, and the state of the symptom.

When a symptom is required by an alert, the state of the symptom is activated, but is dimmed so that you cannot modify it. The state of a required symptom includes an information icon that you can hover over to identify the alert that required this symptom.

## Where You Modify the Policy Symptom Definitions

To modify the policy package of symptoms, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile under the **Policies** section. Click **Add** to add a policy or select the required policy. In the right pane, click **Edit Policy** to edit a policy. In the Create or Edit policies workspace, click **Alerts and Symptoms**. The alert definitions and symptom definitions for the selected object types appear in the workspace.

**Table 1199: Symptom Definitions in the Create or Edit Policies Workspace**

| Option | Description |
|---|---|
| Object Type | Select an object type to view the symptom definitions list by the selected object type. |
| Filters | Limits the list based on the text you type.<br>You can also filter by:<br>• Name<br>• Criticality<br>• Type<br>• State<br>• Local Changes<br>• Unsaved Changes |
| Actions | Select one or more symptom definitions and select activate, deactivate, or inherit to change the state for this policy. |
| Page Size | The number of symptom definitions to list per page. |

| Option | Description |
|---|---|
| Symptom Definitions data grid | Displays information about the symptom definitions for the object types. The full name for Symptom Definition appears in a tooltip when you hover the mouse over the Symptom Definition name.<br>• Symptom Definition. Symptom definition name as defined in the list of symptom definitions in the Content area. Click this name to view the details of the symptom.<br>• State. Symptom definition state, either activated, deactivated, or inherited from the base policy.<br>– <br>Activated. Indicates that a symptom definition will be included.<br>– <br>Activated (Force). Indicates state change due to a dependency.<br>– <br>Deactivated. Indicates that a symptom definition not be included.<br>– <br>Inherited. Indicates that the state of this symptom definition is inherited from the base policy and will be included.<br>– <br>Inherited. Indicates that the state of this symptom definition is inherited from the base policy and will not be included.<br>• Threshold. To change the threshold, you must set the State to **Activated**, set the condition to **Override**, and set the new threshold in the Override Symptom Definition Threshold dialog box.<br>• Type. Type of object to which the alert applies. Type determines whether symptom definitions that apply to HT and DT metrics, properties, events such as message, fault, and metric, and smart early warnings appear in the list.<br>• Criticality. Indicates the criticality.<br>• Adapter. Data source type for which the alert is defined.<br>• Object Type. Object type on which the symptom definition must be evaluated.<br>• Trigger. Static or dynamic threshold, based on the number of symptom definitions, the object type and metrics selected, the numeric value assigned to the symptom definition, the criticality of the symptom, and the number of wait and cancel cycles applied to the symptom definition.<br>• Condition. Activates action on the threshold. When set to Override, you can change the threshold. Otherwise set to default. |

If you do not configure the package, the policy inherits the settings from the selected base policy.

## Groups and Objects details

You can assign your local policy to one or more objects or groups of objects to have VCF Operations analyze those objects according to the settings in your policy. You can trigger alerts when the defined threshold levels are violated, and display the results in your dashboards, views, and reports.

### How the Groups and Objects Workspace Works

When you create a policy, or modify the settings in an existing policy, you apply the policy to one or more objects or groups of objects. VCF Operations uses the settings in the policy to analyze and collect data from the associated objects, and displays the data in dashboards, views, and reports.

### Where You Apply a Policy to Groups and Objects

To apply the policy to an object or groups of objects, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile. Click **Add** to

add a policy or select the required policy. In the right pane, click **Edit Policy** to edit a policy. In the Create or Edit policies workspace, click **Groups and Objects**.

#### Groups and Objects Options

To apply the policy to an object or groups of objects, select the check box for the groups or objects in the workspace.

You can then view the groups and objects associated with the policy. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile. Click **Add** to add a policy or select the required policy. In the right pane, click **Edit Policy** to edit a policy. In the Create or Edit policies workspace, click **Groups and Objects**. Click the **Custom Groups** tab to apply the policy to one or more groups of objects. Click the **Objects** tabs to apply the policy to one or more objects.

For more information about how to create an object group, see the topic called **Custom Object Groups Workspace to Create a New Group**.

For more information about how to create a policy, see Policy Workspace in VCF Operations.

### Real-Time Metrics Details

To collect detailed performance data for effective troubleshooting, you can configure different tiers of metric collection within your monitoring policies. Essentials metrics are always enabled by default and cannot be turned off. If you disable higher-tier collections, like Standard, VCF Operations automatically falls back to collecting only the Essentials metrics. This section is displayed only if the real-time metrics component is deployed.

For more information on the different time-intervals of metric collection, see Real-time Data in the Workbench.

#### Enabling or Disabling 2-Second (ESX Top) Metric Collection

> **Note:**
>
> Before you can enable or disable 2-Second (ESX Top) metric collection, you must deploy and enable Real-Time metrics. For more information, see Deploy Real-Time Metrics.

To collect high-resolution data at a 2-second interval, you must activate the ESX Top category for host objects.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile.
2. Click **Add** to add a policy or you can select a policy and click **Edit Policy** to edit an existing policy.
3. In the policy that you wish to add or edit, scroll down to the tile labeled **Real time metrics**.
4. By default, the **Select Object Type** may be set to **vCenter**. Click the dropdown menu and change it to **Host System**.
5. To Enable: Click the state dropdown menu next to **ESX Top** and select **Activated** in the State column.
6. To Disable: Click the state dropdown menu next to **ESX Top** and select **Deactivated**.
7. Click **SAVE** to apply your policy changes.

You can view 2-second granularity metrics for the selected host from the Troubleshooting Workbench when you select the ESX Top view. The ESX Top view is available for the Virtual Machine resource types. For more information, see Real-time Data in the Workbench.

#### Enabling or Disabling Standard Metric Collection

> **Note:**
>
> Before you can enable or disable Standard metric collection, you must deploy and enable Real-Time metrics. For more information, see Deploy Real-Time Metrics.
>
> **Note:**
>
> Standard metric collection can be enabled only on Large deployment types. For more information, see Start a New VCF Fleet or a New VCF Instance Deployment by Using the VCF Installer Deployment Wizard.

To collect standard metrics, you must activate the Standard category for vCenter objects. Standard metrics are collected at 20 seconds or more time interval.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile.
2. Click **Add** to add a policy or you can select a policy and click **Edit Policy** to edit an existing policy.
3. In the policy that you wish to add or edit, scroll down to the tile labeled **Real time metrics**.
4. Ensure that the **Select Object Type** is set to **vCenter**.
5. To Enable: Click the state dropdown menu next to **Standard** and select **Activated** in the State column.
6. To Disable**:** Click the state dropdown menu next to **Standard** and select **Deactivated**. The metrics which are part of Standard profile are stopped, and they seamlessly move back to the default Essentials collection.
7. Click **SAVE** to apply your policy changes.

For more information on the objects and their time collection interval, download the XLS file here.

# Assigning Policies

The **Policy Assignment** workspace displays all the policies available in your environment.

### Where You Assign Policies

To create and modify policy assignments, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Assignments** tile.

### How the Policy Assignment Workspace Works

You can assign policies to your environment to activate controls, view, and manage your object assignment scope.

| Option | Description |
|---|---|
| Left Pane: Displays the list of policies available in your environment and provides different filters to understand policy distribution. | |
| Inactive/Active | You can filter policies by active or inactive status. A policy is considered to be active only when it is assigned to an object or a custom group. |
| Visualization Type | You can sort active policies based on different criteria.<br>• Fixed: Displays active policies with the sections of the policies fixed and equal to each other.<br>• By Assignments: Displays the policies based on the number of objects assigned.<br>• By Affected Objects: Displays policies based on the affected object count.<br>**Note:**<br>The sorting options are available only for the active policies. |
| Type Policy Name | Type the name of the policy in the search box to filter by policy name. |
| Policies | All the policies available in your environment are displayed in the left pane. Expand the policy card to view the priority, direct assignments, custom groups, affected objects, default assignment, and hierarchy of the policy.<br>**Note:**<br>The default assignments are displayed only for the default policy.<br>When you click on a policy, the objects and custom groups associated with the selected policy are displayed under **Assigned Objects** in the right pane. |

| Option | Description |
|---|---|
| Right Pane: Displays details of the selected policy and allows you to drag and drop objects or custom groups to the selected policy. | |
| Selected Policy | Displays the name of the selected policy. |
| All Objects | Displays all objects and custom groups available in your environment in the **Inventory** and **Custom Groups** tabs respectively.<br>To add new objects to a policy:<br>1. In the right pane, click **All Objects** > **Inventory**.<br>2. Select the objects you want to assign to the policy. You can also search for the objects by typing the object name in the search box.<br>3. Drag the items from the **Inventory** tab and drop them into the policy card on the left pane.<br>4. In the **Assign Objects** window, select one of the following options:<br>a. Only this object: Select this if you want to apply changes only to the selected objects.<br>b. Include child object: Select this if you want to apply changes to the child objects. You can define the depth of change by entering a number in the **Depth** field.<br>**Note:**<br>The maximum value for the Depth field is 10.<br>5. Click **Confirm**.<br>To add custom groups to a policy:<br>1. In the right pane, click **All Objects** > **Custom Groups**.<br>2. Select the custom groups you want to assign to the policy in the **Custom Groups** tab. You can also search for the custom groups by typing the custom group name in the search box.<br>3. Drag the items from the **Custom Groups** tab and drop them into the policy card on the left pane.<br>4. Click **Confirm**. |
| Assigned Objects | Displays the objects and custom groups assigned to the selected policy.<br>• Name: Displays the name of the object/custom group.<br>• Assignment Type: Displays the type of assignment.<br>• Depth: Displays the depth of the child objects that the policy affects. Click the Edit icon to change the depth.<br>• Action: Allows you to delete an object or custom group.<br>You can also drag objects/custom groups from **Assigned Objects** and drop them into the policy card of your choice. |

When multiple policies apply to an object in VCF Operations, the policy with the highest priority is enforced on the object. The only exception is direct assignment: a lower-priority policy is enforced if it is assigned directly to the object.

# Configuring Alerts and Using Actions

Alerts are a way to be watchful of anything that is new or an issue that could be potentially dangerous to your environment. Whenever there is a problem in the environment, the alerts are generated. You can also create new alert definitions so that the generated alerts inform you about the problems in the monitored environment. In VCF Operations , alerts and actions play key roles in monitoring the objects.

You may have several questions while working on alerts. Following are some of the key questions that will help you navigate through the alert documentation.

- Where can I find all my alerts?

- What are the different types of alerts?
- Where can I get more information about alerts?
- How do I create a new alert definition?
- How do I define new symptoms?
- How do I define recommendations for my alerts?
- How can I create notification rules for alerts?
- How do I add my Outbound Plugins?
- How do I create payload templates for my Outbound Plugins?
- How can I export or import Outbound Settings?
- How do I deactivate alerts?
- How do I group alerts?
- What is intelligent alert clustering and how does it help in alert noise reduction?


Actions allow you to make changes to the objects in your environment. When you grant a user access to actions in VCF Operations , that user can take the granted action on any object that VCF Operations manages. For details, see Actions in VCF Operations.

# Understanding Alerts in VCF Operations

This topic provides information on the different types of alerts in VCF Operations, how to access them, and how to view more information about these alerts.

### Types of Alerts

Alerts in VCF Operations are of three types.

| | |
|---|---|
| **Health Alerts** | The health alert list is all the generated alerts that are configured to indicate problems with your environment. You use the health alert list to evaluate, prioritize, and resolve the problems. |
| **Risk Alerts** | The risk alerts list is all the generated alerts that are configured to indicate risk in your environment. Address risk alerts in the near future, before the triggering symptoms that generated the alert negatively affect the health of your environment. |
| **Efficiency Alerts** | The efficiency alerts list is all the generated alerts that are configured to indicate problems with the efficient use of your monitored objects in your environment. Address efficiency alerts to reclaim wasted space or to improve the performance of objects in your environment. |

## Accessing Alerts VCF Operations

The **Alerts** or the **Administrative Alerts** page provides a list of alerts generated in VCF Operations. Use the alert list to determine the state of your environment and to begin resolving problems.

### Where You Find the All Alerts Page

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Alerts**.

### Where You Find the Administrative Alerts Page

As an admin, you can view the administrative alerts by clicking the warning icon next to the Alerts menu or in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Alerts**, and then click the **Administrative Alerts** tab. You can view the **Administrative Alerts** page, only if you are a global admin user or if you have administrative privileges assigned to you.

### How the Alerts and Administrative Alerts Pages Work

By default, only active alerts are initially listed, and the alerts are grouped by **Definition**. Review and manage the alerts in the list by selecting the check box next to it. You can select multiple rows in the list using Shift+click, Control+click, and then use the **Actions** button drop-down list to perform an action on the selected alerts(s).

To see the alert details, click the alert name. The alert details appear on the right, including the symptoms triggered by the alert. The system offers recommendations for addressing the alert and link to run the recommendation. A Run Action button may appear in the details. Hover over the button to learn what recommendation is performed if you click the button. Click the small box on the lower left of the alert list to include the **Suggested Fix** column in the data grid. You can filter by alerts that have the Run option activated and perform the recommended task to address the alert from the Alerts data grid.

Click the name of the object on which the alert was generated to see the object details, and access additional information relating to metrics and events.

If you migrated alerts from a previous version of VCF Operations, the alerts are listed with a cancelled status and alert details are not available.

### All Alerts and Administrative Alerts Options

The alert options include toolbar and data grid options. Use the toolbar options to sort the alert list and to cancel, suspend, or manage ownership. Use the data grid to view the alerts and alert details.

Select an alert from the list to activate the Actions menu:

**Table 1200: Actions Menu**

| Option | Description |
|---|---|
| Cancel Alert | Cancels the selected alerts. If you configure the alert list to display only active alerts, the canceled alert is removed from the list.<br>Cancel alerts when you do not need to address them. Canceling an alert does not cancel the underlying condition that generated it. Canceling alerts is effective if the alert is triggered by fault and event symptoms, because these symptoms are triggered again only if subsequent faults or events occur on the monitored objects. If the alert was generated based on metric or property symptoms, the alert is canceled only until the next collection and analysis cycle. If the violating values are still present, the alert is generated again. |
| Delete Canceled Alerts | Delete canceled (inactive) alerts by doing a group selection or by individually selecting alerts. The option is deactivated for active alerts. |
| Suspend | Suspend an alert for a specified number of minutes.<br>You suspend alerts when you are investigating an alert and do not want the alert to affect the health, risk, or efficiency of the object while you are working. If the problem persists after the elapsed time, the alert is reactivated and it will again affect the health, risk, or efficiency of the object.<br>The user who suspends the alert becomes the assigned owner. |
| Assign to | Assign the alert to a user. You can search for a specific username and click **Save** to assign the alert to the selected user. |
| Take Ownership | As the current user, you make yourself the owner of the alert. You can only take ownership of an alert, you cannot assign ownership. |
| Release Ownership | Alert is released from all ownership. |
| Go to Alert Definition | Switches to the Alert Definitions page, with the definition for the previously selected alert displayed. |

| Option | Description |
|---|---|
| Deactivate... | Provides two options to deactivate the alert:<br>**Note:**<br>To activate the Deactivate option, select **Definition** from the **Group By** drop-down list, and click on the name of the Alert Definition Group.<br>• Deactivate the alert in all policies: This deactivates the alert for all objects for all the policies.<br>• Deactivate alert in selected policies: This deactivates the alert for objects having the selected policy. |
| Open an external application | Actions you can run on the selected object.<br>For example, Open Virtual Machine in vSphere Client. |

**Table 1201: Group By Options**

| Option | Description |
|---|---|
| None | Alerts are not sorted into specific groupings. |
| Definition | Group alerts by definition, that is, group like alerts together. This is the default option. |
| Scope | Group alerts by scope. You can search for alerts within the selected scope. |
| Time | Group alerts by time triggered. You can also group by 1 hour, 4 hours, Today and Yesterday, days of current week, Last week and Older. |
| Criticality | Group alerts by criticality. Values are, from the least critical: Info/ Warning/Immediate/Critical. See also Criticality in the "All Alerts Data Grid Options" table, below. |
| Object Type | Group alerts by the type of object that triggered the alert. For example, group alerts on hosts together. |

**Table 1202: Quick Filters (Alert)**

Alist of fields that can be filtered on by clicking on the funnel icon in the filters bar.

| Quick Filters | Descriptions |
|---|---|
| Filtering options | Limit the list of alerts to those matching the filters you choose.<br>For example, you might have chosen the Time option in the Group By menu. Now you can choose Status -> Active in the Quick Filters menu, and the All Alerts/Administrative Alerts page displays only the active alerts, ordered by the time they were triggered. |
| Options (see also the Group By table for more filter definitions) | |
| Alert id | ID given for an alert. |
| Alert | Name of the alert definition that generated the alert. |
| Owner | Name of operator who owns the alert. |
| Impact | Health, risk, or efficiency indicates the impact category/area. |
| Alert Type | Types of alerts such as Storage, Network, Administrator, and so on helps you find specific alerts. |

| Quick Filters | Descriptions |
|---|---|
| Alert Subtype | Additional information about the type of alert that is triggered on a selected object. This helps you categorize the alerts in a detailed level other than Alert Type, so that you can assign certain types of alerts to specific system administrators. For example, Availability, Performance, Capacity, Compliance, and Configuration. |
| Status | Current state of the alert.<br>Possible values include Active or Canceled. |
| Criticality | The level of importance of the alert in your environment.<br>The level is based on the level assigned when the alert definition was created, or on the highest symptom criticality, if the assigned level was **Symptom Based**.<br>The possible values include:<br>• Critical<br>• Immediate<br>• Warning<br>• Information |
| Triggered On | Name of the object for which the alert was generated, and the object type, which appears in a tooltip when you hover the mouse over the object name.<br>Click the object name to view the object details tabs where you can begin to investigate any additional problems with the object. |
| Control State | State of user interaction with the alert. Possible values include:<br>• Open. The alert is available for action and has not been assigned to a user.<br>• Assigned. The alert is assigned to the user who is logged in when that user clicks **Take Ownership**.<br>• Suspended. The alert was suspended for a specified amount of time. The alert is temporarily excluded from affecting the health, risk, and efficiency of the object. This state is useful when a system administrator is working on a problem and does not want the alert to affect the health status of the object. |
| Object Type | Type of object on which the alert was generated. |
| Created On | Date and time when the alert was generated. |
| Updated On | Date and time when the alert was last modified.<br>An alert is updated whenever one of the following changes occurs:<br>• Another symptom in the alert definition is triggered.<br>• Triggering symptom that contributed to the alert is canceled. |
| Canceled On | Date and time when the alert canceled for one of the following reasons:<br>• Symptoms that triggered the alert are no longer active. Alert is canceled by the system.<br>• Symptoms that triggered the alert are canceled because the corresponding symptom definitions are deactivated in the policy that is applied to the object.<br>• Symptoms that triggered the alert are canceled because the corresponding symptom definitions were deleted.<br>• Alert definition for this alert is deactivated in the policy that is applied to the object.<br>• Alert definition is deleted.<br>• User canceled the alert. |

| Quick Filters | Descriptions |
|---|---|
| Action | Choose **Yes** to filter based on alerts that have the **Run** option activated. Choose **No** to filter based on alerts that have the **Run** option deactivated. |

The Alerts data grid provides the list of generated alerts used to resolve problems in your environment. An arrow in each column heading orders the list in ascending or descending order.

**Table 1203: Alerts and Administrative Alerts Data Grid**

| Option | Description |
|---|---|
| Criticality | Criticality is the level of importance of the alert in your environment. Color coded icon where the shape of the icon represents the type (Health, Risk, Efficiency), and the colour the criticality.<br>The level is based on the level assigned when the alert definition was created, or on the highest symptom criticality, if the assigned level was **Symptom Based**.<br>The possible values include:<br>• Critical<br>• Immediate<br>• Warning<br>• Information |
| Alert | Name of the alert definition that generated the alert.<br>Click the alert name to display the alert details to the right. |
| Triggered On | Name of the object for which the alert was generated, and the object type, which appears in a tooltip when you hover the mouse over the object name.<br>Click the object name to view the object details tabs where you can begin to investigate any additional problems with the object. |
| Created On | Date and time when the alert was generated. |
| Status | Current state of the alert. Values are Active or Inactive. |
| Alert Type | Describes the type of alert that triggered on the selected object, and helps you categorize the alerts so that you can assign certain types of alerts to specific system administrators. For example, Application, Virtualization/Hypervisor, Hardware, Storage, Network, Administrative, and Findings. |
| Alert Subtype | Describes additional information about the type of alert that triggered on the selected object, and helps you categorize the alerts to a more detailed level than Alert Type, so that you can assign certain types of alerts to specific system administrators. For example, Availability, Performance, Capacity, Compliance, and Configuration. |
| Importance | Displays the priority of the alert. The importance level of the alert is determined using a smart ranking algorithm. |
| Suggested Fix | Displays the recommendation to address the alert.<br>This column is not displayed by default. Click the small box on the lower left of the alert list to include the **Suggested Fix** column in the data grid. |
| Action | Click this button to perform the recommendation to address the alert.<br>This column is not displayed by default. Click the small box on the lower left of the alert list to include the **Action** column in the data grid. |

## Viewing Alert Information

When you click an alert from the alerts list, the alert information appears on the right. View the alert information to see the symptoms which triggered the alert, recommendations to fix the underlying issue, and troubleshoot the cause of the alert.

**Different ways to view Alert information**

- In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Alerts**.
- In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory**. Select a group, custom data center, application, or inventory object. Click the object and then the **Alerts** tab.
- In the VCF Operations navigation bar at the top, click the **Search** icon and locate the object of interest. Click the object and then the **Alerts** tab.

The alert description is hidden when you open the alert information. Click **View Description** to see the description of the alert. View the time stamp of when the alert started, and when it was updated, below the alert title.

**Alert Details Tab**

| Section | Description |
|---|---|
| Recommendations | View recommendations for the alert. Click **<** or **>** to cycle through the recommendations. To resolve the alert, click the **Run Action** button if it appears. |
| Other Recommendations | Collapse the section to view additional recommendations. See the links in the **Need More Information?** section to view additional metrics, events, or other details that appear as a link. |
| Alert Basis | |
| Active Only | This option is activated by default. When activated, all active symptoms/conditions that were met for the alert are displayed. When deactivated, all the symptoms/conditions of an alert are displayed. |
| Symptoms | View the symptoms that triggered the alert. Collapse each symptom to view additional information. |
| Conditions | View the conditions that triggered the alert. Collapse each condition to view additional information. |
| Notes | Enter your notes about the alert and click **Submit** to save. |
| Close | Click the X icon to close the alert details tab. |

**Related Alerts Tab**

The **Related Scope** displayed on the right, shows the objects that are one level above and one level below the object on which the alert was triggered. This topology is fixed. You cannot change the scope in the **Related Alerts** tab.

On the right, you can see the following:

- If the same alert was triggered on the object in the past 30 days. This helps you understand if this is a recurring problem or something new.
- If the same alert was triggered on other peers in the same environment, in the past 30 days. This helps you do a quick peer analysis to understand if others are impacted with the same problem.
- All the alerts triggered in the current topology. This helps you investigate if there are other alerts upstream or downstream in the environment which are impacting the health of the object.

**Potential Evidence Tab**

See the **Potential Evidence** tab for potential evidences around the problem, and to arrive at the root cause. This tab displays events, property changes, and anomalous metrics potentially relevant to the alert. The time range and the scope are fixed. To modify the scope or the time range and investigate further, click **Launch Workbench**. This runs the troubleshooting workbench.

The time range that is displayed in the potential evidence tab is two hours and thirty minutes before the alert was triggered. VCF Operations looks for potential evidences in this time range.

## Intelligent Alerts

Every enterprise could have multiple tools that monitor various aspects of their data center operations around the clock. This could cause an alert flooding situation where multiple alerts are generated by a single monitoring tool or multiple tools for the same problem. As a result, IT administrators must sift through thousands of alerts to filter out the noise and focus on key issues, thus, increasing the sheer volume of the alerts and posing an alert storm or alert noise resulting in teams being unable to identify the most critical alerts. Alert flooding happens because monitoring tools lack the intelligence to understand that all alerts depict the same problem.

Machine Learning (ML) helps automate the management of complex systems that contain thousands of objects like VMs, Hosts, and Datastores, through monitoring millions of metrics, huge volumes of logs, and application traces, to capture a high-resolution image of the entire stack.

VCF Operations through intelligent alert clustering helps eliminate business downtime that happens because of a lack of faster troubleshooting abilities and solving critical problems over multiple objects.

### Where You Find the Intelligent Alerts Tab

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Alerts**.

### How Intelligent Alert Clustering Works

Intelligent alerts, also known as alert clusters in VCF Operations, groups related alerts together based on their creation time and their topology distance. This approach provides a more organized and efficient method for troubleshooting, compared to dealing with individual alerts arising from the same underlying issue. Alerts clustering is done based on DBScan algorithm. DBScan (Density-Based Spatial Clustering of Applications with Noise) is an unsupervised clustering Machine Learning algorithm that attempts to group data points closely packed into artificial clusters. In the context of VCF Operations, DBScan has been tailored into a streaming algorithm with specific parameters configured such as minimum points is set to five, time difference is set to fifteen minutes, and topology distance is set to one, for considering only direct children and parents. Two main views, Intelligent alert lifetime and Objects topology, are provided for alert cluster troubleshooting.

The **Intelligent Alert** tab displays the list of alert clusters in the left pane. Click any alert cluster to view the details in the right pane.

| Option | Description |
|---|---|
| Filters | You can filter the alert clusters by their status. Select **Active** or **Inactive** from the **Status** drop-down list and click **Apply**. |
| Alert Cluster | The alert cluster card displays the following:<br>• **Status**: Displays if the status of the alert cluster is active or inactive.<br>• **Object**: Displays the name of the object to which the cluster is assigned.<br>• **Alert Chart**: Displays the number of alerts and objects along with the criticality of the alert. Hover over the graph to view the details.<br>• **Start time**: Displays the time when the first cluster that satisfies the clustering condition is identified.<br>• **End time**: Displays the time when the cluster no longer qualifies to be an alert cluster.<br>Click the alert cluster to view the details in the right pane. |
| Object | Name of the root object. |
| Start time/End time | The start time of the alert cluster is the time when the first cluster that satisfies the clustering condition is identified. The end time of the alert cluster is the time when the cluster no longer qualifies to be an alert cluster. |

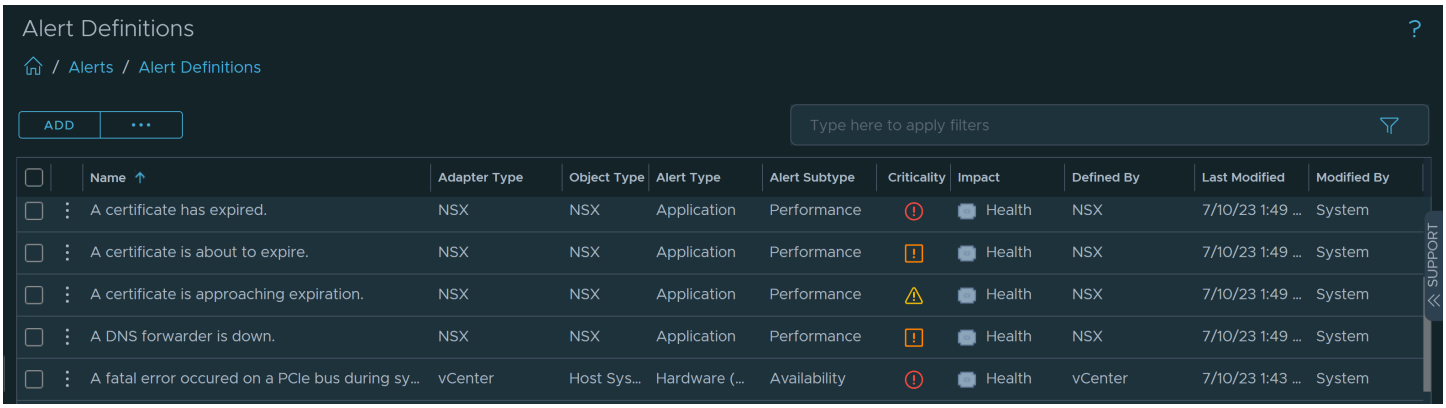
| Option | Description |
|---|---|
| Alerts/Objects | Select **Alerts** to view the graphical representation of alerts in a specific period.<br>Select **Objects** to view the object-relationship chart for an alert cluster. Hover over the object and click **Details** to open the Summary page of the object. |
| How it Started | Click **How it Started** to view the lifetime of an alert cluster. Each bubble displays alerts and objects, hover over the bubble to view more details. |
| Troubleshoot | Click this to launch the Troubleshooting workbench for further troubleshooting. |
| Graph Chart | The graph chart displays the number of alerts by time, for the selected alert cluster.<br>Click the chart legend to filter alerts by:<br>• Criticality<br>  – Critical<br>  – Immediate<br>  – Warning<br>  – Info<br>• Objects<br>Click the Calendar icon to view past alerts by selecting the **Range** or by selecting a date in the **From** and **To** fields. |
| Group By | You can group alerts by:<br>• Definition<br>• Scope<br>• Time<br>• Criticality<br>• Object Type |
| Filters | You can filter alerts by:<br>• Alert ID<br>• Alert name<br>• Owner<br>• Impact<br>• Alert Type<br>• Alert Subtype<br>• Status |

## Alert Definitions in VCF Operations

Alert definitions are a combination of symptoms and recommendations that you combine to identify problem areas in your environment and generate alerts on which you can act for those areas. You can then respond to the alerts with effective solutions that are provided in the recommendations.

## Where You Find Alert Definitions

To manage your alert definitions, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Alerts Definiton** tile under the **Alerts** section.




| Option | Description |
|---|---|
| Toolbar options | Use the toolbar options to manage your alert definitions.<br>• Add. Add an alert definition.<br>Click the horizontal ellipsis to perform the following actions.<br>• Delete. Remove the selected definition.<br>• Export. Downloads the alert definition.<br>• Import. Allows you to import alert definitions. To import:<br>– Click the **Import** option from the horizontal ellipsis.<br>– Click **Browse** and select the file to import.<br>– Select if you want to Overwrite or Skip the file in case of a conflict.<br>– Click **Import** to import the alert definition, and click **Done**.<br>Click on the vertical ellipsis against an individual alert definition to perform the following actions.<br>• Edit. Modify the selected definition.<br>• Delete. Remove the selected definition.<br>• Clone. Create a copy of the selected definition so that you can customize it for your needs.<br>• Export. Downloads the alert definition. |
| Filters | Limits the list of alerts to those matching the filter you create.<br>You can also sort on the columns in the data grid. |
| **Data Grid Columns** | |
| Name | Name of the alert definition, which is also the name of the alert that appears when the symptoms are triggered. |
| Adapter Type | Adapter that manages the selected base object type. |
| Object Type | Base object type against which the alert is defined. |
| Alert Type | Metadata that is used to classify the alert when it is generated.<br>You define the value on the Alert Impact page of the workspace. |
| Alert Subtype | Subcategory of the alert type and is the metadata that is used to classify the alert when it is generated.<br>You define the value on the Alert Impact page of the workspace. |

| Option | Description |
|---|---|
| Criticality | Severity of the alert when it is generated. The criticality includes the following possible values:<br>• Symptom. Alert is configured to display symptom based criticality.<br>• Critical<br>• Immediate<br>• Warning<br>• Info |
| Impact | Alert is configured to affect the Health, Risk, or Efficiency badge. |
| Defined by | Indicates who added the alert definition. The alert can be added by an adapter, a user, or the VCF Operations system. |
| Last Modified | Displays the date on which the alert was last modified. |
| Modified By | Displays the user who last modified the alert. |

Predefined alerts are provided in VCF Operations as part of your configured adapters. Use Alert Definitions to manage your VCF Operations alert library, and to add or modify the definitions.

**Modifying Alert Definitions**

If you modify the alert impact type of an alert definition, any alerts that are already generated will have the previous impact level. Any new alerts will be at the new impact level. If you want to reset all the generated alerts to the new level, cancel the old alerts. If they are generated after cancellation, they will have the new impact level.

**Symptoms in Alert Definitions**

Symptom definitions evaluate conditions in your environment that, if the conditions become true, trigger a symptom and can result in a generated alert. You can add symptom definitions that are based on metrics or super metrics, properties, message events, fault events, or metric events. You can create a symptom definition as you create an alert definition or as an individual item in the appropriate symptom definition list.

When you add a symptom definition to an alert definition, it becomes a part of a symptom set. A symptom set is the combination of the defined symptom with the argument that determines when the symptom condition becomes true.

An alert definition comprises one or more symptom sets. If an alert definition requires all of the symptom sets to be triggered before generating an alert, and only one symptom set is triggered, an alert is not generated. If the alert definition requires only one of several symptom sets to be triggered, then the alert is generated even though the other symptom sets were not triggered.

**Recommendations in Alert Definitions**

Recommendations are the remediation options that you provide to your users to resolve the problems that the generated alert indicates.

When you add an alert definition that indicates a problem with objects in your monitored environment, add a relevant recommendation. Recommendations can be instructions to your users, links to other information or instruction sources, or VCF Operations actions that run on the target systems.

## Creating Alert Definitions

The alert definition process includes adding symptoms that trigger an alert and recommendations that help you resolve the alert. The alert definitions you create with this process are saved to your VCF Operations Alert Definition Overview list and actively evaluated in your environment based on your configured policies.

You can create or reuse existing symptoms and recommendations while defining an alert definition. If you create symptoms and recommendations, you add them to the definition, and they are added to the symptom and recommendations content libraries for future use. You also activate policies and select notifications for the alerts.

1. To create or edit your alert definitions, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Alerts Definiton** tile.
2. Click **Add** to add a definition, or click the vertical ellipsis and select **Edit** to edit the selected definition.

3. In the **Alert** tab, enter details of the alert.

| Option | Description |
|---|---|
| Name | Name of the alert as it appears when the alert is generated. |
| Description | Description of the alert as it appears when the alert is generated. Provide a useful description for your users. |
| For Tenants | Marks an alert definition as tenant-facing when VCF Operations is integrated with VCF Automation. This exposes the generated alerts within the VCF Automation tenant portal, allowing tenant users to monitor alerts relevant to their own workloads, such as VMs and Region Quotas. After the checkbox is selected, the Base Object Type picker automatically restricts available options to tenant-relevant objects, filtering out infrastructure-level entities such as ESX hosts, datastores, and clusters, that are not relevant to tenant users.<br>Displayed when VCF Operations is integrated with VCF Automation. |
| .Base Object Type | The object type against which the alert definition is evaluated and the alert is generated.<br>The drop-down menu includes all of the object types in your environment. You can define an alert definition based on one object type. |
| Impact | Under Advanced Settings, select the badge that is affected if the alert is generated.<br>You can select a badge based on the impact of the alert.<br>• Health<br>• Risk<br>• Efficiency |
| Criticality | Severity of the alert that is communicated as part of the alert notification.<br>Select one of the following values.<br>• Info. Informational purposes only. Does not affect badge color.<br>• Warning. Lowest level. Displays yellow.<br>• Immediate. Medium level. Displays orange.<br>• Critical. Highest level. Displays red.<br>• Symptom Based. In addition to alert criticality, each symptom includes a defined criticality. Criticality of the alert is determined by the most critical of all of the triggered symptoms. The color is dynamically determined accordingly. It you negate symptoms, the negative symptoms to not contribute to the criticality of a symptom-based alert. |
| Alert Type and Subtype | Select the type and subtype of alert.<br>This value is metadata that is used to classify the alert when it is generated, and the information is carried to the alert, including the alert notification.<br>You can use the type and subtype information to route the alert to the appropriate personnel and department in your organization. |

| Option | Description |
|---|---|
| Wait Cycle | The symptoms included in the alert definition remain triggered for this number of collection cycles before the alert is generated.<br>The value must be 1 or greater.<br>This setting helps you adjust for sensitivity in your environment. The wait cycle for the alert definition is added to the wait cycle for the symptom definitions. In most definitions you configure the sensitivity at the level of symptom level and configure the wait cycle of alert definition to 1. This configuration ensures that after all of the symptoms are triggered at the desired symptom sensitivity level, the alert is immediately triggered. |
| Cancel Cycle | The symptoms are cancelled for this number of collection cycles after which the alert is cancelled.<br>The value must be 1 or greater.<br>This setting helps you adjust for sensitivity in your environment. The cancel cycle for the alert definition is added to the cancel cycle for the symptom definitions. In most definitions you configure the sensitivity at the level of symptom level and configure the wait cycle of the alert definition to 1. This configuration ensures that after all of the symptom conditions disappear after the desired symptom cancel cycle, the alert is immediately canceled. |

4. Click **Next** to add symptom definitions.
5. In the **Symptoms/Conditions**tab, drag the selected symptom/condition in to the left pane. Use the workspace on the left to specify whether all or any of the symptoms/conditions or symptom/condition sets must be true to generate an alert. As you add one or more symptoms, you create a symptom expression. If this expression is evaluated as true, then the alert is generated. You can similarly define one or more conditions for your alert, and when the conditions are met, the alert is generated. You can view the alert in the All Alerts page.

**Table 1204: Add Symptoms/Conditions Selection Options**

| Option | Description |
|---|---|
| Defined On | Object that the symptom evaluates.<br>As you create alert definitions, you can select or define symptoms for the base object type and for related object types, based on the object relationship hierarchy. The following relationships are object types as they relate to the alert definition base object type.<br>• Self. A base object type for the alert definition. For example, host system.<br>• Descendant. An object type that is at any level below the base object type, either a direct or indirect child object. For example, a virtual machine is a descendant of a host system.<br>• Ancestor. An object type that is one or more levels higher than the base object type, either a direct or indirect parent. For example, a data center and a vCenter are ancestors of a host system.<br>• Parent. An object type that is in an immediately higher level in the hierarchy from the base object type. For example, a data center is a parent of a host system.<br>• Child. An object type that is one level below the base object type. For example, a virtual machine is a child of a host system. |
| Filter by Object Type | Available only when you select a Defined On value other than Self.<br>Limits the symptoms to those that are configured for the selected object type based on the selected Defined On relationship. |

| Option | Description |
| --- | --- |
| Conditions tab | |
| Select Specific Object | Select a specific object based on its object type, adapter type, policy, collection state, and status. |
| Search | Search the metrics based on object type. |
| Conditions list | List of metrics for the selected object type. To configure a condition, drag it into the left workspace. |
| Log based condition | Create a log based condition without creating a log based symptom first. To configure a condition, drag it into the left workspace. You can combine metric based condition and log based condition to create an advanced alert trigger set.<br>Displayed if Log Management is installed. |
| Symptoms tab | |
| Select Symptom | Select the type of symptom definition that you are adding for the current Defined On object type.<br>• Metric / Property. Add symptoms that use metric and property symptoms. These metrics are based on the operational or performance values, and configuration properties that collects from target objects in your environment.<br>• Message Event. Add symptoms that use message event symptoms. These symptoms are based on events received as messages from a component of VCF Operations or from an external monitored system through the system's REST API.<br>• Fault Event. Add symptoms that use fault symptoms. These symptoms are based on events that monitored systems publish. VCF Operations correlates a subset of these events and delivers them as faults. Faults are intended to signify events in the monitored systems that affect the availability of objects in your environment.<br>• Logs. Select the log based symptoms which you have defined after integration with Log management . You can combine metric and log based symptoms in one alert definition. If you do not want to create a Log based symptom for the alert, you can use a log based Condition.<br>Dsiplayed when Log Management is deployed.<br>• Smart Early Warning. Add a symptom that uses a defined condition that is triggered when the number of anomalies on an object is over the trending threshold. This symptom represents the overall anomalous behavior of the object. Anomalies are based on VCF Operations analysis of the number of applicable metrics that violate the dynamic threshold that determines the normal operating behavior of the object. This symptom is not configurable. You either use it or you do not use it. |
| Create New Symptom | If symptoms that you need for your alert do not exist, you can create them.<br>Opens the symptoms definition dialog box.<br>Not available for Smart Early Warning symptoms, which are predefined in the system. |

| Option | Description |
|---|---|
| All Filters | Filter the list of symptom definitions.<br>• Symptom. Type text to search on the name of the symptom definitions. For example, to display all symptom definitions that have efficiency in their name, type `Efficiency`.<br>• Defined By. Type text to search for the name of the adapter that defines the symptom definitions. For example, to display all symptom definitions provided by the vCenter Adapter, type `vCenter`. To display only user-defined symptom definitions, type the search term `User`.<br>To clear a filter, click the 'x' that appears next to the filter name. |
| Symptoms list | List of existing symptoms for the selected object type. To configure a symptom, drag it into the left workspace.<br>To combine symptoms that are based on multiple levels in the hierarchy, select the new Defined On level and Filter by Object Type before you select and drag the new symptom to the workspace. |

Use the workspace to configure the interaction of the symptoms, symptom sets, and conditions.

**Table 1205: Symptom Sets in the Alert Definition Workspace**

| Option | Description |
|---|---|
| Trigger alert when {operator} of the symptom sets are true | Select the operator for all of the added symptom/condition sets. Available only when you add more than one symptom/condition set.<br>• All. All of the symptom/condition sets must be true before the alert is generated. Operates as a Boolean AND.<br>• Any. One or more of the symptom/condition sets must be true before the alert is generated. Operates as a Boolean OR. |
| Symptoms | The symptom/condition sets comprise an expression that is evaluated to determine if an alert should be triggered.<br>To add one or more symptoms from the symptom list to an existing symptom set, drag the symptom from the list to the symptom set. To create a new symptom set for the alert definition, drag a symptom to the landing area outlined with a dotted line. |

| Option | Description |
| --- | --- |
| Symptom sets | Add one or more symptoms to the workspace, define the points at which the symptom sets are true, and specify whether all or any of the symptoms in the symptom set must be true to generate the alert.<br>A symptom set can include one or more symptoms/conditions, and an alert definition can include one or more symptom/condition sets.<br>If you create a symptom set where the Defined On object is Self, you can set the operator for multiple symptoms in the symptom set.<br>If you create a symptom set where the Defined On object is a relationship other than Self, you can set the operator and modify the triggering threshold. To configure the symptom set criteria, you set the options.<br>• Value operator. Specifies how the value you provide in the value text box is compared to a number of related objects to evaluate the symptom/condition set as true.<br>• Value text box. Number of objects of the specified relationship, based on the value type, that are required to evaluate the symptom/condition set as true.<br>• Value type. Possible types include the following items:<br>– Count. Exact number of related objects meet the symptom/condition set criteria.<br>– Percent. Percentage of total related objects meet the symptom/condition set criteria.<br>– Any. One or more of the related objects meet the symptom/condition set criteria.<br>– All. All of the related objects meet the symptom/condition set criteria.<br>• Symptom set operator. Operator applied between symptoms/conditions in the symptom set.<br>– All. All of the symptoms/conditions must be true before the alert is generated. Operates as a Boolean AND.<br>– Any. One or more of the symptoms/condition must be true before the alert is generated. Operates as a Boolean OR.<br>When you include a symptom in a symptom set, the condition must become true to trigger the symptom set. However, you might want to configure a symptom set where the absence of a symptom condition triggers a symptom. To use the absence of the symptom condition, click the vertical ellipsis on the left of the symptom name and select **Invert Symptom**.<br>Although you can configure symptom criticality, if you invert a symptom, it does not have an associated criticality that affects the criticality of generated alerts. |

**Table 1206: Conditions in the Alert Definition Workspace**

| Option | Description |
|---|---|
| Alert is triggered when {operator} of the sets are true | Select the operator for all of the added condition sets. Available only when you add more than one condition set.<br>• All. All of the condition sets must be true before the alert is generated. Operates as a Boolean AND.<br>• Any. One or more of the condition sets must be true before the alert is generated. Operates as a Boolean OR. |
| Conditions | The condition sets comprise an expression that is evaluated to determine if an alert should be triggered.<br>• Condition. Determines how the value you specify in the value text box is compared to the current value of the metric or property when the condition is evaluated.<br>• Value. Value that specifies the threshold.<br>• Criticality level. Severity of the symptom/condition when it is triggered.<br>• Wait Cycle. The trigger condition should remain true for this number of collection cycles before the symptom/condition is triggered. The default value is 1, which means that the symptom/condition is triggered in the same collection cycle when the condition became true.<br>**Note:**<br>You cannot edit the wait cycle while defining conditions for Properties and Population.<br>• Cancel Cycle. The symptom/condition is canceled after the trigger condition is false for this number of collection cycles after which the symptom/condition is cancelled. The default value is 1, which means that the symptom/condition is canceled in the same cycle when the condition becomes false.<br>**Note:**<br>You cannot edit the cancel cycle while defining conditions for Properties and Population.<br>To add one or more conditions from the condition list to an existing symptom/condition set, drag the condition from the list to the symptom/condition set. |

6. Click **Next** to add recommendations.
7. In the **Recommendations** tab, drag the selected recommendation in to the left pane. Use the workspace on the left to to change the priority order.

**Table 1207: Add Recommendations Options in the Alert Definition Workspace**

| | |
|---|---|
| Create New Recommendation | If recommendations that you need to resolve the symptoms of the problem do not exist, you can create them. |
| All Filters | Filter the list of recommendations.<br>• Description. Type text to search on the name of the recommendation. For example, to display all recommendations that have memory in their name, type `Memory`.<br>• Defined By. Type text to search for the name of the adapter that defines the recommendation. For example, to display all recommendations provided by the vCenter Adapter, type `vCenter`.<br>To clear a filter, click the double arrow icon that appears next to the filter name. |
| Quick filter (Name) | Limits the list based on the text you enter. |

| | |
|---|---|
| List of available recommendations. | List of existing recommendations that you can drag to the workspace.<br>Recommendations are instructions and, where possible, actions that assist you with resolving alerts when they are triggered. |
| Recommendation workspace | Add one or more recommendations to the workspace.<br>If you add more than one recommendation, you can drag the recommendations to change the priority order. |

8. Click **Next** to activate policies.
9. In the **Policies** tab, you can view and select only active policies.

   **Note:**
   If you create an alert without enabling any policies, then the alert remains inactive.
10. Click **Create** to create the alert. The new alert appears in the list of alert definitions.

## Symptom Definitions in VCF Operations

Symptoms are conditions that indicate problems in your environment. You can define symptoms in VCF Operations and add them to alert definitions so that you know when a problem occurs with your monitored objects.

As data is collected from your monitored objects, the data is compared to the defined symptom condition. If the condition is true, then the alert is triggered.

To define symptoms, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Symptom Definiton** tile under the **Alerts** section.

You can define symptoms based on:

- Metric/Property Symptom
- Message Events Symptoms
- Fault Symptoms
- Logs Symptoms

The symptoms defined in your environment are managed in the Symptom Definitions. When the symptoms that are added to an alert definition evaluate as true, an alert is triggered.

**Define Symptoms to Cover All Possible Severities and Conditions**

Use a series of symptoms to describe incremental levels of concern. For example, `Volume nearing capacity limit` might have a severity value of Warning while `Volume reached capacity limit` might have a severity level of Critical. The first symptom is not an immediate threat. The second symptom is an immediate threat.

## Defining Symptoms for Alerts

The symptoms defined in your environment are managed in the **Symptom Definitions** page.

You can create a symptom definition while creating an alert definition or from the **Symptom Definitions** page.

You can define symptoms based on:

- Metric/Property
- Message Events
- Fault
- Logs

## Metric/Property Symptoms

Metric/Property symptoms are based on the operational/performance values or the configuration properties that VCF Operations collects from target objects in your environment.

**Metric Symptom Definitions**

You can configure the symptoms to evaluate static thresholds or dynamic thresholds. You define symptoms based on metrics so that you can create alert definitions that let you know when the performance of an object in your environment is adversely affected.

**Static Thresholds**

Metric symptoms that are based on a static threshold compare the currently collected metric value against the fixed value you configure in the symptom definition.

For example, you can configure a static metric symptom where, when the virtual machine CPU workload is greater than 90, a critical symptom is triggered.

**Dynamic Thresholds**

Metric symptoms that are based on dynamic thresholds compare the currently collected metric value against the trend identified by VCF Operations , evaluating whether the current value is above, below, or generally outside the trend.

For example, you can configure a dynamic metric symptom where, when the virtual machine CPU workload is above the trended normal value, a critical symptom is triggered.

The Metric Symptom Definitions is a list of the metric-based symptoms defined in your VCF Operations environment. You use the information in the list to evaluate the defined metric threshold triggering states and determine if you want to add, edit, or clone symptoms.

**Property Symptom Definitions**

You define symptoms based on properties so that you can create alert definitions that let you know when changes to properties on your monitored objects can affect the behavior of the objects in your environment.

The Property Symptom Definitions is a list of the property-based symptoms in your VCF Operations environment. You use the information in the list to evaluate the defined property triggering states and determine whether to add, edit, or clone symptoms.

### Where You Find Metric/Property Symptoms

To manage symptoms based on metrics/properties, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Symptom Definitons** tile. From the **Symptom Definitions** page, click the **Metric / Property** tab.

You can also define symptoms as you are defining alerts in the Alert Definition Workspace.

**Table 1208: Metric/Property Symptoms Options**

| Option | Description |
|---|---|
| Toolbar options | Use the toolbar options to manage your symptoms. You can select multiple symptoms using Ctrl+click or Shift+click.<br>• Add. Add a symptom definition.<br>Click the horizontal ellipsis to perform the following actions on multiple symptom definitions or click the vertical ellipsis against a symptom definition to perform the action on a single symptom definition.<br>• Edit. Modify the selected symptom definition. Any changes you make affect the alert definitions that include this symptom. You cannot edit a symptom that manages a badge.<br>• Delete. Remove the selected symptom definition. You cannot delete an alert that is used in an alert definition. To delete a symptom, you must first remove it from the alert definitions in which it is used. You cannot delete a symptom that manages a badge.<br>• Clone. Create a copy of the selected symptom definition.<br>• Export. Downloads the symptom definition.<br>• Import. Allows you to import symptom definitions. This option is available only when you click the horizontal ellipsis. To import:<br>– Click the **Import** option from the horizontal ellipsis.<br>– Click **Browse** and select the file to import.<br>– Select if you want to Overwrite or Skip the file in case of a conflict.<br>– Click **Import** to import the symptom definition, and click **Done**. |
| Filter options | Limits the list based on the text you type.<br>You can also sort on the columns in the data grid. |
| Name | Descriptive name of the symptom. |
| Criticality | Severity of the symptom when it is triggered. |
| Object Type | Base object type against which the symptom is defined. |
| Metric Name | Text string that is used as a reference key for the metric. You can use the metric key to locate additional information about how the system statistics are derived from the metric. |
| Operator | Operator used to compare the current value to the threshold value, and trigger the symptom. |
| Value | Text string that is the compared value for the property. |
| Defined By | Indicates whether the symptom was created by a user or provided with a solution adapter. |
| Last Modified | Displays the date on which the symptom was last modified. |
| Modified By | Displays the name of the user who last modified the symptom. |

## Defining Metric/Property Symptoms

A metric symptom is triggered when a metric is compared to the configured static or dynamic thresholds, and the symptom condition is evaluated as true in VCF Operations .

## Defining Metric Symptoms

If the symptom is based on a static threshold, the metric is compared based on the configured operator and the provided numeric value. If the symptom is based on a dynamic threshold, the metric is compared based on whether the current value is above, below, or abnormal compared to the calculated trend value.

1. To define symptoms based on metrics, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Symptom Definitons** tile. From the **Symptom Definitions** page, click the **Metric / Property** tab. Click **Add** to define a metric-based symptom in the workspace.
2. Enter the following details.

| Option | Description |
|---|---|
| Base Object Type | Object against which the symptom is evaluated.<br>Based on the select object type, the list of available metrics displays only the metrics applicable to the object type. |
| Symptom Type | Select **Metrics** from the Symptom Type drop-down list. |
| Select Specific Object | If a metric or supermetric is not listed in the common metric or supermetric list, based on the selected based object type, use Select Object to inspect the metrics or supermetrics of a selected object so that you can locate the property that you must use to create the symptom. Even though you select a metric or supermetric for a specific object, the symptom definition is applicable to all objects with that metric or supermetric in your environment. |
| Search | Use a word search to limit the number of items that appear in the list. |
| Metric list | List of metrics for the selected base object type.<br>Click and drag the metric to the left pane. |
| Symptom Definition workspace<br>You can define symptoms based on static or dynamic thresholds. | |
| Threshold | Determines if the symptom is static or dynamic.<br>• Static thresholds are fixed values that trigger symptoms as true. You can configure one threshold for each symptom. You can also create multiple symptoms for multiple thresholds.<br>For example, configure one symptom where the CPU use is greater than 90 percent and another where the CPU usage is less than 40 percent. Each is a separate symptom and can be added individually to an alert definition.<br>• Dynamic thresholds are based on VCF Operations trended data where the triggering value is determined through the analytics. If the current value of the metric does not fall in the trended range, the symptom is triggered. |

| Option | Description |
| --- | --- |
| Static Threshold configuration options | If you select Static Threshold, configure the options for this threshold type.<br>• Symptom name. Name of the symptom as it appears in the symptom list when configuring an alert definition, as it appears when the alert is generated, and when viewing triggered symptoms.<br>• Condition. Determines how the value you specify in the value text box is compared to the current value of the metric when the symptom is evaluated.<br>• Value. Value that the condition evaluates.<br>• Criticality. Severity of the symptom when it is triggered.<br>• Wait Cycle. The trigger condition should remain true for this number of collection cycles before the symptom is triggered. The default value is 1, which means that the symptom is triggered in the same collection cycle when the condition became true.<br>• Cancel Cycle. The symptom is canceled after the trigger condition is false for this number of collection cycles after which the symptom is cancelled. The default value is 1, which means that the symptom is canceled in the same cycle when the condition becomes false.<br>• Evaluate on instanced metrics. Select this check box so that the system evaluates the object level symptom as well as the instance level symptom. For example, for CPU usage, when the check box is not selected, the symptom is triggered based on the object's CPU usage. However, if you select the check box, the system also evaluates CPU usage of each of the cores. If any of the cores is found to be crossing the threshold, the symptom is triggered.<br>• Exclude the following instances of the metric. To exclude specific instanced metrics from the symptom, drag the metric instances from the left pane. If you cannot locate the metric instance you want to exclude, you can search for it in another object that uses the metric by clicking **Select Specific Object** next to the search box. |

| Option | Description |
|---|---|
| Dynamic Threshold configuration options | If you select Dynamic Threshold, configure the options for this threshold type.<br>• Symptom name. Name of the symptom as it appears in the symptom list when configuring an alert definition, as it appears when the alert is generated, and when viewing triggered symptoms.<br>• Condition. Relationship of the current value to trended range based on the following options:<br>– Above Threshold. If current value is above trended range, the symptom is triggered.<br>– Below Threshold. If the current value is below the trended range, the symptom is triggered.<br>– Abnormal. If the current value is either above or below the trended range, the symptom is triggered.<br>• Criticality. Severity of the symptom when it is triggered.<br>• Evaluate on instanced metrics. Select this check box so that the system evaluates the object level symptom as well as the instance level symptom. For example, for CPU usage, when the check box is not selected, the symptom is triggered based on the object's CPU usage. However, if you select the check box, the system also evaluates CPU usage of each of the cores. If any of the cores is found to be crossing the threshold, the symptom is triggered.<br>• Exclude the following instances of the metric. To exclude specific instanced metrics from the symptom, drag the metric instances from the left pane. If you cannot locate the metric instance you want to exclude, you can search for it in another object that uses the metric by clicking **Select Object** next to the **Metrics** field. |

3. Click **Save**.

## Defining Property Symptom

A property symptom is triggered when the defined threshold is compared with the current property value and the comparison is evaluated as true.

1. To define symptoms based on properties, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Symptom Definitons** tile. From the **Symptom Definitions** page, click the **Metric / Property** tab. Click **Add** to define a property-based symptom in the workspace.
2. Enter the following details.

| Option | Description |
|---|---|
| Base Object Type | Object against which the symptom is evaluated.<br>Based on the selected object type, the list of available properties displays only the properties applicable to the object type. |
| Symptom Type | Select **Properties** from the Symptom Type drop-down list. |
| Select Specific Object | If a property is not listed in the common properties list, based on the selected based object type, use Select Object to inspect the properties of a selected object so that you can locate the property that you must use to create the symptom. Even though you select a property for a specific object, the symptom definition is applicable to all objects with that property in your environment. |
| Search | Use a word search to limit the number of items that appear in the list. |

| Option | Description |
|---|---|
| Property list | List of properties for the selected base object type.<br>Click and drag the property to the left pane. |
| Symptom Definition workspace<br>The properties are configured values that are compared to the value you specify. You can configure a single property symptom or add multiple symptoms.<br>For example, if you need an alert when a particular property, such as Memory Hot Add, is no longer at the value required, you can configure a symptom and add it to an alert definition. | |
| Property | Configure the options:<br>• Symptom name. Name of the symptom as it appears in the symptom list when configuring an alert definition, as it appears when the alert is generated, and when viewing triggered symptoms.<br>• Condition. Determines how the value you specify in the value text box is compared to the current value of the property for an object when the symptom definition is evaluated.<br>• Value. Value that the condition evaluates.<br>• Criticality. Severity of the symptom when it is triggered.<br>• Wait Cycle. The trigger condition should remain true for this number of collection cycles before the symptom is triggered. The default value is 1, which means that the symptom is triggered in the same collection cycle when the condition became true.<br>• Cancel Cycle. The symptom is canceled after the trigger condition is false for this number of collection cycles after which the symptom is cancelled. The default value is 1, which means that the symptom is canceled in the same cycle when the condition becomes false.<br>• Evaluate on instanced properties. Select this check box so that the system evaluates the object level symptom as well as the instance level symptom. For example, for memory usage, when the check box is not selected, the symptom is triggered based on the object's memory usage. However, if you select the check box, the system also evaluates memory usage of each of the cores. If any of the cores is found to be crossing the threshold, the symptom is triggered.<br>• Drop instances to exclude. To exclude specific instanced properties from the symptom, drag the property instances from the right pane. If you cannot locate the property instance you want to exclude, you can search for it in another object that uses the property by clicking **Select Specific Object** next to the search box. |

3. Click **Save**.

## Message Event Symptoms

Message event symptoms are based on events received as messages from a component of VCF Operations or from an external monitored system through the system's REST API. You define symptoms based on message events to include in alert definitions that use these symptoms. When the configured symptom condition is true, the symptom is triggered.

The adapters for the external monitored systems and the REST API are inbound channels for collecting events from external sources. Adapters and the REST server both run in the system. The external system sends the messages, and VCF Operations collects them.

You can create message event symptoms for the supported event types. The following list is of supported event types with example events.

- System Performance Degradation. This message event type corresponds to the EVENT_CLASS_SYSTEM and EVENT_SUBCLASS_PERFORM_DEGRADATION type and subtype in the VCF Operations API SDK.

- Change. The VMware adapter sends a change event when the CPU limit for a virtual machine is changed from unlimited to 2 GHz. You can create a symptom to detect CPU contention issues as a result of this configuration change. This message event type corresponds to the EVENT_CLASS_CHANGE and EVENT_SUBCLASS_CHANGE type and subtype in the VCF Operations API SDK.
- Environment Down. The VCF Operations adapter sends an environment down event when the collector component is not communicating with the other components. You can create a symptom that is used for internal health monitoring. This message event type corresponds to the EVENT_CLASS_ENVIRONMENT and EVENT_SUBCLASS_DOWN type and subtype in the VCF Operations API SDK.
- Notification. This message event type corresponds to the EVENT_CLASS_NOTIFICATION and EVENT_SUBCLASS_EXTEVENT type and subtype in the VCF Operations API SDK.

**Where You Find Message Event Symptoms**

To manage symptoms based on message events, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Symptom Definitons** tile. From the **Symptom Definitions** page, click the **Message Event** tab.

You can also define symptoms as you are defining alerts in the Alert Definition Workspace.

The Message Event Symptom Definitions is a list of the message event-based symptoms defined in your VCF Operations environment. You use the information in the list to evaluate the defined message events and to determine if you want to add, edit, or clone symptoms.

| Option | Description |
|---|---|
| Toolbar options | Use the toolbar options to manage your symptoms. You can select multiple symptoms using Ctrl+click or Shift+click.<br>• Add. Add a symptom definition.<br>Click the horizontal ellipsis to perform the following actions.<br>• Edit. Modify the selected symptom definition. Any changes you make affect the alert definitions that include this symptom. You cannot edit a symptom that manages a badge.<br>• Delete. Remove the selected symptom definition. You cannot delete an alert that is used in an alert definition. To delete a symptom, you must first remove it from the alert definitions in which it is used. You cannot delete a symptom that manages a badge.<br>• Clone. Create a copy of the selected symptom definition.<br>• Export and Import. Export the file as xml from one VCF Operations so that you can import the file on another instance. When you import the file, if you encounter a conflict, you can override the existing file or not import the new file. |
| Filter options | Limits the list based on the text you type.<br>You can also sort on the columns in the data grid. |
| Name | Descriptive name of the symptom. |
| Adapter Type | Adapter type for which the symptom is configured. |
| Object Type | Base object type against which the symptom is defined. |
| Event Type | Defined event classification type. |
| Operator | Operator used to compare the message from the incoming event against the event message specified in the symptom. |
| Event Message | Text string that is compared to the message in the incoming event using the specified operator. |
| Criticality | Severity of the symptom when it is triggered. |
| Defined By | Indicates whether the symptom was created by a user or provided with a solution adapter. |

| Option | Description |
|---|---|
| Last Modified | Displays the date on which the symptom was last modified. |
| Modified By | Displays the name of the user who last modified the symptom. |

## Defining Message Events

You can define message event systems in VCF Operations so that you can create one or more of the symptoms that you can add to an alert definition. A message event symptom is triggered when a message in an incoming event matches the text string in the symptom, based on the specified operator.

1. To define symptoms based on message events, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Symptom Definitons** tile.
2. Select the **Message Event** tab, and then click **Add**.

   **Note:**
3. Enter the following details.

**Table 1209: Symptoms Workspace Options for Message Events**

| Option | Description |
|---|---|
| Based Object Type | Object against which the symptom is evaluated. |
| Symptom Type | Displays the symptom type as **Message Event**. |
| Select the Type of Event | Select the type of incoming event against which you are matching the events as they arrive. The incoming event must contain the following type and subtype combinations.<br>• System Degradation<br>• Change<br>• Environment<br>• Notification<br>• Data Availability<br>• Collector Down<br>• Object Error<br>Click and drag the message event to the left pane. |
| Symptom Definition workspace<br>The Message Event text string is compared to the message in the incoming event by using the specified operator. You can configure a single message event symptom or add multiple symptoms.<br>For example, the VMware adapter sends a change event when the CPU limit for a virtual machine was changed from unlimited to 2 GHz. You can create a symptom to detect CPU contention issues as a result of this configuration change. | |
| Message Event | Configure the options:<br>• Symptom name. Name of the symptom as it appears in the symptom list when configuring an alert definition, as it appears when the alert is generated, and when viewing triggered symptoms.<br>• Condition. Determines how the value you specify in the value text box is compared to the current value of the property for an object when the symptom definition is evaluated.<br>• Value. Value that the condition evaluates.<br>• Criticality. Severity of the symptom when it is triggered. |

4. Click **Save**.

# Fault Symptoms

Fault symptoms are based on events published by monitored systems. VCF Operations correlates a subset of these events and delivers them as faults. Faults are intended to signify events in the monitored systems that affect the availability of objects in your environment. You define symptoms based on faults to include in alert definitions that use these symptoms. When the configured symptom condition is true, the symptom is triggered.

You can create fault symptoms for the supported published faults. Some object types have multiple fault definitions from which to choose, while others have no fault definitions.

If the adapter published fault definitions for an object type, you can select one or more fault events for a given fault while you define the symptom. The symptom is triggered if the fault is active because of any of the chosen events. If you do not select a fault event, the symptom is triggered if the fault is active because of a fault event.

## Where You Find Fault Symptoms

To manage symptoms based on fault message events, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Symptom Definitons** tile. From the **Symptom Definitions** page, click the **Fault** tab.

You can also define symptoms as you are defining alerts in the Alert Definition Workspace.

The Fault Symptom Definitions is a list of the fault-based symptoms defined in your VCF Operations environment. You use the information in the list to evaluate the defined fault message events and to determine whether to add, edit, or clone symptoms.

| Option | Description |
|---|---|
| Toolbar options | Use the toolbar options to manage your symptoms. You can select multiple symptoms using Ctrl+click or Shift+click.<br>• Add. Add a symptom definition.<br>Click the horizontal ellipsis to perform the following actions.<br>• Edit. Modify the selected symptom definition. Any changes you make affect the alert definitions that include this symptom. You cannot edit a symptom that manages a badge.<br>• Delete. Remove the selected symptom definition. You cannot delete an alert that is used in an alert definition. To delete a symptom, you must first remove it from the alert definitions in which it is used. You cannot delete a symptom that manages a badge.<br>• Clone. Create a copy of the selected symptom definition.<br>• Export and Import. Export the file as xml from one VCF Operations so that you can import the file on another instance. When you import the file, if you encounter a conflict, you can override the existing file or not import the new file. |
| Filter options | Limits the list based on the text you type.<br>You can also sort on the columns in the data grid. |
| Name | Descriptive name of the symptom. |
| Adapter Type | Adapter type for which the symptom is configured. |
| Object Type | Base object type against which the symptom is defined. |
| Fault | Selected fault based on object type. |
| Defined By | Indicates whether the symptom was created by a user or provided with a solution adapter. |
| Last Modified | Displays the date on which the symptom was last modified. |
| Modified By | Displays the name of the user who last modified the symptom. |

### Defining Fault Symptoms

You can define fault symptoms in VCF Operations that are based on events published by the monitored systems, so that you can add one or more symptoms to an alert definition. A fault symptom is triggered when a fault is active on the base object because of the occurrence of any of the fault events selected in the symptom definition.

1. To define symptoms based on fault message events, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Symptom Definitons** tile.
2. Select the **Fault** tab and then, click **Add**.
3. Enter the following details.

| Option | Description |
|---|---|
| Based Object Type | Object against which the symptom is evaluated. |
| Symptom Type | Displays the symptom type as **Fault**. |
| Fault Definitions | Select the fault definition for the selected base object type. Some object types do not have fault definitions, and other types have multiple definitions.<br>Click and drag the fault definition to the left pane. |
| Symptom Definition workspace<br>The fault events are published events from monitored systems. You can configure a single fault event symptom or add multiple symptoms.<br>For example, if your base object is host and you drag the Hardware sensor fault for unknown type fault definition, you then select one of two text strings indicating a fault. | |
| Fault Symptom | Configure the options:<br>• Symptom name. Name of the symptom as it appears in the symptom list when configuring an alert definition, as it appears when the alert is generated, and when viewing triggered symptoms.<br>• Value. Select one or more fault events that activate the fault. If you do not select a string, then any of the provided strings are evaluated.<br>• Criticality. Severity of the symptom when it is triggered. |

4. Click **Save**.

## Logs Symptoms

You can create symptoms in VCF Operations based on events ingested by Log management. You can then configure alerts to use the log based symptoms. Previously you had to create and manage alerts in VMware Aria Operations for Logs and VMware Aria Operations separately. Now, VCF Operations provides a unified experience to define log based symptoms and alerts. The prerequisite is the integration between Log management and VCF Operations. For more details, see Configuring and Analyzing Logs.

When you define a symptom in VCF Operations, an alert definition is created in Log management. You must define the query, trigger conditions and the objects on which these conditions must be met. Since Log management does not have a concept of objects, the object scope for the symptom definition is limited to vCenter objects. To define symptoms for other types of objects, use the Log Fallback Object option.

When the alert is triggered in Log management, an event is generated in VCF Operations. VCF Operations processes this event based on the corresponding symptom definition and triggers the alert.

> **Note:**
>
> It can take up to two minutes for the alert definition to be created in Log management after you define a log based symptom in VCF Operations. You cannot modify the alert definition in Log management but can modify it in VCF Operations. For more information, see, Creating Alert Definitions.

To define log based symptoms, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Symptom Definitons** tile. Select the **Logs** tab. Click the **ADD** button to define a Logs based symptom.

**Note:**

If the list of of symptoms is empty and the **Add** button is not available, verify that the prerequisite of integration between Log management and VCF Operations is met.

| Option | Description |
|---|---|
| Base Object Type | Since the alert definition is triggered in Log management , only the following limited set of objects can be selected. They are:<br>• vCenter<br>– Host System<br>– vCenter<br>– Virtual Machine<br>If you want to create a symptom definition for other object types, then select:<br>• Container<br>– Log Fallback Object |
| Symptom Type | Drag and drop the Logs symptom type to the left workspace of the page to continue to the definition. |
| Symptom Trigger Condition | 1. Provide a name for the symptom.<br>2. Define the criticality of the event based on the function and condition.<br>3. Select a time period and group the symptom by static or extracted fields.<br>4. Select a query from the drop-down list or enter a text to search for to filter by.<br>5. Optionally, select a partition to include in the scope of the search.<br>6. Optionally, add more filters.<br>7. Click **SAVE** |

Once you define a symptom, you can create an alert for the same object for which you created the symptom. To define an alert, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations** and then click the **Alerts Definitions** tile. For more information, see, Creating Alert Definitions.

## Understanding Negative Symptoms for VCF Operations Alerts

Alert symptoms are conditions that indicate problems in your environment. When you define an alert, you include symptoms that generate the alert when they become true in your environment. Negative symptoms are based on the absence of the symptom condition. If the symptom is not true, the symptom is triggered.

To use the absence of the symptom condition in an alert definition, you negate the symptom in the symptom set.

All defined symptoms have a configured criticality. However, if you negate a symptom in an alert definition, it does not have an associated criticality when the alert is generated. If the symptom is triggered because the condition is true, the symptom criticality will be the same as the configured criticality.

When negative symptoms are triggered and an alert is generated, the effect on the criticality of the alert depends on how the alert definition is configured.

The following table provides examples of the effect negative symptoms have on generated alerts.

**Table 1210: Negative Symptoms Effect on Generated Alert Criticality**

| Alert Definition Criticality | Negative Symptom Configured Criticality | Standard Symptom Configured Criticality | Alert Criticality When Triggered |
|---|---|---|---|
| Warning | One Critical Symptom | One Immediate Symptom | Warning. The alert criticality is based on the defined alert criticality. |
| Symptom Based | One Critical Symptom | One Warning Symptom | Warning. The negative symptom has no associated criticality and the criticality of the standard symptom determines the criticality of the generated alert. |
| Symptom Based | One Critical Symptom | No standard symptom included | Info. Because an alert must have a criticality and the negative alert does not have an associated criticality, the generated alert has a criticality of Info, which is the lowest possible criticality level. |

# Recommendations in VCF Operations

Recommendations are probable solutions for an alert generated. You can create a library of recommendations that include instructions to your environment administrators or actions that they can run to resolve an alert.

Recommendations provide your network engineers or virtual infrastructure administrators with information to resolve alerts.

Depending on the knowledge level of your users, you can provide more or less information, including the following options, in any combination.

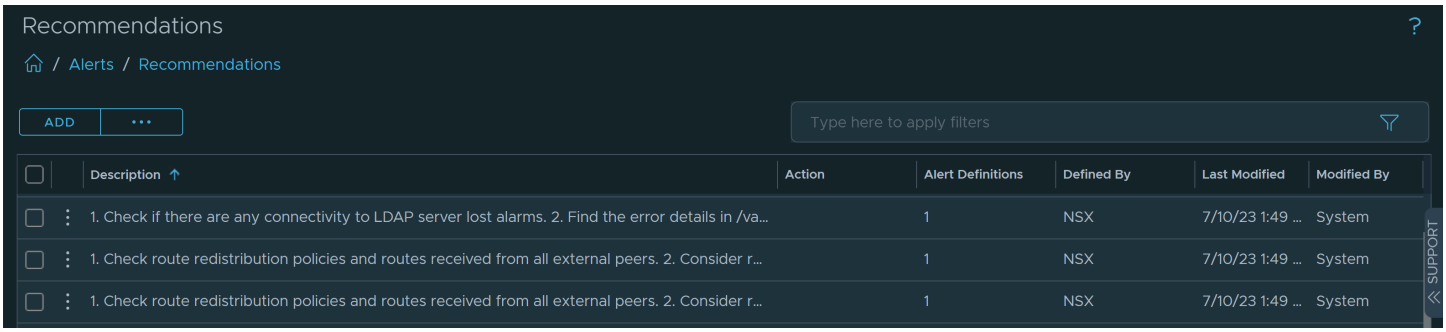
- One line of instruction.
- Steps to resolve the alert on the target object.
- Hyperlink to a Web site, runbook, wiki, or other source.
- Action that makes a change on the target object.

When you define an alert, provide as many relevant action recommendations as possible. If more than one recommendation is available, arrange them in priority order so that the solution with the lowest effect and highest effectiveness is listed first. If no action recommendation is available, add text recommendations. Be as precise as possible when describing what the administrator should do to fix the alert.

### Where You Find Recommendations

To define recommendations, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Recommendations** tile.

You can also define recommendations when you create an alert definition.




| Option | Description |
|---|---|
| Toolbar options | Use the toolbar options to manage your recommendations.<br>• Add. Add a recommendation.<br>Click the horizontal ellipsis to perform the following actions.<br>• Edit. Modify the selected recommendation.<br>• Delete. Remove the selected recommendation.<br>• Clone. Create a copy of the selected recommendation so that you can create a new recommendation that uses the current one.<br>• Export. Downloads the recommendations.<br>• Import. Allows you to import recommendations. To import:<br>– Click the **Import** option from the horizontal ellipsis.<br>– Click **Browse** and select the file to import.<br>– Select if you want to Overwrite or Skip the file in case of a conflict.<br>– Click **Import** to import the recommendation, and click **Done**. |
| Quick Filter | Limits the list based on the text you type.<br>You can also sort on the columns in the data grid. |
| Description | Displays the recommendation text that is provided when the alert is generated.<br>Click this link to view the Details Page. On this page, you can view the alert definitions assigned for a particular recommendation. To remove the selected recommendation from all alert definitions, click **Edit** and then, in the **Edit Recommendation** page, click **Remove from all**. |
| Action | If the recommendation includes running an action, the name of the action is displayed. |
| Alert Definitions | Displays the number of alert definitions assigned for a particular recommendation. |
| Defined By | Indicates whether the recommendation was created by a user or provided with a solution adapter. |
| Last Modified | Displays the date on which the recommendation was last modified. |
| Modified By | Displays the name of the user who last modified the recommendation. |

## Defining Recommendations for Alerts

You can create recommendations that are solutions to alerts generated. The recommendations are intended to ensure that your network operations engineers and virtual infrastructure administrators can respond to alerts as quickly and

accurately as possible. A recommendation can contain links to useful Web sites or local runbooks, instructions as text, or actions that you can initiate from VCF Operations.

1. To define recommendations, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Recommendations** tile.
2. Click **Add** and enter the following details.

| Option | Description |
|---|---|
| Description | Enter the description of what must be done to resolve the triggered alert.<br>The description can include steps a user must take to resolve the alert or it might be instructions to notify a virtual infrastructure administrator.<br>This is a text field. |
| Create a hyperlink | To create a hyperlink, enter a description, select the text, and click the Hyperlink icon to make the text a hyperlink to a Website or local wiki page. |
| Action (Optional) | |
| Adapter Type | Select an adapter type from the drop-down list to narrow down the list of actions displayed in the Actions field. |
| Action | You can add an action as a method to resolve a triggered symptom or a generated alert. Actions must already be configured.<br>You must provide text in the text box to describe the action before you can save the recommendation. |

3. Click **Save**.

These actions, named `Delete Unused Snapshots for Datastore Express` and `Delete Unused Snapshots for VM Express` appear. However, they can only be run in the user interface from an alert whose first recommendation is associated with this action. You can use the REST API to run these actions.

The following actions are also not visible except in the alert recommendations:

- `Set Memory for VM Power Off Allowed`
- `Set CPU Count for VM Power Off Allowed`
- `Set CPU Count and Memory for VM Power Off Allowed`

These actions are intended to be used to automate the actions with the `Power Off Allowed` flag set to true.

## Notifications in VCF Operations

Notifications are alert notifications that meet the filter criteria in the notification rules before they are sent outside VCF Operations. You can configure notification rules for the supported outbound alerts so that you can filter the alerts that are sent to the selected external system.

You can use the notifications list to manage your rules and then use the notification rules to limit the alerts that are sent to the external system. To use notifications, the supported outbound alert plug-ins must be added and running.

With notification rules, you can limit the data that is sent to the following external systems.

- Standard Email. You can create multiple notification rules for various email recipients based on one or more of the filter selections. If you add recipients but do not add filter selections, all the generated alerts are sent to the recipients.
- SNMP Trap. You can configure VCF Operations to log alerts on an existing SNMP Trap server in your environment.
- Log File. You can configure log alerts to a file on each of your VCF Operations nodes.

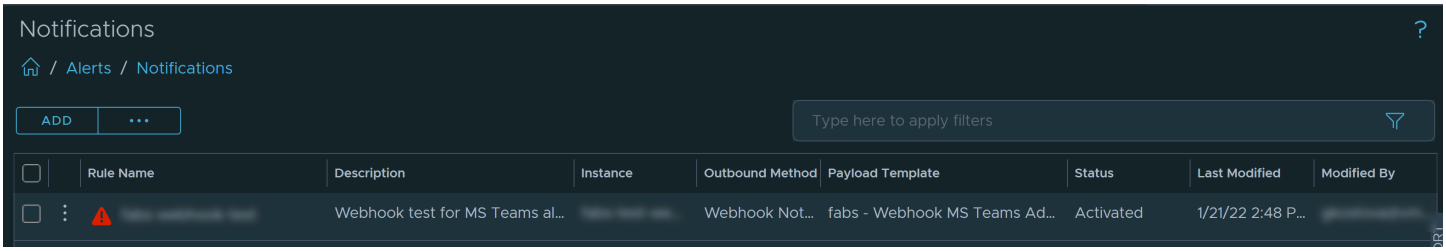
You configure notification options to specify which alerts are sent out for the Standard Email, SNMP, and Log File outbound alert plug-ins. For the other plug-in types, all the alerts are sent when the target outbound alert plug-in is activated.

The most common outbound alert plug-in is the Standard Email plug-in. You configure the Standard Email plug-in to send notifications to one or more users when an alert is generated that meets the criteria you specify in the notification settings.

## Where You Find Notifications

To manage your notifications, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Notifications** tile.

**Note:**
To use notifications, the supported outbound alert plug-ins must be added and running.




| Option | Description |
|---|---|
| Toolbar options | Use the toolbar options to manage your notification rules.<br>• Add. Opens the Add Rule dialog box where you configure the filtering options for the notification rule.<br>Click the horizontal ellipsis to perform the following actions.<br>• Delete. Removes the selected rule.<br>• Deactivate or Activate. Deactivates or activates the selected rule(s).<br>• Export or Import. Export the selected notifications to a ".xml" file so that you can import it on another VCF Operations instance. |
| Quick Filter (Action Name) | Limits the list to actions matching the filter. You can filter by:<br>• Rule Name<br>• Instance<br>• Status<br>• Modified By |
| Rule Name | Name you assigned when you created the notification rule. Click the vertical ellipsis to perform the following actions.<br>• Edit. Allows you to edit the selected rule.<br>• Clone. Allows you to clone an existing notification rule and edit the attributes of the cloned notification rule. You can create multiple alert notification rules so that you can send the same alert notifications to different outbound settings.<br>**Note:**<br>You can clone only one alert notification rule at a time.<br>• Delete. Removes the selected rule.<br>• Deactivate or Activate. Deactivates or activates the selected rule.<br>• Export. Downloads the notification setting. |
| Description | Description of the notification rule. |
| Instance | Name of the configured outbound alert instance for the notification rule.<br>Instances are configured as part of the outbound alerts and can indicate different email servers or sender addresses for alert notifications. |
| Outbound Method | Displays the type of the outbound method that is configured. |
| Payload Template | Displays the payload template that is used. |
| Status | Displays if the rule is activated or not. |
| Email Address | If the rule is for standard email notifications, the alert recipient email addresses are listed. |
| Object Name | If the rule specifies a notification for a particular object, the object name is listed. |

| Option | Description |
|---|---|
| Children | If the rule specifies a notification for a particular object and selected child objects, the child object types are listed. |
| Last Modified | Displays the date on which the rule was last modified. |
| Modified By | Displays the name of the user who last modified the rule. |

## Creating Notification Rules for Alerts

You add, manage, and edit your notification rules in VCF Operations. To send notifications to a supported system, you must configure and activate the settings for outbound alerts.

Before you can create and manage your notification rules, you must configure the outbound alert plug-in instances. For details on configuring outbound plug-ins, see Adding Outbound Notification Plug-Ins in VCF Operations.

You use the Notifications page to manage your alert notification rules. The rules determine which VCF Operations alerts are sent to the supported target systems.

Notification rules are filters that limit the data sent to external systems by using outbound alert plug-ins that are supported, configured, and running. Rather than sending all alerts to all your email recipients, you can use notification rules to send specific alerts. For example, you can send health alerts for virtual machines to one or more of your network operations engineers. You can send critical alerts for selected hosts and clusters to the virtual infrastructure administrator for those objects. Before you can create and manage notification rules, you must configure the outbound alert plug-in instances.

1. To manage your notifications, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Notifications** tile. On the toolbar, click **Add** to add a rule, or click the vertical ellipsis and select **Edit** to edit the selected rule.
2. Enter the following notification details.

| Option | Description |
|---|---|
| Name | Name of the rule that you use to manage the rule instance. |
| Description | Description of the rule. |
| Notification Status | Either activate or deactivate a notification setting. Deactivating a notification will stop the alert notification for that setting and activating it will activate the alert notification. |
| Advanced Settings | |
| Notification Type | Select **Alert** from the drop-down menu.<br>**Note:**<br>Select **Action** as your Notification Type if you want to create Workload Placement (WLP) Action based notification. For details, see Creating Notification Rules for Notification Type 'Action'. |

3. Click **Next**.
4. Define criteria for the notification rule.

| Option | Description |
|---|---|
| Object Scope | |
| Criteria | Object Type, Object, Tags, Applications, and Tiers for which you are filtering the alert notifications.<br>After you select the type, you select the specific instance. For example, if you select **Object**, you then select the specific object by name and determine whether to include any child objects. |
| Alert Scope | |

| Option | Description |
|---|---|
| Category | Alert Types/Subtypes, Alert Impact, or Alert Definition that triggers the alert.<br>After you select the criteria, you can configure the specific selections associated with the criteria. For example, if you select **Alert Definition**, you then select the alert definition that limits the data to alerts with this definition. You can select multiple alert definitions as conditions for a notification to trigger. |
| Criticality | Defined criticality of the alert that results in the data being sent to an external system. For example, if you select **Critical**, then the data that is sent to the external system must also be labeled as critical. |
| Control State | State of the alert, either opened, assigned, or suspended. |
| Notify On | |
| Status | Current state of the alert, either canceled, updated, or new. |
| Notification Heartbeat | |
| Heartbeat | Set this to **Active** if you want to send repeat notifications for the active alerts. The frequency of the notification depends on the collection interval set for the adapter whose object is being evaluated. By default, this checkbox is not selected.<br>**Note:**<br>Setting this option to **Active** can cause a potential surge in the notifications that are generated. |
| Advanced Filters: By Collector | |
| Collector/Group | Select a collector or group if you want to receive notifications for the objects that receive data from the selected collector/group. |
| **Note:**<br>If you do not define any alert filters in the Define Criteria tab, then the notification will be sent for all the alerts without applying any conditions for the object scope, alert scope, or alert state. | |

5. Click **Next**.
6. Select the outbound method that you want to use to send your notification.

| Option | Description |
|---|---|
| Outbound Method | • Select Plug-In Type: Type of plugin. Select one of the outbound alert plug-in types: Log File Plugin, Standard Email Plugin, SNMP Trap Plugin, Webhook Notification Plugin, Slack Plugin, and Service-Now Notification Plugin.<br>• Select Instance: Select the configured instance for the type of plug-in.<br>• Create New Instance: You can also create a new outbound instance for the plug-in type you select.<br>For details, see Adding Outbound Notification Plug-Ins in VCF Operations. |

7. Click **Next**.

8. Select the payload template.

| Option | Description |
|---|---|
| Payload Template | Select the payload template that you want to include in the notification. Each plug-in has its default template and you can select the default template if no customization is required. The template includes additional information about the alert or the object that is displayed in the notification. You can also customize your payload for a Webhook Notification Plugin. For details on creating payload templates, see Creating Payload Templates for Outbound Plugins. |
| The values in this tab differ based on the outbound plug-in you have selected in the previous step. | |
| Outbound Method -Standard Email Plugin | If you are configuring notifications for standard email, you can add recipients and associated information.<br>• Recipient(s). Enter the email addresses of the individuals to whom you are sending email messages that contain alert notifications. If you are sending to more than one recipient, use a semicolon (;) between addresses.<br>• Cc Recipients. Enter the email addresses of the individuals that have to be cc'd for the email.<br>• Bcc Recipients. Enter the email addresses of the individuals that have to be bcc'd for the email.<br>• Notify again. Number of minutes between notifications messages for active alerts. Leave the text box empty to send only one message per alert.<br>• Max Notifications. Number of times to send the notification for the active alert. Leave the text box empty to send only one message per alert.<br>• Delay to notify. Number of minutes to delay before sending a notification when a new alert is generated. For example, if the delay is 10 minutes and a new alert is generated, the notification is not sent for 10 minutes. If the alert is canceled in those 10 minutes, the notification is not sent. The notification delay reduces the number of notifications for alerts that are canceled during that time.<br>• Description. Enter the text to include in the email message. For example, the Attention Host Management team. |

| Option | Description |
| --- | --- |
| Outbound Method - Service-Now Notification Plugin | If you are configuring notifications for a Service-Now notification plug-in, you can add instances and associated information.<br>• Caller. Enter the name of the person who reported the incident or who is affected by the incident.<br>• Category. Specify the category to which the incident belongs.<br>• Sub Category. Specify the sub-category to which the incident belongs.<br>• Business Service. Specify the business service of the incident.<br>• Contact Type. Enter the contact type.<br>• State. Enter the incident state in digits.<br>• Resolution Code. Enter the resolution code for the incident.<br>• Resolution notes. Enter the resolution notes for the incident.<br>• On hold reason. Enter the reason as to why the incident is on hold.<br>• Impact. Set the incident impact in digits. Impact measures the business criticality of the affected service.<br>• Urgency. Set urgency for the incident in digits. Urgency defines the number of days taken to resolve an incident.<br>• Priority. Enter the priority for the incident. Priority defines the sequence in which the incident must be resolved.<br>• Assignment Group. Enter the assignment group for the incident.<br>• Assigned To. Enter the details of the person to whom the incident is assigned.<br>• Severity. Set the severity for the incident in digits.<br>• Upon Approval. Specify the next steps to be taken upon incident approval.<br>• Problem. Enter the details of the related problem if it exists.<br>• Cause by change. Enter the change request which triggered the incident.<br>• Change Request. Enter the details for the related change list if it exists. |
| Outbound Method - Slack Plugin | If you are configuring notifications for a Slack plugin, add the Webhook URL of Slack. For example, the Webhook URL is in the format: `https://hooks.slack.com/services/T00000000/B00000000/XXXXXXXXXXXXXXXXXXXXXXXX`.<br>Create and authorize an app within Slack to obtain the Webhook URL. For details on creating and authorizing an app within Slack, refer to the Slack Documentation.<br>Once you have created the notification rule, the alerts are displayed within that particular Slack channel with a link to the alert. Click the link to view the details of the alert on the Object Summary page. |

9. Click **Next** to test the notification
10. In the **Test Notification** tab, click **Initiate Process** to initiate the notification configuration validation process.
11. If you activate the **Filter the alert definitions and objects based on criteria outlined in the Define Criteria section** option, then the alert definitions and objects displayed below are based on the criteria outlined in the **Define Criteria** section.
12. Select an alert definition and an object for validation.
13. Click **Validate Configuration**.

– Notification Validation Steps: View the steps involved in validating the notification configuration. The steps also indicate an error, if any. The validation steps differ based on the selected Outbound method.

| Outbound Method | Validation Steps |
|---|---|
| Log File Plugin | • Validate Criteria<br>• Permissions<br>• File Created |
| Standard Email Plugin | • Validate Criteria<br>• Establish Endpoint Connection<br>• Certificates<br>• Authentication<br>• Send Notification<br>**Note:**<br>The Standard Email Plugin does not provide a response that can be validated. |
| SNMP Trap Plugin | • Validate Criteria<br>• Establish Endpoint Connection<br>• Send Notification<br>**Note:**<br>The SNMP Trap Plugin does not provide a response that can be validated. |
| Webhook Notification Plugin | • Validate Criteria<br>• Establish Endpoint Connection<br>• Certificates<br>• Authentication<br>• Send Notification<br>• Endpoint Receives Notification |
| Slack Plugin | • Validate Criteria<br>• Establish Endpoint Connection<br>• Authentication<br>• Send Notification<br>• Endpoint Receives Notification |
| ServiceNow Notification Plugin | • Validate Criteria<br>• Establish Endpoint Connection<br>• Authentication<br>• Send Notification<br>• Endpoint Receives Notification |

– Response: The Response tab in the right pane displays if the test passed successfully or if there were any errors.
– Body: Displays the content of the notification.

14. Click **Create** to create the notification rule. You can view the rule you created under **Alerts** > **Notifications**.

## Creating Notification Rules for Notification Type 'Action'

You can create and manage Workload Placement (WLP) Action based notification.

You can configure actions for which you want to receive notifications. Once configured, notifications are sent indicating the successful, failed, or timed out WLP action.

**Note:**
The notifications are sent once the WLP Virtual Machine movement task is completed irrespective of its status.

Before you can create and manage your notification rules, you must configure the Webhook Notification Plugin. For details, see Add a Webhook Notification Plugin for Outbound Instance.

To create notification rules for WLP action:

1. in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Notifications** tile. On the toolbar, click **Add** to add a rule, or click the vertical ellipsis and select **Clone** to clone the selected rule.

   **Note:**
   You cannot change the **Notification Type** while editing a selected notification rule,
2. Enter the following notification details.

| Option | Description |
|---|---|
| Name | Name of the rule that you use to manage the rule instance. |
| Description | Description of the rule. |
| Notification Status | Either activate or deactivate a notification setting. Deactivating a notification will stop the notification for that setting and activating it will activate the notification. |
| Advanced Settings | |
| Notification Type | Select **Action** from the drop-down menu. |

3. Click **Next**.
4. Define criteria for the notification rule.

| Option | Description |
|---|---|
| Object Scope: Select set of Objects for which you want to receive notifications. | |
| Criteria | Select the Criteria as **Object** from the drop-down menu.<br>Search for a specific object by name and determine if you want to include any child or descendant objects, and then add one or more child/descendant objects.<br>The action triggers on ANY of the selected objects: |
| Notify On | |
| Status | Select the action status for which you want to receive notification. You can receive notification for **Succeeded**, **Failed**, and **Timed Out** statuses. |

5. Click **Next**.
6. Select the outbound method that you want to use to send your notification.

| Option | Description |
|---|---|
| Outbound Method | • By default, the outbound method supported is **Webhook Notification Plugin**.<br>• Select Instance: Select the configured instance for the Webhook plug-in.<br>• Create New Instance: You can also create a new outbound instance for the Webhook plug-in type. For details, see Add a Webhook Notification Plugin for Outbound Instance. |

7. Click **Next**.

8. Select the payload template.

| Option | Description |
|---|---|
| Payload Template | Select the Webhook payload template that you want to include in the notification. There is a Default WLP Action Webhook Template and you can select the default template if no customization is required. You can also customize your payload for a Webhook Notification Plugin. For details on creating payload templates, see Creating Payload Templates for Outbound Plugins. |

9. Click **Next** to test the notification
10. In the **Test Notification** tab, click **Initiate Process** to initiate the notification configuration validation process.
11. If you activate the **Filter the objects based on criteria outlined in the Define Criteria section** option, then the objects displayed below are based on the criteria outlined in the **Define Criteria** section.
12. Select an object for validation.
13. Click **Validate Configuration**.
    - Notification Validation Steps: View the steps involved in validating the notification configuration. The steps also indicate an error, if any. The validation steps differ based on the selected Outbound method.

| Outbound Method | Validation Steps |
|---|---|
| Webhook Notification Plugin | • Validate Criteria<br>• Establish Endpoint Connection<br>• Certificates<br>• Authentication<br>• Send Notification<br>• Endpoint Receives Notification |

    - Response: The Response tab in the right pane displays if the test passed successfully or if there were any errors.
    - Body: Displays the content of the notification.
14. Click **Create** to create the notification rule. You can view the rule you created under **Alerts** > **Notifications**.

## Outbound Settings in VCF Operations

You use the Outbound Settings to manage your communication settings so that you can send information to users or applications outside of VCF Operations.

You manage your outbound options from this page, including adding or editing outbound plug-ins, and turning the configured plug-ins on or off. When activated, the plug-in sends a message to users as email notifications, or sends a message to other applications.

Outbound plug-in settings determine how the supported external notification systems connect to their target systems. You configure one or more instances of one or more plug-in types so that you can send data about generated notifications outside of VCF Operations.

You configure each plug-in with the required information, including destination locations, hosts, ports, user names, passwords, instance name, or other information that is required to send notifications to those target systems. The target systems can include email recipients, log files, or other management products.

Some plug-ins are included with VCF Operations, and others might be added when you add a management pack as a solution.

The configuration options vary depending on which plug-in you select from the **Plug-In Type** drop-down menu.

To add outbound notification plug-in, see Adding Outbound Notification Plug-Ins in VCF Operations.

## Where You Find Outbound Settings

To manage your outbound settings, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Outbound Settings** tile.

| Option | Description |
|---|---|
| Toolbar options | Use the toolbar options to manage your Outbound Plug-Ins.<br>• Add. Opens the Outbound Plug-In dialog box where you configure the connection options for the instance.<br>Select an existing plugin and click the vertical ellipsis to perform the following actions.<br>• Edit. Modify the Outbound Plug-In instance details.<br>• Delete. Removes the selected plug-in instance.<br>• Activate or Deactivate. Starts or stops the plug-in instance. Deactivating an instance allows you to stop sending the messages configured for the plug-in without removing the configuration from your environment.<br>• Export. Downloads the outbound settings.<br>• Import. Allows you to import outbound settings. To import:<br>– Click the **Import** option from the horizontal ellipsis.<br>– Click **Browse** and select the file to import.<br>– Select if you want to Overwrite or Skip the file in case of a conflict.<br>– Click **Import** to import outbound settings, and click **Done**. |
| Instance Name | Name that you assigned when you created the plug-in instance. Click the vertical ellipsis to perform the following actions.<br>• Edit. Allows you to edit the selected payload template.<br>• Delete. Removes the selected payload template.<br>• Activate or Deactivate. Starts or stops the plug-in instance. Deactivating an instance allows you to stop sending the messages configured for the plug-in without removing the configuration from your environment.<br>• Export. Downloads the outbound settings. |
| Plug-In Type | Type of configured plug-in for the plug-in instance. The types of plug-ins vary depending on the solutions you have added to your environment.<br>The most common plug-in types include standard email, SNMP trap, and log file. |
| Status | Specifies whether the plug-in is currently running. |

## List of Outbound Plugins

VCF Operations provides outbound plug-ins. This list includes the name of the plug-in and whether you can filter the outbound data based on your notification settings.

If the plug-in supports configuring notification rules, then you can filter the messages before they are sent to the target system. If the plug-in does not support notifications, all messages are sent to the target system, and you can process them in that application.

If you installed other solutions that include other plug-in options, they appear as a plug-in option with the other plug-ins.

Messages and alerts are sent only when the plug-in is activated.

**Table 1211: Notification Support for Outbound plug-ins**

| Outbound plug-in | Configure Notification Rules |
|---|---|
| Log File plug-in | Yes<br>To filter the log file alerts, you can either configure the file named `TextFilter.xml` or configure the notification rules. |
| Standard Email plug-in | Yes |
| SNMP Trap plug-in | Yes |
| Webhook Notification Plugin | Yes |
| Slack plug-in | Yes |
| Service-Now Notification plug-in | Yes |

## Configuring HTTP Proxy for Outbound Settings

You can configure an HTTP proxy for outbound communication over HTTP/HTTPS protocol in VCF Operations. Once the HTTP proxy is configured for an outbound setting, all the corresponding outbound HTTP(S) communication must happen through that proxy.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Outbound Settings** tile. Click the **HTTP Proxy for Outbound Settings** tab.
2. Click **Add**.

**Table 1212: HTTP Proxy Options**

| Options | Description |
|---|---|
| Proxy Name | Name of the HTTP proxy server. |
| Proxy Host/ IP | The IP address of the HTTP proxy. |
| Proxy Port | Port number used to connect to the HTTP proxy server. |
| Proxy Username | Username of the HTTP proxy server. |
| Proxy Password | Password for the HTTP proxy server username. |

3. Click **Save**.
   The HTTP proxy setting is added.

## Importing and Exporting HTTP Proxy for Outbound Settings

You can import or export the HTTP Proxy settings for outbound plugins.

### Importing HTTP Proxy for Outbound Settings

1. From the **HTTP Proxy for Outbound Settings** tab, click the horizontal ellipsis, and then click **Import** to import the outbound proxy settings.
2. Click **Browse** to select the outbound proxy setting and enter the **Encryption Key**.
3. In case of a conflict, click **Overwrite HTTP Proxy for Outbound Settings** to delete the existing proxy setting and proceed. Optionally, click **Skip HTTP Proxy for Outbound Settings** to cancel the import.

   **Note:**
   A conflict happens when you try to import a proxy with a name that already exists in the proxy list. Proxy names are unique and cannot be repeated.
4. Click **Import**.

The HTTP proxy import process begins.

**Exporting HTTP Proxy for Outbound Settings**

1. From the **HTTP Proxy for Outbound Settings** tab, select the HTTP Proxy you want to export, click the horizontal ellipsis, and then click **Export**.
2. Enter a new password in the **Setup a new password to export data** field.
3. Re-enter the password in the **Repeat a password** field.
4. Click **Export**.

The outbound proxy settings data gets exported in the `.json` format. The passphrase entered at the time of export is used to encrypt the sensitive information. The same passphrase must be used at the time of import.

## Adding Outbound Notification Plug-Ins in VCF Operations

You add outbound plug-in instances so that you can notify users about alerts or capture alert data outside of VCF Operations.

You can configure one or more instances of the same plug-in type if you need to direct alert information to multiple target systems.

### Add a Standard Email Plug-In for VCF Operations Outbound Alerts

You add a Standard Email Plug-In so that you can use Simple Mail Transfer Protocol (SMTP) to email VCF Operations alert notifications to your virtual infrastructure administrators, network operations engineers, and other interested individuals.
Ensure that you have an email user account that you can use as the connection account for the alert notifications. If you choose to require authentication, you must also know the password for this account.

1. In the VCF Operations navigation bar at the top, click **Home** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Outbound Settings** tile.
2. Click **Add**, and from the **Plug-In Type** drop-down menu, select **Standard Email Plugin**.
   The dialog box expands to include your SMTP settings.
3. Enter an **Instance Name**.
   This is the name that identifies this instance that you select when you later configure notification rules.
4. Configure the SMTP options appropriate for your environment.

| Option | Description |
|---|---|
| **Use Secure Connection** | Activates secure communication encryption using SSL/TLS. If you select this option, you must select a method in the **Secure Connection Type** drop-down menu.<br>**Note:**<br>In VCF Operations, the Standard Email Plugin always uses the 'starttls' command when you select the **Use Secure Connection** checkbox and select **TLS** as the **Secure Connection Type**. |
| **Requires Authentication** | Activates authentication on the email user account that you use to configure this SMTP instance. If you select this option, you must provide a password for the user account. |
| **SMTP Host** | URL or IP address of your email host server. |
| **SMTP Port** | Default port SMTP uses to connect with the server. |
| **Secure Connection Type** | Select either SSL/TLS as the communication encryption method used in your environment from the drop-down menu. You must select a connection type if you select Use Secure Connection.<br>**Note:**<br>In VCF Operations, the Standard Email Plugin always uses the 'starttls' command when you select the **Use Secure Connection** checkbox and select **TLS** as the **Secure Connection Type**. |

| Option | Description |
|---|---|
| **Sender Email Address** | Email address that appears on the notification message. |
| **Sender Name** | Displayed name for the sender email address. |
| **Credential Type** | Select the **Credential Type** from the list. If your Endpoint URL does not need any authentication, then select **No Credential** from the **Credential Type** list.<br>You can select **Basic Authentication** or **OAuth Authentication**. |
| **Credential** | Add or edit the **Credential** details. Click the plus icon to enter the details of the new credentials in the **Create New Credential** pane, and click **Save**.<br>This field appears only when you select **Basic Authentication** or **OAuth Authentication** as the **Credential Type**.<br>**Basic Authentication**<br>Enter the Name, User Name, and Password.<br>**OAuth Authentication**<br>Enter the following details:<br>• Name: Enter a name for the authentication.<br>• Grant Type: Select either **Client Credentials** or **Password Credentials**.<br>• Authentication URL: Enter the URL from where the access token can be retrieved.<br>• Client ID: Enter the client ID for the authentication URL.<br>• Client Secret: Enter the client secret for the authentication URL.<br>• User Name: Enter the user name for the authentication URL.<br>**Note:**<br>This field appears only when the grant type is **Password Credentials**.<br>• Password: Enter the password for the authentication URL.<br>**Note:**<br>This field appears only when the grant type is **Password Credentials**.<br>• Scope: Enter the labels to specify the access token. The labels specify the permissions that the access tokens will have.<br>• Send Credentials: Select either **In auth header** or **In body**.<br>– In authauthorization header: Sends the Client ID and Client Secret in the header.<br>– In body: Sends the Client ID and Client Secret in the payload body.<br>**Note:**<br>This field appears only when the grant type is **Client Credentials**.<br>• Access Token Path: Enter your access token path.<br>• Validity Token Path: To keep track of when the token is going to expire, enter the validity token path and select the format from the drop-down list. You can choose one of the following formats:<br>• Second<br>• Milisecond<br>• Absolute Time |

| Option | Description |
| --- | --- |
| | • Header Name: Enter a header name. By default, the header name is 'Authorization'.<br>• Prefix: Enter a prefix. By default, the prefix is 'Bearer'.<br>• HTTP Proxy: Select your HTTP Proxy.<br><br>**Note:**<br><br>Starting VCF Operations 8.14, you could only view, add, modify, or delete credentials that you created or were assigned to you. You could view unassigned credentials only if you had the required permissions. When you upgrade to VCF Operations 8.16.1, you can deactivate the **Credential Ownership Enforcement** option from Global Settings to be able to modify credentials created and owned by others. For more information, see List of Global Settings. |
| **Receiver Email Address** | Receiver's email address. |

5. Click **Save**.
6. To start the outbound alert service for this plug-in, select the instance in the list and click **Activate** on the toolbar.

This instance of the Standard Email Plug-In for outbound SMTP alerts is configured and running.

Create notification rules that use the Standard Email Plug-In to send a message to your users about alerts requiring their attention.

## Add a Log File Plug-In for VCF Operations Outbound Alerts

You add a Log File plug-in when you want to configure VCF Operations to log alerts to a file on each of your nodes. If you installed VCF Operations as a multiple node cluster, each node processes and logs the alerts for the objects that it monitors. Each node logs the alerts for the objects it processes.

Ensure that you have write access to the file system path on the target nodes.

All alerts are added to the log file. You can use other applications to filter and manage the logs.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Outbound Settings** tile.
2. Click **Add**, and from the **Plug-In Type** drop-down menu, select **Log File**.
   The dialog box expands to include your log file settings.
3. In the **Alert Output Folder** text box, enter the folder name.
   If the folder does not exist in the target location, the plug-in creates the folder in the target location. The default target location is: `/usr/lib/vmware-vcops/common/bin/`.
4. Click **Save**.
5. To start the outbound alert service for this plug-in, select the instance in the list and click **Activate** on the toolbar.

This instance of the log file plug-in is configured and running.

When the plug-in is started, the alerts are logged in the file. Verify that the log files are created in the target directory as the alerts are generated, updated, or canceled.

**Sample Log File Plug-In Output**
Here is a sample log file plug-in output.

```
AlertId :: 9fb52c9c-40f2-46a7-a005-01bf24ab75e6

AlertStatus :: Active

AlertControlState :: Open
```

```
AlertGenerateTime :: Wed May 06 06:26:05 UTC 2020 (UTC = 1588746365585)

AlertUpdateTime :: Wed May 06 06:26:05 UTC 2020 (UTC = 1588746365585)

AlertMessage :: 9027

AlertSummaryLink :: https://10.27.82.96/ui/index.action#/object/all/1b852a3c-bbdf-41df-
a64d-b40af9673b89/alertsAndSymptoms/alerts/9fb52c9c-40f2-46a7-a005-01bf24ab75e6

AlertType :: Storage - Performance

AlertCriticality :: 4

AffectedResourceId :: 1b852a3c-bbdf-41df-a64d-b40af9673b89

AffectedResourceName :: JNJ_6nodes_Large_HA_4_10.27.83.44

AffectedResourceKind :: VirtualMachine

AffectedResourceParentsNames ::
 VM Entity Status:PoweredOn:all
 DistributedVirtualPortgroup:VM-Network-VLAN-820
 VM Entity Status:PoweredOn:vc_evn-hs1-vc.company.com
 VMFolder:Discovered virtual machine
 HostSystem:evn1-hs1-0808.company.com

AffectedResourceAdapterInstanceResourceName ::
 CompanyAdapter Instance:vc_evn-hs1-vc.company.com

AlertOwner ::

Anomalies ::
 VirtualMachine:JNJ_6nodes_Large_HA_4_10.27.83.44 - [virtualDisk:Aggregate of all
 instances|totalWriteLatency_average] - HT above 30.5647619047619 > 25
 VirtualMachine:JNJ_6nodes_Large_HA_4_10.27.83.44 - [virtualDisk:Aggregate of all
 instances|totalWriteLatency_average] - HT above 30.5647619047619 > 15
 VirtualMachine:JNJ_6nodes_Large_HA_4_10.27.83.44 - [virtualDisk:Aggregate of all
 instances|totalWriteLatency_average] - HT above 30.5647619047619 > 30

Health ::
4.0
Risk ::
2.0
Efficiency ::
1.0
KPIFiring ::

AlertTrigger ::
 Resource                                          Message Info                    Alarm
 Reason  Probability             Prediction Time
 VirtualMachine:JNJ_6nodes_Large_HA_4_10.27.83.44  HT above 30.5647619047619 > 30  HT
 above      Unable to retrive value  Unable to retrive value

AlertRootCause ::
null
AlertRootCauseDetails :: null

AlertName :: Virtual machine disk I/O write latency is high

AlertDescription ::
Virtual machine disk I/O write latency is high
```

## Add an SNMP Trap Plug-In for VCF Operations Outbound Alerts

You add an SNMP Trap plug-in when you want to configure VCF Operations to log alerts on an existing SNMP Trap server in your environment.

Ensure that you have an SNMP Trap server configured in your environment, and that you know the IP address or host name, port number, and community that it uses.

You can provide filtering when you define a Notification using an SNMP Trap destination.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Outbound Settings** tile.
2. Click **Add**, and from the **Plug-In Type** drop-down menu, select **SNMP Trap Plugin**.
   The dialog box expands to include your SNMP trap settings.
3. Enter an **Instance Name**.
4. Configure the SNMP trap settings appropriate to your environment.

| Option | Description |
|---|---|
| **Destination Host** | IP address or fully qualified domain name of the SNMP management system to which you are sending alerts. |
| **Port** | Port used to connect to the SNMP management system. Default port is 162. |
| **Community** | Text string that allows access to the statistics. SNMP Community strings are used only by devices that support SNMPv1 and SNMPv2c protocol. |
| **Username** | User name to configure SNMP trap settings in your environment. If the user name is specified, SNMPv3 is considered as the protocol by the plugin.<br>If left blank, SNMPv2c is considered as the protocol by the plugin.<br>**Note:**<br>SNMP uses User Datagram Protocol (UDP) as its transport protocol. |
| **Authentication Protocol** | Authentication algorithms available are SHA-224, SHA-256, SHA-384, SHA-512. |
| **Authentication Password** | Authentication password. |
| **Privacy Protocol** | Privacy algorithms available are AES192, AES256. |
| **Privacy Password** | Privacy password. |
| **Engine ID** | Engine ID serves as an identifier for the agent. It is used with a hashing function to generate localized keys for authentication and encryption of SNMP v3 messages.<br>It is mandatory to specify the Engine ID when configuring the SNMP Trap plugin. If you do not add the Engine ID and save the SNMP Trap plugin instance, the field is auto-generated the next time you edit the settings. |
| **Varbinds Encoding** | Varbinds Encoding allows you to define character encoding when SNMP notifications are sent in languages other than English. Character encoding is used to convert textual data to binary format. The receiving endpoint decodes the received binary data into text accordingly.<br>By default, the encoding is set to ISO-8859-1. You can select the following encoding option from the list.<br>• UTF-8 |

| Option | Description |
| --- | --- |
| | **Note:**<br>The newly added encoding option will be effective only when the SNMP servers receiving the notifications support the encoding. |

5. Click **Test** to validate the connection.

   **Note:**
   The Community and Username options are mutually exclusive. Define either one of them to avoid an error. If you add a user name, you can optionally define the Authentication Protocol and Authentication Password followed by the Privacy Protocol and Privacy Password. The privacy protocol and its password cannot be defined independent of the authentication protocol and its password.

This instance of the SNMP Trap plug-in is configured and running.

When the plug-in is added, configure notifications for receiving the SNMP traps. For more information, see Creating Notification Rules for Alerts.

### Add a Service-Now Notification Plug-In for Outbound Alerts

You add a Service-Now Notification plug-in when you want to integrate Service Now ticketing system with VCF Operations. Service Now creates an incident whenever an alert is triggered in VCF Operations.

Ensure that you have log in credentials for Service-Now.

Ensure that you are assigned with IT Infrastructure Library (ITIL) role in Service Now.

Using Service-Now Notification Plug-In you can send alert notifications to the Service Now ticketing system to create incidents. The incident includes information like the Caller, Category, Subcategory, Business Service, and other attributes related to alerts.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Outbound Settings** tile.
2. Click **Add** and from the **Plug-In Type** drop-down menu, select **Service-Now Notification Plug-in**.
   The dialog box expands to include your plug-in instance settings.
3. Enter an **Instance Name**.
4. Enter the Service Now URL.
   `https://dev22418.service-now.com/`
5. Enter a value for the Connection Count.
   The connection count represents the maximum number of open connections allowed per node in VCF Operations.
6. Optional: Select your **HTTP Proxy**.
7. Select the **Credential Type** from the list.

   **Note:**
   If your Endpoint URL does not need any authentication, then select **No Credential** from the **Credential Type** list.
8. Add or edit the **Credential** details. Click the plus icon to enter the details of the new credentials in the **Create New Credential** pane, and click **Save**. This field appears only when you select **Basic Authentication** as the **Credential Type**.

   **Note:**
   When you upgrade to the latest version of VCF Operations, all credentials get unassigned. The VCF Operations administrator must assign the credentials from the **Orphan and Unassigned** page. For more information, see the topic, Managing Orphaned and Unassigned Content.

9. To verify the specified paths, credentials, and permissions, click **Test**.
10. Click **Save**.

This instance of the Service-Now Notifications plug-in is configured and running.

When the plug-in is added, configure notifications for creating incidents in the Service-Now ticketing system. For more information, see Creating Notification Rules for Alerts.

### Notifications - Add a Slack Plugin for Outbound Notifications

You can add a Slack plug-in to forward alerts and configure multiple notification rules with different slack channels. The Slack plug-in allows you to receive pre-formatted alert details with alert fields and helps you run VCF Operations using alert links to troubleshoot further.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Outbound Settings** tile.
2. Click **Add** and from the **Plugin Type** drop-down menu, select **Slack Plugin**.
   The dialog box expands to include your plug-in instance settings.
3. Enter an **Instance Name**.
4. Enter a value for the **Connection Count**.
   The connection count represents the maximum number of open connections allowed per node in VCF Operations.
5. Optional: Select your **HTTP Proxy**.
6. To verify the specified paths, credentials, and permissions, click **Test**.
7. Click **Save**.

This instance of the Slack plugin is configured and running.

When the plugin is added, configure notifications for different slack channels. For more information, see Creating Notification Rules for Alerts.

### Add a Webhook Notification Plugin for Outbound Instance

You can integrate Webhook with any endpoint REST API and configure outbound payload.

Ensure that you have login credentials for Webhook.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Outbound Settings** tile.
2. Click **Add** and from the **Plugin Type** drop-down menu, select **Webhook Notification Plugin**.
   The dialog box expands to include your plugin instance settings.
3. Enter an **Instance Name**.
4. Enter the Webhook URL.

   **Note:**
   For notifications that reference webhook outbound instances, the URL that you enter here serves as a base URL that is combined with the endpoint URL suffix defined in related webhook payload templates.
5. Enter a value for the **Connection Count**.
   The connection count represents the maximum number of open connections allowed per node in VCF Operations.

6. Optional: Select your **HTTP Proxy**.
7. Select the **Credential Type** from the list.

   **Note:**
   If your Endpoint URL does not need any authentication, then select **No Credential** from the **Credential Type** list.
8. Add or edit the **Credential** details. Click the plus icon and enter the details of the new credentials in the right pane, and click **Save**.

   **Note:**
   When you upgrade to the latest version of VCF Operations, all credentials get unassigned. The VCF Operations administrator must assign the credentials from the **Orphan and Unassigned** page. For more information, see the topic, Managing Orphaned and Unassigned Content.

   The fields in the **Create New Credential** pane appear based on the **Credential Type** you select.

| Credential Type | Fields |
|---|---|
| Basic Authentication | Enter the Name, User Name, and Password. |
| Bearer Token | Enter the Name of the credential and the Token. |
| OAuth Authentication | Enter the following details:<br>• Name: Enter a name for the authentication.<br>• Grant Type: Select either **Client Credentials** or **Password Credentials**.<br>• Authentication URL: Enter the URL from where the access token can be retrieved.<br>• Client ID: Enter the client ID for the authentication URL.<br>• Client Secret: Enter the client secret for the authentication URL.<br>• User Name: Enter the user name for the authentication URL.<br>**Note:** This field appears only when the grant type is **Password Credentials**.<br>• Password: Enter the password for the authentication URL.<br>**Note:** This field appears only when the grant type is **Password Credentials**.<br>• Scope: Enter the labels to specify the access token. The labels specify the permissions that the access tokens will have.<br>• Send Credentials: Select either **In auth header** or **In body**.<br>– In auth header: Sends the Client ID and Client Secret in the header.<br>– In body: Sends the Client ID and Client Secret in the payload body.<br>**Note:** This field appears only when the grant type is **Client Credentials**.<br>• Access Token Path: Enter your access token path.<br>• Validity Token Path: To keep track of when the token is going to expire, enter the validity token path and select the format from the drop-down list. You can choose one of the following formats:<br>– Second<br>– Milisecond<br>– Absolute Time<br>• Header Name: Enter a header name. By default, the header name is 'Authorization'.<br>• Prefix: Enter a prefix. By default, the prefix is 'Bearer'.<br>• HTTP Proxy: Select your HTTP Proxy. |
| Certificate | Enter the following details:<br>• Name: Enter the name of the certificate.<br>• Certificate: Enter the certificate in the X.509 format.<br>• Certificate Key: Enter the private key. The formats supported are Open SSL, PKCS1, and PKCS8 |

| Credential Type | Fields |
|---|---|
| API Key | Enter the Name, API Key, and the API Value. |

9. To verify the specified paths, credentials, and permissions, click **Test**.

   **Note:**

   - The **Test** feature does not currently support the Custom Templates. If the template contains a custom headers, the test might fail.

10. Click **Save**.

This instance of the Webhook Notification plugin is configured and running.

### Sample Email Alert

Here is a sample email for a newly created alert.

```
Alert Definition Name: Node is experiencing swapping due to memory pressure
Alert Definition Description: Node is experiencing swapping due to memory pressure
Object Name : VMware Aria Operations Node-VMwareAria Cluster Node
Object Type : vC-Ops-Node
Alert Impact: risk
Alert State : warning
Alert Type : Application
Alert Sub-Type : Performance
Object Health State: info
Object Risk State: warning
Object Efficiency State: info
Control State: Open
Symptoms:
SYMPTOM SET - self
```

| Symptom Name | Object Name | Object ID | Metric | Message Info |
|---|---|---|---|---|
| Node swap usage at Warning level | VCF Operations Node-VMware Aria Cluster Node | 50ec874a-2d7d-4e78-98b1-afb26fd67e58 | Swap\|Workload | 59.183 > 30.0 |

```
Recommendations:
Notification Rule Name: rule1
Notification Rule Description:
Alert ID : badc2266-935d-4fb9-8594-e2e71e4866fc
VCOps Server - VMwareAriaClusterNode

Alert details(link)
```

## Exporting and Importing Outbound Settings

As a VCF Operations admin, you can backup the content before upgrading, export all the outbound plugin configurations, and import it into a different VCF Operations instance. You can also export the content from VCF Operations on-prem to VCF Operations.

**Note:**
Any user with "Manage" Outbound Settings permission can export and import outbound plugin configurations.

1. Export an outbound setting.
   a) In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Outbound Settings** tile.
   b) Select the outbound settings that you want to export and click the horizontal ellipses and select **Export**.
   c) Setup a new password to export data. The password should be at least 14 characters long.
   d) Click **Export**.

      The outbound setting data is exported in the .json format. A password is used to encrypt the data in the file using the AES algorithm with a 128-bit key. Use the same password while importing this file.
2. Import an outbound setting.

   **Note:**
   Before importing the outbound setting, ensure that you have exported the outbound plugin configurations.
   a) In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Outbound Settings** tile.
   b) Click the horizontal ellipses and select **Import**.
   c) Click **Browse** to select the `.json` file and enter the password that you had set while exporting the content.
   d) If there is a conflict while importing the content, you can either overwrite the existing outbound settings or skip the import, which is the default.
   e) Click **Import** to import outbound settings to the destination setup.

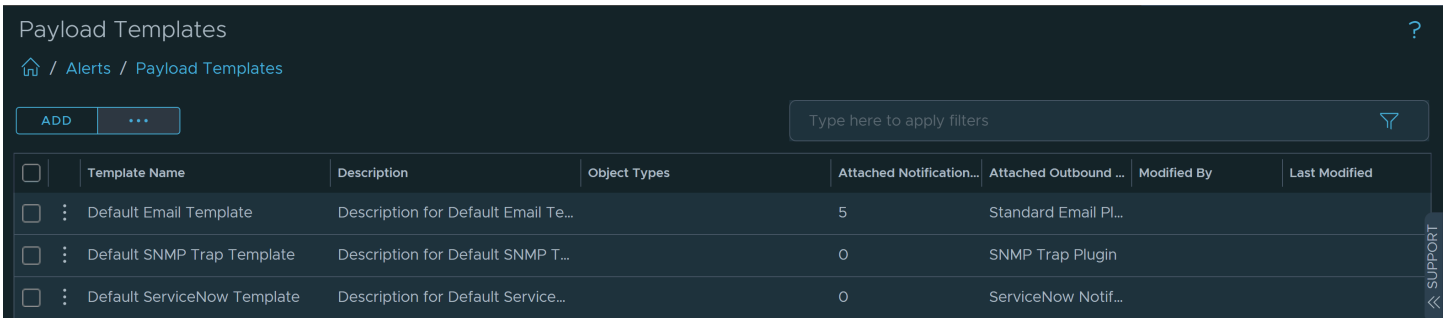
   **Note:**
   While importing outbound settings on VCF Operations, Cloud Proxy configurations will be excluded.

## Payload Templates in VCF Operations

Payload is an essential information in the data block that you send or receive from the server. Use the **Payload Templates** page to view the list of payload templates available for each plug-in in VCF Operations. You can add, manage, and edit your payload templates from this page. Default payload templates are provided for each plug-in type.

## Where You Find Payload Templates

To manage your payload templates, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Payload Templates** tile.




| Option | Description |
|---|---|
| Toolbar options | Use the toolbar options to manage your notification rules.<br>• Add. Use the Create Payload Template dialog box to create new payload templates.<br>Click the horizontal ellipsis to perform the following actions.<br>• Delete. Removes the selected payload template.<br>• Export. Downloads the payload template.<br>**Note:**<br>Export and delete actions are not supported for the default payload templates available for each plug-in.<br>• Import. Allows you to import payload templates. To import:<br>– Click the **Import** option from the horizontal ellipsis.<br>– Click **Browse** and select the file to import.<br>– Select if you want to Overwrite or Skip the import in case of a conflict.<br>– Click **Import** to import the payload template, and then click **Done**. A message with the number of imported and skipped files will appear. |
| Quick Filter | Limits the list based on the text you type. It considers only text from templates Name column.<br>You can also sort columns in the data grid. You can view a blue arrow next to the column according to which sorting was performed, pointing up or down based on the sorting order (ascending or descending). |
| Template Name | Name of the payload template.<br>Click the vertical ellipsis to perform the following actions.<br>• Edit. Allows you to edit the selected payload template.<br>• Clone. Clones the selected payload template.<br>• Delete. Removes the selected payload template.<br>• Export. Downloads the payload template.<br>•<br>**Note:**<br>Edit, Delete, and Export actions are not supported for the default payload templates available for each plug-in. |
| Description | Description of the payload template. |
| Object Types | Base object type against which the payload template is defined, if any. |
| Attached Notification Rules | Notification rule attached to the payload template. |
| Attached Outbound Methods | Outbound plugin type attached to the payload template. |
| Modified By | Name of the last person to modify the payload template. |
| Last Modified | Date on which the payload template was last modified. |

## Creating Payload Templates for Outbound Plugins

You can create a payload template for an outbound plugin of your choice in VCF Operations.

Use payload templates to configure the payload of an email, and customize the subject line, and email body. You can enable your own input properties and different payloads for updated and canceled alerts.

You can customize payloads only for Standard Email Plug-in and Webhook Notification Plug-in.

1. In the VCF Operations navigation bar at the top, click **Operations** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Payload Templates** tile. On the toolbar, click **Add** to create a new payload template.
2. In the **Details** tab, enter the basic details of the payload template.

| Option | Description |
|---|---|
| Name | Provide a name for the payload template. |
| Description | Enter a description for the payload template. |
| Outbound Method | Outbound plugin for which you want to create a new payload template.<br>Select one of the outbound alert plug-in types: Log File Plugin, Standard Email Plugin, SNMP Trap Plugin, Webhook Notification Plugin, Slack Plugin, and Service-Now Notification Plugin. |
| Advanced Settings | |
| Notification Type | • Select **Alert** to configure an alert notification.<br>• Select **Action** to configure notification for WLP VM movement task.<br>**Note:**<br>The **Action** option appears only when you select **Webhook Notification Plugin** in the **Outbound Method** field. You can configure notification for type 'Action' only for the Webhook Notification Plugin. |

3. Click **Next**.
4. In the **Object Content** tab, define the object details that you want to include in the notifications.

   **Note:**
   To create a payload template, it is mandatory to add the object type for all the outbound plugin types except for the Standard Email Plugin and Webhook Notification Plugin.

| Option | Description |
|---|---|
| Add Object Type | • Select an object type from the list. Once you select the object type, define self metrics/properties, ancestors, descendants, ancestor metrics/properties, and descendant metrics/properties associated with the object type that you want to include in the notification.<br>From the right pane, double-click or drag the metrics and properties into the **Add Metrics and Properties** box. You can select up to 30 metrics and properties.<br>**Note:**<br>For Action Type of Notifications, only Virtual Machine is available for selection.<br>• Define the ancestor/descendant object type along with the corresponding metrics and properties.<br>Select the **ancestor/descendant** from the drop-down menu and from the right pane, double-click or drag the corresponding metrics and properties into the left pane.<br>**Note:**<br>For SNMP Trap Plugin, you can only select ancestor objects but you cannot add metrics/properties or define descendant details.<br>The information that you define here will be included in the alert notification for all the plug-ins. However, for a Webhook Notification Plugin and Standard Email Plugin, the information will be included only when you define the values in the **Payload Details** tab. |

5. Click **Create** to create the new payload template or click **Next** if you are creating a payload template for a Standard Email Plugin or a Webhook Notification Plugin.
6. In the **Payload Details** tab, enter the payload details that you want to include in the notification.

   **Note:**
   This tab is available only when you are creating a payload template for a Standard Email Plugin or a Webhook Notification Plugin.

| Option | Description |
|---|---|
| Do you want to add template input properties? | Select **Yes** to add input properties and enter the **Key**, **Type**, **Display Name**, and **Description** of the input property. Otherwise, select **No**.<br>**Note:**<br>The input properties are specific to your endpoint. Once you define the input properties in the template, you must provide the appropriate values in each rule where this template will be used. |
| Do you want different payload details for new, updated, and canceled alerts? | Select **Yes** to define different payload details for new, updated, and canceled alerts. Otherwise, select **No**.<br>**Note:**<br>This field does not appear when the **Notification Type** is **Action**. |
| The following fields appear while creating a payload template for the Standard Email Plugin. | |
| Subject | Enter a subject for the email notification. |
| Body | Enter the content for the email notification. You can also search for parameters in the right pane. Click the copy icon next to the parameter to copy the parameter and you can paste the parameter in the email body.<br>You can use the options in the toolbar to edit, format, and highlight the email content.<br>**Note:**<br>You can set up different email content for new, updated, and canceled alerts. |
| The following fields appear while creating a payload template for the Webhook Notification Plugin. | |
| Endpoint URL | Enter the endpoint URL. The endpoint URL will be appended to the base URL provided in the related webhook outbound instance.<br>**Note:**<br>The entire URL is encoded. However, there is an exception to use the character '/' in the URL. |
| Content Type | Select the content type for the payload. |
| Custom Headers | Enter the **HTTP Custom Header Name** and **Value**. Click the plus icon to add multiple custom headers.<br>**Note:**<br>For webhook payloads using token-based authentication, add an Authorization Header in the format required by the endpoint. |
| HTTP Method | Select the HTTP method of request. |

| Option | Description |
|---|---|
| Payload of the request | Payload for the selected plug-in type. It displays information based on the selected metrics, properties, ancestors, and object types. You can search for parameters in the right pane. Click the copy icon next to the parameter to copy the parameter and you can paste the parameter in the **Payload of the request** box. |

7. Click **Create**.

   Once the payload template is created, you can view it in the **Payload Templates** page. After selecting a payload template in the notification rule, you can view the payload template details in the **Notifications** page.

# Managing Alert Groups in VCF Operations

For easy and better management of alerts, you can arrange them as a group as per your requirement.

It is complicated to identify a problem in large environments as you receive different kind of alerts. To manage alerts easily, group them by their definitions.

For example, there are 1000 alerts in your system. To identify different types of alerts, group them based on their alert definitions. It is also easy to detect the alert having the highest severity in the group.

When you group alerts, you can view the number of times the alerts with the same alert definition are triggered. By grouping alerts, you can perform the following tasks easily and quickly:

- Find the noisiest alert: The alert that has triggered maximum number of times is known as the noisiest alert. Once you find it, you can deactivate it to avoid further noise.
- Filter alerts: You can filter alerts based on a substring in alert definitions. The result shows the group of alerts that contain the substring.

**Note:**

- If you cancel or deactivate an alert group, the alerts are not canceled instantly. It might take some time if the group is large.
- Only one group can be expanded at a time.
- The number next to the group denotes the number of alerts in that particular group.
- The criticality

  sign
  indicates the highest level of severity of an alert in a group.

## Grouping Alerts in VCF Operations

You can group alerts by time, criticality, definition, and object type.

To group alerts:

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Alerts**.
2. Select from the various options available from the **Group By** drop-down menu.

## Deactivating Alerts in VCF Operations

In an alerts group, you can deactivate an alert by a single click.

To deactivate an alert:

1. In the VVCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Alerts**.
2. From the **Group By** drop-down, select **Definition**, and click on the name of the Alert Definition Group.
3. From the data grid, click **Actions** > **Deactivate**.

   You can deactivate the alerts by two methods:

   - Deactivate Alert in All Policies: Deactivates the alert for all the objects for all the policies.

- Deactivate Alert in Selected Policies: Deactivates the alert for the objects having the selected policy.

## Alert Definition Best Practices

As you create alert definitions for your environment, apply consistent best practices so that you optimize alert behavior for your monitored objects.

### Alert Definitions Naming and Description

The alert definition name is the short name that appears in the following places:

- In data grids when alerts are generated
- In outbound alert notifications, including the email notifications that are sent when outbound alerts and notifications are configured in your environment

Ensure that you provide an informative name that clearly states the reported problem. Your users can evaluate alerts based on the alert definition name.

The alert definition description is the text that appears in the alert definition details and the outbound alerts. Ensure that you provide a useful description that helps your users understand the problem that generated the alert.

### Wait and Cancel Cycle

The wait cycle setting helps you adjust for sensitivity in your environment. The wait cycle for the alert definition goes into effect after the wait cycle for the symptom definition results in a triggered symptom. In most alert definitions you configure the sensitivity at the symptom level and configure the wait cycle of alert definition to 1. This configuration ensures that the alert is immediately generated after all of the symptoms are triggered at the desired symptom sensitivity level.

The cancel cycle setting helps you adjust for sensitivity in your environment. The cancel cycle for the alert definition goes into affect after the cancel cycle for the symptom definition results in a cancelled symptom. In most definitions you configure the sensitivity at the symptom level and configure the cancel cycle of alert definition to 1. This configuration ensures that the alert is immediately cancelled after all of the symptoms conditions disappear after the desired symptom cancel cycle.

### Create Alert Definitions to Generate the Fewest Alerts

You can control the size of your alert list and make it easier to manage. When an alert is about a general problem that can be triggered on a large number of objects, configure its definition so that the alert is generated on a higher level object in the hierarchy rather than on individual objects.

As you add symptoms to your alert definition, do not overcrowd a single alert definition with secondary symptoms. Keep the combination of symptoms as simple and straightforward as possible.

You can also use a series of symptom definitions to describe incremental levels of concern. For example, `Volume nearing capacity limit` might have a severity value of Warning while `Volume reached capacity limit` might have a severity level of Critical. The first symptom is not an immediate threat, but the second one is an immediate threat. You can then include the Warning and Critical symptom definitions in a single alert definition with an Any condition and set the alert criticality to be Symptom Based. These settings cause the alert to be generated with the right criticality if either of the symptoms is triggered.

### Avoid Overlapping and Gaps Between Alerts

Overlaps result in two or more alerts being generated for the same underlying condition. Gaps occur when an unresolved alert with lower severity is canceled, but a related alert with a higher severity cannot be triggered.

A gap occurs in a situation where the value is <=50% in one alert definition and >=75% in a second alert definition. The gap occurs because when the percentage of volumes with high use falls between 50 percent and 75 percent, the first problem cancels but the second does not generate an alert. This situation is problematic because no alert definitions are active to cover the gap.

**Actionable Recommendations**

If you provide text instructions to your users that help them resolve a problem identified by an alert definition, precisely describe how the engineer or administrator should fix the problem to resolve the alert.

To support the instructions, add a link to a wiki, runbook, or other sources of information, and add actions that you run from VCF Operations on the target systems.

## Create a Simple Alert Definition

While troubleshooting, you can now quickly create an alert for a particular object type or a metric in a quick and efficient way.

You can create a simple alert definition from the following locations.

- In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Workbench** and select the metric for which you want to create an alert. You can create an alert from the **Potential Evidence** or the **Metrics** tab.
- In the VCF Operations navigation bar at the top, click **Observe** and in the left navigation pane, click **Alerts**. Select an alert and click the **Potential Evidence** tab.
- From the Home page, you can search for the specific object type or metric in the search bar.

1. Click the drop-down menu available in the right side of the widget and select the **Create an Alert Definition** option.

   **Note:**
   You cannot create alert definitions for Badge, Time, and Capacity Remaining metrics.

2. In the Create Alert Definition page, enter the **Name** and **Description** of the alert.
3. Set thresholds, criticality, and the number of wait cycles. Click **Show Advanced Settings** to set Wait Cycle and Cancel Cycle.

   **Note:**
   The Object Type or Metric/Property are pre-selected and cannot be edited.

4. Click **Create**.

   The new alert is created and the policy the object belongs to and its children policies are activated for the alert.

## Create a New Alert Definition

Based on the root cause of the problem, and the solutions that you used to fix the problem, you can create a new alert definition for VCF Operations to alert you. When the alert is triggered on your host system, VCF Operations alerts you and provides recommendations on how to solve the problem.

To alert you before your host systems experience critical capacity problems, and have VCF Operations notify you of problems in advance, you create alert definitions, and add symptom definitions to the alert definition.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Alert Definitions** tile.
2. Enter `capacity` in the search text box.

   Review the available list of capacity alert definitions. If a capacity alert definition does not exist for host systems, you can create one.

3. Click **Add** to create a capacity alert definition for your host systems.
   a) In the alert definition workspace, for the Name and Description, enter `Hosts - Alert on Capacity Exceeded`.
   b) For the Base Object Type, select **vCenter Adapter > Host System**
   c) Under **Advanced Settings**, select the following options.

| Option | Selection |
|---|---|
| Impact | Select **Risk**. |
| Criticality | Select **Immediate**. |
| Alert Type and Subtype | Select **Application : Capacity**. |
| Wait Cycle | Select **1**. |
| Cancel Cycle | Select **1**. |

   d) In the **Symptoms/Conditions** workspace, select the following options.

| Option | Selection |
|---|---|
| Defined On | Select **Self**. |
| Symptom Definition Type | Select **Metric / Property**. |
| Quick filter (Name) | Enter `capacity`. |

   e) From the Symptom Definition list, click **Host System Capacity Remaining is moderately low** and drag it to the left pane.

      In the Symptoms pane, make sure that the Base object exhibits criteria is set to **All** by default.
   f) For Add Recommendations, enter `virtual machine` in the quick filter text box.
   g) Click **Review the symptoms listed and remove the number of vCPUs from the virtual machine as recommended by the system**, and drag it to the recommendations area in the left pane.

      This recommendation is set to Priority 1.
4. Click **Save** to save the alert definition.

   Your new alert appears in the list of alert definitions.

You have added an alert definition to have VCF Operations alert you when the capacity of your host systems begins to run out.

## Create a Log Based Alert Definition

Use this example to create a log based alert definition. Log based alerts are new in VCF 9.0.

For more information on log symptoms, see Log Symptoms.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Alert Definitions** tile.
2. Click **Add** to create a new log alert definition for your host systems. For example, create an alert definition for Cluster Performance - Failed vMotions.
   a) In the alert definition workspace, for the Name, enter `Cluster Performance - Failed vMotions`.
   b) For the Base Object Type, select **Host System**.
3. Click **NEXT**.
4. In the **Symptoms/Conditions** step, do the following.
   a) Drag and drop **Add Log Condition**.
   b) Select **Cluster Performance - Failed vMotions** from the **Filter By** drop-down list.
   c) Change the time period to six hours.
5. Click **NEXT**. Do not add any recommendation. Click **NEXT** again.
6. In the **Policies** step, select the default policy.
7. Click **CREATE**.

In the Alert Definitions table, sort by the last modified column to view the newly created Log based alert definition.

> **Note:**
>
> Some and log-based alert definitions may contain expressions that cannot be converted into the VCF Operations UI-supported format. Attempting to interpret or partially render these queries may lead to incorrect behavior or confusion.
>
> Some complex saved queries and log-based alert definitions may not appear in the UI due to parsing limitations. These items are skipped when their expressions cannot be converted into the VCF Operations UI-supported format.

# Create an Alert Definition for Department Objects

As a virtual infrastructure administrator, you are responsible for the virtual machines and hosts that the accounting department uses. You can create alerts to manage the accounting department objects.

You received several complaints from your users about delays when they are using their accounting applications. Using VCF Operations, you identified the problem as related to CPU allocations and workloads. To better manage the problem, you create an alert definition with tighter symptom parameters so that you can track the alerts and identify problems before your users encounter further problems.

Using this scenario, you create a monitoring system that monitors your accounting objects and provides timely notifications when problems occur.

## Add Description and Base Object to Alert Definition

To create an alert to monitor the CPUs for the accounting department virtual machines and monitor host memory for the hosts on which they operate, you begin by describing the alert.

When you name the alert definition and define alert impact information, you specify how the information about the alert appears in VCF Operations. The base object is the object around which the alert definition is created. The symptoms can be for the base object and for related objects.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Alert Definitions** tile.
2. Click **Add** to add a definition.
3. Type a name and description.

   In this scenario, type `Acct VM CPU early warning` as the alert name, which is a quick overview of the problem. The description, which is a detailed overview, should provide information that is as useful as possible. When the alert is generated, this name and description appears in the alert list and in the notification.

4. From the **Base Object Type** drop-down menu, expand **vCenter Adapter** and select **Host System**.

   This alert is based on host systems because you want an alert that acts as an early warning to possible CPU stress on the virtual machines used in the accounting department. By using host systems as the based object type, you can respond to the alert symptom for the virtual machines with bulk actions rather than responding to an alert for each virtual machine.

5. Click **Advanced Settings** and configure the metadata for this alert definition.
   a) From the **Impact** drop-down menu, select **Risk**.

      This alert indicates a potential problem and requires attention in the near future.
   b) From the **Criticality** drop-down menu, select **Immediate**.

      As a Risk alert, which is indicative of a future problem, you still want to give it a high criticality so that it is ranked for correct processing. Because it is designed as an early warning, this configuration provides a built-in buffer that makes it an immediate risk rather than a critical risk.
   c) From the **Alert Type and Subtype** drop-down menu, select **Performance** under **Virtualization/Hypervisor**.
   d) To ensure that the alert is generated during the first collection cycle after the symptoms become true, set the **Wait Cycle** to `1`.
   e) To ensure that the an alert is removed as soon as the symptoms are no longer triggered, set the **Cancel Cycle** to `1`.

      The alert is canceled in the next collection cycle if the symptoms are no long true.

   These alert impact options help you identify and prioritize alerts as they are generated.

You started an alert definition where you provided the name and description, selected host system as the base object type, and defined the data that appears when the alert generated.

Continue in the workspace, adding symptoms to your alert definition. See Add a Virtual Machine CPU Usage Symptom to the Alert Definition.

## Add a Virtual Machine CPU Usage Symptom to the Alert Definition

To generate alerts related to CPU usage on your accounting virtual machines, you add symptoms to your VCF Operations alert definition after you provide the basic descriptive information for the alert. The first symptom you add is related to CPU usage on virtual machines. You later use a policy and group to apply alert to the accounting virtual machines.

Begin configuring the alert definition. See Add Description and Base Object to Alert Definition.

This scenario has two symptoms, one for the accounting virtual machines and one to monitor the hosts on which the virtual machines operate.

1. In the **Alert Definition Workspace** window, after you configure the **Name and Description**, **Base Object Type**, and **Alert Impact**, click **Next** and configure the symptoms.
2. Begin configuring the symptom set related to virtual machines CPU usage.
   a) From the **Select Symptom** drop-down menu, select **Metric / Property**.
   b) From the **Defined On** drop-down menu, select **Child**.
   c) From the **Filter by Object Type** drop-down menu, select **Virtual Machine**.
   d) Click **Create New** to open the **Add Symptom Definition** workspace window.
3. Configure the virtual machine CPU usage symptom in the **Add Symptom Definition** workspace window.
   a) From the **Base Object Type** drop-down menu, expand **vCenter Adapter** and select **Virtual Machine**.

      The collected metrics for virtual machines appears in the list.
   b) In the metrics list **Search** text box, which searches the metric names, type `usage`.
   c) In the list, expand **CPU** and drag **Usage (%)** to the workspace on the left.
   d) From the threshold drop-down menu, select **Dynamic Threshold**.

      Dynamic thresholds use VCF Operations analytics to identify the trend metric values for objects.

e) In the **Symptom Definition Name** text box, type a name similar to `VM CPU Usage above trend`.
f) From the criticality drop-down menu, select **Warning**.
g) From the threshold drop-down menu, select **Above Threshold**.
h) Leave the **Wait Cycle** and **Cancel Cycle** at the default values of 3.

This Wait Cycle setting requires the symptom condition to be true for 3 collection cycles before the symptom is triggered. This wait avoids triggering the symptom when there is a short spike in CPU usage.

i) Click **Save**.

The dynamic symptom, which identifies when the usage is above the tracked trend, is added to the symptom list.

4. In the **Alert Definition Workspace** window, drag **VM CPU Usage above trend** from the symptom definition list to the symptom workspace on the left.

   The Child-Virtual Machine symptom set is added to the symptom workspace.

5. In the symptoms set, configure the triggering condition so that when the symptom is true on half of the virtual machines in the group to which this alert definition is applied, the symptom set is true.
   a) From the value operator drop-down menu, select **>**.
   b) In the value text box, enter `50`.
   c) From the value type drop-down menu, select **Percent**.

You defined the first symptom set for the alert definition.

Add the host memory usage symptom to the alert definition. See Add a Host Memory Usage Symptom to the Alert Definition.

## Add a Host Memory Usage Symptom to the Alert Definition

To generate alerts related to CPU usage on your accounting virtual machines, you add a second symptom to your VCF Operations alert definition after you add the first symptom. The second symptom is related to host memory usage for the hosts on which the accounting virtual machines operate.

Add the virtual machine CPU usage symptom. See Add a Virtual Machine CPU Usage Symptom to the Alert Definition.

1. In the **Alert Definition Workspace** window, after you configure the **Name and Description**, **Base Object Type**, and **Alert Impact**, click **Next**.
2. Configure the symptom related to host systems for the virtual machines.
   a) From the **Select Symptom** drop-down menu, select **Metric / Property**.
   b) From the **Defined On** drop-down menu, select **Self**.
   c) Click **Create New** to add new symptom.
3. Configure the host system symptom in the **Add Symptom Definition** workspace window.
   a) From the **Base Object Type** drop-down menu, expand **vCenter Adapters** and select **Host System**.
   b) In the metrics list, expand **Memory** and drag **Usage (%)** to the workspace on the left.
   c) From the threshold drop-down menu, select **Dynamic Threshold**.

      Dynamic thresholds use VCF Operations analytics to identify the trend metric values for objects.

   d) In the **Symptom Definition Name** text box, enter a name similar to `Host memory usage above trend`.
   e) From the criticality drop-down menu, select **Warning**.
   f) From the threshold drop-down menu, select **Above Threshold**.
   g) Leave the **Wait Cycle** and **Cancel Cycle** at the default values of 3.

      This Wait Cycle setting requires the symptom condition to be true for three collection cycles before the symptom is triggered. This wait avoids triggering the symptom when a short spike occurs in host memory usage.

   h) Click **Save**.

   The dynamic symptom identifies when the hosts on which the accounting virtual machines run are operating above the tracked trend for memory usage.

   The dynamic symptom is added to the symptom list.

4. In the **Alert Definition Workspace** window, drag **Host memory usage above trend** from the symptoms list to the symptom workspace on the left.

   The Self-Host System symptom set is added to the symptom workspace.

5. On the Self-Host System symptom set, from the value type drop-down menu for **This Symptom set is true when**, select **Any**.

   With this configuration, when any of the hosts running accounting virtual machines exhibit memory usage that is above the analyzed trend, the symptom condition is true.

6. At the top of the symptom set list, from the **Match {operator} of the following symptoms** drop-down menu, select **Any**.

   With this configuration, if either of the two symptom sets, virtual machine CPU usage or the host memory, are triggered, an alert is generated for the host.

You defined the second symptom set for the alert definition and configured how the two symptom sets are evaluated to determine when the alert is generated.

Add recommendations to your alert definition so that you and your engineers know how to resolve the alert when it is generated. See Add Recommendations to the Alert Definition.

## Add Recommendations to the Alert Definition

To resolve a generated alert for the accounting department's virtual machines, you provide recommendations so that you or other engineers have the information you need to resolve the alert before your users encounter performance problems.

Add symptoms to your alert definition. See Add a Host Memory Usage Symptom to the Alert Definition.

As part of the alert definition, you add recommendations that include actions that you run from VCF Operations and instructions for making changes in vCenter that resolve the generated alert.

1. In the **Alert Definition Workspace** window, after you configure the **Name and Description**, **Base Object Type**, **Alert Impact**, and **Add Symptom Definitions**, click **Next** and add the recommended actions and instructions.
2. Click **Create New Recommendation** and select an action recommendation to resolve the virtual machine alerts.
   a) In the **Description** text box, enter a description of the action similar to `Add CPUs to virtual machines.`
   b) From the **Actions** drop-down menu, select **Set CPU Count for VM**.
   c) Click **Create**.
3. Click **Create New Recommendation** and provide an instructive recommendation to resolve host memory problems similar to this example.
   ```
   If this host is part of a DRS cluster, check the DRS settings to verify that the load balancing setting are configured correctly. If necessary, manually vMotion the virtual machines.
   ```
4. Click **Create**.
5. Click **Create New Recommendation** and provide an instructive recommendation to resolve host memory alerts.
   a) Enter a description of the recommendation similar to this example.
   ```
   If this is a standalone host, add more memory to the host.
   ```

b) To make the URL a hyperlink in the instructions, copy the URL to your clipboard.
c) Highlight the text in the text box and click the hyperlink icon.
d) Paste the URL in the **Create a hyperlink** text box and click **OK**.
e) Click **Create**.

6. In the **Alert Recommendation Workspace**, drag **Add CPUs to virtual machines**, **If this host is part of a DRS cluster**, and the **If this is a standalone host** recommendations from the list to the recommendation workspace in the order presented.
7. Click **Next** to select policies and view notifications.
8. Click **Create**.

You provided the recommended actions and instructions to resolve the alert when it is generated. One of the recommendations resolves the virtual machine CPU usage problem and the other resolves the host memory problem.

Create a group of objects to use to manage your accounting objects. See Create a Custom Accounting Department Group.

## Create a Custom Accounting Department Group

To manage, monitor, and apply policies to the accounting objects as a group, you create a custom object group.

Verify that you completed the alert definition for this scenario. See Add Recommendations to the Alert Definition.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Custom Groups** tile under Logical Groupings.
2. Click **Add** to create a new custom group.
3. Type a name similar to `Accounting VMs and Hosts`.
4. From the **Group Type** drop-down menu, select **Department**.
5. From the **Policy** drop-down menu, select **Default Policy**.
   When you create a policy, you apply the new policy to the accounting group.
6. In the Define membership criteria area, from the **Select the Object Type that matches the following criteria** drop-down menu, expand **vCenter Adapter**, select **Host System**, and configure the dynamic group criteria.
   a) From the criteria drop-down menu, select **Relationship**.
   b) From the relationships options drop-down menu, select **Parent of**.
   c) From the operator drop-down menu, select **contains**.
   d) In the **Object name** text box, enter `acct`.
   e) From the navigation tree drop-down list, select **vSphere Hosts and Clusters**.
   You created a dynamic group where host objects that are the host for virtual machines with acct in the virtual machine name are included in the group. If a virtual machine with acct in the object name is added or moved to a host, the host object is added to the group.
7. Click **Preview** in the lower-left corner of the workspace, and verify that the hosts on which your virtual machines that include acct in the object name appear in the **Preview Group** window.
8. Click **Close**.
9. Click **Add another criteria set**.
   A new criteria set is added with the OR operator between the two criteria sets.

10. From the **Select the Object Type that matches the following criteria** drop-down menu, expand **vCenter Adapter**, select **Virtual Machine**, and configure the dynamic group criteria.
    a) From the criteria drop-down menu, select **Properties**.
    b) From the **Pick a property** drop-down menu, expand **Configuration** and double-click **Name**.
    c) From the operator drop-down menu, select **contains**.
    d) In the **Property value** text box, enter `acct`.

    You created a dynamic group where virtual machine objects with acct in the object name are included in the group that depends on the presence of those virtual machines. If a virtual machine with acct in the name is added to your environment, it is added to the group.
11. Click **Preview** in the lower-left corner of the workspace, and verify that the virtual machines with acct in the object name are added to the list that also includes the host systems.
12. Click **Close**.
13. Click **OK**.

    The Accounting VMs and Hosts group is added to the Groups list.

You created a dynamic object group that changes as virtual machines with acct in their names are added, removed, and moved in your environment.

Create a policy that determines how VCF Operations uses the alert definition to monitor your environment. See Create a Policy for the Accounting Alert.

## Create a Policy for the Accounting Alert

To configure how VCF Operations evaluates the accounting alert definition in your environment, you configure a policy that determines behavior so that you can apply the policy to an object group. The policy limits the application of the alert definition to only the members of the selected object group.

- Verify that you completed the alert definition for this scenario. See Add Recommendations to the Alert Definition.
- Verify that you created a group of objects that you use to manage you accounting objects. See Create a Custom Accounting Department Group.

When an alert definition is created, it is added to the default policy and activated, ensuring that any alert definitions that you create are active in your environment. This alert definition is intended to meet the needs of the accounting department, so you deactivate it in the default policy and create a new policy to govern how the alert definition is evaluated in your environment, including which accounting virtual machines and related hosts to monitor.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definitions** tile.
2. Click **Add**.
3. Type a name similar to `Accounting Objects Alerts Policy` and provide a useful description similar to the following example.

   ```
   This policy is configured to generate alerts when
   Accounting VMs and Hosts group objects are above trended
    CPU or memory usage.
   ```
4. Select **Default Policy** from the **Start with** drop-down menu.
5. On the left, click **Customize Alert / Symptom Definitions** and deactivate all the alert definitions except the new Acct VM CPU early warning alert.
   a) In the Alert Definitions area, click **Actions** and select **Select All**.

      The alerts on the current page are selected.
   b) Click **Actions** and select **Deactivate**.

      The alerts indicate Deactivated in the State column.

c) Repeat the process on each page of the alerts list.
d) Select **Acct VM CPU early warning** in the list, click **Actions** and select **Activate**.
The Acct VM CPU early warning alert is now activated.

6. On the left, click **Apply Policy to Groups** and select **Accounting VMs and Hosts**.
7. Click **Save**.

You created a policy where the accounting alert definition exists in a custom policy that is applied only to the virtual machines and hosts for the accounting department.

Create an email notification so that you learn about alerts even you when you are not actively monitoring VCF Operations. See Configure Notifications for the Department Alert.

## Configure Notifications for the Department Alert

To receive an email notification when the accounting alert is generated, rather than relying on your ability to generally monitor the accounting department objects in VCF Operations, you create notification rules.

- Verify that you completed the alert definition for this scenario. See Add Recommendations to the Alert Definition.
- Verify that standard email outbound alerts are configured in your system. See Add a Standard Email Plug-In for VCF Operations Outbound Alerts.

Creating an email notification when accounting alerts are triggered is an optional process, but it provides you with the alert even when you are not currently working in VCF Operations.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Notifications** tile.
2. Click **Add** to add a notification rule.
3. Configure the communication options.
   a) In the **Name** text box, type a name similar to `Acct Dept VMs or Hosts Alerts`.
   b) From the **Select Plug-In Type** drop-down menu, select **StandardEmailPlugin**.
   c) From the **Select Instance** drop-down menu, select the standard email instance that is configured to send messages.
   d) In the **Recipient(s)** text box, type your email address and the addresses of other recipients responsible for the accounting department alerts. Use a semicolon between recipients.
   e) Leave the **Notify again** text box blank.
   If you do not provide a value, the email notice is sent only once. This alert is a Risk alert and is intended as an early warning rather than requiring an immediate response.
   You configured the name of the notification when it is sent to you and the method that is used to send the message.
4. In the Filtering Criteria area, configure the accounting alert notification trigger.
   a) From the **Notification Trigger** drop-down menu, select **Alert Definition**.
   b) Click **Select Alert Definitions**.
   c) Select **Acct VM CPU early warning** and click **Select**.
5. Click **Save**.

You created a notification rule that sends you and your designated engineers an email message when this alert is generated for your accounting department alert definition.

Create a dashboard with alert-related widgets so that you can monitor alerts for the accounting object group. See Create a Dashboard to Monitor Department Objects.

## Create a Dashboard to Monitor Department Objects

To monitor all the alerts related to the accounting department object group, you create a dashboard that includes the alert list and other widgets. The dashboard provides the alert data in a single location for all related objects.

Create an object group for the accounting department virtual machines and related objects. See Create a Custom Accounting Department Group.

Creating a dashboard to monitor the accounting virtual machines and related hosts is an optional process, but it provides you with a focused view of the accounting object group alerts and objects.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then click **Create**.
2. In the Dashboard Configuration definition area, type a tab name similar to `Accounting VMs and Hosts` and configure the layout options.
3. Click **Widget List** and drag the following widgets to the workspace.
   - **Alert List**
   - **Efficiency**
   - **Health**
   - **Risk**
   - **Top Alerts**
   - **Alert Volume**

   The blank widgets are added to the workspace. To change the order in which they appear, you can drag them to a different location in the workspace.
4. On the Alert List widget title bar, click **Edit Widget** and configure the settings.
   a) In the **Title** text box, change the title to `Acct Dept Alert List`.
   b) For the **Refresh Content** option, select **On**.
   c) Type `Accounting` in the **Search** text box and click **Search**.
      The Accounting value corresponds to the name of the object group for the accounting department virtual machines and related hosts.
   d) In the filtered resource list, select the **Accounting VMs and Hosts** group.
      The Accounting VMs and Hosts group is identified in the Selected Resource text box.
   e) Click **OK**.

   The Acct Dept Alert List is now configured to display alerts for the Accounting VMs and Hosts group objects.
5. Click **Widget Interactions** and configure the following interactions.
   a) For Acct Dept Alert List, leave the selected resources blank.
   b) For Top Alerts, Health, Risk, Efficiency, and Alert Volume select **Acct Dept Alert List** from the **Selected Resources** drop-down menu.
   c) Click **Apply Interactions**.

   With the widget interaction configured in this way, the select alert in the Acct Dept Alert List is the source for the data in the other widgets. When you select an alert in the alert list, the Health, Risk, and Efficiency widgets display alerts for that object, Top Alerts displays the topic issues affecting the health of the object, and Alert Volume displays an alert trend chart.
6. Click **Save**.

You created a dashboard that displays the alerts related to the accounting virtual machines and hosts group, including the Risk alert you created.

## Monitoring and Responding to Alerts

Alerts indicate a problem in your environment. Alerts are generated when the collected data for an object is compared to alert definitions for that object type and the defined symptoms are true. When an alert is generated, you are presented with the triggering symptoms, so that you can evaluate the object in your environment, and with recommendations for how to resolve the alert.

Alerts notify you when an object or group of objects are exhibiting symptoms that are unfavorable for your environment. By monitoring and responding to alerts, you stay aware of problems and can react to them in a timely fashion.

Generated alerts drive the status of the top-level badges, Health, Risk, and Efficiency.

In addition to responding to alerts, you can generally respond to the status of badges for objects in your environment.

You can take ownership of an alert or assign alerts to other VCF Operations users.

## Monitoring Alerts in VCF Operations

You can monitor your environment for generated alerts in several areas in VCF Operations. The alerts are generated when the symptoms in the alert definition are triggered, letting you know when the objects in your environment are not operating within the parameters you defined as acceptable.

Generated alerts appear in many areas of VCF Operations so that you can monitor and respond to problems in your environment.

### Alerts

Alerts are classified as Health, Risk, or Efficiency. Health alerts indicate problems that require immediate attention. Risk alerts indicate problems that must be addressed shortly, before the problems become immediate health problems. Efficiency alerts indicate areas where you can reclaim wasted space or improve the performance of objects in your environment.

You can monitor the alerts for your environment in the following locations.

- Alerts
- Health
- Risk
- Efficiency

You can monitor alerts for a selected object in the following locations.

- Alert Details, including the **Summary**, **Timeline**, and **Metric Charts** tabs
- **Summary** tab
- **Alerts** tab
- **Events** tab
- Custom dashboards
- Alert notifications

### Working with Alerts

Alerts indicate a problem that must be resolved so that triggering conditions no longer exist and the alert is canceled. Suggested resolutions are provided as recommendations so that you can approach the problem with solutions.

As you monitor alerts, you can take ownership, suspend, or manually cancel alerts.

When you cancel an alert, the alert and any symptoms of type message event, or metric event are canceled. You cannot manually cancel other types of symptoms. If a message event symptom or metric event symptom triggered the event, then the symptom is no longer triggered. If a metric symptom or property symptom triggered the alert, a new alert might be created for the same conditions in the next few minutes.

The correct way to remove an alert is to address the underlying conditions that triggered the symptoms and generated the alert.

### Migrated Alerts

If you migrated alerts from a previous version of VCF Operations, the alerts are listed in the overview with a canceled status, but alert details are not available.

# Actions in VCF Operations

Actions are the ability to update objects or read data about objects in monitored systems, and are commonly provided in VCF Operations as part of a solution. The actions added by solutions are available from the object Actions menu, list and view menus, including some dashboard widgets, and can be added to alert definition recommendations.

The possible actions include read actions and update actions.

The read actions retrieve data from the target objects.

The update actions modifies the target objects. For example, you can configure an alert definition to notify you when a virtual machine is experiencing memory issues. Add an action in the recommendations that runs the Set Memory for Virtual Machine action. This action increases the memory and resolves the likely cause of the alert.

To see or use the actions for your vCenter objects, you must activate actions in the vCenter Adapter for each monitored vCenter instance. Actions can only be viewed and accessed if you have the required permissions.

## Actions, Modified Objects, and Object Levels in VCF Operations Actions

The list of actions includes the name of the action, the objects that each one modifies, and the object levels at which you can run the action. You use this information to ensure that you correctly apply the actions as alert recommendations and when the actions are available in the **Actions** menu.

**Actions and Modified Objects**

VCF Operations actions make changes to objects in your managed vCenter instances.

When you grant a user access to actions in VCF Operations, that user can take the granted action on any object that VCF Operations manages.

**Action Object Levels**

The actions are available when you work with different object levels, but they modify only the specified object. If you are working at the cluster level and select **Power On VM**, all the virtual machines in the cluster for which you have access permission are available for you to run the action. If you are working at the virtual machine level, only the selected virtual machine is available.

| Action | Modified Object | Object Levels |
|---|---|---|
| Rebalance Container | Virtual Machines | • Data Center<br>• Custom Data Center |
| Delete Idle VM | Virtual Machines | • Clusters<br>• Host Systems<br>• Virtual Machines |
| Set DRS Automation | Cluster | • Clusters |
| Move VM | Virtual Machine | • Virtual Machines |
| Power Off VM | Virtual Machine | • Clusters<br>• Host Systems<br>• Virtual Machines |
| Shut Down Guest OS for VM | Virtual Machine<br>VMware Tools must be installed and running on the target virtual machines to run this action. | • Clusters<br>• Host Systems<br>• Virtual Machines |
| Power On VM | Virtual Machine | • Clusters<br>• Host Systems<br>• Virtual Machines |
| Delete Powered Off VM | Virtual Machine | • Clusters<br>• Host Systems<br>• Virtual Machines |
| Set Memory for VM<br>and<br>Set Memory for VM Power Off Allowed | Virtual Machine | • Clusters<br>• Host Systems<br>• Virtual Machines |
| Set Memory Resources for VM | Virtual Machine | • Clusters<br>• Host Systems<br>• Virtual Machines |

| Action | Modified Object | Object Levels |
|---|---|---|
| Set CPU Count for VM<br>and<br>Set CPU Count for VM Power Off Allowed | Virtual Machine | • Clusters<br>• Host Systems<br>• Virtual Machines |
| Set CPU Resources for VM | Virtual Machine | • Clusters<br>• Host Systems<br>• Virtual Machines |
| Set CPU Count and Memory for VM<br>and<br>Set CPU Count and Memory for VM Power Off Allowed | Virtual Machine | • Clusters<br>• Host Systems<br>• Virtual Machines |
| Delete Unused Snapshots for VM | Snapshot | • Clusters<br>• Host Systems<br>• Virtual Machines |
| Delete Unused Snapshots for Datastore | Snapshot | • Clusters<br>• Datastores<br>• Host Systems |
| Execute Script | Virtual Machine | • Virtual Machine |
| Get Top Processes | Virtual Machine | • Virtual Machine |
| Apply Guest User Mapping | vCenter Server | • vCenter Server<br>**Note:**<br>This action is deprecated and will be removed in the next release. |
| Clear Guest User Mapping | vCenter Server | • vCenter Server<br>**Note:**<br>This action is deprecated and will be removed in the next release. |
| Export Guest User Mapping | vCenter Server | • vCenter Server<br>**Note:**<br>This action is deprecated and will be removed in the next release. |
| Configure Included Services | Service Discovery Adapter Instance | • Service Discovery Adapter Instance<br>**Note:**<br>This action is deprecated and will be removed in the next release. |

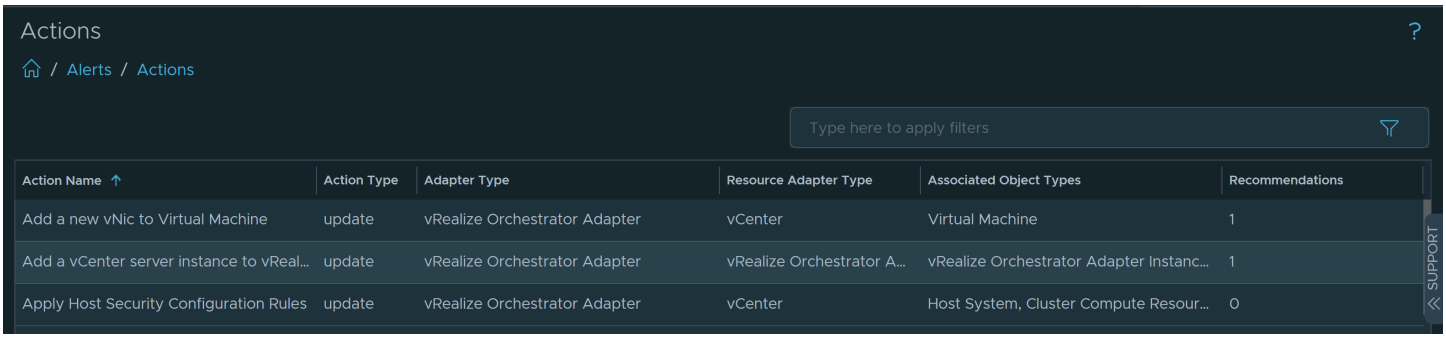
## Viewing Actions List in VCF Operations

Actions are the method you use to configuration changes on managed objects that you initiate from VCF Operations. These actions are available to add to alert recommendations.

Actions are defined to run on the target object from different object levels, allowing you to add actions as recommendations for alert definitions that are configured for different base objects. The Actions page is a list of actions available in your environment.

### Where You Find the Actions List

To view the available actions, in the VCF Operations navigation bar at the top, click **Operation** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Actions** tile.




| Option | Description |
|---|---|
| Filter options | Limits the list based on the text you type.<br>You can also sort on the columns in the data grid. |
| Action Name | Name of the action. Duplicate names indicate that the action name is provided by more than one adapter or has more than one associated object.<br>Click this link to view Details page. On the Details Page, you can click on Recommendations to view the associated recommendations. |
| Action Type | Type of action that the action performs, either read or update.<br>• Update actions make changes to the target objects.<br>• Read actions retrieve data from the target objects. |
| Adapter Type | Name of the configured adapter that provides the action. |
| Resource Adapter Type | Adapter that provides the action. |
| Associated Object Types | Indicates the object level at which the action instance runs. |
| Recommendations | Indicates whether the action is used in at least one recommendation. |

The actions `Delete Unused Snapshots for Datastore Express` and `Delete Unused Snapshots for VM Express` appear. However, these actions can only be run in the user interface from an alert whose first recommendation is associated with this action. You can use the REST API to run these actions.

The following actions are also not visible except in the alert recommendations:

- `Set Memory for VM Power Off Allowed`
- `Set CPU Count for VM Power Off Allowed`
- `Set CPU Count and Memory for VM Power Off Allowed`

These actions are intended to be used to automate the actions with the `Power Off Allowed` flag set to true.

## Actions Supported for Automation

Recommendations can identify ways to remediate problems indicated by an alert. Some of these remediations can be associated with actions defined in your VCF Operations instance. You can automate several of these remediation actions for an alert when that recommendation is the first priority for that alert.

You activate actionable alerts in your policies. By default, automation is deactivated in policies. To configure automation for your policy, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definition** tile. Then, to edit a policy, access the **Alert / Symptom Definitions** workspace, and select **Local** for the **Automate** setting in the Alert / Symptom Definitions pane.

When an action is automated, you can use the **Automated** and **Alert** columns in **Recent Tasks** under **Administration** > **Control Panel** to identify the automated action and view the results of the action.

- VCF Operations uses the **automationAdmin** user account to trigger automated actions. For these automated actions that are triggered by alerts, the Submitted By column displays the **automationAdmin** user.
- The Alert column displays the alert that triggered the action. When an alert is triggered that is associated to the recommendation, it triggers the action without any user intervention.

The following actions are supported for automation:

- Delete Powered Off VM
- Delete Idle VM
- Move VM
- Power Off VM
- Power On VM
- Set CPU Count And Memory for VM
- Set CPU Count And Memory for VM Power Off Allowed
- Set CPU Count for VM
- Set CPU Count for VM Power Off Allowed
- Set CPU Resources for VM
- Set Memory for VM
- Set Memory for VM Power Off Allowed
- Set Memory Resources for VM
- Shut Down Guest OS for VM

#### Roles Needed to Automate Actions

To automate actions, your role must have the following permissions:

- Create, edit, and import policies in **Administration** > **Configurations** > **Policy Definition**.
- Create, clone, edit, and import alert definitions in **Administartion** > **Configurations** > **Alert Definitions**.
- Create, edit, and import recommendation definitions in **Administration** > **Configurations** > **Recommendations**.

> **Important:**
> You set the permissions used to run the actions separately from the alert and recommendation definition. Anyone who can modify alerts, recommendations, and policies can also automate the action, even if they do not have permission to run the action.

For example, if you do not have access to the Power Off VM action, but you can create and modify alerts and recommendations, you can see the Power Off VM action and assign it to an alert recommendation. Then, if you automate the action in your policy, VCF Operations uses the `automationAdmin` user to run the action.

#### Example Action Supported for Automation

For the Alert Definition named `Virtual machine has chronic high CPU workload leading to CPU stress`, you can automate the action named `Set CPU Count for VM`.

When CPU stress on your virtual machines exceeds a critical, immediate, or warning level, the alert triggers the recommended action without user intervention.

## Integration of Actions with VCF Automation

VCF Operations restricts actions on data centers and custom data centers that contains VCF Automation managed child objects such as, cluster compute resources, hosts, and VMs.

You can turn on or turn off the actions on VCF Automation managed objects by modifying the **Operational Actions** from the respective vCenter in Cloud Accounts or by creating a new role with limited action ability on VCF Automation managed objects.

### Actions Determine Whether Objects Are Managed

Actions check the objects in the VCF Automation managed resource container to determine which objects are being managed by VCF Automation.

Actions such as Rebalance Container check the child objects of the data center container or custom data center container to determine whether the objects are managed by VCF Automation. If the objects are being managed, the action does not appear on those objects.

## Working with Actions That Use Power Off Allowed

Some of the actions provided with VCF Operations require the virtual machines to shut down or power off, depending on the configuration of the target machines, to run the actions. You should understand the impact of the Power Off Allowed option before running the actions so that you select the best options for your target virtual machines.

### Power Off and Shut Down

The actions that you can run on your vCenter instances include actions that shut down virtual machines and actions that power off virtual machines. It also includes actions where the virtual machine must be in a powered off state to complete the action. Whether the VM is shut down or powered off depends on how it is configured and what options you select when you run the action.

The shut-down action shuts down the guest operating system and then powers off the virtual machine. To shut down a virtual machine from VCF Operations, the VMware Tools must be installed and running on the target objects.

The power off action turns off the VM without regard for the state of the guest operating system. In this case, if the VM is running applications, your user might lose data. After the action is finished, for example, modifying the CPU count, the virtual machine is returned to the power state it was in when the action began.

### Power Off Allowed and VMware Tools

For the actions where you are increasing the CPU count or the amount of memory on a VM, some operating systems support the actions if the Hot Plug is configured on the VM. For other operating systems, the virtual machine must be in a powered off state to change the configuration. To accommodate this need where the VMware Tools is not running, the Set CPU Count, Set Memory, and Set CPU Count and Memory actions include the Power Off Allowed option.

If you select Power Off Allowed, and the machine is running, the action verifies whether VMware Tools is installed and running.

- If VMware Tools is installed and running, the virtual machine is shut down before completing the action.
- If VMware Tools is not running or not installed, the virtual machine is powered off without regard for the state of the operating system.

If you do not select Power Off Allowed and you are decreasing the CPU count or memory, or the hot plug is not activated for increasing the CPU count or memory, the action does not run and the failure is reported in Recent Tasks.

### Power Off Allowed When Changing CPU Count or Memory

When you run the actions that change the CPU count and the amount of memory, you must consider several factors to determine if you want to use the Power Off Allowed option. These factors include whether you are increasing or decreasing the CPU or memory and whether the target virtual machines are powered on. If you increase the CPU or memory values, whether hot plug is activated also affects how you apply the option when you run the action.

How you use Power Off Allowed when you are decreasing the CPU count or the amount of memory depends on the power state of the target virtual machines.

**Table 1213: Decreasing CPU Count and Memory Behavior Based On Options**

| Virtual Machine Power State | Power Off Allowed Selected | Results |
|---|---|---|
| On | Yes | If VMware Tools is installed and running, the action shuts down the virtual machine, decreases the CPU or memory, and powers the machine back on.<br>If VMware Tools is not installed, the action powers off the virtual machine, decreases the CPU or memory, and powers the machine back on. |
| On | No | The action does not run on the virtual machine. |
| Off | Not applicable. The virtual machine is powered off. | The action decreases the value and leaves the virtual machine in a powered off state. |

How you use Power Off Allowed when you are increasing the CPU count or the amount of memory depends on several factors, including the state of the target virtual machine and whether hot plug is activated. Use the following information to determine which scenario applies to your target objects.

If you are increasing the CPU count, you must consider the power state of the virtual machine and whether CPU Hot Plug is activated when determining whether to apply Power Off Allowed.

**Table 1214: Increasing CPU Count Behavior.**

| Virtual Machine Power State | CPU Hot Plug Activated | Power Off Allowed Selected | Results |
|---|---|---|---|
| On | Yes | No | The action increases the CPU count to the specified amount. |
| On | No | Yes | If VMware Tools is installed and running, the action shuts down the virtual machine, increases the CPU count, and powers the machine back on.<br>If VMware Tools is not installed, the action powers off the virtual machine, increases the CPU count, and powers the machine back on. |
| Off | Not applicable. The virtual machine is powered off. | Not required. | The action increases the CPU count to the specified amount. |

If you are increasing the memory, you must consider the power state of the virtual machine, whether Memory Hot Plug is activated, and whether there is a Hot Memory Limit when determining how to apply Power Off Allowed.

**Table 1215: Increasing Memory Amount Behavior**

| Virtual Machine Power State | Memory Hot Plug Activated | Hot Memory Limit | Power Off Allowed Selected | Results |
|---|---|---|---|---|
| On | Yes | New memory value = hot memory limit | No | The action increases the memory the specified amount. |
| On | Yes | New memory value > hot memory limit | Yes | If VMware Tools is installed and running, the action shuts down the virtual machine, increases the memory, and powers the machine back on.<br>If VMware Tools is not installed, the action powers off the virtual machine, increases the memory, and powers the machine back on. |
| On | No | Not applicable. The hot plug is not activated. | Yes | If VMware Tools is installed and running, the action shuts down the virtual machine, increases the memory, and powers the machine back on.<br>If VMware Tools is not installed, the action powers off the virtual machine, increases the memory, and powers the machine back on. |
| Off | Not applicable. The virtual machine is powered off. | Not applicable. | Not required | The action increases the memory the specified amount. |

## Running Actions from VCF Operations

The actions available in VCF Operations allow you to modify the state or configuration of selected objects in vCenter from VCF Operations. For example, you might need to modify the configuration of an object to address a problematic resource issue or to redistribute resources to optimize your virtual infrastructure.

The most common use of the actions is to solve problems. You can run them as part of your troubleshooting procedures or add them as a resolution recommendation for alerts.

When you grant a user access to actions in VCF Operations, that user can take the granted action on any object that VCF Operations manages.

When you are troubleshooting problems, you can run the actions from the center pane Actions menu. Alternatively, you can run them from the toolbar on list views that contain the supported objects.

When an alert is triggered, and you determine that the suggested action is the most likely way to resolve the problem, you can run the action on one or more objects.

## Run Actions from Toolbars in VCF Operations

When you run actions in VCF Operations, you change the state of vCenter objects. You run one or more actions when you encounter objects where the configuration or state of the object is affecting your environment. These actions allow you to reclaim wasted space, adjust memory, or conserve resources.

- Verify that the vCenter Adapter is configured to run actions for each vCenter instance. See Configure a vCenter Serve Cloud Account in *VCF Operations Configuration Guide.* .
- Verify that the vCenter Adapter is configured to run actions for each vCenter instance. See the *VCF Operations Configuration Guide.* .
- Ensure that you understand how to use the power-off-allowed option if you are running Set CPU Count, Set Memory, and Set CPU Count and Memory actions. See Working with Actions That Use Power Off Allowed section in *VCF Operations Configuration Guide.* .
- Ensure that you understand how to use the power-off-allowed option if you are running Set CPU Count, Set Memory, and Set CPU Count and Memory actions. See the section Working With Actions That Use Power Off Allowed in the VCF Operations Information Center.

This procedure for running actions is based on the VCF Operations**Actions** menus and is commonly used when you are troubleshooting problems. The available actions depend on the type of objects with which you are working. You can also run actions as alert recommendations.

1. Select the object in the Inventory page inventory tree or select one or more objects it in a list view.
2. Click **Actions** on the main toolbar or in an embedded view.
3. Select one of the actions.

   If you are working with a virtual machine, only the virtual machine is included in the dialog box. If you are working with clusters, hosts, or datastores, the dialog box that appears includes all objects.
4. To run the action on the object, select the check box and click **OK**.

   The action runs and a dialog box appears that displays the task ID.
5. To view the status of the job and verify that the job finished, click **Recent Tasks** or click **OK** to close the dialog box.

   The Recent Tasks list appears, which includes the task you just started.

**Rebalance Container Action**
When the workload in your environment becomes imbalanced, you can move the workload across your objects to rebalance the overall workload. The container for the rebalance action can be a data center or a custom data center, and the objects that are moved are the virtual machines in the suggested list provided by the action.

### DRS Must be Activated on Clusters

Your vCenter instance must have a cluster that passes a DRS-activated check for the Rebalance Container action to appear in the Actions drop-down menu.

To get the Rebalance Container action from a custom data center or data center, and the related alerts, you must have the following:

- A vCenter Adapter configured with the actions activated for each vCenter instance
- A vCenter instance with at least one cluster that is DRS-activated.

If your cluster does not have DRS fully automated, the Rebalance Container action notifies you that one or more clusters under the selected container do not have DRS set to fully automated.

To ensure that the Rebalance Container action is available in your environment, you must add DRS. Then, wait one collection cycle for the Rebalance Container action to appear.

### You Must Have Access to All Objects in the Container

If you have access to all objects in a cluster, data center, or custom data center, you can run the Rebalance Container action to move virtual machines to other clusters. When you do not have access to all of the objects in the container, the Rebalance Container action is not available.

## How the Rebalance Container Action Works

If two data centers are experiencing extreme differences in workload - one high and one low - use the Rebalance Container action to balance the workload across those objects. For example, if the CPU demand on a host in one data center exceeds its available CPU capacity, critical pressure occurs on the host. To identify the cause of stress, monitor the CPU demand. Some virtual machines on each host might be experiencing high CPU demand, whereas others might be experiencing a low demand.

The Rebalance Container action moves all affected objects in the suggested list provided by the action to balance the workload. If you do not want to act on the entire set of objects to resolve the problem with workload, you can use the Move VM action to move an individual object.

> **Important:**
> Do not attempt to move virtual machines that are members of a vApp, because the vApp can become nonfunctional. Instead, add affinity rules for these virtual machines to keep them together so that the Move VM and Rebalance Container actions will ignore them.

When workloads become imbalanced, the following alerts can trigger on data centers and custom data centers. These alerts are deactivated by default in the policies.

- Custom data center has unbalanced workload
- Data center has unbalanced workload

When the workloads on hosts in a data center or custom data center differ significantly, click **Operate** > **Alerts** and verify whether the alert triggered. For example, to verify whether the alert triggered on a custom data center, check the alert named `Custom data center has unbalanced workload`. You can click the alert to view the causes of the alert and identify the source of the imbalance problem on the **Summary** tab.

To display the recommendations about the objects to move so that you can rebalance the workload, click the **Rebalance Container** action on the **Summary** tab. The recommendations indicate that you move one or more virtual machines to another host. When you click **OK**, a pop-up message provides a link to track the status of the action in **Recent Tasks**.

The action moves the virtual machines identified in the recommendation to the host machine that has a low workload or stress. You can view the status of the action in the list of recent tasks in **Administration** > **Control Panel**, and then click the **Recent Tasks** tab. You can also use the vSphere Web Client to view the status of the action and the performance for the host.

After the action runs and VCF Operations performs several collection cycles, view the workload on the data center to confirm that the workload was rebalanced and that the alert is gone.

## Where You Run the Action

You can run the Rebalance Container action from the Actions menu for a data center or custom data center, or you can provide it as a suggested action on an alert.

For the supported objects and object levels, this action is available in the following locations in VCF Operations:

- In configured alert recommendations.
- In the Object List and Topology Graph dashboard widgets.

## Action Recommendations

Review the following information about the hosts and virtual machines to ensure that you are submitting the action for the correct objects.

| Option | Description |
|---|---|
| Virtual Machine | Name of the virtual machine on the host that is experiencing an excessive workload. |
| Source Cluster | Name of the cluster on which the virtual machine is running. |
| Datastores | Datastore associated with the virtual machine. |

| Option | Description |
|---|---|
| Destination Cluster | Cluster where the virtual machine is to be moved. DRS selects the host automatically. |
| Reason | Describes the action to be taken and the reason why the move is suggested. For example, the recommendation is to move part of the workload on the cluster to another cluster to reduce the imbalance in CPU demand. |
| Parent vCenter | Identifies the vCenter Server adapter associated with the affected cluster. |

After you click **OK**, the next dialog box provides the task ID and a link to the task list.

**Table 1216: Task ID Dialog Box**

| Option | Description |
|---|---|
| Recent Tasks | To view the status of the job and verify that the job finished, click **Recent Tasks**. |
| OK | To close the dialog box without further action, click **OK**. |

### Delete Idle VM Action

The Delete Idle VM action in VCF Operations removes from your vCenter instances those selected virtual machines that are in an idle state. Use this action to reclaim redundant resources.

#### How the Action Works

The Delete Idle VM action removes from your vCenter instances those virtual machines that are powered on, but that are in an idle state.

#### Where You Run the Action

For the supported objects and object levels, this action is available in the following locations in VCF Operations:

- Embedded just below the top menu.
- In configured alert recommendations.
- In the Object List and Topology Graph dashboard widgets.

#### Action Menu Items

Review the following information about the virtual machines to ensure that you are submitting the action for the correct objects.

| Menu Items | Description |
|---|---|
| Name | Name of the virtual machine as it appears in the environment inventory. |
| Host | Name of the host on which the virtual machine is running. |
| Parent vCenter | Parent vCenter instance where the virtual machine resides. |

After you click **Begin Action**, the next dialog box provides the task ID and a link to the task list.

**Table 1217: Task ID Dialog Box**

| Option | Description |
|---|---|
| Recent Tasks | To view the status of the job and verify that the job finished, click **Recent Tasks**. |
| OK | To close the dialog box without further action, click **OK**. |

### Set DRS Automation Action

You can monitor and configure the vSphereDistributed Resource Scheduler (DRS) automation rules from VCF Operations. DRS monitors and allocates the resources in your environment, and balances the computing capacity across your hosts and virtual machines.

#### How the Action Works

The Set DRS Automation action monitors and configures DRS automation rules. With the Set DRS Automation action, you can activate and deactivate DRS.

If VCF Automation manages any of the virtual machines in your environment, the Set DRS Automation action is not available for that object.

#### Where You Run the Action

For the supported objects and object levels, this action is available in the following locations in VCF Operations:

- Embedded just below the top menu.
- In configured alert recommendations.
- In the Object List and Topology Graph dashboard widgets.

#### Action Menu Items

To ensure that you are submitting the correct action for the correct objects, review the following information about the clusters.

| Menu Items | Description |
|---|---|
| Name | Name of the cluster in the vCenter instance. |
| Automation Level | Level of DRS automation. When DRS is fully automated on the selected cluster, you can run the Set DRS Automation action. |
| Migration Threshold | Recommendations for the migration level of virtual machines. Migration thresholds are based on DRS priority levels, and are computed based on the workload imbalance metric for the cluster. |
| Parent vCenter | Parent vCenter instance where the cluster resides. |

After you click **Begin Action**, the next dialog box provides the task ID and a link to the task list.

**Table 1218: Task ID Dialog Box**

| Option | Description |
|---|---|
| Recent Tasks | To view the status of the job and verify that the job finished, click **Recent Tasks**. |
| OK | To close the dialog box without further action, click **OK**. |

### Execute Custom Scripts

To troubleshoot particular processes, you can upload a script or run a command to receive specific information. You can view the standard output or standard error as applicable.

### Where You Run the Action

For supported objects and object levels, n the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory**, and then select the relevant VM from the Inventory tree. This action is available from the **Actions** menu just below the top menu in VCF Operations.

### Prerequisites

- VMware Tools must be installed and running on the VM.
- vCenter is activated with the successful discovery of VMs.
- The VM must be powered on and connected.
- When you perform the Execute Custom Script action and you need to use `command/utilities`, for those commands that need a sudo user password provision, the full path of `command/utility` must be added to the NOPASSWD commands list.

The configured vCenter user must have the following privileges:

- key: VirtualMachine.GuestOperations.Execute, Localization: Guest operations -> Guest operation program execution
- key: VirtualMachine.GuestOperations.Query, Localization: Guest operations -> Guest operation queries
- The ESX instance that hosts the VMs where scripts should be run, must have HTTPS access to port 443 from the collector node on which the vCenter adapter instance is configured.
- The provided user in the script must have read and write privileges to the temp directory. For Windows systems, the path can be taken from the environment variable `TEMP`. For Linux systems, it is `/tmp` and/or `/var/tmp`.

### Action Options

Enter the VM credentials to authenticate even when the VM guest OS authentication status is "Success". You can run a script by entering it directly or by uploading a script file by optionally providing arguments.

| Option | Description |
|---|---|
| Upload File | Use this option to browse and upload the script that you want to run. |
| File | Browse and upload the script file. |
| Args | List the arguments in the script. |
| Command | Select the option and enter a command in the text box. |
| Timeout | Script execution timeout on VMs. Script execution continues even if the dialog box is closed. You can verify the status from **Recent Tasks**. |
| Execute | Runs the script or command. |
| stdout | Displays the standard output. |
| stderr | Displays errors, if any. |

## Get Top Processes Action

The Get Top Processes action is used for troubleshooting process issues and resource issues related to the applications of the virtual machine.

### How the Action Works

The Get Top Processes action, provides the status of top 10 processes for the selected virtual machine. You can troubleshoot issues related to the resources that are affecting the applications in the virtual machine.

By default, the details of top 10 processes are displayed for the selected virtual machine. You can change the number of processes and view the details for top N processes where N is between 1-100. You have the option to view the processes based on CPU and Memory.

The Get Top Processes action is run on both Windows virtual machine and Linux virtual machine. You can view the summary information for the commands only in a Linux virtual machine.

### Where You Run the Action

For supported objects and object levels, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory**, and then select the relevant VM from the Inventory tree. This action is available from the **Actions** menu just below the top menu in VCF Operations.

### Prerequisites

- VMware Tools must be installed and running on the VM.
- Service discovery is activated with the successful discovery of VMs.
- The VM must be powered on and connected.

### Action Options

You must enter the VM credentials to authenticate when the VM is monitored in a credential-less mode or when the VM is monitored in a credential-based mode where the user is not authenticated. To ensure that you are taking the right action, review the following information.

| Option | Description |
|---|---|
| Number of Processes | Displays the number of processes for which the details are displayed. |
| Refresh | Displays new data about processes, when you change the value for the number of processes. |
| Command | Displays the name of the application |
| PID | Displays the process ID. |
| CPU | Displays the CPU usage in percentage for Linux VMs.<br>Displays the CPU usage in seconds for Windows VMs. The count starts when you start the operating system in the VM . |
| Mem (%) | Displays the Memory usage in KB. |
| User | Displays the user name. |
| Status | Displays the process status. It can be in one these states:<br>• For Linux - I, R, S<br>• For Windows - Unknown, Running, and Sleeping |
| Run | Displays data about the specified numbers of processes. |

## Move Virtual Machine Action

You can use the Move VM action to move virtual machines from one host and datastore to another host and datastore to balance the workload in your environment.

### How the Action Works

When you initiate this action, the **Move VM** wizard opens and scopes the possible destinations. You select the destination host and datastore from the list of available destinations.

To see all destinations, you must have view access to the following object types:

- Scope object, which includes a vCenter, data center, custom data center, or cluster.
- Host in the scope object.
- Datastore in the host.

The destinations include combinations of objects for the move, such as a specific host and datastore, or a different host with the same datastore. You select one of the available combinations. If your environment includes many destination objects, such as many hosts or datastores, enter text in the filter text box to search for specific destination objects.

The **All Filters** option helps you to move the VM according to the following action option:

- Destination Host
- Destination Datastore
- Will it Fit
- VM PowerOff Required
- Affinity Rules

VCF Operations uses vSphere DRS rules that you define in vCenter to help determine good placement decisions for your virtual machines in the move action. The Affinity Rules column indicates whether those rules are violated by the Move VM action.

> **Important:**
> Do not attempt to move virtual machines that are members of a vApp, because the vApp can become nonfunctional. Instead, add affinity rules for these virtual machines to keep them together so that the Move VM and Rebalance Container actions will ignore them.

To initiate the action, you click the **Begin Action** button.

When you finish the wizard, VCF Operations displays a dialog box to indicate that the action has started. To track the status of the action, click the link in the dialog box and view the state of the action in **Recent Tasks**.

### Moving Virtual Machines is Not Allowed Across Data Centers

When you attempt to use the **Move VM** action to move a virtual machine across data centers, VCF Operations must be able to identify the matching network and storage objects for the destination data center. Network objects include VMware virtual switches and distributed virtual switches. Storage objects include datastores and datastore clusters.

Moving a virtual machine across data centers requires VCF Operations to move the virtual machine files and change the virtual machine network configuration. VCF Operations does not currently move the virtual machine files across datastores, nor does it change the virtual machine network configuration. As a result, VCF Operations does not allow you to move virtual machines across data centers.

When you use the **Move VM** action, be aware of the following behavior:

- If you select a single virtual machine, VCF Operations displays the data center where the virtual machine resides.
- If you select multiple virtual machines, but those virtual machines do not share a common data center, the **Move VM** action does not display the data centers, and the **Move VM** action does not appear in the actions menu.

### Where You Run the Action

For the supported objects and object levels, this action is available in the following locations in VCF Operations:

- Embedded just below the top menu.
- In configured alert recommendations.
- In the Object List and Topology Graph dashboard widgets.

### Action Options

Review the following information about the virtual machines to ensure that you are submitting the action for the correct objects.

| Option | Description |
|---|---|
| Priority | Indicates the priority of the proposed move destination. When the action is automated, the proposed destination with priority of 1 is automatically selected. |
| Destination Host | Name of the host to which the virtual machine will be moved. |
| Current CPU Workload | Amount of CPU in GHz available on the host. |
| Current Memory Workload | Amount of memory in GB available on the host. |
| Destination Datastore | Datastore to which the virtual machines storage will be moved. |
| Current Disk Space Workload | Amount of disk space available on the datastore. |

| Option | Description |
|---|---|
| Will it fit | Calculated estimation of whether the virtual machine fits on the selected destination. |
| VM Power Off Required | When set to `No`, the action does not power off the virtual machine before the move. When set to `Yes`, the action powers off the virtual machine before the move takes place, and powers on the virtual machine after the move is complete. If VMware Tools is installed, a guest OS shutdown is used to power off the virtual machine. |
| Affinity Rules | Indicates whether vSphere DRS rules exist, as defined in vCenter. For example, a rule might exist to keep virtual machines together, and another rule might exist to separate virtual machines.<br>This column indicates the following status.<br>• Empty. vSphere DRS rules are not defined.<br>• Green check mark. The move of virtual machines does not violate affinity rules.<br>• Red circle with bar. The move of virtual machines does break affinity rules. If you choose to break the affinity rules, you must resolve any problems manually. |
| Affinity Rule Details | Identifies the virtual machine and the vSphere DRS rule name as defined in vCenter. |

After you click **OK**, the next dialog box provides the task ID and a link to the task list.

**Table 1219: Task ID Dialog Box**

| Option | Description |
|---|---|
| Recent Tasks | To view the status of the job and verify that the job finished, click **Recent Tasks**. |
| OK | To close the dialog box without further action, click **OK**. |

### Power Off Virtual Machine Action

The Power Off VM action in VCF Operations stops one or more selected virtual machines that are in a powered on state. You power off a virtual machine when you are managing resources and reclaiming wasted space.

#### How the Action Works

The Power Off VM action turns off the virtual machine. If VMware Tools is installed and running, the guest operating system is shut down before the machine is powered off. If VMware Tools is not installed and running, the virtual machine is powered off regardless of the state of the guest operating system. In this case, use this action only when you are powering off virtual machines where stopping the guest operating system does not adversely affect the installed applications.

If the target virtual machine is already powered off, the recent task status reports success on the machine, even though the state of the virtual machine did not change.

#### Where You Run the Action

For the supported objects and object levels, this action is available in the following locations in VCF Operations:

- Embedded just below the top menu.
- In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Inventory Management** tile. Select an object in the list.
- In configured alert recommendations.
- In the Object List and Topology Graph dashboard widgets.

### Action Options

Review the following information about the virtual machines to ensure that you are submitting the action for the correct objects.

| Option | Description |
|---|---|
| Selected objects | Check box indicates whether the action is applied to the object. To not run the action on one or more objects, deselect the associated check boxes. This option is available when two or more objects are selected. |
| Name | Name of the virtual machine as it appears in the environment inventory. |
| Power State | Indicates whether the virtual machine is powered on or powered off. |
| Idle VM | Indicates whether the virtual machine is considered to be in the idle state based on the configured idle virtual machine metric. Possible values include:<br>• false. The virtual machine is active.<br>• true. The virtual machine is idle.<br>• unknown. VCF Operations does not have the data required to calculate the idle metric. |
| Idle VM Percentage | Calculated threshold of the idle virtual machine percentage based on the configured reclaimable wasted space policy. |
| CPU Usage Percentage | Calculated threshold of the virtual machine CPU percentage based on the metric named `cpu \| usage_average`. |
| Host | Name of the host on which the virtual machine is running. |
| Adapter Instance | Name of the VMware Adapter as it is configured in VCF Operations. The adapter manages the communication with the vCenter instance. |

After you click **OK**, the next dialog box provides the task ID and a link to the task list.

**Table 1220: Task ID Dialog Box**

| Option | Description |
|---|---|
| Recent Tasks | To view the status of the job and verify that the job finished, click **Recent Tasks**. |
| OK | To close the dialog box without further action, click **OK**. |

### Shut Down Guest Operating System for Virtual Machine Action

The Shut Down Guest OS for VM action shuts down the guest operating system and powers off the virtual machine. You shut down a virtual machine when you are managing resources and reclaiming wasted space.

### How the Action Works

The Shut Down Guest OS for VM action checks that VMware Tools, which is required, is installed on the target virtual machines, then shuts down the guest operating system and powers off the virtual machine. If VMware Tools is not installed or installed but not running, the action does not run and the job is reported as failed in **Recent Tasks**.

If the target virtual machine is already powered off, the recent task status reports success on the machine, even though the state of the virtual machine did not change.

### Where You Run the Action

For the supported objects and object levels, this action is available in the following locations in VCF Operations:

- Embedded just below the top menu.
- In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Inventory Management** tile. Select an object in the list.
- In configured alert recommendations.
- In the Object List and Topology Graph dashboard widgets.

### Action Options

Review the following so you can be sure you are taking the right action.

| Option | Description |
|---|---|
| Selected objects | Check box indicates whether the action is applied to the object. To not run the action on one or more objects, deselect the associated check boxes. This option is available when two or more objects are selected. |
| Name | Name of the virtual machine as it appears in the environment inventory. |
| Power State | Indicates whether the virtual machine is powered on or powered off. |
| Idle VM | Indicates whether the virtual machine is considered to be in the idle state based on the configured idle virtual machine metric.<br>Possible values include:<br>• false. The virtual machine is active.<br>• true. The virtual machine is idle.<br>• unknown. VCF Operations does not have the data required to calculate the idle metric. |
| Idle VM Percentage | Calculated threshold of the idle virtual machine percentage based on the configured reclaimable wasted space policy. |
| Host | Name of the host on which the virtual machine is running. |
| Adapter Instance | Name of the VMware Adapter as it is configured in VCF Operations. The adapter manages the communication with the vCenter instance. |

After you click **OK**, the next dialog box provides the task ID and a link to the task list.

**Table 1221: Task ID Dialog Box**

| Option | Description |
|---|---|
| Recent Tasks | To view the status of the job and verify that the job finished, click **Recent Tasks**. |
| OK | To close the dialog box without further action, click **OK**. |

### Reboot Guest OS for Virtual Machine Action

The Reboot Guest OS for VM action reboots the guest operating system and the virtual machine. You reboot a virtual machine while managing resources or when you have new updates or configuration changes to your virtual machine.

### How the Action Works

The Reboot Guest OS for VM action checks that VMware Tools, which is required, is installed on the target virtual machines, then reboots the guest operating system and the virtual machine. If VMware Tools is not installed or installed but not running, the action does not run, and the job is reported as failed in **Recent Tasks**.

### Where You Run the Action

For the supported objects and object levels, this action is available in the following locations in VCF Operations:

- Embedded just below the top menu.
- In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory**, and select an object in the list.
- In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory**, select an object, click the **Environment** tab, and select an object in the list view.
- In configured alert recommendations.
- In the Object List and Topology Graph dashboard widgets.

### Action Options

Review the following so you can be sure you are taking the right action.

| Option | Description |
|---|---|
| Selected objects | Check box indicates whether the action is applied to the object. To not run the action on one or more objects, deselect the associated check boxes. This option is available when two or more objects are selected. |
| Name | Name of the virtual machine as it appears in the environment inventory. |
| Host | Name of the host on which the virtual machine is running. |
| Parent vCenter | The adapter instance as configured in VCF Operations. The adapter manages the communication with the vCenter instance. |

After you click **OK**, the next dialog box provides the task ID and a link to the task list.

**Table 1222: Task ID Dialog Box**

| Option | Description |
|---|---|
| Recent Tasks | To view the status of the job and verify that the job finished, click **Recent Tasks**. |
| OK | To close the dialog box without further action, click **OK**. |

## Power on Virtual Machine Action

To start one or more virtual machines that are in a powered off state, use the Power On VM action. You power on a virtual machine so that you can shift resources. For example, power on a machine so that you can use it, run applications, or verify that actions that were run on already powered down machines contribute to improved performance.

### How the Action Works

The Power On VM action powers on virtual machines that are powered off. The action does not affect virtual machines that are currently powered on.

If the target virtual machine is already powered on, the task status reports success for the machine even though the state of the virtual machine did not change.

### Where You Run the Action

For the supported objects and object levels, this action is available in the following locations in VCF Operations:

- Embedded just below the top menu.
- In configured alert recommendations.
- In the Object List and Topology Graph dashboard widgets.

### Action Options

To ensure that you are taking the right action, review the following information.

| Option | Description |
|---|---|
| Selected objects | Check box indicates whether the action is applied to the object. To not run the action on one or more objects, deselect the associated check boxes. This option is available when two or more objects are selected. |
| Name | Name of the virtual machine as it appears in the environment inventory. |
| Power State | Indicates whether the virtual machine is powered on or powered off. |
| Host | Name of the host on which the virtual machine is running. |
| Adapter Instance | Name of the VMware Adapter as it is configured in VCF Operations. The adapter manages the communication with the vCenter instance. |

After you click **OK**, the next dialog box provides the task ID and a link to the task list.

**Table 1223: Task ID Dialog Box**

| Option | Description |
|---|---|
| Recent Tasks | To view the status of the job and verify that the job finished, click **Recent Tasks**. |
| OK | To close the dialog box without further action, click **OK**. |

## Delete Powered Off Virtual Machine Action

The Delete Powered Off VM action in VCF Operations removes selected virtual machines that are in a powered off state from your vCenter instances. Use this action to reclaim redundant resources.

### How the Action Works

The Delete Powered Off VM action removes virtual machines from the vCenter instances. If the virtual machine is powered on, the action does not delete the virtual machine.

### Where You Run the Action

For the supported objects and object levels, this action is available in the following locations in VCF Operations:

- Embedded just below the top menu.
- In the VCF Operations navigation bar at the top, click **Operation** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Inventory Management** tile. Select an object in the list.
- In configured alert recommendations.
- In the Object List and Topology Graph dashboard widgets.

## Action Options

To ensure that you are submitting the action for the right objects, review the following information.

| Option | Description |
|---|---|
| Selected objects | Check box indicates whether the action is applied to the object. To not run the action on one or more objects, deselect the associated check boxes. This option is available when two or more objects are selected. |
| Name | Name of the virtual machine as it appears in the environment inventory. |
| Power State | Indicates whether the virtual machine is powered on or powered off. |
| Disk Space | Amount of disk space currently consumed by the virtual machine. |
| Snapshot Space | Amount of disk space currently consumed by the virtual machine snapshots. |
| Memory (MB) | Amount of memory allocated to the virtual machine. |
| CPU Count | Number of CPUs currently configured for the virtual machine. |
| Host | Name of the host on which the virtual machine is running. |
| Adapter Instance | Name of the VMware Adapter as it is configured in VCF Operations. The adapter manages the communication with the vCenter instance. |

After you click **OK**, the next dialog box provides the task ID and a link to the task list.

**Table 1224: Task ID Dialog Box**

| Option | Description |
|---|---|
| Recent Tasks | To view the status of the job and verify that the job finished, click **Recent Tasks**. |
| OK | To close the dialog box without further action, click **OK**. |

### Set Memory for Virtual Machine Action

The Set Memory for VM action in VCF Operations is used to add or remove memory on virtual machines. You increase the memory to address performance problems or decrease the memory to reclaim resources.

## How the Action Works

The Set Memory for VM action perform several tasks. The action determines the power state of the target virtual machines, takes a snapshot when you request it and powers off the machine if necessary and you request it. As well, the action changes the memory to the new value, and returns the virtual machines their original power states.

An alternative form of the Set Memory for Virtual Machine action is available for automation. This action can run when the virtual machine is powered on or off.

Use this version of the action if the automated action has permission to power off the virtual machine, and hot add of memory is not activated on the virtual machine. With hot add activated, you can add memory, but you cannot remove it.

This version of the action would be required if a virtual machine is powered on and the amount of memory must be reduced.

This version of the action has the Power Off Allowed flag set to true. You can select this Power Off Allowed version of the action when you create or edit alerts and associate the alert with a recommendation. When the Power Off Allowed version of this action is automated, you do not select this version of the action.

If Hot Plug is activated on the virtual machines, then power off is not required. If power off is required and VMware Tools is installed, then the virtual machines are shut down before they are powered off.

### Where You Run the Action

For the supported objects and object levels, this action is available in the following locations in VCF Operations:

- Embedded just below the top menu.
- In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Inventory Management** tile. Select an object in the list.
- In configured alert recommendations.
- In the Object List and Topology Graph dashboard widgets.

### Action Options

Review the following information about the virtual machines to ensure that you are submitting the action for the correct objects.

| Option | Description |
|---|---|
| Selected objects | Check box indicates whether the action is applied to the object. To not run the action on one or more objects, deselect the associated check boxes. This option is available when two or more objects are selected.<br>If you modify a value, the check box is selected. The check box must be selected to activate the OK button. |
| Name | Name of the virtual machine as it appears in the environment inventory. |
| New CPU | Number of CPUs when the action is completed. If the value is less than 1 or a value not supported for the virtual machine in vCenter, and the virtual machine is powered on and Hot Add is not activated, the number of CPUs does not change and Recent Tasks shows the action as failed. If the virtual machine is powered off when you submit an unsupported value, the task reports success, but the virtual machine will fail when you run a power on action.<br>The value that appears is the calculated suggested size. If the target virtual machine is new or offline, this value is the current number of CPUs. If VCF Operations has been monitoring the virtual machine for six or more hours, depending on your environment, the value that appears is the CPU Recommended Size metric. |
| Current CPU | Number of configured CPUs. |
| Power State | Indicates whether the virtual machine is powered on or powered off. |
| Power Off Allowed | If selected, the action shuts down or powers off the virtual machine before modifying the value. If VMware Tools is installed and running, the virtual machine is shut down. If VMware Tools is not installed or not running, the virtual machine is powered off without regard for the state of the operating system.<br>In addition to whether the action shuts down or powers off a virtual machine, you must consider whether the object is powered on and what settings are applied.<br>See Working with Actions That Use Power Off section in *VCF Operations Configuration Guide.* . |

| Option | Description |
|---|---|
| Snapshot | Creates a snapshot before changing the number of CPUs. Use this option if you need a snapshot to which you can revert the virtual machine if the action does not produce the expected results.<br>The name of the snapshot is supplied in the Recent Tasks messages for the action.<br>If the CPU is changed with CPU Hot Plug activated, then the snapshot is taken with the virtual machine is running, which consumes more disk space. |
| Host | Name of the host on which the virtual machine is running. |
| Adapter Instance | Name of the VMware Adapter as it is configured in VCF Operations. The adapter manages the communication with the vCenter instance. |

After you click **OK**, the next dialog box provides the task ID and a link to the task list.

**Table 1225: Task ID Dialog Box**

| Option | Description |
|---|---|
| OK | To close the dialog box without further action, click **OK**. |
| Recent Tasks | To view the status of the job and verify that the job finished, click **Recent Tasks**. |

### Set Memory Resources for Virtual Machine Action

The Set Memory Resources for VM action is used to modify the memory reservation and memory limit on virtual machines. You modify the memory reservation and limit to manage resources in your environment, either to reclaim unused resources or to ensure that your virtual machines have the resources they need to run efficiently.

#### How the Action Works

The Set Memory Resources for VM action determines how memory resources are allocated to the virtual machine. The reservation value is the minimum amount of guaranteed memory allocated for the virtual machine. The limit is the maximum amount of memory that the virtual machine can consume.

The reservation and limit values in vCenter are set in megabytes. VCF Operations calculates and reports on memory in kilobytes. When you run this action, the values are presented in kilobytes so that you can implement recommendations from VCF Operations.

To run the action, all options must be configured in the dialog box for the objects on which you are running the action. If you are changing one option to a new value, but not another option, ensure that the option that you do not want to change is configured with the current value.

#### Where You Run the Action

For the supported objects and object levels, this action is available in the following locations in VCF Operations:

- Embedded just below the top menu.
- In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Inventory Management** tile. Select an object in the list.
- In configured alert recommendations.
- In the Object List and Topology Graph dashboard widgets.

## Action Options

To ensure that you are submitting the action for the right objects, review the following information.

| Option | Description |
|---|---|
| Selected objects | Check box indicates whether the action is applied to the object. To not run the action on one or more objects, deselect the associated check boxes. This option is available when two or more objects are selected.<br>If you modify a value, the check box is selected. The check box must be selected to activate the OK button. |
| Name | Name of the virtual machine as it appears in the environment inventory. |
| New Resv (KB) | Amount of memory in kilobytes reserved for the virtual machine when the action is finished. The new reservation value must be less than or equal to the new limit value unless your new limit is unlimited (-1).<br>The reservation supports the following possible values:<br>• If you set the value to 0, the virtual machine is allocated only the currently configured amount of RAM.<br>• If you add or remove reserved memory, the value must be evenly divisible by 1024. |
| Current Resv (KB) | Amount of memory in kilobytes that is configured as the guaranteed memory for the virtual machine. |
| New Limit (KB) | Maximum amount of memory in kilobytes that the virtual machine can consume when the action is completed.<br>The limit supports the following possible values:<br>• If you set the value to 0, then the maximum memory is no greater than the allocated reservation amount.<br>• If you set the value to -1, then the virtual machine memory is unlimited.<br>• It you increase or decrease the limit, the value must be evenly divisible by 1024. |
| Current Limit (KB) | Maximum amount of memory that the virtual machine is currently allowed to consume. |
| Host | Name of the host on which the virtual machine is running. |
| Adapter Instance | Name of the VMware Adapter as it is configured in VCF Operations. The adapter manages the communication with the vCenter instance. |

After you click **OK**, the next dialog box provides the task ID and a link to the task list.

**Table 1226: Task ID Dialog Box**

| Option | Description |
|---|---|
| Recent Tasks | To view the status of the job and verify that the job finished, click **Recent Tasks**. |
| OK | To close the dialog box without further action, click **OK**. |

### Set CPU Count for Virtual Machine Action

The Set CPU action modifies the number of vCPUs on a virtual machine. You increase the number of CPUs to address performance problems or decrease the number of CPU to reclaim resources.

## How the Action Works

The Set CPU Count action shuts down or powers off the target virtual machines. If you are decreasing the CPU count, the action is required. This action creates a snapshot if you request it, changes the number of vCPUs based on the new CPU count you provided, and returns the virtual machines to their original power states.

An alternative form of the Set CPU Count for Virtual Machine action is available for automation. This action can run when the virtual machine is powered on or off.

Use this version of the action if the automated action has permission to power off the virtual machine, and hot add of memory is not activated on the virtual machine. With hot add activated, you can add CPUs, but you cannot remove them.

This version of the action is required if a virtual machine is powered on and the number of CPUs must be reduced.

This version of the action has the Power Off Allowed flag set to true. You can select this Power Off Allowed version of the action when you create or edit alerts and associate the alert with a recommendation. When the Power Off Allowed version of this action is automated, you do not select this version of the action.

If Hot Plug is activated on the virtual machines, then power off is not required. If power off is required and VMware Tools are installed, then the virtual machines are shut down before they are powered off.

## Where You Run the Action

For the supported objects and object levels, this action is available in the following locations in VCF Operations:

- Embedded just below the top menu.
- In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Inventory Management** tile. Select an object in the list.
- In configured alert recommendations.
- In the Object List and Topology Graph dashboard widgets.

## Action Options

Review the following information about the virtual machines to ensure that you are submitting the action for the correct objects.

| Option | Description |
|---|---|
| Selected objects | Check box indicates whether the action is applied to the object. To not run the action on one or more objects, deselect the associated check boxes. This option is available when two or more objects are selected.<br>If you modify a value, the check box is selected. The check box must be selected to activate the OK button. |
| Name | Name of the virtual machine as it appears in the environment inventory. |
| New CPU | Number of CPUs when the action is completed. If the value is less than 1 or a value not supported for the virtual machine in vCenter, and the virtual machine is powered on and Hot Add is not activated, the number of CPUs does not change and Recent Tasks shows the action as failed. If the virtual machine is powered off when you submit an unsupported value, the task reports success, but the virtual machine will fail when you run a power on action.<br>The value that appears is the calculated suggested size. If the target virtual machine is new or offline, this value is the current number of CPUs. If VCF Operations has been monitoring the virtual machine for six or more hours, depending on your environment, the value that appears is the CPU Recommended Size metric. |
| Current CPU | Number of configured CPUs. |

| Option | Description |
|---|---|
| Power State | Indicates whether the virtual machine is powered on or powered off. |
| Power Off Allowed | If selected, the action shuts down or powers off the virtual machine before modifying the value. If VMware Tools is installed and running, the virtual machine is shut down. If VMware Tools is not installed or not running, the virtual machine is powered off without regard for the state of the operating system.<br>In addition to whether the action shuts down or powers off a virtual machine, you must consider whether the object is powered on and what settings are applied.<br>See Working with Actions That Use Power Off section in *VCF Operations Configuration Guide*. . |
| Snapshot | Creates a snapshot before changing the number of CPUs. Use this option if you need a snapshot to which you can revert the virtual machine if the action does not produce the expected results.<br>The name of the snapshot is supplied in the Recent Tasks messages for the action.<br>If the CPU is changed with CPU Hot Plug activated, then the snapshot is taken with the virtual machine is running, which consumes more disk space. |
| Host | Name of the host on which the virtual machine is running. |
| Adapter Instance | Name of the VMware Adapter as it is configured in VCF Operations. The adapter manages the communication with the vCenter instance. |

After you click **OK**, the next dialog box provides the task ID and a link to the task list.

**Table 1227: Task ID Dialog Box**

| Option | Description |
|---|---|
| OK | To close the dialog box without further action, click **OK**. |
| Recent Tasks | To view the status of the job and verify that the job finished, click **Recent Tasks**. |

### Set CPU Resources for Virtual Machine Action

The Set CPU Resources for VM action is used to modify the CPU reservation and CPU limit on virtual machines. You modify the CPU reservation and limit to manage workload demands in your environment.

#### How the Action Works

The Set CPU Resources for VM action determines how CPU resources can be allocated to the virtual machines. The reservation limit is the minimum amount of guaranteed CPU resources allocated to the virtual machine. The limit is the maximum amount of CPU resources that the virtual machine can consume.

To run the action, all options where you configure a value must contain a value for the objects that you want to change. If you are changing one option to a new value, but not another option, ensure that the option that you are not changing is configure with the current value.

#### Where You Run the Action

For the supported objects and object levels, this action is available in the following locations in VCF Operations:

- Embedded just below the top menu.

- In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Inventory Management** tile. Select an object in the list.
- In configured alert recommendations.
- In the Object List and Topology Graph dashboard widgets.

**Action Options**

To ensure that you are submitting the action for the right objects, review the following information.

| Option | Description |
|---|---|
| Selected objects | Check box indicates whether the action is applied to the object. To not run the action on one or more objects, deselect the associated check boxes. This option is available when two or more objects are selected.<br>If you modify a value, the check box is selected. The check box must be selected to activate the OK button. |
| Name | Name of the virtual machine as it appears in the environment inventory. |
| New Resv (MHz) | Amount of CPU resources in megahertz reserved for the virtual machine when the action is finished. The new reservation value must be less than or equal to the new limit value unless your new limit is unlimited (-1).<br>The reservation supports the following possible values:<br>• If you set the value to 0, the virtual machine is allocated only the configured CPU consumption level.<br>• If you add or removed reserved CPU consumption, supply a positive integer unless you set the value to 0. |
| Current Resv (MHz) | Amount of CPU resources that is configured as the guaranteed CPU resources for the virtual machine. |
| New Limit (MHz) | Maximum amount of CPU consumption in megahertz that the virtual machine can consume when the action is completed.<br>The limit supports the following possible values:<br>• If you set the value to 0, the maximum CPU consumption is not greater than the allocated reservation amount.<br>• If you set the value to -1, then the virtual machine CPU consumption is unlimited.<br>• If you add or remove CPU consumption limits, supply a positive integer, unless you set the value to 0 or -1. |
| Current Limit (MHz) | Maximum amount of CPU that the virtual machine can consume. |
| Host | Name of the host on which the virtual machine is running. |
| Adapter Instance | Name of the VMware Adapter as it is configured in VCF Operations. The adapter manages the communication with the vCenter instance. |

After you click **OK**, the next dialog box provides the task ID and a link to the task list.

**Table 1228: Task ID Dialog Box**

| Option | Description |
|---|---|
| Recent Tasks | To view the status of the job and verify that the job finished, click **Recent Tasks**. |
| OK | To close the dialog box without further action, click **OK**. |

### Set CPU Count and Memory for Virtual Machine Action

The Set CPU Count and Memory for VM action is used to add or remove CPUs and memory on virtual machines with only one power off of the virtual machines to perform the combined actions. You modify the CPU and memory to address performance problems or to reclaim resources.

#### How the Action Works

The Set CPU Count and Memory action powers off the target virtual machines. The action also creates a snapshot when requested and changes the number of vCPUs and memory based on the new CPU count and memory values you provided. As well, the action returns the virtual machines their original power states.

An alternative form of the Set CPU Count and Memory for Virtual Machine action is available for automation. This version of the action has the Power Off Allowed flag set to true so that the action is available for automation and can run when the virtual machine is in the powered on state. You can select the Power Off Allowed version of the action when you create or edit alerts and associate the alert with a recommendation. When the Power Off Allowed version of this action is automated, you do not select this version of the action.

If Hot Plug is activated on the virtual machines, then power off is not required. If power off is required and VMware Tools are installed, then the virtual machines are shut down before they are powered off.

To run the action, all options where you configure a value must contain a value for the objects that you want to change. If you are changing one option to a new value, but not another option, ensure that the option that you are not changing is configure with the current value.

#### Where You Run the Action

For the supported objects and object levels, this action is available in the following locations in VCF Operations:

- Embedded just below the top menu.
- In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Inventory Management** tile. Select an object in the list.
- In configured alert recommendations.
- In the Object List and Topology Graph dashboard widgets.

#### Action Options

Review the following information about the virtual machines to ensure that you are submitting the action for the correct objects.

| Option | Description |
|---|---|
| Selected objects | Check box indicates whether the action is applied to the object. To not run the action on one or more objects, deselect the associated check boxes. This option is available when two or more objects are selected.<br>If you modify a value, the check box is selected. The check box must be selected to activate the OK button. |
| Name | Name of the virtual machine as it appears in the environment inventory. |

| Option | Description |
|---|---|
| New CPU | Number of CPUs when the action is completed. If the value is less than 1 or a value not supported for the virtual machine in vCenter, and the virtual machine is powered on and Hot Add is not activated, the number of CPUs does not change and Recent Tasks shows the action as failed. If the virtual machine is powered off when you submit an unsupported value, the task reports success, but the virtual machine will fail when you run a power on action.<br>The value that appears is the calculated suggested size. If the target virtual machine is new or offline, this value is the current number of CPUs. If VCF Operations has been monitoring the virtual machine for six or more hours, depending on your environment, the value that appears is the CPU Recommended Size metric. |
| Current CPU | Number of configured CPUs. |
| Power State | Indicates whether the virtual machine is powered on or powered off. |
| Power Off Allowed | If selected, the action shuts down or powers off the virtual machine before modifying the value. If VMware Tools is installed and running, the virtual machine is shut down. If VMware Tools is not installed or not running, the virtual machine is powered off without regard for the state of the operating system.<br>In addition to whether the action shuts down or powers off a virtual machine, you must consider whether the object is powered on and what settings are applied.<br>See Working with Actions That Use Power Off section in *VCF Operations Configuration Guide*. . |
| Snapshot | Creates a snapshot before changing the number of CPUs. Use this option if you need a snapshot to which you can revert the virtual machine if the action does not produce the expected results.<br>The name of the snapshot is supplied in the Recent Tasks messages for the action.<br>If the CPU is changed with CPU Hot Plug activated, then the snapshot is taken with the virtual machine is running, which consumes more disk space. |
| Host | Name of the host on which the virtual machine is running. |
| Adapter Instance | Name of the VMware Adapter as it is configured in VCF Operations. The adapter manages the communication with the vCenter instance. |

After you click **OK**, the next dialog box provides the task ID and a link to the task list.

**Table 1229: Task ID Dialog Box**

| Option | Description |
|---|---|
| OK | To close the dialog box without further action, click **OK**. |
| Recent Tasks | To view the status of the job and verify that the job finished, click **Recent Tasks**. |

### Delete Unused Snapshots for Virtual Machine Action

The Delete Unused Snapshots for Virtual Machines action in VCF Operations deletes snapshots that are older than the specified age from your datastores. Deleting unused snapshots reclaims wasted space in your environment.

### How the Action Works

The Delete Unused Snapshots for Virtual Machine action comprises two dialog boxes. The first dialog box allows you to select the snapshot age criteria, which must be greater than one day. The second step allows you to select the snapshots to delete, and runs the Delete Unused Snapshots for Virtual Machine action.

The number of days that you specify for each virtual machine is the age of the snapshots based on the creation date. The Delete Unused Snapshots for Virtual Machine action retrieves the snapshot and displays the snapshot name, space consumed, and location so that you can evaluate the snapshots before you delete them.

When you click **Begin Action**, VCF Operations displays a dialog box to indicate that the action has started. To track the status of the action, click the link in the dialog box and view the state of the action in **Recent Tasks**.

### Where You Run the Action

For the supported objects and object levels, this action is available in the following locations in VCF Operations:

- Embedded just below the top menu.
- In configured alert recommendations.
- In the Object List and Topology Graph dashboard widgets.

### Action Options

To ensure that you are submitting the action for the right objects, review the following information.

You first retrieve snapshots based on age, then select the snapshots to delete.

**Table 1230: Retrieve Snapshots**

| Option | Description |
|---|---|
| Name | Name of the virtual machine on which you are running the Delete Unused Snapshots for VM action. |
| Days Old | Age of the snapshots to be deleted. This action retrieves snapshots for the virtual machine that are older than one day. |
| Host | Name of the host with which the virtual machine is associated. |
| Parent vCenter | Name of the VMware Adapter as it is configured in VCF Operations. The adapter manages the communication with the vCenter instance. |

Select the snapshots to delete.

**Table 1231: Delete Snapshots**

| Option | Description |
|---|---|
| Selected objects | Check box indicates whether the action is applied to the object. To not run the action on one or more objects, deselect the associated check boxes. This option is available when two or more objects are selected. |
| VM Name | Name of the virtual machine from which the snapshot was created. |
| Snapshot Name | Name of the snapshot in the datastore. |
| Snapshot Space (MB) | Number of megabytes consumed by the snapshot. |
| Snapshot Create Time | Date and time when the snapshot was created. |
| Snapshot Age | Age of the snapshot in days. |
| Datacenter Name | Name of the data center with which the datastore is associated. |
| Datastore Name | Name of the datastore where the snapshot is managed. |
| Host Name | Name of the host with which the datastore is associated. |

After you click **OK**, the next dialog box provides the task ID and a link to the task list.

**Table 1232: Task ID Dialog Box**

| Option | Description |
|---|---|
| Recent Tasks | To view the status of the job and verify that the job finished, click **Recent Tasks**. |
| OK | To close the dialog box without further action, click **OK**. |

The Delete Unused Snapshots action creates a job for the retrieve snapshots action, and a job for the delete snapshots action.

### Delete Unused Snapshots for Datastore Action

The Delete Unused Snapshots for Datastore action in VCF Operations deletes snapshots that are older than the specified age from your datastores. Deleting unused snapshots reclaims wasted space in your environment.

#### How the Action Works

The Delete Unused Snapshots for Datastore action comprises two dialog boxes. The first dialog box allows you to select the snapshot age criteria, which must be greater than one day. The second step allows you to select the snapshots to delete, and runs the Delete Unused Snapshots for Datastore action.

The number of days that you specify for each datastore is the age of the snapshots based on the creation date. The Delete Unused Snapshots dialog box provides details regarding snapshot name, space consumed, and location so that you can evaluate the snapshots before you delete them.

When you click **Begin Action**, VCF Operations displays a dialog box to indicate that the action has started. To track the status of the action, click the link in the dialog box and view the state of the action in **Recent Tasks**.

#### Where You Run the Action

For the supported objects and object levels, this action is available in the following locations in VCF Operations:

- Embedded just below the top menu.
- In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Inventory Management** tile. Select an object in the list.
- In configured alert recommendations.
- In the Object List and Topology Graph dashboard widgets.

#### Action Options

To ensure that you are submitting the action for the right objects, review the following information.

You first retrieve snapshots based on age, then select the snapshots to delete.

**Table 1233: Retrieve Snapshots**

| Option | Description |
|---|---|
| Name | Name of the datastore on which you are running the delete snapshot action. |
| Days Old | Age of the snapshots to be deleted. This action retrieves snapshots for the datastore that are older than one day. |
| Host | Name of the host with which the datastore is associated. |
| Parent vCenter | Name of the VMware Adapter as it is configured in VCF Operations. The adapter manages the communication with the vCenter instance. |

Select the snapshots to delete.

**Table 1234: Delete Snapshots**

| Option | Description |
|---|---|
| Selected objects | Check box indicates whether the action is applied to the object. To not run the action on one or more objects, deselect the associated check boxes. This option is available when two or more objects are selected. |
| Datastore Name | Name of the datastore where the snapshot is managed. |
| Snapshot Name | Name of the snapshot in the datastore. |
| Snapshot Space (MB) | Number of megabytes consumed by the snapshot. |
| Snapshot Create Time | Date and time when the snapshot was created. |
| Snapshot Age | Age of the snapshot in days. |
| Datacenter Name | Name of the data center with which the datastore is associated. |
| Host Name | Name of the host with which the datastore is associated. |
| VM Name | Name of the virtual machine from which the snapshot was created. |

After you click **OK**, the next dialog box provides the task ID and a link to the task list.

**Table 1235: Task ID Dialog Box**

| Option | Description |
|---|---|
| Recent Tasks | To view the status of the job and verify that the job finished, click **Recent Tasks**. |
| OK | To close the dialog box without further action, click **OK**. |

The Delete Unused Snapshots action creates a job for the retrieve snapshots action, and a job for the delete snapshots action.

**Power On, Power Off, and Reboot Actions**

You can run Power On, Power Off, and Reboot actions for AWS, Azure, and GCP using VCF Operations. These actions allow you to manage your objects and respond quickly to alerts. You can power on an instance to start collecting data, run applications, or verify the actions that were run on already powered-down machines. You can power off or reboot an instance to manage resources and reclaim wasted space.

- To run actions for AWS, ensure your account ID has the following privileges: **ec2:StartInstances**, **ec2:StopInstances** , and **ec2:RebootInstances**.
- To run actions for Azure, ensure you have one of the following privileges: **contributor**, **owner**, **virtual machine contributor**, and **avere contributor**.
- To run actions for GCP, ensure you have one of the following privileges: **owner**, **compute admin**, and **Compute Instance Admin (v1)**.

1. To perform actions on a single instance:
   a) In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Integrations**.
   b) In the **Accounts** tab, click the vertical ellipses against the AWS, Microsoft Azure, or GCP Cloud Account, and then select **Object Details**.

      The Object Browser page is displayed.

c) Locate the EC2 Instance, Azure Virtual Machine, or CE Instance and click **Actions**.

2. Optional: To perform actions on multiple instances at once:
   a) From the left menu click **Inventory**.
   b) Navigate to the AWS, Microsoft Azure, or GCP instance and filter by object type.
   c) Select multiple objects and click the **Actions** icon.
3. Select one of the following actions.

   The Actions vary based on the instance you are using.

   **Note:**
   To start an instance that is in a powered-off state, use the Power On action. The Power Off and Reboot actions appear only when the instance is in the Power On state.

| Accounts | Actions |
|---|---|
| AWS: EC2 Instance | • Power On Instance<br>• Power Off Instance<br>• Reboot Instance<br><br>**Note:**<br>These actions are available only if you activate actions while configuring an AWS account. For more information, see the "Add a Cloud Account for AWS" topic, in the *VCF Operations Configuration Guide*. |
| Microsoft Azure: Azure Virtual Machine | • Power On VM<br>• Power Off VM<br>• Reboot VM<br><br>**Note:**<br>These actions are available only if you activate actions while configuring a Microsoft Azure account. For more information, see the "Add a Cloud Account for Microsoft Azure" topic, in the *VCF Operations Configuration Guide*. |
| GCP: CE Instance | • Power On CE Instance<br>• Power Off CE Instance<br>• Reboot CE Instance<br><br>**Note:**<br>These actions are available only if you activate actions while configuring a GCP account. For more information, see the "Configuring Google Cloud Platform" topic, in the *VCF Operations Configuration Guide*. |

4. Click **Begin Action** and then, click **OK**.

Click **Recent Tasks** under **Administration** > **Control Panel** to verify the status of the action.

## Troubleshoot Actions in VCF Operations

If you are missing data or cannot run actions from VCF Operations, review the troubleshooting options.

Verify that your vCenter Adapter is configured to connect to the correct vCenter instance, and configured to run actions.

Verify that your vCenter Adapter is configured to connect to the correct vCenter instance, and configured to run actions.

**Actions Do Not Appear on Object**
An action might not appear on an object, such as a host or virtual machine, because VCF Automation is managing that object.

Actions such as Rebalance Container might not appear in the drop-down menu when you view the actions for your data center.

- If a data center is managed by VCF Automation, actions do not appear.
- If a data center is not managed by VCF Automation, you can act on the virtual machines that VCF Automation is not managing.

When VCF Automation manages the child objects of a data center or custom data center container, the actions that are normally available on those objects do not appear. They are not available because the action framework excludes actions on objects that VCF Automation manages. You cannot turn on or turn off the exclusion of actions on objects that VCF Automation manages. This behavior is normal.

If you removed the VCF Automation adapter instance, but did not select the **Remove related objects** check box, the actions are still disabled.

Make actions available on the objects in your data center or custom data center in one of two ways. Either confirm that VCF Automation is not managing the objects, or perform the steps in this procedure to remove the VCF Automation adapter instance.

1. To allow actions on an object, go to your VCF Automation instance.
2. Perform the action in VCF Automation, such as to move a virtual machine.

**Missing Column Data in Actions Dialog Boxes**
Data is missing for one or more objects in an Actions dialog box, making it difficult to determine if you want to run the action.

When you run an action one or more objects, some of the fields are empty.

There are two possible causes: 1) the VMware vSphere adapter has not collected the data from the vCenter instance that manages the object. 2) the current VCF Operations user does not have privileges to view the collected data for the object.

1. Verify that VCF Operations is configured to collect the data.
2. Verify that you have the privileges necessary to view the data.

**Missing Column Data in the Set Memory for VM Dialog Box**
The read-only data columns do not display the current values, which makes it difficult to specify properly a new memory value.

Current (MB) and Power State columns do not display the current values, which are collected for the managed object.

The adapter responsible for collecting data from the vCenter on which the target virtual machine is running has not run a collection cycle and collected the data. This omission can occur when you recently created an VMware adapter instance for the target vCenter and initiated an action. The VMware vSphere adapter has a five-minute collection cycle.

1. After you create a VMware adapter instance, wait an extra five minutes.
2. Rerun the **Set Memory for VM** action.

   The current memory value and the current power state appear in the dialog box.

**Host Name Does Not Appear in Action Dialog Box**
When you run an action on a virtual machine, the host name is blank in the action dialog box.

When you select virtual machine on which to run an action, and click the **Action** button, the dialog box appears, but the Host column is empty.

Although your user role is configured to run action on the virtual machines, you do not have a user roll that provides you with access to the host. You can see the virtual machines and run actions on them, but you cannot see the host data for the virtual machines. VCF Operations cannot retrieve data that you do not have permission to access.

You can run the action, but you cannot see the host name in the action dialog boxes.

## Monitor Recent Task Status

The Recent Task status includes all the tasks initiated from VCF Operations. You use the task status information to verify that your tasks finished successfully or to determine the current state of tasks.

You ran at least one action as part of an alert recommendation or from one of the toolbars. See Run Actions from Toolbars in VCF Operations.

You can monitor the status of tasks that are started when you run actions, and investigate whether a task finished successfully.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then select the **Recent Tasks** tile.
2. To determine if you have tasks that are not finished, click the **Status** column and sort the results.

| Option | Description |
|---|---|
| `In Progress` | Indicates running tasks. |
| `Completed` | Indicates finished tasks. |
| `Failed` | Indicates incomplete tasks on at least one object when started on multiple objects. |
| `Maximum Time Reached` | Indicates timed out tasks. |

3. To evaluate a task process, select the task in the list and review the information in the **Details of Task Selected** pane.

   The details appear in the Messages pane. If the information message includes `No action taken`, the task finished because the object was already in the requested state.
4. To view the messages for an object when the task included several objects, select the object in the Associated Objects list.

   To clear the object selection so that you can view all the messages, press the space bar.

Troubleshoot tasks with a status of `Maximum Time Reached` or `Failed` to determine why a task did not run successfully. See Troubleshoot Failed Tasks.

**Recent Tasks in VCF Operations**
The status of the tasks that were recently initiated from VCF Operationsappears in the Recent Task list. You can determine whether a task is finished, still in process, or failed.

### How Recent Tasks Work

The Recent Tasks page reports on logged task events, and the log entries appear in the messages area so that you can troubleshoot failed tasks.

### Where You View Recent Tasks

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then select the **Recent Tasks** tile.

### Recent Task Options

Review the information in the task list to determine if a task is completed or if you must troubleshoot a failed task. To see the details about a task, select the task in the list and review the associated objects and task messages.

**Table 1236: Task List**

| Option | Description |
|---|---|
| Export | Exports the selected task to an XML file.<br>The exported information, which includes the messages, is useful when you are troubleshooting a problem. |
| Edit Properties | Determines how long the recent task data is retained in your system.<br>Set the number of days that VCF Operations keeps the data, after which it is purged from the system. The default value is 90 days. |
| Status drop-down menu | Filters the list based on the status value. |

| Option | Description |
| --- | --- |
| All Filters | Filters the list based on the selected column and the provided values. You can filter the tasks based on the following criteria.<br>• Task<br>• Started Time<br>• Completed Time<br>• Automated<br>• Object Name<br>• Object Type<br>• Event Source<br>• Source Type<br>• Submitted By<br>• Task ID |
| Filter (Object Name) | Limits the tasks in the list to those that match the entered string. The search is based on a partial entry. For example, if you enter `vm`, objects such as `vm001` and `acctvm_east` are included. |
| Task | Name of the task.<br>For example, Set CPU Count for VM. |
| Status | State of the task.<br>Possible states include the following values:<br>• Completed. Task completed successfully on the target objects.<br>• In Progress. Task is running on the target objects.<br>• Failed. Task failed to run on the target objects. If the task started, the reasons for failure might include a faulty script, a script timed out, or actions are not taken. If the task did not start and immediately reports as failed, the reasons might include that the task was not able to start or the script was not found. If the task was not initiated on the target object, it might have failed because of communication or authentication errors.<br>• Maximum Time Reached. Task is running past the amount of time that is the default or configured value. To determine the status, you must troubleshoot the initiated action.<br>• Not Dispatched. The action adapter was not found.<br>• Started. Task is initiated on the object.<br>• Unknown. An error occurred while running the action, but the error was not captured in the task logs. To investigate this status further, check the VCF Operations support logs for the vCenter Adapter, available in the **Administration** area, and check the target system. |
| Started Time | Date and time when the task started. |
| Completed Time | Date and time when the task finished.<br>A completed date does not appear if the task failed or if the maximum timeout is reached. |
| Automated | Indicates whether the action in the task list is automated, indicated by `Yes` or `No`. |
| Object Name | Object on which the task was started. |
| Object Type | Type of object on which the task was started. |

| Option | Description |
|---|---|
| Event Source | The UUID or the name of the event that triggered the action automatically. when an event is triggered that is associated to the recommendation, it triggers the action without the user intervention.<br>For example, you can automate Alert recommendations that have an associated action. Automation is disabled by default. You configure automation in the **Override Alert / Symptom Definitions** area of a policy when you create or edit the policy in the **Policy Definition** tile.<br>An administrator who has the **Automation** role has permission to automate actions in the **Override Alert / Symptom Definitions** area of the policy workspace. |
| Source Type | Authentication source that the user who started the task used when accessing VCF Operations. |
| Submitted By | Name of the user who initiated the task. This column displays the automationAdmin user account for automated actions that are triggered by alerts. |
| Task ID | ID generated when the task, which included one or more actions, was started.<br>The task ID is unique for the task for each adapter. If a task includes tasks that ran using two adapters, you see two task IDs.<br>If the task is a delete snapshot action, two task IDs are generated. One ID is for the retrieve snapshots based on date task, and the other ID is for the delete selected snapshots task. |

The Associated Objects are the objects on which the selected task ran.

**Table 1237: Associated Objects for Selected Task Details**

| Option | Description |
|---|---|
| Object Name | Detailed list of objects that are included in the task selected in the task list.<br>If the task ran on only one object, the list includes one object.<br>If the task ran on multiple objects, each object is listed on a separate row. |
| Object Type | Type of object for each object name. |
| Status | Current state of the task. |
| The following are columns displayed for a Workload Optimization task | |
| Source Cluster | The source Cluster from which the VM is moved during Workload Optimization. |
| Destination Cluster | The destination Cluster to which the VM is moved during Workload Optimization. |
| Source Host | The source Host from which the VM is moved during Workload Optimization. |
| Destination Host | The destination Host to which the VM is moved during Workload Optimization. |
| Source Datastore | The source Datastore from which the VM is moved during Workload Optimization. |
| Destination Datastore | The destination Datastore to which the VM is moved during Workload Optimization. |
| Completion Date | The date when the task was completed. |

The Messages are the log of the task as it ran. If the task does not finish successfully, use the logs to identify problems.

**Table 1238: Messages for Selected Task Details**

| | |
|---|---|
| Severity drop-down menu | Limits the messages based on the Severity value. |
| Filter (Message) | Limits the message in the list to those that match the entered string.<br>The search is based on a partial entry. For example, if you enter `id`, then messages that contain `Task ID` and the phrase `did not complete` are included. |
| Severity | Message level in the logs.<br>The severity includes the following values:<br>• All. Displays all the messages.<br>• Error. Messages generated during a task failure.<br>• Warning. Messages generated as warning when the task is in progress.<br>• Information. Messages added to logs as the task is processed. |
| Time | Date and time the entry was added to the log. |
| Message | Text of the log entry.<br>Use the information in the message to determine why a task failed, and to begin to troubleshoot and resolve the failure.<br>The messages appear with the most recent entry at the top of the list if you do not sort the columns. |

## Troubleshoot Failed Tasks

If tasks fail to run in VCF Operations, review the Recent Tasks page and troubleshoot the task to determine why it failed.

This information is a general procedure for using the information in Recent Tasks to troubleshoot problems identified in the tasks.

**Determine If a Recent Task Failed**
The Recent Tasks provide the status of action tasks initiated from VCF Operations. If you do not see the expected results, review the tasks to determine if your task failed.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then select the **Recent Tasks** tile.
2. Select the failed task in the task list.
3. In the Messages list, locate the occurrences of `Script Return Result: Failure` and review the information between this value and `<-- Executing:[script name] on {object type}.`

   `Script Return Result` is the end of action run and `<-- Executing` indicates the beginning. The information provided includes the parameters that are passed, the target object, and unexpected exceptions that you can use to identify the problem.

**Troubleshooting Maximum Time Reached Task Status**
An action task has a `Maximum Time Reached` status and you do not know the status of the task.

The Recent Tasks list indicates that a task had a status of `Maximum Time Reached`.

The task is running past the amount of time that is the default or configured value. To determine the latest status, you must troubleshoot the initiated action.

The task is running past the amount of time that is the default or configured value for one of the following reasons:

- The action is exceptionally long running and did not finish before the threshold timeout was reached.
- The action adapter did not receive a response from the target system before reaching the timeout. The action might have completed successfully, but the completion status was not returned to VCF Operations.
- The action did not start correctly.

- The action adapter might have an error and be unable to report the status.

To determine whether the action completed successfully, check the state of the target object. If it did not complete, continue investigating to find the root cause.

**Troubleshooting Set CPU or Set Memory Failed Tasks**

An action task for Set CPU Count or Set Memory for VM has a `Failed` status in the recent task list because power off is not allowed.

The Recent Tasks list indicates that a Set CPU Count, Set Memory, or Set CPU and Memory task has a status of `Failed`. When you evaluate the Messages list for the selected task, you see this message.

```
Unable to perform action. Virtual Machine found
  powered on, power off not allowed.
```

When you increase the memory or CPU count, you see this message.

```
Virtual Machine found powered on, power off not allowed, if hot add is
  enabled the hotPlugLimit is exceeded.
```

You submitted the action to increase or decrease the CPU or memory value without selecting the **Allow Power Off** option. When you ran the action where a target object is powered on and where **Memory Hot Plug** is not activated for the target object in vCenter, the action fails.

1. Either activate **Memory Hot Plug** on your target virtual machines in vCenter or select **Allow Power Off** when you run the Set CPU Count, Set Memory, or Set CPU and Memory actions.
2. Check your hot plug limit in vCenter.

**Troubleshooting Set CPU Count or Set Memory with Powered Off Allowed**

A Set CPU Count, Set Memory, or a Set CPU Count and Set Memory action indicates that the action failed in Recent Tasks.

When you run an action that changes the CPU count, the memory, or both, the action fails. It fails even though Power Off Allowed was selected, the virtual machine is running, and the VMware Tools are installed and running.

The virtual machine must shut down the guest operating system before it powers off the virtual machine to make the requested changes. The shutdown process waits 120 seconds for a response from the target virtual machine, and fails without changing the virtual machine.

1. To determine if it has jobs running that are delaying the implementation of the action, check the target virtual machine in vCenter.
2. Retry the action from VCF Operations.

**Troubleshooting Set CPU Count and Memory When Values Not Supported**

If you run the Set CPU Count or Set Memory actions with an unsupported value on a virtual machine, the virtual machine might be left in an unusable state. That outcome requires you to resolve the problem in vCenter.

You cannot power on a virtual machine after you successfully run the Set CPU Count or Set Memory actions. When you review the messages in Recent Tasks for the failed Power On VM action, you see messages stating that the host does not support the new CPU count or new memory value.

Because of the way that vCenter validates changes in the CPU and memory values, you can use the VCF Operations actions to change the value to an unsupported amount. This change can happen when you run the action when the virtual machine is powered off.

If the object was powered on, the task fails, but rolls back any value changes and powers the machine back on. If the object was powered off, the task succeeds and the value is changed in vCenter. However, the target object is left in a state where you cannot power it on using either actions or the vCenter without manually changing the CPU or memory to a supported value.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then select the **Recent Tasks** tile.
2. In the task list, locate your failed Power On VM action, and review the messages associated with the task.
3. Look for a message that indicates why the task failed.
   For example, if you ran a Set CPU Count action on a powered off virtual machine to increase the CPU count from 2 to 4, but the host does not support 4 CPUs. The Set CPU tasks reported that it completed successfully in recent tasks. However, when you attempt to power on the virtual machine, the tasks fails. In this example, the message is `Virtual machine requires 4 CPUs to operate, but the host hardware only provides 2.`
4. Click the object name in the Recent Task list.
   The main pane updates to display the object details for the selected object.
5. Click the **Actions** menu on the toolbar and click **Open Virtual Machine in vSphere Client**.
   The vSphere Web Client opens with the virtual machine as the current object.
6. In the vSphere Web Client, click the **Manage** tab and click **VM Hardware**.
7. Click **Edit**.
8. In the Edit Settings dialog box, change the CPU count or memory to a supported value and click **OK**.
   You can now power on the virtual machine from the Web client or from VCF Operations.

**Troubleshooting Set CPU Resources or Set Memory Resources When the Value Is Not Supported**
If you run the Set CPU Resources action with an unsupported value on a virtual machine, the task fails and an error appears in the Recent Task messages.

The Recent Tasks list indicates that a Set CPU Resource or Set Memory Resource action has a state of `Failed`. When you evaluate the Messages list for the selected task, you see a message similar to the following examples.

```
RuntimeFault exception, message:[A specified parameter was not correct.
 spec.cpuAllocation.reservation]
```

```
RuntimeFault exception, message:[A specified parameter was not correct.
 spec.cpuAllocation.limits]
```

You submitted the action to increase or decrease the CPU or memory reservation or limit value with an unsupported value. For example, if you supplied a negative integer other than -1, which sets the value to unlimited, vCenter cannot make the change and the action failed.

Run the action with a supported value.

The supported values for reservation include 0 or a value greater than 0. The supported values for limit include -1, 0, or a value greater than 0.

**Troubleshooting Set CPU Resources or Set Memory Resources When the Value Is Too High**
You run the Set CPU Resources or Set Memory Resources action and the task fails with an error appearing in the Recent Tasks messages. The reason might be that you entered a value that is greater than the value that your vCenter instance supports.

The Recent Tasks list indicates that a Set CPU Resource or Set Memory Resource action has a state of `Failed`. When you evaluate the Messages list for the selected task, you see messages similar to the following examples.

If you are working with Set CPU Resources, the information message is similar to the following example, where 1000000000 is the supplied reservation value.

```
Reconfiguring the Virtual Machine Reservation to:[1000000000] Mhz
```

The error message for this action is similar to this example.

```
RuntimeFault exception, message:[A specified parameter was not correct: reservation]
```

If you are working with Set Memory Resources, the information message is similar to the following example, where 1000000000 is the supplied reservation value.

```
Reconfiguring the Virtual Machine Reservation to:[1000000000] (MB)
```

The error message for this action is similar to this example.

```
RuntimeFault exception, message:[A specified parameter was not correct.
 spec.memoryAllocation.reservation]
```

You submitted the action to change the CPU or memory reservation or limit value to a value greater than the value supported by vCenter, or the submitted reservation value is greater than the limit.

Run the action using a lower value.

**Troubleshooting Set Memory Resources When the Value Is Not Evenly Divisible by 1024**
If you run the Set Memory Resources action with a value that cannot convert from kilobytes to megabytes, the task fails and an error appears in the Recent Task messages.

The Recent Tasks list indicates that a Set Memory Resource action has a state of `Failed`. When you evaluate the Messages list for the selected task, you see a message similar to the following example.

```
Parameter validation;[newLimitKB] failed conversion to (MB, (KB)[2000] not evenly
 divisible by 1024.
```

Because vCenter manages memory reservations and limit values in megabytes, but VCF Operations calculates and reports on memory in kilobytes, you must provide a value in kilobytes that is directly convertible to megabytes. To do that, the value must be evenly divisible by 1024.

Run the action where the reservation and limit values are configured with supported values.

The supported values for reservation include 0 or a value greater than 0 that is evenly divisible by 1024. The supported values for a limit include -1, 0, or a value greater than 0 that is evenly divisible by 1024.

**Troubleshooting Failed Shut Down VM Action Status**
A shutdown VM action task has a `Failed` status in the Recent Task list.

The Shut Down VM action did not run successfully.

The Recent Tasks list indicates that a Shut Down VM action has a task status of `Failed`. When you evaluate the Messages list for the selected job, you see `Failure: Shut down confirmation timeout`.

The shutdown process involves shutting down the guest operating system and powering off the virtual machine. The wait time is 120 seconds to shut down the guest operating system. If the guest operating system does not shut down in this time, the action fails because the shutdown action is not confirmed.

To determine why the guest operating system did not shut down in the allotted time, check its status in vCenter.

**Troubleshooting VMware Tools Not Running for a Shutdown VM Action Status**
A Shutdown VM action task has a `Failed` status in the Recent Task list and the Message indicates that VMware Tools were required.

The Shutdown VM action did not run successfully.

The Recent Tasks list indicates that a Shutdown VM action has a tasks status of `Failed`. When you evaluate the Messages list for the selected job, you see `VMware Tools: Not running (Not installed)`.

The Shutdown VM action requires that VMware Tools is installed and running on the target virtual machines. If you ran the action on more than one object, then VMware Tools was not installed, or installed but not running, on at least one of the virtual machines.

In the vCenter instance that manages the virtual machine that failed to run the action, install and start VMware Tools on the affected virtual machines.

**Troubleshooting Failed Delete Unused Snapshots Action Status**
A Delete Unused Snapshots action task has a `Failed` status in the Recent Task list.

The Delete Unused Snapshots action did not run successfully.

The Recent Tasks list indicates that a Delete Unused Snapshots action has a task status of `Failed`. When you evaluate the Messages list for the selected job, you see this message.

```
Remove snapshot failed, response wait expired after:[120] seconds,
  unable to confirm removal.
```

The delete snapshot process involves waiting for access to datastores. The wait time is 600 seconds to access the datastore and delete the snapshot. If the delete request is not passed to the datastore in that time, the action does not finish the delete snapshot action.

1. To determine if the snapshot was deleted, check its status in vCenter .
2. If it was not, submit the delete snapshot request at a different time.

# Managing Inventory

You can configure, discover, and manage objects in your environment. You can also view a detailed list of the inventory in your VCF Operations environment where the objects are grouped logically.

## Enhanced Search Capability

The search function on the top supports locating named objects, dashboards, alerts, and so on, in the system. The search function attempts to match or partially match any string you enter; additional capabilities help you to go swiftly to the item you want.

### Where you Find Search

The search function appears on all the pages of VCF Operations in the top menu.

### How Search Works

You start your search by typing in the search bar. VCF Operations displays matching categories or objects.

The search function supports several common categories you can employ to find the item you seek quickly, as follows:

- Object
- Metric
- Feature
- Dashboard
- Report
- View
- Alert definition
- Symptom definition
- Recommendation
- Notification
- I.P.
- Super metric

What this means is that in addition to entering a traditional search phrase, for example, a simple string - "VM" - you can also enter one of the listed categories followed by a string or a name. You can then search for objects within the category. For the Object, View and Dashboard categories, the system displays the object in view mode.

If you want quickly to locate a specific dashboard, for example, start typing "dash..." into the search field. The system offers the search term Dashboards. Select the term using the cursor and then enter the dashboard name or part of the name and press Enter. The system finds the dashboard you want, with editing functions available.

Similarly, you can type "alert" or simply "a" in the search field and the system offers Alert Definition. Select the term and enter part of an alert message, for example, "unbalanced." The system returns the alert, "Cluster has an unbalanced workload," presented in the Alert Definition Workspace where you can edit it.

> **Note:**
> You can type virtual machine in the search bar to list all the virtual machines associated with the host.

**Feature Search**

You can now search for a feature in VCF Operations. Select **Feature** from the drop down list and type the feature name. You can type the feature name or simply part of the feature name. For example, you can type 'licenses' or simply 'li' in the search field. In addition to the feature name, the navigation path to the feature is also displayed.

> **Note:**
>
> Feature names are available for search based on license and user permissions.

**Extend User Search for Alert Assignment**

In VCF Operations now you can search for a user and assign alerts to that user using the following filter options.

- User name
- First Name
- Last Name
- Email Address

VCF Operations displays the details of the searched user in search results, you can use the details to assign the alert to specific user.

**Metric Search**

You can create a search query using a metric, property, object type name, super metric, or instance metric name. Enter the metric name or select the metric, property, or object type from the list of suggestions. The suggestions also display a list of recent searches that match the text you enter in the search bar. You can also delete recent searches. Create a simple query using the metric name and the object type or add conditions and operators to your query to build a more complex query depending on your requirement. Conditions and operators allow you to narrow down the search results. Once you have created the query, click enter to view the search results. The search results are based on the values you select. For more information on how to create queries, see Searching for Metrics, Properties, or Object Types Using Queries.

## Searching for Metrics, Properties, or Object Types Using Queries

The search queries will run across your entire deployment and find all types of objects based on the specified search terms used in a search query. Additionally, VCF Operations provides suggestions to build the search queries, which include recent searches that match the typed text in the global search bar. You can also find specific types of objects using conditions in the search query.

You can create a simple search query starting with a metric, property, or object type, or you can build a more complex query using different conditions to find all types of entities in VCF Operations. Metric queries can be divided into two parts, the metric search query and the metric search result. After you create a metric search query, click **Enter** to view the Metric Search Results page.

You can also use instance metrics and super metrics to search for objects.

1. In the VCF Operations navigation bar at the top, click the **Search** bar and then select **Metric**.

   > **Note:**
   > When you click the cursor in the search bar, a list of suggestions with the metric and property names and the object types appear. You can select a metric, property, or object type from this list. If you hover over the metric and property names you can view the name of the object types that the metric or property belongs to. If you enter an entity in the search bar, the suggestions change and display options based on your entry. You can start with a metric name and then select the object type it belongs to, or start with an object type and add conditions.

2. Create a simple query with a metric or property name. For example, enter `CPU|Usage %` in the search bar.

   If you start a metric search query with a metric name, you must always mention the object type it belongs to.

   > **Note:**
   > If the selected metric belongs to a single object type, then the object type will be selected by default.

3. The object types change based on the metric you enter. Select `Virtual Machine` as the object type.

   A simple metric search query is created as follows: `CPU|Usage % of Virtual Machine`. A green banner with the `Press enter to see the results or continue expanding the query.` message appears.

4. Optional: Click **Enter** to view the results of this simple search query or continue to expand your search query to narrow down the search results.

   **Note:**
   If your query is incomplete, has a syntax error, exceeds the character limit of 300, or has more than five different metric names, the search query will not work. In such cases a red banner appears explaining the issue. For example, in case of a syntax error, the `The query contains syntax error(s). Please modify it to get suggestions.` message appears.

5. The metric search query allows you to search for objects based on the conditions you use. Expand the search query with the **where** and **childOf** conditions.

   a) Add a **where** condition to filter the search results. You can combine the where conditions with logical operators to form complex metric search queries.

   **Note:**
   For string metrics the condition values are case insensitive.

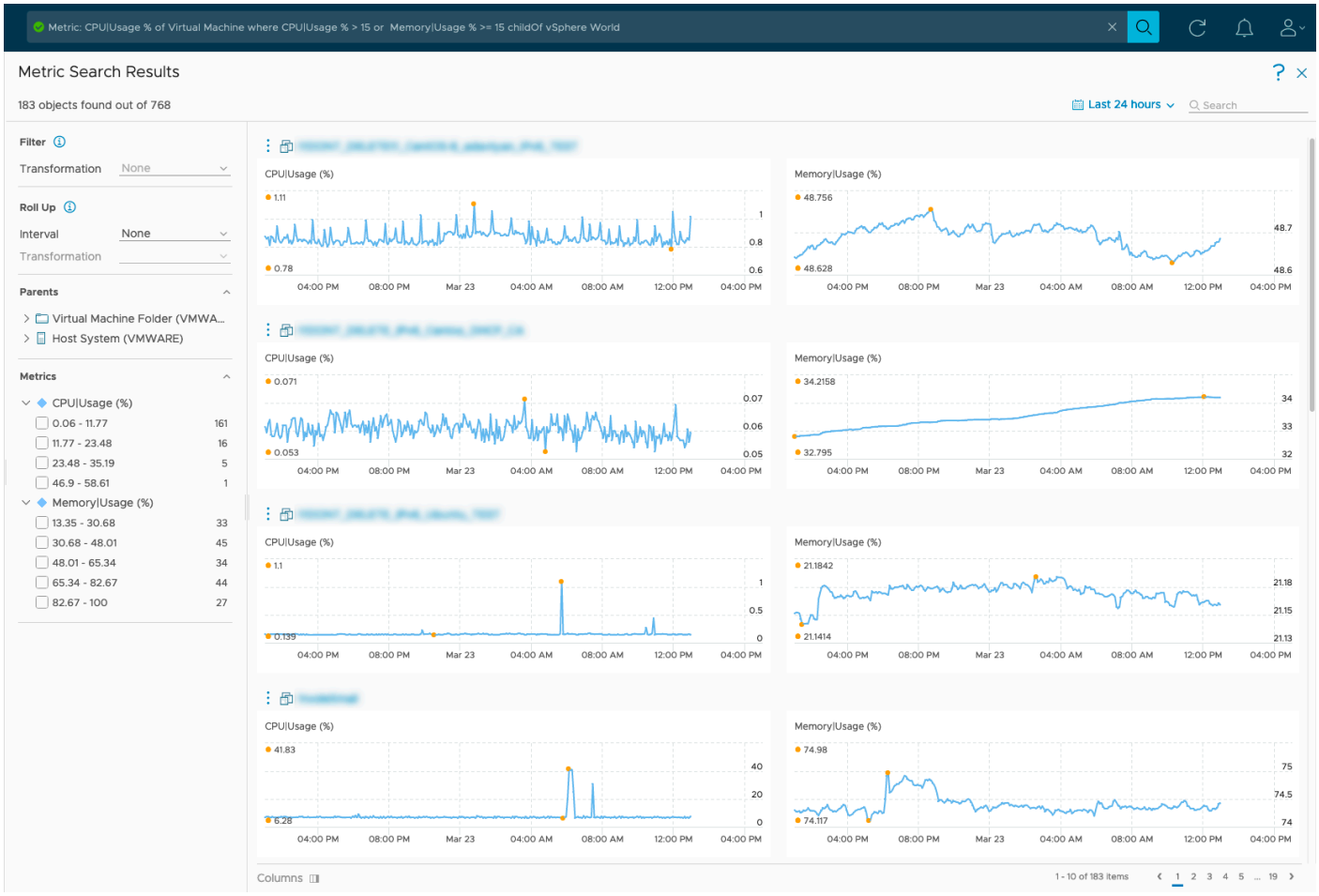
| Operator | Examples |
|---|---|
| **Numeric Operators** | |
| > | `CPU\|Usage % of Virtual Machine where CPU\|Usage % > 15%` |
| >= | `CPU\|Usage % of Virtual Machine where CPU\|Usage % >= 15` |
| < | `CPU\|Usage % of Virtual Machine where CPU\|Usage % < 15` |
| <= | `CPU\|Usage % of Virtual Machine where CPU\|Usage % <= 15%` |
| = | `CPU\|Usage % of Virtual Machine where CPU\|Usage % = 15` |
| != | `CPU\|Usage % of Virtual Machine where CPU\|Usage % != 15` |
| + | `CPU\|Usage % of Virtual Machine where (CPU\|Usage % + Memory\|Usage %) > 15` |
| - | `CPU\|Usage % of Virtual Machine where (CPU\|Usage % - Memory\|Usage %) > 15` |
| * | `CPU\|Usage % of Virtual Machine where (CPU\|Usage % * Memory\|Usage %) > 15` |
| / | `CPU\|Usage % of Virtual Machine where (CPU\|Usage % + Memory\|Usage %)/2 > 15` |
| and | `CPU\|Usage % of Virtual Machine where CPU\|Usage % > 15 and Memory\|Usage % > 15` |
| or | `CPU\|Usage % of Virtual Machine where CPU\|Usage % > 15 or Memory\|Usage % > 15` |
| **String Operators** | |

| Operator | Examples |
| --- | --- |
| equals | `CPU\|Demand MHz of Virtual Machine where Configuration\|Name equals 'Centos'` |
| not equals | `CPU\|Demand MHz of Virtual Machine where Configuration\|Name notEquals 'Centos'` |
| contains | `CPU\|Demand MHz of Virtual Machine where Configuration\|Name contains 'Centos'` |
| notContains | `CPU\|Demand MHz of Virtual Machine where Configuration\|Name notContains 'Centos'` |
| startsWith | `CPU\|Demand MHz of Virtual Machine where Configuration\|Name startsWith 'Centos'` |
| notStartsWith | `CPU\|Demand MHz of Virtual Machine where Configuration\|Name notStartsWith 'Centos'` |

b) Use the **childOf** condition at the end of your query to select an ancestor of the object type to further narrow down your search results. Select the object name, for example, select `vSphere World`.

A complex search query is created as follows: `CPU|Usage % of Virtual Machine where CPU|Usage % > 15 or Memory|Usage % > 15 childOf vSphere World`

6. Click **Enter** to view the metric search results.



The search result is displayed on the Metric Search Results page.

7. The metric search results page displays the total number of objects that match the query. You can use the filter options on this page to further drill down the results and find the required information.

| Option | Description |
|---|---|
| Filter: Transformation | The Transformation filter applies to the actual query and not the results as it applies to objects before the search query runs and cannot be used on the results.<br>Determines what calculation method is applied to the raw data. You can convert the metric values of the search results using the transformation option and also select the type of transformation:<br>• Minimum. The minimum value of the metric over the selected time range.<br>• Maximum. The maximum value of the metric over the selected time range.<br>• Average. The mean of all the metric values over the selected time range.<br>• Sum. The sum of the metric values over the selected time range.<br>• First. The first metric value for the selected time range.<br>• Last. The last value of a metric within the selected time range.<br>• Current. The last available value of a metric if it was last updated not before five collection cycles were complete, otherwise, it is null.<br>**Note:**<br>This option is available only if there is a where condition and the number of objects for the mentioned object type do not exceed the limit of 200 for not more than one month. |
| Roll Up | The roll up filter is used to change data visualization based on the selected time interval and transformation. You can select one of the available options ranging from hour, day, week, month, quarter, or year. For example, if you select Hour as the interval and the Maximum as the transformation, then the system displays maximum values for each one-hour interval. |
| Parents | The parent objects grouped by type for the found objects that the metrics in the query belong to. You can use this option to further narrow down the results. |
| Metric | Displays the metrics value distribution. The metric values are divided into 5 ranges and the number of objects which have the metric lying in that range is available. |
| Time Settings | Use the time settings to select the time range for the query.<br>The number of objects based on metrics in the selected time range is also displayed. You can select a time range to view the information. You can set a time range for a past period or set a future date for the end of the time period. |
| Page Navigation | The page navigation option appears when the results resource count exceeds 10. Use the page navigation to continue viewing search results displayed on the next pages. |
| Columns | If the query contains more than two metrics, you can view the search results of two metrics at a time. Use the columns icon to select which two metrics search results are displayed. |

To further troubleshoot objects, click the vertical ellipsis and select troubleshoot to go to the Workbench page. For more information, see Starting the Troubleshooting Workbench.

# Configuring Objects

Using the power of object management - including metrics and alerts - you can monitor objects, applications, and systems that must stay up and running. Some metrics and alerts are prepackaged into dashboards and policies; others you combine into custom tools

VCF Operations discovers objects in your environment and makes them available to you. With the information that VCF Operations provides, you can quickly access and configure any object. For example, you can determine if a datastore is connected or providing data, or you can power on a virtual machine.

## Object Discovery

Its ability to monitor and collect data on objects in your systems environment makes VCF Operations a critical tool in maintaining system uptime and ensuring ongoing good health for all system resources from virtual machines to applications to storage - across physical, virtual, and cloud infrastructures.

Following are examples of objects that can be monitored.

- vCenter
- Virtual machines
- Servers/hosts
- Compute resources
- Resource pools
- Data centers
- Storage components
- Switches
- Port groups
- Datastores

### Adapters – Key to Object Discovery

VCF Operations collects data and metrics from objects using adapters, that are the central components of management packs. You can customize adapter instances for your virtual environment using cloud accounts and other accounts. VCF Operations uses cloud accounts to manage the communication and integration with other products, applications, and functions.

- Cloud Accounts - You can configure cloud adapter instances and collect data from cloud solutions that are already installed in your cloud environment from the cloud accounts page.
- Other Accounts - You can view and configure native management packs and other solutions that are already installed and configure adapter instances from the other accounts page.
- Repository - You can activate or deactivate native management packs and add or upgrade other management packs from the Repository page.

The screenshot displays the list of available solutions in VCF Operations . You must first Activate the solution before adding and configuring the accounts.

For complete information on configuring management packs and adapters, see Integrating Data Sources with VCF Operations

When you create a new adapter instance, it begins discovering and collecting data from the objects designated by the adapter, and notes the relationships between them. Now you can begin to manage your objects.

### Workload Management Inventory Objects

VCF Operations discovers the following workload management objects and their child objects using the vCenter adapter:

- vSphere Kubernetes

- vSphere Pods
- VSphere Namespace

Clusters activated with vSphere IaaS control plane are called Supervisor Clusters. Kubernetes control plane is created inside the hypervisor layer. This layer contains specific objects that enable the capability to run Kubernetes workloads within ESX. After a Supervisor is created, vSphere Namespaces can be created within the Supervisor. A vSphere Namespace sets the resource boundaries where vSphere Pods, VMs, and VKS clusters can run.

> **Note:**
> If a Supervisor Cluster spans multiple Compute Clusters, a Namespaces resource will be created for each Compute Cluster Resource and related to the Supervisor Cluster.

To understand the vSphere Tanzu Kubernetes architecture, see the *VMware vSphere with Tanzu* product documentation.

Workload management objects are excluded from the following workflows:

- Compliance
- Reclaim
- Rightsizing
- Workload optimization

## About Objects

Objects are the structural components of your mission-critical IT applications: virtual machines, datastores, virtual switches and port groups are examples of objects.

Because downtime equals cost - in unused resources and lost business opportunities - it's crucial that you successfully identify, monitor and track objects in your environment. The goal is to proactively isolate, troubleshoot and correct problems even before users are aware that anything is wrong.

When a user actually reports an issue, the solution should be quick and comprehensive.

For a complete list of objects that can be defined in VCF Operations refer to Object Discovery.

VCF Operations gives you visibility into objects including applications, storage and networks across physical, virtual and cloud infrastructures through a single interface that relates performance information to positive or negative events in the environment.

### Managing Objects

When you monitor a large infrastructure, the number of objects and corresponding metrics in VCF Operations grows rapidly, especially as you add solutions that extend dynamic monitoring and alerts to more parts of your infrastructure. VCF Operations gives you ample tools to stay abreast of events and issues.

### Adding Objects and Configuring Object Relationships

VCF Operations automatically discovers objects and their relationships once you create an adapter instance. You have the added ability to manually add any objects that you want monitored and to configure object relationships using abstract concepts rather than the connections recorded by VCF Operations. Where VCF Operations might discover the cla ssic parent-child relationships between objects, you can create relationships between objects that might not normally be related. For example, you could configure all the datastores supporting a company department to be related.

When objects are related, a problem with one object appears as an anomaly on related objects. So object relationships can help you to identify problems in your environment quickly. The object relationships that you create are called custom groups.

### Custom Groups

To create an automated management system you need some way to organize objects so that you can quickly gain insights. You can achieve a high level of automation using custom groups. You have multiple options for tailoring group attributes to support your monitoring strategy.

For example, you can designate a group either to be static or to be updated automatically with membership criteria that you designate. Consider a non-static group of all virtual machines that are powered on and have OS type Linux. When you power on a new Linux VM, it is automatically added to the group and the policy is applied.

For additional flexibility, you can also specify individual objects to be always included or excluded from a given custom group. Or you can have a different set of alerts and capacity calculations for your production environment versus your testing environments.

**Managing Applications**

VCF Operations allows you to create containers or objects that can contain a group of virtual machines or other objects in different structural tiers. This new application can then be managed as a single object, and have health badges and alarms aggregated from the child objects of the group.

For example, the system administrator of an online training system might request that you monitor components in the Web, application and database tiers of the training environment. You build an application that groups related training objects together in each tier. If a problem occurs with one of the objects, it is highlighted in the application display and you can investigate the source of the problem

**The Power of Object Management**

Using the power of object management, including metrics and alerts - some prepackaged into dashboards and policies, others that you combine into custom monitoring tools - you'll keep a close watch on the objects, applications and systems that must stay up and running.

## Managing Objects in Your Environment

An object is the individual managed item in your environment for which VCF Operations collects data, such as a router, switch, database, virtual machine, host, and vCenter instances.

The system requires specific information about each object. When you configure an adapter instance, VCF Operations performs object discovery to start collecting data from the objects with which the adapter communicates.

An object can be a single entity, such as a database, or a container that holds other objects. For example, if you have multiple Web servers, you can define a single object for each Web server and define a separate container object to hold all of the Web server objects. Groups and applications are types of containers.

Categorize your objects using tags, so that you can easily find, group, or filter them later. A tag type can have multiple tag values. You or VCF Operations assigns objects to tag values. When you select a tag value, VCF Operations displays the objects associated with that tag. For example, if a tag type is Lifecycle and tag values are Development, Test, Pre-production, and Production, you might assign virtual machine objects VM1, VM2, or VM3 in your environment to one or more of these tag values, depending on the virtual machine function.

### Adding an Object to Your Environment

You might want to add an object by providing its information to VCF Operations. For example, some solutions cannot discover all the objects that might be monitored. For these solutions, you must either use manual discovery or manually add the object.

Verify that an adapter is present for the object you plan to add. See Integrating Data Sources with VCF Operations.

Verify that an adapter is present for the object you plan to add. See the Getting Started with *VCF Operations* .

> **Note:**
> Objects added to VCF Operations via API will require an OSI license per object.

When you add an individual object, you provide specific information about it, including the kind of adapter to use to make the connection and the connection method. For example, a vSAN adapter does not know the location of the vSAN devices that you want to monitor.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Inventory Management** tile.
2. From the **Objects** tab, on the toolbar, click the plus sign.
3. Use the topic menus to reveal all fields and provide the required information.

| Option | Description |
|---|---|
| **Display name** | Enter a name for the object. For example, enter `vSAN-Host1`. |
| **Description** | Enter any description. For example, enter `vSAN-Host monitored with vSAN adapter` |
| **Adapter type** | Select an adapter type. For example, select **vSAN Adapter**. |
| **Adapter instance** | Select an adapter instance. |
| **Object type** | Select an object type. For a vSAN adapter, you might select vSAN-Host. When you select the object type, the dialog box selections change to include information you provide so that VCF Operations connect with the selected object type. |
| **Host IP address** | Enter the host IP. For example, enter the IP address of vSAN-Host1. |
| **Port number** | Accept the default port number or enter a new value. |
| **Credential** | Select the Credential, or click the plus sign to add new login credentials for the object. |
| **Collection interval** | Enter the collection interval, in minutes. For example, if you expect the host to generate performance data every 5 minutes, set the collection interval to 5 minutes. |
| **Dynamic Thresholding.** | Accept the default, Yes. |

4. Click **OK** to add the object.

vSAN-Host1 appears in the Inventory as a host object type for the vSAN adapter type.
When you add an individual object, VCF Operations does not begin collecting metrics for the object until you turn on data collection. See Inventory : List of Objects.

For each new object, VCF Operations assigns tag values for its collector and its object type. Sometimes, you might want to assign other tags. See Creating and Assigning Tags.

For each new object, VCF Operations assigns tag values for its collector and its object type. Sometimes, you might want to assign other tags.

**Configuring Object Relationships**
VCF Operations shows the relationship between objects in your environment. Most relationships are automatically formed when the objects are discovered by an installed adapter. In addition, you can use VCF Operations to create relationships between objects that might not normally be related.

Objects are related physically, logically, or structurally.

- Physical relationships represent how objects connect in the physical world. For example, virtual machines running on a host are physically related.
- Logical relationships represent business silos. For example, all the storage objects in an environment are related to one another.
- Structural relationships represent a business value. For example, all the virtual machines that support a database are structurally related.

Solutions use adapters to monitor the objects in your environment so that physical relationship changes are reflected in VCF Operations. To maintain logical or structural relationships, you can use VCF Operations to define the object relationships. When objects are related, a problem with one object appears as an influence on related objects. So object relationships can help you to identify problems in your environment quickly.

Apart from the parent-child relationship, you can also define new relationships in VCF Operations. The relationship between objects in your environment can be one-to-many, many-to-one, or one-one, the relationship can be defined in horizontal , vertical, or diagonal levels.

### Creating and Assigning Tags

A large enterprise can have thousands of objects defined in VCF Operations. Creating object tags and tag values makes it easier to find objects and metrics. With object tags, you select the tag value assigned to an object and view the list of objects that are associated with that tag value.

A tag is a type of information, for example, Adapter Types. Adapter Types is a predefined tag. Tag values are individual instances of that type of information. For example, when the system discovers objects using the vCenter Adapter, it assigns all the objects to the vCenter Adapter tag value under the Adapter Types tag.

You can assign any number of objects to each tag value, and you can assign a single object to tag values under any number of tags. You typically look for an object by looking under its adapter type, its object type, and possibly other tags.

If an object tag is locked, you cannot add objects to it. VCF Operations maintains locked object tags.

#### Predefined Object Tags

VCF Operations includes several predefined object tags. It creates values for most of these tags and assigns objects to the values.

For example, when you add an object, the system assigns it to the tag value for the collector it uses and the kind of object that it is. VCF Operations creates tag values if they do not already exist.

If a predefined tag has no values, there is no object of that tag type. For example, if no applications are defined, the applications tag has no tag values.

Each tag value appears with the number of objects that have that tag. Tag values that have no objects appear with the value zero. You cannot delete the predefined tags or tag values.

**Table 1239: Predefined Tags**

| Tag | Description |
|---|---|
| Collectors (Full Set) | Each defined collector is a tag value. Each object is assigned to the tag value for the collector that it uses when you add the object to VCF Operations. The default collector is VCF Operations Collector. |
| Applications (Full Set) | Each defined application is a tag value. When you add a tier to an application, or an object to a tier in an application, the tier is assigned to that tag value. |
| Maintenance Schedules (Full Set) | Each defined maintenance schedule is a tag value, and objects are assigned to the value when you give them a schedule by adding or editing them. |
| Adapter Types | Each adapter type is a tag value, and each object that uses that adapter type is given the tag value. |
| Adapter Instances | Each adapter instance is a tag value, and each object is assigned the tag value for the adapter instance or instances through which its metrics are collected. |
| Object Types | Each type of object is a tag value, and each object is assigned to the tag value for its type when you add the object. |
| Recently Added Objects | The last day, seven days, 10 days, and 30 days have tag values. Objects have this tag value as long as the tag value applies to them. |
| Object Statuses | Tag value assigned to objects that are not receiving data. |
| Collection States | Tag value assigned to indicate the object collection state, such as collecting or not collecting. |

| Tag | Description |
|---|---|
| Health Ranges | Good (green), Warning (yellow), Immediate (orange), Critical (red), and Unknown (blue) health statuses have tag values. Each object is assigned the value for its current health status. |
| Entire Enterprise | The only tag value is Entire Enterprise Applications. This tag value is assigned to each application. |
| Licensing | Tag values are License Groups found under **Subscriptions** > **Legacy Licenses** > **License Groups**. Objects are assigned to the license groups during VCF Operations installation. |
| Untag | Drag an object to this tag to delete the tag assignment. |

### Add an Object Tag and Assign Objects to the Tag

An object tag is a type of information, and a tag value is an individual instance of that type of information. If the predefined object tags do not meet your needs, you can create your own object tags to categorize and manage objects in your environment. For example, you can add a tag for cloud objects and add tag values for different cloud names. Then you can assign objects to the cloud name.

Become familiar with the predefined object tags.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Inventory Management** tile.
2. Click the **Manage Tags** icon above the list of tags.
3. Click the **Add New Tag** icon to add a new row and type the name of the tag in the row.
   For example, type `Cloud Objects` and click **Update**.
4. With the new tag selected, click the **Add New Tag Value** icon to add a new row and type the name of the value in the row.
   For example, type `Video Cloud` and click **Update**.
5. Click **OK** to add the tag.
6. Click the tag to which you want to add objects to display the list of object tag values.
   For example, click **Cloud Objects** to display the Video Cloud object tag value.
7. Drag objects from the list in the right pane of the Inventory onto the tag value name.
   You can press Ctrl+click to select multiple individual objects or Shift+click to select a range of objects.
   For example, if you want to assign data centers that are connected through the vCenter Adapter, type `vCenter` in the search filter and select the data center objects to add.

### Use a Tag to Find an Object

The quickest way to find an object in VCF Operations is to use tags. Using tags is more efficient than searching through the entire object list.

Tag values that can also be tags are Applications and Object Types. For example, the Object Types tag has values for each object that is in VCF Operations, such as Virtual Machine, which includes all the virtual machine objects in your environment. Each of these virtual machines is also a tag value for the Virtual Machine tag. You can expand the tag value list to select the value for which you want to see objects.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Inventory Management** tile.
2. In the tag list in the center pane, click a tag for an object with an assigned value.
   When you click a tag, the list of values expands under the tag. The number of objects that is associated with each value appears next to the tag value.
   A plus sign next to a tag value indicates that the value is also a tag and that it contains other tag values. You can click the plus sign to see the subvalues.

3. Select the tag value.

   The objects that have that tag value appear in the pane on the right. If you select multiple tag values, the objects in the list depend on the values that you select.

| Tag Value Selection | Objects Displayed |
|---|---|
| More than one value for the same tag | The list includes objects that have either value. For example, if you select two values of the Object Types tag, such as Data Center and Host System, the list shows objects that have either value. |
| Values for two or more different tags | The list includes only objects that have all of the selected values. For example, if you select two values of the Object Types tag, such as Data Center and Host System, and you also select an adapter instance such as vC-1 of the vCenter Adapter instance tag, only Data Center or Host System objects associated with vC-1 appear in the list. Data Center or Host System objects associated with other adapter instances do not appear in the list, nor do objects that are not Data Center or Host System objects. |

4. Select the object from the list.

**Manage Object Tags Workspace**

A large enterprise can have thousands of objects. When objects are assigned to a tag, and you choose to display objects with that tag value, the objects are easier to find on the Inventory list.

**Where You Find Manage Object Tags**

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Inventory Management** tile.

Click the **Manage Tags** icon above the list of tags in the middle pane.

**Manage Object Tags Options**

The Manage Object Tags screen appears with previously created tags listed. In the left pane, you add tags. In the right pane, you add tag values.

- Click **Add a New Tag** and type a new tag name, or select a tag to delete.
- For the selected tag, click **Add a New Tag Value** and type a new tag value name, or select a tag value to delete.
- For the GEO Location tag, you can add and delete a new tag value in the left pane.

**Manage Object Type Tags Workspace**

Every object in your environment is of a particular object type. You use Manage Object Type Tags to control the object type tags displayed.

**How Manage Object Type Tags Works**

For every adapter instance installed, VCF Operations discovers objects in your environment and starts collecting data from those objects.

**Where You Find Manage Object Type Tags**

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Inventory Management** tile. Click the **Manage Object Type Tags** icon above the list of tags.

**Manage Object Type Tags Options**

Depending on the number of adapters installed , there may be hundreds of object type tags. The Manage Object Type Tags options allow you to turn on or off the tags listed.

- Type a filter word to show the object type tags with the word.
- Name lists all the object type tags.
- To toggle the display of an object type tag, select the check box in the Show Tag column of its row.

**Inventory : List of Objects**
VCF Operations discovers objects in your environment for each adapter instance and lists them. From the complete list of all the objects in your environment, you can quickly access and configure any object. For example, you can check if a datastore is connected or providing data, or you can power on a virtual machine.

## How the List Works

Objects appear in a data grid. To find a particular object, you can sort a column in the grid or search for a filter word. In addition to sorting and searching, assigning objects to object tags makes it easier to find objects and metrics.

## Where You Find the List

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Inventory Management** tile. The system lists all the objects in your environment.

## Inventory List Options

The center pane includes object tag options. The **Object** tab in the right pane includes toolbar options for all of the objects in your environment.

**Table 1240: Object Tag Toolbar Options from the Center Pane**

| Option | Description |
|---|---|
| Collapse all | Closes all the tag group selections. |
| Deselect All | Tags remain selected until deselected. Use this option to deselect all tags. |
| Manage Tags | Add a tag or tag value. See Manage Object Tags Workspace. |
| Manage Object Type Tags | There might be many object type tags. Use this option to choose the object type tags to display. See Manage Object Type Tags Workspace. |

**Note:**
If there are more than ten objects under an object tag in the centre pane, you can search for a descendant object using the Search option. If there are more than five hundred objects in a section, use the **View More** button under the last object that is displayed, to view the rest of the objects. Use the **Type to see more** button to find objects if there are more than one thousand objects.

Use the toolbar options to manage objects from the **Objects** Tab.

- Filter options limit the list to objects matching the filter. Filter options include ID, Name, Description, Maintenance Schedule, Adapter Type, Object Type, and Identifiers.
- Select the object to manage from the list. If an object tag is selected, only objects of the selected tag value are listed. Column headings help you to identify the object. See Object List Widget.

**Table 1241: Inventory Toolbar Options from the Object Tab**

| Option | Description |
|---|---|
| Action | Perform an action on the selected object. Available actions depend on the object type. For example, Power on VM applies to the selected virtual machine. See Actions in VCF Operations. |
| Open in external application | If an adapter includes the ability to link to another application for information about the object, click the button to access a link to the application. For example, Open Virtual Machine in a vSphere Client or Search for VM logs in Log management. |

| Option | Description |
|---|---|
| Edit object | Edit the selected object. For example, add or change the maintenance schedule for a virtual machine. If multiple objects of the same type are selected, common identifiers for the object type are editable. For example, change the VM entity name of multiple datastores with a single edit. See Manage Objects Workspace. |
| Add object | VCF Operations discovers objects for most adapters. For adapters that do not support autodiscovery for all objects, the objects are manually added. See Manage Objects Workspace. |
| Discover Objects | Perform an IP scan to discover objects associated with a particular adapter. See Discover Objects Workspace. |
| Delete object | Remove the object from the list. |
| Start maintenance | Take the object offline for maintenance. See Manage Maintenance Schedules for Your Object Workspace. |
| End maintenance | Terminate the maintenance period and put the selected object back online. |
| Clear Selections | Clear all object selections. |
| Select All | Select all objects displayed. |
| Show Detail | Display the **Summary** tab of the selected object. |
| Per page | The number of objects to list per page. |

Manage Objects Workspace

To collect data from an object, you might need to add an object or edit an existing object in your environment. For example, you might need to add objects for an adapter that does not support autodiscovery, or change the maintenance schedule of an existing object.

### Where You Find Manage Objects

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Inventory Management** tile. Click the plus sign to add an object or the edit icon to edit the selected object.

Items that appear in the window depend on the object that you are editing. Not all options can be changed.

**Table 1242: Manage Objects Add or Edit Options**

| Options | Description |
|---|---|
| Display name | Name of the object. Use only letters and numbers. Do not use nonalphanumeric characters or spaces. |
| Description | (Optional) For informational purposes only. |
| Adapter Type | If you are editing an object, you cannot change the adapter type. |
| Adapter Instance | If you are editing an object, you cannot change the adapter instance. |
| Object Type | If you are editing an object, you cannot change the object type. More configuration options might appear, depending on the object type. |
| Dynamic Thresholding | On by default, to activate dynamic thresholding and early warning smart alerts. See VCF Operations Dynamic Thresholds |

Discover Objects Workspace

If VCF Operations does not discover objects after an adapter instance is configured, use manual discovery. Discovering objects is more efficient than adding objects individually.

> **Note:**
> You use discovery to define objects for embedded adapters. VCF Operations discovers objects that use external adapters.

### Where You Find Discover Objects

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Inventory Management** tile.

### Discover Objects

The Discoveries section of the `describe.xml` file for the adapter might include parameters for discovery information. The `describe.xml` file is in the `conf` sub folder of the adapter, for example `xyz_adapter3/conf/describe.xml`.

| Options | Description |
|---|---|
| Collector | Collector that VCF Operations uses to discover objects. Only the VCF Operations Collector is added during installation. |
| Adapter Type | Adapter type for the objects to discover. |
| Adapter Instance | Adapter instance of the selected adapter type. |
| Discovery Info | Selection depends on the adapter type. For example, for a vCenter adapter, the Discovery Info selection adds an option to discover objects of a particular object type. |
| Only New Objects | On by default, to omit objects that are already discovered. |

### Object Type

Based on your selection of Adapter Type, Adapter Instance, and Discovery Info, the Object Type values change. The available adapter types are NSX-T, vCenter Adapter, vSAN Adapter.

NSX-T Object Types:

- Edge Cluster
- Firewall Section
- Groups
- Load Balancer Pool
- Load Balancer Service
- Load Balancer Virtual Server
- Logical Router
- Logical Switch
- Router Service
- Transport Node
- Transport Zone

vCenter Object Types:

- Cluster Compute Resource
- Datastore
- Datacenter
- Folder
- Host System
- Resource Pool
- Virtual Machine

vSAN Object Types:

- Cache Disk
- Capacity Disk
- vSAN Cluster
- vSAN Disk Group
- vSAN Fault Domain
- vSAN Witness Host

Discovery Results List

When you use the Discover Objects feature to manually discover objects in your environment, VCF Operations lists the objects of the specified object type. You can choose the objects to monitor.

**Where You Find Discovery Results**

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Inventory Management** tile.

After you make selections in the Discover Objects Workspace, click **OK**. With the default setting, VCF Operations displays only newly discovered objects. See Discover Objects Workspace.

**Table 1243: Object Types**

| Options | Description |
|---|---|
| Object Type | Discovered object types of the Object Type selected on the Discover Objects Workspace. |
| Object Count | Number of objects of the object type. |
| Import | When selected, imports the object type. Option is active and selectable for newly discovered object types. |
| Collect | When selected, imports the object type and starts collecting data. Option is active and selectable for newly discovered object types. |
| Credential | If the object type requires a login credential to collect data from the object., the value is **True**. |

Double-click the Object Type to display a list of objects to monitor.

**Table 1244: Objects**

| Options | Description |
|---|---|
| Object | Objects of the selected type that exist in the environment for the adapter. For example, the vCenter adapter discovers objects in the vCenter system. |
| Import | When selected, imports the object but does not start collecting data. Option is active and selectable for newly discovered objects that do not exist in the VCF Operations environment . |
| Exists | Indicates that the object exists in the VCF Operations environment. |
| Collect | When selected, imports the object and starts collecting data. Option is active and selectable for newly discovered objects that do not exist in the VCF Operations environment. |

Manage Maintenance Schedules for Your Object Workspace

You use maintenance mode to take an object offline. Many objects in your environment might be intentionally taken offline. For example, you might deactivate a server to update software. If VCF Operations collects metrics when the object is offline, it might generate incorrect alerts that affect the data for the object's health. When an object is in maintenance mode, VCF Operations does not collect metrics from the object and does not generate alerts for it.

### How Maintenance Schedules Work

If an object undergoes maintenance at fixed intervals, you can create a maintenance schedule and assign it to the object. For example, you can put an object into maintenance mode from midnight until 3 a.m. every Tuesday night. You can also manually put an object in maintenance mode, either indefinitely or for a specified period of time. These methods are not mutually exclusive. You can put an object in maintenance mode or take it out of maintenance mode, even if it has an assigned maintenance schedule.

### Where You Find Manage Maintenance Schedules

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Inventory Management** tile. Click **Start Maintenance** from the tool bar in the **Objects** tab.

**Table 1245: Manage Maintenance Schedules Options**

| Options | Description |
|---|---|
| I will come back and end maintenance myself. | Maintenance mode starts for the selected object when you click **OK**. You must manually end maintenance mode for this object. |
| End maintenance in | Type the number of minutes that the object is in maintenance mode. |
| End maintenance on | Click the calendar icon, and select the date that maintenance mode ends. |

Define Custom Property Workspace

In VCF Operations, you can define custom properties to collect and store operational data related to different objects. The custom property can be either a string or a numeric. You can assign custom properties to any subset of objects irrespective of the adapter kind and resource kind. You can use a mouse click, search filter, or a tag selector to select the correct object.

### Where You Find Add/Edit Custom Property

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Inventory Management** tile. Click **Add/Edit Custom Property** from the tool bar in the **Objects** tab.

**Table 1246: Add/Edit Custom Property**

| Options | Description |
|---|---|
| Property Name | Select or enter a property name. |
| Type | Select the property type from the drop-down menu. |
| Value | Enter a value for the property. |

You can assign the custom properties defined in this page to the Custom Object Groups and New Groups.

For more information, see Custom Object Groups Workspace to Create a New Group.

## Managing Custom Object Groups in VCF Operations

A custom object group is a container that includes one or more objects. VCF Operations uses custom groups to collect data from the objects in the group, and report on the data collected.

### Why Use Custom Object Groups?

You use groups to categorize your objects and have the system collect data from the groups of objects and display the results in dashboards and views according to the way you define the data to appear.

You can create static groups of objects, or dynamic groups with criteria that determine group membership as VCF Operations discovers and collects data from new objects added to the environment.

VCF Operations provides commonly used object group types, such as World, Environment, and Licensing. The system uses the object group types to categorize groups of objects. You assign a group type to each group so that you can categorize and organize the groups of objects that you create.

### Types of Custom Object Groups

When you create custom groups, you can use rules to apply dynamic membership of objects to the group, or you can manually add the objects to the group. When you add an adapter, the groups associated with the adapter become available in VCF Operations.

- Dynamic group membership. To dynamically update the membership of objects in a group, define rules when you create a group. VCF Operations adds objects to the group based on the criteria that you define.
- Mixed membership, which includes dynamic and manual.
- Manual group membership. From the inventory of objects, you select objects to add as members to the group.
- Groups associated with adapters. Each adapter manages the membership of the group. For example, the vCenter Server adapter adds groups such as datastore, host, and network, for the container objects in the vSphere inventory. To modify these groups, you must do so in the adapter.

Administrators of VCF Operations can set advanced permissions on custom groups. Users who have privileges to create groups can create custom groups of objects and have VCF Operations apply a policy to each group to collect data from the objects and report the results in dashboards and views.

When you create a custom group, and assign a policy to the group, the system uses the criteria defined in the applied policy to collect data from and analyze the objects in the group. VCF Operations reports on the status, problems, and recommendations for those objects based on the settings in the policy.

> **Note:**
> Only custom groups defined explicitly by users can be exported from or imported to VCF Operations. Users are able to export or import multiple custom groups. Once an import function has been executed, the user must check to determine if a policy or policies should be associated with the imported group. Export-import operations are available for user defined (created explicitly by user) custom groups only.

### How Policies Help VCF Operations Report On Object Groups

When you apply a policy to an object group, VCF Operations uses threshold settings, metrics, super metrics, attributes, properties, alert definitions, and symptom definitions that you activated in the policy to collect data from the objects in the group, and report the results in dashboards and views.

When you create a new object group, you have the option to apply a policy to the group.

- To associate a policy with the custom object group, select the policy in the group creation wizard.
- To not associate a specific policy with the object group, leave the policy selection blank. The custom object group will be associated with the default policy. If the default policy changes, this object group will be associated with the new default policy.

VCF Operations applies policies in priority order, as they appear on the Active Policies tab. When you establish the priority for your policies, VCF Operations applies the configured settings in the policies according to the policy rank order to analyze and report on your objects. To change the priority of a policy, you click and drag a policy row. The default policy

is always kept at the bottom of the priority list, and the remaining list of active policies starts at priority 1, which indicates the highest priority policy. When you assign an object to be a member of multiple object groups, and you assign a different policy to each object group, VCF Operations associates the highest ranking policy with that object.

## Object Group Types in VCF Operations

An object group type is an identifier that you apply to a specific group of objects in your environment to categorize them. You can add new group types, and apply them to groups of objects so that VCF Operations can collect data from the object group and display the results in the dashboards and views.

### How the Group Types Work

Use group types to categorize your objects so that the system can apply policies to them to track, and display specific status, such as alerts, workload, faults, risk, and so on.

When you create a new group type, VCF Operations adds it to the existing list of group types, and creates a new folder with the name of your group type in the Environment Custom Groups list.

When you create a new group of objects, you assign a group type to that group of objects. You add objects from the inventory trees to your custom group, then create your dashboard, add widgets to the dashboard, and configure the widgets to display the data collected from the objects in the group. You can then monitor and manage the objects.

You can apply a group type to a group of objects that you create manually, or to object groups that you cannot modify, such those added by adapters. Each adapter that you add to VCF Operations adds one or more static groups of objects to group the data received from the adapter sources.

The list of group types appears in the Content area under Group Types. The custom object groups appear in the Environment area under Custom Groups.

### Where You Create and Modify a Group Type

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory** and then click **Custom Group and Datacenters**. Click **Group Types** next to custom groups. You can add, edit, delete, and select groups from the group types page.

### Group Type Options

You can add, edit, or delete group types. You cannot edit group types that are created by adapters.

## Groups Tab on the Environment Overview Pane

Groups are containers that can contain any number and type of objects in your environment. VCF Operations collects data from the objects in the group and displays the results in dashboards and views that you define.

### How Groups Work

Groups are installed with VCF Operations, created by an adapter, or created by a user. Based on the group criteria, you can use groups to organize your environment and monitor all objects in the group together. You can also assign policies to groups and make group membership dynamic.

For example, if you have a set of vSphere hosts and you do not want to generate alerts when the host goes into maintenance mode, you can put the vSphere hosts in a group and assign a policy that includes a maintenance schedule setting. During the maintenance period, VCF Operations ignores any metrics for those objects and does not generate any alerts. After the maintenance period ends, VCF Operations returns to monitoring the objects and generates alerts if an outage occurs.

### Where You Find Custom Groups

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Custom Groups** tile.

### Custom Group Options

Click **ADD** button to add a group. You can only edit, clone, or delete a user-created group. You cannot modify groups installed with VCF Operations or by an adapter.

You can click the **Horizontal Ellipses** to import or export the custom group. The Groups data grid displays an overview of the state of each group. You can use the All Filters option to sort the custom groups based on Name, ID, Group Type, and Description columns.

To sort the list of custom groups based on columns, click the column heading of the following columns:

- Name
- Group Type
- Health
- Description
- Members Count

**Table 1247: Group Data Grid Options**

| Option | Description |
|---|---|
| Name | Select the group name to display a summary of the group. Select to the right of the name to edit, clone, or delete the group. |
| Summary | Criticality of the health of any group. Click a group with a red, orange, or yellow criticality to get more details about potential problems with objects in the group. |
| Members Count | Displays the number of members in the selected group. |
| Policy | Displays the policy associated with the selected group. |
| Dynamic Membership | Displays whether the group is static or dynamic. The available options are true and false. |
| Defined by | Displays who has defined the attributes of the group. The available options are:<br>• System<br>• User Defined<br>• Management Pack |

## Custom Object Groups Workspace

You can create and edit custom groups of objects to have VCF Operations collect data from the objects and display the results in the dashboards and views so that you can monitor your objects and take action on them when problems occur.

### How the Custom Groups Workspace Works

When you create a new object group, you define a meaningful group name, and select the group type. To associate the custom object group with a policy for analysis, you select the policy in the group creation wizard. You can leave the policy selection blank to not associate a policy with the object group. When the policy selection is blank, the custom object group is associated with the policy that is designated as the default policy.

You select the object types, and determine whether membership in the object group is static, dynamic, or a combination of static and dynamic membership.

- To create a static object group, you add objects to the group. You do not include criteria for object membership.
- To create a dynamic object group that VCF Operations updates based on specific criteria, you select the object type and define membership criteria for the group based on metrics, relationships, and properties.

When you add objects to a custom object group, a new folder appears in the Custom Groups navigation pane on the left, and includes the member objects.

### Where You Create and Modify Object Groups

To create or modify static or dynamic object groups, or object groups that have a combination of static and dynamic membership, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click

**Administration** > **Configurations**, and then click the **Custom Groups** tile. The **Custom Groups** page displays a list of custom object groups, and the object groups for adapters added to VCF Operations.

To edit existing groups, select a group, and then click the vertical ellipsis and select **Edit**.

Custom Object Groups Workspace to Create a New Group

You can create a new object group, define custom properties, assign a group type and objects to the group. When you create the group, you can assign a policy, or leave the policy selection blank to apply the default policy. VCF Operations collects data from the objects in the group based on the settings in the policy that is associated with the group. The results appear in the dashboards and views.

## Where You Assign Custom Group Type, Policy, and Membership

To assign the group type, policy, and membership, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Custom Groups** tile. Click **Add** to add a new group. In the New Group workspace, you can define the membership criteria, and select the objects to include or exclude.

To associate a policy with the custom object group, select the policy in the group creation wizard. To not associate a specific policy with the object group, leave the policy selection blank. The custom object group will be associated with the default policy. If the default policy changes, this object group will be associated with the new default policy.

**Table 1248: New Group Workspace**

| Option | Description |
|---|---|
| **Custom Group Tab** | |
| Name | Meaningful name of the object group. |
| | Meaningful description for the object group. |
| Group Type | Categorization for the object group. New custom groups appear in a dedicated folder in the Custom Groups navigation pane on the left. |
| Policy | Assigns a policy to one or more groups of objects to have VCF Operations analyze the objects according to the settings in your policy, trigger alerts when the defined thresholds are violated, and display the results in dashboards, views, and reports. You can assign a policy to the group when you create the group, or you can assign it later from the edit custom group wizard or from the policies area. |
| Keep group membership up to date | For dynamic object groups, VCF Operations can discover objects that match the criteria for the group membership according to the rules that you define, and update the group members based on the search results. |
| **Define Membership Criteria Tab** | |

| Option | Description |
| --- | --- |
| Membership Criteria | Defines the criteria for a dynamic object group and has VCF Operations keep the object membership of the group current.<br>• Object Type drop-down menu. Selects the type of objects to add to the group, such as virtual machines.<br>• Metrics, Relationship, and Properties criteria drop-down menu. Defines the criteria for VCF Operations to apply to collect data from the selected objects.<br>– Metrics. An instance of a data type, or attribute, that varies based on the object type. A metric is used as measurement criteria to collect data from objects. For example, you can select system attributes as a metric, where an attribute is a type of data that VCF Operations collects from objects.<br>– Relationship. Indicates how the object is related to other objects. For example, you can require a virtual machine object to be a child object that contains a certain word in the vSphere Hosts and Clusters navigation tree.<br>– Properties. Identifies a configuration parameter for the object. For example, you can require a virtual machine to have a memory limit that is greater than 100KB.<br>– Object Name. Specifies the name of the object. For example, you can create custom group which contains object with specific names.<br>– Tag. Specifies a tag for the object. For example, you can create membership criteria based on specific tags.<br>**Note:**<br>For the Metrics and Properties criteria drop-down menu, the metrics presented by default are a subset of available metrics. If the desired metric is not represented, use the **Select Object** button in the title bar of the data selection tree to re-filter the displayed list.<br>• You can use the filtering option to define the membership criteria for a group. The available options are:<br>– is - Displays the result of the typed text. Example - "!", displays all results equal to the symbol "!"<br>– is not - Displays the result of the typed text. Example - "1_node", displays all results not equal to "1_node"<br>– contains -Displays the result of the typed text. Example - "An", displays all results that contain "An"<br>– does not contain - Displays the result of the typed text. Example - "An", displays all results that does not contain "An"<br>– starts with - Displays the result of the typed text. Example - "!!", displays all results that starts with "!!"<br>– ends with - Displays the result of the typed text. Example - "N", displays all results that ends with "N"<br>– does not start with - Displays the result of the typed text. Example - "N", displays all results that does not start with "N"<br>– does not end with - Displays the result of the typed text. Example - "S", displays all results that does not end with "S"<br>– matches regular expression - Displays the result if it matches the regular expression. Example .*8\.\d*<br>– does not match regular expression Displays the result if it does not match the regular expression. Example .*8\.\d*<br>• Add. Includes another metric, relationship, or property for the object type.<br>• Remove. Deletes the selected object type from the membership criteria, or delete the selected metric, relationship, or property type from the criteria for the object type.<br>• Add another criteria set. Adds another object type to add to the group. For example, you might want to create a single object group to track vCenter instances and Host Systems. |
| **Objects to Include/Exclude Tab** | |

| Option | Description |
|---|---|
| Objects To Always Exclude | Determine which objects to exclude from the group every time VCF Operations collects data from the objects, regardless of the membership criteria. The objects that you include override the criteria that you define for membership. In previous versions of VCF Operations, these objects were called a denylist.<br>• **Select Objects Pane > Inventory Tab**. Displays the list of available object groups. Click on an object group to view the objects in the right pane. You can select some or all the objects in the object group and drag and drop it to the left pane for exclusion from the custom group.<br>• **Select Objects Pane > Custom Group Tab**. Displays the list of available custom groups. Click on a custom group and drag and drop it to the left pane for exclusion from the custom group.<br>• From the left pane, select the objects you want to delete and click **Remove** to delete the objects that you have added for exclusion from the custom group. |
| Objects To Always Include | Determine which objects to include in the group every time VCF Operations collects data from the objects, regardless of the membership criteria. The objects that you include override the criteria that you define for membership. In previous versions of VCF Operations, these objects were called an allowlist.<br>• **Select Objects Pane > Inventory Tab**. Displays the list of available object groups. Click on an object group to view the objects in the right pane. You can select some or all the objects in the object group and drag and drop them to the left pane for inclusion into the custom group.<br>• **Select Objects Pane > Custom Group Tab**. Displays the list of available custom groups. Click on a custom group and drag and drop it to the left pane for inclusion into the custom group.<br>• From the left pane, select the objects you want to delete and click **Remove** to delete the objects that you have added for inclusion to the custom group. |
| **Assign Custom Properties Tab** | |
| Assign Custom Properties | In VCF Operations, you can define custom properties to collect and store operational data related to different objects. The custom property can be either a string or a numeric. You can assign the newly defined custom properties to new groups or existing groups.<br>• Property Name. Select or specify a name for the custom property.<br>• Type. Select the type of custom property from the drop-down menu.<br>The custom property can either be a string or a numeric.<br>• Value. Specify a custom property value, which should be assigned to this custom property when an object is added to the group.<br>• Reset Value. Specify a custom property value, which should be assigned to this custom property when an object leaves the group.<br>• **Reset**. Resets the custom property to a non-zero value.<br>• **Remove**. Removes the custom property from the group.<br>• **Add Another Custom Property**. Adds another custom property to the group. |

- Use the **Previous** and **Next** buttons to navigate between the tabs.
- Use the **Next** button to move to the next tab.
- Use **Create** button to save and create the custom group.
- Use the **Preview** button to preview the list of objects in the group and to verify that the criteria you defined is applicable to the group of objects. If the criteria that you defined is valid, the preview displays applicable objects. If the criteria is not valid, the preview does not display any objects.

## Inventory Details and Tabs

The organization of the tabs and options in VCF Operations provides a built-in workflow that you can use when you work with objects in your environment.

The tabs, **Summary**, **Metrics**, **Alerts**, **Topology**, **Capacity**, and so on, provide progressive levels of detail about the selected object. As you work through the tabs, starting with the high level **Summary** and **Alerts** tabs, you see the general state of an object. The **Environment** tab provides you with specific data about the object.

As you monitor objects in your environment, you discover which tabs provide the information that you need when you are investigating problems.

## Inventory Page and Inventory Detailed View

You can view details of vCenter instances and VCF instances and from the **Inventory** page. You can also view a detailed list of the inventory in your VCF Operations environment where the objects are grouped logically. When you select an object from the detailed view, you see information such as summary information, metrics, alerts, capacity, and other details specific to the selected object.

### Inventory Page

To access the **Inventory** page, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory**.

> **Note:**
> If you see the Inventory detailed view, click **Basic View** at the top in the **Inventory** panel.

The **Inventory** page lists all the vCenter instances that have been configured. At the top of the page, you can view environment details such as the number of VCF accounts, vCenter accounts, data centers, clusters, hosts, VMs, and datastores.

**Table 1249: Options in the Data Grid of the Inventory Page**

| Option | Description |
|---|---|
| vCenter Name | Displays the names of the vCenter instances that are configured. |
| Type | Displays all account types. |
| Data centers | Displays the number of data centers in the environment. |
| vSphere Clusters | Displays the number of vSphere clusters that have been configured. |
| Hosts | Displays the number of hosts in the environment. |
| Virtual Machines | Displays the number of VMs in the environment. |
| Datastores | Displays the number of datastores in the environment. |
| Collection State | Displays the collecting state such as Collecting, Not Collection, and so on. |

### Inventory Detailed View

The Inventory detailed view is a panel that displays logical grouping/hierarchies for the vCenter and and VCF Instances that are configured in the VCF Operations environment. You can view either a tree or list view and then select the object to view object related information.

To access the **Inventory Detailed View**, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane click **Inventory**, and then click on a vCenter instance from the data grid or click **Detailed View** from the top. You will see the **Inventory** panel with the Inventory detailed view.

**Table 1250: Options from the Inventory Detailed View**

| Option | Description |
|---|---|
| Tree view and List view | Toggle between the options to view the inventory in either a tree view or a list view. When you select a certain object your view is scoped around that object.<br>In the list view, you can see all the descendant and ancestor objects of the selected object. When you double click on any of the objects in the list view, you change the scope and the new object becomes the primary object of the view. You can select the number of ancestors or descendants for the object you want to view from the **Levels** option. |
| Hosts and Clusters | Select a host or cluster object to view the object details. |
| Storage | Select a Storage object to view the object details. |
| Networking | Select a Networking object to view the object details. |
| Custom Group and Datacenters | Select an object from Custom Groups, Custom Datacenters, or Business Applications to view the object details. |
| Integrations | Select an object from Others and All Objects to view the object details. |

**Note:**
If there are more than ten objects under a section in the Inventory detailed view, you can search for an object using the Search option. If there are more than thousand objects in a section, use the **View More** button under the last object that is displayed, to view the rest of the objects. Use the **Type to see more** button to find objects if there are more than two thousand objects.

## Evaluating Object Information Using Badge Alerts and the Summary Tab

The Summary tab that is associated with the other object tabs summarizes Heath, Risk, and Efficiency badge alerts for the selected object and displays the top alerts that lead to the current state.

Use this tab as an overview of alerts for an object, object group, or application - to evaluate the effect that alerts are having on an object and to begin troubleshooting problems. For more detail on the badge Alerts, click **Badge Alerts**, further to the right on the tool bar.

### Badge Alert Types

The Health, Risk, and Efficiency badge states are based on the impact of the generated alerts for the selected object.

- Health alerts indicate problems that affect the health of your environment.
- Risk alerts indicate problems that are not immediate threats.
- Efficiency alerts tell you where you can improve performance or reclaim resources.

### Alerts for an Object or an Object Group

For a single object, the Top alerts are the alerts generated for the object. The Top Alerts for Children are the alerts generated for any child or other descendant objects in the currently selected navigation hierarchy. For example, if you are working with a host object in the vSphere Host and Clusters navigation hierarchy, children can include virtual machines and datastores.

Object groups can include one object type, such as hosts, or multiple objects types, such as hosts, virtual machines, and datastores. When you are working with object groups, all the group member objects are children of the group container. The most critical generated alerts for the member objects appear as Top Alerts for Children.

For an object group, the only Top Alerts that might be generated are the predefined group population alerts. If the average health is above the Warning, Immediate, or Critical threshold, a group population alert considers the health of all group

members and is triggered. If a group population alert is generated, the alert affects the badge score and color. If a group population alert is not generated, then the badges are green. This behavior is because an object group is a container for other objects.

#### Summary Tab and Related Hierarchies

The alerts that appear on the **Summary** tab for an object can vary depending on the currently selected hierarchy in the Related Hierarchies in the left pane.

Depending on the selected hierarchy, you see different alerts and relationships on the **Summary** tab for an object. The current focus object name is on the center pane title bar, but the children alerts depend on the relationships that the highlighted hierarchy defined in the Related Hierarchies list in the upper left pane. For example, if you are working with a host object relative to virtual machines in the Hosts and Clusters hierarchy, then children commonly include virtual machines and datastores. But if you are working with the same host as a member of an object group, then any alerts on virtual machines that are also members of the group do not appear. The alerts do not appear because the host and the virtual machines are considered children of the group and peers among each other. In this example, the focus of the **Summary** tab is the host in the context of the group, not the Hosts and Clusters hierarchy.

#### Summary Tab Evaluation Techniques

You can evaluate the state of objects, starting with the **Summary** tab, by selecting an object or object group, click the alerts on the **Summary** tab, and resolve the problems that the alert indicates.

## Summary Tab

The Summary tab provides an overview of the state of the selected object, group, or application. Use this tab to evaluate the impact that alerts are having on the object and use the information to begin troubleshooting problems.

#### How the Summary Tab Works

Based on the object selected, the following summary tabs are displayed:

- VM Summary Tab
- Datastore Summary Tab
- Host Summary Tab
- Cluster Summary Tab
- Custom Group and Container Summary Tab

#### Where to Find the Summary Tab

- In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory** and then select a group, custom data center, application, or inventory object.

#### Understanding the Summary Tab

**Table 1251: Summary Tab Options**

| Option | Description |
|---|---|
| Troubleshoot | Start the Troubleshooting Workbench with the current object in context. |
| Object Summary | This widget displays the details of the selected object. It also displays the number of resources associated with the selected object. |

| Option | Description |
|---|---|
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the labels of the alert. |
| Consumer | Gives the number of active VMs for the selected object. You can also view the usage details for the virtual machine, CPU, and Memory. |
| Provider | Gives the details of available resources for the selected object. You can view the number of hosts and capacity remaining for CPU, RAM, Storage. |
| Cluster | Displays the cluster details of the selected object. |
| Datastore | Displays the datastore details of the selected object. |

**Datastore Summary Tab**

The Datastore Summary tab provides an overview of the state of the selected datastore. For the selected object, the Datastore Summary tab displays the alerts and metrics as they affect the health, risk, or efficiency. Use this tab to evaluate the impact that alerts are having on the datastore and use the information to begin troubleshooting problems.

**Understanding the Datastore Summary Tab**

**Table 1252: Datastore Summary Tab Options**

| Option | Description |
|---|---|
| Troubleshoot | Start the Troubleshooting Workbench with the current object in context. |
| Object Summary | This widget displays the details of the selected object. The widget also displays the number of resources associated with the selected object. |
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the labels of the alert. |
| Time Remaining | This widget displays the number of days remaining till the projected resource utilization crosses the threshold for the usable capacity. |
| Capacity Remaining | This widget displays the unused capacity of your virtual environment to accommodate new virtual machines. |
| Utilization | This widget is used to find out the trends in capacity used by a selected datastore as against the total capacity available. |
| Performance | This widget displays the summary metrics about the overall performance of the object. Click each metric to see the expanded chart. |

| Option | Description |
|---|---|
| Configuration | This widget displays the configuration details for the selected datastore object. |

### Host Summary Tab

The Host Summary tab provides an overview of the state of the selected host. For the selected object, the Host Summary tab displays the alerts and metrics as they affect the health, risk, or efficiency. Use this tab to evaluate the impact that alerts are having on the host and use the information to begin troubleshooting problems.

**Understanding the Host Summary Tab**

**Table 1253: Host Summary Tab Options**

| Option | Description |
|---|---|
| Troubleshoot | Start the Troubleshooting Workbench with the current object in context. |
| Object Summary | This widget displays the details of the selected object. The widget also displays the number of resources associated with the selected object. |
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the labels of the alert. |
| Time Remaining | This widget displays the number of days remaining till the projected resource utilization crosses the threshold for the usable capacity. |
| Capacity Remaining | This widget displays the unused capacity of your virtual environment to accommodate new virtual machines. |
| Utilization | This widget is used to find out the trends in capacity used by a selected datastore as against the total capacity available. |
| Performance | This widget displays the summary metrics about the overall performance of the object. Click each metric to see the expanded chart. |
| Configuration | This widget displays the hardware, CPU, and Network configuration details of the host. |
| | |
| Ping Statistics | This widget displays the availability of end points that exist in your environment and other details. For more details, see Ping Account in VCF Operations. |
| .Mapped Datastore | Provides storage trends for the following datastore types -<br>• vSAN<br>• VMFS<br>• NFS<br>• VMware vSphere Virtual Volumes (vVols)<br>This tells you which datastore is running out of capacity. |

### VM Summary Tab

The VM Summary tab provides an overview of the state of the selected VM. For the selected object, the VM Summary tab displays the alerts and metrics as they affect the health, risk, or efficiency. Use this tab to evaluate the impact that alerts are having on the VM and use the information to begin troubleshooting problems.

**Understanding the VM Summary Tab**

**Table 1254: VM Summary Tab Options**

| Option | Description |
|---|---|
| Troubleshoot | Start the Troubleshooting Workbench with the current object in context. |
| Object Summary | This widget displays the details of the selected object. The widget also displays the number of resources associated with the selected object. |
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the labels of the alert. |
| Time Remaining | This widget displays the number of days remaining till the projected resource utilization crosses the threshold for the usable capacity. |
| Capacity Remaining | This widget displays the unused capacity of your virtual environment to accommodate new virtual machines. |
| Utilization | This widget is used to find out the trends in capacity used by a selected datastore as against the total capacity available. |
| Performance | This widget displays the summary metrics about the overall performance of the object. Click each metric to see the expanded chart. |
| Configuration | This widget displays the virtual hardware, resource allocation, tools, and Network configuration details of the virtual machine. |
| Mapped Datastore | Provides storage trends for the following datastore types -<br>• vSAN<br>• VMFS<br>• NFS<br>• VMware vSphere Virtual Volumes (vVols)<br>This tells the provisioned and used storage for this VM in a particular datastore. |

### Cluster Summary Tab

The Cluster Summary tab provides an overview of the state of the selected cluster. For the selected object, the Cluster Summary tab displays the alerts and metrics as they affect the health, risk, or efficiency. Use this tab to evaluate the impact that alerts are having on the cluster and use the information to begin troubleshooting problems.

## Understanding the Cluster Summary Tab

### Table 1255: Cluster Summary Tab Options

| Option | Description |
|---|---|
| Troubleshoot | Start the Troubleshooting Workbench with the current object in context. |
| Object Summary | This widget displays the details of the selected object. The widget also displays the number of resources associated with the selected object. |
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the labels of the alert. |
| Time Remaining | This widget displays the number of days remaining till the projected resource utilization crosses the threshold for the usable capacity. |
| Capacity Remaining | This widget displays the unused capacity of your virtual environment to accommodate new virtual machines. |
| Virtual Machine Remaining | This widget displays the remaining virtual machines in the cluster. To see the details of the remaining virtual machines, click the Virtual Machine Remaining card. |
| Memory Tiering Nudge | Eligible clusters show a nudge that encourages you to navigate to the Capacity Optimization page and and enable memory tiering under the Resource Effeciency tab. For more information, see Using Memory Tiering for Resource Efficiency. |
| Utilization | This widget is used to find out the trends in capacity used by a selected datastore as against the total capacity available. |
| Performance | This widget displays the summary metrics about the overall performance of the object. Click each metric to see the expanded chart. |
| Configuration | This widget displays the configuration details of the cluster. |
| GPU Information | This widget is only visible if the cluster has GPU configured hosts. It displays the following GPU information:<br>• Number of GPUs: The total number of GPUs in the cluster.<br>• Compute Utilization: The compute utilization percentage of a GPU<br>• Memory Usage: The percentage of memory currently in use out of the total available memory. |
| Metadata | This widget displays the metadata details of the cluster. |
| Mapped Datastore | Provides storage trends for the following datastore types -<br>• vSAN<br>• VMFS<br>• NFS<br>• VMware vSphere Virtual Volumes (vVols)<br>This tells the provisioned and used storage for this VM in a particular datastore. |

### vCenter Server and Data Center Summary Tab

The vCenter Server and data center Summary tab provides an overview of the state of the selected data center or vCenter. For the selected object, the vCenter or data center Summary tab displays the alerts as they affect the health, risk, or efficiency. Use this tab to evaluate the impact that alerts are having on the vCenter or data center and use the information to begin troubleshooting problems.

#### Understanding the vCenter Server and Data Center Summary Tab

| Option | Description |
|---|---|
| Troubleshoot | Start the Troubleshooting Workbench with the current object in context. |
| Object Summary | This widget displays the details of the selected object. The widget also displays the number of resources associated with the selected object. |
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the labels of the alert. |
| Consumer | Gives the number of active VMs for the selected object. You can also view the usage details for the virtual machine, CPU, and Memory. |
| Provider | Gives the details of available resources for the selected object. You can view the number of hosts and capacity remaining for CPU, RAM, Storage. |
| vSphere Distributed Switch Name | Displays the details of the vSphere distributed switch. |
| Metadata | Displays the metadata details of the data center. |
| Cluster | Displays the cluster details of the selected object. |
| Datastore | Displays the datastore details of the selected object. |

### Resource Pool Summary Tab

The Resource Pool Summary tab provides an overview of the state of the resources in the resource pool. For the selected resource, the Resource Pool Summary tab displays the alerts and metrics as they affect the health, risk, or efficiency. Use this tab to evaluate the impact that alerts are having on the resource pool and use the information to begin troubleshooting problems.

#### Understanding the Resource Pool Summary Tab

| Option | Description |
|---|---|
| Troubleshoot | Start the Troubleshooting Workbench with the current object in context. |
| Object Summary | This widget displays the details of the selected object. The widget also displays the number of resources associated with the selected object. |

| Option | Description |
|---|---|
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the labels of the alert. |
| Utilization | This widget is used to find out the trends in capacity used by the selected resource pool as against the total capacity available. |
| Performance | This widget displays the summary metrics about the overall performance of the object. Click each metric to see the expanded chart. |
| Resource Pool | This widget lists the resource pool name, cpu status, and memory status of the resources that are part of the corresponding resource pool. |

### Custom Group and Container Summary Tab

The Custom Group and Container Summary tab provides an overview of the state of the selected group or a container. For the selected object, the Custom Group and Container Summary tab displays the alerts and metrics as they affect the health, risk, or efficiency. Use this tab to evaluate the impact that alerts are having on the group or a container and use the information to troubleshoot the problems.

**Understanding the Custom Group and Container Summary Tab**

**Table 1256: Custom Group and Container Summary Tab Options**

| Option | Description |
|---|---|
| Recommended Actions | This widget displays the health status for the selected object and its descendants. It also displays recommendations to solve problems in an instance.<br>The badges provide a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the badge. |

### Workload Management Activated Cluster Summary Tab

The Workload Management activated cluster is a cluster with Kubernetes activated, running on vSphere (also called Supervisor cluster). It hosts a type of resource pool called Namespaces. The Workload Management Enabled Cluster Summary tab provides an overview of the state of the selected cluster.

## Understanding the Cluster Summary Tab

### Table 1257: Workload Management Enabled Cluster Summary Tab Options

| Option | Description |
|---|---|
| Troubleshoot | Start the Troubleshooting Workbench with the current object in context. |
| Object Summary | This widget displays the details of the selected object. The widget also displays the number of resources associated with the selected object and whether the Workload Management is activated or deactivated. |
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the badge. |
| Time Remaining | This widget displays the number of days remaining till the projected resource utilization crosses the threshold for the usable capacity. |
| Capacity Remaining | This widget displays the unused capacity of your virtual environment to accommodate new virtual machines. |
| Virtual Machine Remaining | The virtual machine remaining number is based on the average profile. The virtual machine remaining numbers are calculated when you activate one or more custom profiles from the policy. The overall virtual machine remaining is based on the most constrained profile. |
| Utilization | This widget is used to find out the trends in capacity used by a selected cluster as against the total capacity available.<br>The key utilization indicators are:<br>• CPU Capacity Usage<br>• Memory Usage<br>• Memory Balloon<br>• Disk Total IOPS<br>• Disk Total Throughput<br>• Network Usage Rate |
| Performance | This widget displays the summary metrics about the overall performance of the object. It displays the latest value and a trend line of the various key performance indicators in a color that indicates its health based on the symptom associated with the metrics. Click each metric to see the expanded chart.<br>The key performance indicators are:<br>• Max VM Memory Contention<br>• Worst Consumer Disk Latency<br>• Consumers with Memory Contention<br>• Consumers with CPU Ready<br>• Physical Network Packets Dropped<br>• Virtual Network Packets Dropped |
| Configuration | This widget displays the hardware, CPU, and Network configuration details of the host. |

| Option | Description |
|---|---|
| Namespaces | Lists the configuration status, current version and Kubernetes status of the namespaces in the cluster. |

**Namespace Summary Tab**

A namespace sets the resource boundaries where vSphere Pods and Tanzu Kubernetes clusters created by using the Tanzu Kubernetes Grid Service can run. The Namespace summary tab provides an overview of the state of the selected Namespace.

**Understanding the Namespace Summary Tab**

**Table 1258: Namespace Summary Tab Options**

| Option | Description |
|---|---|
| Troubleshoot | Start the Troubleshooting Workbench with the current object in context. |
| Object Summary | This widget displays the details of the selected object. The widget also displays the number of resources associated with the selected object. |
| Active Alerts | This widget provides a visual indicator of the alert status based on the alert type.<br>To see the alerts for the object, click the badge . |
| Utilization | This widget is used to find out the trends in capacity used by a selected namespace as against the total capacity available.<br>The key utilization indicators are:<br>• CPU Usage<br>• Consumed Memory |
| Performance | This widget displays the summary metrics about the overall performance of the object. It displays the latest value and a trend line of the various key performance indicators in a color that indicates its health based on the symptom associated with the metrics. Click each metric to see the expanded chart.<br>The key performance indicators are:<br>• Worst Consumer CPU Ready<br>• Worst Consumer Memory Contention<br>• Consumers with Memory Contention<br>• Consumers with CPU Ready |
| Configuration | This widget displays the following configuration details about the Namespaces:<br>• Configuration status<br>• Virtual Machines<br>• Number of Tanzu Kubernetes clusters<br>• Pods |

**vSphere Pod Summary Tab**

vSphere Pods run containers without needing to customize a Kubernetes cluster. You can deploy vSphere Pods directly on ESX hosts It hosts a type of resource pool called Namespace. The vSphere Pod Summary tab provides an overview of the state of the vSphere Pods.

## Understanding the vSphere Pod Summary Tab

### Table 1259: vSphere Pod Tab Summary Options

| Option | Description |
|---|---|
| Troubleshoot | Start the Troubleshooting Workbench with the current object in context. |
| Object Summary | This widget displays the details of the selected object. The widget also displays the number of resources associated with the selected object. |
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the labels of the alert. |
| Time Remaining | This widget displays the number of days remaining until the projected resource utilization crosses the threshold for the usable capacity. |
| Capacity Remaining | This widget displays the unused capacity of your virtual environment to accommodate new virtual machines. |
| Utilization | This widget is used to find out the trends in capacity used by a selected vSphere Pod as against the total capacity available.<br>The key utilization indicators are:<br>• CPU Usage<br>• Free Memory<br>• Guest Page in Rate per second<br>• Virtual Disk Total IOPS<br>• Virtual Disk Total Throughput |
| Performance | This widget displays the summary metrics about the overall performance of the object. It displays the latest value and a trend line of the various key performance indicators in a color that indicates its health based on the symptom associated with the metrics. Click each metric to see the expanded chart.<br>The key performance indicators are:<br>• CPU Queue<br>• Disk Queue<br>• CPU Ready<br>• CPU Co-stop<br>• Memory Contention<br>• Virtual Disk Total Latency<br>• Network Transmitted Packets Dropped |
| Configuration | This widget displays the hardware, CPU, and Network configuration details of the host. |

### Summary Tabs for VCF

The summary tab of VCF provides account details, overall SDDC inventory of the the organization and their the key components, aggregated metrics, alerts, and so on.

VCF Summary Tab

The VCF Summary tab provides organization details, the overall inventory of the organization including key components, alerts, and so on.

### Where To View VCF Summary

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory**. In the **Inventory panel (Detailed View)**, click **Integrations** > **All Objects** > **VMware Cloud Foundation**.

**Table 1260: VCF Summary Options**

| Option | Description |
|---|---|
| Object Summary | This widget displays the details of the selected object. The widget also displays the number of resources associated with the selected object. |
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Critical and Immediate alerts indicate issues that require immediate attention.<br>• Warning alerts indicate that you must look into issues shortly.<br>• Info alerts indicate that you can reclaim resources. |
| Consumer | Gives the number of active VMs for the selected object. You can also view the usage details for the virtual machine, vCPU, RAM, and Provision. |
| Provider (Usable Capacity) | Gives the details of available resources for the selected object. You can view the number of hosts and capacity remaining for vCPU, RAM, and Storage. |
| Configuration Maximums | This widget is used to find out the VCF limits and your consumption against those limits. It displays the total number of management workload domains, VI workload domains, ESX hosts, vCenter instances, and NSX clusters per domain manager. |
| Topology | Gives a graphical representation of objects related to VCF. Click each object to have an expanded view of the object details. |
| Domain Summary | Displays the domain name, type, cluster, virtual machine, datastore details, tanzu k8s cluster, namespace, and pods. |

## VCF Domain Summary Tab

The VCF Domain Summary tab provides details on the overall inventory of the domain including key components, domain health, topology, alerts, and so on.

### Where To View VCF Domain Summary Tab

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory**. In the **Inventory panel (Detailed View)**, click **Integrations** > **All Objects** > **VMware Cloud Foundation**. Click the **Domain** object type and select any one of the domains in the list.

**Note:**

The VCF Domain Summary tab displays the total number of domains which includes both configured and unconfigured domains.

**Table 1261: Domain Summary Options**

| Option | Description |
|---|---|
| Object Summary | Displays the details of the selected object. The widget also displays the number of resources associated with the selected object. |
| Active Alerts | Displays a visual indicator of the alert status for the following alert types.<br>• Critical and Immediate alerts usually require immediate attention.<br>• Warning alerts indicate that you must look into any problems shortly.<br>• Info alerts indicate that you can reclaim resources. |
| Consumer | Displays the number of active VMs for the selected object. You can also view the usage details for the virtual machine, vCPU, RAM, and Provision. |
| Provider (Usable Capacity) | Displays the details of available resources for the selected object. You can view the number of hosts and capacity remaining for vCPU, RAM, Storage. |
| Topology | Displays a graphical representation of objects related to the domains. Click each object to have an expanded view of the object details. |
| Distributed Switch | Dsiplays an overview of the state of the distributed switches. It displays the distributed switch name, version, total number of hosts, the maximum number of ports, and number of used ports. |
| Cluster Summary | Displays an overview of the state of the existing clusters. It displays the cluster name, ESX host, virtual machine, capacity remaining, time remaining, and VM remaining. |
| Datastore Summary | Displays an overview of the state of the existing datastores. It displays the datastore name, capacity, virtual machine, capacity remaining, and time remaining. |

VCF World Summary Tab

The VCF World Summary tab provides details on the VCF World inventory of the organization including key components, recommended actions for the VCF organization, and the health of the object types at the world level.

**Where To View the World Summary Tab**

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory**. In the **Inventory panel (Detailed View)**, click **Integrations** > **All Objects** > **VMware Cloud Foundation** > **VCF World**. Select the **VCF World** object type.

**Table 1262: World Summary Options**

| Option | Description |
|---|---|
| Recommended Actions | Displays details such as object type, number of accounts, and the recommended actions for each account.<br>This widget provides a visual indicator of the alert status for the following alert types.<br>• Critical and Immediate alerts usually require immediate attention.<br>• Warning alerts indicate that you must look into any problems shortly.<br>• Normal and Unknown alerts indicate alerts you can look at later. |
| World | Displays the health badge for the VCF World object type. You can view the average health of all group members, the compliance, efficiency, risk, workload, and the total count of critical, immediate, and normal compliance. |
| Active Alerts | Displays the visual indicator of the alert status for the following alert types. You can view the name of the object and alert description, alert type, and subtype. The time when the alert was generated, the suggested fix, and when the action was taken upon the alert. |
| Information | Displays the account information of the selected object. |

### Summary Tabs for vSAN

The summary tabs for vSAN provide the account details and an overview of the state of the selected vSAN object and it's key components like health, capacity, alerts.

vSAN Cluster Summary Tab

The vSAN Cluster tab provides an overview of the state of the selected vSAN cluster. For the selected object, the vSAN cluster tab displays the alerts, time remaining, capacity remaining, utilization, configuration, and metrics as they affect the health, risk, or efficiency. You can use this tab to evaluate the impact that alerts are having on the vSAN cluster and use that information to begin troubleshooting problems.

#### Where To View vSAN Cluster Summary Page

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory** > **Inventory Panel (Detailed View)**, and then click the **Integrations** icon. Select **All Objects**, expand **vSAN Adapter** > **vSAN Cluster**, and then select a specific **vSAN Cluster**.

**Note:**
The vSAN Cluster Summary tab displayed depends on the vSAN cluster type.

## vSAN OSA HCI Cluster Summary Tab

### Table 1263: vSAN OSA HCI Cluster Summary Tab Options

| Option | Description |
|---|---|
| Cluster Summary | This widget displays the details of the selected object. The widget also displays the number of resources associated with the selected object. |
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the labels of the alert. |
| Capacity Summary | This widget displays the physical capacity of your virtual environment to accommodate new workloads. |
| Critical Alerts | This widget displays alerts associated with the health score of the vSAN cluster. |
| vSAN Health Score | This widget displays the health score of the vSAN cluster. |
| Health Score Trend | This widget is used to find the trends in the health score of the vSAN cluster. You can view daily, weekly, and monthly health score trends. |
| What-if Free Capacity for Workload based on Chosen Policy | This widget displays the estimated capacity for a future workload. This estimate depends on the storage policy that is chosen for additional storage. |
| Utilization | This widget is used to find out the trends in capacity used by a selected vSAN cluster as against the total capacity available. |
| Contention | This widget displays the memory contention details of the vSAN cluster. |

## vSAN ESA HCI Summary Tab

### Table 1264: vSAN ESA HCI Summary Tab Options

| Option | Description |
|---|---|
| Workbench | Start the Workbench with the current object in context. |
| Object Summary | This widget displays the details of the selected object. The widget also displays the number of resources associated with the selected object. |
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the labels of the alert. |

| Option | Description |
|---|---|
| Capacity Summary | This widget displays the total effective capacity, used effective capacity, free effective capacity of your virtual environment in gigabytes, when auto-RAID policy is enabled in the cluster.<br>**Note:**<br>If the auto-RAID policy is disabled, it displays the related physical capacity information.<br>It also displays the remaining capacity percentage to monitor your cluster's storage capacity to accommodate new workloads. |
| vSAN Health Score | This widget displays the health score of the vSAN cluster. |
| Health Score Trend | This widget is used to find the trends in the health score of the vSAN cluster. You can view daily, weekly, and monthly health score trends. |
| Critical Alerts | This widget displays alerts associated with the health score of the vSAN cluster. |
| What-if Free Capacity for Workload based on Chosen Policy | This widget displays the estimated capacity for a future workload. This estimate depends on the storage policy that is chosen for additional storage. |
| Utilization | This widget displays the aggregated IO performance of this vSAN cluster |
| Contention | This widget provides a snapshot of the potential bottlenecks by aggregating network reliability and disk latency/errors. |

## vSAN Compute Cluster Summary Tab

### Table 1265: vSAN Compute Cluster Summary Tab Options

| Option | Description |
|---|---|
| Troubleshoot | Start the Troubleshooting Workbench with the current object in context. |
| Object Summary | This widget displays the details of the selected object. The widget also displays the number of resources associated with the selected object. |
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the labels of the alert. |
| Time Remaining | This widget displays the number of days remaining till the projected resource utilization crosses the threshold for the usable capacity. |
| Capacity Summary | This widget displays the total effective capacity, used effective capacity, free effective capacity of your virtual environment in gigabytes. It also displays the remaining capacity percentage to monitor your cluster's storage capacity to accommodate new virtual machines. |
| Remote Datastores | Displays the remote datastores used by vSAN compute.<br>**Note:**<br>vSAN compute cluster remotely mounts the datastore of vSAN Max. |

| Option | Description |
|---|---|
| vSAN Health Score | This widget displays the health score of the vSAN cluster. |
| Health Score Trend | This widget is used to find the trends in the health score of the vSAN cluster. You can view daily, weekly, and monthly health score trends. |

## vSAN ESA Max Cluster Summary Tab

**Table 1266: vSAN Max ESA Tab Options**

| Option | Description |
|---|---|
| Troubleshoot | Start the Troubleshooting Workbench with the current object in context. |
| Object Summary | This widget displays the details of the selected object. The widget also displays the number of resources associated with the selected object. |
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the labels of the alert. |
| Client Clusters | This widget displays total number of VMs using the vSAN Max cluster for each client cluster. |
| Capacity Summary | This widget displays the total effective capacity, used effective capacity, free effective capacity of your virtual environment in gigabytes. It also displays the remaining capacity percentage to monitor your cluster's storage capacity to accommodate new virtual machines. |
| vSAN Health Score | This widget displays the health score of the vSAN cluster. |
| Health Score Trend | This widget is used to find the trends in the health score of the vSAN cluster. You can view daily, weekly, and monthly health score trends. |
| Critical Alerts | This widget displays alerts associated with the health score of the vSAN cluster. |
| Effective Free Space for Workload based on Chosen Policy | This widget displays the estimated capacity for a future workload, that is based on the selected storage policy. |
| Utilization | This widget is used to find out the trends in capacity used by a selected vSAN cluster as against the total capacity available. |
| Contention | This widget displays the memory contention details of the vSAN cluster. |

### vSAN Disk Group Summary Tab

The vSAN Disk Group Summary tab provides an overview of the state of the selected vSAN Disk Group. For the selected object, the vSAN Disk Group tab displays the alerts, time remaining, capacity remaining, utilization, configuration, and metrics as they affect the health, risk, or efficiency. You can use this tab to evaluate the impact that alerts are having on the vSAN Disk Group and use that information to begin troubleshooting problems.

**Where To View vSAN Cluster Disk Group Summary**

From the left menu, click **Inventory** > **Inventory Panel (Detailed View)** > **Integrations** > **All Objects** > **vSAN Adapter** > **vSAN Disk Group**, and then select a **Disk Group** object type.

**Table 1267: vSAN Cluster Disk Group Summary Options**

| Option | Description |
|---|---|
| Troubleshoot | Start the Troubleshooting Workbench with the current object in context. |
| Object Summary | This widget displays the details of the selected object. The widget also displays the number of resources associated with the selected object. |
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the labels of the alert. |
| Time Remaining | This widget displays the number of days remaining till the projected resource utilization crosses the threshold for the usable capacity. |
| Capacity Remaining | This widget displays the unused capacity of your virtual environment to accommodate new virtual machines. |
| Utilization | This widget is used to find out the trends in capacity used by a selected vSAN cluster disk group as against the total capacity available. |
| Contention | This widget displays the memory contention details of the vSAN cluster. |
| Resync | This widget displays the throughput and latency details for the vSAN cluster disk group. |

vSAN Capacity Disk Summary Tab

The vSAN Capacity Disk tab provides an overview of the state of the selected vSAN capacity disk. For the selected object, the vSAN capacity disk tab displays the alerts, time remaining, capacity remaining, utilization, configuration, and metrics as they affect the health, risk, or efficiency. You can use this tab to evaluate the impact that alerts are having on the vSAN capacity disk and use that information to begin troubleshooting problems.

**Table 1268: vSAN Capacity Disk Summary Tab Options**

| Option | Description |
|---|---|
| Troubleshoot | Start the Troubleshooting Workbench with the current object in context. |
| Object Summary | This widget displays the details of the selected object. The widget also displays the number of resources associated with the selected object. |

| Option | Description |
|---|---|
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the labels of the alert. |
| Time Remaining | This widget displays the number of days remaining till the projected resource utilization crosses the threshold for the usable capacity. |
| Capacity Remaining | This widget displays the unused capacity of your virtual environment to accommodate new virtual machines. |
| Utilization | This widget is used to find out the trends in capacity used by a selected capacity disk as against the total capacity available. |
| Contention | This widget displays the memory contention details for the selected capacity disk. |

vSAN Cache Disk Summary Tab

The vSAN Cache Disk tab provides an overview of the state of the selected vSAN cache disk. For the selected object, the vSAN cache disk tab displays the alerts, time remaining, capacity remaining, utilization, configuration, and metrics as they affect the health, risk, or efficiency. You can use this tab to evaluate the impact that alerts are having on the vSAN cache disk and use that information to begin troubleshooting problems.

**Table 1269: vSAN Cache Disk Summary Tab Options**

| Option | Description |
|---|---|
| Troubleshoot | Start the Troubleshooting Workbench with the current object in context. |
| Object Summary | This widget displays the details of the selected object. The widget also displays the number of resources associated with the selected object. |
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the labels of the alert. |
| Time Remaining | This widget displays the number of days remaining till the projected resource utilization crosses the threshold for the usable capacity. |
| Capacity Remaining | This widget displays the unused capacity of your virtual environment to accommodate new virtual machines. |
| Utilization | This widget is used to find out the trends in capacity used by a selected vSAN cache disk as against the total capacity available. |
| Contention | This widget displays the memory contention details for the selected cache disk. |

vSAN Cluster Fault Domain Summary Tab

The vSAN cluster fault domain summary tab provides details about CPU, CPU Cores, Memory, Disc Space and Alerts associated with the fault domain of the vSAN cluster.

**Where To View vSAN Cluster Fault Domain Summary**

On the menu, click **Inventory** > **Inventory Panel (Detailed View)** > **Integrations** > **All Objects** > **vSAN Adapter** > **vSAN and Storage Devices** > **vSAN Cluster** > **Fault Domain**.

You can also view relationship details and heat map details for the selected vSAN fault domain. The relationship section provides information about the relationship between the objects in your vSAN cluster. The heat map helps you to identify potential problems for the objects in your vSAN fault domain.

vSAN World Summary Tab

The vSAN World summary tab provides an overview of the state of all the vSAN clusters in your environment. For the selected object, the vSAN World Summary tab displays details about capacity remaining, cluster health score, vCenter, Datacenter, ESX host, vSAN Capacity, Space Efficiency, alerts, and metrics as they affect the health, risk, or efficiency of the cluster. You can use this tab to identify the vSAN ESA (Express Storage Architecture) clusters or non vSAN ESA clusters in your environment. You can evaluate the impact of alerts on vSAN clusters and use that information to begin troubleshooting problems.

**Where To View vSAN World Summary Page**

From the left menu, click **Inventory** > **Inventory Panel (Detailed View)** > **Integrations** > **All Objects** > **vSAN Adapter** > **vSAN World**, and then select a **vSAN World** object type.

**Table 1270: vSAN World Summary Tab Options**

| Option | Description |
|---|---|
| Troubleshoot | Start the Troubleshooting Workbench with the current object in context. |
| Object Summary | This widget displays the details of the selected object, you can also view the number of resources associated with the selected object. Displays the total number of vSAN clusters and the number of each vSAN cluster type. It also displays the total number of ESX hosts, virtual machines, and vSAN Datastore at the vSAN World level. |
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the labels of the alert. |

| Option | Description |
|---|---|
| vSAN Cluster Summary | vSAN World covers all the vSAN clusters in the environment. The cluster information is provided in a tabular format with the following columns:<br>• vSAN Cluster: Name of the vSAN cluster.<br>• Cluster Health Score: Displays the overall heath score of the cluster.<br>• vCenter: Displays the vCenter instance associated with the cluster.<br>• Datacenter: Displays the datacenter used by the cluster.<br>• vSAN Type: Displays the vSAN cluster type. For more information, see vSAN Cluster Summary Tab.<br>• ESX host: Displays the number of ESX hosts used in each cluster.<br>• vSAN Capacity: Displays the total capacity in gigabytes.<br>• Capacity Remaining: Displays the remaining capacity in each cluster in gigabytes.<br>• Space Efficiency: Displays the space efficiency feature the cluster has subscribed to during configuration.<br>• Stretched Cluster: Displays if the stretched cluster option is activated or deactivated. |

vSAN Adapter Summary Tab

The vSAN Adapter summary tab provides an overview of the state of all the vSAN adapters in your environment. For the selected object, the vSAN adapter Summary tab displays details about capacity remaining, cluster health score, vCenter, Datacenter, ESX host, vSAN Capacity, Space Efficiency, alerts, and metrics as they affect the health, risk, or efficiency of the adapter. You can evaluate the impact of alerts on vSAN adapters and use that information to begin troubleshooting problems.

**Where To View vSAN Adapter Summary Page**

From the left menu, click **Inventory** > **Inventory Panel (Detailed View)** > **Integrations** > **All Objects** > **vSAN Adapter** > **vSAN Adapter Instance**, and then select the **Adapter Instance** object type.

**Table 1271: vSAN Adapter Summary Tab Options**

| Option | Description |
|---|---|
| Troubleshoot | Start the Troubleshooting Workbench with the current object in context. |
| Object Summary | This widget displays the details of the selected object, you can also view the number of resources associated with the selected object. The vSAN ESA (Express Storage Architecture) column indicates whether the cluster is ESA activated or not. |
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the labels of the alert. |

| Option | Description |
|---|---|
| vSAN Cluster Summary | Each vSAN adapter instance can have one or more clusters. The cluster information is provided in a tabular format with the following columns:<br>• vSAN Cluster: Name of the vSAN cluster.<br>• Cluster Health Score: Displays the overall heath score of the cluster.<br>• Datacenter: Displays the datacenter used by the cluster.<br>• vSAN Type: Displays the vSAN cluster type. For more information, see vSAN Cluster Summary Tab.<br>• ESX host: Displays the number of ESX hosts used in each cluster.<br>• vSAN Capacity: Displays the total capacity in gigabytes.<br>• Capacity Remaining: Displays the remaining capacity in each cluster in gigabytes.<br>• Space Efficiency: Displays the space efficiency feature the cluster has subscribed to during configuration.<br>• Stretched Cluster: Displays if the stretched cluster option is activated or deactivated. |

vSAN Storage Pool Summary Tab

The vSAN Storage Pool summary tab provides an overview of the state of all the vSAN Storage pool in your environment. For the selected object, the vSAN Storage Pool Summary tab displays details about capacity remaining, vCenter, Datacenter, ESX host, vSAN Capacity, Space Efficiency, alerts, and metrics as they affect the health, risk, or efficiency of the cluster.You can evaluate the impact of alerts on the Storage Pool and use that information to begin troubleshooting problems.

**Where To View vSAN Storage Pool Summary Page**

From the left menu, click **Inventory** > **Inventory Panel (Detailed View)** > **Integrations** > **All Objects** > **vSAN Adapter** > **vSAN Storage Pool**, and then select a **Storage Pool** object type.

**Table 1272: vSAN Storage Pool Summary Tab Options**

| Option | Description |
|---|---|
| Troubleshoot | Start the Troubleshooting Workbench with the current object in context. |
| Object Summary | This widget displays the details of the selected object. The widget also displays the number of resources associated with the selected object. |
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the labels of the alert. |
| Time Remaining | This widget displays the number of days remaining till the projected resource utilization crosses the threshold for the usable capacity. |
| Capacity Remaining | This widget displays the unused capacity of your virtual environment to accommodate new virtual machines. |
| Contention | This widget displays the memory contention details of the vSAN Storage Pool. |

vSAN ESA Disk Summary Tab

The vSAN ESA Disk Summary tab provides an overview of the state of the selected vSAN ESA cluster. For the selected object, the vSAN ESA cluster tab displays the alerts, time remaining, capacity remaining, utilization, configuration, and metrics as they affect the health, risk, or efficiency. You can use this tab to evaluate the impact that alerts are having on the vSAN ESA cluster and use that information to begin troubleshooting problems.

**Where To View vSAN ESA Disk Summary Page**

From the left menu, click **Inventory** > **Inventory Panel (Detailed View)** > **Integrations** > **All Objects** > **vSAN Adapter** > **vSAN ESA Disk**, and then click a **vSAN ESA Disk** object type.

**Table 1273: vSAN ESA Disk Summary Tab Options**

| Option | Description |
|---|---|
| Troubleshoot | Start the Troubleshooting Workbench with the current object in context. |
| Object Summary | This widget displays the details of the selected object. The widget also displays the disk details like, size, vendor, model, queue depth, and disk type. |
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the labels of the alert. |
| Time Remaining | This widget displays the number of days remaining till the projected resource utilization crosses the threshold for the usable capacity. |
| Capacity Remaining | This widget displays the unused capacity of your virtual environment to accommodate new virtual machines. |
| Utilization | This widget is used to find out the trends in capacity used by a selected vSAN cluster as against the total capacity available. |
| Contention | This widget displays the memory contention details of the vSAN cluster. |

**Business Application Summary Tab**

The Business Application Summary tab provides an overview of the selected Business Application. Each card on the page provides additional context such as overview, active alerts, KPIs, and object relationships between members of the Business Application.

**Understanding the Business Application Summary Tab**

**Table 1274: Business Application Summary Tab Options**

| Option | Description |
|---|---|
| Troubleshoot button | Launch the Troubleshooting Workbench with the current Business Application in context. |
| Object Summary card | Displays the details of the current Business Application.<br>• Source<br>• Application Tag<br>• Environment<br>• Business Criticality<br>• Description |
| Active Alerts card | Displays a summary of active alerts divided by **Self** and **All**, which are triggered on the Business Application and the child objects.<br>For more details about the alerts by object, see the relationship diagram, or the relationship table section. |
| Relationship view diagram | This section provides two types of relationship diagram views:<br>• Application View. This is the same relationship diagram which you can view in the Business Application page.<br>• Infrastructure View. Displays a graph of the Virtual Machines in the Business Application and their vSphere infrastructure.<br>You can switch between the horizontal view and vertical view by toggling the orientation icon on the right of the relationship view diagram.<br>Move your cursor over an object in the relationship diagram to view more details in a transient pop-up dialog box which appears. |
| Relationship view table | Displays the following details for the objects listed in the Business Application relationship diagram:<br>• Name<br>• Object Type<br>• Capacity Remaining<br>• Alerts |

## Evaluating Metric Information

The **Metrics** tab provides a relationship map and user-defined metric charts. The topological map helps you evaluate objects in the context of their place in your environment topology. The metric charts are based on the metrics for the selected object that you think helps identify the possible cause of a problem in your environment.

Although you might be investigating problems with a single object, for example, a host system, the relationship map allows you to see the host in the context of parent and child objects. It also works as a hierarchical navigation system. If you double-click an object in the map, that object becomes the focus of the map. The available metrics for the object become active in the lower-left pane.

> **Note:**
> The yellow diamond icon next to the metric indicates dynamic threshold breach, the blue diamond icon next to the metric indicates the metric value is within threshold.

You can also build your own set of metric charts. You select the objects and metrics that provide you with a detailed view of changes to different metrics for a single object, or for related objects over time.

Where available, the **Metrics** tab provides pre-defined sets of metrics to help you when looking at a specific aspect of an object. For example, if you have a problem with a host, access the most relevant information about the host by looking at the metrics displayed in the pre-defined lists. You can edit these groups of metrics, and create additional groups, by dragging and dropping metrics and properties from the All Metrics and All Properties lists.

**Where You Find the Metrics Tab**

- In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory** and then select a group, custom data center, application, or inventory object.

## Create Metric Charts When You Troubleshoot a Virtual Machine Problem

You create a custom group of metric charts when you troubleshoot a problem with a virtual machine so that you can compare different metrics. The level of detail that you can create using the **Metrics** tab, can contribute significantly to your effort to find the root cause of a problem.

As an administrator investigating a performance problem with a virtual machine, you determined that you must see detailed charts about the following reported symptoms.

- Guest file system overall disk space usage reaching critical limit
- Guest partition disk space usage

The following method of evaluating problems using the **Metrics** tab is provided as an example for using VCF Operations and is not definitive. Your troubleshooting skills and your knowledge of the particulars of your environment determine which methods work for you.

1. Enter the name of the virtual machine in the **Search** text box on the menu bar.
   In this example, the virtual machine name is `sales-10-dk`.
2. Click the **Metrics** tab.
3. In the relationship topology map, click the virtual machine, **dk-new-10**.
   The metrics list, located in the left of the center pane, displays virtual machine metrics.
4. On the chart toolbar, click **Date Control** and select a time that is on or before the symptoms were triggered.
5. Add metric charts to the display area for the virtual machine.
   a) In the metric list, select **Guest Files System Stats** > **Total Guest File System Free (GB)** and double-click the metric name.
   b) To add the guest partition, for example, C:\, select **Guest Files System Stats** > **C:** > **Guest File System Free (GB)** and double-click the metric name.
   c) To add disk space for comparison, select **Disk Space** > **Capacity Remaining (%)** and double-click the metric name.
6. Compare the charts.
   You can see a decrease in the file system free space, and that the virtual machine disk space capacity remaining is decreasing at a steady rate. You determine that you must add disk space to the virtual machine. However, you do not know if the datastore can support the change to the virtual machine.
7. Add the datastore capacity chart to the charts.
   a) In the topology map, double-click the host.
      The topology map refreshes with the host as the focus object.
   b) Click the datastore.
   c) In the metric list, which is updated to display datastore metrics, select **Capacity** > **Available Space (GB)** and double-click the metric name.
8. To determine if sufficient capacity is available on the datastore to support increasing the disk space on the virtual machine, review the datastore capacity chart.

You know that you must increase the size of the virtual disk on the virtual machine.
Expand the virtual disk on the virtual machine and assign it to stressed partitions. Click **Actions**, on the object title bar, and view the virtual machine in the vSphere Web Client.

## Troubleshooting with the Metrics Tab

The **Metrics** tab provides a relationship graph and metric charts. The relationship graph helps you evaluate objects in the context of their place in your environment topology. Metric charts are based on the metrics for the active map object that you think can help you identify the cause of a problem.

### How the Metrics Tab Works

You can double-click any object in the graph and view the specific parent-child objects for the focus object. If you point to an object icon, you can see the health, risk, and efficiency details. You can also click the **Alerts** link for the number of generated alerts. Click the purple icon to view the child relationships of the object. If you double-click an object icon, the selected object becomes the focus of the map. The graph is updated for the selected object, and the metrics list shows only the metrics for the selected object.

Using the metrics list, you create charts based on metrics that you think can help you investigate problems. You customize the charts to evaluate the data in detail. To save the configured charts, you create a dashboard using the toolbar option.

Where available, the metrics list also displays pre-defined groups of metrics that contain the most relevant metrics for the selected object. You can edit these groups, and create your own customized groups of metrics by dragging and dropping metrics and properties from the All Metrics and All Properties lists.

### Where You Find the Metrics Tab

- In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory** and then select a group, custom data center, application, or inventory object.
- Alternatively, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory** and then use the hierarchies in the left pane to locate the objects that you want.

### Edit Metric State

The edit metric state option allows you activate, deactivate, or inherit metric state, DT state and KPI state. You can use the All Filter option to search and edit the metrics based on the following criteria.

- Name
- State
- Instanced State
- DT State
- KPI State
- Show Only Supermetrics
- Local Changes
- Unsaved Changes

### Metrics Options

The options include the graph toolbar, the metric selector options, the metric charts toolbar, and the toolbar on each chart.

**Table 1275: Relationship Map Toolbar Options**

| Option | Description |
|---|---|
| Back to initial object | Returns the map to original object if you double-clicked on an icon to examine another object. |
| Vertical/Horizontal | Displays a vertical or horizontal view of the graph or tree view. |
| Hide Text/Show Text | Hides or displays the object names. |
| Standard View/Fit View | The **Standard View** option fixes the view to a specific zoom level. The **Fit View** option adjusts the graph or tree view to fit the screen. |

**Table 1276: Specific Functions from the Relationship Map**

| | |
|---|---|
| Hide Node | Hover your cursor over an object and click the **Hide Node** icon to hide the specific node. |
| Show Peers/Hide Peers | Hover your cursor over the selected object in the topology pane and select **ShowPeers/Hide Peers** to view or hide other objects of the same type that exist under the parent object. |
| Pagination/Next Page/Previous Page | At the bottom of the topological view you can view the page number and also move to the next/previous page. |
| Filter | At the bottom of the topological view you can use the filter option to search for objects. |
| Arrows | Use the arrows on each object to view relationships of each object. |

The chart options are used to limit the metric list.

**Table 1277: Metric Chart Selector**

| Option | Description |
|---|---|
| Show collecting metrics | Updates the list to display only the currently collected metrics for the object. |
| Show previewable super metrics | Updates the list to display super metrics for the object.<br>**Note:**<br>The super metrics only appear if the super metric is associated with the object, see Create a Super Metric topic in *VCF Operations Configuration Guide*. . |
| Actions | Click the **Actions** icon to configure metric groups. Verify that you hold the PowerUser or administrator role.<br>• **Add Group**. To add metrics or properties to the group, expand any of the metric groups, and drag one or more metrics to the group.<br>• **Remove Group(s)**. To remove one or more groups.<br>• **Rename Group**. To enter a new name for the group.<br>• **Remove Metric(s) from Group(s)**. To remove one or more metrics or properties from one or more groups, hold down the Ctrl key, and select the metrics or properties that you want to remove. |
| Search | Use a word search to limit the number of items that appear in the list. |
| Time Range | Filters the metrics to show only the ones that have received data in the selected time range. |
| Metric list | Double-click a metric to populate the chart window. To populate the chart window with a separate chart for each of the metrics in the group, double-click a metric group. |

To visualize the specific metric data over time, and compare the results for different metrics, select different combinations of options.

**Table 1278: Metric Chart Toolbar**

| Option | Description |
|---|---|
| Split Charts | Displays each metric in a separate chart. |
| Stacked Chart | Consolidates all charts into one chart. This chart is useful for seeing how the total or sum of the metric values vary over time. To view the stacked chart, ensure that the split chart option is turned off. |
| Y Axis | Shows or hides the Y-axis scale. |
| Metric Chart | Shows or hides the line that connects the data points on the chart. |
| Trend Line | Shows or hides the line and data points that represents the metric trend. The trend line filters out metric noise along the timeline by plotting each data point relative to the average of its adjoining data points. |
| Dynamic Thresholds | Shows or hides the calculated dynamic threshold values for a 24-hour period. |
| Show Entire Period Dynamic Thresholds | Shows or hides dynamic thresholds for the entire time period of the graph. |
| Anomalies | Shows or hides anomalies. Time periods when the metric violates a threshold are shaded. Anomalies are generated when a metric crosses a dynamic or static threshold, either above or below. |
| Show Data Point Tips | Shows or hides the data point tooltips when you hover the mouse over a data point in the chart. |
| Zoom All Charts | Resizes all the charts that are open in the chart pane based on the area captured when you use the range selector.<br>You can switch between this option and **Zoom the View**. |
| Zoom the View | Resizes the current chart when you use the range selector. |
| Pan | When you are in zoom mode, allows you to drag the enlarged section of the chart so that you can view higher or lower, earlier or later values for the metric. |
| Show Data Values | Activates the data point tooltips if you switched to a zoom or pan option. **Show Data Point Tips** must be activated. |
| Refresh Charts | Reloads the charts with current data. |
| Date Controls | Opens the date selector.<br>Use the date selector to limit the data that appears in each chart to the time period you are examining. |
| Generate Dashboard | Saves the current charts as a dashboard. |
| Remove All | Removes all the charts from the chart pane, allowing to you begin constructing a new set of charts. |

Manage individual charts with the toolbar options.

**Table 1279: Individual Metric Charts Toolbar**

| Option | Description |
|---|---|
| Navigation | If an adapter includes the ability to link to another application for information about the object, click the button to access a link to the application. |
| Correlation | Runs metric correlation using the following options:<br>**Self-Metrics**: Runs metric correlation on all metrics for the selected object, to find metrics of similar, or opposite behavioral change for the same time period. The instanced metrics are not assessed in the self-metrics correlation method.<br>**Peers**: Runs metric correlation on the same metrics for all peer objects, to find the same metrics with behavioral changes within peer objects. Peer objects are the direct child objects of the parent for the selected objects. The child objects have the same object type.<br>**Note:**<br>The correlation results only appears if there are at least 11 data points and the time range is within the three months period to run the metric correlation.<br>**Scope**: Runs metric correlation on all metrics for the selected object with the selected scope, to find metrics of similar, or opposite behavioral change for the same time period. The instanced metrics are not assessed in the scope correlation method.<br>After you run the correlation, the results are displayed in the **Corr elation** window. By default, only the first 10 results for correlated metrics are displayed. To view the full list, click **Show More**.<br>You can zoom in to view the correlated metrics and also pin them so that they appear in the preview section of the **All Metrics** tab.<br>**Note:**<br>During the correlation process, some metrics are left out. For example, the badge and VCF Operations generated metrics. By default, the instanced metrics are omitted, except those in the **Aggregate of all instances** group. |
| Save a Snapshot | Creates a PNG file of the current chart. The image is the size that appears on your screen.<br>You can retrieve the file in your browser's download folder. |
| Save a Full Screen Snapshot | Downloads the current graph image as a full-page PNG file, which you can display or save.<br>You can retrieve the file in your browser's download folder. |
| Create an Alert Definition | Allows you to create an alert for an object type or metric in a quick and easier way. For details, see Create a Simple Alert Definition section in *VCF Operations Configuration Guide.* . |
| Download comma-separated data | Creates a CSV file that includes the data in the current chart.<br>You can retrieve the file in your browser's download folder. |

| Option | Description |
|---|---|
| Scales | You can choose a scale for a stacked chart.<br>• Select **Linear** to view a chart in which the Y-axis scale increases in a linear manner. For example, the Y-axis can have ranges from 0 to 100, 100 to 200, 200 to 300, and so on.<br>• Select **Logarithmic** to view a chart in which the Y-axis scale increases in a logarithmic manner. For example, the Y axis can have ranges from 10 to 20, 20 to 300, 300 to 4000, and so on. This scale gives a better visibility of minimum and maximum values in the chart when you have a large range of metric values.<br>**Note:**<br>If you select a logarithmic scale, the chart does not display data points for metric values less than or equal to 0, which leads to gaps in the graph.<br>• Select **Combined** to view overlapping graphs for the metrics. The chart uses individual scales for each graph instead of using a relative scale, and displays a combined view of the graphs.<br>• Select **Combined by Unit** to view a chart that groups the graphs for similar metric units together. The chart uses a common scale for the combined graphs. |
| Move Down | Moves the chart down one position. |
| Move Up | Moves the chart up one position. |
| Close | Deletes the chart. |
| Vertical resize | Resizes the height of a graph in the chart. |
| **Remove** icon next to each metric name in a stacked chart | Removes the graph for the metric from the chart. |

## Investigating Object Alerts

The **Alerts** tab provides a list of generated alerts for the currently selected object. When you are working with objects, reviewing and responding to generated alerts on the **Alert** tab helps you manage problems in your environment.

The alerts notify you when a problem occurs in your environment based on configured alert definitions. Object alerts are useful to you as an investigative tool in two ways. They can provide you with early notification about problems in your environment before a user calls you to report a problem. As well, object alerts can provide information about the object that you can use when troubleshooting general or reported problems.

As you review the **Alerts** tab, you can add ancestors and descendants to the list to broaden your view of the alerts. You can see if alerts on the current object affect other objects. Conversely, you can examine how problems reflected in alerts on other objects affect the current object.

Depending on the practices and workflows of your infrastructure operations team, you can use the object **Alerts** tab to manage generated alerts on individual objects.

- Take ownership of alerts so that your team knows that you are working to resolve the problem.
- Suspend an alert so that is temporarily excluded from affecting the Health, Risk, or Efficiency state of the object while you investigate the problem.
- Cancel alerts that you know are a result of a deliberate action. For example, a network card is removed from a host for replacement. Also cancel alerts that are known issues that you cannot resolve currently because of resource constraints. Canceling an alert that is generated because of only message event or metric event symptoms cancels the alert permanently. If the underlying metric or property condition remains true, canceling an alert that is generated because of metric, super metric, or property symptoms can result in the alert being regenerated . It is only effective to cancel alerts generated because of message event or metric event symptoms.

Investigating and resolving alerts helps you provide the best possible environment to your customers.

## Alerts Tab

The Alerts tab is a list of all the alerts generated for the selected object, group, or application. Use the alerts list to evaluate the number of generated alerts for the object so that you can begin resolving them.

### How the Alerts Tab Works

All the active alerts for the selected object appear in the list. By default, the system groups the alerts by Definition. You can select multiple rows in the list using Shift+click, Control+click. Modify the filter if you want to see inactive alerts.

Manage the alerts in the list using the toolbar options. Click the **alert name** to see the alert details for the affected object. The alert details appear on the right, including the symptoms triggered with the alert. The system offers recommendations for addressing the alert and links to additional information. A **Run Action** button might appear in the details. Point to the button to learn what recommendation is performed if you click the button. To return to the list view, click the **X** at the top right of the alert details.

To see the object details, click the **Summary** Tab.

### Where You Find the Alerts Tab

- In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory**, and then select a group, custom data center, application, or inventory object. Click the **Alerts** tab.

### Alerts Display Options

The alert options include toolbar and data grid options. Use the toolbar options to sort the alert list and to cancel, suspend, or manage ownership. Additional toolbar options enable you to review parent and child alerts related to the alert you are reviewing. Use the data grid to view the alerts and alert details.

**Table 1280: Actions Menu**

| Option | Description |
|---|---|
| Actions menu | Select an alert from the list to turn on the Actions menu, then select an option from the menu. |
| Menu Options: | |
| Cancel Alert | Cancels the selected alerts. If you configure the alert list to display only active alerts, the canceled alert is removed from the list.<br>You cancel alerts when you do not need to address them. Canceling the alert does not cancel the underlying condition that generated the alert. Canceling alerts is effective if the alert is generated by triggered fault and event symptoms because these symptoms are triggered again only when subsequent faults or events occur on the monitored objects. If the alert is generated based on metric or property symptoms, the alert is canceled only until the next collection and analysis cycle. If the violating values are still present, the alert is generated again. |
| Delete Canceled Alerts | Delete canceled (inactive) alerts by making a group selection or by individually selecting alerts. You cannot delete active alerts. |
| Suspend | Suspend an alert for a specified number of minutes.<br>You suspend alerts when you are investigating an alert and do not want the alert to affect the health, risk, or efficiency of the object while you are working. If the problem persists after the elapsed time, the alert is reactivated and it will again affect the health, risk, or efficiency of the object.<br>The user who suspends the alert becomes the assigned owner. |
| Assign to | Assign the alert to a user. You can search for a specific username and click **Save** to assign the alert to the selected user. |

| Option | Description |
|---|---|
| Take Ownership | As the current user, you make yourself the owner of the alert. You can only take ownership of an alert, you cannot assign ownership. |
| Release Ownership | Alert is released from all ownership. |
| Go to Alert Definition | Switches to the Alert Definitions page, with the definition for the previously selected alert displayed. |
| Deactivate | Offers two options for disabling the alert:<br>Deactivate the alert in all policies: this deactivates the alert for all objects for all the policies.<br>Deactivate Alert in Selected Policies: this deactivates the alert for objects having the selected policy. This method works only for objects with alerts. |
| Open an external application | Actions you can run on the selected object.<br>For example, Open Virtual Machine in vSphere Client. |

**Table 1281: Include Menu**

| Options | Description |
|---|---|
| Self | The selected object. |
| Parents <options> | Displays the alerts for the ancestors of the selected object. Parents in this instance include the parents, grandparents, and so on, of the object. For example, the parents of a host are a folder, storage pod, cluster, and data center instance. |
| Children <options> | Displays the alerts for the descendants of the selected object. Children in this instance include the children and grandchildren of the object. For example, the descendants of a host are datastores, resources pools, and virtual machines. |
| Peer | Shows or hides alerts for objects with the same kind of the impacted object which also share the same parent. |

**Table 1282: Group By Options**

| Option | Description |
|---|---|
| None | Alerts are not sorted into specific groupings. |
| Time | Group alerts by time triggered. |
| Criticality | Group alerts by criticality. Values are, from the least critical: Info/ Warning/Immediate/Critical. See also Criticality in the "All Alerts Data Grid Options" table, below. |
| Definition | Group alerts by definition, that is, group like alerts together. Selected by default. |
| Object Type | Group alerts by the type of object that triggered the alert. For example, group alerts on hosts together. |
| Scope | Group alerts by scope. You can search for alerts within the selected scope. |

**Table 1283: Quick Filters**

| | |
|---|---|
| Filtering options | The advanced search and filter lets you search for a symptom by:<br>• Alert ID<br>• Alert<br>• Owner<br>• Impact<br>• Alert Type<br>• Alert Subtype<br>• Status<br>• Criticality<br>• Triggered On<br>• Control State<br>• Object Type<br>• Created On<br>• Updated On<br>• Canceled On<br>• Action |

**Table 1284: Alerts Data Grid**

| Option | Description |
|---|---|
| Criticality | Criticality is the level of importance of the alert in your environment. The alert criticality appears in a tooltip when you hover the mouse over the criticality icon.<br>The level is based on the level assigned when the alert definition was created, or on the highest symptom criticality, if the assigned level was **Symptom Based**.<br>The possible values include:<br>• Critical<br>• Immediate<br>• Warning<br>• Information |
| Alert | Name of the alert definition that generated the alert.<br>Click the alert name to view the alert details tabs where you can begin troubleshooting the alert. |
| Created On | Date and time when the alert was generated. |
| Status | Current state of the alert.<br>Possible values include Active or Canceled. |
| Alert Type | Describes the type of alert that triggered on the selected object, and helps you categorize the alerts so that you can assign certain types of alerts to specific system administrators. For example, Application, Virtualization/Hypervisor, Hardware, Storage, Network, Administrative, and Findings. |
| Alert Subtype | Describes additional information about the type of alert that triggered on the selected object, and helps you categorize the alerts to a more detailed level than Alert Type, so that you can assign certain types of alerts to specific system administrators. For example, Availability, Performance, Capacity, Compliance, and Configuration. |
| Importance | Displays the priority of the alert. The importance level of the alert is determined using a smart ranking algorithm. |

### Alert Details Tab

| Section | Description |
|---|---|
| Recommendations | View recommendations for the alert. Click **<** or **>** to cycle through the recommendations. To resolve the alert, click the **Run Action** button if it appears. |
| Other Recommendations | Collapse the section to view additional recommendations. See the links in the **Need More Information?** section to view additional metrics, events, or other details that appear as a link. |
| Alert Basis | |
| Active Only | This option is activated by default. When activated, all active symptoms/conditions that were met for the alert are displayed. When deactivated, all the symptoms/conditions of an alert are displayed. |
| Symptoms | View the symptoms that triggered the alert. Collapse each symptom to view additional information. |
| Conditions | View the conditions that triggered the alert. Collapse each condition to view additional information. |
| Notes | Enter your notes about the alert and click **Submit** to save. |
| Close | Click the X icon to close the alert details tab. |

### Related Alerts Tab

The **Related Scope** displayed on the right, shows the objects that are one level above and one level below the object on which the alert was triggered. This topology is fixed. You cannot change the scope in the **Related Alerts** tab.

On the right, you can see the following:

- If the same alert was triggered on the object in the past 30 days. This helps you understand if this is a recurring problem or something new.
- If the same alert was triggered on other peers in the same environment, in the past 30 days. This helps you do a quick peer analysis to understand if others are impacted with the same problem.
- All the alerts triggered in the current topology. This helps you investigate if there are other alerts upstream or downstream in the environment which are impacting the health of the object.

### Potential Evidence Tab

See the **Potential Evidence** tab for potential evidences around the problem, and to arrive at the root cause. This tab displays events, property changes, and anomalous metrics potentially relevant to the alert. The time range and the scope are fixed. To modify the scope or the time range and investigate further, click **Launch Workbench**. This runs the troubleshooting workbench.

The time range that is displayed in the potential evidence tab is two hours and thirty minutes before the alert was triggered. VCF Operations looks for potential evidences in this time range.

## Symptoms Display

The symptoms display includes all the symptoms triggered for the current object. Use the symptom list to identify problems with an object so that you can resolve alerts generated for the object.

### How the Symptoms Work

The list is the active triggered symptoms for an object, either as part of a generated alert or as a triggered symptom that is not included in an alert. This complete symptom list is useful for identifying problems that occur on an object but are not currently included in your alert definitions.

Click a symptom in the list to display the symptom details. An arrow in each column heading enables you to order the list in ascending or descending order. You can select multiple rows in the list using Shift+click, Control+click.

**Where You Find the Symptoms Display**

- In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory** and then select a group, custom data center, application, or inventory object. Click the **object** to display the object's **Summary** tab. Click the **Symptoms** subtab in the **Alerts** tab.
- In the menu, select **Search** and locate the object of interest. Click the **object** to display the object's **Summary** tab. Click the **Symptoms** subtab in the **Alerts** tab.

**Table 1285: Symptoms Data Grid**

| Option | Description |
|---|---|
| Criticality | Criticality is the level of importance of a symptom in your environment.<br>The level is based on the same level assigned when the symptom was created. The possible values include:<br>• Critical<br>• Immediate<br>• Warning<br>• Information |
| Symptom | Name of the triggered symptom. |
| Status | Current state of the symptom.<br>Possible values are Active or Inactive. |
| Created On | Date and time when the alert was generated. |
| Canceled On | Date and time when the symptom was canceled. |
| Information | Information about the triggering condition for the symptom, including the trend and current value.<br>The sparkline displays a range of data that includes six hours before the symptom update time and one hour after the update time. |

**Table 1286: Actions Menu**

| Option | Description |
|---|---|
| Actions menu | Select the **Go To Symptom Definition** option to open the **Symptom Definitions** page in the **Configure** > **Alerts** path. |

**Table 1287: Include**

| Option | Description |
|---|---|
| Self | Shows or hides symptoms for the current object. |
| Peer | Shows or hides symptoms for objects like the impacted object. |
| Parents <options> | Shows or hides symptoms for the parent, grandparent, and so on, objects of the current object. |
| Children <options> | Shows or hides the symptoms for the descendants of the impacted object. |

**Table 1288: Filters**

| | |
|---|---|
| Filtering options | The advanced search and filter lets you search for a symptom by:<br>• Symptom<br>• Status<br>• Criticality<br>• Triggered On<br>• Created On<br>• Canceled On |

**Table 1289: Group By Options**

| Option | Description |
|---|---|
| None | Symptoms are not sorted into specific groupings. |
| Time | Group Symptoms by time triggered. |
| Criticality | Group Symptoms by criticality. Values are, from the least critical: Info/Warning/Immediate/Critical. |
| Definition | Group Symptoms by definition, that is, group like Symptoms together. The default. |
| Object Type | Group Symptoms by the type of object that triggered the alert. For example, group Symptoms on hosts together. |
| Scope | Group Symptoms by scope. You can search for Symptoms within the selected scope. |

## Events Display

An event is any change to an object defined by a change in the metrics for that object. You can compare changes to an object with symptoms and other data to identify a possible cause for a generated alert.

### How the Events Display Works

The Events tab is a list of all the events generated for the selected object, group, or application based on messages send by either external sources or by internal self troubleshooting system.

The following vCenter activities are some of the activities that generate VCF Operations events:

- Powering a virtual machine on or off
- Creating a virtual machine
- Installing VMware Tools on the guest OS of a virtual machine
- Adding a newly configured ESX system to a vCenter system

Depending on alert definitions, these events might generate alerts.

You might monitor the same virtual machines with other applications that provide information to VCF Operations, with the adapters for those applications configured to provide change events. In this instance, the **Events** tab includes certain change events that occur on the monitored objects. These change events might provide further insight into the cause of problems that you are investigating.

### Where You Find the Events Display

- In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory** and then select a group, custom data center, application, or inventory object. Click the **object** to display the object's **Summary** tab. Click the **Events** subtab in the **Alerts** tab.

- In the VCF Operations navigation bar at the top, select **Search** and locate the object of interest. Click the **object** to display the object's **Summary** tab. Click the **Events** subtab in the **Alerts** tab.

**Table 1290: Include**

| Option | Description |
|---|---|
| Self | Shows or hides events for the current object. |
| Peer | Shows or hides events for objects like the impacted object. |
| Parents <options> | Shows or hides events for the parent, grandparent, and so on, objects of the current object. |
| Children <options> | Shows or hides the events for the descendants of the impacted object. |

**Table 1291: Events Data Grid**

| Option | Description |
|---|---|
| Criticality | Criticality is the level of importance of the event in your environment. The event criticality appears in a tooltip when you hover the mouse over the criticality icon.<br>The level is based on the level assigned when the event definition was created, or on the highest symptom criticality, if the assigned level was **Symptom Based**.<br>The possible values include:<br>• Critical<br>• Immediate<br>• Warning<br>• Information |
| Object Name | Name of the event definition that generated the event. Click the event name to view the event details tabs where you can begin troubleshooting the event. |
| Event Type | Describes the type of event that triggered on the selected object, and helps you categorize the event so that you can assign certain types of events to specific system administrators. For example, Application, Virtualization/Hypervisor, Hardware, Storage, Network, Administrative, and Findings. |
| Information | Describes additional information about the type of event that triggered on the selected object, and helps you categorize the events to a more detailed level. |
| Created On | Date and time when the event was generated. |
| Cancelled On | Date and time when the event was cancelled. |

**Table 1292: Filters**

| | |
|---|---|
| Filtering options | The advanced search and filter lets you search for a event by:<br>• Symptom<br>• Status<br>• Criticality<br>• Triggered On<br>• Created On<br>• Canceled On |

**Table 1293: Group By Options**

| Option | Description |
|---|---|
| None | Events are not sorted into specific groupings. |
| Time | Group events by time triggered. |
| Criticality | Group events by criticality. Values are, from the least critical: Info/ Warning/Immediate/Critical. |
| Definition | Group events by definition, that is, group like events together. |
| Object Type | Group events by the type of object that triggered the alert. For example, group events on hosts together. |
| Scope | Group events by scope. You can search for events within the selected scope. |

## Timeline Display

The generated alerts, triggered symptoms, and change events for the current object over time appear on the **Timeline** display. You use the timeline to identify common trends over time that are contributing to the status of objects in your environment.

### How the Timeline Display Works

The Timeline works by combining the data for alerts, symptoms and events and displaying it in a unified view. By default only active items are shown, however you can customize the data for each independently tab by using the filters to the top right and from the **INCLUDE** menu, which allows you to use parent, child and peer options to examine the object in a broader environmental context, as it is not always effective to investigate a problem on an individual object by looking only at the object. This context often reveals unexpected influences or consequences for the problem.

The timeline is a tool that provides you a graphical view of patterns. If the system triggers a symptom and then cancels it at various intervals over time, you can compare the event to other changes to the object or to the related objects. These changes might be the root cause of the problem.

The view is limited to 700 alerts, symptoms, and events accordingly. If your timeline includes more than this number, you can use the filters in each tab to remove data from the timeline until it contains data that you find useful for your investigation.

### Where You Find the Events Timeline

- In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory** and then select a group, custom data center, application, or inventory object. Click the **object** to display the object's **Summary** tab. Click the **SHOW TIMELINE** button in the **Alerts** tab.
- In the VCF Operations navigation bar at the top, select **Search** and locate the object of interest. Click the **Alert** tab, then to display the object's **Summary** subtab. Click the **SHOW TIMELINE** button in the **Alerts** tab

**Table 1294: Filter By**

| Option | Description |
|---|---|
| Filter By | Filter By Menu uses the data from Timeline view to display controls for further filtering the data.<br>It can contain alert types and subtypes, event types and criticality levels. Additionally it its possible to hide entire sections like Alerts, Symptoms or Events. |

**Table 1295: Time Range**

| Option | Description |
|---|---|
| Range | Select a time range. The default is **Last 24 hours.** |
| From - To | Select a time range by inputting a combination of date and time for the **From** and **To** fields. |

## Topology Tab

The Topology tab allows you to view the relationship of the objects in a topological view.

**Table 1296: Topology View Toolbar Options**

| Option | Description |
|---|---|
| Back to initial object | Returns the map to original object if you double-clicked on an icon to examine another object. |
| View Tree | Displays a tree view of the relationships. |
| View Graph | Displays a graph view of the relationships. |
| Applications | Displays relationships that are of the type 'application'. |
| View Additional Links | Displays additional links for the applications. |
| Vertical/Horizontal | Displays a vertical or horizontal view of the graph or tree view. |
| Hide Text/Show Text | Hides or displays the object names. |
| Standard View/Fit View | **Standard View**: Fixes the view to a specific zoom level.<br>**Fit View**: Adjusts the graph or tree view to fit the screen. |
| Settings | • **Parent Depth**: Displays up to five levels of parent objects for the selected object depending on your selection.<br>• **Child Depth**: Displays up to five levels of child objects for the selected objects, depending on your selection.<br>• **Page Size**: Sets the number of objects per page. |

**Table 1297: Specific Functions from the Topology Grid View**

| | |
|---|---|
| Hide Node | Hover your cursor over an object and click the **Hide Node** icon to hide the specific node. |
| Show Peers/Hide Peers | Hover your cursor over the selected object in the topology pane and select **ShowPeers/Hide Peers** to view or hide other objects of the same type that exist under the parent object. |
| Pagination/Next Page/Previous Page | At the bottom of the topological view you can view the page number and also move to the next/previous page. |
| Filter | At the bottom of the topological view you can use the filter option to search for objects. |
| Arrows | Use the arrows on each object to view relationships of each object. |

## Environment Tab

VCF Operations discovers the descendants of the selected object and lists them. The Environment tab provides you with specific data about the object.

## How the Environment Tab Works

Descendant objects of the object you selected appear in the datagrid of the **Environment** tab. To find a particular object, you can sort a column in the grid or search for a word in the filter, or use the filter options.

## Where You Find the Environment Tab

From the left menu, click **Inventory** and then click a vCenter instance from the datagrid. From the **Inventory** panel, select an object and then click the **Environment** tab.

Use the toolbar options to manage objects.

- Filter options limit the list to objects matching the filter. Filter options include All, ID, Internal ID, Name, Description, Adapter Type, Object Type, and Identifiers.
- Select the object to manage from the list.

### Table 1298: Toolbar Options

| Option | Description |
|---|---|
| Actions | Perform an action on the selected object. Available actions depend on the object type. For example, Power on VM applies to the selected virtual machine. |
| Open in external application | If an adapter includes the ability to link to another application for information about the object, click the button to access a link to the application. For example, Open Virtual Machine in a vSphere Client or Search for VM logs in Log management. |
| Edit Object | Edit the selected object. For example, add or change the maintenance schedule for a virtual machine. If multiple objects of the same type are selected, common identifiers for the object type are editable. For example, change the VM entity name of multiple datastores with a single edit. |
| Add Object | VCF Operations discovers objects for most adapters. For adapters that do not support autodiscovery for all objects, the objects are manually added. See Manage Objects Workspace. |
| Discover Objects | Perform an IP scan to discover objects associated with a particular adapter. See Discover Objects Workspace. |
| Delete object | Remove the object from the list. |
| Start maintenance | Take the object offline for maintenance. See Manage Maintenance Schedules for Your Object Workspace. |
| End maintenance | Terminate the maintenance period and put the selected object back online. |
| Clear Selections | Clear all object selections. |
| Select All | Select all objects displayed. |
| Add/Edit Custom Property | Define or add custom properties. |
| Go to Details | Display the **Summary** tab of the selected object. |
| Page Size | The number of objects to list per page. |

### Table 1299: Datagrid Options

| Option | Description |
|---|---|
| Name | Displays the name of the object. |
| Adapter Type | Displays the name of the adapter. |

| Option | Description |
|---|---|
| Object Type | Displays the type of object. |
| Policy | Displays the policy associated with the object. |
| Collection State | Displays the collection state of an adapter instance for each object. |
| Collection Status | Displays the collection status of an adapter instance of each object. |

# Configuring Automation Jobs

You can automate jobs to perform certain actions as per a schedule. You can create and manage automation jobs from the Automation Central page. Scheduling of jobs allows you to perform actions without manual supervision. For example, you can automate jobs to run during a maintenance window, which could be outside of working hours.

## Automation Central

Automation Central is where you can create jobs to automate optimization actions which reclaim or rightsize VMs. You can also schedule additional actions such as powering on, powering off, and rebooting VMs and instances. Once you set up recurring jobs, you can track and obtain reports on them. You can customize jobs so that they only run based certain on parameters. For example, if you choose to delete a snapshot as an action, you can specify how old the snapshot must be before it is deleted. You can also schedule jobs from the Reclaim and Rightsizing pages, where you configure the job in the context of a recommendation provided by VCF Operations.

### Actions that you can Automate

You can automate the following actions using Automation Central:

| | |
|---|---|
| **Reclaim** | • Delete old snapshots. You can include idle and powered off VMs for reclamation.<br>• Delete idle VMs<br>• Poweroff Idle VMs<br>• Delete powered off VMs |
| **Rightsize** | • Downsize oversized VMs. You can include idle VMs when resizing.<br>• Scale-up undersized VMs. |
| **Other** | Schedule additional actions such as powering on, powering off, and rebooting VMs and instances. The additional actions are based on the public cloud (GCP, AWS, Microsoft Azure) or vCenter adapter. |

### Where you find Automation Central

After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Automation Central**.

### How Automation Central Works

In the Automation Central page you see a list of upcoming jobs and a calendar under the **Schedule** tab. The calendar displays all the jobs that are scheduled for the current month. You can move between months to see more scheduled jobs.

| | |
|---|---|
| **View the Summary of Scheduled Jobs** | When you click on a date in the calendar, you see a summary of the job. The summary displays the frequency of the job, the type of job, and if the job is activated or deactivated. You can click **Preview** to see more details about the job, or click **Deactivate All Recurrences** to deactivate the job. To edit the job, click the **Edit** link. |
| **View a Report of Jobs** | View reclamation and rightsizing reports. The reclamation report displays graphical and numerical data on the total cost saving, CPUs reclaimed, memory reclaimed and storage reclaimed for different time periods. The rightsizing reports displays graphical and numerical data on the CPUs downsized, memory downsized, CPU oversized, and memory upsized for different time periods. |

| | |
|---|---|
| **View Job History** | You can also view the history of configured jobs which have run. Click the **History** tab above the calendar to see the job name, and job details in a tabular format. Use the search to search for a Job or VM. Click the drop-down in the search box to perform an advanced search.<br><br>The job history page only displays the status of jobs. For detailed information about the task, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel** > **Recent Tasks**. Failed recurring jobs are triggered during the next run.<br><br>If VM Power Off is not allowed, the action fails and is indicate as such in the logs. |
| **View Configured Jobs** | The **Jobs** tab is where you see a list of configured jobs. For each job, clicking the ellipses icon brings up a menu from where you can edit, delete, clone or deactivate the job. If a job that you created is not visible in the list, search for it by entering the name in the search box. Alternatively, you can check by the job status by clicking the drop-down in the search box for advanced search options. |

### Prerequisites to Run Actions

Actions are run through VMware tools. vCenter privileges required to run each actions is documented in topic, Privileges Required for Configuring a vCenter Account. Automation Central calls vCenter APIs with the credentials supplied to the vCenter adapter in VCF Operations.

Automation Central is available for users who have the advanced license and above. You must have the necessary permission to schedule and manage jobs in the **Automation Central** page. An administrator can manage these permissions in the **Roles** tab available when you click **Operate** in the top navigation bar, and in the left navigation pane, click **Administration** > **Control Panel** > **Access Control**. Make sure that the **Manage Job Schedules** permission is selected in the **Assign Permissions** section, under Operations. This is available when you expand the **Automation Central** category.

To schedule a job of a specific type, you must have permission to run the corresponding action defined under **Administration** > **Global Inventory** > **Actions** in the **Roles** tab when you navigate to **Administration** > **Control Panel** > **Access Control**.

**Operational Actions** must be enabled in the vCenter cloud adapter instance to run actions from Automation Central.

To use alternate user credentials for actions, see the topic, Configuring a vCenter Account in VCF Operations.

### Troubleshooting Automation Jobs

To see the logs for scheduled jobs, click **Operate** in the top navigation bar, and in the left navigation pane, click **Administration** > **Control Panel** > **Support Logs** > **Others** and look for the following logs:

- actionScheduler-.log
- actions-data-.log

Or, click **Operate** in the top navigation bar, and in the left navigation pane, click **Administration** > **Control Panel** > **Support Logs** > **Analytics** and search for action execution in analytics-.log.

## Create Job from Automation Central

Create a job to schedule an action to be performed automatically. You can select the type of action you want to perform, and then select the scope of the action. You can filter the scope based on attributes and metrics. Every action has a configuration option, which enables you to control the execution of the job based on conditions.

1. After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Automation Central**. In the Automation Central page, click **Add Job**.

   The **Create New Job** page opens. This page displays the following cards:

   - Reclaim
   - Rightsize
   - Billing . This card is available when you when you have VMware Cloud Director configured in VCF Operations. For more information, see the topic, Scheduling Bill Generation Using Automation Central.
   - Actions

2. Click on a card based on the type of job you want to create.
3. In the first step of the wizard, specify the following properties to create the action:

| Property | Description |
|---|---|
| **Name** | Specify a name for the job. This is displayed in the calendar. |
| **Description** | Provide a description for the job. |
| **Action Configuration** | Select this option for scheduling VM optimization actions. The choices are:<br><br>**Reclaim**:<br><br>1. Delete old snapshots. When you select this option, you must specify at least one of the following:<br>– The snapshot age in days.<br>– The snapshot size in MB using the operators from the drop-down list.<br>– The snapshot name, using one of the operators from the drop-down list.<br>– Select the checkboxes to reclaim idle and powered off VMs.<br><br>**Note:**<br>The criteria for snapshot deletion is a combination of all the options you choose.<br>2. Delete idle VMs. When you select this option, you must specify the idle time duration in days.<br>3. Power off idle VMs. When you select this option, you must specify the idle time duration in days.<br>4. Delete powered off VMs. When you select this option, you must specify the power off duration in days.<br><br>**Note:**<br>The time definition for powered off VMs, idle VMs, and snapshots is configured **Capacity** > **Reclaim** > page. Click the **Settings** link on the page to edit or view the reclamation settings. Ensure that you have reviewed the settings in the Reclaimation page before configuring the job here.<br><br>**Rightsize**:<br><br>1. Downsize oversized VMs. When you select this option, you must select the rightsize resource type, and the downsizing limit amount. You can select the checkbox to include idle VMs in the rightsizing.<br>2. Scale-up undersized VMs. When you select this option, you must select the rightsize resource type.<br><br>**Note:**<br><br>VCF Operations does not check if the VM hot add/ remove setting is enabled. If the VM power off is not allowed, then the action will fail. You will find the error in Recent tasks and in the logs.<br><br>**Actions**<br><br>• **Adapter Type**: Select a public cloud or vCenter adapter from the drop-down list. Options are, vCenter, GCP, AWS, Microsoft Azure. |

| Property | Description |
|---|---|
| | • **Object Type**: Select the object from the drop-down list on which you want the action to be performed. Options are, Virtual Machine, Host System, Cluster Compute Resource.<br>• **Action**: Start typing the name of the action, or select an action from the drop-down list. The options are as follows: |

| Adapter Type | Available Action |
|---|---|
| vCenter | • Power Off VM<br>• Power On VM<br>• Reboot Guest OS For VM<br>• Suspend VM |
| GCP | • Power Off CE Instance<br>• Power On CE Instance<br>• Reboot CE Instance |
| AWS | • Power EC2 Instance<br>• Power On EC2 Instance<br>• Reboot EC2 Instance |
| Microsoft Azure | • Power Off Azure Virtual Machine<br>• Power On Azure Virtual Machine<br>• Reboot Azure Virtual Machine |

**Note:**
If the number of resources for a job is ten or less, then the job runs at once. If the number of resources is more than ten, then the jobs run in groups of ten, in parallel.

4. Click **Next**.
5. In the second step of the wizard, select scope of the object which contains the VMs on which VCF Operations must run the automation jobs.

   **Note:**

   - When you select scope for **Additional Actions**, the objects that you drag and drop must be either of the same object type, or the ancestor object (that will serve as a container) of the object type you have selected in the previous step. For example, if you selected **Host System** in the previous step, you need to select either a Host System object or an object that is higher in hierarchy (Cluster Compute Resources, Datacenter etc).
   - VCF Operations ignores object types which do not match the object type that you selected in the previous step. Invalid object types can be object of descendant object types, objects from a different management pack/ solution, or from unrelated object hierarchy.

   a) Drag and drop the object type to the left pane.
   b) To delete members, click the X icon.
6. After you have dragged & dropped the objects for the action, click **Preview Scope** at the bottom right corner to see on which objects the jobs will potentially be executed. This list may be different after you apply some filter from the **Filter Criteria** page.

   **Note:**
   The final objects list on which the actions will be run can be seen by clicking the **Preview Scope** button for the relevant job in the **Schedule** tab.

7. Click **Next**.
8. In the **Filter Criteria** step of the wizard, you can create rules using which VCF Operations filters objects on which the jobs are run. If you do not create a rule, all the VMs within the scope of the selected object type are affected.
   a) In the **Set Filter Criteria** section, create rules using a combination of metrics, relationship, property, object name or tag values and logical expressions.
   b) You can add more than one criteria by clicking either the ADD ANOTHER CRITERIA SET or ADD buttons. When you click ADD ANOTHER CRITERIA SET, VCF Operations creates an OR logical expression between the criteria sets. When you click ADD button, VCF Operations creates an AND logical expression between the criteria.
9. Click **Next**.
10. In the **Schedule** step of the wizard, configure a schedule for the action to run.

| Property | Description |
|---|---|
| **Start Date** | Set a start date which is on, or after the current date, for the job to start. Select a date from the calendar displayed on the page.<br>**Note:**<br>There will be a delay of up to five minutes for a job to run. The actual start time for an action also depends on the action itself and the number of target objects involved. |
| **Time zone** | For the start time and date to be calculated for the job, select a time zone. Choices are:<br>• Browser - based on the current location reported by the browser.<br>• Host - based on the current location of the host machine.<br>• GMT based time zone - Based on a location calculated as per UTC+0. |
| **Start Time** | Select a start time for the job. The drop-down menu provides options in five minute intervals, starting closest to your current browser reported time. |
| **Recurrence** | Set a recurrence for the job. The choices are:<br>• **One-Time**. No further configuration is possible.<br>• **Daily**. Set the recurrence in days. Set how frequently it must run by providing a value in **Run Every** field.<br>• **Weekly**. Select the days of the week when you want the job to run, by clicking the day of the week abbreviation.<br>• **Monthly**. Select the months when you want the job to run, by clicking the month abbreviation. By default, all the months are selected. For the months that you select, you can configure the job to run:<br>– On specific days of the month, by clicking the number. Or, the last day of the month, without specifying the exact date, by clicking **Last**.<br>– On the specific number of the week (first, second, third, fourth or the last) in the month, combined with the specific day of the week.<br>You can make multiple selections for each of the options in the drop down menu.<br>For the **Daily**, **Weekly**, **Monthly** options, you can set when the job must end based on a date, or the number of occurrences. |

   a) In the **Notifications** section, select the **Receive Updates on Job via Email** check box to receive notifications two hours before the job is set to run and after it has been executed. For the email to be sent, you must also select the

email outbound plugin from the drop down menu, and enter the email address to which the email must be sent. If you have not created an email outbound plugin, see the topic, Outbound Settings in VCF Operations.

**Note:**
You can send updates via email to only one email ID.

11. Click **Create** to complete the steps in the wizard and create the job.

## Create Job from Reclaim or Rightsizing

You can create an automation job based on the recommendation provided by VCF Operations in the Reclaim or Rightsizing pages. You cannot create an automation job outside of the context provided by VCF Operations here. Use Automation Central for that.

1. Do one of the following:
   a) After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity**. Click **Optimize** and then click the **Reclaim** tab.
   b) After you log in to VCF Operations, from the navigation bar at the top, click **Manage** and in the left navigation pane, click **Capacity**. Click **Optimize** and then click the **Rightsizing** tab.
2. In the Reclaim or Rightsizing pages, do the following
   a) Click the data center that you want to optimize.
   b) In the table heading that displays, Select the types of VMs that you want to optimize.
   c) Click the name of a listed cluster to show its VM list.
   d) Select the checkbox next to the VM that you want to optimize.
   e) Click **SCHEDULE ACTION**.
3. In the **Create Schedule Job** dialog box that opens, configure the following parameters:

| Property | Description |
|---|---|
| **Job Name** | Provide a name for the job. This information is displayed in the calendar in the Automation Central page. |
| **Job Description** | Provide a description for the job. |
| **Start Date** | From the date picker, select a date when the automation job should start. |
| **Time of Day** | • From the time picker, select the start time for the job.<br>• From the drop down, select the time zone when the time that you selected is valid for. |
| **Receive updates on Job via Email** | Select this checkbox if you have an email server configured, and you want to receive email notification about the status of the job. Notifications are sent two hours before the execution of the job. |
| **Notification Method** | If you selected the previous option, select the email outbound plugin from the drop down menu, and enter the email address to which the email must be sent. |

4. Click **Create**.

The automation job is created, and is available in the Automation Central page. From there, you can preview, edit, or delete the job.

**Note:**
The scope is not editable since the objects on which the job should be run have been already selected from the Reclaim/Rightsize page.

# Configuring and Analyzing Logs

VCF Operations provides capabilities to collect, parse, modify, process, and analyze logs collected from various VCF components. The prerequisite is the deployment of the log management component. VCF provides an integrated user experience for log analysis. You can view and analyze logs from the VCF Operations console.

# Overview of Log Management

Log Management has evolved from Aria Operations for Logs. Log Management is now integrated with VCF Operations.

Log Management component architecture has evolved significantly across recent releases to deliver tighter integration, higher performance, and simplified operations.

- Aria Operations for Logs (Version 8.x): The legacy, standalone appliance-based solution originally deployed as part of the Aria Suite.
- VCF Operations 9.0: Logs experience was converged into VCF Operations while the logs platform remained separate. This release rebranded the standalone solution and introduced native integration with VMware Cloud Foundation, delivering drastically improved scale and performance.
- VCF Operations 9.1: The latest architecture that comes with fully integrated Logs platform. Log Management is now a built-in, containerized core service running natively within the VCF Management Services Platform. The entire logging user experience, including deployment lifecycles, centralized configuration, access control, and alerting, is fully consolidated into VCF Operations.

# Log Collection Architecture

The centralized log collection architecture offers benefits such as granular configuration and cross-data center collection across various VCF components. To enable this capability, the Log Management component should be installed. Installaing the Log Management component is a Day-N operation, done via the lifecycle process, which automatically integrates it with VCF Operations. VCF Operations provides centralized management for log collection and supports diagnostic features that rely on these logs.

### Overview of Log Management Architecture

Log Management replaces Aria Operations for Logs, providing a comparable set of features and capabilities to what was previously offered. Log Management offers greater scalability than Aria Operations for Logs.

Log Management adopts OpenSearch and focuses on enhanced built-in security, better handling of resiliency during failures, and the ability to scale to much larger cluster sizes than was possible with Aria Operations for Logs.

Log Management runs within VCF management services. There is no longer a separate user interface for Log Management. All the functionality is available within VCF Operations. Content packs have been eliminated and replaced by content integrated into management packs. All lifecycle functions for Log Management, such as deployment, patching, upgrading, and resizing persistent volumes, are standardized based on VCF management services workflows. A break-glass UI provides access to the stand-alone interface, for emergency scenarios when VCF Operations is not accessible.

The deployment architecture of Log Management consists of three main service types within a VCF management services cluster: the gateway pod (Envoy), the log processor service, and the log store. The Envoy pod acts as the gateway, responsible for accepting all traffic from various log sources and communicating with external services. The log processor service is the basic engine of the platform, interacting with the log store (OpenSearch) and handling ingestion, querying, and all log management activities. The log store uses OpenSearch to index and persist all log data, including configuration data.

The Envoy pod acts as the gateway, responsible for accepting all traffic from various log sources and communicating with external services. In HA mode, the gateway is deployed with a single instance per VIP/FQDN. In non-HA mode, a single gateway handles all VIPs. Traffic ingestion flows from the gateway to the log processor and finally to the log store. The log processor implements rules for masking, forwarding, filtering, and creating with partitions, while the log store handles the indexing of log messages, persisting data, and implementing alerts and queries.

### Centralized Configuration Management

The log collection configuration is stored in the Log Management cluster, but updates are always initiated through VCF Operations. If you have remote VCF instances and need to send log data to the main, central Log Management deployment, you can deploy a Unified Cloud Proxy. The Unified Cloud Proxy acts as a secure log forwarder that aggregates and routes data from your remote environments to the primary cluster.

Irrespective of whether logs are forwarded via the Unified Cloud Proxy, or sent directly to the Log Management cluster, the interface for providing centralized log collection is through VCF Operations for the components which follow the log standard introduced in 9.0.

### Unified Cloud Proxy

Unified Cloud Proxy plays a key role in collecting logs from multiple data centers. It works as follows:

1. VCF Operations requests credentials.
   a. VCF Operations requests credentials from Log Management for the log forwarder.
   b. This request includes the Cloud Proxy ID.
2. Credential transmission.
   a. The credentials and Log Management FQDN (Fully Qualified Domain Name) are sent to the Cloud Proxy.
   b. This data is encrypted.
3. Log forwarder setup.
   a. The log forwarder decrypts the payload and establishes a connection with Log Management.
4. Configuration updates.
   a. Any future configuration updates are sent via this established connection.

For more information about the types of Cloud Proxy deployment, see the topic, Collecting Data with Cloud Proxy in VCF Operations.

## Getting Started with Log Management

Log Management (formerly VCF Operations for Logs, and before that, VMware Aria Operations for Logs) collects, analyzes, and visualizes logs across your VMware Cloud Foundation fleet.

As of VCF 9.1, Log Management is a built-in, containerized core service that runs natively within the VCF Management Services Platform and is fully integrated into VCF Operations. There is no longer a separate logs UI for normal use.

### What Log Management Provides

- A centralized log collection framework for vCenter, ESX, NSX, VCF Automation, VCF Operations, VCF Operations HCX, VCF Operations for networks, Identity Broker, SDDC Manager, and other VCF management services components.
- An architecture built on OpenSearch, giving greater scale and resiliency than the legacy Aria Operations for Logs appliance.
- Log search, filtering, saved queries, charts, log comparison, and log-based alerts.
- Log processing controls - masking, filtering, forwarding, and partitioning with configurable retention.
- Agents for Linux (filelog, journald) and Windows (via Fluent Bit) log collection, plus native syslog and CFAPI ingestion.
- Role-based access control down to which logs a user can see (data sets/scopes).
- A Log Management Health Overview dashboard, plus a drill-down dashboard for advanced troubleshooting.
- Log Assist, for generating and uploading support bundles to Broadcom Support.
- A standalone “break-glass” UI for emergency access if the VCF Operations console itself is unreachable.

### Choose a Deployment Model

Log Management is added to your VCF fleet as a Day-2 operation, and at deployment time you choose one of three models.

| Model | Best for | Key attributes |
|---|---|---|
| Simple | Test/dev environments | Single replica; smallest footprint; relies on VCF management services and vSphere HA to restart on failure; no application-level HA |

| Model | Best for | Key attributes |
|---|---|---|
| High Availability | Default for production/fleet deployments | 3–19 replica cluster behind an integrated load balancer; full-to-partial service maintained on node failure |
| Multi-Instance | Large-scale or bandwidth-limited/ segregated environments | One VCF Operations–integrated instance plus additional independently managed instances (each Simple or HA); non-integrated instances run standalone on Aria Operations for Logs 8.18 |

All three are deployed in the same way (through VCF Operations), and all can scale up (bigger nodes) or out (more replicas) later, but never back down.

## Before You Deploy Log Management

| | |
|---|---|
| **Timing** | Log Management is not part of the day-zero install. Deploy it after your initial VCF/vSphere Foundation provisioning is complete. |
| **Networking** | Decide your FQDN(s) and IP address(es) up front. At least one Virtual IP (VIP) + FQDN is required at deployment; more can be added later, but your SSL certificate must include every VIP's FQDN in the SAN field before you can create additional VIPs. |
| **Capacity** | Log Management runs on the VCF management services runtime cluster in your fleet's first VCF Instance, so that cluster needs spare CPU/memory/IP-pool capacity.<br><br>VCF management services auto-scales *out* to accommodate Log Management, but scaling *up* (bigger node size) should be scheduled in a maintenance window. |
| **Storage** | Log data is stored on persistent volumes in the management services runtime cluster's default datastore. Plan datastore capacity and IOPS for your ingestion rate and retention. To use other datastores, migrate log store volumes with vSphere CNS. See Migrating Log Management log store container volumes in vSphere. |
| **Sizing** | Estimate Events Per Second (EPS) as roughly 250 × number of ESX hosts (adjust for high-volume sources like firewall logging), plus expected connections and retention needs. This drives your size/replica choice. Use the VCF Operations Sizing Estimator for sizing requirements. |
| **Retention/archive planning** | Decide whether you need long-term archiving. If so, you need an NFSv3 share or S3-compatible endpoint (up to 5 external storage locations). |
| **Backups** | Configure an SFTP location so Log Management's configuration is backed up automatically as part of VCF management services. |
| **Security** | FIPS mode is on by default. Hosts with CPUs that don't support FIPS cipher acceleration may see reduced ingestion capacity. |

## Deploy Log Management

For more information, see the topic, Deploy Log Management.

## Connect Your Log Sources

- Log collection is on automatically for components that are hosted on VCF management services (VCF Automation, Identity Broker, Log Management itself) and cannot be disabled.
- For the following components, you must explicitly activate log collection on that component's adapter (**Administration** > **Integrations** ).
  - vCenter
  - NSX
  - VCF Operations HCX
  - VCF Operations for networks
- Choose whether each source sends logs directly to the Log Management VIP or via a Cloud Proxy (recommended for remote sites as it buffers and optimizes traffic, but always round-robins across all VIPs).

- Version notes: vCenter 9.1.x uses the new Logs Agent (supports custom filtering/fields). Earlier vCenter versions fall back to rsyslog without that support. Windows hosts need Fluent Bit, configured to forward to the Log Management VIP on port 9543.

For more information, see Configuring Log Sources for Log Management.

### Configure Core Log Processing Settings

All under **Operate** > **Administration** > **Configurations**

| | |
|---|---|
| **Log Partitions** | Segment logs by filter criteria with per-partition retention (2 days to limitations of the storage) and archive configuration. Audit and Default partitions always exist; up to 9 custom ones can be added. For more information, see Configuring Log Partitions. |
| **Log Masking** | Redact sensitive fields/text matching a filter. For more information, see Log Masking. |
| **Log Filtering** | Drop unwanted logs before indexing (useful when a source cannot filter itself, for example, syslog). For more information, see Log Filtering. |
| **Log Forwarding** | Relay filtered/tagged logs to up to 10 external syslog/log-management destinations. For more information, see Log Forwarding. |
| **External Storage** | Register NFS or S3 targets used for archiving, import, and export. For more information, see Configuring External Storage. |
| **Integrated Load Balancer** | Manage VIPs/FQDNs; add more when you need extra ingest capacity or want to tag logs by source. For more information, see Integrated Load Balancer. |
| **Agents and Agent Groups** | Manage Linux/Windows collector agents, authentication secrets, and grouped configuration pushes. For more information, see Overview of Log Management Agents. |

### Analyze and Troubleshoot with Logs

- The **Logs** page (**Operate** > **Logs**) is where you search, filter, build fields, save queries, and chart results.
- **Compare Logs** lets you line up results from up to four queries side by side.
- Logs also surface per-object in **Inventory** > **Logs**and in the **Workbench**, so you can correlate log entries with metrics during an investigation.
- You can define log-based alerts from saved queries, and combine log symptoms with metric symptoms in the same alert definition.
- Several core components (vCenter, ESX, NSX, HCX, Operations for Networks, VCF Operations, Identity Broker, VCF Automation) now emit a standardized, structured log format, which simplifies both manual troubleshooting and automated parsing.

### Control Access to Logs

Log access control is unified into standard VCF Operations RBAC:

1. Define a Data Set (**Access Control** > **Scopes** ) — a set of log filters (for example, “hostname contains x”).
2. Check role permissions (Access Control > Roles) — make sure that the assigned role has permissions that you need (Log Processing, Log Forwarding, Log Collection, and so on).
3. Assign the Scope and Role to a user or group.

A Scope can independently carry an **Object Scope** (which infrastructure objects are visible) and a **Log Data Set** (which log data is visible). Assigning one does not automatically restrict the other.

### Monitor the Health of Log Management

Use the **Log Management Health Overview** dashboard (**Operate** > **Dashboards** > **All**) as your first stop for both active alerts and general health checks. It is organized into four areas:

- Ingestion,
- Retention
- Availability
- Operation

Each of these dashboards has guided remediation steps. A disabled-by-default drill-down dashboard exposes per-component (gateway/processor/store) metrics for deeper troubleshooting, and a companion VCF management services cluster health dashboard helps you rule out resource contention on the underlying platform.

**Get Support When You Need It**

- If the VCF Operations console itself is down, use the standalone break-glass U at the Log Management FQDN to search logs directly.
- From that same standalone UI, you can generate a cluster-wide support bundle.
- For day-to-day support cases, use Log Assist (requires Connected Mode licensing and a Cloud Proxy with outbound access to `eapi.broadcom.com`) to auto-generate and upload log bundles plus diagnostic findings straight to a Broadcom support case.

**If You are Upgrading from an Older Logs Product**

Log Management is never an in-place upgrade. It is always a fresh 9.1 deployment plus a configuration import:

- From an existing 9.0 VCF Operations for Logs instance: credentials are auto-detected and imported.
- From Aria Operations for Logs 8.18.x: you enter the FQDN/credentials manually.
- After import, you get a 90-day query federation window against the old cluster, an on-demand Data Migration Utility, and a separate Log Data Transfer tool for historical data.
- Custom content (dashboards, alerts, queries) does not migrate automatically. Export it as a Content Pack and run it through the CLI conversion utility to produce a Management Pack.

## What's New in Log Management

Log Management brings several enhancements to how you can handle logs from different VCF components.

- Monitoring
  - Detection and surfacing of issues based on logs, alerts, and notifications.
  - Visual analytics provided through dashboards and reports.
- Troubleshooting
  - Log pattern detection and pattern trends.
  - Log compare.
  - Capabilities to detect outliers or the absence of logs.
- Security Operations (Log Administration):
  - Data set definition scope.
  - Merging the data set within the VCF Operations scope, which aligns with user role, group creation, and user role assignment.

    Access control, which allows for data set restrictions based on a log scope to define what you can or cannot access. Roles have also been enhanced to include log pages and other related aspects.
  - Role-based access control (RBAC).
  - Log segregation and retention.
- Log Processing Rules
  - Log Masking: Masking critical fields or text for security sensitivity. This requires providing the field or text to mask, the pattern on which to apply the masking (for example, the entire field or a portion), the mask value, and the filter criteria.
  - Log Forwarding: Forwarding a select set of logs or security audit logs based on filter criteria to a central security product or third-party endpoint. This can also forward tags.
  - Log Filtering: Logs are filtered out or dropped based on a filter criteria from the source. This is useful when the sender cannot filter the logs themselves, for example, with syslogs.
  - Log Archival: Archiving logs for compliance and regulatory needs and primary storage offloading. S3 Endpoint support for archive destinations is a new capability.
  - Log Partition: Logical segregation of logs based on various criteria such as retention, routing, or filtering. You can define the retention period (from 2 days to forever) for any specific partition.
  - Audit Record partition for audit logs.

- Logs Page. Workflow-related changes and fine-tuning were brought in to make the page more usable and easier to consume.
  - Saved queries, direct text search, and additional filters, including partition selection and a time range.
  - A quick link to Log Compare is available.
  - An export function allows exporting query results in JSON format to external storage (NFS) or local storage (if the log size is 20,000 or fewer logs) for consumption or analysis.
  - You can view the distribution of data for any field (for example, app name) within the selected time range.
  - A fields library allows you to view all existing fields, including system-defined indexed fields and discovered non-indexed fields, which customers can choose to index for performance reasons. You can also rediscover fields to get rid of old ones.
  - Logs migrated from a previous version are detected, and a time window for accessing those logs is retained in the current release, accessible under a legacy log integrated partition. You are encouraged to reconfigure log sources through the VCF adapter and vCenter integrations.
- Global Settings Updates
  - New settings in the Global Settings page for agent authentication and for event cluster management. As part of security hardening settings, by default, SSL ingestions are enabled for CFAPI and Syslog.

## Deploy and Upgrade Log Management

Log Management is not an in-place upgrade from Aria Operations for Logs. The process starts with a new installation of the Log Management component via the VCF Operations console. The Log Management component is not part of the day-zero installation flow, so you must deploy it after the initial VCF provisioning and installing day-zero components. During the installation wizard, you must provide at least one FQDN and corresponding IP address. The wizard also presents an option to import settings from Aria Operations for Logs and while importing settings from VCF Operations for logs.

### Configuration and Settings Migration

A key part of the upgrade is migrating configurations from the Aria Operations for Logs 8.18.x and VCF Operations for Logs to the new Log Management installation.

- From 9.0**:** If you are upgrading from VCF 9.0 with an existing integrated VCF Operations for Logs instance, the system automatically detects the old credentials via the service registry. These settings are then automatically imported to the new Log Management, eliminating the need for manual input of the address and credentials
- From 8.18.x**:** If you are upgrading from Aria Operations for Logs 8.18.x, then you must manually enter the FQDN and admin credentials of the old Aria Operations for Logs instance in the import flow.
- Configurations Migrated**:** During this import flow, the system pulls specific configurations from the Aria Operations for Logs and adds them to Log Management, including log filters, log masking, and log forwarder rules. Existing log collection configurations are also integrated into Log Management.
- Report Generation: After the installation and configuration import are complete, a PDF report is generated. This report lists all the configurations that were successfully pulled from the legacy Aria Operations for Logs and added to Log Management. It also includes a list of configurations that cannot be migrated automatically (for example, Datasets) with instruction on how to move those over post-import.

### Data Access and Migration

The upgrade process focuses on configuration migration but provides a mechanism to access old data:

| | |
|---|---|
| **Query Federation** | After installation, Log Management creates an integration with Aria Operations for Logs. This integration is used to provide federation of queries for a period of time, specifically 90 days after the Log Management installation. During this window, you can query your Aria Operations for Logs data from the Logs page in VCF Operations. If you select a legacy Aria Operations for Logs partition, Log Management forwards the query to the legacy instance and fetches the data. |
| **Data Migration Utility** | There is an on-demand data migration utility that you can use anytime you want to migrate log data from Aria Operations for Logs to Log Management. This is a long-running process due to the potential volume of log data. |

### Post-Upgrade Behavior

Following the VCF provisioning, when you navigate to any log capability page in VCF Operations before Log Management is installed, you see a notice prompting you to install the Log Management component.

For upgrade scenarios, specifically from 9.0, where partial integration existed, the day-zero pages are shown to enforce the installation of Log Management. Until the installation is complete, no log data is available in VCF Operations, and accessing log capabilities will redirect you to the installation flow. After the Log Management component is successfully installed, it will appear in the component list, and the option to deploy it will no longer be available, as there is one-to-one mapping between VCF Operations and Log Management.

As part of Day-N operations, you can scale up (vertical, increasing size) and scale out (horizontal, increasing replicas) the cluster. Scale-down operations (decreasing size or number of replicas) are not possible using the VCF Operations UI. You can also scale up the log store storage size as a Day N operation.

### How to Deploy or Upgrade

- For information on how to deploy Log Management, see Deploy Log Management.
- For information on how to upgrade to Log Management, see Upgrade to Log Management 9.1.
- To copy historical logs from VMware Aria Operations for Logs 9.0, integrated VCF Operations for Logs 9.0 to Log Management, see Log Data Transfer.
- To configure log forwarding from standalone single/multiple Aria Operations for Logs 8.18.x instances to Log Management, see Configuring Log Forwarding.

## Log Data Transfer

You can transfer log data for up to 90 days from Aria Operations for Logs 8.x and VCF Operations for Logs 9.0 and to VCF 9.1. The migrated logs are stored in the Log Management cluster. You must deploy Log Management before you can proceed.

### Why Transfer Log Data

VCF does not permit integration of VCF Operations with Aria Operations for Logs. If you had previously integrated Aria Operations for Logs and Aria Operations, and you have upgraded to VCF, then you must use the Log Data Transfer feature to transfer ingested logs to Log Management.

When you set up operations-logs as a log forwarding destination in Aria Operations for Logs, logs which are already ingested by Aria Operations for Logs are not forwarded to VCF Operations. To forward already ingested logs, use the Log Data transfer functionality in VCF Operations.

### How to Transfer Log Data

After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**. Under **Content and Support** section, click **Logs Data Transfer**. If you have not set up any transfer, click the **INITIATE TRANSFER** button and configure the authentication details about the VMware Aria Operations for logs 8.x or VCF Operations for Logs 9.x appliance. You can select the duration of logs to transfer from the drop-down. Transfer logs from the last 10 days to start with.

After you initiate the task, you can check the status of the transfer from the **Logs Data Transfer** card at any time.

**Note:**

- Not all data buckets may be copied unless Log Management has enough free disk space. It is recommended to run this check manually prior to starting the migration to guarantee that all logs from the VMware Aria Operations for logs 8.x or VCF Operations for Logs 9.x instance are migrated to VCF Operations. Before starting the migration, know the location of the live storage in VCF Operations, and the volume of logs stored in the live storage of the old instance.
- Data for any selected interval is fully re-imported. Check the audit record to avoid redundant transfers.
- All source nodes of the Aria Operations for Logs instance must have the same SSH credentials as each other, and be reachable via port 22 from the Log Management nodes, otherwise the migration is unsuccessful.

## Log Partitions and Searching Older Partitions

Log Management provides with you with options to create partitions and query them. You can create multiple custom partitions using specific filtering criteria to segregate log data and perform targeted queries against individual partitions. For upgrade scenarios where a legacy cluster FQDN is specified, the system supports querying from legacy partitions for up to 90 days. Finally, you can query re-imported archive data. To query archived logs, the data must first be re-imported into the platform using the archive import workflow, which creates a searchable archive partition available for selection. Querying is performed on one partition at a time.

### Aria Operations for Logs or VCF Operations for Logs 9.x Logs Partitions

In a brownfield deployment, you upgrade from an existing Aria Operations for Logs or VCF Operations for Logs 9.x instance. You can connect the newly installed Log Management instance with a VMware Aria Operations for logs 8.x or VCF Operations for Logs 9.x instance to the new Log cluster instance as part of the upgrade workflow, and not later on. After this configuration is complete, the **Logs** page in VCF Operations displays a banner which serves as a notification that the connection with the VMware Aria Operations for Logs 8.x or VCF Operations for Logs 9.x instance will expire in 90 days.

You must complete all data and settings, such as datasets, before 90 days. After 90 days, VCF Operations will not be able to query data from older versions.

The connection with Aria Operations for Logs makes it possible to search for logs in legacy partitions directly from the **Logs** page in VCF Operations. This connection with VMware Aria Operations for Logs 8.x or VCF Operations for Logs 9.x instance will automatically be deleted after 90 days, at which point the banner will be hidden, and these legacy partitions will no longer be available for searching. For more information, see Configuring Log Partitions.

### Archive Locations

You can import logs from from archives in a configured external stroage location (S3 or NFS), and search against these locations. This can be found in the **Log Partition** tab under **Administration** > **Configurations** > **Log Processing**. To query these archived logs, you must use the archive import workflow to re-import the data into the platform. Once re-imported, the data resides in a searchable archive partition that can be selected for querying. Note that only one imported log archive can be active for querying at a time. To search a different archive, you must purge the current one and repeat the import process.

You can restore the archived information back to Log Management. VCF Operations allows for restoring archived logs not only from Log Management archive locations but also from Aria Operations for Logs archived locations. It is recommended to import the logs with a small time range because if the import logs fill the storage more than 80%, the task fails. When importing from Aria Operations for Logs archives, only NFS storage is supported. You must provide the mounted path and specify the time range of the archived data that you wish to import.

Initiating an import generates a task that can be monitored under Log Tasks which can be found under **Administration** > **Control Panel** > **Log Import/Export Tasks**. Once the logs have been successfully imported, you can search against this archived data from the **Logs** page using the **Import Partition** drop-down menu. By applying the archived logs option in this section, you can view all the logs that were imported from the specified time range. For more information, see Archive Log Import.

## Configuring Log Partitions

Log partitions enable the segmentation of log data through defined filter criteria and support specific retention and archival policies. This segmentation facilitates efficient log management by allowing for different retention periods and ingestion orders across multiple custom partitions. Separate partitions can be configured to archive to specific NFS or S3 locations. Archiving lets you shorten retention in the log store, which reduces storage use and improves query performance, while the older data stays available in the archive. Each partition's retention policy ensures that logs are either kept until storage reaches the 80% threshold and then are automatically purged. From the **Log Partitions** page, you can create partitions, edit, clone, activate, deactive, and delete partitions.

The order of evaluation for these rules is top-to-bottom. Partitions are processed based on an ingestion order, which you can modify. Since the first matching partition rule takes precedence, you can change the order to ensure logs flow to the

desired partition first. The default partitions, such as Audit records and Default partition cannot be modified. The Audit records partition is always the first filter covered and the default partition the last.

1. After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**. Then click the **Log Processing** card.
2. Click the **Log Partition** tab.
3. The Log Partitions interface presents a tabular view of defined partitions with key attributes.
4. Click **ADD** to add a new Log partition. Or click the vertical ellipses to edit an existing partition. Enter the following values:

| Option | Description |
|---|---|
| **Name** | Provide a name for the partition |
| **Retention Period** | Set the retention period in days. Use 0 for indefinite retention (storage-limited retention). The minimum retention period for index partitions is 1 day and maximum periods vary by partition type.<br><br>Once the specified retention period (set in days) is met, the system automatically removes the logs from the disk. If the retention period is set to zero, logs are kept indefinitely and not automatically removed based on time. Regardless of the set retention period, logs may be removed earlier if the disk reaches a certain capacity threshold. In this case, the system starts deleting the oldest data first to ensure it can continue serving new requests. |
| **Enable Configuration** | Select the checkbox to enable the configuration. This deterimines if the log partition is active or not. |
| **Enable Archive** | Select the checkbox to enable archiving. If you enable archiving, you must enter the NFS or S3 location where the log data is archived.<br><br>Logs are archived the next time the archiving process runs. They are not held until the retention period expires. The archiving process runs whenever the index is rotated, which is every 24 hours, or sooner if a large volume of logs is ingested. |
| **Source** | If you have enabled archiving in the previous step, you can select the external storage. Specify the host source and storage path for long-term storage if archiving is required. In the text box, enter the NFS or S3 location where you want to store the archived data, in folder structure format that your organization maintains. S3 storage is recommended to archive the logs. Or, select the location from tghe dro-down list oif the sources are already defined.<br><br>The storage path must be a valid mount path. The host must be defined in **Configurations** > **External storage**.<br><br>Before configuring an NFS share for import, export, or archive operations, ensure the target directory has the correct ownership and permissions and the path specified is relative to the mount point, and does not start with a leading "/".<br><br>To allow the system to write data successfully, the mounted NFS directory and its subdirectories require specific permissions.<br><br>• Ownership: The directory must be created by the `root` user.<br>• Minimum Permission Level: `447` (Read/Write/Execute for Owner and Others).<br><br>If your NFS directory does not meet these requirements, update them using the following commands on your NFS server:<br><br>`sudo -i` |

| Option | Description |
|---|---|
| | `chmod -R 447 /path/to/your/directory` |
| **Storage Path** | Folder to put the archives, starting from the root of the selected source (provided without a leading "/") |
| **Filtering** | Add one or more filters to refine the logs that you want to store in the log partition. Filters can include key-value matches or Exists/Does not exist on fields such as hostname, appname, and static indexed fields.<br><br>**Note:**<br><br>The logic that is used in the filter preview is different from the logic that is used when filtering in the **Logs** page. For more information, see Log Management Filters.<br><br>**Note:**<br><br>The filtering option is not available when you edit the Audit and Default partitions. |

5. Click **PREVIEW** to preview the log partition or **CREATE** to save it.

The log partition configuration appears in the **Log Partition** tab with information. To modify a single row, click the vertical ellipses. For bulk activation, deactivation or deletion, select multiple rows and click the horizontal ellipses.

**Note:**

For information on Importing Log Archives, see Log Partitions and Searching Older Partitions.

## Migrating Log Management Log Store Container Volumes in vSphere

You can migrate Log Management log store persistent container volumes from one datastore to another using the vSphere Cloud Native Storage (CNS) migration feature. In VMware Cloud Foundation (VCF) and VMware vSphere Foundation (VVF) environments, runtime services cluster VMs are placed on a single datastore. If you face capacity limitations, like max node size, you can migrate the log volumes using the workaround that is provided here. You should note the limitations and reapply the workaround in case of scale and DR actions. Relocating these volumes across multiple target datastores balances IOPS while maintaining continuous log ingestion, data integrity, and platform functionality.

Before migrating log store volumes, ensure the following conditions are met:

- Environment: A VCF Operations 9.1 environment with Log Management deployed on a VCF Management Services (VCFMS) cluster is active and healthy.
- Storage Accessibility: At least two datastores (the source datastore and one or more target datastores) must be fully accessible from all ESX hosts underlying the VCF Management services.
- Capacity: Target datastores must have sufficient free capacity to accommodate the migrated persistent volumes.
- Permissions: Your vSphere Client user account must possess administrative permissions to view and migrate container volumes.
- Establish Baselines:
  - Record the baseline mapping of your current datastore layout and identify the specific volume IDs and sizes backing your log store.
  - Execute a baseline query in the the **Logs** page. Document the sample entries and total result counts for comparison.
  - Ensure active log ingestion streams (via agents and syslog) are operating smoothly.

Limitations and Constraints:

- Only single block volumes can be migrated.

- PMEM (Persistent Memory) datastores are not supported as migration targets.
- The VCF Management services worker node hosting the persistent volume will be locked during the CNS relocation process, meaning no other infrastructure control operations can be performed on it until the move completes.

1. Log in to the vSphere Client.
2. Navigate to the datacenter, cluster, or source datastore hosting your infrastructure.
3. Select the Monitor tab and click **Cloud Native Storage** > **Container Volumes**.
4. To easily isolate the log management volumes, click **Add Filter** and filter by the Kubernetes component label:

   ```
   Label: app.kubernetes.io/component:log-store
   ```

5. Select the specific log store persistent volume (PV) you intend to move and click **Migrate**.
6. In the **Migrate Volume** dialog box, select your preferred target datastore.
7. Review and acknowledge any compatibility warnings (such as storage policy mismatches or datastore inaccessibility), then click **Migrate**.
8. Monitor the migration progress under the **Recent Tasks** pane in the vSphere Client until the task completes successfully.

Once the vSphere migration task finishes, verify data integrity and workload continuity:

- In the Container Volumes view, confirm that the volume's location has updated to the target datastore and its Health Status returns to Accessible or normal.
- Browse the target datastore files to ensure the virtual disk backing the volume exists within the proper First Class Disk (FCD) directory or VM folder.
- Verify that the matching Kubernetes PersistentVolumeClaim (PVC) status remains Bound and shows no active pod scheduling errors.
- In the **Logs** page, re-run your baseline search query to verify that no log data was lost or corrupted during the migration window.

Important Operational Considerations

- **Scale-Out Operations:** During any post-migration cluster scale-out operations, newly added log-store node volumes will automatically attach to the default platform-managed services Datastore. Administrators must manually repeat this migration procedure if they choose to redistribute new volumes to preferred datastores.
- **Disaster Recovery (DR):** If a DR event occurs and the Log Management component is restored from a backup into a fresh VCF Management Services cluster environment, all custom disk placements are reset to the default single VCF Management Services Datastore configuration. You must execute this volume relocation procedure again after the DR environment is online to clear potential storage hotspots.

# Configuring Log Sources for Log Management

You enable Log analysis for vCenter, NSX, VCF Operations HCX, VCF Operations for networks, VCF Operations when you configure the adapter, and for Identity Broker, and VCF Automation when you deploy the component. You then fine-tune the log collection based on logical service grouping, granular log level filtering, and custom tagging.

### Log Collection for VMware Cloud Foundation Components

When you activate log collection on the vCenter Server Cloud Account or VMware Cloud Foundation pages, Log Management automatically begins collecting vCenter events, tasks, and alarms, alongside VMware ESX and vSphere Supervisor logs.

To collect logs for the vCenter instance, you must explicitly activate vCenter log collection on the **Log Sources** page.

The method used to collect vCenter logs depends on the version you are running:

- vCenter 9.1 and later: Log collection is performed using the Logs Agent, which supports advanced log processing.
- Earlier versions of vCenter**:** Log collection relies on rsyslog. When using **rsyslog**, custom filtering and extracted fields are not applied to the collected logs.

The following table outlines the VMware Cloud Foundation (VCF) components that support log collection and their required minimum versions.

| Component | Version Supported |
|---|---|
| VMware vCenter | 9.1 and later |
| VMware Cloud Foundation Operations | All versions |
| VMware NSX | All versions |
| VMware Cloud Foundation Operations for Networks | 9.1 and later |
| VMware HCX | 9.1 and later |

## General Configuration Features

The following configuration options are standard across most supported components:

**Logical Service Groups** Logs are now grouped into logical services rather than individual log paths. You can enable or disable these groups based on your requirements.

> **Note:**
>
> Audit Logs cannot be disabled and are always collected once log collection is activated.

**Log Level Filtering** You can filter logs by severity. The default setting is Info, but this can be adjusted (for example, to Debug) to increase or decrease log granularity.

**Custom Fields (Tagging)** You can define custom fields to tag logs originating from specific files. These tags are appended to the logs and are visible on the Analyze page for easier filtering.

**Reset to Default** This option is available to restore the original configuration if needed.

## Component Specific Log Collection Configuration

Adding a VMware Cloud Foundation account and enabling log collection for each instance of these adapters is recommended over adding the adapters and enabling log collection directly. For more information, see Configuring a Standalone VCF Account in VCF Operations.

**vCenter** You enable Log collection when the adapter is configured or after adding VMware Cloud Foundation. To enable log collection using the adapter, see Configuring a vCenter Account in VCF Operations.

**NSX** You enable log collection when the adapter is configured or after adding VMware Cloud Foundation. To enable log collection using the adapter, see Configuring the NSX Account in VCF Operations.

**VCF Operations HCX** You enable log collection when the adapter is configured.

**VCF Operations for networks** You enable log collection when the adapter is configured.

**VCF Operations** Log collection is enabled by default.

**Identity Broker** Log collection is enabled after the component is deployed in Fleet Lifecycle. For more information, see Deploy Identity Broker as a Day-N Operation.

**VCF Automation** Log collection is enabled after the component is deployed in Fleet Lifecycle. For more information, see Deploying VCF Automation as a Day-N Operation.

> **Note:**
>
> For NSX, the system does not report an active status. When log collection is functioning correctly for NSX, the status displays as **Received**.

## Managing the Logs which are Collected

After you enable log collection, you can manage the types of logs which the adapter collects. For more information, see Managing Collection from Log Sources.

**Upgrade Scenarios**

Log Collection configuration is not automatically updated after migration from a previous version of VCF Operations for logs, or Aria Operations for Logs.

The following configuration settings have their status migrated as part of the Log Management upgrade:

| Configuration Setting | Aria Operaions for Logs 8.18 | VCF Operations for logs 9.0 | Note |
|---|---|---|---|
| Agent Group | Yes | Yes | |
| Alerts | No | No | Report Only |
| Data Partitions | Yes | Yes | |
| Extracted Fields | Yes | Yes | Only custom extracted fields |
| Log Collection Configuration | No | Yes | Only migrated for Log Management 9.x |
| Log Filters | Yes | Yes | |

## Managing Collection from Log Sources

You can manage the types of logs that are collected, logical service grouping, granular log level filtering, and custom tagging.

Verify that you have the **manage** permission to configure log sources. You can configure permissions in the **Access Control** card when you navigate to **Administration** > **Control Panel**. Go to the **Roles** tab and select or deselect the permissions when you add or edit a role.

Verify that the Log Management component is installed. See the topic, Deploy Log Management for more information.

**Note:**

The logs generated by Log Management are also collected, but it cannot be modified or configured.

**Note:**

VCF Operations supports a configuration export and import functionality which allows transferring the configuration from one VCF Operations instance to another. The log collection configuration for the vCenter adapter instances is not included in the export and import.

1. To manage configurations, go to **Administration** > **Configurations**. Click the **Log Collection** card. This opens the **Log Sources** tab.
2. You see the following list of log sources:
   - vCenter
   - NSX
   - VCF Operations HCX
   - VCF Operations for Networks
   - VCF Operations
   - VCF Identity Provider
   - VCF Automation
   - VCF services runtime
   - SDDC Manager

   **Note:**

   For VCF Identity Providers, VCF Automation, and VCF services runtime, the log collection is always enabled and you cannot edit any settings.
3. For the individual components, manage the log collection.
   - Click the vertical ellipsis next to vCenter to edit the following log collection settings.

- For vCenter:

  **Note:**

  To transfer logs from VMware Aria Operations for Logs 8.x and VCF Operations for Logs 9.x, see Log Data Transfer.

| Option | Description |
|---|---|
| **Collect Logs directly in Logs Cluster** | Dropdown to select the VIP to use for log ingestion. |
| **vCenter Server events and alarms collection** | Select the checkbox to enable the collection of vCenter specific events and alarms. |
| **vCenter Log Collection** | Select the checkbox to activate collection of vCenter logs. |
| **Log Filters** | **Log Level :** Dropdown to filter logs by severity (for example, Trace, Info, Debug).<br>**Service Groups:** Toggle to enable or disable specific logical groupings of log paths. |
| **Fields** | Options to **Add New Field** or **Clear All**. Used to tag incoming logs with custom metadata for easier filtering. |
| **Log Types** | For each vCenter service, toggle the log collection and display the path to the log files being collected.<br>**Note:**<br>Audit log collection is mandatory and cannot be disabled. |

- For ESX, edit the following log collection settings:

| Option | Description |
|---|---|
| **Edit log collection configuration** | Select the checkbox to edit the further configuration options |
| **ESX logs** | Select the checkbox to activate collection. |
| **ESX Audit Logs** | Select the checkbox to activate collection. |
| **Syslog Protocol** | Select the protocol over which sylog is collection. Default is TCP over SSL. |

- For vSphere Supervisor, edit the following log collection settings:

| Option | Description |
|---|---|
| **Edit Log collection configuration** | Select the checkbox to edit the further configuration options |
| **Log Filters** | Select an option from the drop-down list. Logs will include the selected level and higher in severity. |
| **Fields** | Add fields with predefined values to events, alarms and logs coming from vSphere for easier querying. |
| **Service Collection** | For each vSphere Supervisor service, toggle log collection and the file path. |

4. For NSX, VCF Operations HCX. And VCF Operations for networks edit the following log collection settings:

| Option | Description |
|---|---|
| **Collector Configuration** | Select the destination cluster for NSX logs. |
| **Log Collection configuration** | Toggle to collect the logs configuration |
| **NSX Log Collection** | Checkbox to activate or deactive collection of logs for the NSX adapter. |
| **Fields** | Add or manage custom fields. |

| Option | Description |
|---|---|
| **Predefined Log Filters** | Log Level: Select an option from the dropdown to filter logs by severity (for example, Trace, Info, Debug). |
| **Service Collection** | For each NSX service, toggle log collection and the file path.<br>**Note:**<br>Audit log collection is mandatory and cannot be disabled. |

5. For VCF Operations, edit the following log collection settings:

| Option | Description |
|---|---|
| **VCF Operations Log Collection** | Toggle to activate the collection.<br>**Note:**<br>For fresh setups, this is enabled automatically and cannot be fully deactivated. |
| **Log Filters** | Log Level: Select an option from the dropdown to filter logs by severity (for example, Trace, Info, Debug). |
| **Service Collection** | For each VCF Operations service, toggle log collection and the file path.<br>**Note:**<br>Audit log collection is mandatory and cannot be disabled. |

6. For SDDC Manager, log collection settings are enabled when you add the VCF Adapter instance. You can override or edit the following options:

| Option | Description |
|---|---|
| **Collector configuration** | Select the logs cluster to which you want to collect the logs in the **Collect Logs directly in Logs Cluster** setting. |
| **SDDC Manager Log Collection** | Activate or deactivate the SDDC Manager Log Collection. |
| **Log Filters** | Log Level: Select an option from the dropdown to filter logs by severity (for example, Trace, Info, Debug). |
| **Fields** | Add or manage custom fields for log filtering. |
| **Service Collection** | For each service, toggle the log collection and the file path.<br>**Note:**<br>Audit log collection is mandatory and cannot be disabled. |

7. Click **SAVE** to save your changes, or **RESET TO DEFAULT** to revert your changes.

# Overview of Log Standardization

By introducing a unified log standard, VCF enhances log usability, simplifies troubleshooting, and improves automation capabilities. The adoption of industry standards and structured logging ensures that logs are more consistent, easy to interpret, and valuable for your organization, your support teams, and your automation processes. The log standard introduced in VCF is a significant step toward greater observability, system efficiency, and proactive problem resolution.

### Why a New Log Standard?

Logging serves as a key source of information for system management, troubleshooting, and automation. Global support teams rely on log statements to diagnose issues, optimize performance, and enhance automation processes. In previous versions, VMware products such as vCenter, vSphere, Aria Operations for Networks, have followed their own logging

standards, often based on the RFC 5424 Syslog Protocol. Though functional, this approach introduced inconsistencies in log formats and patterns, making interpretation more complex and reducing the efficiency of automated log analysis.

To address these challenges, VCF introduces a standardized log format. This new format aligns with industry best practices, including OpenTelemetry principles. It is designed to improve log consistency across components, simplify troubleshooting, and facilitate automation-driven log consumption. In VMware Cloud Foundation 9.0, three components follow a unified format, offering greater interoperability. In future versions, more components will adopt the standardized log format. This reduces the complexity of integrating new tools or migrating to new VCF versions since updates will follow this standardized approach.

### Objectives of Log Standardization

VCF benefits from standardizing logs because of the following reasons:

- Unified Logging Standard: Establishing a single log statement standard across the VMware Cloud Foundation platform while incorporating industry standards.
- Enhanced Automation and Parsing: Enabling automated log parsing and interpretation to improve operational efficiency.
- Ease of Configuration: There is now a single place to enable or disable log collection as well as configure its granularity. The approach is applicable for cross-datacenter VCF platforms by utilizing log forwarder capability within the cloud proxy.

### Impact of Log Standardization on Existing Workflows and Automation

- Automation Adjustments: If you use automated workflows that are tied to logging data, then you may need to adjust your processes for the components using the new log standard. Since the standardized format introduces specific fields (for example, timestamps, user actions, and component details), any custom scripts or automation routines relying on previous formats will require updating to the new structure.
- Dashboard and Alert Configuration: If you have dashboards or alert definitions that are connected to log data, initial reconfiguration is necessary. You can combine log based symptoms with metric based symptoms, ensuring that crucial insights are not missed due to format discrepancies. You can build log based widgets with the existing VCF Operations dashboards. This functionality is new in VCF 9.0.

### Scope of Log Standardization

**Logs Within Scope**

The standardization effort covers log statements that are generated by VCF components that document system operations. This includes:

- Application Logs: These logs capture events, user interactions, errors, and warnings generated by VCF applications. An example is the vCenter logs stored at `/var/log/vmware`.
- System Syslog Transmissions: Standardized messages from network devices, servers, and applications, adhering to the syslog protocol. For instance,VCF Operations for networks logs syslog messages when a PAM session closes.
- Syslog Transmission of Audit Records: Many organizations collect and retain audit records for compliance and security. The new standard applies to transmitting these records via syslog but does not mandate or modify how VCF components store them internally.

**Logs Out of Scope**

Certain types of logs are excluded from standardization due to their unique characteristics and existing industry adherence:

- Transaction/Database Logs: These logs track changes within a database, such as SQL transactions, and are essential for data integrity and disaster recovery.
- Third-Party Logs: Logs generated by external software like Kafka, PostgreSQL, RabbitMQ, and Kubernetes, which already conform to independent standards.
- Network Logs: While network logs provide critical information about IP traffic, they are categorized under system logs and remain outside the current standardization effort.

## Using the Chart to Analyze Logs

The chart in the **Logs** page lets you perform visual analysis on the results of your query.

Charts represent graphical snapshots of the results of log search queries. The chart type can be changed and adjusted to represent the query in different ways, based on user preferences.

You can control the aggregation level of the chart. The **COUNT OF EVENTS** function is selected by default.

**Where you find the Logs Page**

Click **Operate** in the top navigation bar, and in the left navigation pane, click **Logs**. Select the **Logs** tab.

## Changing the Chart Display

You can change how charts look in the **Logs** page.

| Task | Procedure |
|---|---|
| Change the time range of a chart | In the **Logs** page, you can switch the period displayed in the chart. |
| Change the granularity of a chart | In the **Logs** page, you can switch between different time ranges for each point represented on the chart. The available ranges depend on the time range specified for the query. |

## Chart Types

Click **Operate** in the top navigation bar, and in the left navigation pane, click **Logs**. You can select different chart types to change the way the data is visualized. The options are available above the chart display.

Different chart types require different aggregation functions, the use of time series, and group-by fields.

| Chart Type | Aggregation Function | Time Series Requirement | Group-by Field Requirement |
|---|---|---|---|
| Column | Any | Time series | N/A |
| Line | Any | Time series | N/A |
| Area | Any | Time series | N/A |
| Bar | Any | Non-time series | At least one field |
| Pie | Count or Unique Count | Non-time series | At least one field |
| Bubble | Any | Non-time series | Two fields |
| Scalar | Any | Non-time series | N/A |
| Table | Any | Any | N/A |

### Aggregation Function

VCF Operations provides several aggregation functions.

**Change the Type**

You can change the aggregation and grouping of query results displayed in the chart to graphically analyze log events.

Verify that you are logged in to the VCF Operations web user interface with a role that has the relevant permissions. The number of drop-down menus that you see under the chart depends on the selected aggregation function.

1. Use the drop-down menus to the top-left of the chart to change the aggregation function and grouping type.
   - To view the number of events over time, select the **Time series** button.
   - To view only event values, select the **Non-time series** button and select at least one field.
2. Click **APPLY**.

**Table 1300: Change Aggregation and Grouping of Query Results**

| Type | Field | Description |
|---|---|---|
| Count of events | Events only | Creates a chart of the number of events for a specific query. |
| Unique count | Any field | Creates a chart of the number of unique values for a field. |
| Min | Numeric fields only | Creates a chart of the minimum value for a field. |
| Maximum | Numeric fields only | Creates a chart of the maximum value for a field. |
| Average | Numeric fields only | Creates a chart of the average value for a field. |
| Std dev | Numeric fields only | Creates a chart of the standard deviation for a field's values. |
| Sum | Numeric fields only | Creates a chart of the sum of values for a field. |
| Variance | Numeric fields only | Creates a chart of the variance for the values of a field. |

You can modify the way you view the query results.

| View | Description |
|---|---|
| To group query results by specific field values | From the drop-down menu in the chart, select **Non-time series** and a field under **Group By**. |
| To view the number of events for a field. | For example, the number of events per host, select select **Non-time series**, **Count of Events** and **Group By by hostname**. |
| To view a stacked bar chart for a field with groupings over time. | Select both the **Time series** check box and the field check box under **Group By**. |

## Aggregation and Grouping in the Chart

The following table contains examples to illustrate aggregation and grouping in charts.

**Table 1301: Example Aggregation and Grouping in the Chart**

| Selection in the First Drop-Down Menu | Selection in the Second Drop-Down Menu | Time series selection | Result |
|---|---|---|---|
| **Count of events** | **Time series** | **Time series** | The chart displays a bar chart with the number of events for the current query over time. |
| **Average** | **vmw_op_latency (VMware - vSphere)** | **Time series** | The chart displays a line chart with average value of operations latency over time. |

| Selection in the First Drop-Down Menu | Selection in the Second Drop-Down Menu | Time series selection | Result |
|---|---|---|---|
| **Count** | **vmw_esx_problem**<br>**Note:**<br>The vmw_esx_problem field does not appear by default. You must query the vmw_esx_problem field (Add Filter > vmw_esx_problem 'exists') so that vmw_esx_problem appears in the drop-down menu. | **Non-time series** | The chart displays a bar chart of the number of events for containing the vmw_esx_problem field. |
| **Count** | **Time series**, **vmw_esx_problem** | **Time series** | The chart displays a bar chart grouped by vmw_esx_problem over time. |

## Viewing Logs, Queries and Fields

In the **Logs** page, you can search for log messages in context, and collect and analyze log feeds from within VCF Operations. You can view log-related metrics for troubleshooting. You can also dynamically extract fields from log messages based on customized queries. For more information, see Log Fields.

### The Logs Page

After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Logs**. You can view logs, create queries, and fields. In the Logs tab of the **Logs** page, you can search and filter log events, and create queries to extract events based on timestamp, text, source, and fields in log events. Besides presenting the log stream, VCF Operations presents charts of the query results. You can save these charts to look them up later in the **Dashboards** page. You can access the **Logs** page if your user account is associated with a role that has the appropriate permission. You can view or edit settings in the **Logs** page, depending on the sub-categories and corresponding access levels selected for the permission.

#### Viewing the Logs for an Object

When VCF Operations is integrated with Log Management, you can view the logs for a selected supported object from the **Logs** tab in the inventory. You can troubleshoot a problem in your environment by correlating the information in the logs with the metrics.

### Where You Find the Logs Tab

After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Inventory**. Click the **Logs** tab. The **Logs** tab is available after you have configured Log Management in VCF Operations.

#### Log Queries

Queries in VCF Operations can retrieve and summarize events. Once created, you can compare queries.

You can create and save queries from the **Logs** page. To see saved queries click **Operate** in the top navigation bar, and in the left navigation pane, **Administration** > **Configurations**. Then, click the **Saved Queries** card. You can export query results, share your queries with other users, and can save, delete, rename, and load existing queries. You can take snapshots of queries and can save them to dashboards.

A query consists of one or more of the following:

| | |
|---|---|
| **Keywords** | Complete, or full-text, alphanumeric, hyphenated, and/or underscore matches. |
| **Globs** | Complete, or full-text, alphanumeric, hyphenated, and/or underscore matches. |
| **Regular expressions** | Sophisticated string pattern matching based on Java regular expressions. |
| **Field operations** | Keyword, regular expression, and pattern matches applied to extracted fields. |

| | |
|---|---|
| **Aggregations** | Functions that are applied to one or more subgroups of the results. |

VCF Operations supports the following types of queries:

- Message. A query made up of keywords, regular expressions and/or field operations.
- Regular expression or field. A query made up of keywords and/or regular expressions.
- Aggregation. A query made up of a function, one or more groupings, and any number of fields.

You can define custom alerts in VCF Operations and trigger them from scheduled queries of any type.

**Note:**

Some saved queries may contain expressions that cannot be converted into the VCF Operations UI-supported format. Attempting to interpret or partially render these queries may lead to incorrect behavior or confusion.

Some complex saved queries and log-based alert definitions may not appear in the UI due to parsing limitations. These items are skipped when their expressions cannot be converted into the VCF Operations UI-supported format. Saved queries do not save the aggregated query (chart view).

## Best Practices for Creating Search Queries

You can enter queries by using the search bar, or by entering filters.

Use the search bar to refine the results for events. While you can use a filter instead of the search bar, it is often easier to understand a query that uses the search bar over an equivalent filter. The best practice is to use the search bar instead of an equivalent filter when possible.

A filter allows you to create queries by using a regular expression, a field, Logical OR and AND operations, or a combination of search bar and filter queries.

When you create queries by using the search bar and filters, the following best practices apply:

- Ensure queries are not environment specific.
- When constructing a query, use keywords when possible, when keywords are not sufficient use globs, and when globs are not sufficient use regular expressions. Keyword queries are the least resource intensive type of query. Globs are a simplified version of regular expression and are the next least resource intensive type of query. Regular expressions are the most expensive type of query.
- Provide as many keywords as possible when using regular expressions or fields. If a regular expression includes a logical OR, for example `this|that`, do not include keywords. VCF Operations is optimized to perform keyword queries prior to regular expressions to minimize regular expression overhead.

### Log Fields

Fields are a powerful way to add structure to unstructured events and allow the manipulation of both the textual and visual representation of data. You can view the distribution of data for a specific field and access the fields library to view indexed and non-indexed fields. You can choose to index non-indexed fields for performance reasons.

Saved fields are available in the **Fields Library** card. To see the **Fields Library** card, click **Operate** in the top navigation bar, and in the left navigation pane, **Administration** > **Configurations**. You can export or import extracted fields from one VCF Operations instance to another.

## Using Fields in the Logs Page

Click **Operate** in the top navigation bar, and in the left navigation pane, click **Logs**. The Logs page displays a stream of log messages. You can extract any part of a log message that might be applicable to a query or aggregation by selecting the part of the message and clicking **Extract Field**. Fields are a type of regular expression query and are useful for complex pattern matching so you do not need to know, remember, or learn complicated regular expressions.

You can use Fields in different ways including aggregations and filters. Aggregations allow you to apply functions and groupings to fields. Filters allow you to perform operations over fields.

You can create new Fields by extracting them from the logs stream. After you select a part of the message and right-click, and then select **Extract Field**, you see the **Create Field** flyout section on the right side. Enter the values and click **SAVE** to create a Field.

| Field Context Value | Definition |
|---|---|
| Field Name | Only use alphanumeric characters. Ensure that all characters are lower case and use underscores instead of spaces as this makes fields easier to view. |
| Field Type | Select an option from the drop-down list. |
| Regex before value | Include as many keywords as possible. If this field is empty or only contains special characters, then the Regex after value must include keywords. |
| Regex after value | |
| Pre Context | Specifies the exact log text or pattern that immediately precedes the target value. While simple Regex is supported, it is highly recommended to use exact text strings to prevent partial query results.<br>Ensure a clear delimiter separates this context from the target value. Supported delimiters include spaces and the following characters: `` ` ~ ! @ # $ % ^ & * ( ) + = [ ] { } . `` |
| Post Context | Specifies the exact log text or pattern that immediately follows the target value. Like the Pre Context, prefer exact text over complex Regex to ensure accuracy.<br>Ensure a clear delimiter separates the target value from this context. Supported delimiters include spaces and the following characters: `` ` ~ ! @ # $ % ^ & * ( ) + = [ ] { } . `` |
| Keywords | Click **Additional Context** to add one or more keywords, which are separated by space, that appear within events containing the field. |
| Add Filter | A static field, operator, and a potential value that appears within events containing the field.<br>It is common to use this in conjunction with the VCF Operations agent and tags for events that do not contain keywords. |
| Available For | Select the users that can use this Field. |

## Extracted Fields and Regular Expression Handling

To ensure optimal performance and stability, the logs platform enforces strict handling of extracted fields. If an extracted field contains a complex or malformed regular expression (Regex) pattern, VCF Operations cannot process it. Instead, VCF Operations automatically drops the invalid field.

This strict validation applies to all extracted fields in your environment, including:

- Custom extracted fields that you create.
- Out-of-the-box extracted fields that are defined within installed Management Packs.

If you notice that specific extracted fields are missing from your log search results, or certain alerts that rely on extracted fields are failing to trigger, then VCF Operations may have dropped the extracted fields during processing because their underlying Regex patterns were too complex or malformed.

To verify if the logs platform dropped your extracted fields, review the system logs for specific warning messages. Look for log entries marked with a **WARNING** severity level that match the following format:

```
Unable to extract fields:["<field_name_1>", "<field_name_2>"]
```

## Field Governance in the Fields Library

Fields can be manually added from the endpoints sending the logs, such as a FluentD agent. When a log is ingested, Log Management discovers the field, but does not index it by default. Indexing a field makes it quickly searchable in the **Logs** page. Searching by a non-indexed field takes a long time, though it is still searchable. The rediscover option removes a field from the database. This is an option when you want to remove the indexing manually. You can decide which fields

are indexed, thus saving storage and index bloat. This helps maintain capacity limits to prevent performance degradation. If the field continuously appears in subsequent logs, it is discovered again as a non-indexed field.

A Field can be:

- Indexed or Non-Indexed
- System Defined or Discovered

| | |
|---|---|
| **System Defined and Indexed Fields** | • These are well-known fields that come when the Log Management component is installed, and is required for proper functioning. This includes Fields for diagnostics, findings, and audit trails.<br>• The system marks these Fields as indexed by default.<br>• Approximately 77 indexed Fields are included by default.<br>• System-defined Fields cannot be rediscovered. |
| **Discovered and Non-Indexed Fields** | • When a Field which is not predefined in the system is ingested from an agent, it is marked as non-indexed and discovered. |

You can manage Fields by clicking the vertical ellipsis next to a field in the **Fields Library**. To see the **Fields Library** card, click **Operate** in the top navigation bar, and in the left navigation pane, **Administration** > **Configurations**.

- Indexing a Discovered Field**:** You can choose to index a discovered field by clicking **Index Field.** You see a confirmation message stating that indexing a field can reduce storage size and affect performance.
- Rediscovering a Field**:** For discovered fields, you can choose to rediscover a field by clicking **Rediscover**. This action removes the field from the index. If the same field continues to be ingested, it will be rediscovered but will remain non-indexed. You can then choose whether to index it again.

## Best Practices for Fields

In addition to the various components that make up a field, several best practices apply.

- Only create fields for regular expression patterns. If a field can be queried using keyword queries, or will only ever return a single value, then use keyword queries instead of a pre-defined field. If a field will only return two values then consider constructing individual queries instead of extracting a field. Fields are meant to add structure to unstructured data as well as provide a way to query over specific parts of an event.
- Only create fields for regular expression patterns that return a fraction of the total events. Fields that will match most events and/or return a very large number of results are not a good candidate for field extraction. The regular expression will need to be applied to a large quantity of events resulting in a resource intensive operation. If possible add additional keywords to reduce the number of results returned and optimize the query.
- If a field contains keywords within regular expression syntax, then add such keywords as a filter without regular expression syntax. For example, if the value or the context of a field contains keywords within regular expression syntax such as `this|that`, then add the keywords as a text filter to optimize the query like `text contains this, that`.
- Use of the additional context with one or more keywords is recommended over complex regular expressions in the before or after context.
- Add additional context to all extracted fields in order to optimize query performance.
- Extracted fields must be specific because the extraction is executed only on the target logs. You must use ASCII characters in the name of extracted field.

### Searching and Filtering Log Events

Click **Operate** in the top navigation bar, and in the left navigation pane, click **Logs**. You can perform search and filter operations under **ANALYZE LOGS**.

> **Note:**
>
> The logic that is used in the filter preview is different from the logic that is used when filtering in the **Logs** page. For more information, see Log Management Filters.

## Performing a Search

To find only events that contain the specified keywords, enter any complete keywords, globs, or phrases in the search text box.

You can search for log events that match certain values of specific fields. Using quoted text in the main search field matches exact phrases. Entering a space in the main search field is a logical AND operator. Search uses only full tokens. For example, searching for "err" does not find "error" as a match.

> **Note:**
>
> The field name for an ingested event must begin only with a letter or an underscore character and contain only letters, digits, or the underscore character.

> **Note:**
>
> If searching for **esx** within a larger string, the query must be formatted as `esx` or `esx*` depending on the position of the sub-token.

Within a single-row filter, add multiple search queries by typing each query and hitting the Enter key. This behaves like OR filters. For example, select **hostnamecontains** and type `127.0.0.1` hit enter and then type `127.0.0.2` and hit enter again. The search returns events with the host name 127.0.0.1 or 127.0.0.2.

> **Note:**
>
> The **text contains** filter treats each word or phrase that you type before hitting enter as a complete keyword.
>
> Queries with fields using the internal query language syntax names, for example, `from` or `in`, are not able to be processed and should not be used.

You can combine multiple field filters by creating a filter row for each field. You can toggle the operator that is applied to multiple-row filters.

- To apply the AND operator, select **All**.
- To apply the OR operator, select **Any**.

> **Note:**
>
> Regardless of the toggle value, the operator for comma-separated values within a single filter row is OR.

You can use globs in search terms. For example, `vm*` or `vmw?re`.

- For 0 or more than two characters, use *.
- For only one character, use ?.

> **Note:**
>
> Globs can be used as the first character of a search term. For example, you can use *.10.11.12. However, using *10.11.12 does not work.

You can save the current query to load it at a later stage.

## Filter Log Events by Time Range

You can filter log events to view only the events for a certain period. You do that by selecting one of the predefined periods. To set the initial and final point of the time range, select **CUSTOM**.

Search Log Events by Field Operations
You can use the list of existing fields to search log events with specific values for the selected field.

Verify that you are logged in to the web user interface as a user associated with a role that has the relevant permissions. The relevant permissions are View and/or Manage for Logs.

> **Note:**
>
> VCF Operations indexes complete alphanumeric, hyphenated, and underscore characters.

1. After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Logs**.
2. Click **Add Filter**.
3. In the filter row under the search text box, use the first drop-down menu to select any field defined within VCF Operations .

   For example, **hostname**, **text**, **source**, and so on. To query logs from an existing index partition, select the partition from the **PARTITION** drop-down list on the top right. Selecting a partition lists a specific subset of events based on the partition filter and renders quick results.

   The list contains all defined fields that are available statically, in content packs, and in custom content.

   Fields are sorted by name, except for the **text** and **partition** fields. Because **text** is a special field that refers to the message text, **text** appears at the top of the list, and is selected by default. Because **partition** is also a special field that refers to index partitions, **partition** appears after the **text** field in the list.

   **Note:**

   Numeric fields contain additional operators that string fields do not: **=**, **>**, **<**, **>=**, **<=**.
4. In the filter row under the search text box, use the second drop-down menu to select the operation to apply to the field selected in the first drop-down menu.

   For example:

   - Select **is** or **is not**. These filters match the full name. Select the partition from the drop-down menu. Using **is** for a partition matches all the events stored in the specified index partition. Using **is not** for the partition matches all the events that are not stored in the specified index partition.
   - Select **contains**. The **contains** filter matches full tokens: searching for "err" will not find "error" as a match. Using **contains** for the partition matches glob patterns in all existing index partitions.
5. In the text box to the right of the filter drop-down menu, type the value that you want to use as a filter.

   You can list multiple values by hitting enter after typing each value. The operator between these values is `OR`.

   **Note:**

   The text box is not available if you select the **Exists** or **Does not Exist** operator in the second drop-down menu.
6. Optional: To add more filters, click **Add Filter**.

   **Note:**

   You can add only one filter when you select a partition. However, after adding a field and selecting a partition, you can add more filters using other fields.

   A toggle button appears with the All/Any option.
7. Optional: For multiple filter rows, select the operator between filters.

| Option | Description |
|---|---|
| **All** | Select to apply the `AND` operation between filter rows |
| **Any** | Select to apply the `OR` operation between filter rows |

   By default, **All** is selected.

   **Note:**

   The partition filter is considered as a supplemental field. The partition filter is combined with filters containing other fields using the AND operator. However, you can select the OR operator to combine filters with non partition fields.

8. Click **SAVE QUERY** and give a name and description.
9. Select the query from the **Choose Saved Query** drop-down list to run the query.

**Note:**

Querying will be slow if the field is not indexed. For more information on field governance, see Log Fields.

**Search for a Group of Hosts that Have a Common String in Their Names**

Assume that you have several hosts that have a host with the following name: w1-stvc-205-prod3, and another host that is called w1-stvc-206-prod5.

To find all logs for both hosts, create the following query.

1. Leave the search text box empty.
2. Define the query by performing steps 3 to 6.
3. Select **hostname** from the field drop-down menu.
4. Select **Starts With** from the operator drop-down menu.
5. Type in the value **w1-stvc** text box. Alternatively, you can use the contains operator, but then you must use a glob in the search value. In this example, you must type w1-stvc-* in the value text box.
6. Click **SAVE QUERY**. Give a name and description.
7. Select the saved query from the **Choose Saved Query** drop-down list.

Examples of Search Queries

Click **Operate** in the top navigation bar, and in the left navigation pane, click **Logs**. You can use these examples when building your queries.

**Note:**

VCF Operations indexes complete alphanumeric, hyphen, and underscore characters.

**Query for all heartbeat events reported by the ESX hostd process yesterday between 9-10am**

To query for all heartbeat events reported by the ESX hostd process:

1. In the search text box, type `heartbeat*`.
2. Define a filter.
   a. Select **appname** from the first drop-down menu.
   b. Select **contains** from the second drop-down menu.
   c. Type `hostd` in the value text box.
3. Define the time range.
   a. In the **Time Range** drop-down menu select **Custom**.
   b. In the first text box, enter yesterday's date and 9am.
   c. In the second text box, enter yesterday's date and 10am.
4. Save the query and run it.

**Search for a Group of Hosts that Have a Common String in Their Names**

Assume that you have several ESX hosts with similar hostnames, for example one with the following name: w1-stvc-205-prod3, and another with the following name: w1-stvc-206-prod5.

To find all logs for both ESX hosts, create the following query.

1. Leave the search text box empty.
2. Define the filter.
   a. Select **hostname** from the field drop-down menu.
   b. Select **starts with** from the operator drop-down menu.
   c. Type `w1-stvc` in the value text box.
   Alternatively, you can use the **contains** operator, but then you must use a glob in the search value. In this example, you must type `w1-stvc-*` in the value text box.
3. Save the query and run it.

**Query for all errors reported by vCenter tasks, events, and alarms**

To query for all errors reported by vCenter tasks, events, and alarms:

1. In the search text box, type `error`.
2. Define a filter.
   a. Select **vc_event_type** from the first drop-down menu.
   b. Select the **exists** operator from the second drop-down menu.
3. Save the query and run it.

Examples of Regular Expressions
You can type regular expressions in text boxes for field values to extract fields from log events.

The expressions that you type must use the Java regular expressions syntax.

**Table 1302: Additional Examples of Regular Expressions**

| Regular Expression | Description |
|---|---|
| [xyz] | x, y, or z |
| (info\|warn\|error) | info, warn, or error |
| [a-z] | A lowercase letter |
| [^a-z] | Not a lowercase letter |
| [a-z]+ | One or more lowercase letters |
| [a-z]* | Zero or more lowercase letters |
| [a-z]? | Zero or one lowercase letter |
| [a-z] {3} | Exactly three lowercase letters |
| [\d] | A digit |
| \d+$ | One or more digits followed by end of message |
| [0-5] | A number from 0 to 5 |
| \w | A word character (letter, digit, or underscore) |
| \s | White space |
| \S | Any character except white space |
| [a-zA-Z0-9]+ | One or more alphanumeric characters |
| ([a-z] {2,} [0-9] {3,5}) | Two or more letters followed by three to five numbers |

Compare Logs
While troubleshooting, you might have to analyze logs from multiple systems that interact with each other. For a specific time interval, you can run multiple queries with different query criteria to search for logs from various systems, and compare the logs. You can also compare logs that are based on the same query criteria but ingested at different times.

1. After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Logs**. Go to **COMPARE LOGS**.

   **Tip:**

   You can also compare logs from the **Analyze Logs** tab:

   - Do either of the following:
     – Enter a query and in the corner of the page, and click the compare icon. Add more queries to compare logs.
     – Enter a query and in the upper-right corner of the page, click the pin icon. Pin multiple queries to the pinboard and then click **Compare Logs**.
2. In the **COMPARE LOGS** tab, enter details for the first query.
   - Enter a name for the query.

- Select the indexed partition on which you want to run the query and click **Apply**. If you do not select an indexed partition, VCF Operations searches for logs in all the indexed partitions.
- Enter a start date and time for querying the logs. VCF Operations uses the start date and time and the duration in the upper-right corner of the page to calculate the time range for the query. When the query runs, it searches for logs within the time range.
- Add one or more filters for more specific query results.

3. Click **Add Query** to add another query and enter the details described in step 2. You can compare logs for up to four queries.

   **Tip:**
   You can copy an existing query by clicking the copy icon next to the query name.

4. In the bottom-right of the query section, enter the common duration for the added queries. VCF Operations uses the start date and time in a query and the common duration to calculate the time range for the query. When a query runs, it searches for logs within the time range.

   **Note:**
   You can enter a maximum duration of 29 days, 23 hours, and 59 minutes.

5. Click **Compare**.

The log comparison is displayed as a stacked line chart and the log results appear side by side in query tabs under the chart. You can select a query tab on the left side and a query tab on the right side to compare the log results.
By default, the chart is based on the logs corresponding to the selected query tabs. However, to view the chart for all the queries, in the upper-right corner of the chart, click **Show All**.

The chart is plotted based on the common time range for the queries. To view the specific time at which a log was ingested, you can hover over the data point in the chart.

**Information in Log Events**
VCF Operations collects and analyzes all types of machine-generated log data, including application logs, configuration files, messages, performance data, and system state dumps.

You can connect VCF Operations to everything in your environment, including operating systems, applications, storage, firewalls, network devices for enterprise-wide visibility using log analytics.

When VCF Operations is configured and ready to collect logs, there are several ways you can ingest log data including:

- vSphere Integration — VCF Operations can integrate with vSphere components to automatically collect events, tasks and alarms from vCenter and logs from ESX hosts.
- Agents — VCF Operations has collection agents available to send filelog, windows events and journald generated logs to VCF Operations.
- Syslog — VCF Operations can ingest data from any source via syslog. Just set the VCF Operations server as your syslog destination.
- CFAPI — Events are sent in their original format to VCF Operations using cfapi. Events sent over cfapi do not have to follow the guidelines of a syslog event and are not modified to comply with the syslog RFC.

Each event contains the following information.

| Type | Description |
|---|---|
| Timestamp | The time when the event occurred |
| Source | Where the event originated. This could be the originator of the syslog messages such as an ESX host or a forwarder such as a syslog aggregation. |
| Text | The raw text of the event. |
| Fields | A name-value pair extracted from the event. Fields are delivered to the server as static fields only when an agent uses the CFAPI protocol. |

**Note:**

VCF Operations is not responsible for the content of the log messages from the individual component. If you have a question about the log contents, contact the team that generated the log message.

Event Types Grouping
VCF Operations summarizes a large number of individual events into a smaller number of broad event types. VCF Operations uses machine learning to group similar events together, with each group showing the approximate number of events in the group. Grouping events helps identify the most communicative events and the most quiet ones, both of which are critical for troubleshooting.

The **Event Types** tab in the **Analyze** page, under the search bar, provides an aggregated view of the events for the given time range of the query. An event in a group is selected as the representative event. You can click the **Expand** link under each representative event to view the events in the group.As a result of grouping events, an Event Type is assigned to each event. An appropriate *event_type* field is created, which you can further use in regular queries. VCF Operations does not document the exact mechanism for grouping events. It tries to automatically detect groups of similar events based on the number of common parts that the events have. For example, let us consider the following events:

- `[2019-05-20 06:41:24.291+0000] ["SearchWorker-thread-12999"/10.113.164.150 INFO] [com.company.product.analytics.distributed.LogSearchWorkerService] [Worker fully completed query (token=5f6e5e1faf93e4ce) in 11 msec]`
- `[2019-05-20 06:41:24.284+0000] ["SearchWorker-thread-11961"/10.113.164.167 INFO] [com.company.product.analytics.distributed.SearchWorkerService] [Worker fully completed query (token=3b247b2ba6057c47) in 24 msec]`

These events have eight common parts - timestamp, thread name, host IP, logging level, class name, message text, token number, and duration. Now, let us consider the following events:

- `[2019-05-20 06:41:24.291+0000] ["LogSearchWorker-thread-12999"/10.113.164.150 INFO] [com.vmware.loginsight.analytics.distributed.LogSearchWorkerService] [Worker finished search (wait=59500 token=5f6e5e1faf93e4ce) in 12 msec]`
- `[2019-05-20 06:41:20.136+0000] ["AliasStudentStudyPool-thread-1"/192.168.110.24 INFO] [com.vmware.loginsight.analytics.alias.AliasStudent] [looking for alias due to rule DatastoreFromVmFileSystem]`

These events only have three common parts - timestamp, host IP, and logging level. In addition to grouping events together VCF Operations identifies useful fields in each event of the group, known as smart fields. Each smart field appears within the representative event as a hyperlink with a drop-down menu icon next to it. You can click the icon to view a histogram for the values of the field or to define an extracted field based on the smart field.

Analyze Event Trends
You can analyze log events for trends and anomalies.

1. After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Logs**.
2. Construct and run your query by using the search text box and applying filters.
3. Click the **Event Trends** tab.

   VCF Operations compares your query to the same time period immediately before and displays the results.
4. Select the period and sorting from the drop-down menus.

# Configuring Log Masking

You can add a configuration to mask sensitive information in all logs or logs that match the filter criteria you provide. You must provide the field or text to mask, the pattern for masking (entire or partial field), the mask value, and a filter criteria.

You can provide rules to mask sensitive information and preview the masking rule which highlights the part of the log that would be masked.

**Note:**

- A log mask configuration is applied only to the logs that are ingested after you create and activate the configuration.
- A log mask configuration can be created by providing **Field Name** and the filters selected from the static fields list.

1. After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**. Then click the **Log Processing** card.
2. Click the **Log Masking** tab.
3. The Log Masking tab displays existing filtering configurations in a list with the following key UI elements:

| Option | Description |
|---|---|
| **Name** | The log filter configuration name. |
| **Field Name** | The field that you have masked in the logs. |
| **Selector** | Regex selector for the field value. |
| **Mask value** | Value to replace the masked content of the specified fields. |
| **Status** | Shows if the log mask is activated or deactivated. |
| **Vertical Ellipses** | For the selected log mask, you can edit, clone, activate, deactivate or delete the log filter. |

4. Click **ADD**.
5. In the **Name** field, enter a unique name for the log mask configuration.
6. In the **Field Name** drop-down menu, select the field that you want to mask in the logs.
7. In the **Selector** text box, enter the regex selector for the field value, which indicates the part of the field that you want to mask. You must express this value as a capture group in the regex. Capture groups are identified with enclosed parentheses (). You can have multiple capture groups inside a selector. To mask all the content for a specified field, you can set the selector as `(.*)` or `(^[\s\S]$)`.
8. In the **Mask Value** text box, enter a value to replace the masked content of the specified fields, the default value for which is an empty string.
9. Select the **Enable Configuration** check box to enable log masking.
10. Click **Add Filter** to define the logs for which you want to mask information. If you do not add a filter, all the logs are masked. To see the results of your filter, click **PREVIEW**. Set filter conditions. You can add multiple filter conditions.

    **Note:**

    The logic that is used in the filter preview is different from the logic that is used when filtering in the **Logs** page. For more information, see Log Management Filters.

    - Contains
    - Does not contain
    - Starts with
    - Does not start with
    - Matches Regex
    - Exists
    - Does not exist
11. Click **PREVIEW** to preview the log mask or **CREATE** to save it.

The log masking configuration appears in the **Log Masking** tab. Any masking rule created via API may appear here. Click the chevron to see more information about the log mask. To modify a single row, click the vertical ellipsis next to the mask

you want to modify. For bulk activation, deactivation, or deletion, select multiple rows and click the horizontal ellipsis next to the add button.

## Configuring Log Filtering

Filtering log events allows you to control which log messages are ingested, forwarded, or dropped based on defined criteria. This capability enables focused logging, improves performance by reducing noise, and enhances security by excluding irrelevant or sensitive logs. This is useful when endpoints are unable to filter logs, allowing for the removal of unwanted data before indexing. Filtering at this stage still consumes resources from the log management Virtual IP (VIP), and this processing overhead should be factored into overall scale and performance considerations.

1. After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**. Then click the **Log Processing** card.
2. Click the **Log Filtering** tab.
3. The Log Filtering tab displays existing filtering configurations in a list with the following key UI elements:

| Option | Description |
|---|---|
| **Name** | The log filter configuration name. |
| **Status** | Shows if the log filter is activated or deactivated. |
| **Filters** | Filter the items in the table by name and activated or deactivated status. |
| **Vertical Ellipses** | For the selected log filter, you can edit, clone, activate, deactivate or delete the log filter. |
| **ADD** | Define new log filters. |

4. Click **ADD** to add a new filter. You can add up to a maximum of 10 filters.
5. Provide the following information:

| Option | Description |
|---|---|
| **Name** | Enter a unique name for the log filter configuration. |
| **Enable Configuration** | The log filter configuration is activated by default. To deactivate the configuration, click the **Enabled** toggle button. |

6. Click **ADD FILTER**. Set filter conditions. You can add multiple filter conditions.
   - Contains
   - Does not contain
   - Starts with
   - Does not start with
   - Matches Regex
   - Exists
   - Does not exist
7. Click **PREVIEW** to preview the filter or **CREATE** to save it.

   **Note:**

   The logic that is used in the filter preview is different from the logic that is used when filtering in the **Logs** page. For more information, see Log Management Filters.

The log filter configuration appears in the **Log Filtering** tab with information about the drop filter and whether it is activated. Click the chevron to see more information about the filter. To modify a single row, click the vertical ellipses. For bulk activation, deactivation, or deletion, select multiple rows and click the horizontal ellipses.

## Configuring External Storage

Configure NFS or S3 external storage locations for exporting logs, archiving logs, and importing archives. You must define a location before you can perform any of these operations. In the **External Storage** page, you can add, edit, and delete

locations and view their settings. Click the chevron against an entry to see more details. Click the vertical ellipsis to edit or delete an entry, or to import an external archive.

- Before you configure an NFS location as external storage, ensure that an NFS export is already set up and reachable from your VCF Operations deployment:
  - If you are using vSAN as the backing storage for your NFS export, see Enable vSAN File Service and Create a vSAN File Share to create an NFSv3 or NFSv4.1 export.
  - If you are using a third-party NAS or NFS server, set up the export according to your vendor's documentation.
- Before configuring an NFS share for import, export, or archive operations, ensure the target directory has the correct ownership and permissions and the path specified is relative to the mount point, and does not start with a leading "/".
- Before adding an NFS share or S3 location, ensure that all IPs in the VCF Management services Node pool are allowed by the NFS server or S3 endpoint, and by any firewalls between them, because log management traffic can originate from any of those IPs.
- To allow the system to write data successfully, the mounted NFS directory and its subdirectories require specific permissions. If the directory does not have the correct permissions, the log management service fails to start.
  - Ownership: The directory must be created by the root user.
  - Minimum Permission Level: 447 — read-only for Owner and Group, and read/write/execute for Others. This is the minimum permission level that VCF Operations requires to write log data successfully. You can also use a more permissive setting such as 777, but 447 is the minimum.

  If your NFS directory does not meet these requirements, update them using the following commands on your NFS server:

```
sudo -i
chmod -R 447 /path/to/your/directory
```

- Ensure that adequate free disk space is available in the external storage.

**Note:**

S3 is preferred to NFS for archive locations. Unlike S3, if an NFS filesystem is unavailable when the log management service is starting, the service will fail to start.

1. After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**. Click the **External Storage** card.
2. Click **Add New Storage**. You can add up to 5 external storage locations.
3. Under **Storage Type**, select either NFS or S3 from the drop-down list.
4. For S3, enter the following details:

| Option | Description |
|---|---|
| **Name** | The name to use for this external storage location. It is used to select the location from drop-down lists in Log Management. |
| **AccessKey** | The access key ID of the account or user that Log Management uses to connect to the S3 bucket. |
| **SecretKey** | The secret access key that goes with the access key ID. |
| **Bucket Name** | The name of the S3 bucket where the log data is stored. The bucket must already exist. |
| **Endpoint** | The URL of the S3 endpoint. The endpoint must use a certificate issued by a trusted Certificate Authority. Self-signed certificates are not supported. |
| **Server Side Encryption** | Specifies whether the S3 service encrypts the archived data at rest. |
| **Canned acl** | Determines the level of access granted to objects stored in your snapshot repository. The default is private, which means only the owner has full control. Change this setting only if your |

| Option | Description |
|---|---|
| | organization requires specific access permissions for shared or public access. |
| **Region** | The region of the bucket, in the format US_WEST_1. The S3 endpoint must use one of these AWS regions. |
| **Path Style Access** | Specifies whether to address the bucket in the URL path (`endpoint/bucket`) instead of in the host name (`bucket.endpoint`). Select this option if your S3-compatible storage doesn't support virtual-hosted-style addressing. |

**Note:**

You must use a certificate which is issued by a trusted Certificate Authority for S3. The self-signed certificate is not supported.

**Note:**

The S3 endpoint must support chunked transfer encoding, because Log Management uses it to upload data.

**Note:**

S3 is preferred to NFS for archive locations. Unlike S3, if an NFS filesystem is unavailable when the log management service is starting, the service will fail to start.

5. For NFS, enter the following details.
   - Host Name
   - Host Address. For example: `nfs://192.30.10.27:/mnt/logs`
6. Click **Validate** to test endpoint and credential validity.
7. Click **SAVE** to add new storage, or **Cancel** to abort.
8. After making changes, click **APPLY** to apply them. When you click Continue, the service is restarted and the changes are applied. Any other change you make also requires the service to be restarted.

   **Note:**

   Log Management is unavailable for up to a few minutes during the restart. During that time logs whcih are not stored by a Cloud Proxy with Data Persistence may be lost.

Troubleshooting

9. Follow these troubleshooting steps for Storage issues

**Table 1303: Troubleshooting NFS Storage Issues**

| Issue | Checklist for Troubleshooting |
|---|---|
| NFS Server Connectivity | Verify the NFS server is reachable |
| Export Path Validation | Ensure that the specified path exactly matches what is exported or shared by the NFS server. Do not specify a subfolder within the share. |
| NFS Unreachable Recovery | If an NFS server becomes unreachable and pods are stuck in ContainerCreating/Completed/Error state, perform a manual recovery procedure. |
| Write Permissions | Verify the NFS export path has write permissions for the user/group that the pods will use. |
| NFS Version Compatibility | Use `showmount -e <nfs-host>` to validate connectivity and export availability. This does not work with NFSv4. Verify NFS v3 or v2 is enabled. |

**Table 1304: Troubleshooting S3 Storage Issues**

| Issue | Checklist for Troubleshooting |
|---|---|
| Endpoint Format Validation | The endpoint must be a valid URI (starting with `http://` or `https://`), include a hostname, and optionally a port. Check for missing protocols or malformed URLs/hostnames. |
| Bucket Existence and Access | Confirm the S3 bucket exists and credentials are valid. Ensure the IAM user/role has minimum permissions: `s3:ListBucket` and `s3:HeadBucket`. |
| Write and Delete Permissions | Validation by creating and deleting a test object. If validation fails, verify the IAM policy includes `s3:PutObject` and `s3:DeleteObject` permissions for the bucket. |
| Canned ACL Configuration | Invalid Canned ACL values cause validation failure. Use the default private ACL if uncertain. |
| Region and Path-Style Access | Verify the region matches the S3 service region (AWS) or is appropriate for your S3-compatible service. For non-AWS S3, set `pathStyleAccess` to true if needed. |

# Log Import and Export

You can export logs to extract specific datasets resulting from an analysis or search query.

### Why Export Logs

- Data Extraction: You can search for specific logs within the Logs page and export the results for external use.
- Reporting: You can export event results and chart data, which is useful for offline reporting or external auditing.
- External Storage Archival: Up to 20K chart data logs can be exported to the local machine for long-term retention or backup. Event data can be exported to an external storage.

### External Storage Requirement

You can export logs to an external storage target, such as **NFS** or **S3**.

> **Note:**
>
> Imports operate at the index level, not at the individual log entry level. If an index's data range overlaps with the requested import interval even partially, the entire bucket is imported. As a result, some logs outside the requested interval may be included in the import.

## Exporting Logs and Chart Data

You can export log search results and chart data from VCF Operations for offline analysis, auditing, or archival purposes. You can initiate exports directly from the Analyze page and monitor the progress of these tasks from the Configurations menu.

To export datasets containing more than 20,000 results, you must configure an external storage path. Direct downloads to the local file system are limited to 20,000 results.

1. Click **Operate** in the top navigation bar, and in the left navigation pane, click **Logs**.
2. Use the search bar and filters to isolate the specific log events you want to export.
3. Go to the Stream tab below the chart.
4. Click the **EXPORT** button, and from the dropdown menu select **Chart Data** or **Events Results**.
5. If you selected **Events Results,** in the **Events Results** dialog box, configure the export settings:

| Option | Description |
|---|---|
| **Format** | Select JSON or CSV |

| Option | Description |
|---|---|
| **Task Name** | Enter a unique name to identify the export task. |
| **Select Storage Path** | Choose the destination for the export file:<br>• **Local:** Select this option for smaller datasets (up to 20,000 results) to download the file directly to your computer.<br>• **External Storage:** Select this option for larger datasets. You must specify the **Source** (Host S3 or NFS) and **Storage Path**. If no external storage is configured, click the **CREATE NEW** link. For more information, see Configuring External Storage.<br>Before configuring an NFS share for import, export, or archive operations, ensure the target directory has the correct ownership and permissions and the path specified is relative to the mount point, and does not start with a leading "/".<br>To allow the system to write data successfully, the mounted NFS directory and its subdirectories require specific permissions.<br>• Ownership: The directory must be created by the `root` user.<br>• Minimum Permission Level: `447` (Read/Write/Execute for Owner and Others).<br>If your NFS directory does not meet these requirements, update them using the following commands on your NFS server:<br>`sudo -i`<br>`chmod -R 447 /path/to/your/directory`<br>**Note:**<br>If no host is currently assigned, click the link provided in the dialog to assign a host or create a new configuration. |

6. If you selected **Chart Data**, select the desired file format. Options are CSV and JSON.
7. Click **EXPORT.**
8. A confirmation message appears stating the export task has been successfully added to the queue.

You can view the status of initiated export tasks to verify their completion.

1. Click **Operate** in the top navigation bar, and in the left navigation pane, click **Administration** and then **Control Panel**. Select the **Log Import/Export Tasks** card.
2. Review the task list. The table provides details including **Name**, **Owner**, **Storage Path**, **Export Format**, and current **Status**. You can cancel tasks which are in the queue.

## Archive Log Import

You can import logs from archived locations and search against these locations. This preserves old logs that might otherwise be removed from an index partition after the retention period. You can import archives from both the Aria Operations for Logs, and VCF Operations for Logs. To initiate an import task, it is mandatory to have external storage (NFS or S3) integrated.

### Before You Begin

Ensure that you have configured an external storage. For more information, see Configuring External Storage. You can import archive logs from the External Storage page as well.

1. After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**. Then click the **Log Processing** card.

2. Click the **Log Partitions** tab and then scroll to the bottom of the page to the **Imported Log Archives** section.
3. Click **INITIATE NEW LOG IMPORT**.
4. Enter the following details:

| Option | Description |
|---|---|
| Name | Specify a unique name for the import job. |
| Operation Type | From the drop-down list, select the type of import operation. Select **import** to import logs from log management. Select **legacy import** to import logs from Aria Operations for Logs. |
| Source | Specify that will execute the import. If no host is selected, configure the host first. Legacy Import supports only NFS storage. |
| Storage Path | Enter the full file system path (e.g., an NFS or SMB share) where the archive log files reside. This must be accessible from the assigned host, matching the file system convention used when logs were archived. |
| Collection Period | Specify the start and end dates (in **MM/DD/YYYY** format) covering the period of logs you wish to import. This narrows down the ingestion to logs within the given time frame |

5. Click **IMPORT LOGS**.

After the import task is completed, you can search against the restored data from the Logs page by selecting the **Import** section and the **archived logs** section in the drop-down menu. You can import only archive.

## Configuring Log Forwarding

You can configure a log management server to forward incoming log events to a syslog API target. A maximum of 10 forwarding rules can be created. You can forward logs to external services.

### Before You Begin

**Note:**

- Verify that the destination can handle the number of logs that are forwarded. If the destination cluster is much smaller than the forwarding instance, some logs might be dropped.
- Use log forwarding to send filtered or tagged logs to one or more remote destinations such as log management or syslog or both. Log forwarding can be used to support existing logging tools such as SIEM and to consolidate logging over different networks such as DMZ or WAN.
- Log forwarders can be standalone or clustered, but a log forwarder is a separate instance from the remote destination. Instances configured for log forwarding also store logs locally and can be used to query data.
- The network bandwidth, the destination availability, and performance and stability of the environment may impact the reliability of Log Forwarding feature.
- During the upgrade process, any existing log forwarder rules from the Aria Operations for Logs or VCF Operations for Logs are pulled and added into log management.
- Forwarding to Aria Operations for Logs 8.1x is supported.

### Configure Log Forwarding

1. After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**. Then click the **Log Forwarding** card.
2. Click **ADD**.

3. Enter the following details:

| Option | Description |
|---|---|
| Name | A unique name for the new destination, Host, The IP address, or fully qualified domain name. |
| Enable Configuration | Enable or disable the log forwarding configuration. |
| Host | The IP address or fully qualified domain name.<br>**CAUTION:**<br>A forwarding loop is a configuration in which a Log Management cluster forwards logs to itself, or to another cluster, which then forwards the logs back to the original cluster. Such a loop might create an indefinite number of copies of each forwarded log.<br>When creating forwarding destinations, take care not to create indirect forwarding loops. |
| Protocol | When logs are forwarded using **Syslog**, the log's original source is lost and the receiver can record the message's source as the Log Management forwarder's IP address or hostname.<br>When logs are forwarded using **Raw**, the behavior is similar to syslog, but syslog RFC-compliance is not ensured. RAW forwards a log exactly the way it is received, without a custom syslog header added by Log Management. The RAW protocol is useful for third-party destinations, because they expect syslog events in their original form.<br>**Note:**<br>• The source field might have different values depending on the protocol selected on the Log Forwarder:<br>a. For syslog and RAW, the source is the Log Forwarder's Log Management instance IP address. |
| Use SSL to secure your connection | You can optionally secure the connection with SSL for the ingestion API or syslog. If the SSL certificate provided by the forwarding destination is untrusted, you can accept the certificate when you test or save this configuration.<br>All log forwarder configurations must use certificates that include a valid SAN in order to establish proper communication in SSL-enabled mode.<br>To verify your configuration, click **VALIDATE CONNECTION**. This sends test logs to the destination. |
| Custom Fields | Custom fields, also called tags, are user-defined key-value pairs that can be configured log forwarders to categorize and enrich incoming log messages with additional metadata for filtering and analysis purposes. |
| Forward Complementary fields | Complementary fields are additional log message fields beyond the core syslog headers (hostname, appname, procid, msgid) that are optionally forwarded in structured data elements when the this setting is enabled on a log forwarder. |
| Transport Protocol | Select a transport protocol for syslog. You can select UDP or TCP. |

4. To control which logs are forwarded, click **Add Filter**.
   - Contains
   - Does not contain
   - Starts with
   - Does not start with

- Matches Regex
- Exists
- Does not exist

Click **PREVIEW** to preview the log partition or **CREATE** to save it.

**Note:**

The logic that is used in the filter preview is different from the logic that is used when filtering in the **Logs** page. For more information, see Log Management Filters.

**Note:**

Log filtering takes precedence over log forwarding. If there are any filtering rules, the logs matching to that rules are not forwarded.

## Log Management Filters

The filter preview in the log processing pages, and the filters that you use in the **Logs** page do not have a one-to-one correspondence by name. However, you can select operators that produce similar results for both formats.

This difference is important when you use the **PREVIEW** option after you add filters in the following Log Processing pages and use **ADD FILTER** in the **ANALYZE LOGS** tab of the Logs page:

- **Log Masking**
- **Log Filtering**
- **Log Forwarding**
- **Log Partitions**

### Matching Logic Differences Between Log Filtering and Log Processing Filter Preview

When you compare log search using filters in the Logs page with filter preview in the log processing pages, you will observe differences in how search strings are matched. This behavioral difference occurs because Logs page search ues Query Domain Specific Language (DSL) which has a strict token-based matching, whereas log processing filters use flexible full-text matching.

| | |
|---|---|
| **Token-Based Matching (Query DSL)** | In Query DSL, most operators (with the exception of regular expressions) evaluate text at the token level. The system divides the log text into individual tokens separated by specific delimiters. The following characters act as token separators: `~ ! @ # $ % ^ & * ( ) + = [ ] { } \ \| ' : ; / ? , . < > " \t \n Space`<br><br>For a query to return a match in Query DSL, the search criteria must align perfectly with the defined token boundaries. |
| **Full-Text Matching (Log Processing)** | Unlike Query DSL, the DSL used in the log processing tabs (Log Masking, Log Filtering, Log Partition, and Log Forwarding) evaluates the full text string as a single entity rather than breaking it into tokens. This approach is more flexible and successfully matches partial words or substrings, even if they do not align with token boundaries. |

### Matching Behavior Examples

The following examples illustrate how the different matching engines evaluate the same log text and query.

- Target Log Text: `Filter Logic In|VCFOps`

- Generated Tokens: `Filter`, `Logic`, `In`, `VCFOps`

| UI Query | Query DSL Result (Logs page filter) | Log Processing filter | Reason |
|---|---|---|---|
| text contains `Logic In` | Match | Match | The query phrase exactly matches complete, individual tokens (`Logic` and `In`), and also exists as a substring in the full text. |
| text contains `Logic I` | No Match | Match | The letter `I` is a partial string and does not match the full token `In`. Therefore, Query DSL rejects it. However, the log processing engine detects `Logic I` as a valid substring within the full text. |

Because of this architectural difference, you might experience inconsistent behavior when testing preview in the log processing pages, and when using the logs filter in the Logs page, particularly when your search criteria includes partial words or substrings that do not align with standard token boundaries.

## Overview of Log Management Agents

A Log Management agent collects logs from log files and forwards them to a Log Management server or any third-party syslog destination.

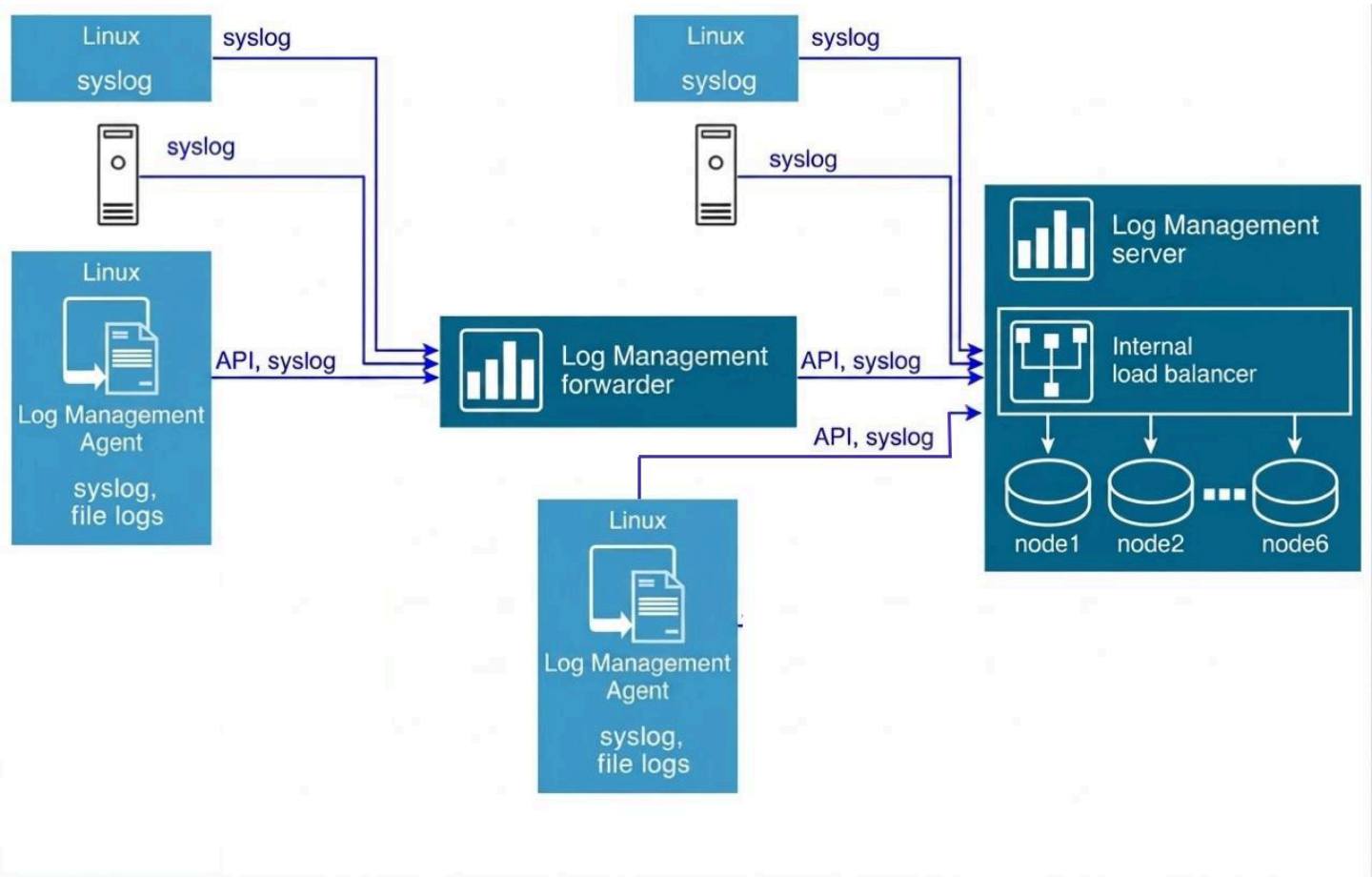
Agents support syslog and the Log Management ingestion API (cfapi protocol) and can be used with Linux. You configure agents through their web interface, through the liagent.ini file on the server and client side, or as part of installation.

Agents include the following features:

- Single or group deployment
- Automatic upgrade
- Parsing that operates on log messages and extracts structured data. You can configure parsers for FileLog and WinLog collectors or both.
- Support for multi-line messages
- Native support for several log rotation schemes
- An extensive ingestion API that includes client-side compression, encryption, and the ability to add metadata to logs

The Log Management server supports centralized configuration management and creation and management of groups of agents.

The following figure shows the elements of an agent deployment configuration.




A Log Management log forwarder is a dedicated instance of a Log Management server whose primary job is to forward logs to a remote destination. Normally, a server instance used as a forwarder is not used for query. The forwarder uses an internal load balancer and is otherwise structured like a Log Management server.

Agents write their own operation logs. For Linux, the path for the operation log is `/var/log/loginsight-agent/liagent_*.log`. Log files are rotated when an agent is restarted or when the file reaches a size of 10 MB. A combined limit of 50 MB of files is kept in rotation. You cannot collect agent logs with the Log Management agent itself.

Agents are used for real-time log collection. Use the Log Management importer to import historic log collections, including support bundles.

On Linux systems, the agent runs as a daemon and starts immediately after installation. The Log Management Linux agent collects logs from log files on Linux machines and forwards them to Log Management servers or syslog destinations. Debian, Red Hat, and Linux binary installation packages are available.

Use the following commands to install the Linux packages:

- For deb:`dpkg -i <deb_package>`
- For RPM: `rpm -i <rpm_package>`

In both cases, you must have appropriate permission to install the packages.

**System and Operating System Requirements**

Log Management Agents support installation on Linux operating systems that satisfy the following core system component dependencies:

- OpenSSL: Version 3.0 or higher
- systemd: Version 235 or higher

**Supported Operating System Versions**

| Operating System | Supported Versions |
|---|---|
| Red Hat Enterprise Linux (RHEL) | RHEL 9, RHEL 10 |
| SUSE Linux Enterprise Server (SLES) | SLES 15 SP7, SLES 16 |
| Ubuntu | Ubuntu 22.04 LTS, 24.04 LTS, 26.04 LTS |
| Debian | Debian 12, Debian 13 |
| VMware Photon OS | Photon 4.0 and higher |

## Configuring Agents

You can configure agents from within VCF Operations. Agents poll the Log Management component regularly to determine if new configurations are available. You can manage log collection agents to forward logs from vCenter instances and other components to VCF Operations. This includes downloading agent media, managing authentication secrets, monitoring agent status, and configuring agent groups.

For more information about secrets and agent authentication, see Agent Authentication for Log Ingestion.

**Note:**

Auto upgrade for the agents that are sending logs through Cloud Proxy Log Forwarder is not supported.

1. After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**.
2. Click the **Log Collection** tile and go to the **Agent** tab.
3. Click **DOWNLOAD AGENT**.
4. Download the agent.
   a) Navigate to the Broadcom support portal to download the Log Agent file. The supported formats are RPM and Debian.
   b) Import the file to your appliance and start the installation.
   c) Allow a few minutes for the system to detect the agent after deployment.

   Manage Agent Secrets

   Agents require a shared secret/token to authenticate with VCF Operations. You must generate this secret and copy it immediately, as it is displayed only once.
5. Select the **SECRETS** tab.
6. Click **GENERATE SECRET**.
7. In the **New Secret Creation** dialog, enter a **Secret name** (for example, `IP_FQDN_OF_Agent Appliance`).
8. Click **CREATE**.
9. The system displays a **Secret created successfully!** message. Click **COPY** to save the **Secret Value** immediately.
10. Click the confirmation dialog box to close it.
11. Navigate to the `/etc/liagent.ini` file in the appliance where you installed the agent. In the `server` section, under hostname, add `secret=COPIED_TOKEN` and save the configuration file.

    Revoke a Secret

12. In the **SECRETS** tab, locate the secret that you wish to remove.
13. Click **Revoke** next to the secret name.
14. A confirmation dialog appears. Review the warning indicating when the secret is permanently removed from the system.
15. Click **REVOKE** to confirm. The status updates to **Revoked**, and the revocation date will be displayed.

    Monitor Agent Status

    You can monitor the health and activity of your connected agents.
16. Navigate to the **Agents** tab.
17. After you download and install the agent on a virtual machine, the agents will appear in a centralized list. This list provides detailed information about the virtual machine, the agent, and its current status. Review the agent list, which provides the following details:

    **Note:**

    For VCF agents, agent configuration coming from log management is not applied.

- **IP Address**. IP address of the agent VM.
- **Hostname**. Hostmame of the agent VM.
- **Version**. Version of the agent installed.
- **OS**. OS of the agent VM.
- **Secret Name**. The secret is displayed here. You can revoke this by clicking the vertical ellipses against the row.
- **VCF Component**. All log collection components which contain embedded agent has VCF component =yes value. These agents are not configurable.
- **Last Active**: The time elapsed since the agent last communicated.
- **Events Sent**: Total number of events forwarded.
- **Events Sent/Sec**: Current throughput rate.
- **Events Dropped**. Events dropped by the agent.
- **Uptime**. Agent uptime.
- **Status**: Indicates if the agent is connected or disconnected

## Configuring Agent Group

An Agent Group allows you to apply configurations to a specific subset of agents. For example, you might group all Log Management. You can create a group of agents that are configured with the same parameters.

**Note:**

VCF log agents deployed on VCF components cannot be added to agent groups. If you need to send logs to an external log management solution, use the log forwarding mechanism instead. For more information, see Configuring Log Forwarding.

1. After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**. Then click the **Log Collection** tile.
2. Click the **Agent Group** tab.
3. Click **ADD**.
4. In the **Creation** step of the wizard, provide a unique **Name** and **Description** for the agent group.
5. Select **Auto update for all Agents** if you want to provide the same settings for all the agents. Enabling this allows automatic updates. When enabled, all the agents listed under the configuration can pull updates from the underlying log platform and apply them to their virtual machines.

   **Note:**

   There can be agents that do not belong to any agent group. For such agents, you must decide whether the agents should be auto-updated or not. This behavior is controlled by a global auto-update flag. The flag is a dynamic

configuration, defaults to false, and can be managed from the Agents page using the Auto Upgrade Agents Group toggle.

6. Specify one or more filters for the agent group. To create a filter, click **ADD FILTER** and specify a field name, an operator, and a value. Choose one of the following fields to filter on:
   - **IP address**
   - **Hostname**
   - **Version**
   - **OS**
7. Select an operator from the drop-down menu and specify a value.
   - Contains -
   - Does not Contain
   - Starts with
   - Does not Start with

   a) Multiple filters are treated as AND operands, and multiple values of the same filter are treated as OR operands. Filters can contain wildcards, such as * and ?. For example, you can select the OS filter contains and specify the value windows to identify all your Windows agents for configuration.
8. Use the **Preview** button to preview the Agents that match the filtering criteria.
9. Click **NEXT**.
10. In the Configuration step of the wizard, click the radio button against **Agent Configuration**.
    - New Agent Group
    - Select from Templates. You can add an agent group that was defined as part of a VCF Operations Management pack to your active groups and apply an agent configuration to the group.
11. Click **CONFIGURE** to specify the agent configuration values. Alternatively, select **Code View** to provide configuration in the form of code.

The agent configuration is applied within 5 minutes.

**Note:**

An agent can technically belong to multiple agent groups; however, in such cases, the applied configuration is non-deterministic. No guarantees are provided regarding which group's configuration will be used, so you must avoid adding agents to multiple groups.

## Set up the Windows system to collect logs in VCF Operations

You can install the supported version of Fluent Bit on a Windows Server 2022 and Windows Server 2025 instance to collect Windows Event Logs, including System, Security, and Application logs. Once configured, Fluent Bit forwards these logs to Log Management using nested JSON.

- Verify that you have an instance running Windows Server 2022 or Windows Server 2025 and possess administrative access to the target server.
- Ensure there is outbound network connectivity from the server to the Log Management VIP on TCP port 9543.
- Download the custom 64-bit supported version of Fluent Bit MSI installer from the VCF Solutions Catalog.
- Obtain the required Log Management CA certificate.

1. Install the Fluent Bit binary using the custom MSI installer. You can double-click the downloaded MSI file and follow the interactive wizard prompts, which installs the software to the following directory: `C:\Program Files\fluent-bit\`.

Alternatively, for automated deployments, open PowerShell and run the silent install command: `msiexec /i fluent-bit-4.2.3-win64.msi /quiet /norestart`.

2. Create a dedicated directory for Fluent Bit log output by running the `New-Item -ItemType Directory -Path "C:\Program Files\fluent-bit\logs" -Force` command in PowerShell.
3. Navigate to the configuration folder at `C:\Program Files\fluent-bit\conf\` and rename the default `fluent-bit.conf` file to `fluent-bit.conf.old` to create a backup.
4. Create a new `fluent-bit.conf` file to specifically target Windows Event Logs and the Log Management destination. Insert the following configuration, ensuring you update the placeholder values (`<PATH_TO_LOG_FILE>`, `<PATH_TO_OFFSET_DB_FILE>`, `<YOUR_CUSTOM_FIELD>`, `<YOUR_CUSTOM_VALUE>`, `<LOG_MANAGEMENT_FQDN>`, and `<PATH_TO_CA_CERTIFICATE>`) with your actual environment details:

```
Ini, TOML
[SERVICE]
    Flush 5
    Daemon Off
    Log_Level info
    Log_File <PATH_TO_LOG_FILE>

[INPUT]
    Name winevtlog
    Channels Application, System, Security
    Interval_Sec 1
    DB <PATH_TO_OFFSET_DB_FILE>

[FILTER]
    Name    modify
    Match   *
    Rename  Message text
    Add <YOUR_CUSTOM_FIELD> <YOUR_CUSTOM_VALUE>

[OUTPUT]
    Name http
    Match *
    Host <LOG_MANAGEMENT_FQDN>
    Port 9543
    URI /api/v2/events
    Format json
    Json_date_key timestamp
    Json_date_format epoch_ms
    Header Content-Type application/json
    tls On
    tls.verify On
    tls.ca_file <PATH_TO_CA_CERTIFICATE>
```

   **Note:**

   For testing purposes only, you can temporarily set `tls.verify Off` to disable certificate verification. Do not use this setting in a production environment.

5. Open PowerShell as an administrator and register Fluent Bit as a Windows Service so that it starts automatically after a system reboot. Run the following command:

```
New-Service fluent-bit -BinaryPathName '"C:\Program Files\fluent-bit\bin\fluent-bit.exe" -c "C:\Program
 Files\fluent-bit\conf\fluent-bit.conf"' -StartupType Automatic -Description "This service runs Fluent
 Bit, a log collector that enables real-time processing and delivery of log data to centralized logging
 systems."
```

6. Start the newly registered service and verify its running status by executing the `Start-Service fluent-bit` and `Get-Service fluent-bit` commands.

   ```
   Start-Service fluent-bit
   Get-Service fluent-bit
   ```

   Verify that the Status is `Running`.

7. Verify that your Windows Server can successfully communicate with the Log Management server. In PowerShell, run the `Test-NetConnection ComputerName "FQDN" -Port 9543` command (substituting your actual FQDN) and confirm that the expected result is `TcpTestSucceeded: True`.
8. After confirming network connectivity, log in to the VCF Operations UI and navigate to **Operate** > **Logs**. Add a filter for `hostname contains [Your-Server-Name]` and confirm that new logs appear from the `winevtlog` source.

Upgrade

9. Back up your live config (at a minimum, the `conf\fluent-bit.conf` or `conf\fluent-bit.yaml`, plus `parsers.conf / plugins.conf` if you use them).
10. If Fluent Bit runs as a Windows service, stop it first (`Stop-Service fluent-bit` in PowerShell, or `sc stop fluent-bit`, depending on how it was set up).
11. Run the new installer. In many environments, this replaces the binaries under the same install directory.
12. Restore config files from backup.
13. Start the service again and confirm the logs.

## Agent Authentication for Log Ingestion

Agent authentication secures the HTTP ingestion path for log data by introducing an optional authentication mechanism using secrets and tokens. This feature supports both the Log Management agent and 3rd-party agents such as Fluentd or FluentBit. By default, authentication is enabled but optional. The server accepts both authenticated and unauthenticated logs. However, you can enforce authentication globally, rejecting any log ingestion requests that do not possess a valid token.

### Why Agent Authentication

| | |
|---|---|
| **Enhanced Security** | Prevents unauthorized log injection into the logging infrastructure. |
| **Flexible Integration** | Works with native Log Management agents and standard 3rd-party log shippers. |
| **Granular Control** | Secrets can be managed, revoked, and tracked individually. |

### Prerequisites & Limitations

Before configuring authentication, ensure the following requirements are met:

- Agent Version**:** Log Management agent version 9.1 or higher is required.
- Protocol: Authentication applies only to HTTP/HTTPS ingestion APIs. Legacy Syslog ingestion remains unauthenticated.

> **Note:**
>
> Windows agents are not supported.

### Managing Secrets

Secrets serve as the initial credentials used to authenticate an agent. You must generate a secret before configuring the agent. You can generate a secret in VCF Operations, or using REST API. For large-scale or automated deployments, use the REST API. For information on how to generate a secret using VCF Operations, see Configuring Agents.

### Configuring the Native Log Management Agent

1. Locate the `liagent.ini` agent configuration file on the target machine.
2. Open the file and locate the `[server]` section.
3. Add the `secret` property with the secret value generated from the Agent configuration page:

```
[server]
hostname=log-server-address
proto=cfapi
secret=<your_generated_secret>
```

4. Save the file.

The agent automatically exchanges this secret for a token and handle renewal when the token expires.

### Configuring Third-party Agents

Third-party agents do not have a built-in mechanism to exchange secrets for tokens. This process must be performed manually or using an external script. Refer to the FluentBit or FluentD documentation for more information.

### Enforcing Authentication

By default, authentication is optional. Unauthenticated logs are accepted, but if an agent attempts to authenticate with an invalid key, it is rejected.

To enforce strict authentication and reject all unauthenticated traffic, enable the setting in the Global Settings. Once enabled, any ingestion request without a valid Bearer token is dropped. Ensure that all agents are correctly configured before enabling this setting. For more information, see List of Global Settings.

### Revoking the Secret

If a secret is compromised or an agent is decommissioned, you can revoke the secret to block access. Revoked secrets are retained in the list for approximately 7 days for audit purposes before being permanently deleted. During this time, the name cannot be reused.

## Log Statement Standardization Format

VMware Cloud Foundation comprises various components each of which issue log messages.

In VCF , the following components use the standard format for their application log messages:

- vCenter
- ESX
- NSX
- VCF Operations HCX
- VCF Operations for networks
- VCF Operations
- Identity Broker
- VCF Automation

This standard addresses a long-standing problem of VCF components using component-specific formats, in some cases, multiple formats for the same component. These different formats complicate troubleshooting due to such factors as using different time stamp formats and recording information in different parts of the log message. The different formats also make it more difficult to automatically parse log messages.

Standardizing the log message format addresses both problems. It will make it easier to manually troubleshoot problems using the log statements and process the log statements programmatically. This standard is based on the ESX log formats for application log files that was introduced in ESX 8.0. If automatic parsing of log messages was developed for the earlier versions of the above noted components, this parsing logic may need to be updated to correctly parse the log statements compliant with the standard.

Application log messages are written into files that are maintained by components on their local file systems. The log statement standard applies to log statements issued by software modules authored by the VCF division and not to third-party software modules such as Apache Tomcat. In 9.1, centralized log collection transmits only ESX logs via syslog and these logs are already standardized

## Log Statement Standard

The log statement standard is described below using Augmented Backus-Naur Form (ABNF) notation. The standard describes the format of an application log message. This section summarizes the key elements.

In the standard, an application log message comprises three parts that are separated by spaces and terminated by a newline. The line below lists the format.

```
HEADER <space> STRUCTURED-DATA <space> MSG
```

> **Note:**
>
> HEADER and STRUCTURED-DATA must be in every log statement while the MSG part may be omitted.

The following log message is an example of one that is compliant with the standard:

```
2025-06-05T05:21:29.840Z INFO api 25572 [ops@4413 threadId=”639”
threadName=”127.0.0.1-8010-exec-26”] Completed POST /diagnostics/findings
```

The list below maps this example back to the format of a log message:

- HEADER: `2025-06-05T05:21:29.840Z INFO api 25572`
- STRUCTURED-DATA: `[ops@4413 threadId=”639” threadName=”127.0.0.1-8010-exec-26"]`
- MSG: `Completed POST /diagnostics/findings`

This example illustrates three key principles of the log message format.

First, the first two parts of the log message format (HEADER and STRUCTURED-DATA) contain sub-parts separated by spaces. For example, the header includes four sub-parts (timestamp, severity, app-name (application name), and proc-id (process ID). The meaning and format of the sub-parts are defined in the table below using Augmented Backus-Naur Form (ABNF) notation.

Second, the timestamps comply to the format defined in the RFC 5424 standard and, in addition, always use UTC/GMT. In earlier VCF versions, the lack of a uniform timestamp made it difficult to manually or programmatically order log messages. The log standard addresses this problem by mandating that all timestamps be in this format.

Finally, every log message that is compliant with the standard includes the sub-part `<string>@4413` in the STRUCTURED-DATA part. In the above example, the string is `ops`. This string reports the component that issued the log statement. In the example above, the log message was issued by VCF Operations. It is abbreviated as `ops`. The formal name for this sub-part is SD-ID. Its format is defined in the table below. The standard requires all compliant log messages to contain the SD-ID. This fact can be used in automatic parsing of log messages. Log messages that include the SD-ID are compliant with the standard. Log messages that do not include SD-ID may be partially compliant or not at all compliant. Parsing logic that is written for the standard will correctly parse compliant messages and may not parse non compliant messages.

If automatic parsing of log messages was developed for the earlier versions of the above noted components, this parsing logic may need to be updated to correctly parse the log statements compliant with the standard. Follow this guidance for reviewing existing parsing logic:

- Review existing automation logic
- If the logic is parsing log statements from components other than the three listed in this section, no action is required.
- If the logic is parsing log statements from the components that are mentioned above, search the application log files of the component for examples of the log messages of interest. Generally, the MSG part of the log message did not change. So, start by searching for the message ignoring any timestamps or other parts of the original log line.
- Review the log message examples found in the previous step. If the message includes the SD-ID, review the parsing logic for any changes.

To create a parser that can programmatically process the VCF application log message format, use the ABNF notation given below. Parser generators can be used to generate a parser that understands the format. Finally, as already noted,

the standard is based on the ESX 8.0 standard. There are a few differences between these standards, which can be understood by comparing the VCF log ABNF to the ESX ABNF for log files that the `vmsyslogd` service manages. An existing parser that understands this ESX log format can be modified and applied for parsing log messages compliant to the standard.

### VCF Application Log Message Format

Application logs are written by a component to its local file system.

| Parameter | Value |
|---|---|
| **SP** | `ASCII space %x20` |
| **LOG-MSG** | `HEADER SP STRUCTURED-DATA [SP MSG] %0A` |
| **HEADER** | `TIMESTAMP SP SEVERITY SP APP-NAME SP PROC-IDENTIFIER` |
| **APP-NAME** | `1*48PRINTUSASCII` |
| **PROC-IDENTIFIER** | `NILVALUE / 1*DIGITS` |
| **TIMESTAMP** | `FULL-DATE "T" UTC-TIME`<br>Complies to RFC 5424 with the requirement for UTC/GMT formatting. |
| **FULL-DATE** | `DATE-FULLYEAR "-" DATE-MONTH "-" DATE-MDAY` |
| **DATE-FULLYEAR** | `4DIGIT` |
| **DATE-MONTH** | `2DIGIT ; 01-12` |
| **DATE-MDAY** | `2DIGIT ; 01-28, 01-29, 01-30, 01-31 based on month/year` |
| **UTC-TIME** | `TIME-HOUR ":" TIME-MINUTE ":" TIME-SECOND [TIME-SECFRAC] "Z"` |
| **TIME-HOUR** | `2DIGIT ; 00-23` |
| **TIME-MINUTE** | `2DIGIT ; 00-59` |
| **TIME-SECOND** | `2DIGIT ; 00-59` |
| **TIME-SECFRAC** | `"." 3*6DIGIT`<br>Providing fractional seconds is considered very important. Millisecond resolution is the minimum allowed (3 digits) when they are added. |
| **SEVERITY** | `[ LINE-MARKER ] SEVERITY-STRING` |
| **SEVERITY-STRING** | `"EMERGENCY" / "ALERT" / "CRITICAL" / "ERROR" / "WARNING" / "NOTICE" / "INFO" / "DEBUG"` |
| **LINE-MARKER** | `"-" \| "+"` |
| **NILVALUE** | `"-"` |
| **STRUCTURED-DATA** | `1*SD-ELEMENT` |
| **SD-ELEMENT** | `"[" SD-ID [*(SP SD-PARAM)] "]"` |
| **SD-PARAM** | `PARAM-NAME "=" %d34 PARAM-VALUE %d34` |
| **SD-ID** | `SD-COMP-ID "@4413"` |
| **SD-COMP-ID** | `1*27PRINTUSASCII` |
| **PARAM-NAME** | `1*32PRINTUSASCII ; characters '"', '\' and ']' MUST be escaped.` |
| **PARAM-VALUE** | `UTF-8-STRING; ; except '', SP, ']', %d34 (")` |
| **MSG** | `UTF-8-STRING` |

# Log Based Alerts

You can create alerts in VCF Operations for events triggered by log symptoms. You can create a query directly in the Log Symptom definition, or select a list of saved queries. Alerts are triggered when incoming logs match the defined alert criteria and contain log entries corresponding to the alert's configured object type.

For more information, see Create a Log Based Alert Definition.

# Integrated Load Balancer

Integrated Load Balancer allows you to manage the IP and DNS addresses of their log management deployment. You can add multiple Virtual IP (VIP) addresses and corresponding FQDNs. You can also configure dual-stack (IPv4/IPv6) deployments by including IPv6 addresses (AAAA records). You can attach up to 5 static fields to VIPs. Logs flowing or ingested through that VIP will have those fields attached, making them searchable via those fields.

To ensure full functionality of Virtual IPs (VIPs), updating the certificate to include all VIP and FQDN entries is required, before a VIP can be created. This is necessary not only to prevent SSL host verification issues but also to satisfy system validation during the addition of new entries.

1. After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**. Then click the **Log Collection** tile.
2. Click the Integrated Load Balancer tab.
3. You list a list of Virtual IPs, their FQDN, and configuration status.
4. To add a new virtual IP, click **CREATE NEW VIRTUAL IP**. Optionally, click the vertical ellipsis next to the existing virtual IP and click **Edit**.

   **Note:**

   You cannot edit the default Virtual IP and FQDN. However, you can add field values to tag the default virtual IP.
5. Provide or edit the following information.

   Note: While HTTP-based logs should use the FQDN, syslog messages can be sent directly to the Virtual IP address.

| Option | Description |
|---|---|
| **Virtual IP** | Provide the virtual IP address that will be used by the load balancer to accept log messages. |
| **FQDN** | Provide the FQDN (Fully Qualified Domain Name). |
| **Fields (Optional)** | Provide one more field names and values to tag logs received by this VIP. For (TCP) syslog-based ingestion field tagging is not supported. |

6. Click **CREATE**.

# Log Administration

Log Administration is facilitated by the data set capability, which defines restrictions on which logs a user can access. This involves applying a scope, which can be restricted to specific log data sets (example, logs containing "application name x"), to a user or user group. The roles include log pages and other aspects.

## Defining a Data Set

You can define a data set to provide users access to specific content.

1. After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Control Panel** under **Administration.** Go to **Access Control** and then to the **Scopes** tab. Click the **ADD** or **EDIT** button, and then scroll down to the **Assign Log Data Set** section.
2. Click **Add Filter**.
3. Use the first drop-down menu to select a field defined within VCF Operations, to filter on.

   For example, **hostname**.

   The list contains static fields only and excludes fields that are extracted, user shared, and fields created through event_type filters.
4. Use the second drop-down menu to select the operation to apply to the field selected in the first drop-down menu.

   For example, select **contains**. The **contains** filter matches full tokens: searching for the string `err` does not result in `error` as a match.
5. In the filter box to the right of the filter drop-down menu, enter the value that you want to use as a filter.

   You can use multiple values. The operator between these values is OR.

   **Note:**

   The box is not available if you select the **exists** or **does not exist** operator in the second drop-down menu.
6. Optional: To add more filters, click **Add Filter**.
7. Optional: To verify that the filter behavior is what you want, click **PREVIEW**, which opens a window to check the logs corresponding to filter.
8. Optional: Click **SAVE**.

## User Management and Role-Based Access Control

Role-based access control lets you manage users and the tasks that they can perform.

### What are Roles

Roles are sets of permissions required to perform particular tasks. VCF Operations has a set of predefined roles. Additionally, you can create custom roles as part of defining security policies, and grant the roles to users. To change the permissions and tasks associated with a custom role, you can update the role settings. The updated settings take effect for all users associated with the role.

- To allow a user to perform a task, you grant the role to the user.
- To prevent a user from performing a task, you revoke the role from the user.

Managing roles for each user is based on their user login account. You can assign multiple roles to a user, in which case the role permissions are merged.

Users who cannot view or access certain objects or cannot perform certain operations were not assigned the roles with the permissions to do so.

### Role-Based Access Control

In VCF 9.1, the log management functionality has been converged into the VCF Operations role based access control. As part of this integration, the legacy log-specific access controls have been migrated to the centralized VCF Operations roles.

This unification allows administrators to manage users, groups, roles, and scopes for both infrastructure monitoring and log management from a single control panel.

For more information, see Access Control in VCF Operations.

| | |
|---|---|
| **Users** | You can control the access and actions of each user by granting or revoking roles to or from the login account of the user. |

| | |
|---|---|
| **Permissions** | Specific permissions control access to log features. If a user's role lacks a specific permission, the corresponding feature tile, APIs, or page in the UI will not be visible to them. For example, tiles such as Log Processing, Log Forwarding, Log Collection, and External Storage are controlled by permissions.<br><br>You can manage permissions in the **Access Control** card by clicking **Operate** in the top navigation bar and in the left navigation pane, clicking **Administration** > **Control Panel**. Go to the **Roles** tab and edit a role to select or deselect the permissions appropriately. For more information, go to the topic, Add or Edit Roles and Assign Permissions. |
| **Data Sets** | Data sets consist of a set of filters. You can use data sets to provide users with access to specific content by associating a data set with scopes. The logs that the user sees in the Logs page, log based widgets and saved queries, is dependent on the data set assigned to them. |
| **Roles** | Roles determine what a user can do within the platform. A role is a collection of specific permissions (privileges).<br><br>• Built-in Roles**:** VCF Operations includes built-in roles, such as **Administrator** (full permissions) and **Read-only** (limited permissions).<br>• **Custom Roles:** Administrators can create custom roles by selecting a specific subset of permissions to tailor access for different user personas. |
| **Scopes** | Scopes determine what data a user can see. In VCF Operations 9.1, a Scope is a container that can include two distinct types of visibility configurations:<br><br>• **Object Scope:** Defines which infrastructure objects (example, Clusters, VMs) are visible to the user in the inventory.<br>• **Log Data Set:** Defines filters for the raw log data available to the user (example, Text contains 'error'). |

In VCF Operations 9.1, Object Scopes and Log Data Sets are independent. Selecting a set of objects (example, Cluster A) in the Object Scope does not automatically restrict log visibility to only logs coming from Cluster A.

- Log visibility is determined solely by the filters defined in the **Assign Log Data Set** section of the Scope.
- It is possible to assign a Log Data Set to a user without assigning an Object Scope. In this scenario, the user will see the filtered logs but may not see the related infrastructure resources in the inventory.

## Configuration Workflow

1. Define the Scope (Data Sets)
   a. Navigate to Access Control > Scopes.
   b. Create a new Scope or edit an existing one.
   c. Define Log Filters: Under the **Assign Log Data Set** section, add specific filters to define which logs belong to this scope.
   d. (Optional) Define Object Scope: Select the infrastructure objects (Clusters, VMs) relevant to this scope.

   For more information, see Defining a Data Set.
2. Define the Role (Permissions). Ensure the role assigned to the user has the necessary log-related privileges.
   a. Navigate to Access Control > Roles
   b. Select a built-in role or create a new custom role.
   c. Verify Log Permissions: Ensure permissions such as **Log Processing**, **Log Forwarding**, or **Log Collection** are checked if the user is required to manage those configurations.

   For more information, see, Add or Edit Roles and Assign Permissions.
3. Assign to User or Group
   a. Navigate to Access Control > User Accounts (or User Groups).
   b. Create a new user or select an existing one.
   c. Assign Role by selecting the role that you configured previously.
   d. Assign Scope by selecting the scope that you configured previously.

   When you log in to VCF Operations, from the navigation bar at the top and click **Operate** and in the left navigation pane click **Logs**. You will only see logs that match the criteria defined in the assigned Log Data Set in the Logs page. When you go to **Administration** > **Configurations**, or **Control Panel**, you will only see the tiles for which they have been assigned permissions.

# Converting Content Packs to Management Packs

The Log Content Pack to VCF Management Pack converter is a utility designed to address the change in content packaging following the introduction of Log Management in VCF Operations. This tool allows you to convert any third-party and custom content packs into Management Packs, ensuring they remain usable within VCF Operations, which only supports Management Packs.

### Purpose of the Tool

- Converting any third-party Content Pack into a VCF Operations compatible Management Pack.
- Converting any custom Content Pack created into a Management Pack.

### How to use the Tool

**Note:**

Ensure that you have Java Runtime Environment (JRE) 17 or higher installed on the machine where you intend to run the conversion tool. You can run the following command to verify the version of Jave that is installed on the machine:

```
java -version
```

**Converting the Content Pack**

1. Download the conversion utility, and the README file wich contains the instructions. The utility can be found in the **Drivers and Tools** tab here, after you log in to the support.broadcom.com portal.
2. Log in to the Aria Operations for Logs or VCF Operations for logs and export the content pack you intend convert and install in VCF Operations.
3. Extract the downloaded utility zip file, which contains the JAR file and the README text file.
4. Go to the command line and execute the command provided in the README file to convert the exported content pack into a management pack. A new folder containing the converted pack file is created.

**Post-Conversion**

1. You must enable the setting. `Unsigned Pack File Installation` in VCF Operations. This is to prevent the error message, `No signature found on pack file` which is displayed during installation. Log in to the VMware Cloud Foundation Operations Administration UI and activate the **Allow Unsigned PAK Installation** under Administrator Settings.
2. Log in to VCF Operations and in navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Integrations**. Click the **Repository** tab.
3. Click **ADD** and then click **BROWSE** to add the newly created management pack file.
4. Install the pack file even if it is already installed and ignore signature checking.
5. Upload the PAK file.
6. Accept the EULA (End-User License Agreement).
7. Proceed with the installation, which will take some time to complete.

**Post-Installation**

1. Once the installation has completed, verify that the management pack is available under the **Repository** tab.
2. The management pack will be a content-only management pack, meaning it will not have any metrics and the source is content pack converter tool.
3. Under the content tab, you can see the dashboards, alerts, and symptom definitions that originated from the original content pack.

# Converting Custom Log Alerts to Management Packs

You can export custom, user-defined alerts from Aria Operations for Logs 8.18, execute the command-line conversion utility, and install the resulting Management Pack within VCF Operations 9.1. Executing this transition preserves legacy monitoring intelligence. To prevent unauthorized modification of platform monitoring files, the custom management pack installation requires administrative privileges and a temporary modification of the security posture of the VCF Operations cluster.

Ensure that the following prerequisites are met before you begin.

- You have the administrator credentials for Aria Operations for Logs 8.18. This is required to access and export custom configurations.
- You have the administrator credentials for the VCF Operations 9.1 Administrator UI. This is required to manage security profiles and install files. For more information, see Accessing Logs when the VCF Operations Console is Inaccessible.
- Java Runtime Environment (JRE) or Java Development Kit (JDK) installed on the local machine, and the permission to run Windows, macOS, or Linux command-line terminal.
- Active registration for the Broadcom Support Portal.

1. Log in to the legacy Aria Operations for Logs 8.18 instance using administrator credentials.
2. Navigate to the **Alerts** section and verify that all custom, user-defined alerts are configured and saved.
3. Click the **Content Pack** tab in the main navigation menu.
4. Select **My Content** from the left-hand pane. Click the **Settings** gear icon located next to **My Content** at the top of the screen.
5. Click **Export**.

   Complete the mandatory fields in the export window. Verify that the custom alerts are selected along with their associated extracted fields (such as the Namespace). Complete the export to download the Content Pack file to the local administrative workstation.
6. Log in to the Broadcom Support Portal and navigate to the **Drivers and Tools** tab, locate the log management section, and download the conversion utility that matches your Java version — `cp-to-mp-convertor-jar-with-dependencies-jdk17.jar` for Java 17 through 24, or `cp-to-mp-convertor-jar-with-dependencies-jdk25.jar` for Java 25 and later. Also, download the **README** file.
7. Open a command prompt or terminal on the administrative workstation containing the downloaded JAR file. Grant permissions to execute the downloaded archive by running one of the following commands:
   a) `chmod +x cp-to-mp-convertor-jar-with-dependencies-jdk17.jar`
   b) `chmod +x cp-to-mp-convertor-jar-with-dependencies-jdk25.jar`
8. Execute the conversion command using Java to generate a .pak file. The examples below use the Java 17 build; substitute cp-to-mp-convertor-jar-with-dependencies-jdk25.jar if you downloaded the Java 25 build. Choose the syntax option that aligns with your migration goals:
   a) Option A: Single Content Pack Conversion. Run this command to convert a specific Content Pack file:

   ```
   java -jar cp-to-mp-convertor-jar-with-dependencies-jdk17.jar -i <path-to-content-pack-file> -o <output-directory>

   The utility creates a new output directory matching the name of the source Content Pack and generates a corresponding .pak file within it.
   ```

   a) Option B: Bulk Content Pack Conversion. Run this command to convert all files within a directory simultaneously:

   ```
   java -jar cp-to-mp-convertor-jar-with-dependencies-jdk17.jar -f <directory-containing-several-content-packs> -o <output-directory>

   The utility processes all Content Packs in the source folder and generates individual output directories for each generated .pak file.
   ```

9. Open a web browser and navigate to the VCF Operations Administrator UI by typing: `https://<VCF-OPS-FQDN>/admin`. Replace `<VCF-OPS-FQDN>` with the correct value and use the local administrator credentials.
10. Navigate to **Administrator Settings** and click **Security Settings**. Click **ENABLE UNSIGNED PAK INSTALLATION** to permit the deployment of custom, unsigned management packs.
11. Navigate to the **Software Install** workspace. Click **Install Software Update**.
12. Upload the custom `.pak` file that you generated. Check the box labeled **Ignore the PAK File Signature Checking** and follow the remaining on-screen prompts to complete the installation wizard.
13. Open a web browser, navigate to the VCF Operations UI, and log in with standard operating credentials. Navigate to **Operate** > **Configuration** > **Alert Definitions**.
14. Search for and verify that the converted custom alerts from Aria Operations for Logs 8.18 are listed and active.

# Log Management Health Overview Dashboard

The Log Management Health Overview dashboard is available to assist you in determining the appropriate course of action when a system alert is triggered. The Log Management Health Overview dashboard is used as a secondary function to assess the overall operation of the system when no alarms are present. The primary intent for this dashboard is to provide additional insights when an alert is raised, helping you understand the underlying condition that triggered the alert.

### Where You Find the Dashboard

After you log in to VCF Operations, from the navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** > **All** and in the **VCF Operations** group, select the **Log Management Health Overview** dashboard.

### Dashboard Structure

The dashboard is structured around four main categories that correspond to four alerts related to logs. The dashboard has a separate section for each of these four key aspects of the system.

| | |
|---|---|
| **Ingestion** | This relates to the process of logs being published to the gateway, sent to the log processor, and then to the log store. |
| **Retention** | This covers the log store indexing the data and then storing it on a disk. |
| **Availability** | This category covers any conditions that could cause the log service to become unavailable. |
| **Operation** | This focuses on asynchronous operations, specifically archiving (writing logs to external storage) and forwarding (configuring the system to send logs to another endpoint). |

### How to Use the Dashboard

The recommended flow for troubleshooting an alert is as follows:

1. Identify the Alert: The dashboard summarizes the number of critical and warning alerts and lists the triggered alerts.
2. Review Symptoms: Multiple conditions can be grouped under a single alert (for example, an ingestion problem). Click on the alert to view the specific symptoms that have been triggered (for example, "ingest queue capacity is greater than 10%" or "dropped events exceed 5%").
3. Find Remediation Steps: Return to the main dashboard and open the corresponding section (for example, the "Ingestion" section for an ingestion alert). This section describes each condition and provides remediation steps for diagnosis.
4. Review Data and Guidance: Within the relevant section, for each condition, data is presented to illustrate the current and recent behavior.
   a. Dial: Shows the current behavior.
   b. Chart: Shows the historical behavior.
5. Read the guidance provided in the text and follow the suggested steps, which may include reviewing other alerts (for example, checking the Availability alert if an ingestion queue capacity issue is observed).
6. Check Knowledge Base (KB): If the suggested remediation actions do not seem to apply, review the Knowledge Base article that is linked or mentioned.

The dashboard can be used to assess the overall operation of the system even when no alarms are present. In a healthy state, you would expect to see no red indicators. By reviewing the widgets, a user can gain comfort that the system is operating as expected.

The dashboard provides information such as:

- The current ingestion rate.
- The response time of the network.
- The status of retention problems.
- Confirmation that services and pods are up and running, and the log store health is green.

**Advanced Troubleshooting**

For advanced troubleshooting, there is a drill down dashboard that can be activated. The drill down dashboard is disabled by default. To activate it, go to Manage, search for *log*,and enable the disabled dashboard. This dashboard is laid out in the same style as the Log Management Health Overview dashboard, but it allows access to metrics about each individual component of the solution.

The log system architecture has three primary components: the gateway, the log processor, and the log store. The drill down dashboard is structured around these components (referred to as ingestion and retention services). Users can click on a component, see the listed instances, and then view the related metrics for a specific instance.

| | |
|---|---|
| **VCF management services Cluster Health** | If troubleshooting based on the logs dashboard suggests the VCF management services VMs may be overloaded, you can review the separate companion dashboard for the VCF management services itself. This page is an overview of the health of the VCF management services and the services running on it. |
| **Information** | This dashboard presents utilization metrics and a table listing the availability and memory usage of all services in the VCF management services cluster, including ops logs. It also details storage usage, backup, gateway information, and node details. |

You should start with the application health perspective (the logs dashboard) and then drill down to lower layers of the system, such as the VCF management services health page, to investigate potential issues like overloaded VCF management services VMs. For example, if CPU utilization is high but all applications are behaving well, it does not necessarily indicate a problem.

## Accessing Logs when the VCF Operations Console is Inaccessible

Log Management is integrated as a built-in, containerized core service running natively within the VCF Management Services Platform (VMSP). The VCF Operations console serves as the primary administrative interface for searching logs, running queries, analyzing trends, and configuring forwarding destinations. If the primary VCF Operations console is offline, unreachable, or undergoing maintenance, you can access local logs directly using the standalone Log Management emergency UI.

- The standalone Log Management FQDN or IP address that is configured during deployment is known and reachable.
- The local system administrator password that is created during VMSP runtime deployment is retrieved.

> **Note:**
>
> The standalone emergency interface uses the local **admin@vsp.local** system-level account. Bypassing the VCF Operations identity broker must be restricted to emergency troubleshooting and avoided in standard production workflows.

1. Open a web browser and navigate to the standalone Log Management FQDN: `https://<Log-Management-FQDN>`
2. At the login screen loads, enter the following administrative credentials:
   a) Username: `admin@vsp.local`
   b) **Password**: The password that is assigned to the **vmware-system-user** during VMSP platform deployment.
3. After you sign in, you can see the standalone interface, where you can click the **Analyze Logs** tab to perform direct diagnostic searches and verify platform status.

# Network Operations

Network Inventory provides you with an overview of your VCF Operations network resources and quick access to the VCF Operations for networks console.

You can access the **Network Inventory** page from VCF Operations by navigating to **Operate** > **Overview** > **Network**. The page includes several useful widgets that provide information on your deployments networking status and resources.

> **Note:**
>
> The scope of information that is provided in**Network Inventory** is limited to inventory, alert, and NSX health information if you have not deployed the VCF Operations for networks appliance. To deploy the appliance, click **Install VCF Operations for Networks** on the top right of the page. For more information, see Deploying VCF Operations for networks. For information on upgrading the VCF Operations for networks, see Upgrading Management Components.

| Widget | Description |
|---|---|
| Networking Inventory | Includes the networking objects in your inventory such as NSX instances, transport nodes, edge clusters, and others. You can use the provided links to navigate to the objects.<br><br>**Note:**<br><br>The count of NSX instances that are shown on the NSX inventory includes both NSX Adapter instances under NSX and VMware Cloud Foundation accounts. |
| Network Alerts | Includes a metric overview of the alerts in your environment. |
| NSX Managers and VPCs | Includes information on the NSX Managers and, if applicable, the Virtual Private Clouds (VPCs) associated with them. For more information on NSX Projects and VPCs, go to NSX Projects and VPCs. |
| NSX Health | Includes information any NSX Managers, transport nodes, or edge nodes experiencing issues. |
| Business Applications with Flows | Includes information that is sourced from VCF Operations for networks used for flow analysis your applications. By selecting the links included in the widget, you navigate to VCF Operations for networks where you can provided flow data to analyze and troubleshoot your applications. For more information on application discovery based on flows, go to Working with Application Discovery. |
| Traffic Summary | Includes an overview of the flow traffic in your environment for the last 24 hours. |

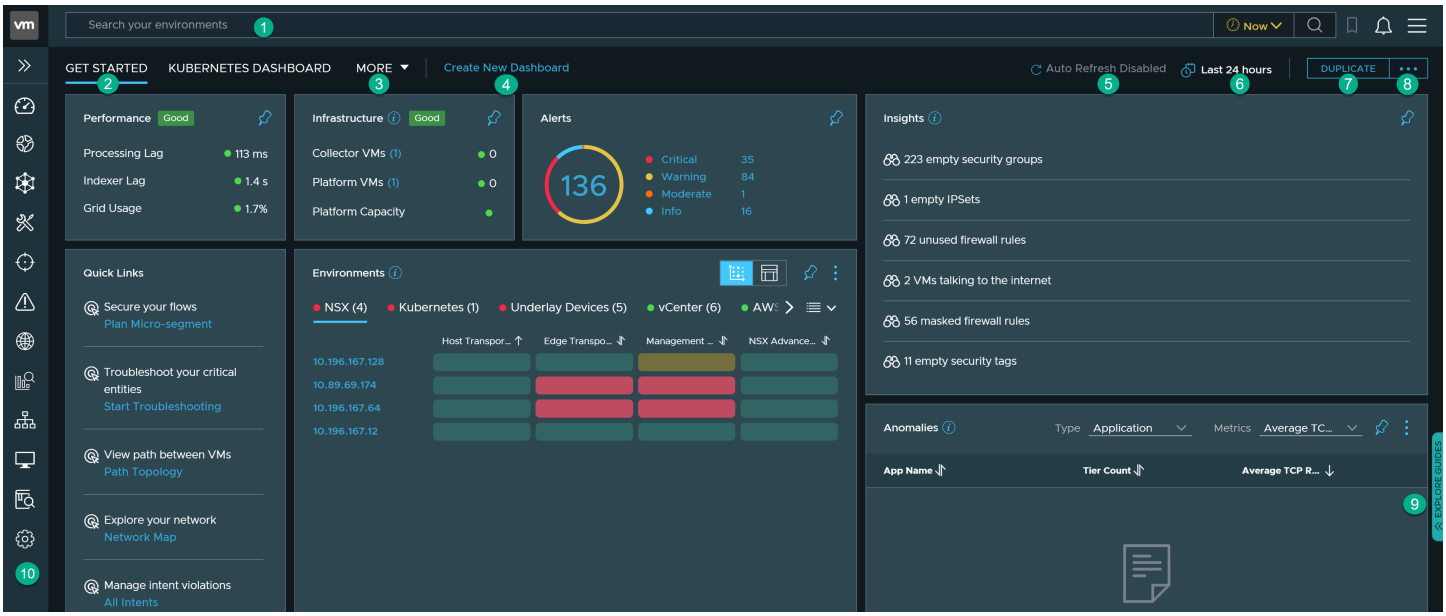
## About VCF Operations for networks

VCF Operations for networks delivers intelligent operations for software-defined networking and security. It helps customers build an optimized, highly-available, and secure network infrastructure across multi-cloud environments.

VCF Operations for networks accelerates micro-segmentation planning and deployment, enables visibility across virtual and physical networks, and provides operational views to manage and scale the VMware NSX deployments.

# VCF Operations for networks Homepage

The homepage enables you to create, view, and manage dashboards in VCF Operations for networks.




The homepage contains several elements:

| Elements | Description |
|---|---|
| Search bar (1) | Enables you to search across your entire data center network and its corresponding entities. Based on your requirement, you can perform search as per the following time line options:<br>• **Presets**: Using this option, you can narrow down your search results for presets such as last week, last 3 days, and last 24 hours.<br>• **At**: Using this option, you can narrow down your search results for a particular date and time.<br>• **Custom**: Using this option, you can search for data between a particular time interval. |
| Get Started (2) | Provides a high-level overview of your entire data center. The **Get Started** dashboard is available by default on the VCF Operations for networks homepage. |
| More (3) | If you have more than one dashboard, the **More** drop-down menu is displayed. This drop-down menu enables you to manage dashboards and arrange the order in which the dashboards are displayed on the homepage.<br>**Note:**<br>You can display a minimum of one dashboard and a maximum of five dashboards on the homepage. |
| Create New Dashboard (4) | Enables you to create a new dashboard and access it from the homepage. |
| Auto Refresh (5) | Enables you to change the auto refresh time frame at a global level. |

| Elements | Description |
|---|---|
| Last 24 hours (6) | Displays the dashboard data for the last 24 hours from the current time. Based on your requirement, you can view the dashboard data as per the following time line options:<br>• **Presets**: View the dashboard data for the last week, last 3 days, and last 24 hours.<br>• **At**: View the dashboard data at a particular date and time.<br>• **Custom**: View the dashboard data between a particular time interval. |
| Duplicate (7) | Enables you to duplicate the selected dashboard.<br>**Note:**<br>You cannot duplicate dashboards such as Kubernetes. |
| More options (8) | Enables you to edit, share, delete, and download the selected dashboard as PDF.<br>**Note:**<br>The options differ based on the dashboard you are accessing. For example, you cannot share or delete the **Get Started** and Kubernetes dashboards. |
| Explore Guides (9) | Shows guided workflows to perform some of the most important tasks in VCF Operations for networks. |
| Navigation pane (10) | Enables you to access key product features such as **Plan & Assess**, **Intent**, and **Entities** without having to type a search query. |

## The Get Started Dashboard

The **Get Started** dashboard is a system-defined dashboard that provides a high-level overview of what is happening in your entire data center. It also provides quick access to the important components of your data center.

The dashboard contains several widgets such as:

| Widgets | Description |
|---|---|
| Performance | Indicates the health of platform VMs, collector VMs, and the platform capacity.<br>**Note:**<br>If metric usage exceeds platform capacity, you will receive an alert with a recommended resolution to increase capacity. |
| Infrastructure | Displays the platform capacity. |
| Alerts | Displays the alerts sorted by the severity. |
| Insights | Highlights critical or behavioral changes in the network at the current time. This section also highlights entities such as applications, firewall rules, and VMs that need immediate attention. |
| Quick Links | Shows useful quick links. |
| Environments | Shows a heat map view indicating the health of entities associated with each environment type. |
| Anomalies | Shows error metrics for applications, VMs, and VMware NSX-T Edge Transport nodes. |

**Note:**
You cannot share, edit, or delete the **Get Started** dashboard. However, you can duplicate the dashboard and save the dashboard as a PDF file.

### VCF Operations for networks Security Compliance

VCF Operations for networks is a pre-hardened and secured appliance. VCF Operations for networks is hardened against well-established and industry-accepted security benchmarks, for example: CIS. Additionally, VMware security team has put internal controls to minimize the security footprint.

VCF Operations for networks also follows all security and privacy best practices defined by VMware Security Engineering, Communications, and Response (vSECR) team.

For any critical known vulnerabilities that require immediate action, VMware releases security advisories for those vulnerabilities with step-by-step instructions on how to keep the appliance secure. In general, all known vulnerabilities are fixed in every release. Use the latest version of the appliance to take advantage of these security fixes.

## Working with Data Sources

VCF Operations for networks allows you to add various data sources to gather information about your data center.

### Adding a Data Source in VCF Operations for networks

You must add data sources into VCF Operations for networks to gather data of your data center.

To add a data source, go to **Settings** > **Accounts and Data Sources** > **Add Source**.

After adding a data source, you see the **Accounts and Data Sources** page where data sources are categorized based on the following options:

- All: Displays all the available data sources.
- With Problems: Displays the data sources where VCF Operations for networks has found one or more problems.
- With Recommendations: Displays auto generated recommendations from VCF Operations for networks for the data sources that require additional information.

Each data source displays the following details

| Properties | Description |
|---|---|
| Type | Displays the type of data source. |
| Name | Displays the name of the data source. |
| Identifier | Displays IP address or FQDN details of data source. |
| Last Metric Collection | Displays the time stamp of the last metric data collection. |
| Last Configuration Collection | Displays the time stamp of the last configuration data collection. |
| Local Tags | Displays all the Working with Local Tags associated with the data source. |
| Polling Interval | Displays the polling interval that is defined for the data source. |
| Discovered Entities | Displays the number of VMs that are discovered for that data source.<br>**Note:**<br>The Discovered Entities column is populated only if the data source is VMware vCenter or AWS. |
| Flows | Displays the count of flows for the data source. |
| Data Collection | Indicates if the date collection on the data source is enabled or not. |

**Important:**
While adding a data source, do not add the same data source more than once using different IP addresses or FQDNs.

To select the columns that you want to view in the table, you can click the **Manage Table Columns** icon on the table header. You can view the details of maximum five columns at a time.

To edit or delete a data source, you can click the **More Options** icon next to the data source. For the data sources that support customizable polling interval, you can also use the **Collect Now** option to run the configuration data collection process.

VCF Operations for networks provides the following functions to enable easy access to the information of data sources.

- You can search a data source by name, IP address, or by the collector VM name by using the search bar located above the column headers.
- You can filter information by different data sources in the **Type** column. You can also filter data sources based on data collection in the **Data Collection** column.
- You can sort information by clicking the **sort** icon on any column.

VCF Operations for networks enables you to add different types of VMware products and other third party data sources ranging from public cloud, containers, firewalls, routers and switches, and load balancers.

## Supported Products and Versions

VCF Operations for networks supports several VMware and third party products and their different versions.

For the list of all VMware products and its supported versions in VCF Operations for networks, see the VMware Product Interoperability Matrix.

The following table lists all the third party devices and its associated operating systems supported in VCF Operations for networks.

| Data Source Version/Model | Supported Operating System (OS) and Versions | Supports Network Assurance and Verification feature (Network Map and Intents) |
|---|---|---|
| Amazon Web Services (Enterprise License Only) | Not Applicable | No |
| Arista 7050QX-32S, 7280TR | Arista EOS 4.26.*, 4.27* | Yes |
| Azure Subscription | Not Applicable | No |
| Brocade VDX 6740, VDX 6940, MLX, MLXe | NetIron 6.0.*, 6.2.* | Yes<br>Only on Brocade MLX |
| Check Point Firewall | Check Point GAiA R80.40, R81.10 | Yes |
| Cisco ACI | Fabric OS and iNXOS 3.2, 4.2.3, 4.2(7f), 5.2 (4d), 6.0(1g) | Yes<br>Only on Cisco ACI 4.2 (7f), 5.2 (4d) and 6.0(1g). |
| Cisco ASA X Series | ASA OS 9.8.1, 9.8(4), 9.12(4), 9.16(4) | Yes |
| Cisco Catalyst 3000, 4500, 6500, 9300, 9500 | IOS-XE 17.03 and IOS 12.1, 12.2, 12.4, 15.2.<br>• Cisco Catalyst 9500 is supported on IOS XE 17.6.4(MD) only.<br>• Cisco Catalyst 6500 is supported on IOS 15 only. | Yes |
| Cisco Nexus 2000***, 3000, 5000, 6000, 7000, 9000 | NX-OS 7.3, 8.2, 8.4, 9.2, 9.3, 10.3(1) | Yes<br>Only on Nexus 2000 (N2K-C2348UPQ-10GE), 5000, 7000, 9000 |

| Data Source Version/Model | Supported Operating System (OS) and Versions | Supports Network Assurance and Verification feature (Network Map and Intents) |
|---|---|---|
| Cisco UCS (Unified Computing System) Series B blade servers, Chassis, Fabric Interconnect, FEX | Cisco UCS Manager and NX-OS 3.2, 4.1, 4.2, 4.2(2a). | Yes<br>Only on Cisco UCS 4.1 and 4.2. |
| Cisco ASR 1K | Cisco IOS XE 15.5(3)S8, 15.5(3)S10, 16.06.05, 16.12.5, 17.9.4(MD). | Yes |
| Cisco ISR 4K | Cisco IOS XE 15.5(3)S9, 16.06.05, 17.4.1a, 17.9.4(MD) | Yes |
| Cisco ASR 9901 and 9910 | Cisco IOS XR 6.1.3, 6.5.1 | Yes |
| Dell S5248F-ON, FORCE10 MXL 10, FORCE10 S6000, S4048, PowerConnect 8024, Z9100, Dell S5200 | Dell Networking Operating System (DNOS) 10. | Yes<br>Only on Dell S5200 (OS10). |
| Fortinet FortiManager | FortiOS 6.4, 7.0 | No |
| F5 BIG - IP LTM | TMOS 13.1, 14.1, 15.1.4, 16.1 | Yes |
| HP Virtual Connect Manager 4.41, HP OneView 3.0 | | No |
| HPE 10508 | Comware | No |
| Huawei Cloud Engine 6800, 7800, 8800 | VRP 8.18 | No |
| Infoblox | Network Identity Operating System (NIOS) 8.0, 8.1, 8.2 | No |
| Juniper EX3300, QFX 51xx Series, 5K1, 10K | Junos 20.2 without QFabric | Yes<br>Only on Juniper QFX |
| Mellanox SN2010, SN2100, SN2410, SN2700, SN3420, SN3700c, and SN4600c. | Onyx 3.8.1304, 3.8.2306 | No |
| OpenShift 3.11 and 4.10.<br>**Note:**<br>You must add OpenShift version 4.x as individual kubernetes cluster using kube-config file. | Redhat Enterprise Linux | No |
| Palo Alto Networks Panorama | PanOS 9.1, 10.1, 10.2, 11.0 | Yes |

**Note:**

- *** VCF Operations for networks does not support Active-Active FEX (VPC) configuration for Cisco Nexus 2000 in Network Assurance and Verification.
- In addition to the mentioned supported OS versions, VCF Operations for networks supports the minor releases associated with major releases. For example, as Cisco IOS 15.2 is supported, all minor releases of this major release, such as 15.2(1), 15.2(2), etc., are also supported. However, newer major releases, such as 15.9, are not supported.
- In addition to the devices mentioned in the table, the following VMware products also supports Network Assurance and Verification feature (Network Map and Intents) in the latest version of VCF Operations for networks:
  - VMware NSX Manager
    - Distributed and Edge firewall
    - Multiple VTEPs
  - VMware vCenter
    - VMware vSphere Distributed Switch (VDS)
- You can get information about the required ports for all data sources supported in VCF Operations for networks at VMware Ports and Protocols.
- For third-party devices such as switches, routers, and firewalls, VCF Operations for networks uses SSH connections for fetching the configuration data. To learn about supported algorithms for such SSH connections, see Encryption Algorithms and Ciphers.

**Encryption Algorithms and Ciphers**

VCF Operations for networks supports several encryption algorithms and ciphers for data sources.

The following table lists the encryption algorithms and ciphers that are supported for data sources in Supported Products and Versions.

| Data sources | Encryption Algorithms and Ciphers |
|---|---|
| Arista switches<br>Brocade VDX switches<br>Check Point Firewall<br>Cisco ASA V Series<br>Cisco ASR and ISR<br>Cisco Catalyst 3000, 4500, 6500, 9300<br>Cisco Nexus N5000, N7000, N9000<br>Cisco UCS Fabric interconnect<br>Dell Force 10 S6000, PowerConnect 8024, S4048, S4810<br>Dell OS 10<br>HPE 5940, 10500<br>HP Virtual Connect Manager<br>Huawei Cloud Engine<br>Juniper switches<br>Mellanox | Key exchange algorithms:<br>• curve25519-sha256<br>• curve25519-sha256@libssh.org<br>• ecdh-sha2-nistp256<br>• ecdh-sha2-nistp384<br>• ecdh-sha2-nistp521<br>• diffie-hellman-group-exchange-sha256<br>• diffie-hellman-group16-sha512<br>• diffie-hellman-group18-sha512<br>• diffie-hellman-group14-sha256<br>Ciphers:<br>• aes128-ctr<br>• aes192-ctr<br>• aes256-ctr<br>• aes128-gcm@openssh.com<br>• aes256-gcm@openssh.com<br>MACs:<br>• hmac-sha2-256-etm@openssh.com<br>• hmac-sha2-512-etm@openssh.com<br>• hmac-sha1-etm@openssh.com<br>• hmac-sha2-256<br>• hmac-sha2-512<br>• hmac-sha1<br>Host key algorithms:<br>• ecdsa-sha2-nistp256<br>• ecdsa-sha2-nistp384<br>• ecdsa-sha2-nistp521<br>• rsa-sha2-512<br>• rsa-sha2-256<br>• rsa-sha2-512<br>• ssh-rsa |
| Brocade MLX<br>Dell Force 10 MXL 10 | Key exchange algorithms:<br>• diffie-hellman-group-exchange-sha1<br>• diffie-hellman-group1-sha1<br>• diffie-hellman-group14-sha1<br>Ciphers:<br>• blowfish-cbc<br>• blowfish-ctr<br>• 3des-cbc<br>• 3des-ctr<br>• aes128-cbc<br>• aes128-ctr<br>• aes192-cbc<br>• aes192-ctr<br>• aes256-cbc<br>• aes256-ctr<br>MACs:<br>• hmac-md5<br>• hmac-md5-96<br>• hmac-sha1<br>• hmac-sha1-96<br>Host key algorithms:<br>• ssh-rsa |
| | **Note:**<br>VCF Operations for networks only supports ssh-rsa algorithms with 1024 bit keys. ssh-rsa algorithms with 2048 bit keys are not supported. |

## Supported VMware Managers

You can add VMware Managers into VCF Operations for networks and collect the details.

**Note:**

If a data source is deleted and then added again, allow for a five-hour buffer period.

### Add VMware vCenter

You can add VMware vCenter servers as data source in VCF Operations for networks.

- Your predefined roles in the VMware vCenter must have the **Global.Settings** privilege assigned at the root level. Also, you must propagate this privilege to the children roles.

  **Note:**
  When you add a role in the VMware vCenter, by default the role is created with three system-defined privileges: **System.Anonymous**, **System.Read**, and **System.View**. These privileges are not visible in the vSphere Client but are used to read certain properties of some managed objects. See the *vSphere Web Services API documentation* for more information.
- To configure and use IPFIX, you must have following VMware vCenter privileges:
  - **Distributed switch:** Modify, Port configuration operation and IPFIX Operation.
  - **dvPort group:** Modify, Policy operation and IPFIX Operation.

  **Note:**
  IPFIX is supported on the following VMware ESX versions:

  - 5.5 Update 2 (Build 2068190) and later.
  - 6.0 Update 1b (Build 3380124) and later.
  - VMware VDS 5.5 and later.
- To identify the VM to VM path, you must install VMware tools on all the VMs in the data center.

To know more about roles in VMware vCenter, see the *vSphere Security* guide.

Multiple VMware vCenter instances can be added to VCF Operations for networks to start monitoring data.

You can also add VMware vCenter data sources directly from VCF Operations by navigating to **Administration** > **Integrations**. To activate data sync between the VMware vCenter data source and VCF Operations for networks, select the data source, activate the **Network and Flow** option under **Network Operations**, and select a data collector. You can also enable NetFlow collection with the **Enable NetFlow (IPFIX) on this vCenter** option.

**Note:**

To enable Antrea IPFIX for collection of Supervisor, namespace, and VKS cluster flows, select **Enable Antrea (IPFIX) on this vCenter**.

Migration, editing, activating, deactivating data sources managed in VCF Operations is restricted.. If you are upgrading to VCF 9 from a previous version, the data source sync from VCF Operations for networks to VCF Operations does not happen automatically and must done explicitly by the user to see the data correctly on the **Network Inventory** dashboard.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under VMware Managers, select **VMware vCenter**.
4. Provide the following details:

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu.<br>**Note:**<br>You can add maximum 10 VMware vCenter Server per collector. |
| **IP Address/FQDN** | Enter the IP address ( IPv4 address or IPv6 address) or fully qualified domain name of the VMware vCenter. |

| Option | Action |
|---|---|
| | **Note:**<br>You must add the data source using either IP address or FQDN. Do not add the data source using both IP address and FQDN. |
| **Username** | Enter the user name with the following privileges:<br>• **Distributed Switch**: Modify<br>• **dvPort group**: Modify |
| **Password** | Enter the password for VCF Operations for networks software to access the VMware vCenter system. |

5. Click **Validate**.

   If the number of VMs discovered exceeds the capacity of the platform or a collector node or both, the validation fails. You will not be allowed to add a data source until you increase the brick size of the platform or create a cluster.

   The specified capacity for each brick size with and without flows is as follows:

| Brick Size | VMs | State of Flows |
|---|---|---|
| Large | 6k | Enabled |
| Large | 10k | Deactivated |
| Medium | 3k | Enabled |
| Medium | 6k | Deactivated |

6. Select **Enable Netflow (IPFIX) on this vCenter** to activate IPFIX.

   For more information on IPFIX, see IPFIX Configuration on VDS and DVPG.

   **Note:**

   - You cannot enable IPFIX on DVPGs with connected NSX-T Edges. After NSX-T is added to VCF Operations for networks, the system automatically detects and excludes these DVPGs.
   - vCenter IPFIX cannot be enabled for DVPGs managed by NSX
   - Overriding the **Switch IP address option** is not supported in VCF Operations for networks. VCF Operations for networks uses the host IP in flow records to configure flow enrichment.
   - Enabling uplink DVPGs is not supported.

7. Add advanced data collection sources to your VMware vCenter system.
8. Optional: In the **Nickname** text box, enter a nickname.
9. Optional: In the **Tags (Optional)** key-value pair text box, enter a key and a value.

   A key-value pair could be any text. For example, you can use `Layer Access` as key-value pair where `layer` is the `key` and access is the value.

   – To apply the tag in all the associated entities, click the **Apply above tag operations to all associated entities** check box. When you select this option, the tag gets applied to all associated entities. For more details about the associated entities, see Working with Local Tags.

     If you clear the **Apply above tag operations to all associated entities** check box, the assigned tag is removed from all the associated entities.

10. Optional: In the **Notes** text box, add a note if necessary.
11. Click **Submit**.

The VMware vCenter system appear on the homepage.

### Add VMware NSX Manager

NSX is designed to address the emerging application frameworks and architectures that have heterogeneous endpoints and technology stacks. In addition to vSphere, these environments may also include other hypervisors, containers, bare metal, and public clouds.

VCF Operations for networks supports NSX deployments where the VMs are managed by VMware vCenter.

#### Considerations

- VCF Operations for networks supports only the NSX setups in which VMware vCenter manages the ESX hosts.
- VCF Operations for networks supports NSGroups, NSX-T Firewall Rules, IPSets, NSX-T Logical Ports, NSX-T Logical Switches, NSX-T distributed firewall IPFIX flows, Segment, Group, and Policy Based VPN.
- VCF Operations for networks supports only NSX-T deployments. Support for NSX-V is removed starting with 9.0. If NSX-T is used instead of NSX, then only those entities are displayed. The same logic applies to the entities such as firewall rules, IPSets, and logical switches.
- With NSX-T 2.4 release, VCF Operations for networks supports NSX Declarative Policy Management which simplifies and automate network and security configurations through outcome-driven policy statements.

  **Note:**

  Micro-segmentation for Security Group is done based on NSX Policy data. But in case there is no corresponding NSX Policy Group, the standalone NS Group is included in the Micro-segmentation analysis.
- Ensure to add only local users in an NSX-T Federation setup.
- Verify that the user has both **Auditor** and **Network/Security Admin roles** are added for use with the data source.

#### Prerequisites

Here are the prerequisites for adding an NSX-T Manager as a data source:

- You must have at least the **Read only** privilege.
- You must add all the VMware vCenter associated with the NSX-T Manager as data sources in VCF Operations for networks.

  **Note:**
  If you add the NSX-T Manager before adding VMware vCenter, then VCF Operations for networks takes around four hours to stabilize.
- Ensure that there are no logical switches in the exclusion list in the Distributed Firewall (DFW). If there are any logical switches in this list, then the flows are not reported for any VMs attached to these logical switches.

#### Adding NSX-T Manager directly in VCF Operations

You can also add NSX-T data sources directly from VCF Operations by navigating to **Administration** > **Integrations**. To enable data sync between the NSX data source and VCF Operations for networks, select the data source, activate the **Network and Flow** option under **Network Operations**, and select a data collector. You can further expand data collection by enabling the **Enable IPFIX** and **Enable latency metric collection** options.

#### Procedure

1. Go to **Settings** > **Accounts and Data Sources** > **Add Source**.
2. Under **VMware Manager**, select **VMware NSX-T Manager**.
3. Provide the user credentials.

| Option | Action |
|---|---|
| Collector VM | Select a collector VM from the drop-down menu. |
| IP Address/FQDN | Enter the IPv4 management address or the FQDN details.<br>**Note:**<br>Currently, VCF Operations for networks does not support IPv6 NSX-T management addresses. |

<table>
<tr><th>Option</th><th>Action</th></tr>
<tr><td>Authentication Method</td><td>Select the authentication method from the drop-down menu.
<table>
<tr><td>Username/Password</td><td>Enter the user name and the password.</td></tr>
<tr><td>Certificate (Principal Identity)</td><td>• Certificate: Click Browse and upload the principal identity certificate.<br>• Private Key: Click Browse and upload the principal identity private key.<br>Note:<br>VCF Operations for networks does not support encrypted principal identity private key.</td></tr>
</table></td></tr>
</table>

**Note:**

- If you have more than one management node in a single NSX-T deployment, you must add only one node as a data source in VCF Operations for networks or use Virtual IP (VIP) (of those nodes). If you add more than one management node, then VCF Operations for networks might not function properly.
- Ensure that you use VIP when you add NSX-T as a data source. If you add a management node IP instead of a VIP, and later if you want to add a VIP or another management node IP, then you have to delete the existing data source to add the new VIP or the Management IP.
- Ensure that each management node in the cluster is reachable from the collector.
- If IPFIX is not required, the user must be a local user with audit level permissions. But if IPFIX is required, the user must have one of the following permissions: **enterprise_admin**, **network_engineer**, or **security_engineer**.

**Note:**
You must add the data source using either IP address or FQDN. Do not add the data source using both IP address and FQDN.

4. Click **Validate**.
5. Select **Enable IPFIX**to update the IPFIX settings on NSX-T. By selecting this option, VCF Operations for networks receives DFW IPFIX flows from NSX-T. For more information on enabling IPFIX, see Enable NSX-T IPFIX.

   **Note:**

   - DFW IPFIX is not supported in the Standard Edition of NSX-T.
   - From VMware Cloud Foundation 9.0 or later, if the DFW license is missing or if DFW is deactivated, VCF Operations for networks receives Switch IPFIX flows from NSX-T.

6. If you want to collect latency metrics data, select **Enable latency metric collection** check box. If you select this option, VCF Operations for networks receives latency metrics such as VTEP - VTEP, vNIC - pNIC, pNIC - vNIC, vNIC - vNIC from NSX-T.

   **Note:**

   - This option is available only for NSX-T 2.5 and later.
     - VTEP - VTEP is available from NSX-T 2.5 and later.
     - vNIC - pNIC, pNIC - vNIC, vNIC - vNIC are available from NSX-T 3.0.2 and later.
   - To enable latency metric collection, you must have **enterprise_admin** permission.
   - Ensure that the port 1991 is open on the collector to receive the latency data from the ESX node.

7. In the **Tags (Optional)** key-value pair text box, enter a key and a value.

Key-value pair could be any text. For example, you can use `Layer Access` as key-value pair where `layer` is the `key` and access is the value.

- To apply the tag in all the associated entities, click the **Apply above tag operations to all associated entities** check box. For more details about the associated entities, see Working with Local Tags.

  If you clear the **Apply above tag operations to all associated entities** check box, the assigned tag is removed from all the associated entities.

8. In the Nickname text box, enter a nickname.
9. In the Notes (Optional) text box, add a note if necessary.
10. Click **Submit**.

**Note:**

For NSX-T version 4.x and later, VCF Operations for networks doesn't support VM-VM path for federation deployment and VCF Operations for networks,Assurance and Verification.

## Examples for Queries

Here are some examples for queries related to NSX-T:

**Table 1305: Queries for NSX-T**

| Queries | Search Results |
|---|---|
| `NSX-T Manager where VC Manager=10.197.53.214` | NSX-T Manager where this particular VC Manager has been added as the compute manager. |
| `NSX-T Logical Switch` | Lists all the NSX-T Logical switches present in the instance of VCF Operations for networks. including the details on whether it is a system-created or a user-created switch. |
| `NSX-T Logical Ports where NSX-T Logical Switch = 'DB-Switch'` | Lists the NSX-T logical ports belonging to that particular NSX-T logical switch, DB-Switch. |
| `VMs where NSX-T Security Group = 'Application-Group'`<br>Or<br>`VMs where NSGroup = 'Application-Group'` | Lists all the VMs in that particular security group, Application-Group. |
| `NSX-T Firewall Rule where Action='ALLOW'` | Lists all the NSX-T Firewall Rules which have their action set as ALLOW. |
| `NSX-T Firewall Rule where Destination Security Group = 'CRM-Group'` | Lists the firewall rules where the CRM-Group is the Destination Security Group. The results include both Direct Destination Security Groups and Indirect Destination Security Groups. |
| `NSX-T Firewall Rule where Direct Destination Security Group = 'CRM-Group'` | Lists the firewall rules where the CRM-Group is the Destination Security Group. The results include only the Direct Destination Security Groups. |
| `VMs where NSX-T Logical Port = 'App_Port-Id-1'` | Lists all the VMs which have that particular NSX-T Logical Port. |
| `NSX-T Transport Zone` | Lists the VLAN and the overlay transport zone and the respective details associated with it including the type of the transport node.<br>**Note:**<br>VCF Operations for networks does not support KVM as a data source. |
| `NSX-T Router` | Lists the TIER 1 and TIER 0 routers. Click the router shown in the results to view more details associated with it including the NSX-T Edge Cluster and the HA mode. |

**Table 1306: Queries for NSX Policy**

| | |
|---|---|
| `NSX Policy Segment` | Lists all the NSX Policy Segments present in the instance of VCF Operations for networks. |
| `NSX Policy Manager` | Lists all the NSX Policy Manages present in the instance of VCF Operations for networks. |
| `NSX Policy Group` | Lists all the NSX Policy Groups present in the instance of VCF Operations for networks. |
| `NSX Policy Firewall` | Lists all the NSX Policy Firewalls present in the instance of VCF Operations for networks. |
| `NSX Policy Firewall Rule` | Lists all the NSX Policy Firewall Rules present in the instance of VCF Operations for networks. |
| `NSX Policy Firewall Rule where Action = 'ALLOW'` | Lists all the NSX Policy Firewall Rules which have their action set as ALLOW. |
| `NSX Policy Based VPN` | Lists all the NSX Policy Based VPNs present in the instance of VCF Operations for networks. |

**Note:**
If NSX-T 2.4 and VMware Cloud on AWS are added as data sources in your VCF Operations for networks, then to get the NSX entities, you must add `SDDC type = ONPREM` filter in your query. For example, `NSX Policy Based VPN where Tier0 = ‘’ and SDDC Type = ‘ONPREM’.`

## Support for NSX-T Metrics

The following table displays the VCF Operations for networks entities that support the NSX-T metrics currently and the widgets that display these metrics on the corresponding entity dashboards.

| Entities | Widgets on the Entity Dashboard | Supported NSX-T Metrics |
|---|---|---|
| Logical Switch | Logical Switch Packet Metrics<br>Logical Switch Byte Metrics | `Multicast and Broadcast Rx`<br>`Multicast and Broadcast Tx`<br>`Unicast Rx`<br>`Unicast Tx`<br>`Dropped Rx`<br>`Dropped Tx`<br>`Rx Packets (Total)`<br>`Tx Packets (Total)` |
| Logical Port | Logical Port Packet Metrics<br>Logical Port Byte Metrics | `Multicast and Broadcast Rx`<br>`Multicast and Broadcast Tx`<br>`Unicast Rx`<br>`Unicast Tx`<br>`Rx Packets (Total)`<br>`Tx Packets (Total)` |
| Router Interface | Router Interface Metrics | `Rx Packets`<br>`Tx Packets`<br>`Dropped Rx Packets`<br>`Dropped Tx Packets`<br>`Rx Bytes`<br>`Tx Bytes` |
| Firewall Rule | Firewall Rule Metrics | `Hit Count`<br>`Flow Bytes`<br>`Flow Packets` |

Here are some sample queries for NSX Metrics:

- `nsx-t logical switch where Rx Packet Drops > 0`
  This query lists all the logical switches where the count of the dropped received packets is greater than 0.
- `nsx-t logical port where Tx Packet Drops > 0`
  This query lists all the logical ports where the count of the dropped transmitted packets is greater than 0.
- `top 10 nsx-t firewall rules order by Connection count`
  This query lists the top 10 firewall rules based on the connection count(`Hit Count`).

  **Note:**

  Metric data from the real-time metrics component will continue to be collected even if the associated vCenter license of the NSX data source expires.

### Security Planning for NSX-T

To plan security for the NSX-T network, you can select the scope as **NSXT Layer2 Network** and use the following query:

```
plan NSX-T Layer2 Network '<NAME_OF_NSX_T_LOGICAL_SEGMENT>'
```

You can also obtain the same result by performing the following steps:

1. Select **Plan & Assess** > **Security Planning** from the left navigation from.
2. Select either **NSX-T L2 Network** or **NSX Policy Segment** as the scope from the drop-down menu.

   **Note:**
   NSX related entities such as **NSX-T L2 Network** and **NSX Policy Segment** are available in the scope. You can use these NSX related entities for security planning.

## NSX Projects and VPCs

VCF Operations for networks provides Enterprise administrators with complete visibility to NSX Projects and VPCs included in data sources.

### What are NSX Projects and VPCs

An NSX Project enables multi-tenancy for NSX network and security objects where each project is analogous to a tenant. A NSX VPC is an additional layer of tenancy that you can configure within an existing project. Both NSX Projects and VPCs are primarily configured and managed in NSX Manager by an administrator. For more information on NSX Projects and VPCs and how to add them in your NSX Manager environment, go to NSX Multi-tenancy.

**Note:**

VCF Operations for networks supports the discovery of Projects and VPCs created in NSX 9.0 and later

### NSX Projects and VPCs in VCF Operations for networks

VCF Operations for networks can now identify and extract data associated with Projects and VPCs included in the NSX-T Manager data source added to your environment. New entities, flow analytics, metrics, and network map topology options are added to VCF Operations for networks and used in Project and VPC specific dashboards and updated data included in existing NSX dashboards. You can receive a general overview of the Projects and VPCs from the **NSX Managers and VPCs** widget on the **Network Operations** page in the VCF Operations Console. By selecting **View Details** and expanding the entry for a given data source, you can view the Projects and VPCs associated with it. Clicking on the entry for a Project or VPC opens the dashboard page for that resource.

**Note:**

You can also view these dashboards from VCF Operations for networks.

You can use the data provided from Projects and VPCs to perform more granular analysis and troubleshooting of your NSX data sources. For example, you can use the uplink and downlink data on the VPC dashboard to troubleshooting connectivity issues.

Current limitations and specifics for VCF Operations for networks VPC usage include:

- Network map path search does not take into account the uplink named teaming policy associated with VLAN backed segments.
- VPC external IP NAT metrics are not captured or displayed in VCF Operations for networks, as these metrics are not exposed by NSX.
- In the Transit Gateway (TGW) dashboard, under router interface metrics, both TGW uplink and VPC uplink router interfaces are displayed. The metrics for the VPC uplink router interface correspond to the TGW downlink metrics.
- Only one service interface to NAT rule association is utilized and displayed in VCF Operations for networks,.

**Note:**

For VCF Operations for networks 9.0. network map path search does not take into account the uplink named teaming policy associated with VLAN backed segments.

**Disk Requirement for Projects and VPCs in VCF Operations for networks**

To support the efficient use of this feature, customers are recommended to increase the disk capacity of their VCF Operations for networks appliance by at least 500 GB if they are running at full capacity.

### Add VMware HCX

You can add VCF Operation HCX as a data source in VCF Operations for networks.

- You must have **Enterprise Administrator** access.
- You must add all the VCF Operations HCX Managers, including both source sites and destination sites, as data sources in VCF Operations for networks. You must also add the corresponding vCenter and NSX instances,

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **VMware Managers**, click **VMware HCX**.
4. In the **Add a New VMware HCX Account or Source** page, provide the required information.

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |

5. Click **Validate**.
6. Optional: In the **Nickname** text box, enter a nickname.
7. Optional: In the **Notes** text box, add a note if necessary.
8. Click **Submit**.

## Supported VMware Cloud Accounts

You can add VMware Cloud on AWS - vCenter account and NSX Manager account in VCF Operations for networks to monitor your VMware Cloud on AWS network.

**Note:**
VCF Operations for networks supports many types of NSX Manager accounts such as VMC on AWS NSX Manager, Azure VMware Solution NSX Manager, and Oracle Cloud VMware Solution NSX Manager.

To add VMware Cloud on AWS - vCenter account and NSX Manager account in VCF Operations for networks as data source, you must Manually Configure Data Sources for VMware Cloud on AWS.

### Manually Configure Data Sources for VMware Cloud on AWS

To monitor your VMware Cloud on AWS environment, you must set up the VCF Operations for networks collector to collect data from VMware Cloud on AWS, and add VMware Cloud on AWS -vCenter and VMware Cloud on AWS - NSX Manager as data sources.

### Preparing VCF Operations for networks to add VMware Cloud on AWS

You must setup VCF Operations for networks collector and configure firewall rules in VMware Cloud on AWS for communication and data collection.

### Set up VCF Operations for networks Collector for VMware Cloud on AWS

You must set up the VCF Operations for networks collector to collect data from VMware Cloud on AWS.

Deploy a VCF Operations for networks collector within every SDDC that needs to be added as a data source.

**Important:**

- Using a VCF Operations for networks collector deployed in one VMware Cloud on AWS SDDC to collect data from another VMware Cloud on AWS SDDC is not supported.
- You must deploy the VCF Operations for networks collector on a native VMware Cloud on AWS segment. Deploying a collector on extended segments is not supported.

1. Log in to VCF Operations for networks.
2. Navigate to **Settings** > **Infrastructure and Support** > **Add Collector VM**.
3. Copy the contents of the shared secret.

   You will require this during the deployment of VCF Operations for networks Collector OVA.
4. Deploy the VCF Operations for networks Collector OVA in the Compute Resource Pool of the VMware Cloud on AWS vCenter.

   Use the shared secret that you have generated.

   **Note:**
   For a single node SDDC in VMware Cloud on AWS, CPU resource reservation for the collector VM must be at least 1251 MHz.
5. Start the collector VM and follow the wizard to pair the collector with the VCF Operations for networks platform.

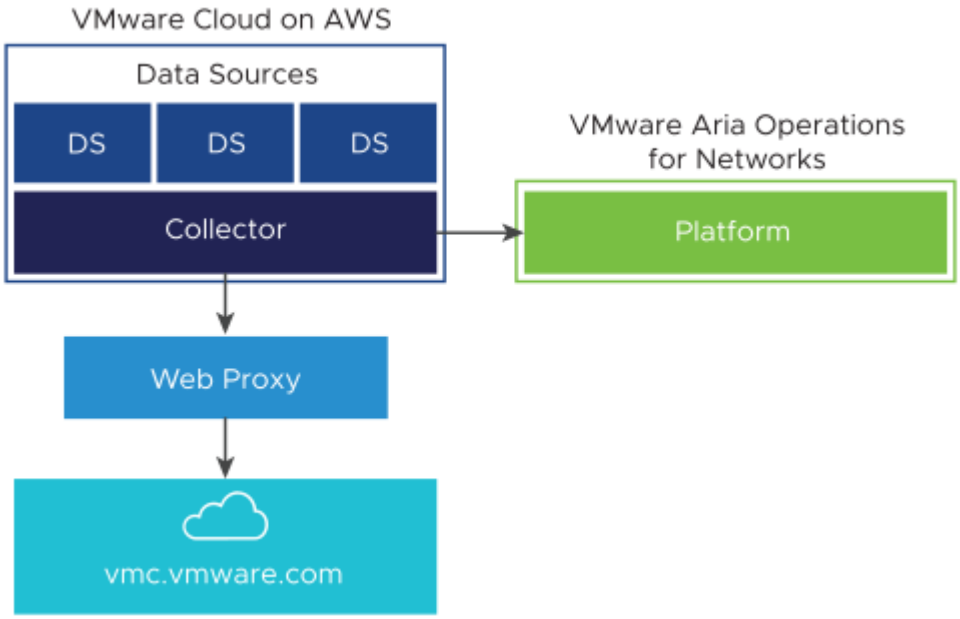
   **Note:**
   During this step, you must configure Web proxies and Firewall rules for VMware Cloud on AWS.
6. Validate that the collector pairs successfully with the platform.

### Configuring Web Proxies for VMware Cloud on AWS

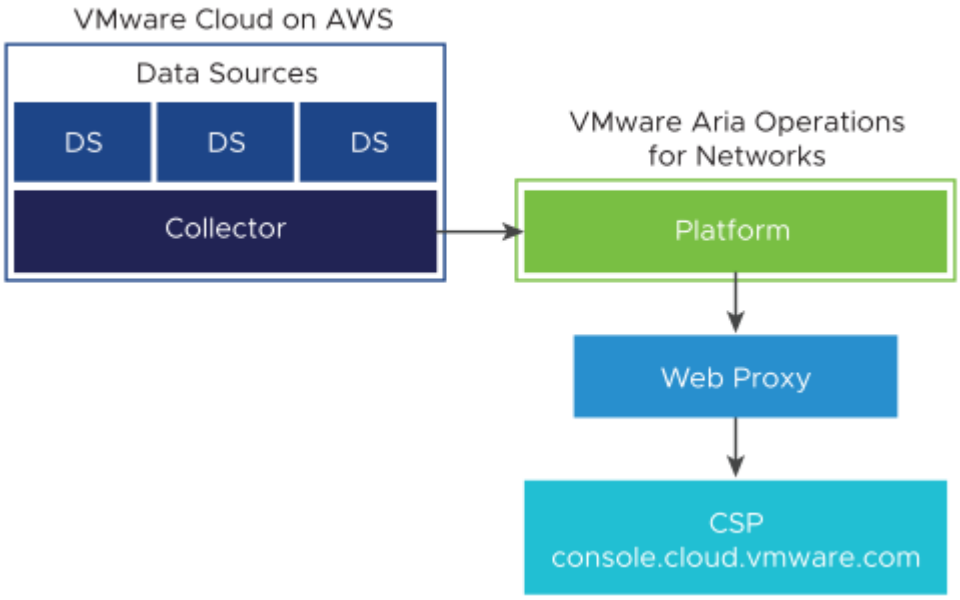
Depending on the scenario you are using, you must configure web proxies to connect to other entities (data sources, cloud service platforms, and so on) when the connection is made over the Internet.

#### Web Proxy Configuration Scenarios

- Scenario 1: Web Proxy not configured.
  You do not need any web proxy configuration.
- Scenario 2: When the Collector is in VMware Cloud on AWS and the Collector is using the web proxy to connect to `vmc.vmware.com`.
  You must enable the web proxy through the user interface (UI). To know how to configure web proxy, see Configuring Web Proxies.




- Scenario 3: When the Platform is on-premises and the Platform is using a web proxy to connect to CSP.
  You must use CLI to enable web proxy (HTTP anonymous and basic authentication web proxy only) in the Platform. To know about enabling web proxy using CLI, see set-web-proxy.




Create VMware Cloud on AWS Firewall Rules for VCF Operations for networks

You must create VMware Cloud on AWS groups and firewall rules to build communication with VCF Operations for networks.

**Prerequisites:**

- Deploy VCF Operations for networks platform and collector (for on-premise) or get the valid subscription (for cloud service).
- You must have required privileges. See Add VMware Cloud on AWS - vCenter and Add VMware Cloud on AWS NSX Manager.
- Deploy a VMware Cloud on AWS software-defined data center (SDDC) 1.8 and later with NSX-T networking.
- For the port requirements of incoming traffic, see VMware Ports and Protocols.
- Ensure you allow the HTTPS port 443 in the firewall to communicate between the Collector and the following domain:

  - `*.vmc.vmware.com`
- Ensure you allow the HTTPS port 443 in the firewall to communicate between the Platforms and the following domain:
  - `console.cloud.vmware.com`

**Configure Firewall Rules for communication between VCF Operations for networks platform and collector**

Configuring firewall rules in VMware Cloud on AWS includes:

- Creating a VMware Cloud on AWS group for VCF Operations for networks collector.
  a. Log in to VMware Cloud on AWS at https://vmc.vmware.com.
  b. On the **Networking & Security** tab, click **Inventory > Groups**.
  c. On the **Groups** card, click **COMPUTE GROUPS**, then click **ADD GROUP** and give the group a **Name** and an optional **Description**.
  d. Click **Set Members** to open the **Select Members** page.
  e. Provide the VCF Operations for networks collector VM details.
  You use this group in the firewall rules that you create later to allow communication between VMware Cloud on AWS NSX Manager and VCF Operations for networks.
- Create a firewall rule.
  a. Log in to the VMC Console at https://vmc.vmware.com.
  b. On the **Networking & Security** tab, click **Gateway Firewall**.
  c. On the **Gateway Firewall** card, click **Compute Gateway**, then click **ADD RULE** and give the new rule a **Name**.
  d. Enter the parameters for the new rule.
    - **Sources**: Enter the name of the VMware Cloud on AWS group containing the VCF Operations for networks collector IP address.
    - **Destinations**: Select **Any**.
    - **Services**: Select **HTTPS**, **DNS**, **DNS-UDP**, **NTP**, **ICMP**.
    - **Action**: Select **Allow**.
    - **Applied To**: Select **Internet Interface**.
    - **Logging**: Enable logging if required. Else this field is unchanged.
    The new rule is enabled by default. Slide the toggle to the left to deactivate it.
  e. Click **Publish**.

**Configure Firewall Rules for communication between collector and NSX Manager, and collector and vCenter**

1. Log in to the VMC Console at https://vmc.vmware.com.
2. On the **Networking & Security** tab, click **Gateway Firewall**.
3. On the **Gateway Firewall** card, click **Management Gateway**, then click **ADD RULE** and give the new rule a **Name**.
4. Enter the parameters for the new rule.
   - **Sources**: Enter the name of the VMware Cloud on AWS group containing the VCF Operations for networks collector IP address.
   - **Destinations**: Select **System Defined Groups**, search for NSX Manager, and then select the NSX Manager entry.
   - **Services**: Select **HTTPS (443)**.
   - **Action**: Select **Allow**.
   - **Logging**: Enable logging if required.
   By default, the new rule is enabled. Slide the toggle to deactivate it.
5. Click **Publish**.
6. Perform the same steps to configure a rule for the VMware vCenter Server.

   **Note:**
   Ensure to select VMware vCenter for the **Destinations** field in Step 4.

## Add VMware Cloud on AWS - vCenter

You can add VMware Cloud on AWS - vCenter as a data source in VCF Operations for networks.

- Obtain the credentials to add VMware Cloud on AWS - vCenter as a data source.
  - To obtain credentials for VMware Cloud on AWS, VMware Cloud on AWS Outposts, and VMware Cloud On Dell EMC, do the following steps:
    a. Log in to your VMware Cloud Services console, do one of the following depending on the service you are using:
       - For VMware Cloud on AWS, under **My Services**, click **VMware Cloud on AWS**.
       - For VMware Cloud on AWS Outposts, under **My Services**, click **VMware Cloud on AWS Outposts**.
       - For VMware Cloud On Dell EMC, under **My Services**, click **VMware Cloud On Dell EMC**.
    b. Click the name of the desired SDDC.
    c. Select the **Settings** tab, perform the following tasks:
       - Expand the **vCenter FQDN** panel and copy or note down the **vCenter FQDN**.
       - Expand the **Default vCenter User Account** panel and copy or note down the default user name (cloudadmin@vmc.local) and the default password.
  - To obtain credentials for Azure VMware Solution, Google Cloud VMware Engine, and Oracle Cloud VMware Solution, access the respective cloud consoles.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **VMware Cloud (VMC)**, select **VMC - vCenter**.
4. On the **Add a VMware Cloud (VMC) - VMware vCenter** page, provide the following details:

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter vCenter FQDN that you have retrieved from VMware Cloud Services. |
| **Username** | Enter the user name that you retrieved from the VMware Cloud Services. |
| **Password** | Enter the password that you retrieved from the VMware Cloud Services. |

5. Click **Validate**.
6. Optional: In the **Nickname** text box, enter a nickname.
7. Optional: In the **Notes** text box, add a note if necessary.
8. Click **Submit**.

### Add VMware Cloud on AWS NSX Manager

You can add VMware Cloud on AWS NSX Manager as a data source in VCF Operations for networks.

- Add VMware Cloud on AWS - vCenter
- Generate API tokens.
  To generate API tokens, you must have **Administrator** and **NSX Cloud Admin** service roles.
- Ensure that the platform node has direct internet access. Also, the firewall rules must allow the platform to connect to the internet. To know about firewall rules, see Create VMware Cloud on AWS Firewall Rules for VCF Operations for networks.
  If you are using a web proxy to access internet, you must configure the web proxy correctly. To know about web proxy configuration, see Configuring Web Proxies for VMware Cloud on AWS.
- To use all the available VCF Operations for networks features and to enable the DFW IPFIX on VMware Cloud on AWS NSX Manager, you must have the following privileges:

| Service | Organization Role | Service Role | Allowed Actions |
|---|---|---|---|
| vCenter on AWS | Org Member | **Administrator** and **NSX Cloud Admin** | Add Data Source, Enable IPFIX. |

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **VMware Cloud (VMC)**, click **VMC on AWS NSX Manager**.
4. In the **Add a VMware Cloud (VMC) on AWS NSX Manager** page, do the following:

| Option | Action |
|---|---|
| **VMC vCenter** | Select the corresponding VMware Cloud on AWS vCenter. |
| **Collector VM** | The System selects the collector automatically based on the selection of the VMware vCenter.<br>**Note:**<br>You must add the NSX Manager to the same collector VM as of the corresponding VMware Cloud on AWS vCenter. |
| **IP Address/FQDN** | Provide the IP address or the FQDN details.<br>**Note:**<br>You get the IP address from **Support** tab of vCenter on AWS SDDC Console. |
| **CSP Refresh Token** | Enter the API token that you have generated. |
| **Web Proxy (Optional)** | Select a web proxy from the **Web Proxy (Optional)** drop-down menu.<br>**Note:**<br>The **Web Proxy (Optional)** is visible only if you have configured a web proxy in VCF Operations for networks. |

5. Click **Validate**.
6. To collect IPFIX flows for DFW, select **Enable DFW IPFIX** check-box.

   **Note:**
   If you get error messages while enabling DFW IPFIX, see Unable to Enable DFW IPFIX troubleshooting topic.
7. Optional: In the **Nickname** text box, enter a nickname.
8. Optional: In the **Notes** text box, add a note if necessary.
9. Click **Submit**.

Add VMC on Dell EMC NSX Manager

You can add VMware Cloud on AWS on Dell EMC NSX Manager as a data source in VCF Operations for networks.

- Add VMware Cloud on AWS - vCenter
- Generate API tokens.
  To generate API tokens, you must have **Administrator** and **NSX Cloud Admin** service roles.
- Ensure that the platform node has direct internet access. Also, the firewall rules must allow the platform to connect to the internet. To know about firewall rules, see Create VMware Cloud on AWS Firewall Rules for VCF Operations for networks.
  If you are using a web proxy to access internet, you must configure the web proxy correctly. To know about web proxy configuration, see Configuring Web Proxies for VMware Cloud on AWS.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. In the **VMware Cloud (VMC)** category, select **VMC on Dell EMC NSX Manager**.
4. On the **Add Account or Source** page, do the following:

| Option | Action |
|---|---|
| **VMC vCenter** | Select the corresponding VMC vCenter. |
| **Collector VM** | The System selects the collector automatically based on the selection of the VMware vCenter.<br>**Note:**<br>You must add the NSX Manager to the same collector VM as of the corresponding VMware vCenter. |
| **IP Address/FQDN** | Provide the IP address or the FQDN details.<br>**Note:**<br>To get the IP address, you must contact VMC on Dell EMC Support. |
| **CSP Refresh Token** | Enter the API token that you have generated. |
| **Web Proxy (Optional)** | Select a web proxy from the **Web Proxy (Optional)** drop-down menu.<br>**Note:**<br>The **Web Proxy (Optional)** is visible only if you have configured a web proxy in VCF Operations for networks. |

5. Click **Validate**.
6. To collect IPFIX flows for DFW, select **Enable DFW IPFIX** check box.

   **Note:**
   If you get error messages while enabling DFW IPFIX, see Unable to Enable DFW IPFIX troubleshooting topic.
7. Optional: In the **Nickname** text box, enter a nickname.
8. Optional: In the **Notes** text box, add a note if necessary.
9. Click **Submit**.

### Add VMC on AWS Outposts NSX Manager

You can add VMC on AWS Outposts NSX Manager as a data source in VCF Operations for networks.

- Add VMware Cloud on AWS - vCenter.
- Generate API tokens.
  For vCenter, you must have **Administrator** and **NSX Cloud Admin** service roles assigned to you.
- Ensure that the platform node has a direct internet access. Also, the firewall rules must allow the platform to connect to the internet. To know about firewall rules, see Create VMware Cloud on AWS Firewall Rules for VCF Operations for networks.
  If you are using a web proxy to access internet, you must configure the web proxy correctly. To know about web proxy configuration, see Configuring Web Proxies for VMware Cloud on AWS.
- To use all the available VCF Operations for networks features and to enable the DFW IPFIX on VMware Cloud on AWS Policy Manager, you must have the following privileges:

| Service | Organization Role | Service Role | Allowed Actions |
|---|---|---|---|
| vCenter on AWS | Org Member | **Administrator** and **NSX Cloud Admin** | Add data source and enable IPFIX. |

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **VMware Cloud (VMC)**, click **VMC - AWS Outposts NSX Manager**.
4. In the **Add Accounts or Source** page, do the following:

| Option | Action |
|---|---|
| **VMC vCenter** | Select the corresponding VMware Cloud on AWS vCenter. |
| **Collector VM** | The System selects the collector automatically based on your selection of the VMware vCenter.<br>**Note:**<br>You must add the NSX Manager to the same collector VM as of the corresponding VMware vCenter. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details.<br>**Note:**<br>You get the IP address from **Support** tab of vCenter on AWS SDDC Console. |
| **CSP Refresh Token** | Enter the API token that you have generated. |
| **Web Proxy (Optional)** | Select a web proxy from the **Web Proxy (Optional)** drop-down menu.<br>**Note:**<br>The **Web Proxy (Optional)** is visible only if you have configured a web proxy in VCF Operations for networks. |

5. Click **Validate**.
6. To enable the distributed firewall to send IPFIX flow records to the collector, select the **Enable DFW IPFIX** check box.

   **Note:**
   If you get error messages while enabling DFW IPFIX, see Unable to Enable DFW IPFIX troubleshooting topic.
7. Optional: In the **Nickname** text box, enter a nickname.
8. Optional: In the **Notes** text box, add a note if necessary.
9. Click **Submit**.

Once the data collection starts, you can gain insight into VMC SDDCs on AWS Outposts. To learn more, see Viewing VMC on AWS Outposts SDDC Details.

### Add Azure VMware Solution NSX Manager

You can add Azure VMware Solution NSX Manager as a data source in VCF Operations for networks.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. In the **VMware Cloud (VMC)** category, select **Azure VMware Solution NSX Manager**.
4. In the **Add Accounts or Source** page, do the following:

| Option | Action |
|---|---|
| **Collector VM** | The System selects the collector automatically based on your selection of the VMware vCenter.<br>**Note:**<br>You must add the NSX Manager to the same collector VM as of the corresponding VMware vCenter. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |

| Option | Action |
|---|---|
| **Password** | Enter the password. |

5. Click **Validate**.
6. To collect IPFIX flows for DFW, select **Enable DFW IPFIX** check box from the **Additional Information** section.

   **Note:**

   - If you get error messages while enabling DFW IPFIX, see Unable to Enable DFW IPFIX troubleshooting topic.
   - If you don't need IPFIX, you must be a local user with audit level permissions. But if you need IPFIX, you must have one of the following permissions: **enterprise_admin**, **network_engineer**, **security_engineer**, or **cloudadmin**.
   - The **cloudadmin** role does not have write access to NSX IPFIX MP API: `/api/v1/ipfix/collectorconfigs/<id>`
   - As this API is not allowed for **cloudadmin**, VCF Operations for networks will no longer support IPFIX flow collection for logical switches created using **Advanced Networking** tab in NSX.
7. Optional: In the **Nickname** text box, enter a nickname.
8. Optional: In the **Notes** text box, add a note if necessary.
9. Click **Submit**.

### Add Google Cloud VMware Engine NSX Manager

You can add Google Cloud VMware Engine NSX Manager as a data source in VCF Operations for networks.

Add VMware Cloud on AWS - vCenter.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. In the **VMware Cloud (VMC)** category, select **Google Cloud VMware Engine NSX Manager**.
4. On the **Add Accounts or Source** page, do the following:

| Option | Action |
|---|---|
| **Collector VM** | The System selects the collector automatically based on your selection of the VMware vCenter.<br>**Note:**<br>You must add the NSX Manager to the same collector VM as of the corresponding VMware vCenter. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |

5. Click **Validate**.
6. To collect IPFIX flows for DFW, select **Enable DFW IPFIX** check box from the **Additional Information** section.

   **Note:**

   - If you get error messages while enabling DFW IPFIX, see Unable to Enable DFW IPFIX troubleshooting topic.
   - If you don't need IPFIX, you must be a local user with audit level permissions. But if you need IPFIX, you must have one of the following permissions: **enterprise_admin**, **network_engineer**, **security_engineer**, or **cloudadmin**.
   - The **cloudadmin** role does not have write access to NSX IPFIX MP API: `/api/v1/ipfix/collectorconfigs/<id>`
   - As this API is not allowed for **cloudadmin**, VCF Operations for networks will no longer support IPFIX flow collection for logical switches created using **Advanced Networking** tab in NSX.

7. Optional: In the **Nickname** text box, enter a nickname.
8. Optional: In the **Notes** text box, add a note if necessary.
9. Click **Submit**.

Add Oracle Cloud VMware Solution NSX Manager

You can add Oracle Cloud VMware Solution NSX Manager as a data source in VCF Operations for networks.

Add VMware Cloud on AWS - vCenter.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. In the **VMware Cloud (VMC)** category, select **Oracle Cloud VMware Solution NSX Manager**.
4. On the **Add Accounts or Source** page, do the following:

| Option | Action |
|---|---|
| **Collector VM** | The System selects the collector automatically based on your selection of the VMware vCenter.<br><br>**Note:**<br>You must add the NSX Manager to the same collector VM as of the corresponding VMware vCenter. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |

5. Click **Validate**.
6. To collect IPFIX flows for DFW, select **Enable DFW IPFIX** check box from the **Additional Information** section.

   **Note:**

   - If you get error messages while enabling DFW IPFIX, see Unable to Enable DFW IPFIX troubleshooting topic.
   - If you don't need IPFIX, you must be a local user with audit level permissions. But if you need IPFIX, you must have one of the following permissions: **enterprise_admin**, **network_engineer**, **security_engineer**, or **cloudadmin**.
   - The **cloudadmin** role does not have write access to NSX IPFIX MP API: `/api/v1/ipfix/collectorconfigs/<id>`
   - As this API is not allowed for **cloudadmin**, VCF Operations for networks will no longer support IPFIX flow collection for logical switches created using **Advanced Networking** tab in NSX.

7. Optional: In the **Nickname** text box, enter a nickname.
8. Optional: In the **Notes** text box, add a note if necessary.
9. Click **Submit**.

## Supported AWS Outposts VMware Solution

In VCF Operations for networks, you can add your data sources that are deployed in the VMware Cloud on AWS on AWS Outposts Solution.

To gain insights about your data sources that are deployed in the VMware Cloud on AWS on AWS Outposts Solution, you must add the following data sources in VCF Operations for networks:

**Note:**
VCF Operations for networks does not support native AWS Outposts components that are attached to your VMware Cloud on AWS on AWS Outposts Solution SDDC.

1. Add VMware Cloud on AWS - vCenter
2. Add VMC on AWS Outposts NSX Manager
3. Add VMware HCX

**What to do Next**

Once the data collection starts, you can view the data source related information in VCF Operations for networks. To know about entity details, see Viewing Entity Details.

## Supported Azure VMware Solution

In VCF Operations for networks, you can add your data sources that are deployed in Azure VMware Solution (AVS).

> **Note:**
> VCF Operations for networks does not support native Azure components that are attached to your AVS SDDC.

To gain insights about your data sources that are deployed in Azure VMware Solution (AVS), you must add the following data sources in VCF Operations for networks:

1. Add VMware Cloud on AWS - vCenter
2. Add Azure VMware Solution NSX Manager
3. Add VMware HCX

**What to do Next**

Once the data collection starts, you can view the data source related information in VCF Operations for networks. To know about entity details, see Viewing Entity Details.

## Supported Oracle Cloud VMware Solution

In VCF Operations for networks you can add your data sources that are deployed in the Oracle Cloud VMware Solution.

> **Note:**
> VCF Operations for networks does not support native Oracle Cloud components that are attached to your Oracle Cloud VMware Solution SDDC.

To gain insights about your data sources deployed in the Oracle Cloud VMware Solution, you must add the following data sources in VCF Operations for networks:

1. Add VMware Cloud on AWS - vCenter
2. Add Oracle Cloud VMware Solution NSX Manager
3. Add VMware HCX

**What to Do Next**

Once the data collection starts, you can view the data source related information in VCF Operations for networks. To know about entity details, see the Viewing Entity Details.

## Supported Google Cloud VMware Engine

In VCF Operations for networks, you can add your data sources that are deployed in the Google Cloud VMware Engine.

> **Note:**
> VCF Operations for networks does not support native Google Cloud components that are attached to your Google Cloud VMware Engine SDDC.

To gain insights about your data sources deployed in Google Cloud VMware engine, you must add the following data sources in VCF Operations for networks .

1. Add VMware Cloud on AWS - vCenter
2. Add Google Cloud VMware Engine NSX Manager
3. Add VMware HCX

### What to Do Next

Once the data collection starts, you can view the data source related information in VCF Operations for networks. To know about entity details, see Viewing Entity Details.

## Supported VMware Cloud on Dell EMC Solution

In VCF Operations for networks, you can add your data sources that are deployed in vCenter on Dell EMC.

### Configuring VMware Cloud on Dell EMC in VCF Operations for networks

To get insights about data sources deployed in vCenter on Dell EMC, you must add the following data sources in VCF Operations for networks:

1. VMware vCenter
2. VMC on Dell EMC NSX Manager
3. VCF Operation HCX

| Sl. No. | Action | Procedure |
|---|---|---|
| 1 | Generate a shared secret for adding a collector. | Generate a Shared Secret |
| 2 | Deploy VCF Operations for networks collector in vCenter on Dell EMC. | Setting up Network Insight Collector (OVA)<br>Make sure to deploy a VCF Operations for networks collector within every SDDC that is added as a data source.<br>Also, for a single node SDDC in VMware Cloud on AWS, make sure the CPU resource reservation for the collector VM must be at least 1251 MHz.<br>**Important:**<br>During this step, you must configure Web proxies and Firewall rules for VMware Cloud on AWS.<br>When you deploy a VCF Operations for networks collector in vCenter on Dell EMC, the following scenarios are not supported:<br>• Using a VCF Operations for networks collector deployed in a VMware Cloud on AWS SDDC to collect data from another VMware Cloud on AWS SDDC.<br>• Deploying a collector on extended segments. You must deploy the VCF Operations for networks collector on a native VMware Cloud on AWS segment. |
| 3 | Add the VMware vCenter server deployed in VMware Cloud on Dell EMC in VCF Operations for networks. | Add VMware Cloud (VMC) - vCenter |
| 4 | Generate an API token from vCenter services. | To generate API tokens, you must have **VMC on Dell EMC Administrator** and **VMC on Dell EMC Cloud Admin role for NSX** service roles. |
| 5 | Add vCenter on Dell EMC NSX Manager in VCF Operations for networks. | Add VMC on DELL EMC NSX Manager |
| 6 | (Optional) Add VMware HCX in VCF Operations for networks. | Add VMware HCX |

### Viewing Entity Details

Once the data collection starts, you can view the data source related information in VCF Operations for networks.

### Generate a Shared Secret

You can generate and import the VCF Operations for networks collector virtual appliance.

Generate a shared secret and import the VCF Operations for networks collector virtual appliance:

1. Log into the VCF Operations for networks UI.
2. Select **Infrastructure and Support**. **Settings** > **Infrastructure and Support**.
3. Scroll down and click **Add Collector VM**.
4. Copy the shared secret and click **Done**.

## Supported Containers

You can add containers into VCF Operations for networks.

### Add Tanzu Kubernetes Grid Integrated Edition

You can add Tanzu Kubernetes Grid Integrated Edition as a data source and fetch your cluster details in VCF Operations for networks.

- You must ensure that:
  - The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.
  - The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
- You must have the **pks.clusters.admin** privilege.
- You must add the corresponding NSX-T Manager.

> **Note:**
> VCF Operations for networks supports only VMware Tanzu Kubernetes Grid Integrated Edition out of all the VMware Tanzu Kubernetes Grid offerings.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under Containers, select **TKGI**.
4. On the Add a new TKGI Account or Source page, provide the following details:

| Field Name | Description |
|---|---|
| NSX-T Manager | Select the NSX-T Manager that supports the underlying networking for the Tanzu Kubernetes Grid Integrated Edition deployment. |
| Collector VM | VCF Operations for networks automatically selects the corresponding collector VM associated with the chosen NSX-T Manager.<br>**Note:**<br>The collector VMs that are added as a NetFlow collector are not available in the list. |
| API Hostname (FQDN) | Enter the FQDN details of the Tanzu Kubernetes Grid Integrated Edition API server. |
| Username | Enter the Tanzu Kubernetes Grid Integrated Edition user name that has access to the clusters.<br>**Note:**<br>The user must have the `pks.clusters.admin` privileges. |

| Field Name | Description |
|---|---|
| Password | Enter the password.<br><br>**Note:**<br>Currently passwords having special characters like `&, (, ), |, <, >, ` are not supported. |

5. Click **Validate**.

   You see the `Validation Successful` message.

6. Optional: In the **Nickname** text box, enter a nickname.

7. Optional: In the **Tags (Optional)** key-value pair text box, enter a key and a value.

   A key-value pair could be any text. For example, you can use `Layer Access` as key-value pair where `layer` is the `key` and access is the value.

   – To apply the tag in all the associated entities, click the **Apply above tag operations to all associated entities** check box. When you select this option, the tag gets applied to all associated entities. For more details about the associated entities, see Working with Local Tags.

     If you clear the **Apply above tag operations to all associated entities** check box, the assigned tag is removed from all the associated entities.

8. Optional: In the **Notes** text box, add a note if necessary.

9. Click **Submit**.

   If you see the `One or more Kubernetes Cluster Master hosts are not reachable from Collector VM` error message, run the following commands on the collector VM:

   a) `tkgi login -a` *`TKGI_API_Server`* `- u` *`username`* `-p` *`password`* `-k`
   b) `tkgi clusters`

      Ensure that the cluster status is `succeeded`.

   c) `tkgi cluster` *`Kubernetes_Cluster_Name`*
   d) `telnet` *`Kubernetes_Master_HostKubernetes_Master_port`*

      Verify the primary host is able to connect.

   e) Repeat step c and step d for all Kubernetes cluster that were discovered from step b.

**Add Kubernetes**

You can add Kubernetes as a data source and fetch your Kubernetes Cluster details into VCF Operations for networks.

- You must ensure that:
  – The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.
  – The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
- You have cluster admin role with read permissions.
- You have added NSX-T Manager in VCF Operations for networks.

**Note:**
The Kubernetes Cluster and the corresponding NSX-T Manager must be added to the same collector VM.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under Containers, select **Kubernetes**.
4. In the Add a new Kubernetes Account or Source page, provide the following details:

| Field Name | Description |
|---|---|
| NSX-T Manager | Select the NSX-T Manager that supports the underlying networking for Kubernetes. |
| Collector VM | VCF Operations for networks automatically selects the corresponding collector VM associated with the chosen NSX-T Manager.<br>**Note:**<br>The collector VMs that are added as a NetFlow collector are not available in the list. |
| Kubeconfig | Click **Browse** and upload the Kubernetes configuration file that has Kubernetes cluster details. For more information about the format of the Kubeconfig configuration file, refer to the Kubernetes documentation.<br>**Note:**<br>The user configured in the Kubeconfig file must have the **List** and **Watch** privileges. |

5. Click **Validate**.
   You see the `Validation Successful` message.
6. Optional: In the **Nickname** text box, enter a nickname.
7. Optional: In the **Notes** text box, add a note if necessary.
8. Click **Submit**.

VCF Operations for networks can now fetch the Kubernetes cluster details.

Go to the Kubernetes Dashboard and view the details, see Viewing Kubernetes Details.

**Add OpenShift**
You can add OpenShift as a data source and fetch your OpenShift details into VCF Operations for networks.

You must ensure that:

- The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
- The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.
- You have added NSX-T Manager in VCF Operations for networks.

**Note:**
OpenShift and the corresponding NSX-T Manager must be added to the same collector VM.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under Containers, select **OpenShift**.
4. On the Add a new OpenShift Account or Source page, provide the following details:

| Field Name | Description |
|---|---|
| NSX-T Manager | Select the NSX-T Manager that supports the underlying networking for OpenShift. |
| Collector VM | VCF Operations for networks automatically selects the corresponding collector VM associated with the chosen NSX-T Manager.<br>**Note:**<br>The collector VMs that are added as a NetFlow collector are not available in the list. |
| Kubeconfig | Click **Browse** and upload the Kubernetes configuration file that has Kubernetes cluster details. For more information about the format of the Kubeconfig configuration file, refer to the Kubernetes documentation.<br>**Note:**<br>The user configured in the Kubeconfig file must have the **List** and **Watch** privileges. |

5. Click **Validate**.
6. Optional: In the **Nickname** text box, enter a nickname.
7. Optional: In the **Notes** text box, add a note if necessary.
8. Click **Submit**.

VCF Operations for networks can now fetch the OpenShift details.

See the details on the Viewing Kubernetes Details.

## Supported Firewalls

VCF Operations for networks supports the following firewalls.

### Add Palo Alto Networks Panorama

You can add Palo Alto Networks Panorama as a data source in VCF Operations for networks.

You must ensure that:

- The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
- The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.
- You have **admin role** with XML API access. For more details, see Palo Alto Firewall.

In the **Panorama** UI, perform the following steps to add an admin role for XML API.

1. Select **Panorama** > **Admin Roles**.
2. Click **Add** to add a new admin role.
3. In the **The Admin Role Profile** window, enter the name to the role and select **Panorama**.
4. Click the **Web UI** tab and deactivate all entries.
5. Click the **XML API** tab and deactivate all entries, except **Configuration** and **Operational Requests**.
6. Click **OK** to close the window.
   The new admin role appears in the list.

7. Click **Commit**.
8. Assign this role to an administrator account or create a new user and assign this role to the new user.

**Note:**
VCF Operations for networks does not currently fetch local Palo Alto Network policies that are directly defined in the devices.

**Note:**
VCF Operations for networks does not support the Palo Alto Panorama integration with multiple NSX managers.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **Firewalls**, click **Palo Alto Networks Panorama**.
4. In the **Add a New Palo Alto Networks Panorama Account or Source** page, provide the required information.

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |

5. Click **Validate**.
6. Define the polling interval for the configuration data collection. You can set the polling interval from 10 minutes to 7 days.
   - **Preset** - Select the interval time from the predefined time set.
   - **Custom Interval** - Set a value and select **Minutes**, **Hours**, and **Days**.
   - **Fixed Schedule** - Select the days and set the time to schedule the interval.
7. Optional: In the **Nickname** text box, enter a nickname.
8. Optional: In the **Tags (Optional)** key-value pair text box, enter a key and a value.

   A key-value pair could be any text. For example, you can use `Layer Access` as key-value pair where `layer` is the `key` and access is the value.

   – To apply the tag in all the associated entities, click the **Apply above tag operations to all associated entities** check box. When you select this option, the tag gets applied to all associated entities. For more details about the associated entities, see Working with Local Tags.

   If you clear the **Apply above tag operations to all associated entities** check box, the assigned tag is removed from all the associated entities.
9. Optional: In the **Notes** text box, add a note if necessary.
10. Click **Submit**.

VCF Operations for networks supports the following features of Palo Alto firewall:

- Neighbor Discovery (LLDP)
- Layer 3 Routing (IPv4)
- Virtual Router
- Virtual Wire (Supported only in network assurance and verification)
- Security Policy
- Security Zone
- Virtual System (Not supported in network assurance and verification)

  **Note:**
  VCF Operations for networks does not support inter-vsys routing of Palo Alto firewalls.
- High Availablility. This feature has limited support and is not supported in network assurance and verification.

  **Note:**
  VIP interfaces are not supported.

### Add Check Point Management Server

VCF Operations for networks supports Check Point Security Manager (SmartCenter) and the Check Point Multi-Domain Security (MDS) Management Servers.

- You must ensure that:
  - The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.
  - The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
- VCF Operations for networks requires read-only privileges for the Web-API access for fetching most of the Check Point data. There are few exceptions as follows:
  - If a non-VSX physical gateway is attached to the management server, the user should have read-write access privileges for the Web API. This is required to fetch the gateway routes for using the `run script` Web API for the VM-VM path computation.
  - If a VSX gateway is attached to the management server, the user should have the SSH access with the same password. In addition, the user should have access to the CLI command `vsx_util view_vs_conf`. This command is used to fetch the VSX gateway routes for the VM-VM path computation.
  - For MDS server IP as data-source, the user should have the Web API access to all domains including the MDS domain and the global domain. It is required to fetch rules, policy packages and other data from all the domains.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under the **Firewall** group, click **Check Point Management Server**.
4. In the **Add a New Check Point Management Server Account or Source** page, provide the required information.

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details.<br>**Note:**<br>If you are adding the Check Point MDS Management server, you must provide the IP of the MDS server. You cannot add the domain management server IP of an MDS server as an individual data source. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |

5. Click **Validate**.
6. Define the polling interval for the configuration data collection. You can set the polling interval from 10 minutes to 7 days.
   - **Preset** - Select the interval time from the predefined time set.
   - **Custom Interval** - Set a value and select **Minutes**, **Hours**, and **Days**.
   - **Fixed Schedule** - Select the days and set the time to schedule the interval.
7. Optional: In the **Nickname** text box, enter a nickname.
8. Optional: In the **Tags (Optional)** key-value pair text box, enter a key and a value.

   A key-value pair could be any text. For example, you can use `Layer Access` as key-value pair where `layer` is the `key` and access is the value.

   - To apply the tag in all the associated entities, click the **Apply above tag operations to all associated entities** check box. When you select this option, the tag gets applied to all associated entities. For more details about the associated entities, see Working with Local Tags.

     If you clear the **Apply above tag operations to all associated entities** check box, the assigned tag is removed from all the associated entities.

9. Optional: In the **Notes** text box, add a note if necessary.
10. Click **Submit**.

VCF Operations for networks supports the following features of Check Point firewall:

- VLAN
- Layer 3 Routing (IPv4)
- Virtual Wire (in network assurance and verification)
- Security Policy
- Security Zone (in network assurance and verification)
- NAT (Network Address Translation). This feature is not supported in network assurance and verification.
- Cluster (Not supported in network assurance and verification)
- Virtual System (Not supported in network assurance and verification)

**Note:**
VCF Operations for networks does not support Check Point Virtual Switch and Virtual Router.

**Configuring and Managing Dropped Flow Alerts for Palo Alto and Check Point Firewalls in VCF Operations**
You can use VCF Operations to handle any dropped flow alerts that you receive from your Palo Alto or Check Point firewalls.

- Verify that log management is added as a data source in VCF Operations for networks.
- Verify that log management is configured as Syslog server for the firewalls. For more information, see Update Syslog server for Check Point and Palo Alto firewalls.
- Verify that port 514 and port 1514 are open. These ports are used to communicate with log management.

When log management reports a dropped flow that is received from a Check Point or a Palo Alto firewall device VCF Operations sends an alert.

1. Download and install the VMware Cloud Foundation Content Pack for VCF Operations for networks from the VCF Solutions Catalog.
2. To check for received dropped flow alerts, navigate to **Operate** > **Alerts** and expand **Log Fallback Object**.
3. To identify the cause for the dropped flow, select the alert, and select **analyze logs** from the **Recommendations** section.

   The **Analyze Logs** tab of the **Logs** page opens. The **Stream** tab contains the details of the dropped flow alert, which you can use to troubleshoot the issue.

**Add Cisco ASA**
You can add Cisco ASA as a data source in VCF Operations for networks.

- You must ensure that:
  - The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.
  - The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
- The user account has the privilege to enter the enable mode by using enable mode password. The user password must be same as the password used for the enable mode of Cisco ASA.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **Firewall** group, click **Cisco ASA**.
4. In the **Add a New Cisco ASA Account or Source** page, provide the required information.

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |

| Option | Action |
|---|---|
| | **Note:**<br>The user should have the enable mode privilege to set terminal length to `0`, and for switching security context. |
| **Password** | Enter the password.<br>**Note:**<br>Ensure that you enter the same password as the one that you used for the enable mode of Cisco ASA. |

5. Click **Validate**.
6. To enable a richer data collection, click **Use SNMP (recommended for richer data collection)** check box, and do the following:
   a) From the **SNMP Version** drop-down menu, select an SNMP version. Depending on which version of SNMP you have selected, do the following:

| Option | Action |
|---|---|
| 2c | In the **Community String** text box, enter a community string. |
| 3 | 1. Enter **Username**, Context Name.<br>2. From the **Authentication Type** drop-down, select an authentication type and enter the **Authentication Password**.<br>3. From the **Privacy Type** drop-down, select a privacy type and enter the **Privacy Password**. |

7. Define the polling interval for the configuration data collection. You can set the polling interval from 10 minutes to 7 days.
   - **Preset** - Select the interval time from the predefined time set.
   - **Custom Interval** - Set a value and select **Minutes**, **Hours**, and **Days**.
   - **Fixed Schedule** - Select the days and set the time to schedule the interval.
8. Optional: In the **Nickname** text box, enter a nickname.
9. Optional: In the **Tags (Optional)** key-value pair text box, enter a key and a value.
   A key-value pair could be any text. For example, you can use `Layer Access` as key-value pair where `layer` is the `key` and access is the value.
   – To apply the tag in all the associated entities, click the **Apply above tag operations to all associated entities** check box. When you select this option, the tag gets applied to all associated entities. For more details about the associated entities, see Working with Local Tags.
   If you clear the **Apply above tag operations to all associated entities** check box, the assigned tag is removed from all the associated entities.
10. Optional: In the **Notes** text box, add a note if necessary.
11. Click **Submit**.

VCF Operations for networks supports the following features of Cisco ASA:

- VLAN
- Layer 3 Routing (IPv4)
- VRF (Virtual routing and forwarding)
- Security Policy
- Security Zone
- Global Policy
- Multiple Context (Not supported in network assurance and verification)

**Add a Fortinet FortiManager**
In VCF Operations for networks, you can add Fortinet FortiManager as a data source.

You must verify the following information:

- The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
- The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.
- You have at least the **Restricted User** role with access to all ADOMs and policy packages.
- You have the **rpc-permit read-write** access enabled from Command Line Interface (CLI).
  To configure the **rpc** permission, use the following command in FortiManager CLI:

```
config system admin user
edit "<administrator name>"
set rpc-permit [none | read | read-write ]
end
```

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **Firewalls**, select **Fortinet FortiManager**.
4. On the **Add a New Fortinet FortiManager Account or Source** page, enter the required information:

| Option | Action |
|---|---|
| **Collector VM** | Select the collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP Address or the FQDN details. |
| **Username** | Enter the user name you want to use for this data source. |
| **Password** | Enter the password. |

5. Click **Validate**.
6. Define the polling interval for the configuration data collection. You can set the polling interval from 10 minutes to 7 days.
   - **Preset** - Select the interval time from the predefined time set.
   - **Custom Interval** - Set a value and select **Minutes**, **Hours**, and **Days**.
   - **Fixed Schedule** - Select the days and set the time to schedule the interval.
7. Optional: In the **Nickname** text box, enter a nickname.
8. Optional: In the **Tags (Optional)** key-value pair text box, enter a key and a value.
   A key-value pair could be any text. For example, you can use `Layer Access` as key-value pair where `layer` is the `key` and access is the value.
   – To apply the tag in all the associated entities, click the **Apply above tag operations to all associated entities** check box. When you select this option, the tag gets applied to all associated entities. For more details about the associated entities, see Working with Local Tags.
   If you clear the **Apply above tag operations to all associated entities** check box, the assigned tag is removed from all the associated entities.
9. Optional: In the **Notes** text box, add a note if necessary.
10. Click **Submit**.

VCF Operations for networks supports the following features of Fortinet FortiGate firewall:

- Layer 3 Routing (IPv4)
- Security Policy
- Security Zone
- NAT (Network Address Translation)
- Virtual Domain

## Supported Converged Infrastructures

VCF Operations for networks supports adding a few third party converged infrastructures.

### Add Cisco UCS Manager

You can add Cisco UCS Manager as a data source in VCF Operations for networks.

You must verify the following information:

- The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
- The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.

**Note:**

Cisco Virtual Machine Fabric Extender (VM-FEX) is not supported in VCF Operations for networks.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **Converged Infrastructures**, select **Cisco UCS Manager**.
4. In the Add a Cisco UCS Account or Source page, provide the following details:

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |

5. Click **Validate**.
6. To enable a richer data collection, click the **Use SNMP (recommended for richer data collection)** check box, and do the following:
   a) From the **SNMP Version** drop-down menu, select an SNMP version. Depending on the version of SNMP that you selected, do the following:

| Option | Action |
|---|---|
| 2c | In the **Community String** text box, enter a community string. |
| 3 | 1. Enter **Username**, Context Name.<br>2. From the **Authentication Type** drop-down, select an authentication type and enter the **Authentication Password**.<br>3. From the **Privacy Type** drop-down, select a privacy type and enter the **Privacy Password**. |

7. Optional: In the **Nickname** text box, enter a nickname.
8. Optional: In the **Tags (Optional)** key-value pair text box, enter a key and a value.

   A key-value pair could be any text. For example, you can use `Layer Access` as key-value pair where `layer` is the `key` and access is the value.

   – To apply the tag in all the associated entities, click the **Apply above tag operations to all associated entities** check box. When you select this option, the tag gets applied to all associated entities. For more details about the associated entities, see Working with Local Tags.

   If you clear the **Apply above tag operations to all associated entities** check box, the assigned tag is removed from all the associated entities.

9. Optional: In the **Notes** text box, add a note if necessary.
10. Click **Submit**.

VCF Operations for networks supports the following features of Cisco UCS Manager:

- Neighbor Discovery (CDP and LLDP). This feature is supported only in network assurance and verification.
- Layer 2 Switching
- VLAN
- Fabric Extender (FEX)
- End Host Mode
- Blade Server
- Rack Server (Not supported in network assurance and verification)

**Add HP Virtual Connect Manager**

You can add HP Virtual Connect Manager as a data source in VCF Operations for networks.

You must verify the following information:

- The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
- The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **Converged Infrastructures**, click **HP Virtual Connect Manager** and provide the required information.

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |

4. Click **Validate**.
5. Optional: In the **Nickname** text box, enter a nickname.
6. Optional: In the **Notes** text box, add a note if necessary.
7. Click **Submit**.

VCF Operations for networks supports the following features of HP Virtual Connect Manager:

- Enclosure / Chassis
- Blade Server
- Interconnect
- Uplink Port
- VLAN

**Add HP One View Manager**

You can add HP One View Manager as a data source in VCF Operations for networks.

You must verify the following information:

- The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
- The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **Converged Infrastructures**, click **HP One View Manager** and provide the required information.

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |

4. Click **Validate**.
5. Optional: In the **Nickname** text box, enter a nickname.
6. Optional: In the **Notes** text box, add a note if necessary.
7. Click **Submit**.

VCF Operations for networks supports the following features of HP One View Manager:

- Enclosure / Chassis
- Blade Server
- Interconnect
- Network Set

## Supported Routers and Switches

VCF Operations for networks supports several third party routers and switches.

**Add Arista Switch SSH**
You can add Arista Switch SSH as a data source in VCF Operations for networks.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **Routers & Switches**, select **Arista Switch SSH** .
4. In the **Add a New Arista Switch SSH Account or Source** page, provide the following details:

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details.<br>**Note:**<br>You must enter the same IP/FQDN that you have used in VMware NSX Manager to configure this switch. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |

5. Click **Validate**.
6. To enable a richer data collection, click **Use SNMP (recommended for richer data collection)** check box, and do the following:
   a) From the **SNMP Version** drop-down menu, select an SNMP version. Depending on which version of SNMP you have selected, do the following:

| Option | Action |
|---|---|
| 2c | In the **Community String** text box, enter a community string. |
| 3 | 1. Enter **Username**, Context Name.<br>2. From the **Authentication Type** drop-down, select an authentication type and enter the **Authentication Password**.<br>3. From the **Privacy Type** drop-down, select a privacy type and enter the **Privacy Password**. |

7. Define the polling interval for the configuration data collection. You can set the polling interval from 10 minutes to 7 days.
8. Optional: In the **Nickname** text box, enter a nickname.
9. Optional: In the **Tags (Optional)** key-value pair text box, enter a key and a value.
   A key-value pair could be any text. For example, you can use `Layer Access` as key-value pair where `layer` is the `key` and access is the value.
   – To apply the tag in all the associated entities, click the **Apply above tag operations to all associated entities** check box. When you select this option, the tag gets applied to all associated entities. For more details about the associated entities, see Working with Local Tags.
   If you clear the **Apply above tag operations to all associated entities** check box, the assigned tag is removed from all the associated entities.
10. Optional: In the **Notes** text box, add a note if necessary.
11. Click **Submit**.

VCF Operations for networks supports the following features of Arista Switch:

- Neighbor Discovery (LLDP)
- Layer 2 Switching
- VLAN
- Layer 3 Routing (IPv4)
- VRF (Virtual routing and forwarding)
- Redundancy (VARP and VRRP)
- Access Control List (in network assurance and verification)
- EVPN-VXLAN
- NSX Controller Hardware VTEP (Not supported in network assurance and verification)

**Add a Brocade Switch**
You can add Brocade switches as a data source in VCF Operations for networks.

You must verify the following information:

- The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
- The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.

You must have the following permissions:

- **Read-only user**: VCF Operations for networks requires read-only third-party device credentials (passwords) to access the SSH data. This user needs to have the **Network Operator** role. These passwords are added when configuring the third-party data provider in VCF Operations for networks user interface.

- **Read-only SNMP user**: VCF Operations for networks uses SNMP to collect metrics information from third-party devices (switches, routers, firewalls, and so on). The third-party devices must have SNMP enabled and be accessible from VCF Operations for networks.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **Routers & Switches**, click the Brocade switch that you want to add and provide the required information.

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |

**Note:**
To know about the supported products and versions, see Supported Products and Versions.

4. Click **Validate**.
5. To enable a richer data collection, click **Use SNMP (recommended for richer data collection)** check box, and do the following:
   a) From the **SNMP Version** drop-down menu, select an SNMP version. Depending on which version of SNMP you have selected, do the following:

| Option | Action |
|---|---|
| 2c | In the **Community String** text box, enter a community string. |
| 3 | 1. Enter **Username**, Context Name.<br>2. From the **Authentication Type** drop-down, select an authentication type and enter the **Authentication Password**.<br>3. From the **Privacy Type** drop-down, select a privacy type and enter the **Privacy Password**. |

6. Define the polling interval for the configuration data collection. You can set the polling interval from 10 minutes to 7 days.
7. Optional: In the **Nickname** text box, enter a nickname.
8. Optional: In the **Notes** text box, add a note if necessary.
9. Click **Submit**.

VCF Operations for networks supports the following features of Brocade switches:

- Brocade MLX
  - Neighbor Discovery (LLDP)
  - Layer 2 Switching
  - VLAN
  - Layer 3 Routing (IPv4)
  - VRF (Virtual routing and forwarding)
  - Access Control List (in network assurance and verification)
- Brocade VDX 6740 and VDX 6940
  - Neighbor Discovery (LLDP)
  - Layer 2 Switching
  - VLAN
  - Layer 3 Routing (IPv4)
  - VRF
  - Redundancy (VRRP)

### Add a Cisco ASR 9901/9910 Router

You can add Cisco ASR 9901/9910 routers as a data source in VCF Operations for networks.

You must verify the following information:

- The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
- The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.

You must have the following permissions:

- **Read-only user**: VCF Operations for networks requires read-only third-party device credentials (passwords) to access the SSH data. This user needs to have the **Network Operator** role. These passwords are added when configuring the third-party data provider in VCF Operations for networks user interface.
- **Read-only SNMP user**: VCF Operations for networks uses SNMP to collect metrics information from third-party devices (switches, routers, firewalls, and so on). The third-party devices must have SNMP enabled and be accessible from VCF Operations for networks.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **Routers & Switches**, click **Cisco ASR 9901/9910**.
4. In the **Add a New Cisco ASR 9901/9910 Account or Source** page, provide the required information.

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |

   **Note:**
   Ensure you are using the correct supported versions. To know about the supported products and versions, see Supported Products and Versions.

5. Click **Validate**.
6. Optional: To enable a richer data collection, click **Use SNMP (recommended for richer data collection)** check box, and do the following:
   a) From the **SNMP Version** drop-down menu, select an SNMP version. Depending on which version of SNMP you have selected, do the following:

| Option | Action |
|---|---|
| 2c | In the **Community String** text box, enter a community string. |
| 3 | 1. Enter **Username**, Context Name.<br>2. From the **Authentication Type** drop-down, select an authentication type and enter the **Authentication Password**.<br>3. From the **Privacy Type** drop-down, select a privacy type and enter the **Privacy Password**. |

7. Define the polling interval for the configuration data collection. You can set the polling interval from 10 minutes to 7 days.
8. Optional: In the **Nickname** text box, enter a nickname.
9. Optional: In the **Tags (Optional)** key-value pair text box, enter a key and a value.

   A key-value pair could be any text. For example, you can use `Layer Access` as key-value pair where `layer` is the `key` and access is the value.

- To apply the tag in all the associated entities, click the **Apply above tag operations to all associated entities** check box. When you select this option, the tag gets applied to all associated entities. For more details about the associated entities, see Working with Local Tags.

  If you clear the **Apply above tag operations to all associated entities** check box, the assigned tag is removed from all the associated entities.

10. Optional: In the **Notes** text box, add a note if necessary.
11. Click **Submit**.

VCF Operations for networks supports the following features of the Cisco ASR 9901/9910 Router:

- Neighbor Discovery (CDP and LLDP)
- Layer 3 Routing (IPv4)
- VRF (Virtual routing and forwarding)
- Access Control List (in network assurance and verification)
- MPLS - BGP VPNv4

**Add a Cisco ASR 1000 Series Router**

You can add a Cisco ASR 1000 series router as a data source in VCF Operations for networks.

- You must verify the following information:
  - The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
  - The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.
- You must have the following permissions:
  - **SSH user**: VCF Operations for networks requires an SSH account with a privilege level 15 to get the CLI data. You can have a privilege level lower than 15 but the SSH account must be able to execute all show commands that VCF Operations for networks needs like `show running-config view full`.

    For the privilege levels less than 15, you must configure the following command to run the show command `show running-config view full`:

    ```
    privilege exec level <privilege_level> show running-config view full
    ```

    For example: `privilege exec level 10 show running-config view full`.
  - **Read-only SNMP user**: VCF Operations for networks uses SNMP to collect metrics information from third-party devices (switches, routers, firewalls, and so on). The third-party devices must have SNMP enabled and be accessible from VCF Operations for networks.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **Routers & Switches**, click the **Cisco ASR 1000**
4. In the **Add a New Cisco ASR 1000 Account or Source** page provide the required information.

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |

**Note:**

Ensure you are using the correct supported versions. To know about the supported products and versions, see Supported Products and Versions.

5. Click **Validate**.
6. To enable a richer data collection, click **Use SNMP (recommended for richer data collection)** check box, and do the following:
   a) From the **SNMP Version** drop-down menu, select an SNMP version. Depending on which version of SNMP you have selected, do the following:

| Option | Action |
|---|---|
| 2c | In the **Community String** text box, enter a community string. |
| 3 | 1. Enter **Username**, Context Name.<br>2. From the **Authentication Type** drop-down, select an authentication type and enter the **Authentication Password**.<br>3. From the **Privacy Type** drop-down, select a privacy type and enter the **Privacy Password**. |

7. Define the polling interval for the configuration data collection. You can set the polling interval from 10 minutes to 7 days.
   - **Preset** - Select the interval time from the predefined time set.
   - **Custom Interval** - Set a value and select **Minutes**, **Hours**, and **Days**.
   - **Fixed Schedule** - Select the days and set the time to schedule the interval.
8. Optional: In the **Nickname** text box, enter a nickname.
9. Optional: In the **Notes** text box, add a note if necessary.
10. Click **Submit**.

VCF Operations for networks supports the following features of the Cisco ASR 1000 Series Router:

- Neighbor Discovery (CDP and LLDP)
- Layer 3 Routing (IPv4)
- VRF (Virtual routing and forwarding)
- Redundancy (HSRP and VRRP)
- Access Control List (in network assurance and verification)

**Add a Cisco Catalyst Switch**

You can add Cisco Catalyst switches as a data source in VCF Operations for networks.

- You must verify the following information:
  - The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
  - The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.
- You must have the following permissions:
  - **SSH user**: VCF Operations for networks requires an SSH account with a privilege level 15 to get the CLI data. You can have a privilege level lower than 15 but the SSH account must be able to execute all show commands that VCF Operations for networks needs like `show running-config view full`.

    For the privilege levels less than 15, you must configure the following command to run the show command `show running-config view full`:

    ```
    privilege exec level <privilege_level> show running-config view full
    ```

    For example: `privilege exec level 10 show running-config view full`.
  - **Read-only SNMP user**: VCF Operations for networks uses SNMP to collect metrics information from third-party devices (switches, routers, firewalls, and so on). The third-party devices must have SNMP enabled and be accessible from VCF Operations for networks.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **Routers & Switches**, click the Catalyst switch you want to add and provide the required information.

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |

   **Note:**
   Cisco Catalyst 9300 Series switches that are part of the StackWise virtual environment must be added as a single switch in VCF Operations for networks.

4. Click **Validate**.
5. To enable a richer data collection, click **Use SNMP (recommended for richer data collection)** check box, and do the following:
   a) Form the **SNMP Version** drop-down menu, select an SNMP version. Depending on which version of SNMP you have selected, do the following:

| Option | Action |
|---|---|
| 2c | In the **Community String** text box, enter a community string. |
| 3 | 1. Enter **Username**, Context Name.<br>2. From the **Authentication Type** drop-down, select an authentication type and enter the **Authentication Password**.<br>3. From the **Privacy Type** drop-down, select a privacy type and enter the **Privacy Password**. |

6. Define the polling interval for the configuration data collection. You can set the polling interval from 10 minutes to 7 days.
   - **Preset** - Select the interval time from the predefined time set.
   - **Custom Interval** - Set a value and select **Minutes**, **Hours**, and **Days**.
   - **Fixed Schedule** - Select the days and set the time to schedule the interval.
7. Optional: In the **Nickname** text box, enter a nickname.
8. Optional: In the **Tags (Optional)** key-value pair text box, enter a key and a value.
   A key-value pair could be any text. For example, you can use `Layer Access` as key-value pair where `layer` is the `key` and access is the value.
   – To apply the tag in all the associated entities, click the **Apply above tag operations to all associated entities** check box. When you select this option, the tag gets applied to all associated entities. For more details about the associated entities, see Working with Local Tags.
   If you clear the **Apply above tag operations to all associated entities** check box, the assigned tag is removed from all the associated entities.
9. Optional: In the **Notes** text box, add a note if necessary.
10. Click **Submit**.

VCF Operations for networks supports the following features of Cisco Catalyst Switch:

- Neighbor Discovery (CDP and LLDP)
- Layer 2 Switching
- VLAN
- Layer 3 Routing (IPv4)
- VRF (Virtual routing and forwarding)

- Redundancy (HSRP and VRRP)
- Access Control List (in network assurance and verification)
- MPLS - BGP VPNv4

**Add a Cisco ISR 4000 Series Router**

You can add a Cisco ISR 4000 series router as a data source in VCF Operations for networks.

- You must verify the following information:
  - The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
  - The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.
- You must have the following permissions:
  - **SSH user**: VCF Operations for networks requires an SSH account with a privilege level 15 to get the CLI data. You can have a privilege level lower than 15 but the SSH account must be able to execute all show commands that VCF Operations for networks needs like `show running-config view full`.

    For the privilege levels less than 15, you must configure the following command to run the show command `show running-config view full`:

    ```
    privilege exec level <privilege_level> show running-config view full
    ```

    For example: `privilege exec level 10 show running-config view full`.
  - **Read-only SNMP user**: VCF Operations for networks uses SNMP to collect metrics information from third-party devices (switches, routers, firewalls, and so on). The third-party devices must have SNMP enabled and be accessible from VCF Operations for networks.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **Routers & Switches**, click the **Cisco ISR 4000**.
4. In the **Add a New Cisco ISR 4000 Account or Source** page provide the required information.

| Option | Action |
|---|---|
| **Collector (Proxy) VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |
| **Web Proxy (Optional)** | Select a web proxy from the drop-down menu.<br>**Note:**<br>The **Web Proxy (Optional)** is visible only if you have configured a web proxy in VCF Operations for networks. |

**Note:**

Ensure you are using the correct supported versions. To know about the supported products and versions, see Supported Products and Versions.

5. To enable a richer data collection, click **Use SNMP (recommended for richer data collection)** check box, and do the following:
   a) From the **SNMP Version** drop-down menu, select an SNMP version. Depending on which version of SNMP you have selected, do the following:

| Option | Action |
|---|---|
| 2c | In the **Community String** text box, enter a community string. |
| 3 | 1. Enter **Username**, Context Name.<br>2. From the **Authentication Type** drop-down, select an authentication type and enter the **Authentication Password**.<br>3. From the **Privacy Type** drop-down, select a privacy type and enter the **Privacy Password**. |

6. Define the polling interval for the configuration data collection. You can set the polling interval from 10 minutes to 7 days.
   - **Preset** - Select the interval time from the predefined time set.
   - **Custom Interval** - Set a value and select **Minutes**, **Hours**, and **Days**.
   - **Fixed Schedule** - Select the days and set the time to schedule the interval.
7. Click **Validate**.
8. Optional: In the **Nickname** text box, enter a nickname.
9. Optional: In the **Notes** text box, add a note if necessary.
10. Click **Submit**.

VCF Operations for networks supports the following features of the Cisco ISR 4000 Series Router:

- Neighbor Discovery (CDP and LLDP)
- Layer 2 Switching
- VLAN
- Layer 3 Routing (IPv4)
- VRF (Virtual routing and forwarding)
- Redundancy (HSRP and VRRP)
- Access Control List (in network assurance and verification)

**Add a Cisco Nexus Switch**

You can add Cisco Nexus switches as a data source in VCF Operations for networks.

You must verify the following information:

- The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
- The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.

You must have the following permissions:

- **Read-only user**: VCF Operations for networks requires read-only third-party device credentials (passwords) to access the SSH data. This user needs to have the **Network Operator** role. These passwords are added when configuring the third-party data provider in VCF Operations for networks user interface.
- **Read-only SNMP user**: VCF Operations for networks uses SNMP to collect metrics information from third-party devices (switches, routers, firewalls, and so on). The third-party devices must have SNMP enabled and be accessible from VCF Operations for networks.

  **Note:**
  If you have a Nexus device configured with multiple Virtual Device Contexts (VDC) then you need to individually specify VDC associated IP address as separate data sources.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **Routers & Switches**, click the Cisco Nexus switch that you want to add.
4. In the Add a new Cisco N5K Account or Source page, provide the following details:

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |

**Note:**
To know about the supported products and versions, see Supported Products and Versions.

5. Click **Validate**.
6. To enable a richer data collection, click **Use SNMP (recommended for richer data collection)** check box, and do the following:
   a) From the **SNMP Version** drop-down menu, select an SNMP version. Depending on which version of SNMP you have selected, do the following:

| Option | Action |
|---|---|
| 2c | In the **Community String** text box, enter a community string. |
| 3 | 1. Enter **Username**, Context Name.<br>2. From the **Authentication Type** drop-down, select an authentication type and enter the **Authentication Password**.<br>3. From the **Privacy Type** drop-down, select a privacy type and enter the **Privacy Password**. |

7. Define the polling interval for the configuration data collection. You can set the polling interval from 10 minutes to 7 days.
   - **Preset** - Select the interval time from the predefined time set.
   - **Custom Interval** - Set a value and select **Minutes**, **Hours**, and **Days**.
   - **Fixed Schedule** - Select the days and set the time to schedule the interval.
8. Optional: In the **Nickname** text box, enter a nickname.
9. Optional: In the **Tags (Optional)** key-value pair text box, enter a key and a value.
   A key-value pair could be any text. For example, you can use `Layer Access` as key-value pair where `layer` is the `key` and access is the value.
   – To apply the tag in all the associated entities, click the **Apply above tag operations to all associated entities** check box. When you select this option, the tag gets applied to all associated entities. For more details about the associated entities, see Working with Local Tags.
   If you clear the **Apply above tag operations to all associated entities** check box, the assigned tag is removed from all the associated entities.
10. Optional: In the **Notes** text box, add a note if necessary.
11. Click **Submit**.

VCF Operations for networks supports the following features of the Cisco Nexus Switch:

- Neighbor Discovery (CDP and LLDP)
- Layer 2 Switching
- VLAN
- Layer 3 Routing (IPv4)
- VRF (Virtual routing and forwarding)

- Redundancy (HSRP and VRRP)
- Access Control List (in network assurance and verification)
- Fabric Extender (FEX)
- BGP-EVPN (not supported in network assurance and verification)
- Gateway Load-Balancing protocol (GLBP). This feature is not supported in network assurance and verification.

**Add a Dell Switch**

You can add Dell switches as a data source in VCF Operations for networks.

You must verify the following information:

- The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
- The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.

You must have the following permissions:

- **Read-only user**: VCF Operations for networks requires read-only third-party device credentials (passwords) to access the SSH data. This user needs to have the **Network Operator** role. These passwords are added when configuring the third-party data provider in VCF Operations for networks user interface.
- **Read-only SNMP user**: VCF Operations for networks uses SNMP to collect metrics information from third-party devices (switches, routers, firewalls, and so on). The third-party devices must have SNMP enabled and be accessible from VCF Operations for networks.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **Routers & Switches**, click the Dell devices that you want to add and provide the following details:

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |

   **Note:**
   If you want to add Dell S4048, Z9100, or S6000 devices in VCF Operations for networks as a data source and if that Dell device is running on OS 10, then you must select **Dell OS10** as an account or a source type while adding that Dell device as a data source in VCF Operations for networks.

4. Click **Validate**.
5. To enable a richer data collection, click **Use SNMP (recommended for richer data collection)** check box, and do the following:
   a) From the **SNMP Version** drop-down menu, select an SNMP version. Depending on which version of SNMP you have selected, do the following:

| Option | Action |
|---|---|
| 2c | In the **Community String** text box, enter a community string. |
| 3 | 1. Enter **Username**, Context Name.<br>2. From the **Authentication Type** drop-down, select an authentication type and enter the **Authentication Password**.<br>3. From the **Privacy Type** drop-down, select a privacy type and enter the **Privacy Password**. |

6. Define the polling interval for the configuration data collection. You can set the polling interval from 10 minutes to 7 days.
   - **Preset** - Select the interval time from the predefined time set.

- **Custom Interval** - Set a value and select **Minutes**, **Hours**, and **Days**.
- **Fixed Schedule** - Select the days and set the time to schedule the interval.

7. Optional: In the **Nickname** text box, enter a nickname.
8. Optional: In the **Tags (Optional)** key-value pair text box, enter a key and a value.

   A key-value pair could be any text. For example, you can use `Layer Access` as key-value pair where `layer` is the `key` and access is the value.

   – To apply the tag in all the associated entities, click the **Apply above tag operations to all associated entities** check box. When you select this option, the tag gets applied to all associated entities. For more details about the associated entities, see Working with Local Tags.

     If you clear the **Apply above tag operations to all associated entities** check box, the assigned tag is removed from all the associated entities.
9. Optional: In the **Notes** text box, add a note if necessary.
10. Click **Submit**.

For Dell OS10 Switches, Enable telemetry on Dell OS10 switches.

VCF Operations for networks supports the following features of the following Dell switches:

- Dell Froce10 MXL 10
  – Neighbor Discovery (LLDP)
  – Layer 2 Switching
  – VLAN
  – Layer 3 Routing (IPv4)
  – Redundancy (VRRP)
- Dell OS10 (S4048, S5200)
  – Neighbor Discovery (LLDP)
  – Layer 2 Switching
  – VLAN
  – Layer 3 Routing (IPv4)
  – VRF (Virtual routing and forwarding)
  – Redundancy (VRRP)
- Dell PowerConnect 8024
  – Neighbor Discovery (LLDP)
  – Layer 2 Switching
  – VLAN
  – Layer 3 Routing (IPv4)
- Dell S4048 (OS9)
  – Neighbor Discovery (LLDP)
  – Layer 2 Switching
  – VLAN
  – Layer 3 Routing (IPv4)
  – Redundancy (VRRP)
- Dell S6000 and Dell Z9100
  – Neighbor Discovery (LLDP)
  – Layer 2 Switching
  – VLAN
  – Layer 3 Routing (IPv4)
  – VRF
  – Redundancy (VRRP)

### Enabling Telemetry on Dell OS10 Switches

You can enable streaming telemetry on the Dell OS10 switches to integrate buffer statistics and tracking on Dell switches.

After adding Dell OS10 switches as a data source in VCF Operations for networks , you can enable streaming telemetry on the switches to send buffer related information to the VCF Operations for networks collector.

The VCF Operations for networks collector ingests this telemetry information and extracts the `per-port egress buffer peak utilization` metric. This metric can be used to diagnose network congestion at a particular switchport in the network, and to identify application performance issues.

If the peak buffer utilization exceeds a threshold of 10 MB, VCF Operations for networks raises the `Switchport Buffer Threshold Exceeded Event` alert.

Here is an example on how to configure streaming telemetry on a Dell OS10 switch:

```
telemetry
enable
!
destination-group dg03
 destination Aria-Ops-for-Networks Collector IP 50000
!
subscription-profile sp03
 sensor-group buffer 15000
 destination-group dg03
 encoding gpb
 transport grpc no-tls
 source-interface ethernet1/1/1
end
```

For more information about configuring telemetry, see the Dell EMC User Guide.

**Add a Huawei 6800/7800/8800 Series Switch**

VCF Operations for networks supports multiple series of Huawei Cloud Engine switches.

You must verify the following information:

- The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
- The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.

You must have at least Read-Only permissions.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **Routers & Switches**, select **Huawei 6800/7800/8800 Series** and provide the required details.

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |

4. Click **Validate**.
5. To enable a richer data collection, click **Use SNMP (recommended for richer data collection)** check box, and do the following:
   a) From the **SNMP Version** drop-down menu, select an SNMP version. Depending on which version of SNMP you have selected, do the following:

| Option | Action |
|---|---|
| 2c | In the **Community String** text box, enter a community string. |
| 3 | 1. Enter **Username**, Context Name.<br>2. From the **Authentication Type** drop-down, select an authentication type and enter the **Authentication Password**.<br>3. From the **Privacy Type** drop-down, select a privacy type and enter the **Privacy Password**. |

6. Define the polling interval for the configuration data collection. You can set the polling interval from 10 minutes to 7 days.
   - **Preset** - Select the interval time from the predefined time set.
   - **Custom Interval** - Set a value and select **Minutes**, **Hours**, and **Days**.
   - **Fixed Schedule** - Select the days and set the time to schedule the interval.
7. Optional: In the **Nickname** text box, enter a nickname.
8. Optional: In the **Tags (Optional)** key-value pair text box, enter a key and a value.
   A key-value pair could be any text. For example, you can use `Layer Access` as key-value pair where `layer` is the `key` and access is the value.
   – To apply the tag in all the associated entities, click the **Apply above tag operations to all associated entities** check box. When you select this option, the tag gets applied to all associated entities. For more details about the associated entities, see Working with Local Tags.
   If you clear the **Apply above tag operations to all associated entities** check box, the assigned tag is removed from all the associated entities.
9. Optional: In the **Notes** text box, add a note if necessary.
10. Click **Submit**.

You can use the following features of VCF Operations for networks with Huawei devices or routers.

- VM-VM path
- VM Underlay topology
- Huawei Router or Switch dashboard
- Metrics: Switch port and router interface metrics
- Dashboards
  – Huawei Router or Switch
  – Router Interfaces
  – Port Channels
  – Switch Ports
  – Routes
- High availability: supports M-LAG (Multi-Chassis Link Aggregation) and VRRP (Virtual Router Redundancy Protocol)
- Searches
  – VRF (Virtual routing and forwarding) of Huawei
  – Router Interface of Huawei
  – Switch port of Huawei
  – Port Channel of Huawei
  – Routes in Huawei
- Huawei NetStream data monitoring

VCF Operations for networks supports the following features of Huawei routers:

- Neighbor Discovery (LLDP)
- Layer 2 Switching
- VLAN
- Layer 3 Routing (IPv4)
- VRF (Virtual routing and forwarding)
- Redundancy (VRRP)

**Add an HPE 5940/10508 Switch**

You can add HPE 5940/10508 switches as a data source in VCF Operations for networks.

You must verify the following information:

- The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
- The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.

You must have the following permissions:

- **Read-only user**: VCF Operations for networks requires read-only third-party device credentials (passwords) to access the SSH data. This user needs to have the **Network Operator** role. These passwords are added when configuring the third-party data provider in VCF Operations for networks user interface.
- **Read-only SNMP user**: VCF Operations for networks uses SNMP to collect metrics information from third-party devices (switches, routers, firewalls, and so on). The third-party devices must have SNMP enabled and be accessible from VCF Operations for networks.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **Routers & Switches**, click **HPE 5940/10508**.
4. In the **Add a New HPE 5940/10508 Account or Source** page, provide the required information.

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |

5. Click **Validate**.
6. To enable a richer data collection, click **Use SNMP (recommended for richer data collection)** check box, and do the following:
   a) From the **SNMP Version** drop-down menu, select an SNMP version. Depending on which version of SNMP you have selected, do the following:

| Option | Action |
|---|---|
| 2c | In the **Community String** text box, enter a community string. |
| 3 | 1. Enter **Username**, Context Name.<br>2. From the **Authentication Type** drop-down, select an authentication type and enter the **Authentication Password**.<br>3. From the **Privacy Type** drop-down, select a privacy type and enter the **Privacy Password**. |

7. Define the polling interval for the configuration data collection. You can set the polling interval from 10 minutes to 7 days.
   - **Preset** - Select the interval time from the predefined time set.
   - **Custom Interval** - Set a value and select **Minutes**, **Hours**, and **Days**.

- **Fixed Schedule** - Select the days and set the time to schedule the interval.

8. Optional: In the **Nickname** text box, enter a nickname.
9. Optional: In the **Tags (Optional)** key-value pair text box, enter a key and a value.

   A key-value pair could be any text. For example, you can use `Layer Access` as key-value pair where `layer` is the `key` and access is the value.

   – To apply the tag in all the associated entities, click the **Apply above tag operations to all associated entities** check box. When you select this option, the tag gets applied to all associated entities. For more details about the associated entities, see Working with Local Tags.

   If you clear the **Apply above tag operations to all associated entities** check box, the assigned tag is removed from all the associated entities.

10. Optional: In the **Notes** text box, add a note if necessary.
11. Click **Submit**.

VCF Operations for networks supports the following features of the HPE 5940/10508 switch:

- Neighbor Discovery (LLDP)
- Layer 3 Routing (IPv4)
- VRF (Virtual routing and forwarding)

**Add a Juniper Switch**

You can add Juniper switches as a data source in VCF Operations for networks.

You must verify the following information:

- The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
- The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under the Routers & Switches, select **Juniper EX/QFX**.
4. In the Add a new Juniper EX/QFX Account or Source page, provide the following details:

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |

5. Click **Validate**.
6. To enable a richer data collection, click **Use SNMP (recommended for richer data collection)** check box, and do the following:

   a) From the **SNMP Version** drop-down menu, select an SNMP version. Depending on which version of SNMP you have selected, do the following:

| Option | Action |
|---|---|
| 2c | In the **Community String** text box, enter a community string. |
| 3 | 1. Enter **Username**, Context Name.<br>2. From the **Authentication Type** drop-down, select an authentication type and enter the **Authentication Password**.<br>3. From the **Privacy Type** drop-down, select a privacy type and enter the **Privacy Password**. |

7. Define the polling interval for the configuration data collection. You can set the polling interval from 10 minutes to 7 days.
   - **Preset** - Select the interval time from the predefined time set.
   - **Custom Interval** - Set a value and select **Minutes**, **Hours**, and **Days**.
   - **Fixed Schedule** - Select the days and set the time to schedule the interval.
8. Optional: In the **Nickname** text box, enter a nickname.
9. Optional: In the **Tags (Optional)** key-value pair text box, enter a key and a value.

   A key-value pair could be any text. For example, you can use `Layer Access` as key-value pair where `layer` is the `key` and access is the value.

   – To apply the tag in all the associated entities, click the **Apply above tag operations to all associated entities** check box. When you select this option, the tag gets applied to all associated entities. For more details about the associated entities, see Working with Local Tags.

   If you clear the **Apply above tag operations to all associated entities** check box, the assigned tag is removed from all the associated entities.
10. Optional: In the **Notes** text box, add a note if necessary.
11. Click **Submit**.

VCF Operations for networks supports the following features of Juniper switch:

- Neighbor Discovery (LLDP)
- Layer 2 Switching
- VLAN
- STP (Spanning Tree Protocol ). This feature is supported only in network assurance and verification.
- Layer 3 Routing (IPv4)
- VRF (Virtual routing and forwarding)
- Redundancy (VRRP). This feature is supported only in network assurance and verification.
- Access Control List (in network assurance and verification)
- EVPN (Ethernet VPN). This feature is not supported in network assurance and verification.

**Add a Mellanox Switch**

You can add SN2010, SN2100, SN2410, SN2700, SN3420, SN3700c, and SN4600c Mellanox switches as a data source in VCF Operations for networks.

You must verify the following information:

- The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
- The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **Routers & Switches**, click **Mellanox SN2100/SN2410** .
4. In the **Add a New Mellanox Account or Source** page, provide the required information.

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |

5. Click **Validate**.
6. To enable a richer data collection, click **Use SNMP (recommended for richer data collection)** check box, and do the following:
   a) From the **SNMP Version** drop-down menu, select an SNMP version. Depending on which version of SNMP you have selected, do the following:

| Option | Action |
|---|---|
| 2c | In the **Community String** text box, enter a community string. |
| 3 | 1. Enter **Username**, Context Name.<br>2. From the **Authentication Type** drop-down, select an authentication type and enter the **Authentication Password**.<br>3. From the **Privacy Type** drop-down, select a privacy type and enter the **Privacy Password**. |

7. Define the polling interval for the configuration data collection. You can set the polling interval from 10 minutes to 7 days.
   - **Preset** - Select the interval time from the predefined time set.
   - **Custom Interval** - Set a value and select **Minutes**, **Hours**, and **Days**.
   - **Fixed Schedule** - Select the days and set the time to schedule the interval.
8. Optional: In the **Nickname** text box, enter a nickname.
9. Optional: In the **Tags (Optional)** key-value pair text box, enter a key and a value.
   A key-value pair could be any text. For example, you can use `Layer Access` as key-value pair where `layer` is the `key` and access is the value.
   – To apply the tag in all the associated entities, click the **Apply above tag operations to all associated entities** check box. When you select this option, the tag gets applied to all associated entities. For more details about the associated entities, see Working with Local Tags.
   If you clear the **Apply above tag operations to all associated entities** check box, the assigned tag is removed from all the associated entities.
10. Optional: In the **Notes** text box, add a note if necessary.
11. Click **Submit**.

VCF Operations for networks supports the following features of Mellanox switch:

- Neighbor Discovery (LLDP)
- Layer 2 Switching
- VLAN
- Layer 3 Routing (IPv4)
- VRF (Virtual routing and forwarding)
- M-LAG (Multi-Chassis Link Aggregation), where the Peer IP is shown
- MAGP (Multi-active Gateway Protocol)

**Add a New Generic Router or Switch**
You can add a generic router or switch to VCF Operations for networks by uploading its device configuration file. VCF Operations for networks uses the information in the device configuration file to provide insight for the router or switch and show the router or switch in the VM-VM path.

- You must verify that the data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.
- You must create the device configuration file in `.zip` format using the SDK provided by the VCF Operations for networks. To create a device configuration file, see https://github.com/vmware/network-insight-sdk-generic-datasources.
  The device configuration file contains information about entities, such as router interfaces, routes, switch ports, VRFs, switch device information, and so on.

**Note:**
You must upload a new configuration file whenever there is a configuration change. VCF Operations for networks does not automatically collect data from the generic router or switches.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under Routers & Switches, click **Generic Routers & Switches**.
4. On the **Add a New Generic Router/Switch** page, modify the required information.

| Option | Action |
|---|---|
| **Collector VM** | Select a Collector VM from the drop-down menu. |
| **Device Configuration File** | Select and upload the configuration file `(.zip)` created using the SDK.<br>**Note:**<br>After you upload a device configuration file in VCF Operations for networks, you cannot modify the information of the uploaded device configuration file. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |

5. Click **Validate**.
6. Optional: In the **Nickname** text box, enter a nickname.
7. Optional: In the **Tags (Optional)** key-value pair text box, enter a key and a value.

   A key-value pair could be any text. For example, you can use `Layer Access` as key-value pair where `layer` is the `key` and access is the value.

   – To apply the tag in all the associated entities, click the **Apply above tag operations to all associated entities** check box. When you select this option, the tag gets applied to all associated entities. For more details about the associated entities, see Working with Local Tags.

   If you clear the **Apply above tag operations to all associated entities** check box, the assigned tag is removed from all the associated entities.
8. Optional: In the **Notes** text box, add a note if necessary.
9. Click **Submit**.

### Edit a Generic Router or Switch

In VCF Operations for networks, you can modify the configuration of an existing generic router or switch by uploading new configuration file.

Create a device configuration file in `.zip` format using the SDK provided by the VCF Operations for networks. A device configuration file contains information about entities like Router Interfaces, Routes, Switch Ports, VRFs, Switch Device Information, and so on. To create a device configuration file, see https://github.com/vmware/network-insight-sdk-generic-datasources.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click the three dots icon next to the generic router or the generic switch data source you want to edit and click **Edit**.
3. Click **Replace File** and upload the new device configuration file.
4. Optional: To view your uploaded device configuration file, click **Upload History**.

   You can view, download, and delete the last five uploaded device configuration files.

5. Click **Validate**.
6. Optional: In the **Nickname** text box, change the nickname.
7. Click **Submit**.

## Supported Other Data Sources

VCF Operations for networks supports other data sources like Cisco ACI.

### Add Cisco ACI

You can add Cisco ACI (Application Centric Infrestructure) as a data source in VCF Operations for networks. This feature is available only for the enterprise license users.

- You must verify the following information:
  - The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
  - The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.
- To connect to the APIC controller REST API over HTTPS, you must have access to all the tenants and have the read-only privilege.
- For SNMP, you must have the read-only permission.
- To authenticate Cisco ACI, you must have a local user role with the following privileges:
  - Security Domain: all
  - Role: admin
  - Access: read

  For details on how to create a local user in Cisco ACI, see the Access, Authentication, and Accounting section in the *Cisco APIC Security Configuration Guide*.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **Others**, select **Cisco ACI**.
4. In the **Add a new Cisco ACI Account or Source** page, provide the following details:

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Provide the IP address of an APIC controller in the cluster.<br>**Note:**<br>You do not have to add the individual switches in the ACI fabric. |
| **Username** | Enter the user name.<br>**Note:**<br>• If the user is a domain user, then you must add `apic:` before the user name. For example, if the user name is **user1** and if the user belongs to the domain **domain1** then specify the user name as `apic:domain1\\user1`. The domain name is case-sensitive.<br>• If you want to use a local user account when default authentication method is set to TACACS, then you must add `apic#fallback\\` before the user name. For example, if the local user name is user2, then specify the user name as `apic#fallback\\user2`. |
| **Password** | Enter the password. |

5. Click **Validate**.
6. To enable a richer data collection, click **Use SNMP (recommended for richer data collection)** check box, and do the following:
   a) From the **SNMP Version** drop-down menu, select an SNMP version. Depending on which version of SNMP you have selected, do the following:

| Option | Action |
|---|---|
| 2c | In the **Community String** text box, enter a community string. |
| 3 | 1. Enter **Username**, Context Name.<br>2. From the **Authentication Type** drop-down, select an authentication type and enter the **Authentication Password**.<br>3. From the **Privacy Type** drop-down, select a privacy type and enter the **Privacy Password**. |

7. Define the polling interval for the configuration data collection. You can set the polling interval from 10 minutes to 7 days.
   - **Preset** - Select the interval time from the predefined time set.
   - **Custom Interval** - Set a value and select **Minutes**, **Hours**, and **Days**.
   - **Fixed Schedule** - Select the days and set the time to schedule the interval.
8. Optional: In the **Nickname** text box, enter a nickname.
9. Optional: In the **Tags (Optional)** key-value pair text box, enter a key and a value.
   A key-value pair could be any text. For example, you can use `Layer Access` as key-value pair where `layer` is the `key` and access is the value.
   – To apply the tag in all the associated entities, click the **Apply above tag operations to all associated entities** check box. When you select this option, the tag gets applied to all associated entities. For more details about the associated entities, see Working with Local Tags.
   If you clear the **Apply above tag operations to all associated entities** check box, the assigned tag is removed from all the associated entities.
10. Optional: In the **Notes** text box, add a note if necessary.
11. Click **Submit**.

Cisco ACI is added as a data source in VCF Operations for networks.

**Note:**

VCF Operations for networks supports the following features of Cisco ACI:

- Neighbor discovery (CDP and LLDP)
- Endpoint group (EPG)
- L3Out EPG
- EPG gateway
- Bridge domain
- VRF
- IPv4 routing
- Contract (in network assurance and verification)

The following features are not supported:

- Fabric extender (FEX)
- IPv6
- Layer 3 EVPN
- Multi-pod
- Multi-site

### Supported Cisco ACI Contracts

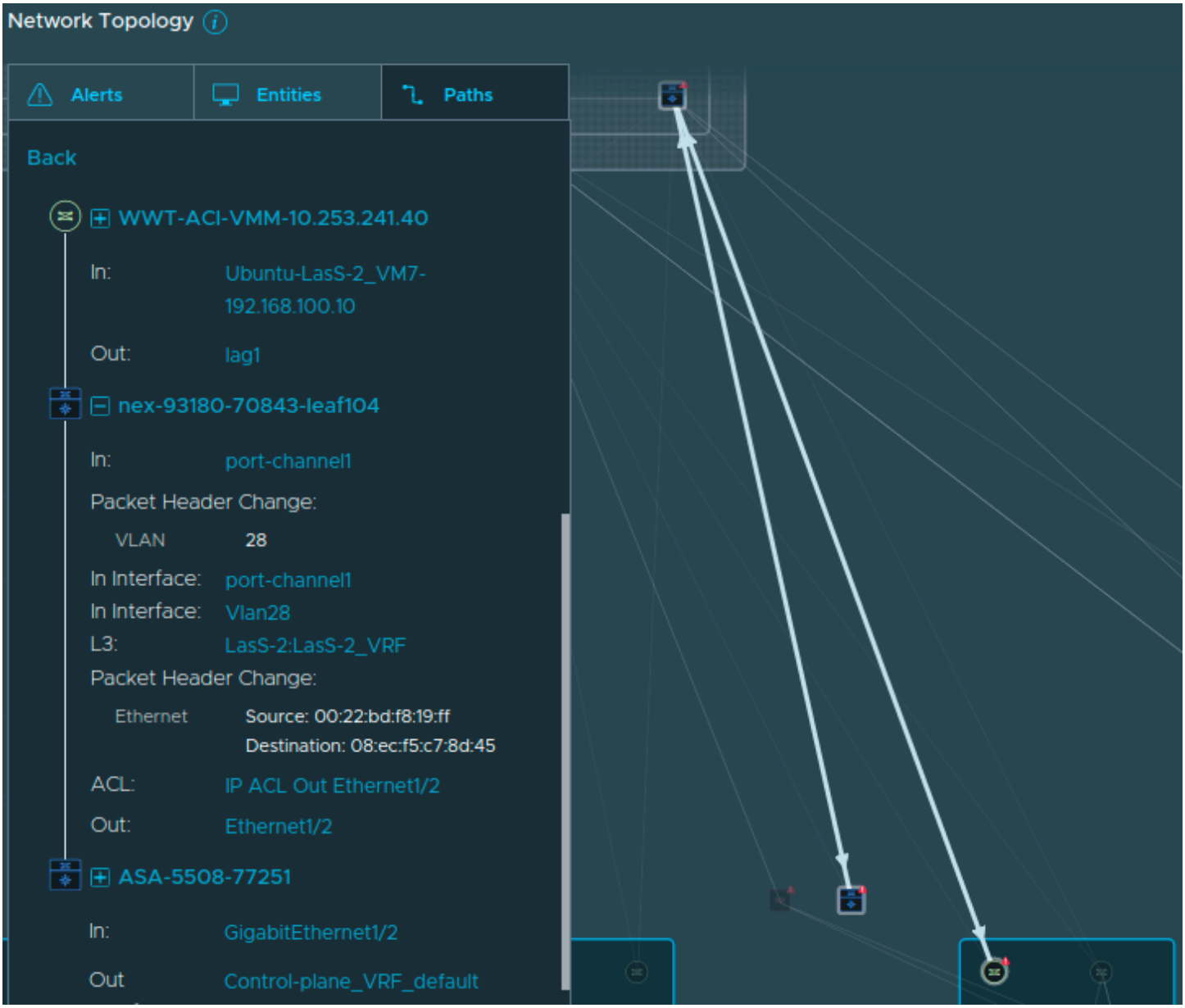
You use Cisco ACI contracts to control traffic flow within the Cisco ACI fabric between EPGs. These contracts are configured between EPGs, or between EPGs and L3out.

When you add a Cisco ACI setup as a data source, VCF Operations for networks supports the following Cisco ACI contracts:

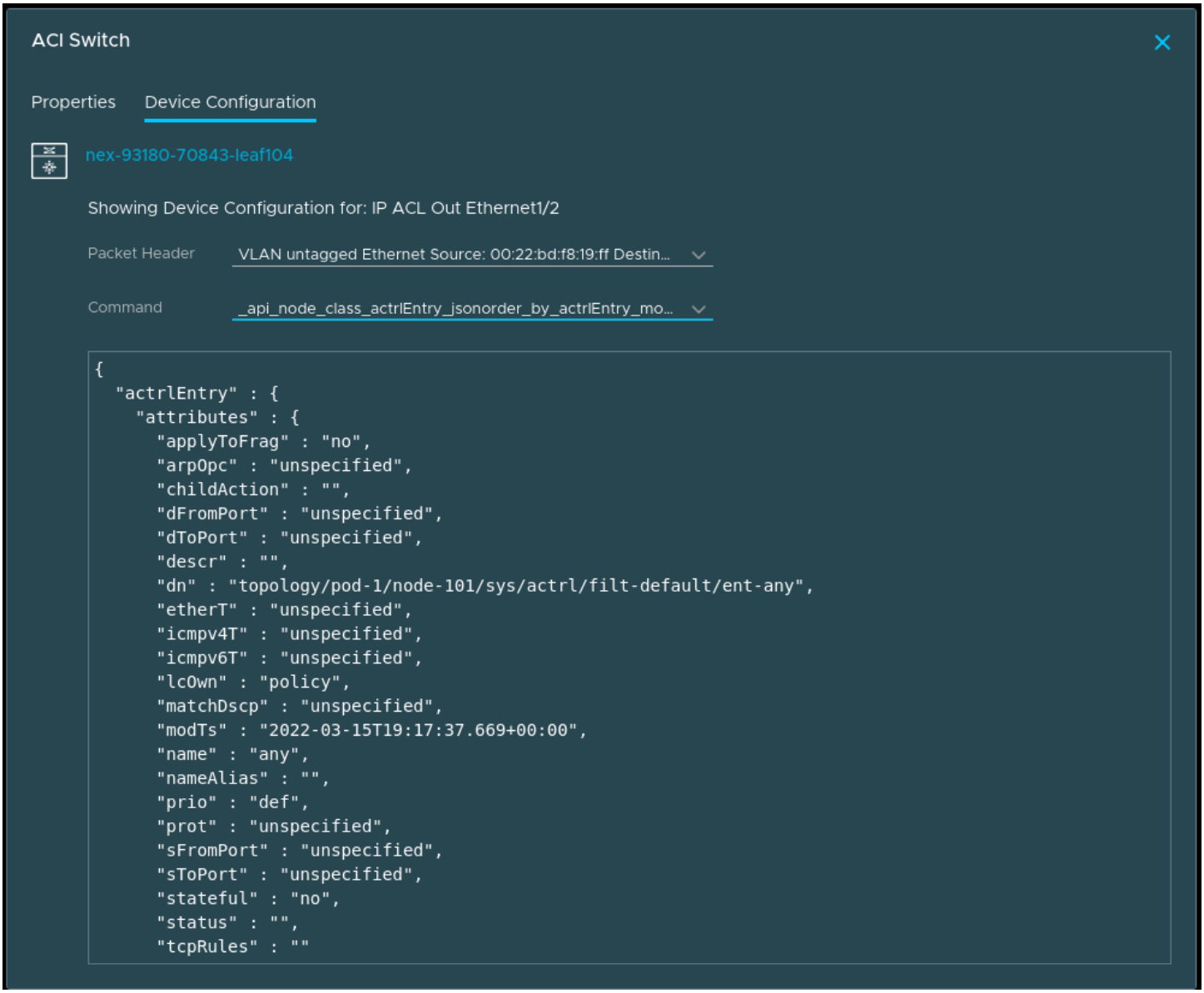
- Intra-EPG Contract
- Intra-EPG Isolation
- VRF Unenforced Mode
- Intra-VRF Contract
- Inter-VRF Contract (Scope - Global)
- Inter-VRF Contract (Scope - Tenant)
- Intra-App Profile Contract
- Inter-App Profile Contract
- Contract Inheritance
- Taboo Contract
- vzAny Contract
- East-West Traffic via Common Tenant
- Preferred Group
- Application EPG to L3Out EPG Contract in VRF
- Application EPG to L3Out EPG Contract between VRF
- Intra Ext-EPG Contract
- Intra Ext-EPG Isolation
- L3Out to L3Out Contract
- Contract between L3Out in common tenant and another VM in different tenant

You see these contracts applied on the Network Map when you search for the traffic traversing Cisco ACI switches.

An example of an access control contract on Cisco ACI:




An example of the device configuration of Cisco ACI:




The following contracts are not supported in VCF Operations for networks:

- Out-of-Band Contract
- Policy Based Redirect
- Stateful Filter in Contract

To learn more about Cisco ACI contracts, see the Cisco ACI Contract Guide.

## Supported Physical Flow Collectors

VCF Operations for networks supports addition of physical flow collectors and enables configuration of switches to push sFlows and NetFlows records to the collector.

### Add a Physical Flow Collector for NetFlow and sFlow

You can add a physical flow collector and configure the switches to send sFlow and NetFlow records to the collector.

The collector VM that is used for NetFlow or sFlow is a dedicated collector. It cannot be used for any other data source.

**Note:**

- Do not use a physical flow collector to collect VeloCloud Edge flow records.

- VCF Operations for networks listens to sFlow records on port 6343 of the collector appliance. Therefore, make sure to send sFlow records to port 6343 of the collector.

If any other data source is added to the collector, it is not available as a Physical Flow Collector for sFlow and NetFlow.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **Flows**, select **Physical Flow Collector (Netflow, sFlow)**.
   sFlows are accepted only on a physical collector.
4. On **Add a new Physical Account or Source** page, click **Add Collector VM**.
5. The **Add a new Operations for Networks Data Collector virtual appliance** window is displayed. Click **Copy** to copy the shared secret to import the collector service ova in your VMware vCenter.
6. Click **Done**.
7. Optional: In the **Nickname** text box, enter a nickname.
8. Optional: In the **Notes** text box, add a note if necessary.
9. Click **Submit**.

**Note:**
VCF Operations for networks collects the packet samples for sFlow and therefore cannot show the complete metrics for the flows.

Configure the switches to push the flows to the Physical Flow Collector. For more information, see Flow Support for Physical Servers topic.

- Define the destination (Collector IP address that you added in VCF Operations for networks).
- Set the port for the flow collector.
- Assign poll interval.

**Note:**
The procedure to configure the switch depends on the type of switch that you want to configure. For more information, see the respective switch documentation.

## Supported Log Servers

VCF Operations for networks supports addition of VCF Operations for logs. This feature is available only for the enterprise license users.

**Configure VCF Operations for logs**
You must add the VCF Operations for logs data source in VCF Operations for networks to get the dropped flow notifications for the Check Point and Palo Alto firewalls.

To configure the VCF Operations for logs dat source, you must do the following:

- Add the VCF Operations for logs data source
- Install Content Pack and Enable Alerts
- Configure Webhook
- Update Syslog server for Check Point and Palo Alto firewalls

### Add VCF Operations for logs

VCF Operations for logs collects syslog messages from Check Point and Palo Alto firewalls. After you add the VCF Operations for logs data source in VCF Operations for networks, the dropped flow notifications for these firewall devices are shown in VCF Operations for networks.

Ensure that you have the API user permissions to install, configure, and manage the content pack.

Install the content pack and enable alerts.

**Note:**

VCF Operations for networks supports addition of VCF Operations for logs 8.0 to 9.0. Make sure you add the supported version of VCF Operations for logs as a data source in VCF Operations for networks. For VCF Operations for logs 9.1 support, please see Viewing Dropped Flow Alert for Check Point and Palo Alto.

When the firewall devices are configured to send syslog messages to VCF Operations for logs and when any policy is affected on these devices, VCF Operations for logs filters deny/drop action messages and sends notification to VCF Operations for networks. VCF Operations for networks consumes these notifications and creates dropped flow events with firewall and flow details.

1. Create or reuse a VCF Operations for logs user with access to the APIs of VCF Operations for logs.
2. Go to **Settings** > **Accounts and Data Sources**.
3. Click **Add Source**.
4. Click **Operations for Logs** under **Log Servers**.
5. On the **Add a New Operations for Logs Server Account or Source** page, click **Instructions** next to the page title. A pop-up window appears providing the prerequisites for adding the VCF Operations for logs data source and the instructions to enable the Webhook URL on VCF Operations for logs.
6. Enter the required details.

| Name | Description |
|---|---|
| Collector VM | Select the IP address of the data collector that you have deployed for the data collection process. |
| IP Address / FQDN | Enter the IP address or the FQDN of the data source. |
| User Name | Enter the user name you want to use for a particular data source. |
| Password | Enter the password for the data source. |
| Authentication Provider | Select the respective authentication provider for the credentials that you have provided. |

7. After the data source is created, a pop-up window is displayed providing the webhook URL and the steps that have to be performed to enable this URL on VCF Operations for logs. Copy the Webhook URL.

   **Note:**
   The Webhook URL, which is generated after the addition of the data source, is used in VCF Operations for logs.
8. Log in to VCF Operations for logs with the credentials that were used for adding this data source. Enable alerts in the VCF Operations for logs application and select the preconfigured Webhook. To ensure that the integration is successful, click **Send Test Alert**.

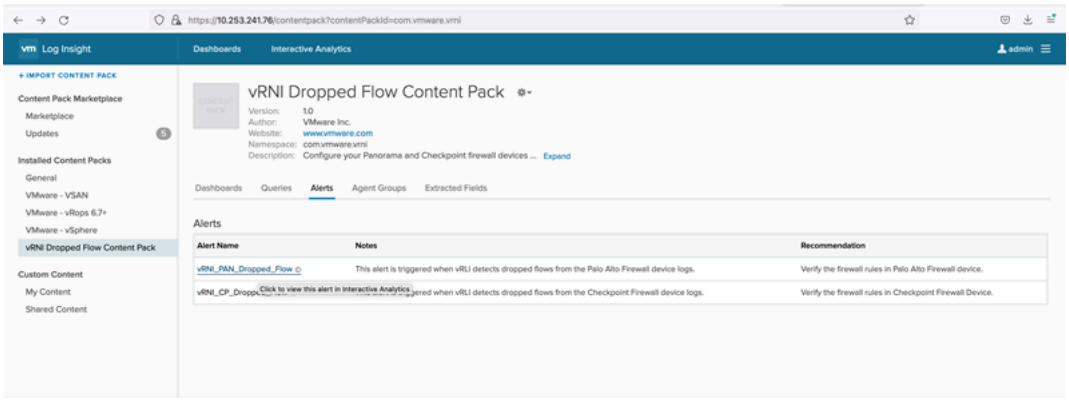
### Install Content Pack and Enable Alerts

Install the content pack before you add VCF Operations for logs data source in VCF Operations for networks.

1. Log in to VCF Operations for logs and from the left pane, click **Content Packs** > **Import Content Pack**.
2. On the **Import Content Pack** page, browse for **Operations for Networks Dropped Flow Content Pack** and click **Import**.

   **Note:**
   VCF Operations for networks supports VCF Operations for logs version 8.0 to 8.8. Make sure you add the supported version of VCF Operations for logs as a data source in VCF Operations for networks.

- After you have installed the content pack and have added the VCF Operations for logs data source in VCF Operations for networks, you can enable the alerts to send information.
- If you have installed the content pack for the first time, view and enable alerts in the **Operations for Networks Dropped Flow Content Pack** page. Click **Send Test Alert** to ensure that the integration is successful.




Configure Webhook

You can configure the Webhook serve in VCF Operations for logs.

1. Log in to VCF Operations for logs and from the left pane, select **Webhook** under **Alerts**.
2. On the **Webhook** page, click **New Webhook** to add a new webhook.
3. Enter the required details.

| Name | VCF Operations for networks Dropped Flow Alert |
|---|---|
| Endpoint | Custom |
| Webhook URL | Copied from Operations for Networks add source page. For example, `https://webhook_user:**********@vrni-appliance/webhooks/loginsight/alert` |
| Content Type | JSON |
| Action | Post |
| Webhook Payload | `{`<br>`"AlertType": 1,`<br>`"AlertName": "${AlertName}",`<br>`"SearchPeriod": ${SearchPeriod},`<br>`"HitOperator": ${NumHits},`<br>`"messages": ${messages}`<br>`}`<br>**Note:**<br>Ensure to add the Webhook payload exactly as given above. |

4. Update the `/etc/hosts` file to connect VCF Operations for networks with the Webhook server.

   **Note:**
   You must have the `sudo/root` permission to update the `/etc/hosts` file.

   For example,

```
ssh root@10.253.241.105
VMware Cloud Foundation Operations for Logs
root@10.253.241.105's password:
Last login: Tue Jun 14 03:31:14 UTC 2022 from 172.31.1.73 on pts/0
Last login: Tue Jun 14 09:15:08 2022 from 172.31.1.46
root@vRLI-8 [ ~ ]# cat /etc/hosts
# Begin /etc/hosts (network card version)
# End /etc/hosts (network card version)
# VAMI_EDIT_BEGIN
# Generated by Studio VAMI service. Do not modify manually.
127.0.0.1   vRLI-8 localhost
```

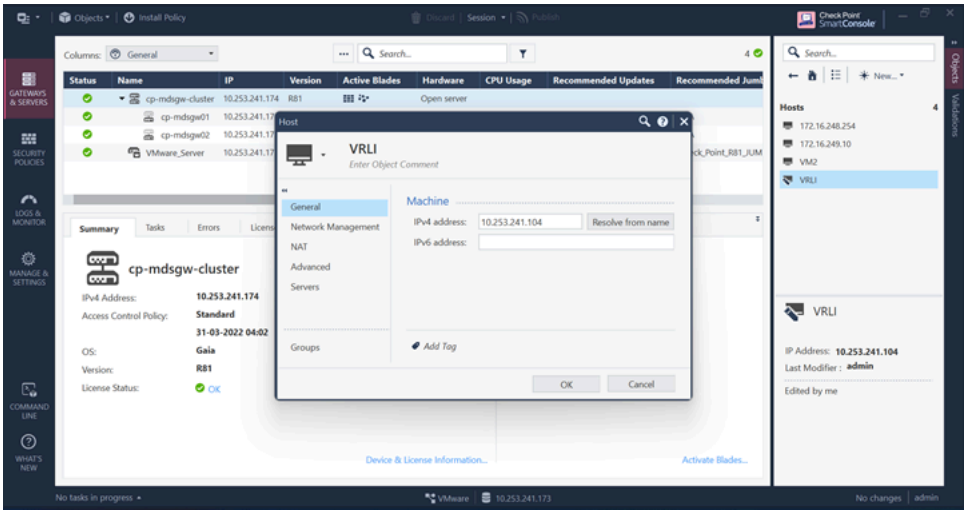
```
10.253.241.206 vrni-appliance
::1    vRLI-8 localhost ipv6-localhost ipv6-loopback
# VAMI_EDIT_END
root@vRLI-8 [ ~ ]#
```

Update Syslog server for Check Point and Palo Alto firewalls

To configure new syslog server for a Check Point firewall, refer to the Check Point page for Working with Syslog Servers.

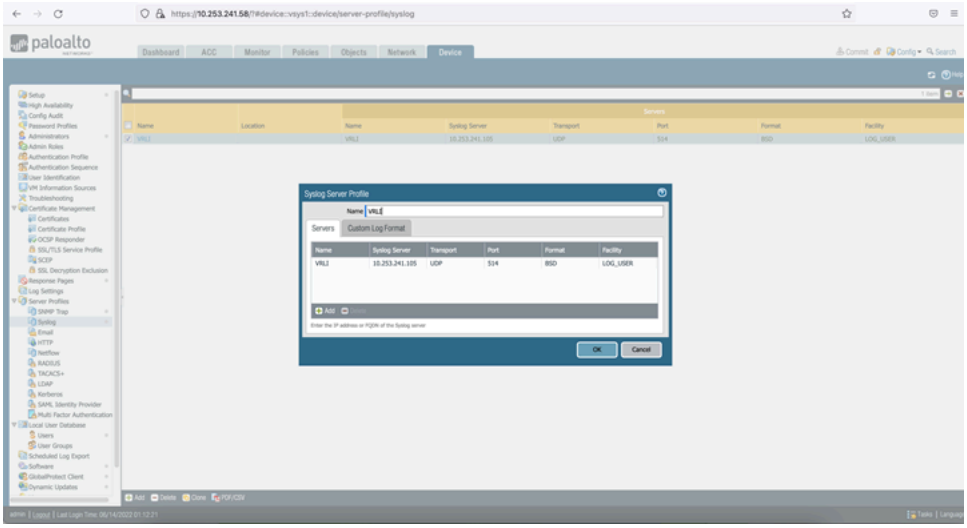
To update the existing syslog server:

1. Open **Smart Console** and click **Object Categories** > **Network Object** > **Hosts**.
2. Select and update the appropriate host.




To configure VCF Operations for logs as the syslog server for a Palo Alto firewall:

1. Login to the Palo Alto web user interface and click **Device** > **Server Profile** > **Syslog**.
2. Update the existing server or click **Add** to add a new server.
3. After adding or updating the server, save the changes.




## Supported DNS Data Providers

VCF Operations for networks supports DNS providers like Infoblox.

### Add Infoblox

VCF Operations for networks allows you to add Infoblox Grid as a DNS data provider in VCF Operations for networks.

You must verify the following information:

- The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
- The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.

Infoblox DNS offers an advanced solution to manage and control DNS. It uses Infoblox Grid to ensure that the DNS is highly available throughout the network. The DNS data from Infoblox is used only for enriching the flows where either the source or the destination IP addresses are associated with the physical devices.

The Infoblox DNS data co-exists with the DNS data that is imported by using CSV.

Infoblox does not need a dedicated collector. If you have configured an Infoblox DNS data source on a collector, you can use the same collector to configure other data sources on it.

**Considerations**

- VCF Operations for networks supports only single-grid mode for Infoblox in the current release.
- Only A Records are supported in the current release. Shared A Records are not supported currently.
- The DNS enrichment is supported only for the IP addresses that are marked as physical in the current release.
- If there are multiple FQDNs for a single physical IP address, all FQDNs are returned.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **DNS**, click **Infoblox**.
4. On the Add a new Infoblox Account or Source page, provide the following details:

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |

5. Click **Validate**.

   **Note:**
   Ensure that you have the `API Privilege` to access the Infloblox APIs.

6. Optional: In the **Nickname** text box, enter a nickname.
7. Optional: In the **Notes** text box, add a note if necessary.
8. Click **Submit**.

The Infoblox DNS data source is added to the environment.

## Supported Load Balancers

VCF Operations for networks supports addition of load balancers as data source.

**Add F5 BIG-IP**

VCF Operations for networks supports the router and load balancer functionalities of F5 BIG-IP. The features like VM-VM path, high availability, VRFs, Routes, Router Interfaces, Switch Ports, Port Channels, Switch Port metrics, VRF Dashboard, Switch Dashboard, Router dashboard, and Assurance and Verification (Load Balancing and Source Network Address Translation in path search, and intents) are supported. For search on the F5 BIG IP entities, use the query string `F5 BIG-IP Data Source`. VCF Operations for networks does not support LLDP neighbors or the neighboring devices in the VM-VM path.

- You must verify the following information:
  - The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.

  - The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.
- You must have the following permissions:
  - The **Guest** role or **Read Only** permissions with **access to all partitions**.
  - Access to F5 BIG-IP REST API.

    **Note:**

    - VCF Operations for networks does not require **root** user privilege.

    **Note:**
    VCF Operations for networks supports only Local Traffic Module (LTM). Enabling other modules can prevent VCF Operations for networks to collect data from the device.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **Load Balancers**, click **F5 BIG-IP**.
4. In the **Add a New F5 BIG-IP Account or Source** page, provide the required information.

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |

5. Click **Validate**.
6. Optional: To enable a richer data collection, click the **Use SNMP (recommended for richer data collection)** check box, and do the following:
   a) From the **SNMP Version** drop-down menu, select an SNMP version. Depending on which version of SNMP you have selected, do the following:

| Option | Action |
|---|---|
| 2c | In the **Community String** text box, enter a community string. |
| 3 | 1. Enter **Username**, Context Name.<br>2. From the **Authentication Type** drop-down, select an authentication type and enter the **Authentication Password**.<br>3. From the **Privacy Type** drop-down, select a privacy type and enter the **Privacy Password**. |

**Note:**
Ensure that you configure SNMP on the F5 BIG-IP UI console.

1. Log in to F5.
2. Navigate to **System** > **SNMP**.
3. Go to **SNMP** > **Agent** > **Access (v1, v2c)**.
4. Enter the community string.
5. Enter the source IP address.
6. Select the **Read Only** access.
7. Click **Finished**.

7. Optional: In the **Nickname** text box, enter a nickname.
8. Optional: In the **Tags (Optional)** key-value pair text box, enter a key and a value.

   A key-value pair could be any text. For example, you can use `Layer Access` as key-value pair where `layer` is the `key` and access is the value.

  - To apply the tag in all the associated entities, click the **Apply above tag operations to all associated entities** check box. When you select this option, the tag gets applied to all associated entities. For more details about the associated entities, see Working with Local Tags.

    If you clear the **Apply above tag operations to all associated entities** check box, the assigned tag is removed from all the associated entities.

9. Optional: In the **Notes** text box, add a note if necessary.
10. Click **Submit**.

Send flow records from F5 to VCF Operations for networks

**Note:**
To send sFLow data to VCF Operations for networks, you must follow the recommendations mentioned in https://support.f5.com/csp/article/K15193.

VCF Operations for networks supports the following features of F5 BIG-IP:

- Neighbor Discovery (LLDP)
- VLAN
- STP (Spanning Tree Protocol ). This feature is supported only in network assurance and verification.
- Layer 3 Routing (IPv4)
- Route Domain
- Virtual Server
- Load Balance Pool
- NAT (Network Address Translation). This feature is supported only in network assurance and verification.
- Partition

**Add VMware Avi Load Balancer**
If you are an enterprise license user, you can add Avi Load Balancer as a data source in VCF Operations for networks.

- You must verify the following information:
  - The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
  - The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.
- You must have access to Avi Load Balancer APIs.
- You must you must have at least **Read** access to the following groups and resources:

| Group | Resource |
|---|---|
| Application | • Virtual Service<br>• Pool<br>• Pool Group |
| Operations | • Alert Config<br>• Alert |
| Infrastructure | • Cloud<br>• Service Engine<br>• Service Engine Group<br>• Virtual Routing Framework (VRF) Context |
| Administration | • System Settings<br>• Controller |
| Accounts | • Users<br>• Roles<br>• Tenant |
| Profiles | Analytics Profiles |

In Avi Load Balancer, the authorization of a user account is decided based the combination of roles and tenants. For more information, see the Avi Load Balancer documentation.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under **Load Balancers**, select **Avi Load Balancer**.
4. On the Add a new Avi Load Balancer Account or Source page, provide the following details:

| Option | Action |
|---|---|
| **Collector VM** | Select a collector VM from the drop-down menu. |
| **IP Address/FQDN** | Enter the IP address or the FQDN details of the Avi Load Balancer Cluster or Controller. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |

5. Click **Validate**.
6. Optional: In the **Nickname** text box, enter a nickname.
7. Optional: In the **Tags (Optional)** key-value pair text box, enter a key and a value.

   A key-value pair could be any text. For example, you can use `Layer Access` as key-value pair where `layer` is the `key` and access is the value.

   – To apply the tag in all the associated entities, click the **Apply above tag operations to all associated entities** check box. When you select this option, the tag gets applied to all associated entities. For more details about the associated entities, see Working with Local Tags.

     If you clear the **Apply above tag operations to all associated entities** check box, the assigned tag is removed from all the associated entities.
8. Optional: In the **Notes** text box, add a note if necessary.
9. Click **Submit**.

After adding the data source, it takes some time to display the Avi Load Balancer details such as the integration with NSX.

To perform a search on Avi Load Balancer entities, use the query string `NSX-ALB Data Source`.

VCF Operations for networks supports the following features of Avi Load Balancer:

- Cluster
- Controller
- Tenant
- Cloud
- Service Engine
- Service Engine Group
- Virtual Service
- Pool
- Pool Member
- Pool Group
- Analytics Profile
- VRF (Virtual routing and forwarding)
- Alert

## Supported CMDBs

VCF Operations for networks supports addition of CMDBs like ServiceNow.

## CMDB Concepts

Basically, a CMDB consists of:

- Configuration item: An entity or a component in a system. Example, a computer, a switch, a service, an application, a server, or a VM.
- Relationship: a link or a type of communication between configuration items. Example: depends on, runs on, exchanges data.

Each configuration item has a defined schema.

- Configuration Item Class: Each configuration item must be associated with a class, which defines its properties.
- Relationship class: Defines the type of relation between configuration items.

You can extend both the classes to add additional properties or customize the properties.

ServiceNow supports application service, which is a set of interconnected applications and hosts that offer a service. ServiceNow allows you to create an application service manually, by using an API, or can automatically discover by Service Mapping. All these applications are stored in ServiceNow CMDB.

When you add a ServiceNow datasource into VCF Operations for networks, VCF Operations for networks fetches the configurations items and the relationships from the ServiceNow CMDB configuration file.

VCF Operations for networks fetches data in regular intervals, by default.

- The complete data fetch happens every 12 hours, which fetches all the records of the classes defined the CMDB configuration. Also, the complete fetch happens when you add or update the datasource.
- The delta fetch happens every 2 minutes, which fetches all new, modified, and deleted records of the classes defined in the CMDB configuration. Approximately, VCF Operations for networks takes around 12 minutes to reflect these details on the user interface.

VCF Operations for networks fetches the class hierarchy and the relationship types during complete fetch only.

Default values for Limitations

| Limit Name | Description | Default Value | Impact for Exceeding the Limit |
|---|---|---|---|
| `maxAppsPerDataSource` | Maximum applications per datasource. | 5000 | The data source stops fetching data with an error on datasource and alerts page and the applications are not updated. |
| `maxTiersPerApp` | Maximum tiers that can be stored per application. | 150 | The applications are not updated until number of tiers are reduced to fit in limit. |
| `maxMembersPerApp` | Maximum members that can be stored per application. | 5000 | The applications are not updated until number of members are reduced to fit in limit. |
| `maxGraphTraversalStackSize` | Maximum size of stack used in graph traversal. | 10000 | The application will not get created and throws `SizeLimitExceededException`. |
| `maxResponseAppCount` | Maximum apps that can be returned in API response. | 5000 | Only the number of applications that fits the limit are returned and UI shows error. |

### Add a ServiceNow Account

You can add ServiceNow as a data source in VCF Operations for networks and fetch your application and tier details. The ServiceNow Configuration Management Database (CMDB) provides you with full visibility of software and hardware infrastructure and the relation between them in your data center. This visibility helps you in managing your inventory.

- You must verify the following information:
  - The data source and its version is supported in VCF Operations for networks. For more information, see Supported Products and Versions.
  - The data source follows VMware Ports and Protocols supported in VCF Operations for networks. For more information, see VMware Ports and Protocols.
- You must have one of the following privileges to your role:

- **Read only admin role** for all the privileges
- **Read only itil role** for the minimum necessary privileges
- Access to the following tables/APIs:

| APIs | Table | Permission Type |
|---|---|---|
| /stats.do | -- | Required |
| /api/now/v1/cimodel/cmdb_ci/fullclasshierarchy | -- | Required |
| /api/now/v2/table/cmdb_rel_type | cmdb_rel_type | Required |
| /api/now/v2/table/cmdb_ci | cmdb_ci | Required |
| /api/now/v2/table/cmdb_ci_service_discovered | cmdb_ci_service_discovered | Required |
| /api/now/v2/table/cmdb_rel_ci | cmdb_rel_ci | Required |
| /api/now/v2/table/cmdb_ci_translation_rule | cmdb_ci_translation_rule | Required |
| /api/now/v2/table/cmdb_ci_cluster_node | cmdb_ci_cluster_node | Required |
| /api/now/v2/table/cmdb_ci_vm_instance | cmdb_ci_vm_instance | Required |
| /api/now/v2/table/cmdb_ci_cluster | cmdb_ci_cluster | Required |
| /api/now/v2/table/cmdb_ci_lb_service | cmdb_ci_lb_service | Required |
| /api/now/v2/table/cmdb_ci_vmware_instance | cmdb_ci_vmware_instance | Required |
| /api/now/v2/table/cmdb_ci_network_adapter | cmdb_ci_network_adapter | Required |
| /api/now/v2/table/cmdb_ci_appl | cmdb_ci_appl | Required |
| /api/now/v2/table/cmdb_ci_spkg | cmdb_ci_spkg | Required |
| /api/now/v2/table/cmdb_ci_computer | cmdb_ci_computer | Required |
| /api/now/v2/table/cmdb_ci_database | cmdb_ci_database | Required |
| /api/now/v2/table/cmdb_ci_qualifier_manual_connection | cmdb_ci_qualifier_manual_connection | Required |
| /api/now/v2/table/cmdb_ci_endpoint | cmdb_ci_endpoint | Required |
| /api/now/v2/table/sys_audit_delete | sys_audit_delete | Optional |
| /api/now/v2/table/sys_user | sys_user | Optional |
| /api/now/v2/table/sys_properties | sys_properties | Optional |

**Note:**
If you do not have access to tables/APIs marked as **Optional**, then VCF Operations for networks runs the application discovery at every 12 hours. And if you have access to tables/APIs marked as **Optional**, VCF Operations for networks runs application discovery at every 30 minutes.

With ServiceNow integration, VCF Operations for networks can discover applications available in ServiceNow CMDB to enable you to directly add them into VCF Operations for networks.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click **Add Source**.
3. Under CMDBs, select **ServiceNow**.
4. On the Add a new ServiceNow Account or Source page, provide the following details:

| Option | Action |
|---|---|
| **Collector (Proxy) VM** | Select a collector VM from the drop-down menu. |

| Option | Action |
|---|---|
| **IP Address/FQDN** | Enter the IP address or the FQDN details. |
| **Username** | Enter the user name. |
| **Password** | Enter the password. |
| **Web Proxy (Optional)** | Select a web proxy from the drop-down menu.<br>**Note:**<br>The **Web Proxy (Optional)** is visible only if you have configured a web proxy in VCF Operations for networks. |

5. Click **Validate**.
6. To add a customized CMDB configuration, do the following:
   a) Select **Customize CMDB configuration**.
   b) Click **download** to download the default configuration file.
   c) Update the file properties. See, Customizing the CMDB Configuration.
   d) On the Add Data Source page, browse to select the updated JSON file.

   **Note:**
   If you add customize CMDB configuration, then VCF Operations for networks validates all the newly added entries for **Required** privileges.
7. Optional: In the **Nickname** text box, enter a nickname.
8. Optional: In the **Notes** text box, add a note if necessary.
9. Click **Submit**.

After you add a ServiceNow data source, VCF Operations for networks discovers the applications available in the ServiceNow CMDB, which you add into VCF Operations for networks. For more information, see Discover Applications.

Default CMDB Configuration File

VCF Operations for networks supports ServiceNow customizations using the configuration file in the JSON format.

```
{
  "fetchOnlyApprovedApplications": false,
  "nameBasedSearchForVm": false,
  "ignoreWorkloadCheck": false,
  "ciGroup": [
    {
      "name": "applicationClasses",
      "value": [
        "cmdb_ci_service_discovered"
      ],
      "valueType": "CI_CLASS",
      "systemGenerated": true,
      "expandCIClass": true
    },
    {
      "name": "relationshipTypeClasses",
      "value": [
        "*"
      ],
      "valueType": "CI_VALUE",
      "systemGenerated": true,
      "expandCIClass": false
    },
    {
      "name": "workloadRelationshipTypeClasses",
      "value": [
```

```
      "Hosted on::Hosts",
      "Instantiates::Instantiated by",
      "Runs on::Runs",
      "Virtualized by::Virtualizes"
    ],
    "valueType": "CI_VALUE",
    "systemGenerated": true,
    "expandCIClass": false
  },
  {
    "name": "workloadCIClasses",
    "value": [
      "cmdb_ci_computer",
      "cmdb_ci_vm_instance",
      "cmdb_ci_vmware_instance"
    ],
    "valueType": "CI_CLASS",
    "systemGenerated": true,
    "expandCIClass": true
  },
  {
    "name": "relationClasses",
    "value": [
      "cmdb_rel_ci"
    ],
    "valueType": "CI_CLASS",
    "systemGenerated": true,
    "expandCIClass": true
  },
  {
    "name": "ignoredCIClasses",
    "value": [
      "cmdb_ci_vcenter_server_obj"
    ],
    "valueType": "CI_VALUE",
    "systemGenerated": true,
    "expandCIClass": true
  },
  {
    "name": "ignoredTierCIClasses",
    "value": [
    ],
    "valueType": "CI_VALUE",
    "systemGenerated": true,
    "expandCIClass": true
  },
  {
    "name": "trackedCIClasses",
    "value": [
      "cmdb_ci_appl",
      "cmdb_ci_cluster",
      "cmdb_ci_cluster_node",
      "cmdb_ci_database",
      "cmdb_ci_lb_service",
      "cmdb_ci_spkg",
      "cmdb_ci_qualifier_manual_connection",
      "cmdb_ci_endpoint",
```

```
        "cmdb_ci_network_adapter",
        "cmdb_ci_translation_rule"
      ],
      "valueType": "CI_CLASS",
      "systemGenerated": true,
      "expandCIClass": true
    }
  ],
  "traversalRule": [
    {
      "fromNode": [
        "applicationClasses"
      ],
      "toNode": [
        "trackedCIClasses",
        "workloadCIClasses"
      ],
      "relationship": [
        "relationshipTypeClasses"
      ],
      "priority": 5
    },
    {
      "fromNode": [
        "trackedCIClasses",
        "workloadCIClasses"
      ],
      "toNode": [
        "trackedCIClasses",
        "workloadCIClasses"
      ],
      "relationship": [
        "relationshipTypeClasses"
      ],
      "priority": 3
    }
  ],
  "traversalStopRule": [
    {
      "fromNode": [
        "trackedCIClasses",
        "workloadCIClasses"
      ],
      "toNode": [
        "applicationClasses"
      ],
      "relationship": [
        "relationshipTypeClasses"
      ],
      "priority": 5
    }
  ],
  "associationRule": [
    {
      "fromNode": [
        "trackedCIClasses",
        "workloadCIClasses"
```

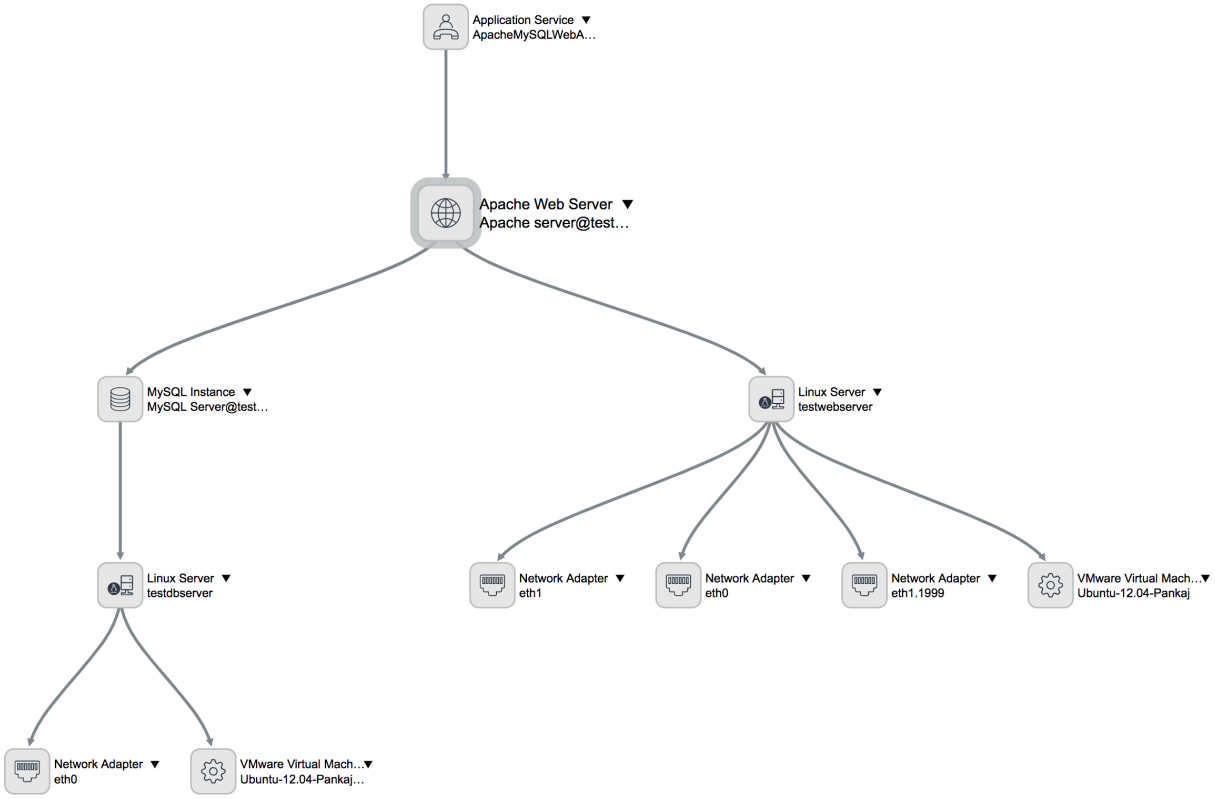
```
        ],
        "toNode": [
          "workloadCIClasses"
        ],
        "relationship": [
          "workloadRelationshipTypeClasses"
        ],
        "priority": 5
      }
    ]
  }
```

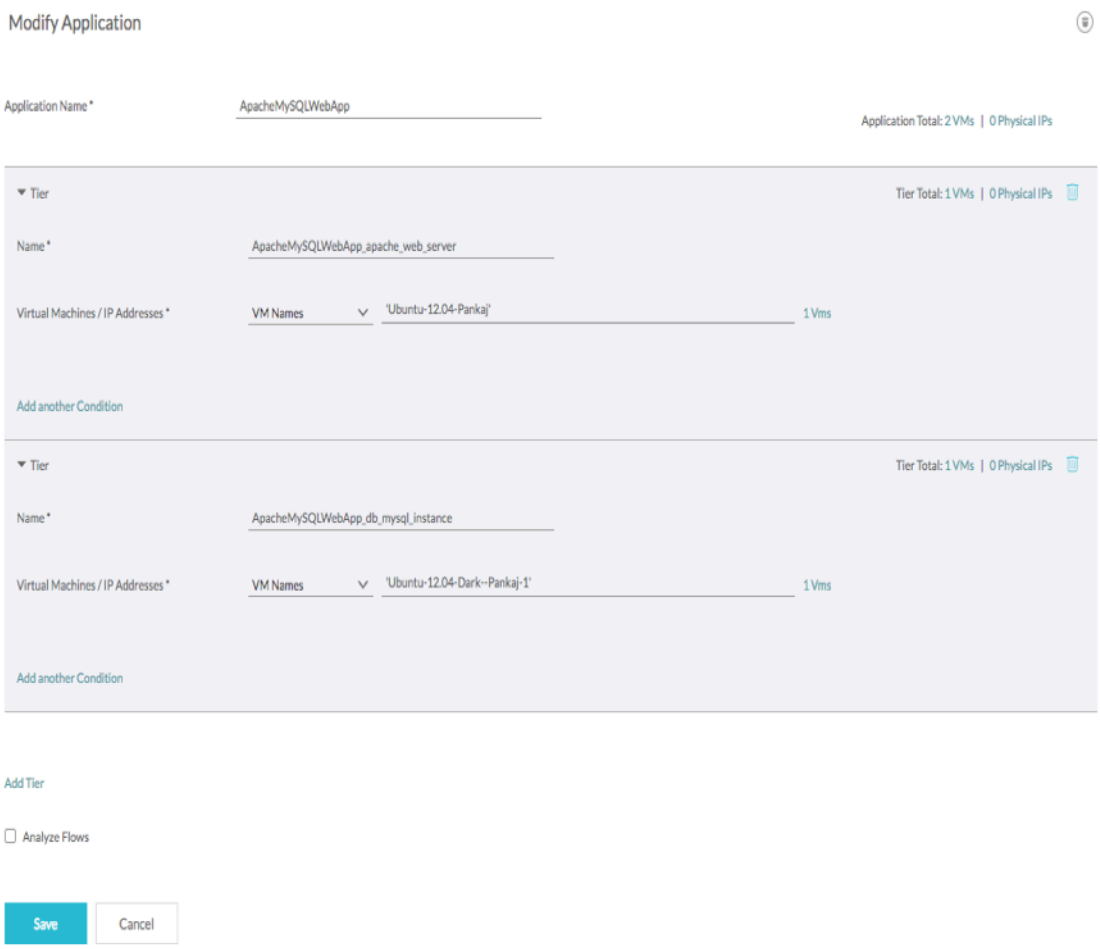
VCF Operations for networks When the configuration change occurs, might take 30 minutes for full data fetch and for recomputation of all applications.

**An example of a ServiceMap and discovered application using default CMDB configuration**

**The updated page on VCF Operations for networks for adding an application**

Using a default CMBU configuration enables VCF Operations for networks to discover the applications in ServiceNow.






## Customizing the CMDB Configuration

To support various customizations, the ServiceNow and VCF Operations for networks integration supports a generic configuration. The CMDB configuration must be in the JSON format.

The configuration includes:

- the configuration items
- the relation between the configuration items
- the rules for the dependency graph traversal.

You can customize the CMDB configuration based on your implementations.

**Note:**
When you change the configuration, a complete fetch happens and all the applications are recomputed. So, this process might take at least 30 minutes to appear on the Discovered Application Dashboard.

| Field Name | Description |
|---|---|
| `fetchOnlyApprovedApplications` | Allows the boolean value to fetch only approved applications from ServiceNow. By default, the value is set to `False`. |
| `nameBasedSearchForVm` | Allows the boolean value to indicate whether to create a custom VM search criteria with the VM name if the ServiceNow VM is not present in VCF Operations for networks. If the value is set to `True`, then a custom VM name criteria is created and count will be reflected when corresponding VM is detected in VCF Operations for networks without recomputing the application. This can be used when you create the dependency graphs or the service map manually, without using Service Mapping. By default, the value is set to `False`. |
| `ignoreWorkloadCheck` | Allows a boolean value to indicate whether to add an entity to the tier even when an associated workload entity does not exist. This can be used when you create the dependency graphs or the service map manually, without using Service Mapping and when relationships are not defined till the workload layer. By default, the value is set to `False`. |
| `ciGroup` | Defines configuration items and relationships to fetch from ServiceNow. This field allows the following properties:<br>• `Name`: Name for the configuration item group<br>• `Value`: List of ServiceNow class names that are part of this group.<br>• `ValueType`: Allows `CI_CLASS` (the class name to fetch) and `CI_VALUE`.<br>— `CI_CLASS` - to fetch the class.<br>— `CI_VALUE`<br>**Note:**<br>VCF Operations for networks always fetches `applicationClasses`, `workloadCIClasses`, `trackedCIClasses`, `workloadCIClasses`, and `relationClasses`.<br>• `systemGenerated`: Allows the boolean value to indicate whether the class is a user-defined class or a default class.<br>• `expandCIClass` - Allows the boolean field to indicate whether to fetch the subclasses of the configuration item class listed in `Value`. |

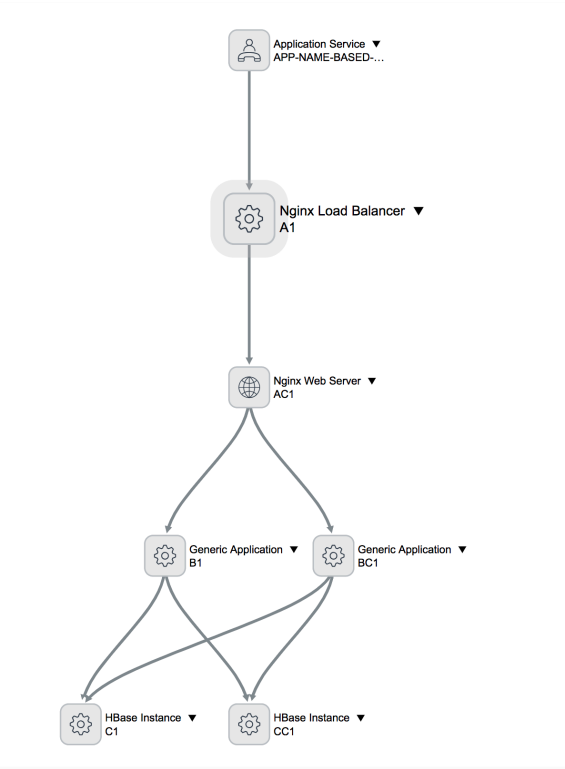
| Field Name | Description |
|---|---|
| `Rules for graph traversal` | Supports three types of traversal rules:<br>• `traversalRule`: All allowed or valid traversals.<br>• `traversalStopRule`: Traversals that are not allowed.<br>**Note:**<br>The rules in `traversalStopRule` have higher priority than the rules in `traversalRule`.<br>• `associationRule`: Traversals that are allowed for the associated workload with entity.<br>Properties of a rule:<br>• `fromNode`: List of `ciGroup` that are the source of the traversal.<br>• `toNode`: List of `ciGroup` that are the destination of traversal.<br>• `relationship`: List of `ciGroup` that have a relationship in a type of traversal.<br>• `priority`: If a `ciGroup` matches two rules, then the rule for the `ciGroup` is set based on the `priority`. Greater the priority number higher the priority value. |
| `applicationClasses` | Lists all entry point configuration item classes for the graph traversal. These classes represent the configuration item types which are used as application classes in the CMDB.<br>The default configuration uses `cmdb_ci_service_discovered` class. This class represents applications created by the ServiceMapping feature of ServiceNow. |
| `workloadCIClasses` | Lists all the configuration items that host either a software-based service or an operating system like Linux Server, Windows Server. Example, VMs, AWS instances, Physical Servers.<br>Typically, workload configuration items are placed towards the end of the dependency graph. Tiers are not created for the configuration item classes that are mentioned in this group.<br>The default configuration contains the following configuration item classes:<br>• `cmdb_ci_computer`: Represents all compute related configuration items. This is a super class for all Linux and Windows Servers.<br>• `cmdb_ci_vm_instance`: Represents virtual compute entities like VMs and AWS instances.<br>• `cmdb_ci_vmware_instance`: Represents VMware VMs. |
| `trackedCIClasses` | Lists all configuration items that can be part of the dependency graphs, but are not `applicationClass` or `workloadCIClass`. The configuration items in this group are required for the graph to complete from `applicationClasses` to `workloadCIClasses`. VCF Operations for networks creates tiers for all the classes mentioned in `trackedCIClasses`, unless the class is mentioned under `ignoredTierCiClasses`. |
| `relationshipTypeClasses` | Lists all related configuration items represented by relation configuration items classes or relation types.<br>The default configuration uses * to fetch all relation types. |
| `workloadRelationshipTypeClasses`: | lists relation types which typically represent the relations with workload entities. Following are the relations supported by default in ServiceNow:<br>• `Hosted on::Hosts`<br>• `Instantiates::Instantiated by`<br>• `Runs on::Runs`<br>• `Virtualized by::Virtualizes` |

| Field Name | Description |
|---|---|
| `ignoredCiClasses` | Lists all the configuration items that VCF Operations for networks must ignore to fetch from ServiceNow CMDB.<br>This is useful while fetching a super class, to ignore the unnecessary subclasses.<br>By default, `cmdb_ci_vcenter_server_obj` is listed under `ignoredCiClasses` as vCenter are not required for the application discovery. |
| `ignoredTierCiClasses` | Lists all the configuration items for which tiers must not be created. |

An Example of Discovering Applications Without Workload Relations

Here is a customized CMDB configuration file in which `nameBasedSearchForVm` is defined to discover the applications, where `cmdb_ci_service_discovered` class is the entry point and the workload relations are not defined.

## Topology




## Customized CMDB Configuration File

```
{
  "fetchOnlyApprovedApplications": false,
  "nameBasedSearchForVm": true,
  "ignoreWorkloadCheck": true,
  "ciGroup": [
    {
      "name": "applicationClasses",
      "value": [
        "cmdb_ci_service_discovered"
      ],
      "valueType": "CI_CLASS",
      "systemGenerated": true,
```

```
      "expandCIClass": true
    },
    {
      "name": "relationshipTypeClasses",
      "value": [
        "*"
      ],
      "valueType": "CI_VALUE",
      "systemGenerated": true,
      "expandCIClass": false
    },
    {
      "name": "workloadRelationshipTypeClasses",
      "value": [
        "Hosted on::Hosts",
        "Instantiates::Instantiated by",
        "Runs on::Runs",
        "Virtualized by::Virtualizes"
      ],
      "valueType": "CI_VALUE",
      "systemGenerated": true,
      "expandCIClass": false
    },
    {
      "name": "workloadCIClasses",
      "value": [
        "cmdb_ci_computer",
        "cmdb_ci_vm_instance",
        "cmdb_ci_vmware_instance"
      ],
      "valueType": "CI_CLASS",
      "systemGenerated": true,
      "expandCIClass": true
    },
    {
      "name": "relationClasses",
      "value": [
        "cmdb_rel_ci"
      ],
      "valueType": "CI_CLASS",
      "systemGenerated": true,
      "expandCIClass": true
    },
    {
      "name": "ignoredCIClasses",
      "value": [
        "cmdb_ci_vcenter_server_obj"
      ],
      "valueType": "CI_VALUE",
      "systemGenerated": true,
      "expandCIClass": true
    },
    {
      "name": "ignoredTierCIClasses",
      "value": [
        "cmdb_ci_qualifier_manual_connection",
        "cmdb_ci_endpoint"
```

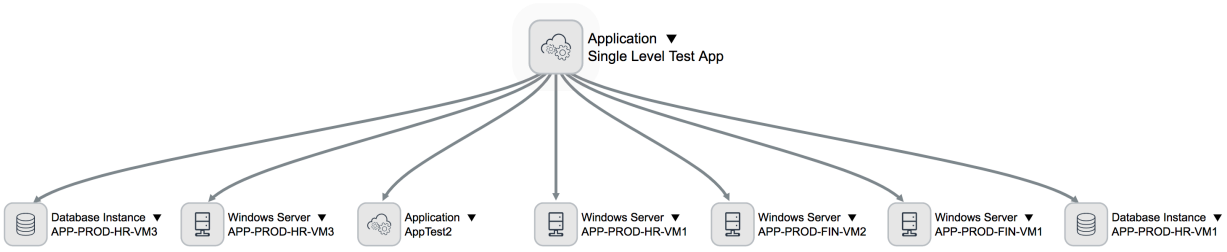
```
      ],
      "valueType": "CI_VALUE",
      "systemGenerated": true,
      "expandCIClass": true
    },
    {
      "name": "trackedCIClasses",
      "value": [
        "cmdb_ci_appl",
        "cmdb_ci_cluster",
        "cmdb_ci_cluster_node",
        "cmdb_ci_database",
        "cmdb_ci_lb_service",
        "cmdb_ci_spkg",
        "cmdb_ci_qualifier_manual_connection",
        "cmdb_ci_endpoint",
        "cmdb_ci_network_adapter",
        "cmdb_ci_translation_rule"
      ],
      "valueType": "CI_CLASS",
      "systemGenerated": true,
      "expandCIClass": true
    }
  ],
  "traversalRule": [
    {
      "fromNode": [
        "applicationClasses"
      ],
      "toNode": [
        "trackedCIClasses",
        "workloadCIClasses"
      ],
      "relationship": [
        "relationshipTypeClasses"
      ],
      "priority": 5
    },
    {
      "fromNode": [
        "trackedCIClasses",
        "workloadCIClasses"
      ],
      "toNode": [
        "trackedCIClasses",
        "workloadCIClasses"
      ],
      "relationship": [
        "relationshipTypeClasses"
      ],
      "priority": 3
    }
  ],
  "traversalStopRule": [
    {
      "fromNode": [
        "trackedCIClasses",
```

```
        "workloadCIClasses"
      ],
      "toNode": [
        "applicationClasses"
      ],
      "relationship": [
        "relationshipTypeClasses"
      ],
      "priority": 5
    }
  ],
  "associationRule": [
    {
      "fromNode": [
        "trackedCIClasses",
        "workloadCIClasses"
      ],
      "toNode": [
        "workloadCIClasses"
      ],
      "relationship": [
        "workloadRelationshipTypeClasses"
      ],
      "priority": 5
    }
  ]
}
```

An Example of Discovering Single Level Applications

Here is a customized CMDB configuration file in which `nameBasedSearchForVm` is defined to discover the single level applications, where `cmdb_ci_service_discovered` class is the entry point and the workload relations are not defined.

## Topology




## Customized CMDB Configuration File

```
{
  "fetchOnlyApprovedApplications": false,
  "nameBasedSearchForVm": true,
  "ignoreWorkloadCheck": true,
  "ciGroup": [
    {
      "name": "applicationClasses",
      "value": [
        "cmdb_ci_appl"
      ],
```

```
      "valueType": "CI_CLASS",
      "systemGenerated": true,
      "expandCIClass": false
    },
    {
      "name": "relationshipTypeClasses",
      "value": [
        "*"
      ],
      "valueType": "CI_VALUE",
      "systemGenerated": true,
      "expandCIClass": false
    },
    {
      "name": "workloadRelationshipTypeClasses",
      "value": [
        "Hosted on::Hosts",
        "Instantiates::Instantiated by",
        "Runs on::Runs",
        "Virtualized by::Virtualizes"
      ],
      "valueType": "CI_VALUE",
      "systemGenerated": true,
      "expandCIClass": false
    },
    {
      "name": "workloadCIClasses",
      "value": [
        "cmdb_ci_computer",
        "cmdb_ci_vm_instance",
        "cmdb_ci_vmware_instance"
      ],
      "valueType": "CI_CLASS",
      "systemGenerated": true,
      "expandCIClass": true
    },
    {
      "name": "relationClasses",
      "value": [
        "cmdb_rel_ci"
      ],
      "valueType": "CI_CLASS",
      "systemGenerated": true,
      "expandCIClass": true
    },
    {
      "name": "ignoredCIClasses",
      "value": [
        "cmdb_ci_vcenter_server_obj"
      ],
      "valueType": "CI_VALUE",
      "systemGenerated": true,
      "expandCIClass": true
    },
    {
      "name": "ignoredTierCIClasses",
      "value": [
```

```
        "cmdb_ci_qualifier_manual_connection",
        "cmdb_ci_endpoint"
      ],
      "valueType": "CI_VALUE",
      "systemGenerated": true,
      "expandCIClass": true
    },
    {
      "name": "trackedCIClasses",
      "value": [
        "cmdb_ci_appl",
        "cmdb_ci_cluster",
        "cmdb_ci_cluster_node",
        "cmdb_ci_database",
        "cmdb_ci_lb_service",
        "cmdb_ci_spkg",
        "cmdb_ci_qualifier_manual_connection",
        "cmdb_ci_endpoint",
        "cmdb_ci_network_adapter",
        "cmdb_ci_translation_rule"
      ],
      "valueType": "CI_CLASS",
      "systemGenerated": true,
      "expandCIClass": true
    }
  ],
  "traversalRule": [
    {
      "fromNode": [
        "applicationClasses"
      ],
      "toNode": [
        "trackedCIClasses",
        "workloadCIClasses"
      ],
      "relationship": [
        "relationshipTypeClasses"
      ],
      "priority": 5
    },
    {
      "fromNode": [
        "trackedCIClasses",
        "workloadCIClasses"
      ],
      "toNode": [
        "trackedCIClasses",
        "workloadCIClasses"
      ],
      "relationship": [
        "relationshipTypeClasses"
      ],
      "priority": 3
    }
  ],
  "traversalStopRule": [
    {
```

```
        "fromNode": [
          "trackedCIClasses",
          "workloadCIClasses"
        ],
        "toNode": [
          "applicationClasses"
        ],
        "relationship": [
          "relationshipTypeClasses"
        ],
        "priority": 5
      }
    ],
    "associationRule": [
      {
        "fromNode": [
          "trackedCIClasses",
          "workloadCIClasses"
        ],
        "toNode": [
          "workloadCIClasses"
        ],
        "relationship": [
          "workloadRelationshipTypeClasses"
        ],
        "priority": 5
      }
    ]
  }
```

### FAQs on Add ServiceNow as a Data Source

You find all the common frequently asked questions related to adding ServiceNow as a data source in this topic.

#### How to Assign Privileges to a Role in ServiceNow for adding in VCF Operations for networks?

1. Create a Role in ServiceNow UI.
2. Click **System Administrator** > **Elevate Roles** > **security_admin** > **OK**.
3. Go to **Access Control (ACL)**, and click **New**.
4. For API access, do the following:

| Option | Action |
|---|---|
| **Type** | Select **REST_Endpoint** from the drop-down menu. |
| **Operation** | Select **execute** from the drop-down menu. |
| **Name** | From the drop-down menu select the API entry that you want to add. For list of APIs, see Add a ServiceNow Account topic. |

5. For tables, do the following:

| Option | Action |
|---|---|
| **Type** | Select **record** from the drop-down menu. |
| **Operation** | Select **read** from the drop-down menu. |
| **Name** | From the first drop-down menu, select the table entry that you want to add. From the second drop-down menu, select **None**. For list of tables, see Add a ServiceNow Account topic. |

6. In the **Requires role** table, select the role you want to add the access control.
7. Click **Submit**.

**Do I need to Add all the APIs/Tables Mentioned in the Add a ServiceNow Account Topic?**

To run application discovery every 30 minutes, you must provide all the required privileges. However, if you have access to the following table, you can add the data source in VCF Operations for networks and **VALIDATE**:

| APIs | Table | Permission Type |
|---|---|---|
| /stats.do | -- | Required |
| /api/now/v1/cimodel/cmdb_ci/fullclasshierarchy | -- | Required |
| /api/now/v2/table/cmdb_rel_type | cmdb_rel_type | Required |
| /api/now/v2/table/cmdb_ci | cmdb_ci | Required |
| /api/now/v2/table/cmdb_ci_service_discovered | cmdb_ci_service_discovered | Required |
| /api/now/v2/table/cmdb_rel_ci | cmdb_rel_ci | Required |

VCF Operations for networks validates the required privileges and shows any missing privileges as an error. You can check the missing privileges, and then get access to the missing privileges and **VALIDATE** again.

**Not able to find `/api/now/v1/cimodel/cmdb_ci/fullclasshierarchy` in ServiceNow Access Control (ACL), How to add the API in ServiceNow ACL?**

1. Create a Role in ServiceNow UI.
2. Click **System Administrator** > **Elevate Roles** > **Security Admin** > **OK**.
3. Go to **Access Control (ACL)**, and click **New**.
4. For API access, do the following:

| Option | Action |
|---|---|
| **Type** | Select **REST_Endpoint** from the drop-down menu. |
| **Operation** | Select **execute** from the drop-down menu. |
| **Name** | In the first drop-down menu enter `CMBD Model API Read`. From the second drop-down menu, select **None**. |

5. In the **Requires role** table, select the role you want to add the access control.
6. Click **Submit**.

**How to make any role read-only in ServiceNow?**

To make a role read-only, you must assign **snc_read_only** to the role. For details, see Read-only role topic in ServiceNow Documentation.

**I have all the necessary privileges, still I am getting `Missing privileges` message, what to do?**

In some of the tables you must add another ACL with the same table to provide access to all the attributes by selecting `*` form the second **Name** drop-down menu. Also, ensure you have both the ACLs in the system to get the necessary privileges.

## Bulk Add Devices as Data Sources

You can bulk upload network devices, such as routers, switches, load balancers, and firewalls in VCF Operations for networks and add those uploaded devices as data sources.

To add multiple network devices at once, you need to add the necessary information related to the network devices in a CSV file and upload the file in VCF Operations for networks.

Download the sample `.csv` file from VCF Operations for networks and add the details mentioned in the `.csv` file. The sample `.csv` file contains all the fields that you need to use to bulk upload your network devices, along with examples for each field.

The `.CSV` file must have information in the following fields:

**Table 1307: CSV Fields**

| Field Name | Description | Importance |
|---|---|---|
| **datasource_type** | The type or category of the data source.<br>**Note:**<br>See the sample `.csv` for accepted values. | Mandatory |
| **ip** | IP address of the data source.<br>You need to add either IP address or FQDN. | Mandatory |
| **fqdn** | FQDN of the data source.<br>You need to add either IP address or FQDN. | Mandatory |
| **usersname** | User name of the device. | Mandatory |
| **password** | Password of the device. | Mandatory |
| **nickname** | Any nick name that you want to give to the device. | Mandatory |
| **notes** | Any additional info that you want to add as a note. | Optional |
| **snmp_version** | SNMP version that you want to use.<br>Valid entries are v2c, v3. | Optional |
| **snmp_community_string** | SNMP v2 community string.<br>If you use SNMP Version as v2c, **snmp_community_string** is a mandatory field. | Optional |
| **snmp_username** | SNMP user name.<br>If you use SNMP version as v3, **snmp_username** is a mandatory field. | Optional |
| **snmp_context_name** | Context name for the SNMP. | Optional |
| **snmp_auth_type** | SNMP Authentication Type.<br>You can leave this field blank if SNMP Authentication Type is not available.<br>Valid entries are MD5, SHA. | Optional |
| **snmp_auth_password** | SNMP Authentication Password.<br>If you use SNMP Version as v3 and authentication type is MD5 or SHA, **snmp_auth_password** is a mandatory field. | Optional |
| **snmp_privacy_type** | SNMP Privacy Type.<br>You can leave this field blank if SNMP privacy type is not available.<br>Valid entries are DES, 3DES, AES128, AES192, or AES256. | Optional |
| **snmp_privacy_password** | SNMP Privacy password. Required only if SNMP Version is v3 and privacy type is present. | Optional |
| **polling_interval_in_mins** | Polling interval details in minutes. | Mandatory |
| **collector_ip** | IP address of the collector that you want to use for the device. | Mandatory |

To bulk add the network devices, do the following steps:

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. In the **Data Source** tab, from the drop-down menu located in the top-right corner, select **Bulk add devices**.
3. In the **Add physical devices in bulk from file** page, click download sample file.

   **Note:**

   You must update all the required field in the `.CSV` file.
4. Click **Browse** and upload the `.CSV` file that has all the details related to the network devices you want to add as data sources.
5. Click **Upload & Add** button.

   If there are any issues in the `.CSV` file, you see all the relevant error listed in a table format. You can also export the errors as a `.CSV` file.

   Correct all the issues and upload the `.CSV` file again to add your devices as data sources.

## Working with Network Discovery

When you have several network devices in your data center, discovering them and adding them manually to VCF Operations for networks as data sources might be time-consuming and inefficient.

VCF Operations for networks automatically discovers the network devices in your data center and provides the options to add these discovered devices as data sources or export their details.

### Discover Network Devices Using Network Auto Discovery

You can discover network devices such as routers, switches, firewalls, and F5 load balancer within your network. Once you have discovered these devices in your network, you can then add them to VCF Operations for networks to get deeper visibility into them.

**Discover Network Devices in VCF Operations for networks**
You can discover supported routers, switches, firewalls, and F5 Load Balancer.

You must verify the network devices supported in VCF Operations for networks. For more information, see Supported Products and Versions.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click the **Device Auto Discovery** tab and select **Discovery Profiles**.
3. To create a new discovery profile, click **Create Discovery Profile**.
4. Define the scope based on which you want to discover the devices. You can discover devices either using the seed IP or the IP range.
   1. To discover devices based on seed IP, from the **Discovery Method** drop-down, select **Seed IP**.
      Here, the devices are recursively discovered using the CDP and LLDP protocols up to the discovery depth that you define.
   2. To discover devices based on the IP range, from the **Discovery Method** drop-down, select **Scan IP Range**.
      The scan IP Range defines the range of IP addresses where network discovery starts a scan.

      **Note:**
      Network auto discovery supports only a maximum of 4096 IP addresses per collector.

   While adding the IP addresses, you can perform the following actions:

   - To add a single IP address, specify the IP address.
   - To add multiple IP addresses, use commas.
   - To add an IP range, specify the IP entries with the **-** or **/** range separator.
   - To add an IP range and exclude few IP addresses, click **Add Exceptions**, enter the exception IPs.

5. Provide appropriate access credentials and click **Next**. Supported access credential types are SSH, SNMP v2, SNMPv3, and APIs.

   To discover F5 Big-IP Load Balancer devices, you must enter your API credentials.

6. Provide a name for saving your discovery profile and set the discovery job schedule.

   1. You can schedule discovery jobs to:
      – Run once at the specified date and time.
      – Or, run periodically such as daily, weekly, monthly, or at your defined custom intervals.
   2. Provide a name for saving your discovery profile.

7. Click **Submit**.

You can bulk add the discovered network devices as data sources to VCF Operations for networks. See Bulk Add Discovered Devices as Data Sources for more information.

**Bulk Add Discovered Devices as Data Sources**
You can bulk add the discovered devices as data sources in VCF Operations for networks.

1. From the left navigation pane, go to **Settings** > **Accounts and Data Sources**.
2. Click the **Device Auto Discovery** tab and select **Discovered Devices**.

   The discovered devices are divided in the following tabs:

   - **All**: Displays all the discovered devices.
   - **Ready to Add**: Displays all the devices that are ready to be added as data sources.
   - **Needs Attention**: Displays the devices that have one or more problems and need remediation.
   - **Excluded**: Displays the devices which cannot be added as data sources. You can also select devices from any of the other tabs and add it to the excluded list. These devices can later be removed from the excluded list and added as data sources.

3. Select one or more devices from the **All** or **Ready to Add** tab and click the **Add As Source** button.
4. To bulk edit information of the devices before adding them as data sources, select the devices, click the three dots icon next to **Add As Source** and click **Bulk Edit and Add**.

   You can edit the following fields.

   - **Collector VM**: From the drop-down, select:
     – **No Change** to retain your existing collector.
     – Or, **Override existing values** to enter a new values.
   - **Autoinferred Device Name**: From the drop-down, select:
     – **No Change** to retain your existing device name.
     – **Append to existing values** to enter new values at the end of the existing values.
     – Or, **Prepend to existing values** to enter new values at the beginning of the existing values.
   - **Access Credentials**: From the drop-down, select:
     – **No Change** to retain your existing credentials.
     – Or, **Override existing values** to provide a new credentials that works across all the data sources.
   - **Config Polling Intervals**: From the drop-down, select:
     – **No Change** to retain the previously configured polling interval.
     – Or, **Override existing values** to provide a new polling interval for the VCF Operations for networks platform to collect the configuration data from the selected data sources
   - **SNMP**: From the drop-down, select:
     – **No Change** to retain your existing SNMP credentials.
     – Or, **Override existing values** to enter a new values.
   - **Notes**: From the drop-down, select:
     – **No Change** to retain your existing notes.
     – Or, **Override existing values** to enter a new note.

   Click **Next**.

In the next screen, review the devices and add them as data sources.

Bulk operations may take some time to complete. If the bulk add operation is successful, you will see a notification for the same.
If the operation fails for one or more data sources, the details of the devices and the details due to which the operation failed are displayed.

You can fix the errors and retry the operation on the failed devices.

### Discover Network Devices by Running CLI Commands

You can generate and run scripts on the VCF Operations for networks Collector CLI to discover devices in your network automatically and export the details of discovered devices to a `.csv` file.

**Note:**

Discovering network devices by running CLI commands will be deprecated in a future release of VCF Operations for networks.

You can discover the following devices by running the CLI commands:

- Cisco switches
- Arista switches
- Juniper switches
- Dell switches
- F5 Big-IP
- Brocade switches
- Palo Alto Networks
- Check Point Firewall
- Cisco ASA

1. Log in to the VCF Operations for networks Collector CLI as a **Support** user.
   Provide your CLI login credentials using the Secure Socket Shell (SSH) client.
2. Switch to a **Ubuntu** user.
   ```
   ub
   ```
3. Run the following command to generate the Python script in your present working directory.
   ```
   java -cp "/home/ubuntu/build-target/common-utils/tools-0.001-SNAPSHOT.jar"
    com.vnera.tools.DiscoveredDevicesTool -w script.py
   ```
   You use this script to perform network discovery.
4. Run the following command to generate the sample credentials file in your present working directory.
   ```
   java -cp "/home/ubuntu/build-target/common-utils/tools-0.001-SNAPSHOT.jar"
    com.vnera.tools.DiscoveredDevicesTool -s sample_creds.csv
   ```
   You use the sample credentials file to access the devices in your network.

   Here is an example of the sample credentials file:

   ```
   "credential_name","connection_type", "username", "password", "snmp_version",
   "community_string", "snmp_user", "snmp_context", "snmp_auth_type", "snmp_auth_password",
   "snmp_privacy_type", "snmp_privacy_password"
   "SSH1","SSH","admin","admin","","","","","","","","","
   "API1","API","admin","admin","","","","","","","","","
   "SNMP1","SNMP","","","v2c","community","","","","","","","
   "SNMP2","SNMP","","","v3","","user","context","authtype","authpass","privacytype","privacypass"
   ```
5. Update the sample credentials file with your SSH, SNMP v2c, SNMPv3, or API credentials.

   You can enter multiple credentials in the file by providing a unique `credential_name` for each entry.

To discover F5 Big-IP load balancers, you must enter your API credentials.

**Note:**
You cannot discover other network devices using your API credentials.

6. Run the `help` command on the Python script to view the options supported by the script.

```
python3 script.py --help
```

You can use these options to authenticate your credentials and perform network discovery.

| Option | Description |
|---|---|
| --host | The IP address of the VCF Operations for networks platform node1. |
| --deployment type | The type of deployment on which you are performing this operation. Accepted values are `onprem` and `saas`.<br>The default value is `onprem`. |
| --username | The user name of the platform user.<br>**Note:**<br>You must always enclose the username in single quotes. |
| --password | The password of the platform user.<br>**Note:**<br>You must always enclose the password in single quotes. |
| --domain_type | The domain type to authenticate user credentials. Accepted values are `LOCAL` and `LDAP`. |
| --domain_value | If the domain type is LDAP, provide your LDAP domain name. |
| --proxy_ips | The IP address of VCF Operations for networks Collector. |
| --credentials_file_path | The location of the credentials file which contains the SSH, SNMP, or API credentials. |
| --network_ranges | The IP address range to perform network discovery on. You can specify the IP addresses in the network range format or the CIDR range format. |
| --seeds | The seed IP addresses or the seed FQDN of network devices to begin the network discovery process.<br>**Note:**<br>When you run the network discovery script, you begin the network discovery process with given seed IP addresses as the starting point.<br>Each seed's neighbor devices are discovered based on industry protocols such as CDP and LLDP. Each discovered device is then used as a seed to recursively discover other devices in your network until all devices in your network are discovered or the seed depth is reached. |

| Option | Description |
|---|---|
| --seed_depth | The depth of discovering network devices.<br>**Note:**<br>The seed depth indicates how many levels of neighbors must be explored from the seed IP address. By default, the depth of the seed is 0; the depth of the seed's neighbors is 1; the depth of the seed's neighbors' neighbors is 2 and so on.<br>If you enter the seed depth as 2, you can discover devices of the seed, the seed's neighbors, and the seed's neighbor's neighbors. |
| --ignored_list | The list of IP addresses to exclude from network discovery. You can specify the IP addresses in the following formats:<br>• Individual IP address<br>• FQDN<br>• Network range<br>• CIDR range |
| --download_discovered_devices | The option to download the previously discovered devices. Enter `True` or `False`. |
| --out | The name of the CSV output file for successful network discovery. The default name of the output file is `auto_gen_discovered_devices.csv`. |
| --clear_all | The option to erase details of the discovery such as discovery job, credential profile, job result, and discovered devices. Enter `True` or `False`. |
| --wait_time_in_secs | The maximum wait time (in seconds) to discover devices. |

7. Run the Python script with one of the following options:

| Choice | Example |
|---|---|
| **Provide the network range for discovery by using the --network_ranges option.** | `python3 script.py --deployment_type onprem --username 'user' --password 'pwd' --domain_type LOCAL --proxy_ips 10.xx.196.xxx --out discovered_devices.csv --credentials_file_path sample_creds.csv --network_ranges 10.xxx.xx.158/24` |
| **Provide the seed address and the seed depth for network discovery by using --seeds and --seed_depth options.** | `python3 script.py --deployment_type onprem --username 'user' --password 'pwd' --domain_type LDAP --domain_value vmware.com --proxy_ips 10.xx.196.xxx --out discovered_devices.csv --credentials_file_path sample_creds.csv --seeds 10.xxx.xx.xxx --seed_depth 2` |

**Note:**

You must run the Python script with your updated credentials file that includes your SSH, SNMP v2c, SNMPv3, or API credentials.

The python script only discovers devices that are within the reachability scope of the collector network. It does not discover devices that are available on the network, but not accessible from the collector network.

8. Based on the status of the operation, do one of the following:

| Status | Action |
|---|---|
| **The operation succeeds.** | Open the CSV file. An example of the CSV file is as follows: |

| Status | Action |
| --- | --- |
| | "DataSourceType","IP","fqdn","Username","Password","CSPRefres<br>"CentralCliEnabled","IPFixEnabled","SwitchType","ParentvCente<br>"snmp_version","snmp_community_string","snmp_username","snmp_<br>"snmp_auth_type","snmp_privacy_type","ProxyIP"<br>"DataSource1","","Domain_name","username","<pwd>","",<br>"nick_name_discovered_devices_1","","","","","","","","","","<br>"DataSource2","10.1xx.xx.1x","","username","pwd","",<br>"<br>nick_name_discovered_devices_2","","","","","","","","","","<br> |
| **The operation fails or times out.** | Run the Python script with the --clear_all option to clear the details of the discovered devices and start a new network discover operation.<br>For example:<br>`python3 script.py --deployment_type onprem --username 'user' --password 'pwd' --proxy_ips 10.xxx.xxx.xx --clear_all true` |

You can bulk add the discovered network devices as data sources to VCF Operations for networks by using the `operations-for-networks-sdk-python` SDK. See Bulk Add Data Sources Using the Python SDK for more information.

## Editing and Migrating Data Sources

You can edit, migrate, and delete data sources from VCF Operations for networks.

### Bulk Add Data Sources Using the Python SDK

You can bulk add network devices as data sources to VCF Operations for networks by using the `network-insight-sdk-python` SDK.

- You must have access to the VCF Operations for networks platform CLI.
- You must have the CSV file containing the list of network devices in the following format:

```
"DataSourceType","IP","fqdn","Username","Password","CSPRefreshToken","NickName",
"CentralCliEnabled","IPFixEnabled","SwitchType","ParentvCenter","IsVMC",
"snmp_version","snmp_community_string","snmp_username","snmp_password",
"snmp_auth_type","snmp_privacy_type","ProxyIP"
"DataSource1","","Domain_name","username","<pwd>","",
"nick_name_discovered_devices_1","","","","","","","","","","","","10.xx.xxx.xx"
"DataSource2","10.1xx.xx.1x","","username","pwd","",
" nick_name_discovered_devices_2","","","","","","","","","","","","10.xx.xx.xxx"
```

  You can create the file manually, or discover the devices in your network by performing network discovery. See Discover Network Devices by Running CLI Commands for more information.

You can add the following data sources only:

- Cisco switches
- Arista switches
- Juniper switches
- Dell switches
- F5
- Brocade switches
- Palo Alto Networks
- Check Point Firewall
- Cisco ASA

1. Log in to the VCF Operations for networks platform CLI as a **support** user.
2. Switch to a **Ubuntu** user. Run the following command:

   ```
   ub
   ```

3. Navigate to the `/home/ubuntu` directory.

   ```
   cd /home/ubuntu
   ```

4. Download the `network-insight-sdk-python` SDK from GitHub.

   ```
   wget https://github.com/vmware/network-insight-sdk-python/archive/refs/heads/master.zip
   ```

5. Unzip the contents of the SDK and delete the ZIP file.

   ```
   unzip master.zip && rm master.zip
   ```

6. Export the Python path to `/home/ubuntu/network-insight-sdk-python-master/swagger_client-py2.7.egg`.

   ```
   export PYTHONPATH=/home/ubuntu/network-insight-sdk-python-master/swagger_client-py2.7.egg
   ```

7. Navigate to the `examples` directory within the SDK. This folder contains the `datasource_addition.py` script.

   ```
   cd /home/ubuntu/network-insight-sdk-python-master/examples
   ```

8. Run the `datasource_addition.py` script to bulk add network devices to VCF Operations for networks. The supported options are:

| Option | Required | Description |
|---|---|---|
| --deployment_type | Required | The type of deployment on which you are performing the bulk addition operation. Accepted values are `onprem` and `saas`. The default value is `onprem`. |
| --platform_ip | Required | The IP address of the VCF Operations for networks platform node1 where you want to add the devices. |
| --username | Required | The user name of the platform user.<br>**Note:**<br>You must always enclose the username in single quotes. |
| --password | Required | The password of the platform user.<br>**Note:**<br>You must always enclose the password in single quotes. |
| --data_sources_csv | Required | The location of the CSV file which contains the details of the devices. |
| --domain_type | Optional | The domain type to authenticate user credentials. Accepted values are `LOCAL` and `LDAP`. |
| --domain_value | Optional | If the domain type is LDAP, provide your LDAP domain name. |

   Here are a few examples.

   ```
   python datasource_addition.py --deployment_type onprem --platform_ip vrni.example.com --username
    'user@vmware' --password 'mypassword' --data_sources_csv /home/ubuntu/discovered_devices.csv
   python datasource_addition.py --deployment_type onprem --platform_ip vrni.example.com --username
    'user@vmware.com' --password 'mypassword' --domain_type LDAP --domain_value vmware.com --
   data_sources_csv /home/ubuntu/discovered_devices.csv
   ```

You can track the result of the network discovery operation on the CLI. If the bulk addition operation fails for one or more devices, you can rectify the errors and reattempt the operation. See Common Data Source Errors for more information.

View the data sources added to VCF Operations for networks by navigating to the **Settings** > **Accounts and Data Sources** page.

### Bulk Edit Data Sources

In VCF Operations for networks, you can enable, deactivate, edit, or delete multiple data sources at once.

You must have added more than one data source into VCF Operations for networks. For more information, see Adding a Data Source in VCF Operations for networks and Bulk Add Data Sources Using the Python SDK.

1. Go to **Settings** > **Accounts and Data Sources**.
2. In the **Accounts and Data Sources** page, select the data sources on which you want to perform bulk operations.
3. From the **Actions** drop-down menu, select the bulk operation that you want to perform.

   **Note:**
   Bulk operations are asynchronous in nature, which means that when a bulk operation is in progress, updates to the selected data sources are temporarily deactivated till the operation completes.

| Operation | Action |
|---|---|
| Enable | Enables data collection on the selected data sources. |
| Disable | Deactivates data collection on the selected data sources. |
| Edit | Allows you to edit the following options:<br>• **Access Credentials** - Edit only the user name and password of the selected data sources.<br>• **Config polling interval** - Edit the interval to collect configuration data from the data sources.<br>• **Notes** - Add additional information that the user wants to add, limited to 150 characters.<br>In the **Edit Sources** page, update the required parameters and click **Submit**.<br>**Note:** Only the common options across the selected data sources are displayed for bulk update.<br>• From the **Access Credentials** drop-down menu, select:<br>– **No Change** to retain your existing credentials.<br>– Or, **Override existing values** to provide a new user name and password that works across all the data sources. After entering the credentials, click **Validate** to validate your credentials against one of the data sources. A successful validation hints that the credentials are most likely true for all the selected data sources.<br>• From the **Config Polling Interval** drop-down menu, select:<br>– **No Change** to retain the previously configured polling interval.<br>– Or, **Override existing values** to provide a new polling interval for the VCF Operations for networks platform to collect the configuration data from the selected data sources.<br>• From the **Notes** drop-down menu, select:<br>– **No Change** to retain your existing notes.<br>– Or, **Override existing values** to enter a new note. |

| Operation | Action |
|---|---|
| Delete | Deletes the selected data sources from VCF Operations for networks.<br>If the selected data sources have a dependency on one or more data sources, do one of the following actions:<br>• Delete the dependent data sources before deleting the selected data sources.<br>• Or, delete the dependent data sources along with the selected data sources. |

Bulk operations may take some time to complete. After the operation completes, if the operation is successfully, VCF Operations for networks displays a notification indicating that the operation succeeded for all the selected data sources.

If the operation fails for one or more data sources, VCF Operations for networks displays an error notification indicating the status of the operation. Click the **View Details** button to view the following details:

- List of data sources for which the operation succeeded.
- List of data sources for which the operation failed, and the reason for failure.

You can fix the errors and retry the operation on the failed data sources.

### Migrate Data Sources to a Different Collector

If a collector VM is down or deleted, you can add a new collector VM and migrate data source from the old collector VM to the new collector VM.

1. Go to **Settings > Infrastructure and Support > Overview and Updates > Collector VMs** section, click the edit icon next to the collector name in the list.

   If the collector VM is down, you see the error message. The Edit Collector page lists all the data sources added to the collector.
2. To migrate a data source, click **Migrate** for a particular data source.
3. Enter the required information of the new collector on the Edit Account or Source page.

**Table 1308:**

| Fields | Description |
|---|---|
| Collector VM | Name of the new collector VM to which the data source has to be migrated |
| IP Address | Pre-filled IP/FQDN address of data source |
| Username | Username for the data source |
| Password | Password for the data source |

4. Click **Validate** and then click **Submit**.

   The data source is then deleted in the old collector VM and is added to the new collector VM.
5. Once the migration is successful, you will see the new collector VM against the data source in the **Enabled** column in the **Accounts and Data Sources** page.

   **Note:**

   - If you are migrating VMware vCenter to another collector VM, then ensure that you migrate the corresponding NSX Manager also to the same collector VM.
   - When you migrate NSX Manager to another collector VM, the child data providers such as NSX controller and NSX Edge are also migrated to the new collector VM.

### Delete a Data Source from VCF Operations for networks

If you do not want to view data from a data source or a data source is not in use, you can delete the data source from VCF Operations for networks.

> **Note:**
> If any data source is no longer available in your environment, you must delete that data source form VCF Operations for networks.

1. Sign in to the VCF Operations for networks web console.
2. Go to **Settings** > **Accounts and Data Sources**.
3. Click the **Delete Data Source** icon next to the data source you want to delete.
   The VCF Operations for networks prompts you for confirmation.
4. Click **Yes**.

> **Note:**
> After you remove a data source from the system, you can add the same data provider back only after two or more hours.

**Unregister a Customer**
To unregister or delete a customer from VCF Operations for networks, you must first delete all the collectors of that customer and then contact VMware Support to remove that customer from VCF Operations for networks.

## Configuring VCF Operations for networks Settings

You can configure various aspects of VCF Operations for networks from the **Settings** page.

### VCF Operations for Networks System Recommendations and Requirements

For optimum performance, you must match the minimum recommendations for the deployment.

#### Recommendations for the Platform Deployment

**Table 1309: Specifications for Platform Brick Size**

| Brick Size | Cores required for 2.1 GHz CPU | Cores required for 2.3 GHz CPU | Cores required for 2.6 GHz CPU | RAM | Disk |
|---|---|---|---|---|---|
| Small | 5 | 5 | 4 | 16 GB | 1 TB |
| Medium | 10 | 9 | 8 | 32 GB | 1 TB |
| Large | 15 | 14 | 12 | 48 GB | 1 TB |
| Extra Large | 20 | 18 | 16 | 64 GB | 2 TB |
| 2 Extra Large | 30 | 27 | 24 | 128 GB | 3 TB |

> **Note:**
> - The Small brick size has limited functionality and must be used for evaluation purposes only. Do not use this brick size in production environments.
> - The reservation for the CPU speed and RAM for each node must be 100% of the value specified above.
> - If the disk in any of the platform nodes exceeds 95% of the capacity, VCF Operations for networks UI will not be accessible.
> - To match your setup to all the specifications, you might have to add the resources (RAM, Disk, CPU). See Broadcom knowledge base article 53550 and Increase the Brick Size of Your Setup.

**Table 1310: Non-Cluster Deployment - Maximum Capacity**

| Brick Size | *Number of VMs (K = Thousand) | Flows per Day (M = Million) | Total Flows (M = Million) | Flow Planning (M = Million) | Metric Capacity (metric points per day) | Number of Network Rules for Network Map and Intents (K = Thousand) | Number of VMs for Flow Based Application Discovery | Number of UI Entities in the Network Map Topology |
|---|---|---|---|---|---|---|---|---|
| Medium | 4K | 1M | 4M | 2M | 1 | Not Applicable | Not Supported | Not Supported |
| Large | 6K | 2M | 8M | 4M | 1.1 | Not Applicable | Not Supported | Not Supported |
| Extra Large | 10K | 2M | 8M | 4M | 2.5 | 170K | 3K | 5K |
| 2 Extra Large | 15K | 3M | 12M | 4M | 3.9 | 170K | 3K | 5K |

**Note:**

- The Network Verification and Assurance (Network Map and Intents) and Flow Based Application Discovery (Flows) features are available only for Extra Large brick size only.
- * The count of VMs and edges mentioned in the table is the maximum individual limit for a single deployment. So, if you have edges in your setup, you might have to reduce the VM count.
- The count of VMs includes the templates on the VMware vCenter as well.
- Total Flows is the maximum count of flows the system can store for the retention period.
- Flow Planning is the total flows for which the system can perform security planning.

**Table 1311: Cluster Deployment - Maximum Capacity**

K=Thousand

M=Million

| Scenario | Brick Size | Cluster Size | Number of VMs | Flows per Day | Total Flows | Flow Planning | Metric Capacity (metric points per day) | Network Rules for Maps and Intents | Number of VMs for Flow Based Application Discovery | Number of UI Entities in Network Map Topology |
|---|---|---|---|---|---|---|---|---|---|---|
| Scenario 1 | Large | 3 | 10K | 2M | 8M | 4M | 1.7 | Not Applicable | Not Supported | 5K |
| | Extra Large | 3 | 18K | 6M | 24M | 6M | 3.8 | 170K | 3K | 5K |
| | Extra Large | 5 | 30K | 10M | 40M | 10M | 6.3 | 170K | 3K | 5K |
| | Extra Large | 7 | 58K | 12M | 48M | 10M | 8.9 | 170K | 3K | 5K |
| | Extra Large | 10 | 100K | 15M | 60M | 10M | 12.7 | 170K | 3K | 5K |
| | Extra Large | 15 | 150K | 20M | 80M | 10M | 18.7 | 170K | 3K | 5K |
| | 2 Extra Large | 3 | 27K | 9M | 36M | 6M | 5.8 | 170K | 3K | 5K |
| | 2 Extra Large | 5 | 45K | 15M | 60M | 10M | 9.7 | 170K | 3K | 5K |
| | 2 Extra Large | 10 | 150K | 22.5M | 90M | 10M | 19.5 | 170K | 3K | 5K |

| Scenario | Brick Size | Cluster Size | Number of VMs | Flows per Day | Total Flows | Flow Planning | Metric Capacity (metric points per day) | Network Rules for Maps and Intents | Number of VMs for Flow Based Application Discovery | Number of UI Entities in Network Map Topology |
|---|---|---|---|---|---|---|---|---|---|---|
| | 2 Extra Large | 15 | 225K | 30M | 120M | 10M | 29.2 | 170K | 3K | 5K |
| Scenario 2 | Extra Large | 3 | 12K | 3M | 12M | 4M | 3.8 | 2M | 12K | 100K |
| | Extra Large | 5 | 18K | 6M | 24M | 6M | 6.3 | 5M | 24K | 100K |
| | Extra Large | 7 | 30K | 10M | 40M | 10M | 8.9 | 5M | 24K | 100K |
| | Extra Large | 10 | 72K | 13M | 52M | 10M | 12.7 | 5M | 24K | 100K |
| | Extra Large | 15 | 130K | 18M | 72M | 10M | 18.7 | 5M | 24K | 100K |
| | 2 Extra Large | 3 | 18K | 4.5M | 18M | 4M | 5.8 | 2M | 12K | 100K |
| | 2 Extra Large | 5 | 27K | 9M | 36M | 6M | 9.7 | 5M | 24K | 100K |
| | 2 Extra Large | 10 | 108K | 19.5M | 78M | 10M | 19.5 | 5M | 24K | 100K |
| | 2 Extra Large | 15 | 195K | 27M | 108M | 10M | 29.2 | 5M | 24K | 100K |

**Note:**

- Metric capacity is calculated using a 31 day retention period and recommended disk allocation for the brick size. Actual capacity can vary if parameters are adjusted.
- Scenario 1 focuses on VMs and flows while running automated Application Discovery and Network Assurance and Verification features with limited capacity. Scenario 2 is for an environment that needs to run automated Application Discovery and Network Assurance and Verification feature with full capacity.
- The Network Verification and Assurance (Network Map and Intents) and Flow Based Application Discovery (Flows) features are available only for Extra Large brick size only.
- The number of UI entities means the sum of nodes and edges supported in the Network Map topology.
- The number of rules includes all forwarding entries, including layer 3, layer 2, access control, and NAT.
- * The count of VMs and edges mentioned in the table is the maximum individual limit for a single deployment. So, if you have edges in your setup, you might have to reduce the VM count.
- The number of VMs includes the templates on the VMware vCenter as well.
- Cluster size is the total number of nodes in the cluster.
- Total Flows is the count of flows in the system for the retention period.
- The query to determine the Total Flows is `count of flows in last 31 days`, assuming the retention period as 31 days.
- Flow Planning is the total flows for which the system can perform security planning.
- VCF Operations for networks supports maximum 10,000 security groups and 10,000 IPSets for a cluster deployment of 10 XL.
- The following cluster configurations are deprecated:
  - 3 L cluster: users should migrate to a 3XL cluster.
  - 7 XL cluster: both scenario 1 and 2 are not supported.
  - 15 XL cluster: users should migrate to a 15 XXL cluster.

## Recommendation for the Collector Deployment

**Table 1312: Specifications for Collector Brick Size**

| Brick Size | Cores required for 2.1 GHz CPU | Cores required for 2.3 GHz CPU | Cores required for 2.6 GHz CPU | RAM | Disk |
|---|---|---|---|---|---|
| Small | 2 | 2 | 2 | 6 GB | 250 GB |
| Medium | 5 | 5 | 4 | 12 GB | 250 GB |
| Large | 10 | 9 | 8 | 16 GB | 250 GB |
| Extra Large | 10 | 9 | 8 | 24 GB | 250 GB |
| 2X Large | 20 | 18 | 16 | 48 GB | 350 GB |

**Note:**

The reservation for the CPU speed and RAM for each node must be 100% of the value specified above.

**Table 1313: Collector Deployment - Maximum Capacity**

| Collector Size | Number of VMs (K = Thousand) | Flows per Day (M = Million) | Flow count in 4 days (M = Million) | Number of Edges for VMware SD-WAN (K = Thousand) |
|---|---|---|---|---|
| Medium | 4K | 2.5M | 3.25M | 4K |
| Large | 10K | 5M | 6.5M | 6K |
| Extra Large | 35K | 10M | 13M | 10K |
| 2X Large | 45K | 17M | 22M | 10K |

**Note:**

- The count of VMs and edges that are mentioned in the table is the maximum individual limit for a single deployment. So, if you have edges in your setup, you might have to reduce the VM count.
- The count of VMs includes the templates on the VMware vCenter as well.
- For a single deployment with more than one collector, the limitation on the total flows across collectors is based on the capacity of the platform.

## Other Requirements and Considerations

- While editing a data source collector, other data source fields cannot be edited.
- During a fresh installation or upgrade to VCF Operations for networks in VCF 9, the support password will be equivalent to the root password. To modify the root password, log in to the appliance through SSH using the support user credentials, elevate to root using the `sudo su` command, and then update the root file using the `passwd` command.
- The network verification and assurance and flow based application discovery features are available for deployments using the Extra Large brick size.
- The maximum time skew between the platform nodes has to be lesser than 30 seconds.
- The availability of the NTP service is critical to system operations. Ensure that you do not reboot the platform node or the collector node when the NTP service is not available.
- When the existing compute resources are completely used by the other processes on the platform, VCF Operations for networks crashes and does not recover automatically. If the services fail to recover, reboot the platform node.
- If the network latency between platform node and upgrade server is greater than 500ms, the VCF Operations for networks upgrade might encounter an error. So, the network latency must be less than 500ms.
- The recommended network latency between platform VMs for optimal performance is up to 3ms. The system performance might degrade beyond the limit.
- The recommended network latency between platform and collector VMs for optimal performance is up to 150ms. The system performance might degrade beyond this limit.

- The recommended disk latency for optimal performance is up to 5ms. If the disk latency is greater than 5ms, the system performance degrades.
- The recommended disk IOPS is 7500.
- For optimum performance, the recommended bandwidth usage between the collector and the platform is 15 Megabytes per second (MBps) or 120 Megabits per second (Mbps).

  **Note:**

  These bandwidth recommendations are made considering the largest collector and the largest platform brick sizes mentioned in this topic.
- For Network Map, the maximum supported firewall rules per VMware NSXManager (including of DFW and edge rules) is 5000.
- To use the Assurance and Verification feature in VCF Operations for networks, you must enable Link Layer Discovery Protocol (LLDP) and Cisco Discovery Protocol (CDP) (if supported) on the network devices.
- As IPv6 addresses occupy more space in memory and disk, enabling IPv6 flow processing can reduce the platform and the collector capacity by up to 20%.

### Supported TLS Versions

VCF Operations for networks supports Transport Layer Security (TLS) versions 1.2 and 1.3.

### Supported Web Browser

- Google Chrome: The latest two versions.
- Mozilla Firefox: The latest two versions.

### Recommendations to Support High Availability

You can customize vSphere HA options to enable vSphere high availability.

- **Host Failure** - Restart VMs
- **Host Isolation**- Deactivated
- **Guest not heartbeating**- Deactivated

## Increase the Brick Size of Your Setup

To match your requirements, you can change the brick size of your appliance from `MEDIUM` to `LARGE` or `LARGE` to `EXTRA-LARGE`.

To increase the brick size of your setup, do the following as necessary:

- Increase the disk size. For more information, see the Increase the Disk Size topic.
- Increase RAM, total vCPU and corresponding reservation. To increase RAM, total vCPU and corresponding reservation, do the following steps that are relevant to your setup:

| Option | Description |
|---|---|
| **For a single node platform or fresh independent OVA** | 1. Log in to VMware vCenter.<br>2. Shutdown the Platform VM using the **Shut Down Guest OS** option.<br>**Note:**<br>Do not use the **Power Off** option.<br>3. Increase RAM, total vCPU and corresponding reservation of the VM as required to match the target brick size.<br>4. **Power On** the Platform VM. |
| **For a cluster setup** | 1. Log in to VMware vCenter.<br>2. Shutdown the Platform VM using the **Shut Down Guest OS** option in the reverse chronological order. For example: Shut down from Node 3 to Node 1. |

| Option | Description |
| --- | --- |
| | **Note:**<br>Do not use the **Power Off** option.<br>After you shut down each Platform node, you must wait for 4-5 minutes for services to stabilize for the remaining Platform nodes.<br>3. Increase RAM, total vCPU and corresponding reservation as required to match the target brick size.<br>4. **Power On** the Platform VMs in the chronological order. For example: Restart from Node 1 to Node 3. |
| **For a collector** | 1. Log in to VMware vCenter.<br>2. Shutdown the Collector VM using the **Shut Down Guest OS** option.<br>**Note:**<br>Do not use the **Power Off** option.<br>3. Increase RAM, total vCPU and corresponding reservation of the VM as required to match the target brick size.<br>4. **Power On** the Collector VM. |

**Note:**
After you increase the brick size of the platform VM, you must reconfigure the platform VM.

### Increase the Disk Size

In VCF Operations for networks, you can add additional disk space to a VCF Operations for networks Platform node.

Ensure that the nodes are in the powered-on state, and you have the correct privileges.

If the disk usage on the Platform are high, adding more disk space helps you prevent any unwanted service interruption.

**Note:**
Do not expand the existing disks available in the VM. You must always add a new disk in the VM.

**Note:**
Do not reboot any VMs until VCF Operations for networks detects the new disk and the new disk space is added in the `/var` partition.

1. Log in to VMware vCenter using the web client with correct privileges.
2. Right-click the VM, and click **Edit Settings**.
3. Add a disk of required size.

   A cron job runs every 5 minutes and checks for any new disks added to the VM, then automatically expands the required partition.

Wait for 10-15 minutes for VCF Operations for networks to detect the disk, and then check and ensure that the additional disk size is visible in the `/var` partition.

In case troubleshooting is required, the cron job script location is: `/home/ubuntu/build-target/common-utils/cron.5m/run-expandpart.sh`. You can find the related log file at `/home/ubuntu/logs/disk_util.log`.

## Create a Support Bundle

You can create a support bundle that collects diagnostic information such as product-specific logs, configuration files of your setup. When you raise a support request, VMware Technical Support uses this information to troubleshoot your setup issues.

1. On the Settings page, click **Infrastructure and Support** > **Support**.
2. Select the platform VMs and the collector VMs for which you want to create the support bundle.

   To select all VMs, click the check box in the header of the platform VMs and the collector VMs tables.

3. Click **Create Support Bundle**.
4. Click **Yes** to confirm creation of a new support bundle.
   VCF Operations for networks takes some time to complete the creation of the bundle.

A new support bundle is created displaying date and time. To initiate the download of support bundle, click the **Download** link next to the respective VM.

**Note:**

- The support bundle creation on a medium sized system can take in excess of fifteen minutes.
- Only two support bundles can be present at one given time. So, while creating a new one, if there are already two support bundles present, the older one is deleted.

Attach the support bundle to your service request for VMware to access the details.

## Viewing System Health

In VCF Operations for networks, you can view the health status of your system. The health of your system is determined by process lag, indexer lag, and grid usage.

If all these parameters are in the green state, your system health is good. If any one of these three parameters is in the red state, your system health is bad. The overall health indicator displays the number of irregularities, with a Red light. The individual irregularities are listed with their details, when the number of problems against overall health, is clicked on. In case of normal functioning, the health indicator shines a Green light. The **Health** indicator turns red if any of the following malfunctioning events occur:

- If the collector stops collecting flow data
- If platform stops processing data due to some reason; for example, insufficient disk space
- If search indexer lags behind, resulting in outdated search result

**Note:**
VCF Operations for networks might not detect an out-of-sync system clock sometimes. If the clock is not in sync with NTP, some services may become unhealthy or stop working.

On the **Settings** page, click **Infrastructure and Support** > **Infrastructure and Updates**.

The health of your system is displayed in the **System Health** section.

**Note:**
If your system health is in the bad state for more than six hours, you must contact the VCF Operations for networks support.

### Viewing System Usage

In VCF Operations for networks, you can view and track the use of resources and see how these entities have changed over time.

The counts of entities used are tracked as metrics and these metrics are tracked once every 24 hours. These metrics have a data retention period of up to 13 months or up to the data retention interval set in the **Data Management** tab of the **Settings** page. The trending metric graphs show the counts of VMs, hosts, applications, firewall rules, daily flows, and weekly flows.

To view the metric graphs:

1. Go to **Settings** > **Infrastructure and Support** > **Infrastructure and Updates**.
2. On the Infrastructure and Updates page, click **Go to Operations for Networks System Dashboard**.
3. You can see the metric graphs under the **System Usage** section of the Overview page.

## Understanding Capacity for Collector and Platform Load

VCF Operations for networks provides the approximate capacity and load information of a collector node and a platform. This limits-based information helps you to prevent the performance and experience issues later.

### Understanding Capacity

There are two kinds of capacity:

- VM capacity: It is defined as the number of discovered VMs that a node or a setup can handle.
- Flow capacity: It is defined as the number of flows that a node or a setup can handle.

The capacity is defined as follows:

- Single platform with one or more collector nodes: The capacity of a collector node or the platform is the number of discovered VMs that it can handle without the degradation of performance.
- Cluster setup: The capacity of the platform in a cluster setup is the aggregation of all the capacities of all the platform nodes while the capacity of collector nodes is considered at the level of an individual node.

### Accessing the Capacity Information

You can view **VM Capacity** and **Flow Capacity** on the **Infrastructure and Support** > **Infrastructure and Updates** page.

For every collector node listed under Collector VMs, only the VM capacity information is provided.

> **Note:**
> When the number of discovered VMs from the data sources across the deployment exceed the capacity of either the system or the collector or both, you will not be allowed to trigger the upgrade.

To view the discovered VMs for a data source:

1. In the **Accounts and Data Sources** page, you can see the number of VMs that have been discovered for a particular data source which is already added and currently active. This column will have a value only if the data source is vCenter or AWS source.

   > **Note:**
   > The discovered VM count includes placeholder and template VMs. So it can be different from the count of VMs in the product.

## View Platform Capacity

In VCF Operations for networks, you can view the platform capacity and usage.

In the **Platform Capacity**, you see information about VMs, Active Flows, All Flows, and Devices. To view the **Platform Capacity**, do the following:

On the **Settings** page, click **Infrastructure and Support** > **Infrastructure and Updates**.

In the **Infrastructure and Updates** page, you can see the **Platform Capacity**.

### Resolving Platform Capacity Problems

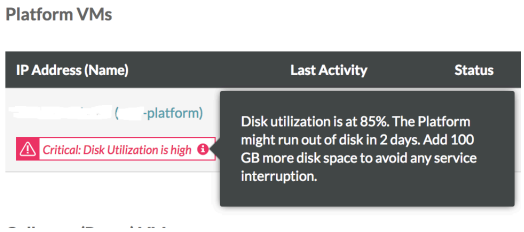
Use the information provided in the following table to resolve platform capacity-related problems.

| Error Text | Cause | Resolution |
|---|---|---|
| `Capacity configuration is mismatched.` | There is a change in the license or the brick size of the setup. | Click **RECONFIGURE**. |
| `System usage has breached the capacity.` | Your system usage has crossed the stipulated platform capacity. | You must do, one of the following:<br>• Click the **RECONFIGURE** button (if available).<br>• If you do not see the **RECONFIGURE** button, then you must add more resources to increase capacity by doing one of the following:<br>– Increase the brick size (if you are using medium or large brick).<br>– Create cluster (if you are using single node deployment).<br>– Expand the cluster (if you are using clustered deployment).<br>To know about creating and expanding clusters, see Creating and Expanding Clusters. |
| `Capacity reconfiguration failed.` | The problem can occur because of a various technical reasons. | Contact VMware Support. |

## Manage Disk Utilization

If the disk utilization is high for a platform or a collector, VCF Operations for networks triggers an alert to warn the user to manage the disk space. VCF Operations for networks also provides a recommendation of how much more disk space is to be added.

To view the alerts related to disk utilization, navigate to the platform dashboard or the collector dashboard. You can also view these alerts in the corresponding collector or the platform section in the **Infrastructure and Support** > **Infrastructure and Updates** page.




To add a hard disk to a node, perform the following steps:

> **Note:**
> Do not expand the existing hard disk.

1. Log in to VMware vCenter through the Web client with sufficient privileges.
2. Right-click the node and select **Edit settings**.
3. Add the hard disk as per recommendations provided in the alert.
   VCF Operations for networks takes a few minutes to detect the appliance and add it to the `/var` partition.

## View the Platform and Collector Details

In VCF Operations for networks, you can view the details of each platform node and collector node.

1. To view the details of a particular platform node, go to **Settings** > **Infrastructure and Support** > **Infrastructure and Updates** > **Platform VMs** and click its name from the list.

   The VCF Operations for networks platform dashboard appears.

2. To view the details of a particular collector node, go to **Settings** > **Infrastructure and Support** > **Infrastructure and Updates** > **Collector VMs** and click its name from the list.

   The VCF Operations for networks collector dashboard appears.

## Configure Data Retention Interval

In VCF Operations for networks, you can specify the duration for which you want to retain your data.

> **Note:**
> VCF Operations for networks supports configurable data management on an enterprise license only. In the advanced license edition, the data retention defaults to 1 month.

The data is divided into the following categories:

**Table 1314:**

| Category | Minimum Value | Maximum Value |
|---|---|---|
| Alerts | 1 month | 13 months |
| Entities and Configuration Data | 1 month | 3 months |
| Metrics | 1 month | 13 months |
| Flows | 1 month | 1 month |
| Miscellaneous Data | NA | 100 GB of additional disk space |

> **Note:**
> For all the categories, the minimum value is the default value.

Different policies can be configured and controlled for each category. You can configure the policy as per your requirement.

To configure data management:

1. Go to **Settings** and click **Data Management**.
2. When you log in for the first time, this page shows the default data.
3. Click **Edit** on the top-right corner of the page to change the data retention period for various categories of data.
4. Click **Submit**. The Updated information is recorded in the database.

> **Note:**
> The retention period for low-resolution metrics is longer than the high-resolution metrics.

## Configuring IP Properties and Subnets

In VCF Operations for networks, you can configure different IP properties for better security planning and identification.

### Import the DNS mapping file

You can upload your DNS mapping file on VCF Operations for networks to provide information about flows between physical devices. The supported file formats are the Bind and CSV file format. Ensure that you have placed these files in a single ZIP file.

> **Note:**
> VCF Operations for networks does not support password-protected ZIP files.

1. Go to **Settings** > **IP Properties and Subnets** > **Physical IP and DNS Mapping**.
2. Click **Upload** to upload your DNS mapping file. After you select and upload the file, click **Validate**. The number of DNS records is displayed after validation.

   The **Upload** operation removes any existing DNS mappings and replaces them with the mappings that are being imported.

   The DNS Mapping file consists of the following fields:

   - Host Name
   - IP Address (IPv4 or IPv6 address)
   - Domain Name

## Configure Mapping Between Subnet and a VLAN

You can define a mapping between subnet and a VLAN.

You can use this mapping for the following:

- Enriching the information about the IP entities that are learned from physical to physical flows by adding the source and destination subnets and the Layer2 networks associated with the flow.
- Planning the network topology based on the subnet and VLAN for physical addresses.

1. Go to **Settings** > **IP Properties and Subnets** > **Physical Subnets and VLANs**.

   This page lists all the subnets and the associated VLAN IDs.
2. Click **Add** to add the subnet and VLAN information.
3. Click **Submit**.
4. After defining the mapping information, you can only edit the VLAN ID that is associated with the subnet.You cannot change to the subnet CIDR associated with the VLAN ID. To edit a subnet associated with the VLAN ID, delete the subnet and create a new subnet VLAN mapping with the updated values.

   When the subnet-VLAN mapping information is updated, a new VLAN is created for the specified VLAN ID and the subnet information is associated with this VLAN.
5. To delete the subnet-VLAN ID mapping, click **Delete**.

   **Note:**
   All VLAN creation, updating, and deletion operations does not happen immediately after the subnet and VLAN mappings are created. It takes some time for the changes to be propagated and the corresponding VLAN to be created or modified.

## Configuring East-West and North-South IPs

In VCF Operations for networks, you can override IP addresses, IP ranges, and subnets as either private IPs (East-West IPs) or public IPs (North-South IPs).

### East-West IPs and North-South IPs

The IP addresses that are within the range of RFC1918 standard and the IP addresses of the VMs in VCF Operations for networks are considered as East-West IPs or private IPs.

The IP addresses that are outside RFC1918 are considered as North-South IPs or public IPs.

VCF Operations for networks supports both IPv4 and IPv6 addresses.

### Configuring East-West and North-South IPs

1. Go to **Settings** > **IP Properties and Subnets** > **Custom IPs** and click **Customize**.
2. In the **East-West IPs** text box, enter custom IPs that you want to override as private IPs. Custom IPs can be IP addresses, IP ranges, or subnets.
3. In the **North-South IPs** text box, enter custom IPs that you want to override as public IPs.

4. Click **Save.**

While adding the IP addresses, you can do the following actions:

- To add a single IP address, specify the IP address and press Enter.
- To add multiple IP addresses, use commas.
- To add an IP range or a subnet, specify the IP entries and press Enter.
- To add an IP range or a subnet and exclude a few IP addresses, click **Add Exceptions**, enter the exception IPs, and click **Done**.

  **Note:**
  You can add IP addresses, IP ranges, and subnets as exceptions. The IP addresses that you add as exceptions in the East-West IPs text box are considered as North-South IPs and vice versa.

## Configuring Alerts and Notifications

In VCF Operations for networks, you can configure various types of alerts and notifications. VCF Operations for networks creates an alert whenever the system meets a preset rule.

### Viewing and Editing Alert Definitions

In VCF Operations for networks, you can view various alert definitions and list of alerts related to the alert definitions.

VCF Operations for networks has the following types of alert definitions:

- **Problems**: Logical or physical problems detected by VCF Operations for networks in the environment. A problem can be something involving any entity, property, or metric. For example, a router interface failing or percent of packet drops of an interface crossing a certain threshold can be detected as a problem.
- **Changes**: Changes that are detected by VCF Operations for networks in the environment for an entity, or properties and metrics of an entity. Create, update, and delete qualifies as a change.
- **Others**: Alerts that are generated by external data sources, and not defined by VCF Operations for networks. For example, NSX-T alerts, Kubernetes alerts, and so on.

To view the alert definitions, on the **Settings** page, click **Alerts and Notification** > **Alert Definitions**.

All the alert definitions of type Problem are displayed in the **Problems** tab.

**Note:**
All the alerts of type **Problem** are also logged into syslog.

All the alert definitions of type Changes are displayed in the **Changes** tab.

All the alert definitions of type Others are displayed in the **Others** tab.

When you point the mouse on each alert, you can see the name, the description, alert tags for that alert.

You can perform the following tasks on the alerts:

- Edit an alert
- Deactivate an alert.
- Perform bulk edit, enable, and deactivate

#### Edit Alert Definitions

You can edit alert definitions and set notifications for the preferred system events.

1. Click the edit icon on the **Action** column for the problem you want to edit.
2. Change the severity.

   **Note:**
   Applicable only for Problem type - Alert Definition.

3. Add or remove event tags if required.

   **Note:**
   Applicable only for Problem type - Alert Definition.

4. Select Include/Exclude entities if you want the alert to be enabled or deactivated for selected entities.

   - To create inclusion rules:

     a. Select **Include/Exclude entities** check box.
     b. Click **INCLUDE**.
     c. Specify the entities which you want to include from the drop-down menu.
        You can specify multiple entities in the inclusion rule by adding multiple conditions by clicking **ADD CONDITION** button.

   - To create exclusion rules:

     a. Select **Include/Exclude entities** check box.
     b. Click **EXCLUDE**.
     c. Specify the entities which you want to exclude from the drop-down menu.

   **Note:**

   - You can create multiple rules in both inclusion and exclusion lists by clicking **ADD CONDITION** button.
   - You can also specify **Custom search** by writing your own query to include or exclude entities.

   **Note:**
   Applicable only for Problem type - Alert Definition.

5. To get alert notification, do the following:

| Option | Action |
|---|---|
| **If you do not have an email server configured** | Click **Configure Mail Server**. To know how to configure a mail server, see Configure Mail Server. |
| **If you have already configured an email server** | Select **Email immediately**, **Email daily digest**, or both to specify the frequency at which you want to receive the emails.<br>If you select **Email immediately**, you get email notification immediately when an alert is raised.<br>If you select **Email daily digest**, you get email notification at a pre-select time every day for all the alerts that are raised in the system. |
| **If you do not have an SNMP trap configured** | Click **Configure SNMP Trap**. To know how to configure an SNMP trap, see Configure SNMP Trap Destination. |
| **If you have already configured an SNMP trap configured** | Select one or more SNMP trap destinations from **Send SNMP Trap to** drop-down menu. You can select up to four SNMP trap destinations. |

6. Click **SUBMIT**.

**Perform a Bulk Edit on an Alert**

In VCF Operations for networks, you can edit multiple alert definitions at once.

1. In the **Alert Definitions** page, select the alerts that you want to edit.
   When you select multiple alerts, the options **ENABLE**, **DISABLE**, and **EDIT** appear above the list.

2. Click **EDIT**, and make the required changes.

| Option | Action |
|---|---|
| **Severity** | Select an option from the drop-down menu.<br>You see the following options:<br>• **No change**: Default value |

| Option | Action |
|---|---|
| | • **Override existing values**: Overwrites the existing values and sets the new value as per your selection. |
| **Tags** | Select an option from the drop-down menu.<br>You can overwrite existing tags, add new tags, or remove existing tags. |
| **Email immediately** | Select an option from the drop-down menu.<br>If you select **Enable**, you get email notification immediately when an alert is raised. |
| **Email daily digest** | Select an option from the drop-down menu.<br>If you select **Enable**, you get email notification at a pre-select time every day for all the alerts that are raised in the system. |
| **SNMP Trap** | Select an option from the drop-down menu.<br>You can enable or deactivate SNMP traps. |

3. Click **SUBMIT**.

**Deactivate an Alert**
You can deactivate alerts as needed to stop receiving notifications when certain conditions are met.

To deactivate an alert:

1. Select the alert in the **Open Problems** section of the Homepage. Or, enter **Problems** in the search bar and select an alert from the list.
2. In the more option (three vertical dots), click **Archive**.
3. Select **Disable all alerts of this type in future for** and select an entity or all entities.
4. Click **SAVE**.

> **Note:**
> The changes made in severity, tags, or inclusion/exclusion rules will reflect for the future alerts. The existing alerts continue to show the old configuration.

**Alert Limitations**
This section provides the limitations for the various alerts.

**Distributed Firewall Rule Masked by Preceding Rule alert Limitation**

This alert has the following limitations:

- This alert is supported only for NSX-T distributed and Edge firewall rules, and NSX-T on VMware Cloud on AWS firewall rules. Other firewall vendors are not supported.
- The following firewall rule properties are currently supported for masking computation:
  - Source
  - Destination
  - Applied To
  - Service protocol and Port ranges
  - Packet type
  - Layer-7 application IDs
- Rules with source or destination inversion are not supported.
- Deactivated rules are ignored.
- Rules with security groups containing excluded members directly or indirectly in Source/Destination or Applied To is not supported.
- The masking computation for Source, Destination, and Applied To properties are based on the static membership and IP range overlap of member IPSets. Dynamic membership of a security group are not considered for masking.

## Create a Search-based Alert

In VCF Operations for networks, you can define an alert through the search. Whenever you perform search, you get a option to create an alert based on your search result.

All the alerts created by users are categorized as Search-based alerts. Search-based alerts were previously know as user defined events.

> **Note:**
> If you are an existing VCF Operations for networks user, and you have configured user-defined events in your VCF Operations for networks, then when you upgrade to VCF Operations for networks version 6.1, you see all your user-defined events under the **Alert Definitions** page. Based on the event type, you can find your events under either **Problem** or **Change** tab in the **Alert Definitions** page.

1. Click

   located in upper-right corner on the search result window.
2. In the **Define Alert** page, do the following:

| Option | Action |
|---|---|
| **Name** | Enter a unique name for the alert. |
| **Search query** | Shows the search criteria for the alert. |
| **Generate alert when** | Select when you want to generate the alert. |
| **Definition type** | Select an alert type. |
| **Severity** | This field specifies the severity of the alert. You can set it to the following values:<br>• Critical<br>• Moderate<br>• Warning<br>• Info |

3. To configure how you want to send the notification, select **Alert Notification** check box and do the following:

| Option | Action |
|---|---|
| **Email frequency** | Select how frequently you want to get the notifications. |
| **Email addresses** | Enter an email address when you want to get the notifications. |
| **SNMP trap destination** | Select trap destinations from the drop-down.<br>You can select maximum four trap destinations. |

4. Click **SUBMIT**.

### Editing Search-based Alerts

In VCF Operations for networks, you can edit a search-based alert definitions.

You can edit the following options:

| Option | Action |
|---|---|
| Name | Provide a unique name. |
| Generate alert when | Select when you want to generate the alert. |
| Definition type | Select a alert type. |
| Severity | This field specifies the severity of the alert. You can set it to the following values:<br>• Critical<br>• Moderate<br>• Warning<br>• Info |

| Option | Action |
|---|---|
| Alert Notification | Configure how you want to send the notification. |

You can edit or delete the alert. While editing it, you can specify the email address and the frequency of the email notification.

## Viewing Platform Health Alerts

The Platform Health Alerts page is your one-stop page to view all the alerts that provide details on the overall health of the system. These alerts might have occurred on a data source or a node in the infrastructure. You can also view these alerts through search.

| Field | Description |
|---|---|
| Name | The name of the alert. |
| Severity | The severity of the alert. You cannot change the severity of the alert. |
| Type | The type of alert. The alert denotes a problem or a change. |
| Notifications | The types of notifications that are sent. The notifications can be sent as an email, an SNMP trap, or both. |

## Alerts List

Here is a list of all the out of the box (previously know as system defined) alerts in VCF Operations for networks. To receive notification about any of these alerts, you must enable notification for that particular alert.

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.110013 | SnapshotNotBuildingEvent | Critical | Unable to load latest network map | Unable to load latest network map |
| 1.3.6.1.4.1.6876.100.1.0.100001 | vmwNSXVLatencyNoDataEvent | Warning | NSXVLatencyNoDataEvent | Network Latency Collection Stopped |
| 1.3.6.1.4.1.6876.100.1.0.100051 | vmwVMCVMLimitExceededEvent | Critical | VMCVMLimitExceededEvent | Number of VMs in VMC SDDC exceeds limit.<br>**Note:** This alert is deprecated in VCF Operations for networks 6.1. |
| 1.3.6.1.4.1.6876.100.1.0.100052 | vmwVMCHostLimitExceededEvent | Critical | VMCHostLimitExceededEvent | Number of Hosts in VMC SDDC exceeds limit.<br>**Note:** This alert is deprecated in VCF Operations for networks 6.1. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.1510 | vmwKubernetesBaseEvent | Moderate | System Alerts from Kubernetes | Alert reported by kubernetes cluster |
| 1.3.6.1.4.1.6876.100.1.0.20001 | vmwEntityDiscoveryChangeEvent | Info | Discovery | This alert is raised when any new entity is discovered |
| 1.3.6.1.4.1.6876.100.1.0.20002 | vmwEntityPropertiesChangeEvent | Info | Configuration Change | This alert is raised when any property of the entity is changed |
| 1.3.6.1.4.1.6876.100.1.0.20003 | vmwFirewallNotInstalledOnHostEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.20004 | vmwHostWithStaleFirewallRulesEvent | Warning | Firewall rule table mismatched between host and NSX Manager | The distributed firewall rule table differs between the host and NSX Manager. |
| 1.3.6.1.4.1.6876.100.1.0.20005 | vmwIpAddressChangeEvent | Info | IP address change | This alert is raised when IP Address of VM is changed |
| 1.3.6.1.4.1.6876.100.1.0.20006 | vmwL2GatewayAnomalyEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.20007 | vmwL2NetworkAddressAnomalyEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.20008 | vmwL2NetworkDiameterExceededEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.20009 | vmwL2NetworkUplinkMissingEvent | Info | Uplink not found for distributed virtual port group | VxLan does not have an uplink on specified Host |
| 1.3.6.1.4.1.6876.100.1.0.20010 | vmwL2NetworkWithNoVMsEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.20011 | vmwLayer2NetworkDiameterChangedEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.20012 | vmwMTUMismatchEvent | Warning | MTU mismatched between VTEP and physical switch port | An MTU mismatch was found in the path between a VTEP and its physical switch port |
| 1.3.6.1.4.1.6876.100.1.0.20013 | vmwNetworkIsolationEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.20014 | vmwNoPathEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.20015 | vmwSpoofguardDisabledEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.20018 | vmwVMotionEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.20019 | vmwVMWithDisconnectedVnicsEvent | N/A | N/A | N/A |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.20020 | vmwVMWithMulipleVnicsOnDifferentVxlansEvent | N/A | N/A | vm %s is connected to more than one vxlans [%s] |
| 1.3.6.1.4.1.6876.100.1.0.20021 | vmwVMWithMulipleVnicsOnSameL2Event | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.20022 | vmwVMWithNoIpAddressEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.20023 | vmwVTEPMissingEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.20024 | vmwL2Event | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.20025 | vmwMembershipChangeEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.20026 | vmwSecurityGroupMembershipChangeEvent | Info | Security Group VM membership change | This alert is raised when membership of Security Group is changed |
| 1.3.6.1.4.1.6876.100.1.0.20027 | vmwFirewallRuleMembershipChangeEvent | Info | Firewall Rule VM membership change | This alert is raised when membership of Firewall Rule is changed |
| 1.3.6.1.4.1.6876.100.1.0.20028 | vmwVlanMembershipChangeEvent | Info | VLAN VM membership change | This alert is raised when membership of Vlan is changed |
| 1.3.6.1.4.1.6876.100.1.0.20029 | vmwVxlanMembershipChangeEvent | Info | VXLAN VM membership change | This alert is raised when membership of Vxlan is changed |
| 1.3.6.1.4.1.6876.100.1.0.20030 | vmwDeleteChangeEvent | Info | Delete change | This alert is raised when any entity is deleted |
| 1.3.6.1.4.1.6876.100.1.0.20031 | vmwVtepFailedPingEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.20034 | vmwEmptySearchStreamChangeEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.20035 | vmwSearchStreamMembershipChangeEvent | N/A | User Defined Change Alert | User defined Change Alert |
| 1.3.6.1.4.1.6876.100.1.0.20036 | vmwEmptySearchStreamProblemEvent | N/A | User-defined zero result issue | User defined Problem when search result is empty |
| 1.3.6.1.4.1.6876.100.1.0.20037 | vmwSearchStreamMembershipProblemEvent | N/A | User-defined change issue | User defined Problem when search result changes |
| 1.3.6.1.4.1.6876.100.1.0.20038 | vmwOspfConfigurationMismatchEvent | Moderate | OSPF area ID mismatched between DLR and Edge router | The OSPF area ID differs on connected router interfaces. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.20039 | vmwServiceVMNotHealthyEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.20040 | vmwServiceVMNotPoweredOnEvent | Critical | NSX infrastructure VM not powered on | The NSX infrastructure VM found to be in powered off state. Services provided by it may get impacted. The NSX infrastructure includes Controller cluster |
| 1.3.6.1.4.1.6876.100.1.0.20041 | vmwServiceVMHighCPUUsageEvent | Warning | High CPU reported for NSX infrastructure VM | An NSX infrastructure VM is experiencing high CPU. This condition may lead to a service disruption. |
| 1.3.6.1.4.1.6876.100.1.0.20042 | vmwServiceVMHighMemoryUsageEvent | Warning | High memory usage reported for NSX infrastructure VM | An infrastructure VM is experiencing a high memory condition. This condition may lead to an NSX service disruption. |
| 1.3.6.1.4.1.6876.100.1.0.20043 | vmwServiceVMHighDiskUsageEvent | Warning | High disk usage reported for NSX infrastructure VM | Most of the allocated disk space for an infrastructure VM has been consumed. The infrastructure VM may become inaccessible or lead to a service disruption. |
| 1.3.6.1.4.1.6876.100.1.0.20050 | vmwIPSetPropertiesChangeEvent | Info | IP Set properties change | This alert is raised when any property of IPSet is changed |
| 1.3.6.1.4.1.6876.100.1.0.20051 | vmwFirewallRulePropertiesChangeEvent | Info | Firewall Rule properties change | This alert is raised when any property of Firewall Rule is changed |
| 1.3.6.1.4.1.6876.100.1.0.20052 | vmwSecurityGroupPropertiesChangeEvent | Info | Security Group properties change | This alert is raised when any property of Security Group is changed |
| 1.3.6.1.4.1.6876.100.1.0.20053 | vmwIPSetMembershipChangeEvent | Info | IP Set membership change | This alert is raised when membership of IPSet is changed |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.20054 | vmwFirewallRuleMaskEvent | Warning | Firewall rule masked by preceding rule | A distributed firewall rule is masked by one or more preceding rules. This condition may indicate a configuration error |
| 1.3.6.1.4.1.6876.100.1.0.20056 | vmwSecurityMembershipChangeEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.20057 | vmwSecurityTagPropertiesChangeEvent | Info | Security Tag properties change | This alert is raised when any property of Security Tag is changed |
| 1.3.6.1.4.1.6876.100.1.0.20058 | vmwSecurityTagMembershipChangeEvent | Info | Security Tag VM membership change | This alert is raised when membership of Security Tag is changed |
| 1.3.6.1.4.1.6876.100.1.0.20059 | vmwHostDatastoreChangeEvent | Info | Datastores of Host changed | This alert is raised when Host Datastore is changed |
| 1.3.6.1.4.1.6876.100.1.0.20060 | vmwVMDatastoreChangeEvent | Info | Datastores of VM changed | This alert is raised when VM Datastore is changed |
| 1.3.6.1.4.1.6876.100.1.0.20061 | vmwVMSnapshotChangeEvent | Info | Snapshots of VM changed | This alert is raised when Snapshot of VM is changed |
| 1.3.6.1.4.1.6876.100.1.0.20062 | vmwVMVirtualDiskChangeEvent | Info | Virtual Disks of VM changed | This alert is raised when Virtual Disk of VM is changed |
| 1.3.6.1.4.1.6876.100.1.0.20063 | vmwIPSetDefinitionMismatchEvent | Info | IPSet definition mismatched between NSX Managers | An IPSet with the same name and different scopes is defined in two NSX Managers. This condition may indicate a configuration error. |
| 1.3.6.1.4.1.6876.100.1.0.20064 | vmwSegmentMismatchEvent | Info | Segment ID ranges overlap between two NSX Managers | The VXLAN segment ID ranges defined in different NSX Managers have overlapping ranges |
| 1.3.6.1.4.1.6876.100.1.0.20065 | vmwVtepEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.20066 | vmwVtepConfigurationFaultEvent | N/A | N/A | N/A |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.20067 | vmwDLRNetworksNotReachableEvent | Critical | DLR networks unreachable from NSX Edge or external router | One or more DLR networks cannot be reached from the uplink interface on the NSX Edge router. This condition suggests either an OSPF configuration error on the Edge-router/ DLR or route not configured on uplink router. |
| 1.3.6.1.4.1.6876.100.1.0.20068 | vmwVtepSubnetMismatchEvent | Moderate | VTEP IP subnet mismatched between host(s) and NSX prepared cluster | One or more host VTEPs' IP addresses are not on the same subnet as other VTEPs in the same cluster. This condition may result in network connectivity issues |
| 1.3.6.1.4.1.6876.100.1.0.20069 | vmwVtepCountMismatchEvent | Critical | Host's VTEP count mismatched with cluster | The host's VTEP count doesn't match the VTEP count of other hosts in the same cluster. VMs on this host connected to a logical switch may not be able to communicate. |
| 1.3.6.1.4.1.6876.100.1.0.20070 | vmwEdgeNetworksNotReachableEvent | Moderate | NSX Edge network unreachable from uplink router | One or more networks connected to an NSX Edge router are not reachable from an uplink router. |
| 1.3.6.1.4.1.6876.100.1.0.20089 | vmwNiInfraChangeEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.20090 | vmwDataSourceEnabledChangeEvent | Info | Data source has been enabled | This alert is raised when data source is enabled |
| 1.3.6.1.4.1.6876.100.1.0.20091 | vmwDataSourceDisabledChangeEvent | Info | Data source has been deactivated | This alert is raised when data source is deactivated |
| 1.3.6.1.4.1.6876.100.1.0.20092 | vmwDataSourceCreatedEvent | Info | Data source has been added | This alert is raised when a data source is added |
| 1.3.6.1.4.1.6876.100.1.0.20093 | vmwPlatformCpuCoreChangeEvent | Info | Platform CPU Cores Change | This alert is raised when CPU cores on platform is changed |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.20094 | vmwPlatformDiskChangeEvent | Info | Platform Disk Change | This alert is raised when Disk on platform is changed |
| 1.3.6.1.4.1.6876.100.1.0.20095 | vmwPlatformMemoryChangeEvent | Info | Platform Memory Change | This alert is raised when memory on platform is changed |
| 1.3.6.1.4.1.6876.100.1.0.20096 | vmwPlatformRebootedEvent | Info | Platform Rebooted | This alert is raised when platform is rebooted |
| 1.3.6.1.4.1.6876.100.1.0.20097 | vmwProxyCpuCoreChangeEvent | Info | Proxy CPU Cores Change | This alert is raised when CPU cores on collector is changed |
| 1.3.6.1.4.1.6876.100.1.0.20098 | vmwProxyDiskChangeEvent | Info | Proxy Disk Change | This alert is raised when Disk on collector is changed |
| 1.3.6.1.4.1.6876.100.1.0.20099 | vmwProxyMemoryChangeEvent | Info | Proxy Memory Change | This alert is raised when memory on collector is changed |
| 1.3.6.1.4.1.6876.100.1.0.20100 | vmwProxyRebootedEvent | Info | Proxy Rebooted | This alert is raised when collector is rebooted |
| 1.3.6.1.4.1.6876.100.1.0.20101 | vmwNIClusterChangeEvent | Info | Cluster has been expanded | This alert is raised when a platform is added to system |
| 1.3.6.1.4.1.6876.100.1.0.20102 | vmwNISystemProxyChangeEvent | Info | Proxy has been added/removed | This alert is raised when proxy is added or removed |
| 1.3.6.1.4.1.6876.100.1.0.20103 | vmwNIClusterCreateEvent | Info | Cluster has been created | This alert is raised when a cluster is created |
| 1.3.6.1.4.1.6876.100.1.0.30001 | vmwThresholdExceededEventCpuReady | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.30002 | vmwThresholdExceededEventCpuCoStop | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.30003 | vmwThresholdExceededEventDiskCommandAbortRule | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.30004 | vmwThresholdExceededEventIODeviceLatencyRule | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.30005 | vmwThresholdExceededEventIOKernelLatencyRule | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.30006 | vmwThresholdExceededEventMemorySwapInRule | N/A | N/A | N/A |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.30007 | vmwThresholdExceededEventMemorySwapOutRule | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.30008 | vmwThresholdExceededEventNetworkRxDropRule | Warning | Receive packet drops detected on host interface | A threshold-crossing amount of receive-side packet drops have been detected on a host interface. |
| 1.3.6.1.4.1.6876.100.1.0.30009 | vmwThresholdExceededEventNetworkTxDropRule | Warning | Transmit packet drops detected on host interface | A threshold-crossing amount of transmit-side packet drops have been detected on a host interface. |
| 1.3.6.1.4.1.6876.100.1.0.30010 | vmwAWSRegionSGLimitEvent | Critical | AWS Security Groups available with AWS Region. | AWS Security Groups available with AWS Region. |
| 1.3.6.1.4.1.6876.100.1.0.30011 | vmwAWSVPCSGLimitEvent | Critical | AWS Security Groups available with AWS VPC. | AWS Security Groups available with AWS VPC. |
| 1.3.6.1.4.1.6876.100.1.0.30012 | vmwAWSSGInboundRuleLimitEvent | Critical | Inbound Rules available with AWS Security Group. | Inbound Rules available with AWS Security Group. |
| 1.3.6.1.4.1.6876.100.1.0.30013 | vmwAWSSGOutboundRuleLimitEvent | Critical | Outbound Rules available with AWS Security Group. | Outbound Rules available with AWS Security Group. |
| 1.3.6.1.4.1.6876.100.1.0.30014 | vmwAWSInterfaceSGLimitEvent | Critical | AWS Security Groups available with AWS Interface. | AWS Security Groups available with AWS Interface. |
| 1.3.6.1.4.1.6876.100.1.0.30100 | vmwPacketDropEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.30101 | vmwSwitchPortPacketDropEvent | Warning | Packets dropped on switch port | Considerable count of packet drops detected at specified switch port |
| 1.3.6.1.4.1.6876.100.1.0.30102 | vmwRouterInterfacePacketDropEvent | Warning | Packets dropped on NSX Edge Gateway interface | A threshold-crossing amount of packet drops have been detected on the vnic interface of an NSX Edge Gateway. |
| 1.3.6.1.4.1.6876.100.1.0.30103 | vmwVnicPacketDropEvent | Warning | Packets dropped at VM | A threshold-crossing amount of packet drops have been detected on a VM interface. |
| 1.3.6.1.4.1.6876.100.1.0.30104 | vmwVTEPUnderlayPacketDropEvent | Moderate | VTEP Underlay Packets Drop | Considerable count of packets drop detected in VTEP underlay |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.30105 | vmwPnicUnderlyingSwitchPortPacketDropEvent | Warning | Packets dropped at PNIC underlying switch port | A threshold-crossing amount of packet drops have been detected on the switch port associated with the specified physical NIC. |
| 1.3.6.1.4.1.6876.100.1.0.30106 | vmwDevicePacketDropEvent | Warning | Packet drops detected on hardware gateway port | A threshold-crossing amount of packet drops have been detected on the specified device. |
| 1.3.6.1.4.1.6876.100.1.0.30110 | vmwSwitchPortUptimeThresholdRecededEvent | Warning | SwitchPortUptimeThresholdRecededEvent | Up time receded |
| 1.3.6.1.4.1.6876.100.1.0.30111 | SwitchPortOperationalDownEvent | Warning | Switch Port is operationally down | Switch Port is operationally down. |
| 1.3.6.1.4.1.6876.100.1.0.30112 | RouterInterfaceOperationalDownEvent | Warning | Router interface is operationally down | Router interface is operationally down. |
| 1.3.6.1.4.1.6876.100.1.0.30116 | UnderlayDeviceFanMalFunctionEvent | Warning | Underlay device fan removed or not working | Underlay device fan removed or not working alert. |
| 1.3.6.1.4.1.6876.100.1.0.30117 | UnderlayDeviceTemperatureThresholdExceededEvent | Warning | Underlay device temperature threshold exceeded | Underlay device temperature threshold exceeded alert. |
| 1.3.6.1.4.1.6876.100.1.0.30118 | UnderlayDeviceFexFanMalFunctionEvent | Warning | FEX offline | Fex fan removed or not working alert. |
| 1.3.6.1.4.1.6876.100.1.0.30119 | UnderlayDeviceFexPsMalFunctionEvent | Warning | Fex power supply removed or not working | Fex power supply removed or not working alert. |
| 1.3.6.1.4.1.6876.100.1.0.30120 | UnderlayDeviceModuleMalFunctionEvent | Warning | Underlay device module removed or not working | Underlay device module removed or not working alert. |
| 1.3.6.1.4.1.6876.100.1.0.30121 | UnderlayDevicePsMalFunctionEvent | Warning | Underlay device power supply removed or not working | Underlay device power supply removed or not working alert. |
| 1.3.6.1.4.1.6876.100.1.0.30122 | UnderlayDeviceBfdSessionRemovedEvent | Warning | Underlay device BFD session deleted | Underlay device BFD session deleted alert. |
| 1.3.6.1.4.1.6876.100.1.0.30123 | UnderlayDeviceLldpNeighbourRemovedEvent | Warning | Underlay device LLDP neighbour removed | Underlay device LLDP neighbour removed alert |
| 1.3.6.1.4.1.6876.100.1.0.30203 | vmwThresholdExceededEventDatastoreFreeSpaceWarning | N/A | N/A | N/A |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.30204 | vmwThresholdExceededEventDatastoreFreeSpaceCritical | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.30205 | vmwThresholdExceededEventDatastoreReadLatency | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.30206 | vmwThresholdExceededEventDatastoreWriteLatency | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.35001 | vmwDistributedFirewallApplyHostEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.35002 | vmwDistributedFirewallApplyVMEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.35003 | vmwNsxEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.35004 | vmwFeatureImpactedEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.35221 | vmwNSXComponentEvent | Critical | NSX management service not running | An NSX management appliance service is down |
| 1.3.6.1.4.1.6876.100.1.0.35222 | vmwNSXBackupEvent | Info | NSX Manager backups not configured | NSX Manager backups are not configured. Proper backup of all NSX components is crucial to restore the system to its working state in the alert of a failure |
| 1.3.6.1.4.1.6876.100.1.0.35223 | vmwNSXBackupAuditLogExcludedEvent | Info | Audit logs excluded from NSX Manager backup | Audit logs are currently excluded from backup |
| 1.3.6.1.4.1.6876.100.1.0.35224 | vmwNSXUnsecureBackupEvent | Info | NSX Manager backups not configured for SFTP | Secure FTP is not currently being used for backups |
| 1.3.6.1.4.1.6876.100.1.0.35225 | vmwNSXBackupSystemEventsExcludedEvent | Info | System alerts excluded from NSX Manager backup | System alerts are currently excluded from backup |
| 1.3.6.1.4.1.6876.100.1.0.35226 | vmwNSXBackupNotScheduledEvent | Info | Scheduled NSX Manager backups not enabled | Scheduled backups of the environment have not been configured |
| 1.3.6.1.4.1.6876.100.1.0.35227 | vmwNSXBackupNotRecordedEvent | Info | NSX Manager backup not recorded | A backup of the environment has not been performed. Proper backup of all NSX components is crucial to restore the system to its working state in the alert of a failure |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.35228 | vmwNSXNtpServerEvent | Info | NTP server not configured for NSX Manager | No NTP server is configured on the NSX Manager |
| 1.3.6.1.4.1.6876.100.1.0.35229 | vmwNSXSysLogServerEvent | Info | Syslog server not configured for NSX Manager | There is no syslog server configured on the NSX Manager. Syslog data is useful for troubleshooting and reviewing data logged during installation and configuration |
| 1.3.6.1.4.1.6876.100.1.0.35230 | vmwControllerSysLogServerEvent | Info | Syslog server not configured for NSX Controller | No syslog server is configured for the NSX Controller. Syslog data is useful for troubleshooting and reviewing data logged during installation and configuration |
| 1.3.6.1.4.1.6876.100.1.0.35231 | vmwNSXIpV6EnabledEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.35232 | vmwNSXOspfNeighborDownEvent | Warning | One or more OSPF neighbors unreachable from NSX Edge router | One or more OSPF neighbors connected to the NSX Edge are down |
| 1.3.6.1.4.1.6876.100.1.0.36022 | vmwClusterFeatureVersionMismatchEvent | Info | NSX feature version mismatched with ESX cluster version | The NSX feature version of the prepared cluster doesn't match with that of NSX Manager. |
| 1.3.6.1.4.1.6876.100.1.0.36023 | vmwHostFeatureVersionMismatchEvent | Info | NSX feature version mismatched between host and cluster | The host's fabric status resource feature version is not same as that of the cluster or NSX Manager |
| 1.3.6.1.4.1.6876.100.1.0.36024 | vmwFeatureVersionMismatchEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.36025 | vmwHostFeatureEnabledMismatchEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.36026 | vmwHostFeatureInstalledMismatchEvent | Info | Network fabric feature status mismatched between host and cluster | The network fabric feature status for a host does not match the status of other hosts in the cluster. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.36027 | vmwHostVtepNotFoundEvent | Critical | No VTEP found on prepared host | A host in a cluster prepared for NSX is missing at least one VTEP. VMs on this host connected to any logical switch may not be able to communicate. |
| 1.3.6.1.4.1.6876.100.1.0.36028 | vmwHostVtepDisconnectedEvent | Warning | Host's VTEP is administratively deactivated | The host's VTEP has been deactivated and is in a not connected state. |
| 1.3.6.1.4.1.6876.100.1.0.36029 | vmwHostVtepEvent | Critical | Host Vtep Disconnected | Host Vtep Disconnected |
| 1.3.6.1.4.1.6876.100.1.0.36030 | vmwClusterHostsVtepMTUMismatchEvent | Warning | VTEP MTU mismatched between host and NSX prepared cluster | The VTEP MTU does not match between a host and the NSX prepared cluster. |
| 1.3.6.1.4.1.6876.100.1.0.36031 | vmwFeatureUnhealthyEvent | Warning | Network fabric feature status in errored state | The installed NSX feature is reported for some problem by NSX Manager. |
| 1.3.6.1.4.1.6876.100.1.0.36032 | vmwEdgeHANotConfiguredEvent | Info | NSX Edge high availability not enabled | High Availability is not enabled on the NSX Edge |
| 1.3.6.1.4.1.6876.100.1.0.36033 | vmwEdgeInterfacesDownEvent | Warning | One or more NSX Edge logical router interfaces are down | One or more NSX Edge interfaces are down. |
| 1.3.6.1.4.1.6876.100.1.0.36041 | vmwModuleUnhealthyEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.36042 | vmwModuleNotLoadedEvent | Critical | NSX VIB or host module not detected on host | At least one of NSX VIB or host module was not detected on the host |
| 1.3.6.1.4.1.6876.100.1.0.36043 | vmwModuleNetworkConnectionFailureEvent | Critical | Message bus and/or control plane connection not established between NSX Manager and host | The message bus and/or control plane agent daemon on this Host is experiencing connection failures with NSX Controller or NSX Manager |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.36044 | vmwHostNetworkControlPlaneMismatchEvent | Moderate | Logical switch table mismatched between host and NSX controller | Logical switch information between the primary NSX controller and all hosts where the logical switch is used does not match. This alert may indicate an error condition is occurring after a sharding change. |
| 1.3.6.1.4.1.6876.100.1.0.36045 | vmwHostNetworkControlPlaneConnectionFailureEvent | Critical | Host control plane to controller connection not established for one or more logical switches | The connection between the control plane agent on an NSX host and its primary NSX Controller for one or more logical switches is not established. This condition can result in stale information on the host and NSX Controllers. |
| 1.3.6.1.4.1.6876.100.1.0.36046 | vmwHostNetworkControlPlaneNotSyncedEvent | Moderate | Logical networking out of sync between host and NSX Controller | The logical switching and routing information on a host is not in sync with the NSX Controller information. To confirm this condition is occurring |
| 1.3.6.1.4.1.6876.100.1.0.36047 | vmwNSXControllerClusterMajorityEvent | Moderate | No NSX controller majority | Some of the NSX Controllers in the cluster are not communicating with the NSX Manager |
| 1.3.6.1.4.1.6876.100.1.0.36048 | vmwNSXControllersVMOnSameHostEvent | Info | All controller VMs deployed to same host | All NSX Controllers in the cluster are deployed to the same host |
| 1.3.6.1.4.1.6876.100.1.0.36049 | vmwVxLanRangeExhaustEvent | Warning | VXLAN segment ID range reaching exhaustion | More than 90% of VXLAN segment IDs have been used |
| 1.3.6.1.4.1.6876.100.1.0.36050 | vmwNSXFirewallDefaultAllowAllRulesEvent | Info | All traffic allowed by default firewall rule | The distributed firewall is configured to allow all traffic by default |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.36051 | vmwLogicalRouterNoUplinkEvent | Info | NSX DLR deployed without an uplink interface | The NSX DLR does not have an uplink interface configured |
| 1.3.6.1.4.1.6876.100.1.0.36052 | vmwEdgeNotHAEvent | Info | NSX Edge is configured but not highly available | When two Edge VMs have been configured for Edge high availability |
| 1.3.6.1.4.1.6876.100.1.0.36053 | vmwEdgeNotDeployedEvent | Info | NSX Edge deployment failed | An NSX Edge has failed to deploy. This condition may indicate that an NSX Edge has been configured without actually being deployed. |
| 1.3.6.1.4.1.6876.100.1.0.36054 | vmwEcmpIsEnabledAndStatefulServicesAreUpEvent | Info | NSX Edge configured with both ECMP and stateful edge services | A firewall |
| 1.3.6.1.4.1.6876.100.1.0.36055 | vmwLogicalRouterDeployedOnEcmpEdgeHostEvent | Info | NSX DLR deployed to same host as one or more NSX ECMP Edges | The NSX distributed logical router control VM is deployed to the same host as one or more NSX Edges that are configured for ECMP. |
| 1.3.6.1.4.1.6876.100.1.0.36056 | vmwEdgeMissingInterfaceOSPFAreaMappingEvent | Info | Missing NSX Edge interface to OSPF area mapping | OSPF is enabled on the NSX Edge |
| 1.3.6.1.4.1.6876.100.1.0.36057 | vmwOspfInsecureAuthRouterEvent | Info | Insecure authentication used in one or more OSPF areas | One or more OSPF areas on the NSX Edge Services Gateway or DLR are not configured to use MD5 authentication |
| 1.3.6.1.4.1.6876.100.1.0.36058 | vmwNSXControllersDeployedCountEvent | Info | Incorrect number of deployed NSX Controllers | There are fewer than three controllers deployed |
| 1.3.6.1.4.1.6876.100.1.0.36059 | vmwNSXControllerNotActiveCountEvent | Moderate | Less than three active NSX Controllers | There are fewer than three active controllers |
| 1.3.6.1.4.1.6876.100.1.0.36060 | vmwNSXControllerEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.36061 | vmwNSXEcmpEdgeDownEvent | Info | One or more NSX Edges in the ECMP Cluster are currently down | One or more NSX Edges in the ECMP Cluster are currently down |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.36062 | vmwNSXMajorityEcmpEdgesDownEvent | Warning | Majority of NSX Edges in the ECMP Cluster are currently down | Majority of NSX Edges in the ECMP Cluster are currently down |
| 1.3.6.1.4.1.6876.100.1.0.36063 | vmwNSXAllEcmpEdgesDownEvent | Critical | All NSX Edges in the ECMP Cluster are currently down | All NSX Edges in the ECMP Cluster are currently down |
| 1.3.6.1.4.1.6876.100.1.0.36064 | vmwNSXEdgeMtuMismatchEvent | Info | The MTU configured on one or more interfaces on the Edge does not match the MTU on the next hop router | The MTU configured on one or more interfaces on the Edges in the same Layer 2 network does not match |
| 1.3.6.1.4.1.6876.100.1.0.36065 | vmwNSXEdgeSplitBrainEvent | Critical | Both NSX Edge HA VMs in active state | Both VMs of Edge HA are in active state. The most common issue is split-brain |
| 1.3.6.1.4.1.6876.100.1.0.36066 | vmwVirtualDistributedRoutingEvent | Warning | VDR port not found on Host for VxLan routing | VDR port is not found on the Host for the specified VxLan |
| 1.3.6.1.4.1.6876.100.1.0.36067 | vmwNSXEdgeBGPNeighbourDownEvent | Critical | One or more BGP neighbours are not in established state | One or more BGP neighbours are not in established state. |
| 1.3.6.1.4.1.6876.100.1.0.37001 | vmwAnalyticsEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.37002 | vmwAnalyticsOutlierEvent | N/A | Outlier violation alert | Alert generated as a result of the anomalous entity |
| 1.3.6.1.4.1.6876.100.1.0.37003 | vmwAnalyticsThresholdEvent | Critical | Threshold violation alert | Alert generated as a result of the specified metric crosses the upper or lower bound specified in the configuration |
| 1.3.6.1.4.1.6876.100.1.0.38001 | vmwVMCEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.40000 | SwitchPortRxNetworkUsageThresholdExceededEvent | Warning | PNIC Rx Network Utilization is high | PNIC utilization exceeds the threshold value of 75%. |
| 1.3.6.1.4.1.6876.100.1.0.40001 | vmwCriticalHostNotAccessibleEvent | Critical | Host with Infrastructure VMs cannot be accessed | Host with Infrastructure VMs cannot be accessed |
| 1.3.6.1.4.1.6876.100.1.0.40002 | SwitchPortTxNetworkUsageThresholdExceededEvent | Warning | PNIC Tx Network Utilization is high | PNIC utilization exceeds the threshold value of 75%. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.40003 | SwitchPortRxNetworkUsageThresholdExceededCriticalEvent | Critical | PNIC Rx Network Utilization is high | PNIC utilization exceeds the threshold value of 90%. |
| 1.3.6.1.4.1.6876.100.1.0.40004 | SwitchPortTxNetworkUsageThresholdExceededCriticalEvent | Critical | PNIC Tx Network Utilization is high | PNIC utilization exceeds the threshold value of 90%. |
| 1.3.6.1.4.1.6876.100.1.0.40005 | ManagementHostHighCPUUsageEvent | Warning | Management Host CPU Usage | Host CPU utilization exceeds the threshold value of 75%. |
| 1.3.6.1.4.1.6876.100.1.0.40006 | ManagementHostHighCPUUsageCriticalEvent | Critical | Management Host CPU Usage Critical alert | Host CPU utilization exceeds the threshold value of 90%. |
| 1.3.6.1.4.1.6876.100.1.0.40007 | HostUnHealthyEvent | Critical | Host is not healthy | Host is not healthy |
| 1.3.6.1.4.1.6876.100.1.0.40008 | FileSystem Disk Available of NSX Controller is Low | Critical | FileSystem Disk Available of NSX Controller is Low | NSX Controller file system disk available percentage less than threshold value. |
| 1.3.6.1.4.1.6876.100.1.0.40011 | EdgeVmVnicRxPacketDropEvent | Critical | Receive packets are getting dropped on the VNIC of Edge VM. | Packets are getting dropped on the VNIC of Edge VM. This may impact the network traffic related to management nodes. |
| 1.3.6.1.4.1.6876.100.1.0.40012 | EdgeVmVnicTxPacketDropEvent | Critical | Transmit packets are getting dropped on the VNIC of Edge VM. | Packets are getting dropped on the VNIC of Edge VM. This may impact the network traffic related to management nodes. |
| 1.3.6.1.4.1.6876.100.1.0.40013 | ManagementHostPnicRxPacketDropEvent | Critical | Receive packets are getting dropped on the PNIC of the management host. | Packets are getting dropped on the pNIC of the management host. This may impact the network traffic related to management nodes. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.40014 | ManagementHostPnicTxPacketDropEvent | Critical | Transmit packets are getting dropped on the PNIC of the management host. | Packets are getting dropped on the pNIC of the management host. This may impact the network traffic related to management nodes. |
| 1.3.6.1.4.1.6876.100.1.0.40015 | ManagementHostConnectionStateDisconnectedEvent | Critical | Management Host is disconnected | vCenter has suspended management of the host. This might be due to - explicitly disconnecting the host, or trying to connect to a host which is in not responding state, or host license has expired. This results in vCenter deactivating HA on the host and hence, VMs on the host are therefore not failed over in the alert of a host isolation. |
| 1.3.6.1.4.1.6876.100.1.0.568 | vmwArkinApplicationMemberLimitEvent | Info | Application Membership limit exceeded | Number of members in application exceeds supported limit |
| 1.3.6.1.4.1.6876.100.1.0.70000 | vmwGenericNSXSystemEvent | Moderate | System Alerts from NSX (warning) | NSX System alerts with high or major severity |
| 1.3.6.1.4.1.6876.100.1.0.70001 | vmwFilterConfigApplyOnHostFailedEvent | Warning | Distributed firewall update for host vNIC failed to be applied | A distributed firewall configuration update failed to be applied to a vNIC on an NSX prepared host. |
| 1.3.6.1.4.1.6876.100.1.0.70002 | vmwRulesetLoadOnHostFailedEvent | Warning | Distributed firewall update failed to be applied to host | A distributed firewall ruleset was not applied to a host. |
| 1.3.6.1.4.1.6876.100.1.0.70003 | vmwConfigUpdateOnHostFailedEvent | Warning | Distributed firewall configuration update failed | Firewall configuration update to an NSX host has timed out. The host is not synced with the latest firewall configuration version. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.70004 | vmwSpoofguardConfigUpdateOnHostFailedEvent | Info | SpoofGuard configuration update failed | An update to the SpoofGuard configuration failed for the host. |
| 1.3.6.1.4.1.6876.100.1.0.70005 | vmwApplyRuleToVnicFailedEvent | Warning | Distributed firewall rule not applied to host vNIC | A distributed firewall rule was not applied to a host's vNIC. |
| 1.3.6.1.4.1.6876.100.1.0.70006 | vmwContainerConfigUpdateOnVnicFailedEvent | Warning | Distributed firewall container update failed on host | Network and security container information used with the vDefend distributed firewall or with Service Composer failed to be updated on an NSX host. |
| 1.3.6.1.4.1.6876.100.1.0.70007 | vmwSpoofguardApplyToVnicFailedEvent | Info | SpoofGuard initial configuration failed | The SpoofGuard configuration was not applied to the specified vNIC on the host. |
| 1.3.6.1.4.1.6876.100.1.0.70008 | vmwHostMessagingConfigurationFailedEvent | Warning | Host messaging configuration update failed | A configuration update pushed to hosts over the NSX messaging channel did not complete. |
| 1.3.6.1.4.1.6876.100.1.0.70009 | vmwHostMessagingConnectionReconfigurationFailedEvent | Warning | Host messaging connection re-config failed | Updated information on the host messaging channel failed to be sent to the NSX host. |
| 1.3.6.1.4.1.6876.100.1.0.70010 | vmwHostMessagingConfigurationFailedNotificationSkippedEvent | Warning | Host messaging channel between host and NSX Manager failed to re-establish | NSX Manager attempted to re-establish the message bus channel when the prepared host connected back to vCenter. This connection again failed |
| 1.3.6.1.4.1.6876.100.1.0.70011 | vmwHostMessagingInfrastructureDownEvent | Warning | Host messaging infrastructure down on host | Two or more messaging-channel heartbeat messages between NSX Manager and an NSX host were missed. |
| 1.3.6.1.4.1.6876.100.1.0.70012 | vmwEdgeVMNotRespondingEvent | Moderate | NSX Edge to NSX Manager heartbeat failed | An NSX Edge VM is not responding to health check by NSX Manager |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.70013 | vmwEdgeUnhealthyEvent | Critical | NSX Edge VM not in active/self state | NSX Edge VM is reporting a problematic state and might not be functioning correctly. |
| 1.3.6.1.4.1.6876.100.1.0.70014 | vmwEdgeVMCommunicationFailureEvent | Critical | NSX Manager to Edge VM communication failure | Communication failure detected between NSX Manager and Edge VM. |
| 1.3.6.1.4.1.6876.100.1.0.70015 | vmwNSXEdgeEvent | N/A | N/A | N/A |
| 1.3.6.1.4.1.6876.100.1.0.71000 | vmwOtherCriticalNSXEvent | Critical | System Alerts from NSX (Critical) | NSX System alerts with critical severity. |
| 1.3.6.1.4.1.6876.100.1.0.80001 | vmwPanNsxNotInRegisteredStateEvent | Critical | Palo Alto Panorama not registered with NSX Manager | Panorama is not in a registered state with NSX Manager. |
| 1.3.6.1.4.1.6876.100.1.0.80002 | vmwPanNsxDynamicUpdateDelayedEvent | Warning | Panorama dynamic membership definition update delayed | Panorama dynamic membership definition update from NSX Manager is delayed. This condition may indicate a network connectivity issue or an issue with NSX Manager's NetX service. |
| 1.3.6.1.4.1.6876.100.1.0.80003 | vmwPanDeviceInDisconnectedStateEvent | Warning | Palo Alto service VM not connected to Panorama | A service VM or device for Palo Alto Networks is in a not connected state with Panorama |
| 1.3.6.1.4.1.6876.100.1.0.80004 | vmwPanNsxServiceApplianceViewMismatchEvent | Critical | Service VM's status mismatched between Panorama and NSX Manager | Mismatch in service appliance information between NSX Manager and Panorama. |
| 1.3.6.1.4.1.6876.100.1.0.80005 | vmwPanNsxFabricAgentNotFoundOnHostEvent | Critical | NSX Fabric Agent not found on Host | Security Fabric Agent not reported by NSX for a Host where as the Cluster is prepped |
| 1.3.6.1.4.1.6876.100.1.0.80006 | vmwPanNsxServiceVMNotFoundOnHostEvent | Critical | Palo Alto service VM not found on host | A Palo Alto Networks' security appliance VM was not found on a host in an NSX prepared cluster. |
| 1.3.6.1.4.1.6876.100.1.0.80100 | vmwCheckpointEvent | N/A | N/A | N/A |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80102 | vmwCheckpointNsxFabricAgentNotFoundOnHostEvent | Critical | CheckpointNsxFabricAgentNotFoundOnHostEvent | Security Fabric Agent not reported by NSX for a Host where as the Cluster is prepped |
| 1.3.6.1.4.1.6876.100.1.0.80103 | vmwCheckpointNsxServiceVMNotFoundOnHostEvent | Critical | CheckpointNsxServiceVMNotFoundOnHostEvent | A Check Point' security appliance VM was not found on a host in an NSX prepared cluster. |
| 1.3.6.1.4.1.6876.100.1.0.80104 | vmwCheckpointGatewaySicStatusNotCommunicatingEvent | Critical | CheckpointGatewaySicStatusNotCommunicatingEvent | A service VM or gateway for Check Point does not have 'communicating' SIC status |
| 1.3.6.1.4.1.6876.100.1.0.80105 | vmwCheckpointNsxServiceApplianceViewMismatchEvent | Critical | Service VM's status mismatched between Check Point and NSX Manager | Mismatch in service appliance information between NSX Manager and Check Point. |
| 1.3.6.1.4.1.6876.100.1.0.80200 | NSXTEvent | N/A | NSX-T System alert | Alarm/Event generated by NSX-T platform |
| 1.3.6.1.4.1.6876.100.1.0.80201 | NSXTVcNotAddedEvent | Warning | One or more vCenter instances not added as data source in VCF Operations for networks | NSX-T has one or more Compute Managers which are not added as data sources in VCF Operations for networks with same IP or FQDN. |
| 1.3.6.1.4.1.6876.100.1.0.80202 | NSXTStandaloneHostsEvent | Warning | One or more Fabric Nodes are added as standalone hosts in NSX-T | One or more Fabric Nodes are added as standalone hosts in NSX-T. Virtual Machines on those hosts will not be visible in VCF Operations for networks. |
| 1.3.6.1.4.1.6876.100.1.0.80203 | vmwNSXTSystemEvent | N/A | System Alerts from NSX-T | Alarm generated by NSX-T platform |
| 1.3.6.1.4.1.6876.100.1.0.80205 | NSXTNoUplinkConnectivityEvent | Warning | NSX-T Tier-1 logical router disconnected | NSX-T Tier-1 logical router is disconnected from Tier-0 router. Networks under this router are not reachable from outside and vice versa. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80206 | NSXTRoutingAdvertisementEvent | Warning | Routing advertisement deactivated | Routing advertisement is deactivated for NSX-T Tier-1 logical router. Networks under this router are not reachable from outside. |
| 1.3.6.1.4.1.6876.100.1.0.80207 | NSXTManagerConnectivityDownEvent | Critical | NSX-T Edge Node has no manager connectivity | NSX-T Edge Node has lost manager connectivity. |
| 1.3.6.1.4.1.6876.100.1.0.80208 | NSXTControllerConnectivityDegradedEvent | Warning | Controller connectivity degraded for NSX-T Edge Node | NSX-T Edge Node is not able to communicate with one or more controllers. |
| 1.3.6.1.4.1.6876.100.1.0.80209 | NSXTControllerConnectivityDownEvent | Critical | NSX-T Edge Node has no controller connectivity | NSX-T Edge Node is not able to communicate with any of the controllers. |
| 1.3.6.1.4.1.6876.100.1.0.80210 | NSXTMtuMismatchEvent | Warning | MTU mismatch between NSX-T Tier-0 and uplink switch/router | The MTU configured on interfaces of Tier-0 logical router does not match with the interfaces of uplink switch/router from same L2 network. This can impact the network performance. |
| 1.3.6.1.4.1.6876.100.1.0.80211 | NSXTExcludedVmFlowEvent | Info | One or More VMs excluded from NSX-T DFW Firewall. | One or more VMs are not protected by NSX-T DFW firewall. VCF Operations for networks will not receive IPFIX flows for these VMs. |
| 1.3.6.1.4.1.6876.100.1.0.80212 | NSXTDoubleVlanTaggingEvent | Warning | Uplink Vlan misconfiguration | Communication is disrupted because VLAN on uplink port of Tier 0 router is different than VLAN on the external gateway. |
| 1.3.6.1.4.1.6876.100.1.0.80213 | NSXTNoTzAttachedOnTnEvent | Warning | No transport zone is attached to the transport node. | No transport zone attached to the transport node. VMs might lose connectivity because of this. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80214 | NSXTVtepDeleteEvent | Warning | No VTEP available on the transport node. | All vteps are deleted from the transport node. VMs might lose connectivity because of this. |
| 1.3.6.1.4.1.6876.100.1.0.80215 | vmwDuplicateL3SwitchEvent | Critical | Same Switch or Router added | Same Switch or Router is added with different IPs. VM to VM path may not be generated. |
| 1.3.6.1.4.1.6876.100.1.0.80216 | vmwLBPoolMemberDownEvent | Critical | Pool Member Down | This alert is raised when pool member of load balancer is down. To know the pool members which are down, search 'Pool Member where state = DISABLED' |
| 1.3.6.1.4.1.6876.100.1.0.80217 | vmwLBPoolDownEvent | Critical | Pool Down | This alert is raised when pool of load balancer is down |
| 1.3.6.1.4.1.6876.100.1.0.80218 | vmwLBPoolEmptyEvent | Critical | Pool Empty | This alert is raised when pool of load balancer is empty. To know pools which are empty, search 'Pool where PoolMembers Count = 0' |
| 1.3.6.1.4.1.6876.100.1.0.80219 | vmwLBPoolMemberVMDownEvent | Critical | VM of Pool Member Down | This alert is raised when VM associated with pool member of load balancer is down |
| 1.3.6.1.4.1.6876.100.1.0.80220 | vmwLBVirtualServerDisableEvent | Critical | Virtual Server of Load Balancer Deactivated | This alert is raised when virtual server of load balancer is deactivated |
| 1.3.6.1.4.1.6876.100.1.0.80221 | vmwLBServiceNodeIPNotFoundEvent | Critical | IP of Service Node not found | This alert is raised when no NIC associated with IP of service node of load balancer is found |
| 1.3.6.1.4.1.6876.100.1.0.80222 | vmwLBServiceNodeMultipleNicFoundEvent | Critical | Multiple NIC of Service Node found | This alert is raised when multiple NICs associated with IP of service node of load balancer are found |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80223 | NSXTSwitchIpfixEnabledEvent | Warning | NSX-T Switch IPFIX is enabled and has a collector profile pointing to one of the collectors. | VCF Operations for networks does not support the IPFIX flow data from NSX-T Switches. It is configured to send IPFIX data to one of the VCF Operations for networks Collector VM. It may have corrupted existing flow data in the system. |
| 1.3.6.1.4.1.6876.100.1.0.80224 | NSXTStandaloneHostsWithoutVcEvent | Critical | vCenter managing one or more fabric nodes in NSX-T is not added as a data source in VCF Operations for networks | vCenter managing one or more fabric nodes in NSX-T is not added as a data source in VCF Operations for networks. Virtual Machines on those hosts will not be visible in VCF Operations for networks. |
| 1.3.6.1.4.1.6876.100.1.0.80225 | NSXTControllerNodeToControlClusterConnectivityEvent | Critical | NSX-T controller node has no control cluster connectivity | NSX-T controller node has lost control cluster connectivity. |
| 1.3.6.1.4.1.6876.100.1.0.80226 | NSXTControllerNodeToMgmtPlaneConnectivityEvent | Critical | NSX-T controller node has no management plane connectivity | NSX-T controller node has lost management plane connectivity. |
| 1.3.6.1.4.1.6876.100.1.0.80227 | NSXTMPNodeToMgmtClusterConnectivityEvent | Critical | NSX-T management node has no management cluster connectivity | NSX-T management node has lost management cluster connectivity. |
| 1.3.6.1.4.1.6876.100.1.0.80228 | NSXTHostNodePnicStatusDownEvent | Warning | NSX-T Host Transport Node Pnic Status is 'Down'. | NSX-T Host Transport Node Pnic Status is 'Down'. |
| 1.3.6.1.4.1.6876.100.1.0.80229 | NSXTHostNodePnicStatusDegradedEvent | Warning | NSX-T Host Transport Node Pnic Status is 'Degraded' | NSX-T Host Transport Node Pnic Status is 'Degraded'. |
| 1.3.6.1.4.1.6876.100.1.0.80230 | NSXTHostNodePnicStatusUnknownEvent | Warning | NSX-T Host Transport Node Pnic Status is 'Unknown'. | NSX-T Host Transport Node Pnic Status is 'Unknown'. |
| 1.3.6.1.4.1.6876.100.1.0.80231 | NSXTHostNodeTunnelStatusDownEvent | Warning | NSX-T Host Transport Node Tunnel Status is 'Down'. | NSX-T Host Transport Node Tunnel Status is 'Down'. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80232 | NSXTHostNodeTunnelStatusDegradedEvent | Warning | NSX-T Host Transport Node Tunnel Status is 'Degraded'. | NSX-T Host Transport Node Tunnel Status is 'Degraded'. |
| 1.3.6.1.4.1.6876.100.1.0.80233 | NSXTHostNodeTunnelStatusUnknownEvent | Warning | NSX-T Host Transport Node Tunnel Status is 'Unknown'. | NSX-T Host Transport Node Tunnel Status is 'Unknown'. |
| 1.3.6.1.4.1.6876.100.1.0.80234 | NSXTHostNodeStatusDownEvent | Warning | NSX-T Host Transport Node Status is 'Down'. | NSX-T Host Transport Node Status is 'Down'. |
| 1.3.6.1.4.1.6876.100.1.0.80235 | NSXTHostNodeStatusDegradedEvent | Warning | NSX-T Host Transport Node Status is 'Degraded'. | NSX-T Host Transport Node Status is 'Degraded'. |
| 1.3.6.1.4.1.6876.100.1.0.80236 | NSXTHostNodeStatusUnknownEvent | Warning | NSX-T Host Transport Node Status is 'Unknown'. | NSX-T Host Transport Node Status is 'Unknown'. |
| 1.3.6.1.4.1.6876.100.1.0.80237 | NSXTEdgeNodePnicStatusDownEvent | Critical | NSX-T Edge Transport Node Pnic Status is 'Down'. | NSX-T Edge Transport Node Pnic Status is 'Down'. |
| 1.3.6.1.4.1.6876.100.1.0.80238 | NSXTEdgeNodePnicStatusDegradedEvent | Critical | NSX-T Edge Transport Node Pnic Status is 'Degraded'. | NSX-T Edge Transport Node Pnic Status is 'Degraded'. |
| 1.3.6.1.4.1.6876.100.1.0.80239 | NSXTEdgeNodePnicStatusUnknownEvent | Critical | NSX-T Edge Transport Node Pnic Status is 'Unknown'. | NSX-T Edge Transport Node Pnic Status is 'Unknown'. |
| 1.3.6.1.4.1.6876.100.1.0.80240 | NSXTEdgeNodeTunnelStatusDownEvent | Critical | NSX-T Edge Transport Node Tunnel Status is 'Down'. | NSX-T Edge Transport Node Tunnel Status is 'Down'. |
| 1.3.6.1.4.1.6876.100.1.0.80241 | NSXTEdgeNodeTunnelStatusDegradeEvent | Critical | NSX-T Edge Transport Node Tunnel Status is 'Degraded'. | NSX-T Edge Transport Node Tunnel Status is 'Degraded'. |
| 1.3.6.1.4.1.6876.100.1.0.80242 | NSXTEdgeNodeTunnelStatusUnknownEvent | Critical | NSX-T Edge Transport Node Tunnel Status is 'Unknown'. | NSX-T Edge Transport Node Tunnel Status is 'Unknown'. |
| 1.3.6.1.4.1.6876.100.1.0.80243 | NSXTEdgeNodeStatusDownEvent | Critical | NSX-T Edge Transport Node Status is 'Down'. | NSX-T Edge Transport Node Status is 'Down'. |
| 1.3.6.1.4.1.6876.100.1.0.80244 | NSXTEdgeNodeStatusDegradedEvent | Critical | NSX-T Edge Transport Node Status is 'Degraded'. | NSX-T Edge Transport Node Status is 'Degraded'. |
| 1.3.6.1.4.1.6876.100.1.0.80245 | NSXTEdgeNodeStatusUnknownEvent | Critical | NSX-T Edge Transport Node Status is 'Unknown'. | NSX-T Edge Transport Node Status is 'Unknown'. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80246 | NSXTHostNodeMgmtConnectivityStatusDownEvent | Warning | NSX-T Host Node has no manager connectivity | Desynchronization between NSX Manager's State of connectivity with Host Transport Nodes |
| 1.3.6.1.4.1.6876.100.1.0.80247 | NSXTEdgeNodeCtlrConnectivityStatusUnknownEvent | Critical | Controller connectivity for NSX-T Edge Node is Unknown. | NSX-T Edge Node Controller connectivity is Unknown. |
| 1.3.6.1.4.1.6876.100.1.0.80248 | NSXTHostNodeCtlrConnectivityStatusDownEvent | Warning | NSX-T Host Node has no controller connectivity | NSX-T Host Node is not able to communicate with any of the controllers. |
| 1.3.6.1.4.1.6876.100.1.0.80249 | NSXTHostNodeCtlrConnectivityStatusDegradedEvent | Warning | Controller connectivity degraded for NSX-T Host Node | NSX-T Host Node is not able to to communicate with one or more controllers. |
| 1.3.6.1.4.1.6876.100.1.0.80250 | NSXTHostNodeCtlrConnectivityStatusUnknownEvent | Warning | Controller connectivity for NSX-T Host Node is Unknown. | NSX-T Host Node Controller connectivity is Unknown. |
| 1.3.6.1.4.1.6876.100.1.0.80252 | NSXTLogicalSwitchAdminStatusDownEvent | Warning | NSX-T Logical Switch Admin Status is 'Down' | NSX-T Logical Switch Admin Status is 'Down' |
| 1.3.6.1.4.1.6876.100.1.0.80253 | NSXTLogicalPortOperationalStatusDownEvent | Critical | NSX-T Logical Port Operational Status is 'Down' | NSX-T Logical Port Operational Status is 'Down'. This could cause a communication failure between two virtual interfaces (VIFs) that are connected to the same logical switch, for example, you cannot ping one VM from another. |
| 1.3.6.1.4.1.6876.100.1.0.80254 | NSXTLogicalPortOperationalStatusUnknownEvent | Warning | NSX-T Logical Port Operational Status is 'Unknown' | NSX-T Logical Port Operational Status is 'Unknown'. This could cause a communication failure between two virtual interfaces (VIFs) that are connected to the same logical switch, for example, you cannot ping one VM from another. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80255 | NSXTComputeManagerConnectionStatusNotUpEvent | Warning | NSX-T Compute Manager Connection Status in not up | NSX-T Compute Manager Connection status is not up |
| 1.3.6.1.4.1.6876.100.1.0.80256 | NSXTClusterBackUpDisabledEvent | Warning | NSX-T Manager backup is not scheduled. | NSX-T Manager backup is not scheduled |
| 1.3.6.1.4.1.6876.100.1.0.80257 | NSXTDFWFirewallDisabledEvent | Critical | NSX-T DFW Firewall is deactivated. | Distributed Firewall is deactivated in the NSX-T Manager |
| 1.3.6.1.4.1.6876.100.1.0.80258 | NSXTLogicalPortReceivedPacketDropEvent | Warning | NSX-T Logical Port Received Packets are getting dropped. | Received packets are getting dropped on the NSX-T Logical Port and associated entities might get affected |
| 1.3.6.1.4.1.6876.100.1.0.80259 | NSXTLogicalPortTransmittedPacketDropEvent | Warning | NSX-T Logical Port Transmitted Packets are getting dropped. | Transmitted packets are getting dropped on the NSX-T Logical Port and associated entities might get affected |
| 1.3.6.1.4.1.6876.100.1.0.80260 | NSXTLogicalSwitchReceivedPacketDropEvent | Warning | NSX-T Logical Switch Received Packets are getting dropped | Received packets are getting dropped on the NSX-T Logical Switch and associated entities might get affected |
| 1.3.6.1.4.1.6876.100.1.0.80261 | NSXTLogicalSwitchTransmittedPacketDropEvent | Warning | NSX-T Logical Switch Transmitted Packets are getting dropped | Transmitted packets are getting dropped on the NSX-T Logical Switch and associated entities might get affected |
| 1.3.6.1.4.1.6876.100.1.0.80262 | NSXTRxPacketDropOnMPNicEvent | Warning | Received packets are dropping on NSX-T Management Node's network interface | Received packets are getting dropped on NSX-T Management Node's network interface. This may impact the network traffic related to NSX-T management cluster. |
| 1.3.6.1.4.1.6876.100.1.0.80263 | NSXTRxPacketDropOnEdgeTnNicEvent | Critical | Received packets are dropping on NSX-T Edge Node's network interface | Received packets are getting dropped on NSX-T Edge Node's network interface. This may impact the network traffic of edge cluster. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80264 | NSXTRxPacketDropOnHostTnNicEvent | Warning | Received packets are dropping on NSX-T Host Node's network interface | Received packets are getting dropped on NSX-T Host Node's network interface. This may impact the network traffic on ESX host. |
| 1.3.6.1.4.1.6876.100.1.0.80265 | NSXTTxPacketDropOnMPNicEvent | Warning | Transmitted packets are dropping on NSX-T Management Node's network interface | Transmitted packets are getting dropped on NSX-T Management Node's network interface. This may impact the network traffic related to NSX-T management cluster. |
| 1.3.6.1.4.1.6876.100.1.0.80266 | NSXTTxPacketDropOnEdgeTnNicEvent | Critical | Transmitted packets are dropping on NSX-T Edge Node's network interface | Transmitted packets are getting dropped on NSX-T Edge Node's network interface. This may impact the network traffic of edge cluster. |
| 1.3.6.1.4.1.6876.100.1.0.80267 | NSXTTxPacketDropOnHostTnNicEvent | Warning | Transmitted packets are dropping on NSX-T Host Node's network interface | Transmitted packets are getting dropped on NSX-T Host Node's network interface. This may impact the network traffic on ESX host. |
| 1.3.6.1.4.1.6876.100.1.0.80301 | vmwHardwareVTEPMismatchEvent | Critical | HardwareVTEPMismatchEvent | Hardware Gateway Bindings Mismatch |
| 1.3.6.1.4.1.6876.100.1.0.80302 | vmwHardwareVTEPPortDownEvent | Critical | HardwareVTEPPortDownEvent | Hardware Gateway Port Down |
| 1.3.6.1.4.1.6876.100.1.0.80402 | NSXTMPNodeServiceCmInventoryStatusEvent | Warning | CM Inventory Serivce has stopped running | CM Inventory Service status has turned to stopped. |
| 1.3.6.1.4.1.6876.100.1.0.80402 | NSXTMPNodeServiceCmInventoryStatusEvent | Critical | CM Inventory Serivce has stopped | One of the Services of the NSX-T Management Node, namely CM Inventory Service has stopped running. |
| 1.3.6.1.4.1.6876.100.1.0.80403 | NSXTMPNodeServiceControllerStatusEvent | Warning | Controller Service has stopped running. | Controller Service status has turned to stopped. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80403 | NSXTMPNodeServiceControllerStatusEvent | Critical | Controller Service has stopped | One of the Services of the NSX-T Management Node, namely Controller Service has stopped running. |
| 1.3.6.1.4.1.6876.100.1.0.80404 | NSXTMPNodeServiceDataStoreStatusEvent | Warning | DataStore Service has stopped running. | DataStore Service status has turned to stopped. |
| 1.3.6.1.4.1.6876.100.1.0.80404 | NSXTMPNodeServiceDataStoreStatusEvent | Critical | DataStore Service has stopped | One of the Services of the NSX-T Management Node, namely DataStore Service has stopped running. |
| 1.3.6.1.4.1.6876.100.1.0.80405 | NSXTMPNodeServiceHttpStatusEvent | Warning | HTTP Service has stopped running. | HTTP Service status has turned to stopped. |
| 1.3.6.1.4.1.6876.100.1.0.80405 | NSXTMPNodeServiceHttpStatusEvent | Critical | HTTP Service has stopped | One of the Services of the NSX-T Management Node, namely HTTP Service has stopped running. |
| 1.3.6.1.4.1.6876.100.1.0.80406 | NSXTMPNodeServiceInstallUpgradeEvent | Warning | Install Upgrade Service has stopped running. | Install Upgrade Service status has turned to stopped. |
| 1.3.6.1.4.1.6876.100.1.0.80406 | NSXTMPNodeServiceInstallUpgradeEvent | Warning | Install Upgrade Service has stopped | One of the Services of the NSX-T Management Node, namely Install Upgrade Service has stopped running. |
| 1.3.6.1.4.1.6876.100.1.0.80407 | NSXTMPNodeServiceLiagentStatusEvent | Warning | Liagent service has stopped running. | Liagent Service status has turned to stopped. |
| 1.3.6.1.4.1.6876.100.1.0.80407 | NSXTMPNodeServiceLiagentStatusEvent | Warning | Liagent service has stopped | One of the Services of the NSX-T Management Node, namely LI Agent Service has stopped running. |
| 1.3.6.1.4.1.6876.100.1.0.80408 | NSXTMPNodeServiceManagerStatusEvent | Warning | Manager Service has stopped running. | Manager Service status has turned to stopped. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80408 | NSXTMPNodeServiceManagerStatusEvent | Critical | Manager Service has stopped | One of the Services of the NSX-T Management Node, namely Manager Service has stopped running. |
| 1.3.6.1.4.1.6876.100.1.0.80409 | NSXTMPNodeServiceMgmtPlaneBusStatusEvent | Warning | Management Plane Service has stopped running. | Management Service status has turned to stopped. |
| 1.3.6.1.4.1.6876.100.1.0.80409 | NSXTMPNodeServiceMgmtPlaneBusStatusEvent | Warning | Management Plane Service has stopped | One of the Services of the NSX-T Management Node, namely Management Plane Bus Service has stopped running. |
| 1.3.6.1.4.1.6876.100.1.0.80410 | NSXTMPNodeServiceMigrationCoordinatorStatusEvent | Warning | Migration Co-ordinator Service has stopped running. | Migration Co-ordinator Service status has turned to stopped. |
| 1.3.6.1.4.1.6876.100.1.0.80410 | NSXTMPNodeServiceMigrationCoordinatorStatusEvent | Warning | Migration Co-ordinator Service has stopped | One of the Services of the NSX-T Management Node, namely Migration Co-ordinator Service has stopped running. |
| 1.3.6.1.4.1.6876.100.1.0.80411 | NSXTMPNodeServiceNodeMgmtStatusEvent | Warning | Node Management Service has stopped running. | Node Management Service status has turned to stopped. |
| 1.3.6.1.4.1.6876.100.1.0.80411 | NSXTMPNodeServiceNodeMgmtStatusEvent | Critical | Node Management Service has stopped | One of the Services of the NSX-T Management Node, namely Node Management Service has stopped running. |
| 1.3.6.1.4.1.6876.100.1.0.80412 | NSXTMPNodeServiceNodeStatsStatusEvent | Warning | Node Statistics Service has stopped running. | Node Statistics Service status has turned to stopped. |
| 1.3.6.1.4.1.6876.100.1.0.80412 | NSXTMPNodeServiceNodeStatsStatusEvent | Critical | Node Statistics Service has stopped | One of the Services of the NSX-T Management Node, namely Node Statistics has stopped running. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80413 | NSXTMPNodeServiceNSXMessageBusStatusEvent | Warning | Message Bus Service has stopped running. | Message Bus Client Service status has turned to stopped. |
| 1.3.6.1.4.1.6876.100.1.0.80413 | NSXTMPNodeServiceNSXMessageBusStatusEvent | Warning | Message Bus Service has stopped | One of the Services of the NSX-T Management Node, namely Message Bus Service has stopped running. |
| 1.3.6.1.4.1.6876.100.1.0.80414 | NSXTMPNodeServiceNSXPlatformClientStatusEvent | Warning | Platform Client Service has stopped running. | Platform Client Service status has turned to stopped. |
| 1.3.6.1.4.1.6876.100.1.0.80414 | NSXTMPNodeServiceNSXPlatformClientStatusEvent | Critical | Platform Client Service has stopped | One of the Services of the NSX-T Management Node, namely Platform Client Service has stopped running. |
| 1.3.6.1.4.1.6876.100.1.0.80415 | NSXTMPNodeServiceNSXUpgradeAgentStatusEvent | Warning | Upgrade Agent Service has stopped running. | Upgrade Service status has turned to stopped. |
| 1.3.6.1.4.1.6876.100.1.0.80415 | NSXTMPNodeServiceNSXUpgradeAgentStatusEvent | Warning | Upgrade Agent Service has stopped | One of the Services of the NSX-T Management Node, namely Upgrade Agent Service has stopped running. |
| 1.3.6.1.4.1.6876.100.1.0.80416 | NSXTMPNodeServiceNTPStatusEvent | Warning | NTP Service has stopped running. | NTP Service status has turned to stopped. |
| 1.3.6.1.4.1.6876.100.1.0.80416 | NSXTMPNodeServiceNTPStatusEvent | Critical | NTP Service has stopped | One of the Services of the NSX-T Management Node, namely NTP Service has stopped running. |
| 1.3.6.1.4.1.6876.100.1.0.80417 | NSXTMPNodeServicePolicyStatusEvent | Warning | Policy Service has stopped running. | Policy Service status has turned to stopped. |
| 1.3.6.1.4.1.6876.100.1.0.80417 | NSXTMPNodeServicePolicyStatusEvent | Critical | Policy Service has stopped | One of the Services of the NSX-T Management Node, namely Policy Service has stopped running. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80418 | NSXTMPNodeServiceSearchStatusEvent | Warning | Search Service has stopped running. | Search Service status has turned to stopped. |
| 1.3.6.1.4.1.6876.100.1.0.80418 | NSXTMPNodeServiceSearchStatusEvent | Critical | Search Service has stopped | One of the Services of the NSX-T Management Node, namely Search Service has stopped running. |
| 1.3.6.1.4.1.6876.100.1.0.80419 | NSXTMPNodeServiceSNMPStatusEvent | Warning | SNMP Service has stopped running. | SNMP Service status has turned to stopped. |
| 1.3.6.1.4.1.6876.100.1.0.80419 | NSXTMPNodeServiceSNMPStatusEvent | Warning | SNMP Service has stopped | One of the Services of the NSX-T Management Node, namely SNMP Service has stopped running. |
| 1.3.6.1.4.1.6876.100.1.0.80420 | NSXTMPNodeServiceSSHStatusEvent | Warning | SSH Service has stopped running. | SSH Service status has turned to stopped. |
| 1.3.6.1.4.1.6876.100.1.0.80420 | NSXTMPNodeServiceSSHStatusEvent | Critical | SSH Service has stopped | One of the Services of the NSX-T Management Node, namely SSH Service has stopped running. |
| 1.3.6.1.4.1.6876.100.1.0.80421 | NSXTMPNodeServiceSyslogStatusEvent | Warning | Syslog Service has stopped running. | Syslog Service status has turned to stopped. |
| 1.3.6.1.4.1.6876.100.1.0.80421 | NSXTMPNodeServiceSyslogStatusEvent | Critical | Syslog Service has stopped | One of the Services of the NSX-T Management Node, namely Syslog Service has stopped running. |
| 1.3.6.1.4.1.6876.100.1.0.80422 | NSXTMPNodeServiceTelemetryStatusEvent | Warning | Telemetry Service has stopped running. | Telemetry Service status has turned to stopped. |
| 1.3.6.1.4.1.6876.100.1.0.80422 | NSXTMPNodeServiceTelemetryStatusEvent | Warning | Telemetry Service has stopped | One of the Services of the NSX-T Management Node, namely Telemetry Service has stopped running. |
| 1.3.6.1.4.1.6876.100.1.0.80423 | NSXTMPNodeServiceUIServiceStatusEvent | Warning | UI Service has stopped running. | UI Service status has turned to stopped. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80423 | NSXTMPNodeServiceUIServiceStatusEvent | Critical | UI Service has stopped | One of the Services of the NSX-T Management Node, namely UI Service has stopped running. |
| 1.3.6.1.4.1.6876.100.1.0.80424 | NSXTMPNodeServiceClusterManagerStatusEvent | Critical | Cluster Manager Service has stopped | One of the Services of the NSX-T Management Node, namely Cluster Manager Service has stopped running. |
| 1.3.6.1.4.1.6876.100.1.0.80501 | vmwIndexerLagEvent | Critical | Indexer Lag Alert | Recent data is still being indexed. Search results may be inaccurate. |
| 1.3.6.1.4.1.6876.100.1.0.80502 | vmwIPFIXFlowDPPausedEvent | Critical | IPFIX Flow Data Source Paused alert | Ipfix Flow Data Source has been paused due to large number of flows. |
| 1.3.6.1.4.1.6876.100.1.0.80503 | vmwGridProcessingStoppedEvent | Critical | Grid Processing Stopped | Grid processing has stopped. |
| 1.3.6.1.4.1.6876.100.1.0.80504 | vmwUnableToSendEmailsEvent | Critical | Unable to Send Emails message | Unable to send email message. |
| 1.3.6.1.4.1.6876.100.1.0.80505 | vmwSMTPNotConfiguredEvent | Critical | SMTP Not Configured | SMTP Not Configured |
| 1.3.6.1.4.1.6876.100.1.0.80506 | vmwSNMPNotConfiguredEvent | Critical | SMTP Not Configured | No SNMP Targets are configured. |
| 1.3.6.1.4.1.6876.100.1.0.80507 | vmwReindexingInProgressEvent | Critical | Reindexing in Process | Data is currently being re-indexed. Search service will be available after this migration activity is complete. |
| 1.3.6.1.4.1.6876.100.1.0.80508 | vmwNodesVersionMismatchEvent | Critical | Nodes Version Mismatch | Nodes Version Mismatch detected |
| 1.3.6.1.4.1.6876.100.1.0.80509 | vmwNotAllServicesRunningEvent | Critical | Not All Services Running | One or more essential services are not running. |
| 1.3.6.1.4.1.6876.100.1.0.80510 | vmwNotAllServicesHealthyEvent | Critical | Not All Services Healthy | One or more essential services are not healthy. |
| 1.3.6.1.4.1.6876.100.1.0.80511 | vmwExpandPartitionFailedEvent | Critical | Expand Partition Failed | Failed to expand disk partition. |
| 1.3.6.1.4.1.6876.100.1.0.80512 | vmwDiskCleanupFailedEvent | Critical | Disk Cleanup Failed | Disk Cleanup service is unhealthy. |
| 1.3.6.1.4.1.6876.100.1.0.80513 | vmwVacuumFailedEvent | Critical | Vacuum Failed | PostgreSQL Vacuum service is unhealthy. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80514 | vmwConfigStoreCleanupFailedEvent | Critical | Config Store Cleanup Failed | Data Retention (Config Store Maintenance) service is unhealthy. |
| 1.3.6.1.4.1.6876.100.1.0.80515 | vmwHBaseRetentionToolFailedEvent | Critical | HBASE Retention Tool Failed | Data Retention (Metric Retention Configuration) service is unhealthy. |
| 1.3.6.1.4.1.6876.100.1.0.80516 | vmwMetricStoreUpdaterFailedEvent | Critical | Metric Store Updater Failed | Data Retention (Metric Store Maintenance) service is unhealthy. |
| 1.3.6.1.4.1.6876.100.1.0.80517 | vmwCollectorLagEvent | Critical | Collecter Lag | Last data collection on collector was more than threshold ago |
| 1.3.6.1.4.1.6876.100.1.0.80518 | vmwCollectionLagEvent | Critical | Collection Lag | Last data collection on data source was more than threshold ago |
| 1.3.6.1.4.1.6876.100.1.0.80519 | vmwGridProcessingLagEvent | Critical | Grid Processing Lag | Grid processing is lagging by more than threshold ago |
| 1.3.6.1.4.1.6876.100.1.0.80520 | vmwConnectionErorEvent | Critical | Connection Error | Error connecting to data source |
| 1.3.6.1.4.1.6876.100.1.0.80521 | vmwNodeNotActiveEvent | Critical | Node Not Active | Node Not Active |
| 1.3.6.1.4.1.6876.100.1.0.80522 | vmwHighDiskUtilizationEvent | Critical | High Disk Utilization | High Disk Utilization |
| 1.3.6.1.4.1.6876.100.1.0.80523 | vmwIndexingAbortedEvent | Critical | Indexing Aborted | Indexing Aborted |
| 1.3.6.1.4.1.6876.100.1.0.80524 | vmwUpgradeFailedEvent | Critical | Upgrade Failed | Upgrade Failed |
| 1.3.6.1.4.1.6876.100.1.0.80525 | vmwFlowProcessingSuspendedEvent | Critical | Flow Processing Suspended | Flow processing suspended |
| 1.3.6.1.4.1.6876.100.1.0.80526 | vmwLargeSdmsDroppedEvent | Critical | Data Processing Error | Large SDMS Dropped |
| 1.3.6.1.4.1.6876.100.1.0.80527 | vmwApplianceNotConfiguredEvent | Critical | Appliance Not Configured | Collector VM configuration incomplete. |
| 1.3.6.1.4.1.6876.100.1.0.80531 | vmwFdbConfigStoreCleanupFailedEvent | | FDB_CONFIG_STORE_CLEANUP_FAILED_EVENT | FDB Config Store Cleanup Failed |
| 1.3.6.1.4.1.6876.100.1.0.80531 | vmwDiskAllocationInsufficientEvent | Info | DISK_ALLOCATION_INSUFFICIENT_EVENT | Disk Not Configured |
| 1.3.6.1.4.1.6876.100.1.0.80601 | vmwFailedEvent | Critical | Data Source Failed | Data Source Failed |
| 1.3.6.1.4.1.6876.100.1.0.80602 | vmwTimeoutEvent | Critical | Data Source Timeout | Data Source Timeout |
| 1.3.6.1.4.1.6876.100.1.0.80603 | vmwConnectionRefusedEvent | Critical | Connection Refused | Connection Refused |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80605 | vmwIncorrectConnectionStringEvent | Critical | Incorrect Connection String | Incorrect Connection String |
| 1.3.6.1.4.1.6876.100.1.0.80606 | vmwInvalidCredentialsEvent | Critical | Invalid Credentials | Invalid Credentials |
| 1.3.6.1.4.1.6876.100.1.0.80608 | vmwUnknownHostEvent | Critical | Unknown Host | Unknown Host |
| 1.3.6.1.4.1.6876.100.1.0.80609 | vmwSNMPConnectionInvalidEvent | Critical | Invalid SNMP Connection | Invalid SNMP Connection |
| 1.3.6.1.4.1.6876.100.1.0.806100012 | vmwPwdAuthModeDisabledAristaEvent | Critical | Password Authentication Deactivated | Password Authentication Deactivated |
| 1.3.6.1.4.1.6876.100.1.0.806100018 | vmwUnsupportedNSXVersionEvent | Critical | Unsupported NSX Version | Unsupported NSX Version |
| 1.3.6.1.4.1.6876.100.1.0.80611 | vmwFailedCredsEncryptEvent | Critical | Credential Encryption Failed | Credential Encryption Failed |
| 1.3.6.1.4.1.6876.100.1.0.80612 | vmwPwdAuthModeDisabledEvent | Critical | Password Authentication Mode Deactivated | Password Authentication Mode Deactivated |
| 1.3.6.1.4.1.6876.100.1.0.80613 | vmwInsufficientPrivilegesEvent | Critical | Insufficient Privileges | Insufficient Privileges |
| 1.3.6.1.4.1.6876.100.1.0.8061313 | vmwFlowCollectionErrorEvent | Critical | Flow Collection Error | Flow Collection Error |
| 1.3.6.1.4.1.6876.100.1.0.8061314 | vmwAWSThrottlingExceptionEvent | Critical | AWS Throttling Exception | AWS Throttling Exception |
| 1.3.6.1.4.1.6876.100.1.0.8061315 | vmwAWSFlowLogAccessDeniedExceptionEvent | Critical | AWS Flow Log Access Denied Exception | AWS Flow Log Access Denied Exception. This alert is raised when user does not have necessary permissions required to collect flow logs. |
| 1.3.6.1.4.1.6876.100.1.0.80614 | vmwNotFoundEvent | Critical | Not Found Alert | Not Found |
| 1.3.6.1.4.1.6876.100.1.0.80616 | vmwInvalidConfigEvent | Critical | Invalid Data Source Config | Invalid Data Source Config |
| 1.3.6.1.4.1.6876.100.1.0.80617 | vmwWarnConfigEvent | Critical | Invalid Data Source Config | Invalid Data Source Config |
| 1.3.6.1.4.1.6876.100.1.0.80618 | vmwUnexpectedDSTypeOrVersionEvent | Critical | Unexpected Datasource Type or Version | Unexpected Datasource Type or Version |
| 1.3.6.1.4.1.6876.100.1.0.80619 | vmwNSXControllerNotFoundEvent | Critical | NSX Controller Not Found | NSX Controller Not Found |
| 1.3.6.1.4.1.6876.100.1.0.80620 | vmwHostNotReachableEvent | Critical | Host Not Reachable | Host Not Reachable |
| 1.3.6.1.4.1.6876.100.1.0.80621 | vmwInvalidResponseFromDatasourceEvent | Critical | Invalid Response From Data Source | Invalid Response From Data Source |
| 1.3.6.1.4.1.6876.100.1.0.80622 | vmwDataProviderNotRunningEvent | Critical | Data Source Not Running | Data Source Not Running |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80623 | vmwPrimaryNSXNotAddedEvent | Critical | Primary NSX Not Added | Primary NSX Not Added |
| 1.3.6.1.4.1.6876.100.1.0.80624 | vmwHostnameResolutionErrorEvent | Critical | Hostname Resolution Error | Hostname Resolution Error |
| 1.3.6.1.4.1.6876.100.1.0.80625 | vmwNumVMsOrHostsNotFoundEvent | Critical | Number of Vms or Hosts Not Found | Number of Vms or Hosts Not Found |
| 1.3.6.1.4.1.6876.100.1.0.80626 | vmwNSXIPFIXStatusMismatchEvent | Critical | NSX IPFIX Status Mismatch | NSX IPFIX Status Mismatch |
| 1.3.6.1.4.1.6876.100.1.0.80627 | vmwFlowPhysicalNodeEvent | Critical | Flow Physical Node Alert | Flow Physical Node |
| 1.3.6.1.4.1.6876.100.1.0.80628 | vmwNotEmptyNodeEvent | Critical | Not Empty Node Alert | Not Empty Node |
| 1.3.6.1.4.1.6876.100.1.0.80629 | vmwUnsupportedNSXTVersionEvent | Critical | Unsupported NSXT Version | Unsupported NSXT Version |
| 1.3.6.1.4.1.6876.100.1.0.80630 | vmwComputeManagersNotFoundEvent | Critical | Compute Managers Not Found | Compute Managers Not Found |
| 1.3.6.1.4.1.6876.100.1.0.80631 | vmwComputeManagersNotAddedEvent | Critical | Compute Managers Not Added | Compute Managers Not Added |
| 1.3.6.1.4.1.6876.100.1.0.80632 | vmwUnsupportedLogInsightVersionEvent | Critical | VCF Operations for logs version not supported | VMware VCF Operations for logs version not supported |
| 1.3.6.1.4.1.6876.100.1.0.80633 | vmwUnsupportedVRNIContentPackVersionEvent | Critical | VCF Operations for networks Content Pack version not supported | VCF Operations for networks Content Pack version not supported |
| 1.3.6.1.4.1.6876.100.1.0.80634 | vmwVRNIContentPackNotInstalledEvent | Critical | VCF Operations for networks Content Pack not found in VCF Operations for logs | VCF Operations for networks Content Pack not found in VCF Operations for logs |
| 1.3.6.1.4.1.6876.100.1.0.80635 | vmwWebhookNotEnabledOnAlertEvent | Critical | Webhook not enabled on a VCF Operations for networks alert | Webhook not enabled for one or more alerts of VCF Operations for networks Content Pack in VCF Operations for logs |
| 1.3.6.1.4.1.6876.100.1.0.80636 | vmwIncorrectWebhookConfiguredOnAlertEvent | Critical | Incorrect webhook URL configured on a VCF Operations for logs alert | Incorrect webhook configuration found for one or more alerts of VCF Operations for networks Content Pack in VCF Operations for logs |
| 1.3.6.1.4.1.6876.100.1.0.80637 | vmwWebhookNotRunningEvent | Critical | Webhook is not running on the Collector(Proxy) VM | Webhook is not running on the Collector(Proxy) VM |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80638 | vmwInfobloxRecordLimitExceededEvent | Critical | The number of records from Infoblox exceeds current limit | The number of records from Infoblox exceeds current limit |
| 1.3.6.1.4.1.6876.100.1.0.80639 | vmwIncorrectInfobloxCredentialEvent | Critical | Incorrect Infoblox Credentials | Infoblox credentials are invalid or the user does not have the "API Privilege" to access Infoblox data |
| 1.3.6.1.4.1.6876.100.1.0.80640 | vmwUnsupportedInfobloxVersionEvent | Critical | Unsupported Infoblox Version | The version of NIOS is not supported. |
| 1.3.6.1.4.1.6876.100.1.0.80641 | vmwUnknownInfobloxVersionEvent | Critical | Unknown Infoblox Version | The version of NIOS cannot be determined. |
| 1.3.6.1.4.1.6876.100.1.0.80642 | vmwNoDVSAvailableEvent | Critical | IPFIX cannot be enabled | IPFIX cannot be enabled since no DVS found |
| 1.3.6.1.4.1.6876.100.1.0.80643 | vmwVCNotOnSameProxyEvent | Critical | NSX Manager and the vCenter DataSource not on the same Collector VM | NSX Manager and the associated vCenter DataSources is not on the same Collector VM. |
| 1.3.6.1.4.1.6876.100.1.0.80644 | vmwNSXTIPFixNoCollectorProfileEvent | Critical | NSX-T IPFIX No Collector Profile | NSXT IPFIX No Collector Profile |
| 1.3.6.1.4.1.6876.100.1.0.80645 | vmwNSXTIPFixNoNewCollectorProfileCanBeAddedEvent | Critical | NSX-T IPFIX no new Collector Profile can be added | NSXT IPFIX no new Collector Profile can be added |
| 1.3.6.1.4.1.6876.100.1.0.80646 | vmwNSXTIPFixNoIPFixProfileEvent | Critical | NSX-T IPFIX No Ipfix Profile | NSXT IPFIX No Ipfix Profile |
| 1.3.6.1.4.1.6876.100.1.0.80647 | vmwNSXTIPFixIPFixProfilePriorityNotZeroEvent | Critical | NSX-T IPFIX Ipfix Profile Priority not zero | NSXT IPFIX Ipfix Profile Priority not zero |
| 1.3.6.1.4.1.6876.100.1.0.80648 | vmwNSXTIPFixCollectorAndIPFixProfileMismatchEvent | Critical | NSX-T IPFIX Collector and Ipfix Profile Mismatch | NSXT IPFIX Collector and Ipfix Profile Mismatch |
| 1.3.6.1.4.1.6876.100.1.0.80649 | vmwNSXTIPFixPortIncorrectEvent | Critical | NSX-T IPFIX Collector Port Incorrect | Collector Port in Collector Profile is incorrect |
| 1.3.6.1.4.1.6876.100.1.0.80650 | vmwNSXTIPFixDFWStatusNotEnabledEvent | Critical | NSX-T IPFIX DFW is not Enabled | NSX-T IPFIX DFW is not Enabled |
| 1.3.6.1.4.1.6876.100.1.0.80651 | vmwPolicyManagerNoDfwIPFixProfile | Critical | DFW IPFIX Profile absent on NSX Policy Manager | DFW IPFIX Profile is not found on NSX Policy Manager |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80652 | vmwPolicyManagerVrniDfwIPFixCollectorAbsent | Critical | VCF Operations for networks IPFIX Collector configuration is absent on NSX Policy Manager | VCF Operations for networks IPFIX Collector IP and port are not present in DFW IPFIX Collector Profile on NSX Policy Manager. |
| 1.3.6.1.4.1.6876.100.1.0.80653 | vmwDatasourceIdentificationChangedEvent | Info | Identity information for Data Source changed | The data source identity information such as certificate or the key has changed. |
| 1.3.6.1.4.1.6876.100.1.0.80654 | vmwPKSKubernetesUnknownHostEvent | Critical | Kubernetes Cluster API Servers not reachable | One or more Kubernetes Cluster's Kube Config file in Tanzu Kubernetes Grid Integrated Edition is not valid. |
| 1.3.6.1.4.1.6876.100.1.0.80655 | vmwKubernetesInsufficientPrivilegesEvent | Critical | Kubernetes Cluster Service Accounts do not have sufficient privileges | One or more Kubernetes Cluster Service Accounts do not have sufficient privileges. |
| 1.3.6.1.4.1.6876.100.1.0.80657 | vmwUANIFileNotProvidedEvent | Critical | File required by Generic Routers & Switches Data Source not provided | File required by Generic Routers & Switches Data Source not provided |
| 1.3.6.1.4.1.6876.100.1.0.80658 | vmwUANIFileDoesNotExistEvent | Critical | File required by Generic Routers & Switches Data Source does not exist | File required by Generic Routers & Switches Data Source does not exist |
| 1.3.6.1.4.1.6876.100.1.0.80659 | vmwNSXTLatencyNotEnabledEvent | Critical | NSXT_LATENCY_NOT_ENABLED_EVENT | NSX-T latency collection is not enabled |
| 1.3.6.1.4.1.6876.100.1.0.80660 | vmwNSXTLatencyMoreBFDProfileEvent | | NSXT_LATENCY_MORE_BFD_PROFILE_EVENT | |
| 1.3.6.1.4.1.6876.100.1.0.80662 | vmwNSXTLatencyCollectorMismatchEvent | Critical | NSXT_LATENCY_COLLECTOR_MISMATCH_EVENT | NSX-T latency collector not configured |
| 1.3.6.1.4.1.6876.100.1.0.80663 | vmwBigIpInsufficientShellAccessEvent | Critical | BIGIP_INSUFFICIENT_SHELL_ACCESS_EVENT | No access to shell |
| 1.3.6.1.4.1.6876.100.1.0.80664 | vmwBigIpInsufficientPartitionAccessEvent | Critical | BIGIP_INSUFFICIENT_PARTITION_ACCESS_EVENT | Insufficient Partition Access |
| 1.3.6.1.4.1.6876.100.1.0.80665 | vmwBigIpInsufficientRoleEvent | Critical | BIGIP_INSUFFICIENT_ROLE_EVENT | Insufficient Role |
| 1.3.6.1.4.1.6876.100.1.0.90001 | vmwVeloCloudEdgeDownEvent | Warning | VeloCloud Edge is not healthy | Edge State of VeloCloud Edge is not connected. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.90002 | vmwVeloCloudLinkDownEvent | Warning | VeloCloud Link is not healthy | Link State of VeloCloud Edge is not connected. |
| 1.3.6.1.4.1.6876.100.1.0.90003 | vmwVeloCloudEdgeL7ApplicationLostPacketEvent | Critical | Packet Drop Alert | Packet Drop Alert on Edge for Application. This is raised when the total packets are more than 1000 and packet drop percentage is greater than 3% and the edge QoE value is less than 9. |
| 1.3.6.1.4.1.6876.100.1.0.90004 | vmwVeloCloudEdgeL7ApplicationRetransmittedPacketEvent | Warning | Retransmitted Packet Alert. | Retransmitted Packet Alert on Edge for Application. This alert is raised when total packets are more than 1000 and retransmitted packet percentage is greater than 5% and the edge QoE value is less than 9. |
| 1.3.6.1.4.1.6876.100.1.0.90005 | vmwVeloCloudLinkLostPacketEventTx | Critical | VeloCloud Link Upstream Packet Loss exceeds threshold. | VeloCloud Link Tx Packet Loss. |
| 1.3.6.1.4.1.6876.100.1.0.90006 | vmwVeloCloudEdgeLocationNotSetEvent | Info | VeloCloud Edge location is not set. | VeloCloud Edge location is not set. |
| 1.3.6.1.4.1.6876.100.1.0.90007 | vmwVeloCloudLinkDegradedVoiceQoeEvent | Critical | VeloCloud Link Voice QOE degraded. | VeloCloud Link Voice QOE Degraded. |
| 1.3.6.1.4.1.6876.100.1.0.90008 | vmwVeloCloudLinkDegradedVideoQoeEvent | Critical | VeloCloud Link Video QOE degraded. | VeloCloud Link Video QOE Degraded. |
| 1.3.6.1.4.1.6876.100.1.0.90009 | vmwVeloCloudLinkDegradedTransQoeEvent | Critical | VeloCloud Link Transactional QOE degraded. | VeloCloud Link Transactional QOE Degraded. |
| 1.3.6.1.4.1.6876.100.1.0.90010 | vmwVeloCloudEdgeDegradedVoiceQoeEvent | Critical | VeloCloud Edge Voice QOE degraded. | VeloCloud Edge Voice QOE degraded. |
| 1.3.6.1.4.1.6876.100.1.0.90011 | vmwVeloCloudEdgeDegradedVideoQoeEvent | Critical | VeloCloud Edge Video QOE degraded. | VeloCloud Edge Video QOE degraded. |
| 1.3.6.1.4.1.6876.100.1.0.90012 | vmwVeloCloudEdgeDegradedTransQoeEvent | Critical | VeloCloud Edge Transactional QOE degraded. | VeloCloud Edge Transactional QOE degraded. |
| 1.3.6.1.4.1.6876.100.1.0.90013 | vmwVeloCloudLinkLostPacketEventRx | Critical | VeloCloud Link Downstream Packet Loss exceeds threshold. | VeloCloud Link RX Packet Loss. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.90014 | vmwVeloCloudTunnelDegradedVoiceQoeEvent | Critical | VeloCloud Tunnel Voice QOE Degraded. | VeloCloud Tunnel Voice QOE Degraded. |
| 1.3.6.1.4.1.6876.100.1.0.90015 | vmwVeloCloudTunnelDegradedVideoQoeEvent | Critical | VeloCloud Tunnel Video QOE Degraded. | VeloCloud Tunnel Video QOE Degraded. |
| 1.3.6.1.4.1.6876.100.1.0.90016 | vmwVeloCloudTunnelDegradedTransQoeEvent | Critical | VeloCloud Tunnel Transaction QOE Degraded. | VeloCloud Tunnel Transaction QOE Degraded. |
| 1.3.6.1.4.1.6876.100.1.0.974 | vmwNSXIApplianceAvailableEvent | Info | Security Intelligence Appliance Available | This event is raised when VCF Operations for networks detects that Security Intelligence appliance is available for an NSX data source. |
| 1.3.6.1.4.1.6876.100.1.0.975 | vmwNSXISubscriptionCreateFailedEvent | Critical | Failed to enable data collection from Security Intelligence | This event is raised when VCF Operations for networks is unable to create a subscription with the Security Intelligence Appliance. |
| 1.3.6.1.4.1.6876.100.1.0.976 | vmwNSXISubscriptionDeleteFailedEvent | Critical | Security Intelligence cleanup failed | This event is raised when VCF Operations for networks is unable to delete a subscription with the Security Intelligence Appliance. |
| 1.3.6.1.4.1.6876.100.1.0.977 | vmwNSXISubscriptionUpdateFailedEvent | Critical | Failed to initiate data collection from Security Intelligence | This event is raised when VCF Operations for networks is unable to update a subscription with the Security Intelligence Appliance. |
| 1.3.6.1.4.1.6876.100.1.0.978 | vmwNSXISSLHandshakeFailedEvent | Critical | Security Intelligence SSL Handshake Failed | This event is raised when the SSL Handshake between VCF Operations for networks Collector and Security Intelligence Appliance fails. |
| 1.3.6.1.4.1.6876.100.1.0.80539 | UNABLE_TO_SEND_SNMP_TRAPS_EVENT | Critical | Unable to Send SNMP Traps | Unable to Send SNMP Traps. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80540 | STREAM_PROCESSING_FAILED_EVENT | Critical | Realtime stream processing failure | Real-time stream processing has failed. |
| 1.3.6.1.4.1.6876.100.1.0.20018 | VMotionEvent | Info | VMotion | This alert is raised when a VM moves from one host to another through live migration. |
| 1.3.6.1.4.1.6876.100.1.0.20018 | VMCSDDCTGWConnectivityFailedEvent | Critical | VMC SDDC VTGW Connectivity Failure | This alert is raised when SDDC within SDDC Group loses connectivity with VTGW |
| 1.3.6.1.4.1.6876.100.1.0.110014 | HSRPConfigurationEvent | Critical | HSRP/VRRP Configuration Error | Identifies HSRP/VRRP configuration errors in the network. |
| 1.3.6.1.4.1.6876.100.1.0.90502 | HcxIXApplianceEncryptionTunnelStatusEvent | Critical | Hybrid Interconnect Service Encryption Tunnel Status Alert | Hybrid Interconnect Service Encryption Tunnel Status is not up. |
| 1.3.6.1.4.1.6876.100.1.0.90503 | HcxNEServiceStatusEvent | Critical | HCX Network Extension Service Status Alert | Network Extension Service is not up. |
| 1.3.6.1.4.1.6876.100.1.0.90504 | HcxNEApplianceEncryptionTunnelStatusEvent | Critical | Network Extension Service Encryption Tunnel Status Alert | Network Extension Service Encryption Tunnel Status is not up |
| 1.3.6.1.4.1.6876.100.1.0.90505 | HcxIXApplianceServicePipelineStatusEvent | Critical | Hybrid Interconnect Service Pipeline Status Alert | Hybrid Interconnect Service Pipeline Status is not up. |
| 1.3.6.1.4.1.6876.100.1.0.90506 | HcxNEApplianceServicePipelineStatusEvent | Critical | Network Extension Service Pipeline Status Alert | Network Extension Service Pipeline Status is not up. |
| 1.3.6.1.4.1.6876.100.1.0.90507 | HcxIXApplianceServiceTransportStatusEvent | Critical | Hybrid Interconnect Service Transport Status Alert | Hybrid Interconnect Service Transport Status is not up. |
| 1.3.6.1.4.1.6876.100.1.0.90508 | HcxNEApplianceServiceTransportStatusEvent | Critical | Network Extension Service Transport Status Alert | Network Extension Service Transport Status is not up. |
| 1.3.6.1.4.1.6876.100.1.0.90509 | HcxIXApplianceServiceSystemStateEvent | Critical | Hybrid Interconnect Service System State Alert | Hybrid Interconnect Service System State is not good. |
| 1.3.6.1.4.1.6876.100.1.0.90510 | HcxNEApplianceServiceSystemStateEvent | Critical | Network Extension Service System State Alert | Network Extension Service System State is not good. |
| 1.3.6.1.4.1.6876.100.1.0.90512 | HcxBMServiceStatusEvent | Critical | HCX Bulk Migration Service Status Alert | Bulk Migration Service is not up. |
| 1.3.6.1.4.1.6876.100.1.0.90513 | HcxVMotionServiceStatusEvent | Critical | HCX VMotion Service Status Alert | VMotion Service is not up. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.90514 | HcxIXServiceNotRunningEvent | Critical | Hybrid Interconnect Service Not Running Alert | Hybrid Interconnect Service is not running. |
| 1.3.6.1.4.1.6876.100.1.0.90515 | HcxNEServiceNotRunningEvent | Critical | Network Extension Service Not Running Alert | Network Extension Service is not running. |
| 1.3.6.1.4.1.6876.100.1.0.90516 | HcxWanOptServiceNotRunningEvent | Critical | WAN Optimisation Service Not Running Alert | WAN Optimisation Service is not running |
| 1.3.6.1.4.1.6876.100.1.0.80542 | CHANGE_EVENTS_DROPPED_EVENT | Warning | Realtime Processing Suppressed | This alerts is raised when realtime processing has stopped due to high system load. Here, some updates can be delayed. |
| 1.3.6.1.4.1.6876.100.1.0.80636 | INCORRECT_WEBHOOK_CONFIGURED_ON_ALERT_EVENT | Critical | Incorrect webhook URL configured on a VCF Operations for logs alert | Incorrect webhook configuration found for one or more alerts of VCF Operations for networks Content Pack in VCF Operations for logs. |
| 1.3.6.1.4.1.6876.100.1.0.80632 | UNSUPPORTED_LOG_INSIGHT_VERSION_EVENT | Critical | VCF Operations for logs version not supported | VCF Operations for logs version is not supported. |
| 1.3.6.1.4.1.6876.100.1.0.80633 | UNSUPPORTED_vRNI_CONTENT_PACK_VERSION_EVENT | Critical | VCF Operations for networks Content Pack version not supported | VCF Operations for networks Content Pack version is not supported. |
| 1.3.6.1.4.1.6876.100.1.0.80635 | WEBHOOK_NOT_ENABLED_ON_ALERT_EVENT | Critical | Webhook not enabled on a Operations for Networks alert | Webhook not enabled for one or more alerts of VCF Operations for networks Content Pack in VCF Operations for logs. |
| 1.3.6.1.4.1.6876.100.1.0.80637 | WEBHOOK_NOT_RUNNING_EVENT | Critical | Webhook is not running on the Collector(Proxy) VM | Webhook is not running on the Collector(Proxy) VM. |
| 1.3.6.1.4.1.6876.100.1.0.80634 | vRNI_CONTENT_PACK_NOT_INSTALLED_EVENT | Critical | VCF Operations for networks Content Pack not found in VCF Operations for logs | VCF Operations for networks Content Pack not found in VCF Operations for logs. |

| OID | Alert Parameter Name | Default Severity | UI Name | Alert Name |
|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80050 | PanLogInsightDroppedFlowEvent | Critical | Dropped Flows Detected on Palo Alto Device | Dropped flows are detected on Palo Alto device through VCF Operations for logs. |
| 1.3.6.1.4.1.6876.100.1.0.80274 | NSXTRouterInterfaceRPFPacketDropEvent | Warning | RPF Packets are getting dropped at NSX-T Router interface | RPF drop ratio exceeded 1% |
| 1.3.6.1.4.1.6876.100.1.0.80543 | HIGH_DISK_IO_UTILIZATION_EVENT | Critical | High Disk IO Utilization | High disk imput/ output utilization. |
| 1.3.6.1.4.1.6876.100.1.0.80051 | CPLogInsightDroppedFlowEvent | Critical | Dropped Flows Detected on Check Point Device | Dropped flows are detected on Check Point device through VCF Operations for logs. |
| 1.3.6.1.4.1.6876.100.1.0.90517 | HcxApplianceAllTunnelDownEvent | Critical | All tunnels down on HCX Appliance Alert | No tunnels are in working state on Hcx Appliance. |
| 1.3.6.1.4.1.6876.100.1.0.90519 | HCX MTU Mismatch Alert | Warning | HCX MTU Mismatch Alert | This alert is raised when the uplink networks for a service mesh have different MTU values. |
| 1.3.6.1.4.1.6876.100.1.0.90518 | HcxConnectionRuleViolationEvent | Warning | HCX Connection Rule Firewall Rule Violated Alert | This alert is raised when the VMware HCX connection rules violate the firewall rules of a datacenter. |
| 1.3.6.1.4.1.6876.100.1.0.90511 | HcxWanOptServiceStatusEvent | Critical | HCX WAN Optimization Service Status Alert | VMware WAN optimization is not up. |
| 1.3.6.1.4.1.6876.100.1.0.90520 | HcxApplianceUnhealthyTunnelsEvent | Warning | Unhealthy tunnels on HCX Appliance | A few tunnels on HCX Appliance are unhealthy. |
| 1.3.6.1.4.1.6876.100.1.0.90521 | HcxManagerVersionMismatchEvent | Critical | Version mismatch between paired HCX Managers | The paired HCX Managers are running different software versions. |
| 1.3.6.1.4.1.6876.100.1.0.90522 | HcxApplianceVersionMismatchEvent | Critical | Version mismatch between paired HCX Appliances | The paired HCX Appliances are running different software versions. |
| 1.3.6.1.4.1.6876.100.1.0.80275 | NSXTIPFixMultipleCollectorConfigurationEvent | Warning | Multiple collectors configured in the NSX Policy DFW IPFIX Collector Profile. | Multiple collectors are configured in the NSX Policy DFW IPFIX Collector Profile. |

## NSX-T Events

VCF Operations for networks raises several self-computed NSX-T alerts. In addition, all NSX-T generated system alerts (for NSX-T versions 2.2 to 2.5) and NSX-T Alarms (for NSX-T version 3.0 onwards) are also visible in VCF Operations for networks.

**Note:**
Starting from 6.3 release, a few events are optimized for quick notification and are delivered under 5 minutes. See the Optimized for quick notification column in the table.

**Table 1315: VCF Operations for networks Computed NSX-T Events**

| OID | Event Name | Default Severity | UI Name | Description | Optimized for quick notification? |
|---|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.40000 | SwitchPortRxNetworkUsageThresholdExceededEvent | Warning | PNIC Rx Network Utilization is high | PNIC utilization exceeds the threshold value of 75%. | No |
| 1.3.6.1.4.1.6876.100.1.0.40001 | vmwCriticalHostNotAccessibleEvent | Critical | Host with Infrastructure VMs cannot be accessed | Host with Infrastructure VMs cannot be accessed | No |
| 1.3.6.1.4.1.6876.100.1.0.40002 | SwitchPortTxNetworkUsageThresholdExceededEvent | Warning | PNIC Tx Network Utilization is high | PNIC utilization exceeds the threshold value of 75%. | No |
| 1.3.6.1.4.1.6876.100.1.0.40003 | SwitchPortRxNetworkUsageThresholdExceededCriticalEvent | Critical | PNIC Rx Network Utilization is high | PNIC utilization exceeds the threshold value of 90%. | No |
| 1.3.6.1.4.1.6876.100.1.0.40004 | SwitchPortTxNetworkUsageThresholdExceededCriticalEvent | Critical | PNIC Tx Network Utilization is high | PNIC utilization exceeds the threshold value of 90%. | No |
| 1.3.6.1.4.1.6876.100.1.0.40005 | ManagementHostHighCPUUsageEvent | Warning | Management Host CPU Usage Event | Host CPU utilization exceeds the threshold value of 75%. | No |
| 1.3.6.1.4.1.6876.100.1.0.40006 | ManagementHostHighCPUUsageCriticalEvent | Critical | Management Host CPU Usage Critical Event | Host CPU utilization exceeds the threshold value of 90%. | No |
| 1.3.6.1.4.1.6876.100.1.0.40007 | HostUnHealthyEvent | Critical | Host is not healthy | Host is not healthy | No |
| 1.3.6.1.4.1.6876.100.1.0.40008 | FileSystem Disk Available of NSX Controller is Low | Critical | FileSystem Disk Available of NSX Controller is Low | NSX Controller file system disk available percentage less than threshold value. | No |
| 1.3.6.1.4.1.6876.100.1.0.40011 | EdgeVmVnicRxPacketDropEvent | Critical | Receive packets are getting dropped on the VNIC of Edge VM. | Packets are getting dropped on the VNIC of Edge VM. This may impact the network traffic related to management nodes. | No |

| OID | Event Name | Default Severity | UI Name | Description | Optimized for quick notification? |
|---|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.40012 | EdgeVmVnicTxPacketDropEvent | Critical | Transmit packets are getting dropped on the VNIC of Edge VM. | Packets are getting dropped on the VNIC of Edge VM. This may impact the network traffic related to management nodes. | No |
| 1.3.6.1.4.1.6876.100.1.0.40013 | ManagementHostPnicRxPacketDropEvent | Critical | Receive packets are getting dropped on the PNIC of the management host. | Packets are getting dropped on the pNIC of the management host. This may impact the network traffic related to management nodes. | No |
| 1.3.6.1.4.1.6876.100.1.0.40014 | ManagementHostPnicTxPacketDropEvent | Critical | Transmit packets are getting dropped on the PNIC of the management host. | Packets are getting dropped on the pNIC of the management host. This may impact the network traffic related to management nodes. | No |
| 1.3.6.1.4.1.6876.100.1.0.80205 | NSXTNoUplinkConnectivityEvent | Warning | NSX-T Tier-1 logical router disconnect event | NSX-T Tier-1 logical router is disconnected from Tier-0 router. Networks under this router are not reachable from outside and vice versa. | No |
| 1.3.6.1.4.1.6876.100.1.0.80206 | NSXTRoutingAdvertisementEvent | Warning | Routing advertisement deactivated | Routing advertisement is deactivated for NSX-T Tier-1 logical router. Networks under this router are not reachable from outside. | No |
| 1.3.6.1.4.1.6876.100.1.0.80207 | NSXTManagerConnectivityDownEvent | Critical | NSX-T Edge Node has no manager connectivity | NSX-T Edge Node has lost manager connectivity. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80208 | NSXTControllerConnectivityDegradedEvent | Warning | Controller connectivity degraded for NSX-T Edge Node | NSX-T Edge Node is not able to communicate with one or more controllers. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80209 | NSXTControllerConnectivityDownEvent | Critical | NSX-T Edge Node has no controller connectivity | NSX-T Edge Node is not able to communicate with any of the controllers. | Yes |

| OID | Event Name | Default Severity | UI Name | Description | Optimized for quick notification? |
|---|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80210 | NSXTMtuMismatchEvent | Warning | MTU mismatch between NSX-T Tier-0 and uplink switch/router | The MTU configured on interfaces of Tier-0 logical router does not match with the interfaces of uplink switch/ router from same L2 network. This can impact the network performance. | No |
| 1.3.6.1.4.1.6876.100.1.0.80211 | NSXTExcludedVmFlowEvent | Info | One or More VMs excluded from NSX-T DFW Firewall. | One or more VMs are not protected by NSX-T DFW firewall. VCF Operations for networks will not receive IPFIX flows for these VMs. | No |
| 1.3.6.1.4.1.6876.100.1.0.80212 | NSXTDoubleVlanTaggingEvent | Warning | Uplink Vlan misconfiguration | Communication is disrupted because VLAN on uplink port of Tier 0 router is different than VLAN on the external gateway. | No |
| 1.3.6.1.4.1.6876.100.1.0.80213 | NSXTNoTzAttachedOnTnEvent | Warning | No transport zone is attached to the transport node. | No transport zone attached to the transport node. VMs might lose connectivity because of this. | No |
| 1.3.6.1.4.1.6876.100.1.0.80214 | NSXTVtepDeleteEvent | Warning | No VTEP available on the transport node. | All vteps are deleted from the transport node. VMs might lose connectivity because of this. | No |
| 1.3.6.1.4.1.6876.100.1.0.80225 | NSXTControllerNodeToControlClusterConnectivityEvent | Critical | NSX-T controller node has no control cluster connectivity | NSX-T controller node has lost control cluster connectivity. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80226 | NSXTControllerNodeToMgmtPlaneConnectivityEvent | Critical | NSX-T controller node has no management plane connectivity | NSX-T controller node has lost management plane connectivity. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80227 | NSXTMPNodeToMgmtClusterConnectivityEvent | Critical | NSX-T management node has no management cluster connectivity | NSX-T management node has lost management cluster connectivity. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80246 | NSXTHostNodeMgmtConnectivityStatusDownEvent | Warning | NSX-T Host Node has no manager connectivity | Desynchronization between NSX Manager's State of connectivity with Host Transport Nodes | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80247 | NSXTEdgeNodeCtlrConnectivityStatusUnknownEvent | Critical | Controller connectivity for NSX-T Edge Node is Unknown. | NSX-T Edge Node Controller connectivity is Unknown. | Yes |

| OID | Event Name | Default Severity | UI Name | Description | Optimized for quick notification? |
|---|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80248 | NSXTHostNodeCtlrConnectivityStatusDownEvent | Warning | NSX-T Host Node has no controller connectivity | NSX-T Host Node is not able to communicate with any of the controllers. | No |
| 1.3.6.1.4.1.6876.100.1.0.80249 | NSXTHostNodeCtlrConnectivityStatusDegradedEvent | Warning | Controller connectivity degraded for NSX-T Host Node | NSX-T Host Node is not able to to communicate with one or more controllers. | No |
| 1.3.6.1.4.1.6876.100.1.0.80250 | NSXTHostNodeCtlrConnectivityStatusUnknownEvent | Warning | Controller connectivity for NSX-T Host Node is Unknown. | NSX-T Host Node Controller connectivity is Unknown. | No |
| 1.3.6.1.4.1.6876.100.1.0.80228 | NSXTHostNodePnicStatusDownEvent | Warning | NSX-T Host Transport Node Pnic Status is 'Down'. | NSX-T Host Transport Node Pnic Status is 'Down'. | No |
| 1.3.6.1.4.1.6876.100.1.0.80229 | NSXTHostNodePnicStatusDegradedEvent | Warning | NSX-T Host Transport Node Pnic Status is 'Degraded' | NSX-T Host Transport Node Pnic Status is 'Degraded'. | No |
| 1.3.6.1.4.1.6876.100.1.0.80230 | NSXTHostNodePnicStatusUnknownEvent | Warning | NSX-T Host Transport Node Pnic Status is 'Unknown'. | NSX-T Host Transport Node Pnic Status is 'Unknown'. | No |
| 1.3.6.1.4.1.6876.100.1.0.80237 | NSXTEdgeNodePnicStatusDownEvent | Critical | NSX-T Edge Transport Node Pnic Status is 'Down'. | NSX-T Edge Transport Node Pnic Status is 'Down'. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80238 | NSXTEdgeNodePnicStatusDegradedEvent | Critical | NSX-T Edge Transport Node Pnic Status is 'Degraded'. | NSX-T Edge Transport Node Pnic Status is 'Degraded'. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80239 | NSXTEdgeNodePnicStatusUnknownEvent | Critical | NSX-T Edge Transport Node Pnic Status is 'Unknown'. | NSX-T Edge Transport Node Pnic Status is 'Unknown'. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80231 | NSXTHostNodeTunnelStatusDownEvent | Warning | NSX-T Host Transport Node Tunnel Status is 'Down'. | NSX-T Host Transport Node Tunnel Status is 'Down'. | No |
| 1.3.6.1.4.1.6876.100.1.0.80232 | NSXTHostNodeTunnelStatusDegradedEvent | Warning | NSX-T Host Transport Node Tunnel Status is 'Degraded'. | NSX-T Host Transport Node Tunnel Status is 'Degraded'. | No |
| 1.3.6.1.4.1.6876.100.1.0.80233 | NSXTHostNodeTunnelStatusUnknownEvent | Warning | NSX-T Host Transport Node Tunnel Status is 'Unknown'. | NSX-T Host Transport Node Tunnel Status is 'Unknown'. | No |
| 1.3.6.1.4.1.6876.100.1.0.80240 | NSXTEdgeNodeTunnelStatusDownEvent | Critical | NSX-T Edge Transport Node Tunnel Status is 'Down'. | NSX-T Edge Transport Node Tunnel Status is 'Down'. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80241 | NSXTEdgeNodeTunnelStatusDegradeEvent | Critical | NSX-T Edge Transport Node Tunnel Status is 'Degraded'. | NSX-T Edge Transport Node Tunnel Status is 'Degraded'. | Yes |

| OID | Event Name | Default Severity | UI Name | Description | Optimized for quick notification? |
|---|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80242 | NSXTEdgeNodeTunnelStatusUnknownEvent | Critical | NSX-T Edge Transport Node Tunnel Status is 'Unknown'. | NSX-T Edge Transport Node Tunnel Status is 'Unknown'. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80234 | NSXTHostNodeStatusDownEvent | Warning | NSX-T Host Transport Node Status is 'Down'. | NSX-T Host Transport Node Status is 'Down'. | No |
| 1.3.6.1.4.1.6876.100.1.0.80235 | NSXTHostNodeStatusDegradedEvent | Warning | NSX-T Host Transport Node Status is 'Degraded'. | NSX-T Host Transport Node Status is 'Degraded'. | No |
| 1.3.6.1.4.1.6876.100.1.0.80236 | NSXTHostNodeStatusUnknownEvent | Warning | NSX-T Host Transport Node Status is 'Unknown'. | NSX-T Host Transport Node Status is 'Unknown'. | No |
| 1.3.6.1.4.1.6876.100.1.0.80243 | NSXTEdgeNodeStatusDownEvent | Critical | NSX-T Edge Transport Node Status is 'Down'. | NSX-T Edge Transport Node Status is 'Down'. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80244 | NSXTEdgeNodeStatusDegradedEvent | Critical | NSX-T Edge Transport Node Status is 'Degraded'. | NSX-T Edge Transport Node Status is 'Degraded'. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80245 | NSXTEdgeNodeStatusUnknownEvent | Critical | NSX-T Edge Transport Node Status is 'Unknown'. | NSX-T Edge Transport Node Status is 'Unknown'. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80252 | NSXTLogicalSwitchAdminStatusDownEvent | Warning | NSX-T Logical Switch Admin Status is 'Down' | NSX-T Logical Switch Admin Status is 'Down' | No |
| 1.3.6.1.4.1.6876.100.1.0.80253 | NSXTLogicalPortOperationalStatusDownEvent | Critical | NSX-T Logical Port Operational Status is 'Down' | NSX-T Logical Port Operational Status is 'Down'. This could cause a communication failure between two virtual interfaces (VIFs) that are connected to the same logical switch, for example, you cannot ping one VM from another. | No |
| 1.3.6.1.4.1.6876.100.1.0.80254 | NSXTLogicalPortOperationalStatusUnknownEvent | Warning | NSX-T Logical Port Operational Status is 'Unknown' | NSX-T Logical Port Operational Status is 'Unknown'. This could cause a communication failure between two virtual interfaces (VIFs) that are connected to the same logical switch, for example, you cannot ping one VM from another. | No |
| 1.3.6.1.4.1.6876.100.1.0.80255 | NSXTComputeManagerConnectionStatusNotUpEvent | Warning | NSX-T Compute Manager Connection Status in not up | NSX-T Compute Manager Connection status is not up | Yes |

| OID | Event Name | Default Severity | UI Name | Description | Optimized for quick notification? |
|---|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80256 | NSXTClusterBackUpDisabledEvent | Warning | NSX-T Manager backup is not scheduled. | NSX-T Manager backup is not scheduled | No |
| 1.3.6.1.4.1.6876.100.1.0.80257 | NSXTDFWFirewallDisabledEvent | Critical | NSX-T DFW Firewall is deactivated. | Distributed Firewall is deactivated in the NSX-T Manager | No |
| 1.3.6.1.4.1.6876.100.1.0.80258 | NSXTLogicalPortReceivedPacketDropEvent | Warning | NSX-T Logical Port Received Packets are getting dropped. | Received packets are getting dropped on the NSX-T Logical Port and associated entities might get affected | No |
| 1.3.6.1.4.1.6876.100.1.0.80259 | NSXTLogicalPortTransmittedPacketDropEvent | Warning | NSX-T Logical Port Transmitted Packets are getting dropped. | Transmitted packets are getting dropped on the NSX-T Logical Port and associated entities might get affected | No |
| 1.3.6.1.4.1.6876.100.1.0.80260 | NSXTLogicalSwitchReceivedPacketDropEvent | Warning | NSX-T Logical Switch Received Packets are getting dropped | Received packets are getting dropped on the NSX-T Logical Switch and associated entities might get affected | No |
| 1.3.6.1.4.1.6876.100.1.0.80261 | NSXTLogicalSwitchTransmittedPacketDropEvent | Warning | NSX-T Logical Switch Transmitted Packets are getting dropped | Transmitted packets are getting dropped on the NSX-T Logical Switch and associated entities might get affected | No |
| 1.3.6.1.4.1.6876.100.1.0.80262 | NSXTRxPacketDropOnMPNicEvent | Warning | Received packets are dropping on NSX-T Management Node's network interface | Received packets are getting dropped on NSX-T Management Node's network interface. This may impact the network traffic related to NSX-T management cluster. | No |
| 1.3.6.1.4.1.6876.100.1.0.80263 | NSXTRxPacketDropOnEdgeTnNicEvent | Critical | Received packets are dropping on NSX-T Edge Node's network interface | Received packets are getting dropped on NSX-T Edge Node's network interface. This may impact the network traffic of edge cluster. | No |
| 1.3.6.1.4.1.6876.100.1.0.80264 | NSXTRxPacketDropOnHostTnNicEvent | Warning | Received packets are dropping on NSX-T Host Node's network interface | Received packets are getting dropped on NSX-T Host Node's network interface. This may impact the network traffic on ESX host. | No |

| OID | Event Name | Default Severity | UI Name | Description | Optimized for quick notification? |
|---|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80265 | NSXTTxPacketDropOnMPNicEvent | Warning | Transmitted packets are dropping on NSX-T Management Node's network interface | Transmitted packets are getting dropped on NSX-T Management Node's network interface. This may impact the network traffic related to NSX-T management cluster. | No |
| 1.3.6.1.4.1.6876.100.1.0.80266 | NSXTTxPacketDropOnEdgeTnNicEvent | Critical | Transmitted packets are dropping on NSX-T Edge Node's network interface | Transmitted packets are getting dropped on NSX-T Edge Node's network interface. This may impact the network traffic of edge cluster. | No |
| 1.3.6.1.4.1.6876.100.1.0.80267 | NSXTTxPacketDropOnHostTnNicEvent | Warning | Host Node's network interface | Transmitted packets are getting dropped on NSX-T Host Node's network interface. This may impact the network traffic on ESX host. | No |
| 1.3.6.1.4.1.6876.100.1.0.80402 | NSXTMPNodeServiceCmInventoryStatusEvent | Warning | CM Inventory Service has stopped running | CM Inventory Service status has turned to stopped. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80403 | NSXTMPNodeServiceControllerStatusEvent | Critical | Controller Service has stopped running. | Controller Service status has turned to stopped. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80404 | NSXTMPNodeServiceDataStoreStatusEvent | Critical | DataStore Service has stopped running. | DataStore Service status has turned to stopped. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80405 | NSXTMPNodeServiceHttpStatusEvent | Critical | HTTP Service has stopped running. | HTTP Service status has turned to stopped. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80406 | NSXTMPNodeServiceInstallUpgradeEvent | Critical | Install Upgrade Service has stopped running. | Install Upgrade Service status has turned to stopped. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80407 | NSXTMPNodeServiceLiagentStatusEvent | Critical | Liagent service has stopped running. | Liagent Service status has turned to stopped. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80408 | NSXTMPNodeServiceManagerStatusEvent | Critical | Manager Service has stopped running. | Manager Service status has turned to stopped. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80409 | NSXTMPNodeServiceMgmtPlaneBusStatusEvent | Critical | Management Plane Service has stopped running. | Management Service status has turned to stopped. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80410 | NSXTMPNodeServiceMigrationCoordinatorStatusEvent | Critical | Migration Co-ordinator Service has stopped running. | Migration Co-ordinator Service status has turned to stopped. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80411 | NSXTMPNodeServiceNodeMgmtStatusEvent | Critical | Node Management Service has stopped running. | Node Management Service status has turned to stopped. | Yes |

| OID | Event Name | Default Severity | UI Name | Description | Optimized for quick notification? |
|---|---|---|---|---|---|
| 1.3.6.1.4.1.6876.100.1.0.80412 | NSXTMPNodeServiceNodeStatsStatusEvent | Warning | Node Statistics Service has stopped running. | Node Statistics Service status has turned to stopped. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80413 | NSXTMPNodeServiceMessageBusStatusEvent | Warning | Message Bus Service has stopped running. | Message Bus Client Service status has turned to stopped. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80414 | NSXTMPNodeServicePlatformClientStatusEvent | Warning | Platform Client Service has stopped running. | Platform Client Service status has turned to stopped. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80415 | NSXTMPNodeServiceUpgradeAgentStatusEvent | Warning | Upgrade Agent Service has stopped running. | Upgrade Service status has turned to stopped. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80416 | NSXTMPNodeServiceNTPStatusEvent | Warning | NTP Service has stopped running. | NTP Service status has turned to stopped. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80417 | NSXTMPNodeServicePolicyStatusEvent | Warning | Policy Service has stopped running. | Policy Service status has turned to stopped. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80418 | NSXTMPNodeServiceSearchStatusEvent | Warning | Search Service has stopped running. | Search Service status has turned to stopped. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80419 | NSXTMPNodeServiceSNMPStatusEvent | Warning | SNMP Service has stopped running. | SNMP Service status has turned to stopped. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80420 | NSXTMPNodeServiceSSHStatusEvent | Warning | SSH Service has stopped running. | SSH Service status has turned to stopped. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80421 | NSXTMPNodeServiceSyslogStatusEvent | Warning | Syslog Service has stopped running. | Syslog Service status has turned to stopped. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80422 | NSXTMPNodeServiceTelemetryStatusEvent | Warning | Telemetry Service has stopped running. | Telemetry Service status has turned to stopped. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80423 | NSXTMPNodeServiceUIServiceStatusEvent | Warning | UI Service has stopped running. | UI Service status has turned to stopped. | Yes |
| 1.3.6.1.4.1.6876.100.1.0.80424 | NSXTMPNodeServiceClusterManagerStatusEvent | Critical | Cluster Manager Service has stopped | One of the Services of the NSX-T Management Node, namely Cluster Manager Service has stopped running. | Yes |

## NSX-T System Events

Here is the list of NSX-T 2.2 to 2.5 events supported in VCF Operations for networks. The Object ID (OID) for all these NSX-T system events is `1.3.6.1.4.1.6876.100.1.0.80203`.

Starting from 6.3 release, all NSX-T system events are optimized for quick notification and are delivered before 5 minutes.

**Table 1316: NSX-T System Events**

| Event Name | Description |
|---|---|
| vmwNSXPlatformSysCpuUsage | CPU Usage on both manager and edge appliances (NSX-T 2.2). |
| vmwNSXPlatformSysDiskUsage | Disk Space Usage on both manager and edge appliance for /var/log partition (NSX-T 2.2). |
| vmwNSXPlatformSysMemUsage | Memory Usage on both manager and edge appliance (NSX-T 2.2). |
| vmwNSXPlatformSysConfigDiskUsage | Disk Usage for Manager and Edge Appliances for /config partition (NSX-T 2.4). |
| vmwNSXPlatformSysVarDumpDiskUsage | Disk Usage for Manager and Edge Appliances for /var/dump partition (NSX-T 2.5). |
| vmwNSXPlatformSysRepositoryDiskUsage | Disk Usage for Manager and Edge Appliances for /repository partition (NSX-T 2.5). |
| vmwNSXPlatformSysRootDiskUsage | Disk usage for Manager and Edge appliances for root partition (NSX-T 2.5). |
| vmwNSXPlatformSysTmpDiskUsage | Disk usage for Manager and Edge appliances for tmp partition (NSX-T 2.5). |
| vmwNSXPlatformSysImageDiskUsage | Disk Usage for Manager and Edge appliances for /image partition (NSX-T 2.5). |
| vmwNSXDhcpPoolUsageOverloadedEvent | DHCP pool overloaded/normal (NSX-T 2.5). |
| vmwNSXDhcpPoolLeaseAllocationFailedEvent | DHCP pool lease allocation failed/succeeded (NSX-T 2.5). |
| vmwNSXPlatformPasswordExpiryStatus | Password expiration for manager (NSX-T 2.4). |
| vmwNSXPlatformCertificateExpiryStatus | Certificate expiration for manager (NSX-T 2.4). |
| vmwNSXRoutingBgpNeighborStatus | BGP neighbor status (NSX-T 2.2). |
| vmwNSXVpnTunnelState | VPN Tunnel up/down (NSX-T 2.2). |
| vmwNSXVpnL2TunnelStatus | L2 VPN Session up/down (NSX-T 2.2). |
| vmwNSXVpnIkeSessionStatus | IKE Session up/down (NSX-T 2.2). |
| vmwNSXDnsForwarderStatus | DNS Forwarder Status (NSX-T 2.4). |
| vmwNSXClusterNodeStatus | Cluster Node status (NSX-T 2.4). |
| vmwNSXFabricCryptoStatus | Edge crypto mux driver failed/passed Known_Answer_Tests(KAT) (NSX-T 2.4). |
| Manager Disk Utilization is not OK | |
| BGP Neighbor down | Need an alert when the BGP neighbor is down. |
| BGP Neighbor Up | Clear Alarm when a neighbor comes up. |
| Storage usage over X | Alarm for Storage over X - Event is raised for all appliance VM (MP, CCP) or transport nodes (edge, host). |
| Memory usage over X | Alarm for Memory over X - Event is raised for all appliance VM (MP, CCP) or transport nodes (edge, host). |
| CPU usage over X | Alarm for CPU over X - Event is raised for all appliance VM (MP, CCP) or transport nodes (edge, host). |

## NSX-T System Alarms

In addition to NSX-T events, all NSX-T 3.0 alarms are displayed as NSX-T system events in VCF Operations for networks 5.2 and later. You can see the complete list alarms generated by NSX-T at: *https://nsx-t_ip_address/nsx/#/app/home/alarms/alarm-definitions*. Here, replace *nsx-t_ip_address* with the IP address of your NSX-T instance.

> **Note:**
> Starting from 6.3 release, all NSX-T system alarms are optimized for quick notification and are delivered before 5 minutes.

## Kubernetes Events

Here is the list of Kubernetes alerts supported in VCF Operations for networks. The Object ID (OID) for all Kubernetes alerts is `1.3.6.1.4.1.6876.100.1.0.1510`.

| Event Name | Severity | Description |
|---|---|---|
| FailedToCreateContainer | Critical | Failed to create container |
| FailedToStartContainer | Critical | Failed to start container |
| PreemptContainer | Warning | Preempting other pods. |
| BackOffStartContainer | Warning | Back-off restarting failed the container. |
| ExceededGracePeriod | Warning | Container runtime did not stop the pod within specified grace period. |
| FailedToKillPod | Warning | Failed to stop a pod. |
| FailedToCreatePodContainer | Moderate | Failed to create a pod container. |
| FailedToMakePodDataDirectories | Moderate | Failed to make pod data directories. |
| NetworkNotReady | Warning<br>Critical | Network is not ready. |
| FailedScheduling | Critical | Unable to schedule pod |
| FailedToPullImage | Warning<br>Critical | Failed to pull the image. |
| FailedToInspectImage | Warning | Failed to inspect the image. |
| ErrImageNeverPullPolicy | Warning | The image's NeverPull Policy is violated. |
| ImagePullBackOff | Critical | Container image pull failed, kubelet is backing off image pull |
| ImageInspectError | Warning | Unable to inspect image |
| ErrImagePull | Critical | Image pull error |
| ErrImageNeverPull | Critical | Required Image is absent on host and PullPolicy is NeverPullImage |
| RegistryUnavailable | Critical | Getting http error when pulling image from registry |
| InvalidImageName | Critical | Unable to parse the image name |
| KubeletSetupFailed | Moderate | Kubelet setup failed. |
| FailedAttachVolume | Critical | Failed to attach volume. |
| FailedMountVolume | Critical | Failed to mount volume. |
| VolumeResizeFailed | Warning | Failed to expand/reduce volume. |
| FileSystemResizeFailed | Warning | Failed to expand/reduce file system. |
| FailedMapVolume | Critical | Failed to map a volume. |
| WarnAlreadyMountedVolume | Warning | Volume is already mounted. |
| ContainerGCFailed | Warning | Container garbage collection failed. |
| ImageGCFailed | Warning | Image garbage collection failed. |
| FailedNodeAllocatableEnforcement | Warning | Failed to enforce System Reserved Cgroup limit. |
| FailedCreatePodSandBox | Warning | Failed to create pod sandbox. |
| FailedStatusPodSandBox | Warning | Failed pod sandbox status. |
| InvalidDiskCapacity | Moderate | Invalid disk capacity. |
| FreeDiskSpaceFailed | Moderate | Free disk space failed. |
| ContainerUnhealthy | Critical | Container is unhealthy. |

| Event Name | Severity | Description |
|---|---|---|
| ContainerProbeWarning | Warning | Container Probe succeeded with a warning. |
| FailedSync | Warning | Pod sync failed. |
| FailedValidation | Warning | Failed pod configuration validation. |
| FailedPostStartHook | Warning | Handler failed for pod start. |
| FailedPreStopHook | Warning | Handler failed for pre-stop. |
| NodeNotReady | Critical | Node is not ready. |
| NodeNotSchedulable | Critical | Node is not schedulable. |
| NodeRebooted | Moderate | Node rebooted. |

## Configure Mail Server

In VCF Operations for networks, you can configure a mail server to receive alert notifications through mail.

To configure a mail server:

1. Go to **Settings** > **Alerts and Notification** > **Mail Server** > **CONFIGURE**.
2. In the **Configure Mail Server** page, enter appropriate values:

| Option | Action |
|---|---|
| Sender Email | Enter an email address where you want to get the notifications. |
| SMTP Hostname/IP Address | Enter a Hostname or IP address of the SMTP server. |
| Encryption | From the drop-down menu, select an encryption type.<br>The following encryption options are available: None, Use SSL, and Support TLS. |
| SMTP Port Number (Optional) | Port number of the SMTP server (default 25). |

**Note:**
To use a Gmail server as the choice of email server, additional configuration settings as listed on Google Support are required.

Optionally, for additional security, select the Authentication check box, and enter the user name and password.

3. Click **SUBMIT** to complete the configuration.

## Configure SNMP Trap Destination

In VCF Operations for networks, you can configure maximum four Simple Network Management Protocol (SNMP) trap agents to receive notifications. The product supports the v2c and v3 versions of SNMP:

1. Go to **Settings** > **Alerts and Notification** > **SNMP Trap Destinations** > **ADD DESTINATION**.
2. In the **Add SNMP Trap Destination** page, in the **Version** drop-down box, select **SNMPv2c** or **SNMPv3** protocol.

   **Note:**
   SNMP v2c protocol does not require authentication. SNMP v3 protocol supports authentication.
3. In the **Destination IP Address/FQDN** text box, enter the IP address of the SNMP agent or the Fully Qualified Domain Name (FQDN).
4. In the **Destination Port** text box, enter a port number of the SNMP agent.

5. Depending on the SNMP version, perform one of the following:

| Option | Action |
|---|---|
| For SNMP v2c | In the **Community String** text box, enter a community string. |
| For SNMP v3 | 1. In the **Username** text box, enter the name of the user you have created in the SNMP agent.<br>2. Select the **Use Authentication** check box.<br>3. Select an authentication protocol, and then enter the password you had set for the particular user in the SNMP agent.<br>4. Select the **Use Privacy**check box, and select a privacy protocol and enter a privacy phrase respectively. |

6. In the **Nickname** field, enter a nick name.
7. To verify whether the configuration is correctly done, click **SEND TEST TRAP**, and then check whether the trap has been sent to the SNMP agent.
8. Click **SUBMIT**.

#### Delete an SNMP Trap Destination

You can delete an SNMP Trap Destination from VCF Operations for networks. If you have multiple SNMP Trap Destinations, then when you delete an SNMP Trap Destination, you can migrate all the notification related to that trap destination to another available trap destination.

1. Click the **Delete** icon next to the trap destination you want to delete.

   A **Confirm Action** pop-up opens.
2. If you want to migrate the alert from the current trap destination to another trap destination, click **Select multiple destinations** drop-down and select the trap destination where you want to migrate the alerts.
3. Click **CONFIRM**.

## Configuring Identity and Access Management

In VCF Operations for networks, you can create users, or configure the access of LDAP users and VMware Identity Manager users. You can also assign different roles to the users.

### User Roles

VCF Operations for networks supports three types of user roles for a user. Users can access VCF Operations for networks features based on the assigned roles.

- **Admin:** An administrator has complete access.
- **Member:** A member user has limited access.
- **Auditor:** An auditor has read-only access, and restricted from all create, add, edit, or delete actions. Users can only view the state.

**Table 1317: Supported features for each role**

| Page | Actions | Admin | Member |
|---|---|---|---|
| [Settings] Logs: Audit Logs | View: Audit Logs Page/Tab | Allowed | -NA- |
| | Enable/Disable: Personally Identifiable Information | Allowed | -NA- |
| | View/Filter: Audit logs | Allowed | -NA- |
| | Export as CSV | Allowed | -NA- |
| [Settings] Logs: Syslog Configuration | View: Syslog Configuration Page/Tab | Allowed | -NA- |

| Page | Actions | Admin | Member |
|---|---|---|---|
| | Enable/Disable Syslog | Allowed | -NA- |
| | Add: Syslog Server | Allowed | -NA- |
| | Edit/Delete: Syslog Servers | Allowed | -NA- |
| | View: Syslog Servers | Allowed | -NA- |
| | View: Source Server Mapping | Allowed | -NA- |
| | Edit: Source Server Mapping | Allowed | -NA- |
| [Settings] About | View Details about Product (Name, Version, Service Tag) | Allowed | Allowed |
| [Settings] System Configuration | View: System Configuration Page/Tab | Allowed | -NA- |
| | View: User Session Timeout | Allowed | -NA- |
| | Edit: User Session Timeout | Allowed | -NA- |
| | View: Data Source Certificate Validation | Allowed | -NA- |
| | Edit: Data Source Certificate Validation | Allowed | -NA- |
| | View: Google Maps Api Key | Allowed | -NA- |
| | Edit: Google Maps Api Key | Allowed | -NA- |
| [Settings] My Preferences | View/Edit: My Preferences | Allowed | Allowed |
| [Settings] License and Usage | View: License and Usage Page/Tab | Allowed | -NA- |
| | View: License Details | Allowed | -NA- |
| | Add/Validate: License Key | Allowed | -NA- |
| | Delete: License Key | Allowed | -NA- |
| | Option: "Want to manage data sources" | Allowed | -NA- |
| | Option (Link to Accounts and Data Source Page): "ADD Data Source to Current Usage" | Allowed | -NA- |
| [Settings] SNMP Trap Destinations | View: SNMP Trap Destination Page/Tab | Allowed | -NA- |
| | View: List of existing SNMP Destinations (with number of alerts configured) | Allowed | -NA- |
| | View: List of alerts configured for each SNMP Destination | Allowed | -NA- |
| | Add/Edit/Delete/Migrate/Send-TestTRAP: SNMP Destinations | Allowed | -NA- |
| [Settings] Mail Server | View: Mail Server Page/Tab | Allowed | -NA- |
| | View: Mail Server existing configuration | Allowed | -NA- |
| | Add/Edit/Delete: Mail Server Config | Allowed | -NA- |
| | Send Test Email | Allowed | -NA- |
| [Settings] Identity & Access Management | View: Identity & Access Management Page/Tab | Allowed | -NA- |
| [Settings] Identity & Access Management: LDAP | View: LDAP Page/Tab | Allowed | -NA- |
| | View: LDAP existing Configuration | Allowed | -NA- |
| | Add/Edit/Delete: LDAP Configuration | Allowed | -NA- |
| [Settings] Identity & Access Management: VIDM | View: VIDM Page/Tab | Allowed | -NA- |
| | View: VIDM existing Configuration | Allowed | -NA- |
| | Add/Edit/Delete: VIDM Configuration | Allowed | -NA- |
| | Toggle: VIDM Configuration | Allowed | -NA- |
| [Settings] Identity & Access Management: User Management | View: User Management Page/Tab | Allowed | -NA- |
| | View: Local/LDAP/VIDM users | Allowed | -NA- |
| | Add/Edit/Delete: Local User | Allowed | -NA- |
| | Add/Edit/Delete: LDAP User | Allowed | -NA- |

| Page | Actions | Admin | Member |
|---|---|---|---|
| | Add/Edit/Delete: VIDM User | Allowed | -NA- |
| [Settings] Events | View: Events Page/Tab | Allowed | Allowed |
| [Settings] Events: System Events | View: System Events Page/Tab | Allowed | Allowed |
| | View: List of System Events | Allowed | Allowed |
| | Edit: System Events | Allowed | Allowed |
| | Enable/Disable: System Events | Allowed | Allowed |
| | Bulk Edit/Enable/Disable: System Events | Allowed | Allowed |
| [Settings] Events: Platform Health Events | View: Platform Health Events Page/Tab | Allowed | Allowed |
| | View: List of Platform Health Events | Allowed | Allowed |
| | Edit: Platform Health Events | Allowed | Allowed |
| | Bulk Edit: Platform Health Events | Allowed | Allowed |
| [Settings] Events: User-Defined Events | View: User-Defined Events Page/Tab | Allowed | Allowed |
| | View: List of User-Defined Events | Allowed | Allowed |
| | Edit/Delete: User-Defined Events | Allowed | Allowed |
| | Enable/Disable: User-Defined Events | Allowed | Allowed |
| [Settings] IP Properties and Subnets | View: IP Properties and Subnets Page/Tab | Allowed | -NA- |
| [Settings] Physical IP and DNS Mapping | View: Physical IP and DNS Mapping Page/Tab | Allowed | -NA- |
| | View: Last Imported Physical IP and DNS Mapping | Allowed | -NA- |
| | Download: Physical IP and DNS Mapping file | Allowed | -NA- |
| | Upload/Replace: Physical IP and DNS Mapping | Allowed | -NA- |
| | Delete: Existing Physical IP and DNS Mapping | Allowed | -NA- |
| [Settings] Physical Subnets and VLAN | View: Physical Subnets and VLAN Page/Tab | Allowed | -NA- |
| | View: Existing list of Physical Subnets and VLAN configured | Allowed | -NA- |
| | Add/Edit/Delete: Physical Subnets and VLAN | Allowed | -NA- |
| [Settings] East-West IPs | View: East-West IPs Page/Tab | Allowed | -NA- |
| | View: Existing East-West IP tags | Allowed | -NA- |
| | Add/Update/Delete: East-West IP tags | Allowed | -NA- |
| [Settings] North-South IPs | View: North-South IPs Page/Tab | Allowed | -NA- |
| | View: Existing North-South IP tags | Allowed | -NA- |
| | Add/Update/Delete: North-South IP tags | Allowed | -NA- |
| [Settings] Accounts and Data Sources | View: Accounts and Data Sources Page/Tab | Allowed | -NA- |
| | View: Existing Data Sources | Allowed | -NA- |
| | Add/Edit/Delete: Data Sources | Allowed | -NA- |
| | Enable/Disable: Existing Data Sources | Allowed | -NA- |
| [Settings] Data Management | View: Data Management Page/Tab | Allowed | -NA- |
| | View: Data Retention Interval details | Allowed | -NA- |
| | Edit: Data Retention Interval details | Allowed | -NA- |
| [Settings] Infrastructure and Support | View: Infrastructure and Support Page/Tab | Allowed | -NA- |
| [Settings] Infrastructure and Support: Infrastructure and Updates | View: Infrastructure and Updates Page/Tab | Allowed | -NA- |
| | View: Infrastructure and Updates Details | Allowed | -NA- |
| | Enable/Disable: Online Update Status | Allowed | -NA- |
| | View Details/Start Upgrade: Online Update | Allowed | -NA- |
| | View: Offline Update | Allowed | -NA- |
| | Upload: Offline Bundle | Allowed | -NA- |

| Page | Actions | Admin | Member |
|---|---|---|---|
| | View: System Health | Allowed | -NA- |
| | View: Platform VMs | Allowed | -NA- |
| | Create Cluster | Allowed | -NA- |
| | Download: Support Bundle | Allowed | -NA- |
| | View: Collector VMs | Allowed | -NA- |
| | Add/Edit/Delete: Collector VMs | Allowed | -NA- |
| [Settings] Infrastructure and Support: Support | View: Support Page/Tab | Allowed | -NA- |
| | View: Product Support Details | Allowed | -NA- |
| | Enable/Disable: Support Tunnel | Allowed | -NA- |
| | View: Customer Experience Improvement Program | Allowed | -NA- |
| | Edit: Customer Experience Improvement Program | Allowed | -NA- |
| | Create: Support Bundle | Allowed | -NA- |
| | Download: Support Bundles | Allowed | -NA- |
| [Settings] Templates | View: Templates Page/Tab | Allowed | Allowed |
| [Settings] Templates: Property Templates | View: Property Templates Page/Tab | Allowed | Allowed |
| | View: Existing Property Templates | Allowed | Allowed |
| | Clone/Edit/Delete: Existing Property Templates | Allowed | Allowed (Self created) |
| [Settings] Templates: App Discovery Templates | View: App Discovery Templates Page/Tab | Allowed | Allowed |
| | View: Existing App Discovery Templates | Allowed | Allowed |
| | Clone/Edit/Delete: Existing App Discovery Templates | Allowed | Allowed (Self created) |
| [Dashboard] Plan & Assess | View: Plan & Assess Tab | Allowed | Allowed |
| [Dashboard] Plan & Assess: Security Planning | View: Security Planning Page (Micro-Segments, Traffic Distribution, Top Ports by Bytes) | Allowed | Allowed |
| | Analyse: Security Planning | Allowed | Allowed |
| | Pin widgets | Allowed | Allowed |
| | Assessment Report | Allowed | Allowed |
| | Wheel/List View: Micro Segments | Allowed | Allowed |
| | Export as CSV | Allowed | Allowed |
| [Dashboard] Plan & Assess: PCI Compliance | View: PCI Compliance Page/Tab | Allowed | Allowed |
| | Assess: PCI Compliance | Allowed | Allowed |
| | Pin widgets/ Create Notifications | Allowed | Allowed |
| | Export CSV/PDF | Allowed | Allowed |
| | Help | Allowed | Allowed |
| [Dashboard] Plan & Assess: Applications | View: Applications Page/Tab | Allowed | Allowed |
| | Add: Applications | Allowed | Allowed |
| | Edit/Delete: Existing Applications | Allowed | Allowed |
| | Export | Allowed | Allowed |
| Application Discovery | View: Discover Tab | Allowed | Allowed |
| | Discover Applications | Allowed | Allowed |
| [Dashboard] Analytics | View: Analytics Page/Tab | Allowed | Allowed |
| [Dashboard] Analytics: Outliers | View: Outliers Page/Tab | Allowed | Allowed |

| Page | Actions | Admin | Member |
|---|---|---|---|
| | View: Existing Outliers Configurations | Allowed | Allowed |
| | Add/Edit/Delete: Existing Outlier Configuration | Allowed | Allowed |
| | Enable/Disable: Existing Outlier Configuration | Allowed | Allowed |
| | Pin widget | Allowed | Allowed |
| [Dashboard] Analytics: Thresholds | View: Thresholds Page/Tab | Allowed | Allowed |
| | View: Existing Thresholds Configurations | Allowed | Allowed |
| | Add/Edit/Delete: Existing Thresholds Configuration | Allowed | Allowed |
| | Enable/Disable: Existing Thresholds Configuration | Allowed | Allowed |
| | Pin widget | Allowed | Allowed |
| [Dashboard] Analytics: Flow Insights | View: Flow Insights Page/Tab | Allowed | Allowed |
| | Analyze: Flow Insights | Allowed | Allowed |
| | Pin widgets | Allowed | Allowed |
| | Export as CSV/ Maximize/ Help | Allowed | Allowed |
| Saved Searches | View: Default Saved Search | Allowed | Allowed |
| | Add/Delete: New Saved Search | Allowed | Allowed |

**Add Local Users**

VCF Operations for networks allows you to add users and assign a role to each user.

1. Go to **Settings** > **Identity & Access Management** > **User Management** .
2. On the User Management page, select the **Local Users** tab.
3. Click **ADD USER** and provide the required details.

| Properties | Description |
|---|---|
| Name | Enter the name of the user. |
| Email (Login ID) | Enter your email or login ID if any. |
| Role | Select the role from drop-down list. |
| Password | Enter the password. |
| Re-enter new password | Re-enter the password for confirmation. |

4. Click **Add User** to save the user information.

**Assign Roles to LDAP Users**

You can assign roles to any LDAP users to allow access to VCF Operations for networks.

Configure LDAP.

1. Go to **Settings** > **Identity & Access Management**.
2. Click **User Management** and select the **LDAP Users** tab.
3. Click **ADD USER**.
4. Provide the login ID of the user to whom you want to assign a role.
5. Select a role from the list. For details, see Configure User Management.
6. Click **ADD USER**.

Configure Lightweight Directory Access Protocol (LDAP)

To allow the LDAP users to log into VCF Operations for networks, you must configure the LDAP service in the VCF Operations for networks platform.

You must have the **Administrator** privilege.

**Note:**
VCF Operations for networks supports the LDAP and LDAPS protocols.

1. Log in to VCF Operations for networks and click **Settings**.
2. Under **Identity & Access Management**, select **LDAP**.
3. Click **Configure**.
4. Provide the following information.

| Field | Description |
|---|---|
| **Domain** | Enter the domain name. This is typically the last part of the user email address after the '@' sign. Example: For a user logging in as johndoe@example.com, this field is `example.com` |
| **LDAP Host URLs** | Enter the host name. You can specify multiple LDAP Host URLs separated by commas. The URLs can be of type IPv4, IPv6, or FQDN.<br>Examples:<br>• IPv4: (ldap://192.168.1.1:389)<br>• IPv6: (ldap://[fc00:192:168:20::e]:389)<br>• FQDN: (ldap://host:port/ldaps://host:port) |
| **Group based access control** | Select this option to configure a group and provide a role to the members of that group.<br>1. Under**Base DN**, type the Base DN, the point from which the server starts searching for users.<br>2. Provide the search attribute.<br>3. Under **Group DN**, select the role of the user for each group. If you select the administrator role for a particular group, then all the members of that group have the administrator privilege. Similarly, if you select the member role for a particular group, then all the members of that group have the member privilege. If this option is not selected, then the group setting is used to assign the privileges. But other valid LDAP users who do not belong to the groups that you have added can log in to the product.<br>4. Click **Add more** to add groups in the inclusion list.<br>5. Select the **Restrict access to members of the above groups only** option to allow access to the users only from the LDAP groups (direct or inherited membership) that you have added. |
| **Username** | User with the necessary rights to log in using the settings provided. |
| **Password** | Password of the user. |

5. Click **SUBMIT**.
   After configuration, you see the **LDAP** details you have configured.

**Import Users from VMware Identity Manager**
You can import VMware Identity Manager user accounts to allow them to use VCF Operations for networks and assign them the roles.

Configure VMware Identity Manager.

**Note:**
If you are switching your authentication method to VMware Identity Manager from LDAP and the domain used in VMware Identity Manager is the same as the LDAP authentication source, you must delete all LDAP users and settings before you switch to VMware Identity Manager.

1. Go to **Settings** > **Identity & Access Management**.
2. Click **User Management** and select the **VMware Identity Manager Users** tab.
3. Click **ADD USER** and provide the required details.

| Field Name | Description |
|---|---|
| Domain Name | Enter the VMware Identity Manager domain name for import. |
| Search Users/Groups | Enter a search string, and select the user account from the autocomplete list. You can either select a single user or select a user group. If you select a group, all the members in the group can access to VCF Operations for networks. |
| Role | Assign a role to the user account. For more information, see Configure User Management. |

4. Click **Add User**.

**Note:**

- If you have selected a group, all the members in the group are assigned the same role. If you want to assign a different role to a specific user in the group, you must add the user individually, and assign the required role. For example, to assign the **Administrator** role only to *user1* in *Mygroup*, perform the following steps:
  1. Add *Mygroup* and assign the **Member** role.
  2. Add *user1* and assign the **Administrator** role.

  The role assigned to the user overwrites the role assigned to the user as a part of group.
- If a user belongs to multiple groups with different roles, the highest privilege role is assigned to the user. For example, if a user belongs to *Group A* that has the **Administrator** role, and also belongs to *Group B* and *Group C* that have the **Member** role, the user inherits the **Administrator** role.

The VMware Identity Manager users or group members can log in to VCF Operations for networks and use the features based on the assigned role.

Configure VMware Identity Manager

Administrators can authorize VMware Identity Manager users for accessing VCF Operations for networks features based on their roles.

Register VCF Operations for networks as an OAuth client to the VMware Identity Manager host. For more information see the VMware Workspace ONE Access Documentation.

1. Log in to VCF Operations for networks and click **Settings**.
2. Under **Identity & Access Management**, select **VMware Identity Manager**.
3. Click **Configure**.
4. Provide the following information.

| Parameter | Description |
|---|---|
| VMware Identity Manager Appliance | The fully qualified domain name (FQDN) of the VMware Identity Manager host. |
| OAuth Client ID | The ID that is created when registering VCF Operations for networks to the VMware Identity Manager host. |
| OAuth Client Secret | The secret that is created when registering VCF Operations for networks to the VMware Identity Manager host. |

| Parameter | Description |
|---|---|
| SHA-256 Thumbprint | This is an optional field. The certificate thumbprint of the VMware Identity Manager host. For more information, see Obtain the Certificate Thumbprint from a vIDM Host. |

5. Click **Submit**.

   After configuration, you see the VMware Identity Manager appliance and the client details you have configured.

6. Click the toggle button to enable or deactivate VMware Identity Manager. If you deactivate, you cannot use the VMware Identity Manager authentication in VCF Operations for networks.

Obtain the Certificate Thumbprint from the VMware Identity Manager Host

For the SSL certificate validation, you can obtain the SHA-256 thumbprint from VMware Identity Manager host.

1. To get the SSL/TLS certificate, run the following command:

   ```
   openssl s_client -connect <FQDN of vIDM host>:443
   ```

   Copy the Server Certificate starting from `-----BEGIN CERTIFICATE-----` to `-----END CERTIFICATE-----` into a file called `cert.pem` and save the file.

2. To get the thumbprint, run the following command:

   ```
   openssl x509 -fingerprint -noout -sha256 -in cert.pem
   ```

You see the thumbprint in the following format:

```
SHA256
Fingerprint=3D:E8:4C:CD:19:D6:AD:23:30:86:E4:A1:72:D5:22:08:F9:72:6D:D3:E7:6E:99:32:C8:C7:3D:F8:
```

Copy the thumbprint and paste it in the configure VMware Identity Manager page.

## Configuring Logs

In VCF Operations for networks, you can view and configure different types of logs.

### View and Export Audit Logs

Audit logs capture administrative actions carried out in the system. These actions are regular CRUD operations, and login/logout alerts. The audit logs capture the actions from the API, the user interface, and the CLI.

- The audit log feature is always on.
- VCF Operations for networks supports the UTC format in the audit logs.
- The audit log is integrated with the syslog. You can configure the syslog collector to collect all the audit logs.
- You can export all the audit log data in a CSV file.

Currently, the following administrative actions are not captured in audit logs:

- SSH login logs. You can find the SSH logs in `/var/log/auth.log`.
- Changes in Physical IP and DNS Mapping.
- Changes in Physical Subnets and VLANS.

1. Go to **Settings** > **Logs** > **Audit Logs**.
2. The following details are shown on the **Audit Logs** page:

| Information | Description |
|---|---|
| `Date & Time` | Timestamp of the actual action performed. |
| `IP Address` | IP address of the client from which the connection is established such as the CLI or the browser. |
| `User Name` | User who is performing the action. |

| Information | Description |
|---|---|
| Object Type | Object on which the action is being performed. |
| Operation | Different actions that the user performs on the object. |
| Object Identifier | Unique identifier for that particular object on which the action is being performed. |
| Response | Indicator for success or failure of the operation |
| Details | Details of the settings that have been changed such as the nickname or a property. |

3. To allow the collection of information when the user logs in through a browser or CLI, enable **Allow collection of Personally Identifiable Information**. This option is deactivated by default.

   **Note:**
   The IP Address and the User Name columns are blank if this option is deactivated.

4. Click **Export as CSV** to export the audit log data in the CSV format.

### Setup Syslog Configuration

You can configure remote syslog servers for VCF Operations for networks by using the **Syslog Configuration** page.

While every collector can potentially have a different remote syslog server, all the platform servers in a cluster use the same remote syslog server.

In the current release, the VCF Operations for networks problem alerts and platform or collector syslogs are sent to the remote syslog server.

Currently, VCF Operations for networks supports only UDP for communication between VCF Operations for networks servers and remote syslog servers. So ensure that your remote syslog servers are configured to accept syslog traffic over UDP.

To configure syslogs:

1. In the **Settings** page, under **Logs**, click **Syslog Configuration**.
   The **Syslog Configuration** page has the configured syslog servers and their mappings to the virtual appliances listed. If you are accessing this page the first time, then the syslog is deactivated by default and the list of servers on this page does not appear.
2. To add a syslog server:
   a. Click **Add Server**.
   b. Enter the IP Address, nickname, and port number of the server. The standard port number for UDP is 514.
      The IP address can be an IPv4 address, IPv6 address, or a fully qualified domain name.
   c. To test the configuration, click **Send Test Log**.
   d. Click **Submit**.
   e. If it is the first server that you have added, then enable syslog at the top of the page.
3. To map the server to platforms, alerts, or collectors:
   a. Under **Source - Server Mapping**, individually edit the server for all platforms, alerts, or collectors.
   b. Select the syslog server.
   c. If you do not want to enable syslog, select the **No server** option.
   d. Click **Submit**.

   **Note:**
   After you make the changes, it can take a few minutes for them to be effective.

## Configuring Web Proxies

In VCF Operations for networks, you can configure web proxies to connect to other entities (data sources, cloud service platforms, and so on) when the connection is made over the Internet.

You can add, edit, and delete web proxies in VCF Operations for networks.

**Note:**
This feature is supported for the following data sources:

| Data Sources | Web Proxy (HTTP) | Web Proxy (HTTPS) |
|---|---|---|
| Service Now | Yes | Yes |
| Amazon Web Services | Yes | Yes |
| Microsoft Azure | Yes | No |
| VMware Cloud on AWS - NSX Manager | Yes | Yes |

**Note:**
To enable a web proxy, you must edit the data source and select a web proxy from the **Web Proxy (Optional)** drop-down menu.

## Supported Web Proxies

In VCF Operations for networks, you can add the following types of web proxies:

- HTTP and HTTPS unauthenticated web proxy
- HTTP and HTTPS authenticated web proxy with the basic authentication
- HTTP web proxy with the NTLM authentication supported through CLI only.

**Note:**
If you set NTLM authentication-based web proxy using CLI on a platform VM, then the platform VM cannot communicate with Cloud Service Providers (CSP). This is because currently, we do not support communication with CSP using NTLM web proxy.

## Add a Web Proxy

1. Go to **Settings** > **Web Proxies** > **ADD WEB PROXY**.
2. In the **Add Web Proxy** window, do the following:

| Option | Action |
|---|---|
| Type | Select a web proxy type from the drop-down menu. |
| IP Address/FQDN | Enter the IP/FQDN of the web proxy you want to add. |
| Port | Enter the port number used by the web proxy. |
| Nickname | Enter a nick name for the web proxy. |

3. Select the **Use credentials** check box, and provide a username, password, and authentication method.
4. Click **SUBMIT**.

## Edit and Delete a Web Proxy

To Edit a web proxy, do the following:

1. **Settings** > **Web Proxies**.
2. In the **Web Proxies** page, click the edit icon next to the web proxy you want to edit, and modify the information as required.
3. Click **SUBMIT**.

**Note:**
You cannot edit web proxies added using CLI.

To delete a web proxy, do the following:

1. Click the delete icon next to the web proxy you want to delete.

**Note:**
Make sure that the web proxy you want to delete has zero **Connected Entities**. If you have one or more **Connected Entities**, then you must migrate the entities to a different web proxy before moving to the next step.

2. In the **Confirm Action** window, click **DELETE**.

**Collector Migration for Web Proxies**

Before migrating collector to version 6.1, if you set a web proxy on the collector using CLI, and if you set the flag `isHttpProxyEnabled = TRUE`, then all the data sources which were reachable through the web proxy, work seamlessly after the migration to 6.1 as the system auto applies the web proxy details to these data source configurations.

**Note:**
During Collector migration to version 6.1, if you set a web proxy on the collector using CLI, and if you set the flag `isHttpProxyEnabled = FALSE` then Amazon Web Services and Microsoft Azure data sources may not work after migration. You have to select a web proxy for both Amazon Web Services and Microsoft Azure data sources from the **Accounts and Data Sources** page. See Supported Public Cloud Accounts for details.

## VCF Operations for networks License

The capabilities of VCF Operations for networks are dependent on the license applied to your vCenter instance.

Starting with version 9, product components, including VCF Operations for networks, are licensed automatically after you license the vCenter instance connected to the component. You no longer license VCF Operations for networks, manually. For more information, see Licensing Overview.

**Note:**

After upgrading to VMware vSphere 9, the vSphere deployment will be in evaluation mode. As a result, the deployment becomes unsupported in VCF Operations for networks, and data collection is paused. After adding a valid license to your vSphere deployment becomes supported again but data collection must be turned back on manually.

## Configure Auto-Refresh Interval

In VCF Operations for networks, you can configure auto-refresh interval for entity pages and dashboards.

VCF Operations for networks provides the auto-refresh feature for the entity dashboards and dashboards. The dashboard refreshes automatically once in every *n* minutes specified on the header bar.

You can specify the time interval for which you want all your dashboards to perform an auto refresh. After the specified time interval (*n* minutes), all the open widgets on the dashboard will reload automatically.

**Note:**

- You cannot change the auto-refresh time interval for a particular dashboard.
- Auto-refresh is paused if you select a past time interval in timeline slider.

You can pause auto-refresh if you do not require it for a particular dashboard. On the header bar, set **Pause** to **ON**. The auto-refresh counter resets once you set **Pause** to **OFF**.

If you are viewing a dashboard and if another user is making changes to it such as changing the layout of the dashboard, the auto-refresh feature not only updates the content but also refreshes the entire dashboard. This occurs only if sharing and collaboration exists between you and the other user.

1. On the **Settings** page, click **My Preferences**. Or on the respective dashboard, click **Modify** next to Auto-Refresh in the header bar.
2. Click **Edit** to change the time interval for auto-refresh. Select the time interval from the drop-down menu. Click **Save**.
3. To deactivate the auto-refresh option, select **Disabled** from the drop-down menu. All the dashboards are deactivated from refreshing automatically if you select this option.

## Enabling Support for IPv6 Flow Processing

VCF Operations for networks collects and processes IPv6 flows from VMware vCenter and NSX data sources.

IPv6 is designated by the Internet Engineering Task Force (IETF) as the successor to IPv4 providing the following benefits:

- Increased address length. The increased address space resolves the problem of address exhaustion and eliminates the need for network address translation. IPv6 uses 128-bit addresses compared with the 32-bit addresses used by IPv4.
- Ability for improved address autoconfiguration of nodes.

1. On the **Settings** page, click **System Configuration**.

   **Note:**
   The **System Configuration** tab is visible only to the `admin user`.

2. Switch the **IPv6 Flow Processing** toggle to true.

   **Note:**

   As IPv6 addresses occupy more space in memory and disk, enabling IPv6 flow processing can reduce the platform and the collector capacity by up to 20%.

After you enable this feature, VCF Operations for networks takes a few minutes to start collecting and processing IPv6 flows.

To view the list of IPv6 flows, from the left navigation, go to **Entities** > **Flows** and filter flows by **Flow Type**.

## Configure User Session Timeout

By default, the user session timeout is set to `15` minutes. You can modify this value according to your preference.

1. On the **Settings** page, click **System Configuration**.

   **Note:**
   The **System Configuration** tab is visible only to the `admin user`.

2. To change your preference for the **User Session Timeout**, click **EDIT**.
3. Drag the slider bar to set the timeout value for the session. The value ranges from `15` minutes to `24` hours.
4. You can also view the details on who modified the timeout value and when in the **Last Modified** field.
5. Click **Submit**. The Success message appears to confirm that the updated session duration will be effective from the next login.

   **Note:**
   The new value for the user session timeout will come into effect only after you log out and log in again.

## Add Google Maps API Key

To get the map view of your SD-WAN deployment, you must add a Google Maps API key in VCF Operations for networks.

Ensure the following:

- You are a member of Google Cloud Platform and billing is enabled in your account.
- You have the Google map API access key for the following API:
  - Maps JavaScript API
  - Geocoding API

  To get the API key, see Get an API key procedure in *Google Maps Platform documentation*.
- You have restricted the API key to prevent any misuse. To know more, see Restrict the API key in *Google Maps Platform documentation*.

1. On the **Settings** page, click **System Configuration**.
2. In the **Google Maps API Key**, enter the API key and click **Save**.

## VCF Operations for networks and FIPS

In VCF Operations for networks enforces FIPS 140-2 validated cryptographic modules for all internal and external connections. FIPS mode is always enabled and cannot be disabled.

FIPS 140-2 is a U.S. and Canadian government standard that specifies security requirements for cryptographic modules. VCF Operations for networks uses FIPS 140-2 validated cryptographic modules for all internal and external connections. The objective of introducing FIPS support in VCF Operations for networks is to simplify compliance with security requirements in regulated environments..

> **Note:**
>
> Starting with version 9.1, new instances of VCF Operations for networks use FIPS-approved cryptographic modules by default. No additional configuration is required to enable FIPS compliance. Upon upgrading, all appliances that were not previously fully FIPS compliant are automatically brought into compliance.

VCF Operations for networks uses the following validated modules:

| FIPS Object Module | Version | Certificate |
|---|---|---|
| VMware's OpenSSL | 3.0.8 | Certificate #4013 |
| BC-FJA (Bouncy Castle FIPS Java API) | version 2.0.0 | Certificate #37009 |

You can find more information about the cryptographic modules that VMware has validated against the FIPS 140-2 standard from here: https://www.vmware.com/security/certifications/fips.html.

## Configure Security Certificate Validation

When you add a data source, or configure LDAP and VMware Identity Manager (vIDM) in VCF Operations for networks, all the certificates related to that data source, LDAP, or vIDM are validated. In VCF Operations for networks, you can configure how you want to validate and accept the certificates.

There are two ways you can configure certificate validation: **Automatic Acceptance** and **Manual Acceptance**. In **Automatic Acceptance**, system automatically accepts all the detected certificate changes, and in **Manual Acceptance**, system show you notification to accept the certificate manually.

1. Go to **Settings** > **System Configurations**.
2. From **Security Certificate Validation** drop-down menu, select one of the validation methods:
   - **Automatic Acceptance**
   - **Manual Acceptance**
3. Click **Save**.

> **Note:**
>
> If you change the certificate validation method from **Manual Acceptance** to **Automatic Acceptance**, you must accept all the available certificate changes manually before you change the certificate validation method.
>
> If you change the **Data Source Certificate Validation** from **Manual Acceptance** to **Automatic Acceptance** without accepting the pending detected certificate changes, then you have to delete all the data source with pending certificate and add them again to get the insight about those data sources.
>
> The security certification validation setting is not applicable to the following data sources: vCenter, NSX-T, and VCF Operations for logs .
>
> Certificate acceptance for these data sources is handled automatically. If automatic acceptance fails, users must manually accept the certificate by navigating to the **Accounts and Data Sources** page.

For IPV6-enabled vCenter, NSX-T, and VCF Operations for logs data sources, automatic certificate acceptance is not supported. In these cases, users must manually accept the certificate by navigating to **Settings** > **Accounts and Data Sources**.

When the certificate of a non-VIP node in an NSX-T cluster is updated using the VCF Operations Console, users are required to manually accept the new certificate through the same path.

### Accept a Data Source Certificate Manually

If you have configured **Security Certificate Validation** as **Manual Acceptance**, then you must accept the new certificate (HTTPS certificate or SSH public key) for each data source whenever the system detects any change in certificate.

When you add a data source in VCF Operations for networks, all the certificates (HTTPs certificates or SSH public keys) related to that data source are automatically added as a trust on first use. After you add the data source in VCF Operations for networks, every time there is any change in the certificate, the system validates the certificate.

If you have configured **Data Source Certificate Validation** as **Manual Acceptance**, whenever VCF Operations for networks detects any changes in the certificate, it stops the data source and shows you an alert message notification to accept the certificate manually. When you accept the certificate, the system starts the data source. You can also see the certificate change alert in the **Accounts and Data Sources** page. Use this procedure to accept the certificate.

In the **Data source certificate update available** alert message notification, click **REVIEW**.

| | |
|---|---|
| **If the number of certificates update available is up to two:** | 1. You see a **Data Source Certificate** window with current certificate and new certificate details.<br>2. Review the new certificate and click **ACCEPT**. |
| **If the number of certificates update available is more than two:** | 1. In the **Accounts and Data Source** page, you see the `Certificate update available. Click here to review and accept` message under the data source where the certificate update is available.<br>2. Click **Click here to review and accept** for the data source you want to review and accept the updated certificate.<br>3. You see a **Data Source Certificate** window with the current certificate and new certificate details.<br>4. Review the new certificate and click **ACCEPT**. |

When you accept the new certificate, you see `Successfully updated the certificate` message.

### Accept vIDM and LDAP Certificate Manually

If you have configured **Security Certificate Validation** as **Manual Acceptance**, then you need to accept the new certificate (TLS/SSL X.509) to configure the vIDM or LDAP successfully.

When you configure the vIDM or LDAP in VCF Operations for networks, and configure the **Security Certificate Validation** as **Manual Acceptance**, you see an alert message notification to accept the certificate. You also see the alert message notification whenever the system detects any changes in the security certificates.

**Note:**
Only an **Admin user** can accept a security certificate. If you do not accept the certificate, the vIDM or LDAP users cannot access VCF Operations for networks.

You can review and accept the security certificate when you see the security alert message notification. To accept the certificate, do the following:

- For vIDM, when you see the **VMware Identity Manager security certificate update** alert message notification, click **ACCEPT**.
- For LDAPS, when you see the **LDAP Security certificate update** alert message notification, click **ACCEPT**.
- For LDAPS, If the number of certificates update available is more than two, you see `Certificate update available since <date>. Click here to review and accept` in the LDAP tab.
  a. Click **Click here to review and accept** for the LDAPS server you want to review and accept the updated certificate.

b. Review the certificate, and click **ACCEPT**.

## Join or Leave the Customer Experience Improvement Program

VCF Operations for networks participates in VMware's Customer Experience Improvement Program (CEIP). This program provides with information that enables VMware to improve its products and services, correct problems, and advise you on how best to deploy and use the products.

As part of the CEIP, VMware collects the technical information about your organization's use of the VMware products and services regularly using your organization's VMware license keys. This information does not personally identify any individual.

Details regarding the data collected through CEIP and the purposes for which it is used by VMware are set forth in the Trust & Assurance Center at VMware Products in CEIP Program.

You can join or leave the Customer Experience Improvement Program (CEIP) for VCF Operations for networks.

1. Navigate to **Settings** > **Infrastructure and Support** > **Support**.
2. Click **Edit** next to **Customer Experience Improvement Program**.
   The **Customer Experience Improvement Program** window opens.
3. To join CEIP, click the **Join the VMware Customer Experience Improvement Program** check box.
   This action activates CEIP and sends data to https://vmware.com.
4. To leave CEIP, deselect the **Join the VMware Customer Experience Improvement Program** check box.
5. Click **Submit**.

## Configure Login Banners

You can configure Login Banners in VCF Operations for networks to display any text, or you can require that the user click a check box, for example, to indicate that they accept terms and conditions. You can enable and deactivate the login banners, and you can require that users click an explicit consent check box.

1. On the Settings page, click **System Configuration**.
2. Click **Edit** in the **Login Banners** row and configure the login banners.

| Option | Description |
|---|---|
| Display Banners | Click the toggle button to enable or deactivate login banners. You can change the other fields only if you enable this option. |
| Login Message Banner | In the **Description** box, enter the message that you want the users to see when they login to VCF Operations for networks. |
| User Consent Form | Enter the title and enter the content message you want to display. This requires that the users agree to the consent before logging in to VCF Operations for networks. |

3. Click **Submit**.
   The login banners are configured and displayed when a user logs in to VCF Operations for networks.

## CLI Commands

The following section includes various useful Command Line Interface (CLI) commands you can use to configure VCF Operations for networks. To use the CLI commands, log in to the VCF Operations for networks appliance as a *consoleuser*.

The following is a table including useful keyboard shortcuts you can use when applying CLI commands:

| Keyboard Shortcut | Function | Example |
|---|---|---|
| `ctrl+u` | To undo a complete line (to provide the same effect as `ctrl+c` which has some other specific interpretation in CLI) | `(cli) modify-password system --user foo`<br>Using `ctrl+w` in the preceding example results in:<br>`(cli)` |
| `ctrl+w` | To undo words | `(cli) modify-password system --user foo`<br>Using `ctrl+u` in the preceding example results in:<br>`(cli)modify-password system --user` |

### `appliance-reboot`

This reference includes examples for syntax information for the `appliance-reboot` command.

| Command | Description | Usage | Examples |
|---|---|---|---|
| `appliance-reboot` | Reboots the appliance | `appliance-reboot [-h] {now}` | `appliance-reboot now` |

Syntax

| Flag | Description |
|---|---|
| `positional arguments: {now}` | Used to avoid the accidental invocation. |

## backup-restore

This reference includes examples for the `backup-restore` command.

For more information about backup and restore and how to run the process, see About Backing up and Restoring of Deployment Data.

| Command | Description | Usage | Examples |
|---|---|---|---|
| `backup-restore` | • Configures backup<br>• Updates Configured backup<br>• Fetches existing backup configuration<br>• Deletes existing backup configuration<br>• Displays backup Status<br>• Displays usage details of the `backup` command | backup-restore backup --action {add\|update\|get\|delete\|status\|help} | • `backup-restore backup --action add --pathconfig-file-path`<br>• `backup-restore backup --action update --pathconfig-file-path`<br>• `backup-restore backup --action get`<br>• `backup-restore backup --action delete`<br>• `backup-restore backup --action status`<br>• `backup-restore backup --action help` |
| `backup-restore` | • Configures restore<br>• Fetches existing restore configuration<br>• Deletes existing restore configuration<br>• Displays restore Status<br>• Displays usage details of the `restore` command | backup-restore restore --action {add\|get\|delete\|status\|help} | • `backup-restore restore --action add --pathconfig-file-path`<br>• `backup-restore restore --action get`<br>• `backup-restore restore --action delete`<br>• `backup-restore restore --action status`<br>• `backup-restore restore --action help` |

### `change-network-settings`

| Command | Description | Usage | Examples |
|---|---|---|---|
| `change-network-settings` | Change network configuration of the appliance. You must use the VM console through the vCenter UI to perform this operation.<br>**Note:**<br>The appliance is rebooted in the end. | `change-network-settings` | `change-network-settings` |

Syntax

| Flag | Description |
|---|---|
| `none` | NA |

**disk-usage**

This reference includes examples for syntax information for the `disk-usage` command.

| Command | Description | Usage | Examples |
|---|---|---|---|
| `disk-usage` | Displays current disk partition disk usage statistics. | `disk-usage` | `disk-usage` |

## finalize-IP-change

This reference includes information for the `finalize-IP-change` command.

| Command | Description | Usage | Examples |
|---|---|---|---|
| `finalize-IP-change` | Execute the final set of configuration changes after IP change. You must run this command on one of the platforms after completing the IP change workflow (`update-IP-change`) | `finalize-IP-change` | `finalize-IP-change` |

**help**

This reference includes examples for syntax information for the `help` command.

| Command | Description | Usage | Examples |
|---|---|---|---|
| `help` | Displays a list of commands with a short description about their use. | To view the list of commands without short description, use the `-a` flag. | `help`<br>`help -a` |

## hostname

This reference includes examples for the `hostname` command.

| Command | Description | Usage | Example |
|---|---|---|---|
| `hostname show` | Shows the current host name of the VCF Operations for networks | `hostname show` | `hostname show` |
| host name update | Updates the host name of VCF Operations for networks collector or platform VMs. | `host name update` | `host name update --help hostname update --hosts <new_hostname>`<br>`hostname update --hosts platform1:<new_hostname1> platform2:<new_hostname2> platform3:<new_hostname3>` |

**Table 1318: Syntax**

| Flag | Description |
|---|---|
| --dryrun | Validates the new hostname without updating the hostname in VCF Operations for networks.<br>Example:<br>`hostname update --dryrun --hosts <new_hostname>`<br>`hostname update --dryrun --`<br>`hosts platform1:<new_hostname1>`<br>`platform2:<new_hostname2>`<br>`platform3:<new_hostname3>` |
| --hosts | Updates the hostname of your VMs in VCF Operations for networks..<br>Example:<br>`hostname update --hosts <new_hostname>`<br>`hostname update --hosts platform1:<new_hostname1>`<br>`platform2:<new_hostname2>`<br>`platform3:<new_hostname3>` |
| --help | Shows all the supported options.<br>Example:<br>`hostname update --help` |

## aria-logs

This reference includes examples for the syntax information for the `aria-logs` command.

**Table 1319: Command**

| Command | Description |
|---|---|
| `aria-logs` | Configures and pushes various logs to the VCF Operations for logsserver. |

### Syntax

The `aria-logs` command supports the `show` flag. This flag displays the currently configured VCF Operations for logs agent and its status.

## logout

This reference includes examples for syntax information for the `logout` command.

| Command | Description | Usage | Examples |
|---|---|---|---|
| `logout` | Exits from the interactive shell of the CLI. | | |

### log-trace

This reference includes examples for syntax information for the `log-trace` command.

| Command | Description | Usage | Examples |
|---|---|---|---|
| `log-trace` | Displays a list of all log files and also component-specific logs that help during debugging. | `log-trace [-h] {list,grep,follow,display}` | `log-trace list`<br>`log-trace follow /var/log/arkin/arkin_cli_install.log`<br>`log-trace grep "ERROR" /var/log/arkin/arkin_cli_install.log`<br>`log-trace display /var/log/arkin/arkin_cli_install.log 20 (to display last 20 lines)` |

Syntax

| Flag | Description |
|---|---|
| `list` | Lists all log files that can be displayed and also lists the component-specific logs.<br>Example:<br>`log-trace list` |
| `follow` | Usage:<br>`log-trace [-h] follow <file name>`<br>Example:<br>`log-trace follow /var/log/arkin/arkin_cli_install.log` |
| `grep` | Usage:<br>`usage: log-trace grep [-h] pattern filename`<br>Example:<br>`log-trace grep "ERROR" /var/log/arkin/arkin_cli_install.log` |
| `display` | Usage:<br>`usage: log-trace display [-h] filename numlines`<br>Example:<br>`log-trace display /var/log/arkin/arkin_cli_install.log 20 (to display last 20 lines)` |

### modify-password

You can use the `modify-password` command to reset the password of the UI or the system user.

| Command | Description | Usage | Examples |
|---|---|---|---|
| `modify-password` | Reset the password of the UI or the system user<br>`(cli) modify-password [-h]` | `modify-password [-h] {system, ui}` | `modify-password system`<br>`modify-password ui` |

**Table 1320: Syntax**

| Flag | Description |
|---|---|
| `positional arguments: {system, ui}`<br><br>`optional arguments: -h, --help` | Example invocations:<br>`modify-password ui [-h] --user USER`<br>`modify-password system [-h] --user {support,consoleuser}` |

### nslookup

This reference includes examples for syntax information for the `nslookup` command.

| Command | Description | Usage | Examples |
|---|---|---|---|
| `nslookup` | Queries the Internet name servers interactively. | `nslookup FQDN` | `nslookup example.com` |

Syntax

| Flag | Description |
|---|---|
| `fqdn` | Server name to query such as `example.com`. |

Related Commands

| Command | Description |
|---|---|
| `ping` | Pings a machine. |
| `telnet` | Tests whether a specific TCP port is open. |
| `traceroute` | Prints route IP packets sent to a remote host. |

### ntp

This reference includes examples for syntax information for the `ntp` command.

| Command | Description | Usage | Examples |
|---|---|---|---|
| `ntp show` | To show the current configuration and status | | |
| `ntp sync` | To force the sync | | |
| `ntp set` | To reset the servers | `ntp set [--ip-fqdn] [--secure]` | `ntp set --ip-fqdn "a.b.me,x.y.z.local"`<br>`ntp set --ip-fqdn "a.b.me,x.y.z.local" --secure` |
| `ntp diagnose` | To print debug data | | |

**Syntax**

Positional arguments for the `ntp set` command:

| Flag | Description |
|---|---|
| `ip-fqdn` | IP addresses or FQDN details of the NTP servers. |
| `secure` | (Optional) If the IP addresses or FQDN details are of secure NTP servers. |

### `ping`

This reference includes examples for syntax information for the `ping` command.

| Command | Description | Usage | Examples |
|---|---|---|---|
| `ping` | Pings a host. | `ping host` | `ping example.com` |

Syntax

| Flag | Description |
|---|---|
| `host` | IP address or FQDN of the host to ping. |

Related Commands

| Command | Description |
|---|---|
| `nslookup` | Queries Internet name servers interactively. |
| `telnet` | Tests whether a specific TCP port is open. |
| `traceroute` | Prints route IP packets sent to a remote host. |

### `telnet`

This reference includes examples for the syntax information for the `telnet` command.

| Command | Description | Usage | Examples |
|---|---|---|---|
| `telnet` | Test whether a specific TCP port is open. This command is for the interactive communication with the host on a particular port number.<br>`PORT` is an optional parameter. You must provide a host other than `localhost` or 127.0.0.1. | `telnet host port` | `telnet 192.168.0.59 80`<br>`telnet example.com 80` |

Syntax

| Flag | Description |
|---|---|
| `host` | IP address or FQDN of the host to ping. |
| `port` | Port number of the host. |

Related Commands

| Command | Description |
|---|---|
| `nslookup` | Queries Internet name servers interactively. |
| `telnet` | Tests whether a specific TCP port is open. |
| `traceroute` | Prints route IP packets sent to a remote host. |

### `secure-tunnel`

This reference includes examples for the syntax information for the `secure-tunnel` command.

To prevent unwanted cyberattack on the insecure inter-platform communication, VCF Operations for networks provides option to secure the inter-platform communication using IPsec based VPN.

When you enable the secure communication, each platform nodes gets an internal IP addressed assigned from a `/24` subnet. By default, VCF Operations for networks uses `192.168.10.0/24` subnet. You can change the default value of subnet when enabling the secure communication. The subnet must not conflict with any other IP address used for configuring VCF Operations for networks. For example, you must not use IPs from `/24` subnet as an SNMP or an SMTP host configured in VCF Operations for networks.

**Note:**
After you enable the secure communication, you cannot change the subnet. To change the subnet again, you must disable the secure communication and enable it again.

**Note:**
Enabling the secure communication operation can take 15 minutes or more depending size of the cluster. To avoid session time out, you must use the VM console through VMware vCenter UI to run this command.

| Command | Description | Use | Examples |
|---|---|---|---|
| `secure-tunnel` | Enable, disable, check status, assign subnet of secure tunnel | `secure-tunnel {-e,-d,-s, -sub SUBNET}` | `secure-tunnel -e -sub 198.168.0.0/24`<br>`secure-tunnel -d`<br>`secure-tunnel -s` |

**Syntax**

| Flag | Description |
|---|---|
| `-e \| -d \|-s` | Enable secure tunnel, disable secure tunnel, or check status of the secure tunnel |
| `-sub <SUBNET>` | /24 subnet address to use as internal IPs of platforms |

### `services`

This reference includes examples for the syntax information for the `services` command.

| Command | Description | Usage | Examples |
|---|---|---|---|
| `services` | Starts, stops, or restarts VCF Operations for networks services. | `services {start,stop,restart} {all,launcher-service,nginx,ntp}` | `services start all`<br>`services start nginx`<br>`services stop nginx`<br>`services restart ntp` |

Syntax

| Flag | Description |
|---|---|
| `start \| stop \| restart` | Starts, stops, or restarts the appliance service based on the parameter that you pass. |
| `all \| launcher-service \| nginx \| ntp` | Initiates the action for a particular service or for all services. |

### set-banner

This reference includes examples for the syntax information for the `set-banner` command.

| Command | Description | Usage | Examples |
|---|---|---|---|
| `set-banner` | Sets the `ssh` banner.<br>Sets the post-login banner for `consoleuser`. | `set-banner [-h] {support,consoleuser} banner` | `set-banner support "Text"`<br>`set-banner consoleuser "Text"` |

Syntax

| Flag | Description |
|---|---|
| `positional arguments:`<br>`{support,consoleuser}`<br>`banner`<br><br>`Quoted text with \n for new line` | Example invocations:<br><br>`(cli) set-banner support "This is a ssh banner. \nRestricted access only!!"`<br>`Successfully set banner!`<br>`(cli) set-banner consoleuser "This is a post-login banner for consoleuser"`<br>`Successfully set banner!`<br>`(cli) logout`<br><br>`# ssh support@10.126.55.193`<br>`This is a ssh banner.`<br>`Restricted access only!!`<br>`support@10.126.55.193's password:`<br><br>`# ssh consoleuser@10.126.55.193`<br>`This is a ssh banner.`<br>`Restricted access only!!`<br>`consoleuser@10.126.55.193's password:`<br>`This is a post-login banner for consoleuser`<br>`(cli)` |

### set-proxy-shared-secret

This reference includes examples for the syntax information for the `set-proxy-shared-secret` command.

| Command | Description | Examples |
|---|---|---|
| `set-proxy-shared-secret` | Configures the platform IP only on the collector VM. | `set-proxy-shared-secret {secret}` |

Syntax

| Flag | Description |
|---|---|
| `secret` | Secret key that the platform interface generates. |

### `setup`

This reference includes examples for the syntax information for the `setup` command.

| Command | Description | Usage | Examples |
|---|---|---|---|
| `setup` | Used for first time appliance configuration. After deployment when VM is powered on for first time user must use this command to initialize the appliance through VM console. | `setup` | `setup` |

Syntax

| Flag | Description |
|---|---|
| - | No parameters |

### `set-web-proxy`

This reference includes examples for the syntax information for the `set-web-proxy` command.

| Command | Description | Usage | Examples |
|---|---|---|---|
| `set-web-proxy` | Enables or disables the Web proxy or shows the status of the Web proxy. | `set-web-proxy [--ip IP] [--port PORT] {enable,disable,show}` | `set-web-proxy enable --ip-fqdn "x.y.local" --port 8080 --authtype AUTHTYPE` |

Syntax

| Flag | Description |
|---|---|
| `--ip` | IP address to enable the Web proxy. |
| `--port` | Port number to enable the Web proxy. |
| `enable \| disable \| show` | Enables or disables or displays the configuration settings. |
| `--authtype` | This is optional parameter. Supported authentication types are NTLM or basic.<br>**Note:**<br>HTTPS web proxy is not supported yet. |

### show-service-status

This reference includes examples for the syntax information for the `show-service-status` command.

| Command | Description | Usage | Examples |
|---|---|---|---|
| `show-service-status` | Shows the status of all VCF Operations for networks services running on the platform or collector VMs. | `show-service-status {debug, cluster}` | `PLATFORM-`<br>`(cli) show-service-status`<br>`ElasticSearch is running.`<br>`HMaster is running.`<br>`HRegionServer is running.`<br>`Kafka is running.`<br>`QuorumPeerMain is running.`<br>`QuorumPeerMain is running.`<br>`ResourceManager is running.`<br>`NodeManager is running.`<br>`restapilayer is running.`<br>`SaasListener is running.`<br>`SamzaAppMaster is running.`<br>`SamzaContainer is running.`<br>`Launcher is running.`<br>`postgresql is running.`<br>`nginx is running.`<br>`patroni is running.`<br>`QuorumPeerMain is running.`<br>`All required services are running.`<br>`QuorumPeerMain is running.`<br>`QuorumPeerMain is running.`<br>`QuorumPeerMain is running.`<br>`QuorumPeerMain is running.`<br>`PROXY-`<br>`(cli) show-service-status`<br>`CollectorMain is running.`<br>`IpfixProcessor is running.`<br>`postgresql is running.`<br>`nginx is running.`<br>`nfcapd is running.` |

Syntax

| Flag | Description |
|---|---|
| `debug \| cluster` | `debug` - displays the health of the services<br>cluster - displays the service status of each node in the cluster |

### show-version

This reference includes examples for the syntax information for the `show-version` command.

| Command | Description |
|---|---|
| `Show-version` | Displays the current version. |

**Table 1321: Usage**

| Command | Description |
|---|---|
| `show-version` | Shows the current version. |
| `show-version history` | Shows upgrade and patch history. |

### `show-config`

This reference includes examples for the syntax information for the `show-config` command.

| Command | Description | Usage | Examples |
|---|---|---|---|
| `show-config` | Shows the CLI configurations. Displays the CLI console banner, network configurations, and OVA parameters. | `show-config` | `(cli) show-config`<br>`network configurations:`<br>`IP Address:`<br>`192.168.21.55`<br>`Netmask: 255.255.255.0`<br>`Gateway: 192.168.21.1`<br>`DNS nameservers: 4.2.2.4`<br>`8.8.8.8`<br>`Domain Search list:`<br>`arkin.net`<br>`NTP servers:`<br>`0.pool.ntp.org`<br>`1.pool.ntp.org`<br>`2.pool.ntp.org`<br>`3.pool.ntp.org` |

### show-connectivity-status

This reference includes examples for the syntax information for the `show-connectivity-status` command.

| Command | Description | Examples |
|---|---|---|
| `show-connectivity-status` | Displays the connectivity status with all systems. | `(cli) show-connectivity-status`<br>`Platform VM`<br>`Deployment UUID: U12xx08ba-d62e-4745-b73c-9eef6ca9f0b1`<br>`Deployment ID: DXX11XX`<br>`Instance ID: IXXX11X`<br>`Created On: Fri May 19 11:47:23 UTC 2017 (1495194443)`<br>`IP Address: 10.126.102.231`<br>`Netmask: 255.255.252.0`<br>`Gateway: 10.126.103.253`<br>`DNS nameservers: 10.148.20.5 10.148.20.6`<br>`NTP servers: time.vmware.com`<br>`NTP status: InSync [10.113.60.176/2/377]`<br>`Gateway ping status: Success`<br>`Upgrade connectivity status (svc.ni.vmware.com:443): Passed`<br>`Support connectivity status (support2.ni.vmware.com:443): Enabled`<br>`Logserver connectivity status (log.ni.vmware.com:443): Passed`<br>`Registration connectivity status (reg.ni.vmware.com:443): Passed`<br>`Registration status: Registered` |

### ssh-service

This command enables or disables the SSH service.

| Command | Description | Usage | Examples |
|---|---|---|---|
| `ssh-service` | Enables or disables the SSH service | `ssh-service [-h] {status,enable,disable}` | `ssh-service status`<br>`ssh-service enable`<br>`ssh-service disable` |

**Table 1322: Syntax**

| Flag | Description |
|---|---|
| `enable \| disable \| status` | Enables or disables the service or provides the status of the service. |

### support-bundle

This reference includes examples for syntax information for the `support-bundle` command.

| Command | Description | Usage | Examples |
|---|---|---|---|
| `support-bundle` | Provides support related logs for debugging. | `support-bundle [-h] {list,create,delete,copy}` | |

Syntax

| Flag | Description |
|---|---|
| `list` | Lists all support bundles created so far.<br>Example:<br>`support-bundle list` |
| `create` | Creates a support bundle to be shared for debugging.<br>Example:<br>`support-bundle create` |
| `delete` | Deletes specific or all bundle files.<br>Example:<br>`To delete a specific bundle file:`<br>`support-bundle delete --bundle DOAUAAO-IOAUACW-07252016-162328`<br>`To delete all bundle files:`<br>`support-bundle delete` |
| `copy` | SCP to any host for sharing with the support team.<br>Example:<br>`support-bundle copy --user support --ip 192.168.1.1 --port 22 --bundle DOAUAAO-IOAUACW-07252016-162328 --targetpath /tmp` |

### `tool-manager`

This reference includes examples for syntax information for the `tool-manager` command.

| Command | Description |
|---|---|
| `tool-manager` | Helps to run the tool |

**Table 1323: Usage**

| Command | Description |
|---|---|
| `tool-manager copy scp --host <IP> --user <username> --path <tool bundle file absolute path>` | Copy the tool bundle to VCF Operations for networks Platform/ collector |
| `tool-manager run --name <tool name>` | Run the tool |
| `tool-manager run [-h] --name <tool name>` | |
| `tool-manager copy url` | |

**Table 1324: Example**

| Command |
|---|
| `tool-manager copy scp --host 192.168.21.18 --user ubuntu --path /home/ubuntu/patch-online-upgrade-3.0.0_201609151002.bundle` |
| `tool-manager run --name patch-online-upgrade-3.0.0_201609151002` |
| `tool-manager copy url https://bundle-hosting-url.com/patch.bundle` |

Syntax

| Flag | Description |
|---|---|
| `--host` | IP address, where upgrade bundle is downloaded from VMware download portal. Upgrade bundle should be downloaded on any Linux machine from where the bundle can be scp to VCF Operations for networks platform or collector. |
| `--user` | User, who can be used to authenticate the Linux machine. |
| `--path` | Source path where the bundle is downloaded from VMware portal on Linux Machine. |
| `--port` | (Optional) - In case ssh port on Linux machine running other than port 22. |

### `traceroute`

This reference includes examples for syntax information for the `traceroute` command.

| Command | Description | Usage | Example |
|---|---|---|---|
| `traceroute` | Prints route IP packets sent to a remote host for the host that you provide as the parameter. | `traceroute host` | `traceroute examples.com` |

Syntax

| Flag | Description |
|---|---|
| `host` | Name of the host such as `example.com`. |

Related Commands

| Command | Description |
|---|---|
| `nslookup` | Queries Internet name servers interactively. |
| `telnet` | Tests whether a specific TCP port is open. |
| `traceroute` | Prints route IP packets sent to a remote host. |

### `update-IP-change`

This reference includes examples for syntax information for the `update-IP-change` command.

| Command | Description | Usage | Examples |
|---|---|---|---|
| `update-IP-change` | Update Platform configuration after IP Address change on other Platforms in a cluster | `update-IP-change [-h] old_ip new_ip` | `update-IP-change 192.168.1.10 192.168.1.100` |

Syntax

| Flag | Description |
|---|---|
| `old_ip` | Old IP of Platform |
| `new_ip` | New IP of Platform |

### `pair-collector`

This reference includes examples for the syntax information for the `pair-collector` command.

| Command | Description |
|---|---|
| `pair-collector` | Checks the current pairing status with Platform or updates Platform's IP/FQDN if it is changed.<br>**Note:**<br>You cannot point an existing Collector to a new Platform. |

**Table 1325: Syntax**

| Flag | Description |
|---|---|
| `set-platform` | Resets updated Platform IP/FQDN. |
| `platform-pair-status` | Shows Platform and Collector pairing status. |

**Table 1326: Example**

| Command | Sample Query |
|---|---|
| For ONPREM: `pair-collector set-platform --ip-or-fqdn <IP address>` | `pair-collector set-platform --ip-or-fqdn 192.168.10.12` |
| For SaaS: `pair-collector set-platform --ip-or-fqdn <FQDN>` | `pair-collector set-platform --ip-or-fqdn platform.exampledomain.com` |
| `pair-collector platform-pair-status` | `pair-collector platform-pair-status` |

## Creating and Expanding Clusters

### Create Clusters

You can create clusters from the **Settings** > **Infrastructure and Support** > **Infrastructure and Updates** page.

**Prerequisites**

At least two additional platforms are required. The additional platform VMs should be deployed and powered on.

**To create cluster**

1. Click **CREATE CLUSTER** for **Platform VMs**.
2. On the **Create Cluster** page, enter the following information:
   - **IP Address**: Enter the IP address of the new platform that you want to add. VCF Operations for networks supports both IPv4 and IPv6 addresses.
   - **Password**: Enter the support user password of the platform VM. If you have not changed the password yet, then refer the *Default Login Credentials* section in *VCF Operations for networks Installation Guide* for the password.
3. To keep adding more platforms, click **Add more** and enter the IP address and the support user password.
4. Click **SUBMIT**. Click **Yes**.
5. After creating a cluster, the user needs to log in to the product again.

**Note:**

- The **CREATE CLUSTER** option is enabled only when the platform is of large brick size. All platforms should be of large brick to create cluster.
- To expand clusters, refer the *Expanding a Cluster* section in the *VCF Operations for networks Installation Guide*.

### Expand or Scale Out Clusters

Once the cluster is created, you can expand or scale out the cluster by adding more platform nodes to it.

**Note:**

- Before you scale up the cluster, you must take a backup of the cluster. For more information about the backup and restore process, see the KB 55829.
- You must perform the expand cluster operation only from Platform 1 (P1) node.

1. Go to **Settings** > **Infrastructure and Support** > **Infrastructure and Updates**.
2. On the **Infrastructure and Updates** page, click **EXPAND CLUSTER** for **Platform VMs**.
3. The IP addresses of the VMs that are part of the cluster already are listed on the **Expand Cluster** page. To add one or more nodes to the existing cluster, provide the IP address (IPv4 or IPv6 address) of the node and the support user password.

   **Note:**

   - Currently, VCF Operations for networks supports 15 nodes in an existing cluster. Once the limit is reached, the **Add more** button is deactivated.
   - Ensure that all the new nodes do not have license keys applied (non-provisioned) and are reachable through SSH.
   - Ensure that you have taken a backup of the existing platform VMs before you go ahead with the cluster expansion.
4. Click **SUBMIT**.

   The step-by-step progress is displayed.
5. Once the cluster expansion link is completed, a message indicating success is displayed.

   While the cluster expansion is in progress, the application cannot be used for any other operation.

If you are on the Deployment Scenario 2, then you must reconfigure your platform. To reconfigure, on the **Infrastructure and Updates** page, under in **Platform Capacity** section, click the **RECONFIGURE** button.

**Note:**
The **RECONFIGURE** button appears only when your platform needs reconfiguration.

**Note:**

For information on deployment options and the other system requirements, see System Recommendations and Requirements.

## Configuring VCF SSO for VCF Operations for networks

To use VCF Single-Sign On (SSO) in VCF Operations for networks, you must configure it manually.

As an administrator, you can configure SSO to streamline access to VCF Operations for networks. After configuration, you can select SSO as you log in method.

1. Log in to VCF Operations for networks with your local account.
2. To begin creating a SSO configuration, click **Configure**.

   The **Identity and Access Management** page opens.

3. Select the VCF SSO option.
4. Enter the identity broker issuer address, client ID, and client secret details.
5. Click **Test Connection**.
6. To finish configuring SSO, click **Submit**.

After configuring SSO, you can now log in using this streamline method by selecting the SSO configuration from the **Login Method** drop-down. For more information on configuring VCF SSO, see *Configuring VCF Single-Sign On* in the fleet management documentation.

**Note:**

If the IP address is changed, it is recommended that you regenerate the SSO configuration and save the change. Additionally, if a user is added to SSO group and then that group is deleted, the user can no longer log in to VCF Operations for networks. To restore access, the administrator must restore the deleted user role.

## Network Assessment and Value Analysis

Powered by VCF Operations for networks, the **Network Assessment and Value** dashboard in VCF Operations allows you to evaluate and optimize your networking capabilities by adopting VCF networking.

- To use the **Flows** tab, verify that you have installed VCF Operations for networks. Using the **Metrics** tab does not require installing VCF Operations for networks.
- Verify that Network Flow Collection is configured.

By performing a network assessment analysis on your vCenter networking capabilities, you can receive an overview of the projected benefits of adopting VCF networking capabilities. The benefits of the network assessment include:

- Visualizing the Capital Expenditures (CapEx) and Operational Expenditures (OpEx) savings you can make by switching to VCF networking.
- Quantify the Total Cost of Ownership (TCO) of your networking infrastructure.
- Show the simplification and optimization benefits of VCF networking.

This assessment focuses on deployments with only virtual networking (VDS). Existing VPC networking configuration is not taken into account in the assessment.

1. Navigate to **Operate** > **Network Inventory** and select **Network Assessment and Value** from the **Virtual Networking** widget.

   Alternitavely, you can also access this feature in VCF Operations by navigating to **Manage** > **Plan** > **What-If Analysis**.
2. Select **Start Network Assessment**.
3. Select the vCenter instance which includes your networking resources and click **Run Assessment**.

   **Note:**

   If you have multiple vCenter instances, you must run a seperate assessment for each instance.

You are shown the **Network Assessment and Valu**e dashboard. The two tabs on the dashboard, **Metrics** and **Flows**, show an assessment of your current vCenter networking capabilities and the projected improvements that you can expect if you migrate to VPC networking. Data included in the assessment include:

- The mobility score of your network infrastructure, which indicates the projected capability to move networking workloads across your data center.
- Compute and memory utilization of your network infrastructure.
- The TCO of your vCenter networking infrastructure and the potential savings of switching to VCF networking.
- Visualization of your current vCenter infrastructure, including VDSs, number of hosts, and clusters and projected infrastructure converted to NSX VPCs.

When using the dashboard, the following functionality must be considered:

- The Projected Mobility Score is always 100% (not computed), because VPC subnets span across all clusters in the domain.

- If the current CPU and Memory Utilization is under 50% and there is host reduction opportunity through workload rearrangement, then projected values are set to 50%. Otherwise, the values remain as they are currently.
- The projected host count is computed at the individual cluster level. Host reduction opportunity across clusters because of a improved Mobility Score is not considered in the projected host count and Potential Savings calculation.
- Low PNIC Bandwidth Utilization might be displayed as 0% because of a value rounding off.
- Unlike other What-If Analysis items, Network Assessment and Value results cannot be saved.

### VPC Planning

After performing a network assessment, you can begin preparing your network infrastructure for migration to VPCs by performing a VPC planning operation.

You must perform a separate VPC planning operation for each vCenter instance in your environment.

1. Navigate to **Operate** > **Network Inventory** and select **VPC Planning** from the **Virtual Networking** widget.

   Alternatively, you can also access this feature in VCF Operations by navigating to **Manage** > **Plan** > **What-If Analysis**.
2. Select your vCenter instance and click **Proceed**.
3. Select a Virtual Distributed Switch (VDS) from your vCenter instance and click **Proceed** to perform the VPC planning operation.

   **Note:**

   You can view detailed information for each Distributed Virtual Port Group (DVPG) in your VDS by selecting the entry in the **Number of Port Groups** column.
4. After the operation is complete, you are shown a list of the recommended VPCs, the suggested subnet, Gateway CIDR IPv4 address, and number of VMs in that subnet.

   For more information on VMs in the suggested subnet, select the entry in the **Number of VMs** column.
5. To export the results of the VPC planning operation, click **Export to .CSV File**.

## About Backing up and Restoring of VCF Operations for Networks Deployment Data

You can take a backup of the VCF Operations for networks data and restore data in a new deployment when the existing setup is unrecoverable. And also, you can replicate the setup configuration on multiple deployments.

Starting from the 6.0 release, VCF Operations for networks enables you to take a backup of the following data:

- Configuration defined under **Settings** option on the UI
  - CEIP status
  - Data manangement (retention interval)
  - IP Properties and Subnet configuration
  - Identity and Access management
  - Syslog configuration
  - Personally Identifiable Information status in Audit logs
  - Mail Server and SNMP configuration
  - System configurations like User Session Timeout, Certificate Validation, Google Maps API key.
  - User preferences like dark theme, dashboard auto refresh.
- Datasources added to the setup, except for Physical Flow Collector (Netflow, sFlow).
- Alerts which includes system alerts, user-defined alerts, platform health

You can store the backup on the local storage, which can store up to five latest backup files. And also you can upload the backup to SSH or FTP servers.

**Note:**

- VCF Operations for networks takes a backup of the above mentioned configuration data only, and not the complete setup data, like flows or application information.

- You can track any failure in the backup or restore process on the **Infrastructure** page (**Settings** > **Infrastructure and Support** > **Overview and Updates** > **Infrastructure** > **Overall Health section** ).
- You can also configure an alert notification for the backup failure under Platform Health Alerts.

## Backup and Restore of VCF Operations for Networks

In VCF Operations for networks, you can store the backup on the local storage (up to five latest backup files), SSH or FTP servers. You can start the backup immediately or schedule a periodic backup.

- The backup and restore setups must have identical versions.
- To avoid any inconsistencies, the restore setup must be a fresh deployment. Take the snapshots of the VMs before you run the restore commands.
- Only **Admin** and **Console** users can configure backup and restore through public APIs and CLI commands respectively. The **Auditor** user can only see if any backup and restore configuration exists on the setup.

1. Start the backup process.

| Method | Description |
|---|---|
| **Using CLI command** | `backup-restore backup --action add --path <config-file-path>`<br><br>Create a backup configuration file on one of the platform nodes under `/home/ubuntu`. To create the file, you must login as **Support** user.<br><br>**Note:**<br>To run the CLI commands, you must log in as a **Console** user on the same platform where you created the backup configuration file. |
| **Using API** | See **settings/backup** on the API Reference page.<br><br>**Note:**<br>You must provide the configuration as the JSON body in the API request. |

**Table 1327: Configurations for the backup process**

| Scenario | Description | File Configuration |
|---|---|---|
| Configuration to trigger backup | Creates the backup file on local file server in the default directory: `/var/lib/backup-restore` or, any user specified backup directory | `{"backup_file_server_type":"LOCAL"}`<br>or<br>`{"backup_file_server_type": "LOCAL",`<br>`"local_file_server":`<br>`{`<br>`"backup_directory": "backup_directory_path"`<br>`}`<br>`}` |
| | Creates the backup on FTP server in the user specified backup directory | `{`<br>`"backup_file_server_type": "FTP",`<br>`"ftp_file_server":`<br>`{`<br>`"server_address": "IP address",`<br>`"port":port_number,`<br>`"username": "username",`<br>`"password": "password",`<br>`"backup_directory": "backup_directory_path"`<br>`}`<br>`}` |

| Scenario | Description | File Configuration |
|---|---|---|
| | Creates the backup on SSH server in the user specified backup directory | <pre>"backup_file_server_type": "SSH",<br>"ssh_file_server":<br>{<br>"server_address": "*IP address*",<br>"port":*port_number*,<br>"username": "*username*",<br>"password": "*password*",<br>"backup_directory": "backup_directory_path"<br>}</pre> |
| Configuration to schedule the backup process | Immediate Backup | <pre>"schedule_now": true,</pre> |
| | Daily Backup<br>The permissible value for `hour` : 0-23 The permissible value for `minute` : 0-59 | <pre>"backup_schedule":<br>{<br>"schedule_period": "DAILY",<br>"hour": 16,<br>"minute": 51<br>},</pre> |
| | Weekly Backup<br>The permissible value for `hour` : 0-23 The permissible value for `minute` : 0-59 The permissible value for `day_of_week` : 1(Sunday) - 7 (Saturday) | <pre>"backup_schedule":<br>{<br>"schedule_period": "WEEKLY",<br>"hour": 16,<br>"minute": 56,<br>"day_of_week": 3<br>},</pre> |
| Backup only few configurations | In the `data_filter` parameter of the configuration file, set the configuration to `true` to backup selected configurations. To skip the backup of a particular configuration, set the value to `false`.<br>**Note:**<br>If you do not specify the `data_filter` parameter, VCF Operations for networks backs up all the configurations. | <pre>"data_filter":<br>{<br>"snmp": true,<br>"smtp": true,<br>"web_proxy": false,<br>"data_sources": false,<br>"events": false,<br>"syslog": true,<br>"ldap": true,<br>"vidm": true,<br>"user_data": true,<br>"physical_subnet_vlan": true,<br>"physical_ip_dns_mapping": true,<br>"system_configuration": true,<br>"east_west_ip": false,<br>"north_south_ip": true,<br>"data_management": true,<br>"online_update_status": true,<br>"ceip_status": true,<br>"audit_logs_pii_status": false<br>}</pre> |

An example of the configuration file for a weekly backup (every Tuesday) on a FTP server with selected configurations.

```
{
"backup_schedule":
  {
      "schedule_period": "WEEKLY",
      "hour": 16,
      "minute": 56,
```

```
        "day_of_week": 3
    },
  "backup_file_server_type": "FTP",
  "ftp_file_server":
    {
        "server_address": "10.125.48.146",
        "port": 22011,
        "username": "testuser",
        "password": "password!",
        "backup_directory": "/backup"
    },
  "data_filter":
    {
        "snmp": true,
        "smtp": true,
        "web_proxy": false,
        "data_sources": false,
        "events": false,
        "syslog": true,
        "ldap": true,
        "vidm": true,
        "user_data": true,
        "physical_subnet_vlan": true,
        "physical_ip_dns_mapping": true,
        "system_configuration": true,
        "east_west_ip": false,
        "north_south_ip": true,
        "data_management": true,
        "online_update_status": true,
        "ceip_status": true,
        "audit_logs_pii_status": false
    }
  }
```

2. Trigger the restore process.

   The sequence for restore is as follows:

| Method | Description |
|---|---|
| **Using CLI commands** | `backup-restore restore --action add --path <config-file-path>`<br><br>Create a restore configuration file on one of the platform nodes under `/home/ubuntu`. To create the file, you must login as **Support** user.<br><br>**Note:**<br>To run the CLI commands, you must log in as a **Console** user on the same platform where you created the restore configuration file. |
| **Using API** | See **settings/restore** on the API Reference page.<br><br>**Note:**<br>You must provide the configuration as the JSON body in the API request. |

**Table 1328: Configurations for the restore process**

| Scenario | Description | File Configuration |
|---|---|---|
| Configuration to trigger restore | Restores from the local file server | { "backup_file_server_type": "LOCAL", "local_file_server": { "backup_directory": "backup_directory_path", "backup_file_name": "file_name.tar" } } |
| | Restores from the FTP file server | { "backup_file_server_type": "FTP", "ftp_file_server": { "server_address": "*IP address*", "port":*port_number*, "username": "*username*", "password": "*password*", "backup_directory": "backup_directory_path" "backup_file_name": "file_name.tar" } } |

| Scenario | Description | File Configuration |
|---|---|---|
| | Restores from SSH file server | "backup_file_server_type": "SSH",<br>"ssh_file_server":<br>{<br>"server_address": "*IP address*",<br>"port":*port_number*,<br>"username": "*username*",<br>"password": "*password*",<br>"backup_directory": "backup_directory_path"<br>"backup_file_name": "*file_name.tar*"<br>} |
| Restore only few configurations | In the data_filter parameter of the configuration file, set the configuration to true to restore selected configurations. To skip the restore of a particular configuration, set the value to false.<br>**Note:** If you do not specify the data_filter parameter, VCF Operations for networks restores all the configurations. | "data_filter":<br>{<br>"snmp": true,<br>"smtp": true,<br>"web_proxy": false,<br>"data_sources": false,<br>"events": false,<br>"syslog": true,<br>"ldap": true,<br>"vidm": true,<br>"user_data": true,<br>"physical_subnet_vlan": true,<br>"physical_ip_dns_mapping": true,<br>"system_configuration": true,<br>"east_west_ip": false,<br>"north_south_ip": true,<br>"data_management": true,<br>"online_update_status": true,<br>"ceip_status": true,<br>"audit_logs_pii_status": false<br>} |

**Note:**
If you do not specify the collector_mapping parameter, all the restored data sources will be mapped to a random collector on the restore setup.

```
"collector_mapping": {
"default_collector": "default_collector_ip",
"mappings": [
{
  "source": "source_collector_ip",
  "destination": "destination_collector_ip"
}
]
}
```

If you specify collector_mapping, all data sources mapped to source_collector_ip on the backup setup will be mapped to destination_collector_ip on the restore setup. If mapping is not specified for any collector on the backup setup, the corresponding data sources will be mapped to the default_collector on the restore setup. You can specify mapping for multiple source collectors to destination collectors.

An example of the configuration file to restore from a FTP server with selected configurations.

```
"backup_file_server_type": "FTP",
"ftp_file_server":
  {
      "server_address": "10.120.18.146",
```

```
        "port": 22011,
        "username": "testuser",
        "password": "Password!",
        "backup_directory": "/backup"
        "backup_file_name": "vrni_backup-19801-10.79.197.0-2020-08-26-06.56.44.tar"
    },
  "collector_mapping":
    {
        "default_collector": "10.120.135.2",
        "mappings":
            [
            {
            "source": "10.134.135.3",
            "destination": "10.120.2.120"
            }
            ]
    },
  "data_filter":
    {
        "snmp": true,
        "smtp": true,
        "web_proxy": false,
        "data_sources": false,
        "events": false,
        "syslog": true,
        "ldap": true,
        "vidm": true,
        "user_data": true,
        "physical_subnet_vlan": true,
        "physical_ip_dns_mapping": true,
        "system_configuration": true,
        "east_west_ip": false,
        "north_south_ip": true,
        "data_management": true,
        "online_update_status": true,
        "ceip_status": true,
        "audit_logs_pii_status": false
    }
  }
```


## VCF Operations Integration

With VCF Operations, you can see the VCF Operations for networks alerts in VCF Operations. And also, you can see the network information in VCF Operations from VCF Operations for networks.

VCF Operations consumes a set of VCF Operations for networks APIs and displays lists of alerts on the Alerts Dashboard. You can identify the VCF Operations for networks alerts with the `vrni-` prefix in the alert names. In addition, you can see the entity on which the alert was triggered.

You can use the **Launch-in-context** option from the entity pages, such as VMs, Hosts, and NSX-T, so you can view the dashboard of that particular entity. This enables you to see the network health and to debug a network issue.

For information about supported VCF Operations versions, see VMware Product Interoperability Matrix.

> **Note:**
> You must add the VCF Operations user into VCF Operations for networks and provide at least **Member** privileges for the user to use this feature in VCF Operations.

# Configuring Flows in VCF Operations for networks

You can use VCF Operations for networks to configure flows to transmit various network data.

## Frequently Asked Questions for IPFIX

Familiarize yourself with core concepts of using IPFIX in VCF Operations for networks.

### What are the VCF Operations for networks IPFIX configuration limits and requirements?

For information on the VCF Operations for networks IPFIX configuration, see VCF Operations for Networks System Recommendations and Requirements.

### What communication protocols are supported?

VCF Operations for networks IPFIX supports the TCP and UDP protocols. Other protocols, such as ICMP, are not supported.

### Does VCF Operations for networks support de-duplication and aggregation of flows from multiple sources?

Yes, de-duplication and aggregation are supported. The same data reported from multiple sources is de-duplicated and the best metric among all the sources is considered for flow analysis. Different attributes from multiple sources are also aggregated and shown in the flow information.

### What happens if the TCP handshake Is unsuccessful?

Handshakes are only performed for TCP, UDP does not require this validation. If the handshake is unsuccessful, the flow is rejected. Currently, VCF Operations for networks only checks for VDS and NSX flows. Other flows do not report TCP flags and as a result, the check cannot be completed. If VDS reports the TCP handshack as unsuccessful for a specific flow, the NSX or physical counterpart is also rejected.

### What are the outbound flow limits?

The number of connections a client can make in VCF Operations for networks is limited. Flows are dropped if this limit is exceeded. This is a protection mechanism to make sure any non-flow entities, such as scanners which scan every machine in the environment, do not overwhelm the total flow limit.

| Protocol Type | Limit for Allowed Flows | Limit for Denied Flows |
|---|---|---|
| TCP | 1000 | 50 |
| UDP | 500 | 50 |

### What are the inbound flow limits?

Limit on the number of external flows in the environment. The inbound limit covers the number of external flows in the environment. This limit ensures that the external flows do not overwhelm the overall flow limit in VCF Operations for networks . When this limit is exceeded, new incoming flows are aggregated with the following client IPs:

- For allowed external flows: 240.240.240.240
- For denied external flows: 240.240.240.241

| Limit Type | Limit for Allowed Flows | Limit for Denied Flows |
|---|---|---|
| Limit per service endpoint | 5000 | 50 |
| Global limit | 500000 | 100000 |

### What can cause dropped flows?

Dropped flows are flows that are not picked up by the VCF Operations for networks collector. Flows can be dropped for the following reasons:

- The wrong reporting point is used. For NSX flows, the flow can be reported from multiple places. VCF Operations for networks only processes flow data reported from either source or destination of the flow. All other flow records are ignored.
- The TCP handshake is unsuccessful.
- The outbound or inbound limits are reached.
- Incompatibility between the flow and collector. For example, non-physical flows are directed to a dedicated physical collector.

### Why is port information aggregated?

Port aggregation is built in to aggregate the ephemeral port flows, such as dynamic FTP, Oracle, MS-RPC, and others. Aggregation helps in reducing the number of flows in the system and provides an summarized view for a large number of flows that are essentially for the same service.

### How does port aggregation work?

Port aggregation is based on the following workflow:

1. For the first three days of having a destination IP be recorded, VCF Operations for networks aggregates the destination ports of that IP in buckets of 10K and starts building a port profile for the IP.
2. After the intial three day period, port aggregation is started in ranges where the port density is high. High port density is selected as this is representative of the ephemeral port opening pattern. The ranges can be dyanmic in size (100, 1000, or 10K) and are created depending on how many ports are opened and how widespread they are in the specific aggregation range. This allows high-port flows to be reported with no aggregation where there is no bulk port open activity happening. It also allows the application of dynamic aggregation.

   **Note:**

   The aggregation range is specific for each server IP address.
3. The profile is continually updated in a time-decayed manner to account for new ports opening up or older ports are not used anymore.

### Why do I see port aggregation for a port where there is no bulk activity?

VCF Operations for networks performs a blanket aggregation of port 1024 when it gets reported for the first time. This aggregation is done to reduce the number of flows in the system initially. After the port profile is built, this aggregation is removed, if there is no bulk port opening activity happening.

### When is a flow closed?

A specific flow entry is closed when VCF Operations for networks does not receive any IPFIX data for the 5-tuple parameters (protocol, source IP, source port, destination IP, and destination port) after a time duration of 25 hours.

### Can a splitter be used to send flows to multiple collectors?

Yes, if the collector limit is reached, a splitter can be used. You must configure the splitter to ensure that the flow for a 4-tuple parameters (source IP, source port, destination IP, and destination port) is sent to a specific collector. Sending the same 4-tuple to different collectors, can cause incorrect metric data to be shown.

### How long does it take to see the flow information after adding a data source?

It usually takes three or four hours for flow information for a newly added data source to start being displayed.

### Why do I see stale firewall rules on flows?

Duplicate flows can be caused by the following scenarios:

- When a firewall rule changes from rule 1 to rule 2. When the change occurs, VCF Operations for networks does not immediately remove the old firewall rule. There is a time window of six hours before the old rule is removed. So, for a maximum of six hours, you can see both rules. The scope of this temporary duplication can increase if more firewall rules are changed in a short amount of time.
- For NSX-T Federation, multiple rules can be applied to flow. For example, one by the Global Manager and another by the Local Manager.

### How is the client and server for a flow determined?

The client and server are dependent on the communication protocol which is used.

For TCP:

- DFW: the directionality is already determined by DFW, and the source and destination of the flow are set corresponding to the client and the server
- VDS: for a TCP connection, the information about the SYN packet sent by the client is reported in a separate IPFIX flow record by the VDS. This IPFIX flow record is used to decide the client and server for a connection. There might be a situation when VCF Operations for networks does not receive the IPFIX packet. This can happen if this IPFIX packet is lost (IPFIX is sent over UDP), or if VCF Operations for networks identifies a long-lived flow which can be terminated because of a period of inactivity. In this case, VCF Operations for networks determines the client and server for a connection by using the source and destination port numbers. The client port is assumed to be greater than the server port.
- OTHERS: for a TCP connection, if the SYN packet sent by the client is reported in a separate IPFIX flow record by the exporter then same approach as VDS is used. Otherwise, VCF Operations for networks determines the client and server for a connection by using the source and destination port numbers. The client port is assumed to be greater than the server port.

For UDP:

- DFW: the directionality is already determined by DFW, and the source and destination of the flow are set corresponding to the client and the server by DFW.
- OTHERS: if only one direction of IPFIX flow records is found, then consider that direction as the flow direction (source is the client and the destination is server) Otherwise, VCF Operations for networks determines the client and server for a connection by using the source and destination port numbers. The client port is assumed to be greater than the server port.

### What is the flow count for a data source page and how is it determined?

The numbers shown on the data source page show the unique flows (4 tuples) seen from the data source in last 24 hours. The value is updated regularly (every ~5 minutes). The number can change since it only considers flows in the moving window of the last 24 hours. The count should match the results of the following query: `'count of flows where reporter = <Data Source>'`

### How are session counts measured?

If the flow is inactive for more than 2 minutes, then VCF Operations for networks considers that the session is completed for that 5-tuple. This is done at the 5-tuple level. VCF Operations for networks shows the flow in the UI with an aggregated 4-tuple generated from the 5-tuple (the source port is dropped). If the source port is changed frequently, each 5-tuple is considered as a separate session. Session counts across a 5-tuple are aggregated to a single 4-tuple.

### What is the overall limit on the number of flows?

The overall flow limits are documented in VCF Operations for Networks System Recommendations and Requirements. If VCF Operations for networks exceeds these limits, the oldest flows are overwritten. Addtionally, if the flow limit is exceeded on the collector side, then the flow data is dropped on the collector itself until the number of flows goes back to the defined limit. Exceeding flow limits can cause gaps for existing flow data or missed data on new flows.

**Are flows that are dropped or rejected by DFW supported?**

Yes, flows dropped or rejected by DFW are supported.

**Why am I not receiving flows even after activating flow collections on the data source or receiving flows after flow collection is deactivated?**

In VCF Operations for networks 9.1 and later, it might take 10 minutes to reflect configuration changes. This delay is valid for both newly configured flow collections and flow collections that are recently deactivated.

## Enabling IPFIX Configuration

IPFIX is an IETF protocol for exporting flow information.

A flow is defined as a set of packets transmitted in a specific timeslot, and sharing the same 5-tuple values - source IP address, source port, destination IP address, destination port, and protocol. The flow information may include properties such as timestamps, packets/bytes count, Input/Output interfaces, TCP Flags, VXLAN ID, Encapsulated flow information, and so on.

> **Note:**
>
> VCF Operations for networks does not display geolocation information included in flows. Contact Broadcom support for assistance.

### IPFIX Configuration on VDS and DVPG

A Virtual Destributed Switch (VDS) in a vSphere environment can be configured to export flow information using IPFIX

Flow monitoring has to be activated on all port groups attached to the VDS. If the packets arrive on port X of a VDS and exit from port Y, a corresponding flow record is sent if flow monitoring is enabled on port Y.

To analyze the complete information of any session, the IPFIX data about packets in both the directions is required.

> **Note:**
>
> Starting with VCF 9.1, activating uplink DVPG connections is no longer supported. The relevant option in the IPFIX configuration for vCenter data centers is deactivated.

VCF Operations for networks collector VM has built-in collector/receiver for IPFIX flow information. You can enable the IPFIX information collection in the VMware vCenter data source settings at various levels of granularity.

**Enable IPFIX Configuration on VDS and DVPG**
To use IPFIX, you must activate the relevant option for your VMware vCenter data source.

1. Select the **Enable NetFlow (IPFIX) on this vCenter** check box when you are adding VMware vCenter in VCF Operations for networks.

   > **Note:**
   >
   > To enable Antrea IPFIX for collection of Supervisor, namespace, and VKS cluster flows, select **Enable Antrea (IPFIX) on this vCenter**.

   You see list of all the available VDS.
2. Select the VDS for which you want to enable IPFIX from the list of available VDS in VMware vCenter.
3. A notification icon is displayed for the VDS where one of the hosts has unsupported version of ESX. If VCF Operations for networks has detected that IPFIX is already configured for a VDS with some other IP address apart from VCF

Operations for networks collector VM, then it displays the **Override** button. Click **Override** to view the list of DVPGs under that VDS.

4. The list of available DVPGs for the selected VDS is displayed. All the DVPGs are selected by default. Turn **Manual Selection** on to select specific DVPGs for which you want to enable IPFIX. Select the desired DVPGs and click **Submit**.

## Using Antrea IPFIX for VKS Flows

By using Antrea IPFIX, you can collect and analyze data about VMware vSphere Kubernetes Service (VKS) flows from your data sources.

- Verify that VCF Operations for networks is deployed.
- Verify that your vCenter data source version is 9.1 or later.
- Verify that your vCenter data source is not added to multiple VCF Operations for networks deployments or that the Antrea IPFIX management services have a different collector assigned to them from the collector currently configured in VCF Operations.
- Verify that your VKS version is 3.5.1 or later.
- Verify that the `AntreaIPFIX` property is activated in the Antrea configuration for each VKS cluster you want to export flows from. To activate the property, you must set its value to `true` in either the `AntreaConfig` (VKS 3.5.1) or the `AddonConfig` (VKS 3.6.0 or later) file. The following is a sample code for activating the property on a VKS 3.6.0 cluster:

```
apiVersion: addons.kubernetes.vmware.com/v1alpha1
kind: AddonConfig
metadata:
  name: ${clustername}-antrea
spec:
  addonConfigDefinitionRef:
    name: antrea.tanzu.vmware.com.2.5.0---vmware.1-tkg.1
    namespace: vmware-system-vks-public
  clusterName: ${clustername}
  values:
    antreaIPFIX:
      enable: true
```

- Verify that you have enabled **Enable Antrea (IPFIX) on this vCenter** in your vCenter data source in VCF Operations.

**Note:**

After enabling the IPFIX option in your data source, you can aslo verify that flow collection is enabled by navigating to the Supervisor cluster in vCenter. Navigate to **Network** > **Management Proxy Configuration** and verify that the **netops-vks-api-proxy** and **netops-antrea-flows** management services are available.

VCF Operations for networks allows you to perform an analysis on VKS flows, such as Kubernetes clusters, pods, and services. You can use this flow information to analyze and troubleshoot your networking infrastructure. For more information on VKS, see Managing vSphere Kubernetes Service Clusters and Workloads.

**Note:**

Antrea IPFIX flow collection currently has the following limitations:

- Supervisor VDS Networking stack is not supported.
- Flow collection is stopped if the corresponding NSX data source is not added to VCF Operations for networks.
- The vCenter and NSX data sources associated with the VKS clusters must be added to the same VCF Operations for networks collector.
- Pod-to-Service flows where the destination VIP is the load balancer IP and not the cluster IP are not supported.
- IPv6 flows are not supported.
- External-to-Pod flows are not supported.

1. In VCF Operations, navigate to **Operate** > **Flows**.
2. Under **Scope**, you can select the scope of the flow analysis.
3. If you defined to scope of the flow analysis, select the type of VKS flow data, for example, **Kubernetes Cluster**, which you want to add to the analysis.

   The **Network Policy** type includes the Antrea cluster network policy, Antrea network policy, and Kubernetes network policy flows.
4. In the search box, select the VKS flow entities that you want to analyze.

   To view the relevant flows for the selected entities, click **View All Flows**.
5. To collect the flow data for the selected entity types, click **Analyze**.

The flow data is presented in the **Top Talkers**, **Recent Flows**, **Network Performance**, **Pod Network Traffic, Network Policy**, and **Node Throughput** tabs.

## NSX IPFIX Configuration

NSX IPFIX provides network monitoring data similar to that provided by physical devices and gives administrators a clear view of virtual network conditions.

NSX virtualizes the network by allowing the network administrator the ability to decouple the network from physical hardware. This functionality makes it easy to grow and shrink the network as needed and making the network transparent to the applications traversing it.

By using NSX IPFIX in a virtualized network, the network administrators gain visibility into the virtual overlay network. It provides visibility on the VTEP that is encapsulating the packet, and the details of the VM that generated the inter-host traffic on an NSX Logical Switch (VXLAN).

The distributed firewall implements stateful tracking of flows. As these tracked flows go through a set of state changes, IPFIX can be used to export data about the status of that flow.

The tracked alerts include flow creation, flow denial, flow update, and flow teardown. The denied alerts are exported as syslogs.

> **Note:**
>
> If vCenter is added as compute manager to NSX, VDS does not export TCP flags in IPFIX flow records. VCF Operations for networks drops such flows.

Starting with VCF 9.1, VDS IPFIX supports TCP retransmission count and round-trip time (RTT) flow metrics for NSX. For more information, see Flow Metrics.

### Enable NSX-T IPFIX

In VCF Operations for networks, you can view DFW or Switch IPFIX flows. For this, you must enable IPFIX flows in the NSX-T data source.

- Ensure that you have any one of the following privileges:
  - `enterprise_admin`
  - `network_engineer`
  - `security_engineer`
- If you are using Distributed Firewall:
  - Ensure that the Distributed (DFW) firewall is enabled.
  - Ensure that priority 0 is available for the VCF Operations for networks IPFIX profile. If there is another IPFIX profile with priority 0, then you have to change it to some other value.

    > **Note:**
    >
    > From VMware Cloud Foundation 9.0 or higher, if the DFW license is missing or if DFW is deactivated, then Switch IPFIX flows will be the default.
- Verify that DFW is activated on each node of your cluster.

**Note:**

To receive DFW IPFIX flows, you must have an applicable security license enabled. For more information, see the vDefend documentation.

To enable NSX-T IPFIX in VCF Operations for networks:

Select **Enable IPFIX** when adding or editing an NSX-T Manager data source.

After you enable IPFIX, VCF Operations for networks creates its own Operations for Networks Collector profile and Operations for Networks IPFIX profile on NSX-T. Ensure that you do not modify any of these profiles.

After enabling IPFIX on NSX-T, if the flows are not seen in VCF Operations for networks, then the following events may occur:

- Operations for Networks Collector Profile is not registered in the NSX-T Manager.
- Operations for Networks IPFIX Profile is not registered in the NSX-T Manager.
- Operations for Networks IPFIX Profile port number has changed.
- Operations for Networks Collector Profile does not match in the Operations for Networks IPFIX profile in the NSX-T Manager.

  **Note:**
  To resolve all the above issues, enable NSX-T IPFIX again.
- Operations for Networks IPFIX Profile priority is not zero in the NSX-T Manager.
  To resolve this issue, log into NSX-T Manager and set the priority of Operations for Networks IPFIX Profile to zero.
- Operations for Networks Collector IP cannot be added in existing Operations for Networks Collector Profile in the NSX-T Manager.
  Delete one of the collectors from the Operations for Networks Collector Profile in the NSX-T Manager and re-enable NSX-T IPFIX from data source page.
- If you are using Distributed Firewall, then check if Distributed Firewall is deactivated in NSX-T Manage.
  Log into NSX-T Manager and enable the DFW firewall.

## Flow Support for Physical Servers

VCF Operations for networks supports the device that sends the NetFlow data of versions v5, v7, and v9. If the DNS Mapping and Subnet-VLAN mapping information is provided, VCF Operations for networks can enrich the NetFlow data with DNS Domains, DNS Host Names, Subnets, and Layer 2 networks. This feature is available for the Enterprise License users only.

To configure NetFlow in VCF Operations for networks, perform the following steps:

1. Add a Physical Flow Collector for NetFlow and sFlow.
2. Configure the NetFlow collector in the physical devices to receive the flows.
3. Importing the DNS Mapping.
4. Configuring the Physical Subnets and VLANs.

### Configure a NetFlow Collector in a Physical Device

In VCF Operations for networks, you can configure physical devices manually to send the NetFlow information to the VCF Operations for networks NetFlow collector.

To configure most of the physical devices, perform the following steps:

1. Create a flow record.
   The required fields for a flow record are:
   - Mark the following fields as `Match`.
     - `ipv4 protocol`
     - `ipv4 source address`
     - `ipv4 destination address`

  - transport source-port
  - transport destination-port
  - interface input
- Mark the following fields as `Collect`.
  - direction
  - counter bytes
  - counter packets
  - timestamp sys-uptime first
  - timestamp sys-uptime last
- Mark the following field as `Match` or `Collect`. If not, skip it.
  - transport tcp flags

2. Create a flow exporter.
   - Provide VCF Operations for networks NetFlow collector IP and port 2055.
3. Configure the flow cache as follows:
   - Active timeout: 30 seconds
   - Inactive timeout: 60 seconds
4. Create the flow monitor using the created flow record and flow exporter.
5. Configure the monitor on each interface.

The sample steps to configure the physical devices are provided in the following sections:

- Configuring a Netflow Collector for Cisco 4500 Series Switches
- Configuring a Netflow Collector for Cisco Nexus 1000v Switches
- Configuring a Netflow Collector for Cisco Nexus 9000 Series Switches

**Note:**
The steps may vary from version to version and device to device.

**Configuring a Netflow Collector for Cisco 4500 Series Switches**
In VCF Operations for networks, you can configure a netflow collector for Cisco 4500 series switches.

1. To create the flow record

```
configure terminal
flow record netflow-original
match ipv4 protocol
match ipv4 source address
match ipv4 destination address
match transport source-port
match transport destination-port
match interface input
collect transport tcp flags
collect counter bytes
collect counter packets
collect timestamp sys-uptime first
collect timestamp sys-uptime last
End
```

2. To create the flow exporter

```
configure terminal
flow exporter e1
destination <PROXY_IP>
transport udp 2055
end
```

3. To create the flow monitor

```
configure terminal
```

```
flow monitor m1
record netflow-original
exporter e1
end
```

4. To configure the timeouts

```
configure terminal
cache timeout inactive 30
cache timeout active 60
end
```

5. To configure the flow monitor for each interface on the ingress mode and the egress mode or at least the ingress mode

```
configure terminal
interface <INTERFACE_NAME>
ip flow monitor m1 unicast input
end
```

**Configuring a Netflow Collector for Cisco Nexus 1000v Switches**

In VCF Operations for networks, you can configure a netflow collector for Cisco Nexus 1000v switches.

1. To configure timeouts

```
configure terminal
Active timeout 60
Inactive timeout 15
end
```

2. To configure the exporter

```
configure terminal
flow exporter <EXPORTER_NAME>
destination <PROXY_IP>
transport udp 2055
source <VSM_IP_OR_SUBNET>
end
```

3. To configure the flow monitor for each interface:

```
configure terminal
flow monitor <MONITOR_NAME>
record netflow-original
exporter <EXPORTER_NAME>
end
```

4. To configure the flow monitor for each interface on the ingress mode and the egress mode or at least the ingress mode.

```
configure terminal
port-profile type vethernet <IF_NAME>
ip flow monitor <MONITOR_NAME> input
ip flow monitor <MONITOR_NAME> output
.
.
end
```

**Configuring a Netflow Collector for Cisco Nexus 9000 Series Switches**

In VCF Operations for networks, you can configure a netflow collector for Cisco 9000 series switches.

Here are some of the sample device commands for Cisco Nexus 9000:

1. To enable the NetFlow feature

```
configure terminal
```

```
feature netflow
end
```

2. To create flow record

```
configure terminal
flow record vrni-record
match ipv4 protocol
match ipv4 source address
match ipv4 destination address
match transport source-port
match transport destination-port
match interface input
collect transport tcp flags
collect counter bytes
collect counter packets
collect timestamp sys-uptime first
collect timestamp sys-uptime last
End
```

3. To create flow exporter

```
configure terminal
flow exporter vrni-exporter
destination <PROXY_IP>
transport udp 2055
version 9
source <INTERFACE_NAME>
end
```

4. To create the flow monitor for each interface

```
configure terminal
flow monitor vrni-monitor
record vrni-record
exporter vrni-exporter
end
```

5. To configure timeouts

```
configure terminal
cache timeout inactive 30
cache timeout active 60
end
```

6. To configure the flow monitor for each interface on the ingress mode and the egress mode or at least the ingress mode

```
configure terminal
interface <INTERFACE_NAME>
ip flow monitor vrni-monitor input
end
```

## Enriching Flows and IP Endpoints

You can import the DNS mapping and the subnet-VLAN mapping information through the UI.

The flow information is enriched with the following types of information based on the import of the DNS data and the specification of subnet-VLAN mappings.

- Source DNS Domain
- Source DNS Host Name
- Destination DNS Domain

- Destination DNS Host Name
- Source L2 Network
- Source Subnet network
- Destination L2 Network
- Destination Subnet network

The IP Endpoint information is enriched with the following types of information based on the import of the DNS data and the specification of subnet-VLAN mappings.

- DNS Domain
- DNS Host Name
- FQDN
- L2 Network
- Subnet network

For more information on enriching flows through the DNS information, refer Import the DNS mapping file.

For more information on enriching flows through the Subnet-VLAN mapping, refer Configure Mapping Between Subnet and a VLAN.

**Note:**

- The DNS mapping and subnet information are enhanced only for the physical IPs. No subnet or DNS mapping information is associated with any vNIC.
- The information is enriched only for flows that have been seen by VCF Operations for networks after this information has been imported.

### Searching Physical to Physical Flows

You can search for the physical to physical flows based on the following attributes:

- Source DNS Host
- Destination DNS Host
- Source DNS Domain
- Destination DNS Domain
- Source Subnet Network
- Destination Subnet Network

You can search for Physical-Physical flows based on the following attributes. A few examples of flow search query using the enriched DNS and Subnet-VLAN mapping information are as follows:

```
bytes,Dns Domain,Dns Host,l2 network of flows where flow type = 'Physical-Physical'
```

```
bytes,Dns Domain,Dns Host,l2 network of flows where flow type = 'Source is VM' and flow type = 'Destination is Physical'
```

```
bytes,Dns Domain,Dns Host,l2 network of flows where flow type = 'Source is Internet' and flow type = 'Destination is Physical'
```

## Viewing Blocked and Protected Flows

The NSX-IPFIX integration enables the visibility of the blocked and protected flows in the system.

The basic filters in the **Security Planning** page are as follows:

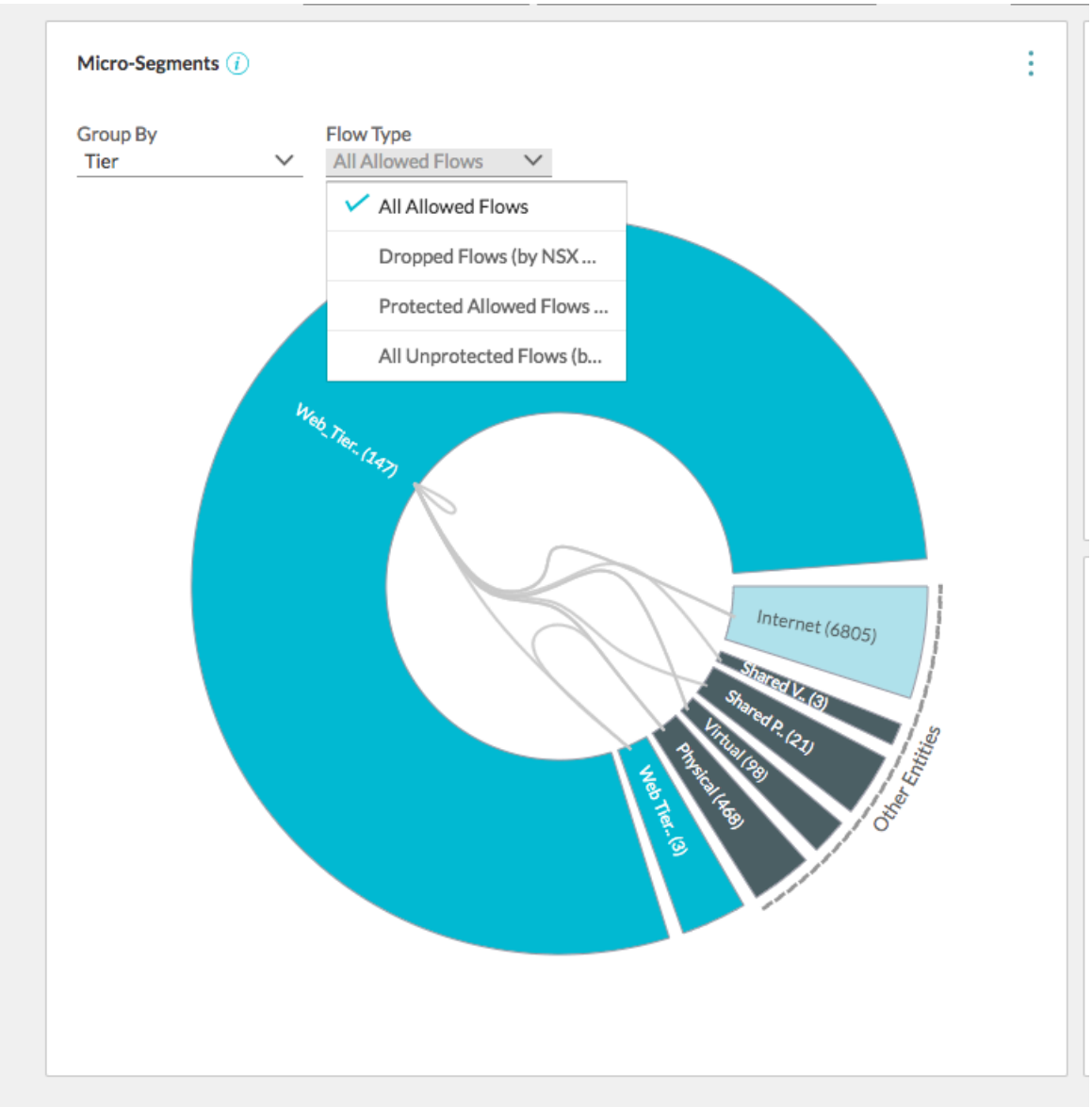
- **All allowed flows**: To see all the flows where firewall rules are set as **Allowed**, select this option. The **All allowed flows** option is the default option.
- **Dropped flows**: This option helps to detect the dropped flows and planning the security in a better way.
- **Protected allowed flows**: This option helps to detect all the flows which have a rule other than of the type `any(source)any(dest)any(service)allow` associated with it. Such flows are known as protected flows.

- **All unprotected flows**: This option helps to detect all the flows that have the default rules of the type `any(source)any(dest)any(service)allow`. Such flows are known as unprotected flows.

The firewall rules are visible only for the allowed and unprotected flows.

For example, if you are in the planning phase and you want to see the allowed flows in the system, perform the following steps:

1. On the Micro-Segmentation Planning page, for a particular group, select **All Allowed Flows** from the drop-down menu.
2. Click the dropped flows in the topology diagram to see the corresponding recommended firewall rules.
3. Implement those firewall rules by exporting them into the NSX manager.




## Network Address Translation (NAT)

VCF Operations for networks supports Static NAT (SNAT), Dynamic NAT (DNAT), reflexive rules in the flows, and the VM-VM Path for NSX-T Edges, Fortinet, and Check Point.

The NAT flow support in VCF Operations for networks is as follows:

- VCF Operations for networks supports the nested NAT hierarchy for NSX for vSphere and NSX-T, and for physical devices, VCF Operations for networks supports the single hierarchy (DNAT) for Fortinet only.

- VCF Operations for networks supports the edges and the tier routers with NAT-defined uplinks.

  **Note:**
  The NAT rules on the NSX Edge version 5.5 or the previous versions are not supported.
- VCF Operations for networks supports SNAT rules with range. However, DNAT must be one-to-one mapping between the destination and translated IP addresses (Parity with NSX for vSphere).
- For Check Point, NAT rules both auto or manually generated are supported for both the source and the destination as network, network-group, or address-range.

**Queries**

To view NAT rules, use the following queries:

- To view all the NAT rules in NSX-T, use the `NSX-T Edge NAT Rule` query.
- To view all the NAT rules in Fortinet, use the `Fortinet NAT Rule` query.
- To view all the NAT rules in Check Point, use the `Check Point NAT Rule` query.
- To view all the NAT rules, use the `NAT Rule` query.

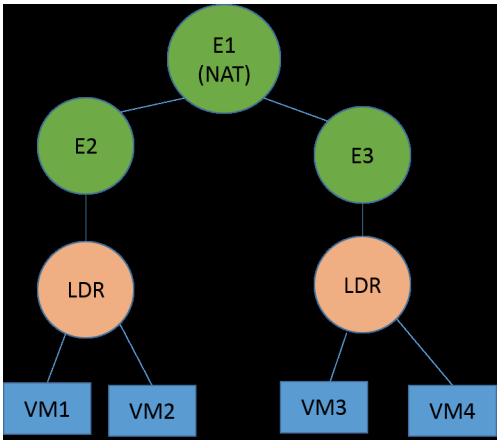
**Consideration**

- VCF Operations for networks does not support the following use cases:
  - In NSX-T, NAT rules can be applied at the service level. For example, in NSX-T, L4 ports set is a type of service and the associated protocols can be TCP or UDP. So in the VM-VM path, the service level details are not supported.
  - Any port level translation is not supported.
  - The SNAT match destination address and the DNAT match source address are not supported. Use the SNAT match destination address as the destination IP address when you specify the SNAT rule. Use the DNAT match source address as the source IP address when you specify the DNAT rule. For example, if there is a destination IP address mentioned in the SNAT rule, VCF Operations for networks applies the SNAT rule irrespective of whether the packet has the destination address as the destination IP address.
  - NSX-T Edge firewall has implications for the data path when enabled with the NAT service on the same logical router. If a flow matches both NAT and Edge firewall, the NAT lookup result takes precedence over firewall. So the firewall is not applied to that flow. If the flow matches only a firewall rule, then the firewall lookup result is honored for that flow.
  - Service translation is not supported.
  - vSEC NAT is not supported.

## NAT Flow Support - Examples

This section consists of a few examples for the supported NAT flow in VCF Operations for networks.

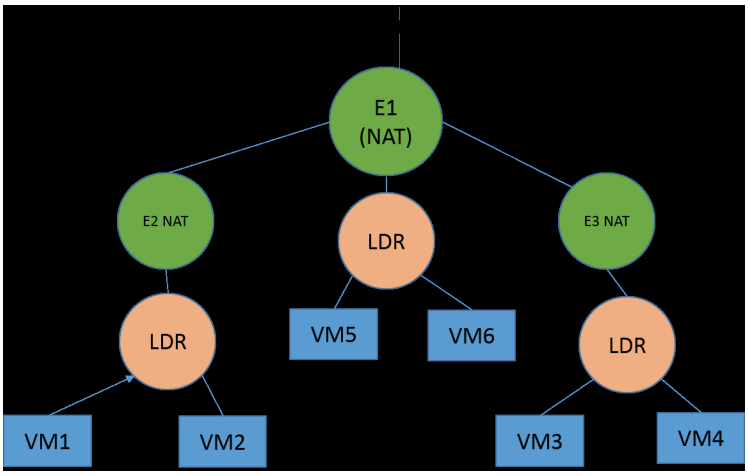
**Example 1**



In this topology, E2, E3, LDRs, VMs (VM1, VM2, VM3, VM4) are part of the NAT domain E1. Anything above E1 such as uplink of E1 is part of the default NAT domain. This topology consists of the following:

The flow from VM1 to VM2 and VM2 to VM1 is reported in VCF Operations for networks. Similarly the flow from VM3 to VM4 and VM4 to VM3 is reported.

**Example 2**




This topology consists of the following:

- VM1 and VM2 are part of the E2 domain.
- VM3 and VM4 are part of the E2 domain.
- The E2 NAT domain and the E3 NAT domain are child domains of the E1 NAT domain.
- E1 is the single child of the default NAT domain.
- VM5 and VM6 are part of the E1 NAT domain.

From example 2, the following flows are reported in VCF Operations for networks:

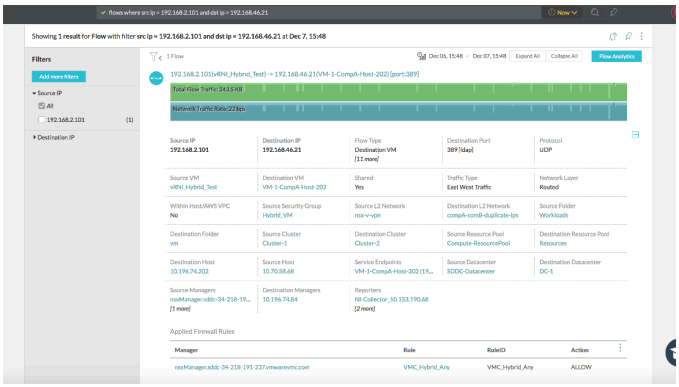
- Flow from VM5 to VM6
- Flow from (VM1, VM2) to (VM3, VM4).

## VMware Cloud on AWS Flows

If you have enabled IPFIX on the data source in the **Settings** page, you can view the flow count and the last collection time.

You can search for any particular flow and get the details associated with the entities. For example, you can view the policy segment and the policy group information in `Source L2 Network` and `Source Security Group` respectively. You can also view the policy firewall rule attached to flow.

VCF Operations for networks supports the hybrid flows over the VPN. The flow information is enriched with the source and the destination entities.



**Note:**
If you have upgraded VMware Cloud on AWS from 1.8 to 1.9 version, you might see the flows twice on the UI.

## Create AWS VPC Flow Log

With Virtual Private Cloud (VPC) Flow Logs, you can capture information about the IP traffic going to and from network interfaces in your VPC.

You can create flow logs through the AWS portal.

1. Sign in to the AWS console.
2. In the **Find Service**text box, enter and select `CloudWatch`.
3. Go to **Logs** > **Action** > **Create log group**.
   The **Create log group** window appears.
4. In the **Create Group Name** field, enter a group name and click **Create log group**.

   **Note:**
   You should set the **Retention setting** to **1 day**. VCF Operations for networks does not retrieve data older than one day, and setting the retention to 1 day, saves on AWS expenses.
5. In the left navigation pane, click **Service** and then enter and select `VPC`.
6. In the **VPC Dashboard** page, click **Your VPCs**.
7. Select the VPC that you want to modify, and click **Flow Logs** > **Create flow log**.
8. In the **Create flow log** window, configure the flow log:

| Option | Action |
|---|---|
| **Filter** | Select one of the following: **Accept**, **Reject**, or **All**. |
| **Destination** | Select **Send to CloudWatch Logs**. |
| **Destination log group** | Select the log group you created. |

9. Click **Set Up Permissions**.
   The system opens the **VPC Flow Logs is requesting permission to use resources in your account** page.
10. Create an IAM role.
    a) In the **VPC Flow Logs is requesting permission to use resources in your account** page, in the **IAM Role**, select **Create a new IAM Role**.
    b) In the **Role Name** text box, enter a role name.
    a) Click **Allow**.
11. On the **Create flow log** page, in the **IAM role** drop-down, select the role you created.
12. Click **Create**

Flow log starts publishing on the selected log group. For more information about VPC Flow Log, see the AWS documentation at https://docs.aws.amazon.com/vpc/latest/userguide/flow-logs.html#create-flow-log.

## Sending Flow Records from F5 To VCF Operations for networks Collectors

To send the flow records, you must do the following:

| Sl No. | Task | Link |
|---|---|---|
| 1 | Create a pool of IPFIX collectors to receive the IPFIX log messages from the BIG-IP system. | Create a pool of IPFIX Collectors |
| 2 | Create a log destination to format the logs in IPFIX templates. | Create an IPFIX Log Destination |
| 3 | Create a log publisher to send logs to a specified log destinations. | Create a Log Publisher |
| 4 | Create an iRule to send the flow information to the configured VCF Operations for networks collector. | Create iRules |
| 5 | Add the iRule to a virtual server configuration so that the iRule parses all of the virtual server's network traffic. | Add the iRule To a Virtual Server |
| 6 | If the collector VM is unreachable from F5, you must create a route entry for the collector to send the flow records. | Create a Route Entry |

## Create a pool of IPFIX Collectors

The BIG-IP system sends IPFIX log messages to a pool of IPFIX collectors.

1. Log in to the F5 console.
2. Click **Main** > **Local Traffic** > **Pools** > **Pool Lists** > **Create**.
   The **New Pool** screen opens.
3. In the **Name** text box, enter a unique name for the pool.
4. In **Health Monitors** section, select **gateway_icmp** and move it in the **Active** box.
5. In the **New Member** section, configure the collector IP address and click **Add**.

| Option | Action |
|---|---|
| **Node Name** | Enter the Collector IP address. |
| **Service Port** | 2055 |

6. Click **Finished**.

## Create an IPFIX Log Destination

Create a log destination to format the logs in IPFIX templates. After the formatting, these logs are sent to the IPFIX collector.

1. In the F5 console, click **Main** > **System** > **Logs** > **Configuration** > **Log Destinations** > **Create**.
   The **Log Destinations** screen appears.
2. In the **Name** text box, enter a unique name.
3. In the **Type** list, click **IPFIX**.
4. Configure the **IPFIX Settings**.

| Option | Action |
|---|---|
| **Protocol** | Click **Netflow V9**. |
| **Pool Name** | Click the pool name you created in the previous step. |

5. Click **Finished**.

### Create a Log Publisher

To send the logs to a specified log destination, you need to create a log publisher .

1. In the F5 console, click **Main** > **System** > **Logs** > **Configuration** > **Log Publishers** > **Create**.
   The **Log Publishers** screen appears.
2. In the **Name** field, enter a unique name.
3. In the **Destination** box, select the log destination you created previously from the **Available** box, and move it in the **Selected** box.
4. Click **Finished**.

### Create iRules

To send the flow information to the configured VCF Operations for networks collector, you must create two iRules. One iRule for the TCP protocol and another iRule for the UDP protocol.

1. In the F5 console, click **Main** > **Local Traffic** > **iRules** > **iRule List** > **Create**.

   Local Traffic

   The **New iRule** screen appears.
2. In the **Name** text box, enter a unique name.
3. In the **Definition** text box, enter the TCP rules for the TCP protocol and the UDP rule for the UDP protocol. For information on the rules, see iRules for the TCP and UDP Protocols.
   Ensure the iRule points to the publisher created previously.
4. Click **Finished**.

#### iRules for the TCP and UDP Protocols

**TCP Rule**

Use the following rule to create the iRule for TCP protocol:

> **Note:**
> Ensure the iRule points to the Log Publisher created previously.

```
when RULE_INIT {
  set static::http_rule1_dest ""
  set static::http_rule1_tmplt ""
}


# CLIENT_ACCEPTED event to initiate IPFIX destination and template
when CLIENT_ACCEPTED {
  set start [clock clicks -milliseconds]
  if { $static::http_rule1_dest == ""} {
    # open the logging destination if it has not been opened yet
    set static::http_rule1_dest [IPFIX::destination open -publisher /Common/<Log Publisher>]
  }
  if { $static::http_rule1_tmplt == ""} {
    # if the template has not been created yet, create the template
    set static::http_rule1_tmplt [IPFIX::template create "flowStartMilliseconds \
                                                          sourceIPv4Address \
                                                          sourceIPv6Address  \
                                                          destinationIPv4Address \
                                                          destinationIPv6Address  \
```

```
                                                    sourceTransportPort \
                                                    destinationTransportPort \
                                                    protocolIdentifier \
                                                    octetTotalCount \
                                                    packetTotalCount \
                                                    octetDeltaCount \
                                                    packetDeltaCount \
                                                    postNATSourceIPv4Address \
                                                    postNATSourceIPv6Address  \
                                                    postNATDestinationIPv4Address \
                                                    postNATDestinationIPv6Address  \
                                                    postNAPTSourceTransportPort \
                                                    postNAPTDestinationTransportPort \
                                                    postOctetTotalCount \
                                                    postPacketTotalCount \
                                                    postOctetDeltaCount \
                                                    postPacketDeltaCount \
                                                    flowEndMilliseconds \ "]

  }
  set rule1_msg1 [IPFIX::msg create $static::http_rule1_tmplt]
}

# SERVER_CONNECTED event to initiate flow data to VCF Operations for networks and populate 5 tuples
when SERVER_CONNECTED {
  set client_closed_flag 0
  set server_closed_flag 0
  IPFIX::msg set $rule1_msg1 flowStartMilliseconds $start
  IPFIX::msg set $rule1_msg1 protocolIdentifier [IP::protocol]

  # Clientside
  if { [clientside {IP::version}] equals "4" } {
    # Client IPv4 address
    IPFIX::msg set $rule1_msg1 sourceIPv4Address [IP::client_addr]
    # BIG-IP IPv4 VIP address
    IPFIX::msg set $rule1_msg1 destinationIPv4Address [clientside {IP::local_addr}]
  } else {
    # Client IPv6 address
    IPFIX::msg set $rule1_msg1 sourceIPv6Address [IP::client_addr]
    # BIG-IP IPv6 VIP address
    IPFIX::msg set $rule1_msg1 destinationIPv6Address [clientside {IP::local_addr}]
  }
  # Client port
  IPFIX::msg set $rule1_msg1 sourceTransportPort [TCP::client_port]
  # BIG-IP VIP port
  IPFIX::msg set $rule1_msg1 destinationTransportPort [clientside {TCP::local_port}]

  # Serverside
  if { [serverside {IP::version}] equals "4" } {
    # BIG-IP IPv4 self IP address
    IPFIX::msg set $rule1_msg1 postNATSourceIPv4Address [IP::local_addr]
    # Server IPv4 IP address
    IPFIX::msg set $rule1_msg1 postNATDestinationIPv4Address [IP::server_addr]
  } else {
    # BIG-IP IPv6 self IP address
    IPFIX::msg set $rule1_msg1 postNATSourceIPv6Address [IP::local_addr]
    # Server IPv6 IP address
```

```
    IPFIX::msg set $rule1_msg1 postNATDestinationIPv6Address [IP::server_addr]
  }
  # BIG-IP self IP port
  IPFIX::msg set $rule1_msg1 postNAPTSourceTransportPort [TCP::local_port]
  # Server port
  IPFIX::msg set $rule1_msg1 postNAPTDestinationTransportPort [TCP::server_port]
}

# SERVER_CLOSED event to collect IP pkts and bytes count on serverside
when SERVER_CLOSED {
  set server_closed_flag 1
  # when flow is completed, BIG-IP to server REQUEST pkts and bytes count
  IPFIX::msg set $rule1_msg1 octetTotalCount [IP::stats bytes out]
  IPFIX::msg set $rule1_msg1 packetTotalCount [IP::stats pkts out]
  # when flow is completed, server to BIG-IP RESPONSE pkts and bytes count
  IPFIX::msg set $rule1_msg1 octetDeltaCount [IP::stats bytes in]
  IPFIX::msg set $rule1_msg1 packetDeltaCount [IP::stats pkts in]
    IPFIX::destination send $static::http_rule1_dest $rule1_msg1
}

# CLIENT_CLOSED event to collect IP pkts and bytes count on clientside
when CLIENT_CLOSED {
  set client_closed_flag 1
  # when flow is completed, client to BIG-IP REQUEST pkts and bytes octetDeltaCount
  IPFIX::msg set $rule1_msg1 postOctetTotalCount [IP::stats bytes in]
  IPFIX::msg set $rule1_msg1 postPacketTotalCount [IP::stats pkts in]
  # when flow is completed, BIG-IP to client RESPONSE pkts and bytes count
  IPFIX::msg set $rule1_msg1 postOctetDeltaCount [IP::stats bytes out]
  IPFIX::msg set $rule1_msg1 postPacketDeltaCount [IP::stats pkts out]
  # record the client closed time in ms
  IPFIX::msg set $rule1_msg1 flowEndMilliseconds [clock click -milliseconds]
    # send the IPFIX log
    IPFIX::destination send $static::http_rule1_dest $rule1_msg1
}
```

## UDP Rule

Use the following rule to create the iRule for UDP protocol:

> **Note:**
> Ensure the iRule points to the Log Publisher created previously.

```
when RULE_INIT {
  set static::http_rule1_dest ""
  set static::http_rule1_tmplt ""
}


# CLIENT_ACCEPTED event to initiate IPFIX destination and template
when CLIENT_ACCEPTED {
  set start [clock clicks -milliseconds]
  if { $static::http_rule1_dest == ""} {
    # open the logging destination if it has not been opened yet
    set static::http_rule1_dest [IPFIX::destination open -publisher /Common/<Log Publisher>]
  }
  if { $static::http_rule1_tmplt == ""} {
    # if the template has not been created yet, create the template
    set static::http_rule1_tmplt [IPFIX::template create "flowStartMilliseconds \
```

```
                                                    sourceIPv4Address \
                                                    sourceIPv6Address  \
                                                    destinationIPv4Address \
                                                    destinationIPv6Address  \
                                                    sourceTransportPort \
                                                    destinationTransportPort \
                                                    protocolIdentifier \
                                                    octetTotalCount \
                                                    packetTotalCount \
                                                    octetDeltaCount \
                                                    packetDeltaCount \
                                                    postNATSourceIPv4Address \
                                                    postNATSourceIPv6Address  \
                                                    postNATDestinationIPv4Address \
                                                    postNATDestinationIPv6Address  \
                                                    postNAPTSourceTransportPort \
                                                    postNAPTDestinationTransportPort \
                                                    postOctetTotalCount \
                                                    postPacketTotalCount \
                                                    postOctetDeltaCount \
                                                    postPacketDeltaCount \
                                                    flowEndMilliseconds \ "]

  }
  set rule1_msg1 [IPFIX::msg create $static::http_rule1_tmplt]
}

# SERVER_CONNECTED event to initiate flow data to VCF Operations for networks and populate 5 tuples
when SERVER_CONNECTED {
  set client_closed_flag 0
  set server_closed_flag 0
  IPFIX::msg set $rule1_msg1 flowStartMilliseconds $start
  IPFIX::msg set $rule1_msg1 protocolIdentifier [IP::protocol]

  # Clientside
  if { [clientside {IP::version}] equals "4" } {
    # Client IPv4 address
    IPFIX::msg set $rule1_msg1 sourceIPv4Address [IP::client_addr]
    # BIG-IP IPv4 VIP address
    IPFIX::msg set $rule1_msg1 destinationIPv4Address [clientside {IP::local_addr}]
  } else {
    # Client IPv6 address
    IPFIX::msg set $rule1_msg1 sourceIPv6Address [IP::client_addr]
    # BIG-IP IPv6 VIP address
    IPFIX::msg set $rule1_msg1 destinationIPv6Address [clientside {IP::local_addr}]
  }
  # Client port
  IPFIX::msg set $rule1_msg1 sourceTransportPort [UDP::client_port]
  # BIG-IP VIP port
  IPFIX::msg set $rule1_msg1 destinationTransportPort [clientside {UDP::local_port}]

  # Serverside
  if { [serverside {IP::version}] equals "4" } {
    # BIG-IP IPv4 self IP address
    IPFIX::msg set $rule1_msg1 postNATSourceIPv4Address [IP::local_addr]
    # Server IPv4 IP address
    IPFIX::msg set $rule1_msg1 postNATDestinationIPv4Address [IP::server_addr]
```

```
  } else {
    # BIG-IP IPv6 self IP address
    IPFIX::msg set $rule1_msg1 postNATSourceIPv6Address [IP::local_addr]
    # Server IPv6 IP address
    IPFIX::msg set $rule1_msg1 postNATDestinationIPv6Address [IP::server_addr]
  }
  # BIG-IP self IP port
  IPFIX::msg set $rule1_msg1 postNAPTSourceTransportPort [UDP::local_port]
  # Server port
  IPFIX::msg set $rule1_msg1 postNAPTDestinationTransportPort [UDP::server_port]
}

# SERVER_CLOSED event to collect IP pkts and bytes count on serverside
when SERVER_CLOSED {
  set server_closed_flag 1
  # when flow is completed, BIG-IP to server REQUEST pkts and bytes count
  IPFIX::msg set $rule1_msg1 octetTotalCount [IP::stats bytes out]
  IPFIX::msg set $rule1_msg1 packetTotalCount [IP::stats pkts out]
  # when flow is completed, server to BIG-IP RESPONSE pkts and bytes count
  IPFIX::msg set $rule1_msg1 octetDeltaCount [IP::stats bytes in]
  IPFIX::msg set $rule1_msg1 packetDeltaCount [IP::stats pkts in]
    IPFIX::destination send $static::http_rule1_dest $rule1_msg1
}

# CLIENT_CLOSED event to collect IP pkts and bytes count on clientside
when CLIENT_CLOSED {
  set client_closed_flag 1
  # when flow is completed, client to BIG-IP REQUEST pkts and bytes octetDeltaCount
  IPFIX::msg set $rule1_msg1 postOctetTotalCount [IP::stats bytes in]
  IPFIX::msg set $rule1_msg1 postPacketTotalCount [IP::stats pkts in]
  # when flow is completed, BIG-IP to client RESPONSE pkts and bytes count
  IPFIX::msg set $rule1_msg1 postOctetDeltaCount [IP::stats bytes out]
  IPFIX::msg set $rule1_msg1 postPacketDeltaCount [IP::stats pkts out]
  # record the client closed time in ms
  IPFIX::msg set $rule1_msg1 flowEndMilliseconds [clock click -milliseconds]
    # send the IPFIX log
    IPFIX::destination send $static::http_rule1_dest $rule1_msg1
}
```

## Add the iRule To a Virtual Server

1. In the F5 console, click **Main** > **Local Traffic** > **Virtual Server** > **Virtual Server List**.
   The **Virtual Server List** screen appears.

2. Select the server you want to add the iRule.
3. Click **Resources** tab, and in the iRule section click **Manage**.
4. Select the TCP and UDP iRules that you created previously and move the iRules from **Available** box to **Enable** box.
5. Click **Finished**.

### Create a Route Entry

The collector VM must be reachable from F5. If the collector VM is unreachable from F5, you must create a route entry for the collector.

To check if the collector VM is reachable from F5, you must run the following command: `ping <collector-ip> -I <virtual interface>` from the Command Line Interface (CLI). If the collector is unreachable form F5, you must create a route entry for the collector.

For example,

```
admin@(localhost)(cfg-sync Standalone)(Active)(/Common)(tmos)# ping 10.153.191.116 -I VLAN301
PING 10.153.191.116 (10.153.191.116) from 10.115.30.50 VLAN301: 56(84) bytes of data.
From 10.115.30.50 icmp_seq=1 Destination Host Unreachable
From 10.115.30.50 icmp_seq=2 Destination Host Unreachable
```

1. In the F5 console, click **Main** > **Network** > **Routes** > **Add**.
   The **New Route** screen appears.
2. In the **Properties** section, configure the route entries to send the flow records from F5 to the VCF Operations for networks collector through the virtual server.

## Viewing Kubernetes and Tanzu Kubernetes Grid Integrated Edition Scoping and Flow Information

You can perform scoping of container entities and view the flow information in VCF Operations for networks.

#### Tanzu Kubernetes Grid Integrated Edition and Kubernetes Flow Information

VCF Operations for networks supports the following flow types for Kubernetes entities.

- VM to Kubernetes Pod
- Kubernetes Pod to Pod
- Destination is Kubernetes Pod
- Source is Kubernetes Pod

You can use these flow types to search for a particular Kubernetes entities.

For example, `flows where flow type = x` where *x* is one of the flow types

VCF Operations for networks can provide flow information such as metrics, time-series and relations for all entities, which includes the container source and destination details and its entities details.

Also, you can view the top talkers by Kubernetes Cluster, Namespace, Service and Node on the Flow Analytics Dashboard.

#### Planning and Micro-segmentation of Kubernetes Entities

You can plan for a specific Kubernetes entity type by selecting Kubernetes Cluster, Kubernetes Service, Kubernetes Namespace, or Kubernetes Node as the scope and Micro-Segments in the Plan Security page. Also, you can plan or analyze data for the application and define grouping based on Kubernetes entities to view the application flow information.

Also, you can export the recommended firewall rules related to Kubernetes entities in the YAML format from Micro-Segments in the Plan Security page.

**Note:**
You cannot export the application scope in the YAML format if it contains VMs or VM members. If the application contains only container entities, exporting to YAML format is available.

# Working with Guided Network Troubleshooting

In VCF Operations for networks, you can use Guided Network Troubleshooting to troubleshoot supported entities from a single page. VCF Operations for networks creates a dependency graph for the entity you want to troubleshoot and shows you possible anomalies that can influence performance and functionality.

In the Guided Network Troubleshooting, when you select an entity to troubleshoot, VCF Operations for networks creates a dependency graph with all other related entities, shows all the relevant and correlated metrics and alerts of the entities in the dependency graph, and highlights all the possible anomalies in the dependency graph and the correlated metrics.

**Note:**
For each tenant, VCF Operations for networks supports a maximum of 5,000 troubleshooting incidents. Also, you can only view dependency graph for a time range within the last 30 days.

## Start a New Troubleshooting Incident

To find the root cause of an abnormality or issue, you can start a troubleshooting incident when you see any abnormality or problems against an entity in VCF Operations for networks. Troubleshooting incidents are also useful to visualize or analyze the dependencies for any entity.

**Note:**
You can also lunch a troubleshooting incident from Ensemble. From Ensemble, click **Troubleshoot** button for the entity you want to start a troubleshooting incident. Ensemble is specifically designed to discover, curate, and manage applications. For more information, see the *Using and Managing VMware Ensemble*.

1. From the left navigation pane, select **Troubleshoot** > **Start New Incident**.
   You see the **Start Troubleshooting** window.
2. From the **Scope** drop-down menu, select the entity you want to troubleshoot.
3. From the **Time Range** drop-down menu, select a **Preset** time range or set a **Custom** time range.
4. In the **Incident Name** text box, enter a name for the incident..
   Based on your scope selection, the system automatically generates an incident name. But you can change or update the automatically generated incident name as per your requirement.
5. Click **Start New**.
   When you select a scope, you see the last 20 incidents that were most recently modified and related to your selected scope at the bottom of the window.
6. Optional: To open an in-progress incident, click the incident you want to view.

A new **Troubleshoot** tab opens on your browser.

## Viewing Troubleshooting Incident Page

The **Troubleshoot** page provides all the relevant and required information you might need to troubleshoot an entity using Guided Network Troubleshooting in VCF Operations for networks.

On this page, you see the following sections:

| Section | Details |
|---|---|
| **Troubleshooting Insights** | You see:<br>• List of the top three troubleshooting insights which contain troubleshooting hints.<br>You can click the **Analysis** button to view the anomaly details. |
| **Troubleshooting Information** | You see:<br>• Information related to the incident, such as status, root cause metrics, closing remarks, created by, created at, and so on.<br>• Information related to the entity that you are troubleshooting, such as start entity name, traffic and flows on the entity.<br>• Any past incidents created for the entity. |
| **Relationships** | You see the dependency graph of the selected entity for which you are troubleshooting.<br>The dependency graph shows all the related entities based on network, storage, and compute relations. You can select the entities in the dependency graph, and view information related to each selected entity.<br>In the dependency graph, all the related entities are grouped in various categories, such as by VMs, Host, Flows, and so on.<br>You can click the plus icon to expand a group. You can also use sort by and group by function to see entities in a particular order in a group.<br>The system highlights the path and entities identified with the anomalies in red. |
| **Metrics** and **Alerts** | You see all the metrics and alerts relevant to the entity you have selected in the **Relationship** widget.<br>You see all the anomalies in a metrics highlighted by<br>( )<br>for quick detection.<br>You can view report based on the analysis type you have selected. Also, you can filter metrics and alerts based on various parameters, such as metric name, entity name, and so on.<br>If you find a metric that might be a potential root cause, you can click to mark the metric.<br>If you find a metric which you want to mark for your reference, you can click to flag the metric. |
| **Notes** | You see a section for notes to add your findings, insights, or other information that you want to remember.<br>You can use this section to collaborate with other users, as anyone who opens the troubleshooting session can see your notes. |

You can also modify the scope of the analysis by using **Modify** button. When you click the **Modify** button located at the top-right corner, you see **Troubleshooting Scope** widget.

| | |
|---|---|
| **Troubleshooting Scope** | In the **Troubleshooting Scope**, you see the following:<br>• You see the selected time range for troubleshooting.<br>You can also change the selected time range for which you want the system to perform troubleshooting.<br>• You see the selected analysis type and level.<br>The single metric performs troubleshooting on the single metric time-series, and the multi-metrics perform troubleshooting on two correlated time-series metrics.<br>• You can also see the sensitivity level. The sensitivity level determines how the system detects anomalies. The higher the level you select, the more the number of anomalies you see.<br>You can select one of the following sensitivity level: low, medium, or high. |

## Find Root Cause by Using Troubleshooting Incident

You can start a troubleshooting incident to find a root cause for any abnormality or problem that you have observed on an entity in VCF Operations for networks.

To find root causes in the **Troubleshoot** page, you can do the following steps.

- In the **Troubleshooting Insights** widget, view the list of top three troubleshooting insights which contain troubleshooting hints.
  - Click the **Analysis** button to view insight which includes metric details and potential anomalies.
- In the **Relationship** widget, select the related entities marked in red to view metrics details related to that entity to find potential anomalies or root cause.
  - You can drill down all the related entities that are marked as red, one by one, and view metrics reated to the entity to find the problem you are facing in your environment. Click (+) to expand each entity group.

    If required, you can also click other entities in the relationship graph and view the related metrics for your troubleshooting investigation.
- In the metric, you can click **Analysis** to view correlated metric analysis.
  - While investigating the root cause you can use filter to find specific metrics and alerts of your interest. All the anomalies in a metrics highlighted by ( ).

    You can also flag a metric and mark a metric as root cause.
- Use the Notes sections to add your thoughts and observations. Anyone who opens this session can see your notes.
- When your investigation is complete, you can close the session by clicking **Close Session** button.

  You can close a session as resolved or unresolved. Resolved status implies you have found a root cause and unresolved status implies that you did not find any root cause. You can also add remarks while closing a session to note down any comments.

  **Note:**
  You can work on a closed incident irrespective of the resolution. You can also change the status to resolved or unresolved. But, you cannot reopen a closed incident.

## Managing Incidents

On the **Troubleshooting Incidents** page, you can manage all the available incidents, start a new incident, and view a summary of all the existing incidents.

In the **Troubleshooting Incidents** page, you see the following sections:

| Section | Details |
|---|---|
| **Summary** | You see:<br>• A summary of all the incidents, including status, entity count for in-progress incidents, and count of top root causes. |
| **Incidents** | You see:<br>• A list of all the incidents, their status, and other important information.<br>You can edit, share, or delete any incidents using the corresponding **Edit**, **Share**, and **Delete** button available in the action column.<br>**Note:**<br>Only admin users can delete incidents created by other users. Member users can delete only the incidents that they create. Audit users cannot delete or create any incidents. |

In this page, you can also create a new incident using **Start New** button.

# Entities

## Viewing Entity Details

The entity pages provide a comprehensive outlook of entities that are present in your data center. This information can range from detailed topologies to show relationships with other entities of your data center to detailed metrics about an entity.

Think of your entire data center as being composed of entities and their relationships. For example, a virtual machine is an entity, and the virtual machine is part of a host which is another entity. Here, the virtual machine and the host share a relationship with each other. VCF Operations for networks provides visibility and information of such entities and relationships, and numerous other entities and relationships that are part of your data center.

Each entity on VCF Operations for networks has a dedicated entity page. This page is a collection of widgets and each widget shows specific information related to the entity. The information provided is both real time and historical, and an exhaustive list of metrics and properties for the entity.

You can expand the **Entities** option on the left navigation panel to view all the entities on VCF Operations for networks. If you want to see more information about entities, then click **Profile** > **Help** on the top-right corner of the page.

### Time Range Widget

The time range widget enables you to view the state of the entity at a particular time in the past. It also provides a bird's eye view of alerts that were detected across a selected time range.

To access the time range widget, click

next to the entity name on top of the page, and select the time range that you want to view.

## Viewing VCF Operations for networks System (NI System) Details

The VCF Operations for networks System (NI System) page provides a snapshot of all the information related to the system. To access the VCF Operations for networks System page:

- On the **Infrastructure and Support** page, click **View Details** next to **Overview**. The NI System page appears.
- Provide `NI System` as the search query to view the VCF Operations for networks System page.

The NI System page is divided into three sections:

- Overview: This section consists of information on the key properties, the data sources, the problems that are open, and all the changes and the problems related to the system. View the details of each data source by clicking it.
- Alerts: This section lists all the problems and changes in the system, data sources, platforms, and the collectors.
- Platforms and Collectors: This section lists all the platforms and the collectors associated with the system. To view more details about any platform or collector, click it.

## Viewing Platform VM details

The **Platform VM** page provides a snapshot of the properties, changes, and problems of a particular platform node.

In the **Platform VM** page, you see:

- Important information about the selected platform node, such as name, IP address, CPU cores, memory, the last upgraded time, and the version.
- Open problems that are associated with the platforms and are open.
- The list of alerts related to the selected platform node.
- The graphical representation of the metrics such as CPU Usage, Memory Usage, and Data Disk Usage.

## Viewing Collector VM Details

The **Collector VM** page provides a snapshot of the properties, changes, and problems of a particular collector node.

In the **Collector VM** page, you see:

- Important information about the selected platform node, such as name, IP address, CPU cores, memory, the last upgraded time, and the version.
- Number of open problems related to the collector and the problem details.
- Number of open problems related to the data sources and the problem details.
- List of the changes that occurred in the data source in the last seven days.
- Details of the data sources and the NetFlow reporters available in the collector. The number of flows are shown for each NetFlow reporter. For data sources, the number of flows and the discovered VMs are shown.
- The graphical representation of the metrics such as CPU Usage, Memory Usage, and Data Disk Usage

## Viewing the VMware vCenter Manager Assessment Dashboard

The **VMware vCenter Manager** page provides a snapshot of all the VMs, VLANs, hosts connected to the **VMware vCenter Manager**, total numbers of traffic, and East-West traffic. You also see various metrics/widgets such as Top talking VMs, Network Optimization with NSX, Path Optimization with NSX, Application Centric Networking with NSX, and Crown Jewel Analysis.

### Generate NSX Assessment Report

You can get a network assessment and optimization report if you do not have NSX Manager added as a data source in VCF Operations for networks.

1. Open a VMware vCenter Manager dashboard.
2. On the top-right corner, click **Generate Report**.
   Open the generated PDF which contains the network assessment and optimization report.

### Application Troubleshooting Use Case

In VCF Operations for networks, you can get an end to end visibility of your applications, where you create applications, set threshold metrics, view flows in and out of applications, view detailed physical path, and even troubleshoot any problem or incident.

After you have added all your data sources, let's explore how you can manage and troubleshoot your application in more details:

| Goal | Description | Actions |
|---|---|---|
| Create Application | After you add your data sources, you can either discover your application or create/ define applications. | • To learn how to create an application, see Create an Application Manually<br>• To discover existing application, see Discover Applications. |
| Add threshold analytic to monitor application performance | To know when the network performance of an application drops below certain threshold, you can add threshold limit to your applications.<br>For example, you can set a threshold for round-trip time (RTT) in your application. Whenever, the RTT cross the threshold, you get an alert. | To learn how to configure thresholds, see Configure Thresholds and Alerts. |
| Search for flows to view network routing and flows details | When you get a threshold violation alerts, you can search for the flows to understand more details about the flow and to track the network path. | To learn about search queries, see Working with Search Queries. |
| See VM-VM path | When you search for flow and click on the flows that you like to view, a dashboards opens. You can view VM-VM path and VM underlay path to understand the flow better. | To learn about VM-VM path, see Network Visibility. |
| Open Network Map | To see the detailed physical path, you can go to the Network Map page and use the path search option.<br>The path details give you an idea of what components in the path need further investigation. | To learn about the Network Map and path search, see Working with the Network Map. |
| Troubleshoot an incident | After you have an idea what could be the problem, you can open a troubleshooting incident which will guide you to further investigate and find the route cause of your problem. | To learn about troubleshooting incident, see Working with Guided Network Troubleshooting |

**Latency Troubleshooting Use Case**

In VCF Operations for networks, you can get visibility into the latency, set a threshold to get notified when the latency violates the configured threshold limits, and identify the root cause of the high latency.

In VCF Operations for networks, you can get details about the following latency-related metrics:

| Types | Descriptions | Access the latency metrics |
|---|---|---|
| **vNIC to vNIC latency** | **The time taken by the packet to travel from the source VM's vNIC to the destination VM's vNIC on the same ESX host or across ESX hosts.** | **See Latency Metrics.**<br>**To monitor these metrics, you can do the following:**<br>**- Create a custom dashboard to monitor these metrics**<br>**- Set a threshold limit to get alert when there is a threshold violation.**<br>**- Create search-based alert for the metrics that are not supported in threshold metrics.** |
| **vNIC to PNIC and PNIC to vNIC latency** | **The time taken by the packet to travel from the source VM's vNIC to the PNIC of the host transport node of the source VM.** | |
| **PNIC to PNIC latency (VTEP to VTEP latency)** | **The time taken by the packet to travel from the PNIC/VTEP of the host of the source VM to the PNIC/VTEP of the host of the destination VM over the underlay network connecting the two hosts.** | |
| **TCP RTT & Retransmission counts for IPFIX Flows** | **NSX-T DFW IPFIX flow records include additional information about the TCP Round Trip time for establishing the flow and the TCP packet retransmission count to compensate for the high RTT or drops. These are in the form of retr ansmissionCount and averageLatency fields associated with the flow between a pair of IP addresses. vRNI enriches the flow information with VM, Hosts, Applications, E-W/N-S nature of traffic etc. and aggregates them and presents the max and average values for these metrics in association with the flows.** | |

## Troubleshoot Latency

When you gain visibility on the latency metrics, you find the root cause of the problem and solve the latency problem.

Let's explore how you can troubleshoot latency using VCF Operations for networks:

1. Run the following search query to get the top 10 vNIC-vNIC pairs with high latency:
   `netopa.interfacePairLatency.absolute.maximum.microsecond, netopa.interfacePairLatency.absolute.average.microsecond,vm1, vm2, virtualInterface1, virtualInterface2 of interface pair info where host is set and virtualInterface1Type != VMKNIC and netopa.interfacePairLatency.absolute.maximum.microsecond > 0 order by netopa.interfacePairLatency.absolute.maximum.microsecond limit 10`
   The search query lists the top 10 vNIC-vNIC pairs with high latency.
   The overall vNIC-vNIC Latency consists of the following components:
   - Source VM to source PNIC (vNIC - PNIC)
   - Source VTEP to destination VTEP (VTEP - VTEP)
   - Destination PNIC to destination VM (PNIC - vNIC).
2. Identify which of these three components is responsible for the high vNIC-vNIC latency.
   This helps to pinpoint the component responsible and investigate further.
   To determine the latency between the above-mentioned components, we need to run the following queries:
   - Source VM to source PNIC (vNIC - PNIC):
     `netopa.interfacePnicLatency.absolute.maximum.microsecond, netopa.interfacePnicLatency.absolute.average.microsecond, pnic, virtualInterface, vm of interface pnic pair info where host = '<source host name>' and vm = '<source VM name>'`
     or

```
vNIC PNIC Latency, max vNIC PNIC Latency, pnic, virtualInterface, vm of interface
pnic pair info where host = '<source host name>' and vm = '<source VM name>'
```

You get the latency of the first component.

– Source VTEP to destination VTEP (VTEP - VTEP):

```
netopa.interfacePairLatency.absolute.maximum.microsecond,
netopa.interfacePairLatency.absolute.average.microsecond, virtualInterface1,
virtualInterface2, host1, transportNode1, host2, transportNode2 of interface pair
info where virtualInterface1Type = VMKNIC and host1 = '<source host name>'
```

**Tip:**

You get the source host and destination host names by running the initial query to get the vNIC-vNIC latency.

– Destination PNIC to destination VM (PNIC - vNIC):

```
netopa.pnicInterfaceLatency.absolute.maximum.microsecond,
netopa.pnicInterfaceLatency.absolute.average.microsecond, pnic, virtualInterface,
vm of interface pnic pair info where host = '<destination host name>' and vm =
'<destinationvm name>'
```

Or

```
PNIC vNIC Latency, max PNIC vNIC Latency,pnic, virtualInterface, vm of interface pnic
pair info where host = '<destination host name>' and vm = '<destinationvm name>'
```

Based on the result of the above queries you can determine which component is responsible for high vNIC-vNIC latency.

## Viewing VMware vCenter Data Source Details

The **VMware vCenter Data Source** page provides a snapshot of the properties, changes, and problems of a particular data source.

In the **VMware vCenter Data Source** page, you see:

- Important information about the selected VMware vCenter data source, such as IP Address/FQDN, Collector Name, Enabled, number of discovered VMs, IPFIX Enabled Status, and so on.
- All the open problems associated to the data source.
- All the changes and the problems encountered in a particular data source in the last seven days.

## Viewing PCI Compliance Details

The **PCI Compliance** page is available only for the Enterprise License users.

### Access PCI Compliance

1. Go to **Plan & Assess** > **PCI Compliance**.
2. The **PCI Compliance** window appears. Select the required scope, the corresponding entity, and the duration for which you require the data. Click **Assess**.
3. The **PCI Compliance** page appears.

### PCI Compliance page details

The **PCI Compliance** page helps in assessing compliance against the PCI requirements only in the NSX environment. These requirements are mentioned under the first pin in the dashboard. The rest of the pins in the dashboard that provide data for the assessment of these requirements are as follows:

- Network flow diagram: It shows the data flow, firewalls, connections, and other details associated with a network.
- Flows: It lists the flows that you view in the network flow diagram.
- Clear text protocol flows based on the destination port: The traffic that flows on certain ports are in clear text. This pin displays the clear text protocol flows based on a particular destination port.

- Virtual machines in scope: It shows the virtual machines in the scope that you have selected in the query. This pin shows the outgoing rules, incoming rules, and security groups for virtual machines in that scope.
- Security groups of virtual machines: It lists the security groups of the virtual machines.
- Virtual machine count by Security Groups: You can view the list of the virtual machines in a security group by clicking Count in this pin.
- Virtual machine count by Security Tags: You can view the list of virtual machines with security tags by clicking Count in this pin.
- Firewall rules applied on internal traffic : You can view the firewall rules for the traffic between the virtual machines within the selected scope.
- Firewall rules applied on incoming traffic: You can view the firewall rules for the traffic that is coming from a virtual machine outside the scope to the virtual machine within the selected scope.
- Firewall rules applied on outgoing traffic: You can view the firewall rules for the traffic that is going to a virtual machine outside the scope from the virtual machine within the selected scope.
- Security tag membership changes: The changes related to the membership for security tags are shown in this pin.
- Security group membership changes: The changes related to the membership of a security group are shown in this pin.
- Firewall rule changes: The changes related to any firewall rule is listed in this pin.

  **Note:**
  If NSX has nested security groups, then the scope of PCI Compliance should be other than security group.

**Export as PDF**

VCF Operations for networks enables you to create and export the information on the PCI Compliance dashboard as a PDF report.

1. In the PCI Compliance dashboard, click **Export as PDF** on the right top side of the page. The Export to PDF window appears.
2. The Export to PDF window lists all the widgets and their respective properties available on the PCI Compliance dashboard. Select the widgets and the properties that you want to export.

   **Note:**

   - You have to select at least one property.
   - The maximum number of properties that you can select is 20.
   - The maximum number of entries in the list view that can be exported is 100.
   - Certain widgets do not allow you to select the properties. In such instances, specify only the number of entries.
3. Provide a title for the PDF report.

   **Note:**

   - The maximum number of characters in the title is 200.
   - The maximum number of pages that can be generated in the report are 50.
4. Click **Preview**. You can see the preview of the complete report.
5. Click **Export PDF**.

## Viewing Kubernetes Details

You can use the Kubernetes dashboard to get a quick overview of your Kubernetes or Tanzu Kubernetes Grid Integrated Edition deployments in VCF Operations for networks.

You see details about,

- Top talking Clusters and Namespaces based on the flows
- Overview of Kubernetes cluster entities such as, count of Namespaces, Pods, services and Nodes
- Kubernetes clusters added in VCF Operations for networks
- List of containers images running on Pods and count of Pods for each container image

- List of new pods discovered, its count, namespace and cluster details.

In addition, you can click on the count of various Kubernetes entities on the dashboard to the see the list view and go to details of that particular entity.

**Table 1329: Kubernetes Entity Dashboard**

| Dashboard | Description |
|---|---|
| Cluster Dashboard | You get the deployment details at the cluster level, which includes<br>• cluster overview that includes count of namespaces, services, pods and nodes in the deployment.<br>• the list of top namespaces based on flows.<br>• Interaction between namespaces. |
| Namespace Dashboard | You get the cluster namespace details such as,<br>• the namespace overview that includes count of pods, services and nodes that are in that particular namespace.<br>• the list of top talking services based on flows.<br>• Service interactions in the namespace.<br>• Network traffic by packets and bytes. |
| Service Dashboard | You see the details of the Kubernetes services, like<br>• the service overview that includes the count of:<br>— open alerts in 24 hours<br>— incoming & outgoing flows in 24 hours<br>— pods<br>— nodes on which the service is deployed.<br>• connectivity between the kubernetes components and NSX-T.<br>• count of active nodes and pods for a specific period.<br>• service interaction in the namespace.<br>• network traffic by packets and bytes. |
| Pods Dashboard | You see the details such as,<br>• the cluster, namespace, and node that the pod belongs<br>• Network traffic between pods based on packets and bytes |
| Nodes Dashboard | You see details such as,<br>• list of namespaces details<br>• list of services<br>• list of container pods<br>• Network traffic between nodes based on packets and bytes |

**Note:**

- VCF Operations for networks collects Kubernetes cluster details from Tanzu Kubernetes Grid Integrated Edition every 10 minutes.
- VCF Operations for networks collects all objects (Namespace, Node, Pod, Service) from Kubernetes cluster every 4 hours. However, if there is any change in the Kubernetes objects, VCF Operations for networks runs the Watch API and updates the changes immediately.
- Tanzu Kubernetes Grid Integrated Edition does not provide details about the Kubernetes primary nodes.
- VCF Operations for networks provides the details of the clusters that are in successfully created state only.

## Common Alerts or Error Messages

- `Data Source not reachable` - Ping the IP/FQDN of the Tanzu Kubernetes Grid Integrated Edition from the collector virtual machine to ensure Tanzu Kubernetes Grid Integrated Edition is reachable.
- `Kubernetes Cluster API Servers not reachable`-Ensure that all the Kubernetes Cluster API Servers are reachable from the collector virtual machine.

## Viewing Load Balancer Details

Load Balancer page summarizes all the information of the virtual servers and the pools that are created on the load balancer.

You see,

- list of virtual servers along with its problems on the load balancer
- list of pools on the load balancer and their associated problems
- Alerts associated with the load balancer
- list of flows, count and its network traffic on different destination IPs.
- the properties of the load balancer that provides information such as the vendor, type, serial number, virtual servers, pools.

## Viewing VM Details

You can use the VM page to get a detailed overview of your VMs available in VCF Operations for networks.

In the VM page, you see the following sections:

| Section | Details |
|---|---|
| **Overview** | You see,<br>• VM details.<br>• topology information.<br>• various configuration parameters.<br>• security-related parameters.<br>• VM to Internet path. |
| **Neighbors** | You see,<br>• the graphical view of various metric properties in comparison with the neighbor VMs<br>• the list of VMs belongs to the same host. |
| **Alerts** | You see the list of alerts related to the selected VM. |
| **Flows** | You see the list of flows which are either originated or trying to reach the selected VM for which firewall action is allowed and denied. Also, you see list of the flows that are originated or trying to reach a VM with App-Id info. |

| Section | Details |
| --- | --- |
| **Metrics** | You see,<br>• the metrics information related to the selected VM.<br>• information about the network usage of ports in the path to ToR.<br>• information about all the metrics properties.<br>• Input - output metrics information.<br>• the virtual disk space.<br>• the datastore performance<br>**Note:** You cannot see the datastore metrics of a VM if it is hosted on vSAN datastore.<br>• the latency metric collection details.<br>**Note:** To see virtual infrastructure latency, the port 1991 must be open on the collector to receive latency data from the ESX host.<br>• the network utilization metrics.<br>**Note:** The metrics support 20 seconds granularity and these metrics are retained only for 48 hours. |

## Viewing NSX Manager Details

You can use the **NSX Manager** page to get a detailed overview of your NSX Manager available in VCF Operations for networks.

### How to access the NSX Manager page

To access this page, search for `NSX Manager where SDDC Type = 'VMC'` and in the search result list, click the **NSX Manager** page you want to view.

### Overview

In the **NSX Manager** page, you see the following section:

**Table 1330:**

| Section | Details |
|---|---|
| **Overview** | You see:<br>• NSX policy entities overview details.<br>• Modified entities in last 24hours.<br>• Top flows by rule.<br>• List of routers.<br><br>**Note:**<br>The number of entities shown in the **NSX Policy Entities Overview** widget and the **Entities in last 24 hours** widget can differ. If some entities discovered in the last 24 hours were deleted, then the number of entities shown in the **Entities in last 24 hours** widget can be greater than the number of entities shown in the **NSX Policy Entities Overview** widget. |
| **Top Talkers** | You see:<br>• Top talking entities in your environment. |
| **Network Traffic and Alerts** | You see:<br>• Network traffic and alerts overview details.<br>• List of alerts. |

## Viewing VMware NSX Manager Details

You can use the **VMware NSX-T Manager** page to get an overview of your VMware NSX-T Manager available in VCF Operations for networks.

To access this page, search for `NSX-T Manager` and in the search result list, click the entity you want to view.

### Overview

In the NSX Manager page, you see:

| Section | Details |
|---|---|
| **Overview** | You see:<br>• A summary of your NSX-T Manager, including the alert chart, number of firewall rules, IPSET, transport zones, applications, and unprotected flows, and flows volume in last 24 hours.<br>• List of properties, firewall rules by number of hits, top flows by rule, and compute managers.<br>• The topology details. Topology provides a contextual view of the entities and also shows alerts associated with the entities.<br>• Top flows by App-Id. |
| **Alert** | You see:<br>• A list of various alerts and analytics threshold alerts. |
| **Flows** | You see:<br>• Various flow analytics. |
| **Metrics** | You see:<br>• NSX-T Management node health details.<br><br>**Note:**<br>The NSX-T Management node health details is available only for NSX-T version 2.4.0 and later. |

## Viewing NSX-T Management Node Details

You can use the **NSX-T Management Node** page to get an overview of your VMware NSX-T Management node details available in VCF Operations for networks.

To access this page, search for `NSX-T Management Node` and in the search result list, click the entity you want to view.

### Overview

In this page, you see:

| Section | Details |
|---|---|
| **Overview** | You see:<br>• A summary of your NSX-T management node, including properties details, system metrics, and service status. |
| **Event** | You see:<br>• A list of various events. |
| **Interface Stats** | You see:<br>• Various interface stats, including received packets, transmitted packets, received dropped packets, transmitted dropped packets and so on. |
| **System Stats** | You see:<br>• Various system stats, including system load, system usage, and file system usage. |

## Viewing Switch Port Details

You can use the Switch Port page to get a detailed overview of the switch ports available in VCF Operations for networks.

To access this page, search for `Switch Port` on the VCF Operations for networks home page. The search result lists all the switch ports on VCF Operations for networks. Click the entity for which you want to view further details.

| Section | Details |
|---|---|
| Overview | You see:<br>• Key configurations and runtime properties of the switch port.<br>• A list of VMs connected to the switch port. |
| Alerts | You see the problems and changes associated with the switch port. |

| Section | Details |
| --- | --- |
| Metrics | You see the following switch port metrics:<br>• **Network Rx Rate** - The average rate at which data was received during the time interval.<br>• **Network Traffic Rate** - The average of received and transmitted traffic during the time interval.<br>• **Network Tx Rate** - The average rate at which data was transmitted during the time interval.<br>• **Dropped Rx Packets** - The average ratio of dropped received packets to total received packets.<br>• **Dropped Tx Packets** - The average ratio of dropped transmitted packets to total transmitted packets.<br>• **Interface Peak Buffer Utilization** - Peak buffer occupancy per port during the time interval.<br>• **uRPF Drop Packets** -The number of packets dropped at the switch port due to unicast reverse-path forwarding (uRPF).<br>**Note:**<br>The number of dropped packets is displayed in the metrics only if you are viewing the switch port details of a Cisco ASR 1K device and if the uRPF feature is enabled on the device. |

## Viewing Router Interface Details

You can use the Router Interface page to get a detailed overview of your router interfaces available in VCF Operations for networks.

To access this page, search for `Router Interface` on the VCF Operations for networks home page. The search result lists all the router interfaces on VCF Operations for networks. Click the entity for which you want to view further details.

| Section | Details |
| --- | --- |
| Overview | You see:<br>• Key configurations and runtime properties of the router interface.<br>• A list of VMs on the same network as this router interface.<br>• A list of VMs on the same gateways as this router interface.<br>• A list of VMs on same the network, but different gateways.<br>• A list of other router interfaces which are on same network. |
| Alerts | You see the problems and changes associated with the router interface. |

| Section | Details |
|---|---|
| Metrics | You see the following router interface metrics:<br>• **Rx packets** - The average number of received packets during the time interval.<br>• **Network Max Tx Rate** - The maximum rate at which the data is transmitted during the time interval.<br>• **Network Max Rx Rate** - The maximum rate at which the data is received during the time interval.<br>• **Tx Packets** - The average number of transmitted packets during the time interval.<br>• **Dropped Rx Packets** - The average ratio of dropped received packets to total received packets.<br>• **Dropped Tx Packets** - The average ratio of dropped transmitted packets to total transmitted packets.<br>• **Rx Bytes** -The total number of bytes received over the router interface.<br>• **Tx Bytes** -The total number of bytes transmitted over the router interface.<br>• **uRPF Drop Packets** -The number of packets dropped at the router interface due to unicast reverse-path forwarding (uRPF).<br>**Note:**<br>The number of dropped packets is displayed in the metrics only if you are viewing the router interface details of an NSX device and if the uRPF feature is enabled on the device. |

## Viewing NSX-T Transport Details

You can use the **NSX-T Transport Node** page to get an overview of your transport node details available in VCF Operations for networks. You can view both host node details and Edge node details in VCF Operations for networks.

### NSX-T Transport Node page where the node type is Host

To access this page, search for `NSX-T Transport Node where Node Type = 'HostNode'` and in the search result list, click the entity you want to view.

### Overview

In this page, you see:

| Section | Details |
|---|---|
| **Overview** | You see:<br>• A summary of your host transport node, including the alert chart, incoming traffic, outgoing traffic, internal traffic, number of network interfaces, and number of total VMs.<br>• Properties details, transport node status, PNIC statistics in last 24 hours, TEP statistics in last 24 hours, and system metrics in last 24 hours.<br>**Note:**<br>The system metrics is available only for NSX-T version 2.4.0 and later. |
| **Alert** | You see:<br>• A list of various alerts. |
| **Latency** | You see:<br>• TEP to TEP latency details. |

| Section | Details |
|---|---|
| **Interface Stats** | You see:<br>• Various interface stats, including received packets, transmitted packets, received dropped packets, transmitted dropped packets and so on. |
| **System Stats** | You see:<br>• Various system stats, including system load, system usage, and file system usage.<br>**Note:**<br>The **System Stats** is available only for NSX-T version 2.4.0 and later. |
| **Flows** | You see:<br>• Top VMs by flows (in last 24 hours) and top rules by flows (in last 24 hours). |

**NSX-T Transport Node page where the node type is Edge**

To access this page, search for `NSX-T Transport Node where Node Type = 'EdgeNode'` and in the search result list, click the entity you want to view.

**Overview**

In this page, you see:

| Section | Details |
|---|---|
| **Overview** | You see:<br>• A summary of your Edge transport node, including the alert chart, number of network interfaces, tier0 service routers, tier1 service routers, and routes.<br>• Properties details, transport node status, uplink statistics in last 24 hours, TEP statistics in last 24 hours, and system metrics in last 24 hours |
| **Alert** | You see:<br>• A list of various alerts. |
| **NAT Stats** | You see:<br>• Various NAT stats, including NAT Rule Statistics, top NAT rules by total bytes, top NAT rules by total packets, and top NAT rules by session count. |
| **Interface Stats** | You see:<br>• Various interface stats, including received packets, transmitted packets, received dropped packets, transmitted dropped packets and so on. |
| **System Stats** | You see:<br>• Various system stats, including system load, system usage, and file system usage. |

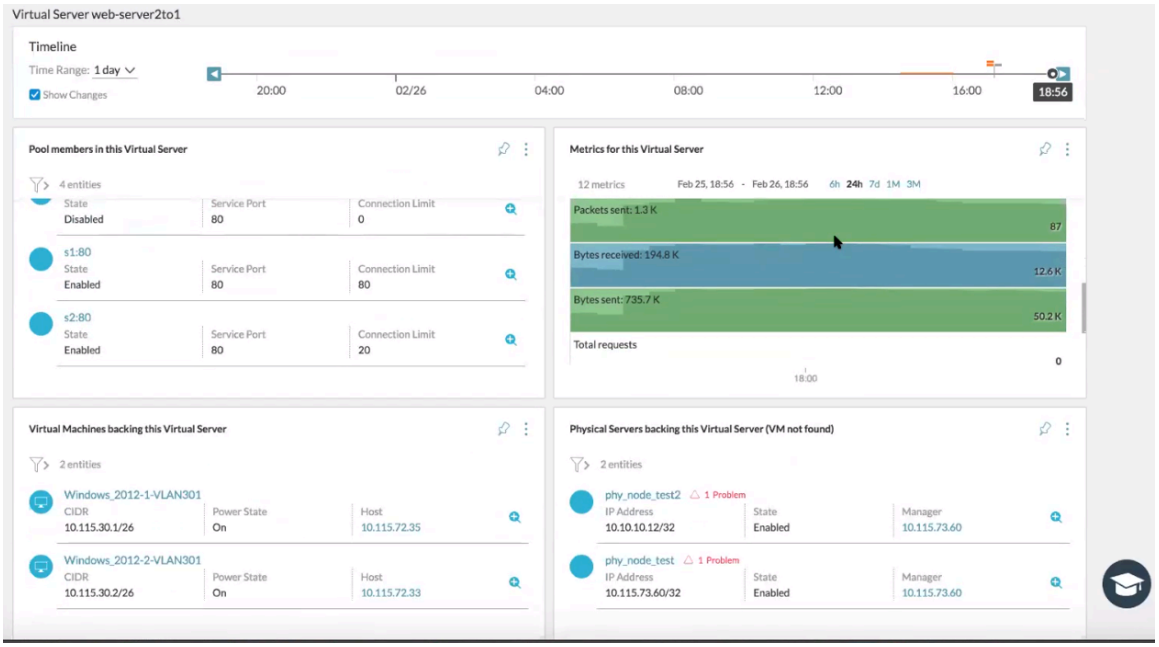
## Viewing Virtual Server Details

The Virtual Server page displays the virtual server metrics, problems, and change alerts.

You see,

- The list of all pool members in the virtual server and its details, along with an alert for any problem.

- The list of virtual machines
- The list of physical servers
- The list of problem alerts that are associated with the virtual server .
- The list of metrics that are related to the virtual server, such as:
  - Connections (count, duration)
  - Network metrics (packets and bytes received or sent)
  - CPU usage
- The top flows for the pool members used by the virtual server.
- The virtual server properties that provides information about the load balancer IP address, network traffic, service port.




To view the path topology associated with the load balancer, you can use the following query:

*client VM name* to *Virtual server IP*.

If there are multiple virtual servers on different service ports, you see the list under the **Select a Destination VM** section. To see the VM to virtual server path, you can select a server from the list and click **Show Path**.

To see a set of VMs on the Virtual Server window, you can click the virtual server on the VM path topology and click **View Path** to see the path from virtual server to the selected VM.

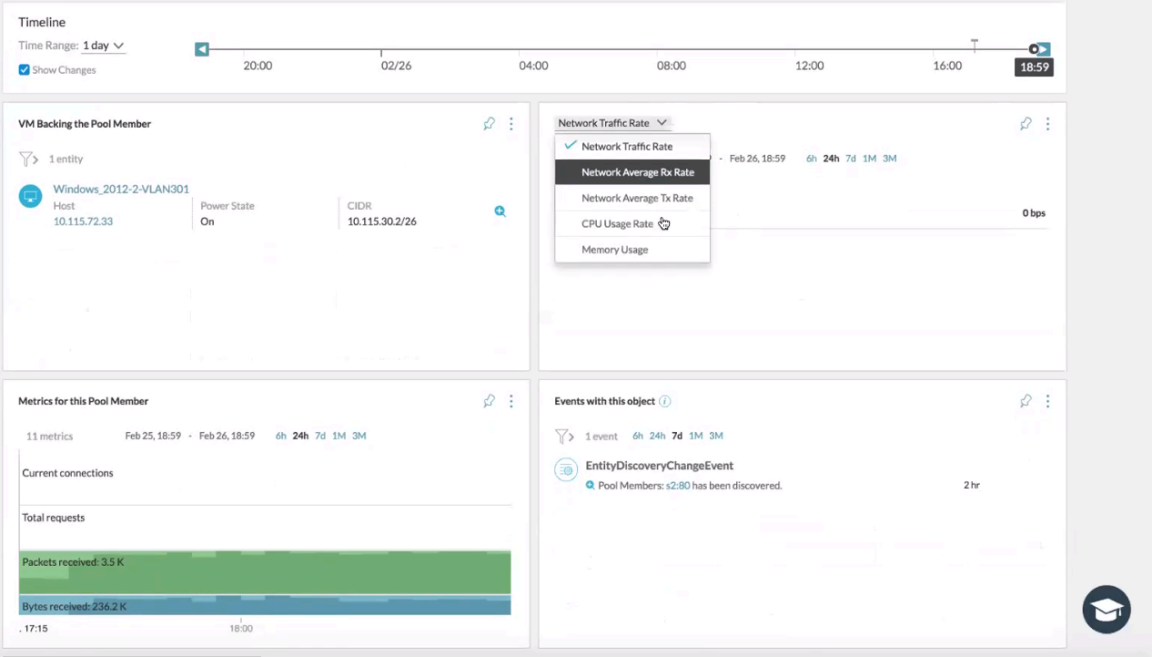
## Viewing Pool Members Details

The Pool Member page provides insight about the pool members, metrics, and alerts associated with the pool member.

You see:

- The list of virtual machines and additional details about the VM
- The list of metrics that are related to the pool member, such as:
  - Connections (count, duration, age )
  - Network metrics (packets and bytes received or sent)
  - CPU usage
- The pool member properties that provide information about the load balancer, node, status, service port.

The Pool Member page also allows you to compare the metrics of the pool member with the metrics of the VM. For example, memory and CPU usage, Network traffic, and so on.




## Viewing Microsoft Azure Details

You can use the **Microsoft Azure** page to get a quick overview of your Azure environment details in VCF Operations for networks.

### How to access

To access this page, on the search bar, enter `Azure`.

> **Note:**
>
> Starting with version 9.1.1, adding new **Microsoft Azure** data sources is not supported. **Microsoft Azure** data sources configured before version 9.1.1 are disabled. Data collection for these data sources has stopped, and they cannot be enabled. Dashboards only display historical data was collected from previously configured **Microsoft Azure** data sources.

### Overview

In this page, you see:

- List of subscriptions
- List of virtual machines
- List of network interfaces, virtual networks, subnets, route tables, and routes
- List of network security groups, application security groups, and NSG rules.

You can also click the entities on this page to the see more detailed insight about the particular entity.

In addition to the **Microsoft Azure** page, you can see the insight about the following Azure entities:

**Table 1331: Azure Entity details**

| Entity name | Description |
|---|---|
| **Azure Application Security Group** | You see:<br>• List of properties, alerts, associated VMs, and associated VMs for last 24 hours.<br>• List of incoming NSG rules and outgoing NSG rules.<br>• List of allowed flows, denied flows, flows for last 24 hours. |
| **Azure Data Source** | You see:<br>• List of properties, alerts, and metrics. |
| **Azure NSG Rules** | You see:<br>• List of properties, alerts, and metrics. |
| **Azure Network Interface** | You see:<br>• List of properties, alerts, and metrics. |
| **Azure Network Security Group** | You see:<br>• List of properties, alerts, NICs, and subnet.<br>• List of outbound rules and inbound rules.<br>• List of allowed flows, denied flows, flows for last 24 hours. |
| **Azure Route** | You see:<br>• List of properties, alerts, and metrics. |
| **Azure Route Table** | You see:<br>• List of properties, alerts, and metrics. |
| **Azure Subnet** | You see:<br>• List of properties, alerts, VMs, NICs, and custom routes.<br>• List of NSG rules. |
| **Azure Subscription** | You see:<br>• List of properties and alerts.<br>• List of virtual machines.<br>• List of NICs, virtual networks, and route table<br>• List of network security groups, application security groups, and NSG Rules. |
| **Azure Virtual Machine** | You see:<br>• List of properties, alerts, NICs, associated Application Security Groups (ASGs).<br>• List of incoming NSG rules and outgoing NSG rules.<br>• List of allowed flows and denied flows. |
| **Azure Virtual Network** | You see:<br>• List of properties, alerts, VMs, VMs created in last 24 hours, associated ASGs, associated ASGs for last 24 hours, subnets, and route tables.<br>• List of allowed flows, denied flows, and flows for last 24 hours. |

## Viewing Internet Service Provider Details

You can use the **Internet Service Provider** page to Internet Service Provider(ISP) visibility across all the edges of an enterprise?.

To access this page, search `Internet Service Provider` and in the search result list, click the entity you want to view.

| Section | Details |
|---|---|
| **Overview** | You see:<br>• List of VeloCloud Edges, quality of experience reports on Edges and Links, packets details, latency details, and flow packets details. |

## Viewing VMC SDDC Details

You can use the **VMC SDDC** page to get an overview of your VMC on AWS SDDC in VCF Operations for networks.

### How to access the VMC SDDC page?

To access this page, search for `VMC SDDC` and in the search result list, select an entity you want to view.

### Overview

In the **VMC SDDC** page, you see:

| Section | Details |
|---|---|
| **Overview** | You see an overview of your NSX Entities and the count of Direct Connect Interface (VIFs) present within SDDC.<br>You see entities in last 24 hours, Top flows by rule which also shows the firewall manager responsible for the firewall rule, list of routers, and properties details.<br>You also can see the top 25 VMs based on the resource pool, metrics, and aggregate type. |
| **Microsegmentation** | You see the graphical representation of flows between entities. |
| **Network Traffic and alerts** | You see an overview of the network traffic and the list of all alerts associated with the SDDC entities discovered by VCF Operations for networks alerts. |
| **Top Talkers** | You can see the top talking entities in VMC SDDC. |
| **Metrics** | You see the Tier0 Router interface metrics, such as Rx Packets, Tx Packets, Dropped Rx Packets, and so on for Public, Cross-VPC, and Direct Connect interfaces.<br>You see Network Utilization metrics.<br>You see Entity Usages metrics which show metrics related to various distributed firewall rules. The Entity Usages metrics also shows the alert threshold and the maximum value by colored dash lines.<br>**Note:**<br>The metrics support 20 seconds granularity and these metrics are retained only for 48 hours. |

### VMC SDDC Limits

The following messages are displayed when the VMC SDDC exceeds the network and security maximum limits:

| Message | Severity | Alert Name |
|---|---|---|
| Number of IPsec VPN tunnels created per SDDC | Info | VMware Cloud Threshold Alert |

| Message | Severity | Alert Name |
|---|---|---|
| Number of logical segments per SDDC | Critical | |
| Number of Management Gateway firewall rules | Critical | |
| Number of Compute Gateway firewall rules | Critical | |
| Number of grouping objects (security groups) | Critical | |
| Number of private virtual interfaces attached to one SDDC | Info | |
| Number of distributed firewall rules per section group | Critical | |
| Number of distributed firewall rules across all categories | Critical | |
| Number of distributed firewall Sections per categories | Critical | |
| Number of distributed firewall sections across all categories | Critical | |

**Related Topic**

VMC SDDC Search Queries

You can search the VMware Cloud on AWS SDDC ( Software-defined data center) entity details in VCF Operations for networks.

## Viewing VMC on AWS Outposts SDDC Details

The **VMC on AWS Outposts SDDC** page provides a detailed overview of an SDDC hosted within AWS facilities.

### How to access the VMC on AWS Outposts SDDC page?

To access this page, do one of the following:

- Search for `VMC on AWS Outposts SDDC` and in the search result list, select an SDDC you want to view.
- Select **VMC on AWS Outposts** on the **Environments** widget of the homepage. From the list, select the SDDC you want to view.

### Overview

In the **VMC on AWS Outposts SDDC** page, you see:

| Section | Details |
|---|---|
| **Overview** | You see the following information:<br>• The number of entities within the SDDC which includes number of VMs, groups, firewall rules, and policy based VPNs.<br>• Important configuration and runtime properties of the SDDC.<br>• Entities discovered in the last 24 hours.<br>• The top firewall rules based on flow traffic.<br>• The top 25 VMs based on a selected metric and aggregate type. |
| **Microsegmentation** | You see the graphical representation of flows between entities. |
| **Network Traffic and alerts** | You see an overview of the network traffic and the list of alerts related to the SDDC. |
| **Top Talkers** | You see the top talking VMs in the SDDC. |

| Section | Details |
| --- | --- |
| **Metrics** | You see the following metrics:<br>• Tier0 router interface metrics such as Rx Packets, Tx Packets, Dropped Rx Packets for public, cross-VPC, and direct connect interfaces.<br>• Network utilization metrics.<br>**Note:**<br>The network utilization metric supports 20 seconds granularity and its retention interval is 48 hours.<br>• Entity usage metrics on various distributed firewall rules. |

## Viewing Arista Hardware Gateway and Arista Hardware Gateway Binding Details

You can view the **Arista Hardware Gateway** and **Arista Hardware Gateway Binding** page to get an overview of your Arista Hardware Gateways.

### How to access the Arista Hardware Gateway page?

To access the **Arista Hardware Gateway** page, search for `Arista Hardware VTEP` and in the search result list, click the entity you want to view.

To access the **Arista Hardware Gateway Binding** page, search for `Arista Hardware Gateway Binding` and in the search result list, click the entity you want to view.

### Overview

In the **Arista Hardware Gateway** page, you see:

- List of alerts
- List of key properties
- List of Arista hardware gateway bindings.

In the **Arista Hardware Gateway Binding** page, you see:

- List of alerts
- List of properties.

## Viewing Avi Load Balancer Details

You can use the Avi Load Balancer entity pages to get detailed insight of Avi Load Balancer entities such as cluster, data source, tenant, controller, pool, virtual service, service engine, and analytics profile.

### How to Access Avi Load Balancer Entity Pages?

To access Avi Load Balancer entity pages:

1. Enter `NSX-ALB` in the Search bar.
2. In the search result, click the entity type that you want to view, and press **Enter**.
3. On the respective entity type page, click the name of the entity for which you want to view the details.

Alternatively, select **Entities** on the left navigation pane and enter `ALB` or `NSX-ALB` in the entity search bar.

## Avi Load Balancer Entity Details Pages

| Entity Name | Description |
|---|---|
| **NSX-ALB Cluster** | You see the deployment details at the cluster level, which includes:<br>• Summary of open alerts<br>• Count of Tenants, Clouds, Service Engines, Service Engine Groups, Virtual Services, Pools and Pool Members in the deployment.<br>• List of key properties<br>• List of Controllers<br>• List of alerts encountered in the last seven days<br>• Health of Virtual Services<br>• Graphical representation of the following metrics:<br>– Controller Metrics<br>– Service Engine Metrics<br>– Virtual Service Metrics<br>– Pool Metrics |
| **NSX-ALB Data Source** | You see the details of the data source which includes:<br>• List of key properties<br>• List of open problems<br>• List of all the changes and the problems encountered in the last seven days |
| **NSX-ALB Controller** | You see the details of the controller which includes:<br>• List of key properties<br>• List of alerts encountered in the last seven days<br>• Graphical representation of the CPU Usage Metrics<br>**Note:**<br>You have to be a super user to view this metrics. |
| **NSX-ALB Tenant** | You see the details of the tenant which includes:<br>• Summary of open alerts<br>• Count of Service Engines, Service Engine Groups, Virtual Services, Pools and Pool Members in the deployment.<br>• List of key properties of the tenant<br>• List of alerts encountered in the last seven days<br>• Graphical representation of the following metrics:<br>– Virtual Service Metrics<br>– Service Engine Metrics<br>– Pool Metrics |
| **NSX-ALB Cloud** | You see the deployment details at the Cloud level which includes:<br>• Summary of open alerts<br>• Count of Service Engines, Service Engine Groups, Virtual Services, Pools and Pool Members in the deployment.<br>• List of key properties<br>• List of open alerts encountered in the last seven days<br>• Graphical representation of the following metrics:<br>– Service Engine Metrics<br>– Virtual Service Metrics<br>– Pool Metrics |

| Entity Name | Description |
|---|---|
| **NSX-ALB Service Engine Group** | You see the details of the Service Engine Group which includes:<br>• Summary of open alerts<br>• Count of Service Engines, Virtual Services, Pools and Pool Members in the deployment.<br>• List of key properties<br>• List of alerts encountered in the last seven days<br>• Graphical representation of the following metrics:<br>– Virtual Service Metrics<br>– Service Engine Metrics<br>– Pool Metrics |
| **NSX-ALB Service Engine** | You see the details of the Service Engine which includes:<br>• Summary of open alerts<br>• Count of Virtual Services, Pools, and Pool Members<br>• List of alerts encountered in the last seven days<br>• Graphical representation of the following metrics:<br>– Virtual Service Metrics<br>– Service Engine Metrics<br>– Pool Metrics |
| **NSX-ALB Virtual Service** | You see the details of the Virtual Service which includes:<br>• List of key properties<br>• List of Pool Members<br>• List of VMs associated with the Virtual Service<br>• List of Pool Members associated with this Virtual Service by Sum of Connections<br>• List of alerts encountered in the last seven days<br>• Graphical representation of the Virtual Service Metrics |
| **NSX-ALB Pool** | You see the details of the pool which includes:<br>• List of key properties<br>• List of Pool Members associated with the pool<br>• List of VMs in the pool<br>• List of alerts encountered in the last seven days<br>• Graphical representation of the Pool Metrics |
| **NSX-ALB Pool Member** | You see the details of the pool member which includes:<br>• List of key properties<br>• Details of the VM that is backing the Pool Member<br>• List of alerts encountered in the last seven days<br>• Graphical representation of VM and Pool Metrics |
| **NSX-ALB Pool Group** | You see the details of the Pool Group which includes:<br>• List of key properties<br>• List of alerts encountered in the last seven days<br>• Graphical representation of the Pool Metrics |
| **NSX-ALB Analytics Profile** | You see the detailed analytics of Avi Load Balancer which includes:<br>• List of key properties<br>• List of alerts encountered in the last seven days<br>• List of Virtual Services |

## Viewing Cisco Nexus Device Details

You can use the **Cisco Nexus Device** page to get an overview of your cisco nexus device available in VCF Operations for networks.

**Overview**

In this page, you see:

- Performance monitoring metrics.

  **Note:**
  To get more insight about each metric, click the individual metric value.
- Alert list.
- Properties details.
- List of switch ports, switch port peers, and VMs connected to port.
- Switch port metrics.

## Viewing Micro-Segmentation Details

You can analyze the flows by selecting scope and segment them accordingly based on entities such as VLAN/VXLAN, Security Groups, Application, Tier, Folder, Subnet, Cluster, virtual machine (VM), Port, Security Tag, Security Group, and IPSet.

The micro-segmentation page provides the analysis details with the topology diagram. This page consists of the following sections:

- Micro-Segments: This widget provides the diagram for topology planning. You can select the type of group and flows. Based on your inputs, you can view the corresponding topology planning diagram.
- Traffic Distribution: This widget provides the details of the traffic distribution in bytes.
- Top Ports by Bytes: This widget lists the top 100 ports that record the highest traffic. The metrics for the flow count and the flow volume are provided. You can view the flows for a particular port by clicking the count of flows corresponding to that port.

To access the micro-segmentation page:

1. In the left navigation panel, click **Security > Plan Security**.
2. Select the scope, subscope, and the duration for which you want to plan and analyze. Click **Analyze**.

   The micro-segmentation page appears.

   **Note:**
   The micro-segmentation wheel view can show upto 600 nodes and 6000 edges. If the limit exceeds, you see the `Too many micro-segments to analyse. Please select a different entity or micro-segmentation criteria` error.

## Viewing VMware HCX Manager Details

You can view the VMware HCX Manager details deployed per VMware vCenter.

To access the VMware HCX Manager page, from the left navigation pane, go to **Environments** and select **VMware HCX Manager**. Alternatively, search **HCX Manager** to list all the VMware HCX Managers. VCF Operations for networks also supports search for all the VMware HCX entities.

The VMware HCX Manager page displays the list of sites, click any of the sites to open the VMware HCX Manager details page of that site.

In the VMware HCX Manager details page you see the following sections.

| Section | Details |
| --- | --- |
| Overview | • In the **Summary** widget, you see the total number of alerts, appliances, services meshes, L2 extensions, tunnels, and applications communicating over the L2 extension in that site.<br>Click the total number on any of these categories to see the associated entities and its details.<br>• In the VMware HCX Appliance Topology widget:<br>  • The **Interconnect** tab shows the sites paired with the current VMware HCX Manager site. Here, the current VMware HCX Manager site is shown at the top and the sites paired are shown at the bottom.<br>  Hover over the connection between the sites to see information about the source appliance, destination appliance, unhealthy tunnels, and the total number to tunnel entities.<br>  If any tunnels have any alerts raised, you can click the number of unhealthy tunnels to view those details. Click the total number of tunnels to see the tunnel entity details.<br>  • The **Network Extension** tab shows the topology of how the networks are extended from the current VMware HCX Manager site to the paired sites.<br>  This topology shows the network extension appliance pairs between the sites connected via tunnels, the L2 networks that are extended via these appliances, and the VMs that belong to these L2 networks. Hover over any of the appliance, L2 networks, VMs, or tunnels to view more details.<br>  **Note:**<br>  You can sometimes see appliances that not connected to any appliances on the paired sites. This occurs when the VMware HCX data source for the paired site of that appliance is not added to VCF Operations for networks.<br>  • You can also filter the topology map to show the entities (with their connections) that have critical alerts.<br>In the **Interconnect** tab and the **Network Extension** tab, click on the site to expand it and only one paired site can be in the expanded state.<br>**Note:**<br>Only the sites that are added as VMware HCX data sources in VCF Operations for networks are shown in these tabs. |
| Alerts | You see the list of various alerts raised based on the health of the VMware HCX entities and any misconfigurations between the paired VMware HCX sites. |

| Section | Details |
| --- | --- |
| Flows | VCF Operations for networks identifies the flows going over the VMware HCX network extension tunnels and enriches them with the information of the VMware HCX L2 extension entity.<br>• **Top Talking Entities Over L2 Extensions**: In this widget, you can see the top talking entities like VM pairs, L2 networks, VMware HCX L2 extensions, etc over the network extension tunnels that belong to the current VMware HCX Manager.<br>• **Flows Per L2 Extensions**: In this widget, you see the number of flows and the related flow metrics for each L2 extension entity in the current VMware HCX Manager.<br>• **Tromboning Flows Per L2 Extension**: In this widget, you see the tromboning flows that traverse from the remote site to the local site and then back to the remote site via the network extension tunnels.<br>This helps you to identify if you want to enable Mobility Optimized Networking on the participating L2 extensions so that the flows from VMs in the remote sites can directly traverse over the remote gateway.<br>• **Flows Per Appliance Pair**: In this widget, you see the flows over the network extension tunnels and related metrics grouped by the network extension appliance pairs. |
| Metrics | You see the metrics details of the VMware HCX appliances VMs and hosts, tunnels, and appliance VM network utilization. |

In addition to the VMware HCX Manager details page, you can also see the insights about the VMware HCX entities. You can access the VMware HCX entity pages from the VMware HCX Manager details page or directly search for them in the VCF Operations for networks search.

### VMware HCX Site

You see the list of properties and alerts of the selected VMware HCX Site.

### VMware HCX Service Mesh

VMware HCX service mesh is a group of VMware HCX services deployed between two site.

In the **Flows** tab of this VMware HCX Service Mesh entity page, VCF Operations for networks identifies the flows going over the HCX network extension tunnels and enriches them with the information of the VMware HCX L2 extension entity.

- **Top Talking Entities Over L2 Extensions**: In this widget, you can see the top talking entities like VM pairs, L2 networks, VMware HCX L2 extensions, etc over the network extension tunnels that belong to the current VMware HCX Manager.
- **Flows Per L2 Extensions**: In this widget, you see the number of flows and the related flow metrics for each L2 extension entity in the current VMware HCX Manager.
- **Tromboning Flows Per L2 Extension**: In this widget, you see the tromboning flows that traverse from the remote site to the local site and then back to the remote site via the network extension tunnels.
  This helps you to identify if you want to enable Mobility Optimized Networking on the participating L2 extensions so that the flows from VMs in the remote sites can directly traverse over the remote gateway.
- **Flows Per Appliance Pair**: In this widget, you see the flows over the network extension tunnels and related metrics grouped by the network extension appliance pairs.

In this entity page you also see:

- List of properties and alerts.
- List of all the VMware HCX service and tunnel entities under that service mesh.

Click any entity to view more details.

### VMware HCX Service

VMware HCX service is a part of the service mesh that is responsible for the VMware HCX features like interconnect, network extension, bulk migrations, etc.

In this entity page, you see a list of all the properties and alerts of the selected VMware HCX service.

### VMware HCX Appliances

VMware HCX appliance is a VM that runs the VMware HCX services. VCF Operations for networks supports VMware HCX interconnect appliance, VMware HCX network extension appliance, and the VMware HCX WAN optimisation appliance.

In the **Flows** tab of this VMware HCX Appliances entity page, VCF Operations for networks identifies the flows going over the HCX network extension tunnels and enriches them with the information of the VMware HCX L2 extension entity.

- **Top Talking Entities Over L2 Extensions**: In this widget, you can see the top talking entities like VM pairs, L2 networks, VMware HCX L2 extensions, etc over the network extension tunnels that belong to the current VMware HCX Manager.
- **Flows Per L2 Extensions**: In this widget, you see the number of flows and the related flow metrics for each L2 extension entity in the current VMware HCX Manager.
- **Tromboning Flows Per L2 Extension**: In this widget, you see the tromboning flows that traverse from the remote site to the local site and then back to the remote site via the network extension tunnels.
  This helps you to identify if you want to enable Mobility Optimized Networking on the participating L2 extensions so that the flows from VMs in the remote sites can directly traverse over the remote gateway.
- **Flows Per Appliance Pair**: In this widget, you see the flows over the network extension tunnels and related metrics grouped by the network extension appliance pairs.

In this entity page, you also see a list of properties, alerts, and metrics for the selected VMware HCX appliance.

### VMware HCX Tunnels

VMware HCX tunnel is an overlay tunnel between the VMware HCX appliances.

In this entity page, you see a list of properties, alerts, and metrics for the selected VMware HCX tunnel entity.

### VMware HCX L2 Extension

VMware HCX L2 extension stretches an L2 network from one site to another.

In the **Flows** tab of this VMware HCX L2 Extension entity page, VCF Operations for networks identifies the flows going over the HCX network extension tunnels and enriches them with the information of the VMware HCX L2 extension entity.

- **Top Talking Entities Over L2 Extensions**: In this widget, you can see the top talking entities like VM pairs, L2 networks, VMware HCX L2 extensions, etc over the network extension tunnels that belong to the current VMware HCX Manager.
- **Flows Per L2 Extensions**: In this widget, you see the number of flows and the related flow metrics for each L2 extension entity in the current VMware HCX Manager.
- **Tromboning Flows Per L2 Extension**: In this widget, you see the tromboning flows that traverse from the remote site to the local site and then back to the remote site via the network extension tunnels.
  This helps you to identify if you want to enable Mobility Optimized Networking on the participating L2 extensions so that the flows from VMs in the remote sites can directly traverse over the remote gateway.
- **Flows Per Appliance Pair**: In this widget, you see the flows over the network extension tunnels and related metrics grouped by the network extension appliance pairs.

In this entity page, you also see a list of properties and applications with flows over the VMware HCX L2 extension.

### VMware HCX Network Profile

In this entity page, you see a list of properties and alerts for the selected VMware HCX network profile entity

### VMware HCX Compute Profile

In this entity page, you see a list of properties and alerts for the selected VMware HCX compute profile.

You can set analytics threshold events on the HCX Tunnel and the HCX Appliance Metrics. For more information about analytics thresholds, see Working with Analytics.

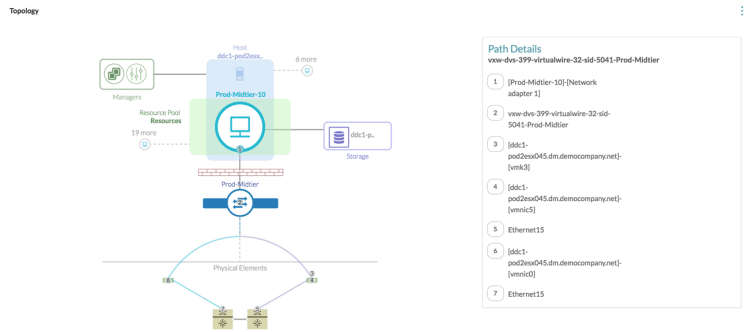
You can also use Guided Network Troubleshooting to troubleshoot VMware HCX entities from a single page. To troubleshoot VMware HCX entities, in the VMware HCX Manager details page or in any of the VMware HCX entity page, click the **Troubleshoot** icon located near the top-right corner.

## Viewing Entity Topology

The topology provides a comprehensive pictorial view of the entity.

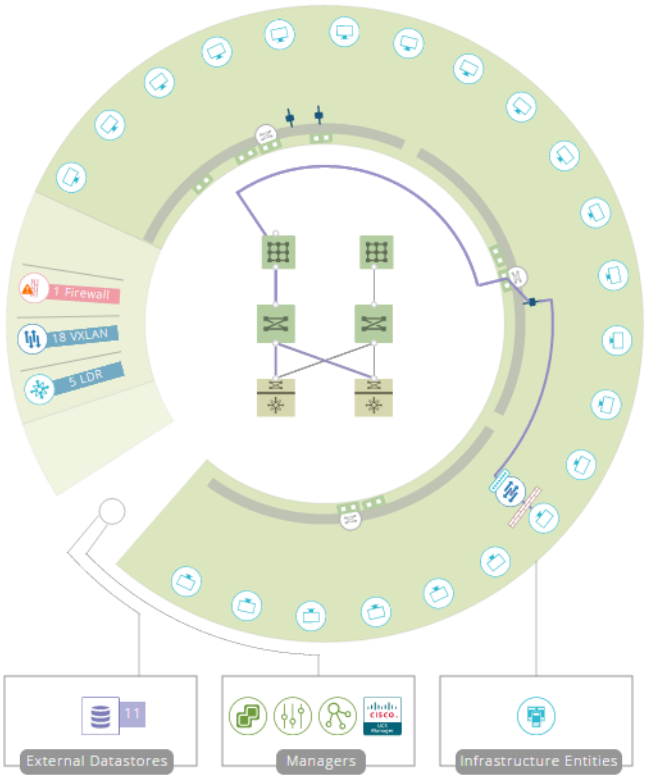
### Virtual Machine Topology

The virtual machine topology provides a comprehensive view of a singular virtual machine in relation to the rest of your data center.




### Host Topology

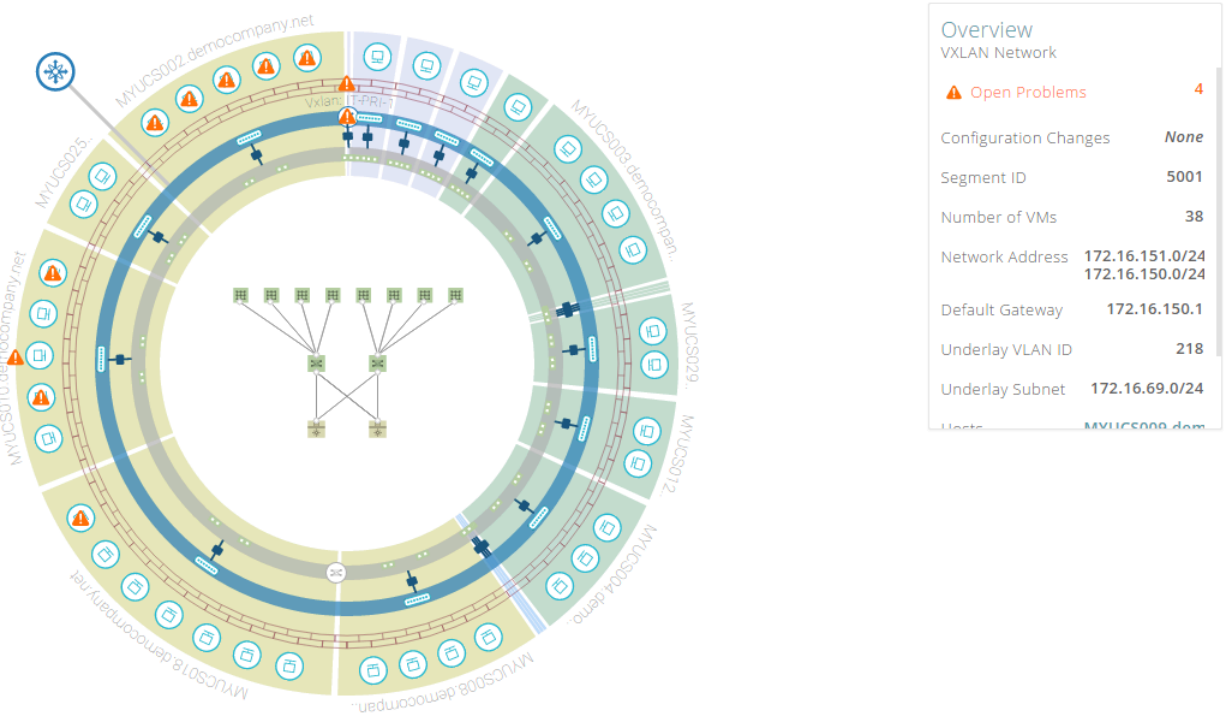
The host topology shows how VMs of a particular host are connected to the virtual and physical components of your data center and also how the host itself is connected with your data center.




## VXLAN Topology

Virtual eXtensible Local Area Network (VXLAN) overlay networking technology is an industry standard that is developed by VMware jointly with the major networking vendors.

The VXLAN topology is an innovative visualization that gives you an overview of the selected VXLAN. The following diagram elucidates the various components that make up the visualization:



**Note:**
Both virtual and physical components can be visualized in this manner.

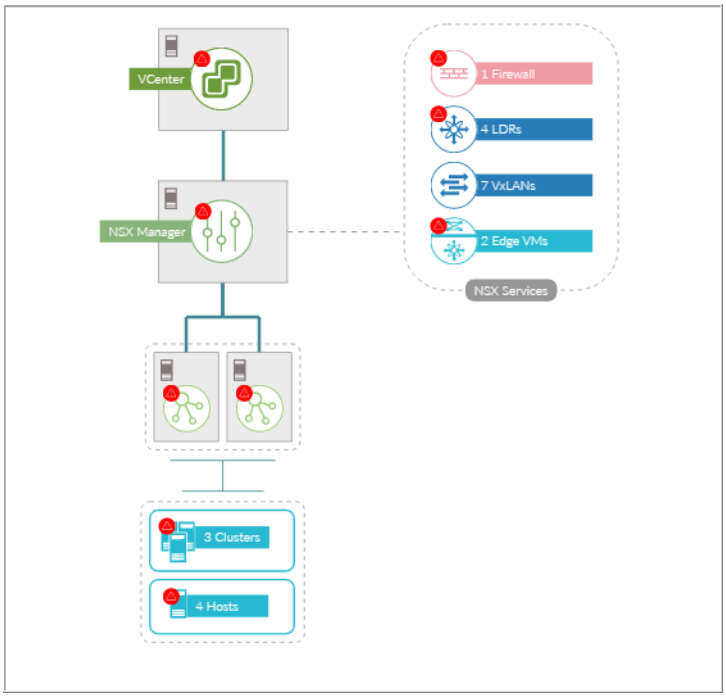
## VLAN Topology

Virtual LANs (VLANs) enable a single physical LAN segment to be further segmented so that groups of ports are isolated from one another as if they were on physically different segments.

The VLAN topology is constructed in a similar manner as the VXLAN topology.

## NSX Manager Topology

The NSX Manager topology shows the components that are associated with the NSX Manager.




**Viewing Audit Information of NSX objects in VCF Operations for networks**
VCF Operations for networks can capture an audit information of NSX objects quickly from the NSX-T Manager. The information includes the user name who created or modified the NSX object, when the operation happened and the operation details on the object.

If you have enabled audit logs in NSX-T Manager, VCF Operations for networks can collect the audit details for some of the NSX-T.

**NSX-T**

List of NSX-T objects for which VCF Operations for networks collects audit details.

**Note:**
The audit information is not available for the VMC Policy entities.

- NSGroup
- NSService
- NSServiceGroup
- NSFirewallRule

**Note:**
The audit information is not available for the Delete alert of the NSFirewallRule.

- IPSet
- NSX Policy Group
- NSX Policy Firewall Rule

**Note:**
If you are using NSX-T version 4.x and later, the following entities are not available and search wont show any result; you must update your user defined dashboards and user defined alerts using the corresponding policy entities:

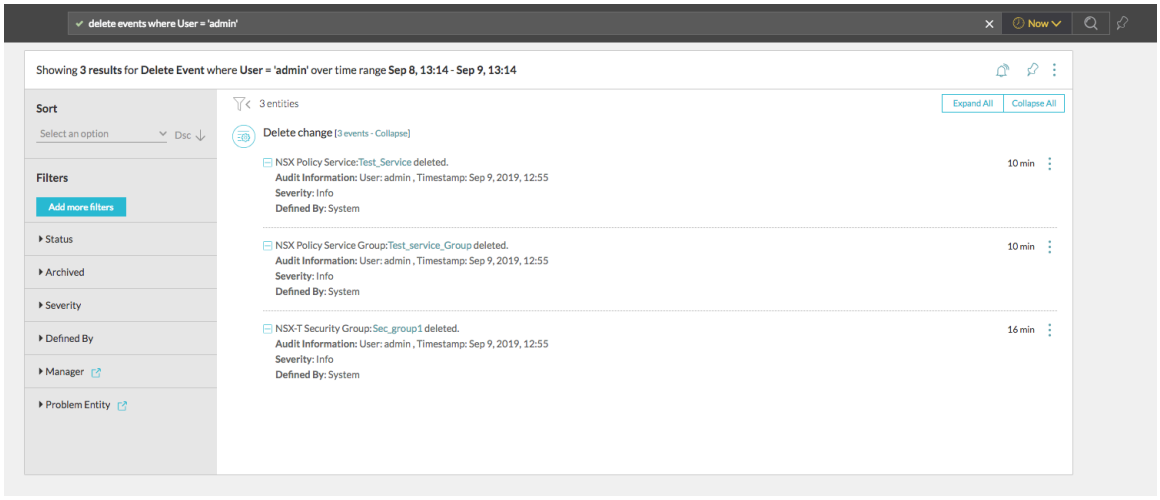
| Entities | Corresponding Policy Entities |
|---|---|
| NSGroup | NSX Policy Group |
| NSService | Policy Service entry |
| NSService Group | Policy service |
| NSX-T Firewall | NSX Policy Firewall |
| NSX-T Firewall Rule | NSX Policy Firewall Rule |
| NSX-T IPSet | IPSet |

The audit details of the NSX-T objects are captured for the Discovery, Property Change and the Delete alerts:

- `Discovery`

- `Properties Change`

- `Delete`



**Note:**
The Delete alerts are not available on the entity dashboard. However, you can search the alert to see the audit information.

**Sample queries to see audit information**

- `alerts where user = username`
- `discovery alerts where user = username`
- `delete alerts where user = username`
- `change alerts where user = username`

# Working with Pins and Dashboards

You can pin any widget on the VCF Operations for networks user interface to a new dashboard or an existing dashboard to access it quickly and easily.

## Pins

The information on each entity page is separated into pins. All the entity pages are made up of pins and each pin contains a specific bit of information related to the entity.

Pins have the following features:

- You can maximize the view of any pin using the **More options** button and also view more information about the pin using the **Help** option.
- Pins can also contain filters so that you can drill down on the data that is displayed on the pin.
- Many pins also contain the **Export as CSV** option so that you can export the data present in the pin in CSV format. You can select the specific properties and the number of CSV rows you want to export in the dialog box that is displayed.

  **Note:**
  The export as CSV feature for the flow data takes more than 30 minutes for 180,000 flows when all fields are selected.
- You can go back to the original page from the pinned page. To go to the original page, on the saved pin, click **More options** > **Go to original page**.

To add a pin, click the **Pin** icon, select the dashboard, and click **PIN**. You can find your saved pins on the **All Dashboards** page.

### Types of Pins

There are many types of pins in VCF Operations for networks. Most of the pins that are available in the software can be categorized into the following:

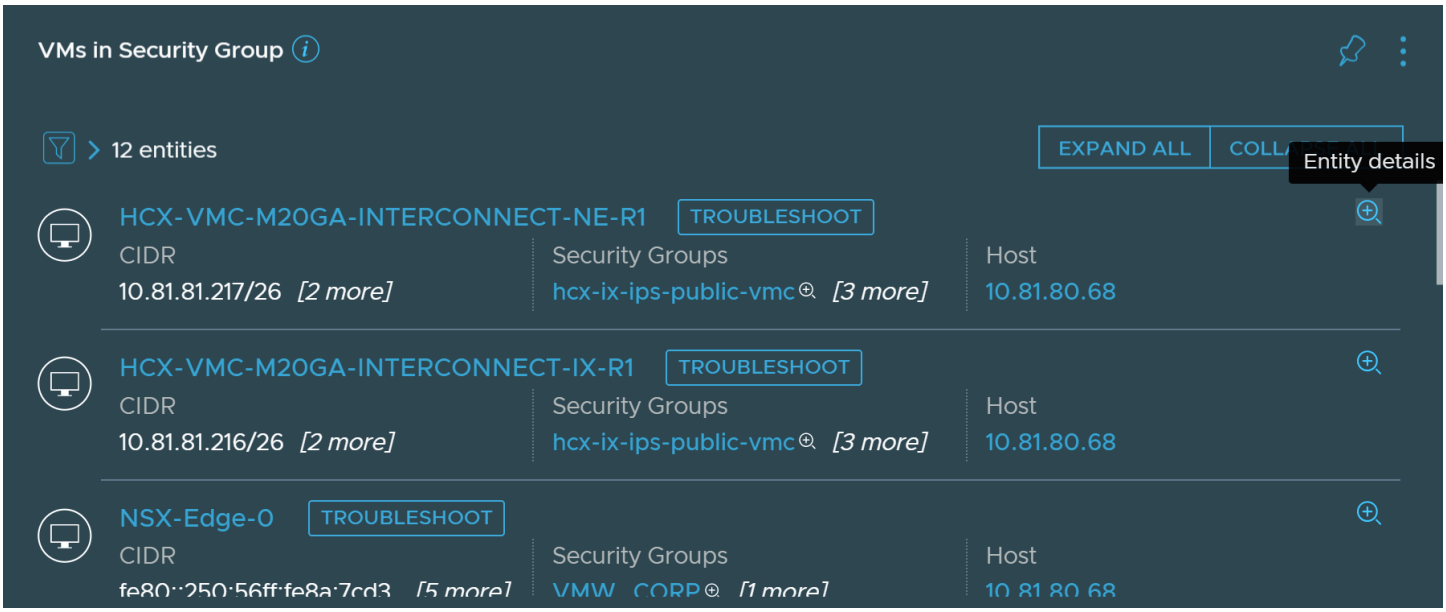
**Metrics Pins**
Metrics pins show important metrics pertaining to the selected entity.

A metrics pin uses the line graph to display data about the selected entity. You can point anywhere on the line graph to get informative tool tips. You can also co-relate multiple entities or two metrics graphs at the same time.

**Entity List View Pins**
The entity list view pins display a list of entities that are grouped by a common theme. The list shows important attributes per entity.

You can see more attributes of a particular entity by clicking the magnify icon on the far right. You can also click the entity name to navigate to the respective entity page.

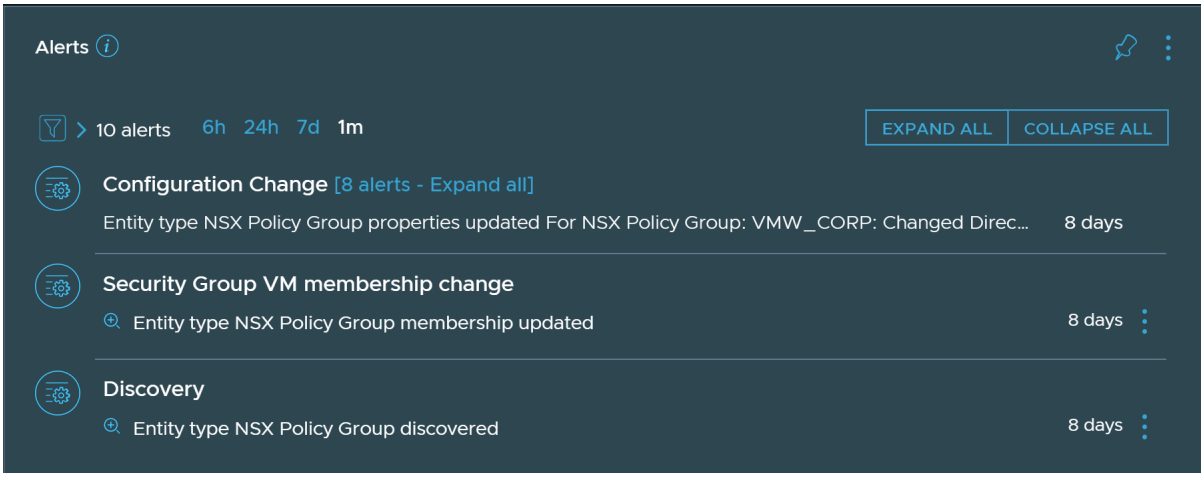
Like other pins, the filter icon houses various facets to filter the list. An example of the entry list view pin is VMs by security groups. By default, this pin shows the VMs that are part of the same security group. You can also filter VMs by properties such as VXLAN, OS hostname, vendor ID, and datastore.




**Alert View List Pins**
The alerts list view pins provide a list of alerts in the chronological order for a particular entity or group of entities.

You can change how far back in time (from now) must the pin show alerts by using the available presets or entering in a custom date/time. Other filter options such as **Alert Status** and **Alert Type** can be selected by clicking the filter icon.

Here, the alerts related to an NSX policy group are displayed. You can expand an alert and click the entity name to navigate to the entity page and view alerts related to that entity. By clicking the filter icon you can filter the alerts based on their status and type.




You can search for the alerts by using the alerts search query. You can search for open or closed alerts with queries such as open alerts or closed alerts. You can also search for problems with the same modifiers.

## Dashboards

You can create a dashboard or pin any widget from any page on a dashboard to make it easier to access and share data. Using dashboards, you can organize and present information from multiple sources in one location, and in real time.

**Note:**

Pinboards are deprecated from VCF Operations for networks version 6.9. They are replaced with dashboards.

When you upgrade to VCF Operations for networks version 6.9, your existing pinboards are automatically migrated and saved as dashboards.

Currently, VCF Operations for networks doesn't support dashboard backup and restore features. So, you cannot take backup of the dashboard data and restore the data in a new deployment.

**Viewing All Dashboards**

To view all dashboards in VCF Operations for networks, on the navigation pane, click **Dashboard** > **All Dashboards**.

You can edit, delete, share, or duplicate dashboards based on the access granted.

**Note:**
If you are an admin user, the **All Dashboards** page displays two tabs:

- The **My Dashboards** tab displays all the dashboards created by other users accessing the VCF Operations for networks instance.
- The **Admin View** tab displays dashboards created by the administrator.

**Note:**
It takes some time for the dashboard to appear in the **All Dashboards** page after it is created.

## Create a Dashboard

In VCF Operations for networks, you create a dashboard to quickly access information that is critical to your business operations.

To create a dashboard, do one of the following:

- a. Click the **Create New Dashboard** option on the homepage or on the **All Dashboards** page.
  b. Enter the name and description for the new dashboard.
  c. From the **Suggested Widgets** section, drag widgets and drop them to the dashboard. Or, click the **Add** button on the widget to add the widget to the dashboard.
  d. Click **Save**.
- a. Click the pin icon on the widget that you want to pin.
  b. Click **Create New Dashboard** on the **Pin to Dashboard** pop-up window.
  c. Enter the name and description for the new dashboard.
  d. Click **Create and Pin**.

**Note:**

- The dashboard name must be unique across VCF Operations for networks.
- The dashboard name must not have more than 100 characters.
- You can use only letters, numbers, and spaces while naming a dashboard.
- The maximum number of entries you can see in the **Recently Modified** list is 15.
- The maximum number of dashboards that you can create across all users is 500. The maximum number includes custom dashboards, shared dashboards, and default dashboards.
- The maximum number of pins allowed per dashboard is 20.

You can perform the following actions on the dashboard:

- Share the dashboard with any other VCF Operations for networks users.
- Edit the name and description of the dashboard.
- You can modify, and rearrange the pins on a dashboard. Their positions are persisted.
- Duplicate and delete the dashboard.
- Export the information on the dashboard as a PDF report.

## Updating a Dashboard

You can add, delete, and organize widgets on the update dashboard page.

You can click the **Edit** button on the respective dashboard page to update the dashboard.

On the update dashboard page, you can:

- Update the dashboard name and description.
- Rearrange widgets in the dashboard by clicking the drag icon on the widget.
- Add new widgets to the dashboard.
- Delete existing widgets from the dashboard.

After updating the dashboard page, you can either click **Save** to save the latest changes, or click **Reset all** to reset the dashboard to the last saved state.

### Sharing a Dashboard

You can share the dashboards that you create with other users or groups (LADP/AD/vIDM). An admin user can view and delete any dashboard.

The following are the features of sharing and collaboration of dashboards:

If you have created a dashboard, you can view, edit, or delete it irrespective of you being an admin or a member user.

| Dashboard Owner | Shared With | Privilege | Possible Actions |
|---|---|---|---|
| Admin | Admin | View and Edit | View, Edit, Delete |
| | Admin | View only | View, Delete |
| | Member | View and Edit | View, Edit |
| | Member | View only | View |
| Member | Admin | View and Edit | View, Edit, Delete |
| | Admin | View only | View, Delete |
| | Member | View and Edit | View, Edit |
| | Member | View only | View |

**Note:**
The admin user can delete any dashboard.

To share a dashboard:

1. Click **More Options** on the dashboard that you want to share.
2. Click **Share**.

   **Note:**
   You can also share a dashboard from the **All Dashboards** page.
3. By default, the link sharing is enabled. You can share the link of a dashboard with any users or groups (LDAP/AD/vIDM).
4. You can add the users with whom you want to share the dashboard. You can specify the privileges such as `view` and `view and edit` to a particular user.

   **Note:**

   Users that only have view priviliges cannot share the dashboard with other users.
5. Click **Save** to save the share and collaboration changes that you have made.

You can view the sharing and collaboration information in the **Shared** column of the particular dashboard on the **All Dashboards** page.

## Manage Dashboards on the Home Page

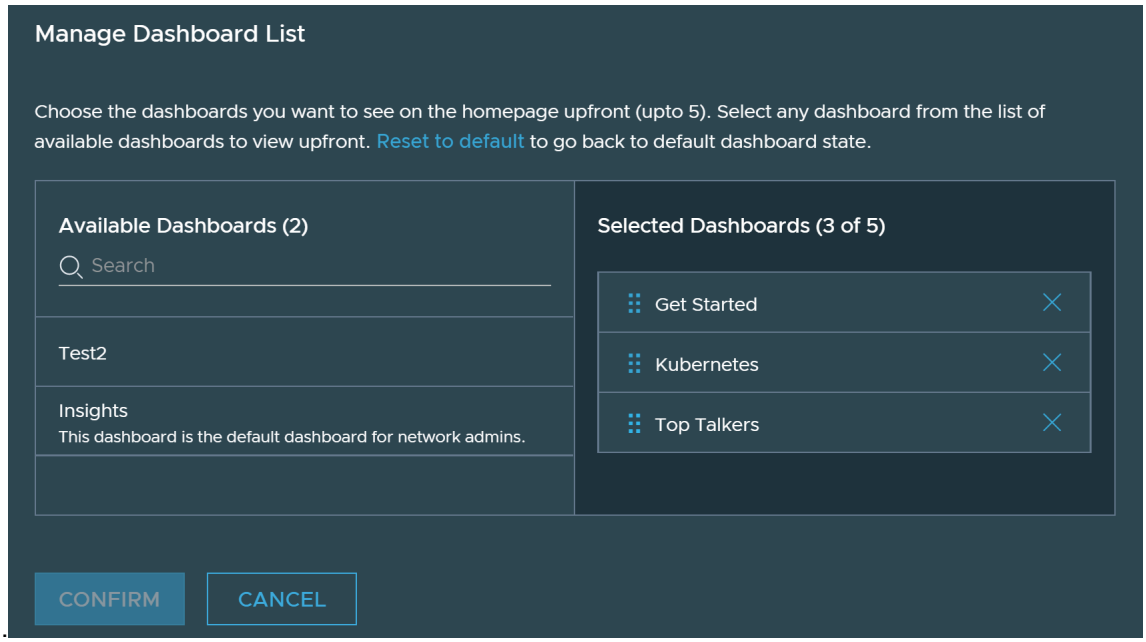
You can choose the dashboards you want to view on the homepage:

You must have created at least one dashboard in VCF Operations for networks.

1. Navigate to the homepage.
2. Click the **More** drop-down menu and select **Manage Dashboard List**.

   **Note:**
   The **More** drop-down menu is displayed only if you have created one or more dashboards in VCF Operations for networks.

   The **Manage Dashboard List** window is displayed.




3. On **Available Dashboards**, hover over a dashboard and click **>** to move the dashboard to **Selected Dashboards**.

   **Note:**
   You can display upto five dashboards on the homepage.

4. Optional: To rearrange the order of dashboards on the **Selected Dashboard** list, use the drag icon before the dashboard name.
5. Click **Confirm** to save changes.

   If you want to revert the changes and restore dashboards to their default state, click **Reset to default**.

## To Duplicate a Dashboard

1. From the left navigation pane, go to **Dashboards** > **All Dashboards**.

   A list of existing dashboards is displayed.
2. To duplicate a dashboard, click the **More Options** icon and select the **Delete** button.
3. On the pop-up window, enter the name and description of the new dashboard and click **Duplicate**.

   **Note:**
   The name of the dashboard is mandatory. The **Duplicate** button is deactivated till you enter the name.

4. If you are trying to duplicate a dashboard that is shared, then you can opt to retain the source dashboard users and their permissions. Select **Keep source dashboard users and permissions** if you want to retain them.

   **Note:**
   If the dashboard that you want to duplicate is shared with you with read-only access, you do not see the **Keep source dashboard users and permissions** option.

   The user who duplicates a dashboard becomes the owner of the new dashboard.

### Delete a Dashboard

1. From the left navigation pane, go to **Dashboards** > **All Dashboards**.
   A list of existing dashboards is displayed.
2. To delete a dashboard, click the **More Options** icon and select the **Delete** button.
3. On the **Confirm Action** pop-up window, click **Delete**.

   **Note:**
   You can delete a dashboard only if:

   - You created the dashboard.
   - You have edit access to the shared dashboard.

   You cannot delete a dashboard that is shared with you with read-only access.

## Working with Local Tags

VCF Operations for networks enables you to add custom local tags for supported data sources.

Local tags are always a key-value pair where you can add a unique key and a value. The local tags help you organize and find your data source faster and better. The local tags has the following characteristics:

- Tags are always key-value pairs.
- Keys must be unique with only one value.
- Tags are extended properties of entities.
- Tags are meant to be consumed with search.
- Tag key can be of maximum 63 characters and value can be of maximum 255 characters.
- Maximum 25 tags can be added per data source.
- Tags can be used only for plain text search.

You can add the local tags only in VCF Operations for networks. When you select **Apply tag operations to all associated entities** check box, the tags get applied to all associated entities related to that data source.

The following are the supported data sources and the associated entities for local tags:

| Data Source | Associated Entities |
|---|---|
| VMware vCenter | Host |
| VMware NSX-T | • NSXT Transport Node<br>• NSXT MP Node<br>• NVDS<br>• NSXT Router Device<br>• Virtual Host Switch<br>• Virtual Host Router |
| Cisco UCS Manager | • Blade servers<br>• Fex<br>• Fabric interconnect switches |
| ACI | Leaf and spine devices |
| Palo Alto Networks | • Pan Physical Device<br>• Pan VM Device |

| Data Source | Associated Entities |
|---|---|
| Checkpoint management server | Check point gateways |
| Cisco Nexus | N2K device |
| • Arista 7050QX-32S, 7280TR<br>• Cisco ASR 1K<br>• Cisco ISR 4K<br>• Cisco ASR 9K<br>• Cisco Catalyst 3000, 4500, 6500, 9300<br>• Dell OS10<br>• Force 10<br>• Juniper<br>• Mellanox<br>• All routers and switches with a single manager | L3 Switch |
| F5 BIG-IP | F5 Big IP |
| Cisco ASA | L3 Switch |

# Load Balancer Support in VCF Operations for networks

Load balancing allows you to distribute inbound application traffic across multiple back-end destinations, including deployments in public or private clouds. It is therefore a necessity to have the concept of a collection of back-end destinations.

In addition to the load balancing support, VCF Operations for networks also provides the following advantages:

- Enables to identify whether the application servers are physical servers or the virtual machines.
- Allows you to debug or troubleshoot the problems easily by providing visibility into the application server (host or VM) information such as configuration, performance, flows.
- Provides visibility into physical or virtual networking components in an application where the load is distributed.
- Raises alerts for any issue in the environment and also helps to detect the reason for the issue. For example, application is not responding because the service node VM is down.
- Provides end to end flow visibility.

  **Note:**
  VCF Operations for networks does not support end to end flow visibility in Avi Load Balancer. However, it supports the VM-VM path.

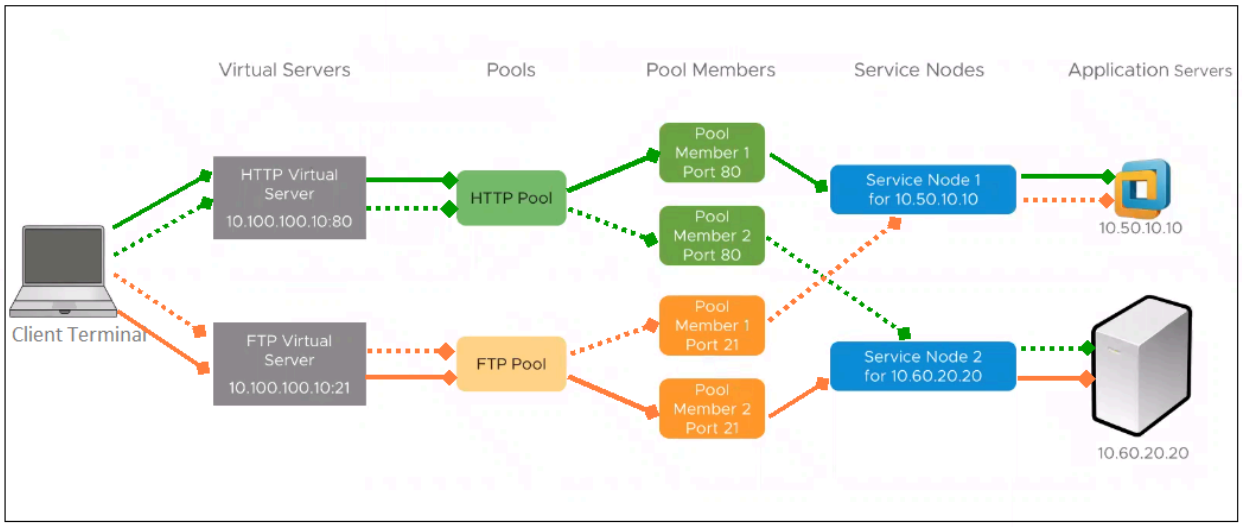
VCF Operations for networks supports the following load balancers:

- F5
- VMware Avi Load Balancer

## F5 as a Load Balancer

VCF Operations for networks supports the F5 load balancing feature by discovering and representing the required F5 components, such as service nodes, pool members, pools, and virtual servers.

### Overview of a F5 Load Balancer and its Components




- Application Servers - The machines where the applications are hosted. For example, if you have a web server, your server runs on application servers (physical or a virtual server).
- Service Nodes - F5 represents the application servers as service nodes. Service node have the same IP address or FQDN of the application server. Each service node can have multiple applications.
- Pool Members - A logical entity. Each application in a service node is represented by a pool member, which has the same IP address or FQDN of the service node. To identify different applications, the pool members embed the port number with the IP address of the Service Nodes.
- Pools - All pool members that serve one application are grouped as a pool.
- Virtual Servers - A public facing IP address of the application. Clients that want to use an application connect to the virtual server IP address (For example, 10.100.100.10) and port number (80 or 21).
- Client Terminal - The connection starts from a client terminal, such as a desktop, laptop, mobile device, or virtual machine.

The client request connects to the virtual server, which decides the pool members based on the pool. Pool members, then, forward the request to the application server (VM or physical server).

> **Note:**
> A single application server can serve multiple requests from different ports and different service nodes.

## Overview of VMware Avi Load Balancer

Avi Load Balancer performs load balancing and delivers application services such as application analytics, predictive autoscaling, and micro-segmentation.

### Platform Overview

The Avi Load Balancer platform includes the following components:

- Controller cluster. A single point of management and control of the Avi Load Balancer system. It is usually deployed as a redundant three-node cluster and has full visibility across the environments. It automates the deployment and management of the service engines.
- Service engine. Receives and executes instructions from the Controller. The Service engines perform load balancing and manages client and server-facing network interactions.
- Virtual service. Advertises an IP address and ports to the external world and listens for client traffic.

### Supported Load Balancers

Avi Load Balancer supports the following load balancers:

- Native Load Balancer, also referred to as L2 Load balancer

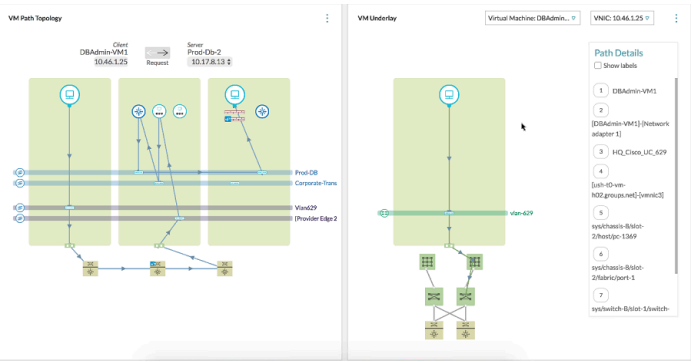
- Equal-Cost Multi-Path (ECMP) Load Balancer

In Avi Load Balancer, virtual services can be scaled across one or more service engines using either native (L2 punting) or BGP-based load balancing techniques. Virtual services in a Native Load Balancer can be scaled upto four service engines. Virtual services in an ECMP Load Balancer can be scaled upto sixty four service engines.

# Network Visibility

## Path Topology

The path topology draws a detailed connection that exists between any two virtual machines in your environment.




The topology involves both Layer 3 and Layer 2 components. You can view this topology using the search query `vm_name_1` to `vm_name_2`. If a path exists, the VM-to-VM path visualization proceeds to populate all the components that exist between `vm_name_1` to `vm_name_2` and also draws an animated path. If the routers are physical, then they are shown outside the boundary. To see physical routers, you must enable Link Layer Discovery Protocol (LLDP) and, if it is supported, Cisco Discovery Protocol (CDP) on the network devices.

In the path topology, you see the VM-to-VM path between the source and the destination. If the default path is not configured between the VMs, an error message appears to inform that the path is not defined or the router interface is not found.

For Kubernetes, the Path topology displays the path for the following scenarios:

- Kubernetes Service to Kubernetes Service
- Kubernetes Service to Kubernetes Pod
- Kubernetes Pod to Kubernete Pod

> **Note:**
> The path involving physical devices is not supported.

The **Path Via Load Balancer** option lists all the load balancers that are used in between the path from the selected source and the destination VM. To see the path between the VMs via a particular load balancer, select the load balancer name from the list. If you hover the mouse on load balancer component on the path topology, you see the following details:

- Virtual Server name
- Load Balancer IP address
- Port number
- Load Balancer Algorithm
- The default gateway that was taken from the load balancer.

You can also see the routing components on the path topology.

If you hover your mouse on any of the routers, edges, or LDRs that are involved in the path, the complete routing or NAT information is shown.

The VM Underlay section that is on the right side of the VM Path topology shows the underlay information of the VMs involved and their connectivity to the top of the rack switches and the ports involved. For Kubernetes entities, VM Underlay displays the VM or the Kubernetes node information on which the Pod resides.

In the VM underlay section, the components are labeled if you select **Show labels** under **Path Details.** In this section, the drop-down list at the top shows the endpoint VMs and the active VMs at the edges. For each edge VM, the neighboring drop-down menu shows the ingress and the egress interface IP addresses. Based on the selection, the underlay path for that particular interface is shown.

You can also reverse the path direction using the arrows on top of the topology map.

The topology map gives more visibility regarding the ports involved in the VM-VM path. In the **Path Details** section, the name of the actual port channel is shown.

For NSX-T version 4.x and later, VCF Operations for networks does no support VM-VM path for federation deployments.

**Note:**
There is no complete visibility for layer 2 on the physical front. If a packet is traversing from one switch to another, there maybe multiple switches involved. But the topology does not show the switches in the underlay network.

## AWS VM-VM Path

The VM-VM path for AWS provides the path visibility between the on-premises VMs and the AWS EC2 instances.

Currently, VCF Operations for networks supports the following scenarios:

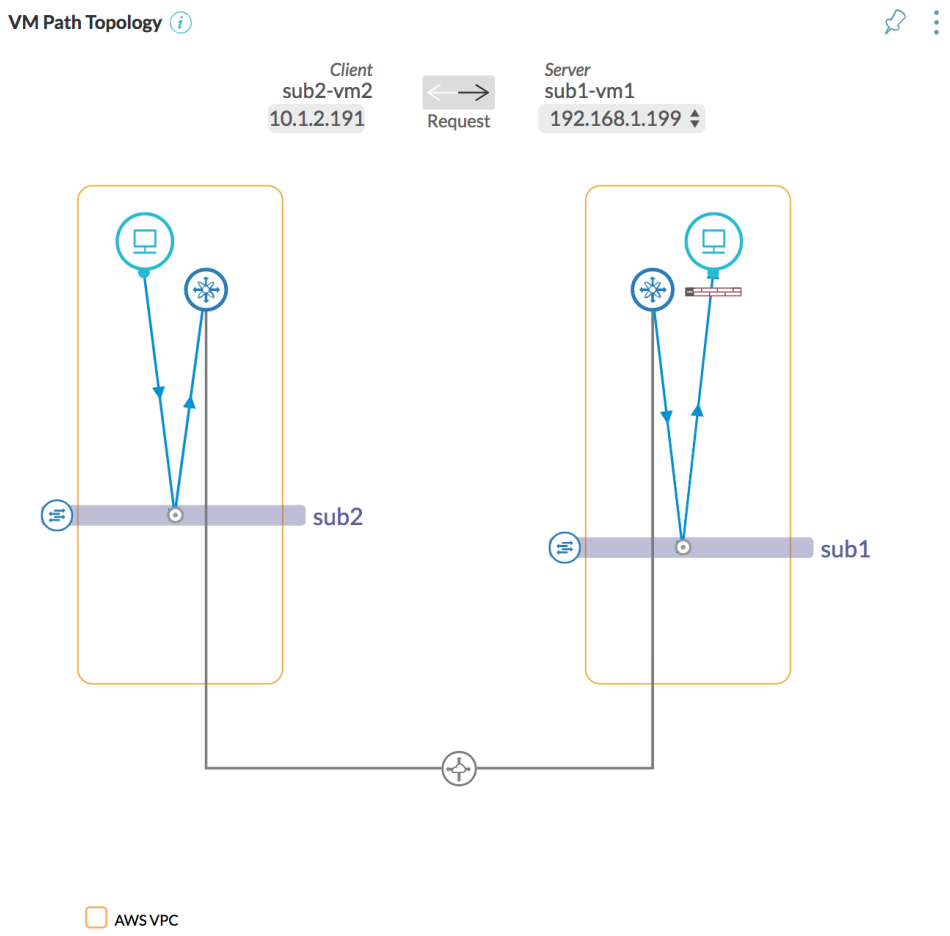
- AWS intra-VPC VM-VM path: This scenario involves the communication between the VMs of the same subnet or different subnets in a particular VPC.
- AWS inter-VPC VM-VM path through the peering connection: This scenario involves the communication between the VM of one VPC to the VM of another VPC through a peering connection.
- AWS VM to Internet: The VM in a VPC communicates to Internet through the Internet Gateway.
- AWS VM to the data center VM through AWS VPN connection: In this scenario, the VM in a VPC communicates to the VM in a data center through the AWS VPN connection. VCF Operations for networks supports SDDC and NSX-T data centers for this scenario.

  **Note:**

  - The hybrid path topology to NSX-T data centers works only when the NSX-T edge routers are configured with a public IP address.
  - VCF Operations for networks does not support the VM underlay topology for AWS.

  **Note:**

An example of the AWS VM-VM path for the AWS inter-VPC VM-VM path through the peering connection is as follows:



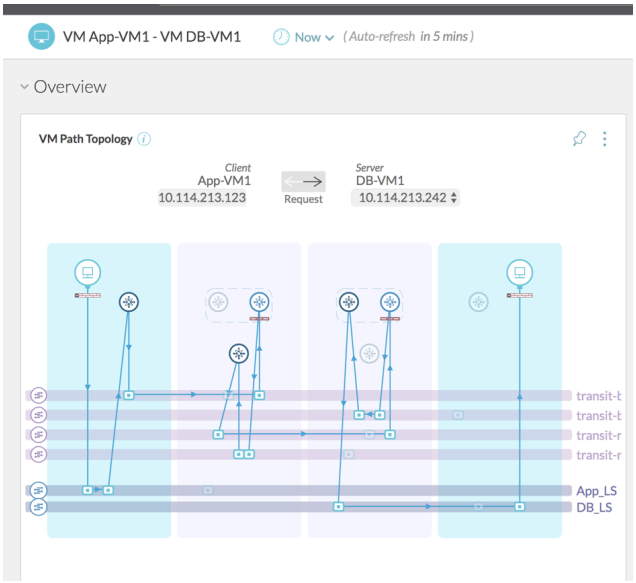

You can view the properties of the peering connection by pointing to its icon in the VM-VM path.

You can search the following entities concerning the AWS VM-VM path:

- AWS Subnet
- AWS Route Table
- AWS Virtual Private Gateway
- AWS Internet Gateway
- AWS VPN Connection
- AWS VPC Peering Connection

## NSX-T VM-VM Path

An example for VM-VM path for NSX-T is as follows:




The blue color represents the host node and the grey color represents the edge node. The icons used in the VM path topology are listed on the right side of the screen along with the labels under Path Details. The distributed routers are shown in the same color irrespective of their tiers. The color of the service router in the topology diagram changes as per the associated tier. All the Tier 1 components are shown at the same level and all the Tier 0 components are shown at a different level. In NSX-T, the edge firewalls are depicted in the diagram.

## VCF Operation HCX VM-VM Path

In VCF Operations for networks, you can view the VM-VM path over a network extended by VCF Operation HCX Network Extension (HCX-NE) service.

VCF Operations for networks supports the following VM-VM path scenarios for VCF Operation HCX Network Extension:

Here are a few examples of VCF Operation HCX VM-VM paths:

### Single Site Network Extension (NE) Scenarios

- The source VM is on a local site and the destination VM is on a remote site. The VCF Operation HCX L2E extends the L2 network of the source VM to the remote site, and the destination VM is on that same extended L2 network on the remote site. So, logically, both the VMs are on the same L2 network.
- The source VM is on a different L2 network on a local site and the VCF Operation HCX L2E extends to a different L2 network from a local site to a remote site. And the destination VM is on this extended L2 network on a remote site. So, logically, the source VM and destination VM are on different L2 network.
- The source and the destination VMs are on a remote site, but both the VMs are on different extended L2 networks extended from the local site to the remote site.
- Both the source VM and the destination VM are on a remote site, and the source VM is on an extended L2 network but the destination VM is on a non-extended L2 network native to the remote site. Also, there is a different L3 connectivity between these sites.
- Path to internet via NE: The VCF Operation HCX L2E extends a L2 network from a local site to a remote and the source VM is on this extended L2 network on the remote site. The source VM is communicating to the internet.

**Figure 443: The source and the destination VM on different sites.**

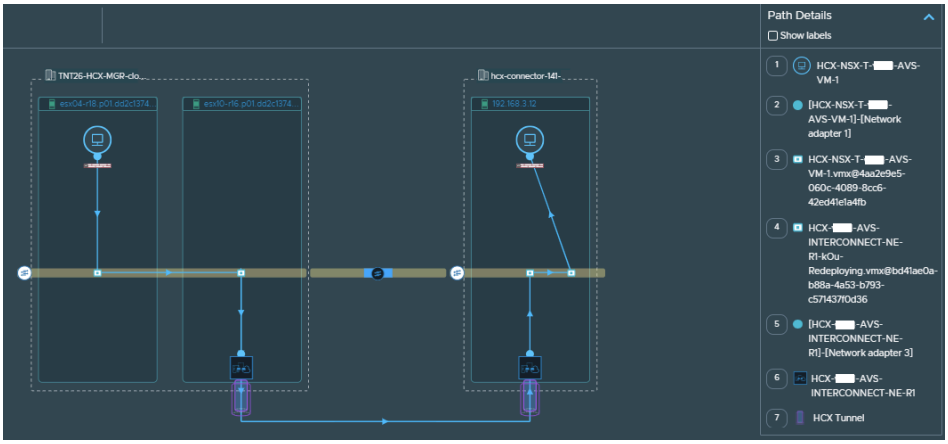



**Figure 444: The source and the destination VM on different remote sites**

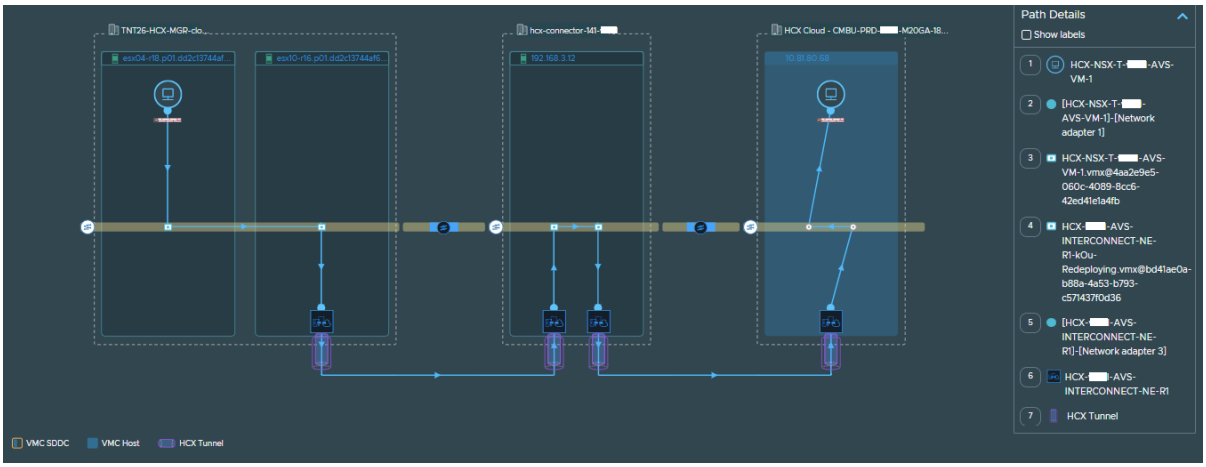
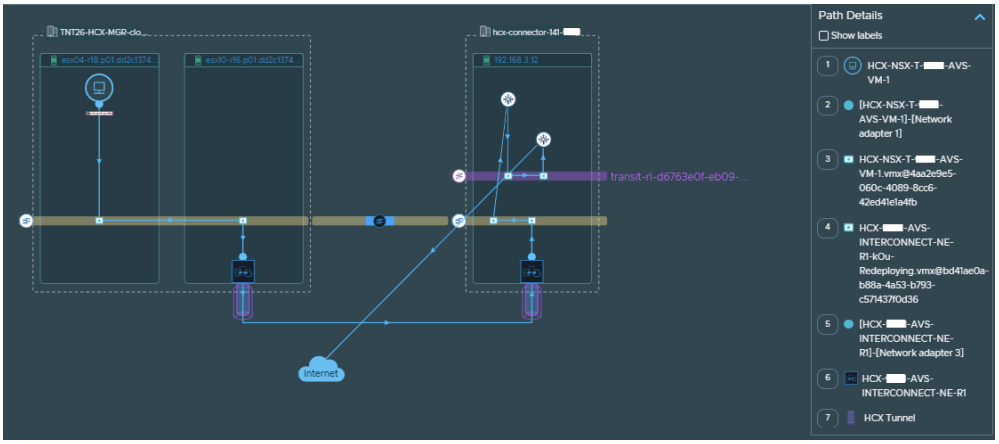



**Figure 445: VMs communicating to the Internet over the VCF Operation HCX Network Extension**




## Multi-site Network Extension (NE) Scenarios

V (One to Many) Network Extension: In this topology, the local site is paired to two remote sites, and a L2 network is extended to the two connected remote sites.

- In this scenario, the source VMs and the destination VMs are on the different remote sites on this same extended L2 network. So,logically, the VMs are on same L2 network.
- In this scenario, different L2 networks are extended to each of the connected remote sites. The source VM is on one remote site on the extended L2 network and destination VM is on another remote site on the extended L2 network. Logically, the VMs are on different L2 networks.

Daisy Chain Network Extension: In this topology, a local site is paired to a remote site and this remote site is paired to another remote site.

- A L2 network is extended to the first remote site and the same L2 network is extended to the next remote site. Source VM is on local site and Destination VM is on the second remote site. Logically, the VMs are on same L2 network.

### Unsupported Paths

Currently, we don not support the following VM-VM paths for VCF Operation HCX:

- Underlay paths
- Path with an endpoint as VCF Operation HCX Network Extension (HCX-NE) appliance.

## Support for NAT in VCF Operations for networks

VCF Operations for networks supports the VM-VM Path for NSX for vSphere, NSX-T Edges, Fortinet, and Check Point.

### VM-VM Path

A sample VM-VM path through NAT is as follows:

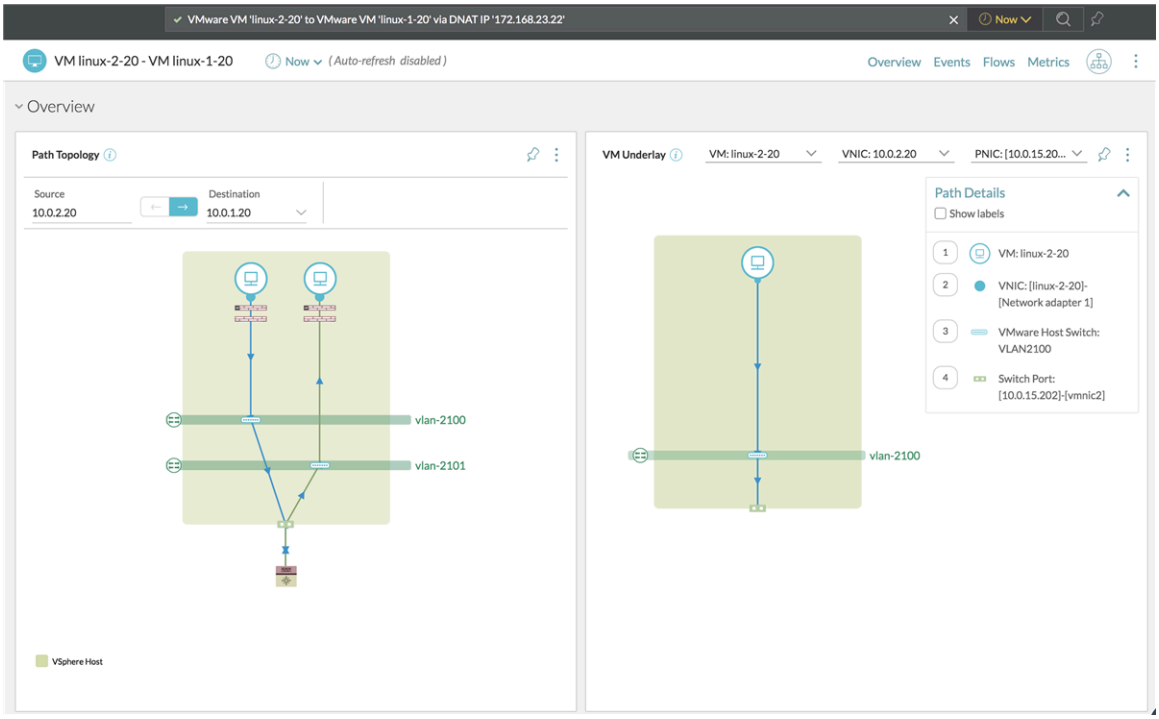
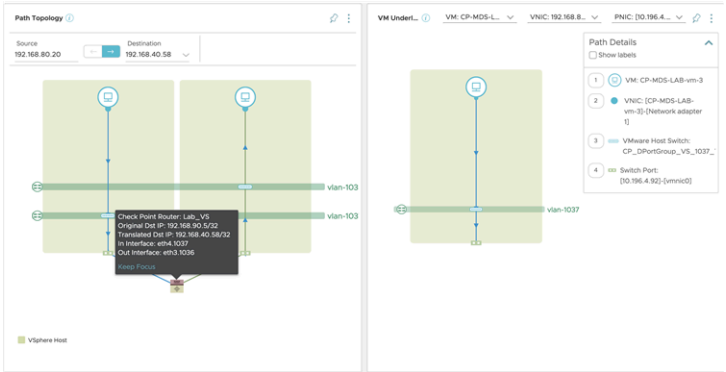



**Figure 446: VM-VM path through Check Point NAT**



**Queries**

To view VM-VM path through NAT, use the following queries:

- If destination VM is behind a Fortinet and Check Point router, and configured with NAT, then use `VMware VM '<name of the VM>' to VMware VM '<name of the VM>' via DNAT` query.
- If destination VM is behind an NSX for vSphere or NSX-T Edge and configured with NAT, then use `VMware VM '<name of the VM>' to VMware VM '<name of the VM>'` query.

**Considerations**

- For the VM-VM path with the NSX-T logical routers where the NAT service is enabled, VCF Operations for networks doesn't show the NSX-T edge firewall rules correctly for such a path.

## Arista Hardware VTEP VM-VM Path

In VCF Operations for networks, you can view a hardware VTEP in the VM-VM path.

Currently, VCF Operations for networks supports the following scenarios:

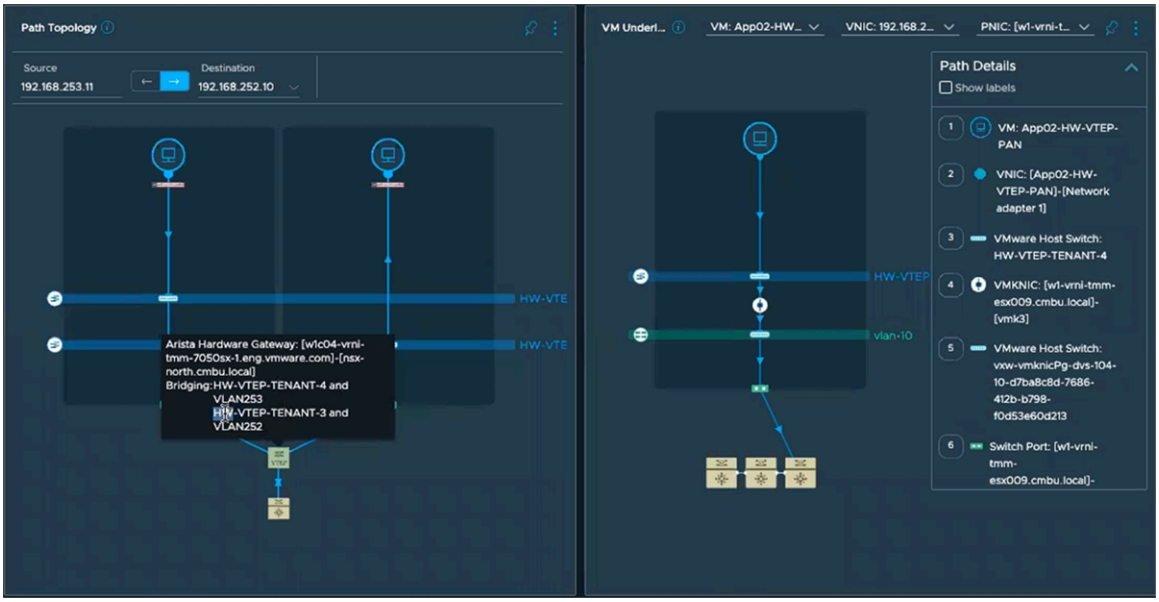
- VM-VM path through hardware VTEP when the source and the destination VMs are located in a different VXLAN and a different host.
- VM-VM path through hardware VTEP when the source and the destination VMs are located in the same host but different VXLAN.
- Hardware VTEP in the VM underlay topology when the switch is directly connected to the host.

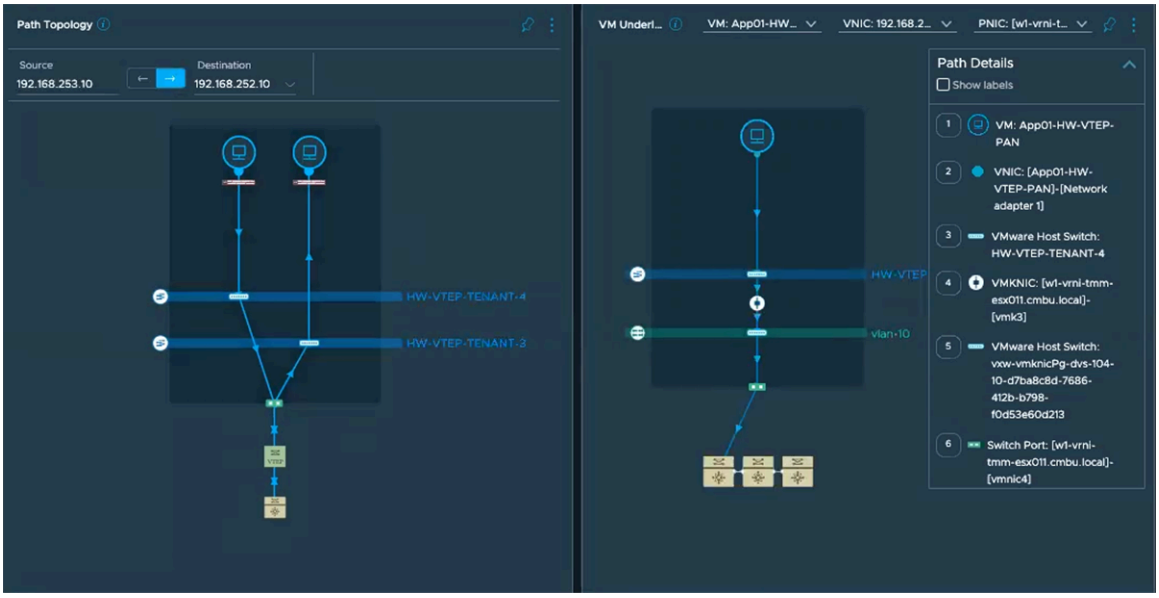
> **Note:**
> When you add the Arista Switch SSH in VCF Operations for networks as a data source, you must use the same IP/FQDN that you have used in the VMware NSX Manager to configure Arista Switch SSH. Otherwise, you do not see the hardware VTEP in the VM-VM path.

You can also view the hardware VTEP in VM topology and in the VM to Internet path if a hardware VTEP is available between the VM and Internet.

VM-VM path through the hardware VTEP when the source and destination VMs are located in the same VXLAN is unsupported.

Here are a few examples of VM-VM path through hardware VTEP:

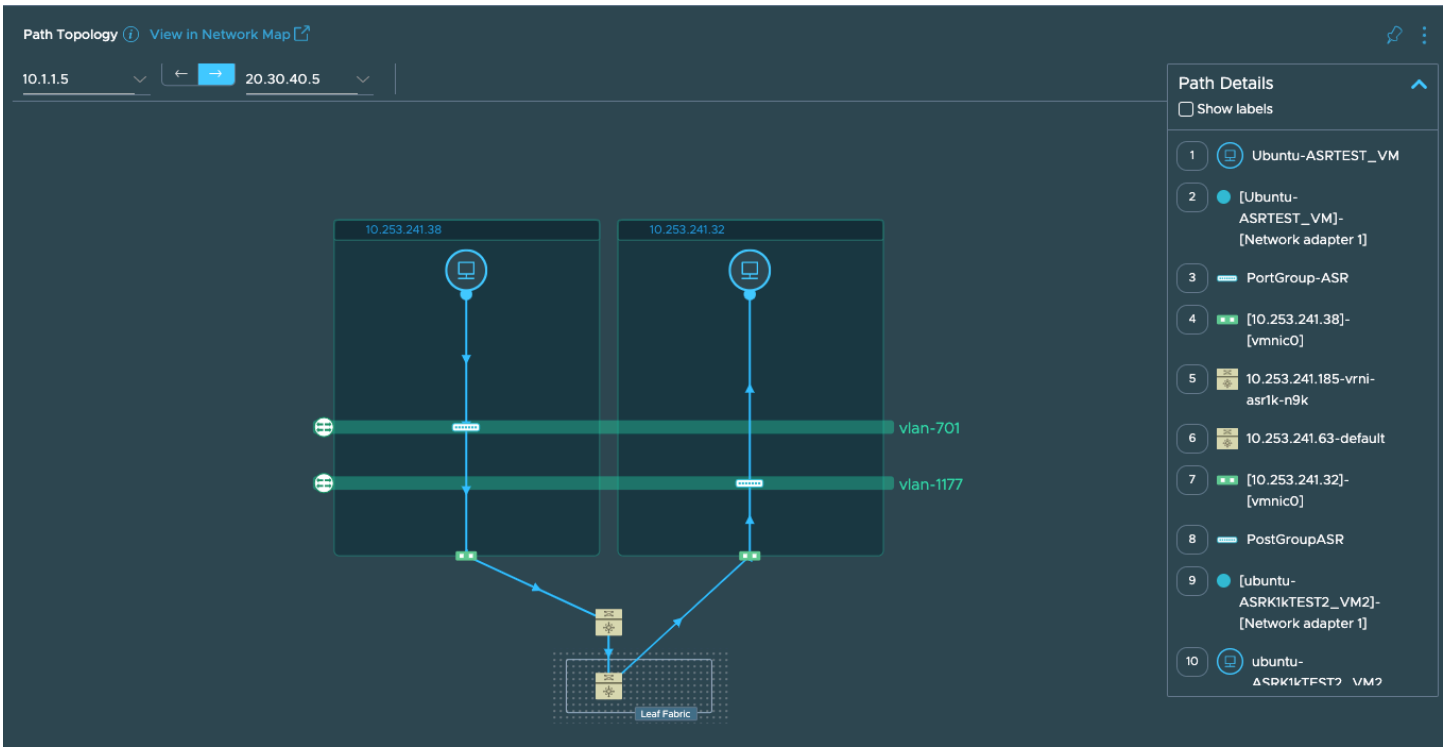





## Cisco ASR 1000 VM-VM Path

In VCF Operations for networks, you can view the VM-VM path through a Cisco ASR 1000.

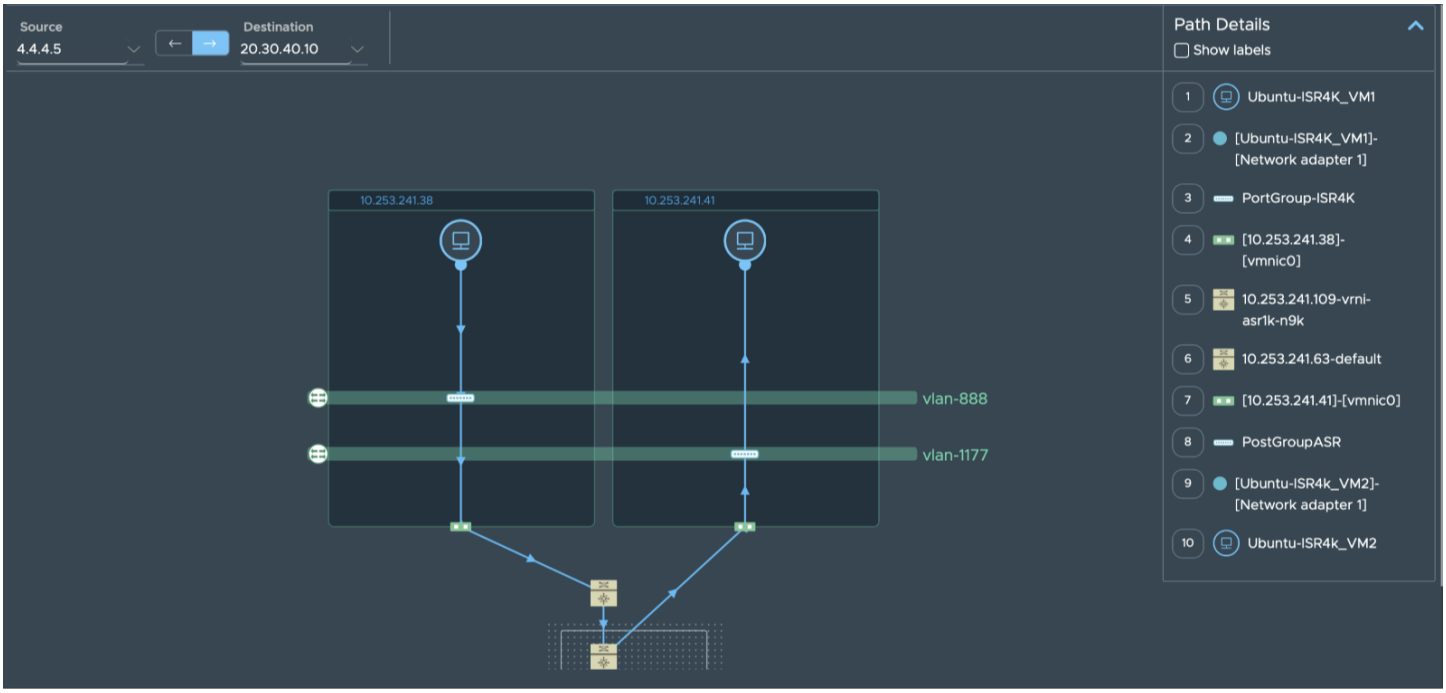
A sample VM-VM path for the Cisco ASR 1000 is as follows:




## Cisco ISR 4000 VM-VM Path

In VCF Operations for networks, you can view the VM-VM path through Cisco ISR 4000.

A sample VM-VM path for the Cisco ISR 4000 is as follows:



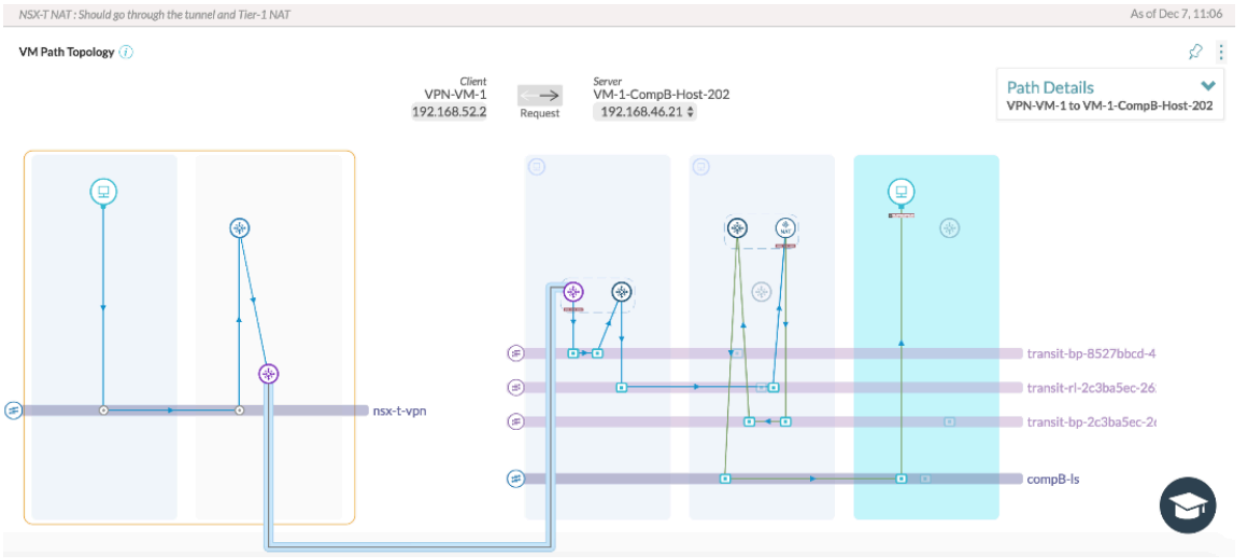

## VMware Cloud (VMC) VM-VM Path

VCF Operations for networks supports the following hybrid paths in VMware Cloud on AWS:

- VMware Cloud on AWS and VMware Cloud on AWS
- VMware Cloud on AWS and NSX-T
- VMware Cloud on AWS and AWS
- Intra VMware Cloud on AWS

For all VMs present in VMware Cloud on AWS, the underlay information is shown only until the segment on which the VM lies because the underlying physical elements of the network are abstracted out by VMware Cloud on AWS and no visibility is present at that level.

A sample VMware Cloud on AWS and NSX-T VM-VM path is as follows:



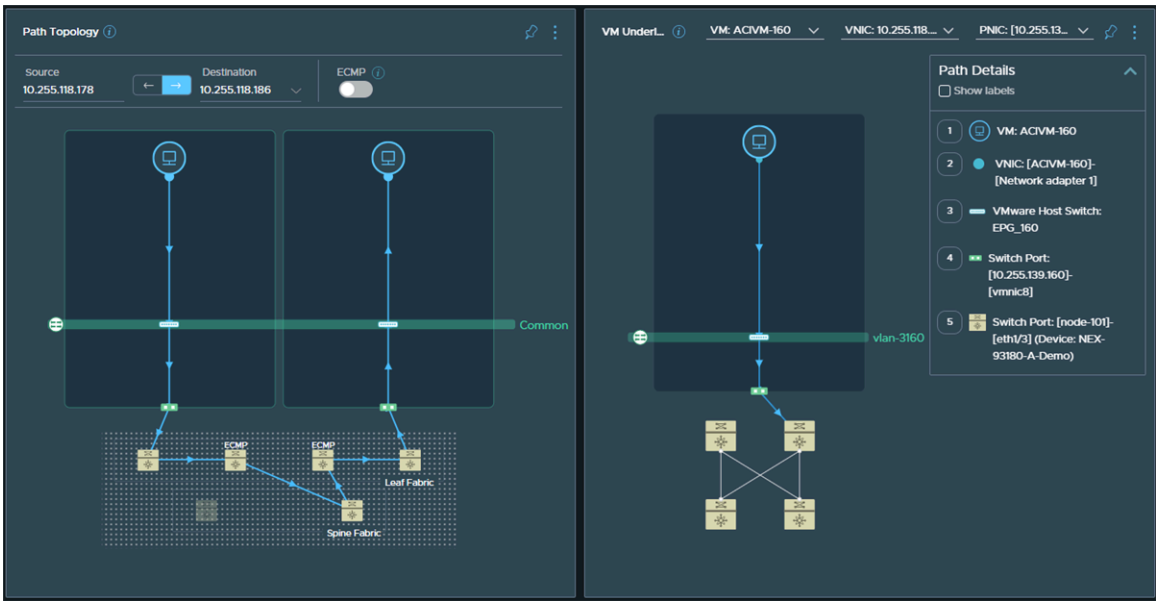

The dark blue line depicts the tunnel.

## Cisco ACI VM-VM Path

In VCF Operations for networks, you can view the VM-VM path through Cisco ACI.

A sample VM-VM path for the Cisco ACI is as follows:

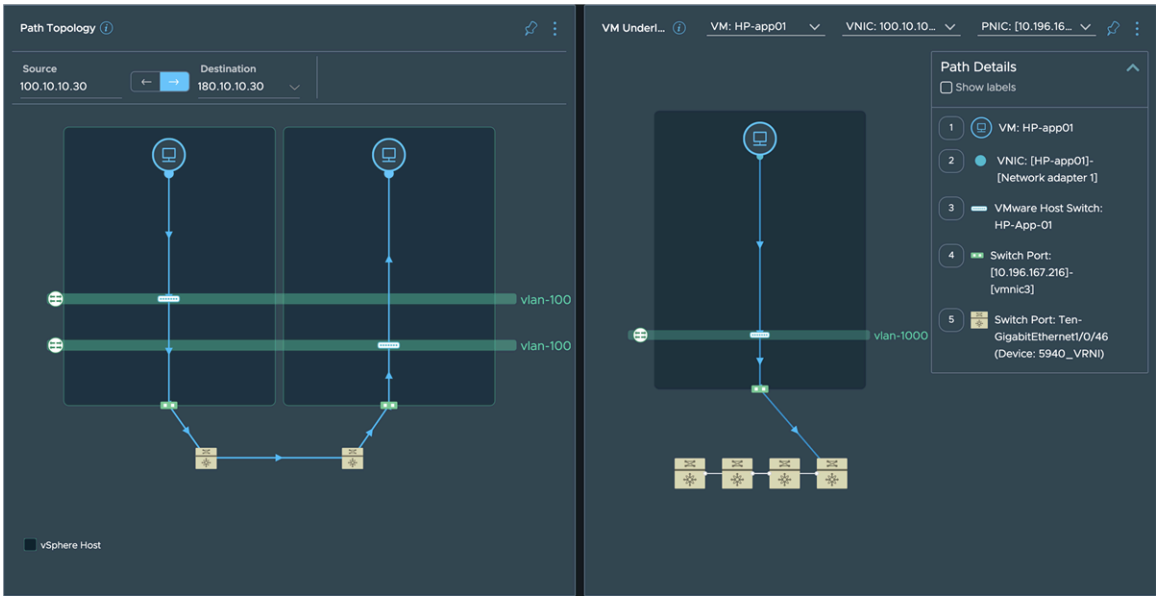
> **Note:**
> VCF Operations for networks shows the VM-VM path through the leaf and spine switches if the Cisco ACI APIs provide the switch level details. If not, VCF Operations for networks shows a single Cisco ACI VRF for the entire fabric instead of the leaf and spine switches in the VM to VM path.




## HPE VM-VM Path

In VCF Operations for networks, you can view the VM-VM path through supported HPE switches.

A sample VM-VM path through an HPE switch is as follows:



## Avi Load Balancer VM-VM Path

The VM-VM path provides the path visibility between the Avi Load Balancer Virtual Services in your environment.

Before viewing the path topology, learn about the core components of the Avi Load Balancer platform. For more information, see the Overview of VMware Avi Load Balancer topic.

**Note:**

VCF Operations for networks does not support vNIC - vNIC path in Avi Load Balancer.

NSX or VMware vCenter manages the network of Avi Load Balancer pool members. Therefore, you can view the following VM-VM paths only:

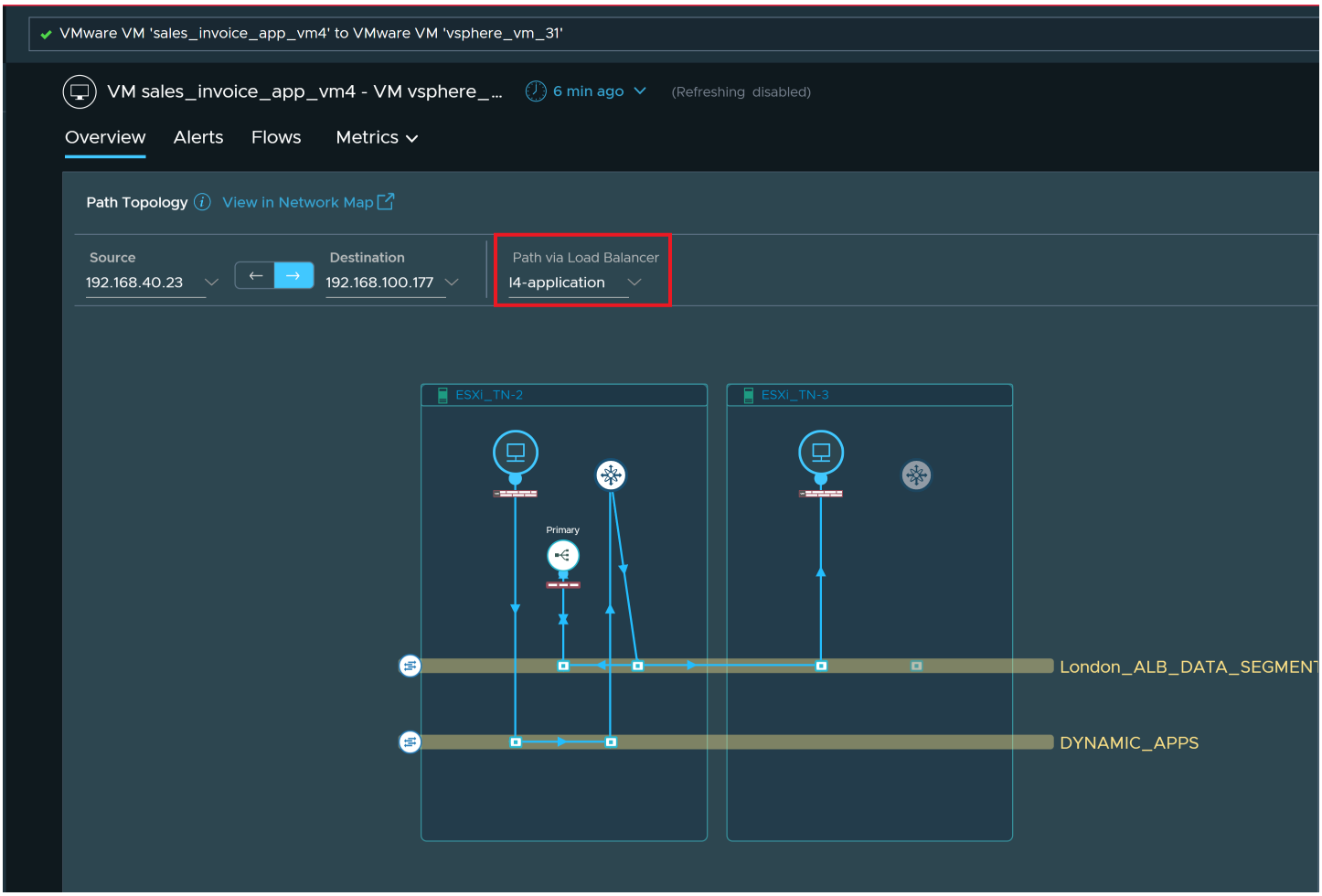
- NSX and Avi Load Balancer and VMware vCenter
- vCenter and Avi Load Balancer
- VMware Cloud on AWS NSX-T and VMware vCenter and Avi Load Balancer

### VM-VM Path Through a Virtual Service

To see the VM-VM path through a virtual service, perform the following steps on the Path Topology page:

1. Select the **Source** and **Destination** VMs.
2. Select the Avi Load Balancer virtual service from the **Path via Load Balancer** drop-down menu.

An example of VM-VM path through an Avi Load Balancer virtual service is as follows:



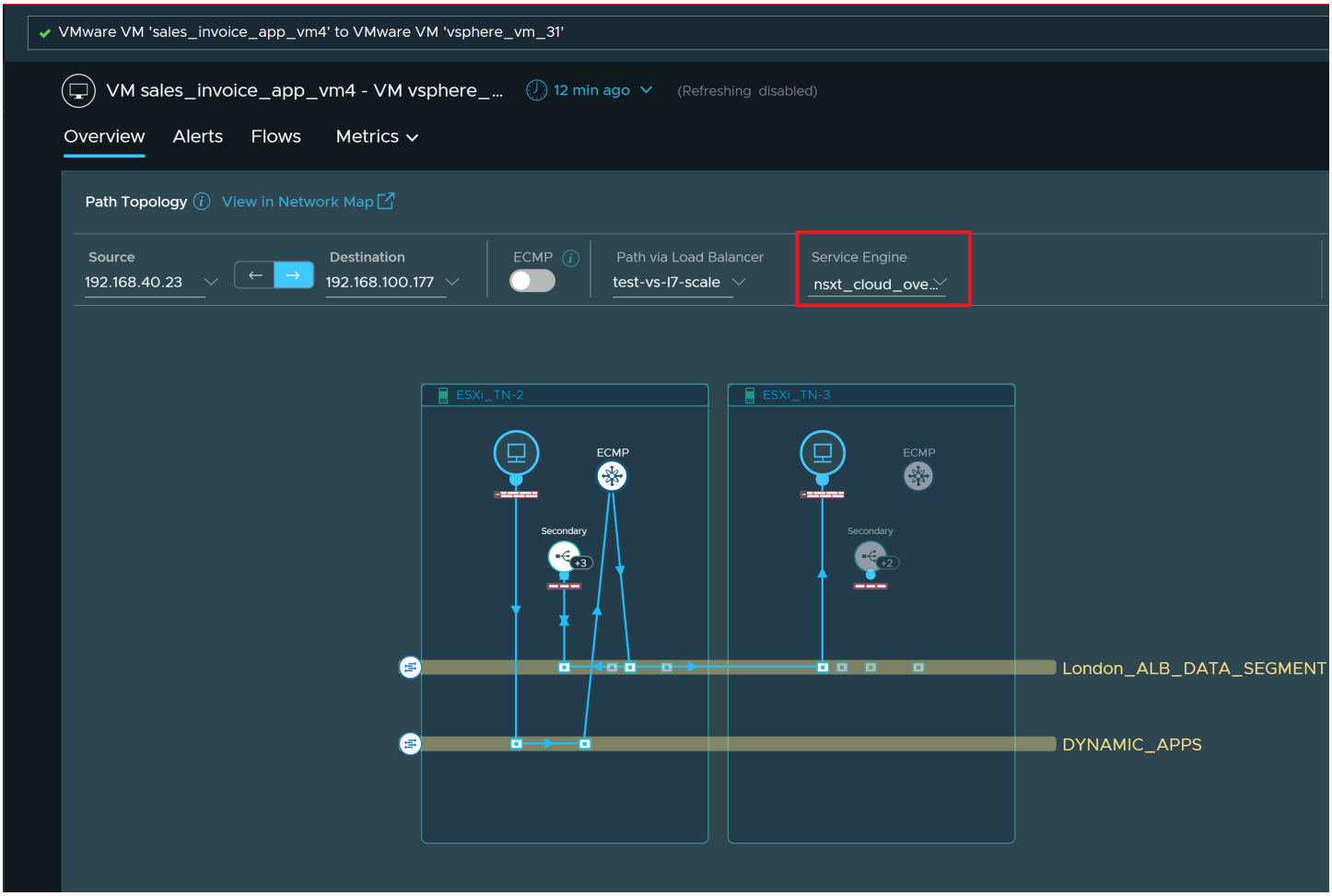
## VM-VM Path Through a Virtual Service That Uses Multiple Service Engines

When a virtual service is scaled across multiple service engines, by default, VCF Operations for networks displays the VM-VM path through the primary service engine.

To see the VM-VM path through a secondary service engine, perform the following steps:

1. Select the **Source** and **Destination** VMs.
2. Select the Avi Load Balancer virtual service from the **Path via Load Balancer** drop-down menu.
3. Select the secondary service engine from the **Service Engine** drop-down menu.

An example of the VM-VM path through a secondary service engine is as follows:




In this example, **+3** indicates that the traffic of the virtual service in ESX_TN-2 is distributed to three service engines, and **+2** indicates that the traffic of the virtual service in ESX_TN-3 is distributed to two service engines.

You can click the number on respective hosts to see the list of service engines associated with that host. You can also select a different service engine from the list to see the VM-VM path through the selected service engine.

> **Note:**
> To see the VM-VM path through all the Service Engines, turn on the **ECMP** toggle. For more information, see Support for Equal-Cost Multi-Path (ECMP) Route.

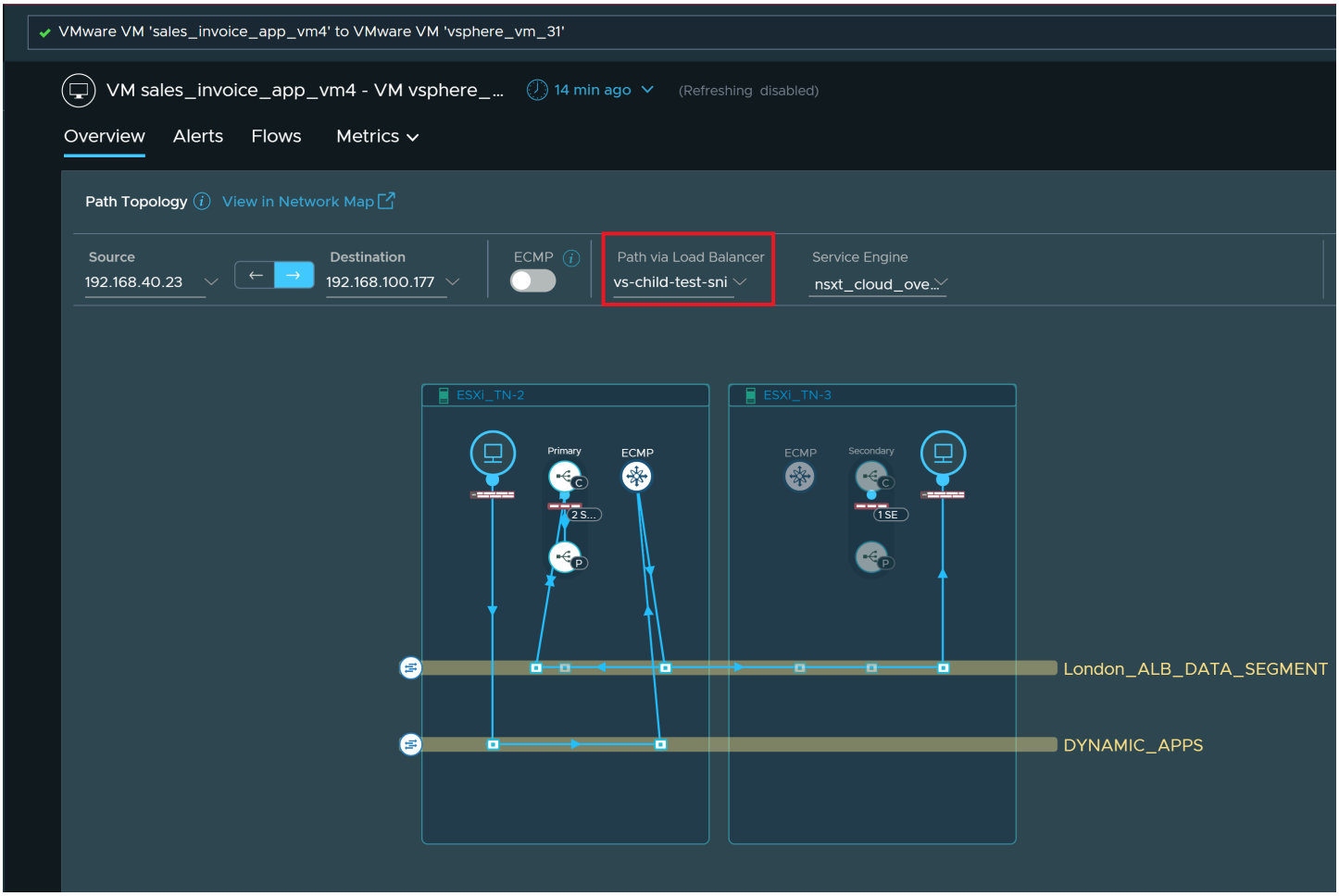
## VM-VM Path Through a Virtual Service That Uses Server Name Indication

Server Name Indication (SNI) is a method of virtual hosting multiple domain names for a single virtual IP.

In Avi Load Balancer, SNI hosting is implemented as parent and child virtual services. For more information, see the VMware Avi Load Balancer documentation.

If you have configured your virtual service to use SNI, select the child virtual service (through which you want to view the VM-VM path) from the **Path via Load Balancer** drop-down menu.

An example of a VM-VM path through a child virtual service is as follows:

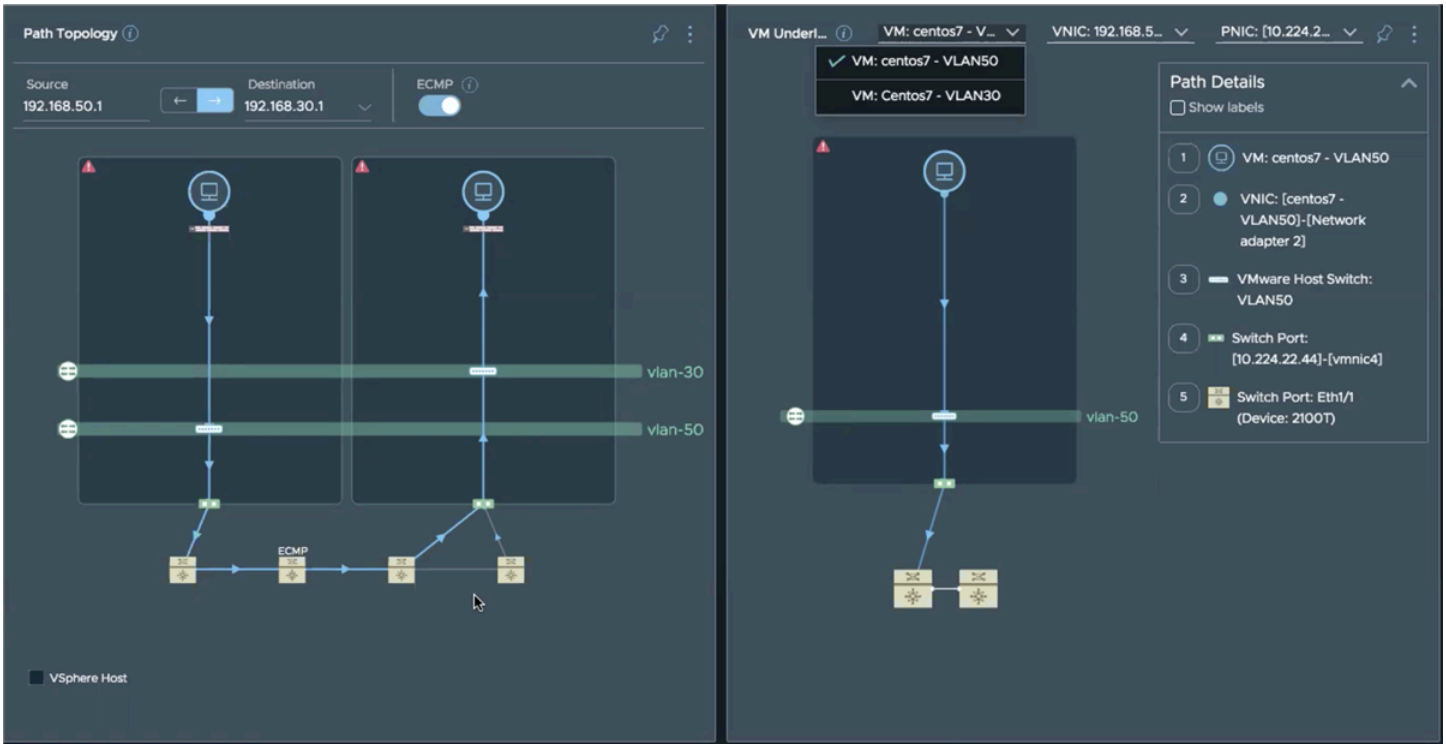



In this example, **2SE** indicates that the traffic of the virtual service in ESX_TN-2 is distributed to two service engines, and **1SE** indicates that the traffic of the virtual service in ESX_TN-3 is handled by a single service engine. You can click **SE** on respective hosts to see the list of service engines associated with the host. You can also select a different service engine from the list to view the VM-VM path through the selected service engine.

## Mellanox VM-VM Path

In VCF Operations for networks, you can view the VM-VM path through supported Mellanox switches.

A sample VM-VM path through a Mellanox switch is as follows:

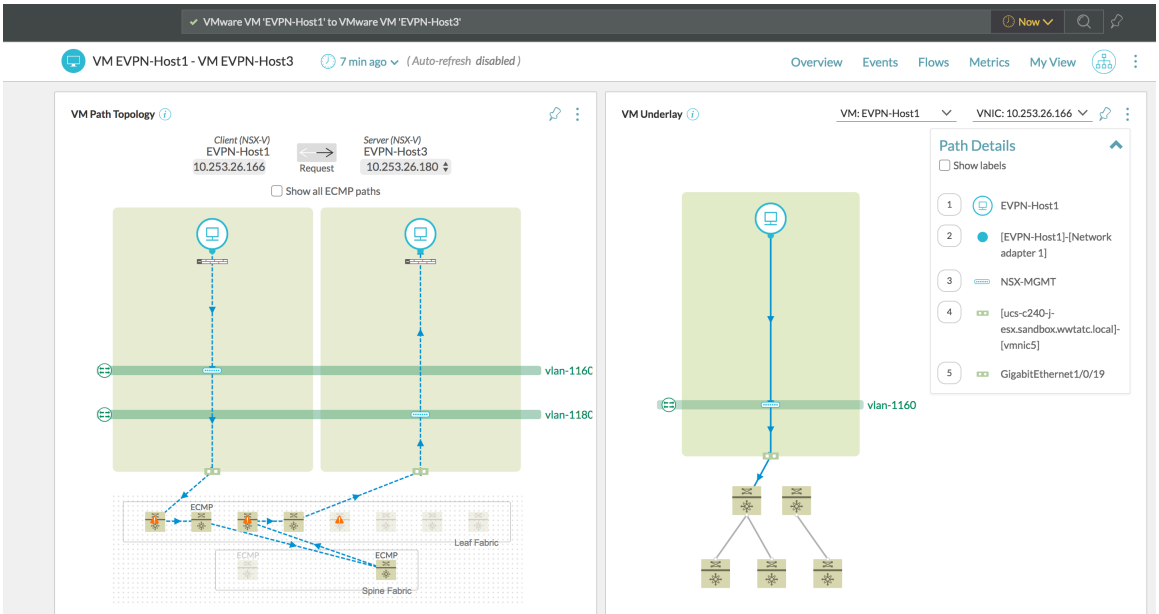



## Support for the Cisco BGP-EVPN Mode

VCF Operations for networks supports the fabric of Cisco 9000 switches configured in the Cisco BGP-EVPN configuration mode for the Enterprise edition only. VCF Operations for networks does not support the switch models other than Cisco Nexus 9000 with the Cisco BGP-EVPN configuration.

Each Cisco Nexus 9000 switch that is a part of the fabric is individually added as a data source. To view all the spine or leaf switches in the fabric, use the `switches where role is set` query.

A sample VM-VM path for the Cisco BGP-EVPN mode is as follows:



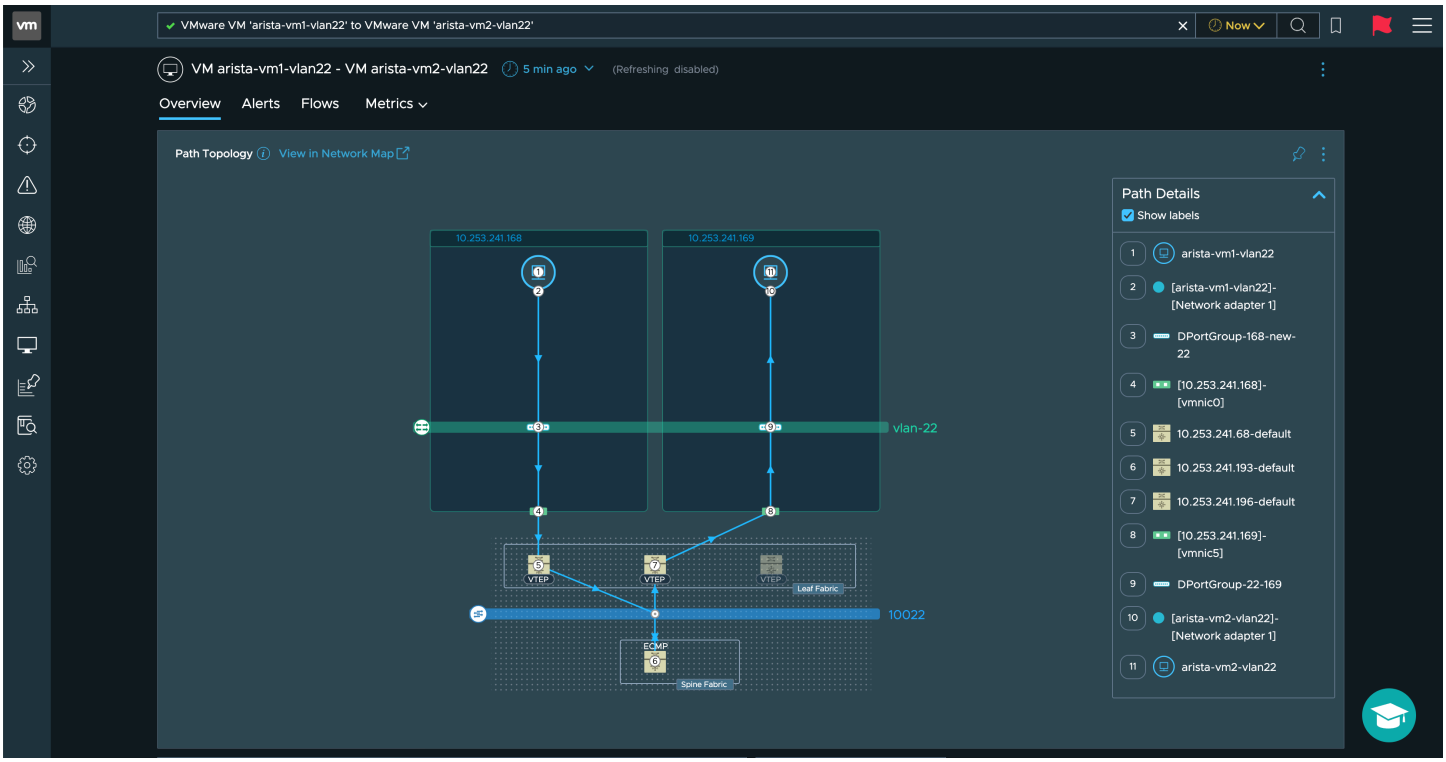
## Support for Arista BGP EVPN

VCF Operations for networks supports Arista Layer 2 and Layer 3 Ethernet VPN (EVPN) in the asymmetric Integrated Routing and Bridging (IRB) mode.

In the asymmetric IRB mode, EVPN allows the routing operation to occur closer to the end host by integrating the routing and bridging functionality directly onto the VXLAN Tunnel Endpoint (VTEP).

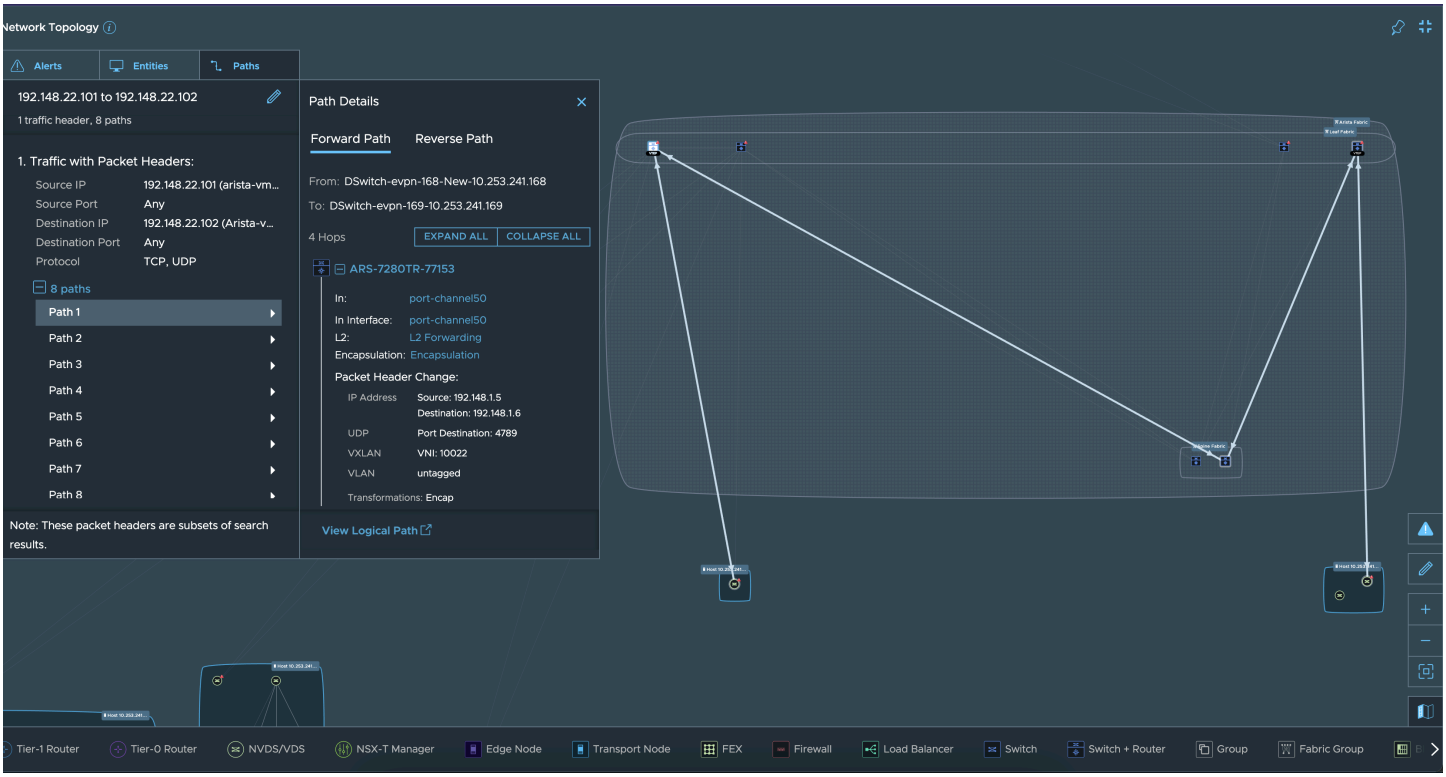
When you view the VM-VM path for Arista switches, VCF Operations for networks also displays the spine and leaf switches of the Arista fabric. Where, the leaf switches are responsible for encapsulation and decapsulation of data packets, and the spine switches are responsible for the BGP routing.

A sample VM-VM path of Arista BGP EVPN (same VLAN) with spine and leaf fabrics is as follows:

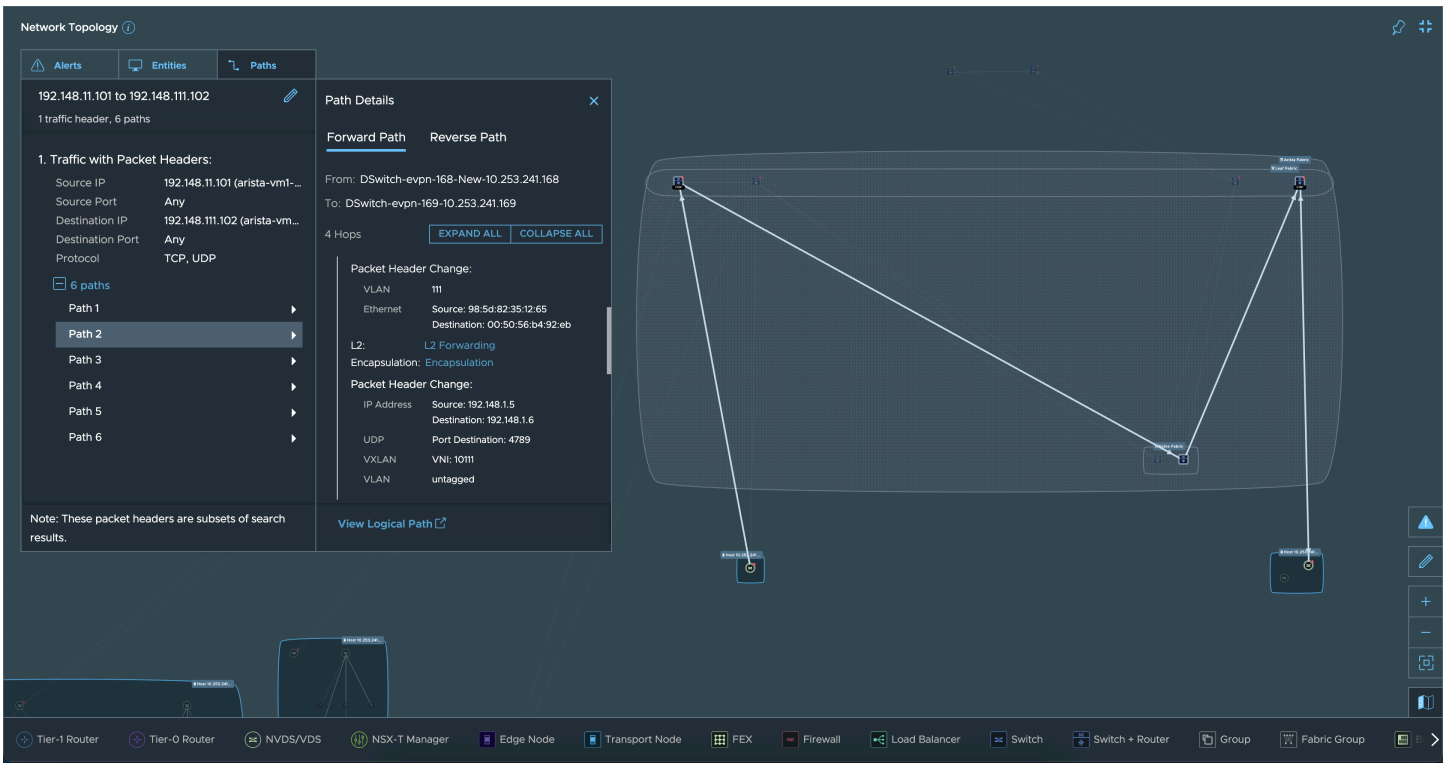



The spine and leaf switches are also displayed when you view the VM-VM path for Arista BGP EVPN on the **Network Map** page.

A sample VM-VM path of Arista BGP EVPN (same VLAN) on the **Network Map** page is as follows:




A sample VM-VM path of Arista BGP EVPN (inter-VLAN) on the **Network Map** page is as follows:

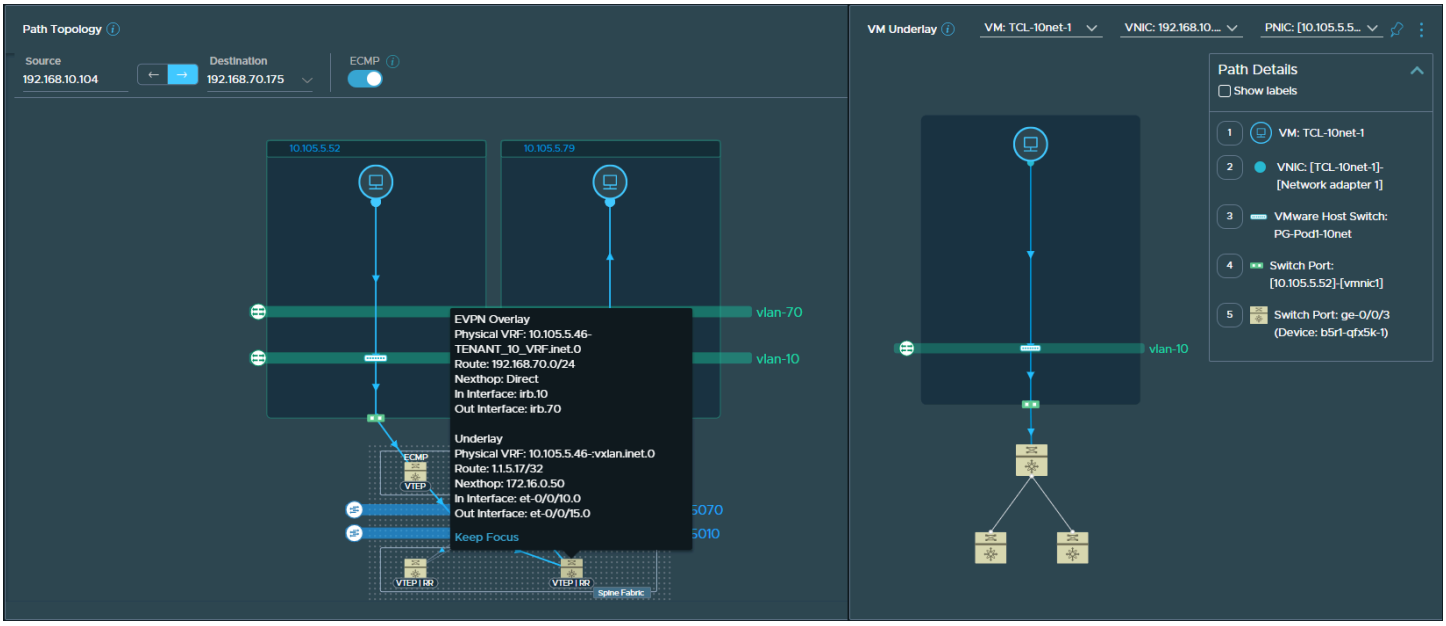



## Support for the Juniper EVPN CRB

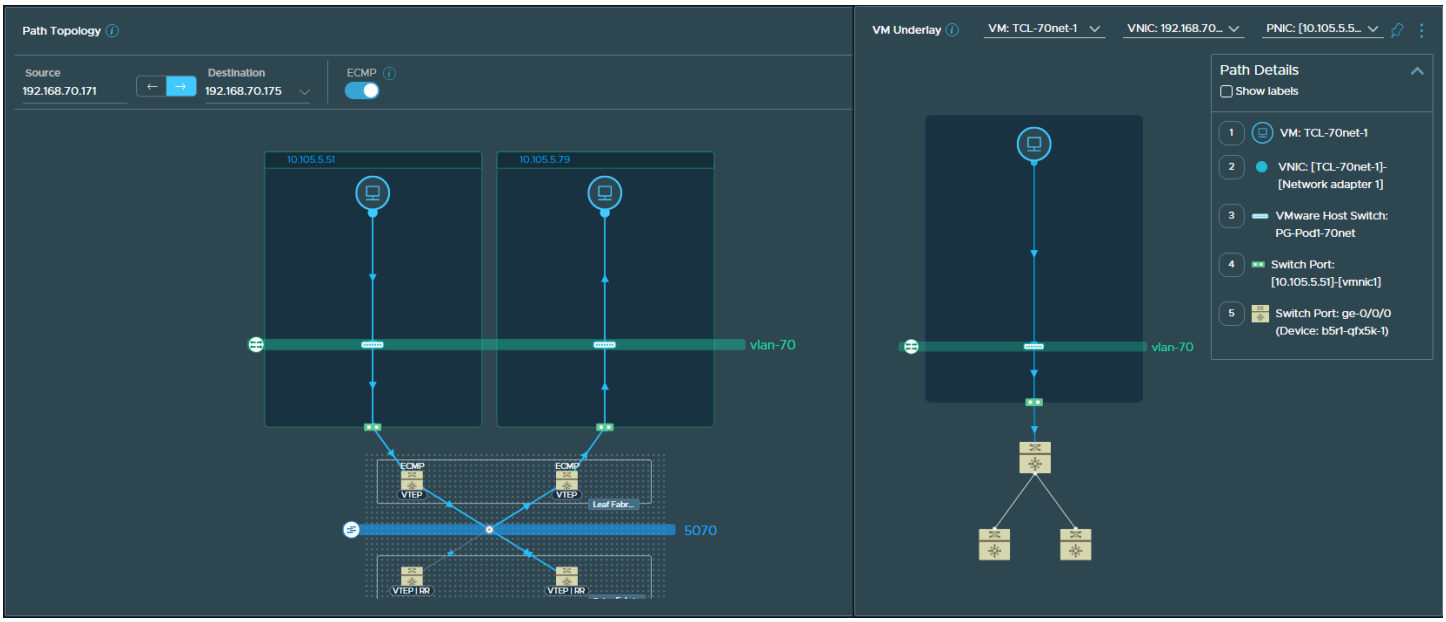
VCF Operations for networks supports the Juniper EVPN Centrally Routed Bridging (CRB) overlay in Asymmetric mode. In CRB, routing happens at the central location with a two layer leaf-spine fabric where the Anycast Gateway (AGW) is

configured on the spine. The AGW is used by the VMs to send out the packets. Here, host is connected to the leaf where the VLAN to VXLAN bridging happens.

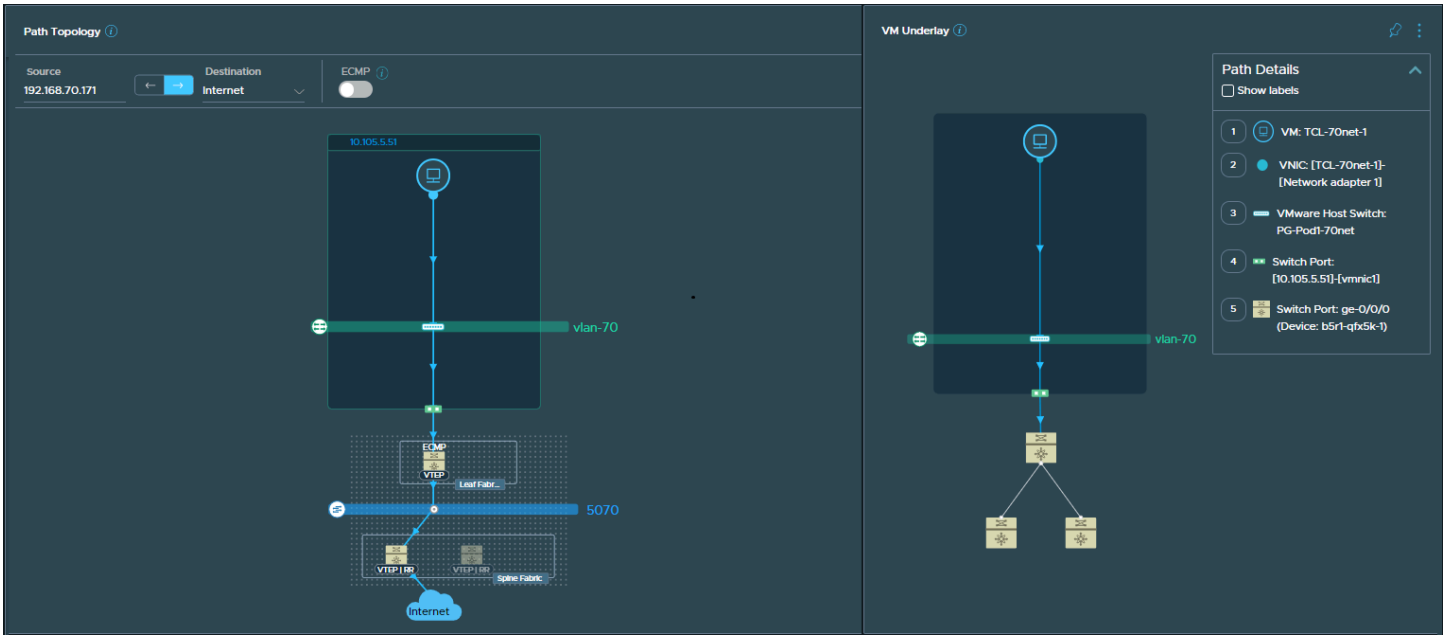
A sample VM-VM path for the Juniper EVPN CRB Inter-VLAN is as follows:




A sample VM-VM path for the Juniper EVPN CRB Same VLAN is as follows:




A sample VM to Internet path for the Juniper EVPN CRB is as follows:

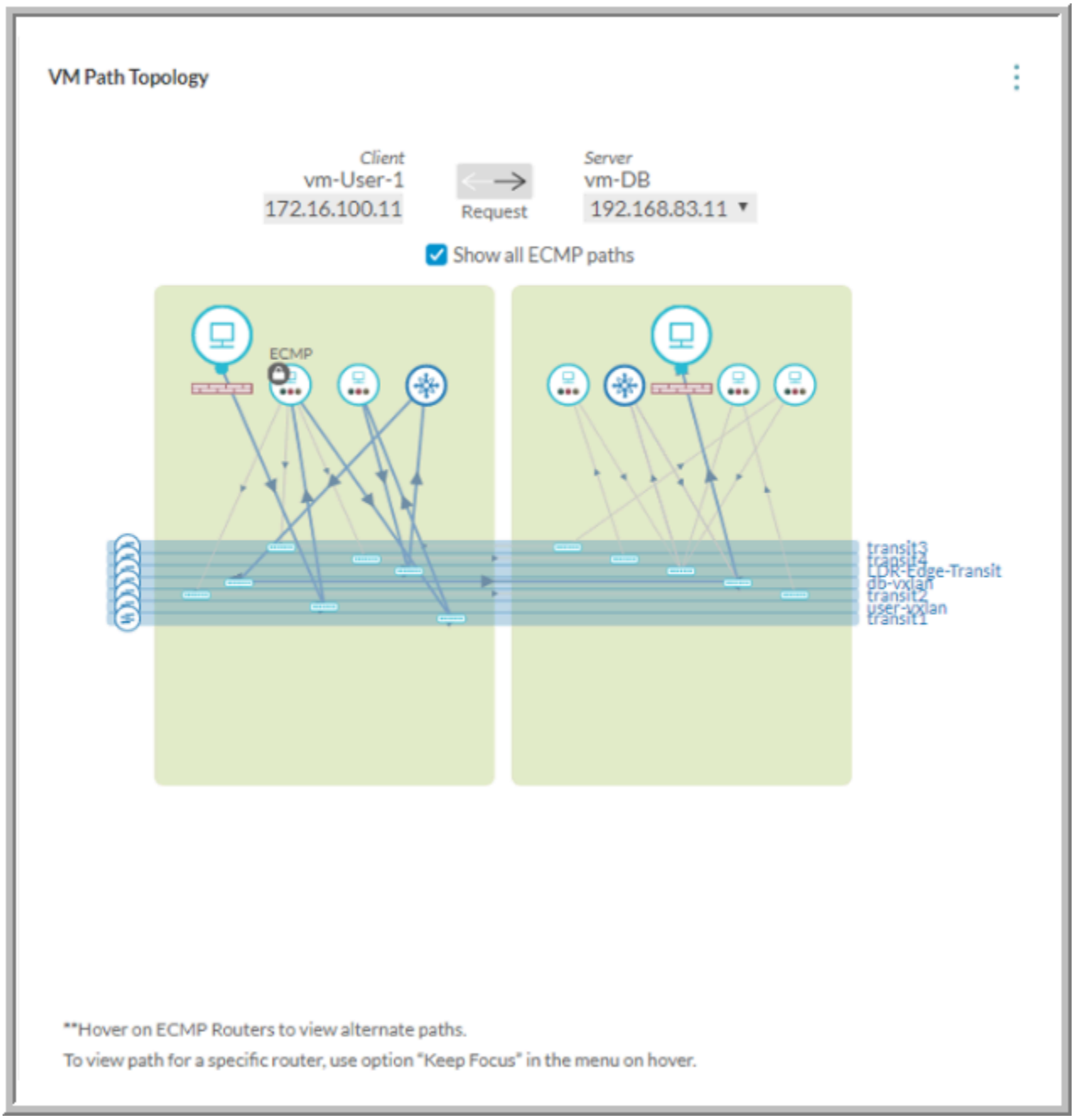



## Support for Equal-Cost Multi-Path (ECMP) Route

VCF Operations for networks provides ECMP support in the VM-VM path.

The VM-VM path shows the following information on ECMP:

- The multiple ECMP paths from source to destination
- The routers on which ECMP occurs
- The possible outgoing paths for a given router (VRF)
- The route for the possible path




In the preceding figure, you can see the ECMP-enabled routers. If you point over them, the additional paths are shown. Also, you can create a path by selecting and locking the routers as per your requirement. If you want to view all the ECMP paths between the two VMs, select the **Show all ECMP paths** option in the topology diagram.

If you want to view the path for a particular router, point on the router and click **Keep Focus**. The paths specific to the router is shown.

### Support for the L2 Bridges

The L2 or the VLAN bridges create a single broadcast domain from multiple VLANs. In the previous releases, if the VM-VM path involved an L2 bridge between two or more VLANs, the VM-VM path did not work. From this release onwards, VCF Operations for networks supports L2 bridging. Currently, this feature is supported only for the Cisco ASA routers.

## View Information on BGP Neighbors

In VCF Operations for networks you can see information on BGP neighbors. You can view the BGP neighbors of an NSX edge or a logical router.

1. Enter `Router where bgp= 'Enable'` in the search bar and press **Enter**.
2. Expand the particular router from list to see the details.

   You can view the following information under BGP neighbors for NSX-T:

   - `IP Address`
   - `Remote AS`
   - `Keep Alive Time`
   - `Hold Down Time`
   - `Status`

   **Note:**

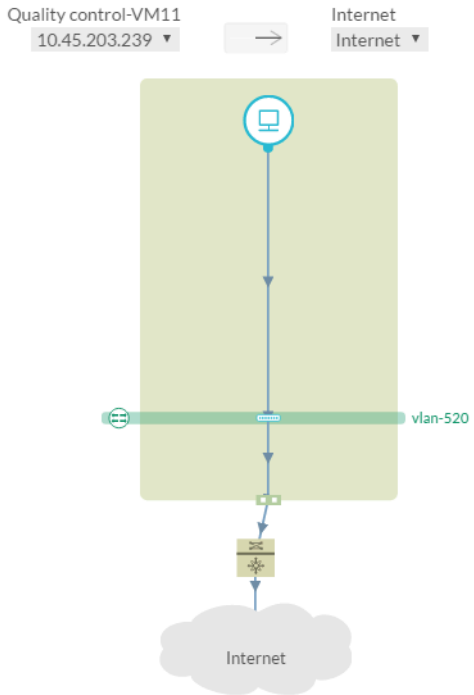
   - If the information regarding the neighbors is not fetched, then the `Status` is shown as `Unknown`.
   - If the `Status` is not `Established.up`, then the `One or more BGP neighbors are not in established state` alert is raised for that edge. You can also view this alert when you search for `problems`.
3. Optional: To view Routers where BGP status is deactivated, search for `Router where bgp= 'Disabled'`.

## Path to Internet

For each virtual machine present in your environment, VCF Operations for networks shows you how the VM is connected to the Internet by using an animated path in the **Path to Internet** pin.

The path populates all the components (both virtual and physical) that exist between the virtual machine and the Internet. It draws an animated path that connects each component in a sequence. The path direction can also be reversed by using the arrows situated above the visualization.

Point your mouse pointer to the entity icons to get their addressable names. Click an icon on the path to display a summarized account of its primary attributes. You can also maximize the pin to see the path details.



# Working with Network Maps and Intents

## Working with the Network Map

Using the network map, you can view a detailed network model of the data sources that are added to VCF Operations for networks.

**Note:**

- The network map is available only for Enterprise and NAV license users who use XL brick.
- Also, network map is supported for a few data sources only. For more information, see Supported Products and Versions.

To access this page, on the VCF Operations for networks home page, click **Network Map**.

The network map contains several elements.

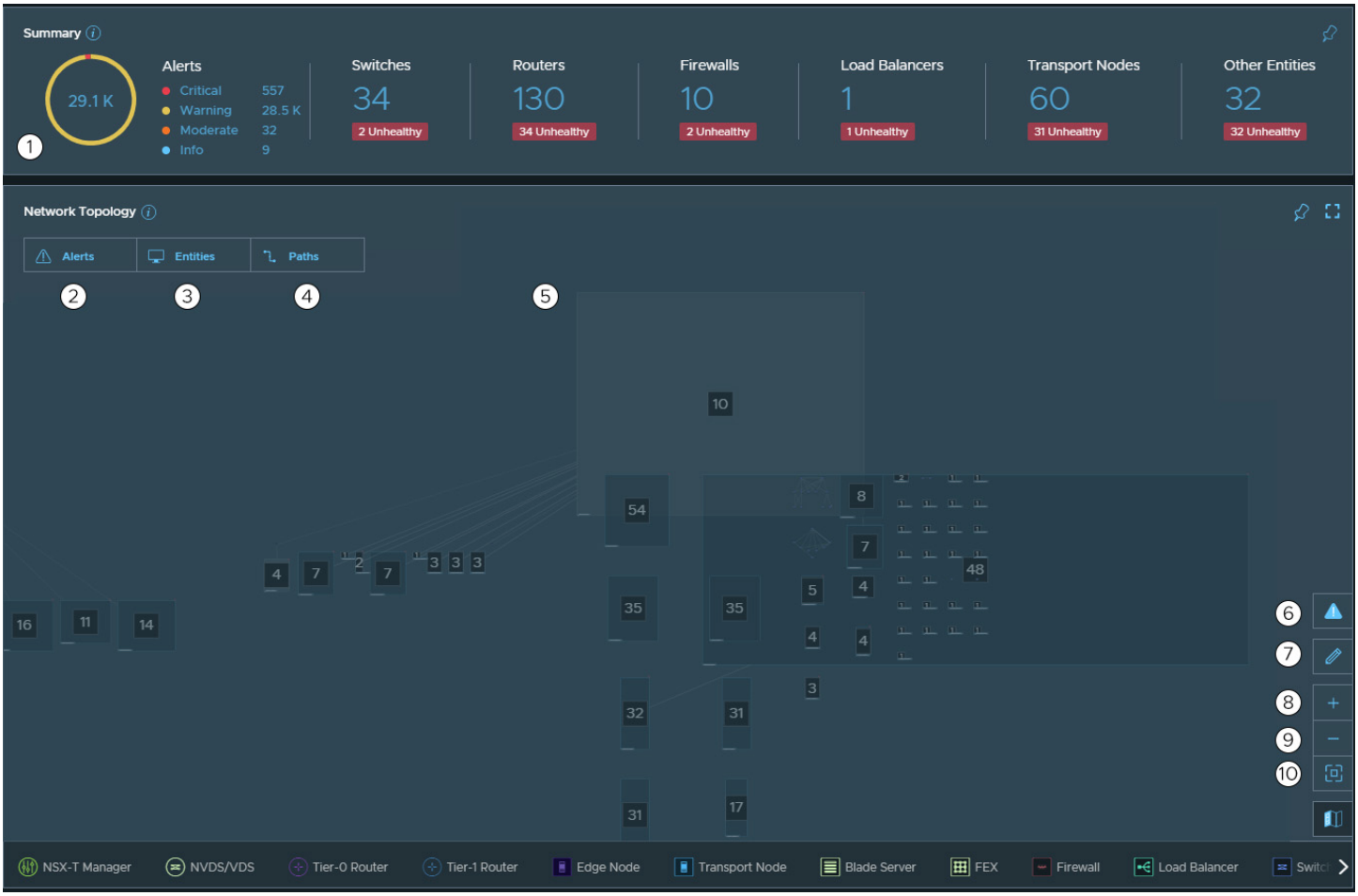


**Table 1332: Elements**

| Elements | Description |
|---|---|
| (1) **Summary** | A summary of your data center which includes an alert chart, switches, routers, firewalls, load balancers, transport nodes, and other entities. |
| (2) **Alerts** | The list of available alerts in your network when you click the **Alerts** tab.<br>You can also search for alerts using the search bar that is available in the **Alerts** tab. To get more details about an alert, click **+** (if available). To see related alerts, click **Expand all**. |
| (3) **Entities** | Entities and groups of entities in your network . All the virtual entities are shown in logical groups. For more information on entities, see Viewing Entity Details. |
| (4) **Paths** | The paths between specified entities. |
| (5) **Network Topology** | An end-to-end interactive topology map of your network including physical and virtual devices. For details, see Network Topology Map. |
| (6) **Alerts** | Hide or show the alerts on the network topology map. |
| (7) **Edit** | Rearrange entities and groups, create groups, and delete groups. To learn about how to create and delete a group, see Add, Edit, or Delete a Group. |
| (8) **Zoom In/Zoom Out** | Zoom in or zoom out of map by clicking the (+) or the (-) button. |
| (9) **Fit to screen** | Rearrange the map to fit the current screen. |
| (10) **Legend** | Show or hide the meaning of each of the icons used in the network topology. |

## Network Topology Map

The network topology map is a map of your entire network at a point in time. VCF Operations for networks constructs this map based on the data collected from all the data sources that are available in VCF Operations for networks and that support network map.

VCF Operations for networks does the following actions:

- Automatically arranges all the virtual entities in different groups for better management.
- Shows the parts of the network affected by alerts so that you can quickly troubleshoot issues.
- Allows you to zoom the network map to see additional details such as group name, entity name, details of an entity, and so on.

## Entities on the Network Map

To see the list of entities and groups of entities available in your network, click **Entities**. All the virtual entities are shown in logical groups. The list of entities can include NSX virtual entities, physical switches, physical firewalls, and so on. For details, see Supported Products and Versions.

To see details about the entities or the groups, click that entity or group. When you click an entity or group, you can see the details of that entity or group and the location of that entity or group in the topology map. You can click ( ) to view more details about that entity and click **Back to summary** to return to the entities list.

You can also search entities or IP addresses in your network using the search bar available in the **Entities** tab. For samples queries, see Search Queries for Entities.

## Auto-layout Map

In the **Network Topology**, you can auto arrange selected entities or your entire map to best fit your screen.

To auto-arrange your **Network Topology**, in **Network Topology** page, click **Edit button** > **Continue** > **Auto-layout map**. You can either auto-arrange entire network topology or a few selected entities.

> **Note:**
> Read the instruction on the UI carefully while using the auto-layout feature.

## Network Map Scope

In the network map, you can view a subset of the map using the **Network Map Scope**; this option enables you to view a more focused version of the network map.

With the **Network Map Scope**, you create a subset of the network map by selecting the entities you are interested in, and the network map creates a network topology only with those selected entities. You can perform the following activities in the network map with **Network Map Scope**:

- Create simple scopes by selecting your choice of entities.
- Edit, modify, or delete existing scopes.
- Toggle between All entities and user-defined scopes using **Scope** drop-down menu.
- Create multiple simple scopes as per your requirement. You can create up to 10 scopes in the network map.

To create a simple scope, do the following:

1. In the **Network Map** page, select **New Scope** > **Create Scope**.
2. In the **Network Map Scope** window, from the **Scope by** drop-down menu, select one of the following:
   – **Entities**: Select entities you want to view in the network map.
   All the entities are segregated in two major categories, virtual entities and physical entities. Each major category has multiple different entity types.

   > **Note:**
   > In the simple scope, you can select entity categories. You cannot select individual entities from the **Scope** drop-down menu.

   > **Note:**
   > The **Network Map Scope** considers only VMware overlay components such as the Edge transport node, the host transport node, and so on as virtual entities, and all third-party virtual appliances, such as F5 BIG-IP VM, Palo Alto VM, Cisco NX-OSv, and so on, as physical entities and grouped under the physical entities.
   – **Groups**: Select groups or subgroups you want to view in the network map. You can also search for groups and sub-groups and select the groups or subgroups you want to view in the network map.
3. Click **Apply**.
   You see a preview of the network map for your scope.
4. Click **Save Scope** located at the top-right corner.
5. If the preview is not per your requirement, click **Discard**.
6. Add a name for your scope, and click **Confirm**.

**Figure 447: Network Map Scope**

You can also edit, delete, or duplicate saved scopes. To edit, delete, or duplicate a saved scope, do the following:

1. In the **Network Map** page, from the **Scope** drop-down located on the top-left corner, select **Manage Saved Scopes**.

2. In the **Network Map: Saved Scopes** page, select the saved scope you want to modify, and do the following:
   - To edit a saved scope, click the **Edit** button, make the required changes and click **Save Changes**.
   - To delete a saved scope, click the **Delete** button and confirm your action.
   - To duplicate a saved scope, click the **Duplicate** button, make the required changes (if necessary), and click **Apply**.

### Search Queries for Entities

Here are some of the sample search queries that you can use in the search bar of the **Entities** tab:

| Search Queries | Description |
|---|---|
| device bb-ny-1 | Shows devices whose name contains 'bb-ny-1'. |
| device 'bb-ny-1' | Shows devices whose name is exactly 'bb-ny-1'. |
| device ny | Shows devices whose name contains 'ny'. |
| abcd | Shows entities whose name contains 'abcd'. |
| 'abcd' | Shows entities whose name is exactly 'abcd'. |
| '1.1.1.1' | Shows entities whose manager IP address is 1.1.1.1. |
| 1.1.1.1 | Shows all the entities whose manager IP address contains 1.1.1.1. For example 1.1.1.1, 21.1.1.12, or 31.1.1.12. |
| device where manager = 1.2.3.4 | Shows device whose manager IP address = 1.2.3.4. |
| host switch | Shows a list of host switches. |
| host switch where name = 'DSwitch-1-localhost' | Shows host switch whose name = DSwitch-1-localhost. |
| host switch where name like 'vds-switch1' | Shows a list of host virtual switches whose name contains 'vds-switch1'. |
| interfaces | Shows a list of all interfaces. |
| interface 'fastethernet0/0' | Shows interfaces whose name contains 'fastethernet0/0'. |
| interface where mac address = 'c401.1cbe.0000' | Shows a list of interfaces whose MAC address contains 'c401.1cbe.0000'. |
| interface where name like vlan | Shows a list of interfaces whose name contain keyword vlan. |
| router interface where ip address = 192.168.10.1 | Shows a list of router interfaces whose IP address contain 192.168.10.1. |
| interface where name like 'ethernet' and start vlan = 10 | Shows a list of interfaces whose name contain 'ethernet' and whose VLAN range start from VLAN 10. |
| router interface where ip address = 192.168.1.0/24 | Shows a list of interfaces whose IP address falls within the given subnet 192.168.1.0/24. |
| 00:01:02:03:04:05 | Shows a list of interfaces whose MAC address is 00:01:02:03:04:05. |
| host router | Shows a list of all virtual routers available in all the hosts. |
| host router where name like 'tier0' | Shows a list of all virtual routers whose name contain 'tier0' |
| port channel where administrative status = 'down' | Shows a list of all port channels whose administrative status is down. |
| network device group | Shows a list of all user created groups and auto inferred hosts as groups. |

### Paths on a Network Map

To find paths between specified entities -

1. In the **Network Topology** section, click **Paths**.
2. Enter the source and destination VMs , and click **Show Paths**.

**Note:**
Powered off VMs are not supported while performing a path search on the Network Map page. If you manually enter a powered off VM name in the **Source** or **Destination** field, path search will display the `Unknown Source` or `Unknown Destination` error message.

To know more about path search, see Use Advanced Path Search.

VCF Operations for networks also supports subnet to subnet path search on network maps.When you click **SHOW PATHS**, the system searches the model of your network available in VCF Operations for networks to find all possible paths matching the search parameters. The results are grouped according to traffic headers. Each traffic header description contains a list of paths that the packets with those headers can follow. There can also be more than one path. For example, you can see multiple paths if your network uses multipath forwarding. You can see a maximum of 25 results in the list. If necessary, you can narrow your search parameters to see all results.

Click **Path N** (N = numerical digit, example: 1, 2, 3, and so on) to see **Path Details**. In the **Path Details**, you can see the forward path, reserve path, hops details and so on.

### Path Search Queries

Following are some sample path search queries:

| From | To | Expected Result |
|---|---|---|
| `switch port where routedportips = 61.0.1.1` | `switch port where routedportips = 61.0.1.2` | Shows paths from routed port with IP 61.0.1.1 to routed port with IP 61.0.1.2 |
| `interface 'ethernet1' where Device = 'R1-arista'` | `interface 'ethernet1' where Device = 'R2-arista'` | Shows paths from interface ethernet1 of R1-arista to interface ethernet1 of R2-arista |
| `61.0.1.1` | `61.0.1.2` | Shows paths from 61.0.1.1 to 61.0.2.1 where both IPs are of routed ports |
| `61.0.1.0/24` | `63.0.1.0/24` | Shows all paths from subnet 61.0.1.0/24 to subnet 63.0.1.0/24. |
| `61.0.1.10` | `63.0.1.0/24` | Shows paths from IP address 61.0.1.10 to subnet 63.0.1.0/24. |

## Use Advanced Path Search

With Advanced Path Search, you can add more filters in your path search.

1. On the VCF Operations for networks home page, click **Network Map**.
2. In the **Network Map** page, click **Paths**.
3. In the **Source** and **Destination** text boxes, enter an entity query or IP address that you want to search. In general, you can use any query in the **Source** and **Destination** text boxes. For sample queries, see Search Queries for Entities.

   Searches can match multiple entities, so that you can search for paths between a set of sources and a set of destinations.

   **Note:**

   - Powered off VMs are not supported while performing a path search on the Network Map page. If you manually enter a powered off VM name in the **Source** or **Destination** field, path search will display the `Unknown Source` or `Unknown Destination` error message.
   - You cannot use a Cisco Nexus 2000 series fabric extender devices as a source or a destination as this device cannot generate or consume traffic.
4. Optional: To find the paths that traverse certain intermediate entities, click **Add a Hop**.

   You can add multiple hops.

5. To add more option in your path search, click the **Advanced Options** check box and provide the required information.

| Option | Action |
|---|---|
| **Packet Headers at Source** | Select one of the following:<br>• **Source Port**: Source transport port.<br>• **Destination Port**: Destination transport port.<br>• **Protocol**: Transport protocol. |
| **Path Type** | Select one of the following:<br>• **Primary**: To see the primary paths.<br>• **Backup**: To see the backup paths.<br>• **All**: To see all available types of paths. |
| **Path Direction** | Select one of the following:<br>• **Forward**: To see paths only in the forward direction.<br>• **Bidirectional**: To see the full round-trip path from source to destination and back. |
| **Path Status** | Select one of the following:<br>• **All**: To see both reachable and blocked paths.<br>• **Reachable**: To see paths for which packets reach the destination.<br>• **Blocked**: To see paths for which packets are blocked before reaching the destination. For example, because they are dropped by a firewall, dropped at a routing black hole, or are caught in a loop. |

6. Click **SHOW PATHS**.

## Add, Edit, or Delete a Group

In the **Network Map** page, you can group multiple entities together for the better management and visibility. In the **Network Map** page, groups define a set of multiple entities. A group can also contain other groups. Groups are used to organize the entity list and to provide a context to the visualization in the network map. You can also use groups when searching for entities and paths.

VCF Operations for networks automatically creates certain types of groups. For example, the system groups virtual entities inside the same physical host. You can also add, edit and delete your own groups.

**Note:**
You cannot edit or delete system-generated groups.

1. In the **Network Map** page, click **Edit** in the lower-right corner.
2. After you click **Edit**, you can do the following:

| Option | Action |
|---|---|
| **Add a group** | To add a group, do the following:<br>1. Click the **Add Group** icon located in the lower-right corner.<br>2. In the **Create Group** window, in the **Name** text box, enter a group name.<br>3. From the **Entities** drop-down, select all the entities that you want to add in the group.<br>**Note:**<br>• You can select entities using device name, groups, IP address, IP range or subnet, and custom search.<br>• You can use combination of these options to find entities.<br>• You can also select entities in the **Network Topology** to create groups. You can only select entities which are not part of any previous group. You can not select entities which are part of any existing groups, but you can select a top-level group to create a new group.<br>• The custom search enables you to find entities using SQL like queries. |

| Option | Action |
|---|---|
| | • If you use the IP range/subnet or custom search to create a group, the group can change size dynamically. This happens because, based on your search criteria, devices get added or removed automatically to the group when any data sources get added or removed in VCF Operations for networks.<br>• When using the IP range/subnet or custom search option to create groups, if an entity matches more than one group, then the entity gets added to the group whose name comes first in alphabetical order.<br>4. From the **Color** drop-down, select a color for the group, and click **Submit**. |
| **Edit a group** | To edit a group, do the following:<br>1. Select the user-defined group that you want to edit, and click **Edit Group**.<br>2. In the **Edit Group** window, do the required changes.<br>3. Click **Submit**.<br>**Note:**<br>You can only edit user-defined groups. You cannot edit system-generated groups. |
| **Delete a group** | To delete a group, do the following:<br>1. Select the group you want to delete, and click **Delete Group**.<br>2. In the **Confirm Action** window, click **Confirm**. |

## Create and Delete Groups by Using a CSV File

You can use the network map group tool to create and delete groups by adding a CSV file.

To use a CSV file to create a group in your network map, the file must include the following three mandatory columns:

- `Member`: Includes the member, which can be a device or subgroup, you want to add. The member can be identified through a name or IP address. The device or subgroup must be part of the network map. If the member is a subgroup, it must be a top-level group available in the network map. The member must not be part of any other group. If the member is a device, it should be an orphan device, not part of any existing group.

  **Note:**
  If the IP address is used as the identification method for a member, it is expected to be the manager IP address. For managed devices, such as Panorama or Check Point, the expected IP address is the device IP address. If the member is a ESX host group, either group name or host IP address can be specified as the member.
- `GroupName`: Includes the name of the new group that you want to create. It must not be part of the network map.
- `Color`: Includes the color coding for the group. If the color you want to add is spelled with two words, the words should be separated by a whitespace. For example, Lemon Yellow. Supported colors include Black, Blue, Brown, Cyan, Gold, Green, Grey, Lemon Yellow, Light Grey, Light Red, Magenta, Mustard, Navy Blue, Neon Green, Neon Pink, Olive Green, Orange, Purple, Red, Sea Green, Sky Blue, and Violet.

The following are two example CSV files. The first example includes device members, while the second includes subgroup members.

```
Member,GroupName,Color
d1,g1,Red
d2,g1,Red
d3,g3,Lemon Yellow
d4,g3,Lemon Yellow
d5,g4,Green
10.10.10.10,g5,Cyan
```

```
Member,GroupName,Color
g1,g2,Blue
g2,g3,Blue
g3,g4,Cyan
g5,g6,Red
g7,g8,Magenta
```

1. Log in to the VCF Operations for networks command line interface.
2. Optional: To validate your new group before creating it, you can run the network map group tool command with the `-v` argument.

   To validate a group with subgroup members, for example, run the `network_map_group_tool -path <pathToFile> -m group -v` command.
3. To create a group with subgroup members, run the `network_map_group_tool -path <pathToFile> -m group` command.
4. To create a group with device members, run the `network_map_group_tool -path <pathToFile> -m device`.
5. To delete all groups specified in a given CSV file, run the `network_map_group_tool -d -path <pathToFile>` command.
6. To delete all user defined groups, run the `network_map_group_tool -d`.

### Troubleshooting Network Map Related Errors

The table lists errors you might encounter when using the **Network Map** with the cause and resolution steps.

| Error Text | Cause | Resolution |
|---|---|---|
| `Path search and intent verification may not be available in Network Map.` | Network-wide rule count has exceeded the current platform capacity. | You must switch your deployment to a cluster of bigger size. Contact VMware support. |
| `The network map and intent verification are unavailable due to insufficient Operations for Networks platform resources.` | Network-wide rule count has exceeded the current platform capacity. | You must switch your deployment to a cluster of bigger size. Contact VMware support. |
| `No data to show.` | No supported data source added. | You must add supported data sources and wait for 30 minutes. |
| `No data to show.` | Supported data source is present, but deactivated. | You must enable collection from supported data sources and wait for 30 minutes. |
| `Path search and intent verification may not be available for some devices.` | Number of rules set by one or more devices has exceeded the supported limit. Or, one or more devices have encountered processing errors. | You must contact VMware support. |
| `Showing only physical network entities on the map.` | Number of Network Map elements have exceeded the supported limit. | You must contact VMware support. |
| `Encountered an error.` | Can be due to various reasons. | You must contact VMware support. |

## Working with Intents

An intent is a desired property of your network. An intent can describe a business policy or an architectural goal. You can use intents to design, manage, and to verify networks based on business policies. Intents enable prediction of network behavior in advance.

VCF Operations for networks provides a powerful set of system defined intents out-of-the-box. You can also configure custom user-defined intents from a menu of intent templates.

Periodically, VCF Operations for networks verifies whether each intent is satisfied, using a model of all possible network-wide behavior. If an intent is violated, it triggers an alert.

**Note:**
This feature is available only for Enterprise License users using XL brick.

## Viewing Intent Details

VCF Operations for networks classifies intents as either system-defined or user-defined.

System-defined intents are provided out-of-the-box. You can clone a system-defined intent to create a user-defined copy, which you can then edit to change its parameters. System-defined intents are enabled by default, but you can deactivate them from the **Manage Intents** page.

You can define and configure new intents in VCF Operations for networks. The intents that you define are called user-defined intents. You can edit, delete, and duplicate these intents. You can also select multiple user-defined intents and enable or deactivate them at once.

Every 30 minutes, VCF Operations for networks verifies whether each intent is met in your network.

- If the system determines that the intent's specified conditions are met at that time, it marks the intent status as `Pass`.
- If the intent's specified conditions are not met, the system raises an alert corresponding to this problem and marks the intent status as `Fail`. If the alerts associated with an intent are closed or archived, then the intent status becomes `Pass`.
- If the system determines that one or more entities in the intent configuration are no longer available, it marks the intent status as `Invalid` and displays the `Entity not available` warning message with the intent name.
  Also, when you edit the configuration parameters of the invalid intent, the system highlights the invalid entities in red.

  **Note:**

  VCF Operations for networks does not process invalid intents. That is, if an intent is invalid, VCF Operations for networks does not verify whether the conditions defined in the intent are met in your network.

  To validate the intent, you must delete or replace the invalid entities in the intent configuration.

You can access the **Intents** page from the left navigation pane. You can view the count of all intents and the count of each individual type on this page. You can click each count to filter and view the details of specific types of intents.

VCF Operations for networks supports many different types of intents for a range of purposes. These intent types are grouped into three broad categories:

- Device Health
- Network Health
- STIG

  **Note:**
  STIG intents are supported for the following devices only:

  - Cisco ASA, Cisco ASR 1000, Cisco Catalyst, Cisco ISR 4000, and Cisco Nexus
  - Juniper EX and QFX, Palo Alto

You can filter the intents based on the intent type or intent category. You can view the count of alerts raised by each intent, and you can click this count to see the alert details.

You can also view the intent status (Pass or Fail) and the severity (Critical, Moderate, Warning, Info) of each available intent.

You can also search intents in the list based on their names.

## Define an Intent

You can define and configure new user-defined intents in VCF Operations for networks. After defining an intent, you can edit, delete, duplicate, enable, or deactivate the intent at any time.

**Note:**
VCF Operations for networks supports a maximum of 1,000 intents. If you have 1,000 intents in your environment, you cannot define any more intents. You must delete existing intents to define any new intents.

1. On the left navigation pane, click **Manage Intents**.
2. On the Intents page, click **DEFINE INTENT**.
3. Expand a category under **Intent Type** and click **Select** on the type of intent you want to create.

   **Note:**
   You cannot edit the intent type after creating the intent. However, you can use the **Change Intent Type** option to change the type while you are in the process of defining the intent.
4. Enter the intent parameters which includes the intent name, the intent severity, intent tags, and notes (optional).

   The parameter list displays additional options based on the intent type you select.
5. Optional: Select the **Include/Exclude entities** check box to limit the defined intent to certain entities.
   a) Choose an action depending on your requirement:

| Action | Procedure |
|---|---|
| Include | To add the entities on which the intent should apply, perform the following steps:<br>1. Click **Include**.<br>2. From the drop-down menu, do one of the following actions:<br>— Select a predefined entity and enter your search query.<br>— Select **Custom query** and enter a custom search query.<br>If you select **L3 Switch Manager** or **Custom search**, you can use **Add exception** to exclude certain entities from the query result.<br>For example, if you use a custom query to select all the Node Ids that are lesser than 3, you can add an exception on Node Id 0. Then, the defined intent applies on Node Ids 1 and 2. |
| Exclude | To add the entities on which the defined intent should not apply, perform the following steps:<br>1. Click **Exclude**.<br>2. From the dropdown menu, do one of the following actions:<br>— Select a predefined entity and enter your search query.<br>— Select **Custom query** and enter a custom search query.<br>If you select **L3 Switch Manager** or **Custom search**, you can use **Add exception** to apply the intent on one or more excluded entities.<br>For example, if you use the custom query to select Node Ids that are lesser than 3, you can add an exception on Node Id 1. Then, the defined intent applies on all Node Ids except ports 0 and 2. |

   b) To add another condition, Click **Add Condition**.
6. Under **Alert Notification**, select how you want to get the alert notifications.

   You get the alert notification on the configured email address or SNMP Trap destination. You can select multiple check boxes as per your requirement.
7. Click **Submit**.

   You can view the new intent on the Intents page.

When you create an intent, VCF Operations for networks immediately verifies whether the conditions defined in the intent are met in your network. If the intent is violated, the system raises an alert corresponding to this problem and marks the intent status as Fail.

## Supported Intent Types

VCF Operations for networks supports the following intent types.

| Group (Category) | Intent Type | Name | UI Name | Severity | Virtual/Physical | Description |
|---|---|---|---|---|---|---|
| STIG | Account Password Protection | Account Not Password Protected | Administrative account access is not password protected on the following devices. | High | Physical | The network device must be password protected for administrative access. |
| | Console Access Password Protection | Console Access Not Password Protected | Console port access is not password protected on the following devices. | High | Physical | The network device must require authentication for console access. |
| | Default Password Existence | Default Password Existence | Default manufacturer password is used on the following devices. | High | Physical | The network device must not have any default manufacturer passwords. |
| | Management Connection Password Protection | Management Connection Not Password Protected | Management port access is not password protected on the following devices. | High | Physical | The network device must require authentication prior to establishing a management connection for administrative access. |
| | Plaintext Password Visibility | Plaintext Password Visibility | Plaintext passwords are visible on the following devices. | High | Physical | The network device must not have plaintext passwords. |
| Network Health | Duplex Mismatch | Duplex Mismatch | Duplex configuration does not match on the following ports. | Critical | Physical, Virtual | Port duplex configuration of the ports on each link should match. |
| | Duplicate IP Address | Duplicate IP Address | Duplicate IP address has been configured on the following interfaces. | Critical | Physical | Duplicate IP address should not be configured on multiple interfaces. |
| | Duplicate MAC Address | Duplicate MAC Address | Duplicate MAC address has been configured on the following interfaces. | Critical | Physical | Duplicate MAC address should not be configured on multiple interfaces. |
| | HSRP/VRRP Configuration Error | HSRP/VRRP Configuration Error | HSRP Configuration contains the following error. | Critical | Physical | Check for HSRP/VRRP configuration mismatch between Active and Standby. |
| | Loop Detection | Loop Detection | Network contains the following loop. | Critical | Physical, Virtual | Network should be loop free. |
| | Native VLAN Mismatch | Native VLAN Mismatch | Native VLAN configuration does not match on the following ports. | Critical | Physical | Native VLAN configuration of the ports on each link should match. |
| | Native VLAN Tagging Mismatch | Native VLAN Tagging Mismatch | Native VLAN Tagging does not match on the following ports. | Critical | Physical | Native VLAN tagging of the ports on each link should match. |
| | Port Channel Member Mismatch | Port Channel Member Mismatch | Port channel member ports should not connect to non-member ports on linked devices. | Critical | Physical | Port channel member ports should not connect to non-member ports on linked devices. |
| | Port Mode Mismatch | Port Mode Mismatch | Port mode configuration does not match on the following ports. | Critical | Physical | Port mode configuration should match on the ports of each link. |

| Group (Category) | Intent Type | Name | UI Name | Severity | Virtual/Physical | Description |
|---|---|---|---|---|---|---|
| | Reachability | Reachability Failure | Network endpoints should be reachable. | Critical | Physical, Virtual | Network endpoints should be reachable. |
| | STP Path Cost Method Inconsistency | STP Path Cost Method Consistency | Inconsistent STP path cost methods have been configured on the following switches. | Moderate | Physical | STP path cost calculation methods should be consistent among switches. |
| | Segmentation | Segmentation Failure | Network endpoints should be segmented. | Critical | Physical, Virtual | Network endpoints should be segmented.<br>**Note:** Segmentation Intent verifies that specified source cannot communicate with destination, even using spoofed source IP addresses. |
| | Trunk VLAN Mismatch | Trunk VLAN Mismatch | Allowed VLANs configuration does not match on the following trunk ports. | Critical | Physical, Virtual | Allowed VLANs configuration should match on the ports of each trunk link. |
| Device Health | HSRP/VRRP Active STP Root Colocation | HSRP/VRRP Active STP Root Colocation | HSRP/VRRP Active should be colocated with STP Root, if both protocols are enabled. | Moderate | Physical | HSRP/VRRP Active is not colocated with the following STP Root. |
| | MTU Mismatch | MTU Mismatch | MTU configuration of the ports on each link should match. | Moderate | Physical, Virtual | MTU configuration of the ports on each link should match. |

**Note:**

- STIG intents are supported for the following devices only:
  - Cisco ASA, Cisco ASR 1000, Cisco Catalyst, Cisco ISR 4000, and Cisco Nexus
  - Palo Alto
- The STP Path Cost Method Inconsistency intent is supported only for Cisco Catalyst and Cisco Nexus devices.
- If a device is configured with port channel sub interfaces (that are associated with different VLANs), or configured with port mode and allowed VLANs that are configured at the port channel level, such configurations are considered while performing intent analysis for the following devices only:
  - Arista switches
  - Dell EMC PowerSwitch S5200 (running on OS10)

## Supported Application Intent Types

In VCF Operations for networks, you can view various types of application intents.

The supported application intents are:

| Group (Category) | Intent Type | Severity | Virtual/Physical | Description |
|---|---|---|---|---|
| Application | Application Flow Health | Critical | Physical, Virtual | Monitors flow health of application at tier level. |

| Group (Category) | Intent Type | Severity | Virtual/Physical | Description |
|---|---|---|---|---|
| | Application Flow Utilization | Critical | Physical, Virtual | Monitors flow utilization of application at tier level. |

**Note:**

- Once you add the application intents, it can take upto 26 hours for the intent violations to get generated.
- VCF Operations for networks removes all system generated application threshold configurations and the alerts related to those threshold configuration.
- For applications that had threshold enabled before VCF Operations for networks version 6.8 upgrade, VCF Operations for networks automatically enables intents for these applications after the upgrade.
- The applications that did not have any thresholds enabled before the VCF Operations for networks version 6.8 upgrade, those applications are moved to the intents exclusion list after the upgrade.

# Supported Firewalls

VCF Operations for networks supports several firewall.

## Cross vCenter NSX

In a cross-vCenter NSX environment, you can have multiple VMware vCenter servers, each of which must be paired with its own NSX Manager.

One NSX Manager is assigned the role of primary NSX Manager, and the others are assigned the role of secondary NSX Manager. The primary NSX Manager is used to deploy a universal controller cluster that provides the control plane for the cross-vCenter NSX environment. The secondary NSX Managers do not have their own controller clusters. The primary NSX Manager can create universal objects, such as universal logical switches. These objects are synchronized to the secondary NSX Managers by the NSX Universal Synchronization Service. You can view these objects from the secondary NSX Managers, but you cannot edit them there. You must use the primary NSX Manager to manage universal objects. The primary NSX Manager can be used to configure any of the secondary NSX Managers in the environment.

The following Universal objects are supported:

- Universal LDR
- Universal Transport Zone
- Universal Logical Switch
- Universal Firewall Rule
- Universal Security Group
- Universal IPSets
- Universal Service
- Universal Service Groups
- Universal Segment Range

## Palo Alto Firewall

VCF Operations for networks displays Palo Alto physical devices in VM-VM path, if the device is available in path.

When you click the device icon in the VM-VM path, you see basic information such as interfaces, routes, and applicable Palo Alto firewall rules.

You can create and register multiple service definitions in Panorama with a given NSX manager. When different ESX clusters have workloads that require the VM Series firewall to handle traffic differently, then multiple service definitions are created. Each service definition has an associated device group from which the policies are picked.

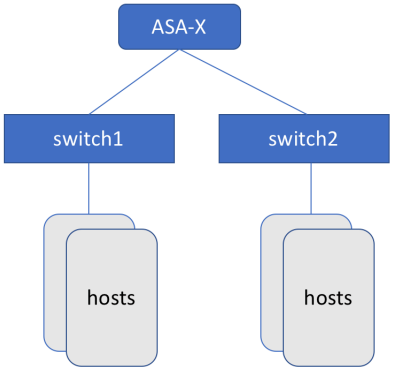
VCF Operations for networks supports Palo Alto policies defined both local to the device, and through Panorama. However, it does not support Inter-VSYS routing.

## Cisco ASA Firewall

VCF Operations for networks supports Cisco ASA firewall.

The features for Cisco ASA firewall are as follows:

- VCF Operations for networks supports only Cisco ASA-X series.
- VCF Operations for networks does not support Firepower modules.
- Currently, VCF Operations for networks supports Cisco ASA operating system version 9.4.
- VCF Operations for networks does not support the cluster deployment of Cisco ASA.
- VCF Operations for networks does not support the high availability of Cisco ASA.
- VCF Operations for networks does not support Cisco ASA if it is directly connected to the host. A topology that is similar to the following example is supported:




- Cisco ASA access rules of only `Extended` type are supported. Other access rule types like `Standard`, `WebType`, `EtherType`, and so on are not supported.
- The Cisco ASA firewall in the VM-to-VM path does not display applicable access rules if the firewall is configured in the `Transparent` mode.

You can perform a query for all the Cisco ASA entities that are supported by VCF Operations for networks.

| Entities in Cisco ASA | Keywords | Sample Queries |
|---|---|---|
| Security Context | ASA Firewall<br>ASA Security Context | `asa firewall where access group = <>` |
| Access Rule | ASA Access Rule | `asa access rule where source ip = <>`<br>`asa access rule where destination ip = '192.168.2.2'`<br>`asa access rule where port = <>`<br>`asa access rule where interface = <>` |
| Access Group | ASA Access Group | `asa access group where interface = <>` |
| Network Object / Network Object Group | ASA Network Object<br>ASA Network Object Group | `asa network object where ip address = <>`<br>`asa network object group where ip address = <>` |
| Service Object / Service Object Group | ASA Service Object<br>ASA Service Object Group | `asa service object where port = <>`<br>`asa service where protocol = <>`<br>`asa service object group` |

## Check Point Firewall

The Check Point Management Server should accept API access from the Collector IP address.

You can set up the access from Check Point SmartConsole application. Go to **Manage & Settings** > **Blades**, and in the **Management API Setting** window, select **All IP addresses**.

If Check Point MDS is added as data-source, VCF Operations for networks fetches data from all the user-defined domains and the global domain.

VCF Operations for networks uses Check Point public Web API for fetching the data from the Check Point management server. If the VSX gateway is attached to the management server, we use SSH-based CLI commands to fetch the VSX-managed Virtual System VS routing table to support display of the VS gateway in the VM-VM path.

You can perform a query for all the Check Point entities that are supported by VCF Operations for networks. All the entities are prefixed by `Check Point`.

**Table 1333: Sample Queries for Check Point**

| Entities in Check Point | Keywords | Queries |
|---|---|---|
| IPset | `Check Point Address Range`<br>`Check Point Network` | `vm where Address Range = <>`<br>`vm where Address Range = <>`<br>`Check Point Address Range where Translated VM = <>` |
| Grouping | `Check Point Network Group` | `Check Point Network Group where Translated VM = <>`<br>`vm where Network Group = <>` |
| Service/ Service Group | `Check Point Service`<br>`Check Point Service Group` | `Check point service where Port = <>`<br>`Check point service where protocol = <>` |
| Access Layer | `Check Point Access Layer` | `Check Point Policy where Access Layer = <>` |
| Domain | `Check Point Domain` | `check point domain where ip address = <>`<br>`check point policy where domain = <>`<br>`check point access layer where domain = <>` |
| Gateways and Gateway Cluster | `Check Point Gateway`<br>`Check Point Gateway Cluster` | `Check Point Gateway Cluster where Policy Package = <>` |
| Policy Package | `Check Point Policy package` | `Check Point Policy where Policy Package = <>`<br>`Check Point Policy Package where Rule = <>` |
| Policy | `Check Point Policy` | `Check point policy where source ip = <> and Destination IP = <>`<br>`Rule where source ip = <> and Destination IP = <> (will display other rules- nsx, redirect along with check point policies in the system)` |

If the Check Point device is present in the VM-VM path, you can see the device's physical gateways and virtual system in the VM topology. When you click the device icon, you can see basic information such as interfaces, routes, and applicable Check Point firewall rules.

> **Note:**
> For the VM-VM path, VCF Operations for networks does not support the VSX cluster containing Virtual Switch and Virtual Router.

Here are some scenarios for which the system alerts are generated for Check Point:

- The NSX fabric agent is not found on the ESX for the Check Point gateway.
- The Check Point service VM is not found.
- The Check point gateway `sic` status is not communicating.
- The discovery and update alerts features for the Check Point entities like address range, networks, policies, groups, policy package, service, service group, and so on.

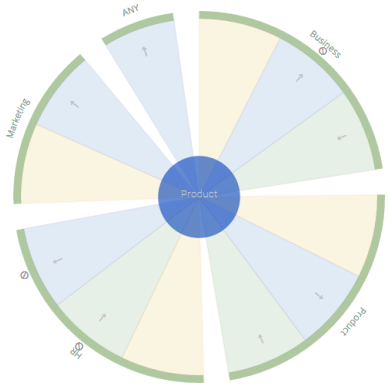
## Security Groups

Security Groups are a set of groups that are managed through a common set of permissions.

The Security Group topology has the following two views:

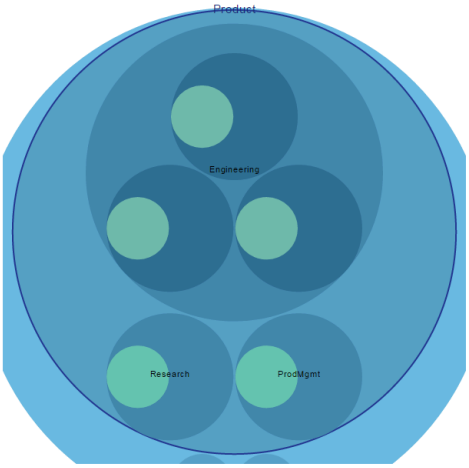
### Firewall View

The Security Group firewall topology displays the relation between the selected Security Group and other Security Groups by showcasing the firewall rules that are applicable between the Security Groups.




### Container View

The Security Group container topology displays how the Security Group is structured with respect to its parent Security Groups or children (Security Groups or other entities).




## Policy-Based VPN

VCF Operations for networks supports policy-based VPN in VMware Cloud on AWS and NSX-T. The following scenarios are supported for the policy-based VPN:

- VPN tunnel between the VMware Cloud on AWS public IP address and the NSX-T/AWS public IP address

- VPN tunnel from the VMware Cloud on AWS public IP address and the corporate firewall public IP address to a 1:1 NAT between the corporate firewall public IP address and the internal NSX Edge

  **Note:**
  VCF Operations for networks does not support the scenario of the VPN tunnel from the VMware Cloud on AWS ending on a corporate firewall and no NAT configured with the internal NSX edge.

**Policy-Based VPN Entities**

VCF Operations for networks fetches data for the `L3 VPN Session` entity which is the actual VPN configured in the data center.

Here are the search terms for the policy-based VPN entities:

**Table 1334:**

| Search terms | Description |
|---|---|
| `Policy based VPN` | All policy-based VPN sessions for VMware Cloud on AWS and NSX-T |
| `VMC Policy based VPN` | VMware Cloud on AWS policy-based VPN sessions |
| `NSX-T Policy based VPN` | NSX-T policy-based VPN sessions |
| `NSX Policy based VPN` | NSX policy-based VPN sessions |

## vDefend Distributed Firewall Inactive Rules

VCF Operations for networks supports the visibility of the vDefend distributed firewall rules for which there have been no flows for some time. These rules are known as inactive rules. Such rules use memory heap and can cause security issues. To monitor these inactive rules, VCF Operations for networks provides the following two widgets in the **Security** dashboard:

**Note:**
To view the Security dashboard, enter **Security** in the search bar.

- Unused vDefend Firewall Rule: This widget lists all the NSX firewall rules where no flow is reported on the given time. You can also use the following search query to retrieve these rules:
  `nsx firewall rule where flow is not set`

  **Note:**

  Ensure that you have enabled vDefend Distributed Firewall IPFIX for the specified time.

## Fortinet Firewall

In VCF Operations for networks, you can view insight about Fortinet firewall.

VCF Operations for networks supports the following Fortinet entities -

- Fortinet Manager
- Fortinet ADOM - Fortinet Administrative Domain details
- Fortinet VDOM - Fortinet Virtual Domain details. VCF Operations for networks supports only Flow-based filtering is supported. Transparent mode is not supported.
- Fortinet Address - List of ADOM specific Addresses. VCF Operations for networks support ipmask, iprange, and NSX fabric connectors.
- Fortinet Address Groups - List of ADOM specific address groups
- Fortinet Dynamic Addresses - List of ADOM specific dynamic addresses (VDOM Mapped Addresses)
- Fortinet Dynamic Address Groups - List of ADOM specific dynamic address groups (VDOM Mapped Address Groups)

- Fortinet Dynamic Interfaces - List of ADOM specific dynamic interfaces.
- Fortinet Zones - List of ADOM specific zones.
- Fortinet Services - List of manual and auto generated services for each ADOM.
- Fortinet Service Groups - List of service groups for each ADOM.
- Fortinet Policy - Fortinet Policies for each ADOM. We currently support only IPv4 policies, Fortinet Global Header Policies, and Fortinet Global Footer policies.
- Fortinet Policy Packages - List of Policy packages. The policy packages name also contain the path to the policy package preceding the name of package.
- Fortinet Devices - List of Fortinet devices that are associated with the FortiManager.
- Fortinet Device Groups - List of Fortinet Device Groups specified by the user.

The followings are not supported:

- VM to VM path for physical devices in transparent mode.
- Advanced (non-IP based) policy properties like User, User Group, Application, and Security Profile.

## Export Firewall Rules by Running CLI Commands

You can export the firewall rules in VCF Operations for networks to a CSV file.

1. Log in to the VCF Operations for networks Platform CLI as a **Support** user.
   Provide your CLI login credentials using the SSH client.
2. Switch to a **Ubuntu** user.
   ```
   ub
   ```
3. Invoke the RDB tool.
   ```
   rdb
   ```
4. Export the firewall rules in VCF Operations for networks to a CSV file using the `run_fw_rule_export -p` command.
   ```
   run_fw_rule_export -p <new csv file with directory path>
   ```
   For example, run the following command to export all firewall rules to a file named `firewallRules_rdb.csv` file in the `tmp` directory:
   ```
   run_fw_rule_export -p /tmp/firewallRules_rdb.csv
   ```
5. Navigate to the directory in which you have saved the CSV file.
   ```
   cd /<directory path>
   ```
6. Change the owner and group of the CSV file to `support`.
   ```
   sudo chown support:support <file name>
   ```
   For example, run the following command to change the owner and group of the `firewallRules_rdb.csv` file to `support`:
   ```
   sudo chown support:support ./firewallRules_rdb.csv
   ```
7. Copy the CSV file to your local machine.
   For example, you can run the `scp` command to copy the CSV file to a mac machine.
   ```
   scp support@10.79.191.187:/tmp/firewallRules_rdb.csv ./
   ```
8. Open the CSV file on your local machine.
   An example CSV file with the firewall rules:

| A | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|
| Rule Name | Sequence | Manager | Entity Type | Firewall Type | Sources | Source Vms | Source SG Members |
| Default LR Layer3 | 1 | 10 | NSX-T Firewall Rule | Edge Firewall | Any | | |
| To_test_tenant2_ | 2 | 10 | NSX-T Firewall Rule | Edge Firewall | Any | | |
| default_blacklist_r | 3 | 10 | NSX-T Firewall Rule | Distributed Firewall | Any | | |
| direct-host-rule-in | 4 | 10 | NSX-T Firewall Rule | Distributed Firewall | Any | | |
| fw_VM2-ten2-hos | 5 | 10 | NSX-T Firewall Rule | Distributed Firewall | | | |
| direct-host-rule-ou | 6 | 10 | NSX-T Firewall Rule | Distributed Firewall | direct-host-ipset | | |
| TEST1 | 7 | 10 | NSX-T Firewall Rule | Distributed Firewall | Any | | |
| Between tenants p | 8 | 10 | NSX-T Firewall Rule | Distributed Firewall | Tenant_Two | | tenant2-ls |
| 16_Aug_Rule_with | 9 | 10 | NSX-T Firewall Rule | Distributed Firewall | IP_Of_VM-web-VM-1-Host-212 | | |
| 16_Aug_Rule2 | 10 | 10 | NSX-T Firewall Rule | Distributed Firewall | ls-app-tenant1 | app-VM1-host133 | |
| 16_Aug_Rule1 | 11 | 10 | NSX-T Firewall Rule | Distributed Firewall | ubuntu-133-2/ubuntu-133-2.vm | web-VM-3-host133 | |

# Working with Micro-Segmentation

VCF Operations for networks provides planning and recommendations for implementing the micro-segmentation security. It helps the user to manage and scale the VMware NSX deployments quickly and confidently.

## Analyzing the Application

The micro-segmentation planning topology shows all the flows that are present in your environment by dividing the flows into segments.

In VCF Operations for networks, a flow is a 4-tuple. It includes:

- Source IP
- Destination IP
- Destination port
- Protocol

You can view the data in two formats: Micro-segment wheel View and the Grid View

### Analyze Flow to Perform Security Planning

With **Security Planning**, you can analyze flows to view segmentations and traffic flow patterns in your environment for better security planning.

1. In the left navigation pane, click **Plan & Assess** > **Security Planning**.
   You see **Security Planning** window.
2. From the **Scope** drop-down menu, select one of the following options:

| Option | Action |
|---|---|
| **All flows** | Select to analyze all the flows available in your environment. |
| **Entities** | Select to analyze flows of a particular entity.<br>Click **Entities** and from the drop-down menu select the entity type.<br>You can also use the **Search entity by name** text box to find and select multiple entities.<br>When you click the **Search entity by name** text box, you can see the name of all the available entities for the selected entity type. |
| **Between entities** | Select to analyze flows between two entities.<br>1. From the **Entity type 1** drop-down menu, select an entity type. Also, you can use **Search entity by name** text box to enter the entities names.<br>You can enter multiple entities in the **Search entity by name** text box. |

| Option | Action |
|---|---|
| | 2. From the **Entity type 2** drop-down menu, select an entity type. Also, you can use **Search entity by name** text box to enter the entities names.<br>You can enter multiple entities in the **Search entity by name** text box. |
| **Flows matching properties** | Select to analyze flows that match a specific condition. You can add multiple conditions by clicking **ADD CONDITION**. |
| **Custom search** | Select to analyze flows that match a custom search query. |

3. From the **Duration** drop-down menu, select the time frame for which you want to analyze the flows. The system analyzes the flows that occurred between that specific time frame.
4. From the **Ports** drop-down menu, select one of the following:

| Option | Action |
|---|---|
| **All** | Select to analyze all ports available for the selected scope. |
| **Include Ports** | Select to include specific ports that you want to analyze. Enter the names of the ports that you want to analyze in the **Search Ports** text box. |
| **Exclude Ports** | Select to exclude specific ports that you do not want to analyze. Enter the names of the ports that you do not want to analyze in the **Search Ports** text box. |

   **Note:**
   If you have selected **Flows matching properties** and **Custom search** in **Scope**, you do not see **Ports** drop-down menu.

5. The **Group by** drop-down menu is auto-filled depending on what you select as **Scope**.
   If necessary, you can change the **Group by** by selecting appropriate option from drop-down menu.
6. Click **ANALYZE**.

## Viewing Micro-Segmentation And Flow Data in Wheel View

In the Segments Wheel view, the blue lines denote the outgoing flows, the yellow lines denote the incoming flows, and the green lines denote the flows that are bidirectional.

Click any of the segments to view its details, and to modify the scope, click **Modify Scope** in the top panel.

The VMs that are outside the selected scope are grouped as **Other Entities** in the micro-segmentation planning topology.

For better visibility and ease of use, you can only see 10 segments at a time. If you have more than 10 segments, then the extra segments are hidden under one large segment located next to **Other Entities**. To see the list of hidden segments point to the large segment next to **Other Entities** and you see the list of hidden segments.

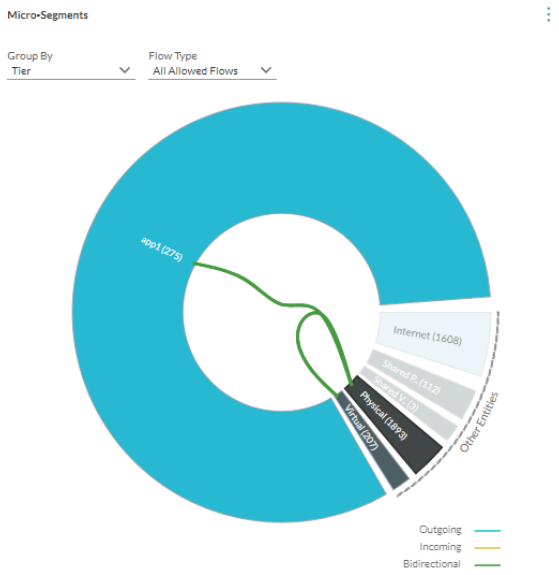
You can also hide the visible segments. To hide a segment point to the segment you want to hide, and click **HIDE**. To unhide a hidden segment, point to the large segment next to **Other Entities**, you see the list of hidden segments, click **SHOW** for the segment you want to unhide.

**Note:**
If you have more the 10 visible segments, then you can not unhide the hidden segments.

You can also analyze the flows by creating subgroups as per Physical, Other Virtual, and Internet categories.

| Group By | Also show groups for |
|---|---|
| VLAN/VXLAN | All |
| Application | Physical |
| ✓ Tier | Virtual |
| Subnet | Internet |
| Folder | ✓ None |
| Cluster | |
| VM | |
| Port | |
| Security Tag | |
| Security Group | |
| IPSet | |
| VPC | |

Each group is expanded into a wedge. In the following topology, the wedge for **Physical group** is seen.




The Flows pin shows that the flows for different time intervals separated by ports. You can either view all the flows or view the flows between two entities. You can filter the flows by Allowed and Blocked flows. You can view flows by either Total

Bytes or by Allowed Session Count. For the flows that are protected by a firewall, a Protected by Firewall sign is used to denote that the flows in that port that are protected by a firewall.

The planning for a scope such as an entire data center or a cluster selects flows that have VMs or Physical Servers (identified by the Physical IPs) as the source or the destination.

A topology has two distinct zones:

- Internal: This zone includes the VMs or the IP addresses in the scope.
- External: This zone includes the VMs or the IP addresses that are out of scope but talk to the VM or IP addresses in the internal zone. The external zone consists of the following wedges:
  - DC Virtual: It includes the source or the destination data center internal VMs that are talking to VMs or IP addresses in the internal zone and are not hosting any well-known shared services such as LDAP, NTP, and so on.
  - Shared Virtual: It includes the destination data center internal VMs hosting well-known shared services such as LDAP, NTP, and so on to which the VMs or IP addresses in the internal zone are talking.
  - DC Physical: It includes the source or the destination data center internal physical IP addresses that are talking to VMs or IP addresses in the internal zone and are not hosting any well-known shared services like LDAP, NTP, and so on.
  - Shared Physical: It includes the destination data center internal Physical IP addresses hosting well-known shared services such as LDAP, NTP, and so on to which the VMs or IP addresses in the internal zone are talking.
  - Internet: It includes the source or the destination data center external VMs or the physical IP addresses that are talking to the VMs or IP addresses in the internal zone.

**Note:**

- Data center Internal implies RFC 1918 designated IPs by default + any overrides defined in E-W settings.
- Data center External implies non-RFC 1918 designated IPs by default + any overrides defined in N-S settings.

## View Micro-segmentation And Flow Data in Grid View

VCF Operations for networks enables you to see the communication between objects in a tabular or grid view.

1. Navigate to **Security** > **Plan Security** and click

2. Select a value for **Group By** option, for example `VMs`, `Application`, `Security Groups` to see the corresponding details in the tabular format.

| Field Name | Description |
|---|---|
| Source Object | Name of the source |
| Destination Object | Name of the destination |
| Related Flows | Count of communication or flows between the source and destination<br>Click on the count value to see the related flow details. |
| Sum of Bytes | Aggregated number of bytes between all the flows |
| Max of Traffic Rate | Maximum traffic rate observed among all related flows |
| Count of Sessions | Number of active sessions for the particular flow |

**Note:**

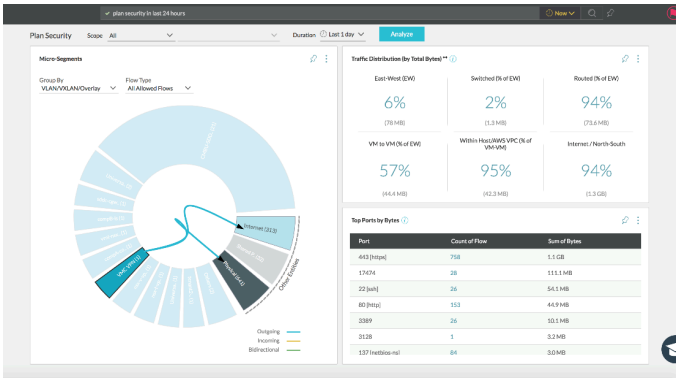
- You can click on each column header to sort the data in ascending or descending order.
- You can hide the field from the table view, click the more icon next on the field header and deselect the field name.

3. In addition, you can perform several actions on the grid view page.
   - In the filters pane on the left side of the screen, you can perform the following actions:
     - Select an individual source or destination to filter the flows that are related to selected source or destination object.
     - Select the firewall action to see the allowed flows or the dropped flows.
     - Select the protection status to see the flow status.
   - Click **Add more filters** to add additional filters.
   - To export the tabular data in a CSV format, click the more option on top of the table, and select **Export as CSV**.

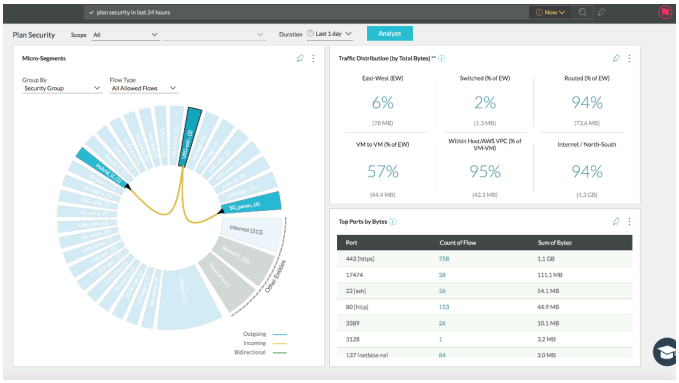
### VMware Cloud on AWS: Planning and Micro-Segmentation

You can plan for a specific VMware Cloud on AWS segment by selecting **VMC Segment** as the scope in the **Plan Security** page.

For the policy segments, use the `VLAN/VXLAN/Overlay` clause in the group.




For the policy groups, use the `Security Group` clause in the group.




## Working with Crown Jewel Analysis

In VCF Operations for networks, you can perform crown jewel analysis to identify possible lateral movement opportunities for an adversary based on existing flow paths between various entities in your network.

You can choose to add your most critical VMs and physical IPs as crown jewels. After you add crown jewels in VCF Operations for networks, you can analyze and view existing flow related information, relationships between the entities based on flows, number of entities directly connected to your crown jewels, total incoming traffic flow to your crown jewels, and so on.

## Add a Crown Jewel

When you add crown jewels in VCF Operations for networks, you can analyze and view existing flow related information, relationships between the entities based on flows, number of entities directly connected to your crown jewels, total incoming traffic flow to your crown jewels, and so on.

1. From left navigation page, click **Plan & Assess** > **Crown Jewel Analysis** > **Add Crown Jewels**.

   **Note:**
   If any crown jewels exist in your environment in VCF Operations for networks, then you can add crown jewels from the **Manage Crown Jewels** page.

2. In the **Add Crown Jewels** page, select VMs or IPs you want to add as crown jewels.

   You can add multiple VMs or IPs as crown jewels. You can add either VMs and IPs that are auto populated by the system or use the custom VM or the custom IP search to add VMs and IPs as crown jewels.

   **Note:**
   You can add up to 25 entities as crown jewels.

3. Click **Submit**.

You can Perform Crown Jewel Analysis or Managing Crown Jewels.

## Perform Crown Jewel Analysis

Perform crown jewel analysis to get insights about the possible reachability of your crown jewels due to lateral movements by an adversary. The reachability score gives an estimate of how far your crown jewels are from various entities and possibly how many lateral moves are required by an adversary to reach your crown jewels. Also, using crown jewels analysis, you get firewall rules suggestions to strengthen the security of your crown jewels assuming that the current state of flows is legitimate.

Verify that you have added at least one crown jewel in VCF Operations for networks.

1. From left navigation page, click **Plan & Assess** > **Crown Jewel Analysis**.
2. In the **Crown Jewel Analysis** window, select the crown jewels you want to analyze.

   **Note:**
   You can select up to 10 crown jewels for analysis.

3. Select **Scope** and **Duration** for your analysis.
4. Click **Analyze**.

The **Crown Jewels Analysis** page shows the total connected entities to the crown jewels, reachability score, incoming flows, total incoming traffic, and so on.

You can toggle between topology view and grid view. Click

to toggle between grid view and topology view.

The reachability score gives an estimate of how far your crown jewels are from various entities and possibly how many lateral moves are required by an adversary to reach your crown jewels. For example, reachability score one estimates that the crown jewel is one step away from the entities at node level 1.

You can also generate a PDF version of the analysis report. Click the **Generate VNA Report** to generate a PDF version of the analysis report.

## Managing Crown Jewels

You can add, delete, analyze crown jewels from the **Manage Crown Jewels** page.

In the **Manage Crown Jewels** page, you see a list of all crown jewels available in VCF Operations for networks. You can search crown jewels, add crown jewels, and preform crown jewel analysis from this page.

# Working with Applications

An application is a collection of tiers. Each tier in an application is a collection of VMs, Kubernetes services, and physical IPs.

You can create a group of tiers and visualize traffic or flows between:

- Applications
- Tiers of the same application
- Shared services
- Internet

You can create or add an application into VCF Operations for networks in three ways:

- Manually on the User Interface
- Public API
- Application Discovery

**Note:**

When you upgrade to VCF Operations for networks version 6.8, application alerts raised due to analytics threshold configuration are closed automatically. If the alerts persist, they are raised as intent alerts by the system after 24 hours of the upgrade.

## Create an Application Manually

You can manually create an application in the VCF Operations for networks user interface.

1. On the left navigation pane, click **Applications** > **Add Applications** > **Add Manually**.
2. In the **Application Name** text box, enter a name for the application you want to create.
3. In the **Tier/Deployment** section, enter a unique name.
4. In the **Member** field, perform the following steps:
   a) Select a condition from the drop-down menu to create a tier.

      You can define a condition based on any of the following properties:

      - VM properties
      - Location of VMs (application, cluster, folders).
      - Kubernetes services (service name, cluster IP address, namespace, cluster IP, or service labels).

      To learn more about the supported properties while defining a condition, see List of Supported Properties.

      To create tiers for the physical IPs based on enriched DNS, Subnet, or VLAN, see Creating Tiers for Physical IPs.

      To search a specific Kubernetes service that has the same name, same IP, or the same tag in multiple clusters, use custom search.
   b) Enter or select the value that you want to add to the tier.

      For entering multiple values, use comma after individual values.
5. Optional: To add another condition, click **Add another Condition**.
6. Optional: To create another tier under an application, click **Add Tier/Deployment**.

   You can create multiple tiers for an application.

   The application creates the tiers and shows the count of VMs, physical IPs, and Kubernetes services matching the conditions.
7. To receive alerts on intent violations, select the **Enable Application Aware Network Health Intents** check box.

   The system monitors the network health of the application and generates alerts if the application health intents are violated.

8. Optional: To preview the flow analysis, on the **Preview Micro-Segments** widget, click **Update view**.
   You see the micro-segment view for the application.
9. Click **Save**.

You can see the application listed on the **All Applications** page. To learn more, see Viewing All Applications .

### Creating Tiers for Physical IPs

While creating an application, you can select **Custom IP Search** from the drop-down list to create tiers for the physical IPs based on the enriched fields. For more information on the enriched fields, refer Enriching Flows and IP Endpoints.

The enriched DNS, Subnet, VLAN information can be used in specifying tiers as follows :

- Web
  `Query: IP Endpoint where Subnet Network = '172.16.101.0/24'`
- App
  `Query: IP Endpoint where Dns Domain = app.example.com`
- DB
  `Query: IP Endpoint where L2 Network = 'vlan-102'`
- Common Services
  `Query: IP Endpoint where Dns Domain = svc.example.com`

### List of Supported Properties

When creating or editing an application, VCF Operations for networks supports specific members.

The following lists shows the supported VM properties for VM members in VCF Operations for networks:

- AWS VPC
- Account
- Application
- Availability Zone
- Azure Application Security Group
- ASG Count
- Cluster
- DVPG
- Datacenter
- Folder
- Host
- IP Address
- IPSet
- IPSet count
- Incoming Port
- L2 Network
- NSGroup
- NSX
- NSX Manager
- NSX Policy Group
- NSX-T IPSet
- Name
- Network Address
- Operating System
- Other Names
- Outgoing Port
- Region

- Resource Pool
- SDDC
- SDDC Group
- Security Group
- Security Group Count
- Security Tag
- Subnet
- Subscription
- Tag
- Tag Key
- Tier
- VLAN
- VXLAN
- vCenter

The following lists shows the supported IP endpoint properties for IP endpoint members in VCF Operations for networks:

- Application
- IP Address
- Incoming port
- L2 Network
- Outgoing port
- Subnet Network
- Tier

The following lists shows the supported Kubernetes properties for Kubernetes services members in VCF Operations for networks:

- Cluster IP
- Cluster Network Address
- External IP
- External Network Address
- Kubernetes Cluster
- Kubernetes Namespace
- Label
- LoadBalancer IP
- LoadBalancer Network Address
- Name
- Node Port
- Port
- Port Name
- Selectors
- Target Port
- Type

## Working with Application Discovery

When you have several applications or when you have multiple tiers in an application, creating applications using the public APIs or the user interface becomes a long process. VCF Operations for networks auto-discovers applications and tiers and enables you to add them automatically, which reduces a lot of manual effort.

In VCF Operations for networks, you can discover applications based on:

- Flows
- VM Attributes (Names, Tags, Security Tags and Security Groups)
- ServiceNow

- Add Manually

  **An example of the application discovery construct by VM Attributes**

  You can discover application using VM Attributes which includes Names, Tags, Security Tags and Security Groups.

  Consider a scenario where you have:

  - Added VMware vCenter server as a data source.
  - Added four VMs in your data center - VM1, VM2, VM3, and VM4.
  - Defined tags (key-value) that defines the application names to which each VMs belong.
  - Defined tags (key-value) that defines the tier to which each VMs belong.

  For example, see the table:

| VM Name | Key-value tags |
|---|---|
| VM1 | • Application Name: MyApplication1<br>• Application Tier: App |
| VM2 | • Application Name: MyApplication1<br>• Application Tier: Web |
| VM3 | • Application Name: MyApplication2<br>• Application Tier: App |
| VM4 | • Application Name: MyApplication2<br>• Application Tier: Web |

**Discovering Applications based on VM Attributes (Tags)**

VCF Operations for networks, you can define a grouping criteria for application discovery for these tags.

> **Note:**
>
> In VCF Operations for networks, NSX tags are displayed as VM tags. Therefore, by default, NSX tags are also supported in application discovery.

In this example, based on the defined tags and grouping criteria, VCF Operations for networks discovers two applications (MyApplication1 and MyApplication2) with two tiers (App and Web) and its related VMs.

| Application | Tier | VM |
|---|---|---|
| MyApplication1 | App | VM1 |
| MyApplication1 | Web | VM2 |
| MyApplication2 | App | VM3 |
| MyApplication2 | Web | VM4 |

**Discovering Applications and Tiers based on VM Attributes (VM Names)**

If the VM names are defined in the following format:

```
ApplicationName : Tier : VMName.
```

For example,

```
MyApplication1 : App : VM1
MyApplication1 : Web : VM2
MyApplication2 : App : VM3
MyApplication2 : Web : VM4
```

When you use the following regular expression (regex), VCF Operations for networks discovers two applications.

- App Regex: `(.*)_(.*)_.*-.*`

- Tier Regex: `(.*)_(.*)_(.*)-.*`

| Application | Tiers and its VMs |
|---|---|
| MyApplication1 | • App and MyApplication1 : App : VM1<br>• Web and MyApplication1 : Web : VM2 |
| MyApplication2 | • App and MyApplication2 : App : VM3<br>• Web and MyApplication2 : Web : VM4 |

**Note:**

Inconsistently defined VM names cannot be grouped for application discovery.

**Discovering Applications Based on Flows**

VCF Operations for networks uses machine learning and statistical analysis to discover applications automatically and group the VMs into their respective applications and tiers. VCF Operations for networks analyzes the network flow between the VMs and groups the VMs by identifying the application and the tier boundaries between these VMs.

The system analyzes the flows on multiple dimensions such as density of the flows between VMs, common open ports between VMs, number of incoming against outgoing connections, configuration data from load balancers, and so on, and groups VMs with high similarity across these dimensions into a unique application. The process repeats again, but only among the VMs that are already grouped into an application, and to group VMs into their respective tiers.

**Note:**

- To discover applications based on flows, you must match the following minimum requirements:
  - Have at least 10 VMs discovered in VCF Operations for networks.
  - Have at least 100 VM-VM flows.
  - Use a platform of XL brick size.
  - To learn about the configuration requirements, see System Recommendations and Requirements.
  - If your configuration does not meet the minimum system requirement, you can either scale up your platform cluster or reconfigure your brick size. To learn about how to scale up the platform cluster, see Expand or Scale Out Clusters. To learn about reconfiguring the brick size, see KB 83374.

  To improve the accuracy of flow-based application discovery, you must add load balancers to VCF Operations for networks. To learn about the supported load balancers, see Supported Load Balancers.

In addition to VCF Operations for networks, you can perform flow-based application discovery from VCF Operations by navigating to **Infrastructure Operations** > **Analyze** and selecting the **Flows** tab.

**Discovering Applications Based on Flows by Uploading a CSV File**

In VCF Operations for networks, you can use the Discovery Options feature to upload a CSV file to discover applications. The CSV file can either be exported from the Configuration Management Data Base (CMDB) or manually created from other tools to improve the Flow based application and tier discovery. The inputs of the CSV file are verified using the Flow based application discovery algorithm and then used to discover applications. The uploaded CSV file can also be used to name applications and tiers.

This feature can enhance the discovery of applications where only some of the application definitions are known, and the application definitions may not be completely accurate. Flow based application discovery can still add VMs or remove VMs using the inputs of the CSV file depending on the actual Flow pattern at the time of discovery.

## Discover Applications

You can discover existing applications based on tags, ServiceNow, names, flows, and advanced properties, and add them into VCF Operations for networks.

1. To discover existing applications in your environment, from the left navigation panel, click **Applications** > **Add Applications**
2. You can discover applications in multiple ways. Select the preferred tab and perform the related steps.

| Tab | Description |
|---|---|
| **Flows** | Use flow based application discovery if you want to discover applications by analyzing the traffic patterns and network flows in your environment.<br>Flow based application discovery is automated and does not require any user input to start the discovery.<br>To learn more, see Discover Applications using Flows. |
| **VM Attributes** | Use VM Attributes based application discovery if your VMs follow a logical naming pattern, or if you have added tags for virtual machines in your VMware vCenter, Azure, AWS, or VMware NSX environments, or if you want to discover applications using a combination of VM names, security groups, tags from VMware vCenter, Azure, AWS, or VMware NSX environments.<br>To learn more, see Discover Applications using Names. |
| **ServiceNow** | Use ServiceNow based application discovery if you have mapped your applications in ServiceNow.<br>To learn more, see Discover Applications using ServiceNow. |
| **Add Manually** | Use add manually to create an application manually by adding tiers and members.<br>To learn more, see Create an Application Manually . |

**Note:**
The following limitations apply when discovering applications in VCF Operations for networks:

| Objects | Maximum limit | Impact if the limit exceeds |
|---|---|---|
| Applications on the map view | 1000 | Applications are not displayed on the map view. |
| Applications on the tabular view | 3000 | Only 3000 applications are displayed on the table view. |
| Saved applications | 10k | You cannot add a new application. |
| Total tiers across applications | 30k | You cannot add a new tier. |
| Tiers per application | 150 | You cannot add a new tier. |
| Members per tier | 5k | You cannot add a new tier. You can only update the existing applications to reduce tiers and members. |
| Members per application | 5k | The application page does not show any useful information such as microsegmentation, metrics, and analytics. To gain complete visibility of the application, you must reduce the application members. Also, VCF Operations for networks raises the following alert:`Application Membership limit exceeded`. |
| Flows per application | 500k | The application topology does not load on the respective application page. |

### Discover Applications using Flows

In VCF Operations for networks, you can discover applications based on flows.

Flow based application discovery uses machine learning algorithms to auto-discover your applications and application boundaries by analyzing the traffic patterns and network flows in your environment. To learn how to discover applications based on flows, see Discovering Applications Based on Flows.

You can also use flow based application discovery by adding applications from VCF Operations by navigating to **Operate** > **Workloads** > **Business Applications** and selecting **Add Applications** > **Based on Flows**.

1. From the left navigation panel, go to **Add Applications** > **Flows**.

   You see all the available applications based on flows. You also see application boundaries and tier boundaries between VMs based on the network flow communication between the VMs.

2. To modify the scope and the naming preference, click **Edit Discovery** and select the duration and the scope of application discovery.

   To learn more about modifying the scope, see Modify Application Scope.

3. To upload a CSV file with pre-existing application definitions, click **Discovery Options**.
   a) Click **Browse** and upload the CSV file.

      The CSV file must have the VM to application name, and the tier name mapping, as the same file is used for application and tier naming.

      **Note:**
      You can also provide regular expressions in the `VM Name` column of the CSV file. For example, to identify all the VM names starting with `MySQL-` for an application, you can provide `MySQL-.*` as the VM name.

   b) Click **Next**.

      VCF Operations for networks validates the uploaded CSV file and shows any missing information as an error followed by a table that lists the information that is missing with the respective row numbers. You can also export the missing information table as a CSV file.

4. Select the naming preference and the order in which you want to name applications and tiers. To learn more, see Set up Application and Tier Naming Preference.

   **Note:**
   VCF Operations for networks recommends the use of naming preferences to create logical and meaningful names for your applications and tiers.

5. Click **Submit**.

6. To pause or stop the auto discovery of flows on the **Discover Applications** page, click the **More Options** icon next to **Edit Discovery**, and select **Pause discovery** or **Stop discovery**.

   **Note:**

   If you click **Stop discovery**, you loose all the discovered applications.

7. Click **Granularity** to group the discovered applications at various levels based on number of tiers and VMs. You can select one of the following **Granularity** levels:

   - **Coarse**: The application group includes more tiers and VMs.
   - **Medium**: The application group includes less tiers and VMs.
   - **Fine**: The application group includes much fewer tiers and VMs.

   In the tabular view of discovered applications, you can see the **Confidence** column which describes the accuracy level of the application grouping. High confidence level indicates that the grouping accuracy is high and requires less user intervention.

View the applications discovered using flows. To learn more, see Viewing Discovered Applications.

### Discover Applications using Tags

You can perform tag based application discovery if you have added tags for virtual machines in your VMware vCenter, Azure, AWS, or VMware NSX environments.

Add tags for virtual machines in your VMware vCenter, Azure, AWS, or VMware NSX environments. To learn how to discover applications based on tags, see Discovering Applications based on VM Attributes (Tags).

1. From the left navigation panel, go to **Applications** > **Discover**.
2. Select the **Tags** tab.
3. Define the scope based on which applications and tiers have to be discovered.
   - Select **All VMs** to see a list of all VMs from all the data sources that are added in VCF Operations for networks.
   - Select **Manual Selection** and filter the VMs based on your requirements such as account, data center, manager, and so on.

     For example, this query limits the discovery of applications to specific regions:

     ```
     VMs where Region in ( 'Central US' , 'East US' )
     ```
4. Group applications using the key-value combination of tags.
   a) Enter a key for the tag.
   b) Optional: Enter a value for the respective key.

   For example, if you enter the key as `Department` and select **Any Value**, VCF Operations for networks will create application boundaries based on all values of the `Department` tag.
5. Group application tiers using the key-value combination of tags.
   a) Enter a key for the tag.
   b) Optional: Enter a value for the respective key.

   For example, if you enter the key as `Role` and select **Any Value**, VCF Operations for networks will create distinct tiers of an application based on all values of the `Role` tag.
6. Optional: Click **Found *count* Applications** to see the list of application names, the VM names and the number of VMs that matches the specified criteria.
7. Optional: Click **Unclassified VMs** to see a list of VMware vCenter, AWS, or Azure VMs that are not following the specified name pattern or tag pattern. You can edit these VMs on the respective environments to fix the name or the tag criteria.
8. Optional: Select the **Save changes to** option to create a template or update an existing template.

   **Note:**
   If you are an admin user, you can update all templates. If you are a member user, you can only edit the templates that you had created.
9. Click **Discover**.

You see the tabular and the hexagonal map view of all applications that matches the criteria.

View the applications discovered by tags. To learn more, see Viewing Discovered Applications.

**Discover Applications using Names**
You can perform name based application discovery if your VMs and Kubernetes services follow logical naming patterns.

Add logical naming patterns to virtual machines in your VMware vCenter, Azure, AWS, or VMware NSX environments.To learn how to discover applications and tiers based on names, see Discovering Applications and Tiers based on VM Attributes (VM Names).

1. From the left navigation panel, go to **Applications** > **Discover**.
2. Select the **Names** tab.
3. Optional: If you have already defined a template using which you want to perform application discovery, select the template from the **App discovery templates** drop-down menu. The application discovery starts automatically.
4. Define the scope based on which you want to discover applications and tiers.
   - Select **All VMs** to see a list of all VMs from all the data sources that are added in VCF Operations for networks.

- Select **Manual Selection** and filter the VMs based on the your requirement like account, datacenter, manager, and so on.

  For example, this query limits the discovery of applications to specific regions:

  ```
  VMs where Region in ( 'Central US' , 'East US' )
  ```

5. Group applications by VM names using regular expressions based pattern matching.
   - Enter a regular expression of your choice.
   - Select **Pattern Builder** or **Advanced** to manually enter a Regular Expression.
     a. Select the reference VM name or select a VM from the list to build a pattern or the regular expression based on the VM name.
     b. Click a position or a group to construct a pattern.

     **Note:**

     After selecting a group, if you select a character or position, VCF Operations for networks ignores your group selection for building the pattern and vice versa.

     Based on your selections, you see the pattern appearing on the screen. And also, you see the list of applications that match the pattern and the count of VMs in the respective applications and the VM names in the respective applications..
     c. Click **Submit**.
6. Optional: Click **Found *count* Applications** see the list of application names, VM names, and the number of VMs that matches the regular expression.
7. Optional: Click **Unclassified VMs** to see a list of VMs that are not following the specified name pattern.
8. Optional: Select the **Save changes to** option to create a template or update an existing template.

   **Note:**
   If you are an admin user, you can update all templates. If you are a member user, you can only edit the templates that you had created.
9. Click **Discover**.

View the applications discovered by names. To learn more, see Viewing Discovered Applications.

**Discover Applications using ServiceNow**
You can perform application discovery based on ServiceNow if you have mapped your applications in the ServiceNow data source that is added to VCF Operations for networks.

You must have added ServiceNow as a data source in VCF Operations for networks.

1. From the left navigation panel, go to **Applications** > **Discover**.
2. Select the **ServiceNow** tab.

You see the applications available on ServiceNow.

View the applications discovered from ServiceNow. To learn more, see Viewing Discovered Applications.

When working with ServiceNow, be aware of the following limitations:

- SNOW app discovery count is limited to now more than 5K apps.
- Traffic, Tag, and ML-based apps are not discovered.
- If there is no dependency map for the SNOW app, it is not discovered.
- If the VMs on which apps run are deleted or the relevant vCenter instance is not added in VCF Operations for networks, the apps are not discovered.
- VCF Operations for networks does not support button-to-top discovery.

**Discover Applications using Advanced Properties**
You can perform an advanced application discovery to discover applications using a combination of VM names and security group tags from your VMware vCenter, Azure, AWS, and NSX environments.

1. From the left navigation panel, go to **Application** > **Add Applications**.
2. Select the **Advanced** tab.
3. Optional: If you have already defined a template using which you want to perform application discovery, select the template from the **App discovery templates** drop-down menu. The application discovery starts automatically.
4. Define the scope based on which you want to discover applications and tiers.
   - Select **All VMs** to see a list of all VMs from all the data sources that are added in VCF Operations for networks.
   - Select **Manual Selection** and filter the VMs based on your requirement like account, data center, manager, and so on.

     For example, this query limits the discovery of applications to specific regions:

     ```
     VMs where Region in ( 'Central US' , 'East US' )
     ```
5. Group applications by different VM properties using regular expressions based pattern matching.
   - Enter a regular expression of your choice.
   - Click **Pattern Builder** to define patterns using groups or position of characters in strings.
     a. To build a pattern or the regular expression based on the VM name, select the default VM name or select a VM from the list.
     b. Click on a position or a group to construct a pattern.

     **Note:**

     After selecting a group, if you select a character or position, VCF Operations for networks ignores your group selection for building the pattern and vice versa.

     Based on your selections, you see the pattern appearing on the screen. You also see the list of applications that match the pattern, the count of VMs in the respective applications, and the VM names in the respective applications.
     c. Click **Submit**.
6. Optional: Click **Found *count* Applications** to see the list of application names, VM names, and the number of VMs that matches the regular expression.
7. Optional: Click **Unclassified VMs** to see a list of VMs that are not following the specified name pattern.
8. Optional: Select **Save changes to** to create a template or update an existing template.

   **Note:**
   If you are an admin user, you can update all templates. If you are a member user, you can only edit the templates that you had created.
9. Click **Discover**.

View the discovered applications. To learn more, see Viewing Discovered Applications.

## Viewing Discovered Applications

When you perform application discovery using tags, names, ServiceNow, flows, or advanced properties, VCF Operations for networks displays the discovered applications that match the criteria in a tabular view and a map view.

The map and the tabular view are interactive. If you click an application in the tabular view, the hexagon representing the application gets highlighted or focused on the map view, displaying shared services and network connections with other applications.

### Tabular View of Applications

In the tabular view, you can do the following:

- View details such as the application name, count of tiers and VMs, and status. If you discover applications using flows, you also see the **Confidence** column indicating the accuracy of discovered applications.

- Click the **Save** icon next to the application to save the application.
- Select one or more applications and click the **Actions** drop-down menu to save selected applications.
- Select one or more applications and click the **Actions** drop-down menu to merge or save applications.
  Select two or more applications, click the **Actions** drop-down menu, and select the **Merge & Save** option to merge selected applications and save them as a single application. You do this action only if your application is discovered as multiple applications and you want to merge them back to a single application.

  **Note:**
  You can merge a maximum of 20 applications.
- If you discover applications using ServiceNow or flows, you can export one or more applications in `.csv` format.

**Map View of Applications**

In the map view, you can do the following:

- Hover over a hexagon to see details such as the application name, discovered VM count, and the tier count.
- On hover, click **Save Application** to save the application.
- Zoom in and zoom out, or move the map to see the applications.
- See applications with unprotected flows.
- See applications talking to the internet.
- See applications using shared services.
- See applications with problems.

**Viewing Discovered Applications by Criteria**

Use the following options that are available on the top of the page to sort, filter, and search discovered applications:

| Option | Description |
|---|---|
| **Granularity** | Displays applications based on size. Fine granularity displays smaller applications and coarse granularity displays only larger applications.<br>**Note:**<br>You see this option only if you have discovered applications using flows. |
| **Sort by** | Sorts applications by name, tiers, and VMs.<br>If you have discovered applications using flows, you can also sort your applications by **Confidence**. |
| **Show** | Displays top 10, 20, 50, or 100 applications on the map and the table view. |
| Search apps, tiers, or VMs | Searches applications by name, tiers, or members. |

## Save Discovered Applications

When you perform an application discovery using tags, names, ServiceNow, flows, or advanced properties, VCF Operations for networks displays all the discovered applications that match the criteria in a table view and a map view. You can save one or more applications from the map view or the table view.

Ensure you have already discovered applications in VCF Operations for networks.

You can choose to save a single application, or save multiple applications in the tabular view.

1. Select the applications you want to save.
   - On the tabular view, select **SAVE** next to the application that you want to save.
   - On the tabular view, select the application that you want to save and click **ACTIONS** > **SAVE**.

- To save multiple applications, on the tabular view, select the check box next to the applications you want to save and click **ACTIONS** > **SAVE**.
- On the map view, hover the mouse on a hexagon and click **Save Application**.

2. In the **Save Application** window, validate the information or modify the existing information such as application name, tier name, or tier members.
3. If you modify the members of the tier, click **Update view** to update the view and analyse the latest flows.

   The micro-segmentation wheel shows the updated view of the application.
4. Click **Save**.

View details of the application by navigating to the application page. To learn more, see Viewing Application Details.

Or, click **Actions** > **Export as CSV** to export the application details in the `.csv` format.

You can define the application count and the fields that you want to export. The application name and the tier name fields are repeated based on the member count (one row per member). The fields that are irrelevant to the application are left empty.

> **Note:**
> You can export the details of applications discovered through flows and ServiceNow to the `.csv` format without saving the applications.

## Merge Discovered Applications

In VCF Operations for networks, you can merge multiple discovered applications into one or more applications. You can merge applications if VCF Operations for networks discovers an application as multiple smaller applications.

Ensure you have already discovered applications in VCF Operations for networks.

1. In the **Discover** tab, from the applications list table, select the applications that you want to merge, and click **ACTIONS** > **Merge & Save**.
2. In the **Merge & Save Applications** window, in the **Application Name** text box, provide a unique name for the application.

   In this page, you can see the application summary which includes the number of tiers, VMs, physical IPs, and services. Also, you see the name of the applications that you are merging.
3. Review the Tier / Deployment details. If needed, you can rename the tiers.

   The application tiers are created based on the number of tiers available in the original applications that you are merging. For example, if you are merging two applications which have one tier each, then in the **Merge & Save Applications** window, you see two tiers. And the system creates a default name for tiers by combining the original application name and tier name. For example, if you are merging two applications, named: App1 and App2, and each of the application consists of one tier, named t1 and t2 respectively. The in the merged application, you see two tiers, named App1_t1 and App2_t2.
4. Deselect the **Enable Application Aware Network Health Intents** check box if you do not want to receive intent based alerts for this application.

   By default, this check box is selected.
5. Optional: To preview the flow analysis, click **Update view** in the **Preview Micro-Segments** widget.
6. Click **Save**.
7. To exit the **Merge & Save Applications** window, click **Cancel**.

## Set up Application and Tier Naming Preference

When you are performing flow-based application discovery, VCF Operations for networks provides options to set up the naming preference and allows you to choose the order of naming your applications and tiers. Setting up naming preferences helps you create logical and meaningful names for your applications and tiers.

By default, VCF Operations for networks names applications and tiers based on the names of the VMs that are part of applications and tiers. If the VM names do not reflect the application or tier then application and tier names do not help you to find the actual applications or tiers.

Therefore, when you perform flow-based application discovery, you can set the naming preference for your applications and tiers. You can also activate or deactive naming preferences, and choose the order of naming the applications and tiers. VCF Operations for networks automatically renames the existing applications and tiers, and names the newly discovered applications and tiers, as per your set naming preference and order.

**Procedure**

1. On the **Discover Applications** page, in the **Scope** bar of the **Flows** tab, click **Edit Discovery**.
2. In the **Edit Discovery** window, click **Application and Tier Naming**.
3. Select the naming preference you want to add.
   You can select one or more of the following naming preferences and arrange them in the order they must be considered:
   - Upload CSV
   - Security tags
   - Security groups
   - Load balancer
   - VM names
   - VM tags

   To arrange the order of the naming preferences, you must use the drag icon.
   VCF Operations for networks renames your applications based on the preferences and the order you select. You can see how your selection changes the application names in the **Tip** section.
4. If you are naming your applications and tiers based on VM names, you can also use regular expressions. You can select the **Regex** option in the **Status** column of **VM Names**, and in the **Application Naming** and **Tier Naming** sections, do one of the following:
   – Select **Regex** and provide a regular expression using which you want to name your applications and tiers.
   – Click **Pattern Builder**, select a sample VM to identify the pattern, choose the position of characters or groups to create the pattern, and click **Submit**.
5. When you select the **Upload CSV** option, the names of VMs, applications, and tiers are taken from the CSV file that was uploaded while discovering applications based on flows.
6. Click **SUBMIT**.

If applications or tiers do not match your selected naming preference, then the system renames applications or tiers based on the VM names.

## Modify Application Scope

By default, VCF Operations for networks analyzes flows for a period of seven days to discover flow based applications. If you want to change the default preference, then you can change the duration and scope of the discovery as per your requirement.

1. On the **Discover Applications** page, in the **Scope** bar of the **Flows** tab, click **Edit Discovery**.
2. In the **Edit Discovery** window, select **Discovery Scope**.
3. From the **Scope** drop-down menu, select your preferred scope. If you want to discover applications based on entities, select **Entities** from the drop-down menu, select an entity type, and provide the entity names.
4. From the **Duration** drop-down menu, select your preferred duration.
5. Click **Submit**.

### Update Application Membership

When VCF Operations for networks detects any changes in the membership of a saved application, it prompts you to update the application.

For flow based application discovery and ServiceNow auto-discovery, VCF Operations for networks discovers and updates applications periodically. If VCF Operations for networks detects any updates or changes in the membership of the saved applications, then you see an **Update available** button on the application for which an update is available. You can update the applications one by one or you can bulk update all the applications for which an update is available.

1. In the **All Applications** > **List** tab, click the **Update available** button next to the application that is having membership changes.
2. In the **Update Application** dialog box, review the changes, and click **Update**.
3. To bulk update multiple application, click the **Update All** button located above the application list.

### Troubleshooting Application Discovery Related Errors

The table lists errors you might encounter while working **Applications** with the cause and resolution steps.

| Error Text | Cause | Resolution |
|---|---|---|
| `The current cluster configuration does not support flow based application discovery (FBAD).` | Your platform does not meet the minimum system requirements. | You must the expand cluster. See Expand or Scale Out Clusters. |
| `The current cluster configuration does not support application discovery due to the large number of VM to VM flows or VMs.` | Your current cluster configuration cannot handle the total numbers of VMs and VM to VM flows. | You must reduce the scope or reconfigure your cluster to support the large number of VMs and VM to VM paths.<br>To learn how to reconfigure clusters, see KB 83347.<br>To learn how to expand clusters, see Expand or Scale Out Clusters.<br>. |

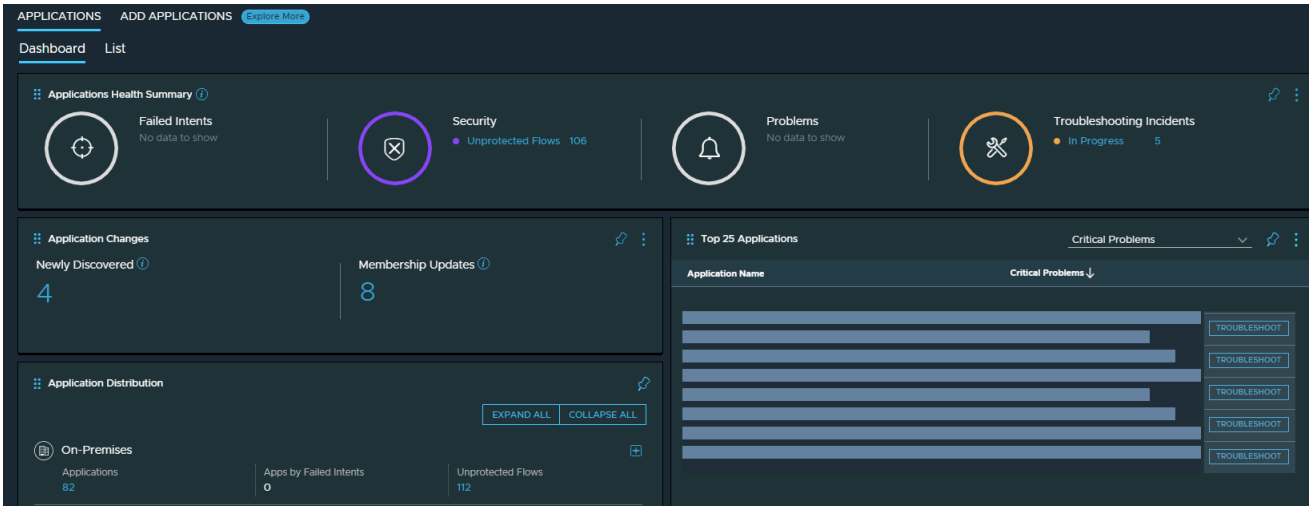
## Viewing All Applications

The **Applications** page provides an end-to-end visibility of applications in VCF Operations for networks.

To access the **Applications** page, from the navigation pane, go to **Applications**.

On this page, you can:

- View the overall health of applications.
- View applications categorized by different alerts
- Start one or more troubleshooting incidents
- Click the **Discover** button to perform application discovery
- Click the **Add** button to add applications manually.



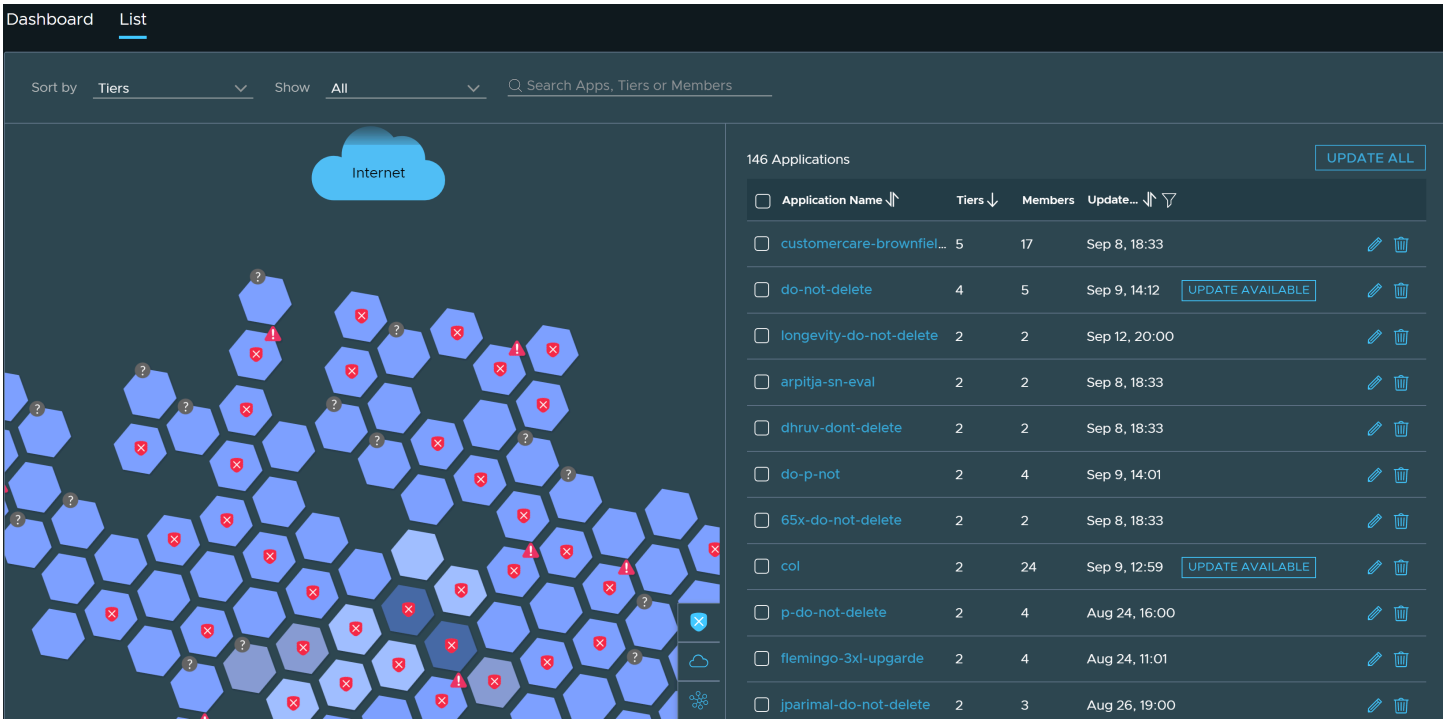

## Dashboard Tab

The dashboard tab provides a complete overview of applications in VCF Operations for networks. The following table describes different elements of the tab:

| Elements | Description |
|---|---|
| **Applications Health Summary** | Shows applications grouped by alerts, security, failed intents, and troubleshooting incidents. To learn more or troubleshoot issues, click the actionable information within the widget. |
| **Application Changes** | Shows applications modified in the last 24 hours. You can:<br>• Click the number associated with **Newly Discovered** to view and save newly discovered applications.<br>• Click the number associated with **Membership Updates** to view and update applications having membership changes. |
| **Application Distribution** | Shows applications grouped by on-premises, cloud, and hybrid environments. You can expand each section to view details such as failed intents and unprotected flows. |
| **Top 25 Applications** | Displays the top 25 applications based on:<br>• Critical problems<br>• Outgoing traffic<br>• Incoming traffic<br>• Failed intents<br>• Retransmission rate<br>• Round trip time<br>• Traffic from internet<br>• Traffic to internet<br>You can click an application to open the respective application page.<br>If you are viewing the top 25 applications based on critical problems, or intent violations, you can click the **Troubleshoot** button next to the application to start a troubleshooting incident. |

## List Tab

The list tab displays applications in a table and a map view.


Dashboard
List
Sort by
Tiers
Show
All
Search Apps, Tiers or Members
Internet
146 Applications
UPDATE ALL
Application Name
Tiers
Members
Update...
customercare-brownfiel...
5
17
Sep 8, 18:33
do-not-delete
4
5
Sep 9, 14:12
UPDATE AVAILABLE
longevity-do-not-delete
2
2
Sep 12, 20:00
arpitja-sn-eval
2
2
Sep 8, 18:33
dhruv-dont-delete
2
2
Sep 8, 18:33
do-p-not
2
4
Sep 9, 14:01
65x-do-not-delete
2
2
Sep 8, 18:33
col
2
24
Sep 9, 12:59
UPDATE AVAILABLE
p-do-not-delete
2
4
Aug 24, 16:00
flemingo-3xl-upgarde
2
4
Aug 24, 11:01
jparimal-do-not-delete
2
3
Aug 26, 19:00

The map and the tabular view are interactive. If you click an application in the tabular view, the hexagon representing the application gets highlighted or focused on the map view, displaying shared services and network connections with other applications.

| View | Description |
|---|---|
| **Table** | The tabular view displays details such as application name, count of members, count of tiers and updated at.<br>You can do the following:<br>• Hover the mouse on the value in the **Members** column to see the individual count of VMs, physical IPs, and services.<br>• Click an application to open the respective application page and view the details of that application.<br>To learn more, see Viewing Application Details.<br>• Update all applications by clicking the **Update All** button on top of the table view.<br>• Edit or delete an application by clicking respective icons next to the application.<br>• Select one or more applications to delete or export in the `.csv` format.<br>• Update applications individually by clicking the **Update Available** button next to the application.<br>To learn more, see Update Application Membership. |
| **Map** | On the map view, you can:<br>• Hover over a hexagon to see application details such as name, discovered VM count, and the tier count.<br>• On hover, click **View Details** to open the respective application page.<br>• Zoom in or zoom out of map by clicking the (+) or the (-) button.<br>• View applications with unprotected flows and problems.<br>• View applications talking to the internet.<br>• View applications using shared services.<br>• Click the lines connecting an application to the internet to view flow details such as:<br>– Count of source and destination flows<br>– Count of unprotected source flows<br>– Count of unprotected destination flows<br>**Note:**<br>If there are no flows or members associated with the application, a question mark is displayed on the hexagon. |

## Viewing Application Details

The application page provides a complete visibility of an application in VCF Operations for networks. The application page also allows you to troubleshoot problems and view analytics.

To view the details of an application, from the navigation pane, go to **Applications** > **List** and select an application from the list.

The following table describes different components of the application page:

| Component | Description |
|---|---|
| **Overview** | You see the overview of the application which includes:<br>• The application summary which provides information about alerts, incoming and outgoing traffic, flows, intents associated with the application, countries associated with the application, VMs, physical IPs, Kubernetes services counts, and so on.<br>• The **Application Topology** which displays the end-to-end interactive topology map of your application.<br>• Alerts related to the application.<br>• **Application Members Manager** which manages entities (such as VMware vCenter and NSX) for all application members. |
| **What's New** | • You see what's new in the last 24 hours which includes:<br>— Incoming and outgoing traffic count<br>— Dropped flows<br>— Analytics alerts<br>— New members<br>— New unprotected members<br>• External services accessed by the application in the last 24 hours.<br>• Services accessed by the application using the internet in the last 24 hours.<br>• Application ports used in the last 24 hours. |
| **Metric** | You see the following metrics:<br>• VM metric information that represents network rate, CPU, memory, and disk information.<br>• Kubernetes metrics |
| **Flows** | You see the following traffic flows or flow analytics associated with the application in the last 24 hours:<br>• Top talkers by flows<br>• Top application flows by rule<br>• Top incoming traffic by App-Id<br>• Top outgoing traffic by App-Id |
| **Microsegmentation** | You see:<br>• Contextual flows between entities, which provides data of different flow types such as all allowed and dropped flows, protected and unprotected flows by vDefend Distributed Firewall.<br>• What's new in the last 24 hours |
| **Intents** | You see the following options from the drop-down menu:<br>• **Intent Definitions**: Displays the metrics that are monitored to verify whether the conditions defined in the intents are met for the application.<br>• **Intent Violations**: Displays the intent violation alerts associated with the application. |

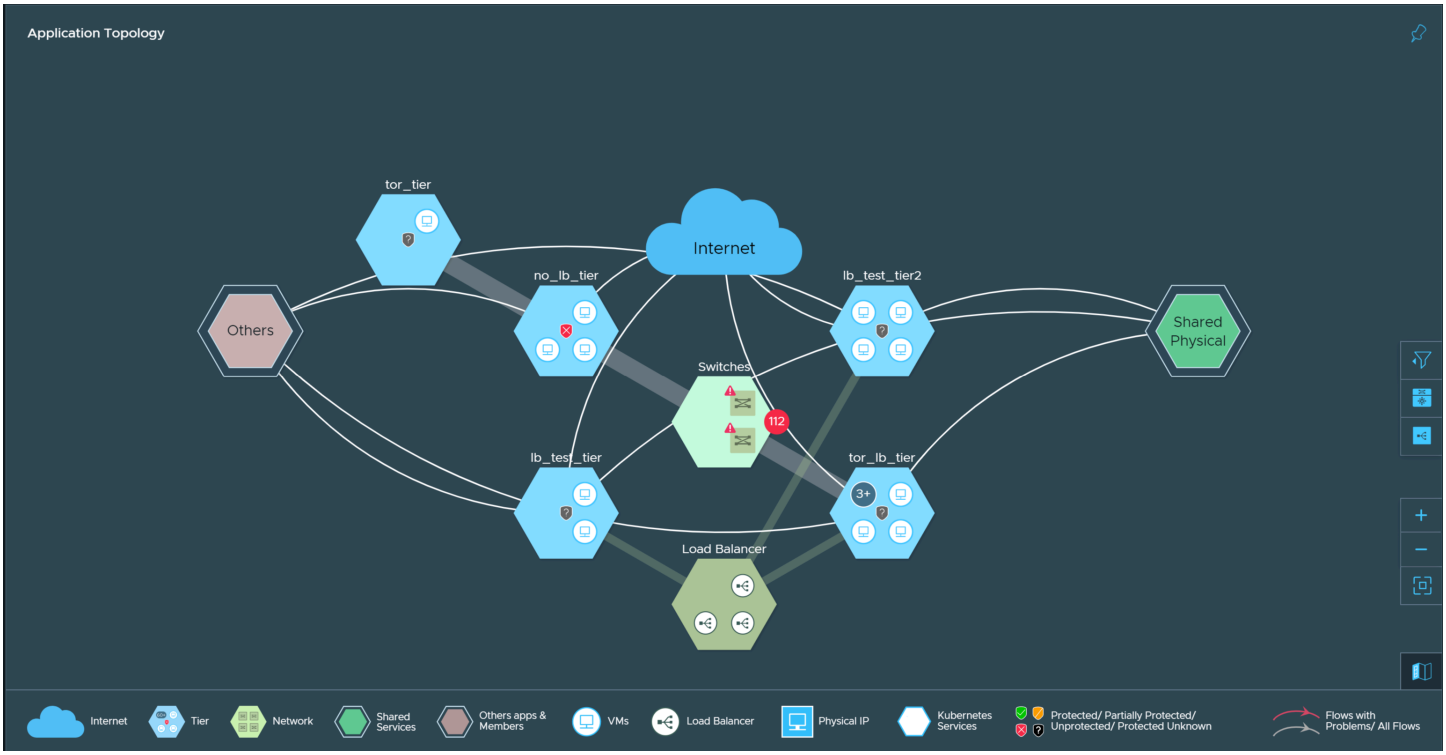
To start a troubleshooting incident from the application page, click the **Troubleshoot** button on the top right corner of the application dashboard.

Click the vertical ellipsis next to the **Troubleshoot** button to modify the application or export to PDF.

## Application Topology Map

The application topology map is shown on the Overview tab. It displays the traffic or the flows between the tiers of the same application and between applications.

- Tier overview
- List of members of the application (VMs, Kubernetes services, and physical IPs).
- Flows between tiers, internet, and shared services.
- The physical IPs that the application depends on or uses.
- Shared services
- Other applications with which this application talks to.
- List of load balancers in the application.




In the application topology map, hexagons represent tiers, load balancers, switches, shared services, and other network entities. You can hover over the hexagon and click **View Details** to view the details of that network entity.

The thin lines that run between two hexagons represent the flows between those two network entities. You can click anywhere on the flow to view details such as service endpoints of the flows, total traffic of the flow, and so on.

Similarly, if the virtual machines of a tier are configured behind one or more load balancers, you can select the thick lines that runs between the load balancer hexagon and the tier hexagon to view the details of virtual services, pools, and flows that are shared between that tier and the load balancers.

You can click the **Pin widget** icon at the upper-right corner of the topology map to either pin this topology map to an existing dashboard, or create a dashboard.

You can use the filter and the zoom options at the lower-right corner of the topology map to:

- View all flows, unprotected flows, or only degraded flows in the topology.
- Show or hide the switches associated with the application.
- Show or hide the load balancers associated with the application.
- Zoom in or zoom out of map
- Rearrange the map to fit the current screen
- Show or hide the legend which indicates the representation of each icon in the application topology.

## Working with Analytics

In VCF Operations for networks, using **Analytics**, you can detect outliers, configure threshold, and get flow analytics.

# Viewing Flow Insight Details

The **Flow Insights** page provides an insight into data centers, devices, and flows. It is a context-based page as it performs analysis based on the entities, flows, and the time range that you select.

To access the Flow Insight page, see Access Flow Insight page.

Alternatively, you can search for `Flows` and in the search result page, click **Flow Insight**.

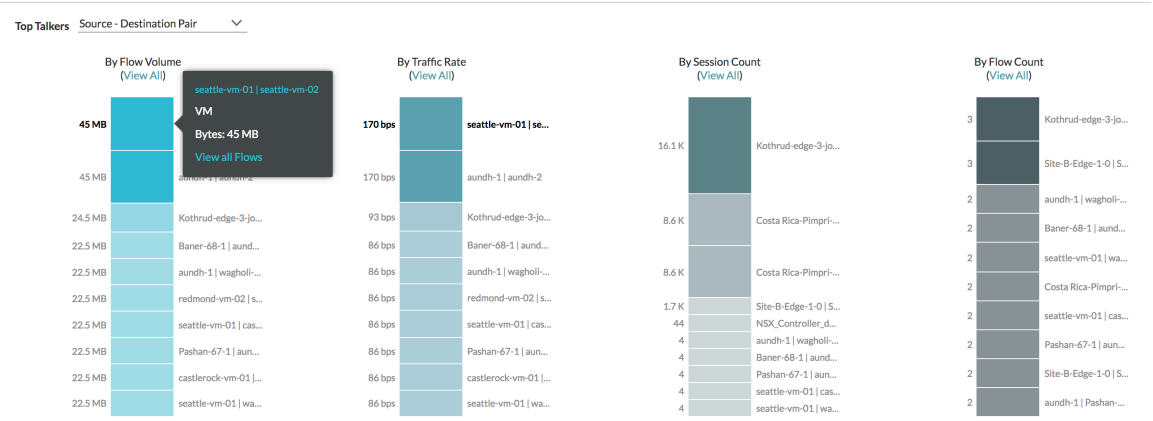
The various sections in the Flow Analytics Dashboard are:

- Top Talkers
- What's New
- Network Performance
- Outliers

## Top Talkers

This section helps you to recognize which entities are talking the most in your environment. You can select different kinds of entities such as Source-Destination pair, VM, Cluster, L2 Network, Subnet. This widget lists the top 10 talkers in the entity category that you select. It helps the customer to plan for network optimization. The metrics that are represented by bars in this widget are as follows:

- By Flow Volume: Indicates the traffic volume.
- By Traffic Rate: Indicates the rate of traffic.
- By Session Count: Indicates the number of sessions.
- By Flow Count: Indicates the number of flows




**Note:**

- If a VM appears in one or more metrics, when you point to that VM in a bar, it will also be highlighted in other bars.
- When you click a VM in the metrics bar, the complete list of flows coming to this VM is shown.
- When you select VM as the entity in the Top Talkers list, all the flows related to this VM irrespective of it being the source or destination is shown. If you select Source VM in the list, then only the flows coming from this VM are considered.
- If you are considering the physical flows, you can select either Source IP or Destination IP.
- After you select the Source-Destination pair and point on the metric bar, if you click the link in the tool tip, the corresponding dashboard appears. For example, for a VM in Sourec-Destination pair, the VM-VM path dashboard appears.
- For a flow group view or a flow entity projection or a flows group query, you cannot see the **Flow Analytics** button.

## What's New

This section helps you to track what services and entities are discovered in the data center in the selected time range. The widget in this section are as follows:

- New Virtual Machines Accessing Internet: Lists the new VMs that access Internet.
- New Internet Services Accessed: Lists the new Internet services discovered in the environment.
- New Internal Services Accessed: Lists the new intranet services that are discovered and accessed from the Internet endpoint.
- New Internal/E-W Services Accessed: Lists the services that are exposed and accessed by the machines within a data center
- New Services with Blocked Flows: Lists services that have blocked flows. This section is populated only for IPFIX.
- New Firewall Rule Hits: Lists the new firewall rules that are brought into effect. This section is populated only for IPFIX.
- New App-Ids: Lists the new application IDs. This section is populated only for Security Intelligence.

## Network Performance

In this section, you can find and visualize the abnormal flows for the various ranges of TCP Round Trip Time (RTT) values based on the selected criteria.

**Note:**
VCF Operations for networks shows average TCP RTT metrics at 5 minutes granularity for last 24 hours only.

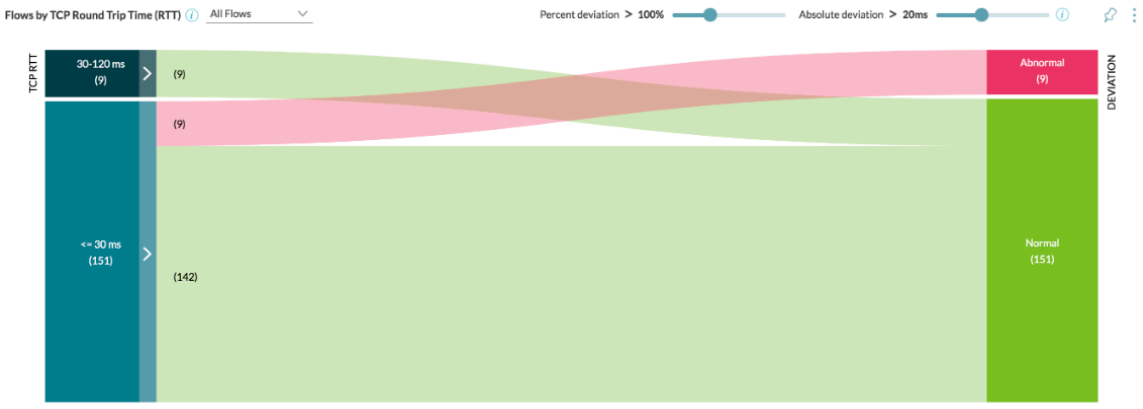
If the flow deviation percentage is 100 percent and absolute deviation is 20 milliseconds (ms), then VCF Operations for networks considers that flow as an abnormal flow.

In the visualization, the left side shows the different range of TCP RTT and the right sides shows the normal and abnormal deviation range. Based on the values of percentage deviation and absolute deviation, flows are connected from left (TCP RTT) to right (DEVIATION). You can analyze the following types of flows:

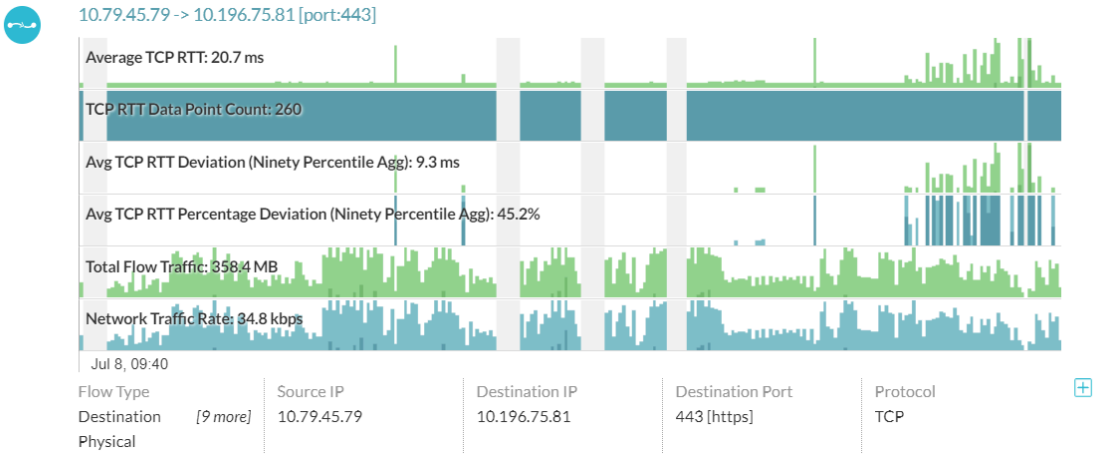
- Inter-Host
- Intra-Host
- Internet
- All Flows

You can also change the percent deviation and absolute deviation based on your requirement.

In the following example, there are two different ranges of TCP RTT, one is less than equal to 30 ms and the other is 30-120 ms. You can find there are total 151 flows comes under the less than equal to 30 ms TCP RTT range. Out of the 151 flows, 9 flows are shown as an abnormal flow.




To get more insight about the TCP RTT distribution information and counts of flows, click the colored line in the visualization. In the following example, you can see the detail insight about the TCP RTT distribution information and counts of flows:

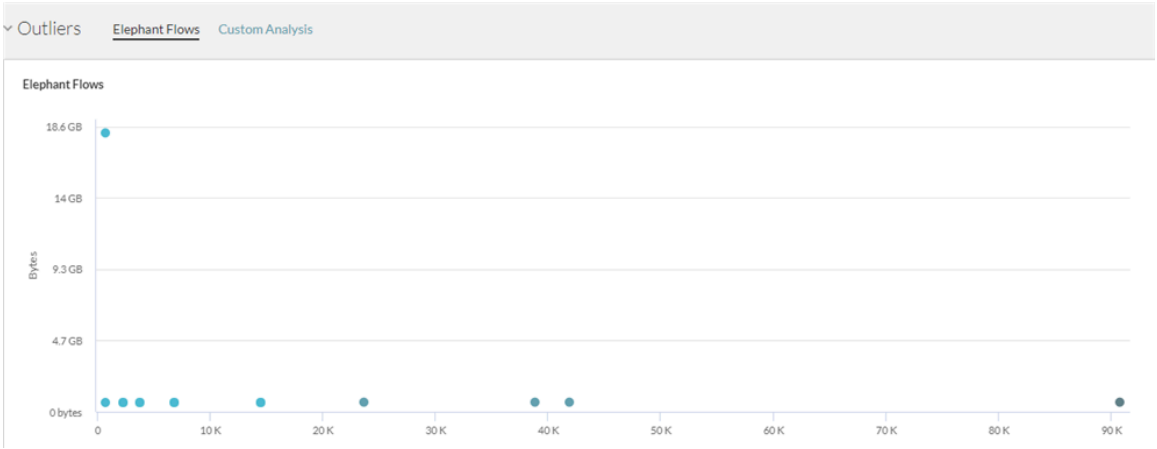



## Outliers

This section helps you to track and analyze related data. It consists of the following sections:

- Elephant Flows: This section helps to identify the flows which have small count of sessions and high throughput versus flows which have large count of sessions and small throughput. Typically, the flows with the large session counts and small throughput are also referred as mice flows. The analysis is based on the ratio of bytes to the number of sessions. Each dot in the graph represents multiple flows. When you point to a dot, you can see the list of flows. To view the details of a particular flow, click that flow in the list.
- Custom Analysis: This section allows you to visualize the flow data on 2 dimensions of your choice. It helps in analyzing the data to find the outliers in various ways.

  **Note:**
  The metrics represented in this section are the approximate values and not the exact values.




## Access Flow Insight page

You can access flow insight from left navigation pane.

1. In the left navigation pane, click **Analytics** > **Flow Insights**.
   You see **Flow Insight** window.
2. From the **Scope** drop-down menu, select one of the following options:

| Option | Action |
|---|---|
| **All flows** | Select to analyze all the flows available in your environment. |
| **Entities** | Select to analyze flows of a particular entity. |

| Option | Action |
|---|---|
| | Click **Entities** and from the drop-down menu select the entity type.<br>You can also use the **Search entity by name** text box to find and select multiple entities.<br>When you click the **Search entity by name** text box, you can see the name of all the available entities for the selected entity type. |
| **Between entities** | Select to analyze flows between two entities.<br>1. From the **Entity type 1** drop-down menu, select an entity type. Also, you can use **Search entity by name** text box to enter the entities names.<br>You can enter multiple entities in the **Search entity by name** text box.<br>2. From the **Entity type 2** drop-down menu, select an entity type. Also, you can use **Search entity by name** text box to enter the entities names.<br>You can enter multiple entities in the **Search entity by name** text box. |
| **Flows matching properties** | Select to analyze flows that match a specific condition. You can add multiple conditions by clicking **ADD CONDITION**. |
| **Custom search** | Select to analyze flows that match a custom search query. |

3. From the **Duration** drop-down menu, select the time frame for which you want to analyze the flows. The system analyzes the flows that occurred between that specific time frame.
4. From the **Ports** drop-down menu, select one of the following:

| Option | Action |
|---|---|
| **All** | Select to analyze all ports available for the selected scope. |
| **Include Ports** | Select to include specific ports that you want to analyze. Enter the names of the ports that you want to analyze in the **Search Ports** text box. |
| **Exclude Ports** | Select to exclude specific ports that you do not want to analyze. Enter the names of the ports that you do not want to analyze in the **Search Ports** text box. |

**Note:**
If you have selected **Flows matching properties** and **Custom search** in **Scope**, you do not see **Ports** drop-down menu.

5. Click **ANALYZE**.

## Analytics - Outlier Detection

VCF Operations for networks offers outlier detection based on the metrics associated with the flows defined over the VMs and physical IP addresses. These VMs/IPs should have similar traffic patterns so that a classification of a particular VM/IP as an outlier is of value. For example, the VMs, which belong to the same tier of an application, generally perform the same function for the application, such as the VMs of an SQL database serving requests for a web application. For these kind of VMs, the number of requests received, the amount of traffic sent out, the session count, and so on go through a series of similar variations.

Through outlier detection, VCF Operations for networks enables you to detect a particular VM which might be experiencing very different traffic pattern compared to other VMs/IPs in the group. For example, if the VM is sending or receiving much higher/lower traffic compared to the rest of the group. It could be because of a wrongly configured load balancer, DDOS attack, and so on. VCF Operations for networks classifies such VMs/IPs as outliers. By looking at these outliers, the user easily knows about this unexpected behaviour and takes appropriate actions.

## How to Detect the Outlier VMs

1. On the sidebar, click **Analytics**. Click **Outlier**.
2. Click **Add** to add a configuration.
3. In the **Analytics/Configure** page, provide the following details for the configuration:

**Table 1335:**

| Field | Description |
|---|---|
| Name | Name of the configuration |
| Scope | Name of the group that defines the VMs and the IPs for which the analysis needs to be done. You can select Application Tier or Security Group as the scope.<br>If you select Application Tier, provide the name of the application and the tier separately. The number of VMs and Physical IPs that are defined for the tier is shown next to the name of the tier.<br>If you select **Security Group**, provide the name of the Security Group.<br>**Note:**<br>The current limit for the number of VMs and Physical IPs in a tier is 200. Choose a tier or a security group with VMs and Physical IPs less than this limit. The scope should also contain a minimum of 3 VMs/Physical IPs.<br>You can view the micro segmentation for the selected configuration by clicking **View Micro-Segments**. |
| Detection Type | Currently, VCF Operations for networks enables you to detect the outlier in the system. |
| Metric | The detection is based on this flow metric. You can select the following options:<br>• **Bytes**<br>• **Packets**<br>• **Sessions**<br>• **Traffic Rate** |
| Traffic Direction | You can select **Outgoing**, **Incoming**, or **Both** as the traffic direction. If you select **Both**, then you can specify Incoming or Outgoing in the preview of the configuration. |
| Traffic Type | You can select **Internet**, **East-West**, or All based on the requirement. |
| Destination Ports | You can either select all ports detected on the flows discovered on the selected scope or manually enter the destination ports of your choice. If you select **All Ports**, the number of the destination ports is shown. If you select **Manually enter ports**, then enter the ports in the autocomplete text box, the analysis would be restricted to only these ports<br>**Note:**<br>The current limit for the number of ports is 20. |
| Sensitivity | It is a measure of the sensitivity of the detection and reporting that you require. The default value is **Medium**. |

| Field | Description |
|---|---|
| Preview | This section provides a preview of the particular configuration based on the inputs and parameters that you have provided. Specify the ports and the traffic direction if you have selected Both for Traffic Direction before. You will be able to identify the outlier VM in the graph. |

**Note:**

- The outlier is detected by evaluating the data available in last 24 hours.
- You need a continuous flow of IPFIX data to detect the outlier.

4. Click **Submit** to create the analytics configuration.
5. Once the application is created, it is available in the list view of the applications in the Analytics Configurations page. Click that particular application to see a dashboard associated with it.

## Analytics: Static and Dynamic Thresholds

You can configure thresholds and receive alerts based on aberrations in the behavior of the entities.

You can configure two types of thresholds:

- Static threshold: If a particular metric value goes beyond or below the configured value, then a static-threshold-based alert is generated.
- Dynamic threshold: If the threshold is determined by the system based on the analysis of the historical data, an alert is generated in case this threshold is violated. The data is analyzed for a period of 7 days before any alert is generated. The process of creating a baseline is restricted to 21 days of the historic data and the older metric values are not considered to create a baseline for the new metric values.

VCF Operations for networks generates an alert immediately after a threshold is violated.

### Configure Thresholds and Alerts

You can add a threshold configuration and gets alerts for the configured threshold.

To configure the analytics-associated thresholds and alerts:

1. On the Home page, in the left navigation pane, click **Analytics** > **Thresholds** > **Add**.
   The **Threshold - Add configuration** page opens.
2. In the **Name** text box, enter a unique name for the configuration.
3. From the **Scope** drop-down menu, select the entities that are to be considered for analysis. For example the entities can be VMs, Flows, etc.
   a) To identify a subset of entities that are to be considered for analysis, enter one or more criteria in the **Select criteria** text box.
4. In the **Condition** section, set a condition to create an alert.
   Based on the condition you set, the system decides if the threshold is violated.
   a) In the **aggregared over** text box, select the grouping of the entity. You can set a threshold on a cumulative metric by aggregating the data over a group of entities.
   b) In the **when** text box, enter the value for which you are checking the threshold.
   c) Configure a static or a dynamic threshold.
      - To configure a static threshold, select either of the following threshold conditions from the list:
        – **exceeds threshold**
        – **drops below**
        – **is outside range**

- To configure the dynamic threshold, select **deviates from the past behavior**. Select the sensitivity based on your requirement of reporting.

d) Enter the upper bound and lower bound range.

When you enter the `Upper Bound` or the `Lower Bound` range, ensure that you enter the value based on the unit of measure for the selected condition. The following conversion values are for your reference:

- `1 Kbps= 1000 bps`
- `1 Mbps= 1000 kbps`
- `1 Gbps = 1000 mbps`
- `1 KB=1024 B`
- `1 MB=1024 KB`
- `1 GB = 1024 MB`

When you set the threshold, you can view the associated graph at the top of the page. The pink bar denotes the VMs or the flows violating the threshold. You can view the list of the entities that have violated the threshold and the entities that are within the threshold in the system.

5. From the **Minimum Violation Duration** drop-down menu, select a time duration after which VCF Operations for networks should trigger alerts if the violation lasts longer than the selected time duration.

   For example, If you set **Minimum Violation Duration** as 15 minutes, you get alerts only if the violations stays active over 15 minutes.

6. Configure the severity of the alert.
7. In the **Notification** section, configure when and who can receive notifications.

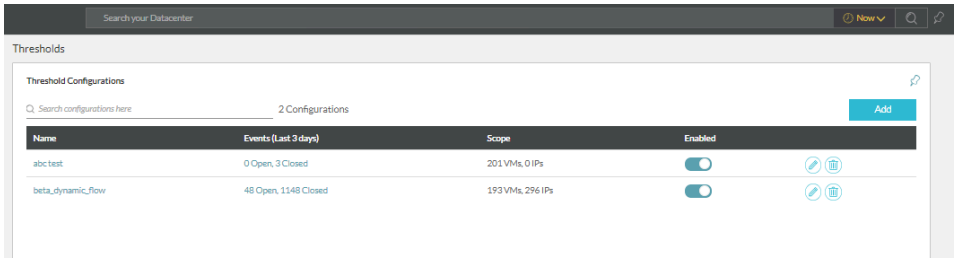
   **Note:**
   Select **Send SNMP Trap** only if you have configured SNMP traps on your system.

8. Click **Submit** to create the threshold configuration.

## View the Threshold Configuration Page

Once you have added a threshold configuration, you can view its details on the **Threshold Configuration** page.

1. On the left navigation pane, click **Analytics**. Click **Thresholds**.

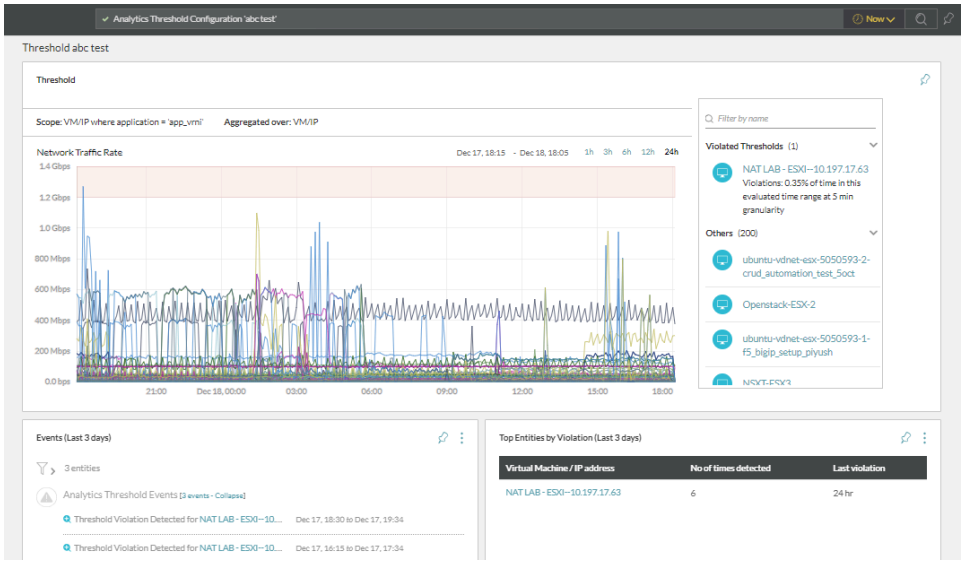



2. The following details for a threshold configuration are provided:

   - `Name`
   - `Alerts`
   - `Scope`

   If the configuration is deactivated, then the alert for the violation of that particular threshold is not generated. You can also search for any particular threshold configuration on this page.

3. Click the desired threshold configuration from the list to view the dashboard for that particular configuration.



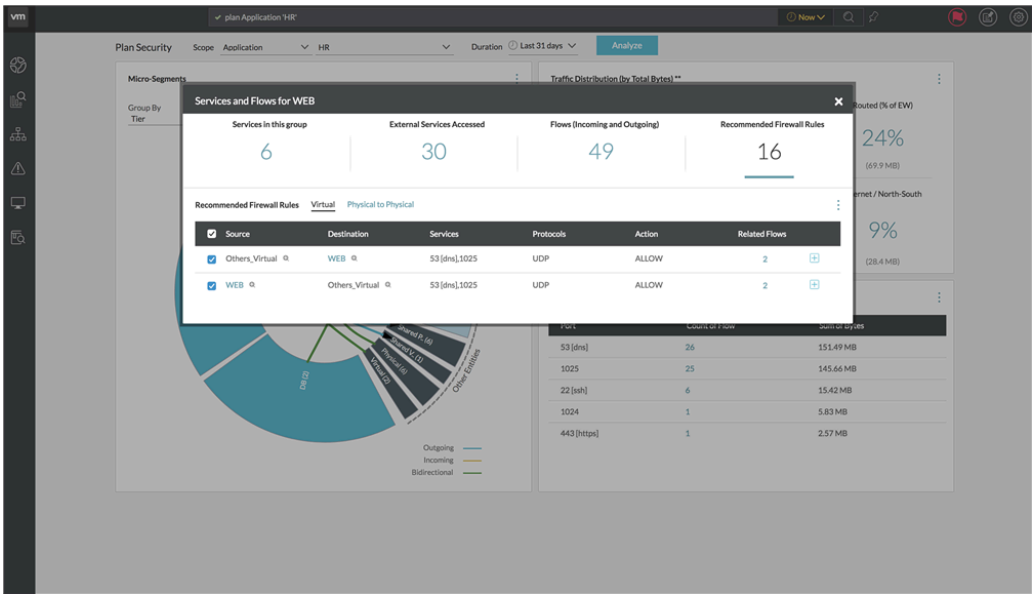

You can view the following widgets on the dashboard:

- Graph: The threshold graph helps you detect the entities that have violated the thresholds.
- Alerts: This widget provides the list of alerts that have been generated for violated thresholds for the last three days.
- Top Entities by Violation: This widget lets you know the top entities that have been the cause of aberrations for the last three days.

## Viewing Recommended Firewall Rules

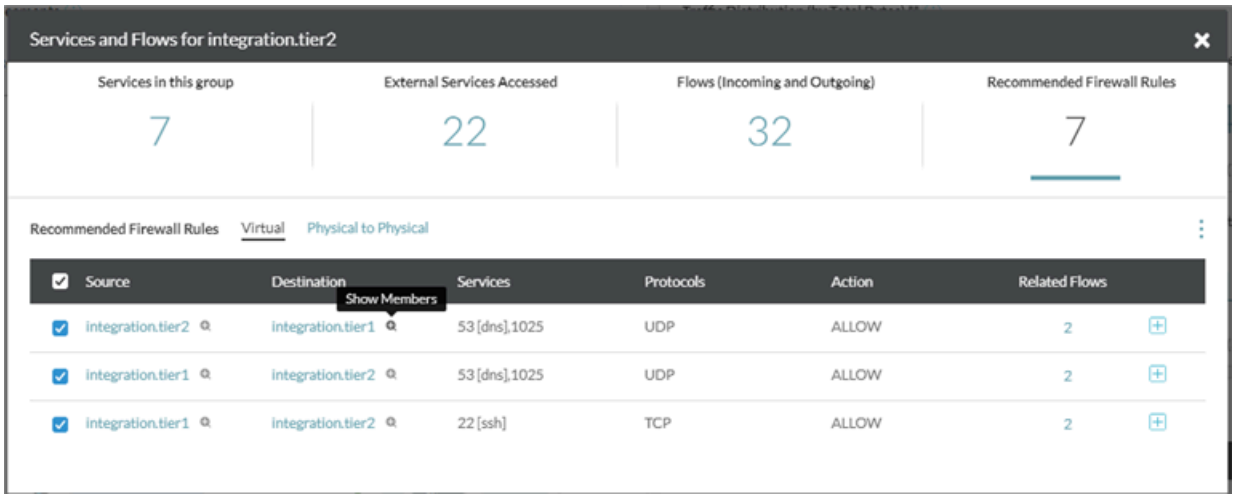
On the **Security Planning** page, when you click the wedge or the edge in the **Micro-Segments** topology diagram, you can view the list of the services, flows, and recommended firewall rules for that particular segment. Click **Recommended Firewall Rules** to view the rules defined on it. The members of the source or the destination are listed under the following types of rules:

- Physical to Physical: This tab lists all the rules associated with the physical and Internet IPs. The rules can be for physical-physical, physical-Internet, Internet-physical, or Internet-Internet entities.
- Virtual: This tab lists all rules where at least one of the endpoints is a VM.




For each firewall rule, the following details are available:

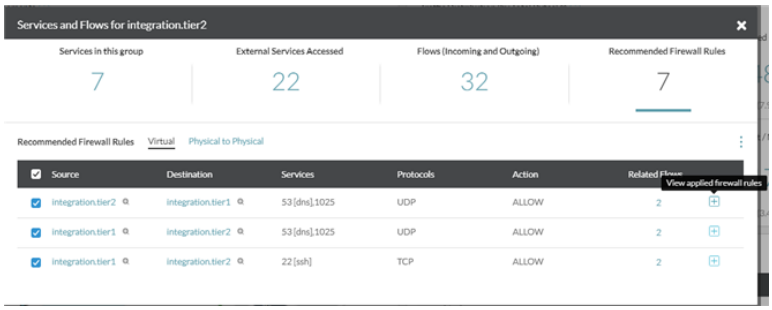
- Show members of the group: Click the + sign next to the name of the entity to see the members of the group.




**Note:**

- The members are not shown for the groups belonging to the Internet category.
- If a security group has both virtual and physical IPs, the physical and the Internet IPs are not shown in the list of the members of that particular group.
- The member Kubernetes services are shown under the **Kubernetes Services** tab.
- If the member count or the entry is zero for **Virtual Machine**, **Physical & Internet IPs**, or **Kubernetes Services** the tab is not visible.

- Source
- Destination
- Services
- Protocols
- Action
- Related Flows: Click the number of the related flows to see the list of flows with the corresponding flow information.
- View Applied Firewall Rules: Click the + sign next to the **Related Flows** column to view the applied firewall rules corresponding to the similar sets of flows.




You can export the recommended rules as XML or CSV based on your requirement.

**Note:**
You can export recommended rules related to Kubernetes objects in the YAML format also.

Refer to Exporting Rules for more information on these artifacts.

## Recommended Firewall Rule to Secure Vulnerable OS

Use the following procedure to get recommended firewall rule to secure vulnerable OS:

1. Go to **Applications** > **All Applications** > **Add**.

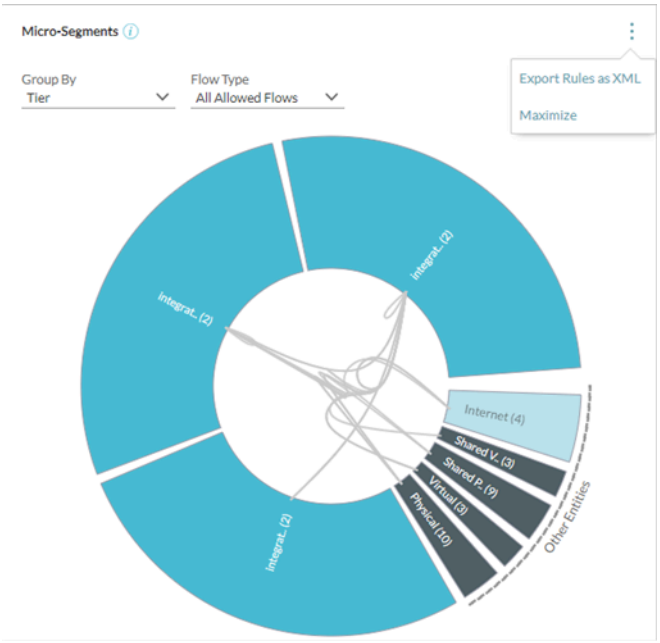
2. In the **Add Application** page, do the following actions:

| Options | Actions |
|---|---|
| **Application Name** | Type a name for the Application |
| **Tier / Deployment** | • Provide a unique name for the tier.<br>• In the **Member** drop-down, select **Custom VM Search** and in the **Search your application environment** text box add the following condition:<br>`in the qualifier put the matching criteria as: Operating System like 'Microsoft Windows Server 2003' or Operating System like 'Microsoft Windows Server 2008' or Operating System like 'Red Hat Enterprise Linux 6' or Operating System like 'Red Hat Enterprise Linux 5' or Operating System like 'SUSE Linux Enterprise 10'` |

3. Click **Save**.
4. Go to **Plan & Assess** > **Security Planning**.
5. In the **Scope** drop-down, select **Application** and the name of application you created.
6. In **Duration** drop-down, select **Last 7 days**.
7. To get the recommended firewall rules, click **Analyze**

## Exporting Rules

You can export all the rules as XML for the entire topology. You can find this menu item in the **Micro-Segmentation Planning** page as follows:




The Export as XML option is available only for the following entities:

- Security Group
- Application Tier

If the planning scope spans a single NSX Manager only, the generated artifacts contain the XML files corresponding to the recommended services and firewall rules. If the planning scope spans multiple NSX managers, the generated artifacts contain the XML files corresponding to the recommended services, IPsets, security groups, and the firewall rules.

The following are the placeholder artifacts for security groups:

- `SG-Others_Internet.xml`
- `SG-Other.xml`

You can export all the rules as XML or CSV for a particular wedge or edge depicted in the topology diagram.

**Note:**
You can export recommended rules related to Kubernetes objects in the YAML format also.

## NSX DFW Universal Artifacts

It is easy to manage objects in universal security groups across the various VMware vCenter and NSX deployments. VCF Operations for networks supports the generation and the import of the universal artifacts for the Application and Tier groups only. With the universal security groups, it becomes easy to deploy and manage the firewall rules easily in the cross VMware vCenter scenarios. Ensure that you import the universal artifacts on the primary NSX manager. You can manage the membership of the universal security group only through the primary NSX manager.

A universal security group can consist of:

- Other universal groups
- Universal IP sets
- Universal Security Tag

When you export the rules as XML, in addition to the NSX manager specific folders, a universal folder is created which consists of the NSX DFW universal artifacts. The corresponding universal security groups, universal IP sets, universal security tags, and universal DFW firewall rules are created after importing the NSX DFW universal artifacts.

**Note:**

- The universal security tag is supported in only active-standby mode.
- The universal IP set is supported in both active-active and active-standby modes.

You can create universal IP set or universal security tag based on your requirement. If you create the universal security tag, then you can map the application VM to the security tag. Else, the universal IP set is used.

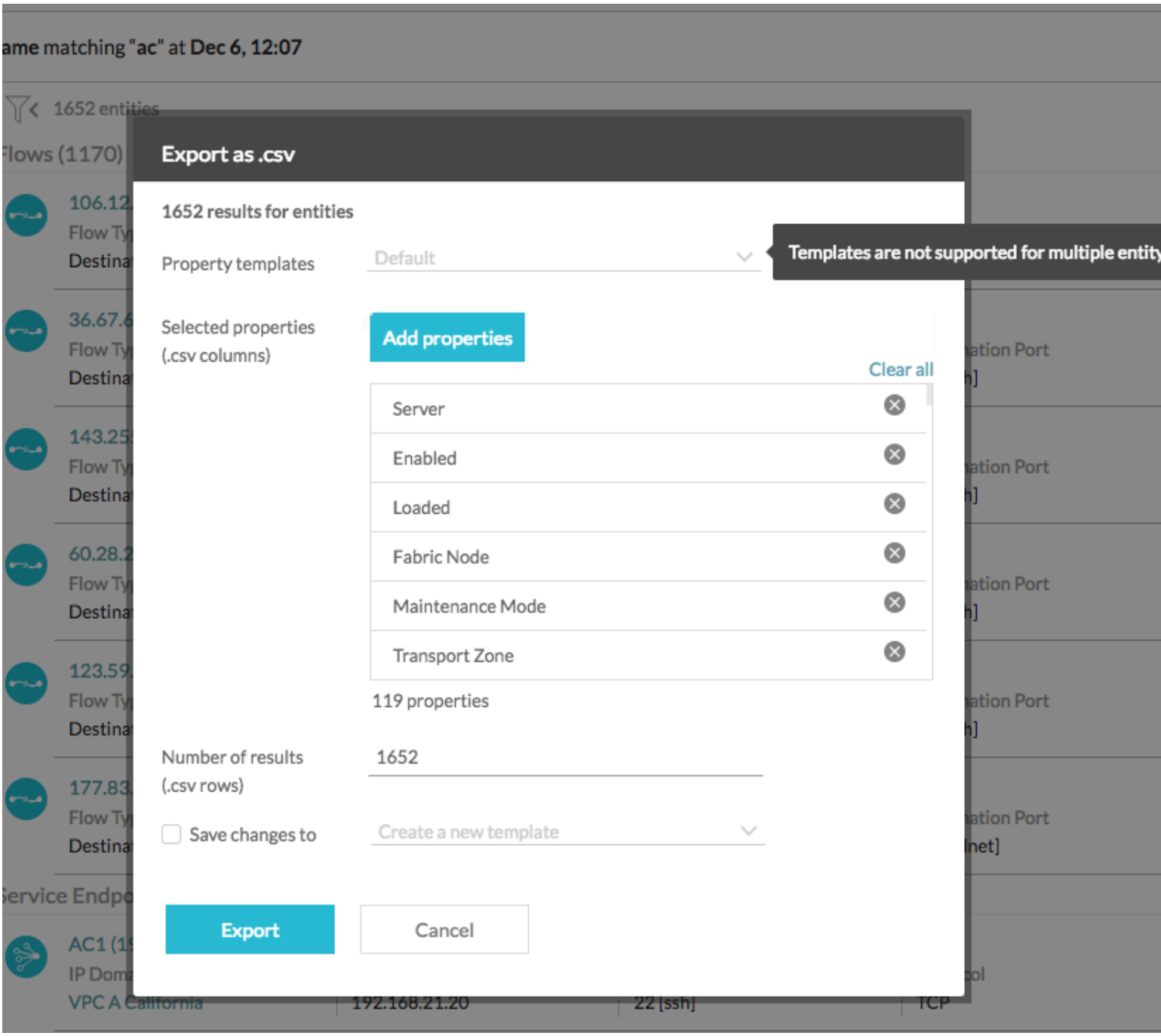
You can use the following flags in the import tool:

| Flag Name | Description |
|---|---|
| `-uni` | To import artifacts from the universal folder. |
| `-utag` | To import the universal artifacts with the universal security tags in the membership of the universal security groups. |
| `-log` | To create rules in which logging is enabled.<br>**Note:**<br>This flag is not specific to universal option. |

## Save the Configuration for CSV Export as Property Template

While exporting the data from widgets in the CSV files, you can save the combination of the properties (or columns) that you want to export in the property templates. These property templates are enabled for the CSV export when the results belong to a single entity type. If you search with a keyword that lists

the multiple entity types, they cannot save the combination of the properties in the property templates.




When you open the CSV export modal, you will see the default property selections for the search results (based on the entity type). You can change this list of the selected properties and save the new configuration for your future reference. Alternatively, you can also load or open a pre-saved property template from the **Templates** section on the CSV export modal. When you change the value, you will see the selected properties for the selected property template.

Once you make changes to the selected properties for the export, you can create a property template from the CSV export modal or edit an existing property template. This template is of the same entity type as that of the current search results.

You can view the list of the existing property templates in the system by navigating to the **Settings -> Property Templates** page. The list on the **Property Templates** page shows the existing templates with details such as entity type, last updated, and number of properties. You can edit or delete property templates from the **Property Templates** page. You can edit the property template except change its name.

## Export and Apply Kubernetes Network Policies

You can export the recommended network policy rules related to Kubernetes objects in the YAML format. VCF Operations for networks supports exporting to YAML format for the group by Namespace and group by Service topologies only.

- Add the Kubernetes cluster as a data source
- Add VMware PKS as a Data Source

1. To export the recommended rules to YAML format, on the Plan Security model, select your Kubernetes cluster for which you wish to plan security, and perform one of the steps.
   - Expand more options in the Micro-Segments widget and select **Export Rules as YAML**, or
   - Select a node on the Micro-Segments wheel view, click on the count of Recommended Firewall Rules, expand more options and select **Export Rules as YAML**.

   VCF Operations for networks downloads a ZIP file named with the Kubernetes Network Policies and a timestamp associated with it. When you unzip the file, you see the following five CSV files and also multiple folders depending on the number of clusters. Each folder will contain multiple YAML files for the cluster.

| File Name | Description |
|---|---|
| `network-policy-others-ipaddress.csv` | Contains the IP addresses of the physical servers and virtual machine with which the services or namespaces are communicating. |
| `recommended-namespace-labels-to-add.csv` | Contains the labels to be attached to the pods associated with the namespace.<br>Example:<br>• **Cluster** - pdk8s<br>• **Namespace** - sock-shop<br>• **Label** - sock-shop-pdk8s |
| `recommended-service-labels-to-add.csv` | Contains the labels to be attached to the pods associated with the service.<br>Example:<br>• **Cluster** - pdk8s<br>• **Namespace** - sock-shop<br>• **Service** - front-end<br>• **Label** - Service:front-sock-shop-pdk8s<br>• **Cluster** - pdk8s<br>• **Namespace** - sock-shop<br>• **Service** - user<br>• **Label** - Service:user-sock-shop |
| `recommended-network-policy.csv` | Contains all the rules recommended by VCF Operations for networks. |
| `exported-network-policy-rule-names.csv` | Lists all the network policies exported based on the recommended rules. |

2. To apply the service labels, perform the following steps:
   a) Run the following Kubernetes CLI command.

   `kubectl edit deployment` *`service-name`* `-n` *`namespace-name`*

   `kubectl edit deployment` *`redis-primary`* `-n` *`guestbook`*

   The deployment file of the service opens.

b) In the service label list, append the label which has been suggested in the CSV file, to the labels mentioned in the spec section of service deployment.

3. To apply the namespace labels, perform the following steps:
   a) Run the following Kubernetes CLI command.
      `kubectl edit namespace namespace-name`
      `kubectl edit namespace guestbook`
      The deployment file of the namespace opens.
   b) In the metadata , append the label which has been suggested in the CSV file, to the labels mentioned in the `spec` section of namespace deployment.
4. Run the following command to verify whether the labels are applied to the pods.
   `kubectl get pods -n namespace-name--show-labels`
   `kubectl get pods guestbook--show-labels`
   See the labels in the result view.

   **Note:**
   The labels are not reflected on Pods, when you apply on Namespace.
5. To create the network policies, copy the YAML files from the respective cluster folder to another folder and run either of the commands:
   - `kubectl apply -f <folder-name>/` - to apply all the firewall rules together.
   - `kubectl apply -f <folder-name>/<firewall-rule>.yaml` - to apply firewall rules one by one

## Working with Search Queries

VCF Operations for networks provides a robust search for all the entities in your environment.

VCF Operations for networks provides a list of pre-defined list of search queries out-of-the-box under **Help** > **Useful Searches**. You can refer or use these queries as a syntax to structure the search query, and modify them as per your requirement. Each of these queries are associated with a description and a set of tags. You can also filter these search queries based on a keyword or a specific tag. When searching for a query, you get alternate search suggestions if the search fails to show results for that query.

To run the query, you can click on a query in the list. When you select a query that does not require a parameter value, the query would be executed directly. If a query requires a parameter value, the homepage of VCF Operations for networks opens in a new tab which shows the query with the sample value. You can replace them with the actual values like host name, application name, that exists in your setup.

The **Supported Properties** option on the **Help** page displays the list of entities, which when expanded displays all the supported properties, description, sample queries and tags of the particular entity. You can filter the properties based on the tags.

You can use the **View Search Query** option on the widget to see the search queries that are associated with each widget on the pinboards. Also, VCF Operations for networks enables you to copy the query, which you can modify and execute on the search bar. If multiple sub-widgets exists in a widget (for example, an Overview widget on a dashboard), you can see the queries for each sub-widget that are associated with the respective name.

**Note:**
As of now, the **View Search Query** option is available for the only following:

- NSX Manager Details
- Flow Insights Details
- Application Details
- Home Page
- VM Details
- Host Details
- NSX-T Manager topology
- NSX-T Host Transport Node

- NSX-T Edge Transport Node
- NSX Policy Manager Details

Here are some of the terms that can help you with the search feature in VCF Operations for networks:

- Entities: A data center consists of physical and logical building blocks such as host, virtual machine, switch, router, NSX Manager and so on. The instances of these blocks are entities.
- Property: An entity consists of multiple properties. A property can be either a configuration property or a metric property.
  a. Configuration Property: An entity can be described by its configuration properties. A configuration property can be either integer or real value or a string or a boolean value.
     - Name, CPU cores, and operating system for virtual machines
     - Name, and number of virtual machines for hosts
  b. Metric Property: Any property which measures a particular characteristic of an entity is a metric property. The values of metric properties are captured at regular intervals of time. CPU usage, memory usage, and network usage for virtual machines are some examples of metric properties.
- Aggregate Functions: They can be used in the search queries to compute the total number of instances of a particular entity type or maximum property of an entity. VCF Operations for networks supports following aggregation functions.
  a. `sum`
  b. `max`
  c. `min`
  d. `avg`

When you search for entities, the software displays the entities that match your search query on the **Results** page.

For each search query, the search bar suggests you the next term that you can use to narrow down your search results. For example, when you enter the term **vm**, the search bar displays a possible list of terms that you can add to your existing term to narrow down your search results. The search bar also validates each search query. A check mark denotes a valid search query and a cross mark denotes an invalid search query.

## Save and Delete Search Queries

VCF Operations for networks allows you to run a search query and save the query for later use. It also allows you to delete the saved searches.

**Note:**

- VCF Operations for networks provides the following default saved searches:
  – All Flows
  – Applications
  – Azure
  – Kubernetes Dashboard
  – Top Trends
  – NSX
- You cannot save or delete the default saved searches.
- You cannot save an invalid search query.
- The saved searches are user specific and the default saved searches are available to all the users.

1. To save a query, run a search and click the bookmark icon next to the search bar.

   You see the bookmark icon highlighted to ensure that the the query is saved. You can find the search listed under **Saved Searches** in the left navigation pane. To see all the queries that are saved, click **Saved Searches** > **Manage saved searches**.

2. To delete the saved search, click on the bookmark icon again and click **Delete** on the confirm action dialog box.

   You can also delete the saved search from the **Manage saved searches** window.

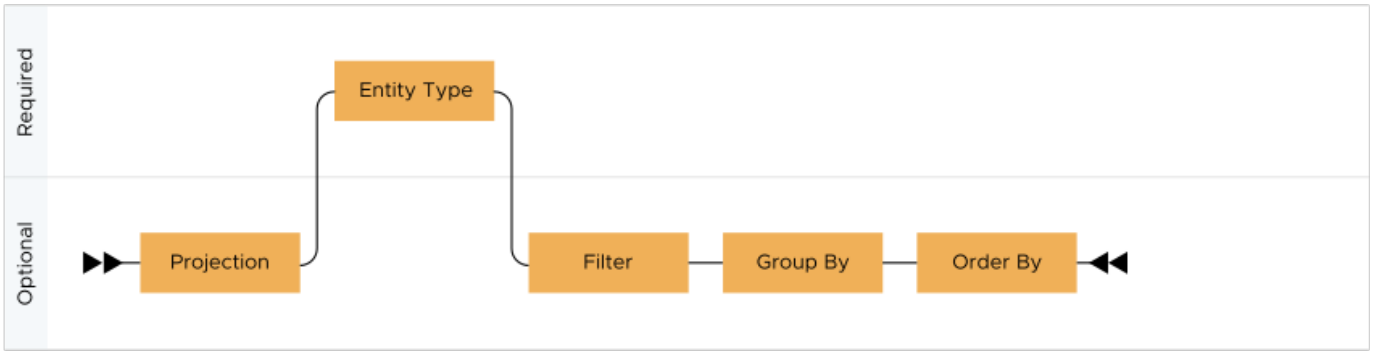
3. To delete multiple saved search queries together,
   a) In the left navigation pane, click **Saved Searches** > **Manage saved searches**.
   b) Select the queries that you want to delete.
   c) Click the **Delete** option.
   d) Confirm the delete.

## Search Queries

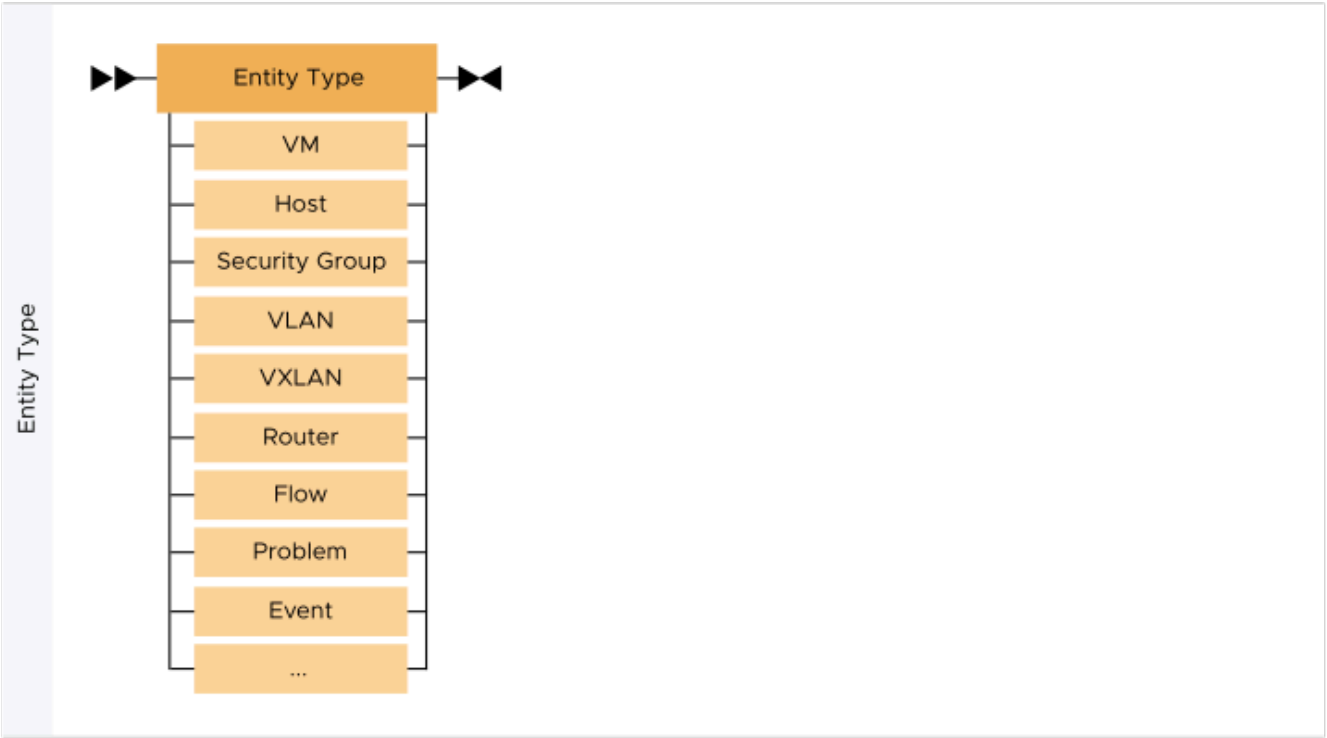
The search queries runs searches across your entire deployment and enables you to find all types of entities based on the specified search term. Additionally, VCF Operations for networks provides suggestions to build the queries, which includes saved searches and recent searches that match the typed text in the global search bar.

Search queries can be divided into following categories:

1. **Structured Queries**
   A structured query consists of the following components:




   – **Entity Type**: An entity type represents the type of object that we want to search. And it can be either in a singular form or in a plural form. The entity type is mandatory in a structured query.

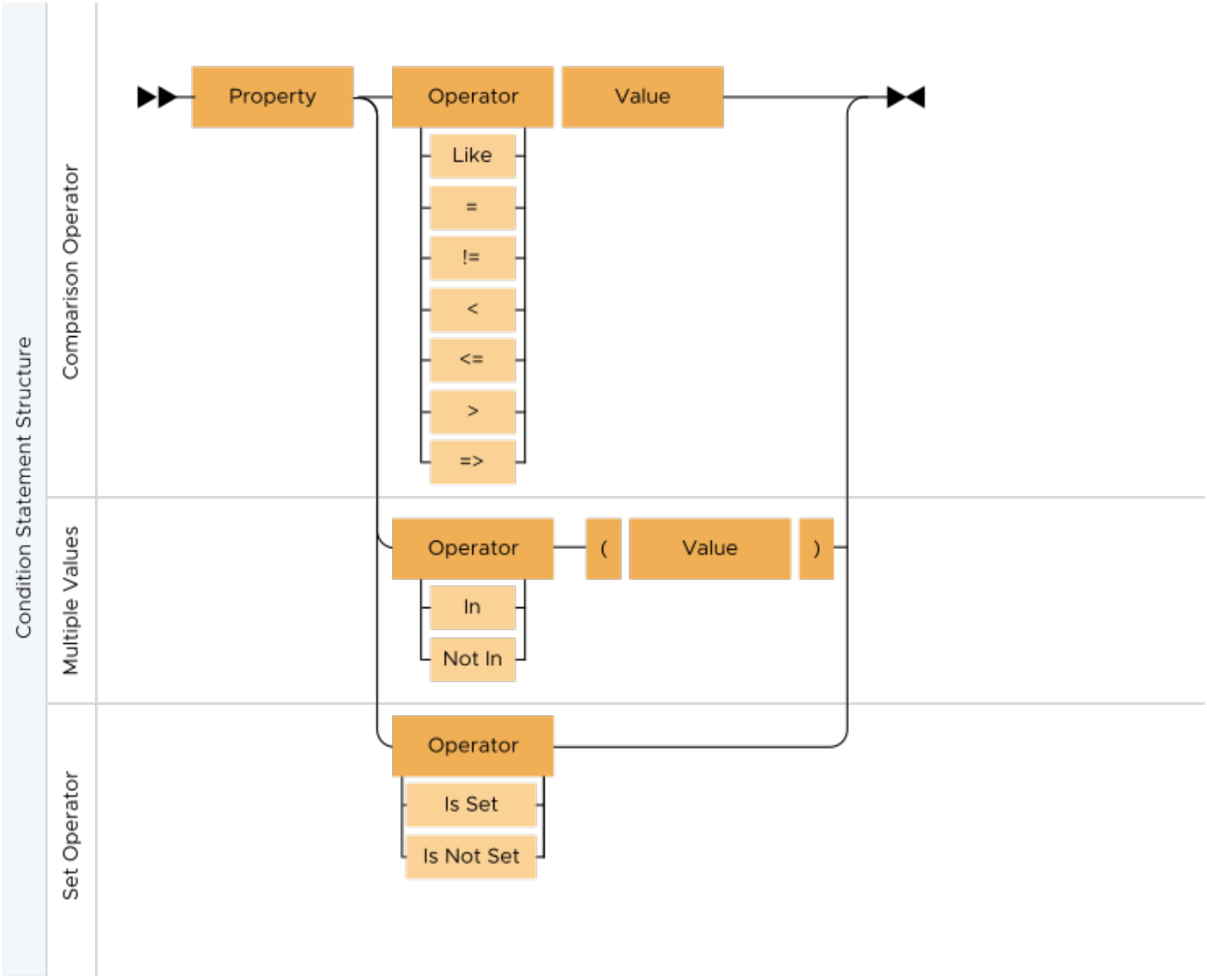



   Here are some examples:
   a. `Virtual machines`
   b. `Hosts`

c. `Flows`
d. `MTU Mismatch Alerts`
e. `Problems`

– **Filters**: The syntax for filter is as follows:

The syntax for condition is as follows:



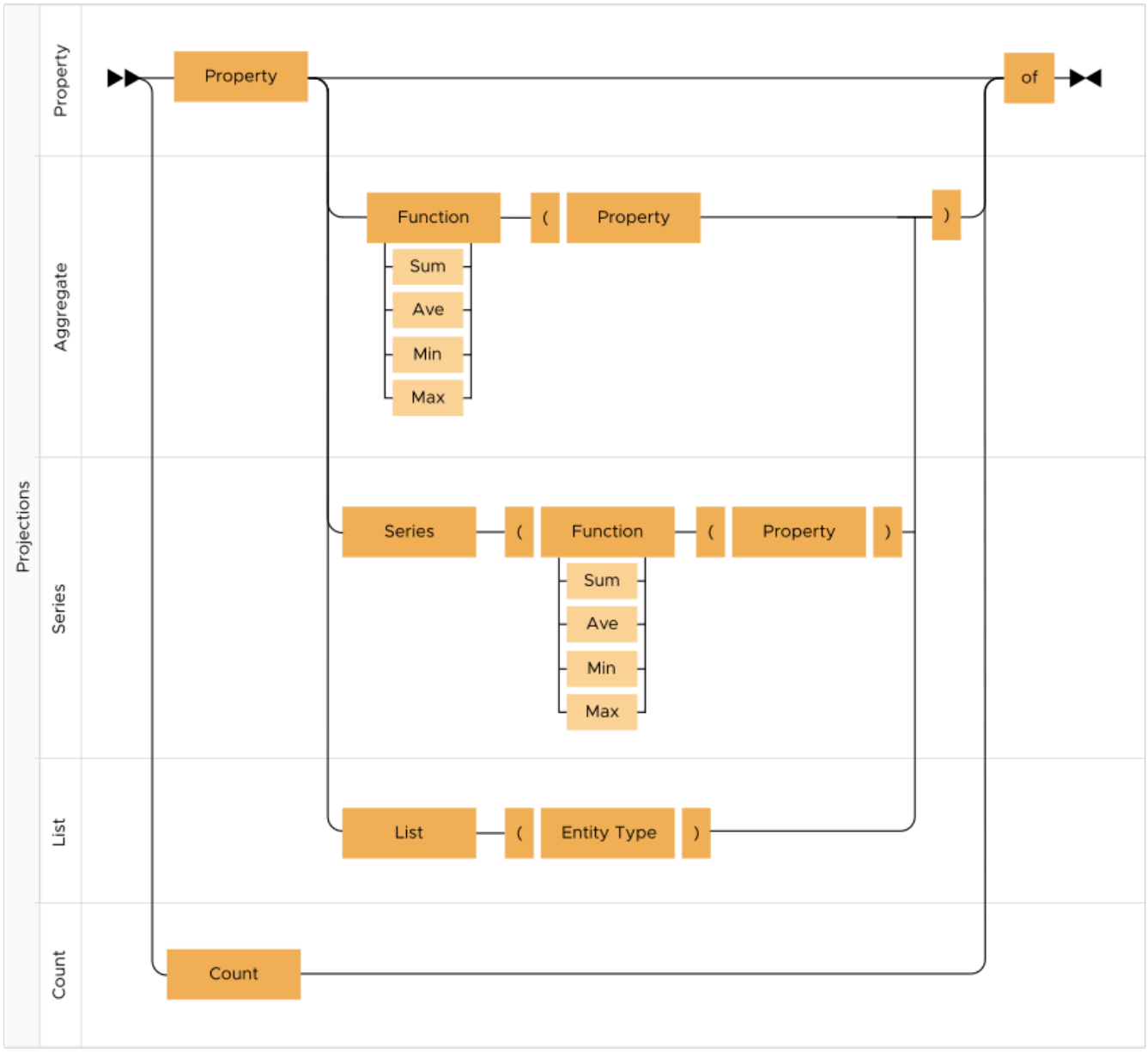

A filter clause can be used to filter search results. The condition in a filter clause consists of property, comparison operator, and value. The conditions can be combined with logical operators to form complex conditions. Here is a list of the operators that you can use:

| Operator | Examples |
|---|---|
| = | `flows where source ip address = '10.16.240.0/24'`<br>`flows where flow type = 'Source is VM'` |
| != | `vms where ip address != '10.17.0.0/16'` |

| Operator | Examples |
|---|---|
| > | `vms where memory > 4096 mb` |
| < | `vms where cpu usage rate < 70%` |
| >= | `vms where memory >= 4096 mb` |
| <= | `vms where cpu usage rate <= 70%` |
| like | `vms where name like 'app'` |
| not like | `vms where name not like 'app'` |
| in | `flows where port in (22, 23, 80, 443)`<br>`vm where ip address in (192.168.91.11, 192.168.91.10)` |
| not in | `flows where port not in (22, 23, 80, 443)`<br>`vm where ip address not in (192.168.91.11, 192.168.91.10)` |
| is set | `vms where firewall rule is set` |
| is not set | `vms where firewall rule is not set` |
| () | `flows where (src tier = 'App' and destination tier = 'DB') OR (destination tier = 'App' and source tier = 'DB')` |
| and | `flows where src tier = 'App' and destinationtier = 'DB'` |
| or | `flows where flow type = 'Source is VMKNIC' or flow type = 'Destination is VMKNIC'` |
| matches | `vm where name matches '.*'`<br>`vm where name matches 'a.*'`<br>`vm where name matches '[a-z]vm-delta[0-9]'` |
| not matches | `vm where name not matches '.*'`<br>`vm where name not matches 'a.*'`<br>`vm where name not matches '[a-z]vm-delta[0-9]'` |
| nested 'in' operator | `vm where in (vm where name = 'x')`<br>`vm where in (vm of host where name = 'x')`<br>`vm where host in (host of vm where name = 'x')`<br>`vm where name in (name of vm where name = 'x')` |

**Projections**: A projection clause in a query decides what fields must be displayed from the filtered entities. This is an optional clause. If the projection clause is not specified, then the default set of fields is shown in the search results. A projection clause can contain any one of the following items:

1. Property
2. Count
3. List
4. Aggregation
5. Series




a. **Property**: When entities are searched by an entity type, default set of properties are shown in the search results. Using projections, we can select the fields that should appear in the search results. For example, `os of vms` lists all virtual machines with `OS property` in the search results.
   Here are some more examples:

   - `cpu cores of vms`
   - `source ip address of flows`

   If a metric property is used, a graph is displayed for each entity with the metric property as `y-axis` and time as `x-axis`.
b. **Count**: The count query can be used to compute the number of objects of an entity type. Here are some examples:
   - `count of vms`
   - `count of hosts`
   - `count of flows`
c. **List**: A list operator is helpful if the filter condition cannot be applied on the entity that you fetch.
   For example:

`List(host) of vms where memory <= 2gb`

This query fetches list of hosts, whereas the filter condition is applied on virtual machines. For example,

`List(ip address)of vms where cpu cores = 1`

**Aggregate functions**: An aggregate function allows you to calculate a single value from a numerical `config` or `metric` property. The search query language supports the following aggregate functions:

- `max`
- `sum`
- `min`
- `avg`

Here are some examples:

- `sum(memory) of hosts`
- `sum(memory), sum(cpu cores) of vms`
- `sum(bytes) of flows`

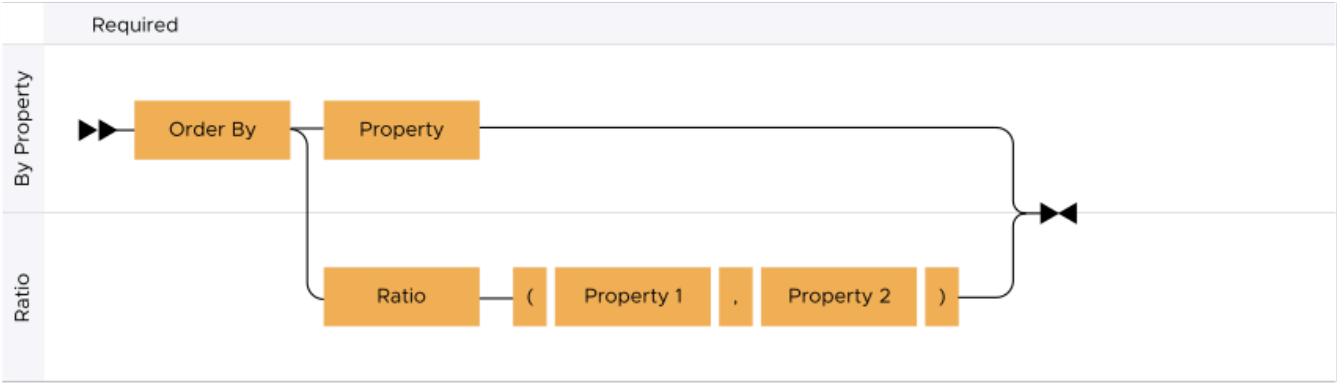
d. **Series**: A series operator is used to perform aggregation on the metric properties. For example:

`series(avg(cpu usage)) of vms where cpu cores = 4`

This query displays graph containing average cpu usage of all virtual machines with 4 cpu cores. Here are some examples:

- `series(sum(network usage)) of vms where name like 'app'`
- `series(sum(memory usage)) of vms where name like 'db'`
- `series(avg(cpu usage)), series(avg(memory usage)) of vms`

– **Ordering**: The search results can be sorted using `order by` clause. Only one field is allowed in `order by` clause. Results are sorted in descending order by default.




Here are some examples:

a. `vms order by cpu cores`

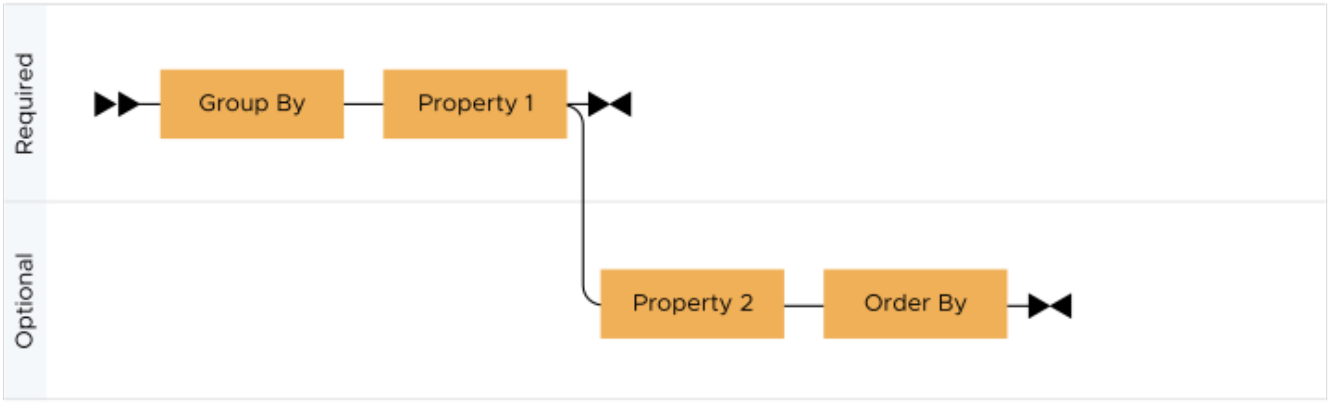
b. `vms order by cpu cores asc`

c. `flows order by bytes`

The `limit` clause can be used to limit the number of results. This must be preceded by the `order by` clause. For example:

`vms order by memory limit 5`

– **Grouping**: The entities can be grouped by a property. When entities are grouped by a property, by default, the number of results in each group are shown. By adding a projection, sum/max/min of any property can be computed . Adding `order by` clause sorts the results. If `order by` or `projection` clause is present in a query, then the aggregation function must be present.




`sum(bytes) of flows group by dest vm`

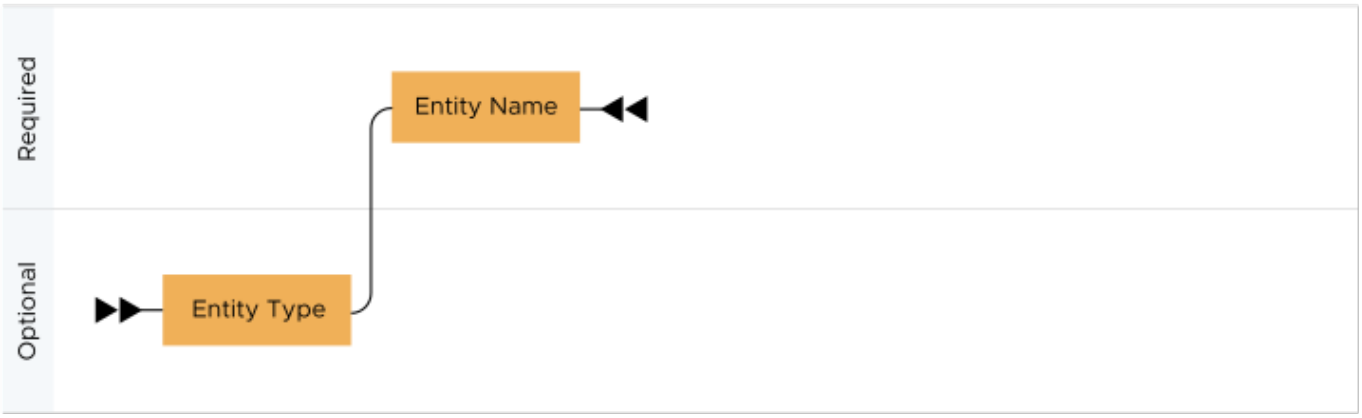
This query is valid as the query has aggregation function in the projection clause. A query such as `bytes of flows group by dest vm` is invalid as there is no aggregation function in the projection clause.

Here are some examples:

a. `vms group by host`

b. `sum (bytes) of flows group by dest vm order by sum(bytes)`

2. **Entity Queries**




**Search by entity type**: All entities of an entity type can be listed by searching the entity type.

Examples: `vms` , `hosts`, `flows`, `nsx managers`

a. **Search by entity name**

- Search by full name: If the full name of an entity is known, it can be searched by enclosing the name in single quotes.

  Examples: `'prod-68-1'`, `'app1-72-1'`

- Search by partial name: Search by a single word or multiple words fetches all the entities matching the input words.

  Examples: `prod`, `app1`

**Note:**

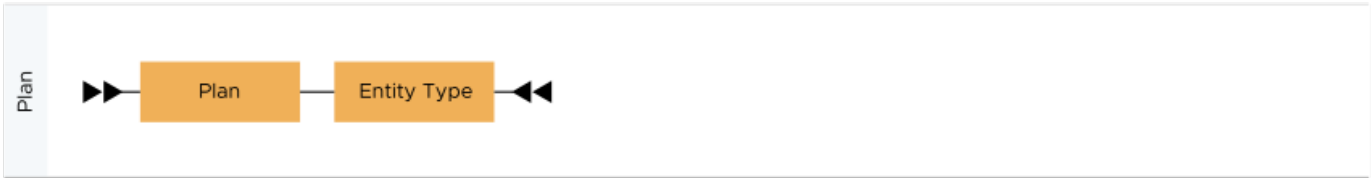
If input contains keywords or entity types, then it may be processed as a search query.

- Search by entity type and name: If both the name and the type of an entity are known, it can be searched by querying entity type and entity name together.

  Example: The search query `'vm app1'` returns all VMs containing `app1`.

3. **Planning Queries**

These queries can be used to plan the security of the data center by analyzing flows.

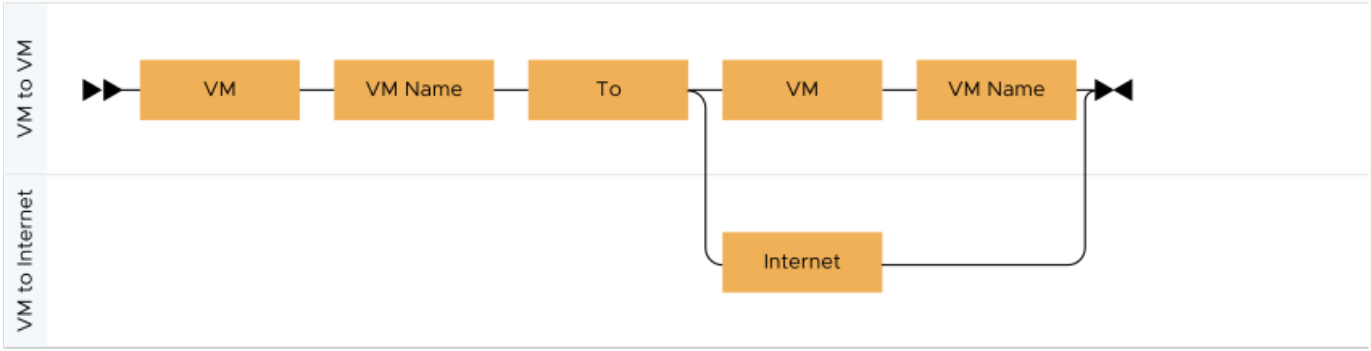



Examples:

1. `plan securitygroup1`
2. `plan host1`
3. `plan security`

4. **Path Queries**

These queries can be used to show the path between two VMs or the path from VM to Internet.




Examples:

1. `Vm 'vm1' to Vm 'vm2'`
2. `VM 'vm1' to Internet`

**Note:**

- The search queries are not case-sensitive.
- The entity types or the configuration properties can have synonyms. For example, entity type `'virtual machine'` has synonym `'vm'`.

## Azure Search Queries

You can search Azure entity details in VCF Operations for networks.

Here are some sample search queries:

| Azure Entities | Sample Queries |
|---|---|
| Microsoft Azure | `Azure` |
| Azure Application Security Group | `Azure Application Security Group where Azure Virtual Network = 'Test-vnet2'` |
| Azure Data Source | `Azure Data Source` |
| Azure NSG Rule | `Azure NSG Rule where Action = 'ALLOW'` |
| Azure Network Interface | `Azure Network Interface where Azure Virtual Network = 'Test-vnet2'` |
| Azure Network Security Group | `Azure Network Security Group where Subscription = 'vRNI-dev'` |
| Azure Route | `Azure Route where Route Table = 'TestRouteTable'` |

| Azure Entities | Sample Queries |
|---|---|
| Azure Route Table | `Azure Route Table where Azure Virtual Network = 'aks-vnet-28255566'` |
| Azure Subnet | `Azure Subnet where Azure Virtual Network = 'vrni-01-vnet'` |
| Azure Subscription | `Azure Subscription` |
| Azure Virtual Machine | `Azure Virtual Machine where Azure Application Security Group = 'TestASG'` |
| Azure Virtual Network | `Azure Virtual Network where Azure Peer Virtual Network = 'vrni-01-vnet'` |

## Search Queries for Cisco ACI Entities

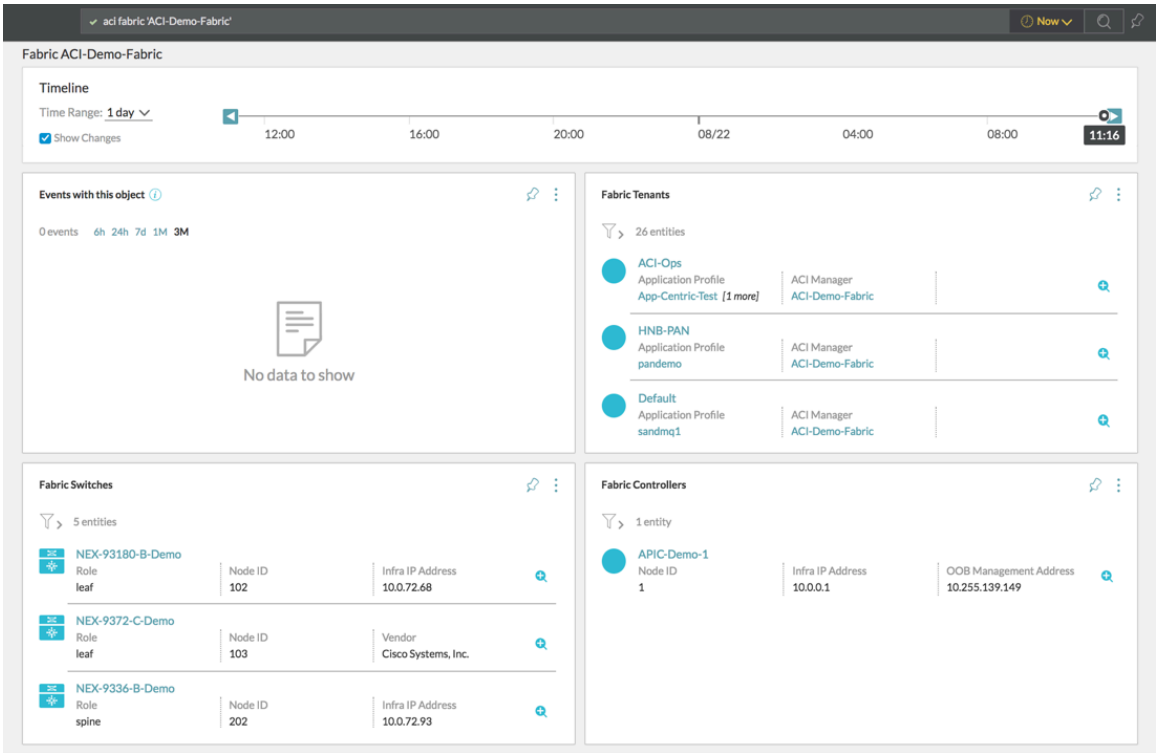
Here is a list of some of the Cisco ACI entities on which you can perform a search:

**Note:**
The entities are prefixed by `aci`.

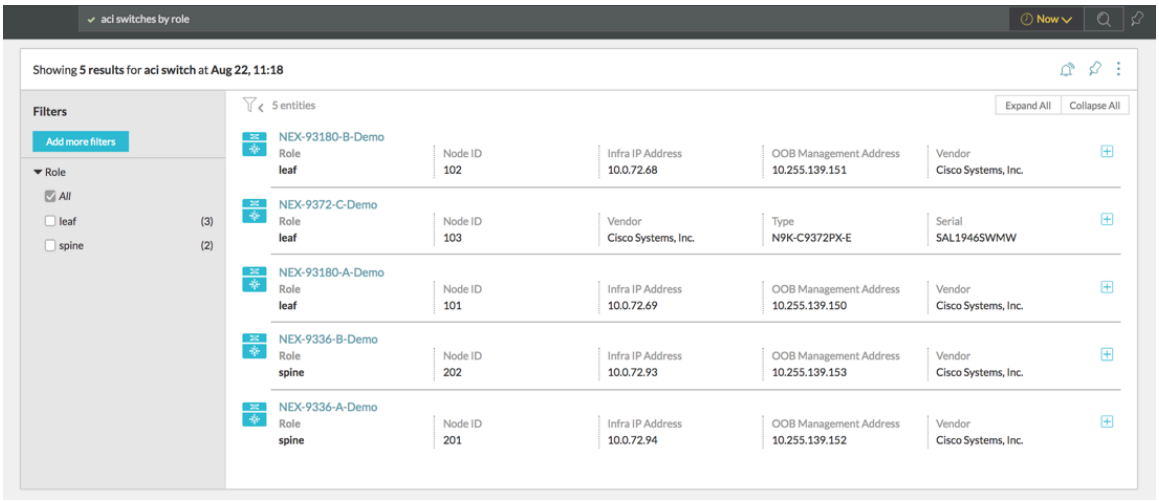
- `aci application profile`
- `aci bridge domain`
- `aci contract`
- `aci endpoint group`
- `aci fabric`
- `aci switch`
- `aci tenant`
- `aci zoning filter`
- `aci zoning rule`

Here are some sample search queries:

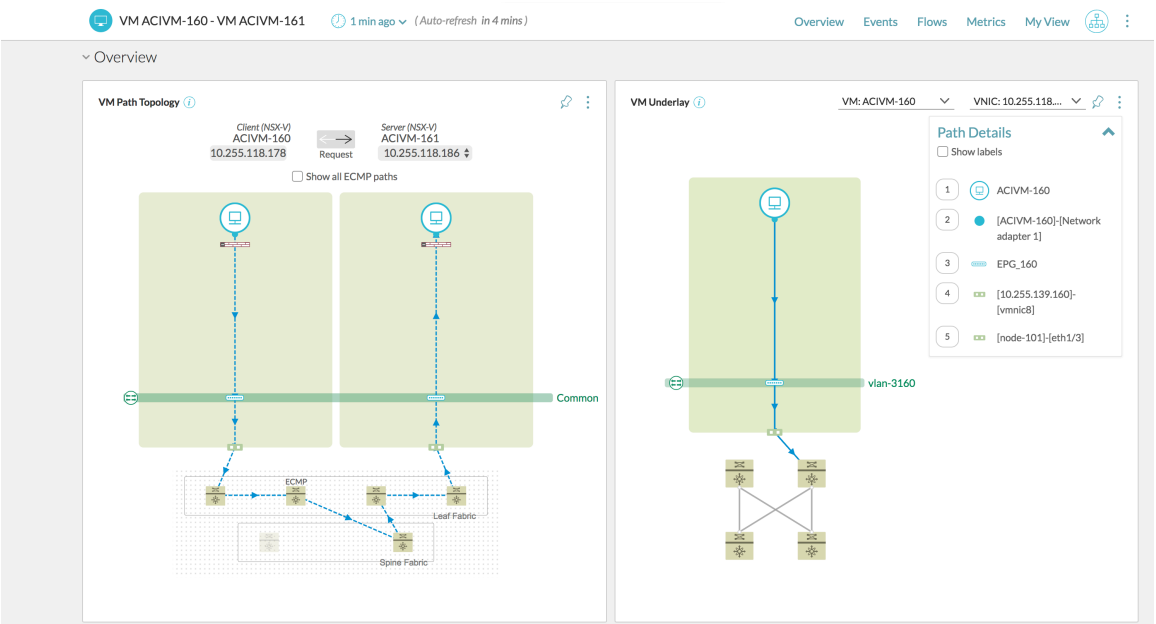
- `aci fabric 'ACI-Demo-Fabric'`: This query retrieves information on the tenants, switches, and controllers in the ACI fabric.



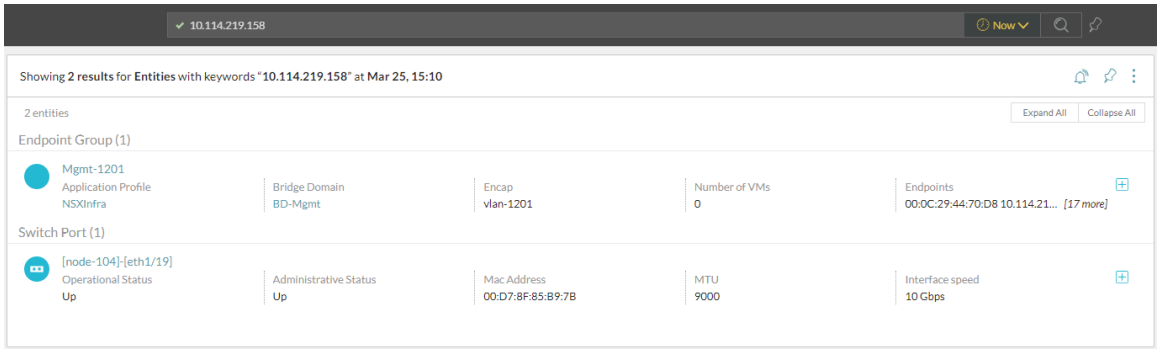
- `aci switches by role`: This query retrieves information on the various leaf switches or the spine switches in the ACI fabric.
  From the switch list, click a switch name to get more details about it.



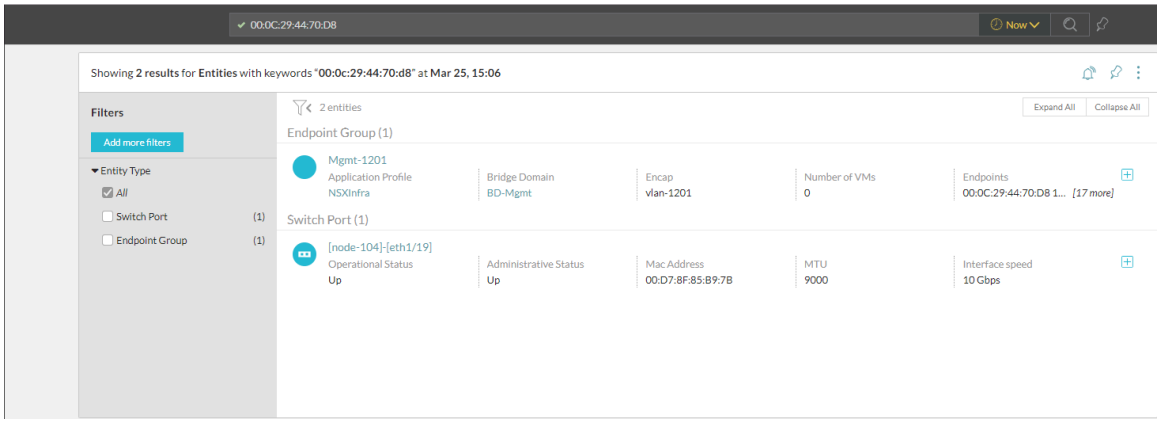

- `aci endpoint group`: This query retrieves a list of the endpoint groups with the associated VMs, bridge domains, and VRFs.
- `aci application profile 'Production'`: This query retrieves the application profile of Production with the contained endpoint groups and VMs.
- `VMware VM 'ACIVM-160' to VMware VM 'ACIVM-161'` : This query shows the VM-VM path between the two VMs.



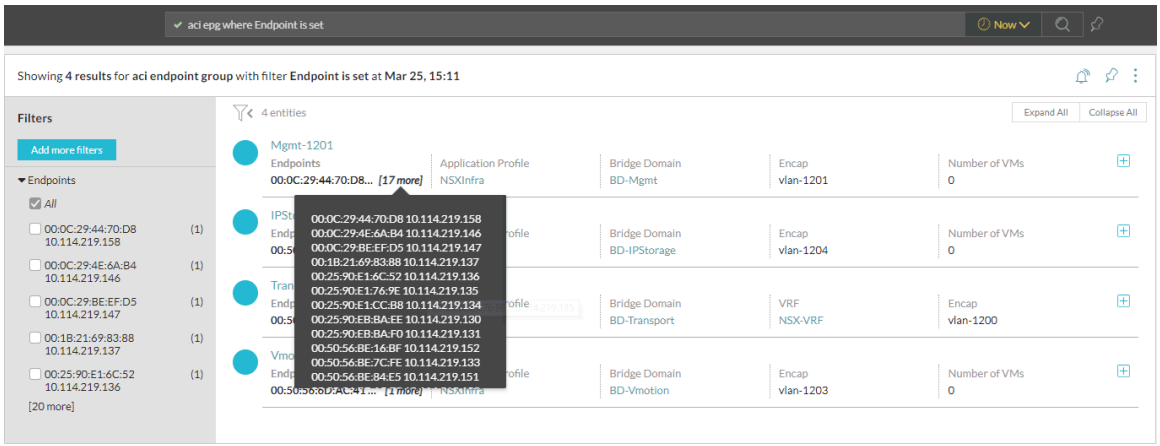

- You can search with IP address to get the port, end point group, and bridge domain details.



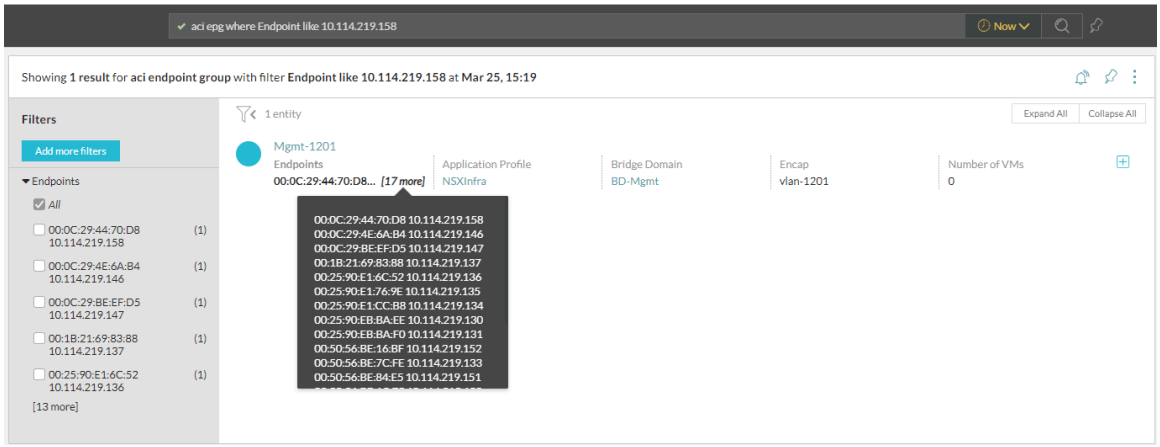

- You can search with Mac address to get the port, end point group, and bridge domain details.




- You can search for an end point group and get the list of associated endpoints.




- You can search for an endpoint.



## Palo Alto Search Queries

You can search Palo Alto entity details in VCF Operations for networks.

Here are some sample search queries:

| Entities | Queries |
|---|---|
| Palo Alto Address | `Palo Alto address where vm = <>`<br>`VM where Address = <>` |
| Palo Alto Address Group | `Palo Alto address group where Translated VMs = <>`<br>`VM where address group = <>` |
| Palo Alto Device | `Palo Alto Device where Version = <>`<br>`Palo Alto Device where connected = true`<br>`Palo Alto Device where family = 'PA-5060'` |
| Palo Alto Physical Device | `Palo Alto Physical Device where model = 'PA-5060'` |
| Palo Alto VM Device | `Palo Alto VM Device where model = 'PA-VM'` |
| Palo Alto Device Group | `Palo Alto Device Group where device = <>`<br>`Palo Alto Device Group where address = <>`<br>`Palo Alto Device Group where address group = <>` |
| Palo Alto Service | `Palo Alto service where Port = <>`<br>`Palo Alto service where Protocol = <>` |
| Palo Alto Service Group | `Palo Alto service group where Member = <>` |
| Palo Alto Policy | `Palo Alto Policy where Source vm = <> and Destination vm = <>`<br>`Palo Alto Policy where Source IP = <> and Destination IP = <>` |
| Palo Alto firewall | `Palo Alto firewall where Rule = <>` |
| Palo Alto Zone | `Palo Alto Zone where device = <>` |
| Palo Alto Virtual System | `Palo Alto Virtual System where Device = <>`<br>`Palo Alto Virtual System where Device Group = <>` |

**Note:**
Other than the queries, you can also use facets to analyze the search results.

## Fortinet Search Queries

You can search Fortinet entity details in VCF Operations for networks.

Here are some sample search queries:

| Fortinet Entities | Sample Queries |
|---|---|
| Fortinet Policy Package | `Fortinet Policy Package where Domain Manager = 'ADOM_NAME'` |
| Fortinet Policy | `Fortinet Policy where Source IP = '10.0.0.15'` |
| Fortinet Address | `Fortinet Address where Address Type = 'ipmask'` |

| Fortinet Entities | Sample Queries |
|---|---|
| Fortinet Dynamic Address | `Fortinet Dynamic Address where Domain Manager = 'ADOM_NAME'` |
| Fortinet Dynamic Address Group | `Fortinet Dynamic Address Group where Domain Manager = 'ADOM_NAME'` |
| Fortinet service | `Fortinet Service where port = 5900` |
| Fortinet service Group | `Fortinet Service Group where Manger = '10.0.15.101'` |
| Fortinet ADOM | `Fortinet ADOM where Manager ID = '10.0.15.101'` |
| Fortinet VDOM | `Fortinet VDOM where Domain Manager = 'ADOM_NAME'` |
| Fortinet Dynamic Interface | `Fortinet Dynamic Interface where Domain Manager = 'ADOM_NAME'` |

## Enriching Flows with the Infoblox DNS Data

VCF Operations for networks supports two sources of DNS information:

- Imported CSV file
- Infoblox DNS

**Note:**
If there is a conflict between Infoblox DNS and the CSV file, the information from Infoblox DNS takes precedence.

You can use various search queries to find out more about the source of DNS entries in a flow.

**Table 1336:**

| Keyword | Sample Search Query | Description |
|---|---|---|
| DNS Provider | `Flows where DNS Provider='Infoblox'` | Provides the list of flows in which the DNS data is obtained from Infoblox. |
| DNS Provider | `Flows where DNS Provider='CSV'` | Provides the list of flows in which the DNS data is obtained from CSV. |
| Source DNS Provider | `Flows where Source DNS Provider='Infoblox'` | Provides the list of flows in which the DNS provider for the source IP address is Infoblox. |
| Destination DNS Provider | `Flows where Destination DNS provider='Infoblox'` | Provides the list of flows in which the DNS provider for the destination IP address is Infoblox. |

## Common Search Queries for Kubernetes Entities

You can search for Kubernetes entities details in VCF Operations for networks.

**Common Queries**

- Search Flows : `flows where` *`Kubernetes Object = Object name`*
  Example: flows where *`Kubernetes Cluster = 'Production'`*
- View the service scale: `kubernetes pods group by Kubernetes Services`
- View the node load: `kubernetes Pods group by Kubernetes Node`
- View the node health: `MemoryPressure and PIDPressure and DiskPressure and Ready of Kubernetes Node`

- View flow compliance: `flows from Kubernetes Object` *`name of the object`* `to Kubernetes Object` *`name of the object`*
  Example: `flows from Kubernetes Namespace'`*`PCI`*`' to Kubernetes Namespace'`*`Non-PCI`*`'`
- View the Path topology:
  - Kubernetes service *`service name`* to Kubernetes service *`service name`*
  - Kubernetes service *`service name`* to Kubernetes pod *`pod name`*
  - Kubernetes pod *`pod name`* to Kubernetes pod *`pod name`*

**Table 1337: Queries on Kubernetes Object**

| Kubernetes Object | Query | Description |
|---|---|---|
| Namespace | • kubernetes namespace where L2 Networks = '*a*'<br>• list(Kubernetes Node) of Kubernetes Pod where Kubernetes Namespace = '*a*' | • Return the Kubernetes namespace where it is connected to L2 Network 'a'<br>• Return the list of Kubernetes nodes where Kubernetes namespace is 'a' |
| Pod | • NSX-T Logical port where connectedto.modelKey in (modelKey of kubernetes nodes) order by Tx Packets desc<br>• NSX-T Logical port where connectedto.modelKey in (modelKey of kubernetes pods) and Rx Packet Drops > 0<br>• new kubernetes pod in last 1 hour | • Return the list of logical ports which are connected to a node based on transferred packets in descending order<br>• Return the list of logical ports which are connected to Kubernetes pods and Rx dropped packets > 0<br>• New Kubernetes pods discovered in last one hour |
| Services | • kubernetes pods where kubernetes services is not set<br>• kubernetes pods group by Kubernetes Services, Kubernetes Cluster | • List of Kubernetes pods that does not have a service<br>• Number of pods running on each service |
| Nodes | • kubernetes nodes where Ready != 'True'<br>• kubernetes node where Virtual Machine = 'vm-a' | • List of unhealthy Kubernetes nodes<br>• Kubernetes node that is part of 'vm-a' virtual machine |
| Flows | • flows where kubernetes service is set<br>• flows where source kubernetes node = 'a' | • List of flows where either a source or a destination Kubernetes service exists<br>• List of flows where source Kubernetes node = 'a' or destination Kubernetes node = 'a' |

**Table 1338: Additional Queries**

| Entity/Components | Query | Description |
|---|---|---|
| Applications that have Kubernetes entities | application where virtual member = ‘service-a’ | List of all applications where Kubernetes service ‘service-a’ is a member |
| | application where virtual member = ‘service-a’ and virtual member.Kubernetes Namespace = ‘namespace-b’ | List of all applications where Kubernetes service ‘service-a’ and Kubernetes namespace is ‘namespace-b’ is a member |
| | tier where virtual member = ‘service-a’ and virtual member.Kubernetes Namespace = ‘namespace-b’ | List of all tiers where Kubernetes service ‘service-a’ and Kubernetes namespace is ‘namespace-b’ is a member |
| | count of applications where Virtual Member in (kubernetes services) | The number of applications where member is of type Kubernetes service |
| | count of applications where virtual member in (kubernetes services where Kubernetes Namespace = ‘sock-shop’) | The number of applications where member is of type Kubernetes service which is under Kubernetes namespace ‘sock-shop’ |

| Entity/Components | Query | Description |
|---|---|---|
| | list(virtual member) of applications where Name = ‘app-1’ and virtual member.Kubernetes Cluster is set | List of all the Kubernetes services as members for the application ‘app-1’ |
| Metrics | nsx-t logical port where (ConnectedTo in (Kubernetes Pods where kubernetes cluster is set)) and Rx Packet Drops > 0 group by ConnectedTo order by max(Rx Packet Drops) | Rx Packet drops that are grouped based on Kubernetes pod |
| | nsx-t logical port where (ConnectedTo in (Kubernetes Nodes where kubernetes cluster is set)) and Rx Packet Drops > 0 group by ConnectedTo order by max(Rx Packet Drops) | Rx Packet drops that are grouped based on Kubernetes node |
| | nsx-t logical switch where Rx Packet Drops > 0 and Tag like ‘ncp/project:’ order by Rx Packet Drops | Rx Packet drops that are grouped based on Kubernetes namespace |
| | nsx-t logical switch where Rx Packet Drops > 0 and Tag like ‘ncp/project:<namespace name>’ | Rx Packet drops that are specific to a Kubernetes namespace |
| | nsx-t logical port where (ConnectedTo in (Kubernetes Pods where kubernetes cluster is set)) and Rx Packet Drops > 0 group by ConnectedTo.Kubernetes service order by max(Rx Packet Drops) | Rx Packet drops that are grouped based on Kubernetes services |
| | flows where firewall action = ‘DROP’ group by Kubernetes Service | Drop flows that are grouped based on Kubernetes services |
| | flows where firewall action = ‘DROP’ group by source Kubernetes Namespace | List of all dropped flows that are grouped based on Kubernetes namespace |
| Kubernetes alerts | Kubernetes alerts where Problem Entity = ‘<pod/ namespace/node Name>’ | List of all Kubernetes alerts for the given Kubernetes entity. Kubernetes entity would be either pod or namespace or node |
| | Kubernetes alerts where alert code = ‘ImagePullBackOff’ in last 24 hours | List of Kubernetes alerts of type ‘ImagePullBackOff’ in last 24 hours |
| | Kubernetes alerts where problem entity.Kubernetes Cluster = ‘<cluster-a>’ | List of all Kubernetes alerts for the given cluster |

## Sample Search Queries Related to Load Balancer

To filter or search data related to the load balancer, you can use search queries.

- `vm where lbServiceNodes is set` - Lists all the VMs that are hosting an application where the load is distributed.
- `vm where lbServiceNodes is set and PowerState !='POWEREDON'` - Lists all the VMs that are hosting a load balanced application, but currently non-functional.
- `pool member where state = 'DISABLED'` - Lists all pool members that are deactivated.
- `Count of Pool Members where Service Port = '80'` - Provides the count of all pool members for a particular type of service that are running on port 80.
- `service node where virtual machine is not set` - Lists all service nodes that are using physical servers as application servers. Or, shows the VMware vCenter Server that is hosting multiple VMs, and is not added in VCF Operations for networks.

## Search Queries for NSX Firewall Rules

You can search for NSX Firewall rules in VCF Operations for networks.

**Table 1339: NSX Firewall Rules Queries**

| Search Query | Description |
|---|---|
| `VM where incoming rules.Source Any` | View rules with any source (can combine with a specific port). |
| `Firewall rule where action = allow and service any = true` | View firewall rules that allow any ports. |
| `Firewall Rule Masked Alert` | View the list of unused firewall rules. |
| `New firewall rules in last 24 hours` | View the firewall rules created in the last 24 hours. |
| `New firewall rules in last 7 days` | View the firewall rules created in the last 7 days. |
| `New firewall rules in last 30 days` | View the firewall rules created in the last 30 days. |
| `Firewall rule where flow is not set` | View the list of all inactive firewall rules. |
| `Flow group by firewall rule` | View the count of flows hitting each firewall rule. |
| `Security group where Indirect Incoming Rules is not set and Indirect Outgoing Rules is not set and Direct Incoming Rules is not set and Direct Outgoing Rules is not set` | View the security group that is not used. |
| `Ipset where Indirect Incoming Rules is not set and Indirect Outgoing Rules is not set and Direct Incoming Rules is not set and Direct Outgoing Rules is not set` | View the IPSet that is not used. |
| `Flow where rule id in (1011, 1012, 1013)` | Flows hitting a specific rule ID. |
| `Flow where application = app1` | Flows hitting the application. |

- Unused firewall rules
- Firewall rule Masking rule alert

## Avi Load Balancer Search Queries

You can search Avi Load Balancer entity details and alerts in VCF Operations for networks.

Here are some sample search queries on Avi Load Balancer entities:

| Avi Load Balancer Entities | Sample Queries | Description |
|---|---|---|
| Data Source | `NSX-ALB Data Source where Problem = 'Data Source is not reachable'` | Shows a list of Avi Load Balancer data sources that are not reachable from the collector VM |
| Cluster | `NSX-ALB Cluster where Controller Count < 3` | Shows a list of Avi Load Balancer clusters having less than three controllers |
| Controller | `NSX-ALB Controller group by Manager where CPU Usage Rate > 50 %` | Shows a list of Avi Load Balancer clusters where the CPU usage rate of the controller is greater than 50% |
| Virtual Service | • `NSX-ALB Virtual Service group by Analytics Profile`<br>• `NSX-ALB Virtual Service group by Name order by Active connections`<br>• `NSX-ALB Virtual Service where Operational Status = 'Down'` | • Shows a list of virtual services that are grouped by analytics profile<br>• Shows a list of virtual services that are grouped by active connections<br>• Shows a list of virtual services that are shutdown |

| Avi Load Balancer Entities | Sample Queries | Description |
|---|---|---|
| Service Engine | `NSX-ALB Service Engine where Vnic in (Vnic where Tx Packet Drops > 100)` | Shows a list of service engines where the packet drops of vNIC is greater than 100 |
| Pool Group | `NSX-ALB Pool Group where Cloud = 'nsxt_cloud_overlay'` | Shows a list of pool groups associated with the Cloud `'nsxt_cloud_overlay'`. |
| Pool | `NSX-ALB Pool where Tenant = 'admin'` | Shows a list of server pools associated with the tenant `'admin'`. |
| Analytics Profile | `NSX-ALB Analytics Profile group by Tenant, Manager where Tenant = 'admin'` | Shows a list of analytics profiles that are grouped by the cluster and tenant `'admin'` |

Here are some sample queries on Avi Load Balancer alerts:

| Sample Queries | Description |
|---|---|
| `Alerts where Alert Tags = 'NSX-ALB'` | Shows a list of Avi Load Balancer system alerts |
| `Alerts where Alert Tags = 'NSX-ALB' and Problem Entity = 'nsxt_cloud_overlay-Avi-se-azjyc'` | Shows a list of alerts associated with the entity `'nsxt_cloud_overlay-Avi-se-azjyc'` |
| `Alerts where Alert Tags = 'NSX-ALB' group by Problem Entity` | Shows a list of alerts grouped by Avi Load Balancer entities |

## VMC SDDC Search Queries

You can search the VMware Cloud on AWS SDDC ( Software-defined data center) entity details in VCF Operations for networks.

Here are few sample search queries:

| VMC SDDC entities | Sample Queries | Description |
|---|---|---|
| NSX Manager | `vmc sddc where NSX Manager` | Shows NSX Manager associated with the VMware Cloud on AWS SDDC. |
| NSX Manager FQDN | `vmc sddc where NSX Manager Fqdn` | Shows the NSX Manager FQDN for the VMware Cloud on AWS SDDC. |
| NSX Manager Private Ip | `vmc sddc where NSX Manager Private Ip` | Shows the NSX Manager private IP address for the VMware Cloud on AWS SDDC. |
| NSX Manager Public Ip | `vmc sddc where NSX Manager Public Ip` | Shows the NSX Manager public IP address for the VMware Cloud on AWS SDDC. |
| Name | `vmc sddc where Name` | Shows the name of the VMware Cloud on AWS SDDC. |
| Org Id | `vmc sddc where Org Id` | Shows the organization ID to which the SDDC belongs. |
| Org Name | `vmc sddc where Org Name` | Shows the organization name to which the SDDC belongs. |
| Region | `vmc sddc where Region` | Shows the AWS region where the SDDC is located. |
| VC FQDN | `vmc sddc where VC FQDN` | Shows the VMware vCenter FQDN for the VMC SDDC. |

| VMC SDDC entities | Sample Queries | Description |
|---|---|---|
| VC Manager | `vmc sddc where VC Manager` | Shows the VMware vCenter Manager associated with the VMware Cloud on AWS SDDC. |
| VC Private Ip | `vmc sddc where VC Private Ip` | Shows the private IP address of the VMware vCenter for the VMware Cloud on AWS SDDC. |
| VC Public Ip | `vmc sddc where VC Public Ip` | Shows the public IP address of the VMware vCenter for the VMware Cloud on AWS SDDC |
| Vendor ID | `vmc sddc where Vendor ID` | Shows the ID of the SDDC. |

**Related Topic**

Viewing VMC SDDC Details

You can use the **VMC SDDC** page to get an overview of your VMC on AWS SDDC in VCF Operations for networks.

## VMware NSX-T Manager Search Queries for BGP and ECMP

You can search the VMware NSX-T Manager entity details in VCF Operations for networks.

Here are few sample search queries:

| VMware NSX-T Manager Entities | Sample Queries |
|---|---|
| BGP Neighbor | • BGP Neighbors of a given NSX-T Manager:<br>`list(BGP Neighbor) of NSX-T Router where Manager = 'NSX-T Manager Name'`<br>• BGP Neighbors of a given NSX-T Router:<br>`list(BGP Neighbor) of NSX-T Router 'Router Name'`<br>• BGP Neighbors of a given NSX-T Transport Node:<br>`list(BGP Neighbor) of NSX-T Service Router where Active Node = 'Transport Node Name'`<br>• BGP Neighbor with non active BGP Status:<br>`list(BGP Neighbor) of BGP Connection where BGP Status != 'Active'`<br>• BGP Neighbor with a specific AS number:<br>`BGP Neighbor where Local AS = '64500` |
| BGP Connection | • All BGP Connection of a BGP Neighbor:<br>`BGP Connection where BGP Neighbor = 'BGP Neighbor IP Address'`<br>• All BGP Connection with a specific BGP Status:<br>`BGP Connection where BGP Status != 'ACTIVE'`<br>• All BGP Connections of an NSX-T Router or NSX-T Manager:<br>`BGP Connection where Logical Router = 'Router Name' or Manager = 'NSX-T Manager Name'`<br>• All BGP Connections with a specific Remote AS:<br>`BGP Connection where Remote AS = '12334'` |

| VMware NSX-T Manager Entities | Sample Queries |
|---|---|
| Routes tagged with BGP Neighbor | • BGP Learned Routes for a given NSX-T Manager:<br>`Route where BGP Neighbor = 'BGP Neighbor IP Address' and Vrf.manager = 'NSX-T Manager Name'`<br>• BGP Learned Routes on a given Router:<br>`Route where BGP Neighbor = 'BGP Neighbor IP Address' and vrf = 'Router Name'`<br>• BGP Learned Routes from a BGP Neighbor with a Specific BGP Status:<br>`Route where BGP Neighbor in (list(BGP Neighbor) of BGP Connection where BGP Status != 'Active')`<br>**Note:**<br>VCF Operations for networks stores the FIB routes from the NSX-T Edge nodes. These search queries, return the FIB routes which are learned from the BGP Neighbor details. |
| ECMP Status | • NSX-T Router with ECMP enabled:<br>`NSX-T Router where ECMP = 'Enabled' and manager = 'NSX-T Manager Name'`<br>• NSX-T Edge having ECMP enabled:<br>`NSX-T Transport Node where ECMP = 'Enabled'`<br>• NSX-T Edge Cluster where an ECMP Router is deployed:<br>`NSX-T Edge Cluster where ECMP = 'Enabled'`<br>**Note:**<br>An NSX-T router is ECMP enabled when it is deployed with the HA of active-active. |

## AWS Account Search Queries

You can search AWS account entity details in VCF Operations for networks.

Here are few sample search queries:

| AWS Entities | Sample Queries |
|---|---|
| AWS EC2 | `AWS EC2 where Security Group = '' and Account = ''` |
| AWS interface | `AWS Interface where Network address like '' order by Manager` |
| AWS Internet Gateway | `AWS Internet Gateway where AWS VPC = ''` |
| AWS Route | `AWS Route where Account = '' group by Name` |
| AWS Route Table | `AWS Route Table where AWS Subnet = ''` |
| AWS Security Group | `AWS Security Group where Group Name = ''` |
| AWS Subnet | `AWS Subnet where Region = '' group by Vendor ID`<br>`AWS Subnet where Availability Zone = ''` |
| AWS VPC | `sum(bytes) of flows where Source AWS VPC = '' and Destination AWS VPC = ''` |
| AWS VPC Peering Connection | `AWS VPC Peering Connection where Acceptor AWS VPC = ''` |
| AWS VPN Connection | `AWS VPN Connection where AWS Virtual Private Gateway = ''` |

| AWS Entities | Sample Queries |
|---|---|
| AWS Virtual Private Gateway | `AWS Virtual Private Gateway where AWS VPC = ''` |

All the AWS accounts in your organization that are added to the primary account are known as Linked Accounts. For more information, see Primary and Linked AWS Accounts and Add a Primary AWS Account.

Here are few sample queries for linked AWS account discovery:

| AWS Entities | Sample Queries |
|---|---|
| AWS Linked Account | `AWS Linked Account` |
| AWS Direct Connect Gateway | `AWS DX Gateway where ASN = ''` |
| AWS TGW Attachment | `AWS TGW Attachment where Entity Type = 'AWS TGW Attachment'` |

### VMware Cloud on AWS for AWS Entities

Here are the entities related to VMware Cloud on AWS NSX Policy Manager:

- `NSX Policy Manager Data Source`
- `NSX Policy Manager`
- `NSX Policy Firewall`
- `NSX Policy Firewall Rule`
- `NSX Policy Segment`
- `NSX Policy Based VPN`
- `NSX Policy Group`

**Note:**
If NSX-T 2.4 and VMware Cloud on AWS are added as data sources in your VCF Operations for networks, then to get the VMware Cloud on AWS entities, you must add `SDDC type = VMC` filter in your query. For example, to list the Policy Based VPNs for VMware Cloud on AWS, enter `NSX Policy Based VPN where Tier0 = ‘’ and SDDC Type = ‘VMC’.`

Some sample search queries related to the VMware Cloud on AWS entities are:

- `VMs where L2 Network = '' (L2 Network -> NSX Policy Segment)`
- `NSX Policy Based VPN where Tier0 = ''`
- `NSX Policy Based VPN where Local Network = '' (Local Network of Policy Based VPN Rule)`
- `NSX Policy Based VPN where Remote Network = '' (Remote Network of Policy Based VPN Rule)`
- `NSX Policy Group where Translated VM = ''`
- `VM where NSX Policy Group = ''`

**Note:**

- NSX Policy Manager does not support child groups or IPSETS. Hence all the searches like `NSX Policy firewall rule where Indirect __________ = ''` or `NSX Policy group where Indirect _____= ''` are deactivated.

### Search Queries for HCX Mobility Optimized Networking

You can search for HCX Mobility Optimized Networking in VCF Operations for networks

**Table 1340: HCX Mobility Optimized Networking Queries**

| Search Query | Description |
|---|---|
| `VM where MON = 'true'` | View the list of HCX Mobility Optimized Networking enabled VMs. |
| `HCX L2 Extensions where MON = 'true'` | View the list of HCX Mobility Optimized Networking enabled HCX L2 Extensions. |
| `Flows where flow type = 'HCX Mobility Optimized Flow'` | View the list of HCX Mobility Optimized Networking optimized flows. |

## Advanced Queries

Here are some examples for advanced queries:

**Flow Queries for Communication Patterns**

- Total Traffic across data centers or sites (DCI link use)
  `sum(bytes) of flows where ( Dst Manager = 'abc' AND src manager = 'cba') OR ( Dst Manager = 'cba' AND src manager = 'abc')`
- Total VTEP traffic
  - `sum(bytes) of flows where Flow Type = 'Src is VTEP' or flow type = 'Dst is VTEP' VTEP traffic grouped by VMKNIC`
  - `sum(bytes) of flows where Flow Type = 'Src is VTEP' or Flow Type = 'Dst is VTEP' group by ip`
- Other Management Traffic
  `flows where Flow Type = 'Source is VMKNIC' or Flow Type = 'Destination is VMKNIC'`
- Flows for extended L2 network
  `flows where flow type = 'Extended L2 Network' and Destination IP Address = 10.172.13.14`

**Flow Queries for Aggregation and Grouping**

- Total Internet traffic by Source VM
  `sum(bytes) of flows where Flow Type = 'Internet' group by src vm`
- Top ports by total bytes
  `sum(bytes) of flow group by port order by sum(bytes)`
- Top subnet pairs by routed traffic volume
  `sum(bytes) of flow where Flow Type = 'Routed' group by Source Subnet Network, destination subnet network order by sum(bytes)`
- Total VM by total pair bytes
  `sum(bytes) of flows group by src vm , dest vm order by sum(bytes)`
- Top Server VM/Port by total bytes
  `sum(bytes) of flows group by dest vm , port order by sum(bytes)`

**Flow Queries for Capacity Estimation and Sizing**

- Total bytes of all `vm-internet/internet-vm` traffic grouped by ESX ( Palo Alto Service VM sizing)
  `sum(bytes) of flows where flow type = 'internet' and (flow type = ' src is vm ' OR flow type = 'destination is vm ') group by host order by sum(bytes)`
- Aggregated traffic series for matching flows ( Palo Alto Service VM sizing)
  `series( sum(byte rate)) of flows where host = 'ddc1-pod2esx012.dm.democompany.net' and (Flow Type = 'Source is VM' OR flow type = 'Destination is VM')`

**Useful Queries for Application**

- VMs in a given application
  `VM where application = 'CRM'`
- Routed Flows from a given application
  `Flows where source application = CRM and Flow Type = 'Routed'`
- Flows between two tiers (one-way)
  `Flows where src tier = 'App' and Destination Tier = 'DB'`
- Flows between two tiers (one-way)
  `Flows where ( src tier = 'App' and destination Tier = 'DB') OR (destination tier = 'App' and source tier = 'DB')`

**Useful Queries for VM and ESX**

- Properties of Prod -Midtier-1 VM (MAC, IP, host, and so on)
  `CPU Usage Rate, Network Rate, Memory Usage Rate, mac address, ip , vxlan , host of vm 'Quality control-VM26'`
- Network segments having the highest VM count
  `vm group by l2 network`
- Datastores have highest VM count
  `vm group by datastore`
- Hosts by vSphere version
  `host group by version`
- Hosts by vSphere Builds
  `host group by OS`
- All VMs on all host/blade slotted in a particular UCS chassis (Nested Query)
  `vm where host in (host where Blade like 'sys/chassis-1')`

**Useful Queries: General Capacity**

- Number of Datacenters:
  `count of datacenter`
- Number of clusters
  `count of cluster`
- Number of Hosts
  `count of host`
- Number of VMs
  `count of vm`
- Number of Networks
  `count of vlan`

**Useful Queries: Routes**

- VNIs by Primary controller
  `vxlan group by Primary Controller`
- Routes for Provider edge 3
  `routes where vrf = 'Provider Edge 3'`
- Routes of DMZ DLR
  `NextHop Router of routes where VRF = 'LDR-DMZ'`
- Routes having the given router as next hop
  `routes where NextHop Router = 'California-Edge'`

**Useful Queries: Firewall Rules**

- Firewall rules between two VMs

```
firewall rules from 'Prod-Midtier-1' to 'Prod-Db-1'
```

- Rules with have `ANY` source

  ```
  firewall rules where Service Any = true
  ```
- VMs for a given rule

  ```
  vm where Firewall Rule = 'Prod MidTier to Prod DB - DBService '
  ```
- Firewall rules where any port is allowed

  ```
  firewall rule where action = allow and service any = true
  ```
- Flows hitting a particular firewall rule

  ```
  flows where firewall rule = 'Admin to Prod and Lab - SSH'
  ```
- Denied flows in the system

  ```
  flows where firewall action = deny
  ```
- View gateway firewall

  ```
  Firewall Rule where firewall type = 'GatewayFirewall'
  ```
- View distributed firewall

  ```
  Firewall Rule where firewall type = 'Distributed Firewall'
  ```

**Useful Queries: General Traffic Patterns**

- East-West and North-South traffic count, switched traffic count, routed traffic count, and VM to VM traffic count

  ```
  plan security in last 7 days
  ```

**Useful Queries: Traffic from a security lens**

- Top talkers VMs details

  ```
  top 7 vm group by name, Vlan order by sum(Total Network Traffic) in last 7 days
  ```
- Networks that carry the most traffic

  ```
  top 7 vlan group by Vlan id, vm count order by sum(Total Network Traffic) in last 7
  days
  ```
- Networks where most of the communication is within the VLAN (not crossing a physical firewall or L3 boundary)

  ```
  top 7 flow where Flow Type = 'Switched' group by Subnet Network order by sum(Bytes) in
  last 7 days
  ```
- Networks where most of the communication is across VLAN (may be causing bottleneck problems at physical firewall)

  ```
  top 7 flow where Flow Type = 'Routed' group by Source Subnet Network, Destination
  Subnet Network order by sum(Bytes) in last 7 days
  ```
- VMs that talks outside the country

  ```
  top 7 flow where Destination Country != 'United States' group by Source VM, Destination
  Country order by sum(Bytes) in last 7 days
  ```
- Data stores experiencing the most storage latencies

  ```
  avg(Read Latency), avg(Write Latency) of top 7 vm group by Datastore, vlan order by
  avg(Write Latency) in last 7 days
  ```

**Useful Queries: Compliance/Vulnerabilities**

- Vulnerable OSs details

  ```
  vm where Operating System like 'Microsoft Windows Server 2003' or Operating System like
  'Microsoft Windows Server 2008' or Operating System like 'Red Hat Enterprise Linux 6'
  or Operating System like 'Red Hat Enterprise Linux 5' or Operating System like 'SUSE
  Linux Enterprise 10' group by vlan, Operating System
  ```
- Vulnerable OS Count

  ```
  count of vm where Operating System like 'Microsoft Windows Server 2003' or Operating
  System like 'Microsoft Windows Server 2008' or Operating System like 'Red Hat
  Enterprise Linux 6' or Operating System like 'Red Hat Enterprise Linux 5' or Operating
  System like 'SUSE Linux Enterprise 10'
  ```
- Total attack surface due to Old OSs

```
vm where vlan in (vlan of vm where os in ('Microsoft Windows Server 2003', 'Microsoft Windows Server 2008', 'Red Hat Enterprise Linux 6', 'Red Hat Enterprise Linux 5', 'SUSE Linux Enterprise 10')) group by Vlan
count of vm where vlan in (vlan of vm where os in ('Microsoft Windows Server 2003', 'Microsoft Windows Server 2008', 'Red Hat Enterprise Linux 6', 'Red Hat Enterprise Linux 5', 'SUSE Linux Enterprise 10'))
```

**Note:**
To get recommended firewall rule for the vulnerable OS, see Recommended Firewall Rule to Secure Vulnerable OS.

## Time Control

Time-control allows you to run a search query within the context of a selected time or time range. You can select from a list of presets such as last 24 hours, last 3 days, and so on. You can also specify a particular date and time using the **At** option or even a range using the **Between** option.

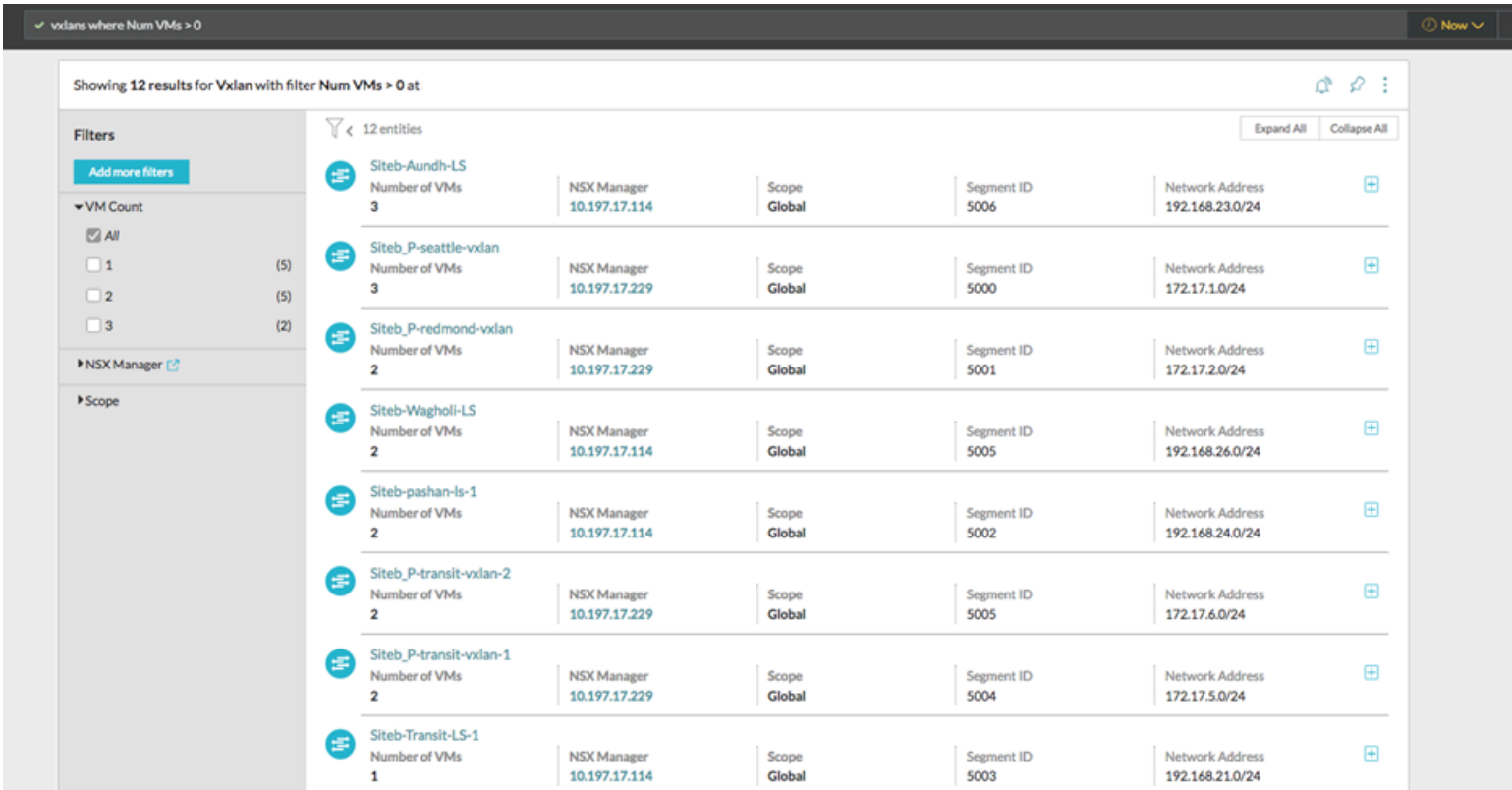
## Search Results

The search results page provides a detailed list of concerned entities that match a particular search. The page itself provides numerous information that ranges from the list of entities, their corresponding properties, and facets to filter the search results to refine your search.

You can also expand or collapse each entry in the search results to view more information about a particular entry. You can also create a notification for each search.

**Note:**
You can point to a particular property in the search results and also in the entity pages to view a tool tip containing more information about that property.

The following graphic shows the search results for the VXLANs where `num vms > 0` search query for a time from the past.



## Filters

When you get the search results, click **Add more filters** on the left pane as per your requirements. You can view a series of filter categories that you can use to narrow down the search results. The number of available filters for each category is mentioned in a small box beside the category.

View the available filters for that category (along with a short explanation for each filter) and click to apply that filter. You can also use the filter search box to search for a particular filter and VCF Operations for networks automatically shows the filters that match your search query and you can click to apply that filter.

Each filter has several properties to refine the search results. When you select a filter property from one of the filters, then the selected property is highlighted in the search results.

## VMware vCenter Tags

VCF Operations for networks provides VMware vCenter tags for search and planning.

You can perform a search of VMs based on the VMware vCenter tags and custom attributes. For example, you can use the following query for search VMs by using tags:

```
vm where tag = '{keyname}:{value}'
```

Every tag belongs to a category. In this example, the keyname is the category to which the tag belongs and value is the name of the tag.

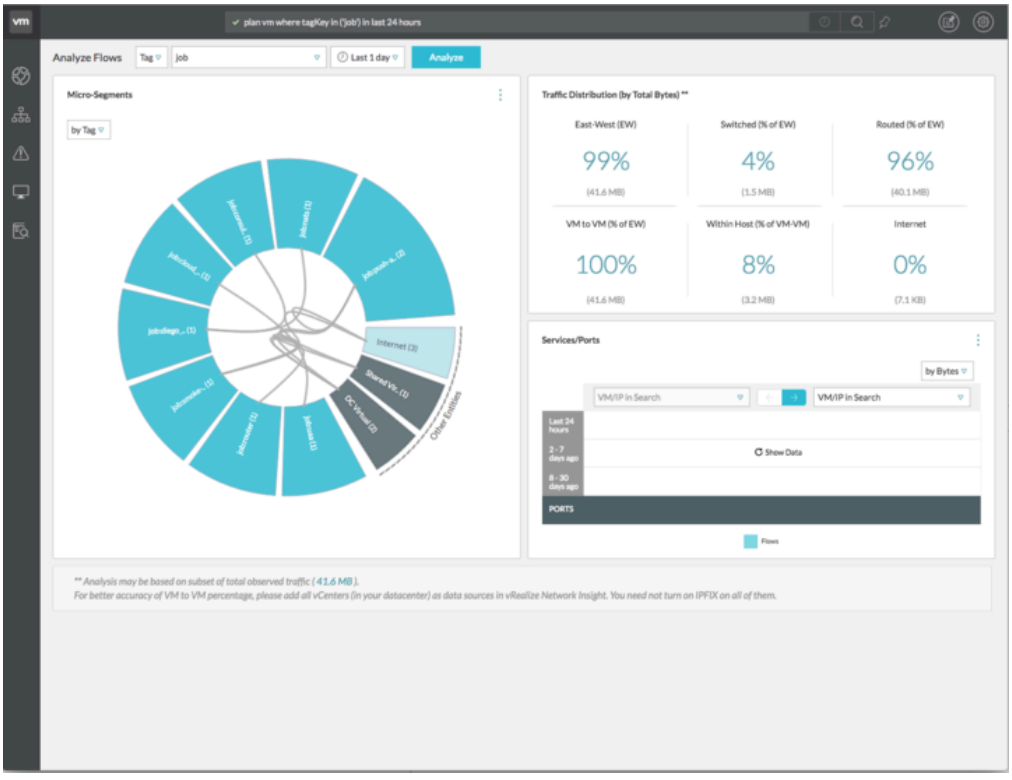
You can also provide an alternate name to a VM by using VMware vCenter tags or custom attributes by using the `name` key. This alternate name is shown as the `other names` property. It is also possible to search and make path queries using the alternate name.

For example, the following queries are supported:

```
vm "other-name-1"
        vm "other-name-1" to vm "other-name-2"
```

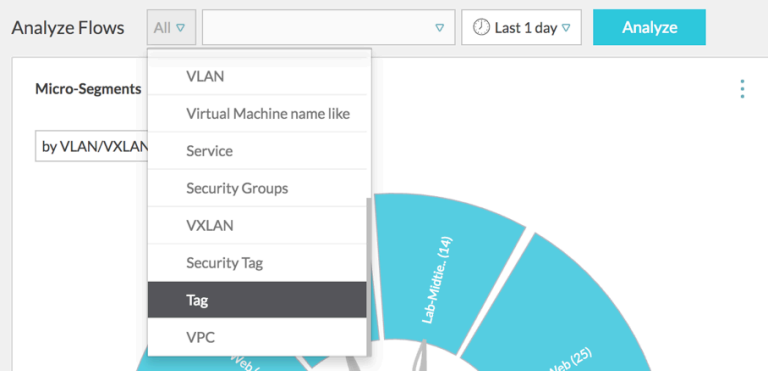
In this example, `other-name-1` and `other-name-2` are custom attributes with the `name` key or tags belonging to the `name` category.

You can also analyze the flows in the network by using the VMware vCenter tags as shown in the figure.

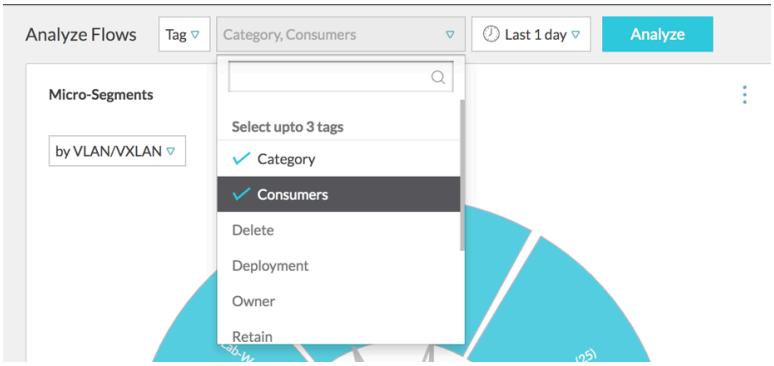


### Using VMware vCenter Tags

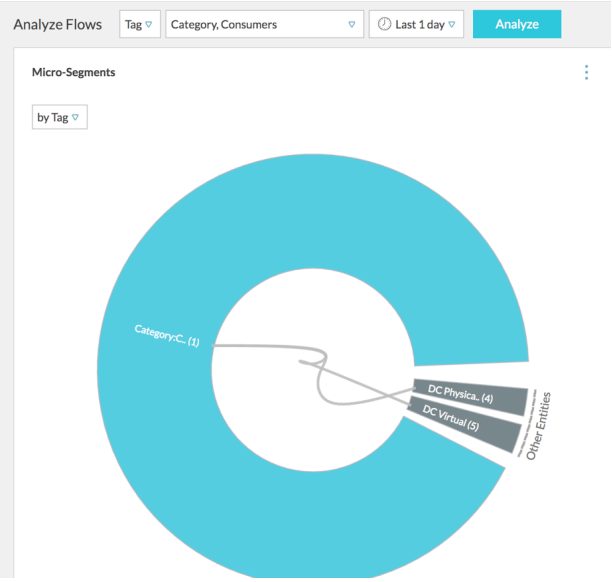
To use the VMware vCenter tags, select the **Tag** option from the **Analyze Flows** drop-down menu.




You can also select up to three tags at this level. After you select the tag, click **Analyze**.




In **Group by Criteria**, **Tag** is selected.




## Working with the Streaming Databus

The databus is a framework for sharing the data stored in VCF Operations for networks with other applications using APIs. With the databus, you can share high volumes of data with low latency in near real-time.

VCF Operations for networks databus provides the following benefits:

- Provides a way to programmatically consume high-volume data in near real-time.

- Works on a change-based consumption model which gives databus a distinct advantage over traditional API calls, especially in the context of high-volume data like flows and metrics.

## Use Cases

With VCF Operations for networks databus, you can use the data for various purposes. Following are a couple of examples of how you can use the databus:

| Use case | Description | Steps |
|---|---|---|
| Retain flows for a longer duration | In general, flow retention in VCF Operations for networks is one month.<br>With the databus, you can retain flows for a longer duration. | 1. Set up a subscriber URL to receive HTTPS/HTTP data.<br>2. Ensure you are aware of the required security information. For more information, see the *Know about the Security and SSL Certificate* section.<br>3. Register a subscriber to subscribe flows to the message group.<br>4. Data starts streaming. For more information on message format, see the *Message Format Received from the Databus* section.<br>5. To learn more about the databus capabilities, see Understanding the Databus Terminologies and Capabilities. |
| Sending alerts to various destinations | By defaults, you can send alerts to SNMP / SMTP destinations.<br>With databus, you can send alerts to various other destinations, such as Slack, ServiceNow, PagerDuty, and so on.<br>A few examples of alerts that you can get using databus are system alerts for password expiry, VMware vCenter disconnected, and so on. | 1. Set up a subscriber URL to receive HTTPS/HTTP data.<br>2. Ensure you are aware of the required security information. For more information, see the *Know about the Security and SSL Certificate* section.<br>3. Register a subscriber to subscribe flows to the message group.<br>4. Data starts streaming. For more information on message format, see the *Message Format Received from the Databus* section.<br>5. To learn more about the databus capabilities, see Understanding the Databus Terminologies and Capabilities. |
| Streaming metrics out of VCF Operations for networks | Polling APIs for metrics is a multi-step process that includes getting entity IDs, periodically calling the APIs, and filter out duplicate information.<br>The databus makes this entire process very efficient; you subscribe to the metrics group or sub-group of your choice, and the metrics directly gets stream out to the endpoint as the metrics get processed. | 1. Set up a subscriber URL to receive HTTPS/HTTP data.<br>2. Ensure you are aware of the required security information. For more information, see the *Know about the Security and SSL Certificate* section.<br>3. Register a subscriber to subscribe flows to the message group.<br>4. Data starts streaming. For more information on message format, see the *Message Format Received from the Databus* section.<br>5. To learn more about the databus capabilities, see Understanding the Databus Terminologies and Capabilities. |

## Supported Message Groups

The databus streaming is a better alternative to APIs if you are looking to poll information periodically. And to receive information from databus, you must subscribes to message group(s). A message group is a specific set of data that

you can subscribe from the databus. Each message group can have up to 10 subscribers, except for Flows and Metrics message groups which can have only one subscriber per message group. Also, the Metrics message group can have only one subscriber per message group.

VCF Operations for networks supports the following message groups:

| Message Group | Sub-message Group | Message Details |
|---|---|---|
| Metrics | vms-metrics | Metric, interval, time stamp, unit, entity type, and points. |
| | hosts-metrics | Metric, interval, time stamp, unit, entity type, and points. |
| | flows-metrics | Metric, interval, time stamp, unit, entity type, and points. |
| | nics-metrics | Metric, interval, time stamp, unit, entity type, and points. |
| | switchports-metrics | Metric, interval, time stamp, unit, entity type, and points. |
| | nsxt-metrics | Metric, interval, time stamp, unit, entity type, and points. |
| Applications | N/A | Entity ID, entity type, name, created by, creation time, last modified by, last modified time, source, tiers, and discovery info. |
| Flows | N/A | Source VM, destination VM, source IP, destination IP, port, source folders, destination folders, protocol, source IP sets, destination IP sets, source security groups, destination security groups, traffic type, source security tags, destination security tags, source VM tags, destination VM tags, within host, firewall action, firewall rule ID, source managers, destination managers, and flow tag. |
| Problems | N/A | Entity ID, entity type, name, anchor entities, message, event tags, admin state, archived, event time epoch ms, event type, and severity. |

**Table 1341: Configuration-related Message Group**

| Message Group | Message Details |
|---|---|
| VMs | Entity ID, name, entity type, IP addresses, default gateway, default gateways,VNICs, source firewall rules, destination firewall rules, VM UUID, manager UUID, cluster, resource pool, host, vendor ID, vCenter manager, folders, datastore, data center, source inversion rules, destination inversion rules, CPU count, memory, OS full name, and HCX info. |
| Hosts | Entity ID, name, entity type, VMKNICs, cluster, vCenter manager, VM count, datastore, service tag, vendor ID, maintenance mode, connection state, CPU count, memory, and manager UUID. |
| NICs | Entity Id, name, entity type, VLAN, and VM. |

| Message Group | Message Details |
|---|---|
| Switchports | Entity ID, name, entity type, operational status, administrative status, sub type, device, and manager. |

**Tip:**
A sub-message group provides more fine-grained information. All subscribers to a particular message group receive the same information.

The data published over databus is categorized into different message groups. As a user you can subscribe to one or more message groups, and data related to those message groups starts streaming to HTTP or HTTPS endpoints.

Use the following steps to configure the databus:

### 1. Know about the Security and the SSL Certificate

While using the databus, you must be aware of the following security information:

| Environment | Details |
|---|---|
| On-prem | • HTTP/HTTPS as a transport protocol - When using databus for on-prem environment, you can get data from the databus over HTTP / HTTPS endpoint. To ensure that the data is encrypted during transport and verification of subscriber authenticity, use HTTPS.<br>• Firewall Rules - To ensure only VCF Operations for networks can send data to the subscriber, you need to enable the firewall rules. All the VCF Operations for networks platform nodes in a cluster send the data to the subscriber, so you must configure the firewall rules to allow traffic from all the platform nodes.<br>• SSL Certificates - The SSL certificates used by the subscribers must be signed by well-known CAs. If the SSL certificates are self-signed or signed by internal CAs of the organization and not trusted by Java, you must import the CA certificate to the Java trust store of VCF Operations for networks. Contact VMware support to get help on how to import the Java trust store.<br>**Note:**<br>The CA certificate that is imported to the trust store will be trusted for all outgoing SSL traffic from VCF Operations for networks platforms. |
| SaaS | • HTTPS as a transport protocol - When using databus for a SaaS environment, you can get data from the databus only over HTTPS endpoints.<br>• CSP token for message authentication - Each databus message from VCF Operations for networks (SaaS) contains an authorization header with a JWT token issued by CSP. You as a subscriber can verify the authenticity of the token by validating the token with the public key from CSP (`https://console.cloud.vmware.com/csp/gateway/am/api/auth/token-public-key`). You can verify the authenticity of the token to ensure that the client ID in the token is the client ID of the VCF Operations for networks databus application registered on CSP. You get the client ID details from VCF Operations for networks Databus developers.<br>• SSL Certificates - The SSL certificates used by the subscribers must be signed by well-known CAs. By default the SSL certificates that VCF Operations for networks (SaaS) issues are signed by JDK. VCF Operations for networks (SaaS) doesn't support the import of custom CA certificate. |

### 2. Register, Update, or Delete a Subscriber

- To register a subscriber, you must do an API call using the POST method. For details, see the API Reference.
  If the registration is successful, the message response contains a subscriber ID. You can use the subscriber ID to delete a subscription or update a subscription. After successful subscriber registration, it may take up to five minutes for the subscriber to start receiving messages.
- To update a subscriber, you must do an API call using the PUT method. For details, see the API Reference.

  **Note:**
  You can only update the Subscriber URL. You cannot update the `message group`.
- To delete a subscriber, you must do an API call using the DELETE method. For details, see the API Reference.
  After successful subscriber deletion, it may take up to five minutes for the data publishing to stop.

### 3. Message Format Received from the Databus

The messages received as part of various message groups are a list of multiple messages and have a common high-level structure. Each message in the list is a cloud-event formatted object. To learn more about cloud events, see cloudevents.

All the message has two sets of specific fields:

- Common structure - which is common across all the message group. For example:

```
 "id": "14cfd27c-76d8-43e3-9fe9-f53bf69c0c5e",
     "type": "Application",
     "specversion": "1.0",
     "source": "CAESBkFCQ0RFRg==",
     "messageGroup": "applications",
     "status": "CREATE",
```

- Message group-specific structure - which is unique for every message group. For example, the following is the data object within an application message which is unique to the application message group:

```
    "data": {
      "entity_id": "1:561:0",
      "entity_type": "APPLICATION",
      "name": "app1",
      "created_by": "test@vnera.com",
      "creation_time": 1637303575039,
      "last_modified_by": "test@vnera.com",
      "last_modified_time": 1637303575039,
      "source": "MANUAL",
      "tiers": [
        {
          "entity_id": "1:562:1",
          "entity_type": "TIER",
          "name": "app1_t1"
       }
      ]
    }
```

The structure of the objects within databus messages is in line with the schema of public API responses, except for the metric where a subset is in use. The message structure is backward compatible across different versions of VCF Operations for networks. In the later versions, new fields may get added.

The following table lists the names of the different types of messages for every message group:

| Message groups | Type or Schema of the message |
|---|---|
| Alerts (problem events only) | Entity Message List |
| Applications | Application Message List |
| flows | Flow Message |

| Message groups | Type or Schema of the message |
| --- | --- |
| metrics | The structure of the metrics data is different from the public APIs to make the messages compact.<br><pre>[<br>  {<br>  "id": "string",<br>  "type": "string",<br>  "specversion": "string",<br>  "source": "string",<br>  "message_group": "metrics",<br>  "data": {<br>  "metric": "string",<br>  "interval": 0,<br>  "timestamp": 0,<br>  "unit": "string",<br>  "entity_type": "string",<br>  "points": [<br>  {<br>  "entity_id": "string",<br>  "value": 0<br>  }<br>  ]<br>  }<br>  }<br>]</pre> |

For more information on the above objects and message groups, see the API Reference.

## Understanding the Databus Terminologies and Capabilities

The databus provides various capabilities, and understanding these capabilities helps you use the databus more efficiently.

### Subscriber

A subscriber is an endpoint that registers with the databus to receive the data from the message group. You cannot have multiple subscribers with the same message group and the same Subscriber URL.

### Subscriber URL

An URL where you receive the data from the databus. VCF Operations for networks databus sends all the messages from any message group to the subscriber URL. You can also use a subscriber URL to subscribe to multiple message groups.

> **Note:**
> Databus starts streaming date from the moment you subscribe to any message group and will continue until you unsubscribe. The system handles any temporary loss of communication through inbuilt mechanisms and data streaming resumes once the issue is resolved. You don't need to take any action.

### Messages Ordering

The messages sent over the databus are generally ordered by the timestamps when the event or the object is stored in VCF Operations for networks. But, in certain scenarios these ordering guarantees cannot be met. For reliable ordering of the messages, the subscribers must make use of the timestamp field in the Databus messages and sequence the message based on the timestamp.

### Near Real-time Streaming

In the databus, there is a delay between data getting processed in VCF Operations for networks and the databus streaming the data out. The delay could range from three seconds up to one minute. This delay varies based on the message groups and is subject to the pipeline being clear.

### Load Balancer to Scale Data Processing

While receiving data from VCF Operations for networks databus, you can use a load balancer to horizontally scale the data processing service. To use a load balancer, you must use the load balancer URL as the subscriber URL during the databus registration process. A load balancer can distribute the data to a horizontally scalable data processing services.

### Heartbeat

To help troubleshoot the communication channel between VCF Operations for networks databus and the subscriber endpoint, VCF Operations for networks sends hourly heartbeat messages to every subscriber. If a URL is specified in more than one subscription, then the subscriber receives one heartbeat message for every subscription.

Example:

```
{
  "id": "c2214eea-68a6-4e5a-9e4b-ff5886a42cd3",
  "type": "HEARTBEAT",
  "specversion": "1.0",
  "source": "CL58EgdESzVDSEE1",
  "message_group": "vms",
  "status": "HEARTBEAT",
  "timestamp": 1681897371522
}
```

### Message batching

VCF Operations for networks databus inherently supports batching where several messages are grouped together in a batch and sent to the subscriber. By default, the maximum size of a batch is two megabytes. When size of a single message is more than two megabytes then the message is trimmed.

> **Note:**
> The batch creation creates an additional delay which is less than a minute.

Example of a trimmed message:

```
[
{
"id": "elfcdccb-c06a-419b-90bd-7618e8453f87",
"type": "Application",
"specversion": "1.0",
"source": "CNaBARIHRElXRlgxRg==",
"message_group": "applications",
"status": "UPDATE_TRIMMED",
"data": {
"entity_id": "16598:561:716945762559381432", "entity_type": "Application",
"name": "My-3Tier-App"
}
]
```

### Delivery Guarantee

A message delivery is considered successful if the subscriber responds with an HTTP code other than the 503 error code.

If a subscriber doesn’t receive a message, but VCF Operations for networks doesn't get a 503 error code, VCF Operations for networks considers the message as delivered and does not try to send the message again, and the message might get lost. We have introduced this mechanism to ensure that a continuous retry does not overwhelm

your system. For example, If your production environment is down, continuous retries would be very noisy. In this case, sending an HTTP response other than 503 is a way for you to indicate that you do not want to retry.

In VCF Operations for networks, the messages are sourced by different services and then get added into a disk-based queue. The disk-based queue has an expiry of two hours.

Any message if not delivered to the subscriber within two hours, those messages get deleted from the queue in a FIFO manner, and doesn't get delivered to the subscriber. Also, if the volume of the messages in VCF Operations for networks is beyond what can be reliably forwarded by the system, messages may get dropped. When the databus subscriber is unreachable, the system tries to send messages until it delivers the message successfully. A back-off mechanism is added in version 6.9, where the system retries message delivery in exponential time.

#### Filtering

When you register a subscriber, the subscriber gets all the messages of that message group. VCF Operations for networks databus does not support Filtering of messages. If you don't want all the messages in a message group, you can discard the messages based on the filtering conditions before further processing. You can also deactivate alerts from the VCF Operations for networksUI.

#### Correlating messages with Other Objects

In certain scenarios, the messages received by the databus contain contextual information about a few other related entities without the full details of those entities. But, the message contains a specific identifier such as an `entity_id`. To get details about those related entities, you must subscribe to those entities using APIs. For example, subscribing to a host message group can help to correlate the hosts of the events received from the problems message group.

## Planning Disaster Recovery for VCF Operations for networks

VMware Live Recovery is a disaster recovery automation software that provides policy-based management, non-disruptive testing, and automated orchestration. To protect your VCF Operations for networks, VMware Live Recovery automates every aspect of applying a disaster recovery plan to accelerate recovery and eliminate the risks involved when using a manual process.

The prerequisites for the disaster recovery operation for VCF Operations for networks are as follows:

- Ensure that you have installed and configured vSphere Replication.
- VMware Live Recovery should be deployed and configured on both the protected and the recovery sites.
- Ensure that the site pairing is configured properly from within the VMware Live Recovery UI before proceeding with the creation of the recovery plan and other components.
- VMware vSphere Replication should be enabled for each of the protected nodes of the VCF Operations for networks setup in context. While enabling VMware vSphere Replication, provide sufficient RPO considering the VCF Operations for networks node size and usage so that minimum data loss is expected to incur during a disaster.
- Ensure that you create a separate protection group for VCF Operations for networks. For small and non-distributed deployments, ensure that all the VMs are in the same protection group. For distributed deployments, it is recommended that you place all the platforms in a single protection group so that it is easy to recover. You can place the collectors in different protection groups.
- Create a recovery plan and add the protection groups containing VCF Operations for networks VMs to this plan. Ensure that the protection group containing the platform nodes get the higher precedence. In the recovery plan, ensure that the primary platform node is placed in a higher priority group than the other platform nodes.
- Currently, any type of IPv4 customization with VMware Live Recovery is not supported

It is recommended that you migrate or recover VCF Operations for networks VMs to an identical network configuration. Also as per the VMware Live Recovery recommendation, you can perform test run periodically to ensure the existing plan works with underlying infrastructure and the configured RPO limit.

- Migrate or recover VCF Operations for networks VMs to an identical network configuration.
  If the recovery site is configured to have the same network configuration as the protected site and a mapping is created between the identical networks, configure all replicated VCF Operations for networks virtual machines to be

started with the same IPs, because these VMs are the protected nodes. The recovered system will become operational after the planned migration or disaster recovery has finished successfully.

- Do not specify any IP customization for a recovery plan when the recovery site does not have the same network as that of the protected site. In this scenario, VMware Live Recovery is used for the recovery of the appliance VMs. For configuring network post recovery, manually assign the network settings as follows:

1. Run `change-network-settings` command on all platforms to change the IP.

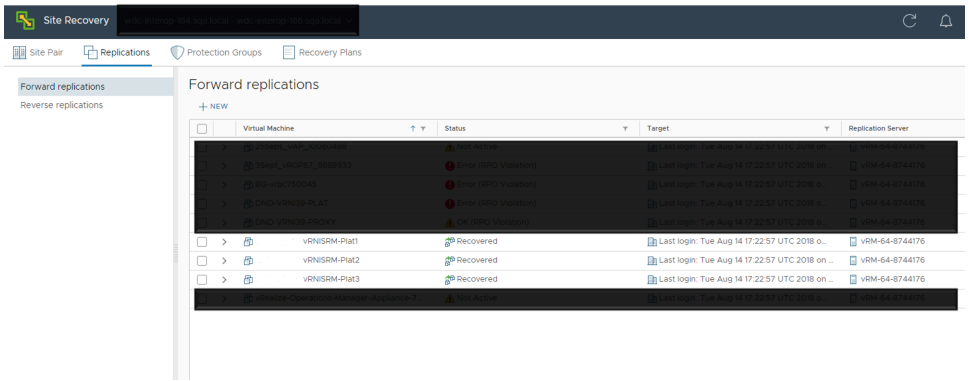
   **Note:**
   Ensure that the platform VM reboots successfully before you run the next command.
2. Run `update-IP-change` command on all the nodes.
   For example, in a 3-node cluster, to update IP of platform1 from IP1 to IP2, run the `update-IP-change` command on platform2 and platform3 with IP1 and IP2 as the arguments. To update IP of platform2, run the `update-IP-change` command on platform1 and platform3.
3. Run `finalize-IP-change` command on platform1.
4. Run the `show-connectivity-status` command on all the collectors and search for `Platform_VM_IP/URL` to identify the associated platform nodes.
5. Run `pair-collector set-platform --ip-or-fqdn platform-newIP` on all collectors.
   For example, if the CollectorA is associated to platform2, run `pair-collector set-platform --ip-or-fqdn platform2-newIP`.
6. Check the service status. If some of the services on the platform nodes are not running, reboot the nodes in the recommended order.

   **Note:**
   For more information on the commands mentioned above, see *VCF Operations for networks Command Line Reference Guide*.

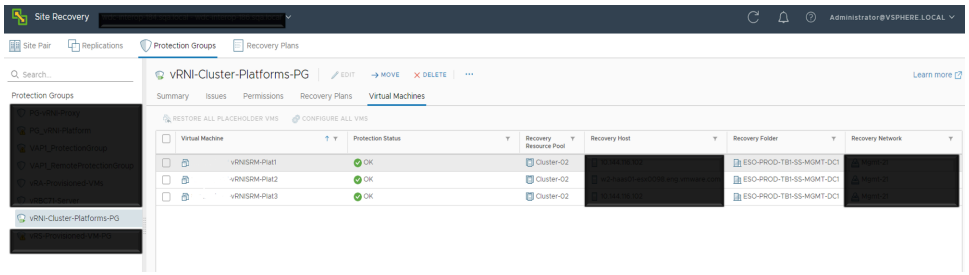
## Sample Disaster Recovery Scenario

Here are the steps for a sample scenario for VCF Operations for networks Disaster Recovery (DR):

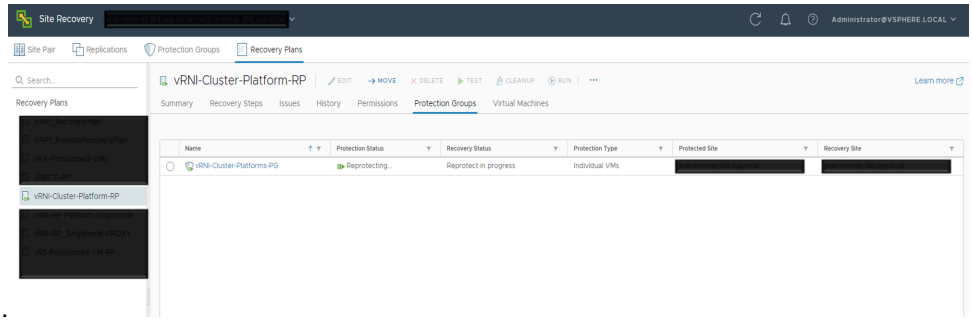
1. Ensure that VMware Live Recovery is configured and up in both the protected and the recovery sites.
2. Configure replication for each of the VCF Operations for networks nodes that are to be protected. While configuring the replication, provide adequate Recovery Point Objectives (RPO) time for the VCF Operations for networks instance. For example, if it is a VCF Operations for networks deployment with a single platform and collector nodes (medium size), then RPO of 45 minutes is good. But if it is a cluster with nodes having bricks of large size, then the adequate RPO should be provided. The snapshot interval configuration is specific to the user environment and requirements.



3. Create a protection group. Include the VMs that you want to protect under a specific protection group.




4. Create the recovery plan where you include the respective protection groups.




5. Perform test recovery. This is to ensure that your recovery plan works as expected.
6. VMware Live Recovery recommends that users perform planned migration at regular intervals to validate the integrity of the existingDR plan.
7. Suppose the recovery site has a network configuration that forces the VCF Operations for networks VMs to come up with the new IPs. Recover the VCF Operations for networks VMs with a recovery plan that assumes no network change for the recovered VMs. Once the recovery of the VMs is reported as a success, in VCF Operations for networks, assign new IP addresses manually to the VCF Operations for networks nodes, apply new certificates, and re-initialize the cluster.
8. As IPv4 customization with VMware Live Recovery is not supported currently, as a work around you can perform DR with VCF Operations for networks assuming as if there is no network change.

   To manually assign the network settings:

   1. Run the `change-network-settings` to change the IP simultaneously on all platforms.
   2. For every platform whose IP is changed, run `update-IP-change` command on all the other nodes.
      For example, for the IP change of platform1 from IP1 to IP2, run the `update-IP-change` command on platform2 and platform3 with IP1 and IP2 as the arguments. For the IP change of platform2, run the `update-IP-change` command on platform1 and platform3.
   3. Run `finalize-IP-change` command once on platform1 after step1 and step 2 are completed.
   4. Run `pair-collector set-platform --ip-or-fqdn <with-updated-ip-of-Platform1>` on the collector node.
   5. Check the service status. If some of the services on the platform nodes are not running, reboot the nodes in the recommended order.

## VMware Transit Connect Support in VCF Operations for networks

An SDDC group uses VMware Managed Transit Gateway (VTGW) to provide high-bandwidth, low-latency connections between SDDCs in the group and to other VPCs in the same region. You can also add a Direct Connect Gateway (DXGW) to provide centralized connectivity to your on-premises SDDCs.

An SDDC group is a logical entity designed to simplify management of your organization's VMware Cloud on AWS resources at scale. Collecting SDDCs into an SDDC group provides a number of benefits to an organization with multiple SDDCs whose workloads need a high-bandwidth, low-latency connection to each other. All network traffic between group members travels over a VTGW network. Routing between compute networks of all SDDCs in a group is managed automatically by VTGW as subnets are added and deleted.

The data source requirements to support VMware Transit Connect are:

| Data Source | Entites | Permissions |
|---|---|---|
| VMC NSX | SDDC group, VTGW, SDDC attachments, VTGW route table, and VTGW routes | Administator |
| AWS account with native VPCs | AWS TGW attachment | EC2 API permissions |
| AWS account with DXGW | DXGW and DXGW associations | EC2 API permissions and Direct Connect permissions |

VTGW support in VCF Operations for networks includes:

- VMC SDDC Group as a search query to view the SDDC group details in the VMC SDDC Group page.
- VMware Transit Gateway as a search query to the view the VTGW details in the VMware Transit Gateway page.
- Search and view details of the related entities like VTGW routes, route tables, and VTGW attachments.
- Flows for:
  - SDDC to SDDC via VTGW.
  - SDDC to native AWS VPC via VTGW.

VTGW support in VCF Operations for networks does not include:

- Flows between VMC SDDC and on-premise SDDC via VTGW and Direct Connect Gateway.
- VM-VM path visualization between VMs communicating over VTGW.
- Native AWS Transit Gateway.

## View Flows over VMware Managed Transit Gateway

You can view the list of all flows that are going over the VTGW, which enables you to view the traffic and helps you to analyze and understand the utilization level of VTGW.

To see all the flows via VTGW, you can run the following query:

```
flows where tgw = ''
```

You can see the flows from SDDC to SDDC, SDDC to VPC, and VPC to SDDC via VTGW using the following flow types:

- SDDC-SDDC via TGW
- SDDC-VPC via TGW
- VPC-SDDC via TGW

> **Note:**
> Flows are not supported between VMC SDDC and on-premise datacenters via VTGW and Direct Connect Gateway.

### VTGW Search Queries

You can search on VTGW entities in VCF Operations for networks.

**Table 1342: Search Queries**

| Description | Query |
|---|---|
| Get a list of flows over VTGW | `flows where tgw = '<tgw-name>'` |
| Get a list of flows between VMC SDDCs over VTGW | `flows where tgw = '<tgw-name>' and flow type = 'SDDC-SDDC via TGW'` |
| Get a list of flows from VMC SDDC to Native AWS VPC over VTGW | `flows where tgw = '<tgw-name>' and flow type = 'SDDC-VPC via TGW'` |
| Get a list of flows from Native AWS VPC to VMC SDDC over VTGW | `flows where tgw = '<tgw-name>' and flow type = 'VPC-SDDC via TGW'` |

| Description | Query |
|---|---|
| Get a list of max egress traffic volume between SDDCs over VTGW | `max(series(sum(bytes))) of flows where tgw = '<tgw-name>' and flow type = 'SDDC-SDDC via TGW' group by Source VMC SDDC` |
| Get a list of max egress traffic volume between VMC SDDC and AWS VPC over VTGW | `max(series(sum(bytes))) of flows where tgw = '<tgw-name>' and flow type = 'SDDC-VPC via TGW' group by Source VMC SDDC` |
| Get a list of max ingress traffic volume between VMC SDDC and AWS VPC over VTGW | `max(series(sum(bytes))) of flows where tgw = '<tgw-name>' and flow type = 'VPC-SDDC via TGW' group by Destination VMC SDDC` |

## Viewing VMC SDDC Group details

You can use the VMC SDDC group page to get an overview of your VMC on AWS SDDC group.

### How to access the VMC SDDC Group page?

To access this page, search for `VMC SDDC Group` and in the search result list, select an entity you want to view.

### Overview

In the **VMC SDDC Group** page, you see:

| Section | Details |
|---|---|
| **Overview** | You see an overview of your VMC SDDC group entities and the counts of VMware Managed Transit Gateway, Direct Connect Gateway, Direct Connect Interface, and so on present within the VMC SDDC group.<br>The **VTGW Route Table** shows the VMware Managed Transit Gateway route details of the entities in that VMC SDDC group. |
| **Flows** | You see an overview of the bandwidth and the traffic usage over the VMware Managed Transit Gateway. |
| **Network Traffic and alerts** | You see an overview of the network traffic and the list of all alerts associated with the VMC SDDC group entities discovered by VCF Operations for networks alerts. |

## Viewing VMware Transit Gateway Details

You can use the VMware Transit Gateway page to get an overview of the VMware Managed Transit Gateway entities.

### How to access the VMware Transit Gateway page?

To access this page, search for `VMware Transit Gateway` and in the search result list, select an entity you want to view.

### Overview

In the **VMware Transit Gateway** page, you see:

| Section | Details |
|---|---|
| **Overview** | You see an overview of the selected entity, with its configuration and runtime property details.<br>The **VTGW Route Table** shows the route details of the selected entity. |
| **Flows** | You see an overview of the bandwidth and the network traffic usage over the VMware Managed Transit Gateway. |
| **Alerts** | You see an overview of the list of all alerts associated with the VMware Managed Transit Gateway entities discovered by VCF Operations for networks alerts. |

## Direct Connect Support in VCF Operations for networks

Direct Connect is a mechanism to provide data transfer connection between an on-premise location and public cloud services. VCF Operations for networks supports the Direct Connect feature for VMware Cloud on AWS.

Direct Connect support enables you to:

- Identify flows that pass over Direct Connect between on-premise data center and VMware Cloud on AWS SDDC.
- Run the flow analytics to understand the flow bandwidth and the packet rate.
- View the detailed path topology between virtual machines that communicate over Direct Connect.
- View details about Direct Connect and associated alerts.

### Direct Connect data fetch mechanism

VCF Operations for networks fetches Direct Connect information by using the VMware Cloud on AWS NSX APIs. Therefore, to get the Direct Connect information, you must add the VMware Cloud on AWS related data sources (VMware vCenter and NSX Manager).

> **Note:**
> You do not have to add an AWS account for Direct Connect support. However, you must add the following Data Sources:
>
> - VMware Cloud on AWS vCenter
> - VMware Cloud on AWS NSX Manager
> - On-premise VMware vCenter
> - On-premise NSX or On-premise NSX Manager
> - On-premise DC Perimeter Physical Router (Supported routers include Cisco N9k and Cisco ASR 9k)
> - DX Colo Physical Router (Supported routers include Cisco N9k and Cisco ASR 9k)
>
> These data sources are required by VCF Operations for networks to construct VM-VM path topology and to enrich the IPFIX flows with Direct Connect information.

### What data is collected with Direct Connect support

- Direct Connect related configuration details in the VMware Cloud on AWS SDDC.
- Advertised and learnt subnets for Direct Connect at the SDDC level.
- Configuration information of Direct Connect Interfaces (VIFs) associated with the SDDC.
- Flows reported by Distributed Firewall (DFW) in VMware Cloud on AWS.

> **Note:**
>
> - The NetFlow enablement is not required on the colocation routers.

- Route based VPN is not supported for Direct Connect. So, even if you have enabled the Use VPN as backup to Direct Connect option, VPN backup fails.
- The metrics, and the advertised or learnt subnets information is not available at the individual VIF level.

**Direct Connect Entities**

- *VMware Cloud on AWS Direct Connect:* This is the parent entity for all Direct Connect entities in VCF Operations for networks, which models the configuration information of Direct Connect within the VMware Cloud on AWS SDDC.
- *Direct Connect Interface*: This models the AWS Direct Connect VIF information provided by VMware Cloud on AWS. This entity enables the exchange of advertised and learnt routes between VMware Cloud on AWS and on-premise data center.

## Viewing VMC Direct Connect Details

You can see the **VMC Direct Connect** page for an overview of its properties and the entities associated with Direct Connect based on the information collected from VMware Cloud on AWS.

**Table 1343: Direct Connect Dashboard**

| Section | Details |
|---|---|
| Properties | Primary properties of Direct Connect, which includes the associated SDDC, local ASN, learned and the advertised routes, failed advertised routes. |
| Direct Connect Interfaces | The list of all Direct Connect Virtual Interfaces that are associated with the Direct Connect |
| Network Traffic and Alerts | The network traffic statistics for Direct Connect connections. You get detailed information about total data volume, packet count and bandwidth between datacenters over Direct Connect.<br>The list of alerts associated with Direct Connect. |
| Top Talkers | You can see the top talking entities in VMC SDDC that are communicating over Direct Connect. |

## Viewing Flows over Direct Connect

You can view the list of all flows that are running on the Direct Connect, which enables to view the traffic over Direct Connect. This helps you to analyze and understand the utilization level of Direct Connect.

When you search with the `Flows where connection = Direct Connection_ID` query, you see the list of flows that pass over the Direct Connect, and also information such as the bandwidth utilization, and network traffic rate for a particular Direct Connect. Update this line - Under the Flow Type, you can see whether the flow is on VPN, Direct Connect, or on the hybrid network.

To see the Direct Connect flows only, you can run the following query:

```
flows where flow type = Direct Connect group by Connection
```

To see the count of flows and the data volume on each Direct Connect connection, run the following query:

```
max(series(sum(Bytes)))of Flows where flow type = Direct Connect and group by Connection
```

To see the count of flows and the packet count on each Direct Connect Interface, run the following query:

```
max(series(sum(packets)))of Flows where flow type = Direct Connect and group by Connection
```

For additional queries, see Direct Connect Search Queries.

## Direct Connect Search Queries

You can search on VMware Cloud on AWS Direct Connect and Direct Connect Interface entities in VCF Operations for networks.

**Table 1344: Search Queries**

| Description | Query |
|---|---|
| Get a list of VMware Cloud on AWS Direct Connect entities based on which you can filter the information | `VMC Direct Connect where` |
| Get the VMware Cloud on AWS Direct Connect list view | `VMC Direct Connect` |
| Get the maximum data volume over Direct Connect | `max(series(sum(bytes))) of flows where connection = 'Connection-ID' and flow type ='Different Dc' and source vm is set and destination vm is set and flow type = 'Direct Connect'` |
| Get the maximum packets over Direct Connect | `max(series(sum(packets))) of flows where connection = 'Connection-ID' and flow type ='Different Dc' and source vm is set and destination vm is set and flow type = 'Direct Connect'` |
| Get the maximum packets over Direct Connect to the Internet | `max(series(sum(packets))) of flows where connection = 'Connection-ID' and flow type ='Destination is internet' and flow type = 'Direct Connect'` |
| Get the maximum data volume over Direct Connect to the Internet | `max(series(sum(bytes))) of flows where connection = 'Connection-ID' and flow type = 'Destination is internet' and flow type = 'Direct Connect'` |
| Get the maximum packets between data centers over Direct Connect | `max(series(sum(packets))) of flows where connection = 'Connection-ID' and flow type ='Different Dc' and source vm is set and destination vm is set group by Source Dc, Destination Dc and flow type = 'Direct Connect'` |
| Get the maximum data volumes between data centers over Direct Connect | `max(series(sum(bytes))) of flows where connection = '64638-10.63.229.131' and flow type ='Different Dc' and source vm is set and destination vm is set group by Source Dc, Destination Dc and flow type = 'Direct Connect'` |

# VCF Operations for networks General FAQs

The VCF Operations for networks FAQs guide provides the user with frequently asked questions about VCF Operations for networks.

### Intended Audience

This information is intended for users who work with VCF Operations for networks.

### How do I Create a Support Bundle?

Refer to the Broadcom knowledge base article 343485.

### How to Create a read-only Admin Role in Palo Alto Networks Panorama for XML API access?

To add an **Admin** role for XML API access:

1. Select **Panorama** > **Admin Roles**.
2. Click **Add** to add a new Admin Role to open Admin Role Profile dialog box
3. In Admin Role Profile dialog box
   a. Give a name to the role (for example, `api-only-admin`)
   b. Select the **Role** as **Panorama**
   c. Deactivate all entries in the Web UI tab
   d. Enable all entries except **Commit** in the XML API tab
   e. Click **OK** to close the dialog box, a new **Admin Role** appears in the list with the name
   f. Click **Commit** to commit the changes to Panorama
4. Assign this **Admin** role to an administrative account.

### I see an event/error on Data Source as "The data source identity information such as a certificate or the key has changed." What does it mean?

VCF Operations for networks has received a new certificate from a data source which is not the same as the one stored in the product. You can accept the new certificate by navigating to the data sources page and accepting the new certificate for that data source.

### What is the limit for importing DNS records in VCF Operations for networks?

The limits for importing DNS records are:

- Infobox DNS Data source: You can import 900K records from a single data source.
- Manual DNS records import: You can import DNS records using multiple `.csv` or Bind files packaged as a zip file. There is no limit to the number of records that you can import, however there are upload limits as follows:
  – Number of files in a single zip file – 25
  – Max size of single zip file – 10 MB.

## Installation and Configuration

### What are the resource requirements for VCF Operations for networks?

Refer to the VCF Operations for Networks System Recommendations and Requirements topic for Resource Requirements.

### What happens if I enter the incorrect key during VCF Operations for networks collector OVA deployment?

The secret key is not validated during VCF Operations for networks collector OVA deployment. The deployment completes even with an incorrect secret key. However, pairing can fail and VCF Operations for networks collector does not show up as detected on VCF Operations for networks UI.

To correct the shared secret, log in to VCF Operations for networks collector CLI and run the `set-proxy-shared-secret` command to set the correct secret key. This command replaces the old key with the new one, and therefore, VCF Operations for networks platform detects the VCF Operations for networks collector and pairs up.

### How do I configure DNS after VCF Operations for networks collector OVA is deployed?

Log in to VCF Operations for networks collector CLI, and run the change-network-settings command. This interactive command will provide the user an option to add or modify DNS after which the VCF Operations for networks collector will be reconfigured with the new DNS.

If any of the network parameters is not configured correctly, use the change-network-settings command to modify the network configuration parameters.

**How do I find out VCF Operations for networks collector VM IP from the UI?**

Go to Settings page and select VCF Operations for networks Infrastructure menu option. The IP address of both, VCF Operations for networks Platform and VCF Operations for networks collector VMs is displayed.

**What should I do if VCF Operations for networks collector is not detected in 5 minutes after deploying VCF Operations for networks collector OVA?**

Log in to VCF Operations for networks collector using `consoleuser` (refer to the VCF Operations for networks Command-Line Interface Guide) and verify the following:

- Verify VCF Operations for networks platform pairing status with VCF Operations for networks collector using the CLI `show-connectivity-status` .
- If the pairing stating is shown `Passed`, open the platform UI in a new browser window and login to check status.
- If the pairing status is showing `Failed`, then the shared secret key specified during the VCF Operations for networks collector OVA deployment could be wrong. To fix this problem, use the `set-proxy-shared-secret` command to set the correct secret key. This command replaces the old key with the new one, and therefore, VCF Operations for networks platform can detect VCF Operations for networks collector.
- If the `show-connectivity-status` shows network reachability to VCF Operations for networks Platform as **Failed**, then verify whether VCF Operations for networks platform is reachable from VCF Operations for networks collector VM using the `ping` command.
- If it is not reachable, then verify if NTP, DNS, gateway, and other network parameters are configured correctly using `show-config` command.
- If any of the network parameters is not configured correctly, use the `setup` command to modify the network configuration parameters.

**What should I do if I forget my login credentials?**

- If you are the UI local user: Contact VCF Operations for networks UI administrator to reset the credentials for you.
- If you are the administrator: Reset the VCF Operations for networks UI credentials using CLI `modify-password`.
- If you are working on VCF Operations for networks versions prior to 9.0, contact support.

**How do I change the login password?**

To change the login password:

1. On the VCF Operations for networks standalone UI, go to **Settings** > **My Preferences** > **Change Password**.
2. On the **Change Password** page, fill in the required information and click **Save**.

**What do I do if I get the login screen before detecting the VCF Operations for networks collector VM?**

- This behavior is expected when the browser is refreshed or URL is opened in a new window before detecting the collector.
- Log in by using the credentials set during license activation for the `admin@local` username.

**Does VCF Operations for networks support multiple vCenter/NSX instances?**

Yes, VCF Operations for networks supports multiple vCenter and NSX Manager.

**What is port aggregation and what is the mechanism to do it?**

Port aggregation is built in to aggregate the ephemeral port flows – like dynamic FTP, Oracle, MS-RPC etc. This helps in reducing the number of flows in system and provide an aggregated view for large number of flows that are essentially for the same service.

The mechanism to do it is as follows:

- For first three days of noticing a `destination_ip`, we will aggregate destination ports on that particular IP in buckets of 10K and start building a port-profile for that IP (build a port-profile per destination IP).

- After three days, once we have built a profile, we will start aggregating port ranges where the port density is high (reflect ephemeral port opening pattern). The ranges themselves will be dynamic in size such as 100, 1,000, 10,000, and will be created depending on how many ports are being opened and how widespread they are in the given range of aggregation.

  **Note:**
  This decision happens independently for each server IP address.
- This will enable high-port flows to be reported with no aggregation where there is no bulk port open activity happening and also let dynamic aggregation to be applied where such activity is happening.
- The profile is continually updated in a time-decayed manner to account for new ports opening up or older ones being not used any more.

**How do I change the IP Address, Gateway, or Netmask after VCF Operations for networks OVA is deployed?**

To change VCF Operations for networks platform/collector network settings, log in to CLI and run the `change-network-settings` command. This interactive command will provide the user an option to modify the IP address, gateway, netmask, and so forth after which the VCF Operations for networks appliance is reconfigured with new details.

**Note:**

- This task must be done using VM console session as the appliance reboots in the end.
- If VCF Operations for networks Platform IP is modified and it is paired with proxies then on each collector VM run this CLI command:

  ```
  pair-collector set-platform --ip-or-fqdn <New_Platform_IP>
  ```
- For more information on changing IP address of any collector or platform node using the CLI commands, see How to change the IP of any platform node or a collector in a cluster?

**How do I change from an Evaluation License to a VCF License?**

Refer to the Add and Change License section in the VCF Operations for networks User Guide.

## Adding a vCenter Account

**What if I am getting a "Request timed out" message while adding vCenter using IP address?**

- Verify that the vCenter Server IP address is reachable from the VCF Operations for networks collector VM.
- Log in to VCF Operations for networks collector CLI and use the ping to ensure that IP is reachable and telnet to ensure that the vCenter is reachable on port 443.
- If vCenter is reachable, then retry adding.
- If IP address is not reachable, then verify whether the gateway is correctly configured from VCF Operations for networks collector VM using command `show-config`.
- If gateway is incorrect, then correct it using the `setup` command by logging in to CL with `consoleuser`.

**What if I am getting a "IP/FQDN is invalid" message while adding vCenter ?**

- Verify whether provided IP/FQDN for vCenter is correct.
- Verify whether FQDN is reachable from VCF Operations for networks collector VM using `ping` command.
- If it is not reachable, then verify if the DNS is configured correctly on VCF Operations for networks collector VM using `nslookup FQDN` and `show-config` command.
- If DNS is incorrect, then correct it using the `setup` command by logging in to CL with `consoleuser`.

**What privileges are required to add a vCenter instance in VCF Operations for networks?**

VCF Operations for networks requires the VMware vCenter credentials with the following privileges:

- System.Anonymous
- System.Read
- Global.Settings

- System.View

**What VC privileges are required to resolve the 'User does not have required privileges.' error when enabling Netflow/IPFIX on the source page?**

VCF Operations for networks requires the vCenter credentials with the following privileges to enable Netflow/IPFIX:

- Distributed Switch: Modify
- dvPort group: Modify

Please make sure that provided vCenter user have permission on vCenter root folder and all of its child entities e.g all folders and all datacenters.

**How frequently is the data fetched from the environment?**

VCF Operations for networks collector fetches data every 10 minutes from vCenter, NSX, AVI (NSX ALB), and HCX.

**How soon the analysis of data will start after adding the vCenter instance?**

Analysis of data starts right away after adding a vCenter instance. Product UI will show partial picture of data within few minutes which can take two hours to get complete.

> **Note:**
> Flow traffic data changes continuously and include at least 24 hours of data in its analysis.

**How do I clean up Netflow/IPFIX settings in vCenter if I have deleted the VCF Operations for networks appliance?**

- Using VMware vSphere Web Client: Go to **Home** > **Networking** > **VDS (Name)** > **Netflow** Settings. Remove VCF Operations for networks collector IP from Collector settings.
- Using VMware vSphere Windows Client: Go to **Home > Inventory > Networking > VDS (Name) > Edit** Settings. Remove VCF Operations for networks collector IP from Collector settings in Netflow tab. This step is required to be done for each VDS for which Netflow/ IPFIX is enabled.

**How do I clean up Netflow/IPFIX configuration in VCF Operations for networks?**

In VCF Operations for networks standalone UI, go to **Settings** > **Accounts and DataSources**, and select the target vCenter. Click the vertical ellipsis and click **Edit**. Navigate to the additional information section and uncheck **Netflow/ IPFIX**T. Click **Submit** to submit the data source and save the changes. This removes Netflow/ IPFIX configuration that is done by VCF Operations for networks.

**How much time does it take to show correct firewall rules in the VM-VM path after adding VMware NSX in VCF Operations for networks?**

After you add VMware NSX Manager in VCF Operations for networks, it can take up to 26 hours to compute the VM to the firewall rule relationship.

**Why I am unable to see the PNIC in the VM-VM path after adding vCenter in VCF Operations for networks?**

Normally, it takes around two hours for VCF Operations for networks to compute the VM-VM path after you add a vCenter instance in VCF Operations for networks as a data source. But in some rare situations, it can take around 8-10 hours to show the PNIC correctly in the VM-VM path after you add the VMware vCenter in VCF Operations for networks.

## Micro-Segmentation and Network Traffic Flows

**What do the numbers in the Traffic Distribution Pin represent?**

The percentage gives an overview of the traffic distribution based on flow analysis.

**Table 1345:**

| Traffic | Description |
|---|---|
| East-West (EW) | East-West traffic as the percentage of the traffic of the total group |
| Switched (% of EW) | Switched traffic as the percentage of East-West traffic |
| Routed (% of EW) | Routed traffic as the percentage (%) of East-West traffic |
| Within Host (% of VM-VM) | Traffic with source and destination on same host as percentage of virtual machine to virtual machine traffic |
| VM to VM (% of EW) | Virtual machine to virtual machine traffic as percentage of East-West traffic |
| Internet | Internet traffic as percentage of the traffic of the total group |

**How are ports aggregated in flows?**

Port aggregation is built in to aggregate the ephemeral port flows - like dynamic FTP, Oracle, MS-RPC etc. This helps in reducing the number of flows in system and provide an aggregated view for large number of flows that are essentially for the same service. The mechanism to do this is as follows:

- For first three days of noticing a destination_ip , we aggregate dst ports on that IP in buckets of 10K and start building a port-profile for that IP.
- Once three days are over - and we have built a profile that can be used with confidence - we will start aggregating port ranges where the port density is high (in other words - reflect ephemeral port opening pattern). The ranges themselves will be dynamic in size - 100, 1,000, 10,000 and will be created depend on how many ports are being opened and how widespread they are in the given range of aggregation.
- This will enable high-port flows to be reported with no aggregation where there is no bulk port open activity happening; and also let dynamic aggregation to be applied where such activity is happening.
- The profile is continually updated in a time-decayed manner to account for new ports opening up or older ones being not used any more.

**What does the 240.240.240.240 IP address signify in VCF Operations for networks?**

240.240.240.240 is a place holder IP address in VCF Operations for networks. This IP address is used if there are very large number of IP addresses (> 5000) hitting some particular IP. All further incoming Internet IPs (5001th onwards) with this placeholder IP 240.240.240.240 can be replaced for that service end point.

This is to limit the number of flows in the system, as publicly exposed service that log each internet client individually could result in very large number of flows - which would result in increased system load.

For all the flows that have been replaced with this placeholder IP, all the metrics are aggregated on the corresponding flow with this IP address, so there is no loss of statistics at an aggregate level.

All the destination IP for the flows reported in the flows view are shown as originating from 240.240.240.240 are actually being hit by large count of internet IP ( > 5000).

**Which are the protocols supported while configuring flows in VCF Operations for networks?**

The TCP and UDP protocols are supported while configuring flows in VCF Operations for networks.

**Does VCF Operations for networks handle de-duplication and aggregation of flow attributes from multiple sources?**

Yes, in VCF Operations for networks, the same data reported from multiple sources is de-duplicated and the best metric among all the sources is considered. Also, different attributes from multiple sources are aggregated and shown on the flow.

> **Note:**
> The Reporter field in the flow shows the reporter of the flow.

#### Why can't we see the VM names in the source or destination of the flows?

Some of the reasons for a VM (Source, Destination) not being attached to a flow are:

- The source or the destination vCenter data source is not added
- The IP is physical IP or Internet IP or Kubernetes POD or Container IP
- If the vmtools is not installed or active on the VM and the flow is not reported from the VDS
- If there are multiple nics assigned with the same IP and there is no NATing involved

#### Why do we see multiple Firewall rules on a flow?

When a firewall rule is changes from R1 to R2, VCF Operations for networks shows the new firewall rule (R2) but does not immediately remove the old firewall rule (R1). It takes up to six hours for the old rule to be removed. During this time period, before the old rule is removed, both the rules are displayed. If this happens multiple times (for example, R1 is changed to R2 and then R2 is changed to R3 before the old rule is removed), you will see multiple firewall rules for a single flow.

#### What is the flow count for a data source page and how is it calculated?

The numbers shown on the data source page are the unique flow counts (four tuples) seen from the data source in last 24 hours and the value is updated regularly (every 5 minutes). The flow count number might change, as it only considers the flows of last 24 hours. The count must match the result for the query: `'count of flows where reporter = '`.

## Data Management and Processing

#### What is the default retention period?

The default retention period is one month. Retention can be configured for non-flow data (configuration, event, and metric data) beyond one month from **Settings** > **Data Management**. Flow data retention is fixed at one month and cannot be changed.

#### How data is handled on the collector?

All data on the collector, including flow data is converted to SDM (Self Describing Message) before being sent to the platform. This includes all configuration, inventory, and metric data from any data source. If the platform is unreachable, or sending the data fails, it is written to disk on the collector VM (under **/var/BLOB_STORE**) until it can be sent.

#### When will data start to purge on the collector?

For processed data: 20 GB of disk space is allocated to store data on disk, split equally between inventory and metric data. When this space fills up, the oldest data is deleted to make room for new data. For raw flow data: Disk space is allocated to store raw flow data — 15 GB for single-collector deployments, or up to 45 GB in deployments using secondary collectors. As this space fills, older flow files are deleted to make room for new ones.

#### How is data deleted when the allocated limit is breached?

Both processed data and raw flow records are deleted in the order that they were received — the oldest data is always deleted first.

#### When will new data stop being processed on the collector?

Inventory and metric data are always processed as long as collector services are healthy. Flow data processing may pause if the active flow limit is reached — either on the specific collector or across the platform overall.

#### If the connection between platform and collector is lost, and no data has been purged, will all data reconcile once reconnected?

Yes, all data stored on disk on the collector is sent to the platform once connectivity is restored, and should reconcile completely — except where data loss conditions exist on the platform side (see below).

#### What conditions can cause data loss on the platform?

Flow data may stop being sent to the platform if the platform-wide active flow limit is reached. Metric data, including flow metrics, may be dropped if there is a processing lag of more than 8 hours on the platform.

#### In what order is data sent when the connection is re-established?

The oldest data is always sent first.

#### Is data stored on disk on the collector even when there is no connectivity problem?

If there is no communication issue, data is sent directly to the platform without being written to disk. Data is only written to disk after a failed attempt to send it.

#### How does the collector know which flow data was already processed if there is an interruption?

The collector tracks the last flow data that it successfully processed, so it can resume from where it left off after any interruption.

#### What is the maximum size of a single data message, and how would I know if one is dropped?

There is a 15 MB limit on the size of a single data message. If the platform drops a message for exceeding this limit, an alert is raised so you are aware of it.

## IPFIX

#### What is IPFIX?

IPFIX is an IETF protocol for exporting flow information. A flow is defined as a set of packets transmitted in a specific timeslot, and sharing 5-tuple values - source IP address, source port, destination IP address, destination port, and protocol. The flow information may include properties such as timestamps, packets/bytes count, Input/output interfaces, TCP Flags, VXLAN ID, Encapsulated flow information and so on. This is often referred to as Netflow. However, IPFIX is the standard IETF protocol.

#### What flow information is exported by the VDS?

A VDS in vSphere environment can be configured to export flow information using IPFIX. Enable flow monitoring on all the port groups attached to the VDS. If packets arrive on port X of a VDS and exit from port Y, a corresponding flow record is emitted if flow monitoring is enabled on port Y. The direction of every flow record is set as "Egress".

#### How does VCF Operations for networks use IPFIX?

VCF Operations for networks uses VMware VDS IPFIX to collect network traffic data. Every session has two paths. For example: Session A↔C has A→C packets and C→A packets. To analyze the complete information of any session, IPFIX data about packets in both the directions is required. Refer following diagram where VM-A is connected to DVPG-A and is talking to VM-C. Here DVPG-A will only provide data about the C→A packets, and DVPG-Uplink will provide data about A→C packets. To get the complete information of A's traffic, IPFIX should be enabled on DVPG-A, DVPG-uplink.

#### How do I troubleshoot VCF Operations for networks Flow Collection?

1. Please ensure that the specific VDS and its DVPGs and Uplink properties has Netflow monitoring **Enabled** and the collector IP address is that of VCF Operations for networks collector.
2. IPFIX Netflow packets are getting dropped in between by a firewall (NSX, Virtual or Physical). Please ensure that the Netflow packets destined for UDP port 2055 on VCF Operations for networks collector IP is allowed by any firewall that may be present in the route between ESXi Host and the VCF Operations for networks Collector.
3. The ESXi host has ceased to send IPFIX Netflow packets. The ESXi host backs off sending the Netflow packets after some time if UDP port 2055 is not reachable. This may happen due to the firewall dropping the packets.

4. The VCF Operations for networks collector is not reachable by ESXi Host due to network routing problem. Ensure that the proper routroute existse exist between ESXi Host and the VCF Operations for networks collector.

**When is a service considered shared?**

| Protocol | Port |
|---|---|
| DNS | 53 |
| Bootpc | 68 |
| Kerberos | 88 |
| Pop3 | 110 |
| sunrpc | 111 |
| NTP | 123 |
| map | 143 |
| Imap3 | 220 |
| SMTP | 25 |
| LDAP | 389 |
| IGMPv3Lite | 465 |
| syslog | 514 |
| Submission | 587 |
| syslog-conn | 601 |
| LDAPS | 636 |
| IMAPS | 993 |
| POP3S | 995 |
| NFS | 2049 |
| MSFT-GC | 3268 |
| MSFT-GC-SSL | 3269 |

## Clustering - General

**Can a collector VM be clustered?**

No. Clustering of collector VMs is not supported.

**Does VCF Operations for networks need a load balancer like VCF Operations for logs?**

VCF Operations for networks clustering functions as a scale-out mechanism rather than a high availability solution; a failure of the primary node causes the entire service to become unavailable.

**What happens if the connection between remote collector & platform goes down?**

In case the connection between platform and collector VM is down, the collector VM stores data locally (depending on the disk space) and sends it whenever it is connected again.

**Can VCF Operations for logs be integrated with VCF Operations for networks?**

Yes, VCF VCF Operations for logs can be integrated with VCF Operations for networks for direct log collection using the VCF Operations UI.

### What happens when a node reboots?

If a collector node reboots, there is data loss during the downtime. For more information, see the Broadcom Knowledge Base article 314428.

### How to change the IP of any platform node or a collector in a cluster?

In a cluster, you can change the IP of any collector or platform node using the CLI commands.

For more information on updating network settings and IP addresses, see the following Broadcom Knowledge Base articles:

- For Standard/Standalone Deployments: 370647
- For Clustered Deployments & Multi-Node Platform Migrations: 423336

  **Note:**
  The appliance reboots at the end of the process. So you must perform these steps on the VM console.

- To change the collector IP, run the `change-network-settings` command.
- To change the platform IP,
  a. Run the `change-network-settings` command to change the IP.

     **Note:**
     Ensure that the platform VM reboots successfully before you run the next command.
  b. For every platform whose IP is changed, run `update-IP-change` command on all the other nodes.
     For example, for the IP change of platform1 from IP1 to IP2, run the `update-IP-change` command on platform2 and platform3 with IP1 and IP2 as the arguments. For the IP change of platform2, run the `update-IP-change` command on platform1 and platform3.
  c. Run `finalize-IP-change` command on platform1 after step1 and step 2 are completed.
  d. Run the `show-connectivity-status` command on a collector and search for `Platform VM IP/URL` to identify if it is associated to this platform.

e. Run `pair-collector` to reflect new platform IP on associated collectors.

| Scenarios | Procedure |
|---|---|
| In a 3-Node cluster, only the platform2 IP is changed. No collector shows associated with it. | 1. Run the `change-network-settings` to change the IP on platform2.<br>**Note:**<br>Ensure that the platform VM reboots successfully before you run the next command.<br>2. Run `update-IP-change` command on all the other nodes, except platform2.<br>For example, for the IP change of platform2, run the `update-IP-change` command on platform1 and platform3.<br>3. Run `finalize-IP-change` command on platform1 after step1 and step 2 are completed. |
| In a 3-Node cluster, platform1 and platform2 IPs are changed. | 1. Run `change-network-settings` on platform1.<br>2. Run `change-network-settings` on platform2.<br>**Note:**<br>Ensure that the platform VM reboots successfully before you run the next command.<br>3. Run `update-IP-change platform1-oldIP platform1-newIP` on platform2 and platform3.<br>4. Run `update-IP-change platform2-oldIP platform2-newIP` on platform1 and platform3.<br>5. Run `finalize-IP-change` command on platform1.<br>6. Run the `show-connectivity-status` command on all the collectors and search for `Platform VM IP/URL` to identify the associated platform nodes.<br>7. Run `pair-collector set-platform --ip-or-fqdn platform-newIP` on collectors that are associated with platfrom1 and platform2.<br>For example, if the CollectorA is associated to platform2, rest are associated to platform3, run `pair-collector set-platform --ip-or-fqdn platform2-newIP`.<br>8. Run `pair-collector set-platform --ip-or-fqdn platform-newIP` on collectors that are associated with platfrom1 and platform2.<br>For example, if the CollectorA is associated to platform2, rest are associated to platform3, run `pair-collector set-platform --ip-or-fqdn platform2-newIP`. |

| Scenarios | Procedure |
| --- | --- |
| In a multi-node cluster setup, all the platform and collector IP are changed. | 1. Run `change-network-settings` command on all platforms to change the IP.<br>**Note:**<br>Ensure that the platform VM reboots successfully before you run the next command.<br>2. Run `update-IP-change` command on all the nodes.<br>For example, in a 3-node cluster, to update IP of platform1 from IP1 to IP2, run the `update-IP-change` command on platform2 and platform3 with IP1 and IP2 as the arguments. To update IP of platform2, run the `update-IP-change` command on platform1 and platform3.<br>3. Run `finalize-IP-change` command on platform1.<br>4. To change the collector IP, run the `change-network-settings` command.<br>5. Run the `show-connectivity-status` command on all the collectors and search for `Platform_VM_IP/URL` to identify the associated platform nodes.<br>6. Run `pair-collector set-platform --ip-or-fqdn platform-newIP` on all collectors.<br>For example, if the CollectorA is associated to platform2, run `pair-collector set-platform --ip-or-fqdn platform2-newIP`. |

**How much disk space is needed on platform1?**

Platform1 requires more disk space compared to other nodes in the cluster as some of the configuration data is stored on Platform1 only.

**What happens if any of the nodes ran out of disk space?**

The UI starts showing error messages when disk space on any particular platform node reaches a certain threshold. Add more disk space to the platform node by logging in to VMware vCenter. The platform/collector VM must be in powered-on state while adding disk space. For more information, see Increase the Disk Size.

**What is the replication factor of data in the platform cluster?**

The replication factor is 2 for all persisted data in the platform cluster. There is no replication on the collector side.

**How to change the hostname of a platform or a collector node?**

You can update the hostname of a platform or a collector VM using the `hostname update` command. Updating the hostname helps you easily identify the collector VM or the platform VM in your data center and align the naming convention of the VM with other VMs in your data center.

To change the hostname of platform VMs or collector VMs, perform these steps:

1. Log in to the respective VM console.
2. Switch to a Ubuntu user.
3. Run the `hostname show` command to view the present hostname of VCF Operations for networks collector or platform VMs.
   ```
   hostname show
   ```
4. To view the supported options, run the following command:
   ```
   hostname update --help
   ```

**Note:**

You use the `hostname update` command to update the hostname.

| Option | Description |
|---|---|
| --dryrun | This option validates the new hostname without updating the hostname in VCF Operations for networks. |
| --hosts | This option updates the hostname of your VMs in VCF Operations for networks. |

5. Run the `hostname update --dryrun` command to validate the new hostname.
   Syntax:
   - To validate the hostname of a platform or a collector VM in a single node VCF Operations for networks instance, run the following command:
     `hostname update --dryrun --hosts <new_hostname>`
   - To validate hostnames of platform VMs in a cluster, run the following command:
     `hostname update --dryrun --hosts platform1:<new_hostname1> platform2:<new_hostname2> platform3:<new_hostname3>`

     **Note:**

     - Use the following guidelines for hostnames:
       - Use only alphanumeric characters, period ("."), and hyphens("-") in the hostname.
       - Use less than 64 characters.
       - Do not start or end the hostname with a period or a hyphen.
6. Run the `hostname update --hosts` command to update the hostname.
   Syntax:
   To update the hostname of a platform or a collector VM in a single node VCF Operations for networks instance, run:
   `hostname update --hosts <new_hostname>`
   To update the hostname of one or more platform VMs in a cluster, run the following command:
   `hostname update --hosts platform1:<new_hostname1> platform2:<new_hostname2> platform3:<new_hostname3>`

   **Note:**

   After changing the hostname, it takes up to an hour for the hostname to reflect on the VCF Operations for networks UI.

## Clustering - Installing and Configuring

### Do all platform VMs have to be on the same L2/L3 segment?

No. However, it is best to keep all platform nodes on a common network with low latencies between nodes. This is because many of the distributed components replicate data among the nodes and high latencies can cause system performance and stability issues. For more information, see VCF Operations for Networks System Recommendations and Requirements.

### What happens if there is a failure during the cluster creation process?

- Cluster creation for single node: Before converting a single node to a multi-node cluster, take a cold snapshot of the original platform, the new platform, and all deployed collectors. If a failure occurs, delete the secondary platform nodes and recover the primary platform and collector virtual machines from the snapshot.

**What happens if there is a failure during the cluster expansion process?**

Similar to the cluster creation scenario, take snapshots of the existing cluster and the newly deployed nodes (all platform nodes and collectors) and perform the steps mention in the following Broadcom Knowledge Base article KB article 314428.

**What happens to the existing data and configuration when I expand the single node deployment to a cluster?**

All data and configuration is maintained without any change. The data will be accessible after cluster creation.

**Can you have platform VM in different regions?**

No, we require the Platform nodes to co-located be in the same site. The collector servers can be geo-distributed.

The collector communicates with vCenter over HTTPS to pull inventory, configuration, VM topology, and push that data to the platform. You only need one collector for each vCenter instance. Collectors do not need to be assigned for each ESXi host or cluster—even if that vCenter manages hundreds of remote sites.

IPFIX data is exported toward the collector over UDP port. Every exporter (ESXi host) must have a working network path to the collector. If hosts are in remote sites with high latency or bandwidth constraints, sending flows across WAN links to a central collector can be inefficient.

**Can platform hosted on vSAN Stretch clusters (2 Data centers)?**

Yes, vSAN clusters within the same or across data centers would still ensure certain IO performance like local storage. For more information, see VCF Operations for Networks System Recommendations and Requirements.

**Can we host cluster nodes on different vSAN Clusters?**

Yes, Different nodes of a Platform cluster could be hosted on different underlying datastores.

## Clustering - Scaling

**Can I extend an already created cluster?**

Yes, extending a cluster is supported up to 15 nodes.

**What happens if a non-primary platform VM becomes unavailable?**

Grid processing halts; however, UI and API access remain available through the primary node. The system retains previously processed and stored data. Extended node unavailability can result in the permanent loss of incoming data. Upon node restoration, the system processes only the most recent data.

**What kind of load balancing is supported?**

External load balancers are not supported. The system internally load balances API calls directed to the platform, whether accessed directly or through the user interface. The collector manages load balancing for collector-to-platform communication, routing traffic to the platform cluster using a round-robin method.

**Will creating a platform cluster increase network bandwidth consumption between collector and platform?**

Collector-to-platform traffic is distributed across the platform cluster. Aggregate bandwidth consumption from the collector remains constant. However, as the cluster forms and processes additional data, bandwidth requirements increase proportionally.

**What is the frequency of data transmission between collector VM to platform VM?**

The collector VM sends data to the platform VM continuously.

**Does any optimization of data take place in the collector VM?**

Various deduplications, compressions, reduction, or batching steps happen in the collector VM. When the connection between the platform VM and the collector VM is down, the collector VM stores data locally (depending on the disk space) and sends it whenever the connection is restored.

**Is there any optimization done for network bandwidth?**

Yes, various deduplication/ compressions/ reduction/ batching steps happen on the collector VM.

**Is clustering possible on the collector?**

No. Clustering is not possible on the collector.

**How does vCenter send traffic to the collector?**

vCenter instances do not send traffic to the collector. The collector connects to configured vCenter instances to fetch inventory and metric data. VDS exports IPFIX NetFlow data to the collector via UDP.

### Clustering - Deployment

**What is Platform1 and why do I need to remember this node?**

The system designates the node that is used to initiate the cluster creation process as Platform1. Perform all external integrations and backend operations exclusively from Platform1. Platform1 also helps to change network across cluster nodes from CLI.

**How do I access the UI after scaling out the cluster?**

The UI can be accessed from any platform node.

**Can I use a platform node which is deployed in different data center for creating clusters?**

Nodes within a cluster exchange data continuously. To prevent latency, a cluster must consist of platform nodes deployed in the same data center.

**What happens to data on existing platform when I scale out the platform node?**

The data on an existing platform node is preserved and distributed across all nodes in a cluster.

**Do the number of collector VMs and data sources, including physical devices, determine the required platform cluster size?**

The size of the platform cluster is determined by the number of VMs, number of flows, number of network rules, and applications, etc. For more information, see VCF Operations for Networks System Recommendations and Requirements.

**Does VCF Operations for networks support platform cluster distributed across 2 data centers for HA reasons?**

No. The platform Cluster does not support splitting across data centers. All Platform Cluster VMs should be in the same site. Platform Cluster does not support HA today.

## Metric Definitions in VCF Operations for networks

To display different types of metrics on the user interface, VCF Operations for networks collects data from multiple data sources. Each piece of data collected is called a metric observation or a metric value.

VCF Operations for networks displays the following metrics on the user interface:

| Metric Type | Description | Data Sources Supporting the Metric |
|---|---|---|
| Entity metrics | Entity metrics display information such as resource utilization and performance of entities in your network. They include metrics such as host metrics, VM metrics, datastore metrics, and so on.<br>To learn more, see Entity Metrics | All data sources added to VCF Operations for networks except ServiceNow, Infoblox, and VCF Operations for logs data sources. |
| Flow metrics | Flow metrics display the traffic flow data collected from flow-enabled data sources on your network.<br>The flow data from the data sources is:<br>• Derived and aggregated to show different flow metrics such as total bytes, traffic rate, and so on.<br>• Aggregated on the tier level to show application tier metrics such as average drop flows, total internet packet, and so on.<br>To learn more, see Flow Metrics. | • VMware vCenter<br>• NSX Manager<br>• Netflow collector<br>• Physical flow collector |
| Health metrics | Health metrics represent the overall health of the VCF Operations for networks instance. These metrics are useful in diagnosing health problems related to the VCF Operations for networks instance.<br>To learn more, see Health Metrics. | VCF Operations for networks Collector and Platform |
| Latency metrics | Latency metrics display information about the latency of data packets traveling between the following devices:<br>• Interface pNIC pairs<br>• Interface pairs<br>These metrics are useful in monitoring the latency of the network to diagnose and troubleshoot performance bottlenecks.<br>To learn more, see Latency Metrics. | NSX-T Manager |
| VM network utilization metrics | The VM network utilization metrics display the performance of the VM based on network utilization. These metrics are critical to track network activity and throughput.<br>In VCF Operations for networks, these metrics are displayed at 20 second resolution to showcase the peaks and troughs of network utilization.<br>**Note:**<br>VM network utilization metrics are displayed at 20 seconds granularity only if you are viewing the metric chart within the last 48 hours.<br>Beyond 48 hours, the metric resolution is displayed based on the selected time range.<br>See VM Network Utilization Metrics to learn more. | VMware vCenter |

| Metric Type | Description | Data Sources Supporting the Metric |
|---|---|---|
| Edge transport node metrics | Starting with version 9, you can now view enhanced metrics taken from the the edge transport nodes of your NSX environment. This includes metrics retrieved in five minute intervals and more granular metrics retrieved every 20 seconds. Granular metrics can be separated in four categories:<br>• Node System metrics<br>• DP CPU Core metrics<br>• Node Process metrics<br>• Memory Pool metrics | NSX Manager |

## Comprehending Metric Data

You can view your metric data at different levels of granularity, based on the time range. For example, you can choose a one day view, which can be useful when troubleshooting. Or, choose a broader time range (for example, 30 days) so that you can see trends over time.

When you view metric data for different time ranges, you see different values indicated in the metric chart. This difference is because VCF Operations for networks displays different resolutions of data for different time ranges.

The default time range of any metric in VCF Operations for networks is 1 day and the resolution is 5 minutes. As the metric resolution becomes coarser, the metric data is down-sampled. That is, the resolution of the data is reduced. This is performed to create better metric graph visualizations by retaining only the important characteristics of the metric.

Time range and metric resolution for entity metrics:

| Time Range | Metric Resolution |
|---|---|
| 0 hours – 3.5 days | 5 minutes |
| 3.5 days – 15 days | 30 minutes |
| 15 days – 60 days | 2 hours |
| Beyond 60 days | 8 hours |

Time range and metric resolution for flow metrics:

| Time Range | Metric Resolution |
|---|---|
| 0 hours - 24 hours (within last 24 hours) | 5 minutes |
| 0 hours - 24 hours (beyond last 24 hours) | 30 minutes |
| 1 day – 15 days | 30 minutes |
| 15 days – 31 days | 2 hours |

Time range and metric resolution for special VM network utilization metrics:

| Time Range | Metric Resolution |
|---|---|
| 0 hours – 6 hours (within last 48 hours) | 20 seconds |
| 0 hours – 6 hours (beyond last 48 hours) | 5 minutes |
| 6 hours – 3.5 days | 5 minutes |
| 3.5 days – 15 days | 30 minutes |
| 15 days – 60 days | 2 hours |
| Beyond 60 days | 8 hours |

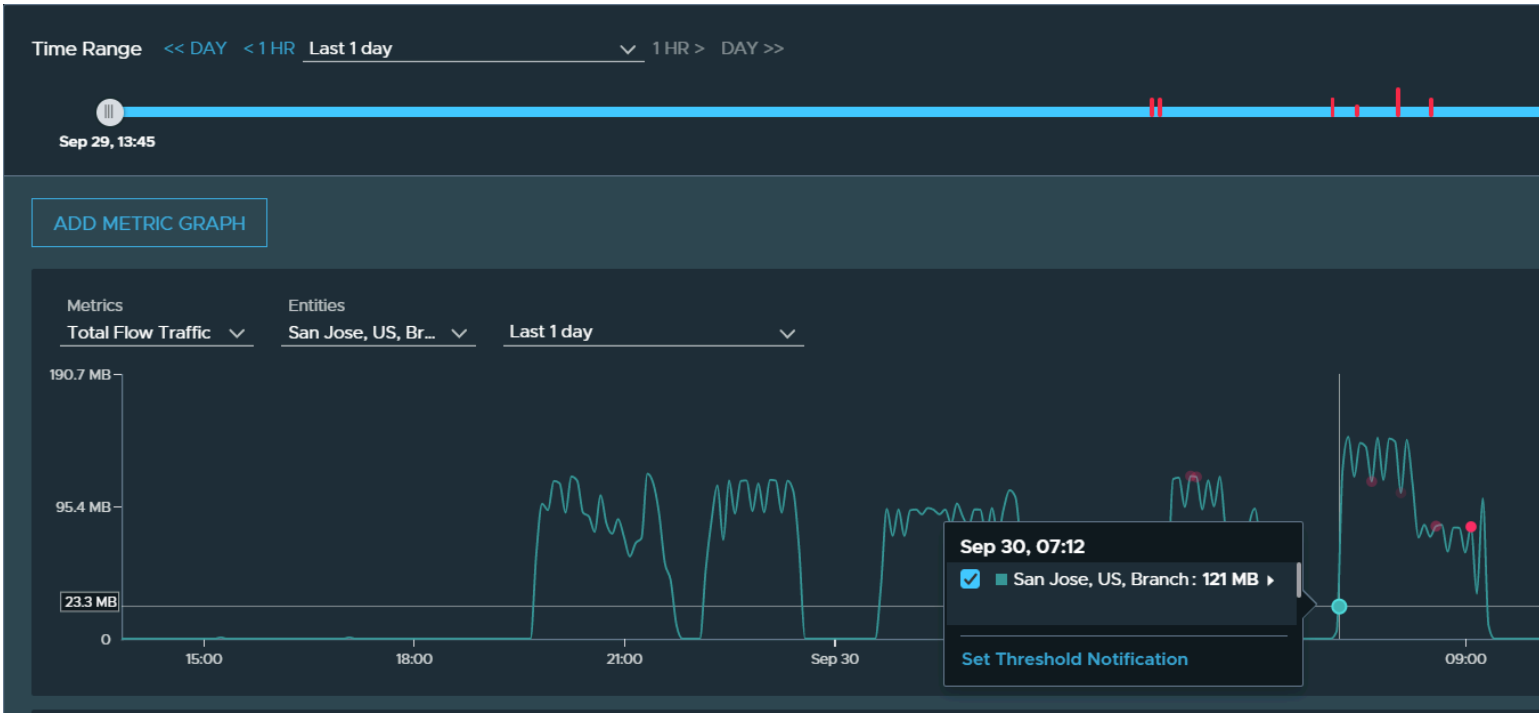
## Viewing Metric Data

In VCF Operations for networks, the metric data of an entity is displayed as line graph on the metric chart. You can point anywhere on the line graph to get informative tool tips. You can also co-relate multiple entities or two metrics graphs at the same time.

You can modify the time range of a metric by either using the slider or entering in a custom date and time. You can also adjust the time range for each entity within a metric.

You can select different metrics, entities, and time from the drop-down menu that is available in every metric chart. You can also adjust the view to a row view or a column view by using the view button in the top-right corner of the metrics chart. You can also export individual metrics graph in `CSV` format.

Here is an example metrics chart that shows the total flow of traffic in a VMware SD-WAN Edge device.




## Setting the Retention Interval

In VCF Operations for networks, you can choose how long the system retains metric data before automatically deleting it by setting the retention interval for each metric.

To set the retention interval of a metric, navigate to the **Settings** > **Data Management** page. To learn more, see Configure Data Retention Interval.

The minimum and maximum retention intervals defined by the system is as follows. Based on your business requirement, you can choose an interval between these values.

| Metric Type | Resolution | Minimum Retention | Maximum Retention |
|---|---|---|---|
| Entity metrics | 5 minutes | 1 month | 13 months |
| | 30 minutes | 1 month | 13 months |
| | 2 hours | 1 month | 13 months |

| Metric Type | Resolution | Minimum Retention | Maximum Retention |
|---|---|---|---|
| | 8 hours | 1 month | 13 months |
| Flow metrics<br>**Note:** You cannot edit the retention interval of flow metrics on the VCF Operations for networks UI. | 5 minute | 24 hours | 24 hours |
| | 30 minutes | 1 month | 1 month |
| | 2 hours | 1 month | 1 month |
| Health metrics | 5 minutes | 1 month | 13 months |
| | 30 minutes | 1 month | 13 months |
| | 2 hours | 1 month | 13 months |
| | 8 hours | 1 month | 13 months |
| VM network utilization metrics | 20 second<br>**Note:** You cannot edit the retention interval of VM network utilization metrics with 20 seconds resolution on the VCF Operations for networks UI. | 2 days | 2 days |
| | 5 minutes | 1 month | 13 months |
| | 30 minutes | 1 month | 13 months |
| | 2 hours | 1 month | 13 months |
| | 8 hours | 1 month | 13 months |

**Note:**

The retention interval you choose has a direct impact on disk space.

Reducing the log retention period deletes all logs older than the new retention period specified. For example, changing the retention period of the entity metrics from 6 months to 1 month deletes all logs older than a month. Therefore, consider backing up your log files before reducing the retention period.

Increasing the log retention period adds new logs to the collection. Therefore, check the available disk space and consider upgrading your disk space when increasing the log retention period.

## Entity Metrics

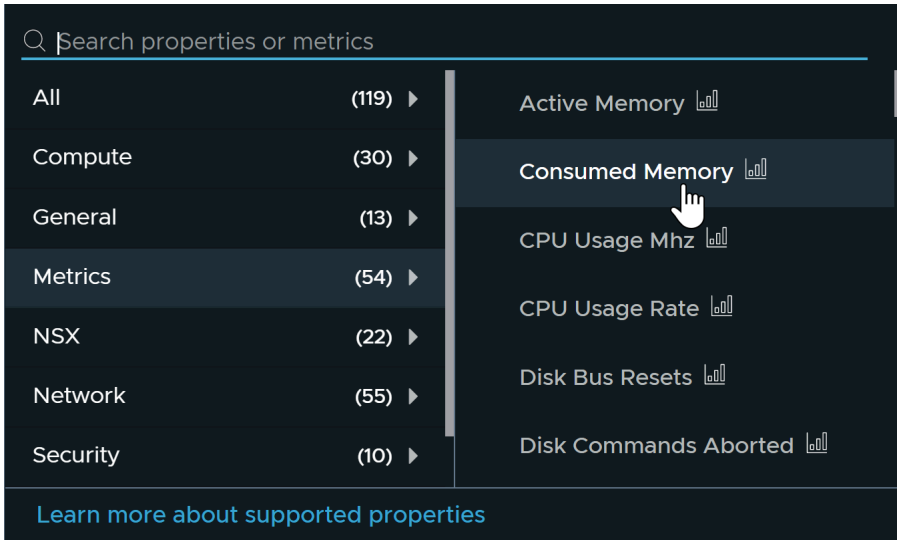
VCF Operations for networks collects entity metrics from data sources in your environment such as VMware vCenter, NSX to display information such as resource utilization, and performance.

To learn about the retention period of entity metrics, see Metric Definitions.

### Viewing Entity Metrics

To view the entity metrics on the user interface, do the following:

1. Enter the entity name on the search bar.
2. On the search result page, click **Add More Filters** under the **Filters** option.
3. Select **Metrics**. The entity related metrics are displayed to the right.
4. Select the metric you want to view.




## Application Metrics

VCF Operations for networks collects network and flow metrics for applications.

To view the application metrics on the user interface, do the following:

1. Enter `applications where name is set` on the search bar.
2. On the search result page, click **Add More Filters** under the **Filters** option.
3. Select **Metrics**. The application metrics are displayed to the right.
4. Select the metric you want to view.

The following metrics are available for each application in your VCF Operations for networks environment.

| Metric Name | Metric API Name | Description |
|---|---|---|
| App/Tier TCP Retransmitted Packets | net.tcpRetransmitedPackets.delta.summation.number | Total number of retransmitted packets in all the flows of the application. |
| Median TCP RTT (ms) | net.p50FlowAvgTcpRtt.absolute.average.microsecond | Median round-trip time of all TCP flows associated with an application. |
| Network Packet Count | net.packetsTotal.delta.summation.number | Total number of packets transmitted and received from an application. |
| Network Rx Rate (kbps) | net.received.rate.average.kiloBitsPerSecond | Average rate at which data is received across all endpoints of the application. |
| Network Tx Rate (kbps) | net.transmitted.rate.average.kiloBitsPerSecond | Average rate at which data is transmitted from all endpoints in the application. |
| Rx Flow Drops | flow.droppedRx.delta.summation.number | Total number of dropped incoming flows to total number of incoming flows. |
| TCP Retransmitted Packet Ratio (%) | net.tcpRetransmitedPackets.ratio.average.percent | Percentage of retransmitted packets in all the flows associated with an application. |
| Total Rx Bytes | net.totalRx.delta.summation.bytes | Total number of bytes received by the application. |
| Total Tx Bytes | net.totalTx.delta.summation.bytes | Total number of bytes transmitted by the application. |
| Tx Flow Drops | flow.droppedTx.delta.summation.number | Total number of dropped outgoing flows to total number of outgoing flows. |

## AWS Virtual Private Cloud Metrics

VCF Operations for networks collects flow metrics for AWS Virtual Private Cloud (VPC) instances.

To view AWS VPC metrics on the user interface, do the following:

1. Enter `AWS Virtual Private Cloud` on the search bar.
2. On the search result page, select an AWS VPC of your choice.
3. On the respective AWS VPC page, select the **Metrics** tab.

| Metric Name | Metric API Name | Description |
|---|---|---|
| Outbound Expire Flow Drop Count | flow.fiveTupleExpireOutbound.delta.summation.number | Number of five tuples dropped because of reaching the maximum outbound limit for source IP addresses in flows. |
| Wrong Collector Flow Drop Count | flow.rawRecordReporterWrongCollector.delta.summation.number | Number of raw flow records dropped as the flow records were reported on a wrong type of VCF Operations for networks collector. |

## Datastore Metrics

VCF Operations for networks collects datastore metrics to learn about storage space use and to check metric trends to identify spikes or abnormal growth.

| Metric Name | Metric API Name | Decription |
|---|---|---|
| Capacity | datastore.capacity.absolute.latest.bytes | Total storage capacity of the datastore. |
| Committed Space | datastore.committed.absolute.latest.bytes | Storage space on the datastore that is used by the virtual machine to store disks, logs, snapshots, and configuration files. |
| Free Space | datastore.freespace.absolute.latest.bytes | Free space available on the datastore. |
| Free Space Percent (%) | datastore.freespace.absolute.latest.percent | Ratio of free storage space to the total storage capacity on the datastore. |
| Provisioned Space | datastore.provisioned.absolute.latest.bytes | Storage capacity allocated for all VMs on the datastore. |
| Provisioned Space Percent (%) | datastore.provisioned.absolute.latest.percent | Ratio of the provisioned space to the total storage capacity for the datastore. |
| Read IOPS | datastore.numberReadAveraged.rate.average.number | Average number of read commands issued per second on the datastore. |
| Read Latency (ms) | datastore.weightedTotalReadLatency.absolute.average.millisecond | Average read latency over a period of 24 hours. |
| Read Rate (kbps) | datastore.read.rate.average.kiloBytesPerSecond | Read throughput averaged over a period of 24 hours. |
| RW IOPS | datastore.numberRWAveraged.rate.average.number | Average number of read and write commands issued per second over a period of 24 hours. |
| RW Rate (kbps) | datastore.usage.rate.average.kiloBytesPerSecond | Read and write throughput averaged over a period of 24 hours. |
| Uncommitted Space | datastore.uncommitted.absolute.latest.bytes | Additional storage space that is potentially used by the VM on the datastore. This storage space is needed for thin-provisioned VMs to allow lazily allocated disks to expand or storage for swap. |
| Used Space | datastore.usedspace.absolute.latest.bytes | Storage capacity used by VMs on the datastore. |
| Used Space Percent (%) | datastore.usedspace.absolute.latest.percent | Ratio of used storage capacity to the total storage capacity on the datastore. |

| Metric Name | Metric API Name | Decription |
|---|---|---|
| Write IOPS | datastore.numberWriteAveraged.rate.average.number | Average number of write commands issued per second on the datastore. |
| Write Latency (ms) | datastore.weightedTotalWriteLatency.absolute.average.millisecond | Average write latency over a period of 24 hours. |
| Write Rate (kbps) | datastore.write.rate.average.kiloBytesPerSecond | Write throughput averaged over a period of 24 hours. |

## Device Interface Service Metrics

The device interface service metrics provide information about the traffic of the interface of a device accessing a particular service.

| Metric Name | Metric API Name | Decription |
|---|---|---|
| Bytes | net.serviceTotalBytes.delta.summation.bytes | Total incoming and outgoing traffic from a device interface accessing a service. |
| Egress Bytes | net.serviceEgress.delta.summation.bytes | Total outgoing traffic from a device interface accessing a service. |
| Ingress Bytes | net.serviceIngress.delta.summation.bytes | Total incoming traffic to a device interface accessing a service. |
| Low Priority Bytes | net.serviceLowPriorityTotalBytes.delta.summation.bytes | Total incoming and outgoing traffic of low priority flows of a device interface accessing a service. |
| Low Priority Egress Bytes | net.serviceLowPriorityEgress.delta.summation.bytes | Total outgoing traffic of low priority flows from a device interface accessing a service. |
| Low Priority Ingress Bytes | net.serviceLowPriorityIngress.delta.summation.bytes | Total incoming traffic of low priority flows to a device interface accessing a service. |

## Host Metrics

VCF Operations for networks collects many metrics for hosts, including CPU usage, datastore, memory, network, and disk metrics.

### CPU Usage Metrics for Hosts

These metrics provide information about the CPU use on the host.

| Metric Name | Metric API Name | Description | Supported by Databus |
|---|---|---|---|
| CPU Co-stop Time (%) | cpu.costop.delta.summation.millisecond | Percentage of time the host is ready, but unable to run due to co-scheduling constraints. | |
| CPU Ready Time (%) | cpu.ready.delta.summation.millisecond | Percentage of time the VM was waiting in line to use the CPU on the host. | |
| CPU Run Time (ms) | cpu.run.delta.summation.millisecond | Total time spent in `run` state by all VMs of this host. | |
| CPU Usage (MHz) | cpu.usagemhz.rate.average.megaHertz | Average CPU use in megahertz. | Yes |

| Metric Name | Metric API Name | Description | Supported by Databus |
|---|---|---|---|
| CPU Usage Rate (%) | cpu.usage.rate.average.percent | Percentage of the CPU used out of its total capacity. | Yes |
| CPU Wait Time (ms) | cpu.wait.delta.summation.millisecond | Total CPU time spent in wait state such as CPU idle, CPU swap wait, and CPU I/O wait. | |
| Uptime (s) | sys.uptime.absolute.latest.second | Total time elapsed since the last system startup. | |

**Network Metrics for Hosts**

These metrics provide information about the network performance of the host.

| Metric Name | Metric API Name | Description | Supported by Databus |
|---|---|---|---|
| Dropped Rx Packets | net.droppedRx.delta.summation.number | Average ratio of the number of dropped Rx packets to the total number of Rx packets. | Yes |
| Dropped Tx Packets | net.droppedTx.delta.summation.number | Average ratio of the number of dropped Tx packets to the total number of Tx packets. | Yes |
| Max IO Latency | datastore.maxTotalLatency.absolute.latest.millisecond | Highest latency value across all datastores connected to the host. | |
| Max Network Traffic Rate (Kbps) | net.usage.rate.maximum.kiloBitsPerSecond | Maximum data transmission rate on the host over any five minute interval in the selected duration. Maximum network rate includes both transmitted and received data. | Yes |
| Max Network Tx Rate (Kbps) | net.transmitted.rate.maximum.kiloBitsPerSecond | Maximum data transmission rate on the host over any five minute interval in the selected duration. | Yes |
| Max Packet Drop Ratio (%) | net.packetDrop.ratio.maximum.percent | Maximum ratio of total packets dropped to total number of packets received over any five minute interval in the selected duration. | Yes |
| Max Packet Drops | net.droppedTotal.delta.maximum.number | Maximum number of dropped packets over any five minute interval in the selected duration. | yes |
| Multicast Rx Packets | net.multicastRx.delta.summation.number | Total number of multicast packets received across all pNICs (physical network adapters) during the sampling interval. | Yes |
| Multicast Tx Packets | net.multicastTx.delta.summation.number | Total number of multicast packets transmitted across all pNICs during the sampling interval. | Yes |

| Metric Name | Metric API Name | Description | Supported by Databus |
|---|---|---|---|
| Network Max Rx Rate (Kbps) | net.received.rate.maximum.kiloBitsPerSecond | Maximum network data receive rate on the host over any five minute interval in the selected duration. | Yes |
| Network Packet Count | net.packetsTotal.delta.summation.number | Total number of packets transmitted and received during the interval across all pNICs. | Yes |
| Network Packet Drop Percent (%) | net.packetDrop.ratio.average.percent | Ratio of total number of dropped packets to total number of packets. | Yes |
| Network Packet Drops | net.droppedTotal.delta.summation.number | Total number of packet drops on the host. | Yes |
| Network Rx Rate (Kbps) | net.received.rate.average.kiloBitsPerSecond | Average rate at which data was received during the time interval. | Yes |
| Network Traffic Rate (Kbps) | net.usage.rate.average.kiloBitsPerSecond | Average rate of received and transmitted traffic during the time interval. | Yes |
| Network Tx Rate (Kbps) | net.transmitted.rate.average.kiloBitsPerSecond | Average rate at which data was transmitted during the time interval. | Yes |
| Outbound Expire Flow Drop Count | flow.fiveTupleExpireOutbound.delta.summation.number | Number of five tuples dropped due to the number of source IP addresses of flows reaching the outbound limit.<br>The outbound limit for TCP flows is 1000 (allowed) and 50 (denied), and the outbound limit of UDP flows is 500 (allowed) and 50 (denied).<br>This is a protection mechanism to make sure that the entities do not overload the total flow limit. | |
| Rx Broadcast Packets | net.broadcastRx.delta.summation.number | Number of broadcast packets received during a specified time interval. | Yes |
| Rx Broadcast Ratio | net.broadcastByRx.ratio.average.percent | Percent of broadcast packets received during a specified time interval. | Yes |
| Rx Packet Drop Percent (%) | net.packetDropRx.ratio.average.percent | Ratio of total number of dropped Rx packets to total number of Rx packets. | Yes |
| Rx Packets | net.packetsRx.delta.summation.number | Average number of received packets during the time interval. | Yes |
| Total Disk Latency (ms) | disk.totalLatency.absolute.average.millisecond | Average total latency (sum of kernel and device latency) for both read and write commands. | |
| Total Network Traffic | net.usage.delta.summation.kiloBytes | Sum of data transmitted and received for all the NIC instances of the host. | Yes |

| Metric Name | Metric API Name | Description | Supported by Databus |
|---|---|---|---|
| Tx Broadcast Packets | net.broadcastTx.delta.summation.number | Number of broadcast packets transmitted during a specified time interval. | Yes |
| Tx Broadcast Ratio | net.broadcastByTx.ratio.average.percent | Average percent of broadcast packets transmitted during a specified time interval. | Yes |
| Tx Packet Drop Percent (%) | net.packetDropTx.ratio.average.percent | Ratio of total number of dropped Tx packets to total number of Tx packets. | Yes |
| Tx Packets | net.packetsTx.delta.summation.number | Average number of transmitted packets during the time interval. | Yes |
| Unicast Rx Packets | net.unicastRx.delta.summation.number | Total number of unicast packets received by the host. | Yes |
| Unicast Tx Packets | net.unicastTx.delta.summation.number | Total number of unicast packets transmitted by the host. | Yes |
| Wrong Collector Flow Drop Count | flow.rawRecordReporterWrongCollector.delta.summation.number | Number of raw flow records dropped as they were reported on a wrong type of VCF Operations for networks collector.<br>Physical flows must be reported to physical proxies. Other flows must be reported to non-physical proxies. | |

## Datastore Metrics for Hosts

These metrics provide information about the datastore use of the host.

| Metric Name | Metric API Name | Description |
|---|---|---|
| Read IOPS | datastore.numberReadAveraged.rate.average.number | Average number of read commands issued per second during the collection interval. |
| Read Latency (ms) | datastore.totalReadLatency.absolute.average.millisecond | Average latency of each read operation issued by the link through which the host is connected to the datastore. |
| Read Rate (Kbps) | datastore.read.rate.average.kiloBytesPerSecond | Average number of kilobytes read by the link through which the host is connected to the datastore. |
| RW IOPS | datastore.numberRWAveraged.rate.average.number | Average number of read and write commands issued per second during the collection interval. |
| RW Rate (Kbps) | datastore.usage.rate.average.kiloBytesPerSecond | The aggregated input rates and output rates of the link through which the host is connected to the datastore. |
| Write IOPS | datastore.numberWriteAveraged.rate.average.number | Average number of write commands issued per second during the collection interval. |
| Write Latency (ms) | datastore.totalWriteLatency.absolute.average.millisecond | Average latency of each write operation issued by the link through which the host is connected to the datastore. |

| Metric Name | Metric API Name | Description |
|---|---|---|
| Write Rate (Kbps) | datastore.write.rate.average.kiloBytesPerSecond | Average number of kilobytes written by the link through which the host is connected to the datastore. |

## Memory Metrics for Hosts

These metrics provide information about memory use and allocation on the host.

| Metric Name | Metric API Name | Description | Supported by Databus |
|---|---|---|---|
| Memory Balloon | mem.vmmemctl.absolute.average.kiloBytes | The amount of memory currently claimed by the balloon driver.<br>This metric indicates that the host is taking memory from less needful VMs, and allocating it to those VMs where there is an increased memory requirement. | |
| Memory State (0-3) | mem.state.absolute.latest.number | Overall state of the memory.<br>The value is an integer between 0 (high) and 3 (low). | |
| Memory Swap Used | mem.swapused.absolute.average.kiloBytes | The amount of swap memory currently in use.<br>A large amount of swap memory does not indicate a performance problem. Check the swap rates to verify if the guest is actively in need of more memory than that is available. | |
| Memory Swap-In Rate (Kbps) | mem.swapinRate.rate.average.kiloBytesPerSecond | Rate at which the memory is swapped from disk to active memory during the time interval. | |
| Memory Swap-Out Rate (Kbps) | mem.swapoutRate.rate.average.kiloBytesPerSecond | Rate at which the memory is being swapped from active memory to the disk during the time interval. | |
| Memory Usage Rate (%) | mem.usage.absolute.average.percent | Percentage of memory used from the available machine memory. | Yes |
| Reserved Memory (KB) | mem.reservedCapacity.absolute.average.megaBytes | Amount of reserved memory by powered-on VMs and the memory used for running VMware vSphere services on the host. | |
| Overhead Memory (KB) | mem.overhead.absolute.average.kiloBytes | Memory overhead reported by the host.<br>Memory overhead is the total memory allocated to powered on VMs beyond its reserved memory and the memory used for running vSphere services on the host. | |
| Consumed Memory (KB) | mem.consumed.absolute.average.kiloBytes | Total memory consumed by the VMs and VMware vSphere services in the host. | |

| Metric Name | Metric API Name | Description | Supported by Databus |
|---|---|---|---|
| Active Memory (KB) | mem.active.absolute.average.kiloBytes | Total memory actively used by the host.<br>Total memory includes memory used by VMs and VMware vSphere services running on the host. The VMkernel estimates the total memory based on recently touched memory pages. | Yes |

## Disk Metrics for Hosts

These metrics provide information about disk use of the host.

| Metric Name | Metric API Name | Description |
|---|---|---|
| Average Read Latency (ms) | disk.deviceReadLatency.absolute.average.millisecond | Average latency of each read operation issued to complete a Small Computer System Interface (SCSI) command from the physical device. |
| Average Write Latency (ms) | disk.deviceWriteLatency.absolute.average.millisecond | Average latency of each write operation issued to complete a SCSI command from the physical device. |
| Disk Bus Resets | disk.busResets.delta.summation.number | Number of SCSI-bus reset commands issued during the collection interval. |
| Disk Commands Aborted | disk.commandsAborted.delta.summation.number | Number of SCSI commands aborted during the collection interval. |
| Disk Device Latency (ms) | disk.deviceLatency.absolute.average.millisecond | Average amount of time taken to complete a SCSI command from the physical device. |
| Disk Kernel Latency (ms) | disk.kernelLatency.absolute.average.millisecond | Average amount of time spent by the VMkernel processing each SCSI command. |

| Metric Name | Metric API Name | Description |
|---|---|---|
| Disk Queue Depth | disk.queueDepth.absolute.latest.number | Maximum queue depth during the collection interval. |
| Disk Queue Latency | disk.queueLatency.absolute.average.millisecond | Average amount of time spent in the VMkernel queue, per SCSI command, during the collection interval. |
| Disk Reads (KB) | disk.read.rate.average.kiloBytesPerSecond | Average number of reads through this link. |
| Disk Writes (KB) | disk.write.rate.average.kiloBytesPerSecond | Average number of writes through this link. |
| Read IOPS | disk.numberReadAveraged.rate.average.number | Average number of reads issued to complete a SCSI command through this link. |
| RW IOPS | disk.numberRWAveraged.rate.average.number | Average number of reads or writes issued to complete a SCSI command through this link. |
| Total Disk Latency (ms) | disk.totalLatency.absolute.average.millisecond | Average amount of time taken during the collection interval to process a SCSI command issued by the guest operating system to the VM. |
| Write IOPS | disk.numberWriteAveraged.rate.average.number | Average number of writes issued to complete a SCSI command through this link. |

## HCX Tunnel Metrics

The HCX tunnel metrics provide information about the network performance of HCX tunnels.

| Metric Name | Metric API Name | Decription |
|---|---|---|
| Dropped Rx Packets | net.droppedRx.delta.summation.number | Number of dropped Rx packets in the selected duration. |
| Dropped Tx Packets | net.droppedTx.delta.summation.number | Number of dropped Tx packets in the selected duration. |

| Metric Name | Metric API Name | Decription |
|---|---|---|
| Max Network Traffic Rate (kbps) | net.usage.rate.maximum.kiloBitsPerSecond | Maximum network usage rate on the HCX tunnel over any five minute interval in the selected duration. |
| Max Network Tx Rate (kbps) | net.transmitted.rate.maximum.kiloBitsPerSecond | Maximum data transmission rate on the HCX tunnel over any five minute interval in the selected duration. |
| Network Max Rx Rate (kbps) | net.received.rate.maximum.kiloBitsPerSecond | Maximum network data receive rate on the HCX tunnel over any five minute interval in the selected duration. |
| Network Packet Count | net.packetsTotal.delta.summation.number | Total number of packets transmitted and received during the time interval. |
| Network Rx Rate (kbps) | net.received.rate.average.kiloBitsPerSecond | Average rate at which the data is received during the time interval. This metric represents the bandwidth of the network. |
| Network Traffic Rate (kbps) | net.usage.rate.average.kiloBitsPerSecond | Average rate of received and transmitted traffic during the time interval. |
| Network Tx Rate (kbps) | net.transmitted.rate.average.kiloBitsPerSecond | Average rate at which the data is transmitted during the time interval. This metric represents the bandwidth of the network. |
| Rx Packet Errors | net.errorRx.delta.summation.number | Number of received packet errors during the time interval. |
| Rx Packet Per Second | net.ppsRx.rate.average.number | Number of packets received per second during the time interval. |
| Rx Packets | net.packetsRx.delta.summation.number | Average number of received packets during the time interval. |
| Total Network Traffic | net.usage.delta.summation.kiloBytes | Sum of data transmitted and received during the time interval. |
| Total Rx Bytes | net.totalRx.delta.summation.bytes | Sum of bytes received during the time interval. |
| Total Tx Bytes | net.totalTx.delta.summation.bytes | Sum of bytes transmitted during the time interval. |
| Tx Packet Errors | net.errorTx.delta.summation.number | Number of transmitted packet errors during the time interval. |
| Tx Packet Per Second | net.ppsTx.rate.average.number | Number of packets transmitted per second during the time interval. |
| Tx Packets | net.packetsTx.delta.summation.number | Average number of transmitted packets during the time interval. |

### Kubernetes Service Metrics

Kubernetes service metrics provide information about the number of pods and nodes deployed for a particular service.

| Metric Name | Metric API Name | Description |
|---|---|---|
| Nodes Count | sys.kubernetes.service.nodes.absolute.latest.number | Number of Kubernetes nodes deployed for a service in a five minute interval. |
| Pods Count | sys.kubernetes.service.pods.absolute.latest.number | Number of Kubernetes pods deployed for a service in a five minute interval. |

## Metrics for NSX Policy Manager

VCF Operations for networks collects many metrics for NSX policy manager including the firewall metrics.

| Metric Name | Metric API Name | Description |
|---|---|---|
| Distributed Firewall Application Rules Count | usage.distributedFirewallApplicationRuleCount.absolute.maximum.number | Number of distributed firewall rules of category application. |
| Distributed Firewall Emergency Rules Count | usage.distributedFirewallEmergencyRuleCount.absolute.maximum.number | Number of distributed firewall rules of category emergency. |
| Distributed Firewall Environment Rules Count | usage.distributedFirewallEnvironmentRuleCount.absolute.maximum.number | Number of distributed firewall rules of category environment. |
| Distributed Firewall Ethernet Rules Count | usage.distributedFirewallEthernetRuleCount.absolute.maximum.number | Number of distributed firewall rules of category Ethernet. |
| Distributed Firewall Infrastructure Rules Count | usage.distributedFirewallInfrastructureRuleCount.absolute.maximum.number | Number of distributed firewall rules of category infrastructure. |

## Metrics for NSX-T Dropped Packets

VCF Operations for networks collects flow metrics for NSX-T Dropped Packets.

| | | |
|---|---|---|
| Blocked Rx Packets | net.blockedRx.delta.summation.number | Number of blocked received packets due to standby interface. |
| Blocked Tx Packets | net.blockedTx.delta.summation.number | Number of blocked transmitted packets due to standby interface. |
| Duplicate Address Detected Dropped Rx Packets | net.dadDroppedRx.delta.summation.number | Total number of received detected duplicate address for blocked packets in the selected duration. |
| Duplicate Address Detected Dropped Tx Packets | net.dadDroppedTx.delta.summation.number | Total number of transmitted detected duplicate address in blocked packets for the selected duration. |
| Destination Unsupported Dropped Rx Packets | net.destUnsupportedDroppedRx.delta.summation.number | Number of received packets that were dropped during the collection interval. |
| Destination Unsupported Dropped Tx Packets | net.destUnsupportedDroppedTx.delta.summation.number | Number of transmitted packets that were dropped because the destination address (such as an IP address or MAC address) was unreachable or invalid. |
| Firewall Dropped Rx Packets | net.firewallDroppedRx.delta.summation.number | Number of received packets that were dropped by the firewall. |
| Firewall Dropped Tx Packets | net.firewallDroppedTx.delta.summation.number | Number of transmitted packets dropped due to non-firewall during the selected interval of router interface. |

| | | |
|---|---|---|
| Fragmentation Dropped Rx Packets | net.fragNeededDroppedRx.delta.summation.number | Number of received packets dropped due to fragmentation needed. |
| Fragmentation Dropped Tx Packets | net.fragNeededDroppedTx.delta.summation.number | Number of transmitted packets dropped due to fragmentation needed. |
| IPSec Dropped Rx Packets | net.ipsecDroppedRx.delta.summation.number | Number of received packets dropped by IPSec. |
| IPSec Dropped Tx Packets | net.ipsecDroppedTx.delta.summation.number | Number of transmitted packets dropped by IPSec. |
| IPSec Security Association Absence Dropped Rx Packets | net.ipsecNoSaDroppedRx.delta.summation.number | Number of received packets dropped due to IPSec no security association. |
| IPSec Security Association Absence Dropped Tx Packets | net.ipsecNoSaDroppedTx.delta.summation.number | Number of transmitted packets dropped due to IPSec no security association. |
| IPSec VTI Absence Dropped Rx Packets | net.ipsecNoVtiDroppedRx.delta.summation.number | Number of received packets dropped due to no VTI is associated with IPSec. |
| IPSec VTI Absence Dropped Tx Packets | net.ipsecNoVtiDroppedTx.delta.summation.number | Number of transmitted packets dropped due to no VTI is associated with IPSec. |
| IPSec Policy Block Dropped Rx Packets | net.ipsecPolicyBlockDroppedRx.delta.summation.number | Number of received packets dropped due to IPSec policy blocked. |
| IPSec Policy Block Dropped Tx Packets | net.ipsecPolicyBlockDroppedTx.delta.summation.number | Number of transmitted packets dropped due to IPSec policy blocked. |
| IPSec Policy Error Dropped Rx Packets | net.ipsecPolicyErrorDroppedRx.delta.summation.number | Number of received packets dropped due to IPSec policy error. |
| IPSec Policy Error Dropped Tx Packets | net.ipsecPolicyErrorDroppedTx.delta.summation.number | Number of transmitted packets dropped due to IPSec policy error. |
| IPV6 Dropped Rx Packets | net.IPV6DroppedRx.delta.summation.number | Number of received IPV6 packets dropped. |
| IPV6 Dropped Tx Packets | net.IPV6DroppedTx.delta.summation.number | Number of transmitted IPV6 packets dropped. |
| Kernel Nic Interface Dropped Rx Packets | net.KNIDroppedRx.delta.summation.number | Number of received packets dropped by DPDK kernal NIC interface. |
| Kernel Nic Interface Dropped Tx Packets | net.KNIDroppedTx.delta.summation.number | Number of transmitted packets dropped by DPDK kernal NIC interface. |
| Unsupported L4 Port Dropped Rx Packets | net.L4PortUnsupportedDroppedRx.delta.summation.number | Number of received packets dropped due to unsupported L4 port. |
| Unsupported L4 Port Dropped Tx Packets | net.L4PortUnsupportedDroppedTx.delta.summation.number | Number of transmitted packets dropped due to unsupported L4 port. |
| Malformed Dropped Rx Packets | net.malformedDroppedRx.delta.summation.number | Number of received packets dropped due to malformed. |

| | | |
|---|---|---|
| Malformed Dropped Tx Packets | net.malformedDroppedTx.delta.summation.number | Number of transmitted packets dropped due to malformed. |
| ARP Absence Dropped Rx Packets | net.noARPDroppedRx.delta.summation.number | Number of received packets dropped due to the absence of ARP. |
| ARP Absence Dropped Tx Packets | net.noARPDroppedTx.delta.summation.number | Number of transmitted packets dropped due to the absence of ARP. |
| Linked Port Absence Dropped Rx Packets | net.noLinkedDroppedRx.delta.summation.number | Number of received packets dropped due to the absence of linked port. |
| Linked Port Absence Dropped Tx Packets | net.noLinkedDroppedTx.delta.summation.number | Number of transmitted packets dropped due to the absence of linked port. |
| Insufficient Memory Dropped Rx Packets | net.noMemoryDroppedRx.delta.summation.number | Number of received packets dropped due to insufficient memory. |
| Insufficient Memory Dropped Tx Packets | net.noMemoryDroppedTx.delta.summation.number | Number of transmitted packets dropped due to insufficient memory. |
| Receiver Absence Dropped Rx Packets | net.noReceiverDroppedRx.delta.summation.number | Number of received packets dropped due to absence of receiver. |
| Receiver Absence Dropped Tx Packets | net.noReceiverDroppedTx.delta.summation.number | Number of transmitted packets dropped due to absence of receiver. |
| Route Absence Dropped Rx Packets | net.noRouteDroppedRx.delta.summation.number | Number of received packets dropped due to absence of route. |
| Route Absence Dropped Tx Packets | net.noRouteDroppedTx.delta.summation.number | Number of transmitted packets dropped due to absence of route. |
| Non IP Dropped Rx Packets | net.nonIPDroppedRx.delta.summation.number | Number of non IP received packets dropped. |
| Non IP Dropped Tx Packets | net.nonIPDroppedTx.delta.summation.number | Number of non IP transmitted packets dropped. |
| Unsupported Protocol Dropped Rx Packets | net.protoUnsupportedDroppedRx.delta.summation.number | Number of received packets dropped due to unsupported protocol. |
| Unsupported Protocol Dropped Tx Packets | net.protoUnsupportedDroppedTx.delta.summation.number | Number of transmitted packets dropped due to unsupported protocol. |
| Redirect Dropped Rx Packets | net.redirectDroppedRx.delta.summation.number | Number of redirect received packets dropped. |
| Redirect Dropped Tx Packets | net.redirectDroppedTx.delta.summation.number | Number of redirect transmitted packets dropped. |
| Service Insert Dropped Rx Packets | net.serviceInsertDroppedRx.delta.summation.number | Number of service insert received packets dropped. |
| Service Insert Dropped Tx Packets | net.serviceInsertDroppedTx.delta.summation.number | Number of service insert transmitted packets dropped. |
| TTL Exceeded Dropped Rx Packets | net.ttlExceededDroppedRx.delta.summation.number | Number of time to live exceeded received packets dropped. |

| | | |
|---|---|---|
| TTL Exceeded Dropped Tx Packets | net.ttlExceededDroppedTx.delta.summation.number | Number of time to live exceeded transmitted packets dropped. |
| Non Firewall Dropped Rx Packets | net.nonFirewallDroppedRx.delta.summation.number | Number of received packets dropped due to non-firewall during the selected interval of router interface. |
| Non Firewall Dropped Tx Packets | net.nonFirewallDroppedTx.delta.summation.number | Number of transmitted packets dropped due to non-firewall during the selected interval of router interface |
| Non Firewall Tx Packet Drop Percent | net.nonFirewallDroppedRx.ratio.average.percent | Average ratio of transmitted packet drops due to non firewall to the total transmitted packets. |
| Firewall Rx Packet Drop Percent | net.firewallDroppedRx.ratio.average.percent | Average ratio of received packet drops due to firewall to the total received packets. |
| Max Firewall Rx Packet Drop Percent | net.firewallDroppedRx.ratio.maximum.percent | Percentage of maximum received packets dropped due to firewall to the total number of received packets over 5 minutes interval in the selected duration. |
| Max Non Firewall Rx Packet Drop Percent | net.nonFirewallDroppedRx.ratio.maximum.percent | Percentage of maximum received packets dropped due to non firewall to the total number of received packets over 5 minutes interval in the selected duration. |
| Firewall Tx Packet Drop Percent | net.firewallDroppedTx.ratio.average.percent | Average ratio of transmitted packet drops due to firewall to the total transmitted packets. |
| Non Firewall Tx Packet Drop Percent | net.nonFirewallDroppedTx.ratio.average.percent | Average ratio of transmitted packet drops due to non firewall to the total transmitted packets. |
| Max Firewall Tx Packet Drop Percent | net.firewallDroppedTx.ratio.maximum.percent | Percentage of maximum transmitted packets dropped due to firewall to the total number of transmitted packets in the selected duration. |
| Max Non Firewall Tx Packet Drop Percent | net.nonFirewallDroppedTx.ratio.maximum.percent | Percentage of maximum transmitted packets dropped due to non firewall to the total number of transmitted packets in the selected duration. |

## Metrics for Avi Load Balancer

VCF Operations for networks collects many metrics for Avi Load Balancer, including controller metrics, pool metrics, pool member metrics, service engine metrics, and virtual service metrics.

## Controller Metrics

| Metric Name | Metric API Name | Description |
|---|---|---|
| CPU Usage Rate (%) | cpu.usage.rate.average.percent | Average percentage of the CPU used out of its total capacity.<br>**Note:**<br>This metric provides the host's view of the CPU usage and not the guest operating system view.<br>It is the average CPU utilization over all available virtual CPUs in the virtual machine. |
| Disk Usage Rate | disk.used.absolute.latest.percent | Average percentage of the disk used out of its total capacity. |
| Memory Usage Rate | mem.usage.absolute.average.percent | Average percentage of the memory used out of the total configured memory. |

## Pool and Pool Member Metrics

| Metric Name | Metric API Name | Description |
|---|---|---|
| Current Connections | net.lbPoolCurrentConnections.absolute.latest.number | Number of current connections processed by this pool during the time interval. |
| Health Score | nsxalb.healthScore.absolute.latest.number | The score indicating the health of the pool. |
| Lossy Connections | nsxalb.lossyConnections.delta.summation.number | Number of lossy connections during the time interval between the service engine and the server. |
| Network Packet Count | net.packetsTotal.delta.summation.number | Total number of received and transmitted packets processed by this pool during the time interval. |
| Network Rx Rate (kbps) | net.received.rate.average.kiloBitsPerSecond | Average rate of received packets processed by this pool during the time interval. |
| Network Traffic Rate (kbps) | net.usage.rate.average.kiloBitsPerSecond | Average rate of received and transmitted packets processed by this pool during the time interval. |
| Network Tx Rate (kbps) | net.transmitted.rate.average.kiloBitsPerSecond | Average rate of transmitted packets processed by this pool during the time interval. |
| Rx Packets | net.packetsRx.delta.summation.number | Total number of received packets processed by this pool during the time interval. |
| Server Responses | nsxalb.l7ServerResponse.delta.summation.number | Total number of server HTTP responses handled by the pool during the time interval. |
| Tx Packets | net.packetsTx.delta.summation.number | Total number of transmitted packets processed by this pool during the time interval. |

## Service Engine Metrics

| Metric Name | Metric API Name | Description |
|---|---|---|
| CPU Usage Rate (%) | cpu.usage.rate.average.percent | Average percentage of the CPU used out of its total capacity.<br>**Note:**<br>This metric provides the host's view of the CPU usage and not the guest operating system view.<br>It is the average CPU utilization over all available virtual CPUs in the virtual machine. |
| Dropped Rx Packets | net.droppedRx.delta.summation.number | Average number of received packets in Avi Load Balancer virtual services and Avi Load Balancer non-virtual services dropped during the selected duration. |
| Health Score | nsxalb.healthScore.absolute.latest.number | The score indicating the health of the service engine. |
| Memory Usage Rate (%) | mem.usage.absolute.average.percent | Average percentage of the memory used out of the total configured memory. |
| Network Packet Count | net.packetsTotal.delta.summation.number | Total number of received and transmitted packets processed by all the vNICs of the service engine (VM) during the selected duration. |
| Network Rx Rate (kbps) | net.received.rate.average.kiloBitsPerSecond | Average rate of received packets across all the VNICs of the service engine. |
| Network Traffic Rate (kbps) | net.usage.rate.average.kiloBytesPerSecond | Average rate of received and transmitted packets across all the VNICs of the service engine. |
| Network Tx Rate (kbps) | net.transmitted.rate.average.kiloBitsPerSecond | Average rate of transmitted packets across all the VNICs of the service engine. |
| Network Tx Rate (kbps) | net.transmitted.rate.average.kiloBytesPerSecond | Average rate of transmitted packets across all the VNICs of the service engine. |
| Rx Packets | net.packetsRx.delta.summation.number | Number of packets received by all the vNICs of the service engine (VM) during the selected duration. |
| Session Count | net.connection.delta.summation.number | Number of connections to the service engine during the time interval.<br>The number of connections include the dropped connections and the connections lost due to policy drops. |
| Tx Packets | net.packetsTx.delta.summation.number | Number of packets transmitted by all the vNICs of the service engine during the selected duration. |

**Virtual Service Metrics**

| Metric Name | Metric API Name | Description |
|---|---|---|
| Client Responses | nsxalb.l7ClientResponses.delta.summation.number | Number of HTTP responses sent to clients during the time interval. |
| Current Connections | net.lbVirtSvrCurrentConnections.absolute.latest.number | Number of connections completed during the time interval. |
| Health Score | nsxalb.healthScore.absolute.latest.number | The score indicating the health of the virtual service. |
| Network Packet Count | net.packetsTotal.delta.summation.number | Total number of received and transmitted packets during the time interval. |
| Network Rx Rate (kbps) | net.received.rate.average.kiloBitsPerSecond | Average rate of received packets during the time interval. |
| Network Traffic Rate (kbps) | net.usage.rate.average.kiloBitsPerSecond | Average rate of received and transmitted packets during the time interval. |
| Network Tx Rate (kbps) | net.transmitted.rate.average.kiloBitsPerSecond | Average rate of transmitted packets during the time interval. |
| New Connections | net.newConnections.delta.summation.number | Number of new connections established during the time interval. |
| Rx Packets | net.packetsRx.delta.summation.number | Number of received packets during the time interval. |
| SSL Connections | nsxalb.sslConnections.delta.summation.number | Number of new SSL transactions during the time interval excluding session reuse and errored connection attempts. |
| Tx Packets | net.packetsTx.delta.summation.number | Number of transmitted packets during the time interval. |

## Port Channel Metrics

The port channel metrics provide information about the network performance of the port channel.

| Metric Name | Metric API Name | Description |
|---|---|---|
| Discarded Rx Packets | net.discardedRx.delta.summation.number | Number of discarded inbound packets. |
| Discarded Tx Packets | net.discardedTx.delta.summation.number | Number of discarded outbound packets. |
| Dropped Rx Packets | net.droppedRx.delta.summation.number | Number of received packets that were dropped during the collection interval. |
| Dropped Tx Packets | net.droppedTx.delta.summation.number | Number of transmitted packets that were dropped during the collection interval. |
| FCS Errors | net.errorFcs.delta.summation.number | Number of frames transmitted or received with a bad checksum (CRC value) in the Ethernet frame.<br>Frame check sequence errors indicate the frames of data corrupted during transmission. These frames are dropped and not propagated to other ports. |
| Giant Errors | net.errorGiants.delta.summation.number | Number of giant errors over the network during the selected duration. |
| Interface Rx Utilization (%) | net.rx.utilization.absolute.latest.percent | Interface utilization percentage for the received traffic. |

| Metric Name | Metric API Name | Description |
|---|---|---|
| Interface Tx Utilization (%) | net.tx.utilization.absolute.latest.percent | Interface utilization percentage for the transmitted traffic. |
| Interface Utilization (%) | net.utilization.absolute.latest.percent | Interface utilization percentage for received and transmitted traffic. |
| Max Network Traffic Rate (kbps) | net.usage.rate.maximum.kiloBitsPerSecond | Maximum network usage rate over any five minute interval in the selected duration. |
| Max Network Tx Rate (kbps) | net.transmitted.rate.maximum.kiloBitsPerSecond | Maximum network data transmit rate over any five minute interval in the selected duration. |
| Max Packet Drop Ratio (%) | net.packetDrop.ratio.maximum.percent | Maximum five minute interval ratio of total dropped packets to total number packets. |
| Max Packet Drops | net.droppedTotal.delta.maximum.number | Maximum number of dropped Rx and Tx packets over any five minute interval in the selected duration. |
| Multicast Rx Packets | net.multicastRx.delta.summation.number | Number of multicast packets received in the selected duration. |
| Multicast Tx Packets | net.multicastTx.delta.summation.number | Number of multicast packets transmitted in the selected duration. |
| Network Align Errors | net.errorAlign.delta.summation.number | Number of alignment errors detected over the network in the selected duration.<br>Network alignment errors occur if a frame arrives with part of an octet missing, and having an FCS error. |
| Network Carrier Sense Errors | net.errorCarrierSense.delta.summation.number | Number of carrier sense errors detected over the network in the selected duration.<br>Network carrier sense errors occur when an interface cannot transmit a frame because it did not detect a carrier. |
| Network Deferred Tx Errors | net.errorDeferredTx.delta.summation.number | Number of deferred transmission errors detected over the network in the selected duration. |
| Network Excessive Collision Errors | net.errorExcessiveCollisions.delta.summation.number | Number of excessive collision errors detected over the network in the selected duration.<br>Network excessive collision error occurs when a station receives 16 consecutive collisions while attempting to transmit a single frame. In such situations, the frame is dropped due to the excessive collisions. |
| Network Late Collision Errors | net.errorLateCollisions.delta.summation.number | Number of late collision errors detected over the network in the selected duration. |
| Network Long Frame Errors | net.errorFrameTooLongs.delta.summation.number | Number of network long frame errors detected over the network in the selected duration. |
| Network Max Rx Rate | net.received.rate.maximum.kiloBitsPerSecond | Maximum rate at which the data is received during the time interval. |
| Network Multi Collision Errors | net.errorMultiCollisionFrames.delta.summation.number | Number of multi collision errors detected over the network in the selected duration. |
| Network Packet Count | net.packetsTotal.delta.summation.number | Total number of packets transmitted and received during the time interval. |
| Network Packet Drop Percent | net.packetDrop.ratio.average.percent | Average ratio of total number of dropped packets to total number of packets. |
| Network Packet Drops | net.droppedTotal.delta.summation.number | Total number of packets dropped during the time interval. |

| Metric Name | Metric API Name | Description |
|---|---|---|
| Network Receive Errors | net.errorRcv.delta.summation.number | Number of receive errors received on the physical interface in the selected duration.<br>Receive errors occur when there is excessive output rates of traffic, or if the receive buffer is full. |
| Network Rx Int Mac Errors | net.errorRxIntMac.delta.summation.number | Number of frames for which reception on a particular interface has failed due to an internal MAC sublayer receive error. |
| Network Rx Rate | net.received.rate.average.kiloBitsPerSecond | Average rate at which data is received during the time interval. |
| Network Single Collision Errors | net.errorSingleCollisionFrames.delta.summation.number | Number of single collision errors detected over the network in the selected duration. |
| Network SQE Test Errors | net.errorSQETests.delta.summation.number | Number of signal quality test errors detected over the network in the selected duration. |
| Network Symbol Errors | net.errorSymbol.delta.summation.number | Number of symbol errors detected over the network in the selected duration. |
| Network Traffic Rate | net.usage.rate.average.kiloBitsPerSecond | Average rate of received and transmitted traffic during the time interval. |
| Network Tx Intmac Errors | net.errorTxIntMac.delta.summation.number | Number of frames for which transmission on a particular interface fails due to an internal MAC sublayer transmit error. |
| Network Tx Rate | net.transmitted.rate.average.kiloBitsPerSecond | Average rate of transmitted traffic during the time interval. |
| Network Undersize Errors | net.errorUnderSize.delta.summation.number | Number of undersize errors detected over the network in the selected duration.<br>This error indicates a bad frame generated by the connected device. |
| Network Xmit Errors | net.errorXmit.delta.summation.number | Number of Xmit errors detected over the network in the selected duration.<br>Xmit errors occur when there is excess input rates of traffic, or if the transmit buffer is full.<br>Increase speed and move the link partner to full-duplex to minimize the occurance of this error. |
| Out Queue Length | net.outQLen.absolute.average.number | Number of output packets waiting in the queue. |
| Rx Broadcast Packets | net.broadcastRx.delta.summation.number | Number of broadcast packets received during the time interval. |
| Rx Broadcast Ratio (%) | net.broadcastByRx.ratio.average.percent | Percentage of broadcast packets received during the time interval. |
| Rx Jumbo Packets | net.jumboRx.delta.summation.number | Number of jumbo packets received during the time interval. |
| Rx Packet Drop Percent (%) | net.packetDropRx.ratio.average.percent | Average ratio of total number of dropped Rx packets to total number of Rx packets. |

| Metric Name | Metric API Name | Description |
|---|---|---|
| Rx Packet Errors | net.errorRx.delta.summation.number | Number of received packet errors detected over the network in the selected duration.<br>For packet-oriented interfaces, Rx packet error indicates the number of inbound packets that contained errors preventing them from being deliverable to a higher-layer protocol.<br>For character-oriented or fixed-length interfaces, Rx packet error indicates the number of inbound transmission units that contained errors preventing them from being deliverable to a higher-layer protocol. |
| Rx Packets | net.packetsRx.delta.summation.number | Number of packets received during the interval. |
| Total Network Traffic | net.usage.delta.summation.kiloBytes | Sum of data transmitted and received during the time interval. |
| Tx Broadcast Packets | net.broadcastTx.delta.summation.number | Number of broadcast packets transmitted during a specified time interval. |
| Tx Broadcast Ratio | net.broadcastByTx.ratio.average.percent | Average percent of broadcast packets transmitted during a specified time interval. |
| Tx Jumbo Packets | net.jumboTx.delta.summation.number | Number of jumbo packets transmitted during the time interval. |
| Tx Packet Drop Percent | net.packetDropTx.ratio.average.percent | Ratio of total number of dropped Tx packets to total number of Tx packets. |
| Tx Packet Errors | net.errorTx.delta.summation.number | Number of transmitted packet errors detected over the network in the selected duration.<br>For packet-oriented interfaces, Tx packet errors indicate the number of outbound packets that were not transmitted because of errors.<br>For character-oriented or fixed-length interfaces, Tx packet errors indicate the number of outbound transmission units that were not transmitted because of errors. |
| Tx Packets | net.packetsTx.delta.summation.number | Number of packets transmitted during the time interval. |
| Unicast Rx Packets | net.unicastRx.delta.summation.number | Total number of unicast packets received during the time interval. |
| Unicast Tx Packets | net.unicastTx.delta.summation.number | Total number of unicast packets transmitted during the time interval. |
| Unknown Rx Packets | net.unknownRx.delta.summation.number | Total number of unknown packets received during the time interval.<br>For packet-oriented interfaces, unknown Rx packets indicate the number of packets received through the interface that were discarded because of an unknown protocol or an unsupported protocol.<br>For character-oriented or fixed-length interfaces that support protocol multiplexing, unknown Rx packets indicate the number of transmission units received through the interface that were discarded because of an unknown protocol or an unsupported protocol. |

## NetFlow Reporter Metrics

NetFlow reporter metrics provide information about the flow records dropped at the VCF Operations for networks collector.

| Metric Name | Metric API Name | Description |
|---|---|---|
| Outbound expire flow drop count | flow.fiveTupleExpireOutbound.delta.summation.number | Number of five tuples dropped due to outbound limit hit for Source IP addresses in Flows. |
| Wrong collector flow drop count | flow.rawRecordReporterWrongCollector.delta.summation.number | Number of raw flow records dropped as these were reported on the wrong type of VCF Operations for networks collector. |

## Metrics for NSX-T Entities

When you add a VMware NSX Manager as a datasource, VCF Operations for networks collects metrics from NSX entities that are associated with the deployment to troubleshoot problems in your environment.

### Metrics for NSX-T Firewall Rule

NSX firewall rules is a set of customizable rules that protect the system against network threats.

The following metrics provide information about the traffic processed by NSX firewall rules:

| Metric Name | Metric API Name | Description |
|---|---|---|
| Flow Packets | net.packet.delta.summation.number | Number of packets processed by the NSX firewall rule. |
| Hit Count | net.hit.delta.summation.number | Number of hits received by the NSX firewall rule. |
| Session Count | net.connection.delta.summation.number | Number of sessions processed by the NSX firewall rule. |
| Total Bytes | net.byte.delta.summation.number | Number of bytes processed by the NSX firewall rule. |

### Metrics for NSX-T Edge Node

NSX edge node metrics provide information about the usage of NSX Edge nodes.

Starting VCF 9.1, NSX-T Edge monitoring metrics are not available on the VCF Operations for networks UI. The metrics are available on the NSX-T Edge dashboards in **Operate** > **VCF Health** in VCF Operations. For more information, see NSX Health in VCF Operations Health and Diagnostics.

**Table 1346: General Metrics**

| Metric Display Name | Metric API Name | Export Frequency | Description |
|---|---|---|---|
| Admin State | adminState | 5 minutes | Administrative state of the edge cluster, which can be one of the following: Up, Down |
| BFD Tunnel Down Count | net.bfd.tunnel.down.count.absolute.latest.number | 5 minutes | Total number of Bidirectional Forwarding Detection (BFD) tunnels in down state. |
| BFD Tunnel Up Count | net.bfd.tunnel.up.count.absolute.latest.number | 5 minutes | Total number of Bidirectional Forwarding Detection (BFD) tunnels in Up state. |

| Metric Display Name | Metric API Name | Export Frequency | Description |
|---|---|---|---|
| Controller Connectivity | controlConnectionStatus | 5 minutes | Status of the control connection between edge node and controller node, which can be one of the following: Up, Down |
| CPU Cores | nCPUs | 5 minutes | The number of CPU cores in the edge node. |
| CPU Usage | sys.loadFifteenMinutes.rate.latest.percent | 5 minutes | Percentage of CPU usage in the edge node. |
| DataPath IPC Thread Status | edgeDPThreadStatus | 5 minutes | The datapath thread status, which can be one of the following: Healthy, Unhealthy |
| Disk Used | disk.node.usage.absolute.latest.kiloBytes | 5 minutes | Kilobytes of disk space used across all partitions. |
| DP Memory Usage Rate | mem.edgeDP.usage.absolute.average.percent | 20 seconds | Percentage of RAM used by the edge datapath. |
| Edge Agent Status | edgeAgentStatus | 5 minutes | Status of the Edge Agent, which can be one of the following: Up, Down |
| Edge OPS Agent Status | nsxOpsAgentStatus | 5 minutes | Status of the Ops Agent, which can be one of the following: Up, Down |
| Fifteen Minute Load Average | sys.loadFifteenMinutes.absolute.maximum.number | 20 seconds | The average system load over the past 15 minutes, as returned by the getloadavg() API method. System load is defined as the number of processes in the system run queue. |
| Five Minute Load Average | sys.loadFiveMinutes.absolute.maximum.number | 20 seconds | The average system load over the past 5 minutes, as returned by the getloadavg() API method. System load is defined as the number of processes in the system run queue. |
| HA State | nodeRole | 5 minutes | The High Availability (HA) state of the edge node in the edge cluster, which can be one of the following: Disabled, Offline, Discover, StateSync, Inactive, Active. "Active" represents the healthy state. |
| Maintenance Mode | maintenanceMode | 5 minutes | Status of the edge maintenance mode. |
| Management Connection Status | mgmtConnectionStatus | 5 minutes | Status of the management connection, which can be one of the following: Up, Down |
| Memory Cache | mem.cache.absolute.latest.kiloBytes | 20 seconds | Number of bytes used by the edge memory cache. |
| Memory Usage | mem.usage.absolute.latest.kiloBytes | 20 seconds | Total amount of usable RAM in bytes, defined as physical RAM minus a number of reserved bits and the kernel binary code. |
| Memory Usage Rate | mem.usage.absolute.average.percent | 20 seconds | Percentage of memory in use over total memory, per process. |
| One Minute Load Average | sys.loadOneMinute.absolute.maximum.number | 20 seconds | The average system load over the past 1 minute, as returned by the getloadavg() API method. System load is defined as the number of processes in the system run queue. |

| Metric Display Name | Metric API Name | Export Frequency | Description |
|---|---|---|---|
| PNIC / Bond Status | pnicStatus | 5 minutes | The pNIC status, which can be one of the following: Up, Down, Degraded, Unknown |
| PNIC Down Count | net.pnic.down.count.absolute.latest.number | 5 minutes | The pNIC down count. |
| PNIC Up Count | net.pnic.up.count.absolute.latest.number | 5 minutes | The number of physical NICs (pNICs) that are up. |
| Service Status | serviceStatus | 5 minutes | Collection of the following statuses: Datapath Config Channel, Datapath Status Channel, Routing Status Channel, Routing Status |
| Status | overallStatus | 5 minutes | The roll-up status of the pNIC, management connection, control connection, tunnel, and agent statuses, which can be one of the following: Up, Down, Degraded, Unknown |
| Swap Memory Usage | mem.swapused.absolute.latest.kiloBytes | 20 seconds | The amount of swap memory in use. |
| System Uptime | sys.uptime.absolute.latest.millisecond | 5 minutes | Number of milliseconds that a system has been running continuously without interruption or reboot. |
| Tier0 Routing Status | tier0RoutingStatus | 5 minutes | The overall routing status. |
| Total Disk Space | totalDiskSpace | 5 minutes | The total disk space for all partitions allocated by the edge node. |
| Total Memory | totalMemory | 20 seconds | Total amount of RAM used. |
| Total Swap | totalSwapMemory | 20 seconds | The total size of swap memory. |
| Tunnel Status | tunnelStatus | 5 minutes | The tunnel status, which can be one of the following: Up, Down, Degraded, Unknown |
| VTEP State | vtepState | 5 minutes | The VTEP state of the edge cluster, which can be one of the following: Up, Down |

**Table 1347: GPU Core Metrics**

| Metric Display Name | Metric API Name | Export Frequency | Description |
|---|---|---|---|
| Core CPU Usage | net.edge.service.cpuCore.usage.rate.average.percent | 5 minutes | The percentage of time spent by all processes, per non-dataplane service core. |
| Edge Dataplane CPU Core Mega Flow Cache Hit Rate Percent | edge.dataplane.cpu.core.megaflowcache.hit.rate.average.percent | 5 minutes | Percentage of hits in the mega flow cache, per datapath CPU core. |
| Edge Dataplane CPU Core Micro Flow Cache Hit Rate Percent | edge.dataplane.cpu.core.microflowcache.hit.rate.average.percent | 5 minutes | Percentage of hits in the micro flow cache, per datapath CPU core. |
| Edge Dataplane CPU Core Usage Rate (%) | edge.dataplane.cpu.core.usage.rate.average.percent | 20 seconds | Percentage of utilized cycles per datapath CPU core. |
| Edge Dataplane CPU Core Usage Rate Rx | edge.dataplane.cpu.core.ppsRx.usage.rate.average.number | 5 minutes | Number of datapath-received packets per second, per datapath CPU core. |
| Edge Dataplane CPU Core Usage Rate Tx | edge.dataplane.cpu.core.ppsTx.usage.rate.average.number | 5 minutes | Number of datapath-transmitted packets per second, per datapath CPU core. |

| Metric Display Name | Metric API Name | Export Frequency | Description |
|---|---|---|---|
| Mega Flow Cache Entry Added Count | net.edgeDP.cpuCore.megaflowCache.insertions.absolute.average.number | 5 minutes | Average number of new entires added to the mega flow cache, per datapath CPU core. |
| Mega Flow Cache Hits | net.edgeDP.cpuCore.megaflowCache.hit.absolute.average.number | 5 minutes | Number of packets forwarded through the mega flow cache, per datapath CPU core. |
| Mega Flow Cache Max Entries Seen | net.edgeDP.cpuCore.megaflowCache.maxSeen.rate.average.percent | 5 minutes | Percentage of maximum number of entries seen in the mega flow cache, per datapath CPU core. |
| Mega Flow Cache Missed Packets | net.edgeDP.cpuCore.megaflowCache.dont.cache.absolute.latest.number | 5 minutes | Number of packets that cannot be cached in the mega flow cache, per datapath CPU core. |
| Mega Flow Cache Packets Passed to Next Layer | net.edgeDP.cpuCore.megaflowCache.missed.absolute.maximum.number | 5 minutes | Maximum number of packets passed to the next layer due to cache miss in the mega flow cache, per datapath CPU core. |
| Mega Flow Cache Skipped Packets | net.edgeDP.cpuCore.megaflowCache.skipped.absolute.latest.number | 5 minutes | Number of packets skipped by mega flow cache due to unsupported type or disabled cache, per datapath CPU core. |
| Micro Flow Cache Entry added Count | net.edgeDP.cpuCore.microflowCache.insertions.absolute.average.number | 5 minutes | Average number of new entries added to the micro flow cache, per datapath CPU core. |
| Micro Flow Cache Hits | net.edgeDP.cpuCore.microflowCache.hit.absolute.average.number | 5 minutes | Number of packets forwarded through the micro flow cache, per datapath CPU core. |
| Micro Flow Cache Max Entries Seen | net.edgeDP.cpuCore.microflowCache.maxSeen.rate.average.percent | 5 minutes | Percentage of max entries seen in the micro flow cache, per datapath CPU core. |
| Micro Flow Cache Missed Packets | net.edgeDP.cpuCore.microflowCache.dont.cache.absolute.latest.number | 5 minutes | Number of packets that cannot be cached in the micro flow cache, per datapath CPU core. |
| Micro Flow Cache Packets Passed to Next Layer | net.edgeDP.cpuCore.microflowCache.missed.absolute.maximum.number | 5 minutes | Number of packets passed to the next layer due to cache miss in the micro flow cache, per datapath CPU core. |
| Micro Flow Cache Skipped Packets | net.edgeDP.cpuCore.microflowCache.skipped.absolute.latest.number | 5 minutes | Number of packets skipped due to unsupported type or disabled cache, per datapath CPU core. |
| Packets Processed | net.edgeDP.cpuCore.packetsProcessedPerCore.absolute.latest.number | 5 minutes | Number of packets processed per second, per datapath CPU core, from NICs, KNIs, and other cores. |
| Packets Queued Per Port Per Queue | net.edgeDP.cpuCore.packetsPerPortPerQueue.absolute.latest.number | 5 minutes | Number of packets queued per datapath CPU core, per port, per queue. |

### Table 1348: Edge File System Metrics and Memory Pool Metrics

| Metric Display Name | Metric API Name | Export Frequency | Description |
|---|---|---|---|
| Disk Available Percentage | disk.available.rate.latest.percent | 5 minutes | Percentage of total disk space on the edge node that is currently unused and available for storage. |
| Disk Read Time | disk.read.wait.time.absolute.average.millisecond | 5 minutes | The average time in milliseconds that read requests wait on the block device per partition. |

| Metric Display Name | Metric API Name | Export Frequency | Description |
|---|---|---|---|
| Disk Write Time | disk.write.wait.time.absolute.average.millisecond | 5 minutes | The average time in milliseconds that write requests wait on the block device per partition. |
| IO Mode | fileSystemsIOMode | 5 minutes | Storage mode per partition. Possible storage modes are read, write, or both read/write. |
| Used | disk.usage.absolute.latest.kiloBytes | 5 minutes | Total kilobytes used by file system, per partition. |
| Datapath Memory Pool Usage Percent | mem.pool.usage.absolute.latest.percent | 20 seconds | Percentage of memory allocated to the datapath that is currently being used to store and process network packets. |

**Table 1349: Management Kernel Metrics**

| Metric Display Name | Metric API Name | Export Frequency | Description |
|---|---|---|---|
| Dropped Rx Packets | net.droppedRx.delta.summation.number | 5 minutes | Number of received packets dropped during the sampling interval. |
| Dropped Tx Packets | net.droppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped during the sampling interval. |
| Rx Packet Drop Rate | net.packetDropReceived.ratio.average.percent | 5 minutes | Average ratio of dropped received packets to total received packets. |
| Rx Packet Errors | net.errorRx.delta.summation.number | 5 minutes | For packet-oriented interfaces, the number of inbound packets that contained errors preventing their delivery to a higher-layer protocol. For character-oriented or fixed-length interfaces, the number of inbound transmission units that contained errors preventing them from delivery to a higher-layer protocol. |
| Rx Packet Per Second | net.ppsRx.rate.average.number | 5 minutes | Average rate of packets received per second. |
| Rx Packets | net.packetsRx.delta.summation.number | 5 minutes | Number of packets received during the sampling interval. |
| Total Packet Per Second | net.ppsTotal.rate.average.number | 5 minutes | Average rate of total packets (received + transmitted) per second. |
| Total Rx Bytes | net.totalRx.delta.summation.bytes | 5 minutes | Total number of received bytes. |
| Total Tx Bytes | net.totalTx.delta.summation.bytes | 5 minutes | Total number of transmitted bytes. |
| Tx Packet Drop Rate | net.packetDropTransmitted.ratio.average.percent | 5 minutes | Average ratio of dropped transmitted packets to total transmitted packets. |

| Metric Display Name | Metric API Name | Export Frequency | Description |
|---|---|---|---|
| Tx Packet Errors | net.errorTx.delta.summation.number | 5 minutes | For packet-oriented interfaces, the number of outbound packets that could not be transmitted due to errors. For character-oriented or fixed-length interfaces, the number of outbound transmission units that could not be transmitted due to errors. |
| Tx Packet Per Second | net.ppsTx.rate.average.number | 5 minutes | Average rate of packets transmitted per second. |
| Tx Packets | net.packetsTx.delta.summation.number | 5 minutes | Number of packets transmitted by all the virtual network interface controllers (vNICs) of the VM during the sampling interval. |

**Table 1350: Node Process Metrics**

| Metric Display Name | Metric API Name | Export Frequency | Description |
|---|---|---|---|
| CPU Usage Rate | cpu.usage.rate.average.percent | 20 seconds | Percentage of process CPU time over total system CPU time. The percentage could exceed 100 when you have multiple CPU cores. This metric is collected per process. |
| Memory Usage Rate | mem.usage.absolute.average.percent | 20 seconds | Percentage of memory in use over total memory, per process. |

**Table 1351: Physical Interface Metrics**

| Metric Display Name | Metric API Name | Export Frequency | Description |
|---|---|---|---|
| Dropped Tx Packets | net.droppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped during the sampling interval on the physical interface. |
| Lookup Failure Rx Drop PPS | net.lookUpFailureRx.delta.summation.number | 5 minutes | Number of received packet drops per second due to lookup failures on the datapath physical interface. |
| MBUF Packet Alloc Failures | net.rxNoMbuf.delta.summation.number | 5 minutes | The number of received packet drops per second on the atapath physical interface caused by Memory Buffer (MBUF) allocation failures. |
| Rx Misses | net.rxMisses.delta.summation.number | 5 minutes | Total number of received packet misses during the sampling interval, per datapath physical interface. This metric is available only for NSX Edge transport node interfaces. |
| Rx Packet Drop Rate | net.packetDropReceived.ratio.average.percent | 5 minutes | Average ratio of dropped received packets to total received packets. |

| Metric Display Name | Metric API Name | Export Frequency | Description |
|---|---|---|---|
| Rx Packet Errors | net.errorRx.delta.summation.number | 5 minutes | For packet-oriented interfaces, the number of inbound packets that contained errors preventing their delivery to a higher-layer protocol. For character-oriented or fixed-length interfaces, the number of inbound transmission units that contained errors preventing them from delivery to a higher-layer protocol. |
| Rx Packet Per Second | net.ppsRx.rate.average.number | 5 minutes | Average rate of packets received per second. |
| Rx Packets | net.packetsRx.delta.summation.number | 5 minutes | Number of packets received through the physical interface during the sampling interval. |
| Throughput | net.nicThroughput.delta.summation.number | 5 minutes | NIC throughtput (rx+tx) per datapath physical interface. |
| Total Packet Per Second | net.ppsTotal.rate.average.number | 5 minutes | Average rate of total packets (received + transmitted) per second. |
| Total Rx Bytes | net.totalRx.delta.summation.bytes | 5 minutes | Total number of received bytes on the physical interface. |
| Total Tx Bytes | net.totalTx.delta.summation.bytes | 5 minutes | Total number of transmitted bytes through the physical interface. |
| Tx Packet Drop Rate | net.packetDropTransmitted.ratio.average.percent | 5 minutes | Average ratio of dropped transmitted packets to total transmitted packets. |
| Tx Packet Errors | net.errorTx.delta.summation.number | 5 minutes | For packet-oriented interfaces, the number of outbound packets that could not be transmitted due to errors. For character-oriented or fixed-length interfaces, the number of outbound transmission units that could not be transmitted due to errors. |
| Tx Packet Per Second | net.ppsTx.rate.average.number | 5 minutes | Average rate of packets transmitted per second. |
| Tx Packets | net.packetsTx.delta.summation.number | 5 minutes | Number of packets transmitted through the physical interface during the sampling interval. |

**Table 1352: Service Router Metrics**

| Metric Display Name | Metric API Name | Export Frequency | Description |
|---|---|---|---|
| BGP Neighbour Down Count | net.bgp.neighbor.down.count.absolute.latest.number | 5 minutes | Number of Border Gateway Protocol (BGP) neighbors identified as Down. |
| BGP Neighbour Up Count | net.bgp.neighbor.up.count.absolute.latest.number | 5 minutes | Number of Border Gateway Protocol (BGP) neighbors identified as Up. |
| OSPF Neighbor Down Count | net.ospf.neighbor.down.count.absolute.latest.number | 5 minutes | Number of Open Shortest Path First (OSPF) neighbors identified as Down. |
| OSPF Neighbor Up Count | net.ospf.neighbor.up.count.absolute.latest.number | 5 minutes | Number of Open Shortest Path First (OSPF) neighbors identified as Up. |

**Table 1353: Uplink Router Metrics**

| Metric Display Name | Metric API Name | Export Frequency | Description |
|---|---|---|---|
| ARP Absence Dropped Rx Packets | net.noARPDroppedRx.delta.summation.number | 5 minutes | Number of received packets dropped due to the absence of ARP on the Tier-0 uplink. |
| ARP Absence Dropped Tx Packets | net.noARPDroppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped due to absence of ARP on the Tier-0 uplink. |
| Blocked Rx Packets | net.blockedRx.delta.summation.number | 5 minutes | Number of blocked received packets due to a standby interface, on the Tier-0 uplink. |
| Blocked Tx Packets | net.blockedTx.delta.summation.number | 5 minutes | Number of transmitted packets blocked by standby interface, on the Tier-0 uplink. |
| Destination Unsupported Dropped Rx Packets | net.destUnsupportedDroppedRx.delta.summation.number | 5 minutes | Number of packets dropped on the Tier-0 uplink, due to an unsupported destination such as multicast/loopback/ reserved address. |
| Destination Unsupported Dropped Tx Packets | net.destUnsupportedDroppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped on the Tier-0 uplink, because destination is not supported. |
| Dropped Rx Packets | net.droppedRx.delta.summation.number | 5 minutes | Number of received packets dropped on the Tier-0 uplink during the sampling interval. |
| Dropped Tx Packets | net.droppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped on the Tier-0 uplink during the sampling interval. |
| Duplicate Address Detected Dropped Rx Packets | net.dadDroppedRx.delta.summation.number | 5 minutes | Number of received packets dropped on the Tier-0 uplink due to duplicate IP detection. |
| Duplicate Address Detected Dropped Tx Packets | net.dadDroppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped on the Tier-0 uplink due to duplicate IP detection. |
| Firewall Dropped Rx Packets | net.firewallDroppedRx.delta.summation.number | 5 minutes | Number of received packet drops on the Tier-0 uplink due to firewall. |
| Firewall Dropped Tx Packets | net.firewallDroppedTx.delta.summation.number | 5 minutes | Number of transmitted packet drops on the Tier-0 uplink due to firewall. |
| Fragmentation Dropped Rx Packets | net.fragNeededDroppedRx.delta.summation.number | 5 minutes | Number of received packets dropped on the Tier-0 uplink due to a need for fragmentation. |
| Fragmentation Dropped Tx Packets | net.fragNeededDroppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped on the Tier-0 uplink due to a need for fragmentation. |
| Insufficient Memory Dropped Rx Packets | net.noMemoryDroppedRx.delta.summation.number | 5 minutes | Number of received packets dropped on the Tier-0 uplink due to insufficient memory. |
| Insufficient Memory Dropped Tx Packets | net.noMemoryDroppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped on the Tier-0 uplink due to insufficient memory. |
| IPSec Dropped Rx Packets | net.IPsecDroppedRx.delta.summation.number | 5 minutes | Number of received packets dropped by IPsec input on the Tier-0 uplink. |
| IPSec Dropped Tx Packets | net.IPsecDroppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped by IPsec input on the Tier-0 uplink. |

| Metric Display Name | Metric API Name | Export Frequency | Description |
|---|---|---|---|
| IPSec Policy Block Dropped Rx Packets | net.IPsecPolicyBlockDroppedRx.delta.summation.number | 5 minutes | Number of received packets dropped on the Tier-0 uplink due to blocking by IPsec policy. |
| IPSec Policy Block Dropped Tx Packets | net.IPsecPolicyBlockDroppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped on the Tier-0 uplink due to blocking by IPsec policy. |
| IPSec Policy Error Dropped Rx Packets | net.IPsecPolicyErrorDroppedRx.delta.summation.number | 5 minutes | Number of received packets dropped on the Tier-0 uplink due to IPsec policy error. |
| IPSec Policy Error Dropped Tx Packets | net.IPsecPolicyErrorDroppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped on the Tier-0 uplink due to IPsec policy error. |
| IPSec Security Association Absence Dropped Rx Packets | net.IPsecNoSaDroppedRx.delta.summation.number | 5 minutes | Number of received packets dropped on the Tier-0 uplink by IPsec missing a Security Association (SA). |
| IPSec Security Association Absence Dropped Tx Packets | net.IPsecNoSaDroppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped on the Tier-0 uplink by IPsec missing a Security Association (SA). |
| IPSec VTI Absence Dropped Rx Packets | net.IPsecNoVtiDroppedRx.delta.summation.number | 5 minutes | Number of received packets dropped on the Tier-0 uplink by IPsec input missing a Virtual Tunnel Interface (VTI). |
| IPSec VTI Absence Dropped Tx Packets | net.IPsecNoVtiDroppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped on the Tier-0 uplink by IPsec input missing a Virtual Tunnel Interface (VTI). |
| IPV6 Dropped Rx Packets | net.IPV6DroppedRx.delta.summation.number | 5 minutes | Number of received IPv6 packet drops on the Tier-0 uplink. |
| IPV6 Dropped Tx Packets | net.IPV6DroppedTx.delta.summation.number | 5 minutes | Number of transmitted IPv6 packet drops on the Tier-0 uplink. |
| Kernel Nic Interface Dropped Rx Packets | net.KNIDroppedRx.delta.summation.number | 5 minutes | Number of received packets dropped on the Tier-0 uplink by the DPDK kernel NIC interface. |
| Kernel Nic Interface Dropped Tx Packets | net.KNIDroppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped on the Tier-0 uplink by the DPDK kernel NIC interface. |
| Linked Port Absence Dropped Rx Packets | net.noLinkedDroppedRx.delta.summation.number | 5 minutes | Number of received packets dropped on the Tier-0 uplink due to the absence of a linked port. |
| Linked Port Absence Dropped Tx Packets | net.noLinkedDroppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped on the Tier-0 uplink due to the absence of a linked port. |
| Malformed Dropped Rx Packets | net.malformedDroppedRx.delta.summation.number | 5 minutes | Number of received packets dropped on the Tier-0 uplink due to malformed packet. |
| Malformed Dropped Tx Packets | net.malformedDroppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped on the Tier-0 uplink due to malformed packet. |
| Non IP Dropped Rx Packets | net.nonIPDroppedRx.delta.summation.number | 5 minutes | Number of received packets dropped on the Tier-0 uplink because interface allows only IP packets. |
| Non IP Dropped Tx Packets | net.nonIPDroppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped on the Tier-0 uplink because interface allows only IP packets. |

| Metric Display Name | Metric API Name | Export Frequency | Description |
|---|---|---|---|
| Receiver Absence Dropped Rx Packets | net.noReceiverDroppedRx.delta.summation.number | 5 minutes | Number of received packets dropped on the Tier-0 uplink due to absence of receiver. |
| Receiver Absence Dropped Tx Packets | net.noReceiverDroppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped on the Tier-0 uplink due to absence of receiver. |
| Redirect Dropped Rx Packets | net.redirectDroppedRx.delta.summation.number | 5 minutes | Number of redirect RX packets dropped. |
| Redirect Dropped Tx Packets | net.redirectDroppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped due to exceeded time to live. |
| Route Absence Dropped Rx Packets | net.noRouteDroppedRx.delta.summation.number | 5 minutes | Number of received packets dropped on the Tier-0 uplink, due to the absence of a route. |
| Route Absence Dropped Tx Packets | net.noRouteDroppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped on the Tier-0 uplink, due to the absence of a route. |
| Rx Packets | net.packetsRx.delta.summation.number | 5 minutes | Number of packets received through the Tier-0 uplink during the sampling interval. |
| Service Insert Dropped Rx Packets | net.serviceInsertDroppedRx.delta.summation.number | 5 minutes | Number of service insert RX packets dropped on the Tier-0 uplink. |
| Service Insert Dropped Tx Packets | net.serviceInsertDroppedTx.delta.summation.number | 5 minutes | Number of service insert TX packets dropped on the Tier-0 uplink. |
| Total Rx Bytes | net.totalRx.delta.summation.bytes | 5 minutes | Total number of bytes received through the Tier-0 uplink. |
| Total Tx Bytes | net.totalTx.delta.summation.bytes | 5 minutes | Total number of bytes transmitted through the Tier-0 uplink. |
| TTL Exceeded Dropped Rx Packets | net.ttlExceededDroppedRx.delta.summation.number | 5 minutes | Number of received packets dropped on the Tier-0 uplink due to exceeded time to live (TTL). |
| TTL Exceeded Dropped Tx Packets | net.ttlExceededDroppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped on the Tier-0 uplink due to exceeded time to live (TTL). |
| Tx Packets | net.packetsTx.delta.summation.number | 5 minutes | Number of packets transmitted through the Tier-0 uplink during the sampling interval. |
| Unsupported L4 Port Dropped Rx Packets | net.L4PortUnsupportedDroppedRx.delta.summation.number | 5 minutes | Number of received packets dropped on the Tier-0 uplink due to an unsupported L4 port. |
| Unsupported L4 Port Dropped Tx Packets | net.L4PortUnsupportedDroppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped on the Tier-0 uplink due to an unsupported L4 port. |
| Unsupported Protocol Dropped Rx Packets | net.protoUnsupportedDroppedRx.delta.summation.number | 5 minutes | Number of received packets dropped due to unsupported protocol. |
| Unsupported Protocol Dropped Tx Packets | net.protoUnsupportedDroppedTx.delta.summation.number | 5 minutes | Number of transmitted packets dropped due to an unsupported protocol. |
| uRPF Drop Packets | net.droppedRPFRx.delta.summation.number | 5 minutes | Number of packets dropped on the Tier-0 uplink due to violations of the Unicast Reverse Path Forwarding (uRPF) security mechanism. |

### Metrics for NSX-T Manager

VCF Operations for networks collects flow metrics for NSX manager.

| Metric Name | Metric API Name | Description |
|---|---|---|
| Outbound Expire Flow Drop Count | flow.fiveTupleExpireOutbound.delta.summation.number | Number of five tuples dropped because of reaching the maximum outbound limit for source IP addresses in flows. |
| Wrong Collector Flow Drop Count | flow.rawRecordReporterWrongCollector.delta.summation.number | Number of raw flow records dropped as the flow records were reported on a wrong type of VCF Operations for networks collector. |

### Metrics for NSX-T Management Node

VCF Operations for networks collects NSX management node metrics to monitor the performance of NSX management nodes and troubleshoot issues, if any.

| Metric Name | Metric API Name | Description |
|---|---|---|
| CPU Usage (%) | sys.loadFifteenMinutes.rate.latest.percent | Percentage of CPU used out of its total capacity in the time interval. |
| Fifteen Minute Load Average | sys.loadFifteenMinutes.absolute.maximum.number | Average system load over an interval of fifteen minutes. |
| Five Minute Load Average | sys.loadFiveMinutes.absolute.maximum.number | Average system load over a period of five minutes. |
| Memory Cache | mem.cache.absolute.latest.kiloBytes | Total memory used by the system for disk caching in the time interval. |
| Memory Usage | mem.usage.absolute.latest.kiloBytes | Total amount of memory used by the system, including swap memory, in the time interval. |
| Memory Usage Rate (%) | mem.usage.absolute.latest.percent | Average percentage of the memory used, including swap memory, out of the total available memory. |
| One Minute Load Average | sys.loadOneMinute.absolute.maximum.number | Average system load in a minute. |
| Swap Memory Usage | mem.swapused.absolute.latest.kiloBytes | Amount of swap memory used by the system in the time interval. |
| System Uptime (ms) | sys.uptime.absolute.latest.millisecond | Total time elapsed since the last system startup. |

### Metrics for NSX-T Logical Switch

NSX logical switch metrics provide information about the traffic processed by NSX logical switches.

| Metric Name | Metric API Name | Description |
|---|---|---|
| Dropped Rx Packets | net.droppedRx.delta.summation.number | The number of received packets dropped to total received packets. |
| Dropped Tx Packets | net.droppedTx.delta.summation.number | The number of transmitted packets dropped to total transmitted packets. |
| Multicast & Broadcast Rx Bytes | net.multicastBroadcastRx.delta.summation.bytes | Total number of received multicast & broadcast bytes. |
| Multicast & Broadcast Rx Packets | net.multicastBroadcastRx.delta.summation.number | Total number of received multicast & broadcast packets. |

| Metric Name | Metric API Name | Description |
|---|---|---|
| Multicast & Broadcast Tx Bytes | net.multicastBroadcastTx.delta.summation.bytes | Total number of transmitted multicast & broadcast bytes. |
| Multicast & Broadcast Tx Packets | net.multicastBroadcastTx.delta.summation.number | Total number of transmitted multicast & broadcast packets. |
| Rx Packet Drop Rate (%) | net.packetDropReceived.ratio.average.percent | Average ratio of received packets dropped to total received packets. |
| Rx Packets | net.packetsRx.delta.summation.number | Total number of received Packets during the time interval |
| Rx Unicast Packets | net.unicastPacketsRx.delta.summation.number | Average number of unicast packets received during the time interval. |
| Total Rx Bytes | net.totalRx.delta.summation.bytes | Sum of bytes received during the time interval. |
| Total Tx Bytes | net.totalTx.delta.summation.bytes | Sum of bytes transmitted during the time interval. |
| Tx Packet Drop Rate (%) | net.packetDropTransmitted.ratio.average.percent | Average ratio of transmitted packets dropped to total transmitted packets. |
| Tx Packets | net.packetsTx.delta.summation.number | Total number of packets transmitted during the time interval. |
| Tx Unicast Packets | net.unicastPacketsTx.delta.summation.number | Average number of unicast packets transmitted during the time interval. |
| Unicast Rx Bytes | net.unicastRx.delta.summation.bytes | Total volume of unicast packets received over the time interval, expressed in bytes. |
| Unicast Tx Bytes | net.unicastTx.delta.summation.bytes | Total volume of unicast packets transmitted over the time interval, expressed in bytes. |

### Metrics for NSX-T Logical Port

NSX logical port metrics provide information about the traffic processed by NSX logical ports.

| Metric Name | Metric API Name | Description |
|---|---|---|
| Dropped Rx Packets | net.droppedRx.delta.summation.number | The number of received packets dropped to total received packets. |
| Dropped Tx Packets | net.droppedTx.delta.summation.number | The number of transmitted packets dropped to total transmitted packets. |
| Interface Rx Utilization (%) | net.rx.utilization.absolute.latest.percent | Interface utilization percentage for the received traffic. |
| Interface Tx Utilization (%) | net.tx.utilization.absolute.latest.percent | Interface utilization percentage for the transmitted traffic. |
| Interface Utilization (%) | net.utilization.absolute.latest.percent | Average interface utilization percentage for received and transmitted traffic. |
| Multicast & Broadcast Rx Bytes | net.multicastBroadcastRx.delta.summation.bytes | Total number of received multicast & broadcast bytes. |
| Multicast & Broadcast Rx Packets | net.multicastBroadcastRx.delta.summation.number | Total number of received multicast & broadcast packets. |
| Multicast & Broadcast Tx Bytes | net.multicastBroadcastTx.delta.summation.bytes | Total number of transmitted multicast & broadcast bytes. |

| Metric Name | Metric API Name | Description |
|---|---|---|
| Multicast & Broadcast Tx Packets | net.multicastBroadcastTx.delta.summation.number | Total number of transmitted multicast & broadcast packets. |
| Rx Packet Drop Rate (%) | net.packetDropReceived.ratio.average.percent | Average ratio of received packets dropped to total received packets. |
| Rx Packets | net.packetsRx.delta.summation.number | Total number of received Packets during the time interval |
| Rx Unicast Packets | net.unicastPacketsRx.delta.summation.number | Average number of unicast packets received during the time interval. |
| Total Rx Bytes | net.totalRx.delta.summation.bytes | Sum of bytes received during the time interval. |
| Total Tx Bytes | net.totalTx.delta.summation.bytes | Sum of bytes transmitted during the time interval. |
| Tx Packet Drop Rate (%) | net.packetDropTransmitted.ratio.average.percent | Average ratio of transmitted packets dropped to total transmitted packets. |
| Tx Packets | net.packetsTx.delta.summation.number | Total number of packets transmitted during the time interval. |
| Tx Unicast Packets | net.unicastPacketsTx.delta.summation.number | Average number of unicast packets transmitted during the time interval. |
| Unicast Rx Bytes | net.unicastRx.delta.summation.bytes | Total volume of unicast packets received over the time interval, expressed in bytes. |
| Unicast Tx Bytes | net.unicastTx.delta.summation.bytes | Total volume of unicast packets transmitted over the time interval, expressed in bytes. |

**Metrics for NSX-T Transport Node**

VCF Operations for networks collects NSX transport node metrics to monitor the performance of NSX transport nodes and troubleshoot issues, if any.

**NAT Metrics**

NAT metrics provide information about the total traffic consumed by the service router for the NAT (Network Address Translation) rules configured on the NSX transport node.

| Metric Name | Metric API Name | Description |
|---|---|---|
| Flow Packet Count | flow.totalPackets.delta.summation.number | Sum of packets processed by all the NAT rules from all logical routers which are present on the transport node. |
| Session Count | net.connection.delta.summation.number | Sum of connections of all NAT rules from all logical routers which are present on the transport node. |
| Total Traffic | flow.totalBytes.delta.summation.bytes | Sum of bytes of all NAT rules from all logical routers which are present on the transport node. |

## Node Metrics

Node metrics provide information about the CPU usage and the system load of the NSX transport node.

| Metric Name | Metric API Name | Description |
|---|---|---|
| CPU Usage (%) | sys.loadFifteenMinutes.rate.latest.percent | Percentage of CPU used out of its total capacity in the time interval. |
| Fifteen Minute Load Average | sys.loadFifteenMinutes.absolute.maximum.number | Average system load over an interval of fifteen minutes. |
| Five Minute Load Average | sys.loadFiveMinutes.absolute.maximum.number | Average system load over a period of five minutes. |
| Memory Cache | mem.cache.absolute.latest.kiloBytes | Total memory used by the system for disk caching in the time interval. |
| Memory Usage | mem.usage.absolute.latest.kiloBytes | Total amount of memory used by the system, including swap memory, in the time interval. |
| Memory Usage Rate (%) | mem.usage.absolute.latest.percent | Average percentage of the memory used, including swap memory, out of the total available memory. |
| One Minute Load Average | sys.loadOneMinute.absolute.maximum.number | Average system load in a minute. |
| Swap Memory Usage | mem.swapused.absolute.latest.kiloBytes | Amount of swap memory used by the system in the time interval. |
| System Uptime (ms) | sys.uptime.absolute.latest.millisecond | Total time elapsed since the last system startup. |

### Metrics for NSX-T Edge NAT Rule

NSX Edge NAT rules metrics provide information about the traffic processed by NAT rules of NSX Edge devices.

| Metric Name | Metric API Name | Description |
|---|---|---|
| Flow Packets | net.packet.delta.summation.number | Total number of flow packets processed by the NSX Edge NAT rule. |
| Session Count | net.connection.delta.summation.number | The total number of sessions handled by the NSX Edge NAT rule. |
| Total Bytes | net.byte.delta.summation.number | Total number of bytes processed by the NSX Edge NAT rule. |

### Metrics for NSX-T Router

VCF Operations for networks collects NSX router metrics to monitor the NAT (Network Address Translation) rules configured on the router.

| Metric Name | Metric API Name | DesMetrics for NSX-T Logical Portcription |
|---|---|---|
| Flow Packets | net.packet.delta.summation.number | Sum of packets of all the NAT rules which are present on the logical router. |
| Session Count | net.connection.delta.summation.number | Sum of connections of all the NAT rules which are present on the logical router. |
| Total Bytes | net.byte.delta.summation.number | Sum of bytes of all the NAT rules which are present on the logical router. |

## Router Interface Metrics

The router interface metrics provide information about the network performance of the router interface.

| Metric Name | Metric API Name | Description |
|---|---|---|
| Discarded Rx Packets | net.discardedRx.delta.summation.number | Number of discarded inbound packets. |
| Discarded Tx Packets | net.discardedTx.delta.summation.number | Number of discarded outbound packets. |
| Dropped Rx Packets | net.droppedRx.delta.summation.number | Number of received packets that were dropped during the collection interval. |
| Dropped Tx Packets | net.droppedTx.delta.summation.number | Number of transmitted packets that were dropped during the collection interval. |
| FCS Errors | net.errorFcs.delta.summation.number | Number of frames transmitted or received with a bad checksum (CRC value) in the Ethernet frame.<br>Frame check sequence errors indicate the frames of data corrupted during transmission. These frames are dropped and not propagated to other ports. |
| Giant Errors | net.errorGiants.delta.summation.number | Number of giant errors (errors caused by packet having size greater than MTU of the interface) over the network during the selected duration. |
| Interface Rx Utilization (%) | net.rx.utilization.absolute.latest.percent | Interface utilization percentage for the received traffic. |
| Interface Tx Utilization (%) | net.tx.utilization.absolute.latest.percent | Interface utilization percentage for the transmitted traffic. |
| Interface Utilization (%) | net.utilization.absolute.latest.percent | Interface utilization percentage for received and transmitted traffic. |
| Max Network Traffic Rate (kbps) | net.usage.rate.maximum.kiloBytesPerSecond | Maximum network usage rate over any five minute interval in the selected duration. |
| Max Network Tx Rate (kbps) | net.transmitted.rate.maximum.kiloBytesPerSecond | Maximum network data transmit rate over any five minute interval in the selected duration. |
| Max Packet Drop Ratio (%) | net.packetDrop.ratio.maximum.percent | Maximum five minute interval ratio of total dropped packets to total number of packets. |
| Max Packet Drops | net.droppedTotal.delta.maximum.number | Maximum number of dropped Rx and Tx packets over any five minute interval in the selected duration. |
| Multicast Rx Packets | net.multicastRx.delta.summation.number | Number of multicast packets received in the selected duration. |
| Multicast Tx Packets | net.multicastTx.delta.summation.number | Number of multicast packets transmitted in the selected duration. |
| Network Align Errors | net.errorAlign.delta.summation.number | Number of alignment errors detected over the network in the selected duration.<br>Network alignment errors occur if a frame arrives with part of an octet missing, and has an FCS error. |
| Network Carrier Sense Errors | net.errorCarrierSense.delta.summation.number | Number of carrier sense errors detected over the network in the selected duration.<br>Network carrier sense errors occur when an interface cannot transmit a frame because it did not detect a carrier. |
| Network Deferred Tx Errors | net.errorDeferredTx.delta.summation.number | Number of deferred transmission errors detected over the network in the selected duration. |

| Metric Name | Metric API Name | Description |
|---|---|---|
| Network Excessive Collision Errors | net.errorExcessiveCollisions.delta.summation.number | Number of excessive collision errors detected over the network in the selected duration.<br>Network excessive collision errors occur when a station receives 16 consecutive collisions while attempting to transmit a single frame. In such a situation, the frame is dropped due to the excessive collisions. |
| Network Late Collision Errors | net.errorLateCollisions.delta.summation.number | Number of late collision errors detected over the network in the selected duration. |
| Network Long Frame Errors | net.errorFrameTooLongs.delta.summation.number | Number of network long frame errors detected over the network in the selected duration. |
| Network Max Rx Rate | net.received.rate.maximum.kiloBytesPerSecond | The maximum rate at which the data is received during the time interval. |
| Network Multi Collision Errors | net.errorMultiCollisionFrames.delta.summation.number | Number of multi collision errors detected over the network in the selected duration. |
| Network Packet Count | net.packetsTotal.delta.summation.number | Total number of packets transmitted and received during the time interval. |
| Network Packet Drop Percent | net.packetDrop.ratio.average.percent | Average ratio of total number of dropped packets to total number of packets. |
| Network Packet Drops | net.droppedTotal.delta.summation.number | Total number of packets dropped during the time interval. |
| Network Receive Errors | net.errorRcv.delta.summation.number | Number of receive errors received on the physical interface in the selected duration.<br>Receive errors occur when there is excessive output rates of traffic, or if the receive buffer is full. |
| Network Rx Int Mac Errors | net.errorRxIntMac.delta.summation.number | Number of frames for which reception on a particular interface has failed due to an internal MAC sublayer receive error. |
| Network Rx Rate | net.received.rate.average.kiloBytesPerSecond | Average rate at which data is received during the time interval. |
| Network Single Collision Errors | net.errorSingleCollisionFrames.delta.summation.number | Number of single collision errors detected over the network in the selected duration. |
| Network SQE Test Errors | net.errorSQETests.delta.summation.number | Number of signal quality test errors detected over the network in the selected duration. |
| Network Symbol Errors | net.errorSymbol.delta.summation.number | Number of symbol errors detected over the network in the selected duration. |
| Network Traffic Rate | net.usage.rate.average.kiloBytesPerSecond | Average rate of received and transmitted traffic during the time interval. |
| Network Tx Intmac Errors | net.errorTxIntMac.delta.summation.number | Number of frames for which transmission on a particular interface failed due to an internal MAC sublayer transmit error. |
| Network Tx Rate | net.transmitted.rate.average.kiloBytesPerSecond | Average rate of transmitted traffic during the time interval. |
| Network Undersize Errors | net.errorUnderSize.delta.summation.number | Number of undersize errors detected over the network in the selected duration.<br>Undersize errors indicate bad frames generated by connected devices.<br>. |

| Metric Name | Metric API Name | Description |
|---|---|---|
| Network Xmit Errors | net.errorXmit.delta.summation.number | Number of transmit errors detected over the network in the selected duration.<br>Xmit errors occur when there is excess input rates of traffic, or if the transmit buffer is full.<br>To minimize the occurrence of this error, you must increase the speed of the interface and move the link partner to full-duplex. |
| Out Queue Length | net.outQLen.absolute.average.number | Number of output packets waiting in the queue. |
| Rx Broadcast Packets | net.broadcastRx.delta.summation.number | Number of broadcast packets received during the time interval. |
| Rx Broadcast Ratio (%) | net.broadcastByRx.ratio.average.percent | Percentage of broadcast packets received during the time interval. |
| Rx Jumbo Packets | net.jumboRx.delta.summation.number | Number of jumbo packets received during the time interval. |
| Rx Packet Drop Percent (%) | net.packetDropRx.ratio.average.percent | Average ratio of total number of dropped Rx packets to total number of Rx packets. |
| Rx Packet Errors | net.errorRx.delta.summation.number | Number of received packet errors detected over the network in the selected duration.<br>For packet-oriented interfaces, Rx packet error indicates the number of inbound packets that contained errors preventing them from being delivered to a higher-layer protocol.<br>For character-oriented or fixed-length interfaces, Rx packet error indicates the number of inbound transmission units that contained errors preventing them from being delivered to a higher-layer protocol. |
| Rx Packets | net.packetsRx.delta.summation.number | Number of packets received during the time interval. |
| Total Network Traffic | net.usage.delta.summation.kiloBytes | Sum of data transmitted and received during the time interval. |
| Tx Broadcast Packets | net.broadcastTx.delta.summation.number | Number of broadcast packets transmitted during the time interval. |
| Tx Broadcast Ratio | net.broadcastByTx.ratio.average.percent | Average percent of broadcast packets transmitted during the time interval. |
| Tx Jumbo Packets | net.jumboTx.delta.summation.number | Number of jumbo packets transmitted during the time interval. |
| Tx Packet Drop Percent | net.packetDropTx.ratio.average.percent | Ratio of total number of dropped Tx packets to total number of Tx packets. |
| Tx Packet Errors | net.errorTx.delta.summation.number | Number of transmitted packet errors detected over the network in the selected duration.<br>For packet-oriented interfaces, Tx packet errors indicate the number of outbound packets that were not transmitted because of errors.<br>For character-oriented or fixed-length interfaces, Tx packet errors indicate the number of outbound transmission units that were not transmitted because of errors. |

| Metric Name | Metric API Name | Description |
|---|---|---|
| Tx Packets | net.packetsTx.delta.summation.number | Number of packets transmitted during the time interval. |
| Unicast Rx Packets | net.unicastRx.delta.summation.number | Total number of unicast packets received during the time interval. |
| Unicast Tx Packets | net.unicastTx.delta.summation.number | Total number of unicast packets transmitted during the time interval. |
| Unknown Rx Packets | net.unknownRx.delta.summation.number | Total number of unknown packets received during the time interval.<br>For packet-oriented interfaces, unknown Rx packets indicate the number of packets received through the interface that were discarded because of an unknown protocol or an unsupported protocol.<br>For character-oriented or fixed-length interfaces that support protocol multiplexing, unknown Rx packets indicate the number of transmission units received through the interface that were discarded because of an unknown protocol or an unsupported protocol. |
| Total Rx Bytes | net.totalRx.delta.summation.bytes | Sum of bytes received during the time interval. |
| Total Tx Bytes | net.totalTx.delta.summation.bytes | Sum of bytes transmitted during the time interval. |
| uRPF Drop Packets | net.droppedRPFRx.delta.summation.number | Number of packets dropped due to unicast reverse-path forwarding (uRPF) |
| uRPF Packet Drop Ratio | net.droppedRPFRx.rate.average.percent | Average ratio of dropped uRPF packets to total Rx packets. |

## Switch Port Metrics

Switch port metrics provide information about the network performance of switch ports.

### Network Metrics

These metrics provide information about the network performance of the switch port.

| Metric Name | Metric API Name | Description | Supported by Databus |
|---|---|---|---|
| Average Egress Latency | net.latency.absolute.average.nanosecond | Average end-to-end latency of successful requests made to a switch port. | |
| Carrier Losses Detected | net.txcarrier.delta.summation.number | Number of transmitted packets that experienced loss of carriers per host port over a five minute interval.<br>This metric is captured only for the switch ports of NSX-T host transport nodes. | |

| Metric Name | Metric API Name | Description | Supported by Databus |
|---|---|---|---|
| Collisions Detected | net.txcolls.delta.summation.number | Number of transmitted packets that experienced Ethernet collisions on a host over a five minute interval. A non-zero value of this field indicates possibility of network congestion.<br>This metric is captured only for the switch ports of NSX-T host transport nodes. | |
| Discarded Rx Packets | net.discardedRx.delta.summation.number | Number of discarded inbound packets.<br>Packets are discarded to free up the buffer space. | |
| Discarded Tx Packets | net.discardedTx.delta.summation.number | Number of discarded outbound packets.<br>Packets are discarded to free up the buffer space. | |
| Dropped Rx Packets | net.droppedRx.delta.summation.number | Number of Rx packets dropped during the transmission interval. | Yes |
| Dropped Tx Packets | net.droppedTx.delta.summation.number | Number of Tx packets dropped during the transmission interval. | Yes |
| FCS Errors | net.errorFcs.delta.summation.number | Number of frames transmitted or received with a bad checksum (CRC value) in the Ethernet frame.<br>Frame check sequence (FCS) errors indicate the frames of data corrupted during transmission. These frames are dropped and not propagated to other ports. | |
| Giant Errors | net.errorGiants.delta.summation.number | Number of giant errors over the network during the selected duration. | |
| Host Latency Per pNIC (ms) | netopa.pnicLatency.absolute.average.microsecond | Time taken by the data to pass from one host pNIC on a network to another. | Yes |
| Interface Peak Buffer Utilization | net.peakBufferUtil.absolute.maximum.bytes | Peak buffer occupancy per switch port during the time interval. | |
| Interface Rx Utilization (%) | net.rx.utilization.absolute.latest.percent | Interface utilization percentage for the received traffic. | |
| Interface Status UP (ms) | net.interfaceStatus.delta.summation.millisecond | Interface uptime in milliseconds. | |
| Interface Tx Utilization (%) | net.tx.utilization.absolute.latest.percent | Interface utilization percentage for the transmitted traffic. | |
| Interface Utilization (%) | net.utilization.absolute.latest.percent | Interface utilization percentage for the transmitted and received traffic. | |

| Metric Name | Metric API Name | Description | Supported by Databus |
|---|---|---|---|
| Max Host Latency Per pNIC | netopa.pnicLatency.absolute.maximum.microsecond | Maximim time taken by the data to pass from one host pNIC on a network to another. | Yes |
| Max Network Traffic Rate (Kbps) | net.usage.rate.maximum.kiloBitsPerSecond | Maximum network usage rate on the switch port over any five minute interval in the selected duration. | |
| Max Network Tx Rate (Kbps) | net.transmitted.rate.maximum.kiloBitsPerSecond | Maximum network data transmission rate over any five minute interval in the selected duration. | |
| Max Packet Drop Ratio (%) | net.packetDrop.ratio.maximum.percent | Maximum five minute interval ratio of total dropped packets to total number of packets. | |
| Max Packet Drops | net.droppedTotal.delta.maximum.number | Maximum number of dropped packets over any five minute interval in the selected duration. | |
| Maximum Egress Latency | net.latency.absolute.maximum.nanosecond | Maximum end-to-end latency of successful requests made to a switch port. | |
| Maximum Total Network Rate (Kbps) | net.total.rate.maximum.kiloBitsPerSecond | Maximum network usage rate over any five minute interval in the selected duration. | |
| Min Host Latency Per pNIC (ms) | netopa.pnicLatency.absolute.minimum.microsecond | Minimum time taken by the data to pass from one host pNIC on a network to another. | |
| Minimum Egress Latency | net.latency.absolute.minimum.nanosecond | Minimum end-to-end latency of successful requests made to a switch port. | |
| Multicast Rx Packets | net.multicastRx.delta.summation.number | Number of multicast packets received in the selected duration. | |
| Multicast Tx Packets | net.multicastTx.delta.summation.number | Number of multicast packets transmitted in the selected duration. | |
| Network Align Errors | net.errorAlign.delta.summation.number | Number of alignment errors detected over the network in the selected duration.<br>Network alignment errors occur if a frame arrives with part of an octet missing, and having an FCS error. | |
| Network Carrier Sense Errors | net.errorCarrierSense.delta.summation.number | Number of carrier sense errors detected over the network in the selected duration. | |
| Network Deferred Tx Errors | net.errorDeferredTx.delta.summation.number | Number of deferred transmission errors detected over the network in the selected duration. | |

| Metric Name | Metric API Name | Description | Supported by Databus |
|---|---|---|---|
| Network Excessive Collision Errors | net.errorExcessiveCollisions.delta.summation.number | Number of excessive collision errors detected over the network in the selected duration. | |
| Network Late Collision Errors | net.errorLateCollisions.delta.summation.number | Number of late collision errors detected over the network in the selected duration. | |
| Network Long Frame Errors | net.errorFrameTooLongs.delta.summation.number | Number of network long frame errors detected over the network in the selected duration. | |
| Network Max Rx Rate (Kbps) | net.received.rate.maximum.kiloBitsPerSecond | Maximum rate at which the data is received during the time interval. | |
| Network Maximum Rx Rate (Kbps) | net.rx.rate.maximum.kiloBitsPerSecond | Maximum rate at which the data is received during the time interval. | |
| Network Maximum Tx Rate (Kbps) | net.tx.rate.maximum.kiloBitsPerSecond | Maximum rate at which the data is transmitted during the time interval. | |
| Network Multi Collision Errors | net.errorMultiCollisionFrames.delta.summation.number | Number of multi collision errors detected over the network in the selected duration. | |
| Network Packet Count | net.packetsTotal.delta.summation.number | Total number of packets transmitted and received from the switch port. | |
| Network Packet Drop Percent (%) | net.packetDrop.ratio.average.percent | Average ratio of total number of dropped packets to total number of packets. | |
| Network Packet Drops | net.droppedTotal.delta.summation.number | Total number of dropped packets during the time interval. | |
| Network Receive Errors | net.errorRcv.delta.summation.number | Number of receive errors on the switch port during the time interval.<br>Receive errors occur when there is excessive output rates of traffic, or if the receive buffer is full. | |
| Network Rx Int Mac Errors | net.errorRxIntMac.delta.summation.number | The number of frames for which reception on a particular interface fails due to an internal MAC sublayer receive error. | |
| Network Rx Rate (Kbps) | net.received.rate.average.kiloBitsPerSecond | Average rate at which data is received during the time interval. This metric represents the bandwidth of the network. | Yes |
| Network Single Collision Errors | net.errorSingleCollisionFrames.delta.summation.number | Number of single collision errors detected over the network in the selected duration. | |
| Network SQE Test Errors | net.errorSQETests.delta.summation.number | Number of signal quality test errors detected over the network in the selected duration. | |

| Metric Name | Metric API Name | Description | Supported by Databus |
|---|---|---|---|
| Network Symbol Errors | net.errorSymbol.delta.summation.number | Number of symbol errors detected over the network in the selected duration. | |
| Network Traffic Rate (Kbps) | net.usage.rate.average.kiloBitsPerSecond | Average rate of received and transmitted traffic during the time interval. | Yes |
| Network Tx Intmac Errors | net.errorTxIntMac.delta.summation.number | Number of frames for which transmission on a particular interface fails due to an internal MAC sublayer transmit error. | |
| Network Tx Rate (Kbps) | net.transmitted.rate.average.kiloBitsPerSecond | The average rate at which data was transmitted during the time interval. | Yes |
| Network Undersize Errors | net.errorUnderSize.delta.summation.number | Number of undersize errors detected over the network in the selected duration.<br>This error is an indication of a bad frame generated by the connected device. | |
| Network Xmit Errors | net.errorXmit.delta.summation.number | Number of Xmit errors detected over the network in the selected duration.<br>Xmit errors occur when there is excess input rates of traffic, or if the transmit buffer is full.<br>The transmit buffer becomes full because of excessive collisions and 10 MB ports. You must increase the speed of the interface and move the link partner to full-duplex to minimize the occurrence of Xmit errors. | |
| Out Queue Length | net.outQLen.absolute.average.number | Number of output packets waiting in the queue. | |
| Rx Broadcast Packets | net.broadcastRx.delta.summation.number | Number of broadcast packets received during the specified time interval. | |
| Rx Broadcast Ratio (%) | net.broadcastByRx.ratio.average.percent | Percentage of broadcast packets received during the time interval. | |
| Rx Frame Errors | net.rxframes.delta.summation.number | Number of frame errors received during the time interval.<br>Rx frame errors can occur due to various reasons such as noise or interference in the communication channel, incorrect packet formatting, or issues with the network hardware. | |
| Rx Jumbo Packets | net.jumboRx.delta.summation.number | Number of jumbo packets received during the time interval. | |

| Metric Name | Metric API Name | Description | Supported by Databus |
|---|---|---|---|
| Rx Packet Drop Percent (%) | net.packetDropRx.ratio.average.percent | Average ratio of total number of dropped Rx packets to total number of Rx packets. | |
| Rx Packet Errors | net.errorRx.delta.summation.number | Number of received packet errors detected over the network in the selected duration.<br>For packet-oriented interfaces, an Rx packet error indicates the number of inbound packets that contained errors preventing them from being deliverable to a higher-layer. | |
| Rx Packet Per Second | net.ppsRx.rate.average.number | Average rate at which packets are received per second. | |
| Rx Packets | net.packetsRx.delta.summation.number | Number of packets received during the time interval. | |
| Switch Port Uptime (%) | net.interfaceStatus.rate.average.percent | Interface uptime percentage. | |
| Total Network Traffic (Kb) | net.usage.delta.summation.kiloBytes | Sum of data transmitted and received during the time interval. | |
| Total Packet Per Second | net.ppsTotal.rate.average.number | Average rate of total packets (received packets and transmitted packets) per second. | |
| Total Rx Bytes | net.totalRx.delta.summation.bytes | Total number of bytes received by the switch port. | |
| Total Tx Bytes | net.totalTx.delta.summation.bytes | Total number of bytes transmitted by the switch port. | |
| Tx Broadcast Packets | net.broadcastTx.delta.summation.number | Number of broadcast packets transmitted during the time interval. | |
| Tx Broadcast Ratio (%) | net.broadcastByTx.ratio.average.percent | Average percent of broadcast packets transmitted during a specified time interval. | |
| Tx Jumbo Packets | net.jumboTx.delta.summation.number | Number of jumbo packets transmitted during the time interval. | |
| Tx Packet Drop Percent (%) | net.packetDropTx.ratio.average.percent | Ratio of total number of dropped Tx packets to total number of Tx packets. | |
| Tx Packet Errors | net.errorTx.delta.summation.number | Number of transmitted packet errors detected over the network in the selected duration.<br>For packet-oriented interfaces, Tx packet errors indicate the number of outbound packets that were not transmitted because of errors. | |
| Tx Packet Per Second | net.ppsTx.rate.average.number | Number of packets transmitted per second during the time interval. | |

| Metric Name | Metric API Name | Description | Supported by Databus |
|---|---|---|---|
| Tx Packets | net.packetsTx.delta.summation.number | Total number of packets transmitted during the time interval. | |
| Unicast Rx Packets | net.unicastRx.delta.summation.number | Total number of unicast packets received during the time interval. | |
| Unicast Tx Packets | net.unicastTx.delta.summation.number | Total number of unicast packets transmitted during the time interval. | |
| Unknown Rx Packets | net.unknownRx.delta.summation.number | Total number of unknown packets received during the time interval.<br>For packet-oriented interfaces, unknown Rx packets indicate the number of packets received through the interface that were discarded because of an unknown protocol or an unsupported protocol.<br>For character-oriented or fixed-length interfaces that support protocol multiplexing, unknown Rx packets indicate the number of transmission units received through the interface that were discarded because of an unknown protocol or an unsupported protocol. | |
| uRPF Drop Packets | net.droppedRPFRx.delta.summation.number | Number of packets dropped at the switch port due to unicast reverse-path forwarding (uRPF). | |
| uRPF Packet Drop Ratio (%) | net.droppedRPFRx.rate.average.percent | Average ratio of dropped uRPF packets to total Rx packets. | |
| RX Missed Error | net.pnic.receiveMissedErrors.absolute.latest.number | The number of total RX missed errors. | |
| RX Ring Buffer Usage Percentage | net.pnicRx.ringBuffer.usage.absolute.average.percent | The average percentage of RX ring buffer capacity used. | |
| TX Ring Buffer Usage Percentage | net.pnicTx.ringBuffer.usage.absolute.average.percent | The avarage percentage of TX ring buffer capacity used. | |
| RX Ring Buffer Current Capacity | net.pnicRx.committed.absolute.latest.bytes | Currently configured Rx Ring buffer capacity in bytes. | |
| TX Ring Buffer Current Capacity | net.pnicTx.committed.absolute.latest.bytes | Currently configured Tx Ring buffer capacity in bytes. | |

### Edge Data Plane CPU Core Metrics

These metrics provide information about the usage of the Edge Data Plane CPU cores that are connected to the switch ports.

| Metric Name | Metric API Name | Description |
|---|---|---|
| Edge Dataplane.CPU Core Micro Flow Cache Hit Percent | edge.dataplane.cpu.core.microflowcache.hit.rate.average.percent | Average percentage of micro flow cache hit rate of the Edge dataplane CPU core . |
| Edge Dataplane.CPU Core Mega Flow Cache Hit Percent | edge.dataplane.cpu.core.megaflowcache.hit.rate.average.percent | Average percentage of mega flow cache hit rate of the Edge dataplane CPU core |

## Service Node Metrics

Service node metrics provide information about the performance of a particular service node.

| Metric Name | Metric API Name | Decription |
|---|---|---|
| Bytes Received | net.lbServiceNodeBytesIn.delta.summation.bytes | Total number of bytes received by the specified node address from the server. |
| Bytes Sent | net.lbServiceNodeBytesOut.delta.summation.bytes | Total number of bytes sent to the server from the specified node address. |
| Current Connections | net.lbServiceNodeCurrentConnections.delta.summation.number | Total number of ongoing connections between the server and the specified node address. |
| max connections | net.lbServiceNodeMaxConnections.delta.summation.number | Maximum number of connections between the server and the specified node address. |
| Packets Received | net.lbServiceNodePacketsIn.delta.summation.number | Total number of packets received by the specified node address from the server. |
| Packets Sent | net.lbServiceNodePacketsOut.delta.summation.number | Total number of packets sent to the server from the specified node address. |
| Total Connections | net.lbServiceNodeTotalConnections.delta.summation.number | Total number of connections between the server and the specified node address. |
| Total Requests | net.lbServiceNodeTotalRequests.delta.summation.number | Total number of requests going through the specified node address. |

## Tier Metrics

VCF Operations for networks collects network and flow related metrics for Tiers.

| Metric Name | Metric API Name | Description |
|---|---|---|
| App Tier TCP Internet Retransmitted Packet Ratio | net.tcpRetransmitedInternetPackets.ratio.average.percent | Average percentage of retransmitted internet packets to total number of internet flows associated with the tier. |
| App/Tier TCP Retransmitted Packets | net.tcpRetransmitedPackets.delta.summation.number | Total number of retransmitted packets to the total number of transmitted packets associated with the tier. |
| App/Tier TCP Retransmitted Packets Internet | net.tcpRetransmitedPacketsInternet.delta.summation.number | Total number of retransmitted internet packets to the total number of transmitted internet packets associated with the tier. |
| Avg East-West RTT | net.eastWestTcpRtt.absolute.average.microsecond | Max of Avg Round trip time tier-east west traffic |
| Avg Internet RTT | net.internetTcpRtt.absolute.average.microsecond | Max of Avg Round trip time tier-internet traffic |
| Dropped Flows | net.droppedFlows.count.absolute.latest.number | Total number of dropped flows associated with the tier to the total number of flows. |
| East-West ReTransmitted Ratio | net.tcpRetransmitedEastWestPackets.ratio.average.percent | Max of Avg Retransmission rate-east west traffic |
| Internet Dropped Flows | net.droppedInternetFlows.count.absolute.latest.number | Total number of dropped internet flows associated with the tier to the total number of internet flows. |
| Median TCP RTT | net.p50FlowAvgTcpRtt.absolute.average.microsecond | Median round trip time of all the TCP flows associated with the tier. |
| Median TCP RTT Internet Flows | net.p50InternetFlowAvgTcpRtt.absolute.average.microsecond | Median round trip time of all the internet TCP flows associated with the tier. |
| Network Packet Count | net.packetsTotal.delta.summation.number | Total packets received and transmitted from a tier in the selected duration. |
| Network Rx Rate | net.received.rate.average.kiloBitsPerSecond | Average rate at which data is received across all endpoints in the tier. |
| Network Tx Rate | net.transmitted.rate.average.kiloBitsPerSecond | Average rate at which data is transmitted across all endpoints in the tier. |
| p95 TCP RTT | net.p95FlowAvgTcpRtt.absolute.average.microsecond | p95 latency round trip time of all the TCP flows associated with the tier. |
| Rx Flow Drops | flow.droppedRx.delta.summation.number | Total number of dropped incoming flows to a tier. |
| Rx Internet Flow Drops | flow.droppedRxInternet.delta.summation.number | Total number of dropped incoming internet flows to a tier. |
| TCP Retransmitted Packet Ratio | net.tcpRetransmitedPackets.ratio.average.percent | Percentage of retransmitted packets in all the flows associated with the tier. |

| Metric Name | Metric API Name | Description |
|---|---|---|
| Total Internet Packets | net.packetsTotalInternet.delta.summation.number | Total number of internet packets received and transmitted from the tier. |
| Total Rx Bytes | net.totalRx.delta.summation.bytes | Sum of bytes received during the time interval. |
| Total Tx Bytes | net.totalTx.delta.summation.bytes | Sum of bytes transmitted during the time interval. |
| Tx Flow Drops | flow.droppedTx.delta.summation.number | Total number of dropped outgoing flows to total number of outgoing flows. |
| Tx Internet Flow Drops | flow.droppedTxInternet.delta.summation.number | Total number of dropped outgoing internet flows to total number of outgoing internet flows. |

## Virtual Connect Manager Metrics

VCF Operations for networks collects flow metrics for Virtual Connect Managers.

| Metric Name | Metric API Name | Description |
|---|---|---|
| Outbound Expire Flow Drop Count | flow.fiveTupleExpireOutbound.delta.summation.number | Number of five tuples dropped because of reaching the maximum outbound limit for source IP addresses in flows. |
| Wrong Collector Flow Drop Count | flow.rawRecordReporterWrongCollector.delta.summation.number | Number of raw flow records dropped as they were reported on a wrong type of VCF Operations for networks collector. |

## VLAN Metrics

Virtual local area network (VLAN) metrics provide information about network the performance of the VLAN.

| Metric Name | Metric API Name | Description |
|---|---|---|
| Dropped Rx Packets | net.droppedRx.delta.summation.number | Number of dropped Rx packets in the selected duration. |
| Dropped Tx Packets | net.droppedTx.delta.summation.number | Number of dropped Tx packets in the selected duration. |
| Max Network Traffic Rate | net.usage.rate.maximum.kiloBitsPerSecond | Maximum network usage rate over any five minute interval in the selected duration. |
| Max Network Tx Rate | net.transmitted.rate.maximum.kiloBitsPerSecond | Maximum data transmission rate over any five minute interval in the selected duration. |
| Max Packet Drop Ratio | net.packetDrop.ratio.maximum.percent | Maximum ratio of total packets dropped to total number of packets received over any five minute interval in the selected duration. |
| Max Packet Drops | net.droppedTotal.delta.maximum.number | Maximum number of dropped packets over any five minute interval in the selected duration. |

| Metric Name | Metric API Name | Description |
|---|---|---|
| Multicast Rx Packets | net.multicastRx.delta.summation.number | Total number of multicast packets received during the sampling interval. |
| Multicast Tx Packets | net.multicastTx.delta.summation.number | Total number of multicast packets transmitted during the sampling interval. |
| Network Max Rx Rate | net.received.rate.maximum.kiloBitsPerSecond | Maximum network data receive rate on the host over any five minute interval in the selected duration. |
| Network Packet Count | net.packetsTotal.delta.summation.number | Total number of packets transmitted and received during the time interval. |
| Network Packet Drop Percent | net.packetDrop.ratio.average.percent | Ratio of total number of dropped packets to total number of packets. |
| Network Packet Drops | net.droppedTotal.delta.summation.number | Total number of packet drops. |
| Network Rx Rate | net.received.rate.average.kiloBitsPerSecond | Average rate at which data was received during the time interval.<br>This metric represents the bandwidth of the network. |
| Network Traffic Rate | net.usage.rate.average.kiloBitsPerSecond | Average rate of received and transmitted traffic during the time interval. |
| Network Tx Rate | net.transmitted.rate.average.kiloBitsPerSecond | Average rate at which data was transmitted during the time interval.<br>This metric represents the bandwidth of the network. |
| Rx Broadcast Packets | net.broadcastRx.delta.summation.number | Number of broadcast packets received during a specified time interval. |
| Rx Broadcast Ratio | net.broadcastByRx.ratio.average.percent | Percentage of broadcast packets received during a specified time interval. |
| Rx Packet Drop Percent | net.packetDropRx.ratio.average.percent | Ratio of total number of dropped Rx packets to total number of Rx packets on all interfaces that are part of the VLAN. |
| Rx Packets | net.packetsRx.delta.summation.number | Average number of received packets during the time interval. |
| Total Network Traffic | net.usage.delta.summation.kiloBytes | Sum of data transmitted and received during the time interval. |
| Tx Broadcast Packets | net.broadcastTx.delta.summation.number | Number of broadcast packets transmitted during a specified time interval. |
| Tx Broadcast Ratio | net.broadcastByTx.ratio.average.percent | Average percent of broadcast packets transmitted during a specified time interval. |
| Tx Packet Drop Percent | net.packetDropTx.ratio.average.percent | Ratio of total number of dropped Tx packets to total number of Tx packets on all interfaces that are part of the VLAN. |

| Metric Name | Metric API Name | Description |
|---|---|---|
| Tx Packets | net.packetsTx.delta.summation.number | Average number of transmitted packets during the time interval. |
| Unicast Rx Packets | net.unicastRx.delta.summation.number | Total number of unicast packets received by the host. |
| Unicast Tx Packets | net.unicastTx.delta.summation.number | Total number of unicast packets transmitted by the host. |

### Virtual Disk Metrics

Virtual disk metrics provide information about the performance of the virtual disk.

| Metric Name | Metric API Name | Description |
|---|---|---|
| Read IOPS | virtualDisk.numberReadAveraged.rate.average.number | Average number of read commands issued per second on the virtual disk. |
| Read Latency (ms) | virtualDisk.totalReadLatency.absolute.average.millisecond | Read latency in milliseconds averaged over a period of 24 hours. |
| Read Rate (kbps) | virtualDisk.read.rate.average.kiloBytesPerSecond | Read throughput averaged over a period of 24 hours. |
| RW IOPS | virtualDisk.numberRWAveraged.rate.average.number | Average number of read and write commands issued on the virtual disk over a period of 24 hours. |
| RW Rate (kbps) | virtualDisk.usage.rate.average.kiloBytesPerSecond | Read and write throughput averaged over a period of 24 hours. |
| Used Space | virtualDisk.used.absolute.latest.bytes | Total space used by the virtual disk. |
| Write IOPS | virtualDisk.numberWriteAveraged.rate.average.number | Average number of write commands issued per second on the virtual disk. |
| Write Latency (ms) | virtualDisk.totalWriteLatency.absolute.average.millisecond | Write latency in milliseconds averaged over a period of 24 hours. |
| Write Rate (kbps) | virtualDisk.write.rate.average.kiloBytesPerSecond | Write throughput in milliseconds averaged over a period of 24 hours. |

### VM Datastore Metrics

Virtual Machine Datastore metrics provide information about the performance of the link that connects the datastore to the VM.

| Metric Name | Metric API Name | Decription |
|---|---|---|
| Committed Space | datastore.committed.absolute.latest.bytes | Storage space on the datastore that is commited for this link. |
| Read IOPS | datastore.numberReadAveraged.rate.average.number | Average number of read commands issued per second during the collection interval. |
| Read Latency (ms) | datastore.totalReadLatency.absolute.average.millisecond | Average latency of each read operation issued by the link through which the host is connected to the datastore. |

| Metric Name | Metric API Name | Decription |
|---|---|---|
| Read Rate (Kbps) | datastore.read.rate.average.kiloBytesPerSecond | Average number of kilobytes read by the link through which the host is connected to the datastore. |
| RW IOPS | datastore.numberRWAveraged.rate.average.number | Average number of read and write commands issued per second during the collection interval. |
| RW Rate (Kbps) | datastore.usage.rate.average.kiloBytesPerSecond | The aggregated input rates and output rates of the link through which the host is connected to the datastore. |
| Write IOPS | datastore.numberWriteAveraged.rate.average.number | Average number of write commands issued per second during the collection interval. |
| Write Latency (ms) | datastore.totalWriteLatency.absolute.average.millisecond | Average latency of each write operation issued by the link through which the host is connected to the datastore. |
| Write Rate (Kbps) | datastore.write.rate.average.kiloBytesPerSecond | Average number of kilobytes written by the link through which the host is connected to the datastore. |

## Virtual Machine Metrics

VCF Operations for networks collects CPU usage, datastore, memory, and network metrics for VMs.

### CPU Usage Metrics

CPU usage metrics provide information about the CPU use on the VM.

| Metric Name | Metric API Name | Description | Supported by Databus |
|---|---|---|---|
| CPU Co-Stop Rate (%) | cpu.costop.rate.average.percent | Average percentage of time spent by the vCPUs in the `CoStop` state due to co-vCPU scheduling contention. | |
| CPU Co-Stop Rate vCPU (%) | cpu.costop_vcpu.rate.average.percent | Average percentage of time spent by a vCPU in the `CoStop` state due to co-vCPU scheduling contention. | |
| CPU Co-Stop Time | cpu.costop.delta.summation.millisecond | Time spent in the `CoStop` state by all vCPUs of a VM. | |
| CPU Ready Rate (%) | cpu.ready.rate.average.percent | Percentage of time that the VM was waiting in line to use the CPU on the host.<br>The CPU ready time depends on the number of VMs on the host and their CPU loads. | Yes |
| CPU Ready Rate vCPU (%) | cpu.ready_vcpu.rate.average.percent | Percentage of time per vCPU that the VM was waiting in line to use the CPU on the host.<br>The CPU ready time depends on the number of VMs on the host and their CPU loads. | |

| Metric Name | Metric API Name | Description | Supported by Databus |
|---|---|---|---|
| CPU Ready Time | cpu.ready.delta.summation.millisecond | Amount of time a VM is ready to use the CPU but was unable to schedule physical CPU time. The VM cannot schedule physical CPU time when all the VMware vSphere ESX host CPU resources are busy. | |
| CPU Run Rate (%) | cpu.run.rate.average.percent | Average percentage of time spent in the `Run` state by all vCPUs of this VM. | |
| CPU Run Rate vCPU (%) | cpu.run_vcpu.rate.average.percent | Average percentage of time spent by a vCPU in the `Run` state. | |
| CPU Run Time | cpu.run.delta.summation.millisecond | Amount of time spent in the `Run` state by all vCPUs of this VM. | |
| CPU Usage (MHz) | cpu.usagemhz.rate.average.megaHertz | Amount of actively used vCPU. The CPU Usage metric is associated with the host and not the guest operating system. | |
| CPU Usage Rate (%) | cpu.usage.rate.average.percent | Average percentage of actively used vCPU to the total available CPU. The CPU Usage metric is associated with the host and not the guest operating system. CPU usage rate indicates the average CPU utilization over all available vCPUs in the VM. For example, if a VM with one vCPU is running on a host that has four physical CPUs and the CPU usage is 100%, the VM is using a 100% of one physical CPU. | Yes |
| CPU Wait Rate (%) | cpu.wait.rate.average.percent | Percentage of CPU time spent in the `Wait` state such as CPU idle, CPU swap wait, and CPU I/O wait. | |
| CPU Wait Rate vCPU (%) | cpu.wait_vcpu.rate.average.percent | Percentage of CPU time per vCPU spent in the `Wait` state such as CPU idle, CPU swap wait, and CPU I/O wait. | |
| CPU Wait Time | cpu.wait.delta.summation.millisecond | Amount of CPU time spent in the `Wait` state such as CPU idle, CPU swap wait, and CPU I/O wait. | |

## Network Metrics

Network metrics provide information about the network performance of the VM.

| Metric Name | Metric API Name | Description | Supported by Databus |
|---|---|---|---|
| Dropped Rx Packets | net.droppedRx.delta.summation.number | Number of dropped Rx packets to the total number of Rx packets. | Yes |
| Incomplete TCP Session Count | net.incomplete.sessionCount.delta.summation.number | Total number of incomplete TCP sessions received at the VM from other VMs or IP addresses during the time interval.<br>**Note:**<br>This metric is available only for Enterprise License users having a deployment of XXL brick Size Collector.<br>If you scale up the brick size of your existing collector to XXL, you must restart the `nfcapd` service to start collecting the Incomplete TCP Session Count metric.<br>Run the following command on the Collector CLI:<br>`sudo service nfcapd restart` | Yes |
| Dropped Tx Packets | net.droppedTx.delta.summation.number | Number of dropped Tx packets to the total number of Tx packets. | Yes |
| Max Network Traffic Rate (Kbps) | net.usage.rate.maximum.kiloBitsPerSecond | Maximum data transmission rate on the VM over any five minute interval in the selected duration. Maximum network rate includes both transmitted and received data. | Yes |
| Max Network Tx Rate (Kbps) | net.transmitted.rate.maximum.kiloBitsPerSecond | Maximum data transmission rate on the VM over any five minute interval in the selected duration. | |
| Max Packet Drop Ratio (%) | net.packetDrop.ratio.maximum.percent | Maximum ratio of total packets dropped to total number of packets received over any five minute interval in the selected duration. | |
| Max Packet Drops | net.droppedTotal.delta.maximum.number | Maximum number of dropped packets over any five minute interval in the selected duration. | Yes |

| Metric Name | Metric API Name | Description | Supported by Databus |
|---|---|---|---|
| Multicast Rx Packets | net.multicastRx.delta.summation.number | Total number of multicast packets received across all pNICs (physical network adapters) during the sampling interval. | |
| Multicast Tx Packets | net.multicastTx.delta.summation.number | Total number of multicast packets transmitted across all pNICs during the sampling interval. | Yes |
| Network Max Rx Rate (Kbps) | net.received.rate.maximum.kiloBitsPerSecond | Maximum network data receive rate on the VM over any five minute interval in the selected duration. | |
| Network Packet Count | net.packetsTotal.delta.summation.number | Total number of packets transmitted and received during the interval across all pNICs. | Yes |
| Network Packet Drop Percent (%) | net.packetDrop.ratio.average.percent | Ratio of total number of dropped packets to total number of packets. | Yes |
| Network Packet Drops | net.droppedTotal.delta.summation.number | Total number of packet drops on the VM. | Yes |
| Network Rx Rate (Kbps) | net.received.rate.average.kiloBitsPerSecond | Average rate at which data was received during the time interval. | Yes |
| Network Traffic Rate (Kbps) | net.usage.rate.average.kiloBitsPerSecond | Average rate at which data was received and transmitted during the time interval. | Yes |
| Network Tx Rate (Kbps) | net.transmitted.rate.average.kiloBitsPerSecond | Average rate at which data was transmitted during the time interval. | Yes |
| Rx Broadcast Packets | net.broadcastRx.delta.summation.number | Total number of broadcast packets received during the time interval. | |
| Rx Broadcast Ratio (%) | net.broadcastByRx.ratio.average.percent | Percentage of broadcast packets received during the time interval. | |
| Rx Packet Drop Percent (%) | net.packetDropRx.ratio.average.percent | Ratio of total number of dropped Rx packets to total number of Rx packets. | |
| Rx Packets | net.packetsRx.delta.summation.number | Average number of received packets during the time interval. | Yes |
| Total Network Traffic | net.usage.delta.summation.kiloBytes | Sum of data transmitted and received across all vNICs of the VM. | Yes |
| Total Packets Per Second | net.ppsTotal.rate.average.number | Average rate of received and transmitted packets per second. | |
| Tx Broadcast Packets | net.broadcastTx.delta.summation.number | Number of broadcast packets transmitted during a specified time interval. | |

| Metric Name | Metric API Name | Description | Supported by Databus |
|---|---|---|---|
| Tx Broadcast Ratio (%) | net.broadcastByTx.ratio.average.percent | Average percentage of broadcast packets transmitted during the time interval. | |
| Tx Packet Drop Percent (%) | net.packetDropTx.ratio.average.percent | Ratio of total number of dropped Tx packets to total number of Tx packets. | |
| Tx Packets | net.packetsTx.delta.summation.number | Average number of transmitted packets during the time interval. | Yes |
| Unicast Rx Packets | net.unicastRx.delta.summation.number | Total number of unicast packets received by the host. | |
| Unicast Tx Packets | net.unicastTx.delta.summation.number | Total number of unicast packets transmitted by the host. | |
| Uptime | sys.uptime.absolute.latest.second | Total time elapsed since the last system startup. | |
| Uncommitted Space | storage.uncommitted.absolute.latest.bytes | Uncommitted space of VM storage. | |
| Snapshot Size | snapshot.used.absolute.latest.bytes | The size of the VM snapshot. | |

## Datastore Metrics

Datastore metrics provide information about the datastore use.

| Metric Name | Metric API Name | Decription |
|---|---|---|
| Read IOPS | datastore.numberReadAveraged.rate.average.number | Average number of read commands issued per second during the collection interval. |
| Max IO Latency | datastore.maxTotalLatency.absolute.latest.millisecond | Highest latency value across all datastores used by the VM. |
| Read Latency (ms) | datastore.totalReadLatency.absolute.average.millisecond | Average latency of each read operation issued by this VM on its datastore. |
| Read Rate (Kbps) | datastore.read.rate.average.kiloBytesPerSecond | Read throughput of datastores connected to the VM averaged over 24 hours. |
| RW IOPS | datastore.numberRWAveraged.rate.average.number | Average number of read and write commands sent from the VM to the connected datastores. |
| RW Rate (Kbps) | datastore.usage.rate.average.kiloBytesPerSecond | Read and write throughput of datastores connected to the VM averaged over 24 hours. |
| Write IOPS | datastore.numberWriteAveraged.rate.average.number | Average number of write commands sent from the VM to the connected datastores. |
| Write Latency | datastore.totalWriteLatency.absolute.average.millisecond | Average latency of each write operation issued by this VM on its datastore. |
| Write Rate (Kbps) | datastore.write.rate.average.kiloBytesPerSecond | Write throughput of datastores connected to the VM averaged over 24 hours. |

## Memory Metrics for Hosts

Memory metrics provide information about memory use and allocation on the virutal machine.

| Metric Name | Metric API Name | Decription | Supported by Databus |
|---|---|---|---|
| Memory Balloon | mem.vmmemctl.absolute.average.kiloBytes | Average memory of the guest physical memory allocated to `vmmemctl` for this VM. `vmmemctl` is a VMware exclusive memory-management driver that controls ballooning. | |
| Memory Swap-In Rate (Kbps) | mem.swapinRate.rate.average.kiloBytesPerSecond | Rate at which the memory is swapped from disk to active memory during the time interval. | |
| Memory Swap-Out Rate (Kbps) | mem.swapoutRate.rate.average.kiloBytesPerSecond | Rate at which the memory is swapped from active memory to the disk during the time interval. | |
| memory Swapped | mem.swapped.absolute.average.kiloBytes | Amount of guest physical memory swapped out to the VM's swap file by the VMkernel. The swapped memory stays on the disk until the VM needs it. This metric refers to VMkernel swapping and not to guest OS swapping. | |
| Memory Usage Rate (%) | mem.usage.absolute.average.percent | Percentage of memory used from the available machine memory. | Yes |
| Overhead Memory | mem.overhead.absolute.average.kiloBytes | Amount of memory allocated to a VM beyond its reserved amount.<br>The VM requires overhead memory to power on.<br>Overhead memory includes the space reserved for the VM frame buffer and various virtualization data structures.<br>It depends on the number of vCPUs and the configured memory for the guest operating system. | |
| Consumed Memory | mem.consumed.absolute.average.kiloBytes | Amount of host memory used by the VM for its guest operating system's physical memory.<br>Consumed memory does not include the overhead memory.<br>It only includes shared memory and memory that might be reserved, but not used. That is, `consumed memory = memory granted - memory saved from page sharing`. | |

| Metric Name | Metric API Name | Decription | Supported by Databus |
|---|---|---|---|
| Active Memory | mem.active.absolute.average.kiloBytes | Amount of guest physical memory that is in use by the VM.<br>Active memory is estimated using VMkernel statistical sampling. It represents the actual amount of memory the VM needs. Therefore, the unused memory can be swapped out or ballooned without any performance impact. | |

### Snapshot Metrics

The snapshot metric provide information about the VM snapshot.

| Metric Name | Metric API Name | Decription |
|---|---|---|
| Snapshot Size | snapshot.used.absolute.latest.bytes | The size of the VM snapshot. |

## Virtual Network Interface Card Metrics

Virtual network interface card (vNIC) metrics provide information about the performance of the virtual network interface card.

| Metric Name | Metric API Name | Description |
|---|---|---|
| Dropped Rx Packets | net.droppedRx.delta.summation.number | Number of received packets dropped during the collection interval. |
| Dropped Tx Packets | net.droppedTx.delta.summation.number | Number of transmitted packets dropped during the collection interval. |
| Interface Rx Utilization (%) | net.rx.utilization.absolute.latest.percent | Interface utilization percentage for the received traffic. |
| Interface Tx Utilization (%) | net.tx.utilization.absolute.latest.percent | Interface utilization percentage for the transmitted traffic. |
| Interface Utilization (%) | net.utilization.absolute.latest.percent | Interface utilization percentage for received and transmitted traffic. |
| Max Network Traffic Rate (Kbps) | net.usage.rate.maximum.kiloBytesPerSecond | Maximum total network usage rate over a five minute interval. |
| Max Network Tx Rate (Kbps) | net.transmitted.rate.maximum.kiloBytesPerSecond | Maximum network data transmit rate over five minute interval. |
| Max Packet Drop Ratio (%) | net.packetDrop.ratio.maximum.percent | Maximum five minute interval ratio of total dropped packets to total number packets. |
| Max Packet Drops | net.droppedTotal.delta.maximum.number | Maximum number of dropped packets over any five minute interval in the selected duration. |
| Multicast Rx Packets | net.multicastRx.delta.summation.number | Number of multicast packets received during the time interval. |

| Metric Name | Metric API Name | Description |
|---|---|---|
| Multicast Tx Packets | net.multicastTx.delta.summation.number | Number of multicast packets transmitted during the time interval. |
| Network Max Rx Rate (Kbps) | net.received.rate.maximum.kiloBytesPerSecond | Maximum network data receive rate over any five minute interval. |
| Network Packet Count | net.packetsTotal.delta.summation.number | Total number of packets transmitted and received during the time interval. |
| Network Packet Drop Percent (%) | net.packetDrop.ratio.average.percent | Average ratio of dropped packets to total number packets. |
| Network Packet Drops | net.droppedTotal.delta.summation.number | Total number of packet drops. |
| Network Rx Rate (Kbps) | net.received.rate.average.kiloBytesPerSecond | Average rate at which data was received during the time interval.<br>This metric represents the bandwidth of the network. |
| Network Rx Rate (Kbps) | net.received.rate.average.kiloBytesPerSecond | Average rate at which data was received during the time interval.<br>This metric represents the bandwidth of the network. |
| Network Traffic Rate (Kbps) | net.usage.rate.average.kiloBytesPerSecond | Average rate of received and transmitted traffic during the time interval. |
| Network Tx Rate (Kbps) | net.transmitted.rate.average.kiloBytesPerSecond | Average rate at which data was transmitted during the interval.<br>This metric represents the bandwidth of the network. |
| Rx Broadcast Packets | net.broadcastRx.delta.summation.number | Number of broadcast packets received during the time interval. |
| Rx Broadcast Ratio (%) | net.broadcastByRx.ratio.average.percent | Ratio of received broadcast packets to total number of received packets. |
| Rx Packet Drop Percent (%) | net.packetDropRx.ratio.average.percent | Average ratio of dropped received packets to total received packets. |
| Rx Packets | net.packetsRx.delta.summation.number | Number of packets received during the time interval. |
| Total Network Traffic | net.usage.delta.summation.kiloBytes | Sum of data transmitted and received during the time interval. |
| Tx Broadcast Packets | net.broadcastTx.delta.summation.number | Number of broadcast packets transmitted during the time interval. |
| Tx Broadcast Ratio (%) | net.broadcastByTx.ratio.average.percent | Ratio of transmitted broadcast packets to total transmitted packets. |

| Metric Name | Metric API Name | Description |
|---|---|---|
| Tx Packet Drop Percent (%) | net.packetDropTx.ratio.average.percent | Average ratio of dropped transmitted packets to total transmitted packets. |
| Tx Packets | net.packetsTx.delta.summation.number | Number of packets transmitted during the time interval. |
| Unicast Rx Packets | net.unicastRx.delta.summation.number | Total number of unicast packets received during the time interval. |
| Unicast Tx Packets | net.unicastTx.delta.summation.number | Total number of unicast packets transmitted during the time interval. |

## Virtual Network Interface Metrics

Virtual network interface (Vmknic) metrics provide information about the performance of the virtual network interface that is used by the VMKernel.

| Metric Name | Metric API Name | Description |
|---|---|---|
| Carrier Losses Detected | net.txcarrier.delta.summation.number | Number of carrier losses detected at the vmknic. |
| Collisions Detected | net.txcolls.delta.summation.number | Number of collisions detected at the vmknic. |
| Dropped Rx Packets | net.droppedRx.delta.summation.number | Average ratio of the number of dropped Rx packets to the total number of Rx packets. |
| Dropped Tx Packets | net.droppedTx.delta.summation.number | Average ratio of the number of dropped Tx packets to the total number of Tx packets. |
| Maximum Total Network Rate | net.total.rate.maximum.kiloBitsPerSecond | Maximum total network rate over any five minute interval in the selected duration. |
| Network Maximum Rx Rate | net.rx.rate.maximum.kiloBitsPerSecond | Maximum network data receive rate on the network interface over any five minute interval in the selected duration. |
| Network Maximum Tx Rate | net.tx.rate.maximum.kiloBitsPerSecond | Maximum network data transmission rate on the network interface over any five minute interval in the selected duration. |
| Network Packet Count | net.packetsTotal.delta.summation.number | Total number of packets transmitted and received during the time interval. |
| Network Packet Drops | net.droppedTotal.delta.summation.number | Total number of packet drops on the vmknic. |
| Rx Frame Errors | net.rxframes.delta.summation.number | Number of frame errors received over the network in the selected duration. |
| Rx Packet Drop Rate | net.packetDropReceived.ratio.average.percent | Average ratio of dropped Rx packets to the total number of Rx packets. |

| Metric Name | Metric API Name | Description |
|---|---|---|
| Rx Packet Errors | net.errorRx.delta.summation.number | Number of received packet errors detected over the network in the selected duration. |
| Rx Packet Per Second | net.ppsRx.rate.average.number | Average rate at which packets are received per second. |
| Rx Packets | net.packetsRx.delta.summation.number | Number of packets received over the time interval. |
| Total Packet Per Second | net.ppsTotal.rate.average.number | Average number of packets received and transmitted per second. |
| Total Rx Bytes | net.totalRx.delta.summation.bytes | Total number of bytes received by the vmknic. |
| Total Tx Bytes | net.totalTx.delta.summation.bytes | Total number of bytes transmitted by the vmknic. |
| Tx Packet Drop Rate | net.packetDropTransmitted.ratio.average.percent | Average ratio of dropped Tx packets to total Tx packets. |
| Tx Packet Errors | net.errorTx.delta.summation.number | Number of transmitted packet errors detected over the network in the selected duration. |
| Tx Packet Per Second | net.ppsTx.rate.average.number | Average rate at which packets are transmitted per second. |
| Tx Packets | net.packetsTx.delta.summation.number | Number of packets transmitted over the time interval. |

## Flow Metrics

VCF Operations for networks collects the flow data from various data sources in your network that has flow export capabilities, such as VMware vSphere Distributed Switch and VMware vDefend Distributed Firewall, and calculates traffic information.

To view the flow metrics on the user interface, do the following:

1. Enter `Flows` on the search bar.
2. On the search result page, click **Add More Filters** under the **Filters** option.
3. Select **Metrics**. The metrics are displayed to the right.
4. Select the metric you want to view.

**Table 1354: Flow Metrics**

| Metric Name | Metric API Name | Description | Supported by Databus |
|---|---|---|---|
| Active Sessions Count | flow.openSessionCount.absolute.latest.number | The total number of active or open sessions that are aggregated into this flow.<br>A session consists of 5 tuples (source IP, destination IP, source port, destination port, and protocol).<br>The aggregated flow consists of 4 tuples (source IP, destination IP, destination port, and protocol).<br>For a client server session, incoming and outgoing sessions are aggregated into a single flow. | |
| Average TCP RTT (ms) | flow.nsxTcpRtt.absolute.average.microsecond | Average TCP RTT of the flow within the specified duration. If no duration is specified, the default is last 24 hours. | |
| Destination Bytes | flow.dstBytes.delta.summation.bytes | Total number of bytes sent by the destination of the flow. | Yes |
| Flow Packet Count | flow.totalPackets.delta.summation.number | Total number of packets exchanged between the source and the destination of a flow.<br>A flow consists of 4 tuples (source IP, destination IP, destination port and protocol) and is an aggregation of sessions identified by 5 tuples (source IP, destination IP, source port, destination port and protocol).<br>For a client server connection, incoming and outgoing sessions are aggregated into a single flow. | Yes |
| Maximum TCP RTT | flow.nsxTcpRtt.absolute.maximum.microsecond | Maximum TCP RTT of the flow within the specified duration. If no duration is specified, the default is last 24 hours. | |
| Packet Loss | flow.packetDrop.ratio.average.percent | Ratio of the number of packets dropped to the total number of packets transmitted. | |
| Retransmitted Packet Ratio (%) | flow.retransmittedPacketsTx.ratio.average.percent | Ratio of the number of packets retransmitted to the total number of packets transmitted. | |
| Sessions | flow.allowedSessionCount.delta.summation.number | The total number of established sessions that have been aggregated into a flow.<br>A session consists of 5 tuples (source IP, destination IP, source port, destination port and protocol). The aggregated flow consists of 4 tuples (source IP, destination IP, destination port and protocol).<br>For a client server session, incoming and outgoing sessions are aggregated into a single flow. | Yes |
| Source Bytes | flow.srcBytes.delta.summation.bytes | Total number of bytes sent by the source of the flow. | Yes |
| TCP Retransmission Ratio (%) | flow.tcpRetransmittedPackets.ratio.maximum.percent | Ratio of TCP retransmitted packets to total number of TCP transmitted packets. | Yes |

| Metric Name | Metric API Name | Description | Supported by Databus |
|---|---|---|---|
| Total Traffic | flow.totalBytes.delta.summation.bytes | Total number of bytes exchanged between the source and destination of a flow. A flow consists of 4 tuples (source IP, destination IP, destination port and protocol) and is an aggregation of sessions identified by 5 tuples (source IP, destination IP, source port, destination port and protocol).<br>For a client server connection, incoming and outgoing sessions are aggregated into a single flow. | Yes |
| Traffic Rate | flow.totalBytesRate.rate.average.bitsPerSecond | Average of incoming and outgoing traffic. | Yes |

To learn about the retention period of flow metrics, see Metric Definitions.

## Health Metrics

VCF Operations for networks collects health metrics from the Collector, Platform, and system to diagnose and monitor health problems related to the VCF Operations for networks instance.

Health metrics are of the following types:

- Node metrics
- Service metrics
- System metrics

### Viewing Health Metrics

You can view the node and service metrics on VCF Operations for networksPlatform and Collector pages. Where as, you can view the system metrics only on the **System Dashboard** page.

To view node and service metrics:

1. Enter `Platform` or `Collector` on the search bar.
2. On the search results page, select a Platform or Collector to view the health metrics available for that entity.

To view the system metrics, Enter `System Dashboard` on the search bar.

### Node Metrics

Node metrics provide information about the memory, disk IO, and CPU utilization of a node.

| Metric Name | Metric API Name | Descrip |
|---|---|---|
| Memory Usage | CUSTOM_METRIC[level:node]vRNI.used.memory.percentage.rate.average.number | Percent configur |
| Data Disk Usage | CUSTOM_METRIC[level:node]vRNI.used.disk.percentage.rate.average.number | Percent given no |
| CPU Usage | CUSTOM_METRIC[level:node]vRNI.cpu.utilization.percentage.rate.average.number | Percent availabl |

### Service Metrics

Service metrics indicate if a service is running. In the metric API name, you must replace `<service_name>` with your service name (such as IpfixProcessor, ElasticSearch, Kafka) to view the state of the service.

| Metric Name | Metric API Name | Descrip |
|---|---|---|
| Service Uptime | CUSTOM_METRIC[service.name:<service_name>]vRNI.service.uptime.rate.average.number | Binary i for netw • 0 de heal • 1 de is no • 2 de |

### System Metrics

System metrics provide information about the performance and usage of the overall VCF Operations for networks system.

| Metric Name | Metric API Name | Descrip |
|---|---|---|
| Processing Lag | grid.messageAge.absolute.latest.millisecond | The lag grid. High pro showing |
| Grid Usage | grid.busy.absolute.latest.percent | The cap network The utili the proc in poten lag. Adding and dele count ar the proc |
| Indexer Lag | grid.indexerLag.absolute.latest.millisecond | The lag High inc showing |
| VM Count | vRNI.internal.tenants.usage.vm.count.absolute.maximum.number | Total co |
| Host Count | vRNI.internal.tenants.usage.host.count.absolute.maximum.numb | Total co |
| Application Count | vRNI.internal.tenants.usage.application.count.absolute.maximum.number | Total co system. |
| Daily Flow Count | vRNI.internal.tenants.usage.flow.daily.count.absolute.maximum.number | Total co |
| Weekly Flow Count | vRNI.internal.tenants.usage.flow.weekly.count.absolute.maximum.number | Total co days. |
| Firewall Rule Count | vRNI.internal.tenants.usage.firewallRule.count.absolute.maximum.number | Total co |

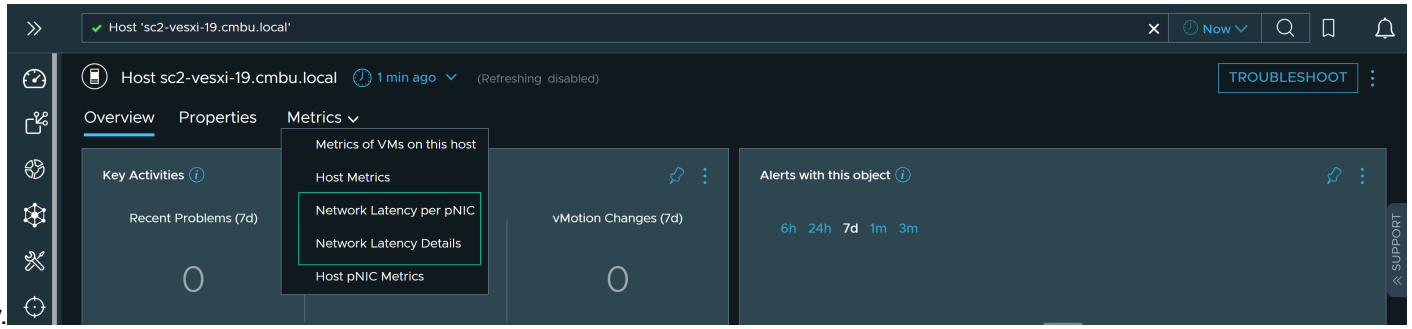
## Latency Metrics

Latency metrics provide information about the latency of a network. You can monitor the latency to diagnose and troubleshoot performance bottlenecks in your network.

### Viewing Latency Metrics

To view the latency metrics on the user interface, do the following:

1. Enter `Host`, `VMware VM`, or `NSX-T transport node` on the search bar. Latency metrics are available only for these entities.
2. On the search results page, click the entity you want to view.
3. On the entity page, select the **Metrics** tab to view the latency metrics available for that entity.




**Note:**

To view latency metrics, you must have selected the **Enable latency metric collection** check box when adding NSX managers as data sources in VCF Operations for networks.

## Interface Pair Metrics

Interface pair metrics provide information regarding the latency of the data packets traveling between the source and destination VXLAN tunnel endpoints (VTEPs) or virtual network interface cards (vNICs).

| Metric Name | Metric API Name | Description |
|---|---|---|
| Max Interface Latency | netopa.interfacePairLatency.absolute.maximum.microsecond | Maximum round-trip time taken by the data packet to travel from VTEP to VTEP or from vNIC to vNIC. |
| Min InterfaceLatency | netopa.interfacePairLatency.absolute.minimum.microsecond | Minimum round-trip time taken by the data packet to travel from VTEP to VTEP or from vNIC to vNIC. |
| Interface Latency | netopa.interfacePairLatency.absolute.average.microsecond | Average round-trip time taken by the data packet to travel from VTEP to VTEP or from vNIC to vNIC. |

## Interface pNIC Pair Info Metrics

Interface Physical Network Interface Card (pNIC) pair info metrics provide information about the following network latency statistics:

- pNIC to vNIC
- vNIC to pNIC

| Metric Name | Metric API Name | Description |
|---|---|---|
| Max pNIC vNIC Latency | netopa.pnicInterfaceLatency.absolute.maximum.microsecond | Maximum round-trip time taken by the data packet to travel from pNIC to vNIC. |
| Max vNIC pNIC Latency | netopa.interfacePnicLatency.absolute.maximum.microsecond | Maximum round-trip time taken by the data packet to travel from vNIC to pNIC. |

| Metric Name | Metric API Name | Description |
|---|---|---|
| Min pNIC vNIC Latency | netopa.pnicInterfaceLatency.absolute.minimum.microsecond | Minimum round-trip time taken by the data packet to travel from pNIC to vNIC. |
| Min vNIC pNIC Latency | netopa.interfacePnicLatency.absolute.minimum.microsecond | Minimum round-trip time taken by the data packet to travel from vNIC to pNIC. |
| pNIC vNIC Latency | netopa.pnicInterfaceLatency.absolute.average.microsecond | Average round-trip time taken by the data packet to travel from pNIC to vNIC. |
| vNIC pNIC Latency | netopa.interfacePnicLatency.absolute.average.microsecond | Average round-trip time taken by the data packet to travel from vNIC to pNIC. |

### VM Network Utilization Metrics

VCF Operations for networks collects VM network utilization metrics to display the performance of VMs based on network utilization.

VM network utilization metrics track network activity and throughput of VMs. To view these metrics on the user interface, do the following:

1. Enter `VMware VM` on the search bar.
2. On the search result page, select a VM of your choice.
3. On the respective VM page, select **Network Utilization** under the **Metrics** tab.

| Metric Name | Metric API Name | Description | Supported by Databus |
|---|---|---|---|
| Rx Packet Per Second | net.ppsRx.rate.average.number | Average rate at which packets are received per second. | |
| Tx Packet Per Second | net.ppsTx.rate.average.number | Average rate at which packets are transmitted per second. | |
| Total Packet Per Second | net.ppsTotal.rate.average.number | Average number of packets received and transmitted per second. | |
| Network Traffic Rate (Kbps) | net.usage.rate.average.kiloBitsPerSecond | Average rate at which the network is utilized for transmitting and receiving packets across all vNICs (virtual network interface controller) during the interval. | Yes |
| Network Rx Rate (Kbps) | net.received.rate.average.kiloBitsPerSecond | Average rate at which data is received across the virtual machine's vNIC (virtual network interface controller) during the interval.<br>This metric represents the bandwidth of the network. | Yes |
| Network Tx Rate (Kbps) | net.transmitted.rate.average.kiloBitsPerSecond | Maximum network data rate (transmitted and received rates) during the selected duration. | Yes |

To learn about the retention period of VM network utilization metrics, see Metric Definitions.

## Troubleshooting and Known Limitations

## Common Data Source Errors

When you add a data source, you can come across several errors. This table contains the list of common errors with the cause and resolution for each.

**Table 1355:**

| Error Text | Cause | Resolution |
|---|---|---|
| Invalid Response from Data Source | VCF Operations for networks collector was unable to process the information received from the Data Source as the information was not in the expected format. | In some data providers this problem is observed intermittently and might go away in the next polling cycle. If it occurs consistently, contact support. |
| Data Source is not reachable from collector VM | Data source IP address on SSH/REST (port 22 or 443) is either not reachable from the VCF Operations for networks collector VM or the data source is not responding. This error occurs while adding the data source. | Verify connectivity to the data source from VCF Operations for networks collector VM on port 22 or 443. Make sure data source is up and running and the firewall is not blocking connection from VCF Operations for networks collector VM to the data source. |
| No NSX Controller found | An NSX Controller has been selected in the NSX Manager data source page but there is no NSX Controller installed. | Install an NSX Controller on NSX Manager and then select NSX Controller check box on the NSX Manager data source page. |
| Data source type or version mismatch | Provided data source IP Address/FQDN is not of selected data source type. | Verify that provided data source IP Address/FQDN is of selected data source type and version is supported by VCF Operations for networkst |
| Error connecting to data source | VCF Operations for networks collector VM is unable to connect to the data source. This error occurs after adding the data source. | Verify connectivity to the data source from VCF Operations for networks collector VM on port 22 or 443. Make sure that the data source is up and running and firewall is not blocking connection from VCF Operations for networks collector VM to the data source. |
| Not found | VCF Operations for networks collector VM is not found. | Check if pairing is done between VCF Operations for networks collector VM and VCF Operations for networks Platform VM. |
| Insufficient privileges to enable IPFix | The user who is trying to enable IPFIX in vCenter does not have the required privileges. | Provide adequate privileges to the user. For more information, see Add VMware vCenter. |
| IP/FQDN is invalid | The IP/FQDN provided on the data source page is not valid or does not exist. | Provide valid IP/FQDN address. |
| No data being received | VCF Operations for networks Platform VM is not receiving data from VCF Operations for networks collector VM for that data source. | Contact Support. |
| Invalid credentials | Provided credentials are invalid. | Provide the correct credentials. |
| Connection string is invalid | The IP/FQDN provided on data source page is not in proper format | Provide valid IP/FQDN address. |
| Recent data may not be available, due to processing lag | VCF Operations for networks Platform VM is overloaded and lagging behind in processing data. | Contact support. |

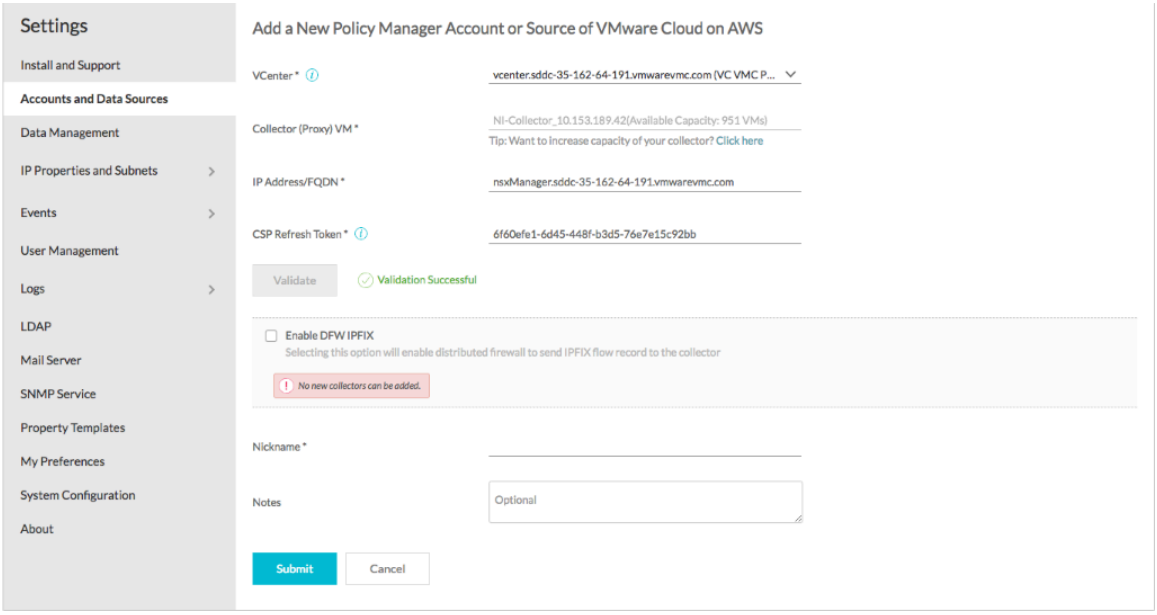
| Error Text | Cause | Resolution |
|---|---|---|
| Request timed out, please try again | VCF Operations for networks could not complete request in specified time. | Try again. If the issue is not fixed, then contact support. |
| Failed for unknown reason, please retry or contact support | Request failed for some unknown reason. | Try again. If the issue is not fixed, then contact support. |
| Password authentication for SSH needs to be enabled on device | SSH login using password is deactivated on the device added | Enable password authentication for SSH on the device being added for monitoring. |
| SNMP connection error | Error connecting to the SNMP port | Verify if SNMP is configured correctly on the target device. |

## Unable to Enable DFW IPFIX

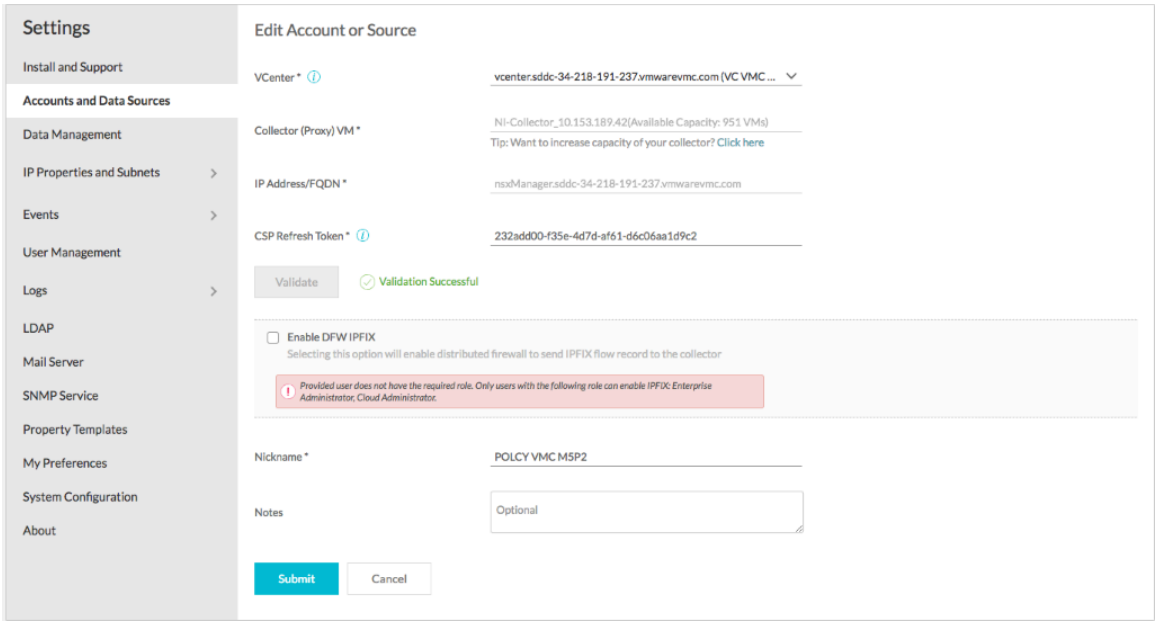
VCF Operations for networks does not allow you to enable DFW IPFIX.

While adding a policy manager or source of VMware Cloud on AWS, when you attempt to enable DFW IPFIX, you might see the following error messages:

- `No New collectors can be added.`
- `Provided user does not have the required role. Only users with the following role can enable IPFIX: Cloud Administrator.`
- VMware Cloud on AWS supports only four collectors to its DFW IPFIX collector profile. So, when the existing profile already has four collectors, you see the `No New collectors can be added` message.




- The user does not have the write permission. Only users with **Cloud Administrator** role can perform the write operation on the VMware Cloud on AWS policy manager.




- To add a new collector, you must:
  - Delete an existing collector, or
  - Create a new profile, or
- To avoid or fix the user role issue, perform one of the following steps:
  - Assign the **Cloud Administrator** role to the user, or
  - Log in as user with **Cloud Administrator** role.

## User Account Locked After Failed Login Attempts

You are not able to log in to VCF Operations for networks.

After several failed login attempts your account is locked, and you are not able to log in VCF Operations for networks user interface (UI).

After five consecutive failed login attempts, the system temporarily locks the account for 15 minutes. Each subsequent failed login attempt adds an additional 15 minutes to the lockout duration. If you keep trying to log in after the initial lockout, your account can get locked for a maximum of one day.

You can use the new password that you have received from the admin to log in to the VCF Operations for networks UI.

## Known Limitations

View the unsupported list of features or actions.

The following list contains features or actions that are currently not supported:

- If the collector is in shutdown state during a scheduled discovery operation, you see a double dash `(--)` in the last discovery status and discovery time.
- For virtual entities like VMs, VMware vCenter, NSX, and so on, the following intents don't work:
  - Duplicate IP Address
  - Duplicate MAC Address
  - HSRP/VRRP Active STP Root Colocation
  - HSRP/VRRP Configuration Error
  - Port Channel Member Mismatch
  - STP Path Cost Method Inconsistency
  - All STIG intents

- For Cisco Bridge Virtual Interface (BVI), the Duplicate IP Address intent might trigger false positive alerts for all the BVI member interfaces which share the same IP address.
- VCF Operations for networks supports only Per VLAN Spanning Tree (PVST) protocol for HSRP/VRRP Active STP Root Colocation intent.
- For Cisco ASA devices, the Default Password Existence intent doesn't raise an alert even though the default password is in use. This is because passwords in Cisco ASA configuration is unavailable in plain text and a same password carries different encrypted values for different users.
- In the Application Summary widget, the Intents count shows as two even when no intent is set for applications.
- In the VeloCloud Edge Summary widget, the Intents count shows as two even when no intent is set for VMware SD-WAN.
- End-to-end path search involving Cisco ACI and VMware NSX-T data sources does not work in the network map.
- The autocomplete feature is not supported for IPv6 addresses.
- Cisco ASA redundant interfaces are not included in Port Channel Member Mismatch intent check as the redundant interface to member interface mapping is not supported in VCF Operations for networks.
- If you sort firewall rules using Comment filter and if that rule comment contains double quote (""), then you see `Invalid query format in "Firewall Rule Comment is set"` as double quote ("") in a firewall rule comment is not supported in VCF Operations for networks.
- In the Network Map, when you search for Path that involves Cisco ASA BVI interfaces, you see an incomplete path as Cisco ASA BVI interfaces are not supported in Network Map.
- VCF Operations for networks currently shows only first 25 alerts per intent.
- For Cisco ASR1000, the following intents are not supported:
  - STP Path Cost Method Inconsistency
  - HSRP/VRRP Active And STP Root Co-location
- Data collection does not collect the inventory from Kubernetes clusters managed by Tanzu Kubernetes Grid Integrated Edition when the last action is in failed state.
- Deactivate the Tanzu Kubernetes Grid Integrated Edition data source in VCF Operations for networks while upgrading the Tanzu Kubernetes Grid Integrated Edition and re-enable it after the upgrade is successful.
- NSX-T Federation behavior with Advanced license:

  NSX-T Federation is not supported with advanced license. Hence, VCF Operations for networks continues to treat NSX-T federated objects as local objects and due to this, you may see the following behavior:
  - As stretched networking is not supported, duplicate flows can be displayed.
  - VM-VM path does not consider stretched entities and an *unknown path* message is displayed when VMs are across sites.
    - If the networking for the VMs within the site is managed by the VMware NSX-T Federation Local manager (LMs), it works as expected.
    - If the networking for a site is managed by the VMware NSX-T Federation Global Managers (GMs) (like the Layer 2 network created by GM for a specific site), then even within site VM-VM path does not work.
  - Global entities cannot be searched.
  - You can see a reference to a global entity in some other Policy entities. If you click this entity, a *Feature Not Supported* message is displayed.
  - GMs can not be added as a data source in VCF Operations for networks. All the VMware NSX-T Federation related data is fetched from the LMs.
- NSX-T Federation behavior for an advance license user upgrading from an earlier version of VCF Operations for networks to advance license version 6.4 and above:
  - The global entities that were searchable before the upgrade, will not be searchable after the upgrade.
  - After the upgrade, site tagged flows are deleted after a certain duration of time and duplicate flows can appear.
  - Some entities can have stale references to the Global Entities, for example, policy groups can have references to global groups, or local firewall rules can have references to the global firewall rules. Clicking these references displays a *standard not supported by license* message.
- In the Network Map, path search does not work if VMware NSX Edge VM is connected to NVDS.
- For devices that do not support CDP and LLDP protocols, Port Channel Member Mismatch intent does not generate alerts if:
  - Neighbor devices with inconsistent port-channel configuration make one or more member ports inactive

  - There is no ARP table entry pointing to an interface that is not a member of any port-channel, but its neighbor is member of a port-channel
- The Port Mode Mismatch intent might generate false positive alerts for Cisco ASA devices if VLAN sub-interfaces are configured at port channels.
- If Panorama fails to commit configuration changes to a Palo Alto gateway, VCF Operations for networks continues to use the latest configuration present in Panorama, which may not be in sync with the policies enforced by the Palo Alto gateway.
- The paths via tunnels if the ECMP (Equal Cost Multi Path routing) is enabled, may overlap with each other.
- In case of Federation, vCenter must be added as a compute manager in VMware NSX-T for the global entity enrichment to work properly.

  VMware NSX-T Federation Global Managers (GMs) can not be added as a data source in VCF Operations for networks. All the VMware NSX-T Federation related data is fetched from the VMware NSX-T Federation Local manager (LMs).
- Some complex VM-VM paths in a VMware NSX-T Federation setup may exhibit cross-lines. Use the path labels to understand the path topologies for such paths.
- Firewall masking event does not work for the Global firewall rule and the Applied To field is missing for Global Gateway Firewall Rules.
- ServiceNow changed the data-modelling and storing of data for the applications discovered by the Service Mapping plugin with the Discoverable by Service Mapping configuration option. This change is not supported in VCF Operations for networks for ServiceNow based application discovery. VCF Operations for networks still supports applications discovered by Service Mapping with the Manual Endpoint configuration option.
- Global Rules spanning across site do not show up in VM-VM path.
- Invalid credentials error can occur on the VMware NSX-T data source if the VMware NSX-T added to VCF Operations for networks is upgraded.

  **Workaround**: Re-enter the credentials.
- When HSRP/VRRP interfaces at two devices are configured with the same matching Virtual IP address but have the protocol Group ID misconfigured, the HSRP/VRRP Configuration Error intent misclassifies that as "2 Active Master interfaces", instead of classifying it as 'Master interface doesn't know Standby'.
- When the content of an open alert changes, VCF Operations for networks does not show the new value in the alert details.
- In the Network Map, the entity details window of the distributed switches does not show alerts that are originated from intents.
- You cannot see the unprotected flows for the Kubernetes service in the Micro-Segmentation Planning page.
- With the release of vSphere 7.x, some VCF Operations for networks features stop to work in 6.x versions due to vSphere Tanzu. For more information, see the KB article 78492.
- Spanning Tree Protocol (STP) support in VCF Operations for networks is only limited to Cisco Per-VLAN Spanning Tree Protocol (PVSTP).
- VCF Operations for networks recommends configuring unique FQDN hostnames on host entities (like UCS Blades or servers) for proper interpretation of underlying host entities. Absence of unique FQDN hostnames may lead to incorrect association of VMs to underlying similar hostnames.
- Unable to change the IP address of a Federation instance after it is configured in Federation.

  **Workaround**Unpair the instance and reconfigure it with the required IP address.
- If a Federation instance is in the disconnected state, the Unpair option is deactivated.
- For some time the values of the Federation page Key Performance Indicators (KPI) are inconsistent when compared to the values in the entity details page or the actual instance. This is because the polling frequencies of Federation page KPIs are different.
- VCF Operations for networks does not support Federation as a search query and the time range selection for this query. Federation page always shows the present data.
- In the Federation page, the deployment widget shows new changes after one minute as the response time for the Deployments widget is cached for one minute.
- Any Federation instance that is unreachable for more than 24 hours, is not shown in the Environment widget of the Federation page.
- Path search in Network Map does not return any results if Check Point VRRP virtual IP is used as the source address.

- In the Federation page, the Platform Capacity section includes unknown flows with internet flows and east-west flows for the deployed instances.
- Flow based link inference algorithm of VCF Operations for networks used by Network Assurance and Verification (NAV) can show incorrect links between a pair of devices when unsupported devices are present between the pair. These incorrect links can cause Loop Detection intent to generate false alerts.
- Known limitations for IPv6 addresses:
  - Online upgrade and support tunnel for pure IPv6 and dual-stack setup is not certified.
  - VCF Operations for networks does not support the change of IP family from IPv4 to IPv6 and conversely.
  - VCF Operations for networks does not support webhooks for IPv6 and dual-stack.
- When you add VMware NSX-T Manager using the Certificate (Principal Identity) authentication method, Security Intelligence integration does not work with the principal identity user.
- VCF Operations for networks fails to collect data for some `show` commands using SSH from devices (such as Arista switches). The fail happens if `enable` password is configured for the device and the password prompt shows up when the user account attempts to enter the `enable` mode.

## Create a Migration Plan and Export Migration Waves

You can use VCF Operations for networks to create a migration plan for your workloads. After creating your migration plan, you export it as part of migration waves and perform the migration in VCF Operation HCX.

To enhance your search and filtering capabilities, you can include more VM attributes in a CSV file you upload in VCF Operations for networks. In addition to the VM attributes you want to use, this file must contain the mandatory `vc_ip` and `vm_moref` attributes. The `vc_ip` attribute value is the IP address of the vCenter instance where the VM is hosted and the `vm_moref` attribute is the ID of the VM in the vCenter instance. To retrieve these attributes:

1. From VCF Operations for networks, select your VMs and from the top-right menu click **Export as CSV**.
2. In the **Selected properties (CSV columns)** field, click **Add properties**.
3. Search for and add the `Vendor ID` and `Manager` properties.
4. Click **Export**.
5. Open the exported CSV file. The `vCenter` column contains the value for the `vc_ip` property and the `Vendor ID` column contains the value for the `vm_moref` property.

Use the comprehensive search, filtering, and flow analysis capabilities of VCF Operations for networks to identify and select the VMs for your workload migration. The creation of the migration plan consists of:

- Creating the scope for your workload migration. Optionally, you can upload a CSV file with more VM attributes to further enhance the scoping process.
- Adding the workload of your migration scope to migration waves.
- Organizing your migration waves into mobility groups.

After creating your migration plan in VCF Operations for networks, you export the migration waves by commiting them which imports them toVCF Operation HCX where the actual workload migration is performed.

> **Note:**
>
> The following procedure outlines how you can create a migration plan in VCF Operations for networks, you export the migration to VCF Operation HCX, You can also perform this action from VCF Operations by navigating to **Launchpad** > **Workload Mobility**.

1. From the left navigation panel, navigate to **Plan and Access** > **Migration Planning** and click **Create Plan**.
2. Add more VM attributes for your migration plan.
   a) From the **Migration Attributes** widget, select **Upload.**
   b) Select your CSV file.
   c) Verify that you are familiar with the content rules and that your CSV file includes the mandatory `vc_ip` and `vm_moref` attributes and click **Upload and Proceed**.

      After uploading the CSV file, you can see a breakdown of the total number of identified VMs, how many of them match the new attributes, and if there are any errors. The newly added attributes are available as filters in all views for VMs and migration planning.
3. Enter a name for your migration plan and click **Proceed**.
4. Define the migration scope.
   a) Filter the VMs you want to add to your migration plan by using the available attributes or add more attributes from the **Filters** tab.
   b) After selecting the VMs you want to add to the migration plan, click **Add to Scope**.
   c) Optional: By selecting **Review Network Dependency**, you can view any out-of-scope VMs identified as dependencies and can add them to the current migration scope.
5. Create migration waves.

   **Note:**

   The following procedure outlines the process of creating migration waves and groups manually. You can also generate the waves and groups in your plan automatically by selecting **Auto Generate Waves and Groups**. You can also start the process of automatic generation even if you have selected the manual generation process. After automatic generation is complete, you are sent to the **Summary** page.

   a) Select the VMs you want to add to a migration wave.

      **Note:**
      You can only add VMs belonging to the same vCenter instance in a specific migration wave. Parts of your workload from different vCenter instances must be included in a separate migration wave.
   b) Select **Add to Migration Wave**.
   c) Enter a name for the migration wave, add the destination HCX manager, and click **Proceed**.

      **Note:**

      Selecting a destination HCX manager during this step is only available if you have added that HCX manager as a data source in VCF Operations for networks.
6. Organize your migration waves into mobility groups.
   a) Select the VMs from your migration wave that you want to add to a mobility group.

      **Note:**
      You can add the VMs from a single migration wave to multiple mobility groups.
   b) Select **Create Mobility Group**.
   c) Enter a name for the mobility group and click **Proceed**.
   d) Repeat for all of your migration waves.
7. On the **Summary** page, you see an overview of your migration plan and the option to make edits to your waves and groups before commiting the plan to VCF Operation HCX.

   **Note:**

   You can remove the automatically generated migration waves from your plan by selecting **Remove Auto Generated Waves and Groups. If you have munally updated the content of these waves, they will no longer be considered automatically generated and the above option used to removed them will disappear.**

8. When you are ready to export a migration wave to VCF Operation HCX, navigate to the migration wave entry under **Migration Waves** and click **Commit**.

   The **Commit** option is only available if you have added a destination HCX manager. If you have not added an HCX manager in the previous step, you can select the migration wave and add it now. Alternatively, you can manually export the migration wave by clicking the vertical elipsis menu to the left of the migration wave and selecting **Download Wave**.

   After clicking **Commit**, the migration wave is sent to VCF Operation HCX. Depending on the validation results, the wave status is marked as either **Commited** or **Failed**. Failed migration waves include an error message describing the encountered issue and the option to re-commit the relevant wave. The status for migration waves for which the workload migration in VCF Operation HCX is in progress is marked as **Executing**.

9. Repeat for all of your migration waves.

You have created a migration plan and exported your migration waves.

Perform the workload migration in VCF Operation HCX. For more information on VCF Operation HCX, go to the About Migrating Mobility Groups from Migration Waves.

## Planning Application Migration to VMware Cloud on AWS using VCF Operations for networks

Using VCF Operations for networks, you can assess your on-premise environment for application migration to VMware Cloud on AWS.

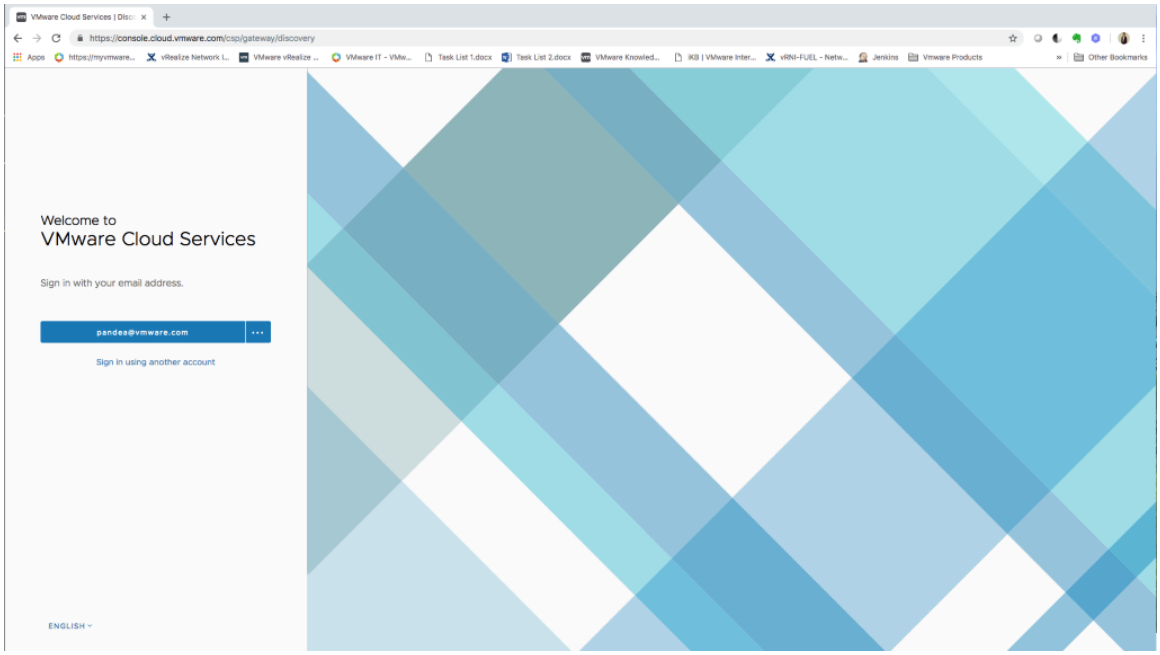
| Steps | Procedure | Description |
|---|---|---|
| Step 1 | Setting up your Environment | • Accept the End User License Agreement (EULA).<br>a. Create a VMware user account or log in to the VMware account.<br>b. Update the registration form.<br>New users will receive an email to activate their account.<br>c. Accept the VMware terms and EULA.<br>• Download the OVA files.<br>a. Log in to the VMware Product Download page at https://my.vmware.com/group/vmware/home.<br>b. Search for VCF Operations for networks.<br>c. Download the latest VCF Operations for networks platform and collector OVA files.<br>• Prepare for installation.<br>a. Verify the System Recommendations and Requirements.<br>b. Verify the Supported products and versions |
| Step 2 | Deployment | 1. Deploy the VCF Operations for networks platform OVA file.<br>2. Generate a shared secret,<br>3. Deploy the VCF Operations for networks collector OVA file.<br>4. Deploy VMware Cloud on AWS. |

| Steps | Procedure | Description |
|---|---|---|
| Step 3 | Data Source Addition | 1. Log in to VCF Operations for networks.<br>2. Add VMC vCenter<br>3. Add NSX Manager<br>4. Add VMware HCX<br>**Note:**<br>VCF Operations for networks automatically detects if you are in the process of migrating your data center using VMware HCX and generates notifications to guide you in the following scenarios:<br>• Adding cloud data sources.<br>• Gaining visibility of your hybrid cloud environments.<br>• Monitoring your cloud workloads.<br>• Gaining visibility of your migrated workloads.<br>You can click the links on the notifications to add your cloud data sources, access your on-premises and cloud environments, or gain network visibility. |
| Step 4 | Model Application | • Analyze application dependencies.<br>a. Create an Application<br>b. Creating Tiers for Physical IPs<br>c. Analyzing the Application<br>d. VMC: Planning and Micro-Segmentation<br>• Analyze recommended firewall rules<br>• Perform search<br>• Create and use dashboards |

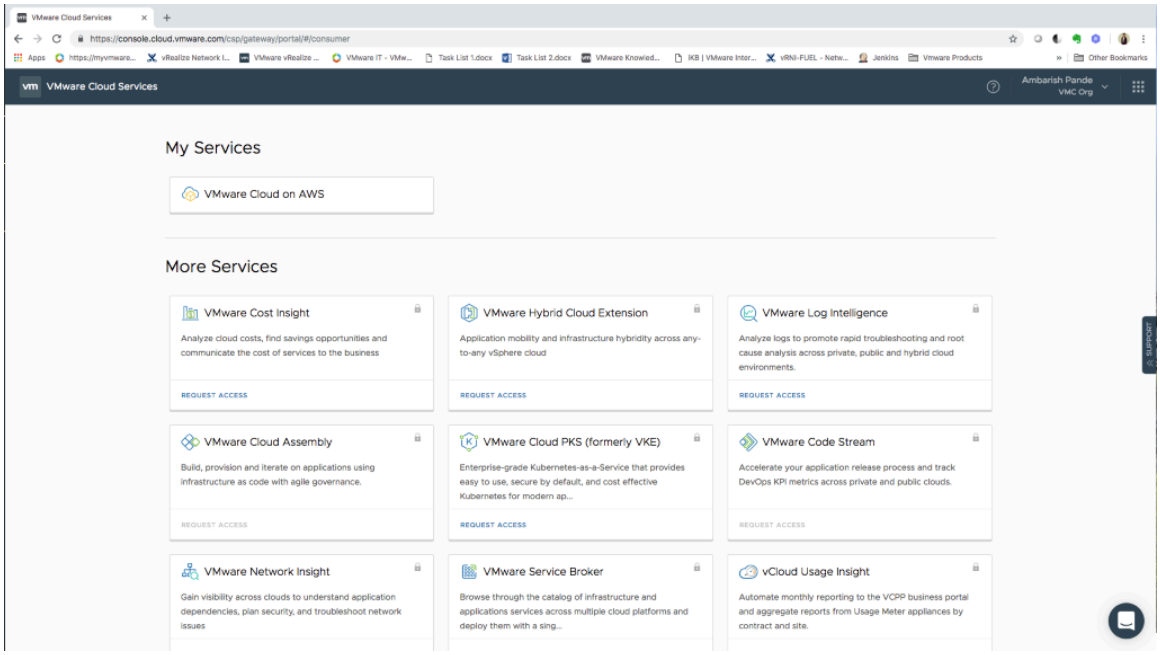
## How do I obtain the CSP Refresh Token for NSX Manager

For adding a VMware Cloud on AWS NSX Manager as a Data Source in to VCF Operations for networks, you need a refresh token.

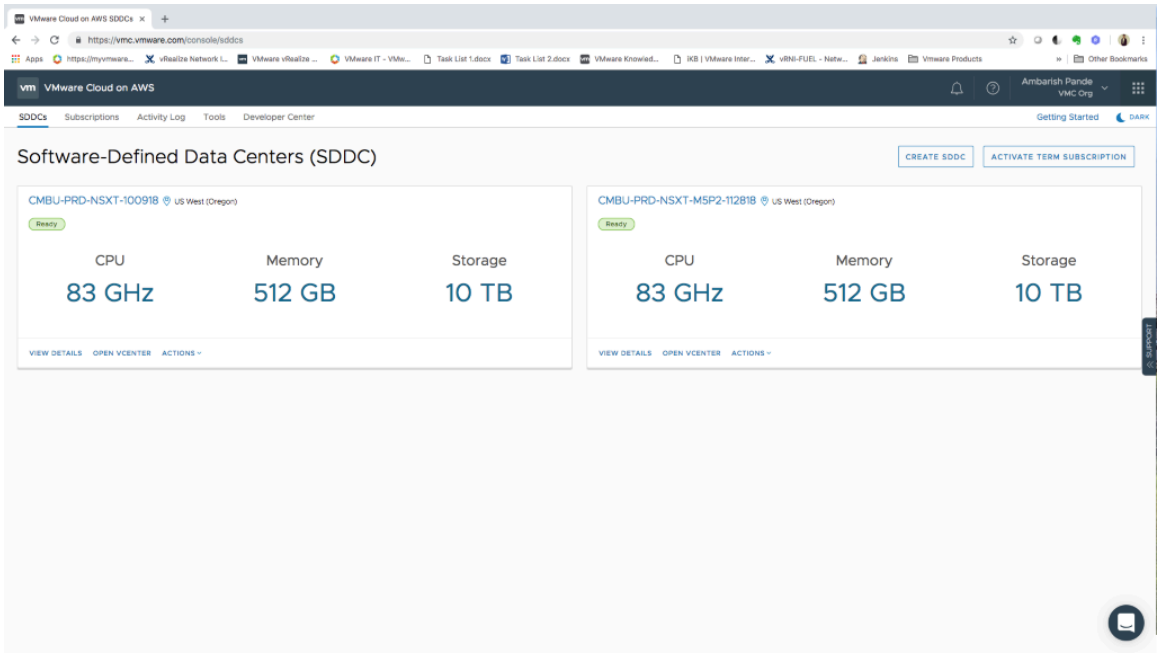
1. Log in to the vCenter services console.



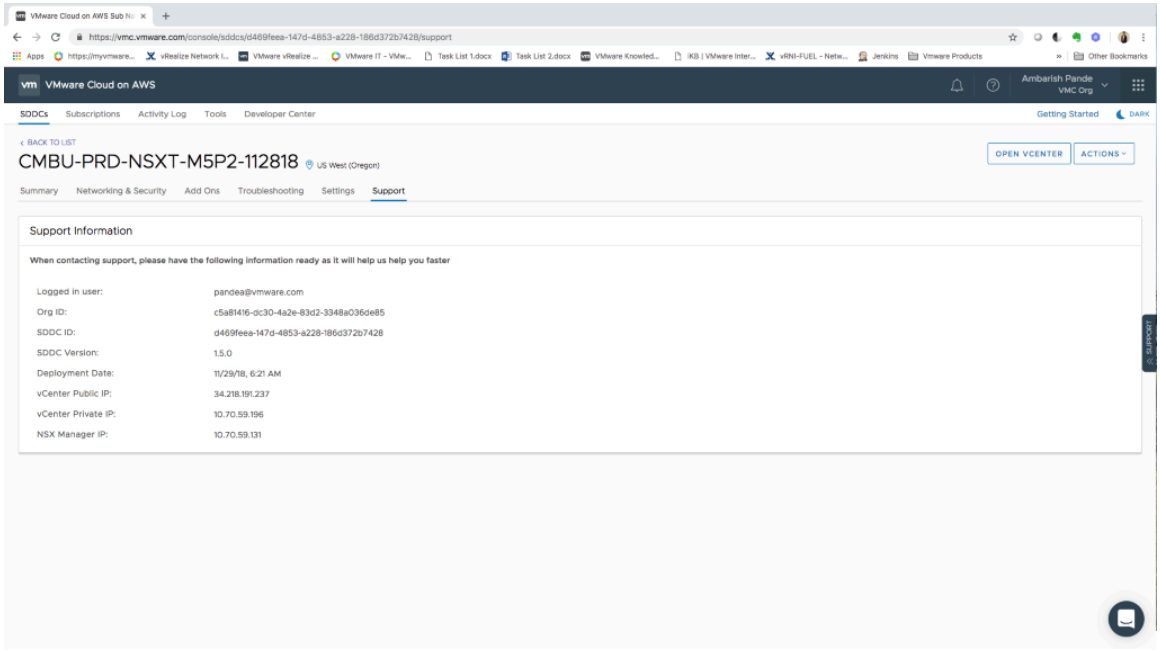

2. Under My Services, click VMware Cloud on AWS.



3. Select the desired Software-Defined Data Center (SDDC).

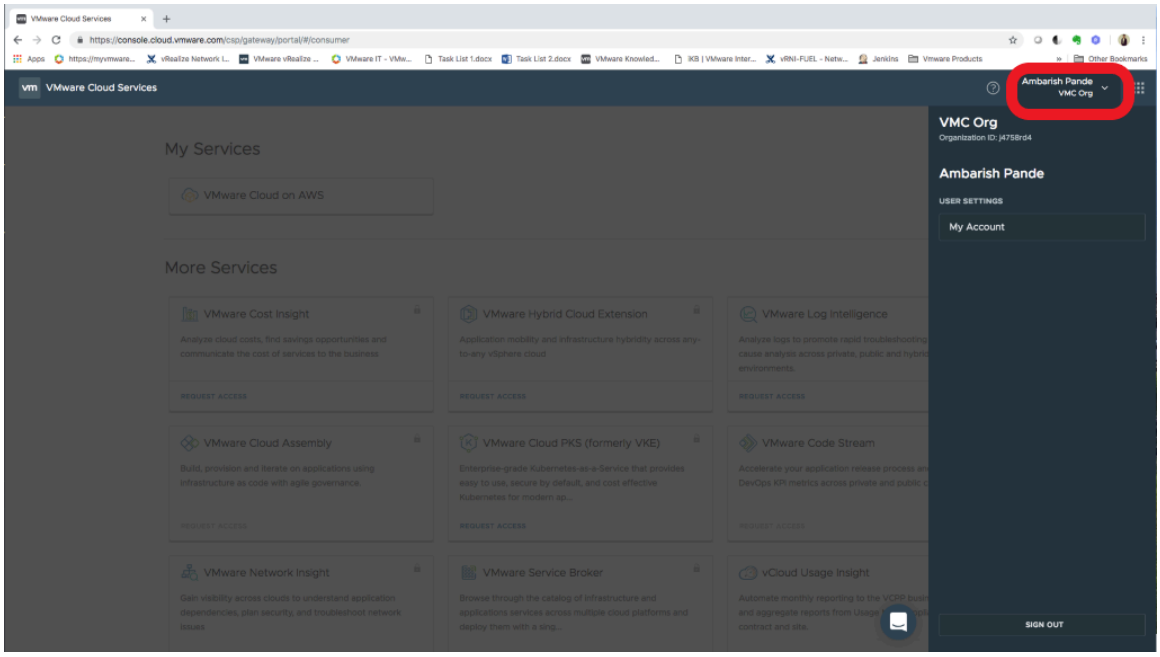



4. Click the **Support** tab.




5. Make a note of the NSX Manager IP address.
6. Click on the organization name on the top banner.

   **Note:**
   Ensure that the organization resides in the selected SDDC.




7. Generate the API token.

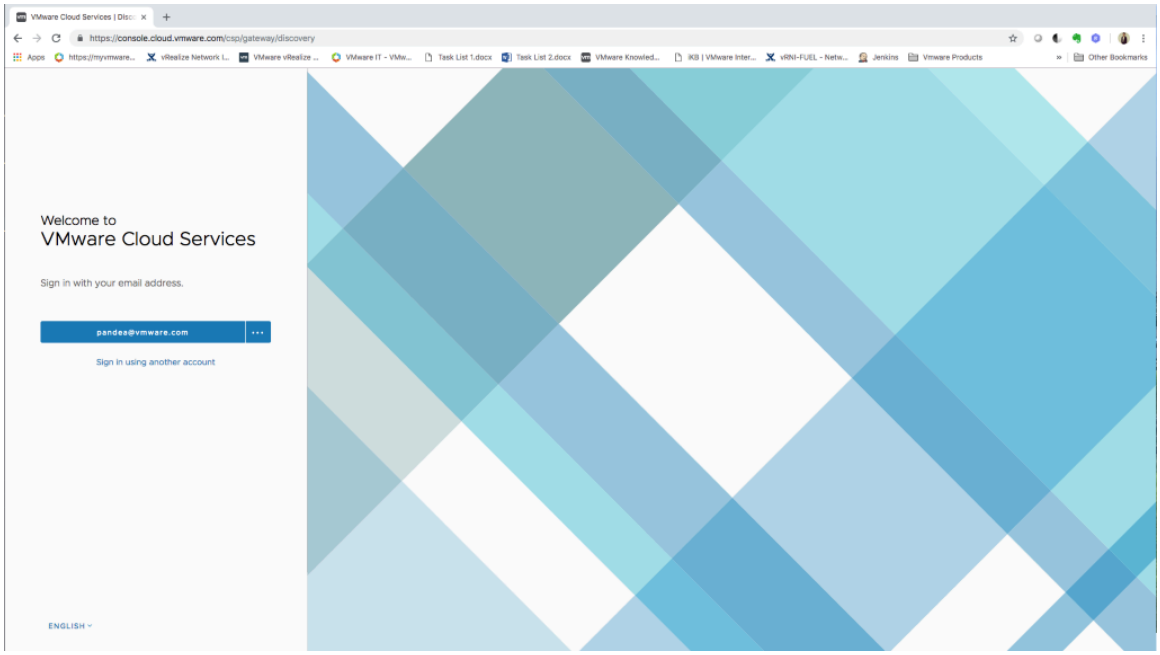
   **Note:**
   To generate the API token, you must have the **Administrator** and the **NSX Cloud Admin** privileges.

You can use this token for authenticating all VMware Cloud on AWS SDDCs on the organization.

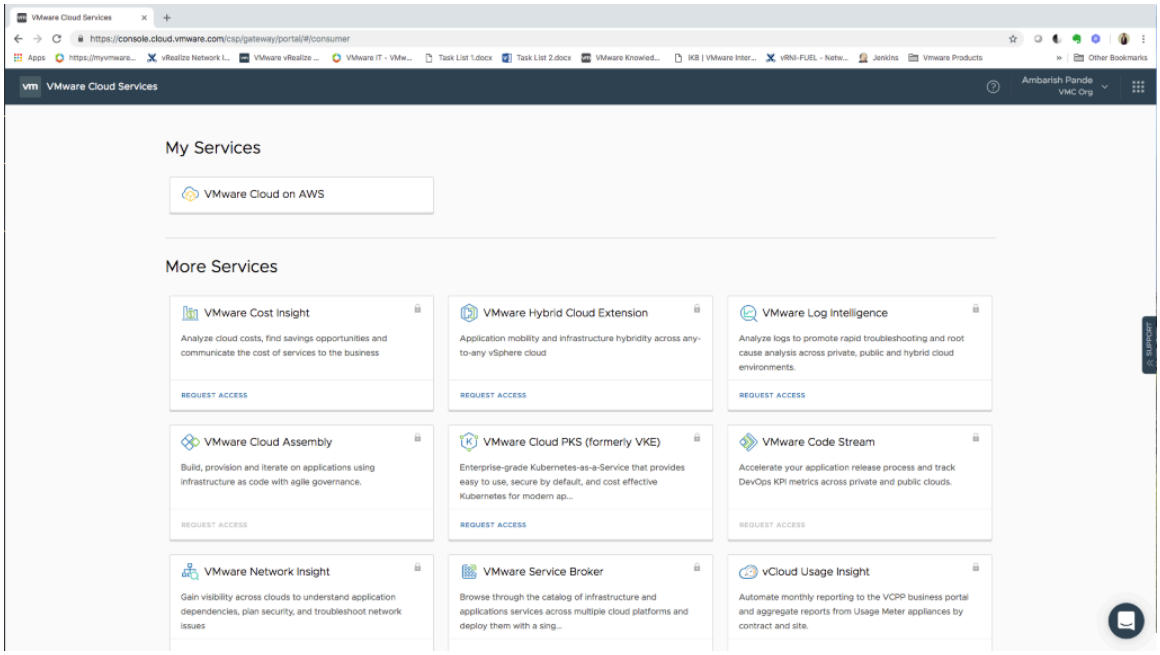
## How Do I Obtain VMware vCenter Credentials

To add a VMware vCenter data source into VCF Operations for networks, you need the VMware vCenter credentials.

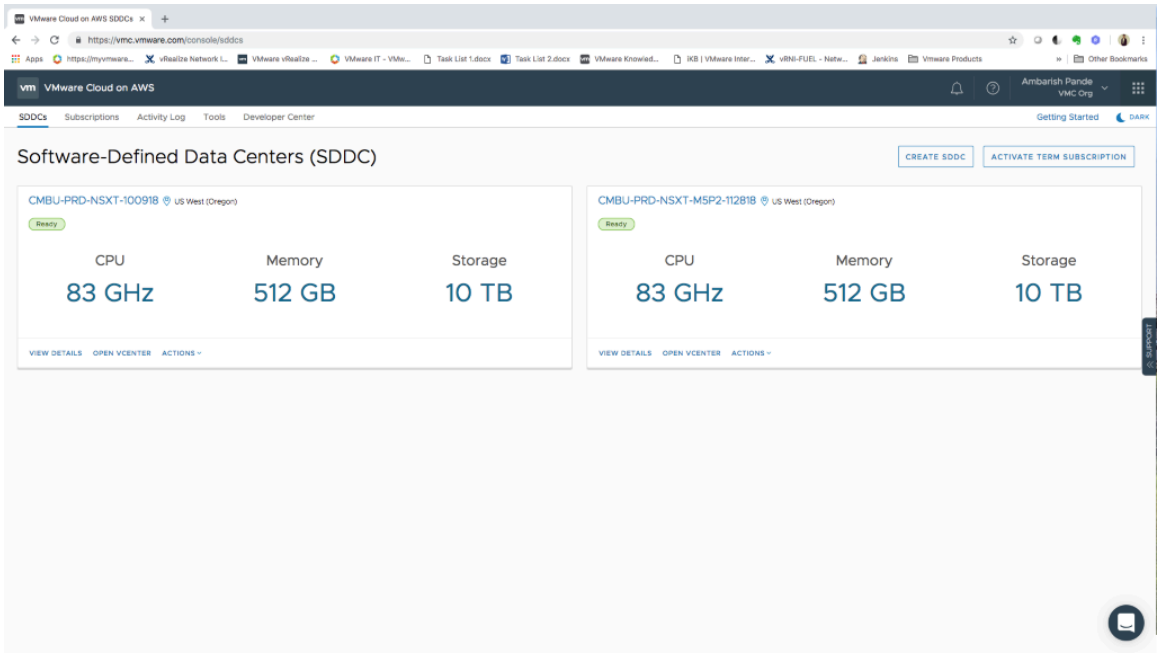
1. Log in to the vCenter services console.



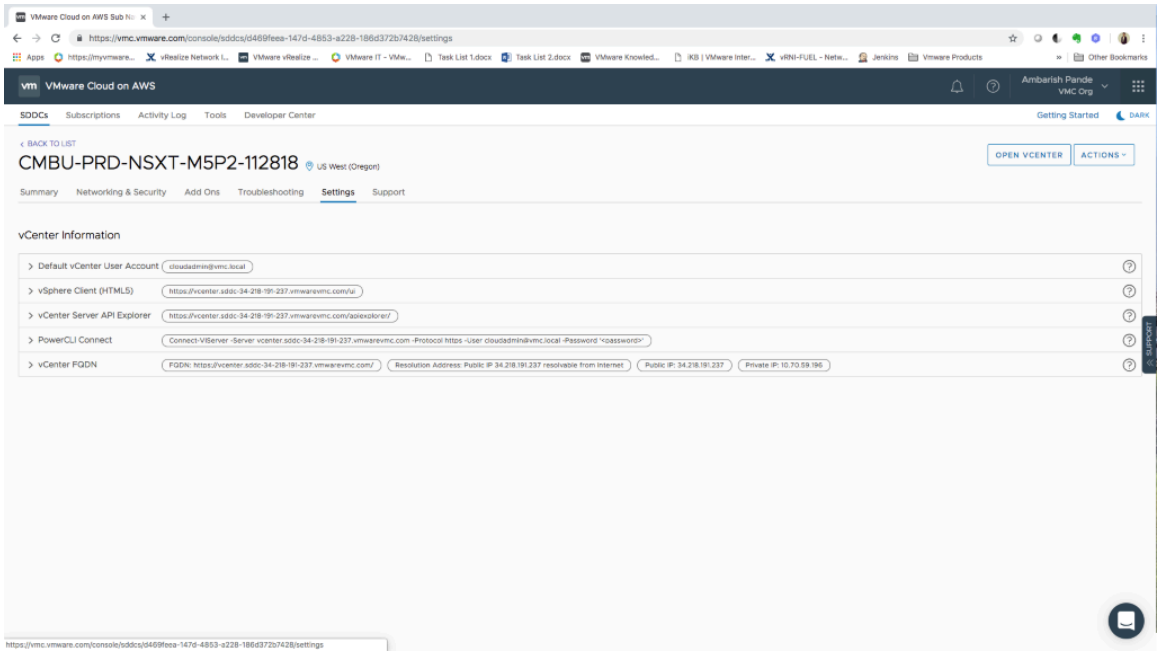
2. Under **My Services**, click **VMware Cloud on AWS**.



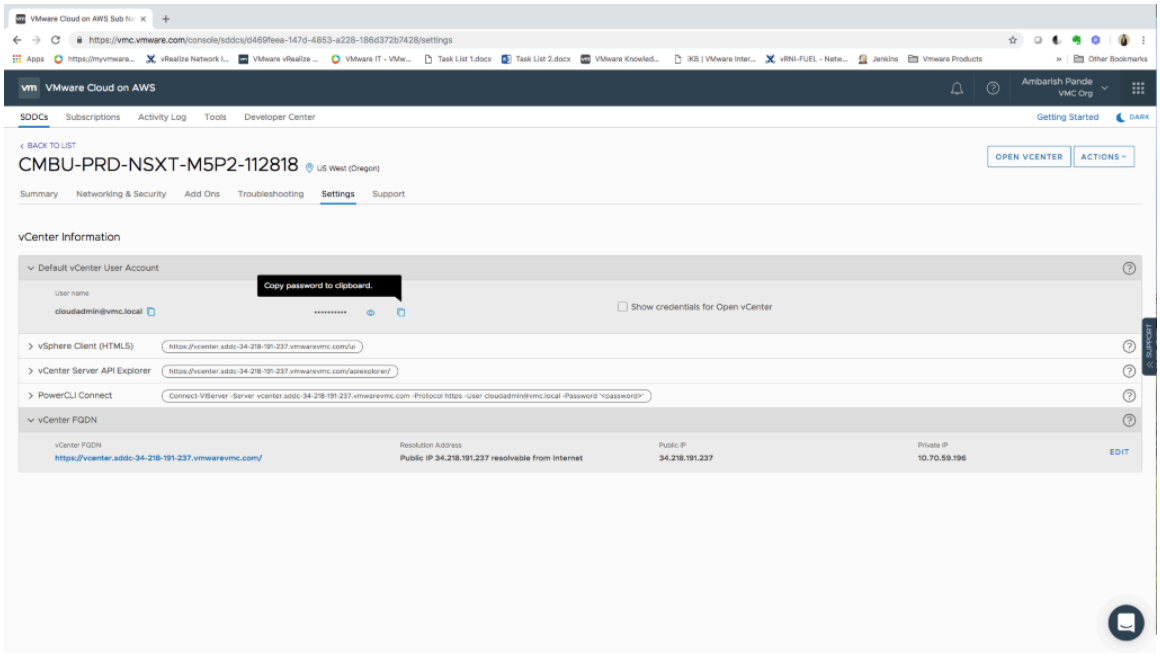

3. Select the desired Software-Defined Data Center (SDDC).



4. Select the **Settings** tab.




5. Expand **vCenter FQDN** and make a note of the vCenter FQDN details.
6. Expand **Default vCenter User account**, and copy the user name and password.




## Compute Gateway Firewall Rule

When communicating with the VCF Operations for networks platform; the collector requires HTTPS port 443 to be open for outgoing traffic.

Following VMware hosted URLs are accessed by the collector through the firewall:

- *.vmwareidentity.com
- gaz.csp-vidm-prod.com
- *.vmware.com
- *.ni-onsaas.com

Additionally, NTP and DNS traffic should be allowed for the correct functioning of the VCF Operations for networks or VCF Operations for networks collector.

Create a firewall rule with the following details:

- Name: An appropriate descriptive name
- Source: The name of the VMware Cloud on AWS Group containing the collector IP address.
- Destination: Select **ANY**
- Services – Select **HTTPS, DNS, DNS-UDP, NTP, ICMP**
- Action – **Allow**
- Applied To – **Internet Interface**
- Logging – Enable logging, if required.

# Using VCF Operations Diagnostics

Discover and remediate known issues, and monitor the operational state of the private cloud infrastructure in VCF Operations.


- How Diagnostics Helps You Keep a Healthy VCF Software Stack
- See Known Issues in Your Environment and Apply Fixes with Findings
- Overwriting the Diagnostics Configuration
- Monitor Your Environment and Prevent Issues with VCF Health
- Use Log Assist to Accelerate Your Support Cases


## How Diagnostics Helps You Keep a Healthy VCF Software Stack

With VCF Operations Health and Diagnostics, you can quickly diagnose and address issues to minimize their impact and improve operational efficiency, security, and reliability of your VCF platform.

VCF Operations Health and Diagnostics facilitates your job as a VCF private cloud infrastructure administrator, providing an in-product capability to discover, troubleshoot, and remediate known and potential issues. VCF Operations Health and Diagnostics builds on diagnostic tools and operational data to address the top drivers of service support requests to help you avoid or resolve such issues yourself. With health and diagnostics data, you can act immediately as it provides recommendations on how to resolve known issues or narrow down and even isolate possible root causes for emerging issues. If you cannot find sufficient information to resolve an issue, you can reach out to support. But you do so with proper context and using automated log bundling with Log Assist. The complexity and size of some VCF infrastructure might pose a challenge to collect and act on all operational data. VCF Operations Health and Diagnostics makes it easy by uniting all metrics in a single pane of glass and at the same time providing means of self-support on known issues. VCF Operations Health and Diagnostics solves the challenge in three ways:

- **Findings**, where you discover known issues and can apply fixes that are based on recommendations, such as best practices, knowledge base articles (KBs), and VMware Security Advisory (VMSA) notifications. Starting with VCF Operations health and diagnostics 9.1, you can also use a set of public APIs that allow you to consume findings in your own tracking and monitoring systems.
- **VCF Health**, where you proactively monitor the operational state of all components of the VCF software stack. **VCF Health** helps you prevent issues that are not related to product bugs, but are caused by factors such as:
  - Availability/Performance: issues with physical devices: servers, storage, or network components.
  - Misconfigurations: incorrect settings affecting functionality.
  - Security
  - Need for optimization
  - Lack of immediate observability: issues such as expired certificates or NTP drifts leading to integration issues.
- **Log Assist**, where you automatically generate log bundles along with other relevant context and transfer to the Broadcom support portal for a speedy resolution of issues.

VCF Operations Health and Diagnostics also helps you with identifying and resolving product bugs. With VCF Operations health and diagnostics 9.1, you can:

- Self-diagnose connectivity, availability, and performance issues impacting ESX hosts.

- Troubleshoot vSphere High Availability issues, such as failure to restart protected VMs.
- Avoid disk size breaches when VMs have a large number of snapshots.
- Track the CPU and Memory utilization on cluster level and CPU, Memory, and Disk utilization on node level for VCF services runtime.

With **Findings** and **VCF Health** combined, you can detect and remediate issues, resulting in generating a lower number of support tickets, avoiding downtime, and reducing the mean time to resolution.

For example, if you see slowness in migration tasks in your environment, you can troubleshoot vSphere vMotion operations, looking at the precopy time, duration, and bandwidth. You can also check CPU usage, CPU contention, and memory utilization, and see error logs in a single screen, without changing context and tools.

**Note:**

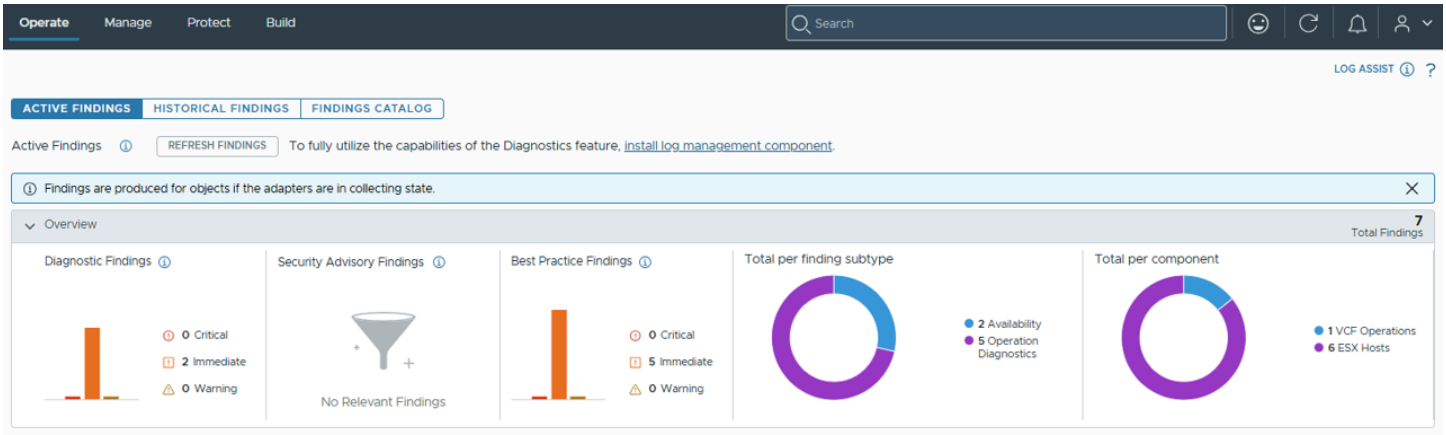
For a comprehensive experience with the diagnostics workflow, you must register your environment in Connected Mode to use the Log Assist capability. You need an Internet connection on the client browser to access KB links, troubleshooting guides, and documentation.

## What are Findings in Diagnostics

**Findings** in VCF Operations Health and Diagnostics helps you find and resolve known issues in your VCF software stack by using a list of log and property-based findings, also called diagnostic rules, or signatures.

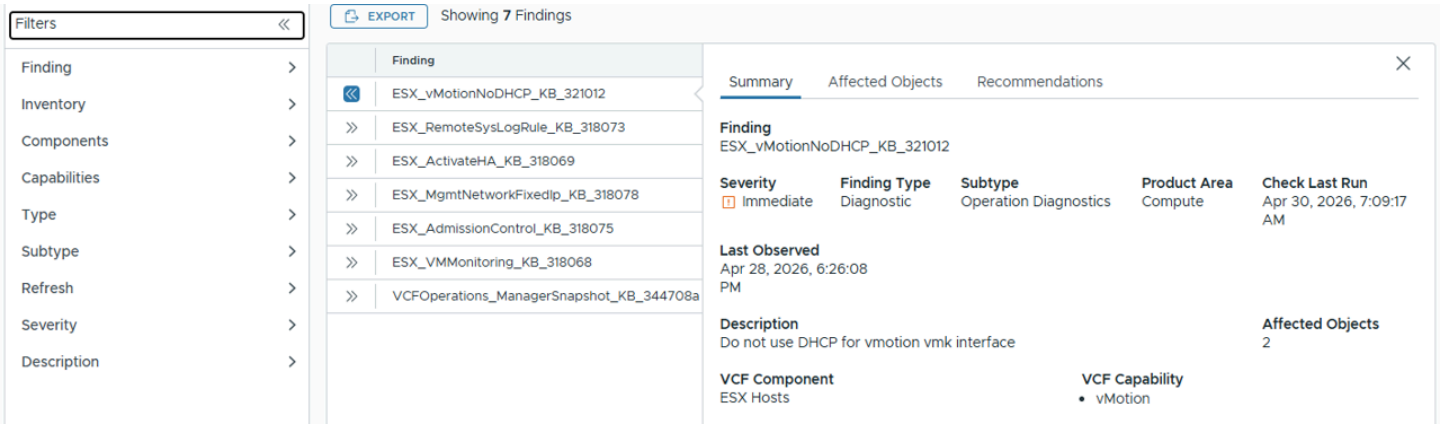
Findings result from scanning your VCF software stack for known product bugs, security notifications from VMware Security Advisory (VMSA), and actionable best practice recommendations.

**Figure 448: Diagnostic Findings Summary View**




You see what are the affected objects, details of the finding, the recommendation for resolution, and usually a link to a KB.

**Figure 449: Findings Details Grid**




The **Findings** capability in health and diagnostics also integrates with Log management to provide more depth and flexibility in troubleshooting. You review the available findings and decide on the appropriate next steps to keep your VCF platform up-to-date and healthy.

VCF Operations Health and Diagnostics works with more than 900 property and log-based rules, which are updated by the Diagnostics Content Management Pack. You can see a list of all signatures in the **Findings Catalog**.

For more information, read See Known Issues in Your Environment and Apply Fixes with Findings.

### Findings Rules

Log-based rules detect an actual occurrence of a known issue in your environment. Property-based rules indicate that the issue is present in the current build and could occur or have already occurred. All finding rules relate to specific build numbers. Depending on the finding, you can decide what corrective actions are applicable to your environment and how urgent it is to apply such actions. An example of corrective action is to apply a patch or upgrade.

For more information, read See Known Issues in Your Environment and Apply Fixes with Findings.

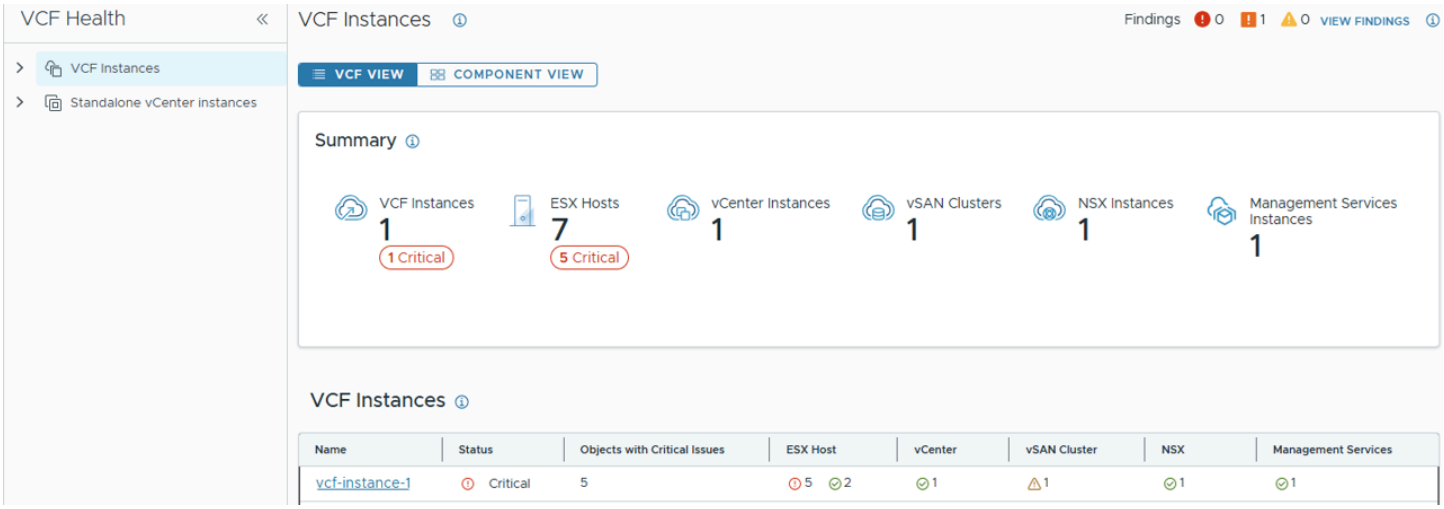
## What is VCF Health in Diagnostics

**VCF Health** is a comprehensive, high-level overview of your VCF deployment, enabling you to proactively monitor, diagnose, and remediate the operational state of your environment.

**VCF Health** is designed to reduce downtime, improve visibility, and increase operational resilience by offering actionable insights into VCF software stack, affecting compute, networking, storage, and the management layer.

The structure of **VCF Health** allows a drill-down to the hierarchy from a VCF instance to its management and workload domains, vCenter instances, NSX instances, vSAN, and ESX hosts. **VCF Health** only works on environments that are installed by using the VCF Installer and provide the level of centralized data that health and diagnostics collects, such as certificates and configurations. This data is not available if you manually add components. With **VCF Health**, you can reduce downtime or the mean time to resolution by identifying failures of underlying infrastructure such as ESX hosts or remediating emerging performance issues.

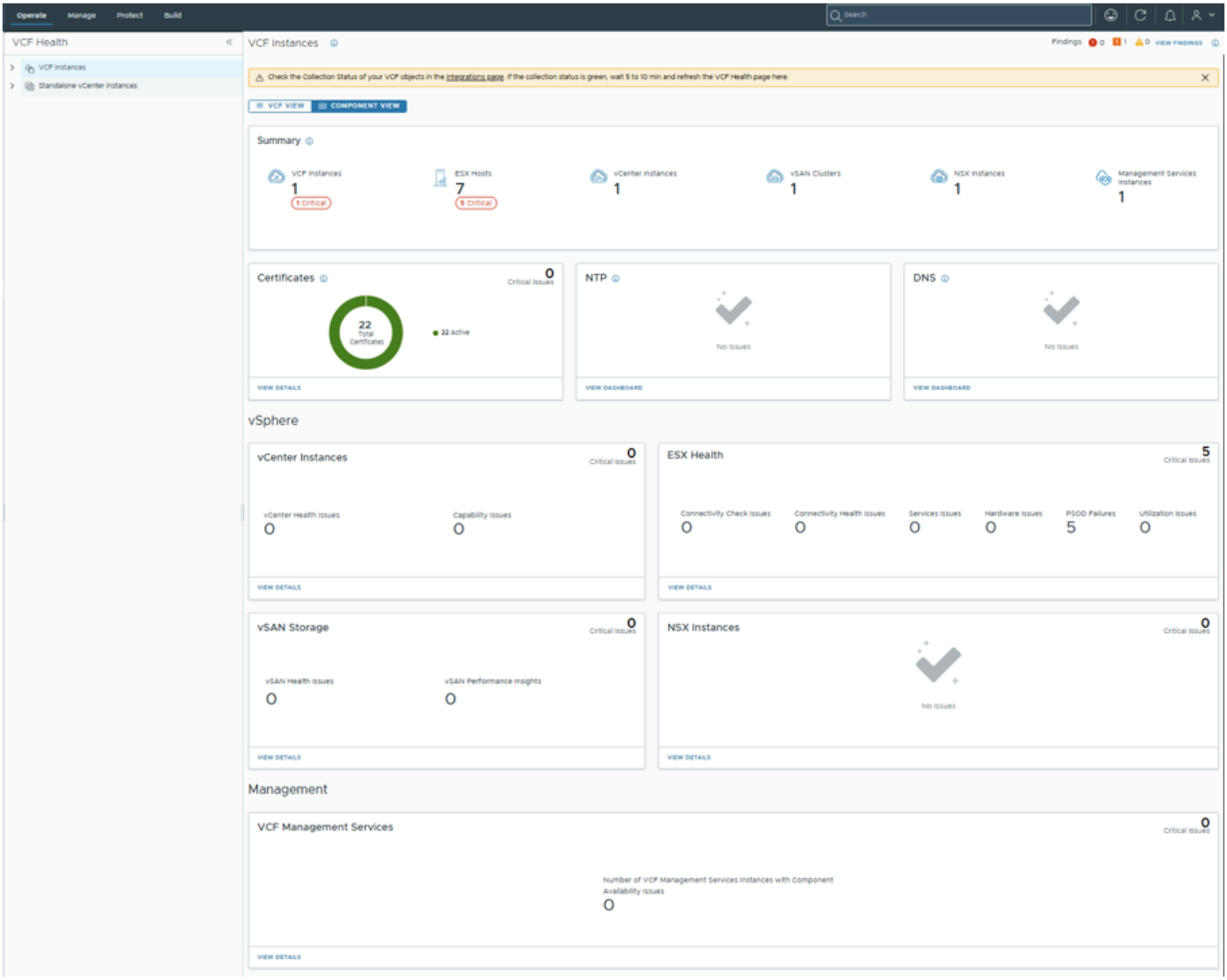
**Figure 450: VCF Health Summary View**




For example, if you see expiring certificates or notice NTP drift or DNS sync, you can avoid networking and configuration issues. Or if you see a trend of failing tasks with virtual machine snapshots or vSphere vMotion migrations: you can handle the underlying cause for the emerging negative trend to prevent issues from occurring.

From the **Component** view, you see a holistic overview, summarizing the health of all objects within the environment.

**Figure 451: VCF Health Component View**




## How to Set up and Maintain Diagnostics

Complete VCF Operations health and diagnostics for your VCF software stack by installing and configuring Log management and Real-time metrics, and setting up integration between components.

The **Findings** capability has no license dependency and works on both VCF and VMware vSphere Foundation. Anything that applies to VCF applies to vSphere Foundation as well, within the scope of your environment. You can apply only diagnostic capabilities relevant to your setup. For example, you must take care of the NSX integration only if you use NSX in your environment. You also do not need to have an SDDC Manager in your environment for **Findings** to work.

The management pack for VCF Operations health and diagnostics is automatically installed and the integration does not require manual configuration. Some VCF platform components need installation or integration by using adapters to become active and start collecting diagnostics data.

### Setup

- After the initial installation of VCF management services, make sure you install log management and real-time metrics. See Deploy Log Management and Deploy Real-Time Metrics.
- In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, select **Administration** > **Configurations** > **Log Management** to configure log collection into log management from VCF inventory objects, such as vCenter instances.
- When you install VCF by using the VCF Installer, you get by default one unified cloud proxy that facilitates log collection and the Log Assist capability. You use a unified cloud proxy to collect logs into the log cluster and to collect, and upload support bundles to the Broadcom Support Portal. For more information, see Set up Log Assist and Collecting Data with Cloud Proxy in VCF Operations.
- Make sure that you have connectivity between components in your environment. The networking port that is required between VCF Operations and endpoints in ESX , vCenter, NSX, and SDDC Manager, is SSL 443. For Log Assist, the connection is between the cloud proxy and the respective component, not from the cluster node.
- For log management, the required port is 9543. The minimum bandwidth is 200 KB/s.
- Generation of diagnostic findings depends on the following adapters:
  - VcfAdapter (VMware Cloud Foundation)
  - VMWARE (vCenter Adapter)
  - NSXTAdapter (NSX)
  - CASAdapter (VCF Automation for VM Apps Organization, formerly vRealize Automation Adapter)
  - VCF 9.1 adds an adapter for the VCF services runtime component. With the data that the adapter provides, you can self-diagnose connectivity, availability, and performance issues impacting your VCF components that are part of the VCF management services**.**

    **Note:**

    Diagnostic findings are produced for VCF components only if their adapters are in collecting state. You see property and log-based findings only for inventory objects that have an active cloud proxy and are in green Collection Status.
- For findings related to earlier versions of VCF Automation, you must:
  - add and activate the VCF Automation for VM Apps Organization adapter from the **Repository** at **Operate** > **Administration** > **Integrations**.
  - Create an account.
  - Add the account to a **Default Cloud Proxy** group.

### Scalability

Each VCF instance that you deploy gets one cloud proxy by default, but adding more cloud proxies to segregate workloads helps with accelerating computation times.

### Performance

The Diagnostics rule engine depends on I/O and requires SSD type of storage, preferably sub-millisecond access times, such as NVMe. Slower disks might lead to an increase of computation times linear to the performance of the disk.

### Update Cadence

Update cadence depends on the urgency of hot fixes and the number of accrued signatures. Usually comes along with a maintenance release of VCF Operations for which you see notifications in **Build** > **Lifecycle**. Starting from VCF 9.1, updates of the Diagnostics Content Management Pack might be released between maintenance releases as well.

The other updates are security-related and come in the form of a VMware Security Advisory (VMSA).

**Record-keeping**

Log-based findings expire 36 hours after the issue was first detected in the logs. Property-based findings are removed from the system when the issue is resolved. VCF platform private cloud infra admins can export findings in tabular format from the **Findings** tab to share with the other teams or keep for auditing purposes.

**Roles**

You use Role-Based Access Control to manage access to health and diagnostics. Normally, VCF platform private cloud infra admins manage health and diagnostics and reach out to respective teams, such as storage or networking, for further investigations outside of the VMware software stack.

## VCF Operations Health

Along with VCF Operations health and diagnostics , you can track the health and resiliency of your VCF Operations environment in a dedicated dashboard.

VCF 9.1 adds the **VCF Operations Health** dashboard to provide visibility on the health and resiliency of your VCF Operations platform, log management, and real-time metrics. In the VCF Operations navigation bar at the top, click **Operate**, in the left navigation pane select **Dashboards**, and expand **All** to get to the **VCF Operations Health**, **Log Management Health Overview**, and **Real-time metrics Health** dashboards.

In the VCF Operations Platform health section, you see the health of your VCF platform, including supporting services and integrated components, such as:

1. VCF Operations cluster - a high-level summary on the state of the cluster: if vSphere High Availability is active, the number of nodes, cloud proxies, and adapter instances in the cluster.
2. Nodes and cloud proxies - a list of all nodes and cloud proxies that are associated with the VCF Operations cluster. Shows the high-level summary of availability, number of vCPUs associated with them, and the percentage of available storage in databases and storage logs.
3. Structure - shows the interactive topology of cluster, nodes, cloud proxies, and their associated services.
4. Details of the selected object - shows detailed metrics that are associated with an object you select: either the cluster, its nodes, a cloud proxy, or any of the services.
5. License server - shows a high-level summary of the license server availability along with percentage usage of CPU, Disk, and Memory.
6. About and How To Use widgets to help with your onboarding.
7. Links to **Log Management Health Overview** and **Real-time metrics Health** dashboards for more information on the VCF services runtime cluster.

The VCF Operations Health dashboard replaces the following self-help dashboards:

- Self Performance Details
- Self Services Communications
- Self Services Summary
- Self Cluster Statistics
- Self Health
- Self Troubleshooting

## See Known Issues in Your Environment and Apply Fixes with Findings

Findings collect data from all VCF components to help you identify existing or potential issues in your environment.

Findings are derived from properties and product logs data from vCenter, ESX, NSX, SDDC Manager, VCF Operations, vSAN, and VCF Automation. Findings help you identify potential problems or inform you about existing issues in your environment, and provides steps to resolve them.

## Active Findings

In the **Findings** screen, you see **Active Findings** that shows all the findings that are discovered within the last 24 hours. Checks against property-based signatures run every four hours scanning vCenter, ESX, NSX, SDDC Manager, VCF Operations, vSAN, and VCF Automation. Within 24 hours, whenever a condition in your setup matches a signature, you see it in the **Active Findings** grid. When you expand a finding in the grid, you see:

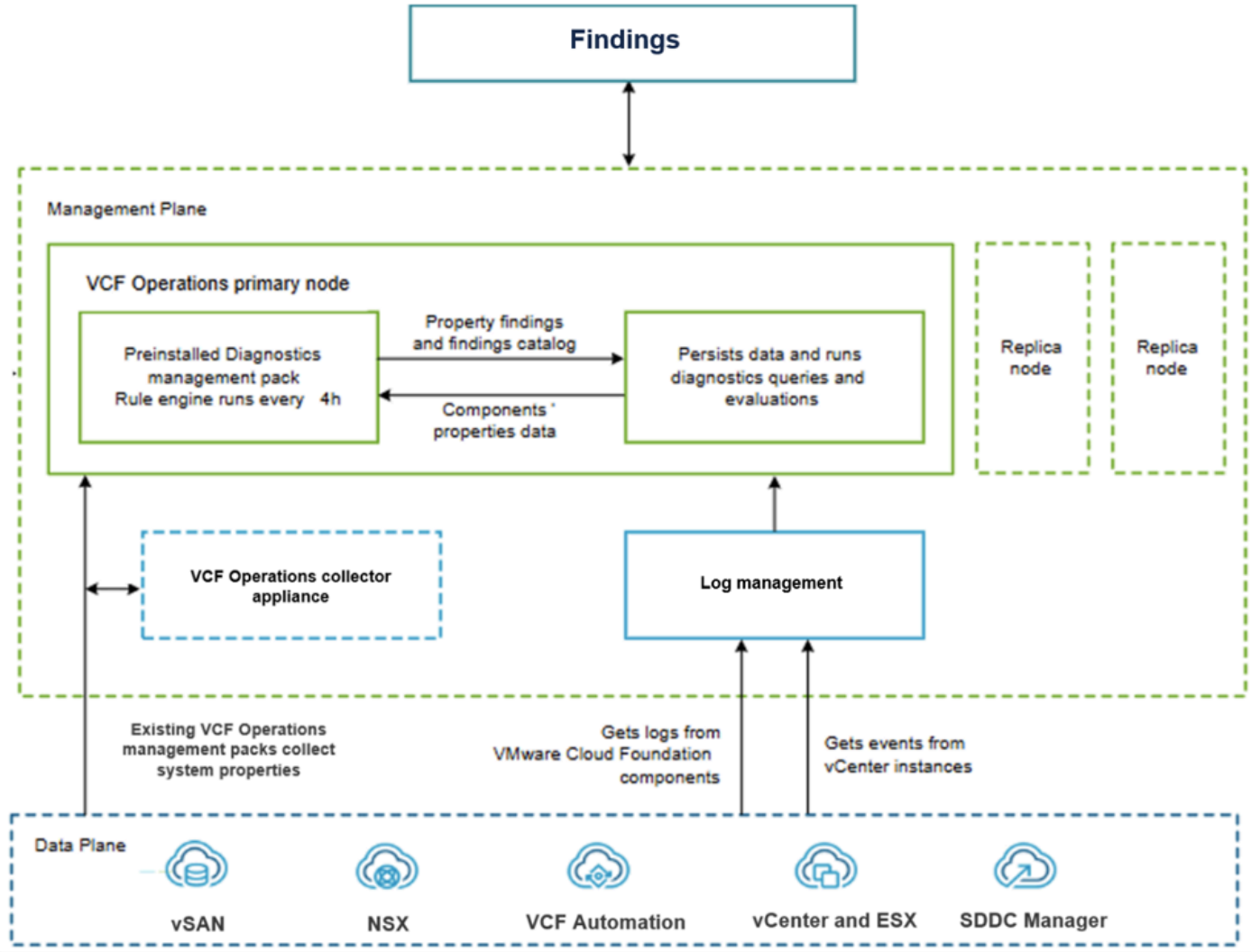
- a summary and description of the identified issue
- the affected objects with links that take you to an object summary page
- recommendations on how to handle the issue, either specific steps or a KB

The severity of a finding is defined by the following criteria:

- Critical: Findings that can cause possible data corruption, environmental outage, significant performance impact, or a moderate to important security vulnerability within your environment.
- Immediate: Findings impacting the functionality of your environment that requires immediate attention but are not critical.
- Warning: Findings that are an advanced notice of a potential issue that might result in loss of data or an error state within your environment.

In the four-hour window of checking against property-based signatures, if you see null results, then your environment is healthy.

**Figure 452: Architecture Diagram and Data Flow of Findings**




In addition to the scheduled property-based scans, within the 24-hour window of **Active Findings**, you can run on-demand log-based checks by using the **Refresh Findings** option. With that option, you can narrow down the scope of your monitoring to a specific component, capability, or functionality. To run a log scan, you must have Log management installed and integrated in your environment. For more information, see How to Set up and Maintain VCF Operations Health and Diagnostics. VCF Operations Health and Diagnostics also provides configuration findings, but they are part of the operational state workflow in VCF Health.

You can trigger **Refresh Findings** for one or more specific management appliances or ESX hosts. This capability is important for three reasons:

- Some log scans might time out, which can cause some hosts to be skipped from assessment in the final result. If you troubleshoot a problem with such hosts or a cluster comprising such hosts, a refresh guarantees that the final result includes all known issues in your system.
- Some troubleshooting flows include checks also implemented as findings, for example, known issues with vSphere vMotion. With a refresh, you do not need to scan all inventory, but limit to specific objects.
- You can use **Refresh Findings** to troubleshoot upgrade failures.

In the **Check Last Run** column, you see the date and time when automated scans were completed or a manual scan for log-based findings was triggered.

In the **Last Observed** column, you see when findings were observed or occurred. For more details, see KB 375104.

> **Note:**
>
> Diagnostic findings for all objects are displayed on the **Findings** page even if the object no longer exists in your environment. This behavior is due to the global settings of VCF Operations that retain historical objects.

Log-based rules detect an actual occurrence of a known issue while property-based rules state that the issue is present in the build or configuration and might occur, or has occurred. In previous releases, all signatures were related to specific build numbers. With VCF 9.0 and later, property-based rules are derived from the configuration of objects such as the installed network drivers and ESX host configuration, regardless of the specific version. Depending on the findings, you can decide if corrective actions are applicable to your environment and how urgent it is to apply such actions. Sometimes, the corrective action is to apply a patch or upgrade, which requires planning, and you cannot immediately apply the recommendations. Findings are categorized by severity, component, type, and capability. In **Findings**, you see the number of affected objects for each issue, which helps in assessing the impact of the findings, prioritizing their remediation.

You can review and store the results in the **Active Findings** grid in tabular format by using the **Export** button. The export function collects all available data, including automatic findings, such as the **Active Findings**, and on-demand findings, such as logs when you use **Refresh Findings** or **Historical Findings**. In the file `vcf_operations_diagnostic_findings_summary.csv` that you get from the export package, you see the three categories under the **Refresh Type** column with values `Auto`, `Manual Active`, and `Manual Historical` respectively.

### VMSA Findings

To facilitate safeguarding your infrastructure, starting with health and diagnostics 9.1.1, all security-related findings for which Broadcom issues a VMware Security Advisory (VMSA), are collected in a separate tab. You can see the VMSA-related findings in the **Active Findings** tab along with all other findings, but in the **VMSA Findings** tab these are highlighted to facilitate immediate action. In the details grid, you see the same columns as in the **Active Findings** tab, but when you expand a row, you can see a summary of the vulnerability, a list of links to the affected objects in the inventory, and the recommendations you need to follow to resolve it.

### Historical Findings

If you investigate an issue that has occurred beyond the 24-hour window of active findings, you can use the **Historical Findings** option. By using this option, you can evaluate log-based findings for a specified time in the past and for the relevant inventory. With **Historical Findings**, you can also check if an active finding that you have fixed based on the recommendations has been fully addressed. You can also see what specific fix you used for a given issue.

> **Note:**
>
> **Refresh Findings** and **Generate Findings** are active only if you configure log management or if a log scan is already in progress.

You can also export the **Historical Findings** report for auditing purposes or for sharing with other teams, or subject matter experts. The **Export** button is on the **Active Findings** tab. The export operation captures historical logs as well under the `Manual Historical` category in the **Refresh Type** column in the file `vcf_operations_diagnostic_findings_summary.csv` that you get from the export package.

**Note:**

You can run only one Historical Findings query at a time. Current results overwrite previous reports. For new installations or migrated systems, active and historical findings might not immediately show results.

### Findings Catalog

Click the **Findings Catalog** tab to see the complete list of signatures that are being checked in your environment at any given time.

**Findings Catalog** replaces KB 371661 as the central reference point for available signatures.

### Log Assist

Starting with VCF Operations 9.0, **Findings** provide a shortcut to the **Log Assist** capability. **Log Assist** extends the log exports functionality with the diagnostics capability. You can use **Log Assist** to automatically upload the support bundles and diagnostic findings to your existing support ticket for troubleshooting by Broadcom Support.

## Diagnostics Public REST API

VCF Operations Health and Diagnostics provides a set of public APIs that allow you to consume findings in your own tracking and monitoring systems.

Starting with VCF Operations health and diagnostics 9.1, you can use a set of public APIs to customize data from **Findings**. The API allows you to integrate diagnostic findings into your internal reporting and monitoring workflows such as risk management or ticketing. The VCF Operations Health and Diagnostics API allows you to centralize monitoring of real-time findings or get data for trend analyses. The VCF Operations Health and Diagnostics API provides 2 authenticated, filtered, sorted, and paginated endpoints, `Active Findings`, and `Affected Objects`. With these endpoints, you can query findings and their respective affected objects, which allow you to integrate findings into your systems or any external systems you already use. The Diagnostics API uses the underlying VCF Operations infrastructure and is compliant with all existing public APIs.

### Authentication

For authentication, you use the existing authentication endpoint in VCF Operations and running a `POST`request to `/api/auth/token/acquire` which returns a response providing validation and a token, which you can use to make requests to the public API of VCF Operations.

Example:

```
{   "token": "the-token-value",   "validity": 1739285135796, //token validity as timestamp   "expiresAt":
 "Tuesday, March 11, 2027 at 2:45:35 PM Coordinated Universal Time", //expiry date   "roles": [] }
```

After acquiring the token, you can authenticate consecutive requests by including an `Authorization` header.

Example:

```
Authorization: OpsToken the-token-value
```

### Authorization

The token used to access any of these APIs has to be issued for a user with the following privileges: `environment.vcfhealth.view` and `vcf_diagnostics.view`.

### Findings

1. Get paginated findings:
   You can stream all your findings to an authenticated endpoint that is paginated and uses filters to narrow down the returned results, excluding `Affected Objects` to reduce the response size. You get results for the `Affected Objects` for a finding from a separate endpoint.
   Request:

Finding API requests use `POST` and the request body is an object with a single field: `filters`.

Example:

```
POST /api/diagnostics/findings/query Query:
page: int, Page number for the data pagination, 0 based.
pageSize: int, Page size, the maximum number of values returned.
sortBy: string, The field to sort on. Default value: SEVERITY.
sortOrder: string. The order for the sort. Possible values: ASCENDING, DESCENDING. Default value:
 DESCENDING.

Body: {   "filter": {     "resourceIds": [       "some-resource-id"     ], //filter by the UUID of the
 resource that generated the finding
"resourceKinds": ["VMwareAdapter Instance", "CAS_HEALTH_SERVICE",     ], //filter by resource kinds
 "adapterKinds": ["VMWARE",    ], //filter by adapter kinds
"capabilities": ["Snapshot", "VM Operations"], //filter by capabilities
"categories": ["Availability", "Operation Diagnostics" ], //filter by categories
"severities": ["Warning" ], //filter by severities
"ruleUuids": ["some-uuid", "another-uuid" ], //filter by rule UUIDs
"refreshTypes": ["Manual","Auto" ], //filter by refresh types
"findingTypes": ["Diagnostic", "Security Advisory"], //filter by finding types
"fromOccurrenceTime": 1234567890 //ts in millis indicating the earliest time of occurrence for the
 returned findings. This will return only findings that have occurred since the specified time until the
 current time. } }
```

Response:

The response is an object with a single field: `findings`.

Example:

```
{ "pageInfo": { "totalCount": 39, "page": 0, "pageSize": 1000 },
"findings": [     {     "ruleUuid": "1234-abcd", // the rule UUID for this finding
"ruleName": "some-rule-name", // this will be populated from the current ruleId field, but will be
 presented as rule name as this is more fitting name for the public facing API
"ruleDescription":
"Rule Description",
"severity": "critical", // severity of the finding, possible values are: none, info, warning, immediate,
 critical, automatic
"category": "Availability", // category of the finding, possible values are: Security Advisory,
 Availability, Operation Diagnostics, Pre-Upgrade checks, Compatibility, Product Interop, Performance,
 Upgrade, Troubleshoot, Security, Deployment Integrity
"capabilities": [ "VMOperations" ], // capabilities for the finding, possible values are: vMotion, DRS,
 HA, Snapshot, Workload Provisioning, Workload Domain Provisioning, Deployment Integrity, Certificates,
 VC_Resources, VC_Services, DNS, NTP, VM Operations, Other
"refreshMode": "Manual", // refresh mode of the finding, possible values are: Manual, Auto
 "affectedObjectsCount": 15, // the count of the affected objects associated with this finding
 "findingType": "Diagnostic", // type of the finding, possible values are: Diagnostic, Security Advisory,
 Best Practice, Compliance
"lastObservedTimeInMillis": 1708705894946     }   ] }
```

2. Get affected objects for a single finding:

The endpoint uses a Rule UUID retrieved from the previous request as a path parameter to return affected objects for that specific rule.

Request:

The request uses `POST`.

Example:

```
POST /api/diagnostics/findings/{ruleUuId}/affectedobjects/query Query:  page: int, Page number for the data
 pagination, 0 based. pageSize: int, Page size, the maximum number of values returned. Body: {   "filter":
 { "resourceIds": ["some-resource-id"], //filter by the UUID of the resource that generated the finding
```

```
"fromOccurrenceTime": 1234567890 //ts in millis indicating the earliest time of occurrence for the returned affected objects. This will return only affected objects that have occurred since the specified time until the current time. } }
```

Response:

The response is an object with a single field `affectedObjects` containing the current page of affected objects.

Example:

```
{ "pageInfo": { "totalCount": 39, "page": 0, "pageSize": 1000 }, "affectedObjects": [ { "name": "object-1", "resourceId": "some-resource-id", "adapterKind": "VMWARE", "resourceKind": "VirtualMachine" "checkTime": 1712955466761, "occurrenceTime": 1712955466761 } ] }
```

## Overwriting the Diagnostics Configuration

By using a JSON solution configuration file, you can manage some parameters of VCF Operations health and diagnostics . You can deactivate rules or set defaults for some tasks, depending on your environment.

With VCF Operations health and diagnostics, you can consider sizing limits for your environment. If you use the full capabilities of a VCF platform, you aim to avoid excessive use of resources for the cloud proxies. Best practice is to segregate workloads to optimize performance and computation time of health and diagnostics. This means dividing an environment by some logical grouping, such as by geography or business function. You can manage the limit of the number of objects that are part of such workloads and can set up a cloud proxy for each workload. You see an alert for the threshold of the max object count that you must consider when segregating workloads. This alert gives you a decision point whether to segregate workloads or increase the limit if it makes sense for your setup. For more information on how to use the JSON solution configuration file, see KB 369791. For sizing considerations, see KB 397782.

## Monitor Your Environment and Prevent Issues with VCF Health

With VCF Health, private cloud infrastructure administrators can monitor the entire VCF stack in depth and breadth that goes beyond the capacity of any individual VCF component or capability.

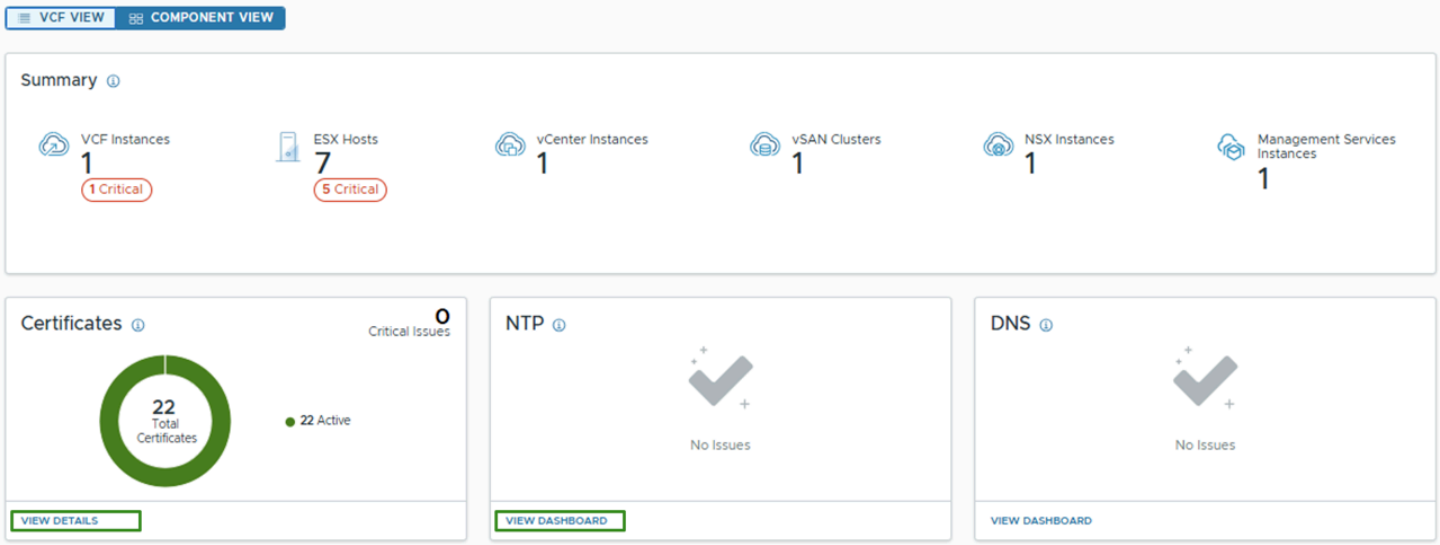
VCF Health provides a suite of monitoring capabilities that help private cloud infrastructure administrators to ensure the optimal performance, availability, and stability within private cloud environments, across all VCF instances. With VCF Health, you gain immediate insights into the overall health and status of the private cloud infrastructure and you can proactively identify and resolve issues.

**VCF Health** gives two starting points for monitoring:

- **VCF View**, which allows a drill-down in the inventory tree from a VCF instance to its management and workload domains, vCenter instances, NSX instances, vSAN, and ESX hosts..
- **Component View**, which provides a holistic view summarizing all objects in your environment.

Based on data from support requests, **VCF Health** screens are structured to help you with the most critical steps in the maintenance and management of the VCF platform. For example, tracking certificates, NTP sync, and DNS reverse lookup from operational point of view or tracking the performance of vCenter instances from infrastructure point of view. You can drill down to each tile that summarizes a VCF operational metric to an individual dashboard or panel in VCF Operations by following the **View Dashboard** or **View Details** link.

**Figure 453: Links for Drill-down from the VCF Health Tiles**




With VCF Operations health and diagnostics 9.1, the list of tracked VCF components and capabilities extends to include vSAN, NSX, ESX, VCF services runtime, and vSphere High Availability. VCF Operations health and diagnostics 9.1 adds new metrics to the existing reporting for vSphere vMotion, VM Operations, vCenter, and Snapshots.

> **Note:**
>
> VCF Health provides data only on active inventory objects. If you delete a resource, all related events are deleted immediately.

vCenter is the key object for monitoring and reporting as it is the key building block in a VCF platform. **VCF Health** also covers the key support request-generating areas in a VCF environment: health of ESX hosts, workload operations, such as deployments and updates of virtual machines, virtual machines snapshots, and vSphere vMotion tasks.

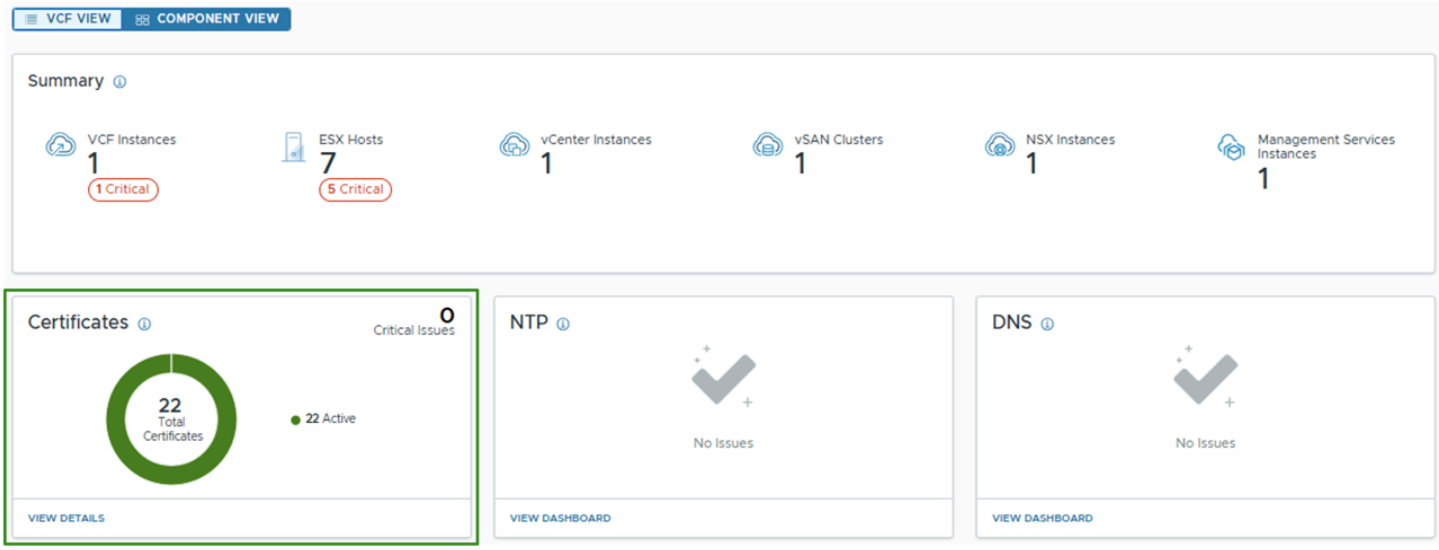
## Certificates Health in Diagnostics

VCF 9.0 and later provides centralized certificate management to monitor and manage all certificates across all VCF components.

Certificate issues are reported for objects that have a VCF 9.0 or later entitlement. A VCF domain is entitled if the vCenter in the domain is of version 9.0 or later and has a non-expired 9.0 license.

Clicking **View Details** on the **Certificates** tile takes you to the **VCF Management** screen under **Manage** > **Fleet Management** > **Certificates**, where you can take pending action on certificates as necessary: renew, import, or replace a certificate, or activate auto-renewal.

**Figure 454: Certificates Tile in VCF Health**




**Note:**

To see all certificates, data collection of each VCF Operations health and diagnostics integration must be in a healthy state.

Click any of the certificates to open the certificate details pane to see the particulars for this certificate. Use certificate management to monitor the status and expiration of the following certificate categories:

- Platform Certificate
- Root Certificate
- STS Certificate
- TLS Certificate

After you add a new certificate or update your certificate, it takes approximately 10 minutes for the certificate information to be reflected on the certificates page.

For more information, see *Managing Certificates in VMware Cloud Foundation*.

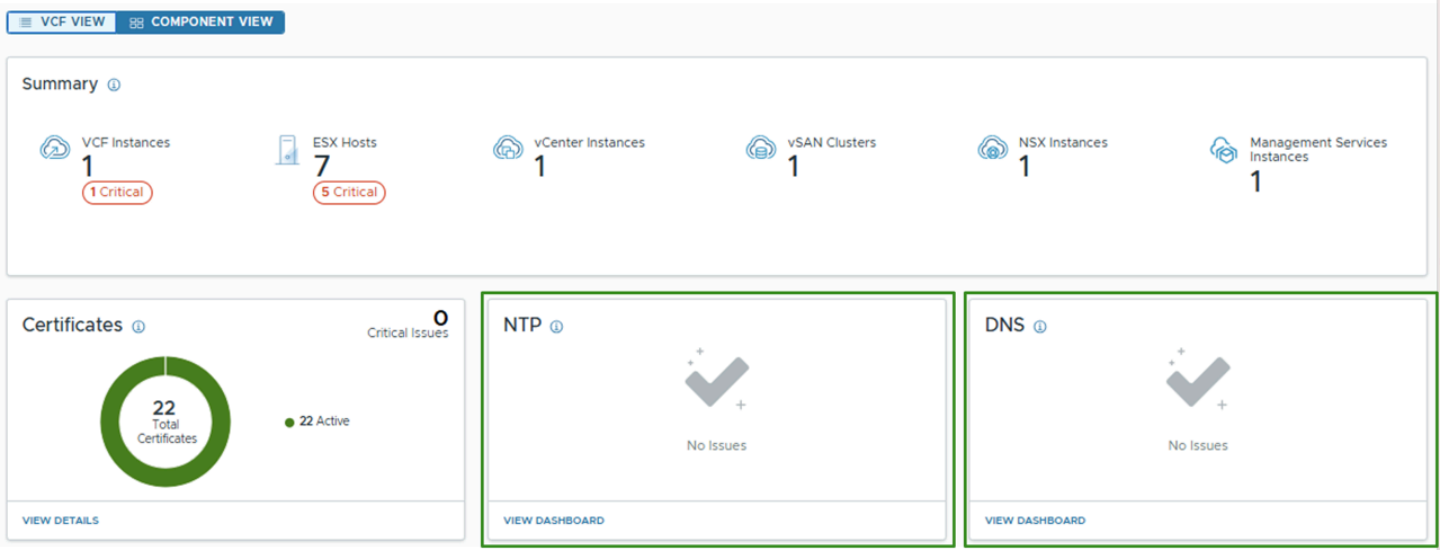
## NTP and DNS Health in Diagnostics

Monitor NTP and DNS health to avoid issues that NTP drift and DNS forward and reverse lookups might cause, such as failing vSphere vMotion tasks.

You need to monitor NTP and DNS settings over time to avoid issues that NTP drift and DNS reverse lookup might cause, such as failing vSphere vMotion tasks. For example, if you do not set up a DNS forward and reverse lookup, in some environments you get some connections labeled as insecure. As a result, certain integrations might break with errors

that no secure connection can be established. With VCF Operations health and diagnostics, you can see if NTP is not configured or is out of sync, as well as the DNS lookup issues for each object in your environment.

**Figure 455: NTP and DNS Tiles in VCF Health**




## NTP

On the NTP tile, you see if NTP is not configured on some components. You can use the dashboard that is linked to the tile to check which NTP server you must use to configure NTP. You can verify that components within each VCF deployment are configured with the correct NTP servers.

The count of configured components only includes components that are accessible. You also see the NTP server drift, which is computed for each component as follows:

1. Fetching the current time from the component
2. Subtracting the current time of the Cloud Proxy performing this check
3. Adding the result to the difference already accrued in the Cloud Proxy time drift from its NTP server

For example, the current time from the component minus the current time of the cloud proxy equals 35 seconds. We add 35 seconds to the difference already accrued in the time drift of the cloud proxy, say another 35 seconds. The result is a time drift larger than a minute and you see it on the tile. This calculation executes every 5 minutes for vCenter and every six hours for ESX and NSX.

> **Note:**
>
> You might see some transient discrepancies between **VCF Health** and the **VMware Infra Health Overview** dashboard that shows NTP configuration data. The Infra Health dashboard gets data from the SDDC Manager every 24 hours, while Health and Diagnostics gets data from the individual components in maximum six hours.

**DNS**

On the DNS tile, you see if DNS is not configured on some components. You can use the dashboard that is linked to the tile to check which DNS server you must use to configure DNS. You can verify that components within each VCF deployment are configured with the correct DNS servers.

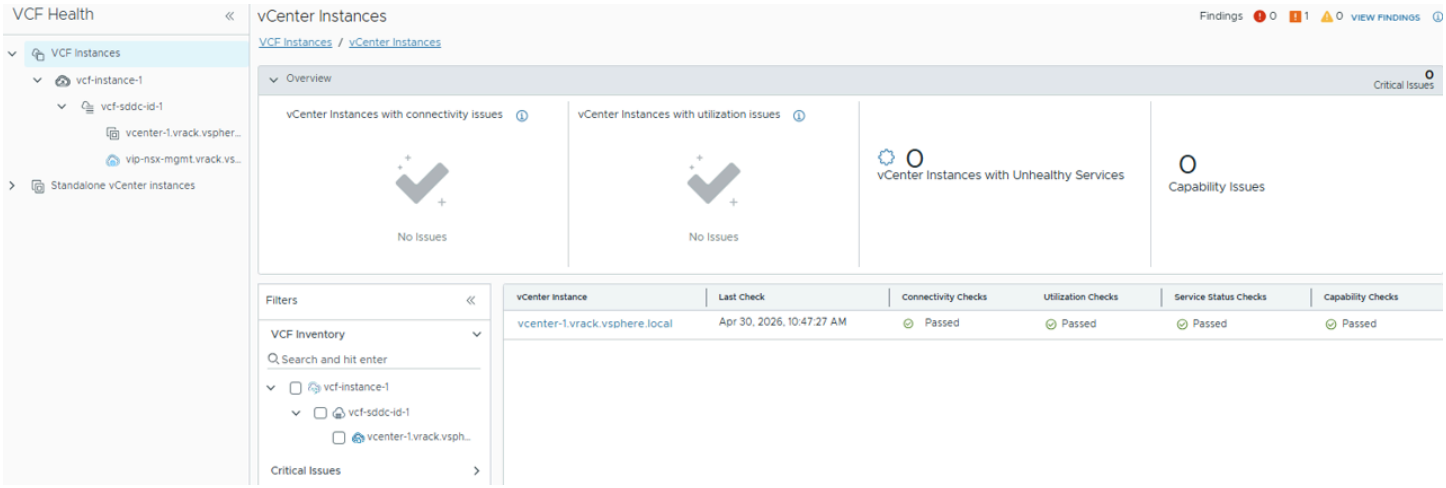
The count of configured components only includes components that are accessible. You also see a count of failed forward and reverse lookups, evaluated based on the DNS server with which the VCF Operations appliance is configured. The DNS lookups report failure if the PTR or A record lookups do not return a value.

## vCenter Health in Diagnostics

With **VCF Health**, you can track data for connectivity, utilization, and services in a vCenter system that you cannot derive from any individual tool for vCenter monitoring.

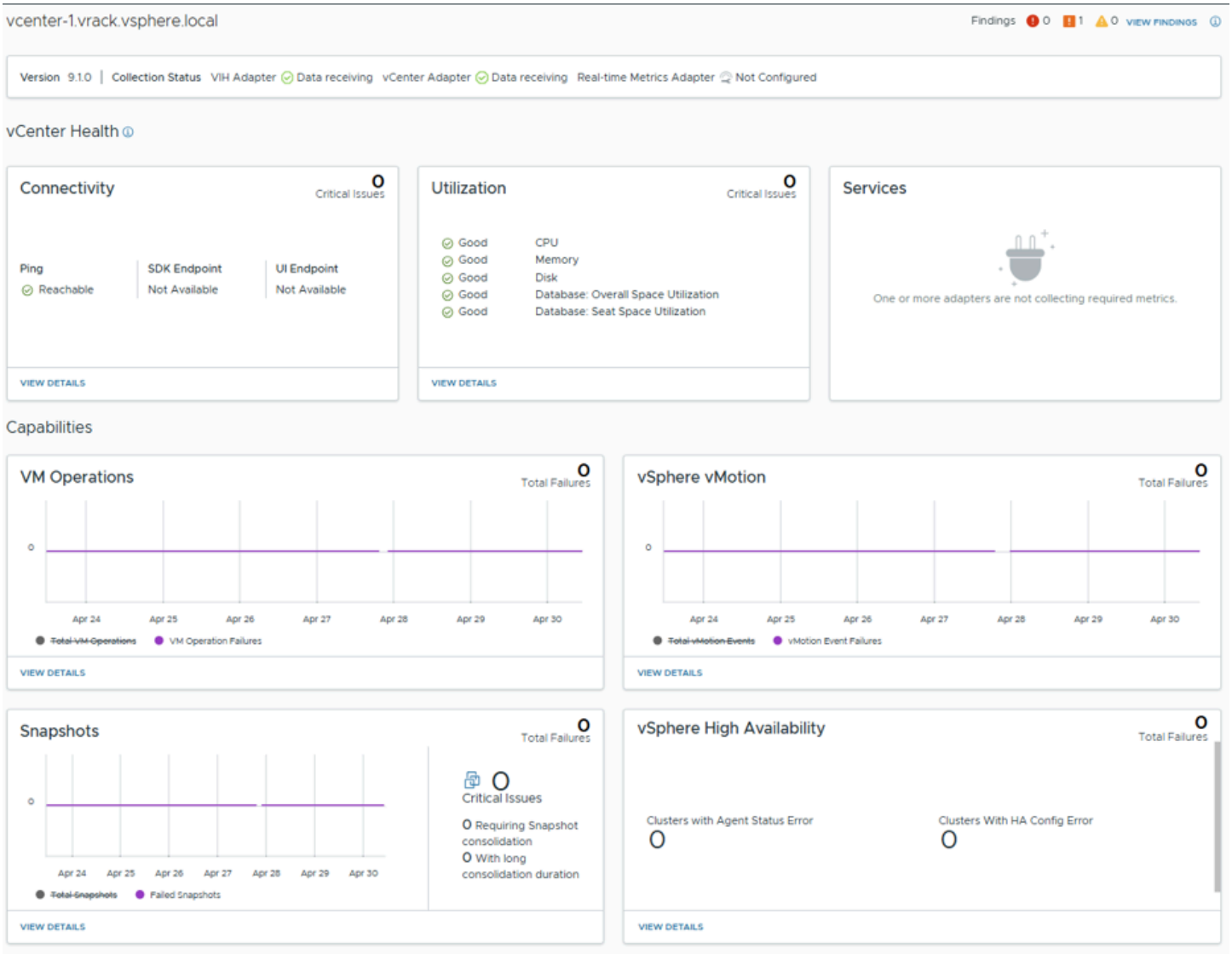
vCenter is the key object for monitoring and reporting as it is the key building block in a VCF platform. **VCF Health** helps you with that task and adds value by providing data for connectivity, utilization, and services that you cannot derive from any individual tool.

**Figure 456: vCenter Instances View in VCF Health**




Given the complexity and size of VCF deployments, testing multiple vCenter instances for reachability might be challenging. Especially if you want to confirm all aspects - networking, user interface, and API, since one does not guarantee the rest. For example, API calls to a vCenter instance might be successful, but the vSphere Client might not be reachable. Such cases might indicate issues with the vCenter services or the vCenter database.

**Figure 457: Individual vCenter Instance Level View**




## Connectivity

In the **Connectivity** screen, you see the connectivity trend over the last seven days. The data tracks the reachability of vCenter instances by network, UI, and API.

VCF Operations Health and Diagnostics 9.1 adds endpoint monitoring, replacing the ping reachable/unreachable check to the vSphere Client and APIs. The SDK/API endpoint metrics track any internal or external integration of your environment that uses API calls. The UI endpoint metrics track the vSphere Client endpoint availability, focused on vCenter. Availability by ping and UI is scored from 0 to 100 points. The check includes the `HTTP 503 Service Unavailable` server error response, and the `HTTP 429 Too Many Requests`, when API session limits are reached.

Pings to a vCenter instance check if the vCenter virtual machine is powered on and available online. Pings to a vCenter instance also check if all vCenter services are active and if the vSphere Client is responsive. Since many of the VCF integrations also occur by using APIs, VCF Health checks reachability by API.

You can drill down data for each reachability alternative and can investigate any unexpected sustained outages by reviewing findings for known issues or can use recommendations for general troubleshooting.

**Note:**

Data on the **List of failed API for SDK/UI endpoint availability** table depends on the log management adapter. Make sure that the adapter is in a connected and receiving state.

**Utilization**

In the **Utilization** screen, health and diagnostics measures thresholds for CPU, Memory, Disk, and Database space usage, so you can easily identify issues that might result from a memory leak or a scalability issue, and to proactively resolve issues before they occur.

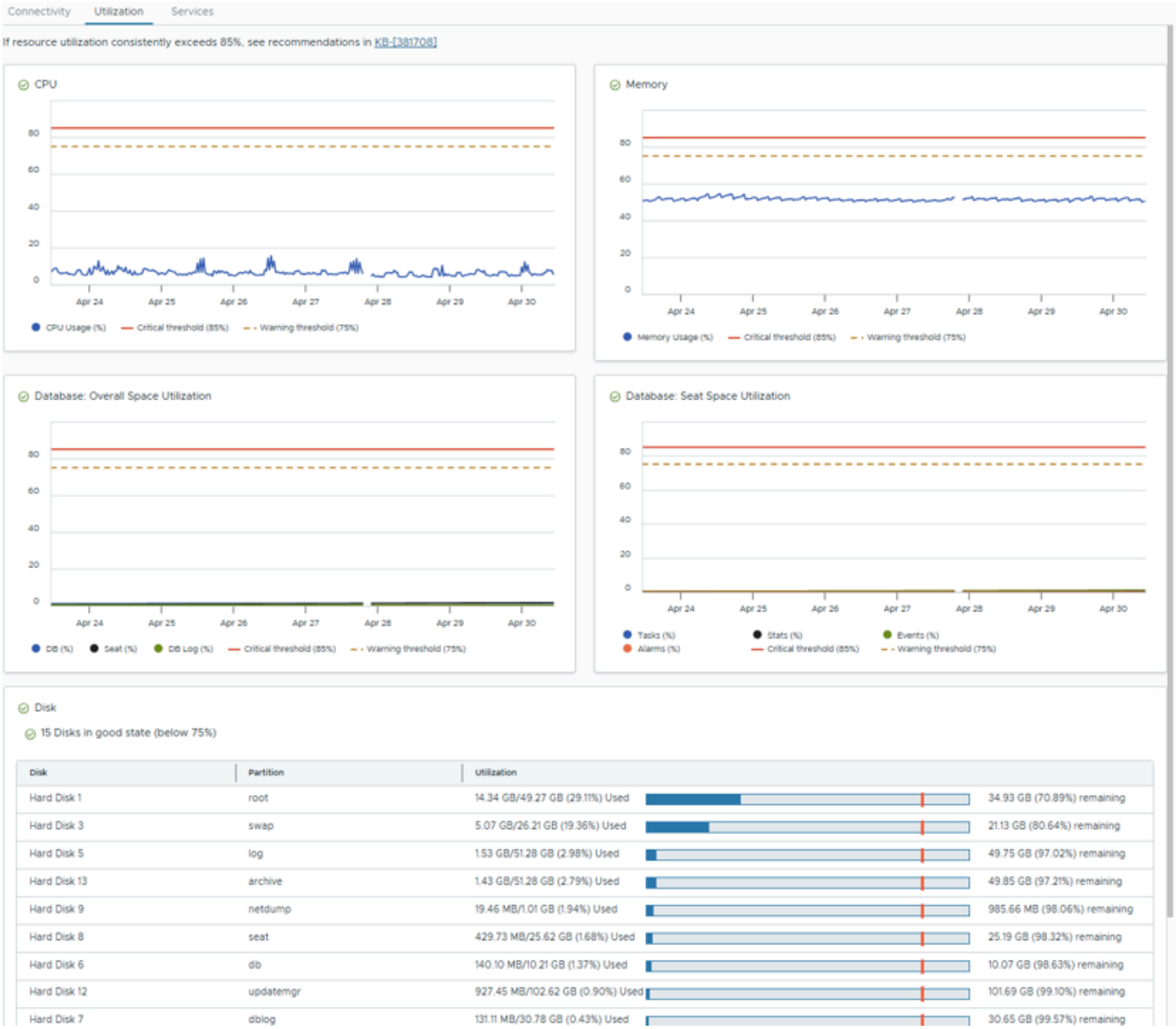
VCF Operations Health and Diagnostics 9.1 enhances the usability of the **Utilization** screen by providing details in condensed charts that give all information in one layer. You no longer have to drill down to see details in multiple charts and layers.

**Note:**

VCF Health supports the CPU, Memory, and Disk metrics on the **Utilization** dashboard under **vCenter Health** only for environments that migrate from vCenter 8.0 Update 3 and later.

The panel of each metric gives you a seven-day overview of the utilization against a warning threshold of 75% and a critical threshold of 85%. Charts are similar to this sample:

**Figure 458: CPU utilization panel**




The utilization of the vCenter Database is presented in two charts: one for the overall utilization of the vCenter DB, DB logs, and Stats, Events, Alarms, Tasks (SEAT), and another for SEAT alone, with a breakdown for each category.

The **Disk** chart is different from the other charts in the **Utilization** screen, because it provides a breakdown of the current utilization of all virtual disks in the selected vCenter instance and prompts immediate action, unlike the other charts, where you see historic results and act retroactively.

Click the **Check Recommendations** button in each metric panel to get guidance on remediation steps.

### Services

In the **Services** screen, you see the health and run state of the critical vCenter services over a period of seven days, which allows you to take immediate action on any currently unhealthy service or follow the health trend to monitor if any service is repeatedly unhealthy, or unhealthy for a prolonged period. You can troubleshoot by using the KB article available in this screen.

With VCF Operations health and diagnostics 9.1, the list of services covers all 17 critical services for vCenter, including some of its related capabilities such as the vSphere Supervisor. For more information, see KB 381709.

VCF Operations Health and Diagnostics 9.1 provides data that allows you to investigate the warning and critical states. For example, see a Memory utilization chart for the specific service that helps you realize that a certain memory limit is reached. Each service has its own set of metrics.

> **Note:**
>
> For vCenter 9.1, a Critical status is reported when the vpxd service memory, connectivity, or thread utilization exceeds 85%. Warning status is between 75% and 85%. Checks run on every 5 minutes. In earlier versions, the health status of the vpxd service was collected periodically from the vCenter Appliance Management Interface.

Status of Healthy, Warning, and Critical is based on the memory and connection state of services. The health status of vCenter services is calculated differently in VCF Operations for health and diagnostics, based on data from the Real-time metrics adapter, than in vCenter. You might see different statuses when looking at an individual service in the vSphere Client or the Virtual Appliance Management Interface.

You can drill down to see the health graph for each service.

> **Note:**
>
> The health charts depend on the Real-time metrics adapter. Make sure that the adapter is in a receiving and connected state.

### Capabilities

On the vCenter instance level, you can follow through with monitoring specific aspects of infrastructure and performance, covering:

- Virtual machine (VM) operations
- vSphere vMotion
- Snapshots
- vSphere High Availability

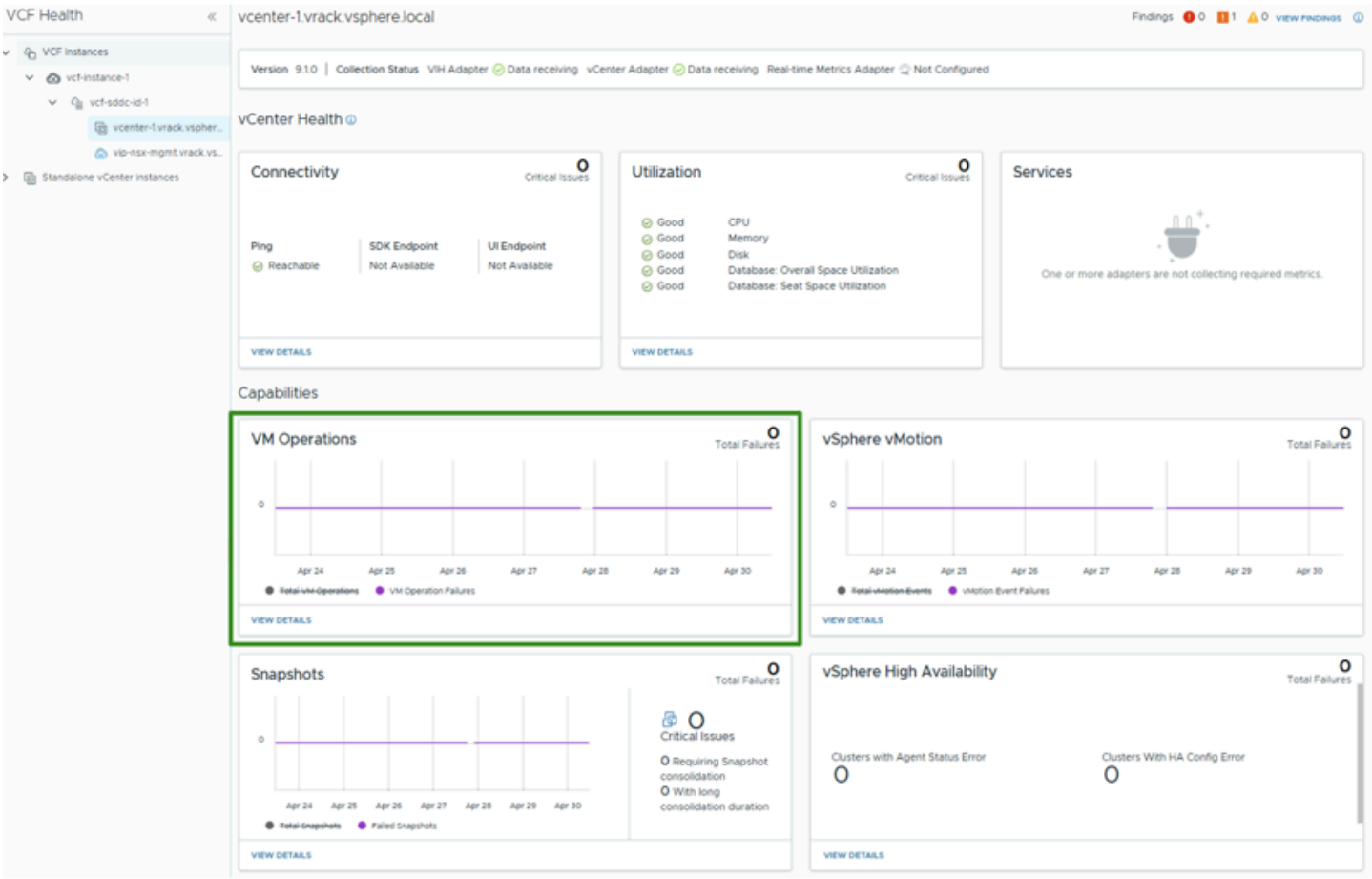
In each tile, you see an overview with data for the past seven days. You also see a ranking in each capability for the highest number of incidents, or most impactful issues, or objects with the highest number of failures.

## Virtual Machine Operations Health in Diagnostics

With VCF Operations health and diagnostics, you can track data from all tasks to create, delete, power, and reconfigure virtual machines in a vCenter instance.

**VM Operations** under **Capabilities** on vCenter instance level replaces the legacy **Workload Operations** tile in Diagnostic Findings 5.2 and later. **VM Operations** covers the entire span of tasks to create, delete, power, and reconfigure virtual machines in a vCenter instance.

**Figure 459: VM Operations Tile in VCF Health**




The VM Operations data allows you to catch issues before they occur during large or automated deployments. Often workload operational tasks are not run by a VCF admin, but from VCF Automation agents, or API tools, which create and destroy VMs in large numbers.

For example, from VM Operations you can spot when a blueprint that an automation workflow uses is failing, which might prevent production-down scenarios. With VM Operations, you see the balance between VM provisioning requests that succeed and fail, which gives you a reliable means of identifying such potential issues. In addition, you can drill down to see the distribution of failures by cause or type of initiator. You also get specific recommendations on the type of operation, such as Reconfigure, which further helps you troubleshoot existing or to catch emerging issues.

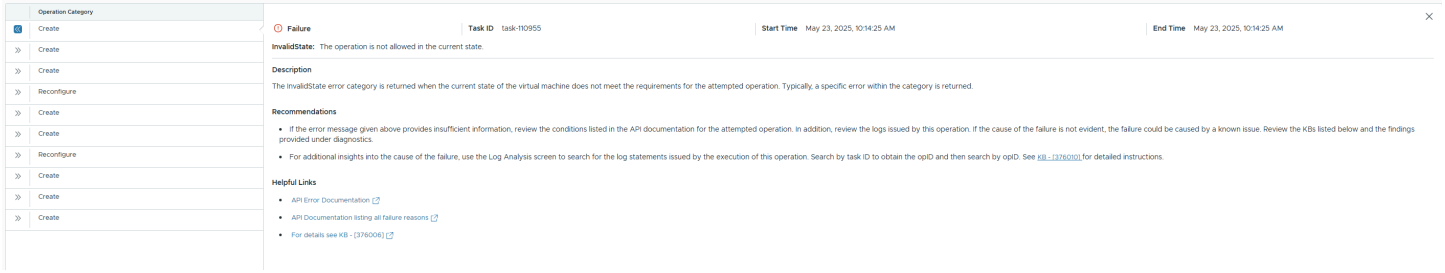
VM Operations monitors vCenter tasks and reports such that fail with context details and the respective error message, so you can proactively troubleshoot a potential issue before it occurs. To help you better understand of root cause of a failure, every report includes a link to the vCenter API documentation for the API that was invoked.

You can also see the distribution of VM Operation failures for the last seven days, the main reasons for failures, and the main initiators with failures.

VCF Operations Health and Diagnostics 9.1 adds the **VM Operations Task ID** parameter that allows a drill-down to particular task. Starting from VCF Diagnostics 9.1, you can also use the **Error Stack** panel to analyze the logs upon a VM failure, in the same screen.

When you expand a row in the grid, you see specific recommendations and links to API docs and KBs that help you identify errors:

**Figure 460: VM Operations Details Screen**




VM Operations covers the following operations, as displayed in the vSphere Client:

Create Operations Failures:

- Apply DRS clone recommendation
- Create New Virtual Machine
- Clone vApp
- Clone Virtual Machine or Template
- Deploy OVF
- Mark a template as Virtual Machine
- Register Virtual Machine

Power Operation Failures:

- Apply DRS power on recommendation
- Power on, off, and suspend virtual machine
- Initiate guest OS shutdown
- Start, stop, and suspend vApp
- Initiate guest OS standby
- Hard stop virtual machine
- Reset Virtual Machine
- Reboot Guest

Delete Operation Failures:

- Delete Virtual Machine
- Unregister and Delete Virtual Machine
- Delete and unregister vApp

Reconfigure Operation Failures:

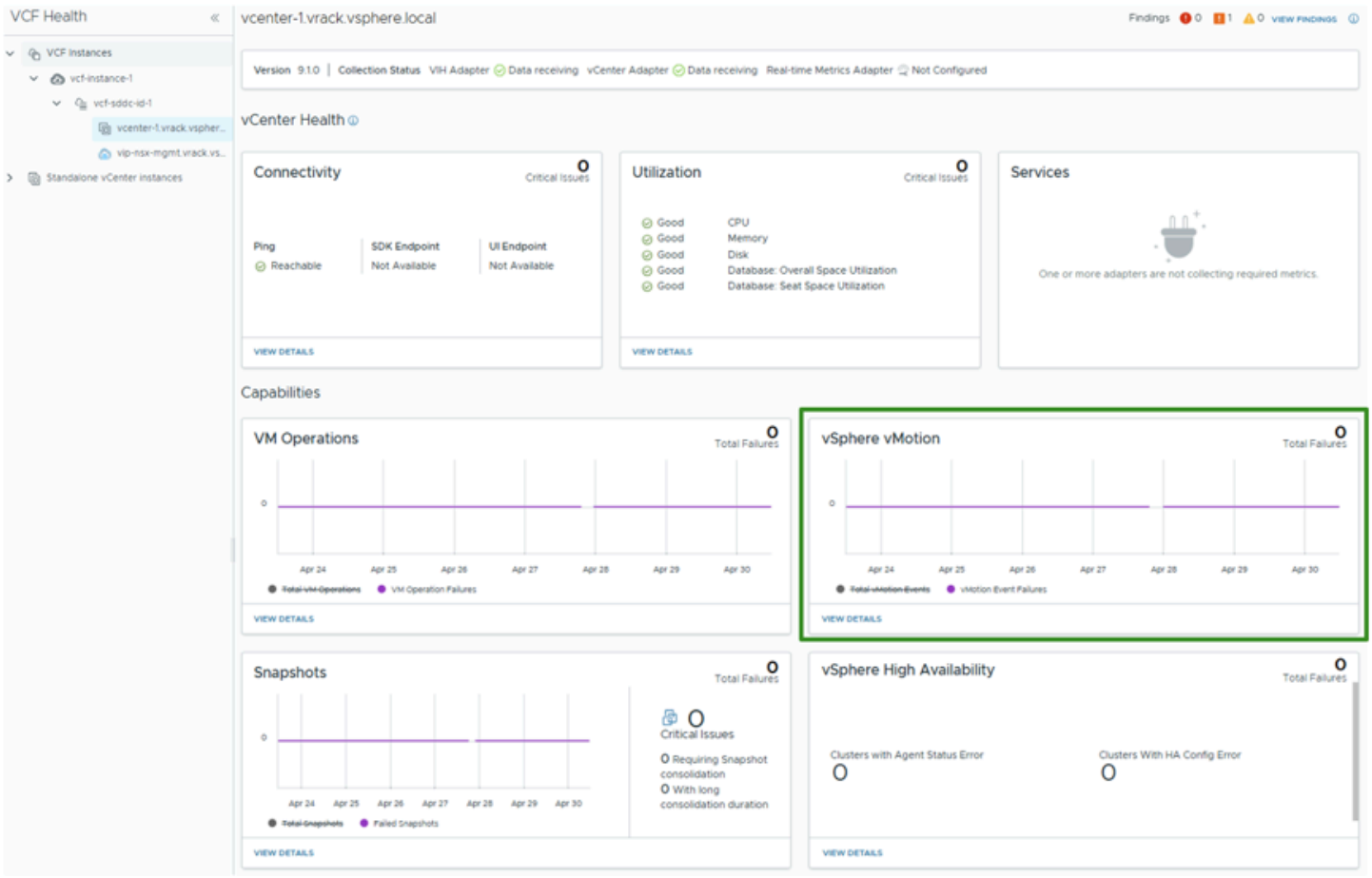
- Reconfigure Virtual Machine
- Attach or detach a virtual disk
- Update VM Hardware version
- Guest OS operations, such as VMware Tools and Guest OS Customization

## vSphere vMotion Health in Diagnostics

With vMotion Health, you can directly see the root cause of vSphere vMotion failures instead of running multiple checks or calling support.

In the **vSphere vMotion** tile under **Capabilities** on vCenter instance level, you see data from vCenter events and log statements that are collected and analyzed to help you troubleshoot possible failures. The span of vCenter monitoring under **Tasks and Events** is relatively short and vSphere vMotion failure events roll over quickly. You might not identify an issue before you see an impact on the operations, such as applications slowing down or failing. vMotion Health provides you with this capability at a glance, allowing you to see critical issues as they emerge.

**Figure 461: vSphere vMotion Tile in VCF Health**




Critical issues indicate definite failure of vMotion tasks. If such an issue affects either the source or destination host, or the connection between them, VM migrations fail. Warning issues indicate a possible failure, depending on the VMNICs, configuration, and connectivity.

> **Note:**
>
> If you see no vMotion operations available on the screen, confirm that you have configured log management to collect events from your vCenter instances.
>
> If you see no time or bandwidth values available for successful vMotion operations, check if log management receives events from vCenter instances.

In the **vMotion** screen, you not only get all data in real time, but also see breakdown by status, failed or success, and by reason, such as **Invalid state** or **Invalid Power state.** You also see a description of the possible issue and recommendations for resolution. In the success scenario, you see data for the pre-copy time, pre-copy bandwidth, and switchover time. These metrics help you see how well your infrastructure is configured and provide insight into why a vMotion could take longer than expected.

For example, an increase in the pre-copy time and switchover time compared to earlier vMotion tasks of the same VM indicates a possible issue with your physical network and infrastructure: it might be a physical switch too busy or a NIC driver dropping packets. Or, it could be that the vMotion task was run while the VM was issuing many writes or the source/destination hosts were overloaded at the time.

For failure scenarios, you see the type of failure, such as timeout, and an explanation of this failure type. For failure types that can result in a vMotion task timing out, VCF Health provides a troubleshooting flow.

From the cluster level in the details grid, you see the Configuration and Connection checks, which create an inventory of issues on several categories. These checks evaluate both the source and destination host that is involved in the task, including VMkernel port groups, network connectivity, and jumbo frames. You can expand each of the checks and can see details. For example, you might see that the root cause for a failure are MTU of different sizes, 9000 and 1500.

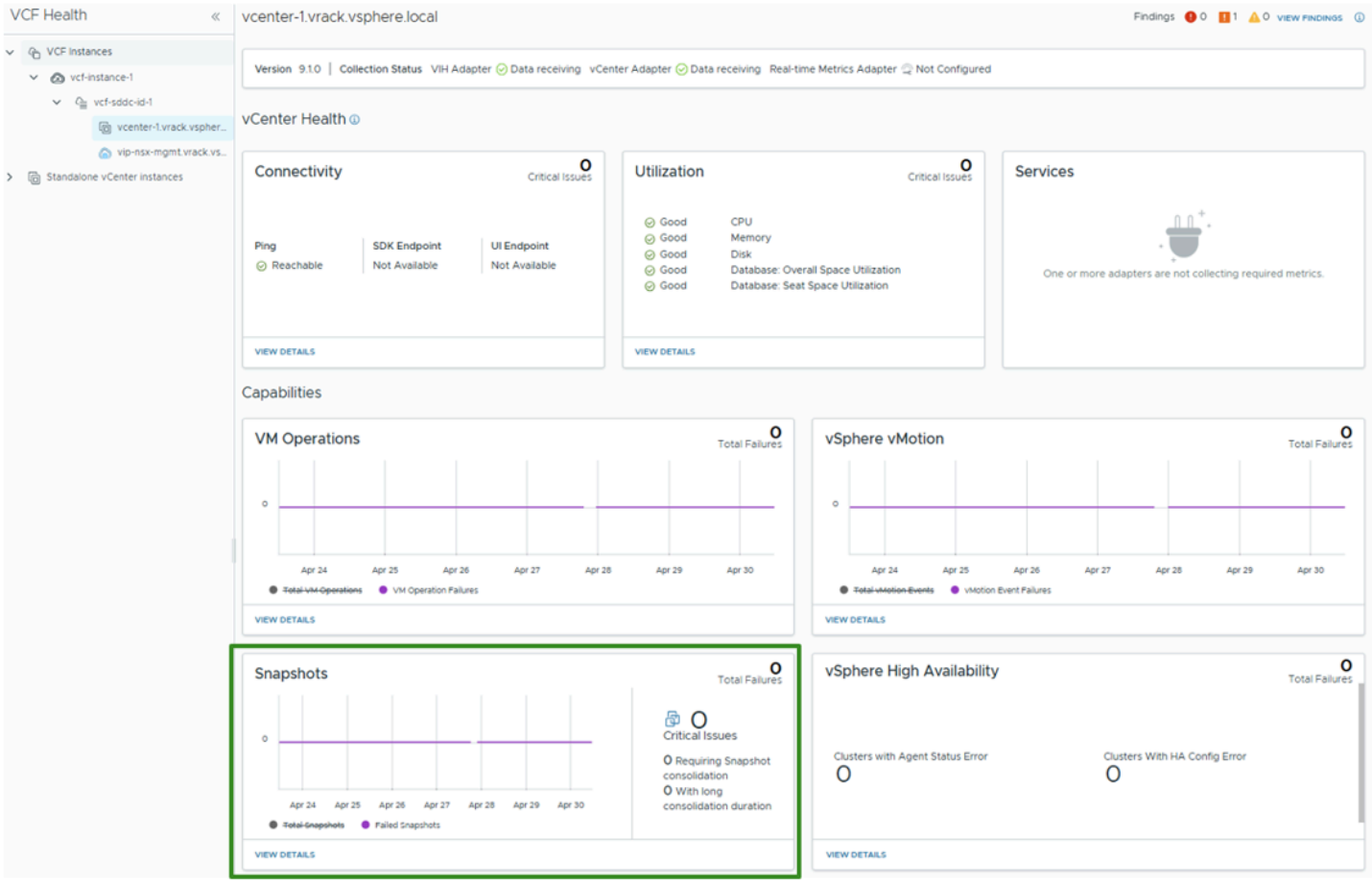
Other important vMotion health checks are available in the **Findings** tab. If the troubleshooting flow reports no issue, review the manual and auto-refreshed findings on vSphere vMotion. After the troubleshooting is complete, you can see the details on recommended steps to fix the parameter after you click **Check Results**. You can export the results or can rerun the troubleshooting.

## Virtual Machine Snapshots Health in Diagnostics

With VCF Operations health and diagnostics, you can track long-running snapshot delete, create, and consolidation operations, and details on failed snapshots, or disks in need of consolidation.

VCF Health monitors vCenter snapshot tasks of every virtual machine, VMs that need consolidation, and VMs with too many or too large snapshots. Similar to vSphere vMotion tasks, you cannot consistently track snapshot operations in the vCenter **Tasks and Events** because they are rolling over fast, and you come to know about an underlying issue with snapshots too late. Snapshots in production environments are not best practice, but any VM backup software uses snapshots and such use drives a serious number of support requests.

**Figure 462: Snapshots Tile in VCF Health**




In the **Overview** panel of the **Snapshots** tile under **Capabilities** on vCenter level, you see a list of critical issues that are related to snapshots and the following data for the last seven days:

- Ratio of failed snapshots
- VMs with most failed snapshots
- Top initiators of failed snapshots
- A chart for the deletion and consolidation time.

### Snapshot Operations

In the **Snapshot Operations** screen, you see the status of ongoing snapshot delete, create, and consolidate operations, and details on snapshots that have failed. You can filter out snapshot operations that run for 4 or more hours. Such operations might be a business problem and might require to consider a cancellation of the operation. To see data latency of the source data stores, click **Check** under the **Check Latency** column of the detail grid. You can see relevant data such as the actual data store latency, I/Os for the VM, and the overall write I/O load for the data store. Assess if networking, storage, or system contention are increasing the time to complete the operation. If networking or storage issues are suspected, you might need to involve your infrastructure team or vendor.

For snapshot failures, you see the error that vCenter reports. You can expand each row in the data grid to see details on the error and recommendations for troubleshooting. Recommendations include KBs and links to API documentation, similar to the other detail grids in **VCF Health**.

> **Note:**
>
> The list of snapshot failures does not include snapshot delete operations that fail during the second step of a delete operation. In the first step, snapshots are removed from the snapshot tree, and the second, snapshot data is consolidated to the parent snapshot. If a snapshot delete fails in the second step, vCenter intentionally reports the task as successful and issues events to warn about the failure. The **Snapshot Operations** screen does not report this type of failure. When a failure occurs in this second step, such VMs show in the **VMs that need consolidation** screen.
>
> If the underlying datastore is using native snapshots, long-running snapshot delete and consolidate operations are not expected. Native snapshots do not involve the movement of data when deleting snapshots.

For failed events, VCF Operations health and diagnostics 9.1 adds the task ID, which is linked to the respective vCenter instance and provides all details on the failed task in the same screen. In addition, the details screen includes the **Fault stack trace** section, where you directly see the error logs. This section saves you logging in to a shell of an ESX host and using ESXCLI commands to get the logs, which requires training and experience. Even if you need to contact the support team, you can do so with the logs and details ready.

### VMs needing consolidation

In the **VMs needing consolidation** screen, you see the allocation of VMs on datastores and the free space so that you can decide whether to run a consolidation.

Snapshot delete operations can fail if the space on a datastore is not sufficient to consolidate the disks. You can use the **VMs needing consolidation** screen to determine the storage used by a VM on each of its datastores. Space can be freed up on a datastore by consolidating VMs with smaller footprints first, deleting all snapshots, or moving the disks to other datastores. Act before the free space falls below 200 MB.

When consolidation is initiated, the task might take long, depending on the size of the disk and the data store performance. Consolidation tasks might also affect latency and some tasks that take seconds might turn into minutes. In the **VMs needing consolidation** tab, you can see such conditions and can prevent them from aggravating. For example, you can:

- Reduce the number of users accessing an application,
- Schedule a consolidation window and inform users not to access the app,
- Move to other VMs from the data store.

The view also adds the **Max data to consolidate** parameter for the running consolidation so that you make sure to allocate the necessary space and avoid memory outages.

### VMs with Snapshots

VMs with too many or too large snapshots can cause performance and storage issues.

You might not know that a VM runs on a snapshot: for example, when a backup software takes care of snapshots. Snapshot files can also significantly increase in size with every snapshot taken and by the time of consolidation, the size might become huge. The unhandled volume of such automatic snapshots can affect the performance of the VM and other VMs in the same datastore. To provide visibility on such scenarios, VCF Operations health and diagnostics 9.1 adds tracking of the number of snapshots per VM, the pace of growth in size, and the biggest VMs with snapshots.

In the **Overview** panel of the **VMs with snapshots** tab, you see VMs with fast-growing snapshot sizes, the biggest VMs in terms of snapshot size, and VMs with more than three snapshots. These metrics help you prevent a situation when a datastore goes out of space due to unattended snapshot issues.

In the details grid, you get all metrics that you need to decide how to handle VMs with large snapshots that might affect your environment.

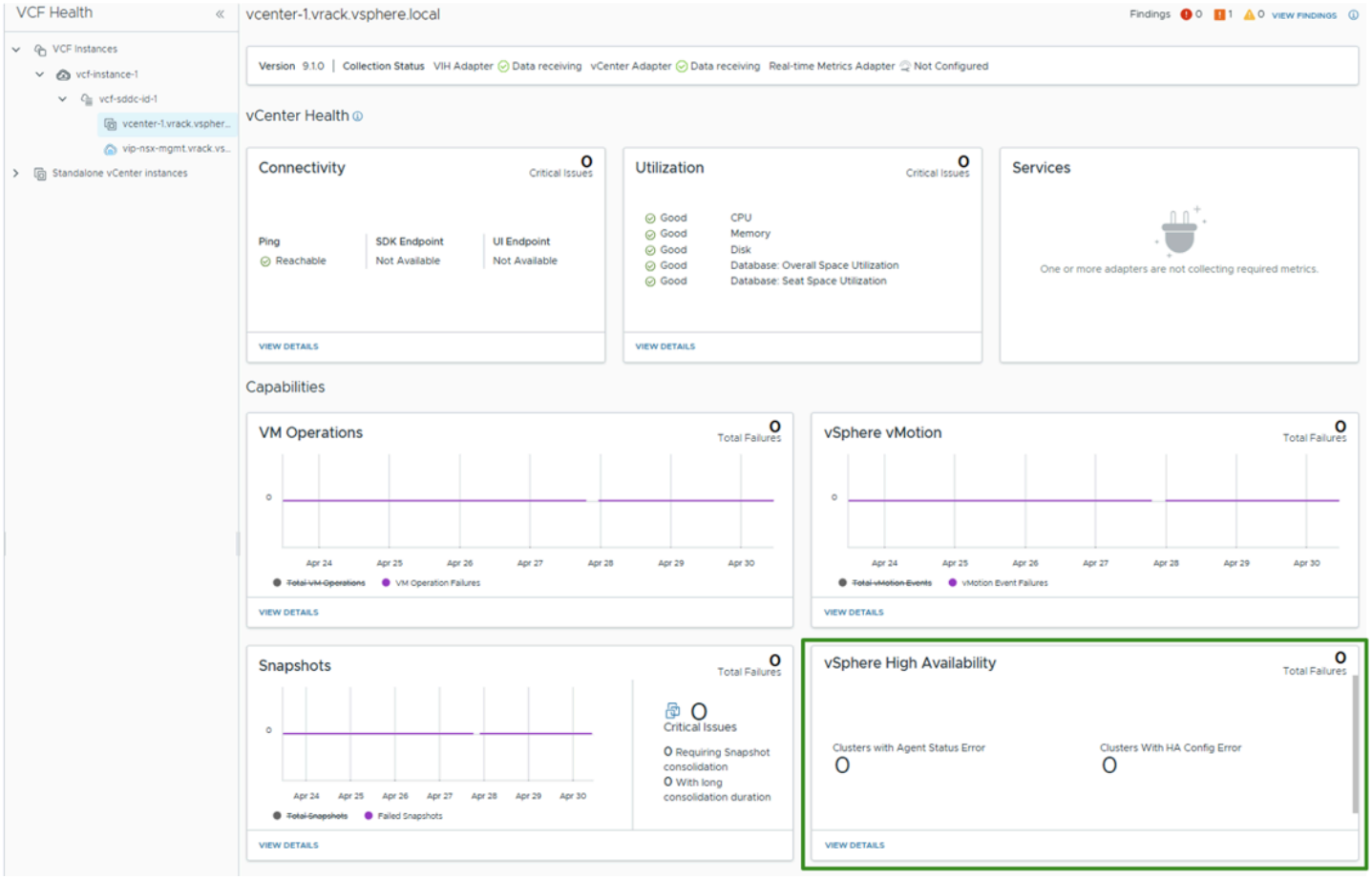
## vSphere High Availability Health in Diagnostics

vSphere High Availability metrics help proactive identification of issues and preventive resolution from a view that lists all high availability clusters with their operational state.

In the **vSphere High Availability** tile under **Capabilities** on vCenter instance level, you see all configuration issues and High Availability agents with errors.

You can drill down to specific configuration issues or agent errors per cluster by expanding the details grid on the landing page.

**Figure 463: vSphere High Availability Tile in VCF Health**

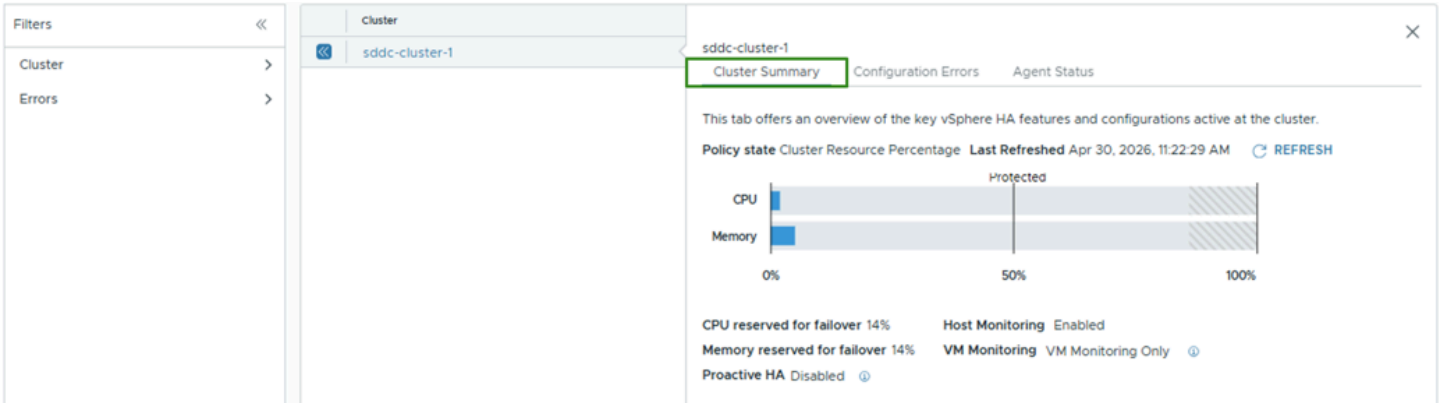



In the detail view per cluster, under the **Cluster Summary** tab, you can see a summary of the cluster in terms of CPU and Memory reservations, and policies.

**Note:**

The CPU and Memory chart data depends on the HA configuration policy. You can see the current policy in the **Admission Control Policy** field on the cluster level in the vSphere Client, before you drill down to the details view.

**Figure 464: Cluster Summary View**




In the other tabs, **Configuration Errors** and **Agent Status**, you see cluster and host-specific configuration issues, and agent status, along with recommendations for resolution.

> **Note:**
>
> Diagnostics track a curated list of critical HA configuration issues that require immediate attention. In the vSphere Client, you might see more HA configuration errors but they might not be critical.

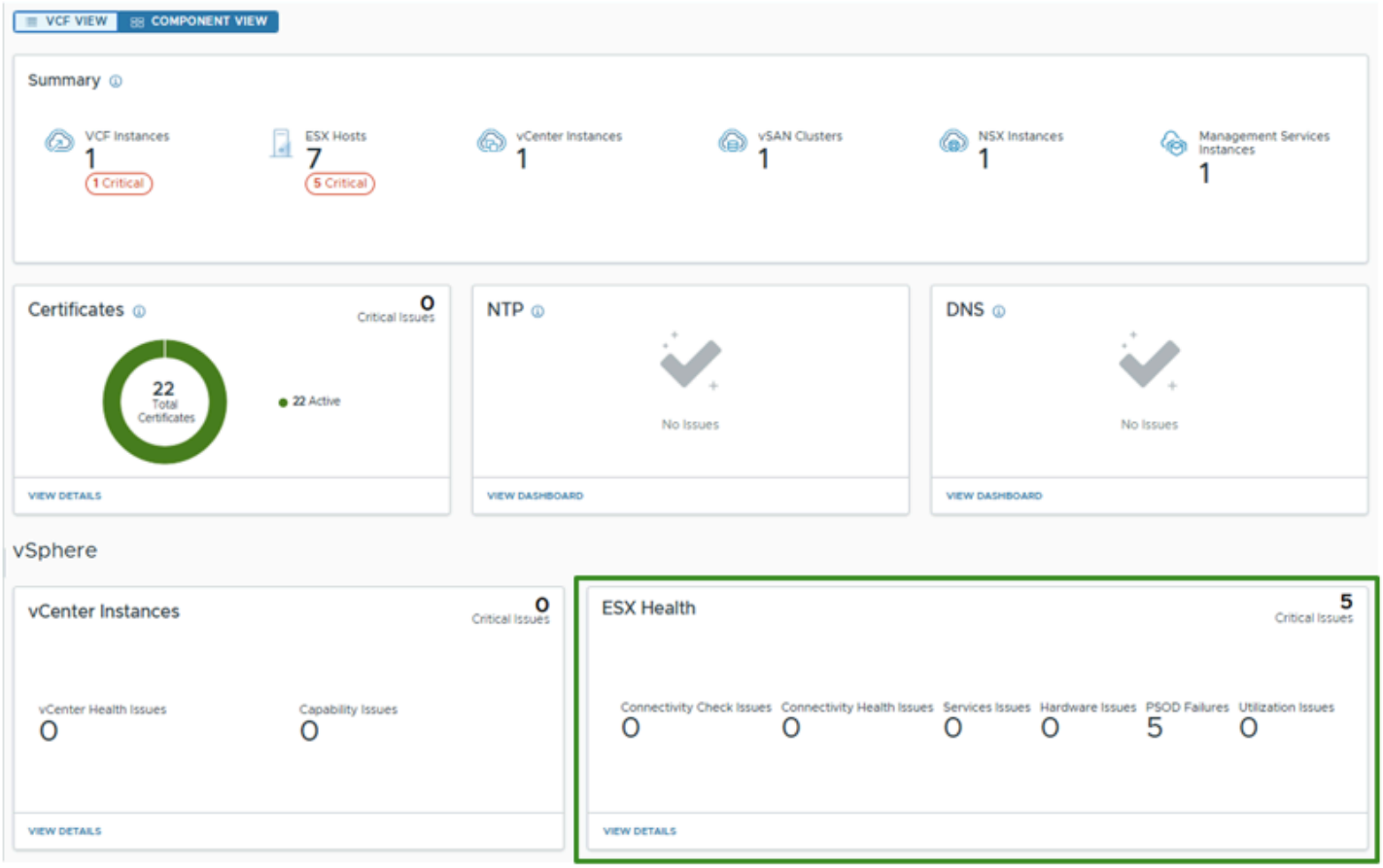
In both sections, you can use the **REFRESH** button to get the latest list of issues and respective recommendations, and the **DOWNLOAD OUTPUT** button to export the list.

## ESX Health in Diagnostics

With VCF Operations health and diagnostics 9.1, you can monitor all ESX hosts in your environment in a single screen, diagnose all aspects of operation and performance, and proactively resolve issues.

VCF Operations Health and Diagnostics 9.1 adds a gateway to monitor the operational state of ESX hosts across your environment. You can track the connectivity and availability of ESX hosts and the state of critical services in a single screen.

**Figure 465: ESX Tile in VCF Health**




ESX metrics are available in several layers of progressive disclosure. You start from a landing page with a summary of the critical issues for the entire fleet. Next, you can see a breakdown by metric on the level of a VCF instance or domain. Finally, you can get to detail on specific checks for capabilities and services for each ESX host.

The **ESX Health** tile on the VCF instance and domain levels summarizes the critical issues that impact availability and connectivity of ESX hosts, which are grouped in the following categories:

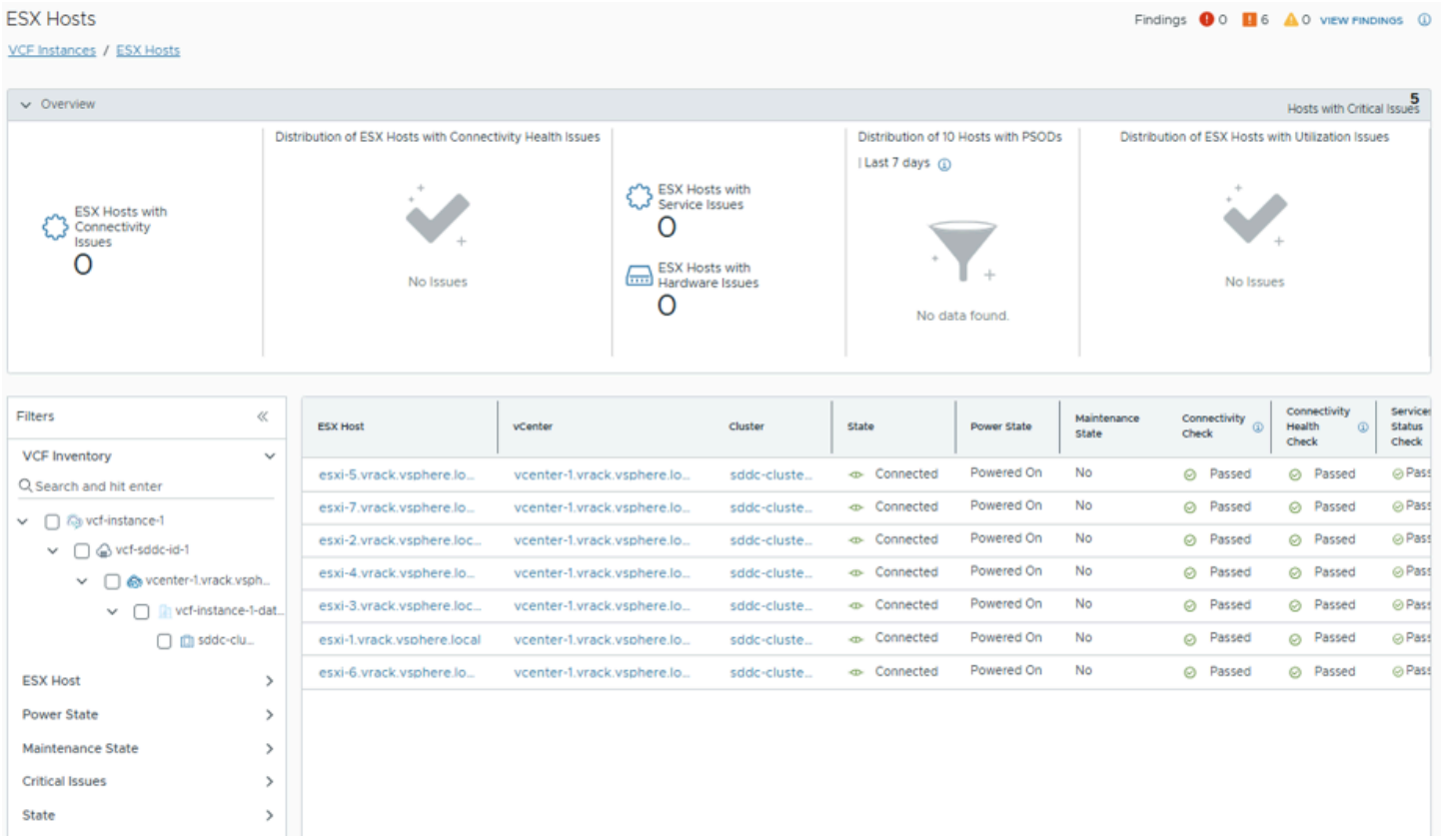
- Reachability
- Connectivity
- Services
- Hardware (Memory and SSD)
- Purple Screen of Diagnostic Failures (PSOD)
- Utilization

For each category, you can drill down to see health details, investigate the root cause, and get guidance for resolving the issues that you identify.

When you click **View Details** on the ESX tile on the **Component View** on a VCF instance level or on a domain-level page, you end up on the **ESX Hosts** page. This page lists all hosts in the selected instance or domain, and summarizes results from checks that run every 5 minutes for each ESX health category. You can see a summary of the critical issues, but also a breakdown of the results from the checks for each host in the details grid below the **Overview** panel.

The **Filters** menu in the left-hand side of the details grid mirrors the columns in the grid to allow you quickly getting to specific insights.

**Figure 466: ESX Details View**




From the host-level page, you have three levels of drill-down:

- When you select an ESX host, you get a details page, where you see the breakdown of issues per category or service. You also see an NSX-specific panel, and separate tabs for health-related, and network-related checks and statuses.
- You can drill down to health-related and network-related checks and statuses:
  - From the **Health** tab, you see groups of ESX-related checks that return either a system status, such as `Expired` for certificates, or an operational status, such as `Critical` for the hostd service, which indicates that the hostd process is down on the given host.

    The ESX-related checks are grouped as follows:
    - Connectivity: Checks the connection state between ESX hosts and vCenter, VCF Operations, and NSX.
    - Connectivity Health: Checks certificates, NTP, and DNS health to avoid issues that expired ceritifcates, NTP drift, and DNS reverse lookup might cause, such as failing vSphere vMotion tasks.
    - Services Status: Checks the three key daemons that work to manage compute and storage in your environment.

      **Note:**

      To get data for the Services Status section, you need to obtain an API session token by making a POST request to the vCenter REST API endpoint `https://<vcenter-ip-or-fqdn>/api/session`. Using that token, you can connect to ESX hosts and get the health report for each host for each service. The report is generated on each ESX host, and the API is exposed by the ESX hosts. These reports do not exist in vCenter and require a separate connection to ESX hosts. You can use a Cloud Proxy to handle restrictions such as firewalls, but such proxies also needs to first get a vCenter token to ensure connectivity.
    - Hardware Status: Checks the memory, SSD storage, and hardware sensors health.

VCF Operations health and diagnostics 9.1 also tracks the most impactful operational state, which is the failure of ESX hosts with a purple diagnostic screen. In the **PSOD (Purple Screen of Diagnostic) Failure** tile, you can see a **Warning** flag for PSOD events in the last seven days or a **Critical** flag, which indicate three or more PSODs in the last 15 days or invalid core dump target configuration. For the core dump target, either a file or partition is missing or is set at less than the minimum size that you see in the tile.

**Note:**

Always set up a scratch partition when installing ESX. The scratch partition collects the core dumps on persistent memory, as opposed to RAM, and allows better traceability of PSOD events.

If the core dump target configuration is not valid, you see the PSOD Status as Critical.

When you click on **View Details** under the **PSOD (Purple Screen of Diagnostic) Failure** tile, you can see the PSOD stack trace and history. From the PSOD details tile, you can also navigate to **Findings** to check for an underlying known issue.

  - From the **Network Stats**, you see charts of the Host CPU and Datapath CPU usage over the past seven days with a threshold of 85% utilization that triggers a Critical status.
- For operational issues, each section in the host-specific view provides either:
  - A KB to troubleshoot or resolve an unhealthy or critical status;
  - A link to a details page, where you see the current status, historic status, description of the status if unhealthy, and recommendations for resolution.

ESX metrics also provide a series of data that is important from the perspective of NSX:

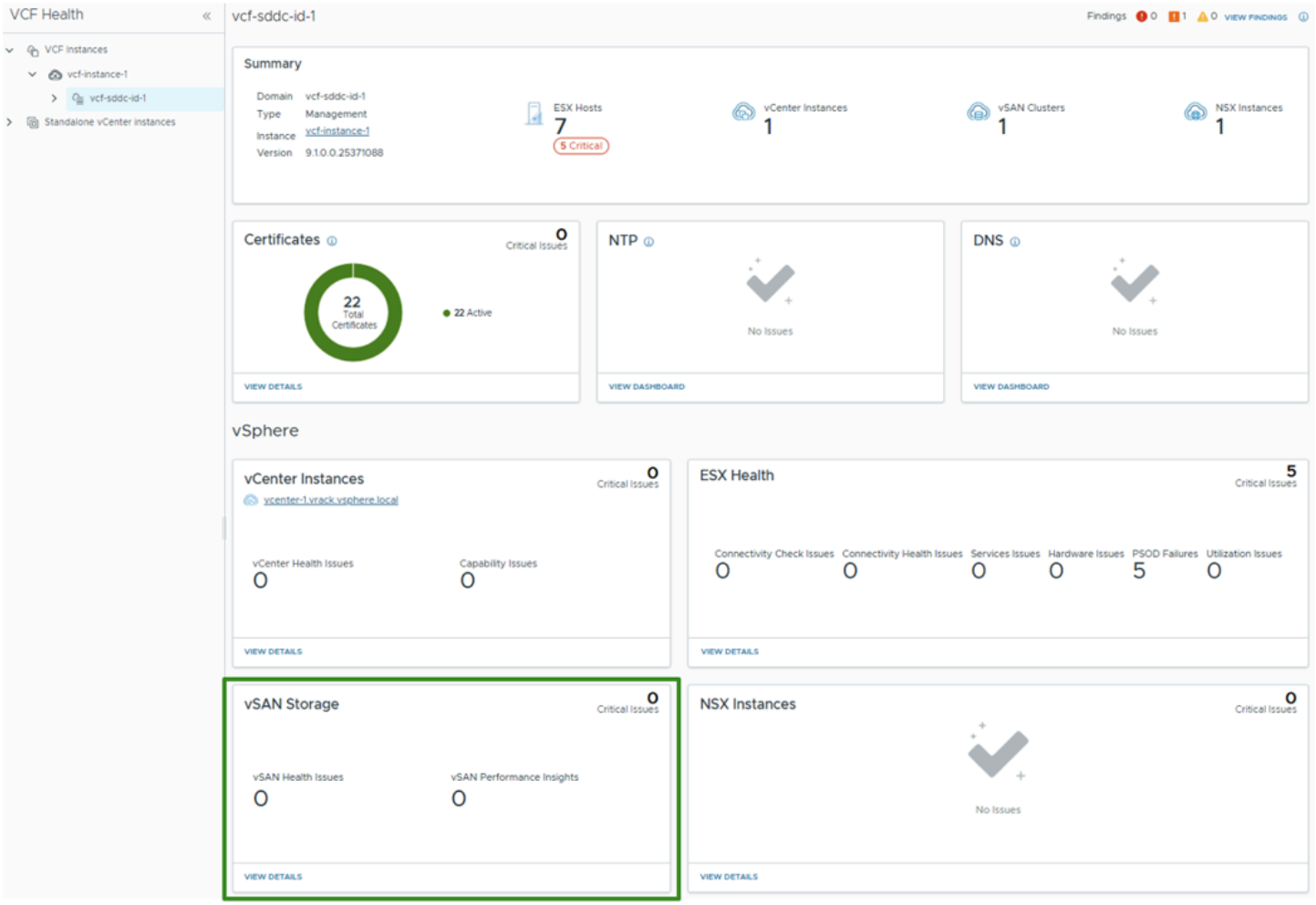
- The details grid on the **ESX Hosts** landing page adds the **Tunnels** column that tracks critical issues with NSX tunnels.
- In the **Summary** panel of the domain-level page, you see NSX-related data, such as running state of NSX tunnels, PNIC/Bond, and TEP IP.
- The **Network Utilization** tab of the ESX landing page adds a section for PNIC Network Statistic, the physical network tile of ESX hosts. This tab provides interface statistics, such as bits per second, packets per second, and dropped packets percentage.

## vSAN Health in Diagnostics

VCF Operations Health and Diagnostics 9.1 gives you a single pane of glass for any vSAN-related monitoring, issue detection, and troubleshooting tasks.

VCF Operations Health and Diagnostics 9.1 builds on the **Storage Operations** capability, which is a centralized monitoring console that is configured to observe storage across VCF deployments. **Storage Operations** removed the limitation that vSAN performance and health metrics were only available in vCenter instances. Collecting vSAN data required logging in to each vCenter instance. Such a task was difficult, but also unreliable, because in this case you do not find the most urgent issues first. VCF Operations Health and Diagnostics 9.1 gives you all operational data and insights at a glance. You can see the actual health status of all storage clusters across the fleet at any time, and can act on the most urgent ones without a delay. And you can drill down from the high-level metrics to specific clusters and vCenter instances from a single screen.

**Figure 467: vSAN Tile in VCF Health**




In VCF Operations health and diagnostics 9.1, **Storage Operations** is integrated with the **VCF Health** capability to add value by combining health scores and metrics with the means of accessing vCenter instances from the same screen. With the combined capability, you can spot a negative health condition in the **vSAN Health** screen, follow up on the root cause, and take guided remediation steps in the vCenter instance.

In VCF 9.1, you have three entry points to vSAN storage metrics. Under **VCF Health**, you can get to the vSAN storage metrics from either the VCF fleet level view or the vCenter level view. You can also get to the vSAN tile in **VCF Health** from **Storage Operations**.

- From **Operate** > **Overview** > **Storage**, you see a summary of the critical and warning health issues, and the count of important performance insights for vSAN clusters.

- From the VCF fleet level, under **Operate** > **VCF Health** > **VCF Instances**, you have two entry points: the **VCF View** and **Components View** respectively:
  - In the details grid of the **VCF View**, when you select an instance with issues that are highlighted in the **vSAN Cluster** column, you get to the **vSAN Storage** view.
    - The **Health** view lists all vSAN clusters for the given VCF instance and provides a details grid with row for each error on each cluster. When you expand a row, you see details on the issue, description, risk if no action is taken, and the impact that the issue has or might have on other issues in your environment.

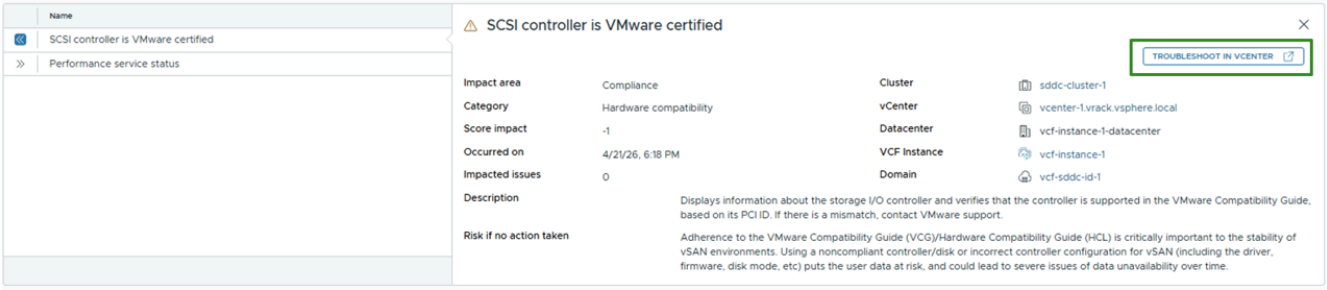
      **Note:**

      In the **Health** tab, you can set the sequence and number of columns in the details grid by using the **Manage Columns** button. For example, the **Score Impact** column is hidden by default, but you can make it available by managing the columns.

      You can click the **Troubleshoot in vCenter** button to get into the vSphere Client, where you can follow the prescribed remediation directly in the vCenter instance managing the vSAN cluster.

      **Figure 468: Link for Troubleshooting in the vSphere Client**

      You can click the **Troubleshoot in vCenter** button to get into the vSphere Client, where you can follow the prescribed remediation directly in the vCenter instance managing the vSAN cluster.

    - The **Performance Insights** view, you can drill down the inventory to check for specific insights, severity, and type over a given period. Similar to the **Health** tab, when you expand a row, you see details on the issue, description, charts, and a link to the vCenter instance, where you can take corrective action.
  - In the **Component View**, you see a summary of the availability, performance, capacity, and compliance issues across your fleet. When you click **View Details**, you get to the **vSAN Storage** view and can follow up on the health or performance issues respectively.
- When you drill down to the **vCenter Health** view from the VCF instance view, you see a summary of health and performance issues on the vSAN storage for the specific vCenter instance.

For more information, see Monitoring Storage Operations, Configuring a vCenter Account in VCF Operations, and Configuring a vSAN Account.

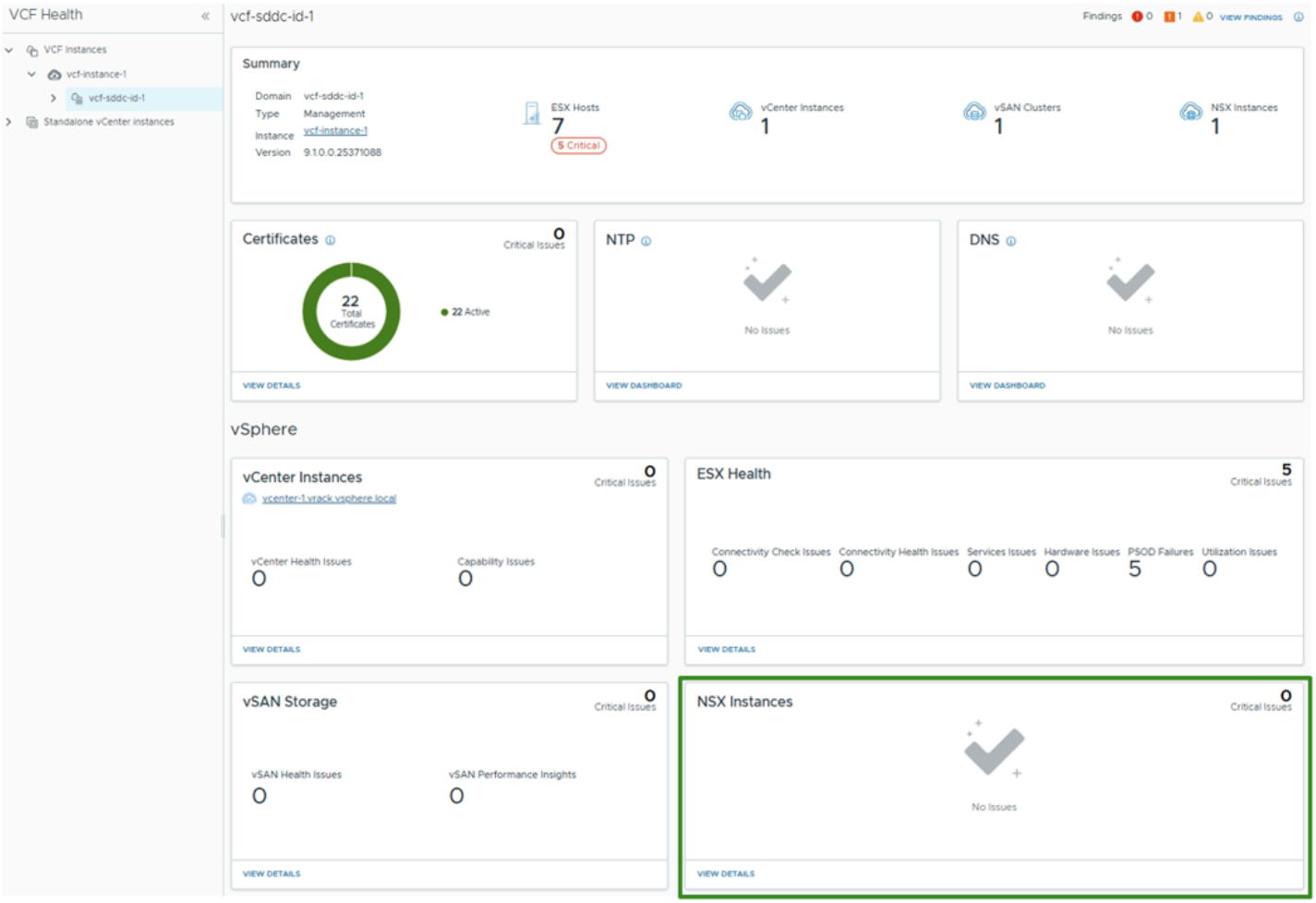
## NSX Health in Diagnostics

VCF Operations Health and Diagnostics 9.1 provides the key metrics for NSX building blocks that help you track points of failure to prevent overutilization and optimize performance in your system.

VCF Operations Health and Diagnostics 9.1 adds a tile with NSX metrics to answer the most important question in infrastructure operations - what is wrong and how to fix it. NSX-related data covers all operational aspects of the underlying infrastructure and appliances, and leads you to resolution such as balancing loads or moving around workloads. NSX metrics in VCF Operations health and diagnostics 9.1 are helpful to both infrastructure admins and NSX

experts. Infrastructure admins can holistically monitor the performance of the VCF environment, while NSX experts can dig into data for troubleshooting and creating action plans based on the real-time metrics.

**Figure 469: NSX Tile in VCF Health**




VCF Operations Health and Diagnostics 9.1 replaces the use of different screens or tools just to get data and adds further value by directly providing actionable insights. The **NSX Instances** tile in **VCF Health** gives you a summary of the critical issues, which require immediate attention across NSX building blocks, and is an instant starting point for troubleshooting.

NSX data is available from two paths:

1. **Operate** > **VCF Health** > **VCF Instances** > **Component View**for infrastructure admins, who need a high-level view of NSX in parallel to other VCF components.

2. **Operate** > **Overview** > **Network**, for networking admins who directly need access to NSX status.

You can drill down to all network-related operational and performance details on component and infrastructure level from both starting points..

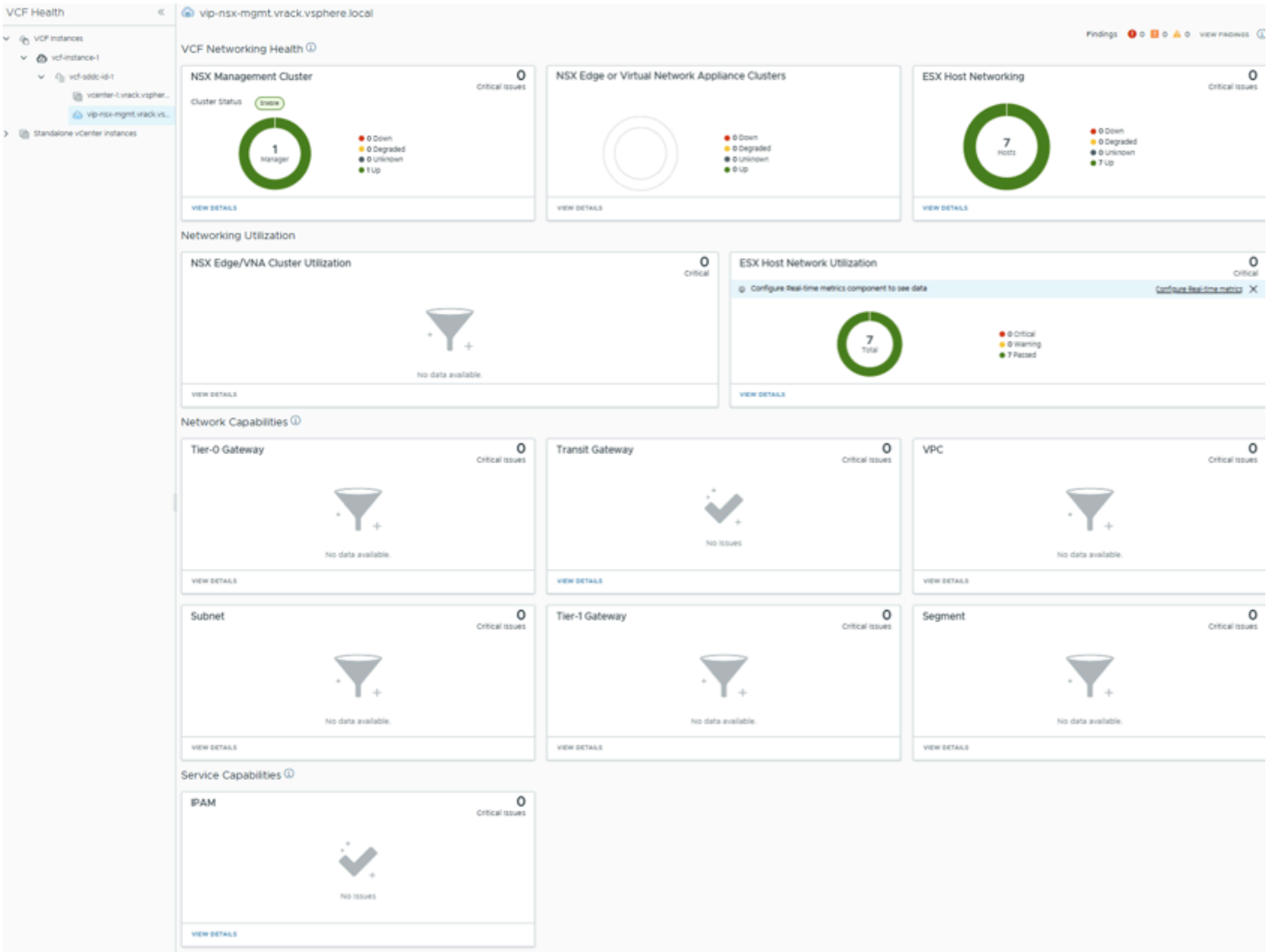
The NSX critical issues are split in four categories: NSX Manager, Edge/Virtual Network Assessment (VNA), ESX hosts networking, and Capabilities.

You get to the NSX tile from the**Component View**in either:

- **VCF Instances**, where you see the critical issues count for each category summarized across the entire fleet
- Individual VCF Instance views, where the summary is narrowed down to the specific deployment.

Clicking **View Details** on the **NSX Instances** tile takes you to the details page, listing NSX instances in scope. You see a summary of the most urgent issues and their type, such as application or capability issues, or NSX Manager issues. A pie chart for each of the metrics provides better visualization for the most urgent or wide-spread ones.

**Figure 470: NSX Details View**



When you drill down to a specific NSX instance, you get to the **VCF Networking Health** screen, which tracks four areas for the selected instance:

- VCF Networking Health
- Networking Utilization
- Network Capabilities
- Services Capabilities

> **Note:**
>
> An NSX instance might stretch across multiple workload domains and the same NSX instance tile appears in each of the domains that use that NSX instance.

### VCF Networking Health

**VCF Networking Health** provides you with data on the operational state, such as cluster and node status, and adds the infrastructure aspect: you get data for underlying networking issues of ESX hosts.

**View Details** on the **NSX Management Cluster** tile under **VCF Networking Health,** takes you to a screen that gives the following information for the management cluster:

- Overview: provides operational statuses for the management node and underlying ESX hosts
- Utilization: tracking CPU, Memory, Disk, and Current Usage Per Partition
- Services: operational and performance status for NSX-related services
- API Metrics, summarizing API requests and response statistics.

**View Details** on the **NSX Edge or Virtual Network Appliance Clusters** tile under **VCF Networking Health,** takes you to a screen that gives operational and performance status for NSX Edge or VNA Cluster.

**View Details** on the **ESX Host Networking** tile under **VCF Networking Health,** takes you to the **ESX Hosts** detail view under the ESX tile in **VCF Health**.

On ESX host level, compute and network details are combined in a single view. ESX and NSX track similar metrics, but apart from the common issues, such as connectivity status, the ESX landing page also shows NSX-specific metrics such as Tunnels and Utilization. The ESX landing page provides an NSX-specific summary of the NSX versions, tunnels, PNICs, and Tunnel End Point (TEP) IPs in your environment.

If you want to drill down to utilization metrics, in the **ESX Hosts** landing page you get to the **Network Utilization** tab. This tab gives you a visual representation of historic data on the key performance indicators per host. For example, when your network is dropping packets, you can quickly check if the condition comes from a resource perspective due to a specific utilization issue.

- In the **Interface Stats** tab, you also have the key metrics that are related to packet drops, such as packets per second, bits per second, or dropped packets percentage. For each metric in the **Statistics** column, data is provided in five-minute intervals and the **View Details** button provides time series data over a period of time.
- In the **Resources** tab, you can see if the cause for current networking issues is the CPU utilization of the ESX host.

### Networking Utilization

If you want to focus on troubleshooting network-related utilization issues, the **Networking Utilization** section combines the **NSX Edge/ VNA Cluster Utilization** and **ESX Host Network Utilization** views.

When you click **View Details** on the **NSX Edge/ VNA Cluster Utilization** tile, you can get to a specific NSX edge or VNA cluster view. When you drill down to a particular cluster, you can see the specific status and configuration details, and the underlying ESX host information.

The **Network Utilization** tab provides edge datapath and service metrics. The datapath metrics provide usage details around datapath CPU, memory pools, interface statistics, and and insight to NSX Capabilities such as T0, T1, and VPCs configured to use the edge node. Service metrics provide information about services CPU, memory, and associated Load Balancer usage.

When you drill down into the edges, you can bridge the operational state with performance issues. If an edge is down, you can see what is the cause. For example, you can see that an edge is down because of a connectivity issue or an issue with the underlying ESX host. This association can be found on the overview tab for easy navigation.

### Network Capabilities

In the **Network Capabilities**, you see details on:

- Tier-0 Gateway
- Tier-1 Gateway
- Transit Gateway
- VPC
- Subnet
- Segment

Each tile in this panel tracks the Realization Status and Runtime State of the capabilities and gives you an immediate entry point to troubleshoot critical issues.

### Services Capabilities

VCF Operations health and diagnostics 9.1 also adds a view to track details on the IP address management (IPAM). You see where each IPAM is used, the realization status, and the percentage of usage so that you can quickly spot issues and act to resolve them.

Some scenarios, related to each screen, with recommended resolutions, starting from the NSX instance level:

**Table 1356: NSX Management Cluster**

| Problem scenario | Potential causes | Diagnostics screen | Remediation |
|---|---|---|---|
| Management plane service is degraded<br>• NSX Manager status is down or degraded<br>• NSX Services are down or degraded<br>• Management and control plane communication between NSX Managers and ESX hosts is down or degraded | High memory or CPU on an the NSX Manager instance | **NSX Networking Health** > **NSX Manager Cluster** > **View Details** > **Utilization** | High NSX Manager CPU: KB 375209<br>High NSX Manager Memory: KB 375215 |
| | Degraded control or management channel | **NSX Networking Health** > **NSX Manager Cluster** > **View Details** > **Underlying ESX Host Information** | Follow up on the information for the specific ESX host in the **ESX Host Networking** tile. |
| Unavailable API on NSX Manager | API rate limit has been reached | **NSX Networking Health** > **NSX Manager Cluster** > **View Details** > **API Metrics** | Based on the data for API requests, either lower the concurrent requests or increase the rate limit. |

### Table 1357: NSX Edge Cluster

| Problem scenario | Potential causes | Diagnostics screen | Remediation |
|---|---|---|---|
| Dropped north/south packets<br>High throughput | Service router (SR) on NSX edge is impacted, high datapath CPU or interface is down | Review drops:<br>**Networking Utilization** > **NSX Edge/VNA Cluster Utilization** > **View Details** > **NSX Edge or VNA Cluster** > **NSX Edge** > **Network Utilization** > **Datapath Metrics** > **Interface Statistics**<br>Review CPU:<br>**Networking Utilization** > **NSX Edge/VNA Cluster Utilization** > **View Details** > **NSX Edge or VNA Cluster** > **NSX Edge** > **Network Utilization** > **Datapath Metrics** > **Datapath CPU** | Interface statistics: KB 324167.<br>High Datapath CPU: https://knowledge.broadcom.com/external/article/330483 |
| | Degraded ESX host. | To find the ESX host on which the edge is running:<br>**Networking Utilization** > **NSX Edge/VNA Cluster Utilization** > **View Details** > **NSX Edge or VNA Cluster** > **NSX Edge** > **Underlying ESX Host** | Follow up on the information for the specific ESX host in the **ESX Host Networking** tile. |
| Degraded performance for network services (Load Balancer or VPN) | High services CPU and/or Datapath CPU | Review Datapath CPU:<br>**Networking Utilization** > **NSX Edge/VNA Cluster Utilization** > **View Details** > **NSX Edcge or VNA Cluster** > **NSX Edge** > **Network Utilization** > **Datapath Metrics** > **Datapath CPU**<br>Review Services CPU:<br>**Networking Utilization** > **NSX Edge/VNA Cluster Utilization** > **View Details** > **NSX Edcge or VNA Cluster** > **NSX Edge** > **Network Utilization** > **Service Metrics** > **Services CPU** | High Datapath CPU: KB 330483<br>High Services CPU: KB 368272 |
| Degraded performance for network services (NAT) | High Datapath CPU and/or high NAT sessions | Review Datapath CPU:<br>**Networking Utilization** > **NSX Edge/VNA Cluster Utilization** > **View Details** > **NSX Edcge or VNA Cluster** > **NSX Edge** > **Network Utilization** > **Datapath Metrics** > **Datapath CPU**<br>Review NAT sessions:<br>**Networking Utilization** > **NSX Edge/VNA Cluster Utilization** > **View Details** > **NSX Edcge or VNA Cluster** > **NSX Edge** > **Network Utilization** > **Datapath Metrics** > **NAT** | High Datapath CPU: KB 330483 |

**Table 1358: ESX Hosts**

| Problem scenario | Potential causes | Diagnostics screen | Remediation |
|---|---|---|---|
| Dropped east/west packets | High vCPU<br>Reaching Network Capacity | Review overall drops:<br>**Networking Utilization** > **ESX Host Networking** > **View Details** > **ESX Host** > **Network Utilization** > **Interface Stats**<br>See either of the **Aggregated VNIC Statistics**, **VMKernal NIC Statistics**, or **PNIC Statistics** charts. | Network Performance Monitoring for ESX Hosts: KB 419008 |
| | High vCPU<br>Reaching Network Capacity | Review CPU:<br>**Networking Utilization** > **ESX Host Networking** > **View Details** > **ESX Host** > **Network Utilization** > **Resources**<br>See the **vCPU and Datapath CPU** charts. | Network Performance Monitoring for ESX Hosts: KB 419008<br>High ESX CPU: KB 395309 |
| | High vCPU<br>Reaching Network Capacity | Review drops:<br>**Networking Utilization** > **ESX Host Networking** > **View Details** > **ESX Host** > **Network Utilization** > **Interface Stats**<br>See either the **Aggregated VNIC Statistics** or **VMKernal NIC Statistics** charts. | Network Performance Monitoring for ESX Hosts: KB 419008 |
| | High VM Contention | **Networking Utilization** > **ESX Host Networking** > **View Details** > **ESX Host** > **Network Utilization** > **Resources**<br>See the **Max Ready Time for vCPU** chart. | Network Performance Monitoring for ESX Hosts: KB 419008 |

## VCF Management Services Health in Diagnostics

With VCF Operations health and diagnostics 9.1, you can monitor the health of VCF management services, the group of components that run on the VCF services runtime component.

With VCF 9.1, the following components are deployed as part of VCF management services:

- Real-time metrics, which provides real-time data analytics
- Log management, which is the VCF log analysis solution
- Salt master, which provides centralized management of configuration drift and compliance
- Real-time metrics store, which helps to store data at scale and short interval metrics.
- Software depot, the unified depot that consolidates all VCF component binaries and metadata
- Fleet lifecycle, which provides a centralized and automated management of multiple VCF instances
- Identity broker, the dedicated authentication solution for the VCF components
- SDDC lifecycle, which handles automated updates and essential maintenance tasks
- Telemetry, which collects configuration, performance, and feature usage data

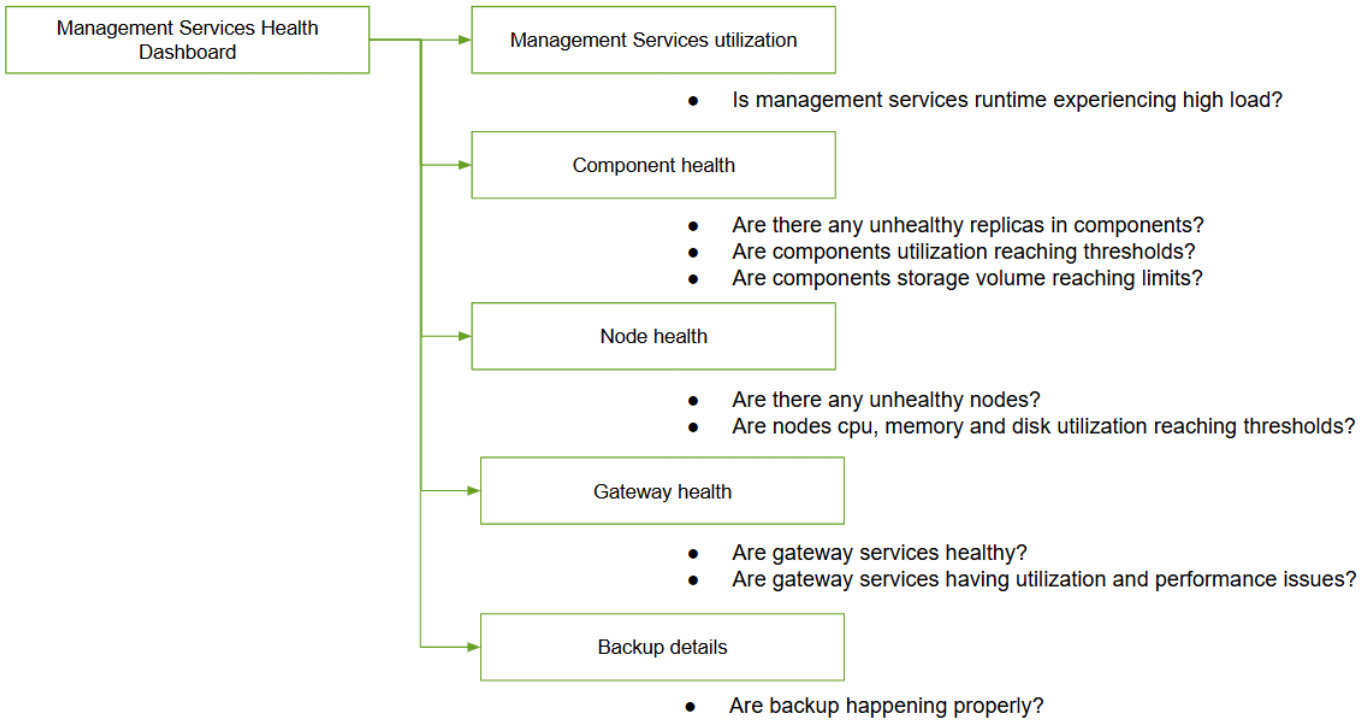
With VCF Operations health and diagnostics 9.1, you can self-diagnose scaling, performance, connectivity, and availability issues impacting theReal-time metrics, Log management, Identity broker, Fleet lifecycle, and Salt master components that run on VCF services runtime component.

**Note:**

The term VCF management services refers to the group of management components deployed in each VCF instance that runs on the VCF services runtime component, which provides a standard and common runtime.

**Figure 471: Structure of the VCF management services metrics and health in VCF Operations health and diagnostics**

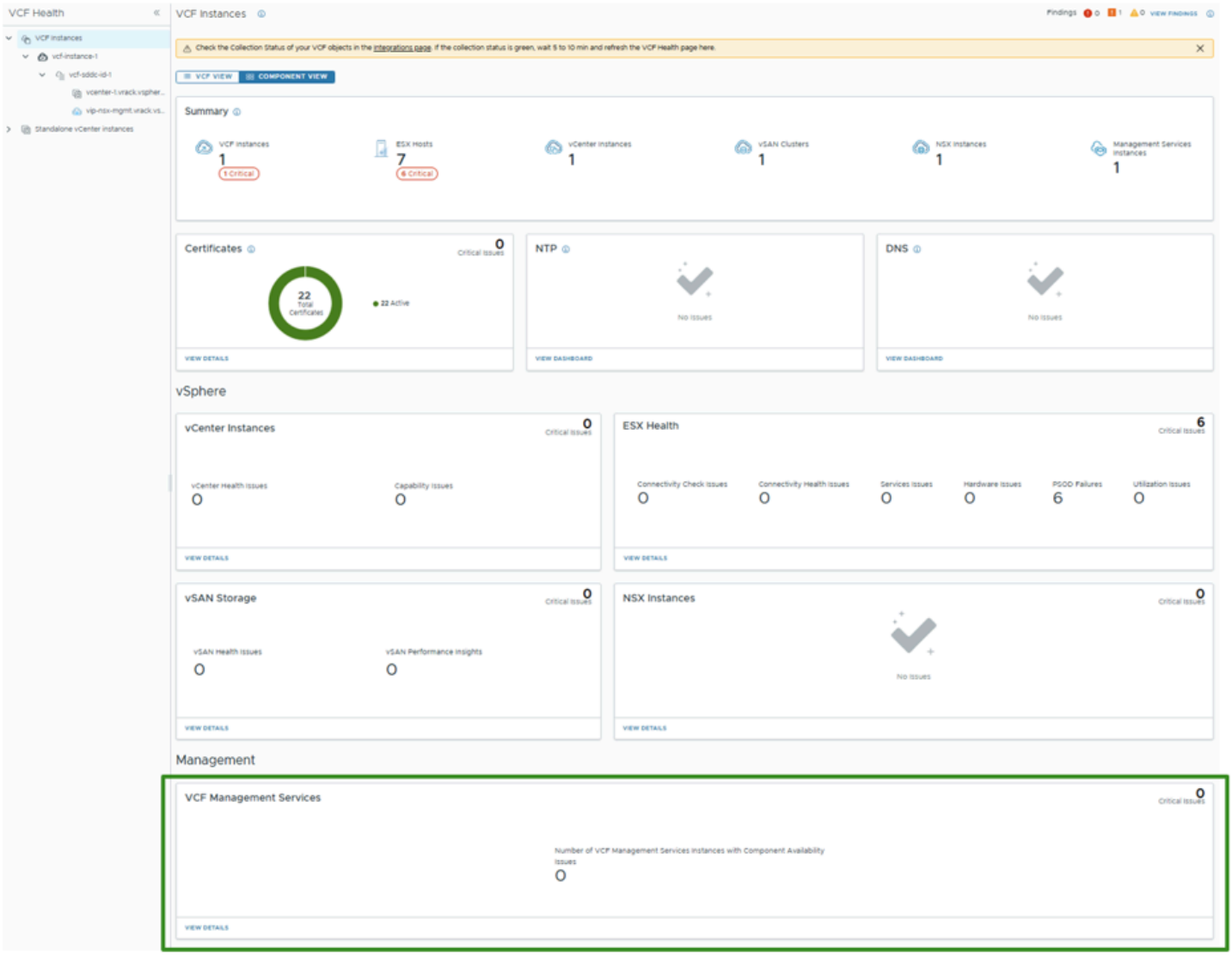
Structure of the VCF management services metrics and health in VCF Operations health and diagnostics




From the **VCF View**, you can see the summary of critical issues for either services that enable components running on VCF services runtime or the respective components running on it. From the**VCF View**, you can drill down to VCF instances, VCF domains, and vCenter instances.

In the **Management** tile of the **Component View**, you see the number of VCF instances with component availability issues. From the **Component View**, click **View Details** to go to the **VCF Management Services** page.

**Figure 472: VCF Management Services Tile in VCF Health**




The **VCF Management Services** dashboard provides health metrics for components including services runtime nodes and services, focusing on essential issues such as utilization and availability.

In the **Overview** panel of the **VCF Management Services** page, you see the number of issues that need immediate attention from either component or services runtime perspective. You drill down to specific metrics per component, node, or service in the **Components** panel.

In the **Management Services Utilization Overview** panel under **VCF Management Services**, you can see the overall CPU and memory utilization details focusing on scale and performance issues. The scale-related metrics are static and you get recommendations on thresholds and actions by KBs.

In the **Components** panel, you can track the availability and memory utilization for each component and the storage utilization of storage volumes.

In the **Backup Details** panel, you see the status of backup jobs for the past 24 hours.

> **Note:**
>
> When you first run a backup of a VCF instance, you see the task as complete, but you do not immediately see a record in the **Backup Details** panel. It takes two completed tasks to trigger the count of backup jobs in the **Backup Details** panel.

In the **Gateway Services** panel, you can also track the utilization of the underlying services for the containerized environment and see the health of each service, based on its specific performance parameters. For example, request sizes, request/response sizes, and latency.

In the **Nodes** panel, you see the breakdown by CPU, memory, and disk utilization per node, and the health status.

See some scenarios, related to each screen, with recommended resolutions:

| Problem Scenario | Panel | Potential Causes | Recommended Action |
|---|---|---|---|
| Component is down due to an infrastructure issue<br>or<br>Component is degraded due to overutilization | **Management Services Utilization Overview**<br>(Monitoring the combined CPU and memory usage across all nodes in the services runtime cluster.) | The overall utilization of the cluster is consistently exceeding pre-defined operational thresholds of 90% for Warning or 95% for Critical status. Such conditions typically come from an increase in workloads such as API calls, processing of monitoring data, Day-N actions, or logs forwarding. Indicates that the current cluster might require more resources. | Resize the services runtime cluster by using fleet lifecycle.<br>Add more resources (CPU/ memory) to the existing nodes or scale out by adding new nodes to the cluster to accommodate the increased load. |
| | **Components**<br>(Availability column: status check on the operational state of individual components) | Operations: A patch, upgrade, or other maintenance procedure such as scaling is in progress. During such operations, components might be temporarily restarted, leading to a down status.<br>Service Instability: When no maintenance operations are running and the component status remains down for multiple monitoring cycles, typically 5 minutes per cycle. This suggests an unexpected failure or deadlock within the service. | Check Maintenance: Check if a patch or upgrade is in progress. If a maintenance job is running, the temporary unavailability is expected.<br>Restart Services: If the component is continuously down for an extended period, restart the affected services by using fleet lifecycle to attempt recovery. If the issue persists, collect logs and create a support ticket. |
| | **Components**<br>(Memory Utilization column: tracks the threshold for each component) | Components are consistently processing workloads that exceed their current capacity and you see memory utilization spiking. | If this behavior is consistent, not a temporary peak:<br>• If the component has a VCF Operations dashboard, such as for logs management, see the dashboard for memory limits and other remediation.<br>• Allocate more memory/CPU to the component by using fleet lifecycle. |

| Problem Scenario | Panel | Potential Causes | Recommended Action |
|---|---|---|---|
| | **Nodes**<br>(Utilization and health of the nodes) | Potential causes for component downtime.<br>Node failures can result from resource constraints, such as high memory, CPU, or disk pressure. Other reasons can be underlying infrastructure problems, including connectivity issues with vSAN, vCenter clusters, or networking.<br>For example, a vCenter cluster in which the nodes are running, might have insufficient resources. Or the persistent volumes that back stateful services might be inaccessible due to an APD condition. | Check the component utilization. Assess underlying infrastructure utilization through the vCenter, vSAN, or NSX Health tiles.<br>• If the component has a VCF Operations dashboard such as the logs management dashboard, refer to the component-specific dashboards for memory limits and other remediations. |
| Component is running out of space | **Components**<br>(Volume column: Monitoring the disk space utilization for persistent storage volumes used by specific components, such as databases and log storage. | The storage volume allocated to a component is nearing the available capacity due to the accumulated size of the workload data (database size, retained logs, metrics, binaries). If the volume exceeds the available capacity, it can impact the availability or can degrade the performance of the component. | Resize the storage volume. Increase the disk capacity for the affected component by using fleet lifecycle.<br>• If the component has a VCF Operations dashboard such as the logs management dashboard, refer to the component-specific dashboards for memory limits and other remediations. |
| Component is facing a latency issue | **Gateway Services**<br>(Check for lower-than-expected network response time or significant delay when interacting with the management services.) | This condition can be due to several factors:<br>1. An increase in the workload that a component is processing. High request volume can lead to queue buildup and delayed processing, manifesting as network latency.<br>2. Issues in the underlying network infrastructure that might be impacting response times. | If the Gateway service is down, you can use the API to restart the service.<br>If the slowdown is consistent with high utilization, it might be necessary to add more capacity to handle the increased load. You either:<br>1. Increase the size of the VCF services runtime cluster.<br>2. Add virtual IPs (VIPs) to the component for more load balancing of incoming requests across the available cluster nodes.<br>If the slowdown is not related to increased load, look for outstanding issues with the virtual or physical networking layer. |

| Problem Scenario | Panel | Potential Causes | Recommended Action |
|---|---|---|---|
| Component backup fails or does not run | **Backup Details**<br>(The scheduled or on-demand backup operation for the management services component has failed to complete successfully.) | SFTP Reachability: The configured Secure File Transfer Protocol (SFTP) target server is not reachable. Network connectivity issue, firewall blocking, incorrect credentials, or SFTP service down might be the reason. Or authentication is failing to the SFTP server.<br>Storage Capacity: The SFTP storage location is full, preventing the backup archive from being written.<br>Internal Service Issues: An internal service issue within the management component or the backup process itself cause a failure. | Review Task Details: See the task details for the failed backup job in the management interface to check for immediate error messages.<br>Check the SFTP server: Confirm that the SFTP server is reachable and has sufficient disk space. If a backup fails due to an authentication failure, update the password for the VCF services runtime in the backup settings in VCF Operations.<br>Check for component health: If the cause is not readily apparent from task details or logs, collect error information and raise a support ticket. |

For more information, see KB 417255 and Performing Day-N Actions on Components.

## Use Log Assist to Accelerate Your Support Cases

To accelerate resolution of support cases, you can use Log Assist to automatically generate log bundles and attach diagnostic findings data.

Log Assist provides a streamlined mechanism for creating and securely transferring support bundles to Broadcom Support, accelerating the resolution of technical issues related to VCF. Log Assist automates processes to eliminate the manual effort that is usually required to collect and upload logs and diagnostic information. The automated gathering of pertinent technical data allows Broadcom Technical Support to diagnose and resolve issues faster and more efficiently.

Sometimes, in **Findings** you might not see a known issue reflecting a problem in your environment. Or you cannot proactively identify the issue by drilling down to the VCF infrastructure metrics that **VCF Health** provides. In such cases, you can reach out to Broadcom support, but you do so with context and automatically generated log bundles by using Log Assist.

Log Assist adds both property and log-based findings that are active at the time of downloading the diagnostic report to your automatically generated log bundles. A link to Log Assist is available from **Findings**.

You can also add log-based findings that you generate by using the **Refresh Findings** workflow in **Findings** and the **Export** functionality. Broadcom Support uses the diagnostic findings data to identify known issues that might contribute to the issue in the support case. For more details, see KB 375820.

**Note:**

Log Assist does not perform cleanups. You can manually remove support bundles when no longer necessary.

After an upgrade to VCF Operations 9.1 or later, make sure Log management is also on the same version. Otherwise, you do not see it in the Log Assist inventory and Log Assist cannot collect bundles from Log management.

**Important:**

To use the online functionalities of Log Assist and send log bundles, you must provide access from the could proxy in your environment to the eapi.broadcom.com URL. For more information, see Public URLs Required for Online Functionalities for VCF and vSphere Foundation.

Before generating a log bundle for the selected inventory object, Log Assist runs several connectivity checks:

1. Connectivity from your environment to the assigned cloud proxy.
2. Connectivity from the assigned cloud proxy to the selected component.
3. Connectivity of your environment to the Broadcom Support Portal to make sure that you can upload the log bundles.

4. Validates that the permissions of the integration are sufficient for downloading a log bundle.

Upon failure, you see error-specific messages and directions for resolution.

For the log bundle type that Log Assist generates for each of the supported VCF components, see the following table:

| VCF Components that Log Assist Supports | Log Bundle Type |
|---|---|
| VCF Operations cluster | FULL support bundle |
| vCenter | Standard log bundle that you get with the `GenerateLogBundles_Task` method in the `DiagnosticManager` service interface in vCenter |
| ESX | Standard log bundle from the Direct Console User Interface (DCUI) |
| NSX | Uses the `DEFAULT` content filter with no core dumps and audit logs, log age limit = 5 days, on transport or management nodes |
| Cloud Proxies | Standard log bundle from **Operate** > **Administration** > **Control Panel** > **Support Bundles** |
| SDDC Manager | API logs + SDDC Manager logs, no explicit manifests |
| VCF services runtime | Default log bundle with `type: vsp` and all `dataTypes`<br>The `lookBackWindow` for all VCF services runtime-related log bundle types is 14 days. |
| Log management | Default log bundle with `type: ops-logs` and `dataType: service-logs, service-heapdumps` |
| Identity broker | Default log bundle with `type: vidb` and `dataType: service-logs, service-heapdumps` |

**Note:**

Each unified Cloud Proxy that you use with Log Assist supports up to five parallel support bundles at a time. If you select more components, they are grouped in a queue. The speed of the support bundle upload depends on the uplink bandwidth.

The upload of log bundles to Broadcom support servers does not require external FTP clients.

With VCF Operations health and diagnostics 9.1, Log Assist adds the following options:

- Generate bundles for selected VCF Operations nodes, not only the cluster-level bundle option.
- Transfer log bundles from Log management and Identity broker under VCF services runtime.

**Note:**

The support bundles for VCF services runtime cannot transfer when the management cluster is down.

## Set up Log Assist

Starting with VCF 9.1.1, Log Assist is enabled by default and you do not need to activate it on the Cloud Proxies page.

Starting with VCF 9.1.1, Log Assist is active by default on all cloud proxies in your environment.

**Important:**

You still need to assign objects from the Inventory for each cloud proxy, as described in step 4 in this procedure.

For VCF versions earlier than VCF 9.1.1, to start using Log Assist you must make sure that your environment is registered in Connected Mode, activate Log Assist on a cloud proxy, and assign Inventory objects to the cloud proxy.

- In the VCF Operations navigation bar at the top, click **Manage** and in the left navigation pane, select **Licenses** > **Registration** and make sure you register your licenses in **Connected Mode**.

- See the topic Collecting Data with Cloud Proxy in VCF Operations on how to add a unified cloud proxy to your environment. The legacy cloud proxy capability, which in VCF 9.0 and later is defined as the Classic type, does not work with Log Assist.
- Provide access from the could proxy in your environment to the `eapi.broadcom.com` URL. For more information, see Public URLs Required for Online Functionalities for VCF and vSphere Foundation.
- Verify the connectivity between the cloud proxy and objects from the VCF Inventory that you assign to Log Assist. The required connection is between the cloud proxy and the respective component, not to the cluster node.

After you activate Log Assist on a cloud proxy, you can assign Inventory objects to Log Assist as described in Step 4. You can map different Inventory objects to specific could proxies.

1. In the VCF Operations navigation bar at the top, select **Operate** > **Administration** > **Cloud Proxies**.
2. Select a cloud proxy. Click the ellipsis next to the check mark, and select **Activate Log Assist**.

   The Log Assist adapter only runs in standalone mode, no groups are supported.
3. (Optional) If you want to activate Log Assist on all cloud proxies:
   a) Select the checkbox on top.
   b) Click the ellipsis next to the **Add** button.
   c) Select **Activate Log Assist**.
4. Go to **Operate** > **Administration** > **Configurations** > **Log Assist Assignments** and assign objects from the Inventory to the Log Assist-enabled cloud proxy by drag-and-drop.

## Attach Logs and Diagnostic Findings to Support Tickets with Log Assist

You can generate a log bundle for any selected inventory object, link and upload it to your support case with Broadcom.

With Log Assist, you can:

- Generate a log bundle on demand
- Link the bundle to a support case
- Attach diagnostic findings to your support case

1. Log in to the Broadcom Support Portal with your credentials.
2. Select the component for which you file a support case and create the case.

   With creating a case, you get a case number. Keep this number for further reference in your support workflow. If you get support from a Broadcom partner, you should note the partner ID along with the case number.
3. Upload data to the case.
   a) Log in to VCF Operations.
   b) In the **Control Panel**, navigate to the **Log Assist** card.
   c) Click **Prepare a Transfer**.

      In the new screen, the **Select an inventory object** tab is automatically selected.
   d) Select the affected object from the VCF inventory tree.

      You can select objects from the VCF inventory tree up to the ESX host level. You cannot select an individual virtual machine. Dimmed check boxes indicate that the object has no mapping to a unified collector. To configure the mapping, go to **Infrastructure Operations** > **Configurations** > **Log Assist Assignment** and drag and drop the required object. After assignment is complete, continue with your cloud transfer.
   e) Click **Next** and wait for the connectivity validation.

      The check might fail due to intermittent connectivity issues and you can click **RE-VALIDATE** to get the latest connectivity status for your environment. If the check fails again, you must troubleshoot your network connectivity and click **RE-VALIDATE** when you resolve the underlying issue.

      The connectivity check might also fail due to missing required privileges for the object. To fix privileges and connectivity, see KBs 375817 and 375818.

4. On the **Transfer to Support Portal** tab, enter your case number in the **Support case ID** field.
5. Click **Transfer**.

   If the transfer is successful, you return to the **Operate** page.
6. In case a transfer fails, you can see the errors for specific objects and can retry the task.

   Log Assist is designed to handle intermittent network or infrastructure issues automatically before requiring user intervention.

   If a log bundle operation or transfer fails, Log Assist automatically retries the task up to three times. If the issue persists, the task is marked as failed.

   VCF Operations Health and Diagnostics 9.1 adds the **Retry** button to facilitate manual retries of failed log bundle operations.

   a) In the **Log Assist** page, under **Transfer History**, expand the row for the current **Support Case** and look at the **Activity Feed**, where you see a breakdown of the successfully completed and failed log bundle operations.
   b) Click the **Retry** button to re-run only the failed tasks that you see in the **Activity Feed**.

      VCF Operations Health and Diagnostics 9.1 simplifies the retry process by pre-populating the Support Case ID and inventory selections from the original request, so you only re-run the failed tasks.

# Troubleshooting with the Workbench

The **Workbench** is where teams can collaborate, build a knowledge base of issue investigations which can be saved for future reference, and enhance troubleshooting by viewing real time data. You can view data with 20-second granularity for real-time performance evaluation by default in the **Real Time Analysis** tab, selecting the **Overview** sub-tab. For Host System and Virtual Machine object types, an additional ESX Top view is available that provides 2-second granularity data. To enable this view, activate the ESX Top category under Real time metrics for ESX hosts being investigated, in the policy settings.

## What Does the Troubleshooting Workbench Do

The **Workbench** page is where you investigate your VCF stack comprising of the following components:

- vCenter
- ESX
- vSAN
- NSX

The new observability engine, introduced in VCF 9.1, collects metrics in real time, provides alerts and notifications, analyzes and summarizes rich data, provides data enrichment, recommends actions and provides user-driven enrichment. The real-time data component must be installed, and is not available by default.

The **Workbench** page provides:

- Unified visibility into all operational data
- Real-time data at 20-second, and 2-second granularity for mission-critical workloads, containers, and applications hosted on ESX.
- Summary dashboard for selected objects
- Top N data for VM, Host, Cluster, Datacenter type of objects
- Quick access to metrics and alerts
- SSH access and Script execution
- Collaboration tools and note taking
- Log Analysis
- Network Flow Analysis

## Where You Find the Troubleshooting Workbench Home Page

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Workbench**.

The **Workbench** home page displays a search bar, a list of troubleshooting sessions, and recent searches. You can open a session to find potential evidence for your problems.

### How Workbench Home Page Works

All troubleshooting sessions that are active in the current login are displayed in the **Workbench** home page. The Workbench now organizes investigations into two distinct sections:

| | |
|---|---|
| **Draft Sessions** | Your active investigations are automatically preserved as drafts. Any changes made to the scope, time settings, and notes are retained upon logout. When you log back in, you can resume your work from the Draft Sessions section, where your previous context is restored. |
| **Saved Sessions** | Investigations that you explicitly save are stored in the Saved Sessions section. These sessions permanently retain your specific scope, time parameters, and notes for future reference. |

## Starting the Troubleshooting Workbench

The Troubleshooting Workbench is where you perform advanced troubleshooting tasks on an alert that triggered on an object. You can investigate both known and unknown issues in VCF Operations .

### Where You Find the Troubleshooting Workbench

You can start the troubleshooting session with an alert in context from the alert information page, or you can search for an object and start the troubleshooting session to investigate known or unknown issues related to the object.

- To start the Workbench with an alert in context, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Alerts**.
- To start the Workbench with an alert in context, but for a specific object, select a group, custom data center, application, or inventory object from the **Inventory**. Click the object and then the **Alerts** tab. Click the **TROUBLESHOOT** button on the top right.
- On the **Workbench** page to investigate known or unknown issues with an object in context, search for the object.

The Workbench navigation bar provides quick access to the following tabs for comprehensive investigation:

- Potential Evidence
- Real Time Analysis (containing Overview, Top N, and Custom Query)
- Logs
- Flows
- Metrics
- Alerts
- Tools
- Notes

## Potential Evidence in Workbench

You look for potential evidences of a problem within a specific scope and time range in the **Potential Evidence** tab. The Selected Scope control on the left of the **Workbench** page is where you vary the scope.

You can vary the scope in the following ways:

- You can select only the object that you are investigating, or include several upstream and downstream relationships by increasing the scope. As you increase the scope, more objects are displayed in the inventory tree. The number of objects in scope should not exceed 300.
- You can select a custom scope to include objects of your choice. Click **Custom** to open an interactive window where you use the pointer to visually rearrange your objects, view relationships and add peers to modify the relationships. To see details about the object, place the pointer for a few seconds above the object. You can reset a custom scope to start all over again.
- You can use the drop-down menu to narrow down the type of objects displayed.

The default time range is two hours before and thirty minutes after the alert triggered when the context is alert based, or when the context is object based, the default time range is last 6 hours. You can select a different time range, up to 7 days, using the date and time controls.

The time range selector offers a split view:

- Presets: Quickly select from Last hour, Last 6 hours, Last 12 hours, Last 24 hours, Last 7 days, Today, Last day, Last week
- Custom Range: Manually define a specific Start date/time and End date/time for precise analysis. The custom selection date range should not exceed 7 days.

The potential evidence is based on Events, Property Changes, and Anomalous Metrics which are displayed on the right of the Workbench change in the **Potential Evidence** tab. Information in these sections is displayed as cards.

| | |
|---|---|
| **Events** | Displays events, based on a change in the metrics. Events for metrics that have breached the usual behavior, and major events that have occurred within the selected scope and time are displayed. The cards are based on dynamic thresholds for a metric, which is calculated based on historical and incoming data. |
| **Property Changes** | Displays important configuration changes that occurred within the selected scope and time. Both single and multiple property changes are displayed. For multiple property changes, you can view the latest and previous changes. |
| **Anomalous Metrics** | Metrics which have shown drastic changes within the selected scope and time. Ranks the results based on the degree of change. The most recent anomalous metric based on a time-sliced comparison in the current time range is given the highest weightage. |

You can explore more details about any of the cards that are displayed in the Workbench by clicking the card pop-out option. You can close a card and it is no longer displayed in the Workbench. To load the cards again, click **Go** in the **Time Range**.

## Real-time Data in the Workbench

The **Real Time Analysis** tab in the Workbench page helps you visualize near real-time data, offering live performance data streaming at 20-second granularity. The **Real Time Analysis** is the central hub for performance analysis and is divided into three sub-tabs, namely, **OVERVIEW**, **TOP N**, and **CUSTOM QUERY**. **OVERVIEW** and **TOP N** sub-tabs in the **Workbench** page help you visualize near real-time data, offering performance data streaming at 20 seconds (or more) and 2-second granularity and covering **esxtop** use cases. The Real-time functionality offers high-granularity metrics, interactive charts, date range pickers, and the ability to add chart screenshots to notes. The Real-time metrics component of VCF works on supported objects in the Inventory.

### Real-time Troubleshooting

The **OVERVIEW** sub-tab under the **Real Time Analysis** tab provides instant visibility into system performance metrics across ESX, virtual machines (VMs), NSX Transport nodes, Edge (for transport nodes object type objects), and other vSphere components. It is designed to help you diagnose issues such as sudden performance drops, CPU spikes, or memory overuse.

The tab applies the observability engine that streams real-time performance data at 20 seconds (or more) and 2-second granularity for selected objects within a defined scope and category. You can view, filter, and interact with performance data to isolate problems in real time.

The key features of the real-time troubleshooting in the **OVERVIEW** sub-tab are as follows:

- High-Granularity Data. The performance metrics refreshed every 20 seconds (or more) and 2 seconds for **esxtop** subset of metrics in the **OVERVIEW** sub-tab. For more information how to change the metric collection to 2-second interval, see Real-Time Metrics Details.
- Custom Time Range Selection. You can focus on specific time periods for detailed analysis.
- Multiple Views. You can switch between summary, CPU, or targeted troubleshooting views.
- With interactive charts, you can do the following:
  - Zoom: Zoom in on a single view or use **Zoom All Charts** to apply changes globally.

  - Focus Mode: **Enable Single Series Focus Mode** to isolate specific metric lines for clearer analysis.
  - Use the **Chart height** slider to adjust the vertical size of graphs (Min to Max) for better visibility.
  - Legends: Toggle **Show Chart Legends** to identify data series without cluttering the view.
- Policy-Based Data Collection: Supports configurable data collection, retention, and roll-up policies.
- With chart actions, you can do the following:
  - Refresh data instantly.
  - Save charts as PDF.
  - Copy chart data to clipboard.
  - Add screenshots or notes directly to the Notes section.
- Context Switching: You can connect via SSH to run command-line utilities for deeper investigation on ESX hosts. These commands provide real-time statistics on CPU, memory, disk, and network utilization.

**Note:**

Before you get started with the **Real Time Analysis** tab, you must Deploy Real-Time Metrics.

## Reporting Intervals for Resource Types

The frequency at which performance data is collected and reported depends on the specific resource type and the source of the metrics.

| Resource Type | Reporting Interval | Details |
|---|---|---|
| Host System and Virtual Machine | 2 seconds | High-granularity collection for a limited set of approximately 21 specific metrics. |
| | 20 seconds | Standard real time performance interval. |
| | 5 minutes | Default PerformanceManager historical interval. |
| Cluster and Datacenter | 60 seconds | Metrics sourced directly from the VPXD Stats Registry. |
| | 5 minutes | Standard relevant PerformanceManager metrics. |
| NSX (Transport Node) | 60 seconds | High-frequency metric availability. |
| | 5 minutes | Standard monitoring frequency. |

## Top N Troubleshooting

Located within the **Real Time Analysis** tab, the **TOP N** sub-tab, helps you identify and analyze the top resource consumers for VM, Host, Cluster, Datacenter, NSX and vSAN type of objects, without using command-line tools. Top N metrics now feature explicit labeling. This ensures that the most resource-intensive objects are easily identifiable within the charts. The system displays child metrics, offering deeper visibility into the performance of underlying sub-components and object hierarchies.

It presents the Top N descendant objects of the same type based on selected performance metrics, simplifying the detection of resource bottlenecks and abnormal activity.

The key features of the Top N tab are as follows:

- Configurable Visualization: The **Visualise Top N** panel lets you drill down to different levels of the Inventory to see the objects with the worst metrics or performance in the given hierarchy or object over the selected period. You can refine your view as follows:
  - Metric Selection: Choose the specific metric you wish to analyze (for example, CPU Contention).
  - Visualise Limits: toggle between viewing the **Top 5**, **Top 10**, or **All objects** to focus on the most critical offenders.
  - Multi-Metric Support: Click **Add Another Metric** to compare different performance indicators simultaneously.
- Interactive Charts:
  - Zoom in or reset zoom for comparative views.

  - Reload data to refresh metrics.
  - Export charts as PDF, copy to clipboard, or add to notes.
- Custom Time and Value Ranges: Enables detailed comparisons of performance values over selected periods.

**Note:**

2-second interval metrics are not available in the **TOP-N** sub-tab.

## 2-Second Granularity

The **Workbench** page offers high-resolution (2-second) metrics in the **OVERVIEW** sub-view. Once the **ESX Top** policy is activated for your Host Systems, the Troubleshooting Workbench provides access to 2-second granularity data. However, the availability and behavior of this high-resolution data vary depending on the specific view you are using within the **OVERVIEW** sub-tab.

In the **OVERVIEW** sub-tab, you can view 2-second data for specific objects.

- Supported Objects**:** High-resolution data is available when you select a Host System or a Virtual Machine from the scope.
- Enabling the View**:** You must explicitly select the **ESX Top** view from the view dropdown menu to see 2-second granularity.
- Time Range Constraint**:** High-resolution data is resource-intensive. If you select a time range greater than 6 hours while in the **ESX Top** view, the system will display an error message rather than showing the data. You must reduce the time range to 6 hours or less to view these metrics.

The **Top N** sub-tab behavior differs from the Overview. Data in the **Top N** view is always displayed at 20-second granularity, regardless of whether the **ESX Top** policy is enabled or if the metric supports higher resolution. This design choice prevents clutter and confusion by maintaining a consistent step window for ranked lists.

When using the **Custom Query** sub-tab to analyze metrics with `esxtop: true` enabled, you may encounter duplicate data lines if not filtered correctly.

By default, if you query a metric that has both standard and high-resolution data, the chart will display two concurrent lines: one for 2-second data and one for 20-second data. This can make the chart difficult to read and may show conflicting spikes due to aggregation differences.

To view only the 2-second granularity data, you must filter your PromQL query using the **ESX_TOP** feature flag.

When building your query, ensure you add a filter for `Feature equal to ESX_TOP` (syntax: for example mem.tier.consumed.MEMTYPE{feature='ESX_TOP'}) to exclude the standard 20-second data stream.

## Custom Query

The **Custom Query** sub-tab, located within the **Real Time Analysis** tab, allows you to execute specific queries using PromQL (Prometheus Query Language) to retrieve granular performance data that may not be covered by standard dashboards. This tool is essential for deep-dive analysis where specific, custom metric correlations are required.

Use the **Add Query** input field to enter and run your PromQL query. To assist you in constructing valid queries without autocomplete, the **Query Builder Help** section provides three tabbed resources below the input field.

**Note:**

This field does not provide autocomplete or suggestions. You must enter a complete PromQL query exactly as you want it to run. Ensure you have the exact metric names and syntax ready before typing.

When you click **Need help? Use the quick Query Builder**, the query builder displays the following tabs.

**Functions**

Lists all supported PromQL functions available for your queries.

- Usage**:** Browse this list to find mathematical or aggregation functions such as `sum`, `min`, `max`, `avg`, and `topk`.
- Description**:** Each function includes a brief description of its logic (for example, `sum` calculates the Sum of values across all time series).
- Copy**:** Click the copy icon next to a function name to paste it into your query string.

| | |
|---|---|
| **Metrics** | Provides a searchable catalog of all available performance metrics in the system.<br>• Search: Use the search bar to find specific metrics (for example, IOPS or Packet Drop).<br>• Reference: This is critical for finding the exact internal name of a metric (for example, `LWD_SNAP_SES_TOTAL_WRITE_TIME`) since there is no autocomplete.<br>• Copy: Locate the metric you need and click the copy icon to add it to your query. |
| **Sample Queries** | Offers pre-built templates to help you understand syntax and structure. These samples demonstrate how to combine metrics and functions correctly. You can copy these samples and replace the metric names with those relevant to your specific investigation.<br>• Sample query for host system `cpu.ready.HOST`<br>• Sample query for host system filtered for 2 sec granularity, after the 2 sec policy is configured for the selected host: `cpu.ready.HOST{feature='ESX_TOP'}`<br>• Sample query for host system filtered for 20 sec granularity `cpu.ready.HOST{feature='TROUBLESHOOTING'}` |

## Alerts and Metrics in the Workbench

When you pin a metric, it appears in the **Metrics** tab of the **Workbench** page. You can perform further investigation on the metric in the Metrics tab. You can compare the pinned metrics with other metrics that are displayed in the tab. You can close the pinned metrics and browse other metrics for specific objects. Similarly, the **Alerts** and **Events** tabs are where you investigate the potential evidences further. You can filter and group alerts. If you want to focus on the alerts for a specific object in your selected scope, you can clear all the alerts and then click the object in the scope.

## Collaboration in the Workbench

You can add notes and save troubleshooting sessions for further investigation. Investigations can be marked as resolved, shared with other users or teams, or deleted if no longer needed. You can add comments and screenshots. The collaborative feature allows seamless handover of investigations between users.

### Saving and Sharing Sessions

When an issue arises with the resource, administrators typically begin their investigation on the **Workbench** page by adjusting the scope and time settings to help identify the root cause. However, investigations may be interrupted by higher-priority tasks or require an extended period to complete, risking the loss of progress.

The Save and Share feature lets you preserve your current troubleshooting state and resume it later. Additionally, a sharing feature enables you to invite colleagues into the same troubleshooting session, creating a shared space for joint investigation.

Each troubleshooting session includes a status indicator that reflects its current progress through the investigation workflow:

| Status | Description | Behavior |
|---|---|---|
| Draft | Unsaved session | Changes made to the scope, time settings, and notes do not persist after logout. |
| In-Progress | Saved session with ongoing investigation | The session is persisted and can be accessed later. Investigation is actively ongoing. |
| Resolved | Issue identified and documented | The session is marked as complete. The investigation findings have been documented, and the issue is considered closed. |

Other users that you collaborate with can view saved investigations with the following details:

- Troubleshooting Name

- Object Name
- Object Type
- Status
- Created On
- Created By

Collaborative sessions can be edited, further shared and deleted.

### Saving Notes

From the **Notes** tab, you can add notes, including text and screenshots to document findings, ensuring all collaborators have access to the same context and inside.

When viewing metric charts, click the vertical ellipsis on any widget to access the following options:

| | |
|---|---|
| **Copy to clipboard** | The copied image is formatted for internal use within the Notes feature. Pasting into external applications is not supported. |
| **Add to Notes** | Saves the chart directly to the session's Notes tab |

The Notes section has a maximum size of 400 KB. Prioritize key charts and remove outdated notes to stay within the limit.

The following options are available to manage Notes:

| | |
|---|---|
| **Add** | Create notes with text or chart attachments |
| **Edit** | Update existing notes |
| **Delete** | Remove outdated notes to keep the workspace clean. In shared sessions, all notes are visible to collaborators, ensuring shared context and preserving a full audit trail of the investigation. |

## Scripting and SSH Access in the Workbench

The **Tools** tab in the **Workbench** page provides you with an SSH-based access terminal (SSH Access) allowing you to establish secure shell connections to your infrastructure components. This feature enables you to execute command-line tools, run diagnostics, and troubleshoot issues directly from VCF Operations.

### SSH Access

SSH Access is available for the following resource types:

- Virtual Machine
- Host System
- vCenter.

> **Note:**
>
> For Virtual Machine resource types, VMware Tools is required to open SSH connections to the endpoints

### Prerequisites

Before using SSH Access: -

| | |
|---|---|
| **Permissions** | You must have the "SSH Access" privilege assigned to your role. |
| **Network** | Target system must be reachable from VCF Operations or VCF Operations Collector (Cloud proxy) |
| **SSH Service** | SSH must be enabled on the target<br>• ESX: SSH service must be running<br>• vCenter: SSH must be enabled on the appliance<br>• VMs: SSH server must be installed and running |
| **Credentials** | Valid credentials for the target system. For Virtual machine the credential provided should be an administrator account. |

## Connection Workflow

The SSH Access feature utilizes a two-step wizard to establish a secure connection to your resources.

**Step 1: Connect to Terminal**

1. Select a supported resource (Virtual Machine, Host System, or vCenter) from the inventory tree. Or Navigate to Workbench and type the resource name in the search box
2. Navigate to **Tools** > **SSH Access**.
3. Enter the required credentials (refer to the **Authentication Options** section below for specific requirements).
4. Click **Connect**.

**Step 2: Terminal Access**

Once the connection is established, you will have full interactive shell access to the target system.

## Authentication Options

Authentication requirements vary based on the type of resource you are connecting to.

**Virtual Machines**

Virtual Machines utilize a two-tier authentication approach:

1. Admin Credentials (Validation)
   - Username**:** Enter the username for the admin account. *(Note: This provides credentials for the endpoint guest OS).*
   - Password**:** Enter the password associated with the admin account.
2. SSH Credentials (Access)
   - Use same credentials: Select this option to reuse the admin credentials provided above for the SSH session.
   - Paste Key**:** Select this option to provide an SSH private key for authentication.
3. Port**:** Enter the SSH port number (typically 22).

**Host Systems and vCenter**

Host Systems and vCenter resources utilize a single-tier authentication approach:

1. Admin Credentials (Validation and Access)
   - Username**:** Enter the username for the endpoint.
   - Provide one of the following access details:
     - Password**:** Enter the SSH password.
     - Paste Key: Provide an SSH private key.
2. Port**:** Enter the SSH port number (typically 22).

## Terminal Capabilities

The **SSH access** page allows you to execute command-line tools to diagnose network connectivity and system performance. Commands are categorized by their output behavior.

Once connected, you have full interactive shell access. The commands available depend on the target system:

**ESX Host Commands**

- `esxtop` - Real-time performance monitoring
- `esxcli` - ESX command-line interface
- `vim-cmd` - Virtual machine management
- `vmkping` - Network connectivity testing
- `tcpdump-uw` / `pktcap-uw` - Packet capture

**vCenter Commands**

- `service-control` - Service management
- `vpxd` - vCenter service
- Standard Linux commands (`top`, `tail`, `grep`, and so on.)

**Virtual Machine Commands**

- Depends on guest OS (Linux or Windows)
- Full access to installed tools and utilities

## Scripting

You can execute tailored diagnostic scripts directly on Virtual Machines (VMs).

1. Scope the Target: Select a specific VM to narrow the diagnostic scope.
2. Execute: Navigate to the **Tools** tab and select **EXECUTE SCRIPT**.

## Flows in the Workbench

The troubleshooting workbench includes a Flows tab which displays network flows collected from VCF Operations for networks.

### Supported Content Types

The troubleshooting workbench collects flow information about the following content types:

- VM
- Host
- Cluster
- Datacenter
- Folder
- Project
- VPC
- Security group
- Transport node

Flow information about other content types is not collected and displayed in the troubleshooting workbench.

### Requirements

To view network flow information in the troubleshooting workbench, verify that the following requirements are met:

- VCF Operations for networks is installed and paired with VCF Operations.
- Flow collection is activated in your data sources.
- The data sources are added as integrations in VCF Operations.
- Navigate to the VCF Operations Integrations page and validate that there are no issues with the connectivity between VCF Operations and VCF Operations for networks.
- Navigate your data sources and validate that there are no issues with flow collection.

## Logs in the Workbench

The Workbench provides a tab to view Logs. The Log management component must be installed to view Logs. When you click **Operate** in the top navigation bar, and in the left navigation pane, click **Logs** and select the **Logs** tab, you will see all the collected logs which are collected. The **Logs** tab in Workbench shows the logs of the specific object which is monitored. For example, if you collected the vCenters logs, you see all the vCenter logs in the Logs tab. While in Workench you see object-specific logs, such us specific ESX logs. You can also compare logs from the **Compare Log** tab.

For more information on Log Management, see Configuring and Analyzing Logs.

# Configuring Green Score

Green Score is used to measure energy efficiency in digital operations through data center virtualization. Green Score is comprised of multiple parameters for effective measurement of the score. Virtualization plays a crucial role in optimizing resource utilization, helping reduce hardware footprints in the data centers and further enables savings in power consumption and floor space.

You can configure Green Score at the Organization level and for physical data centers.

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# Configuring Green Score Data at the Organization Level

Virtualization plays a crucial role in data center consolidation and thereby helps in the reduction of hardware footprint in the data centers. Virtualization enables lower power consumption and floor space usage in data centers, which can improve the overall energy efficiency and decrease related carbon emissions. Use this page to configure Green Score data in VCF Operations. Using Green Score, organizations can track where they are in their Green Score journey and how they are progressing over time.

**Note:**
Ensure that the following permissions are assigned:

- Capacity > Green Score > Change Organization Settings
- Capacity > Green Score > View Green Score Page

## Configuring Green Score for the First Time

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Green Score**. From the **Green Score** page, click **Configure** from the Green Score panel. Enter the details in the **Organization** tab.

**Table 1359: Organization Details**

| Option | Description |
|---|---|
| **Green Score Component Input - Virtualization Ratio(%)** | |
| Compute Virtualization | Enter the percentage of compute services that are virtual. Default value is 80%. The default values are based on the VMware estimate of the most common numbers. |
| Network Virtualization | Enter the percentage of network services that are virtual. Default value is 30%. The default values are based on the VMware estimate of the most common numbers. |
| Storage Virtualization | Enter the percentage of storage technologies that are virtual. Default value is 40%. The default values are based on the VMware estimate of the most common numbers. |
| Desktop Virtualization | Enter the percentage of desktops that are virtual. Default value is 20%. The default values are based on the VMware estimate of the most common numbers. |
| **Power Sources (%)** | |
| Add Power Source | Click this option to add the share of power sources that you use. The total should add up to 100%.<br>The power sources are coal, oil, natural gas, hydroelectric, solar photovoltaic, wind, biomass, and nuclear.<br>The default value for coal is 79.2% and the default value for hydroelectric is 20.8%. |
| **Average Age of Hardware in years (as of today)** | |
| Servers (years) | Enter the average age of the servers in your organization. Default age is two years. The default values are based on the VMware estimate of the most common numbers. The value changes dynamically each day (For example, if the average server age is 2 years on day 1, the average server age will change to 2.5 years after 6 months automatically). |

| Option | Description |
|---|---|
| Storage (years) | Enter the average age of storage in your organization. Default age is two years. The default values are based on the VMware estimate of the most common numbers. The value changes dynamically each day (For example, if the average server age is 2 years on day 1, the average server age will change to 2.5 years after 6 months automatically). |
| Network (years) | Enter the average age of network in your organization. Default age is three years. The default values are based on the VMware estimate of the most common numbers. The value changes dynamically each day (For example, if the average server age is 2 years on day 1, the average server age will change to 2.5 years after 6 months automatically). |
| Desktop (years) | Enter the average age of the desktops in your organization. Default age is five years. The default values are based on the VMware estimate of the most common numbers. The value changes dynamically each day (For example, if the average server age is 2 years on day 1, the average server age will change to 2.5 years after 6 months automatically). |

After you click **Save**, the Green Score data is calculated. Calculation of the Green Score data may take some time.

### Green Score Details

Green Score is calculated based on five components, which are: Workload Efficiency, Resource Utilization, the extent of Virtualization, Power Source, and Hardware Efficiency. Each of these components has varying levels of impact on the energy usage within a data center and hence has a different weightage when contributing to a common Green Score.

- **Workload Efficiency**: Contributes to 22.5% weightage Efficiency = Non Wastage / (Wastage + Non Wastage)
- **Resource Utilization**: Contributes to 12.5% weightage based on the utilization of hardware from the absolute capacity without considering buffers and High Availability.
- **Virtualization**: Contributes to 15% weightage.
- **Power Source**: Contributes to 37.5% weightage. Scoring is based on the carbon intensity (defined here: https://www.eia.gov/tools/glossary/?id=C) of the power source used.
- **Hardware Efficiency**: Contributes to 12.5% weightage, based on the assumption that newer generation hardware will be more energy efficient.

**Table 1360: Green Score Details and Dependencies**

| Name | Description |
|---|---|
| Green Score Meter | A higher score indicates a greater Green Score. Green Score is calculated based on the different values calculated for workload efficiency, resource utilization, virtualization, power source, and hardware efficiency. |
| Green Score Trend | Indicates the Green Score trend of the company. You can view daily, weekly, and monthly trends. |
| **Optimized Demand** | |

| Name | Description |
| --- | --- |
| Workload Efficiency | Measures the wastage in the data center and calculates a score based on various waste sources. The workload efficiency value is calculated based on different metrics, such as powered off VMs, oversized VMs, idle VMs, snapshots, and orphaned disks. The metrics are as follows:<br>Total Consumption CPU = (vCenter) CPU\|vCPUs Allocated on all Powered On Consumers<br>Total Consumption Memory (GB) = (vCenter) Memory\|Host Usage (KB)<br>Total Consumption Disk (TB) = (vCenter) Disk Space\|Total Provisioned Disk Space (GB)<br>Oversized VM\|vCPU = (VM) Summary\|Oversized\|Virtual CPUs and Oversized VM\|Memory (GB) = (VM) Summary\|Oversized\| Memory (KB) - These metrics are used to calculate workload efficiency only for powered on VMs. This means that vCPUs and memory of powered off and oversized VMs are excluded from the calculation.<br>Idle VM\|Disk (TB) = (Cluster) Reclaimable\|Idle VMs\|Disk Space (GB)<br>Idle VM\|vCPU = (Cluster) Reclaimable\|Idle VMs\|CPU (vCPUs)<br>Idle VM\|Memory (GB) = (Cluster) Reclaimable\|Idle VMs\|Memory (KB)<br>Powered Off VM\|Disk (TB) = (Cluster) Reclaimable\|Powered Off VMs\|Disk Space (GB)<br>Snapshot\|Disk (TB) = (Cluster) Reclaimable\|VM Snapshots\|Disk Space (GB)<br>Orphaned VMDK\|Disk (TB) = (Datastore) Reclaimable\|Orphaned Disks\|Disk Space (GB) |
| **Lean Operations** | |
| Resource Utilization | Provides a value based on server and storage utilization. The score is based on the utilization of hardware from absolute capacity without considering buffers and HA. Here are the metrics used:<br>Total ESX Hosts CPU Utilization (GHz) = (vCenter) CPU\|Usage (MHz)<br>Total ESX Hosts CPU Capacity (GHz) = (vCenter) CPU\|Total Capacity (MHz)<br>Total ESX Hosts Memory Utilization (TB) = (vCenter) Memory\|Host Usage (KB)<br>Total ESX Hosts Memory Capacity (TB) = (vCenter) Memory\|Total Capacity (KB) Organization/Physical<br>Total datastores capacity = (vCenter) Disk Space\|Total Capacity (GB)<br>Total datastores utilization = (vCenter) Disk Space\|Utilization (GB)<br>Total RDM = Sum((VM) Virtual Disk \ Configured Size) |
| Virtualization | Virtualization score is calculated based on the virtualization percentage of compute, storage, network, and desktop entered in the **Organization Details** page. The formula is a weighted average of virtualization percentages with 40%, 30%, 20%, and 10% corresponding weights.<br>**Note:**<br>At the organization level, Green Score is calculated based on physical data centers that are configured for Green Score. In addition, Green Score calculation at the organization level also includes vCenter instances that are not associated with any physical data center. |

| Name | Description |
|---|---|
| **Energy Efficient Infrastructure** | |
| Power Source | The score is based on the carbon intensity (defined here: https://www.eia.gov/tools/glossary/?id=C) of the power source used. For example, traditional coal based power sources will have a carbon intensity rate of 888 g/kWh while wind based power source will have a carbon intensity of 26 only (Reference : WNA Report - Comparison of Lifecycle Greenhouse Gas Emissions of Various Electricity Generation Sources).<br>**Note**: A 100% badge score indicates zero carbon emissions, and a 0% badge score indicates full amount of possible carbon emissions.<br>Each value is calculated based on the share of the power sources entered in the **Organization Details** page, after which the average value is calculated. The value for each of the power resources is calculated as follows:<br>Green Factor Shares (100%) Adjusted Score (Green Factor * Shares)<br>Coal = 17% (Green Factor)<br>Oil = 31% (Green Factor)<br>Natural Gas = 53% (Green Factor)<br>Solar Photovoltaic = 92% (Green Factor)<br>Biomass = 96% (Green Factor)<br>Nuclear = 97% (Green Factor)<br>Hydroelectric = 98% (Green Factor)<br>Wind = 98% (Green Factor)<br>**Note:**<br>Shares are user inputs in the **Organization Details** page.<br>**Note:**<br>At the organization level, Green Score is calculated based on physical data centers that are configured for Green Score. In addition, Green Score calculation at the organization level also includes vCenter instances that are not associated with any physical data center. |
| Hardware Efficiency | The hardware efficiency badge is based on the assumption that newer generation hardware is more energy efficient than the older generation hardware - by being able to run more workloads on the same sized newer hardware and also the ability of newer hardware to optimize energy consumption. The efficiency of hardware which includes server, storage, network, and desktop is based on the average age of the hardware entered in the **Organization Details** page. For example, if you had entered the age of your servers as 2 years, then the hardware efficiency for servers would be 80%, because the useful life duration is considered as 10 years.<br>**Note:**<br>At the organization level, Green Score is calculated based on physical data centers that are configured for Green Score. In addition, Green Score calculation at the organization level also includes vCenter instances that are not associated with any physical data center. |

### Power Consumption Chart

The power consumption chart displays the cumulative power consumption for the last 30 days. It is calculated using the power usage metrics and is the sum of the last month's daily power consumption. When you click the Power Consumption option, you can also view the **Power Consumption** trend which can be daily, weekly, or monthly.

### Estimated Carbon Footprint Chart

This estimated carbon footprint calculation represents the amount of Scope 2 greenhouse gas emissions, expressed as carbon dioxide equivalents, emitted indirectly as a result of datacenter operations during a given period.

### Environmental Impact

The environmental impact converts the greenhouse gas emissions numbers from power consumed into an equivalent number of carbon emissions emitted from the number of smartphones charged, as per the calculations based on Greenhouse Gas Equivalencies Calculator. For detailed info on this please refer to https://www.epa.gov/energy/greenhouse-gases-equivalencies-calculator-calculations-and-references#smartphones.

For information about Green Score dashboards, see Green Score Dashboards.

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## Configuring Green Score Data for Physical Data Centers

Virtualization plays a crucial role in data center consolidation and thereby helps in the reduction of hardware footprint in the data centers. Virtualization enables lower power consumption and floor space usage in data centers, which can improve the overall energy efficiency and decrease related carbon emissions. Use this page to configure Green Score data in VCF Operations for physical data centers that have been added to a vCenter account or to a VCF account and configured for Green Score. Using Green Score, organizations can track where they are in their Green Score journey and how they are progressing over time.

> **Note:**
>
> Ensure that the following permissions are assigned:
>
> - Capacity > Green Score > Change Organization Settings
> - Capacity > Green Score > View Green Score Page

### Configuring Green Score for the First Time for Physical Data Centers

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Green Score**.
2. From the **Green Score** page, click **Configure** from the Green Score panel. Enter the details in the **Organization** tab.

   > **Note:**
   > You must configure Green Score data at the Organization level and only then will you be able to configure Green Score for physical data centers. For information about configuring Green Score at the Organization level, see Configuring Green Score Data at the Organization Level.
3. Click the **Save** button in the **Organization** tab to save the Organization level details and to activate the **Physical Data Centers** tab.
4. From the **Physical Data Centers** tab, click the **Add Physical Data Center** button to add a physical data center to a vCenter Server account or to a VCF account if you have not previously added a physical data center from the **Administration** > **Control Panel** > **Physical Data Center** page. For more information about adding physical data centers, see Adding Physical Data Centers in VCF Operations.

5. When you create a physical data center, the Green Score configuration page for that physical data center is displayed and you can fill in all the details.

   **Note:**
   If you have already added a physical data center from the **Administration** > **Physical Data Center** page, a list of physical data centers appears on the **Physical Data Centers** tab. You can select each of the physical data centers from the vertical ellipses called **Add Green Score Configuration**.
6. Select a physical data center from the Physical Data Centers list and then enter the details in the **Physical Data Centers** tab.

**Table 1361: Physical Data Center Details**

| Option | Description |
|---|---|
| **Green Score Component Input - Virtualization Ratio(%)** | |
| Compute Virtualization | Enter the percentage of compute services that are virtual. Default values for the physical data center Green Score configuration are inherited from Organization level inputs. |
| Network Virtualization | Enter the percentage of network services that are virtual. Default values for the physical data center Green Score configuration are inherited from Organization level inputs. |
| Storage Virtualization | Enter the percentage of storage technologies that are virtual. Default values for the physical data center Green Score configuration are inherited from Organization level inputs. |
| Desktop Virtualization | Enter the percentage of desktops that are virtual. Default values for the physical data center Green Score configuration are inherited from Organization level inputs. |
| **Power Sources (%)** | |
| Add Power Source | Click this option to add the share of power sources that you use. The total should add up to 100%.<br>The power sources are coal, oil, natural gas, hydroelectric, solar photovoltaic, wind, biomass, and nuclear.<br>The default value for coal is 79.2% and the default value for hydroelectric is 20.8%. |
| **Average Age of Hardware in years (as of today)** | |
| Servers (years) | Enter the average age of the servers for your physical data center. Default values for the physical data center Green Score configuration are inherited from Organization level inputs. . The value changes dynamically each day (For example, if the average server age is 2 years on day 1, the average server age will change to 2.5 years after 6 months automatically). |
| Storage (years) | Enter the average age of the servers for your physical data center. Default values for the physical data center Green Score configuration are inherited from Organization level inputs. The value changes dynamically each day (For example, if the average server age is 2 years on day 1, the average server age will change to 2.5 years after 6 months automatically). |
| Network (years) | Enter the average age of the servers for your physical data center. Default values for the physical data center Green Score configuration are inherited from Organization level inputs. The value changes dynamically each day (For example, if the average server age is 2 years on day 1, the average server age will change to 2.5 years after 6 months automatically). |

| Option | Description |
|---|---|
| Desktop (years) | Enter the average age of the servers for your physical data center. Default values for the physical data center Green Score configuration are inherited from Organization level inputs. The value changes dynamically each day (For example, if the average server age is 2 years on day 1, the average server age will change to 2.5 years after 6 months automatically). |

After you click **Save**, the Green Score data is calculated. Calculation of the Green Score data may take some time.

## Green Score Details

Green Score is calculated based on five components, which are: Workload Efficiency, Resource Utilization, the extent of Virtualization, Power Source, and Hardware Efficiency. Each of these components has varying levels of impact on the energy usage within a data center and hence has a different weightage when contributing to a common Green Score.

- **Workload Efficiency**: Contributes to 22.5% weightage Efficiency = Non Wastage / (Wastage + Non Wastage).
- **Resource Utilization**: Contributes to 12.5% weightage based on the utilization of hardware from the absolute capacity without considering buffers and High Availability.
- **Virtualization**: Contributes to 15% weightage.
- **Power Source**: Contributes to 37.5% weightage – scoring based on the carbon intensity (defined here: https://www.eia.gov/tools/glossary/?id=C) of the power source used.
- **Hardware Efficiency**: Contributes to 12.5% weightage, based on the assumption that newer generation hardware will be more energy efficient.

**Table 1362: Green Score Details and Dependencies**

| Name | Description |
|---|---|
| Physical Data Centers (list) | Displays the configured physical data centers in a list. Select the physical data center for which you want to see Green Score details in a map view or from a list using the options at the top-right corner.<br>To reset to the Organizational level, click the **X** button for the selected physical data center. |
| Physical Data Centers by Green Score | Displays the number of all physical data centers. At the bottom of the panel, you can view the number physical data centers that are configured (green in color) and not configured (gray in color) for Green Score. For configured physical data centers, you can view the range of Green Score values applicable to the configured physical data center. |
| Green Score Meter | A higher score indicates a greater Green Score. Green Score is calculated based on the different values calculated for workload efficiency, resource utilization, virtualization, power source, and hardware efficiency. |
| Green Score Trend | Indicates the Green Score trend of the company. You can view daily, weekly, and monthly trends. |
| **Optimized Demand** | |

| Name | Description |
|---|---|
| Workload Efficiency | Measures the wastage in the data center and calculates a score based on various waste sources. The workload efficiency value is calculated based on different metrics, such as powered off VMs, oversized VMs, idle VMs, snapshots, and orphaned disks. The metrics are as follows:<br>Total Consumption CPU = (vCenter) CPU\|vCPUs Allocated on all Powered On Consumers<br>Total Consumption Memory (GB) = (vCenter) Memory\|Host Usage (KB)<br>Total Consumption Disk (TB) = (vCenter) Disk Space\|Total Provisioned Disk Space (GB)<br>Oversized VM\|vCPU = (VM) Summary\|Oversized\|Virtual CPUs and Oversized VM\|Memory (GB) = (VM) Summary\|Oversized\|Memory (KB) - These metrics are used to calculate workload efficiency only for powered on VMs. This means that vCPUs and memory of powered off and oversized VMs are excluded from the calculation.<br>Idle VM\|Disk (TB) = (Cluster) Reclaimable\|Idle VMs\|Disk Space (GB)<br>Idle VM\|vCPU = (Cluster) Reclaimable\|Idle VMs\|CPU (vCPUs)<br>Idle VM\|Memory (GB) = (Cluster) Reclaimable\|Idle VMs\|Memory (KB)<br>Powered Off VM\|Disk (TB) = (Cluster) Reclaimable\|Powered Off VMs\|Disk Space (GB)<br>Snapshot\|Disk (TB) = (Cluster) Reclaimable\|VM Snapshots\|Disk Space (GB)<br>Orphaned VMDK\|Disk (TB) = (Data store) Reclaimable\|Orphaned Disks\|Disk Space (GB) |
| **Lean Operations** | |
| Resource Utilization | Provides a value based on server and storage utilization. The score is based on the utilization of hardware from absolute capacity without considering buffers and HA. Here are the metrics used:<br>Total ESX Hosts CPU Utilization (GHz) = (vCenter) CPU\|Usage (MHz)<br>Total ESX Hosts CPU Capacity (GHz) = (vCenter) CPU\|Total Capacity (MHz)<br>Total ESX Hosts Memory Utilization (TB) = (vCenter) Memory\|Host Usage (KB)<br>Total ESX Hosts Memory Capacity (TB) = (vCenter) Memory\|Total Capacity (KB) Organization/Physical<br>Total datastores capacity = (vCenter) Disk Space\|Total Capacity (GB)<br>Total datastores utilization = (vCenter) Disk Space\|Utilization (GB)<br>Total RDM = Sum((VM) Virtual Disk \ Configured Size) |
| Virtualization | Virtualization score is calculated based on the virtualization percentage of compute, storage, network, and desktop entered in the corresponding physical data center's configuration. The formula is a weighted average of virtualization percentages with 40%, 30%, 20%, and 10% corresponding weights. |
| **Energy Efficient Infrastructure** | |

| Name | Description |
| --- | --- |
| Power Source | The score is based on the carbon intensity (defined here: https://www.eia.gov/tools/glossary/?id=C) of the power source used. For example, traditional coal based power sources will have a carbon intensity rate of 888 g/kWh while wind based power source will have a carbon intensity of 26 only (Reference : WNA Report - Comparison of Lifecycle Greenhouse Gas Emissions of Various Electricity Generation Sources.)<br>**Note**: A 100% badge score indicates zero carbon emissions and a 0% badge score indicates full amount of possible carbon emissions.<br>Each value is calculated based on the share of the power sources entered in the **Organization Details** page, after which the average value is calculated. The value for each of the power resources is calculated as follows:<br>Green Factor Shares (100%) Adjusted Score (Green Factor * Shares)<br>Coal = 17% (Green Factor)<br>Oil = 31% (Green Factor)<br>Natural Gas = 53% (Green Factor)<br>Solar Photovoltaic = 92% (Green Factor)<br>Biomass = 96% (Green Factor)<br>Nuclear = 97% (Green Factor)<br>Hydroelectric = 98% (Green Factor)<br>Wind = 98% (Green Factor)<br>**Note:**<br>Shares are user inputs in the corresponding physical data center's page. |
| Hardware Efficiency | The hardware efficiency badge is based on the assumption that newer generation hardware is more energy efficient than the older generation hardware - by being able to run more workloads on the same sized newer hardware and also the ability of newer hardware to optimize energy consumption. The efficiency of hardware which includes server, storage, network, and desktop is based on the average age of the hardware entered in the **Organization Details** page. However, you can change and input ages that are specific to the physical data center. For example, if you had entered the age of your servers as 2 years, then the hardware efficiency for servers would be 80%, because the useful life duration is considered as 10 years. |

### Power Consumption Chart

The power consumption chart displays the cumulative power consumption for the last 30 days. It is calculated using the power usage metrics and is the sum of the last month's daily power consumption. When you click the Power Consumption option, you can also view the **Power Consumption** trend which can be daily, weekly, or monthly.

### Estimated Carbon Footprint Chart

This estimated carbon footprint calculation represents the amount of Scope 2 greenhouse gas emissions, expressed as carbon dioxide equivalents, emitted indirectly as a result of datacenter operations during a given period.

### Environmental Impact

The environmental impact converts the greenhouse gas emissions numbers from power consumed into an equivalent number of carbon emissions emitted from the number of smartphones charged, as per the calculations based on Greenhouse Gas Equivalencies Calculator. For detailed info on this please refer to https://www.epa.gov/energy/greenhouse-gases-equivalencies-calculator-calculations-and-references#smartphones.

For information about Green Score dashboards, see Green Score Dashboards.

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# Green Score Dashboards

Using the Green Score dashboards in VCF Operations, you can view the estimated reduction in carbon footprint achieved using virtualization; you can view clusters and data centers and compare them based on estimated $CO_2$ emissions or power consumption; you can identify idle VMs; and identify old hardware, such as compute and storage.

The following Green Score dashboards are added to the predefined VCF Operations dashboards:

- Energy Efficiency with Virtualization Dashboard
- Energy Efficient Clusters Dashboard
- Environmental Impact of Idle VMs Dashboard
- Energy Efficient Infrastructure Dashboard

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## Energy Efficiency with Virtualization Dashboard

Use this dashboard to view information on how much power consumption you have already saved with Virtualization.

### How to Use the Dashboard

- The banner at the top of the dashboard is a sample image, and you can upload your company logo, tag line, and so on.
- The **Optimization by Virtualization** widget provides details of power saved for the vSphere World object in percent and compares the power consumption of a vCenter environment before and after virtualization assuming that 100W is the power consumption for a low-range server before virtualization.
- Use the $CO_2$ **Emissions Avoided (Tonnes)** widget to view the estimated sum of carbon emissions avoided for the last 24 hours. $CO_2$ emission per kWh is assumed as 0.672 kg (based on reference values from Greenhouse Gas Equivalencies Calculator).
- Using the **Power Consumption** widget, you can view estimated power consumed before and after virtualization. Power consumption before virtualization is based on the assumption that a low-range server consumes 100W. Hence, power consumption before virtualization is calculated using the count of VMs, which is the number of physical servers if not virtualized, at 100W per server.
- Using the $CO_2$ **Emission** widget, you can view estimated $CO_2$ emissions before and after virtualization. $CO_2$ emission per kWh is assumed as 0.672 kg (based on reference values from Greenhouse Gas Equivalencies Calculator). The value is calculated in Kilogram (kg).
- The $CO_2$ **Emission Chart** and the **Power Consumption Chart** provide estimated $CO_2$ emissions (in kg) for the last 24 hours and power consumption (in kWh) for the last 24 hours, respectively.

### Custom Values

- **Predefined Values**
  - Power consumption of a small server (1 socket, 10 cores, 32 GB RAM) = 0.1 kW (Assumption used in calculating power consumption before virtualization).
  - $CO_2$ emission per kWh = 0.672 kg (Reference values from Greenhouse Gas Equivalencies Calculator).
- You can add a custom property for $CO_2$ emission, which can be applied per cluster compute resource.
  Custom values of $CO_2$ emission per kWh can be added to each cluster compute resource by creating a custom property with the name $CO_2$ **Emission** and type as **Numeric** and the relevant custom values.

  **Note:**
  If you do not define custom properties, predefined values for $CO_2$ emission are used.

### Points to Note

Your actual savings may be different. The following are not included:

- Physical buildings and land. With virtualization, you can decrease your hardware footprint by having less physical racks, and therefore have a potentially lower carbon footprint.
- Network equipment. Fewer number of physical servers results in lesser network ports. Because firewall, load balancers, IDS, and IPS can be virtual machines, you have less equipment.
- Other components like Uninterrupted Power Supply (UPS), lighting, cooling, and labor.

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## Energy Efficient Clusters Dashboard

Use this dashboard to view clusters and data centers and compare them based on estimated $CO_2$ emissions or power consumption. You can then identify the most energy efficient cluster to provision workloads. You can compare the power consumption of each compute component in the data center, showcase all the compute components with the lowest power consumption, compare physical data centers based on power consumption, and compare hardware models to find optimal power consumption.

### How to Use the Dashboard

As larger clusters consume more power than smaller clusters, the total power consumption cannot be used to determine a cluster to provision the next workload. Hence Power Efficiency is calculated based on the power consumption per GHz of CPU usage.

- The banner at the top of the dashboard is a sample image, and you can upload your company logo, tag line, and so on.
- Use the **Top-10 Energy Efficient Clusters by Power Efficiency** table to view clusters that are "power efficient." Power efficiency is calculated as power usage per GHz of CPU Usage. Power efficiency alone cannot be used as the criteria to provision a VM, capacity must also be available in the cluster. View the Capacity Remaining% and Time Remaining information for each cluster to make an informed decision.
- Use the **Top-10 Energy Efficient Clusters by Power Consumption** table to view clusters that consume less power. Use this table when you compare clusters based on the total power consumption. Larger clusters will consume more power than smaller clusters, but may still not be the most efficient cluster based on power efficiency. Power consumption alone cannot be used as the criteria to provision a VM, capacity must also be available in the cluster. View the Capacity Remaining% and Time Remaining information for each cluster to make an informed decision.

- The goal of the **Energy Efficient Clusters** dashboard is to help you identify the most energy efficient cluster to provision the next VM. However, the most energy efficient cluster may not have enough capacity to provision a VM, and hence you also need to check capacity before you decide on the most energy efficient cluster to provision. Use relevant metrics for the selected cluster which can help you identify a target cluster. The metrics are: Power Consumption of the Cluster (kWh), CPU Usage% in the cluster, and Memory usage% in the cluster. Use the associated widgets to view the metrics. The metrics are available for the last 30 days to help you identify the target energy efficient cluster to provision the next VM.
- Use the **Heatmap of vSphere Clusters based on Power Consumption** to view all the clusters based on Power Consumption in kWh. The size of each cluster in the heatmap is determined by the number of VMs in the cluster.
- When you add geo tags to clusters or physical data centers, those objects can be mapped in the **Geo** widget. Ensure that you add geo tags to the relevant objects.

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## Environmental Impact of Idle VMs Dashboard

Use this dashboard to identify idle VMs in the data center, power off/delete idle VMs to reduce power consumption and improve data center efficiency, and identify remediation plans to offset the damage caused by those idle VMs which cannot be removed.

### How to Use the Dashboard

- The banner at the top of the dashboard is a sample image, and you can upload your company logo, tag line, and so on.
- The **Cluster's Idle VM** table provides a list of clusters with reclaimable memory, vCPUs, and disk space from idle VMs. You can also view the total resources that can be reclaimed from idle VMs. Click a cluster row to view the relevant Green Score metrics.
- Use the **Relevant Metrics of Selected Cluster** widget to view metrics of the selected cluster, such as power wasted by idle VMs (Wh) and estimated $CO_2$ emissions by idle VMs (kg).
- The **Wasted Power from Idle VMs** heatmap allows you to view vSphere clusters configured to display wasted power from idle VMs. Power in watt-hour (Wh) is used to build the heatmap. A threshold value of 800 Wh changes the heatmap to red.
- The $CO_2$ **Emission from Idle VMs** heatmap allows you to view vSphere clusters configured to display estimated $CO_2$ Emission from idle VMs. $CO_2$ emission in kilogram (kg) is used to build the heatmap. A threshold value of 750 kg changes the heatmap to red.

### Custom Values

- **Predefined Values**
  - $CO_2$ emission per kWh = 0.672 kg (Reference values from Greenhouse Gas Equivalencies Calculator).
- Custom values for $CO_2$ emission can be applied per cluster compute resource by adding a custom property. Custom values of $CO_2$ emission per kWh can be added to each cluster compute resource by creating a custom property with the name $CO_2$ **Emission** and type as **Numeric** and the relevant custom values.

  **Note:**
  If no custom properties are defined, predefined values for $CO_2$ emission are used.

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### Energy Efficient Infrastructure Dashboard

Use this dashboard to identify old hardware, such as compute and storage, in the data center and replace them with new generation hardware components that are more power efficient. You can also use this dashboard to reduce the overheads and buffers and identify smaller clusters that have higher overheads. The aim is to run with fewer overheads and buffer, without compromising on performance.

**How to Use the Dashboard**

- Smaller clusters have a relatively higher overhead. A cluster with two nodes has 50% overhead, while a cluster with 10 nodes has only 10% overhead. Clusters with lesser capacity require more hosts and hence consume more electricity. The **Small Clusters** table lists clusters that meet one of the following criteria:
  - <=4 nodes
  - <=120 CPU cores and < 1 TB memory

  Click a cluster row to view the context of the selected cluster. An empty widget indicates that the defined efficiency goals are met.
- Advancements in technology help ESX hosts to deliver higher efficiency. ESX hosts can deliver more CPU and memory capacity, often with low power requirements. The **Ageing Compute Hardware** table lists ESX hosts that meet one of the following criteria:
  - ESX version 6 or later
  - <=40 CPU cores and < 256 GB of memory

  Click a cluster row to view the context of the selected cluster. An empty widget indicates that the defined efficiency goals are met.
- Just like compute hardware, newer storage hardware is typically more power efficient than older storage hardware. The **Ageing Storage Hardware** table lists datastores that meet the following criteria:
  - VMFS version 5 or older.
  - Not a local datastore.

  Click a cluster row to view the context of the selected cluster. An empty widget indicates that the defined efficiency goals are met.

  If you have goals that are different from those defined, you can modify the criteria of the widgets by updating the filters.

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# Configuring Dashboards and Widgets

Dashboards present a visual overview of the performance and state of objects in your virtual infrastructure. Widgets are the visual components used to build dashboards, displaying information about attributes, resources, applications, or overall processes within your environment.

# Dashboards in VCF Operations

Dashboards present a visual overview of the performance and state of objects in your virtual infrastructure. You use dashboards to determine the nature and timeframe of existing and potential issues with your environment. You create dashboards by adding widgets to a dashboard and configuring them.

VCF Operations collects performance data from monitored software and hardware resources in your enterprise and provides predictive analysis and real-time information about problems. The data and analysis are presented through alerts, in configurable dashboards, on predefined pages, and in several predefined dashboards.

- You can start with several predefined dashboards in VCF Operations.
- You can create extra ones that meet your specific needs using widgets, views, badges, and filters to change the focus of the information.
- You can clone and edit the predefined dashboards or start from scratch.
- To display data that shows dependencies, you can add widget interactions in dashboards.
- You can provide role-based access to various dashboards for better collaboration in teams.

**Table 1363: Features**

| Features | Description |
|---|---|
| Manage | You can also manage dashboards by navigating to the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. Click **Dashboards** > **Manage** from the centre pane. |
| Create | Use this option to create a dashboard. See Create and Configure Dashboards. |
| Search | You can search for a dashboard across the **Favorites**, **Recents**, and **All** folders in the **Dashboards** pane in the centre. |
| Favorites | You can mark a dashboard as a favorite using the **Favorite** icon at the top of each dashboard. All the dashboards that you have marked as a favorite, are listed under the Favorites folder in the **Dashboards** pane in the centre. |
| Recents | The dashboards are listed in the order in which you select them, with the most recent dashboard that you selected, appearing at the top. Up to ten dashboards can be displayed as Recent dashboards.<br>If you do not pin the dashboard and log out of the user interface, on logging back in, the dashboard is removed from the Recents folder. |
| Shared | If you have shared the dashboard, the shared icon is displayed against the dashboard name. |
| All | Lists the dashboard folders and the dashboards that are activated. You can use this menu for quick navigation through your dashboards. To view the dashboards that are listed in the panel under **All**, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All**. You can also search for dashboards using keywords and letters. |

| Features | Description |
|---|---|
| Actions | Available dashboard actions, such as edit, delete, Set as Dashboards Home, and Add to Product Home. These actions are applied directly to the dashboard that you are on.<br>**Set as Dashboards Home**: Adds the dashboard to the **Favorites** list. To remove the dashboard from the **Favorites** list, select **Actions** > **Unset as Dashboards Home**.<br>**Add to Product Home**: Adds the dashboard as a tab in the **Home** page. You can also reorder the tabs in the **Home** page using drag and drop. You can also set this option by clicking the **Home** icon at the top right side of the dashboard. To remove the dashboard from the Home page, select **Actions** > **Remove from Product Home**.<br>**Note:**<br>You can add up to 5 dashboards to the **Home** page. |
| Dashboard Time | The dashboard time panel is activated by default on all predefined and user-created dashboards. Using this option, you can select a time for the widgets in the dashboard. The default time is 6 hours. The pre-defined time/day options in the panel are 1 hour, 6 hours, 24 hours, or 7 days. You can also set a customized time option.<br>To activate widgets to use the dashboard time, select ***Date Controls/Time Range*** > **Dashboard Time** from the widget toolbar. Some widgets have **Dashboard Time** as the default option. For example, **Metric Chart**, **View**, **Rolling View**, **Sparkline**, **Health Chart**, and **Mashup Chart** widgets.<br>Dashboard time persists if:<br>• You activate a widget in a dashboard to use the dashboard time and then log out and log back in, or<br>• You activate a widget in a dashboard to use the dashboard time, and you export and then import the dashboard into another instance of VCF Operations. |

## Accessing Predefined Dashboards

You can access some of the useful, predefined dashboards from the **Dashboards** home page.

To access these dashboards, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All**.

The dashboards are categorized as follows: Availability, Configuration, Inventory, Performance, Capacity, and Cost to name a few. To easily access some of the useful, predefined dashboards under these categories, click on the drop-down button against the selected category and click on the specific dashboard.

## Types of Dashboards

You can use the predefined dashboards or create your own custom dashboard in VCF Operations.

See Using Predefined Dashboards for Quick Insights for more information.

### Custom Dashboards

You can create dashboards that meet your environment needs in VCF Operations.

For information about creating a dashboard, see Create and Configure Dashboards.

## Create and Configure Dashboards

To view the status of all objects in VCF Operations, create a dashboard by adding widgets or views. You can create and modify dashboards and configure them to meet your environment needs.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**.
2. Click **Dashboards** > **Create** from the centre pane.
3. Complete the following steps to:
   a) Enter a name for the dashboard.
      Dashboard Name
   b) Add widgets or views to the dashboard.
      Widget or View List Details
   c) Configure widget interactions.
      Widget and View Interactions Details
   d) Create dashboard navigation.
      Dashboard Navigation Details
4. Click **Save**.
5. Click **Actions** > **Edit Dashboard** to modify the dashboard.

### Dashboard Name

The name and visualization of the dashboard as it appears on the VCF Operations Home page.

#### Where You Add a Name in a Dashboard

To create your dashboard, in the VCF Operations navigation bar at the top, click **Operations** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **Create**. Enter a name in the **New Dashboard** field.

To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operations** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All**. Click on the dashboard you want to edit and select **Actions** > **Edit Dashboard**.

If you use a forward slash while entering a name, the forward slash acts as a group divider and creates a folder with the specified name in the dashboards list if the name does not exist. For example, if you name a dashboard `clusters/hosts`, the dashboard is named `hosts` under the group `clusters`.

#### Related Concepts

Widget Interactions

Widget interactions are the configured relationships between widgets in a dashboard where one widget provides information to a receiving widget. When you are using a widget in the dashboard, you select data on one widget to limit the data that appears in another widget, allowing you to focus on a smaller subset data.

Widget Definitions List

A widget is a pane on a dashboard that contains information about configured attributes, resources, applications, or the overall processes in your environment. Widgets can provide a holistic, end-to-end view of the health of all the objects and applications in your enterprise. If your user account has the necessary access rights, you can add and remove widgets from your dashboards.

### Widget or View List Details

VCF Operations provides a list of widgets or views that you can add to your dashboard to monitor specific metrics and properties of objects in your environment.

## Where You Add Widgets or Views to a Dashboard

To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **Create** to add a dashboard.

To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All**. Click on the dashboard you want to edit and select **Actions** > **Edit Dashboard**.

## How to Add Widgets or Views to a Dashboard

In the widgets list panel, you see a list of predefined VCF Operations widgets or views that are most commonly used. Click **Show More** to view all the widgets and views. Click **Add Section** to add sections to the dashboard. You can drag and drop widgets under each section. While creating a dashboard, you can provide a name, description, edit, delete, collapse, and expand the sections in a dashboard. If you add sections and do not add widget/s to the sections, by default the widgets are displayed under the last section in the dashboard.

To add a widget or view to the dashboard workspace in the upper panel, you can:

- Drag the widget or view to the dashboard workspace or a section in the dashboard workspace in the upper panel, or
- Click on a widget or view from the widgets list panel, view a brief description of the widget or view from the pop up box and then click **Add to Dashboard**, or
- Double-click on the widget or view.

To locate a widget or view, you can enter the name or part of the name of a widget or view in the **Filter** > **Name** option. For example, when you enter `top`, the list is filtered to display the Top Alerts, Top-N, and Topology Graph widgets. You can then select the widget you require. You can also filter by either widgets or views by selecting **Filter** > **Show** to add a widget or view to the dashboard. Drag the widget or view to the dashboard workspace in the upper panel.

> **Note:**
> The following widgets are deprecated. Usage of these widgets is discouraged as they will be phased out in future releases. Deprecated widgets are marked with a yellow triangle.
>
> - Current Policy
> - Weather Map
> - Anomalies
> - DRS Cluster Settings
> - Efficiency
> - Environment Status
> - Risk
> - Environment
> - Container Overview
> - Faults

Most widgets or views must be configured individually to display information. For more information about how to configure each widget, see Widgets in VCF Operations.

## How to Arrange Widgets or Views in a Dashboard

You can modify your dashboard layout to suit your needs. By default, the first widgets or views that you add are automatically arranged horizontally wherever you place them.

- To position a widget or a view, drag the widget or view to the desired location in the layout. Other widgets and views automatically rearrange to make room.
- To resize a widget or a view, drag the bottom-right corner of the widget or the view.
- To maximize or minimize a widget or a view, use the maximize and minimize options in the top-right corner.

### Related Concepts

Widget Interactions

Widget interactions are the configured relationships between widgets in a dashboard where one widget provides information to a receiving widget. When you are using a widget in the dashboard, you select data on one widget to limit the data that appears in another widget, allowing you to focus on a smaller subset data.

Widget Definitions List

A widget is a pane on a dashboard that contains information about configured attributes, resources, applications, or the overall processes in your environment. Widgets can provide a holistic, end-to-end view of the health of all the objects and applications in your enterprise. If your user account has the necessary access rights, you can add and remove widgets from your dashboards.

## Widget and View Interactions Details

You can connect widgets and views so that the information they show depends on each other.

### Where You Create Widget and View Interactions

To create interactions for widgets or views in a dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **Create** to add a dashboard. From the toolbar, turn on the **Interaction Mode** toggle.

To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All**. Click on the dashboard you want to edit and select **Actions** > **Edit Dashboard**.

Internal interactions are interactions defined between widgets in a single dashboard. External interactions are interactions defined between a widget in one dashboard and widget/s in another dashboard.

### How to Create and Remove Widget Interactions

The list of available interactions depends on the widgets or views in the dashboard. Widgets and views can provide, receive, and can both provide and receive interactions at the same time.

To create interactions, turn on the **Interaction Mode** toggle. Click a provider plug and drag to the receiver. You can also apply interactions from receiver to provider plugs. For more information about how interactions work, see Widget Interactions.

To remove interactions, turn off the **Interaction Mode** toggle. You can also click the provider plug and select **Remove Interaction** > **<widget name>**.

**Note:**
You can create up to 25 widget interactions in a dashboard.

### How to Easily View Widget Interactions

To view widget interactions from a dashboard, click on an object from a provider widget that has widget interactions defined. A window appears with the external and internal widget interaction details. Clicking on an external interaction takes you to the external dashboard. Click on the internal interaction to view the details in the receiver widget.

**Related Concepts**

Widget Interactions

Widget interactions are the configured relationships between widgets in a dashboard where one widget provides information to a receiving widget. When you are using a widget in the dashboard, you select data on one widget to limit the data that appears in another widget, allowing you to focus on a smaller subset data.

Widget Definitions List

A widget is a pane on a dashboard that contains information about configured attributes, resources, applications, or the overall processes in your environment. Widgets can provide a holistic, end-to-end view of the health of all the objects and applications in your enterprise. If your user account has the necessary access rights, you can add and remove widgets from your dashboards.

## Dashboard Navigation Details

You can apply sections or context from one dashboard to another. You can connect widgets and views to widgets and views in the same dashboard or to other dashboards to investigate problems or better analyze the provided information.

### Where You Add Another Dashboard

To create dashboard navigation to a dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **Create** to add a dashboard. In the dashboard workspace, turn on the **Interaction Mode** toggle. From the **Select Another Dashboard** drop-down menu, select the dashboard to which you want to navigate.

To edit your dashboard, fin the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All**. Click on the dashboard you want to edit and select **Actions** > **Edit Dashboard**.

### How Dashboard Navigation Works

You can create dashboard navigation only for provider widgets and views. The provider widget or view sends information to the destination widget or view. When you create dashboard navigation, the destination widgets or views are filtered based on the information type they can receive.

### How to Add Dashboard Navigation to a Dashboard

The list of available dashboards for navigation depends on the available dashboards and the widgets and views in the current dashboard. To add navigation, you can drag from a sender widget interaction plug to a receiver widget interaction plug. You can select more than one applicable widget or view.

> **Note:**
> If a dashboard is unavailable for selection, it is unavailable for dashboard navigation.

The Dashboard Navigation icon
(
) appears in the top menu of each widget or view when a dashboard navigation is available.

After you have set widget interaction in the provider dashboard, the widget and menu bar are highlighted and two arrows appear in the top-left corner of the widget. After you have set widget interaction, clicking the object in the provider widget takes you to the receiver widget of the navigated dashboard.

**Related Concepts**

Widget Interactions

Widget interactions are the configured relationships between widgets in a dashboard where one widget provides information to a receiving widget. When you are using a widget in the dashboard, you select data on one widget to limit the data that appears in another widget, allowing you to focus on a smaller subset data.

Widget Definitions List

A widget is a pane on a dashboard that contains information about configured attributes, resources, applications, or the overall processes in your environment. Widgets can provide a holistic, end-to-end view of the health of all the objects and applications in your enterprise. If your user account has the necessary access rights, you can add and remove widgets from your dashboards.

## Manage Dashboards

You can select dashboards individually or as a group and perform several actions.

To manage your dashboards, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **Manage**. Use the options from the horizontal ellipsis next to the **Add** option.

All the dashboards are listed on this page. You can filter the dashboards based on the name of the dashboard, the dashboard folder, activated dashboards, shared dashboards, or the dashboard owner. You can click **Add** to create a dashboard. For information about creating a dashboard, see Create and Configure Dashboards.

You can select a dashboard from the list, click the vertical ellipsis against each dashboard, and select the various options such as edit, delete, clone, and deactivate a dashboard. You can also change ownership of dashboards and export the dashboard. By default, the list of dashboards is sorted by name and all the columns can be sorted.

> **Note:**
> A wrench icon appears when the data in an imported dashboard depends on the existence of one or more adapters that are currently not present. The wrench icon disappears if the required data in an imported dashboard appears in VCF Operations after configuration.
>
> Imported dashboards regardless of used data, remain stuck and include a wrench icon if the dashboard that is stuck (with the wrench icon), already exists.

**Datagrid Options**

| Column Names | Description |
|---|---|
| Name | Displays the name of the dashboard. |
| Folder | Lists the folder to which each dashboard belongs. |
| Description | Displays the description of the dashboard. |
| Activated | Displays whether the dashboard is activated or not. You can also activate and deactivate the dashboard. |
| URL | Displays whether the dashboard is shared externally. For dashboards that have been shared, click to view the shared links. |
| Shared | Displays whether the dashboard is shared internally. Click the icon to view and edit the groups to which the dashboard has been shared from the **Group Sharing** dialog box.<br>From the **Group Sharing** dialog box, to share dashboard edit privileges with the group, you can click the **Allow Editing** check box or click the pencil icon if a dashboard is shared with a group.<br>**Note:**<br>Users who are part of the user group to whom the dashboard has been shared, will see an unlocked or locked icon in the **Shared** column. The icon is unlocked when the user with whom the dashboard is shared can edit it, and the icon is locked when the dashboard cannot be edited. |
| Editable | Displays if the dashboard is editable. |
| Owner | Displays the owner of the dashboard. |
| Report Usage | Displays the number of reports where the dashboard is used. Click the number in the column to view the name of the report/s. Click the report name to navigate to the report template in edit mode. |
| Last Modified | Displays the date the dashboard was last modified. |
| Modified By | Displays the user who last modified the dashboard. |

You can select more than one dashboard and perform a set of options by clicking the horizontal ellipsis next to the **Add** option.

**Table 1364: Dashboards Options**

| Option | Description | Usage |
|---|---|---|
| Delete | Deletes a dashboard. | |
| Activate | Activates a dashboard that was previously deactivated. | |
| Deactivate | Deactivates a dashboard. | |
| Change Ownership | Assigns a new owner to the dashboard. | After you assign a dashboard to a new owner, the dashboard is no longer displayed as one of your dashboards.<br>When you transfer a dashboard that was previously shared with user groups, information about the shared user groups and group hierarchy is retained. |
| Export | When you export a dashboard, VCF Operations creates a dashboard file in JSON format. | You can export a dashboard from one VCF Operations instance and import it to another.<br>To export a dashboard, select the dashboard that you want to export, and click **Export** from the horizontal ellipsis. |
| Import | A PAK or JSON file that contains dashboard information from VCF Operations. | You can import a dashboard that was exported from another VCF Operations instance.<br>To import a dashboard:<br>1. Click the **Import** option from horizontal ellipsis.<br>2. Click **Browse** and select a Dashboard ZIP, PAK, or JSON file to import.<br>3. Select if you want to **Overwrite** or **Rename** the file in case of a conflict.<br>4. Click **Import** to import the dashboard, and click **Done**. |
| Auto-rotate Dashboards | Changes the order of the dashboard tabs on VCF Operations home page. | You can configure VCF Operations to switch from one dashboard to another. For more information, see Auto-Rotate Dashboards. |
| Manage Summary Dashboards | Provides you with an overview of the state of the selected object, group, or application. | You can change the **Summary** tab with a dashboard to get information specific to your needs. For more information, see Manage Summary Dashboards |
| Manage Dashboard Folders | Groups dashboards in folders. | You can create dashboard folders to group the dashboards in a way that is meaningful to you. For more information, see Manage Dashboard Folders. |
| Manage Dashboard Sharing | Makes a dashboard available to other users or user groups. | You can share a dashboard or dashboard template with one or more user groups. For more information, see Share Dashboards with Users. |

The dashboard list depends on your access rights.

## Manage Summary Dashboards

The **Summary** tab provides you with an overview of the state of the selected object, group, or application. You can change the **Summary** tab with a dashboard to get information specific to your needs.

### Where You Configure a Summary Tab Dashboard

To manage the summary dashboards, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **Manage**. Click the horizontal ellipsis next to the **Add** option and select **Manage Summary Dashboards**.

### How You Manage the Summary Dashboards

**Table 1365: Manage Summary Dashboards Toolbar Options**

| Option | Description |
|---|---|
| Use Default | Click to use VCF Operations default **Summary** tab. |
| Assign a Dashboard | Click to view the Dashboard List dialog box that lists all the available dashboards. |
| Adapter Type | Adapter type for which you configure a summary dashboard. |
| Filter | Use a word search to limit the number of adapter types that appear in the list. |

To change the Summary tab for an object, select the object in the left panel, click the **Assign a Dashboard** icon. Select a dashboard for it from the All Dashboards dialog box and click **OK**. From the Manage Summary Dashboards dialog box click **Save**. You see the dashboard that you have associated to the object type when you navigate to the **Summary** tab of the object details page.

## Auto-Rotate Dashboards

You can change the order of the dashboard tabs on your home page. You can configure VCF Operations to switch from one dashboard to another. This feature is useful if you have several dashboards that show different aspects of your enterprise's performance and you want to look at each dashboard in turn.

### Where You Configure Auto-Rotation of a Dashboard

To reorder and configure a dashboard switch, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **Manage**. Select **Auto-rotate Dashboards** from the horizontal ellipsis next to the **Add** option.

### How You Reorder the Dashboards

The list shows the dashboards as they are ordered. Drag the dashboards up and down to change their order on the home page.

### How You Configure an Automatic Dashboard Rotation

1. Double-click a dashboard from the list to configure.
2. From the Rotation drop-down menus, select **On**.
3. Select the time interval in seconds.
4. Select the dashboard to switch and click **Update**.
5. Click **Save** to save your changes.

On the home page, the current dashboard will switch to the dashboard that is defined after the specified time interval.

## Manage Dashboard Folders

You can create dashboard folders to group the dashboards in a way that is meaningful to you.

#### Where You Manage Dashboard Folders

To manage the dashboard folders, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **Manage**. Click the horizontal ellipsis next to the **Add** option and click **Manage Dashboard Folders**.

#### How You Manage the Dashboard Folders

**Table 1366: Manage Dashboard Folders Options**

| Option | Description |
|---|---|
| Folders | A hierarchy tree with all the available group folders. |
| Dashboards List | A list with all available dashboards. |

To create a dashboard folder, click **New Folder** in the **Folders** pane and enter the name of the folder. If you want to create a folder under another folder, select a parent folder under which you want to create the child folder, then click **New Folder**. To add a dashboard, drag one from the dashboards list to the selected folder in the **Folders** pane.

You can delete folders and/or detach dashboards from a folder, by selecting one or more folders and dashboards from the **Folders** pane and by clicking **Actions** > **Delete**.

You can rename a folder by selecting a single folder from the **Folders** pane and by clicking **Actions** > **Rename**.

### Share Dashboards with Users

You can share a dashboard with one or more user groups. When you share a dashboard, it becomes available to all the users in the user group that you select. The dashboard appears the same to all the users who share it. If you edit a shared dashboard, the dashboard changes for all users. Other users can only view a shared dashboard. They cannot change it.

#### Where You Share a Dashboard From

To share a dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **Manage**. Click the horizontal ellipsis next to the **Add** option and click **Manage Dashboard Sharing**.

**Table 1367: Dashboard Sharing Options**

| Option | Description |
|---|---|
| All Dashboards | Link to view all the available dashboards that you can share. The dashboards are displayed on the right side in the dashboards list. |
| User Groups | Lists the available user groups that you can share a dashboard with. The list includes the **Everyone** group. |
| Dashboard List | List of shared dashboards with the selected user group or all the available dashboards that you can share, if no user group is selected. |

#### Manage Dashboard Sharing

To share a dashboard, navigate to the dashboard in the list of dashboards and drag it to the group to share it with, on the left.

To stop sharing a dashboard with a group, click that group on the left panel, navigate to the dashboard in the right panel, and click **Stop Sharing** above the list.

**Table 1368: Datagrid Options**

| Column Name | Description |
|---|---|
| Name | Displays the name of the dashboard. |
| Folder | Displays the folder where the dashboard is located. |
| Description | Displays the description of the dashboard. |
| Activated | Displays whether the dashboard is activated for viewing. |
| Shared | Displays if the dashboard is shared with user groups. |
| Editable | Click the check box to share dashboard edit privileges with the specific group. |
| Owner | Displays the owner of the dashboard. |
| Last Modified | Displays the date the dashboard was last modified. |
| Modified By | Displays who modified the dashboard. |

## Dashboards Actions and Options

You can change the order of the dashboard tabs, configure VCF Operations to switch from one dashboard to another, create dashboard folders to group the dashboards in a way that is meaningful to you, share a dashboard or dashboard template with one or more user groups, and transfer selected dashboards to a new owner.

### Options for Sharing Dashboards

You can share predefined or custom dashboards using URLs, emails, and by copying the code to embed the dashboard into confluence or other internal official web pages. You can also assign and unassign a dashboard to specific user groups and export the dashboard configuration details.

When you use a non-authenticated shared URL, as a user you can open the dashboard in a new browser session. If you have already logged into VCF Operations in another session, you are redirected to this dashboard and the user authentication permissions apply. To ensure that the non-authenticated URL opens the intended dashboard, as a user you must log out from all existing user sessions.

The dashboard shared with the URL opens in a page where you can access all the widgets within the dashboard and you can interact with the given widgets at the same time. A non-authenticated dashboard however, does not allow you to browse to other areas of VCF Operations.

Dashboard sharing can only be applied to Groups with a VCF Operations Standard Edition license.

#### Where You Can Access the Options to Share Dashboards

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards**, and then click on an existing dashboard and then click the **Share Dashboard** icon in the top-right corner.

**Table 1369: Options in the Share Dashboard Dialog Box**

| Option | Description |
|---|---|
| URL | Allows you to copy the tiny URL for the selected dashboard.<br>• Set the expiry period for the link to **1 day**, **1 week**, **1 month**, **3 Months**, or **Never Expire**.<br>• Click **Copy Link** to copy the link to a new window from where you can view the dashboard.<br>**Note:**<br>• As a user, if you open a shared link and you are logged into VCF Operations, you are navigated to your default dashboard, instead of viewing the shared one.<br>• As a user, if you log in to the same IP that was shared with you previously, you cannot access the page with the same browser.<br>• As a user, ensure that you have the following permission: **Dashboards > Dashboard Management > Share (Public)**.<br>You can stop sharing a dashboard you had previously shared. To stop sharing a dashboard, click the **Unshare Link** option and enter the URL of the dashboard that you want to stop sharing and click **Unshare**.<br>Authentication is not required to view the shared dashboard. |
| Email | Allows you to send an email with the URL details of the dashboard, to a specific person.<br>• Set the expiry period for the link to **1 day**, **1 week**, **1 month**, **3 months**, or **Never Expire**.<br>• Configure an SMTP instance. See Add a Standard Email Plug-In for VCF Operations Outbound Alerts.<br>• Enter an email address and click the **Send Email** button to send an email with the URL details of the dashboard.<br>Authentication is not required to view the shared dashboard. |
| Embed | Provides an embedded code for the dashboard. You can use this code to embed the dashboard in relevant confluence pages that your company executives routinely use and analyze.<br>• Set the expiry period for the link to **1 day**, **1 week**, **1 month**, **3 Months**, or **Never Expire**.<br>**Note:**<br>• If you embed a dashboard in the **Text** widget, the widget does not display any data.<br>• When you open an HTML/confluence page with an embedded dashboard from the same browser that you have logged into VCF Operations, the dashboard does not load.<br>Authentication is not required to view the shared dashboard. |
| Groups | Allows you to assign and unassign a dashboard to specific user groups.<br>• Select the group to which you want to grant dashboard access from the drop-down menu.<br>• To share dashboard edit privileges with the group, select the **Allow Editing** check box if a dashboard is shared with a group.<br>• Click **Include**. You can include more than one group.<br>• From the label, select the cross mark to unassign the dashboard.<br>Log in to VCF Operations to view the shared dashboard. |
| Export | Allows you to export the dashboard configuration details.<br>Log in to VCF Operations to export/import a dashboard. |

### Manage Widgets in Dashboards

You can replicate widgets multiple times in a dashboard by using the copy and paste functionality.

Navigate to the dashboard from which you want to copy widgets. Select **Actions** > **Edit Dashboard**. Select one or more widgets that you want to copy by clicking the title of the widget and then select **Actions** > **Copy Widget(s)**. Click **Actions** > **Paste Widget(s)** to paste one or more widgets in the same dashboard.

To paste one or more widgets into another dashboard, exit the edit screen of the dashboard by selecting **Cancel**. Navigate to the dashboard to which you want to paste one or more widgets and select **Actions** > **Edit Dashboard** and then **Actions** > **Paste Widget(s)**.

## Widgets in VCF Operations

Widgets are the panes on your dashboards. You add widgets to a dashboard to create a dashboard. Widgets display information about attributes, resources, applications, or the overall processes in your environment.

You can configure widgets to reflect your specific needs. The available configuration options vary depending on the widget type. You must configure some of the widgets before they display any data. Many widgets can provide or accept data from one or more widgets. You can use this feature to set the data from one widget as filter and display related information on a single dashboard.

### Widget Interactions

Widget interactions are the configured relationships between widgets in a dashboard where one widget provides information to a receiving widget. When you are using a widget in the dashboard, you select data on one widget to limit the data that appears in another widget, allowing you to focus on a smaller subset data.

#### How Interactions Work

If you configured interactions between widget at the dashboard level, you can then select one or more objects in the providing widget to filter the data that appears in the receiving widget, allowing you to focus on data related to an object.

To use the interaction option between the widgets in a dashboard, you configure interactions at the dashboard level. If you do not configure any interactions, the data that appears in the widgets is based on how the widget is configured.

When you configure widget interaction, you specify the providing widget for the receiving widget. For some widgets, you can define two providing widgets, each of which can be used to filter data in the receiving widget.

For example, if you configured the Object List widget to be a provider widget for the Top-N widget, you can select one or more objects in the Object List widget and the Top-N displays data only for the selected objects.

For some widgets, you can define more than one providing widget. For example, you can configure the Metric Chart widget to receive data from a metrics provider widget and an objects providing widget. In such case, the Metric Chart widget shows data for any object that you select in the two provider widgets.

**Related reference**

Dashboard Navigation Details

You can apply sections or context from one dashboard to another. You can connect widgets and views to widgets and views in the same dashboard or to other dashboards to investigate problems or better analyze the provided information.

Dashboard Name

The name and visualization of the dashboard as it appears on the VCF Operations Home page.

Widget and View Interactions Details

You can connect widgets and views so that the information they show depends on each other.

Widget or View List Details

VCF Operations provides a list of widgets or views that you can add to your dashboard to monitor specific metrics and properties of objects in your environment.

# Configuration Files

You can create configuration files to upload SVG content and define topological hierarchies in dashboards through supported widgets. You can also create configuration files that display adapter kind, resource kind, and the associated metrics that can be displayed in dashboards using supported widgets.

## Configuration Files for Widget Metric Configuration

You can create an XML file that displays the adapter type, resource kind, and the associated metrics that can be displayed in a widget and dashboard.

### How Configuration Files Work

From the **Configuration Files** page, you can create a XML configuration file with the adapter type, object type, and metric details to be displayed in a dashboard. The final output is an XML file. The supported widgets are Metric Chart, Property List, Rolling View Chart, Scoreboard, Sparkline Chart, and Topology Graph. To use the metric configuration, you must set the widget Self Provider to **Off** and create a widget interaction with a provider widget.

### Where You Find Widget Metric Configuration File

To manage the configuration files, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Widget Metric Configuration** tile. To add a configuration file, click the **Add** button. You see the **Create Configuration File** page.

**Table 1370: New Configuration File Page Options**

| Option | Description |
|---|---|
| Name | Enter a name for the configuration file. |
| Description | Enter a description for the configuration file. |
| Containing Folder | Select the folder under which you want to store the new configuration file. If you have not created a folder, you can select the **User Defined** folder if it appears. The **System Defined** folder will not appear for selection. |
| Text box | You can define the adapter type, object type, and metrics.<br>From the **Adapter Kind** drop down above the text box, select the adapter type. From the **Resource Kind** drop down above the text box, select the object type. From the **Metric** drop down above the text box, select a metric.<br>Use the orange **Format XML** button to format the XML content. Click **Save**. You see the configuration file under the selected folder. You can preview the XML that was created. |

**Table 1371: Widget Metric Configuration Toolbar and Data Grid Options**

| Option | Description |
|---|---|
| Filter | Click the filter icon to filter the configuration files using the following criteria: Name, Description, Last Modified, and Modified By. |
| **Options from the horizontal ellipsis** | |
| Edit | Select a configuration file to edit the contents of the configuration file. |
| Delete | Select one or more configuration files if you want to delete them. You cannot delete configuration files under the **System Defined** folder. |

| Option | Description |
|---|---|
| Clone | Select a configuration file to clone. You can edit the configuration file options in the **New Configuration File** page and click **Save** |
| Move | Select one or more configuration files if you want to move them to another folder. You can also move configuration files using the drag and drop option. You cannot move a configuration file to or from the **System Defined** folder. |
| Export | Select one or more configuration files if you want to download them. |
| Import | Select this option if you want to import configuration files. To import:<br>• Click the **Import** option from the horizontal ellipsis.<br>• Click **Browse** and select the file to import.<br>• Select if you want to Overwrite or Skip the file in case of a conflict.<br>• Click **Import** to import the configuration file, and click **Done**. |
| Create New Folder | Enter a name for the folder and click **OK**. You can create a new folder for meaningful grouping. |
| Rename Folder | Select a folder created by a user and rename the folder. You cannot rename the **System Defined** and **User Defined** folders. |
| **Data Grid Options** | |
| Name | Lists the configuration files that have been created. The **System Defined** folder contains configuration files that are predefined. The **User Defined** folder contains orphan configuration files that were created by users but not assigned to a folder. |
| In Use | Indicates whether the configuration file is in use in a dashboard and supported widget.<br>**Yes**: Indicates that the configuration file is used in a dashboard. Hover and click on the green tick mark to view the name of the dashboard and the widget that uses this configuration file.<br>**No**: Indicates that the configuration file is not used in a dashboard. |
| Description | Displays the description entered while creating/editing the configuration file. |
| Last Modified | Displays when the configuration file was last updated. |
| Modified By | Displays who last modified the configuration file. For example, *admin* or *maintenanceAdmin*. |

## Configuration Files for Text Widget Content

You can upload an SVG in Text widgets that are utilized in different dashboards.

### How Configuration Files Work

From the **Configuration Files** page, you can upload an SVG file to be displayed in a Text widget.

### Where You Find Text Widget Configuration File

To manage the configuration files, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Text Widget Content** tile. To add a configuration file, click the **Add** button. You see the **Text Widget Content** page.

**Table 1372: New Configuration File Page Options**

| Option | Description |
|---|---|
| Name | Enter a name for the configuration file. |
| Description | Enter a description for the configuration file. |
| Containing Folder | Select the folder under which you want to store the new configuration file. If you have not created a folder, you can select the **User Defined** folder if it appears. The **System Defined** folder will not appear for selection. |
| Text box | From your system, using an XML editor, open the SVG file to be uploaded. Copy the text from the SVG file into the text box. Use the orange **Format XML** button to format the XML content. Click **Save**. You see the configuration file under the selected folder. You can preview the SVG that was uploaded.<br>**Note:**<br>Ensure that the text in the text box starts with `<svg`. |

**Table 1373: Text Widget Toolbar and Data Grid Options**

| Option | Description |
|---|---|
| Filter | Click the filter icon to filter the configuration files using the following criteria: Name, Description, Last Modified, and Modified By. |
| **Options from the horizontal ellipsis** | |
| Edit | Select a configuration file to edit the contents of the configuration file. |
| Delete | Select one or more configuration files if you want to delete them. You cannot delete configuration files under the **System Defined** folder. |
| Clone | Select a configuration file to clone. You can edit the configuration file options in the **New Configuration File** page and click **Save**. |
| Move | Select one or more configuration files if you want to move them to another folder. You can also move configuration files using the drag and drop option. You cannot move a configuration file to or from the **System Defined** folder. |
| Export | Select one or more configuration files if you want to download them. |
| Import | Select this option if you want to import configuration files. To import:<br>• Click the **Import** option from the horizontal ellipsis.<br>• Click **Browse** and select the file to import.<br>• Select if you want to Overwrite or Skip the file in case of a conflict.<br>• Click **Import** to import the configuration file, and click **Done**. |
| Create New Folder | Enter a name for the folder and click **OK**. You can create a new folder for meaningful grouping. |
| Rename Folder | Select a folder created by a user and rename the folder. You cannot rename the **System Defined** and **User Defined** folders. |
| **Data Grid Options** | |

| Option | Description |
|---|---|
| Name | Lists the configuration files that have been created. The **System Defined** folder contains configuration files that are predefined. The **User Defined** folder contains orphan configuration files that were created by users but not assigned to a folder. |
| In Use | Indicates whether the configuration file is in use in a Text widget and dashboard.<br>**Yes**: Indicates that the configuration file is used in a **Text** widget. Hover and click on the green tick mark to view the name of the dashboard and the Text widget that uses this configuration file.<br>**No**: Indicates that the configuration file is not used in a dashboard. |
| Description | Displays the description entered while creating/editing the configuration file. |
| Last Modified | Displays when the configuration file was last updated. |
| Modified By | Displays who last modified the configuration file. For example, *admin* or *maintenanceAdmin*. |

## Management Packs Configuration

You can create an XML file for specific adapter kinds with additional threshold details such as configuration limits, name, type, and value. The configuration file can be used in a widget and dashboard.

### How Configuration Files Work

From the **Management Packs Configuration** page, you can create a XML file with adapter specific details with additional thresholds to be displayed in dashboards and widgets. The final output is an XML file. The final output is an XML file. The supported widgets are Metric Chart, Property List, Rolling View Chart, Scoreboard, Sparkline Chart, and Topology Graph.

### Where You Find Resource Kind Metric Configuration File

To manage the configuration files, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Management Packs Configuration** tile. To add a configuration file, click the **Add** button. You see the **Create Configuration File** page.

**Table 1374: New Configuration File Page Options**

| Option | Description |
|---|---|
| Name | Enter a name for the configuration file. |
| Description | Enter a description for the configuration file. |
| Containing Folder | Select the folder under which you want to store the new configuration file. If you have not created a folder, you can select the **User Defined** folder if it appears. The **System Defined** folder will not appear for selection. |
| Text box | You can define the adapter kind and thresholds such as configuration limits, name, type, and value.<br>Select the tags above the text box to add to the configuration file.<br>Use the orange **Format XML** button to format the XML content.<br>Click **Save**. You see the configuration file under the selected folder. You can preview the XML that was created. |

**Table 1375: Solutions Configuration Toolbar and Data Grid Options**

| Option | Description |
|---|---|
| Filter | Click the filter icon to filter the configuration files using the following criteria: Name, Description, Last Modified, and Modified By. |
| **Options from the horizontal ellipsis** | |
| Edit | Select a configuration file to edit the contents of the configuration file. |
| Delete | Select one or more configuration files if you want to delete them. You cannot delete configuration files under the **System Defined** folder. |
| Clone | Select a configuration file to clone. You can edit the configuration file options in the **New Configuration File** page and click **Save** |
| Move | Select one or more configuration files if you want to move them to another folder. You can also move configuration files using the drag and drop option. You cannot move a configuration file to or from the **System Defined** folder. |
| Export | Select one or more configuration files if you want to download them. |
| Import | Select this option if you want to import configuration files. To import:<br>• Click the **Import** option from the horizontal ellipsis.<br>• Click **Browse** and select the file to import.<br>• Select if you want to Overwrite or Skip the file in case of a conflict.<br>• Click **Import** to import the configuration file, and click **Done**. |
| Create New Folder | Enter a name for the folder and click **OK**. You can create a new folder for meaningful grouping. |
| Rename Folder | Select a folder created by a user and rename the folder. You cannot rename the **System Defined** and **User Defined** folders. |
| **Data Grid Options** | |
| Name | Lists the configuration files that have been created. The **System Defined** folder contains configuration files that are predefined. The **User Defined** folder contains orphan configuration files that were created by users but not assigned to a folder. |
| Description | Displays the description entered while creating/editing the configuration file. |
| Last Modified | Displays when the configuration file was last updated. |
| Modified By | Displays who last modified the configuration file. For example, *admin* or *maintenanceAdmin*. |

## Configuration Files for the Topology Widget

You can create an XML file with topology hierarchies to be displayed in a Topology widget.

### How Configuration Files Work

From the **Configuration Files** page, you can create a XML file with topology hierarchies to be displayed in a Topology widget. The final output is an XML file.

## Where You Find Resource Kind Metric Configuration File

To manage the configuration files, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Topology Widget Configuration** tile. To add a configuration file, click the **Add** button. You see the **Create Configuration File** page.

### Table 1376: New Configuration File Page Options

| Option | Description |
|---|---|
| Name | Enter a name for the configuration file. |
| Description | Enter a description for the configuration file. |
| Containing Folder | Select the folder under which you want to store the new configuration file. If you have not created a folder, you can select the **User Defined** folder if it appears. The **System Defined** folder will not appear for selection. |
| Text box | You can define the topology hierarchies.<br>Select the tags above the text box to add to the configuration file. Use the orange **Format XML** button to format the XML content. Click **Save**. You see the configuration file under the selected folder. You can preview the XML that was created. |

### Table 1377: Topology Widget Configuration Toolbar and Data Grid Options

| Option | Description |
|---|---|
| Filter | Click the filter icon to filter the configuration files using the following criteria: Name, Description, Last Modified, and Modified By. |
| **Options from the horizontal ellipsis** | |
| Edit | Select a configuration file to edit the contents of the configuration file. |
| Delete | Select one or more configuration files if you want to delete them. You cannot delete configuration files under the **System Defined** folder. |
| Clone | Select a configuration file to clone. You can edit the configuration file options in the **New Configuration File** page and click **Save** |
| Move | Select one or more configuration files if you want to move them to another folder. You can also move configuration files using the drag and drop option. You cannot move a configuration file to or from the **System Defined** folder. |
| Export | Select one or more configuration files if you want to download them |
| Import | Select this option if you want to import configuration files. To import:<br>• Click the **Import** option from the horizontal ellipsis.<br>• Click **Browse** and select the file to import.<br>• Select if you want to Overwrite or Skip the file in case of a conflict.<br>• Click **Import** to import the configuration file, and click **Done**. |
| Create New Folder | Enter a name for the folder and click **OK**. You can create a new folder for meaningful grouping. |
| Rename Folder | Select a folder created by a user and rename the folder. You cannot rename the **System Defined** and **User Defined** folders. |
| **Data Grid Options** | |

| Option | Description |
|---|---|
| Name | Lists the configuration files that have been created. The **System Defined** folder contains configuration files that are predefined. The **User Defined** folder contains orphan configuration files that were created by users but not assigned to a folder. |
| In Use | Indicates whether the configuration file is in use in a dashboard and topology widget.<br>**Yes**: Indicates that the configuration file is used in a dashboard. Hover and click on the green tick mark to view the name of the dashboard and the widget that uses this configuration file.<br>**No**: Indicates that the configuration file is not used in a dashboard. |
| Description | Displays the description entered while creating/editing the configuration file. |
| Last Modified | Displays when the configuration file was last updated. |
| Modified By | Displays who last modified the configuration file. For example, *admin* or *maintenanceAdmin*. |

## Widget Definitions List

A widget is a pane on a dashboard that contains information about configured attributes, resources, applications, or the overall processes in your environment. Widgets can provide a holistic, end-to-end view of the health of all the objects and applications in your enterprise. If your user account has the necessary access rights, you can add and remove widgets from your dashboards.

**Table 1378: Summary of Widgets**

| Widget Name | Description |
|---|---|
| Alert List | Shows a list of alerts for the objects that the widget is configured to monitor. If no objects are configure, the list displays all alerts in your environment. |
| Alert Volume | Shows a trend report for the last seven days of alerts generated for the objects it is configured to monitor. |
| Anomalies | Shows a chart of the anomalies count for the past 6 hours. |
| Anomaly Breakdown | Shows the likely root causes for symptoms for a selected resource. |
| Capacity Remaining | Shows a percentage indicating the remaining computing resources as a percent of the total consumer capacity. It also displays the most constrained resource. |
| Container Details | Shows the health and alert counts for each tier in a single selected container. |
| Container Overview | Shows the overall health and the health of each tier for one or more containers. |
| Current Policy | Shows the highest priority policy applied to a custom group. |
| Data Collection Results | Shows a list of all supported actions specific for a selected object. |
| DRS Cluster Settings | Shows the workload of the available clusters and the associated hosts. |
| Efficiency | Shows the status of the efficiency-related alerts for the objects that it is configured to monitor. Efficiency is based on generated efficiency alerts in your environment. |
| Environment | Lists the number of resources by object or groups them by object type. |
| Environment Overview | Shows the performance status of objects in your virtual environment and their relationships. You can click an object to highlight its related objects and double-click an object to view its Resource Detail page. |
| Environment Status | Shows statistics for the overall monitored environment. |
| Faults | Shows a list of availability and configuration issues for a selected resource. |
| Forensics | Shows how often a metric had a particular value, as a percentage of all values, within a given time period. It can also compare percentages for two time periods. |

| Widget Name | Description |
|---|---|
| Health | Shows the status of the health-related alerts for the objects that it is configured to monitor. Health is based on generated health alerts in your environment. |
| Health Chart | Shows health information for selected resources, or all resources that have a selected tag. |
| Heat Map | Shows a heat map with the performance information for a selected resource. |
| Mashup Chart | Brings together disparate pieces of information for a resource. It shows a health chart and metric graphs for key performance indicators (KPIs). This widget is typically used for a container. |
| Metric Chart | Shows a chart with the workload of the object over time based on the selected metrics. |
| Metric Picker | Shows a list of available metrics for a selected resource. It works with any widget that can provide resource ID. |
| Object List | Shows a list of all defined resources. |
| Object Relationship | Shows the hierarchy tree for the selected object. |
| Object Relationship (Advanced) | Shows the hierarchy tree for the selected objects. It provides advanced configuration options. |
| Property List | Shows the properties and their values of an object that you select. |
| PromQL Viewer | Uses the Prometheus Query Language to identify real time data to be visualized. |
| Recommended Actions | Displays recommendations to solve problems in your vCenter instances. With recommendations, you can run actions on your data centers, clusters, hosts, and virtual machines. |
| Risk | Shows the status of the risk-related alerts for the objects that it is configured to monitor. Risk is based on generated risk alerts in your environment. |
| Rolling View Chart | Cycles through selected metrics at an interval that you define and shows one metric graph at a time. Miniature graphs, which you can expand, appear for all selected metrics at the bottom of the widget. |
| Scoreboard | Shows values for selected metrics, which are typically KPIs, with color coding for defined value ranges. |
| Scoreboard Health | Shows color-coded health, risk, and efficiency scores for selected resources. |
| Sparkline Chart | Shows graphs that contain metrics for an object . If all the metrics in the Sparkline Chart widget are for an object that another widget provides, the object name appears at the top right of the widget. |
| Tag Picker | Lists all defined resource tags. |
| Text Display | Reads text from a Web page or text file and shows the text in the user interface. |
| Time Remaining | Shows a chart of the Time Remaining values for a specific resource over the past 7 days. |
| Top Alerts | Lists the alerts most likely to negatively affect your environment based on the configured alert type and objects. |
| Top-N | Shows the top or bottom N number metrics or resources in various categories, such as the five applications that have the best or worst health. |
| Topology Graph | Shows multiple levels of resources between nodes. |
| View | Shows a defined view depending on the configured resource. |
| Weather Map | Uses changing colors to show the behavior of a selected metric over time for multiple resources. |
| Workload | Shows workload information for a selected resource. |
| Workload Pattern | Shows a historical view of the hourly workload pattern of an object. |

For more information about the widgets, see the VCF Operations help.

**Related reference**

Dashboard Navigation Details

You can apply sections or context from one dashboard to another. You can connect widgets and views to widgets and views in the same dashboard or to other dashboards to investigate problems or better analyze the provided information.

Dashboard Name

The name and visualization of the dashboard as it appears on the VCF Operations Home page.

Widget and View Interactions Details

You can connect widgets and views so that the information they show depends on each other.

Widget or View List Details

VCF Operations provides a list of widgets or views that you can add to your dashboard to monitor specific metrics and properties of objects in your environment.

## Alert List Widget

The Alert List widget is a list of alerts for the objects it is configured to monitor. You can create one or more alert lists in VCF Operations for objects that you add to your custom dashboards. The widget provides you with a customized list of alerts on objects in your environment.

### How the Alert List Widget and Configuration Options Work

You can add the Alert List widget to one or more custom dashboards and configure it to display data that is important to different dashboard users. The data that appears in the widget is based on the configured options for each widget instance. You edit an Alert List widget after you add it to a dashboard. The changes you make to the options create a custom alert list to meet the needs of the dashboard users.

### Where You Find the Alert List Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Alert List Widget Toolbar Options

On the title bar of the widget, click the **Show Toolbar** icon to access the toolbar options.

| Option | Description |
|---|---|
| Dashboard Navigation | Actions you can run on the selected alert.<br>For example, you use the option to open a vCenter, data center, virtual machine, or in the vSphere Web Client, allowing you to directly modify an object for which an alert was generated and fix any problems. |
| Reset Interaction | Returns the widget to its initial configured state and undoes any interactions selected in a providing widget.<br>Interactions are usually between widgets in the same dashboard, or you can configure interactions between widgets on different dashboards. |
| Perform Multi-Select Interaction | If the widget is a provider for another widget on the dashboard, you can select multiple rows and click this button. The receiving widget then displays only the data related to the selected interaction items.<br>Use Ctrl+click for Windows, or Cmd+click for Mac OS X, to select multiple individual objects or Shift+click to select a range of objects, and click the icon to activate the interaction. |
| Display Filtering Criteria | Displays the object information on which this widget is based. |
| Select Date Range | Limits the alerts that appear in the list to the selected date range. |

| Option | Description |
|---|---|
| Cancel Alert | Cancels the selected alerts. If you configure the alert list to display only active alerts, the canceled alert is removed from the list.<br>You cancel alerts when you do not need to address them. Canceling the alert does not cancel the underlying condition that generated the alert. Canceling alerts is effective if the alert is generated by triggered fault and event symptoms because these symptoms are triggered again only when subsequent faults or events occur on the monitored objects. If the alert is generated based on metric or property symptoms, the alert is canceled only until the next collection and analysis cycle. If the violating values are still present, the alert is generated again. |
| Suspend | Suspend an alert for a specified number of minutes.<br>You suspend alerts when you are investigating an alert and do not want the alert to affect the health, risk, or efficiency of the object while you are working. If the problem persists after the elapsed time, the alert is reactivated and it will again affect the health, risk, or efficiency of the object.<br>The user who suspends the alert becomes the assigned owner.<br>**Note:**<br>You can cancel or retrigger the alert, if it is still active when its suspension period has ended, by rerunning the automated actions connected to the alert. In this case, you can suppress cancelation and update on all instances of an alert on an object. To activate this option, open the property file `/usr/lib/vmware-vcops/user/conf/analytics/advanced.properties` and add `retriggerExpiredSuspendedActiveAlerts = true` to the property file, and restart the VCF Operations analytics service or the VCF Operations cluster. |
| Take Ownership | As the current user, you make yourself the owner of the alert.<br>You can only take ownership of an alert, you cannot assign ownership. |
| Release Ownership | Alert is released from all ownership. |
| Group By | Group alerts by the options in the drop-down menu. |
| Filter | Locate data in the widget. |

**Table 1379: Group By Options**

| Option | Description |
|---|---|
| None | Alerts are not sorted into specific groupings. |
| Time | Group alerts by time triggered. The default. |
| Criticality | Group alerts by criticality. Values are, from the least critical: Info/ Warning/Immediate/Critical. See also Criticality in the Alert List Widget Data Grid table. |
| Definition | Group alerts by definition, that is, group like alerts together. |
| Object Type | Group alerts by the type of object that triggered the alert. For example, group alerts on hosts together. |

## Alert List Widget Data Grid Options

The data grid provides information on which you can sort and search.

Expand the grouped alerts to view the data grid.

| Option | Description |
|---|---|
| Criticality | Criticality is the level of importance of the alert in your environment. The alert criticality appears in a tooltip when you hover the mouse over the criticality icon.<br>The level is based on the level assigned when the alert definition was created, or on the highest symptom criticality, if the assigned level was **Symptom Based**. |
| Alert | Description of the alert. |
| Triggered On | Name of the object for which the alert was generated. |
| Created On | Date and time when the alert was generated. |
| Status | Current state of the alert. |
| Alert Type | Alert type is assigned when you create the alert definition. It helps you categorize and route the alert to the appropriate domain administrator for resolution.<br>The possible values include:<br>• Application<br>• Virtualization/Hypervisor<br>• Hardware (OSI)<br>• Storage<br>• Network |
| Alert Sub-Type | Alert subtype is assigned when you create the alert definition. It helps you categorize and route the alert to the appropriate domain administrator for resolution.<br>The possible values include:<br>• Availability<br>• Performance<br>• Capacity<br>• Compliance<br>• Configuration |
| Importance | Displays the priority of the alert. The importance level of the alert is determined using a smart ranking algorithm. |

## Alert List Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

The **Input Transformation** section provides options to transform the input for the widget.

The **Output Filter** section provides options to restrict the widget data based on the selected filter criteria.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| **Input Data** | |
| Objects | Select objects on which you want to base the widget data.<br>1. Click the **Add New Objects** icon and select objects in the pop-up window. The selected objects appear in a list in this section.<br>While selecting objects, you can use the **Filter** text box to search for objects. You can also expand the **Tag Filter** pane on the left hand side to select one or more object tag values. A list of objects with the selected tag values appears. If you select more than one value for the same tag, you can choose objects that have any of the tags applied. If you select more than one value for different tags, you can choose only the objects that have all the tags applied.<br>2. Optionally, select objects from the list and click the **Remove Selected Objects** icon to remove the selected objects.<br>Click the **Select All** icon to select all the objects in the list.<br>Click the **Clear Selection** icon to clear your selection of objects in the list. |
| All | If you select this option, the widget data is based on all the objects in your environment. The following sections provide options to refine the objects for the widget data. |
| **Input Transformation** | |
| Relationship | Transform the input for the widget based on the relationship of the objects. For example, if you select the **Children** check box and a **Depth** of **1**, the child objects are the transformed inputs for the widget. |
| **Output Filter** | |

| Option | Description |
| --- | --- |
| Basic | Pick tags to refine the widget data. The widget data is based on the objects that have the picked tags applied. If you pick more than one value for the same tag, the widget includes objects that have any of the tags applied. If you pick more than one value for different tags, the widget includes only the objects that have all the tags applied.<br>If the objects have an input transformation applied, you select tag values for the transformed objects. |
| Advanced | Refine the widget data further based on the filter criteria for object types. The widget data is based on the objects for the filtered object types.<br>If the objects have a tag filter applied in the **Basic** subsection, you define filter criteria for the object types of the objects with tag filter applied. If the objects with tag filter applied do not belong to any of the object types in this filter criteria, the widget skips this filter and includes all the objects with tag filter applied.<br>If the objects have an input transformation applied, you define filter criteria for the object types of the transformed objects.<br>1. In the first drop-down menu, select an object type.<br>2. In the second drop-down menu, select the option based on which you want to define the filter criteria. For example, if you select **Metrics** for the **Datacenter** object type, you can define a filter criteria based on the value of a specific metric for data centers.<br>3. In the drop-down menus and text boxes that appear, select or enter values to filter the objects.<br>4. To add more filter criteria, click **Add**.<br>5. To add another filter criteria set, click **Add another criteria set**. |
| Alert Related | A group of filters limits the alerts that appear in this alert list to those that meet the selected criteria.<br>If the objects on which the alerts are based have an input transformation applied, you define filters for the alerts based on the transformed objects.<br>You can configure the following filters:<br>• Status. Select one or more alert states to include in the list.<br>• Criticality. Select one or more levels of criticality.<br>• Control State. Select one or more control states to include in the list.<br>• Impact. Select one or more alert badges to include in the list.<br>• Actions.<br>• Alert Type. Select the subtype in the type list. This value was assigned when you configured the alert definition.<br>• Alert Definition. Drag and drop the alert definitions to the left pane from the Alert Definitions list. Click **OK** to filter by the selected Alert Definitions.<br>The selections you make for each filter are displayed as labels for easy visibility. |

## Alert Volume Widget

The Alert Volume widget is a trend report for the last seven days of alerts generated for the objects it is configured to monitor in VCF Operations. You can create one or more alert volume widgets for objects that you add to your dashboards. The alert volume provides you with a customized trend report on objects that helps you identify changes in alert volume, indicating a problem in your environment.

## How the Alert Volume Widget and Configuration Options Work

You can add the Alert Volume widget to one or more custom dashboards and configure it to display data that is important to different dashboard users. The data that appears in the widget is based on the configured options for each widget instance. The changes you make to the options create a custom widget to meet the needs of the dashboard users.

## Where You Find the Alert Volume Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

## Alert Volume Widget Display Options

The Alert Volume widget displays a trend chart, symptoms by criticality, and active alerts.

| Option | Description |
|---|---|
| Trend chart | Volume of critical, immediate, and warning symptoms for the configured objects. |
| Symptoms by criticality | Number of symptoms for each criticality level. |
| Active Alerts | Number of active alerts. Alerts can have more than one triggering symptom. |

## Alert Volume Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |

| Option | Description |
|---|---|
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| **Input Data** | |
| Object | Search for objects in your environment and select the object on which you are basing the widget data. You can also click the **Add Object** icon and select an object from the object list. You can use the **Filter** text box to refine the object list and the **Tag Filter** pane to select an object based on tag values. |

## Anomalies Widget

The Anomalies widget displays the anomalies for a resource for the past 6 hours at time intervals you set.

The Anomalies widget shows or hides time periods when the metric violates a threshold that is configured. The widget color indicates the criticality of the violation.

### Where You Find the Anomalies Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Anomalies Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | • On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| **Input Data** | |
| Object | Search for objects in your environment and select the object on which you are basing the widget data. You can also click the **Add Object** icon and select an object from the object list. You can use the **Filter** text box to refine the object list and the **Tag Filter** pane to select an object based on tag values. |

## Anomaly Breakdown Widget

The Anomaly Breakdown widget shows the likely root causes for symptoms for a selected resource.

### How the Anomaly Breakdown Widget and Configuration Options Work

### Where You Find the Anomaly Breakdown Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Anomaly Breakdown Widget Display Options

The Anomaly Breakdown widget displays scores, volume, and a list of anomaly metrics.

| Option | Description |
|---|---|
| Score | Anomaly value. |
| Volume | VCF Operations full set metric count for the selected object in the specified time range. |
| Anomaly Metrics List | List of alarms for the selected object in the specified time range. |

### Anomaly Breakdown Widget Toolbar Options

On the title bar of the widget, click the **Show Toolbar** icon to access the toolbar options.

| Option | Description |
|---|---|
| Show Bar Details | If the widget is displaying data for multiple objects, you can select a row and click this button to view the list of alarms for the selected object. |
| Perform Multiple Interaction | If the widget is a provider for another widget on the dashboard, you can select multiple rows and click this button. The receiving widget then displays only the data related to the selected interaction items.<br>Use Ctrl+click for Windows, or Cmd+click for Mac OS X, to select multiple individual objects or Shift+click to select a range of objects, and click the icon to activate the interaction. |

### Anomaly Breakdown Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

The **Output Filter** section provides options to restrict the widget data based on the selected filter criteria.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |

| Option | Description |
|---|---|
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| Mode | Display a single object or multiple objects. |
| Show | Select the number of objects to display in multiple objects mode. |
| **Input Data** | |
| Object | Search for objects in your environment and select the object on which you are basing the widget data. You can also click the **Add Object** icon and select an object from the object list. You can use the **Filter** text box to refine the object list and the **Tag Filter** pane to select an object based on tag values. |
| **Output Filter** | |
| Basic | Pick tags to refine the widget data. The widget data is based on the objects that have the picked tags applied. If you pick more than one value for the same tag, the widget includes objects that have any of the tags applied. If you pick more than one value for different tags, the widget includes only the objects that have all the tags applied. |

## Container Details Widget

The Container Details widget displays graphs that show a summary of child objects, metrics, and alerts of an object in the inventory.

### How the Container Details Widget and Configuration Options Work

The Container Details widget treats objects from the inventory as containers and objects. Containers are objects that contain other objects. The widget lists the containers and shows the number of containers, objects, metrics, and alerts of the observed object. The widget also displays the alerts of each container and an icon links to its child objects. For example, if you select from the inventory a host that contains three objects such as, two virtual machines and one datastore, the Container Details widget displays summary information with three containers, two objects that are the child objects of the two virtual machines, and the number of alerts for the host and the number of metrics for the child objects of the host. The widget also lists each of the three containers, with the number of alerts for each object. Clicking an object in the graph takes you to the object details page. When you point to the icon next to the object, a tool tip shows the name of the related resource and its health. For example, when you point to the icon next to a virtual machine, the tool tip shows a related datastore and its health. Clicking the icon takes you to the object detail page of the related object, which is the datastore following the example.

You edit a container details widget after you add it to a dashboard. You can configure the widget to take information from another widget in the dashboard and to analyze it. When you select **Off** from the Self Provider option and set source and receiver widgets in the **Widget Interactions** menu during editing of the dashboard, the receiver widget shows information about an object that you select from the source widget. For example, you can configure the Container Details widget to display information about an object that you select from the Object Relationship widget in the same dashboard.

## Where You Find the Container Details Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

## Container Details Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| Mode | You can change the size of the graph using the Compact or Large buttons. |
| **Input Data** | |

| Option | Description |
|---|---|
| Object | Search for objects in your environment and select the object on which you are basing the widget data. You can also click the **Add Object** icon and select an object from the object list. You can use the **Filter** text box to refine the object list and the **Tag Filter** pane to select an object based on tag values. |

## Capacity Remaining Widget

The Capacity Remaining widget displays a percentage indicating the remaining computing resources as a percent of the total consumer capacity. It also displays the most constrained resource.

### Where You Find the Capacity Remaining Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Capacity Remaining Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |

| Option | Description |
|---|---|
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| **Input Data** | |
| Object | Search for objects in your environment and select the object on which you are basing the widget data. You can also click the **Add Object** icon and select an object from the object list. You can use the **Filter** text box to refine the object list and the **Tag Filter** pane to select an object based on tag values. |

## Container Overview Widget

The Container Overview widget gives a graphical presentation of the health, risk, and efficiency of an object or list of objects in the environment.

### How the Container Overview Widget and Configuration Options Work

The Container Overview widget displays the current status, the status for a previous time period of the health, risk, and the efficiency of an object or list of objects. You can configure the widget to display information for one or more objects that you are interested in when you select the **Object** mode during configuration of the widget. The widget displays information for all objects from an object type or types when you select the **Object Type** mode during configuration of the widget. You can open the object detailed page of each object in the data grid when you click the object.

You edit a container overview widget after you add it to a dashboard. You can configure the widget to display information about an object or to display information about all objects from an object type by using the **Object** or **Object Type** mode. The configuration options change depending on your selection of mode.

### Where You Find the Container Overview Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Container Overview Widget Toolbar Options

On the title bar of the widget, click the **Show Toolbar** icon to access the toolbar options.

The toolbar contains icons that you can use to get more information about other widgets or dashboards.

| Option | Description |
|---|---|
| Perform Multi-Select Interaction | If the widget is a provider for another widget on the dashboard, you can select multiple rows and click this button. The receiving widget then displays only the data related to the selected interaction items. Use Ctrl+click for Windows, or Cmd+click for Mac OS X, to select multiple individual objects or Shift+click to select a range of objects, and click the icon to activate the interaction. |
| Filter | You can filter the objects in the data grid. |

| Option | Description |
|---|---|
| Dashboard Navigation | You can explore information from another dashboard.<br>**Note:**<br>This toolbar icon exists when you configure the widget to interact with a widget from another dashboard.<br>Use **Dashboard Navigation** menu during dashboard configuration to configure the widgets to interact.<br>When you select an object from an object data grid and click the toolbar icon, it takes you to a related dashboard. For example, you can configure the widget to send information to a Topology Graph widget that is on another dashboard, for example dashboard 1. When you select a VM from the data grid, click **Perform Multi-Select Interaction** , click **Dashboard Navigation** and select **Navigate** > **dashboard 1**. It takes you to dashboard 1, where you can observe selected VM and objects related to it. |

## Container Overview Widget Data Grid Options

The data grid provides information on which you can sort and search.

| Option | Description |
|---|---|
| Name | Name of the object |
| Health | Shows information about the health parameter.<br>Status displays the badge of the current health status of an object. You can check the status in a tool tip when you point to the badge.<br>Last 24 Hours displays the statistic of health parameter for last 24 hours. |
| Risk | Shows information about the risk parameter.<br>Status displays the badge of the current risk status of an object. You can check the status in a tool tip when you point to the badge.<br>Last Week displays the statistics of the health parameter for the last week. |
| Efficiency | Shows information about the efficiency parameter.<br>Status displays the badge of the current efficiency status of an object. You can check the status in a tool tip when you point to the badge.<br>Last Week displays statistic of the efficiency parameter for the last week. |

## Container Overview Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Mode | Use **Object** to select an object from the environment to observe.<br>Use **Object Type** to select the type of the objects to observe. |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| **Input Data** | |
| Object | Select objects on which you want to base the widget data.<br>1. Click the **Add New Objects** icon and select objects in the pop-up window. The selected objects appear in a list in this section.<br>While selecting objects, you can use the **Filter** text box to search for objects. You can also expand the **Tag Filter** pane on the left hand side to select one or more object tag values. A list of objects with the selected tag values appears. If you select more than one value for the same tag, you can choose objects that have any of the tags applied. If you select more than one value for different tags, you can choose only the objects that have all the tags applied.<br>2. Optionally, select objects from the list and click the **Remove Selected Objects** icon to remove the selected objects.<br>Click the **Select All** icon to select all the objects in the list.<br>Click the **Clear Selection** icon to clear your selection of objects in the list. |

| Option | Description |
| --- | --- |
| Object Type | Select an object type in your environment on which you want to base the widget data.<br>1. Click the **Add Object Type** icon to search for and add an object type.<br>When you search for object types, you can filter the types in the list by selecting a type from the **Adapter Type** drop-down menu or by using the **Filter** text box.<br>2. Optionally, select the object type from the list and click the **Delete Object Type** icon to remove the selected object type. |

## Current Policy Widget

The Current Policy widget displays the active operational policy that is assigned to your object or object group. VCF Operations uses the assigned policy to analyze your objects, control the data that is collected from those objects, generate alerts when problems occur, and display the results in the dashboards.

### How the Current Policy Widget and Configuration Options Work

You add the Current Policy widget to a dashboard so that you can quickly see which operational policy is applied to an object or object group. To add the widget to a dashboard, you must have access permissions associated with the roles assigned to your user account.

The configuration changes that you make to the widget creates a custom instance of the widget that you use in your dashboard to identify the current policy assigned to an object or object group. When you select an object on the dashboard, the policy applied to the object appears in the Current Policy widget, with an embedded link to the policy details. To display the inherited and local settings for the applied policy, click the link.

### Where You Find the Current Policy Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Current Policy Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options.<br>For example, to view the policy applied to each object that you select in the Object List widget, select **Off** for Self Provider. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| **Input Data** | |
| Object | Search for objects in your environment and select the object on which you are basing the widget data. You can also click the **Add Object** icon and select an object from the object list. You can use the **Filter** text box to refine the object list and the **Tag Filter** pane to select an object based on tag values. |

## Data Collection Results Widget

The Data Collection Result widget shows a list of all supported actions specific for a selected object. The widget retrieves data specific to a selected object actions and uses the action framework to run data collection actions.

### How the Data Collection Results Widget and Configuration Options Work

You can add the Data Collection Results widget to one or more custom dashboards and configure it to display data that is important to different dashboard users. The data that appears in the widget is based on the configured options for each widget instance.

The Data Collection Results widget is a receiver of a resource or metric ID. It can interact with any resource or metric ID that provides widgets such as Object List and Metric Picker. To use the widget, you must have an environment that contains the following items.

- A vCenter Adapter instance
- A VCF Operations for Horizon View Adapter
- A VCF Operations for Horizon View Connection Server

You edit a Data Collection Result widget after you add it to a dashboard. The changes you make to the options create a custom widget to meet the needs of the dashboard users.

## Where You Find the Data Collection Results Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

## Data Collection Results Widget Toolbar Options

On the title bar of the widget, click the **Show Toolbar** icon to access the toolbar options.

| Option | Description |
|---|---|
| Results | Shows all finished and currently running actions for the selected object. |
| Choose Action | Shows a list with all supported actions specific for the selected object. The selected object is a result of widget interactions. |

## Data Collection Results Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget updates only when you open the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |

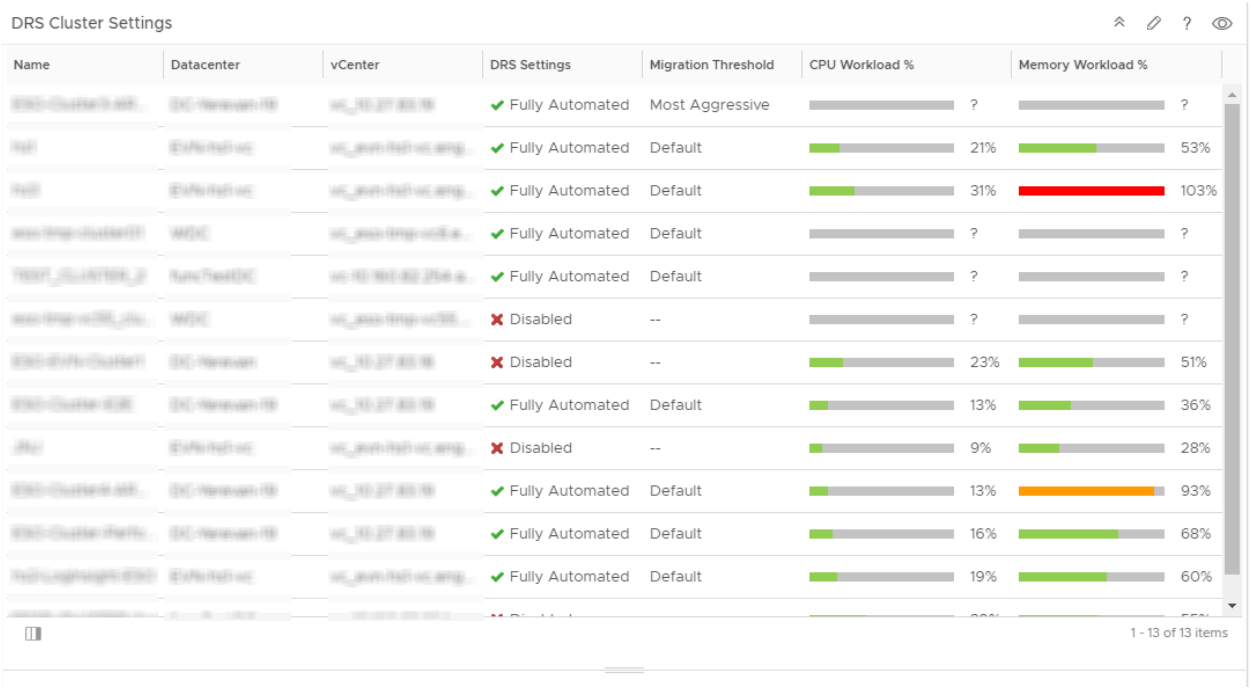
| Option | Description |
|---|---|
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| **Input Data** | |
| Config | Specifies self provider choice and selection of a resource instance. |
| Selected Object | When you select an object, this text box is populated by the object. |
| Start new data collection on interaction change | Indicates whether to start a new data collection action when the object selection changes in the source widget. |
| Objects | List of objects in your environment that you can search or sort by column so that you can locate the object on which you are basing the data that appears in the widget. |
| Defaults | Specifies the default data collection action selected for each object type. |
| Object Types | List of object types in your environment that you can search or sort by column so that you can locate the object type on which you are basing the data that appears in the widget. You can filter the types in the list by selecting a type from the **Adapter Type** drop-down menu or by using the **Filter** text box. |
| Default Data Collection Action | This panel is populated by the object type that you select in the object types list.<br>You can select only one default data collection action for an object type. |

## DRS Cluster Settings Widget

The DRS Cluster Settings widget displays the workload of the available clusters and the associated hosts. You can change the Distributed Resource Scheduler (DRS) automation rules for each cluster.

### How the DRS Cluster Settings Widget and Configuration Options Work

You can view CPU workload and memory workload percentages for each of the clusters. You can view CPU workload and memory workload percentages for each host in the cluster by selecting a cluster in the data grid. The details are displayed in the data grid below. You can set the level of DRS automation and the migration threshold by selecting a cluster and clicking **Cluster Actions** > **Set DRS Automation**.




You edit a DRS Cluster Settings widget after you add it to a dashboard. To configure the widget, click the edit icon at the upper-right corner of the widget window. You can add the DRS Cluster Settings widget to one or more custom dashboards and configure it to display data that is important to different dashboard users. The data that appears in the widget is based on the configured options for each widget instance.

The DRS Cluster Settings widget appears on the dashboard named vSphere DRS Cluster Settings, which is provided with VCF Operations.

## Where You Find the DRS Cluster Settings Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

## DRS Cluster Settings Widget Toolbar Options

On the title bar of the widget, click the **Show Toolbar** icon to access the toolbar options.

| Option | Description |
|---|---|
| Cluster Actions | Limits the list to actions that match the cluster you select. |
| Show | The drop-down menu displays the parent vCenter instances where the clusters reside. You can also view the data centers under each parent vCenter instance. Select a parent vCenter to view the workload of the available clusters in the data grid.<br>The default setting displays the clusters across all vCenters. |

| Option | Description |
|---|---|
| Filter | Filters the data grid by name, data center, vCenter, DRS settings, and migration threshold. |

### DRS Cluster Settings Widget Data Grid Options

The data grid provides information on which you can sort and search.

| Option | Description |
|---|---|
| Name | Displays the names of the clusters in the selected parent vCenter instance. |
| Datacenter | Displays the data centers that belong to each cluster. |
| vCenter | Displays the parent vCenter instance where the cluster resides. |
| DRS Settings | Displays the level of DRS automation for the cluster.<br>To change the level of DRS automation for the cluster, select **Cluster Actions** > **Set DRS Automation** from the toolbar. You can change the automation level by selecting an option from the drop-down menu in the Automation Level column. |
| Migration Threshold | Recommendations for the migration level of virtual machines. Migration thresholds are based on DRS priority levels, and are computed based on the workload imbalance metric for the cluster. |
| CPU Workload % | Displays the percentage of CPU in GHz available on the cluster. |
| Memory Workload % | Displays the percentage of memory in GB available on the cluster. |

### DRS Cluster Settings Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |

## Efficiency Widget

The efficiency widget is the status of the efficiency-related alerts for the objects it is configured to monitor. Efficiency alerts in VCF Operations usually indicate that you can reclaim resources. You can create one or more efficiency widgets for objects that you add to your custom dashboards.

### How the Efficiency Widget and Configuration Options Work

You can add the efficiency widget to one or more custom dashboards and configure it to display data that is important to the dashboard users.

The state of the badge is based on your alert definitions. Click the badge to see the **Summary** tab for objects or groups configured in the widget. From the **Summary** tab, you can begin determining what caused the current state. If the widget is configured for an object that has descendants, you should also check the state of descendants. Child objects might have alerts that do not impact the parent.

If the **Badge Mode** configuration option is set to **Off**, the badge and a chart appears. The type of chart depends on the object that the widget is configured to monitor.

- A population criticality chart displays the percentage of group members with critical, immediate, and warning efficiency alerts generated over time, if the monitored object is a group.
- A trend line displays the efficiency status of the monitored object over time if the object does not provide its resources to any other object, or where no other object depends on the monitored object's resources. For example, if the monitored object is a virtual machine or a distributed switch.
- A pie chart displays the reclaimable, stress, and optimal percentages for the virtual machines that are descendants of the monitored object for all other object types. You use the chart to identify objects in your environment from which you can reclaim resources. For example, if the object is a host or datastore.

If the **Badge Mode** is set to **On**, only the badge appears.

Edit an efficiency widget after you add it to a dashboard. The changes you make to the options create a custom widget that provides information about an individual object, a custom group of objects, or all the objects in your environment.

### Where You Find the Efficiency Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

## Efficiency Widget Display Options

The Efficiency widget displays an efficiency badge. The widget also displays an efficiency trend when not in badge mode.

| Option | Description |
|---|---|
| Efficiency Badge | Status of the objects configured for this instance of the widget.<br>Click the badge to open the **Alerts** tab for the object that provides data to the widget. |
| Efficiency Trend | Displays a chart, depending on the selected or configured object. The charts vary, depending on whether the monitored object is a group, a descendent object, or an object that provides resources to other objects. The chart appears only if the **Badge Mode** configuration option is off. If the **Badge Mode** is on, only the badge appears. |

## Efficiency Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |

| Option | Description |
|---|---|
| Badge Mode | Determines whether the widget displays only the badge, or the badge and a weather map or trend chart.<br>Select one of the following options:<br>• On. Only the badge appears in the widget.<br>• Off. The badge and a chart appear in the widget. The chart provides additional information about the state of the object. |
| **Input Data** | |
| Object | Search for objects in your environment and select the object on which you are basing the widget data. You can also click the **Add Object** icon and select an object from the object list. You can use the **Filter** text box to refine the object list and the **Tag Filter** pane to select an object based on tag values. |

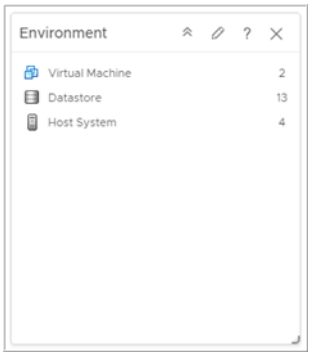
## Environment Widget

The Environment widget displays the resources which collects data. You can create one or more lists in VCF Operations for the resources that you add to your custom dashboards.

### How the Environment Widget and Configuration Options Work

The Environment widget lists the number of resources by object or groups them by object type. You can add the Environment widget to one or more custom dashboards and configure it to display data that is important to different dashboard users. The data that appears in the widget is based on the configured options for each widget instance.

You edit an Environment widget after you add it to a dashboard. The changes you make to the options help create a custom widget to meet the needs of the dashboard users.




### Where You Find the Environment Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Environment Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| **Input Data** | |
| Object | Search for objects in your environment and select the object on which you are basing the widget data. You can also click the **Add Object** icon and select an object from the object list. You can use the **Filter** text box to refine the object list and the **Tag Filter** pane to select an object based on tag values. |

## Environment Overview Widget

The Environment Overview widget displays the health, risk, and efficiency of resources for a given object from the managed inventory.

### How the Environment Overview Widget and Configuration Options Work

You can add the Environment Overview widget to one or more custom dashboards.

The widget displays data for objects from one or several types. The data that the widget displays depends on the object type and category that you selected when you configured the widget.

The objects in the widget are ordered by object type.

The parameters for the health, risk, and efficiency of an object appear in a tool tip when you point to the object.

When you double-click an object on the Environment Overview widget, you can view detailed information for the object.

To use the Environment Overview widget, you must add it to the dashboard and configure the data that appears in the widget. You must select at least one badge and an object. Additionally, you can select an object type.

The Environment Overview widget has basic and advanced configuration options. The basic configuration options are activated by default.

To use all features of the Environment Overview widget, you must change the default configuration of the widget. Log in to the VCF Operations machine and set `skittlesCustomMetricAllowed` to `true` in the `web.properties` file. The `web.properties` file is located in the `/usr/lib/vmware-vcops/user/conf/web` folder. The change is propagated after you use the `service vmware-vcops-web restart` command to restart the UI.

You must use the **Badge** tab to select the badge parameters that the widget shows for each object. You must use the **Config** tab to select an object or object type. To observe a concrete object from the inventory, you can use the **Basic** option. To observe a group of objects or objects from different types, you must use the **Advanced** option.

### Where You Find the Environment Overview Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Environment Overview Widget Toolbar Options

On the title bar of the widget, click the **Show Toolbar** icon to access the toolbar options.

The toolbar contains icons that you can use to get more information about badges.

| Option | Description |
|---|---|
| Badge | You can select a Health, Risk, or Efficiency badge for objects that appear in the widget. The tool tip of a badge shows the standard name of the badge. |
| Status | You can filter objects based on their badge status and their state. |
| Sort | You can sort objects by letter or by number. |

### Environment Overview Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |

| Option | Description |
| --- | --- |
| Selected Object | Object that is the basis for the widget data.<br>To populate the text box, select **Config** > **Basic** and select an object from the list. |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| Badge | Defines a parameter to observe. You can select or deselect Health, Risk, and Efficiency parameters using check boxes. Default configuration of the widget selects all badges.<br>Select at least one badge parameter. |
| Config | **Basic**<br>List of objects in your environment that you can search or sort by column so that you can locate the object on which you are basing the data that appears in the widget. |

| Option | Description |
| --- | --- |
| | **Advanced**<br>You can use Object Types to select a type of the objects to observe information about health, risk, and efficiency. Double-click the object type to select it.<br>Use the **Adapter Type** drop-down menu to filter the objects types based on an adapter.<br>You can use the **Use vSphere Default** button to observe the main vSphere object types.<br>To remove an object type from the list, click **Remove Selected** next to **Use vSphere Default**.<br>You can use the **Object Type Categories** menu to select a group or groups of object types to observe.<br>You can use the Object tree to select an object to filter the displayed objects. For example, to observe a datastore of a VM, double-click **Datastore** from the **Object Types** menu to select it. Click the datastore when it is in the list of object types, and find the VM in the object tree and select it. To return to your previous configuration of the widget, click **Datastore** from the list of object types and click **Deselect All** in the object tree window.<br>The metrics tree and badge data grids are available configuration options only if the default configuration of the widget is changed. To use these configuration options, log in to the VCF Operations machine and set `skittlesCustomMetricAllowed` to `true` in the `web.properties` file. The `web.properties` file is located in the `/usr/lib/vmware-vcops/user/conf/web` folder. |

## Environment Status Widget

The Environment Status widget displays the statistics for the overall monitored environment.

### How the Environment Status Widget and Configuration Options Work

You customize the output of the widget by choosing a category such as Objects, Metrics, Applications, Alerts, Analytics, and Users. You can filter the data by using the tags tree from **Select which tags to filter** in the configuration window.

You edit an environment status widget after you add it to a dashboard. To configure the widget, click the pencil at the right corner of the widget window. You must select at least one type of information from **OBJECTS**, **METRICS**, **APPLICATIONS**, **ALERTS**, **ANALYTICS**, **USERS** categories for the widget to display. By default, the widget displays statistics information about all objects in the inventory. You can use the Select which tags to filter option to filter the information. The widget can interact with other widgets in the dashboard, taking data from them and displaying statistics . For example, you can have a Object List widget , which is the source of the data and an Environment Status widget, which is the destination. If you select objects and perform a multiselection interaction from the Object List widget, the Environment Status widget results are updated based on the selections you made in the Object List.

### Where You Find the Environment Status Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a

widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Environment Status Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

The **Input Transformation** section provides options to transform the input for the widget.

The **Output Data** section provides options to select object types on which you are basing the widget data.

The **Output Filter** section provides options to restrict the widget data based on the selected filter criteria.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard.<br>The widget is also updated when it is in interaction mode. For example, when an item is selected in the provider widget, the content of the Environment Status widgets is refreshed. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| **Input Data** | |

| Option | Description |
|---|---|
| Objects | Select objects on which you want to base the widget data.<br>1. Click the **Add New Objects** icon and select objects in the pop-up window. The selected objects appear in a list in this section.<br>While selecting objects, you can use the **Filter** text box to search for objects. You can also expand the **Tag Filter** pane on the left hand side to select one or more object tag values. A list of objects with the selected tag values appears. If you select more than one value for the same tag, you can choose objects that have any of the tags applied. If you select more than one value for different tags, you can choose only the objects that have all the tags applied.<br>2. Optionally, select objects from the list and click the **Remove Selected Objects** icon to remove the selected objects.<br>Click the **Select All** icon to select all the objects in the list.<br>Click the **Clear Selection** icon to clear your selection of objects in the list. |
| All | If you select this option, the widget data is based on all the objects in your environment. The following sections provide options to refine the objects for the widget data. |
| **Input Transformation** | |
| Relationship | Transform the input for the widget based on the relationship of the objects. For example, if you select the **Children** check box and a **Depth** of **1**, the child objects are the transformed inputs for the widget. |
| **Output Data** | |
| Objects | The widget shows summarized information about the objects in your environment. You can filter the information that appears in self provider mode when you select an object from Select which tag to filter. You can select what type of information to include in the summary of resources. For example, if you select **Adapter Types** > **Container** from Select which tag to filter and click **Objects** and **Objects Collecting** , the widget displays the number of containers and collecting containers. |
| Metrics | The widget shows summarized information about available metrics. You can filter the information that appears in self provider mode when you select an object from Select which tag to filter. You can select what type of information to include in the summary of metrics. |
| Applications | The widget shows summarized information about available applications. You can filter the information that appears in self provider mode when you select an object from Select which tag to filter. You can select what type of information to include in the summary of applications. |
| Alerts | The widget shows summarized information about alerts in your environment. You can filter the information that appears in self provider mode when you select an object from Select which tag to filter. You can select what type of information to include in the summary of alerts. |

| Option | Description |
|---|---|
| Analytics | The widget shows summarized information about the analytics plug-ins. You can filter the information that appears in self provider mode when you select an object from Select which tag to filter. You can select what type of information to include in the summary of analytics. |
| Users | The widget shows the number of users defined in VCF Operations. Select **Administration** > **Access Control** > **User Accounts**. |
| **Output Filter** | |
| Basic | Pick tags to refine the widget data. The widget data is based on the objects that have the picked tags applied. If you pick more than one value for the same tag, the widget includes objects that have any of the tags applied. If you pick more than one value for different tags, the widget includes only the objects that have all the tags applied.<br>If the objects have an input transformation applied, you select tag values for the transformed objects. |
| Advanced | Refine the widget data further based on the filter criteria for object types. The widget data is based on the objects for the filtered object types.<br>If the objects have a tag filter applied in the **Basic** subsection, you define filter criteria for the object types of the objects with tag filter applied. If the objects with tag filter applied do not belong to any of the object types in this filter criteria, the widget skips this filter and includes all the objects with tag filter applied.<br>If the objects have an input transformation applied, you define filter criteria for the object types of the transformed objects.<br>1. In the first drop-down menu, select an object type.<br>2. In the second drop-down menu, select the option based on which you want to define the filter criteria. For example, if you select **Metrics** for the **Datacenter** object type, you can define a filter criteria based on the value of a specific metric for data centers.<br>3. In the drop-down menus and text boxes that appear, select or enter values to filter the objects.<br>4. To add more filter criteria, click **Add**.<br>5. To add another filter criteria set, click **Add another criteria set**. |

## Faults Widget

The Faults widget displays detailed information about faults experienced by an object

The Faults widget configuration options are used to customize each instance of the widget that you add to your dashboards.

### Where You Find the Faults Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you

want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Faults Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| **Input Data** | |
| Object | Search for objects in your environment and select the object on which you are basing the widget data. You can also click the **Add Object** icon and select an object from the object list. You can use the **Filter** text box to refine the object list and the **Tag Filter** pane to select an object based on tag values. |

## Forensics Widget

The Forensics widget shows how often a metric has a particular value as a percentage of all values, within a given time period. It can also compare percentages for two time periods.

## How the Forensics Widget and Configuration Options Work

You can add the Forensics widget to one or more custom dashboards and configure it to display data that is important to different dashboard users. The data that appears in the widget is based on the configured options for each widget instance.

You edit the Forensics widget after you add it to a dashboard. The changes you make to the options create a custom widget to meet the needs of the dashboard users.

## Where you Find the Forensics Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

## Forensics Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |

| Option | Description |
|---|---|
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| Percentile | Indicates how much data is above or below the specific value. For example, it indicates that 90% of the data is more than 4 when a vertical line occurs on the value 4. |
| **Input Data** | |
| | Select metrics on which you want to base the widget data. You can select an object and pick its metrics.<br>1. Click the **Add New Metrics** icon to add metrics for the widget data. Select an object to view its metric tree and pick metrics for the object. The picked metrics appear in a list in this section.<br>The metric tree shows common metrics for several objects when you click the **Show common metrics** icon.<br>While selecting objects for which you want to pick metrics, you can use the **Filter** text box to search for objects. You can also expand the **Tag Filter** pane on the left hand side to select one or more object tag values. A list of objects with the selected tag values appears. If you select more than one value for the same tag, you can choose objects that have any of the tags applied. If you select more than one value for different tags, you can choose only the objects that have all the tags applied.<br>2. Optionally, select metrics from the list and click the **Remove Selected Metrics** icon to remove the selected metrics.<br>Click the **Select All** icon to select all the metrics in the list.<br>Click the **Clear Selection** icon to clear your selection of metrics in the list. |

## Heatmap Widget

The Heatmap widget contains graphical indicators that display the current value of two selected attributes of objects of tag values that you select. In most cases, you can select only from internally generated attributes that describe the general operation of the objects, such as health or the active anomaly count. When you select a single object, you can select any metric for that object.

### How the Heatmap Widget and Configuration Options Work

You can add the Heatmap widget to one or more custom dashboards and configure it to display data that is important to the dashboard users.

The Heatmap widget has a General mode and an Instance mode. The General mode shows a colored rectangle for each selected resource. In the Instance mode, each rectangle represents a single instance of the selected metric for an object.

You can click a color or the size metric box in the bottom of the Heatmap widget to filter the display of cells in the widget. You can click and drag the color filter to select a range of colors. The Heatmap widget displays cells that match the range of colors.

When you point to a rectangle for an object, the widget shows the resource name, group-by values, the current values of the two tracked attributes, virtual machine details, the metric name, and the value of the color. Click **Show Sparkline** to view the value.

You edit a Heatmap widget after you add it to a dashboard. The changes you make to the options create a custom widget that provides information about an individual object, a custom group of objects, or all the objects in your environment.

### Where You Find the Heatmap Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Heatmap Widget Toolbar Options

On the title bar of the widget, click the **Show Toolbar** icon to access the toolbar options.

| Option | Description |
|---|---|
| Dashboard Navigation | Actions you can run on the selected alert.<br>For example, you use the option to open a vCenter, data center, virtual machine, or in the vSphere Web Client, allowing you to directly modify an object for which an alert was generated and fix any problems. |
| Group Zoom | You can roll-up non-significant resources with similar characteristics into groups to obtain only the relevant data among the thousands of resources in the system. The roll-up method improves performance and decreases the memory usage. The roll-up box encompasses the average color and the sum of the sizes of all the resources. You can view all the resources by zooming in the roll-up box. |
| Show/Hide Text | Show or hide the cell name on the heatmap rectangle. |
| Show Details | If you configure the Heatmap widget as a provider to another widget, such as the Metric Chart widget , you can double-click a rectangle to select that object for the widget. If the widget is in Metric mode, double-clicking a rectangle selects the resource associated with the metric and provides that resource to the receiving widget. Optionally, you can select a cell from the heatmap and click the **Show Details** icon to see details about the cell. |
| Reset Interaction | Returns the widget to its initial configured state and undoes any interactions selected in a providing widget. |
| Reset Zoom | Resets the heatmap display to fit in the available space. |
| Heatmap Configuration Drop-down | Select from a list of predefined heatmaps. |

### Heatmap Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

The **Input Transformation** section provides options to transform the input for the widget.

The **Output Data** section provides options to select object types on which you are basing the widget data.

The **Output Filter** section provides options to restrict the widget data based on the selected filter criteria.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| **Input Data** | |
| Objects | Select objects on which you want to base the widget data.<br>1. Click the **Add New Objects** icon and select objects in the pop-up window. The selected objects appear in a list in this section.<br>While selecting objects, you can use the **Filter** text box to search for objects. You can also expand the **Tag Filter** pane on the left hand side to select one or more object tag values. A list of objects with the selected tag values appears. If you select more than one value for the same tag, you can choose objects that have any of the tags applied. If you select more than one value for different tags, you can choose only the objects that have all the tags applied.<br>2. Optionally, select objects from the list and click the **Remove Selected Objects** icon to remove the selected objects.<br>Click the **Select All** icon to select all the objects in the list.<br>Click the **Clear Selection** icon to clear your selection of objects in the list. |
| All | If you select this option, the widget data is based on all the objects in your environment. The following sections provide options to refine the objects for the widget data. |
| **Input Transformation** | |
| Relationship | Transform the input for the widget based on the relationship of the objects. For example, if you select the **Children** check box and a **Depth** of **1**, the child objects are the transformed inputs for the widget. |

| Option | Description |
|---|---|
| **Output Data** | |
| Configurations | List of saved heatmap configuration options. You can create a configuration and save it in the list. From the options on the right, you can also delete, clone, and reorder the configurations. |
| Name | Name of the widget. |
| Group by | First-level grouping of the objects in the heatmap. |
| Then by | Second-level grouping of the objects in the heatmap. |
| Relational Grouping | After you select the Group by and Then by objects, select the **Relational Grouping** check box to reorganize the grouping of the objects, and to relate the objects selected in the Group by text box with the objects selected in the Then by text box. |
| Mode | **General mode** The widget shows a colored rectangle for each selected resource. The size of the rectangle indicates the value of one selected attribute. The color of the rectangle indicates the value of another selected attribute.<br>**Instance mode** Each rectangle represents a single instance of the selected metric for a resource. A resource can have multiple instances of the same metric. The rectangles are all the same size. The color of the rectangles varies based on the instance value. You can use instance mode only if you select a single resource kind. |
| Object Type | Object that is the basis for the widget data. |
| Size by | An attribute to set the size of the rectangle for each resource. Resources that have higher values for the Size By attribute have larger areas of the widget display. You can also select fixed-size rectangles. In most cases, the attribute lists include only metrics that VCF Operations generates. If you select a resource kind, the list shows all the attributes that are defined for the resource kind. |
| Color by | An attribute to set the color of the rectangle for each resource. |
| Solid Coloring | Select this option to use solid colors instead of a color gradient. By default, the widget assigns red color for high value, brown color for intermediate value and green color for low value. Click the color box to set a different color for the values. You can add up to seven color thresholds by clicking color range. |
| Color | Shows the color range for high, intermediate and low values. You can set each color and type minimum and maximum color values in the **Min Value** and **Max Value** text boxes. By default, green indicates a low value and red indicates the high end of the value range. You can change the high and low values to any color and set the color to use for the midpoint of the range. You can also set the values to use for either end of the color range, or let VCF Operations define the colors based on the range of values for the attribute.<br>If you leave the text boxes blank, VCF Operations maps the highest and lowest values for the **Color By** metric to the end colors. If you set a minimum or maximum value, any metric at or beyond that value appears in the end color. |
| **Output Filter** | |

| Option | Description |
|---|---|
| Basic | Pick tags to refine the widget data. The widget data is based on the objects that have the picked tags applied. If you pick more than one value for the same tag, the widget includes objects that have any of the tags applied. If you pick more than one value for different tags, the widget includes only the objects that have all the tags applied.<br>If the objects have an input transformation applied, you select tag values for the transformed objects. |
| Advanced | Refine the widget data further based on the filter criteria for object types. The widget data is based on the objects for the filtered object types.<br>If the objects have a tag filter applied in the **Basic** subsection, you define filter criteria for the object types of the objects with tag filter applied. If the objects with tag filter applied do not belong to any of the object types in this filter criteria, the widget skips this filter and includes all the objects with tag filter applied.<br>If the objects have an input transformation applied, you define filter criteria for the object types of the transformed objects.<br>1. In the first drop-down menu, select an object type.<br>2. In the second drop-down menu, select the option based on which you want to define the filter criteria. For example, if you select **Metrics** for the **Datacenter** object type, you can define a filter criteria based on the value of a specific metric for data centers.<br>3. In the drop-down menus and text boxes that appear, select or enter values to filter the objects.<br>4. To add more filter criteria, click **Add**.<br>5. To add another filter criteria set, click **Add another criteria set**. |

## Health Widget

The Health widget is the status of the health-related alerts for the objects it is configured to monitor in VCF Operations. Health alerts usually require immediate attention. You can create one or more health widgets for different objects that you add to your custom dashboards.

### How the Health Widget and Configuration Options Work

You can add the Health widget to one or more custom dashboards and configure it to display data that is important to the dashboard users. The information that it displays depends on how the widget is configured.

The state of the badge is based on your alert definitions. Click the badge to see the **Summary** tab for objects or groups configured in the widget. From the **Summary** tab, you can begin determining what caused the current state. If the widget is configured for an object that has descendants, you should also check the state of descendants. Child objects might have alerts that do not impact the parent.

If the **Badge Mode** configuration option is set to **Off**, the badge and a chart appears. The type of chart depends on the object that the widget is configured to monitor.

- A trend line displays the health status of the monitored object if the object does not provide its resources to any other object. For example, if the monitored object is a virtual machine or a distributed switch.
- A weather map displays the health of the ancestor and descendant objects of the monitored object for all other object types. For example, if the monitored object is a host that provides CPU and memory to a virtual machine.

If the **Badge Mode** is set to **On**, only the badge appears.

You edit a Health widget after you add it to a dashboard. The changes you make to the options create a custom widget that provides information about an individual object, a custom group of objects, or all the objects in your environment.

## Where You Find the Health Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

## Heath Widget Display Options

The Health widget displays a health badge. The widget also displays a health trend when not in badge mode.

| Option | Description |
|---|---|
| Health Badge | Status of the objects configured for this instance of the widget.<br>Click the badge to open the **Alerts** tab for the object that provides data to the widget.<br>If the **Badge Mode** option is off, a health weather map or trend chart appears for the object. Whether the map or chart appears depends on the object type. The health weather map displays tool tips for up to1000 objects. |
| Health Trend | Displays a chart, depending on the selected or configured object. The charts vary, depending on whether the monitored object is a group, a descendent object, or an object that provides resources to other objects. The chart appears only if the **Badge Mode** configuration option is off. If the **Badge Mode** is on, only the badge appears. |

## Heath Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |

| Option | Description |
|---|---|
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| Badge Mode | Determines whether the widget displays only the badge, or the badge and a weather map or trend chart.<br>Select one of the following options:<br>• On. Only the badge appears in the widget.<br>• Off. The badge and a chart appear in the widget. The chart provides additional information about the state of the object. |
| **Input Data** | |
| Object | Search for objects in your environment and select the object on which you are basing the widget data. You can also click the **Add Object** icon and select an object from the object list. You can use the **Filter** text box to refine the object list and the **Tag Filter** pane to select an object based on tag values. |

## Health Chart Widget

The Health Chart widget displays Health, Risk, Efficiency, or custom metric charts for selected objects. You use the widget to compare the status of similar objects based on the same value or name.

### How the Health Chart Widget and Configuration Options Work

You can add the Health Chart widget to one or more custom dashboards and configure it to display data that is important to the dashboard users. The information that it displays depends on how the widget is configured.

If the widget is configured to display Health, Risk, or Efficiency, the chart values are based on the generated alerts for the selected alert type for the selected objects.

If the widget is configured to display custom metrics, chart values are based on the metric value for the configured time period.

You edit the Health Chart widget after you add it to the dashboard. The changes you make to the options create a custom widget with the selected charts.

The charts are based either on Health, Risk, or Efficiency alert status, or you can base them on a selected metric. You can include a single object, multiple objects, or all objects of a selected type.

To view the value of the object at a particular time, point your cursor over the chart. A date range and metric value tool tip appear.

A context drop-down menu for each chart can be accessed at the top-right corner after the last metric value.

For each chart, you can view the minimum, maximum, and last metric values. The values are displayed at the top-right corner of each chart. Each of the values is preceded by an appropriate icon of the same color as the state of the metric value.

If there is not enough space to view the metric values, a blue information icon is displayed. Point your cursor over the icon to view the metric value details.

### Where You Find the Health Chart Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Health Chart Widget Toolbar Options

On the title bar of the widget, click the **Show Toolbar** icon to access the toolbar options.

| Option | Description |
|---|---|
| Date Controls | Use the date selector to limit the data that appears in each chart to the time period you are examining.<br>Select **Dashboard Time** to activate the dashboard time panel. The option chosen in the dashboard time panel is effective. The default time is 6 hours.<br>**Dashboard Time** is the default option. |

### Health Chart Widget Graph Selector Options

The graph selector options determine how individual data appears in the graph.

| Option | Description |
|---|---|
| Close | Deletes the chart. |
| Save a snapshot | Creates a PNG file of the current chart. The image is the size that appears on your screen. You can retrieve the file in your browser's download folder. |
| Save a full screen snapshot | Downloads the current graph image as a full-page PNG file, which you can display or save. You can retrieve the file in your browser's download folder. |
| Download comma-separated data | Creates a CSV file that includes the data in the current chart. You can retrieve the file in your browser's download folder. |
| Units | Select the units in which the widget displays data. This option is visible when you select a custom source of data in the widget configuration. |

### Health Chart Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

The **Input Transformation** section provides options to transform the input for the widget.

The **Output Filter** section provides options to restrict the widget data based on the selected filter criteria.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| Order By | Determines how the object charts appear in the widget.<br>You can order them based on value or name, and in ascending or descending order. |
| Chart Height | Controls the height of all charts. Choose from three possible choices - Small, Medium, Large. Default is Medium. |
| Pagination number | Number of charts that appears on a page.<br>If you prefer scrolling through the charts, select a higher number. If you prefer to page through the results, select a lower number. |
| Auto Select First Row | Determines whether to start with the first row of data. |
| Metric | Determines the source of the data.<br>• Health, Risk, or Efficiency. The displayed charts are based on one of these alert badges.<br>• Custom. The displayed charts are based on the selected metric and use either alert symptom state colors or the selected custom color. You can select a unit for the custom metric from the drop-down menu or choose to allow the widget to automatically pick a unit.<br>If you apply custom colors, enter the value in each box that is the highest or lowest value that should be that color. You can select a unit for the metric. |
| Metric Unit | Select a unit for the custom metric. |

| Option | Description |
| --- | --- |
| Show | Select one or more of the following items to display in the widget:<br>• Select **Object Name** to display the name of the object in the widget.<br>• Select **Metric Name** to display the name of the metric in the widget. |
| **Input Data** | |
| Objects | Select objects on which you want to base the widget data.<br>1. Click the **Add New Objects** icon and select objects in the pop-up window. The selected objects appear in a list in this section.<br>While selecting objects, you can use the **Filter** text box to search for objects. You can also expand the **Tag Filter** pane on the left hand side to select one or more object tag values. A list of objects with the selected tag values appears. If you select more than one value for the same tag, you can choose objects that have any of the tags applied. If you select more than one value for different tags, you can choose only the objects that have all the tags applied.<br>2. Optionally, select objects from the list and click the **Remove Selected Objects** icon to remove the selected objects.<br>Click the **Select All** icon to select all the objects in the list.<br>Click the **Clear Selection** icon to clear your selection of objects in the list. |
| All | If you select this option, the widget data is based on all the objects in your environment. The following sections provide options to refine the objects for the widget data. |
| **Input Transformation** | |
| Relationship | Transform the input for the widget based on the relationship of the objects. For example, if you select the **Children** check box and a **Depth** of **1**, the child objects are the transformed inputs for the widget. |
| **Output Filter** | |
| Basic | Pick tags to refine the widget data. The widget data is based on the objects that have the picked tags applied. If you pick more than one value for the same tag, the widget includes objects that have any of the tags applied. If you pick more than one value for different tags, the widget includes only the objects that have all the tags applied.<br>If the objects have an input transformation applied, you select tag values for the transformed objects. |

| Option | Description |
| --- | --- |
| Advanced | Refine the widget data further based on the filter criteria for object types. The widget data is based on the objects for the filtered object types.<br>If the objects have a tag filter applied in the **Basic** subsection, you define filter criteria for the object types of the objects with tag filter applied. If the objects with tag filter applied do not belong to any of the object types in this filter criteria, the widget skips this filter and includes all the objects with tag filter applied.<br>If the objects have an input transformation applied, you define filter criteria for the object types of the transformed objects.<br>1. In the first drop-down menu, select an object type.<br>2. In the second drop-down menu, select the option based on which you want to define the filter criteria. For example, if you select **Metrics** for the **Datacenter** object type, you can define a filter criteria based on the value of a specific metric for data centers.<br>3. In the drop-down menus and text boxes that appear, select or enter values to filter the objects.<br>4. To add more filter criteria, click **Add**.<br>5. To add another filter criteria set, click **Add another criteria set**. |

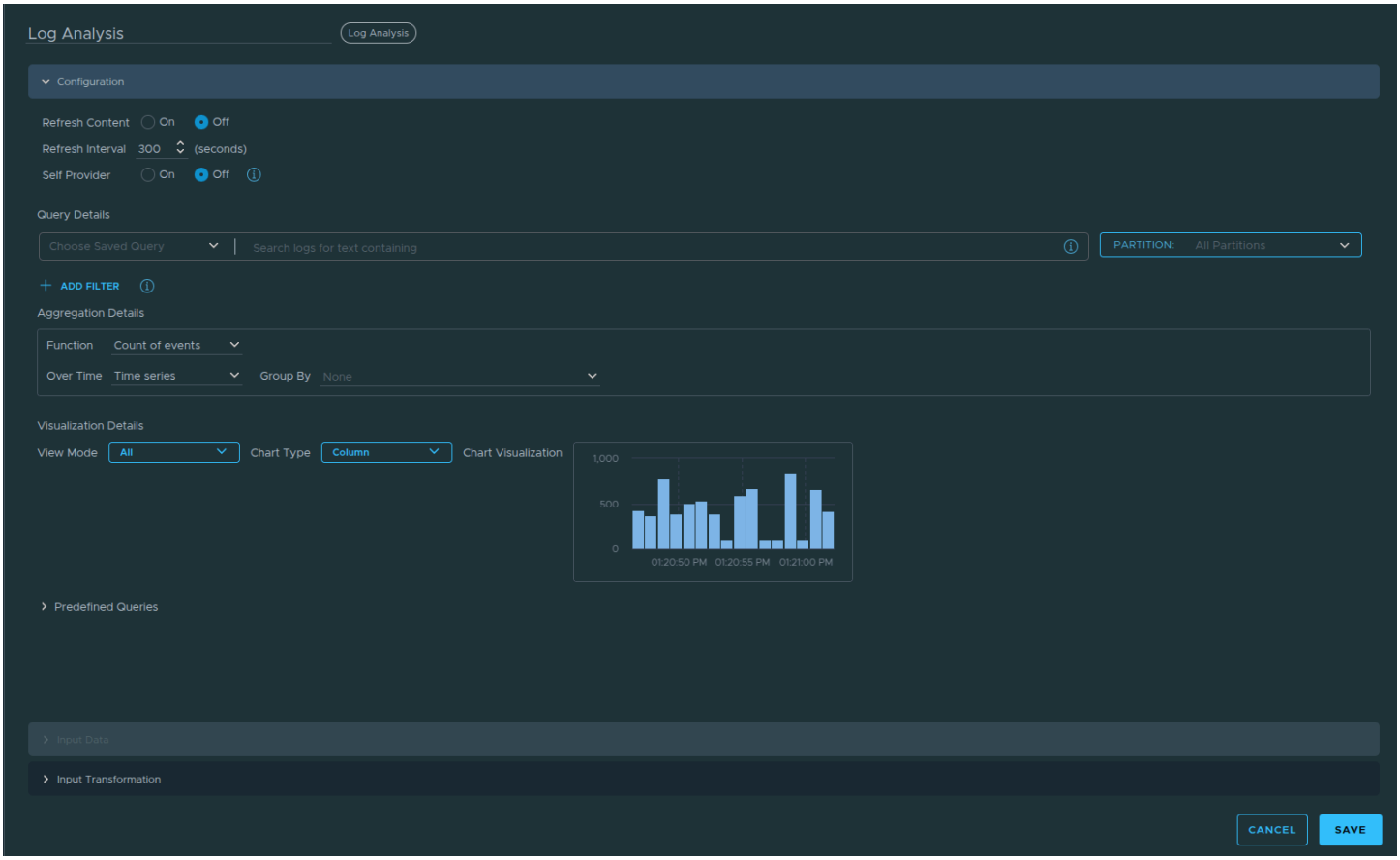
## Log Analysis Widget

The Log Analysis widget displays the logs after you have integrated VCF Operations and Log management. You can filter logs based on chart and message queries.

The following screenshot shows the log analysis widget.

**Figure 473: Log management widget**

Log management widget in VCF Operations console.




You can view, filter, and search the logs that are displayed.

## Where You Find the Log Analysis Widget

Create a new dashboard with the Log Analysis widget. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All**to see your configured dashboards. Cick **Create**. The widget panel in the dashboard page shows a list of available widgets. If you cannot find the Log Analysis widget, click **SHOW MORE**. Drag the Log Analysis widget to the dashboard workspace in the upper panel. In the title bar of the widget, click the **Edit Widget** icon to configure the widget.

## Log Analysis Widget Configuration Options

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |

| Option | Description |
|---|---|
| Self Provider | • On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| Query Details | • Choose saved chart and message queries from the drop down list. You can create and save queries from the **Infrastructure Operations** > **Analyze** page.<br>• Search for logs using keywords.<br>• Select the index partition from the drop-down list. |
| Add Filter | Click **ADD FILTER** to add more filters. |
| Aggregation Details | Customize the visual representation of events by using aggregation queries. |
| Visualization Details | Change the view mode and chart type from the drop-down lists. A visualization displays how the chart will look. The aggregation details determines what types of visualization you can choose. You can also choose to display event stream, event types or event trends. |
| Predefined Queries | Click the **+** icon to add a list of predefined queries. |
| **Input Data** | |
| Object | Search for objects in your environment and select the object on which you are basing the widget data. You can also click the **Add Object** icon and select an object from the object list. You can use the **Filter** text box to refine the object list and the **Tag Filter** pane to select an object based on tag values. |

**Note:**
Do not use the log analysis form widget for reporting. The widget requires user inputs to show results. The widget is interactive and performs cross-product queries.

## Mashup Chart Widget

The Mashup Chart widget shows disparate pieces of information for a resource. It shows a health chart and metric graphs for key performance indicators (KPIs).

### How the Mashup Chart Widget and Configuration Options Work

The Mashup Chart widget contains charts that show different aspects of the behavior of a selected resource. By default, the charts show data for the past six hours.

The Mashup Chart widget contains the following charts.

- A Health chart for the object, which can include each alert for the specified time period. Click an alert to see more information, or double-click an alert to open the Alert Summary page.
- Metric graphs for any or all the KPIs for any objects listed as a root cause object. For an application, this chart shows the application and any tiers that contain root causes. You can select the KPI to include by selecting **Chart Controls** > **KPIs** on the widget toolbar. Any shared area on a graph indicates that the KPI violated its threshold during that time period.

The metric graphs reflect up to five levels of resources, including the selected object and four child levels.

You edit a Mashup Chart widget after you add it to a dashboard. The changes you make to the options create a custom widget to meet the needs of the dashboard users.

## Where You Find the Mashup Chart Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

## Mashup Chart Widget Toolbar Options

On the title bar of the widget, click the **Show Toolbar** icon to access the toolbar options.

The toolbar contains icons that you can use to change the view.

| Option | Description |
|---|---|
| Filters | Filter data based on criticality, status, and alert type. |
| Event Filters | Filter based on the type of event such as, change, notification, and fault. |
| Date Controls | Use the date selector to limit the data that appears in each chart to the time period you are examining.<br>Select **Dashboard Time** to activate the dashboard time panel. The option chosen in the dashboard time panel is effective. The default time is 6 hours.<br>**Dashboard Time** is the default option. |
| Dashboard Navigation | You can navigate to another dashboard when the object under consideration is also available in the dashboard to which you navigate. |

## Mashup Chart Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |

| Option | Description |
|---|---|
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| **Input Data** | |
| Object | Search for objects in your environment and select the object on which you are basing the widget data. You can also click the **Add Object** icon and select an object from the object list. You can use the **Filter** text box to refine the object list and the **Tag Filter** pane to select an object based on tag values. |

## Metric Chart Widget

You can use the Metric Chart widget to monitor the workload of your objects over time. The widget displays data based on the metrics that you select.

### How the Metric Chart Widget and Configuration Options Work

You can add the Metric Chart widget to one or more custom dashboards and configure it to display the workload for your objects. The data that appears in the widget is based on the configured menu items for each widget instance.

You edit the Metric Chart widget after you add it to a dashboard. The changes you make to the menu items create a custom widget with the selected metrics that display the workload on your objects.

To select metrics, you can select an object from the object list, then select the metrics. Or, you can select a tag from the object tag list to limit the object list, then select an object. You can configure multiple charts for the same object or multiple charts for different objects.

To use the metric configuration, which displays a set of metrics that you defined in an XML file, the dashboard and widget configuration must meet the following criteria:

- The dashboard **Widget Interaction** menu items are configured so that another widget provides objects to the target widget. For example, an Object List widget provides the object interaction to a chart widget.
- The widget **Self Provider** options are set to **Off**.
- The custom XML file in the **Metric Configuration** drop-down menu is in the `/usr/lib/vmware-vcops/tools/opscli` directory and has been imported into the global storage using the import command.

### Where You Find the Metric Chart Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Metric Chart Widget Toolbar Options

On the title bar of the widget, click the **Show Toolbar** icon to access the toolbar options.

The toolbar contains icons that you can use to change the view of the graphs.

| Option | Description |
|---|---|
| Split Charts | Displays each metric in a separate chart. |
| Stacked Chart | Consolidates all charts into one chart. This chart is useful for seeing how the total or sum of the metric values vary over time. To view the stacked chart, ensure that the split chart option is turned off. |
| Dynamic Thresholds | Shows or hides the calculated dynamic threshold values for a 24-hour period. |
| Show Entire Period Dynamic Thresholds | Shows or hides dynamic thresholds for the entire time period of the graph. |
| Static Thresholds | Shows or hides the threshold values that have been set for a single metric. |
| Anomalies | Shows or hides anomalies. Time periods when the metric violates a threshold are shaded. Anomalies are generated when a metric crosses a dynamic or static threshold, either above or below. |
| Trend Line | Shows or hides the line and data points that represents the metric trend. The trend line filters out metric noise along the timeline by plotting each data point relative to the average of its adjoining data points. |
| Show Data Values | Activates the data point tooltips if you switched to a zoom or pan option. **Show Data Point Tips** must be activated. |
| Zoom All Charts | Resizes all the charts that are open in the chart pane based on the area captured when you use the range selector.<br>You can switch between this option and **Zoom the View**. |
| Zoom the View | Resizes the current chart when you use the range selector. |
| Pan | When you are in zoom mode, allows you to drag the enlarged section of the chart so that you can view higher or lower, earlier or later values for the metric. |
| Zoom to Fit | Resets the chart to fit in the available space. |
| Remove All | Removes all the charts from the chart pane, allowing to you begin constructing a new set of charts. |
| Refresh Charts | Reloads the charts with current data. |
| Date Controls | Opens the date selector.<br>Use the date selector to limit the data that appears in each chart to the time period you are examining.<br>Select **Dashboard Time** to activate the dashboard time panel. The option chosen in the dashboard time panel is effective. The default time is 6 hours.<br>**Dashboard Time** is the default option. |
| Generate Dashboard | Saves the current charts as a dashboard. |

| Option | Description |
|---|---|
| Save as PDF | Saves the current metric chart as a PDF file.<br>If you have added many metrics for different objects to a metric chart, you can quickly download and share the PDF file with another user or VM owner. You can create ad-hoc reports when analyzing metrics. You can also add notes to the metrics to provide context to the user. |

### Metric Chart Widget Graph Selector Options

The graph selector options determine how individual data appears in the graph.

| Option | Description |
|---|---|
| Close | Deletes the chart. |
| Save a snapshot | Creates a PNG file of the current chart. The image is the size that appears on your screen. You can retrieve the file in your browser's download folder. |
| Download comma-separated data | Creates a CSV file that includes the data in the current chart.<br>You can retrieve the file in your browser's download folder. |
| Save a full screen snapshot | Downloads the current graph image as a full-page PNG file, which you can display or save.<br>You can retrieve the file in your browser's download folder. |
| Units | You can display the data with dots or as a percentage. |
| Thresholds | You can choose to show/hide **Critical**, **Immediate**, and **Warning** thresholds in the current chart. |
| Scales | You can choose a scale for a stacked chart.<br>• Select **Linear** to view a chart in which the Y axis scale increases in a linear manner. For example, the Y axis can have ranges from 0 to 100, 100 to 200, 200 to 300, and so on.<br>• Select **Logarithmic** to view a chart in which the Y axis scale increases in a logarithmic manner. For example, the Y axis can have ranges from 10 to 20, 20 to 300, 300 to 4000, and so on. This scale gives a better visibility of minimum and maximum values in the chart when you have a large range of metric values.<br>**Note:**<br>If you select a logarithmic scale, the chart does not display data points for metric values less than or equal to 0, which leads to gaps in the graph.<br>• Select **Combined** to view overlapping graphs for the metrics. The chart uses individual scales for each graph instead of using a relative scale, and displays a combined view of the graphs.<br>• Select **Combined by Unit** to view a chart that groups the graphs for similar metric units together. The chart uses a common scale for the combined graphs. |
| Move Down | Moves the chart down one position. |
| Move Up | Moves the chart up one position. |

You can take the following actions on the Metric Chart graph.

| Option | Description |
|---|---|
| Y Axis | Shows or hides the Y-axis scale. |
| Chart | Shows or hides the line that connects the data points on the chart. |
| Data Point Tips | Shows or hides the data point tooltips when you hover the mouse over a data point in the chart. |
| Zoom by X | Enlarges the selected area on the X axis when you use the range selector in the chart to select a subset of the chart. You can use **Zoom by X** and **Zoom by Y** simultaneously. |
| Zoom by Y | Enlarges the selected area on the Y axis when you use the range selector in the chart to select a subset of the chart. You can use **Zoom by X** and **Zoom by Y** simultaneously. |

| Option | Description |
|---|---|
| Zoom by Dynamic Thresholds | Resizes the Y axis of the chart so that the highest and the lowest values on the axis are the highest and the lowest values of the dynamic threshold calculated for this metric. |
| Vertical resize | Resizes the height of a graph in the chart. |
| **Remove** icon next to each metric name in a stacked chart | Removes the graph for the metric from the chart. |

**Metric Chart Configuration Options**

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

The **Input Transformation** section provides options to transform the input for the widget.

The **Output Data** section provides options to select object types on which you are basing the widget data.

The **Output Filter** section provides options to restrict the widget data based on the selected filter criteria.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| **Input Data** | |

| Option | Description |
|---|---|
| Metrics | Select metrics on which you want to base the widget data. You can select an object and pick its metrics.<br>1. Click the **Add New Metrics** icon to add metrics for the widget data. Select an object to view its metric tree and pick metrics for the object. The picked metrics appear in a list in this section.<br>The metric tree shows common metrics for several objects when you click the **Show common metrics** icon.<br>While selecting objects for which you want to pick metrics, you can use the **Filter** text box to search for objects. You can also expand the **Tag Filter** pane on the left hand side to select one or more object tag values. A list of objects with the selected tag values appears. If you select more than one value for the same tag, you can choose objects that have any of the tags applied. If you select more than one value for different tags, you can choose only the objects that have all the tags applied.<br>2. Optionally, select metrics from the list and click the **Remove Selected Metrics** icon to remove the selected metrics.<br>Click the **Select All** icon to select all the metrics in the list.<br>Click the **Clear Selection** icon to clear your selection of metrics in the list.<br>Optionally, you can customize a metric and apply the customization to other metrics in the list.<br>1. Double-click a metric box in the list to customize the metric and click **Update**.<br>You can use the **Box Label** text box to customize the label of a metric box.<br>You can use the **Unit** text box to define a measurement unit of each metric.<br>You can use the **Color Method** option to define a coloring criteria for each metric. If this option is set to **Custom**, you can enter color values in the **Yellow**, **Orange**, and **Red** text boxes. You can also set coloring by symptom definition. If you do not want to use color, select **None**.<br>For example, to view the remaining memory capacity of a VM, select **Virtual Machine** as an object type, expand the **Memory** from the metric tree and double-click **Capacity Remaining(%)**. Define a meaningful label name and measurement unit to help you when you observe the metrics. You can select **Custom** from the **Color Method** drop-down menu and specify different values for each color, for example 50 for **Yellow**, 20 for **Orange**, and 10 for **Red**.<br>2. Select a metric and click the **Apply to All** icon to apply the customization for the selected metric to all the metrics in the list. |
| Objects | Select objects on which you want to base the widget data.<br>1. Click the **Add New Objects** icon and select objects in the pop-up window. The selected objects appear in a list in this section.<br>While selecting objects, you can use the **Filter** text box to search for objects. You can also expand the **Tag Filter** pane on the left hand side to select one or more object tag values. A list of objects with the selected tag values appears. If you select more than one value for the same tag, you can choose objects that have any of the tags applied. If you select more than one value for different tags, you can choose only the objects that have all the tags applied.<br>2. Optionally, select objects from the list and click the **Remove Selected Objects** icon to remove the selected objects.<br>Click the **Select All** icon to select all the objects in the list.<br>Click the **Clear Selection** icon to clear your selection of objects in the list. |
| All | If you select this option, the widget data is based on all the objects in your environment. The following sections provide options to refine the objects for the widget data. |
| **Input Transformation** | |
| Relationship | Transform the input for the widget based on the relationship of the objects. For example, if you select the **Children** check box and a **Depth** of **1**, the child objects are the transformed inputs for the widget. |
| **Output Data** | |
| Empty drop-down menu | Specifies a list with attributes to display. |

| Option | Description |
|---|---|
| | Select metrics on which you want to base the widget data. You can select an object and pick its metrics.<br>1. Click the **Add New Metrics** icon to add metrics for the widget data. Select an object to view its metric tree and pick metrics for the object. The picked metrics appear in a list in this section.<br>The metric tree shows common metrics for several objects when you click the **Show common metrics** icon.<br>While selecting objects for which you want to pick metrics, you can use the **Filter** text box to search for objects. You can also expand the **Tag Filter** pane on the left hand side to select one or more object tag values. A list of objects with the selected tag values appears. If you select more than one value for the same tag, you can choose objects that have any of the tags applied. If you select more than one value for different tags, you can choose only the objects that have all the tags applied.<br>2. Optionally, select metrics from the list and click the **Remove Selected Metrics** icon to remove the selected metrics.<br>Click the **Select All** icon to select all the metrics in the list.<br>Click the **Clear Selection** icon to clear your selection of metrics in the list.<br>Optionally, you can customize a metric and apply the customization to other metrics in the list.<br>1. Double-click a metric box in the list to customize the metric and click **Update**.<br>You can use the **Box Label** text box to customize the label of a metric box.<br>You can use the **Unit** text box to define a measurement unit of each metric.<br>You can use the **Color Method** option to define a coloring criteria for each metric. If this option is set to **Custom**, you can enter color values in the **Yellow**, **Orange**, and **Red** text boxes. You can also set coloring by symptom definition. If you do not want to use color, select **None**.<br>For example, to view the remaining memory capacity of a VM, select **Virtual Machine** as an object type, expand the **Memory** from the metric tree and double-click **Capacity Remaining(%)**. Define a meaningful label name and measurement unit to help you when you observe the metrics. You can select **Custom** from the **Color Method** drop-down menu and specify different values for each color, for example 50 for **Yellow**, 20 for **Orange**, and 10 for **Red**.<br>2. Select a metric and click the **Apply to All** icon to apply the customization for the selected metric to all the metrics in the list. |
| **Output Filter** | |
| | Refine the widget data further based on the filter criteria for object types. The widget data is based on the objects for the filtered object types.<br>If the objects have an input transformation applied, you define filter criteria for the object types of the transformed objects.<br>1. In the first drop-down menu, select an object type.<br>2. In the second drop-down menu, select the option based on which you want to define the filter criteria. For example, if you select **Metrics** for the **Datacenter** object type, you can define a filter criteria based on the value of a specific metric for data centers.<br>3. In the drop-down menus and text boxes that appear, select or enter values to filter the objects.<br>4. To add more filter criteria, click **Add**.<br>5. To add another filter criteria set, click **Add another criteria set**. |

## Metric Picker Widget

The Metric Picker widget displays a list of available metrics for a selected object.

### How the Metric Picker Widget and Configuration Options Work

With the Metric Picker widget, you can check the list of the object's metrics. To select an object to pick its metrics, you use another widget as a source of data, for example, Topology Graph widget. To set a source widget that is on the same dashboard, you use the Widget Interactions menu when you edit a dashboard. To set a source widget that is on another dashboard, use the **Dashboard Navigation** menu when you edit a dashboard that contains the source widget. You can also search for objects using tags.

You edit a Metric Picker widget after you add it to a dashboard. The changes you make to the options create a custom chart to meet the needs of the dashboard users.

### Where You Find the Metric Picker Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Metric Picker Widget Toolbar Options

On the title bar of the widget, click the **Show Toolbar** icon to access the toolbar options.

The toolbar contains icons that you can use to change the view of the graphs.

| Option | Description |
|---|---|
| Show common metrics | Filter based on common metrics. |
| Show collecting metrics | Filter based on collecting metrics. |
| Metrics or Properties | Filter based on metrics or property metrics. |
| Time Range | Filter based on selected time range. |
| Search | Search for dashboards, views, and network IP addresses using tags. |

### Metric Picker Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

| Option | Action |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |

| Option | Action |
|---|---|
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |

## Object List Widget

The Object List widget displays a list of the objects available in the environment.

### How the Object List Widget and Configuration Options Work

The Object List widget displays a data grid with objects in the inventory. The default configuration of the data grid appears in Object List Widget Options section. You can customize it by adding or removing default columns. You can use the **Additional Column** option to add metrics when you configure the widget.

You edit an Object List widget after you add it to a dashboard. Configuration of the widget helps you observe parent and child objects. You can configure the widget to display the child objects of an object selected from another widget, for example, another Object List or Object Relationship widget, in the same dashboard.

Click the legend at the bottom of the widget to filter the objects based on threshold. Point your cursor over any of the boxes to view tooltips.

### Where You Find the Object List Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Object List Widget Toolbar Options

On the title bar of the widget, click the **Show Toolbar** icon to access the toolbar options.

| Option | Description |
|---|---|
| Action | Selects from a set of actions specific for each object type. To see available actions, select an object from the list of objects and click the toolbar icon to select an action. For example, when you select a datastore object in the graph, you can select **Delete Unused Snapshots for Datastore**. |
| Dashboard Navigation | Navigates you to the object. For example, when you select a datastore from the list of objects and click **Dashboard Navigation**, you can open the datastore in vSphere Web Client. |
| Reset Grid Sort | Returns the list of resources to its original order. |

| Option | Description |
|---|---|
| Reset Interaction | Returns the widget to its initial configured state and undoes any interactions selected in a providing widget.<br>Interactions are usually between widgets in the same dashboard, or you can configure interactions between widgets on different dashboards. |
| Object Detail | Select an object and click this icon to show the Object Detail page for the object. |
| Perform Multi-Select Interaction | If the widget is a provider for another widget on the dashboard, you can select multiple rows and click this button. The receiving widget then displays only the data related to the selected interaction items.<br>Use Ctrl+click for Windows, or Cmd+click for Mac OS X, to select multiple individual objects or Shift+click to select a range of objects, and click the icon to activate the interaction. |
| Display Filtering Criteria | Displays the object information on which this widget is based. |
| Page Size | |
| Filter | Locate data in the widget.<br>You can search for objects or filter the list based on the values of the metrics or properties in the additional columns of the Configuration section. |

## Object List Widget Data Grid Options

The data grid provides a list of inventory objects on which you can sort and search.

| Option | Description |
|---|---|
| ID | Unique ID for each object in the inventory, randomly generated and produced by VCF Operations. |
| Name | Name of the object in the inventory. |
| Description | Displays the short description of the object given during creation of the object |
| Adapter Type | Shows the adapter type for each object. |
| Object Type | Displays the type of the object in the inventory. |
| Policy | Displays policies that are applied to the object. To see policy details and create policy configurations, in the menu click **Administration**, and then in the left pane click **Policies**. |
| Creation Time | Displays the date, time, and time zone of the creation of an object that was created in the inventory. |
| Identifier 1 | Can contain the custom name of the object in the inventory or default unique identifier, depending on the type of inventory object. For example, My_VM_1 for a VM in the inventory, or 64-bit hexadecimal value for VCF Operations Node. |
| Identifier 2 | Can contain the abbreviation of an object type and the unique decimal number or parent instance, depending on the type of the object. For example, vm-457 for a VM and an IP address for VCF Operations Node. |
| Identifier 3 | Can contain a unique number identifying an adapter type. For example, 64-bit hexadecimal value for vCenter Adapter |
| Identifier 4 | Additional unique identifiers for the object. This option varies and depends on the adapter type that the object uses. |

| Option | Description |
|---|---|
| Identifier 5 | Additional unique identifiers for the object. This option varies and depends on the adapter type that the object uses. |
| Object Flag | Displays a badge icon for each object. You can see the status when you point to the badge. |
| Collection State | Displays the collection state of an adapter instance of each object. You can see the name of the adapter instance and its state in a tool tip when you point to the state icon. To manage an adapter instance to start and stop collection of data, in the menu, click **Ad ministration**, and then in the left pane click **Inventory**. |
| Collection Status | Displays the collection status of the adapter instance of each object. You can see the name of the adapter instance and its status in a tool tip when you point to the status icon. To manage an adapter instance to start and stop collection of data, in the menu, click **Administration**, and then in the left pane click **Inventory**. |
| Relevance | Displays the user interest on objects based on the number of clicks. The relevance is determined using a system-wide ranking algorithm that rates the object with most clicks as most relevant object. |
| Internal ID | Unique number that VCF Operations uses to identify the object internally. For example, the internal ID appears in log files used for troubleshooting. |

## Object List Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

The **Input Transformation** section provides options to transform the input for the widget.

The **Output Filter** section provides options to restrict the widget data based on the selected filter criteria.

The **Additional Columns** section provides options to select metrics that are displayed as additional columns in the widget.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |

| Option | Description |
|---|---|
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| Auto Select First Row | Determines whether to start with the first row of data. |
| **Input Data** | |
| Objects | Select objects on which you want to base the widget data.<br>1. Click the **Add New Objects** icon and select objects in the pop-up window. The selected objects appear in a list in this section.<br>While selecting objects, you can use the **Filter** text box to search for objects. You can also expand the **Tag Filter** pane on the left hand side to select one or more object tag values. A list of objects with the selected tag values appears. If you select more than one value for the same tag, you can choose objects that have any of the tags applied. If you select more than one value for different tags, you can choose only the objects that have all the tags applied.<br>2. Optionally, select objects from the list and click the **Remove Selected Objects** icon to remove the selected objects.<br>Click the **Select All** icon to select all the objects in the list.<br>Click the **Clear Selection** icon to clear your selection of objects in the list. |
| All | If you select this option, the widget data is based on all the objects in your environment. The following sections provide options to refine the objects for the widget data. |
| **Input Transformation** | |
| Relationship | Transform the input for the widget based on the relationship of the objects. For example, if you select the **Children** check box and a **Depth** of **1**, the child objects are the transformed inputs for the widget. |
| **Output Filter**<br>**Note:**<br>If there are more than ten objects under a section in the inventory tree, you can search for an object using the search option. If there are more than thousand objects in a section, use the **View More** button under the last object that is displayed to view the rest of the objects. | |

| Option | Description |
| --- | --- |
| Basic | Pick tags to refine the widget data. The widget data is based on the objects that have the picked tags applied. If you pick more than one value for the same tag, the widget includes objects that have any of the tags applied. If you pick more than one value for different tags, the widget includes only the objects that have all the tags applied.<br>If the objects have an input transformation applied, you select tag values for the transformed objects. |
| Advanced | Refine the widget data further based on the filter criteria for object types. The widget data is based on the objects for the filtered object types.<br>If the objects have a tag filter applied in the **Basic** subsection, you define filter criteria for the object types of the objects with tag filter applied. If the objects with tag filter applied do not belong to any of the object types in this filter criteria, the widget skips this filter and includes all the objects with tag filter applied.<br>If the objects have an input transformation applied, you define filter criteria for the object types of the transformed objects.<br>1. In the first drop-down menu, select an object type.<br>2. In the second drop-down menu, select the option based on which you want to define the filter criteria. For example, if you select **Metrics** for the **Datacenter** object type, you can define a filter criteria based on the value of a specific metric for data centers.<br>3. In the drop-down menus and text boxes that appear, select or enter values to filter the objects.<br>4. To add more filter criteria, click **Add**.<br>5. To add another filter criteria set, click **Add another criteria set**. |
| **Additional Columns** | |
| Empty drop-down menu | Specifies a list with attributes to display. |
| | Add metrics based on object types. The selected metrics are displayed as additional columns in the widget.<br>1. Click the **Add New Metrics** icon to add metrics based on object types. The metrics that you add appear in a list in this section.<br>While selecting object types for which you want to pick metrics, you can filter the object types by adapter type to pick an object type. On the metrics pane, click the **Select Object** icon to select an object for the object type. Pick metrics of the selected object from the metric tree.<br>For example, you can select the **Datacenter** object type, click the **Select Object** icon to display the list of data centers in your environment, and pick metrics of the selected data center.<br>2. Optionally, you can double-click a metric box in the list to customize the label of the metric and click **Update**. |

## Object Relationship Widget

The Object Relationship widget displays the hierarchy tree for the selected object. You can create one or more hierarchy trees in VCF Operations for the selected objects that you add to your custom dashboards.

## How the Object Relationship Widget and Configuration Options Work

You can add the Object Relationship widget to one or more custom dashboards and configure it to display data that is important to different dashboard users. The data that appears in the widget is based on the configured options for each widget instance.

You edit an Object Relationship widget after you add it to a dashboard. The changes you make to the options help create a custom widget to meet the needs of the dashboard users.

## Where You Find the Object Relationship Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

## Object Relationship Widget Toolbar Options

On the title bar of the widget, click the **Show Toolbar** icon to access the toolbar options.

| Option | Description |
|---|---|
| Dashboard Navigation | You can navigate to another dashboard when the object under consideration is also available in the dashboard to which you navigate. To be able to navigate to another dashboard, configure the relevant option when you create or edit the dashboard. |
| Badge | Displays the Health, Risk, or Efficiency alerts on the objects in the relationship map. You can select a badge for objects that appear in the widget. The tool tip of a badge shows the object name, object type, and the name of the selected badge with the value of the badge. You can only select one badge at a time. |
| Zoom to fit | Resets the chart to fit in the available space. |
| Pan | Click this icon and click and drag the hierarchy to show different parts of the hierarchy. |
| Show values on point | Shows or hides the data point tooltips when you hover the mouse over a data point in the chart. |
| Zoom the view | Click this icon and drag to outline a part of the hierarchy. The display zooms to show only the outlined section. |
| Display Filtering Criteria | Shows the filtering settings for the widget in a pop-up window. |
| Zoom in | Zooms in on the hierarchy. |
| Zoom out | Zooms out on the hierarchy. |
| Reset to Initial Object | If you change the hierarchy of the initial configuration or the widget interactions, click this icon to return to the initial resource. Clicking this icon also resets the initial display size. |
| Object Detail | Select an object and click this icon to show the Object Detail page for the object. |
| Show Alerts | Select the resource in the hierarchy and click this icon to show alerts for the resource. Alerts appear in a pop-up window. You can double-click an alert to view its Alert Summary page. |

## Object Relationship Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

The **Output Filter** section provides options to restrict the widget data based on the selected filter criteria.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| Auto Zoom to Fixed Node Size | You can configure a fixed zoom level for object icons in the widget display.<br>If your widget display contains many objects and you always need to use manual zooming, this feature is useful because you can use it to set the zoom level only once. |
| Node Size | You can set the fixed zoom level at which the object icons display. Enter the size of the icon in pixels.<br>The widget shows object icons at the pixel size that you configure. |
| **Input Data** | |
| Object | Search for objects in your environment and select the object on which you are basing the widget data. You can also click the **Add Object** icon and select an object from the object list. You can use the **Filter** text box to refine the object list and the **Tag Filter** pane to select an object based on tag values. |
| **Output Filter** | |

| Option | Description |
|---|---|
| Basic | Pick tags to refine the widget data. The widget data is based on the objects that have the picked tags applied. If you pick more than one value for the same tag, the widget includes objects that have any of the tags applied. If you pick more than one value for different tags, the widget includes only the objects that have all the tags applied. |
| Advanced | Refine the widget data further based on the filter criteria for object types. The widget data is based on the objects for the filtered object types.<br>If the objects have a tag filter applied in the **Basic** subsection, you define filter criteria for the object types of the objects with tag filter applied. If the objects with tag filter applied do not belong to any of the object types in this filter criteria, the widget skips this filter and includes all the objects with tag filter applied.<br>1. In the first drop-down menu, select an object type.<br>2. In the second drop-down menu, select the option based on which you want to define the filter criteria. For example, if you select **Metrics** for the **Datacenter** object type, you can define a filter criteria based on the value of a specific metric for data centers.<br>3. In the drop-down menus and text boxes that appear, select or enter values to filter the objects.<br>4. To add more filter criteria, click **Add**.<br>5. To add another filter criteria set, click **Add another criteria set**. |

## Object Relationship (Advanced) Widget

The Object Relationship (Advanced) widget displays a graph or tree view, or a hexagon view that depicts the parent-child relationship of the selected object. It provides advanced configuration options. You can create a graph or tree view, or a hexagon view in VCF Operations for the selected objects that you add to your custom dashboards.

### How the Object Relationship (Advanced) Widget and Configuration Options Work

You can add the **Object Relationship (Advanced)** widget to one or more custom dashboards and configure it to display data that is important to different dashboard users. The data that appears in the widget is based on the configured options for each widget instance.

You can edit an **Object Relationship (Advanced)** widget after you add it to a dashboard. The changes you make to the options help create a custom widget to meet the needs of the dashboard users.

There are two modes, the Classic view mode is a hierarchy tree view, and relationships between objects are represented by corresponding colors. The Hexagon view mode is displayed as a ring around the hexagon grid, with each segment representing all the criticalities in the current grid.

You can double-click any object in the graph or tree view, or hexagon view and see the specific parent-child objects for the focus object. If you point your cursor over an object icon, you see the health, risk, and efficiency details. You can also click the **Alerts** link for the number of generated alerts.

In the hexagon view, you can see the total object count. The parent-child view is listed on the left side as you click on each object. You can view the criticality of the objects as depicted by the color by hovering your cursor over the hexagon or ring around the hexagon.

### Where You Find the Object Relationship (Advanced) Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Object Relationship (Advanced) Widget Toolbar Options

On the title bar of the widget, click the **Show Toolbar** icon to access the toolbar options.

| Options | Description |
|---|---|
| Dashboard Navigation | You can navigate to another dashboard when the object under consideration is also available in the dashboard to which you navigate. To navigate to another dashboard, configure the relevant option when you create or edit the dashboard. |
| Back to Initial Object | If you change the hierarchy of the initial configuration or the widget interactions, click this icon to return to the initial resource. Clicking this icon also resets the initial display size. |
| View Tree/View graph<br>(Displayed when you select the Classic view mode) | Displays a tree or graph view of the relationships. |
| Applications<br>(Displayed when you select the Classic view mode) | Displays relationships that are of the type 'application'. |
| Vertical/Horizontal<br>(Displayed when you select the Classic view mode) | Displays a vertical or horizontal view of the graph or tree view. |
| Hide Text/Show Text<br>(Displayed when you select the Classic view mode) | Hides or displays the object names. |
| Standard View/Fit View | The **Standard View** option fixes the view to a specific zoom level<br>The **Fit View** option adjusts the graph or tree view to fit the screen. |
| View Additional Links | Displays additional links for the applications. |
| Quick Filter | Enter the name of an object that you want to see in the graph or tree view or in the hexagon view. |
| View Parents/View Children<br>(Displayed when you select the Hexagon view mode) | Click the arrows on the left side of the grid to view the parents or children for the selected object, in the hexagon. |

**Table 1380: Specific Functions from the Object Relationship (Advanced) Grid View**

| Function | Description |
|---|---|
| Filter By Type | The filtering button appears in the upper right corner of the grid in Classic mode if any 'Relationship Type' defined by its corresponding color, exists.<br>You can filter by the type of relationship between parent and child. For example, filter by Runson, Application, Location, and so on. Each relationship type is associated with a color and the connections in the tree view between the parent-child objects are depicted in the specific color. |
| Hide Node | Hover your cursor over an object and click the **Hide Node** icon to hide the specific node. |

| Function | Description |
|---|---|
| Show Peers/Hide Peers | Hover your cursor over the selected object in the grid view and select **ShowPeers/Hide Peers** to view or hide other objects of the same type that exist under the parent object. |
| Pagination/Next Page/Previous Page | At the bottom of the grid view you can view the page number and also move to the next/previous page. |
| Filter | At the bottom of the grid view you can use the filter option to search for objects. |
| Arrows | Use the arrows on each object to view relationships of each object. |

### Object Relationship (Advanced) Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

The **Output Filter** section provides options to restrict the widget data based on the selected filter criteria.

| Option | Description |
|---|---|
| Name | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |

| Option | Description |
|---|---|
| View Mode | Select either Classic or Hexagon.<br>If you choose Classic, you can view a graph or tree, that depicts the parent-child relationship of the selected object after you configure the widget.<br>If you select Hexagon, you see a hexagon view that depicts the parent-child relationship of the selected object after you configure the widget. |
| Inventory trees | Select an existing predefined traversal spec for the initial object relationship graph or tree view. |
| Relationship Types<br>(Displayed when you select the Classic view mode) | Select the relationship type you want displayed in the dashboard. Each type of relationship type is associated with a color.<br>**Note:**<br>The relationship type filter you select here takes precedence over the 'Filter By Type' option you see in the datagrid of the widget. |
| Parents Depth<br>(Displayed when you select the Classic view mode) | Select the depth of parent objects to be displayed. |
| Children Depth<br>(Displayed when you select the Classic view mode) | Select the depth of child objects to be displayed. |
| Page Size<br>(Displayed when you select the Classic view mode) | Select the number of objects to list per page. |
| **Input Data** | |
| Object | Search for objects in your environment and select the object on which you are basing the widget data. You can also click the **Add Object** icon and select an object from the object list. You can use the **Filter** text box to refine the object list and the **Tag Filter** pane to select an object based on tag values. |
| **Output Filter** | |
| Basic | Pick tags to refine the widget data. The widget data is based on the objects that have the picked tags applied. If you pick more than one value for the same tag, the widget includes objects that have any of the tags applied. If you pick more than one value for different tags, the widget includes only the objects that have all the tags applied. |
| Advanced | Refine the widget data further based on the filter criteria for object types. The widget data is based on the objects for the filtered object types.<br>If the objects have a tag filter applied in the **Basic** subsection, you define filter criteria for the object types of the objects with tag filter applied. If the objects with tag filter applied do not belong to any of the object types in this filter criteria, the widget skips this filter and includes all the objects with tag filter applied.<br>1. In the first drop-down menu, select an object type.<br>2. In the second drop-down menu, select the option based on which you want to define the filter criteria. For example, if you select **Metrics** for the **Datacenter** object type, you can define a filter criteria based on the value of a specific metric for data centers.<br>3. In the drop-down menus and text boxes that appear, select or enter values to filter the objects.<br>4. To add more filter criteria, click **Add**.<br>5. To add another filter criteria set, click **Add another criteria set**. |

## Property List Widget

You can use the Property List widget to view the properties of objects and their values.

### How the Property List Widget and Configuration Options Work

To observe the properties of objects in the Property List widget, you can select object property metrics when you configure the widget itself (Self Provider mode activated). Alternatively, you can select objects or object property metrics from another widget (Self Provider mode deactivated). You can also view a default or custom set of properties by selecting a preconfigured XML file in the Metric Configuration drop-down menu of the widget configuration window.

You edit a Property List widget after you add it to a dashboard. You can configure a widget to receive data from another widget by selecting **Off** for Self Provider mode. When the widget is not in Self Provider mode, it displays a set of predefined properties and their values of an object that you select on the source widget. For example, you can select a host on a Topology widget and observe its properties in the Property List widget. To configure the Property List as a receiver widget that is on the same dashboard, use the **Widget Interactions** menu when you edit a dashboard. To configure a receiver widget that is on another dashboard, use the **Dashboard Navigation** menu when you edit a source dashboard.

### Where You Find the Property List Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Property List Widget Data Grid Options

The data grid provides information on which you can sort and search.

| Option | Description |
|---|---|
| Object Name | Name of the object, whose properties you observe. You can sort the properties by object name. To open the Object Details page, click an object name. |
| Property Name | Name of the property. You can sort the properties by property name. |
| Value | Value of the property. You can sort the properties by value. |

### Property List Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

The **Input Transformation** section provides options to transform the input for the widget.

The **Output Data** section provides options to select object types on which you are basing the widget data.

The **Output Filter** section provides options to restrict the widget data based on the selected filter criteria.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget. If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| Visual Theme | Select a predefined visual style for each instance of the widget. The options are: Original and Compact. |
| Show Metric Full Name | You can choose to view the full name of the metrics. The options are: On and Off. |
| **Input Data** | |

| Option | Description |
| --- | --- |
| Metrics | Select metrics on which you want to base the widget data. You can select an object and pick its metrics.<br>1. Click the **Add New Metrics** icon to add metrics for the widget data. Select an object to view its metric tree and pick metrics for the object. The picked metrics appear in a list in this section.<br>The metric tree shows common metrics for several objects when you click the **Show common metrics** icon.<br>While selecting objects for which you want to pick metrics, you can use the **Filter** text box to search for objects. You can also expand the **Tag Filter** pane on the left hand side to select one or more object tag values. A list of objects with the selected tag values appears. If you select more than one value for the same tag, you can choose objects that have any of the tags applied. If you select more than one value for different tags, you can choose only the objects that have all the tags applied.<br>2. Optionally, select metrics from the list and click the **Remove Selected Metrics** icon to remove the selected metrics.<br>Click the **Select All** icon to select all the metrics in the list.<br>Click the **Clear Selection** icon to clear your selection of metrics in the list.<br>You can define measurement units for the metrics in the list. Double-click a metric box in the list, select a measurement unit in the **Unit** drop-down menu, and click **Update**. |
| Objects | Select objects on which you want to base the widget data.<br>1. Click the **Add New Objects** icon and select objects in the pop-up window. The selected objects appear in a list in this section.<br>While selecting objects, you can use the **Filter** text box to search for objects. You can also expand the **Tag Filter** pane on the left hand side to select one or more object tag values. A list of objects with the selected tag values appears. If you select more than one value for the same tag, you can choose objects that have any of the tags applied. If you select more than one value for different tags, you can choose only the objects that have all the tags applied.<br>2. Optionally, select objects from the list and click the **Remove Selected Objects** icon to remove the selected objects.<br>Click the **Select All** icon to select all the objects in the list.<br>Click the **Clear Selection** icon to clear your selection of objects in the list. |
| All | If you select this option, the widget data is based on all the objects in your environment. The following sections provide options to refine the objects for the widget data. |
| **Input Transformation** | |
| Relationship | Transform the input for the widget based on the relationship of the objects. For example, if you select the **Children** check box and a **Depth** of **1**, the child objects are the transformed inputs for the widget. |
| **Output Data** | |
| Empty drop-down menu | Specifies a list with attributes to display. |

| Option | Description |
|---|---|
| | 1. Click the **Add New Metrics** icon to add metrics based on object types. The metrics that you add appear in a list in this section.<br>While selecting object types for which you want to pick metrics, you can filter the object types by adapter type to pick an object type. On the metrics pane, click the **Select Object** icon to select an object for the object type. Pick metrics of the selected object from the metric tree.<br>For example, you can select the **Datacenter** object type, click the **Select Object** icon to display the list of data centers in your environment, and pick metrics of the selected data center.<br>2. Optionally, you can define measurement units for the metrics and properties in the list. Double-click a metric or properties box in the list, select a measurement unit in the **Unit** drop-down menu, and click **Update**.<br>3. You can use the **Color Method** option to define a coloring criteria for each metric. If this option is set to **Custom**, you can enter color values in the **Yellow**, **Orange**, and **Red** text boxes. You can also set coloring by symptom definition. If you do not want to use color, select **None**. |
| **Output Filter** | |
| | Refine the widget data further based on the filter criteria for object types. The widget data is based on the objects for the filtered object types.<br>If the objects have a tag filter applied in the **Basic** subsection, you define filter criteria for the object types of the objects with tag filter applied. If the objects with tag filter applied do not belong to any of the object types in this filter criteria, the widget skips this filter and includes all the objects with tag filter applied.<br>If the objects have an input transformation applied, you define filter criteria for the object types of the transformed objects.<br>1. In the first drop-down menu, select an object type.<br>2. In the second drop-down menu, select the option based on which you want to define the filter criteria. For example, if you select **Metrics** for the **Datacenter** object type, you can define a filter criteria based on the value of a specific metric for data centers.<br>3. In the drop-down menus and text boxes that appear, select or enter values to filter the objects.<br>4. To add more filter criteria, click **Add**.<br>5. To add another filter criteria set, click **Add another criteria set**. |

## PromQL Viewer Widget

The PromQL Viewer widget uses the Prometheus Query Language to identify real-time data to be visualized. As an admin, you can modify PromQL queries to create advanced queries and transformations. You can also create a PromQL query by browsing and choosing relevant metrics from the Query Builder Help. You can then validate the syntax of the query.

### How the PromQL Viewer Widget and Configuration Options Work

#### Prerequisites

- Ensure that the real-time metrics component has been installed before you configure the PromQL Viewer widget. For more information, see Deploy Real-Time Metrics.
- Ensure that you have deployed a VCF Instance. For more information, see Start a New VCF Fleet or a New VCF Instance Deployment by Using the VCF Installer Deployment Wizard.

The PromQL Viewer widget displays operational insights using PromQL. As an admin you can visualize metrics using PromQL.

You can edit the PromQL Viewer widget on the Home dashboard, or on another dashboard where you add the widget. With the widget configuration options, you can assign a new name to the widget, set the refresh content, set the refresh interval, and so on.

### Where You Find the PromQL Viewer Widget

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the PromQL Viewer widget to the dashboard workspace in the upper panel.

### PromQL Viewer Widget Toolbar Options

On the title bar of the widget, click the **Show Toolbar** icon to access the toolbar options.

| Option | Description |
|---|---|
| Date Controls | Use the date selector to limit the data that appears in each chart to the time period you are examining.\<br>Select **Dashboard Time** to activate the dashboard time panel. The option chosen in the dashboard time panel is effective. The default time is 6 hours. **Dashboard Time** is the default option. |
| Single series focus mode | The single series focus mode highlights one trend at a time for more straightforward analysis.<br>When this option is activated, hovering your cursor over a point makes that series the sole one displayed, allowing the chosen point and line to stand out distinctly. Additionally, you can select one or more legends to retain only those lines on the chart, concealing all other series until you choose to deselect them.<br>If the option is not enabled, when you click on an legend from the list, it will be deselected. |
| Show Chart Legend | Shows or hides the legend chart. |
| Refresh | Activate or deactivate the automatic refreshing of data in this widget. |
| Chart Height | Resize the height of the chart. |

### PromQL Viewer Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |

| Option | Description |
|---|---|
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| **Output Data** | |
| Source | • If the widget is in self provider mode, from the drop-down list, select the domain as source, and then select if the object type is vCenter/vSAN or NSX. The default selection is vCenter/vSAN.<br>• If the widget is not in self provider mode, select the VCF Instance as the source. |
| Query | Enter a complete PromQL query exactly as you want it to run. As there are no autocomplete suggestions, ensure that the entire expression is complete.<br>Ensure that the query is relevant for the selected object. If the query is not relevant, the chart will not display any data. |
| Need Help? Use the Quick Query Filter | Click the **Need Help? Use the Quick Query Filter** link to use the Query Builder Help. From the Query Builder Help, you can copy the relevant Function, Metric, or a Sample Query and paste it in the **Query** field. |
| Validate | Click **Validate** to validate the syntax of the query, |

**PromQL Queries**

The PromQL Viewer widget allows you to run raw PromQL queries directly against the metric data within your VCF Operations environment. This topic covers the core concepts, supported features, syntax, and limitations of PromQL queries.

**Procedure:**

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**.
2. Click **Create**, and add the **PromQL Viewer** widget.
3. Enter your PromQL query in the query field.
4. Select your time range and click **Execute**.

**Points to Note:**

- **Start simple:** Test your metric selector before adding functions or aggregations.
- **Verify label names:** Label names are case-sensitive.
- **Use appropriate time ranges:** Large ranges may impact performance.

## Core Concepts

| Concept | Description | Example |
|---|---|---|
| Instant Vector | A set of time series with a single sample value at a specific point in time. | `cpu.capacity.usage.HOST` |
| Range Vector | A set of time series with a range of samples over a time window. | `cpu.capacity.usage.HOST[5m]` |
| Label Matchers | Filters that restrict which time series are returned. | `cpu.capacity.usage.HOST{HOST='host-17'}` |
| Scalar | A simple numeric value. | 42, 3.14 |
| Subquery | A query evaluated over a range with a specified step. | `cpu.capacity.usage.HOST[30m:5m]` |

## Selectors

1. **Instant Vector Selectors.** Allows you to select a set of time series and a single sample value for each at a specific point in time.
    1. **Simple Match.** Select a metric across all time series: `cpu.capacity.usage.HOST`.
    2. **With Label Matchers.** Filter time series using label conditions. VCF 9.1 supports four matcher types:

| Matcher | Operator | Description | Example |
|---|---|---|---|
| Equality | **=** | Label equals value. | `cpu.capacity.usage.HOST{HOST='host-17'}` |
| Non-equality | **!=** | Label does not equal value. | `cpu.capacity.usage.HOST{HOST!='host-17'}` |
| Regex match | **=~** | Label matches regex pattern.<br>**Note:**<br>Regex matches are fully anchored. A match of `HOST=~"foo"` is treated as `HOST=~"^foo$"`. | `cpu.capacity.usage.HOST{HOST=~'host-1.*'}` |
| Regex non-match | **!~** | Label does not match regex. | `cpu.capacity.usage.HOST{HOST!~'host-1.*'}` |

**Note:**

You can combine multiple matchers with commas: `cpu.capacity.usage.HOST{HOST!='host-17', vc_ip='127.0.0.1'}`.

2. **Range Vector Selectors.** Selects a set of time series containing all the data points that are found within a specified duration of time (for example, `[5m]`) back from the current instant.

**Note:**

Range vector selectors are not allowed as outermost selector in the range queries.

a. **Simple Match.** Append a duration in brackets to select samples over a time window: `cpu.capacity.usage.HOST[5m]`.
b. **With label matchers.** `cpu.capacity.usage.HOST{HOST='host-17'}[5m]`

### Supported duration units

| Unit | Meaning |
|---|---|
| ms | milliseconds |
| s | seconds |

| Unit | Meaning |
|---|---|
| m | minutes |
| h | hours |
| d | days |
| w | weeks |

**Note:**

Common durations: `1m`, `5m`, `30m`, `1h`, `24h`, `7d`

## Operators

1. **Arithmetic Operators**. Execute scalar or vector-to-vector math by automatically matching pairs with identical label sets.

| Operator | Description | Example |
|---|---|---|
| **+** | Addition | cpu.usage.HOST + 100 |
| **-** | Subtraction | mem.total.HOST - mem.used.HOST |
| ***** | Multiplication | cpu.usage.HOST * 2 |
| **/** | Division | mem.used.HOST / mem.total.HOST |
| **%** | Modulo | cpu.usage.HOST % 10 |
| **^** | Exponentiation | cpu.usage.HOST ^ 2 |

Examples:
- # Scalar arithmetic: `5 * 2`
- # Vector multiplied by scalar: `cpu.capacity.usage.HOST{HOST='host-17'} * 2`
- # Vector arithmetic (element-wise): `cpu.capacity.usage.HOST{HOST='host-17'} * memory.usage.HOST{HOST='host-17'}`

2. **Comparison Operators.** Compare values and optionally return boolean results.

| Operator | Description |
|---|---|
| **==** | Equal |
| **!=** | Not equal |
| **>** | Greater than |
| **<** | Less than |
| **>=** | Greater than or equal |
| **<=** | Less than or equal |

Using the `bool` modifier:
When comparing two scalars, you must use the `bool` modifier to return 0 or 1:
Examples:
# Scalar vs scalar (bool required): `5 > bool 3`
# Vector vs scalar (bool optional - filters by default, returns 0/1 with bool): `cpu.capacity.usage.HOST > bool 80`
# Vector vs vector: `cpu.capacity.usage.HOST == memory.usage.HOST`

3. **Logical Operators.** Combine instant vectors using set operations.

| Operator | Description | Example |
|---|---|---|
| `and` | Intersection (series present in both) | `cpu.usage.HOST and net.usage.HOST` |
| `or` | Union (series present in either) | `cpu.usage.HOST or net.usage.HOST` |
| `unless` | Complement (left minus right) | `cpu.usage.HOST unless net.usage.HOST` |

**Note:**

Logical operators work only between instant vectors, not scalars.

## Aggregation Operators

Aggregations reduce multiple time series into fewer series.

### Supported Aggregations

| Operator | Description |
|---|---|
| `sum` | Sum of values |
| `min` | Minimum value |
| `max` | Maximum value |
| `avg` | Average value |
| `count` | Count of series |
| `topk` | Top K series by value |
| `bottomk` | Bottom K series by value |
| `topn_sum` | Top N series by sum *(VCF extension)* |
| `bottomn_sum` | Bottom N series by sum *(VCF extension)* |

### Syntax:

- Basic aggregation: `sum(cpu.capacity.usage.HOST)`
- With `by` clause (group by specific labels):

```
sum by (host) (cpu.usage.HOST)
# or
sum(cpu.usage.HOST) by (host)
```

- With without clause (exclude specific labels from grouping):

```
sum without (host, host_ip) (cpu.usage.HOST)
# or
sum(cpu.usage.HOST) without (host, host_ip)
```

### Examples:

- sum
  - `sum(cpu.capacity.usage.HOST)`
  - `sum by (host) (cpu.usage.HOST)`
  - `sum without (host, host_ip) (cpu.usage.HOST)`
- min/max/ avg/count
  - `min(cpu.capacity.usage.HOST)`
  - `max by (host) (cpu.capacity.usage.HOST)`
  - `avg without (host) (cpu.capacity.usage.HOST)`
  - `count(cpu.capacity.usage.HOST) by (host)`

- topk/bottomk
  - # Top 3 series by value: `topk(3, cpu.capacity.usage.HOST)`
  - # Top 3 per group: `topk(3, cpu.capacity.usage.HOST) by (HOST)`
  - # Bottom 3 excluding certain labels: `bottomk(3, cpu.capacity.usage.HOST) without (VM)`
- topn_sum/bottomn_sum
  - # Top 3 series by sum of values: `topn_sum(3, cpu.usage.HOST)`
  - # Bottom 3 series by sum: `bottomn_sum(3, cpu.usage.HOST)`

    **Note:**

    `topn_sum` and `bottomn_sum` are VCF-specific extensions (not part of standard PromQL) and do not support `by` or `without` clauses.

**Functions**

1. **Over-Time Functions.** Transform a range vector into an instant vector by calculating a single aggregate value for each individual time series over the specified time window.

| Function | Description | Example |
|---|---|---|
| avg_over_time | Average value over the range. | `avg_over_time(cpu.usage.HOST[5m])` |
| min_over_time | Minimum value over the range. | `min_over_time(cpu.usage.HOST[1h])` |
| max_over_time | Maximum value over the range. | `max_over_time(cpu.usage.HOST[30m])` |
| sum_over_time | Sum of values over the range. | `sum_over_time(cpu.usage.HOST[24h])` |
| count_over_time | Count of samples over the range. | `count_over_time(cpu.usage.HOST[7d])` |
| last_over_time | Most recent value in the range. | `last_over_time(cpu.usage.HOST[5m])` |

   **Example with label matcher:** `avg_over_time(cpu.capacity.usage.HOST{HOST='host-17'}[5m])`
2. **Rate and Counter Functions**

| Function | Description | Example |
|---|---|---|
| rate | Per-second average rate of increase. | **rate(http.requests.total[5m])** |
| irate | Instant rate using the last two data points. | `irate(http.requests.total[5m])` |
| resets | Number of counter resets in the range. | `resets(http.requests.total[5m])` |

   **Examples:**
   - `rate(cpu.capacity.usage.HOST{host='host-17'}[5m])`
   - `irate(cpu.capacity.usage.HOST[5m])`
   - `resets(mem.physical.total.HOST[5m])`
3. **Math Functions.** Functions performing mathematical operations.

| Function | Description | Example |
|---|---|---|
| abs | Absolute value | `abs(cpu.capacity.usage.HOST)` |
| ceil | Round up to nearest integer | `ceil(cpu.capacity.usage.HOST)` |
| floor | Round down to nearest integer | `floor(cpu.capacity.usage.HOST)` |

   **Examples:**
   - `abs(cpu.capacity.usage.HOST{host='host-17'})`
   - `ceil(cpu.capacity.usage.HOST)`
   - `floor(cpu.capacity.usage.HOST{host='host-17'})`

4. **Absence Functions.** Detect missing data.

| Function | Description | Example |
|---|---|---|
| absent | Returns 1 if the instant vector is empty | `absent(cpu.capacity.usage.HOST)` |
| absent_over_time | Returns 1 if no samples exist in the range | `absent_over_time(cpu.usage.HOST[5m])` |

**Examples:**

- `absent(cpu.capacity.usage.HOST{HOST='nonexistent'})`
- `absent_over_time(cpu.capacity.usage.HOST{HOST='host-17'}[5m])`

## Subqueries

Subqueries evaluate an expression over a time range with a specified resolution step.

**Syntax:**

```
<expression>[<range>:<step>]
```

**Note:**

The **<step>** (resolution) is optional. If omitted, it defaults to the global evaluation interval.

**Examples:**

- # Evaluate metric over 30 minutes with 5-minute steps: `cpu.usage.HOST{host='host-15'}[30m:5m]`
- # Function result as subquery: `count_over_time(cpu.usage.HOST{host='host-15'}[5m])[30m:1m]`

**Important:**

Subqueries only support instant queries in VCF 9.1. See the limitations listed in the table called **Limitations in VCF 9.1**.

## Limitations in VCF 9.1

The following PromQL features are not supported in VCF Operations 9.1.

| Limitation | Description |
|---|---|
| Subqueries in range queries | Subqueries only work with instant queries. |
| Nested subqueries | Cannot nest a subquery inside another function. For example, `sum_over_time(cpu.usage.HOST[30m:5m])` is not supported |
| Unary operators | `-metric` and `+metric prefix` operators are not supported. |
| Vector matching modifiers | `on`, `ignoring`, `group_left`, and `group_right` are not supported. |
| Some aggregation operators | `group`, `stddev`, `stdvar`, `count_values`, and `quantile` are not implemented. |
| Some functions | `delta`, `deriv`, `idelta`, `changes`, `clamp`, `clamp_min`, `clamp_max`, `round`, `sqrt`, `ln`, `log2`, `log10`, `exp`, `sgn`, `sort`, `sort_desc`, `scalar`, `vector`, `time`, `timestamp`, `histogram_*`, `holt_winters`, `predict_linear`, `label_join`, `label_replace`, `present_over_time`, `stddev_over_time`, `stdvar_over_time`, `quantile_over_time`, trigonometric functions, and date/time functions are not implemented |
| Offset modifier | The `offset` modifier is not fully supported |
| @ modifier | The `@` modifier for overriding evaluation time is not supported |
| Aggregation on name label | Using `name` in `by` or `without` clauses is not supported. |

### Troubleshooting

Here is a list of common errors.

| Error | Cause | Solution |
|---|---|---|
| "Empty or null metric name" | The query has no metric name. | Ensure your query starts with a valid metric name |
| "Unsupported function" | Using a function not implemented in VCF 9.1. | See the Limitations section documented above. |
| "Matrix selector not supported as outermost operation" | The range vector is used without a function in a range query. | Wrap in a function, such as `last_over_time(metric[5m])` |
| "Subquery not supported for nested queries" | A subquery used inside another expression. | Use the range selector with the step given in query params. |
| "Grouping operations are not supported" | Using `on`, `ignoring`, and so on. | Remove vector matching modifiers |
| "Unary operation is not supported" | Using `-metric` or `+metric`. | Use `0 - metric` or just `metric` instead. |
| "Comparisons between scalars must use BOOL modifier" | Scalar comparison without `bool`. | Add `bool` keyword: `5 > bool 3` |

## Recommended Actions Widget

The Recommended Actions widget displays recommendations to solve problems in your vCenter instances. With recommendations, you can run actions on your data centers, clusters, hosts, and virtual machines.

### How the Recommended Actions Widget and Configuration Options Work

The Recommended Actions widget appears on the Home dashboard, and displays the health status for the objects in your vCenter instance. At a glance, you can see how many objects are in a critical state, and how many objects need immediate attention.

From the Recommended Actions widget, you can focus in on problems further by, for example, clicking an object where the alerts triggered, and by clicking an individual alert.

You can edit the Recommended Actions widget on the Home dashboard, or on another dashboard where you add the widget. With the widget configuration options, you can assign a new name to the widget, set the refresh content, and set the refresh interval.

The Recommended Actions widget includes a selection bar, a summary pane, a toolbar for the data grid, and alert information for your objects in a data grid.

### Where You Find the Recommended Actions Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Recommended Actions Widget Selection Bar and Summary Pane

| Option | Description |
|---|---|
| Scope | Allows you to select an instance of vCenter, and a data center in that instance. |
| Object tabs | Displays the object types with the number of objects affected in parentheses. You can display the actions for virtual machines, host systems, clusters, vCenter instances, and datastores. |
| Badge | Select the Health, Risk, or Efficiency badge to display alerts on your objects. Health alerts require immediate attention. Risk alerts require attention in the immediate future. Efficiency alerts require your input to reclaim wasted space or to improve the performance of your objects. For each badge, you can view critical, immediate, and warning alerts.<br>• Health Status. With the Health badge selected, displays the number of affected objects and a summary of their health based on the alerts that triggered on the object. Lists the objects that have the worst health, and the number of alerts that triggered on each object.<br>• Risk Status. With the Risk badge selected, displays the number of affected objects and a summary of their risk based on the alerts that triggered on the object. Lists the objects that have the highest, and the number of alerts that triggered on each object.<br>• Efficiency Status. With the Efficiency badge selected, displays the number of affected objects. Lists the objects that have the lowest efficiency based on the alerts that triggered on the object, and the number of alerts that triggered on each object. |
| Search filter | Narrows the scope of the objects that appear. Enter a character or a number to search and display an object. When a filter is active, the name of the filter appears below the Search filter text box. |

### Recommended Actions Widget Toolbar Options

The toolbar allows you to address an alert, and to filter the alert list.

| Option | Description |
|---|---|
| Cancel Alert | Cancels the selected alert.<br>You cancel alerts when you do not need to address them. Canceling the alert does not cancel the underlying condition that generated the alert. Canceling alerts is effective if the alert is generated by triggered fault and event symptoms because these symptoms are triggered again only when subsequent faults or events occur on the monitored objects. If the alert is generated based on metric or property symptoms, the alert is canceled only until the next collection and analysis cycle. If the violating values are still present, the alert is generated again. |
| Suspend | Suspends an alert for a specified number of minutes.<br>You suspend alerts when you are investigating an alert and do not want the alert to affect the health, risk, or efficiency of the object while you are working. If the problem persists after the elapsed time, the alert is reactivated and it will again affect the health, risk, or efficiency of the object.<br>The user who suspends the alert becomes the assigned owner. |
| Quick Filter | Narrows the search to one of the available filter types. For example, you can display all alerts that are related to the Compliance Alert Subtype. |

### Recommended Actions Widget Data Grid Options

The data grid displays the alerts that triggered on your objects. To resolve the problems indicated by the alerts, you can link to the alerts and the objects on which the alerts triggered.

| Option | Description |
|---|---|
| Criticality | Criticality is the level of importance of the alert in your environment. The alert criticality appears in a tooltip when you hover the mouse over the criticality icon.<br>The level is based on the level assigned when the alert definition was created, or on the highest symptom criticality, if the assigned level was **Symptom Based**. |
| Actionable | When an alert has an associated action, you can run the action on the object to resolve the alert. |

| Option | Description |
|---|---|
| Suggested Fix | Describes the recommendation to resolve the problem. For example, for Compliance alerts, the recommendation instructs you to use the *vSphere Hardening Guide* to resolve the problem.<br>You can view other available recommendations and their associated actions, if any, to resolve the problem when you click the drop-down menu. |
| Name | Name of the object for which the alert was generated, and the object type, which appears in a tooltip when you hover the mouse over the object name.<br>Click the object name to view the object details tabs where you can begin to investigate any additional problems with the object. |
| Alert | Name of the alert definition that generated the alert.<br>Click the alert name to view the alert details tabs where you can begin troubleshooting the alert. |
| Alert Type | Describes the type of alert that triggered on the selected object, and helps you categorize the alerts so that you can assign certain types of alerts to specific system administrators. For example, Application, Virtualization/ Hypervisor, Hardware, Storage, and Network. |
| Alert Subtype | Describes additional information about the type of alert that triggered on the selected object, and helps you categorize the alerts to a more detailed level than Alert Type, so that you can assign certain types of alerts to specific system administrators. For example, Availability, Performance, Capacity, Compliance, and Configuration. |
| Time | Date and time that the alert triggered. |
| Alert ID | Unique identification for the alert. This column is hidden by default. |

### Recommended Actions Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |

## Risk Widget

The risk widget is the status of the risk-related alerts for the objects it is configured to monitor. Risk alerts in VCF Operations usually indicate that you should investigate problems in the near future. You can create one or more risk widgets for objects that you add to your custom dashboards.

### How the Risk Widget and Configuration Options Work

You can add the risk widget to one or more custom dashboards and configure it to display data that is important to the dashboard users.

The state of the badge is based on your alert definitions. Click the badge to see the **Summary** tab for objects or groups configured in the widget. From the **Summary** tab, you can begin determining what caused the current state. If the widget is configured for an object that has descendants, you should also check the state of descendants. Child objects might have alerts that do not impact the parent.

If the Badge Mode configuration option is set to Off, the badge and a chart appear. The type of chart depends on the object type that the widget is configured to monitor.

- A population criticality chart displays the percentage of group members with critical, immediate, and warning risk alerts generated over time, if the monitored object is a group.
- A trend line displays the risk status of the monitored object for all other object types.

If the Badge Mode is set to On, only the badge appears.

You edit a risk widget after you add it to a dashboard. The changes you make to the options create a custom widget that provides information about an individual object, a custom group of objects, or all the objects in your environment.

### Where You Find the Risk Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Risk Widget Display Options

The Risk Widget displays a risk badge. The widget also displays a risk trend chart when not in badge mode.

| Option | Description |
|---|---|
| Risk Badge | Status of the objects configured for this instance of the widget. Click the badge to open the **Alerts** tab for the object that provides data to the widget. |
| Risk Trend | Displays a chart, depending on the selected or configured object. The charts vary, depending on whether the monitored object is a group, a descendent object, or an object that provides resources to other objects. The chart appears only if the **Badge Mode** configuration option is off. If the **Badge Mode** is on, only the badge appears. |

### Risk Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| Badge Mode | Determines whether the widget displays only the badge, or the badge and a weather map or trend chart.<br>Select one of the following options:<br>• On. Only the badge appears in the widget.<br>• Off. The badge and a chart appear in the widget. The chart provides additional information about the state of the object. |
| **Input Data** | |
| Object | Search for objects in your environment and select the object on which you are basing the widget data. You can also click the **Add Object** icon and select an object from the object list. You can use the **Filter** text box to refine the object list and the **Tag Filter** pane to select an object based on tag values. |

## Rolling View Chart Widget

The Rolling View Chart widget cycles through selected metrics at an interval that you define and shows one metric graph at a time. Miniature graphs, which you can expand, appear for all selected metrics at the bottom of the widget.

## How the Rolling View Chart Widget and Configuration Options Work

The Rolling View Chart widget shows a full chart for one selected metric at a time. Miniature graphs for the other selected metrics appear at the bottom of the widget. You can click a miniature graph to see the full graph for that metric, or set the widget to rotate through all selected metrics at an interval that you define. The key in the graph indicates the maximum and minimum points on the line chart.

You edit a Rolling View Chart widget after you add it to a dashboard. The changes you make to the options create a custom chart to meet the needs of the dashboard users.

## Where You Find the Rolling View Chart Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

## Rolling View Chart Widget Toolbar Options

On the title bar of the widget, click the **Show Toolbar** icon to access the toolbar options.

The toolbar contains icons that you can use to change the view of the graphs.

| Option | Description |
|---|---|
| **Trend Line** | Shows or hides the line and data points that represents the metric trend. The trend line filters out metric noise along the timeline by plotting each data point relative to the average of its adjoining data points. |
| **Dynamic Thresholds** | Shows or hides the calculated dynamic threshold values for a 24-hour period. |
| **Show Entire Period Dynamic Thresholds** | Shows or hides dynamic thresholds for the entire time period of the graph. |
| **Anomalies** | Shows or hides anomalies. Time periods when the metric violates a threshold are shaded. Anomalies are generated when a metric crosses a dynamic or static threshold, either above or below. |
| **Zoom to Fit** | Changes all graphs to show the entire time period and value range. |
| **Zoom the view** | Click this icon and drag to outline a part of the hierarchy. The display zooms to show only the outlined section. |
| **Pan** | Click this icon and click and drag the hierarchy to show different parts of the hierarchy. |
| **Show Data Values** | After you click the **Show data point tips** icon to retrieve the data, click this icon and point to a graphed data point to show its time and exact value. In non-split mode, you can hover over a metric in the legend to show the full metric name, the names of the adapter instances (if any) that provide data for the resource to which the metric belongs, the current value, and the normal range. If the metric is currently alarming, the text color in the legend changes to yellow or red, depending on your color scheme. Click a metric in the legend to highlight the metric in the display. Clicking the metric again toggles its highlighted state. |
| **Date Controls** | Use the date selector to limit the data that appears in each chart to the time period you are examining.<br>Select **Dashboard Time** to activate the dashboard time panel. The option chosen in the dashboard time panel is effective. The default time is 6 hours.<br>**Dashboard Time** is the default option. |

## Rolling View Chart Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

The **Input Transformation** section provides options to transform the input for the widget.

The **Output Data** section provides options to select object types on which you are basing the widget data.

The **Output Filter** section provides options to restrict the widget data based on the selected filter criteria.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| Auto Transition Interval | Time interval for a switch between charts in the widget. |
| **Input Data** | |

| Option | Description |
| --- | --- |
| Metrics | Select metrics on which you want to base the widget data. You can select an object and pick its metrics.<br>1. Click the **Add New Metrics** icon to add metrics for the widget data. Select an object to view its metric tree and pick metrics for the object. The picked metrics appear in a list in this section.<br>The metric tree shows common metrics for several objects when you click the **Show common metrics** icon.<br>While selecting objects for which you want to pick metrics, you can use the **Filter** text box to search for objects. You can also expand the **Tag Filter** pane on the left hand side to select one or more object tag values. A list of objects with the selected tag values appears. If you select more than one value for the same tag, you can choose objects that have any of the tags applied. If you select more than one value for different tags, you can choose only the objects that have all the tags applied.<br>2. Optionally, select metrics from the list and click the **Remove Selected Metrics** icon to remove the selected metrics.<br>Click the **Select All** icon to select all the metrics in the list.<br>Click the **Clear Selection** icon to clear your selection of metrics in the list.<br>You can define measurement units for the metrics in the list. Double-click a metric box in the list, select a measurement unit in the **Unit** drop-down menu, and click **Update**. |
| Objects | Select objects on which you want to base the widget data.<br>1. Click the **Add New Objects** icon and select objects in the pop-up window. The selected objects appear in a list in this section.<br>While selecting objects, you can use the **Filter** text box to search for objects. You can also expand the **Tag Filter** pane on the left hand side to select one or more object tag values. A list of objects with the selected tag values appears. If you select more than one value for the same tag, you can choose objects that have any of the tags applied. If you select more than one value for different tags, you can choose only the objects that have all the tags applied.<br>2. Optionally, select objects from the list and click the **Remove Selected Objects** icon to remove the selected objects.<br>Click the **Select All** icon to select all the objects in the list.<br>Click the **Clear Selection** icon to clear your selection of objects in the list. |
| All | If you select this option, the widget data is based on all the objects in your environment. The following sections provide options to refine the objects for the widget data. |
| **Input Transformation** | |
| Relationship | Transform the input for the widget based on the relationship of the objects. For example, if you select the **Children** check box and a **Depth** of **1**, the child objects are the transformed inputs for the widget. |
| **Output Data** | |
| Empty drop-down menu | Specifies a list with attributes to display. |

| Option | Description |
|---|---|
| | Add metrics based on object types. The objects corresponding to the selected metrics are the basis for the widget data.<br>1. Click the **Add New Metrics** icon to add metrics based on object types. The metrics that you add appear in a list in this section.<br>While selecting object types for which you want to pick metrics, you can filter the object types by adapter type to pick an object type. On the metrics pane, click the **Select Object** icon to select an object for the object type. Pick metrics of the selected object from the metric tree.<br>For example, you can select the **Datacenter** object type, click the **Select Object** icon to display the list of data centers in your environment, and pick metrics of the selected data center.<br>2. Optionally, you can define measurement units for the metrics in the list. Double-click a metric box in the list, select a measurement unit in the **Unit** drop-down menu, and click **Update**. |
| **Output Filter** | |
| | Refine the widget data further based on the filter criteria for object types. The widget data is based on the objects for the filtered object types.<br>If the objects have a tag filter applied in the **Basic** subsection, you define filter criteria for the object types of the objects with tag filter applied. If the objects with tag filter applied do not belong to any of the object types in this filter criteria, the widget skips this filter and includes all the objects with tag filter applied.<br>If the objects have an input transformation applied, you define filter criteria for the object types of the transformed objects.<br>1. In the first drop-down menu, select an object type.<br>2. In the second drop-down menu, select the option based on which you want to define the filter criteria. For example, if you select **Metrics** for the **Datacenter** object type, you can define a filter criteria based on the value of a specific metric for data centers.<br>3. In the drop-down menus and text boxes that appear, select or enter values to filter the objects.<br>4. To add more filter criteria, click **Add**.<br>5. To add another filter criteria set, click **Add another criteria set**. |

## Scoreboard Widget

The Scoreboard widget shows the current value for each metric of the objects that you select.

### How the Scoreboard Widget and Configuration Options Work

Each metric appears in a separate box or as a gauge chart depending on the mode you choose. The value of the metric determines the color of the box or the gauge chart. You define the ranges for each color when you edit the widget. You can customize the widget to use a sparkline chart to show the trend of changes of each metric. If you point to a box, the widget shows the source object and metric data. Icons in the box indicate the level of criticality.

You edit a Scoreboard widget after you add it to a dashboard. The widget can display metrics of the objects selected during editing of the widget or selected on another widget. When the Scoreboard widget is not in Self Provider mode, it shows metrics defined in a configuration XML file that you select in the Metric Configuration. It shows 10 predefined metrics if you do not select an XML file or if the type of the selected object is not defined in the XML file.

For example, you can configure the Scoreboard widget to use the sample Scoreboard metric configuration and to receive objects from the Topology Graph widget. When you select a host on a Topology Graph widget, the Scoreboard widget shows the workload, memory, and CPU usage of the host.

To set a source widget that is on the same dashboard, you must use the Widget Interactions menu when you edit a dashboard. To set a source widget that is on another dashboard, you must use the Dashboard Navigation menu when you edit the source dashboard.

## Where You Find the Scoreboard Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

## Scoreboard Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

The **Input Transformation** section provides options to transform the input for the widget.

The **Output Data** section provides options to select object types on which you are basing the widget data.

The **Output Filter** section provides options to restrict the widget data based on the selected filter criteria.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |

| Option | Description |
|---|---|
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options.<br>When the Scoreboard widget is not in self-provider mode, it shows metrics defined in a configuration XML file that you select in the Metric Configuration. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| View Mode | Select either Classic or Gauge.<br>If you choose Classic, you can view the metric in a separate box. The value of the metric determines the color of the box.<br>If you select Gauge, you can view a gauge chart of the thresholds of the metric based on the values that you set for each color.<br>You can set the color and maximum value from the datagrid. |
| Round Decimals | Select the number of decimal places to round the scores that the widget displays. |
| Box Columns | Select the number of columns that appear in the widget. |
| Layout Mode | Select a Fixed Size or Fixed View layout. |
| Fixed Size<br>Fixed View | Use these options to customize the size of the box for each object. |
| Old metric values | Select **Show** if you want the widget to show the previous value of the metric, if the current value is not available. Select **Hide** to hide the previous value of the metric, if the current value is not available. |
| Visual Theme | Select a predefined visual style for each instance of the widget. |
| Max Scores Count | Use these menus to customize the format of the scores that the widget displays. |
| Show | Select one or more of the following items to display in the widget:<br>• Select **Object Name** to display the name of the object in the widget.<br>• Select **Metric Name** to display the name of the metric in the widget.<br>• Select **Metric Unit** to display the metric unit in the widget.<br>• Select **Sparkline** to display the Sparkline chart for each metric. |
| Period Length | Select a length of time for the statistic information that the sparkline chart displays. |
| Show DT | Select an option to show or hide the dynamic threshold for the sparkline chart. |
| **Input Data** | |

| Option | Description |
| --- | --- |
| Metrics | Select metrics on which you want to base the widget data. You can select an object and pick its metrics.<br>1. Click the **Add New Metrics** icon to add metrics for the widget data. Select an object to view its metric tree and pick metrics for the object. The picked metrics appear in a list in this section.<br>The metric tree shows common metrics for several objects when you click the **Show common metrics** icon.<br>While selecting objects for which you want to pick metrics, you can use the **Filter** text box to search for objects. You can also expand the **Tag Filter** pane on the left hand side to select one or more object tag values. A list of objects with the selected tag values appears. If you select more than one value for the same tag, you can choose objects that have any of the tags applied. If you select more than one value for different tags, you can choose only the objects that have all the tags applied.<br>2. Optionally, select metrics from the list and click the **Remove Selected Metrics** icon to remove the selected metrics.<br>Click the **Select All** icon to select all the metrics in the list.<br>Click the **Clear Selection** icon to clear your selection of metrics in the list.<br>Optionally, you can customize a metric and apply the customization to other metrics in the list.<br>1. Double-click a metric box in the list to customize the metric and click **Update**.<br>You can use the **Box Label** text box to customize the label of a metric box.<br>You can use the **Unit** text box to define a measurement unit of each metric.<br>You can use the **Color Method** option to define a coloring criteria for each metric. If this option is set to **Custom**, you can enter color values in the **Yellow**, **Orange**, and **Red** text boxes. You can also set coloring by symptom definition. If you do not want to use color, select **None**.<br>For example, to view the remaining memory capacity of a VM, select **Virtual Machine** as an object type, expand the **Memory** from the metric tree and double-click **Capacity Remaining(%)**. Define a meaningful label name and measurement unit to help you when you observe the metrics. You can select **Custom** from the **Color Method** drop-down menu and specify different values for each color, for example 50 for **Yellow**, 20 for **Orange**, and 10 for **Red**.<br>You can use the **Link to** option to add links to external and internal pages. Internal links open in the same tab. External links open in a new tab. Examples of external links are URLs whose hostname does not match with the current VCF Operations instance hostname. Internal links are URLs whose hostname matches the current VCF Operations instance hostname or starts with *index.action*.<br>2. Select a metric and click the **Apply to All** icon to apply the customization for the selected metric to all the metrics in the list. |

| Option | Description |
| --- | --- |
| Objects | Select objects on which you want to base the widget data.<br>1. Click the **Add New Objects** icon and select objects in the pop-up window. The selected objects appear in a list in this section.<br>While selecting objects, you can use the **Filter** text box to search for objects. You can also expand the **Tag Filter** pane on the left hand side to select one or more object tag values. A list of objects with the selected tag values appears. If you select more than one value for the same tag, you can choose objects that have any of the tags applied. If you select more than one value for different tags, you can choose only the objects that have all the tags applied.<br>2. Optionally, select objects from the list and click the **Remove Selected Objects** icon to remove the selected objects.<br>Click the **Select All** icon to select all the objects in the list.<br>Click the **Clear Selection** icon to clear your selection of objects in the list. |
| All | If you select this option, the widget data is based on all the objects in your environment. The following sections provide options to refine the objects for the widget data. |
| **Input Transformation** | |
| Relationship | Transform the input for the widget based on the relationship of the objects. For example, if you select the **Children** check box and a **Depth** of **1**, the child objects are the transformed inputs for the widget. |
| **Output Data** | |
| Empty drop-down menu | Specifies a list with attributes to display. |

| Option | Description |
| --- | --- |
| | Add metrics based on object types. The objects corresponding to the selected metrics are the basis for the widget data.<br>1. Click the **Add New Metrics** icon to add metrics based on object types. The metrics that you add appear in a list in this section.<br>While selecting object types for which you want to pick metrics, you can filter the object types by adapter type to pick an object type. On the metrics pane, click the **Select Object** icon to select an object for the object type. Pick metrics of the selected object from the metric tree.<br>For example, you can select the **Datacenter** object type, click the **Select Object** icon to display the list of data centers in your environment, and pick metrics of the selected data center.<br>2. Optionally, select metrics from the list and click the **Remove Selected Metrics** icon to remove the selected metrics.<br>Click the **Select All** icon to select all the metrics in the list.<br>Click the **Clear Selection** icon to clear your selection of metrics in the list.<br>Optionally, you can customize a metric and apply the customization to other metrics in the list.<br>1. Double-click a metric box in the list to customize the metric and click **Update**.<br>You can use the **Box Label** text box to customize the label of a metric box.<br>You can use the **Unit** text box to define a measurement unit of each metric.<br>You can use the **Max Value** option to define the maximum value to be displayed in the gauge chart. To set the maximum value, you can either enter an absolute max value or select a metric and the current value of the metric is considered the maximum value in the gauge chart.<br>You can use the **Color Method** option to define a coloring criteria for each metric. If this option is set to **Custom**, you can enter color values in the **Yellow**, **Orange**, and **Red** text boxes. You can also set coloring by symptom definition. If you do not want to use color, select **None**.<br>**Note:**<br>For the gauge view, you can use only the **Custom** option.<br>For example, to view the remaining memory capacity of a VM, select **Virtual Machine** as an object type, expand the **Memory** from the metric tree and double-click **Capacity Remaining(%)**. Define a meaningful label name and measurement unit to help you when you observe the metrics. You can select **Custom** from the **Color Method** drop-down menu and specify different values for each color, for example 50 for **Yellow**, 20 for **Orange**, and 10 for **Red**.<br>You can use the **Link to** option to add links to external and internal pages. Internal links open in the same tab. External links will open in a new tab. Examples of external links are URLs whose hostname does not match with the current VCF Operations instance hostname. Internal links are URLs whose hostname matches the current VCF Operations instance hostname or starts with *index.action*.<br>2. Select a metric and click the **Apply to All** icon to apply the customization for the selected metric to all the metrics in the list. |

| Option | Description |
|---|---|
| **Output Filter** | |
| | Refine the widget data further based on the filter criteria for object types. The widget data is based on the objects for the filtered object types.<br>If the objects have a tag filter applied in the **Basic** subsection, you define filter criteria for the object types of the objects with tag filter applied. If the objects with tag filter applied do not belong to any of the object types in this filter criteria, the widget skips this filter and includes all the objects with tag filter applied.<br>If the objects have an input transformation applied, you define filter criteria for the object types of the transformed objects.<br>1. In the first drop-down menu, select an object type.<br>2. In the second drop-down menu, select the option based on which you want to define the filter criteria. For example, if you select **Metrics** for the **Datacenter** object type, you can define a filter criteria based on the value of a specific metric for data centers.<br>3. In the drop-down menus and text boxes that appear, select or enter values to filter the objects.<br>4. To add more filter criteria, click **Add**.<br>5. To add another filter criteria set, click **Add another criteria set**. |

## Scoreboard Health Widget

The Scoreboard Health widget displays color-coded health, risk, efficiency, and custom metrics scores for objects that you select.

### How the Scoreboard Health Widget and Configuration Options Work

The icons for each object are color coded to give a quick indication of the state of the object. You can configure the widget to display the scores of common or specific metrics of the object. You can use the symptom state color code or you can define your criteria to color the images. If you configure the widget to show the metric for objects that do not have this metric, those objects have blue icons.

You can double-click an object icon to show the Object Detail page for the object. When you point to the icon, a tool tip shows the name of the object and the name of the metric.

You edit a Scoreboard Health widget after you add it to a dashboard. To configure the widget, click the pencil at the upper-right corner of the widget window. The widget can display metrics of the objects that you select when you edit the widget, or that you select on another widget. For example, you can configure the widget to show the CPU workload of an object that you select on the Topology Graph widget. To set a source widget that is on the same dashboard, you must use the Widget Interactions menu when you edit a dashboard. To set a source widget that is on another dashboard, you must use the Dashboard Navigation menu when you edit the source dashboard.

### Where You Find the Scoreboard Health Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a

widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Scoreboard Health Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The **Configuration** section provides general configuration options for the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| Image Type | Select an image type for the metrics. |
| Metric | Select the default or custom metric. |
| Pick Metric | Active only when you select **Custom** from the **Metric** menu.<br>Use to select a custom metric for the objects that the widget displays. Click **Pick Metric** and select an object type from the Object Type pane.<br>Use the Metric Picker pane to select a metric from the metric tree and click **Select Object** to check the objects from the type that you select on the Object Types pane. |
| Use Symptom state to color chart | Select to use the default criteria to color the image. |
| Custom ranges | Use to define custom criteria to color the image. You can define a range for each color. |

| Option | Description |
|---|---|
| **Input Data** | |
| | Select objects on which you want to base the widget data.<br>1. Click the **Add New Objects** icon and select objects in the pop-up window. The selected objects appear in a list in this section.<br>While selecting objects, you can use the **Filter** text box to search for objects. You can also expand the **Tag Filter** pane on the left hand side to select one or more object tag values. A list of objects with the selected tag values appears. If you select more than one value for the same tag, you can choose objects that have any of the tags applied. If you select more than one value for different tags, you can choose only the objects that have all the tags applied.<br>2. Optionally, select objects from the list and click the **Remove Selected Objects** icon to remove the selected objects.<br>Click the **Select All** icon to select all the objects in the list.<br>Click the **Clear Selection** icon to clear your selection of objects in the list. |

## Sparkline Chart Widget

The Sparkline Chart widget displays graphs that contain metrics for an object in VCF Operations. You can use VCF Operations to create one or more graphs that contain metrics for objects that you add to your custom dashboards.

### How the Sparkline Chart Widget and Configurations Options Work

If the metrics in the Sparkline Chart are for an object that another widget provides, the object name appears at the top right of the widget. If you select a metric when you edit the widget configuration, the widget uses the metric and its corresponding object as the source for dashboard interactions. The line in the graphs represents the average value of the selected metric for the specified time period. The boxed area in the graph represents the dynamic threshold of the metric.

Point to a graph in the Sparkline Chart widget to view the value of a metric in the form of a tool tip. You can also view the maximum and minimum values on a graph. The values are displayed as orange dots.

You can add the Sparkline Chart widget to one or more custom dashboards and configure it to display data that is important to different dashboard users. The data that appears in the widget is based on the configured options for each widget instance.

The metrics shown in sparkline widget is the current value, to view the average value you can use transformations in list views or distribution charts to calculate an average. Another way to get to an average value is to double click on the sparkline to open the metric chart, click and drag to select a range, keep the mouse button depressed and hover for a few seconds, you should see a popup that has average value.

### Where You Find the Sparkline Chart Widget

The widget might be included on any of your custom dashboards. On the menu, click **Dashboards** to display a list of dashboards in the left pane.

### Sparkline Chart Widget Toolbar Options

On the title bar of the widget, click the **Show Toolbar** icon to access the toolbar options.

The toolbar contains icons that you can use to change the view of the graphs.

| Option | Description |
|---|---|
| Dashboard Navigation | You can navigate to another dashboard when the object you select is also available in the dashboard to which you want to navigate. |
| Refresh | Refreshes the widget data. |
| Time Range | Select the range for the time period to show on the graphs. You can select a period from the default time range list or select start and end dates and times.<br>Select **Dashboard Time** to activate the dashboard time panel. The option chosen in the dashboard time panel is effective. The default time is 6 hours.<br>**Dashboard Time** is the default option. |
| Remove All | Removes all graphs. |

## Sparkline Chart Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

The **Input Transformation** section provides options to transform the input for the widget.

The **Output Data** section provides options to select object types on which you are basing the widget data.

The **Output Filter** section provides options to restrict the widget data based on the selected filter criteria.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |

| Option | Description |
| --- | --- |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| Show Object Name | You can view the name of the object before the metric name in the Sparkline Chart widget.<br>• **On**. Displays the name of the object before the metric name in the widget.<br>• **Off**. Does not display the name of the object in the widget. |
| Column Sequence | Select the order in which to display the information.<br>• **Graph First**. The metric graph appears in the first column in the widget display.<br>• **Label First**. The metric label appears in the first column in the widget display. |
| Show DT | Select an option to show or hide the dynamic threshold for the sparkline chart. |
| **Input Data** | |

| Option | Description |
| --- | --- |
| Metrics | Select metrics on which you want to base the widget data. You can select an object and pick its metrics.<br>1. Click the **Add New Metrics** icon to add metrics for the widget data. Select an object to view its metric tree and pick metrics for the object. The picked metrics appear in a list in this section.<br>The metric tree shows common metrics for several objects when you click the **Show common metrics** icon.<br>While selecting objects for which you want to pick metrics, you can use the **Filter** text box to search for objects. You can also expand the **Tag Filter** pane on the left hand side to select one or more object tag values. A list of objects with the selected tag values appears. If you select more than one value for the same tag, you can choose objects that have any of the tags applied. If you select more than one value for different tags, you can choose only the objects that have all the tags applied.<br>2. Optionally, select metrics from the list and click the **Remove Selected Metrics** icon to remove the selected metrics.<br>Click the **Select All** icon to select all the metrics in the list.<br>Click the **Clear Selection** icon to clear your selection of metrics in the list.<br>Optionally, you can customize a metric and apply the customization to other metrics in the list.<br>1. Double-click a metric box in the list to customize the metric and click **Update**.<br>You can use the **Box Label** text box to customize the label of a metric box.<br>You can use the **Unit** text box to define a measurement unit of each metric.<br>You can use the **Color Method** option to define a coloring criteria for each metric. If this option is set to **Custom**, you can enter color values in the **Yellow**, **Orange**, and **Red** text boxes. You can also set coloring by symptom definition. If you do not want to use color, select **None**.<br>For example, to view the remaining memory capacity of a VM, select **Virtual Machine** as an object type, expand the **Memory** from the metric tree and double-click **Capacity Remaining(%)**. Define a meaningful label name and measurement unit to help you when you observe the metrics. You can select **Custom** from the **Color Method** drop-down menu and specify different values for each color, for example 50 for **Yellow**, 20 for **Orange**, and 10 for **Red**.<br>2. Select a metric and click the **Apply to All** icon to apply the customization for the selected metric to all the metrics in the list. |

| Option | Description |
| --- | --- |
| Objects | Select objects on which you want to base the widget data.<br>1. Click the **Add New Objects** icon and select objects in the pop-up window. The selected objects appear in a list in this section.<br>While selecting objects, you can use the **Filter** text box to search for objects. You can also expand the **Tag Filter** pane on the left hand side to select one or more object tag values. A list of objects with the selected tag values appears. If you select more than one value for the same tag, you can choose objects that have any of the tags applied. If you select more than one value for different tags, you can choose only the objects that have all the tags applied.<br>2. Optionally, select objects from the list and click the **Remove Selected Objects** icon to remove the selected objects.<br>Click the **Select All** icon to select all the objects in the list.<br>Click the **Clear Selection** icon to clear your selection of objects in the list. |
| All | If you select this option, the widget data is based on all the objects in your environment. The following sections provide options to refine the objects for the widget data. |
| **Input Transformation** | |
| Relationship | Transform the input for the widget based on the relationship of the objects. For example, if you select the **Children** check box and a **Depth** of **1**, the child objects are the transformed inputs for the widget. |
| **Output Data** | |
| Empty drop-down menu | Specifies a list with attributes to display. |

| Option | Description |
| --- | --- |
| | Add metrics based on object types. The objects corresponding to the selected metrics are the basis for the widget data.<br>Click the **Add New Metrics** icon to add metrics for the widget data. Select an object to view its metric tree and pick metrics for the object. The picked metrics appear in a list in this section.<br>The metric tree shows common metrics for several objects when you click the **Show common metrics** icon.<br>While selecting objects for which you want to pick metrics, you can use the **Filter** text box to search for objects. You can also expand the **Tag Filter** pane on the left hand side to select one or more object tag values. A list of objects with the selected tag values appears. If you select more than one value for the same tag, you can choose objects that have any of the tags applied. If you select more than one value for different tags, you can choose only the objects that have all the tags applied.<br>Optionally, you can customize a metric and apply the customization to other metrics in the list.<br>1. Double-click a metric box in the list to customize the metric and click **Update**.<br>You can use the **Box Label** text box to customize the label of a metric box.<br>You can use the **Unit** text box to define a measurement unit of each metric.<br>You can use the **Color Method** option to define a coloring criteria for each metric. If this option is set to **Custom**, you can enter color values in the **Yellow**, **Orange**, and **Red** text boxes. You can also set coloring by symptom definition. If you do not want to use color, select **None**.<br>For example, to view the remaining memory capacity of a VM, select **Virtual Machine** as an object type, expand the **Memory** from the metric tree and double-click **Capacity Remaining(%)**. Define a meaningful label name and measurement unit to help you when you observe the metrics. You can select **Custom** from the **Color Method** drop-down menu and specify different values for each color, for example 50 for **Yellow**, 20 for **Orange**, and 10 for **Red**.<br>2. Select a metric and click the **Apply to All** icon to apply the customization for the selected metric to all the metrics in the list. |
| **Output Filter** | |
| | Refine the widget data further based on the filter criteria for object types. The widget data is based on the objects for the filtered object types.<br>If the objects have an input transformation applied, you define filter criteria for the object types of the transformed objects.<br>1. In the first drop-down menu, select an object type.<br>2. In the second drop-down menu, select the option based on which you want to define the filter criteria. For example, if you select **Metrics** for the **Datacenter** object type, you can define a filter criteria based on the value of a specific metric for data centers.<br>3. In the drop-down menus and text boxes that appear, select or enter values to filter the objects.<br>4. To add more filter criteria, click **Add**.<br>5. To add another filter criteria set, click **Add another criteria set**. |

## Tag Picker Widget

The Tag Picker widget lists all available object tags.

### How the Tag Picker Widget and Configuration Options Work

With the Tag Picker widget, you can check the list of the object tags. You can use the widget to filter the information that another widget shows. You can select one or more tags from the object tree or search for tags, and the destination widget displays information about the objects with this tag. For example, you can select **Object Types** > **Virtual Machine** on the Tag Picker widget to observe statistic information about the VMs on the Environment Status widget.

You edit a Tag Picker widget after you add it to a dashboard. To configure the widget, click the pencil in the upper right of the widget window. You can configure the Tag Picker widget to send information to another widget on the same dashboard or on another dashboard. To set a receiver widget that is on the same dashboard, use the **Widget Interactions** menu when you edit a dashboard. To set a receiver widget that is on another dashboard, use the **Dashboard Navigation** menu when you edit a source dashboard. You can configure two Tag Picker widgets to interact when they are on different dashboards.

### Where You Find the Tag Picker Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Tag Picker Widget Toolbar Options

On the title bar of the widget, click the **Show Toolbar** icon to access the toolbar options.

| Option | Description |
|---|---|
| Collapse All | Close all expanded tags and tag values. |
| Deselect All | Remove all filtering and view all objects in the widget. |
| Tag Picker | Select an object from your environment. |
| Dashboard Navigation | **Note:**<br>Appears on the source widget and when the destination widget is on another dashboard.<br>Use to explore the information on another dashboard. |

### Tag Picker Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |

## Text Display Widget

You can use the Text Display widget to show text in the user interface. The text appears in the Text Display widget on the dashboard.

The Text Display widget can read text from a Web page or text file. You specify the URL of the Web page or the name of the text file when you configure the Text widget. To use the Text Display widget to read text files you must set a property in the *web.properties* file to specify the root folder that contains the file.

You can enter content in the Text Display widget in plain text or rich text format based on the view mode that you configure. Configure the Text Display widget in HTML view mode to display content in rich text format. Configure the Text Display widget in Text mode to display content in plain text format.

The Text Display widget can display websites that use the HTTPS protocol. The behavior of the Text Display widget with websites that use HTTP, depends on the individual settings of the websites.

> **Note:**
> If the webpage that you are linking to has `X-Frame-Options` set to `sameorigin`, which denies rendering a page in an iframe, the Text Display widget cannot display the contents of the webpage.

### How the Text Display Widget Configuration Options Work

You can configure the widget in the Text view mode or HTML view mode. In the HTML view mode, you can click **Edit** in the widget and use the rich text editor to add content.

If you configure the widget to use Text view mode, you can specify the path to the directory that contains the files to read or you can provide a URL. The content in the URL will be shown as text. If you do not specify a URL or text file, you can add content in the widget. Double-click the widget and enter content in plain text.

You can also use command-line interface (CLI) commands to add file content to the Text Display widget.

- To view a list of parameters, run the `file -h|import|export|delete|list txtwidget` command.
- To import text or HTML content, run the `import txtwidget input-file [--title title] [--force]` command.
- To export the content to the file, run the `export txtwidget all|title[{,title}] [output-dir]` command.
- To delete imported content, run the `delete txtwidget all|title[{,title}]` command.
- To view the titles of the content, run the `list txtwidget` command.

## Where You Find the Text Display Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

## Text Display Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |

| Option | Description |
|---|---|
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| View mode | Display text in text or rich text format. You can configure the widget in HTML view mode only when the **URL** and **File** fields are blank. |
| URL | Enter the URL. |
| File | Navigate to the file that contains the source text file by clicking the **Select** button.<br>To add, edit, and remove source text files, go to the **Text Widget Content** tile in the Configuration page. From the left menu, click **Operations** > **Configurations**, and then click the **Text Widget Content**tile from the VCF Operations user interface. |
| Test | Validates the correctness of the text file or URL that you enter. |

## Time Remaining Widget

The Time Remaining widget displays how much time remains before the resources of the object are exhausted.

VCF Operations calculates the percentage by object type based on historical data for the pattern of use for the object type. You can use the time remaining percentage to plan provisioning of physical or virtual resources for the object or rebalance the workload in your virtual infrastructure.

### Where You Find the Time Remaining Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Time Remaining Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| **Input Data** | |
| Object | Search for objects in your environment and select the object on which you are basing the widget data. You can also click the **Add Object** icon and select an object from the object list. You can use the **Filter** text box to refine the object list and the **Tag Filter** pane to select an object based on tag values. |

## Top Alerts Widget

Top alerts are the alerts with the greatest significance on the objects it is configured to monitor in VCF Operations. These are the alerts most likely to negatively affect your environment and you should evaluate and address them.

### How the Top Alerts Widget and Configuration Options Work

You can add the top alerts widget to one or more custom dashboards and configure it to display data that is important to different dashboard users. The data that appears in the widget is based on the configured options for each widget instance.

You edit a top alerts widget after you add it to a dashboard. The changes you make to the options help create a custom widget to meet the needs of the dashboard users.

### Where You Find the Top Alerts Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Top Alerts Widget Display Options

The Top Alerts widget includes the short description of alerts configured for the widget. The alert name opens a secondary window from which you can link to the alert details. In the alert details, you can begin resolving the alerts.

| Option | Description |
|---|---|
| Alert name | Name of the generated alert. Click the name to open the alert details. |
| Alert description | Number of affected objects, and the number of recommendations and the best recommendation to resolve the alert. |

### Top Alerts Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

The **Input Transformation** section provides options to transform the input for the widget.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |

| Option | Description |
|---|---|
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| Impact Badge | Select the badge for which you want alerts to appear.<br>The affected badge is configured when you configure the alert definition. |
| Number of Alerts | Select the maximum number of alerts to display in the widget. |
| **Input Data** | |
| Object | Search for objects in your environment and select the object on which you are basing the widget data. You can also click the **Add Object** icon and select an object from the object list. You can use the **Filter** text box to refine the object list and the **Tag Filter** pane to select an object based on tag values. |
| **Input Transformation** | |
| Relationship | Transform the input for the widget based on the relationship of the objects. For example, if you select the **Children** check box and a **Depth** of **1**, the child objects are the transformed inputs for the widget. |

## Top-N Widget

The Top-N widget displays the top n results from analysis of an object or objects that you select.

### How the Top-N Widget and Configuration Options Work

You can select an object when you configure the Top-N widget or you can select an object on another widget. The widget shows an analysis of the applications, alerts, and metrics of an object and its child objects depending on how you configure the widget. The widget can show an analysis of the current values or values over a period of time. You can receive detailed information about each object on the widget. When you double-click an object, the Object Detail page appears.

You can configure a widget to receive data from another widget by selecting **Off** for Self Provider. You can configure a widget to display results from analysis of an object that you select on the source widget.

For example, you can select a host on a Topology widget and observe the metric analysis of the virtual machines on the host. To set a receiver widget that is on the same dashboard, use the **Widget Interactions** menu when you edit a dashboard. To set a receiver widget that is on another dashboard, use the **Dashboard Navigation** menu when you edit a source dashboard.

### Where You Find the Top-N Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a

widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Top-N Widget Toolbar Options

On the title bar of the widget, click the **Show Toolbar** icon to access the toolbar options.

The toolbar contains icons that you can use to change the view of the graphs.

| Icon | Description |
|---|---|
| Dashboard Navigation | Takes you to a predefined object. For example, when you select a datastore from the data grid and click **Dashboard Navigation**, you can open the datastore in the vSphere Web Client. |
| Select Date Range | Limits the alerts that appear in the list to the selected date range.<br>Select **Dashboard Time** to activate the dashboard time panel. The option chosen in the dashboard time panel is effective. The default time is 6 hours. |
| Object details | Select an object and click this icon to show the Object Detail page for the object. |
| Display Filtering Criteria | Shows the filtering settings for the widget in a pop-up window. |
| Percentile | Filters and displays objects based on the percentile entered while configuring the widget. You can change the percentile using the drop-down option.<br>**Note:**<br>The Percentile option is activated in the toolbar of the widget after you configure the widget with the Percentile option. To configure the widget with the Percentile option, navigate to **Configurations** > **Top-N Options** > **Metric Analysis** > **Percentile**. |

### Top-N Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

The **Input Transformation** section provides options to transform the input for the widget.

The **Output Data** section provides options to select object types on which you are basing the widget data.

The **Output Filter** section provides options to restrict the widget data based on the selected filter criteria.

The **Additional Columns** section provides options to select metrics that are displayed as additional columns in the widget.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |

| Option | Description |
| --- | --- |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| Redraw Rate | Set the redraw rate. |
| Bars Count | Select the number of top results. |
| Round Decimals | Select the number of decimals to round the scores displayed in the widget. |
| Filter old metrics | Select or deselect whether the analysis includes old metric values. |
| Application Health and Performance | • Top Least Healthy. The top n results from an analysis of the object or objects that are the least healthy.<br>• Top Most Healthy. The top n results from an analysis of the object or objects that are the most healthy.<br>• Top Most Volatile. The sorted list of values based on the standard deviation of values for several alerts over time.<br>Select the criteria for analysis of the objects. |
| Alert Analysis | Select the criteria for analysis of the alerts. |
| Metric Analysis | If you select this option, you must select a metric in the **Output Data** section.<br>• Top Highest Utilization. A list of objects with similar object types that have the highest utilization on configuring usage metrics like CPU usage and memory usage.<br>• Top Lowest Utilization. A list of objects with similar object types that have the lowest utilization on configuring usage metrics like CPU usage and memory usage.<br>• Top Abnormal States. The objects are ordered by the duration of all alarms that are triggered on the selected metric for a selected interval.<br>• Top Highest Volatility. The sorted list of values based on the standard deviation of values for several alerts over time.<br>• Percentile. Objects are filtered based on the percentile entered.<br>Select the criteria for analysis of the metric that you select from the metric tree. |
| **Input Data** | |

| Option | Description |
| --- | --- |
| Objects | Select objects on which you want to base the widget data.<br>1. Click the **Add New Objects** icon and select objects in the pop-up window. The selected objects appear in a list in this section.<br>While selecting objects, you can use the **Filter** text box to search for objects. You can also expand the **Tag Filter** pane on the left hand side to select one or more object tag values. A list of objects with the selected tag values appears. If you select more than one value for the same tag, you can choose objects that have any of the tags applied. If you select more than one value for different tags, you can choose only the objects that have all the tags applied.<br>2. Optionally, select objects from the list and click the **Remove Selected Objects** icon to remove the selected objects.<br>Click the **Select All** icon to select all the objects in the list.<br>Click the **Clear Selection** icon to clear your selection of objects in the list. |
| All | If you select this option, the widget data is based on all the objects in your environment. The following sections provide options to refine the objects for the widget data. |
| **Input Transformation** | |
| Relationship | Transform the input for the widget based on the relationship of the objects. For example, if you select the **Children** check box and a **Depth** of **1**, the child objects are the transformed inputs for the widget. |
| **Output Data** | |
| | Select an object type in your environment on which you want to base the widget data.<br>1. Click the **Add Object Type** icon to search for and add an object type.<br>When you search for object types, you can filter the types in the list by selecting a type from the **Adapter Type** drop-down menu or by using the **Filter** text box.<br>2. Optionally, select the object type from the list and click the **Delete Object Type** icon to remove the selected object type.<br>If the objects have an input transformation applied, the transformed objects are the basis for the widget data. |
| Metric | Select a common metric or a metric for the selected object type in the list. The metric is the basis for the widget data. |
| Label | Type in a name that displays as a label for the metric.<br>You can add a label if you have selected **Metric Analysis** > **Top Highest Utilization** or **Metric Analysis** > **Top Lowest Utilization** as Top-N options in the **Configuration** section. |
| Unit | You can define measurement units for the metrics. Select a measurement unit in the **Unit** drop-down menu.<br>You can add a unit if you have selected **Metric Analysis** > **Top Highest Utilization** or **Metric Analysis** > **Top Lowest Utilization** as Top-N options in the **Configuration** section. |
| Maximum | Specify the maximum value based on which the bar size is calculated.<br>You can add a maximum value if you have selected any of the options under **Metric Analysis**. |

| Option | Description |
| --- | --- |
| Color Method | You can use the **Color Method** option to define a coloring criteria for each metric. If this option is set to **Custom**, you can enter color values in the **Yellow**, **Orange**, and **Red** text boxes. If you do not want to use color, select None.<br>You can add color thresholds if you have selected **Metric Analysis** > **Top Highest Utilization**, **Metric Analysis** > **Top Lowest Utilization**, or **Metric Analysis** > **Percentile** as Top-N options in the **Configuration** section. |
| **Output Filter** | |
| Basic | Pick tags to refine the widget data. The widget data is based on the objects that have the picked tags applied. If you pick more than one value for the same tag, the widget includes objects that have any of the tags applied. If you pick more than one value for different tags, the widget includes only the objects that have all the tags applied.<br>If the objects have an input transformation applied, you select tag values for the transformed objects. |
| Advanced | Refine the widget data further based on the filter criteria for object types. The widget data is based on the objects for the filtered object types.<br>If the objects have a tag filter applied in the **Basic** subsection, you define filter criteria for the object types of the objects with tag filter applied. If the objects with tag filter applied do not belong to any of the object types in this filter criteria, the widget skips this filter and includes all the objects with tag filter applied.<br>If the objects have an input transformation applied, you define filter criteria for the object types of the transformed objects.<br>1. In the first drop-down menu, select an object type.<br>2. In the second drop-down menu, select the option based on which you want to define the filter criteria. For example, if you select **Metrics** for the **Datacenter** object type, you can define a filter criteria based on the value of a specific metric for data centers.<br>3. In the drop-down menus and text boxes that appear, select or enter values to filter the objects.<br>4. To add more filter criteria, click **Add**.<br>5. To add another filter criteria set, click **Add another criteria set**. |
| **Additional Columns** | |
| | Add metrics based on object types. The selected metrics are displayed as additional columns in the widget.<br>1. Click the **Add New Metrics** icon to add metrics based on object types. The metrics that you add appear in a list in this section.<br>While selecting object types for which you want to pick metrics, you can filter the object types by adapter type to pick an object type. On the metrics pane, click the **Select Object** icon to select an object for the object type. Pick metrics of the selected object from the metric tree.<br>For example, you can select the **Datacenter** object type, click the **Select Object** icon to display the list of data centers in your environment, and pick metrics of the selected data center.<br>2. Optionally, you can double-click a metric box in the list to customize the label of the metric and click **Update**. |

## Topology Graph Widget

The Topology Graph widget gives a graphical presentation of objects and their relationships in the inventory. You can customize each instance of the widget in your dashboard.

### How the Topology Graph Widget and Configuration Options Work

The Topology Graph widget helps you explore all nodes and paths connected to an object from your inventory. Connection between the objects might be a logical, physical, or network connection. The widget can display a graph that shows all of the nodes in the path between two objects, or that shows the objects related to a node in your inventory. You select the type of graph in the Exploration Mode when you configure the widget. You can select the levels of exploration between nodes in the displayed graph by using **Relationship** check boxes when you edit the widget. The widget displays all object types in the inventory by default, but you can select object types to view by using the Object View list during the configuration process. Double-clicking an object on the graph takes you to a detailed page about the object.

### Where You Find the Topology Graph Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Topology Graph Widget Toolbar Options

On the title bar of the widget, click the **Show Toolbar** icon to access the toolbar options.

| Option | Description |
|---|---|
| Action | Use to select from predefined actions for each object type. To see available predefined actions, select an object in the graph and click the toolbar to select an action. For example, when you select a datastore object in the graph, you can click **Delete Unused Snapshots for Datastore** to apply this action to the object. |
| Dashboard Navigation | Takes you to a predefined object. For example, when you select a datastore from the graph and click **Dashboard Navigation**, you can open the datastore in the vSphere Web Client. |
| Pan | Use to move the entire graph. |
| Show values on point | Provides a tool tip with parameters when you point to an object in the graph. |
| Zoom in | Zooms in the graph. |
| Zoom out | Zooms out the graph. |
| Hierarchical View | Use to switch to hierarchical view. Hierarchical view is activated only for Node Exploration mode and with selected inventory tree. |
| Graph View | Use to switch to graph view. |
| Object Detail | Select an object and click this icon to show the Object Detail page for the object. |

| Option | Description |
|---|---|
| Expand Node | Selects which object types related to your object to show on the graph. For example, if you select a virtual machine from the graph and click **Expand Node** toolbar icon and select **Host System**, the host on which the virtual machine is located is added to the graph. |
| Hide Node(s) | Use to remove a given object from the graph |
| Reset To Initial Object | Use to return to the initially displayed graph and configured object types. |
| Explore Node | Use to explore a node from a selected object in the graph. For example, if the graph displays a connection between a VM, a host, and a datastore, and you want to check the connection of the host with the other objects in the inventory, you can select the host and click **Explore Node**. |
| Status | Use to select objects based on their status or their state. |

### Topology Graph Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |

| Option | Description |
|---|---|
| Exploration Mode | Use **Node Exploration mode** to observe a selected object from an object list and the objects related to it. For example, if you select a virtual machine and select node exploration mode, the widget shows the host where the VM is placed and the datastore storing the files of the VM.<br>Use **Path Exploration mode** to observe the relation between two objects. You must select them from the Select First Object list and the Select Second Object list. For example, if you select to explore the path between a VM and a vCenter, the graph shows you both objects and all nodes in the path between the VM and server as datastore, datastore cluster, and data center.<br>**Important:**<br>To select object view is mandatory for the widget to start working in path exploration mode. |
| Show Paths | Use **All** to observe connections between a node and nodes related to it as well as connections between the nodes. For example, if you are using node exploration mode and you select to observe a VM and all objects types, the graph shows a VM connected to its datastore and host and the connection between the host and datastore. |
| | Use **Discovered Only** to observe directly related nodes. For example, if you are using node exploration mode and you select to observe a VM and all objects types, the graph will shows the VM connected to its datastore and to its host, but without the connection between the host and datastore. |
| Configuration File | The default configuration includes parent and child relationship. Drop-down options depend on the installed Solutions. You can add a new type of relationship to the Relationship pane. |
| Metric Configuration | Specifies a list with attributes to display. |
| Layout | Select whether you want a graph view or hierarchical view for the topology graph. |
| Tree type | For a hierarchical layout, select whether you want a tree type view. |
| **Input Data** | |
| Selected object | From the object list, select an object on which you want to base the widget data. |
| Degree of separation | Available only when node exploration mode is selected. Use to define the levels of exploration in node exploration mode. The lowest degree configuration shows only directly related nodes rather than higher degrees that show the inventory in details. |
| Select First Object | Available only in path exploration mode. Select the first object from the object list. |
| Select Second Object | Available only in path exploration mode. Select the second object from the object list. |
| Object view | Use to select which types of objects to observe in the graph. |
| Relationship | Select the type of relationship between objects to observe in the graph, respectively the details about your inventory . The common relationships for all objects are parent and child, but the list of relationships can vary depending on added solutions to VCF Operations. |

## View Widget

The View widget provides the VCF Operations view functionality into your dashboard.

### How the View Widget and Configuration Options Work

A view presents collected information for an object in a certain way depending on the view type. Each type of view helps you to interpret metrics, supermetrics, properties, alerts, policies, and data from a different perspective.

You can add the View widget to one or more custom dashboards and configure it to display data that is important to the dashboard users. List views can send interactions to other widgets.

### Where You Find the View Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

You can export the view as a CSV file for any view type.

### View Widget Toolbar Options

The View widget toolbar depends on the displayed view type.

| Option | Description |
|---|---|
| Export as CSV | You can export the view as a CSV file for any view type. |
| Open in External Application | Ability to link to another application for information about the object. For example, you have a List view with VMs. You can select any VM and select **Open in External Application** to open the VM in vSphere Web Client. |

| Option | Description |
|---|---|
| Time Settings | Use the time settings to select the time interval of data transformation. These options are available for all view types, except Image.<br>• Relative Date Range. Select a relative date range of data transformation.<br>• Specific Date Range. Select a specific date range of data transformation.<br>• Absolute Date Range. Select a date or time range to view data for a time unit such as a complete month or a week. For example, you can run a report on the third of every month for the previous month. Data from the first to the end of the previous month is displayed as against data from the third of the previous month to the third of the current month.<br>The units of time available are: Hours, Days, Weeks, Months, and Years.<br>The locale settings of the system determine the start and end of the unit. For example, weeks in most of the European countries begin on Monday while in the United States they begin on Sunday.<br>• Dashboard Time. Select this option to activate the dashboard time panel. The option chosen in the dashboard time panel is effective. The default time is 6 hours. |
| Items per page | You can set the number of results that appear in the widget. Available for **List** view only. |
| Roll up interval | The time interval at which the data is rolled up. |
| Actions | An action on the selected object. Depends on the object type. |
| Filter | Limits the list to objects for a specific host, data center, and so on. You can drill-down in the hierarchical level. Available for **List**, **Trend**, and **Distribution** types of Views. |
| Filter by name | Limits the list to objects of a specific name. Available for **List** view only. |

### View Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget.

The **Output Data** section provides options to select object types on which you are basing the widget data.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |

| Option | Description |
|---|---|
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| **Input Data** | |
| Inventory trees | Select an existing predefined traversal spec to pick an object for the widget data. |
| Object | In self-provider mode, click the **Add Object** icon to select an object from the object list. The object list is displayed based on the inventory tree selection. You can also search for the object in this text box. |
| **Output Data** | |
| | A list of defined views available for the selected object is displayed.<br>You can create, edit, delete, clone, export, and import views directly from the View widget configuration options.<br>For more information, see Views in VCF Operations. |
| Auto Select First Row | Determines whether to start with the first row of data for list type views. |
| Show | Select one or more of the following items to display in the widget:<br>• To display the list of legends in the widget, select **Legend**.<br>• To display the name of the labels in the widget, select **Labels**. |

## Weather Map Widget

The Weather Map widget provides a graphical display of the changing values of a single metric for multiple resources over time. The widget uses colored icons to represent each value of the metric. Each icon location represents the metric value for particular resources. The color of an icon changes to show changes in the value of the metric.

### How the Weather Map Widget and Configuration Options Work

You can add the Weather Map widget to one or more custom dashboards and configure it to display data that is important to different dashboard users. The data that appears in the widget is based on the configured options for each widget instance.

Watching how the map changes can help you understand how the performance of the metric varies over time for different resources. You can start or stop the display using the **Pause** and **Play** options at the bottom of the map. You can move

the slider forwards or backwards to a specific frame in the map. If you leave the widget display and return, the slider remains in the same state.

The map does not show the real-time performance of the metrics. You select the time period, how fast the map refreshes, and the interval between readings. For example, you might have the widget play the metric values for the previous day, refreshing every half second, and have each change represent five minute's worth of metric values.

To view the object that an icon represents, click the object.

### Where You Find the Weather Map Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Weather Map Widget Toolbar Options

On the title bar of the widget, click the **Show Toolbar** icon to access the toolbar options.

The toolbar contains the icons that you can use to view the graph.

| Icon | Description |
|---|---|
| **Pause** and **Play** | Start or stop the display. The icon remains in the same state if you leave the widget display and return. |
| **Display Filtering Criteria** | View the current settings for the widget, including the current metric. |

### Weather Map Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Output Data** section provides options to select object types on which you are basing the widget data.

The **Output Filter** section provides options to restrict the widget data based on the selected filter criteria.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |

| Option | Description |
|---|---|
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Redraw Rate | An interval at which cashed data is refreshed based on newly collected data.<br>For example, if you set metric history to `Last 6 hours` and image redraw rate to `15 minutes`, and data is collected every 5 minutes, the data collected during 10 minutes will not be calculated at the 15 minutes.<br>For example, if you set metric history to `Last 6 hours` and image redraw rate to `15 minutes`, and data is collected every 5 minutes, the data collected during 10 minutes will not be calculated at the 15 minutes. |
| Metric History | Select the time period for the weather map, from the previous hour to the last 30 days. |
| Metric Sample Increment | Select the interval between metric readings. For example, if you set this option to one minute and set the Metric History to one hour, the widget has a total of 60 readings for each metric. |
| Group by | Select a tag value by which to group the objects. |
| Sort by | Select **Object name** or **Metric value** to set the way to sort the objects. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| Frame Transition Interval | Select how fast the icons change to show each new value. You can select the interval between frames and the number of frames per second (fps). |
| Start Over Delay | The number of seconds for the display to remain static when it reaches the end of the Metric History period, the most current readings, before it starts over again from the beginning. |
| Color | Shows the color range for high, intermediate, and low values. You can set each color and type minimum and maximum color values in the **Min Value** and **Max Value** text boxes.<br>If you leave the text boxes blank, VCF Operations maps the highest and lowest values for the **Color By** metric to the end colors.<br>If you set a minimum or maximum value, any metric at or beyond that value appears in the end color. |
| **Output Data** | |

| Option | Description |
|---|---|
| | Select an object type in your environment on which you want to base the widget data.<br>1. Click the **Add Object Type** icon to search for and add an object type.<br>When you search for object types, you can filter the types in the list by selecting a type from the **Adapter Type** drop-down menu or by using the **Filter** text box.<br>2. Optionally, select the object type from the list and click the **Delete Object Type** icon to remove the selected object type. |
| Metric | Select a common metric or a metric for the selected object type in the list. The metric will be the basis for the widget data. The object corresponding to the metric is the selected object for the widget. |
| **Output Filter** | |
| Basic | Pick tags to refine the widget data. The widget data is based on the objects that have the picked tags applied. If you pick more than one value for the same tag, the widget includes objects that have any of the tags applied. If you pick more than one value for different tags, the widget includes only the objects that have all the tags applied. |
| Advanced | Refine the widget data further based on the filter criteria for object types. The widget data is based on the objects for the filtered object types.<br>If the objects have a tag filter applied in the **Basic** subsection, you define filter criteria for the object types of the objects with tag filter applied. If the objects with tag filter applied do not belong to any of the object types in this filter criteria, the widget skips this filter and includes all the objects with tag filter applied.<br>1. In the first drop-down menu, select an object type.<br>2. In the second drop-down menu, select the option based on which you want to define the filter criteria. For example, if you select **Metrics** for the **Datacenter** object type, you can define a filter criteria based on the value of a specific metric for data centers.<br>3. In the drop-down menus and text boxes that appear, select or enter values to filter the objects.<br>4. To add more filter criteria, click **Add**.<br>5. To add another filter criteria set, click **Add another criteria set**. |

## Workload Widget

The Workload widget displays data indicating how hard a selected resource is working.

The Workload widget displays a graph depicting how hard the object that you selected is working. The Workload widget reports data on CPU usage, Memory usage, Disk I/O, and Network I/O.

### Where You Find the Workload Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the

left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### About Datastore Metrics for Virtual SAN

The metric named `datastore|oio|workload` is not supported on Virtual SAN datastores. This metric depends on `datastore|demand_oio`, which is supported for Virtual SAN datastores.

The metric named `datastore|demand_oio` also depends on several other metrics for Virtual SAN datastores, one of which is not supported.

- The metrics named `devices|numberReadAveraged_average` and `devices|numberWriteAveraged_average` are supported.
- The metric named `devices|totalLatency_average` is not supported.

As a result, VCF Operations does not collect the metric named `datastore|oio|workload` for Virtual SAN datastores.

### Workload Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |

| Option | Description |
|---|---|
| **Input Data** | |
| Object | Search for objects in your environment and select the object on which you are basing the widget data. You can also click the **Add Object** icon and select an object from the object list. You can use the **Filter** text box to refine the object list and the **Tag Filter** pane to select an object based on tag values. |

## Workload Pattern Widget

The Workload Pattern widget displays a historical view of the hourly workload of an object.

### Where You Find the Workload Pattern Widget

The widget might be included on any of your custom dashboards. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All** to see your configured dashboards.

To customize the data that appears in the dashboard widget, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**. To create your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **Create**. To edit your dashboard, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the center pane click **Dashboards** > **All**. Click the dashboard that you want to edit and select **Actions** > **Edit Dashboard**. Toggle between the **Views** and **Widgets** option to view and add a widget or view to the dashboard. The widgets list panel displays a list of all the predefined widgets. Drag the widget to the dashboard workspace in the upper panel.

### Workload Pattern Widget Configuration Options

On the title bar of the widget, click the **Edit Widget** icon to configure the widget.

The configuration options are grouped into one or more sections. You can select the objects on which you want to base the widget data and refine the objects in the following sections. Each section filters the objects further and pushes the filtered objects to the next section. The widget data is based on the objects that are the output of the last section.

The **Configuration** section provides general configuration options for the widget.

The **Input Data** section provides options to specify input for the widget. This section appears when the widget is in self provider mode.

| Option | Description |
|---|---|
| Title | Enter a custom title that identifies this widget from other instances that are based on the same widget template. |
| Widget Description | Enter a custom description for the widget. The description you enter is displayed as a tooltip against the name of the widget in the dashboard. |
| **Configuration** | |
| Refresh Content | Activate or deactivate the automatic refreshing of the data in this widget.<br>If not activated, the widget is updated only when the dashboard is opened or when you click the **Refresh** button on the widget in the dashboard. |
| Refresh Interval | If you activate the **Refresh Content** option, specify how often to refresh the data in this widget. |

| Option | Description |
|---|---|
| Self Provider | Indicates whether the objects for which data appears in the widget are defined in the widget or provided by another widget.<br>• On - You define the objects for which data appears in the widget.<br>• Off - You configure other widgets to provide the objects to the widget using the dashboard widget interactions options. |
| Details URL | Enter an external URL or an internal URL. The **View Details** button that appears at the bottom of the associated dashboard, links to the URL that you enter in the **Details URL** field. For example, you can enter an external URL such as `https://google.com/` or an internal URL such as `/vcf-operations/ui/inventory` that links to the **Inventory** page in VCF Operations. |
| **Input Data** | |
| Object | Search for objects in your environment and select the object on which you are basing the widget data. You can also click the **Add Object** icon and select an object from the object list. You can use the **Filter** text box to refine the object list and the **Tag Filter** pane to select an object based on tag values. |

# Using Predefined Dashboards for Quick Insights

VCF Operations includes a broad set of simple to use, but customizable dashboards to get you started with monitoring your VMware environment. The predefined dashboards address several key questions including how you can troubleshoot your VMs, the workload distribution of your hosts, clusters, and datastores, the capacity of your data center, and information about the VMs. You can also view log details.

Each set of dashboards is complemented with a series of out-of-the-box customizable alerts and reports to assist with your operational awareness. Alerts, reports, and dashboards, each have a purpose with minimal overlap. Several activities that are carried out using alerts should be carried out using dashboards. Reports should be kept to a minimum as they are not interactive and do not provide timely information.

The following table details how alerts, dashboards, and reports are complimentary.

VCF Operations dashboards support a concept we call insight. Insight complements alerts but does not replace it. Alerts miss the larger picture and only see what is triggered. For one object that reaches the threshold, there might be many just beneath the threshold. The objects below the threshold are called insight.

Alerts might auto-close if the symptoms disappear. Managing alerts is not the same as minimizing alerts. Minimizing alerts is about preventing alerts.

**Working with Predefined Dashboards**

To access the predefined dashboards, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Dashboards** > **All**.

Any customization you make is overwritten during upgrade and as a result, it is recommended that you back up your dashboards before an upgrade.

## Application Monitoring Dashboards

Use the **Linux OS discovered by Telegraf** dashboard and the **Windows OS discovered by Telegraf** dashboard to view the Linux and Windows OS objects discovered on all types of machines such as Virtual Machines, Azure Virtual Machine, EC2 Instance, and physical servers.

### Linux OS discovered by Telegraf Dashboard

Use the **Linux OS discovered by Telegraf** dashboard to view detailed information related to Linux OS objects. You can monitor CPU, disk, and network metric usage. You can also view object relationships.

**Table 1381: Widgets in the Linux OS discovered by Telegraf Dashboard**

| Widget | Description |
|---|---|
| Linux Operating Systems | Displays a list Linux OS objects that are discovered on all types of machines. Click on the Linux OS object for which you want to view details. The other widgets are populated with relevant details. |
| Process | Displays Linux Process metrics for the selected Linux OS object. |
| CPU | Displays CPU metrics for the selected Linux OS object. |
| Memory | Displays Memory metrics for the selected Linux OS object. |
| Memory Swap File | Displays Memory Swap File metrics for the selected OS object. |
| CPU: Individual CPU Performance | Displays all CPU usage metrics for the selected Linux OS object. |
| Disk: Individual Partition Capacity | Displays the instance name, total capacity, free capacity, and percentage of capacity that is used. |
| Disk: Individual Disk Performance | Displays the Individual disk performance metrics for the selected Linux OS object. |
| Related Objects | Displays a detailed object relationship of the Linux OS objects. |

## Windows OS discovered by Telegraf Dashboard

Use the **Windows OS discovered by Telegraf** dashboard to view detailed information related to Windows OS objects. You can monitor CPU, disk, and network metric usage. You can also view object relationships.

**Table 1382: Widgets in the Windows OS discovered by Telegraf Dashboard**

| Widget | Description |
|---|---|
| Windows Operating Systems | Displays a list Windows OS objects that are discovered on all types of machines. Click on the Windows OS object for which you want to view details. The other widgets are populated with relevant details. |
| Associated VM | Displays the VM version, CPU, memory, and disk space details. |
| System | Displays System metrics for the selected Windows OS object. |
| CPU | Displays CPU metrics for the selected Windows OS object. |
| Memory | Displays Memory metrics for the selected Windows OS object. |
| Windows: Memory Rate | Displays Memory Rate metrics for the selected Windows OS object. |
| CPU: Individual CPU Usage | Displays all CPU usage metrics for the selected Windows OS object. |
| Disk: Individual Partition Capacity | Displays the instance name, total capacity, free capacity and percentage of capacity that is used. |
| Network: Individual Adapter Usage | Displays the individual network adapter usage metrics for the selected Windows OS object. |
| Related Objects | Displays a detailed object relationship of the Windows OS objects. |

# Availability Dashboards

Availability covers the uptime of the object now and the uptime trend over time. The availability of hybrid clouds should be tracked at both the provider and consumer layers to understand the availability of the environment. These dashboards show the current uptime and the uptime percentage over the past month.

## VM Availability Dashboard

Use the **VM Availability** dashboard to calculate the availability of the Guest OS. The availability of the Guest OS is calculated because the Guest OS might not be running even when the VM is powered on. There are two layers of Availability, that is, the Consumer layer and the Provider layer. This dashboard covers the Consumer layer. You can view VMs in the selected data center, uptime trend for a selected cluster, and so on.

### Design Considerations

The **VM Availability** dashboard helps you check the availability (uptime in percentage) of VMs, as availability is typically part of the services provided by the IaaS provider.

This dashboard does not check the application uptime because it is possible that the application such as, a database, or a web server, is down while the underlying Windows or Linux is up. Generally, the service provided by the IaaS team is only for Windows or Linux. For information on the application, use the network ping or application-specific agent such as application monitoring.

### How to Use the Dashboard

- In the **Datacenters** widget, click any data center from the list.
  - To view the overall information, click the **vSphere World** object.
  - The other widgets are automatically updated once you click any data center.
  - Create a filter that reflects your class of service for this widget. Group by the class of services such as gold, silver, and bronze and default the selection to Gold. In this way, the monitoring is not cluttered with less critical workloads and you can focus on the important VMs. You can achieve this by creating a VCF Operations custom group for each class of service.
- The **VMs by Uptime in the last 30 days** widget displays the average uptime of VMs grouped by their availability. The bucket distribution helps you cater to a wide array of environments. If you are monitoring only production VMs where the uptime is expected to be near 100% all the time, edit the bucket to meet your operational needs.
  - The VMs in the **Selected Datacenter** widget display all the VMs that are currently deployed to the data center. The average uptime is displayed for the last month. For a production VM, expect this number to be 100% or closer to 100%.

    **Note:**
    The Services column will be blank unless Service Discovery is activated and the services/processes are discovered on a specific virtual machine.
  - The VMs column includes all VMs including the powered off VMs.
- Click any VM in the **VMs by Uptime in the last 30 days** widget to view the details in the **VM in the Selected VM Powered On Status**, **Selected VM Uptime Trend**, and **Selected Cluster Uptime Trend** widgets.
  - The **Selected VM Uptime Trend** widget displays the selected VM's Guest Tool Uptime (%) across the last 30 days.
- The **Guest OS: Services** widget displays the service state over time and the process or services running inside the Guest OS. If Guest OS services or processes are discovered inside a VM, their availability is analyzed. This requires the Service Discovery.
- The ESX **Host(s) where the VM has run** widget displays the historical migration of the VM. This can be useful in determining the cause of a VM downtime.

### Points to Note

- The metric only tracks the availability of VMware Tools and not the entire Guest OS. If VMware Tools is not up, it assumes the Guest OS to be down. You can check that this is not a false negative by adding a few line charts that display the evidence of activity. A good counter is IO counters such as Disk IOPS, Disk Throughput, and Network Transmit Throughput, because IO requires CPU processing. CPU usage is not a reliable counter as the work by VMkernel on the VM is charged to the CPU counters.
- VCF Operations exhibits a new ping adapter. This allows you to enhance the accuracy of the uptime measurement by creating a super metric that adds the ping information or by checking the process using an agent, such as application monitoring.
- Add a property widget that lists the selected VM properties to give you more context about the VM. In a large environment, the VM name alone might not provide enough context.

## vSphere Availability Dashboard

There are two layers of Availability, that is, the Consumer layer and the Provider layer. The **vSphere Availability** dashboard covers the Provider layer. This dashboard includes a cluster and not an ESX host because the cluster is operationally a single compute provider. This dashboard considers the N+1 design, where the cluster can withstand one host failure. Logically, a cluster with fewer hosts has a higher risk.

### Design Considerations

The **vSphere Availability** dashboard helps you analyze and report the uptime, as availability is typically part of the official business SLA. It is also often required in the monthly operational summary report.

This dashboard is not designed for live monitoring of the uptime. A NOC style of dashboard is better suited for those use cases. VMware Tools such as Log management must be leveraged as the fault is typically preceded with soft errors.

### How to Use the Dashboard

- The **Clusters** widget lists all the clusters in the environment. It is sorted by the lowest uptime so that the cluster with the lowest uptime in the last one month is displayed.
  - The **Running Hosts** column is color-coded as logically a smaller cluster has a higher risk. A single host failure results in a relatively higher capacity degradation.
  - The **vSAN?** column is hyper-converged, which means both the compute and the storage part is considered.
  - The **Admission Control Policy** column is based on the Cluster Configuration \ DAS Configuration \ Active property. The mapping between the code to name is:
    - -1 : Disabled
    - 0 : Cluster Resource Percentage
    - 1 : Slot Policy (Powered-on VMs)
    - 2 : Dedicated Failover Hosts
  - In a large environment, creating a filter for the list of clusters can make it more manageable. Group by the class of services such as gold, silver, and bronze and default the selection to Gold. In this way, you can easily view your gold clusters.
- Click any cluster from the **Clusters** widget.
  - The cluster uptime is automatically plotted in the **Selected Cluster Uptime Trend** widget. It uses 99%, 99.%, and 99.99% as the threshold for red, orange, and yellow colors respectively.
  - The ESX host details in ESX **in the Selected Cluster** widget are automatically updated. For more context, you can add a property widget that lists the selected ESX host properties.
  - In the ESX **in the Selected Cluster** widget, the **Connected to vCenter** and **Maintenance State** columns are not the average values, as both are string. However, they display the last state in the selected period. This allows you to go back to a specific point in time and view availability at that point.
- The **Datastores not available** widget lists only the datastores with powered off status. This covers both local and shared datastores. To add context, consider adding an extra column such as the data center where it resides, and the datastore types such as NFS and VMFS.
- The **Port Group Availability** widget lists port groups that currently have an uptime of less than 100%. To add context, consider adding an extra column such as the data center where it resides, number of used ports, and the maximum number of ports.
- For more context, you can add a property widget that lists the selected object properties. Multiple tables can drive the same property widget, but the object type must be the same.
- In a large environment, you can create a filter for this dashboard. Group by the class of services such as gold, silver, and bronze and default the selection to Gold. In this way, the monitoring is not cluttered with less critical workloads.
- In the ESX **in the Selected Cluster** widget, the **Connected to vCenter** and **Maintenance State** columns are not the average values, as both are string. However, they display the last state in the selected period. This allows you to go back to a specific point in time and view availability at that point.

**Points to Note**

- You can add vCenter Server and NSX components availability. This requires the VMware SDDC Health Monitoring Solution.

## Ping Overview Dashboard

Use the Ping Overview dashboard to configure the ping functionality and verify the availability of end points that exist in your virtual environment. The ping functionality is configured at the adapter instance for IP addresses, group of IP addresses, and FQDN. You can view ping adapter details like, latency distribution and packet loss distribution in this dashboard.

**Customizations Available for Your Use**

For more context, you can add a property widget that lists the selected object properties. Multiple tables can drive the same property widget, but the object type must be the same.

> **Note:**
> The FQDN names are checked for validity, the FQDN validation relies on RFC1034 and RFC1123, and only top level domains of the internet are validated. The `.local` domain is not supported as it does not fall into the list of top-level domains in the Domain Name System (DNS) of the Internet.

**Widget Information**

- Latency distribution - You can use this widget to see the objects that are experiencing high latency.
- Packet Loss Distribution – You can use this widget to see the objects that are experiencing high packet loss.
- Ping Targets – You can use this widget to view the list of ping targets grouped by their FQDN. Latency and packet loss information is also displayed for the ping objects.
- Breakdown by Source Initiator – You can use this widget to view the List of ping statistics by the source (ping initiator). You can ping the target from multiple locations, to determine if the issue is network-related or server-related.

## vSAN Health Dashboard

Use the vSAN Health dashboard to monitor the health of your vSAN cluster.

You can use the vSAN Health dashboard to monitor the status of the cluster components, diagnose issues, and troubleshoot problems. The dashboard displays the critical and important alerts that need immediate attention.

**How to Use the Dashboard**

The **vSAN Clusters** widget lists all the clusters in the environment with issues if the cluster health score is below 60 or has any critical alerts.

It shows the cluster names, ESX hosts, resync, and color-coded health and risk of the cluster. vSAN resync is a utilization metric, but its presence can impact performance. If the table is all green, no need to analyze it further.

You can focus on issues that need immediate attention and resolution for your clusters to be up and running. This dashboard displays the alerts triggered in your vSAN cluster and provides insights into the cause of the alerts.

The **Red Alert** column displays the critical alerts in the cluster that need immediate attention. If there are red alerts, click the cluster. The **Alerts** tab displays all the alert lists. Click a particular alert and check the alert details, related alerts, and potential evidence related to that alert. Under **Alert Details**, you can check the recommended remediation to troubleshoot the specific issue. Click the Knowledge Base article for more details.

Click any cluster to view the properties of the selected cluster in the **Property of Selected Cluster** widget. These are the properties that helps you understand the cluster health issues. If something is not there, check the alerts in the alert lists.

Once you select any particular cluster, the alerts in that cluster are listed in the **Alert Lists** widget. Click an alert to view the alert details, related alerts, and potential evidence. It also displays the alert symptoms.

# Capacity Dashboards

Capacity quantifies the resources used, resources remaining, and opportunities to reclaim unused resources. Projections of the demand provide a proactive view of capacity. The **Capacity Dashboards** display capacity in terms of time remaining before capacity is projected to run out, the amount of capacity remaining, the number of VMs that might fit in the remaining capacity, and reclaimable resources that can increase the available capacity.

Capacity management is about balancing demand and supply. It is about meeting demand with the lowest possible cost.

For IaaS or DaaS, capacity management begins before the hardware is deployed. It begins with a business plan that defines what class of service will be provided. Each class of service, for example, gold, silver, bronze is differentiated by the quality of service and covers the availability, for example, 99.99% uptime for Gold, 99.95% uptime for Silver. It also covers performance, for example, 10 ms disk latency for Gold, 20 ms disk latency for Silver, and security or compliance.

The quality incurs cost and in turn drives price. Gold VM is higher per vCPU and per GB of RAM because it has a higher quality of service. A proper pricing model must be planned. If you want your customers to rightsize in advance, then a 64 vCPU VM has to be more than 64x the price of 1 vCPU VM. If the pricing model is a simple straight line, there is no incentive to go small and no penalty if it is over provisioned. In this case, you end up forcing rightsizing in production, which is a costly and time consuming process.

Demand is more than the active load that is consuming your capacity. Since capacity based on utilization is incomplete by itself, the principles displayed in the following figure are considered.

- Latent demand. Many critical VMs are protected with Disaster Recovery. During a Disaster Recovery drill or an actual disaster, this load is consumed.
- Potential demand. Many newly provisioned VMs take time to reach their expected demand. It takes time for the database to reach the full size, the user base to reach the target, and the functionalities to complete. When this is achieved, it results in the increase in demand.
- Unmet demand occurs when the VM or Kubernetes Pod is undersized. The load is running nearly 100% most of the time.
- Excessive demand can wreak havoc in a shared environment. A group of highly demanding VM can collectively impact overall performance of the cluster or datastore.

## Cluster Capacity Dashboard

The **Cluster Capacity** dashboard includes the ESX host and resource pools as they impact cluster capacity.

## Design Considerations

See Capacity Dashboards for common design considerations among all the dashboards for capacity management.

## How to Use the Dashboard

The **Cluster Capacity** dashboard is layered, gradually providing details as you work top-down in the dashboard.

### Overall Analysis

- The three bar charts which are **Clusters by Capacity Remaining**, **Clusters by Time Remaining**, **Clusters by VM Remaining**, summarize the overall situation. The first two charts can be used together to identify when you need to add capacity to address growth. Time remaining uses historical growth in a cluster to forecast when more capacity is needed. This allows you to operate more efficiently by making sure you have enough capacity currently and proactively plan for adding capacity. The third bar chart which is **Clusters by VM Remaining**, provides complete contexts, as different clusters can have different VM sizes.

  For a large environment, a heat map is helpful. The three heat maps are Time Remaining, Capacity Remaining, and VM Remaining. If your cluster sizes are not standardized, create another heat map, and use the number of ESX hosts to show the size difference.

### Cluster Analysis

- The **Clusters Capacity** widget provides a table with details. The number of ESX hosts are color coded as smaller clusters have a relatively higher overhead. Select a cluster from the table to view the capacity details that are automatically displayed.

  **Performance**

  Ensure that the performance of the cluster meets your SLAs.

  **Utilization**

  The next two charts are Memory Workload (%) and CPU Workload (%), that show values relative to your usable capacity. Utilization is displayed for three months and not one week. The daily average is displayed and not the hourly average, so you can focus on the overall trend. For memory, the focus is on consumed memory and not active memory.

  **Allocation**

  You can view the trend of the three which are CPU, disk, and memory components together on the **Overcommit Ratio** chart. In general, your CPU overcommit should be the highest, followed by the disk (because of thin provision). Memory overcommit tends to be near one due to its nature as cache.

  Use the line chart in the **Allocation** widget, to see the trend. The data is averaged hourly.

  In the **VM Count** widget, the trend line of the number of VMs over time is important to spot if there are many newly provisioned VM. If you see that the VMs are increasing but demand remains low, it indicates a sign of potential demand in the future.

  **Reservation**

  Reservation can impact the efficiency of your cluster. Your cluster could be low on capacity because of real workload or just reservation. If your cluster size varies, complement the reservation number by showing a relative value. Once you have a standardized number, you can visualize them on a heat map.
- **ESX Analysis**

  Good cluster capacity does not indicate that there is no issue at the ESX level. Unbalance is a common problem, especially in large clusters and stretched clusters.

  The ESX **Hosts in a Cluster** table displays all the member ESX hosts. You can see the unbalance clearly, thanks to the color code. The color code reflects the unbalance.

  The **99th percentile Performance** column takes the 99th percentile value of the ESX Performance (%) metric.

  Select an ESX host to view the details. Both the **CPU Workload (%)** and the **Memory Workload (%)** trend line charts display if there is a steady demand, cyclical demand, rising demand, or declining demand. The trend is as important as the present value. View trends over a longer time. Utilization is displayed for three months and not one week. The daily average is displayed and not the hourly average. The focus is on memory consumed and not memory active. Memory consumed includes the total memory consumed, so it includes the memory consumed by VMkernel. Both total and usable utilization in terms of memory and CPU are displayed and provides the absolute amount of capacity.
- **VM Analysis**

Use the **VMs in the selected Cluster or Host** table to analyze the cause of the low capacity remaining and which VMs are impacting the infrastructure resources, such as, CPU, memory, and disk space. The table lists either the VMs in the cluster or host. When you select one of the VMs, additional relevant information is displayed.

If there are many large VMs running low on capacity you can stop provisioning until you upsize the existing VMs first.

## Datastore Capacity Dashboard

The **Datastore Capacity** dashboard complements out of the box capacity pages and dashboards. It focuses on storage, provides an overall picture, and highlights the datastores that need attention.

### Design Considerations

See Capacity Dashboards for common design consideration among all the dashboards for capacity management.

### How to Use the Dashboard

The **Datastore Capacity** dashboard is layered, gradually providing details as you work top-down in the dashboard.

- **Overall Analysis**
  The summary banner answer basic questions such as number of datastores, the capacity, number of VMs, and the running VMs.
  The distribution charts are **Shared Datastores by Capacity Remaining** and **Shared Datastores by Time Remaining**.
  There are three heat maps, the primary being the **Remaining Capacity** heat map. The two other heat maps cover used capacity. One of them is designed for an environment that uses datastore clusters. Each box represents a datastore. If you have many datastores, the heat map will group them. You can further see the members. The larger the datastore, the larger the box.
  The **Shared Datastores** table lists the shared datastores. The table provides a summary, displaying all the datastores at a glance. They are grouped by data center. By default, the table is sorted by the least capacity remaining. There are three reclamation opportunities: powered off VM, snapshot, and orphaned VMDK.
- **Datastore Analysis**
  Select a datastore from the **Summary** table. The capacity details are automatically displayed. A snapshot that lasts beyond a few days should be investigated. Orphaned VMDK are the ones that are not associated to any VM.
  For disk space, the total capacity, allocated capacity, and the actual capacity used are displayed.
- **VM Analysis**
  To analyse at the VM level, review the **VMs in the datastore** table. Click on the VM you want to investigate further to see usage over time.
- **Local Datastores**
  The **Local Datastores Capacity** table appears at the end of the dashboard. Avoid running VMs on local data stores, unless the storage requirements can be met with a local disk and does not need vMotion.

### Points to Note

If the underlying LUN is also thin provisioned, add visibility into the physical array. The dashboard does not have datastore clusters. If your environment uses datastore clusters, modify this dashboard or create a new one. In a large environment with many datastores and datastore clusters, add a View List to list the datastore clusters, so you get summary information. Alternatively, create a heat map, listing the datastore clusters.

## ESX Capacity Dashboard

The ESX **Capacity** dashboard supports the **Cluster Capacity** dashboard and is also required for the non-clustered ESX.

### Design Considerations

See Capacity Dashboards for common design considerations among all the dashboards for capacity management.

### How to Use the Dashboard

The **Summary** heat map provides an overall view of the ESX Host capacity, grouped by their clusters.

- Each ESX host is represented by a box, displaying their capacity remaining.
- The ESX host size is made constant for ease of use. If your ESX sizes are not standardized, consider using the number of physical cores or Total CPU GHz to display the difference in sizes. Ensure that the smallest ESX is not too small.
- Wastage is displayed by a new color. Dark gray indicates wastage as capacity is not used. The performance problem due to low utilization can be caused by a bottleneck elsewhere.

The ESX **Hosts Capacity** widget lists all the ESX hosts in your environment, grouped by their parent cluster.

- The standalone ESX are displayed at the bottom under No Group.
- In a large environment with many data centers, you can zoom into a specific vCenter or data center. You can also filter or search for specific ESX hosts matching certain names.
- The **99th Percentile Performance** column takes the 99th percentile value of the ESX Performance (%) metric. To rule out the outlier, the worst performance (which is equivalent to 100th percentile) is not considered. Also, the performance threshold is set to be stringent.

Select one of the ESX hosts from the ESX **Hosts Capacity** widget. All the three line charts automatically display the trend of selected ESX host.

- Displays both total and usable utilization in terms of RAM and CPU.
- Utilization is displayed for three months and not one week. The daily average is displayed and not the hourly average and the focus is on RAM consumed and not RAM active.

### Points to Note

- Add a drill-down to the ESX **Capacity** dashboard. A logical place to initiate this drill-down is in the **Cluster Capacity List** widget. Link this widget into the table of ESX host in the destination dashboard.
- A technology refresh is often used to address the capacity shortage. Consider adding a property widget that displays the hardware model and specification to help you determine the age of the hardware.

## VM Capacity Dashboard

The **VM Capacity** dashboard helps you analyze the capacity of all the VMs with the ability to analyze each VM.

### Design Considerations

See Capacity Dashboards for common design considerations among all the dashboards for capacity management.

### How to Use the Dashboard

- **Overall Analysis**

  The **Datacenters** table lists all your data centers. vSphere World is included so you can see all the VMs from all the data centers. Unlike infrastructure objects, there are potentially tens of thousands of VMs. The charts will take longer to refresh if you select vSphere World.

  The **VMs by Capacity** widget, groups the VMs by capacity remaining, while the **VMs by Time Remaining** widget groups the VMs by time remaining. Ideally, you want all of them to be low on capacity remaining, but high on time remaining.

  The heat map provides an additional view by grouping them by cluster. The heat map helps you identify which cluster is at risk (where most of the VMs need more capacity) and which cluster provides extra resources (where most of the VMs are not using their capacity).

  Review the **VMs by Capacity Remaining** heat map. The heat map provides the next level of detail by grouping the VMs by clusters, so you can see which clusters need attention. The VM size is standardized for better visualization.
- **VM Analysis**

  Review the **VMs Capacity in the Selected Datacenter** table that lists all the VMs in the selected data center. The list is sorted by the VM with the least capacity remaining. You can also sort by Time Remaining.

Select a VM from the table. The capacity details are displayed. You can view both the CPU and memory trend over time. Three months data is displayed and is averaged to hourly so you can see the overall trend.

Disk details are displayed using the Guest OS partition. Avoid using VM virtual disk as there may not be a direct mapping to the actual partition.

The recommendation for right-sizing is displayed for both CPU and memory. Unlike a physical server, it is important to right-size a VM. For CPU, the CPU Usage counter is used instead of Demand. For disk, Guest OS partition level is displayed. There is no overall capacity at the VM level because different partitions have different capacity.

- **Relevant Configuration**
  Relevant configuration is automatically displayed to provide context to the VM. Information such as VM owner and business units can be useful in the analysis.

## Reclamation Dashboard

The **Reclamation** dashboard helps you manage various types of reclamation that can be carried out on VMs and datastore. It is designed for both the Capacity team and the Operations team.

### How to Use the Dashboard

- **Overall Analysis**
  The scoreboard provides a summary of the total reclamation.
  You can select either a data center, a cluster, or a datastore. The datastore level is required as orphaned disks do not have VM association, hence it is not related to any cluster. The **Datastore** table only drives the snapshot table.
  **Powered-off VMs Distribution Size**, **Idle VMs Distribution by Memory Footprint**, and **Snapshots Distribution by Size** charts display summary information.
  Adjust the bucket size in the charts to suit your operational requirements. The reclamation potentials are presented as three bar charts, each corresponds to an area you can reclaim:
  - Powered off VMs that are no longer needed contribute to wasted disk usage. Consider deleting them to free up space or moving them to archival storage.
  - Idle VMs are running, but not being used actively. These VMs consume memory that may be used by active VMs. Consider removing these VMs to reduce memory contention.
  - Snapshots are meant to be temporary and can cause performance issues and waste disk space if not deleted after a few days.

  When reviewing each of the three tables observe that they are sorted by the largest reclamation opportunities. This allows you to get the greatest benefit from the least amount of effort. For example, focus on snapshots first, as it does not involve changing the VM. For the powered off VMs, you may want to consider VMs that have been powered off longer and these are more likely to be unneeded. Idle VMs can be challenging to reclaim since they are still running, so you might prioritize powered off VMs before attempting to reclaim idle VMs.
  The **99P CPU Usage** column displays the CPU usage at the 99th percentile during the time period. It is an easy way to check if it is indeed idle.
- **VM Analysis**
  To analyze VMs for reclamation opportunities, select a VM from one of the three tables (Powered Off VM, Idle VMs, or VM Snapshots). The selected VM will populate the widgets with the following details:
  - Powered off over time displays the amount of time a VM has been powered off.
  - CPU Usage over time provides insights into the aggregate CPU usage, including peak usage periods. This way you can validate that an idle VM has not had any brief usage.
  - VM Snapshot over time provides you with an understanding of the age and growth of snapshots on the VM. Watch for fast growing snapshots, as these can quickly consume disk space.
  - Contexts of selected VM is a summary of the VM configuration information.

### Points to Note

To organize your reclaim efforts, it is helpful to create custom groups to make it easier to filter by department or VM owner. This can make it easier to seek approvals and communicate with anyone who may be impacted.

## vSAN Capacity Dashboard

The **vSAN Capacity** dashboard complements the vSphere **Cluster Capacity** dashboard by displaying capacity related to vSAN. To manage vSAN capacity, use both dashboards.

### Design Considerations

The dashboard focuses on vSAN specific metrics, but does not list non-vSAN clusters.

See Capacity Dashboards for common design considerations among all the dashboards for capacity management.

### How to Use the Dashboard

- **Overall Analysis**
  The **Clusters by Capacity Remaining** and **Clusters by Time Remaining** bar charts focus on vSAN disk space and not compute and network.
  Use the **vSAN Clusters** table to view vSAN specific metrics.
- **Cluster Analysis**
  Select a vSAN cluster from the **vSAN Clusters** table. The detailed capacity is automatically displayed.
  The **Utilization** widget displays the utilization for all three elements, as you need to consider all three. Network is not shown as typically it is not a problem.
  Like a physical array, there can be hot spots and imbalance. The **Is Disk Group Space Utilization Balanced** heat map displays individual disk groups.
  Reclaimable storage is a key component of proactive capacity management. You can view details for both VMs and non-VMs.
  You can view the **Dedupe and Compressed** scoreboard for more details in this area.
- **Disk Group Analysis**
  If there is imbalance, you can analyze each disk group. The **Disk Groups Selected in Selected vSAN Cluster** displays all the disk groups in the cluster. Their usage may not be similar, but should not deviate drastically. To view the disk group usage trend, click on a disk group.
- **VM Analysis**
  You can analyze individual VMs in the selected cluster from the **VMs in the selected vSAN Cluster** table and check their usage and snapshot. To view the trend in Usage, click on a VM. In addition, the relevant configurations of the VM is displayed.

## vSAN Stretched Clusters

The vSAN Stretched Clusters dashboard provides an overview of the cluster resources used across vSAN fault domains. Using the stretched clusters dashboard you can monitor the resource consumption at the site level for Preferred Sites and Secondary Sites. You can create custom dashboards for specific vSAN stretched cluster metrics.

### Where to View vSAN Stretched Cluster Objects

From the left menu, click **Operations** > **Dashboards**. From the Dashboards panel, click **All** > **Capacity** > **vSAN Stretched Clusters**.

You can also view the vSAN stretched cluster objects from **Inventory** > **vSAN** > **vSAN and Storage Devices** > **vSAN Clusters**, if the vSAN cluster is a stretched cluster.

The vSAN Stretched Clusters dashboard provides information about CPU Capacity, Cores, Memory Capacity, and Disk Capacity for the Preferred Site and the Secondary Site. You can identify the vSAN stretched clusters running out of capacity looking at the utilization metrics.

## vSAN ESA Capacity Dashboard

The vSAN ESA ( Express Architecture) Capacity dashboard provides an overview of the capacity available across all the vSAN clusters. The dashboard displays the time remaining before capacity is projected to run out, the amount of

capacity remaining, the number of vSAN clusters that might fit in the remaining capacity, and reclaimable resources that can increase the available capacity.

**How to Use the Dashboard**

- **Overall Analysis**
  The **All Clusters by Capacity Remaining** and **All Clusters by Time Remaining** bar charts focuses on the number of vSAN disk against the used disk capacity percentage across vSAN clusters.
- **Cluster Analysis**
  Select a vSAN ESA cluster from the **vSAN Clusters** table. The detailed capacity is automatically displayed.
  The **Utilization** widget displays the utilization for all three elements, as you need to consider all three. Network is not shown as typically it is not a problem.
  The **Is Storage Pool Space Utilization Balanced** heat map displays individual storage pools.
  Reclaimable storage is a key component of proactive capacity management. You can view details for both VMs and non-VMs.
- The **Storage Pools in Selected vSAN Cluster** widget displays the utilization for Disks and Disk Space.
- Like a physical array, there can be hot spots and imbalance. The **Is Physical Disk Space Utilization Balanced** heat map displays individual physical disks.
- **VM Analysis**
  You can analyze individual VMs in the selected cluster from the **VMs in the selected vSAN Cluster** table and check their usage and snapshot. To view the trend in Usage, click on a VM. In addition, the relevant configurations of the VM is displayed.

## vSAN OSA Capacity Dashboard

The vSAN OSA (Original Storage Architecture) Capacity dashboard provides an overview of the capacity available across all the vSAN clusters. The dashboard displays the time remaining before capacity is projected to run out, the amount of capacity remaining, the number of vSAN clusters that might fit in the remaining capacity, and reclaimable resources that can increase the available capacity.

**How to Use the Dashboard**

- **Overall Analysis**
  The **All Clusters by Capacity Remaining** and **All Clusters by Time Remaining** bar charts focuses on the number of vSAN disk against the used disk capacity percentage across vSAN clusters.
- **Cluster Analysis**
  Select a vSAN OSA cluster from the **vSAN Clusters** table. The detailed capacity is automatically displayed.
  The **Utilization** widget displays the utilization for all three elements, as you need to consider all three. Network is not shown as typically it is not a problem.
  Like a physical array, there can be hot spots and imbalance. The **Is Disk Group Space Utilization Balanced** heat map displays individual disk groups.
  Reclaimable storage is a key component of proactive capacity management. You can view details for both VMs and non-VMs.
- **Disk Group Analysis**
  If there is imbalance, you can analyze each disk group. The **Disk Groups Selected in Selected vSAN Cluster** displays all the disk groups in the cluster. Their usage may not be similar, but should not deviate drastically. To view the disk group usage trend, click on a disk group.
- **VM Analysis**
  You can analyze individual VMs in the selected cluster from the **VMs in the selected vSAN Cluster** table and check their usage and snapshot. To view the trend in Usage, click on a VM. In addition, the relevant configurations of the VM is displayed.

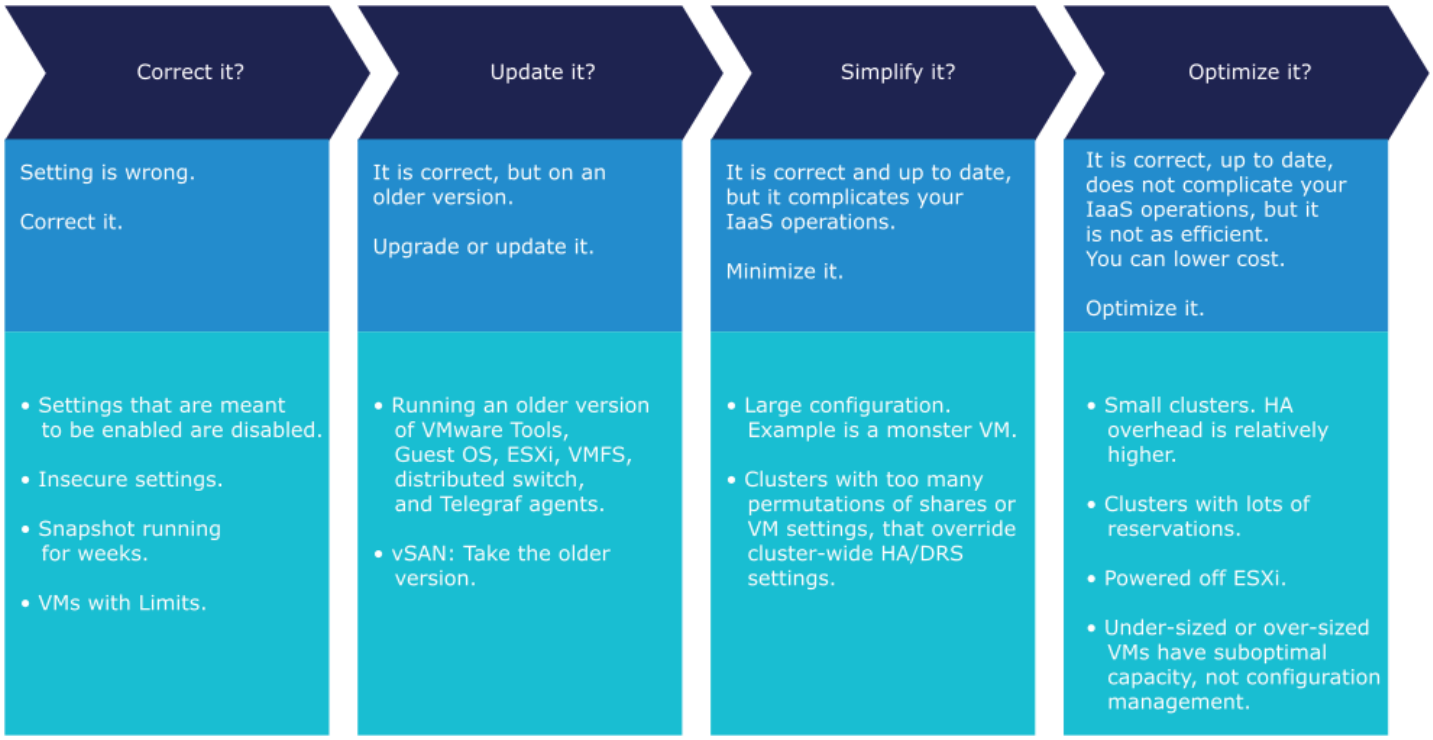
# Configuration Dashboards

As an operations management software, VCF Operations focuses on the impact to day-to-day operations a product has, rather than the feature of the product itself. Products under monitoring, such as vSphere and vSAN, can have features that are related, but have a different impact on operations. For example, vSphere provides Limits, Reservation, and Shares for the VM.

Limits, Shares, and Reservation. As a feature, they are closely related, and appear in the same dialog box and must be learned as one. However, they impact operations differently. The following table describes that in more detail.

| | | | |
|---|---|---|---|
| VM Limit | Impacts the VM | • Should not be used. Right-size instead.<br>• Results in unpredictable performance of the Guest OS. | Incorrect Configuration |
| VM Reservation | Impacts the Infrastructure | • Keep the total amount low, and relative to the total capacity of the cluster.<br>• Absolute value. A 2-GHz reservation is in fact a 2-GHz reservation.<br>• Results in suboptimal infrastructure capacity, as overcommit is not possible. | Suboptimal Configuration |
| VM Share | Impacts the Infrastructure | • Keep the number of variations to below three. One for each class of service.<br>• Relative value. 2000 worth of reservations depends on the value of other VM reservations. Be careful when you move the VM to another cluster, as the relative value changes.<br>• Results in complex operations. It is harder to troubleshoot performance when the dynamic entitlements of each VM fluctates more. | Complex Configuration |

VCF Operations follows the principle that there are different impacts on operations and applies a methodology for looking at configuration. It does not group the settings by features or objects. Rather, it begins with the impact and prioritizes what can be done.



Each operation is unique and as a result, customers run operations differently. What is right for other customers, might not be right for you. Even in the same environment, what is right for a development environment might not be appropriate for a production environment.

The following table lists some of the areas for improvement for the operations in your environment:

### Areas of Improvement

| | Correct it? | Update it? | Simplify it? | Optimize it? |
|---|---|---|---|---|
| **IaaS Consumer:**<br>• Process<br>• Applications<br>• Guest OS<br>• Container<br>• VM | • Java JVM or Database \ memory config too large relative to Guest OS<br>• Guest \ Metric not collecting<br>• Guest \ High TX Broadcast packets<br>• VM \ Tools not installed<br>• VM \ Tools not running<br>• VM \ CPU Limit<br>• VM \ Memory Limit<br>• VM \ Old Snapshot<br>• VM \ On local Datastore | • Guest OS \ Tools<br>• Guest OS \ Windows<br>• Guest OS \ Linux<br>• Guest OS \ Telegraf agent<br>• VM \ Hardware (vmx) | • VM \ Large VM (CPU, RAM, Disk)<br>• VM \ lots of disks, NIC card<br>• VM \ lots of IP address.<br>• VM \ with RDM<br>• VM \ on multiple datastores<br>• VM \ Fault Tolerant<br>• VM \ SRM protected<br>• VM \ Hot Add\|Remove \ CPU<br>• VM \ Hot Add\|Remove \ RAM | • Java JVM or Database \ memory config too small relative to Guest OS<br>• Guest OS \ no visibility<br>• Container \ smaller than the parent VM<br>• VM \ Tools unmanaged<br>• VM \ bigger than the whole ESXi cores.<br>• VM \ bigger than CPU socket.<br>• VM \ Large Snapshot<br>• VM \ Reservation. |
| **IaaS Provider:**<br>• Telegraf<br>• ESXi<br>• Cluster<br>• Datastore & Cluster<br>• Switch and Port Group<br>• Hardware<br>• NSX<br>• vSAN | • ESXi \ vMotion disabled<br>• ESXi \ Disconnected from vCenter<br>• ESXi \ Maintenance Mode<br>• ESXi \ NTP disabled<br>• ESXi \ Standalone<br>• Cluster \ Admission Control disabled<br>• Cluster \ HA disabled<br>• Cluster \ HA Failover %<br>• Cluster \ DRS disabled<br>• Cluster \ DRS manual<br>• Cluster Inconsistency<br>  • BIOS, ESXi: version<br>  • BIOS, ESXi: Power Management<br>  • ESXi Storage Path<br>  • ESXi Hardware<br>• Datastore Cluster inconsistency<br>  • Capacity<br>  • Performance<br>• Datastore \ single path<br>• Datastore \ no path. This is unlikely.<br>• NSX \ no redundancy for Controller, Manager | • ARC \ server<br>• ARC \ agent<br>• ESXi \ hardware<br>• ESXi \ vSphere<br>• ESXi \ 1 Gb NIC.<br>• Server \ not on warranty<br>• vCenter \ version<br>• Datastore \ VMFS version<br>• vSAN \ version<br>• Switch \ version<br>• NSX \ version | • ESXi \ Too many variations. No standard<br>• Cluster \ Many VM Shares (CPU)<br>• Cluster \ Many VM Shares (RAM)<br>• Cluster \ Resource Pools<br>• Cluster \ Stretched compute + storage<br>• Cluster \ 32 nodes or more<br>• Cluster \ VM to Host affinity<br>• Cluster \ Too many storage paths<br>• Datastore \ Shared by >1 cluster WLP uses this<br>• Datastore \ Many paths<br>• Network \ LBT?<br>• Network \ MAC Address change | • ESXi \ low CPU cores count<br>• ESXi \ low RAM size<br>• ESXi \ Powered Off<br>• ESXi \ HT Disabled<br>• ESXi \ 4 socket or higher.<br>• Cluster \ small clusters \ host especially for vSAN<br>• Cluster \ small clusters \ CPU<br>• Cluster \ small clusters \ RAM<br>• Cluster \ EVC Mode<br>• Cluster \ High Reservation<br>• Cluster \ DRS Automation Level<br>• Cluster \ DPM disabled<br>• vSAN \ All Flash: Dedupe disabled<br>• vSAN \ All Flash: Compressed disabled<br>• Datastore \ small<br>• Datastore \ low VM count<br>• Datastore \ no ESXi<br>• Distributed Switch \ unused |

### Design Considerations

The dashboards display configurations that need immediate attention, before displaying the overall configuration. This helps you take measures toward optimizing configuration.

Operations vary among customers, and as a result, it is not possible to design one dashboard to meet every customer's operational needs. A configuration that is important for one customer might not be relevant for another customer. Tailor the dashboard to your unique environment. You can collapse or expand the widgets to allow relevant data to be displayed.

The overall layout is designed to balance ease of use, performance (loading time of the dashboard page), and completeness of configuration check. As a result, not all configuration settings are displayed. Lack of screen real estate is another consideration behind the design.

## Cluster Configuration Dashboard

Use the **Cluster Configuration** dashboard to view the overall configuration of vSphere clusters in your environment, especially configurations that need attention.

### Design Considerations

See the Configuration Dashboards page for common design considerations among all the dashboards for configuration management.

As there are many configurations to be verified, if you have a larger screen, add additional checks as you deem fit, or add legends to the pie-charts.

### How to Use the Dashboard

The **Cluster Configuration** dashboard is organized into sections for ease of use.

- The first section of the dashboard consists of three bar-charts. They correspond to the three main features of vSphere clusters, namely High Availability (HA), Dynamic Resource Scheduler (DRS), and Distributed Power Management (DPM).
  - HA: The best practice is to activate HA admission control. You can specify the admission control policy in the vCenter and the threshold for failover shares.
  - DRS: The best practice is to activate DRS. Envision the vSphere cluster as a single logical computer that balances within itself.
  - DPM: The best practice is to activate DPM in an environment where environmental concern is the top priority or the high peak rarely occurs as most of the time you run very low utilization.
- The second section of the dashboard consists of eight pie-charts. They show the relative distribution of key configurations.
  - Two of the bar-charts cover admission control. You must activate admission control. The pie-charts displays the policy code instead of the policy name, as it is based on the property: `Cluster Configuration | Das Configuration | Active Admission Control Policy`. The mapping between the code to name is:
    - -1 = Disabled
    - 0 = Cluster Resource Percentage
    - 1 = Slot Policy (Powered-on VMs)
    - 2 = Dedicated Failover Hosts
  - There are two bar-charts that cover the HA Failover Share. One for CPU, and one for memory.
  - The next two bar-charts cover DRS settings. You might want to fully automate DRS, which means that there is no operator intervention required for both initial VM placement and subsequent load balancing, but with a moderate migration threshold (value = 3.0). The value ranges from 1.0 to 5.0.
  - There are two pie-charts that show reservation. One for CPU and one for memory. Minimize the total reservation value as it prevents overcommit of resources and hence results in less optimal utilization. Memory reservation can remain and occupy the memory space of the ESX host, even though the VM does not use the memory anymore. Consider the analogy of unused files that you have not opened for months in the c:\ drive of your laptop. They still take up space on the hard disk. Keep the number of distinct shares to below three (or at a minimum), matching the distinct classes of service.
- The third section of the dashboard consists of two bar-charts. They show the absolute distribution of the clusters.
  - The first bar-chart displays the cluster grouped by the number of ESX hosts. Small clusters, defined as having a lower number of ESX hosts, have a higher overhead while large clusters have a higher risk if there are cluster-wide outages. Performance risk is lower, because there are more nodes that DRS can tap on, but if there is an actual problem, troubleshooting can be tougher, because there are more nodes to analyze. For large clusters, have a disaster recovery plan as an unexpected cluster-wide outage can impact many VMs.
- The fourth section of the dashboard lets you drill-down to an individual cluster.
  - A table lists all the clusters with their key configuration. You can export this list as a spreadsheet for further analysis or reporting.
  - Select a cluster. The list of ESX hosts under the cluster, with shares and resource pool information, is automatically filled up.
  - Keep the number of distinct shares to below three (or at a minimum), matching the distinct classes of service. Avoid providing different services to individual VMs as that increases the complexity of the cluster performance.
  - Keep the number of resource pools minimal.
  - Some of the columns are color coded to facilitate quick reviews. Adjust their threshold to either reflect your current situation or your desired ideal state.

**Points to Note**

- The number of buckets in the pie-chart or bar-chart are balanced between the available screen estate, ease of use, and functionality. Modify the buckets to either reflect your current situation or your desired ideal state.
- `No data to display` does not imply that there is something wrong with data collection by VCF Operations. It might signify that none of the objects meet the filtering criteria of the widget, and as a result there is nothing to display.
- In a large environment, create a filter for this dashboard. Group by the class of services such as, gold, silver, and bronze. Default the selection to gold. In this way, your monitoring is not cluttered with less critical workloads.

- To view the content of a slice in a pie-chart or a bucket in a bar-chart, click on it. The list cannot be exported. Clicking an object name, takes you to the object summary page. The page provides key configuration information, with other summary information.

## ESX Configuration Dashboard

Use the ESX **Configuration** dashboard to view the overall configuration of the ESX hosts in your environment, especially the configurations that need attention.

### Design Considerations

See the Configuration Dashboards page for common design considerations among all the dashboards for configuration management.

As there are many configurations to be verified, if you have a larger screen, add additional checks as you deem fit, or add legends to the pie-charts.

### How to Use the Dashboard

The ESX **Configuration** dashboard is organized into sections for ease of use.

- The upper section of the dashboard displays basic ESX configurations that should be standardized for ease of operations.
  - There are six pie-charts that are displayed as one set because there is a relationship between their values. There should be a correlation between them. Ideally, the ESX version, the ESX build, and the BIOS must be identical across all ESX hosts in a cluster. Keep the variations of the hardware model, NIC speed, and storage path minimal. The more complex the pie-chart, the more variants you have. This results in complex operations, that potentially results in higher operating expenses.
  - The configurations should reflect your current architecture standard. Each pie-chart counts the occurrences of a particular value. A large slice signifies that the value is the most common value, and if that is not your current standard, then you must address it.
- The second section of the dashboard displays configurations that are potentially suboptimal.
  - The three bar-charts display various size dimensions of the ESX hosts. The bar-charts are designed to be seen as one set. Ensure that there are a minimal number of variations to reduce complexity.
  - Smaller ESX hosts have a relatively higher overhead, and are limited in running larger VMs. If they have a low core count, they might be using an outdated CPU. Small ESX hosts are more expensive on a per core, per GB, per rack unit basis than larger ones if they occupy the same space. However, a 4-CPU socket ESX host is likely to be too large, resulting in a concentration risk (too many VMs in a single ESX host). Maintain a good balance that balances your budget and risk constraints.
  - Adjust the distribution chart bucket size to fit your environment.
- The third section of the dashboard displays configurations that you might want to avoid.
  - The six bar-charts focus on security, availability, and capacity settings that you can set as a standard. For example, you should consider activating the NTP daemon for a consistent time, which is critical for logging and troubleshooting.
  - The three tables list the actual ESX hosts that are in a non-productive state. They can be on maintenance mode, powered off, or in a disconnected state.
- The last section of the dashboard displays all the ESX hosts in your environment.
  - You can sort the columns and export the results into a spreadsheet for further analysis.
  - Some of the columns are color-coded to facilitate quick reviews. Adjust their threshold to either reflect your current situation or your desired ideal state.

### Points to Note

- The number of buckets in the pie-chart or bar-chart are balanced between the available screen estate, ease of use, and functionality. Modify the buckets to either reflect your current situation or your desired ideal state.

- `No data to display` does not imply that there is something wrong with data collection by VCF Operations. It might signify that none of the objects meet the filtering criteria of the widget, and as a result, there is nothing to display.
- In a large environment, create a filter for this dashboard. Group by the class of services such as, gold, silver, and bronze. Default the selection to gold. In this way, your monitoring is not cluttered with less critical workloads.
- For complete visibility, consider adding physical server monitoring by using the appropriate management pack. For more information, see the following page.

## Network Configuration Dashboard

Use the **Network Configuration** dashboard to view the overall configuration of vSphere distributed switches in your environment, especially for the areas that need your attention.

### Design Considerations

See the Configuration Dashboards page for common design considerations among all the dashboards for configuration management.

### How to Use the Dashboard

The dashboard is organized into two sections for ease of use.

- The first section displays network configurations that need your attention.
  - There are five bar-charts that focus on critical security settings.
  - The last bar-chart displays the version of the vSphere Distribution Switch. Aim to keep the version current, or match your vSphere version.
- The second section provides overall configuration information, with the ability to drill down to a specific switch.
  - Click the row to select a switch from the list.
  - The ESX hosts, port groups, and the VMs on the switch are displayed.
  - Review each of the tables. For the ESX host table, ensure that the settings are consistent.
  - Some of the columns are color-coded to facilitate quick reviews. Adjust their threshold to either reflect your current situation or your desired ideal state.
  - You can sort the columns and export the result into a spreadsheet for further analysis.

### Points to Note

- `No data to display` does not imply that there is something wrong with data collection by VCF Operations. It might signify that none of the objects meet the filtering criteria of the widget, and as a result there is nothing to display.
- For complete visibility, consider adding physical network device monitoring by using the appropriate management pack. For more information, see the following page.
- To view the content of a slice in a pie-chart or a bucket in a bar-chart, click on it. The list cannot be exported. Clicking an object name, takes you to the object summary page. The page provides key configuration information, with other summary information.

## VM Configuration Dashboard

Use the **VM Configuration** dashboard to view the overall configuration of virtual machines in your environment, especially for the areas that need attention.

### Design Considerations

See the Configuration Dashboards page for common design considerations among all the dashboards for configuration management.

As there are many configurations to be verified, if you have a larger screen, add additional checks as you deem fit, or add legends to the pie-charts.

### How to Use the Dashboard

- Click the row to select a data center from the data center table.
  - In a large environment, loading thousands of VMs increases the web page loading time. As a result, the VM is grouped by data center. In addition, it might make sense to review the VM configuration per data center.
  - For a small environment, vSphere World is provided, so you can view all the VMs in the environment.

  The **VM Configuration** dashboard is organized into three sections for ease of use. All the three sections display the VM configuration for the selected data center.
- The first section covers limits, shares, and reservations.
  - Their values can easily become inconsistent among VMs, especially in an environment with multiple vCenter instances.
  - Shares should be mapped to a service level, to provide a larger proportion of shared resources to those VMs who pay more. This means that you should only have as many shares as your service levels. If your IaaS provides gold, silver, and bronze, then you should have only three types of shares.
  - Value of the shares and reservation is relative. If you move a VM from one cluster to another (in the same or different vCenter), you might have to adjust the shares.
  - Reservation impacts your capacity. Memory reservation works differently from CPU reservation, and it is more permanent.
- The second section covers VMware Tools.
  - VMware Tools is a key component of any VM, and should be kept running and up to date.
- The third section covers other key VM configurations.
  - Keep the configurations consistent by minimizing the variants. This helps to reduce complexity.
  - **VM Network Cards** widget. If you suspect that your environment might have a VM with no NIC, consider adding it as a dedicated bucket.
- The last section of the dashboard is collapsed by default.
  - You can view all the VMs with their key configurations.
  - You can sort the columns and export the results into a spreadsheet for further analysis.

### Points to Note

- The number of buckets in the pie-chart or bar-chart are balanced between the available screen estate, ease of use, and functionality. Modify the buckets to either reflect your current situation or your desired ideal state.
- `No data to display` does not imply that there is something wrong with data collection by VCF Operations. It might signify that none of the objects meet the filtering criteria of the widget, and as a result there is nothing to display.
- To view the content of a slice in a pie-chart or a bucket in a bar-chart, click on it. The list cannot be exported. Clicking an object name, takes you to the object summary page. The page provides key configuration information, with other summary information.
- The pie-chart and bar-chart cannot drive other widgets. For example, you cannot select one of the pie-slices or buckets, and expect it to act as a filter to a list or a table.
- You can apply a specific color in a pie-chart or distribution chart for a specific numeric value, but not string value. For example, you cannot apply the color red to the value `Not Installed`.

## vSAN Configuration Dashboard

The **vSAN Configuration** dashboard provides overall configuration details and is useful in large clusters with many vSANs, where you have to follow a certain standard configuration.

### Design Considerations

See Configuration Dashboards for common design considerations among all the dashboards for configuration management.

### How to Use the Dashboard

The **vSAN Configuration** dashboard is organized into three sections for ease of use.

- The first section displays six pie-charts.
  - There are five bar-charts that focus on critical security settings.
  - The last bar-chart shows the version of the vSphere Distribution Switch. Aim to keep the version current, or match your vSphere version.
- The second section displays three bar-charts.
  - The three bar-charts together provide a good overview of the vSAN key capacity configuration. By analyzing the distribution, you can identify if you have capacity configuration that is outside your expectation.
- The last section of the dashboard displays all the vSAN clusters with their key configuration.
  - Some of the columns are color-coded to facilitate quick reviews. Adjust their threshold to either reflect your current situation or your desired ideal state.
  - You can sort the columns and export the result into a spreadsheet for further analysis.

**Points to Note**

- The number of buckets in the pie-chart or bar-chart are balanced between the available screen estate, ease of use, and functionality. Modify the buckets to either reflect your current situation or your desired ideal state.
- To view the content of a slice in a pie-chart or a bucket in a bar-chart, click on it. The list cannot be exported. Clicking an object name, takes you to the object summary page. The page provides key configuration information, with other summary information.

## Workload Management Configuration Dashboard

This dashboard provides a quick configuration summary of all the key objects associated with workload management such as Supervisor Clusters, Namespaces, vSphere Pods and Tanzu Kubernetes clusters. It is essential that the configuration is consistent across all the objects. Configuration drifts may result in inconsistent performance or availability of the applications leveraging workload management Kubernetes constructs.

Use the dashboard to ensure that the configuration is consistent across all objects.

You can view the following widgets in the dashboard.

- **Environment Summary**
- **Supervisor Cluster Versions**
- **Cluster Status**
- **Pod Data**
- **Supervisor Cluster Configuration Summary**
- **Pod Configuration Summary**
- **Kubernetes cluster Configuration Summary**
- **Namespace Configuration Summary**

## Consumer \ Correct it? Dashboard

The **Consumer \ Correct it?** dashboard complements the main VM configuration dashboards by displaying the actual VMs, with their relevant information. The dashboard is designed for vSphere administrators and the platform team, to facilitate follow-up action with the VM owners. The **Consumer \ Correct it?** dashboard is one of the eight dashboards that check the environment for optimization opportunities.

**Design Considerations**

The dashboard is designed to focus on VMs that need attention. Lists are used to keep it simple, and show actual objects. The lists can be tailored using the filter and the custom group. The lists can also be exported for an offline discussion.

The dashboard is extendable, reflecting the reality that different customers have a different set of settings to verify. Since the dashboard layout is a collection of tables (List View), you can extend it by adding more tables. You can add more List View widgets to verify the VM configurations that your operations require.

**How to Use the Dashboard**

The **Consumer\Correct it?** dashboard is a collection of tables (List View), which can be reviewed independently. Click the object name to navigate to the Object Summary page to view more configurations. There can be valid reasons why specific configurations are not followed. It is recommended that you discuss best practices with VMware.

- Tools Widgets:
  - Using VMware Tools has multiple benefits.
  - VCF Operations uses VMware Tools to retrieve Guest OS metrics. Without this, right-sizing VM memory can be inaccurate, because the hypervisor metrics (VM Memory Consumed and VM Memory Active) are not designed to measure Windows or Linux memory utilization. ESX VMkernel does not have visibility into the Guest OS for security reasons.
  - Independent software vendor (ISV) support is the most common reason that VMware Tools is not installed. The ISV vendor might claim that no additional software is installed in their appliance unless they have certified it.
  - If VMware Tools is installed, there might be reasons why the application team deactivates it. The Infrastructure team should inform and educate their application team, and document the technical recommendations about why VMware Tools is recommended to be running all the time.
- CPU Limits and Memory Widgets:
  - It is recommended that you do not use memory and CPU limits as it can result in an unpredictable performance. The Guest OS is not aware of this restriction as it is at the hypervisor level. It is recommended that you shrink the VM instead.
- Guest OS Counters Missing Widget:
  - There is no visibility into the Guest OS performance counters because the requirements are not met. The memory counter is especially important as VM Consumed and VM Active are not replacements for Guest OS counters. See KB 55675 for more details.
- Old Snapshot Widget:
  - Ensure that the snapshot is removed within one day after the change request. If not, it might result in a large snapshot and impact the performance of the VM.

**Points to Note**

- Add a banner summary to the top of this dashboard so that you can verify if there is an incorrect confirmation. Add a scoreboard and select the World object and then collapse all the tables below. Create a super metric for each summary and apply it to the World object.
- In a large environment, create a filter for this dashboard to help you to focus on a segment of the environment. Group it by a class of service such as, gold, silver, and bronze. Default the selection to gold, your most important environment. In this way, your monitoring is not cluttered with less critical workloads.
- There are other VM configurations that maybe relevant to your environment. Review the list of VM settings that you might want to add to this dashboard.
- For context, add a property widget that lists the selected VM properties. In this way, you can check the property of your interest without leaving the screen. Multiple List View widgets can drive the same property widget, so you do not have to create one property widget for each List View.
- If your operations require it, add a list of VMs that do not have these three key performance counters: CPU Run Queue, CPU Context Switch, and Disk Queue Length.

## Consumer \ Optimize it? Dashboard

The **Consumer \ Optimize it?** dashboard complements the main VM configuration dashboard by displaying the actual VMs, with their relevant information. The dashboard is designed for vSphere administrators and the platform team, to facilitate follow-up action with the VM owners. The **Consumer \ Optimize it?** dashboard is one of the eight dashboards that check the environment for optimization opportunities. A suboptimal configuration might not impact performance or increase complexity, but it can be more expensive.

### Design Considerations

The **Consumer \ Optimize it?** dashboard follows the same design considerations specified for the Consumer \ Correct it? Dashboard. The eight Configuration > Review dashboards form an optimization flow and are designed as a set. Use them together, as you go through the optimization review process.

### How To Use the Dashboard

The **Consumer \ Optimize it?** dashboard is a collection of tables (List View), that can be reviewed independently. Click the object name to navigate to the Object Summary page to view more configurations. There can be valid reasons why specific configurations are not followed. It is recommended that you discuss best practices with VMware.

- VM Reservation:
  - VM reservation causes a positive impact on the VM, but a negative impact on the cluster. Total reservation cannot exceed cluster capacity. This creates a suboptimal cluster as VMs do not use the entire assigned memory at the same time.
  - VM reservation places a constraint on the DRS placement and HA calculation. Avoid using reservation as a means to differentiate performance SLA among all the VMs in the same cluster. It is difficult to correlate CPU Ready with CPU Reservation. A VM CPU Ready does not improve two times because you increase its CPU reservation by two times. There is no direct correlation.
- Guest OS visibility:
  - Since your workloads are sharing resources and are over-committed, your operations are easier if you know what is running inside. This helps with monitoring and troubleshooting, resulting in optimal operations.
  - For critical VMs, consider logging the Guest OS, such as Windows and Linux, to capture errors that do not surface as metrics. These errors typically appear as events in the log files or in the event database in the case of Windows. Use Log management to parse Windows events into log entries that can be analyzed.
- Snapshot:
  - Old snapshots tend to be larger. They consume more space and have a higher chance of impacting performance.

### Points to Note

See the **Points to Note** section as specified in the Consumer \ Correct it? Dashboard. This dashboard follows the same design considerations, and as a result shares limitations and customization ideas.

## Consumer \ Simplify it?

The **Consumer \ Simplify it?** dashboard complements the main VM configuration dashboard by displaying the actual VMs, with their relevant information. The dashboard is designed for vSphere administrators and the platform team, to facilitate follow-up action with the VM owners. The **Consumer \ Simplify it?** dashboard is one of the eight dashboards that check the environment for optimization opportunities.

### Design Considerations

The **Consumer \ Simplify it?** dashboard follows the same design considerations specified for the Consumer \ Correct it? Dashboard. The eight Configuration > Review dashboards form an optimization flow and are designed as a set. Use them together, as you go through the optimization review process.

### How to Use the Dashboard

The **Consumer \ Simplify it?** dashboard is a collection of tables (List View), that can be reviewed independently. Click the object name to navigate to the Object Summary page to view more configurations. There can be valid reasons why specific configurations are not followed. It is recommended that you discuss best practices with VMware.

- Large VMs (CPU, Memory, and Disk):
  - A large VM, relative to the underlying ESX host and datastore, requires more careful planning (Day 0) and monitoring (Day 2).

  - Ensure that the VM size does not exceed the size of the underlying ESX host. If your ESX host has CPU hyper-threading, do not count the logical processor. Instead, count the physical core. For best performance, keep it within a (non-uniform memory access) NUMA boundary.
  - During monitoring, verify if the VM is highly utilized. If the VM vCPU count is equal to the ESX cores, and the VM is running at almost full capacity, you might not be able to run other VMs. Large VMs can impact the performance of other VMs, especially if it is given higher shares. Only when the large VM is under-utilized, can the ESX hosts run other VMs.
  - If the number of configured vCPUs on a VM is higher than the number of cores per socket on the ESX, the VM can experience the NUMA effect. If the ESX has more than one physical CPU (socket), cross-NUMA access negatively impacts performance.
  - The larger the VM, the longer the time required to vMotion, Storage vMotion, and backup.
  - For disk space, if the disk is thin-provisioned and under-utilized, you can deploy other VMs in the same datastore. Ensure that the snapshot is tracked closely, as the risk of capacity running out is higher for a large virtual disk.
- VMs with many virtual disks:
  - It is simpler to have a 1:1 mapping between Guest OS partitions and the underlying virtual disk (VMDK or RDM).
  - For performance and capacity, evaluate the disks and partitions. Each virtual disk must be monitored in terms of IOPS, throughput, and latency. Having multiple virtual disks increases the monitoring and troubleshooting need.
  - If the reason for having many virtual disks is performance, identify which counter serves as proof that multiple virtual disks are required. It is possible that the performance required is met by a single virtual disk.
- VM with many IP addresses or NICs:
  - A VM might need multiple networks, such as production, back up, and management. It is recommended that you route the network interfaces through the NSX-Edge VM. A VM that has multiple network interfaces can bridge the network, causing security risks or network problems.
  - A VM that is part of multiple networks can do so with just a single NIC. A single NIC can be configured to access multiple networks, with each interface having their own IP configuration.

#### Points to Note

See the **Points to Note** section as specified in the Consumer \ Correct it? Dashboard. This dashboard follows the same design considerations, and hence shares limitations and customization ideas.

## Consumer \ Update it? Dashboard

The **Consumer \ Update it?** dashboard complements the main VM configuration dashboard by displaying the actual VMs, with their relevant information. The dashboard is designed for vSphere administrators and the platform team, to facilitate follow-up action with the VM owners. The **Consumer \ Update it?** dashboard is one of the eight dashboards that check the environment for optimization opportunities.

#### Design Considerations

The **Consumer \ Update it?** dashboard follows the same design considerations specified for the Consumer \ Correct it? Dashboard. The eight Configuration > Review dashboards form an optimization flow and are designed as a set. Use them together, as you go through the optimization review process.

#### How to Use the Dashboard

The **Consumer \ Update it?** dashboard is a collection of tables (List View), which can be reviewed independently. Click the object name to navigate to the Object Summary page to view more configurations. There can be valid reasons why specific configurations are not followed. It is recommended that you discuss best practices with VMware.

- Outdated Tools Widget:
  - Lists all the VMware Tools versions that are still supported. Tailor the filter to fit your operational needs.
- Outdated VM Hardware Widget:
  - Lists all the VM vmx versions that are not 13, 14, 15, or 16. Tailor the filter to fit your operational needs.
- Outdated Windows and Red Hat Widgets:
  - Lists all the Windows client versions that are not version 10.

  - Lists all the Windows server versions that are not versions 2016 and 2019.
  - Lists all the RHEL versions that are not version 7 or 8.
  - If you run other operating systems like Ubuntu, clone the widget. You can also repurpose the widget if you do not run RHEL and Windows.

#### Points to Note

See the **Points to Note** section as specified in the Consumer \ Correct it? Dashboard. This dashboard follows the same design considerations, and hence shares limitations and customization ideas.

## Provider \ Correct it? Dashboard

The **Provider \ Correct it?** dashboard complements the main vSphere configuration dashboards by displaying the actual vSphere objects, with their relevant information. The dashboard is designed for vSphere administrators and the platform team. The **Provider \ Correct it?** dashboard is one of the eight dashboards that check the environment for optimization opportunities.

#### Design Considerations

The **Provider \ Correct it?** dashboard follows the same design considerations specified in the Consumer \ Correct it? Dashboard. The eight Configuration > Review dashboards form an optimization flow and are designed as a set. Use them together, as you go through the optimization review process.

#### How to Use the Dashboard

The dashboard is organized into three sections for ease of use.

- The first section covers vSphere cluster configurations.
  - A cluster is the smallest logical building block for compute. Consider it as a single computer with physically independent components. As a result, consistency matters.
  - Clusters with DRS set to manual. This means that DRS initiated vMotion does not take place unless the administrator manually approves it. Since DRS calculates every five minutes, your quick approval is required to prevent a change of condition.
  - Clusters with HA disabled. Without high availability provided by the infrastructure, each application must protect itself from an infrastructure failure.
  - Clusters with DRS disabled. DRS focuses on performance and capacity, while HA focuses on availability. Without DRS, you must build a buffer on every ESX host to cope with peak demand.
  - Clusters with Admission Control disabled. Reservation is respected only when Admission Control is activated.
- The second section covers the ESX host configurations.
  - ESX with Network Time Protocol disabled. Logs are a critical component of operations, and are the main source of information in troubleshooting. While troubleshooting performance across objects, the sequence of logs determines which event is the likely root cause, as the oldest event starts the chain of events.
  - A disconnected ESX host indicates that the ESX host is not participating in HA and you cannot migrate any VM on it.
  - An ESX host that is in maintenance mode does not contribute resources to the cluster or the data center if there is a standalone ESX.
- The third section covers ESX host configurations that must be consistent within a cluster.
  - BIOS version and ESX versions.
  - BIOS Power Management, ESX: Power Management. Ideally, should be set to OS controlled. The ESX level should be set to balance level.
  - ESX Storage Path. Ensure that the number of paths and the path policies are identical.
  - ESX hardware specifications. Different specifications can result in inconsistent performances experienced by the VM.

#### Points to Note

- See the **Points to Note** section as specified in the Consumer \ Correct it? Dashboard. This dashboard follows the same design considerations, and as a result, shares limitations and customization ideas.
- If you have a standalone ESX, and you plan to replace it with a clustered ESX host, add a table to list them.
- Based on your security settings, add a table to check the Distributed Switch and Port Group to ensure that security settings such as promiscuous mode, are used correctly.

## Provider \ Optimize it? Dashboard

The **Provider \ Optimize it?** dashboard complements vSphere configuration dashboards by displaying the actual vSphere objects, with their relevant information. The dashboard is designed for vSphere administrators and the platform team. The **Provider \ Optimize it?** dashboard is one of the eight dashboards that checks the environment for optimization opportunities.

#### Design Considerations

The **Provider \ Optimize it?** dashboard follows the same design considerations specified in the Consumer \ Correct it? Dashboard. The eight Configuration > Review dashboards form an optimization flow and are designed as a set. Use them together, as you go through the optimization review process.

#### How to Use the Dashboard

The dashboard is organized into three sections for ease of use.

- The first section covers vSphere cluster configurations:
  - A small cluster has a higher HA overhead when compared to a large one. For example, a three-node cluster has 33% overhead while a 10-node cluster has 10%. For vSAN, a low number of hosts limits the availability option. Your choice of FTT is relatively more limited.
  - Many small clusters result in silos of resources. As a cluster behaves like a single computer, ensure that it has enough CPU cores, CPU GHz, and Memory. For ESX in 2020, it is typical to have 512 GB of RAM. This results in 12 TB of RAM for a 12-node cluster, which is enough for DRS to place many VMs as it balances them.
  - If there is a lot of reservation, add a list for clusters with a relatively high reservation. If your clusters are of different sizes, use a super metric to convert the reservation value to a percentage.
- The second section covers ESX host configurations.
  - Small ESX. A small host faces scalability limits in running a larger VM. While a 2-socket, 32-cores, 128 GB memory ESX can run 30 vCPU, 100 GB RAM VMs, the VM experiences a non-uniform memory access (NUMA) effect.
  - ESX powered off. You can mark the ESX hosts for decommissioning using the custom property feature of VCF Operations. You can then create a separate list, so they are not overlooked.
- The third section cover storage and network.
- Unused network (distributed port group). This is a potential security risk as you might not monitor it.

#### Points to Note

- See the **Points to Note** section as specified in the Consumer \ Correct it? Dashboard. This dashboard follows the same design considerations, and as a result, shares limitations and customization ideas.
- For CPU cores, a change in vSphere licensing means that the ideal core is 32-cores per CPU socket. This maximizes the software license.

## Provider \ Simplify it? Dashboard

The **Provider \ Simplify it?** dashboard complements vSphere configuration dashboards by displaying the actual vSphere objects, with their relevant information. The dashboard is designed for vSphere administrators and the platform team. The **Provider \ Simplify it?** dashboard is one of the eight dashboards that checks the environment for optimization opportunities.

### Design Considerations

The **Provider \ Simplify it?** dashboard follows the same design considerations specified in the Consumer \ Correct it? Dashboard. The eight Configuration > Review dashboards form an optimization flow and are designed as a set. Use them together, as you go through the optimization review process.

### How to Use the Dashboard

- Click the row in the **Clusters** widget to select one of the clusters from the table.
  - A cluster is more complex to operate when it has resource pools, shares, and limits.
- Review the list of resource pools:
  - Ensure that the number of VMs in each resource pool reflects the intended settings for the VM. The resource pool value is divided and shared among the VMs. The more the VMs, the lesser the resources allotted to each VM.
  - Verify if there are VMs who are siblings to the resource pools.
  - Verify if the resource pools are further split into subresource pools.
- Review the CPU Share and Memory Shares pie-charts:
  - Multiple combinations of shares, especially both CPU and memory, makes troubleshooting difficult.
  - Each share must map to exactly one class of service, such as one for gold and one for silver as the shares define the class of service. Shares are also relative, meaning the value depends on the value of sibling objects, such as, resource pool or VM. Ensure that the values are consistent across clusters to avoid unintended consequences while moving the VM to another cluster.
- Review the CPU Reservation and Memory Reservation tables:
  - High total reservation, especially both CPU and memory, complicates the cluster operations as it impacts the HA slot calculation, and limits the DRS choice of placement.
- Click the object name to navigate to the Object Summary page to view more configurations. There can be valid reasons why specific configurations are not followed. It is recommended that you discuss best practices with VMware.

### Points to Note

See the **Points to Note** section as specified in the Consumer \ Correct it? Dashboard. This dashboard follows the same design considerations, and as a result, shares limitations and customization ideas.

## Provider \ Update it? Dashboard

The **Provider \ Update it?** dashboard complements the main vSphere configuration dashboards by displaying the actual vSphere objects, with their relevant information. The dashboard is designed for vSphere administrators and the platform team. The **Provider \ Update it?** dashboard is one of the eight dashboards that checks the environment for optimization opportunities.

As part of operations best practices, keep the infrastructure up to date. Running outdated components that are too far behind the latest version, can cause support problems or upgrade problems. It is common that the fix for the problem is only available in the later versions. Outdated hardware can also result in higher operating costs. Outdated hardware might cost more data center footprint, such as rack space, cooling, and UPS. Refreshing your technology and consolidation are two common techniques to optimize cost.

### Design Considerations

The **Provider \ Update it?** dashboard follows the same design considerations specified in the Consumer \ Correct it? Dashboard. The eight Configuration > Review dashboards form an optimization flow and are designed as a set. Use them together, as you go through the optimization review process.

### How to Use the Dashboards

The **Consumer \ Update it?** dashboard is a collection of tables (List View) that can be reviewed independently. Click the object name to navigate to the Object Summary page to view more configurations. There can be valid reasons why specific configurations are not followed. It is recommended that you discuss best practices with VMware.

- Outdated vSphere Components Widgets:

  - Lists all the vCenter Servers versions that are not 6.7 or 7.0.
  - Lists all the ESX host versions that are not 6.5, 6.7, or 7.0.
  - Lists all the vSAN ESX host versions that are not 6.7 or 7.0. A more stringent filter is applied for vSAN because of a relatively higher maturity in the latest release. From VCF Operations and Log management, there are more counters, properties, and events that improve monitoring and troubleshooting.
  - Lists all the vSphere distributed switches, regardless of the version.
  - You should tailor the filter to fit your operational needs.
- Outdated Server BIOS Widget:
  - Lists all the ESX hosts regardless of the BIOS version. Edit the widget and tailor the filter to fit your operational needs.
- Other than customizing the existing widgets, consider adding the following checks:
  - ESX hosts with outdated hardware, using a filter based on your environment.
  - ESX hosts that are no longer on warranty. Create a custom property to capture the end of warranty.
  - Physical storage arrays with outdated firmware, model, and an expiring warranty.
  - Physical network switch with an outdated OS version and hardware model

    **Note:**
    Install the relevant management pack for the last two points.

**Points to Note**

See the **Points to Note** section as specified in the Consumer \ Correct it? Dashboard. This dashboard follows the same design considerations, and as a result, shares limitations and customization ideas.

# Cost Dashboards

The dashboards in the cost category cater to cloud administrators who are responsible for managing the expenses related to your cloud infrastructure. Using Cost dashboards, you can compare the cost of VMware cloud infrastructure with other cloud platforms. You can analyze the cloud comparison results and identify the opportunities to manage your cloud resources efficiently.

## Consumer Layer

The consumer layer dashboards of VCF Operations helps you to know how a customer can do a deeper analysis of the Return on Investment from the consumer perspective.

The available dashboards for consumers are:

- Chargeback VM Price Dashboard
- Showback VM Cost Dashboard
- Showback vSphere Pod Cost Dashboard

### Chargeback (Business Application Price)

In VCF Operations VMs are analyzed financially from the perspective of cost and price.

| | |
|---|---|
| **Cost** | Cost represents your expenditure to operate the VM for your customer. In VCF Operations, configure Cost Drivers to automatically calculate VM costs based on infrastructure expenses. Cost Drivers encompass server hardware, storage, licenses, applications, maintenance, labor, network, facilities, and additional costs set within VCF Operations. |
| **Price** | Price is what you charge your customer for using the VM. It can be based on the VM's cost or a predefined rate card, including upcharges, profit, and service charges. |

For configuring pricing cards for VCF Automation managed VMs, see the topic, What is Cloud Assembly. For other VMs, the information about configuring pricing cards is in the topic, Add New Pricing Card.

#### How to Use the Dashboard

Select an object in the **Select a Group** widget to view the price of the group. **Price Summary of Selected Group** shows the month to date price of the group. **VM Price Distribution (Top 100)** shows the most expensive VMs in the group. **Powered Off VMs** shows reclaimable VMs and their potential savings. **Idle VMs** shows reclaimable VMs and their potential savings. **VMs with Snapshots** shows reclaimable snapshots and their age. **Price of VMs in the Selected Group** shows the price and configuration of each VM in the selected group.

#### Customizing the Dashboard

The definition of a group is based on several constructs such as Virtual Machine Folder, Custom Groups, VMware Aria Automation Projects, and more. To add more objects for chargeback reporting, edit the **Select a Group** widget and select desired object types under **Output Filter** > **Basic** > **Object Types**.

> **Note:**
> Objects are not displayed correctly if the widget configuration is not complete. You must complete the object list widget configuration before you can edit the widget.

### Chargeback VM Price Dashboard

The chargeback VM price dashboard lets you know how much you must spend to run a VM on behalf of your customer. In VCF Operations, you can configure the cost drivers and let the system automatically determine how much a VM costs based on your infrastructure requirement. Cost Drivers cover server hardware, storage, licenses, application, maintenance, labor, network, facilities, and additional costs configured within VCF Operations.

Price is what you charge your customer for running their VM. The price of a VM can be based on the cost of the VM or based on a rate card that you define. Prices can include up charges, service charges, and others.

#### How to Use the Dashboard

- Select a Group widget displays the price of the group.
- Price Summary of Selected Group shows the month to date price of the group.
- VM Price Distribution (Top 100) shows the most expensive VMs in the group.
- Powered Off VMs shows reclaimable VMs and their potential savings.
- Idle VMs shows reclaimable VMs and their potential savings.
- VMs with Snapshots shows reclaimable snapshots and their age.
- Price of VMs in the Selected Group shows the price and configuration of each VM in the selected group.

### Showback (Business Application Cost)

This dashboard enables showback of VM cost based on groups.

Improve the accuracy of these costs by editing Cost Drivers. To edit Cost Drivers, see the topic, Editing Cost Drivers. Refer to Overview of Cost Drivers if you are not customizing Cost Drivers.

#### How to Use the Dashboard

Select an object in the **Select a Group** widget to view the cost of the group. **Cost Summary (This Month)** shows the month to date cost, potential savings, and projected cost of the group. **VM Cost Distribution (Top 100)** shows the most expensive VMs in the group. **Potential Savings (Top 10)** shows the VMs ranked by their potential savings. **Members of the Group (Select to View Trend)** shows the cost and configuration of each VM in the selected group. **Cost Trend of Selected VM** shows the trend of the VMs cost over time.

#### How to Customize the Dashboard

The definition of a group is based on several constructs such as Virtual Machine Folder, Custom Groups, VMware Aria Automation Projects, and more. To add more objects for showback reporting, edit the **Select a Group** widget and select desired object types under **Output Filter** > **Basic** > **Object Types**.

**Note:**
Objects are not displayed correctly if the widget configuration is not complete. You must complete the object list widget configuration before you can edit the widget.

### Showback VM Cost Dashboard

The Showback VM Cost dashboard provides a quick Showback of the cost associated with the VMs in a group. Based on the Showback you can improve the accuracy of the costs by editing the cost drivers. Cost drivers that are not customized use reference cost, cost driver customization is available only in Advanced or Enterprise edition of VCF Operations.

**How to Use the Dashboard**

- Select an object in the Select a Group widget to view the cost of the group.
- Cost Summary (This Month) shows the month to date cost, potential savings, and projected cost of the group.
- VM Cost Distribution (Top 100) shows the most expensive VMs in the group.
- Potential Savings (Top 10) shows the VMs ranked by their potential savings.
- Members of the Group (Select to View Trend) shows the cost and configuration of each VM in the selected group.
- Cost Trend of Selected VM shows the trend of the VMs cost over time.

### Showback vSphere Pod Cost Dashboard

The Showback vSphere Pod Cost Dashboard provides a quick Showback of the cost associated with the vSphere Pods in a group. Based on the Showback you can improve the accuracy of the costs by editing the cost drivers. Cost Drivers that are not customized use reference cost, cost driver customization is available only in Advanced or Enterprise edition of VCF Operations.

**How to Use the Dashboard**

- Select an object in the Select a Group widget to view the cost of the group.
- Cost Summary (This Month) shows the month to date cost and projected cost of the group.
- vSphere Pod Cost Distribution (Top 100) shows the most expensive vSphere Pods in the group.
- Idle vSphere Pods shows the vSphere Pods that have been identified as potentially idle.
- Members of the Group (Select to View Trend) shows the cost and configuration of each vSphere Pod in the selected group.
- Cost Trend of Selected vSphere Pod shows the trend of the vSphere Pod's cost over time.

## Provider Layer

The provider layer dashboards of VCF Operations, helps you to know how a customer can analyze the Return on Investment for the virtual infrastructure used in the customer’s environment.

The available dashboards for providers are:

- Assess Cost Dashboard
- Datacenter Cost Drivers Dashboard
- Server Hardware Depreciation Dashboard
- Base Rate Analysis Dashboard
- VM Cost versus Price Dashboard
- Reclaimable Hosts Dashboard

### Assess Cost Dashboards

The **Assess Cost** dashboard provides an overview of the scale of your infrastructure in terms of physical capacity available.

**Customizations Available for Your Use**

Certain data centers can be excluded, such as the development data centers that do not have to be expensed, by customizing the views in the widget.

**Widget Information**

- You can view the total cost of ownership per month for the infrastructure and the savings opportunities details, if any, for the infrastructure.
- You can view the details of the division of infrastructure investments across all data centers. The dashboard provides the magnitude of each data center in terms of the number of physical servers and virtual machines. It also provides details about the amount of savings that can be achieved from each of these data centers.
- The dashboard displays data about how you invest across clusters of different quality offered across all vCenter Servers.

## Base Rate Analysis Dashboard

The **Base Rate Analysis** dashboard helps you analyze the cost efficiency of your data center.

**Customizations Available for Your Use**

Certain data centers can be excluded, such as the development data centers that do not have to be expensed, by customizing the views in the widget.

**Widget Information**

- The total cost of ownership is the cost required to run your data center per month. This is derived from the cost drivers.
- The average cost per VM is derived by considering the cost of all the VMs in your environment. The cost of each VM depends on the base rate of the cluster the VM is placed on and its utilization. The base rate of the cluster is computed based on the total cost of ownership and the expected utilization levels of the cluster. Storage base rates are directly obtained from cost drivers.
- If the cluster is running on an allocation-based capacity model, the base rate is derived from the total cost of the cluster and the over-commit ratio. The base rate is indicative of how costly a resource is, on a given cluster.
- A base rate is derived from the total cost and the expected utilization of the cluster.
- A deeper analysis of the base rates can be performed using the CPU, memory, or storage-related widgets, which help rank clusters and datastores relative to their base rates.

## Business Applications Cost vs. Price

The Business Applications showback, chargeback, and cost vs. price dashboard helps analyze the relationship between the cost and pricing of VMs. Use this dashboard to ensure that the chargeback prices for VMs are adequate to cover their operational costs.

**How to Use the Dashboard**

- The **Select a Group** widget lets you select a group of VMs to analyze.
- The **Summary (Month to Date)** widget shows the month to date price and cost.
- The **Members of the Group (Select to View Trend)** shows all VMs in the selected group with their Month to Date Cost, Today's Cost, Month to Date Price, and Today's Price.
- The **Daily Cost and Daily Price** trend chart shows both cost and price over time.

> **Note:**
> Objects are not displayed correctly if the widget configuration is not complete. You must complete the object list widget configuration before you can edit the widget.

## Datacenter Cost Drivers Dashboard

The **Datacenter Cost Drivers** dashboard provides the cost of different data centers in a private cloud.

### Customizations Available for Your Use

Certain data centers can be excluded, such as the development data centers that do not have to be expensed, by customizing the views in the widget.

### Widget Information

- You can select individual data centers to view summary and trends. The summary of the data center costs is grouped into two:
  - Compute. Covers all the costs that are spent on compute related hardware, software, and services.
  - Non-Compute. Covers storage and network.
- Expense trends provide cost variations over a period which indicate infrastructure additions or removal to the data center.
- Cluster expenses indicate the component clusters of a data center that consume the costs. Datastores that represent the storage part of the data center cost are listed alongside.

  **Note:**
  Network costs are mapped directly to ESX hosts and hence are costed under compute as well, as of today. This might change in the future.
- When you select a cluster, you can view the component hosts that the cluster is made up of and their monthly depreciated costs. It also provides details on the purchase cost of the server and how many months until it depreciates completely.

  **Note:**
  Server costs can be suggested out-of-the-box by the system, or can be customized by the user. Depreciation information is not available for servers when the server costs are suggested out-of-the-box by the system. Depreciation information is available for those servers when the server cost is customized by the user.

## Reclaimable Hosts Dashboard

The Reclaimable Hosts dashboard helps you to identify clusters with reclaimable hosts and the potential cost savings from reclaiming the hosts. Reclaimable hosts are identified from the Total Recommended Capacity generated by the AI powered capacity engine in VCF Operations.

### Widget Information

- The Reclaimable Hosts Cost Pie Chart displays the reclaimable host cost distribution for individual clusters in your virtual environment.
- Potential Savings graph depicts the total cost savings (potential) for all the clusters in your virtual environment for a given period.
- The Top 10 Clusters with Reclaimable Hosts displays the number of reclaimable hosts.
- The Top 10 Clusters with Reclaimable Hosts by Cost displays the top 10 reclaimable hosts by cost.

## Server Hardware Depreciation Dashboard

The Server Hardware Depreciation Dashboard helps you to calculate the depreciation value for server hardware which is marked as owned in Cost Drivers. You can configure the depreciation cost settings as per your business requirement.

### Widget Information

- The Server Purchase Cost is the total purchase price of all servers as entered in Cost Drivers.

- Accumulated Depreciation is the amount of server purchase costs that have been depreciated according to purchase date and depreciation settings.
- Remaining Depreciation is the amount of server purchase costs left to be depreciated.
- Number of Fully Depreciated Servers identifies servers that have been fully depreciated. These servers may exhibit higher failure rates or have lower capacity. Use What-If scenarios to model the cost and capacity impact of replacing these servers.

### VM Cost vs. Price Dashboard

The VM Cost vs. Price Dashboard helps you to analyze the relationship between cost and price for virtual machines. You can use this dashboard to ensure the price of VMs for chargeback is sufficient to cover the cost of running virtual machines.

#### How to Use the Dashboard

- Select a Group allows selection of a group of VMs to analyze.
- Summary (Month to Date) shows the month to date price and cost.
- Members of the Group (Select to View Trend) shows all VMs in the selected group with their Month to Date Cost, Today's Cost, Month to Date Price, and Today's Price.
- Daily Cost and Daily Price trend chart shows both cost and price over time.

## Cost Optimization

The cost optimization dashboards are as follows:

- Cost Optimization Dashboard
- Potential Cost Savings Dashboard
- Realized Cost Savings Dashboard
- Total Cost of Ownership Dashboard
- VM Rightsizing Details Dashboard

### Cost Optimization Dashboard

The Cost Optimization dashboard helps you to measure the return on investment if you use VCF Operations to manage your virtual infrastructure. You can track the total cost of ownership of the entire environment along with potential savings and realized savings from recommendations provided, the dashboard helps you quantify the cost efficiency and cost savings over time.

Potential Savings is a summary of all cost savings opportunities identified by VCF Operations. Realized Savings is a summary of cost savings from actions performed that are related to recommendations provided by VCF Operations.

#### Widget Information

- Total Cost of Ownership provides details of the monthly cost of server hardware, licenses, maintenance, facilities, labor, network, storage, and additional costs.
- Average Cost per VM is a good indicator of cost efficiency over time. It is natural for the cost per VM to go up when new capacity is added and trend downwards as additional capacity is consumed. The goal is to reduce the average cost per VM over time.
- Realize Savings Breakdown shows the cost of reclaimed resources from the VM identified by VCF Operations.
- Potential Savings covers the cost savings opportunities identified by VCF Operations.

### Potential Cost Savings Dashboard

The Potential Cost Savings Dashboard helps you to measure the cost saving as reported by VCF Operations. You can evaluate the potential savings to track recommendations and improve cost efficiency over time. The dashboard shows

both cost savings and capacity savings for idle VMs, powered off VMs, VM snapshots, orphaned disks, oversized VMs, and reclaimable hosts.

**Widget Information**

- The Cost Savings Breakdown widget displays potential savings and reclaimable capacity for idle VMs, powered off VMs, VM snapshots, orphaned disks, oversized VMs, and reclaimable hosts. You can also view the allocation changes for the oversized VMs.
- Reclaimable widget provides the metric details for the reclaimable vCPU, reclaimable memory, and reclaimable disk space.
- The Optimization Opportunities Breakdown widget covers the projected costs to improve performance as identified by VCF Operations.
- Allocation Changes for Undersized VMs shows the number of vCPUs and GB of memory to add to undersized VMs.

## Realized Cost Savings Dashboard

The Realized Cost Savings Dashboard helps you to quantify the realized cost savings from actions performed that are related to recommendations provided by VCF Operations. You can analyze the realized savings to track improvements to cost efficiency over time. Realized savings covers powered off VMs that were flagged as idle, deleted VMs that were flagged as idle or powered off, deleted snapshots that were flagged as reclaimable, deleted disks that were flagged as orphaned, oversized VMs that were rightsized, and deleted hosts that were flagged as reclaimable.

**Widget Information**

- Realized Savings covers the cost savings from reclamation opportunities recommended by VCF Operations.
- The Reclaimed Capacity shows the amount of capacity that was reclaimed based on recommendations from VCF Operations.
- Allocation Changes for Oversized VMs shows the number of vCPUs and GB of memory removed from formerly oversized VMs.
- Cost of Deleted VMS shows the cost of all deleted VMs in the past 30 days, shows the cost of all deleted VMs (by cluster) for the past 30 days, and shows the year-to-date cost of all deleted VMs.

## Total Cost of Ownership Dashboard

The Total Cost of Ownership dashboard helps you to understand the total cost of ownership of your environment from multiple perspectives. You can use this dashboard to learn how cost drivers, capacity, and data centers affect the total cost of ownership.

**Widget Information**

- Cost Driver Breakdown widget shows how cost drivers affect the total cost of ownership.
- Cost of Capacity Used and Capacity Remaining widgets shows cost breakdown by the cost of capacity used and the cost of capacity remaining.
- Cost per Datacenter widget shows how the costs broken down per data center.

## VM Rightsizing Details Dashboard

The VM Rightsizing Details dashboard provides an overview of the rightsizing recommendations for Undersized VMs and Oversized VMs. Rightsizing is defined as changing the amount of resources allocated to a VM based on the Recommended Size for a VM. Recommended Size is the maximum projected utilization for the projection period from the current time to 30 days after the warning threshold value for time remaining.

**How to Use the Dashboard**

- Select a Cluster, Datacenter, or World Object.

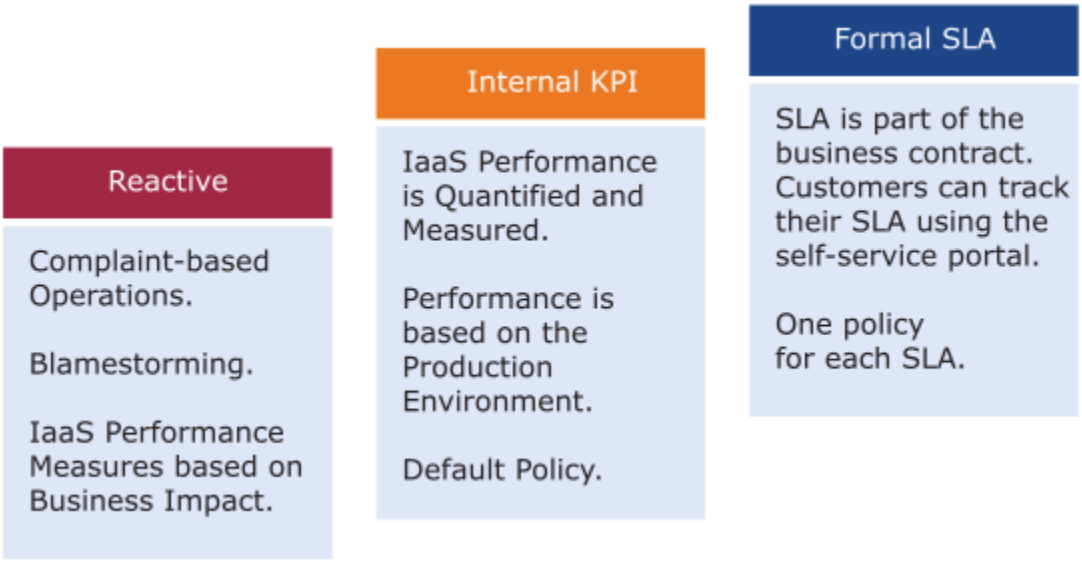
- Select an Undersized VM to view the recommendations.
- Select an Oversized VM to view the recommendations.
- Search for a VM to view the recommendations.

## Performance Dashboards

Performance is about ensuring workloads get the necessary resources. Key Performance Indicators (KPI) can be used to identify performance problems related to workloads. Use these KPIs to define SLAs associated with tiers of service. These dashboards use KPIs to display the performance of workloads at the consumer layer and the aggregate performance of workloads at the provider layer.

SLA is the formal business contract that you have with your customers. Typically, SLA is between the IaaS provider (the infrastructure team) and the IaaS customer (the application team or business unit). Formal SLA needs operational transformation, for example, it requires more than technical changes and you might need to look at the contract, price (not cost), process, and people. KPI covers SLA metrics and additional metrics that provide early warning. If you do not have an SLA, then start with Internal KPI. You must understand and profile the actual performance of your IaaS. Use the default settings in VCF Operations if you do not have your own threshold, as those thresholds have been selected to support proactive operations.

The following graphics depict the above relationship.




### The Three Processes of Performance Management

In performance management, there are three distinct processes.

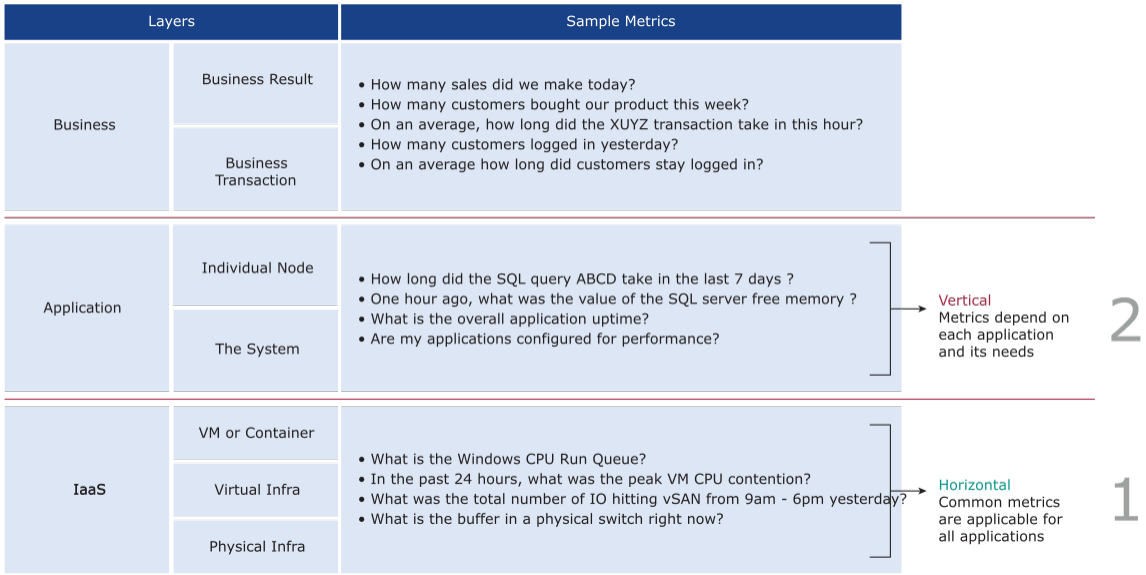
- Planning. Set your performance goals. When you architect a vSAN, you must know how many milliseconds of disk latency you want. 10 milliseconds measured at the VM level (not the vSAN level) is a good start.
- Monitoring. Compare the plan with the actual. Does the reality match what your architecture was supposed to deliver? If not, you must fix it.
- Troubleshooting. When the reality is not according to the plan, you must fix it proactively and not wait for issues and complaints.

To understand what is not healthy for performance management consider the following areas in the given order.

1. Contention: This is the primary indicator.
2. Configuration: Check the version incompatibilities.
3. Availability: Check for soft errors. vMotion stun time, lock up. This requires Log Insight.
4. Utilization: Check this in the end. If the first three parameters are good, you can skip this.

## The Three Layers of Performance Management

There are three main realms of enterprise applications. Each of these realms has its own set of teams. Each team has a set of unique responsibilities and requires the associated skill set. The three realms comprise of Business, Application, and IaaS. Refer to the graphic below to understand the three layers and the typical questions asked on each layer.



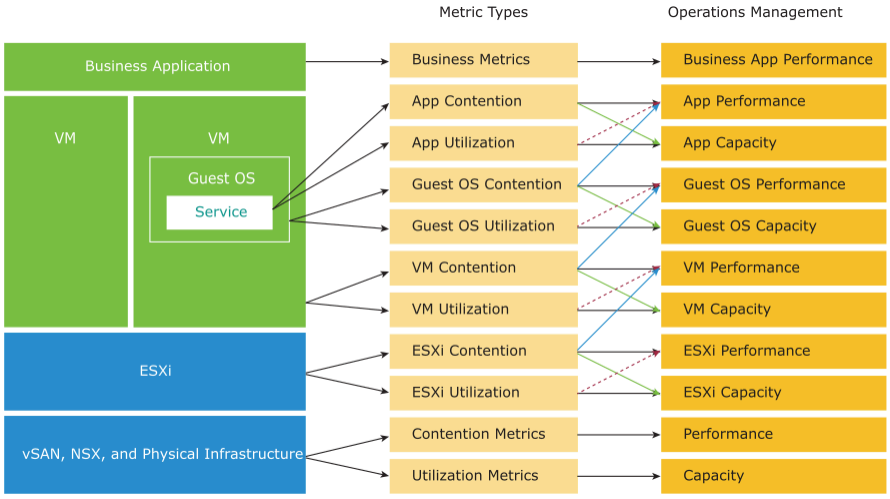

Performance Management is largely an exercise in elimination. The methodology slices each layer and determines if that layer is causing the performance problem. Hence it is imperative to have a single metric to indicate if a particular layer is performing or not. This primary metric is aptly named Key Performance Indicator (KPI).

The upper layer depends on the layer below it, and hence the infrastructure layer is typically the source of contention. As a result, focus on the bottom layer first, as it serves as the foundation for the layer above it. The good part is this layer is typically a horizontal layer, providing a set of generic infrastructure services, regardless of what business applications are running on it.

## The Two Metrics of Performance Management

The primary counter for performance is contention. Most look at utilization, because they fear something wrong might happen if utilization is high. That something is contention. Contention manifests in different forms like, queue, latency, dropped, canceled, and context switch.

However, do not confuse ultra-high utilization indicators as a performance problem. If your ESX host experiences ballooning, compression, and swapping, it does not mean that your VM has a performance problem. You measure the performance of the host by how well it serves its VMs. While performance is related to the ESX host utilization, the performance metric is not based on the utilization, instead it is based on contention metrics.



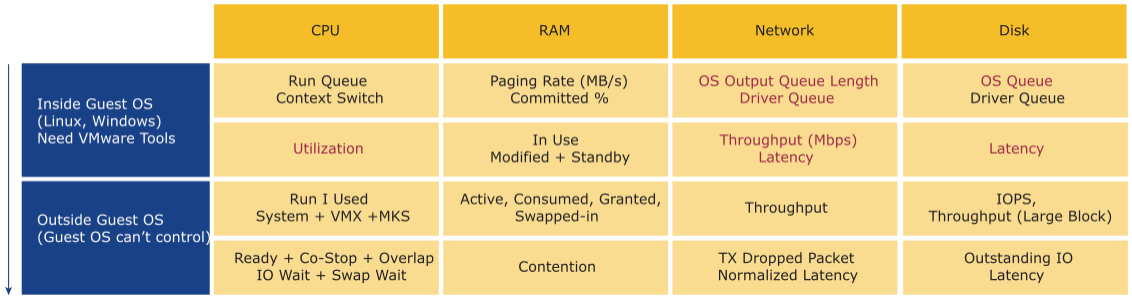

It is possible for VMs in the cluster get affected from poor performance, while the cluster utilization is low. One main reason is cluster utilization looks at the provider layer (ESX), while performance looks at an individual consumer (VM). The following table shows various possible reasons.

| Infra Configuration | VM and Guest OS Configuration |
|---|---|
| ESX Settings<br>• Host and BIOS power management causes Frequency to drop.<br>• HT enabled. It looks like twice the capacity, but it is actually 1.25 X throughput.<br>• ESX - HW compatibility. Driver and firmware are two areas that can impact the performance.<br>• Mismatch of queue depths along the various storage stacks. Must calibrate all the way to the physical array.<br>• vMotion too slow or high stunned time. | VM: Limit, Share, and Reservation<br>• Make sure that no limit is set. CPU ready includes limit.<br>• Make sure that the shares are consistent (as per what the VMs want or you agree to.)<br>• Avoid reservation if possible. This impacts the net available resources for the other VMs. |
| Network<br>• MTU mismatch.<br>• Hops. Especially horse-shoe, or going through multiple ESX. | Size: NUMA effect. VM spanning NUMA nodes. |
| Cluster Settings<br>• Inconsistent configuration among hosts in a cluster. EVC Mode can play a part if the hosts are from different generations.<br>• Resource Pool<br>– Make sure the shares match the number of VMs.<br>– Make sure that no VM is siblings to RP.<br>• VM- Host Affinity.<br>• DRS Setting. | Snapshot. IO is processes 2x.<br>VM drivers. |
| vSAN<br>• The host where the storage was having performance issues. | Windows or Linux process ping pong, process runaway, and OS level queue. |

From the performance management point of view, the vSphere cluster is the smallest logical building block of the resources. While the resource pool and VM Host affinity can provide a smaller slice, they are operationally complex, and they cannot deliver the promised quality of IaaS service. Resource pool cannot provide a differentiated class of service. For example, your SLA states that gold is two times faster than silver because it is charged at 200% more. The resource pool can give gold two times more shares. Whether those extra shares translate into half the CPU readiness cannot be determined up front.

## VM Performance

Since VM is the most important object in vSphere, it warrants an extra explanation. The graphic below lists the counters you should look at.?



| | CPU | RAM | Network | Disk |
|---|---|---|---|---|
| Inside Guest OS (Linux, Windows) Need VMware Tools | Run Queue<br>Context Switch | Paging Rate (MB/s)<br>Committed % | OS Output Queue Length<br>Driver Queue | OS Queue<br>Driver Queue |
| | Utilization | In Use<br>Modified + Standby | Throughput (Mbps)<br>Latency | Latency |
| Outside Guest OS (Guest OS can't control) | Run I Used<br>System + VMX +MKS | Active, Consumed, Granted, Swapped-in | Throughput | IOPS,<br>Throughput (Large Block) |
| | Ready + Co-Stop + Overlap<br>IO Wait + Swap Wait | Contention | TX Dropped Packet<br>Normalized Latency | Outstanding IO<br>Latency |

The KPI counters can get technical for some users, so VCF Operations includeS a starting line to get them started. You can adjust the threshold, once you profile your environment. This profiling is a good exercise, as most customers do not have a baseline.The profiling requires an advanced edition.

| | Metric | Green | Yellow | Orange | Red |
|---|---|---|---|---|---|
| Guest OS Contention | Total CPU Run Queue | 0-5 | > 5 | > 10 | > 20 |
| | CPU Context Switch Rate | 0 - 5K | < 25K | < 100K | > 100K |
| | Total Disk Queue Length | 0 - 25 | > 25 | > 50 | > 100 |
| Guest OS Usage | RAM Free (MB) | > 512 MB | > 256 | > 128 | ≤ 128 |
| | RAM Page-in Rate (KB/s) | 0 - 25K | > 25 K | > 50 K | > 100K |
| VM Contention | CPU Co-Stop (%) | 0 - 2.5% | > 1 | > 3 | > 5 |
| | [SLA] CPU Ready (%) | 0 - 2.5% | > 2.5 | > 5 | > 7.5 |
| | Total CPU Overlap (ms) at VM level | 0 - 1000 | > 1000 | > 2500 | > 5000 |
| | CPU IO Wait | 0 - 1000 | > 1000 | > 2500 | > 5000 |
| | [SLA] RAM Contention (%) | 0 - 1% | > 1 | > 2 | > 4 |
| | [SLA] Disk Latency (ms) | 0 - 10 ms | > 10 | > 20 | > 40 |
| | [SLA] Network TX Dropped Packet | 0 | > 0 | > 1 | > 2 |
| VM Usage | CPU Usage (%) | 0 - 85% | > 85 | > 90 | > 95 |

## Performance Metrics

VCF Operations uses the following threshold for internal KPI.

| IaaS | VM Counter | Threshold |
|---|---|---|
| CPU | Ready | 2.5% |
| RAM | Contention | 1% |
| Disk | Latency | 10 ms |
| Network | TX Dropped Packet | 0 |

The table is an example of a stringent threshold. A high standard for performance is used because it is an internal KPI for the consumption of the infrastructure team. It is not an external formal SLA that is confirmed with the customers. There must be a buffer between the internal KPI and the external SLA so that the operations team receive early warnings and has the time to react before the external SLA is breached. A high standard also works from the mission critical point to view to the development environment. If the standard is set to the least performing environment, then it cannot be applied to the more critical development.

A single threshold is used to keep the operations simple. This means that the performance in production is expected to have a higher score than the development environment. The development environment performance is expected to be worse than the production environment, while everything else is equal. A single threshold helps to explain the difference in Quality of Service (QoS) provided by a different class of service. For example, if you pay less, you get a poor performance and if you pay half the price, expect to get half the performance.

The four elements of IaaS (CPU, RAM, Disk, and Network) as mentioned in the table, are evaluated on every collection cycle. The collection time is set at five minutes as it is an appropriate balance for monitoring. If SLA is based on one minute, it is too close and results in either cost increase or reduction in threshold.

#### Design Considerations

All the performance dashboards share the same design principles. They are intentionally designed to be similar, as it is confusing if each dashboard looks different from one another, considering they have the same objective.

The dashboards are designed with separate two sections: summary and detail.

- The summary section is typically placed at the top of the dashboard to provide the overall picture.
- The detail section is placed below the summary section. It lets you drill down into a specific object. For example, you can get the detailed performance report of any specific VM.

In the detail section, use the quick context switch to check the performance of multiple objects during performance troubleshooting. For example, if you are looking at the VM performance, you can view the VM-specific information and the KPIs without changing screens. You can move from one VM to another and view the details without opening multiple windows.

The dashboard uses progressive disclosure to minimize information overload and ensure the webpage loads fast. Also, if your browser session remains, the interface remembers your last selections.

Many of the performance and capacity dashboards share a similar layout since there is a shared commonality between these pillars of operations.

## Guest OS Performance Profiling Dashboard

Use the **Guest OS Performance Profiling** dashboard to know the actual performance of your environment.

Some counters directly impact the performance of Windows or Linux, the operating systems running inside the VM. These KPIs are outside the control of the hypervisor.

Modern operating systems such as Linux and Windows use memory as cache, since it is faster than a disk. Some counters directly impact the performance of Windows or Linux. These KPIs are outside the control of a hypervisor, which means that the ESX VMkernel cannot control the increase or decrease of the KPI values. The KPI visibility also requires an agent, such as VMware Tools. As a result, they are typically excluded in performance monitoring.

Since they are closer to the applications, it is critical to know their values and establish an acceptable range. The acceptable level of these KPIs among all the VMs in your environment varies. By profiling the actual performance across time and from all VMs, you can establish a threshold that is supported by facts. Since there are 8766 instances of 5 minutes in a month, profiling 1000 VM over a month means you are analyzing 8.8 million datapoints.

#### Design Considerations

The dashboard uses progressive disclosure to minimize information overload and ensures that the webpage loads fast.

In a large environment, loading thousands of VMs increases the loading time of VCF Operations. As a result, the VM is grouped by data center. For a small environment, vSphere World is provided so you can see all the VMs in the environment.

#### How to Use the Dashboard

Select data center from the data centers list. The three tables listing CPU, memory, and disk will show the VMs in the selected data center or vSphere world. Each table shows the highest value in the last one week (2016 datapoints based on five minutes collection cycles), and hence uses the term max as a prefix, for example Max Page-Out/sec or Max Guest OS Disk Queue.

Select any of the VMs in any of the tables. The three line charts are displayed. They are showing data from the same VM to facilitate correlation.

- CPU table widget:
  - The Max CPU Queue column shows the highest number of processes in the queue during the given period. As a best practice, keep the queue below three for each queue. A VM with eight CPUs has eight queues, hence keep this number below 24.

  - The CPU Hyperthreading gives twice the queue as it should as both threads are interspersed in the core pipeline.
  - CPU Context Switch. There is a cost associated with the context switch. There is no guidance for this number, and it varies widely.
- Memory list widget:
  - In memory paging, the modern operating systems (Linux and Windows) use memory as cache, it is much faster than a disk. It proactively pre-fetches pages and anticipates future needs (Windows calls this Superfetch). The rate pages that are being brought in and out can reveal memory performance abnormalities. A sudden change, or one that has sustained over time, can indicate page faults. Page faults indicate that pages are not readily available and must be brought in. If a page fault occurs too frequently, it can impact application performance. While there is no concrete guidance, as it varies by application, you can view a relative size. operating systems typically use 4 KB or 2 MB page sizes.
- Disk list widget:
  - Disk queues are queued IO commands that are not sent to the VM. They have been retained inside the Guest OS (either at a kernel level or a driver level). A high disk queue in the guest OS, accompanied by low IOPS at the VM, can indicate that the IO commands are stuck waiting on processing by Windows/Linux. There is no concrete guidance regarding these IO commands threshold as it varies for different applications. You should view this with the Outstanding Disk IO at the VM layer.

#### Points to Note

- These Guest OS widgets do not appear unless the vSphere pre-requisites are met.
- Once you determine an acceptable threshold for your environment, consider adding thresholds to the table so you can easily view the VMs that exceed a threshold.
- The CPU queue is the sum from all virtual CPUs. A larger VM can tolerate a higher queue as it has more processors. If you want to compare VMs of different sizes, create a super metric that calculates the queue per vCPU. For more information, see Create a Super Metric.
- Group the VM by clusters of the same class (for example, Gold), so you can see the profile for each environment.
- For a smaller environment, consider changing the table from listing data centers to listing clusters.

## Network Top Talkers Dashboard

Use the **Network Top Talkers** dashboard to monitor network demand in your IaaS. In a shared environment, a few VMs generating excessive activity can impact the entire data center. While a single VM might not cause a serious problem, a few of them can.

#### Design Considerations

The **Network Top Talkers** dashboard helps you analyze how hard these VMs hit your IaaS. It classifies the workload into two: short bursts and sustained hits. A short burst lasts for a short period, maybe for a few minutes. A sustained hit can last for an hour and cause serious problems.

The **Network Top Talker** dashboard forms a pair with the **Storage Heavy Hitter** dashboard. To understand the IO demand in your environment, use both of them concurrently.

The **Network Top Talkers** dashboard displays sustained hits that last for an hour, as they can cause serious problems in a shared IaaS environment. You can identify the villain VMs and compare their demands with the capabilities of the underlying IaaS.

#### How to Use the Dashboard

The dashboard shows the current workload. This is the total network load (received and transmitted) from all the vSphere environments monitored by VCF Operations. The idea is to give you an indicator on how hard the overall load is.

- Select a data center from the data centers list.
  - The columns show the number of clusters, ESX hosts, and VMs for each data center. The VM count includes the powered off VM. To only see the running VM count, edit the widget.
  - If you want to see information from all the data centers, select the vSphere world row.
  - Upon selection, the Total Demand Line chart and the Top Talkers tables fill up.

- Total Demand Line Chart
  - The total throughput (received and transmitted) in the selected data center.
  - Displays both, the five minute peak and the hourly average in one line chart. You can click the metric name to hide it.
- Top Talkers Table
  - The table shows the most demanding VM. You can identify the villain VM and compare their demands with the capabilities of the underlying IaaS. Knowing the infrastructure capability is important. For example, an ESX with 2 x10 GB port can theoretically handle 20 GB TX + 20 GB RX as its full duplex.

#### Points to Note

- Understanding high demand helps you monitor IaaS and plan your capacity. IaaS provides four services, CPU, memory, disk, and network. While CPU, memory, and disk are bound, an active VM can consume all your network bandwidth, packet per second capacity, and the storage IOPS capcity. A VM with 4 vCPU and 16 GB memory cannot consume more than this amount, the same applies to disk space. A VM configured with 100 GB disk space cannot consume more than that.
- Network throughput, disk throughput, and disk IOPS can spike as their physical limits are very high per VM. This means that IaaS has enough capacity for all workloads and performs well until the VMs start consuming abnormally high amounts of network and disk bandwidth.

## Storage Heavy Hitters Dashboard

The **Storage Heavy Hitters** dashboard forms a pair with the **Network Top Talkers** dashboard. To understand the IO demands in your environment, use both of them together. If you are using ethernet-based storage, storage traffic runs over the same physical network as your ethernet-based network traffic.

#### Design Considerations

The **Storage Heavy Hitters** dashboard forms a pair with the **Network Top Talkers** dashboard, so they share a consideration behind their design. For more information, see Network Top Talkers Dashboard.

#### How to Use the Dashboard

- See the **Network Top Talkers** dashboard as they have the same design.
  - The main difference between **Storage Heavy Hitters** and **Network Top Talkers** is that the storage IO has two dimensions: IOPS and throughput.
  - Network IO does not have the IOPS dimension as the packet size is identical (1500 bytes being the standard packet, and 9000 bytes being the jumbo frames).
  - Storage IOPs and throughput are related, so use both to gain insight, they should display a similar pattern. If not, that indicates varying block sizes. For example, a throughput spike without an accompanying IOPs spike indicates large block sizes.
- Which VMs hit the storage the hardest.
  - The table shows the most demanding VM. You can identify the villain VM and compare their demands with the capabilities of the underlying IaaS. Knowing the infrastructure capability is important, because different classes of SSD have different IOPS and throughput capabilities.

    After identifying the villain VM, talk to the VM owners if the numbers are excessive during peak hours and identify the reasons behind the excessive usage. You must ensure that they do not create a hot spot. For example, vSAN cluster with > 100 disk can handle numerous IOPS but if the VM objects are only on a few disks, those disks can become a hot spot.

#### Points to Note

- Interpreting IOPs and throughput metrics depends on your underlying physical storage. For visibility into this hardware layer, add physical storage metrics to the dashboard.

# VM Contention Dashboard

The **VM Contention** dashboard is the primary dashboard for VM performance. It is designed for VMware administrators or architects. It can be used for both, monitoring, and troubleshooting. Once you determine that there is a performance issue, use the **VM Utilization** dashboard to see if the contention is caused by high utilization.

## Design Considerations

This dashboard is used as part of your Standard Operating Procedure (SOP). It is designed for daily use, hence the views are set to show data for the last 24 hours. The dashboard provides performance metrics for virtual machines in the selected data center.

To view the common design considerations among all performance management dashboards, see the Performance Dashboards.

For understanding the performance concept of the selected counters and their thresholds, see the Performance Dashboards

## How to Use the Dashboard

- Select a data center from the data center table.
  - For a smaller environment, select vSphere World to see all the VMs from all the data centers.

    **Note:**
    The count of VMs includes the powered off VMs too. To exclude powered off VMs, modify the widget and select the running VM metric.
- The two bar charts are automatically shown.
  - Use them together to get an insight about your CPU readiness and your Memory contention analysis. Analyze how the cluster serves the VMs. For each VM, it picks the worst metric in the last 24 hours. By default, VCF Operations collects data every 5 minutes, so this is the highest value among 288 datapoints. Once it has the value from each VM, the bar charts puts each VM in the respective performance buckets. The threshold in the buckets considers best practices, hence they are color coded.
  - For any critical environment, expect that all the VMs are served well by the IaaS. You must see green on both distribution charts. For development purposes, you can tolerate a small amount of contention in both CPU and Memory.
- VM Performance in selected Data Center.
  - Analyze by data center as performance problems tend to be isolated in a single physical environment. For example, a performance problem in country A typically does not cause a performance problem in country B.
  - The table is sorted by KPI Breach columns, directing your attention to the VMs that are not served well by the IaaS.
  - The table shows the hostnames known by Windows or Linux. This is the name that the application team or VM owner knows, as they might not be familiar with the VM name.
  - The rest of the columns show performance counters. Because the goal is proactive monitoring, the counters are the worst and not the average, during the monitoring period. Because the operations context here is performance, not capacity, the table considers the last 24 hours only. Daily use is encouraged as any activity older than 24 hours is considered irrelevant from a performance troubleshooting viewpoint.
  - The column KPI Breach counts the number of SLA breaches in any given 5 minutes. As a VM consumes four resources of IaaS (CPU, memory, disk, and network), the counter varies from 0–4, with 0 being the ideal. The value 4 indicates that all 4 IaaS services are not delivered. The same threshold is used regardless of class of service, as this is an internal KPI, not an external SLA. Your internal threshold should be more stringent, so that you have a reaction time.
- Select a VM from the table.
  - All the health charts show the KPI of that VM.
  - The health charts display the last value, lowest value, and the peak value. Expect that the peak is within your threshold.

**Points to Note**

- This dashboard uses Guest OS counters and VM counters appropriately. The two layers are distinct layers, and they each provide a unique visibility that the other layers might not give. For example, when the VMkernel de-schedules a VM as it has to process something else (for example, other VM, kernel interrupt). The Guest OS does not know the reason. In fact, it experiences frozen time for that particular vCPU running on the physical core and experiences time jumps when it is scheduled again.
- Guest OS counters logically require VMware Tools.
- The health chart is color coded. Change the settings if it does not suit your environment. If you are unsure of what suitable numbers to set for your environment, profile the metrics. The Guest OS Performance Profiling Dashboard provides an example of how to profile metrics.
- For a smaller environment with one or two data centers, change the filter from data center to cluster. Once you are list a cluster, you can then add the cluster performance (%) metric and sort them in an ascending order. This way the cluster that needs immediate attention is on the top.
- If you have a screen real estate, group the VMs by cluster or by ESX host. This way, you can quickly see if the problem is in a particular cluster or ESX host.
- Change the default timeline from one week to one day as and when required to suit your operations.
- If you navigate a lot to the **VM Utilization** dashboard from this dashboard, add a connection using the dashboard to dashboard navigation feature. For more details, see Dashboard Navigation Details.

## VM Performance Dashboard

Use the **VM Performance** dashboard to find out if a VM has a performance problem. As a first step, when a VM has a problem, verify if other VMs have the same problem. If the problem is widespread the root cause is not with the VM.

**How to Use the Dashboard**

The **VM Performance** dashboard is organized into sections for ease of use.

- Select a data center from the **Datacenters** widget. To find out if there is a performance problem, what the problem is, and the extent of the problem, use the following three bar charts together: **Are VMs facing CPU Ready**, **Are VMs facing Memory Contention**, **Are VMs facing Disk Latency**. Each bar chart analyzes how the VMs are served by the cluster. These bar charts indicate if the VMs are waiting for CPU resources, facing memory contention, or disk latency. For each VM, it picks the worst metric in the last 24 hours. By default, VCF Operations collects data every 5 minutes, so this is the highest value among 288 datapoints (12 x 24 = 288). Once it has the value from each VM, the bar charts put each VM in the respective performance buckets. The threshold in the buckets considers best practices, and hence they are color coded. For each bar chart, you can change the time period to the period of your interest. The maximum number is then displayed. The value is the worst 20-seconds, within the 5-minute collection time period. For your mission-critical environment, you must expect that all the VMs are being served well by the IaaS. If you see green on the distribution charts, you do not have to analyze further.

  For development, you may tolerate a small amount of contention in both CPU and Memory as you need to balance cost.

  You can also change the filter from data center to cluster. If you are listing clusters, you can then add the cluster performance (%) metric and sort them in ascending order. This way the cluster that needs immediate attention is on top.

  You can click on the bar to see the list of VMs under that performance bucket. From there, you can select a VM, and it's KPI is automatically displayed on the lower section of the dashboard.
- **Multiple VM Analysis**

  When you select a data center from the **VMs Performance in selected Datacenter** widget, the table listing all the VMs in the data center is displayed.

  The table is sorted by the KPI Breached column, directing your attention to the VMs that are not served well by the IaaS. The column counts the number of SLA breaches in any given 5-minute period. It is based on the counter `Performance \ Number of KPIs Breached`. As a VM consumes four resources of IaaS (CPU, memory, disk, and network), the counter varies from zero through four, with zero being the ideal. The value four indicates that all four IaaS services are not delivered. The same threshold is used regardless of class of service, as this is an internal KPI, and not an external SLA.

Because the goal is proactive monitoring, as opposed to reactive troubleshooting, the counters show the worst value instead of the average of the monitoring period.

- **Per VM Analysis**
  When you select a VM from the table, the CPU, memory, disk, and network performance charts are automatically displayed, each widget showing the KPIs of that VM.
- **Alerts**
  The relevant alerts are displayed automatically. You can view the settings by editing the widget, and adjust them accordingly to fit your operational needs.
- **Virtual Disks**
  A VM can have many disks, and it is possible that these disks may have different performance levels. The table lists the individual virtual disks and their contention and utilization metrics.
- **Configuration**
  The relevant configuration of the selected VM is displayed. You can customize as appropriate.
- **Relationship**
  From the VM, you can navigate to the parent cluster or datastore. Use the **Relationship** widget to navigate and auto select the associated cluster or datastore.

## VM Utilization Dashboard

The VMware administrator uses the **VM Utilization** dashboard with the **VM Contention** dashboard for managing performance.

### Design Considerations

Use the **VM Utilization** dashboard to identify virtual machines with a high utilization in a selected data center. When utilization exceeds 100%, performance can be negatively impacted especially when a queue develops inside the Windows or Linux operating systems. By default, VCF Operations has a 5-minute collection interval. For 5 minutes, there might be 300 seconds worth of data points. If a spike is experienced for a few seconds, it might not be visible if the remaining 300 seconds is low utilization.

To view the common design considerations among all performance management dashboards, see the Performance Dashboards.

### How to Use the Dashboard

- Select a data center from the data center table.
  - For a smaller environment, select vSphere World to see all the VMs from all the data centers.

    **Note:**
    The count of VMs includes the powered off VMs too. To exclude powered off VMs, modify the widget and select the running VM metric.
- VM Peak CPU Usage (%).
  - There is no peak memory use as it is not applicable. Memory is a form of storage, for example consider a hard disk occupied space. A 90% utilization of the total space is not slower than 10%. This means that the issue is related to capacity issue and not performance.
  - The bar chart is color coded using five colors instead of four. The color gray is introduced to convey any wastage. Resources that are hardly utilized do not signify that the performance is at its peak. It can also mean the opposite. For example, if a VM needs 1+ vCPU, configuring it with 2 CPUs results in better performance instead of configuring it with 128 CPUs.
- VM Peak Utilization.
  - Analyze by data center as performance problems tend to be isolated in a single physical environment. For example, a performance problem in country A typically does not cause a performance problem in country B.
  - The table focuses on peak utilization, because the context is performance and not capacity.
- Select a VM from the table.
  - All the health charts show the KPI of that VM.

  - Compliment the free memory with the memory IOPS or the memory throughput metric. The metrics in a gigabyte measure the space, and not the speed. Memory is a form of storage, so what you must measure is the rate, for example, read-write per second.

#### Points to Note

- The **VM Utilization** dashboard complements the **VM Contention** dashboard. For more information, see the points to note in the VM Contention Dashboard.

## Troubleshoot an Application Dashboard

The VMware Cloud Foundation Operations Application Management Pack provides discovered applications to be managed in VCF Operations. Using the **Troubleshoot an Application** dashboard, users can see the applications and the relevant metrics and alerts for the selected application. The dashboard also displays its relationship to the infrastructure. In the list of metrics, select a metric to see its trend over time.

## Cluster Contention Dashboard

The **Cluster Contention** dashboard is the primary dashboard for vSphere cluster performance. It is designed for VMware administrators or architects. It can be used for both, monitoring and troubleshooting. Once you determine that there is a performance issue, use the **Cluster Utilization** dashboard to see if the contention is caused by high utilization.

#### Design Considerations

This dashboard is used as part of your Standard Operating Procedure (SOP). It is designed for daily use, hence the views are set to show data for the last 24 hours. The dashboard provides performance metrics for virtual machines in the selected data center.

Utilization of the cluster is not shown in the **Cluster Contention** dashboard. You must separate the two concepts: utilization and contention. Performance and capacity are different concepts managed by two separate teams. Both CPU and memory are also shown separately. You can have a problem with one, without any issue in the other. CPU is more common as memory tends to have a lower overcommit ratio.

To view the common design considerations among all performance management dashboards, see the Performance Dashboards.

#### How to Use the Dashboard

- Average Cluster Performance (%).
  - This is the primary KPI for your entire IaaS. It plots how your IaaS is performing every 5 minutes, giving you the trend view of the overall performance.
  - The metric itself is simply the average of the Cluster KPI / Performance (%) metric. This performance metric in turn averages the VM Performance / Number of KPIs Breached metric from all the running VMs in the cluster. Hence a value of 100% indicates that every running VM in the cluster is served well.
  - As this KPI takes into account every running VM in your environment, the number should be steady. The analogy in real life is the stock market index. While individual stocks can be volatile, overall the index should be relatively steady on a 5 minutes by 5 minutes basis.
  - The relative movement of the metric is as important as the absolute value of the metric. Your absolute number might not be as high you want it to be, but if there are no complaints for a long time, then there is no urgent business justification to improve it.
- Clusters Performance.
  - It lists all the clusters, sorted by the least performing cluster in the last one week. You can change this time period.
  - The worst performance shows the lowest number in the time period. As VCF Operations collects data every 5 minutes, there are 12 x 24 x 7 = 2016 data points in a week. This column shows the worst point among these 2016 datapoints.
  - A single number among 2016 datapoints can be an outlier that needs to be complemented with another number sometimes. A logical choice is the average of these numbers. For the average performance to be low, a lot of

criterias have to be low. Waiting for the average causes a delay in your operations, and rise in complaints. For performance monitoring, the 95th percentile is a better summary than the average.
  - Your cluster should function at a 100% and perform its fuctions as planned.
- Select a cluster from the table.
  - All the health charts show the KPI of the selected cluster.
  - For performance, it is important to show both the depth and breadth of the performance problems. A problem that impacts one or two VMs requires a different troubleshooting than a problem that impacts all the VMs in the cluster.
  - The depth is shown by reporting the worst among any VM counter. So the highest value of VM CPU Ready, VM Memory contention, and VM Disk Latency among all the running VMs are shown. If the worst number is good, then you do not need to look at the rest of the VMs.
  - A large cluster with thousands of VMs can have a single VM experiencing poor performance while 99.9% of the VM population is fine. The depth counter might not report that most VMs are fine. It only reports the worst. This is where the breadth counters come in.
  - The breadth counters report the percentage of the VM population that is experiencing performance problem. The threshold is set to be stringent, as the goal is to provide early warning and activate proactive operations.

**Points to Note**

It is possible for VMs in the cluster to suffer from poor performance, while the cluster utilization is low. One main reason is cluster utilization looks at the provider layer (ESX), while performance looks at individual consumer (VM). The following table shows various possible reasons.

| Event | Aware? |
|---|---|
| Power Management | No |
| HT | No |
| Ready | No |
| Co-Stop | No |
| System | No |
| Steal | No |
| IO Wait | No |
| Memory Wait | No |

From the performance management point of view, the vSphere cluster is the smallest logical building block of the resources. While the resource pool and VM Host affinity can provide a smaller slice, they are operationally complex, and they cannot deliver the promised quality of IaaS service. Resource pool cannot provide a differentiated class of service. For example, your SLA states that gold is two times faster than silver because it is charged at 200% more. The resource pool can give gold two times more shares. Whether those extra shares translate into half the CPU readiness cannot be determined up front.

Certain settings such as DRS automation level and the presence of many resource pools can impact performance. Consider adding a property widget to show the relevant property of a selected cluster, and a relationship widget to show resource pools.

For a large environment with many clusters, add a grouping to make the list more manageable. Group it by class of service, so you can focus more on the critical clusters.

## Cluster Performance Dashboard

The **Cluster Performance** dashboard combines the functionality of the **Cluster Contention**, **Cluster Utilization**, ESX **Contention**, and ESX **Utilization** dashboards.

**How to Use the Dashboard**

The **Cluster Performance** dashboard is organized into sections for ease of use.

- **Overall Analysis**
  The **Average Cluster Performance** health chart is green when all the clusters are performing well. If the clusters are unable to serve the VMs well, all the clusters are no longer green, with a few occurrences of red.
  As the chart displays all the clusters, it uses the vSphere World object. This object is the parent of the vCenter object, and so it displays all the clusters from all the vCenter Servers.
  The metric used is `Performance \ Clusters Performance (%)` and is the primary KPI for your entire IaaS. It plots how your IaaS is performing every five minutes, giving you the trend view of overall performance.
- **Multi-Cloud Analysis**
  If the health chart is not green, and you want to find out which clusters are not performing, use the **vSphere Clusters** widget. The table lists all the clusters, starting with the cluster with the lowest performance. By default, the data displayed is from the last 24 hours. The **Worst Performance** column displays the lowest number in the time period. By default, VCF Operations collects data every 5 minutes, so this is the lowest point among 288 datapoints (12 x 24 = 288).
  The **Worst Performance** column displays the lowest performance in the last period, specified under **Time Settings**.
- **Per-Cluster Analysis**
  Select a cluster from the **vSphere Clusters** widget to see the trend over time. After you determine the cluster you want to investigate, review the five scoreboards: CPU, memory, disk, network, and others.
- **VM Shares**
  A common root cause for uneven performance problems is uneven shares. Each slice in the pie chart must correspond to a class of service. If the entire cluster is serving one class, then you should see a simple circle with no slices.
- **Resource Pool Analysis**
  Resource pool is another common reason behind uneven VM performances. A cluster with too many resource pools makes performance management difficult.
  The **Resource Pools in the cluster** provides a table listing all the resource pools.
- **ESX Analysis**
  A cluster is a collection of ESX hosts and the performance can be affected by uneven performance among the member hosts. You can drill down from a cluster to the ESX hosts. The ESX **Hosts in the Selected Cluster** widget lists all the ESX hosts in the cluster, sorted by the worst performance in the last 24 hours. If the table displays values in green, you do not have to analyze further.
  You can change the time period to the period of your interest. The maximum number will be reflected accordingly.
  The table helps you quickly compare the performance of each ESX. You can also see the performance over time, to see a trend.
  Certain settings such as power management and hyper threading can impact performance. The ESX **Hosts in the Selected Cluster** widget displays the relevant property of a selected ESX Host.
- **VM Analysis**
  When you select a cluster or ESX from the **Running VMs in the selected Cluster or ESX widget** the VMs that are running are automatically listed. Use this table to verify if the cluster or host performance problems were caused by VM configuration and usage. It is possible that the VM was not on the same host at the time of problem, due to vMotion. To drill down into a particular VM, select it and click the double arrow before the widget title.
- **Datastore Analysis**
  Use the **Shared datastores in the Cluster** widget to see a list of shared datastores accessible by hosts in the cluster. You can also drill down to the selected datastore.

## Cluster Utilization Dashboard

The VMware administrator uses the **Cluster Utilization** dashboard with the **Cluster Contention** dashboard for performance management.

## Design Considerations

This dashboard supports the **Cluster Contention** dashboard. Use it to identify vSphere clusters with high utilization in a selected data center. When utilization exceeds 100%, performance can be negatively impacted especially when VMs experience a contention. By default, VCF Operations has a 5-minutes collection interval. For five minutes, there may be 300 seconds worth of data points. If a spike is experienced for a few seconds, it may not be visible if the remaining of the 300 seconds is low utilization.

To view the common design considerations among all performance management dashboards, see the Performance Dashboards.

## How to Use the Dashboard

- CPU(%) and Memory (%).
  - Review the CPU and Memory distribution charts for an overview of the CPU and memory utilization of the clusters.
  - The highest metric in the last one week is used. Average or 95th percentile is not used as this is utilization and not contention. High utilization does not mean bad performance.
  - One week is used instead of one day to give you a longer time horizon and covers the weekend. Adjust the timeline as you deem fit for your operations.
  - Expect memory to be higher than CPU, as it is a form of cache. The Memory Consumed counter is used as it is more appropriate than the Memory Active counter.
  - Low utilization can actually indicate bad performance, as not much of real work gets done. The chart uses the dark gray color for low utilization.
- Clusters Utilization.
  - The cluster utilization table lists all the clusters, sorted by the highest utilization in the last one week. If the table displays the green color, then there is no need to analyze further.
  - You can change the time period to the period of your interest. The maximum number is reflected accordingly.
- Select a cluster from the table.
  - All the utilization charts show the key utilization metrics of the selected cluster.
  - For memory, the high utilization counters are explicitly shown, Balloon, Compressed, and Swapped. Notice they exist even though utilization is not even at 90%, indicating high pressure in the past. If you look only at utilization, you might think you are safe.
  - The line charts show both average and highest among ESX hosts in the cluster. The reason is unbalanced and it is not rare. There are many settings that can contribute to it (for example, DRS settings, VM Reservation, VM – Host Affinity, Resource Pool, Stretched Cluster, and Large VMs).
  - The disk IOPS is split into read and write to gain insight into the behavior. Some workload is read oriented, while others are write oriented.
  - The disk throughput is not shown as it sums all the traffic. In reality, each ESX host has its own limit.
  - The vMotion line chart is added, as a high number of vMotion can indicate that the cluster load is volatile, assuming the DRS Automation level is not set to the most sensitive setting.

## Points to Note

- If your operations team have some forms of standardization that utilization should not exceed a certain threshold, you can add the threshold into the line chart. The threshold line helps less technical teams as they can see how the real value compares with the threshold.
- Consider adding a third distribution chart. Show the balloon counter in this third chart, as it complements the consumed counter. If there is no ballooning, a high consumed value is in fact better than a lower value.
- The workload metric can exceed a 100% because it is demand / usable capacity * 100. This can happen if you have four hosts in a cluster with each host running at 100% demand and admission control is set to 50%.
- The **VM Utilization** dashboard complements the **VM Contention** dashboard. For more information, see the points to note in the Cluster Contention Dashboard.

## VM Rightsizing Dashboard

The **VM Rightsizing** dashboard helps you adjust the VM size for optimal performance at the lowest cost. It covers both undersized and oversized scenarios. The dashboard looks at the long term trend and covers both undersized and oversized scenarios. This dashboard is designed for the Capacity and the Operations teams, as rightsizing a VM helps in the day-to-day performance.

### How to Use the Dashboard

- **Overall Analysis**
  The scoreboard provides a summary of the total undersized and oversized CPU and memory.
  You can select either a data center or a cluster. In most cases, rightsizing analysis should be done at the cluster level as VMs typically do not move inter-cluster. The counters are displayed to provide better context. Focus on reclaiming VM capacity in a cluster that is low on capacity remaining.
  The distribution charts that display rightsizing are automatically displayed. Other than the bar charts, the **Undersized VM** and **Oversized VM** tables list the actual VMs.
- **VM Analysis**
  Select a VM to investigate further. The utilization is automatically displayed. VM usage reflects the amount of capacity consumed. This is based on the aggregate vCPU usage at five minute granularity, which provides a clear understanding of capacity used.
  Rightsizing a VM can help improve performance for the VM as well as the cluster. Because of this, metrics that show performance bottlenecks, such as CPU Ready and CPU Run Queue, are provided to help you confirm how rightsizing may result in less contention and better performance.
  Memory utilization is collected from the guest OS via VMTools. If guest OS metrics are not available, then the memory configured value is used instead. Rightsizing memory improves performance by reducing memory ballooning and contention. For example, VMs with over-provisioned memory are more likely to experience ballooning.

## Datastore Performance Dashboard

Use the **Datastore Performance** dashboard to view performance problems related to storage such as high latency, high outstanding IO, and low utilization. This dashboard is designed for both the VMware administrator and the Storage administrator, to foster a closer collaboration between the two teams. Local datastores are treated separately.

### Design Considerations

To view the common design considerations among all performance management dashboards, see the Performance Dashboards.

### How to Use the Dashboard

The **Datastore Capacity** dashboard is layered, gradually providing details as you work top-down in the dashboard.

- **Overall Analysis**
  Select a data center from the **Datacenters** table. The three bar charts which are, **VM Performance**, **Read Performance**, and **Write Performance** provide an overall analysis of the datastore performance in a given vCenter data center or vSphere World. They work together to provide better insight. Just like other performance charts, the value displayed is the worst value during the time period. After you select a data center, if the Worst VM Disk Latency displays `No data to display`, it means that you have no observed latency issues for VM disk performance.
  The **VM Performance** chart displays the kind of latency and how many VMs experience that kind of latency. The **VM Performance** chart is your primary chart as it measures latency at the VM level. The **Read Performance** and **Write Performance** charts measure latency at the datastore level, which means they are the normalized average of all VMs in that datastore. Expect the **VM Performance** chart to be higher than the **Read Performance** and **Write Performance** charts. Read and write latency are displayed separately for better insight.
- **Datastore Analysis**
  The **Datastores Performance** table automatically lists all the shared datastores in the data center or vSphere World. Both the worst (peak) performance and the 95th percentile are displayed. If the latter is close to the peak and it is

also high, then it is a sustained problem. If the latter is low, then the problem is for a short duration. The table is color coded. Select a datastore that you want to troubleshoot. The relevant metrics and configuration are displayed.

- **VM Analysis**
  The list of VMs running in the selected datastore is displayed, with the relevant contention and utilization counters. Select the VM that you want to troubleshoot. The contention and utilization of the VM are automatically displayed. The number is at the VM level. If you suspect one of the virtual disks has high latency, use the counter Peak Virtual Disk Read Latency (ms) and Peak Virtual Disk Write Latency (ms).
- **Relationship**
  From the **Related Clusters and Hosts to selected Datastores** widget, select either an ESX host, a vSphere cluster, or a vSAN cluster. The relevant contention and utilization counters are displayed.

## ESX Contention Dashboard

The ESX **Contention** dashboard is the primary dashboard for managing ESX host performance. The VMware administrator or architect can use it to monitor and troubleshoot any performance issue. If you determine that there is a performance issue, use the **ESX Utilization** dashboard to see if the cause for the contention is high utilization.

### Design Consideration

The ESX **Contention** dashboard complements the Cluster Contention Dashboard, and shares the same design consideration.

This dashboard is used as part of your Standard Operating Procedure (SOP). It is designed to be used daily, hence the views are set to show data in the last 24 hours. The dashboard provides performance metrics for virtual machines in the selected data center.

To view the common design considerations among all performance management dashboards, see the Performance Dashboards.

### How to Use the Dashboard

- ESX CPU Performance and ESX Memory Performance.
  - Review the two distribution charts for an overview of all the ESX host's utilization and memory performance.
  - Both charts are using the percentage of VM facing performance counter and not the worst performance among VM counter because you are looking at the ESX performance and not at the single VM performance. See how it handles all the VMs.
  - The bar chart is color coded. Keep the percentage of the VM population not being served under 10%.
- ESX Hosts Performance.
  - The ESX hosts performance table lists all the ESX hosts, sorted by the worst performance in the last 24 hours. If the table is displays the green color, then there is no need to analyze further. The reason 24 hours is selected instead of one week is that the performance greater than 24 hours are likely to be irrelevant.
  - You can change the time period to the period of your interest. The maximum number is reflected accordingly.
- Select an ESX host from the table.
  - All the health charts show the KPI of the selected cluster.
  - For performance, it is important to show both depth and breadth of a performance problem. A problem that impacts one or two VMs require a different troubleshooting than a problem that impacts all VMs in the cluster.
  - Worst CPU overlap among VMs in the host is included as it indicates a lot of interruptions. A running VM might get interrupted because the VMkernel needs the physical core to run something else. High and frequent numbers of interruptions are not healthy and can impact the VM performance.
  - Expect the network error to be 1% and dropped packet to be 0 most of the times, if not always. If it is not zero, analyze it to see if there are any patterns across all ESX hosts, and bring it up with your network team.

### Points to Note

- Consider adding a third distribution chart and display the CPU co-stop counter in this third chart, as it complements the CPU ready counter. If your environment has relatively slow network and storage IO, you can add IO wait too.

- Unlike the **Cluster Performance** dashboard, there is no average ESX hosts performance (%) at the vSphere World level. The reason is most ESX hosts are part of a cluster and monitoring should be done at the cluster level.
- Certain settings such as power management and hyper threading can impact the performance. Consider adding a property widget to show relevant properties of a selected ESX host.

## ESX Utilization Dashboard

The VMware administrator uses the ESX **Utilization** dashboard with the ESX **Contention** dashboard to manage performance.

### Design Considerations

The ESX **Utilization** dashboard supports the ESX **Contention** dashboard. Use it to identify vSphere clusters with high utilization in a selected data center. When utilization exceeds 100%, performance can be negatively impacted, especially when a VM experiences contention. By default, VCF Operations has a 5-minute collection interval. For 5 minutes, there might be 300 seconds worth of data points. If a spike is experienced for a few seconds, it might not be visible if the remaining 300 seconds is low utilization.

The dashboard complements the Cluster Utilization Dashboard dashboard, by providing the extra details. Hence it has a similar layout.

To view the common design considerations among all performance management dashboards, see the Performance Dashboards.

### How to Use the Dashboard

- ESX Hosts Utilization.
  - It lists all the ESX hosts, sorted by the highest utilization in the last one week. If the table is all displaying the green color, then there is no need to analyze further.
  - You can change the time period to the period of your interest. The maximum number is reflected accordingly.
- Select an ESX host from the table.
  - All the utilization charts display the key utilization metrics of the selected cluster.
  - For memory, the high utilization counters are explicitly shown, for example balloon, compressed, or swapped. You might notice they exist even though utilization is not even at 90%, indicating that there was a high pressure in the past. If you look at only utilization, you might think you are safe.
  - The disk IOPS and the disk throughput are split into read and write to gain an insight into the behavior. Some workload is read oriented, while others are write oriented.
  - The network throughput is split into sent (transmit) and received to gain insight into the behavior. The total usage can be misleading because it sums up the send and receive traffic. In reality the network pipe is one for each direction (due to the full duplex nature of Ethernet), and not shared.

### Points to Note

If your operations team have some forms of standardization that the utilization should not exceed a certain threshold, you can add the threshold into the line chart. The threshold line helps less technical teams as they can see how the real value compares with the threshold. For more information, see the points to note in the ESX Contention Dashboard.

## Network Performance Dashboard

Use the **Network Performance** dashboard to view performance problems related to network such as high latency, frequent retransmit, and many dropped packets. This dashboard is designed for both the VMware administrator and the Network administrator, to foster a closer collaboration between the two teams.

### Design Considerations

To view the common design considerations among all performance management dashboards, see the Performance Dashboards.

The dashboard activates you to drill down from the distributed switch to the ESX host and port groups in the switch, and then to the VM.

### How to Use the Dashboard

- Distributed Switches.
  - The distributed switches table lists all the switches, sorted by the highest packet dropped. The table splits the incoming traffic and the outgoing traffic for better analysis.
  - As the focus is on performance and not capacity, the throughput counters are not shown.
- Select a switch from the distributed switches table.
  - The health chart shows the dropped packet trend over time.
  - It does not narrow down the list of port groups automatically, as the list of port groups are always showing all the port groups in your environment.
  - If necessary, expand the two collapsed widgets. They show the network throughput and broadcast packets. Utilization is also shown so that you can correlate and understand whether the dropped packets are due to higher utilization.
- Port Groups and ESX Hosts in the selected switch.
  - They get listed when you select a switch from the distributed switches table.
  - Just like the distributed switch, you can also see their relevant counts.
- If your environment has unused network switches, you can filter them out from this list, as this dashboard focuses only on performance.

### Points to Note

- vSphere network is by nature distributed. Each ESX contributes to the physical NIC. This represents the physical capacity. Distributed switch and its port groups span across these independent network cards. This makes it harder to define and measure its performance. An unbalance can happen among ESX hosts or physical NIC. In a sense, it is like distributed storage ( vSAN). Capacity management does not apply to a port group, since its upper limit (also known as the physical capacity) can vary by even a minute.
- Latency within a data center should be below 1 millisecond. Use VMware Aria Network Insight to study the latency or the retransmitting problems, caused by moving into the lateral traffic.
- Add a physical network using the appropriate management pack.

Most packets are unicast, between a pair of sender and receiver. If your environment has many VMs sending broadcast packets to everyone and multicast packets to many targets, add a Top-N widget to find out which VMs are sending these packets.

## vSAN Contention Dashboard

The **vSAN Contention** dashboard is the primary dashboard for managing vSAN performance. The VMware administrator or architect can use it to monitor and troubleshoot the vSAN cluster performance. If you determine that there is a performance issue, use the **vSAN Utilization** dashboard to see if the cause for the contention is high utilization.

### Design Considerations

To view the common design considerations among all performance management dashboards, see the Performance Dashboards.

The **vSAN Contention** dashboard complements the ESX Contention Dashboard, and shares the same design consideration. It focuses on the storage and vSAN specific metrics, and does not repeat what is already covered. It does not list any non vSAN cluster.

### How to Use the Dashboard

- vSAN Peak VM Latency, vSAN Peak CPU Ready, vSAN Peak Dropped Packet.
  - Review the three distribution charts for an overview of all the vSAN clusters performance.

  - The vSAN peak VM latency chart shows the distribution of disk latency experienced by all the VMs in the cluster. You should expect most of the VMs to experience latency that matches your expectation. For example, in an all flash systems, the VMs should not have >20 ms disk latency. If your vSAN environment is all flash, you must adjust the distribution bucket to a more stringent set.
  - The vSAN peak CPU ready chart shows if any of the vSAN kernel modules has to wait for CPU. Expect this number to be near 0% and below 1%, as vSAN should not wait for CPU time. vSAN gets higher priority than VM World as it lives in the kernel space.
  - The vSAN peak dropped packet chart shows if any of the vSAN clusters are dropping packet in the vSAN network (not the VM network). vSAN relies on the network to keep the cluster in-sync. This number should be near 0% and less than 1%.
- vSAN Clusters.
  - It lists all the vSAN clusters, sorted by the least performing.
  - It lists all the ESX hosts, sorted by the worst performance in the last 24 hours. If the table is showing all green, then there is no need to analyze further. The reason 24 hours is selected instead of one week is that the performance issues greater than 24 hours are likely to be irrelevant.
  - You can change the time period to the period of your interest. The maximum number is reflected accordingly.
- Select a vSAN cluster from the vSAN clusters table.
  - All the health charts show the KPI of the selected cluster.
  - If you are using SMART, the two heat maps at the bottom of the dashboard provide early warning.

**Points to Note**

- A large vSAN cluster can have many components. Each of these components can have multiple performance metrics. The total number of KPI can reach hundreds of metrics. For example, take a 10 node cluster. It can have 530 counters to check. VCF Operations aggregates them by introducing a set of KPIs. This analysis reduces the number to a more manageable number. The following table shows the KPIs and their formula.

| Name | What it is |
|---|---|
| Max Capacity Disk Latency (ms) | Highest latency among all capacity disks take the worst, not average, as the latency in a single capacity disk is already an average of all its VMs. If there are 50 VMs on the disk and 30 are issuing IO on it, then its average is among 30. |
| Min Disk Group Write Buffer Free (%) | Lowest free capacity among all the disk group write buffers. If this number is low, one of your buffers is not enough. While you want to maximize your cache, a low number is an early warning for capacity management. |
| Max Disk Group Read Cache/Write Buffer Latency (ms) | Each disk has a Read Cache Read Latency, Read Cache Write Latency (for writing into cache), Write Buffer Write Latency, and Write Buffer Read Latency (for de-staging purpose). This takes the highest among all these four numbers and the highest among all disk groups. It is the max of the max because each of the four datapoints is an average of all the VMs on it. |
| Sum Disk Group Errors | Sum of the bus reset + sum of commands canceled among all the disk groups. You must use sum and not get the max as each member should return zero. |
| Count Disk Group Congestion Above 60 | The number of disk groups congestion greater than 60. 60 is hardcoded in the vSAN Management Pack as it is a good starting point. As any congestion above 60 serves an early warning, count how many of such occurrences happen. |
| Max Disk Group Congestion | The highest congestion among all disk groups. A high number indicates that at least one disk group is not performing. |
| Min Disk Group Capacity Free (%) | The lowest free capacity among all disk groups. A low space triggers rebalance. |

| Name | What it is |
|---|---|
| Min Disk Group Read Cache Hit Rate (%) | The lowest hit rate among the disk group read cache. Ensure that this number is high as it indicates that the read is served by cache. |
| Sum vSAN PortGroup Packets Dropped (%) | Sum of all vSAN VMkernel port RX dropped packet + TX dropped packet. You should expect no dropped packet in your vSAN network. |

## vSAN File Services

The VMware administrator uses the **vSAN File Services** dashboard to monitor the file services running in their vSAN environment.

### Design Considerations

This dashboard is designed to complement the vSAN file services management provided by the vCenter. The vCenter is more of an administrative tool, while VCF Operations is more of an operations tool. Each tool performs their specific functions and does not duplicate information.

### How to Use the Dashboard

- File Shares by Used Space and Latency.
  - Review the file shares by used space and latency heat map.
  - It shows all the file shares in your environment.
  - The greater the use (consumption), the greater the box, so you can easily see the most consumed ones.
  - The file shares are colored by latency. You must watch out for boxes with red color.
- vSAN Clusters with File Services activated.
  - It lists all the vSAN clusters with file services activated, giving a convenient view to see which clusters have these settings turned on.
- Select a vSAN cluster from the vSAN clusters with file services activated table.
  - The file servers in the selected vSAN cluster are shown. When you select a file server, it filters the file shares list to show the file shares in the selected file server.
  - The file shares in the selected vSAN cluster are shown. Selecting a file share displays all the relevant KPI on the file share.

### Points to Note

vSAN File Servers and vSAN File Shares are two new objects in Management Pack for vSAN.

## vSAN Performance Dashboard

Use the **vSAN Performance** dashboard along with the **Cluster Capacity** dashboard.

### How to Use the Dashboard

The **vSAN Performance** dashboard is organized into sections for ease of use.

- **Cluster Analysis**

  The **vSAN Clusters** widget lists all the vSAN clusters, starting with cluster with the highest VM disk latency. By default, the widget displays data from the last 24 hours.

  The first column displays if the distribution of disk latency is experienced by all the VMs in the cluster. You can expect a majority of the VMs to experience latency that matches your expectation.

  The second column displays if any of the vSAN kernel modules have to wait for CPU. Expect this number to be near 0% and below 1%, as vSAN should not be waiting for CPU time.

  The third column displays if any of the vSAN clusters are dropping packets in the vSAN network (not the VM network). vSAN relies on the network to keep the cluster in-sync. This number should be near 0% and less than 1%.

Select a cluster to investigate further. The VM latency distribution is automatically displayed.

- **Contention**
  You can view various disk-related contention counters of the cluster from the **Contention** widget.
- **Utilization**
  Contention metrics are complemented by utilization metrics. A large block size can result in high throughput in a relatively low IOPS. If you see a large block size when you are not expecting it, investigate which applications are using it.
- **Disk Groups**
  You can drill down to the disk group level. All the counters are the worst value among the disk groups.
- **Read Cache**
  All the values displayed are the worst values among the read cache of the disk group.
- **Other KPIs**
  A problem in performance can also be caused by non-storage. vSAN resync is a type of utilization metric, but its presence can impact performance.
- **Disk Group Analysis**
  You can drill down to individual disk groups from the **Disk Groups** widget. Ensure that the disk groups are fairly balanced. Select the disk group that you want to analyze. Both the contention and utilization metrics are automatically displayed.
- **Cache Disks**
  You can drill down to cache disks from the **Cache Disks** widget. Ensure that the configuration is consistent. Select the cache disk you want to analyze. Both the contention and utilization metrics are automatically displayed.
- **Capacity Disks**
  You can drill down to capacity disks from the **Capacity Disks** widget. Ensure that the configuration is consistent. Select the capacity disk you want to analyze. Both the contention and utilization metrics are automatically displayed.

## vSAN Utilization Dashboard

The VMware administrator uses the **vSAN Utilization** dashboard with the **vSAN Contention** dashboard to manage performance.

### Design Consideration

The **vSAN Utilization** dashboard supports the **vSAN Contention** dashboard. Use it to identify vSAN clusters with high utilization in a selected data center. When utilization exceeds 100%, performance can be negatively impacted, especially when a VM experiences contention. By default, VCF Operations has a 5-minute collection interval. For 5 minutes, there might be 300 seconds worth of data points. If a spike is experienced for a few seconds, it might not be visible if the remaining 300 seconds is low utilization.

To view the common design considerations among all performance management dashboards, see the Performance Dashboards.

### How to Use the Dashboard

- Clusters Utilization.
  - It lists all the vSAN clusters, sorted by the least performing.
- Select a vSAN cluster from the clusters utilization table.
  - All the health charts show the KPI of selected cluster.
- Disk Groups
  - It lists all the vSAN clusters, sorted by the least performing.
- Select a Disk Group from the disk groups table.
  - All the health charts show the KPI of selected cluster.

**Points to Note**

- The **vSAN Utilization** dashboard complements the **vSAN Contention**. For more information, see the points to note in the vSAN Contention Dashboard.

## vSAN ESA Performance Dashboard

The vSAN ESA (Express Storage Architecture) Performance dashboard provides an overview of the performance issues related to your vSAN clusters. The dashboard displays the read and write latency issues that affect the vSAN clusters present in your environment. You can use this dashboard to identify and troubleshoot performance issues related to Contention and Utilization across VMs that are part of the vSAN cluster.

**How to Use the Dashboard**

The **vSAN ESA Performance** dashboard is organized into sections for ease of use.

- **Cluster Analysis**
  The **vSAN ESA Clusters** widget lists all the vSAN clusters, starting with cluster with the highest VM disk latency. By default, the widget displays data from the last 24 hours.
  The first column displays if the distribution of disk latency is experienced by all the VMs in the cluster. You can expect a majority of the VMs to experience latency that matches your expectation.
  The second column displays if any of the vSAN kernel modules have to wait for CPU. Expect this number to be near 0% and below 1%, as vSAN should not be waiting for CPU time.
  The third column displays if any of the vSAN clusters are dropping packets in the vSAN network (not the VM network). vSAN relies on the network to keep the cluster in-sync. This number should be near 0% and less than 1%.
  Select a cluster to investigate further. The VM latency distribution is automatically displayed.
- **Contention**
  You can view various disk-related contention counters of the cluster from the **Contention** widget.
- **Utilization**
  Contention metrics are complemented by utilization metrics. A large block size can result in high throughput in a relatively low IOPS. If you see a large block size when you are not expecting it, investigate which applications are using it.
- **Configuration**
  You can view the configuration details such as storage type, number of VMs, ESX, Storage Pools, Physical Disks of the selected vSAN cluster.
- **Storage Pools**
  You can view the configuration, contention, and utilization details of the Storage Pool for the selected vSAN cluster.
- **Physical Disks**
  You can view the configuration, contention, and utilization details of the Physical Disks for the selected vSAN cluster.
- **Related Objects to Selected vSAN Cluster**
  You can view the details of the related objects for the selected vSAN cluster.

## vSAN OSA Performance Dashboard

The vSAN OSA (Original Storage Architecture) Performance dashboard provides an overview of the performance issues related to your vSAN clusters. The dashboard displays the read and write latency issues that affect the vSAN clusters present in your environment. You can use this dashboard to identify and troubleshoot performance issues related to Contention and Utilization across VMs that are part of the vSAN cluster.

**How to Use the Dashboard**

The **vSAN Performance** dashboard is organized into sections for ease of use.

- **Cluster Analysis**
  The **vSAN Clusters** widget lists all the vSAN clusters, starting with cluster with the highest VM disk latency. By default, the widget displays data from the last 24 hours.

The first column displays if the distribution of disk latency is experienced by all the VMs in the cluster. You can expect a majority of the VMs to experience latency that matches your expectation.

The second column displays if any of the vSAN kernel modules have to wait for CPU. Expect this number to be near 0% and below 1%, as vSAN should not be waiting for CPU time.

The third column displays if any of the vSAN clusters are dropping packets in the vSAN network (not the VM network). vSAN relies on the network to keep the cluster in-sync. This number should be near 0% and less than 1%.

Select a cluster to investigate further. The VM latency distribution is automatically displayed.

- **Contention**
  You can view various disk-related contention counters of the cluster from the **Contention** widget.
- **Disk Group Analysis**
  You can drill down to individual disk groups from the **Disk Groups** widget. Ensure that the disk groups are fairly balanced. Select the disk group that you want to analyze. Both the contention and utilization metrics are automatically displayed.
- **Utilization**
  Contention metrics are complemented by utilization metrics. A large block size can result in high throughput in a relatively low IOPS. If you see a large block size when you are not expecting it, investigate which applications are using it.
- **Configuration**
  You can view the configuration details such as storage type, number of VMs, ESX, Storage Pools, Physical Disks of the selected vSAN cluster.
- **Compute and Network**
  You can view the compute and network metrics for the selected vSAN cluster.
- **Cache Disks**
  You can drill down to cache disks from the **Cache Disks** widget. Ensure that the configuration is consistent. Select the cache disk you want to analyze. Both the contention and utilization metrics are automatically displayed.
- **Capacity Disks**
  You can drill down to capacity disks from the **Capacity Disks** widget. Ensure that the configuration is consistent. Select the capacity disk you want to analyze. Both the contention and utilization metrics are automatically displayed.
- **Related Objects to Selected vSAN Cluster**
  You can view the details of the related objects for the selected vSAN cluster.

### vSphere Performance Profiling Dashboard

Use the vSphere Performance Profiling dashboard to determine a suitable Service Level Agreement (SLA).

This dashboard displays the performance of all vSphere clusters and their shared datastores. For each VM, the dashboard displays the performance metric at the 99th percentile. For each SLA (CPU, Memory, Disk), the dashboard shows the worst and the average performing VM.

> **Note:**
> Avoid setting the SLAs at a 100th percentile, as the 100th percentile usually indicates an outlier.

## Private AI (GPU) Dashboards

Use the Private AI (GPU) dashboards to monitor and troubleshoot GPU issues in VCF Operations.

To access the dashboards, from the left menu, click **Infrastructure Operations** > **Dashboards and Reports**. Click **Dashboards** and then navigate to **All** > **Private AI**.

The following dashboards are available:

| Dashboard Name | Purpose |
|---|---|
| GPU Equipped Clusters | Use this dashboard for the details related to GPU compute utilization and GPU memory usage at the cluster level, host level, and GPU level. |
| GPU Overview | Use this dashboard to view if any GPUs have high-temperature. This dashboard also highlights GPUs with low to zero usage by analyzing their capacity based on compute and memory utilization. |

### GPU Equipped Clusters Dashboard

Use the GPU Equipped Clusters dashboard to view the the clusters that have GPU activated devices. You can view the GPU device information at the cluster, host, and GPU level.

This dashboard displays the total number of GPU devices, GPU compute usage, memory usage, and memory at the cluster and host level. You can select a GPU to view the cluster level GPU utilization and the host level GPU utilization. You can also view the GPU metrics and properties at the GPU level, and further drill down to get an overview of the GPU temperature and GPU instance metric chart.

### GPU Overview Dashboard

Use the GPU Overview dashboard to view the GPU summary of all the GPUs present in your environment.

This dashboard displays the total number of GPUs and highlights GPUs as per heat map, memory usage (%) and compute utilization (%). You can select a GPU to view its relationship topology, the GPU instance name and properties, and the GPU instance metric chart at the GPU level.

## Service Discovery Dashboards

Using the service discovery dashboards, you can determine the inter-dependencies of virtual machines and the dependencies of each service in the respective virtual machines.

The following service discovery dashboards are added to the predefined VCF Operations dashboards:

- Service Distribution
- Service Relationships
- Service Visibility
- Virtual Machine Relationships

### Service Distribution Dashboard

You can use the dashboard to view the distribution of different services in the selected data center, cluster, or a host system. You can also view known and unknown services including the category and distribution percentage across a vSphere resource.

You can use the dashboard widgets in several ways:

- **Inventory Item**: Use this widget to view a hierarchical representation of objects in the form of badges.
- **Known Services Distribution**: Use this widget to view different services discovered from a selected object.
- **Service Categories**: Use this widget to view the service categories that are discovered by selecting an object from the resource widget.
- **User Defined Services Distribution**: Use this widget to view a list of user-defined services.

### Service Relationships Dashboard

You can use the dashboard to view properties of the service such as the install path, the ports used, and the version. You can also view the relationship between the services that run on other VMs.

You can use the dashboard widgets in several ways:

- **List of Services Discovered**: Use this widget to view the services that have been discovered.
- **Connections from the Selected Services**: Use this widget to view the relationship between the services and the other services running on the VMs.
- **Properties of the Selected Service**: Use this widget to view the properties of the selected services.

### Service Visibility Dashboard

You can use the dashboard to view a list of VMs without service visibility and VMs with user-defined services after you select a vSphere object.

You can use the dashboard widgets in several ways:

- **Inventory Tree**: Use this widget to view a hierarchical representation of objects in the form of badges.
- **Virtual Machines without Service Visibility**: Use this widget to view information about services where discovery has failed.
- **Virtual Machines with User-Defined Services**: Use this widget to view a list of VMs where the user has defined such services.

### Virtual Machine Relationships Dashboard

You can use the dashboard to view a list of VMs with service discovery details such as, status, method, incoming/outgoing connections, and protection groups. When you select a VM, the dashboard displays a list of discovered services on the VM, the relationships of the VMs with other VMs based on the relationships of the discovered service.

You can use the dashboard widgets in several ways:

- **List of virtual machines**: Use this widget to view all the VMs discovered by the vCenter.
- **Node relationship of the selected VM**: Use this widget to view the relationship between the objects.
- **List of Services running in the selected VM**: Use this widget to view all the properties of the selected VM.
- **Connections of Virtual Machines**: Use this widget to view the relationship between one or more VMs

## Skyline Operational Overview Dashboard

The Skyline Operational Overview dashboard displays your inventory summary, high level configuration data organizing by attributes such as vCenter instances, ESX hosts, virtual machines, and VM Guest Tools configurations.

#### Cluster Configuration

Use this dashboard to view the cluster configurations and settings related to VMware Distributed Resource Scheduler (DRS), VMware High Availability (HA), and HA Admission Control Enabled. For example, to view how many hosts have HA or DRS configured, you can click the pie chart to view the cluster information. Click a cluster to view the cluster summary, alerts, and other information.

#### ESX Host Configuration

You can view your ESX host configuration organized by hardware model, BIOS version, and host version. Click the pie chart to view the host information. Click any host to view the cluster summary, alerts, and other information.

#### Virtual Machine Configuration

You can view your virtual machine capacity in terms of CPU, RAM, and disk size. Clicking the pie chart displays the VM information. Click any VM to view the cluster summary, alerts, and other information.

#### VM Guest OS Distribution

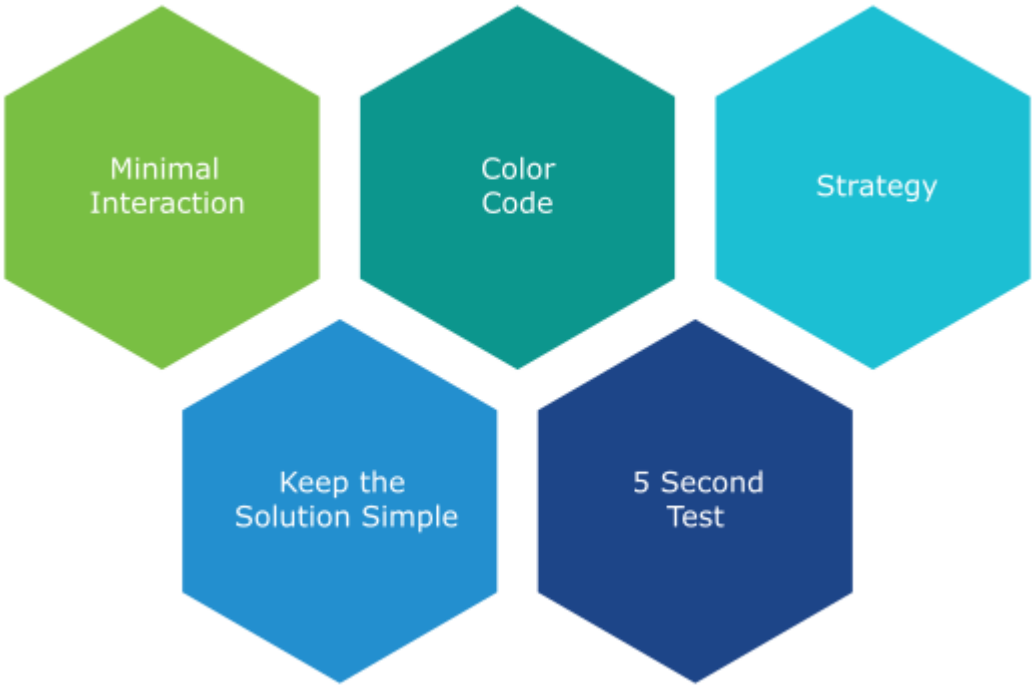
Monitor your guest tool information such as OS distribution, tool version, and tool status. Clicking the pie chart displays the tools information. Click any tool to view the summary, alerts, and other information.

## Dashboard Library

## Executive Summary Dashboards

The requirements of the CIO, Head of Global Infrastructure, and IT Senior Management vary from the requirements of the technical teams. The **Executive Summary** dashboards provide an overall information on capacity and inventory in business terms.

These dashboards allow you to display problems related to budget and resource, and provide visibility to the senior management into the live environment. By doing this, you can prove the need for additional hardware. If there is wastage that has to be reclaimed, you can display where and how large the wastage is using these dashboards. VCF Operations provides two example dashboards to get you started. As each executive might have a unique requirement or preference, the dashboards can be customized accordingly. The five principles displayed in the following figure are used to design the **Executive Summary** dashboards.




- Keeping the interaction, such as clicking, zooming, and sorting to a minimal.
- Use of color codes to have a user interface that is easy to understand.
- Each dashboard answers a specific question and the information is presented in business terms.
- Keep the solution simple and have a portal that is easy to access.
- Ensure that the dashboards are understood within five seconds.

### Capacity Summary Dashboard

The **Capacity Summary** dashboard is used by the Ops team to explain capacity to IT Management. This dashboard works together with the **Inventory Summary** dashboard. The inventory provides details on available resources and what is running on these resources. The capacity provides details on the remaining capacity and time.

#### Design Considerations

See Executive Summary Dashboards for common design considerations among all the dashboards for the IT senior management.

#### How to Use the Dashboard

The **Capacity Summary** dashboard has two sections:

- The top section of the dashboard provides a summary at the vSphere World level.

  - The **VM Growth** widget displays the weekly average of the VM growth and provides holistic visibility of overall growth across all data centers for both running and powered off workloads. If an increase in the VM count is not accompanied by a corresponding increase in utilization, these newly provisioned VMs are likely not yet used.
  - The **Overcommit Ratio** widget highlights the efficiency gained by vSphere virtualization running multiple workloads on a shared infrastructure. Overcommitment has to be further reviewed along with elevated resource contention to understand the impact of performance on VMs competing for resources. In general, Overcommit is required to be financially more economical than the public cloud.

    **Note:**
    VCF Operations uses physical CPU Cores not Logical Cores (Hyper-threading) for all CPU-based capacity calculations.
- The bottom section of the dashboard activates drill down into individual compute or storage capacity.
  - Capacity is split into Compute (vSphere Clusters) and Storage (Data stores) views. The heat map displays capacity by size and color by time remaining. By selecting either Clusters or Data stores, you can further drill down understand the remaining capacity and time (in days).

**Points to Note**

- Capacity remaining is not displayed at the vSphere World level as it can be misleading, especially in a global, or large infrastructure. Clusters also tend to serve a different purpose and they are not interchangeable.
- If you are using both on-prem and external cloud, for example, VMware on AWS, consider splitting the dashboard into two columns.

## Inventory Summary Dashboard

The **Inventory Summary** dashboard is used by the Ops team to explain capacity to IT Management. This dashboard works together with the **Capacity Summary** dashboard. The inventory provides details on available resources and what is running on these resources. The capacity provides details on the remaining capacity and time.

**Design Considerations**

See Executive Summary Dashboards for common design considerations among all the dashboards for the IT senior management.

**How to Use the Dashboard**

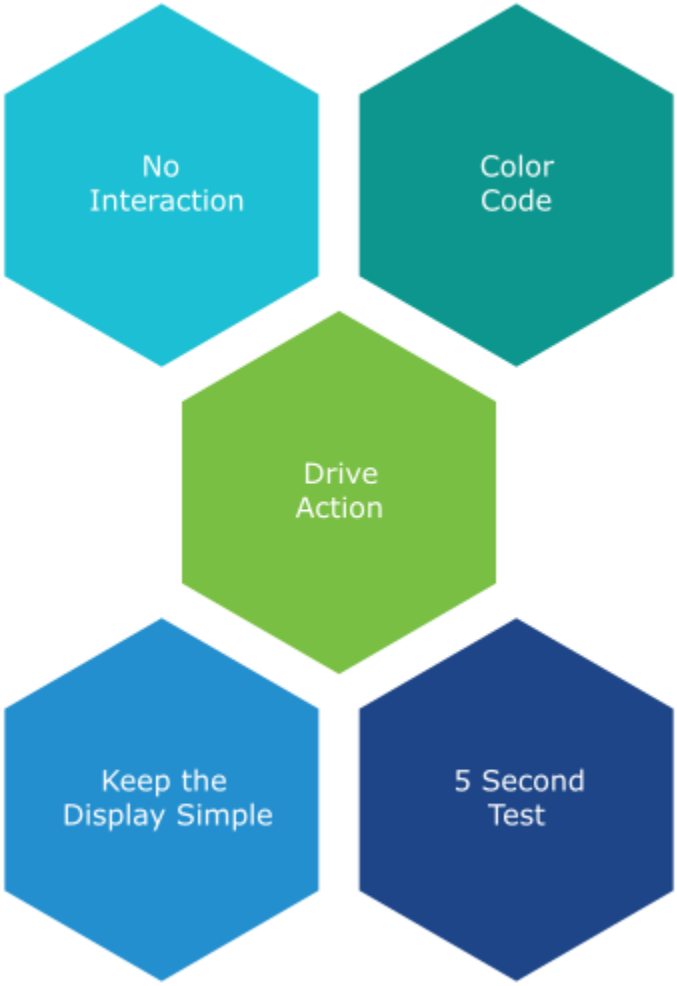
- The **Summary** widget provides a quick view of the key inventory number.
  - The scoreboard is interactive. This widget drives the eight pie charts that are placed at the bottom of the dashboard. Since all the information is at the vSphere World level, clicking any of them will display details of the total inventory.
- Select any data center from the **Datacenters** widget.
  - This widget drives Clusters and Datastores so that you can quickly view what you have in a given data center and related capacity.
  - For a small environment, the vSphere World is displayed so you can view all the VMs in the environment.
  - To sort by any of the columns in the table, click the column title.
- The eight charts in the dashboard provide details of the inventory. They are driven by **Datacenters**, **Compute**, **Storage**, and **Summary** widgets.

**Points to Note**

- Understand the relationship hierarchy in vSphere. For example, Compute (cluster) is not a parent of Storage (datastore), so logically it is not possible to display datastores in a cluster. Data center consists of compute (cluster), network (distributed switch), and storage (datastore).
- Datastores do not drive the pie chart. This is a known limitation in the View widget.
- If your senior management wants to view the largest VM in a given environment, add a Top-N widget to list the top 10 largest consumers so that CPU, memory, and the disk details are highlighted.

## Network Operation Center

A dashboard projected on the large screen serves a different business purpose than a dashboard on your laptop or desktop. It is placed strategically because it displays a time sensitive information. Dashboards complement alerts and cannot replace it. The five principles displayed in the following figure are used to design the predefined **Network Operation Center** dashboards.




- Keeping the interaction, such as clicking, zooming, and sorting to a minimal. Avoid having buttons, use of mouse or keyboard to view data.
- Use of color codes to have a user interface that is easy to understand.
- Displaying content that drives action. Display of live information as the focus is on immediate remediation. Problems that need immediate actions are displayed, for example, stop provisioning of new VM or take action on VMs that abuse the shared infrastructure.
- Display of problems that do not require immediate attention are avoided, for example, increase supply of infrastructure, such as adding hardware.
- Keep the display simple and have a portal that is easy to access.
- Dashboards are designed to display minimal and critical information only.
- Displays of numbers in percentage, with 0% being poor and 100% being perfect. To display utilisation, you can use the following markers:
  - 50% indicates good and balanced utilization. However, the ideal value is 75%
  - 0% indicates wastage
  - 100% indicates high utilization
- Ensure that the dashboards are understood within five seconds.

## Live! Cluster Performance Dashboard

The **Live! Cluster Performance** dashboard provides live information on whether the requests of the VMs are met by their underlying compute clusters. This dashboard focuses on CPU, Memory, and the performance of the clusters. Use this dashboard to view if there is any problem in meeting the demands of the VMs and if there is any unbalance within a cluster. The **Live! Cluster Performance** dashboard is the primary dashboard and it complements the **Live! Cluster Performance** dashboard which is the secondary dashboard. This secondary dashboard displays if the performance problem is caused by high utilization. The primary dashboard answers the question 'Is our IaaS performing?', while the secondary dashboard answers the question 'Is our IaaS working hard?'.

### Design Consideration

The **Live! Cluster Performance** dashboard displays three heat maps. The heat maps complement each other and must be used together. The location of each cluster and ESX hosts within those clusters is identical in all heat maps. The fixed positioning allows you to compare if the problem is caused by memory contention, CPU ready, or CPU co-stop.

The sizes of each cluster and ESX hosts are constant. Variable sizing creates a distraction and can result in small boxes, making it difficult to read.

The focus of the performance is on the population and not on a single VM. This is not a single VM troubleshooting dashboard but a dashboard focusing on infra problem. As the infra counter is mathematically an aggregation of VM counters, you must have a right roll-up strategy. As the goal is to provide an early warning, do not use the average as a roll-up technique. Use the percentage of the population exceeding a threshold. The threshold is set to be stringent to receive an early warning.

### How to Use the Dashboard

Review the heat maps, **Memory Contention**, **CPU Ready**, and **CPU Co-Stop** and see if there is any color other than green.

- Green indicates that almost 100% of the VMs have received the CPU and memory that was requested. The threshold is set such that if the 10% of the VM population does not receive the requested resources, then the heat map turns red.
- Red indicates an early warning. Stringent thresholds are used to activate proactive attention and remediation operations. The heat map can turn red because of the high standard that is applied even when there is no complaint from the VM owner yet.
- The light gray indicates that there is no VM running on the host and the metric is not computing.

View if there is any unbalance.

- There are two types of unbalance, cluster unbalance, and resource type unbalance.
- The ESX hosts are grouped by the cluster, so that the unbalance within a cluster can be easily viewed. Cluster unbalance is a real possibility and it is best monitored and not just assumed.
- If the three heat maps are different, then there is a resource unbalance. For example, if the memory contention is mostly red, but the two CPU heat maps are green, it means you have an unbalance between memory and CPU.
- If a single ESX host displays different color across the three heat maps, it indicates that there is an unbalance between the CPU and memory resources in the host.

For NOC Operator, drill-down by selecting one of the VMs on the heat map.

- The **Trends of Selected ESX Host** widget will automatically display the performance counters. To hide any metric, click the name in the legend.

As part of the deployment, configure auto-rotate among the NOC dashboards. If you want to view one dashboard, then you can remove the VCF Operations menu by using the URL sharing feature. This makes the overall user interface presentable and allows you to focus on the dashboard.

**Points to Note**

- You can add Disk Latency if you have the screen real estate. Use the counter 'Percentage of Consumers facing Disk Latency (%)'. It is a part of a datastore object, not a cluster, as a VM in a cluster can have disks across multiple datastores. Organize this storage performance by data center and not by the cluster.

## Live! Cluster Utilization Dashboard

The **Live! Cluster Utilization** dashboard complements the **Cluster Performance** dashboard. Use this dashboard to view the clusters that are working excessively and are close to their physical limit. This dashboard displays ESX hosts that have CPU or memory saturation that can lead to performance issues for the VMs running on the host.

**Design Considerations**

This dashboard is designed to complement the **Live! Cluster Performance** dashboard and it shares design considerations.

**How to Use the Dashboard**

As this dashboard has an identical design with the **Live! Cluster Performance** dashboard, it has the same usage procedure. Unlike the heat maps in the **Live! Cluster Performance** dashboard, the three heat maps in this dashboard have a different scale, reflecting the different nature of the counters.

Logically, memory is a form of storage. It acts as a cache to disk as it is much faster. A high utilization is better, as it indicates that more data is being cached. The ideal situation is when ESX host Consumed metric is red but ESX host Ballooned metric is green. When Ballooned is red and Consumed is gray, it means that there was high pressure in the past but it is not there anymore. The reason the ballooned stays red is because the ballooned pages were never requested back.

The ballooned memory counter was selected over the swapped or compressed memory counters as it is a better leading indicator. Since all three can co-exist at the same time, they are displayed in the line chart. Ballooned is displayed in absolute amount and not as a percentage, because the higher the size, higher are the chances for it to impact a VM. If you feel using percentage is easier for your operations, create a super metric to translate the value.

The heat map displays Wastage by a new color. The dark gray color indicates that wastage as capacity is not used. The performance problem due to low utilization can be caused by a bottleneck elsewhere.

Analyze if the ESX host is contributing. A light gray box indicates that the host is a part of the cluster but there is no utilization. It is possible for the host to be in the maintenance mode or is powered off.

**Points to Note**

- ESX host chooses to swap over compression if the compression ratio is less than 4x.
- If the ESX host's physical NIC is saturated in your environment, then you can add a Network Throughput heat map.

## Live! Heavy Hitters Dashboard

The **Live! Heavy Hitters** dashboard helps you analyze the misuse of the shared infrastructure. This dashboard displays details of VMs misusing shared infrastructure and if that has caused performance problems to the other VMs. The shared infrastructure includes risks. The cause for excessive load might be attacks, for example, the denial of service, process runaway, or a mass activation of agents. The most demanding VM is the largest. If a handful of VMs is dominating the shared infrastructure, their collective size is displayed on the dashboard.

**Design Considerations**

See the Performance Dashboards page for common design considerations among all the dashboards for performance management.

In a shared environment, it is possible to have a victim-villain problem. In the heat map, the villain VM is the one with the largest box size, while the victim VM is the one with the red box. If a handful of VMs is dominating the shared infrastructure, their collective size will be highly visible on the dashboard.

**How to Use the Dashboard**

- The heat maps, Disk IOPS, Disk Throughput, Network Throughput, and CPU Demand displays the four different loads that can be excessive. The heat maps display the relative value and not the absolute value. A VM does not generate a high load in the absolute term just because it has a large configuration.
- Each heat map has its color threshold, reflecting the nature of the contention metrics used in each of them.
- For NOC Operator, drill-down by selecting one of the VMs on the heat map. All the four line charts are automatically displayed, enabling you to get a complete picture of the selected VM.

**Points to Note**

- Memory is not displayed as it is a form of storage. The memory counters are space utilization and not speed. Think of disk space instead of IOPS. It can cause a capacity problem on the shared ESX host, but not performance problems to other VMs.
- In a large environment, it might be difficult to view a small victim VM. Consider having multiple dashboards and use them interchangeably.

## Live! vSphere VM Changes Dashboard

The **Live! vSphere VM Changes** dashboard displays changes to vSphere VMs as they happen. You can use the dashboard to ensure both the numbers and patterns are as per your expectation.

**How to Use the Dashboard**

The **Overview** widget at the top of the dashboard provides summary and overall high level vSphere VM numbers. Ensure that the numbers match your expectation. You can set thresholds so that it is easier to see if the values change (drop or increase) to a level that you are comfortable with. By default, you can view details of the last 6 hours. This allows you to view trends and patterns in the changes.

There are 3 types of detailed information covering different types of changes. This covers all the possible changes that could happen to a VM, hence providing real time visibility into the movement or volatility. The changes are categorized into three groups for ease of understanding: Inventory, State, and Location. The changes in each widget are displayed in order of importance.

- **Inventory Changes:** The sum of the changes in VM inventory over the last 24 hours. Information is displayed about whether VMs are added or removed as per your expectation, if they are added or removed the right way (cloning vs free style creation vs template based deployment), and why there is a high count of VMs being unregistered or deleted.
- **Location Changes:** Refers to VMs that are moved to another host or datastore. This can be hot or cold migration. Hot migration is displayed first as it might impact performance because it changes both compute and storage. Pay attention to the storage migrations as they take the longest amount of time. If it takes longer than expected, something could be wrong. On the other hand, cold migration is typically an activity that should match the change request as the VMs are shutdown.
- **State Changes:** As reset is the least desired it is displayed first. It is used as the last resort. Suspend and Power Off are least preferred as these actions should be done from within the Guest OS. Expect these numbers to be low. A high number of powered on VMs could lead to high demand.

There are two types of widgets for each of the 3 types of changes:

- The first type of widget is a trend line. In this case the widgets are **Changes in VM Inventory**, **Changes in VM Location**, and **Changes in VM State**. Ensure that both the absolute amount and the pattern match your expectation.
- The second type of widget is a scoreboard showing the present number for each of the changes. In this case the widgets are **Details of VM Inventory Changes**, **Details of VM Location Changes**, and **Details of VM State Changes**. The change is sorted from the least desired (most impact to operations) to the least important.

**Points to Note**

The changes are color coded. You can adjust the thresholds for each change, however it is recommended that you avoid too many variations as it can get confusing. You can also adjust the widget size as required for large displays.

# Software Defined Wide Area Network Dashboard

The Software-Defined Wide Area Network (SD-WAN) dashboard allows you to configure and monitor the services related to VeloCloud and SD-WAN using VCF Operations. Using the SD-WAN dashboard, you can also collect the metrics for VeloCloud Orchestrator and VeloCloud Gateway.

By default the SD-WAN dashboards are deactivated, if you want to know how to activate them, see Manage Dashboards. The following services are discovered using the VeloCloud Orchestrator:

- Java Application
- VeloCloud Orchestrator
- Nginx
- ClickHouse
- MySQL
- Redis
- Network Time Protocol

The following services are discovered using VeloCloud Gateway:

- Network Time Protocol
- VeloCloud Gateway

## Troubleshoot SD-WAN Dashboard

You can use the widgets in Troubleshoot SD-WAN dashboard to monitor and troubleshoot the services and applications associated with the SD-WAN.

You can use the dashboard widget in several ways:

- **Troubleshoot Virtual Machine (VM)**: Use this widget to navigate to a specific VM and troubleshoot the issues.
- **Troubleshoot Orchestrator**: Use this widget to navigate to a specific orchestrator and troubleshoot the issues.
- **Troubleshoot Gateway**: Use this widget to navigate to a specific gateway and troubleshoot the issues.
- **Troubleshoot Application**: Use this widget to navigate to a specific application and troubleshoot the issues.
- **Relationship**: Use this widget to view the services and operating system associated with the VeloCloud Orchestrator.
- **Top Alerts**: Use this widget to view the top alerts associated with the SD-WAN.

## Troubleshoot SD-WAN Gateway Dashboard

You can use the widgets in Troubleshoot SD-WAN Gateway dashboard to monitor and troubleshoot all the services and applications associated with the SD-WAN gateway.

You can use the dashboard widget in several ways:

- **Active Alerts on the Gateway**: Use this widget to view the active alerts for the gateway.
- **Health of Gateway Applications**: Use this widget to view the health status of the applications in the gateway.
- **Examine Operating System**: Use this widget to examine the operating system status.
- **Gateway Summary Status**: Use this widget to view the summary information for the gateway.
- **Gateway Process Status**: Use this widget to view the process information for the gateway.
- **Gateway Resource Metrics**: Use this widget to view the resource metrics associated with the gateway.
- **Parent Host**: Use this widget to view the parent host information.
- **Parent Cluster**: Use this widget to view the parent cluster information.

## Troubleshoot SD-WAN Orchestrator Dashboard

You can use the widgets in Troubleshoot SD-WAN Orchestrator dashboard to monitor and troubleshoot the services and applications associated with the SD-WAN Orchestrator.

You can use the dashboard widget in several ways:

- **Active Alerts on the Orchestrator**: Use this widget to view the active alerts for the Orchestrator.
- **Health of Orchestrator Applications**: Use this widget to view the health status of the applications in the gateway.
- **Examine Operating System**: Use this widget to examine the operating system status.
- **Examing MySQL**: Use this widget to examine the MySQL application.
- **Orchestrator Service Status**: Use this widget to view the service status of the Orchestrator.
- **Redis Status**: Use this widget to view the status of the Redis application.
- **API Check Status**: Use this widget to check the API status.
- **Nginx Status**: Use this widget to check the Nginx status.
- **Parent Host**: Use this widget to view the parent host information.
- **Parent Cluster**: Use this widget to view the parent cluster information.

# VCF Automation Dashboards

The VCF Automation dashboards allow you to track performance, health, utilization, and availability attributes of deployments made to SDDC clouds and public cloud end points. You can also monitor the virtual machines and track the utilization and performance of the VMs in your SDDCs and public cloud accounts..

### VCF Automation Predefined Dashboards

The following VCF Automation dashboards are added to the predefined VCF Operations dashboards:

- Cloud Automation Environment Overview
- Cloud Automation Project Cost Overview
- Cloud Automation Resource Consumption Overview
- Cloud Automation Deployment Overview
- Cloud Automation Top-N Dashboard

## Automation Environment Overview

You can use the widgets in the Cloud Automation Environment Overview dashboard to view the environment details for the vCenter Cloud Zone objects. You can use the Cloud Automation Environment Overview dashboard to view the projects, deployments associated with the vCenter Cloud accounts.

You can use the dashboard widgets in several ways.

- **Environment Summary**: Use this widget to view the SDDC cloud zones, public cloud zones, projects, deployments, blueprints, SDDC VMs and public cloud VM details for the cloud accounts present in your environment.
- **SDDC Cloud Zones**: Use this widget to view the CPU, Disk, Memory, health, risk, and efficiency details for the SDDC cloud zone objects present in your environment.
- **Public Cloud Zones**: Use this widget to view the CPU, Disk, Memory, health, risk, and efficiency details for the public cloud zone objects present in your environment.
- **Project List**: Use this widget to view the total blueprints, cloud zones, deployments, virtual machines, health, risk, efficiency details in your environment.
- **Top Alerts**: Use this widget to view the top alerts in your environment.
- **VM List**: Use this widget to view all the VM details in your environment.
- **Blueprint List**: Use this widget to view the blueprint objects in your environment.
- **Deployment List**: Use this widget to view the blueprint objects deployed in your environment.
- **SDDC Virtual Machines**: Use this widget to view the SDDC VM's resource details.
- **Public Cloud Resources**: Use this widget to view the public cloud resource details.

## Automation SDDC Project Price Overview

You can use the cloud automation SDDC project price overview dashboard to view the project price details associated with each of VMware Cloud on AWS SDDC and public cloud accounts. With the project price overview dashboard, you can view price details for individual projects and find out deployments with the highest cost. .

You can use the dashboard widgets in several ways.

- **Project Cost**: Use this widget to view the project wise cost for compute, storage, and additional resources associated with your cloud environment.
- **Daily Price Over Time**: Use this widget to view the price of individual projects on a day to day basis.
- **Relationship**: Use this widget to view the relationship between the objects and the projects present in your cloud environment.
- **Deployment Price by Selected Project**: Use this widget to view the deployment cost for the selected project in your cloud environment.
- **Deployment With Highest Cost**: Use this widget to view the highest cost associated with projects that are present in your cloud environment.

## Automation SDDC Resource Consumption Overview

You can use the widgets in the Cloud Automation SDDC Resource Consumption Overview dashboard to view the resources consumed by VCF Automation on Cloud Accounts.

You can use the Cloud Automation Resource Consumption Overview dashboard widgets in several ways.

- **Cloud Account**: Use this widget to view all the attributes related to the cloud account.
- **SDDC Cloud Zone**: Use this widget to view all the attributes related to the SDDC cloud zones.
- **Project**: Use this widget to view all the project details associated with your cloud account.
- **Cluster List**: Use this widget to view all the details associated with the clusters in your account.
- **Cluster Utilization**: Use this widget to view the cluster utilization details for the cloud accounts.
- **Deployment Heat Map by Project**: Use this widget to view the heat map for each deployed project in your cloud environment.
- **SDDC Cloud Zone Capacity**: Use this widget to view the memory and storage capacity that is allocated, reserved, and free for each cloud zone object.
- **SDDC Cloud Zone Memory Trend**: Use this widget to view and analyze a seven-day trend for the memory allocated, reserved, and free for the cloud zone.
- **SDDC Cloud Zone Storage Trend**: Use this widget to view and analyze a seven-day trend for the storage allocated, reserved, and free for the cloud zone.

## Automation Deployment Overview

You can use the cloud automation deployment overview dashboard to view the deployment details associated with your cloud environment. You can view details about cloud accounts, cloud zones, projects, and deployments. The dashboard also provides details about the deployment resources and the relationship between deployments and their objects.

You can use the dashboard widgets in several ways.

- **Cloud Account**: Use this widget to know the cloud account details of the selected account.
- **Cloud Zone**: Use this widget to know the cloud zone details like adapter, collection state, policy and object type of the selected account.
- **Project**: Use this widget to know the project details of the selected project.
- **Deployments**: Use this widget to know the deployment details of the cloud accounts in your environment.
- **Deployment Resources**: Use this widget to know the deployment resource details across the cloud accounts in your environment.
- **Object Relationship (Advanced)**: Use this widget to view the relationship between the objects and deployments present in your cloud environment.
- **Deployment Heat Map by Project**: Use this widget to view the heat map for each deployed project in your cloud environment.

- **Deployment Heat Map by Blueprint**: Use this widget to view the heat map for each deployed blueprint in your cloud environment.

## Automation Top-N Dashboard

You can use the widgets in the Cloud Automation Top-N dashboard to view the projects with most critical alerts, to view the blueprint with most deployments, and to view the deployments with the highest cost.

You can use the dashboard widgets in several ways.

- **Project with Most Critical Alerts**: Use this widget to view the projects which has most critical alerts.
- **Top Alerts**: Use this widget to view the top alerts for the projects in your cloud account.
- **Blueprints with Most Deployment**: Use this widget to view the blueprint which has maximum deployments for the cloud account.
- **Relationship**: Use this widget to analyze the relationship between blueprints and deployments, and deployment and cost.

## Project Chargeback

The Project Chargeback dashboard lets you know how much you must spend to run an VCF Automation managed VM on behalf of your customer. In VCF Operations, you can configure the cost drivers and let the system automatically determine how much a VM costs based on your infrastructure requirement. Cost Drivers cover server hardware, storage, licenses, application, maintenance, labor, network, facilities, and additional costs configured within VCF Operations.

Price is what you charge your customer for running their VM. The price of a VM can be based on the cost of the project or based on a rate card that you define. Prices can include up charges, service charges, and others.

### How to Use the Dashboard

- Select a Project widget displays the price of the group. The Select a Deployument widget lets you select deployments under the project, followed by the chargeback metrics.
- Price Summary of Selected Project shows the month to date price of the group.
- VM Price Distribution (Top 100) shows the most expensive VMs in the group.
- Powered Off VMs shows reclaimable VMs and their potential savings.
- Idle VMs shows reclaimable VMs and their potential savings.
- Reclaimable VMs with Snapshots shows reclaimable snapshots and their age.
- Price of VMs in the Selected Deployment shows the price and configuration of each VM in the selected group.

## Project Cost vs. Price

The VM Cost vs. Price Dashboard helps you to analyze the relationship between cost and price for VMs managed by VCF Automation. You can use this dashboard to ensure the price of VMs for chargeback is sufficient to cover the cost of running virtual machines.

### How to Use the Dashboard

- Select a Project allows selection of a group of VMs to analyze. The Select a Deployment widget allows you to select deployments under the project, followed by the price/cost metrics.
- Summary (Month to Date) shows the month to date price and cost.
- VMs of the Selected Deployment (Select to View Trend) shows all VMs in the selected group with their Month to Date Cost, Today's Cost, Month to Date Price, and Today's Price.
- Daily Cost and Daily Price trend chart shows both cost and price over time.

## Project Showback

The Showback VM Cost dashboard provides a quick Showback of the cost associated with the VCF Automation managed VMs in a group. The definition of a group is based on several constructs such as Virtual Machine Folder, Custom Groups,

VMware Aria Automation Projects, and more. Based on the Showback you can improve the accuracy of the costs by editing the cost drivers. Cost drivers that are not customized use reference cost, cost driver customization is available only in Advanced or Enterprise edition of VCF Operations .

**How to Use the Dashboard**

- Select a Project widget displays the price of the group. The Select a Deployument widget lets you select deployments under the project, followed by the showback metrics.
- Cost Summary (This Month) shows the month to date cost, potential savings, and projected cost of the group.
- VM Cost Distribution (Top 100) shows the most expensive VMs in the group.
- Potential Savings (Top 10) shows the VMs ranked by their potential savings.
- VMs of the Selected Deployment (Select to View Trend) shows the cost and configuration of each VM in the selected group.
- Cost Trend of Selected VM shows the trend of the VMs cost over time.

# VCF Operations Dashboards

With VCF Operations dashboards you can monitor and troubleshoot objects in your cloud infrastructure.

The following v VCF Operations dashboards are added to the predefined VCF Operations dashboards:

- VCF Operations Cloud Billing
- VCF Operations Cloud Universal Billing

## VCF Operations Cloud Billing

The VCF Operations Cloud Billing dashboard provides you object billing details of Operating System Instance (OSI) used in your cloud environment.

**How to Use the Dashboard**

- OSIs and Billable Objects widget provides the total count of OSIs and billable objects. You have to update these widgets depending on your subscription limits.
- OSIs Across Object Types widget provides distribution of OSIs across different object types.
- Billable Object Types List widget provides a list of all the object types which are managed by VCF Operations and consume license unit(s).
- OSIs Consumption Across Object Types displays a heat map and maps the magnitude of OSIs consumption for different object types with the relevant heat map colors. The widget also interacts with OSIs Consumption Over Time and displays how OSIs count have been changing for a given object type over a period of time.
- List of Objects widget displays the object details like name, adapter type, object type, policy, creation time, collection state, and collection status. Use the filter option to filter different objects from the list of objects.

**How to Edit OSIs and Billable Objects Widget To Set Correct Color Codes**

1. Click the Edit icon at the top right corner of the widget.
2. Go to **Output Data** section.
3. Double click the row which has column heading Yellow, Orange, or Red.
4. Set the **Color Method** to custom.
5. Enter appropriate values as per your subscription limits.

## VCF Operations Cloud Universal Billing

The VCF Operations Cloud Universal Billing dashboard provides the object billing details based on CPU usage.

#### How to Use the Dashboard

- CPUs and Billable Objects widgets provides the total count of CPUs and Billable Objects. You have to update these widgets based upon your subscription limits.
- CPUs Across Object Types widget provides distribution of CPUs across different object types.
- Billable Object Types List widget provides a list of all the object types which are actively managed by VCF Operations and consume license unit(s).
- CPUs Consumption Across Object Types displays a heatmap and maps the magnitude of CPUs consumption for different object types with heatmap colors. The widget also interacts with CPUs Consumption Over Time and displays how CPUs count has been changing for a given object type over a period of time.

#### How to Edit CPUs and Billable Objects Widget To Set Correct Color Codes

1. Click the Edit icon at the top right corner of the widget.
2. Go to **Output Data** section.
3. Double click the row which has column heading Yellow, Orange, or Red.
4. Set the **Color Method** to custom.
5. Enter appropriate values as per your subscription limits.

# Inventory Dashboards

The three vSphere Inventory dashboards and workload management inventory dashboards cater to the compute, network, and storage aspects of your SDDC. Using these dashboards, you can navigate through the environment and view your inventory and their key metrics at a glance. The Network and Storage dashboards can be shared with the network and storage teams respectively, giving them the necessary visibility, and increasing the collaboration between teams.

#### vSphere inventory dashboards

The vSphere inventory dashboards are built specifically for each role, but they share a common design. They have a similar layout and are used in the same manner. This makes learning easier, especially in smaller environments where the same team manages the full environment.

These dashboards help you answer several key questions:

- What is the topology of your vSphere compute inventory?
- What is the topology of your vSphere storage inventory?
- What is the topology of your vSphere network inventory?

#### Workload Management Inventory Dashboard

This is a unified dashboard for the new workload management objects. It shows the relationships and KPIs for the workload management objects. For example, you can see the topology view from the Tanzu Kubernetes clusters to the physical infrastructure.

## vSphere Compute Inventory Dashboard

You can use the vSphere Compute Inventory Dashboard to browse through the topology of your vSphere compute inventory which includes information related to vSphere world, vCenter, data center, clusters, hosts, virtual machines, properties, and metrics.

You can select an object type to view the properties and metrics related to it. You can also view the clusters, ESX hosts, and virtual machines associated with the object.

You can use the dashboard widgets in several ways.

- **Properties**: View the properties related to an object in the environment.
- **Metrics**: View the metrics related to the object.
- **Clusters**: View the cluster functionality.

- **ESX Hosts**: View the data related to the hosts.
- **Virtual Machines**: View VMs that belong to the object.

## vSphere Network Inventory Dashboard

The vSphere Network Inventory Dashboard allows you to browse through the topology of your vSphere network inventory which includes information related to vSphere world, vCenter, data center, distributed vSwitches, distributed port groups, virtual machines, properties, and metrics.

You can select an object type to view the properties and metrics related to it. You can also view the distributed vSwitches, distributed port groups, virtual machines associated with it.

You can use the dashboard widgets in several ways.

- **Properties**: View the properties related to the object in the environment.
- **Metrics**: View the metrics of the object.
- **Distributed vSwitches**: View details related to the distributed vSwitches.
- **Distributed Port Groups**: View data relevant to distributed port groups.
- **Virtual Machines**: View VMs that belong to the object.

## vSphere Storage Inventory Dashboard

The vSphere Storage Inventory dashboard allows you to browse through the topology of your vSphere storage inventory which includes information related to vSphere world, vCenter, data center, datastore clusters, datastores, virtual machines, properties, and metrics.

You can select an object type to view the properties and metrics related to it. You can also view the datastore clusters, datastores, and virtual machines associated with it.

You can use the dashboard widgets in several ways.

- **Properties**: View the properties related to the object in the environment.
- **Metrics**: View the metrics of the object.
- **Datastore Clusters**: View the datastore cluster functionality.
- **Datastores**: View the datastore functionality.
- **Virtual Machines**: View VMs that belong to the object.

## Workload Management Inventory Dashboard

?The Workload Management Inventory dashboards curates the Kubernetes inventory across all the Workload Management activated vSphere environments and displays it here. This includes an end to end topology map showcasing the health of all the objectes along with upstream and downstream dependencies. Upon clicking any object in the relationship tree, the related inventorty of Supervisor Clusters, Namespaces, Pods, Developer Managed VMs and Tanzu Kubernetes clusters can be viewed and exported from this dashboard.

You can select an object type to view the properties and key metrics related to it.

You can use the dashboard widgets in several ways.

- **Environment Summary**: Provides a summary of the supervisor cluster and the child objects.
- **Relationships**: An interactive canvas where you can view the relationship between the different objects in the workload management inventory.
- **Properties**: View the properties related to the object in the environment.
- **Metrics**: View the metrics of the object.
- **Supervisor Clusters**: View the supervisor cluster functionality.
- **Tanzu Kubernetes cluster**: View the Tanzu Kubernetes cluster functionality.
- **Virtual Machines**: View VMs that belong to the object.
- **vSphere Pods**: View information about vSphere Pods.

## vSphere VM Inventory Dashboard

The vSphere VM Inventory dashboard provides real-time updates on changes to virtual machines, allowing users to monitor both the quantity and patterns of these changes to ensure they align with expectations. To facilitate understanding, the 20 types of changes are categorized into three groups, with changes within each group ordered by importance. For instance, deletions are prioritized over deployments. This tool is specifically designed for the Network Operations Center support team to help them efficiently track and manage VM modifications.

### The vSphere VM Inventory Dashboard Display

The Overview widget provides high-level metrics, displaying data from the last six hours by default to help identify patterns. The Changes in VM Location section tracks movements across ESX hosts and datastores, including vSAN. The Changes in VM States section monitors changes in power or running states, detailing how these changes were executed.

### Customizing the vSphere VM Inventory Dashboard

Adjust the threshold settings to suit your specific needs, but be cautious of setting too many variations as it can become confusing. If you operate with two physical data centers, consider splitting the dashboard into two columns, one for each data center. For those using the dashboard on a desktop, adding a navigation link to the vSphere VM Inventory dashboard can enhance convenience.

# Dashboards in NSX Management Pack

The **NSX Main** dashboard provides an overview of the network objects. It displays the topology of a selected object, how it connects to the elements in the network, and a view of related alerts.

**Table 1383: Widgets in NSX Main Dashboard**

| Widget | Description |
|---|---|
| NSX Instances | Displays the list of environments that are being monitored. When you select an environment in this widget, the other widgets in the **NSX Main** dashboard display data for the selected adapter. |
| Environment Overview | Displays a top-level view of the selected environment and following key components.<br>• NSX Manager<br>• Controller Node<br>• Logical Router<br>• Logical Switch<br>• Load Balancer Virtual Server<br>• Transport Zone |
| Top Alerts | Displays all the open alerts for the selected object in the **Environment Overview** widget. |
| Topology Graph | Displays the topology of the selected object in the **Environment Overview** widget. |

## NSX Inventory Dashboard

The **NSX Inventory** dashboard in VCF Operations provides a holistic view of NSX environment topology. It offers an graphical representation of the interconnections between various NSX components.

This dashboard helps administrators with essential insights into the following components:

- NSX Management Nodes: Displays information related to the status and configuration of NSX management nodes.
- T0 and T1 Routers: Displays details of the t0 and t1 routers including their configurations and associations.
- NSX Load Balancers (LBs) and LB Pools: Displays information about NSX Load Balancers, virtual servers, pool members, and health checks.
- Edge Clusters: Displays data on Edge Clusters including cluster membership and resource consumption.

- Transport Zones: Displays insights into transport zones, offering visibility into network segmentation and connectivity options.
- Edge Nodes and Host Nodes: Monitors Edge Nodes and Host Nodes that are crucial components of the NSX infrastructure.

## Dashboards in VMware Infrastructure Health

Use the VMware Infrastructure Health dashboards to view data related to its certificates, historic trend, health, VM, and versions of all products under the VCF management domain.

**Table 1384: VMware Infrastructure Health Dashboards**

| Dashboard | Description |
|---|---|
| VMware Infrastructure Health Certificate Overview Dashboard | The VMware Infrastructure Health dashboard provides certificate information related to vCenter , NSX Manager nodes, ESX, and Adapter certificate alerts. |
| VMware Infrastructure Health Historic Trend | Use VMware Infrastructure Health health trend dashboard to view the health trend for each component in the cloud management applications.<br>The dashboard provides an overall trend of the health components of the cloud management applications, that are monitored for the last seven days. |
| VMware Infrastructure Health Health Overview | You can use VMware Infrastructure Health dashboard to view and analyze the application-specific problems in the accounts in the VMware Cloud Management Plane. This Dashboard provides the health information for each of the accounts. You can select the component in the cloud management applications from the widgets available in the dashboard. The widgets help in rendering the infrastructure health of that component with its service health and associated configuration alerts, if any. |
| VMware Infrastructure Health VM Optimizer | The VMware Infrastructure Management VM Optimizer dashboard helps you to know how you can reclaim capacity from oversized, idle, powered off VMs and optimize these VMs for performance. |
| VMware Infra VCF BOM Version | The VMware Infra VCF BOM Version dashboard displays all the versions of all products under the VCF management domain. |
| VCF HRM Host Systems | The VCF HRM Host System dashboard displays the health, reports health issues, and allows you to manage the health of the VCF host systems. |
| VCF HRM | The VCF HRM dashboard displays the health, reports health issues, and allows you to manage the health of each application in each VCF management domain. |

## vSphere Dashboards

The vSphere dashboard provides an overview of ESX hosts availability, vCenter appliance availability, and vSphere daily check of the vSphere clusters in your VCF Operations environment. The following vSphere dashboards are added to the predefined VCF Operations dashboards:

- ESX Host Availability
- vCenter Appliance Availability
- vSphere Daily Check

## ESX Host Availability

ESX Host Availability dashboard displays the list of unresponsive ESX hosts, list of hosts in maintainance mode, and list of powered on or unknown power state of the ESX hosts.

Hosts in maintainance mode and power state is an user initiated action. A host becomes becomes unresponsive because of an external factor that vCenter Server is unaware of. This can happen due to several reasons such as a network connectivity issue between the host and vCenter Server.

On the ESX Hosts Availability dashboard, the Unresponsive ESX widget displays the **Power State** as **Unknown** and **Maintainance State** as **notIn Maintainance**.

**How to Use the Dashboard**

- In the Unresponsive ESX widget, click any ESX host from the list.
  - The **Parent Cluster of selected ESX** widget is automatically displayed once you click any host. This widget displays the cluster properties of the selected ESX hosts such as HA enabled, DRS enabled.
  - The ESX **Hosts in the selected cluster** widget is automatically displayed once you click any cluster property tab. This widget displays the number of unresponsive and running hosts. You can identify which hosts has issues and try resolving the issues by accessing the hosts.
- Click any host in the ESX **on Maintenance Mode** or **Power on or Unknown Power State ESX** widget to view the details in the **Parent Cluster of selected ESX** widget.
- Click any cluster property tab in the **Parent Cluster of selected ESX** widget to view the details in the ESX **Hosts in the selected cluster**.

## vCenter Appliance Availability

vCenter Appliance Availability dashboard displays the list of vCenter instances with reachability and availability issues.

If you have hundreds of vCenter instances and need to check their availability and the status of their services, it can be tedious to log in to each vCenter individually to gather this information. This dashboard helps you identify reachability issues and services that are encountering problems.

**How to Use the Dashboard**

The **vCenter Appliance** widget displays the instance name, total ESX hosts, VMs, reachability of each vCenter. To check if there are any issues in the vCenter services, click a vCenter instance in the vCenter Appliance widget. The **Services in the selected vCenter** widget is automatically displayed once you click a vCenter instance. This widget displays the health status of the vCenter services. You can sort the services based on health status according to the following status: Started, Stopped, or Unknown.

Clicking an instance shows you the summary of the instance with metrics, logs, alerts, and other details.

## vSphere Daily Check Dashboard

Use the vSphere daily check dashboard to monitor the live operations of the vSphere clusters in your VCF Operations environment.

Use the vSphere Daily Check dashboard to review the health and performance of the vSphere clusters in your VCF Operations environment. The dashboard displays the critical and important alerts that need immediate attention. You can use this dashboard to prevent or minimize alerts for the next 12 to 24 hours. This dashboard focuses on the daily activities of the environment and provides updates based on changes or alerts in the clusters. The dashboard reflects the following changes in your environment:

- Configuration changes
- Consumption changes: Any unexpected increase or sudden decrease in consumption.
- Supply changes. Any unexpected decrease in supply.
- Dynamic changes: Any VM state, VM location, or VM inventory changes.

Use the dashboard to focus on issues that need immediate attention and resolution for proper functioning of your environment. You can adjust the thresholds and parameters to suit your need and add metrics like error logs to customize the dashboard.

This dashboard complements the alerts triggered in your environment and provides insights into the cause of the alerts and also shows the overall picture. This helps troubleshoot the specific issue without disrupting the whole environment. As part of the daily checks, the following areas are covered:

- Availability: Daily checks cover the availability of powered on VMs and ESX hosts. The numbers reflected should be within the expected range. For example, if a new cluster is added or if several new VMs have been provisioned the day before, the dashboard should reflect the revised available of the VMs and ESX hosts.
- Performance: Daily checks cover the average performance of consumers (VMs) and providers (compute, network, storage). For example, the dashboard monitors the total CPU, memory, disk, and network utilization as any sudden increase or decrease in performance can cause issues.
- Compliance: Daily checks cover the compliance status and focuses on the present status. The present status covers the past 12 to 24 hours.
- Configuration: Daily checks cover the changes in settings and make sure that the settings match the authorized changes executed during the change window.

# Configuring Reports and Views

A report is a scheduled snapshot of views and dashboards. A view helps you to interpret metrics, properties, policies of various monitored objects including alerts, symptoms, and so on

## Reports in VCF Operations

A report is a scheduled snapshot of views and dashboards. You can create reports in VCF Operations to represent objects and metrics. The report can contain a table of contents, cover page, and footer.

With the VCF Operations reporting functions, you can generate a report to capture details related to current or predicted resource needs. You can download the report in a PDF or CSV file format for future and offline needs.

#### Types of Report Templates

There are two types of report templates in VCF Operations.

- Predefined report templates that are out of the box with VCF Operations. For more information, see Accessing Report Templates .
- Custom report templates that you create based on your requirement. For more information, see Create a Report Template.

### Create a Report Template

You create a report to generate a scheduled snapshot of views and dashboards. You can track current resources and predict potential risks to the environment. You can schedule automated reports at regular intervals. The name and description of the report template as they appear in the list of templates on the **Report Templates** tab.

1. To create report templates, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Reports** > **Create**.
2. From the **Create Report Template** page, complete the options in each tab.
3. At the end of each tab, you can go to the previous or next tab. You can also cancel the creation of the report template.
4. After you have added all the details, click **Create** to create the report template.

### Name and Description Tab

**Table 1385: Name and Description Options in the Create Report Template Page**

| Option | Description |
|---|---|
| Name | Name of the template as it appears on the **Report Templates** tab. |
| Description | Description of the template. |

### Report Content Tab

The report template contains views and dashboards. Views present collected information for an object. Dashboards give a visual overview of the performance and state of objects in your virtual infrastructure. You can combine different views and dashboards and order them to suit your needs. The report template contains images of the views/dashboards that are added.

**Table 1386: Views and Dashboards Options and Sections in the Create Report Template Page**

| Option | Description |
|---|---|
| Report Template Structure | To add a view or a dashboard to your report template, select it from the **Views and Dashboards** list in the right pane and drag it to the **Report Template Structure** pane. |
| View and Dashboards | Select **Views** or **Dashboards** to display a list of available views or dashboards that you can add to the template. |
| Filter | Search for views or dashboards based on the following filter criteria.<br>Views: name, type, description, subject, metric, and modified by.<br>Dashboards: name and shared.<br>To see the complete list of views or dashboards, delete the search box contents and press Enter. |
| Vertical Ellipsis > Portrait/Landscape | You can select a portrait or landscape orientation for each view or dashboard from the vertical ellipsis next to the title of the view/dashboard after you drag and drop the view/dashboard to the left pane. |
| Vertical Ellipsis > Colorization | You can activate or deactivate a colorized PDF output for each list view from the vertical ellipsis next to the title of a list view after you drag and drop the list view to the left pane.<br>Available only for list views. |

### Layout and Format Tab

The report template can contain layout options such as a cover page, table of contents, and footer. Formats are the outputs in which you can generate the report.

**Table 1387: Layout and Format Options in the Create Report Template Page**

| Option | Description |
|---|---|
| Cover Page | Can contain an image up to 5 MB.<br>The default report size is 8.5 inches by 11 inches. The image is resized to fit the report front page. |
| Table of contents | Provides a list of the template parts, organized in the order of their appearance in the report. |

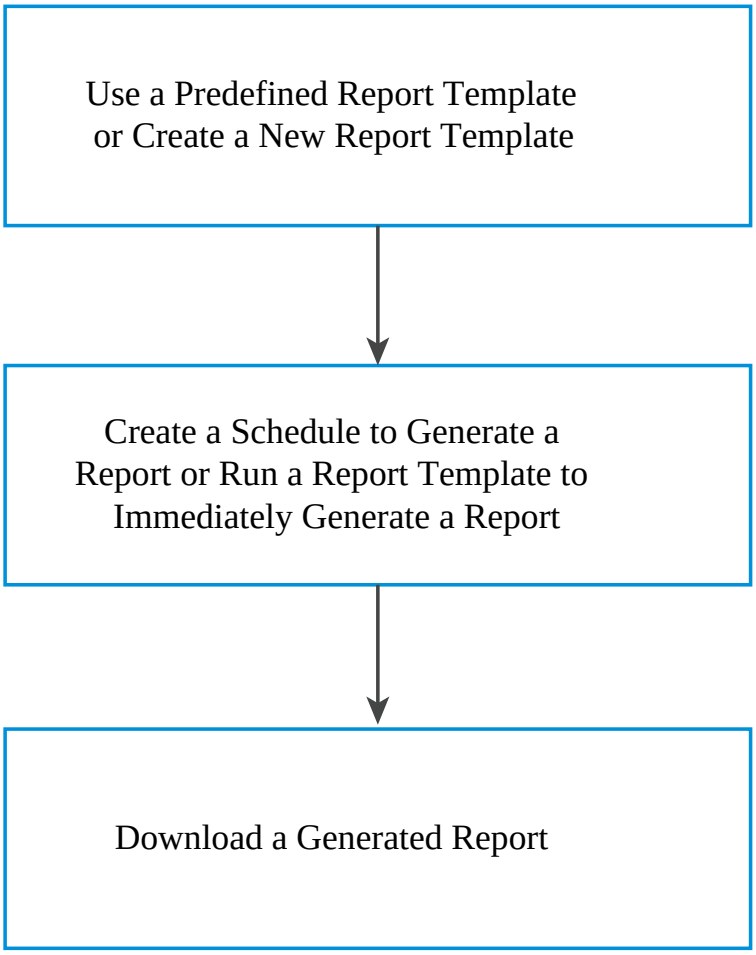
| Option | Description |
|---|---|
| Footer | Includes the date when the report is created, a note that the report is created by VCF Operations, and page number. |
| PDF | With the PDF format, you can read the reports, either on or off line. This format provides a page-by-page view of the reports, as they appear in printed form. |
| CSV | In the CSV format, the data is in a structured table of lists. |

## Manage Report Templates

Using VCF Operations you can manage report templates by running, scheduling, and generating report templates. Additionally, you can create custom report templates, edit existing report templates, download report templates, and so on.

### Reports Workflow

The following flowchart describes a simple workflow for reports in VCF Operations.




## Accessing Report Templates

A report template contains views and dashboards. Views present collected information for an object. Dashboards give a visual overview of the performance and state of objects in your virtual infrastructure. VCF Operations offers several predefined report templates that you can use based on your requirement.

### Where You Can Access Report Templates From

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Reports**. The **Report Templates** tab is displayed on the right.

The listed report templates are user-defined and predefined by VCF Operations. You can order them by template name, description, subject, date they were modified, last run report, or the user who modified them. For each template, you can see the number of generated reports and schedules.

You can filter the reports based on the name of the report template, the subject, and the owner. You can click **Add** to create a report template. For information about creating a report template, see Create a Report Template.

You can select a report template from the list, click the vertical ellipsis against each report template, and select options such as run, edit, schedule, delete, clone, and export a report.

**Table 1388: Predefined Filter Groups**

| Filter Group | Description |
|---|---|
| Name | Filter by the template name. For example, type `my template` to list all reports that contain the `my template` phrase in their name. |
| Subject | Filter by another object. If the report contains more than one view applicable for another type of object, you can filter by the other objects. |
| Owner | Filter by the owner of the report template. |

The maximum number of reports per template is 10. After the tenth report is generated, VCF Operations deletes the oldest report. vSphere users must be logged in until the report generation is complete. If you log out or your session expires, the report generation fails.

### Report Template Actions

You can select more than one report template and perform a set of actions by clicking the horizontal ellipsis next to the **Add** option.

| Option | Description |
|---|---|
| Delete | Deletes the report template. |
| Export | Downloads the report template. |
| Import | Allows you to import a report template by selecting a report template in XML or zip file format.<br>To import a report template:<br>• Click the **Import** option from the horizontal ellipsis.<br>• Click **Browse** and select a report template ZIP or XML file to import.<br>• Select if you want to Overwrite or Skip the file in case of a conflict.<br>• Click **Import** to import the report template, and click **Done**. |
| Change default cover image | Allows you to change the default cover image of the report template. For more information, see Upload a Default Cover Page Image for Reports. |

## Schedule a Report

In VCF Operations, the schedule of a report is the time and recurrence of report generation. To generate a report on a selected date, time, and recurrence, you create a schedule for the report template. You set the email options to send the generated report to your team.

- Download the generated report to verify the output.
- To activate sending email reports, you must have configured Outbound Alert Settings. See Notifications in VCF Operations.

The date range for the generated report is based on the time when the report is generated and not on the time when you schedule the report or when VCF Operations places the report in the queue.

**Note:**
Only users created in VCF Operations can add and edit report schedules.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Reports**.
2. From the **Report Templates** tab, select the relevant report template from the list.
3. Click the vertical ellipsis and select **Schedule**.

   To schedule a report for tenants, click the vertical ellipsis, and select **Schedule for Tenants**.

   **Note:**

   - To view the **Schedule for Tenants** option, click the filter option and select **Yes** against the option **For Tenants**.
   - Scheduled reports are always published to tenants.
4. Select an object and click **Next**.
5. Select the time zone, date, hour, and minutes (in the range of 0, 15, 30, and 45 minutes) to start the report generation.

   VCF Operations generates the scheduled reports in sequential order. Generating a report can take several hours. This process might delay the start time of a report when the previous report takes an extended period of time.
6. From the **Recurrence** drop-down menu, select one of the following options for report generation:

| Option | Description |
|---|---|
| **Daily** | You can set the periodicity in days. For example, you can set report generation to every two days. |
| **Weekly** | You can set the periodicity in weeks. For example, you can set report generation to every two weeks on Monday. |
| **Monthly** | You can set the periodicity in months. |

7. Select the **Email report** check box to send an email with the generated report.

   Email a generated report to a predefined email group or to a network shared location. For more information about how to set up and configure the email options, see Add a Standard Email Plug-In for VCF Operations Outbound Alerts.

   a) In the **Email addresses** text box, enter the email addresses that must receive the report. You can also add email addresses in the CC list and BCC list.
   b) Select an outbound rule.

   An email is sent according to this schedule every time a report is generated.
8. Save a generated report to an external location.

### Editing a Report Schedule

To edit the schedule of a report, click the link in the **Schedules** column against the report template from the **Report Templates** tab, and then from the **Scheduled Reports** dialog box, click **Edit Schedule**. You see the **Scheduled Reports** page.

**Table 1389: Scheduled Reports Toolbar Options**

| Options | Description |
|---|---|
| New Schedule | You can create a schedule for the report. |
| Edit Schedule | You can edit an existing report schedule. |
| Delete Schedule | You can delete an existing report schedule. |
| Transport Report Schedule | You can assign a new owner for the selected report schedule. You can select a target user from the **Transfer Report Schedules** dialog box. |

**Note:**
You can edit, clone, and delete report templates. Before you do, familiarize yourself with the consequences of these actions. When you edit a report template and delete it, all reports generated from the original and the edited templates are deleted. When you clone a report template, the changes that you make to the clone do not affect the source template. When you delete a report template, all generated reports are also deleted.

## Generate and Regenerate a Report

To generate a report, use a predefined or custom report template.

Create a report template.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Reports**.
2. From the **Report Templates** tab on the right, navigate to the relevant report template, click the vertical ellipsis, and select **Run**.

   To generate a report for tenants, click the vertical ellipsis, and select **Run for Tenants**.
3. Select an associated object from the **Select Object** dialog box and click **OK**.

   **Note:**

   To publish the report to tenants, click the the **Publish to Tenants** option.

The report is generated and listed on the **Generated Reports** tab.

**Note:**
To regenerate the selected report, from the **Generated Reports** tab, click the vertical ellipsis against the generated report and select **Run**.

Download the generated report and verify the output.

## Accessing Generated Reports

You can view a list of report templates that have been generated in VCF Operations.

**Where You Can Access Generated Reports From**

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Reports**. The **Generated Reports** tab on the right pane contains all the generated reports. If the report is generated through a schedule, the owner is the user who created the schedule.

**Note:**
The maximum number of reports per template is 10. After the tenth report is generated, VCF Operations deletes the oldest report.

To select a generated report from the list, click the vertical ellipsis against each generated report and select options such as run and delete. You can also select more than one generated report and select the **Delete** button above the data grid to delete a generated report.

You can filter the reports list by adding a filter from the upper-right corner of the panel.

**Table 1390: Predefined Filter Groups**

| Filter Group | Description |
|---|---|
| Report Name | Filter by the report template name. For example, type `my template` to list all reports that contain the `my template` phrase in their name. |
| Template | Filter by the report template. You can select a template from a list of templates applicable for this object. |
| Completion Date/Time | Filter by the date, time, or time range. |
| Subject | Filter by another object. If the report contains more than one view applicable for another type of object, you can filter by that second object. |
| Status | Filter by the status of the report. |

You can download a report in a PDF or CSV format. You define the format that a report is generated in the report template.

If you log in to VCF Operations with vCenter credentials and generate a report, the generated report is always blank.

## Download a Report

To verify that the information appears as expected, you download the generated report.

Generate a report.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Reports**.
2. From the **Generated Reports** tab on the right, click the PDF or the CSV icon in the **Download** column to download the report.

VCF Operations saves the report file.

Schedule a report generation and set the email options, so your team receives the report.

## Publish a Report to Tenants

You can publish a report to the tenant.

**Prerequisites**

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards**, and then in the pane in the centre, click **Reports**.
2. From the **Reports** page, click the **Generated Reports** tab.
3. Click the the generated report and then click the **Publish** button.

The report is published to the tenant in VCF Automation.

## Upload a Default Cover Page Image for Reports

You can upload a common default image for the cover page of reports. You do not have to upload a cover page for each report. The cover pages of predefined reports are modified when you use this option. The cover pages of user-defined reports do not change.

### Where Do You Upload a Default Cover Page Image for Reports

To upload a default cover page for reports, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Reports**. From the **Report Templates** tab on the right side, click the horizontal ellipsis next to the **Add** option and click the **Change default cover image** option.

### How Do You Upload a Default Cover Page Image for Reports

Browse for the image that you want to add to the cover page and click **Save**. You can also use the default product image that is available.

## Views in VCF Operations

VCF Operations provides several types of views. Each type of view helps you to interpret metrics, properties, policies of various monitored objects including alerts, symptoms, and so on, from a different perspective. Views also show information that the adapters in your environment provide.

You can configure VCF Operations views to show transformation, forecast, and trend calculations.

- The transformation type determines how the values are aggregated.
- The trend option shows how the values tend to change, based on the historical, raw data. The trend calculations depend on the transformation type and roll up interval.
- The forecast option shows what the future values can be, based on the trend calculations of the historical data.

You can use VCF Operations views in different areas of VCF Operations.

- To manage all views, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Views** > **Manage**.
- To see the data that a view provides in your dashboard, add the View widget to the dashboard. For more information, see View Widget.

**Table 1391: Views Options**

| Options | Description |
|---|---|
| Manage | You can manage views by navigating to **Home** and in the left navigation pane, click **Views** > **Manage**. |
| Add | Use this option to create a view. See Creating and Configure a View. |
| Search | You can search for a view across the **Recents** and **All** folders in the **Views** panel. |
| Recent | The views are listed in the order in which you select them, with the most recent view that you selected, appearing at the top. Up to ten views can be displayed as Recent views.<br>If you do not pin the view and log out of the user interface, on logging back in, the view is removed from the Recents folder |
| All | Lists the views based on their type. You can use this menu for quick navigation through your views. When you navigate to a view, the views are listed in the **Views** panel under **All**. You can also search for views using keywords and letters. |

### Views and Reports Ownership

The default owner of all predefined views and templates is System. If you edit them, you become the owner. If you want to keep the original predefined view or template, you have to clone it. After you clone it, you become the owner of the clone.

The last user who edited a view, template, or schedule is the owner. For example, if you create a view you are listed as its owner. If another user edits your view, that user becomes the owner listed in the Owner column.

The user who imports the view or template is its owner, even if the view is initially created by someone else. For example, *User 1* creates a template and exports it. *User 2* imports it in back, the owner of the template becomes *User 2*.

The user who generated the report is its owner, regardless of who owns the template. If a report is generated from a schedule, the user who created the schedule is the owner of the generated report. For example, if *User 1* creates a template and *User 2* creates a schedule for this template, the generated report owner is *User 2*.

## Accessing Predefined Views

You can access some of the useful, predefined views from the **Views** home page.

To access these views, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Views**.

The views are categorized as follows: Availability, Capacity, Configuration, Inventory, Performance, and Compliance. To easily access some of the useful, predefined dashboards under these categories, click on the drop-down button against the selected category and click on the specific dashboard.

## Views Overview

A view presents collected information for an object in a certain way depending on the view type. Each type of view helps you to interpret metrics, properties, policies of various monitored objects including alerts, symptoms, and so on, from a different perspective.

### How You Access the Views Page

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Views** > **Manage**.

### Manage and Preview Views

You can preview a view by clicking a view from the **Views** page. Click on a view and add an object if necessary by clicking **Preview source** from the upper-right corner of the specific view.

From the **Views** page you can select a view from the list, click the vertical ellipsis against each view, and select the various options such as edit, delete, clone, and export a view.

You can filter the views based on the name, type, description, subject, metric, and owner. You can click the **Add** option to create a view. For information about creating a view, see Creating and Configure a View.

Views are also categorized and listed in the **Views** panel based on the type of view and subject.

**Table 1392: Filter Groups**

| Filter Group | Description |
|---|---|
| Name | Filter by the view name. For example, type `my view` to list all views that contain the `my view` phrase in their name. |
| Type | Filter by the view type. |
| Description | Filter by the view description. For example, type `my view` to list all views that contain the `my view` phrase in their description. |
| Subject | Filter by the subject. |
| Metric | Filter by a metric. All views using the metric specified will be displayed.<br>You must first select a subject and then select a metric. |
| Modified By | Filter by the user who modified the view. |

**Datagrid Options**

| Column Names | Description |
|---|---|
| Name | Displays the name of the view. |
| Type | Displays the type of view: list, summary, trend, distribution, text, or image. |
| Description | Displays the description of the dashboard. |
| Subject | Displays the base object type for which the view shows information. |
| Dashboard Usage | Displays the number of dashboards where the view is used. Click the number in the column to view the name of the dashboard/s. Click the dashboard name to navigate to the dashboard. |
| Report Usage | Displays the number of reports where the view is used. Click the number in the column to view the name of the report/s. Click the report name to navigate to the report template in edit mode. |
| Last Modified | Displays the date the view was last modified. |
| Modified By | Displays the user who last modified the view. |

### Views Actions

You can select more than one view and perform a set of actions by clicking the horizontal ellipsis next to the **Add** option.

| Option | Description |
|---|---|
| Delete | Deletes the view. |
| Export | Downloads the view. |
| Import | Allows you to import a view by selecting a view in XML or zip file format.<br>To import a view:<br>• Click the **Import** option from the horizontal ellipsis.<br>• Click **Browse** and select a view XML or ZIP file to import.<br>• Select if you want to Overwrite or Skip the file in case of a conflict.<br>• Click **Import** to import the view, and click **Done**. |

## Views and Reports Ownership

The owner of views, reports, or templates might change over time.

The default owner of all predefined views and templates is System. If you edit them, you become the owner. If you want to keep the original predefined view or template, you have to clone it. After you clone it, you become the owner of the clone.

The last user who edited a view, template, or schedule is the owner. For example, if you create a view you are listed as its owner. If another user edits your view, that user becomes the owner listed in the Owner column.

The user who imports the view or template is its owner, even if the view is initially created by someone else. For example, *User 1* creates a template and exports it. *User 2* imports it in back, the owner of the template becomes *User 2*.

The user who generated the report is its owner, regardless of who owns the template. If a report is generated from a schedule, the user who created the schedule is the owner of the generated report. For example, if *User 1* creates a template and *User 2* creates a schedule for this template, the generated report owner is *User 2*.

## Creating and Configure a View

To collect and display information for a specific object, you can create a custom view.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Views** > **Create**.
2. Select one of the following views from the right panel.
   - List View
   - Summary View
   - Trend View
   - Distribution View
   - Text View
   - Image View
3. At the end of each tab in the selected view, you can go to the previous or next tab. You can also cancel the creation of the view.
4. After you have added all the details, click **Create** to create the view.

### List View

List views provide tabular data about specific objects in the monitored environment that correspond to the selected view.

#### Where You Find the List View

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Views** > **Create**. Click **List** from the right page.

#### Name and Configuration Tab

| Option | Description |
|---|---|
| Name | Name of the view as it appears on the Views page. |
| Description | Description of the view. |
| **Settings** | |
| Items per page | Select the number of items per page. Each item is one row and its metrics and properties are the columns. |
| Top result count | Select the top results. Restricts the number of results. For example, if you list all the clusters in a View, selecting 10 in this option displays the top 10 clusters with the relevant information. You can reduce the number of rows for the purposes of reporting. |
| Include Deleted Objects | Select to add deleted objects. |
| Show Objects | Select the type of object you want displayed in the view. You can select **Existing**, **Deleted**, or **All** objects. |
| Show Object Creation Date | Select to display the date the object was created. |
| Make the view available at > Dashboards through the View widget | Select if you want to make the view available in a dashboard. |
| Make the view available at > Report template creation and modification | Select if you want to make the view available in a report template. |
| Hide the view for the selected object types | Select an object type for which you do not want to see this view. For example, you have a list view with the subject <virtual machines>. It is visible when you select any of its parent objects. You add a data center from the list. The view is not visible anymore on the data center level. |

## Data Tab

The data definition process includes adding properties, metrics, policies, or data that adapters provide to a view. These are the items by which VCF Operations collects, calculates, and presents the information for the view.

### How to Add Data to a View

If you selected more than one subject, click the subject for which you add data. Double-click either a metric or a property from the tree in the left panel to add it to the view. For each subject that you select, the data available to add might be different. The Data, Transformation, and Configuration details are displayed.

You can see a live preview of the view type when you select a subject and associated data, and then click **Preview Source**.

| Option | Description |
|---|---|
| Add Subject | Select the base object type for which the view shows information. The subject you specify determines where the view is applicable. If you select more than one subject, the view is applicable for each of them. |
| Group By | You can group the results based on a parent object, by making a selection in the Group By drop-down option. If you generate a report based on the list view for which a group has been specified, the report displays group-based information for the selected object. You can also view summary calculations for the group of objects in the report, along with the total summary results for all the objects. |
| **Breakdown By** | |
| Add interval breakdown | Select this check box to see the data for the selected resources broken down in time intervals.<br>After you select this check box, you can enter a label, specify whether the values have to be sorted in ascending or descending order, and select a breakdown interval for the time range. |
| Add Instance breakdown | Select this check box to see the data for all instances of the selected resources.<br>After you select this check box, you can enter a label and select a metric group to break down all the instances in that group. Deselect **Show non-instance aggregate metric** to display only the separate instances. Deselect **Show only instance name** to display the metric group name and instance name in the instance breakdown column.<br>For example, you can create a view to display CPU usage by selecting the metric **CPU:0\|Usage**. If you add an instance breakdown column, the column **CPU:0\|Usage** displays the usage of all CPU instances on separate rows (0, 1, and so on). To avoid ambiguity, you can change the metric label of **CPU:0\|Usage** to **Usage**. |
| **Data Grid options** | |

| Option | Description |
|---|---|
| **Self** drop down option | Click the drop down button to select an ancestor or descendant of the selected subject. You can select metrics and properties from the data selection tree and then configure them.<br>**Note:**<br>Error messages are displayed in the following cases:<br>• If the same object type is used as an ancestor/ descendant and as the primary object type.<br>• If you do not select a metric for the primary object type. You must select at least one metric.<br>• If the number of related ancestor/descendant object types used in the configuration exceeds a maximum of 3. |
| Add Object Name | If you have selected an ancestor or descendant from the **Self** drop down option, click **Add Object Name** to add the name of the selected descendant or ancestor as a column name in the **Preview Source** pane. |
| Data selection tree (Metrics and Properties) | Select a metric or a property.<br>**Note:**<br>The metrics presented by default are a subset of available metrics. If the desired metric is not represented, use the **Select Object** button in the title bar of the data selection tree to re-filter the displayed list. |
| Data column | Click the metric or property to enter configuration details in the configuration column. |
| Transformation column | Displays the type of transformation applied to the data. |
| Configuration column | |
| Metric name | Default metric name. |
| Metric label | Customizable label as it appears in the view or report. |
| Units | Depends on the added metric or property. You can select in what unit to display the values. For example, for CPU\|Demand (MHz) from the **Units** drop-down menu, you can change the value to Hz, kHz, or GHz. If you select **Auto**, the scaling is set to a meaningful unit. |
| Sort order | Orders the values in ascending or descending order<br>**Note:**<br>Sort order is not activated for ancestor or descendant objects. |

| Option | Description |
| --- | --- |
| Transformation | Determines what calculation method is applied on the raw data. You can select the type of transformation:<br>• Minimum. The minimum value of the metric over the selected time range.<br>• Maximum. The maximum value of the metric over the selected time range.<br>• Average. The mean of all the metric values over the selected time range.<br>• Sum. The sum of the metric values over the selected time range.<br>• First. The first metric value for the selected time range.<br>• Last. The last value of a metric within the selected time range.<br>If you have selected **Last** as the transformation in versions before vRealize Operations 6.7, and the end of specified time range is not before the last five minutes, use the **Current** transformation.<br>• Current. The last available value of a metric if it was last updated not before five collection cycles were complete, otherwise it is null.<br>• Standard Deviation. The standard deviation of the metric values.<br>• Metric Correlation. Displays the value when another metric is at the minimum or maximum. For example, displays the value for memory.usage when cpu.usage is at a maximum.<br>• Forecast. Performs a regressive analysis and predicts future values. Displays the last metric value of the selected range.<br>• Percentile. Calculates the specified percentile for the data range. For example, you can view the 95th percentile, 99th percentile, and so on.<br>• Expression. Allows you to construct a mathematical expression over existing transformations using minus, plus, multiplication, division, unary minus, unary plus, and round brackets. For example, `sum/((max + min)/2)`. You can use the operands of some of the existing transformations such as, max, min, avg, sum, first, last, current. You cannot use standard deviation, forecast, metric correlation, and percentile.<br>You can customize the metric unit label when you select the Expression transformation. For example, some of the metric units available are, vCPUs, Bps, KBps, Mbps, and MBps.<br>• Timestamp: You can choose between Absolute Timestamp or Relative Timestamp.<br>– If applied to a numeric metric/property defined with a time-unit definition, the actual value is converted to a human readable timestamp. The metric value is rounded-off to an hour.<br>– In the remaining cases, a timestamp is displayed when metrics and properties are added or modified. In this case, the behavior is the same as the Timestamp option selected for a non-Timestamp transformation. Applicable for Absolute Timestamp and Relative Timestamp.<br>Available for List view and Minimum, Maximum, Current, First, and Last transformation. |
| Ranges for metric coloring | You can associate colors to metrics by entering a percentage, range, or specific state. For example, you can enter Powered Off in the **Red Bound** field when you select virtual machine as an object. You can set the colors only for views and not for csv or pdf formats. |

| Option | Description |
|---|---|
| Series Roll up | The time interval at which the data is rolled up. You can select one of the available options. For example, if you select Sum as a Transformation and 5 minutes as the roll-up interval, then the system selects 5-minute interval values and adds them.<br>This option is applicable to the Transformation configuration option. |

### Time Settings Tab

Use the time settings to select the time interval of data transformation. These options are available for all view types, except Image.

You can set a time range for a past period or set a future date for the end of the time period. When you select a future end date and no data is available, the view is populated by forecast data. Data is collected based on the browser time.

**Table 1393: Time Settings Options**

| Configuration Option | Description |
|---|---|
| Time Range Mode | In Basic mode, you can select date ranges.<br>In Advanced mode, you can select any combination of relative or specific start and end dates.<br>You can also activate the **Business Hour** option and select business hours/days for weekdays. |
| Relative Date Range | Select a relative date range of data transformation.<br>Available in Basic mode. |
| Specific Date Range | Select a specific date range of data transformation.<br>Available in Basic mode. |
| Absolute Date Range | Select a date or time range to view data for a time unit such as a complete month or a week. For example, you can run a report on the third of every month for the previous month. Data from the first to the end of the previous month is displayed as against data from the third of the previous month to the third of the current month.<br>The units of time available are: **Hours**, **Days**, **Weeks**, **Months**, and **Years**.<br>The locale settings of the system determine the start and end of the unit. For example, weeks in most of the European countries begin on Monday while in the United States they begin on Sunday.<br>Available in Basic mode. |
| Relative Start Date | Select a relative start date of data transformation.<br>Available in Advanced mode. |
| Relative End Date | Select a relative end date of data transformation.<br>Available in Advanced mode. |
| Specific Start Date | Select a specific start date of data transformation.<br>Available in Advanced mode. |
| Specific End Date | Select a specific end date of data transformation.<br>Available in Advanced mode. |
| Currently selected date range | Displays the date or time range you selected. For example, if you select a specific date range from 5/01/2016 to 5/18/2016, the following information is displayed: `May 1, 2016 12:00:00 AM to May 18, 2016 11:55:00 PM`. |

| Configuration Option | Description |
|---|---|
| Business Hours | Select business hours from Monday to Sunday by moving the sliders on the left and right sides to set the start and end time for each day of the week.<br>For example, as a VM owner, you can track the average utilization of VMs over a week (business days), during specified hours of the day (business hours).<br>This option is available for Minimum, Maximum, Average, Sum, and Percentile transformations<br>Available in Advanced mode for List Views.<br>**Note:**<br>Business Hours is not supported for the **Interval Breakdown** option. |

## Filter Tab

The filter option allows you to add additional criteria when the view displays too much information. For example, a List view shows information about the health of virtual machines. From the **Filter** tab, you add a risk metric less than 50%. The view displays the health of all virtual machines with risk less than 50%. For selected criteria you can also apply Business Hours, if the selected transformation type you add as a filter is supported by the business hours functionality.

To add a filter to a view, from an existing or new view dialog box, click the **Filter** tab. Fill in the details for each row and click **Add**. You can activate Business Hours for the metric selected.

Each subject has a separate filter box. For Alerts Roll up, Alert, and Symptom subjects not all applicable metrics are supported for filtering.

**Note:**
For Symptom based views, in the filter, if you select either **Symptom Definition Name**, **Alert Definition Name**, or **Alert Type**, it is recommended that you select a preview source that has a smaller number of objects.

**Table 1394: Filter Add Options**

| Option | Description |
|---|---|
| Add | Adds another criteria to the criteria set. The filter returns results that match all the specified criteria.<br>If you add a filter for an instance metric or property, all the instances of the object for which the criteria is met, will be displayed in the preview screen.<br>For instance metrics or properties, you can filter based on transformations such as, Current, Average, First, Last, Maximum, Minimum, Sum, and Timestamp. |
| Add another criteria set | Adds another criteria set. The filter returns results that match one criteria set or another. |

## Summary Tab

You can add more than one summary row or column and configure each to show different aggregations. In the summary configuration panel, you select the aggregation method and what data to include or exclude from the calculations.

To add a summary row or column to a view, from an existing or new view dialog box, click the **Summary** tab in the right pane. Click the plus sign to add a summary row.

For the Summary view, the summary column shows aggregated information by the items provided on the **Data** tab.

### Previous, Next, Create, and Cancel Options

At the end of each tab, you can go to the previous or next tab. You can also cancel the creation of the view. After you have added all the details, click **Create** to create the view.

## Summary View

Summary views provide tabular data about the use of resources in the monitored environment.

### Where You Find the Summary View

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Views** > **Create**. Click **Summary** from the right page.

### Name and Configuration Tab

| Option | Description |
|---|---|
| Name | Name of the view as it appears on the Views page. |
| Description | Description of the view. |
| **Settings** | |
| Items per page | Select the number of items per page. Each item is one row and its metrics and properties are the columns. |
| Make the view available for > Dashboards through the View widget | Select if you want to make the view available in a dashboard. |
| Make the view available for > Report template creation and modification | Select if you want to make the view available in a report template. |
| Hide the view for the selected object types | Select an object type for which you do not want to see this view. For example, you have a list view with the subject <virtual machines>. It is visible when you select any of its parent objects. You add a data center from the list. The view is not visible anymore on the data center level. |

### Data Tab

The data definition process includes adding properties, metrics, policies, or data that adapters provide to a view. These are the items by which VCF Operations collects, calculates, and presents the information for the view.

### How to Add Data to a View

If you selected more than one subject, click on the subject for which you want to add data. Double-click either a metric or a property from the tree in the left panel to add it to the view. For each subject that you select, the data available to add might be different. The Data, Transformation, and Configuration details are displayed.

You can see a preview of the view type when you select a subject and associated data, and then click **Select preview source**.

| Option | Description |
|---|---|
| Add Subject | Select the base object type for which the view shows information. The subject you specify determines where the view is applicable. If you select more than one subject, the view is applicable for each of them. |
| **Data Grid options** | |
| Data selection tree | Select a metric or property |

| Option | Description |
|---|---|
| Data column | Click the metric or property to enter configuration details in the configuration column. |
| Transformation column | |
| Configuration column | |
| Metric name | Default metric name. |
| Metric label | Customizable label as it appears in the view or report. |
| Units | Depends on the added metric or property. You can select in what unit to display the values. For example, for CPU\|Demand(MHz) from the **Units** drop-down menu, you can change the value to Hz, KHz, or GHz. If you select **Auto**, the scaling is set to a meaningful unit. |
| Sort order | Orders the values in ascending or descending order. |
| Transformation | Determines what calculation method is applied on the raw data. You can select the type of transformation:<br>• Minimum. The minimum value of the metric over the selected time range.<br>• Maximum. The maximum value of the metric over the selected time range.<br>• Average. The mean of all the metric values over the selected time range.<br>• Sum. The sum of the metric values over the selected time range.<br>• First. The first metric value for the selected time range.<br>• Last. The last value of a metric within the selected time range. If you have selected **Last** as the transformation in versions before vRealize Operations 6.7, and the end of specified time range is not before the last five minutes, use the **Current** transformation.<br>• Current. The last available value of a metric if it was last updated not before five collection cycles were complete, otherwise it is null.<br>• Standard Deviation. The standard deviation of the metric values.<br>• Metric Correlation. Displays the value when another metric is at the minimum or maximum. For example, displays the value for memory.usage when cpu.usage is at a maximum.<br>• Forecast. Performs a regressive analysis and predicts future values. Displays the last metric value of the selected range.<br>• Percentile. Calculates the specified percentile for the data range. For example, you can view the 95th percentile, 99th percentile, and so on.<br>• Expression. Allows you to construct a mathematical expression over existing transformations using minus, plus, multiplication, division, unary minus, unary plus, and round brackets. For example, `sum/((max + min)/2)`. You can use the operands of some of the existing transformations such as, max, min, avg, sum, first, last, current. You cannot use standard deviation, forecast, metric correlation, and percentile. You can customize the metric unit label when you select the Expression transformation. For example, some of the metric units available are, vCPUs, Bps, KBps, Mbps, and MBps. |

| Option | Description |
|---|---|
| Series Roll up | The time interval at which the data is rolled up. You can select one of the available options. For example, if you select Sum as a Transformation and 5 minutes as the roll-up interval, then the system selects 5-minute interval values and adds them.<br>This option is applicable to the Transformation configuration option. |

### Time Settings Tab

Use the time settings to select the time interval of data transformation. These options are available for all view types, except Image.

You can set a time range for a past period or set a future date for the end of the time period. When you select a future end date and no data is available, the view is populated by forecast data. Data is calculated based on the browser time.

**Table 1395: Time Settings Options**

| Configuration Option | Description |
|---|---|
| Time Range Mode | In Basic mode, you can select date ranges.<br>In Advanced mode, you can select any combination of relative or specific start and end dates. |
| Relative Date Range | Select a relative date range of data transformation.<br>Available in Basic mode. |
| Specific Date Range | Select a specific date range of data transformation.<br>Available in Basic mode. |
| Absolute Date Range | Select a date or time range to view data for a time unit such as a complete month or a week. For example, you can run a report on the third of every month for the previous month. Data from the first to the end of the previous month is displayed as against data from the third of the previous month to the third of the current month.<br>The units of time available are: **Hours**, **Days**, **Weeks**, **Months**, and **Years**.<br>The locale settings of the system determine the start and end of the unit. For example, weeks in most of the European countries begin on Monday while in the United States they begin on Sunday.<br>Available in Basic mode. |
| Relative Start Date | Select a relative start date of data transformation.<br>Available in Advanced mode. |
| Relative End Date | Select a relative end date of data transformation.<br>Available in Advanced mode. |
| Specific Start Date | Select a specific start date of data transformation.<br>Available in Advanced mode. |
| Specific End Date | Select a specific end date of data transformation.<br>Available in Advanced mode. |
| Currently selected date range | Displays the date or time range you selected. For example, if you select a specific date range from 5/01/2016 to 5/18/2016, the following information is displayed: `May 1, 2016 12:00:00 AM to May 18, 2016 11:55:00 PM`. |

### Filter Tab

The filter option allows you to add additional criteria when the view displays too much information. For example, a view shows information about the health of virtual machines. From the **Filter** tab, you add a risk metric less than 50%. The view displays the health of all virtual machines with risk less than 50%.

To add a filter to a view, from an existing or new view dialog box, click the **Filter** tab. Fill in the details for each row and click **Add**.

Each subject has a separate filter box. For Alerts Roll up, Alert, and Symptom subjects not all applicable metrics are supported for filtering.

**Table 1396: Filter Add Options**

| Option | Description |
|---|---|
| Add | Adds another criteria to the criteria set. The filter returns results that match all the specified criteria.<br>If you add a filter for an instance metric or property, all the instances of the object for which the criteria is met, will be displayed in the preview screen.<br>For instance metrics or properties, you can filter based on transformations such as, Current, Average, First, Last, Maximum, Minimum, Sum, and Timestamp. |
| Add another criteria set | Adds another criteria set. The filter returns results that match one criteria set or another. |

#### Summary Tab

You can add more than one summary row or column and configure each to show different aggregations. In the summary configuration panel, you select the aggregation method and what data to include or exclude from the calculations.

To add a summary row or column to a view, from an existing or new view dialog box, click the **Summary** tab in the right pane. Click the plus sign to add a summary row.

For the Summary view, the summary column shows aggregated information by the items provided on the **Data** tab.

#### Previous, Next, Create, and Cancel Options

At the end of each tab, you can go to the previous or next tab. You can also cancel the creation of the view. After you have added all the details, click **Create** to create the view.

### Trend View

Trend views use historic data to generate trends and forecasts for resource use and availability in the monitored environment.

#### Where You Find the Trend View

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Views** > **Create**. Click **Trend** from the right page.

#### Name and Configuration Tab

| Option | Description |
|---|---|
| Name | Name of the view as it appears on the Views page. |
| Description | Description of the view. |
| **Settings** | |

| Option | Description |
|---|---|
| The maximum plot lines | Enter the maximum number of plot lines. Limits the output in terms of the objects displayed in the live preview of the view type on the left upper pane. The number you set as the maximum number of plot lines determines the plot lines.<br>For example, if you plot historical data and set the maximum at 30 plot lines, then 30 objects are displayed. If you plot historical, trend, and forecast lines, and set the maximum to 30 plot lines, then only 10 objects are displayed as each object has three plot lines. |
| Make the view available for > Dashboards through the View widget | Select if you want to make the view available in a dashboard. |
| Make the view available for > Report template creation and modification | Select if you want to make the view available in a report template. |
| Hide the view for the selected object types | Select an object type for which you do not want to see this view. For example, you have a list view with the subject <virtual machines>. It is visible when you select any of its parent objects. You add a data center from the list. The view is not visible anymore on the data center level. |

## Data Tab

The data definition process includes adding properties, metrics, policies, or data that adapters provide to a view. These are the items by which VCF Operations collects, calculates, and presents the information for the view.

### How to Add Data to a View

If you selected more than one subject, click on the subject for which you want to add data. Double-click the data from the tree in the left panel to add it to the view. For each subject the data available to add, might be different.

You can see a live preview of the view type when you select a subject and associated data, and then click **Select preview source**.

| Option | Description |
|---|---|
| Add Subject | Select the base object type for which the view shows information. The subject you specify determines where the view is applicable. If you select more than one subject, the view is applicable for each of them. |
| **Data Grid options** | |
| Data selection tree | Select a metric or a property. |
| Data column | Click the metric or property to enter configuration details in the configuration column. |
| Transformation column | Displays the type of transformation applied to the data. |
| Configuration column | |
| Metric name | Default metric name. |
| Metric label | Customizable label as it appears in the view or report. |
| Units | Depends on the added metric or property. You can select in what unit to display the values. For example, for CPU\|Demand (MHz) from the Units drop-down menu, you can change the value to Hz, KHz, or GHz. If you select Auto, the scaling is set to a meaningful unit. |

| Option | Description |
|---|---|
| Sort order | Select whether you want to sort data in either ascending or descending order.<br>**Note:**<br>Null values are included. To exclude them, you can apply a filter. |
| Data Series | Select whether to include historical data, trend of historical data, and forecast for future time in the trend view calculations. |
| Ranges for metric coloring | You can associate colors to metrics by entering a percentage, range, or specific state. For example, you can enter Powered Off in the **Red Bound** field when you select virtual machine as an object. You can set the colors only for views and not for csv or pdf formats. |
| Series Roll up | The time interval at which the data is rolled up. You can select one of the available options. For example, if you select Sum as a Transformation and 5 minutes as the roll-up interval, then the system selects 5-minute interval values and adds them.<br>This option is applicable to the Transformation configuration option.<br>Available for all views. |
| Threshold Lines | You can set a threshold for a single metric:<br>• None. You have not set a threshold.<br>• By Symptom Definition. You can set a threshold value based on a symptom definition.<br>• Custom. You can set the threshold value as **Warning**, **Critical**, or **Immediate**. These options are available only for the **Custom** option. |

## Time Settings Tab

Use the time settings to select the time interval of data transformation.

You can set a time range for a past period or set a future date for the end of the time period. When you select a future end date and no data is available, the view is populated by forecast data. Data is calculated based on the browser time.

**Note:**

Trend and forecast calculations in Views are based solely on historical data for the selected resource–metric pair. By default, only the most recent 90 days of historical data are used for these calculations, even if the time range configured in the View is longer. The time range selected in the View controls only the data displayed, not the data used for trend or forecast calculations. Capacity-related settings do not affect trend or forecast calculations in Views and are not configurable for this feature.

**Table 1397: Time Settings Options**

| Configuration Option | Description |
|---|---|
| Time Range Mode | In Basic mode, you can select date ranges.<br>In Advanced mode, you can select any combination of relative or specific start and end dates. |
| Relative Date Range | Select a relative date range of data transformation.<br>Available in Basic mode. |
| Specific Date Range | Select a specific date range of data transformation.<br>Available in Basic mode. |

| Configuration Option | Description |
|---|---|
| Absolute Date Range | Select a date or time range to view data for a time unit such as a complete month or a week. For example, you can run a report on the third of every month for the previous month. Data from the first to the end of the previous month is displayed as against data from the third of the previous month to the third of the current month.<br>The units of time available are: **Hours**, **Days**, **Weeks**, **Months**, and **Years**.<br>The locale settings of the system determine the start and end of the unit. For example, weeks in most of the European countries begin on Monday while in the United States they begin on Sunday.<br>Available in Basic mode. |
| Relative Start Date | Select a relative start date of data transformation.<br>Available in Advanced mode. |
| Relative End Date | Select a relative end date of data transformation.<br>Available in Advanced mode. |
| Specific Start Date | Select a specific start date of data transformation.<br>Available in Advanced mode. |
| Specific End Date | Select a specific end date of data transformation.<br>Available in Advanced mode. |
| Currently selected date range | Displays the date or time range you selected. For example, if you select a specific date range from 5/01/2016 to 5/18/2016, the following information is displayed: `May 1, 2016 12:00:00 AM to May 18, 2016 11:55:00 PM`. |

## Filter

The filter option allows you to add additional criteria when the view displays too much information.

To add a filter to a view, from an existing or new view dialog box, click the **Filter** tab. Fill in the details for each row and click **Add**.

Each subject has a separate filter box. For Alerts Roll up, Alert, and Symptom subjects not all applicable metrics are supported for filtering.

**Table 1398: Filter Add Options**

| Option | Description |
|---|---|
| Add | Adds another criteria to the criteria set. The filter returns results that match all the specified criteria.<br>If you add a filter for an instance metric or property, all the instances of the object for which the criteria is met, will be displayed in the preview screen.<br>For instance metrics or properties, you can filter based on transformations such as, Current, Average, First, Last, Maximum, Minimum, Sum, and Timestamp. |
| Add another criteria set | Adds another criteria set. The filter returns results that match one criteria set or another. |

## Previous, Next, Create, and Cancel Options

At the end of each tab, you can go to the previous or next tab. You can also cancel the creation of the view. After you have added all the details, click **Create** to create the view.

## Distribution View

Distribution views provide aggregated data about resource distribution in the monitored environment. When you add a distribution type of View to a dashboard, you can click a section of the pie chart or on one of the bars in the bar chart to view the list of objects filtered by the selected segment.

### Where You Find the Distribution View

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Views** > **Create**. Click **Distribution** from the right page.

### Name and Configuration Tab

| Option | Description |
|---|---|
| Name | Name of the view as it appears on the Views page. |
| Description | Description of the view. |
| **Configuration** | |
| Visualization | You can view the data as a pie chart, a bar chart, or a donut chart. When you add a distribution type of View to a dashboard, you can click a section of the pie chart, or on one of the bars in the bar chart, or a section of the donut chart to view the list of objects filtered by the selected segment. You can select the display colors for single or multi-colored charts. |
| Coloring | |
| Colorize | The colors of the slices in the pie chart are displayed in the order of the colors in the color palette. |
| Select Color | Select the color that you want the chart to appear in. If there is more than one slice in a pie chart, the colors are chosen sequentially from the color palette. In a bar chart, the bars are all the same color. |
| Distribution Type > Dynamic Distribution | |
| Buckets Count | The number of buckets to use in the data distribution. |
| Buckets Size Interval | The bucket size is determined by the defined interval divided by the specified number of buckets. |
| Buckets > Size > Logarithmic bucketing | The bucket size is calculated to logarithmically increasing sizes. This provides a continuous coverage of the whole range with the specified number of buckets. The base of the logarithmic sizing is determined by the given data. |
| Buckets > Size > Simple Max/Min bucketing | The bucket size is divided equally between the measured min and max values. This provides a continuous coverage of the whole range with the specified number of buckets. |
| Distribution Type > Manual Distribution | Specify the number of buckets and the minimum and maximum values of each bucket.<br>You can also select a color for each defined bucket that you specify. |
| Distribution Type > Discrete Distribution | Specify the number of buckets in which VCF Operations distributes the data.<br>If you increase the number of buckets, you can see more detailed data. |
| Distribution Type > Summary | |
| **Settings** | |

| Option | Description |
|---|---|
| Make the view available for > Dashboards through the View widget | Select if you want to make the view available in a dashboard. |
| Make the view available for > Report template creation and modification | Select if you want to make the view available in a report template. |
| Hide the view for the selected object types | Select an object type for which you do not want to see this view. For example, you have a trend view with the subject <virtual machines>. It is visible when you select any of its parent objects. You add a data center from the list. The view is not visible anymore on the data center level. |

## Data Tab

The data definition process includes adding properties, metrics, policies, or data that adapters provide to a view. These are the items by which VCF Operations collects, calculates, and presents the information for the view.

## How to Add Data to a View

If you selected more than one subject, click the subject for which you add data. Double-click either a metric or a property from the tree in the left panel to add it to the view. For each subject that you select, the data available to add might be different. The Data, Transformation, and Configuration details are displayed.

You can see a live preview of the view type when you select a subject and associated data, and then click **Select preview source**.

| Option | Description |
|---|---|
| Add Subject | Select the base object type for which the view shows information. The subject you specify determines where the view is applicable. If you select more than one subject, the view is applicable for each of them. |
| **Data Grid options** | |
| Data selection tree | Double-click to select a metric or a property. |
| Data column | Click the metric or property to enter configuration details in the Configuration column. |
| Transformation column | Displays the type of transformation applied to the data. |
| Configuration column | |
| Metric name | Default metric name. |
| Metric label | Customizable label as it appears in the view or report. |
| Units | Depends on the added metric or property. You can select in what unit to display the values. For example, for CPU\|Demand (MHz) from the **Units** drop-down menu, you can change the value to Hz, KHz, or GHz. If you select **Auto**, the scaling is set to a meaningful unit. |
| Sort order | Orders the values in ascending or descending order. |

| Option | Description |
| --- | --- |
| Transformation | Determines what calculation method is applied on the raw data. You can select the type of transformation:<br>• Minimum. The minimum value of the metric over the selected time range.<br>• Maximum. The maximum value of the metric over the selected time range.<br>• Average. The mean of all the metric values over the selected time range.<br>• Sum. The sum of the metric values over the selected time range.<br>• First. The first metric value for the selected time range.<br>• Last. The last value of a metric within the selected time range. If you have selected **Last** as the transformation in versions before vRealize Operations 6.7, and the end of specified time range is not before the last five minutes, use the **Current** transformation.<br>• Current. The last available value of a metric if it was last updated not before five collection cycles were complete, otherwise it is null.<br>• Standard Deviation. The standard deviation of the metric values.<br>• Metric Correlation. Displays the value when another metric is at the minimum or maximum. For example, displays the value for memory.usage when cpu.usage is at a maximum.<br>• Forecast. Performs a regressive analysis and predicts future values. Displays the last metric value of the selected range.<br>• Percentile. Calculates the specified percentile for the data range. For example, you can view the 95th percentile, 99th percentile, and so on.<br>• Expression. Allows you to construct a mathematical expression over existing transformations using minus, plus, multiplication, division, unary minus, unary plus, and round brackets. For example, `sum/((max + min)/2)`. You can use the operands of some of the existing transformations such as, max, min, avg, sum, first, last, current. You cannot use standard deviation, forecast, metric correlation, and percentile. You can customize the metric unit label when you select the Expression transformation. For example, some of the metric units available are, vCPUs, Bps, KBps, Mbps, and MBps. |
| Series Roll up | The time interval at which the data is rolled up. You can select one of the available options. For example, if you select Sum as a Transformation and 5 minutes as the roll-up interval, then the system selects 5-minute interval values and adds them.<br>This option is applicable to the Transformation configuration option. |

## Time Settings Tab

Use the time settings to select the time interval of data transformation.

You can set a time range for a past period or set a future date for the end of the time period. When you select a future end date and no data is available, the view is populated by forecast data. Data is calculated based on the browser time

**Table 1399: Time Settings Options**

| Configuration Option | Description |
|---|---|
| Time Range Mode | In Basic mode, you can select date ranges.<br>In Advanced mode, you can select any combination of relative or specific start and end dates. |
| Relative Date Range | Select a relative date range of data transformation.<br>Available in Basic mode. |
| Specific Date Range | Select a specific date range of data transformation.<br>Available in Basic mode. |
| Absolute Date Range | Select a date or time range to view data for a time unit such as a complete month or a week. For example, you can run a report on the third of every month for the previous month. Data from the first to the end of the previous month is displayed as against data from the third of the previous month to the third of the current month.<br>The units of time available are: **Hours**, **Days**, **Weeks**, **Months**, and **Years**.<br>The locale settings of the system determine the start and end of the unit. For example, weeks in most of the European countries begin on Monday while in the United States they begin on Sunday.<br>Available in Basic mode. |
| Relative Start Date | Select a relative start date of data transformation.<br>Available in Advanced mode. |
| Relative End Date | Select a relative end date of data transformation.<br>Available in Advanced mode. |
| Specific Start Date | Select a specific start date of data transformation.<br>Available in Advanced mode. |
| Specific End Date | Select a specific end date of data transformation.<br>Available in Advanced mode. |
| Currently selected date range | Displays the date or time range you selected. For example, if you select a specific date range from 5/01/2016 to 5/18/2016, the following information is displayed: `May 1, 2016 12:00:00 AM to May 18, 2016 11:55:00 PM`. |

## Filter Tab

The filter option allows you to add additional criteria when the view displays too much information.

To add a filter to a view, from an existing or new view dialog box, click the **Filter** tab. Fill in the details for each row and click **Add**.

Each subject has a separate filter box. For Alerts Roll up, Alert, and Symptom subjects not all applicable metrics are supported for filtering.

**Table 1400: Filter Add Options**

| Option | Description |
|---|---|
| Add | Adds another criteria to the criteria set. The filter returns results that match all the specified criteria.<br>If you add a filter for an instance metric or property, all the instances of the object for which the criteria is met, will be displayed in the preview screen.<br>For instance metrics or properties, you can filter based on transformations such as, Current, Average, First, Last, Maximum, Minimum, Sum, and Timestamp. |
| Add another criteria set | Adds another criteria set. The filter returns results that match one criteria set or another. |

#### Previous, Next, Create, and Cancel Options

At the end of each tab, you can go to the previous or next tab. You can also cancel the creation of the view. After you have added all the details, click **Create** to create the view.

### Text View

Text views allows you to insert provided text. The text can be dynamic and contain metrics and properties. You can format text to increase or decrease the font size, change the font color, highlight text, and align text to the left, right, or center. You can also make the selected text appear bold, in italics, or underlined. By default the text view is available only for report template creation and modification. You can change this in the **Visibility** option in the **Name and Configuration** tab.

#### Where You Find the Text View

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Views** > **Create**. Click **Text** from the right page.

#### Name and Configuration Tab

| Option | Description |
|---|---|
| Name | Name of the view as it appears on the Views page. |
| Description | Description of the view. |
| **Settings** | |
| Make the view available for > Dashboards through the View widget | Select if you want to make the view available in a dashboard. |
| Make the view available for > Report template creation and modification | Select if you want to make the view available in a report template. This is the default option. |
| Hide the view for the selected object types | Select an object type for which you do not want to see this view. For example, you have a list view with the subject <virtual machines>. It is visible when you select any of its parent objects. You add a data center from the list. The view is not visible anymore on the data center level. |

#### Data Tab

The data definition process includes adding properties, metrics, policies, or data that adapters provide to a view. These are the items by which VCF Operations collects, calculates, and presents the information for the view.

## How to Add Data to a View

If you selected more than one subject, click the subject for which you add data. Double-click the data from the tree in the left panel to add it to the view. For each subject, the data available to add might be different. The data and configuration details are displayed.

You can see a live preview of the view type when you select a subject and associated data, and then click **Select preview source**.

| Option | Description |
|---|---|
| Add Subject | Select the base object type for which the view shows information. The subject you specify determines where the view is applicable. If you select more than one subject, the view is applicable for each of them. |
| **Data Grid options** | |
| Data selection tree | Select a metric or property. |
| Data column | Click the metric or property to enter configuration details in the Configuration column.<br>You can also enter text to display in the view. |
| **Configuration column** | |
| Metric name | Default metric name. |
| Metric label | Customizable label as it appears in the view or report. |
| Units | Depends on the added metric or property. You can select in what unit to display the values. For example, for CPU\|Demand(MHz) from the **Units** drop-down menu, you can change the value to Hz, KHz, or GHz. If you select **Auto**, the scaling is set to a meaningful unit. |

| Option | Description |
|---|---|
| Transformation | Determines what calculation method is applied on the raw data. You can select the type of transformation:<br>• Minimum. The minimum value of the metric over the selected time range.<br>• Maximum. The maximum value of the metric over the selected time range.<br>• Average. The mean of all the metric values over the selected time range.<br>• Sum. The sum of the metric values over the selected time range.<br>• First. The first metric value for the selected time range.<br>• Last. The last value of a metric within the selected time range. If you have selected **Last** as the transformation in versions before vRealize Operations 6.7, and the end of specified time range is not before the last five minutes, use the **Current** transformation.<br>• Current. The last available value of a metric if it was last updated not before five collection cycles were complete, otherwise it is null.<br>• Standard Deviation. The standard deviation of the metric values.<br>• Metric Correlation. Displays the value when another metric is at the minimum or maximum. For example, displays the value for memory.usage when cpu.usage is at a maximum.<br>• Forecast. Performs a regressive analysis and predicts future values. Displays the last metric value of the selected range.<br>• Percentile. Calculates the specified percentile for the data range. For example, you can view the 95th percentile, 99th percentile, and so on.<br>• Expression. Allows you to construct a mathematical expression over existing transformations using minus, plus, multiplication, division, unary minus, unary plus, and round brackets. For example, `sum/((max + min)/2)`. You can use the operands of some of the existing transformations such as, max, min, avg, sum, first, last, current. You cannot use standard deviation, forecast, metric correlation, and percentile. You can customize the metric unit label when you select the Expression transformation. For example, some of the metric units available are, vCPUs, Bps, KBps, Mbps, and MBps. |
| Series Roll up | The time interval at which the data is rolled up. You can select one of the available options. For example, if you select Sum as a Transformation and 5 minutes as the roll-up interval, then the system selects 5-minute interval values and adds them.<br>This option is applicable to the Transformation configuration option. |

## Time Settings

Use the time settings to select the time interval of data transformation. These options are available for all view types, except Image.

You can set a time range for a past period or set a future date for the end of the time period. When you select a future end date and no data is available, the view is populated by forecast data. Data is calculated based on the browser time.

**Table 1401: Time Settings Options**

| Configuration Option | Description |
|---|---|
| Time Range Mode | In Basic mode, you can select date ranges.<br>In Advanced mode, you can select any combination of relative or specific start and end dates. |
| Relative Date Range | Select a relative date range of data transformation.<br>Available in Basic mode. |
| Specific Date Range | Select a specific date range of data transformation.<br>Available in Basic mode. |
| Absolute Date Range | Select a date or time range to view data for a time unit such as a complete month or a week. For example, you can run a report on the third of every month for the previous month. Data from the first to the end of the previous month is displayed as against data from the third of the previous month to the third of the current month.<br>The units of time available are: **Hours**, **Days**, **Weeks**, **Months**, and **Years**.<br>The locale settings of the system determine the start and end of the unit. For example, weeks in most of the European countries begin on Monday while in the United States they begin on Sunday.<br>Available in Basic mode. |
| Relative Start Date | Select a relative start date of data transformation.<br>Available in Advanced mode. |
| Relative End Date | Select a relative end date of data transformation.<br>Available in Advanced mode. |
| Specific Start Date | Select a specific start date of data transformation.<br>Available in Advanced mode. |
| Specific End Date | Select a specific end date of data transformation.<br>Available in Advanced mode. |
| Currently selected date range | Displays the date or time range you selected. For example, if you select a specific date range from 5/01/2016 to 5/18/2016, the following information is displayed: `May 1, 2016 12:00:00 AM to May 18, 2016 11:55:00 PM`. |

## Filter

The filter option allows you to add additional criteria when the view displays too much information.

To add a filter to a view, from an existing or new view dialog box, click the **Filter** tab. Fill in the details for each row and click **Add**.

Each subject has a separate filter box. For Alerts Roll up, Alert, and Symptom subjects not all applicable metrics are supported for filtering.

**Table 1402: Filter Add Options**

| Option | Description |
|---|---|
| Add | Adds another criteria to the criteria set. The filter returns results that match all the specified criteria.<br>If you add a filter for an instance metric or property, all the instances of the object for which the criteria is met, will be displayed in the preview screen.<br>For instance metrics or properties, you can filter based on transformations such as, Current, Average, First, Last, Maximum, Minimum, Sum, and Timestamp. |
| Add another criteria set | Adds another criteria set. The filter returns results that match one criteria set or another. |

#### Previous, Next, Create, and Cancel Options

At the end of each tab, you can go to the previous or next tab. You can also cancel the creation of the view. After you have added all the details, click **Create** to create the view.

## Image View

Image views allow you to insert a static image. By default the image view is available only for report template creation and modification.

#### Where You Find the Image View

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Views** > **Create**. Click **Image** from the right page.

#### Name and Configuration Tab

| Option | Description |
|---|---|
| Name | Name of the view as it appears on the Views page. |
| Description | Description of the view. |
| **Settings** | |
| Make the view available for > Dashboards through the View widget | Select if you want to make the view available in a dashboard. |
| Make the view available for > Report template creation and modification | Select if you want to make the view available in a report template. This is the default option. |
| Hide the view for the selected object types | Select an object type for which you do not want to see this view. For example, you have a list view with the subject <virtual machines>. It is visible when you select any of its parent objects. You add a data center from the list. The view is not visible anymore on the data center level. |

#### Data Tab

#### How to Add Data to a View

From the **Data** tab, browse and select an image from the left panel to add it to the view.

You can see a live preview of the view type when you select an image and then click **Preview source**.

#### Previous, Next, Create, and Cancel Options

At the end of each tab, you can go to the previous or next tab. You can also cancel the creation of the view. After you have added all the details, click **Create** to create the view.

## Editing, Cloning, and Deleting a View

You can edit, clone, and delete a view. Before you do, familiarize yourself with the consequences of these actions.

#### Edit a View

When you edit a view, all changes are applied to the report templates that contain the view. To edit a view, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Views** > **Manage**. From the **Views** page on the right, select a view, click the vertical ellipsis against the view, and select **Edit**.

#### Clone a View

When you clone a view, the changes that you make to the clone do not affect the source view. To clone a view, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Views** > **Manage**. From the **Views** page on the right, select a view, click the vertical ellipsis against the view, and select **Clone**.

#### Delete a View

When you delete a view, it is removed from all the report templates that contain it. To delete a view, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Views** > **Manage**. From the **Views** page on the right, select a view, click the vertical ellipsis against the view, and select **Delete**.

## Share a View

You might want to present insights into your application or infrastructure performance to a wider audience, using tools such as Microsoft Power BI. By using the view share functionality, you can export a view to an external tool to ensure relevant stakeholders have the latest insights. You can use the provided API end point URL to import view data in a reporting tool. You can also export the configuration details of the view.

#### Where You Can Access the Option to Share Views

From the left menu, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Dashboards** and then in the centre pane click **Views** > **Manage**. From the **Views** page on the right, click an existing view, select a preview source, and then click the **Share View** icon in the top-right corner.

**Table 1403: Options in the Share View Dialog Box**

| Option | Description |
|---|---|
| URL Tab | You can export view data using a URL. The resulting view data includes view configuration data, objects, and metrics metadata, per object metric data, summary and calculated data, and other view specific data. View data usually corresponds to the data shown in the View UI for the specified view configuration. You can copy and paste the URL in other reporting tools for integration with VCF Operations.<br>• Preview Source: You can select the last previewed source.<br>• URL: You can copy the URL for the selected configured view. To use the URL as an authorized user, you need an authorization token.<br>**Note:**<br>The URL provided is the required REST call for fetching data. See the API Programming Guide for details.<br>• Authorization: The authorization token is generated. To generate the authorization token, enter the **Username** and **Password** and click **Generate**. |
| Export Tab | You can download and export the view configuration details. |

# Extensions Management

**Important:**

Extensions is available ONLY for select customers. Reach out to your account team to work with the VCF Operations product team to enable the capabilities.

**Note:** Do NOT directly enable the features as it is not supported.

The following capabilities are part of Extensions:

- **Platform Observability**
  Kubernetes Observability for the Platform Engineering team.
- **VMware Intelligent Assist**
  AI assisted chatbot to help troubleshoot VCF software (ESX, vCenter, and Management Services).
- **Content Builder**
  AI assisted mechanism to build management packs.

# Configuring Super Metrics

The super metric is a mathematical formula that contains one or more metrics or properties. It is a custom metric that you design to help track combinations of metrics or properties, either from a single object or from multiple objects. If a single metric does not inform you about the behavior of your environment, you can define a super metric.

After you define it, you assign the super metric to one or more object types. This action calculates the super metric for the objects in that object type and simplifies the metrics display. For example, you define a super metric that calculates the average CPU usage on all virtual machines, and you assign it to a cluster. The average CPU usage on all virtual machines in that cluster is reported as a super metric for the cluster.

When the super metric attribute is activated in a policy, you can also collect super metrics from a group of objects associated with a policy.

Because super metric formulas can be complex, plan your super metric before you build it. The key to creating a super metric that alerts you to the expected behavior of your objects is knowing your own enterprise and data. Use this checklist to help identify the most important aspects of your environment before you begin to configure a super metric.

**Table 1404: Designing a Super Metric Checklist**

| | |
|---|---|
| Determine the objects that are involved in the behavior to track. | When you define the metrics to use, you can select either specific objects or object types. For example, you can select the specific objects VM001 and VM002, or you can select the object type virtual machine. |
| Determine the metrics to include in the super metric. | If you are tracking the transfer of packets along a network, use metrics that refer to packets in and packets out. In another common use of super metrics, the metrics might be the average CPU usage or average memory usage of the object type you select. |
| Decide how to combine or compare the metrics. | For example, to find the ratio of packets in to packets out, you must divide the two metrics. If you are tracking CPU usage for an object type, you might want to determine the average use. You might also want to determine what the highest or lowest use is for any object of that type. In more complex scenarios, you might need a formula that uses constants or trigonometric functions. |
| Decide where to assign the super metric. | You define the objects to track in the super metric, then assign the super metric to the object type that contains the objects being tracked. To monitor all the objects in a group, activate the super metric in the policy, and apply the policy to the object group. |
| Determine the policy to which you add the super metric. | After you create the super metric, you add it to a policy. For more information, refer to Policy Workspace in VCF Operations. |

**What Else Can You Do with Super Metrics**

- To see the super metrics in your environment, generate a system audit report. For more information, refer to System Audit for VCF Operations.
- To create alert definitions to notify you of the performance of objects in your environment, define symptoms based on super metrics. For more information, refer to Symptom Definitions in VCF Operations.
- Learn about the use of super metrics in policies. For more information, refer to Policy Workspace in VCF Operations.
- Use OPS CLI commands to import, export, configure, and delete super metrics. For more information, refer to the OPS CLI documentation.
- To display metric-related widgets, create a custom set of metrics. You can configure one or more files that define different sets of metrics for a particular adapter and object types. This ensures that the supported widgets are populated based on the configured metrics and selected object type.

## Create a Super Metric

Create a super metric when you want to check the health of your environment, but you cannot find a suitable metric to perform the analysis.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configuration**, and then click the **Super Metrics** tile.
2. Click **Add** .

   The **Create Super Metric** wizard opens.
3. Optional: To edit a super metric, click the vertical ellipsis next to the super metric and select **Edit**. You can also edit the super metric using the **EDIT** option in super metrics page.
4. Enter a meaningful name for the super metric such as `Worst VM CPU Usage (%)` in the **Name** text box.

   **Note:**
   It is important that you have an intuitive name as it appears in dashboards, alerts, and reports. For meaningful names, always use space between words so that it is easier to read. Use title case for consistency with the out of the box metrics and add the unit at the end.

5. Provide a brief summary of the super metric in the **Description** text box.

   **Note:**
   Information regarding the super metric, like why it was created and by whom can provide clarity and help you track your super metrics with ease.

6. From the **Object Types** drop-down list, select the object to associate with the super metric and click **Next**.
7. Create the formula for the super metric.

   For example, to add a super metric that captures the average CPU usage across all virtual machines in a cluster, perform the following steps.

   a) Select the function or operator. This selection helps combine the metric expression with operators and/or functions. In the super metric editor, enter `avg` and select the **avg** function.

      You can manually enter functions, operators, objects, object types, metrics, metrics types, property, and properties types in the text box and use the suggestive text to complete your super metric formula.

      Alternatively, select the function or operator from the **Functions** drop-down menu.

   b) To create a metric expression, enter `Virtual` and select **Virtual Machine** from the object type list.

   c) Add the metric type, enter `usage`, and select the **CPU|Usage (%)** metric from the metric type list.

      **Note:**
      The expression ends with depth=1 by default. If the expression ends with depth=1, that means that the metric is assigned to an object that is one level above virtual machines in the relationship chain. However, since this super metric is for a cluster which is two levels above virtual machine in the relationship chain, change the depth to 2.

      The depth can also be negative, this happens when you need to aggregate the parents of a child object. For example, when aggregating all the VMs in a datastore, the metric expression ends with depth=-1, because VM is a parent object of datastore. But, if you want to aggregate all the VMs at a Datastore Cluster level, you need to implement 2 super metrics. You cannot directly aggregate from VM to Datastore Cluster, because both are parents of a datastore. For a super metric to be valid, depth cannot be 0 (-1+1=0). Hence, you need to create the first super metric (with depth=-1) for the aggregate at the datastore level, and then build the second super metric based on the first (with depth = 1).

      The metric expression is created.

   d) To calculate the average CPU usage of powered on virtual machines in a cluster, you can add the `where` clause. Enter `where=""`.

      **Note:**
      The `where` clause cannot point to another object, but can point to a different metric in the same object. For example, you cannot count the number of VMs in a cluster with the CPU contention metric > SLA of that cluster. The phrase "SLA of that cluster " belongs to the cluster object, and not to the VM object. The right operand must also be a number and cannot be another super metric or variable.

   e) Position the pointer between the quotation marks, enter **Virtual**, and select the **Virtual Machine** object type and the **System|Powered ON** metric type.

   f) To add the numeric value for the metric, enter `==1`.

   g) To view hints and suggestions, click **ctrl+space** and select the objects, object types, metrics, metrics types, property, and properties types to build your super metric formula.

   h) Select **This** option from the drop-down menu.

      If **This** option is selected during the creation of a metric expression, it means that the metric expression is associated to the object for which the super metric is created.

8. Select the unit of the super metrics from the **Unit** drop-down.

   **Note:**
   The super metrics unit configured here can be changed in the metrics charts, widgets, and views.

9. Click **Validate** to verify that the super metric formula has been created correctly.

   After you click the **Preview** button the system selects a random object and displays a metric graph showing values for the current super metric. For example, if you have selected `Host System` in the **Object Types** tab, after you click the **Preview** button it will randomly select a host system object from the list of the available objects and displays the graph

for the selected host. Alternatively, you can also type in the object name in the **Object** text box, and the result will also depend on the pre-selected object type.

a) Expand the **Preview** section.

A metric graph is displayed showing values of the metric collected for the object. Verify that the graph shows values over time.

b) Click **Next**.

The Policies page is displayed.

10. Select the policy which you want to associate with super metric and click **Update**.

The selected policy is applied to the super metric. You can view the super metric you created, the associated object type, and policy on the **Super Metrics** page.

# Enhancing Your Super Metrics

You can enhance your super metrics by using clauses and resource entry aliasing.

### Where Clause

The `where` clause verifies whether a particular metric value can be used in the super metric. Use this clause to point to a different metric of the same object, such as `where=(${metric=metric_group|my_metric} > 0)`.

For example: `count(${objecttype = ExampleAdapter, adaptertype = ExampleObject, metric = ExampleGroup|Rating, depth=2, where =($value==1})`

### IsFresh Function

Use the `isFresh` function in the `where` clause to check if the last value of the metrics is fresh or not.

For every metric published in VCF Operations, the point with the latest publishing time is called as the last point of that metric. The value of that metric's last point is called the last value of that metric. A metric's last point is considered fresh when the time elapsed after the metric's last point is lesser than the estimated publishing interval of that metric.

The `isFresh` function returns true if the last value of the metrics is fresh. For example, in the following scenarios, the function:

- `${this, metric=a|b, where=($value.isFresh())}`, returns the last value of the metric a|b if the last value is fresh.
- `${this, metric=a|b, where=($value == 7 && $value.isFresh())}`, returns the last value of the metric a|b if it is equal to seven and is fresh.
- `${this, metric=a|b, where=(${metric=c|d} == 7 && ${metric=c|d}.isFresh())}`, returns the last value of the metric a|b only if the last value of the metric c|d is equal to seven and is fresh.

### Resource Entry Aliasing

Resource entries are used to retrieve metric data from VCF Operations for computing super metrics. A resource entry is the part of an expression which begins with `$` followed by a `{..} block`. When computing a super metric, you might have to use the same resource entry multiple times. If you have to change your computation, you must change every resource entry, which might lead to errors. You can use resource entry aliasing to rewrite the expression.

The following example, shows a resource entry that has been used twice.

```
(min(${adaptertype=VMWARE, objecttype=HostSystem, attribute= cpu|demand|
active_longterm_load, depth=5, where=($value>=0)}) + 0.0001)/(max(${adaptertype=VMWARE,
objecttype=HostSystem, attribute=cpu|demand|active_longterm_load, depth=5,
where=($value>=0)}) + 0.0001)"
```

The following example shows how to write the expressing using resource entry aliasing. The output of both expressions is the same.

```
(min(${adaptertype=VMWARE, objecttype=HostSystem, attribute= cpu|demand|
active_longterm_load, depth=5, where=($value>=0)} as cpuload) + 0.0001)/(max(cpuload) +
0.0001)"
```

Follow these guidelines when you use resource entry aliasing:

- When you create an alias, make sure that after the resource entry you write `as` and then `alias:name`. For example: **${…} as alias_name**.
- The alias cannot contain the ()[]+-*/%|&!=<>,.?:$ special characters, and cannot begin with a digit.
- An alias name, like all names in super metric expressions, is case-insensitive.
- Use of an alias name is optional. You can define the alias, and not use it in an expression.
- Each alias name can be used only once. For example: `${resource1,…} as r1 + ${resource2,…} as R1`.
- You can specify multiple aliases for the same resource entry. For example: **${…} as a1 as a2**.

#### Conditional Expression ?: Ternary Operators

You can use a ternary operator in an expression to run conditional expressions.

For example: `expression_condition ? expression_if_true : expression_if_false`.

The result of the conditional expression is converted to a number. If the value is not 0, then the condition is assumed as true.

For example: `-0.7 ? 10 : 20` equals 10. `2 + 2 / 2 - 3 ? 4 + 5 / 6 : 7 + 8` equals 15 (7 + 8).

Depending on the condition, either `expression_if_true` or `expression_if_false` is run, but not both of them. In this way, you can write expressions such as, `${this, metric=cpu|demandmhz} as a != 0 ? 1/a : -1`. A ternary operator can contain other operators in all its expressions, including other ternary operators.

For example: `!1 ? 2 ? 3 : 4 : 5` equals 5.

## Exporting and Importing a Super Metric

You can export a super metric from one VCF Operations instance and import it to another VCF Operations instance. For example, after developing a super metric in a test environment, you can export it from the test environment and import it use in a production environment.

If the super metric to import contains a reference to an object that does not exist in the target instance, the import fails. VCF Operations returns a brief error message and writes detailed information to the log file.

1. Export a super metric.
   a) In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configuration**, and then click the **Super Metrics** tile.
   b) Select the super metric to export, click the horizontal ellipsis and then click **Export**.

      VCF Operations creates a super metric file, for example, `SuperMetric.json`.
   c) Download the super metric file to your computer.
2. Import a super metric.
   a) From the left menu, select **Configure** and then click **Super Metrics**.
   b) Click the horizontal ellipsis and then click **Import**.
   c) (Optional). If the target instance has a super metric with the same name as the super metric you are importing, you can either overwrite the existing super metric or skip the import, which is the default.

## Super Metrics Tab

A super metric is a mathematical formula that contains a combination of one or more metrics for one or more objects. With super metrics you can assess information more quickly when you are observing fewer metrics.

#### Where You Configure Super Metrics

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configuration**, and then click the **Super Metrics** tile.

To view the details for a specific super metric, click the super metric from the list. The super metric details are displayed in the right-side panel. The super metric details include the assigned object types, formula and policies activated for the selected super metric.

**Table 1405: Configuration Options for Super Metrics**

| Option | Description |
|---|---|
| Toolbar | Use the toolbar selections to manage super metric options.<br>• Add New Super Metric. Starts the Create Super Metric workspace.<br>• Edit Selected Super Metric. Starts the Create Super Metric workspace.<br>• Clone Selected Super Metric. Duplicates the super metric. Edit the clone or associate it with a different object type.<br>• Delete Selected Super Metric.<br>• Export Selected Super Metric. Exports a super metric to use in another VCF Operations instance. See Exporting and Importing a Super Metric.<br>• Import Super Metric. Imports a super metric to this VCF Operations instance. See Exporting and Importing a Super Metric. |
| Super Metrics list | Configured super metrics listed by name and formula description. |

### Enhancements to the Super Metric Functions

In the earlier implementation of aggregate functions in super metrics, you had to explicitly specify the Adapter Kind and Resource Kind in the formula.

Old Formula

```
count(${adaptertype=VMWARE, objecttype=HostSystem,attribute=badge|health, depth=1})
```

The new implementation of aggregate function provides a way to define a super-metric without explicitly specifying the Resource Kind. You can use "objecttype=*" in the super-metric formula which indicates you to consider all Resource Kinds having the specified attribute.

New Formula

```
count(${adaptertype=VMWARE, objecttype=*,attribute=badge|health, depth=1})
```

**Note:**

The explicit specification of "adaptertype" remains mandatory. However, "*" can be used only to select all Resource Kinds for the given Adapter Kind.

## Manage Super Metric Workspace

You use the Manage Super Metric workspace to create or edit a super metric. The toolbar helps you to build the mathematical formula with the objects and metrics you select.

### Where You Configure Super Metrics

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configuration**, and then click the **Super Metrics** tile.

**Table 1406: Super Metrics Workspace Options**

| Option | Description |
|---|---|
| Super Metric | • Name. The name you give to the super metric.<br>• Description. The textual description you give about the super metric. |
| Object Types Pane | Use this page to associate the super metric with an object type. You can use this list to select the object type with the metrics to measure. The object type selection affects the list of objects, metrics, and attribute types displayed. |
| Formula | Define the formula you want to associate with the super metric. You can preview and validate the formula before you create it. Use drop-down menu to select the metrics to add to the formula.<br>• Functions. Mathematical functions that operate on a single object or group of objects. See Super Metric Functions and Operators.<br>• Operators. Mathematical symbols to enclose or insert between functions. See Enhancing Your Super Metrics.<br>• This Object. Assigns the super metric to the object selected in the Object pane and displays `this` in the formula instead of a long description for the object.<br>• Unformatted. Slide the toggle button to the right to view the unformatted version of the formula, slide the button to the left to view the formula in textual format.<br>• Preview. Shows the super metric in a graph. Look at the graph so that you can verify that VCF Operations is calculating the super metric for the target objects that you selected. |
| Policies | Displays the policies related to the object types you assigned your super metric to. |

## Super Metric Functions and Operators

VCF Operations includes functions and operators that you can use in super metric formulas. The functions are either looping functions or single functions.

### Looping Functions

Looping functions work on more than one value.

**Table 1407: Looping Functions**

| Function | Description |
|---|---|
| avg | Average of the collected values. |
| combine | Combines all the values of the metrics of the included objects in a single metric timeline. |
| count | Number of values collected. |
| max | Maximum value of the collected values. |
| min | Minimum value of the collected values. |
| sum | Total of the collected values. |

**Note:**
VCF Operations 5.x included two sum functions: `sum (expr)` and `sumN (expr, depth)`. VCF Operations 6.x includes one sum function: `sum (expr)`. Depth is set at depth=1 by default. For more information about setting depth, refer to Create a Super Metric.

## Looping Function Arguments

The looping function returns an attribute or metric value for an object or object type. An attribute is metadata that describes the metric for the adapter to collect from the object. A metric is an instance of an attribute. The argument syntax defines the desired result.

For example, CPU usage is an attribute of a virtual machine object. If a virtual machine has multiple CPUs, the CPU usage for each CPU is a metric instance. If a virtual machine has one CPU, then the function for the attribute or the metric return the same result.

**Table 1408: Looping Function Formats**

| Argument syntax example | Description |
|---|---|
| *funct*(${this, metric =*a\|b:optional_instance\|c*}) | Returns a single data point of a particular metric for the object to which the super metric is assigned. This super metric does not take values from the children or parents of the object. |
| *funct*(${this, attribute=*a\|b:optional_instance\|c*}) | Returns a set of data points for attributes of the object to which the super metric is assigned. This super metric does not take values from the child or parent of the object. |
| *funct*(${adaptertype=*adaptkind*, objecttype=*reskind*, resourcename=*resname*, identifiers={id1=*val1*id2=*val2*,…}, metric=*a\|b:instance\|c*}) | Returns a single data point of a particular metric for the *resname* specified in the argument. This super metric does not take values from the children or parents of the object. |
| *funct*(${adaptertype=*adaptkind*, objecttype=*reskind*, resourcename=*resname*, identifiers={*id1=val1, id2=val2,…*}, attribute=*a\|b:optional_instance\|c*}) | Returns a set of data points. This function iterates attributes of the *resname* specified in the argument. This super metric does not take values from the child or parent of the object. |
| *funct*(${adaptertype=*adaptkind*, objecttype=*reskind*, depth=*dep*}, metric=*a\|b:optional_instance\|c*}) | Returns a set of data points. This function iterates metrics of the *reskind* specified in the argument. This super metric takes values from the child (depth > 0) or parent (depth < 0) objects, where *depth* describes the object location in the relationship chain.<br>For example, a typical relationship chain includes a data center, cluster, host, and virtual machines. The data center is at the top and the virtual machines at the bottom. If the super metric is assigned to the cluster and the function definition includes depth = 2, the super metric takes values from the virtual machines. If the function definition includes depth = -1, the super metric takes values from the data center. |
| *funct*(${adaptertype=*adaptkind*, objecttype=*reskind*, depth=*dep*}, attribute=*a\|b:optional_instance\|c*}) | Returns a set of data points. This function iterates attributes of the *reskind* specified in the argument. This super metric takes values from the child (depth > 0) or parent (depth < 0) objects. |

For example, `avg(${adaptertype=VMWARE, objecttype=VirtualMachine, attribute=cpu|usage_average, depth=1})` averages the value of all metric instances with the cpu|usage_average attribute for all objects of type `VirtualMachine` that the vCenter adapter finds. VCF Operations searches for objects one level below the object type where you assign the super metric.

## Single Functions

Single functions work on only a single value or a single pair of values.

**Table 1409: Single Functions**

| Function | Format | Description |
|---|---|---|
| *abs* | abs(x) | Absolute value of x. x can be any floating point number. |
| *acos* | acos(x) | Arccosine of x. |
| *asin* | asin(x) | Arcsine of x. |
| *atan* | atan(x) | Arctangent of x. |
| *ceil* | ceil(x) | The smallest integer that is greater than or equal to x. |
| *cos* | cos(x) | Cosine of x. |
| *cosh* | cosh(x) | Hyperbolic cosine of x. |
| *exp* | exp(x) | e raised to the power of x. |
| *floor* | floor(x) | The largest integer that is less than or equal to x. |
| *log* | log(x) | Natural logarithm (base *x*) of x. |
| *log10* | log10(x) | Common logarithm (base 10) of x. |
| *pow* | pow(x,y) | Raises x to the y power. |
| *rand* | rand() | Generates a pseudo random floating number greater than or equal to 0.0 and less than 1.0. |
| *sin* | sin(x) | Sine of x. |
| *sinh* | sinh(x) | Hyperbolic sine of x. |
| *sqrt* | sqrt(x) | Square root of x. |
| *tan* | tan(x) | Tangent of x. |
| *tanh* | tanh(x) | Hyperbolic tangent of x. |

## Operators

Operators are mathematical symbols and text to enclose or insert between functions.

**Table 1410: Numeric Operators**

| Operators | Description |
|---|---|
| + | Plus |
| - | Subtract |
| * | Multiply |
| / | Divide |
| % | Modulo |
| == | Equal |
| != | Not equal |
| < | Less than |
| <= | Less than, or equal |
| > | Greater than |
| >= | Greater than, or equal |
\|\| | Or |
&& | And |
! | Not |

| Operators | Description |
|---|---|
| ? : | Ternary operator. If/then/else<br>For example: `conditional_expression ? expression_if_condition_is_true : expression_if_condition_is_false`<br>For more information about ternary operators, see Enhancing Your Super Metrics. |
| ( ) | Parentheses |
| [ ] | Use in an array of expressions |
| [x, y, z] | An array containing x, y, z. For example, min([x, y, z]) |

**Table 1411: String Operators**

| String Operators | Description |
|---|---|
| equals | Returns true if metric/property string value is equal to specified string. |
| contains | Returns true if metric/property string value contains specified string. |
| startsWith | Returns true if metric/property string value starts with the specified prefix. |
| endsWith | Returns true if metric/property string value ends with the specified suffix. |
| !equals | Returns true if metric/property string value is not equal to specified string. |
| !contains | Returns true if metric/property string value does not contain specified string. |
| !startsWith | Returns true if metric/property string value does not start with the specified prefix. |
| !endsWith | Returns true if metric/property string value does not end with the specified suffix. |

**Note:**
String operators are valid in 'where' condition only. For example: `${this, metric=summary|runtime|isIdle, where = "System Properties|resource_kind_type !contains GENERAL"}`

# Configuring Administration Settings

After VCF Operations is installed and configured, you can use administration settings to manage your environment. You find most administration settings under the Administration selection of the VCF Operations user interface.

## Modifying Global Settings

The global settings control the system settings for VCF Operations, including data retention and system timeout settings. You set times to delete objects, set timeouts, store historical data, use dynamic threshold and capacity calculations, and determine how vCenter users log in. For automated actions, you can select whether to allow actions to be triggered from alert recommendations automatically. You can modify one or more of the settings to monitor your environment better. These settings affect all your users.

The global settings do not affect metric interactions, color indicators, or other object management behaviors. These behaviors are configured in your policies.

Settings related to managing objects with VCF Operations are available on the **Inventory** page.

### Access Global Settings

From the left menu, click **Administration** > **Global Settings**. The global settings are categorised into Data Retention, Cost/Price, User Access, System Settings, and Network Settings. Select one of the categories to edit the global settings. If you cannot find a settings, type the name of the setting in the search bar. When you edit a setting, the **Save** and **Cancel** buttons appear. Save the setting after your selection to complete the editing process.

### Global Settings Best Practices

Most of the settings pertain to how long VCF Operations retains the collected and processed data. The default values are common retention periods. You might need to adjust the time periods based on your local policies or disk space.

## List of Global Settings

The global settings determine how VCF Operations retains data, keeps connection sessions open, and other settings. These are system settings that affect all users. Some of these settings are not editable. The global settings are grouped into four types.

### Settings in the Data Retention Category

**Table 1412: Global Setting Default Values and Descriptions**

| Setting | Default Value | Description |
|---|---|---|
| Action History | 14 days | Number of days to retain the historical record of remediation actions. The data is purged from the system after the specified number of days. |
| Deleted Objects | 168 hours | This setting determines the number of hours for which objects in the inventory must be retained when:<br>• The objects are deleted from an adapter data source or server before deleting them from VCF Operations<br>• A cloud account is deleted from VCF Operations but the check box to delete related objects is not selected in the **Delete All Accounts** dialog box.<br>An object deleted from an adapter data source is identified by VCF Operations as not existing and VCF Operations can no longer collect data about the object. Whether VCF Operations identifies deleted objects as not existing depends on the adapter. This feature is not implemented in some adapters.<br>For example, if the retention time is 360 hours and a virtual machine is deleted from a vCenter instance, the virtual machine remains as an object in VCF Operations for 15 days before it is deleted.<br>This setting applies to objects deleted from the data source or server, not to any objects you delete from VCF Operations on the Inventory page.<br>You can define the number of hours per object type to retain objects that no longer exist and check for object type overrides. To add individual object types and set up their values, click the **Object Retention Period** icon. You can also edit or delete these object types. |
| Deletion Scheduling Interval | 24 hours | Determines the frequency to schedule deletion of resources. This setting works with the Deleted Objects setting to remove objects that no longer exist in the environment. VCF Operations transparently marks objects for removal that have not existed for the length of time specified under Deleted Objects. VCF Operations then removes the marked objects at the frequency specified under Deletion Scheduling Interval. |

| Setting | Default Value | Description |
|---|---|---|
| Object History | 90 days | Number of days to retain the history of the object configuration, relationship, and property data.<br><br>**Note:**<br><br>The retention decision for each property value is based on the end date of that value instead of the start date. So any property value with an end date within the last 90 days interval (default setting) is retained. For example, if property value A was first published 1 year ago (which is long before the object history retention interval of 90 days) and property value B was published 3 days ago, and if no more points for that property exists, then the end date for the value A will be 3 days ago and for value B now. So both end dates will be within the recent 90 day limit and both values will be retained.<br><br>The configuration data is the collected data from the monitored objects on which the metrics are based. The collected data includes changes to the configuration of the object.<br>The data is purged from the system after the specified number of days. |
| Generated Reports Retention | Deactivated | Number of days to retain generated reports. If deactivated, all the generated reports will be retained.<br>The minimum number of days you can set is one and the maximum number of days you can set is 3600. |
| Generated Tenant Report Retention | Deactivated | Number of days to retain generated tenant reports.<br>The minimum number of days you can set is 30 and the maximum number of days you can set is 90. If deactivated, the generated tenant reports will be retained as per the setting in the **Generated Reports Retention** field.<br>This option is visible only if you have deployed VCF Automation using the VCF Installer Deployment Wizard. |
| Symptoms/Alerts | 45 days | Number of days to retain canceled alerts and symptoms.<br>The alerts and symptoms are either canceled by the system or by a user. |
| Time Series Data Retention | 6 months | Number of months that you want to retain the collected and calculated metric data for the monitored objects. This setting is set to 6 months by default for 5-minutes interval data retention. |
| Additional Time Series Retention | 36 months | The number of months that the roll-up data extends beyond the regular period. The roll-up data is available starting from the end of the regular period and until the end of the roll-up data retention period. If you specify 0 as the value, then this will effectively deactivate the Additional Time Series Data Retention time and only data specified in Time Series Retention is stored. This setting ensures that after 6 months of normal retention for 5 minutes, the seventh month data is rolled up into a one hour roll up. You can set up this option up to 120 months for data roll ups. |
| Deleted Users | 100 days | You can specify the number of days to keep custom content created by a user who has been removed from VCF Operations or by the automatic synchronization of LDAP. For example, the custom dashboards created by a user. |
| Configuration management drift history Data retention | 7 days | The number of days to retain historical data for the configuration drift report. |

| Setting | Default Value | Description |
|---|---|---|
| External Event Based Active Symptoms | Deactivated | The number of days to retain the external event-based active symptoms.<br>**Note:**<br>Before modifying the retention period, ensure that the potential impact on existing alerting workflows has been thoroughly evaluated. If you activate **External Event Based Active Symptoms**, it automatically remediates the associated diagnostic findings and cancels any alerts that rely on those symptoms. This applies to symptoms that are active and unresolved during the configured retention period. |
| Generated Bills | 1440 days | Enter the number of days for which you want to retain the generated bills.<br>You can retain the generated bills to a maximum of 1440 days.<br>You must deploy VCF Automation to view this setting. |
| Real-time metrics Data Retention | 15 days | Duration for which the real-time metrics are retained. |

## Settings in the Cost/Price Category

### Table 1413: Global Setting Default Values and Descriptions

| Setting | Default Value | Description |
|---|---|---|
| Currency | | You can specify the currency unit that is used for cost calculation. Click **Set Currency** and select the currency from the list of currency types. Click the check box to confirm your selection and then click **Set Currency**.<br>You can edit the currency to update the currency for cost calculation.<br>**Note:**<br>Editing the currency has the following effects:<br>• Views, dashboards, and reports display data with the old and new currency and there will be discrepancies if you compare data across a timeline that includes currency change.<br>• Billing schedules get updated if the currency is changed between billing cycles. New scheduled bills are generated with the new currency and the currency change date as the start date. You must manually generate a bill for the timeline before that for the older currency.<br>In Tenant Bills, if the provider generates the bill during the currency change period, tenants will see multiple bills.<br>• The cost and price metrics collected after the update display the new currency value. Older metrics that are already published continue to display the old currency value.<br>• The cost analysis report only displays data with the new currency if the analysis includes a time range with a currency change.<br>To view the currency change timeline, you can view the **Currency Code** property under the Universe object type. |
| Show Cost Savings | Activated | Controls whether Cost savings is displayed in all the pages under 'Optimize'. |
| Cost Calculation | Activated 9:00 PM | Host time to calculate the costs for objects. |

| Setting | Default Value | Description |
|---|---|---|
| Cluster Utilization Ceiling Factor | 5 | Ceiling for Expected Utilization when running on Actual Utilization. |
| Tag Based Costing Metrics | Deactivated | When activated, VCF Operations calculates additional cost metrics per tag. |

## Settings in the User Access Category

**Table 1414: Global Setting Default Values and Descriptions**

| Setting | Default Value | Description |
|---|---|---|
| Allow vCenter users to log in to VCF Operations | Activated | Allows vCenter users to log in to VCF Operations.<br>**Note:**<br>Disabling the VCF Operations extension in vCenter removes its authorization to connect to VCF Operations. Consequently, vCenter users cannot access the appropriate VCF Operations pages. |
| Password Policy | Activated | Allows users to manage user passwords and determine the criteria used for account lockout, password strength, and the password change policy in VCF Operations. You can edit the following settings:<br>• Account Lockout: Indicates whether the account lockout is in effect, and indicates the number of login attempts allowed before the account is locked. The account lockout policy is activated by default.<br>• Password Strength: Indicates whether the policy that requires users to strengthen their password is in effect, and the minimum number of characters required to make a strong password. The password strength policy is activated by default.<br>• Password Change: Indicates how often the password must be changed, and whether users will receive a warning. The account password change policy is activated by default.<br>For more information about VCF Fleet level password policy, see Access Control: Password Policy . |
| Concurrent UI login sessions | Activated | Allows concurrent UI login sessions per user. Once changed, this setting affects the subsequent login sessions. |
| Allow non-imported vIDM user access | Activated | Allows non-imported VMware Identity Manager users to be created automatically as read-only users upon first access. If deactivated, only VMware Identity Manager imported users or users belonging to imported VMware Identity Manager groups will be granted access. |
| Login Message | Deactivated | Allows the user to display the login message when the user launches the login page in VCF Operations. You can set a message that requires explicit consent, for example, terms and conditions or a Standard Mandatory DoD Notice.<br>You can enter the **Title** and content you want to display and enter a button label for users to click to provide their consent. You can add text and images that are copied from an external source and edit it using the formatting options available. You can preview the message before you save it. |

## Settings in the Capacity Optimization Category

Settings in the capacity optimization category display information about powered-off VMs, idle VMs, snapshots, and orphaned disks.

**Table 1415: Global Settings Default Values and Descriptions**

| Setting | Default Value | Description |
|---|---|---|
| Include Powered off VMs for Reclamation | Activated 7 Days | VMs that have been continuously powered off during the defined time period.<br>The total storage capacity used is reclaimable. Total storage reclaimable cost is computed by multiplying storage rate with storage utilization. The direct cost of VM is also attributed. |
| Idle VM Management | | |
| Include Idle VMs for Reclamation | Activated | You can configure the following parameters based on which VCF Operations calculates idle VMs:<br>• **Minimum number of days that the VM has been idle for**: The total number of days the VM has been idle to be classified as a reclaimable idle VM.<br>• **Exclude VM provisioned in last**: The number of days before which the provisioned VMs are not considered in the classification of reclaimable idle. |
| VM Idleness Criteria | 100 MHz of CPU for 100% of Time | Defines the criteria to identify VMs that remain idle during the defined time period.<br>You can set the following values:<br>• MHz of CPU consumption of VMs, and percentage of the time VMs have less than defined MHz of CPU within each day. |
| Include Snapshots for Reclamation | Activated | VM snapshots is older than the entire defined time period.<br>Snapshots of a VM use storage space and such storage is reclaimable. The reclaimable cost is computed by multiplying the storage rate with the reclaimable storage value. |
| Orphaned Disk Management | | |
| Include Orphaned Disks for Reclamation | Activated | Orphaned disks are VMDKs that are associated with a VM which are not in inventory, but still available in a data store. You can configure the minimum number of days for which VMDKs not related to any existing VM will be reported as orphaned and appear under the Orphaned Disks in the Reclaim page. |
| Orphaned Disks Collection Time | 8:00 pm | Host time to collect orphaned disks.<br>**Note:**<br>VCF Operations checks for orphaned VMDKs in the vCenter instances based on the set time. The **Orphaned Disks Collection** and the **Cost Calculation** settings are interrelated. It is recommended to schedule orphaned disks collection before cost calculation so that the cost is calculated based on the recent list of reclaimable orphaned VMDKs. The default value for Orphaned Disks Collection is set to 8:00 PM, and the default value for Cost Calculation is set to 9:00 PM.<br>You can navigate to the **Cost/Price** section under **Administ ration** > **Global Settings** and change the value of the **Cost Calculation** time. |
| Exclude Broadcom appliances from Optimization | Activated | This will exclude appliance VMs from Powered off VM, Idle VM, Oversized VM and Undersized VM. However appliance VMs will be still available in the list of Snapshot VM list |
| Exclusion settings based on time | 7 Days | Minimum number of days of data collection required before VMs can be included for rightsizing optimization |

## Settings in the System Settings Category

### Table 1416: Global Setting Default Values and Descriptions

| Setting | Default Value | Description |
|---|---|---|
| SSL Syslog | Activated | Enforce SSL for syslog ingestion. Note that SSL is supported for TCP only. |
| SSL API Ingestion | Activated | Enforce SSL for API ingestion in Log Management. |
| Enforced Authenticated Log Ingestion | Deactivated | Enforce authentication for HTTP-based log ingestion (for example Logs Agent, fluentbit, fluentd). For more information, see Agent Authentication for Log Ingestion. |
| Log Event 'Type & Trends' | Activated | Enable detailed event processing for Event Types and Event Trends. Disabling this can improve performance but these views will not display data. |
| Session Inactivity Timeout | 30 minutes | If your connection to VCF Operations is idle for the specified amount of time, you are logged out of the application.<br>You must provide credentials to log back in. |
| Dynamic Threshold Calculation | Activated | Determines when to calculate normal levels of threshold violation for all objects.<br>If the setting is deactivated, the following area of VCF Operations does not work or are not displayed:<br>• Alert symptom definitions based on dynamic thresholds will not work<br>• Metric charts that display normal behavior are not present<br>Deactivate this setting only if you have no alternative options for managing resource constraints for your VCF Operations system. |
| Compliance Assessment Schedule | 02:00 AM | Set the compliance assessment start time for all enabled compliance benchmarks. VCF Operations starts the evaluation based on the benchmark at this time everyday. |
| In Page Feedback | Activated | When activated, you can send feedback to Broadcom by using the feedback button and VCF Operations can show you optional surveys. This setting requires internet access. |
| Broadcom's Customer Experience Improvement Program ("CEIP") | Activated | Determines whether to participate in the Customer Experience Improvement Program by having VCF Operations send anonymous usage data to https://vmware.com.<br>**Note:**<br>CEIP is activated by default when you install a new VCF Operations instance or when you upgrade VCF Operations.<br>Deactivate this setting only if you want to opt out of CEIP.<br>**Note:**<br>If you deactivate this setting in version 8.14 of VCF Operations, it stays deactivated for future releases. |

| Setting | Default Value | Description |
|---|---|---|
| System access URL | | You can specify the URL that is used to access the system when a load balancer is used. The URL that you enter here is displayed in the outbound notifications and while sharing dashboards.<br>The IP /FQDN of the URL is used to register vCenter and VMware Identity Manager when you configure the vCenter cloud account and VMware Identity Manager accordingly.<br>When you use VMware Identity Manager with a load balancer for VCF Operations, the System access URL is used during the source registration authentication.<br>When you use identity broker with a load balancer for VCF Operations, the System access URL (load balancer IP/FQDN) is used as the redirect destination after authentication. If this URL is configured incorrectly, users will not be redirected back to the load balancer after successful authentication. |
| Docker image default registry URL | | You can specify the URL that is used to pull the docker images for containerized adapters. This URL can be overridden by installed management packs.<br>You can bring your own content using the containerized adapters. For more information, contact `ops-integration-sdk-beta@vmware.com` . |
| Automated Actions | Activated | Determines whether to allow VCF Operations to automate actions. When an alert triggered, the alert provides recommendations for remediation. You can automate an action to remediate an alert when the recommendation is the first priority for that alert. You activate actionable alerts in your policies. |
| Credential ownership enforcement | Activated | This setting enforces credential ownership and allows users to only access credentials they own.<br>Deactivate this option to modify credentials owned by other users. Once deactivated, you can view, or edit credentials created by other users, or delete credentials that are not in use.<br>**Note:**<br>After deactivating, if a user edits a credential owned by another user and then reactivates this option, the ownership of the credential transfers to the user who edited it last.<br>For more information on credentials, see Configuring Credentials in Integrations and Adding Credentials. |
| Activate Standard Certificate Validation | Deactivated | This option activates certificate verification to Test Connection in the Create or Modify AI screen, using a standard verification flow.<br>The option checks CA authority.<br>• Certificate Subject DN<br>• Subject alternative name<br>• Certificate validity period<br>• Revocation list<br>This option also presents dialogs to user if one of those checks fail. It is up to the adapter implementation on how the adapter checks source certificate validity during a normal collection cycle. On a usual scenario, adapters just perform a thumb-print verification. However, in case this flag is activated, Test connection validates certificates in full scale and accepts certificates that are matching all criteria without any user dialogs. |

| Setting | Default Value | Description |
|---|---|---|
| Threshold For Adapter and Outbound Plugin Certificate Expiration Alert | 5 for critical<br>14 for immediate<br>30 for warning | Set the number of days before which the system must raise certification expiration alerts.<br>Alerts related to adapter and outbound plugin certificate expiration are evaluated daily. To change the values of Critical, Immediate, and Warning alerts, click the corresponding icon and move them along the slider. Alternatively, set the values manually.<br>Click **Save** after you make your changes.<br>**Note:**<br>For critical alerts, an alert banner is displayed under **Home** > **Overview** on the day of certification expiry. For example, if the critical alert is set to 10, then on the tenth day, an alert banner is displayed in the Home page. |

### Settings in the Network Settings Category

**Table 1417: Global Setting Default Values and Descriptions**

| Setting | Default Value | Description |
|---|---|---|
| HTTP Proxy | Not Configured | As an admin, you can configure VCF Operations with a network proxy.<br>To configure a HTTP proxy, ensure that you have the IP or hostname, port, and the credentials of your proxy.<br>After you enter the details, test the connection, and save the configuration.<br>**Note:**<br>SSL termination proxy is not supported in VCF Operations. |

## Access Control: Password Policy

You can modify the user passwords and determine the criteria used for account lockout, password strength, and the password change policy in VCF Operations.

### Modify the Password Policy Settings

You can modify the existing password policy settings in VCF Operations. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Global Settings**, and then click **User Access**. In the **User Access** section, navigate to **Password Policy**.

As a best practice, manage the password settings at the fleet level to ensure compliance across all VCF components.

**Table 1418: Access Control Edit Password Policy Settings**

| Option | Description |
|---|---|
| Account Lockout | Modify the settings to lock user accounts.<br>• **Activate Account Lockout Policy**: Activate the policy to lock user accounts. For a super administrator user, the account lockout policy is activated by default and cannot be deactivated. The super administrator user account is locked for approximately one hour, and then unlocked.<br>• **Number of failed login attempts before lockout**: Indicates the number of consecutive failed logins that a user can attempt to log in to VCF Operations before their account is locked. The default number of tries is three.<br>• **Login fail eviction time**: Indicates the time available for the user to complete the login attempts. You can set fail evication time in seconds. |
| Password Strength | Modify the settings required for users to create strong passwords.<br>• **Activate Password Strength Policy**: When selected, activates the policy to require users to strengthen their password.<br>• **Minimum password length**: Indicates the number of characters required for user passwords. The default length is eight characters.<br>• **Passwords must contain numeric**: When selected, users must include a combination of numbers and letters in their passwords.<br>• **Passwords must not match user names**: To ensure security, users are not allowed to use their user name as their password.<br>• **Passwords must contain at least one uppercase and one lowercase letter**: When selected, users must include one or more uppercase and lowercase characters.<br>• **Passwords must contain special characters**: When selected, users must include one or more special characters. Special characters include: !@#$%^&*+= |
| Password Life | Modify the settings required for users to change their password.<br>• **Activate Password Change Policy**: Activate the policy to require users to change their password at specific intervals.<br>• **Passwords expire every 90 days**: Users receive notification five days before the password expires.<br>• **Warn users 5 days prior to expiration**: Indicate when to have VCF Operations notify users that their password will expire. The default is five days before their password expires. |

# Managing Users and Access Control in VCF Operations

As a system administrator you can manage all aspects of user access control. You create user accounts, assign each user to be a member of one or more user groups, and assign roles to each user or user group to set their privileges.

Users must have privileges to access specific features in the VCF Operations user interface and API. Access control is defined by assigning privileges to both users and objects. You can assign one or more roles to users, and allow them to perform a range of different actions on the same types of objects. For example, you can assign a user with the privileges to delete a virtual machine, and assign the same user with read-only privileges for another virtual machine.

### User Access Control

You can authenticate users in VCF Operations in several ways.

- Create local user accounts in VCF Operations.
- Use VMware vCenter users. After the vCenter is registered with VCF Operations, configure the vCenter user options in the VCF Operations global settings to activate a vCenter user to log in to VCF Operations. When logged into VCF Operations, vCenter users access objects according to their vCenter-assigned permissions.
- Add an authentication source to authenticate imported users and user group information from an external identity provider.t

- Use LDAP to import users or user groups from an LDAP server. LDAP users can use their LDAP credentials to log in to VCF Operations.
- Create a single sign-on source and import users and user groups from a single sign-on server. Single sign-on users can use their single sign-on credentials to log in to VCF Operations and vCenter. You can also use Active Directory through single sign-on by configuring the Active Directory through single sign-on and adding the single sign-on source to VCF Operations.

## Users of VCF Operations

Each user has an account to authenticate them when they log in to VCF Operations.

The accounts of local users and LDAP users are visible in the VCF Operations user interface when they are set up. The accounts of vCenter and single sign-on users only appear in the user interface after a user logs in for the first time. Each user can be assigned one or more roles, and can be an authenticated member of one or more user groups.

### Local Users in VCF Operations

When you create user accounts in a VCF Operations instance, VCF Operations stores the credentials for those accounts in its global database, and authenticates the account user locally.

Each user account must have a unique identity, and can include any associated user preferences.

If you are logging in to VCF Operations as a local user, and on occasion receive an `invalid password` message, try the following workaround. In the Login page, change the Authentication Source to **All vCenter**, change it back to **Local Users**, and log in again.

### vCenter Users in VCF Operations

VCF Operations supports vCenter users. To log in to VCF Operations, vCenter users must be valid users in vCenter.

#### Roles and Associations

A vCenter user must have either the vCenter Admin role or one of the VCF Operations privileges, such as PowerUser which assigned at the root level in vCenter, to log in to VCF Operations. VCF Operations uses only the vCenter privileges, meaning the VCF Operations roles, at the root level, and applies them to all the objects to which the user has access. After logging in, vCenter users can view all the objects in VCF Operations that they can already view in vCenter.

#### Logging in to vCenter Instances and Accessing Objects

vCenter users can access either a single vCenter instance or multiple vCenter instances, depending on the authentication source they select when they log in to VCF Operations.

- If users select a single vCenter instance as the authentication source, they have permission to access the objects in that vCenter instance. After the user has logged in, an account is created in VCF Operations with the specific vCenter instance serving as the authentication source.
- If users select **All vCenter Servers** as the authentication source, and they have identical credentials for each vCenter in the environment, they see all the objects in all the vCenter instances. Only users that have been authenticated by all the vCenter Servers in the environment can log in. After a user has logged in, an account is created in VCF Operations with all vCenter instances serving as the authentication source.

VCF Operations does not support linked vCenter instances. Instead, you must configure the vCenter adapter for each vCenter instance, and register each vCenter instance to VCF Operations.

Only objects from a specific vCenter instance appear in VCF Operations. If a vCenter instance has other linked vCenter instances, the data does not appear.

#### vCenter Roles and Privileges

You cannot view or edit vCenter roles or privileges in VCF Operations. VCF Operations sends roles as privileges to vCenter as part of the vCenter Global privilege group. A vCenter administrator must assign VCF Operations roles to users in vCenter.

VCF Operations privileges in vCenter have the role appended to the name. For example, VCF Operations ContentAdmin Role, or VCF Operations PowerUser Role.

#### Read-Only Principal

A vCenter user is a read-only principal in VCF Operations, which means that you cannot change the role, group, or objects associated with the role in VCF Operations. Instead, you must change them in the vCenter instance. The role applied to the root folder applies to all the objects in vCenter to which a user has privileges. VCF Operations does not apply individual roles on objects. For example, if a user has the PowerUser role to access the vCenter root folder, but has read-only access to a virtual machine, VCF Operations applies the PowerUser role to the user to access the virtual machine.

#### Refreshing Permissions

When you change permissions for a vCenter user in vCenter, the user must log out and log back in to VCF Operations to refresh the permissions and view the updated results in VCF Operations. Alternatively, the user can wait for VCF Operations to refresh. The permissions refresh at fixed intervals, as defined in the `$ALIVE_BASE/user/conf/auth.properties` file. The default refreshing interval is half an hour. If necessary, you can change this interval for all nodes in the cluster.

#### Single Sign-On and vCenter Users

When vCenter users log into VCF Operations by way of single sign-on, they are registered on the VCF Operations User Accounts page. If you delete the account of a vCenter user that has logged into VCF Operations by way of single sign-on, or remove the user from a single sign-on group, the user account entry still appears on the User Account page and you must delete it manually.

#### Generating Reports

vCenter users cannot create or schedule reports in VCF Operations.

### Backward Compatibility for vCenter Users in VCF Operations

VCF Operations provides backward compatibility for users of the earlier version of VCF Operations, so that users of vCenter who have privileges in the earlier version in vCenter can log in to VCF Operations.

When you register VCF Operations in vCenter, certain roles become available in vCenter.

- The Administrator account in the previous version of VCF Operations maps to the PowerUser role.
- The Operator account in the previous version of VCF Operations maps to the ReadOnly role.

During registration, all roles in VCF Operations, except for VCF Operations Administrator, Maintenance, and Migration, become available dynamically in vCenter. Administrators in vCenter have all of the roles in VCF Operations that map during registration, but these administrator accounts only receive a specific role on the root folder in vCenter if it is specially assigned.

Registration of VCF Operations with vCenter is optional. If users choose not to register VCF Operations with vCenter, a vCenter administrator can still use their user name and password to log in to VCF Operations, but these users cannot use the vCenter session ID to log in. In this case, typical vCenter users must have one or more VCF Operations roles to log in to VCF Operations.

When multiple instances of vCenter are added to VCF Operations, user credentials become valid for all of the vCenter instances. When a user logs in to VCF Operations, if the user selects all vCenter options during login, requires that the user's credentials are valid for all of the vCenter instances. If a user account is only valid for a single vCenter instance, that user can select the vCenter instance from the login drop-down menu to log in to VCF Operations.

vCenter users who log in to VCF Operations must have one or more of the following roles in vCenter:

- VCF Operations Content Admin Role
- VCF Operationss General User Role 1
- VCF Operations General User Role 2
- VCF Operations General User Role 3
- VCF Operations General User Role 4
- VCF Operations Power User Role
- VCF Operations Power User without Remediation Actions Role
- VCF Operations Read Only Role

For more information about vCenter users, groups, and roles, see the vCenter documentation.

### External User Sources in VCF Operations

You can import user accounts from external authentication sources so that you can use them in your VCF Operations instance.

There are a few types of external user identity sources:

- Lightweight Directory Access Protocol (LDAP): Use the LDAP source if you want to use the Active Directory or LDAP servers as authentication sources. The LDAP source does not support multi-domains even when there is a two-way trust between Domain A and Domain B.
- Single Sign-On (SSO): Use a single sign-on source to perform single sign-on with any application that supports vCenter single sign-on, including VCF Operations.

## Roles and Privileges in VCF Operations

VCF Operations provides several predefined roles to assign privileges to users. You can also create your own roles.

You must have privileges to access specific features in the VCF Operations user interface. The roles associated with your user account determine the features you can access and the actions you can perform.

Each predefined role includes a set of privileges for users to perform, create, read, update, or delete actions on components such as dashboards, reports, administration, capacity, policies, problems, symptoms, alerts, user account management, and adapters.

| | |
|---|---|
| **Administrator** | Includes privileges to all features, objects, and actions in VCF Operations . |
| **PowerUser** | Users have privileges to perform the actions of the Administrator role except for privileges to user management and cluster management. VCF Operations maps vCenter administrators to this role. |
| **PowerUserMinusRemediation** | Users have privileges to perform the actions of the Administrator role except for privileges to user management, cluster management, and remediation actions. |
| **ContentAdmin** | Users can manage all content, including views, reports, dashboards, and custom groups in VCF Operations . |
| **GeneralUser-1 through GeneralUser-4** | These predefined template roles are initially defined as ReadOnly roles. vCenter administrators can configure these roles to create combinations of roles to give users multiple types of privileges. Roles are synchronized to vCenter once during registration. |
| **ReadOnly** | Users have read-only access and can perform read operations, but cannot perform write actions such as create, update, or delete. |

## User Scenario: Manage User Access Control

As a system administrator or virtual infrastructure administrator, you manage user access control in VCF Operations so that you can ensure the security of your objects. Your company just hired a new person, and you must create a user account and assign a role to the account so that the new user has permission to access specific content and objects in VCF Operations.
Verify that the following conditions are met:

- VCF Operations is installed and operating properly, and contains objects such as clusters, hosts, and virtual machines.

- One or more user groups are defined.
- An LDAP authentication source is configured.

In this scenario you will learn how to create user accounts and roles, and assign roles to the user accounts to specify access privileges to views and objects. You will then demonstrate the intended behavior of the permissions on these accounts.

You will create a new user account, named Tom User, and a new role that grants administrative access to objects in the VCF Operations clusters. You will apply the new role to the user account.

Finally, you will import a user account from an external LDAP user database that resides on another machine to VCF Operations, and assign a role to the imported user account to configure the user's privileges.

Create a new role.

### Create a New Role

You use roles to manage access control for user accounts in VCF Operations.

Verify that you understand the context of this scenario. See User Scenario: Manage User Access Control.

In this procedure, you will add a new role and assign administrative permissions to the role.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Access Control** tile.
2. Click the **Roles** tab.
3. Click the **Add** icon on the toolbar to create a role.
   The **Create Role** dialog box appears.
4. For the role name, type `admin_cluster`, then type a description and click **SAVE**.
   The admin_cluster role appears in the list of roles.
5. Click the **admin_cluster** role.
6. In the Details grid below, on the Permissions pane, click the **Edit** icon.
7. Select the **Select all permissions** check box.
8. Click **Update**.
   This action gives this role administrative access to all the features in the environment.

Create a user account, and assign this role to the account.

### Create a User Account

As an administrator you assign a unique user account to each user so that they can use VCF Operations. While you set up the user account, you assign the privileges that determine what activities the user can perform in the environment, and upon what objects.

Create a new role. See Create a New Role.

In this procedure, you will create a user account, assign the `admin_cluster` role to the account, and associate the objects that the user can access while assigned this role. You will assign access to objects in the VCF Operations cluster. Then, you will test the user account to confirm that the user can access only the specified objects.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Access Control** tile.
2. Click the **User Accounts** tab.
3. Click the **Add** icon to create a new user account, and provide the information for this account.

| Option | Description |
|---|---|
| **User Name** | Type the user name to use to log in to VCF Operations. |
| **Password** | Type a password for the user. |

| Option | Description |
|---|---|
| **Confirm Password** | Type the password again to confirm it. |
| **First Name** | Type the user's first name. For this scenario, type `Tom`. |
| **Last Name** | Type the user's last name. For this scenario, type `User`. |
| **Email Address** | (Optional). Type the user's email address. |
| **Description** | (Optional). Type a description for this user. |
| **Deactivate this user** | Do not select this check box, because you want the user to be active for this scenario. |
| **Require password change at next login** | Do not select this check box, because you do not need to change the user's password for this scenario. |

4. Click the **Objects** tab.
5. Select the **admin_cluster** role from the drop-down menu.
6. In the Object Hierarchies list, select the **VCF Operations Cluster** check box.
7. Click **Finish**.

   You created a new user account for a user who can access all the **VCF Operations Cluster** objects. The new user now appears in the list of user accounts.
8. Log out of VCF Operations.
9. Log in to VCF Operations as Tom User, and verify that this user account can access all the objects in the VCF Operations cluster hierarchy, but not other objects in the environment.
10. Log out of VCF Operations.

You used a specific role to assign permission to access all objects in the VCF Operations Cluster to a user account named Tom User.

Import a user account from an external LDAP and assign permissions to the user account.

## Import a User Account and Assign Permissions

You can import user accounts from external sources, such as an LDAP database on another machine, or a single sign-on server, so that you can give permission to those users to access certain features and objects in VCF Operations.

- Configure an authorization source. See Authentication Sources.

1. Log out of VCF Operations, then log in as a system administrator.
2. From the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Access Control** tile.
3. On the toolbar, click the horizontal ellipsis and then click **Import from Source** icon.
4. Specify the options to import user accounts from an authorization source.
   a) On the Import Users page, from the **Import From** drop-down menu, select an authentication source.
   b) In the **Domain Name** drop-down menu, type the domain name from which you want to import users, and click **Search**.
   c) Select the users you want to import, and click **Next**.
   d) On the **Groups** tab, select the user group to which you want to add this user account.
   e) Click the **Objects** tab, select the **admin_cluster** role, and select the **Assign this role to the user** check box.
   f) In the Object Hierarchies list, select the **VCF Operations Cluster** check box, and click **Finish**.
5. Log out of VCF Operations.
6. Log in to VCF Operations as the imported user.
7. Verify that the imported user can access only the objects in the VCF Operations Cluster.

You imported a user account from an external user database or server to VCF Operations, and assigned a role and the objects the user can access while holding this role to the user.

You have finished this scenario.

## Configure a Single Sign-On Source in VCF Operations

As a system administrator or virtual infrastructure administrator, you use single sign-on to activate SSO users to log in securely to your VCF Operations environment.

- Verify that the server system time of the single sign-on source and VCF Operations are synchronized. If you need to configure the Network Time Protocol (NTP), see Configuring the NTP Settings for VMware vSphere Foundation Components.
- Verify that you have access to a Platform Services Controller through the vCenter. See the VMware vSphere Information Center for more details.

After the single sign-on source is configured, users are redirected to an SSO identity source for authentication. When logged in, users can access other vSphere components such as the vCenter without having to log in again.

1. Log in to VCF Operations as an administrator.
2. From the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Authentication Sources** tile.
3. Click **Add**.
4. In the Add Source for User and Group Import dialog box, provide information for the single sign-on source.

| Option | Action |
|---|---|
| **Source Display Name** | Type a name for the import source. |
| **Source Type** | Verify that SSO SAML is displayed. |
| **Host** | Enter the IP address or FQDN of the host machine where the single sign-on server resides. If you enter the FQDN of the host machine, verify that every node in the VCF Operations cluster can resolve the single sign-on host FQDN. |
| **Port** | Set the port to the single sign-on server listening port. By default, the port is set to 443. |
| **User Name** | Enter the user name that can log into the SSO server. |
| **Password** | Enter the password. |
| **Grant administrator role to VCF Operations for future configuration?** | Select **Yes** so that the SSO source is reregistered automatically if you make changes to the VCF Operations setup. If you select |

| Option | Action |
|---|---|
| | **No**, and the VCF Operations setup is changed, single sign-on users will not be able to log in until you manually re-register the single sign-on source. |
| **Automatically redirect to vRealize Operations single sign-on URL?** | Select **Yes** to direct users to the vCenter single-sign on log in page. If you select **No**, users are not redirected to SSO for authentication. |
| **Import single sign-on user groups after adding the current source?** | Select **Yes** so that the wizard directs you to the Import User Groups page when you have completed the SSO source setup. If you want to import user accounts, or user groups at a later stage, select **No**. |
| **Advanced options** | If your environment uses a load balancer, enter the IP address of the load balancer. |

5. Click **Test** to test the source connection, and then click **OK.**

   The certificate details are displayed.

6. Select the **Accept this Certificate** check box, and click **OK**.
7. In the Import User Groups dialog box, import user accounts from an SSO server on another machine.

| Option | Action |
|---|---|
| **Import From** | Select the single sign-on server you specified when you configured the single sign-on source. |
| **Domain Name** | Select the domain name from which you want to import user groups. If Active Directory is configured as the LDAP source in the PSC, you can only import universal groups and domain local groups if the vCenter resides in the same domain. |
| **Result Limit** | Enter the number of results that are displayed when the search is conducted. |
| **Search Prefix** | Enter a prefix to use when searching for user groups. |

8. In the list of user groups displayed, select at least one user group, and click **Next**.
9. In the Roles and Objects pane, select a role from the **Select Role** drop-down menu, and select the **Assign this role to the group** check box.
10. Select the objects users of the group can access when holding this role.

    To assign permissions so that users can access all the objects in VCF Operations, select the **Allow access to all objects in the system** check box.

11. Click **OK.**
12. Familiarize yourself with single-sign on and confirm that you have configured the single sign-on source correctly.
    a) Log out of VCF Operations.
    b) Log in to the vSphere Web Client as one of the users in the user group you imported from the single sign-on server.
    c) In a new browser tab, enter the IP address of your VCF Operations environment.
    d) If the single sign-on server is configured correctly, you are logged in to VCF Operations without having to enter your user credentials.

## Edit a Single Sign-On Source

Edit a single sign-on source if you need to change the administrator credentials used to manage the single sign-on source, or if you have changed the host of the source.

When you configure an SSO source, you specify either the IP address or the FQDN of the host machine where the single sign-on server resides. If you want to configure a new host, that is, if the single sign-on server resides on a different host machine than the one configured when the source was set up, VCF Operations removes the current SSO source, and creates a new source. In this case, you must reimport the users you want to associate with the new SSO source.

If you want to change the way the current host is identified in VCF Operations, for example, change the IP address to the FQDN and the reverse, or update the IP address of the PSC if the IP address of the configured PSC has changed, VCF Operations updates the current SSO source, and you are not required to reimport users.

1. Log in to VCF Operations as an administrator.
2. From the navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Authentication Sources** tile.
3. Select the single sign-on source, click the vertical ellipsis and then click **Edit**.
4. Make changes to the single sign-on source, and click **OK**.
   If you are configuring a new host, the New Single Sign-On Source Detected dialog box appears.
5. Enter the administrator credentials that were used to set up the single sign-on source, and click **OK**.
   The current SSO source is updated.
6. Click **OK** to accept the certificate.
7. Import the users you want to associate with the SSO source.

## Access Control in VCF Operations

Each user must have a unique account with one or more roles assigned to enforce a role-based security when they use VCF Operations. You create a user account, and assign the account to be a member of one or more user groups to allow the user to inherit the roles and scopes associated with the user group.

### Where You Find the Access Control Options

You can manage user accounts and their associated user groups, roles, scopes, and passwords.

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Access Control** tile.

**Table 1419: Access Control Tabs**

| Option | Description |
|---|---|
| User Accounts | Add, edit, remove, or import VCF Operations user accounts from an LDAP database, and manage user roles, their membership in groups, and the scopes assigned for association with the user. Import user accounts from an LDAP database that resides on another machine.<br>vCenter users who are logged in to VCF Operations, either logged in directly or through the vSphere Client, appear in the list of user accounts. |
| User Groups | Add, edit, or remove, or import VCF Operations user groups, update the members in a group and the associated scopes that they can access. Import user groups from an LDAP database or a single sign-on database that resides on another machine.<br>VCF Operations continuously synchronizes the user membership of imported LDAP user groups when the autosync option is enabled in the LDAP configuration. |
| Roles | For users to perform actions in VCF Operations, they must be assigned specific roles. With role-based access, when you assign a role to a user, you are determining not only what actions the user can perform in the system, but also the objects upon which those actions can be performed while holding the role. For example, to import or export a policy, the role assigned to your user account must have the Import or Export permissions enabled for policy management. |
| Scopes | Add, edit, clone, or remove scope associated with users or groups in VCF Operations. Scope lets you limit the access of a user or a set of users to VCF Operations. You can also define the scope for all the objects managed by VCF Operations. |

## Access Control: User Accounts Tab

You can add, edit, or remove VCF Operations user accounts, and import user accounts from an external LDAP database. With access control, you manage roles, the objects a user can access while assigned a specific role, and the membership in user groups.

### Where You Manage User Accounts

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Access Control** tile.

To view the details of the user account configured in VCF Operations, click the user account, the user account details is displayed in the right-side panel. The user account details include the user groups associated with the user account along with the roles and scope of the selected user account.

> **Note:**
> For user accounts only the exclusive roles that is associated with the user account is displayed. The roles inherited from the user groups are not displayed.

**Table 1420: Access Control User Accounts Summary Grid**

| Summary Grid Options | Description |
|---|---|
| User Accounts toolbar | To manage user accounts, use the toolbar options.<br>• Click the **Add** button to add a user account, and provide the details for the user account in the **Account Information** page.<br>• Click the **Vertical Ellipses** to perform any one of the following actions:<br>– **Edit**. Edit the selected user account, and modify the details for the user account in the **Edit User Account** dialog box.<br>– **Export**. Click **Export User Accounts** to export a user account, and provide the details to export the user account in the **Export Authentication Source** dialog box.<br>– **Delete**. Delete a user account.<br>• Click the **Horizontal Ellipses** and select **Delete**. Delete a user account.<br>– **Export**. Click export to export a user account, and provide the details to export the user account in the **Export Authentication Source** dialog box.<br>– Click **Import** to import a user account, and provide the details to import the user account in the **Import User Accounts** dialog box.<br>– Click **Import from source** to import a user account, and provide the details to import the user group in the **Import User** dialog box. |
| User Name | User name, without spaces, that you use to log in to VCF Operations. |
| First Name | User's first name, created when you create the user account. |
| Last Name | User's last name, created when you create the user account. |
| Email | User's email address, created when you create the user account. |
| Description | Description of the user account, defined when you create the user account. This information can identify the type of user and a summary of their access privileges. |
| Imported | Indicates whether the user account is imported or not. |
| Source Type | Indicates whether the user account is a local user, or an external user who is integrated through an external authentication source, such as from LDAP, SSO, AD, OpenLDAP, vCenter. |
| Activated | Indicates whether the user account is activated to use VCF Operations features. An administrator can edit a user account to manually activate it, or deactivate it to prevent user access to VCF Operations. |
| Locked | Indicates whether VCF Operations has locked the user account. For example, a user account can get locked based on the password lockout policy, or if the user enters an incorrect password three times in the span of five minutes. |
| Access All Objects | Indicates whether the user account is allowed to access all the objects that are imported into the VCF Operations instance. |

| Summary Grid Options | Description |
|---|---|
| Modified By | Indicates the last person to update the user account. |
| Last Modified | Indicates the last time the user account was updated. |
| Last Login Time | Indicates the last time the user logged in. |
| Filters | Activates you to search the list of user accounts according to the following criteria:<br>• User Name<br>• First Name<br>• Last Name<br>• Description<br>• Activated<br>• Locked<br>• Access All Objects<br>• Modified By<br>• Last Login Time |

After you add a user account, use the Details grid to view and edit which user accounts are assigned to user groups, and view the permissions assigned to the user account.

**Table 1421: Access Control User Accounts Details Grid**

| Details Grid Options | Description |
|---|---|
| First Name | User's first name, created when you create the user account. |
| Last Name | User's last name, created when you create the user account. |
| Modified By | Displays the name of the user who last updated the user account. |
| Last Modified | Displays the date and time when the user account was last updated. |
| Roles and Scope | Displays the role and scope associated with the selected user.<br>• Roles: Indicates the name of the role or roles assigned to the user.<br>• Scope : Indicates the scope associated with the user account. |
| User Groups | Assigned user groups appear when you click a user in the summary grid. You can then view and modify which user groups the user is associated with. |

### Modify User Accounts and Assign Groups and Permissions

You can add user accounts so that users can access the features of VCF Operations and certain objects in the environment. Or, modify user accounts to change their attributes, deactivate or lock the accounts, or require them to change their password. After you add user accounts, you can assign them to one or more user groups, and assign roles and scopes. Assign the administrators role only to specific users who must access objects and perform actions in the entire environment.

#### Where You Add or Edit User Accounts

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Access Control** tile.
2. In the **User Accounts** tab, click **Add**.
3. To edit a user account, click the vertical ellipsis and select **Edit**. You can also click the **EDIT ACCOUNT** button in the **User Details** page and edit the user account.
   - By default, the user is added to the "Everyone" group. You can add or remove users to or from the user group when you create or edit the user group, . You can assign the permissions to the users by selecting the required role-scope pairs.

- When you add users to a user group, the user inherits the permissions (role-scope pairs) from that group. You also have the option of not assigning permissions to the user, in that case the user actions are restricted, and the user cannot login to VCF Operations.
- You cannot change the user name once you create it. So, select your username appropriately.
- You have read only access for the Super Administrator account. You can change the password for the Super Administrator account from the user interface only.

**Table 1422: Add or Edit Users Accounts- User Details Page**

| User Details Options | Description |
|---|---|
| User Name | User name, without spaces to access the VCF Operations. |
| Password | User's password to access the VCF Operations instance. |
| Confirm Password | Confirmation of the user's password. |
| Require password change at next login | Users can change their password the next time they log in to the VCF Operations instance. |
| Deactivate this user | Deactivate the user account so that a user cannot access the VCF Operations instance. |
| First Name | User's first name, created when you create the user account. |
| Last Name | User's last name, created when you create the user account. |
| Email Address | User's email address, created when you create the user account. |
| Description | Description of the user account, defined when you create the user account. This information can identify the type of user and a summary of their access rights. |

**Assign Roles and Scope**

The Assign Roles and Scope option lets you select a role for each user and then assign a scope for that role.

4. From the **Select Role** drop-down menu, select a role for the user.
5. From the **Select Scope** drop-down menu, assign a scope for the selected user.

   **Note:**
   You can click the + sign to add multiple roles and then assign the required scope for each role.

**Assign User Groups**

The Assign User Group option lets you select user groups that the user will be a member of.

6. Select the user groups for which you want the user to be member of.
7. Click **Save**.

   **Note:**
   You cannot manually add users to groups imported from LDAP and SSO.

**Import User Accounts From Source**

You can import user accounts so that users can access the features of VCF Operations and the objects in the environment. After you import user accounts, you can add them to non-imported user groups, assign roles and scopes to them.

**Where You Import User Accounts**

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Access Control** tile.
2. Click the horizontal ellipsis next to **Add** and then, click **Import from Source** to import user from another source.

**Table 1423: Import Users from a LDAP Source**

| User Details Options | Description |
|---|---|
| Import From | Select the authentication source from the drop-down menu, the options are LDAP host machine, VMware Identity Manager, Active Directory, or Other sources configured to import user accounts. To know more about authentication,see Authentication Sources. |
| User Name | Click **Use alternate credentials** to display the user name of the LDAP source credential used to import user accounts to the VCF Operations instance. |
| Password | Password for the LDAP source credential to import user accounts to the VCF Operations instance. |
| Basic | Use the basic import setting with search option to look for user accounts. |
| Advanced | Displays the advanced import settings.<br>• Group Search Criteria. Search criteria to find LDAP groups. If not included, VCF Operations uses the default search parameters: `(|(objectClass=group)(objectClass=groupOfNames))`<br>• Member Attribute. Name of the attribute for a group object that contains the list of members. If not included, VCF Operations uses member by default.<br>• User Search Criteria. Search criteria to use the member field to find and cache LDAP users. You enter sets of key=value pairs in the form `(|(key1=value1)(key2=value2))`. If not included, VCF Operations searches for each user separately. This operation might take extra time.<br>• Member Match Field. Name of the attribute for a user object to match with the member entry from a group object. If not included, VCF Operations treats the member entry as a distinguished name.<br>• LDAP Context Attributes. Attributes that VCF Operations applies to the LDAP context environment. You enter sets of key=value pairs separated by commas, such as `java.naming.referral=ignore,java.naming.ldap.deleteRDNfalse`. |
| Search String | Enter a search string, and click **Search** to start the search for user accounts. |
| User Name Summary grid | Lists the users available for import. Select the check box for each user to import, or select the **User Name** check box to import all users. User accounts that are already imported to VCF Operations do not appear in the list. |

**Table 1424: Import Users from a VMware SSO Source**

| User Details Options | Description |
|---|---|
| Import From | VMware SSO is configured as the source to import user accounts. |
| Search Prefix | Enter a search string, and click **Search** to start the search for user accounts. |
| User Name Summary grid | Lists the users available for import. Select the check box for each user to import, or select the **User Name** check box to import all users. To appear in the list, the user configuration must be set to primary group in the default domain user group. User accounts that are already imported to VCF Operations do not appear in the list.<br>**Note:**<br>After importing the user, you must edit the user information and assign a role and a scope. |

**Table 1425: Import Users from a VMware Identity Manager Source**

| User Details Options | Description |
|---|---|
| Import From | VMware Identity Manager configured as the source to import user accounts. |
| Search Option | Enter the search option, it can be a domain or search prefix. |
| Domain Name | Enter the domain name for import. |
| Search Prefix | Enter a search string, and click **Search** to start the search for user accounts. |
| User Name Summary grid | Lists the users available for import. Select the check box for each user to import, or select the **User Name** check box to import all users. To appear in the list, the user configuration must be set to primary group in the default domain user group. User accounts that are already imported to VCF Operations do not appear in the list. |

**Table 1426: Import Users from a Single Sign On Source**

| User Details Options | Description |
|---|---|
| Import From | SSO source configured as the source to import user accounts. |
| Domain Name | Select the domain name for import from the drop-down menu. |
| Result Limit | Determines the number of users displayed. |
| Search Prefix | Enter a search prefix, and click **Search** to start the search for user accounts. |
| User Name Summary grid | Lists the users available for import. Select the check box for each user to import, or select the **User Name** check box to import all users. To appear in the list, the user configuration must be set to primary group in the default domain user group. User accounts that are already imported to VCF Operations do not appear in the list. |

3. After you enter the import users details, click **Next**

**Table 1427: Import Users Accounts- Assign Groups Page**

| Assign Groups | Description |
|---|---|
| Assign Groups | Select or deselect the groups associated with the user account. To select or deselect all accounts, click the **Group Name** check box. You cannot add user accounts to groups imported from LDAP. |

4. Click **Finish**.

### Export and Import of User Accounts

You can export User account configurations from one VCF Operations and import into any VCF Operations.

#### Export User Accounts

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click **Access Control** tile.
2. In the **User Accounts** tab, select the user accounts to be exported. Click the horizontal ellipsis next to **Add** and then, click **Export**.
3. You will be prompted to enter a password when you export the user accounts. Enter the password and note it down, you have to use the same password when you import the user accounts.
4. Click **Export**.

The user accounts `.json` file is downloaded to the default download location.

**Import User Accounts**

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click **Access Control** tile.
2. In the **User Accounts** tab, click the horizontal ellipsis next to **Add** and then, click **Import**.
3. Click **Browse** and select the user accounts `.json` file.
4. Enter the same password which you had used during user accounts export.
5. In case of a conflict, select either **Overwrite existing User Accounts** or **Skip User Accounts**.
6. Click **Import**.

**Important Points**

- User scopes will be matched in the target VCF Operations, which means the user scopes references are being exported, then assigned to user while importing only if those scopes exist on the target VCF Operations.
- User roles will be matched in the target VCF Operations, which means user roles references are being exported, then assigned to user while importing only if those roles exist on the target VCF Operations.
- User groups will be matched in the target VCF Operations, which means user group references are being exported, then assigned to user while importing only if those user groups exist on the target VCF Operations.
- Import of the external users fails if their authentication source is not found in the target VCF Operations.
- The traversal specifications are not imported or exported in VCF Operations.
- An error message is displayed with details for failed imports.

## Access Control: User Groups Tab

You can manage the user groups associated with the users and objects in your environment. You can import user groups from an LDAP database that resides on another machine, or from a single sign-on server or from a source in VMware Identity Manager.

**Where You Manage User Groups**

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Access Control** tile.
2. Click the **User Groups** tab.

To view the details of the user groups configured in VCF Operations, click the user group, the user group details are displayed in the right-side panel. The user group details include the user accounts associated with the selected user group along with the roles and scope associated with the user group.

**Table 1428: Access Control User Groups Summary Grid**

| Option | Description |
|---|---|
| User Groups toolbar | To manage user groups, use the toolbar options.<br>• Click the **Add** button to add a user group, and provide the details for the user group in the **Group Information** page.<br>• Click the **Vertical Ellipses** to perform any one of the following actions:<br>– **Edit**. Edit the selected user group, and modify the details for the user group in the **Edit User Group** dialog box.<br>– **Delete**. Delete a user group.<br>– **Clone**. Clone a user group, and enter a name and description for the cloned user group.<br>– **Export**. Click export to export a user group, and provide the details to export the user group in the **Export Authentication Source** dialog box.<br>• Click the **Horizontal Ellipses** and select the available options.<br>– **Delete**. Click delete to Delete a user group.<br>– **Export**. Click export to export a user group, and provide the details to export the user group in the **Export Authentication Source** dialog box.<br>– Click **Import** to import a user group, and provide the details to import the user group in the **Import User Groups** dialog box.<br>– Click **Import from source** to import a user group, and provide the details to import the user group in the **Import User** dialog box. |
| Group Name | Name of the user group. |
| Description | Description of the group, indicating its purpose. |
| Members | Number of members in the group. |
| Group Type | Type of group, either a local user group or a group imported from LDAP. |
| Distinguished Name | Names for LDAP objects, such as domains and users. |
| Access All Objects | Indicates if the user group account is allowed to access all the objects that are imported into the VCF Operations instance. |
| Modified By | Indicates the last person to update the user group. |
| Last Modified | Indicates the last time the user group was updated. |
| Filters | Enables you to search the list of user groups according to the following criteria:<br>• Group Name<br>• Description<br>• Access All Objects<br>• Modified By |

After you select a user group in the summary grid, view details about associated users in the Details pane.

**Table 1429: Access Control User Groups Details Grid**

| Option | Description |
|---|---|
| User Group Name | User Group name, created when you create the user group. |
| Modified By | Displays the name of the user who last updated the user group. |
| Last Modified | Displays the date and time when the user group was last updated. |
| Description | Description of the group, indicating its purpose. |
| Roles and Scope | Displays the role and scope associated with the selected user.<br>• Role: Indicates the name of the role or roles assigned to the user.<br>• Scope: Indicates the scope associated with the user group. |
| User Accounts | Displays the user accounts associated with the selected user group. |

### Add User Groups and Assign Members and Roles

You can view and modify the details for user groups, including users, roles, and scope.

#### Where You Add User Groups

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Access Control** tile.
2. Select the **User Groups** tab and then click **Add**
3. To edit a user group, click the vertical ellipsis and select **Edit**. You can also click the **EDIT GROUP** button in the **Group Details** page and edit the user group.

**Table 1430: Add or Edit Group Information - Name and Description Page**

| Option | Description |
|---|---|
| Group Name | Name of the user group, either created manually, imported from a single sign-on server, or imported from an LDAP database that resides on another machine. |
| Description | Description of the user group, indicating its purpose. |

4. **Assign Roles and Scope**
   The Assign Roles and Scope option lets you select a role for the user group and then assign a scope for each role.
5. From the **Select Role** drop-down menu, select a role for the user group.
6. From the **Select Scope** drop-down menu, assign scope for the selected user group.

   **Note:**
   You can click the + sign to add multiple roles and then assign the required scope for each role.

   **Assign Users**
   The Assign Users option lets you add members to the group based on the membership criteria that you have defined.
7. Select the users who would be the members of this group.
8. Click **Save**.

### Import User Groups From Source

You import user groups from a single sign-on server, VMware Identity Manager, Active Directory, or an LDAP database on another machine so that you can use those groups in VCF Operations.

#### Where You Import User Groups

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Access Control** tile.
2. Select **User Groups** tab, click the horizontal ellipsis next to **ADD** button, and select **Import**

   **Note:**
   You can edit the imported group to assign Roles and Scopes.

   The options displayed in the Import User Groups page depend upon the authentication source you select.

**Table 1431: Import User Groups Page - LDAP, Active Directory, and Others Sources**

| Option | Description |
|---|---|
| Import From | Host machine configured as the source to import the user groups. These options are displayed when the host machine of an LDAP, Active Directory, or Other source is selected. |
| User Name | User name of the source credential to import user groups to the VCF Operations instance. |
| Password | Password for the source credential to import user groups to the VCF Operations instance. |
| Search String | Invoke the search for user groups. |
| Basic | Use the basic import setting with search option to look for user groups. |

| Option | Description |
|---|---|
| Advanced | Displays the advanced import settings.<br>• Group Search Criteria. Search criteria to find LDAP groups. If not included, VCF Operations uses the default search parameters: `(|(objectClass=group)(objectClass=groupOfNames))`<br>• Member Attribute. Name of the attribute for a group object that contains the list of members. If not included, VCF Operations uses member by default.<br>• User Search Criteria. Search criteria to use the member field to find and cache LDAP users. You enter sets of key=value pairs in the form `(|(key1=value1)(key2=value2))`. If not included, VCF Operations searches for each user separately. This operation might take extra time.<br>• Member Match Field. Name of the attribute for a user object to match with the member entry from a group object. If not included, VCF Operations treats the member entry as a distinguished name.<br>• LDAP Context Attributes. Attributes that VCF Operations applies to the LDAP context environment. You enter sets of key=value pairs separated by commas, such as `java.naming.referral=ignore,java.naming.ldap.deleteRDNfalse`. |
| Group Name | Displays a list of user groups. Select the **Group Name** check box to import all the displayed user groups, or select the check box next to each user group that you want to import. |

**Table 1432: Import User Groups Page - Single Sign On Source**

| Option | Description |
|---|---|
| Import From | Host machine configured as the source to import the user groups. |
| Domain Name | User name of the source credential to import user groups to the VCF Operations instance. |
| Result Limit | Determines the number of groups displayed. |
| Search Prefix | Enter a search prefix to narrow your search. |
| Group Name | Displays a list of user groups. Select the **Group Name** check box to import all the displayed user groups, or select the check box next to each user group that you want to import. |

**Table 1433: Import User Groups from a VMware SSO Source**

| User Detail Options | Description |
|---|---|
| Import From | VMware SSO is configured as the source to import user groups. |
| Search Prefix | Enter a search string, and click **Search** to start the search for user groups. |
| Group Name Summary grid | Lists the users available for import. Select the check box for each user group to import, or select the **Group Name** check box to import all groups. User groups that are already imported to VCF Operations do not appear in the list.<br>**Note:**<br>After importing the user group, you must edit the user group information and assign a role and a scope. |

**Table 1434: Import User Groups from a VMware Identity Manager Source**

| User Details Options | Description |
|---|---|
| Import From | VMware Identity Manager configured as the source to import user groups. |
| Domain Name | Enter the domain name for import. |
| Search Prefix | Enter a search string, and click **Search** to start the search for user groups. |

| User Details Options | Description |
|---|---|
| Group Name Summary grid | Lists the users available for import. Select the check box for each user group to import, or select the **Group Name** check box to import all groups. User groups that are already imported to VCF Operations do not appear in the list. |

3. After you enter the import user group details, click **Next**.
4. Click **Finish**.

**Export and Import of User Groups**

You can export User Group configurations from one VCF Operations and import into any VCF Operations.

**Export User Groups**

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click **Access Control** tile.
2. In the **User Groups** tab, select the user groups to be exported. Click the horizontal ellipsis next to **Add** and then, click **Export**.

The user groups `.json` file is downloaded to the default download location.

**Import User Groups**

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click **Access Control** tile.
2. In the **User Groups** tab, click the horizontal ellipsis next to **Add** and then, click **Import**.
3. Click **Browse** and select the user groups `.json` file.
4. In case of a conflict, select either **Overwrite existing User Groups** or **Skip User Groups**.
5. Click **Import**.

**Important Points**

- User group's scopes are matched in the target VCF Operations, which means user group's scopes references are exported, then assigned to the user group while importing only if those scopes exist on the target VCF Operations.
- User group's roles are matched in the target VCF Operations, which means user group's role references are exported, then assigned to the user group while importing only if those roles exist on the target VCF Operations.
- User group's members are matched in the target VCF Operations, which means user group's member references are exported, then assigned to the user group while importing only if those users exist on the target VCF Operations.
- Import of the external user groups fails if their authentication source is not found in the target VCF Operations.
- The traversal specifications are not imported or exported in VCF Operations.
- An error message is displayed with details for failed imports.

## Access Control: Roles Tab

You can assign users-specific roles to perform actions and view features and objects in VCF Operations. With role-based access, users can only perform the actions that their permissions allow.

**Where You Manage User Roles**

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Access Control** tile.
2. Click the **Roles** tab.

To view the details of roles configured in VCF Operations, click the role, the role details are displayed in the right-side panel. The role details include the permissions, user accounts, and user groups associated with the selected role.

You can view and edit details about a role, by selecting a role in the summary grid, and clicking the **Edit** icon in the Roles toolbar.

**Table 1435: Access Control Roles Summary Grid**

| Option | Description |
|---|---|
| Roles toolbar | To manage roles, use the toolbar icons.<br>• Click the **Add** icon. to add a user role, provide the name and description for the role, and assign permissions in the **Role Information** page.<br>**Note:**<br>To assign all the available permissions to the selected role, select the **Select all Permissions** check box.<br>• Click the **Vertical Ellipses** to perform any one of the following actions:<br>– **Edit**. Edit the selected user role, and modify the details for the role in the **Edit Role** dialog box.<br>– **Delete**. Delete a user role.<br>– **Clone**. Clone the selected user role<br>– **Export**. Export a user role.<br>• Click the **Horizontal Ellipses** to perform any one of the following actions:<br>– **Delete**. Click delete to Delete a role.<br>– **Export**. Click export to export a role.<br>– Click **Import** to import a role from the **Import Roles** page. You can import existing roles, overwrite existing roles, and skip roles. |
| Filters | Activates you to search the list of roles according to the following criteria:<br>• Role Name<br>• Role Description<br>• Modified By |

You can view details for the user accounts and user groups associated with a selected role in the Details panes.

**Table 1436: Access Control Roles Details Panes**

| Option | Description |
|---|---|
| Role Name | Name of the role to apply to a specific level of users, such as user for base users or administrator for users with administrative permissions. |
| Description | Description of the role, indicating its purpose. |
| Modified By | Displays the name of the user who last updated the role. |
| Last Modified | Displays the date and time when the role was last updated. |
| Permissions | Displays the permissions assigned to the role according to the categories: Subscriptions, Administration, Automation Central, Configure, Data Sources, Environment, Home, Optimize, Plan, Troubleshoot, and Visualize. Expand the tree of each category to view all the assigned permissions.<br>You can edit the permissions assigned to the role by clicking the **Edit Role** icon.<br>• Click the **Expand All** button to expand the trees of all the categories, and select the check boxes to apply permissions for the selected role. |
| User Accounts | The roles are assigned to the users. You can only view users who have been assigned roles directly, the users belonging to a user group are not displayed here. The information in this pane is based on the data entered when you created the user, or imported with the user.<br>• First Name. Indicates the first name of each user who is assigned this role.<br>• Last Name. Indicates the last name of each user who is assigned this role.<br>• User name , without spaces, that will log in to VCF Operations<br>• Email. Indicates the email address for each user who is assigned this role. |

| Option | Description |
|---|---|
| User Groups | The roles are assigned to the selected user groups.<br>• Group Name: Name of each group that is associated with the selected role.<br>• Members: Number of members in each group. |

### Add or Edit Roles and Assign Permissions

You can add or edit user-specific roles to perform actions and view features and objects in VCF Operations. With role-based access, you can assign roles to users and let them only perform the actions that their permissions allow.

#### Where You Add or Edit User Roles

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**,and then click the **Access Control** tile.
2. Select the **Roles** tab and click **Add**.
3. To edit a role, select the role and then click **Edit Role**. You can also click the **EDIT ROLE** button in the **Role Details** page and edit the role.

**Table 1437: Add or Edit Role Information Page**

| Option | Description |
|---|---|
| Name | Name of the role to apply to a specific level of users, such as user for base users or administrator for users with administrative permissions. |
| Description | Description of the role, indicating its purpose. |
| **Assign Permissions**<br>**Note:**<br>You can set default permissions for the selected role using the **Apply Default Permissions** option. The Apply Default Option is available only when you edit an Out-of-the-box role. | |
| Select all Permissions | Click **Select all Permissions** to apply the permissions for the role you create or select. You can click the arrow next to the categories to set specific permissions to the roles you define. |
| Expand All | Click the **Expand All** button to expand the trees of all the categories and select the check boxes to apply or modify the permissions for the selected role. |
| Collapse All | Click the **Collapse All** button to Close the tree view and display only the high-level categories. |

4. Click **Save**.

### Export and Import of Roles

You can export User Roles configurations from one VCF Operations and import into any VCF Operations.

#### Export User Roles

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Access Control** tile.
2. In the **Roles** tab, select the user roles to be exported. Click the horizontal ellipsis next to **Add** and then, click **Export**.

The roles `.json` file is downloaded to the default download location.

#### Import Roles

Invalid privilege keys of a role are ignored during import.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click **Access Control** tile.
2. In the **Roles** tab, click the horizontal ellipsis next to **Add** and then, click **Import**.
3. Click **Browse** and select the roles `.json` file.

4. In case of a conflict, select either **Overwrite existing Roles** or **Skip Roles**.
5. Click **Import**.

**Important Points**

- Invalid permissions are ignored during import based on target VCF Operations.
- An error message is displayed with details for failed imports.

## Access Control: Scopes Tab

You can add, edit, or remove scope associated with users and groups in VCF Operations. Scope provides you the ability to limit access of a user or a set of users to VCF Operations. The All Objects scope is created by default. Assign the All Objects scope to the user to access all objects discovered by VCF Operations.

**Where to Manage Scope**

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Access Control** tile.
2. Click the **Scopes** tab.

To view the details for a specific scope defined in VCF Operations, click the scope. The scope details are displayed in the right-side panel. The scope details include the user accounts and user groups associated with the scope.

**Table 1438: Access Control Scope Summary Grid**

| Option | Description |
|---|---|
| Scope Toolbar | To manage scopes, use the toolbar options.<br>• Click **Add** to add the scope and provide the name and description for the scope, and assign members to objects in the **Scope Information** page.<br>Select **Object Hierarchies** and then select the object to assign members to. You can select one or more objects.<br>**Note:**<br>For vSphere Storage object hierarchy, select the **Propagate to children** checkbox to include all future discovered objects in the scope.<br>• Click the **Vertical Ellipses** to perform any one of the following actions:<br>– **Edit**. Edit a specific scope and modify the details for the scope in the **Edit Scope Information** page.<br>– **Clone**. Clone a scope and enter a name and description for the cloned scope.<br>– **Delete**. Delete a scope.<br>• Click the **Horizontal Ellipses** and select **Delete**. Delete a scope. |
| Filters | Activates you to search the list of scopes according to the following criteria:<br>• Scope Name<br>• Description<br>• Modified By |

**Table 1439: Access Control Scope Details Grid**

| Option | Description |
|---|---|
| Scope Name | Displays the name of the selected scope. |
| Description | Displays the description of the scope, indicating its purpose. |
| Modified By | Indicates the last person to update the scope. |
| Last Modified | Indicates the time when the scope was last updated. |

| Option | Description |
|---|---|
| User Accounts | Displays the user accounts associated with the selected scope. |
| User Groups | Displays the user groups associated with the selected scope. |

### Modify Scopes and Assign Member Objects

You can add and modify the scope defined in VCF Operations. Scope lets you create a package and associate the objects and object hierarchies to the package. You can select the object hierarchy and assign the required objects to the selected object hierarchy. After you define a scope, you can associate the scope to user accounts and user groups.

#### Where You Add or Edit Scope

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Access Control** tile.
2. Click the **Scopes** tab.
3. In the **Scopes** tab, click **Add**.
4. To edit a scope, select the scope and then click **Edit Scope**. You can also click the **EDIT SCOPE** button in the **Scope Details** grid and edit the scope.

**Table 1440: Access Control Scopes Details Grid**

| Option | Description |
|---|---|
| Scope Name | Name of the scope to apply to a specific user or a set of users, such as user for base users or administrator for users with administrative permissions. |
| Description | Description of the scope, indicating its purpose. |

5. To add or edit the scope, select the required **Object Hierarchy**, and then select the **Object** to be associated.
6. Click **Save**.

After you save the scope, you can assign the scope to user accounts and user groups. You can also assign a log data set. For more information, see Defining a Dataset.

## Authentication Sources

VCF Operations uses authentication sources that activate you to import and authenticate users and user group information that reside on another machine: the Lightweight Directory Access Protocol (LDAP) platform-independent protocol, Active Directory, VMware Identity Manager, VMware SSO, Single Sign-On, and Others.

#### Where You Manage Authentication Sources

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Authentication Sources** tile.

> **Note:**
>
> Native 2FA is not currently supported for local user accounts. Customers requiring MFA on all logins should use single sign-on with an identity provider that enforces MFA for all interactive users, and should restrict local account usage to emergency/break-glass access only, protected by a strong unique password, IP allow-listing, session timeout, and audit logging of local-account logins.

**Table 1441: Authentication Sources Toolbar and Data Grid**

| Option | Description |
|---|---|
| Authentication Sources toolbar | To manage authentication sources, use the toolbar icons.<br>• **Add** icon: Add an authentication source, and provide the information for the source in the Add Source for User and Group Import dialog box.<br>• Click the **Vertical Ellipses** to perform any one of the following actions:<br>— **Edit**. Edit the selected authentication source, and modify the details in the Edit Source dialog box.<br>— **Delete**. Delete an authentication source.<br>— **Synchronize User Groups**. Synchronize users within the groups imported through the selected Active Directory or LDAP authentication source. |
| Source Display Name | Name that you assign to the authentication source. |
| Source Type | Indicates the type of directory services access technology to access the source machine where the authentication database of user accounts resides. Options include:<br>• SSO SAML: An open-standard data format that activates Web browser single sign-on.<br>• VMware Identity Manager: A platform where you can manage users and groups, manage resources and user authentication, and access policies and entitle users to resources.<br>• VMware SSO: Browser based SSO that uses an external identity provider to log into vCenter, VCF Operations, Log management, and VCF Operations Orchestrator without providing credentials again.<br>• Open LDAP: A platform-independent protocol that provides access to an LDAP database on another machine to import user accounts.<br>• Active Directory or Other: Specifies any other LDAP-based directory services, such as Novel or Open DJ, used to import user accounts from an LDAP database on a Linux Mac machine. |
| Host | Name or IP address of the host machine where the user database resides. |
| Port | Port used for the import. |
| Base DN | Base distinguished name for the user search. VCF Operations locates only the users under the Base DN. The Base DN is an elementary entry for an imported user's distinguished name (DN), which is the base entry for the user name without the need for other related information such as the full path to the user account, or the inclusion of related domain components. Although VCF Operations populates the Base DN, an Administrator must verify the Base DN before saving the LDAP configuration. |
| Auto Synchronization | When selected, activates VCF Operations to map imported LDAP users to user groups. |
| Last Synchronized | Date and time that the synchronization last occurred. |

## Configure VMware Single Sign-On for VCF Operations

After installing or upgrading to VCF Operations 8.18, you can configure VCF Operations for VMware Single Sign-On. When you configure VMware Single Sign-On, you use an external identity provider to sign into VCF Operations.

- Ensure that the associated vCenter Server host is configured for VMware Single Sign-On. For more information about configuring a vCenter Server host for VMware Single Sign-On, see *Configure VMware Single Sign-on* in the vSphere Authentication documentation.

  **Note:**

  After configuring VMware Single Sign-On configuration for VCF Operations, you can still log in to VCF Operations with a local account.

1. Log in to VCF Operations with a local account.
2. Configure VMware SSO as an authentication type. For more information, see Authentication Sources and Authentication Sources: Add Authentication Source for User and Group Import.

   **Note:**
   Only one VMware Single Sign-On configuration can exist during any period of time.
3. To allow users or groups to log in to VCF Operations using **VMware SSO** authentication source, import users or groups from the authentication source into VCF Operations. For more information, see Import User Accounts From Source and Import User Groups From Source.

   **Note:**
   You must select a role for each user and assign a scope for each role.
4. A certificate of type **VMware SSO** is created and can be viewed from **Administration** > **Control Panel** > **Certificates**.
5. Users from the VMware SSO authentication source can now log into VCF Operations using VMware SSO. When you log in to using VMware SSO, you will be redirected to an external authentication page. Enter the credentials to log in to VCF Operations.

   **Note:**

   - You can delete and unregister a VMware Single Sign-On authentication source. For more information see, Delete and Deregister a VMware Single Sign-On Authentication Source.
   - You have to re-register the VMware SSO server in certain scenarios. Navigate to the **Authentication Sources** page, click the **Re-register source** link and enter the vCenter Server credentials. Re-register the VMware SSO server in the following two scenarios:
     – If you add or remove nodes from a cluster in VCF Operations, and/or
     – If you have modified the system access URL when a load balancer is used (**Administration** > **Global Settings** > **System Settings** > **System Access URL option**).

## Authentication Sources: Add Authentication Source for User and Group Import

When you import user account information that resides on another machine, you must define the criteria used to import the user accounts from the source machine.

### Where You Add or Edit Authentication Sources

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Authentication Sources** tile.
2. Click **Add**.
3. To edit authentication sources, click **Edit**.

**Table 1442: Authentication Sources Add Source for User and Group Import**

| Option | Description |
|---|---|
| Source Display Name | Name that you assign to the authentication source. |
| Source Type<br><br>**Note:**<br>The option you select in the **Source Type** drop-down box, determines the options available in this dialog box. | Indicates the type of directory services access technology to access the source machine where the database of user accounts resides. There are two types of databases: LDAP and single sign-on. Options include:<br>• SSO SAML: An XML-based standard for a web browser single sign-on that activates users to perform single sign-on to multiple applications.<br>• VMware Identity Manager: A platform where you can manage users and groups, manage resources and user authentication, and access policies and entitle users to resources.<br>• VMware SSO: Browser based SSO that uses an external identity provider to log into vCenter, VCF Operations, Log management, and VCF Operations Orchestrator without providing credentials again.<br>• Open LDAP: A platform-independent protocol that provides access to an LDAP database on another machine to import user accounts.<br>• Other: Specifies any other LDAP-based directory services, such as Novel or OpenDJ, used to import user accounts from an LDAP database on a Linux Mac machine. |

**Table 1443: Authentication Sources Add Source for User and Group Import - Options Available When SSO SAML Is Selected.**

| Name | Description |
|---|---|
| Host | Name or IP address of the host machine where the single sign-on user server resides. |
| Port | The single sign-on listening port. By default this is set to 443. |
| User Name | Name of the user account that can log in to the single sign-on host machine. |
| Password | Password of the user account that can log in to the single sign-on host machine. |
| Grant administrator role to VCF Operations for future configuration? | When you create a single sign-on source, a new VCF Operations user account is created on the single sign-on server.<br>• Select **Yes**, to grant VCF Operations an administrative role so that it can be used to configure the SSO source if changes are made to the VCF Operations setup.<br>• If you select **No** and the VCF Operations setup is changed, SSO users will not be able to log in until you re-register the SSO source. |
| Automatically redirect to VCF Operations single sign-on URL? | After you have configured a single sign-on source, users are redirected to the vCenter SSO server.<br>• Select **Yes**, to redirect users to the single sign-on server for authentication.<br>• If you select **No** users must sign in through the VCF Operations login page. |

| Name | Description |
|---|---|
| Import single sign-on user groups after adding the current source? | When you have set up a single sign-on source, you import users and user groups into VCF Operations so that single sign-on users can access the system with their single sign-on permissions.<br>• If you select **Yes**, the wizard directs you to the Import User Groups page so that you can import user groups when you have finished setting up the SSO source.<br>• If you want to import user accounts, or user groups at a later stage, select **No**. |
| Advanced | If your system uses a load balancer, enter the IP address of the load balancer. |
| Test | Tests whether the host machine can be reached with the credentials provided. |

**Table 1444: Authentication Sources Add Source for User and Group Import - Options Available When VMware SSO  Is Selected.**

| Name | Description |
|---|---|
| Host | Name or IP address of the vCenter host machine on which VMware SSO has been configured. |
| Port | The single sign-on listening port. By default this is set to 443. |
| Tenant | |
| Username | Name of the vCenter user account that can log in to the VMware SSO host machine.<br>**Note:**<br>The user must have the *VcIdentityProviders.Manage* permission assigned. |
| Password | Password of the vCenter user account that can log in to the VMware SSO host machine. |

**Note:**
Only one VMware Single Sign-On configuration can exist during any period of time.

**Table 1445: Authentication Sources Add Source for User and Group Import - Options Available When Open LDAP, Active Directory, and Other Are Selected.**

| Option | Description |
|---|---|
| Integration Mode Basic settings | Applies basic settings to integrate the LDAP import source with the instance of VCF Operations.<br>Use Basic integration mode to have VCF Operations discover the host machine where the LDAP database resides, and set the base distinguished name (Base DN) used to search for users. You provide the name of the domain and the subdomain, which VCF Operations uses to populate the Host and Base DN details, and the name and password of the user who can log in to the LDAP host machine.<br>In Basic mode, attempts to fetch the host and port from the DNS server, and obtain the Global Catalog and domain controllers for the domain, with preference given to SSL/TLS-activated servers.<br>• Domain/Subdomain. Domain information for the LDAP user account.<br>**Note:**<br>To import users and groups from multiple subdomains, use the root domain.com instead of subdomain. Using a subdomain limits the visibility of VCF Operations to groups and users from that specific subdomain.<br>• Use SSL/TLS. When selected, VCF Operations uses the Secure Sockets Layer/Transport Layer Security (SSL/TLS) protocol to provide secure communication when you import users from an LDAP database. You do not need to install the SSL/TLS certificate. Instead, VCF Operations prompts you to view and verify the thumbprint, and accept the LDAP server certificate. After you accept the certificate, the LDAP communication proceeds.<br>• If Active Directory uses a self-signed certificate, then the certificate should contain the **Subject Alternative Name** field. VCF Operations can successfully verify the Active Directory certificate and integrate with Active Directory only if, the host name or the IP address provided in the **Subject Alternative Name** field matches the address of the domain controller on which the certificate is used.<br>• User Name. Name of the user account that can log in to the LDAP host machine.<br>• Reset Password. Reset the password of the user account that can log in to the LDAP host machine.<br>• Automatically synchronize user membership for configured groups. When selected, activates VCF Operations to map imported LDAP users to user groups.<br>• Host. Name or IP address of the host machine where the LDAP user database resides.<br>• Port. Port used for the import. Use port 389 if you are not using SSL/TLS, or port 636 if you are using SSL/TLS, or another port number of your choice. Global Catalog ports are 3268 for non-SSL/TLS, and 3269 for SSL/TLS.<br>• Base DN. Base distinguished name for the user search. VCF Operations locates only the users under the Base DN. The Base DN is an elementary entry for an imported user's distinguished name (DN), which is the base entry for the user name without the need for other related information such as the full path to the user account, or the inclusion of related domain components. Although VCF Operations populates the Base DN, an Administrator must verify the Base DN before saving the LDAP configuration.<br>• Common Name. LDAP attribute used to identify the user name. The default attribute for Active Directory is `userPrincipalName`. |
| Integration Mode Advanced settings | Applies advanced settings to integrate the LDAP import source with the instance of VCF Operations.<br>Use Advanced integration mode to manually provide the host name and base distinguished name (Base DN) to have VCF Operations import users. You provide the name and password of the user who can log in to the LDAP host machine.<br>• Host. Name or IP address of the host machine where the LDAP user database resides.<br>• Use SSL/TLS. When selected, VCF Operations uses the Secure Sockets Layer/Transport Layer Security (SSL/TLS) protocol to provide secure communication when you import users from an LDAP database. You do not need to install the SSL/TLS certificate. Instead, VCF Operations prompts you to view and verify the thumbprint, and accept the LDAP server certificate. After you accept the certificate, the LDAP communication proceeds.<br>• If Active Directory uses a self-signed certificate, then the certificate should contain the **Subject Alternative Name** field. VCF Operations can successfully verify the Active Directory certificate and integrate with Active Directory only if, the host name or the IP address provided in the **Subject Alternative Name** field matches the address of the domain controller on which the certificate is used.<br>• Base DN. Base distinguished name for the user search. VCF Operations will locate only the users under the Base DN. The Base DN is an elementary entry for an imported user's distinguished name (DN), which is the base entry for the user name without the need for other related information such as the full path to the user account, or the inclusion of related |

| Option | Description |
|---|---|
| Search Criteria | Displays the search criteria settings.<br>Although VCF Operations populates part of the search criteria, an Administrator must verify the settings to ensure that the settings are correct according to the properties of the LDAP type.<br>• Group Search Criteria. Search criteria to find LDAP groups. If not included, VCF Operations uses the default search parameters: `(|(objectClass=group)(objectClass=groupOfNames))`<br>• Member Attribute. Name of the attribute for a group object that contains the list of members. If not included, VCF Operations uses member by default.<br>• User Search Criteria. Search criteria to use the member field to find and cache LDAP users. You enter sets of key=value pairs in the form `(|(key1=value1)(key2=value2))`. If not included, VCF Operations searches for each user separately. This operation might take extra time.<br>• Member Match Field. Name of the attribute for a user object to match with the member entry from a group object. If not included, VCF Operations treats the member entry as a distinguished name.<br>• LDAP Context Attributes. Attributes that VCF Operations applies to the LDAP context environment. You enter sets of key=value pairs separated by commas, such as `java.naming.referral=ignore,java.naming.ldap.deleteRDNfalse`. |
| Test | Tests whether the host machine can be reached, with the credentials provided. Although a test of the connection is successful, users who use the search feature must have read permissions in the LDAP source.<br>This test does not verify the accuracy of the Base DN or Common Name entries. |

**Table 1446: Authentication Sources Add Source for User and Group Import - Options Available When VMware Identity Manager Is Selected.**

| Option | Description |
|---|---|
| Host | Name or IP address of the VMware Identity Manager machine where the single sign-on user server resides. |
| Port | The single sign-on listening port. By default this is set to 443. |
| Tenant | This is an optional field. |
| User name | VMware Identity Manager system-domain tenant administrator user name. |
| Password | Password of the VMware Identity Manager system-domain tenant administrator. |
| Redirect IP/ FQDN | This is the IP address of VCF Operations node where a user is redirected after a successful authentication from VMware Identity Manager. By default, this is the IP address of the VCF Operations primary node.<br>**Note:**<br>When the primary replica becomes the primary node on VCF Operations, then VCF Operations administrator has to manually edit the IP address and set it to the IP address of the current primary node. |
| Test | Tests whether the VMware Identity Manager machine can be reached, with the credentials provided. |

## Export and Import of Authentication Sources

You can export Authentication source configurations from one and import into any VCF Operations.

**Export Authentication Sources**

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click **Authentication Sources** tile.
2. In the **Authentication Sources** tab, select the authentication sources to be exported. Click the horizontal ellipsis next to **Add** and then, click **Export**.
3. You are prompted to enter a password when you export authentication sources. Enter the password and note it down, you have to use the same password when you import the authentication sources.
4. Click **Export**.

The authentication sources `.json` file is downloaded to the default download location.

**Import Authentication Sources**

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click **Authentication Sources** tile.
2. In the **Authentication Sources** page, click the horizontal ellipsis next to **Add** and then, click **Import**.
3. Click **Browse** and select the authentication sources `.json` file.
4. Enter the same password which you had used during authentication sources export.
5. In case of a conflict, select either **Overwrite existing Authentication Sources** or **Skip Authentication Sources**.
6. Click **Import**.
7. After the import you must re-accept all untrusted certificates manually.

**Important Points**

- An error message is displayed with details for failed imports.

## Delete and Deregister a VMware Single Sign-On Authentication Source

Deleting the VMware Single Sign-on authentication source automatically removes users and user groups that are associated with the VMware Single Sign-On authentication source.

- A VMware SSO authentication source must be configured.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Authentication Sources** tile.
2. Select the **VMware SSO** authentication source, and then click the vertical ellipsis and select **Delete**.
3. Enter the **User Name** and **Password** of the vCenter Server user account that can log in to the VMware SSO host machine.
4. Select **Force Delete** if you cannot access the single sign-on server. VCF Operations continues to attempt to clean the VMware Single Sign-On registration even when errors occur.
5. Click **OK**.

## Give Administrator Access to AD or LDAP Users

You can assign users from AD or LDAP administrator access to VCF Operations so that you can distribute the workload among domain users.

**Before You Proceed**

- Know the credentials of the AD or LDAP account.
- Create the group of users who will get administrative access in AD or LDAP.
- To assign administrator access to users from AD or LDAP in VCF Operations, you must log in as administrator in VCF Operations and import the user group to whom you would like to grant administrator access from AD or LDAP.

1. In a Web browser, navigate to the master node administration interface: `https://master-node-name-or-ip-address/admin`.

2. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click **Administrator Settings**.
3. Expand the Active Directory/Open LDAP Integration section.
4. Click the **Activate AD/Open LDAP** button to activate the setting.
5. In the Domain/Subdomain setting, provide the FQDN of the domain name, for example, mydomain.com. Do not provide an IP address. This domain name must be DNS-resolvable.
6. Expand Advanced Settings and change the default settings.

**Table 1447: Advanced Settings**

| Setting | Description |
|---|---|
| Host | The host name is populated to Auto by default. You can select a host from the drop-down list. |
| Port | The port number is populated depending on the **Use SSL/TLS** selection in the domain name. |
| Base DN | The Base Distinguished Name for users is populated based on the domain. Optionally, enter the DN from which to start user searches. For example, cn=Users,dc=myCorp,dc=com. |
| Common Name | The common name is populated to `userPrincipalName` by default. You can change this value from the drop-down menu. |

7. Expand Search Criteria and change the default settings.

**Table 1448: Search Criteria Settings**

| Setting | Description |
|---|---|
| Group Search Criteria | The group search criteria is populated to `(|(objectclass=group)(objectclass=groupofnames)(objectclass=groupOfUniqueNames))` by default. |
| User Search Criteria | The user search criteria is populated to `(|(objectclass=user)(objectclass=person)(objectclass=inetOrgPerson)(objectclass=organizationalPerson))` by default. |
| Member Attribute | The member attribute is populated to member by default. |

8. Import the SSL certificate. You can import only one SSL certificate PEM file. The imported SSL certificate PEM file can contain more than one certificate. For more details, see Broadcom knowledge base article 2046591.
9. Click **Select User Group**. A new dialog box opens.

**Table 1449: Select User Group Settings**

| Setting | Description |
|---|---|
| **Credentials** > **Username** | Enter the AD/LDAP username. |
| **Credentials** > **Password** | Enter the AD/LDAP password. |
| Search box | Search for the user group name or distinguished name. |
| Select | Select for the user group name or distinguished name to give administrative access to its users. |

10. Click **TEST** to verify that the AD or LDAP connection works.
11. Click **SAVE**.

After AD or LDAP users get administrative rights, they cannot see the local administrator password, but they can reset it. You cannot add or remove any user from the group or view the list of users from VCF Operations.

AD or LDAP integration in VCF Operations supports all three formats: `"DOMAIN\user_name"`, `"CN=username,DC=domain,DC=com"`, and `"user_name@domain.com"`. The Admin UI only supports the email format.

**Note:**

As an administrator, if you face any issues while assigning users from AD or LDAP administrator access to VCF Operations:

- Ensure that the users have "`distinguishedName`" attribute in the LDAP records.
- Ensure that the users have enough permissions to perform the following LDAP queries:
  - `(|(CN=...)(sAMAccountName=...)(userPrincipalName=...)(mail=...))`. If the user name is given in the DN format (e.g. "`CN=username,DC=domain,DC=com`"), the query is not performed.
  - `&((|(objectclass=group)(objectclass=groupofnames)(objectclass=groupOfUniqueNames))(member=...))`, or custom group search criteria if it was modified.

## Audit Users and the Environment in VCF Operations

At times, you might need to provide documentation as an evidence of the sequence of activities that took place in your VCF Operationsenvironment. Auditing allows you to view the users, objects, and information that is collected. To meet audit requirements, such as for business critical applications that contain sensitive data that must be protected, you can generate reports on the activities of your users, the privileges assigned to users to access objects, and the counts of objects and applications in your environment.

Auditing reports provide traceability of the objects and users in your environment.

| | |
|---|---|
| **User Activity Audit** | Run this report to understand the scope of user activities, such as logging in, actions on clusters and nodes, changes to system passwords, activating certificates, and logging out. |
| **User Permissions Audit** | Generate this report to understand the scope of user accounts and their roles, access groups, and access privileges. |
| **System Audit** | Run this report to understand the scale of your environment. This report displays the counts of configured and collecting objects, the types and counts of adapters, configured and collecting metrics, super metrics, applications, and existing virtual environment objects. This report can help you determine whether the number of objects in your environment exceeds a supported limit. |
| **VC Pricing Migration Audit** | Run this report to understand the details of VC pricing migration. |
| **Chargeback Migration Audit** | Run this report to understand the details of Chargeback migration. |
| **System Component Audit** | Run this report to display a version list of all the components in your environment. |

### Reasons for Auditing Your Environment

Auditing in VCF Operations helps data center administrators in the following types of situations.

- You must track each configuration change to an authenticated user who initiated the change or scheduled the job that performed the change. For example, after an adapter changes an object, which is associated with a specific object identifier at a specific time, the data center administrator can determine the principal identifier of the authenticated user who initiated the change.
- You must track who made changes to your data center during a specific range of time, to determine who changed what on a particular day. You can identify the principal identifiers of authenticated users who were logged in to VCF Operations and running jobs, and determine who initiated the change.
- You must determine which objects were affected by a particular user during a time-specific range of time.
- You must correlate events that occurred in your data center, and view these events overlayed so that you can visualize relationships and the cause of the events. Events can include login attempts, system start up and shutdown, application failures, watchdog restarts, configuration changes of applications, changes to security policy, requests, responses, and status of success.
- You must validate that the components installed in your environment are running the latest version.

## User Activity Audit

The user activity report helps you understand the scope of user activities in yourVCF Operations instance, such as when users logged in, actions they took on clusters and nodes, changes they made to system passwords, when they activated certificates, and when they logged out.

### Where You Audit User Activity

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Audit** tile. The activities that users performed in the environment appear on the page.

**Table 1450: User Activity Audit Actions**

| Option | Description |
|---|---|
| Download | Download the user activity audit information to a report in PDF or XLS format. |
| Configure | Configure the settings to send the user activity log to an external syslog server to meet security auditing requirements.<br>**Note:**<br>Audit events are only forwarded if the underlying log4j2 configuration on the node handling audit logging already defines a logger category for AuditLog with an appender reference. On a standard deployment, this is pre-configured. If enabling forwarding has no effect (no events reach the external syslog server despite this setting being enabled), verify that the node's log4j2 configuration contains the following:<br>1. logger.comintegrienalivecommonauditAuditLog.name = AuditLog<br>2. logger.comintegrienalivecommonauditAuditLog.level = INFO<br>3. logger.comintegrienalivecommonauditAuditLog.appenderRef.AuditLogAppender.ref = AuditLogAppender<br>• Output log to external syslog server: When selected, VCF Operations sends the log to a separate server machine.<br>• IP Address or Host Name: Identification for the syslog server.<br>• Port: VCF Operationsport used to send the audit information to the external server. |
| Date Range | Display the list of user activities performed in the past based on a selected number of hours, days, weeks, months, or years, or between two specific dates and times. |
| Starting Line | Indicates the starting line of the file. 0 is for the first line. -1 or no value indicates that the file has to be displayed from the end. |
| Number of Lines | Specifies the number of lines to be displayed in the search result. For example: If you want to see the first 10 occurrences of a particular chunk of text, enter the number of lines as 10 and the starting line as 0. |
| Filter | Filters the data according to User ID, Subject, Auth Source, Session, Origin, Audit ID, and Result. |

## User Permissions Audit

A user permissions audit report provides an overview of the local users and LDAP imported users in your VCF Operations instance, and a list of groups to which each user belongs. This report helps you understand the scope of the user accounts and their roles, access groups, and access privileges in your environment.
The report displays the access group associated with each local user and the access privileges granted to the user in each access group. This report does not include vCenter users, roles, or privileges.

When a user is a member of a specific user group, the associated access group could provide the user with access to configuration, dashboards, and templates, or to specific navigation areas in the user interface such as Administration. The access rights associated with the access group include actions for each access group, such as the ability to add, edit, or delete dashboards, or to view, configure, or manage objects.

**Where You Audit User Permissions**

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Audit** tile.
2. Click the **User Permissions Audit** tab.

The permissions assigned to users, and their associated access groups and access privileges, appear on the page.

**Table 1451: User Permissions Audit Actions**

| Option | Description |
|---|---|
| Download | Download the user permissions audit information to a report in PDF or XLS format. |

## System Audit for VCF Operations

A system audit report provides an overview of the counts of objects, metrics, super metrics, applications, and custom groups in your VCF Operations instance. This report can help you understand the scale of your environment.

The system audit report displays the types and number of objects that VCF Operations manages. Reported objects include those that are configured and collecting data, the types of objects, object counts for adapters, the metrics that are configured and being collected, super metrics, VCF Operations generated metrics, the number of applications used, and the number of custom groups.

You can use this report to help determine whether the number of objects in your environment exceeds a supported limit.

**Where You Audit the System**

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Audit** tile.
2. Click the **System Audit** tab.

The objects and their associated counts appear in the report.

**Table 1452: System Audit Actions**

| Option | Description |
|---|---|
| Download | Download the system information to a report in PDF or XLS format. |

## System Component Audit

A system component audit report provides a version list of every component installed in the system.

**Where You Audit System Components**

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Audit** tile.
2. Click the **System Component Audit** tab.

A list of components installed in the environment appears on the page.

**Table 1453: System Component Audit Actions**

| Option | Description |
|---|---|
| Download | Display the version information in a new browser window. |

## VC Pricing Migration Audit

The vCenter Pricing Migration Audit provides details of the migration of pricing cards to VCF Operations policies.

### Where You Audit vCenter Pricing Migration Components

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Audit** tile.
2. From the **Audit** page, click **Chargeback Audit tab** > **VC Pricing Migration Audit tab**.

**Table 1454: vCenter Pricing Migration Audit Actions**

| Option | Description |
|---|---|
| Download | Download the vCenter Pricing migration information to a report in PDF or XLS format. |
| Date Range | Display the list of vCenter Pricing migration activities performed in the past based on a selected number of hours, days, weeks, months, or years, or between two specific dates and times. |
| Starting Line | Indicates the starting line of the file . 0 is for the first line. -1 or no value indicates that the file has to be displayed from the end. |
| Number of Lines | Specifies the number of lines to be displayed in the search result. For example: If you want to see the first 10 occurrences of a particular chunk of text, enter the number of lines as 10 and the starting line as 0. |
| Filter | Filters the data according to User ID, User Name, Auth Source, Session, Client IP, Category, and Message. |

## Chargeback Migration Audit

The Chargeback Migration Audit provides details of the migration of VMware Chargeback configurations and content to VCF Operations.

### Where You Audit Chargeback Migration Components

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Audit** tile.
2. From the **Audit** page, click **Chargeback Audit tab** > **Chargeback Migration Audit tab**.

**Table 1455: Chargeback Migration Audit Actions**

| Option | Description |
|---|---|
| Download | Download the Chargeback migration information to a report in PDF or XLS format. |
| Date Range | Display the list of Chargeback migration activities performed in the past based on a selected number of hours, days, weeks, months, or years, or between two specific dates and times. |
| Starting Line | Indicates the starting line of the file . 0 is for the first line. -1 or no value indicates that the file has to be displayed from the end. |
| Number of Lines | Specifies the number of lines to be displayed in the search result. For example: If you want to see the first 10 occurrences of a particular chunk of text, enter the number of lines as 10 and the starting line as 0. |

| Option | Description |
|---|---|
| Filter | Filters the data according to User ID, User Name, Auth Source, Session, Client IP, Category, and Message. |

## User Preferences in VCF Operations

You can configure the user preferences to determine the VCF Operations display options, such as the number of metrics and groups to display and whether to synchronize system time with the host machine.

In the VCF Operations navigation bar at the top, click the

icon, and then click **Preferences**. The user preference settings appear in the dialog box.

**Table 1456: User Preference Settings**

| Option | Description |
|---|---|
| Display | Configure how many metrics and root cause groups to display.<br>• Color scheme: Set the user interface to display in light or dark colors. |
| Language and Time | • Language: Select the language you want VCF Operations to be displayed in. You can select the browser language or select a language from the drop-down list.<br>• Font: Select the font for reports.<br>Synchronize the time used for the VCF Operations instance, and display the updated time when VCF Operations communicates with the host machine.<br>• Browser time. All dates and times displayed in the user interface use the time zone settings of the local browser.<br>• Host time. All dates and times displayed in the user interface use the time zone of the host machine.<br>• Show update time in the application header. Displays the updated time in the top-level header of the VCF Operations user interface. The updated time-stamp appears to the left of the refresh button. Other features, such as dashboards, use the updated time to display data at specific intervals. |
| Product Survey Opt Out | Allows you to opt out of providing regular feedback from the VCF Operations UI. |
| Account | Change the password for the user account. |

# VCF Operations Support Logs for Product UI

Use the expandable tree of VCF Operations log files for troubleshooting issues in the product UI.

### How VCF Operations Support Logs Work

You can browse and load the VCF Operations log files for review and for troubleshooting issues in the product UI.. You can also edit the log file folders, limit the retained log size, and set logging levels.

VCF Operations logs are categorized by cluster node, and log type. All logs are in the UTC formatted date and time. The logging format is as follows:

```
Date/Time+0000, LEVEL, [THREAD/IP Address], [Specific Fields], CLASS - MESSAGE
```

If you have configured a timezone for the VCF Operations VM, the system logs will be in that timezone. The VCF Operations logs will remain in UTC.

### Where You Find VCF Operations Support Logs

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Support Logs** tile.

## Log Viewer Options

Use the toolbar options to control the tree of items and the viewer.

1. Click **Node** and select any component that is listed under the node.
2. Click the gear icon, enter the logging levels and log size.
3. Click **OK**.

> **Note:**
> Not all components have relevant syslog information. Therefore, not all nodes have the configuration option activated.

**Figure 474: Log Options**

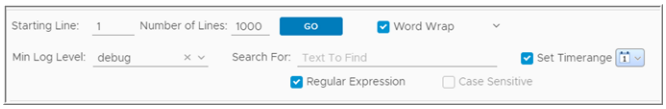



**Table 1457: Log Viewer Toolbar Options**

| Option | Description |
|---|---|
| Group By | Organizes the tree by cluster node or log type. |
| Collapse All | Closes the view of the tree to show only the high-level folders. |
| Edit Properties | For the selected folder, you can limit the log size and set logging levels. |
| Delete Selected File | Deletes the log file. |
| Starting Line | Indicates the starting line of the file .<br>0 is for the first line. -1 or no value indicates that the file has to be displayed from the end. |
| Number of Lines | Specifies the number of lines to be displayed in the search result.<br>For example:<br>If you want to see the first 10 occurrences of a particular chunk of text, enter the number of lines as 10 and the starting line as 0. |
| Min Log Level | If you specify the minimum log level, the logs for that particular log level and higher are shown.<br>For example:<br>If you select **warning**, the logs having the same log level (**warning**) and higher are shown . |
| Text to Find | Enter the specific text that you want to search in the logs. Add the following filters for search, if required:<br>• **Case Sensitive**<br>• **Regular Expression**<br>You can perform the search at various levels:<br>• On a single file: Use this option if you want to search a single log file .<br>• On all the log files of an entity: Use this option if you want to search all the log files of an entity such as a log type or folder.<br>• On all the log files of a node: Use this option if you want to search all the log files that are grouped under a node.<br>The last modified time for any file is found by placing the pointer on the file in the tree. |
| Set Timerange | If you specify a time range, the logs for that particular time range are shown in the search results. |
| Word Wrap | If you select this option, the part of the line that does not fit on the screen is moved to the next line. If you do not select this option, a scroll bar is provided to see the complete line. |

# Cluster Management

VCF Operations includes a central page where you can monitor and manage the nodes in your VCF Operations cluster and the adapters that are installed on the nodes.

## How Cluster Management Works

Cluster management lets you view and change the online or offline state of the overall VCF Operations cluster or the individual nodes. In addition, you can activate or deactivate high availability (HA) and view statistics related to the adapters that are installed on the nodes.

## Where You Find Cluster Management

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Cluster Management** tile.

## Cluster Management Options

The options include cluster-level monitoring and management features.

**Table 1458: Initial Setup Status Details**

| Option | Description |
|---|---|
| Actions | Expand Actions.<br>• Click **Network Time Protocol Settings** to set the global network time protocol settings for your cluster. For more information see Configuring the NTP Settings for VMware vSphere Foundation Components.<br>• Click **Rebalance** to rebalance the adapter, disk, memory, or network load across VCF Operations cluster nodes to increase the efficiency of your environment. |
| Cluster Status | Displays the online, offline, or unknown state of the VCF Operations cluster.<br>Once CA is activated, it displays the status of the two fault domains. |
| High Availability | Indicates whether HA is activated, deactivated, or degraded. |
| Continuous Availability | Indicates whether CA is activated, deactivated, or degraded. |

VCF Operations provides node-level information and a toolbar for taking nodes online or offline.

**Table 1459: Nodes in the VCF Operations Cluster**

| Option | Description |
|---|---|
| Node Name | Machine name of the node.<br>The node that you are logged into displays a dot next to the name. |
| Node Address | Internet protocol (IP) address of the node. Primary and replica nodes require static IP addresses. Data nodes can use DHCP or static IP. |
| Cluster Role | Type of VCF Operations node: primary, data, replica, or witness node (only applicable for CA). |
| Fault Domain | Displays the fault domain a node is associated to in a CA activated cluster.<br>**Note:**<br>This column appears only if CA is activated. |
| Node Pair | Displays which pair the node belongs to. For example, in CA, nodes are added in pairs. If there are four nodes, the column displays whether the node is part of pair one or two.<br>**Note:**<br>This column appears only if CA is activated. |

| Option | Description |
|---|---|
| State | Running, Not Running, Going Online, Going Offline, Inaccessible, Failure, Error |
| Status | Online, offline, unknown, or other condition of the node. |
| Objects in Process | Total environment objects that the node currently monitors. |
| Objects Being Collected | Total environment objects that the node collected. |
| Metrics in Process | Total metrics that the node has discovered since being added to the cluster. |
| Metrics Being Collected | Total metrics the node has collected since being added to the cluster. |
| Version | Displays the VCF Operations software version and the build number installed on the node. |

In addition, there are adapter statistics for the selected node.

**Table 1460: Adapters on Server**

| Option | Description |
|---|---|
| Name | Name that the installing user gave to the adapter. |
| Status | Indication of whether the adapter is collecting data or not. |
| Objects Being Collected | Total environment objects that the adapter currently monitors. |
| Metrics Being Collected | Total metrics that the adapter has collected since being installed on the node. |
| Last Collection Time | Date and time of the most recent data collection by the adapter. |
| Added On | Date and time when the adapter was installed on the node. |

## Cluster Management using the VCF Operations Admin UI

VCF Operations includes a central page where you can monitor and manage the nodes in your VCF Operations cluster and the adapters that are installed on the nodes.

### How Cluster Management Works

You can view and change the online or offline state of the overall VCF Operations cluster or the individual nodes. In addition, you can activate or deactivate high availability (HA) and continuous availability (CA) and view statistics related to the adapters that are installed on the nodes.

### Where You Find Cluster Management

Log in to the VCF Operations administration interface at https://*master-node-name-or-ip-address*/admin.

### Cluster Management Options

The options include cluster-level monitoring and management features.

**Table 1461: Initial Setup Status Details**

| Option | Description |
|---|---|
| Cluster Status | Displays the online, offline, or unknown state of the VCF Operations cluster and provides an option to take the cluster online or offline.<br>If a cluster fails to go offline, click the **Force Take Offline** button to take the cluster offline.<br>**Note:**<br>The Force Take Offline button appears only when the Bring Cluster offline operation fails.<br>You can select to display the reason for taking the cluster offline. Select the **Show reason on maintenance page** check box in the **Take Cluster Offline** dialog box. When you log in to VCF Operations when the cluster is offline, the reason for taking the cluster offline is displayed. |
| High Availability | Indicates whether HA is activated, deactivated, or degraded and provides an option to change that setting. |
| Continuous Availability | Indicates whether CA is activated, deactivated, or degraded and provides an option to change that setting. |

VCF Operations provides node-level information as well as a toolbar for taking nodes online or offline.

**Table 1462: Nodes in the VCF Operations Cluster**

| Option | Description |
|---|---|
| Generate Passphrase | Generate a passphrase that can be used instead of the administrator credentials to add a node to this cluster. |
| Add New Node | Add a new node to the this cluster. You cannot add a witness node. |
| Take Node Online/Offline | You can select the required node and bring it online or offline. You are required to understand the risk involved and provide a valid reason for the action performed when you bring a node online or offline. |
| Remove Node | Remove node from the cluster without any loss of collected data. Data nodes must be removed by shrinking. |
| Reload Nodes | Reload data in the screen. |

| Option | Description |
|---|---|
| Shrink Cluster | This option provides a mechanism to remove a node without having to lose any data. The shrink cluster removes nodes by migrating data from one node to any other node.<br>All the historical data is either moved to the primary node or any other node, which has sufficient disk space.<br>If HA is activated and you have selected the replica node for removal, then you are asked to select another replica node. VCF Operations provides a list of nodes that be a possible candidate to become a replica node.<br>VCF Operations stops collecting data from the removed nodes. However, the data that is available in the removed node is migrated to an existing node. Once the migration is complete, then the removed nodes are deleted with the cluster state as offline.<br>**Note:**<br>VCF Operations cannot move pinned adapters. The adapter instances which were pinned on removed nodes do not move to another collector automatically. You must change the collector before starting the shrink cluster process. |
| Replace Node | You cannot remove one node in a CA cluster, because the nodes are in pairs. Use the replace node option to replace one of the nodes in the CA cluster. The following rules apply:<br>• Replace master node - not allowed<br>• Replace data node -<br>— Single data node - allowed<br>— Data nodes from different FDs - allowed<br>— Pair of data nodes - not allowed<br>— Data nodes from the same FD - not allowed<br>• Replace witness node - allowed |

**Table 1463: Nodes in the VCF Operations Cluster**

| Option | Description |
|---|---|
| Node Name | Machine name of the node.<br>The node that you are logged into displays a dot next to the name. |
| Node Address | Internet protocol (IP) address of the node. Primary and replica nodes require static IP addresses. Data nodes may use DHCP or static IP. |
| Cluster Role | Type of VCF Operations node: primary, data, or replica. |
| State | Powered on, powered off, unknown, or other condition of the node. |
| Status | Online, offline, unknown, or other condition of the node. |
| Objects | Total environment objects that the node currently monitors. |
| Metrics | Total metrics that the node has collected since being added to the cluster. |
| Build | VCF Operations software build number installed on the node. |
| Version | VCF Operations software version installed on the node. |
| Deployment Type | Type of machine on which the node is running: vApp |
| SSH Status | Activate or deactivate the SSH Status. |

In addition, there are adapter statistics for the selected node.

**Table 1464: Adapters on Server**

| Option | Description |
|---|---|
| Name | Name that the installing user gave to the adapter. |
| Status | Indication of whether the adapter is collecting data or not. |
| Objects | Total environment objects that the adapter currently monitors. |
| Metrics | Total metrics that the adapter has collected since being installed on the node. |
| Last Collection Time | Date and time of the most recent data collection by the adapter. |
| Added On | Date and time when the adapter was installed on the node. |

## Monitoring Data Collection

The collection status page provides an overview of the data that is being collected at the cluster level in VCF Operations. You can view the details for each collector and you can also view the adapter instance within the collector. The collection status page also provides recommendations in case of any issues caused by the collection mechanism.

The collection status page summarizes the number of objects and metrics that are getting collected by the VCF Operations collectors and adapter instances. You can view the status of each collector and adapter instance and find issues, if any. If there are anomalies, the collection status page provides recommendations to resolve those issues.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Collection Status** tile.

**Table 1465: Collection Status Overview Options**

| Option | Description |
|---|---|
| Overview | You can view the overall number of objects and metrics that is collected by your cluster. The charts underneath the numbers represent the graphical view of the collected data. Hover over the charts to view their values. |
| Collectors | You can view the total number of collectors, including data collectors in your cluster. If any of the collectors stop working, it is reported here. |
| Adapter Instances | You can view the total number of adapter instances that are receiving data. If any of the adapter instances do not receive data, it is reported here. |

2. Click the **Show Issues Only** check box to filter collectors and adapter instances that have issues.
3. By default, the collection cycle is displayed in the **Topology** structure. Using the topology view, you can clearly view the flow of data from the adapter instances to the collectors, and from the collectors into the cluster.
   Alternatively, you can click the **List View** icon to view the collection cycle in a tabular structure.

**Table 1466: Collection Status Topology View**

| Option | Description |
|---|---|
| Cluster | The VCF Operations cluster collects data using its collectors. Hover over the cluster to view the name and type of the cluster. |
| Collectors | All the collectors that are part of a cluster are listed. You can view the number of objects and metrics being collected. Hover over the collector to view the name and type of the collector. Click the **Expand** icon to view the adapter instances. |

| Option | Description |
|---|---|
| Adapter Instances | All the adapter instances that are part of a collector are listed. You can view the number of objects and metrics that are being collected. Hover over the adapter instance to view the name and type of the adapter. |

4. The collection status of each of the instances are colour coded.

**Table 1467: Collection Status of Adapter Instances**

| Colour | Status | Description |
|---|---|---|
| Green | Collecting. | Resource is receiving data. |
| Grey | Stopped | Resource manually stopped by user. |
| Yellow | Warning | Resource is receiving data but has a problem. You can view the warning message and recommendation on how to resolve it. |
| Red | Failed | Resource fails to collect data due to some problem. |

5. Click the collector to view the details.

| Option | Description |
|---|---|
| Details | |
| Name | Name of the collector. |
| IP Address | Internet protocol (IP) address of the collector. |
| Status | Status of the collector.<br>**Note:**<br>If any of the adapter instances within the collector has an anomaly, the status will reflect as warning. |
| Uptime | Total time elapsed since the collector started receiving data. |
| Creation Date | The day the collector was created. |
| Last HeartBeat | The last heart beat from the collector in the defined interval. |
| Version | The version of the collector. |
| Adapters | Total number of adapter instances in the collector. |
| Performance Details | |
| CPU | The average percentage of CPU used by the collector. |
| Memory | The percentage of memory used by the collector. |
| Data Collection Details | |
| Objects | Number of objects collected. |
| Metrics | Number of metrics collected. |
| Started Adapters | Number of adapter instances receiving data. |
| Threads | Number of threads collector service that is being used. |

6. Click the adapter instance to view its details.

| Option | Description |
|---|---|
| Details | |
| Name | Name of the adapter instance. |

| Option | Description |
|---|---|
| Status | Status of the adapter instance.<br>**Note:**<br>If the adapter instance has a problem, the status will reflect as warning and also display the recommendation to resolve it. |
| Data Collection Details | |
| Objects | Number of objects collected. |
| Metrics | Number of metrics collected. |
| Events | Number of events collected. |
| New Objects | Determines whether new objects are collected. |
| New Metrics | Determines whether new metrics are collected. |
| New Properties | Determines whether new properties are collected. |
| Property Value Changes | Determines if the property values have changed. |
| Relationship Updates | Determines if there are any changes in the relationship. |
| Elapsed Collection Time | Duration of the last collection cycle. |

# Managing Content

As a VCF Operations administrator, you can take regular backups of your custom and the out-of-the-box content to manage your operational or regulatory needs. If there is a wrong edit or if the need to recover data arises, then you can use the recent backup to restore the content or import the content to a different setup. By taking regular backups, you can also upgrade the VCF Operations to the latest build without losing or overriding the custom content.

**Note:**
Any user with the "Content Management" permission can export the content. However, only a admin user or the user assigned to the Administrator role has the privilege to export all the content, including the content owned by other users, for example, custom dashboards.

When you use a VCF single sign-on (SSO) account that has a VCF administrator role, you should assign the correct role for exporting and importing content within the VCF Operations component. To assign the VCF Operations component role to a VCF SSO account, navigate to **Operate** > **Administration** > **Control Panel** > **Access Control**, and then choose the VCF SSO account to assign the appropriate group or role.

## Creating a Backup

You can create regular backups of your custom and the out-of-the-box content in VCF Operations. You can use this backup to restore your content or export the content while setting up another environment.

You can either import content or export content at a given point in time. You can select the content or the configuration type that you want to export. When you export, ensure you export Configuration and Content separately. Some content types have dependencies on other types. The dependencies, when not selected, will not be exported. When you import, you must import the Configuration before you import the Content so that all user related information is imported first. Select all the required types to ensure that are no missing items when importing.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Content Management** tile.
2. In the **Export** tab, select the content and configurations that you want to export. You can take a backup of the following content types and configurations available in VCF Operations.

   **Note:**
   Click **Include out-of-the-box content** under the Content section to include the out-of-the-box content while exporting. Click **Select All** under both Content and Configuration sections to select all the content and configurations respectively, while exporting.

| Content | |
|---|---|
| • Dashboards<br>• Views<br>• Reports<br>• Report Schedules<br>• Configuration Files<br>• Alert Definitions<br>• Symptom Definitions<br>• Log Query Configuration<br>• Log Extracted Fields<br>• Log Agent Groups<br>• Log Templates | • Recommendations<br>• Notifications<br>• Payload Templates<br>• Policies<br>• Custom Groups<br>• Custom Metric Groups<br>• Super Metrics<br>• Compliance Scorecards<br>• Configuration Management Templates |
| Configuration | |
| • Authentication Sources<br>• Users<br>• User Groups<br>• User Roles<br>• Integration Accounts<br>• Http Proxies<br>• Outbound Settings | • Cost Drivers<br>• Custom Services<br>• Service Based Applications<br>• Rule Based Applications<br>• Application Definition Assignments<br>• Custom Profiles<br>• Global Settings |

   **Note:**

   The Log Management options are displayed only when the component is deployed.

3. Click **Export** to create a backup.

   **Note:**
   For configurations such as Integration Accounts, HTTP Proxies, Outbound Settings, Users, and Authentication Sources that have sensitive information, you must set up a new password to export data. The password should be at least 14 characters long.

   The system compresses the content into one ZIP file. After the export is complete, the Download ZIP file link is available in the Export tab.
4. Click the **Download ZIP file** link to download the backup content.

   You can use the downloaded content to restore your content or export it to a different setup.

   **Note:**

   The ZIP file generated by a certain user will not overwrite that of another user, that is, if User A generates the ZIP file after User B, the latter's ZIP file will not be overwritten.

## Importing Content

You can take regular backups of your custom and the out-of-the-box content and import it to a different environment.

- Ensure that you have downloaded the backup ZIP file. For details, see Creating a Backup.

- Ensure that all the users who own the custom dashboards or report schedules are present in the destination setup so that the custom content is assigned to the respective owners when the content is imported. Otherwise, the custom content of the owners who are not present in the destination setup will be skipped while importing the content.

  **Note:**

  You can either import content or export content at a given point in time. When you export, ensure you export Configuration and Content separately. When you import, you must import the Configuration before you import the Content so that all user related information is imported first.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Content Management** tile.
2. Click the **Import** tab and then, click **Browse** to select the downloaded ZIP file with the exported content.

   The data included in the ZIP file is displayed in the 'Data Available' table.
3. If there is a conflict while importing the content, you can select to either **Overwrite existing content** or **Skip item(s)**.

   The import report with the time-stamp is displayed after the import operation is complete. You can view this information under the **Results** section in the same page.
4. For content types with sensitive information, enter the password that you had set while exporting the content.
5. Click **Import**.

   After the import is completed, the content is available in the destination setup.

### Best Practices for Migrating Content

Follow the below practices to ensure that your content is successfully migrated.

- Use the *admin* user account or a user assigned to the Administrator role to export all of the content, including other users' custom content, such as dashboards and report schedules.
- Before importing the content, ensure that the Management Packs to which the content is related is installed on the destination setup.

## Managing Orphaned and Unassigned Content

Dashboards, report schedules, and credentials created by deleted users or that are unassigned in VCF Operations are stored in the Orphaned and Unassigned Content page. As an administrator, you are the new owner of these dashboards, reports schedules, and credentials. You can transfer the ownership of dashboards and report schedules and assign the credentials to new users or to yourself.

#### From Where You Can Transfer Ownership of Dashboards, Report Schedules, and Credentials

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Orphaned and Unassigned** tile.

#### Orphaned and Unassigned Page

You can view a list of all deleted users or unassigned users from the **Deleted Users** panel in the left pane of the **Orphaned and Unassigned** page. Based on your selection in the **Deleted Users** panel, the dashboards, report schedules, and credentials for the deleted user are displayed under the **Dashboard**, **Report Schedules**, **Credentials** tabs in the **Orphaned and Unassigned** page.

As an admin user, you can take ownership, assign ownership, or discard orphaned dashboards, report schedules and credentials, from the **Actions** menu in the **Dashboards**, **Report Schedules**, and **Credentials** tabs. Enter the name or part of the name of a dashboard, report schedule, or credential in the **Filter** option and click **Enter**. The relevant dashboard, report schedule, or credential is displayed.

**Table 1468: Actions Menu Options**

| Actions | Options |
|---|---|
| Take Ownership | You can take ownership of the selected dashboard, report schedule, or credential. |
| Assign Ownership | You can assign a new owner for the selected dashboards or report schedules. You can select a target user from the **Transfer Dashboards/Report Schedule/Credential** dialog box. |
| Discard | You can permanently delete the dashboard, report schedule, and unassigned credentials that are not in use in any adapter instance. You cannot delete an active credential that is being used by an adapter instance even if it is unassigned. For more information on credentials, see Configuring Credentials in Integrations. |

## Adding Physical Data Centers in VCF Operations

In VCF Operations, you can create and manage physical data centers and associate cloud accounts with them.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Physical Data Centers** tile.
2. On the Physical Data Centers page, click **Add**.
   Add Physical Data Centers page opens.
3. Enter a **Name** for the physical data center.
4. Select a location on the map.
   You can use the search bar to find a specific location. You can also zoom in or zoom out for a better view of the world map and find a suitable location.
5. Double-click on the selected location to create the data center.
   The location name and coordinates are reflected in the **Location** field.
6. From the **Associate Cloud Account(s) with this Physical Data Center** section, select the cloud account(s) you want to assign to this physical data center.

   **Note:**

   The Associate Cloud Account(s) with this Physical Data Center section only lists the configured vCenter and VCF accounts. You can assign multiple cloud accounts to a physical data center.
7. Click **Save**.
   The physical data center is created.
8. View the new physical data center on the Physical Data Center page.

| Option | Description |
|---|---|
| Name | Name of the physical data center you entered at the time of creation. |
| Number of associated cloud accounts | Displays the total number of cloud accounts associated with the physical data center. |
| Filter | Filters the list of physical data centers based on Name. |

9. Optional: Click the vertical ellipsis and select **Edit** to edit the physical data center.
10. Optional: Click the vertical ellipsis or the horizontal ellipsis and then select **Delete** to remove a physical data center.

## VCF Operations Maintenance Schedules

Maintenance schedules identify objects that are in maintenance mode at specific times, which prevents VCF Operations from showing misleading data based on those objects being offline or in other unusual states because of maintenance.

Many objects in the enterprise might be intentionally taken offline. For example, a server might be deactivated to update software. If VCF Operations collects metrics when an object is offline, it might generate incorrect anomalies and alerts that affect the data for setting dynamic thresholds for the object attributes. When an object is identified as being in maintenance mode, VCF Operations does not collect metrics from the object or generate anomalies or alerts for it. In addition, VCF Operations cancels any active symptoms and alerts for the object.

If an object undergoes maintenance at fixed intervals, you can create a maintenance schedule and assign it to the object. For example, you can put an object in maintenance mode from midnight until 3 a.m. each Tuesday night. You can also manually put an object in maintenance mode, either indefinitely or for a specified period of time. These methods are not mutually exclusive. You can manually put an object in maintenance mode, or take it out of maintenance mode, even it if has an assigned maintenance schedule. For more information, see Manage Maintenance Schedules for Your Object Workspace.

### How Maintenance Schedules Work

Maintenance schedules require that you select the days and time-of-day when updates or other object maintenance occurs. Note that creating a maintenance schedule does not activate the schedule. A maintenance schedule must be part of a policy before the schedule can take effect. For more information, see Maintenance Schedule Details.

### Where You Find the Maintenance Schedules

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Maintainance Schedules** tile.

Click **Add** or click the **Vertical Ellipses** to edit, or remove items.

**Table 1469: Maintenance Schedule Toolbar Options**

| Option | Description |
|---|---|
| Add | Open a window in which you can select the maintenance schedule settings for a new schedule. |
| Edit | Change the maintenance schedule settings for an existing schedule. |
| Delete | Remove the selected maintenance schedule. |

## Manage Maintenance Schedules

Add or edit a maintenance schedule to take an object offline. VCF Operations does not collect data from an object that is offline.

### Where You Find Manage Maintenance Schedules

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations**, and then click the **Maintainance Schedules** tile.
2. Click **Add** or click the **Vertical Ellipses** to edit, or remove items.

**Table 1470: Manage Maintenance Schedule Add or Edit Options**

| Option | Description |
|---|---|
| Schedule Name | Name that describes the maintenance schedule. |
| Time Zone | Time zone in which you are currently located. |
| Days | Number of days the maintenance period covers. |

| Option | Description |
|---|---|
| Recurrence | Specify a maintenance schedule to run over a selected period.<br>• Once<br>• Daily<br>• Weekly<br>• Monthly |
| Expire after | The number of times the schedule is run. |
| Expire on | The date upon which the schedule stops running. |

## Create a VCF Operations Support Bundle

You can create a VCF Operations support bundle to gather log and configuration files for analysis when troubleshooting VCF Operations issues.

Support bundles require the selection of nodes or the entire cluster, and the desired logging level. When you create a support bundle, VCF Operations gathers files from the selected cluster nodes into ZIP files for convenience.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Support Bundles** tile.
2. Click **ADD**.
3. Select the bundle type:
   - **Light support bundle**
   - **Full support bundle**
4. Select the target cluster nodes.

   Only logs from the selected nodes are included in the support bundle.
5. Click **OK**, and then click **OK** again to confirm support bundle creation.

   Support bundle creation time varies depending on the size of the logs and number of selected nodes.

   The support bundle is created.
6. Once the support bundle is created, use the data grid options to view details.

**Table 1471: Support Bundle Data Grid Options**

| Option | Description |
|---|---|
| Bundle | System-generated identifier for the support bundle. |
| Bundle Type | • Light: Include 24 hours of logs.<br>• Full: Includes all available logs and configuration files. |
| Date and Time Created | Time when support bundle creation began. |
| Status | Progress of support bundle creation. |

7. Click the **Hortizontal Ellipses** to delete, download, or reload support bundles.

   **Note:**

   You can download the support bundle ZIP files for analysis.

   Review the log files for error messages. If troubleshooting assistance is required, send the diagnostic data to VMware Support. Use the toolbar to delete the outdated support bundle to save disk space when the issue is resolved or closed.

## VCF Operations Support Bundles

VCF Operations support bundles contain log and configuration files that help troubleshoot a VCF Operations issue.

### How Support Bundles Work

Support bundles require that you select nodes or the entire cluster, and the level of logging that you want to collect. After VCF Operations creates the support bundle, you download it in ZIP format for analysis.

### Where You Find Support Bundles

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Support Bundles** tile.

### Support Bundle Options

The options include toolbar and data grid options.

You can click **Add** or click the **Hortizontal Ellipses** to delete, download, or reload support bundles.

**Table 1472: Support Bundle Toolbar Options**

| Option | Description |
|---|---|
| Add | Open a dialog box that guides you through the process of creating a support bundle. |
| Delete | Remove the selected support bundle. |
| Download | Download the support bundle in ZIP format. |
| Reload Support Bundles | Refresh the list of support bundles. |

Use the data grid options to view item details.

**Table 1473: Support Bundle Data Grid Options**

| Option | Description |
|---|---|
| Bundle | System-generated identifier for the support bundle. |
| Bundle Type | • Light. Include 24 hours of logs.<br>• Full. Include all available logs and configuration files. |
| Date and Time Created | Time when support bundle creation began. |
| Status | Progress of support bundle creation. |

## VCF Operations Dynamic Thresholds

A threshold marks the boundary between normal and abnormal behavior for a metric. In addition to fixed thresholds, VCF Operations supports dynamic thresholds for a metric, calculated based on historical and incoming data.

### How Dynamic Thresholds Work

By default, dynamic thresholds are refreshed on a regular schedule, but you can recalculate dynamic thresholds outside of the schedule if you want to capture the most recent data.

### Where You Find Dynamic Thresholds

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Dynamic Thresholds** tile.

### Dynamic Threshold Options

The dynamic threshold feature includes options to activate or deactivate the calculation process and to review associated values.

**Table 1474: Dynamic Threshold Options**

| Option | Description |
|---|---|
| Activate | The dynamic threshold calculation process ia activated by default. |
| Deactivate | Stop the dynamic threshold calculation currently in progress. |
| Calculation progress | Percentage completion of the current dynamic threshold calculation. |
| Host time to calculate normal behavior of all objects | Timestamps and metric counts associated with the last dynamic threshold calculation, and the time for the next scheduled calculation. |

# VCF Operations Adapter Redescribe

When VCF Operations redescribes an adapter, VCF Operations finds the adapter files, gathers information about the abilities of the adapter, and updates the user interface with information about the adapter.

### How Adapter Redescribe Works

After installing or updating an adapter, capture the adapter information by having VCF Operations redescribe its adapters.

### Where You Find Adapter Redescribe

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Redescribe** tile.

### Adapter Redescribe Options

The feature includes an option to start the adapter describe process.

**Table 1475: Adapter Redescribe Options**

| Option | Description |
|---|---|
| Redescribe | Start the adapter describe process. |

VCF Operations provides adapter-specific details from the redescribe process.

**Table 1476: Adapter Redescribe Details**

| Option | Description |
|---|---|
| Name | Adapter to which the redescribe process applies. |
| Status | Success, failure, or other condition related to the last redescribe process. |
| Describe Version | Version of `describe.xml` against which the last redescribe process ran. |
| Adapter Version | Version of the adapter against which the last redescribe process ran. |
| Message | Additional details about the last redescribe process. |

# Customizing Icons

Every object or adapter in your environment has an icon representation. You can customize how the icon appears.

VCF Operations assigns a default icon to each object type and adapter type . Taken collectively, object types and adapter types are known as objects in your environment. Icons represent objects in the UI and help you to identify the type of object. For example, in the Topology Graph widget on a dashboard, labeled icons show how objects are connected to one other. You can quickly identify the type of object from the icon.

If you want to differentiate objects, you can change the icon. For example, a virtual machine icon is generic. If you want to pictorially distinguish the data that a vSphere virtual machine provides from the data that a Hypervisor virtual machine provides, you can assign a different icon to each.

## Customize an Object Type Icon

You can use the default icons that VCF Operations provides, or you can upload your own graphics file for an object type. When you change an icon, your changes take effect for all users.

If you plan to use your own icon files, verify that each image is in PNG format and has the same height and width. For best results, use a 256x256 pixel image size.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Icons** tile.
2. Click the **Object Type Icons** tab.
3. Assign the Object Type icon.
   a) Select the object type in the list with the icon to change.
      By default, object types for all adapter types are listed. To limit the selection to the object types that are valid for a single adapter type, select the adapter type from the drop-down menu.
   b) Click the **Upload** icon.
   c) Browse to and select the file to use and click **Done**.
4. Optional: To return to the default icon, select the object type and click the **Assign Default Icons** icon.
   The original default icon appears.

**Related Concepts**

Customizing Icons

Every object or adapter in your environment has an icon representation. You can customize how the icon appears.

## Object Type Icons Tab

VCF Operations obtains data from different sources. Data sources are classified by the type of object or object type. In UI locations where metric data appears for objects, VCF Operations includes an icon to show the object type. To graphically distinguish the different types of objects, you can customize the icon.

### Where You Customize Object Type Icons

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Icons** tile. Click the **Object Type Icons** tab.

**Table 1477: Object Type Icons Options**

| Option | Description |
|---|---|
| Adapter Type | Icons for all adapters are listed by default. To list a subset of the object types that are valid for one type of adapter, select the adapter type. |
| Toolbar options | Manages the selected icon.<br>• **Upload** uploads a PNG file to uniquely identify the object type.<br>• **Assign Default icons** returns the selection to the original icon. |
| Search | Search for objects with a particular name to narrow the selection of object types displayed. |
| Object Type | Name of the type of object. |
| Icon | Pictorial representation of the type of object. |

**Related Concepts**

Customizing Icons

Every object or adapter in your environment has an icon representation. You can customize how the icon appears.

**Related Tasks**

Customize an Object Type Icon

You can use the default icons that VCF Operations provides, or you can upload your own graphics file for an object type. When you change an icon, your changes take effect for all users.

## Customize an Adapter Type Icon

You can use the default icons that VCF Operations provides, or you can upload your own graphics file for an adapter type. When you change an icon, your changes take effect for all users.

If you plan to use your own icon files, verify that each image is in PNG format and has the same height and width. For best results, use a 256x256 pixel image size.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Icons** tile.
2. Click the **Adapter Type Icons** tab.
3. Assign the Adapter Type icon.
   a) Select the adapter type in the list with the icon to change.
   b) Click the **Upload** icon.
   c) Browse to and select the file to use and click **Done**.
4. Optional: To return to the default icon, select the adapter type and click the **Assign Default Icons** icon.

   The original default icon appears.

**Related Concepts**

Customizing Icons

Every object or adapter in your environment has an icon representation. You can customize how the icon appears.

### Adapter Type Icons Tab

Adapters collect and provide data to VCF Operations. Adapters are classified by the type of adapter or adapter kind. To graphically distinguish the different types of adapters, you can customize the icon.

#### Where You Customize Adapter Type Icons

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Control Panel**, and then click the **Icons** tile. Click the **Adapter Type Icons** tab.

**Table 1478: Adapter Type Icons Options**

| Option | Description |
|---|---|
| Toolbar options | Manages the selected icon.<br>• **Upload** uploads a PNG file to uniquely identify the adapter type.<br>• **Assign Default icons** returns the selection to the original icon. |
| Name | Name of the type of adapter. |
| Icon | Pictorial representation of the type of adapter. |

**Related Concepts**

Customizing Icons

Every object or adapter in your environment has an icon representation. You can customize how the icon appears.

**Related Tasks**

Customize an Adapter Type Icon

You can use the default icons that VCF Operations provides, or you can upload your own graphics file for an adapter type. When you change an icon, your changes take effect for all users.

## Allocate More Virtual Memory to VCF Operations

You might need to add virtual memory to keep the VCF Operations process running.

When the VCF Operations virtual machine requests more memory than is available, the Linux kernel might kill the `vcops-analytics` process, and the product might become unresponsive. If that happens, use the reservation feature in vSphere to specify the guaranteed minimum memory allocation for VCF Operations virtual machines.

1. In the vSphere Client inventory, right-click the VCF Operations virtual machine and select **Edit Settings**.
2. Click the **Resources** tab, and select **Memory**.
3. Use the **Reservation** option to allocate more memory.

# Viewing Metrics and Properties

VCF Operations provides definitions for the metrics and properties, defined on objects in your environment.

## Metric Definitions in VCF Operations

Metric definitions provide an overview of how the metric value is calculated or derived. If you understand the metric, you can better tune VCF Operations to display results that help you to manage your environment.

VCF Operations collects data from objects in your environment. Each piece of data collected is called a metric observation or value. VCF Operations uses the VMware vCenter adapter to collect raw metrics. VCF Operations uses the VCF Operations adapter to collect self-monitoring metrics. In addition to the metrics it collects, VCF Operations calculates capacity metrics, badge metrics, and metrics to monitor the health of your system.

VCF Operations collects data from objects in your environment. Each piece of data collected is called a metric observation or value. VCF Operations uses the VMware vCenter® adapter to collect raw metrics. VCF Operations uses the VCF Operations adapter to collect self-monitoring metrics. In addition to the metrics it collects, VCF Operations calculates capacity metrics, badge metrics, and metrics to monitor the health of your system.

All metric definitions are provided. The metrics reported on your system depend on the objects in your environment. You can use metrics to help troubleshoot problems. See Troubleshooting with the Metrics Tab.

All metric definitions are provided. The metrics reported on your system depend on the objects in your environment. You can use metrics to help troubleshoot problems.

### Metrics for vCenter Components

VCF Operations connects to VMware vCenter® instances through the vCenter adapter to collect metrics for vCenter components and uses formulas to derive statistics from those metrics. You can use metrics to troubleshoot problems in your environment.

vCenter components are listed in the `describe.xml` file for the vCenter adapter. The following example shows sensor metrics for the host system in the `describe.xml` file.

```
<ResourceGroup instanced="false" key="Sensor" nameKey="1350" validation="">
    <ResourceGroup instanced="false" key="fan" nameKey="1351" validation="">
        <ResourceAttribute key="currentValue" nameKey="1360" dashboardOrder="1" dataType="float"
 defaultMonitored="false"   isDiscrete="false" isRate="false" maxVal="" minVal=""  unit="percent"/>
        <ResourceAttribute key="healthState" nameKey="1361" dashboardOrder="1" dataType="float"
 defaultMonitored="false"   isDiscrete="false" isRate="false" maxVal="" minVal="" />
    </ResourceGroup>
    <ResourceGroup instanced="false" key="temperature" nameKey="1352" validation="">
        <ResourceAttribute key="currentValue" nameKey="1362" dashboardOrder="1" dataType="float"
 defaultMonitored="false"   isDiscrete="false" isRate="false" maxVal="" minVal="" />
        <ResourceAttribute key="healthState" nameKey="1363" dashboardOrder="1" dataType="float"
 defaultMonitored="false"   isDiscrete="false" isRate="false" maxVal="" minVal="" />
```

```
    </ResourceGroup>
  </ResourceGroup>
```

Each `ResourceAttribute` element includes the name of a metric that appears in the UI and is documented as a Metric Key.

**Table 1479: Sensor Metrics for Host System Cooling**

| Metric Key | Metric Name | Description |
|---|---|---|
| Sensor\|fan\|currentValue | Speed | Fan speed. |
| Sensor\|fan\|healthState | Health State | Fan health state. |
| Sensor\|temperature\|currentValue | Temperature | Host system temperature. |
| Sensor\|temperature\|healthState | Health State | Host system health state. |

## vSphere Metrics

VCF Operations collects CPU use, disk, memory, network, and summary metrics for objects in the vSphere world.

Capacity metrics can be calculated for vSphere world objects. See Capacity Analytics Generated Metrics.

### vSphere World Super Metrics for ROI Dashboard

vSphere world super metrics provide information about the new metrics added to the ROI dashboard.

| Metric Name | Description |
|---|---|
| Cost\|Total Cost of Ownership | This metric shows the total cost cost of ownership with potential savings and optimizations.<br>Key: cost\|total_aggregated_cost |
| Online Capacity Analytics Capacity Remaining Profiles | This metric shows the VMs remaining based on the average VM profile.<br>Key: OnlineCapacityAnalytics\|capacityRemainingProfile |
| Cost\|Server Hardware(Owned) Cost | This metric shows the sum of server hardware depreciated cost having purchase type as Owned across all the vCenters.<br>Key: cost\|total_serverHardware_owned_cost |
| Cost\|Server Hardware(Leased) Cost | This metric shows the sum of server hardware depreciated cost having purchase type as Leased across all the vCenters.<br>Key: cost\|total_serverHardware_leased_cost |
| Cost\|Host OS License cost | This metric shows the sum of host OS license cost across all the vCenters.<br>Key: cost\|total_hostOsl_cost |
| Cost\|Network Cost | This metric shows the sum of network cost across all the vCenters.<br>Key: cost\|total_network_cost |
| Cost\|Maintenance Cost | This metric shows the sum of maintenance cost across all the vCenters.<br>Key: cost\|total_maintenance_cost |
| Cost\|Server Labor Cost | This metric shows the sum of server labor cost across all the vCenters.<br>Key: cost\|total_serverLabor_cost |
| Cost\|Facilities Cost | This metric shows the sum of facilities cost across all the vCenters.<br>Key: cost\|total_facilities_cost |
| Cost\|Additional Cost | This metric shows the sum of additional cost across all the vCenters.<br>Key: cost\|total_additional_cost |
| Cost\|VM Direct Cost | This metric shows sum of direct Cost (VI labor + OS Labor) across all vCenters.<br>Key: cost\|total_vm_direct_cost |

| Metric Name | Description |
|---|---|
| Cost\|Capacity Used Compute Cost | This metric displays the cost of the used compute capacity.<br>Key: cost\|capacity_used\|compute |
| Cost\|Capacity Remaining Compute Cost | This metric displays the cost of the remaining compute capacity.<br>Key: cost\|capacity_remaining\|compute |
| Cost\|Capacity Used Storage Cost | This metric displays the cost of the used storage capacity.<br>Key: cost\|capacity_used\|storage |
| Cost\|Capacity Remaining Storage Cost | This metric displays the cost of the remaining storage capacity.<br>Key: cost\|capacity_remaining\|storage |
| Cost\|Potential Savings Idle VMs | This metric displays the potential savings from Idle VMs.<br>Key: cost\|potential_savings\|idle_vms |
| Cost\|Potential Savings Powered Off VMs | This metric displays the potential savings from powered off VMs.<br>Key: cost\|potential_savings\|poweredOff_vms |
| Cost\|Potential Savings VM Snapshots | This metric displays the potential savings from VM snapshots.<br>Key: cost\|potential_savings\|vm_snapshots |
| Cost\|Potential Savings Orphaned Disks | This metric displays the potential savings from orphaned disks.<br>Key: cost\|potential_savings\|orphaned_disks |
| Cost\|Potential Savings Oversized VMs | This metric displays the potential savings from oversized VMs.<br>Key: cost\|potential_savings\|oversized_vms |
| Cost\|Potential Savings Cost Optimization Opportunities | This metric displays the potential savings from cost optimization opportunities.<br>Key: cost\|potential_savings\|cost_optimization_opportunities |
| Cost\|Total Cost of Ownership | This metric shows the total cost cost of ownership with potential savings and optimizations.<br>Key: cost\|potential_savings\|total_cost_of_ownership |
| Server Purchase Cost | This metric shows the server purchase cost.<br>Key: cost\|server_purchase_cost |
| Accumulated Depreciation | This metric displays the sum of the accumulated depreciation (Depreciation is calculated from the purchase date till current date) of servers across all the vCenters.<br>Key: cost\|accumulatedDepreciation |
| Remaining Depreciation | This metric displays the sum of the remaining depreciation (Remaining Depreciation is calculated from the current date till Depreicated year) of servers across all the vCenters.<br>Key: cost\|accumulatedDepreciation |
| Number of Fully Depreciated Servers | This metric displays the number of fully depreciated servers across all the vCenters.<br>Key: cost\|hardwareTotalCost |
| Reclaimed vCPUs from Idle VMs | This metric displays the number of reclaimable vCPUs from idle VMs.<br>Key: reclaimable\|idle_vms\|cpu |
| Reclaimed Memory from Idle VMs | This metric displays the amount of reclaimable memory from the idle VMs.<br>Key: reclaimable\|idle_vms\|mem |
| Reclaimed Disk Space from Idle VMs | This metric displays the amount of reclaimable disk space from the idle VMs.<br>Key: reclaimable\|idle_vms\|diskspace |
| Reclaimed Disk Space from Powered Off VMs | This metric displays the amount of reclaimable disk space from the powered off VMs.<br>Key: reclaimable\|poweredOff_vms\|diskspace |
| Reclaimed Disk Space from VM Snapshots | This metric displays the amount of reclaimable disk space from the VM Snapshots.<br>Key: reclaimable\|vm_snapshots\|diskspace |
| Reclaimed Disk Space from Orphaned Disks | This metric displays the amount of reclaimable disk space from the orphaned disks.<br>Key: reclaimable\|orphaned_disk\|diskspace |
| Rightsize - vCPUs to Remove from Oversized VMs | This metric displays the number of vCPUs to remove from the oversized VMs.<br>Key: summary\|oversized\|vcpus |

| Metric Name | Description |
|---|---|
| Rightsize - Memory to Remove from Oversized VMs | This metric displays the amount of memory to be removed from the oversized VMs.<br>Key: summary\|oversized\|memory |
| Rightsize - vCPUs to Add from Undersized VMs | This metric displays the number of vCPUs to be added from the undersized VMs.<br>Key: summary\|undersized\|vcpus |
| Rightsize - Memory to Add from Undersized VMs | This metric displays the amount of memory to be added from the undersized VMs.<br>Key: summary\|undersized\|memory |
| Total Storage Cost | This metric displays the sum of storage cost across all vCenters.<br>Key: cost\|totalCost |
| Total Potential Savings | This metric displays the sum of all the potential savings (Idle VMs + Powered off Vms + Snapshot + Orphaned Disks + Oversized VMs).<br>Key: reclaimable\|cost |
| **New vSphere Metrics Added for ROI Dashboard** | |
| Potential Savings from Oversized VMs | This metric displays the sum of all the potentials savings gained from oversized VMs across vcenters.<br>Key: cost\|reclaimableCost |
| Reclaimable Host Cost | This metric displays the reclaimable host cost based on the recommended size.<br>Key: cost\|potential_savings\|total_reclaimable_host_cost |
| Cost\|Potential Increase\|Undersized VMs Cost | This metric displays the rightsizing value for the undersized VMs.<br>Key: cost\|potential_increase\|undersized_vms |
| Cost\|Realized Savings\|Total Realized Savings | This metric displays the total realize savings for VMs across all vCenters.<br>Key: cost\|realized_savings\|total_realized_savings |
| Cost\|Realized Savings\|Idle Savings | This metric displays the total realized savings for idle VMs across all vCenters.<br>Key: cost\|realized_savings\|realized_idle_savings |
| Cost\|Realized Savings\|Powered Off Savings | This metric displays the total realized savings for powered off VMs across all vCenters.<br>Key: cost\|realized_savings\|realized_poweredOff_savings |
| Cost\|Realized Savings\|Snapshot Space Savings | This metric displays the total realized savings for snapshot space across all vCenters.<br>Key: cost\|realized_savings\|realized_snapshotSpace_savings |
| Cost\|Realized Savings\|Oversized Savings | This metric displays the oversized savings across all vCenters.<br>Key: cost\|realized_savings\|realized_oversized_savings |
| Cost\|Realized Savings\|Orphaned Disk Space Savings | This metric displays the amount of disk space saved by orphaned disks across all vCenters.<br>Key: cost\|realized_savings\|realized_orphanedDiskSpace_savings |
| Cost\|Realized Savings\|Reclaimable Host Savings | This metric displays the amount of reclaimable host savings across all vCenters.<br>Key: cost\|realized_savings\|realized_reclaimableHost_savings |
| Compute Realized\|vCPUs from Oversized VMs | This metric displays the number of vCPUs realized across all vCenters.<br>Key: compute_realized\|realized_oversized_vcpus |
| Compute Realized\|Memory from Oversized VMs | This metric displays the amount of memory realized from oversized VMs across all vCenters.<br>Key: compute_realized\|realized_oversized_mem |
| Realized Potential Memory Consumed from Oversized VMs | This metric displays the potential memory consumed from oversized VMs across all vCenters.<br>Key: realized\|realizedPotentialMemConsumed |
| Total Number Of Reclaimable Hosts | This metric displays the total number of reclaimable hosts across all vCenters.<br>Key: metric=cost\|reclaimableHostCost |
| Compute Realized\|vCPUs from Idle VMs | This metric displays the realized vCPUs from idle VMs across all vCenters.<br>Key: compute_realized\|realized_idle_vcpus |
| Compute Realized\|Memory from Idle VMs | This metric displays the amount of memory realized from idle VMs across all vCenters.<br>Key: compute_realized\|realized_idle_mem |

| Metric Name | Description |
|---|---|
| Disk Space Realized\|Idle VMs | This metric displays the amount of disk space realized from idle VMs across all vCenters.<br>Key: storage_realized\|realized_idle_diskSpace |
| Disk Space Realized\|Powered Off VMs | This metric displays the amount of disk space realized from powered off VMs across all vCenters.<br>Key: storage_realized\|realized_poweredOff_diskSpace |
| Disk Space Realized\|VM Snapshots | This metric displays the amount of disk space realized from VM snapshots across all vCenters.<br>Key: storage_realized\|realized_snapshotSpace |
| Disk Space Realized\|Orphaned Disks | This metric displays the amount of disk space realized from orphaned disks across all vCenters.<br>Key: storage_realized\|realized_orphaned_diskSpace |

### CPU Usage Metrics

CPU usage metrics provide information about CPU use.

| Metric Name | Description |
|---|---|
| CPU\|Capacity usage | CPU usages as a percent during the interval.<br>Key: cpu\|capacity_usagepct_average |
| CPU\|CPU contention(%) | This metric shows the percentage of time the VMs in the ESX hosts are unable to run because they are contending for access to the physical CPUs. The number shown is the average number for all VMs. This number is lower than the highest number experienced by the VM most impacted by CPU contention.<br>Use this metric to verify if the host can serve all its VMs efficiently. Low contention means that the VM can access everything it demands to run smoothly. It means that the infrastructure is providing good service to the application team.<br>When using this metric, ensure that the number is within your expectation. Look at both the relative number and the absolute number. Relative means a drastic change in value, meaning that the ESX is unable to serve the VMs. Absolute means that the real value itself is high. Investigate why the number is high. One factor that impacts this metric is CPU Power Management. If CPU Power Management clocks down the CPU speed from 3 GHz to 2 GHz, the reduction in speed is accounted for because it shows that the VM is not running at full speed.<br>This metric is calculated in the following way: cpu\|capacity_contention / (200 * summary\|number_running_vcpus)<br>Key: cpu\|capacity_contentionPct |
| CPU\|Demand (%) | This metric shows the amount of CPU resources a virtual machine might use if there were no CPU contention or CPU limit. This metric represents the average active CPU load for the past five minutes.<br>Keep this number below 100% if you set the power management to maximum.<br>This metric is calculated in the following way: ( cpu.demandmhz / cpu.capacity_provisioned)*100<br>Key: cpu\|demandPct |
| CPU\|Demand (MHz) | This metric shows the amount of CPU resources a virtual machine might use if there were no CPU contention or CPU limit.<br>Key: cpu\|demandmhz |
| CPU\|Demand | CPU demand in megahertz.<br>Key: cpu\|demand_average |
| CPU\|IO wait | IO wait (ms).<br>Key: cpu\|iowait |
| CPU\|number of CPU Sockets | Number of CPU sockets.<br>Key: cpu\|numpackages |

| Metric Name | Description |
|---|---|
| CPU\|Overall CPU Contention | Overall CPU contention in milliseconds.<br>Key: cpu\|capacity_contention |
| CPU\|Provisioned Capacity (MHz) | Capacity in MHz of the physical CPU cores.<br>Key: cpu\|capacity_provisioned |
| CPU\|Provisioned vCPU(s) | Number of provisioned CPU cores.<br>Key: cpu\|corecount_provisioned |
| CPU\|Reserved Capacity (MHz) | Total CPU capacity reserved by virtual machines.<br>Key: cpu\|reservedCapacity_average |
| CPU\|Usage (MHz) | CPU usages, as measured in megahertz, during the interval.<br>• VM - Amount of actively used virtual CPU. This is the host's view of the CPU usage, not the guest operating system view.<br>• Host - Sum of the actively used CPU of all powered on virtual machines on a host. The maximum possible value is the frequency of the two processors multiplied by the number of processors. For example, if you have a host with four 2 GHz CPUs running a virtual machine that is using 4000 MHz, the host is using two CPUs completely: 400 / (4 2000) = 0.50<br>Key: cpu\|usagemhz_average |
| CPU\|Wait | Total CPU time spent in wait state. The wait total includes time spent in the CPU Idle, CPU Swap Wait, and CPU I/O Wait states.<br>Key: cpu\|wait |
| CPU\|Workload (%) | Percent of workload<br>Key: cpu\|workload |

## Memory Metrics

Memory metrics provide information about memory use and allocation.

| Metric Name | Description |
|---|---|
| mem\|Contention (%) | This metric shows the percentage of time VMs are waiting to access swapped memory.<br>Use this metric to monitor ESX memory swapping. A high value indicates that the ESX is running low on memory, and a large amount of memory is being swapped.<br>Key: mem\|host_contentionPct |
| mem\|Machine Demand (KB) | Host memory demand in kilobytes.<br>Key: mem\|host_demand |
| mem\|Provisioned Memory | Provisioned host memory in kilobytes.<br>Key: mem\|host_provisioned |
| mem\|Reserved Capacity (KB) | Total amount of memory reservation used by powered-on virtual machines and vSphere services on the host.<br>Key: mem\|reservedCapacity_average |
| mem\|Usable Memory (KB) | Usable host memory in kilobytes.<br>Key: mem\|host_usable |
| mem\|Host Usage (KB) | Host memory use in kilobytes.<br>Key: mem\|host_usage |
| mem\|Usage/Usable (%) | Memory usage as percentage of total configured or available memory.<br>Key: mem\|host_usagePct |
| mem\|Workload (%) | Percent of workload.<br>Key: mem\|workload |

### Network Metrics

Network metrics provide information about network performance.

| Metric Name | Description |
|---|---|
| net\|Packets Dropped (%) | This metric shows the percentage of received and transmitted packets dropped in the collection interval.<br>Use this metric to monitor the reliability and performance of the ESX network. A high value indicates that the network is not reliable and performance decreases.<br>Key: net\|droppedPct |
| net\|Usage Rate (KB per second) | Sum of the data transmitted and received for all of the NIC instances of the host or virtual machine.<br>Key: net\|usage_average |
| net\|Workload (%) | Percent of workload.<br>Key: net\|workload |

### Disk Metrics

Disk metrics provide information about disk use.

| Metric Name | Description |
|---|---|
| disk\|Total IOPS | Average number of commands issued per second during the collection cycle.<br>Key: disk\|commandsAveraged_average |
| disk\|Usage Rate (KB per second) | Average of the sum of the data read and written for all of the disk instances of the host or virtual machine.<br>Key: disk\|usage_average |
| disk\|Workload (%) | Percent of workload.<br>Key: disk\|workload |

### Summary Metrics

Summary metrics provide information about overall performance.

| Metric Name | Description |
|---|---|
| summary\|Number of Running Hosts | Number of running hosts.<br>Key: summary\|number_running_hosts |
| summary\|Number of Running VMs | This metric shows the number of running VMs at a given point in time. The data is sampled every five minutes.<br>A large number of running VMs might be a reason for CPU or memory spikes because more resources are used in the host. The number of running VMs gives you a good indicator of how many requests the ESX host must juggle. Powered off VMs are not included because they do not impact ESX performance. A change in the number of running VMs can contribute to performance problems. A high number of running VMs in a host also means a higher concentration risk, because all the VMs fail if the ESX crashes.<br>Use this metric to look for a correlation between spikes in the running VMs and spikes in other metrics such as CPU contention, or memory contention.<br>Key: summary\|number_running_vms |
| summary\|Number of Clusters | Total number of clusters.<br>Key: summary\|total_number_clusters |
| summary\|Total Number of Datastores | Total number of datastores.<br>Key: summary\|total_number_datastores |
| summary\|Number of Hosts | Total number of hosts.<br>Key: summary\|total_number_hosts |

| Metric Name | Description |
|---|---|
| summary\|Number of VMs | Total number of virtual machines.<br>Key: summary\|total_number_vms |
| summary\|Total Number of Datacenters | Total number of data centers.<br>Key: summary\|total_number_datacenters |
| summary\|Number VCPUs on Powered on VMs | Number of virtual CPUs on powered-on virtual machines.<br>Key: summary\|number_running_vcpus |
| summary\|Average Running VM Count per Running Host | Average running virtual machine count per running host.<br>Key: summary\|avg_vm_density |
| summary\|Number of Reclaimable Hosts | Displays the number of reclaimable hosts.<br>Key: summary\|total_number_reclaimable_hosts |

### Virtual Machine Operations Metrics for vSphere World

VM operations metrics provide information about the actions performed on VMs. The following are some important points you must know about VM operation metrics for vSphere World.

- VM operations metrics is not collected for custom data centers.
- If you edit a VM settings and do not perform any action, still it is considered as VM reconfigure operation.
- During Revert Snapshot, VMs are powered-off, but this operation is not counted under VM Power-off metric.
- Adding ESX with VMs is not counted under VM Create metric.
- Removing ESX with VMs is not coutned under VM Remove metric.
- VM hardstop operation is not counted under VM Power Off metric.

| Metric Name | Description |
|---|---|
| Inventory | |
| VM Clone | This metric displays the number of clone operations on the virtual machine.<br>Key: Inventory\|VM Clone |
| VM Create | This metric displays the number of create operations on the virtual machine.<br>Key: Inventory\|VM Create |
| VM Delete | This metric displays the number of delete operations on the virtual machine.<br>Key: Inventory\|VM Delete |
| VM Reconfigure | This metric displays the number of reconfigure operations on the virtual machine.<br>Key: Inventory\|VM Reconfigure |
| VM Register | This metric displays the number of register operations on the virtual machine.<br>Key: Inventory\|VM Register |
| VM Template Deploy | This metric displays the number templates deployed on the virtual machine.<br>Key: Inventory\|VM Template Deploy |
| VM Unregister | This metric displays the number of unregister operations on the virtual machine.<br>Key: Inventory\|VM Unregister |
| Location | |
| Storage vMotion | This metric displays the number of migrations with vMotion (datastore change operations for Powered-on VMs).<br>Key: Location\|Storage vMotion |

| Metric Name | Description |
|---|---|
| VM Datastore Change (powered-off VMs) | This metric displays the number of datastore change operations, for powered-off and suspended virtual machines.<br>Key: Location\|VM Datastore Change (powered-off VMs) |
| VM Host and Datastore Change (powered-off VMs) | This metric displays the number of host and datastore change operations, for powered-off and suspended virtual machines.<br>Key: Location\|VM Host and Datastore Change (powered-off VMs) |
| VM Host and Datastore Change (powered-on VMs) | This metric displays the number of host and datastore change operations, for powered-on and suspended virtual machines.<br>Key: Location\|VM Host and Datastore Change (powered-on VMs) |
| VM Host Change (powered-off VMs) | This metric displays the number of host change operations, for powered-off and suspended virtual machines.<br>Key: Location\|VM Host Change (powered-off VMs) |
| vMotion | This metric displays the number of migrations with vMotion (host change operations for powered-on VMs).<br>Key: Location\|vMotion |
| State | |
| VM Guest Reboot | This metric displays the number of reboot operations on the virtual machine guest.<br>Key: State\|VM Guest Reboot |
| VM Guest Shutdown | This metric displays the number of shutdown operations on the virtual machine guest.<br>Key: State\|VM Guest Shutdown |
| VM Power Off | This metric displays the number of power-off operations on the virtual machine.<br>Key: State\|VM Power Off |
| VM Power On | This metric displays the number of power-on operations on the virtual machine.<br>Key: State\|VM Power On |
| VM Reset | This metric displays the number of reset operations on the virtual machine guest.<br>Key: State\|VM Reset |
| VM Standby Guest | This metric displays the number of standby operations on the virtual machine guest.<br>Key: State\|VM Standby Guest |
| VM Suspend | This metric displays the number of suspend operations on the virtual machine.<br>Key: State\|VM Suspend |

## vCenter Server Metrics

VCF Operations collects CPU use, disk, memory, network, and summary metrics for vCenter Server system objects.

vCenter Server metrics include capacity and badge metrics. See definitions in:

- Capacity Analytics Generated Metrics
- Badge Metrics

## CPU Usage Metrics

CPU usage metrics provide information about CPU use.

| Metric Name | Description |
|---|---|
| Capacity Usage (%) | Percent capacity used.<br>Key: cpu\|capacity_usagepct_average |
| CPU Contention (%) | Percent CPU contention.<br>Key: cpu\|capacity_contentionPct |
| Demand (%) | Percent demand.<br>Key: cpu\|demandPct |
| Demand (MHz) | CPU utilization level based on descendant virtual machines utilization. This Includes reservations, limits, and overhead to run the virtual machines.<br>Key: cpu\|demandmhz |
| Demand | CPU Demand.<br>Key: cpu\|demand_average |
| IO Wait (ms) | IO wait time in milliseconds.<br>Key: cpu\|iowait |
| Number of CPU Sockets | Number of CPU sockets.<br>Key: cpu\|numpackages |
| Overall CPU Contention (ms) | Overall CPU contention in milliseconds.<br>Key: cpu\|capacity_contention |
| Provisioned Capacity (MHz) | Provisioned capacity in megahertz.<br>Key: cpu\|capacity_provisioned |
| Provisioned vCPU | Number of provisioned virtual CPU cores.<br>Key: cpu\|corecount_provisioned |
| Reserved Capacity (MHz) | Sum of the reservation properties of the immediate children of the host's root resource pool.<br>Key: cpu\|reservedCapacity_average |
| Usage (MHz) | Average CPU use in megahertz.<br>Key: cpu\|usagemhz_average |
| Wait (ms) | CPU time spent on the idle state.<br>Key: cpu\|wait |
| Overhead | Amount of CPU that is overhead.<br>Key: cpu\|overhead_average |
| Demand without overhead | Value of demand excluding any overhead.<br>Key: cpu\|demand_without_overhead |
| Provisioned Capacity | Provisioned capacity (MHz).<br>Key: cpu\|vm_capacity_provisioned |
| Total Capacity (MHz) | Total CPU resources configured on the descendant ESX hosts.<br>Key: cpu\|capacity_provisioned |
| Usable Capacity (MHz) | The usable CPU resources that are available for the virtual machines after considering reservations for vSphere High Availability (HA) and other vSphere services.<br>Key: cpu\|haTotalCapacity_average |

### Datastore Metrics

Datastore metrics provide information about the datastore.

| Metric Name | Description |
|---|---|
| Outstanding IO requests | OIO for datastore.<br>Key: datastore\|demand_oio |
| Read IOPS | Average number of read commands issued per second during the collection interval.<br>Key: datastore\|numberReadAveraged_average |
| Write IOPS | Average number of write commands issued per second during the collection interval.<br>Key: datastore\|numberWriteAveraged_average |
| Read Throughput (KBps) | Amount of data read in the performance interval.<br>Key: datastore\|read_average |
| Write Throughput (KBps) | Amount of data written to disk in the performance interval.<br>Key: datastore\|write_average |

### Disk Metrics

Disk metrics provide information about disk use.

| Metric Name | Description |
|---|---|
| Total IOPS | Average number of commands issued per second during the collection cycle.<br>Key: disk\|commandsAveraged_average |
| Total Latency (ms) | Average amount of time taken for a command from the perspective of the guest operating system. This metric is the sum of the Kernel Device Command Latency and Physical Device Command Latency metrics.<br>Key: disk\|totalLatency_average |
| Total Throughput (KBps) | Average of the sum of the data read and written for all the disk instances of the host or virtual machine.<br>Key: disk\|usage_average |
| Total queued outstanding operations | Sum of queued operations and outstanding operations.<br>Key: disk\|sum_queued_oio |
| Max Observed OIO | Max observed IO for a disk.<br>Key: disk\|max_observed |

### Disk Space Metrics

Disk space metrics provide information about disk space use.

| Metric Name | Description |
|---|---|
| Total disk space used (KB) | Total disk space used on all datastores visible to this object.<br>Key: diskspace\|total_usage |
| Total disk space (KB) | Total disk space on all datastores visible to this object.<br>Key: diskspace\|total_capacity |
| Total provisioned disk space (KB) | Total provisioned disk space on all datastores visible to this object.<br>Key: diskspace\|total_provisioned |
| Utilization (GB) | Storage space used on the connected vSphere Datastores.<br>Key: diskspace\|total_usage |

| Metric Name | Description |
|---|---|
| Total Capacity (GB) | Total storage space available on the connected vSphere datastores.<br>Key: diskspace\|total_capacity |

### Memory Metrics

Memory metrics provide information about memory use and allocation.

| Metric Name | Description |
|---|---|
| Contention (%) | Percent host memory contention.<br>Key: mem\|host_contentionPct |
| Machine Demand (KB) | Host memory demand in kilobytes.<br>Key: mem\|host_demand |
| ESX System Usage | Memory usage by the VMkernel and ESX user-level services.<br>Key: mem\|host_systemUsage |
| Provisioned Memory (KB) | Provisioned host memory in kilobytes.<br>Key: mem\|host_provisioned |
| Reserved Capacity (KB) | Sum of the reservation properties of the immediate children of the host's root resource pool.<br>Key: mem\|reservedCapacity_average |
| Usable Memory (KB) | Usable host memory in kilobytes.<br>Key: mem\|host_usable |
| Host Usage (KB) | Host memory use in kilobytes.<br>Key: mem\|host_usage |
| Usage/Usable (%) | Percent host memory used.<br>Key: mem\|host_usagePct |
| Contention (KB) | Host contention in kilobytes.<br>Key: mem\|host_contention |
| VM Overhead (KB) | Memory overhead reported by host.<br>Key: mem\|overhead_average |
| Utilization (KB) | Memory utilization level based on the descendant virtual machines utilization. Includes reservations, limits, and overhead to run the Virtual Machines.<br>Key: mem\|total_need |
| Total Capacity (KB) | Total physical memory configured on descendant ESX hosts.<br>Key: mem\|host_provisioned |
| Usable Capacity (KB) | The usable memory resources available for the virtual machines after considering reservations for vSphere HA and other vSphere services.<br>Key: mem\|haTotalCapacity_average |

### Network Metrics

Network metrics provide information about network performance.

| Metric Name | Description |
|---|---|
| Packets Dropped (%) | Percent network packets dropped.<br>Key: net\|droppedPct |
| Total Throughput (KBps) | Sum of the data transmitted and received for all of the NIC instances of the host or virtual machine.<br>Key: net\|usage_average |

| Metric Name | Description |
|---|---|
| Packets Received | Number of packets received in the performance interval.<br>Key: net\|packetsRx_summation |
| Packets Transmitted | Number of packets transmitted in the performance interval.<br>Key: net\|packetsTx_summation |
| Received Packets Dropped | Number of received packets dropped in the performance interval.<br>Key: net\|droppedRx_summation |
| Transmitted Packets Dropped | Number of transmitted packets dropped in the performance interval.<br>Key: net\|droppedTx_summation |
| Data Transmit Rate (KBps) | Average amount of data transmitted per second.<br>Key: net\|transmitted_average |
| Data Receive Rate (KBps) | Average amount of data received per second.<br>Key: net\|received_average |

### Summary Metrics

Summary metrics provide information about overall performance.

| Metric Name | Description |
|---|---|
| Number of Running Hosts | Number of hosts that are on.<br>Key: summary\|number_running_hosts |
| Number of Running VMs | Number of virtual machines that are on.<br>Key: summary\|number_running_vms |
| Number of Clusters | Total number of clusters.<br>Key: summary\|total_number_clusters |
| Total Number of Datastores | Total number of datastores.<br>Key: summary\|total_number_datastores |
| Number of Hosts | Total number of hosts.<br>Key: summary\|total_number_hosts |
| Number of VMs | Total number of virtual machines.<br>Key: summary\|total_number_vms |
| Maximum Number of VMs | Maximum number of virtual machines.<br>Key: summary\|max_number_vms |
| Workload Indicator (%) | Percent workload indicator.<br>Key: summary\|workload_indicator |
| Total Number of data centers | Total number of data centers.<br>Key: summary\|total_number_datacenters |
| Number of Cores on Powered On Hosts | Number of cores on powered-on hosts.<br>Key: summary\|number_powered_on_cores |
| Number VCPUs on Powered on VMs | Number of virtual CPUs on powered-on virtual machines.<br>Key: summary\|number_running_vcpus |
| Average Running VM Count per Running Host | Average running virtual machine count per running host.<br>Key: summary\|avg_vm_density |
| VC Query Time (ms) | vCenter Server query time in milliseconds.<br>Key: summary\|vc_query_time |
| Derived Metrics Computation Time (ms) | Derived metrics computation time in milliseconds.<br>Key: summary\|derived_metrics_comp_time |
| Number of objects | Number of objects.<br>Key: summary\|number_objs |

| Metric Name | Description |
|---|---|
| Number of VC Events | Number of vCenter Server events.<br>Key: summary\|number_vc_events |
| Number of SMS Metrics | Number of SMS metrics.<br>Key: summary\|number_sms_metrics |
| Collector Memory Usage (MB) | Collector memory use in megabytes.<br>Key: summary\|collector_mem_usage |

### Virtual Machine Operations Metrics for vCenter Server

VM operations metrics provide information about the actions performed on VMs. The following are some important points you must know about VM operation metrics for vCenter Server.

- VM operations metrics is not collected for custom data centers.
- If you edit a VM settings and do not perform any action, still it is considered as VM reconfigure operation.
- During Revert Snapshot, VMs are powered-off, but this operation is not counted under VM Power-off metric.
- Adding ESX with VMs is not counted under VM Create metric.
- Removing ESX with VMs is not coutned under VM Remove metric.
- VM hardstop operation is not counted under VM Power Off metric.

| Metric Name | Description |
|---|---|
| Inventory | |
| VM Clone | This metric displays the number of clone operations on the virtual machine.<br>Key: Inventory\|VM Clone |
| VM Create | This metric displays the number of create operations on the virtual machine.<br>Key: Inventory\|VM Create |
| VM Delete | This metric displays the number of delete operations on the virtual machine.<br>Key: Inventory\|VM Delete |
| VM Reconfigure | This metric displays the number of reconfigure operations on the virtual machine.<br>Key: Inventory\|VM Reconfigure |
| VM Register | This metric displays the number of register operations on the virtual machine.<br>Key: Inventory\|VM Register |
| VM Template Deploy | This metric displays the number templates deployed on the virtual machine.<br>Key: Inventory\|VM Template Deploy |
| VM Unregister | This metric displays the number of unregister operations on the virtual machine.<br>Key: Inventory\|VM Unregister |
| Location | |
| Storage vMotion | This metric displays the number of migrations with vMotion (datastore change operations for Powered-on VMs).<br>Key: Location\|Storage vMotion |
| VM Datastore Change (powered-off VMs) | This metric displays the number of datastore change operations, for powered-off and suspended virtual machines.<br>Key: Location\|VM Datastore Change (powered-off VMs) |

| Metric Name | Description |
|---|---|
| VM Host and Datastore Change (powered-off VMs) | This metric displays the number of host and datastore change operations, for powered-off and suspended virtual machines.<br>Key: Location\|VM Host and Datastore Change (powered-off VMs) |
| VM Host and Datastore Change (powered-on VMs) | This metric displays the number of host and datastore change operations, for powered-on and suspended virtual machines.<br>Key: Location\|VM Host and Datastore Change (powered-on VMs) |
| VM Host Change (powered-off VMs) | This metric displays the number of host change operations, for powered-off and suspended virtual machines.<br>Key: Location\|VM Host Change (powered-off VMs) |
| vMotion | This metric displays the number of migrations with vMotion (host change operations for powered-on VMs).<br>Key: Location\|vMotion |
| State | |
| VM Guest Reboot | This metric displays the number of reboot operations on the virtual machine guest.<br>Key: State\|VM Guest Reboot |
| VM Guest Shutdown | This metric displays the number of shutdown operations on the virtual machine guest.<br>Key: State\|VM Guest Shutdown |
| VM Power Off | This metric displays the number of power-off operations on the virtual machine.<br>Key: State\|VM Power Off |
| VM Power On | This metric displays the number of power-on operations on the virtual machine.<br>Key: State\|VM Power On |
| VM Reset | This metric displays the number of reset operations on the virtual machine guest.<br>Key: State\|VM Reset |
| VM Standby Guest | This metric displays the number of standby operations on the virtual machine guest.<br>Key: State\|VM Standby Guest |
| VM Suspend | This metric displays the number of suspend operations on the virtual machine.<br>Key: State\|VM Suspend |

## Disabled Metrics

The following metrics are disabled in this version of VCF Operations. This means that they do not collect data by default.

You can enable these metrics in the Policy workspace. For more information, in VMware Docs search for Collect Metrics and Properties Details.

You can enable these metrics in the Policy workspace. For more information, see Metrics and Properties Details.

| Metric Name | Description |
|---|---|
| Max Observed Number of Outstanding IO Operations | Maximum observed number of outstanding IO operations.<br>Key: datastore\|maxObserved_OIO |
| Max Observed Read Rate | Max observed rate of reading data from the datastore.<br>Key: datastore\|maxObserved_Read |

| Metric Name | Description |
|---|---|
| Max Observed Reads per second | Max observed average number of read commands issued per second during the collection interval.<br>Key: datastore\|maxObserved_NumberRead |
| Max Observed Writes per second | Max observed average number of write commands issued per second during the collection interval.<br>Key: datastore\|maxObserved_NumberWrite |
| Max Observed Write Rate | Max observed rate of writing data from the datastore.<br>Key: datastore\|maxObserved_Write |
| Max Observed Throughput (KBps) | Max observed rate of network throughput.<br>Key: net\|maxObserved_KBps |
| Max Observed Transmitted Throughput (KBps) | Max observed transmitted rate of network throughput.<br>Key: net\|maxObserved_Tx_KBps |
| Max Observed Received Throughput (KBps) | Max observed received rate of network throughput.<br>Key: net\|maxObserved_Rx_KBps |

## Virtual Machine Metrics

VCF Operations collects configuration, CPU use, memory, datastore, disk, virtual disk, guest file system, network, power, disk space, storage, and summary metrics for virtual machine objects.

### GPU Metrics

GPU metrics provide information about the GPU usage and performance.

| Metric Name | Description |
|---|---|
| vGPU:vGPU-id\|vGPU Memory Reservation (KB) | The amount of GPU memory reserved for the assigned vGPUs. |
| vGPU:vGPU-id\|vGPU Memory Used (KB) | The amount of GPU memory being used in kilobytes. |
| vGPU:vGPU-id\|vGPU 3D / Compute utilization (%) | The percentage of time the GPU is actively processing tasks. |
| vGPU:vGPU-id\|vGPU Memory Utilization (%) | The amount of GPU memory being used in percentage. |
| vGPU:vGPU-id\|vGPU Encode Utilization (%) | Encoding is the process of converting raw data into a compressed format. |
| vGPU:vGPU-id\|vGPU Decode Utilization (%) | Decoding is the process of converting compressed format into a raw format. |

### Guest Operating System Metrics

Guest Operating System metrics provide information about the new metrics added to the Guest Operating System.

| Metric Name | Description |
|---|---|
| Guest\|Peak Guest OS Page-out/rate within collection cycle | This metric displays the highest memory page-out rate reported by guest operating system, measured as peak of any 20-second average during the collection interval.<br>**Note:**<br>To collect the guest metrics, ensure that VM Tools is installed and up and running on virtual machine, on vCenter server.<br>Key: guest\|20_sec_peak_page.outRate_latest |

## Metrics for ROI Dashboard

Virtual machine metrics provide information about the new metrics added to the ROI dashboard.

| Metric Name | Description |
|---|---|
| Potential Memory Consumed Reclaimable(GB) | This metric displays the sum of all the reclaimble consumed memory for the virtual machine. |
| Potential CPU Usage Increase (GHz) | This metric displays the potential increase in CPU usage for the virtual machine. |
| Potential Memory Usage Increase (GB) | This metric displays the potential increase in memory usage for the virtual machine. |
| Potential Savings | This metric displays the sum of all the potential savings (Idle VMs + Powered off Vms + Snapshot + Orphaned Disks + Oversized VMs). |
| Potential Cost Increase | This metric displays the potential increase in costs associated with the virtual machine. |

## Application Level Cost Roll up Metrics

The application-level cost roll up in VCF Operations includes few additional metrics at the application level. VCF Operations has introduced Business Application as a new object, the business application object can have Tiers and Applications as its children. The cost roll up option allows you to aggregate all VM costs associated with the application and tiers and publish them at VM level.

**Table 1480: Application Level Cost Roll up Metrics**

| Metric Name | Description |
|---|---|
| Monthly Effected Projected Total Cost | This metric displays the effective virtual machine cost projected for the full month. |
| Effective MTD Cost | This metric displays the month to date effective application cost for the selected virtual machine. |
| Effective Daily Cost | This metric displays the effective daily cost of the application associated with the virtual machine. |

## Configuration Metrics for Virtual Machines

Configuration metrics provide information about virtual machine configuration.

| Metric Name | Description |
|---|---|
| Config\|Thin Provisioned Disk | Thin Provisioned Disk.<br>Key: config\|hardware\|thin_Enabled |
| Config\|Number of CPUs | Number of CPUs for a Virtual Machine.<br>From VCF Operations 6.7 and onwards, this metric is measured in vCPUs instead of cores.<br>Key: config\|hardware\|num_Cpu |
| Config\|Disk Space | Disk space metrics.<br>Key: config\|hardware\|disk_Space |

## CPU Usage Metrics for Virtual Machines

CPU usage metrics provide the information about CPU use.

| Metric Name | Description |
|---|---|
| CPU\|Other Wait (ms) | CPU time spent waiting for IO.<br>Key: cpu\|otherwait |
| CPU\|Overall CPU Contention (ms) | The amount of time the CPU cannot run due to contention.<br>Key: cpu\|capacity_contention |
| CPU\|Reservation Used | CPU Reservation Used.<br>Key: cpu\|reservation_used |
| CPU\|Effective Limit | CPU Effective Limit.<br>Key: cpu\|effective_limit |
| CPU\|Other Wait (%) | Percentage Other Wait.<br>Key: cpu\|otherwaitPct |
| CPU\|Swap wait (%) | Percentage swap waits for CPU.<br>Key: cpu\|swapwaitPct |
| CPU\|Wait (%) | Percentage of the total CPU time spent in wait state.<br>Key: cpu\|waitPct |
| CPU\|System (%) | Percentage CPU time spent on system processes.<br>Key: cpu\|systemSummationPct |
| CPU\|Capacity entitlement (MHz) | CPU entitlement for the VM after considering all limits.<br>Key: cpu\|capacity_entitlement |
| CPU\|Capacity Demand Entitlement (% | Percent capacity demand entitlement.<br>Key: cpu\|capacity_demandEntitlementPct |
| CPU\|CPU Contention (%) | CPU contention as a percentage of 20-second collection interval.<br>Key: cpu\|capacity_contentionPct |
| CPU\|Total Capacity | Provisioned CPU capacity in megahertz.<br>Key: cpu\|vm_capacity_provisioned |
| CPU\|Demand (MHz) | Total CPU resources required by the workloads on the virtual machine.<br>Key: cpu\|demandmhz |
| CPU\|Host demand for aggregation | Host demand for aggregation.<br>Key: cpu\|host_demand_for_aggregation |
| CPU\|Demand (ms) | The total CPU time that the VM might use if there was no contention.<br>Key: cpu\|demand_average |
| CPU\|Demand (%) | CPU demand as a percentage of the provisioned capacity.<br>Key: cpu\|demandPct |
| CPU\|Usage (%) | This metric indicates the percentage of CPU that was used out of all the CPU that was allocated to the VM. CPU usage can indicate when the VM is undersized.<br>Key: cpu\|usage_average |
| CPU\|Usage (MHz) | CPU use in megahertz.<br>Key: cpu\|usagemhz_average |

| Metric Name | Description |
|---|---|
| CPU Workload % | This metric indicates the CPU workload % for the VM, the maximum threshold for this is 80% and the minimum threshold is 20%. If your Maximum line is constantly ~100% flat, you may have a runaway process. If this chart is below or less than 20% all the time for the entire month, then all the large VMs are oversized. This number must hover around 40%, indicating the sizing done was accurate. |
| CPU\|System (ms) | CPU time spent on system processes.<br>Key: cpu\|system_summation |
| CPU\|Ready (%) | This metric indicates the percentage of time in which the VM was waiting in line to use the CPU on the host.<br>A large ready time for a VM indicates that the VM needed CPU resources but the infrastructure was busy serving other VMs. A large ready time might indicate that the host is trying to serve too many VMs.<br>Whenever the CPU ready is larger than 10%, you should check if the host is overloaded, or if the VM really needs all the resources that were allocated to it.<br>Key: cpu\|readyPct |
| CPU\|Extra (ms) | Extra CPU time in milliseconds.<br>Key: cpu\|extra_summation |
| CPU\|Guaranteed (ms) | CPU time that is guaranteed for the virtual machine.<br>Key: cpu\|guaranteed_latest |
| CPU\|Co-stop (%) | Percentage of time the VM is ready to run, but is unable to due to co-scheduling constraints.<br>Key: cpu\|costopPct |
| CPU\|Latency | Percentage of time the VM is unable to run because it is contending for access to the physical CPUs.<br>Key: cpu\|latency_average |
| CPU\|Max Limited | Time the VM is ready to run, but is not run due to maxing out its CPU limit setting.<br>Key: cpu\|maxlimited_summation |
| CPU\|Overlap | Time the VM was interrupted to perform system services on behalf of that VM or other VMs.<br>Key: cpu\|overlap_summation |
| CPU\|Run | Time the VM is scheduled to run.<br>Key: cpu\|run_summation |
| CPU\|Entitlement Latest | Entitlement Latest.<br>Key: cpu\|entitlement_latest |
| CPU\|Total Capacity (MHz) | Total CPU capacity allocated to the virtual machine.<br>Key: cpu\|vm_capacity_provisioned |
| CPU\|Peak vCPU Ready | The highest CPU Ready among the virtual CPUs.<br>Key: cpu\|peak_vcpu_ready |
| CPU\|Peak vCPU Usage | The highest CPU Usage among the virtual CPU, compared with the static configured CPU frequency. A constantly high number indicates that one or more of the CPUs have high utilization.<br>Key: cpu\|peak_vcpu_usage |
| CPU\|20-second Peak CPU System (%) | The highest system CPU, measured as a peak of any 20-second average during the collection interval.<br>Key: cpu\|20-second peak cpu system |

| Metric Name | Description |
|---|---|
| CPU\|20-second Peak vCPU Co-Stop (%) | The highest CPU Co-Stop among any of the vCPU, measured as a peak of any 20-second average during the collection interval.<br>Key: cpu\|20-second peak vcpu co-stop |
| CPU\|20-second Peak vCPU IO-Wait(%) | The highest CPU Other Wait among any of the vCPU, measured as a peak of any 20-second average during the collection interval.<br>Key: cpu\|20-second peak vcpu io-wait |
| CPU\|20-second Peak vCPU Overlap (ms) | The highest CPU Overlap among any of the vCPU, measured as a peak of any 20-second average during the collection interval.<br>Key: cpu\|20-second peak vcpu overlap |
| CPU\|20-second Peak vCPU Ready (%) | The highest CPU Ready among any of the vCPU, measured as a peak of any 20-second average during the collection interval.<br>Key: cpu\|20-second peak vcpu ready |
| CPU\|20-second Peak vCPU Swap Wait (%) | The highest CPU Swap Wait among any of the vCPU, measured as a peak of any 20-second average during the collection interval.<br>Key: cpu\|20-second peak vcpu swap wait |
| CPU \| vCPU Usage Disparity | The absolute gap between the highest vCPU Usage and the lowest vCPU Usage.<br>Key: cpu\|vcpu_usage_disparity |

### CPU Utilization for Resources Metrics for Virtual Machines

CPU utilization for resources metrics provides information about resource CPU use.

| Metric Name | Description |
|---|---|
| rescpu\|CPU Active (%) (*interval*) | The average active time (actav) or peak active time (actpk) for the CPU during various intervals.<br>Key:<br>rescpu\|actav1_latest<br>rescpu\|actav5_latest<br>rescpu\|actav15_latest<br>rescpu\|actpk1_latest<br>rescpu\|actpk5_latest<br>rescpu\|actpk15_latest |
| rescpu\|CPU Running (%) (*interval*) | The average runtime (runav) or peak active time (runpk) for the CPU during various intervals.<br>Key:<br>rescpu\|runav1_latest<br>rescpu\|runav5_latest<br>rescpu\|runav15_latest<br>rescpu\|runpk1_latest<br>rescpu\|runpk5_latest<br>rescpu\|runpk15_latest |

| Metric Name | Description |
|---|---|
| rescpu\|CPU Throttled (%) (*interval*) | Amount of CPU resources over the limit that were refused, average over various intervals.<br>Key:<br>rescpu\|maxLimited1_latest<br>rescpu\|maxLimited5_latest<br>rescpu\|maxLimited15_latest |
| rescpu\|Group CPU Sample Count | The sample CPU count.<br>Key: rescpu\|sampleCount_latest |
| rscpu\|Group CPU Sample Period (ms) | The sample period.<br>Key: rscpu\|samplePeriod_latest |

## Memory Metrics for Virtual Machines

Memory metrics provide information about memory use and allocation.

| Metric Name | Description |
|---|---|
| Mem\|Host Active (KB) | Host active memory use in kilobytes.<br>Key: mem\|host_active |
| Mem\|Contention (KB) | Memory contention in kilobytes.<br>Key: mem\|host_contention |
| Mem\|Contention (%) | Percent memory contention.<br>Key: mem\|host_contentionPct |
| Mem\|Guest Configured Memory (KB) | Guest operating system configured memory in kilobytes.<br>Key: mem\|guest_provisioned |
| Mem\|Guest Active Memory (%) | Percent guest operating system active memory.<br>Key: mem\|guest_activePct |
| Mem\|Guest Non-Pageable Memory (KB) | Guest operating system non-pageable memory in kilobytes.<br>Key: mem\|guest_nonpageable_estimate |
| Mem\|Reservation Used | Memory Reservation Used.<br>Key: mem\|reservation_used |
| Mem\|Effective Limit | Memory Effective Limit.<br>Key: mem\|effective_limit |
| Mem\|Demand for aggregation | Host demand for aggregation.<br>Key: mem\|host_demand_for_aggregation |
| Mem\|Balloon (%) | Percentage of total memory that has been reclaimed via ballooning.<br>Key: mem\|balloonPct |
| Mem\|Guest Usage (KB) | This metric shows the amount of memory the VM uses.<br>Key: mem\|guest_usage |
| Mem\|Guest Demand (KB) | Guest operating system demand in kilobytes.<br>Key: mem\|guest_demand |
| Mem\|Guest Non-Pageable Memory (KB) | Guest operating system non-pageable memory in kilobytes.<br>Key: mem\|host_nonpageable_estimate |
| Mem\|Host Demand (KB) | Memory demand in kilobytes.<br>Key mem\|host_demand |
| Mem\|Host Workload | Host Workload (%).<br>Key: host_workload |

| Metric Name | Description |
|---|---|
| Mem\|Zero (KB) | Amount of memory that is all 0.<br>Key: mem\|zero_average |
| Mem\|Swapped (KB) | This metric shows how much memory is being swapped. Meaning, the amount of unreserved memory in kilobytes.<br>Key: mem\|swapped_average |
| Mem\|Swap Target (KB) | Amount of memory that can be swapped in kilobytes.<br>Key: mem\|swaptarget_average |
| Mem\|Swap In (KB) | Swap-in memory in kilobytes.<br>Key: mem\|swapin_average |
| Mem\|Balloon Target (KB) | Amount of memory that can be used by the virtual machine memory control.<br>Key: mem\|vmmemctltarget_average |
| Mem\|Consumed (KB) | Amount of host memory consumed by the virtual machine for guest memory in kilobytes.<br>Key: mem\|consumed_average |
| Mem\|Overhead (KB) | Memory overhead in kilobytes.<br>Key: mem\|overhead_average |
| Mem\|Swap In Rate (KBps) | Rate at which memory is swapped from disk into active memory during the interval.<br>Key: mem\|swapinRate_average |
| Mem\|Active Write (KB) | Active writes in kilobytes.<br>Key: mem\|activewrite_average |
| Mem\|Compressed (KB) | Compressed memory in kilobytes.<br>Key: mem\|compressed_average |
| Mem\|Compression Rate (KBps) | Compression rate in kilobytes per second.<br>Key: mem\|commpressionRate_average |
| Mem\|Decompression Rate (KBps) | Decompression rate in kilobytes per second.<br>Key: mem\|decompressionRate_average |
| Mem\|Overhead Max (KB) | Maximum overhead in kilobytes.<br>Key: mem\|overheadMax_average |
| Mem\|Zip Saved (KB) | Zip-saved memory in kilobytes.<br>Key: mem\|zipSaved_latest |
| Mem\|Zipped (KB) | Zipped memory in kilobytes.<br>Key: mem\|zipped_latest |
| Mem\|Entitlement | Amount of host physical memory the VM is entitled to, as determined by the ESX schedule.<br>Key: mem\|entitlement_average |
| Mem\|Capacity Contention | Capacity Contention.<br>Key: mem\|capacity.contention_average |
| Mem\|Swap In Rate from Host Cache | Rate at which memory is being swapped from host cache into active memory.<br>Key: mem\|llSwapInRate_average |
| Mem\|Swap Out Rate to Host Cache | Rate at which memory is being swapped to host cache from active memory.<br>Key: mem\|llSwapOutRate_average |
| Mem\|Swap Space Used in Host Cache | Space used for caching swapped pages in the host cache.<br>Key: mem\|llSwapUsed_average |

| Metric Name | Description |
|---|---|
| Mem\|Overhead Touched | Actively touched overhead memory (KB) reserved for use as the virtualization overhead for the VM.<br>Key: mem\|overheadTouched_average |
| Memory\|VM Memory Demand (kb) | Key: mem\|vmMemoryDemand |
| Memory\|Consumed (%) | Key: mem\|consumedPct |
| Mem\|Utilization (KB) | Memory used by the virtual machine. Reflects the guest OS memory required for vSphere and certain VMTools versions or for virtual machine consumption.<br>Key: mem\|vmMemoryDemand |
| Mem\|Total Capacity (KB) | Memory resources allocated to powered on virtual machine.<br>Key: mem\|guest_provisioned |
| Mem\|20-second Peak Contention (%) | The highest Memory Contention, measured as peak of any 20-second average during the collection interval.<br>Key: guest\|20-second_peak_contention |
| Guest\|Peak Guest OS Page-out/rate within collection cycle | This metric shows the highest memory page-out rate reported by guest operating system, measured as peak of any 20-second average during the collection interval.<br>Key: guest\|20_sec_peak_page.outRate_latest |
| Guest\|Needed Memory | Amount of memory needed for the Guest OS to perform optimally. This memory is considered as a cache for the disk and is a little more than the actual used memory.<br>Key: guest\|mem.needed_latest |
| Guest\|Free Memory | Amount of memory that is not used but is readily available. If the cache is high, a low free memory does not mean that the Guest OS needs more memory.<br>Key: guest\|mem.free_latest |
| Guest\|Physical Usable Memory | Amount of memory available to the Guest OS. Meaning, this amount is close to the amount of configured memory to the VM.<br>Key: guest\|mem.physUsable_latest |
| Guest\|20-second Peak Disk Queue Length | The highest Disk Queue Length, measured as peak of any 20-second average during the collection interval.<br>Key: guest\|20-second_peak_disk_queue_length |
| Guest\|20-second Peak Run Queue | The highest Run Queue, measured as peak of any 20-second average during the collection interval.<br>Key: guest\|20-second_peak_run_queue |
| Guest\|20-second Peak CPU Context Switch Rate | The highest CPU Context Switch Rate, measured as peak of any 20-second average during the collection interval.<br>Key: guest\|20-second_peak_cpu_context switch rate |

### Datastore Metrics for Virtual Machines

Datastore metrics provide information about datastore use.

| Metric Name | Description |
|---|---|
| Datastore\|Total IOPS | Average number of commands issued per second during the collection interval.<br>Key: datastore\|commandsAveraged_average |
| Datastore\|Outstanding IO requests | OIO for datastore.<br>Key: datastore\|demand_oio |

| Metric Name | Description |
|---|---|
| Datastore\|Number of Outstanding IO Operations | Number of outstanding IO operations.<br>Key: datastore\|oio |
| Datastore\|Demand | Datastore demand.<br>Key: datastore\|demand |
| Datastore\|Total Latency (ms) | The average amount of time taken for a command from the perspective of a Guest OS. This is the sum of Kernel Command Latency and Physical Device Command Latency.<br>Key: datastore\|totalLatency_average |
| Datastore\|Total Throughput (KBps) | Usage Average (KBps).<br>Key: datastore\|usage_average |
| Datastore\|Used Space (MB) | Used space in megabytes.<br>Key: datastore\|used |
| Datastore\|Not Shared (GB) | Space used by VMs that is not shared.<br>Key: datastore\|notshared |
| Datastore\|Read IOPS | Average number of read commands issued per second during the collection interval.<br>Key: datastore\|numberReadAveraged_average |
| Datastore\|Write IOPS | Average number of write commands issued per second during the collection interval.<br>Key: datastore\|numberWriteAveraged_average |
| Datastore\|Read Throughput (KBps) | This metric shows the amount of data that the VM reads to the datastore per second.<br>Key: datastore\|read_average |
| Datastore\|Read Latency (ms) | Average amount of time for a read operation from the datastore. Total latency = kernel latency + device latency.<br>Key: datastore\|totalReadLatency_average |
| Datastore\|Write Latency (ms) | Average amount of time for a write operation to the datastore. Total latency = kernel latency + device latency.<br>Key: datastore\|totalWriteLatency_average |
| Datastore\|Write Throughput (KBps) | This metric shows the amount of data that the VM writes to the datastore per second.<br>Key: datastore\|write_average |
| Datastore\|Highest Latency | Highest Latency.<br>Key: datastore\|maxTotalLatency_latest |
| Datastore\|Total Latency Max | Total Latency Max (ms).<br>Key: datastore\|totalLatency_max |

## Disk Metrics for Virtual Machines

Disk metrics provide information about disk use.

| Metric Name | Description |
|---|---|
| Disk Space\|vSAN Overhead (GB) | Displays the extra virtual machine disk space used by the vSAN system.<br>Disk Space\|vSAN Overhead (GB) |
| Disk Space\|VMUsedWithoutOverhead(GB) | Displays the virtual machine disk space without vSAN overhead.<br>Disk Space\|Usage Without Overhead (GB) |

| Metric Name | Description |
|---|---|
| Disk Space<br><br>**Note:**<br>The Disk Space metrics are displayed by the virtual machine object only when the vSAN adapter is configured with vCenter. | |
| Disk\|Read IOPS | Average number of read commands issued per second during the collection interval.<br>Key: disk\|numberReadAveraged_average |
| Disk\|Write IOPS | Average number of write commands issued per second during the collection interval.<br>Key: disk\|numberWriteAveraged_average |
| Disk\|Total IOPS | Average number of commands issued per second during the collection interval.<br>Key: disk\|commandsAveraged_average |
| Disk\|Total Throughput (KBps) | Use rate in kilobytes per second.<br>Key: disk\|usage_average |
| Disk\|I/O Usage Capacity | This metric is a function of storage\|usage_average and disk\|workload. Storage\|usage_average is an average over all storage devices. This means that disk\|usage_capacity is not specific to the selected VM or the host of the VM.<br>Key: disk\|usage_capacity |
| Disk\|Number of Outstanding IO Operations | Number of outstanding IO operations.<br>Key: disk\|diskoio |
| Disk\|Queued Operations | Queued operations.<br>Key: disk\|diskqueued |
| Disk\|Demand (%) | Percent demand.<br>Key: disk\|diskdemand |
| Disk \|Total Queued Outstanding Operations | Sum of Queued Operation and Outstanding Operations.<br>Key: disk \|sum_queued_oio |
| Disk\|Max Observed OIO | Max Observed IO for a disk.<br>Key: disk\|max_observed |
| Disk\|Read Throughput KBps) | Amount of data read in the performance interval.<br>Key: disk\|read_average |
| Disk\|Write Throughput (KBps) | Amount of data written to disk in the performance interval.<br>Key: disk\|write_average |
| Disk\|Bus Resets | The number of bus resets in the performance interval.<br>Key: disk\|busResets_summation |
| Disk\|Commands canceled | The number of disk commands canceled in the performance interval.<br>Key: disk\|commandsAborted_summation |
| Disk\|Highest Latency | Highest latency.<br>Key: disk\|maxTotalLatency_latest |
| Disk\|SCSI Reservation Conflicts | SCSI Reservation Conflicts.<br>Key: disk\|scsiReservationConflicts_summation |
| Disk\|Read Latency (ms) | The average amount of time taken for a read from the perspective of a Guest OS. This is the sum of Kernel Read Latency and Physical Device Read Latency.<br>Key: disk\|totalReadLatency_average |

| Metric Name | Description |
|---|---|
| Disk\|Write Latency (ms) | The average amount of time taken for a write from the perspective of a Guest OS. This is the sum of Kernel Write Latency and Physical Device Write Latency.<br>Key: disk\|totalWriteLatency_average |
| Disk\|Total Latency (ms) | The average amount of time taken for a command from the perspective of a Guest OS. This is the sum of Kernel Command Latency and Physical Device Command Latency.<br>Key: disk\|totalLatency_average |

### Virtual Disk Metrics for Virtual Machines

Virtual disk metrics provide information about virtual disk use.

| Metric Name | Description |
|---|---|
| Virtual Disk:<scsi_controller>\|IOPS per GB | Displays the disk IO per second per Gigabyte of storage.<br>Key: virtualDisk:<scsi_controller>\|iopsPerGB |
| Virtual Disk\|Total Throughput | Amount of data read from/written to storage in a second. This is averaged over the reporting period.<br>Key: virtualDisk\|usage |
| VirtualDisk\|Total Latency | Total latency.<br>Key: virtualDisk\|totalLatency |
| VirtualDisk\|Total IOPS | Average number of commands per second.<br>Key: virtualDisk\|commandsAveraged_average |
| VirtualDisk\|Read Requests | Average number of read commands issued per second to the virtual disk during the collection interval.<br>Key: virtualDisk\|numberReadAveraged_average |
| VirtualDisk\|Write Requests | Average number of write commands issued per second to the virtual disk during the collection interval.<br>Key: virtualDisk\|numberWriteAveraged_average |
| VirtualDisk\|Read Throughput (KBps) | Rate of reading data from the virtual disk in kilobytes per second.<br>Key: virtualDisk\|read_average |
| VirtualDisk\|Read Latency (ms) | Average amount of time for a read operation from the virtual disk. Total latency = kernel latency + device latency.<br>Key: virtualDisk\|totalReadLatency_average |
| VirtualDisk\|Write Latency (ms) | Average amount of time for a write operation to the virtual disk. Total latency = kernel latency + device latency.<br>Key: virtualDisk\|totalWriteLatency_average |
| VirtualDisk\|Write Throughput (KBps) | Rate of writing data from the virtual disk in kilobytes per second.<br>Key: virtualDisk\|write_average |
| VirtualDisk\|Bus Resets | The number of bus resets in the performance interval.<br>Key: virtualDisk\|busResets_summation |
| VirtualDisk\|Commands Aborted | The number of disk commands canceled in the performance interval.<br>Key: virtualDisk\|commandsAborted_summation |
| VirtualDisk\|Read Load | Storage DRS virtual disk metric read load.<br>Key: virtualDisk\|readLoadMetric_latest |

| Metric Name | Description |
|---|---|
| VirtualDisk\|Outstanding Read Requests | Average number of outstanding read requests to the virtual disk.<br>Key: virtualDisk\|readOIO_latest |
| VirtualDisk\|Write Load | Storage DRS virtual disk write load.<br>Key: virtualDisk\|writeLoadMetric_latest |
| VirtualDisk\|Outstanding Write Requests | Average number of outstanding write requests to the virtual disk.<br>Key: virtualDisk\|writeOIO_latest |
| VirtualDisk\|Number of Small Seeks | Small Seeks.<br>Key: virtualDisk\|smallSeeks_latest |
| VirtualDisk\|Number of Medium Seeks | Medium Seeks.<br>Key: virtualDisk\|mediumSeeks_latest |
| VirtualDisk\|Number of Large Seeks | Large Seeks.<br>Key: virtualDisk\|largeSeeks_latest |
| VirtualDisk\|Read Latency (microseconds) | Read Latency in microseconds.<br>Key: virtualDisk\|readLatencyUS_latest |
| VirtualDisk\|Write Latency (microseconds) | Write Latency in microseconds.<br>Key: virtualDisk\|writeLatencyUS_latest |
| VirtualDisk\|Average Read request size | Read IO size.<br>Key: virtualDisk\|readIOSize_latest |
| VirtualDisk\|Average Write request size | Write IO size.<br>Key: virtualDisk\|writeIOSize_latest |
| Virtual Disk\|Outstanding IO requests (OIOs) | Key: virtualDisk\|vDiskOIO |
| Virtual Disk\|Used Disk Space (GB) | Key: virtualDisk\|actualUsage |
| Virtual Disk\|Peak Virtual Disk IOPS | The highest disk IO per second among the virtual disks. A constantly high number indicates that one or more virtual disks are sustaining high IOPS.<br>Key: virtualDisk\|peak_vDisk_iops |
| Virtual Disk\|Peak Virtual Disk Read Latency | The highest read latency among the virtual disks. A high number indicates that one or more virtual disks are experiencing poor performance.<br>Key: virtualDisk\|peak_vDisk_readLatency |
| Virtual Disk\|Peak Virtual Disk Write Latency | The highest write latency among the virtual disks. A high number indicates that one or more virtual disks are experiencing poor performance.<br>Key: virtualDisk\|peak_vDisk_writeLatency |
| Virtual Disk\|20-secod Peak Latency (ms) | The highest latency among any of the virtual disk, measured as peak of any 20-second average during the collection interval.<br>Key: virtualDisk\|20-second_peak_latency |
| Virtual Disk\|Peak Virtual Disk throughput | The highest disk throughput among the virtual disks.<br>Key: virtualDisk\|peak_vDisk_throughpu |

### Guest File System Metrics for Virtual Machines

Guest file system metrics provide information about guest file system capacity and free space.

The data for these metrics is only displayed when VMware Tools has been installed on the virtual machines. If VMware Tools is not installed, features dependent on these metrics, including capacity planning for virtual machine guest storage, will not be available.

| Metric Name | Description |
|---|---|
| Guest file system\|Guest File System Capacity (MB) | Total capacity on guest file system in megabytes.<br>Key: guestfilesystem\|capacity |
| Guest file system\|Guest File System Free (MB) | Total free space on guest file system in megabytes.<br>Key: guestfilesystem\|freespace |
| Guest file system\|Guest File System Usage (%) | Percent guest file system.<br>Key: guestfilesystem\|percentage |
| Guest file system\|Guest File System Usage | Total usage of guest file system.<br>From VCF Operations 6.7 and onwards, this metric is measured in GBs.<br>Key: guestfilesystem\|usage |
| Guest file system\|Total Guest File System Capacity (GB) | This metric displays the amount of disk space allocated for the VM.<br>Correlate other metrics with this metric to indicate if changes occur in the disk space allocation for the VM.<br>Key: guestfilesystem\|capacity_total |
| Guest file system\|Total Guest File System Usage (%) | This metric displays the amount of display space being used out of the total allocated disk space.<br>Use his metric to track if the overall usage is stable, or if it reaches the limits. Do not include VMs with a disk space usage of >95% since this might impact your system.<br>Key: guestfilesystem\|percentage_total |
| Guest file system\|Total Guest File System Usage | Total usage of guest file system.<br>Key: guestfilesystem\|usage_total |
| Guest file system\|Utilization (GB) | Storage space used by the Guest OS file systems. The disk space is available only if VM tools are installed and running. If the VM tools are not installed, the disk space capacity is not applicable.<br>Key: guestfilesystem\|usage_total |
| Guest file system\|Total Capacity (GB) | Storage space used by the Guest OS file systems. The disk space is available only if VM tools are installed and running. If the VM tools are not installed, the disk space capacity is not applicable.<br>Key: guestfilesystem\|capacity_total |

### Network Metrics for Virtual Machines

Network metrics provide information about network performance.

| Metric Name | Description |
|---|---|
| Network\|Peak Network Packet per second within collection cycle | This metric shows the highest VM packets per second rate, measured as peak of any 20-second average during the collection interval.<br>Key: net\|20_sec_peak_packetsPerSec |
| Net\|Total Throughput (KBps) | The sum of the data transmitted and received for all the NIC instances of the host or virtual machine.<br>Key: net\|usage_average |

| Metric Name | Description |
|---|---|
| Net\|Data Transmit Rate (KBps) | This metric shows the rate of data being sent by the VM per second.<br>Key: net\|transmitted_average |
| Net\|Data Receive Rate (KBps) | This metric shows the rate of data received by the VM per second.<br>Key: net\|received_average |
| Net\|Packets per second | Number of packets transmitted and received per second.<br>Key: net\|PacketsPerSec |
| Net\|Packets Received | Number of packets received in the performance interval.<br>Key: net\|packetsRx_summation |
| Net\|Packets Transmitted | Number of packets transmitted in the performance interval.<br>Key: net\|packetsTx_summation |
| Net\|Transmitted Packets Dropped | This metric shows the number of transmitted packets dropped in the collection interval<br>Key: net\|dropppedTx_summation |
| Net\|Packets Dropped (%) | Percentage of packets dropped.<br>Key: net\|droppedPct |
| Net\|Packets Dropped | Number of packets dropped in the performance interval.<br>Key: net\|dropped |
| Net\|Broadcast Packets Transmitted | Number of broadcast packets transmitted during the sampling interval.<br>Key: net\|broadcastTx_summation |
| Net\|Broadcast Packets Received | Number of broadcast packets received during the sampling interval.<br>Key: net\|broadcastRx_summation |
| Net\|Multicast Packets Received | Number of multicast packets received.<br>Key: net\|multicastRx_summation |
| Net\|Multicast Packets Transmitted | Number of multicast packets transmitted.<br>Key: net\|multicastTx_summation |
| Net\|VM to Host Data Transmit Rate | Average amount of data transmitted per second between VM and host.<br>Key: net\|host_transmitted_average |
| Net\|VM to Host Data Receive Rate | Average amount of data received per second between VM and host.<br>Key: net\|host_received_average |
| Net\|VM to Host Usage Rate | The sum of the data transmitted and received for all the NIC instances between VM and host.<br>Key: net\|host_usage_average |
| Net\|20-second Peak Usage Rate (KBps) | The higest Usage Rate, measured as peak of any 20 second average during the collection interval.<br>Key: net\|20-second_peak_usage_rate |

## System Metrics for Virtual Machines

System metrics for virtual machines provide general information about the virtual machine, such as its build number and running state.

| Metric Name | Description |
|---|---|
| Sys\|Powered ON | Powered on virtual machines. 1 if powered on, 0 if powered off, -1 if unknown<br>Key: sys\|poweredOn |
| Sys\|OS Uptime | Total time elapsed, in seconds, since last operating system start.<br>Key: sys\|osUptime_latest |

## Power Metrics for Virtual Machines

Power metrics provide information about power use.

| Metric Name | Description |
|---|---|
| Power\|Total Energy Consumed in the collection period (Wh) | Displays the total electricity consumed based on the time interval selected. The default collection cycle is set to 5 mins. You can continue using the default setting or edit it for each adapter instance. For example, if the time interval is set to its default value, the value represents the energy consumed per 5 mins. |
| Power\|Total Power Consumed By VM (Wh) | Displays the total power consumed by a virtual machine in an hour. The data collected is over a period of an hour and published along with the other metrics in VCF Operations. In case of a connectivity or availability issue in VCF Operations or vCenter adapter instance, this hourly metric might not be published and the missed value during this period does not get recalculated. Once the connection is re-established, the next data points get published.<br>**Note:**<br>This metric is deactivated by default. You can activate it from the Policies page. For more information, see Metrics and Properties Details in the *VCF Operations Configuration Guide*. |
| Power\|Power (Watt) | Average power use in watts. |
| Power\|Current Power Consumption Rate (Watt) | The power consumption rate per second, averaged over the reporting period.<br>Key: power\|power_average |
| Power\|(DEP) Energy (Joule) | Total energy consumed in joules.<br>Key: power\|energy_summation |

## Disk Space Metrics for Virtual Machines

Disk space metrics provide information about disk space use.

| Metric Name | Description |
|---|---|
| Diskspace\|Provisioned Space (GB) | Provisioned space in gigabytes.<br>Key: diskspace\|provisioned |
| Diskspace\|Provisioned Space for VM | Provisioned space for VM.<br>Key: diskspace\|provisionedSpace |

| Metric Name | Description |
|---|---|
| Diskspace\|Snapshot Space (GB) | Space used by snapshots.<br>Key: diskspace\|snapshot |
| Diskspace\|Virtual machine used (GB) | Space used by virtual machine files in gigabytes.<br>Key: diskspace\|perDsUsed |
| Diskspace\|Active not shared | Unshared disk space used by VMs excluding snapshot.<br>Key: diskspace\|activeNotShared |

### Storage Metrics for Virtual Machines

Storage metrics provide information about storage use.

| Metric Name | Description |
|---|---|
| Storage\|Total IOPS | Average number of commands issued per second during the collection interval.<br>Key: storage\|commandsAveraged_average |
| Storage\|Contention (%) | Percent contention.<br>Key: storage\|contention |
| Storage\|Read Throughput (KBps) | Read throughput rate in kilobytes per second.<br>Key: storage\|read_average |
| Storage\|Read IOPS | Average number of read commands issued per second during the collection interval.<br>Key: storage\|numberReadAveraged_average |
| Storage\|Total Latency (ms) | Total latency in milliseconds.<br>Key: storage\|totalLatency_average |
| Storage\|Total Usage (KBps) | Total throughput rate in kilobytes per second.<br>Key: storage\|usage_average |
| Storage\|Write Throughput (KBps) | Write throughput rate in kilobytes per second.<br>Key: storage\|write_average |
| Storage\|Write IOPS | Average number of write commands issued per second during the collection interval.<br>Key: storage\|numberWriteAveraged_average |

### Summary Metrics for Virtual Machines

Summary metrics provide information about overall performance.

| Metric Name | Description |
|---|---|
| Summary\|Availability % | Displays the uptime of Guest OS, expressed as a percentage of the collection period.<br>Key: summary\|availability_kpi |
| Summary\|Running | Number of running virtual machines.<br>Key: summary\|running |
| Summary\|Desktop Status | Horizon view desktop status.<br>Key: summary\|desktop_status |

| Metric Name | Description |
|---|---|
| Summary\|Configuration\|Type | Indicates the type of virtual machine object based on which you can identify the type of virtual machine. The valid values for the virtual machine object property are:<br>• default - represents a regular virtual machine<br>• template - represents a powered off virtual machine template.<br>• srm_placeholder - represents a powered on VMware Live Recovery virtual machine.<br>• ft_primary - represents the primary Fault Tolerance virtual machine.<br>• ft_secondary - represents the secondary Fault Tolerance virtual machine.<br>Key: summary\|config\|type |
| Summary\|Guest Operating System\|Guest OS Full Name | Displays the guest operating system name.<br>Key: summary\|guest os full name |
| Summary\|Oversized\|Potential Memory | Displays the oversized potential memory.<br>Key: summary\|oversized\|potentialMemConsumed |
| Summary\|Undersized\|Potential CPU Usage | Displays the undersized potential CPU used.<br>Key: summary\|undersized\|potentialCpuUsage |
| Summary\|Undersized\|Potential Memory | Displays the undersized potential memory used.<br>Key: summary\|undersized\|potentialMemUsage |
| Reclaimable Idle | Boolean flag indicating whether VM is considered as reclaimable because it is in Idle state.<br>Key: summary\|idle |
| Reclaimable Powered Off | Boolean flag indicating whether VM is considered as reclaimable because it is in powered off state.<br>Key: summary\| poweredOff |
| Reclaimable Snapshot Space (GB) | Reclaimable snapshot space.<br>Key: summary\| snapshotSpace |

### Cost Metrics for Virtual Machines

Cost metrics provide information about the cost.

| Metric Name | Description |
|---|---|
| Monthly OS Labor Cost | Monthly operating system labor cost of the virtual machine.<br>Key: cost\|osLaborTotalCost |
| Monthly Projected Total Cost | Virtual machine cost projected for full month.<br>Key: Cost\|monthlyProjectedCost |
| Monthly VI Labor Cost | Monthly virtual infrastructure labor cost of the virtual machine.<br>Key: cost\|viLaborTotalCost |
| MTD Compute Total Cost | Total compute cost (including CPU and memory) of the virtual machine.<br>Key: cost\|compTotalCost |
| MTD CPU Cost | Month to Date Virtual Machine CPU Cost. It is based on utilization. The more the VM uses, the higher its cost.<br>Key: cost\|cpuCost |

| Metric Name | Description |
|---|---|
| MTD Monthly Cost | Month to date direct cost (comprising of OS labor, VI labor and any windows desktop instance license) of the virtual machine. It also comprises of the additional and application cost of the virtual machine.<br>Key: cost\|vmDirectCost |
| MTD Memory Cost | Month to Date Memory Cost of Virtual Machine. It is based on utilization. The more the VM uses, the higher its cost.<br>Key: cost\|memoryCost |
| MTD Storage Cost | Month to date storage cost of the virtual machine.<br>Key: cost\|storageCost |
| MTD Total Cost | Month to date total compute cost (including CPU and memory) of the virtual machine.<br>Key: cost\|monthlyTotalCost |
| Potential Savings | Reclaimable cost of VM for being either idle, powered-off, or having snapshots.<br>Key: cost\|reclaimableCost |
| Cost\|Allocation\|MTD VM CPU Cost (Currency) | Month to Date Virtual Machine CPU Cost computed based on resource overcommit ratio set for its parent cluster in policy.<br>cost\|allocation\|allocationBasedCpuMTDCost |
| Cost\|Allocation\|MTD VM Memory Cost (Currency) | Month to Date Virtual Machine CPU Memory cost computed based on resource overcommit ratio set for its parent cluster in policy.<br>cost\|allocation\|allocationBasedMemoryMTDCost |
| Cost\|Allocation\|MTD VM Storage Cost (Currency) | Month to Date Virtual Machine CPU Storage cost computed based on resource overcommit ratio set for its parent cluster (or datastore cluster) in policy.<br>cost\|allocation\|allocationBasedStorageMTDCost |
| Cost\|Allocation\|MTD VM Total Cost (Currency) | Month to Date Virtual Machine Total Cost is the summation of the CPU Cost, Memory Cost, Storage Cost and Direct Cost, based on overcommit ratios set in policy for the parent cluster or datastore cluster.<br>cost\|allocation\|allocationBasedTotalCost |
| Cost\|Effective Daily Cpu Cost (Currency) | Daily CPU cost of the selected virtual machine. |
| Cost\|Effective Daily Memory Cost (Currency) | Daily Memory cost of the selected virtual machine. |
| Cost\|Effective Daily Storage Cost (Currency) | Daily Storage cost of the selected virtual machine. |
| Cost\|Daily Additional Cost | Daily Additional cost of the selected virtual machine. |
| Cost\|Effective Daily Cost (Currency) | Effective Daily cost is the sum of effective daily CPU cost + effective daily memory cost + effective daily storage cost + daily additional cost. |
| Cost\|Effective MTD Cost (Currency) | Effective MTD cost is the sum of effective daily CPU cost from beginning of month until now + effective daily memory cost from beginning of month until now + effective daily storage cost from beginning of month until now + daily additional cost from beginning of month until now. |

## Virtual Hardware Metrics for Virtual Machines

| Metric Name | Description |
|---|---|
| Configuration\|Hardware\|Number of CPU cores per socket | This metric displays the number of CPU cores per socket. |
| Configuration\|Hardware\|Number of virtual CPUs | This metric displays the number of CPUs in the virtual machine. |
| Configuration\|Hardware\|Number of virtual sockets: | This metric displays the number of virtual sockets in the virtual machine. |
| Configuration\|Hardware\|Memory: | This metric displays the memory used in the virtual machine. |
| Configuration\|CPU Resource Allocation\|Limit | This metric displays the resource allocation limit of the virtual machine. |
| Configuration\|CPU Resource Allocation\|Reservation | This metric displays the reserved resources for the virtual machine. |
| Configuration\|CPU Resource Allocation\|Shares\| | This metric displays the shared resources for the virtual machine. |
| Summary\|Guest Operating System\|Tools Version | This metric displays the tools version of the guest operating system. |
| Summary\|Guest Operating System\|Tools Version Status | This metric displays the status of the tools in the guest operating system. |
| Summary\|Guest Operating System\|Tools Running Status | This metric displays whether the tools are functional in the guest operating system. |
| Guest File System:/boot\|Partition Capacity (GB) | This metric displays the boot partition capacity in the guest file system. |
| Guest File System:/boot\|Partition Utilization (%) | This metric displays the boot partition usage percentage in the guest file system. |
| Guest File System:/boot\|Partition Utilization (GB) | This metric displays the boot partition used in the guest file system. |
| Virtual Disk\|Configured | This metric displays the disk space of the configured virtual disk. |
| Virtual Disk\|Label | This metric displays the disk label of the configured virtual disk. |
| Disk Space\|Snapshot Space | This metric displays the snap shot details of the virtual machine. |
| Network\|IP Address | This metric displays the IP address of the virtual machine. |
| Network\|MAC Address | This metric displays the MAC address of the virtual machine. |

## Disabled Instanced Metrics

The instance metrics created for the following metrics are disabled in this version of VCF Operations. This means that these metrics collect data by default but all the instanced metrics created for these metrics, do not collect data by default.

| Metric Name |
|---|
| Configuration\|Hardware\|Number of virtual CPUs |
| CPU\|Ready (%) |
| CPU\|Usage (MHz) |
| Net\|Broadcast Packets Transmitted |
| Net\|Data Transmit Rate (KBps) |
| Net\|Data Receive Rate (KBps) |
| Net\|Multicast Packets Transmitted |
| Net\|Packets Dropped |
| Net\|Packets Dropped (%) |

| Metric Name |
|---|
| Net\|pnicByteRx_average |
| Net\|pnicByteTx_average |
| Net\|Transmitted Packets Dropped |
| Net\|Usage Rate (KBps) |
| VirtualDisk\|Read IOPS |
| VirtualDisk\|Read Latency (ms) |
| VirtualDisk\|Read Throughput (KBps) |
| VirtualDisk\|Total IOPS |
| VirtualDisk\|Total Latency |
| VirtualDisk\|Total Throughput (KBps) |
| Virtual Disk\|Used Disk Space (GB) |
| VirtualDisk\|Write IOPS |
| VirtualDisk\|Write Latency (ms) |
| VirtualDisk\|Write Throughput (KBps) |
| Datastore\|Outstanding IO requests |
| Datastore\|Read IOPS |
| Datastore\|Read Latency (ms) |
| Datastore\|Read Throughput (KBps) |
| Datastore\|Total IOPS |
| Datastore\|Total Latency (ms) |
| Datastore\|Total Throughput (KBps) |
| Datastore\|Write IOPS |
| Datastore\|Write Latency (ms) |
| Datastore\|Write Throughput (KBps) |
| Disk\|Total IOPS |
| Disk\|Total Throughput (KBps) |
| Disk\|Read Throughput KBps) |
| Disk\|Write Throughput (KBps) |
| Diskspace\|Access Time (ms) |
| Diskspace\|Virtual machine used (GB) |

**Disabled Metrics**

The following metrics are disabled in this version of VCF Operations. This means that they do not collect data by default.

You can enable these metrics in the Policy workspace. For more information, in VMware Docs search for Collect Metrics and Properties Details.

You can enable these metrics in the Policy workspace.

| Metric Name | Description |
|---|---|
| CPU\|50% of Recommended number of vCPUs to Remove | This metric is superseded by the capacity engine. cpu\|numberToRemove50Pct |
| CPU\|Capacity entitlement (mhz) | cpu\|capacity_entitlement |

| Metric Name | Description |
|---|---|
| CPU\|Co-stop (msec) | Use the Co-Stop (%) metric instead of this metric.<br>cpu\|costop_summation |
| CPU\|Demand Over Capacity (mhz) | cpu\|demandOverCapacity |
| CPU\|Demand Over Limit (mhz) | Use Contention (%) metric instead of this metric.<br>cpu\|demandOverLimit |
| CPU\|Dynamic entitlement | cpu\|dynamic_entitlement |
| CPU\|Estimated entitlement | cpu\|estimated_entitlement |
| CPU\|Idle (%) | cpu\|idlePct |
| CPU\|Idle (msec) | cpu\|idle_summation |
| CPU\|Other Wait (msec) | cpu\|otherwait |
| CPU\|Normalized Co-stop (%) | Use the Co-Stop (%) metric instead of this metric.<br>cpu\|perCpuCoStopPct |
| CPU\|Provisioned vCPU(s) (Cores) | cpu\|corecount_provisioned |
| CPU\|Ready (msec) | Choose the Use Ready (%) metric instead of this metric.<br>cpu\|ready_summation |
| CPU\|Recommended Size Reduction (%) | cpu\|sizePctReduction |
| CPU\|Swap Wait (msec) | cpu\|swapwait_summation |
| CPU\|Total Wait (msec) | cpu\|wait |
| CPU\|Used (msec) | cpu\|used_summation |
| CPU\|Wait (msec) | cpu\|wait_summation |
| Datastore I/O\|Max Observed Number of Outstanding IO Operations | datastore\|maxObserved_OIO |
| Datastore I/O\|Max Observed Read Rate (kbps) | datastore\|maxObserved_Read |
| Datastore I/O\|Max Observed Reads per second | datastore\|maxObserved_NumberRead |
| Datastore I/O\|Max Observed Write Rate (kbps) | datastore\|maxObserved_Write |
| Datastore I/O\|Max Observed Writes per second | datastore\|maxObserved_NumberWrite |
| Disk Space\|Not Shared (gb) | diskspace\|notshared |
| Disk Space\|Number of Virtual Disks | diskspace\|numvmdisk |
| Disk Space\|Shared Used (gb) | diskspace\|shared |
| Disk Space\|Total disk space used (gb) | diskspace\|total_usage |
| Disk Space\|Total disk space (gb) | diskspace\|total_capacity |
| Disk Space\|Virtual Disk Used (gb) | diskspace\|diskused |
| Guest File System stats\|Total Guest File System Free (gab) | guestfilesystem\|freespace_total |
| Guest\|Active File Cache Memory (kb) | guest\|mem.activeFileCache_latest |
| Guest\|Context Swap Rate per second | guest\|contextSwapRate_latest |
| Guest\|Huge Page Size (kb) | guest\|hugePage.size_latest |
| Guest\|Page Out Rate per second | guest\|page.outRate_latest |
| Guest\|Total Huge Pages | guest\|hugePage.total_latest |
| Memory\|50% of Reclaimable Memory Capacity (gb) | This metric is superseded by the capacity engine.<br>mem\|wasteValue50PctInGB |
| Memory\|Balloon (kb) | mem\|vmmemctl_average |
| Memory\|Demand Over Capacity | mem\|demandOverCapacity |
| Memory\|Demand Over Limit | mem\|demandOverLimit |

| Metric Name | Description |
|---|---|
| Memory\|Granted (kb) | mem\|granted_average |
| Memory\|Guest Active (kb) | mem\|active_average |
| Memory\|Guest Dynamic Entitlement (kb) | mem\|guest_dynamic_entitlement |
| Memory\|Guest Workload (%) | mem\|guest_workload |
| Memory\|Host Demand with Reservation (kb) | mem\|host_demand_reservation |
| Memory\|Host Dynamic Entitlement (kb) | mem\|host_dynamic_entitlement |
| Memory\|Host Usage (kb) | mem\|host_usage |
| Memory\|Host Workload (%) | mem\|host_workload |
| Memory\|Latency (%) | Use the Memory Contention (%) metric instead of this metric.<br>mem\|latency_average |
| Memory\|Recommended Size Reduction (%) | mem\|sizePctReduction |
| Memory\|Shared (kb) | mem\|shared_average |
| Memory\|Swap Out Rate (kbps) | mem\|swapoutRate_average |
| Memory\|Usage (%) | mem\|usage_average |
| Memory\|Estimated entitlement | mem\|estimated_entitlement |
| Network I/O\|Data Receive Demand Rate (kbps) | net\|receive_demand_average |
| Network I/O\|Data Transmit Demand Rate (kbps) | net\|transmit_demand_average |
| Network I/O\|VM to Host Data Receive Rate (kbps) | net\|host_received_average |
| Network I/O\|VM to Host Data Transmit Rate (kbps) | net\|host_transmitted_average |
| Network I/O\|VM to Host Max Observed Received Throughput (kbps) | net\|host_maxObserved_Rx_KBps |
| Network I/O\|VM to Host Max Observed Throughput (kbps) | net\|host_maxObserved_KBps |
| Network I/O\|VM to Host Max Observed Transmitted Throughput (kbps) | net\|host_maxObserved_Tx_KBps |
| Network I/O\|VM to Host Usage Rate (kbps) | net\|host_usage_average |
| Network\|bytesRx (kbps) | net\|bytesRx_average |
| Network\|bytesTx (kbps) | net\|bytesTx_average |
| Network\|Demand (%) | Use absolute numbers instead of this metric.<br>net\|demand |
| Network\|I/O Usage Capacity | net\|usage_capacity |
| Network\|Max Observed Received Throughput (kbps) | net\|maxObserved_Rx_KBps |
| Network\|Max Observed Throughput (kbps) | net\|maxObserved_KBps |
| Network\|Max Observed Transmitted Throughput (kbps) | net\|maxObserved_Tx_KBps |
| Network\|Packets Received per second | net\|packetsRxPerSec |
| Network\|Packets Transmitted per second | net\|packetsTxPerSec |
| Network\|Received Packets Dropped | net\|droppedRx_summation |
| Storage\|Demand (kbps) | storage\|demandKBps |
| Storage\|Read Latency (msec) | storage\|totalReadLatency_average |
| Storage\|Write Latency (msec) | storage\|totalWriteLatency_average |
| Summary\|CPU Shares | summary\|cpu_shares |
| Summary\|Memory Shares | summary\|mem_shares |
| Summary\|Number of Datastores | summary\|number_datastore |
| Summary\|Number of Networks | summary\|number_network |

| Metric Name | Description |
|---|---|
| Summary\|Workload Indicator | summary\|workload_indicator |
| System\|Build Number | sys\|build |
| System\|Heartbeat | sys\|heartbeat_summation |
| System\|Product String | sys\|productString |
| System\|Uptime (sec) | sys\|uptime_latest |
| System\|vMotion Enabled | vMotion should be enabled for all. It is not necessary to track all VMs every five minutes.<br>sys\|vmotionEnabled |

## Host System Metrics

VCF Operations collects many metrics for host systems, including CPU use, datastore, disk, memory, network, storage, and summary metrics for host system objects.

Capacity metrics can be calculated for host system objects. See Capacity Analytics Generated Metrics.

### GPU Metrics

GPU metrics provide information about the GPU usage and performance.

| Metric Name | Description |
|---|---|
| **Metrics Aggregated at Host Level** | |
| GPU\|Compute Utilization (%) | The compute utilization percentage of a GPU. |
| GPU\|Memory Usage (%) | Memory currently in use as a percentage of total available memory. |
| GPU\|Memory Used (KB) | The amount of GPU memory used in kilobytes. |
| GPU\|Number of GPUs | Number of GPUs. |
| GPU\|Total Memory (KB) | Total memory in kilobytes. |
| **GPU Level Metrics** | |
| GPU\|<GPU-ID>\|Compute Utilization (%) | The compute utilization percentage of a GPU. |
| GPU\|<GPU-ID>\|Memory Usage (%) | Memory currently in use as a percentage of total available memory. |
| GPU\|<GPU-ID>\|Memory Used (KB) | The amount of GPU memory used in kilobytes. |
| GPU\|<GPU-ID>\|Memory Reserved (KB) | The amount of GPU memory reserved in kilobytes. |
| GPU\|<GPU-ID>\|Total Memory (KB) | Total memory in kilobytes. |
| GPU\|<GPU-ID>\|Temperature (Celsius) | The temperature of a GPU in degrees Celsius. |
| GPU\|<GPU-ID>\|Power Used (Watt) | The power used by a GPU in watts. |
| GPU:GPU-id\|GPU Compute Usage (MHz) | Total MHZ currently consumed by workloads (note time delay). |
| GPU:GPU-id\|GPU Compute Limit (MHz) | Total MHZ available on the GPU. |
| GPU:GPU-id\|GPU Compute Demand (MHz) | MHZ requested for a GPU over time. |
| GPU:GPU-id\|GPU Memory Demand (MHz) | KB requested over time. |
| GPU\|Number of vGPU Configured VM-s | Sum of GPU configured VMs. |

### Host System Metrics for ROI Dashboard

Host system metrics provide information about cost saving across vCenters

| Metric Name | Description |
|---|---|
| Cost\|Monthly Additional Total Cost | This metric shows the total sum of additional cost across all the vCenters for an entire month.<br>Key: cost\|additionalTotalCost |

### Configuration Metrics for Host Systems

Configuration metrics provide information about host system configuration.

| Metric Name | Description |
|---|---|
| Configuration\|Hyperthreading\|Active | Displays the hyperthreading status of the host.<br>Key: configuration\|hypwerthreading\|active |
| Configuration\|Hyperthreading\|Available | Displays whether the hyperthreading option is available for this host.<br>Key: configuration\|hypwerthreading\|available |
| Configuration\|Storage Device\|Multipath Info\|Total number of Active Path | Displays the amount of active path information for the storage device<br>Key: configuration\|storagedevice\|multipathinfo\|total numberofActive path |
| Configuration\|Storage Device\|Total number of path | Displays the total number of path for the storage device.<br>Key: configuration\|storagedevice\|total number of path |
| Configuration\|Failover Hosts | Failover Hosts.<br>Key: configuration\|dasConfig\|admissionControlPolicy\|failoverHost |

### Hardware Metrics for Host Systems

Hardware metrics provide information about host system hardware.

| Metric Name | Description |
|---|---|
| Hardware\|Number of CPUs | Number of CPUs for a host.<br>Key: hardware\|cpuinfo\|num_CpuCores |
| Hardware\|ServiceTag | Displays the service tag of the host system.<br>Key: hardware\|servicetag |

### CPU Usage Metrics for Host Systems

CPU usage metrics provide information about CPU use.

| Metric Name | Description |
|---|---|
| CPU\|Capacity Usage (%) | Percent CPU capacity used.<br>Key: cpu\|capacity_usagepct_average |
| CPU\|Usage (%) | Average CPU usage as a percentage.<br>Key: cpu\|usage_average |

| Metric Name | Description |
|---|---|
| CPU\|CPU Contention (%) | This metric indicates the percentage of time the virtual machines in the ESX hosts are unable to run because they are contending for access to the physical CPU(s). This is the average number of all VMs. Naturally, the number will be lower than the highest number experienced by the worst hit VM (a VM that suffers the highest CPU contention).<br>Use this metric to verify if the host is able to serve all of its VMs well.<br>When using this metric, ensure the number is within your expectation. The metric is affected by several factors so you need to watch both relative numbers and absolute numbers. Relative means a drastic change in value. This indicates that the ESX is unable to service its VMs.<br>Absolute means that the real value is high and should be checked. One factor that impacts the CPU contention metric is CPU Power Management. If CPU Power Management clocks down the CPU speed from 3 GHz to 2 GHz that reduction in speed is taken into consideration. This is because the VM is not running at full speed.<br>Key: cpu\|capacity_contentionPct |
| CPU\|Demand (%) | This metric shows the percentage of CPU resources all the VMs would use if there was no CPU contention or any CPU limits set. It represents the average active CPU load for the past five minutes.<br>Keep the number of this metric below 100% if you set Power Management to Maximum.<br>Key: cpu\|demandPct |
| CPU\|Demand (MHz) | CPU demand in megahertz. CPU utilization level based on descendant Virtual Machines utilization. Includes limits and overhead to run Virtual Machines, but not reservations.<br>Key: cpu\|demandmhz |
| CPU\|IO Wait (ms) | IO wait time in milliseconds.<br>Key: cpu\|iowait |
| CPU\|Number of CPU Sockets | Number of CPU sockets.<br>Key: cpu\|numpackages |
| CPU\|Overall CPU Contention (ms) | Overall CPU contention in milliseconds.<br>Key: cpu\|capacity_contention |
| CPU\|Provisioned Capacity (MHz) | Capacity in MHz of the physical CPU cores.<br>Key: cpu\|capacity_provisioned |
| CPU\|Provisioned virtual CPUs | Provisioned virtual CPUs.<br>Key: cpu\|corecount_provisioned |
| CPU\|Total Wait | CPU time spent in idle state.<br>Key: cpu\|wait |
| CPU\|Demand | CPU demand.<br>Key: cpu\|demand_average |
| CPU\|Usage (MHz) | CPU use in megahertz.<br>Key: cpu\|usagemhz_average |
| CPU\|Reserved Capacity (MHz) | The sum of the reservation properties of the (immediate) children of the host's root resource pool.<br>Key: cpu\|reservedCapacity_average |

| Metric Name | Description |
|---|---|
| CPU\|Total Capacity (MHz) | Total CPU capacity in megahertz. Amount of CPU resources configured on the ESX hosts.<br>Key: cpu\|capacity_provisioned |
| CPU\|Overhead (KB) | Amount of CPU overhead.<br>Key: cpu\|overhead_average |
| CPU\|Demand without overhead | Value of demand excluding any overhead.<br>Key: cpu\|demand_without_overhead |
| CPU\|Core Utilization (%) | Percent core utilization.<br>Key: cpu\|coreUtilization_average |
| CPU\|Utilization(%) | Percent CPU utilization.<br>Key: cpu\|utilization_average |
| CPU\|Core Utilization (%) | Core Utilization.<br>Key: cpu\|coreUtilization_average |
| CPU\|Utilization (%) | Utilization.<br>Key: cpu\|utilization_average |
| CPU\|Co-stop (ms) | Time the VM is ready to run, but is unable to due to co-scheduling constraints.<br>Key: cpu\|costop_summation |
| CPU\|Latency (%) | Percentage of time the VM is unable to run because it is contending for access to the physical CPUs.<br>Key: cpu\|latency_average |
| CPU\|Ready (ms) | Time spent in ready state.<br>Key: cpu\|ready_summation |
| CPU\|Run (ms) | Time the virtual machine is scheduled to run.<br>Key: cpu\|run_summation |
| CPU\|Swap wait (ms) | Amount of time waiting for swap space.<br>Key: cpu\|swapwait_summation |
| CPU\|Wait (ms) | Total CPU time spent in wait state.<br>Key: cpu\|wait_summation |
| CPU\|Provisioned Capacity | Provisioned capacity (MHz).<br>Key: cpu\|vm_capacity_provisioned |
| CPU\|Active Host Load For Balance (Long Term) | Active Host Load For Balance (Long Term).<br>Key: cpu\|acvmWorkloadDisparityPcttive_longterm_load |
| CPU\|Active Host Load For Balance (Short Term) | Active Host Load For Balance (Short Term).<br>Key: cpu\|active_shortterm_load |
| CPU\| CPU Model | Displays the host's CPU model.<br>Key: cpu\|cpu model |
| CPU\|Peak CPU Core Usage | The highest CPU Usage among the CPU cores. A constantly high number indicates that one or more physical cores have high utilization.<br>Key: cpu\|peak_cpu_core_usage |

## CPU Utilization for Resources Metrics for Host Systems

CPU utilization for resources metrics provide information about CPU activity.

| Metric Name | Description |
|---|---|
| Rescpu\|CPU Active (%) (*interval*) | Average active time for the CPU over the past minute, past five minutes, and at one-minute, five-minute, and 15-minute peak active times.<br>Key:<br>rescpu\|actav1_latest<br>rescpu\|actav5_latest<br>rescpu\|actav15_latest<br>rescpu\|actpk1_latest<br>rescpu\|actpk5_latest<br>rescpu\|actpk15_latest |
| Rescpu\|CPU Running (%) (*interval*) | Average run time for the CPU over the past minute, past five minutes, past 15 minutes, and at one-minute, five-minute, and 15-minute peak times.<br>Key:<br>rescpu\|runav1_latest<br>rescpu\|runav5_latest<br>rescpu\|runav15_latest<br>rescpu\|runpk1_latest<br>rescpu\|runpk5_latest<br>rescpu\|runpk15_latest |
| Rescpu\|CPU Throttled (%) (*interval*) | Scheduling limit over the past minute, past five minutes, and past 15 minutes.<br>Key:<br>rescpu\|maxLimited1_latest<br>rescpu\|maxLimited5_latest<br>rescpu\|maxLimited15_latest |
| Rescpu\|Group CPU Sample Count | Group CPU sample count.<br>Key: rescpu\|sampleCount_latest |
| Rescpu\|Group CPU Sample Period (ms) | Group CPU sample period in milliseconds.<br>Key: rescpu\|samplePeriod_latest |

## Datastore Metrics for Host Systems

Datastore metrics provide information about datastore use.

| Metric Name | Description |
|---|---|
| Datastore\|Outstanding IO requests | OIO for datastore.<br>Key: datastore\|demand_oio |
| Datastore\|Commands Averaged | Average number of commands issued per second during the collection interval.<br>Key: datastore\|commandsAveraged_average |
| Datastore\|Number of Outstanding IO Operations | Number of outstanding IO operations.<br>Key: datastore\|oio |
| Datastore\|Total Latency (ms) | The average amount of time taken for a command from the perspective of a Guest OS. This is the sum of Kernel Command Latency and Physical Device Command Latency.<br>Key: datastore\|totalLatency_average |

| Metric Name | Description |
|---|---|
| Datastore\|Total Throughput (KBps) | Usage Average (KBps).<br>Key: datastore\|usage_average |
| Datastore\|Demand | Demand.<br>Key: datastore\|demand |
| Datastore\|Storage I/O Control aggregated IOPS | Aggregate number of IO operations on the datastore.<br>Key: datastore\|datastoreIops_average |
| Datastore\|Read IOPS | Average number of read commands issued per second during the collection interval.<br>Key: datastore\|numberReadAveraged_average |
| Datastore\|Write IOPS | Average number of write commands issued per second during the collection interval.<br>Key: datastore\|numberWriteAveraged_average |
| Datastore\|Read Throughput (KBps) | Rate of reading data from the datastore in kilobytes per second.<br>Key: datastore\|read_average |
| Datastore\|Storage I/O Control normalized latency (ms) | Normalized latency in microseconds on the datastore. Data for all virtual machines is combined.<br>Key: datastore\|sizeNormalizedDatastoreLatency_average |
| Datastore\|Read Latency (ms) | Average amount of time for a read operation from the datastore. Total latency = kernel latency + device latency.<br>Key: datastore\|totalReadLatency_average |
| Datastore\|Write Latency (ms) | Average amount of time for a write operation to the datastore. Total latency = kernel latency + device latency.<br>Key: datastore\|totalWriteLatency_average |
| Datastore\|Write Throughput (KBps) | Rate of writing data to the datastore in kilobytes per second.<br>Key: datastore\|write_average |
| Datastore\|Max Queue Depth | Max Queue Depth.<br>Key: datastore\|datastoreMaxQueueDepth_latest |
| Datastore\|Highest Latency | Highest Latency.<br>Key: datastore\|maxTotalLatency_latest |
| Datastore\|Total Latency Max | Total Latency Max (ms).<br>Key: datastore\|totalLatency_max |
| Datastore\|Read Latency | Read Latency.<br>Key: datastore\|datastoreNormalReadLatency_latest |
| Datastore\|Write Latency | Write Latency.<br>Key: datastore\|datastoreNormalWriteLatency_latest |
| Datastore\|Data Read | Data Read.<br>Key: datastore\|datastoreReadBytes_latest |
| Datastore\|Data Read Rate | Data Rate.<br>Key: datastore\|datastoreReadIops_latest |
| Datastore\|Read Load | Storage DRS metric read load.<br>Key: datastore\|datastoreReadLoadMetric_latest |
| Datastore\|Outstanding Read Requests | Outstanding Read Requests.<br>Key: datastore\|datastoreReadOIO_latest |
| Datastore\|Data Written | Data Written.<br>Key: datastore\|datastoreWriteBytes_latest |
| Datastore\|Data Write Rate | Data Write Rate.<br>Key: datastore\|datastoreWriteIops_latest |

| Metric Name | Description |
|---|---|
| Datastore\|Write Load | Storage DRS metric write load.<br>Key: datastore\|datastoreWriteLoadMetric_latest |
| Datastore\|Outstanding Write Requests | Outstanding Write Requests.<br>Key: datastore\|datastoreWriteOIO_latest |
| Datastore\|VM Disk I/O Workload Disparity | Percentage Disk I/O workload disparity among the VMs on the Host.<br>Key: datastore\|vmWorkloadDisparityPc |
| Datastore\|Peak Datastore Read Latency | The highest read latency among the datastores. A high number indicates that one or more datastores are experiencing poor performance.<br>Key: datastore\|peak_datastore_readLatency |
| Datastore\|Peak Datastore Write Latency | The highest write latency among the datastores. A high number indicates that one or more datastores are experiencing poor performance.<br>Key: datastore\|peak_datastore_writeLatency |

### Disk Metrics for Host Systems

Disk metrics provide information about disk use.

| Metric Name | Description |
|---|---|
| Disk\|Total Throughput (KBps) | Average of the sum of the data read and written for all of the disk instances of the host or virtual machine.<br>disk\|usage_average |
| Disk\|I/O Usage Capacity | This metric is a function of storage\|usage_average and disk\|workload. storage\|usage_average is an average over all storage devices. This means that disk\|usage_capacity is not specific to the selected VM or the host of the VM.<br>Key: disk\|usage_capacity |
| Disk\|Total IOPS | Average number of commands issued per second during the collection interval.<br>Key: disk\|commandsAveraged_average |
| Disk\|Total Latency (ms) | The average amount of time taken for a command from the perspective of a Guest OS. This is the sum of Kernel Command Latency and Physical Device Command Latency.<br>Key: disk\|totalLatency_average |
| Disk\|Read IOPS | Average number of read commands issued per second during the collection interval.<br>Key: disk\|numberReadAveraged_average |
| Disk\|Write IOPS | Average number of write commands issued per second during the collection interval.<br>Key: disk\|numberWriteAveraged_average |
| Disk\|Read Throughput (KBps) | Amount of data read in the performance interval.<br>Key: disk\|read_average |
| Disk\|Write Throughput (KBps) | Amount of data written to disk in the performance interval.<br>Key: disk\|write_average |
| Disk\|Bus Resets | The number of bus resets in the performance interval.<br>Key: disk\|busResets_summation |

| Metric Name | Description |
|---|---|
| Disk\|Read Latency (ms) | The average amount of time taken for a read from the perspective of a Guest OS. This is the sum of Kernel Read Latency and Physical Device Read Latency.<br>Key: disk\|totalReadLatency_average |
| Disk\|Write Latency (ms) | The average amount of time taken for a write from the perspective of a Guest OS. This is the sum of Kernel Write Latency and Physical Device Write Latency.<br>Key: disk\|totalWriteLatency_average |
| Disk\|Physical Device Latency (ms) | The average time taken to complete a command from the physical device.<br>Key: disk\|deviceLatency_average |
| Disk\|Kernel Latency (ms) | The average time spent in ESX Server VMKernel per command.<br>Key: disk\|kernelLatency_average |
| Disk\|Queue Latency (ms) | The average time spent in the ESX Server VMKernel queue per command.<br>Key: disk\|queueLatency_average |
| Disk\|Number of Outstanding IO Operations | Number of Outstanding IO Operations.<br>Key: disk\|diskoio |
| Disk\|Queued Operations | Queued Operations.<br>Key: disk\|diskqueued |
| Disk\|Demand | Demand.<br>Key: disk\|diskdemand |
| Disk\|Total Queued Outstanding operations | Sum of Queued Operation and Outstanding Operations.<br>Key: disk\|sum_queued_oio |
| Disk\|Max Observed OIO | Max Observed IO for a disk.<br>Key: disk\|max_observed |
| Disk\|Highest Latency | Highest Latency.<br>Key: disk\|maxTotalLatency_latest |
| Disk\|Max Queue Depth | Maximum queue depth during the collection interval.<br>Key: disk\|maxQueueDepth_average |
| Disk\|SCSI Reservation Conflicts | SCSI Reservation Conflicts.<br>Key: disk\|scsiReservationConflicts_summation |

## Memory Metrics for Host Systems

Memory metrics provide information about memory use and allocation.

| Metric Name | Description |
|---|---|
| Mem\|Contention (%) | This metric is used to monitor ESX memory usage.<br>When the value is high, it means the ESX is using a good percentage of available memory. You may need to add more memory to other memory-related metrics.<br>Key: mem\|host_contentionPct |
| Mem\|Contention (KB) | Host contention in kilobytes.<br>Key: mem\|host_contention |
| Mem\|Host Usage (KB) | Machine usage in kilobytes.<br>Key: mem\|host_usage |
| Mem\|Machine Demand (KB) | Host demand in kilobytes.<br>Key: mem\|host_demand |

| Metric Name | Description |
|---|---|
| Mem\|Overall Memory used to run VMs on Host (KB) | Overall memory used to run virtual machines on the host in kilobytes.<br>Key: mem\|host_usageVM |
| Mem\|Provisioned Memory (KB) | Provisioned memory in kilobytes.<br>Key: mem\|host_provisioned |
| Mem\|Minimum Free Memory (KB) | Minimum free memory.<br>Key: mem\|host_minfree |
| Mem\|Reserved Capacity (%) | Percent reserved capacity.<br>Key: mem\|reservedCapacityPct |
| Mem\|Usable Memory (KB) | Usable memory in kilobytes.<br>Key: mem\|host_usable |
| Mem\|Usage (%) | Memory currently in use as a percentage of total available memory.<br>Key: mem\|host_usagePct |
| Mem\|ESX System Usage | Memory usage by the VMkernel and ESX user-level services.<br>Key: mem\|host_systemUsage |
| Mem\|Guest Active (KB) | Amount of memory that is actively used.<br>Key: mem\|active_average |
| Mem\|Consumed (KB) | Amount of host memory consumed by the virtual machine for guest memory.<br>Key: mem\|consumed_average |
| Mem\|Granted (KB) | Amount of memory available for use.<br>Key: mem\|granted_average |
| Mem\|Heap (KB) | Amount of memory allocated for heap.<br>Key: mem\|heap_average |
| Mem\|Heap Free (KB) | Amount of free space in the heap.<br>Key: mem\|heapfree_average |
| Mem\|VM Overhead (KB) | Memory overhead reported by host.<br>Key: mem\|overhead_average |
| Mem\|Reserved Capacity (KB) | Sum of memory reserved by consumers - Resource Pools (RP) and powered on VMs that are not in Resource Pools. RP reservations at the vCenter level are distributed to the hosts depending on the number of powered on VMs and their entitlement within the RP.<br>Key: mem\|reservedCapacity_average |
| Mem\|Shared (KB) | Amount of shared memory in kilobytes.<br>Key: mem\|shared_average |
| Mem\|Shared Common (KB) | Amount of shared common memory in kilobytes.<br>Key: mem\|sharedcommon_average |
| Mem\|Swap In (KB) | Amount of memory swapped in.<br>Key: mem\|swapin_average |
| Mem\|Swap Out KB) | Amount of memory swapped out.<br>Key: mem\|swapout_average |
| Mem\|Swap Used (KB) | Amount of memory used for swapped space in kilobytes.<br>Key: mem\|swapused_average |
| Mem\|VM kernel Usage (KB) | Amount of memory used by the VM kernel.<br>Key: mem\|sysUsage_average |
| Mem\|Unreserved (KB) | Amount of unreserved memory in kilobytes.<br>Key: mem\|unreserved_average |

| Metric Name | Description |
|---|---|
| Mem\|Balloon (KB) | This metric shows the total amount of memory currently used by the VM memory control. This memory was reclaimed from the respective VMs at some point in the past, and was not returned.<br>Use this metric to monitor how much VM memory has been reclaimed by ESX through memory ballooning.<br>The presence of ballooning indicates the ESX has been under memory pressure. The ESX activates ballooning when consumed memory reaches a certain threshold.<br>Look for increasing size of ballooning. This indicates that there has been a shortage of memory more than once. Look for size fluctuations which indicate the ballooned out page was actually required by the VM. This translates into a memory performance problem for the VM requesting the page, since the page must first be brought back from the disk.<br>Key: mem\|vmmemctl_average |
| Mem\|Zero (KB) | Amount of memory that is all zero.<br>Key: mem\|zero_average |
| Mem\|State (0-3) | Overall state of the memory. The value is an integer between 0 (high) and 3 (low).<br>Key: mem\|state_latest |
| Mem\|Usage (KB) | Host memory use in kilobytes.<br>Key: mem\|host_usage |
| Mem\|Usage (%) | Memory currently in use as a percentage of total available memory.<br>Key: mem\|usage_average |
| Mem\|Swap In Rate (KBps) | Rate at which memory is swapped from disk into active memory during the interval in kilobyte per second.<br>Key: mem\|swapinRate_average |
| Mem\|Swap Out Rate (KBps) | Rate at which memory is being swapped from active memory to disk during the current interval in kilobytes per second.<br>Key: mem\|swapoutRate_average |
| Mem\|Active Write (KB) | Average active writes in kilobytes.<br>Key: mem\|activewrite_average |
| Mem\|Compressed (KB) | Average memory compression in kilobytes.<br>Key: mem\|compressed_average |
| Mem\|Compression Rate (KBps) | Average compression rate in kilobytes per second.<br>Key: mem\|compressionRate_average |
| Mem\|Decompression Rate (KBps) | Decompression rate in kilobytes per second.<br>Key: mem\|decompressionRate_average |
| Mem\|Total Capacity (KB) | Sum of the amount of physical memory configured on ESX hosts of the cluster in KB.<br>Key: mem\|host_provisioned |
| Mem\|Latency | Percentage of time the VM is waiting to access swapped or compressed memory.<br>Key: mem\|latency_average |
| Mem\|Capacity Contention | Capacity Contention.<br>Key: mem\|capacity.contention_average |
| Mem\|Swap In Rate from Host Cache | Rate at which memory is being swapped from host cache into active memory.<br>Key: mem\|llSwapInRate_average |

| Metric Name | Description |
|---|---|
| Mem\|Swap In from Host Cache | Amount of memory swapped-in from host cache.<br>Key: mem\|llSwapIn_average |
| Mem\|Swap Out Rate to Host Cache | Rate at which memory is being swapped to host cache from active memory.<br>Key: mem\|llSwapOutRate_average |
| Mem\|Swap Out to Host Cache | Amount of memory swapped-out to host cache.<br>Key: mem\|llSwapOut_average |
| Mem\|Swap Space Used in Host Cache | Space used for caching swapped pages in the host cache.<br>Key: mem\|llSwapUsed_average |
| Mem\|Low Free Threshold | Threshold of free host physical memory below which ESX begins to reclaim memory from VMs through ballooning and swapping.<br>Key: mem\|lowfreethreshold_average |
| Mem\|VM Memory Workload Disparity | Percentage Memory workload disparity among the VMs on the Host.<br>Key: mem\|vmWorkloadDisparityPct |
| Mem\|Active Host Load For Balance (Long Term) | Active Host Load For Balance (Long Term).<br>Key: mem\|active_longterm_load |
| Mem\|Active Host Load For Balance (Short Term) | Active Host Load For Balance (Short Term).<br>Key: mem\|active_shortterm_load |
| Mem\|Utilization | Memory utilization level based on descendant Virtual Machines utilization. Includes reservations, limits and overhead to run Virtual Machines<br>Key: mem\|total_need |

### Network Metrics for Host Systems

Network metrics provide information about network performance.

| Metric Name | Description |
|---|---|
| Network\|Driver | This metric displays the type of network driver.<br>Key: net\|driver |
| Network\|Speed | This metric displays the network speed.<br>Key: net\|speed |
| Network\|Management Address | This metric displays the management address of the host network.<br>Key: net\|management address |
| Network\|IP Address | This metric displays the IP address of the host network.<br>Key: net\|IPaddress |
| Net\|Packets Transmitted per second | This metric shows the number of packets transmitted during the collection interval.<br>Key: net\|packetsTxPerSec |
| Net\|Packets per second | Number of packets transmitted and received per second.<br>Key: net\|packetsPerSec |
| Net\|Total Throughput (KBps) | The sum of the data transmitted and received for all the NIC instances of the host or virtual machine.<br>Key: net\|usage_average |
| Net\|I/O Usage Capacity | I/O Usage Capacity.<br>Key: net\|usage_capacity |

| Metric Name | Description |
|---|---|
| Net\|Data Transmit Rate (KBps) | Average amount of data transmitted per second.<br>Key: net\|transmitted_average |
| Net\|Data Receive Rate (KBps) | Average amount of data received per second.<br>Key: net\|received_average |
| Net\|Packets Received | Number of packets received in the performance interval.<br>Key: net\|packetsRx_summation |
| Net\|Packets Transmitted | Number of packets transmitted in the performance interval.<br>Key: net\|packetsTx_summation |
| Net\|Broadcast Packets Received | Number of broadcast packets received during the sampling interval.<br>Key: net\|broadcastRx_summation |
| Net\|Broadcast Packets Transmitted | Number of broadcast packets transmitted during the sampling interval.<br>Key: net\|broadcastTx_summation |
| Net\|Error Packets Transmitted | Number of packets with errors transmitted.<br>Key: net\|errorsTx_summation |
| Net\|Multicast Packets Received | Number of multicast packets received.<br>Key: net\|multicastRx_summation |
| Net\|Multicast Packets Transmitted | Number of multicast packets transmitted.<br>Key: net\|multicastTx_summation |
| Net\|FT Throughput Usage | FT Throughput Usage.<br>Key: net\|throughput.usage.ft_average |
| Net\|HBR Throughput Usage | HBR Throughput Usage.<br>Key: net\|throughput.usage.hbr_average |
| Net\|iSCSI Throughput Usage | iSCSI Throughput Usage.<br>Key: net\|throughput.usage.iscsi_average |
| Net\|NFS Throughput Usage | NFS Throughput Usage.<br>Key: net\|throughput.usage.nfs_average |
| Net\|VM Throughput Usage | VM Throughput Usage.<br>Key: net\|throughput.usage.vm_average |
| Net\|vMotion Throughput Usage | vMotion Throughput Usage.<br>Key: net\|throughput.usage.vmotion_average |
| Net\|Unknown Protocol Frames Received | Number of frames with unknown protocol received.<br>Key: net\|unknownProtos_summation |

### System Metrics for Host Systems

System metrics provide information about the amount of CPU that resources and other applications use.

| Metric Name | Description |
|---|---|
| Sys\|Power On | 1 if the host system is powered on, 0 if the host system is powered off, or -1 if the power state is unknown.<br>Key: sys\|poweredOn |
| Sys\|Uptime (seconds) | Number of seconds since the last system startup.<br>Key: sys\|uptime_latest |
| Sys\|Disk Usage (%) | Percent disk use.<br>Key: sys\|diskUsage_latest |

| Metric Name | Description |
|---|---|
| Sys\|Resource CPU Usage (MHz) | Amount of CPU that the Service Console and other applications use.<br>Key: sys\|resourceCpuUsage_average |
| Sys\|Resource CPU Active (1 min. average) | Percentage of resource CPU that is active. Average value during a one-minute period.<br>Key: sys\|resourceCpuAct1_latest |
| Sys\|Resource CPU Active (%) (5 min. average) | Percentage of resource CPU that is active. Average value during a five-minute period.<br>Key: sys\|resourceCpuAct5_latest |
| Sys\|Resource CPU Alloc Max (MHz) | Maximum resource CPU allocation in megahertz.<br>Key: sys\|resourceCpuAllocMax_latest |
| Sys\|Resource CPU Alloc Min (MHz) | Minimum resource CPU allocation in megahertz.<br>Key: sys\|resourceCpuAllocMin_latest |
| Sys\|Resource CPU Alloc Shares | Number of resource CPU allocation shares.<br>Key: sys\|resourceCpuAllocShares_latest |
| Sys\|Resource CPU Max Limited (%) (1 min. average) | Percent of resource CPU that is limited to the maximum amount. Average value during a one-minute period.<br>Key: sys\|resourceCpuMaxLimited1_latest |
| Sys\|Resource CPU Max Limited (%) (5 min. average) | Percentage of resource CPU that is limited to the maximum amount. Average value during a five-minute period.<br>Key: sys\|resourceCpuMaxLimited5_latest |
| Sys\|Resource CPU Run1 (%) | Percent resource CPU for Run1.<br>Key: sys\|resourceCpuRun1_latest |
| Sys\|Resource CPU Run5 (%) | Percent resource CPU for Run5.<br>Key: sys\|resourceCpuRun5_latest |
| Sys\|Resource Memory Alloc Max (KB) | Maximum resource memory allocation in kilobytes.<br>Key: sys\|resourceMemAllocMax_latest |
| Sys\|Resource Memory Alloc Min (KB) | Minimum resource memory allocation in kilobytes.<br>Key: sys\|resourceMemAllocMin_latest |
| Sys\|Resource Memory Alloc Shares | Number of resource memory shares allocated.<br>Key: sys\|resourceMemAllocShares_latest |
| Sys\|Resource Memory Cow (KB) | Cow resource memory in kilobytes.<br>Key: Sys\|resourceMemCow_latest |
| Sys\|Resource Memory Mapped (KB) | Mapped resource memory in kilobytes.<br>Key: ys\|resourceMemMapped_latest |
| Sys\|Resource Memory Overhead (KB) | Resource memory overhead in kilobytes.<br>Key: sys\|resourceMemOverhead_latest |
| Sys\|Resource Memory Shared (KB) | Shared resource memory in kilobytes.<br>Key: sys\|resourceMemShared_latest |
| Sys\|Resource Memory Swapped (KB) | Swapped resource memory in kilobytes.<br>Key: sys\|resourceMemSwapped_latest |
| Sys\|Resource Memory Touched (KB) | Touched resource memory in kilobytes.<br>Key: sys\|resourceMemTouched_latest |
| Sys\|Resource Memory Zero (KB) | Zero resource memory in kilobytes.<br>Key: sys\|resourceMemZero_latest |
| Sys\|Resource Memory Consumed | Resource Memory Consumed Latest (KB).<br>Key: sys\|resourceMemConsumed_latest |

| Metric Name | Description |
|---|---|
| Sys\|Resource File descriptors usage | Resource File descriptors usage (KB).<br>Key: sys\|resourceFdUsage_latest |
| Sys\|vMotion Enabled | 1 if vMotion is enabled or 0 if vMotion is not enabled.<br>Key: sys\|vmotionEnabled |
| Sys\|Not in Maintenance | Not in maintenance.<br>Key: sys\|notInMaintenance |

### Management Agent Metrics for Host Systems

Management agent metrics provide information about memory use.

| Metric Name | Description |
|---|---|
| Management Agent\|Memory Used (%) | Amount of total configured memory that is available for use.<br>Key: managementAgent\|memUsed_average |
| Management Agent\|Memory Swap Used (KB) | Sum of the memory swapped by all powered-on virtual machines on the host.<br>Key: managementAgent\|swapUsed_average |
| Management Agent\|Memory Swap In (KBps) | Amount of memory that is swapped in for the Service Console.<br>Key: managementAgent\|swapIn_average |
| Management Agent\|Memory Swap Out (KBps) | Amount of memory that is swapped out for the Service Console.<br>Key: managementAgent\|swapOut_average |
| Management Agent\|CPU Usage | CPU usage.<br>Key: managementAgent\|cpuUsage_average |

### Storage Adapter Metrics for Host Systems

Storage adapter metrics provide information about data storage use.

| Metric Name | Description |
|---|---|
| Storage Adapter\|Driver | Displays the driver details of the storage adapter.<br>Key: storage adapter\|driver |
| Storage Adapter\|Port WWN | Displays the world wide network port for the storage adapter.<br>Key: storage adapter\|portwwn |
| Storage Adapter\|Total Usage (KBps) | Total latency.<br>Key: storageAdapter\|usage |
| Storage Adapter\|Total IOPS | Average number of commands issued per second by the storage adapter during the collection interval.<br>Key: storageAdapter\|commandsAveraged_average |
| Storage Adapter\|Read IOPS | Average number of read commands issued per second by the storage adapter during the collection interval.<br>Key: storageAdapter\|numberReadAveraged_average |
| Storage Adapter\|Write IOPS | Average number of write commands issued per second by the storage adapter during the collection interval.<br>Key: storageAdapter\|numberWriteAveraged_average |
| Storage Adapter\|Read Throughput (KBps) | Rate of reading data by the storage adapter.<br>Key: storageAdapter\|read_average |

| Metric Name | Description |
|---|---|
| Storage Adapter\|Read Latency (ms) | This metric shows the average amount of time for a read operation by the storage adapter.<br>Use this metric to monitor the storage adapter read operation performance. A high value means that the ESX is performing a slow storage read operation.<br>Total latency is the sum of kernel latency and device latency.<br>Key: storageAdapter\|totalReadLatency_average |
| Storage Adapter\|Write Latency (ms) | This metric shows the average amount of time for a write operation by the storage adapter.<br>Use this metric to monitor the storage adapter write performance operation. A high value means that the ESX is performing a slow storage write operation.<br>Total latency is the sum of kernel latency and device latency.<br>Key: storageAdapter\|totalWriteLatency_average |
| Storage Adapter\|Write Throughput (KBps) | Rate of writing data by the storage adapter.<br>Key: storageAdapter\|write_average |
| Storage Adapter\|Demand | Demand.<br>Key: storageAdapter\|demand |
| Storage Adapter\|Highest Latency | Highest Latency.<br>Key: torageAdapter\|maxTotalLatency_latest |
| Storage Adapter\|Outstanding Requests | Outstanding Requests.<br>Key: storageAdapter\|outstandingIOs_average |
| Storage Adapter\|Queue Depth | Queue Depth.<br>Key: storageAdapter\|queueDepth_average |
| Storage Adapter\|Queue Latency (ms) | The average time spent in the ESX Server VM Kernel queue per command.<br>Key: storageAdapter\|queueLatency_average |
| Storage Adapter\|Queued | Queued.<br>Key: storageAdapter\|queued_average |
| Storage Adapter\|Peak Adapter Read Latency | The highest read latency among the storage adapters. A high number indicates that one or more storage adapters are experiencing poor performance.<br>Key: storageAdapter\|peak_adapter_readLatency |
| Storage Adapter\|Peak Adapter Write Latency | The highest write latency among the storage adapters. A high number indicates that one or more storage adapters are experiencing poor performance.<br>Key: storageAdapter\|peak_adapter_writeLatency |

## Storage Metrics for Host Systems

Storage metrics provide information about storage use.

| Metric Name | Description |
|---|---|
| Storage\|Total IOPS | Average number of commands issued per second during the collection interval.<br>Key: storage\|commandsAveraged_average |
| Storage\|Read Latency (ms) | Average amount of time for a read operation in milliseconds.<br>Key: storage\|totalReadLatency_average |
| Storage\|Read Throughput (KBps) | Read throughput rate in kilobytes.<br>Key: storage\|read_average |

| Metric Name | Description |
|---|---|
| Storage\|Read IOPS | Average number of read commands issued per second during the collection interval.<br>Key: storage\|numberReadAveraged_average |
| Storage\|Total Latency (ms) | Total latency in milliseconds.<br>Key: storage\|totalLatency_average |
| Storage\|Total Usage (KBps) | Total throughput rate in kilobytes per second.<br>Key: storage\|usage_average |
| Storage\|Write Latency (ms) | Average amount of time for a write operation in milliseconds.<br>Key: storage\|totalWriteLatency_average |
| Storage\|Write Throughput (KBps) | Write throughput rate in kilobytes per second.<br>Key: storage\|write_average |
| Storage\|Write IOPS | Average number of write commands issued per second during the collection interval.<br>Key: storage\|numberWriteAveraged_average |

### Sensor Metrics for Host Systems

Sensor metrics provide information about host system cooling.

| Metric Name | Description |
|---|---|
| Sensor\|Fan\|Speed (%) | Percent fan speed.<br>Key: Sensor\|fan\|currentValue |
| Sensor\|Fan\|Health State | Fan health state.<br>Key: Sensor\|fan\|healthState |
| Sensor\|Temperature\|Temp C | Fan temperature in centigrade.<br>Key: Sensor\|temperature\|currentValue |
| Sensor\|Temperature\|Health State | Fan health state.<br>Key: Sensor\|temperature\|healthState |

## Power Metrics for Host Systems

Power metrics provide information about host system power use.

| Metric Name | Description |
|---|---|
| Power\|Total Energy Consumed in the collection period (Wh) | Displays the total electricity consumed based on the time interval selected. The default collection cycle is set to 5 mins. You can continue using the default setting or edit it for each adapter instance. For example, if the time interval is set to its default value, the value represents the energy consumed per 5 mins. |
| Power\|Total Host System Power Consumed in an Hour (Wh) | Displays the the total electricity power consumed in an hour by ESX Host. The data collected is over a period of an hour and published along with the other metrics in VCF Operations. In case of a connectivity or availability issue in VCF Operations or vCenter adapter instance, this hourly metric might not be published and the missed value during this period does not get recalculated. Once the connection is re-established, the next data points get published.<br>**Note:**<br>This metric is deactivated by default. You can activate it from the Policies page. |
| Power\|Power (Watt) | Host power use in watts.<br>Key: power\|power_average |
| Power\|Current Power Consumption Rate (Watt) | The power consumption rate per second, averaged over the reporting period.<br>Key: power\|power_average |
| Power\|Power Cap (Watt) | Host power capacity in watts.<br>Key: power\|powerCap_average |
| Power\|Host Power Capacity Usage – Idle | Power consumed by the host in its idle state. This is the power consumed by the host when there are no VMs in it.<br>Key: power\|capacity.usageIdle_average |
| Power\|(DEP) Energy (Joule) | Total energy consumed in joules.<br>Key: power\|energy_summation |

## Disk Space Metrics for Host Systems

Disk space metrics provide information about disk space use.

| Metric Name | Description |
|---|---|
| Diskspace\|Number of Virtual Disks | Number of virtual disks.<br>Key: diskspace\|numvmdisk |
| Diskspace\|Shared Used (GB) | Used shared disk space in gigabytes.<br>Key: diskspace\|shared |
| Diskspace\|Snapshot | Disk space used by snapshots in gigabytes.<br>Key: diskspace\|snapshot |
| Diskspace\|Virtual Disk Used (GB) | Disk space used by virtual disks in gigabytes.<br>Key: diskspace\|diskused |
| Diskspace\|Virtual machine used (GB) | Disk space used by virtual machines in gigabytes.<br>Key: diskspace\|used |
| Diskspace\|tTotal disk space used | Total disk space used on all datastores visible to this object.<br>Key: diskspace\|total_usage |

| Metric Name | Description |
|---|---|
| Diskspace\|Total disk spacey | Total disk space on all datastores visible to this object.<br>Key: diskspace\|total_capacity |
| Diskspace\|Total provisioned disk space | Total provisioned disk space on all datastores visible to this object.<br>Key: diskspace\|total_provisioned . |
| Diskspace\|Utilization (GB) | Storage space utilized on connected vSphere datastores.<br>Key: diskspace\|total_usage |
| Diskspace\|Workload (%) | Total storage space available on connected vSphere datastores.<br>Key: diskspace\|total_capacity |

## Summary Metrics for Host Systems

Summary metrics provide information about overall host system performance.

| Metric Name | Description |
|---|---|
| Summary\|Number of Running VMs | This metric shows the number of VMs running on the host during the last metric collection time.<br>Large spikes of running VMs might be a reason for CPU or memory spikes as more resources are used in the host.<br>Number of Running VMs gives you a good indicator of how many requests the ESX host must juggle. This excludes powered off VMs as they do not impact ESX performance. A change in this number in your environment can contribute to performance problems. A high number of running VMs in a host also means a higher concentration risk, as all the VMs will become unavailable (or be relocated by HA) if the ESX crashes.<br>Look for any correlation between spikes in the number of running VMs and spikes in other metrics such as CPU Contention/Memory Contention.<br>Key: summary\|number_running_vms |
| Summary\|Maximum Number of VMs | Maximum number of virtual machines<br>Key: summary\|max_number_vms |
| Summary\|Number of vMotions | This metric shows the number of vMotions that occurred in the host in the last X minutes.<br>The number of vMotions is a good indicator of stability. In a healthy environment, this number should be stable and relatively low.<br>Look for correlation between vMotions and spikes in other metrics such as CPU/Memory contention.<br>The vMotion should not create any spikes, however, the VMs moved into the host might create spikes in memory usage, contention and CPU demand and contention.<br>Key: summary\|number_vmotion |
| Summary\|Total Number of Datastores | Total Number of Datastores.<br>Key: summary\|total_number_datastores |
| Summary\|Number of VCPUs on Powered On VMs | Total number of VCPUs of Virtual Machines that are powered on.<br>Key: summary\|number_running_vcpus |

| Metric Name | Description |
|---|---|
| Summary\|Total Number of VMs | Total number of virtual machines.<br>**Note:**<br>This is the total number of VMs excluding VM templates.<br>Key: summary\|total_number_vms |
| Summary\|Number of VM Templates | Number of VM Templates<br>Key: summary\|number_vm_templates |
| Summary\|Consider for Balance | Summary\|Consider for Balance = 1 when the host is Powered On, Connected, not in Maintenance Mode, and not a Failover Host, otherwise it = -1 |

### HBR Metrics for Host Systems

Host-based replication (HBR) metrics provide information about vSphere replication.

| Metric Name | Description |
|---|---|
| HBR\|Replication Data Received Rate | Replication Data Received Rate.<br>Key: hbr\|hbrNetRx_average |
| HBR\|Replication Data Transmitted Rate | Replication Data Transmitted Rate.<br>Key: hbr\|hbrNetTx_average |
| HBR\|Replicated VM Count | Number of replicated virtual machines.<br>Key: hbr\|hbrNumVms_average |

### Cost Metrics for Host Systems

Cost metrics provide information about the cost.

| Metric Name | Description |
|---|---|
| Monthly Maintenance Total Cost | Monthly total cost for maintenance.<br>Key: cost\|maintenanceTotalCost |
| Monthly Host OS License Total Cost | Monthly total cost for the host operating system license.<br>Key: cost\|hostOslTotalCost |
| Monthly Network Total Cost | Monthly total cost for network including cost of NIC cards associated with host.<br>Key: cost\|networkTotalCost |
| Monthly Server Hardware Total Cost | Monthly total cost for server hardware, based on amortized monthly value.<br>Key: cost\|hardwareTotalCost |
| Monthly Facilities Total Cost | Monthly total cost of facilities including real estate, power, and cooling.<br>Key: cost\|facilitiesTotalCost |
| Monthly Server Labor Total Cost | Monthly total cost for the server operating system labor.<br>Key: cost\|hostLaborTotalCost |
| Monthly Server Fully Loaded Cost | Monthly cost for a fully loaded server incorporating all cost driver values attributed to the server.<br>Key: cost\|totalLoadedCost |
| MTD Server Total Cost | Month to date cost for a fully loaded server incorporating all cost driver values attributed to the server.<br>Key: totalMTDCost |

| Metric Name | Description |
|---|---|
| Server Accumulated Depreciation | Month to date accumulated cost for a deprecated server.<br>Key: Cost\|Server Accumulated Depreciation |
| Aggregated Daily Total Cost | Daily aggregate daily total cost of the deleted VM present in the host system.<br>Key: Cost\|aggregatedDailyTotalCost |
| Aggregated Deleted VM Daily Total Cost | Daily aggregate cost of the deleted VM present in the host system.<br>Key: Cost\|aggregatedDeletedVmDailyTotalCost |

## Disabled Instanced Metrics

The instance metrics created for the following metrics are disabled in this version of VCF Operations. This means that these metrics collect data by default but all the instanced metrics created for these metrics, do not collect data by default.

| Metric Name |
|---|
| Datastore\|Outstanding IO requests (OIOs) |
| Datastore\|Read IOPS |
| Datastore\|Read Latency (ms) |
| Datastore\|Read Throughput (KBps) |
| Datastore\|Total Latency (ms) |
| Datastore\|Total Throughput (KBps) |
| Datastore\|unmapIOs_summation |
| Datastore\|unmapsize_summation |
| Datastore\|Write IOPS |
| Datastore\|Write Latency (ms) |
| Datastore\|Write Throughput (KBps) |
| Disk\|Physical Device Latency (ms) |
| Disk\|Queue Latency (ms) |
| Disk\|Read IOPS |
| Disk\|Read Latency (ms) |
| Disk\|Read Throughput (KBps) |
| Disk\|Write IOPS |
| Disk\|Write Latency (ms) |
| Disk\|Write Throughput (KBps) |
| Net\|Data Receive Rate (KBps) |
| Net\|Data Transmit Rate (KBps) |
| Net\|Error Packets Transmitted |
| Net\|Packets Dropped (%) |
| Net\|Packets Transmitted per second |
| Net\|Received Packets Dropped |
| Net\|Transmitted Packets Dropped |
| Net\|Usage Rate (%) |
| Storage Adapter\|Read IOPS |
| Storage Adapter\|Read Latency (ms) |

| Metric Name |
|---|
| Storage Adapter\|Read Throughput (KBps) |
| Storage Adapter\|Write IOPS |
| Storage Adapter\|Write Latency (ms) |
| Storage Adapter\|Write Throughput (KBps) |

## Disabled Metrics

The following metrics are disabled in this version of VCF Operations. This means that they do not collect data by default.

You can enable these metrics in the Policy workspace. For more information, in VMware Docs search for Collect Metrics and Properties Details.

| Metric Name | Key |
|---|---|
| CPU\|Idle (msec) | cpu\|idle_summation |
| CPU\|Used (msec) | cpu\|used_summation |
| Datastore I/O\|Average Observed Virtual Machine Disk I/O Workload | datastore\|vmPopulationAvgWorkload |
| Datastore I/O\|Max Observed Number of Outstanding IO Operations | datastore\|maxObserved_OIO |
| Datastore I/O\|Max Observed Read Rate (kbps) | datastore\|maxObserved_Read |
| Datastore I/O\|Max Observed Reads per second | datastore\|maxObserved_NumberRead |
| Datastore I/O\|Max Observed Write Rate (kbps) | datastore\|maxObserved_Write |
| Datastore I/O\|Max Observed Writes per second | datastore\|maxObserved_NumberWrite |
| Datastore I/O\|Maximum Observed VM Disk I/O Workload | datastore\|vmPopulationMaxWorkload |
| Network I/O\|bytesRx (kbps) | net\|bytesRx_average |
| Network I/O\|bytesTx (kbps) | net\|bytesTx_average |
| Network I/O\|Demand (%) | net\|demand |
| Network I/O\|Error Packets Received | net\|errorsRx_summation |
| Network I/O\|Max Observed Received Throughput (kbps) | net\|maxObserved_Rx_KBps |
| Network I/O\|Max Observed Throughput (kbps) | net\|maxObserved_KBps |
| Network I/O\|Max Observed Transmitted Throughput (kbps) | net\|maxObserved_Tx_KBps |
| Network I/O\|Packets Received per second | net\|packetsRxPerSec |
| Network I/O\|Packets Dropped | net\|dropped |
| Summary\|Workload Indicator | summary\|workload_indicator |
| vFlash Module\|Latest Number of Active Vm Disks | vflashModule\|numActiveVMDKs_latest |
| Net\|Received Packets Dropped | Number of received packets dropped in the performance interval.<br>Key: net\|droppedRx_summation |
| Net\|Transmitted Packets Dropped | Number of transmitted packets dropped in the performance interval.<br>Key: net\|droppedTx_summation |
| Net\|Packets Dropped (%) | This metric shows the percentage of received and transmitted packets dropped during the collection interval.<br>This metric is used to monitor reliability and performance of the ESX network. When a high value is displayed, the network is not reliable and performance suffers.<br>Key: net\|droppedPct |

| Metric Name | Key |
|---|---|
| Diskspace\|Not Shared (GB) | Unshared disk space in gigabytes.<br>Key: diskspace\|notshared |

## Cluster Compute Resource Metrics

VCF Operations collects configuration, disk space, CPU use, disk, memory, network, power, and summary metrics for cluster compute resources.

Cluster Compute Resource metrics include capacity and badge metrics. See definitions in:

- Capacity Analytics Generated Metrics
- Badge Metrics

### GPU Metrics

GPU metrics provide information about the GPU usage and performance.

| Metric Name | Description |
|---|---|
| GPU\|Compute Utilization (%) | The compute utilization percentage of a GPU. |
| GPU\|Memory Usage (%) | Memory currently in use as a percentage of total available memory. |
| GPU\|Memory Used (KB) | The amount of GPU memory used in kilobytes. |
| GPU\|Number of GPUs | Number of GPUs. |
| GPU\|Total Memory (KB) | Total memory in kilobytes. |
| GPU\|Number of vGPU Configured VM-s | Sum of GPU configured VMs. |

### Cluster Metrics for ROI Dashboard

Cluster metrics provide information about the metrics in ROI dashboard.

| Metric Name | Description |
|---|---|
| Total Number Of Reclaimable Hosts | This metric displays the total number of reclaimable hosts across all vCenters.<br>Key: metric=cost\|reclaimableHostCost |
| Total Reclaimable Host Cost | This metric displays the reclaimable host cost based on the recommended size.<br>Key: cost\|reclaimableHostCost |

### Configuration Metrics for Cluster Compute Resources

Configuration metrics provide information about configuration settings.

| Metric Name | Description |
|---|---|
| Configuration\|DAS Configuration\|Admission Control Enabled | DAS configuration admission control enabled.<br>Key: configuration\|dasconfig\|AdministrationControlEnabled |
| Configuration\|DAS Configuration\|Active Admission Control Policy | DAS configuration active admission control policy.<br>Key: configuration\|dasconfig\|activeAdministrationControlPolicy |
| Configuration\|DRS Configuration\|Affinity Rules | Affinity rules for DRS configuration.<br>Key: configuration\|DRSconfiguration\|affinity rules |

| Metric Name | Description |
|---|---|
| Configuration\|DRS Configuration\|Tolerance Imbalance Threshold | Displays the tolerance imbalance threshold for DRS configuration.<br>Key: configuration\|DRSconfiguration\|ToleranceimbalanceThreshold |
| Configuration\|DRS Configuration\|Default DRS behavior | Displays the default DRS configuration behavior.<br>Key: configuration\|DRSconfiguration\|DefaultDRSbehaviour |
| Configuration\|DRS Configuration\|Idle Consumed Memory | Displays the idle memory consumed by DRS configuration.<br>Key: configuration\|DRSconfiguration\|IdleConsumedMemory |
| Configuration\|DRS Configuration\| DRS vMotion Rate | Displays the vMotion rate for the DRS configuration.<br>Key: configuration\|DRSconfiguration\|DRSvMotion Rate |
| Configuration\|DPM Configuration\|Default DPM behavior | Displays the default behavior for the DPM configuration.<br>Key: configuration\|DPMconfiguration\|DefaultDPMbehaviour |
| Configuration\|DPM Configuration\|DPM Enabled | Displays whehter the DPM Configuration is enabled or not.<br>Key: configuration\|DPMConfiguration\|DPMEnabled |
| Configuration\|Failover Level | DAS configuration failover level.<br>Key: configuration\|dasconfig\|failoverLevel |
| Configuration\|Active Admission Control Policy | DAS configuration active admission control policy.<br>Key: configuration\|dasconfig\|activeAdministrationControlPolicy |
| Configuration\|CPU Failover Resources Percent | Percent CPU failover resources for DAS configuration admission control policy.<br>Key: configuration\|dasconfig\|adminissionControlPolicy\|cpuFailoverResourcesPercent |
| Configuration\|Memory Failover Resources Percent | Percent memory failover resources for DAS configuration admission control policy.<br>Key: configuration\|dasconfig\|adminissionControlPolicy\|memoryFailoverResourcesPercent |

## Disk Space Metrics for Cluster Compute Resources

Disk space metrics provide information about disk space use.

| Metric Name | Description |
|---|---|
| DiskSpace\|Snapshot Space | Displays the disk space used by the snapshot.<br>Key: DiskSpace\|snapshot space |
| Diskspace\|Virtual machine used (GB) | Space used by virtual machine files in gigabytes.<br>Key: diskspace\|used |
| Diskspace\|Total disk space used | Total disk space used on all datastores visible to this object.<br>Key: diskspace\|total_usage |
| Diskspace\|Total disk space | Total disk space on all datastores visible to this object.<br>Key: diskspace\|total_capacity |
| Diskspace\|Total provisioned disk space | Total provisioned disk space on all datastores visible to this object.<br>Key: diskspace\|total_provisioned |
| Diskspace\|Virtual Disk Used (GB) | Space used by virtual disks in gigabytes.<br>Key: diskspace\|diskused |
| Diskspace\|Snapshot Space (GB) | Space used by snapshots in gigabytes.<br>Key: diskspace\|snapshot |
| Diskspace\|Shared Used (GB) | Shared used space in gigabytes.<br>Key: diskspace\|shared |
| Diskspace\|Utilization (GB) | Storage space used on the connected vSphere Datastores.<br>Key: diskspace\|total_usage |

| Metric Name | Description |
|---|---|
| Diskspace\|Total Capacity (GB) | Total storage space available on the connected vSphere datastores.<br>Key: diskspace\|total_capacity |

### CPU Usage Metrics for Cluster Compute Resources

CPU usage metrics provide information about CPU use.

| Metric Name | Description |
|---|---|
| CPU\|Allocation\|Usable Capacity after HA and Buffer (vCPUs) | This metric shows the total capacity taking into consideration the over-commit ratio and after subtracting the CPU resources needed for HA and reserved buffer.<br>Key: cpu\|alloc\|usableCapacity |
| CPU\|Capacity Usage | This metric shows the percentage of the capacity used.<br>Key: cpu\|capacity_usagepct_average |
| CPU\|CPU Contention (%) | This metric is an indicator of the overall contention for CPU resources that occurs across the workloads in the cluster. When contention occurs, it means that some of the virtual machines are not immediately getting the CPU resources they are requesting.<br>Use this metric to identify when a lack of CPU resources might be causing performance issues in the cluster.<br>This metric is the sum of the CPU contention across all hosts in the cluster averaged over two times the number of physical CPUs in the cluster to account for hyper-threading. CPU contention takes into account:<br>• CPU Ready<br>• CPU Co-stop<br>• Power management<br>• Hyper threading<br>This metric is more accurate than CPU Ready since it takes into account CPU Co-stop and Hyper threading.<br>When using this metric, the number should be lower than the performance you expect. If you expect performance at 10%, then the number should be lower than 10%.<br>Since this value is averaged across all hosts in the cluster, you might find that some hosts have a higher CPU contention while others are lower. To ensure that vSphere spreads out the running workloads across hosts, consider enabling a fully automated DRS in the cluster.<br>Key: cpu\|capacity_contentionPct |
| CPU\|Demand\|Usable Capacity after HA and Buffer (MHz) | This metric shows the total capacity after subtracting the CPU resources needed for HA and reserved buffer.<br>Key: cpu\|demand\|usableCapacity |
| CPU\|Demand (%) | This metric is an indicator of the overall demand for CPU resources by the workloads in the cluster.<br>It shows the percentage of CPU resources that all the virtual machines might use if there were no CPU contention or CPU limits set. It represents the average active CPU load in the past five minutes.<br>Key: cpu\|demandPct |
| CPU\|Demand (MHz) | Sum of CPU utilization of all virtual machines on this cluster, including limits and VM overhead.<br>Key: cpu\|demandmhz |
| CPU\|Number of CPU Sockets | Number of CPU sockets.<br>Key: cpu\|numpackages |

| Metric Name | Description |
|---|---|
| CPU\|Overall CPU Contention | Overall CPU contention in milliseconds.<br>Key: cpu\|capacity_contention |
| CPU\|Host Provisioned Capacity | Provisioned CPU capacity in megahertz.<br>Key: cpu\|capacity_provisioned |
| CPU\|Provisioned CPUs | Number of Physical CPUs (Cores).<br>Key: cpu\|corecount_provisioned |
| CPU\|Usage (MHz) | Average CPU use in megahertz.<br>Key: cpu\|usagemhz_average |
| CPU\| VM CPU Usage (Mhz) | Sum of CPU usages of all VMs in the cluster. The CPU\|Usage (Mhz) metric value of each VM is taken for summation.<br>Key: cpu\|vm_usagemhz_average |
| CPU\|Demand | CPU Demand.<br>Key: cpu\|demand_average |
| CPU\|Overhead | Amount of CPU overhead.<br>Key: cpu\|overhead_average |
| CPU\|Demand without overhead | Value of demand excluding any overhead.<br>Key: cpu\|demand_without_overhead |
| CPU\|Provisioned Capacity | Provisioned Capacity (MHz).<br>Key: cpu\|vm_capacity_provisioned |
| CPU\|Number of hosts stressed | Number of hosts stressed.<br>Key: cpu\|num_hosts_stressed |
| CPU\|Stress Balance Factor | Stress Balance Factor.<br>Key: cpu\|stress_balance_factor |
| CPU\|Lowest Provider Capacity Remaining | Lowest Provider Capacity Remaining.<br>Key: cpu\|min_host_capacity_remaining |
| CPU\|Workload Balance Factor | Workload Balance Factor.<br>Key: cpu\|workload_balance_factor |
| CPU\|Highest Provider Workload | Highest Provider Workload.<br>Key: cpu\|max_host_workload |
| CPU\|Host workload Max-Min Disparity | Difference of Max and Min host workload in the container.<br>Key: cpu\|host_workload_disparity |
| CPU\|Host stress Max-Min Disparity | Difference of Max and Min host stress in the container.<br>Key: cpu\|host_stress_disparity |
| CPU\|Total Capacity (MHz) | Total CPU resources configured on the descendant ESX hosts.<br>Key: cpu\|capacity_provisioned |
| CPU\|Usable Capacity (MHz) | The usable CPU resources that are available for the virtual machines after considering reservations for vSphere High Availability (HA) and other vSphere services.<br>Key: cpu\|haTotalCapacity_average |

## Disk Metrics for Cluster Compute Resources

Disk metrics provide information about disk use.

| Metric Name | Description |
|---|---|
| Disk\|Total IOPS | Average number of commands issued per second during the collection interval.<br>Key: disk\|commandsAveraged_average |
| Disk\|Total Latency (ms) | Average amount of time taken for a command from the perspective of the guest operating system. This metric is the sum of the Kernel Command Latency and Physical Device Command Latency metrics.<br>Key: disk\|totalLatency_average |
| Disk\|Read Latency (ms) | Average amount of time for a read operation from the virtual disk. The total latency is the sum of Kernel latency and device latency.<br>Key: disk\|totalReadLatency_average |
| Disk\|Write Latency (ms) | The average amount of time taken for a read from the perspective of a Guest OS. This is the sum of Kernel Read Latency and Physical Device Read Latency.<br>Key: disk\|totalWriteLatency_averag |
| Disk\|Read IOPS | Average number of read commands issued per second during the collection interval.<br>Key: disk\|numberReadAveraged_averag |
| Disk\|Total Throughput (KBps) | Average of the sum of the data read and written for all the disk instances of the host or virtual machine.<br>Key: disk\|usage_average |
| Disk\|Write IOPS | Average number of write commands issued per second during the collection interval.<br>Key: disk\|numberWriteAveraged_average |
| Disk\|Read Requests | Amount of data read from the disk during the collection interval.<br>Key: disk\|read_average |
| Disk\|Write Requests | Amount of data written to the disk during the collection interval.<br>Key: disk\|write_average |
| Disk\|Total Queued Outstanding operations | Sum of queued operation and outstanding operations.<br>Key: disk\|sum_queued_oio |
| Disk\|Max Observed OIO | Max observed outstanding IO for a disk.<br>Key: disk\|max_observed |

## Memory Metrics for Cluster Compute Resources

Memory metrics provide information about memory use and allocation.

| Metric Name | Description |
|---|---|
| Mem\|Active Write (KB) | Active writes in kilobytes.<br>Key: mem\|activewrite_average |
| Mem\|Compressed (KB) | Average compression in kilobytes.<br>Key: mem\|compressed_average |
| Mem\|Compression Rate (KBps) | Average compression rate in kilobytes.<br>Key: mem\|compressionRate_average |
| Mem\|Consumed (KB) | Amount of host memory consumed by the virtual machine for guest memory.<br>Key: mem\|consumed_average |

| Metric Name | Description |
|---|---|
| Mem\|Contention (%) | This metric is an indicator of the overall contention for memory resources that occurs across the workloads in the cluster. When contention occurs, it means that some of the VMs are not immediately getting the memory resources that they are requesting.<br>Use this metric to identify when lack of memory resources might be causing performance issues in the cluster.<br>Key: mem\|host_contentionPct |
| Mem\|Contention (KB) | Contention in kilobytes.<br>Key: mem\|host_contention |
| Mem\|Decompression Rate (KBps | Decompression rate in kilobytes.<br>Key: mem\|decompressionRate_average |
| Mem\|Granted (KB) | Amount of memory available for use.<br>Key: mem\|granted_average |
| Mem\|Guest Active (KB) | Amount of memory that is actively used.<br>Key: mem\|active_average |
| Mem\|Heap (KB) | Amount of memory allocated for heap.<br>Key: mem\|heap_average |
| Mem\|Heap Free (KB) | Free space in the heap.<br>Key: mem\|heapfree_average |
| Mem\|Balloon | This metric shows the amount of memory currently used by the virtual machine memory control. It is only defined at the VM level.<br>Key: mem\|vmmemctl_average |
| Mem\|VM Overhead (KB) | Memory overhead reported by host.<br>Key: mem\|overhead_average |
| Mem\|Provisioned Memory (KB) | Provisioned memory in kilobytes.<br>Key: mem\|host_provisioned |
| Mem\|Reserved Capacity (KB) | Sum of memory reservations by consumers such as Resource Pools (RP) and powered on VMs that are not in Resource Pools, aggregated and published on cluster level. RP reservations at the vCenter level are distributed to hosts depending on the number of powered on VMs and their entitlement within the Resource Pool. For more information on resource allocation reservation, share, and limit, see Configuring Resource Allocation Settings.<br>Key: mem\|reservedCapacity_average |
| Mem\|Shared (KB) | Amount of shared memory.<br>Key: mem\|shared_average |
| Mem\|Shared Common (KB) | Amount of shared common memory.<br>Key: mem\|sharedcommon_average |
| Mem\|Swap In (KB) | Amount of memory that is swapped in for the service console.<br>Key: mem\|swapin_average |
| Mem\|Swap In Rate (KBps) | Rate at which memory is swapped from disk into active memory during the interval.<br>Key: mem\|swapinRate_average |
| Mem\|Swap Out (KB) | Amount of memory that is swapped out for the service console.<br>Key: mem\|swapout_average |
| Mem\|Swap Out Rate (KBps) | Rate at which memory is being swapped from active memory into disk during the current interval.<br>Key: mem\|swapoutRate_average |

| Metric Name | Description |
|---|---|
| Mem\|Swap Used (KB) | Amount of memory used for swap space.<br>Key: mem\|swapused_average |
| Mem\|Total Capacity (KB) | Total capacity in kilobytes.<br>Key: mem\|totalCapacity_average |
| Mem\|Unreserved (KB) | Memory available for reservation by VMs and Resource Pools (RP), aggregated and published on cluster level. The reservations are set for VMs and RPs and not for ESX hosts or clusters. For more information on resource allocation reservation, share, and limit, see Configuring Resource Allocation Settings.<br>Key: mem\|unreserved_average |
| Mem\|Usable Memory (KB) | Usable memory in kilobytes.<br>Key: mem\|host_usable |
| Mem\|Usage/Usable | Percent memory used.<br>Key: mem\|host_usagePct |
| Mem\|Host Usage (KB) | Memory use in kilobytes.<br>Key: mem\|host_usage |
| Mem\|Machine Demand | Memory Machine Demand in KB.<br>Key: mem\|host_demand |
| Mem\|ESX System Usage | Memory usage by the VMkernel and ESX user-level services.<br>Key: mem\|host_systemUsage |
| Mem\|Usage (%) | This metric shows the portion of the total memory in all hosts in the cluster that is being used.<br>This metric is the sum of memory consumed across all hosts in the cluster divided by the sum of physical memory across all hosts in the cluster. |
| Memory \|Workload (%) | Demand over usable capacity. Wherever applicable, demand includes limit and contention. |
| Mem\|Usage (KB) | Memory currently in use as a percentage of total available memory.<br>Key: mem\|usage_average |
| Mem\|VM kernel Usage (KB) | Amount of memory that the VM kernel uses.<br>Key: mem\|sysUsage_average |
| Mem\|Zero (KB) | Amount of memory that is all 0.<br>Key: mem\|zero_average |
| Mem\|Number of Hosts Stressed | Number of hosts stressed.<br>Key: mem\|num_hosts_stressed |
| Mem\|Stress Balance Factor | Stress balance factor.<br>Key: mem\|stress_balance_factor |
| Mem\|Lowest Provider Capacity Remaining | Lowest provider capacity remaining.<br>Key: mem\|min_host_capacity_remaining |
| Mem\|Workload Balance Factor | Workload balance factor.<br>Key: mem\|workload_balance_factor |
| Mem\|Highest Provider Workload | Highest provider workload.<br>Key: mem\|max_host_workload |
| Mem\|Host workload Max-Min Disparity | Difference of Max and Min host workload in the container.<br>Key: mem\|host_workload_disparity |
| Mem\|Host stress Max-Min Disparity | Difference of Max and Min host stress in the container.<br>Key: mem\|host_stress_disparity |

| Metric Name | Description |
|---|---|
| Mem\|Utilization (KB) | Memory utilization level based on the descendant virtual machines utilization. Includes reservations, limits, and overhead to run the Virtual Machines.<br>Key: mem\|total_need |
| Mem\|Total Capacity (KB) | Sum of the amount of physical memory configured on ESX hosts of the cluster.<br>Key: mem\|host_provisioned |
| Mem\|Usable Capacity (KB) | Memory resources available after subtracting reservations for vSphere HA (failover host/s) from the cluster's total memory capacity.<br>Hosts that are in the maintenance mode are not included in the calculation.<br>Key: mem\|haTotalCapacity_average |

### Network Metrics for Cluster Compute Resources

Network metrics provide information about network performance.

| Metric Name | Description |
|---|---|
| Net\|Data Receive Rate (KBps) | Average amount of data received per second.<br>Key: net\|received_average |
| Net\|Data Transmit Rate (KBps) | Average amount of data transmitted per second.<br>Key: net\|transmitted_average |
| Net\|Packets Dropped | Number of packets dropped in the performance interval.<br>Key: net\|dropped |
| Net\|Packets Dropped (%) | Percentage of packets dropped.<br>Key: net\|droppedPct |
| Net\|Packets Received | Number of packets received in the performance interval.<br>Key: net\|packetsRx_summation |
| Net\|Packets Transmitted | Number of packets transmitted in the performance interval.<br>Key: net\|packetsTx_summation |
| Net\|Received Packets Dropped | Number of received packets dropped in the performance interval.<br>Key: net\|droppedRx_summation |
| Net\|Transmitted Packets Dropped | Number of transmitted packets dropped in the performance interval.<br>Key: net\|droppedTx_summation |
| Net\|Total Throughput (KBps) | The sum of the data transmitted and received for all the NIC instances of the host or virtual machine.<br>Key: net\|usage_average |
| Network\|Error Packets | Displays the total number of error packets (transmitted and received) from all the ESX in the cluster within a time interval of 20 seconds.<br>Key: Network\|Error Packets |

### Datastore Metrics for Cluster Compute Resources

Datastore metrics provide information about Datastore use.

| Metric Name | Description |
|---|---|
| Datastore\|TotalThroughput | Displays the total throughput for the datastore.<br>Key: datastore\|thorughput |
| Datastore\|Outstanding IO requests | OIO for datastore.<br>Key: datastore\|demand_oio |

| Metric Name | Description |
|---|---|
| Datastore\|Read IOPS | Average number of read commands issued per second during the collection interval.<br>Key: datastore\|numberReadAveraged_average |
| Datastore\|Write IOPS | Average number of write commands issued per second during the collection interval.<br>Key: datastore\|numberWriteAveraged_average |
| Datastore\|Read Throughput (KBps) | Amount of data read in the performance interval.<br>Key: datastore\|read_average |
| Datastore\|Write Throughput (KBps) | Amount of data written to disk in the performance interval.<br>Key: datastore\|write_average |
| Datastore\|Read Latency | Average amount of time taken for a read operation from the datastore.<br>Key: datastore\|ReadLatency |
| Datastore\|Write Latency | Average amount of time taken for a write operation from the datastore.<br>Key: datastore\|WriteLatency |
| Datastore\|Max VM Disk Latency | Maximum amount of time taken to read or write data from a virtual machine.<br>Key: datastore\|MaxVMDiskLatency |
| Datastore\|Outstanding IO Requests (OIOs) | This metric displays the outstanding datastore IO requests.<br>Key: datastore\|OutstandingIORequests |
| Datastore\|Host SCSI Disk Partition | This metric displays the datastore host scsi partition.<br>Key: datastore\|HostSCSIDiskPartition |
| Devices\|Command Aborted | The metric lists the stopped device commands.<br>Key: devices\|CommandAborted |

### Cluster Services Metrics for Cluster Compute Resources

Cluster Services metrics provide information about cluster services.

| Metric Name | Description |
|---|---|
| Cluster Services\|Total Imbalance | Total imbalance in cluster services<br>Key: clusterServices\|total_imbalance |
| ClusterServices\|Effective CPU Resources (MHz) | VMware DRS effective CPU resources available.<br>Key: clusterServices\|effectivecpu_average |
| ClusterServices\|Effective Memory Resources (KB) | VMware DRS effective memory resources available.<br>Key: clusterServices\|effectivemem_average |
| Cluster Services\|DRS Initiated vMotion Count | clusterServices\|number_drs_vmotion |

## Power Metrics for Cluster Compute Resources

Power metrics provide information about power use.

| Metric Name | Description |
|---|---|
| Power\|Total Energy Consumed in the collection period (Wh) | Displays the total electricity consumed based on the time interval selected. The default collection cycle is set to 5 mins. You can continue using the default setting or edit it for each adapter instance. For example, if the time interval is set to its default value, the value represents the energy consumed per 5 mins. |
| Power\|Current Power Consumption Rate (Watt) | The power consumption rate per second, averaged over the reporting period.<br>Key: power\|power_average |
| Power\|Power Cap (Watt) | Average power capacity in watts.<br>Key: power\|powerCap_average |
| Power\|(DEP) Energy (Joule) | Total energy consumed in joules<br>Key: power\|energy_summation |

## Summary Metrics for Cluster Compute Resources

Summary metrics provide information about overall performance.

| Metric Name | Description |
|---|---|
| Summary\|Number of Running Hosts | Number of running hosts.<br>Key: summary\|number_running_hosts |
| Summary\|Number of Running VMs | This metric shows the total number of VMs running on all hosts in the cluster.<br>Key: summary\|number_running_vms |
| Summary\|Number of vMotions | This metric shows the number of vMotions that occurred during the last collection cycle.<br>When using this metric, look for a low number which indicates that the cluster might serve its VMs. A vMotion can impact VM performance during the stun time.<br>Key: summary\|number_vmotion |
| Summary\|Number of Hosts | Total number of hosts.<br>Key: summary\|total_number_hosts |
| Summary\|Total Number of VMs | Total number of virtual machines.<br>**Note:**<br>This shows the total number of VMs excluding VM templates under the datastore.<br>Key: summary\|total_number_vms |
| Summary\|Total Number of Datastores | Total number of datastores.<br>Key: summary\|total_number_datastores |
| Summary\|Number of VCPUs on Powered On VMs | Number of virtual CPUs on powered-on virtual machines.<br>Key: summary\|number_running_vcpus |
| Summary\|Average Running VM Count per Running Host | Average number of running virtual machines per running host.<br>Key: summary\|avg_vm_density |
| Summary\|Cluster Availability (%) | Percentage of hosts powered-on in the cluster.<br>Key: summary\|cluster_availability |
| Summary\|Datastore | Displays the status of the datastore.<br>Key: summary\|datastore |

| Metric Name | Description |
|---|---|
| Summary\|Type | Displays the datastore type.<br>Key: summary\|type |
| Summary\|Is Local | Displays whether the datastore is local or not.<br>Key: summary\|Islocal |
| Summary\|Number of VM Templates | Number of VM templates.<br>Key: summary\|number_vm_templates |
| Summary\|Number of Pods | Number of pods.<br>**Note:**<br>This is published if the cluster is Workload Management enabled or there are pods under the cluster.<br>Key: summary\|total_number_pods |
| Summary\|Number of Namespaces | Number of namespaces.<br>**Note:**<br>This is published if the cluster is Workload Management enabled or there are namespaces under the cluster.<br>Key: summary\|numberNamespaces |
| Summary\|Number Kubernetes Clusters | Number of Kuberntes clusters.<br>**Note:**<br>This is published if the cluster is Workload Management enabled or there are Kuberntes clusters under the cluster.<br>Key: summary\|numberKubernetesClusters |
| Summary\|Number of Developer Managed VMs | Number of developer managed VMs.<br>**Note:**<br>This is published if the cluster is Workload Management enabled or there are developer managed VMs under the cluster.<br>Key: summary\|numberDeveloperManagedVMs |
| Namespaces\|Config Status | Workload Management configuration status.<br>**Note:**<br>This is published if the cluster is Workload Management enabled.<br>Key: namespaces\|configStatus |
| Namespaces\|Kubernetes Status | Kubernetes status.<br>**Note:**<br>This is published if the cluster is Workload Management enabled.<br>Key: namespaces\|kuberntesStatus |

## Reclaimable Metrics for Cluster Compute Resources

Reclaimable metrics provide information about reclaimable resources.

| Metric Name | Description |
|---|---|
| Idle VMs\|CPU (vCPUs) | Number of reclaimable vCPUs of Idle VMs within the cluster.<br>Key: reclaimable\|idle_vms\|cpu |
| Idle VMs\|Disk Space (GB) | Reclaimable disk space of Idle VMs within the cluster.<br>Key: reclaimable\|idle_vms\|disksapce |
| Idle VMs\|Memory (KB) | Reclaimable memory of Idle VMs within the cluster.<br>Key: reclaimable\|idle_vms\|mem |

| Metric Name | Description |
|---|---|
| Idle VMs\|Potential Savings | Potential saving after reclamation of resources of Idle VMs within the cluster.<br>Key: reclaimable\|idle_vms\|cost |
| Powered Off VMs\|Disk Space (GB) | Reclaimable disk space of Powered Off VMs within the cluster.<br>Key: reclaimable\|poweredOff_vms\| diskspace |
| Powered Off VMs\|Potential Savings | Potential saving after reclamation of resources of Powered Off VMs within the cluster.<br>Key: reclaimable\|poweredOff_vms\|cost |
| VM Snapshots\|Disk Space (GB) | Reclaimable disk space of VM Snapshots within the cluster.<br>Key: reclaimable\| vm_snapshots \| diskspace |
| VM Snapshots \|Potential Savings | Potential saving after reclamation of VM Snapshots within the cluster.<br>Key: reclaimable\| vm_snapshots \|cost |

## Cost Metrics for Cluster Compute Resources

Cost metrics provide information about the cost.

| Metric Name | Description |
|---|---|
| Cluster CPU Base Rate | Base rate for Cluster CPU calculated by dividing the monthly total cluster CPU cost by cluster CPU utilization % and CPU cluster capacity (gHZ).<br>Key:cost\|cpuBaseRate |
| Cluster CPU Utilization (%) | Expected CPU utilization that is set by the user in cluster cost page.<br>Key:cost\|cpuExpectedUtilizationPct |
| Cluster Memory Base Rate | Cluster memory base rate calculated by dividing the monthly total cluster memory cost by cluster memory utilization % and memory cluster capacity (GB).<br>Key: cost\|memoryBaseRate |
| Cluster Memory Utilization (%) | Expected memory utilization that is set by the user in cluster cost page.<br>Key: cost\|memoryExpectedUtilizationPct |
| Monthly Cluster Allocated Cost | Monthly cluster allocated cost calculated by subtracting the monthly cluster unallocated cost from the monthly cluster total cost.<br>Key: cost\|allocatedCost |
| Monthly Cluster Total Cost | Fully loaded compute cost of all hosts underneath the cluster.<br>Key: cost\|totalCost |
| Monthly Cluster Unallocated Cost | Monthly cluster unallocated cost calculated by subtracting the monthly cluster allocated cost from the monthly cluster total cost.<br>Key: cost\|unAllocatedCost |
| Monthly Total Cluster CPU Cost | Cost attributed to the cluster CPU from monthly cluster total cost.<br>Key: cost\|totalCpuCost |
| Monthly Total Cluster Memory Cost | Cost attributed to the cluster memory from monthly cluster total cost.<br>Key: cost\|totalMemoryCost |
| MTD Cluster CPU Utilization (GHz) | Month to date CPU utilization of the cluster.<br>Key: cost\|cpuActualUtilizationGHz |

| Metric Name | Description |
|---|---|
| MTD Cluster Memory Utilization (GB) | Month to date memory utilization of the cluster.<br>Key: cost\|memoryActualUtilizationGB |
| Monthly Cluster Allocated Cost (Currency) | The monthly allocated cost of all VMs in a cluster.<br>cost\|clusterAllocatedCost |
| Cost\|Allocation\|Monthly Cluster Unallocated Cost (Currency) | The monthly unallocated is calculated by subtracting the monthly allocated cost from the cluster's cost.<br>cost\|clusterUnAllocatedCost |
| Aggregated Daily Total Cost | Daily aggregate daily total cost of the deleted VM present in the host system.<br>Key: Cost\|aggregatedDailyTotalCost |
| Aggregated Deleted VM Daily Total Cost | Daily aggregate cost of the deleted VM present in the host system.<br>Key: Cost\|aggregatedDeletedVmDailyTotalCost |

### Profiles Metrics for Cluster Compute Resources

Profiles metrics provide information about the profile specific capacity.

| Metric Name | Description |
|---|---|
| Profiles\|Capacity Remaining Profile (Average) | The capacity remaining in terms of fitting the average consumer.<br>Key: Profiles\|capacityRemainingProfile_<profile uuid> |
| Profiles\|Capacity Remaining Profile (<custom profile name>) | Published for custom profiles enabled from policy on Cluster Compute Resource.<br>Key: Profiles\|capacityRemainingProfile_<profile uuid> |

### Capacity Allocation Metrics for Cluster Compute Resources

Capacity allocation metrics provide information about the allotment of capacity, see Capacity Analytics Generated Metrics.

### Virtual Machine Operations Metrics for Clusters

VM operations metrics provide information about the actions performed on VMs. The following are some important points you must know about VM operation metrics for clusters.

- VM operations metrics is not collected for custom data centers.
- If you edit a VM settings and do not perform any action, still it is considered as VM reconfigure operation.
- During Revert Snapshot, VMs are powered-off, but this operation is not counted under VM Power-off metric.
- Adding ESX with VMs is not counted under VM Create metric.
- Removing ESX with VMs is not coutned under VM Remove metric.
- VM hardstop operation is not counted under VM Power Off metric.

| Metric Name | Description |
|---|---|
| Inventory | |
| VM Clone | This metric displays the number of clone operations on the virtual machine.<br>Key: Inventory\|VM Clone |
| VM Create | This metric displays the number of create operations on the virtual machine.<br>Key: Inventory\|VM Create |
| VM Delete | This metric displays the number of delete operations on the virtual machine.<br>Key: Inventory\|VM Delete |

| Metric Name | Description |
|---|---|
| VM Reconfigure | This metric displays the number of reconfigure operations on the virtual machine.<br>Key: Inventory\|VM Reconfigure |
| VM Register | This metric displays the number of register operations on the virtual machine.<br>Key: Inventory\|VM Register |
| VM Template Deploy | This metric displays the number templates deployed on the virtual machine.<br>Key: Inventory\|VM Template Deploy |
| VM Unregister | This metric displays the number of unregister operations on the virtual machine.<br>Key: Inventory\|VM Unregister |
| Location | |
| Storage vMotion | This metric displays the number of migrations with vMotion (datastore change operations for Powered-on VMs).<br>Key: Location\|Storage vMotion |
| VM Datastore Change (powered-off VMs) | This metric displays the number of datastore change operations, for powered-off and suspended virtual machines.<br>Key: Location\|VM Datastore Change (powered-off VMs) |
| VM Host and Datastore Change (powered-off VMs) | This metric displays the number of host and datastore change operations, for powered-off and suspended virtual machines.<br>Key: Location\|VM Host and Datastore Change (powered-off VMs) |
| VM Host and Datastore Change (powered-on VMs) | This metric displays the number of host and datastore change operations, for powered-on and suspended virtual machines.<br>Key: Location\|VM Host and Datastore Change (powered-on VMs) |
| VM Host Change (powered-off VMs) | This metric displays the number of host change operations, for powered-off and suspended virtual machines.<br>Key: Location\|VM Host Change (powered-off VMs) |
| vMotion | This metric displays the number of migrations with vMotion (host change operations for powered-on VMs).<br>Key: Location\|vMotion |
| State | |
| VM Guest Reboot | This metric displays the number of reboot operations on the virtual machine guest.<br>Key: State\|VM Guest Reboot |
| VM Guest Shutdown | This metric displays the number of shutdown operations on the virtual machine guest.<br>Key: State\|VM Guest Shutdown |
| VM Power Off | This metric displays the number of power-off operations on the virtual machine.<br>Key: State\|VM Power Off |
| VM Power On | This metric displays the number of power-on operations on the virtual machine.<br>Key: State\|VM Power On |
| VM Reset | This metric displays the number of reset operations on the virtual machine guest.<br>Key: State\|VM Reset |

| Metric Name | Description |
|---|---|
| VM Standby Guest | This metric displays the number of standby operations on the virtual machine guest.<br>Key: State\|VM Standby Guest |
| VM Suspend | This metric displays the number of suspend operations on the virtual machine.<br>Key: State\|VM Suspend |

### Disabled Metrics

The following metrics are disabled in this version of VCF Operations. This means that they do not collect data by default.

You can enable these metrics in the Policy workspace. For more information, in VMware Docs search for Collect Metrics and Properties Details.

You can enable these metrics in the Policy workspace. For more information, see *Metrics and Properties Details*.

| Metric Name | Key |
|---|---|
| CPU\|Capacity Available to VMs (mhz) | cpu\|totalCapacity_average |
| CPU\|IO Wait (msec) | cpu\|iowait |
| CPU\|Reserved Capacity (mhz) | cpu\|reservedCapacity_average |
| CPU\|Total Wait (msec) | cpu\|wait |
| Datastore I/O\|Max Observed Number of Outstanding IO Operations | datastore\|maxObserved_OIO |
| Datastore I/O\|Max Observed Read Rate (kbps) | datastore\|maxObserved_Read |
| Datastore I/O\|Max Observed Reads per second | datastore\|maxObserved_NumberRead |
| Datastore I/O\|Max Observed Write Rate (kbps) | datastore\|maxObserved_Write |
| Datastore I/O\|Max Observed Writes per second | datastore\|maxObserved_NumberWrite |
| Storage\|Total Usage (kbps) | storage\|usage_average |
| Summary\|Average Provisioned Capacity per Running VM (mhz) | summary\|avg_vm_cpu |
| Summary\|Average Provisioned Memory per Running VM (kb) | summary\|avg_vm_mem |
| Summary\|Average Provisioned Memory per Running VM (kb) | summary\|avg_vm_mem |
| Summary\|Maximum Number of VMs | summary\|max_number_vms |
| Summary\|Workload Indicator | summary\|workload_indicator |
| Network I/O\|Max Observed Received Throughput (KBps) | net\|maxObserved_Rx_KBps |
| Network I/O\|Max Observed Throughput (KBps) | net\|maxObserved_KBps |
| Network I/O\|Max Observed Transmitted Throughput (KBps) | net\|maxObserved_Tx_KBps |
| Diskspace\|Not Shared (GB) | Space used by VMs that is not shared.<br>Key: diskspace\|notshared |

## Resource Pool Metrics

VCF Operations collects configuration, CPU usage, memory, and summary metrics for resource pool objects.

Resource Pool metrics include capacity and badge metrics. See definitions in:

- Capacity Analytics Generated Metrics
- Badge Metrics

### Configuration Metrics for Resource Pools

Configuration metrics provide information about memory and CPU allocation configuration.

| Metric Name | Description |
|---|---|
| Memory Allocation Reservation | Memory Allocation Reservation.<br>Key: config\|mem_alloc_reservation |

### CPU Usage Metrics for Resource Pools

CPU usage metrics provide information about CPU use.

| Metric Name | Description |
|---|---|
| Capacity Demand Entitlement (%) | CPU Capacity Demand Entitlement Percentage.<br>Key: cpu\|capacity_demandEntitlementPct |
| Capacity entitlement (MHz) | CPU Capacity Entitlement.<br>Key: cpu\|capacity_entitlement |
| CPU Contention (%) | CPU capacity contention.<br>Key: cpu\|capacity_contentionPct |
| Demand (MHz) | CPU demand in megahertz.<br>Key: cpu\|demandmhz |
| Overall CPU Contention | Overall CPU contention in milliseconds.<br>Key: cpu\|capacity_contention |
| Usage | Average CPU use in megahertz.<br>Key: cpu\|usagemhz_average |
| Effective limit | CPU effective limit.<br>Key: cpu\|effective_limit |
| Reservation Used | CPU reservation used.<br>Key: cpu\|reservation_used |
| Estimated entitlement | CPU estimated entitlement.<br>Key: cpu\|estimated_entitlement |
| Dynamic entitlement | CPU dynamic entitlement.<br>Key: cpu\|dynamic_entitlement |
| Demand without overhead | Value of demand excluding any overhead.<br>Key: cpu\|demand_without_overhead |

### Memory Metrics for Resource Pools

Memory metrics provide information about memory use and allocation.

| Metric Name | Description |
|---|---|
| Balloon | Amount of memory currently used by the virtual machine memory control.<br>Key: mem\|vmmemctl_average |
| Compression Rate | Compression rate in kilobytes per second.<br>Key: mem\|compressionRate_average |
| Consumed | Amount of host memory consumed by the virtual machine for guest memory.<br>Key: mem\|consumed_average |
| Contention | Machine contention.<br>Key: mem\|host_contentionPct |

| Metric Name | Description |
|---|---|
| Guest usage | Guest memory entitlement.<br>Key: mem\|guest_usage |
| Guest demand | Guest memory entitlement.<br>Key: mem\|guest_demand |
| Contention (KB) | Machine contention in kilobytes.<br>Key: mem\|host_contention |
| Decompression Rate | Decompression rate in kilobytes per second.<br>Key: mem\|decompressionRate_average |
| Granted | Average of memory available for use.<br>Key: mem\|granted_average |
| Guest Active | Amount of memory that is actively used.<br>Key: mem\|active_average |
| VM Overhead | Memory overhead reported by host.<br>Key: mem\|overhead_average |
| Shared | Amount of shared memory.<br>Key: mem\|shared_average |
| Reservation Used | Memory Reservation Used.<br>Key: mem\|reservation_used |
| Dynamic Entitlement | Memory Dynamic Entitlement.<br>Key: mem\|dynamic_entitlement |
| Effective Limit | Memory Effective Limit.<br>Key: mem\|effective_limit |
| Swap In Rate | Rate at which memory is swapped from disk into active memory during the interval.<br>Key: mem\|swapinRate_average |
| Swap Out Rate | Rate at which memory is being swapped from active memory to disk during the current interval.<br>Key: mem\|swapoutRate_average |
| Swapped | Amount of unreserved memory.<br>Key: mem\|swapped_average |
| Usage (%) | Memory currently in use as a percentage of total available memory.<br>Key: mem\|usage_average |
| Zero | Amount of memory that is all zero.<br>Key: mem\|zero_average |
| Zipped (KB) | Latest zipped memory in kilobytes.<br>Key: mem\|zipped_latest |
| Swap In (KB) | Amount of memory swapped in kilobytes.<br>Key: mem\|swapin_average |
| Swap Out (KB) | Amount of memory swapped out in kilobytes.<br>Key: mem\|swapout_average |
| Swap Used | Amount of memory used for swap space in kilobytes.<br>Key: mem\|swapused_average |
| Total Capacity | Total capacity.<br>Key: mem\|guest_provisioned |

### Summary Metrics for Resource Pools

Summary metrics provide information about overall performance.

| Metric Name | Description |
|---|---|
| Number of Running VMs | Number of running virtual machines.<br>Key: summary\|number_running_vms |
| Total Number of VMs | Total number of virtual machines.<br>**Note:**<br>This shows the total number of VMs excluding VM templates.<br>Key: summary\|total_number_vms |
| IO Wait (ms) | IO wait time in milliseconds.<br>Key: summary\|iowait |
| Number of VM Templates | Number of VM Templates.<br>Key: summary\|number_vm_templates |

## Data Center Metrics

VCF Operations collects CPU usage, disk, memory, network, storage, disk space, and summary metrics for data center objects.

Data center metrics include capacity and badge metrics. See definitions in:

- Capacity Analytics Generated Metrics
- Badge Metrics

### Data Center Metrics for ROI Dashboard

Data center metrics provide information about data center savings across vCenters.

| Metric Name | Description |
|---|---|
| Realized Cost Savings | |
| Realized Savings Idle Cost | This metric displays the total realize savings for VMs across all vCenters.<br>Key: cost\|realized_savings\|realizedIdleCost |
| Realized Savings Powered Off Cost (AOA) | This metric displays the total realized savings for powered off VMs across all vCenters.<br>Key: cost\|realized_savings\|realizedPoweredOffCost |
| Realized Savings Snapshot Space Cost (AOA) | This metric displays the snapshots space saved across all vCenters.<br>Key: cost\|realized_savings\|realizedSnapshotSpaceCost |
| Realized Savings Oversized Cost (AOA) | This metric displays the oversized savings across all vCenters.<br>Key: cost\|realized_savings\|realizedOversizedCost |
| Realized Savings Orphaned Disk Space Cost (AOA) | This metric displays the amount of disk space saved by orphaned disks across all vCenters.<br>Key: cost\|realized_savings\|realizedOrphanedDiskSpaceCost |
| Realized Savings Reclaimable Host Cost (AOA) | This metric displays the amount of reclaimable host savings across all vCenters.<br>Key: cost\|realized_savings\|realizedReclaimableHostCost |
| Realized vCPUs from Oversized VMs | This metric displays the number of vCPUs realized across all vCenters.<br>Key: realized\|realizedVCpus |

| Metric Name | Description |
|---|---|
| Compute Realized Memory from Oversized VMs | This metric displays the amount of memory realized from oversized VMs across all vCenters.<br>Key: compute_realized\|realizedOversizedMem |
| Realized Potential Memory Consumed from Oversized VMs | This metric displays the potential memory consumed from oversized VMs across all vCenters.<br>Key: realized\|realizedPotentialMemConsumed |
| Compute Realized vCPUs from Oversized VMs | This metric displays the realized vCPUs from oversized VMs across all vCenters.<br>Key: compute_realized\|realizedOversizedVCpus |
| Compute Realized vCPUs from Idle VMs | This metric displays the realized vCPUs from idle VMs across all vCenters.<br>Key: compute_realized\|realizedIdleVCpus |
| Compute Realized Memory from Idle VMs | This metric displays the amount of memory realized from idle VMs across all vCenters.<br>Key: compute_realized\|realizedIdleMem |
| Disk Space Realized Idle VMs | This metric displays the amount of disk space realized from idle VMs across all vCenters.<br>Key: storage_realized\|realizedIdleDiskSpace |
| Disk Space Realized PoweredOff VMs | This metric displays the amount of disk space realized from powered off VMs across all vCenters.<br>Key: storage_realized\|realizedPoweredOffDiskSpace |
| Disk Space Realized VM Snapshots | This metric displays the amount of disk space realized from VM snapshots across all vCenters.<br>Key: storage_realized\|realizedSnapshotSpace |
| Disk Space Realized Orphaned Disks | This metric displays the amount of disk space realized from orphaned disks across all vCenters.<br>Key: storage_realized\|realizedIdleDiskSpace |
| Realized Savings Total Realized Cost | This metric displays the total realized cost across all vCenters.<br>Key: cost\|realized_savings\|realizedTotalCost |

### CPU Usage Metrics for Data Centers

CPU usage metrics provide information about CPU use.

| Metric Name | Description |
|---|---|
| Capacity Usage (%) | Percent capacity used.<br>Key: cpu\|capacity_usagepct_average |
| CPU Contention (%) | CPU capacity contention.<br>Key: cpu\|capacity_contentionPct |
| Demand (%) | CPU demand percentage.<br>Key: cpu\|demandPct |
| Demand | Demand in megahertz.<br>Key: cpu\|demandmhz |
| Demand (MHz) | CPU utilization level based on descendant virtual machines utilization. This Includes reservations, limits, and overhead to run the virtual machines.<br>Key: cpu\|demandmhz |
| Overhead (KB) | Amount of CPU overhead.<br>Key: cpu\|overhead_average |

| Metric Name | Description |
|---|---|
| Demand without overhead | Value of demand excluding any overhead.<br>Key: cpu\|demand_without_overhead |
| Total Wait | CPU time spent on idle state.<br>Key: cpu\|wait |
| Number of CPU Sockets | Number of CPU sockets.<br>Key: cpu\|numpackages |
| Overall CPU Contention (ms) | Overall CPU contention in milliseconds.<br>Key: cpu\|capacity_contention |
| Host Provisioned Capacity (MHz) | Host provisioned capacity in megahertz.<br>Key: cpu\|capacity_provisioned |
| Provisioned vCPU(s) | Provisioned vCPU(s).<br>Key: cpu\|corecount_provisioned |
| Reserved Capacity (MHz) | The sum of the reservation properties of the (immediate) children of the host's root resource pool.<br>Key: cpu\|reservedCapacity_average |
| Usage | Average CPU usage in megahertz.<br>Key: cpu\|usagemhz_average |
| IO Wait | IO wait time in milliseconds.<br>Key: cpu\|iowait |
| Provisioned Capacity | Provisioned Capacity.<br>Key: cpu\|vm_capacity_provisioned |
| Stress Balance Factor | Stress Balance Factor.<br>Key: cpu\|stress_balance_factor |
| Lowest Provider Capacity Remaining | Lowest Provider Capacity Remaining.<br>Key: cpu\|min_host_capacity_remaining |
| Workload Balance Factor | Workload Balance Factor.<br>Key: cpu\|workload_balance_factor |
| Highest Provider Workload | Highest Provider Workload.<br>Key: cpu\|max_host_workload |
| Host workload Max-Min Disparity | Difference of Max and Min host workload in the container.<br>Key: cpu\|host_workload_disparity |
| Host stress Max-Min Disparity | Difference of Max and Min host stress in the container.<br>Key: cpu\|host_stress_disparity |
| Total Capacity (MHz) | Total CPU resources configured on the descendant ESX hosts.<br>Key: cpu\|capacity_provisioned |
| Usable Capacity (MHz) | The usable CPU resources that are available for the virtual machines after considering reservations for vSphere High Availability (HA) and other vSphere services.<br>Key: cpu\|haTotalCapacity_average |

## Disk Metrics for Data Centers

Disk metrics provide information about disk use.

| Metric Name | Description |
|---|---|
| Total IOPS | Average number of commands issued per second during the collection interval.<br>Key: disk\|commandsAveraged_average |
| Total Latency (ms) | Average amount of time taken for a command from the perspective of the guest operating system. This metric is the sum of the Kernel Latency and Physical Device Latency metrics.<br>Key: disk\|totalLatency_average |
| Total Throughput (KBps) | Average of the sum of the data read and written for all the disk instances of the host or virtual machine.<br>Key: disk\|usage_average |
| Total queued outstanding operations | Sum of queued operations and outstanding operations.<br>Key: disk\|sum_queued_oio |
| Max observed OIO | Max observed IO for a disk.<br>Key: disk\|max_observed |

## Memory Metrics for Data Centers

Memory metrics provide information about memory use and allocation.

| Metric Name | Description |
|---|---|
| Contention (%) | Machine Contention Percentage.<br>Key: mem\|host_contentionPct |
| Machine Demand (KB) | Memory machine demand in kilobytes.<br>Key: mem\|host_demand |
| ESX System Usage | Memory usage by the VM kernel and ESX user-level services.<br>Key: mem\|host_systemUsage |
| Provisioned Memory (KB) | Provisioned host memory in kilobytes.<br>Key: mem\|host_provisioned |
| Reserved Capacity (KB) | Reserved memory capacity in kilobytes.<br>Key: mem\|reservedCapacity_average |
| Usable Memory (KB) | Usable host memory in kilobytes.<br>Key: mem\|host_usable |
| Host Usage | Host memory use in kilobytes.<br>Key: mem\|host_usage |
| Usage/Usable (%) | Percent host memory used.<br>Key: mem\|host_usagePct |
| VM Overhead | Memory overhead reported by host.<br>Key: mem\|overhead_average |
| Stress Balance Factor | Stress Balance Factor.<br>Key: mem\|stress_balance_factor |
| Lowest Provider Capacity Remaining | Lowest Provider Capacity Remaining.<br>Key: mem\|min_host_capacity_remaining |
| Workload Balance Factor | Workload Balance Factor.<br>Key: mem\|workload_balance_factor |

| Metric Name | Description |
|---|---|
| Highest Provider Workload | Highest Provider Workload.<br>Key: mem\|max_host_workload |
| Host workload Max-Min Disparity | Difference of Max and Min host workload in the container.<br>Key: mem\|host_workload_disparity |
| Host stress Max-Min Disparity | Difference of Max and Min host stress in the container.<br>Key: mem\|host_stress_disparity |
| Utilization (KB) | Memory utilization level based on the descendant virtual machines utilization. Includes reservations, limits, and overhead to run the Virtual Machines.<br>Key: mem\|total_need |
| Total Capacity (KB) | Total physical memory configured on descendant ESX hosts.<br>Key: mem\|host_provisioned |
| Usable Capacity (KB) | The usable memory resources available for the virtual machines after considering reservations for vSphere HA and other vSphere services.<br>Key: mem\|haTotalCapacity_average |

### Network Metrics for Data Centers

Network metrics provide information about network performance.

| Metric Name | Description |
|---|---|
| Packets Dropped | Percentage of packets dropped.<br>Key: net\|droppedPct |
| Max Observed Throughput | Max observed rate of network throughput.<br>Key: net\|maxObservedKBps |
| Data Transmit Rate | Average amount of data transmitted per second.<br>Key: net\|transmitted_average |
| Data Receive Rate | Average amount of data received per second.<br>Key: net\|received_average |
| Total Throughput (KBps) | The sum of the data transmitted and received for all the NIC instances of the host or virtual machine.<br>Key: net\|usage_average |

### Storage Metrics for Data Centers

Storage metrics provide information about storage use.

| Metric Name | Description |
|---|---|
| Total Usage | Total throughput rate.<br>Key: storage\|usage_average |

### Datastore Metrics for Data Centers

Datastore metrics provide information about Datastore use.

| Metric Name | Description |
|---|---|
| Outstanding IO requests | OIO for datastore.<br>Key: datastore\|demand_oio |
| Read IOPS | Average number of read commands issued per second during the collection interval.<br>Key: datastore\|numberReadAveraged_average |
| Write IOPS | Average number of write commands issued per second during the collection interval.<br>Key: datastore\|numberWriteAveraged_average |
| Read Throughput (KBps) | Amount of data read in the performance interval.<br>Key: datastore\|read_average |
| Write Throughput (KBps) | Amount of data written to disk in the performance interval.<br>Key: datastore\|write_average |

### Disk Space Metrics for Data Centers

Disk space metrics provide information about disk use.

| Metric Name | Description |
|---|---|
| Virtual machine used | Used virtual machine disk space in gigabytes.<br>Key: diskspace\|used |
| Total disk space used | Total disk space used on all datastores visible to this object.<br>Key: diskspace\|total_usage |
| Total disk space | Total disk space on all datastores visible to this object.<br>Key: diskspace\|total_capacity |
| Total provisioned disk space | Total provisioned disk space on all datastores visible to this object.<br>Key: diskspace\|total_provisioned |
| Shared Used (GB) | Shared disk space in gigabytes.<br>Key: diskspace\|shared |
| Snapshot Space (GB) | Snapshot disk space in gigabytes.<br>Key: diskspace\|snapshot |
| Virtual Disk Used (GB) | Used virtual disk space in gigabytes.<br>Key: diskspace\|diskused |
| Number of Virtual Disks | Number of Virtual Disks.<br>Key: diskspace\|numvmdisk |
| Utilization (GB) | Storage space used on the connected vSphere Datastores.<br>Key: diskspace\|total_usage |
| Total Capacity (GB) | Total storage space available on the connected vSphere datastores.<br>Key: diskspace\|total_capacity |

## Summary Metrics for Data Centers

Summary metrics provide information about overall performance.

| Metric Name | Description |
|---|---|
| Number of Running Hosts | Number of hosts that are ON.<br>Key: summary\|number_running_hosts |
| Number of Running VMs | Number of running virtual machines.<br>Key: summary\|number_running_vms |
| Maximum Number of VMs | Maximum number of virtual machines.<br>Key: summary\|max_number_vms |
| Number of Clusters | Total number of clusters.<br>Key: summary\|total_number_clusters |
| Number of Hosts | Total number of hosts.<br>Key: summary\|total_number_hosts |
| Number of VMs | Total number of virtual machines.<br>Key: summary\|total_number_vms |
| Total Number of Datastores | Total number of datastores.<br>Key: summary\|total_number_datastores |
| Number of VCPUs on Powered On VMs | Total number of VCPUs of virtual machines that are powered on.<br>Key: summary\|number_running_vcpus |
| Workload Indicator | Workload indicator.<br>Key: summary\|workload_indicator |
| Average Running VM Count per Running Host | Average number of running virtual machines per running host.<br>Key: summary\|avg_vm_density |
| WLP | Displays the VM migration trend as part of workload optimization. These metrics are deactivated by default. You must activate them from policies.<br>• Fail Count: Number of failed VM move attempts in the last daily cycle.<br>• Number of runs: The total number of times the WLP was run during the last daily cycle.<br>• Success Count: The number of successful VM moves during the last daily cycle. |

## Reclaimable Metrics for Data Centers

Reclaimable metrics provide information about reclaimable resources.

| Metric Name | Description |
|---|---|
| CPU (vCPUs) | Number of reclaimable vCPUs within the data center.<br>Key: reclaimable\|cpu |
| Disk Space | Reclaimable disk space within the data center.<br>Key: reclaimable\|diskspace |
| Potential Savings | Potential saving after reclamation of resources of all reclaimable VMs (Idle VMs, Powered Off VMs, VM snapshots) within the data center.<br>Key: reclaimable\|cost |
| Memory (KB) | Reclaimable memory within the data center.<br>Key: reclaimable\|mem |

| Metric Name | Description |
|---|---|
| Virtual Machines | Number of VMs having reclaimable resources (Memory, disk space, vCPU) within the data center.<br>Key: reclaimable\|vm_count |
| Idle VMs\|Potential Savings | Potential saving after reclamation of resources of Idle VMs within the data center.<br>Key: reclaimable\|idle_vms\|cost |
| Powered Off VMs\|Potential Savings | Potential saving after reclamation of resources of Powered Off VMs within the data center.<br>Key: reclaimable\|poweredOff_vms\|cost |
| VM Snapshots\|Potential Savings | Potential saving after reclamation of VM snapshots within the data center.<br>Key: reclaimable\|vm_snapshots \|cost |
| Reclaimable\|Orphaned Disks\|Potential Savings (Currency) | Displays the potential savings after reclaimation of disk space by removing orphaned VMDks from all datastores under datacenter.<br>reclaimable\|cost |
| Reclaimable\|Number of Orphaned Disks | Number of reclaimable orphaned disks is the sum of all orphaned disks on it's datastore.<br>reclaimable\|orphaned_disk_count |
| Reclaimable Host Cost | This metric calculates the cumulated reclaimable host cost at cluster level and displays the same metric at data center level.<br>Reclaimable\|Reclaimable Host Cost (Currency) |

## Cost Metrics for Data Centers

Cost metrics provide information about the cost.

**Note:**
The Effective metrics based on the capacity model assigned at the cluster (Demand/Allocation) pick the corresponding metric from underneath the clusters.

| Metric Name | Description |
|---|---|
| Monthly Cluster Aggregated Allocated Cost | This metric displays the sum of the monthly allocated cost for both cluster and unclustered hosts.<br>Key: Cost\|Allocation\|Monthly Cluster Aggregated Allocated Cost |
| Monthly Cluster Aggregated Unallocated Cost | This metric displays the sum of both cluster and unclustered hosts unallocated cost.<br>Key: Cost\|Allocation\|Monthly Cluster Aggregated Unallocated Cost |
| Monthly Datastore Aggregated Allocated Cost | This metric displays the monthly aggregated allocated cost at datastore level.<br>Key: Cost\|Allocation\|Monthly Datastore Aggregated Allocated Cost |
| Monthly Datastore Aggregated Unallocated Cost | This metric displays the monthly aggregated unallocated cost at datastore level.<br>Key: Cost\|Allocation\|Monthly Datastore Aggregated Unallocated Cost |
| Monthly Cluster Effective Aggregated Allocated Cost | This metric displays the Monthly Cluster Effective Aggregated Allocated Cost cost.<br>Key: Cost\|Monthly Cluster Effective Aggregated Allocated Cost |
| Monthly Cluster Effective Aggregated Unallocated Cost | This metric displays the Monthly Cluster Effective Aggregated Unallocated Cost.<br>Key: Cost\|Monthly Cluster Effective Aggregated Unallocated Cost |

| Metric Name | Description |
|---|---|
| Monthly Datastore Effective Aggregated Allocated Cost | This metric displays the Monthly Datastore Effective Aggregated Allocated Cost at datastore level.<br>Key: Cost\|Allocation\|Monthly Datastore Effective Aggregated Allocated Cost |
| Monthly Datastore Effective Aggregated Unallocated Cost | This metric displays the Monthly Datastore Effective Aggregated Unallocated Cost at datastore level.<br>Key: Cost\|Allocation\|Monthly Datastore Effective Aggregated Unallocated Cost |
| **Note:**<br>The eight allocation metrics mentioned above, helps you to calculate the cost when you enable the allocation model for costing. The same set of metrics are available when you enable demand model also. | |
| Monthly Cluster Aggregated Cost | This metric displays the sum of monthly aggregated allocated and unallocated cost for clusters.<br>Key: cost\|clusterCost |
| Monthly Cluster Aggregated Unallocated Cost | This metric displays the sum of the monthly unallocated cost for both cluster and unclustered hosts.<br>Key: Cost\|Monthly Cluster Aggregated Unallocated Cost |
| Monthly Datacenter Aggregated Total Cost | Monthly aggregated total cost for the data center.<br>Key: Cost\|Monthly Datacenter Aggregated Total Cost |
| Monthly Datastore Total Cost | Monthly data store total cost.<br>Key: cost\|totalCost |
| Monthly Datastore Aggregated Allocated Cost | Monthly aggregated allocated cost for the datastore.<br>Key: cost\|aggrDataStoreAllocatedCost |
| Monthly Datastore Aggregated Unallocated Cost | Monthly aggregated unallocated cost for the datastore.<br>Key: cost\|aggrDataStoreUnallocatedCost |
| Monthly VM Aggregated Direct Cost | Month to date aggregated VM direct cost across all the VMs under the data center.<br>Key: cost\|vmDirectCost |

## Virtual Machine Operations Metrics for Data Centers

VM operations metrics provide information about the actions performed on the VMs in the datacenter. The following are some important points you must know about VM operation metrics for data centers.

- VM operations metrics is not collected for custom data centers.
- If you edit a VM settings and do not perform any action, still it is considered as VM reconfigure operation.
- During Revert Snapshot, VMs are powered-off, but this operation is not counted under VM Power-off metric.
- Adding ESX with VMs is not counted under VM Create metric.
- Removing ESX with VMs is not coutned under VM Remove metric.
- VM hardstop operation is not counted under VM Power Off metric.

| Metric Name | Description |
|---|---|
| Inventory | |
| VM Clone | This metric displays the number of clone operations on the virtual machine.<br>Key: Inventory\|VM Clone |
| VM Create | This metric displays the number of create operations on the virtual machine.<br>Key: Inventory\|VM Create |

| Metric Name | Description |
|---|---|
| VM Delete | This metric displays the number of delete operations on the virtual machine.<br>Key: Inventory\|VM Delete |
| VM Reconfigure | This metric displays the number of reconfigure operations on the virtual machine.<br>Key: Inventory\|VM Reconfigure |
| VM Register | This metric displays the number of register operations on the virtual machine.<br>Key: Inventory\|VM Register |
| VM Template Deploy | This metric displays the number templates deployed on the virtual machine.<br>Key: Inventory\|VM Template Deploy |
| VM Unregister | This metric displays the number of unregister operations on the virtual machine.<br>Key: Inventory\|VM Unregister |
| Location | |
| Storage vMotion | This metric displays the number of migrations with vMotion (datastore change operations for Powered-on VMs).<br>Key: Location\|Storage vMotion |
| VM Datastore Change (powered-off VMs) | This metric displays the number of datastore change operations, for powered-off and suspended virtual machines.<br>Key: Location\|VM Datastore Change (powered-off VMs) |
| VM Host and Datastore Change (powered-off VMs) | This metric displays the number of host and datastore change operations, for powered-off and suspended virtual machines.<br>Key: Location\|VM Host and Datastore Change (powered-off VMs) |
| VM Host and Datastore Change (powered-on VMs) | This metric displays the number of host and datastore change operations, for powered-on and suspended virtual machines.<br>Key: Location\|VM Host and Datastore Change (powered-on VMs) |
| VM Host Change (powered-off VMs) | This metric displays the number of host change operations, for powered-off and suspended virtual machines.<br>Key: Location\|VM Host Change (powered-off VMs) |
| vMotion | This metric displays the number of migrations with vMotion (host change operations for powered-on VMs).<br>Key: Location\|vMotion |
| State | |
| VM Guest Reboot | This metric displays the number of reboot operations on the virtual machine guest.<br>Key: State\|VM Guest Reboot |
| VM Guest Shutdown | This metric displays the number of shutdown operations on the virtual machine guest.<br>Key: State\|VM Guest Shutdown |
| VM Power Off | This metric displays the number of power-off operations on the virtual machine.<br>Key: State\|VM Power Off |
| VM Power On | This metric displays the number of power-on operations on the virtual machine.<br>Key: State\|VM Power On |

| Metric Name | Description |
|---|---|
| VM Reset | This metric displays the number of reset operations on the virtual machine guest.<br>Key: State\|VM Reset |
| VM Standby Guest | This metric displays the number of standby operations on the virtual machine guest.<br>Key: State\|VM Standby Guest |
| VM Suspend | This metric displays the number of suspend operations on the virtual machine.<br>Key: State\|VM Suspend |

### Disabled Metrics

The following metrics are disabled in this version of VCF Operations. This means that they do not collect data by default.

You can enable these metrics in the Policy workspace. For more information, in VMware Docs search for Collect Metrics and Properties Details.

You can enable these metrics in the Policy workspace. For more information, see Metrics and Properties Details.

| Metric Name | Key |
|---|---|
| Datastore I/O\|Max Observed Number of Outstanding IO Operations (IOPS) | datastore\|maxObserved_OIO |
| Datastore I/O\|Max Observed Read Rate (KBps) | datastore\|maxObserved_Read |
| Datastore I/O\|Max Observed Reads per second (IOPS) | datastore\|maxObserved_NumberRead |
| Datastore I/O\|Max Observed Write Rate (KBps) | datastore\|maxObserved_Write |
| Datastore I/O\|Max Observed Writes per second (IOPS)0 | datastore\|maxObserved_NumberWrite |
| Max Observed Transmitted Throughput | Max observed transmitted rate of network throughput.<br>Key: net\|maxObserved_Tx_KBps |
| Max Observed Received Throughput | Max observed received rate of network throughput.<br>Key: net\|maxObserved_Rx_KBps |
| Not Shared (GB) | Unshared disk space in gigabytes.<br>Key: diskspace\|notshared |

## Custom Data Center Metrics

VCF Operations collects CPU usage, memory, summary, network, and datastore metrics for custom data center objects.

Custom data center metrics include capacity and badge metrics. See definitions in:

- Capacity Analytics Generated Metrics
- Badge Metrics

### CPU Usage Metrics for Custom Data Centers

CPU usage metrics provide information about CPU use.

| Metric Name | Description |
|---|---|
| Host Provisioned Capacity | Host provisioned capacity (MHz).<br>Key: cpu\|capacity_provisioned |
| Provisioned vCPU(s) | Provisioned vCPU(s).<br>Key: cpu\|corecount_provisioned |

| Metric Name | Description |
|---|---|
| Demand without overhead | Value of demand excluding any overhead.<br>Key: cpu\|demand_without_overhead |
| Number of hosts stressed | Number of hosts stressed.<br>Key: cpu\|num_hosts_stressed |
| Stress Balance Factor | Stress balance factor.<br>Key: cpu\|stress_balance_factor |
| Lowest Provider Capacity Remaining | Lowest provider capacity remaining.<br>Key: cpu\|min_host_capacity_remaining |
| Workload Balance Factor | Workload balance factor.<br>Key: cpu\|workload_balance_factor |
| Highest Provider Workload | Highest provider workload.<br>Key: cpu\|max_host_workload |
| Host workload Max-Min Disparity | Host workload max-min disparity.<br>Key: cpu\|host_workload_disparity |
| Host stress Max-Min Disparity | Difference of max and min host stress in the container.<br>Key: cpu\|host_stress_disparity |
| Demand (MHz) | CPU utilization level based on descendant virtual machines utilization. This Includes reservations, limits, and overhead to run the virtual machines.<br>Key: cpu\|demandmhz |
| Total Capacity (MHz) | Total CPU resources configured on the descendant ESX hosts.<br>Key: cpu\|capacity_provisioned |
| Usable Capacity (MHz) | The usable CPU resources that are available for the virtual machines after considering reservations for vSphere High Availability (HA) and other vSphere services.<br>Key: cpu\|haTotalCapacity_average |

### Memory Metrics for Custom Data Centers

Memory metrics provide information about memory use.

| Metric Name | Description |
|---|---|
| Usable Memory | Usable memory.<br>Key: mem\|host_usable |
| Machine Demand | Memory machine demand in KB.<br>Key: mem\|host_demand |
| Number of hosts stressed | Number of hosts stressed.<br>Key: mem\|num_hosts_stressed |
| Stress Balance Factor | Stress balance factor.<br>Key: mem\|stress_balance_factor |
| Lowest Provider Capacity Remaining | Lowest provider capacity remaining.<br>Key: mem\|min_host_capacity_remaining |
| Workload Balance Factor | Workload balance factor.<br>Key: mem\|workload_balance_factor |
| Highest Provider Workload | Highest provider workload.<br>Key: mem\|max_host_workload |
| Host workload Max-Min Disparity | Host workload max-min disparity.<br>Key: mem\|host_workload_disparity |

| Metric Name | Description |
|---|---|
| Host stress max-min disparity | Host stress max-min disparity.<br>Key: mem\|host_stress_disparity |
| Utilization (KB) | Memory utilization level based on the descendant virtual machines utilization. Includes reservations, limits, and overhead to run the Virtual Machines.<br>Key: mem\|total_need |
| Total Capacity (KB) | Total physical memory configured on descendant ESX hosts.<br>Key: mem\|host_provisioned |
| Usable Capacity (KB) | The usable memory resources available for the virtual machines after considering reservations for vSphere HA and other vSphere services.<br>Key: mem\|haTotalCapacity_average |

### Summary Metrics for Custom Data Centers

Summary metrics provide information about overall performance.

| Metric Name | Description |
|---|---|
| Number of Running VMs | Number of virtual machines that are ON.<br>Key: summary\|number_running_vms |
| Maximum Number of VMs | Maximum number of virtual machines.<br>Key: summary\|max_number_vms |
| Status | Status of the data center.<br>Key: summary\|status |
| WLP | Displays the VM migration trend as part of workload optimization. These metrics are deactivated by default. You must activate them from policies.<br>• Fail Count: Number of failed VM move attempts in the last daily cycle.<br>• Number of runs: The total number of times the WLP was run during the last daily cycle.<br>• Success Count: The number of successful VM moves during the last daily cycle. |

### Network Metrics for Custom Data Centers

Network metrics provide information about network performance.

| Metric Name | Description |
|---|---|
| Usage Rate | The sum of the data transmitted and received for all the NIC instances of the host or virtual machine.<br>Key: net\|usage_average |
| Data Transmit Rate | Average amount of data transmitted per second.<br>Key: net\|transmitted_average |
| Data REceive Rate | Average amount of data received per second.<br>Key: net\|received_average |

## Datastore Metrics for Custom Data Centers

Datastore metrics provide information about datastore use.

| Metric Name | Description |
|---|---|
| Outstanding IO requests | OIO for datastore.<br>Key: datastore\|demand_oio |
| Read IOPS | Average number of read commands issued per second during the collection interval.<br>Key: datastore\|numberReadAveraged_average |
| Write IOPS | Average number of write commands issued per second during the collection interval.<br>Key: datastore\|numberWriteAveraged_average |
| Read Throughput (KBps) | Amount of data read in the performance interval.<br>Key: datastore\|read_average |
| Write Throughput (KBps) | Amount of data written to disk in the performance interval.<br>Key: datastore\|write_average |

## Reclaimable Metrics for Custom Data Centers

Reclaimable metrics provide information about reclaimable resources.

| Metric Name | Description |
|---|---|
| CPU (vCPUs) | Number of reclaimable vCPUs within the custom data center.<br>Key: reclaimable\|cpu |
| Disk Space | Reclaimable disk space within the custom data center.<br>Key: reclaimable\|diskspace |
| Potential Savings | Potential saving after reclamation of resources of all reclaimable VMs (Idle VMs, Powered Off VMs, VM snapshots) within the custom data center.<br>Key: reclaimable\|cost |
| Memory (KB) | Reclaimable memory within the custom data center.<br>Key: reclaimable\|mem |
| Number of Orphaned Disks | Number of reclaimable orphaned disks within the custom data center.<br>reclaimable\|orphaned_disk_count |
| Reclaimable\|Orphaned Disks\|Potential Savings | Potential savings in cost after reclamation of orphaned disks across the custom data center.<br>Key: reclaimable\|orphaned_disk\|cost<br>**Note:**<br>The orphaned disk reclamation feature might not work as expected when VCF Operations monitors multiple vCenters which use shared data stores. |
| Virtual Machines | Number of VMs having reclaimable resources (Memory, disk space, vCPU) within the custom data center.<br>Key: reclaimable\|vm_count |
| Idle VMs\|Potential Savings | Potential saving after reclamation of resources of Idle VMs within the custom data center.<br>Key: reclaimable\|idle_vms\|cost |
| Powered Off VMs\|Potential Savings | Potential saving after reclamation of resources of Powered Off VMs within the custom data center.<br>Key: reclaimable\|poweredOff_vms\|cost |

| Metric Name | Description |
|---|---|
| VM Snapshots\|Potential Savings | Potential saving after reclamation of VM snapshots within the custom data center.<br>Key: reclaimable\|vm_snapshots \|cost |
| Reclaimable\|Orphaned Disks\|Potential Savings (Currency) | Displays the potential savings after reclaimation of disk space by removing orphaned VMDks from all datastores under custom datacenters.<br>reclaimable\|cost |
| Reclaimable\|Number of Orphaned Disks | Number of reclaimable orphaned disks is the sum of the numbers of orphaned disks on it's datastore.<br>reclaimable\|orphaned_disk_count |

### Disk Space Metrics for Custom Data Centers

Disk space metrics provide information about disk use.

| Metric Name | Description |
|---|---|
| Utilization (GB) | Storage space used on the connected vSphere Datastores.<br>Key: diskspace\|total_usage |
| Total Capacity (GB) | Total storage space available on the connected vSphere datastores.<br>Key: diskspace\|total_capacity |

### Disabled Metrics

The following metrics are disabled in this version of VCF Operations. This means that they do not collect data by default.

You can enable these metrics in the Policy workspace. For more information, in VMware Docs search for Collect Metrics and Properties Details.

You can enable these metrics in the Policy workspace. For more information, see Metrics and Properties Details.

| Metric Name | Key |
|---|---|
| Max Observed Throughput | Max observed rate of network throughput.<br>Key: net\|maxObserved_KBps |
| Max Observed Transmitted Throughput | Max observed transmitted rate of network throughput.<br>Key: net\|maxObserved_Tx_KBps |
| Max Observed Received Throughput | Max observed received rate of network throughput.<br>Key: net\|maxObserved_Rx_KBps |
| Max Observed Reads per second | Max observed average number of read commands issued per second during the collection interval.<br>Key: datastore\|maxObserved_NumberRead |
| Max Observed Read Rate | Max observed rate of reading data from the datastore.<br>Key: datastore\|maxObserved_Read |
| Max Observed Writes per second | Max observed average number of write commands issued per second during the collection interval.<br>Key: datastore\|maxObserved_NumberWrite |
| Max Observed Write Rate | Max observed rate of writing data from the datastore.<br>Key: datastore\|maxObserved_Write |
| Max Observed Number of Outstanding IO Operations | Max observer number of outstanding IO operations.<br>Key: datastore\|maxObserved_OIO |

## Storage Pod Metrics

VCF Operations collects datastore and disk space metrics for storage pod objects.

Storage Pod metrics include capacity and badge metrics. See definitions in:

- Capacity Analytics Generated Metrics
- Badge Metrics

### Table 1481: Datastore Metrics for Storage Pods

| Metric Name | Description |
|---|---|
| Read IOPS | Average number of read commands issued per second during the collection interval.<br>Key: datastore\|numberReadAveraged_average |
| Writes per second | Average number of write commands issued per second during the collection interval.<br>Key: datastore\|numberWriteAveraged_average |
| Read Throughput (KBps) | Amount of data read in the performance interval.<br>Key: datastore\|read_average |
| Write Throughput (KBps) | Amount of data written to disk in the performance interval.<br>Key: datastore\|write_average |
| Total Throughput (KBps) | Usage Average.<br>Key: datastore\|usage_average |
| Read Latency | Average amount of time for a read operation from the datastore. Total latency = kernel latency + device latency.<br>Key: datastore\|totalReadLatency_average |
| Write Latency | Average amount of time for a write operation to the datastore. Total latency = kernel latency + device latency.<br>Key: datastore\|totalWriteLatency_average |
| Total Latency (ms) | The average amount of time taken for a command from the perspective of a Guest OS. This is the sum of Kernel Command Latency and Physical Device Command Latency.<br>Key: datastore\|totalLatency_average |
| Total IOPS | Average number of commands issued per second during the collection interval.<br>Key: datastore\|commandsAveraged_average |

### Table 1482: Disk Space Metrics for Storage Pods

| Metric Name | Description |
|---|---|
| Freespace | Unused space available on datastore.<br>Key: diskspace\|freespace |
| Total used | Total space used.<br>Key: diskspace\|disktotal |
| Capacity | Total capacity of datastore.<br>Key: diskspace\|capacity |
| Virtual Machine used | Space used by virtual machine files.<br>Key: diskspace\|used |
| Snapshot Space | Space used by snapshots.<br>Key: diskspace\|snapshot |

## VMware Distributed Virtual Switch Metrics

VCF Operations collects network and summary metrics for VMware distributed virtual switch objects.

VMware Distributed Virtual Switch metrics include badge metrics. See definitions in Badge Metrics.

### Table 1483: Network Metrics for VMware Distributed Virtual Switches

| Metric Name | Description |
|---|---|
| Total Ingress Traffic | Total ingress traffic (KBps).<br>Key: network\|port_statistics\|rx_bytes |
| Total Egress Traffic | Total egress traffic (KBps).<br>Key: network\|port_statistics\|tx_bytes |
| Egress Unicast Packets per second | Egress unicast packets per second.<br>Key: network\|port_statistics\|ucast_tx_pkts |
| Egress Multicast Packets per second | Egress multicast packets per second.<br>Key: network\|port_statistics\|mcast_tx_pkts |
| Egress Broadcast Packets per second | Egress broadcast packets per second.<br>Key: network\|port_statistics\|bcast_tx_pkts |
| Ingress Unicast Packets per second | Ingress unicast packets per second.<br>Key: network\|port_statistics\|ucast_rx_pkts |
| Ingress Multicast Packets per second | Ingress multicast packets per second.<br>Key: network\|port_statistics\|mcast_rx_pkts |
| Ingress Broadcast Packets per second | Ingress broadcast packets per second.<br>Key: network\|port_statistics\|bcast_rx_pkts |
| Egress Dropped Packets per second | Egress dropped packets per second.<br>Key: network\|port_statistics\|dropped_tx_pkts |
| Ingress Dropped Packets per second | Ingress dropped packets per second.<br>Key: network\|port_statistics\|dropped_rx_pkts |
| Total Ingress Packets per second | Total ingress packets per second.<br>Key: network\|port_statistics\|rx_pkts |
| Total Egress Packets per second | Total egress packets per second.<br>Key: network\|port_statistics\|tx_pkts |
| Utilization | Use (KBps).<br>Key: network\|port_statistics\|utilization |
| Total Dropped Packets per second | Total dropped packets per second.<br>Key: network\|port_statistics\|dropped_pkts |
| Percentage of Dropped Packets | Percentage of dropped packets.<br>Key: network\|port_statistics\|dropped_pkts_pct |
| Max Observed Ingress Traffic (KBps) | Max observed ingress traffic (KBps).<br>Key: network\|port_statistics\|maxObserved_rx_bytes |
| Max Observed Egress Traffic (KBps) | Max observed egress traffic (KBps).<br>Key: network\|port_statistics\|maxObserved_tx_bytes |
| Max Observed Utilization (KBps) | Max observed utilization (KBps).<br>Key: network\|port_statistics\|maxObserved_utilization |

**Table 1484: Summary Metrics for VMware Distributed Virtual Switches**

| Metric Name | Description |
|---|---|
| Maximum Number of Ports | Maximum number of ports.<br>Key: summary\|max_num_ports |
| Used Number of Ports | Used number of ports.<br>Key: summary\|used_num_ports |
| Number of Blocked Ports | Number of blocked ports.<br>Key: summary\|num_blocked_ports |

**Table 1485: Host Metrics for VMware Distributed Virtual Switches**

| Metric Name | Description |
|---|---|
| MTU Mismatch | Maximum Transmission Unit (MTU) mismatch.<br>Key: host\|mtu_mismatch |
| Teaming Mismatch | Teaming mismatch.<br>Key: host\|teaming_mismatch |
| Unsupported MTU | Unsupported MTU.<br>Key: host\|mtu_unsupported |
| Unsupported VLANs | Unsupported VLANs.<br>Key: host\|vlans_unsupported |
| Config Out Of Sync | Config Out Of Sync.<br>Key: host\|config_outofsync |
| Number of Attached pNICs | Number of attached physical NICs.<br>Key: host\|attached_pnics |

## Distributed Virtual Port Group Metrics

The vCenter Adapter instance collects network and summary metrics for distributed virtual port groups.

Distributed Virtual Port Group metrics include badge metrics. See definitions in Badge Metrics.

**Table 1486: Network Metrics for Distributed Virtual Port Groups**

| Metric Name | Description |
|---|---|
| Ingress Traffic | Ingress traffic (KBps).<br>Key: network\|port_statistics\|rx_bytes |
| Egress Traffic | Egress traffic (KBps).<br>Key: network\|port_statistics\|tx_bytes |
| Egress Unicast Packets per second | Egress unicast packets per second.<br>Key: network\|port_statistics\|ucast_tx_pkts |
| Egress Multicast Packets per second | Egress multicast packets per second.<br>Key: network\|port_statistics\|mcast_tx_pkts |
| Egress Broadcast Packets per second | Egress broadcast packets per second.<br>Key: network\|port_statistics\|bcast_tx_pkts |
| Ingress Unicast Packets per second | Ingress unicast packets per second.<br>Key: network\|port_statistics\|ucast_rx_pkts |
| Ingress Multicast Packets per second | Ingress multicast packets per second.<br>Key: network\|port_statistics\|mcast_rx_pkts |

| Metric Name | Description |
|---|---|
| Ingress Broadcast Packets per second | Ingress broadcast packets per second.<br>Key: network\|port_statistics\|bcast_rx_pkts |
| Egress Dropped Packets per second | Egress dropped packets per second.<br>Key: network\|port_statistics\|dropped_tx_pkts |
| Ingress Dropped Packets per second | Ingress dropped packets per second.<br>Key: network\|port_statistics\|dropped_rx_pkts |
| Total Ingress Packets per second | Total Ingress packets per second.<br>Key: network\|port_statistics\|rx_pkts |
| Total Egress Packets per second | Total Egress packets per second.<br>Key: network\|port_statistics\|tx_pkts |
| Utilization | Utilization (KBps).<br>Key: network\|port_statistics\|utilization |
| Total Dropped Packets per second | Total dropped packets per second.<br>Key: network\|port_statistics\|dropped_pkts |
| Percentage of Dropped Packets | Percentage of dropped packets.<br>Key: network\|port_statistics\|dropped_pkts_pct |
| Max Observed Ingress Traffic (KBps) | Max observed ingress traffic (KBps).<br>Key: network\|port_statistics\|maxObserved_rx_bytes |
| Max Observed Egress Traffic (KBps) | Max observed egress traffic (KBps).<br>Key: network\|port_statistics\|maxObserved_tx_bytes |
| Max Observed Utilization (KBps) | Max observed utilization (KBps).<br>network\|port_statistics\|maxObserved_utilization |

**Table 1487: Summary Metrics for Distributed Virtual Port Groups**

| Metric Name | Description |
|---|---|
| Maximum Number of Ports | Maximum number of ports.<br>Key: summary\|max_num_ports |
| Used Number of Ports | Used number of ports.<br>Key: summary\|used_num_ports |
| Number of Blocked Ports | The number of blocked ports.<br>Key: summary\|num_blocked_ports |

## Datastore Cluster Metrics

VCF Operations collects profile metrics for the datastore cluster resources.

### Profiles Metrics for Datastore Cluster Resources

Profiles metrics provide information about the profile specific capacity.

| Metric Name | Description |
|---|---|
| Profiles\|Capacity Remaining Profile (Average) | The capacity remaining in terms of fitting the average consumer.<br>Key: Profiles\|capacityRemainingProfile_<profile uuid> |
| Profiles\|Capacity Remaining Profile (<custom profile name>) | Published for custom profiles enabled from policy on Datastore Cluster Resource.<br>Key: Profiles\|capacityRemainingProfile_<profile uuid> |

### Capacity Allocation Metrics for Datastore Cluster Resources

Capacity allocation metrics provide information about the allotment of capacity, see Capacity Analytics Generated Metrics.

## Datastore Metrics

VCF Operations collects capacity, device, and summary metrics for datastore objects.

Capacity metrics can be calculated for datastore objects. See Capacity Analytics Generated Metrics.

### Capacity Metrics for Datastores

Capacity metrics provide information about datastore capacity.

| Metric Name | Description |
|---|---|
| Capacity\|Available Space (GB) | This metric shows the amount of free space that a datastore has available.<br>Use this metric to know how much storage space is unused on the datastore. Try to avoid having too little free disk space in order to accommodate unexpected storage growth on the datastore. The exact size of the datastore is based on company policy.<br>Key: capacity\|available_space |
| Capacity\|Provisioned (GB) | This metric shows the amount of storage that was allocated to the virtual machines.<br>Use this metric to know how much storage space is being used on the datastore.<br>Check the metric trend to identify spikes or abnormal growth.<br>Key: capacity\|provisioned |
| Capacity\|Total Capacity (GB) | This metric shows the overall size of the datastore.<br>Use this metric to know the total capacity of the datastore. Typically the size of the datastore should not be too small. VMFS datastore size has grown over the years as virtualization matures and larger virtual machines are now onboard. Ensure that the size can handle enough virtual machines to avoid datastore sprawl. A best practice is to use 5 TB for VMFS and more for vSAN.<br>Key: capacity\|total_capacity |
| Capacity\|Used Space (GB) | This metric shows the amount of storage that is being used on the datastore.<br>Key: capacity\|used_space |
| Capacity\|Workload (%) | Capacity workload.<br>Key: capacity\|workload |
| Capacity\|Uncommitted Space (GB) | Uncommitted space in gigabytes.<br>Key: capacity\|uncommitted |
| Capacity\|Total Provisioned Consumer Space | Total Provisioned Consumer Space.<br>Key: capacity\|consumer_provisioned |

| Metric Name | Description |
|---|---|
| Capacity\|Used Space (%) | This metric shows the amount of storage that is being used on the datastore.<br>Use this metric to know the percentage of storage space being used on the datastore.<br>When using this metric, verify that you have at least 20% of free storage. Less than this, and you might experience problems when a snapshot is not deleted. If you have more than 50% free storage space, you are not utilizing your storage in the best possible way.<br>Key: capacity\|usedSpacePct |

### Device Metrics for Datastores

Device metrics provide information about device performance.

| Metric Name | Description |
|---|---|
| Devices\|Bus Resets | This metric shows the number of bus resets in the performance interval.<br>Key: devices\|busResets_summation |
| Devices\|Commands Aborted | This metric shows the number of disk commands canceled in the performance interval.<br>Key: devices\|commandsAborted_summation |
| Devices\|Commands Issued | This metric shows the number of disk commands issued in the performance interval.<br>Key: devices\|commands_summation |
| Devices\|Read Latency (ms) | This metric shows the average time taken for a read from the perspective of a guest operating system. This metric is the sum of the Kernel Disk Read Latency and Physical Device Read Latency metrics.<br>Key: devices\|totalReadLatency_averag |
| Devices\|Kernel Disk Read Latency (ms) | Average time spent in ESX host VM Kernel per read.<br>Key: devices\|kernelReadLatency_average |
| Devices\|Kernel Write Latency (ms) | Average time spent in ESX Server VM Kernel per write.<br>Key: devices\|kernelWriteLatency_average |
| Devices\|Physical Device Read Latency (ms) | Average time taken to complete a read from the physical device.<br>Key: devices\|deviceReadLatency_average |
| Devices\|Queue Write Latency (ms) | Average time spent in the ESX Server VM Kernel queue per write.<br>Key: devices\|queueWriteLatency_average |
| Devices\|Physical Device Write Latency (ms) | Average time taken to complete a write from the physical disk.<br>Key: devices\|deviceWriteLatency_average |

## Datastore Metrics for Datastores

Datastore metrics provide information about datastore use.

| Metric Name | Description |
|---|---|
| Datastore\|Total Latency (ms) | This metric shows the adjusted read and write latency at the datastore level. Adjusted means that the latency is taking into account the number of IOs. If your IO is read-dominated, the combined value is influenced by the reads.<br>This is the average of all the VMs running in the datastore. Because it is an average, some VMs logically experience higher latency that the value shown by this metric. To see the worst latency experienced by any VM, use the Maximum VM Disk Latency metric.<br>Use this metric to see the performance of the datastore. It is one of two key performance indicators for a datastore, the other being the Max Read Latency. The combination of Maximum and Average gives better insight into how well the datastore is coping with the demand.<br>The number should be lower than the performance you expect.<br>Key: datastore\|totalLatency_average |
| Datastore\|Total Throughput (KBps) | Average use in kilobytes per second.<br>Key: datastore\|usage_average |
| Datastore\|Read Latency (ms | Average amount of time for a read operation from the datastore. Total latency = kernel latency + device latency.<br>Key: datastore\|totalReadLatency_average |
| Datastore\|Write Latency (ms) | Average amount of time for a write operation to the datastore. Total latency = kernel latency + device latency.<br>Key: datastore\|totalWriteLatency_average |
| Datastore\|Demand | Demand.<br>Key: datastore\|demand |
| Datastore\|Outstanding IO requests | OIO for datastore.<br>Key: datastore\|demand_oio |
| Datastore\|Read IOPS | This metric displays the average number of read commands issued per second during the collection interval.<br>Use this metric when the total IOPS is higher than expected. See if the metric is read or write dominated. This helps determine the cause of the high IOPS. Certain workloads such as backups, anti-virus scans, and Windows updates carry a Read/Write pattern. For example, an anti-virus scan is heavy on read since it is mostly reading the file system.<br>Key: datastore\|numberReadAveraged_average |
| Datastore\|Write IOPS | This metric displays the average number of write commands issued per second during the collection interval.<br>Use this metric when the total IOPS is higher than expected. Drill down to see if the metric is read or write dominated. This helps determine the cause of the high IOPS. Certain workloads such as backups, anti-virus scans, and Windows updates carry a Read/Write pattern. For example, an anti-virus scan is heavy on read since it is mostly reading the file system.<br>Key: datastore\|numberWriteAveraged_average |
| Datastore\|Read Throughput (KBps) | This metric displays the amount of data read in the performance interval.<br>Key: datastore\|read_average |

| Metric Name | Description |
|---|---|
| Datastore\|Write Throughput (KBps) | This metric displays the amount of data written to disk in the performance interval.<br>Key: datastore\|write_average |

### About Datastore Metrics for Virtual SAN

The metric named `datastore|oio|workload` is not supported on Virtual SAN datastores. This metric depends on `datastore|demand_oio`, which is supported for Virtual SAN datastores.

The metric named `datastore|demand_oio` also depends on several other metrics for Virtual SAN datastores, one of which is not supported.

- The metrics named `devices|numberReadAveraged_average` and `devices|numberWriteAveraged_average` are supported.
- The metric named `devices|totalLatency_average` is not supported.

As a result, VCF Operations does not collect the metric named `datastore|oio|workload` for Virtual SAN datastores.

### Disk Space Metrics for Datastores

Disk space metrics provide information about disk space use.

| Metric Name | Description |
|---|---|
| Diskspace\|Number of Virtual Disk | Number of virtual disks.<br>Key: diskspace\|numvmdisk |
| Diskspace\|Provisioned Space (GB) | Provisioned space in gigabytes.<br>Key: diskspace\|provisioned |
| Diskspace\|Shared Used (GB) | Shared used space in gigabytes.<br>Key: diskspace\|shared |
| Diskspace\|Snapshot Space (GB) | This metric shows the amount of space taken by snapshots on a given database.<br>Use this metric to know how much storage space is being used by virtual machine snapshots on the datastore.<br>Check that the snapshot is using 0 GB or minimal space. Anything over 1 GB should trigger a warning. The actual value depends on how IO intensive the virtual machines in the datastore are. Run a DT on them to detect anomaly. Clear the snapshot within 24 hours, preferably when you have finished backing up, or patching.<br>Key: diskspace\|snapshot |
| Diskspace\|Virtual Disk Used (GB) | Virtual disk used space in gigabytes.<br>Key: diskspace\|diskused |
| Diskspace\|Virtual machine used (GB) | Virtual machine used space in gigabytes.<br>Key: diskspace\|used |
| Diskspace\|Total disk space used | Total disk space used on all datastores visible to this object.<br>Key: diskspace\|total_usage |
| Diskspace\|Total disk space | Total disk space on all datastores visible to this object.<br>Key: diskspace\|total_capacity |
| Diskspace\|Total used (GB) | Total used space in gigabytes.<br>Key: diskspace\|disktotal |
| Diskspace\|Swap File Space (GB) | Swap file space in gigabytes.<br>Key: diskspace\|swap |

| Metric Name | Description |
|---|---|
| Diskspace\|Other VM Space (GB) | Other virtual machine space in gigabytes.<br>Key: diskspace\|otherused |
| Diskspace\|Freespace (GB) | Unused space available on datastore.<br>Key: diskspace\|freespace |
| Diskspace\|Capacity (GB) | Total capacity of datastore in gigabytes.<br>Key: diskspace\|capacity |
| Diskspace\|Overhead | Amount of disk space that is overhead.<br>Key: diskspace\|overhead |

## Summary Metrics for Datastores

Summary metrics provide information about overall performance.

| Metric Name | Description |
|---|---|
| Summary\|Number of Hosts | This metric shows the number of hosts that the datastore is connected to.<br>Use this metric to know how many clusters the datastore is attached to.<br>The number should not be too high, as a datastore should not be mounted by every host. The datastore and cluster should be paired to keep operations simple.<br>Key: summary\|total_number_hosts |
| Summary\|Total Number of VMs | This metric shows the number of virtual machines which save their VMDK files on the datastore. If a VM has four VMDKs stored in four datastores, the VM is counted on each datastore.<br>Use this metric to know how many VMs have at least one VMDK on a specific datastore.<br>The number of VMs should be within your Concentration Risk policy.<br>You should also expect the datastore to be well used. If only a few VMs are using the datastore, this is not considered a good use.<br>Key: summary\|total_number_vms |
| Summary\|Maximum Number of VMs | Maximum number of virtual machines.<br>Key: summary\|max_number_vms |
| Summary\|Workload Indicator | Workload indicator.<br>Key: summary\|workload_indicator |
| Summary\|Number of Clusters | This metric shows the number of clusters that the datastore is connected to.<br>Key: summary\|total_number_clusters |
| Summary\|Number of VM Templates | Number of VM Templates.<br>Key: Summary\|Number of VM Templates |

## Template Metrics for Datastores

| Metric Name | Description |
|---|---|
| Template\|Virtual Machine used | Space used by virtual machine files.<br>Key: template\|used |
| Template\|Access Time | Last access time.<br>Key: template\|accessTime |

### Cost Metrics for Datastores

Cost metrics provides information about the cost.

| Metric Name | Description |
|---|---|
| Monthly Disk Space Base Rate | Disk space base rate for datastore displays the cost of 1 GB storage.<br>Key: cost\|storageRate |
| Monthly Total Cost | Monthly total cost, computed by multiplying datastore capacity with monthly storage rate.<br>Key: cost\|totalCost |
| Cost\|Allocation\|Disk Space Base Rate (Currency) | Monthly storage rate for datastore displays the cost of 1 GB storage when the overcommit ratio is set in policy.<br>cost\|storageRate |
| Cost\|Allocation\|Monthly Datastore Allocated Cost(Currency/Month) | Monthly allocated cost as compared to the total cost of the datastore |
| Cost\|Allocation\|Monthly Datastore Unallocated Cost(Currency/Month) | Monthly unallocated cost as compared to the total cost of the datastore. |

### Reclaimable Metrics

Reclaimable metrics provide information about reclaimable resources.

| Metric Name | Description |
|---|---|
| Reclaimable\|Orphaned Disks\|Disk Space (GB) | Summary of storage used by all orphaned VMDKs on the datastore.<br>Key: reclaimable\|orphaned_disk\|diskspace |
| Reclaimable\|Orphaned Disks\|Potential Savings (Currency) | Potential saving after reclamation of storage by removing orphaned VMDks from the datastore.<br>Key: reclaimable\|orphaned_disk\|cost |

### Disabled Instanced Metrics

The instance metrics created for the following metrics are disabled in this version of VCF Operations. This means that these metrics collect data by default but all the instanced metrics created for these metrics, do not collect data by default.

| Metric Name |
|---|
| Devices\|Kernel Latency (ms) |
| Devices\|Number of Running Hosts |
| Devices\|Number of Running VMs |
| Devices\|Physical Device Latency (ms) |
| Devices\|Queue Latency (ms) |
| Devices\|Queue Read Latency (ms) |
| Devices\|Read IOPS |
| Devices\|Read Latency (ms) |
| Devices\|Read Requests |
| Devices\|Read Throughput (KBps) |
| Devices\|Total IOPS |
| Devices\|Total Latency (ms) |

| Metric Name |
|---|
| Devices\|Total Throughput (KBps) |
| Devices\|Write IOPS |
| Devices\|Write Latency (ms) |
| Devices\|Write Requests |
| Devices\|Write Throughput (KBps) |

### Disabled Metrics

The following metrics are disabled in this version of VCF Operations. This means that they do not collect data by default.

You can enable these metrics in the Policy workspace. For more information, in VMware Docs search for Collect Metrics and Properties Details.

You can enable these metrics in the Policy workspace. For more information, see Metrics and Properties Details.

| Metric Name | Key |
|---|---|
| Capacity\|Data Store Capacity Contention (%) | capacity\|contention |
| Datastore I/O\|Demand Indicator | datastore\|demand_indicator |
| Datastore I/O\|Max Observed Number of Outstanding IO Operations | datastore\|maxObserved_OIO |
| Datastore I/O\|Max Observed Read Latency (msec) | datastore\|maxObserved_Read |
| Datastore I/O\|Max Observed Read Latency (msec) | datastore\|maxObserved_ReadLatency |
| Datastore I/O\|Max Observed | datastore\|maxObserved_NumberRead |
| Datastore I/O\|Max Observed Write Latency (msec) | datastore\|maxObserved_Write |
| Datastore I/O\|Max Observed Write Latency (msec) | datastore\|maxObserved_WriteLatency |
| Datastore I/O\|Max Observed Writes per second | datastore\|maxObserved_NumberWrite |
| Datastore\|Demand Indicator | Demand Indicator.<br>Key: datastore\|demand_indicator |
| Diskspace\|Not Shared (GB) | Unshared space in gigabytes.<br>Key: diskspace\|notshared |

## Cluster Compute Metrics for Allocation Model

VCF Operations collects configuration, disk space, CPU use, disk, memory, network, power, and summary metrics for cluster compute resources.

### Cost Metrics for Cluster Compute Resources

Cost metrics provide information about the cost.

| Metric Name | Description |
|---|---|
| Cluster CPU Base Rate | Base rate for Cluster CPU calculated by dividing the monthly total cluster CPU cost by cluster CPU over-commit ratio.<br>Key:Cost\|Allocation\|ClusterCPUBaseRate |
| Cluster Memory Base Rate | Cluster memory base rate calculated by dividing the monthly total cluster memory cost b cost by cluster memory over-commit ratio.<br>Key: Cost\|Allocation\|ClusterMemoryBaseRate |

| Metric Name | Description |
|---|---|
| Monthly Cluster Allocated Cost | Sum of of monthly cluster CPU, Memory, and Storage costs<br>Key: Cost\|Allocation\|MonthlyClusterAllocatedCost |
| Monthly Cluster Unallocated Cost | Monthly cluster unallocated cost calculated by subtracting the monthly cluster allocated cost from the monthly cluster total cost.<br>Key: Cost\|Allocation\| MonthlyClusterUnallocatedCost |
| Monthly Storage Rate | Datastore base rate is calculated by dividing Storage base rate based on utilization by over commit ratio.<br>Key:Cost\|Allocation\|Monthly Storage Rate |

## Virtual Machine Metrics for Allocation Model

VCF Operations collects configuration, disk space, CPU use, disk, memory, network, power, and summary metrics for virtual machine resources.

### Cost Metrics for Virtual Machines

Cost metrics provide information about the cost.

| Metric Name | Description |
|---|---|
| MTD VM CPU Cost | Month to date virtual machine CPU cost.<br>Key: Cost\|Allocation\|MTD VM CPU Cost |
| MTD VM Memory Cost | Month to date virtual machine memory cost.<br>Key: Cost\|Allocation\|MTD VM Memory Cost |
| MTD VM Storage Cost | Month to date storage cost of the virtual machine.<br>Key: Cost\|Allocation\|MTD VM Storage Cost |
| MTD VM Total Cost | Addition of CPU ,Memory ,Storage, and Direct cost.<br>Key: Cost\|Allocation\|MTD VM Total Cost |

## Metrics for Namespace

VCF Operations collects metrics for Namspace through the vCenter adapter and uses formulas to derive statistics from those metrics. You can use metrics to troubleshoot problems in your environment.

**Table 1488: Metrics for Namespace**

| Metric Key | Localized Name | Description |
|---|---|---|
| cpu\|usagemhz_average | CPU\|Usage | Average CPU usage in MHZ. |
| cpu\|demandmhz | CPU\|Demand | Demand(MHz). |
| cpu\|capacity_contentionPct | CPU\|Contention | Percent of time descendant virtual machines are unable to run because they are contending for access to the physical CPU(s). |
| cpu\|effective_limit | CPU\|Effective limit | CPU Effective limit. |
| cpu\|reservation_used | CPU\|Reservation Used | CPU Reservation Used. |
| cpu\|estimated_entitlement | CPU\|Estimated entitlement | CPU Estimated entitlement. |
| cpu\|dynamic_entitlement | CPU\|Dynamic entitlement | CPU Dynamic Entitlement. |
| cpu\|capacity_contention | CPU\|Overall CPU Contention | Overall CPU Contention (ms). |
| cpu\|capacity_demandEntitlementPct | CPU\|Capacity Demand Entitlement | CPU Capacity Demand Entitlement Percentage. |

| Metric Key | Localized Name | Description |
|---|---|---|
| mem\|usage_average | Memory\|Usage | Memory currently in use as a percentage of total available memory. |
| mem\|guest_provisioned | Memory\|Total Capacity | Total Capacity. |
| mem\|active_average | Memory\|Guest Active | Amount of memory that is actively used. |
| mem\|granted_average | Memory\|Granted | Amount of memory available for use. |
| mem\|shared_average | Memory\|Shared | Amount of shared memory. |
| mem\|overhead_average | Memory\|VM Overhead | Memory overhead reported by host. |
| mem\|consumed_average | Memory\|Consumed | Amount of host memory consumed by the virtual machine for guest memory. |
| mem\|host_contentionPct | Memory\|Contention | Machine Contention Percentage. |
| mem\|guest_usage | Memory\|Guest Usage | Guest Memory Entitlement. |
| mem\|guest_demand | Memory\|Guest Demand | Guest Memory Entitlement. |
| mem\|reservation_used | Memory\|Reservation Used | Memory Reservation Used. |
| mem\|effective_limit | Memory\|Effective limit | Memory Effective limit. |
| mem\|swapinRate_average | Memory\|Swap In Rate | Rate at which memory is swapped from disk into active memory during the collection interval. This can impact performance. |
| mem\|swapoutRate_average | Memory\|Swap Out Rate | Rate at which memory is being swapped from active memory to disk during the current interval. |
| mem\|vmmemctl_average | Memory\|Balloon | Amount of memory currently used by the virtual machine memory control. |
| mem\|zero_average | Memory\|Zero | Amount of memory that is all 0. |
| mem\|swapped_average | Memory\|Swapped | Amount of unreserved memory. |
| mem\|zipped_latest | Memory\|Zipped | N/A |
| mem\|compressionRate_average | Memory\|Compression Rate | N/A |
| mem\|decompressionRate_average | Memory\|Decompression Rate | N/A |
| mem\|swapin_average | Memory\|Swap In | Amount of memory swapped in. |
| mem\|swapout_average | Memory\|Swap Out | Amount of memory swapped out. |
| mem\|swapused_average | Memory\|Swap Used | Amount of memory used for swap space. |
| mem\|host_contention | Memory\|Contention | Machine Contention. |
| mem\|dynamic_entitlement | Memory\|Dynamic Entitlement | Memory Dynamic Entitlement. |
| diskspace\|total_usage | Disk Space\|Utilization | Storage space utilized on connected vSphere Datastores. |
| summary\|configStatus | Summary\|Config Status | Workload Management Configuration Status. |
| summary\|total_number_pods | Summary\|Number of Pods | Number of Pods. |
| summary\|numberKubernetesClusters | Summary\|Number of Kubernetes clusters | Number of Kubernetes clusters. |
| summary\|number_running_vms | Summary\|Number of Running VMs | Number of Running VMs. |
| summary\|total_number_vms | Summary\|Total Number of VMs | Total Number of VMs. |
| summary\|iowait | Summary\|IO Wait | IO Wait. |

## Metrics for Tanzu Kubernetes cluster

VCF Operations collects metrics for Tanzu Kubernetes cluster through the vCenter adapter and uses formulas to derive statistics from those metrics. You can use metrics to troubleshoot problems in your environment.

**Table 1489: Metrics for Tanzu Kubernetes clusters**

| Metric Key | Localized Name | Description |
|---|---|---|
| cpu\|usagemhz_average | CPU\|Usage | Average CPU usage in MHZ |
| cpu\|demandmhz | CPU\|Demand | Demand(MHz) |
| cpu\|capacity_contentionPct | CPU\|Contention | Percent of time descendant virtual machines are unable to run because they are contending for access to the physical CPU(s). |
| cpu\|effective_limit | CPU\|Effective limit | CPU Effective limit |
| cpu\|reservation_used | CPU\|Reservation Used | CPU Reservation Used |
| cpu\|estimated_entitlement | CPU\|Estimated entitlement | CPU Estimated entitlement |
| cpu\|dynamic_entitlement | CPU\|Dynamic entitlement | CPU Dynamic Entitlement |
| cpu\|capacity_contention | CPU\|Overall CPU Contention | Overall CPU Contention (ms) |
| cpu\|capacity_demandEntitlementPct | CPU\|Capacity Demand Entitlement | CPU Capacity Demand Entitlement Percentage |
| mem\|usage_average | Memory\|Usage | Memory currently in use as a percentage of total available memory |
| mem\|guest_provisioned | Memory\|Total Capacity | Total Capacity |
| mem\|active_average | Memory\|Guest Active | Amount of memory that is actively used |
| mem\|granted_average | Memory\|Granted | Amount of memory available for use |
| mem\|shared_average | Memory\|Shared | Amount of shared memory |
| mem\|overhead_average | Memory\|VM Overhead | Memory overhead reported by host |
| mem\|consumed_average | Memory\|Consumed | Amount of host memory consumed by the virtual machine for guest memory |
| mem\|host_contentionPct | Memory\|Contention | Machine Contention Percentage |
| mem\|guest_usage | Memory\|Guest Usage | Guest Memory Entitlement |
| mem\|guest_demand | Memory\|Guest Demand | Guest Memory Entitlement |
| mem\|reservation_used | Memory\|Reservation Used | Memory Reservation Used |
| mem\|effective_limit | Memory\|Effective limit | Memory Effective limit |
| mem\|swapinRate_average | Memory\|Swap In Rate | Rate at which memory is swapped from disk into active memory during the collection interval. This can impact performance. |
| mem\|swapoutRate_average | Memory\|Swap Out Rate | Rate at which memory is being swapped from active memory to disk during the current interval |
| mem\|vmmemctl_average | Memory\|Balloon | Amount of memory currently used by the virtual machine memory control |
| mem\|zero_average | Memory\|Zero | Amount of memory that is all 0 |
| mem\|swapped_average | Memory\|Swapped | Amount of unreserved memory |
| mem\|zipped_latest | Memory\|Zipped | N/A |
| mem\|compressionRate_average | Memory\|Compression Rate | N/A |
| mem\|decompressionRate_average | Memory\|Decompression Rate | N/A |

| Metric Key | Localized Name | Description |
|---|---|---|
| mem\|swapin_average | Memory\|Swap In | Amount of memory swapped in |
| mem\|swapout_average | Memory\|Swap Out | Amount of memory swapped out |
| mem\|swapused_average | Memory\|Swap Used | Amount of memory used for swap space |
| mem\|host_contention | Memory\|Contention | Machine Contention |
| mem\|dynamic_entitlement | Memory\|Dynamic Entitlement | Memory Dynamic Entitlement |
| summary\|number_running_vms | Summary\|Number of Running VMs | Number of Running VMs |
| summary\|total_number_vms | Summary\|Total Number of VMs | Total Number of VMs |
| summary\|iowait | Summary\|IO Wait | IO Wait |

## Metrics for vSphere Pods

VCF Operations collects metrics for vSphere Pods through the vCenter adapter and uses formulas to derive statistics from those metrics. You can use metrics to troubleshoot problems in your environment.

**Table 1490: Metrics for vSphere Pods**

| Metric Key | Metric Name | Description |
|---|---|---|
| config\|hardware\|num_Cpu | Configuration\|Hardware\|Number of CPUs | Number of CPUs. It counts both the vSocket and vCore. A VM with 2 vSockets x 4 vCores each has 8 vCPU. |
| config\|hardware\|disk_Space | Configuration\|Hardware\|Disk Space | Disk space metrics |
| config\|hardware\|thin_Enabled | Configuration\|Hardware\|Thin Provisioned Disk | Thin Provisioned Disk |
| config\|cpuAllocation\|slotSize | Configuration\|CPU Resource Allocation\|HA Slot Size | vSphere HA Slot Size for CPU |
| config\|memoryAllocation\|slotSize | Configuration\|Memory Resource Allocation\|HA Slot Size | vSphere HA Slot Size for Memory |
| cpu\|usage_average | CPU\|Usage | CPU Usage divided by VM CPU Configuration in MHz |
| cpu\|usagemhz_average | CPU\|Usage | Amount of actively used virtual CPU. This is the host's view of the CPU usage, not the guest operating system view. |
| cpu\|usagemhz_average_mtd | CPU\|Usage average MTD | Month to date average CPU usage in MHZ |
| cpu\|readyPct | CPU\|Ready | Percentage of CPU the VM is ready to run, but unable due to ESX has no ready physical core to run it. High Ready value impacts VM performance |
| cpu\|capacity_contentionPct | CPU\|Contention | Percentage of time VM is not getting the CPU resource it demanded. Impacted by Ready, Co-Stop, Hyper Threading and Power Management |
| cpu\|corecount_provisioned | CPU\|Provisioned vCPU(s) | Number of CPUs. It counts both the vSocket and vCore. A VM with 2 vSockets x 4 vCores each has 8 vCPU. |
| cpu\|vm_capacity_provisioned | CPU\|Total Capacity | Configured Capacity in MHz, based on nominal (static) frequency of the CPU |

| Metric Key | Metric Name | Description |
|---|---|---|
| cpu\|demandmhz | CPU\|Demand | The amount of CPU resources virtual machine would use if there were no CPU contention or CPU limit. |
| cpu\|demandPct | CPU\|Demand (%) | The percentage of CPU resources virtual machine would use if there were no CPU contention or CPU limit. |
| cpu\|reservation_used | CPU\|Reservation Used | CPU Reserved for the VM. It's guaranteed to be available when the VM demands it. |
| cpu\|effective_limit | CPU\|Effective limit | Limit placed on the VM by vSphere. Avoid using limit as it impacts VM performance |
| cpu\|iowaitPct | CPU\|IO Wait | Percentage of time VM CPU is waiting for IO. Formula is Wait - Idle - Swap Wait. High value indicates slow storage subsystem |
| cpu\|swapwaitPct | CPU\|Swap wait | Percentage of time CPU is waiting on data swap-in. Mapped to vCenter CPU Swap wait |
| cpu\|costopPct | CPU\|Co-stop (%) | Percentage of time the VM is ready to run, but is unable to due to co-scheduling constraints. VM with less vCPU have lower co-stop value. |
| cpu\|system_summation | CPU\|System | CPU time spent on system processes |
| cpu\|wait_summation | CPU\|Wait | Total CPU time spent in wait state |
| cpu\|ready_summation | CPU\|Ready | CPU time spent on ready state |
| cpu\|used_summation | CPU\|Used | CPU time that is used |
| cpu\|iowait | CPU\|IO Wait | IO Wait |
| cpu\|wait | CPU\|Total Wait | CPU time spent on idle state |
| cpu\|capacity_demandEntitlementPct | CPU\|Capacity Demand Entitlement | CPU Capacity Demand Entitlement Percentage |
| cpu\|host_demand_for_aggregation | CPU\|Host Demand For Aggregation | Host demand for aggregation |
| cpu\|dynamic_entitlement | CPU\|Dynamic entitlement | CPU Dynamic entitlement |
| cpu\|capacity_contention | CPU\|Overall CPU Contention | Overall CPU Contention (ms) |
| cpu\|estimated_entitlement | CPU\|Estimated entitlement | CPU Estimated entitlement |
| cpu\|idlePct | CPU\|Idle | % CPU time that is idle |
| cpu\|waitPct | CPU\|Wait | % Total CPU time spent in wait state |
| cpu\|systemSummationPct | CPU\|System | % CPU time spent on system processes |
| cpu\|demandOverLimit | CPU\|Demand Over Limit | Amount of CPU Demand that is over the configured CPU Limit |
| cpu\|demandOverCapacity | CPU\|Demand Over Capacity | Amount of CPU Demand that is over the configured CPU Capacity |
| cpu\|perCpuCoStopPct | CPU\|Normalized Co-stop | Percentage of co-stop time, normalized across all vCPUs |
| cpu\|swapwait_summation | CPU\|Swap Wait | Amount of time waiting on swap. |
| cpu\|costop_summation | CPU\|Co-stop | Time the VM is ready to run, but is unable to due to co-scheduling constraints. |
| cpu\|idle_summation | CPU\|Idle | CPU time that is idle. |
| cpu\|latency_average | CPU\|Latency | Percentage of time the VM is unable to run because it is contending for access to the physical CPUs. |

| Metric Key | Metric Name | Description |
|---|---|---|
| cpu\|maxlimited_summation | CPU\|Max Limited | Time the VM is ready to run, but is not run due to maxing out its CPU limit setting. |
| cpu\|overlap_summation | CPU\|Overlap | Time the VM was interrupted to perform system services on behalf of that VM or other VMs. |
| cpu\|run_summation | CPU\|Run | Time the VM is scheduled to run. |
| cpu\|entitlement_latest | CPU\|Entitlement Latest | Entitlement Latest. |
| cpu\|demandEntitlementRatio_latest | CPU\|Demand-to-entitlement Ratio | CPU resource entitlement to CPU demand ratio (in percents) |
| cpu\|readiness_average | CPU\|Readiness | Percentage of time that the virtual machine was ready, but could not get scheduled to run on the physical CPU. |
| rescpu\|actav1_latest | CPU Utilization for Resources\|CPU Active (1 min. average) | The average active time for the CPU over the past minute |
| rescpu\|actav5_latestswapinRate_average | CPU Utilization for Resources\|CPU Active (5 min. average) | The average active time for the CPU over the past five minutes. |
| rescpu\|actav5_latest | CPU Utilization for Resources\|CPU Active (5 min. average) | The average active time for the CPU over the past five minutes |
| rescpu\|actav15_latest | CPU Utilization for Resources\|CPU Active (15 min. average) | The average active time for the CPU over the past fifteen minutes |
| rescpu\|actpk1_latest | CPU Utilization for Resources\|CPU Active (1 min. peak) | The peak active time for the CPU over the past minute |
| rescpu\|actpk5_latest | CPU Utilization for Resources\|CPU Active (5 min. peak) | The peak active time for the CPU over the past five minutes |
| rescpu\|actpk15_latest | CPU Utilization for Resources\|CPU Active (15 min. peak) | The peak active time for the CPU over the past fifteen minutes |
| rescpu\|runav1_latest | CPU Utilization for Resources\|CPU Running (1 min. average) | The average runtime for the CPU over the past minute |
| rescpu\|runav5_latest | CPU Utilization for Resources\|CPU Running (5 min. average) | The average runtime for the CPU over the past five minutes |
| rescpu\|runav15_latest | CPU Utilization for Resources\|CPU Running (15 min. average) | The average runtime for the CPU over the past fifteen minutes |
| rescpu\|runpk1_latest | CPU Utilization for Resources\|CPU Running (1 min. peak) | The peak active time for the CPU over the past minute |
| rescpu\|runpk5_latest | CPU Utilization for Resources\|CPU Running (5 min. peak) | The peak active time for the CPU over the past five minutes |
| rescpu\|runpk15_latest | CPU Utilization for Resources\|CPU Running (15 min. peak) | The peak active time for the CPU over the past fifteen minutes |
| rescpu\|maxLimited1_latest | CPU Utilization for Resources\|CPU Throttled (1 min. average) | The scheduling limit over the past minute |
| rescpu\|maxLimited5_latest | CPU Utilization for Resources\|CPU Throttled (5 min. average) | The scheduling limit over the past five minutes |
| rescpu\|maxLimited15_latest | CPU Utilization for Resources\|CPU Throttled (15 min. average) | The scheduling limit over the past fifteen minutes |
| rescpu\|sampleCount_latest | CPU Utilization for Resources\|Group CPU Sample Count | The sample CPU count |
| rescpu\|samplePeriod_latest | CPU Utilization for Resources\|Group CPU Sample Period | The sample period |

| Metric Key | Metric Name | Description |
|---|---|---|
| mem\|usage_average | Memory\|Usage | Memory currently in use as a percentage of total available memory |
| mem\|balloonPct | Memory\|Balloon | Percentage of guest physical memory that is currently claimed from the virtual machine through ballooning. This is the percentage of guest physical memory that has been allocated and pinned by the balloon driver. Balloon does not necessarily mean the VM performance is affected. |
| mem\|swapped_average | Memory\|Swapped | Amount of unreserved memory |
| mem\|consumed_average | Memory\|Consumed | Amount of ESX Host memory mapped/consumed by the virtual machine for guest memory |
| mem\|consumed_average_mtd | Memory\|Consumed average MTD | average MTD Amount of host memory consumed by the virtual machine for guest memory |
| mem\|consumedPct | Memory\|Consumed (%) | Amount of host memory consumed by the virtual machine for guest memory. Consumed memory does not include overhead memory. It includes shared memory and memory that might be reserved, but not actually used. |
| mem\|overhead_average | Memory\|Overhead | Amount of overhead memory used by ESX to run the Virtual Machine. |
| mem\|host_contentionPct | Memory\|Contention | Percentage of time the VM has contended for memory. |
| mem\|guest_provisioned | Memory\|Total Capacity | Memory resources allocated to the Virtual Machine |
| mem\|guest_usage | Memory\|Guest Usage | Guest Memory Entitlement |
| mem\|guest_demand | Memory\|Guest Demand | Guest Memory Entitlement |
| mem\|host_demand | Memory\|Host Demand | Memory Demand in KB |
| mem\|reservation_used | Memory\|Reservation Used | Memory Reservation Used |
| mem\|effective_limit | Memory\|Effective limit | Memory Effective limit |
| mem\|vmMemoryDemand | Memory\|Utilization | Amount of memory utilized by the Virtual Machine. Reflects the guest OS memory required (for certain vSphere and VMTools versions) or Virtual Machine consumption |
| mem\|nonzero_active | Memory\|Non Zero Active | Non Zero Active Memory |
| mem\|swapinRate_average | Memory\|Swap In Rate | Rate at which memory is swapped from disk into active memory during the collection interval. This can impact performance. |
| mem\|swapoutRate_average | Memory\|Swap Out Rate | Rate at which memory is being swapped from active memory to disk during the current interval. |
| mem\|compressed_average | Memory\|Compressed | Percentage of total memory that has been compressed by vSphere. If and only if the page is accessed by the Guest OS, will performance be affected. |
| mem\|overheadMax_average | Memory\|Overhead Max | N/A |
| mem\|vmmemctl_average | Memory\|Balloon | Amount of memory currently used by the virtual machine memory control |
| mem\|active_average | Memory\|Guest Active | Amount of memory that is actively used |
| mem\|granted_average | Memory\|Granted | Amount of memory available for use |

| Metric Key | Metric Name | Description |
|---|---|---|
| mem\|shared_average | Memory\|Shared | Amount of shared memory |
| mem\|zero_average | Memory\|Zero | Amount of memory that is all 0 |
| mem\|swaptarget_average | Memory\|Swap Target | Amount of memory that can be swapped |
| mem\|swapin_average | Memory\|Swap In | Amount of memory swapped in |
| mem\|swapout_average | Memory\|Swap Out | Amount of memory swapped out |
| mem\|vmmemctltarget_average | Memory\|Balloon Target | Amount of memory that can be used by the virtual machine memory control |
| mem\|host_dynamic_entitlement | Memory\|Host Dynamic Entitlement | Mem Machine Dynamic Entitlement |
| mem\|host_active | Memory\|Host Active | Machine Active |
| mem\|host_usage | Memory\|Host Usage | Machine Usage |
| mem\|host_contention | Memory\|Contention | Machine Contention |
| mem\|guest_activePct | Memory\|Guest Active Memory | Guest active memory as percentage of configured |
| mem\|guest_dynamic_entitlement | Memory\|Guest Dynamic Entitlement | Guest Memory Dynamic Entitlement |
| mem\|host_demand_reservation | Memory\|Host Demand with Reservation | Memory Demand with Reservation considered in KB |
| mem\|host_nonpageable_estimate | Memory\|Guest Non Pageable Memory | Guest Non Pageable Memory Estimates |
| mem\|guest_nonpageable_estimate | Memory\|Host Non Pageable Memory | Guest Non Pageable Memory Estimates |
| mem\|estimated_entitlement | Memory\|Estimated entitlement | Memory Estimated entitlement |
| mem\|host_demand_for_aggregation | Memory\|Host Demand For Aggregation | Host demand for aggregation |
| mem\|demandOverLimit | Memory\|Demand Over Limit | Amount of Memory Demand that is over the configured Memory Limit |
| mem\|demandOverCapacity | Memory\|Demand Over Capacity | Amount of Memory Demand that is over the configured Memory Capacity |
| mem\|activewrite_average | Memory\|Active Write | N/A |
| mem\|compressionRate_average | Memory\|Compression Rate | N/A |
| mem\|decompressionRate_average | Memory\|Decompression Rate | N/A |
| mem\|zipSaved_latest | Memory\|Zip Saved | N/A |
| mem\|zipped_latest | Memory\|Zipped | N/A |
| mem\|entitlement_average | Memory\|Entitlement | Amount of host physical memory the VM is entitled to, as determined by the ESX schedule. |
| mem\|latency_average | Memory\|Latency | Percentage of time the VM is waiting to access swapped or compressed memory. |
| mem\|capacity.contention_average | Memory\|Capacity Contention | Capacity Contention. |
| mem\|llSwapInRate_average | Memory\|Swap In Rate from Host Cache | Rate at which memory is being swapped from host cache into active memory. |
| mem\|llSwapOutRate_average | Memory\|Swap Out Rate to Host Cache | Rate at which memory is being swapped to host cache from active memory. |
| mem\|llSwapUsed_average | Memory\|Swap Space Used in Host Cache | Space used for caching swapped pages in the host cache. |
| mem\|overheadTouched_average | Memory\|Overhead Touched | Actively touched overhead memory (KB) reserved for use as the virtualization overhead for the VM. |

| Metric Key | Metric Name | Description |
|---|---|---|
| net\|usage_average | Network\|Usage Rate | The sum of the data transmitted and received for all the NIC instances of the host or virtual machine |
| net\|transmitted_average | Network\|Data Transmit Rate | Average amount of data transmitted per second |
| net\|received_average | Network\|Data Receive Rate | Average amount of data received per second |
| net\|droppedTx_summation | Network\|Transmitted Packets Dropped | Number of outgoing packets dropped in the performance interval. Investigate if the number is not 0 |
| net\|droppedPct | Network\|Packets Dropped (%) | Percentage of packets dropped |
| net\|dropped | Network\|Packets Dropped | Number of packets dropped in the performance interval |
| net\|broadcastTx_summation | Network\|Broadcast Packets Transmitted | Total number of broadcast packets transmitted. Investigate further if this number is high |
| net\|multicastTx_summation | Network\|Multicast Packets Transmitted | Number of multicast packets transmitted. Investigate further if this number is high |
| net\|idle | NetworkIidle | N/A |
| net\|usage_capacity | Network\|I/O Usage Capacity | I/O Usage Capacity |
| net\|maxObserved_KBps | Network\|Max Observed Throughput | Max observed rate of network throughput |
| net\|maxObserved_Tx_KBps | Network\|Max Observed Transmitted Throughput | Max observed transmitted rate of network throughput |
| net\|maxObserved_Rx_KBps | Network\|Max Observed Received Throughput | Max observed received rate of network throughput |
| net\|packetsRx_summation | Network\|Packets Received | Number of packets received in the performance interval |
| net\|packetsTx_summation | Network\|Packets Transmitted | Number of packets transmitted in the performance interval |
| net\|demand | Network\|Demand | N/A |
| net\|packetsRxPerSec | Network\|Packets Received per second | Number of packets received in the performance interval |
| net\|packetsTxPerSec | Network\|Packets Transmitted per second | Number of packets transmitted in the performance interval |
| net\|packetsPerSec | Network\|Packets per second | Number of packets transmitted and received per second |
| net\|droppedRx_summation | Network\|Received Packets Dropped | Number of received packets dropped in the performance interval |
| net\|broadcastRx_summation | Network\|Broadcast Packets Received | Number of broadcast packets received during the sampling interval |
| net\|multicastRx_summation | Network\|Multicast Packets Received | Number of multicast packets received |
| net\|bytesRx_average | Network\|bytesRx | Average amount of data received per second |
| net\|bytesTx_average | Network\|bytesTx | Average amount of data transmitted per second |
| net\|host_transmitted_average | Network\|VM to Host Data Transmit Rate | Average amount of data transmitted per second between VM and host |
| net\|host_received_average | Network\|VM to Host Data Receive Rate | Average amount of data received per second between VM and host |
| net\|host_usage_average | Network\|VM to Host Usage Rate | The sum of the data transmitted and received for all the NIC instances between VM and host |

| Metric Key | Metric Name | Description |
|---|---|---|
| net\|host_maxObserved_Tx_KBps | Network\|VM to Host Max Observed Transmitted Throughput | Max observed transmitted rate of network throughput between VM and host |
| net\|host_maxObserved_Rx_KBps | Network\|VM to Host Max Observed Received Throughput | Max observed received rate of network throughput between VM and host |
| net\|host_maxObserved_KBps | Network\|VM to Host Max Observed Throughput | Max observed rate of network throughput between VM and host |
| net\|transmit_demand_average | Network\|Data Transmit Demand Rate | Data Transmit Demand Rate |
| net\|receive_demand_average | Network\|Data Receive Demand Rate | Data Receive Demand Rate |
| disk\|usage_average | Physical Disk\|Total Throughput | Amount of data read from/written to storage in a second. This is averaged over the reporting period |
| disk\|read_average | Physical Disk\|Read Throughput | Amount of data read from storage in a second. This is averaged over the reporting period |
| disk\|write_average | Physical Disk\|Write Throughput | Amount of data written to storage in a second. This is averaged over the reporting period |
| disk\|usage_capacity | Physical Disk\|I/O Usage Capacity | I/O Usage Capacity |
| disk\|busResets_summation | Physical Disk\|Bus Resets | The number of bus resets in the performance interval |
| disk\|commandsAborted_summation | Physical Disk\|Commands Aborted | The number of disk commands stopped in the performance interval |
| disk\|diskoio | Physical Disk\|Number of Outstanding IO Operations | Number of Outstanding IO Operations |
| disk\|diskqueued | Physical Disk\|Queued Operations | Queued Operations |
| disk\|diskdemand | Physical Disk\|Demand | Demand |
| disk\|sum_queued_oio | Physical Disk\|Total Queued Outstanding operations | Sum of Queued Operation and Outstanding Operations. |
| disk\|max_observed | Physical Disk\|Max Observed OIO | Max Observed IO for a disk. |
| disk\|numberReadAveraged_average | Physical Disk\|Read IOPS | Number of read operations per second. This is averaged over the reporting period. |
| disk\|numberWriteAveraged_average | Physical Disk\|Write IOPS | Number of write operations per second. This is averaged over the reporting period. |
| disk\|maxTotalLatency_latest | Physical Disk\|Highest Latency | Highest Latency. |
| disk\|scsiReservationConflicts_summation | Physical Disk\|SCSI Reservation Conflicts | SCSI Reservation Conflicts. |
| disk\|totalReadLatency_average | Physical Disk\|Read Latency | Average amount of time for a read operation by the storage adapter. |
| disk\|totalWriteLatency_average | Physical Disk\|Write Latency | Average amount of time for a write operation by the storage adapter. |
| disk\|totalLatency_average | Physical Disk\|Total Latency | Total Latency. |
| sys\|poweredOn | System\|Powered ON | 1 if the VM is connected (available for management) and powered on, otherwise 0. |
| sys\|osUptime_latest | System\|OS Uptime | Total time elapsed, in seconds, since last operating system boot-up |
| sys\|uptime_latest | System\|Uptime | Number of seconds since system startup |
| sys\|heartbeat_summation | System\|Heartbeat | Number of heart beats from the virtual machine in the defined interval |

| Metric Key | Metric Name | Description |
|---|---|---|
| sys\|vmotionEnabled | System\|vMotion Enabled | 1 if vMotion enabled, 0 if not enabled |
| sys\|productString | System\|Product String | VMware product string |
| sys\|heartbeat_latest | System\|Heartbeat Latest | Number of heartbeats issued per virtual machine during the interval |
| summary\|running | Summary\|Running | Running |
| summary\|desktop_status | Summary\|Desktop Status | Horizon View Desktop Status |
| summary\|poweredOff | Summary\|Reclaimable Powered Off | Powered Off = 1. Not powered off = 0 |
| summary\|idle | Summary\|Reclaimable Idle | Idle = 1. Not idle = 0 |
| summary\|oversized | Summary\|Is Oversized | Oversized = 1. Not oversized = 0 |
| summary\|undersized | Summary\|Is Undersized | Is Undersized |
| summary\|snapshotSpace | Summary\|Reclaimable Snapshot Space | Reclaimable Snapshot Space |
| summary\|oversized\|vcpus | Summary\|Oversized\|Virtual CPUs | Virtual CPUs |
| summary\|oversized\|memory | Summary\|Oversized\|Memory | Memory |
| summary\|undersized\|vcpus | Summary\|Undersized\|Virtual CPUs | Virtual CPUs |
| summary\|undersized\|memory | Summary\|Undersized\|Memory | Memory |
| summary\|metering\|value | Summary\|Metering\|Total price | Total price of the resource(Sum of all price components) |
| summary\|metering\|storage | Summary\|Metering\|Storage price | Price of Storage related components of the resource |
| summary\|metering\|memory | Summary\|Metering\|Memory price | Price of Memory related components of the resource |
| summary\|metering\|cpu | Summary\|Metering\|CPU price | Price of CPU related components of the resource |
| summary\|metering\|additional | Summary\|Metering\|Additional price | Price of additional components of the resource |
| summary\|metering\|partialPrice | Summary\|Metering\|Partial price | Shows whether the calculated price is partial for the resource |
| summary\|workload_indicator | Summary\|Workload Indicator | Workload Indicator |
| summary\|cpu_shares | Summary\|CPU Shares | CPU Shares |
| summary\|mem_shares | Summary\|Memory Shares | Memory Shares |
| summary\|number_datastore | Summary\|Number of Datastores | Number of Datastores |
| summary\|number_network | Summary\|Number of Networks | Number of Networks |
| guestfilesystem\|capacity | Guest File System\|Partition Capacity | Disk space capacity on guest file system partition. |
| guestfilesystem\|percentage | Guest File System\|Partition Utilization (%) | Guest file system partition space utilization in percentage |
| guestfilesystem\|usage | Guest File System\|Partition Utilization | Guest file system partition space utilization |
| guestfilesystem\|capacity_total | Guest File System\|Total Capacity | Disk space capacity on guest file system |
| guestfilesystem\|percentage_total | Guest File System\|Utilization (%) | Guest file system disk space utilization in percentage |
| guestfilesystem\|usage_total | Guest File System\|Utilization | Guest file system disk space utilization |
| guestfilesystem\|freespace | Guest File System\|Guest File System Free | Total free space on guest file system |
| guestfilesystem\|capacity_property | Guest File System\|Guest File System Capacity Property | Total capacity of guest file system as a property |

| Metric Key | Metric Name | Description |
|---|---|---|
| guestfilesystem\|freespace_total | Guest File System\|Total Guest File System Free | Total free space on guest file system |
| guestfilesystem\|capacity_property_total | Guest File System\|Total Capacity Property | Total capacity of guest file system as a property |
| guest\|mem.free_latest | Guest\|Free Memory | Free Memory |
| guest\|mem.needed_latest | Guest\|Needed Memory | Needed Memory |
| guest\|mem.physUsable_latest | Guest\|Physically Usable Memory | Physically Usable Memory |
| guest\|page.inRate_latest | Guest\|Page In Rate per second | Page In Rate per second |
| guest\|page.size_latest | Guest\|Page Size | Page Size |
| guest\|swap.spaceRemaining_latest | Guest\|Remaining Swap Space | Remaining Swap Space |
| guest\|cpu_queue | Guest\|CPU Queue | The number of ready threads queuing in the CPU. Linux includes threads in running state. A number greater than 2 for prolong period indicates CPU core bottleneck. |
| guest\|disk_queue | Guest\|Disk Queue | The number of outstanding requests + IO currently in progress. |
| guest\|contextSwapRate_latest | Guest\|Context Swap Rate per second | Context Swap Rate per second |
| guest\|hugePage.size_latest | Guest\|Huge Page Size | Huge Page Size |
| guest\|hugePage.total_latest | Guest\|Total Huge Pages | Total Huge Pages |
| guest\|mem.activeFileCache_latest | Guest\|Active File Cache Memory | Active File Cache Memory |
| guest\|page.outRate_latest | Guest\|Page Out Rate per second | Page Out Rate per second |
| guest\|disk_queue_latest | Guest\|Disk Queue Latest | The number of outstanding requests + IO currently in progress. |
| virtualDisk\|numberReadAveraged_average | Virtual Disk\|Read IOPS | Number of read operations per second. This is averaged over the reporting period |
| virtualDisk\|numberWriteAveraged_average | Virtual Disk\|Write IOPS | Number of write operations per second. This is averaged over the reporting period |
| virtualDisk\|read_average | Virtual Disk\|Read Throughput | Amount of data read from storage in a second. This is averaged over the reporting period |
| virtualDisk\|totalReadLatency_average | Virtual Disk\|Read Latency | Average amount of time for a read operation by the storage adapter. |
| virtualDisk\|totalWriteLatency_average | Virtual Disk\|Write Latency | Average amount of time for a write operation by the storage adapter. |
| virtualDisk\|write_average | Virtual Disk\|Write Throughput | Amount of data written to storage in a second. This is averaged over the reporting period |
| virtualDisk\|usage | Virtual Disk\|Total Throughput | Amount of data read from/written to storage in a second. This is averaged over the reporting period |
| virtualDisk\|totalLatency | Virtual Disk\|Total Latency | Total Latency |
| virtualDisk\|commandsAveraged_average | Virtual Disk\|Total IOPS | Number of read/write operations per second. This is averaged over the reporting period |
| virtualDisk\|vDiskOIO | Virtual Disk\|Outstanding IO requests | OIO for datastore. |
| virtualDisk\|actualUsage | Virtual Disk\|Used Disk Space | Virtual Disk space usage |
| virtualDisk\|busResets_summation | Virtual Disk\|Bus Resets | The number of bus resets in the performance interval |

| Metric Key | Metric Name | Description |
|---|---|---|
| virtualDisk\|commandsAborted_summation | Virtual Disk\|Commands Aborted | The number of disk commands stopped in the performance interval |
| virtualDisk\|readLoadMetric_latest | Virtual Disk\|Read Load | Storage DRS virtual disk metric read load |
| virtualDisk\|readOIO_latest | Virtual Disk\|Outstanding Read Requests | Average number of outstanding read requests to the virtual disk |
| virtualDisk\|writeLoadMetric_latest | Virtual Disk\|Write Load | Storage DRS virtual disk write load |
| virtualDisk\|writeOIO_latest | Virtual Disk\|Outstanding Write Requests | Average number of outstanding write requests to the virtual disk |
| virtualDisk\|smallSeeks_latest | Virtual Disk\|Number of Small Seeks | Small Seeks |
| virtualDisk\|mediumSeeks_latest | Virtual Disk\|Number of Medium Seeks | Medium Seeks |
| virtualDisk\|largeSeeks_latest | Virtual Disk\|Number of Large Seeks | Large Seeks |
| virtualDisk\|readLatencyUS_latest | Virtual Disk\|Read Latency (microseconds) | Read latency in microseconds |
| virtualDisk\|writeLatencyUS_latest | Virtual Disk\|Write Latency (microseconds) | Write Latency in microseconds |
| virtualDisk\|readIOSize_latest | Virtual Disk\|Average Read request size | Read IO size |
| virtualDisk\|writeIOSize_latest | Virtual Disk\|Average Write request size | Write IO size |
| diskspace\|pod_used | Disk Space\|Pod used | Space used by Pod files |
| diskspace\|provisionedSpace | Disk Space\|Provisioned Space for Pod | Provisioned space for Pod. In thin provisioned, it is the full space allocated (which may not be used yet). |
| diskspace\|notshared | Disk Space\|Not Shared | Space used by VM that is not shared with other VM |
| diskspace\|activeNotShared | Disk Space\|Active not shared | Unshared disk space used by VMs excluding snapshot |
| diskspace\|perDsUsed | Disk Space\|Pod used | Space used by all files of the Pod on the datastore (disks, snapshots, configs, logs, etc). |
| diskspace\|total_usage | Disk Space\|Utilization | Total disk space used on all datastores visible to this object |
| diskspace\|total_capacity | Disk Space\|Total Capacity | Total disk space on all datastores visible to this object |
| diskspace\|diskused | Disk Space\|Virtual Disk Used | Space used by virtual disks |
| diskspace\|snapshot | Disk Space\|Snapshot Space | Space used by snapshots |
| diskspace\|shared | Disk Space\|Shared Used | Shared space used |
| diskspace\|provisioned | Disk Space\|Provisioned Space | Provisioned space |
| diskspace\|snapshot\|used | Disk Space\|Snapshot\|Pod used | Disk space used by the Pod snapshot files. This is the space that can be potentially reclaimed if the snapshot is removed. |
| diskspace\|snapshot\|accessTime | Disk Space\|Snapshot\|Access Time | The date and time the snapshot was taken. |
| storage\|totalReadLatency_average | Storage\|Read Latency | Average amount of time for a read operation. |
| storage\|totalWriteLatency_average | Storage\|Write Latency | Average amount of time for a write operation. |
| storage\|read_average | Storage\|Read Rate | Read throughput rate |
| storage\|write_average | Storage\|Write Rate | Write throughput rate |

| Metric Key | Metric Name | Description |
|---|---|---|
| storage\|usage_average | Storage\|Total Usage | Total throughput rate |
| storage\|numberReadAveraged_average | Storage\|Reads per second | Average number of read commands issued per second during the collection interval |
| storage\|numberWriteAveraged_average | Storage\|Writes per second | Average number of write commands issued per second during the collection interval |
| storage\|commandsAveraged_average | Storage\|Commands per second | Average number of commands issued per second during the collection interval |
| storage\|totalLatency_average | Storage\|Total Latency | Total latency |
| storage\|demandKBps | Storage\|Demand | N/A |
| storage\|contention | Storage\|Contention percentage | N/A |
| cost\|monthlyTotalCost | Cost\|MTD Total Cost | Month To Date Cost of Virtual Machine |
| cost\|monthlyProjectedCost | Cost\|Monthly Projected Total Cost | Virtual Machine cost projected for full month |
| cost\|compTotalCost | Cost\|MTD Compute Total Cost | Month to Date Total Compute Cost (Including CPU and Memory) of Virtual Machine |
| cost\|directCost | Cost\|Monthly Direct Cost | Monthly Direct Cost (comprising of OS Labor, VI Labor and any windows desktop instance license) of Virtual Machine |
| cost\|cpuCost | Cost\|MTD CPU Cost | Month to Date Virtual Machine CPU Cost. It is based on utilization. The more the VM uses, the higher its cost. |
| cost\|memoryCost | Cost\|MTD Memory Cost | Month to Date Memory Cost of Virtual Machine. It is based on utilization. The more the VM uses, the higher its cost. |
| cost\|storageCost | Cost\|MTD Disk Space Cost | Month to Date Disk Space Cost of Virtual Machine |
| cost\|reclaimableCost | Cost\|Potential Savings | Potential Savings |
| cost\|osLaborTotalCost | Cost\|Monthly OS Labor Cost | Operating System Labor Cost of Virtual Machine for full month |
| cost\|viLaborTotalCost | Cost\|Monthly VI Labor Cost | Monthly VI Labor Cost |
| cost\|effectiveTotalCost | Cost\|MTD Effective Total Cost | Month to Date Cost of Virtual Machine considering the allocation and demand model |
| cost\|effectiveProjectedTotalCost | Cost\|Monthly Effective Projected Total Cost | Virtual Machine cost projected for full month considering the allocation and demand model |
| cost\|allocation\|allocationBasedCpuMTD Cost | Cost\|Allocation\|MTD CPU Cost | Month to Date Virtual Machine CPU Cost. It is based on utilization. The more the VM uses, the higher its cost. |
| cost\|allocation\|allocationBasedMemoryM TDCost | Cost\|Allocation\|MTD Memory Cost | Month to Date Memory Cost of Virtual Machine. It is based on utilization. The more the VM uses, the higher its cost. |
| cost\|allocation\|allocationBasedStorageM TDCost | Cost\|Allocation\|MTD Disk Space Cost | Month to Date Disk Space Cost of Virtual Machine |
| cost\|allocation\|allocationBasedTotalMTD Cost | Cost\|Allocation\|MTD Total Cost | Month To Date Cost of Virtual Machine |
| cost\|allocation\|allocationBasedTotalCost | Cost\|Allocation\|Monthly Projected Total Cost | Virtual Machine cost projected for full month |

| Metric Key | Metric Name | Description |
|---|---|---|
| datastore\|demand_oio | Datastore\|Outstanding IO requests | Amount of IO waiting in the queue to be executed. High IO, coupled with high latency, impacts performance. |
| datastore\|numberReadAveraged_average | Datastore\|Read IOPS | Number of read operations per second. This is averaged over the reporting period. |
| datastore\|numberWriteAveraged_average | Datastore\|Write IOPS | Number of write operations per second. This is averaged over the reporting period. |
| datastore\|read_average | Datastore\|Read Throughput | Amount of data read from storage in a second. This is averaged over the reporting period. |
| datastore\|totalReadLatency_average | Datastore\|Read Latency | Average amount of time for a read operation at the datastore level. It's an average of all the VMs in the datastore. |
| datastore\|totalWriteLatency_average | Datastore\|Write Latency | Average amount of time for a write operation by the storage adapter. |
| datastore\|write_average | Datastore\|Write Throughput | Amount of data written from storage in a second. This is averaged over the reporting period. |
| datastore\|totalLatency_average | Datastore\|Total Latency | Normalized Latency, taking into account the read/write ratio. |
| datastore\|usage_average | Datastore\|Total Throughput | Amount of data read from/written to storage in a second. This is averaged over the reporting period. |
| datastore\|commandsAveraged_average | Datastore\|Total IOPS | Number of read/write operations per second. This is averaged over the reporting period. |
| datastore\|used | Datastore\|Used Space | Used Space. |
| datastore\|demand | Datastore\|Demand | Max of datastore "Reads Per Sec", "Writes Per Sec", "Read Rate", "Write Rate", "OIO Per Sec" percentages. |
| datastore\|maxTotalLatency_latest | Datastore\|Highest Latency | Highest Latency. |
| datastore\|totalLatency_max | Datastore\|Total Latency Max | Total Latency Max (ms). |
| datastore\|maxObserved_NumberRead | Datastore\|Max Observed Reads per second | Max observed average number of read commands issued per second during the collection interval. |
| datastore\|maxObserved_Read | Datastore\|Max Observed Read Rate | Max observed rate of reading data from the datastore. |
| datastore\|maxObserved_NumberWrite | Datastore\|Max Observed Writes per second | Max observed average number of write commands issued per second during the collection interval. |
| datastore\|maxObserved_Write | Datastore\|Max Observed Write Rate | Max observed rate of writing data from the datastore. |
| datastore\|maxObserved_OIO | Datastore\|Max Observed Number of Outstanding IO Operations | N/A |

#### Power Metrics for vSphere Pods

Power metrics provide information about the vSphere pods power use.

| Metric Name | Description |
|---|---|
| Power\|Total Energy Consumed in the collection period (Wh) | Displays the total electricity consumed based on the time interval selected. The default collection cycle is set to 5 mins. You can continue using the default setting or edit it for each adapter instance. For example, if the time interval is set to its default value, the value represents the energy consumed per 5 mins. |
| Power\|Current Power Consumption Rate (Watt) | The power consumption rate per second, averaged over the reporting period.<br>Key: power\|power_average |
| Power\|(DEP) Energy (Joule) | Total energy consumed in joules.<br>Key: power\|energy_summation |

## OS and Application Monitoring Metrics

Metrics are collected for operating systems, application services, remote checks, Linux processes, and Windows services.

### Operating System Metrics

Metrics are collected for Linux and Windows operating systems.

#### Linux Platforms

The following metrics are collected for Linux operating systems:

**Table 1491: Metrics for Linux**

| Metric | Metric Category | KPI |
|---|---|---|
| <Instance name>\| Usage Idle | CPU | False |
| <Instance name>\| Usage IO-Wait | CPU | False |
| <Instance name>\|Time Active | CPU | True |
| <Instance name>\|Time Guest | CPU | False |
| <Instance name>\|Time Guest Nice | CPU | False |
| <Instance name>\|Time Idle | CPU | False |
| <Instance name>\|Time IO-Wait | CPU | False |
| <Instance name>\|Time IRQ | CPU | True |
| <Instance name>\|Time Nice | CPU | False |
| <Instance name>\|Time Soft IRQ | CPU | True |
| <Instance name>\|Time Steal | CPU | False |
| <Instance name>\|Time System | CPU | False |
| <Instance name>\|Time User | CPU | True |
| <Instance name>\|Usage Active (%) | CPU | True |
| <Instance name>\|Usage Guest (%) | CPU | False |
| <Instance name>\|Usage Guest Nice (%) | CPU | False |
| <Instance name>\|Usage IRQ (%) | CPU | True |

| Metric | Metric Category | KPI |
|---|---|---|
| <Instance name>\|Usage Nice (%) | CPU | False |
| <Instance name>\|Usage Soft IRQ (%) | CPU | True |
| <Instance name>\|Usage Steal (%) | CPU | False |
| <Instance name>\|Usage System (%) | CPU | True |
| <Instance name>\|Usage User (%) | CPU | True |
| CPU Load1 (%) | CPU Load | False |
| CPU Load15 (%) | CPU Load | False |
| CPU Load5 (%) | CPU Load | False |
| <Instance name>\|IO Time | Disk IO | False |
| <Instance name>\|Read Time | Disk IO | False |
| <Instance name>\|Reads | Disk IO | False |
| <Instance name>\|Write Time | Disk IO | False |
| <Instance name>\|Writes | Disk IO | False |
| <Instance name>\|Disk Free | Disk | False |
| <Instance name>\|Disk Total | Disk | False |
| <Instance name>\|Disk Used (%) | Disk | False |
| Cached | Memory | False |
| Free | Memory | False |
| Inactive | Memory | False |
| Total | Memory | True |
| Used | Memory | True |
| Used Percent | Memory | True |
| Blocked | Processes | True |
| Dead | Processes | False |
| Running | Processes | False |
| Sleeping | Processes | False |
| Stopped | Processes | False |
| Zombies | Processes | False |
| Free | Swap | False |
| In | Swap | False |
| Out | Swap | False |
| Total | Swap | True |
| Used | Swap | True |
| Used Percent | Swap | True |
| Telegraf Availability | None | False |

**Windows Platforms**

The following metrics are collected for Windows operating systems:

### Table 1492: Metrics for Windows

| Metric | Metric Category | KPI |
|---|---|---|
| Idle Time | CPU | False |
| Interrupt Time | CPU | False |
| Interrupts persec | CPU | True |
| Privileged Time | CPU | False |
| Processor Time | CPU | False |
| User Time | CPU | False |
| DPC Time (%) | CPU | False |
| Usage Guest (%) | CPU | False |
| Usage System (%) | CPU | False |
| Usage User (%) | CPU | False |
| Avg. Disk Bytes Read | Disk | False |
| Avg. Disk sec Read | Disk | False |
| Avg. Disk sec Write | Disk | False |
| Avg. Disk Write Queue Length | Disk | False |
| Avg. Disk Read Queue Length | Disk | False |
| Disk Read Time | Disk | False |
| Disk Write Time | Disk | False |
| Free Megabytes | Disk | False |
| Free Space | Disk | False |
| Idle Time | Disk | False |
| Split IO persec | Disk | False |
| Available Bytes | Memory | True |
| Cache Bytes | Memory | False |
| Cache Faults persec | Memory | False |
| Committed Bytes | Memory | True |
| Demand Zero Faults persec | Memory | False |
| Page Faults persec | Memory | True |
| Pages persec | Memory | False |
| Pool Nonpaged Bytes | Memory | True |
| Pool Paged Bytes | Memory | False |
| Transition Faults persec | Memory | False |
| Total (bytes) | Memory | False |
| Used (bytes) | Memory | False |
| Used Percent(%) | Memory | False |
| Bytes Received persec | Network | False |
| Bytes Sent persec | Network | False |
| Packets Outbound Discarded | Network | False |
| Packets Outbound Errors | Network | False |
| Packets Received Discarded | Network | False |

| Metric | Metric Category | KPI |
|---|---|---|
| Packets Received Errors | Network | False |
| Packets Received persec | Network | False |
| Packets Sent persec | Network | False |
| Elapsed Time | Process | False |
| Handle Count | Process | False |
| IO Read Bytes persec | Process | False |
| IO Read Operations persec | Process | False |
| IO Write Bytes persec | Process | False |
| IO Write Operations persec | Process | False |
| Privileged Time | Process | False |
| Processor Time | Process | False |
| Thread Count | Process | False |
| User Time | Process | False |
| Context Switches persec | System | False |
| Processes | System | False |
| Processor Queue Length | System | False |
| System Calls persec | System | False |
| System Up Time | System | False |
| Threads | System | False |
| Used Percent (%) | Swap | False |
| Total (bytes) | Swap | False |
| Telegraf Availability | None | False |

### Application Service Metrics

Metrics are collected for 23 application services.

Here is the list of application services:

- Active Directory
- Apache HTTPD
- Microsoft SQL Server
- PostgreQL
- Microsoft IIS
- MS Exchange
- JBoss Server
- RabbitMQ
- MySQL
- NGINX
- SharePoint
- Oracle WebLogic Server
- Pivotal TC Server
- ActiveMQ
- Apache HTTPD
- Oracle DB
- Cassandra

- HyperV
- MongoDB
- Riak KV
- Network Time Protocol
- WebSphere
- Java Application

### Active Directory Metrics

Metrics are collected for the Active Directory application service.

**Table 1493: Active Directory Metrics**

| Metric Name | Category | KPI |
|---|---|---|
| Database Cache % Hit (%) | Active Directory Database | True |
| Database Cache Page Faults/sec | Active Directory Database | True |
| Database Cache Size | Active Directory Database | False |
| Data Lookups | Active Directory DFS Replication | False |
| Database Commits | Active Directory DFS Replication | True |
| Avg Response Time | Active Directory DFSN | True |
| Requests Failed | Active Directory DFSN | False |
| Requests Processed | Active Directory DFSN | False |
| Dynamic Update Received | Active Directory DNS | False |
| Dynamic Update Rejected | Active Directory DNS | False |
| Recursive Queries | Active Directory DNS | False |
| Recursive Queries Failure | Active Directory DNS | False |
| Secure Update Failure | Active Directory DNS | False |
| Total Query Received | Active Directory DNS | True |
| Total Response Sent | Active Directory DNS | True |
| Digest Authentications | Active Directory Security System-Wide Statistics | True |
| Kerberos Authentications | Active Directory Security System-Wide Statistics | True |
| NTLM Authentications | Active Directory Security System-Wide Statistics | True |
| Directory Services:<InstanceName>\|Base Searches persec | Active Directory Services | False |
| Directory Services:<InstanceName>\|Database adds persec | Active Directory Services | False |
| Directory Services:<InstanceName>\|Database deletes persec | Active Directory Services | False |
| Directory Services<InstanceName>\|Database modifys/sec | Active Directory Services | False |
| Directory Services<InstanceName>\|Database recycles/sec | Active Directory Services | False |
| Directory Services<InstanceName>\|DRA Inbound Bytes Total/sec | Active Directory Services | False |
| Directory Services<InstanceName>\|DRA Inbound Objects/sec | Active Directory Services | False |

| Metric Name | Category | KPI |
|---|---|---|
| Directory Services<InstanceName>\|DRA Outbound Bytes Total/sec | Active Directory Services | False |
| Directory Services<InstanceName>\|DRA Outbound Objects/sec | Active Directory Services | False |
| Directory Services<InstanceName>\|DRA Pending Replication Operations | Active Directory Services | False |
| Directory Services<InstanceName>\|DRA Pending Replication Synchronizations | Active Directory Services | False |
| Directory Services<InstanceName>\|DRA Sync Requests Made | Active Directory Services | False |
| Directory Services<InstanceName>\|DRA Sync Requests Successful | Active Directory Services | False |
| Directory Services<InstanceName>\|DS Client Binds/sec | Active Directory Services | True |
| Directory Services<InstanceName>\|DS Directory Reads/sec | Active Directory Services | False |
| Directory Services<InstanceName>\|DS Directory Searches/sec | Active Directory Services | True |
| Directory Services<InstanceName>\|DS Server Binds/sec | Active Directory Services | True |
| Directory Services<InstanceName>\|DS Threads in Use | Active Directory Services | True |
| Directory Services:<InstanceName>\|LDAP Active Threads | Active Directory Services | False |
| Directory Services:<InstanceName>\|LDAP Client Sessions | Active Directory Services | True |
| Directory Services<InstanceName>\|LDAP Closed Connections/sec | Active Directory Services | False |
| Directory Services<InstanceName>\|LDAP New Connections/sec | Active Directory Services | True |
| Directory Services<InstanceName>\|LDAP Searches/sec | Active Directory Services | True |
| Directory Services<InstanceName>\|LDAP Successful Binds/sec | Active Directory Services | False |
| Directory Services<InstanceName>\|LDAP UDP operations/sec | Active Directory Services | False |
| Directory Services:<InstanceName>\|LDAP Writes/sec | Active Directory Services | False |
| Application Availability | Active Directory | False |

### ActiveMQ Metrics

Metrics are collected for the ActiveMQ application service.

**Table 1494: ActiveMQ Metrics**

| Metric Name | Category | KPI |
|---|---|---|
| Buffer Pool<InstanceName>\|Count | Active MQ | False |
| Buffer Pool<InstanceName>\|Memory Used | Active MQ | False |
| Buffer Pool<InstanceName>\|Total Capacity | Active MQ | False |
| Class Loading\|Loaded Class Count | Active MQ | False |
| Class Loading\|Unloaded Class Count | Active MQ | False |
| Class Loading\|Total Loaded Class Count | Active MQ | False |
| File Descriptor Usage\|Max File Descriptor Count | Active MQ | False |
| File Descriptor Usage\|Open File Descriptor Count | Active MQ | False |
| Garbage Collection<InstanceName>\|Total Collection Count | Active MQ | False |
| Garbage Collection<InstanceName>\|Total Collection Time | Active MQ | False |
| JVM Memory Pool<InstanceName>\|Peak Usage\|Committed Memory | Active MQ | False |
| JVM Memory Pool<InstanceName>\|Peak Usage\|Initial Memory | Active MQ | False |
| JVM Memory Pool<InstanceName>\|Peak Usage\|Maximum Memory | Active MQ | False |
| JVM Memory Pool<InstanceName>\|Peak Usage\|Used Memory | Active MQ | False |
| JVM Memory Pool<InstanceName>\|Usage\|Committed Memory | Active MQ | False |
| JVM Memory Pool<InstanceName>\|Usage\|Initial Memory | Active MQ | False |
| JVM Memory Pool<InstanceName>\|Usage\|Maximum Memory | Active MQ | False |
| JVM Memory Pool<InstanceName>\|Usage\|Used Memory | Active MQ | False |

| Metric Name | Category | KPI |
|---|---|---|
| Application Availability | Active MQ | False |
| Threading\|Thread Count | Active MQ | False |
| Uptime | Active MQ | False |
| UTILIZATION\|Process CpuLoad | Active MQ | False |
| UTILIZATION\|Memory Limit | ActiveMQ Broker | True |
| UTILIZATION\|Memory Percent Usage (%) | ActiveMQ Broker | True |
| UTILIZATION\|Store Limit | ActiveMQ Broker | False |
| UTILIZATION\|Store Percent Usage (%) | ActiveMQ Broker | False |
| UTILIZATION\|Temp Limit | ActiveMQ Broker | False |
| UTILIZATION\|Temp Percent Usage (%) | ActiveMQ Broker | False |
| UTILIZATION\|Total Consumer Count | ActiveMQ Broker | True |
| UTILIZATION\|Total Dequeue Count | ActiveMQ Broker | True |
| UTILIZATION\|Total Enqueue Count | ActiveMQ Broker | True |
| UTILIZATION\|Total Message Count | ActiveMQ Broker | True |
| JVM Memory\|Heap Memory Usage\|Initial Memory | ActiveMQ JVM Memory Usage | False |
| JVM Memory\|Heap Memory Usage\|Committed Memory | ActiveMQ JVM Memory Usage | False |
| JVM Memory\|Heap Memory Usage\|Maximum Memory | ActiveMQ JVM Memory Usage | False |
| JVM Memory\|Heap Memory Usage\|Used Memory | ActiveMQ JVM Memory Usage | False |
| JVM Memory\|Non Heap Memory Usage\|Committed Memory | ActiveMQ JVM Memory Usage | False |
| JVM Memory\|Non Heap Memory Usage\|Initial Memory | ActiveMQ JVM Memory Usage | False |
| JVM Memory\|Non Heap Memory Usage\|Maximum Memory | ActiveMQ JVM Memory Usage | False |
| JVM Memory\|Non Heap Memory Usage\|Used Memory | ActiveMQ JVM Memory Usage | False |
| JVM Memory\|Object Pending FinalizationCount | ActiveMQ JVM Memory Usage | False |
| UTILIZATION\|Process CpuLoad | ActiveMQ OS | False |
| UTILIZATION\|System Cpu Load | ActiveMQ OS | False |
| UTILIZATION\|Consumer Count | ActiveMQ Topic | True |
| UTILIZATION\|Dequeue Count | ActiveMQ Topic | True |

| Metric Name | Category | KPI |
|---|---|---|
| UTILIZATION\|Enqueue Count | ActiveMQ Topic | True |
| UTILIZATION\|Queue Size | ActiveMQ Topic | True |
| UTILIZATION\|Producer Count | ActiveMQ Topic | False |

### Apache HTTPD Metrics

Metrics are collected for the Apache HTTPD application service.

**Note:**
Metrics are collected for the Events MPM. Metrics are not collected for the other MPMs.

**Table 1495: Apache HTTPD Metrics**

| Metric Name | Category | KPI |
|---|---|---|
| UTILIZATION\|Busy Workers | Apache HTTPD | True |
| UTILIZATION\|Bytes Per Req | Apache HTTPD | False |
| UTILIZATION\|Bytes Per Sec | Apache HTTPD | False |
| UTILIZATION\|CPU Load | Apache HTTPD | True |
| UTILIZATION\|CPU User | Apache HTTPD | False |
| UTILIZATION\|Idle Workers | Apache HTTPD | True |
| UTILIZATION\|Request Per Sec | Apache HTTPD | True |
| UTILIZATION\|SCBoard Closing | Apache HTTPD | False |
| UTILIZATION\|SCBoard DNS Lookup | Apache HTTPD | False |
| UTILIZATION\|SCBoard Finishing | Apache HTTPD | False |
| UTILIZATION\|SCBoard Idle Cleanup | Apache HTTPD | False |
| UTILIZATION\|SCBoard Keep Alive | Apache HTTPD | False |
| UTILIZATION\|SCBoard Logging | Apache HTTPD | False |
| UTILIZATION\|SCBoard Open | Apache HTTPD | False |
| UTILIZATION\|SCBoard Reading | Apache HTTPD | False |
| UTILIZATION\|SCBoard Sending | Apache HTTPD | False |
| UTILIZATION\|SCBoard Starting | Apache HTTPD | False |
| UTILIZATION\|SCBoard Waiting | Apache HTTPD | False |
| UTILIZATION\|Total Accesses | Apache HTTPD | False |
| UTILIZATION\|Total Bytes | Apache HTTPD | True |
| UTILIZATION\|Total Connections | Apache HTTPD | False |
| UTILIZATION\|Uptime | Apache HTTPD | True |
| UTILIZATION\|Asynchronous Closing Connections | Apache HTTPD | False |
| UTILIZATION\|Asynchronous Keep Alive Connections | Apache HTTPD | False |
| UTILIZATION\|Asynchronous Writing Connections | Apache HTTPD | False |
| UTILIZATION\|ServerUptimeSeconds | Apache HTTPD | False |
| UTILIZATION\|Load1 | Apache HTTPD | False |
| UTILIZATION\|Load5 | Apache HTTPD | False |
| UTILIZATION\|ParentServerConfigGeneration | Apache HTTPD | False |

| Metric Name | Category | KPI |
|---|---|---|
| UTILIZATION\|ParentServerMPMGeneration | Apache HTTPD | False |
| Application Availability | Apache HTTPD | False |

### Tomcat Server

Metrics are collected for the Tomcat Server application service.

**Table 1496: Tomcat Server**

| Metric Name | Category | KPI |
|---|---|---|
| Buffer Pool<InstanceName>\|Count | Tomcat Server | False |
| Buffer Pool<InstanceName>\|Memory Used | Tomcat Server | False |
| Buffer Pool<InstanceName>\|Total Capacity | Tomcat Server | False |
| Class Loading\|Loaded Class Count | Tomcat Server | False |
| Class Loading\|Total Loaded Class Count | Tomcat Server | False |
| Class Loading\|Unloaded Class Count | Tomcat Server | False |
| File Descriptor Usage\|Max File Descriptor Count | Tomcat Server | False |
| File Descriptor Usage\|Open File Descriptor Count | Tomcat Server | False |
| Garbage Collection:<InstanceName>\|Total Collection Count | Tomcat Server | False |
| Garbage Collection:<InstanceName>\|Total Collection Time | Tomcat Server | True |
| JVM Memory\|Heap Memory Usage\|Committed Memory | Tomcat Server | False |
| JVM Memory\|Heap Memory Usage\|Initial Memory | Tomcat Server | False |
| JVM Memory\|Heap Memory Usage\|Maximum Memory | Tomcat Server | False |
| JVM Memory\|Heap Memory Usage\|Used Memory | Tomcat Server | False |
| JVM Memory\|Non Heap Memory Usage\|Committed Memory | Tomcat Server | False |
| JVM Memory\|Non Heap Memory Usage\|Initial Memory | Tomcat Server | False |
| JVM Memory\|Non Heap Memory Usage\|Maximum Memory | Tomcat Server | False |
| JVM Memory\|Non Heap Memory Usage\|Used Memory | Tomcat Server | False |
| JVM Memory\|Number of Object Pending Finalization Count | Tomcat Server | False |
| JVM Memory\|Pool:<InstanceName>\|Peak Usage\|Committed Memory | Tomcat Server | False |
| JVM Memory\|Pool:<InstanceName>\|Peak Usage\|Initial Memory | Tomcat Server | False |
| JVM Memory\|Pool:<InstanceName>\|Peak Usage\|Maximum Memory | Tomcat Server | False |
| JVM Memory\|Pool:<InstanceName>\|Peak Usage\|Used Memory | Tomcat Server | False |

| Metric Name | Category | KPI |
|---|---|---|
| JVM Memory\|Pool:<InstanceName>\|Usage\| Committed Memory | Tomcat Server | False |
| JVM Memory\|Pool:<InstanceName>\|Usage\|Initial Memory | Tomcat Server | False |
| JVM Memory\|Pool:<InstanceName>\|Usage\| Maximum Memory | Tomcat Server | False |
| JVM Memory\|Pool:<InstanceName>\|Usage\|Used Memory | Tomcat Server | False |
| Process CPU Usage (%) | Tomcat Server | True |
| System CPU Usage (%) | Tomcat Server | True |
| System Load Average (%) | Tomcat Server | True |
| Threading\|Thread Count | Tomcat Server | False |
| Uptime | Tomcat Server | True |
| Application Availability | Tomcat Server | False |
| JSP Count | Tomcat Server Web Module | False |
| JSP Reload Count | Tomcat Server Web Module | False |
| JSP Unload Count | Tomcat Server Web Module | False |
| Servlet:<InstanceName>\|Total Request Count | Tomcat Server Web Module | False |
| Servlet:<InstanceName>\|Total Request Error Count | Tomcat Server Web Module | False |
| Servlet:<InstanceName>\|Total Request Processing Time | Tomcat Server Web Module | False |
| Cache : Hit Count | Tomcat Server Web Module | False |
| Cache : Lookup Count | Tomcat Server Web Module | False |
| Current Thread Count | Tomcat Server Global Request Processor | True |
| Current Threads Busy | Tomcat Server Global Request Processor | True |
| errorRate | Tomcat Server Global Request Processor | False |
| Total Request Bytes Received | Tomcat Server Global Request Processor | False |
| Total Request Bytes Sent | Tomcat Server Global Request Processor | False |
| Total Request Count | Tomcat Server Global Request Processor | True |
| Total Request Error Count | Tomcat Server Global Request Processor | True |
| Total Request Processing Time | Tomcat Server Global Request Processor | False |

### Microsoft IIS Metrics

Metrics are collected for the Microsoft IIS application service.

#### Table 1497: IIS Metrics

| Metric Name | Category | KPI |
|---|---|---|
| HTTP Service Request Queues<InstanceName>AppPool\|CurrentQueueSize | IIS HTTP Service Request Queues | True |
| HTTP Service Request Queues<InstanceName>AppPool\|RejectedRequests | IIS HTTP Service Request Queues | False |
| Web Services<InstanceName> Web Site\|Bytes Received | IIS Web Services | False |

| Metric Name | Category | KPI |
|---|---|---|
| Web Services<InstanceName> Web Site\|Bytes Sent/sec | IIS Web Services | False |
| Web Services<InstanceName> Web Site\|Bytes Total/sec | IIS Web Services | False |
| Web Services<InstanceName> Web Site\|Connection Attempts/sec | IIS Web Services | False |
| Web Services<InstanceName> Web Site\|Current Connections | IIS Web Services | False |
| Web Services<InstanceName> Web Site\|Get Requests/sec | IIS Web Services | False |
| Web Services<InstanceName> Web Site\|Locked Errors/sec | IIS Web Services | False |
| Web Services<InstanceName> Web Site\|Not Found Errors/sec | IIS Web Services | False |
| Web Services<InstanceName> Web Site\|Post Requests/sec | IIS Web Services | False |
| Web Services<InstanceName> Web Site\|Service Uptime | IIS Web Services | False |
| Web Services<InstanceName> Web Site\|Total Bytes Sent | IIS Web Services | False |
| Web Services<InstanceName> Web Site\|Total Get Requests | IIS Web Services | True |
| Web Services<InstanceName> Web Site\|Total Post Requests | IIS Web Services | True |
| Web Services<InstanceName> Web Site\|Total Put Requests | IIS Web Services | False |
| Current File Cache Memory Usage (bytes) | IIS Web Services Cache | False |
| File Cache Hits Percent (%) | IIS Web Services Cache | False |
| Kernel URI Cache Hits Percent (%) | IIS Web Services Cache | False |
| Kernel URI Cache Misses | IIS Web Services Cache | False |
| Total Flushed URIs | IIS Web Services Cache | False |
| URI Cache Hits | IIS Web Services Cache | False |
| URI Cache Hits Percent (%) | IIS Web Services Cache | False |
| URI Cache Misses | IIS Web Services Cache | False |
| ASP.NET<InstanceName>\|Application Restarts | IIS ASP.NET | True |
| ASP.NET<InstanceName>\|Request Wait Time | IIS ASP.NET | True |
| ASP.NET<InstanceName>\|Requests Current | IIS ASP.NET | True |
| ASP.NET<InstanceName>\|Requests Queued | IIS ASP.NET | True |
| ASP.NET<InstanceName>\|Requests Rejected | IIS ASP.NET | True |
| MS.NET<InstanceName>\|Allocated Bytes/sec | MS.NET | True |
| MS.NET<InstanceName>\|Current Queue Length | MS.NET | False |
| MS.NET<InstanceName>\|Finalization Survivors | MS.NET | False |
| MS.NET<InstanceName>\|Gen 0 Collections | MS.NET | False |
| MS.NET<InstanceName>\|Gen 0 heap size | MS.NET | False |

| Metric Name | Category | KPI |
|---|---|---|
| MS.NET<InstanceName>\|Gen 1 Collections | MS.NET | False |
| MS.NET<InstanceName>\|Gen 1 heap size | MS.NET | False |
| MS.NET<InstanceName>\|Gen 2 Collections | MS.NET | False |
| MS.NET<InstanceName>\|Gen 2 heap size | MS.NET | False |
| MS.NET<InstanceName>\|IL Bytes Jitted / sec | MS.NET | False |
| MS.NET<InstanceName>\|Induced GC | MS.NET | False |
| MS.NET<InstanceName>\|Large Object Heap size | MS.NET | False |
| MS.NET<InstanceName>\|No of current logical Threads | MS.NET | True |
| MS.NET<InstanceName>\|No of current physical Threads | MS.NET | True |
| MS.NET<InstanceName>\|No of current recognized threads | MS.NET | False |
| MS.NET<InstanceName>\|No of Exceps Thrown / sec | MS.NET | True |
| MS.NET<InstanceName>\|No of total recognized threads | MS.NET | False |
| MS.NET<InstanceName>\|Percent Time in Jit | MS.NET | False |
| MS.NET<InstanceName>\|Pinned Objects | MS.NET | False |
| MS.NET<InstanceName>\|Stack Walk Depth | MS.NET | False |
| MS.NET<InstanceName>\|Time in RT checks | MS.NET | False |
| MS.NET<InstanceName>\|Time Loading | MS.NET | True |
| MS.NET<InstanceName>\|Total No of Contentions | MS.NET | False |
| MS.NET<InstanceName>\|Total Runtime Checks | MS.NET | True |
| Application Availability | Microsoft IIS | False |

### Java Application Metrics

Metrics are collected for the Java application service.

#### Table 1498: Java Application Metrics

| Metric Name | Category | KPI |
|---|---|---|
| Buffer Pool<InstanceName>\|Count | Java Application | False |
| Buffer Pool<InstanceName>\|Memory Used | Java Application | False |
| Buffer Pool<InstanceName>\|Total Capacity | Java Application | False |
| Class Loading\|Loaded Class Count | Java Application | True |
| Class Loading\|Total Loaded Class Count | Java Application | False |
| Class Loading\|Unloaded Class Count | Java Application | False |
| Garbage Collection<InstanceName>\|Total Collection Count | Java Application | False |
| Garbage Collection<InstanceName>\|Total Collection Time | Java Application | False |
| JVM Memory\|Heap Memory Usage\|Committed Memory | Java Application | False |

| Metric Name | Category | KPI |
|---|---|---|
| JVM Memory\|Heap Memory Usage\|Initial Memory | Java Application | False |
| JVM Memory\|Heap Memory Usage\|Maximum Memory | Java Application | False |
| JVM Memory\|Heap Memory Usage\|Used Memory | Java Application | False |
| JVM Memory\|JVM Memory Pool<InstanceName>\|Peak Usage\|Committed Memory | Java Application | False |
| JVM Memory\|JVM Memory Pool<InstanceName>\|Peak Usage\|Initial Memory | Java Application | False |
| JVM Memory\|JVM Memory Pool<InstanceName>\|Peak Usage\|Maximum Memory | Java Application | False |
| JVM Memory\|JVM Memory Pool<InstanceName>\|Peak Usage\|Used Memory | Java Application | False |
| JVM Memory\|JVM Memory Pool<InstanceName>\|Usage\|Committed Memory | Java Application | False |
| JVM Memory\|JVM Memory Pool<InstanceName>\|Usage\|Initial Memory | Java Application | False |
| JVM Memory\|JVM Memory Pool<InstanceName>\|Usage\|Maximum Memory | Java Application | False |
| JVM Memory\|JVM Memory Pool<InstanceName>\|Usage\|Used Memory | Java Application | False |
| JVM Memory\|Non Heap Memory Usage\|Committed Memory | Java Application | False |
| JVM Memory\|Non Heap Memory Usage\|Initial Memory | Java Application | False |
| JVM Memory\|Non Heap Memory Usage\|Maximum Memory | Java Application | False |
| JVM Memory\|Non Heap Memory Usage\|Used Memory | Java Application | False |
| JVM Memory\|Object Pending Finalization Count | Java Application | False |
| Uptime | Java Application | True |
| Threading\|Thread Count | Java Application | True |
| Process CPU Usage % | Java Application | False |
| System CPU Usage % | Java Application | False |
| System Load Average % | Java Application | False |

### JBoss Server Metrics

Metrics are collected for the JBoss Server application service.

#### Table 1499: JBoss Server Metrics

| Metric Name | Category | KPI |
|---|---|---|
| Buffer Pool<InstanceName>\|Count | Jboss Server | False |
| Buffer Pool<InstanceName>\|Memory Used | Jboss Server | False |
| Buffer Pool<InstanceName>\|Total Capacity | Jboss Server | False |
| Class Loading\|Loaded Class Count | Jboss Server | False |

| Metric Name | Category | KPI |
|---|---|---|
| Class Loading\|Total Loaded Class Count | Jboss Server | False |
| Class Loading\|Unloaded Class Count | Jboss Server | False |
| File Descriptor Usage\|Max File Descriptor Count | Jboss Server | False |
| File Descriptor Usage\|Open File Descriptor Count | Jboss Server | False |
| Http Listener<InstanceName>\|Bytes Received | Jboss Server | False |
| Http Listener<InstanceName>\|Bytes Sent | Jboss Server | False |
| Http Listener<InstanceName>\|Error Count | Jboss Server | False |
| Http Listener<InstanceName>\|Request Count | Jboss Server | False |
| Https Listener<InstanceName>\|Bytes Received | Jboss Server | False |
| Https Listener<InstanceName>\|Bytes Sent | Jboss Server | False |
| Https Listener<InstanceName>\|Error Count | Jboss Server | False |
| Https Listener<InstanceName>\|Request Count | Jboss Server | False |
| Process CPU Usage (%) | Jboss Server | False |
| System CPU Usage (%) | Jboss Server | False |
| System Load Average (%) | Jboss Server | False |
| Threading\|Daemon Thread Count | Jboss Server | False |
| Threading\|Peak Thread Count | Jboss Server | False |
| Threading\|Thread Count | Jboss Server | False |
| Threading\|Total Started Thread Count | Jboss Server | False |
| Uptime | Jboss Server | False |
| UTILIZATION\|Heap Memory Usage | Jboss Server | False |
| Application Availability | Jboss Server | False |
| Garbage Collection<InstanceName>\|Total Collection Count | Jboss JVM Garbage Collector | False |
| Garbage Collection<InstanceName>\|Total Collection Time | Jboss JVM Garbage Collector | False |
| JVM Memory\|Heap Memory Usage\|Committed Memory | Jboss JVM Memory | False |
| JVM Memory\|Heap Memory Usage\|Initial Memory | Jboss JVM Memory | False |
| JVM Memory\|Heap Memory Usage\|Maximum Memory | Jboss JVM Memory | False |
| JVM Memory\|Heap Memory Usage\|Used Memory | Jboss JVM Memory | True |
| JVM Memory\|Non Heap Memory Usage\|Committed Memory | Jboss JVM Memory | False |
| JVM Memory\|Non Heap Memory Usage\|Initial Memory | Jboss JVM Memory | False |
| JVM Memory\|Non Heap Memory Usage\|Maximum Memory | Jboss JVM Memory | False |
| JVM Memory\|Non Heap Memory Usage\|Used Memory | Jboss JVM Memory | False |
| JVM Memory\|Object Pending Finalization Count | Jboss JVM Memory | True |
| UTILIZATION\|Active Count | Jboss Datasource Pool | False |
| UTILIZATION\|Available Count | Jboss Datasource Pool | False |

| Metric Name | Category | KPI |
|---|---|---|
| JVM Memory Pool<InstanceName>\|Collection Usage\|Committed Memory | Jboss JVM Memory Pool | False |
| JVM Memory Pool<InstanceName>\|Collection Usage\|Initial Memory | Jboss JVM Memory Pool | False |
| JVM Memory Pool<InstanceName>\|Collection Usage\|Used Memory | Jboss JVM Memory Pool | False |
| JVM Memory Pool<InstanceName>\|Collection Usage\|Maximum Memory | Jboss JVM Memory Pool | False |
| JVM Memory Pool<InstanceName>\|Peak Usage\| Committed Memory | Jboss JVM Memory Pool | False |
| JVM Memory Pool<InstanceName>\|Peak Usage\| Initial Memory | Jboss JVM Memory Pool | False |
| JVM Memory Pool<InstanceName>\|Peak Usage\| Maximum Memory | Jboss JVM Memory Pool | False |
| JVM Memory Pool<InstanceName>\|Peak Usage\| Used Memory | Jboss JVM Memory Pool | False |
| JVM Memory Pool<InstanceName>\|Usage\| Committed Memory | Jboss JVM Memory Pool | False |
| JVM Memory Pool<InstanceName>\|Usage\|Initial Memory | Jboss JVM Memory Pool | False |
| JVM Memory Pool<InstanceName>\|Usage\|Maximum Memory | Jboss JVM Memory Pool | False |
| JVM Memory Pool<InstanceName>\|Usage\|Used Memory | Jboss JVM Memory Pool | False |

### HyperV Metrics

Metrics are collected for the HyperV application service.

**Table 1500: HyperV Metrics**

| Metric Name | Category | KPI |
|---|---|---|
| VM:Hyper-V Virtual Machine Health Summary\|Health Critical | HyperV | False |
| VM<instanceName>\|Physical Memory | HyperV | False |
| VM<instanceName>Hv VP 0\|Total Run Time | HyperV | False |
| VM<instanceName>\|Bytes Received | HyperV | False |
| VM<instanceName>\|Bytes Sent | HyperV | False |
| VM<instanceName>\|Error Count | HyperV | False |
| VM<instanceName>\|Latency | HyperV | False |
| VM<instanceName>\|Queue Length | HyperV | False |
| VM<instanceName>\|Throughput | HyperV | False |
| CPU<instanceName>\|Idle Time | HyperV | True |
| CPU<instanceName>\|Processor Time | HyperV | True |
| CPU<instanceName>\|User Time | HyperV | True |
| Disk<instanceName>\|Avg Disk Queue Length | HyperV | False |
| Disk<instanceName>\|Idle Time | HyperV | False |

| Metric Name | Category | KPI |
|---|---|---|
| Disk<instanceName>\|Read Time | HyperV | True |
| Disk<instanceName>\|Write Time | HyperV | True |
| Process<instanceName>\|Private Bytes | HyperV | False |
| Process<instanceName>\|Processor Time | HyperV | False |
| Process<instanceName>\|Thread Count | HyperV | False |
| Process<instanceName>\|User Time | HyperV | False |
| System\|Processes | HyperV | False |
| System\|Processor Queue Length | HyperV | False |
| System\|System UpTime | HyperV | False |
| Memory\|Available Bytes | HyperV | False |
| Memory\|Cache Bytes | HyperV | False |
| Memory\|Cache Faults | HyperV | False |
| Memory\|Pages | HyperV | False |
| Network<instanceName>\|Packets Outbound Error | HyperV | False |
| Network<instanceName>\|Packets Received Error | HyperV | False |
| Application Availability | HyperV | False |

### Oracle DB Metrics

Metrics are collected for the Oracle DB application service.

Oracle DB cannot be activated on Linux platforms.

**Table 1501: Oracle DB Metrics**

| Metric Name | Category | KPI |
|---|---|---|
| Utilization\|Active Sessions | OracleDB | True |
| Utilization\|Buffer CacheHit Ratio | OracleDB | False |
| Utilization\|Cursor CacheHit Ratio | OracleDB | False |
| Utilization\|Database Wait Time | OracleDB | False |
| Utilization\|Disk Sort persec | OracleDB | False |
| Utilization\|Enqueue Timeouts Persec | OracleDB | False |
| Utilization\|Global Cache Blocks Corrupted | OracleDB | False |
| Utilization\|Global Cache Blocks Lost | OracleDB | False |
| Utilization\|Library CacheHit Ratio | OracleDB | False |
| Utilization\|Logon persec | OracleDB | True |
| Utilization\|Memory Sorts Ratio | OracleDB | True |
| Utilization\|Rows persort | OracleDB | False |
| Utilization\|Service Response Time | OracleDB | False |
| Utilization\|Session Count | OracleDB | True |
| Utilization\|Session Limit | OracleDB | False |
| Utilization\|Shared Pool Free | OracleDB | False |
| Utilization\|Temp Space Used | OracleDB | False |
| Utilization\|Total Sorts persec | OracleDB | False |
| Utilization\|Physical Read Bytes Persc | OracleDB | False |
| Utilization\|Physical Read IO Requests Persc | OracleDB | False |

| Metric Name | Category | KPI |
|---|---|---|
| Utilization\|Physical Read Total Bytes Persec | OracleDB | False |
| Utilization\|Physical Reads Persec | OracleDB | True |
| Utilization\|Physical Reads Per Txn | OracleDB | False |
| Utilization\|Physical Write Bytes Persc | OracleDB | False |
| Utilization\|Physical Write IO Requests Persc | OracleDB | False |
| Utilization\|Physical Write Total Bytes Persc | OracleDB | False |
| Utilization\|Physical Writes Persc | OracleDB | True |
| Utilization\|Physical Writes Per Txn | OracleDB | False |
| Utilization\|User Commits Percentage | OracleDB | False |
| Utilization\|User Commits Persc | OracleDB | False |
| Utilization\|User Rollbacks Percentage | OracleDB | False |
| Utilization\|User Rollbacks persec | OracleDB | True |
| Utilization\|User Transaction Persec | OracleDB | False |
| Utilization\|Database Time Persc | OracleDB | False |
| Application Availability | Oracle DB | False |

### Cassandra Metrics

Metrics are collected for the Cassandra application service.

#### Table 1502: Cassandra Metrics

| Metric Name | Category | KPI |
|---|---|---|
| Cache<InstanceName>\|Capacity | Cassandra | False |
| Cache<InstanceName>\|Entries | Cassandra | True |
| Cache<InstanceName>\|HitRate | Cassandra | True |
| Cache<InstanceName>\|Requests | Cassandra | True |
| Cache<InstanceName>\|Size | Cassandra | False |
| ClientRequest<InstanceName>\|Failures | Cassandra | False |
| ClientRequest<InstanceName>\|Latency | Cassandra | False |
| ClientRequest<InstanceName>\|Timeouts | Cassandra | False |
| ClientRequest<InstanceName>\|Total Latency | Cassandra | False |
| ClientRequest<InstanceName>\|Unavailables | Cassandra | False |
| CommitLog\|Pending Tasks | Cassandra | False |
| CommitLog\|Total Commit Log Size | Cassandra | False |
| Compaction\|Bytes Compacted | Cassandra | False |
| Compaction\|Completed Tasks | Cassandra | False |
| Compaction\|Pending Tasks | Cassandra | False |
| Compaction\|Total Compactions Completed | Cassandra | False |
| Connected Native Clients | Cassandra | False |
| HeapMemoryUsage\|committed | Cassandra | False |
| HeapMemoryUsage\|init | Cassandra | False |
| HeapMemoryUsage\|max | Cassandra | False |

| Metric Name | Category | KPI |
|---|---|---|
| HeapMemoryUsage\|used | Cassandra | False |
| NonHeapMemoryUsage\|committed | Cassandra | False |
| NonHeapMemoryUsage\|init | Cassandra | False |
| NonHeapMemoryUsage\|max | Cassandra | False |
| NonHeapMemoryUsage\|used | Cassandra | False |
| ObjectPendingFinalizationCount | Cassandra | False |
| Storage\|Exceptions Count | Cassandra | False |
| Storage\|Load Count | Cassandra | False |
| Table<InstanceName>\|Coordinator Read Latency | Cassandra | False |
| Table<InstanceName>\|Live Diskspace Used | Cassandra | False |
| Table<InstanceName>\|Read Latency | Cassandra | False |
| Table<InstanceName>\|Total Diskspace Used | Cassandra | False |
| Table<InstanceName>\|Total Read Latency | Cassandra | False |
| Table<InstanceName>\|Total Write Latency | Cassandra | False |
| Table<InstanceName>\|Write Latency | Cassandra | False |
| ThreadPools<InstanceName>\|Active Tasks | Cassandra | False |
| ThreadPools<InstanceName>\|Currently Blocked Tasks | Cassandra | False |
| ThreadPools<InstanceName>\|Pending Tasks | Cassandra | False |
| Application Availability | Cassandra | False |

### MongoDB Metrics

Metrics are collected for the MongoDB application service.

**Table 1503: MongoDB Metrics**

| Metric Name | Category | KPI |
|---|---|---|
| UTILIZATION\|Active Reads | MongoDB | True |
| UTILIZATION\|Active Writes | MongoDB | True |
| UTILIZATION\|Connections Available | MongoDB | False |
| UTILIZATION\|Connections Total Created | MongoDB | False |
| UTILIZATION\|Current Connections | MongoDB | True |
| UTILIZATION\|Cursor Timed Out | MongoDB | True |
| UTILIZATION\|Deletes Per Sec | MongoDB | False |
| UTILIZATION\|Document Inserted | MongoDB | False |
| UTILIZATION\|Document Deleted | MongoDB | False |
| UTILIZATION\|Flushes Per Sec | MongoDB | False |
| UTILIZATION\|Inserts Per Sec | MongoDB | False |
| UTILIZATION\|Net Input Bytes | MongoDB | False |
| UTILIZATION\|Open Connections | MongoDB | True |
| UTILIZATION\|Page Faults Per Second | MongoDB | False |
| UTILIZATION\|Net Output Bytes | MongoDB | False |

| Metric Name | Category | KPI |
|---|---|---|
| UTILIZATION\|Queries Per Sec | MongoDB | False |
| UTILIZATION\|Queued Reads | MongoDB | True |
| UTILIZATION\|Queued Writes | MongoDB | True |
| UTILIZATION\|Total Available | MongoDB | False |
| UTILIZATION\|Total Deletes Per Sec | MongoDB | False |
| UTILIZATION\|Total Passes Per Sec | MongoDB | False |
| UTILIZATION\|Total Refreshing | MongoDB | False |
| UTILIZATION\|Updates Per Sec | MongoDB | False |
| UTILIZATION\|Volume Size MB | MongoDB | False |
| Application Availability | MongoDB | False |
| UTILIZATION\|Collection Stats | MongoDB DataBases | False |
| UTILIZATION\|Data Index Stats | MongoDB DataBases | True |
| UTILIZATION\|Data Indexes | MongoDB DataBases | False |
| UTILIZATION\|Data Size Stats | MongoDB DataBases | True |
| UTILIZATION\|Average Object Size stats | MongoDB DataBases | False |
| UTILIZATION\|Num Extents Stats | MongoDB DataBases | False |

### MS Exchange Metrics

Metrics are collected for the MS Exchange application service.

**Table 1504: MS Exchange Metrics**

| Metric Name | Category | KPI |
|---|---|---|
| Active Manager Server\|Active Manager Role | MS Exchange | False |
| Active Manager Server\|Database State Info Writes per second | MS Exchange | False |
| Active Manager Server\|GetServerForDatabase Server-Side Calls | MS Exchange | False |
| Active Manager Server\|Server-Side Calls per second | MS Exchange | True |
| Active Manager Server\|Total Number of Databases | MS Exchange | True |
| ActiveSync\|Average Request Time | MS Exchange | True |
| ActiveSync\|Current Requests | MS Exchange | False |
| ActiveSync\|Mailbox Search Total | MS Exchange | False |
| ActiveSync\|Ping Commands Pending | MS Exchange | False |
| ActiveSync\|Requests per second | MS Exchange | True |
| ActiveSync\|Sync Commands per second | MS Exchange | True |
| ASP.NET\|Application Restarts | MS Exchange | False |
| ASP.NET\|Request Wait Time | MS Exchange | True |
| ASP.NET\|Worker Process Restarts | MS Exchange | False |
| Autodiscover Service\|Requests per second | MS Exchange | True |
| Availability Service\|Average Time to Process a Free Busy Request | MS Exchange | True |

| Metric Name | Category | KPI |
|---|---|---|
| Outlook Web Access\|Average Search Time | MS Exchange | True |
| Outlook Web Access\|Requests per second | MS Exchange | False |
| Outlook Web Access\|Current Unique Users | MS Exchange | False |
| Application Availability | MS Exchange | False |
| Performance\|Database Cache Hit (%) | MS Exchange Database | False |
| Performance\|Database Page Fault Stalls per second | MS Exchange Database | True |
| Performance\|I/O Database Reads Average Latency | MS Exchange Database | True |
| Performance\|I/O Database Writes Average Latency | MS Exchange Database | True |
| Performance\|I/O Log Reads Average Latency | MS Exchange Database | False |
| Performance\|I/O Log Writes Average Latency | MS Exchange Database | False |
| Performance\|Log Record Stalls per second | MS Exchange Database | False |
| Performance\|Log Threads Waiting | MS Exchange Database | False |
| Performance\|I/O Database Reads Average Latency | MS Exchange Database Instance | False |
| Performance\|I/O Database Writes Average Latency | MS Exchange Database Instance | False |
| Performance\|Log Record Stalls per second | MS Exchange Database Instance | False |
| Performance\|Log Threads Waiting | MS Exchange Database Instance | False |
| Performance\|LDAP Read Time | MS Exchange Domain Controller | False |
| Performance\|LDAP Search Time | MS Exchange Domain Controller | False |
| Performance\|LDAP Searches Timed Out per minute | MS Exchange Domain Controller | False |
| Performance\|Long Running LDAP Operations per minute | MS Exchange Domain Controller | False |
| Performance\|Connection Attempts per second | MS Exchange Web Server | True |
| Performance\|Current Connections | MS Exchange Web Server | False |
| Performance\|Other Request Methods per second | MS Exchange Web Server | False |
| Process\|Handle Count | MS Exchange Windows Service | False |
| Process\|Memory Allocated | MS Exchange Windows Service | False |
| Process\|Processor Time (%) | MS Exchange Windows Service | True |
| Process\|Thread Count | MS Exchange Windows Service | False |
| Process\|Virtual Memory Used | MS Exchange Windows Service | False |
| Process\|Working Set | MS Exchange Windows Service | False |

### Microsoft SQL Server Metrics

Metrics are collected for the Microsoft SQL Server application service.

**Table 1505: MS SQL Metrics**

| Metric Name | Category | KPI |
|---|---|---|
| CPU<InstanceName>\|CPU Usage (%) | Microsoft SQL Server | False |
| Database IO\|Rows Reads Bytes/Sec | Microsoft SQL Server | False |
| Database IO\|Rows Reads/Sec | Microsoft SQL Server | False |
| Database IO\|Rows Writes Bytes/Sec | Microsoft SQL Server | False |
| Database IO\|Rows Writes/Sec | Microsoft SQL Server | False |

| Metric Name | Category | KPI |
|---|---|---|
| Performance\|Access Methods\|Full Scans per second | Microsoft SQL Server | False |
| Performance\|Access Methods\|Index Searches | Microsoft SQL Server | False |
| Performance\|Access Methods\|Page Splits per second | Microsoft SQL Server | False |
| Performance\|Broker Activation\|Stored Procedures Invoked per second | Microsoft SQL Server | False |
| Performance\|Buffer Manager\|Buffer cache hit ratio (%) | Microsoft SQL Server | True |
| Performance\|Buffer Manager\|Checkpoint Pages/sec | Microsoft SQL Server | True |
| Performance\|Buffer Manager\|Lazy writes per second | Microsoft SQL Server | True |
| Performance\|Buffer Manager\|Page life expectancy | Microsoft SQL Server | True |
| Performance\|Buffer Manager\|Page lookups per second | Microsoft SQL Server | False |
| Performance\|Buffer Manager\|Page reads per second | Microsoft SQL Server | False |
| Performance\|Buffer Manager\|Page writes per second | Microsoft SQL Server | False |
| Performance\|Databases\|Active Transactions | Microsoft SQL Server | True |
| Performance\|Databases\|Data File(s) Size | Microsoft SQL Server | True |
| Performance\|Databases\|Log Bytes Flushed/Sec | Microsoft SQL Server | False |
| Performance\|Databases\|Log File(s) Size | Microsoft SQL Server | False |
| Performance\|Databases\|Log File(s) Used Size | Microsoft SQL Server | False |
| Performance\|Databases\|Log Flush Wait Time | Microsoft SQL Server | False |
| Performance\|Databases\|Log Flushes per second | Microsoft SQL Server | False |
| Performance\|Databases\|Transactions per second | Microsoft SQL Server | False |
| Performance\|Databases\|Write Transactions per second | Microsoft SQL Server | False |
| Performance\|Databases\|XTP Memory Used | Microsoft SQL Server | False |
| Performance\|General Statistics\|Active temp Tables | Microsoft SQL Server | False |
| Performance\|General Statistics\|Logins per second | Microsoft SQL Server | False |
| Performance\|General Statistics\|Logouts per second | Microsoft SQL Server | False |
| Performance\|General Statistics\|Processes Blocked | Microsoft SQL Server | False |
| Performance\|General Statistics\|Temp Tables Creation Rate | Microsoft SQL Server | False |
| Performance\|General Statistics\|User Connections | Microsoft SQL Server | False |
| Performance\|Locks\|Average Wait Time | Microsoft SQL Server | False |
| Performance\|Locks\|Lock Requests per second | Microsoft SQL Server | False |
| Performance\|Locks\|Lock Wait Time | Microsoft SQL Server | True |
| Performance\|Locks\|Lock Waits per second | Microsoft SQL Server | True |
| Performance\|Locks\|Number of Deadlocks per second | Microsoft SQL Server | True |
| Performance\|Memory Manager\|Connection Memory | Microsoft SQL Server | False |
| Performance\|Memory Manager\|Lock Memory | Microsoft SQL Server | False |
| Performance\|Memory Manager\|Log Pool Memory | Microsoft SQL Server | False |

| Metric Name | Category | KPI |
|---|---|---|
| Performance\|Memory Manager\|Memory Grants Pending | Microsoft SQL Server | True |
| Performance\|Memory Manager\|SQL Cache Memory | Microsoft SQL Server | False |
| Performance\|Memory Manager\|Target Server Memory | Microsoft SQL Server | True |
| Performance\|Memory Manager\|Total Server Memory | Microsoft SQL Server | True |
| Performance\|Resource Pool Stats\|internal\|Active memory grant amount | Microsoft SQL Server | False |
| Performance\|Resource Pool Stats\|internal\|CPU Usage Percentage (%) | Microsoft SQL Server | False |
| Performance\|Resource Pool Stats\|internal\|Disk Read Bytes per second | Microsoft SQL Server | False |
| Performance\|Resource Pool Stats\|internal\|Disk Read IO | Microsoft SQL Server | False |
| Wait Stats:<InstanceName>\|Wait Time (ms) | Microsoft SQL Server | False |
| Wait Stats<InstanceName>\|Number of Waiting tasks (ms) | Microsoft SQL Server | False |
| Performance\|Resource Pool Stats\|internal\|Disk Read IO Throttled Per Second | Microsoft SQL Server | False |
| Performance\|Resource Pool Stats\|internal\|Disk Write Bytes per second (Bps) | Microsoft SQL Server | False |
| Performance\|Resource Pool Stats\|internal\|Disk Write IO Throttled per second | Microsoft SQL Server | False |
| Performance\|Resource Pool Stats\|internal\|Used Memory | Microsoft SQL Server | False |
| Performance\|SQL Statistics \| Batch Requests Per Second | Microsoft SQL Server | False |
| Performance\|SQL Statistics \| SQL Compilations per second | Microsoft SQL Server | False |
| Performance\|SQL Statistics \| SQL Re-Compilations per second | Microsoft SQL Server | False |
| Performance\|Transactions \| Free space in tempdb (KB) | Microsoft SQL Server | False |
| Performance\|Transactions \| Transactions | Microsoft SQL Server | False |
| Performance\|Transactions \| Version Store Size (KB) | Microsoft SQL Server | False |
| Performance\|User Settable Counter \| User Counter 0 to 10 | Microsoft SQL Server | False |
| Performance\|Workload Group Stats\|internal\|Active Requests | Microsoft SQL Server | False |
| Performance\|Workload Group Stats\|internal\|Blocked Tasks | Microsoft SQL Server | False |
| Performance\|Workload Group Stats\|internal\|CpU Usage (%) | Microsoft SQL Server | False |
| Performance\|Workload Group Stats\|internal\|Queued Requests | Microsoft SQL Server | False |

| Metric Name | Category | KPI |
|---|---|---|
| Performance\|Workload Group Stats\|internal\|Request Completed/sec | Microsoft SQL Server | False |
| Application Availability | Microsoft SQL Server | False |

There are no metrics collected for Microsoft SQL Server Database.

### MySQL Metrics

Metrics are collected for the MySQL application service.

**Table 1506: MySQL Metrics**

| Metric Name | Category | KPI |
|---|---|---|
| Aborted connection count | MySQL | True |
| Connection count | MySQL | True |
| Event wait average time | MySQL | False |
| Event wait count | MySQL | False |
| Binary Files\|Binary Files Count | MySQL | False |
| Binary Files\|Binary Size Bytes | MySQL | False |
| Global Status\|Aborted Clients | MySQL | False |
| Global Status\|Binlog Cache Disk Use | MySQL | False |
| Global Status\|Bytes Received | MySQL | False |
| Global Status\|Bytes Sent | MySQL | False |
| Global Status\|Connection Errors Accept | MySQL | False |
| Global Status\|Connection Errors Internal | MySQL | False |
| Global Status\|Connection Errors Max Connections | MySQL | False |
| Global Status\|Queries | MySQL | False |
| Global Status\|Threads Cached | MySQL | False |
| Global Status\|Threads Connected | MySQL | False |
| Global Status\|Threads Running | MySQL | False |
| Global Status\|Uptime | MySQL | False |
| Global Variables\|Delayed Insert Limit | MySQL | False |
| Global Variables\|Delayed Insert Timeout | MySQL | False |
| Global Variables\|Delayed Queue Size | MySQL | False |
| Global Variables\|Max Connect Errors | MySQL | False |
| Global Variables\|Max Connections | MySQL | False |
| Global Variables\|Max Delayed Threads | MySQL | False |
| Global Variables\|Max Error Count | MySQL | False |
| InnoDB\|All deadlock count | MySQL | False |
| InnoDB\|Buffer Pool Bytes Data | MySQL | False |
| InnoDB\|Buffer Pool Bytes Data | MySQL | False |
| InnoDB\|Buffer Pool Bytes Dirty | MySQL | False |
| InnoDB\|Buffer Pool Dump Status | MySQL | False |

| Metric Name | Category | KPI |
|---|---|---|
| InnoDB\|Buffer Pool Load Status | MySQL | False |
| InnoDB\|Buffer Pool Pages Data | MySQL | False |
| InnoDB\|Buffer Pool Pages Dirty | MySQL | False |
| InnoDB\|Buffer Pool Pages Flushed | MySQL | False |
| InnoDB\|Buffer pool size | MySQL | True |
| InnoDB\|Checksums | MySQL | False |
| InnoDB\|Open file count | MySQL | False |
| InnoDB\|Row lock average time | MySQL | False |
| InnoDB\|Row lock current waits | MySQL | False |
| InnoDB\|Row lock maximum time | MySQL | False |
| InnoDB\|Row lock time | MySQL | False |
| InnoDB\|Row lock waits | MySQL | True |
| InnoDB\|Table lock count | MySQL | False |
| Performance Table IO Waits\|IO Waits Total Delete | MySQL | False |
| Performance Table IO Waits\|IO Waits Total Fetch | MySQL | False |
| Performance Table IO Waits\|IO Waits Total Insert | MySQL | False |
| Performance Table IO Waits\|IO Waits Total Update | MySQL | False |
| Process List\|Connections | MySQL | False |
| Application Availability | MySQL | False |
| IO waits average time | MySQL Database | False |
| IO waits count | MySQL Database | True |
| Read high priority average time | MySQL Database | False |
| Read high priority count | MySQL Database | False |
| Write concurrent insert average time | MySQL Database | False |
| Write concurrent insert count | MySQL Database | False |

### NGINX Metrics

Metrics are collected for the NGINX application service.

**Table 1507: NGINX Metrics**

| Metric Name | Category | KPI |
|---|---|---|
| HTTP Status Info\|Accepts | Nginx | True |
| HTTP Status Info\|Active connections | Nginx | False |
| HTTP Status Info\|Handled | Nginx | True |
| HTTP Status Info\|Reading | Nginx | False |
| HTTP Status Info\|Requests | Nginx | False |
| HTTP Status Info\|Waiting | Nginx | True |
| HTTP Status Info\|Writing | Nginx | False |
| Application Availability | Nginx | False |

### Network Time Protocol Metrics

Metrics are collected for the Network Time Protocol application service.

#### Table 1508: Network Time Protocol Metrics

| Metric Name | Category | KPI |
|---|---|---|
| NTPD \|delay | Network Time Protocol | True |
| NTPD \| jitter | Network Time Protocol | True |
| NTPD \| offset | Network Time Protocol | True |
| NTPD \| poll | Network Time Protocol | False |
| NTPD \| reach | Network Time Protocol | True |
| NTPD \| when | Network Time Protocol | False |
| Application Availability | Network Time Protocol | False |

### Oracle WebLogic Server Metrics

Metrics are collected for the Oracle WebLogic Server application service.

#### Table 1509: Oracle WebLogic Server Metrics

| Metric Name | Category | KPI |
|---|---|---|
| UTILIZATION\|Process Cpu Load | Oracle WebLogic Server | True |
| UTILIZATION\|System Cpu Load | Oracle WebLogic Server | False |
| UTILIZATION\|System Load Average | Oracle WebLogic Server | False |
| Application Availability | Oracle WebLogic Server | False |
| UTILIZATION\|Collection Time | Weblogic Garbage Collector | True |
| UTILIZATION\|Connections HighCount | Weblogic JMS Runtime | True |
| UTILIZATION\|JMS Servers TotalCount | Weblogic JMS Runtime | False |
| UTILIZATION\|Active Total Count Used | Weblogic JTA Runtime | False |
| UTILIZATION\|Active Transactions TotalCount | Weblogic JTA Runtime | False |
| UTILIZATION\|Transaction Abandoned TotalCount | Weblogic JTA Runtime | True |
| UTILIZATION\|Transaction RolledBack App TotalCount | Weblogic JTA Runtime | True |
| UTILIZATION\|Heap Memory Usage | Weblogic JVM Memory | True |
| UTILIZATION\|Non Heap Memory Usage | Weblogic JVM Memory | False |
| UTILIZATION\|Peak Usage | Weblogic JVM Memory Pool | True |
| UTILIZATION\|Usage | Weblogic JVM Memory Pool | False |
| UTILIZATION\|UpTime | Weblogic JVM Runtime | False |

### Pivotal TC Server Metrics

Metrics are collected for the Pivotal TC Server application service.

**Table 1510: Pivotal TC Server Metrics**

| Metric Name | Category | KPI |
|---|---|---|
| Buffer Pool<InstanceName>\|Count | Pivotal TC Server | False |
| Buffer Pool<InstanceName>\|Memory Used | Pivotal TC Server | False |
| Buffer Pool<InstanceName>\|Total Capacity | Pivotal TC Server | False |
| Class Loading\|Loaded Class Count | Pivotal TC Server | False |
| Class Loading\|Total Loaded Class Count | Pivotal TC Server | False |
| Class Loading\|Unloaded Class Count | Pivotal TC Server | False |
| File Descriptor Usage\|Max File Descriptor Count | Pivotal TC Server | False |
| File Descriptor Usage\|Open File Descriptor Count | Pivotal TC Server | False |
| Garbage Collection:<InstanceName>\|Total Collection Count | Pivotal TC Server | False |
| Garbage Collection:<InstanceName>\|Total Collection Time | Pivotal TC Server | False |
| Process CPU Usage (%) | Pivotal TC Server | True |
| JVM Memory\|Heap Memory Usage\|Committed Memory | Pivotal TC Server | True |
| JVM Memory\|Heap Memory Usage\|Initial Memory | Pivotal TC Server | False |
| JVM Memory\|Heap Memory Usage\|Maximum Memory | Pivotal TC Server | False |
| JVM Memory\|Heap Memory Usage\|Used Memory | Pivotal TC Server | True |
| JVM Memory\|Non Heap Memory Usage\|Committed Memory | Pivotal TC Server | True |
| JVM Memory\|Non Heap Memory Usage\|Initial Memory | Pivotal TC Server | False |
| JVM Memory\|Non Heap Memory Usage\|Maximum Memory | Pivotal TC Server | False |
| JVM Memory\|Non Heap Memory Usage\|Used Memory | Pivotal TC Server | True |
| JVM Memory\|Number of Object Pending Finalization Count | Pivotal TC Server | True |
| JVM Memory\|Pool:<InstanceName>\|Peak Usage\| Committed Memory | Pivotal TC Server | False |
| JVM Memory\|Pool:<InstanceName>\|Peak Usage\| Initial Memory | Pivotal TC Server | False |
| JVM Memory\|Pool:<InstanceName>\|Peak Usage\| Maximum Memory | Pivotal TC Server | False |
| JVM Memory\|Pool:<InstanceName>\|Peak Usage\| Used Memory | Pivotal TC Server | False |
| JVM Memory\|Pool:<InstanceName>\|Usage\| Committed Memory | Pivotal TC Server | False |
| JVM Memory\|Pool:<InstanceName>\|Usage\|Initial Memory | Pivotal TC Server | False |
| JVM Memory\|Pool:<InstanceName>\|Usage\| Maximum Memory | Pivotal TC Server | False |

| Metric Name | Category | KPI |
|---|---|---|
| JVM Memory\|Pool:<InstanceName>\|Usage\|Used Memory | Pivotal TC Server | False |
| Process CPU Usage (%) | Pivotal TC Server | True |
| System CPU Usage (%) | Pivotal TC Server | True |
| Uptime | Pivotal TC Server | True |
| Threading\|Thread Count | Pivotal TC Server | False |
| System Load Average | Pivotal TC Server | False |
| Application Availability | Pivotal TC Server | False |
| Current Thread Count | Pivotal TC Server Thread Pool | False |
| Current Threads Busy | Pivotal TC Server Thread Pool | True |
| Total Request Bytes Received | Pivotal TC Server Thread Pool | False |
| Total Request Bytes Sent | Pivotal TC Server Thread Pool | False |
| Total Request Count | Pivotal TC Server Thread Pool | True |
| Total Request Error Count | Pivotal TC Server Thread Pool | True |
| Total Request Processing Time | Pivotal TC Server Thread Pool | True |
| JSP Count | Pivotal TC Server Web Module | False |
| JSP Reload Count | Pivotal TC Server Web Module | False |
| JSP Unload Count | Pivotal TC Server Web Module | False |

**PostgreSQL**
Metrics are collected for the PostgreSQL application service.

**Table 1511: PostgreSQL**

| Metric Name | Category | KPI |
|---|---|---|
| Buffers\|Buffers Allocated | PostgreSQL | False |
| Buffers\|Buffers Written by Backend | PostgreSQL | True |
| Buffers\|Buffers Written by Background Writer | PostgreSQL | True |
| Buffers\|Buffers Written During Checkpoints | PostgreSQL | True |
| Buffers\|fsync Call Executed by Backend | PostgreSQL | False |
| Checkpoints\|Checkpoints sync time | PostgreSQL | False |
| Checkpoints\|Checkpoints write time | PostgreSQL | False |
| Checkpoints\|Requested checkpoints performed count | PostgreSQL | False |
| Checkpoints\|Scheduled checkpoints performed count | PostgreSQL | False |
| Clean scan stopped count | PostgreSQL | False |
| Application Availability | PostgreSQL | False |
| Disk Blocks\|Blocks Cache Hits | PostgreSQL Database | False |
| Disk Blocks\|Blocks Read | PostgreSQL Database | False |
| Disk Blocks\|Blocks Read Time | PostgreSQL Database | False |
| Disk Blocks\|Blocks Write Time | PostgreSQL Database | False |
| Statistics\|Backends Connected | PostgreSQL Database | False |

| Metric Name | Category | KPI |
|---|---|---|
| Statistics\|Data Written by Queries | PostgreSQL Database | True |
| Statistics\|Deadlocks Detected | PostgreSQL Database | True |
| Statistics\|Queries Cancelled | PostgreSQL Database | True |
| Statistics\|Temp Files Created by Queries | PostgreSQL Database | False |
| Transactions\|Transactions Committed | PostgreSQL Database | True |
| Transactions\|Transactions Rolled Back | PostgreSQL Database | True |
| Tuples\|Tuples Deleted | PostgreSQL Database | True |
| Tuples\|Tuples Fetched | PostgreSQL Database | True |
| Tuples\|Tuples Inserted | PostgreSQL Database | True |
| Tuples\|Tuples Returned | PostgreSQL Database | True |
| Tuples\|Tuples Updated | PostgreSQL Database | True |

### RabbitMQ Metrics

Metrics are collected for the RabbitMQ application service.

**Table 1512: RabbitMQ Metrics**

| Metric Name | Category | KPI |
|---|---|---|
| CPU\|Limit | RabbitMQ | False |
| CPU\|Used | RabbitMQ | True |
| Disk\|Free | RabbitMQ | False |
| Disk\|Free limit | RabbitMQ | False |
| FileDescriptor\|Total | RabbitMQ | False |
| FileDescriptor\|Used | RabbitMQ | False |
| Memory\|Limit | RabbitMQ | False |
| Memory\|Used | RabbitMQ | True |
| Messages\|Acked | RabbitMQ | False |
| Messages\|Delivered | RabbitMQ | False |
| Messages\|Delivered get | RabbitMQ | False |
| Messages\|Published | RabbitMQ | False |
| Messages\|Ready | RabbitMQ | False |
| Messages\|Unacked | RabbitMQ | False |
| Socket\|Limit | RabbitMQ | False |
| Socket\|Used | RabbitMQ | True |
| UTILIZATION\|Channels | RabbitMQ | True |
| UTILIZATION\|Connections | RabbitMQ | True |
| UTILIZATION\|Consumers | RabbitMQ | True |
| UTILIZATION\|Exchanges | RabbitMQ | True |
| UTILIZATION\|Messages | RabbitMQ | True |
| UTILIZATION\|Queues | RabbitMQ | True |
| Application Availability | RabbitMQ | False |

| Metric Name | Category | KPI |
|---|---|---|
| Messages\|Publish in | RabbitMQ Exchange | False |
| Messages\|Publish out | RabbitMQ Exchange | False |
| Consumer Utilisation | RabbitMQ Queue | False |
| Consumers | RabbitMQ Queue | False |
| Memory | RabbitMQ Queue | False |
| Messages\|Ack | RabbitMQ Queue | False |
| Messages\|Ack rate | RabbitMQ Queue | False |
| Messages\|Deliver | RabbitMQ Queue | False |
| Messages\|Deliver get | RabbitMQ Queue | False |
| Messages\|Persist | RabbitMQ Queue | False |
| Messages\|Publish | RabbitMQ Queue | False |
| Messages\|Publish rate | RabbitMQ Queue | False |
| Messages\|Ram | RabbitMQ Queue | False |
| Messages\|Ready | RabbitMQ Queue | False |
| Messages\|Redeliver | RabbitMQ Queue | False |
| Messages\|Redeliver rate | RabbitMQ Queue | False |
| Messages\|Space | RabbitMQ Queue | False |
| Messages\|Unack | RabbitMQ Queue | False |
| Messages\|Unacked | RabbitMQ Queue | False |
| Messages | RabbitMQ Queue | False |

There are no metrics collected for RabbitMQ Virtual Host.

**Riak KV Metrics**
Metrics are collected for the Riak KV application service.

**Table 1513: Riak KV Metrics**

| Metric Name | Category | KPI |
|---|---|---|
| UTILIZATION\|CPU Average | Riak KV | False |
| UTILIZATION\|Memory Processes | Riak KV | False |
| UTILIZATION\|Memory Total | Riak KV | False |
| UTILIZATION\|Node GETs | Riak KV | True |
| UTILIZATION\|Node GETs Total | Riak KV | False |
| UTILIZATION\|Node PUTs | Riak KV | True |
| UTILIZATION\|Node PUTs Total | Riak KV | False |
| UTILIZATION\|PBC Active | Riak KV | True |
| UTILIZATION\|PBC Connects | Riak KV | True |
| UTILIZATION\|Read Repairs | Riak KV | True |
| UTILIZATION\|vNODE Index Reads | Riak KV | True |
| UTILIZATION\|vNODE Index Writes | Riak KV | True |
| Application Availability | Riak KV | False |

### SharePoint Metrics

Metrics are collected for the SharePoint Server application service.

**Table 1514: SharePoint Server Metrics**

| Metric Name | Category | KPI |
|---|---|---|
| Sharepoint Foundation\|Active Threads | SharePoint Server | True |
| Sharepoint Foundation\|Current Page Requests | SharePoint Server | False |
| Sharepoint Foundation\|Executing SQL Queries | SharePoint Server | False |
| Sharepoint Foundation\|Executing Time/Page Request | SharePoint Server | True |
| Sharepoint Foundation\|Incoming Page Requests Rate | SharePoint Server | False |
| Sharepoint Foundation\|Object Cache Hit Count | SharePoint Server | False |
| Sharepoint Foundation\|Reject Page Requests Rate | SharePoint Server | False |
| Sharepoint Foundation\|Responded Page Requests Rate | SharePoint Server | True |
| SQL query executing time | SharePoint Server | False |
| Application Availability | SharePoint Server | False |
| Network\|Received Data Rate | SharePoint Web Server | True |
| Network\|Sent Data Rate | SharePoint Web Server | True |
| Process\|Processor Time (%) | SharePoint Windows Service | False |
| Process\|Threads | SharePoint Windows Service | False |

### WebSphere Metrics

Metrics are collected for the WebSphere application service.

**Table 1515: WebSphere Metrics**

| Metric Name | Category | KPI |
|---|---|---|
| Thread Pool\|Active Count\| Current | Thread Pool | False |
| Thread Pool\|Active Count\|High | Thread Pool | False |
| Thread Pool\|Active Count\|Low | Thread Pool | False |
| Thread Pool\|Active Count\|Lower | Thread Pool | False |
| Thread Pool\|Active Count\|Upper | Thread Pool | False |
| JDBC\|Close Count | JDBC | False |
| JDBC\|Create Count | JDBC | False |
| JDBC\|JDBC Pool Size\|Average | JDBC | False |
| JDBC\|JDBC Pool Size\|Current | JDBC | False |
| JDBC\|JDBC Pool Size\|Lower | JDBC | False |
| JDBC\|JDBC Pool Size\|Upper | JDBC | False |
| Garbage Collection<InstanceName>\|Total Collection Count | WebSphere | False |

| Metric Name | Category | KPI |
|---|---|---|
| Garbage Collection<InstanceName>\|Total Collection Time | WebSphere | False |
| JVM Memory\|Heap Memory Usage\|Committed Memory | WebSphere | False |
| JVM Memory\|Heap Memory Usage\|Initial Memory | WebSphere | False |
| JVM Memory\|Heap Memory Usage\|Maximum Memory | WebSphere | False |
| JVM Memory\|Heap Memory Usage\|Used Memory | WebSphere | False |
| JVM Memory\|Non Heap Memory Usage\|Committed Memory | WebSphere | False |
| JVM Memory\|Non Heap Memory Usage\|Initial Memory | WebSphere | False |
| JVM Memory\|Non Heap Memory Usage\|Maximum Memory | WebSphere | False |
| JVM Memory\|Non Heap Memory Usage\|Used Memory | WebSphere | False |
| JVM Memory\|Number of Object Pending Finalization Count | WebSphere | False |
| JVM Memory\| Pool<InstanceName>\|Peak Usage\|Committed Memory | WebSphere | False |
| JVM Memory\| Pool<InstanceName>\|Peak Usage\|Initial Memory | WebSphere | False |
| JVM Memory\| Pool<InstanceName>\|Peak Usage\|Maximum Memory | WebSphere | False |
| JVM Memory\| Pool<InstanceName>\|Peak Usage\|Used Memory | WebSphere | False |
| JVM Memory\| Pool<InstanceName>\|Usage\| Committed Memory | WebSphere | False |
| JVM Memory\| Pool<InstanceName>\|Usage\| Initial Memory | WebSphere | False |
| JVM Memory\| Pool<InstanceName>\|Usage\| Maximum Memory | WebSphere | False |
| JVM Memory\| Pool<InstanceName>\|Usage\| Used Memory | WebSphere | False |
| Process Cpu Load | WebSphere | False |

| Metric Name | Category | KPI |
|---|---|---|
| System Cpu Load | WebSphere | False |
| System Load Average | WebSphere | False |
| Application Availability | WebSphere | False |

## Windows Service Metrics

Metrics are collected for Windows services.

### Table 1516: Windows Service Metrics

| Metric Name | Category | KPI |
|---|---|---|
| AVAILABILITY\|Resource Availability | Services | False |
| UTILIZATION\|Memory Usage(%) | Services | False |
| UTILIZATION\|CPU Usage(%) | Services | False |

## Linux Process Metrics

Metrics are collected for Linux services.

### Table 1517: Linux Process Metrics

| Metric Name | Category | KPI |
|---|---|---|
| AVAILABILITY\|Resource Availability | Processes | False |
| UTILIZATION\|Memory Usage (%) | Processes | False |
| UTILIZATION\|CPU Usage (%) | Processes | False |
| UTILIZATION\|Number of Processes | Processes | False |

## Remote Check Metrics

Metrics are collected for object types such as HTTP, ICMP, TCP, and UDP.

**HTTP Metrics**
VCF Operations discovers metrics for HTTP remote checks.

**HTTP Metrics**

### Table 1518: HTTP Metrics

| Metric Name | KPI |
|---|---|
| Availability | False |
| Content Length | False |
| Response Code | False |
| Response Time | True |
| Result Code | False |

### ICMP Metrics
VCF Operations discovers metrics for the ICMP object type.

**Table 1519: ICMP Metrics**

| Metric Name | KPI |
|---|---|
| Availability | False |
| Average Response Time | True |
| Packet Loss (%) | False |
| Packets Received | False |
| Packets Transmitted | False |
| Result Code | False |

### TCP Metrics
VCF Operations discovers metrics for the TCP object type.

**Table 1520: TCP Metrics**

| Metric Name | KPI |
|---|---|
| Availability | False |
| Response Time | True |
| Result Code | False |

### UDP Metrics
VCF Operations discovers metrics for the UDP object type.

**Table 1521: UDP Metrics**

| Metric Name | KPI |
|---|---|
| Availability | False |
| Response Time | True |
| Result Code | False |

## VeloCloud Application Service Metrics

Metrics are collected for application services supported by VeloCloud.

### VeloCloud Gateway Metrics
Metrics are collected for the VeloCloud Gateway.

**Table 1522: VeloCloud Gateway Metrics**

| Component | Metrics |
|---|---|
| DPDK | DPDK:mbuf \| pool free |
| NAT | NAT \| Active Flows (%) |
| | NAT \| Active Flows |

| Component | Metrics |
|---|---|
| | NAT \| Active Routes |
| | NAT \| Active Routes Used (%) |
| | NAT \| Connected Peers |
| | NAT \| NAT Entries |
| NTP Server | NTP Server:ntp.ubuntu.com \| offset value |
| Summary | Summary \| Active Tunnels Count (%) |
| | Summary \| Average Packets Dropped |
| | Summary \| Average wMarkDrop |
| | Summary \| BGP Enabled VRFs |
| | Summary \| BGP Neighbors |
| | Summary \| CLR Count |
| | Summary \| Connected Edges |
| | Summary \| NAT |
| | Summary \| SSH Failed Login |
| | Summary \| Unstable Path Percentage |
| | Summary \| VMCP CTRL Drop Count |
| | Summary \| VMCP TX Drop Count |
| VC Queue | VC Queue \| ipv4_bh packet drop |
| VCMP Tunnel | VCMP Tunnel \| ctrl_0 packet drop |
| | VCMP Tunnel \| ctrl_1 packet drop |
| | VCMP Tunnel \| data_0 packet drop |
| | VCMP Tunnel \| data_1 packet drop |
| | VCMP Tunnel \| init packet drop |

### VeloCloud Orchestrator Metrics

Metrics are collected for the VeloCloud Orchestrator.

#### Table 1523: VeloCloud Orchestrator Metrics

| Component | Metrics |
|---|---|
| General | General \| Free Memory (%) |
| | General \| Status |

### Metrics - Ngnix

Metrics are collected for the VeloCloud Ngnix.

#### Table 1524: Ngnix Metrics

| Component | Metrics |
|---|---|
| HTTP Status Info | HTTP Status Info \| Accepts |
| | HTTP Status Info \| Active Connections |
| | HTTP Status Info \| Handled |

| Component | Metrics |
|---|---|
| | HTTP Status Info \| Reading |
| | HTTP Status Info \| Requests |
| | HTTP Status Info \| Waiting |
| | HTTP Status Info \| Writing |

### Metrics - Redis

Metrics are collected for the VeloCloud Redis.

#### Table 1525: Redis Metrics

| Component | Metrics |
|---|---|
| Publish Subscribe. | Publish Subscribe \| Channels |
| Total | Total \| Commands Processed |
| | Total \| Connections Received |
| Used | Used \| CPU |
| | Used \| Memory |
| | Used \| Peak Memory |

### Metrics - ClickHouse

Metrics are collected for the VeloCloud Clickhouse.

#### Table 1526: Clickhouse Metrics

| Component | Metrics |
|---|---|
| Background | Background \| Pool Task |
| Buffer | Buffers \| Allocation (Bytes) |
| | Buffers \| Compressed Read Buffer (Bytes) |
| | Buffers \| Compressed Read Buffer Blocks |
| | Buffers \| IO Allocation (Bytes) |
| | Buffers \| Storage Buffer (Bytes) |
| | Buffers \| Storage Buffer Rows |
| Events | Events \| Context Lock |
| | Events \| Disk Write Elapsed (µs) |
| | Events \| File Open |
| | Events \| Function Execute |
| | Events \| Hard Page Faults |
| | Events \| Lock Readers Wait (µs) |
| | Events \| OS IO wait (ms) |
| | Events \| OS Write (Bytes) |
| | Events \| Query |
| | Events \| Readers Wait (ms) |
| | Events \| Real Time |

| Component | Metrics |
|---|---|
| | Events \| Soft Page Faults (µs) |
| | Events \| System Time (µs) |
| | Events \| User Time (µs) |
| Global Thread | Global \| Global Thread |
| | Global \| Global Thread Active |
| Local Thread | Local \| Local Thread |
| | Local \| Local Thread Active |
| Replicas | Replicas \| Max Absolute Delay |
| | Replicas \| Max Insert In Queue |
| | Replicas \| Max Merge In Queue |
| | Replicas \| Max Queue Size |
| | Replicas \| Max Relative Delay |
| | Replicas \| Total Insert In Queue |
| | Replicas \| Total Merge Queues |
| | Replicas \| Total Queue Size |
| Summary | Summary \| Background Pool Task |
| | Summary \| Dict Cache Requests |
| | Summary \| File Open Writes |
| | Summary \| Merge |
| | Summary \| Number of Databases |
| | Summary \| Number of Distributed Send |
| | Summary \| Number of Tables |
| | Summary \| Read |
| | Summary \| Replicated Checks |
| | Summary \| Storage Buffer Rows |
| | Summary \| Uncompressed Cache Cells |
| | Summary \| Uptime |
| | Summary \| Write |
| | Summary \| Zookeeper Session |
| | Summary \| Zookeeper Watch |
| Write Buffer | Write Buffer \| File Descriptor Write |
| Replicated | Replicated Fetch |
| Memory | Memory Tracking |
| Query | Query Thread |

## Service Discovery Metrics

Service discovery discovers metrics for several objects. It also discovers CPU and memory metrics for discovered services.

### Virtual Machine Metrics

Service Discovery discovers metrics for virtual machines.

**Table 1527: Virtual Machine Metrics**

| Metric Name | Description |
|---|---|
| Guest OS Services\|Total Number of Services | Number of out-of-the-box and user-defined services discovered in the VM. |
| Guest OS Services\|Number of User Defined Services | Number of user-defined services discovered in the VM. |
| Guest OS Services\|Number of OOTB Services | Number of out-of-the-box services discovered in the VM. |
| Guest OS Services\|Number of Outgoing Connections | Number of outgoing connection counts from the discovered services. |
| Guest OS Services\|Number of Incoming Connections | Number of incoming connection counts to the discovered services. |

## Service Summary Metrics

Service discovery discovers summary metrics for the service object. The object is a single service object.

**Table 1528: Service Summary Metrics**

| Metric Name | Description |
|---|---|
| Summary\|Incoming Connections Count | Number of incoming connections. |
| Summary\|Outgoing Connections Count | Number of outgoing connections. |
| Summary\|Connections Count | Number of incoming and outgoing connections. |
| Summary\|Pid | Process ID. |

## Service Performance Metrics

Service discovery discovers performance metrics for the service object. The object is a single service object.

**Table 1529: Service Performance Metrics**

| Metric Name | Description |
|---|---|
| Performance metrics group\|CPU | CPU usage in percentage. |
| Performance metrics group\|Memory | Memory usage in KB. |
| Performance metrics group\|IO Read Throughput | IO read throughput in KBps. |
| Performance metrics group\|IO Write Throughput | IO write throughput in KBps. |

## Service Type Metrics

Service discovery discovers metrics for service type objects.

**Table 1530: Service Type Metrics**

| Metric Name | Description |
|---|---|
| Number of instances | Number of instances of this service type. |

## Calculated Metrics

VCF Operations calculates metrics for capacity, badges, and the health of the system. Calculated metrics apply to a subset of objects found in the `describe.xml` file that describes each adapter.

From data that the vCenter adapter collects, VCF Operations calculates metrics for objects of type:

- vSphere World
- Virtual Machine
- Host System
- Datastore

From data that the VCF Operations adapter collects, VCF Operations calculates metrics for objects of type:

- Node
- Cluster

### Capacity Analytics Generated Metrics

The capacity engine computes and publishes metrics that can be found in the Capacity Analytics Generated group. These metrics help you to plan your resource use based on consumer demand.

#### Capacity Analytics Generated Metrics Group

Capacity analytics uses the capacity engine to analyze historical utilization and generate projected utilization. The engine takes the Demand and Usable Capacity (Total Capacity - HA - buffer) metrics as input and calculates the output metrics that belong to the capacity analytics generated metrics group.

The capacity analytics generated metrics group contains containers and each container contains three output metrics, which are Capacity Remaining, Recommended Size, and Recommended Total Capacity. It also contains the Capacity Remaining Percentage and Time Remaining metrics, which show the most constrained values of the containers.

For the capacity metrics group, full metric names include the name of the resource container. For example, if recommended size metrics are computed for CPU or memory, the actual metric names appear as cpu|demand|recommendedSize or mem|demand|recommendedSize.

**Table 1531: Capacity Metrics Group**

| Metric Name | Description |
|---|---|
| Time Remaining (Day(s)) | The number of days remaining till the projected utilization crosses the threshold for the usable capacity.<br>Key: timeRemaining |
| Capacity Remaining | Capacity remaining is the maximum point between the usable capacity now and the projected utilization for 3 days into the future. If the projected utilization is above 100% of the usable capacity, Capacity Remaining is 0.<br>Key: capacityRemaining |
| Capacity Remaining Percentage (%) | The percentage of Capacity Remaining of the most constrained resource with respect to the usable capacity.<br>Key: capacityRemainingPercentage |
| Recommended Size | The maximum projected utilization for the projection period from the current time to 30 days after the warning threshold value for time remaining. The warning threshold is the period during which the time remaining is green. Recommended Size excludes HA settings.<br>Key: recommendedSize |
| Recommended Total Capacity | The maximum projected utilization for the projection period from the current time to 30 days after the warning threshold value for time remaining. Recommended Total Capacity excludes HA settings.<br>Key: recommendedTotalCapacity |

## Capacity Analytics Generated Allocation Metrics

Capacity allocation metrics provide information about the allotment of capacity for Cluster Compute and Datastore Cluster Resources.

| Metric Name | Description |
|---|---|
| Capacity Analytics Generated\|CPU\|Allocation\|Capacity Remaining (vCPUs) | For vSphere objects published on Cluster Compute Resource only. Capacity Remaining based on overcommit ratio (if configured in effective policy).<br>Key: OnlineCapacityAnalytics\|cpu\|alloc\|capacityRemaining |
| Capacity Analytics Generated\|CPU\|Allocation\|Recommended Total Capacity (Cores) | For vSphere objects published on Cluster Compute Resource only. The recommended level of total capacity, to maintain a green state for time remaining for the given object.<br>Key: OnlineCapacityAnalytics\|cpu\|alloc\|recommendedTotalSize |
| Capacity Analytics Generated\|CPU\|Allocation\|Time Remaining (Day(s)) | For vSphere objects published on Cluster Compute Resource only. The number of days remaining is calculated for both group and container. It calculates the time remaining before the resources run out.<br>Key: OnlineCapacityAnalytics\|cpu\|alloc\|timeRemaining |
| CPU\|Allocation\|Usable Capacity after HA and Buffer (vCPUs) | For vSphere objects published on Cluster Compute Resource only. The usable capacity (total capacity - HA) based on configured overcommit ratio.<br>Key: cpu\|alloc\|usableCapacity |
| Capacity Analytics Generated\|CPU\|Allocation\|Recommended Size (Cores) | For vSphere objects published on Cluster Compute Resource only. The recommended level of usable capacity (total capacity - HA), to maintain a green state for time remaining for the given object.<br>Key: OnlineCapacityAnalytics\|cpu\|alloc\|recommendedSize |
| vRealize Operations Manager Generated Properties\|CPU\| Allocation\|Overcommit Ratio Setting | For vSphere objects published on Cluster Compute Resource only. This property shows the allocation overcommit ratio for CPU provided in effective policy.<br>Key: System Properties\|cpu\|alloc\|overcommitRatioSetting |
| vRealize Operations Manager Generated Properties\|CPU\| Allocation\|Buffer (%) | CPU buffer percent defined by policy setting for allocation based capacity computation.<br>Key: Properties\|cpu\|alloc\|bufferSetting |
| Capacity Analytics Generated\|Memory\|Allocation\|Capacity Remaining (KB) | For vSphere objects published on Cluster Compute Resource only. Capacity Remaining based on overcommit ratio (if configured in effective policy).<br>Key: OnlineCapacityAnalytics\|mem\|alloc\|capacityRemaining |
| Capacity Analytics Generated\|Memory\|Allocation\|Recommended Total Capacity (KB) | For vSphere objects published on Cluster Compute Resource only. The recommended level of total capacity, to maintain a green state for time remaining for the given object.<br>Key: OnlineCapacityAnalytics\|mem\|alloc\|recommendedTotalSize |
| Capacity Analytics Generated\|Memory\|Allocation\|Time Remaining (Day(s)) | For vSphere objects published on Cluster Compute Resource only. The number of days remaining is calculated for both group and container. It calculates the time remaining before the resources run out.<br>Key: OnlineCapacityAnalytics\|mem\|alloc\|timeRemaining |
| Memory\|Allocation\|Usable Capacity (KB) | For vSphere objects published on Cluster Compute Resource only. The usable capacity (total capacity - HA) based on configured overcommit ratio.<br>Key: mem\|alloc\|usableCapacity |

| Metric Name | Description |
|---|---|
| Capacity Analytics Generated\|Memory\|Allocation\|Recommended Size (KB) | For vSphere objects published on Cluster Compute Resource only. The recommended level of usable capacity (total capacity - HA), to maintain a green state for time remaining for the given object.<br>Key: OnlineCapacityAnalytics\|mem\|alloc\|recommendedSize |
| vRealize Operations Manager Generated Properties\|Memory\| Allocation\|Overcommit Ratio Setting | For vSphere objects published on Cluster Compute Resource only. This property shows the allocation overcommit ratio for Memory provided in effective policy.<br>Key: System Properties\|mem\|alloc\|overcommitRatioSetting |
| vRealize Operations Manager Generated Properties\|Memory\| Allocation\|Buffer (%) | Memory buffer percent defined by policy setting for allocation based capacity computation.<br>Key: System Properties\|mem\|alloc\|bufferSetting |
| Capacity Analytics Generated\|Disk Space\|Allocation\|Capacity Remaining (GB) | For vSphere objects published on Cluster Compute Resource and Datastore Cluster Resource. Capacity Remaining based on overcommit ratio (if configured in effective policy).<br>Key: OnlineCapacityAnalytics\|diskspace\|alloc\|capacityRemaining |
| Capacity Analytics Generated\|Disk Space\|Allocation\| Recommended Size (GB) | For vSphere objects published on Cluster Compute Resource and Datastore Cluster Resource. The recommended level of total capacity to maintain a green state for time remaining for the given object.<br>Key: OnlineCapacityAnalytics\|diskspace\|alloc\|recommendedSize |
| Capacity Analytics Generated\|Disk Space\|Allocation\|Time Remaining (Day(s)) | For vSphere objects published on Cluster Compute Resource and Datastore Cluster Resource. The number of days remaining is calculated for both group and container. It calculates the time remaining before the resources run out.<br>Key: OnlineCapacityAnalytics\|diskspace\|alloc\|timeRemaining |
| Disk Space\|Allocation\|Usable Capacity (GB) | For vSphere objects published on Cluster Compute Resource and Datastore Cluster Resource. Usable capacity based on overcommit ratio (if configured in effective policy).<br>Key: diskspace\|alloc\|usableCapacity |
| vRealize Operations Manager Generated Properties\|Disk Space\| Allocation\|Overcommit Ratio Setting | For vSphere objects published on Cluster Compute Resource and Datastore Cluster Resource. This property shows the allocation overcommit ratio for Disk Space provided in effective policy.<br>key: System Properties\|diskspace\|alloc\|overcommitRatioSetting |
| vRealize Operations Manager Generated Properties\|Disk Space\| Allocation\|Buffer (%) | Disk Space buffer percent defined by policy setting for allocation based capacity computation.<br>Key: System Properties\|diskspace\|alloc\|bufferSetting |

## Capacity Analytics Generated Profiles Metrics

Profiles metrics provide information about the profile specific capacity for Cluster Compute, Datastore Cluster, Data Center, Custom Data Center, and vCenter Server resources.

| Metric Name | Description |
|---|---|
| Capacity Analytics Generated\|Capacity Remaining (Profile) | Published on Cluster Compute Resource. Calculated as a minimum of all Profiles\|capacityRemainingProfile_<profile uuid> metrics.<br>Key: OnlineCapacityAnalytics\|capacityRemainingProfile |
| Capacity Analytics Generated\|Capacity Remaining (Profile) | Published on Datastore Cluster Resource. Calculated as a minimum of all Profiles\|capacityRemainingProfile_<profile uuid> metrics.<br>Key: OnlineCapacityAnalytics\|capacityRemainingProfile |
| Capacity Analytics Generated\|Capacity Remaining (Profile) | Published on Data Center, Custom Data Center and vCenter Server Resources. Computed as a sum of OnlineCapacityAnalytics\|capacityRemainingProfile metric of descendant Cluster Compute Resources.<br>Key: OnlineCapacityAnalytics\|capacityRemainingProfile |

## Capacity Demand Model Metrics

Demand model metrics provide information about the usable capacity and projected utilization of resources across VMs, Host Systems, Cluster Compute, Datastore Cluster, Data Center, Custom Data Center, and vCenter Server resources.

| Metric Name | Description |
|---|---|
| Capacity Analytics Generated\|CPU\|Capacity Remaining (MHz) | Published on Virtual Machine. The max point between the usable capacity and the projected utilization between now and three days.<br>Key: OnlineCapacityAnalytics\|cpu\|capacityRemaining |
| Capacity Analytics Generated\|CPU\|Recommended Size (MHz) | Published on Virtual Machine. The recommended level of usable capacity (total capacity - HA) to maintain a green state for the remaining time.<br>Key: OnlineCapacityAnalytics\|cpu\|recommendedSize |
| Capacity Analytics Generated\|CPU\|Time Remaining (Day(s)) | Published on Virtual Machine. The number of days remaining till the projected utilization crosses the threshold for the usable capacity.<br>Key: OnlineCapacityAnalytics\|cpu\|timeRemaining |
| Capacity Analytics Generated\|Disk Space\|Capacity Remaining (GB) | Published on Virtual Machine. The max point between the usable capacity and the projected utilization between now and three days into the future.<br>Key: OnlineCapacityAnalytics\|diskspace\|capacityRemaining |
| Capacity Analytics Generated\|Disk Space\|Recommended Size (GB) | Published on Virtual Machine. The recommended level of usable capacity (total capacity - HA) to maintain a green state for the remaining time.<br>Key: OnlineCapacityAnalytics\|diskspace\|recommendedSize |
| Capacity Analytics Generated\|Disk Space\|Time Remaining (Day(s)) | Published on Virtual Machine. The number of days remaining till the projected utilization crosses the threshold for the usable capacity.<br>Key: OnlineCapacityAnalytics\|diskspace\|timeRemaining |

| Metric Name | Description |
|---|---|
| Capacity Analytics Generated\|Memory\|Capacity Remaining (KB) | Published on Virtual Machine. The max point between the usable capacity and the projected utilization between now and three days into the future.<br>Key: OnlineCapacityAnalytics\|mem\|capacityRemaining |
| Capacity Analytics Generated\|Memory\|Recommended Size (KB) | Published on Virtual Machine. The recommended level of usable capacity (total capacity - HA) to maintain a green state for the remaining time.<br>Key: OnlineCapacityAnalytics\|mem\|recommendedSize |
| Capacity Analytics Generated\|Memory\|Time Remaining (Day(s)) | Published on Virtual Machine. The number of days remaining till the projected utilization crosses the threshold for the usable capacity.<br>Key: OnlineCapacityAnalytics\|mem\|timeRemaining |
| Capacity Analytics Generated\|CPU\|Demand\|Capacity Remaining (MHz) | Published on Host System. The max point between the usable capacity and the projected utilization between now and three days into the future.<br>Key: OnlineCapacityAnalytics\|cpu\|demand\|capacityRemaining |
| vRealize Operations Manager Generated Properties\|CPU\|Demand\|Buffer (%) | CPU buffer percent defined by policy setting for demand based capacity computation.<br>Key: System Properties\|cpu\|demand\|bufferSetting |
| Capacity Analytics Generated\|CPU\|Demand\|Recommended Size (MHz) | Published on Host System. The recommended level of usable capacity (total capacity - HA) to maintain a green state for the remaining time.<br>Key: OnlineCapacityAnalytics\|cpu\|demand\|recommendedSize |
| Capacity Analytics Generated\|CPU\|Demand\|Time Remaining (Day(s)) | Published on Host System. The number of days remaining till the projected utilization crosses the threshold for the usable capacity.<br>Key: OnlineCapacityAnalytics\|cpu\|demand\|timeRemaining |
| Capacity Analytics Generated\|Disk Space\|Demand\|Capacity Remaining (GB) | Published on Host System. The max point between the usable capacity and the projected utilization between now and three days into the future.<br>Key: OnlineCapacityAnalytics\|diskspace\|demand\|capacityRemaining |
| vRealize Operations Manager Generated Properties\|Disk Space\|Demand\|Buffer (%) | Disk Space buffer percent defined by policy setting for demand based capacity computation.<br>System Properties\|diskspace\|demand\|bufferSetting |
| Capacity Analytics Generated\|Disk Space\|Demand\|Recommended Size (GB) | Published on Host System. The recommended level of usable capacity (total capacity - HA) to maintain a green state for the remaining time.<br>Key: OnlineCapacityAnalytics\|diskspace\|demand\|recommendedSize |
| Capacity Analytics Generated\|Disk Space\|Demand\|Time Remaining (Day(s)) | Published on Host System. The number of days remaining till the projected utilization crosses the threshold for the usable capacity.<br>Key: OnlineCapacityAnalytics\|diskspace\|demand\|timeRemaining |
| Capacity Analytics Generated\|Memory\|Demand\|Capacity Remaining (KB) | Published on Host System. The max point between the usable capacity and the projected utilization between now and three days into the future.<br>Key: OnlineCapacityAnalytics\|mem\|demand\|capacityRemaining |
| vRealize Operations Manager Generated Properties\|Memory\|Demand\|Buffer (%) | Memory buffer percent defined by policy setting for demand based capacity computation.<br>Key: System Properties\|mem\|demand\|bufferSetting |

| Metric Name | Description |
|---|---|
| Capacity Analytics Generated\|Memory\|Demand\|Recommended Size (KB) | Published on Host System. The recommended level of usable capacity (total capacity - HA) to maintain a green state for the remaining time.<br>Key: OnlineCapacityAnalytics\|mem\|demand\|recommendedSize |
| Capacity Analytics Generated\|Memory\|Demand\|Time Remaining (Day(s)) | Published on Host System. The number of days remaining till the projected utilization crosses the threshold for the usable capacity.<br>Key: OnlineCapacityAnalytics\|mem\|demand\|timeRemaining |
| Capacity Analytics Generated\|Disk Space\|Usage\|Capacity Remaining (GB) | Published on Datastore. The max point between the usable capacity and the projected utilization between now and three days into the future.<br>Key: OnlineCapacityAnalytics\|diskspace\|total\|capacityRemaining |
| Capacity Analytics Generated\|Disk Space\|Usage\|Recommended Size (GB) | Published on Datastore. The recommended level of usable capacity (total capacity - HA) to maintain a green state for the remaining time.<br>Key: OnlineCapacityAnalytics\|diskspace\|total\|recommendedSize |
| Capacity Analytics Generated\|Disk Space\|Usage\|Time Remaining (Day(s)) | Published on Datastore. The number of days remaining till the projected utilization crosses the threshold for the usable capacity.<br>Key: OnlineCapacityAnalytics\|diskspace\|total\|timeRemaining |
| Capacity Analytics Generated\|CPU\|Demand\|Capacity Remaining (MHz) | Published on Cluster Compute Resource. The max point between the usable capacity and the projected utilization between now and three days into the future.<br>Key: OnlineCapacityAnalytics\|cpu\|demand\|capacityRemaining |
| Capacity Analytics Generated\|CPU\|Demand\|Recommended Size (MHz) | Published on Cluster Compute Resource. The recommended level of usable capacity (total capacity - HA) to maintain a green state for the remaining time.<br>Key: OnlineCapacityAnalytics\|cpu\|demand\|recommendedSize |
| Capacity Analytics Generated\|CPU\|Demand\|Recommended Total Capacity (MHz) | Published on Cluster Compute Resource. The recommended level of total capacity to maintain a green state for the time remaining.<br>Key: OnlineCapacityAnalytics\|cpu\|demand\|<br>recommendedTotalSize |
| Capacity Analytics Generated\|CPU\|Demand\|Time Remaining (Day(s)) | Published on Cluster Compute Resource. The number of days remaining till the projected utilization crosses the threshold for the usable capacity.<br>Key: OnlineCapacityAnalytics\|cpu\|demand\|timeRemaining |
| Capacity Analytics Generated\|Disk Space\|Demand\|Capacity Remaining (GB) | Published on Cluster Compute Resource. The max point between the usable capacity and the projected utilization between now and three days into the future.<br>Key: OnlineCapacityAnalytics\|diskspace\|demand\|<br>capacityRemaining |
| Capacity Analytics Generated\|Disk Space\|Demand\|Recommended Size (GB) | Published on Cluster Compute Resource. The recommended level of usable capacity (total capacity - HA) to maintain a green state for the time remaining.<br>Key: OnlineCapacityAnalytics\|diskspace\|demand\|<br>recommendedSize |
| Capacity Analytics Generated\|Disk Space\|Demand\|Time Remaining (Day(s)) | Published on Cluster Compute Resource. The number of days remaining till the projected utilization crosses the threshold for the usable capacity.<br>Key: OnlineCapacityAnalytics\|diskspace\|demand\|timeRemaining |

| Metric Name | Description |
|---|---|
| Capacity Analytics Generated\|Memory\|Demand\|Capacity Remaining (KB) | Published on Cluster Compute Resource. The max point between the usable capacity and the projected utilization between now and three days into the future.<br>Key: OnlineCapacityAnalytics\|mem\|demand\|capacityRemaining |
| Capacity Analytics Generated\|Memory\|Demand\|Recommended Size (KB) | Published on Cluster Compute Resource. The recommended level of usable capacity (total capacity - HA) to maintain a green state for the time remaining.<br>Key: OnlineCapacityAnalytics\|mem\|demand\|recommendedSize |
| Capacity Analytics Generated\|Memory\|Demand\|Recommended Total Capacity (KB) | Published on Cluster Compute Resource. The recommended level of total capacity to maintain a green state for the time remaining.<br>Key: OnlineCapacityAnalytics\|mem\|demand\|<br>recommendedTotalSize |
| Capacity Analytics Generated\|Memory\|Demand\|Time Remaining (Day(s)) | Published on Cluster Compute Resource. The number of days remaining till the projected utilization crosses the threshold for the usable capacity.<br>Key: OnlineCapacityAnalytics\|mem\|demand\|timeRemaining |
| Capacity Analytics Generated\|Disk Space\|Usage\|Capacity Remaining (GB) | Published on Datastore Cluster. The max point between the usable capacity and the projected utilization between now and three days into the future.<br>Key: OnlineCapacityAnalytics\|diskspace\|total\|capacityRemaining |
| Capacity Analytics Generated\|Disk Space\|Usage\|Recommended Size (GB) | Published on Datastore Cluster. The recommended level of usable capacity (total capacity - HA) to maintain a green state for the time remaining.<br>Key: OnlineCapacityAnalytics\|diskspace\|total\|recommendedSize |
| Capacity Analytics Generated\|Disk Space\|Usage\|Time Remaining (Day(s)) | Published on Datastore Cluster. The number of days remaining till the projected utilization crosses the threshold for the usable capacity.<br>Key: OnlineCapacityAnalytics\|diskspace\|total\|timeRemaining |
| Capacity Analytics Generated\|CPU\|Demand\|Capacity Remaining (MHz) | Published on Datacenter, Custom Datacenter, vCenter. The max point between the usable capacity and the projected utilization between now and three days into the future.<br>Key: OnlineCapacityAnalytics\|cpu\|demand\|capacityRemaining |
| Capacity Analytics Generated\|CPU\|Demand\|Recommended Size (MHz) | Published on Datacenter, Custom Datacenter, vCenter. The recommended level of usable capacity (total capacity - HA) to maintain a green state for the time remaining.<br>Key: OnlineCapacityAnalytics\|cpu\|demand\|recommendedSize |
| Capacity Analytics Generated\|CPU\|Demand\|Recommended Total Capacity (MHz) | Published on Datacenter, Custom Datacenter, vCenter. The recommended level of total capacity to maintain a green state for the time remaining.<br>Key: OnlineCapacityAnalytics\|cpu\|demand\|<br>recommendedTotalSize |
| Capacity Analytics Generated\|CPU\|Demand\|Time Remaining (Day(s)) | Published on Datacenter, Custom Datacenter, vCenter. The number of days remaining till the projected utilization crosses the threshold for the usable capacity.<br>Key: OnlineCapacityAnalytics\|cpu\|demand\|timeRemaining |
| Capacity Analytics Generated\|Disk Space\|Demand\|Capacity Remaining (GB) | Published on Datacenter, Custom Datacenter, vCenter. The max point between the usable capacity and the projected utilization between now and three days into the future.<br>Key: OnlineCapacityAnalytics\|diskspace\|demand\|<br>capacityRemaining |

| Metric Name | Description |
|---|---|
| Capacity Analytics Generated\|Disk Space\|Demand\|Recommended Size (GB) | Published on Datacenter, Custom Datacenter, vCenter. The recommended level of usable capacity (total capacity - HA) to maintain a green state for the time remaining.<br>Key: OnlineCapacityAnalytics\|diskspace\|demand\|recommendedSize |
| Capacity Analytics Generated\|Disk Space\|Demand\|Time Remaining (Day(s)) | Published on Datacenter, Custom Datacenter, vCenter. The number of days remaining till the projected utilization crosses the threshold for the usable capacity.<br>Key: OnlineCapacityAnalytics\|diskspace\|demand\|timeRemaining |
| Capacity Analytics Generated\|Memory\|Demand\|Capacity Remaining (KB) | Published on Datacenter, Custom Datacenter, vCenter. The max point between the usable capacity and the projected utilization between now and three days into the future.<br>Key: OnlineCapacityAnalytics\|mem\|demand\|capacityRemaining |
| Capacity Analytics Generated\|Memory\|Demand\|Recommended Size (KB) | Published on Datacenter, Custom Datacenter, vCenter. The recommended level of usable capacity (total capacity - HA) to maintain a green state for the time remaining.<br>Key: OnlineCapacityAnalytics\|mem\|demand\|recommendedSize |
| Capacity Analytics Generated\|Memory\|Demand\|Recommended Total Capacity (KB) | Published on Datacenter, Custom Datacenter, vCenter. The recommended level of total capacity to maintain a green state for the time remaining.<br>Key: OnlineCapacityAnalytics\|mem\|demand\|recommendedTotalSize |
| Capacity Analytics Generated\|Memory\|Demand\|Time Remaining (Day(s)) | Published on Datacenter, Custom Datacenter, vCenter. The number of days remaining till the projected utilization crosses the threshold for the usable capacity.<br>Key: OnlineCapacityAnalytics\|mem\|demand\|timeRemaining |

## Badge Metrics

Badge metrics provide information for badges in the user interface. They report the health, risk, and efficiency of objects in your environment.

VCF Operations 6.x analyzes badge metric data at five-minute averages, instead of hourly. As a result, you might find that efficiency and risk badge calculations are more sensitive than in previous versions. Badge metrics continue to be published nightly.

**Table 1532: Badge Metrics**

| Metric Name | Description |
|---|---|
| Badge\|Compliance | Badge\|Compliance(%) metric shows the compliance score for the given object based on the number of violated and total number of compliance symptoms calculated with the following formula `[Math.round(100 - (((double)triggeredSymptoms/totalSymptoms)*100))]`. So, this metric shows the compliance score per object based on the violated symptoms (which can be seen under the object's details Compliance tab). This metric should not be mixed up with the compliance score shown for the benchmark in the **Operations** > **Compliance** page which considers compliant/non-compliant vs total objects.Badge\|Compliance(%) metric is used in 4 views: - Compliance \ VM Distribution - Compliance \ vSphere Distributed Port Groups - Compliance \ vSphere ESX Hosts - Compliance \ vSphere VMs, as well as 1 deprecated dashboard's widget - vSphere Security Compliance dashboard's Compliance Summary widget. |
| Badge\|Efficiency | Overall score for efficiency. The final score is between 1-100. Where Green - 100, Yellow - 75, Orange - 50, Red - 25, Unknown: -1. The score is derived from the criticality of alerts in the Efficiency category. |
| Badge\|Health | Overall score for health. The final score is between 1-100. Where Green - 100, Yellow - 75, Orange - 50, Red - 25, Unknown: -1. The score is derived from the criticality of alerts in the Health category. |
| Badge\|Risk | Overall score for risk. The final score is between 1-100. Where Green - 0, Yellow - 25, Orange - 50, Red - 75, Unknown: -1. The score is derived from the criticality of alerts in the Risk category. |

## System Metrics

System metrics provide information used to monitor the health of the system. They help you to identify problems in your environment.

**Table 1533: System Metrics**

| Metric Name | Description |
|---|---|
| vRealize Operations Generated\|Self - Health Score | This metric displays the system health score of self resource. The value ranges from 0 to 100 depending on noise and the number of alarms.<br>Key: System Attributes\|health |
| vRealize Operations Generated\|Self - Metric Count | This metric displays the number of metrics that the adapter generates for the given object. This value does not include the number of metrics generated by VCF Operations, such as, Badge metrics, vRealize Operations Generated metrics and metrics generated by Capacity Engine<br>Key: System Attributes\|all_metrics |
| vRealize Operations Generated\|Total Anomalies | This metric displays the number of active anomalies (symptoms, events, DT violations) on the object and its children.<br>In previous versions of VCF Operations, this metric used to be named vRealize Operations Generated\| Self - Total Anomalies.<br>Key: System Attributes\|total_alarms |
| vRealize Operations Generated\|Full Set - Metric Count | This metric displays the number of metrics that the adapter of the children of the given object generates.<br>Key: System Attributes\|child_all_metrics |

| Metric Name | Description |
|---|---|
| vRealize Operations Generated\|Availability | This metric value is computed based on the adapter instance statuses monitoring the resource. Resource availability is displayed as 0-down, 1-Up, -1-Unknown.<br>Key: System Attributes\|availability |
| vRealize Operations Generated\|Alert Count Critical | This metric displays the number of critical alerts on the object and its children.<br>Key: System Attributes\|alert_count_critical |
| vRealize Operations Generated\|Alert Count Immediate | This metric displays the number of immediate alerts on the object and its children.<br>Key: System Attributes\|alert_count_immediate |
| vRealize Operations Generated\|Alert Count Warning | This metric displays the number of active warning alerts on the object and its children.<br>Key: System Attributes\|alert_count_warning |
| vRealize Operations Generated\|Alert Count Info | This metric displays the number of active info alerts on the object and its children.<br>Key: System Attributes\|alert_count_info |
| vRealize Operations Generated\|Total Alert Count | This metric displays the sum of all alert count metrics.<br>In previous versions of VCF Operations, this metric was named vRealize Operations Generated\|Full Set - Alert Count.<br>Key: System Attributes\|total_alert_count |
| vRealize Operations Generated\|Self-Alert Count | This metric displays the number of all alerts on the object.<br>Key: System Attributes\|self_alert_count |

### log management Generated Metrics

The metrics in the log management Generated group provide information that you can use to observe or troubleshoot VCF Operations for failures and to monitor performance.

When VCF Operations is integrated with log management and metric calculation is enabled, log management calculates the number of logs corresponding to different queries and sends them as metrics to VCF Operations. These metrics are calculated for vCenter objects, host objects, and virtual machine objects. The metrics can be mapped to a VCF Operations object based on the log management field *`vmw_vrops_id`*, which is constructed based on hostname or source fields.

**Table 1534: Log Insight Generated Metrics**

| Metric Name | Description |
|---|---|
| log management Generated\|Error Count | The number of error logs for the selected object.<br>Key: log_insight_generated\|error_count |
| log management Generated\|Total Log Count | The total number of logs for the selected object.<br>Key: log_insight_generated\|total_log_count |
| log management Generated\|Warning Count | The number of warning logs for the selected object.<br>Key: log_insight_generated\|warning_count |

## Self-Monitoring Metrics for VCF Operations

VCF Operations uses the VCF Operations adapter to collect metrics that monitor its own performance. These self-monitoring metrics drive capacity models for VCF Operations objects and are useful for diagnosing problems with VCF Operations.

## Analytics Metrics

VCF Operations collects metrics for the VCF Operations analytics service, including threshold checking metrics.

**Table 1535: Analytics Metrics**

| Metric Key | Metric Name | Description |
|---|---|---|
| ActiveAlarms | Active DT Symptoms | Active DT Symptoms. |
| ActiveAlerts | Active Alerts | Active alerts. |
| PrimaryResourcesCount | Number of primary objects | Number of primary objects |
| LocalResourcesCount | Number of local objects | Number of local objects |
| PrimaryMetricsCount | Number of primary metrics | Number of primary metrics |
| LocalMetricsCount | Number of local metrics | Number of local metrics |
| ReceivedResourceCount | Number of received objects | Number of received objects |
| ReceivedMetricCount | Number of received metrics | Number of received metrics |
| LocalFDSize | Number of forward data entries | Number of locally stored primary and redundant entries in forward data region. |
| LocalPrimaryFDSize | Number of primary forward data entries | Number of locally stored primary entries in forward data region. |
| LocalFDAltSize | Number of alternative forward data entries | Number of locally stored primary and redundant entries in alternative forward data region. |
| LocalPrimaryFDAltSize | Number of alternative primary forward data entries | Number of locally stored primary entries in alternative forward data region. |
| CurrentHeapSize | Current heap size | Current heap size. |
| MaxHeapSize | Max heap size | Max heap size |
| CommittedMemory | Committed memory | Committed memory |
| CPUUsage | CPU usage | CPU usage |
| Threads | Threads | Threads |
| UpStatus | Threads | Threads |

### Overall Threshold Checking Metrics for the Analytics Service

Overall threshold checking captures various metrics for work items used to process incoming observation data. All metrics keys for the overall threshold checking metrics begin with OverallThresholdChecking, as in `OverallThresholdChecking|Count` or `OverallThresholdChecking|CheckThresholdAndHealth|OutcomeObservationsSize|TotalCount`.

**Table 1536: Overall Threshold Checking Metrics for the Analytics Service**

| Metric Key | Metric Name | Description |
|---|---|---|
| Count | Count | Count |
| Duration\|TotalDuration | Total | Total length of duration (ms) |
| Duration\|AvgDuration | Average | Average duration (ms) |
| Duration\|MinDuration | Minimum | Minimum duration (ms) |

| Metric Key | Metric Name | Description |
|---|---|---|
| Duration\|MaxDuration | Maximum | Maximum duration (ms) |
| IncomingObservationsSize\|TotalCount | Total | Total |
| IncomingObservationsSize\|AvgCount | Average | Average |
| IncomingObservationsSize\|MinCount | Minimal | Minimal |
| IncomingObservationsSize\|MaxCount | Maximal | Maximal |
| CheckThresholdAndHealth\|Count | Count | Count |
| CheckThresholdAndHealth\|Duration\|TotalDuration | Total | Total length of duration (ms) |
| CheckThresholdAndHealth\|Duration\|AvgDuration | Average | Average duration (ms) |
| CheckThresholdAndHealth\|Duration\|MinDuration | Minimum | Minimum duration (ms) |
| CheckThresholdAndHealth\|Duration\|MaxDuration | Maximum | Maximum duration (ms) |
| CheckThresholdAndHealth\| OutcomeObservationsSize\|TotalCount | Total | Total |
| CheckThresholdAndHealth\| OutcomeObservationsSize\|AvgCount | Average | Average |
| CheckThresholdAndHealth\| OutcomeObservationsSize\|MinCount | Minimal | Minimal |
| CheckThresholdAndHealth\| OutcomeObservationsSize\|MaxCount | Maximal | Maximal |
| SuperMetricComputation\|Count | Count | Count |
| SuperMetricComputation\|Duration\|TotalDuration | Total | Total length of duration (ms) |
| SuperMetricComputation\|Duration\|AvgDuration | Average | Average duration (ms) |
| SuperMetricComputation\|Duration\|MinDuration | Minimum | Minimum duration (ms) |
| SuperMetricComputation\|Duration\|MaxDuration | Maximum | Maximum duration (ms) |
| SuperMetricComputation\|SuperMetricsCount\| TotalCount | Total | Total |
| SuperMetricComputation\|SuperMetricsCount \| AvgCount | Average | Average |
| SuperMetricComputation\|SuperMetricsCount \| MinCount | Minimal | Minimal |
| SuperMetricComputation\|SuperMetricsCount \| MaxCount | Maximal | Maximal |
| StoreObservationToFSDB\|Count | Count | Count |
| StoreObservationToFSDB\|Duration\|TotalDuration | Total | Total length of duration (ms) |
| StoreObservationToFSDB\|Duration\|AvgDuration | Average | Average duration (ms) |
| StoreObservationToFSDB\|Duration\|MinDuration | Minimum | Minimum duration (ms) |
| StoreObservationToFSDB\|Duration\|MaxDuration | Maximum | Maximum duration (ms) |
| StoreObservationToFSDB\| StoredObservationsSize\|TotalCount | Total | Total |
| StoreObservationToFSDB\| StoredObservationsSize\|AvgCount | Average | Average |
| StoreObservationToFSDB\| StoredObservationsSize\|MinCount | Minimal | Minimal |
| StoreObservationToFSDB\| StoredObservationsSize\|MaxCount | Maximal | Maximal |

| Metric Key | Metric Name | Description |
|---|---|---|
| UpdateResourceCache\|Count | Count | Count |
| UpdateResourceCache\|Duration\|TotalDuration | Total | Total |
| UpdateResourceCache\|Duration\|AvgDuration | Average | Average |
| UpdateResourceCache\|Duration\|MinDuration | Minimum | Minimum |
| UpdateResourceCache\|Duration\|MaxDuration | Maximum | Maximum |
| UpdateResourceCache\| ModifcationEstimateCount\|TotalCount | Total | The number of estimated modifications done during each resource cache object update. |
| UpdateResourceCache\| ModifcationEstimateCount\|AvgCount | Average | Average |
| UpdateResourceCache\| ModifcationEstimateCount\|MinCount | Minimal | Minimal |
| UpdateResourceCache\| ModifcationEstimateCount\|MaxCount | Maximal | Maximal |
| ManageAlerts\|Count | Count | The total number of times the threshold checking work items perform alert updates. |
| ManageAlerts\|Duration\|TotalDuration | Total | The duration for the alert updates operations. |
| ManageAlerts\|Duration\|AvgDuration | Average | Average |
| ManageAlerts\|Duration\|MinDuration | Minimum | Minimum |
| ManageAlerts\|Duration\|MaxDuration | Maximum | Maximum |
| UpdateSymptoms\|Count | Count | The total number of times the threshold checking work items check and build symptoms. |
| UpdateSymptoms\|Duration\|TotalDuration | Total | The duration for the check and build symptoms operation. |
| UpdateSymptoms\|Duration\|AvgDuration | Average | Average |
| UpdateSymptoms\|Duration\|MinDuration | Minimum | Minimum |
| UpdateSymptoms\|Duration\|MaxDuration | Maximum | Maximum |

## Dynamic Threshold Calculation Metrics for the Analytics Service

All metrics keys for the dynamic threshold calculation metrics begin with DtCalculation, as in `DtCalculation|DtDataWrite|WriteOperationCount` or `DtCalculation|DtAnalyze|AnalyzeOperationCount`.

### Table 1537: Dynamic Threshold Calculation Metrics for the Analytics Service

| Metric Key | Metric Name | Description |
|---|---|---|
| DtDataWrite\|WriteOperationCount | Write operation count | Write operation count |
| DtDataWrite\|Duration\|TotalDuration | Total | Total length of duration (ms) |
| DtDataWrite\|Duration\|AvgDuration | Average | Average duration (ms) |
| DtDataWrite\|Duration\|MinDuration | Minimum | Minimum duration (ms) |
| DtDataWrite\|Duration\|MaxDuration | Maximum | Maximum duration (ms) |
| DtDataWrite\|SavedDtObjectCount\|TotalCount | Total | Total |

| Metric Key | Metric Name | Description |
|---|---|---|
| DtDataWrite\|SavedDtObjectCount\|AvgCount | Average | Average |
| DtDataWrite\|SavedDtObjectCount\|MinCount | Minimal | Minimal |
| DtDataWrite\|SavedDtObjectCount\|MaxCount | Maximal | Maximal |
| DtAnalyze\|AnalyzeOperationCount | Analyze Operation Count | Analyze Operation Count |
| DtAnalyze\|Duration\|TotalDuration | Total | Total length of duration (ms) |
| DtAnalyze\|Duration\|AvgDuration | Average | Average duration (ms) |
| DtAnalyze\|Duration\|MinDuration | Minimum | Minimum duration (ms) |
| DtAnalyze\|Duration\|MaxDuration | Maximum | Maximum duration (ms) |
| DtAnalyze\|AnalyzedMetricsCount\|TotalCount | Total | Total |
| DtAnalyze\|AnalyzedMetricsCount\|AvgCount | Average | Average |
| DtAnalyze\|AnalyzedMetricsCount\|MinCount | Minimal | Minimal |
| DtAnalyze\|AnalyzedMetricsCount\|MaxCount | Maximal | Maximal |
| DtDataRead\|ReadOperationsCount | Read Operation Count | Read Operation Count |
| DtDataRead\|Duration\|TotalDuration | Total | Total length of duration (ms) |
| DtDataRead\|Duration\|AvgDuration | Average | Average duration (ms) |
| DtDataRead\|Duration\|MinDuration | Minimum | Minimum duration (ms) |
| DtDataRead\|Duration\|MaxDuration | Maximum | Maximum duration (ms) |
| DtDataRead\|ReadDataPointsCount\|TotalCount | Total | Total |
| DtDataRead\|ReadDataPointsCount\|AvgCount | Average | Average |
| DtDataRead\|ReadDataPointsCount\|MinCount | Minimal | Minimal |
| DtDataRead\|ReadDataPointsCount\|MaxCount | Maximal | Maximal |

**Table 1538: Function Call Metrics for the Analytics Service**

| Metric Key | Metric Name | Description |
|---|---|---|
| FunctionCalls\|Count | Number of function calls | Number of function calls |
| FunctionCalls\|AvgDuration | Average execution time | Average execution time |
| FunctionCalls\|MaxDuration | Max execution time | Max execution time |

## Collector Metrics

VCF Operations collects metrics for the VCF Operations Collector service objects.

**Table 1539: Collector Metrics**

| Metric Key | Metric Name | Description |
|---|---|---|
| ThreadpoolThreadsCount | Number of pool threads | Number of pool threads. |
| RejectedFDCount | Number of rejected forward data | Number of rejected forward data |
| RejectedFDAltCount | Number of rejected alternative forward data | Number of rejected alternative forward data |
| SentFDCount | Number of sent objects | Number of sent objects |
| SentFDAltCount | Number of alternative sent objects | Number of alternative sent objects |

| Metric Key | Metric Name | Description |
|---|---|---|
| CurrentHeapSize | Current heap size (MB) | Current heap size. |
| MaxHeapsize | Max heap size (MB) | Maximum heap size. |
| CommittedMemory | Committed memory (MB) | Amount of committed memory. |
| CPUUsage | CPU usage | CPU usage. |
| Threads | Threads | Number of threads. |
| UpStatus | Up Status | Up Status |

## Controller Metrics

VCF Operations collects metrics for the VCF Operations Controller objects.

**Table 1540: Controller Metrics**

| Metric Key | Metric Name | Description |
|---|---|---|
| RequestedMetricCount | Number of requested metrics | Number of requested metrics |
| ApiCallsCount | Number of API calls | Number of API calls |
| NewDiscoveredResourcesCount | Number of discovered objects | Number of discovered objects |

## FSDB Metrics

VCF Operations collects metrics for the VCF Operations file system database (FSDB) objects.

**Table 1541: FSDB Metrics**

| Metric Key | Metric Name | Description |
|---|---|---|
| StoragePoolElementsCount | Number of storage work items | Number of storage work items |
| FsdbState | Fsdb state | Fsdb state |
| StoredResourcesCount | Number of stored objects | Number of stored objects |
| StoredMetricsCount | Number of stored metrics | Number of stored metrics |

**Table 1542: Storage Thread Pool Metrics for FSDB**

| Metric Key | Metric Name | Description |
|---|---|---|
| StoreOperationsCount | Store operations count | Store operations count |
| StorageThreadPool\|Duration\|TotalDuration | Total | Total number of duration (ms) |
| StorageThreadPool\|Duration\|AvgDuration | Average | Average duration (ms) |
| StorageThreadPool\|Duration\|MinDuration | Minimum | Minimum duration (ms) |
| StorageThreadPool\|Duration\|MaxDuration | Maximum | Maximum duration (ms) |
| StorageThreadPool\|SavedMetricsCount\|TotalCount | Total | Total |
| StorageThreadPool\|SavedMetricsCount\|AvgCount | Average | Average |
| StorageThreadPool\|SavedMetricsCount\|MinCount | Minimal | Minimal |

| Metric Key | Metric Name | Description |
|---|---|---|
| StorageThreadPool\|SavedMetricsCount\|MaxCount | Maximal | Maximal |

## Product UI Metrics

VCF Operations collects metrics for the VCF Operations product user interface objects.

**Table 1543: Product UI Metrics**

| Metric Key | Metric Name | Description |
|---|---|---|
| ActiveSessionsCount | Active sessions | Active sessions |
| CurrentHeapSize | Current heap size | Current heap size. |
| MaxHeapsize | Max heap size | Maximum heap size. |
| CommittedMemory | Committed memory | Amount of committed memory. |
| CPUUsage | CPU usage | Percent CPU use. |
| Threads | Threads | Number of threads. |
| SessionCount | Number of active sessions | Number of active sessions |
| SelfMonitoringQueueSize | Self Monitoring queue size | Self Monitoring queue size |

**Table 1544: API Call Metrics for the Product UI**

| Metric Key | Metric Name | Description |
|---|---|---|
| APICalls\|HTTPRequesterRequestCount | HTTPRequester request count | HTTPRequester request count |
| APICalls\|AvgHTTPRequesterRequestTime | HTTPRequester average request time | HTTPRequester average request time (ms) |
| APICalls\|FailedAuthenticationCount | Failed Authentication Count | Failed Authentication Count |
| APICalls\|AvgAlertRequestTime | Average alert request time | Average alert request time (ms) |
| APICalls\|AlertRequestCount | Alert request count | Alert request count |
| APICalls\|AvgMetricPickerRequestTime | Average metric-picker request time | Average metric-picker request time (ms) |
| APICalls\|MetricPickerRequestCount | Metric picker request count | Metric picker request count |
| APICalls\|HeatmapRequestCount | Heatmap request count | Heatmap request count |
| APICalls\|AvgHeatmapRequestTime | Average HeatMap request time | Average HeatMap request time (ms) |
| APICalls\|MashupChartRequestCount | Mashup Chart request count | Mashup Chart request count |
| APICalls\|AvgMashupChartRequestTime | Average Mashup Chart request time | Average Mashup Chart request time (ms) |
| APICalls\|TopNRequestCount | Top N request count | Top N request count |
| APICalls\|AvgTopNRequestTime | Average Top N request time | Average Top N request time (ms) |
| APICalls\|MetricChartRequestCount | Metric Chart request count | Metric Chart request count |
| APICalls\|AvgMetricChartRequestTime | Average MetricChart request time | Average MetricChart request time (ms) |

## Admin UI Metrics

VCF Operations collects metrics for the VCF Operations administration user interface objects.

**Table 1545: Admin UI Metrics**

| Metric Key | Metric Name | Description |
|---|---|---|
| CurrentHeapSize | Current heap size | Current heap size (MB). |
| MaxHeapsize | Max heap size | Maximum heap size (MB). |
| CommittedMemory | Committed memory | Amount of committed memory (MB) . |
| CPUUsage | CPU usage | CPU usage (%). |
| Threads | Threads | Number of threads. |
| SessionCount | Number of active sessions | Number of active sessions |
| SelfMonitoringQueueSize | Self Monitoring queue size | Self Monitoring queue size |

**Table 1546: API Call Metrics for the Admin UI**

| Metric Key | Metric Name | Description |
|---|---|---|
| APICalls\|HTTPRequesterRequestCount | HTTPRequester request count | HTTPRequester request count |
| APICalls\|AvgHTTPRequesterRequestTime | HTTPRequester average request time | HTTPRequester average request time (ms) |

## Suite API Metrics

VCF Operations collects metrics for the VCF Operations API objects.

**Table 1547: Suite API Metrics**

| Metric Key | Metric Name | Description |
|---|---|---|
| UsersCount | Number of users | Number of users |
| ActiveSessionsCount | Active sessions | Active sessions |
| GemfireClientReconnects | Gemfire Client Reconnects | Gemfire Client Reconnects |
| GemfireClientCurrentCalls | Gemfire Client Total Outstanding | Gemfire Client Total Outstanding |
| CurrentHeapSize | Current heap size | Current heap size (MB) . |
| MaxHeapsize | Max heap size | Maximum heap size (MB) . |
| CommittedMemory | Committed memory | Amount of committed memory (MB). |
| CPUUsage | CPU usage | CPU usage (%) . |
| CPUProcessTime | CPU process time | CPU process time (ms) |
| CPUProcessTimeCapacity | CPU process time capacity | CPU process time capacity (ms) |
| Threads | Threads | Number of threads. |

**Table 1548: Gemfire Client Call Metrics for the Suite API**

| Metric Key | Metric Name | Description |
|---|---|---|
| GemfireClientCalls\|TotalRequests | Total Requests | Total Requests |
| GemfireClientCalls\|AvgResponseTime | Average Response Time | Average Response Time (ms) |
| GemfireClientCalls\|MinResponseTime | Minimum Response Time | Minimum Response Time (ms) |
| GemfireClientCalls\|MaxResponseTime | Maximum Response Time | Maximum Response Time |

| Metric Key | Metric Name | Description |
|---|---|---|
| GemfireClientCalls\|RequestsPerSecond | Requests per Second | Requests per Second |
| GemfireClientCalls\|CurrentRequests | Current Requests | Current Requests |
| GemfireClientCalls\|RequestsCount | Requests Count | Requests Count |
| GemfireClientCalls\|ResponsesCount | Responses Count | Responses Count |

**Table 1549: API Call Metrics for the Suite API**

| Metric Key | Metric Name | Description |
|---|---|---|
| APICalls\|TotalRequests | Total Requests | Total Requests |
| APICalls\|AvgResponseTime | Average Response Time (ms) | Average Response Time (ms) |
| APICalls\|MinResponseTime | Minimum Response Time (ms) | Minimum Response Time (ms) |
| APICalls\|MaxResponseTime | Maximum Response Time | Maximum Response Time |
| APICalls\|ServerErrorResponseCount | Server Error Response Count | Server Error Response Count |
| APICalls\|FailedAuthenticationCount | Failed Authentication Count | Failed Authentication Count |
| APICalls\|FailedAuthorizationCount | Failed Authorization Count | Failed Authorization Count |
| APICalls\|RequestsPerSecond | Requests per Second | Requests per Second |
| APICalls\|CurrentRequests | Current Requests | Current Requests |
| APICalls\|ResponsesPerSecond | Responses per Second | Responses per Second |
| APICalls\|RequestsCount | Requests Count | Requests Count |
| APICalls\|ResponsesCount | Responses Count | Responses Count |

## Cluster and Slice Administration Metrics

VCF Operations collects metrics for VCF Operations Cluster and Slice Administration (CaSA) objects.

**Table 1550: Cluster and Slice Administration Metrics**

| Metric Key | Metric Name | Description |
|---|---|---|
| CurrentHeapSize | Current heap size | Current heap size (MB). |
| MaxHeapsize | Max heap size | Maximum heap size (MB). |
| CommittedMemory | Committed memory | Amount of committed memory (MB). |
| CPUUsage | CPU usage | CPU usage (%) |
| Threads | Threads | Number of threads. |

**Table 1551: API Call Metrics for Cluster and Slice Administration**

| Metric Key | Metric Name | Description |
|---|---|---|
| API Calls\|TotalRequests | Total Requests | Total Requests |
| API Calls\|AvgResponseTime | Average Response Time | Average Response Time (ms) |
| API Calls\|MinResponseTime | Minimum Response Time | Minimum Response Time (ms) |
| API Calls\|MaxResponseTime | Maximum Response Time | Maximum Response Time (ms) |
| API Calls\|ServerErrorResponseCount | Server Error Response Count | Server Error Response Count |
| API Calls\|FailedAuthenticationCount | Failed Authentication Count | Failed Authentication Count |

| Metric Key | Metric Name | Description |
|---|---|---|
| API Calls\|FailedAuthorizationCount | Minimum Response Time | Minimum Response Time (ms) |

## Watchdog Metrics

VCF Operations collects watchdog metrics to ensure that the VCF Operations services are running and responsive.

### Watchdog Metrics

The watchdog metric provides the total service count.

**Table 1552: Watchdog Metrics**

| Metric Key | Metric Name | Description |
|---|---|---|
| ServiceCount | Service Count | Service Count |

### Service Metrics

Service metrics provide information about watchdog activity.

**Table 1553: Metrics for the VCF Operations Watchdog Service**

| Metric Key | Metric Name | Description |
|---|---|---|
| Service\|Enabled | Enabled | Enabled |
| Service\|Restarts | Restarts | Number of times the process has been unresponsive and been restarted by Watchdog. |
| Service\|Starts | Starts | Number of times the process has been revived by Watchdog. |
| Service\|Stops | Stops | Number of times the process has been stopped by Watchdog. |

## Node Metrics

VCF Operations collects metrics for the VCF Operations node objects.

Metrics can be calculated for node objects. See Calculated Metrics.

**Table 1554: Node Metrics**

| Metric Key | Metric Name | Description |
|---|---|---|
| Component Count | Component count | The number of VCF Operations objects reporting for this node |
| PrimaryResourcesCount | Number of primary objects | Number of primary objects |
| LocalResourcesCount | Number of local objects | Number of local objects |
| PrimaryMetricsCount | Number of primary metrics | Number of primary metrics |
| LocalMetricsCount | Number of local metrics | Number of local metrics |
| PercentDBStorageAvailable | Percent disk available /storage/db | Percent disk available /storage/db |
| PercentLogStorageAvailable | Percent disk available /storage/log | Percent disk available /storage/log |

| Metric Key | Metric Name | Description |
|---|---|---|
| FPing stats | Latency\|Average (ms) | |
| FPing stats | Latency\|Maximum (ms) | |
| FPing stats | Latency\|Minimum (ms) | |
| FPing stats | Packet Loss\|Average (%) | Percentage average packet loss between nodes. |
| FPing stats | Packet Loss\|Maximum (%) | Percentage maximum packet loss between nodes. |
| FPing stats | Packet Loss\|Minimum (%) | Percentage minimum packet loss between nodes. |

**Table 1555: Memory Metrics for the Node**

| Metric Key | Metric Name | Description |
|---|---|---|
| mem\|actualFree | Actual Free | Actual Free |
| mem\|actualUsed | Actual Used | Actual Used |
| mem\|free | Free | Free ) |
| mem\|used | Used | Used |
| mem\|total | Total | Total |
| mem\|demand_gb | Estimated memory demand | Estimated memory demand |

**Table 1556: Swap Metrics for the Node**

| Metric Key | Metric Name | Description |
|---|---|---|
| swap\|total | Total | Total |
| swap\|free | Free | Free |
| swap\|used | Used | Used |
| swap\|pageIn | Page in | Page in |
| swap\|pageOut | Page out | Page out |

**Table 1557: Resource Limit Metrics for the Node**

| Metric Key | Metric Name | Description |
|---|---|---|
| resourceLimit\|numProcesses | Number of processes | Number of processes |
| resourceLimit\|openFiles | Number of open files | Number of open files |
| resourceLimit\|openFilesMax | Number of open files maximum limit | Number of open files maximum limit |
| resourceLimit\|numProcessesMax | Number of processes maximum limit | Number of processes maximum limit |

**Table 1558: Network Metrics for the Node**

| Metric Key | Metric Name | Description |
|---|---|---|
| net\|allInboundTotal | All inbound connections | All inbound total |
| net\|allOutboundTotal | All outbound connections | All outbound total |
| net\|tcpBound | TCP bound | TCP bound |

| Metric Key | Metric Name | Description |
|---|---|---|
| net\|tcpClose | TCP state CLOSE | Number of connections in TCP CLOSE |
| net\|tcpCloseWait | TCP state CLOSE WAIT | Number of connections in TCP state CLOSE WAIT |
| net\|tcpClosing | TCP state CLOSING | Number of connections in TCP state CLOSING |
| net\|tcpEstablished | TCP state ESTABLISHED | Number of connections in TCP state ESTABLISHED |
| net\|tcpIdle | TCP state IDLE | Number of connections in TCP state IDLE |
| net\|tcpInboundTotal | TCP inbound connections | TCP inbound connections |
| net\|tcpOutboundTotal | TCP outbound connections | TCP outbound connections |
| net\|tcpLastAck | TCP state LAST ACK | Number of connections in TCP state LAST ACK |
| net\|tcpListen | TCP state LISTEN | Number of connections in TCP state LISTEN |
| net\|tcpSynRecv | TCP state SYN RCVD | Number of connections in TCP state SYN RCVD |
| net\|tcpSynSent | TCP state SYN_SENT | Number of connections in TCP state SYN_SENT |
| net\|tcpTimeWait | TCP state TIME WAIT | Number of connections in TCP state TIME WAIT |

**Table 1559: Network Interface Metrics for the Node**

| Metric Key | Metric Name | Description |
|---|---|---|
| net\|iface\|speed | Speed | Speed (bits/sec) |
| net\|iface\|rxPackets | Receive packets | Number of received packets |
| net\|iface\|rxBytes | Receive bytes | Number of received bytes |
| net\|iface\|rxDropped | Receive packet drops | Number of received packets dropped |
| net\|iface\|rxFrame | Receive packets frame | Number of receive packets frame |
| net\|iface\|rxOverruns | Receive packets overruns | Number of receive packets overrun |
| net\|iface\|txPackets | Transmit packets | Number of transmit packets |
| net\|iface\|txBytes | Transmit bytes | Number of transmit bytes |
| net\|iface\|txDropped | Transmit packet drops | Number of transmit packets dropped |
| net\|iface\|txCarrier | Transmit carrier | Transmit carrier |
| net\|iface\|txCollisions | Transmit packet collisions | Number of transmit collisions |
| net\|iface\|txErrors | Transmit packet errors | Number of transmit errors |
| net\|iface\|txOverruns | Transmit packet overruns | Number of transmit overruns |

### Table 1560: Disk Filesystem Metrics for the Node

| Metric Key | Metric Name | Description |
|---|---|---|
| disk\|fileSystem\|total | Total | Total |
| disk\|fileSystem\|available | Available | Available |
| disk\|fileSystem\|used | Used | Used |
| disk\|fileSystem\|files | Total file nodes | Total file nodes |
| disk\|fileSystem\|filesFree | Total free file nodes | Total free file nodes |
| disk\|fileSystem\|queue | Disk queue | Disk queue |
| disk\|fileSystem\|readBytes | Read bytes | Number of bytes read |
| disk\|fileSystem\|writeBytes | Write bytes | Number of bytes written |
| disk\|fileSystem\|reads | Reads | Number of reads |
| disk\|fileSystem\|writes | Writes | Number of writes |

### Table 1561: Disk Installation Metrics for the Node

| Metric Key | Metric Name | Description |
|---|---|---|
| disk\|installation\|used | Used | Used |
| disk\|installation\|total | Total | Total |
| disk\|installation\|available | Available | Available |

### Table 1562: Disk Database Metrics for the Node

| Metric Key | Metric Name | Description |
|---|---|---|
| disk\|db\|used | Used | Used |
| disk\|db\|total | Total | Total |
| disk\|db\|available | Available | Available |

### Table 1563: Disk Log Metrics for the Node

| Metric Key | Metric Name | Description |
|---|---|---|
| disk\|log\|used | Used | Used |
| disk\|log\|total | Total | Total |
| disk\|log\|available | Available | Available |

### Table 1564: CPU Metrics for the Node

| Metric Key | Metric Name | Description |
|---|---|---|
| cpu\|combined | Combined load | Combined load (User + Sys + Nice + Wait) |
| cpu\|idle | Idle | Idle time fraction of total available cpu (cpu load) |
| cpu\|irq | Irq | Interrupt time fraction of total available cpu (cpu load) |

| Metric Key | Metric Name | Description |
|---|---|---|
| cpu\|nice | Nice | Nice time fraction of total available cpu (cpu load) |
| cpu\|softIrq | Soft Irq | Soft interrupt time fraction of total available cpu (cpu load) |
| cpu\|stolen | Stolen | Stolen time fraction of total available cpu (cpu load) |
| cpu\|sys | Sys | Sys time fraction of total available cpu (cpu load) |
| cpu\|user | User (cpu load) | User time fraction of total available cpu (cpu load) |
| cpu\|wait | Wait (cpu load) | Wait time fraction of total available cpu (cpu load) |
| cpu\|total | Total available for a cpu | Total available for a cpu |
| cpu\|allCpuCombined | Total combined load for all cpus | Total combined load for all cpus (cpu load) |
| cpu\|allCpuTotal_ghz | Available | Available |
| cpu\|allCpuCombined_ghz | Used | Used |
| cpu\|allCpuCombined_percent | CPU usage | CPU usage (%) |

**Table 1565: Device Metrics for the Node**

| Metric Key | Metric Name | Description |
|---|---|---|
| device\|iops | Reads/Writes per second | Average number of read/write commands issued per second during the collection interval. |
| device\|await | Average transaction time | Average transaction time (milliseconds). |
| device\|iops_readMaxObserved | Maximum observed reads per second | Maximum observed reads per second. |
| device\|iops_writeMaxObserved | Maximum observed writes per second | Maximum observed writes per second. |

**Table 1566: Service Metrics for the Node**

| Metric Key | Metric Name | Description |
|---|---|---|
| service\|proc\|fdUsage | Total number of open file descriptors | Total number of open file descriptors. |

**Table 1567: NTP Metrics for the Node**

| Metric Key | Metric Name | Description |
|---|---|---|
| ntp\|serverCount | Configured server count | Configured server count |
| ntp\|unreachableCount | Unreachable server count | Unreachable server count |
| ntp\|unreachable | Unreachable | Is the NTP server unreachable. Value of 0 is reachable, 1 means the server was not reached or did not respond. |

**Table 1568: Heap Metrics for the Node**

| Metric Key | Metric Name | Description |
|---|---|---|
| heap\|CurrentHeapSize | Current heap size | Current heap size |
| heap\|MaxHeapSize | Max heap size | Max heap size |
| heap\|CommittedMemory | Committed Memory | Committed Memory |

## Cluster Metrics

VCF Operations collects metrics for the VCF Operations cluster objects including dynamic threshold calculation metrics and capacity computation metrics.

Metrics can be calculated for cluster objects. See Calculated Metrics.

### Cluster Metrics

Cluster metrics provide host, resource, and metric counts on the cluster.

**Table 1569: Cluster Metrics**

| Metric Key | Metric Name | Description |
|---|---|---|
| HostCount | Number of Nodes in Cluster | Number of Nodes in Cluster |
| PrimaryResourcesCount | Number of primary resources | Number of primary resources |
| LocalResourcesCount | Number of local resources | Number of local resources |
| PrimaryMetricsCount | Number of primary metrics | Number of primary metrics |
| ReceivedResourceCount | Number of received resources | Number of received resources |
| ReceivedMetricCount | Number of received metrics | Number of received metrics |
| Count (VM) | Usage Count | This metric displays how many units of the license capacity is currently used. |
| License Metrics for Cluster Object | | |
| Used (VM) | Usage (%) | This metric displays the percentage of the total license capacity currently used.<br>**Note:**<br>An alert is generated if the license threshold is in any of the following states:<br>• >= 80% - Warning<br>• >= 90% - Immediate<br>• =95% - Catastrophic |
| Days Remaining (day) | Days Remaining | This metric displays the days remaining before the license expires. |
| Capacity | Capacity | Displays the maximum number of units (of the given capacity type) that can be licensed by this license key. |
| Type | Type | Displays the license type for the cluster object. |
| Expiry | Expiration Date | Displays the date when the license expires. |

| Metric Key | Metric Name | Description |
|---|---|---|
| Capacity | Capacity | Displays the maximum number of units (of the given capacity type) that can be licensed by this license key. |
| Disbalance factor (%) | Disbalance factor (%) | Identifies the state of disbalance in a VCF Operations cluster and calculates the disbalance factor (when one or more nodes have a higher density among all the nodes in the cluster). Based on the disbalance percentage of the node, the following alerts are triggered:<br>• Warning: If the disbalance factor metric is equal to or greater than 5%.<br>• Immediate: If the disbalance factor metric is equal to or greater than 7%.<br>• Critical: If the disbalance factor metric is equal to or greater than 10%. |

## DT Metrics

DT metrics are dynamic threshold metrics for the cluster. Non-zero values appear only if metric collection occurs while the dynamic threshold calculations are running.

**Table 1570: DT Metrics for the Cluster**

| Metric Key | Metric Name | Description |
|---|---|---|
| dt\|isRunning | Running | Running |
| dt\|dtRunTime | Running duration | Running duration (ms) |
| dt\|StartTime | Running start time | Running start time |
| dt\|percentage | Percent | Percent (%) |
| dt\|executorCount | Executor Node Count | Executor Node Count |
| dt\|resourceCount | Resource Count | Resource Count |
| dt\|fsdbReadTime | FSDB Read Time | FSDB Read Time (ms) |
| dt\|dtObjectSaveTime | DT Object Save Time | DT Object Save Time (ms) |
| dt\|dtHistorySaveTime | DT History Save Time | DT History Save Time (ms) |
| dt\|executor\|resourceCount | Resource Count | Resource Count |

## Capacity Computation (CC) Metrics

CC metrics are capacity computation metrics for the cluster. Non-zero values appear only if metric collection occurs while the capacity computation calculations are running.

**Table 1571: CC Metrics for the Cluster**

| Metric Key | Metric Name | Description |
|---|---|---|
| cc\|isRunning | Running | Running |
| cc\|runTime | Total Run Time | Total Run Time |
| cc\|startTime | Start time | Start time |
| cc\|finishTime | Finish Time | Finish Time |

| Metric Key | Metric Name | Description |
|---|---|---|
| cc\|totalResourcesToProcess | Total Objects Count | Total Objects Count |
| cc\|progress | Progress | Progress |
| cc\|phase1TimeTaken | Phase 1 Computation Time | Phase 1 Computation Time |
| cc\|phase2TimeTaken | Phase 2 Computation Time | Phase 2 Computation Time |

## Gemfire Cluster Metrics

Gemfire metrics provide information about the Gemfire cluster.

**Table 1572: Gemfire cluster Metrics for the Cluster**

| Metric Key | Metric Name | Description |
|---|---|---|
| GemfireCluster\|System\|AvgReads | Average reads per second | The average number of reads per second for all members |
| GemfireCluster\|System\|AvgWrites | Average writes per second | The average number of writes per second for all members |
| GemfireCluster\|System\|DiskReadsRate | Disk reads rate | The average number of disk reads per second across all distributed members |
| GemfireCluster\|System\|DiskWritesRate | Disk writes rate | The average number of disk writes per second across all distributed members |
| GemfireCluster\|System\|GarbageCollectionCount | Total garbage collection count | The total garbage collection count for all members |
| GemfireCluster\|System\|GarbageCollectionCountDelta | New garbage collection count | The new garbage collection count for all members |
| GemfireCluster\|System\|JVMPauses | JVM pause count | The number of detected JVM pauses |
| GemfireCluster\|System\|JVMPausesDelta | New JVM pause count | The number of new detected JVM pauses |
| GemfireCluster\|System\|DiskFlushAvgLatency | Disk flush average latency | Disk flush average latency (msec) |
| GemfireCluster\|System\|NumRunningFunctions | Number of running functions | The number of map-reduce jobs currently running on all members in the distributed system |
| GemfireCluster\|System\|NumClients | Number of clients | The number of connected clients |
| GemfireCluster\|System\|TotalHitCount | Total hit count | Total number of cache hits for all regions |
| GemfireCluster\|System\|TotalHitCountDelta | New hit count | Number of new cache hits for all regions |
| GemfireCluster\|System\|TotalMissCount | Total miss count | The total number of cache misses for all regions |
| GemfireCluster\|System\|TotalMissCountDelta | New miss count | Number of new cache misses for all regions |
| GemfireCluster\|System\|Member\|FreeSwapSpace | Swap space free | Swap space free (MB) |
| GemfireCluster\|System\|Member\|TotalSwapSpace | Swap space total | Swap space total (MB) |
| GemfireCluster\|System\|Member\|CommittedVirtualMemorySize | Committed virtual memory size | Committed virtual memory size (MB) |
| GemfireCluster\|System\|Member\|SystemLoadAverage | System load average | System load average |

| Metric Key | Metric Name | Description |
|---|---|---|
| GemfireCluster\|System\|Member\|FreePhysicalMemory | Free physical memory | Free physical memory (MB) |
| GemfireCluster\|System\|Member\|TotalPhysicalMemory | Total physical memory | Total physical memory (MB) |
| GemfireCluster\|System\|Member\|CacheListenerCallsAvgLatency | Average cache listener calls latency | Average cache listener calls latency (msec) |
| GemfireCluster\|System\|Member\|CacheWriterCallsAvgLatency | Average cache writer calls latency | Average cache writer calls latency (msec) |
| GemfireCluster\|System\|Member\|DeserializationAvgLatency | Average deserialization latency | Average deserialization latency (msec) |
| GemfireCluster\|System\|Member\|FunctionExecutionRate | Function executions per second | Function executions per second |
| GemfireCluster\|System\|Member\|JVMPauses | Number of JVM pauses | Number of JVM pauses |
| GemfireCluster\|System\|Member\|NumRunningFunctions | Number of running functions | Number of running functions |
| GemfireCluster\|System\|Member\|PutsRate | Puts per second | Puts per second |
| GemfireCluster\|System\|Member\|GetsRate | Gets per second | Gets per second |
| GemfireCluster\|System\|Member\|GetsAvgLatency | Average gets latency | Average gets latency (msec) |
| GemfireCluster\|System\|Member\|PutsAvgLatency | Average puts latency | Average puts latency (msec) |
| GemfireCluster\|System\|Member\|SerializationAvgLatency | Average serialization latency | Average serialization latency (msec) |
| GemfireCluster\|System\|Member\|Disk\|DiskFlushAvgLatency | Flush average latency | Flush average latency (msec) |
| GemfireCluster\|System\|Member\|Disk\|DiskReadsRate | Average reads per second | Average reads per second |
| GemfireCluster\|System\|Member\|Disk\|DiskWritesRate | Average writes per second | Average writes per second |
| GemfireCluster\|System\|Member\|Network\|BytesReceivedRate | Average received bytes per second | Average received bytes per second |
| GemfireCluster\|System\|Member\|Network\|BytesSentRate | Average sent bytes per second | Average sent bytes per second |
| GemfireCluster\|System\|Member\|JVM\|GCTimeMillis | Garbage Collection time | Total amount of time spent on garbage collection |
| GemfireCluster\|System\|Member\|JVM\|GCTimeMillisDelta | New Garbage Collection time | New amount of time spent on garbage collection |
| GemfireCluster\|System\|Member\|JVM\|TotalThreads | Total threads | Total threads |
| GemfireCluster\|System\|Member\|JVM\|CommitedMemory | Committed Memory | Committed Memory (MB) |
| GemfireCluster\|System\|Member\|JVM\|MaxMemory | Max Memory | Max Memory (MB) |
| GemfireCluster\|System\|Member\|JVM\|UsedMemory | Used Memory | Used Memory (MB) |

| Metric Key | Metric Name | Description |
|---|---|---|
| GemfireCluster\|Region\|SystemRegionEntryCount | Entry Count | Entry Count |
| GemfireCluster\|Region\|DestroyRate | Destroys per second | Destroys per second |
| GemfireCluster\|Region\|CreatesRate | Creates per second | Creates per second |
| GemfireCluster\|Region\|GetsRate | Gets per second | Gets per second |
| GemfireCluster\|Region\|BucketCount | Bucket count | Bucket count |
| GemfireCluster\|Region\|AvgBucketSize | Average number of entries per bucket | Average number of entries per bucket |
| GemfireCluster\|Region\|Member\|ActualRedundancy | Actual redundancy | Actual redundancy |
| GemfireCluster\|Region\|Member\|BucketCount | Bucket count | Bucket count |
| GemfireCluster\|Region\|Member\|AvgBucketSize | Average number of entries per bucket | Average number of entries per bucket |
| GemfireCluster\|Region\|Member\|CreatesRate | Creates per second | Creates per second |
| GemfireCluster\|Region\|Member\|GetsRate | Gets per second | Gets per second |
| GemfireCluster\|Region\|Member\|DestroyRate | Destroys per second | Destroys per second |
| GemfireCluster\|Region\|Member\|MissCount | Number of misses count | Number of cache misses |
| GemfireCluster\|Region\|Member\|MissCountDelta | Number of new cache misses | Number of new cache misses |
| GemfireCluster\|Region\|Member\|HitCount | Number of hits count | Number of cache hits |
| GemfireCluster\|Region\|Member\|HitCountDelta | Number of new cache hits | Number of new cache hits |

## Threshold Checking Metrics

Threshold checking metrics check the processed and computed metrics for the cluster.

### Table 1573: Threshold Checking Metrics for the Cluster

| Metric Key | Metric Name | Description |
|---|---|---|
| ThresholdChecking\|ProcessedMetricCount | Number of processed metrics | Number of processed metrics |
| ThresholdChecking\|ProcessedMetricRate | Received metric processing rate (per second) | Received metric processing rate (per second) |
| ThresholdChecking\|ComputedMetricCount | Number of computed metrics | Number of computed metrics |
| ThresholdChecking\|ComputedMetricRate | Computed metric processing rate (per second) | Computed metric processing rate (per second) |

## Memory Metrics

Memory metrics provide memory CPU use information for the cluster.

### Table 1574: Memory Metrics for the Cluster

| Metric Key | Metric Name | Description |
|---|---|---|
| Memory\|AvgFreePhysicalMemory | Average free physical memory | Average free physical memory (GB) |
| Memory\|TotalFreePhysicalMemory | Free physical memory | Free physical memory (GB) |
| Memory\|TotalMemory | Total Available Memory | Total Available Memory (GB) |
| Memory\|TotalUsedMemory | Actual Used Memory | Actual Used Memory (GB) |

| Metric Key | Metric Name | Description |
|---|---|---|
| Memory\|TotalDemandMemory | Memory Demand | Memory Demand (GB) |

### Elastic Memory Metrics

Elastic memory metrics provide reclaimable memory CPU use information for the cluster.

**Table 1575: Memory Metrics for the Cluster**

| Metric Key | Metric Name | Description |
|---|---|---|
| ElasticMemory\|TotalMemory | Total Available Memory | Total Available Memory (GB) |
| ElasticMemory\|TotalUsedMemory | Actual Used Memory | Actual Used Memory (GB) |
| ElasticMemory\|TotalDemandMemory | Memory Demand | Memory Demand (GB) |

### CPU Metrics

CPU metrics provide CPU information for the cluster.

**Table 1576: CPU Metrics for the Cluster**

| Metric Key | Metric Name | Description |
|---|---|---|
| cpu\|TotalCombinedUsage | CPU Load | CPU Load |
| cpu\|TotalAvailable | CPU Available | CPU Available |
| cpu\|TotalAvailable_ghz | Available | Available (GHz) |
| cpu\|TotalUsage_ghz | Used | Used (GHz) |
| cpu\|TotalUsage | CPU usage | CPU usage (%) |

### Disk Metrics

Disk metrics provide available disk information for the cluster.

**Table 1577: Disk Metrics for the Cluster**

| Metric Key | Metric Name | Description |
|---|---|---|
| Disk\|DatabaseStorage\|AvgAvailable | Average node disk available | Average node disk available |
| Disk\|DatabaseStorage\|MinAvailable | Minimum node disk available | Minimum node disk available |
| Disk\|DatabaseStorage\|MaxAvailable | Maximum node disk available | Maximum node disk available |
| Disk\|DatabaseStorage\|TotalAvailable | Available | Available |
| Disk\|DatabaseStorage\|Total | Total | Total |
| Disk\|DatabaseStorage\|TotalUsed | Used | Used |
| Disk\|LogStorage\|AvgAvailable | Average node disk available | Average node disk available |
| Disk\|LogStorage\|MinAvailable | Minimum node disk available | Minimum node disk available |
| Disk\|LogStorage\|MaxAvailable | Maximum node disk available | Maximum node disk available |
| Disk\|LogStorage\|TotalAvailable | Available | Available |
| Disk\|LogStorage\|Total | Total | Total |
| Disk\|LogStorage\|TotalUsed | Used | Used |

## Persistence Metrics

VCF Operations collects metrics for various persistence resources or service groups.

### Activity Metrics

Activity metrics relate to the activity framework.

**Table 1578: Activity Metrics for Persistence**

| Metric Key | Metric Name | Description |
|---|---|---|
| Activity\|RunningCount | Number Running | Number Running |
| Activity\|ExecutedCount | Number Executed | Number Executed |
| Activity\|SucceededCount | Number Succeeded | Number Succeeded |
| Activity\|FailedCount | Number Failed | Number Failed |

### Controller XDB Metrics

Controller metrics relate to the primary database.

**Table 1579: Controller XDB Metrics for Persistence**

| Metric Key | Metric Name | Description |
|---|---|---|
| ControllerXDB\|Size | Size | Size (Bytes) |
| ControllerXDB\|TempDBSize | Temporary DB Size | Temporary DB Size (Bytes) |
| ControllerXDB\|TotalObjectCount | Total Object Count | Total Object Count |
| ControllerXDB\|AvgQueryDuration | Average Query Duration | Average Query Duration (ms) |
| ControllerXDB\|MinQueryDuration | Minimum Query Duration | Minimum Query Duration (ms) |
| ControllerXDB\|MaxQueryDuration | Maximum Query Duration | Maximum Query Duration (ms) |
| ControllerXDB\|TotalTransactionCount | Total Transaction Count | Total Transaction Count |
| ControllerXDB\|LockOperationErrorCount | Lock Operation Error Count | Lock Operation Error Count |
| ControllerXDB\|DBCorruptionErrorCount | DB Corruption Error Count | DB Corruption Error Count |
| ControllerXDB\|DBMaxSessionExceededCount | DB Maximum Sessions Exceeded Count | DB Maximum Sessions Exceeded Count |
| ControllerXDB\|NumberWaitingForSession | Number of operations waiting for a session | Number of operations waiting for a session from the session pool |
| ControllerXDB\|AvgWaitForSessionDuration | Average acquisition time from session pool | Average acquisition time from session pool |
| ControllerXDB\|MinWaitForSessionDuration | Minimum acquisition time from session pool | Minimum acquisition time from session pool |
| ControllerXDB\|MaxWaitForSessionDuration | Maximum acquisition time from session pool | Maximum acquisition time from session pool |
| ControllerXDB\|TotalGetSessionCount | Total requests for a session from the session pool | Total requests for a session from the session pool |
| ControllerXDB\|MaxActiveSessionCount | Maximum Concurrent Session Count | Maximum concurrent session count during the past collection interval. |

## Alarm SQL Metrics

Alarm metrics relate to the persistence of alerts and symptoms.

**Table 1580: Alarm XDB Metrics for Persistence**

| Metric Key | Metric Name | Description |
|---|---|---|
| AlarmSQL\|Size | Size (Bytes) | Size (Bytes) |
| AlarmSQL\|AvgQueryDuration | Average Query Duration (ms) | Average Query Duration (ms) |
| AlarmSQL\|MinQueryDuration | Minimum Query Duration (ms) | Minimum Query Duration (ms) |
| AlarmSQL\|MaxQueryDuration | Maximum Query Duration (ms) | Maximum Query Duration (ms) |
| AlarmSQL\|TotalTransactionCount | Total Transaction Count | Total Transaction Count |
| AlarmSQL\|TotalAlarms | Alarm Total Object Count | Alarm Total Object Count |
| AlarmSQL\|TotalAlerts | Alert Total Object Count | Alert Total Object Count |
| AlarmSQL\|AlertTableSize | Alert Table Size | Alert Table Size |
| AlarmSQL\|AlarmTableSize | Alarm Table Size | Alarm Table Size |

## Key Value Store Database (KVDB)

KVDB metrics relate to the persistence of storing key-value data.

| Metric Key | Metric Name | Description |
|---|---|---|
| KVDB\|AvgQueryDuration | Average Query Duration | Average Query Duration |
| KVDB\|MinQueryDuration | Minimum Query Duration | Minimum Query Duration |
| KVDB\|MaxQueryDuration | Maximum Query Duration | Maximum Query Duration |
| KVDB\|TotalTransactionCount | Total Transaction Count | Total Transaction Count |

## Historical Inventory Service XDB Metrics

Historical inventory service metrics relate to the persistence of configuration properties and their changes.

**Table 1581: Historical XDB Metrics for Persistence**

| Metric Key | Metric Name | Description |
|---|---|---|
| HisXDB\|FunctionCalls\|Count HisXDB\|<br>FunctionCalls | Number of Function calls | Number of Function calls |
| HisXDB\|FunctionCalls\|AvgDuration | Average execution time | Average execution time |
| HisXDB\|FunctionCalls\|MaxDuration | Max execution time | Max execution time |
| HisXDB\|Size | Size | Size (Bytes) |
| HisXDB\|TempDBSize | Temporary DB Size | Temporary DB Size (Bytes) |
| HisXDB\|TotalObjectCount | Total Object Count | Total Object Count |
| HisXDB\|AvgQueryDuration | Average Query Duration | Average Query Duration (ms) |
| HisXDB\|MinQueryDuration | Minimum Query Duration | Minimum Query Duration (ms) |
| HisXDB\|MaxQueryDuration | Maximum Query Duration | Maximum Query Duration (ms) |
| HisXDB\|TotalTransactionCount | Total Transaction Count | Total Transaction Count |
| HisXDB\|LockOperationErrorCount | Lock Operation Error Count | Lock Operation Error Count |
| HisXDB\|DBCorruptionErrorCount | DB Corruption Error Count | DB Corruption Error Count |

| Metric Key | Metric Name | Description |
|---|---|---|
| HisXDB\|DBMaxSessionExceededCount | DB Maximum Sessions Exceeded Count | DB Maximum Sessions Exceeded Count |
| HisXDB\|NumberWaitingForSession | Number of operations waiting for a session | Number of operations waiting for a session from the session pool |
| HisXDB\|AvgWaitForSessionDuration | Average acquisition time from session pool | Average acquisition time from session pool |
| HisXDB\|MinWaitForSessionDuration | Minimum acquisition time from session pool | Minimum acquisition time from session pool |
| HisXDB\|MaxWaitForSessionDuration | Maximum acquisition time from session pool | Maximum acquisition time from session pool |
| HisXDB\|TotalGetSessionCount | Total requests for a session from the session pool | Total requests for a session from the session pool |
| HisXDB\|HisActivitySubmissionCount | HIS activity submission count | Number of Historical Inventory Service activities submitted |
| HisXDB\|HisActivityCompletionCount | HIS activity completion count | Number of Historical Inventory Service activities completed |
| HisXDB\|HisActivityCompletionDelayAvg | HIS activity average completion delay | The average amount of time from activity submission to completion |
| HisXDB\|HisActivityCompletionDelayMax | HIS activity maximum completion delay | The maximum amount of time from activity submission to completion |
| HisXDB\|HisActivityAbortedCount | HIS activity abort count | Number of Historical Inventory Service activities stopped |

## VCF Automation Metrics

VCF Automation collects metrics for objects such as, cloud zone, project, deployment, blueprint, cloud account, user, and cloud automation services world Instance.

### Blueprint Metrics

VCF Automation collects metrics for objects such as blueprint object.

**Table 1582: Blueprint Metrics**

| Property Name | Metrics |
|---|---|
| Summary | VMCount |

### Project Metrics

VCF Automation collects metrics for objects such as project object.

**Table 1583: Project Metrics**

| Property Name | Metrics |
|---|---|
| Summary | VMCount |
| Summary | TotalDeployments |
| Summary | TotalCloudZones |
| Summary | TotalBlueprints |

| Property Name | Metrics |
|---|---|
| Summary | Metering\|Additional price |
| Summary | Metering\|CPU Price |
| Summary | Metering\|Memory price |
| Summary | Metering\|Storage Price |
| Summary | Metering\|Total price |

## Deployment Metrics

VCF Automation collects the metrics for the deployment object.

**Table 1584: Deployment Metrics**

| Property Name | Metrics |
|---|---|
| Summary | Metering\|Additional price |
| Summary | Metering\|CPU Price |
| Summary | Metering\|Memory price |
| Summary | Metering\|Storage Price |
| Summary | Metering\|Total price |
| Summary | Metering\|Partial price |

## Organization Metrics

VCF Automation collects the metrics for the organization object.

**Table 1585: Organization Metrics**

| Property Name | Metrics |
|---|---|
| Summary | TotalBlueprints |
| Summary | TotalProjects |
| Summary | VMCount |
| Summary | TotalDeployments |
| Summary | TotalCloudZones |

## VCF Automation Adapter Metrics

VCF Automation collects the metrics for the adapter object.

**Table 1586: VCF Automation Adapter Metrics**

| Property Name | Metrics |
|---|---|
| Summary | TotalCloudZones |
| Summary | VMCount |
| Summary | TotalDeployments |

| Property Name | Metrics |
|---|---|
| Summary | TotalBlueprints |
| Summary | TotalProjects |

## Cloud Automation Services World Metrics

VCF Automation collects the metrics for the Cloud Automation Services world object.

**Table 1587: Cloud Automation Services World Metrics**

| Property Name | Metrics |
|---|---|
| Summary | TotalDeployments |
| Summary | VMCount |
| Summary | TotalCloudZones |
| Summary | TotalProjects |
| Summary | TotalBlueprints |

## Cloud Automation Services Entity Status Metrics

VCF Automation collects the metrics for the Cloud Automation Services (CAS) entity status object.

**Table 1588: Cloud Automation Services Entity Status Metrics**

| Property Name | Metrics |
|---|---|
| Summary | TotalClusters |

## Cloud Zone Metrics

VCF Automation collects memory, storage, and vCPU limit related metrics at project level on per cloud zone basis.

### Cloud Zone Limits Metrics

You can view the memory, storage, and vCPU limit related metrics defined in VCF Automation in VCF Operations. VCF Operations alerts when the utilisation exceeds configured limits in any of the individual cloud zones.

> **Note:**
> The resource limits are configured in VCF Automation. For vCenter objects, limits for all resources - vCPU, memory, and storage - can be configured. However, for public clouds like AWS, Azure, and GCP only vCPU and memory limits can be configured.

**Table 1589: Cloud Zone Limits Metrics**

| Property Name | Metrics | Description |
|---|---|---|
| Cloud Zone Limits | Memory allocated (KB) | This metric displays the total utilized cloud zone memory. |
| | Memory Limit (KB) | This metric displays the limit set on the cloud zone memory usage. |

| Property Name | Metrics | Description |
|---|---|---|
| | Memory Utilized (%) | This metric displays the percentage of memory used out of the memory limit set. Formula ((Memory allocated (KB)/ Memory Limit (KB)) |
| | Storage allocated (GB) | This metric displays the total storage utilized by the cloud zone. |
| | Storage Limit (GB) | This metrics displays the limit set on the cloud zone storage usage. |
| | Storage Utilized (%) | This metric displays the percentage of storage used out of the storage limit set. Formula (Storage allocated (KB)/ Storage Limit (KB)) |
| | vCPU allocated | This metric displays the total vCPU utilized by the cloud zone. |
| | vCPU Limit | This metrics displays the limit set on the cloud zone vCPU usage. |
| | vCPU Utilized (%) | This metric displays the percentage of vCPU used out of the vCPU limit set. Formula (vCPU allocated (KB)/ vCPU Limit (KB)) |

**Summary Metrics for Cloud Zone Centers**

Summary metrics provide information about the VM migration as part of workload planning.

| Metric Name | Description |
|---|---|
| WLP | Displays the VM migration trend as part of workload optimization. These metrics are deactivated by default. You must activate them from policies.<br>• Fail Count: Number of failed VM move attempts in the last daily cycle.<br>• Number of runs: The total number of times the WLP was run during the last daily cycle.<br>• Success Count: The number of successful VM moves during the last daily cycle. |

## Metrics for vSAN

VCF Operations collects metrics for vSAN objects.

From the left menu, click **Inventory** > **Inventory Panel (Detailed View)** > **Integrations** > **All Objects** > **vSAN Adapter**, and then select one of the vSAN adapter objects listed and click the **Metrics** tab.

### Disk I/O and Disk Space Metrics for vSAN Disk Groups

The VCF Operations collects the metrics you use to monitor the performance of your vSAN disk groups.

Disk I/O metrics for the vSAN disk groups include:

- Disk I/O|Reads Per Second (IOPS)
- Disk I/O|Writes Per Second (IOPS
- Disk I/O|Max Observed Reads Per Second (IOPS)
- Disk I/O|Max Observed Writes Per Second (IOPS)

- Disk I/O|Throughput Read (bps)
- Disk I/O|Throughput Write (bps)
- Disk I/O|Average Read Latency (ms)
- Disk I/O|Average Write Latency (ms)
- Disk I/O|Total Bus Resets
- Disk I/O|Total Commands Aborted per second

The following Disk I/O metrics are disabled by default:

- Disk I/O|Read Count
- Disk I/O|Write Count
- Disk I/O|Average Device Latency
- Disk I/O|Average Device Read Latency
- Disk I/O|Average Device Write Latency
- Disk I/O|Total Number of Errors

Disk space metrics for vSAN disk groups include:

- Disk Space|Capacity (bytes)
- Disk Space|Used (bytes)
- Disk Space|Usage (%)

### Read Cache Metrics for vSAN Disk Groups

The VCF Operations collects metrics and performs capacity trend analysis on a hybrid vSAN read cache. Read Cache metrics are not collected for a vSAN all-flash configuration.

Read cache metrics for the vSAN disk group include:

- Read Cache|Hit Rate (%)
- Read Cache|Miss Rate Ratio
- Read Cache|Reads Per Second (IOPS)
- Read Cache|Read Latency (ms)
- Read Cache|Writes Per Second (IOPS)
- Read Cache|Write Latency (ms)

The following read cache metrics are disabled by default:

- Read Cache|Read I/O Count
- Read Cache|Write I/O Count

### Write Buffer Metrics for vSAN Disk Groups

The VCF Operations collects the metrics you use to monitor the write buffer capacity of your vSAN disk groups.

A reasonably balanced system consumes a significant amount of write buffer. Before placing additional workload on the vSAN, check the write buffer metrics for the vSAN disk group.

- Write Buffer|Capacity (bytes)
- Write Buffer|Free (%)
- Write Buffer|Usage (%)
- Write Buffer|Used (byte)
- Write Buffer|Reads Per Second (IOPS)
- Write Buffer|Read Latency (ms)
- Write Buffer|Writes Per Second (IOPS)
- Write Buffer|Write Latency (ms)

The following write buffer metrics are disabled by default:

- Write Buffer|Read I/O Count
- Write Buffer|Write I/O Count

### Congestion Metrics for vSAN Disk Groups

The VCF Operations collects congestion metrics for the vSAN disk group.

- Congestion| Memory Congestion - Favorite
- Congestion| SSD Congestion - Favorite
- Congestion| IOPS Congestion - Favorite
- Congestion| Slab Congestion
- Congestion| Log Congestion
- Congestion| Comp Congestion

### Cache De-stage Metrics for vSAN Disk Groups

The VCF Operations collects cache de-stage metrics for the vSAN disk groups.

Cache de-stage metrics include:

- Bytes De-stage from SSD
- Zero-bytes De-stage

### Resync Traffic Metrics for vSAN Disk Groups

The VCF Operations collects resync traffic metrics for the vSAN disk groups.

Resync traffic metrics include:

- Read IOPS for Resync Traffic
- Write IOPS for Resync Traffic
- Read Throughput for Resync Traffic
- Write Throughput for Resync Traffic
- Read Latency for Resync Traffic
- Write Latency for Resync Traffic

### Metrics for vSAN Cluster

The VCF Operations collects the metrics you use to monitor the performance of your vSAN cluster.

VCF Operations enhances the capacity calculation for vSAN, using the new slack space provided by the new vSAN API. Cost calculation is still done using the old way which reserves 30% memory for Slack Overhead.

Metrics for vSAN cluster include:

| Component | Metrics |
|---|---|
| Component Limit | • vSAN\|Component Limit\|Component Limit Used (%)<br>• vSAN\|Component Limit\|Total Component Limit<br>• vSAN\|Component Limit\|Used Component Limit |
| Disk Space | • vSAN\|Disk Space\|Disk Space Used (%)<br>• vSAN\|Disk Space\|Total Disk Space (GB)<br>• vSAN\|Disk Space\|Used Disk Space (GB)<br>• vSAN\|Disk Space\|Usable Capacity (GB)<br>• vSAN\|Disk Space\|Oversubscription Capacity (GB)<br>• vSAN\|Disk Space\|Oversubscription Ratio<br>**Note:**<br>The oversubscription metrics are applicable only for the vSAN OSA clusters. |
| Health | vSAN\|Health\|Score |
| Read Cache | • vSAN\|Read Cache\|Read Cache Reserved (%)<br>• vSAN\|Read Cache\|Reserved Read Cache Size (GB)<br>• vSAN\|Read Cache\|Total Read Cache Size (GB) |
| Performance | • vSAN\|Read Cache\|Reads Per Second (IOPS)<br>• vSAN\|Read Cache\|Read Throughput (KBps)<br>• vSAN\|Read Cache\|Average Read Latency (ms)<br>• vSAN\|Read Cache\|Writes Per Second (IOPS)<br>• vSAN\|Read Cache\|Write Throughput (KBps)<br>• vSAN\|Read Cache\|Average Write Latency (ms)<br>• vSAN\|Read Cache\|Congestion<br>• vSAN\|Read Cache\|Outstanding I/O<br>• vSAN\|Read Cache\|Total IOPS<br>• vSAN\|Read Cache\|Total Latency (ms)<br>• vSAN\|Read Cache\|Total Throughput (KBps) |
| Deduplication And Compression Overview | • vSAN\|Deduplication And Compression Overview\|Used Before<br>• vSAN\|Deduplication And Compression Overview\|Used After<br>• vSAN\|Deduplication And Compression Overview\|Savings<br>• vSAN\|Deduplication And Compression Overview\|Ratio |
| Summary | • Summary\|Number of Storage Pools<br>• Summary\|Number of ESA Disks<br>• Summary\|Number of Cache Disks<br>• Summary\|Total Number of Capacity Disks<br>• Summary\|CPU Workload<br>• Summary\|Memory Workload<br>• Summary\|Total Number of Disk Groups<br>• Summary\|Total Active Alerts Count<br>• Summary\|Total Number of VMs<br>• Summary\|Total Number of Hosts<br>• Summary\|vSAN Cluster Capacity Remaining (%)<br>• Summary\|vSAN Cluster Storage Time Remaining<br>• Summary\|vSAN Capacity Disk Used<br>• Summary \| Total vSAN CPU Used (MHz)<br>• Summary \| Max vSAN CPU Ready<br>• Summary \| Worst VM Disk Latency |

| Component | Metrics |
|---|---|
| KPI | • KPI\|Max ESA Disk IOPS<br>• KPI\|Max ESA Disk Latency<br>• KPI\|Min Storage Pool Free Capacity<br>• KPI\|Sum Storage Pool Errors<br>• KPI\|Disk Groups Metrics<br>• KPI\|Sum Host VMKernel Packets Dropped<br>• KPI\|Count Disk Group Congestion Above 50<br>• KPI\|Max Disk Group Congestion<br>• KPI\|Sum Disk Group Errors<br>• KPI\|Min Disk Group Capacity Free<br>• KPI\|Min Disk Group Read Cache Hit Rate<br>• KPI\|Min Disk Group Write Buffer Free<br>• KPI\|Max Disk Group Read Cache/Write Buffer Latency<br>• KPI\|Max Capacity Disk Latency<br>• KPI \| Max Capacity Disk IOPS |
| IO Size | • vSAN \| Performance \| I/O Size (KB)<br>• vSAN \| Performance \| Read I/O Size (KB)<br>• vSAN \| Performance \| Write I/O Size (KB) |
| Resynchronization Status ( Metrics applicable for vSAN 6.7 and later) | • vSAN \| Resync \| Bytes left to resync (bytes)<br>• vSAN \| Resync \| Resyncing Objects |
| What If Analysis | This is an instanced metric<br>• vSAN \| What If \| Effective Free Space (GB) |
| Stretched Cluster | • vSAN\|Stretched Cluster\|Latency Between Sites\|Preferred and Secondary (ms)<br>• vSAN\|Stretched Cluster\|Latency Between Sites\|Preferred and Witness (ms)<br>• vSAN\|Stretched Cluster\|Latency Between Sites\|Secondary and Witness (ms) |
| File Share | • vSAN\|FileServices\|totalShareCount |
| File Service | • vSAN \| File Services \| File Shares Used Disk Space (GB)<br>• vSAN \| File Services \| Root FS Used Disk Space (GB)<br>• vSAN \| File Services \| File Shares Count |
| Slack Space | • vSAN\|Slack Space\|Internal Operations Capacity (GB)<br>• vSAN\|Slack Space\|Host Rebuild Capacity (GB)<br>• vSAN\|Slack Space\|Transient Capacity Used (GB) |

## Metrics for vSAN Enabled Host

The VCF Operations collects the metrics you use to monitor the performance of your vSAN enabled host.

Metrics for a vSAN enabled host include:

| Component | Metrics |
|---|---|
| Component Limit | • vSAN\|Component Limit\|Component Limit Used (%)<br>• vSAN\|Component Limit\|Total Component Limit<br>• vSAN\|Component Limit\|Used Component Limit |
| Disk Space | • vSAN\|Disk Space\|Disk Space Used (%)<br>• vSAN\|Disk Space\|Total Disk Space (GB)<br>• vSAN\|Disk Space\|Used Disk Space (GB) |
| Read Cache | • vSAN\|Read Cache\|Read Cache Reserved (%)<br>• vSAN\|Read Cache\|Reserved Read Cache Size (GB)<br>• vSAN\|Read Cache\|Total Read Cache Size (GB) |
| Performance Metrics | |

| Component | Metrics |
|---|---|
| • Network | • vSAN\|Performance\|Network\|Inbound Packets Loss Rate<br>• vSAN\|Performance\|Network\|Outbound Packets Loss Rate<br>• vSAN\|Performance\|Network\|<vnic>\|Inbound Packets Loss rate (%)<br>• vSAN\|Performance\|Network\|<vnic>\|Outbound Packets Loss Rate (%)<br>• vSAN\|Performance\|Network\|<vnic>\|Inbound Packets Per second<br>• vSAN\|Performance\|Network\|<vnic>\|Outbound Packets Per second<br>• vSAN\|Performance\|Network\|<vnic>\|Throughput Inbound (KBps)<br>• vSAN\|Performance\|Network\|<vnic>\|Throughput Outbound (KBps) |
| • CPU Utilization | • vSAN \| Performance \| CPU \| Ready (%)<br>• vSAN \| Performance \| CPU \| Usage (%)<br>• vSAN \| Performance \| CPU \| Used (MHz)<br>• vSAN \| Performance \| CPU \| Core Utilization (%) (For Hyper-Threading Technology) |
| • PCPU Utilization | • vSAN \| Performance \| PCPU \| Ready (%)<br>• vSAN \| Performance \| CPU \| PCPU Usage (%) |
| • Memory | • vSAN \| Performance \| Memory \| Usage (%)<br>• vSAN \| Performance \| Memory \| Used (GB) |

### Metrics for vSAN Datastore

The VCF Operations collects the metrics you use to monitor the performance of your vSAN datastore.

Datastore I/O metrics for vSAN datastore include:

- Datastore I/O|Reads Per Second (IOPS)
- Datastore I/O|Read Rate (KBps)
- Datastore I/O|Read Latency (ms)
- Datastore I/O|Writes Per Second (IOPS)
- Datastore I/O|Write Rate (KBps)
- Datastore I/O|Write Latency (ms)
- Datastore I/O|Outstanding I/O requests
- Datastore I/O|Congestion
- Capacity | Usable Capacity

### Metrics for vSAN Cache Disk

The VCF Operations collects the metrics you use to monitor the performance of your vSAN cache disk.

Metrics for vSAN cache disk include:

| Component | Metrics |
|---|---|
| Performance | • Performance\|Bus Resets<br>• Performance\|Commands Aborted Per Second<br>The following performance metrics are disabled by default:<br>• Performance\|Device Latency (ms)<br>• Performance\|Device Read Latency (ms)<br>• Performance\|Device Write Latency (ms)<br>• Performance\|Read Requests Per Second<br>• Performance\|Average Reads Per Second<br>• Performance\|Write Requests Per Second<br>• Performance\|Average Writes Per Second<br>• Performance\|Read Rate<br>• Performance\|Write Rate<br>• Performance\|Usage<br>• Performance\|HDD Errors |
| SCSI SMART Statistics<br>**Note:** SMART data collection is disabled by default. To enable SMART data collection, ensure that the `Enable SMART data collection` instance identifier is set to true. For proper data collection, ensure that ESX hosts in your vCenter Server inventory have CIM service enabled and CIM providers for each SMART metric installed. | • SCSI SMART Statistics\|Health Status<br>• SCSI SMART Statistics\|Media Wearout Indicator<br>• SCSI SMART Statistics\|Write Error Count<br>• SCSI SMART Statistics\|Read Error Count<br>• SCSI SMART Statistics\|Power on Hours<br>• SCSI SMART Statistics\|Reallocated Sector Count<br>• SCSI SMART Statistics\|Raw Read Error Rate<br>• SCSI SMART Statistics\|Drive Temperature<br>• SCSI SMART Statistics\|Maximum Observed Drive Temperature<br>• SCSI SMART Statistics\|Drive Rated Max Temperature<br>• SCSI SMART Statistics\|Write Sectors TOT Count<br>• SCSI SMART Statistics\|Read Sectors TOT Count<br>• SCSI SMART Statistics\|Initial Bad Block Count<br>• SCSI SMART Statistics\|Worst Media Wearout Indicator<br>• SCSI SMART Statistics\|Worst Write Error Count<br>• SCSI SMART Statistics\|Worst Read Error Count<br>• SCSI SMART Statistics\|Worst Power-on Hours<br>• SCSI SMART Statistics\|Power Cycle Count<br>• SCSI SMART Statistics\|Worst Power Cycle Count<br>• SCSI SMART Statistics\|Worst Reallocated Sector Count<br>• SCSI SMART Statistics\|Worst Raw Read Error Rate<br>• SCSI SMART Statistics\|Worst Driver Rated Max Temperature<br>• SCSI SMART Statistics\|Worst Write Sectors TOT Count<br>• SCSI SMART Statistics\|Worst Read Sectors TOT Count<br>• SCSI SMART Statistics\|Worst Initial Bad Block Count |
| Capacity | • vSAN\|Health\|Capacity\|Total Disk Capacity (GB)<br>• vSAN\|Health\|Capacity\|Used Disk Capacity (GB) |
| Congestion Health | • vSAN\|Health\|Congestion Health\|Congestion Value |

| Component | Metrics |
| --- | --- |
| Performance | • vSAN\|Performance\|Physical Layer Reads Per Second<br>• vSAN\|Performance\|Physical Layer Writes Per Second<br>• vSAN\|Performance\|Physical Layer Read Throughput (KBps)<br>• vSAN\|Performance\|Physical Layer Write Throughput (KBps)<br>• vSAN\|Performance\|Physical Layer Read Latency (ms)<br>• vSAN\|Performance\|Physical Layer Write Latency (ms)<br>• vSAN\|Performance\|Physical Layer Read Count<br>• vSAN\|Performance\|Physical Layer Write Count<br>• vSAN\|Performance\|Device Average Latency (ms)<br>• vSAN\|Performance\|Guest Average Latency (ms) |

### Metrics for vSAN Capacity Disk

The VCF Operations collects the metrics you use to monitor the performance of your vSAN capacity disk.

Metrics for vSAN capacity disk include:

| Component | Metrics |
|---|---|
| Performance | • Performance\|Bus Resets<br>• Performance\|Commands Aborted Per Second<br>The following performance metrics are disabled by default:<br>•<br>• Performance\|Device Latency (ms)<br>• Performance\|Device Read Latency (ms)<br>• Performance\|Device Write Latency (ms)<br>• Performance\|Read Requests Per Second<br>• Performance\|Average Reads Per Second<br>• Performance\|Write Requests Per Second<br>• Performance\|Average Writes Per Second<br>• Performance\|Read Rate<br>• Performance\|Write Rate<br>• Performance\|Usage<br>• Performance\|HDD Errors |
| SCSI SMART Statistics<br>**Note:**<br>SMART data collection is disabled by default. To enable SMART data collection, ensure that the `Enable SMART data collection` instance identifier is set to true. For proper data collection, ensure that ESX hosts in your vCenter Server inventory have CIM service enabled and CIM providers for each SMART metric installed. | • SCSI SMART Statistics\|Health Status<br>• SCSI SMART Statistics\|Media Wearout Indicator<br>• SCSI SMART Statistics\|Write Error Count<br>• SCSI SMART Statistics\|Read Error Count<br>• SCSI SMART Statistics\|Power on Hours<br>• SCSI SMART Statistics\|Reallocated Sector Count<br>• SCSI SMART Statistics\|Raw Read Error Rate<br>• SCSI SMART Statistics\|Drive Temperature<br>• SCSI SMART Statistics\|Maximum Observed Drive Temperature<br>• SCSI SMART Statistics\|Drive Rated Max Temperature<br>• SCSI SMART Statistics\|Write Sectors TOT Count<br>• SCSI SMART Statistics\|Read Sectors TOT Count<br>• SCSI SMART Statistics\|Initial Bad Block Count<br>• SCSI SMART Statistics\|Worst Media Wearout Indicator<br>• SCSI SMART Statistics\|Worst Write Error Count<br>• SCSI SMART Statistics\|Worst Read Error Count<br>• SCSI SMART Statistics\|Worst Power-on Hours<br>• SCSI SMART Statistics\|Power Cycle Count<br>• SCSI SMART Statistics\|Worst Power Cycle Count<br>• SCSI SMART Statistics\|Worst Reallocated Sector Count<br>• SCSI SMART Statistics\|Worst Raw Read Error Rate<br>• SCSI SMART Statistics\|Worst Driver Rated Max Temperature<br>• SCSI SMART Statistics\|Worst Write Sectors TOT Count<br>• SCSI SMART Statistics\|Worst Read Sectors TOT Count<br>• SCSI SMART Statistics\|Worst Initial Bad Block Count |
| Capacity | • vSAN\|Health\|Total Disk Capacity (GB)<br>• vSAN\|Health\|Used Disk Capacity (GB)<br>• vSAN\|FileServices\|FileSharesUsedDiskSpace<br>• vSAN\|FileServices\|RootFsUsedDiskSpace |
| Congestion Health | vSAN\|Health\|Congestion Value |

| Component | Metrics |
|---|---|
| Performance | • vSAN\|Performance\|Physical Layer Reads Per Second<br>• vSAN\|Performance\|Physical Layer Writes Per Second<br>• vSAN\|Performance\|Physical Layer Read Throughput (KBps)<br>• vSAN\|Performance\|Physical Layer Write Throughput (KBps)<br>• vSAN\|Performance\|Physical Layer Read Latency (ms)<br>• vSAN\|Performance\|Physical Layer Write Latency (ms)<br>• vSAN\|Performance\|Physical Layer Read Count<br>• vSAN\|Performance\|Physical Layer Write Count<br>• vSAN\|Performance\|Device Average Latency (ms)<br>• vSAN\|Performance\|Guest Average Latency (ms)<br>• vSAN\|Performance\|vSAN Layer Reads Per Second<br>• vSAN\|Performance\|vSAN Layer Writes Per Second<br>• vSAN\|Performance\|vSAN Layer Read Latency (ms)<br>• vSAN\|Performance\|vSAN Layer Write Latency (ms)<br>• vSAN\|Performance\|vSAN Layer Read Count<br>• vSAN\|Performance\|vSAN Layer Write Count<br>• vSAN \| Performance \| vSAN Layer Total IOPS |

Properties for vSAN capacity disk include:

- Name
- Size
- Vendor
- Type
- Queue Depth

### Metrics for vSAN Fault Domain Resource Kind

The VCF Operations collects the metrics you use to monitor the performance of your vSAN stretched cluster with fault domain.

Metrics for vSAN fault domain resource kind includes:

- CPU
  - Demand
    - Demand (MHz)
    - Demand without overhead (MHz)
    - Overhead (MHz)
    - Reserved Capacity (MHz)
    - Total Capacity (MHz)
    - VM CPU Usage (MHz)
    - Workload (%)
- Disk Space
  - Demand
    - Workload (%)
- Memory
  - Contention (KB)
  - Demand
    - Host Usage (KB)
    - Machine Demand (KB)
    - Reserved Capacity (KB)
    - Total Capacity (KB)
    - Utilization (KB)

    - Workload (%)
- vSAN
  - Disk Space
    - Total Disk Space (GB)
    - Used Disk Space (GB)

### Metrics for vSAN World

The VCF Operations collects the metrics you use to monitor the performance of your vSAN world.

Metrics for vSAN world include:

- Summary|Total Number of VMs
- Summary|Total Number of Hosts
- Summary|Total IOPS
- Summary|Total Latency
- Summary|Total Number of Clusters
- Summary|Total Number of DiskGroups
- Summary|Total Number of Cache Disks
- Summary|Total Number of Capacity Disks
- Summary|Total Number of Datastores
- Summary|Total vSAN Disk Capacity (TB)
- Summary|Total vSAN Disk Capacity Used (TB)
- Summary|Remaining Capacity (TB)
- Summary|Remaining Capacity (%)
- Summary|Total Savings by Deduplication and Compression (GB)

### Metrics for vSAN File Server

The VCF Operations collects the metrics you use to monitor the performance of your vSAN File Server.

Metrics for vSAN File Server

| Component | Metrics |
|---|---|
| File Server | • vSAN \| Disk Space File Shares Used Disk Space (GB)<br>• vSAN \| Summary \| File Shares Count |

### Metrics for vSAN File Share

The VCF Operations collects the metrics you use to monitor the performance of your vSAN File Share.

Metrics for vSAN File Share

| Component | Metrics |
|---|---|
| Disk Space | • vSAN \| Disk Space \| Used Disk Space (GB) |
| Read Performance | • vSAN \| Performance \| Read Throughput Requested (MBps)<br>• vSAN \| Performance \| Read Throughput Transferred (MBps)<br>• vSAN \| Performance \| Read IOPS<br>• vSAN \| Performance \| Read Latency (ms) |

| Component | Metrics |
| --- | --- |
| Write Performance | • vSAN \| Performance \| Write Throughput Requested (MBps)<br>• vSAN \| Performance \| Write Throughput Transferred (MBps)<br>• vSAN \| Performance \| Write IOPS<br>• vSAN \| Performance \| Write Latency (ms) |

## Capacity Model for vSAN Objects

The capacity model introduced in VCF Operations 6.7 now extends the support for vSAN objects like, vSAN cluster, Fault domains, and Cache/Capacity disks. The Capacity tab provides Time Remaining data for the selected vSAN cluster, Fault domain, Cache/Capacity Disk objects. The information is presented in a graphical format.

### Where You Find the Capacity Tab

In the menu, click **Environment**, then select a group, custom data center, application, or inventory object. The Object details page appears. Click the **Capacity** tab.

The VCF Operations defines the capacity model for the following vSAN resource containers:

- vSAN Cluster
  - Disk Space
- vSAN Fault Domain
  - CPU
  - Memory
  - Disk Space
- vSAN Cache/Capacity Disk
  - Disk Space

### Understanding the Capacity Tab

For the selected vSAN resource, the capacity tab lists the capacity used and Time Remaining until the associated CPU, memory, and disk space resources, respectively, run out.

- If you select the vSAN cluster, the capacity tab lists the capacity used and time remaining until the associated disk space runs out.
- If you select the vSAN Fault Domain, the capacity tab lists the capacity used and time remaining until the associated CPU, memory, and disk space resources run out.
- If you select the vSAN Cache/Capacity Disk Space, the capacity tab lists capacity used and time remaining until the associated disk space runs out.

The available graph depicts - for your choice of CPU, memory, or disk space - the amount of resource used, plotted against time. A line on the graph shows 100 percent usable capacity and a trend line projects how swiftly resource use is approaching 100 percent. The time line shows when the selected resource is to reach capacity.

## Metrics for vSAN Storage Pool

The VCF Operations collects the metrics you use to monitor the performance of your vSAN storage pool.

| Component | Metrics |
|---|---|
| Disk I/O | • Disk I/O\|Physical Layer Read Latency (µs)<br>• Disk I/O\|Physical Layer Write Latency (µs)<br>• Disk I/O\|Physical Layer Read IOPS<br>• Disk I/O\|Physical Layer Write IOPS<br>• Disk I/O\|Physical Layer Read Throughput (KBps)<br>• Disk I/O\|Physical Layer Write Throughput (KBps)<br>• Disk I/O\|Total Bus Resets<br>• Disk I/O\|Total IOPS Aborted<br>• Disk I/O\|Number of Errors |
| Disk Space | • Disk Space\|Capacity (bytes)<br>• Disk Space\|Usage (%)<br>• Disk Space\|Used (bytes)<br>• Disk Space\|Workload(%) |

## Metrics for vSAN ESA Disk

The VCF Operations collects the metrics you use to monitor the performance of your vSAN ESA disk.

| Component | Metrics |
|---|---|
| vSAN Health\|Capacity | • vSAN\|Health\|Capacity\|Total Disk Capacity (GB)<br>• vSAN\|Health\|Capacity\|Used Disk Capacity (GB)<br>• vSAN\|Health\|Capacity\|Used Disk Capacity (%)<br>• Disk I/O\|Physical Layer Write IOPS<br>• Disk I/O\|Physical Layer Read Throughput (KBps)<br>• Disk I/O\|Physical Layer Write Throughput (KBps)<br>• Disk I/O\|Total Bus Resets<br>• Disk I/O\|Total IOPS Aborted<br>• Disk I/O\|Number of Errors |
| vSAN Health\|Congestion | vSAN Health\|Congestion Health\|Congestion Value |
| vSAN Performance | • vSAN\|Performance\|Physical Layer Read IOPS<br>• vSAN\|Performance\|Physical Layer Write IOPS<br>• vSAN\|Performance\|Physical Layer Read Throughput (KBps)<br>• vSAN\|Performance\|Physical Layer Write Throughput (KBps)<br>• vSAN\|Performance\|Physical Layer Read Latency (µs)<br>• vSAN\|Performance\|Physical Layer Write Latency (µs)<br>• vSAN\|Performance\|Physical Layer Read Count<br>• vSAN\|Performance\|Physical Layer Write Count<br>• vSAN\|Performance\|Device Latency (µs)<br>• vSAN\|Performance\|Guest Latency (µs)<br>• vSAN\|Performance\|vSAN Layer Read IOPS<br>• vSAN\|Performance\|vSAN Layer Write IOPS<br>• vSAN\|Performance\|vSAN Layer Total IOPS<br>• vSAN\|Performance\|vSAN Layer Average Read Latency (µs)<br>• vSAN\|Performance\|vSAN Layer Average Write Latency (µs)<br>• vSAN\|Performance\|vSAN Layer Read Throughput (KBps)<br>• vSAN\|Performance\|vSAN Layer Write Throughput (KBps) |

## Metrics in NSX Adapter

The NSX adapter collects metrics for objects within its plug-in.

**Table 1590: Metrics in the NSX On-Premise**

| Resource | Metrics | Metric Keys |
|---|---|---|
| Management Cluster | System Capacity<br>• Max Supported Count<br>• Max Threshold Percentage<br>• Min Threshold Percentage<br>• Usage Count<br>• Usage Count Percentage<br>• Severity | System Capacity Keys<br>• System Capacity\|<Object_Kind>\|MaxSupportedCount<br>• System Capacity\|<Object_Kind>\|MaxThresholdPercentage<br>• System Capacity\|<Object_Kind>\|MinThresholdPercentage<br>• System Capacity\|<Object_Kind>\|UsageCount<br>• System Capacity\|<Object_Kind>\|UsageCountPercentage<br>• System Capacity\|<Object_Kind>\|Severity |
| Transport Node | • CPU\|<br>– CPU Cores<br>– DPDK CPU Cores<br>– DPDK CPU Core Average Usage<br>– DPDK CPU Core Highest Usage<br>– Non-DPDK CPU Core Average Usage<br>– Non-DPDK CPU Core Highest Usage<br>• Memory<br>– Total<br>– Used<br>– Cache<br>– Total Swap<br>– Used Swap | • CPU Metric Keys<br>– Cpu\|Cores<br>– Cpu\|DPDKCores<br>– Cpu\|AvgDpdkCpuCoreUsage<br>– Cpu\|HighDpdkCpuCoreUsage<br>– Cpu\|AvgNonDpdkCpuCoreUsage<br>– Cpu\|HighNonDpdkCpuCoreUsage<br>• Memory metric keys<br>– Memory\|Total<br>– Memory\|Used<br>– Memory\|Cache<br>– Memory\|Total Swap<br>– Memory\|Used Swap |
| | File Systems\|<FileSystemMount>\|Used | FileSystems\|Used |
| | Statistics\|Interface\|<InterfaceID><br>• Received Data (bytes)<br>• Received Packets dropped<br>• Received Packets errors<br>• Received Framing errors<br>• Received Packets<br>• Transmitted Data (bytes)<br>• Transmitted Packets dropped<br>• Transmitted Packets errors<br>• Transmitted carrier losses detected<br>• Transmitted Packets<br>• Transmitted Collisions detected | Statistics Metric Keys<br>• stats\|Interface\|RxData<br>• stats\|Interface\|RxDropped<br>• stats\|Interface\|RxErrors<br>• stats\|Interface\|RxFrame<br>• stats\|Interface\|RxPackets<br>• stats\|Interface\|TxData<br>• stats\|Interface\|TxDropped<br>• stats\|Interface\|TxErrors<br>• stats\|Interface\|TxCarrier<br>• stats\|Interface\|TxPackets<br>• stats\|Interface\|TxColls |

| Resource | Metrics | Metric Keys |
|---|---|---|
| Load Balancer Service | • CPU Usage(%)<br>• Memory Usage(%)<br>• Active Transport Nodes<br>• Standby Transport Nodes<br>• Sessions:<br>– L4Average<br>– L4Current<br>– L4Maximum<br>– L4Total<br>– L7Average<br>– L7Current<br>– L7Maximum<br>– L7Total | • CPU Usage<br>• Memory Usage<br>• Active Transport Nodes<br>• Standby Transport Nodes<br>• Sessions\|L4Average<br>• Sessions\|L4Current<br>• Sessions\|L4Maximum<br>• Sessions\|L4Total<br>• Sessions\|L7Average<br>• Sessions\|L7Current<br>• Sessions\|L7Maximum<br>• Sessions\|L7Total |
| Load Balancer Virtual Server | • Statistics<br>– Bytes\|Inbound Bytes Total<br>– Bytes\|Average Inbound Bytes Per Second<br>– Bytes\|Outbound Bytes Total<br>– Bytes\|Average Outbound Bytes Per Second<br>– Http\|Http Request Rate<br>– Http\|Http Requests<br>– Packets\|Inbound Packets Total<br>– Packets\|Inbound Packets Rate<br>– Packets\|Outbound Packets Total<br>– Packets\|Outbound Packets Rate<br>– Packets\|Dropped<br>• Sessions<br>– Average Current Sessions Per Second<br>– Current Sessions<br>– Maximum Sessions<br>– Dropped Sessions<br>– Total Sessions | • Statistics metric keys<br>– stats\|Bytes\|Inbound<br>– stats\|Bytes\|InboundRate<br>– stats\|Bytes\|Outbound<br>– stats\|Bytes\|OutboundRate<br>– stats\|Http\|RequestRate<br>– stats\|Http\|Requests<br>– stats\|Packets\|Inbound<br>– stats\|Packets\|InboundRate<br>– stats\|Packets\|Outbound<br>– stats\|Packets\|OutboundRate<br>– stats\|Packets\|Dropped<br>• Sessions metric keys<br>• – Sessions\|CurrentRate<br>– Sessions\|Current<br>– Sessions\|Maximum<br>– Sessions\|Dropped<br>– Sessions\|Total |
| Load Balancer Pool | • Statistics<br>– Bytes\|Inbound Bytes Total<br>– Bytes\|Average Inbound Bytes Per Second<br>– Bytes\|Outbound Bytes Total<br>– Bytes\|Average Outbound Bytes Per Second<br>– Http\|Http Request Rate<br>– Http\|Http Requests<br>– Packets\|Inbound Packets Total<br>– Packets\|Inbound Packets Rate<br>– Packets\|Outbound Packets Total<br>– Packets\|Outbound Packets Rate<br>– Packets\|Dropped<br>• Sessions<br>– Average Current Sessions Per Second<br>– Current Sessions<br>– Maximum Sessions<br>– Dropped Sessions<br>– Total Sessions | • Statistics metric keys<br>– stats\|Bytes\|Inbound<br>– stats\|Bytes\|InboundRate<br>– stats\|Bytes\|Outbound<br>– stats\|Bytes\|OutboundRate<br>– stats\|Http\|RequestRate<br>– stats\|Http\|Requests<br>– stats\|Packets\|Inbound<br>– stats\|Packets\|InboundRate<br>– stats\|Packets\|Outbound<br>– stats\|Packets\|OutboundRate<br>– stats\|Packets\|Dropped<br>• Sessions metric keys<br>– Sessions\|CurrentRate<br>– Sessions\|Current<br>– Sessions\|Maximum<br>– Sessions\|Dropped<br>– Sessions\|Total |

| Resource | Metrics | Metric Keys |
|---|---|---|
| Management Services | • Service Monitor Process ID<br>• Service Monitor Runtime state<br>• Service Process ID<br>• Service Runtime State | • ServiceMonitorProcessId<br>• ServiceMonitorRuntimeState<br>• ServiceProcessIds<br>• ServiceRuntimeState |
| Logical Router | Statistics<br>• Received Data (bytes)<br>• Received Packets dropped<br>• Received Packets<br>• Transmitted Data (bytes)<br>• Transmitted Packets dropped<br>• Transmitted Packets | Statistics metric keys<br>• stats\|RxData<br>• stats\|RxDropped<br>• stats\|RxPackets<br>• stats\|TxData<br>• stats\|TxDropped<br>• stats\|TxPackets |
| | Configuration Maximums<br>• Router Port Count<br>• ARP Entries Count<br>• Tier 1 Router Count<br>• Route Map Count<br>• Route Maps\|<RouteMapName:RouteMapId>\| Rule Count<br>• Prefix List Count<br>• IP Prefix Lists\| <IPPrefixListName:IPPrefixListId>\|Prefix List Entries Count | Configuration Maximums metric keys<br>• configMax\|routerPortCount<br>• configMax\|routerArpEntryCount<br>**Note:** Metric applicable for T1 router.<br>• configMax\|tier1RouterCount<br>• configMax\|routeMapCount<br>• configMax\|RouteMaps\|routeMapRuleCount<br>**Note:** Metric applicable for T0 router.<br>• configMax\|prefixListCount<br>• configMax\|IPPrefixLists\|prefixListEntriesCount<br>**Note:** Metric applicable for T0 and T1 router. |
| Logical Switch | Statistics<br>• Inbound Bytes Total<br>• Inbound Bytes Dropped<br>• Inbound Bytes Throughput<br>• Outbound Bytes Total<br>• Outbound Bytes Dropped<br>• Outbound Bytes Throughput<br>• Inbound Packets Total<br>• Inbound Packets Dropped<br>• Inbound Packets Throughput<br>• Outbound Packets Total<br>• Outbound Packets Dropped<br>• Outbound Packets Throughput | Metric keys<br>• stats\|IngressBytes<br>• stats\|IngressBytesDropped<br>• stats\|IngressBytesThroughput<br>• stats\|IngressPackets<br>• stats\|IngressPacketsDropped<br>• stats\|IngressPacketsThroughput<br>• stats\|EgressBytes<br>• stats\|EgressBytesDropped<br>• stats\|EgressBytesThroughput<br>• stats\|EgressPackets<br>• stats\|EgressPacketsDropped<br>• stats\|EgressPacketsThroughput |
| Logical Switch Group | Configuration Maximums<br>• Logical Segment Count | Metric keys<br>• configMax\|LogicalSegmentCount |
| Management Appliances | Management Node Count | Management node count |
| Manager Node | • File Systems\|<FileSystemMount><br>– File System Id<br>– File System Type<br>– Total (KB)<br>– Used(KB)<br>– Used(%) | • File Systems Metric Keys<br>– FileSystems\|<FileSystemMount>\|FileSystemId<br>– FileSystems\|<FileSystemMount>\|Type<br>– FileSystems\|<FileSystemMount>\|Total<br>– FileSystems\|<FileSystemMount>\|Used<br>– FileSystems\|<FileSystemMount>\|usedPercentage |

| Resource | Metrics | Metric Keys |
|---|---|---|
| | Network Interfaces\|<InterfaceID>\|<br>• Received Data\|Bits per second<br>• Received Data\|Cumulative(bytes)<br>• Received Framing Errors\|Cumulative<br>• Received Framing Errors\|Per second<br>• Received Packets\|Cumulative<br>• Received Packets\|Per Second<br>• Received Packets Dropped\|Cumulative<br>• Received Packets Dropped\|Per second<br>• Received Packets Error\|Cumulative<br>• Received Packets Error\|Per second<br>• Transmitted Carrier losses detected\|Cumulative<br>• Transmitted Carrier losses detected\|Per second<br>• Transmitted Collisions detected\|Cumulative<br>• Transmitted Collisions detected\|Per second<br>• Transmitted Data\|Bits per second<br>• Transmitted Data\|Cumulative(bytes)<br>• Transmitted Packets\|Cumulative<br>• Transmitted Packets\|Per second<br>• Transmitted Packets Dropped\|Cumulative<br>• Transmitted Packets Dropped\|Per second<br>• Transmitted Packets errors\|Cumulative<br>• Transmitted Packets errors\|Per second | Network Interface metric keys<br>• Interfaces\|<InterfaceID>\|RxData\|BitsPerSecond<br>• Interfaces\|<InterfaceID>\|RxData\|Cumulative<br>• Interfaces\|<InterfaceID>\|RxFrame\|Cumulative<br>• Interfaces\|<InterfaceID>\|RxFrame\|PerSecond<br>• Interfaces\|<InterfaceID>\|RxPackets\|Cumulative<br>• Interfaces\|<InterfaceID>\|RxPackets\|PerSecond<br>• Interfaces\|<InterfaceID>\|RxDropped\|Cumulative<br>• Interfaces\|<InterfaceID>\|RxDropped\|PerSecond<br>• Interfaces\|<InterfaceID>\|RxErrors\|Cumulative<br>• Interfaces\|<InterfaceID>\|RxErrors\|PerSecond<br>• Interfaces\|<InterfaceID>\|TxCarrier\|Cumulative<br>• Interfaces\|<InterfaceID>\|TxCarrier\|PerSecond<br>• Interfaces\|<InterfaceID>\|TxColls\|Cumulative<br>• Interfaces\|<InterfaceID>\|TxColls\|PerSecond<br>• Interfaces\|<InterfaceID>\|TxData\|BitsPerSecond<br>• Interfaces\|<InterfaceID>\|TxData\|Cumulative<br>• Interfaces\|<InterfaceID>\|TxPackets\|Cumulative<br>• Interfaces\|<InterfaceID>\|TxPackets\|PerSecond<br>• Interfaces\|<InterfaceID>\|TxDropped\|Cumulative<br>• Interfaces\|<InterfaceID>\|TxDropped\|PerSecond<br>• Interfaces\|<InterfaceID>\|TxErrors\|Cumulative<br>• Interfaces\|<InterfaceID>\|TxErrors\|PerSecond |
| | CPU<br>• CPU Cores<br>• DPDK CPU Cores<br>• DPDK CPU Core Average Usage<br>• DPDK CPU Core Highest Usage<br>• Non-DPDK CPU Core Average Usage<br>• Non-DPDK CPU Core Highest Usage | CPU Metric Keys<br>• Cpu\|Cores<br>• Cpu\|DPDKCores<br>• Cpu\|AvgDpdkCpuCoreUsage<br>• Cpu\|HighDpdkCpuCoreUsage<br>• Cpu\|AvgNonDpdkCpuCoreUsage<br>• Cpu\|HighNonDpdkCpuCoreUsage |
| | Memory<br>• Total<br>• Used<br>• Cache<br>• Total Swap<br>• Used Swap | Memory metric keys<br>• Memory\|Total<br>• Memory\|Used<br>• Memory\|Cache<br>• Memory\|TotalSwap<br>• Memory\|UsedSwap |
| Controller Cluster | • Controller Node Count<br>• Cluster Status\|Controller Cluster Status<br>• Cluster Status\|Management cluster Status | Controller cluster metrics keys<br>• Cluster Status\|Controller Node Count<br>• ClusterStatus\|ControllerClusterStatus<br>• ClusterStatus\|ManagementClusterStatus<br>**Note:**<br>These metrics are not collected for NSX version above 2.4 |

| Resource | Metrics | Metric Keys |
|---|---|---|
| Controller Node | • Connectivity Status\|Cluster Connectivity<br>• Connectivity Status\|Manager Connectivity<br>• File System ID<br>• File System Type<br>• Total(KB)<br>• Used(KB)<br>• Used(%)<br>• Network Interfaces\|<InterfaceID>\|<br>• Received Data\|Bits per second<br>• Received Data\|Cumulative(bytes)<br>• Received Framing Errors\|Cumulative<br>• Received Framing Errors\|Per second<br>• Received Packets\|Cumulative<br>• Received Packets\|Per Second<br>• Received Packets Dropped\|Cumulative<br>• Received Packets Dropped\|Per second<br>• Received Packets Error\|Cumulative<br>• Received Packets Error\|Per second<br>• Transmitted Carrier losses detected\| Cumulative<br>• Transmitted Carrier losses detected\|Per second<br>• Transmitted Collisions detected\|Cumulative<br>• Transmitted Collisions detected\|Per second<br>• Transmitted Data\|Bits per second<br>• Transmitted Data\|Cumulative(bytes)<br>• Transmitted Packets\|Cumulative<br>• Transmitted Packets\|Per second<br>• Transmitted Packets Dropped\|Cumulative<br>• Transmitted Packets Dropped\|Per second<br>• Transmitted Packets errors\|Cumulative<br>• Transmitted Packets errors\|Per second | **Note:**<br>These metrics are not collected for NSX version above 2.4<br>• ConnectivityStatus\|ClusterConnectivity<br>• ConnectivityStatus\|ManagerConnectivity<br>• FileSystems\|<FileSystemMount>\|FileSystemId<br>• FileSystems\|<FileSystemMount>\|Type<br>• FileSystems\|<FileSystemMount>\|Total<br>• FileSystems\|<FileSystemMount>\|Used<br>• FileSystems\|<FileSystemMount>\|usedPercentage<br>• Interfaces\|<InterfaceID>\|RxData\|BitsPerSecond<br>• Interfaces\|<InterfaceID>\|RxData\|Cumulative<br>• Interfaces\|<InterfaceID>\|RxFrame\|Cumulative<br>• Interfaces\|<InterfaceID>\|RxFrame\|PerSecond<br>• Interfaces\|<InterfaceID>\|RxPackets\|Cumulative<br>• Interfaces\|<InterfaceID>\|RxPackets\|PerSecond<br>• Interfaces\|<InterfaceID>\|RxDropped\|Cumulative<br>• Interfaces\|<InterfaceID>\|RxDropped\|PerSecond<br>• Interfaces\|<InterfaceID>\|RxErrors\|Cumulative<br>• Interfaces\|<InterfaceID>\|RxErrors\|PerSecond<br>• Interfaces\|<InterfaceID>\|TxCarrier\|Cumulative<br>• Interfaces\|<InterfaceID>\|TxCarrier\|PerSecond<br>• Interfaces\|<InterfaceID>\|TxColls\|Cumulative<br>• Interfaces\|<InterfaceID>\|TxColls\|PerSecond<br>• Interfaces\|<InterfaceID>\|TxData\|BitsPerSecond<br>• Interfaces\|<InterfaceID>\|TxData\|Cumulative<br>• Interfaces\|<InterfaceID>\|TxPackets\|Cumulative<br>• Interfaces\|<InterfaceID>\|TxPackets\|PerSecond<br>• Interfaces\|<InterfaceID>\|TxDropped\|Cumulative<br>• Interfaces\|<InterfaceID>\|TxDropped\|PerSecond<br>• Interfaces\|<InterfaceID>\|TxErrors\|Cumulative<br>• Interfaces\|<InterfaceID>\|TxErrors\|PerSecond |
| Router Service | RouterService | • BGP Neighbor:<BGPNeighborName>\|Service Router State\|Connection State<br>• BGP Neighbor:<BGPNeighborName>\|Advertised Routes\|Transport Nodes:<TransportNodeIP>\|Advertised Route Count<br>• BGP Neighbor:<BGPNeighborName>\|Advertised Routes\|Transport Nodes:<TransportNodeIP>Routes\| ASPath<br>• BGP Neighbor:<BGPNeighborName>\|Advertised Routes\|Transport Nodes:<TransportNodeIP>Routes\| Next Hop |

**Table 1591: Metrics in the NSX on VMware Cloud on AWS**

| Resource | Metrics | Metric Keys |
|---|---|---|
| Logical Router | The following metrics are specify to Tier 0 Router.<br>Statistics \| Interface<br>• Received Data (Bytes)<br>• Received Packets<br>• Received Packets Dropped<br>• Transmitted Data<br>• Transmitted Received Data (Bytes)<br>• Transmitted Received Packets<br>• Transmitted Received Packets Dropped | Stats Metrics<br>Statistics \| Interface<br>• stats\|Interface\|RxData<br>• stats\|Interface\|RxPackets<br>• stats\|Interface\|RxDropped<br>• stats\|Interface\|TxData<br>• stats\|Interface\|TxPackets<br>• stats\|Interface\|TxDropped<br>**Note:**<br>These metrics are only for Tier 0 Router. |
| Firewall Section Group | Configuration Maximums<br>• Distributed Firewall Section Count<br>• Distributed Firewall Rule Count<br>• MGW Gateway Firewall Rule Count<br>• CGW Gateway Firewall Rule Count<br>• Distributed Application Firewall Rule Count<br>• Distributed Application Firewall Section Count<br>• Distributed Environment Firewall Rule Count<br>• Distributed Environment Firewall Section Count<br>• Distributed Infrastructure Firewall Rule Count<br>• Distributed Infrastructure Firewall Section Count<br>• Distributed Emergency Firewall Rule Count<br>• Distributed Emergency Firewall Section Count<br>• Distributed Ethernet Firewall Rule Count<br>• Distributed Ethernet Firewall Section Count<br>**Note:**<br>These metrics are only for NSX on VMware Cloud on AWS. For NSX on-premise, the values for these metrics show zero. | Configuration metric keys<br>• configMax\|MaxDistributedFirewallSections<br>• configMax\|MaxDistributedFirewallRules<br>• configMax\|MaxMGWGatewayFirewallRules<br>• configMax\|MaxCGWGatewayFirewallRules<br>• configMax\|MaxDistributedApplicationFirewallRules<br>• configMax\|MaxDistributedApplicationFirewallSections<br>• configMax\|MaxDistributedEnvironmentFirewallRules<br>• configMax\|MaxDistributedEnvironmentFirewallSections<br>• configMax\|MaxDistributedInfrastructureFirewallRules<br>• configMax\|MaxDistributedInfrastructureFirewallSections<br>• configMax\|MaxDistributedEmergencyFirewallRules<br>• configMax\|MaxDistributedEmergencyFirewallSections<br>• configMax\|MaxDistributedEthernetFirewallRules<br>• configMax\|MaxDistributedEthernetFirewallSections<br>**Note:**<br>These metrics are only for NSX on VMware Cloud on AWS. For NSX on-premise, the values for these metrics is shown as zero. |

| Resource | Metrics | Metric Keys |
|---|---|---|
| Logical Switch Group | Configuration Maximums<br>• Logical Segment Count<br>• Extended Network Count | Metric Keys<br>• configMax\|LogicalSegmentCount<br>• configMax\|ExtendedNetworkcount<br>**Note:**<br>The metric (configMax\|ExtendedNetworkcount) is only for NSX on VMware Cloud on AWS. For NSX on-premise, its value is zero. |

## Green Score Metrics

Green Score metrics are collected for virtual machine, host system, cluster compute resource, vSphere World, and Organization object types.

### Metric Names

| Metric Name | Object Type | Description |
|---|---|---|
| Power\|Total Energy (Wh) | Virtual Machine | Total energy used.<br>Formula:<br>Total Energy (Wh) = Sum(Power\|Energy (Joule))/3600 |
| Power\|Total Energy (Wh) | Host System | Total energy used.<br>Formula:<br>Total Energy (Wh) = Sum(Power\|Energy (Joule))/3600 |
| Sustainability\|$CO_2$ Emission (kg) | Cluster Compute Resource | Estimated carbon dioxide emissions. Calculated as power consumption* $CO_2$ Emission rate<br>Formula:<br>$CO_2$ Emission (kg) = Sum(Host System(Power\|Total Energy(Wh)))/1000 * 0.672 |
| Sustainability\|$CO_2$ Emission before Virtualization (kg) | Cluster Compute Resource | Estimated carbon dioxide emissions before virtualization, assuming that power consumption per physical server is 100W, reflecting a low end hardware specification.<br>Formula:<br>$CO_2$ Emission before Virtualization (kg) = Summary\|Number of Running VMs * 0.1 * 0.672 |
| Sustainability\|$CO_2$ Emission by Idle VMs (kg) | Cluster Compute Resource | Total estimated carbon dioxide emission from all idle VMs. Calculated as $CO_2$ emission rate * Power consumed by Idle VMs, where the rate is set at cluster property / 1000.<br>Formula:<br>$CO_2$ Emission by Idle VMs (kg) = Sum(VM(Power\|Total Energy (Wh)), If VM(Summary\|Reclaimable Idle = 1) * 0.672<br>OR<br>$CO_2$ Emission by Idle VMs (kg) = Power Wasted by Idle VMs (Wh) * 0.672 |

| Metric Name | Object Type | Description |
|---|---|---|
| Sustainability\|Electricity Cost Savings | Cluster Compute Resource | Estimated cost savings by virtualizing workloads. Calculated from the difference between power consumption before virtualization and after virtualization. The electricity cost is defined at the cluster custom property.<br>Formula:<br>Electricity Cost Savings = (Summary\|Number of Running VMs * 0.1 - Sum(Host System(Power\|Total Energy(Wh)))/1000) * 0.108<br>OR<br>Daily Electricity Cost Savings = (Summary\|Number of Running VMs * 0.1 - Power usage (kWh)) * 0.108 |
| Sustainability\|Power usage (kWh) | Cluster Compute Resource | Power usage calculated from all hosts in kWh.<br>Formula:<br>Power usage (kWh) = Sum(Host System(Power\|Total Energy(Wh))/1000 |
| Sustainability\|Power usage per GHz (Wh) | Cluster Compute Resource | Power used per GHz of processing power. This is a metric of how power-efficient your CPUs are.<br>Formula:<br>Power usage per GHz (Wh) = Sum(Host System(Power\|Total Energy(Wh))/CPU\|Usage (MHz)/1000 |
| Sustainability\|Power Wasted by Idle VMs (Wh) | Cluster Compute Resource | Sum of electricity power used by all VMs classified as idle by the system.<br>Formula:<br>Power Wasted by Idle VMs (Wh) = Sum(VM(Power\|Total Energy (Wh)), If VM(Summary\|Reclaimable Idle = 1) |
| Sustainability\|$CO_2$ Emission (kg) | vSphere World | Total estimated carbon dioxide emissions. Calculated as sum of carbon emission from all clusters.<br>Formula:<br>$CO_2$ Emission (kg) = Sum(Cluster(Sustainability\|$CO_2$ Emission (kg)) |
| Sustainability\|$CO_2$ Emission before Virtualization (kg) | vSphere World | Total estimated carbon dioxide emissions before virtualization. Calculated as the sum of $CO_2$ emission from all clusters.<br>Formula:<br>$CO_2$ Emission before Virtualization (kg) = Sum(Cluster(Sustainability\|$CO_2$ Emission before Virtualization (kg))) |
| Sustainability\|$CO_2$ Emission Avoided (t) | vSphere World | Estimated amount of carbon emissions avoided with virtualization. Calculated by difference in values of carbon emissions before and after virtualization and converting value in kg to Tonnes.<br>Formula:<br>$CO_2$ Emission Avoided (t) = ($CO_2$ Emission before Virtualization (kg) - $CO_2$ Emission (kg)) / 1000 |

| Metric Name | Object Type | Description |
|---|---|---|
| Sustainability\|Electricity Cost Savings | vSphere World | Total estimated cost savings by virtualizing workloads at vSphere World. Calculated as the sum of Electricity cost savings.<br>Formula:<br>Electricity Cost Savings = Sum(Cluster(Daily Electricity Cost Savings)) |
| Sustainability\|Power Savings with Virtualization (%) | vSphere World | Percentage of power savings achieved by virtualization. Calculated by Formula = (Power usage before Virtualization - Power usage after Virtualization)/Power usage before Virtualization *100.<br>Formula:<br>Power Savings with Virtualization (%) =( Power usage Before Virtualization (kWh) - Power usage (kWh) ) / Power usage Before Virtualization (kWh) * 100 |
| Sustainability\|Power usage (kWh) | vSphere World | Power usage calculated from all hosts in kWh.<br>Formula:<br>Power usage (kWh) = Sum(Host System(Power\|Total Energy(Wh)))/1000 |
| Sustainability\|Power usage Before Virtualization (kWh) | vSphere World | Power usage assuming that each low range server consumes 0.1 kWh.<br>Formula:<br>Power usage Before Virtualization (kWh) = Summary\|Number of Running VMs * 0.1 |
| Sustainability\|Power Wasted by Idle VMs (Wh) | vSphere World | Sum of electricity power used by all VMs classified as idle by the system.<br>Formula:<br>Power Wasted by Idle VMs (Wh) = Sum((VM(Power\|Total Energy (Wh)), If VM(Summary\|Reclaimable Idle = 1)) |
| Sustainability\|Adjusted Score\|Workload Efficiency (%) | Organization | Indicates efficiency based on resource wastage in the virtual environment.<br>Formula:<br>Workload Efficiency (%) = 100 - (Wastage/ Usage Ratio CPU + Wastage/Usage Ratio Memory + Wastage/Usage Ratio Disk) * 100 / 3 |
| Sustainability\|Adjusted Score\|Hardware Efficiency (%) | Organization | Indicates efficiency of hardware in the environment, based on the age of the hardware with the assumption that newer hardware is more energy efficient than older hardware.<br>Formula:<br>Hardware Efficiency (%) = ((10 - Average Age of Servers)/ 10 * 30%) +((10 - Average Age of Storage)/ 10 * 30%) + ((10 - Average Age of Network)/ 10 * 20%) + ((10 - Average Age of Desktop)/ 10 * 20%)<br>The Average age of hardware variables in the formula is provided in the **Organization Details** page. |

| Metric Name | Object Type | Description |
|---|---|---|
| Sustainability\|Adjusted Score\|Resource Utilization (%) | Organization | Indicates utilization levels of the physical resources in the environment.<br>Formula:<br>Resource Utilization (%) = (Virtualized Server (%) * 44.45%) + (Virtualized Storage (%) * 33.33%) + (Virtualized Network (%) * 22.22%) |
| Sustainability\|Adjusted Score\|Virtualization (%) | Organization | Indicates the extent of virtualization in the environment.<br>Formula:<br>Virtualization (%) = (Server Virtualization (%) * 0.4) + (Storage Virtualization (%) * 0.3) + (Network Virtualization (%) * 0.2) + (Desktop Virtualization (%) * 0.1) |
| Sustainability\|Adjusted Score\|Power Source (%) | Organization | Indicates the sources of power being used in the environment. Assumes usage of renewable energy sources leads to less carbon emissions.<br>Formula:<br>Power Source Efficiency (%) = Sum(Adjusted Scores of each Power Source)<br>Adjusted Scores of Power Source = Power Source Green Factor * Power Source Share in Organization |
| Sustainability\|Hardware Utilization\|Server (%) | Organization | Indicates server utilization levels.<br>Formula:<br>Virtualized Server (%) = ((Total ESX Hosts CPU Utilization (GHz) / Total ESX Hosts CPU Capacity (GHz)) + (Total ESX Hosts Memory Utilization (TB) / Total ESX Hosts Memory Capacity (TB))) / 2 |
| Sustainability\|Hardware Utilization\|Storage (%) | Organization | Indicates storage utilization levels.<br>Formula:<br>Virtualized Storage (%) = (Total datastores utilization + Total RDM) / (Total datastores capacity + Total RDM + Physical Disk Overhead) |
| Sustainability\|Green Score (%) | Organization | A score that indicates how energy efficient your data center is.<br>Formula:<br>Green Score (%) = (Workload Efficiency (%) * 22.5%) + (Resource Utilization (%) * 12.5%) + (Virtualization Adoption (%) * 15%) + (Power Source Efficiency (%) * 37.5%) + (Hardware Power Efficiency (%) * 12.5%) |
| Sustainability\|Power Usage (kWh) | Organization | Sum of the power usage across all the clusters in the environment.<br>Formula:<br>Power Consumption = Sum ( (Cluster) Sustainability\|Power Usage (kWh) ) |
| Sustainability\|$CO_2$ Emission (kg) | Organization | Total estimated carbon emissions in the environment based on power consumed.<br>Formula:<br>$CO_2$ Emission = Power Consumption * $CO_2$ Emission Ratio |

| Metric Name | Object Type | Description |
|---|---|---|
| Sustainability\|Workload Efficiency\|Idle VM Count | Organization | Number of Idle VMs, which are considered idle based on the Reclamation Idleness settings. |
| Sustainability\|Workload Efficiency\| Orphaned Disk Space (GB) | Organization | The aggregated size of all orphaned disks which are reclaimable based on the Orphaned Disk Reclamation settings. |
| Sustainability\|Workload Efficiency\| Oversized VM Count | Organization | Number of Oversized VMs, which are considered oversized by the capacity engine recommendation and are not in the Power Off state. |
| Sustainability\|Workload Efficiency\|Powered Off VM Count | Organization | The number of VMs which have been in the Powered Off state for the last 7 days. The number of days (7) can be configured from the Reclamation Settings. |
| Sustainability\|Workload Efficiency\|Snapshot Disk Space (GB) | Organization | The aggregated size of all snapshots which are reclaimable based on the Snapshot Reclamation settings. |

**Disclaimer:**

The conversion calculators and information shown in this documentation reference public information and have been provided to help translate abstract carbon emissions numbers into easily understandable terms solely for informational purposes. You should not rely on the calculators or information provided herein for any other purpose, including for any regulatory disclosure or diligence purposes. Broadcom may update, upgrade, revise, adjust or otherwise change the features and conversion methodologies at any time. Broadcom has not separately reviewed, approved, or endorsed the public information or third-party websites. No representation or warranty is made by Broadcom as to the accuracy, reasonableness or completeness of the calculations and related information herein.

## Metrics for Policies

Each policy is an object in VCF Operations. This allows you to track every change in the policy and create any dashboard, report, or alert on policy usage distribution.

VCF Operations collects metrics for the object type 'Policy'.

| Metric Key | Metric Name | Description |
|---|---|---|
| NumberOfEffectiveObjects | Number of Affected Objects | Represents the count of objects affected by the effective policy. |
| NumberOfVirtualMachines | Number of Affected vCenter Virtual Machines | Represents the number of vCenter Virtual Machines associated with a specific policy. |
| NumberOfCustomGroups | Number of Assigned Custom Groups | Represents the number of assigned custom groups for a specific policy. |
| NumberOfDefaultAssignedObjects | Number of Default Assigned Objects | Represents the number of objects that are assigned by default for a specific policy. This metric value is 0 for all policies except the default policy. |
| NumberOfDirectAssignedObjects | Number of Directly Assigned Objects | Represents the number of objects that are directly assigned for a specific policy. |
| NumberOfInheritedObjects | Number of Objects Assigned By Scope | Represents the count of inherited objects for a specific policy. |

# Property Definitions in VCF Operations

Properties are attributes of objects in the VCF Operations environment. You use properties in symptom definitions. You can also use properties in dashboards, views, and reports.

VCF Operations uses adapters to collect properties for target objects in your environment. Property definitions for all objects connected through the vCenter adapter are provided. The properties collected depend on the objects in your environment.

You can add symptoms based on properties to an alert definition so that you are notified if a change occurs to properties on your monitored objects. For example, disk space is a hardware property of a virtual machine. You can use disk space to define a symptom that warns you when the value falls below a certain numeric value.

You can add symptoms based on properties to an alert definition so that you are notified if a change occurs to properties on your monitored objects. For example, disk space is a hardware property of a virtual machine. You can use disk space to define a symptom that warns you when the value falls below a certain numeric value. See the *VCF Operations User Guide*.

VCF Operations generates Object Type Classification and Subclassification properties for every object. You can use object type classification properties to identify whether an object is an adapter instance, custom group, application, tier, or a general object with property values *ADAPTER_INSTANCE*, *GROUP*, *BUSINESS_SERVICE*, *TIER*, or *GENERAL*, respectively.

## Properties for vCenter Server Components

The VMware vSphere solution is installed with VCF Operationsand includes the vCenter adapter. VCF Operations uses the vCenter adapter to collect properties for objects in the vCenter system.

vCenter components are listed in the `describe.xml` file for the vCenter adapter. The following example shows the runtime property `memoryCap` or Memory Capacity for the virtual machine in the `describe.xml`.

```
<ResourceGroup instanced="false" key="runtime" nameKey="5300" validation="">
   <ResourceAttribute key="memoryCap" nameKey="1780" dashboardOrder="200" dataType="float"
                      defaultMonitored="true" isDiscrete="false" isRate="false" maxVal=""
                      minVal="" isProperty="true" unit="kb"/>
</ResourceGroup>
```

The `ResourceAttribute` element includes the name of the property that appears in the UI and is documented as a Property Key. `isProperty = "true"` indicates that `ResourceAttribute` is a property.

### vCenter Server Properties

VCF Operations collects summary and event properties for vCenter system objects.

**Table 1592: Summary Properties Collected for vCenter System Objects**

| Property Key | Property Name | Description |
|---|---|---|
| summary\|version | Version | Version |
| summary\|vcuuid | VirtualCenter ID | Virtual Center ID |
| summary\|vcfullname | Product Name | Product Name |

**Table 1593: Event Properties Collected for vCenter System Objects**

| Property Key | Property Name | Description |
|---|---|---|
| event\|time | Last VC Event Time | Last Virtual Center Event Time |
| event\|key | Last VC Event ID | Last Virtual Center Event ID |

**Table 1594: Custom Field Manager Property Collected for vCenter System Objects**

| Property Key | Property Name | Description |
|---|---|---|
| CustomFieldManager\|CustomFieldDef | Custom Field Def | Custom Field Def for vCenter Tagging information at the Adapter level. |

**Table 1595: Compliance Configuration Related Properties for vCenter System Objects**

| Property Key | Property Name | Description |
|---|---|---|
| vc_appliance\|hasAccessSSH | Appliance\|Access SSH | Access SSH |
| vc_appliance\|networkNICs | Appliance\|Number of NICs | NICs |

## Virtual Machine Properties

VCF Operations collects configuration, runtime, CPU, memory, network I/O, and properties about summary use for virtual machine objects. Properties are collected with the first cycle of data collection. Once collected, the next property collection occurs only when there is data change. In case of no data change, no property is collected.

**Table 1596: Properties Collected for Virtual Machines with vSAN Adapter**

| Property Key | Property Name | Description |
|---|---|---|
| **Note:**<br>The following Disk Space properties are displayed by the virtual machine object only when the vSAN adapter is configured with vCenter. | | |
| Virtual Disk:scsi0:0\|Storage Policy\|Cache Reservation | Storage Policy\|Cache Reservation | This property helps you with flash read cache reservation. |
| Virtual Disk:scsi0:0\|Storage Policy\| Checksum Disabled | Storage Policy\| Checksum Disabled | This property checks whether checksum is disabled for the disk object. |
| Virtual Disk:scsi0:0\|Storage Policy\| Encryption Service | Storage Policy\| Encryption Service | This property checks whether the encryption service is used or not. |
| Virtual Disk:scsi0:0\|Storage Policy\|Failures to Tolerate | Storage Policy\| Failures to Tolerate | This property gives the number of host or disk failures the storage object can tolerate. |
| Virtual Disk:scsi0:0\|Storage Policy\|Force Provisioning | Storage Policy\|Force Provisioning | This property checks whether force provisioning is disabled for the disk object. |
| Virtual Disk:scsi0:0\|Storage Policy\|Object Space Reservation | Storage Policy\|Object Space Reservation | This property helps you to check the object space reservation. |
| Virtual Disk:scsi0:0\|Storage Policy\|Replica Preference | Storage Policy\|Replica Preference | This property helps you to identify the fault tolerance method. |
| Virtual Disk:scsi0:0\|Storage Policy\|Site Disaster Tolerance | Storage Policy\|Site Disaster Tolerance | This property gives the number of fault domain failures, the storage object can tolerate. |
| Virtual Disk:scsi0:0\|Storage Policy\|Space Efficiency | Storage Policy\|Space Efficiency | This property helps you with the space efficiency method. |
| Virtual Disk:scsi0:0\|Storage Policy\|Storage Policy in Use | Storage Policy in Use | This property gives the storage policy associated with the virtual disk. |
| Virtual Disk:scsi0:0\|Storage Policy\|Stripe Width | Storage Policy\|Stripe Width | This property gives the number of disk stripes per object. |

### Table 1597: Properties Collected for System

| Property Key | Property Name | Description |
|---|---|---|
| system\|notes | System\|Notes | This property helps you to track the System notes defined in vCenter. |

### Table 1598: VCF Automation Properties Collected for Virtual Machine Objects

| Property Key | Property Name | Description |
|---|---|---|
| vRealize Automation\|Blueprint Name | Blueprint Name | Virtual machines deployed by VCF Automation to be excluded from workload placements. |

### Table 1599: Properties Collected for Virtual Machine Objects to Support VIN Adapter Localization

| Property Key | Property Name | Description |
|---|---|---|
| RunsOnApplicationComponents | Application components running on the Virtual Machine | Application components running on the Virtual Machine |
| DependsOnApplicationComponents | Application components the Virtual Machine depends on | Application components running on other machines that this Virtual Machine depends on. |

### Table 1600: Properties Collected for Guest File Systems

| Property Key | Property Name | Description |
|---|---|---|
| guestfilesystem\|capacity_property | Guest File System stats\|Guest File System Capacity Property | This property is disabled by default. |
| guestfilesystem\|capacity_property_total | Guest File System stats\|Total Guest File System Capacity Property(gb) | This property is disabled by default. |

### Table 1601: Properties Collected for Disk Space Objects

| Property Key | Property Name | Description |
|---|---|---|
| diskspace\|snapshot\|creator | Disk Space\|Snapshot\|Creator | This property is disabled by default. |
| diskspace\|snapshot\|description | Disk Space\|Snapshot\|Description | This property is disabled by default. |

### Table 1602: Configuration Properties Collected for Virtual Machine Objects

| Property Key | Property Name | Description |
|---|---|---|
| config\|ctkEnabled | Configuration\|Changed Block Tracking | This property displays if Change Block Tracking is enabled, if enabled, tracks changes on disk sectors. This helps in performing incremental backups on the VM. |
| Configuration\|Hardware\|Number of CPUs (vCPUs) | Number of CPUs (vCPUs) | This property displays the number of CPUs configured in the VM, the count includes both in the vSocket and the vCore. |

| Property Key | Property Name | Description |
|---|---|---|
| Configuration\|Number of RDMs | Number of RDMs | This property displays the number of RDMs configured in the VM. This property is enabled by default. |
| Configuration\|Number of Virtual Disks | Number of Virtual Disks | This property displays the number of virtual disks configured in the VM, the count includes the RDMs. |
| Configuration\|Hardware\|Number of VMDKs | Number of VMDKs | This property displays the number of VMDKs configured in the VM. This property is enabled by default. |
| Summary\|Is Horizon Managed | Is Horizon Managed | This property displays whether the selected object is managed by Horizon or not. |
| summary\|datastoreClusters | Summary\|Datastore Cluster(s) | This property is applicable only if the VM belongs to a datastore cluster.<br>**Note:**<br>A VM with multiple virtual disks can belong to multiple datastore clusters. |
| Summary\|Configuration\|Number of NICs | Number of NICs | his property displays the number of NICs configured in the VM. |
| config\|name | Name | This property displays the virtual object name |
| config\|guestFullName | Guest OS from vCenter | This property is set by the vCenter during the VM creation. It may differ from the value of the Guest/ |
| config\|hardware\|numCpu | Number of virtual CPUs | Number of virtual CPUs |
| config\|hardware\|memoryKB | Memory | Memory |
| config\|hardware\|thinEnabled | Thin Provisioned Disk | Indicates whether thin provisioning is enabled |
| config\|hardware\|diskSpace | Disk Space | Disk Space |
| config\|cpuAllocation\|reservation | Reservation | CPU reservation |
| config\|cpuAllocation\|limit | Limit | CPU limit |
| config\|cpuAllocation\|shares\|shares | Shares | CPU shares |
| config\|memoryAllocation\|reservation | Reservation | CPU reservation |
| config\|memoryAllocation\|limit | Limit | Limit |
| config\|memoryAllocation\|shares\|shares | Shares | Memory shares |
| config\|extraConfig\|mem_hotadd | Memory Hot Add | Memory Hot Add Configuration |
| config\|extraConfig\|vcpu_hotadd | VCPU Hot Add | VCPU Hot Add Configuration |
| config\|extraConfig\|vcpu_hotremove | VCPU Hot Remove | VCPU Hot Remove Configuration |
| config\|security\|disable_autoinstall | Disable tools auto install (isolation.tools.autoInstall.disable) | Disable tools auto install (isolation.tools.autoInstall.disable) |
| config\|security\|disable_console_copy | Disable console copy operations (isolation.tools.copy.disable) | Disable console copy operations (isolation.tools.copy.disable) |
| config\|security\|disable_console_dnd | Disable console drag and drop operations (isolation.tools.dnd.disable) | Disable console drag and drop operations (isolation.tools.dnd.disable) |

| Property Key | Property Name | Description |
|---|---|---|
| config\|security\|enable_console_gui_options | Enable console GUI operations (isolation.tools.setGUIOptions.enable) | Enable console GUI operations (isolation.tools.setGUIOptions.enable) |
| config\|security\|disable_console_paste | Disable console paste operations (isolation.tools.paste.disable) | Disable console paste operations (isolation.tools.paste.disable) |
| config\|security\|disable_disk_shrinking_shrink | Disable virtual disk shrink (isolation.tools.diskShrink.disable) | Disable virtual disk shrink (isolation.tools.diskShrink.disable) |
| config\|security\|disable_disk_shrinking_wiper | Disable virtual disk wiper (isolation.tools.diskWiper.disable) | Disable virtual disk wiper (isolation.tools.diskWiper.disable) |
| config\|security\|disable_hgfs | Disable HGFS file transfers (isolation.tools.hgfsServerSet.disable) | Disable HGFS file transfers (isolation.tools.hgfsServerSet.disable) |
| config\|security\| disable_independent_nonpersistent | Avoid using independent nonpersistent disks (scsiX:Y.mode) | Avoid using independent nonpersistent disks (scsiX:Y.mode) |
| config\|security\|enable_intervm_vmci | Enable VM-to-VM communication through VMCI (vmci0.unrestricted) | Enable VM-to-VM communication through VMCI (vmci0.unrestricted) |
| config\|security\|enable_logging | Enable VM logging (logging) | Enable VM logging (logging) |
| config\|security\|disable_monitor_control | Disable VM Monitor Control (isolation.monitor.control.disable) | Disable VM Monitor Control (isolation.monitor.control.disable) |
| config\|security\| enable_non_essential_3D_features | Enable 3D features on Server and desktop virtual machines (mks.enable3d) | Enable 3D features on Server and desktop virtual machines (mks.enable3d) |
| config\|security\| disable_unexposed_features_autologon | Disable unexposed features - autologon (isolation.tools.ghi.autologon.disable) | Disable unexposed features - autologon (isolation.tools.ghi.autologon.disable) |
| config\|security\| disable_unexposed_features_biosbbs | Disable unexposed features - biosbbs (isolation.bios.bbs.disable) | Disable unexposed features - biosbbs (isolation.bios.bbs.disable) |
| config\|security\| disable_unexposed_features_getcreds | Disable unexposed features - getcreds (isolation.tools.getCreds.disable) | Disable unexposed features - getcreds (isolation.tools.getCreds.disable) |
| config\|security\| disable_unexposed_features_launchmenu | Disable unexposed features - launchmenu (isolation.tools.ghi.launchmenu.change) | Disable unexposed features - launchmenu (isolation.tools.ghi.launchmenu.change) |
| config\|security\| disable_unexposed_features_memsfss | Disable unexposed features - memsfss (isolation.tools.memSchedFakeSampleStats.disable) | Disable unexposed features - memsfss (isolation.tools.memSchedFakeSampleStats.disable) |
| config\|security\| disable_unexposed_features_protocolhandler | Disable unexposed features - protocolhandler (isolation.tools.ghi.protocolhandler.info.disable) | Disable unexposed features - protocolhandler (isolation.tools.ghi.protocolhandler.info.disable) |
| config\|security\| disable_unexposed_features_shellaction | Disable unexposed features - shellaction (isolation.ghi.host.shellAction.disable) | Disable unexposed features - shellaction (isolation.ghi.host.shellAction.disable) |
| config\|security\| disable_unexposed_features_toporequest | Disable unexposed features - toporequest (isolation.tools.dispTopoRequest.disable) | Disable unexposed features - toporequest (isolation.tools.dispTopoRequest.disable) |
| config\|security\| disable_unexposed_features_trashfolderstate | Disable unexposed features - trashfolderstate (isolation.tools.trashFolderState.disable) | Disable unexposed features - trashfolderstate (isolation.tools.trashFolderState.disable) |
| config\|security\| disable_unexposed_features_trayicon | Disable unexposed features - trayicon (isolation.tools.ghi.trayicon.disable) | Disable unexposed features - trayicon (isolation.tools.ghi.trayicon.disable) |
| config\|security\| disable_unexposed_features_unity | Disable unexposed features - unity (isolation.tools.unity.disable) | Disable unexposed features - unity (isolation.tools.unity.disable) |

| Property Key | Property Name | Description |
|---|---|---|
| config\|security\|disable_unexposed_features_unity_interlock | Disable unexposed features - unity-interlock (isolation.tools.unityInterlockOperation.disable) | Disable unexposed features - unity-interlock (isolation.tools.unityInterlockOperation.disable) |
| config\|security\|disable_unexposed_features_unity_taskbar | Disable unexposed features - unity-taskbar (isolation.tools.unity.taskbar.disable) | Disable unexposed features - unity-taskbar (isolation.tools.unity.taskbar.disable) |
| config\|security\|disable_unexposed_features_unity_unityactive | Disable unexposed features - unity-unityactive (isolation.tools.unityActive.disable) | Disable unexposed features - unity-unityactive (isolation.tools.unityActive.disable) |
| config\|security\|disable_unexposed_features_unity_windowcontents | Disable unexposed features - unity-windowcontents (isolation.tools.unity.windowContents.disable) | Disable unexposed features - unity-windowcontents (isolation.tools.unity.windowContents.disable) |
| config\|security\|disable_unexposed_features_unitypush | Disable unexposed features - unitypush (isolation.tools.unity.push.update.disable) | Disable unexposed features - unitypush (isolation.tools.unity.push.update.disable) |
| config\|security\|disable_unexposed_features_versionget | Disable unexposed features - versionget (isolation.tools.vmxDnDVersionGet.disable) | Disable unexposed features - versionget (isolation.tools.vmxDnDVersionGet.disable) |
| config\|security\|disable_unexposed_features_versionset | Disable unexposed features - versionset (solation.tools.guestDnDVersionSet.disable) | Disable unexposed features - versionset (solation.tools.guestDnDVersionSet.disable) |
| config\|security\|disable_vix_messages | Disable VIX messages from the VM (isolation.tools.vixMessage.disable) | Disable VIX messages from the VM (isolation.tools.vixMessage.disable) |
| config\|security\|enable_vga_only_mode | Disable all but VGA mode on virtual machines (svga.vgaOnly) | Disable all but VGA mode on virtual machines (svga.vgaOnly) |
| config\|security\|limit_console_connection | Limit number of console connections (RemoteDisplay.maxConnection) | Limit number of console connections (RemoteDisplay.maxConnection) |
| config\|security\|limit_log_number | Limit number of log files (log.keepOld) | Limit number of log files (log.keepOld) |
| config\|security\|limit_log_size | Limit log file size (log.rotateSize) | Limit log file size (log.rotateSize) |
| config\|security\|limit_setinfo_size | Limit VMX file size (tools.setInfo.sizeLimit) | Limit VMX file size (tools.setInfo.sizeLimit) |
| config\|security\|enable_console_VNC | Enable access to VM console via VNC protocol (RemoteDisplay.vnc.enabled) | Enable access to VM console via VNC protocol (RemoteDisplay.vnc.enabled) |
| config\|security\|disable_device_interaction_connect | Disable unauthorized removal, connection of devices (isolation.device.connectable.disable) | Disable unauthorized removal, connection of devices (isolation.device.connectable.disable) |
| config\|security\|disable_device_interaction_edit | Disable unauthorized modification of devices (isolation.device.edit.disable) | Disable unauthorized modification of devices (isolation.device.edit.disable) |
| config\|security\|enable_host_info | Enable send host information to guests (tools.guestlib.enableHostInfo) | Enable send host information to guests (tools.guestlib.enableHostInfo) |
| config\|security\|network_filter_enable | Enable dvfilter network APIs (ethernetX.filterY.name) | Enable dvfilter network APIs (ethernetX.filterY.name) |
| config\|security\|vmsafe_cpumem_agentaddress | VMsafe CPU/memory APIs - IP address (vmsafe.agentAddress) | VMsafe CPU/memory APIs - IP address (vmsafe.agentAddress) |
| config\|security\|vmsafe_cpumem_agentport | VMsafe CPU/memory APIs - port number (vmsafe.agentPort) | VMsafe CPU/memory APIs - port number (vmsafe.agentPort) |
| config\|security\|vmsafe_cpumem_enable | Enable VMsafe CPU/memory APIs (vmsafe.enable) | Enable VMsafe CPU/memory APIs (vmsafe.enable) |
| config\|security\|disconnect_devices_floppy | Disconnect floppy drive | Disconnect floppy drive |
| config\|security\|disconnect_devices_cd | Disconnect CD-ROM | Disconnect CD-ROM |

| Property Key | Property Name | Description |
|---|---|---|
| config\|security\|disconnect_devices_usb | Disconnect USB controller | Disconnect USB controller |
| config\|security\|disconnect_devices_parallel | Disconnect parallel port | Disconnect parallel port |
| config\|security\|disconnect_devices_serial | Disconnect serial port | Disconnect serial port |
| config\|faultTolerant | config\|faultTolerant | |

**Note:**
Security properties not collected by default. They are collected only if the *vSphere Hardening Guide* policy is applied to the objects, or if the *vSphere Hardening Guide* alerts are manually enabled in the currently applied policy.

**Table 1603: Runtime Properties Collected for Virtual Machine Objects**

| Property Key | Property Name | Description |
|---|---|---|
| runtime\|memoryCap | Memory Capacity | Memory Capacity |

**Table 1604: CPU Usage Properties Collected for Virtual Machine Objects**

| Property Key | Property Name | Description |
|---|---|---|
| cpu\|limit | CPU limit | CPU limit |
| cpu\|reservation | CPU reservation | CPU reservation |
| cpu\|speed | CPU | CPU Speed |

**Table 1605: Memory Properties Collected for Virtual Machine Objects**

| Property Key | Property Name | Description |
|---|---|---|
| mem\|host_limit | VM Limit | Mem Machine Limit |
| mem\|host_reservation | Memory\|VM Reservation(kb) | This property is disabled by default. |

**Table 1606: Network Properties Collected for Virtual Machine Objects**

| Property Key | Property Name | Description |
|---|---|---|
| net:<vnic_id>\|portGroup | Network:<vnic_id>\|Port Group | This is a property at each virtual NIC level, not at the VM level. A VM with multiple NIC connects to different port groups, with this property you can identify to which port group the vnic blongs and has mapping with. |
| net:<nic_key>\|uptCompatibilityEnabled | Network:<nic_key>\|Direct Path I/O Status | If there is the VM, with cofigured direct path IO - IO directly comes to VM, bypassing the hypervisor. The property shows the status. |
| net\|mac_address | Mac Address | Mac Address |
| net\|ip_address | IP Address | IP Address |
| net\|vnic_label | Network:<ID>\|Label | This property is disabled by default. |
| net\|nvp_vm_uuid | Network I/O\|NVP VM UUID | This property is disabled by default. |
| net\|vnic_type | Network I/O\|Virtual NIC Type | This property is disabled by default. |
| net\|ipv6_address | Network\|IPv6 Address | This property is disabled by default. |

| Property Key | Property Name | Description |
|---|---|---|
| net\|ipv6_prefix_length | Network\|IPv6 Prefix Length | This property is disabled by default. |
| net\|default_gateway | Network\|Network I/O\|Default Gateway | This property is disabled by default. |
| net\|subnet_mask | Network\|Subnet Mask | This property is disabled by default. |

**Table 1607: Summary Properties Collected for Virtual Machine Objects**

| Property Key | Property Name | Description |
|---|---|---|
| summary\|customTag\|customTagValue | Value | Custom Tag Value |
| summary\|tag | vSphere Tag | vSphere Tag Name |
| summary\|parentCluster | Parent Cluster | Parent Cluster |
| summary\|parentHost | Parent Host | Parent Host |
| summary\|parentDatacenter | Parent data center | Parent data center |
| summary\|parentVcenter | Parent vCenter | Parent vCenter |
| summary\|guest\|fullName | Guest OS Full Name | This property is provided by the VMware Tools. It will differ to the value set in vCenter if the Guest OS was upgraded, or if a different Guest OS was installed. |
| summary\|guest\|ipAddress | Guest OS IP Address | Guest OS IP Address |
| summary\|guest\|toolsRunningStatus | Tools Running Status | Guest Tools Running Status |
| summary\|guest\|toolsVersionStatus2 | Tools Version Status | Guest Tools Version Status 2 |
| summary\|guest\|vrealize_operations_agent_id | vRealize Operations Agent ID | An ID to identify a VM in Agent Adapter's world. |
| summary\|guest\| vrealize_operations_euc_agent_id | vRealize Operations Euc Agent ID | An ID to identify a VM in Agent Adapter's world. |
| summary\|config\|numEthernetCards | Number of NICs | Number of NICs |
| summary\|config\|isTemplate | VM Template | Indicates whether it is a VM Template. |
| summary\|runtime\|powerState | Power State | Power State |
| summary\|runtime\|connectionState | Connection State | Connection State |
| summary\|config\|appliance | Appliance | Appliance |
| summary\|config\|productName | Product Name | Product Name |
| summary\|UUID | UUID | Unique UUID instance in vCenter that identifies all the virtual machine instances. |
| summary\|smbiosUUID | SMBIOS UUID | System Management BIOS UUID of a virtual machine. |

**Table 1608: Virtual Disk Properties Collected for Virtual Machine Objects**

| Property Key | Property Name | Description |
|---|---|---|
| virtualDisk:<scsi_id>\|encryptionStatus | Virtual Disk:<scsi_id>\|Encryption Status | This property provides encryption status on SCSI on VDMK layer. The property is per virtual disk and does not apply to RDM disk. |
| virtualDisk:<scsi_controller>\|iopsLimit | Virtual Disk:<scsi_controller>\|IOPS Limit | This property shows the set IOPS limit on virtual disk on SCSI level, when IOPS limit is set beyond the disk. This helps you to understand the current performance state, why the IOPS is not increasing and limited. |

| Property Key | Property Name | Description |
|---|---|---|
| Virtual Disk:scsi0:0\|Configuired Size(GB) | Configuired Size(GB) | This property displays the disk space configured for the virtual disk. |
| Virtual Disk:scsi0:0\|Datastore | Datastore | This property displays the name of the datastore where the scsi disk is present. If an RDM is present, only a pointer to the RDM is present in the datastore. |
| Virtual Disk:scsi0:0\|Disk Mode | Disk Mode | This property determines how a virtual disk is affected by snapshots. The disk mode acts on each individual VMDK, not on the whole VM. The available options are Independent persistent, Persistent, Non-persistent, this property is disabled by default. |
| Virtual Disk:scsi0:0\|SCSI Bus Sharing | SCSI Bus Sharing | This property sets the type of bus sharing for the VM and determines whether to share the bus or not. Depending on the type of bus sharing, the VM can access the same virtual disk simultaneously on the same server or any other server. The available options are None, Physical, Virtual, this property is disabled by default. |
| Virtual Disk:scsi0:0\|SCSI Controller Type | SCSI Controller Type | This virtual storage controller property connects the virtual and physical disks to the VM. The available options are LSI SAS/ PVSCSI, this property is disabled by default. |
| Virtual Disk:scsi0:0\|Virtual Disk Sharing | Virtual Disk Sharing | This property allows VMFS-backed disks to be shared by multiple VMs. The available options are Unspecified, No sharing, Multi-Writer. You can use this option to disable protection for certain cluster aware applications, where the applications ensures that simultaneous write operation from two VMs does not induce data loss. This property is disabled by default. |
| Virtual Disk:scsi0:0\|Virtual Device Node | Virtual Device Node | This property determines the virtual device bus location. The virtual disks are enumerated starting with the first controller. This property is disabled by default. |
| Virtual Disk:scsi0:0\|Is RDM | Is RDM | This property indicates whether the virtual disk is an RDM or not. This property is enabled by default. |
| | | |
| Virtual Disk\|File Name | File Name | This property is disabled by default. |
| Virtual Disk\|Label | Label | This property displays the device label. |
| Virtual Disk:scsi1:1\|Compatibility Mode | Compatibility Mode | This property displays the compatibility mode for the RDMs. The options are physical and virtual. Virtual mode specifies the full virtualization of the mapped device whereas physical mode specifies minimal SCSI virtualization of the mapped device. This property is disabled by default in the base policy. |

**Table 1609: Virtual Disk Properties Collected for POD Objects**

| Property Key | Property Name | Description |
|---|---|---|
| Virtual Disk:scsi0:0\|Virtual Device Node | Virtual Device Node | This property determines the virtual device bus location. The virtual disks are enumerated starting with the first controller. This property is disabled by default. |
| Virtual Disk:scsi0:0\|Virtual Disk Sharing | Virtual Disk Sharing | This property allows VMFS-backed disks to be shared by multiple VMs. The available options are Unspecified, No sharing, Multi-Writer. You can use this option to disable protection for certain cluster aware applications, where the applications ensures that simultaneous write operation from two PODs does not induce data loss. This property is disabled by default. |
| Virtual Disk:scsi0:0\|Disk Mode | Disk Mode | This property determines how a virtual disk is affected by snapshots. The disk mode acts on each individual VMDK, not on the whole POD. The available options are Independent persistent, Persistent, Non-persistent, this property is disabled by default. |
| Virtual Disk:scsi0:0\|SCSI Controller Type | SCSI Controller Type | This virtual storage controller property connects the virtual and physical disks to the POD. The available options are LSI SAS/PVSCSI, this property is disabled by default. |

**Table 1610: Datastore Properties Collected for Virtual Machine Properties**

| Property Key | Property Name | Description |
|---|---|---|
| datastore\|maxObservedNumberRead | Datastore I/O\|Highest Observed Number of Read Requests | |
| datastore\|maxObservedNumberWrite | Datastore I/O\|Highest Observed Number of Write Requests | |
| datastore\|maxObservedOIO | Datastore I/O\|Highest Observed Outstanding Requests | |
| datastore\|maxObservedRead | Datastore I/O\|Highest Observed Read Rate(kbps) | |
| datastore\|maxObservedWrite | Datastore I/O\|Highest Observed Write Rate(kbps) | |

**Table 1611: Compliance Configuration Related Properties for Virtual Machine Objects**

| Property Key | Property Name | Description |
|---|---|---|
| config\|security\|disconnect_devices_virtualhd audiocard | Configuration\|Security\|Virtual HD Audio Card Disconnected | NIL |
| config\|security\|disconnect_devices_virtualah cicontroller | Configuration\|Security\|Virtual AHCI Controller Disconnected | NIL |
| config\|security\|disconnect_devices_virtualen soniq1371 | Configuration\|Security\|Virtual Ensoniq 1371 Disconnected | NIL |

Datastore properties collected for virtual machine objects have been disabled in this version of VCF Operations. This means that they do not collect data by default.

## Host System Properties

VCF Operations collects configuration, hardware, runtime, CPU, network I/O, and properties about summary use for host system objects.

**Table 1612: GPU properties collected for Host System Objects**

| Property Key | Property Name | Description |
|---|---|---|
| gpu\|<GPU-id>\|active_type | GPU\|<GPU-id>\|Active Type | Active Type |
| gpu\|<GPU-id>\|configured_type | GPU\|<GPU-id>\|Configured Type | Configured Type |
| gpu\|<GPU-id>\|device_name | GPU\|<GPU-id>\|Device Name | Device name |
| gpu\|<GPU-id>\|vendor_name | GPU\|<GPU-id>\|Vendor Name | Vendor name |
| gpu\|assignmentPolicy | GPU\|Assignment Policy | Assignment Policy |
| GPU:GPU-id\|GPU Slowdown Temperature (Celcius) | GPU Slowdown Temperature (Celsius) | The slowdown temperature is the thermal threshold at which a GPU starts reducing it's clock speed to lower power consumption and prevent overheating. |
| GPU:GPU-id\|GPU Shutdown Temperature (Celcius) | GPU Shutdown Temperature (Celsius) | The shutdown temperature is the thermal threshold at which a GPU automatically powers off, to protect itself from damage due to overheating. |
| GPU:GPU-id\|GPU Power limit (Max TDP) (W) | GPU Power limit (Max TDP) (Watts) | Maximum power in Watts that this GPU can draw. |

**Table 1613: Configuration Properties Collected for Host System Objects**

| Property Key | Property Name | Description |
|---|---|---|
| config\|name | Name | Name |
| config\|diskSpace | Disk Space | Disk Space |
| config\|network\|nnic | Number of NICs | Number of NICs |
| config\|network\|linkspeed | Average Physical NIC Speed | Average Physical NIC Speed |
| config\|network\|dnsserver | DNS Server | List of DNS Servers |
| config\|product\|productLineId | Product Line ID | Product Line ID |
| config\|product\|apiVersion | API Version | API Version |
| config\|storageDevice\|plugStoreTopology\| numberofPath | Total number of Path | Total number of storage paths |
| config\|storageDevice\|multipathInfo\| numberofActivePath | Total number of Active Path | Total number of active storage paths |
| config\|storageDevice\|multipathInfo\| multipathPolicy | Multipath Policy | Multipath Policy |
| config\|hyperThread\|available | Available | Indicates whether hyperthreading is supported by the server |
| config\|hyperThread\|active | Active | Indicates whether hyperthreading is active |
| config\|ntp\|server | NTP Servers | NTP Servers |
| config\|security\|ntpServer | NTP server | NTP server |
| config\|security\|enable_ad_auth | Enable active directory authentication | Enable active directory authentication |

| Property Key | Property Name | Description |
|---|---|---|
| config\|security\|enable_chap_auth | Enable mutual chap authentication | Enable mutual chap authentication |
| config\|security\|enable_auth_proxy | Enable authentication proxy (UserVars.ActiveDirectoryVerifyCAMCertificate) | Enable authentication proxy (UserVars.ActiveDirectoryVerifyCAMCertificate) |
| config\|security\|syslog_host | Remote log host (Syslog.global.logHost) | Remote log host (Syslog.global.logHost) |
| config\|security\|dcui_access | Users who can override lock down mode and access the DCUI (DCUI.Access) | Users who can override lock down mode and access the DCUI (DCUI.Access) |
| config\|security\|shell_interactive_timeout | Shell interactive timeout (UserVars.ESXiShellInteractiveTimeOut) | Shell interactive timeout (UserVars.ESXiShellInteractiveTimeOut) |
| config\|security\|shell_timeout | Shell timeout (UserVars.ESXiShellTimeOut) | Shell timeout (UserVars.ESXiShellTimeOut) |
| config\|security\|dvfilter_bind_address | Dvfilter bind ip address (Net.DVFilterBindIpAddress) | Dvfilter bind ip address (Net.DVFilterBindIpAddress) |
| config\|security\|syslog_dir | Log directory (Syslog.global.logDir) | Log directory (Syslog.global.logDir) |
| config\|security\|firewallRule\|allowedHosts | Allowed hosts | Allowed hosts in the firewall configuration |
| config\|security\|service\|isRunning | Running | Indicates whether a service is running or not. Services are: Direct Console UI, ESX shell, SSH, or NTP Daemon. |
| config\|security\|service\|ruleSet | Ruleset | Ruleset for each service. |
| config\|security\|service\|policy | Policy | Policy for each service. |
| config\|security\|tlsdisabledprotocols | TLS Disabled Protocols | TLS Disabled Protocols |

**Note:**
Security properties not collected by default. They are collected only if the *vSphere Hardening Guide* policy is applied to the objects, or if the *vSphere Hardening Guide* alerts are manually enabled in the currently applied policy.

**Table 1614: Cost Properties Collected for Host System Objects**

| Property Key | Property Name | Description |
|---|---|---|
| Cost\|Energy Consumed (Joule) | Energy Consumed (Joule) | Displays the energy consumed in Joules. |
| Cost\|Number of Rack Units | Number of Rack Units | Displays the number of rack units in the host. |
| Cost\|OS Categories | OS Categories | Displays the operating system categories in the host. |
| Cost\|IsServerLeased | Is Server Leased | Displays whether the server is leased or not. |
| Cost\|RemainingDepreciationMonths | Remaining Depreciation Months | Displays the remaining number of depreciation months. |
| Cost\|ServerPurchaseCost | Server Purchase Cost | Server Purchase Cost is displayed in the currency format chosen. |
| Cost\|ServerPurchaseDate | Server Purchase Date | Server Purchase Date is displayed |

**Table 1615: Hardware Properties Collected for Host System Objects**

| Property Key | Property Name | Description |
|---|---|---|
| hardware\|bioisReleaseDate | Hardware\|BIOS Release Date | This property displays the release date corresponding to the version of the installed BIOS. |
| hardware\|memorySize | Memory Size | Memory Size |
| hardware\|cpuInfo\|numCpuCores | Number of CPU Cores | Number of CPU Cores |
| hardware\|cpuInfo\|hz | CPU Speed per Core | CPU Speed per Core |
| hardware\|cpuInfo\|numCpuPackages | Number of CPU Packages | Number of CPU Packages |
| hardware\|cpuInfo\|<br>powerManagementPolicy | Active CPU Power Management Policy | Active CPU Power Management Policy |
| hardware\|cpuInfo\|<br>powerManagementTechnology | Power Management Technology | Power Management Technology |
| hardware\|cpuInfo\|biosVersion | BIOS Version | BIOS Version |
| hardware\|vendor | Hardware\|Vendor | Indicates the hardware manufacturer |

**Table 1616: Runtime Properties Collected for Host System Objects**

| Property Key | Property Name | Description |
|---|---|---|
| runtime\|connectionState | Connection State | Connection State |
| runtime\|powerState | Power State | Power State |
| runtime\|maintenanceState | Maintenance State | Maintenance State |
| runtime\|memoryCap | Memory Capacity | Memory Capacity |

**Table 1617: Configuration Manager Properties Collected for Host System Objects**

| Property Key | Property Name | Description |
|---|---|---|
| configManager\|memoryManager\|<br>consoleReservationInfo\|<br>serviceConsoleReserved | Service Console Reserved | Service console reserved memory |

**Table 1618: CPU Usage Properties Collected for Host System Objects**

| Property Key | Property Name | Description |
|---|---|---|
| cpu\|speed | CPU | CPU Speed |
| cpu\|cpuModel | CPU Model | CPU Model |

**Table 1619: Network Properties Collected for Host System Objects**

| Property Key | Property Name | Description |
|---|---|---|
| net:<pnic>\|configuredSpeed | Network:<pnic>\|Configured Speed | This property displays the configured network speed of the network card. If this is higher than actual, the card is not operating at full capacity. |
| net:<pnic>\|speed | Network:<pnic>\|Actual Speed | This property displays the actual operating speed of the network card, which can be lower than its configured capacity due to auto-negotiation. The options are Enabled or Disabled. |
| net\|maxObservedKBps | Highest Observed Throughput | Highest Observed Throughput (KBps) |
| net\|mgmt_address | Management Address | Management Address |
| net\|ip_address | IP Address | IP Address |
| net\|discoveryProtocol\|cdp\|managementIpAddress | Management IP Address | Management IP Address |
| net\|discoveryProtocol\|cdp\|systemName | System Name | System Name |
| net\|discoveryProtocol\|cdp\|portName | Port Name | Port Name |
| net\|discoveryProtocol\|cdp\|vlan | VLAN | VLAN |
| net\|discoveryProtocol\|cdp\|mtu | MTU | Maximum Transmission Unit |
| net\|discoveryProtocol\|cdp\|hardwarePlatform | Hardware Platform | Hardware Platform |
| net\|discoveryProtocol\|cdp\|softwareVersion | Software Version | Software Version |
| net\|discoveryProtocol\|lldp\|systemDescription | System Description | System Description |
| net\|discoveryProtocol\|lldp\|mtu | MTU | Maximum Transmission Unit |
| net\|discoveryProtocol\|lldp\|portDescription | Port Description | Port Description |
| net\|discoveryProtocol\|lldp\|aggregationStatus | Aggregation Status | Aggregation Status |
| net\|discoveryProtocol\|lldp\|managementIpAddress | Management IP Address | Management IP Address |
| net\|discoveryProtocol\|lldp\|systemName | System Name | System Name |
| net\|discoveryProtocol\|lldp\|portName | Port Name | Port Name |
| net\|discoveryProtocol\|lldp\|vlan | VLAN | VLAN |

**Table 1620: System Properties Collected for Host System Objects**

| Property Key | Property Name | Description |
|---|---|---|
| sys\|build | Build number | VMWare build number |
| sys\|productString | Product String | VMWare product string |

**Table 1621: Summary Properties Collected for Host System Objects**

| Property Key | Property Name | Description |
|---|---|---|
| Summary\|Is Horizon Managed | Is Horizon Managed | This property displays whether the selected object is managed by Horizon or not. |
| summary\|version | Version | This property displays theVersion. |
| summary\|hostuuid | Host UUID | This property displays the Host UUID. |
| summary\|evcMode | Current EVC Mode | This property displays the Current EVC Mode. |
| summary\|customTag\|customTagValue | Value | This property displays the Custom Tag Value. |
| summary\|tag | vSphere Tag | This property displays the vSphere Tag Name. |
| summary\|parentCluster | Parent Cluster | This property displays the Parent Cluster. |
| summary\|parentDatacenter | Parent Datacenter | This property displays the Parent Datacenter. |
| summary\|parentVcenter | Parent Vcenter | This property displays the Parent Vcenter. |

**Table 1622: Datastore Properties Collected for Host System Objects**

| Property Key | Property Name | Description |
|---|---|---|
| datastore\|maxObservedNumberRead | Datastore I/O\|Highest Observed Number of Read Requests | |
| datastore\|maxObservedNumberWrite | Datastore I/O\|Highest Observed Number of Write Requests | |
| datastore\|maxObservedOIO | Datastore I/O\|Highest Observed Outstanding Requests | |
| datastore\|maxObservedRead | Datastore I/O\|Highest Observed Read Rate(kbps) | |
| datastore\|maxObservedWrite | Datastore I/O\|Highest Observed Write Rate(kbps) | |
| net\|discoveryProtocol\|cdp\|timeToLive | Network I/O\|Discovery Protocol\| Cisco Discovery Protocol\|Time to Live | |
| net\|discoveryProtocol\|lldp\|timeToLive | Network I/O\|Discovery Protocol\| Link Layer Discovery Protocol\| Time to Live | |

Datastore properties collected for host system objects have been disabled in this version of VCF Operations. This means that they do not collect data by default.

**Table 1623: Storage Path Properties Collected for Host System Objects**

| Property Key | Property Name | Description |
|---|---|---|
| storageAdapter\|port_WWN | Storage Adapter\|Port WWN | The port world wide name for storage adapter. Available for FC adapters only. |

**Table 1624: Compliance Configuration Related Properties for Host System Objects**

| Property Key | Property Name | Description |
|---|---|---|
| config\|security\|password_max_days | Configuration\|Security\|Password Max Days | Password Max Days |
| config\|security\|welcome_message | Configuration\|Security\|Welcome Message Configured | Welcome Message Configured |
| config\|security\|issue | Configuration\|Security\|SSH Connection Banner Message Configured | SSH Connection Banner Message Configured |
| config\|security\|host_client_session_timeout | Configuration\|Security\|Host Client Session Timeout | Host Client Session Timeout |
| config\|security\|has_lockdown_exception_users | Configuration\|Security\|Has Lockdown exception users | Has Lockdown exception users |

## Cluster Compute Resource Properties

VCF Operations collects configuration and summary properties for cluster compute resource objects.

**Table 1625: License Properties for Cluster Objects**

| Property Key | Property Name | Description |
|---|---|---|
| License type | License Type | Displays the license type for the cluster object |
| Expiry | Expiry | Displays the number of days remaining before the license expires the cluster object<br>**Note:**<br>An alert is generated if the license threshold is<br>• >= 80% - Warning<br>• >= 90% - Immediate<br>• =95% - Catastrophic |

**Table 1626: Configuration Properties Collected for Cluster Compute Resource Objects**

| Property Key | Property Name | Description |
|---|---|---|
| config\|name | Name | Name |

**Table 1627: Summary Properties Collected for Cluster Compute Resource Objects**

| Property Key | Property Name | Description |
|---|---|---|
| Summary\|Is Horizon Managed | Is Horizon Managed | This property displays whether the selected object is managed by Horizon or not. |
| summary\|parentDatacenter | Parent data center | This property displays the Parent data center. |
| summary\|parentVcenter | Parent vCenter | This property displays the Parent vCenter. |
| summary\|customTag\|customTagValue | Value | This property displays the Custom Tag Value. |

| Property Key | Property Name | Description |
|---|---|---|
| summary\|tag | vSphere Tag | This property displays the vSphere Tag Name. |

**Table 1628: DR, DAS, and DPM Configuration Properties Collected for Cluster Compute Resource Objects**

| Property Key | Property Name | Description |
|---|---|---|
| configuration\|drsconfig\|enabled | Enabled | Indicates whether DRS is enabled |
| configuration\|drsconfig\|defaultVmBehavior | Default DRS Behavior | Default DRS Behavior |
| configuration\|drsconfig\|affinityRules | Affinity Rules | DRS Affinity Rules |
| configuration\|dasconfig\|enabled | HA Enabled | HA Enabled |
| configuration\|dasconfig\| admissionControlEnabled | Admission Control Enabled | Admission Control Enabled |
| configuration\|dpmconfiginfo\|enabled | DPM Enabled | DPM Enabled |
| configuration\|dpmconfiginfo\| defaultDpmBehavior | Default DPM Behavior | Default DPM Behavior |
| configuration\|infraUpdateHaConfig\| remediation | Cluster Configuration\|HA Configuration\|Remediation | This property displays the Remediation mode taken by vSphere Cluster to deal with host failure. The options available are Quarantine mode, Maintenance mode, Mixed mode. |
| configuration\|drsConfig\| pctIdleMBInMemDemand | Cluster Configuration\|DRS Configuration\|Idle Consumed Memory | |
| configuration\|drsConfig\|targetBalance | Cluster Configuration\|DRS Configuration\|Tolerable imbalance threshold | |

DRS properties are collected for disaster recovery. DAS properties are collected for high availability service, formerly distributed availability service. DPM properties are collected for distributed power management.

## Resource Pool Properties

VCF Operations collects configuration, CPU, memory, and summary properties for resource pool objects.

**Table 1629: Configuration Properties Collected for Resource Pool Objects**

| Property Key | Property Name | Description |
|---|---|---|
| config\|name | Name | Name |
| config\|cpuAllocation\|reservation | Reservation | CPU reservation |
| config\|cpuAllocation\|limit | Limit | CPU limit |
| config\|cpuAllocation\|expandableReservation | Expandable Reservation | CPU expandable reservation |
| config\|cpuAllocation\|shares\|shares | Shares | CPU shares |
| config\|memoryAllocation\|reservation | Reservation | Memory reservation |
| config\|memoryAllocation\|limit | Limit | Memory limit |
| config\|memoryAllocation\| expandableReservation | Expandable Reservation | Memory expandable reservation |
| config\|memoryAllocation\|shares\|shares | Shares | Memory shares |

**Table 1630: CPU Usage Properties Collected for Resource Pool Objects**

| Property Key | Property Name | Description |
|---|---|---|
| cpu\|limit | CPU Limit | CPU Limit |
| cpu\|reservation | CPU reservation | CPU Reservation |
| cpu\|expandable_reservation | CPU expandable reservation | CPU Expandable Reservation |
| cpu\|shares | CPU Shares | CPU Shares |
| cpu\|corecount_provisioned | Provisioned vCPU(s) | Number of CPUs. It counts both the vSocket and vCore. A VM with 2 vSockets x 4 vCores each has 8 vCPU. |

**Table 1631: Memory Properties Collected for Resource Pool Objects**

| Property Key | Property Name | Description |
|---|---|---|
| mem\|limit | Memory limit | Memory limit |
| mem\|reservation | Memory reservation | Memory reservation |
| mem\|expandable_reservation | Memory expandable reservation | Memory expandable reservation |
| mem\|shares | Memory Shares | Memory Shares |

**Table 1632: Summary Properties Collected for Resource Pool Objects**

| Property Key | Property Name | Description |
|---|---|---|
| summary\|customTag\|customTagValue | Value | Custom Tag Value |
| summary\|tag | vSphere Tag | vSphere Tag Name |

## Data Center Properties

VCF Operations collects configuration and summary properties for data center objects.

**Table 1633: Configuration Properties Collected for Data Center Objects**

| Property Key | Property Name | Description |
|---|---|---|
| config\|name | Name | Name |

**Table 1634: Summary Properties Collected for Data Center Objects**

| Property Key | Property Name | Description |
|---|---|---|
| summary\|parentVcenter | Parent Vcenter | Parent Vcenter |
| summary\|customTag\|customTagValue | Value | Custom Tag Value |
| summary\|tag | vSphere Tag | vSphere Tag Name |

## Storage Pod Properties

VCF Operations collects configuration and summary properties for storage pod objects.

**Table 1635: Configuration Properties Collected for Storage Pod Objects**

| Property Key | Property Name | Description |
|---|---|---|
| config\|name | Name | Name |
| config\|sdrsconfig\|vmStorageAntiAffinityRules | VM storage antiaffinity rules | Storage Distributed Resource Scheduler (SDRS) VM anti-affinity rules |
| config\|sdrsconfig\|vmdkAntiAffinityRules | VMDK antiaffinity rules | Storage Distributed Resource Scheduler (SDRS) Virtual Machine Disk (VMDK) anti-affinity rules |

## VMware Distributed Virtual Switch Properties

VCF Operations collects configuration and summary properties for VMware distributed virtual switch objects.

**Table 1636: Configuration Properties Collected for VMware Distributed Virtual Switch Objects**

| Property Key | Property Name | Description |
|---|---|---|
| config\|networkResourceManagementEnabled | Configuration\|Network IO Control | This property shows the status of Network IO Control. Enabled means control over Network IO (using shares, limitations and reservation) are in place on each port group. |
| config\|name | Name | Name |

**Table 1637: Capability Properties Collected for VMware Distributed Virtual Switch Objects**

| Property Key | Property Name | Description |
|---|---|---|
| capability\|nicTeamingPolicy | NIC Teaming Policy | NIC Teaming Policy |

## Distributed Virtual Port Group Properties

VCF Operations collects configuration and summary properties for distributed virtual port group objects.

**Table 1638: Configuration Properties Collected for Distributed Virtual Port Group Objects**

| Property Key | Property Name | Description |
|---|---|---|
| config\|portBinding | Configuration\|Port Binding | This property shows how ports are assigned to virtual machines connected to this distributed port group. Available values: earlyBinding, ephemeral, lateBinding. |
| config\|portAllocation | Configuration\|Port Allocation | This property shows the type of port allocation, such as elastic or fixed. Options are: true=Elastic, false=Fixed. |
| config\|name | Name | Name |
| Configuration\|Uplink | Uplink | Indicates whether the portgroup is uplink portgroup. |

**Table 1639: Summary Properties Collected for Distributed Virtual Port Group Objects**

| Property Key | Property Name | Description |
|---|---|---|
| summary\|active_uplink_ports | Active DV uplinks | Active DV uplinks |

## Datastore Properties

VCF Operations collects configuration, summary, and properties about datastore use for datastore objects.

**Table 1640: Configuration Properties of Datastore and Datastore Cluster Objects**

| Property Key | Property Name | Description |
|---|---|---|
| config\|iormConfigStatus | Configuration\|Storage IO Control Status | Displays the status of Storage IO Control. If enabled control over Disk IO (using shares, limitations and reservation) is in place.Value: True or False. |
| summary\|total_number_datastores | Summary\|Total Number of Datastores | Displays the total number of member datastores in the cluster. |
| summary\|parentVcenter | Summary\|Parent vCenter | Displays the details of the parent vCenter. |
| summary\|parentDatacenter (GB) | Summary\|Parent Datacenter | Displays the details of the parent Datacenter. |

**Table 1641: Capacity Properties Collected for vSAN Datastore Objects**

| Property Key | Property Name | Description |
|---|---|---|
| Capacity\|Available Space (GB) | Available Space | Displays the available disk space in GB. |
| Capacity\|Provisioned (GB) | Provisioned (GB) | Displays the provisioned datastore size in GB. |
| Capacity\|Total Capacity (GB) | Total Capacity (GB) | Displays the total datastore capacity in GB. |
| Capacity\|Total Provisioned Consumer Space (GB) | Total Provisioned Consumer Space (GB) | Displays the total provisioned consumer space in GB. |
| Capacity\|Used Space (GB) | Used Space (GB) | Displays the used disk space in GB. |
| Capacity\|Used Space (%) | Used Space (%) | Displays the used disk space in percentage. |
| Capacity\|Usable Capacity (GB) | Usable Capacity (GB) | Displays the usable disk capacity in GB.<br>**Note:**<br>Earlier the vSAN Datastore base rate was calculated on the basis of Total Capacity of the disk, now the vSAN datastore base rate is calculated based on the usable capacity. |

**Table 1642: Summary Properties Collected for Datastore Objects**

| Property Key | Property Name | Description |
|---|---|---|
| summary\|vmfs_version | VMFS (Virtual Machine File System) Version | Displays the VMFS version number, contains both major version and minor version number.<br>**Note:**<br>The VMFS version property is visible, only when the datastore type is VMFS. |
| summary\|diskCapacity | Disk Capacity | Disk Capacity |
| summary\|isLocal | Is Local | Is local datastore |

| Property Key | Property Name | Description |
|---|---|---|
| summary\|customTag\|customTagValue | Value | Custom Tag Value |
| summary\|accessible | Datastore Accessible | Datastore Accessible |
| summary\|path | Summary\|Path | |
| summary\|scsiAdapterType | Summary\|SCSI Adapter Type | This property is disabled by default. |
| summary\|aliasOf | Summary\|Alias Of | Indicates whether the datastore is an alias of another. The published value is the container ID of the datastore for which it is an alias.<br>**Note:**<br>This property may have 2 values. It's either "none", that means the datastore is not an alias of another datastore, or datastore <containerID> that is the Container ID of the datastore for which this is an alias. |

**Table 1643: Datastore Properties Collected for Datastore Objects**

| Property Key | Property Name | Description |
|---|---|---|
| datastore\|hostcount | Host Count | Host Count |
| datastore\|hostScsiDiskPartition | Host SCSI Disk Partition | Host SCSI Disk Partition |
| * datastore\|maxObservedNumberRead | Datastore I/O\|Highest Observed Number of Read Requests | Disabled |
| * datastore\|maxObservedNumberWrite | Datastore I/O\|Highest Observed Number of Write Requests | Disabled |
| * datastore\|maxObservedOIO | Datastore I/O\|Highest Observed Outstanding Requests | Disabled |
| * datastore\|maxObservedRead | Datastore I/O\|Highest Observed Read Latency | Disabled |
| * datastore\|maxObservedReadLatency | Datastore I/O\|Highest Observed Read Latency | Disabled |
| * datastore\|maxObservedWrite | Datastore I/O\|Highest Observed Write Latency | Disabled |
| * datastore\|maxObservedWriteLatency | Datastore I/O\|Highest Observed Write Latency | Disabled |

**Table 1644: Datastore Properties Collected for vVol Datastore Objects**

| Property Key | Property Name | Description |
|---|---|---|
| storageArray\|modelId | Storage Array\|Model | Storage array model of VVol datastore.<br>**Note:**<br>This property is published for vVol datastores only and is available starting from vCenter version 6.0. |
| storageArray\|name | Storage Array\|Name | Storage array name of vVol datastore.<br>**Note:**<br>This property is published for vVol datastores only and is available starting from vCenter version 6.0. |
| storageArray\|id | Storage Array\|ID | Storage array ID of vVol datastore.<br>**Note:**<br>This property is published for vVol datastores only and is available starting from vCenter version 6.0. |
| storageArray\|vendorId | Storage Array\|Vendor | Storage array vendor of vVol datastore.<br>**Note:**<br>This property is published for vVol datastores only and is available starting from vCenter version 6.0. |
| protocolEndpoints\|name | Protocol Endpoints\|Name | Protocol endpoint's name of vVol datastore.<br>**Note:**<br>This is an instanced property that is published per protocol endpoint instance (e. g. eui.3362663138636633) for vVol datastores only. It is available starting from vCenter version 6.0. |
| protocolEndpoints\|type | Protocol Endpoints\|Type | Protocol endpoint's type of vVol datastore.<br>**Note:**<br>This is an instanced property that is published per protocol endpoint instance (e. g. eui.3362663138636633) for vVol datastores only. It is available starting from vCenter version 6.5. |
| protocolEndpoints\|hosts | Protocol Endpoints\|Hosts | Hosts associated with protocol endpoint of vVol datasore.<br>**Note:**<br>This is an instanced property that is published per protocol endpoint instance (e. g. eui.3362663138636633) for vVol datastores only. It is available starting from vCenter version 6.0. |

Datastore properties marked with an asterisk (*) have been disabled in this version of VCF Operations. This means that they do not collect data by default.

## vSphere Pod Properties

VCF Operations collects summary and event properties for vSphere Pods.

**Table 1645: Summary Properties Collected for vSphere Pod Objects**

| Property Key | Localized Name | Description |
|---|---|---|
| config\|name | Configuration\|Name | Resource name. |
| config\|guestFullName | Configuration\|Guest OS from vCenter | This is the value provided by vCenter. vCenter set it during VM creation. The value may not match the value inside the Guest. |
| config\|version | Configuration\|Version | Virtual Machine Version. |
| config\|createDate | Configuration\|Creation Date | Object Creation Date. |
| config\|numVMDKs | Configuration\|Number of Virtual Disks | Number of Virtual Disks. |
| config\|faultTolerant | Configuration\|Fault Tolerant | Fault tolerance enabled. |
| config\|ft_role | Configuration\|FT Role | Role of the VM in Fault Tolerance Group. |
| config\|ft_peer_vm | Configuration\|FT Peer VM | Peer of the VM in Fault Tolerance Group. |
| config\|hardware\|numCpu | Configuration\|Hardware\|Number of virtual CPUs | Number of virtual CPUs. |
| config\|hardware\|memoryKB | Configuration\|Hardware\|Memory | Memory. |
| config\|hardware\|thinEnabled | Configuration\|Hardware\|Thin Provisioned Disk | Thin Provisioned Disk. |
| config\|hardware\|numCoresPerSocket | Configuration\|Hardware\|Number of CPU cores per socket | Number of CPU cores per virtual socket. |
| config\|hardware\|numSockets | Configuration\|Hardware\|Number of virtual sockets | Number of virtual sockets. |
| config\|hardware\|diskSpace | Configuration\|Hardware\|Disk Space | Disk space metrics. |
| config\|cpuAllocation\|reservation | Configuration\|CPU Resource Allocation\|Reservation | N/A |
| config\|cpuAllocation\|limit | Configuration\|CPU Resource Allocation\|Limit | |
| config\|cpuAllocation\|shares\|shares | Configuration\|CPU Resource Allocation\|Shares\|Shares | |
| config\|memoryAllocation\|reservation | Configuration\|Memory Resource Allocation\|Reservation | |
| config\|memoryAllocation\|limit | Configuration\|Memory Resource Allocation\|Limit | |
| config\|memoryAllocation\|shares\|shares | Configuration\|Memory Resource Allocation\|Shares\|Shares | |
| config\|extraConfig\|mem_hotadd | Configuration\|Extra Configuration\|Memory Hot Add | Memory Hot Add Configuration. |
| config\|extraConfig\|vcpu_hotadd | Configuration\|Extra Configuration\|vCPU Hot Add | vCPU Hot Add Configuration. |
| config\|extraConfig\|vcpu_hotremove | Configuration\|Extra Configuration\|vCPU Hot Remove | vCPU Hot Remove Configuration. |

| Property Key | Localized Name | Description |
|---|---|---|
| config\|extraConfig\|mem_tps_share | Configuration\|Extra Configuration\| VM MEM TPS | N/A |
| config\|security\|disable_autoinstall | Configuration\|Security\| Disable tools auto install (isolation.tools.autoInstall.disable) | |
| config\|security\|disable_console_copy | Configuration\|Security\|Disable console copy operations (isolation.tools.copy.disable) | |
| config\|security\|disable_console_dnd | Configuration\|Security\|Disable console drag and drop operations (isolation.tools.dnd.disable) | |
| config\|security\|enable_console_gui_options | Configuration\|Security\|Enable console GUI operations (isolation.tools.setGUIOptions.enable) | |
| config\|security\|disable_console_paste | Configuration\|Security\|Disable console paste operations (isolation.tools.paste.disable) | |
| config\|security\|disable_disk_shrinking_shrink | Configuration\|Security\| Disable virtual disk shrink (isolation.tools.diskShrink.disable) | |
| config\|security\|disable_disk_shrinking_wiper | Configuration\|Security\| Disable virtual disk wiper (isolation.tools.diskWiper.disable) | |
| config\|security\|disable_hgfs | Configuration\|Security\| Disable HGFS file transfers (isolation.tools.hgfsServerSet.disable) | |
| config\|security\|disable_independent_nonpersistent | Configuration\|Security\|Avoid using independent nonpersistent disks (scsiX:Y.mode) | |
| config\|security\|enable_intervm_vmci | Configuration\|Security\| Enable VM-to-VM communication through VMCI (vmci0.unrestricted) | |
| config\|security\|enable_logging | Configuration\|Security\|Enable VM logging (logging) | |
| config\|security\|disable_monitor_control | Configuration\|Security\| Disable VM Monitor Control (isolation.monitor.control.disable) | |
| config\|security\|enable_non_essential_3D_features | Configuration\|Security\|Enable 3D features on Server and desktop virtual machines (mks.enable3d) | |
| config\|security\|disable_unexposed_features_autologon | Configuration\|Security\|Disable unexposed features - autologon (isolation.tools.ghi.autologon.disable) | |
| config\|security\|disable_unexposed_features_biosbbs | Configuration\|Security\|Disable unexposed features - biosbbs (isolation.bios.bbs.disable) | |

| Property Key | Localized Name | Description |
|---|---|---|
| config\|security\|disable_unexposed_features_getcreds | Configuration\|Security\|Disable unexposed features - getcreds (isolation.tools.getCreds.disable) | |
| config\|security\|disable_unexposed_features_launchmenu | Configuration\|Security\| Disable unexposed features - launchmenu (isolation.tools.ghi.launchmenu.change) | |
| config\|security\|disable_unexposed_features_memsfss | Configuration\|Security\|Disable unexposed features - memsfss (isolation.tools.memSchedFakeSampleStats.disable) | |
| config\|security\|disable_unexposed_features_protocolhandler | Configuration\|Security\| Disable unexposed features - protocolhandler (isolation.tools.ghi.protocolhandler.info.disable) | |
| config\|security\|disable_unexposed_features_shellaction | Configuration\|Security\|Disable unexposed features - shellaction (isolation.ghi.host.shellAction.disable) | |
| config\|security\|disable_unexposed_features_toporequest | Configuration\|Security\|Disable unexposed features - toporequest (isolation.tools.dispTopoRequest.disable) | |
| config\|security\|disable_unexposed_features_trashfolderstate | Configuration\|Security\| Disable unexposed features - trashfolderstate (isolation.tools.trashFolderState.disable) | |
| config\|security\|disable_unexposed_features_trayicon | Configuration\|Security\|Disable unexposed features - trayicon (isolation.tools.ghi.trayicon.disable) | |
| config\|security\|disable_unexposed_features_unity | Configuration\|Security\|Disable unexposed features - unity (isolation.tools.unity.disable) | |
| config\|security\|disable_unexposed_features_unity_interlock | Configuration\|Security\| Disable unexposed features - unity-interlock (isolation.tools.unityInterlockOperation.disable) | |
| config\|security\|disable_unexposed_features_unity_taskbar | Configuration\|Security\| Disable unexposed features - unity-taskbar (isolation.tools.unity.taskbar.disable) | |
| config\|security\|disable_unexposed_features_unity_unityactive | Configuration\|Security\| Disable unexposed features - unity-unityactive (isolation.tools.unityActive.disable) | |
| config\|security\|disable_unexposed_features_unity_windowcontents | Configuration\|Security\| Disable unexposed features - unity-windowcontents (isolation.tools.unity.windowContents.disable) | |
| config\|security\|disable_unexposed_features_unitypush | Configuration\|Security\|Disable unexposed features - unitypush (isolation.tools.unity.push.update.disable) | |

| Property Key | Localized Name | Description |
|---|---|---|
| config\|security\|disable_unexposed_features_versionget | Configuration\|Security\|Disable unexposed features - versionget (isolation.tools.vmxDnDVersionGet.disable) | |
| config\|security\|disable_unexposed_features_versionset | Configuration\|Security\|Disable unexposed features - versionset (solation.tools.guestDnDVersionSet.disable) | |
| config\|security\|disable_vix_messages | Configuration\|Security\|Disable VIX messages from the VM (isolation.tools.vixMessage.disable) | |
| config\|security\|enable_vga_only_mode | Configuration\|Security\|Disable all but VGA mode on virtual machines (svga.vgaOnly) | |
| config\|security\|limit_console_connection | Configuration\|Security\|Limit number of console connections (RemoteDisplay.maxConnection) | |
| config\|security\|limit_log_number | Configuration\|Security\|Limit number of log files (log.keepOld) | |
| config\|security\|limit_log_size | Configuration\|Security\|Limit log file size (log.rotateSize) | |
| config\|security\|limit_setinfo_size | Configuration\|Security\|Limit VMX file size (tools.setInfo.sizeLimit) | |
| config\|security\|enable_console_VNC | Configuration\|Security\| Enable access to VM console via VNC protocol (RemoteDisplay.vnc.enabled) | |
| config\|security\|disable_device_interaction_connect | Configuration\|Security\| Disable unauthorized removal, connection of devices (isolation.device.connectable.disable) | |
| config\|security\|disable_device_interaction_edit | Configuration\|Security\| Disable unauthorized modification of devices (isolation.device.edit.disable) | |
| config\|security\|enable_host_info | Configuration\|Security\|Enable send host information to guests (tools.guestlib.enableHostInfo) | |
| config\|security\|network_filter_enable | Configuration\|Security\| Enable dvfilter network APIs (ethernetX.filterY.name) | |
| config\|security\|vmsafe_cpumem_agentaddress | Configuration\|Security\|VMsafe CPU/memory APIs - IP address (vmsafe.agentAddress) | |
| config\|security\|vmsafe_cpumem_agentport | Configuration\|Security\|VMsafe CPU/memory APIs - port number (vmsafe.agentPort) | |
| config\|security\|vmsafe_cpumem_enable | Configuration\|Security\|Enable VMsafe CPU/memory APIs (vmsafe.enable) | |

| Property Key | Localized Name | Description |
|---|---|---|
| config\|security\|disconnect_devices_floppy | Configuration\|Security\|Disconnect floppy drive | |
| config\|security\|disconnect_devices_cd | Configuration\|Security\|Disconnect CD-ROM | |
| config\|security\|disconnect_devices_usb | Configuration\|Security\|Disconnect USB controller | |
| config\|security\|disconnect_devices_parallel | Configuration\|Security\|Disconnect parallel port | |
| config\|security\|disconnect_devices_serial | Configuration\|Security\|Disconnect serial port | |
| config\|security\|pci_device_configured | Configuration\|Security\|DCUI timeout | |
| runtime\|memoryCap | Runtime\|Memory Capacity | Memory Capacity. |
| cpu\|limit | CPU\|CPU Limit | CPU Limit. |
| cpu\|reservation | CPU\|CPU reservation | CPU Reservation. |
| cpu\|speed | CPU\|CPU | CPU Speed. |
| mem\|host_reservation | Memory\|Host Active | Machine Active. |
| mem\|host_active | Memory\|Host Usage | Machine Usage. |
| net\|mac_address | Network\|Mac Address | N/A |
| net\|ip_address | Network\|IP Address | |
| net\|subnet_mask | Network\|Subnet Mask | |
| net\|ipv6_address | Network\|IPv6 Address | IPv6 Address. |
| net\|ipv6_prefix_length | Network\|IPv6 Prefix Length | IPv6 Prefix Length. |
| net\|default_gateway | Network\|Default Gateway | N/A |
| net\|nvp_vm_uuid | Network\|NVP VM UUID | |
| net\|vnic_type | Network\|Virtual NIC Type | Virtual Machine's network adapter type. |
| net\|vnic_label | Network\|Label | Device label. |
| summary\|UUID | Summary\|UUID | Instance UUID in vCenter that uniquely identify all virtual machine instances. |
| summary\|MOID | Summary\|MOID | Managed object ID in vCenter. This is unique in scope of vCenter. |
| summary\|swapOnlyDatastore | Summary\|Datastore with only swap file | Datastore containing only the swap file and no other files from this VM. |
| summary\|customTag\|customTagValue | Summary\|Custom Tag\|Value | Custom Tag Value. |
| summary\|tag | Summary\|vSphere Tag | vSphere Tag Name. |
| summary\|tagJson | Summary\|vSphere Tag Json | vSphere Tag in Json format. |
| summary\|folder | Summary\|vSphere Folder | vSphere Folder Name. |
| summary\|parentCluster | Summary\|Parent Cluster | Parent Cluster. |
| summary\|parentHost | Summary\|Parent Host | Parent Host. |
| summary\|parentDatacenter | Summary\|Parent Datacenter | Parent Datacenter. |
| summary\|parentNamespace | Summary\|Parent Namespace | Parent Namespace. |
| summary\|parentVcenter | Summary\|Parent vCenter | Parent vCenter. |
| summary\|parentFolder | Summary\|Parent Folder | Parent Folder. |

| Property Key | Localized Name | Description |
|---|---|---|
| summary\|datastore | Summary\|Datastore(s) | Datastore(s). |
| summary\|guest\|fullName | Summary\|Guest Operating System\|Guest OS from Tools | This is the value provided by VMware Tools. This value will differ to the value set in vCenter if the Guest OS was upgraded, or a different Guest OS was installed. |
| summary\|guest\|ipAddress | Summary\|Guest Operating System\|Guest OS IP Address | Guest OS IP Address. |
| summary\|guest\|hostName | Summary\|Guest Operating System\|Hostname | Hostname of the guest operating system, if known. |
| summary\|guest\|toolsRunningStatus | Summary\|Guest Operating System\|Tools Running Status | Guest Tools Running Status. |
| summary\|guest\|toolsVersionStatus2 | Summary\|Guest Operating System\|Tools Version Status | Guest Tools Version Status 2. |
| summary\|guest\|toolsVersion | Summary\|Guest Operating System\|Tools Version | VM tools version installed on guest OS. |
| summary\|guest\|vrealize_operations_agent_id | Summary\|Guest Operating System\|vRealize Operations Agent ID | An ID to identify a VM in Agent Adapter's world. |
| summary\|guest\|vrealize_operations_euc_agent_id | Summary\|Guest Operating System\|vRealize Operations Euc Agent ID | An ID to identify a VM in Agent Adapter's world. |
| summary\|config\|numEthernetCards | Summary\|Configuration\|Number of NICs | Number of NICs. |
| summary\|config\|productName | Summary\|Configuration\|Product Name | Product Name. |
| summary\|config\|appliance | Summary\|Configuration\|Appliance | Appliance. |
| summary\|runtime\|isIdle | Summary\|Runtime\|Idleness indicator | This property indicates whether the monitored instance is idle or not. |
| summary\|runtime\|powerState | Summary\|Runtime\|Power State | Power State. |
| summary\|runtime\|connectionState | Summary\|Runtime\|Connection State | Connection State. |
| summary\|smbiosUUID | SMBIOS UUID | System Management BIOS UUID of a virtual machine.<br>**Note:**<br>The SMBIOS UUID metric for vSphere Pod is disabled by default. You have to enable the metric at the policy level. |
| guestfilesystem\|capacity_property | Guest File System\|Guest File System Capacity Property | Total capacity of guest file system as a property. |
| guestfilesystem\|capacity_property_total | Guest File System\|Total Capacity Property | Total capacity of guest file system as a property. |
| virtualDisk\|datastore | Virtual Disk\|Datastore | Datastore. |
| virtualDisk\|configuredGB | Virtual Disk\|Configured | Virtual Disk configured disk space. |
| virtualDisk\|label | Virtual Disk\|Label | Device Label. |
| virtualDisk\|fileName | Virtual Disk\|File Name | Virtual Disk file name. |

| Property Key | Localized Name | Description |
|---|---|---|
| diskspace\|snapshot\|mor | Disk Space\|Snapshot\|Managed Object Reference | Managed Object Reference. |
| diskspace\|snapshot\|name | Disk Space\|Snapshot\|Name | Snapshot name. |
| diskspace\|snapshot\|numberOfDays | Disk Space\|Snapshot\|Number of Days Old | Number of days since snapshot creation. |
| diskspace\|snapshot\|snapshotAge | Disk Space\|Snapshot\|Age (Days) | Virtual Machine's topmost snapshot age in days. |
| diskspace\|snapshot\|creator | Disk Space\|Snapshot\|Creator | Creator. |
| diskspace\|snapshot\|description | Disk Space\|Snapshot\|Description | Snapshot description. |
| vsan\|policy\|compliance | vSAN\|VM Storage Policies\| Compliance | Compliance status of the VM storage object. |
| datastore\|maxObservedNumberRead | Datastore\|Highest Observed Number of Read Requests | Highest Observed Number of Read Requests. |
| datastore\|maxObservedRead | Datastore\|Highest Observed Read Rate | Highest Observed Read Rate (KBps). |
| datastore\|maxObservedNumberWrite | Datastore\|Highest Observed Number of Write Requests | Highest Observed Number of Write Requests. |
| datastore\|maxObservedWrite | Datastore\|Highest Observed Write Rate | Highest Observed Write Rate (KBps). |
| datastore\|maxObservedOIO | Datastore\|Highest Observed Outstanding Requests | Highest Observed Outstanding Requests. |

**Table 1646: Compliance Configuration Related Properties for vSphere Pod Objects**

| Property Key | Property Name | Description |
|---|---|---|
| config\|security\|disconnect_devices_virtualhd audiocard | Configuration\|Security\|Virtual HD Audio Card Disconnected | NIL |
| config\|security\|disconnect_devices_virtualah cicontroller | Configuration\|Security\|Virtual AHCI Controller Disconnected | NIL |
| config\|security\|disconnect_devices_virtualen soniq1371 | Configuration\|Security\|Virtual Ensoniq 1371 Disconnected | NIL |

## Namespace Properties

VCF Operations collects summary and event properties for Namespace.

**Table 1647: Summary Properties Collected for Namespace Objects**

| Property Key | Localized Name | Description |
|---|---|---|
| config\|name | Configuration\|Name | Resource name |
| config\|resourceLimits\|namespace\|cpu | Configuration\|Resource Limits\| Namespaces\|CPU | CPU |
| config\|resourceLimits\|namespace\|mem | Configuration\|Resource Limits\| Namespaces\|Memory | Memory |
| config\|resourceLimits\|namespace\|diskspace | Configuration\|Resource Limits\| Namespaces\|Disk Space | Disk space metrics |

| Property Key | Localized Name | Description |
|---|---|---|
| config\|resourceLimits\|containers\|cpu_request | Configuration\|Resource Limits\|Containers\|CPU Request | CPU Request Default |
| config\|resourceLimits\|containers\|cpu_limit | Configuration\|Resource Limits\|Containers\|CPU Limit | CPU Limit Default |
| config\|resourceLimits\|containers\|mem_request | Configuration\|Resource Limits\|Containers\|Memory Request | Memory Request Default |
| config\|resourceLimits\|containers\|mem_limit | Configuration\|Resource Limits\|Containers\|Memory Limit | Memory Limit Default |
| config\|objectLimits\|compute\|pod_count | Configuration\|Object Limits\|Compute\|Pods | Number of Pods |
| config\|objectLimits\|compute\|deployment_count | Configuration\|Object Limits\|Compute\|Deployments | Deployments |
| config\|objectLimits\|compute\|job_count | Configuration\|Object Limits\|Compute\|Jobs | Jobs |
| config\|objectLimits\|compute\|daemon_sets | Configuration\|Object Limits\|Compute\|Daemon Sets | Daemon Sets |
| config\|objectLimits\|compute\|replica_sets | Configuration\|Object Limits\|Compute\|Replica Sets | Replica Sets |
| config\|objectLimits\|compute\|replication_controllers | Configuration\|Object Limits\|Compute\|Replication Controllers | Replication Controllers |
| config\|objectLimits\|compute\|stateful_sets | Configuration\|Object Limits\|Compute\|Stateful Sets | Stateful Sets |
| config\|objectLimits\|storage\|config_maps | Configuration\|Object Limits\|Storage\|Config Maps | Config Maps |
| config\|objectLimits\|storage\|secret_count | Configuration\|Object Limits\|Storage\|Secrets | Secrets |
| config\|objectLimits\|storage\|persistent_volume_claim | Configuration\|Object Limits\|Storage\|Persistent Volume Claim | Persistent Volume Claim |
| config\|objectLimits\|network\|services | Configuration\|Object Limits\|Network\|Services | Services |
| summary\|parentDatacenter | Summary\|Parent Datacenter | Parent Datacenter |
| summary\|parentCluster | Summary\|Parent Cluster | Parent Cluster |
| summary\|parentVcenter | Summary\|Parent vCenter | Parent vCenter |
| mem\|limit | Memory\|Memory limit | Memory limit |
| mem\|reservation | Memory\|Memory reservation | Memory reservation |
| mem\|expandable_reservation | Memory\|Memory expandable reservation | Memory Expandable Reservation |
| mem\|shares | Memory\|Memory Shares | Memory Shares |
| cpu\|limit | CPU\|CPU Limit | CPU Limit |
| cpu\|reservation | CPU\|CPU Reservation | CPU Reservation |
| cpu\|expandable_reservation | CPU\|CPU expandable reservation | CPU expandable Reservation |
| cpu\|shares | CPU\|CPU Shares | CPU Shares |
| cpu\|corecount_provisioned | CPU\|Provisioned vCPU(s) | Number of CPUs. It counts both the vSocket and vCore. A VM with 2 vSockets x 4 vCores each has 8 vCPU. |

## Tanzu Kubernetes cluster Properties

VCF Operations collects summary and event properties forTanzu Kubernetes clusters.

**Table 1648: Summary Properties Collected for Tanzu Kubernetes cluster Objects**

| Property Key | Localized Name | Description |
|---|---|---|
| config\|name | Configuration\|Name | Resource name |
| config\|cpuAllocation\|reservation | Configuration\|CPU Resource Allocation\|Reservation | N/A |
| config\|cpuAllocation\|limit | Configuration\|CPU Resource Allocation\|Limit | N/A |
| config\|cpuAllocation\|expandableReservation | Configuration\|CPU Resource Allocation\|Expandable Reservation | N/A |
| config\|cpuAllocation\|shares\|shares | Configuration\|CPU Resource Allocation\|Shares\|Shares | N/A |
| config\|memoryAllocation\|reservation | Configuration\|Memory Resource Allocation\|Reservation | N/A |
| config\|memoryAllocation\|limit | Configuration\|Memory Resource Allocation\|Limit | N/A |
| config\|memoryAllocation\|expandableReservation | Configuration\|Memory Resource Allocation\|Expandable Reservation | N/A |
| config\|memoryAllocation\|shares\|shares | Configuration\|Memory Resource Allocation\|Shares\|Shares | N/A |
| cpu\|limit | CPU\|CPU Limit | CPU Limit |
| cpu\|reservation | CPU\|CPU Reservation | CPU Reservation |
| cpu\|expandable_reservation | CPU\|CPU expandable reservation | CPU expandable Reservation |
| cpu\|shares | CPU\|CPU Shares | CPU Shares |
| cpu\|corecount_provisioned | CPU\|Provisioned vCPU(s) | Number of CPUs. It counts both the vSocket and vCore. A VM with 2 vSockets x 4 vCores each has 8 vCPU. |
| mem\|limit | Memory\|Memory limit | Memory limit |
| mem\|reservation | Memory\|Memory reservation | Memory reservation |
| mem\|expandable_reservation | Memory\|Memory expandable reservation | Memory Expandable Reservation |
| mem\|shares | Memory\|Memory Shares | Memory Shares |
| summary\|parentDatacenter | Summary\|Parent Datacenter | Parent Datacenter |
| summary\|parentNamespace | Summary\|Parent Namespace | Parent Namespace |

## All Folder Properties

VCF Operations collects configuration and summary properties for All Folders.

**Table 1649: Summary Properties Collected for All Folder Objects**

| Property Key | Property Name | Description |
|---|---|---|
| summary\|parentDatacenter | Summary\|Parent Datacenter | This property shows the details of the parent datacenter. |
| summary\|parentVcenter | Summary\|Parent vCenter | This property shows the details of the parent vCenter. |
| summary\|tag | Summary\|vSphere Tag | This property shows the details of the vSphere tag name. |

## Self-Monitoring Properties for VCF Operations

VCF Operations uses the VCF Operations adapter to collect properties that monitor its own objects. These self-monitoring properties are useful for monitoring changes within VCF Operations.

### Analytics Properties

VCF Operations collects properties for the VCF Operations analytics service.

**Table 1650: Properties Collected for Analytics Service Objects**

| Property Key | Property Name | Description |
|---|---|---|
| HAEnabled | HA Enabled | Indicates HA is enabled with a value of 1, disabled with a value of 0. |
| ControllerDBRole | Role | Indicates persistence service role for the controller: 0 – Primary, 1 – Replica, 4 – Client.. |
| ShardRedundancyLevel | Shard redundancy level | The target number of redundant copies for Object data. |
| LocatorCount | Locator Count | The number of configured locators in the system |
| ServersCount | Servers Count | The number of configured servers in the system |

### Node Properties

VCF Operations collects properties for the VCF Operations node objects.

**Table 1651: Configuration Properties Collected for Node Objects**

| Property Key | Property Name | Description |
|---|---|---|
| config\|numCpu | Number of CPU | Number of CPUs |
| config\|numCoresPerCpu | Number of cores per CPU | Number of cores per CPU |
| config\|coreFrequency | Core Frequency | Core Frequency |

**Table 1652: Memory Properties Collected for Node Objects**

| Property Key | Property Name | Description |
|---|---|---|
| mem\|RAM | System RAM | System RAM |

**Table 1653: Service Properties Collected for Node Objects**

| Property Key | Property Name | Description |
|---|---|---|
| service\|proc\|pid | Process ID | Process ID |

## OS and Application Monitoring Properties

Properties are collected for operating systems, application services, remote checks, Linux processes, and Windows services which can be used to create reports, views, and dashboards.

### Guest Information Properties

VCF Operations displays the following guest information properties for all objects created by the OS and Application Monitoring management pack.

- Guest Info
  - Hostname
  - IP
  - OS Name
  - OS Version
  - Telegraf Version

## Service Discovery Properties

VCF Operations displays object properties for service discovery.

### Service Discovery Adapter Instance Properties

VCF Operations displays the following properties for the service discovery adapter instance.

**Table 1654: Service Discovery Adapter Instance Properties**

| Property Name | Description |
|---|---|
| Action Identifier | An FQDN and IP pair of the end point vCenter Server that is used to identify the adapter instance that has to run actions on the vCenter. |

### Virtual Machine Properties

VCF Operations displays the following properties for virtual machines.

**Table 1655: Virtual Machine Properties**

| Property Name | Description |
|---|---|
| Guest OS Services\|Authentication Method | Refers to the VM guest operating system authentication method. The guest operating system can be authenticated either via a common user/password or a guest alias. |
| Guest OS Services\|Discovery Status | Reflects the result of service discovery operation on the VM's guest operating system. |
| Guest OS Services\|Authentication Status | Guest operating system authentication status. |
| Guest OS Services\|Inbound Ports | List of VM inbound ports. These are the ports on which the discovered services are listening. |

| Property Name | Description |
|---|---|
| SRM Info\|Protection Group | Protection group to which the VM belongs. |
| SRM Info\|Recovery Plans | List of recovery plans covering the VM. |

### Services Properties

VCF Operations displays the following properties for services.

**Table 1656: Services Properties**

| Property Name | Description |
|---|---|
| Type | The name of the service type. |
| Install Path | The install path. |
| Ports | List of service listening ports. |
| Virtual Machine | The name of the parent VM. |
| Virtual Machine MOID | The MOID of the VM. |
| Version | Version of the discovered service. |
| Is Application Member | Indicates that the service is a member of the group of services forming an application. |
| Category | Category of the service. |
| Connection Type | If there is a remote process that was connected to one of the listening ports of the given service, then the property's value is set to Incoming. If not, it is set to Outgoing. If there is no connection to another service, then the value of the property is set to N/A. |
| Has Dynamic Port | Indicates whether the service has dynamic ports or not. |
| Status | Indicates the status of the service.<br>**Up:** The service is running.<br>**Down:** The service is unavailable on the monitored VM.<br>**Unavailable:** The service is unavailable on a VM that is not being monitored.<br>**None:** The service is not available within 7 days. |

## Properties for vSAN

VCF Operations displays object properties for vSAN.

### Properties for vSAN Disk Groups

VCF Operations displays the following property for vSAN disk groups:

- vSAN Disk Groups: Configuration|vSAN Configuration
- vSAN Disk Groups: Configuration | Number of Disks

## Properties for vSAN Cluster

The VCF Operations displays the following properties for vSAN cluster.

| Property Name | Description |
|---|---|
| Configuration\|vSAN\|vSAN ESA | Indicates whether vSAN ESA configuration is enabled on the vSAN cluster. |
| Configuration\|vSAN\|Deduplication and Compression Enabled | Indicates whether deduplication and compression is eanbled on the vSAN cluster. |
| Configuration\|vSAN\|Preferred fault domain | Indicates whether the preferred fault domain is not set for the witness host in a vSAN Stretched cluster. |
| Configuration\|vSAN\|Stretched Cluster | Indicates whehter vSAN stretch cluster is enabled or not. |
| Configuration\|vSAN\|vSAN Configuration | Indicates whether the vSAN cluster is configured or not. |
| Configuration\|vSAN\|Encryption | Indicates whether the vSAN cluster is encypted or not. |
| Configuration \| vSAN \| File Service | Indicates whether vSAN File Services is enabled or not. |
| Configuration \| vSAN \| File Service Domain:<domainName> \| DNS Servers | Indicates the IP addresses of DNS servers, which are used to resolve the host names within the DNS domain. |
| Configuration \| vSAN \| File Service Domain:<domainName> \| DNS Suffixes | Indicates the list of DNS suffixes which can resolved by the DNS servers. |
| Configuration \| vSAN \| File Service Domain:<domainName> \| Gateway | Indicated the default gateway IP address for the file service access point. |
| Configuration \| vSAN \| File Service Domain:<domainName> \| Primary IP | Indicates the primary IP address for the file service. |
| Configuration \| vSAN \| File Service Domain:<domainName> \| Subnet Mask | Indicates the subnet mask for the vSAN cluster. |
| Summary \| Type | vSAN Cluster Type |
| Configuration \| vSAN \| File Service Domain:<domainName> \| IP Address :<ipaddress> \| FQDN | Indicates the Full Qualified Domain name (FQDN) to be used with IP address for the vSAN File Server instance. |

## Properties for vSAN Enabled Host

VCF Operations displays the following property for vSAN enabled host.

- Configuration|vSAN Enabled
- Configuration|vSAN|Encryption

## Properties for vSAN Cache Disk

VCF Operations displays the following properties for the vSAN cache disk.

Properties for vSAN include:

| Component | Metrics |
|---|---|
| Configuration | • Configuration Properties\|Name<br>• Configuration Properties\|Size<br>• Configuration Properties\|Vendor<br>• Configuration Properties\|Type<br>• Configuration Properties\|Queue Depth<br>• Configuration\|vSAN\|Encryption<br>• Configuration \| Model |
| SCSI SMART Statistics | • SCSI SMART Statistics\|Media Wearout Indicator Threshold<br>• SCSI SMART Statistics\|Write Error Count Threshold<br>• SCSI SMART Statistics\|Read Error Count Threshold<br>• SCSI SMART Statistics\|Reallocated Sector Count Threshold<br>• SCSI SMART Statistics\|Raw Read Error Rate Threshold<br>• SCSI SMART Statistics\|Drive Temperature Threshold<br>• SCSI SMART Statistics\|Drive Rated Max Temperature Threshold<br>• SCSI SMART Statistics\|Write Sectors TOT Count Threshold<br>• SCSI SMART Statistics\|Read Sectors TOT Count Threshold<br>• SCSI SMART Statistics\|Initial Bad Block Count Threshold |

### Properties for vSAN Capacity Disk

VCF Operations displays the following properties for the vSAN capacity disk.

Properties for vSAN include:

| Component | Metrics |
|---|---|
| Configuration | • Configuration Properties\|Name<br>• Configuration Properties\|Size<br>• Configuration Properties\|Vendor<br>• Configuration Properties\|Type<br>• Configuration Properties\|Queue Depth<br>• Configuration\|vSAN\|Encryption |
| SCSI SMART Statistics | • SCSI SMART Statistics\|Media Wearout Indicator Threshold<br>• SCSI SMART Statistics\|Write Error Count Threshold<br>• SCSI SMART Statistics\|Read Error Count Threshold<br>• SCSI SMART Statistics\|Reallocated Sector Count Threshold<br>• SCSI SMART Statistics\|Raw Read Error Rate Threshold<br>• SCSI SMART Statistics\|Drive Temperature Threshold<br>• SCSI SMART Statistics\|Drive Rated Max Temperature Threshold<br>• SCSI SMART Statistics\|Write Sectors TOT Count Threshold<br>• SCSI SMART Statistics\|Read Sectors TOT Count Threshold<br>• SCSI SMART Statistics\|Initial Bad Block Count Threshold |

### Properties for vSAN File Server

VCF Operations displays the following properties for vSAN file server.

- Configuration | vSAN | Primary
- Configuration | vSAN | FQDN

### Properties for vSAN File Share

VCF Operations displays the following properties for vSAN file share.

- Configuration |vSAN| Domain Name
- Configuration | vSAN| Hard Quota
- Configuration |vSAN| Soft Quota
- Configuration |vSAN | Label|<key>
- Configuration |vSAN | Access Point|<key>
- Configuration | vSAN | Permission:<permission> | Client IP Range
- Configuration | vSAN | Permission:<permission> | Root Squash

### Properties for vSAN Storage Pool

VCF Operations displays the following properties for vSAN Storage Pool.

| Property Name | Description |
|---|---|
| Configuration\|Number of Disks | Displays the total number of vSAN ESA disks in the storage pool. |

### Properties for vSAN ESA Disk

VCF Operations displays the following properties for the vSAN ESA Disk.

| Property Name | Description |
|---|---|
| Configuration\|Model | Displays the model number of the SCSI device. |
| Configuration\|Name | Displays the user configurable name for the SCSI device. |
| Configuration\|Queue Depth | Displays the queue depth of the SCSI device. |
| Configuration\|Size (GB) | Displays the size of SCSI device using the Logial Block Addressing Scheme (number of blocks) x (size of blocks). |
| Configuration\|Type | Displays the type of the SCSI device. |
| Configuration\|Vendor | Displays the vendor for the SCSI device. |
| Configuration \| vSAN \| Encryption | Indicates whether data encryption is enable on vSAN disk. If enabled all the VM data residing on the vSAN disk is encrypted. |

## Properties for Certificate Monitoring

VCF Operations displays the following certificate summary properties.

**Table 1657: Adapter Instance Certificate Summary Properties, published on Adapter Instance Object**

| Property Name | Property Key | Description |
|---|---|---|
| End Date | Certificate Summary:endpointIdentifier\|End Date | End date of the adapter certificate. |
| Start Date | Certificate Summary: endpointIdentifier \| Start Date | Start date of the adapter certificate. |
| Issuer DN | Certificate Summary:endpointIdentifier\| Issuer DN | Distinguished name of the adapter certificate issuer. |
| No. of days to expire | Certificate Summary:endpointIdentifier\|No. of days to expire | Number of days left before the expiration of the adapter certificate. |

**Table 1658: Authentication Source Certificate Summary Properties, published on Universe Object**

| Property Name | Property Key | Description |
|---|---|---|
| End Date | Certificate Summary\|Authentication Sources:authenticationSourceId\|End Date | End date of the authentication source certificate. |
| Start Date | Certificate Summary\|Authentication Sources: authenticationSourceId\|Start Date | Start date of the authentication source certificate. |
| Issuer DN | Certificate Summary\|Authentication Sources:authenticationSourceId\|Issuer DN | Distinguished name of the authentication source certificate issuer. |
| No. of days to expire | Certificate Summary\|Authentication Sources:authenticationSourceId\|No. of days to expire | Number of days left before the expiration of the authentication source certificate |

**Table 1659: Outbound Plugin Certificate Summary Properties, published on Universe Object**

| Property Name | Property Key | Description |
|---|---|---|
| End Date | Certificate Summary\|Outbound Plugins:outboundPluginId\|End Date | End date of the outbound plugin certificate |
| Start Date | Certificate Summary\|Outbound Plugins : outboundPluginId \|Start Date | Start date of the outbound plugin certificate |
| Issuer DN | Certificate Summary\| Outbound Plugins : outboundPluginId \|Issuer DN | Distinguished name of the outbound plugin certificate issuer. |
| No. of days to expire | Certificate Summary\| Outbound Plugins : outboundPluginId \|No. of days to expire | Number of days left before the expiration of the outbound plugin certificate |

## Properties for VCF Automation

VCF Operations displays properties for VCF Automation objects.

Some of the useful properties for project objects deployed through VCF Automation are as follows:

- Project|CustomProperties: Custom properties defined for the project.
- Project|OrganizationID: Organization ID of the project.
- Project|userEmail: Email address of the user for the project.

One of the useful properties for the deployment object is:

- Deployment|User: User associated with the deployment.

One of the useful properties for the cloud zone object is:

- CloudAutomation|ResourceTags: Resource tags associated with the cloud zone.

One of the useful properties for the blueprint object is:

- Blueprint|User: User associated with the blueprint.

One of the useful properties for the CASworkd object is:

- CASWorld|metering|MeteringPolicyId: Metering policy ID associated with the CAS World object.

One of the useful properties for the virtual machine object is:

- Cloud Automation|CustomProperties: Custom properties associated with the virtual machine.

One of the useful properties for Cloud Zone is:

- Cloud Automation|Resource Tags: Resources tags associated with the cloud automation.

## Properties in the NSX

VCF Operations displays the following properties for the NSX adapter.

**Table 1660: Properties in the NSX Adapter**

| Resource | Properties common in NSX and NSX on VMware Cloud on AWS | Properties in NSX on-premise | Properties NSX on VMware Cloud on AWS |
|---|---|---|---|
| Management Cluster | | • NSXT Product Version<br>• Status Summary\|Cluster Status\| Management Cluster Status<br>• Status Summary\|Cluster Status\| Controller Cluster Status<br>• Status Summary\|vIDM Connection Status<br>• Status Summary\|Compute Managers\| <ComputeManagerName>\|Status<br>• Configuration Maximums<br>– Compute Manager count<br>– Prepared vC Cluster count | |
| Firewall Section | Summary<br>• Create Time<br>• Create User<br>• Last Modified Time<br>• Last Modified User<br>• Protection<br>• Revision<br>• System Owned<br>Configuration<br>• Firewall Rule Count Size | Configuration<br>• Firewall Stateful | Configuration<br>• Type<br>• Domain id<br>• Precedence<br>• Category |

| Resource | Properties common in NSX and NSX on VMware Cloud on AWS | Properties in NSX on-premise | Properties NSX on VMware Cloud on AWS |
|---|---|---|---|
| Transport Node<br>**Note:** This object is specific to NSX on-premise and is not available in NSX on VMware Cloud on AWS. | | • Summary<br>– Create Time<br>– Create User<br>– Last Modified Time<br>– Last Modified User<br>– Protection<br>– Revision<br>– System Owned<br>– Summary\|FQDN<br>• Status Summary<br>– Transport Node State<br>– Transport Node Deployment State<br>– LCA Connectivity Status<br>– Management Plane Connectivity Status<br>– Host Node Deployment Status<br>– Management connection Status<br>– Controller connection Status<br>• Load Balancer Usage<br>– Current Small LB services<br>– Current Medium LB services<br>– Current Large LB services<br>– Current Extra Large LB services<br>– Current LB Pools<br>– Current LB Pool Members<br>– Current LB Virtual Servers<br>– Remaining Small LB services<br>– Remaining Medium LB services<br>– Remaining Large LB services<br>– Remaining Extra Large LB services<br>– Remaining LB Pool Members<br>• Tunnels\|<Tunnel-Name>\|Status<br>• File Systems\|<FileSystemMount><br>– Total<br>– Type<br>– File System ID | |
| Load Balancer Service<br>**Note:** This object is specific to NSX on-premise and is not available in NSX on VMware Cloud on AWS. | | • Summary<br>– Create Time<br>– Create User<br>– Last Modified Time<br>– Last Modified User<br>– Protection<br>– Revision<br>– System Owned<br>– LB Service Operational Status | |

| Resource | Properties common in NSX and NSX on VMware Cloud on AWS | Properties in NSX on-premise | Properties NSX on VMware Cloud on AWS |
|---|---|---|---|
| Load Balancer Virtual Server<br>**Note:** This object is specific to NSX on-premise and is not available in NSX on VMware Cloud on AWS. | | • Summary<br>– Create Time<br>– Create User<br>– Last Modified Time<br>– Last Modified User<br>– Protection<br>– Revision<br>– System Owned<br>– LB Virtual Operational State | |
| Load Balancer Pool<br>**Note:** This object is specific to NSX on-premise and is not available in NSX on VMware Cloud on AWS. | | • Summary<br>– Create Time<br>– Create User<br>– Last Modified Time<br>– Last Modified User<br>– Protection<br>– Revision<br>– System Owned<br>– Status | |
| Transport Zone<br>**Note:** This object is specific to NSX on-premise and is not available in NSX on VMware Cloud on AWS. | | Summary<br>• Create Time<br>• Create User<br>• Last Modified Time<br>• Last Modified User<br>• Protection<br>• Revision<br>• Switch Mode<br>• System Owned | |
| Logical Router | • Summary<br>– Create Time<br>– Create User<br>– Last Modified Time<br>– Last Modified User<br>– Protection<br>– Revision<br>– System Owned | • Configuration<br>– Failover Mode<br>– High Availability Mode<br>– Edge Cluster Id<br>– Router Type<br>• Services Enabled<br>– HA Status Per Transport Node\|<TransportNodeID>\|HA Status<br>– Firewall Enabled<br>– Load balancer Enabled<br>– DNS Enabled<br>– L2VPN Enabled<br>– IPSEC VPN Enabled | |

| Resource | Properties common in NSX and NSX on VMware Cloud on AWS | Properties in NSX on-premise | Properties NSX on VMware Cloud on AWS |
|---|---|---|---|
| Router Service | 1. Tier-0 Router Services ? BGP Service<br>— Summary\|BGP Neighbor Count<br>2. Tier-1 Router Services ? NAT Rules<br>— Summary\|NAT Rule Count<br>3. Tier-1 Router Services ? Static Routes<br>— Summary\|Static Route Count | • All logical routers ? Static Routes ? Summary\|Static Route Count<br>• All logical routers ? NAT Rule ? Summary\|NAT Rule Count<br>• Tier 0 ? BGP Service ? Summary<br>— ECMP Status<br>— Status<br>• Tier 0 ? BFD Service ? Summary<br>— Status<br>— BFD Neighbor Count<br>• Tier 0 ? Route Redistribution ? Summary<br>— Status<br>— Redistribution Rule count<br>• Tier 1 ? Route Advertisement ? Summary\|<br>— Route Advertisement Count<br>— Status | |
| Logical Switch | • Summary<br>— Create Time<br>— Create User<br>— Last Modified Time<br>— Last Modified User<br>— Protection<br>— Revision<br>— System Owned | • Summary<br>— Logical Switch State<br>• Configuration<br>— Replication Mode<br>— Admin State<br>— VNI | Configuration<br>• Type |
| Management Appliances<br>**Note:** This object is specific to NSX on-premise and is not available in NSX on VMware Cloud on AWS. | | NSXT API Version | |
| Manager Node<br>**Note:** This object is specific to NSX on-premise and is not available in NSX on VMware Cloud on AWS. | | • NSXT Manager Node Version<br>• Connectivity Status\|Management Plane Connectivity Status | |

| Resource | Properties common in NSX and NSX on VMware Cloud on AWS | Properties in NSX on-premise | Properties NSX on VMware Cloud on AWS |
|---|---|---|---|
| Group | Configuration Maximums\|Count<br>• IP Address Count<br>• Expressions Count<br>• vm Count | Configuration Maximums\|Count\|Tag Count | |
| Edge Cluster<br>**Note:** This object is specific to NSX on-premise and is not available in NSX on VMware Cloud on AWS. | | Summary<br>• Create Time<br>• Create User<br>• Last Modified Time<br>• Last Modified User<br>• Protection<br>• Revision<br>• System Owned<br>• Edge Cluster Member Type | |

## Placement Group Properties

The following properties are available for each Placement Group instance in your VCF Operations environment.

**Table 1661: Placement Group Properties**

| Service | Property |
|---|---|
| Placement Group | State |
| | Strategy |

## Properties for VeloCloud Gateway

VCF Operations displays properties of VeloCloud Gateway objects.

Some of the useful properties for VeloCloud Gateway are as follows:

- Summary | Core Count
- Summary | Gateway Activation Status
- Summary | Gateway Network Interface Errors
- Summary | Gateway Time Zone
- Summary | ICMP Status
- Summary | Is Eth0 DPDK Enabled
- Summary | is Eth1 DPDK Enabled
- Summary | Registration Status
- Summary | VCO IP
- Summary | Version

## Properties for VeloCloud Orchestrator

VCF Operations displays properties of VeloCloud Orchestrator objects.

Some of the useful properties for VeloCloud Orchestrator are as follows:

- General | DR SSH Tunnel Status

- General | Internet Connectivity
- General | IP Address
- General | NTP Time Zone

### Green Score Properties

In the computed metrics, constant values for carbon dioxide emissions and power consumption have been used.

The values are taken from the following links:

- $CO_2$ emission per kWh = 0.672 kg. Reference values from Greenhouse Gas Equivalencies Calculator
- Power consumption of a small server (1 socket, 10 cores, 32 GB RAM) = 0.1 kW (Assumption).

**Table 1662:**

| Property Name | Object Type | Type | Default Value |
|---|---|---|---|
| Electricity Rate | Cluster Compute Resource | Numeric | 0.108 |
| $CO_2$ Emission | Cluster Compute Resource | Numeric | 0.672 |

**Disclaimer:**

The conversion calculators and information shown in this documentation reference public information and have been provided to help translate abstract carbon emissions numbers into easily understandable terms solely for informational purposes. You should not rely on the calculators or information provided herein for any other purpose, including for any regulatory disclosure or diligence purposes. Broadcom may update, upgrade, revise, adjust or otherwise change the features and conversion methodologies at any time. Broadcom has not separately reviewed, approved, or endorsed the public information or third-party websites. No representation or warranty is made by Broadcom as to the accuracy, reasonableness or completeness of the calculations and related information herein.

### Properties for Policies

VCF Operations displays the following properties for the object type 'Policy'.

| Property Key | Property Name | Description |
|---|---|---|
| IsDefaultPolicy | Is Default Policy | This property indicates if a specific policy is the default policy. This property value is False for all policies except the default policy. |
| PolicyPriority | Policy Priority | This property indicates the priority of a specific policy. |

# Extending Monitoring Capabilities

## Installing and Configuring Management Packs in VMware Cloud Foundation Operations from the Solutions Catalog

Integrations in VMware Cloud Foundation Operations expand your ability to monitor and troubleshoot your software-defined data center and third-party products. You can install and configure management packs from the Solutions Catalog to connect physical and virtual layers, providing visibility into performance, capacity, and efficiency.Use these guides to install integrations in connected or disconnected environments. You can access configuration details for specific management packs, including AWS, Kubernetes, ServiceNow, and various databases. If you encounter problems, apply the troubleshooting steps to resolve adapter, installation, and data collection issues.

# Getting Started with Integrations for VMware Cloud Foundation Operations

Integrations in VMware Cloud Foundation (VCF) Operations provide an extensible capability for expanding the monitoring, troubleshooting, and restoring the functionality of your SDDC and other third-party solutions. Administrators can install and configure management packs in a VMware Cloud Foundation Operations instance.

VMware Cloud Foundation Operations natively provides an insight into the virtual layer of your data centers. The Integrations provide the flexibility and customization by allowing you to bring in data from different VMware systems and environments and see the performance, scalability, redundancy, and efficiency of your stack. To see the vertical in that stack, you can use management packs.

You can use management packs to relate the physical and application layer objects to the virtual components. This helps you prioritize visualization and alerts and view your environment in a holistic way. Installing management packs in VCF Operations enables you to have a logical and structured view of your environment where you can find data with ease. This allows for capacity planning, crisis management, cost calculations, and root cause analysis.

## VMware Cloud Foundation Operations for Integrations Index

You can access VCF Operations for Integrations User Guides and Release Notes by browsing through the list on this page. The User Guide provides information on how you can install, configure, and use the VCF Operations for Integrations. Release Notes provide details about new features, known issues, and resolved issues with every release.

### About VMware Cloud Foundation Operations for Integrations

Management packs in VMware Cloud Foundation Operations provide capability for expanding the monitoring, troubleshooting, and remediation functionality of SDDC and third party solutions. Administrators can install and configure management packs in a VMware Cloud Foundation Operations instance.

### Download and Installation

If VCF Operations is connected to Internet, you can install Integrations directly from within the product. Alternatively, you can also download Integrations for VCF Operations from the VCF Solutions Catalog.

### Release Notes

The VMware Cloud Foundation Operations for Integrations Release Notes provide details on what's new with every latest release.

## List of Management Packs

This page contains management packs which are currently supported by VCF Operations. You can also find the list of Knowledge base articles which provides details into Management Packs that have been announced End of Support by Broadcom.

### Supported Management Packs

| Management Pack for | User Guide |
|---|---|
| **Aggregator** | Management Pack for Aggregator |
| **Orchestrator** | Management Pack for Orchestrator |
| **HCX** | Management Pack for VCF Operations HCX |
| **Microsoft SQL Server** | Management Pack for Microsoft SQL Server |
| **MongoDB** | Management Pack for MongoDB |
| **MySQL** | Management Pack for MySQL |
| **Network Devices** | Management Pack for Network Devices |
| **Oracle Database** | Management Pack for Oracle Database |

| Management Pack for | User Guide |
|---|---|
| **PostgreSQL** | Management Pack for PostgreSQL |
| **ServiceNow** | Management Pack for ServiceNow |
| **Avi Load Balancer** | Management Pack for VMware Avi Load Balancer |
| **Telco Cloud Automation Platform** | Management Pack for VMware Telco Cloud Automation Platform |
| **vSphere Replication** | Management Pack for vSphere Replication |
| **SNMP** | Management Pack for SNMP |

**Important:**

**Mandatory Management Pack Upgrades in 9.1.1**

To enforce a stronger security posture, this release resolves critical vulnerabilities in third-party libraries. **All affected Management Packs must be upgraded immediately after product upgrade to ensure continued operations and compatibility.**

Affected Management Packs:

- VMware Cloud Foundation Management Pack for Microsoft SQL Server
- VMware Cloud Foundation Management Pack for MongoDB
- VMware Cloud Foundation Management Pack for MySQL
- VMware Cloud Foundation Management Pack for PostgreSQL
- VMware Cloud Foundation Management Pack for ServiceNow
- VMware Cloud Foundation Management Pack for Oracle Database
- VMware Cloud Foundation Management Pack for Network Devices
- SNMP Management Pack
- VCF Operations Management Pack for Orchestrator
- VMware Cloud Foundation Management Pack for Kubernetes
- VMware Cloud Foundation Management Pack for Aggregator
- VMware Cloud Foundation Management Pack for VCF Operations HCX

**Unsupported Management Packs**

See the management packs that have reached End of General Support.

- End of General Support (EOGS) for VCF Operations Management Packs, Effective September 30, 2025
- End of General Support (EOGS), for Aria Operations Management Pack for Horizon (MP4H), Effective March 28, 2025
- End of General Support (EOGS) for Aria Operations Management Packs, Effective October 1 2024
- End of General Support (EOGS) for Aria Operations Management Packs, Effective December 31, 2023

## Installing and Configuring Integrations in VMware Cloud Foundation Operations

Follow the respective procedure to install your management pack depending on whether your environment can connect to the Internet or not.

### Installing Integrations in a Connected Environment

Installing a Management Pack in a connected environment, can be done from VCF Operations. All integrations are available in Solutions Catalog. For information on compatibility between products, see Product Interoperability Matrix.

1. Log in to the VCF Operations user interface with administrator privileges.
2. In the menu, select **Administration** and in the left pane select **Integrations** > **Solutions Catalog**.
3. The **Available** tab shows all the Management Packs available to be downloaded and installed in VCF Operations.
4. Click **Download** on the specific Management Pack
5. Accept the EULA and download the Management Pack.
6. After successfull download, the Management Pack is available under the **Downloaded** tab.
7. Go to the **Downloaded** tab and click **Install** on the management pack tile.
8. Once the installation is complete, you can start using the management pack by adding Accounts.
9. In the menu, select **Administration** and in the left pane select **Integrations** > **Accounts**.
10. On the **Accounts** tab, click **Add Account**.
11. On the **Accounts Types** page, click the management pack you have installed.
12. Enter a display name and description for the cloud account.
    a) Enter the `Name` for the management pack instance as you want it to appear in VCF Operations.
    b) Enter the `Description` for the management packs with any additional information that helps you manage your instances.
13. Select the `Credential` you want to use to sign on to the environment from the drop-down menu. To add new credentials to access the management pack, click the plus sign and create a credential.
14. Enter `Credential/Group` to manage the management pack on VCF Operations. If you have only one account, select the Default collector group.
15. Enter the connection information as required to configure the instance of the management pack. Each management pack has unique settings, configure the settings as prompted to configure that instance of the management pack. For more information on individual management packs, see List of Management Packs.
16. Click **VALIDATE CONNECTION** to initiate the authentication request to the CSP authentication service.
17. Configure the Advanced Settings for the management pack.
18. Click **ADD**.

Verify that the adapter is configured and collecting data by viewing application-related data.

**Installing Integrations in a Disconnected Environment**
Integrations consist of a PAK file that contains default dashboards for the objects that the management pack identifies. After installation, configure the integration.

All integrations are available in VCF Solutions Catalog. For information on compatibility between products, see VMware Product Interoperability Matrix.

- Download the PAK file from VCF Solutions Catalog.
- Save that PAK file to a temporary folder on your local system.
- Before you install the PAK file or upgrade your VCF Operations instance, clone any customized content to preserve it. Customized content can include alert definitions, symptom definitions, recommendations, and views.

1. Log in to the VCF Operations user interface with administrator privileges.
2. In the menu, select **Administration** and in the left pane select **Integrations** > **Repository**.
3. On the **Repository** tab, click **Add**. The **Add Solution** wizard opens. The wizard includes three pages where you locate and upload a PAK file, accept the EULA and install, and review the installation.
4. Click **Browse** to locate the temporary folder and select the PAK file that you downloaded.
5. To upgrade to the latest version of the management pack:
   a) Select the **Install the PAK** file check box even if it is already installed to upgrade the management pack using the current PAK file even if an older version of the PAK file is already uploaded. Do not overwrite or update the solution alerts, symptoms, recommendations, and policies and leave all user customizations in place.
   b) Select the **Reset Default Content,** overwriting to a newer version provided by this update. User modifications to DEFAULT Alert Definitions, Symptoms, Recommendations, Policy Definitions, Views, Dashboards, Widgets, and Reports is overwritten. If you are installing a product software update, clone or backup the content before you proceed check box to upgrade the management pack using the current PAK file and overwrite the solution default alerts, symptoms, recommendations, and policies with the newer versions provided with the current PAK file.

      A reset overwrites customized content. Clone your customized content before the upgrade.
6. A warning, **The PAK file is unsigned** appears if the PAK file is not signed with a digital signature that VMware provides. The digital signature indicates the original developer or publisher and provides the authenticity of the management pack. If installing a PAK file from an untrusted source is a concern, check with the management pack distributor before proceeding with the installation.
7. Click **Upload**. The upload might take several minutes.
8. Read and accept the EULA, and click **Next**. Installation details appear in the window during the process.

   The installation starts only if all the nodes of the cluster are accessible.
9. Review the installation progress, including the VCF Operations nodes where the adapter was installed. When the installation is complete, click **Finish**.
10. In the menu, select **Administration** and in the left pane select **Integrations** > **Accounts**.
11. On the Accounts tab, click **Add Account**.
12. On the **Accounts Types** page, click the management pack that you have installed.
13. Enter a display name and description for the cloud account.
    a) Enter the `Name` for the management pack instance as you want it to appear in VCF Operations.
    b) Enter the `Description` for the management packs with any additional information that helps you manage your instances.
14. Select the Credential that you want to use to sign on to the environment from the drop-down menu. To add new credentials to access the management pack, click the plus sign and create a credential.
15. Enter `Collector/Group` to manage the management pack on VCF Operations. If you have only one account, select **Default collector group**.
16. Enter the connection information as required to configure the instance of the management pack. Each management pack has unique settings, configure the settings as prompted to configure that instance of the management pack. For more information on individual management packs, see List of Management Packs.
17. Click **VALIDATE CONNECTION** to initiate the authentication request to the CSP authentication service.
18. Configure the **Advanced Settings** for the management pack.
19. Click **ADD**.

Verify that the adapter is configured and collecting data by viewing application-related data.

# Troubleshooting

The troubleshooting section provides guidance on commonly performed activities to troubleshoot problems in VCF Operations Cloud. You can troubleshoot adapter, installation, data collection, cloud proxy and other issues related to VCF Operations Cloud.

## vSphere Management Pack Issues

You can troubleshoot general problems associated with vSphere and vCenter Management Packs, such as credential errors, connection errors, collector group errors that might occur when using VCF Operations Cloud.

### Wrong Credentials

You might be prompted with incorrect credentials error while accessing vCenter.

Solution: Ensure that the specified user credentials are correct and the user has required permissions to access the target vCenter.

> **Note:**
>
> The user credentials are case-sensitive. Ensure that you enter the correct values.

### No Collector Group Defined

Collector group is not defined for vCenter in VCF Operations Cloud.

Solution: In VCF Operations Cloud, deploy a Cloud Proxy and select it as the collector for the target vCenter. For details, refer to the Troubleshooting section in the *VCF Operations Cloud Getting Started Guide*.

### No Connection to Target vCenter

No connection is established between VCF Operations Cloud and vCenter.

Solution: Ensure that you validate the connection once you configure a Cloud account.

> **Note:**
>
> FQDN/IP of the target vCenter must be reachable from all the nodes be it VCF Operations on-prem or Cloud Proxy or VCF Operations Cloud.

### Plug-In not Registered

An error message indicates that the plug-in is not registered with vCenter.

Solution: In Management Pack configurations, navigate to **Manage Registrations**, then unregister and re-register the Management Pack.

### Management Pack Not Collecting Data

vCenter Management Pack might not be collecting all the required data, even though the connection between VCF Operations Cloud and Management Pack is valid.

Solution:

1. Verify minimum collection credentials. For more information, refer the Knowledge Base article.
2. Add `an account with administrative privileges` temporarily to verify the collection.

### Metrics Missing After vCenter Upgrade

You might notice that some of the metrics are missing, after you upgrade vCenter.

Solution: Please check vCenter documentation to verify if the statistic has moved to another level. If that is the case, you must increase the statistics level for said metric on vCenter.

## Management Pack Configuration Issues

You can troubleshoot Management Pack configuration issues like, connection errors, data collection errors, certificate errors and other issues which occur while using VCF Operations.

### Error When Validating Connection

When you try to validate a connection between VCF Operations Cloud and the Management Pack, an error message is displayed.

Solution:

1. Ensure port 443 is open between the collector and vCenter using telnet.
2. Ensure permissions are adequate. Use an account with administrator privileges.
3. Check `collector.log` file for test connection failures.
4. Use `openssl s_client -connect` to check whether the certificate is OK on vCenter.

### Test Connection Works but Collection Never Starts

You might notice that the test connection works but the Management Pack fails to collect the data.

Solution: Check the Management Pack logs for errors related to user permissions or vCenter session errors. Sometimes network or vCenter itself can have issues which prevents the Management Pack from starting.

### Test Connection Fails due to Multiple Certificates

You might observe that the test connection fails if there are multiple certificates.

Solution: Check for wild card certificates in **Administration** > **Certificates** page as VCF Operations Cloud will use these certificates as default.

### Data Collection is Inconsistent

You might observe that the Management Pack starts data collection, but the collected data is inconsistent. Confirm whether this happens to all vCenter instances or just one instance. If the issue occurs on all vCenter instances, the problem is with VCF Operations Cloud or network.

If the issue occurs only on one vCenter instance, check the following.

Solution:

1. Check vCenter query time at Management Pack level.
2. Check Tag query time to ensure tags are not causing delay.
3. Check Management Pack logs for hints as to why the collection might be slow.
4. Confirm latency between vCenter> Remote Collector / Cloud Proxy> VCF Operations Cloud is within supported limits.
5. Confirm vCenter itself is sized correctly and `/var/log/vmware/vapi` has no heap dumps on vCenter. Missing metrics.
6. Check the policy and ensure that the metric is enabled.
7. If looking for instanced metrics, ensure `instanced` is set to enabled for the metric in the policy.
8. Confirm advanced setting in the Management Pack has not disabled any collection type.
9. Ensure user has permissions.
10. Check vCenter MOB and confirm that the metric is available.

## General Management Pack Issues

You can troubleshoot generic Management Pack problems such as, connection errors, collector or collector group errors, certificate errors and other issues which occur while using VCF Operations Cloud.

#### Certificate Expired

VCF Operations Cloud might prompt you with a Certificate Expired or Renew Certificate message when you try to validate the connection.

Solution:

1. Validate the connection to import the new certificate.
2. You might have to remove the existing certificate.

#### Incorrect Credentials

You might be prompted with incorrect credentials error message when you validate the connection between VCF Operations Cloud and the Management Pack.

Solution: Acquire correct credentials for the endpoint.

#### Unreachable Error

Connection unreachable or connection timeout error is displayed when you try to validate the connection between VCF Operations and the Management Pack.

Solution: Validate the Operations/CP is responsive to ping and https requests through `ping <endpoint> and curl -v telnet://<endpoint>:443` to validate connectivity.

> **Note:**
>
> Most end points use port 443 for connection, for more information please refer VCF Operations Cloud documentation.

#### Collector or Collector Group Missing

The Management Pack might report collector or collector group missing error, if the collector is offline or the collector is not assigned to a Management Pack.

Solution: Verify that collector is online and is assigned to the Management Pack.

### Management Pack Collection Cycle Issues

You can troubleshoot vCenter Management Pack collection cycle issues such as high latency issue, end point not processing requests, and other issues which occur while using VCF Operations.

#### High Latency Issue

You might observe high latency between the collector and the end point.

Solution:

Add Remote Collector /Cloud Proxy on site where endpoint resides (Remote Collector max latency 200ms - Cloud Proxy max latency 500ms between Remote Collector /Cloud Proxy and analytics nodes). Ping can show latency between collector and endpoint. ~10-20ms max. This may depend on the amount of data that needs to be collected.

## Integrations Configuration Guides

### Management Pack for Aggregator

The *Management Pack for Aggregator* document provides information about the Aggregator feature that is introduced in VCF Operations.

## Intended Audience

This information is intended for anyone who wants to install the Aggregator. feature to manage distributed deployments of VCF Operations instances.

### Introduction

With Aggregator you can unify your multi-site VCF Operations deployment into a single pane of glass. This single pane of glass provides a holistic view of the child VCF Operations instance. You can instantiate a deployment of VCF Operations with the capability of receiving key metrics for specified objects from other VCF Operations deployment in your environment.

> **Important:**
>
> SDDC Health and Configuration Dashboard is not supported due to the EOGS of SDDC Health management pack and will be removed from the aggregator management pack in the future release.

The management pack allows you to unlock the following use cases:

- Provide a good set of Executive Dashboards for a worldwide company, without the need to have multiple browser tabs open for several VCF Operations instances.
- Provide a summary of performance, capacity, and configuration to Senior Executives and Virtual Infrastructure Administrators across all your vSphere environments.
- Provide a unified view of alerts triggered across the virtual environments into a single pane for making it easier for NOC or Helpdesk to initiate action.
- Ability to create a data warehouse where user selected set of metrics can be stored for data archiving and report use cases.
- Ability to provide summarized views of health and configuration of your Software Defined Data Center stack. This includes core applications such as VMware vCenter, VMware NSX, and VMware vSAN. The solution also covers the management applications such as VCF Operations, Log management, VCF Automation, VCF Business, and VMware Site Recovery Manager.

## Architecture

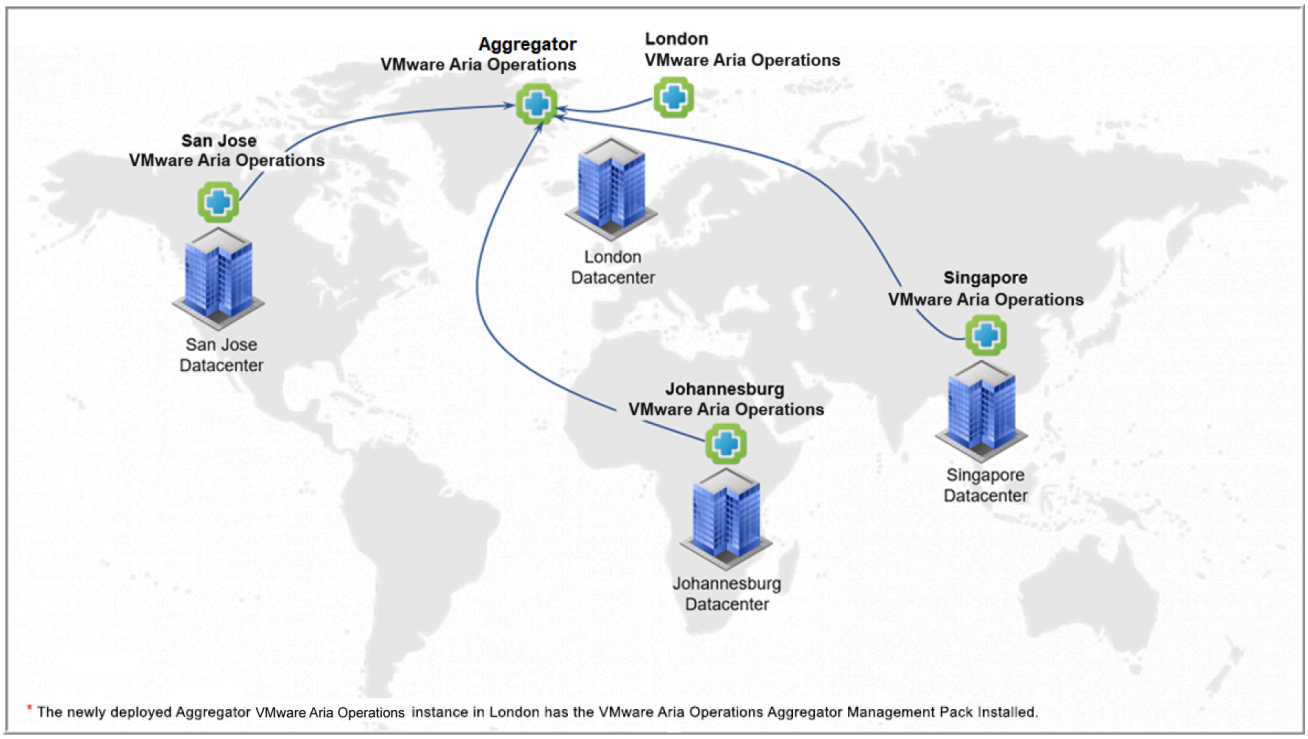
The Management Pack for Aggregator follows a simple server-client architecture. This management pack is installed on the VCF Operations instance, which you have identified as your unified single pane of glass instance. It is highly recommended that this instance is only used for Aggregator. After you deploy VCF Operations, you can use the management pack configuration page to add your existing VCF Operations instances to point to this Aggregator instance.

When configuring the management pack, enter the name of a configuration file that stores the information about the objects and metrics to be fetched from your existing VCF Operations instances into the newly deployed Aggregator instance.

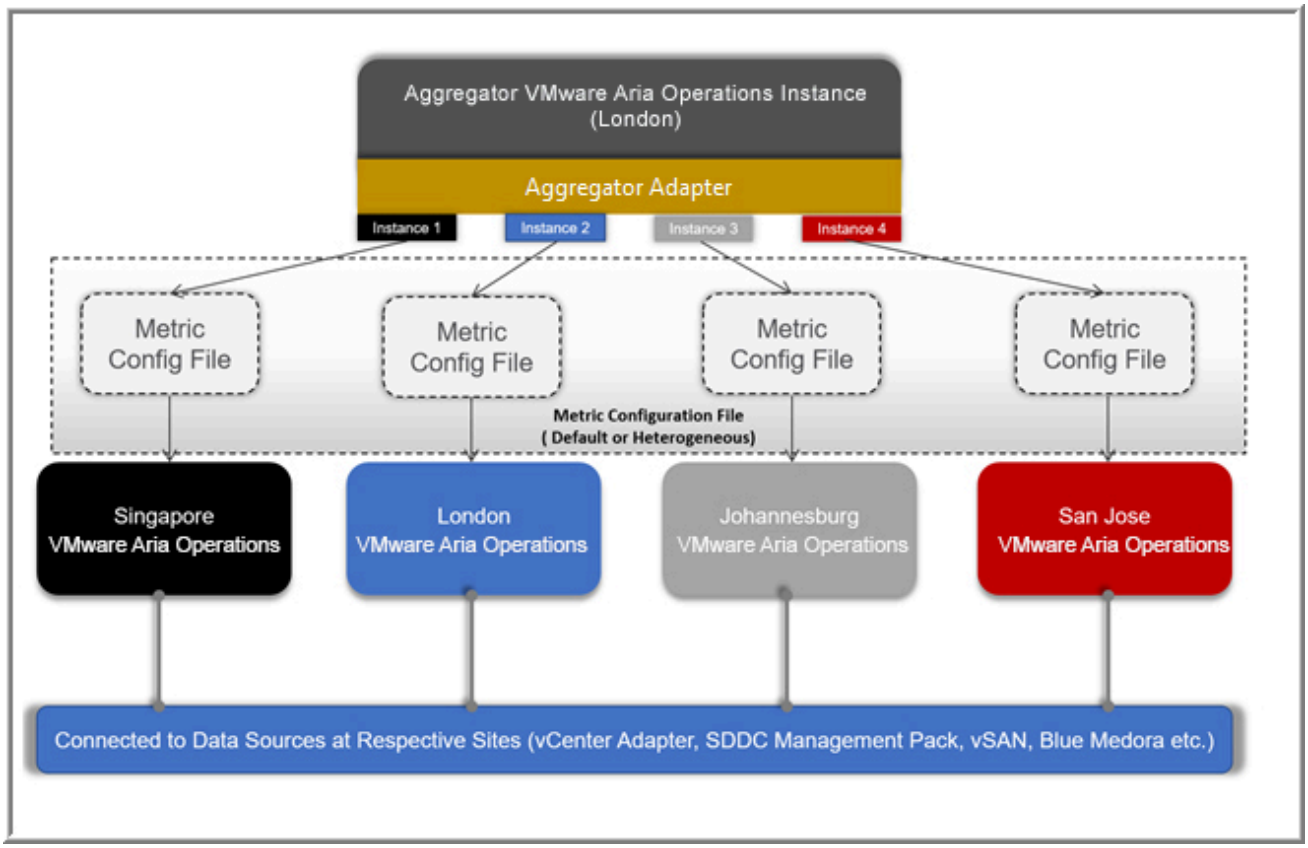
You can plan your architecture to apply the capabilities of Management Pack for Aggregator.

ACME Corp a Global Manufacturing company has virtual environments spanning across the globe to meet their regional business requirements. They are leveraging the VMware Software Defined data center to provide key IT services to line of business. They have an SDDC footprint in Singapore, London, Johannesburg, and San Jose. As a part of their Capacity Management and Performance Troubleshooting requirements they are leveraging VCF Operations to help their Operations and Procurement teams. They currently have an instance of VCF Operations deployed at each of these data center locations. ACME Corporation is looking to leverage the Management Pack for Aggregator to create a single pane of glass across all the data center locations. They want to leverage this single pane for their operations, capacity procurement, and senior executives to provide them with a unified view of their IT landscape. As their London data center has the largest footprint and low latency connectivity to other data centers, they want to host this unified instance in their London data center.

Based on the mentioned requirements, ACME Corp must deploy a new global standalone instance of VCF Operations with the Aggregator Management pack using the following deployment architecture:




Logical Architecture of the Management Pack for Aggregator for ACME Corp.




**Sizing and Collection Time Interval Considerations**
The VCF Operations instance hosting the Aggregator management pack should be sized just like any other VCF Operations instances.

For more information, see the KB article 324340. The following are the best practices.

- Out-of-the-box, Management Pack for Aggregator collects data for objects such as ESX hosts, clusters, datacenters, datastores, and so on. Virtual machine data is not collected. Based on this, most of the environments can be monitored with one Extra Large Node.
- The Aggregator instance of VCF Operations should be used only for collecting data from the Aggregator MP. No other MP on this VCF Operations instance should be configured to collect from any other data sources.
- If you add new objects and metrics to an existing deployment of your Aggregator VCF Operations instance, ensure to run the sizing calculator to meet the sizing requirements.
- With Management Pack for Aggregator, you can either collect data directly from child VCF Operations instances or use a remote collector to optimize the collection process. The default collection cycle of Management Pack for Aggregator is 5 minutes. This time interval works accurately for a centralized VCF Operations collecting data from other child VCF Operations instances that are located within a short distance. You can also use a remote collector for data collection. Otherwise, when centralized VCF Operations collects data directly without a remote collector and latency is long, you can set the collection interval to 15 minutes. This ensures data collection without any gaps.
- As most of the Aggregator use cases are met by collecting data for objects other than virtual machines from your environment, it is recommended to avoid the collection of virtual machine data as this drastically increases the sizing requirement for the Aggregator instance in a large-scale environment.

**Deploying the Standalone Global Instance**

Before you can configure Federation, you must deploy a dedicated VCF Operations instance to serve as the Global Ops Instance

- **Capacity:** Determine if you need a specific size. For most global views, a **Large** or **Extra Large** node is recommended to handle the object count from all child sites.
- **Network:** An IP address and DNS name (e.g., global-ops.corp.local).
- **Media:** The standard VCF Operations OVA file (same as used for Fleet Ops Instances).

This process converts a blank VCF Operations node into your Global Ops Instance.

1. Deploy the OVF/OVA
   a) Log in to your vSphere Client.
   b) Right-click the cluster/host and select **Deploy OVF Template.**
   c) Select the `VCF Operations .ova` file.
   d) **Select Deployment Configuration:** Choose **Large** or **Extra Large** (Federation nodes require significant memory to map objects from all sites).
   e) **Network Mapping:** Connect to a management network that can reach *all* regional VCF Operations instances.
   f) **Customize Template:**
      - Set the IP address, Gateway, DNS, and Timezone.
      - **Crucial:** Ensure the Timezone matches your Fleet Operations Instances (usually UTC is best practice for all).

g) Power on the appliance.

2. **Initial Cluster Setup**
   a) Navigate to https://<Global-IP>/admin in your browser.
   b) Select Add New Nodes"Master Node Configuration**:** You do not need to add Data Nodes or Remote Collectors immediately unless you have an immense environment (100k+ objects).
   c) **Set Admin Password:** Create a strong root and admin password.
   d) **Certificate:** Install a CA-signed certificate immediately to ensure secure HTTPS handshakes with Fleet Ops Instances
   e) **Finish:** Allow the cluster to come online (usually 10-20 minutes).
3. **"Empty State" Verification**
   a) **Do NOT** configure vCenter adapters on this instance.
   b) **Do NOT** install standard management packs (Storage, AWS, etc.).
   c) Why? The Global Ops Instance should be kept "clean." Its only job is to aggregate data from other Ops instances. If you start monitoring local vCenters with it, you will pollute the global view and waste resources.

**License your Global Operations Instance**
For VCF 9.0 and later, Broadcom introduced a centralized licensing model that replaces individual 25-character keys with a unified **License File**. This new workflow uses **VCF Operations** (formerly Aria Operations) as the "Licensing Anchor" for your environment

- **Version:** This procedure applies to **VCF 9.0+**. (VCF 5.x and older still use legacy SDDC Manager keys).
- **Account:** A Broadcom Support Portal account with valid VCF entitlement.
- **BSC Access:** Access to https://vcf.broadcom.com.

You must register your Global Operations instance with the **VCF Business Services Console (BSC)** to generate and apply this license file.

**Method A: Connected Mode (Recommended)**

*Best for environments where the Global Operations instance has direct Internet access.*

1. **Initiate Registration (Global Ops):**
   a) Log in to your Global Ops instance UI.
   b) Navigate to **Administration** > **ManagementLicensing**
   c) Select the **VCF Registration** tab (or "Subscription" depending on the patch level).
   d) Click **Register with BSC**.
   e) Enter your Broadcom Support Portal credentials when prompted.
2. **Assign Entitlements (BSC Portal):**
   a) The browser will redirect or open the **VCF Business Services Console** (vcf.broadcom.com).
   b) You will see your VCF Ops instance listed as "Pending" or "Registered.
   c) Select the instance and click **Manage Entitlements**.
   d) Select the License Entitlement. You must assign at least a 1-core license to the Global Operations Instance.
   e) Click **Save/Activate.**
3. Finalize: Return to the Global Operations UI.
   a) Click **Sync License.**
   b) Status should change to `Active.e`

**Method B: Disconnected Mode (Air-Gapped)**

*Best for high-security environments with no Internet access.*

1. Generate Registration File (On Global Ops)
   a) Log in to the Global Ops UI.
   b) Navigate to **Administration > Management > Licensing**.
   c) Select **Offline Registration**.
   d) Click **Generate Registration File**.
   e) Download the .json or .data file to your local computer.
2. Upload to BSC (On Internet-Connected PC)
   a) Log in to the **VCF Business Services Console** (vcf.broadcom.com).
   b) Go to **License Management > Offline Registration**.
   c) Click **Upload Registration File** and select the file you downloaded in Step 1.
   d) Once processed, select the Entitlements (Cores/TiBs) to assign to this file. You must assign at least a 1-core license.
   e) Click **Generate License File**.
   f) Download the resulting License File (e.g., license.enc or .lic).
3. Import License File (Back to Global Ops)
   a) Return to the Global Ops UI.
   b) In the Offline Registration window, click **Import License File**.
   c) Upload the file generated in Step 2.
   d) Click Activate.

**Install Management Pack for Aggregator**
The Management Pack for Aggregator requires a dedicated instance of VCF Operations on which this solution is installed. The dedicated instance applies the management pack to communicate to your other VCF Operations instances in your environment to collect key metrics.

- Download the Management Pack for Aggregator PAK file from VCF Solutions Catalog.
- Verify that you have at least one VCF Operations node deployed to install the Aggregator management pack.
- It is recommended to install the VMware SDDC Health Monitoring Solution on the child VCF Operations instances, to get an aggregated view of the health of all the SDDC core and management applications installed across your environment.
- Verify that each VCF Operations instance is communicating with the centralized instance and vice-versa.

  **Note:**
  The Aggregator adapter requires the same port to communicate as the web interface. There is no requirement of an additional open port.

1. Log in to the VCF Operations with administrator privileges.
2. In the menu, select **Administration** and in the left pane select **Solutions** > **Repository**.
3. On the **Repository** tab, click **Add/Upgrade**.
4. Browse to locate the temporary folder and select the PAK file.
5. Click **Upload.**
   The upload might take several minutes.
6. Read and accept the EULA, and click **Next**.
7. When the Aggregator management pack is installed, click **Finish**.

**Configure Management Pack for Aggregator**
You can configure adapter instances and add the VCF Operations instances you want to monitor.

- Verify that you have installed Management Pack for Aggregator.

1. On the menu, click **Administration** and in the left pane, click **Solutions** > **Other Accounts**.
2. Under Other Accounts, click **Add Account**.
3. On the Accounts Type page, click the **Aggregator Adapter**.
4. Enter the display name of the adapter.
5. Enter the source VCF Operations host name or IP address.

   **Note:**
   You can either use the host name of IP address of the primary node of your child VCF Operations deployment or use the Load Balancer Virtual IP or DNS Name. Use a short name and do not use FQDN.
6. Enter the **federation_config** in the **Configuration File Name**.

   `federation_config` is the default configuration file shipped with the management pack. For more information, see Configuration Configuration File in the Aggregator Management Pack.
7. Enter the Credential details.
8. Optionally, configure the **Advance Settings**.

   **Note:**

   - By default, the setting **Append 'vROps Host' to Resource name** is set to `True`. This setting appends the child VCF Operations IP name to the resource name. When the setting **Append 'vROps Host' to Resource name** is set to `False`, the suffix will not be appended to the resource name.
   - By default, the child VCF Operations Port is set to 443. The child VCF Operations is also used by the Aggregator adapter to retrieve resources and data.
9. Click **Add**.
10. Repeat the steps from step 1 through step 8 for multiple instances of VCF Operations.
11. You can view the collection status of all the multiple instances of VCF Operations under the solution page.

Configuration File in the Aggregator Management Pack

When you configure an Aggregator adapter instance, you must specify a config file. The config file specifies what kind of information an adapter instance must collect. Out-of-the-box, the Aggregator Management Pack is shipped with config file named as `federation_config`.

The config file retrieves the following information.

- Resources
- Metrics
- Properties

For example, if you want the adapter to collect information like host system and CPU metrics of that object, then this information has to be defined in the config file. When these parameters are added, then the adapter instance starts collecting information pertaining to the required specification. The changes are reflected in the config file only when the adapter instance is restarted.

With Management Pack for Aggregator, you can retrieve a subset of objects from an object type of a child VCF Operations. The adapter provides a mechanism to filter the subset by either using properties or tags.

The tag-based filter used is:

```
<Filter> <Tag category="SOME_CATEGORY" name="SOME_TAG" /> </Filter>
```

The property-based filter used is:

```
<Filter> <Property attrkey="SOME_KEY" value="SOME_VALUE"/> </Filter>
```

**Note:**
You can use only one filter for a given object type, otherwise, the configuration file validation fails.

To create filters for resource kinds, add the filtering element immediately after the resource kind element in the config file. For example:

```
<ResourceKind resourceKindKey="...">
        <Filter>
            <Property attrkey="..." value="..."/>
        </Filter>
        <Metric ..."/>
        <Property .../>
    </ResourceKind>
```

## Types of Config Files Used in the Management Pack for Aggregator

| | |
|---|---|
| **Default Config File** | Management Pack for Aggregator contains a default config file. The Management Pack for Aggregator dashboards are based on these config files. The default config file contains information of vCenter to host level and not VM level. The config file also provides information of vCenter operation adapter objects, VMware SDDC Health Monitoring Solution adapter, and VMware Site Recovery Manager adapter.<br><br>**Note:**<br>The default federation_config file is aligned with the latest VCF Operations. Ensure that the desired metrics/properties from the older versions (which no longer exist in the newer versions) are mentioned in the config file. |
| **Heterogeneous Config File** | You are not limited to using the same config file for all adapter instances when you are configuring multiple adapter instances for a separate child of VCF Operations instance. You can collect different data from different child VCF Operations instances. You can have different level of information in different config files. |

### Locate Configuration File

The default file has pre-defined objects and metrics, which are collected from the child VCF Operations instances.

Verify that you have installed the Management Pack for Aggregator on your environment.

1. Login to VCF Operations as an administrator.
2. In the menu, click **Administration**, and in the left pane click **Configuration** > **Metric Configurations.**
3. Under Metric Configurations, expand **ResskndMetric**.
   You can see the newly installed `federation_config` file in the list.
4. You can view the content of this file by selecting this file from the list.

### Best Practices for Configuration File Management

Follow certain best practices to ensure optimal performance of the Management Pack for Aggregator.

- Do not modify the out-of-the-box shipped configuration file. Create a file, copy the content of the out of the box file and make additions to this new file.
- Do not remove objects or metrics included in the out-of-the-box configuration file as it impacts the content of the out-of-the-box dashboards.
- Once you have created a configuration file, it is required to stop the collection on the Management Pack for Aggregator, change the configuration file name to reflect the newly created file and start the collection again. The Aggregator adapter will now start collecting the additional objects and metrics.
- Version control the configuration files with proper naming conventions and version numbers.
- Avoid adding objects and metrics which are not required to be collected.
- Test your newly created configuration file with a test instance of child VCF Operations instance before applying it to your production systems.
- The collection of Aggregated metrics from the child VCF Operations monitored by the Management Pack for Aggregator is not managed in policies. The addition and removal of metrics and properties is managed directly inside the configuration file. Only the metrics and properties configured in the federation_config file are collected. The collection of configured metrics and properties can be disabled and re-enabled from policy.

## Dashboards in the Management pack for Aggregator

Management pack for Aggregator caters to certain out-of- the-box use cases, which are delivered through dashboards. After the installation and configuration of the Cloud Federation Adapter management pack, you can access these out-of-the-box dashboards.

Verify that you have installed the Management pack for Aggregator.

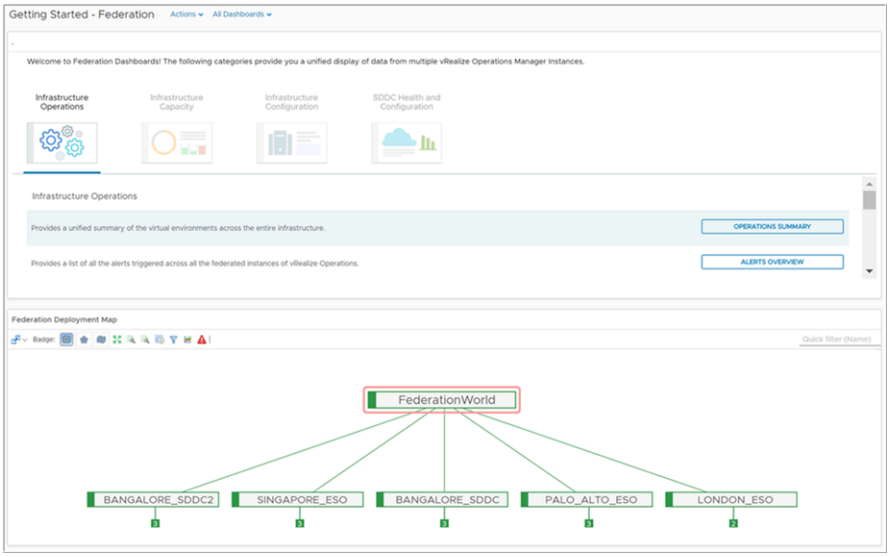
1. To access the dashboards, in the left pane of the VCF Operations, click **Dashboards**.
2. Click **Aggregator** to view all dashboards.
3. From the dashboard list, select **Getting Started- Aggregator** dashboard in the list of all the dashboards and associated use cases.

Getting Started - Aggregator Dashboard

The Getting Started - Aggregator dashboard displays the catalog of Aggregator dashboards and the Aggregator deployment map.

You can step through the multiple categories under Aggregator page to cater to specific use cases. The deployment map on this page provides a topology of your Aggregator Deployment with colors depicting the current health and collection status.

**Figure 475: Getting Started - Aggregator Dashboard**




## Infrastructure Operations

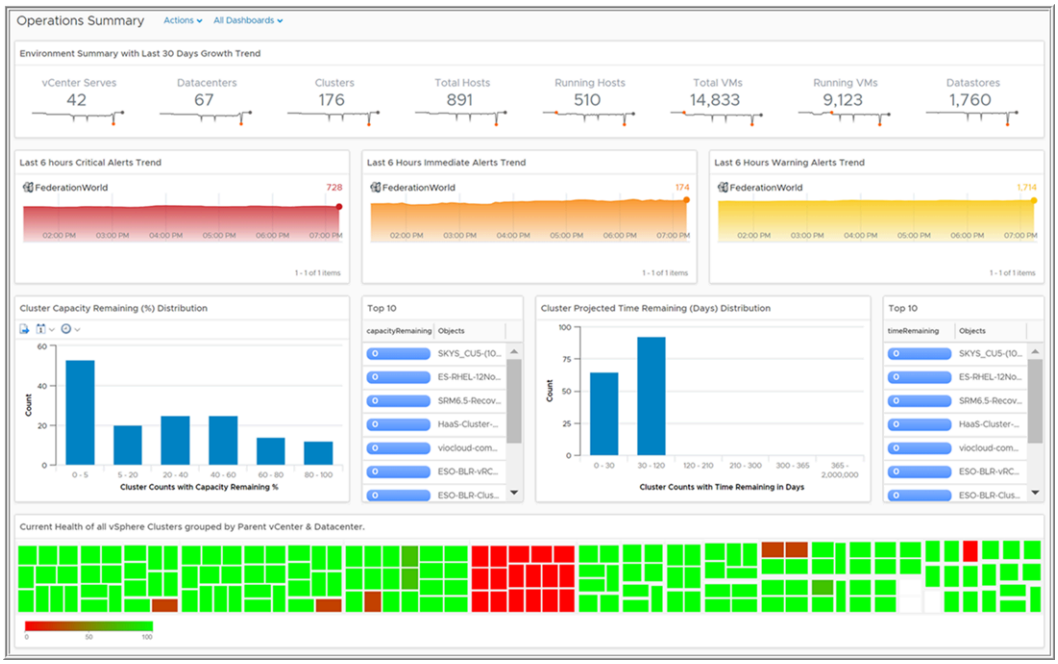
The Getting Started page provides access to the following OOTB categories. The infrastructure category caters to Senior Executives and NOC users by providing them an operations summary and alerts. The dashboards under this category are.

- Operations Summary
- Alerts Overview

Operations Summary Dashboard

The Operations Summary dashboard provides a count of your infrastructure inventory objects across all the VCF Operations instances being monitored by the Management Pack for Aggregator. It provides you the overall alert volume in these environment, along with a summarized view of health and capacity of the vSphere clusters in your environment.

**Figure 476: Operations Summary Dashboard**



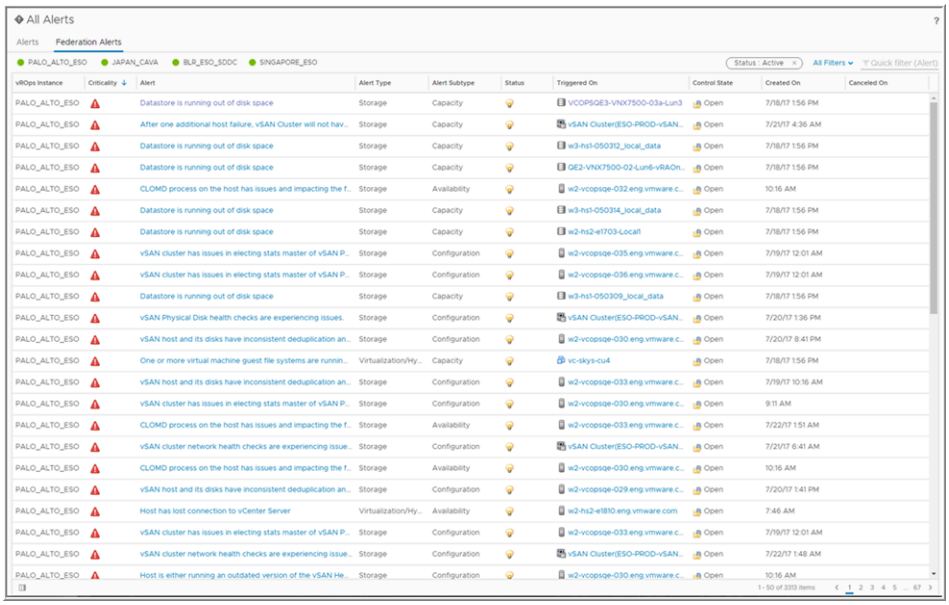

Alerts Overview

The Alerts Overview dashboard provides the alerts page of VCF Operations. On this page, you can not only view the alerts triggered on your VCF OperationsAggregator instance, but also view the alerts triggered in your child VCF Operations instances.

The overview page includes all the alerts triggered on the VCF Operations instance and is not dependent on the objects you are monitoring from the child VCF Operations instance.

This page can help your NOC teams to have a single pane for all the alerts across your environments. To view the details of a specific alert, click on that alert to get directed to the child VCF Operations instance where the alert was originally triggered for troubleshooting purposes.

**Figure 477: Alerts Overview**




Infrastructure Capacity

The Infrastructure Capacity category caters to Senior Executives and NOC users by providing them an overview of overall capacity and reclamation opportunities.

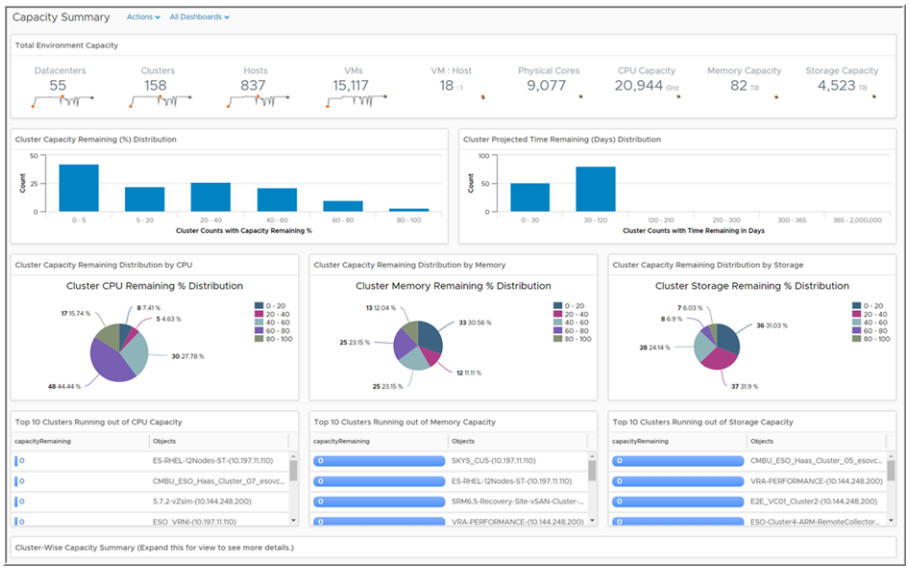
The following dashboards are under the Infrastructure Capacity:

- Capacity Summary
- Capacity Optimization

### Capacity Summary Dashboard

The Capacity Summary Dashboard provides you with a summary of the total physical capacity available across all your environments being monitored by multiple instances of VCF Operations. The dashboard also provides you a list view of all your clusters across your environment with details around inventory, capacity and utilization of those clusters. You can export this list into a CSV file for reporting purposes.

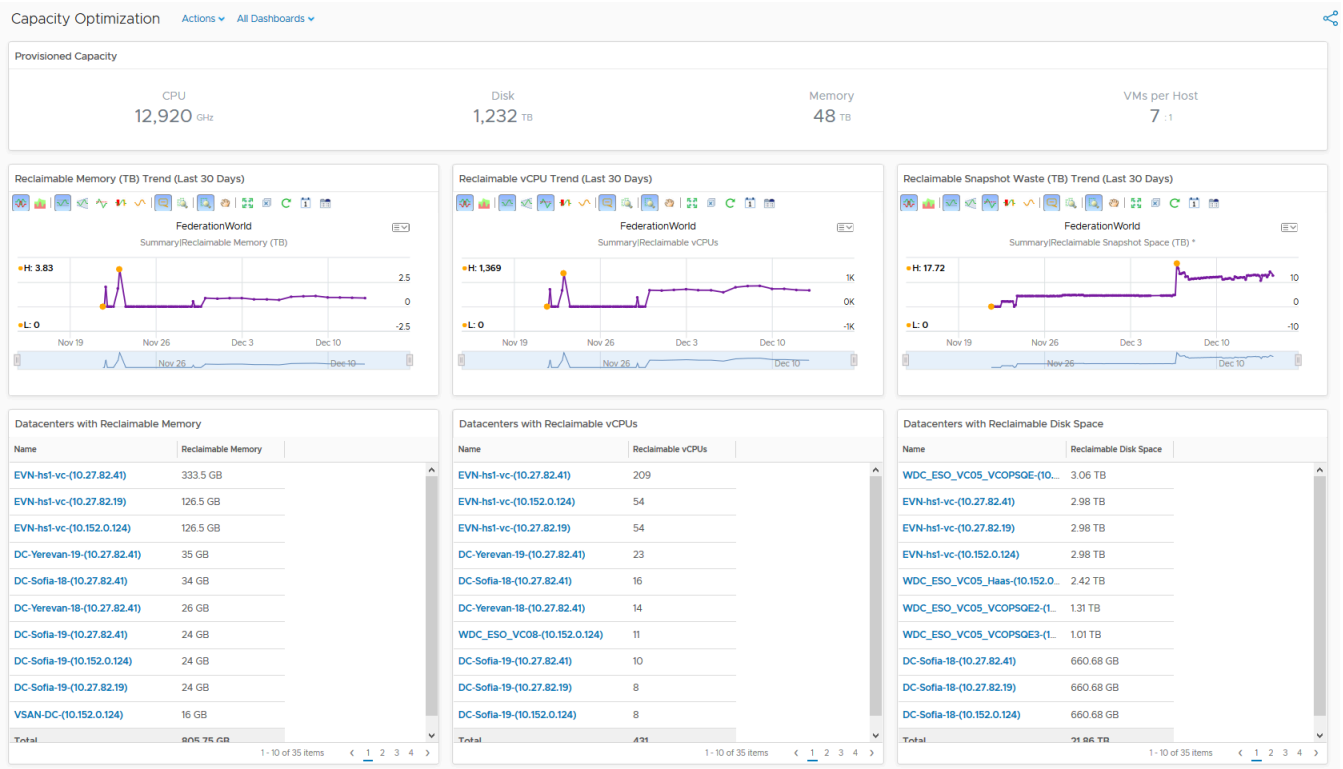
**Figure 478: Capacity Summary Dashboard**




### Capacity Optimization Dashboard

The Capacity Optimization dashboard provides you a quick view of resource optimization opportunities within your virtual infrastructure. This dashboard is focused on improving the efficiency of your multiple instances of VCF Operations environment by reducing the wastage of resources.

**Figure 479: Capacity Optimization Dashboard**




## Infrastructure Configuration

Infrastructure Configuration category caters to Senior Executives and virtual infrastructure admin by providing them the summary of ESX Host and cluster configurations.

The dashboards under Infrastructure Configuration category are:

- vSphere Cluster Configuration
- vSphere Host Configuration

## vSphere Cluster Configuration Dashboard

The vSphere Cluster Configuration dashboard provides a quick overview of your vSphere cluster configurations from various instances of VCF Operations. The dashboard highlights the areas that are important in delivering performance and availability to your virtual machines.

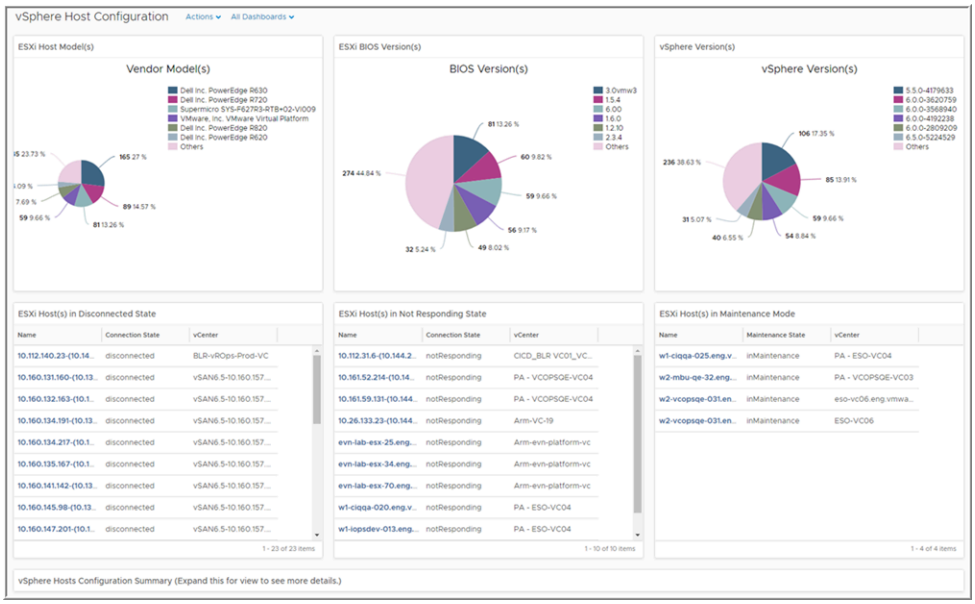
**Figure 480: vSphere Cluster Configuration Dashboard**

The dashboard also highlights if there are vSphere clusters which are not configured for DRS, High Availability (HA), or admission control to avoid any resource bottlenecks or availability issues when a host fails. The heat map in this dashboard helps you to identify if you have hosts where vMotion was not enabled as this may not allow the VMs to move from or to that host. This can cause potential performance issues for the VMs on that host if the host gets too busy.

### vSphere Host Configuration Dashboard

The vSphere Host Configuration dashboard provides an overview of your ESX host configurations, and displays inconsistencies so that you can take corrective action.

**Figure 481: vSphere Host Configuration Dashboard**




The dashboard also measures the ESX hosts against the vSphere best practices and indicates deviations that can impact the performance or availability of your virtual infrastructure. Although you can view this type of data in other dashboards, in this dashboard you can export the ESX configuration view and share it with other administrators.

### SDDC Health and Configuration

This category caters to Senior Executives and virtual infrastructure administrator by providing a summary of health and configuration of the SDDC Management Stack.

Health & Configuration dashboard is available under the SDDC Health and Configuration category.

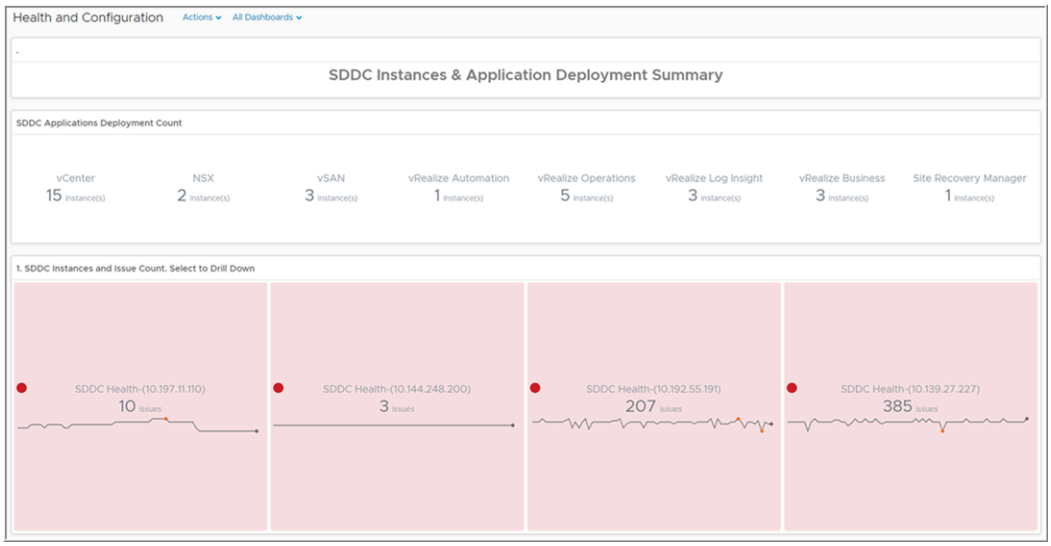
Health and Configuration Dashboard

This dashboard provides health of all the SDDC components. The Aggregator adapter monitors and displays the health of vCenter, VCF Operations health, VCF Automation and other stack components.

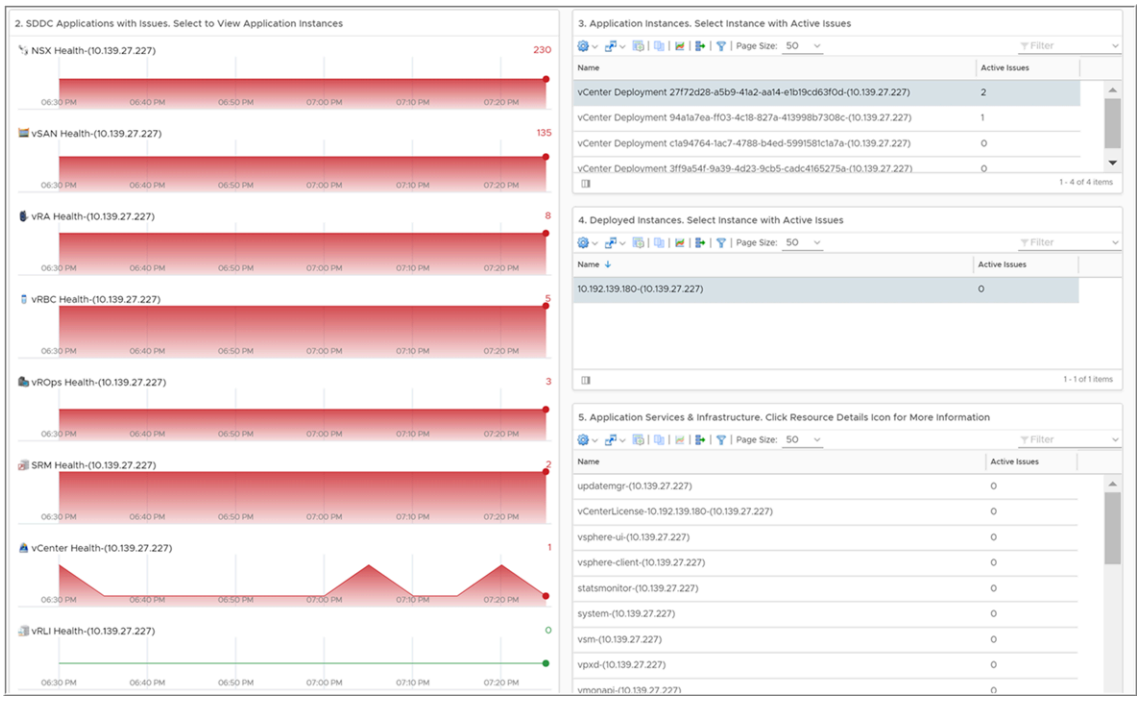
**Note:**

Verify to install and configure the SDDC health adapter installed on the child VCF operations instances to display the health and configuration information in this dashboard.

**Figure 482: Health and Configuration Dashboard**




This dashboard provides information of summary of the number of SDDC components, applications, deployments and its health status. The **SDDC instances and Issue count** widget in the dashboard displays detailed information of which object requires an immediate attention and from which instance the issue has occurred. When you click on the particular VCF Operations child instance, you can drill down to analyze the root cause of an issue.



### Customizing Icons in Management Pack for Aggregator

Every object or adapter in your environment has an icon representation. You can customize how the icon appears.

You can customize the appearance of the icons for your Aggregator objects.

#### Customize Icons

Every object or adapter in your environment has an icon representation. You can customize how the icon appears. You can use familiar or customized icons from vCenter, Management Pack for NSX for vSphere or VMware SDDC Health Monitoring Solution adapter for the Aggregator objects.

If you want to add new icons, then verify that you have an icon on your local machine.

1. In the menu, click **Administration**, and then in the left pane, click **Configuration** > **Icons**
2. Select the adapter type from the **Adapter type** drop-down.
3. Select the Object type and then click the **Assign Default Icons**
4. To assign default icons for a selected object type, click **Yes**.

## Management Pack for Orchestrator

The guide describes the steps to run Orchestrator workflows from VCF Operations.

The Management Pack for Orchestrator guide describes the process of deploying, configuring and managing the Management Pack for Orchestrator in your VCF Operations environment.

### Intended Audience

The information in this guide is intended for VCF Operations administrators.

### Introduction to Management Pack for Orchestrator

#### Introduction to the Orchestrator

You can run Orchestrator workflows from VCF Operations after you install and configure the Management Pack for Orchestrator. You can run workflows to remediate alerts in VCF Operations or automate tasks.

Management Pack for Orchestrator supports the following types of workflows:

- Custom vCenter package workflows.
- User defined vCenter workflows in Orchestrator
- Out of the box (OOTB) workflows which ship with the Management Pack for Orchestrator.

You can map these workflows as actions to alerts via recommendations. A **Run Action** button is displayed when VCF Operations displays the alerts. Using the **Run Action** button, you can execute the workflows.

You can map the user-defined workflows from the Orchestrator after you discover them in VCF Operations. If you have user-defined workflow packages, you can discover the package in VCF Operations and map each workflow in the package to actions. You can also map custom vCenter workflows.

Management Pack for Orchestrator ships with 15 out of the box workflows. This does not include the python based actions. They are automatically mapped to the alerts via recommendations.

You can create a new policy or modify an existing policy for automated workflow execution when the vSphere objects governed by the policy receives alerts. The Reconnect Host workflow is an out of the box workflow which supports automated execution. To automate the workflow execution, you must apply the policy to a custom group called Orchestrator Adapter Managed Objects. This group contains the vSphere objects.

Automated workflow execution is possible on workflows which have only one input parameter. The input parameter is the object name itself. To automate workflows which have more than one input parameter, you can add the input parameter data inside the action script of the workflow because automated workflow execution supports only one input parameter.

### Using the Orchestrator to Scale Up your Infrastructure

You can use the Orchestrator workflows to scale up your infrastructure and quickly act upon capacity related alerts.

For example, your VI Admin receives an alert that a VM is low on memory. If the alert is mapped to the Custom Reconfigure VM workflow with predefined values to increase the RAM in the VM, you can configure the workflow to be automatically executed. When the alert is received, VCF Operations triggers the action to reconfigure the VM and the memory is scaled up in the VM against which the alert is received.

#### Supported Configurations

The Management Pack for Orchestrator supports workflows that are in an external instance of Orchestrator or an instance of Orchestrator which is embedded in an instance of VCF Automation which is connected to VCF Operations.

The Management Pack for Orchestrator supports running of workflows on vCenter instances that are connected to VCF Operations. The instance of Orchestrator may be connected to more vCenter instances but you cannot execute workflows on those instances unless they are connected to VCF Operations.

The VCF Operations may be connected to an external VCF Automation which has an embedded Orchestrator. You can execute Management Pack for Orchestrator workflows from this instance of Orchestrator as well.

> **Note:**
> Since the standalone Management Pack for Orchestrator does not support the VMware Cloud on AWS authentication, you cannot execute the Management Pack for Orchestrator workflows on VMware Cloud on AWS vCenters using Management Pack for Orchestrator on VCF Operations.

#### Overview of Tasks

After you install the Management Pack for Orchestrator, you must perform the required tasks before you can run a workflow and remediate the alert.

| Task | Purpose |
|---|---|
| Configure Management Pack for Orchestrator | The workflows that help you remediate alerts are available in Orchestrator. You must configure the Management Pack for Orchestrator to connect to a external Orchestrator or an instance embedded in VCF Automation.<br>Using dynamic actions, Management Pack for Orchestrator automatically converts all the custom vCenter package workflows to alert actions. |
| Add a vCenter server instance to Orchestrator | You can optionally perform this task to add a vCenter server instance to Orchestrator from VCF Operations. |
| Discover additional Orchestrator packages. This task is optional. | After you configure the Orchestrator adapter, you can discover vCenter user-defined packages or workflows. |
| Map user defined workflows to resources as actions. | You can map user defined workflows to resources, or modify custom workflows that are already mapped to resources before you trigger them as actions.<br>If the user defined workflows are part of a package, you cannot map the package directly. You can only map workflows in the package. |
| Map workflows to alerts. | Map the custom workflows, user defined workflows and out of the box workflows to alerts via recommendations to see the **Run Action** button in an alert. |
| Apply a new policy or modify an existing policy to automate execution of workflows. | Automate the execution of workflows that have predefined parameters by using a policy on the management groups that contain the Orchestrator Adapter Managed Objects. |
| Run a workflow | You can run a workflow from an alert via the **Run Action** button, from the inventory explorer, or from the dashboard. |

## Installing and Configuring the Management Pack for Orchestrator

You can download the Management Pack for Orchestrator from the VMware Solution Exchange website.

### Install the Management Pack

The Management Pack for Orchestrator consists of a PAK file that contains out of the box workflows from the Orchestrator. You can install more than one Management Pack for Orchestrator .

The solution that you downloaded include a PAK file. Save that PAK file to a temporary folder on your local system.

1. Log in to the VCF Operations user interface with admin privileges.
2. From the left menu, click **Data Sources** > **Integrations**.
3. Go to the **Repository** tab in the right pane, and then click **Add**.
4. Browse to locate the temporary folder and select the PAK file.

   For example, `vmware-MPforAutomationOrchestrator-<version#-build#>.pak`.
5. Make the following choices:
   - Select the checkbox against, **Install the PAK file even if it is already installed.**
   - Select the checkbox against, **Reset Default Content, overwriting to a newer version provided by this update. User modifications to DEFAULT Alert Definitions, Symptoms, Recommendations, Policy Definitions, Views, Dashboards, Widgets and Reports will be overwritten. If you are installing a product software update, clone or backup the content before you proceed.**

     **Note:**

     Select these checkboxes if you are updating the Management Pack for Orchestrator from version 3.1 to version 3.2.
6. Click **Upload**.

   The upload might take several minutes.
7. Read and accept the EULA, and then click **Next**.

   Installation details appear in the window during the process.
8. When the installation is finished, click **Finish**.

Configure the Management Pack for Orchestrator to connect to an external Orchestrator or an instance embedded in VCF Automation.

### Configure the Management Pack for Orchestrator

After you install the management pack, add and configure an external Orchestrator or an instance embedded in VCF Automation from which you run workflows.

1. From the left menu, click **Administration** > **Integrations**, and then click **Add**.
2. Select the Management Pack for Orchestrator and click **Configure**.
3. Configure the adapter instance.

| Option | Description |
|---|---|
| Name | The name for the adapter instance. |
| Description | (Optional) The description of the adapter instance. |
| VCF Automation Orchestrator Host | The URL of Orchestrator or VCF Automation instance from where you run workflows. Specify the FQDN.<br>If a load balancer is present, the URL must use the load balancer's hostname. |
| Port | The default port is 443. |

| Option | Description |
|---|---|
| Auto Discovery | Discover objects automatically.<br>• To set automatic discovery for objects, select **True**.<br>• To set the automatic discovery off, select **False**. |
| Credential | To add new credentials click the plus (+) sign.<br>**Credential Kind**.<br>From the drop-down menu, select one of the below:<br>1. Orchestrator Credential<br>2. Automation Credential<br>3. Service Account Refresh Token<br>**Important:**<br>• For standalone Orchestrator, enable Basic Authentication in Orchestrator and use Orchestrator Credential kind.<br>• For Automation 8.x, enable Basic Authentication in Orchestrator and use Automation Credential.<br>• For VCFA 9.0 Provider Organization embedded Orchestrator, use Service Account Refresh Token credentials to configure the adapter.<br>• **Credential name**.<br>The name by which you are identifying the configured credentials.<br>While using Orchestrator Credential or Automation credential as 'Credential Kind', use below:<br>– Username. The username of an Orchestrator user or Automation user.<br>– Password. The password of an Orchestrator user or Automation user.<br>While using 'Service Account Refresh Token', enter the Refresh Token generated using steps mentioned on this topic/page:<br>Create a Service Account Refresh Token<br>**Note:**<br>Verify that you have API access to Orchestrator to execute the workflows. |
| Collectors/Groups | The collector on which the Management Pack for VCF Automation instance runs.<br>• For one collector instance, select default collector group.<br>• For multiple collectors, to distribute the workload and optimize performance, select the collector to manage the adapter process for this instance. |

4. Click **Test Connection** to validate the connection.
5. Click **Save Settings**.

   The adapter instance is added to the list.

Verify that the VCF Operations management pack is fetching workflows.

- In the **Solutions** page, verify that the **Collection State** column and **Collection Status** are green under **Configured Adapter Instances**. The **Collection Status** displays **Data Receiving** after the first collection cycle. The default time for a collection cycle is five minutes.
- From the left menu, click **Environment**, and then click **Inventory**. Verify that the VCF Operations workflows are displayed on the table and sort by the **Adapter Type** column or filter by name of the adapter instance if necessary.
- Optionally, discover additional packages to configure workflows.

Ensure that the instance of VCF Operations where you are configuring the Management Pack for Orchestrator has an instance of vCenter with Orchestrator configured. If this is not configured, an alert is displayed in VCF Operations. You can add a vCenter instance to Orchestrator to remediate this alert.

### Add a vCenter Instance to Management Pack for Orchestrator

An instance of vCenter must be available in the Orchestrator and the same instance must be configured in VCF Operations. If this is not available, you can add a vCenter Instance to Management Pack for Orchestrator from VCF Operations.

The Management Pack for Orchestrator must be configured.

1. From the left menu, click **Environment**, and then click **Inventory**.
2. Select **Adapter Instances**, and then select **VCF Automation Orchestrator Adapter Instance** from the list of adapter instances.
3. Select the Orchestrator adapter instance.

   A list of all the objects are displayed on the right pane.
4. From the right pane, select the Orchestrator to which you want to add a vCenter instance.
5. Click the **Action** icon, and then click **Add a vCenter Instance to VCF Automation Orchestrator**.
6. Configure the properties.
7. Click **Begin Action**.

   A task is added to the task list.

### Upgrade Management Pack

You can upgrade to a newer version of Management Pack for Orchestrator.

1. Log in to the VCF Operations user interface with admin privileges.
2. From the left menu, click **Data Sources** > **Integrations**.
3. Go to the **Repository** tab in the right pane, and then click **Add**.
4. Browse to locate the temporary folder and select the PAK file.
5. Select the **Install the PAK file even if it is already installed** check box.

   This selection reloads the PAK file but retains the custom preferences of the user. Also, this selection does not overwrite or update the solution alerts, symptoms, recommendations, and policies.
6. Select the **Reset Default Content, overwriting to a newer version provided by this update** check box.

   This selection reloads the PAK file and overwrites the existing solution alerts, symptoms, recommendations, and policies with newer versions provided with the current PAK file.
7. Click **Upload**.

   The upload might take several minutes.
8. Read and accept the EULA, and then click **Next**.
9. When the upgrade is complete, click **Finish**.
10. On the Manage Solutions page, click **Test Connection** to validate the connection.

Import the Apply Host Security Configuration Rules Out of the Box (OOTB) workflow to Orchestrator.

### Import Workflow

After you upgrade to the newer version of the Management Pack for Orchestrator, you have to manually import the new Apply Host Security Configuration Rules workflow to Orchestrator.

1. Log in to the VCF Operations instance.
2. From the left menu, click **Environment** > **Inventory**, and then click **Adapter Instances** > **VCF Automation Orchestrator Adapter Instance**.
3. Select the adapter instance.
4. Click **Action** icon, and then select **Import VMware Cloud Foundation Operations Remediation Package to vRO**.
5. Click **Overwrite Package**, and then click **Begin Action** to import the Apply Host Security Configuration Rules workflow to Orchestrator.

**Create a Service Account Refresh Token**
The user can create an Service Account Refresh Token.

1. Login to VCF Automation Provider Management portal.
2. Navigate to Access Control → Service Accounts → Click on New.
3. Enter the Name.
4. Select the required roles. You can create a new role just with Orchestrator Administrator privileges or use existing System Administrator with full access.
5. Generate the Software ID by clicking on the magic-wand icon.
6. Click NEXT. Review the information and click on Finish.
7. The Service Accounts table now reflects the newly created Service Account.
8. Edit the newly created Service account to **disable / deselect the Requires Rotation** Setting and save your changes
9. Expand the service account row to view the Client ID.
10. Create a curl command to perform device authorization by replacing VCFA_TENANT_MANAGER_FQDN and SERVICE_ACCOUT_CLIENT_ID. `curl -k 'https://<VCFA_TENANT_MANAGER_FQDN>/tm/oauth/provider/device_authorization' \ --header 'Content-Type: application/x-www-form-urlencoded' \ --data-urlencode 'client_id=<SERVICE_ACCOUNT_CLIENT_ID>'`
11. Execute the curl command and copy the value of user_code from the response. `{ "device_code": "k81ANxCMOjaRFxeVZL0gzXwS37LhYkSF2pwhH1IzSIA", "user_code": "5MX5-206H", "verification_uri": "https://vcfa.10.161.199.184.vcfd.example.com/provider/access-control/service-accounts", "expires_in": 3600, "interval": 60 }`
12. Under Service Accounts, Click on the Review Access Requests tab.
13. Paste the User Code copied from Step 11, and click on the LOOKUP button. Click on the GRANT button to approve the access request.
14. Copy the value of device_code from the response in Step 13.
15. Create a curl command to generate a Refresh Token by replacing VCFA_TENANT_MANAGER_FQDN, ORGANIZATION_SERVICE_ACCOUT_CLIENT_ID and DEVICE_CODE_FROM_DEVICE_AUTH_RESPONSE in `curl -k 'https://<VCFA_TENANT_MANAGER_FQDN>/tm/oauth/provider/token' \ --header 'Content-Type: application/x-www-form-urlencoded' \ --data-urlencode 'client_id=<ORGANIZATION_SERVICE_ACCOUNT_CLIENT_ID>' \ --data-urlencode 'grant_type=urn:ietf:params:oauth:grant-type:device_code' \ --data-urlencode 'device_code=<DEVICE_CODE_FROM_DEVICE_AUTH_RESPONSE>'`
16. Execute the curl command and copy the refresh_token from the response. `{ "access_token": "eyJraWQiOiJ1cm46dmNsbhcMqLveEA", "token_type": "Bearer", "expires_in": 3600, "refresh_token": "aJ4WNSZWmgwOZeg0M5cY9UUehjvicSRg"}`
17. Use the refresh_token from Step 16 to configure adapter instance.

**Configuring Orchestrator Workflows**

### Package Discovery

By default, Management Pack for Orchestrator discovers custom vCenter workflows and Management Pack for Orchestrator Out of the box (OOTB) workflows. You can extend the workflow capability of the Management Pack for Orchestrator by installing additional packages.

Install and configure the Management Pack for Orchestrator. The Management Pack for Orchestrator must connect to an external adapter before you install packages related to the adapter. In the **Solutions** page, verify that the **Collection State** column and **Collection Status** are green under **Configured Adapter Instances**. The **Collection Status** displays **Data Receiving** after the first collection cycle.

Install additional packages to support workflows related to different adapters.

1. Log in to the VCF Operations instance.
2. From the left menu, click **Environment** > **Inventory**, and then click **Adapter Instances** > **VCF Automation Orchestrator Adapter Instance**.
3. Select a Orchestrator adapter instance.
4. Next to the name of the Orchestrator adapter instance in the right pane, click **Configure Package Discovery** under **Actions**.
5. The **Configure Package Discovery** dialog box opens. Add a list of packages to be discovered in the following format: `<packageName>`
6. Click **Begin Action**.
7. The package discovery task starts. You can view the task ID in the **Configure Package Discovery** dialog box. Click **OK**.
8. To track the progress of the task, go to the **Recent Tasks** section in the **Administration** menu. Package discovery starts after the first content cycle.

### Mapping Workflows to Resources as Actions

You can map custom and user-defined workflows to actions. Repeat the task to add one workflow to more than one resource. You can perform this task for multiple workflows simultaneously within a package by running the same action on a package resource.

You must discover user-defined workflows them before you can add them as an action to a resource.

> **Note:**
>
> You can only map workflows to the supported target resources.

1. From the left menu, click **Environment**, and then click **Inventory**.
2. Click **Adapter Instances**, and then click **VCF Orchestrator Adapter Instance** from the list of adapter instances.
3. Select the Orchestrator adapter instance that contains the workflow that you want to run.

   A list of all the workflows are displayed on the right pane.

   > **Note:**
   >
   > This action is present at the Adapter instance level, package level, and workflow level. The most recommended is the package and workflow level.
4. From the right menu, select the workflow that you want to add to a resource.
5. Click the **Action** icon, and then click **Create/Modify Workflow Action on vCenter Resources**.

   > **Note:**
   >
   > Ensure that there is at least one input parameter in the workflow, otherwise the action fails when it is run.

6. Configure the following properties

| Property | Option |
|---|---|
| Workflow | Select a workflow from the drop-down menu.<br>**Note:**<br>Do not modify out-of-the-box workflows. |
| Available on Resource Type | If the workflow is already configured as an action in a resource type, it is displayed here. |
| Action Target Type | If the workflow is already configured as an action in the target resource type, it is displayed here. |
| Resource Type | Select a resource type from the drop-down menu. |
| Target Resource Type | Select a resource type. The resource type and the target resource type should be the same. |
| Operation | Choose to add or remove the workflow as an action. Options are **Add** and **Remove**. |

   With the Management Pack for Orchestrator 3.2, you can map any custom workflow to the parent resource. The parent object is the Resource Type and the Target Type is the actual vCenter object that the workflow is targeted for. The workflow accepts the target object as its input parameter. For example, if you want to map a workflow “Change Ram” to a host system and run the same workflow on multiple virtual machines under it, then select the Resource Type as **Host System** and the Target Type as **Virtual Machine**.

7. Click **Begin Action**.

   A task is added to the task list.

You can verify that the action is available by clicking the **Action** icon after you select the resource with the required **Adapter Type** and **Object Type** in the right pane. From the list displayed under the **Action** icon, verify that the workflow that you added to the resource is available as an action in the list. You must wait one collection cycle before you can verify. You can monitor the task list to see the status of the task.

### Mapping Actions to Alerts

After you install and configure the Management Pack for Orchestrator, you can map custom and user defined workflows (actions) to alerts via recommendations. After you perform this mapping, you can execute the Orchestrator workflow to remediate issues when VCF Operations displays alerts. Out of the box workflows are automatically mapped to alerts.

Before you map alerts to workflows, review the out of the box workflows and corresponding recommendations and alerts. See Out of the Box Workflows in the *Appendix*.

1. From the left menu, click **Troubleshoot**, and then click **Alert**.
2. Click **Actions**, and then select **Go to Alert Definition**.
3. Select an Alert that you want to modify, and then click **Edit**.

   The **Edit Alert Definition** dialog box opens.
4. Click the **Recommendations** tab.

   The available recommendation is displayed.

5. Click the **Create New Recommendation** icon to add a new recommendation.
6. Enter the description of what must be done to resolve the triggered alert.
7. Select an option from the drop-down list for **Adapter Type**.
8. Select an option from the drop-down list for **Action**, as a method to resolve a triggered symptom or a generated alert. Actions must already be configured in VCF Operations.
9. Click **Create**.
10. Drag and drop the recommendation that you created to the **Recommendations** section in the right pane.
11. Click **Update** to save the new alert definition.

### Viewing Workflows in Packages

View relationship between the Orchestrator and package to view the workflows. The **All Metrics** tab provides a map between the Orchestrator and the package.

Install and configure the Management Pack for Orchestrator and install a package before viewing the workflows in the packages.

1. Log in to the VCF Operations instance.
2. From the left menu, click **Environment** > **Inventory**, and then click **Adapter Instances**.
3. Select the instance of Orchestrator under **vRO Packages and Workflows**.
4. Click the **All Metrics** tab on the right pane.
5. Expand the **Show Object Relationship** section to view the relationship between the Orchestrator instance and packages.
6. Click a package to see a list of workflows in the package.

   By default, Management Pack for Orchestrator displays the following packages:

**Table 1663: Default Management Pack for Orchestrator Packages**

| Package Name | Workflows |
|---|---|
| com.vmware.library.vcenter | vCenter Workflows |
| com.vmware.vrops.oob_content | Default Management Pack for Orchestrator workflows |

### Configuring Automatic Remediation

A policy is a set of rules that you define for VCF Operations to use to analyze and display information about the objects in your environment. You can modify an existing policy or create a new policy and configure it for automatic alert remediation. To automate the workflow execution, you must apply the policy to a custom group called Orchestrator Adapter-Managed Objects. This group contains the vSphere objects.

Ensure that the following prerequisites are met:

- You have mapped custom and user-defined workflows to resources as actions. Automated workflow execution is possible on workflows which have only one input parameter. The only other input parameter is `vrops_alert_id` and it is used to fetch the ID of the alert that triggered the workflow. This input parameter should be of the type String. To automate workflows which have more than one input parameter, you must create a user-defined workflow and predefine the input parameters in the workflow.
- You have mapped the action to an alert through a recommendation.

1. From the left menu, click **Configure**, and then click **Policies**.
2. Select the current default policy. In the **More Actions** tab, click **Edit**.
3. Select **Groups and Objects** tab in the **Default Policy [Edit]** window.
4. Click **Save**.
5. Select the checkbox next to **VCF Orchestrator Adapter Managed Objects**.
6. Select **Alerts and Symptoms** tab in the **Default Policy [Edit]** window.
7. In the **Alert Definitions** pane, filter by the name of the alert to which you have mapped the action that you want to automate.
8. Select the alert from the filter results.
   a) In the **Automate** column, select **Local**.
   b) In the **State** column, select **Enabled**.
9. Click **Save**.

The default policy is modified to automatically run the configured action when mapped alerts are displayed.

### Import Remediation Package to Orchestrator

After you install and configure the Management Pack for Orchestrator, VCF Operations automatically imports the remediation package to Orchestrator after the first collection cycle. If you delete the remediation package in Orchestrator, you see an alert in VCF Operations. You can then re-import the remediation package to Orchestrator.

Verify that you have configured the Management Pack for Orchestrator

1. Log in to the VCF Operations instance.
2. From the left menu, click **Environment** > **Inventory**, and then click **Adapter Instances** > **VCF Orchestrator Adapter Instance**.
3. Select a Orchestrator adapter instance.
4. Next to the name of the Orchestrator adapter instance in the right pane, click **Import VMware Cloud Foundation Operations Remediation Package to vRO** under **Actions**.
5. Choose to overwrite package if required.
6. Click **Begin Action**.

Log on to Orchestrator to verify if the remediation package is available.

## Executing Orchestrator Worfklows

### Running a Workflow from an Alert

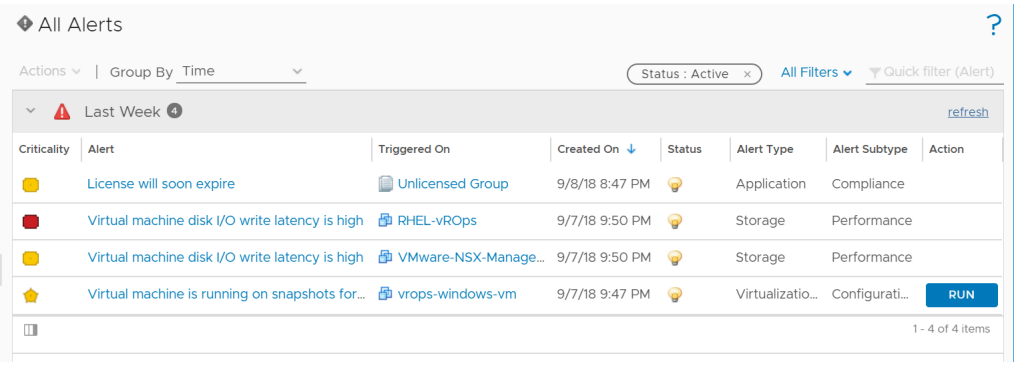
Run a Orchestrator workflow from an alert to resolve issues reported by VCF Operations.

Verify that you have configured the Management Pack for Orchestrator.

You can execute a Orchestrator workflow from VCF Operations.

1. From the left menu, click **Troubleshoot**, and then click **Alerts**.
   The alerts are displayed on the right pane. They may be grouped either by Time, Criticality, Definition, Object Type, or Scope.

2. Select an alert in the **All Alerts** page.
3. Click the **Show Columns** icon on the bottom left. Select the **Action** checkbox to show the actions column.
4. If the alert recommendation has an associated action, click the **Run** button.




You can check if the task has been completed to verify if the workflow has executed. To check the status of the task, go to the **Recent Tasks** section under the **Administration** menu in the left pane.

### Running a Workflow from the Inventory Explorer

Run a Orchestrator workflow from the Inventory Explorer to perform common management tasks.

Verify that you have configured the Management Pack for Orchestrator. You can run a workflow only on unique resource names. If two resources have the same name, the workflow may not run.

1. From the left menu, click **Administration**, and then click **Inventory** on the left pane.
2. Click **Adapter Instances**, and then click **VCF Orchestrator Adapter Instance** from the list of adapter instances.
3. Select the Adapter Instance that contains the workflow that you want to run.
   A list of all the workflows are displayed on the right pane.
4. Click the **Actions** icon and then **Execute Workflow Action** to run the workflow.
5. In the **Execute Workflow Action** dialog box, enter the input parameters to run the workflow. You must provide input parameters in the **Input parameter to run the Workflow** and **vCenter Server IP** columns.

**Table 1664: Input Format for the *Input parameter to run the Workflow* column**

| Workflow Data type | Format for Providing Input Values |
|---|---|
| String/Secure String/VC:XYZ (any String) | Resource name without single or double quotes |
| Number | Any numeric value. For example, you can provide the index number which corresponds to the VM's guest Operating System name for input parameter data type for the guest Operating System identifier. See Using the Input Parameter Data Type for Guest Operating System Identifier in the *Appendix* for more information. |
| Boolean | True or False |
| Array/XYZ (any string) | Input for the array in the following format:<br>`(input1, input2) or [input1, input2] or {input1, input2}` |
| Optional inputs | If you do not wish to specify for any value for an optional input, you just need to provide a comma (,) in place of the value. |

**Note:**
: User input should be ordered with respect to the data type of the workflow input parameter.

- For the **vCenter Server IP** column, select the vCenter server IP from the dropdown. This is only required for the workflows that you run in the vCenter.

6. Click **Begin Action**.

You can check if the task has been completed to verify if the workflow has executed.

### Running an Action from the vCenter Adapter Instance

You can run an action on multiple vCenter resources of the same type from the Inventory Explorer.

Verify that you have configured the Management Pack for Orchestrator.

1. From the left menu, click **Environment**, and then click **Inventory**.
2. Click **vCenter Server**, and then click **vCenter Adapter Instance** from the list of adapter instances.
3. Select the vCenter resources of type Virtual Machine.
4. Click the **Actions** icon, and then select the action that you want to run.
5. Enter the input parameters for the Virtual Machines and click **Begin Action**.

From the left menu, click **Administration**, and then click **Recent Tasks**, to check if the action has been successfully completed.

### Supported Parent Relationships for Workflows

When the workflow is targeted for a Virtual Machine object, the workflow is run for the supported VM parent object types. The following table describes the supported VM parent object types on which the workflow can be run.

| Parent Object | Object Type |
|---|---|
| VMFolder | VirtualMachine |
| ResourcePool | VirtualMachine |
| HostSystem | VirtualMachine |
| ClusterComputeResource | HostSystem |
| HostFolder | HostSystem |
| Datacenter | Datastore |
| HostSystem | Datastore |
| DatastoreFolder | Datastore |
| StoragePod | Datastore |
| ClusterComputeResource | ResourcePool |
| Datacenter | ClusterComputeResource |
| Datacenter | VmwareDistributedVirtualSwitch |
| NetworkFolder | VmwareDistributedVirtualSwitch |
| Datacenter | DistributedVirtualPortgroup |
| VmwareDistributedVirtualSwitch | DistributedVirtualPortgroup |
| NetworkFolder | DistributedVirtualPortgroup |
| VMwareAdapterInstance | Datacenter |
| Datacenter | VMFolder |
| Datacenter | HostFolder |
| Datacenter | NetworkFolder |
| Datacenter | DatastoreFolder |

### Determine the Status of a Workflow After Execution

Recent Tasks provide the status of a workflow after you execute it. View the recent tasks to see if the workflow executed correctly. If any workflow is in the failed/unknown state, you will get a notification in the alerts window with the details of

the resources for which the workflow failed. If there are multiple failures, they are combined, and the alert is cleared in the next collection cycle.

> **Note:**
> If a workflow which was triggered as part of automatic remediation fails, you must run the workflow manually to remediate the alert. Management Pack for VCF Operations will not execute the workflow that was triggered as part of automatic remediation if it fails the first time, even if the alert is still active.

1. From the left menu, click **Administration**, and then click **Recent Tasks**.
2. Select the workflow task from the list of recent tasks. You can view details of the selected task in the bottom pane.

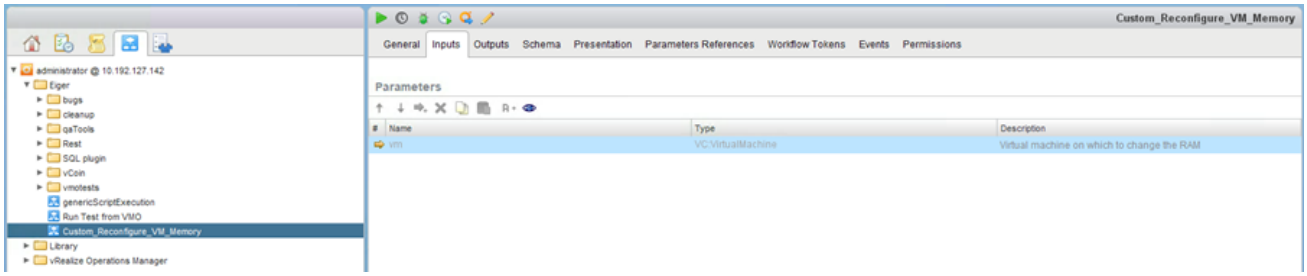
Example: Configuring and Executing Auto Remediation

This example describes the steps to define a custom workflow in Orchestrator and configure auto-remediation of an alert using the custom workflow.

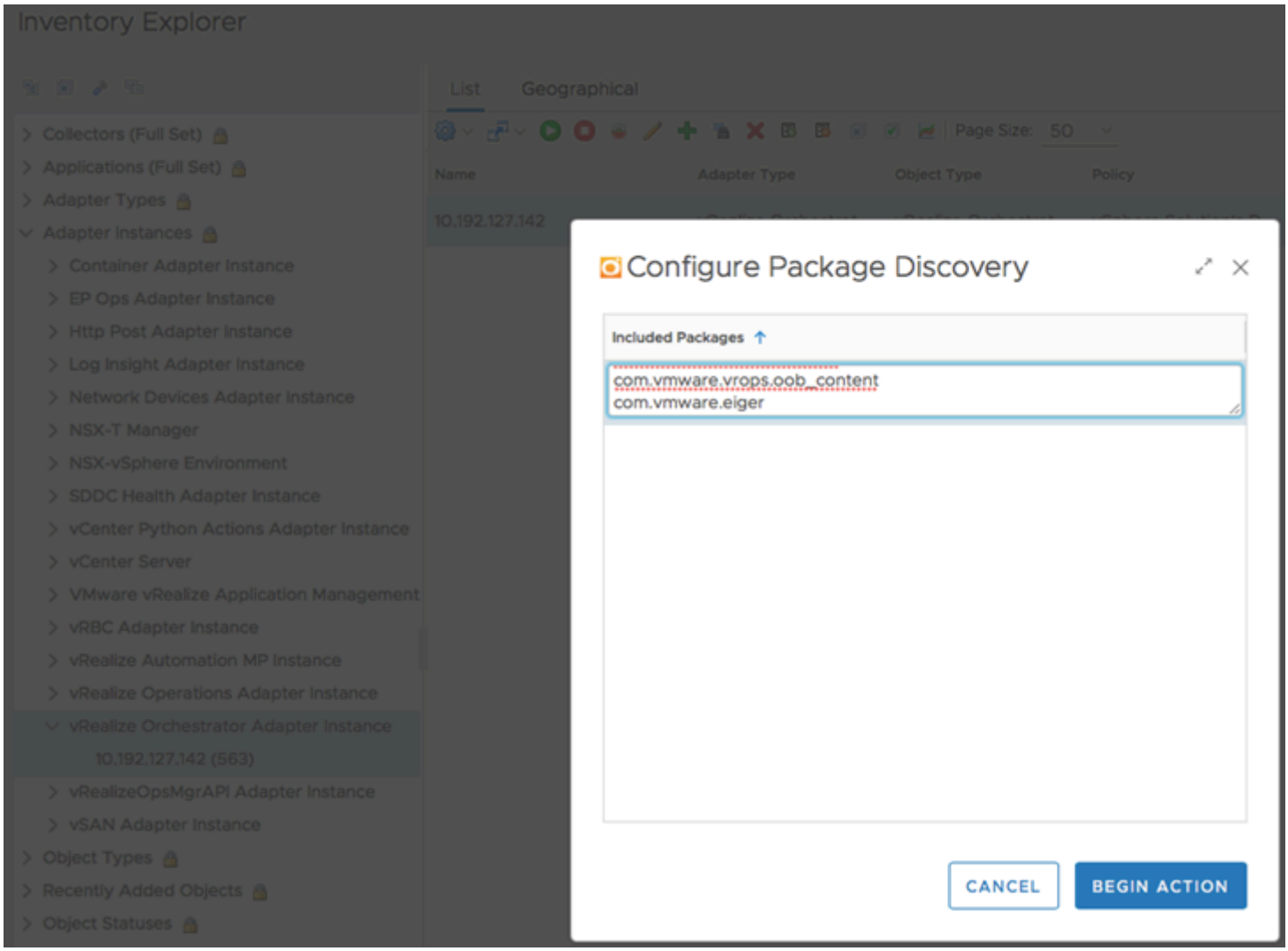
Automated workflow execution is possible on workflows which have only one input parameter. The input parameter should be of type `VC:<VC datatype>` in the Orchestrator workflows.

In this example, we will create a custom Orchestrator workflow where the input parameter name is `vm`.

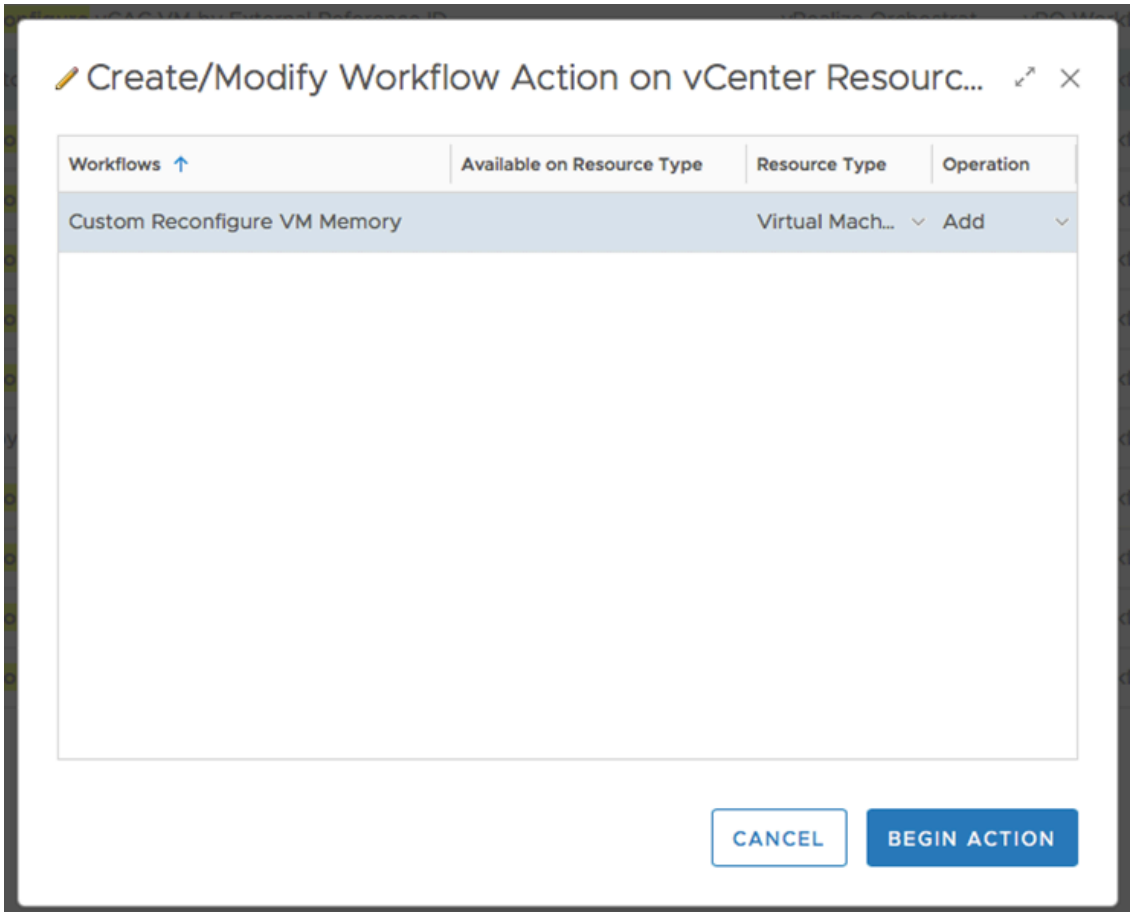
1. Create a custom workflow in Orchestrator. The custom workflow should have one input parameter of type `VC: <VC Datatype>`. In our example, the workflow **Custom Reconfigure VM Memory**, has one input parameter of type `VC:VirtualMachine`, and the name of the input parameter is `vm`.



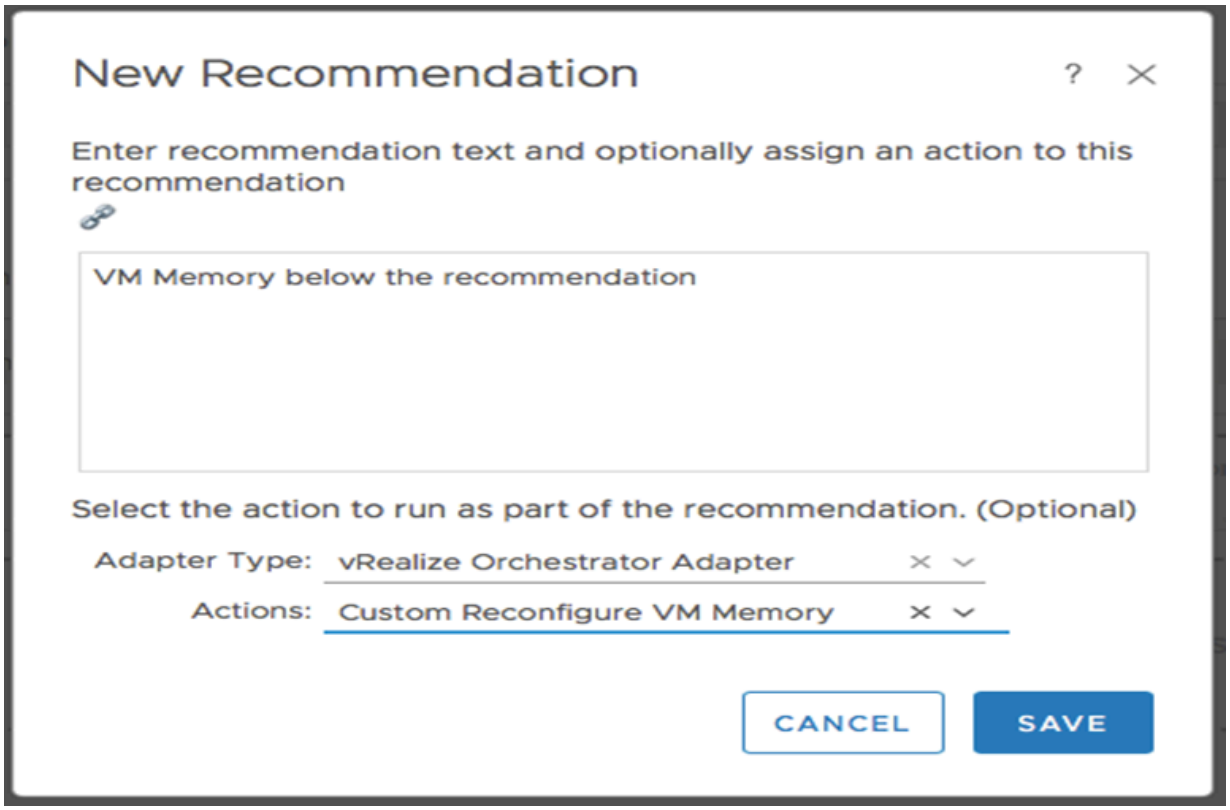
2. Import the custom workflow into VCF Operations.
   a) From the left menu, click **Environment** > **Inventory**, and then click **Adapter Instances** > **VCF Automation Orchestrator Adapter Instance**.
   b) Select a Orchestrator adapter instance.
   c) Next to the name of the Orchestrator adapter instance in the right pane, click **Configure Package Discovery** under **Actions**.
   d) The **Configure Package Discovery** dialog box opens. Add a list of packages to be discovered in the following format: `<packageName>`



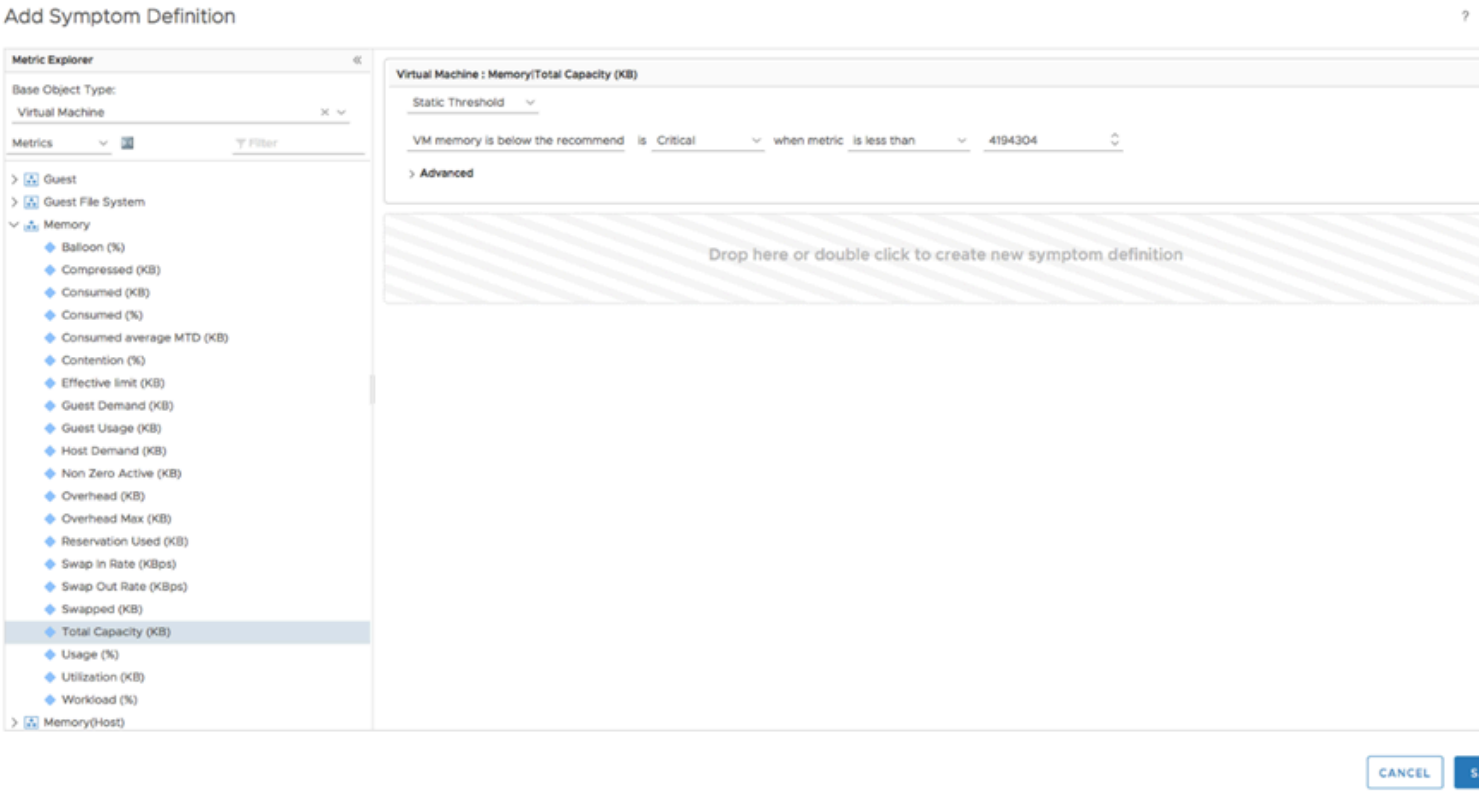
3. Map the custom workflow to a vCenter resource. Here the vCenter resource is of type `VC:VirtualMachine` which should match the Orchestrator workflow.
   a) From the left menu, select **Administration**, and then click **Configuration** > **Inventory Explorer** on the left pane.
   b) Select **Adapter Instances**, and then click **VCF Automation Orchestrator Adapter Instance** from the list of adapter instances.
   c) Select the Orchestrator adapter instance that contains the workflow that you want to run.
   d) From the right pane, select the workflow that you want to add to a resource.
   e) Click the **Action** icon, and then **Create/Modify Workflow Action on vCenter Resources**.



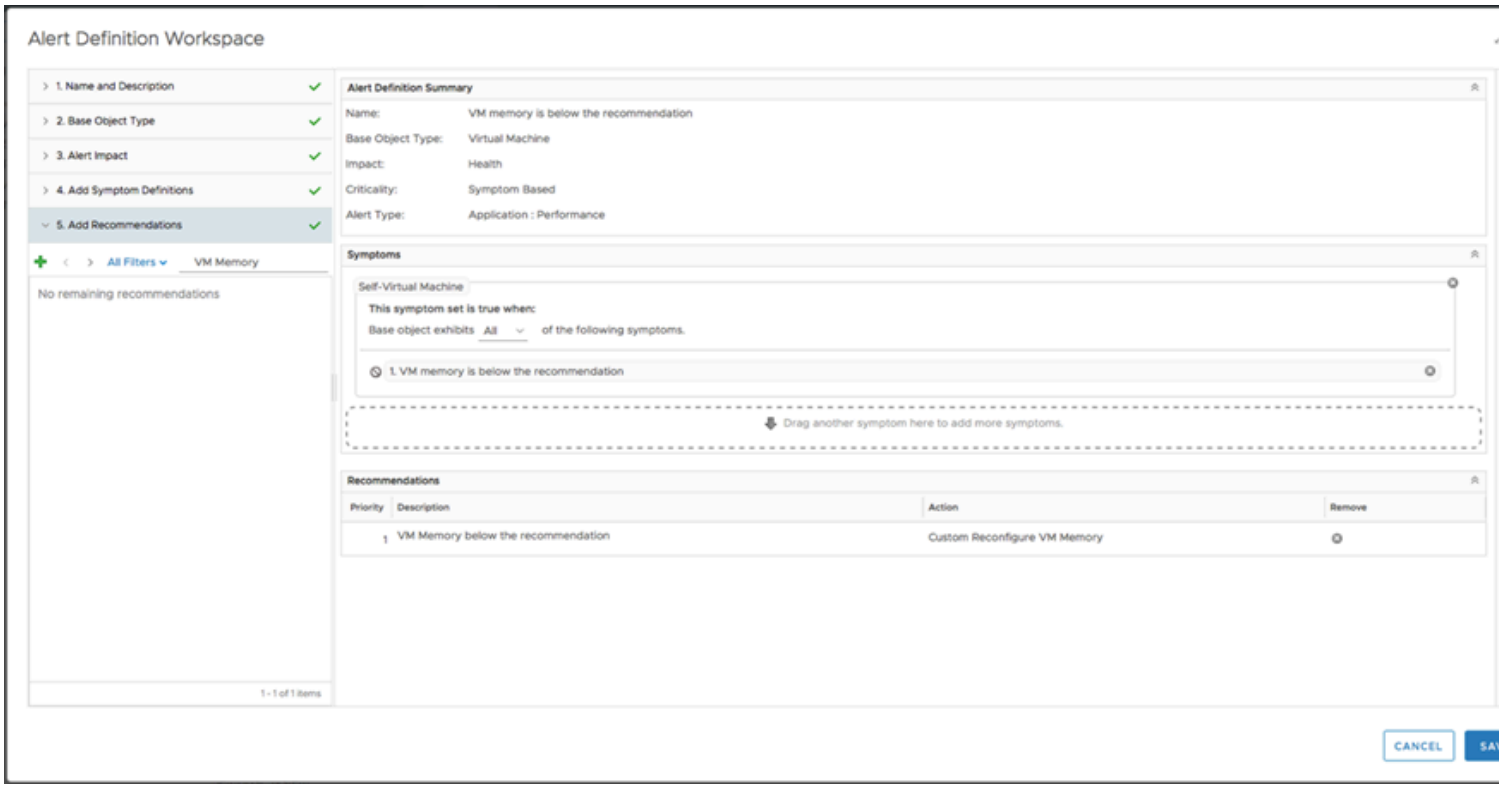
4. Create a custom recommendation. To define recommendations, in the menu, click **Alerts**, and then in the left pane, click **Alert Settings** > **Recommendations**.



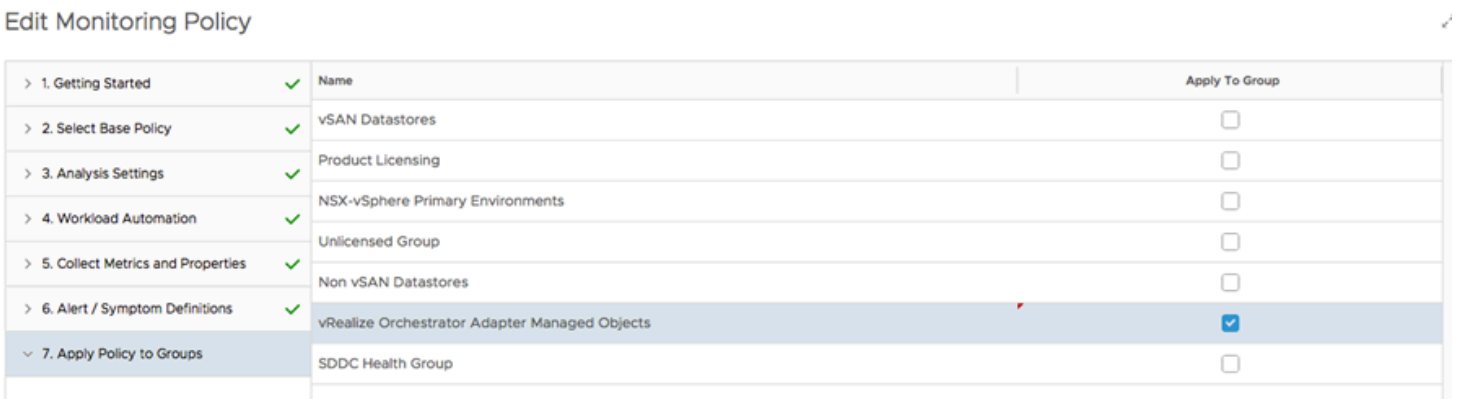

5. Create a symptom which looks for VMs that have memory below 64GB. In the menu, click **Alerts**, and then on the left pane, click **Alert Settings** > **Symptom Definitions**.

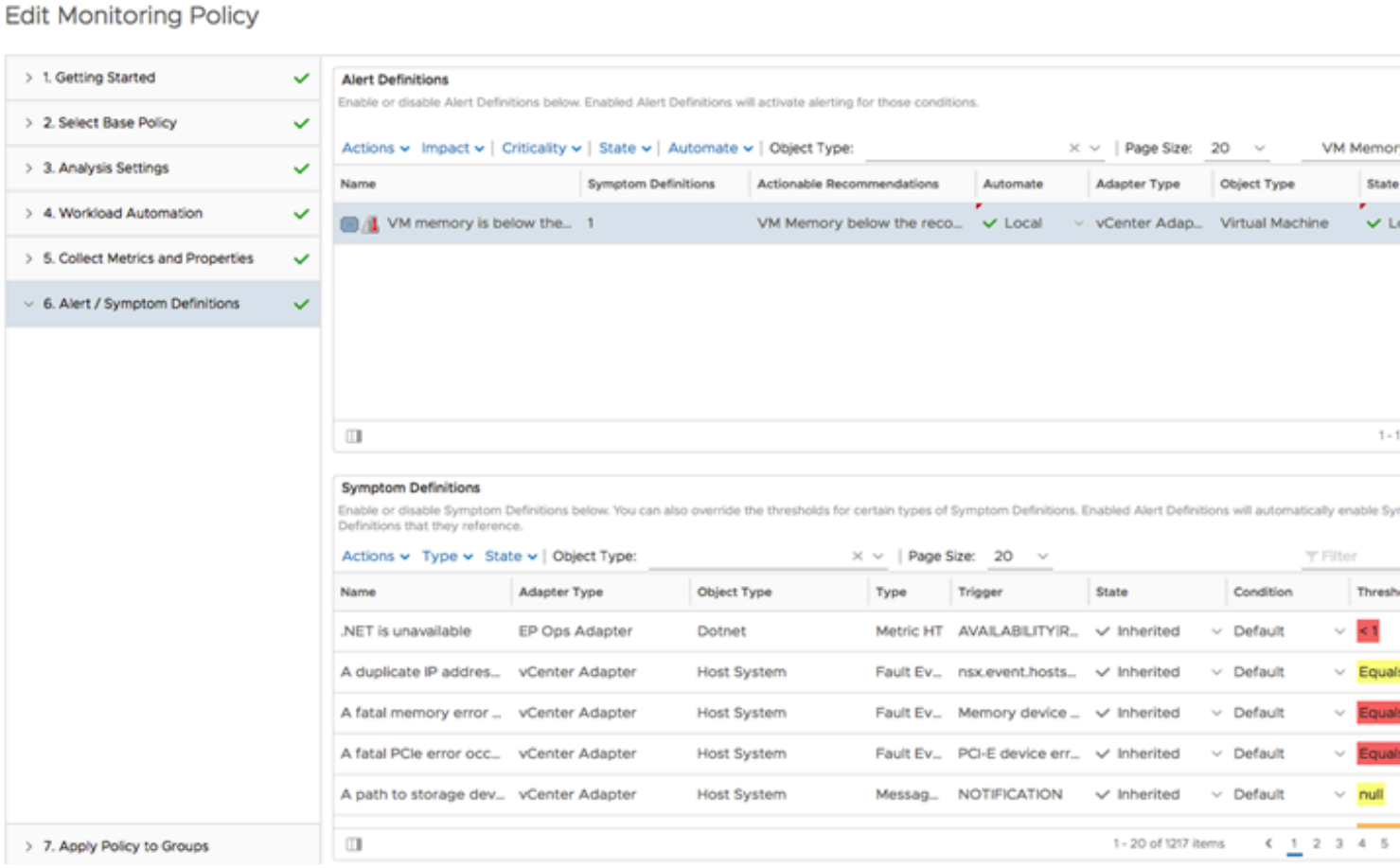


6. Create a custom alert which will invoke the Orchestrator workflow/action automatically. Map the symptom and recommendation to the alert. To create or edit your alert definitions, in the menu, click **Alerts** and then in the left pane, click **Alert Settings** > **Alert Definitions**.

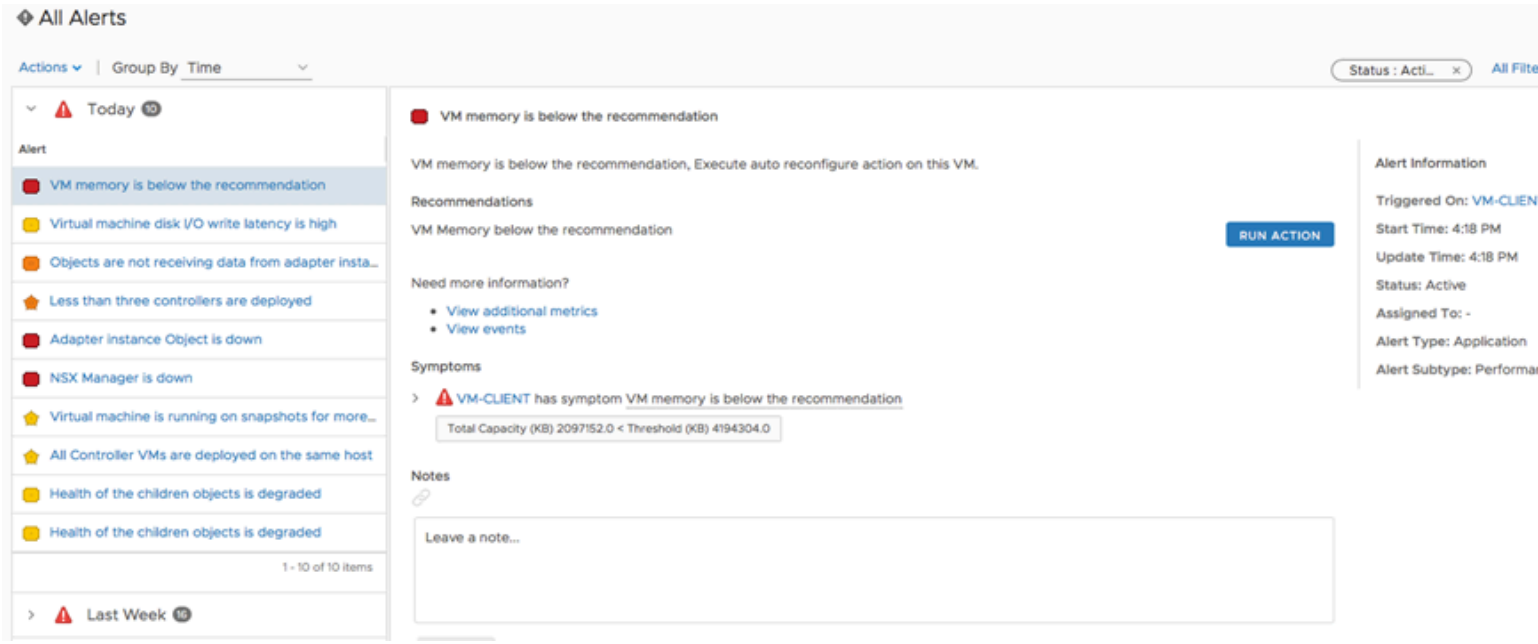


7. Create/Modify the policy and apply the Orchestrator Adapter group.
   a) From the left menu, click **Administration**, and then in the left pane, click **Policies**. Click the **Policy Library** tab, and then click the **Edit Selected Policy** icon to edit a policy.
   b) In the **Automate** and **State** columns, change the setting to **Local**.

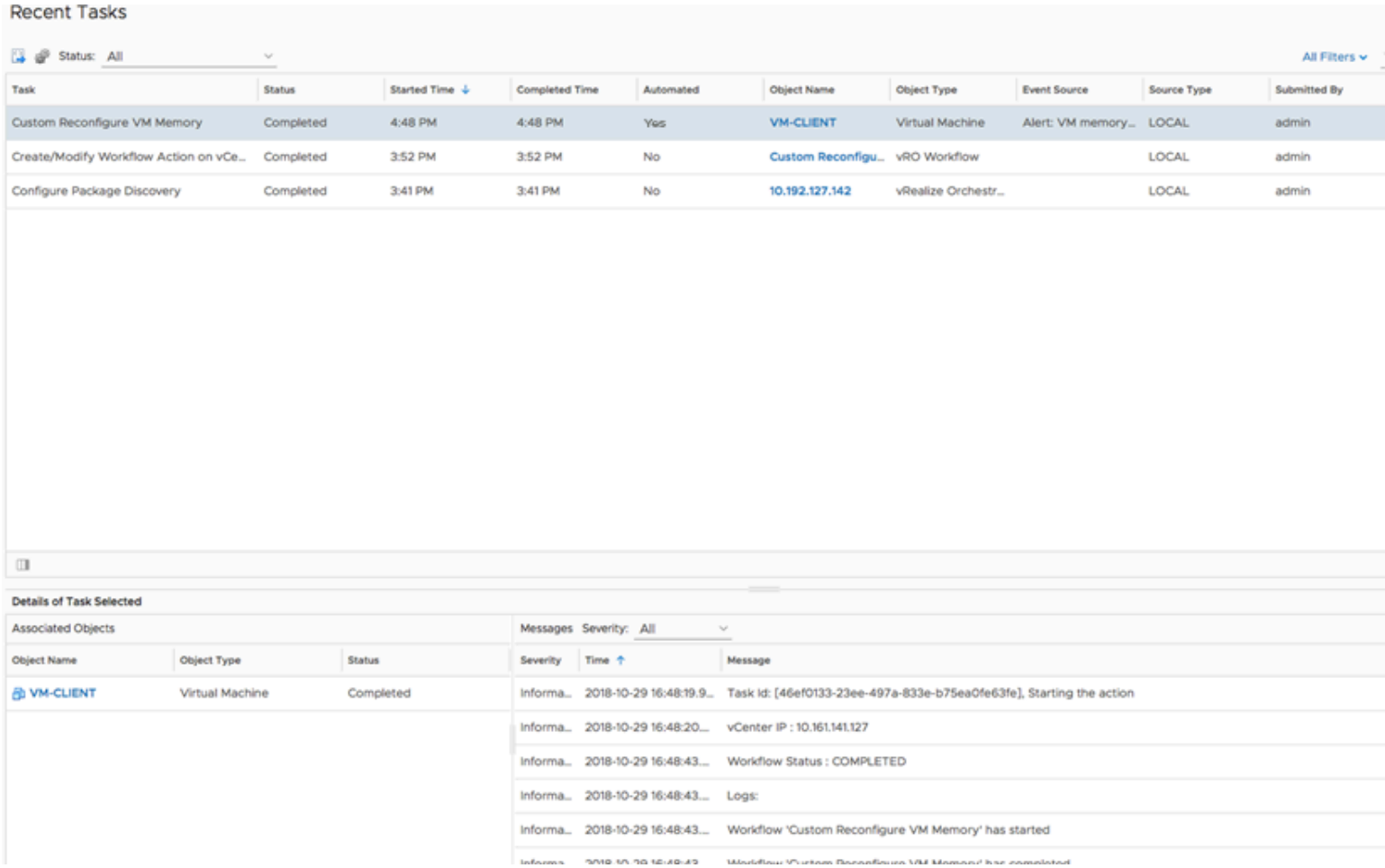



After configuring the custom workflow, VCF Operations will look for VMs that have RAM less than 64GB and generate an alert.




Based on the alert, Management Pack for Orchestrator invokes the custom workflow automatically. The alert remediated in next collection by executing the mapped recommendation. You can verify this in the **Recent Tasks**.

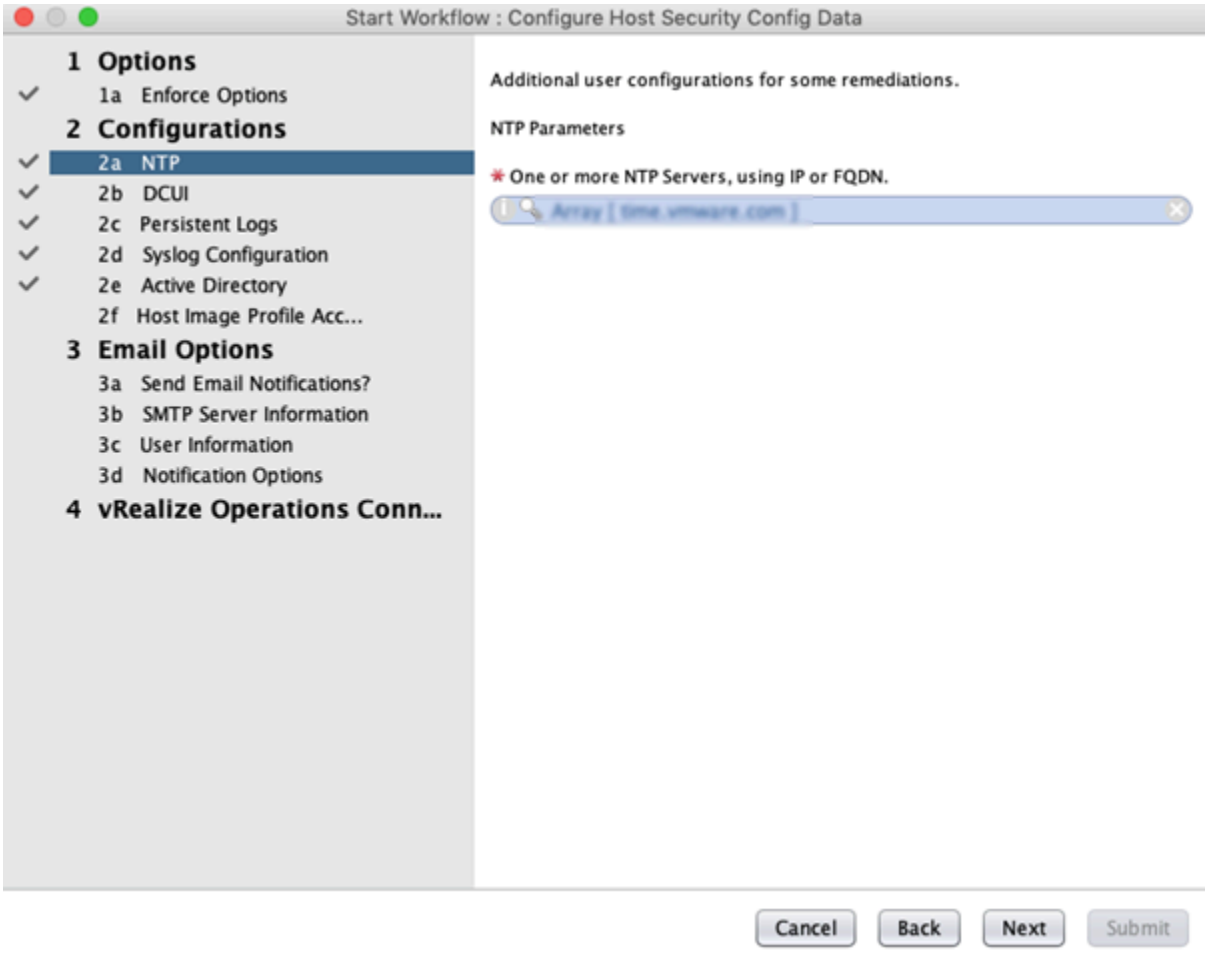



Example: Configuring Automated Compliance Remediation

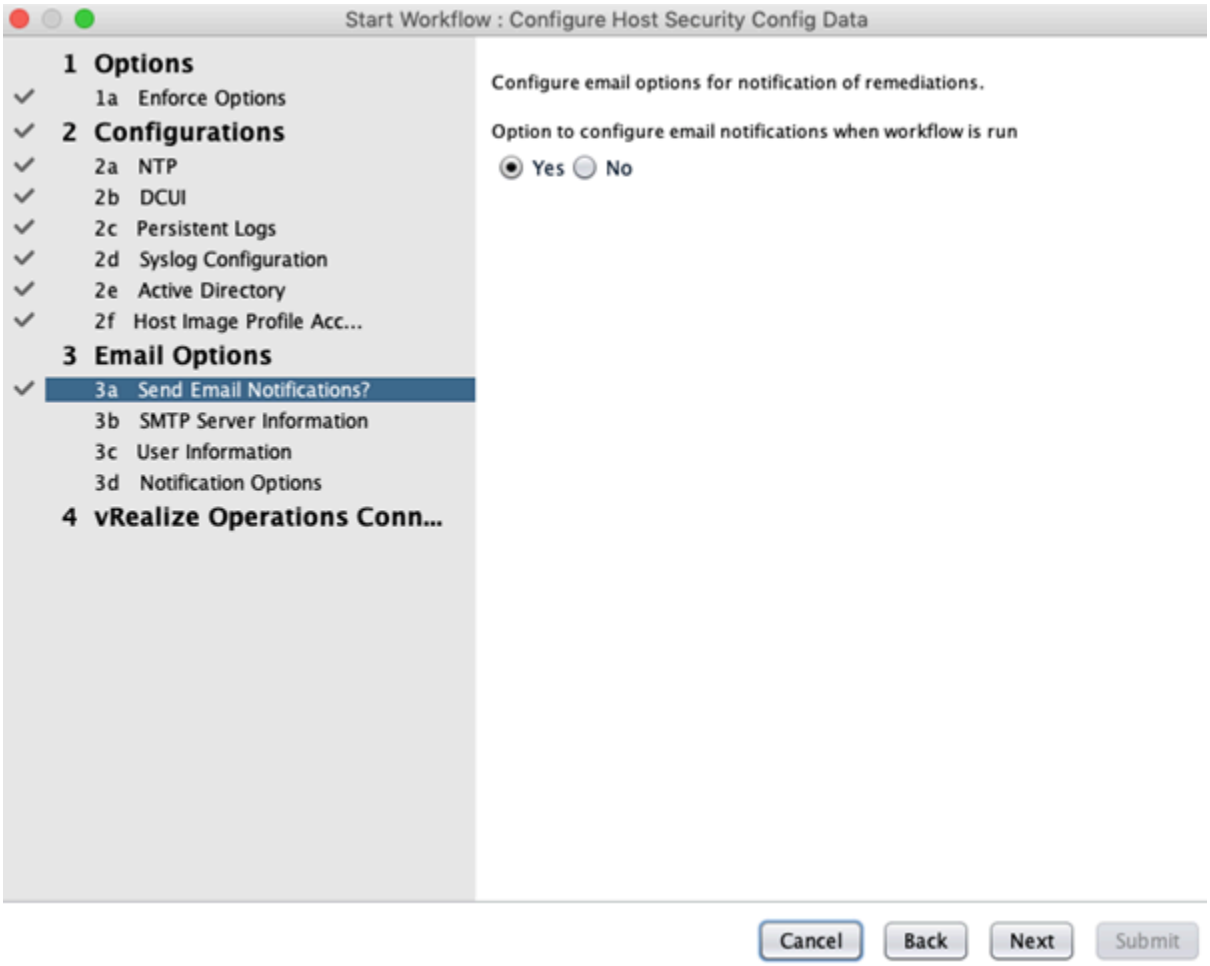
You can configure automated compliance remediation by applying host security configuration rules and setting options to enforce hardening rules.

Install or upgrade to Management Pack for Orchestrator version 3.2.

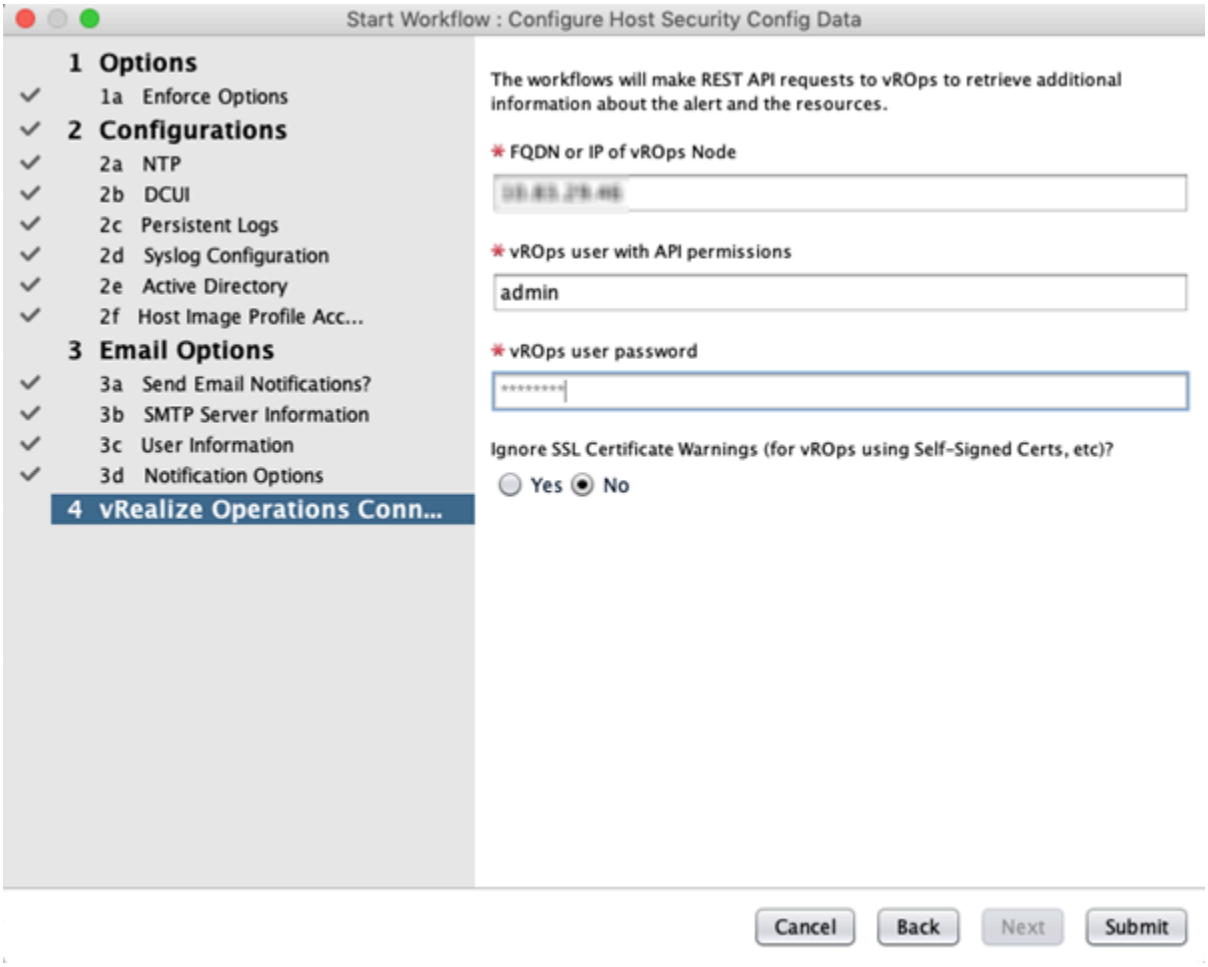
1. Add the *Apply Host Security Configuration Rules* out of the box workflow to the Orchestrator server. For more information, see Import Workflow.
2. Log in to Orchestrator to enforce the hardening rules in Orchestrator.
3. Run the *Configure Host Security Config Data* workflow to configure the **Host Security Hardening Configuration** and **Email Config** attributes.
4. In the **Enforce Options**, you can choose to selectively enforce rules by configuring the configuration elements.
   a) Under Configurations, set the NTP values.



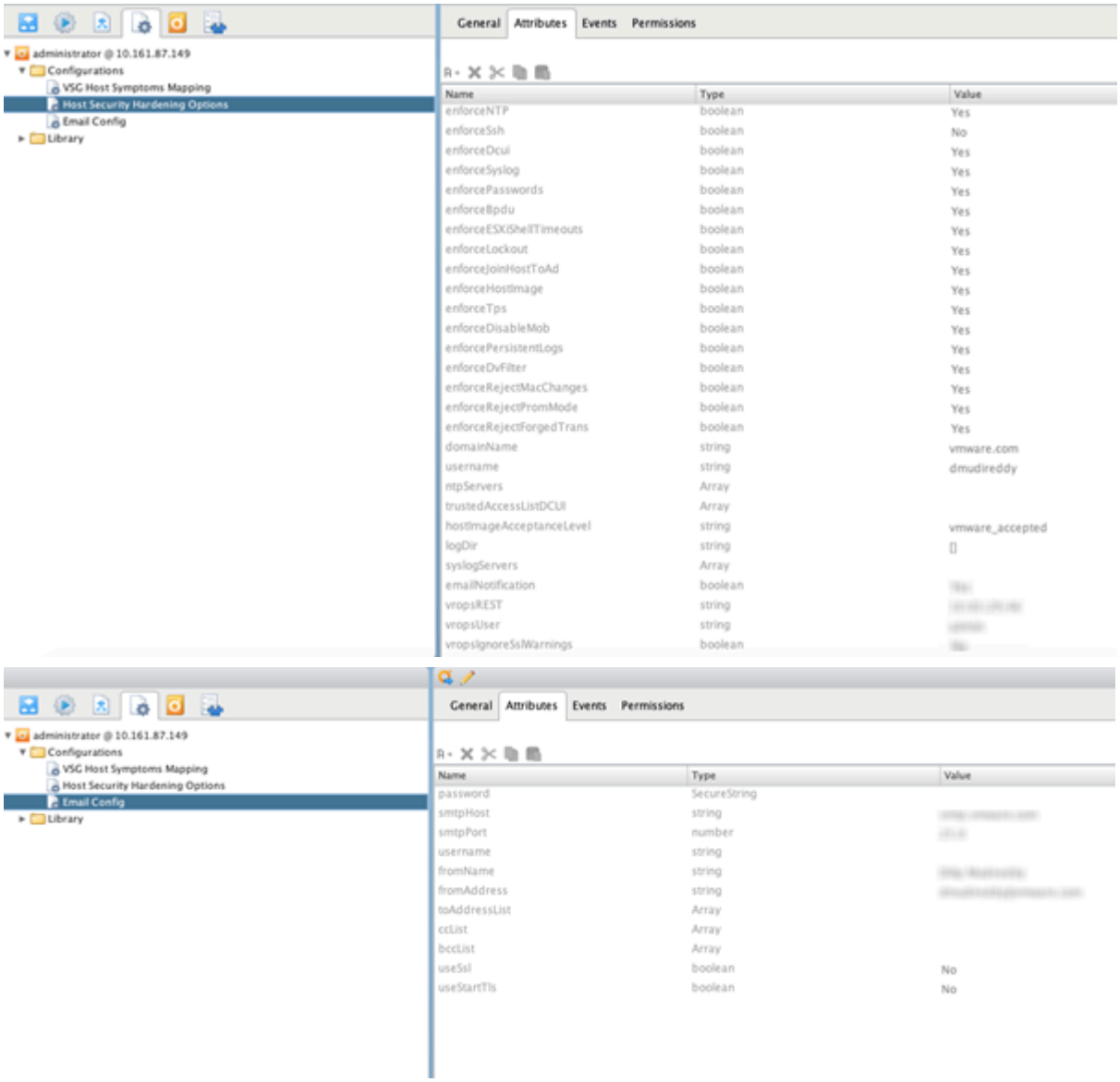
b) Under Email Options, configure **Send Email Notifications?**



c) Under VCF Operations Connections, configure the connection and login credentials.




After executing the workflow, both the **Host Security Hardening Options** and **Email Config** configurational elements are filled with attribute values that are enforced.




5. After setting the options to enforce hardening rules, enforce them automatically through auto remediation feature of Management Pack for Orchestrator
   a) Go to **Configure** > **Policies**.
   b) Select the policy currently active, and then click **Edit Policy** on the right pane.
   c) Select the **Alerts and Symptoms** tab, and then search for the alert, ESX *Host is violating VMware vSphere Security Configuration Guide for vCenter version 6.5 or 6.7*.
   d) Select the alert, and then in the **Automate** column, change the value to **Enabled**.
   e) Select the **Apply Policy to Groups** tab on the left pane, and then select the **VCF Automation Orchestrator Adapter Managed Objects** checkbox in the panel that appears on the right.

When any new ESX *Host is violating VMware vSphere Security Configuration Guide for vCenter version 6.5 or 6.7* alert is triggered, Management Pack for Orchestrator will automatically invoke the action mapped to it. If there is an existing alert, you can manually run the action against that alert. For any new alerts, the action automatically gets triggered in the next collection cycle. The recipient you configured earlier will receive an email after the action is successfully executed.

### Creating Workflow Actions on vCenter Resources for Custom Groups

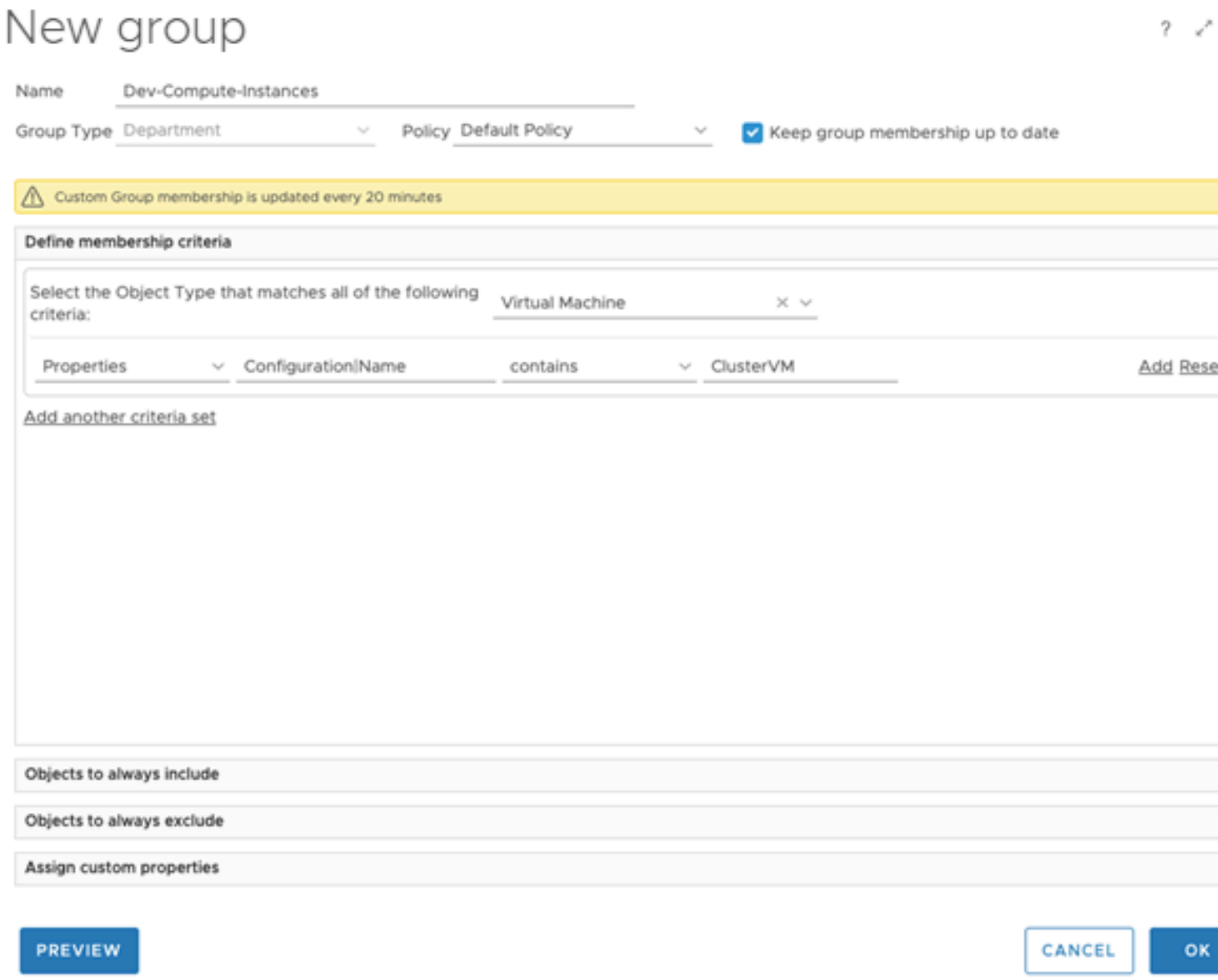
You can run a custom workflow as an action on custom groups. The custom groups can be Resource Type and Target Type which is the actual vCenter object the workflow is targeted for. This workflow accepts an array of vCenter objects as the input parameter.

Example: Scaling Out Group of Virtual Machine Memory

Create and import the custom workflow 'Scale VM Group Memory' in the Management Pack for Orchestrator. This workflow accepts an array of Virtual Machines as the input parameter (Array/VC: Virtual Machine).

The following example is to scale out the group of Virtual Machine memory that is part of the custom group 'Dev-Compute-Instances' and of the Group Type 'Department'.

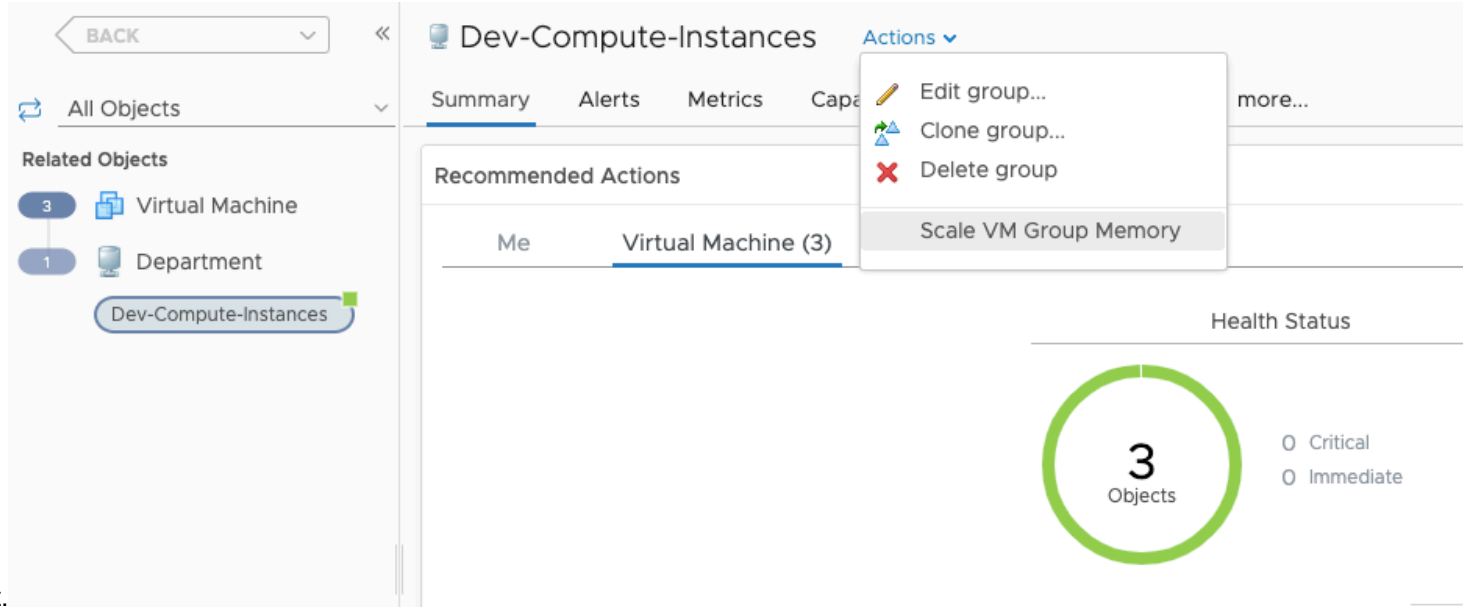
1. Log in to VCF Operations.
2. Click **Environment**, and then click **Custom Groups**.
3. Click **Add**. Provide the following details to create a custom group.
   - **Name**: Provide a name for the group. For example, 'Dev-Compute-Instances'.
   - **Description**: Provide a description for the group name.
   - **Group Type**: Select the type as **Department**.
   - **Policy**: Select the policy as **Default Policy**.
   - **Select the Object Type that matches all of the following criteria**: Select **Virtual Machine** from the list.



**Note:**
All resources must be of the same Object Type and must belong to the same vCenter. In this example, Virtual Machine has to be the Object Type and you cannot select Virtual Machine from multiple vCenters.

4. Click **OK** to create a custom group.
5. From the left menu, click **Environment**, and then click **Inventory**.
6. Select **Adapter Instances**, and then select **VCF Automation Orchestrator Adapter Instance** from the list of adapter instances.
7. Select the Orchestrator adapter instance that contains the workflow that you want to run.
   A list of all the workflows are displayed on the right pane.
8. From the right pane, select the workflow that you want to add to a resource.
9. Click the **Actions** icon and then **Create/Modify Workflow Action on vCenter Resources**.

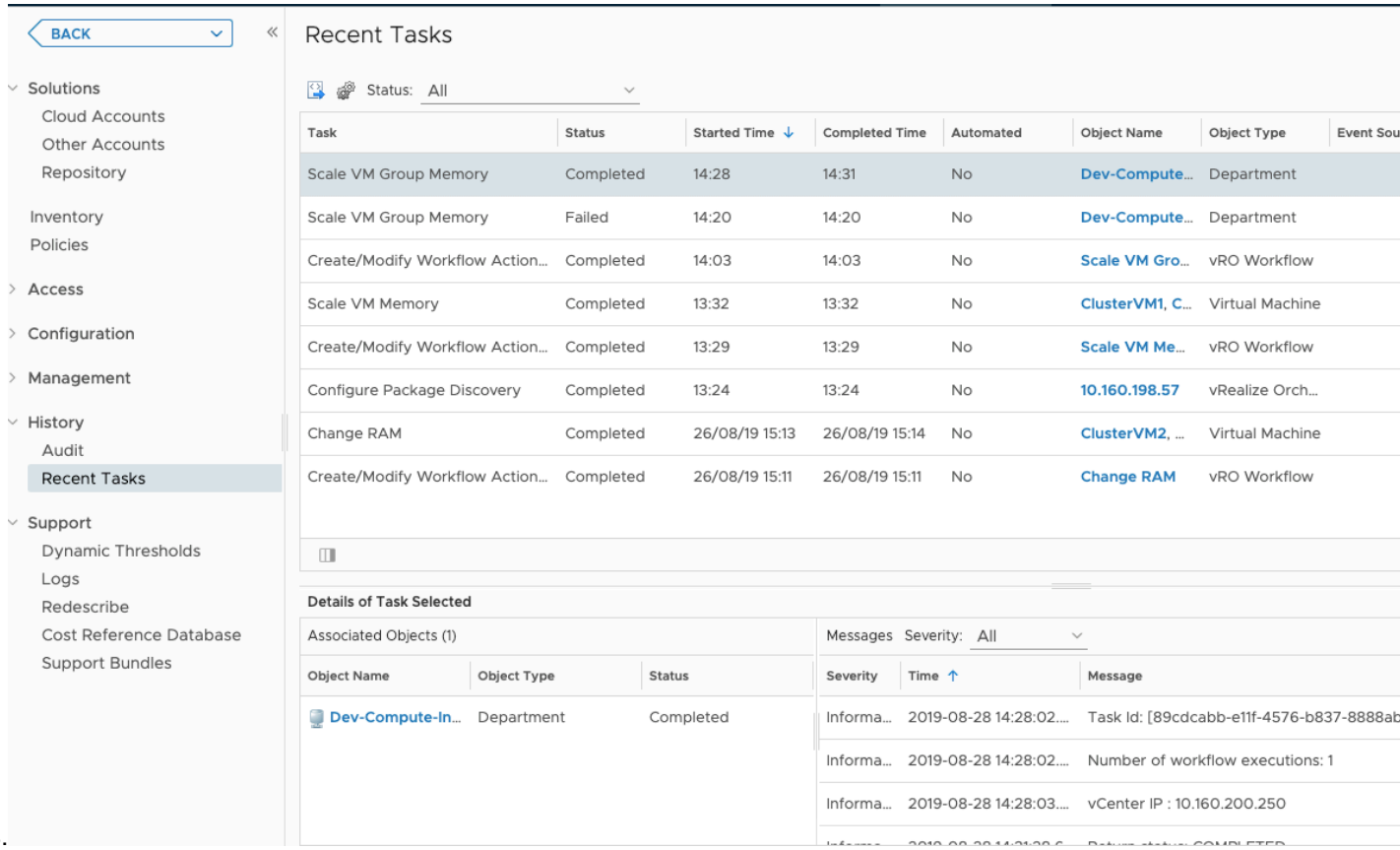
   **Note:**
   Ensure that there is at least one input parameter in the workflow, otherwise the action fails when it is run.
10. Select the **Resource Type** as **Department** and **Target Resource Type** as **Virtual machine**.
11. Click **Begin Action**.
12. From the left menu, click **Environment**, and then in the right pane, select the custom group 'Dev-Compute-Instances' that you created.
13. Click the **Actions** icon, and then select the custom workflow **Scale VM Group Memory** from the list.




   The Virtual Machines are listed as comma-separated values.

14. Enter the memory that you want to scale and click **Begin Action**.

    The workflow is run for the array of Virtual Machines that are part of the listed group.
    To view the status of the workflow, click **Administration**, and then click **Recent**




**Tasks**.

Supported Combinations for Custom Groups

| Resource Type | Target Resource Type |
|---|---|
| Department | VirtualMachine |
| Department | HostSystem |
| Department | Datastore |
| Department | VmwareDistributedVirtualSwitch |
| Department | DistributedVirtualPortgroup |
| Environment | VirtualMachine |
| Environment | HostSystem |
| Environment | Datastore |
| Environment | VmwareDistributedVirtualSwitch |
| Environment | DistributedVirtualPortgroup |
| Function | VirtualMachine |
| Function | HostSystem |
| Function | Datastore |
| Function | VmwareDistributedVirtualSwitch |
| Function | DistributedVirtualPortgroup |
| Location | VirtualMachine |

| Resource Type | Target Resource Type |
|---|---|
| Location | HostSystem |
| Location | Datastore |
| Location | VmwareDistributedVirtualSwitch |
| Location | DistributedVirtualPortgroup |
| Security Zone | VirtualMachine |
| Security Zone | HostSystem |
| Security Zone | Datastore |
| Security Zone | VmwareDistributedVirtualSwitch |
| Security Zone | DistributedVirtualPortgroup |
| Service Level Objective | VirtualMachine |
| Service Level Objective | HostSystem |
| Service Level Objective | Datastore |
| Service Level Objective | VmwareDistributedVirtualSwitch |
| Service Level Objective | DistributedVirtualPortgroup |

### Management Pack for Orchestrator Dashboard

Access the Management Pack for Orchestrator Dashboard

The Management Pack for Orchestrator dashboard in the VCF Operations displays information that helps you monitor workflows.

1. From the left menu of VCF Operations, click Orchestrator Workflows Overview.

   The Orchestrator Workflows Overview dashboard displays the following widgets:

   - Orchestrator Adapter Instances
   - Orchestrator Instance Properties
   - vRO Health
   - Top-10 Workflows
   - Number of Successful and Failed Executions
   - Packages
   - Workflows
   - Number of Execution
   - Workflow Input Parameter

2. Click **Actions** > **Set as Home Page**, to set this dashboard as one of the default dashboards in the **Quick Start** page.

OrchestratorWorkflows Overview Dashboard

The Orchestrator Workflows Overview dashboard displays information about the Orchestrator adapter and workflows.

The Orchestrator Workflows Overview dashboard displays the following widgets by default.

#### Orchestrator Adapter Instances

You can view a list of Orchestrator adapters that you have configured in this widget. The widget displays the name of the adapter, adapter type, object type, policy applied, collection state, and collection status. Click on an adapter instance to view the properties of the adapter in the Orchestrator Instance Properties widget.

#### Orchestrator Instance Properties

Displays the properties of the Orchestrator adapter instances. You can view the adapter instance name and Orchestrator host settings.

**vRO Health**

Displays the status of the health-related alerts for the Orchestrator adapter instance. The state of the badge is based on your alert definitions. Click the badge to see the **Summary** tab for objects or groups configured in the widget. From the **Summary** tab you can begin determining what caused the current state. If the widget is configured for an object that has descendants, you should also check the state of descendants. Child objects might have alerts that do not impact the parent.

**Top 10 Executed Workflows**

Displays a count of the top 10 most frequently run workflows. This count includes workflows that have executed both successfully and unsuccessfully.

**Number of Successful and Failed Executions**

Displays the number of success and failed workflow executions. You can optionally modify the workflow execution metric collection time.

1. Log in to the VCF Operations instance.
2. Navigate to the properties file located in the path: `/usr/lib/vmware-vcops/user/plugins/inbound/VMwareAriaAutomationOrchestratorAdapter3/conf /Orchestrator.properties`.
3. Edit the Orchestrator.properties file and set the value for the key `WORKFLOW_EXEXCUTION_METRIC_COLLECTION_INTERVAL = 720`
   This value affects the **Number of Success and Failed Executions** widget in the Orchestrator Workflows Overview dashboard. The value you specify here decides the rate/frequency at which the workflow execution success/failure metrics will be updated in VCF Operations. The default value is 720 minutes.

**Packages**

Displays a data grid with objects in the inventory. The default configuration of the data grid appears in Object List Widget Options section. You can customize it by adding or removing default columns.

**Workflows**

Displays a data grid with list of Orchestrator workflows.

**Number of Execution**

Displays a metric chart of the workflow that you selected in the **Top-10 Executed Workflows** widget. You can use this information to monitor the workload of your workflows over time

**Workflow Input Parameters**

Displays the input parameters which was provided for executed workflows.

### Troubleshooting in Management Pack for Orchestrator

An Error Occurs When You Log In After Upgrading Orchestrator from 8.7 to 8.8

From Orchestrator 8.8 onwards, the basic authentication of your Orchestrator deployment is disabled by default. In certain use cases you must enable this authentication by setting the `com.vmware.o11n.sso.basic-authentication.enabled` system property.

You can enable basic authentication for your Orchestrator deployment by setting a system property.

1. Log in to the Control Center as **root**.
2. Select **System Properties**.
3. Click **New**.
4. Under **Key**, enter `com.vmware.o11n.sso.basic-authentication.enabled`.

5. Under **Value**, enter `true`.
6. (Optional) Enter a description for the new system property.
7. Click **Add**, and wait for the Orchestrator server to restart.

**Appendix**

## Out of the Box Orchestrator Workflows

The Management Pack for Orchestrator ships with out of the box Orchestrator workflows. These workflows help you remediate alerts in VCF Operations. They are mapped to alerts from recommendations. When an object displays an alert given here, you can run the workflow from the alert.

### Out of the Box Workflows for Custom Recommendations

Out of the box workflows for VM:

- VM Power Operations
- Reconfigure VM
- Move VM
- Manage Snapshots for VM
- Upgrade Guest Tools for VM
- Migrate Virtual Machine with vMotion
- Add a new vNIC to Virtual Machine
- Update CPU Reservation
- Update Memory Reservation
- Remove or Delete unused Template

Out of the box workflows for Host:

- Host Maintenance Mode
- Host Power Operations
- Decomission Host
- Reconnect Host. You can automate this workflow to run when an alert is displayed.
- Apply Host Security Configuration Rules

### Out of the Box Workflows for Default Recommendations

You can map out of the box workflows to recommendations already in VCF Operations. The following table lists the default recommendations and the alerts for which these recommendations are applicable:

| Out of the Box Workflow/Action Name | Alert in VCF Operations | Recommendation in VCF Operations |
|---|---|---|
| Upgrade Guest Tools for VM | Virtual Machine has unsupported VMware Tools. | If the VMware tools version is too new, downgrade it to a supported version. Otherwise, upgrade VMware Tools to the latest version. Select this virtual machine in the vSphere Client or Web Client, then right-click and select "Upgrade VMware Tools". |
| VM Power Operation | vSphere HA failed to restart a network isolated virtual machine | Manually power on the virtual machine. |
| VM Power Operation | vSphere HA cannot perform a failover operation for the virtual machine | You can also try to power on the virtual machine and investigate any returned errors. |

| Out of the Box Workflow/Action Name | Alert in VCF Operations | Recommendation in VCF Operations |
|---|---|---|
| Reconnect Host | Host has lost connection to vCenter Server | Click "Open Host in vSphere Web Client" in the Actions menu at the top of Alert details page to connect to the vCenter managing this host and manually reconnect the host to vCenter Server. After the connection to the host is restored by vCenter Server, the alert will be canceled. |
| Set CPU Count for VM | Virtual Machine CPU usage is at 100% for an extended period of time. | Add more CPU capacity to this virtual machine. |
| Migrate Virtual Machine using vMotion. | Virtual Machine has CPU contention due to memory page swapping in the host | Use vMotion to migrate this virtual machine to a different host or cluster |
| Migrate Virtual Machine using vMotion. | Virtual machine has memory contention due to memory compression, ballooning, or swapping | Use vMotion to migrate this virtual machine to a different host or cluster. |
| Move VM | Virtual machine is demanding more CPU than its entitlement | Move this virtual machine to a different host or cluster |
| Move VM | Virtual machine in a DRS cluster is demanding more CPU than its entitlement | Move this virtual machine to a different host or cluster |
| Move VM | Virtual machine in a cluster is demanding more CPU than its entitlement | Move this virtual machine to a different host or cluster |
| Manage Snapshots for VM | Datastore is running out of disk space. | Delete unused snapshots for datastore |
| Manage Snapshots for VM | Virtual machine disk I/O read latency is high | If the virtual machine has multiple snapshots, delete unused snapshots for VM |
| Manage Snapshots for VM | Virtual machine disk I/O write latency is high | If the virtual machine has multiple snapshots, delete unused snapshots for VM |
| Manage Snapshots for VM | Virtual machine has disk I/O latency problem caused by snapshots | If the virtual machine has multiple snapshots, delete unused snapshots for VM |
| Manage Snapshots for VM | Virtual machine is running on snapshots for more than 2 days | If the virtual machine has multiple snapshots, delete unused snapshots for VM |
| Host Maintenance Mode | Host is in maintenance mode for at least 72 hours | If the time period of maintenance window is expected, you can cancel this alert. Otherwise, investigate further and take the host out of maintenance mode. |
| Host Power Operation | N/A | N/A |
| Decommission Host | N/A | N/A |
| Add a new vNIC to Virtual Machine | N/A | Add an additional NIC to the VM to spread the packet load more evenly |
| Update CPU Reservation | N/A | If the virtual machine CPU reservation is set, Decrease the CPU reservation configuration |
| Update Memory Reservation | N/A | If the virtual machine Memory reservation is set, Decrease the Memory reservation configuration |
| Remove or Delete unused Template | N/A | N/A |

Using the Input Parameter Data Type for Guest Operating System Identifier

When you use a workflow which as an input parameter data type `vc:VirtualMachineGuestOSIdentifier(VMGuestOS)`, you can provide the index number which corresponds to the

VM's guest Operating System name. Use the index number from the table. For example, to execute the Create Custom Virtual Machine workflow to create a DOS Operating System, you provide the input parameter in the following format: `vc:VirtualMachineGuestOSIdentifier(1).`

**Index Numbers for Guest Operating System Identifier**

| Index | Virtual Machine Guest Operating System Name |
|---|---|
| 1 | win2000ProGuest |
| 2 | win2000ServGuest |
| 3 | win2000AdvServGuest |
| 4 | winXPProGuest |
| 5 | winXPPro64Guest |
| 6 | winNetEnterpriseGuest |
| 7 | winNetDatacenterGuest |
| 8 | winNetStandardGuest |
| 9 | winNetWebGuest |
| 10 | winNetBusinessGuest |
| 11 | winNetEnterprise64Guest |
| 12 | winNetDatacenter64Guest |
| 13 | winNetStandard64Guest |
| 14 | winVistaGuest |
| 15 | winVista64Guest |
| 16 | winLonghornGuest |
| 17 | winLonghorn64Guest |
| 18 | windows7Guest |
| 19 | windows7_64Guest |
| 20 | Windows7Server64Guest |
| 21 | windows8Guest |
| 22 | windows8_64Guest |
| 23 | windows8Server64Guest |
| 24 | windows9Guest |
| 25 | windows9_64Guest |
| 26 | windows9Server64Guest |
| 27 | windows2019srv_64Guest |
| 28 | windows11_64Guest |
| 29 | windows2019srvNext_64Guest |
| 30 | windows12_64Guest |
| 31 | windows2022srvNext_64Guest |
| 32 | asianux3Guest |
| 33 | asianux3_64Guest |
| 34 | asianux4Guest |
| 35 | asianux4_64Guest |
| 36 | asianux5_64Guest |

| Index | Virtual Machine Guest Operating System Name |
|---|---|
| 37 | asianux6_64Guest |
| 38 | asianux7_64Guest |
| 39 | asianux8_64Guest |
| 40 | asianux9_64Guest |
| 41 | centosGuest |
| 42 | centos64Guest |
| 43 | centos6Guest |
| 44 | centos6_64Guest |
| 45 | centos7_64Guest |
| 46 | centos8_64Guest |
| 47 | centos9_64Guest |
| 48 | coreos64Guest |
| 49 | debian6Guest |
| 50 | debian6_64Guest |
| 51 | debian7Guest |
| 52 | debian7_64Guest |
| 53 | debian8Guest |
| 54 | debian8_64Guest |
| 55 | debian9Guest |
| 56 | debian9_64Guest |
| 57 | debian10Guest |
| 58 | debian10_64Guest |
| 59 | debian11Guest |
| 60 | debian11_64Guest |
| 61 | oracleLinuxGuest |
| 62 | oracleLinux64Guest |
| 63 | oracleLinux6Guest |
| 64 | oracleLinux6_64Guest |
| 65 | oracleLinux7_64Guest |
| 66 | oracleLinux8_64Guest |
| 67 | oracleLinux9_64Guest |
| 68 | rhel5Guest |
| 69 | rhel5_64Guest |
| 70 | rhel6Guest |
| 71 | rhel6_64Guest |
| 72 | rhel7Guest |
| 73 | rhel7_64Guest |
| 74 | rhel8_64Guest |
| 75 | rhel9_64Guest |
| 76 | fedoraGuest |

| Index | Virtual Machine Guest Operating System Name |
|---|---|
| 77 | fedora64Guest |
| 78 | opensuseGuest |
| 79 | opensuse64Guest |
| 80 | sles10Guest |
| 81 | sles10_64Guest |
| 82 | sles11Guest |
| 83 | sles11_64Guest |
| 84 | sles12Guest |
| 85 | sles12_64Guest |
| 86 | sles15_64Guest |
| 87 | sles16_64Guest |
| 88 | ubuntuGuest |
| 89 | ubuntu64Guest |
| 90 | rockylinux_64Guest |
| 91 | coreos64Guest |
| 92 | otherlinuxguest |
| 93 | otherlinux64guest |
| 94 | other4xLinuxGuest |
| 95 | other4xLinux64Guest |
| 96 | other5xLinuxGuest |
| 97 | other4xLinux64Guest |
| 98 | darwin14_64Guest |
| 99 | darwin15_64Guest |
| 100 | darwin16_64Guest |
| 101 | darwin17_64Guest |
| 102 | darwin18_64Guest |
| 103 | darwin19_64Guest |
| 104 | darwin20_64Guest |
| 105 | darwin21_64Guest |
| 106 | darwin22_64Guest |
| 107 | dosGuest |
| 108 | eComStationGuest |
| 109 | eComStation2Guest |
| 110 | freebsdGuest |
| 111 | freebsd64Guest |
| 112 | solaris10Guest |
| 113 | solaris10_64Guest |
| 114 | solaris11_64Guest |
| 115 | vmwarePhoton64Guest |
| 116 | vmkernel7Guest |

| Index | Virtual Machine Guest Operating System Name |
|---|---|
| 117 | vmkernel8Guest |
| 118 | vmkernel9Guest |

Location of Log Files

Use the log files to view information about the Orchestrator adapter and workflows. You can access the log files from VCF Operations or by accessing the machine where you installed VCF Operations via SSH.

#### Access Log Files from Orchestrator

To access the log files for the collection cycle and status of the Orchestrator adapter, from the left menu, click **Administration**, and then in the left pane, click **Support** > **Logs**. Expand the node and the **COLLECTOR** folder. Expand the **adapters** folder and look for the VCF Automation Orchestrator Adapter_number.log file in the **VCF Automation OrchestratorAdapter** folder.

To access the log files for actions and workflows executed, open the VC Automation OrchestratorAdapter_0.log file.

#### Access Log Files via SSH

After you log in to the VCF Operations machine, access the log files in the following locations:

**Table 1665:**

| File | Location |
|---|---|
| Log location folder for Orchestrator adapter | /storage/log/vcops/log/adapters/VMware Aria Automation OrchestratorAdapter |
| Log for collection cycle and status of Orchestrator adapter | /storage/log/vcops/log/adapters/VMware Aria Automation OrchestratorAdapter/VMware Aria Automation OrchestratorAdapter_Number.log |
| Log for Orchestrator adapater actions and workflows executed | /storage/log/vcops/log/adapters/VMware Aria Automation OrchestratorAdapter/VMware Aria Automation OrchestratorAdapter_0.log |

## Management Pack for VMware Identity Manager

The *Management Pack for VMware Identity Manager* guide provides information about installing and configuring the Management Pack for VMware Identity Manager. The Management Pack for VMware Identity Manager allows you to retrieve alerts and findings from VMware Identity Manager to VCF Operations.

#### Intended Audience

The information in this guide is intended for anyone who wants to install and configure the Management Pack for VMware Identity Manager.

**Installing the Management Pack for VMware Identity Manager**

The Management Pack for VMware Identity Manager consists of a PAK file that contains default dashboards for the objects that the solution identifies.

To install the management pack, download the PAK file from VCF Solutions Catalog to a temporary folder on your local system. For more information, see **Installing and Configuring Integrations in VCF Operations**.

Configure an adapter instance for the management pack.

**Configuring Management Pack for VMware Identity Manager**

After you install the management pack, configure an instance for Management Pack for VMware Identity Manager.

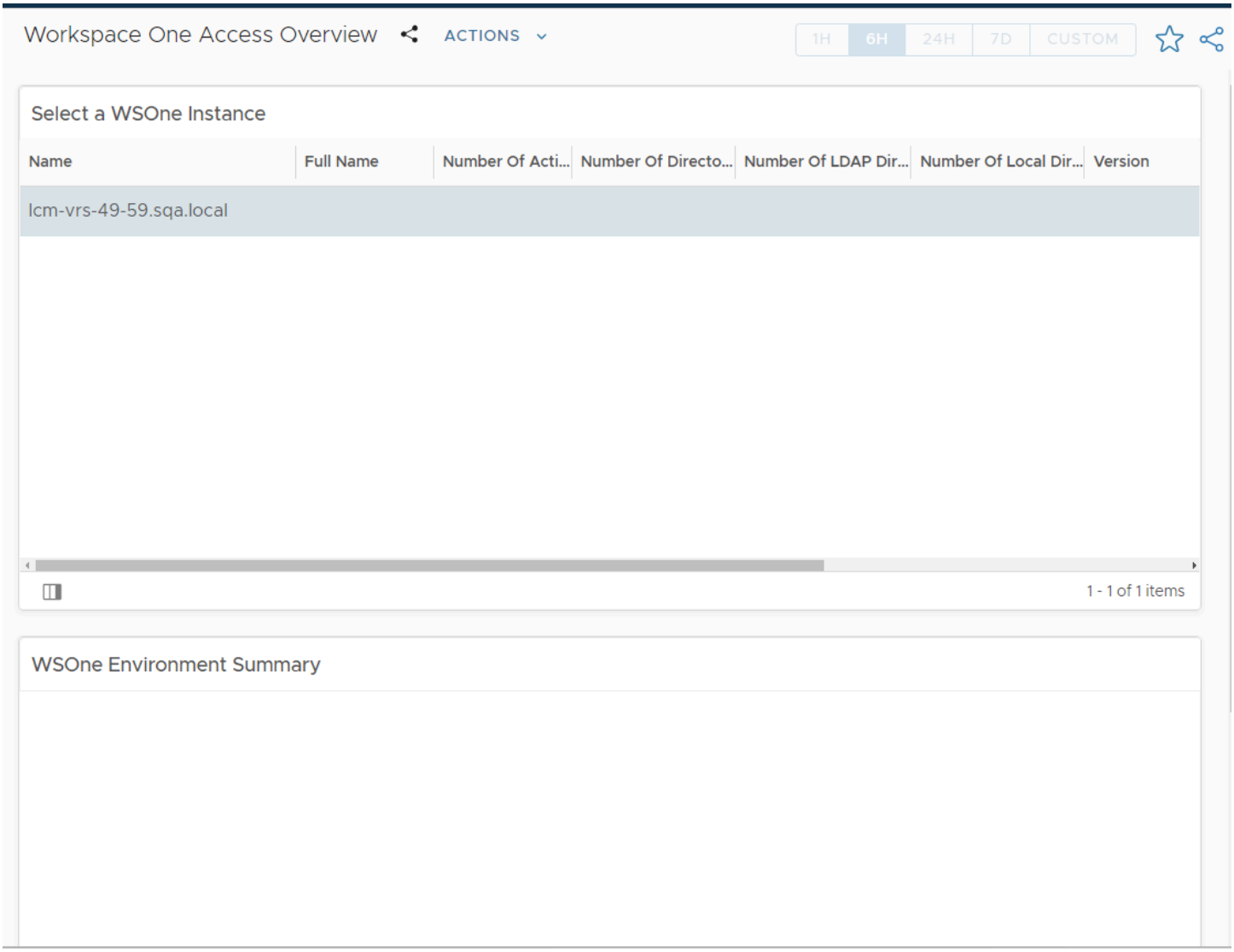
1. In the menu, select Data Sources, and in the left pane select **Integrations > Accounts**.
2. On the Accounts tab, click **Add**.
3. On the Accounts Types page, select **VMware Identity Manager Adapter**.
4. Configure the adapter instance.

| Option | Description |
|---|---|
| **Display Name** | The name for the adapter instance. |
| **Description** | (Optional) The description of the adapter instance. |
| **WSOne Host** | Enter the FQDN, or the IP address. |
| **Credential** | Select the credential you want to use to sign on to the environment from the drop-down menu. To add new credentials to access the VMware Identity Manager environment, click the plus sign.<br>• **Credential Name**. The name by which you are identifying the configured credentials.<br>• **User Name**. VMware Identity Manager local admin of system domain user name.<br>• **Password**. VMware Identity Manager local admin of system domain password. |

5. Enter **Collector/Group** to manage the management pack on VCF Operations. If you have only one account, select Default collector group.
6. Enter the connection information as required to configure the instance of the management pack. Each management pack has unique settings, configure the settings as prompted to configure that instance of the management pack. For more information on individual management packs, see List of Management Packs.
7. Click **VALIDATE CONNECTION** to initiate the authentication request to the CSP authentication service.
8. Click **ADD**.

Verify that the adapter is configured and collecting data by viewing application-related data.

**Dashboards in the Management Pack for VMware Identity Manager**
You can use the **Management Pack for VMware Identity Manager Overview** dashboard to view the environment summary for apps, groups, devices, users, activity, and health for your VMware Identity Manager instance.




**Metrics in the Management Pack for VMware Identity Manager**
The Management Pack for VMware Identity Manager collects metrics for objects within its plug-in.

**Table 1666: VMware Identity Manager Health Metrics**

| Resource | Metric Key | Metric Value |
|---|---|---|
| Workspace ONE Cluster Node | • Node Name Name<br>• Node UUID<br>• Version<br>• Node Status<br>• Node IP<br>• Time Zone<br>• Health | • String<br>• String<br>• String<br>• String<br>• String<br>• String<br>• String |
| Cert Proxy Certificate | • Issuer<br>• Subject<br>• Start Date<br>• End Date<br>• Port | • String<br>• String<br>• String<br>• String<br>• String |

| Resource | Metric Key | Metric Value |
|---|---|---|
| Certificate | • Issuer<br>• Subject<br>• Start Date<br>• End Date<br>• Port | • String<br>• String<br>• String<br>• String<br>• String |
| Cert Proxy | • Status<br>• Summary<br>• Port Type | • String<br>• String<br>• String |
| Disk Space | • Status<br>• Summary | • String<br>• String |
| AirWatch API Server | • Status<br>• Summary | • String<br>• String |
| Connector | • Status<br>• Summary | • String<br>• String |
| Configurator | • Status<br>• Summary | • String<br>• String |
| ACS Health | • Status<br>• Summary | • String<br>• String |
| WSOne FQDN | • Status<br>• Summary | • String<br>• String |
| Port Connectivity | Status | String |
| Database Connection | Status | String |
| Analytics Connection | Status | String |
| Messaging Connection | Status | String |
| EhCache Cluster Diagnostics | Status | String |
| Elasticsearch Health | • Status<br>• Master Node<br>• Indices Count<br>• Docs Count<br>• Unassigned Shared<br>• Cluster Nodes List | • String<br>• String<br>• Integer<br>• Integer<br>• String<br>• String |
| RabbitMQ | • Status<br>• Name<br>• Number of Queue | • String<br>• String<br>• String |
| Application Manager | • Status<br>• Summary | • String<br>• String |
| Directory System | • Number of Synced Users<br>• Domain<br>• Number of Synced Groups<br>• Health<br>• UUID<br>• Name<br>• Type<br>• Number of Alerts<br>• Last Sync Time<br>• Sync Status | • Float<br>• String<br>• Float<br>• Float<br>• String<br>• String<br>• String<br>• Float<br>• String<br>• String |

| Resource | Metric Key | Metric Value |
|---|---|---|
| VMware Workspace ONE Access Adapter | • Health<br>• Total Users<br>• Total Groups<br>• Total Apps<br>• Total Devices<br>• All Activity<br>• Number of Unique User Logins<br>• Total Logins<br>• Users Added<br>• Users Removed<br>• Users Updated<br>• Groups Added<br>• Groups Removed<br>• Groups Updated<br>• Number Running Directories<br>• Number IDAP Directories<br>• Number Active Directories<br>• Number Local Directories<br>• WSOne UUID<br>• WSOne Full Name<br>• Version<br>• Status<br>• Time Zone<br>• Health | • String<br>• Float<br>• Float<br>• Float<br>• Float<br>• Float<br>• Float<br>• Float<br>• Float<br>• Float<br>• Float<br>• Float<br>• Float<br>• Float<br>• Float<br>• Float<br>• String<br>• Float<br>• Float<br>• Float<br>• String<br>• String<br>• String<br>• String |

### Alerts in the Management Pack for VMware Identity Manager

The following alerts are triggered when any of the monitoring resources in VMware Identity Manager displays an unexpected behavior.

**Table 1667: Alerts in the Management Pack for VMware Identity Manager**

| Alert | Description |
|---|---|
| Health of WSOne Instance is degraded. | Triggered when the WSOne VM is down. |
| Health of WSOne ACSHealth-ApplicationDeployment is Low. | Triggered when the WSOne ACSHealth-ApplicationDeployment service is down. |
| Health of Application Deployment is Critically Low. | Triggered when the Application Deployment service is down. |
| WSOne Analytics Connection Test Connection failed. | Triggered when the Analytics service is down. |
| WSOne Database Test connection failed. | Triggered when the Database service is down. |
| Health of WSOne Disk Space is Critically Low. | Triggered when the Disk Space on WSOne VM is low. |
| WSOne Messaging Server Test Connection failed. | Triggered when the Messaging Server service is down. |
| Health of WSOne Application Manager-Application Deployment is Critically Low. | Triggered when the Application Manager-Application Deployment service is down. |
| Health of WSOne AirWatch API Server is Critically low. | Triggered when the Manager-Application service is down. |
| Health of WSOne Certificate is not good , look like it is Expired. | Triggered when WSOne Certificate has expired or deleted. |
| Health of WSOne RabbitMQ Service is Critically Low. | Triggered when the RabbitMQ service is down. |
| Health of WSOnePort Connectivity is Critically Low. | Triggered when the Port Connectivity service is down. |
| Health of FQDN WSOne Server is critically low. | Triggered when the FQDN service is down. |

| Alert | Description |
|---|---|
| Health of WSOne Configurator-ApplicationDeployment is Low. | Triggered when the Configurator-ApplicationDeployment service is down. |
| Health of WSOne Elasticsearch is Red. | Triggered when the Elasticsearch service is down. |
| Health of WSOne Connector-ApplicationDeployment is Critically Low. | Triggered when the Connector-ApplicationDeploymen service is down. |

## Management Pack for VCF Operations HCX

The Management Pack for VCF Operations HCX enables seamless monitoring and centralized management of your HCX Manager instances directly from VMware Cloud Foundation (VCF) Operations.

### Key Features and Benefits

- Simplified Administration: Streamlines daily administrative tasks, including certificate management, password updates, and password policy enforcement.
- Comprehensive Data Visualization: Collects and aggregates critical metrics, change events, and resource topology from across your fleet of HCX Manager instances, visualizing them directly within VCF Operations dashboards.
- Accelerated Troubleshooting: Empowers administrators to rapidly detect, isolate, and resolve performance bottlenecks within HCX Interconnects and active migration workloads.
- Unified User Experience: Delivers a consistent, end-to-end monitoring interface across both your source and destination vSphere environments.

### System Requirements

Before installing and configuring the Management Pack, ensure your system meets the following requirements.

For compatibility between products, please refer to:

VMware Cloud Foundation Operations Interoperability for HCX.

| | VCF Operations HCX Requirements |
|---|---|
| Connection | The Management Pack collects data via a REST API |
| Credentials | vCenter username and password |

### Installing the Management Pack

The Management Pack for VCF Operations HCX is an add-on integration that can be installed in one of the following ways:

- On top of the core VCF Operations platform using a PAK file available through Customer Connect. See, Installing Integrations in a Disconnected Environment.
- Through the the VCF Operations UI. See, Installing and Configuring Management Packs in VMware Cloud Foundation Operations from the Solutions Catalog.

Prerequisites

- Verify that you have access to the latest version of the management pack and the management pack is compatible with your version of VCF Operations.
- Ensure that you have the privileges to install a management pack.

### Configuring the Management Pack

#### Configure HCX Manager User and Privileges

Before registering with VCF Operations, you must create a user with the appropriate local permissions.

1. Log in to the corresponding vCenter Server and create a new user account for the deployed HCX Manager.

2. Add the user account to both the Migration Appliance Administrators and Migration Users groups in vCenter.
   This designated user account is responsible for password and certificate management, as well as the collection of migration metrics from the HCX Manager. Consequently, the assignment of both roles is mandatory.

Alternatively user privileges can be assigned by logging in to the Appliance Management UI, navigating to **Configuration** > **HCX Role Mapping and Edit**, and adding a user name or user groups to the Migration User and Migration Appliance Admin sections.

**Register HCX Manager in VCF Operations**

1. Log in to VCF Operations.
2. Navigate to **Operate** > **Administration** > **Integrations** and select **VCF Operations HCX**.
3. Click **Add Account**.
4. Enter the HCX Manager hostnam**e** and the appropriate user credentials created in the previous section, and click **Add**.
5. Select the appropriate Cloud Proxy.
6. Click **Add**.

After successful registration, the HCX instance should appear in the Certificates and Password screens under its specific domain.

While the Management Pack allows the registration of various HCX Managers, only those imported via VCF Operations are compatible with these automated password and certificate management tools.

**Validating Data Collection**

After you add an adapter instance, you will most likely want to validate the data that it collects in VCF Operations. Wait until one collection cycle is complete and the Collection Status for the adapter instance on the Solutions page shows as "Data Receiving".

1. Select the **Environment** navigation shortcut.
2. Under **Inventory Trees**, select **All Objects**, then expand the **HCX Adapter** objects list.
3. Select a resource from the list, then click the **All Metrics** tab to select metrics and validate values against the data source.

## Using the Management Pack

The Management Pack for VCF Operations HCX includes the following out-of-the-box features, harnessing the power of the VCF Operations user interface:

- Resource Kinds
- Dashboards
- Reports
- Alerts
- Metrics

## Resource Kinds

The Management Pack for VCF Operations HCX creates Resource Kinds and Relationships (associations) for critical resources in your VCF Operations HCX environment.

**Resource Kinds**

Performance, health, and availability data are collected for the following VCF Operations HCX resource kinds:

- Site Connection
- Migration
- vMotion
- Replication Assisted vMotion
- OS Assisted Migration

### Alerts

| Alert Name | Severity |
|---|---|
| Certificate has expired on HCX appliance | Critical |
| Certificate expiring in next 30 days on HCX appliance | Immediate |
| System Certificate expiring in next 30 days on HCX Interconnect (IX) Datapath Appliance / Network Extension (NE) Appliance | Automatic |
| System Certificate expired on HCX Interconnect (IX) Datapath Appliance / Network Extension(NE) Appliance | Automatic |
| Password has expired on HCX appliance | Critical |
| Password expiring in next 30 days on HCX appliance | Immediate |

## Management Pack for Microsoft SQL Server

The **Management Pack for Microsoft SQL Server** is an embedded adapter for VMware Cloud Foundation Operations, collecting performance data from your Microsoft SQL Server environment and providing predictive analytics and real-time information about problems in your infrastructure—all within the VCF Operations user interface.

### Notable Features

The Management Pack for Microsoft SQL Server includes the following notable features:

- **Comprehensive monitoring dashboards**
- **250+ collected metrics** for Microsoft SQL Server resources
- **30+**detailed alerts and notifications for critical and warning conditions
- **Capacity definitions** for connection limit, CPU demand, memory usage, and disk capacity
- **External relationships** from your VMware Virtual Machines to SQL Server Instance and Always On Availability Groups

### How the Management Pack Works

The Management Pack for Microsoft SQL Server connects to your **Microsoft SQL Server instance** via **JDBC** using the **jTDS driver.** SQL queries are executed in order to import health, performance, availability, capacity, and relationships data for your SQL server resources into VCF Operations.

### Getting Started

Wondering where to start? See System Requirements for details.

#### System Requirements for Management Pack for Microsoft SQL Server

Before installing and configuring the Management Pack, ensure your system meets the following requirements.

For compatibility between products, please refer to:

VMware Cloud Foundation Operations Interoperability for Microsoft SQL Server

-OR-

Third Party Solution Interoperability for Microsoft SQL Server

| | Microsoft SQL Server Requirements |
|---|---|
| Connection | **Standalone Configuration:**<br>• Hostname (or IPv4 address) of MS SQL Server instance<br>**Clustered Configuration:**<br>• **Virtual IP (VIP)**, or hostname for the VIP, of MS SQL Server instance<br>Default Instance Port: 1433 (TCP)<br>**Important:**<br>If using dynamic ports, you will want to change the default port to **SQL Server Browser Service UDP port 1434**. |
| Credentials | MS SQL User Name/Password (Active Directory or Local)<br>• See: Assigning MS SQL User Privileges for information about creating the least amount of permissions for a MS SQL monitoring user.<br>• If using SSL, see: Adding an SSL Certificate to the VMware Cloud Foundation Operations Truststore. |

**Important:**

The **CPU Usage** metric for **Microsoft SQL Server Instances** is not available on MS SQL Server **Standard** or **Express** editions.

Assigning MS SQL User Privileges

This topic contains the minimum MS SQL user permissions needed for an MS SQL Server Management Pack monitoring user. An automated script is provided below; it can help you create a local LPU user or can be modified for use with a domain user.

**Important:**

It is recommended to create your LPU prior to installing and configuring the Management Pack. You should use your own MS SQL credentials (login), user, and role names.

**Required Permissions and Role**

To create your MS SQL least-privileged user (LPU), you will need to complete the following general tasks:

1. Grant your MS SQL login credentials the following permissions:
   – **VIEW SERVER STATE**
   – **VIEW DATABASE STATE**
   – **VIEW ANY DEFINITION**
2. Next, to ensure all future-created databases are monitored, create a user that maps to your MS SQL credentials and has the**public** role on the following databases:
   – each**user**database
   – **msdb** database
   – **model** database

**Create Local LPU User Script**

Run the following script to automate creation of a local MS SQL Server LPU user with the permissions/role mentioned above:

```
use master;

CREATE LOGIN sql_lpu WITH PASSWORD = 'your-password';
CREATE USER sql_lpu FOR LOGIN sql_lpu;
```

```
GRANT CONNECT SQL to sql_lpu;
GRANT VIEW SERVER STATE to sql_lpu;
GRANT VIEW DATABASE STATE TO sql_lpu;
GRANT VIEW ANY DEFINITION TO sql_lpu;
GO

GRANT EXEC on [xp_readerrorlog] to sql_lpu
GO

use model;
CREATE USER sql_lpu FOR LOGIN sql_lpu;
GO

use msdb;
CREATE USER sql_lpu FOR LOGIN sql_lpu;
GRANT SELECT ON [dbo].[syssessions] TO sql_lpu
GRANT SELECT ON [dbo].[sysjobhistory] to sql_lpu
GRANT SELECT ON [dbo].[sysjobsteps] TO sql_lpu
GRANT SELECT ON [dbo].[sysjobs_view] TO sql_lpu
GRANT SELECT ON [dbo].[sysjobactivity] TO sql_lpu
ALTER ROLE [SQLAgentReaderRole] ADD MEMBER [sql_lpu]
-- For SQL Server 2008 R2, use: sp_addrolemember [SQLAgentReaderRole], [sql_lpu]


-- Goes through each user database and adds public permissions
DECLARE @name NVARCHAR(MAX)
DECLARE db_cursor CURSOR FOR
SELECT name
FROM MASTER.dbo.sysdatabases
WHERE name NOT IN ('master','model','msdb','tempdb')
OPEN db_cursor
FETCH NEXT FROM db_cursor INTO @name
WHILE @@FETCH_STATUS = 0
BEGIN
    EXECUTE('USE ' + @name + '; CREATE USER sql_lpu FOR LOGIN sql_lpu;' );
    FETCH NEXT FROM db_cursor INTO @name
END
CLOSE db_cursor
DEALLOCATE db_cursor
```

**Tables, Views, and Functions Used by Permission**

Permission: VIEW ANY DATABASE

master.dbo.sysdatabases

sys.databases

Permission: EXEC on [xp_readerrorlog]

master.dbo.xp_readerrorlog

Permission: GRANT SELECT ON [dbo].[sysjobactivity]

msdb.dbo.sysjobactivity

Permission: GRANT SELECT ON [dbo].[sysjobhistory]

msdb.dbo.sysjobhistory

Permission: GRANT SELECT ON [dbo].[sysjobs_view]

msdb.dbo.sysjobs_view

Permission: GRANT SELECT ON [dbo].[sysjobsteps]

msdb.dbo.sysjobsteps

Permission: GRANT VIEW SERVER STATE

sys.all_views

Permission: public

sys.allocation_units

sys.configurations

sys.database_files

sys.partitions

sys.sysusers

Permission: VIEW ANY DEFINITION

sys.availability_groups

sys.availability_replicas

sys.indexes

sys.internal_tables

sys.master_files

sys.objects

sys.syslogins

Permission: VIEW DATABASE STATE

sys.dm_db_index_physical_stats

Permission: VIEW SERVER STATE

sys.dm_db_index_usage_stats

sys.dm_db_missing_index_details

sys.dm_db_missing_index_group_stats

sys.dm_db_missing_index_groups

sys.dm_exec_query_stats

sys.dm_exec_requests

sys.dm_exec_sessions

sys.dm_exec_sql_text

sys.dm_hadr_availability_group_states

sys.dm_hadr_availability_replica_states

sys.dm_hadr_database_replica_states

sys.dm_io_virtual_file_stats

sys.dm_os_buffer_descriptors

sys.dm_os_cluster_nodes

sys.dm_os_performance_counters

sys.dm_os_process_memory

sys.dm_os_schedulers

sys.dm_os_sys_info

sys.dm_os_sys_memory

sys.dm_os_volume_stats

sys.dm_os_wait_stats

sys.dm_resource_governor_resource_pools

sys.sysprocesses

**What to do next**

Adding an SSL Certificate to the VMware Cloud Foundation Operations Truststore

Adding an SSL Certificate to the VMware Cloud Foundation Operations Truststore

For added security, we recommend adding an SSL certificate to the VCF Operations truststore for your MS SQL host server.

> **Note:**
>
> Different levels of security can be configured in the Management Pack. The **least secure configuration** is to set the **SSL Config** Advanced Setting to **No SSL.** The next level of security is to set **SSL Config** to **No Verify,** and the management pack will accept all SSL certificates. For the highest level of security, you can set **SSL Config** to **Verify,** and the Management Pack will have to verify the SSL certificate on your VCF Operations system. See: Creating an Adapter Instance .

**To add the SSL certificate to your VCF Operations truststore:**

1. Obtain the SSL certificate for your MS SQL host server from your Internet browser. Export the certificate as an X.509 Certificate (PEM).
2. Copy the certificate to your VCF Operations machine.
3. Use 'ssh' to log in to the VCF Operations machine as the root user, then run the following command:
   **Linux:**
   $VCOPS_BASE/jre/bin/keytool -import -alias <product_alias> -file /tmp/<certfile> -keystore "$VCOPS_DATA_VCOPS/user/conf/ssl/tcserver.truststore" -storepass `grep ssltruststorePassword /storage/vcops/user/conf/ssl/storePass.properties | sed s/ssltruststorePassword=//` -trustcacerts
   **Windows:**
   %VCOPS_BASE%\jre\bin\keytool -import -alias <product_alias> C:\path\to\certfile -keystore "%VCOPS_DATA_VCOPS%\user\conf\ssl\tcserver.truststore" -storepass <truststore_password> -trustcacerts
   **Parameter Descriptions:**
   – <product_alias> is a unique name for each key that you add (per host)
   – <certfile> is the location where the cert file was saved
4. Run the reboot command to re-start the VCF Operations machine for the changes to take effect.

**Troubleshooting the Management Pack**

Once you have finished Installing and Configuring the Management Pack , check the **Collection State** and **Collection Status** for the adapter instance on the **Solutions** page. If they show as **Not Collecting** or **No Data Receiving**, respectively, the following troubleshooting topics can help you diagnose and correct common problems with your adapter instance.

**Test Connection Errors**

When Creating an Adapter Instance , click **Test Connection** after entering all of your configuration settings to ensure VCF Operations can connect to the specified host.

The following errors are possible:

**Bad Microsoft SQL Server host name**

**Error Message**: Unable to establish a valid connection to the target system. Unknown server Host name "<hostname>". Check connection settings.

**Description:** Ensure the host name (or IPv4 address) provided when Creating an Adapter Instance is correct.

> **Important:**
>
> IPv6 addresses are not supported.

**Bad Microsoft SQL Server user name/password**

**Error Message**:Unable to establish a valid connection to the target system. Login failed for user "<username>". Check connection settings.

**Description:**Ensure the user name and password provided when Creating a Credential is correct.

**Instance/Port Mismatch**

**Error Message**: Unable to establish a valid connection to the target system. Instance on Default Port 1433 is MSSQLSERVER, does not match supplied Instance of NAMEDINSTANCE.

**Description:** Ensure the instance name and port are correct when Creating an Adapter Instance .

> **Note:**
>
> The default instance is MSSQLSERVER; default port is 1433.

**Collection Timeout**

**Error Message**: pollCompletionService method returned null or timed out before the result was returned with timeout set to <timeout length> seconds.

**Description**: The default **Thread Timeout** value of 30s may be insufficient for your environment. Raising it to 240s should resolve any collection issues due to this problem. Note also that raising the timeout does not cause collection to take any longer, it just gives it more time in which to complete. In addition, increasing the **Thread Pool Size** should help performance if you are running into collection timeout problems. You can find these parameters listed here: Creating an Adapter Instance .

## Adapter and Collector Log Files

You can review adapter errors in the adapter and collector log files, which can be viewed in the VCF Operations UI or in an external log viewer.

- **Adapter log files** are located in the $VCOPS_BASE/user/log/adapters/SqlServerAdapter/ folder.
- **Collector log files** are located in the $VCOPS_BASE/user/log/ folder.

To set the base log level for the collector in VMware Cloud Foundation Operations:

1. Navigate to **Administration** > **Support** > **Logs.**
2. Expand the **PRIMARY** folder, then select the **COLLECTOR** folder.
3. Click the **Edit Properties** icon.
4. In the **Root logger level** drop-down menu, select the desired option.

- The default logging level is **warning.**
- To troubleshoot issues, set the logging level to **info.**
- To view detailed messages, including micro steps, queries, and returned results, set the logging level to **debug.**

> **CAUTION:**
>
> If you set the logging level to **debug,** log files can become large very quickly. Set the logging level to **debug** only for short periods of time.

For more information about viewing log files and modifying levels, refer to the VMware Cloud Foundation Operations online help.

### Using the Management Pack

The Management Pack for Microsoft SQL Server includes the following out-of-the-box features, harnessing the power of the VCF Operations user interface:

- Inventory Tree (Traversal Spec)
- Dashboards
- Tags
- Views
- Reports
- Analysis Badges and Capacity Definitions
- Alerts
- Metrics

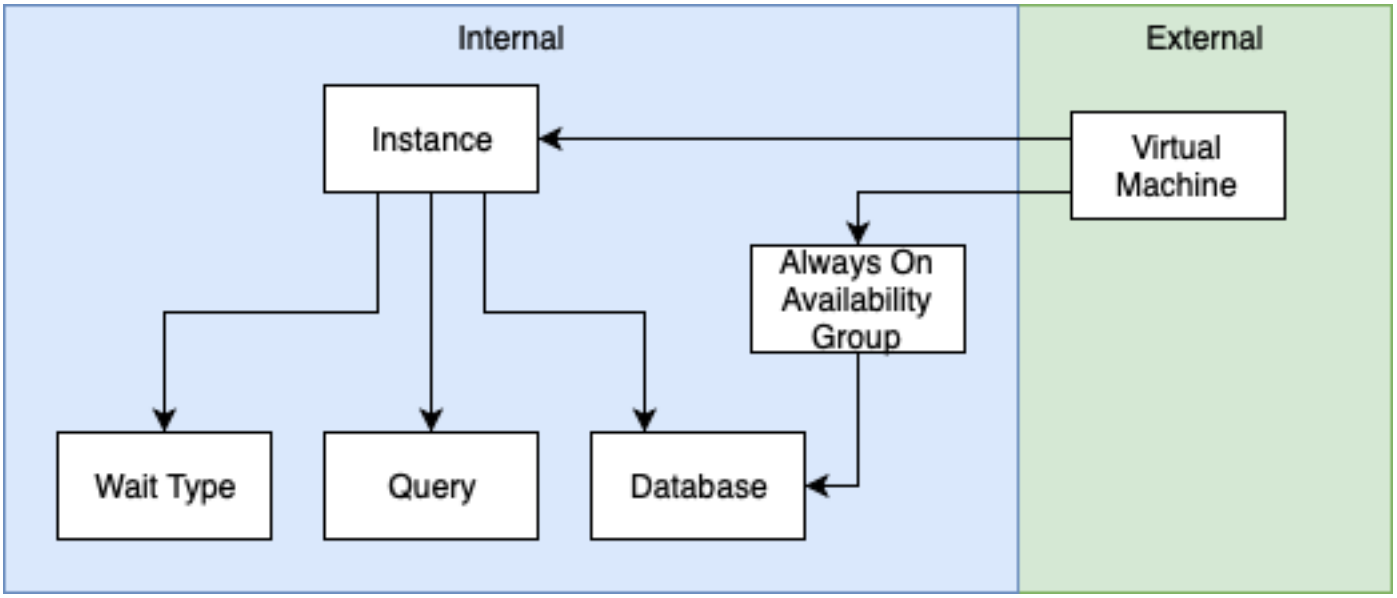
#### Inventory Tree (Traversal Spec), Resources and Relationsips

The Inventory Tree (Traversal Spec) feature in VCF Operations allows you to easily navigate your Microsoft SQL Server environment. The hierarchical structure implicitly shows relationships among resources and enables quick drill-downs to root-cause issues.

To access your Microsoft SQL Server Inventory Tree, select **Environment** from the top navigation bar, then select **Microsoft SQL Server** from the left panel.

**Resources and Relationships**

**Figure 483:**




#### Dashboards

Dashboards are the primary feature for monitoring and troubleshooting Microsoft SQL Server from within VCF Operations.

The following dashboards are included in this release of the Management Pack for Microsoft SQL Server:

- Microsoft SQL Database Drill Down
- Microsoft SQL Infrastructure Drill Down
- Microsoft SQL Server
- Microsoft SQL Query

### Tags

To further customize how your Management Pack dashboards are displayed, the following tags (containers) are currently available for filtering:

- **SQL Clusters** — all Microsoft SQL Servers being monitored that are Failover clusters
- **SQL Server Environment** — all Microsoft SQL Servers being monitored
- **SQL Server Instances** — SQL Server Instances to be used as the root object of the traversal
- **VMware Virtual Machines Hosting SQL Server** — all Virtual Machines on which Microsoft SQL Servers are running

To select a tag, perform the following steps:

> **Important:**
>
> Depending on the dashboard and version of VCF Operations you are using, the steps for selecting a tag may vary slightly.

1. Click the **Content** navigation shortcut.
2. Select **Dashboards** from the navigation pane.
3. Select the dashboard you want to filter, then click **Edit.**
4. Select **Edit Widget** for the specific dashboard widget you want to filter.
5. In the dialog that appears, under **Select which tags to filter**, expand the **Microsoft SQL Server Tag** option and select the desired tag(s).
6. To save your configuration and exit the Edit window, click **Save.**

> **Note:**
>
> If selecting multiple configurations, you can save each one without exiting the Edit window by clicking **Save Configuration.**

### Views

The Management Pack for Microsoft SQL Server creates views that allow the user to view statistics of metrics for various Microsoft SQL Server resources. The views help give a broad picture of the entire system, as opposed to a more in depth view.

The following views are available in the Management Pack:

| View | Type | Description |
|---|---|---|
| MS SQL Queries | List | Lists all metrics on SQL Query resources |
| MS SQL Query Text | Text | SQL Query Text used for Query Analysis dashboard |
| MS SQL Server Database Disk Access | List | Lists Microsoft SQL Server Disk Access Activity |
| MS SQL Server Database Locks | List | Lists Disk Delays for Microsoft SQL Server Databases |
| MS SQL Server Database Overview | List | Lists performance and activity for Microsoft SQL Server Databases |
| MS SQL Server Database Size | List | Lists space usage breakdowns for Microsoft SQL Server Databases |
| MS SQL Server Disk Access | List | Lists Disk Activity information for Microsoft SQL Servers |
| MS SQL Server Locks | List | Lists lock and wait time information for Microsoft SQL Servers |
| MS SQL Server Memory Breakdown | List | Lists breakdown of Microsoft SQL Server Memory |
| MS SQL Server Overview | List | Provides overview list of Microsoft SQL Servers |

| View | Type | Description |
|---|---|---|
| MS SQL Server Size Distribution | Distribution | Provides distribution view of database sizes |
| MS SQL Server Stress | List | Lists stress breakdown of Microsoft SQL Servers |

To access the Management Pack views:

1. Navigate to **Environment** > **All Objects** > **Microsoft SQL Server.**
2. Double-click on the desired object (resource).
3. Select the **Details** tab, then **Views.**

The available views for that resource are listed and can be selected.

### Reports

The Management Pack for Microsoft SQL Server contains out-of-the-box reports that you can easily export and share with key stakeholders in either .pdf or .csv formats.

The following reports are available in the Management Pack:

| Report | Description | Views Used |
|---|---|---|
| MS SQL Query Analysis | Provides a breakdown of query executions, as well as unique query plans and query text | MS SQL Queries |
| MS SQL Server Database Size Distribution | Provides a size distribution breakdown of databases on a server, along with a size distribution list for each database | MS SQL Server Size Distribution, MS SQL Server Database Size |
| MS SQL Server Environment Disk Access | Provides a look at server and database disk access operations | MS SQL Server Disk Access, MS SQL Server Database Disk Access |
| MS SQL Server Environment Locks | Provides a breakdown of server level lock requests and wait times, as well as database delays | MS SQL Server Locks, MS SQL Server Database Locks |
| MS SQL Server Environment Overview | Provides server and database overview and database size distribution | MS SQL Server Database Overview, MS SQL Server Overview, MS SQL Server Database Size |
| MS SQL Server Memory Breakdown | Provides a breakdown of physical memory used, recommended, stolen, as well as maximum and total physical memory | MS SQL Server Memory Breakdown |
| MS SQL Server Stress | Provides a server level stress assessment and resource usage | MS SQL Server Stress |

To access Management Pack reports:

1. Navigate to **Environment** > **All Objects** > **Microsoft SQL Server.**
2. Double-click on the desired object (resource).
3. Select the **Reports** tab, then **Report Templates**.

The available report(s) for the selected resource are listed and can be selected.

To run a selected report, click the **Run Template** icon.

### Analysis Badges and Capacity Definitions

Using the predictive analytics capabilities of VCF Operations Analysis Badges through capacity definitions, the Management Pack for Microsoft SQL Server populates capacities as listed in the table below.

### MS SQL Database Capacities

| Container | Use in Workload |
|---|---|
| **Database Disk Capacity** | |
| Reserved disk space* vs. Effective free disk space<br>*Database disk space usage (including allocated but unused) measured against free disk space on drives the database resides on | yes |

### MS SQL Server Instance Capacities

| Container | Use in Workload |
|---|---|
| **Connection Limit** | |
| Number of connections* vs. Number of possible connections<br>*Connection usage as reported by MS SQL Server (includes active and idle) | yes |
| **CPU Demand** | |
| CPU Usage* vs. Maximum CPU Usage<br>*Usage percent of Host CPU used by MS SQL Server application measured by MS SQL Server | yes |
| **Memory Usage** | |
| Effective Memory* vs. Total Server Memory<br>*Host RAM in use by MS SQL Server application measured by MS SQL Server | yes |
| **Disk Capacity** | |
| Used Disk Space* vs. Total Disk Space<br>*Aggregate Disk usage on all disks the MS SQL Server uses reported by MS SQL Server | yes |

Alerts

The Management Pack for Microsoft SQL Server creates alerts and provides recommended actions based on various symptoms (health status notifications and events) that it detects in your Microsoft SQL Server environment. See the table below for the list of alerts available in the Management Pack.

> **Important:**
>
> Alerts indicated with an asterisk (*) will **not** come through when the **Exclude All Events** Advanced Setting when Creating an Adapter Instance is set to **True.**

| Name | Description | Symptom | Recommendation |
|---|---|---|---|
| MS SQL Server Deprecated Feature* | This alert indicates a deprecated feature is being used on the MS SQL Server. | MS SQL Server Deprecated Feature | To avoid failures due to this feature in future updates, it is recommended to update the feature to the current implementation. |
| MS SQL Server Low Page Life Expectancy* | The actual page life expectancy is lower than the ideal page life expectancy. | MS SQL Server Low Page Life Expectancy | Increase the memory on the SQL Server to allow for a greater page life expectancy. |
| MS SQL Server Average Query CPU Time is High | This alert indicates that the average query CPU time has risen above normal. | Average Query CPU Time is High | |

| Name | Description | Symptom | Recommendation |
|---|---|---|---|
| MS SQL Server Number of Queries That End In Deadlock has Risen | This alert indicates that the number of queries that end in deadlocks has risen above normal. | Number of Queries That End In Deadlock has Risen | |
| MS SQL Server Cluster Owner Is Down | This alert indicates that the owning node of the cluster is down. | MS SQL Server Cluster Owner Is Down | |
| MS SQL Server Cluster Owner Is Paused | This alert indicates that the owning node of the cluster is paused. | MS SQL Server Cluster Owner Is Paused | |
| MS SQL Database Index: Never Accessed Indexes* | This alert indicates that there are indexes that have not been accessed since the SQL Server was started or the database was closed, whichever is shorter. | Never Accessed Indexes | Unused indexes can consume disk space through index storage and consume processing power during index maintenance. Determine if the index is necessary, and, if not, consider removing. |
| MS SQL Database file close to max capacity limit* | This alert indicates that there is a file on the database that is close to reaching its max capacity limit. | Database file close to max capacity limit | |
| MS SQL Database Index: Unused Maintained Indexes* | This alert indicates that there are indexes that are being maintained but have not been read. | MS SQL Database Index: Unused Maintained Indexes | Unused indexes can consume disk space through index storage and consume processing power during index maintenance. Determine if the index is necessary, and, if not, consider removing. |
| MS SQL Database Index: Potentially Inefficient Indexes* | This alert indicates that there are indexes that are being written to more than they are being read and are potentially inefficient. | Potentially Inefficient Indexes | Inefficient indexes have a maintenance cost in terms of space and processing power. Determine if the benefit of the index is greater than the cost of maintaining it. If not, consider removing the index. |
| MS SQL Database Index: Missing Beneficial Index* | This alert indicates that there is a missing beneficial index that is being recommended by MS SQL Server. | Missing Beneficial Index | Performing queries without indexes can be time consuming and cause poor query performance. Consider adding the recommended index. |
| MS SQL Database Index: Highly Fragmented Indexes* | This alert indicates there are indexes that are greater than 30% fragmented. | Highly Fragmented Indexes | Performing queries with fragmented indexes can be time consuming and cause poor query performance. Fragmentation can also cause more disk space to be consumed than is needed. Recommend rebuilding the fragmented index. |
| MS SQL Database Index: Mildly Fragmented Indexes* | This alert indicates there are indexes that are between 15% and 30% fragmented. | Mildly Fragmented Indexes | Performing queries with fragmented indexes can be time consuming and cause poor query performance. Fragmentation can also cause more disk space to be consumed than is needed. Recommend reorganizing the fragmented index. |

| Name | Description | Symptom | Recommendation |
|---|---|---|---|
| MS SQL Database Is In Emergency State | This alert indicates the MS SQL Database is in an Emergency State. | MS SQL Database Is In Emergency State | |
| MS SQL Database Is In Offline State | This alert indicates the MS SQL Database is in an Offline State. | MS SQL Database Is In Offline State | |
| MS SQL Database Is In Non-Critical Non-Online State | This alert indicates the MS SQL Database is in a non-critical non-online state. | MS SQL Database Is In Suspect State<br>MS SQL Database Is In Recovery Pending State<br>MS SQL Database Is In Recovery State<br>MS SQL Database Is In Restoring State | |
| MS SQL Always On Group Database Is In a Not Healthy State | This alert indicates the MS SQL Always On Group Database is in a Not Healthy State. | MS SQL Always On Group Database Is In a Not Healthy State | |
| MS SQL Always On Group Database Is In a Partially Healthy State | This alert indicates the MS SQL Always On Group Database is in a Partially Healthy State. | MS SQL Always On Group Database Is In a Partially Healthy State | |
| MS SQL Always On Group Database Is In a Suspended State | This alert indicates the MS SQL Always On Group Database is in a Suspended State. | MS SQL Always On Group Database Is In a Suspended State | |
| MS SQL Always on Availability Group in Non Healthy State | This alert indicates the MS SQL Always On Availability Group is in a non healthy synchronization state. | MS SQL Always on Availability Group in Non Healthy State | |
| MS SQL Always on Availability Group in Partially Healthy State | This alert indicates the MS SQL Always On Availability Group is in a partially healthy synchronization state. | MS SQL Always on Availability Group in Partially Healthy State | |
| Error reading database log | | An error occurred while processing the log for database <database-name>. If possible, restore from backup. If a backup is not available, it might be necessary to rebuild the log.<br>An error occurred while processing the log for database <database-name>. The log block version <log-version> is unsupported. This server supports log version <log-version> to <log-version>.<br>An error occurred while processing the log for database <database-name>. The log block could not be decrypted. | |
| Error Fetching Page | | Attempt to fetch logical page <#>_PGID in database <database-name> failed. It belongs to allocation unit <allocation-unit> not to <allocation-unit>. | |
| Cannot Find File | | Cannot find file ID <fileID-name> on device <device-name>. | |

| Name | Description | Symptom | Recommendation |
|---|---|---|---|
| Consistency Errors | | Database <database-name>: consistency errors in system catalogs prevent further DBCC processing. | |
| Recovery Error | | An error occurred during recovery, preventing the database <database_name> (<databaseID#>:<instanceID#>) from restarting. Diagnose the recovery errors and fix them, or restore from a known good backup. If errors are not corrected or expected, contact Technical Support. | |
| Recovery Failed | | Restore recovery failed because a problem was detected with data in a temporary file that is needed for recovery. This error can be caused because of a storage device failure or because the restore checkpoint restart file has been moved or deleted. Retry...(truncated). | |
| Invalid File ID | | Could not open File Control Block (FCB) for invalid file ID <file-ID> in database <database-name>. Verify the file location. Execute DBCC CHECKDB. | |
| Consistency-Base I/O Error | | SQL Server detected a logical consistency-based I/O error: <error-name>. It occurred during a <messageID>_MSG of page <page-number>_PGID in database ID <database-ID> at offset <offset> in file <filename> Additional messages in the SQL Server error log or operating system error log may provide more detail. | |
| Operating System Error | | The operating system returned error <error-name> to SQL Server during a <messageID>_MSG at offset <offset> in file <file-name>. Additional messages in the SQL Server error log and operating system error log may provide more detail. This is a severe system-level error condition hat threatens database integrity and must be corrected immediately. Complete a full database consistency check (DBCC CHECKDB). This error can be caused by many factors; for more information, see SQL Server Books Online. | |

| Name | Description | Symptom | Recommendation |
|---|---|---|---|
| Couldn't Allocate space | | Could not allocate space for object <object-name>. <Reason> in database <database-name> because the <file-group-name> filegroup is full. Create disk space by deleting unneeded files, dropping objects in the filegroup, adding additional files to the filegroup, or setting autogrowth on for existing files in the filegroup. | |
| Transaction Log Full | | The transaction log for database <database-name> is full due to <reason>. | |
| Backup Failed | | Backup failed to complete the command <command-name>. Check the backup application log for detailed messages. Backup failed because there is a mismatch in file metadata for file <file-name>. | |
| Could not open file | | Open Failed: Could not open file <file-name> for file number <file-#>. OS error: <error>. | |
| Operating System File Error | | <Error Prefix>: Operating system error <error> occurred while creating or opening file <file-name>. Diagnose and correct the operating system error, and retry the operation. | |
| Database Corruption | | Corruption in database ID <database-ID#>, object ID <object-ID#> possibly due to schema or catalog inconsistency. Run DBCC CHECKCATALOG. | |
| SQL Server Agent is not running | SQL Server Agent is currently not running. | SQL Server Agent is Stopped. | Install and/or start SQL Server Agent. |
| Job Failed | | A job has failed. | |

Metrics

The Management Pack for Microsoft SQL Server collects the following metrics, according to Microsoft SQL Server resource kind.

**Important:**

The CPU Usage metric for Microsoft SQL Server Instances is not available on MS SQL Server **Standard** or **Express** editions.

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| Microsoft SQL Server Tag | Relationships | Virtual Machine Children | string | List of children Virtual Machine resources |
| Microsoft SQL Server Tag | Relationships | Instance Children | string | List of children Instance resources |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| Microsoft SQL Server | Relationships | Instance Children | string | List of children Instance resources |
| Microsoft SQL Server Always On Availability Group | General | Automated Backup Preference | string | Automated backup perference for availability group. |
| Microsoft SQL Server Always On Availability Group | General | Name | string | Name of availability group. |
| Microsoft SQL Server Always On Availability Group | General | Replica ID | string | Replica ID of availability group. |
| Microsoft SQL Server Always On Availability Group | General | Replicas | string | List of replicas being used by availability group. |
| Microsoft SQL Server Always On Availability Group | General | Group ID (Deprecated) | string | |
| Microsoft SQL Server Always On Availability Group | Status | Primary Recovery Health | string | Description of recovery health of the primary replica. |
| Microsoft SQL Server Always On Availability Group | Status | Primary Replica | string | Primary replica on availability group. |
| Microsoft SQL Server Always On Availability Group | Status | Secondary Recovery Health | string | Description of recovery health of the secondary replica. |
| Microsoft SQL Server Always On Availability Group | Status | Synchronization Health | float | Number represenation of rollup of synchronization health of all availability replicas in the availability group. |
| Microsoft SQL Server Always On Availability Group | Status | Synchronization Health Property | string | Description of rollup of synchronization health of all availability replicas in the availability group. |
| Microsoft SQL Server Always On Availability Group | Relationships | Database Children | string | List of child Database resources |
| Microsoft SQL Server Always On Availability Group | Relationships | Virtual Machine Parents | string | List of parent Virtual Machine resources |
| Microsoft SQL Server Always On Availability Group | HADR Replica Server | Availability Group Name | string | Name of availability group. |
| Microsoft SQL Server Always On Availability Group | HADR Replica Server | Group ID | string | Group ID of availability group. |
| Microsoft SQL Server Always On Availability Group | HADR Replica Server | Name | string | Name of replica server. |
| Microsoft SQL Server Always On Availability Group | HADR Replica Server | Availability Mode | string | Description of HADR replica server availablility mode. |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| Microsoft SQL Server Always On Availability Group | HADR Replica Server | Failover Mode | string | Failover mode for HADR replica server. |
| Microsoft SQL Server Always On Availability Group | HADR Replica Server | Always On Availability Group Parents | string | List of parent Always On Availability Group resources |
| Microsoft SQL Server Database | Active Processes | Active Connections | float | Nubmer of active connections to database. |
| Microsoft SQL Server Database | Active Processes | Active Transactions | float | Number of active transactions on database. |
| Microsoft SQL Server Database | Availability States | Suspended Reason | string | If the database is suspended, the reason for the suspended state. |
| Microsoft SQL Server Database | Availability States | Synchronization Health | string | Description of the synchronization_health of the availability database. |
| Microsoft SQL Server Database | Availability States | Synchronization State | string | Description of the data-movement state. |
| Microsoft SQL Server Database | Capacity | Data Space | float | Database data space. |
| Microsoft SQL Server Database | Capacity | Disk Space Available | float | Amount of space available on database. |
| Microsoft SQL Server Database | Capacity | Index Space | float | Database index space |
| Microsoft SQL Server Database | Capacity | Log Size | float | Size of the log area in memory for database. |
| Microsoft SQL Server Database | Capacity | Reserved Space | float | Database reserved space. |
| Microsoft SQL Server Database | Capacity | Reserved Space Unused | float | Database unused reserved space. |
| Microsoft SQL Server Database | Capacity | Size Used | float | Database size. |
| Microsoft SQL Server Database | Capacity | Space Available | float | Percentage of space available on disk used by database. |
| Microsoft SQL Server Database | Capacity | Space Used | float | Percentage of space used on disk used by database. |
| Microsoft SQL Server Database | Capacity | Total Space | float | Total space of disk used by database. |
| Microsoft SQL Server Database | Capacity | Unallocated Space Available | float | Unallocated space available on database. |
| Microsoft SQL Server Database | Capacity | Usable Disk Space | float | The amount of space on disk that is usable by a database, this includes the free space of the entire disk, and any space being used by the database. |
| Microsoft SQL Server Database | Config | Auto Close | string | Shows whether auto close is enabled |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| Microsoft SQL Server Database | Config | Auto Close Enabled | string | Shows whether auto close is enabled. |
| Microsoft SQL Server Database | Config | Auto Create Statistics | string | Shows whether auto create statistics is enabled. |
| Microsoft SQL Server Database | Config | Auto Shrink | string | Shows whether auto shrink is enabled. |
| Microsoft SQL Server Database | Config | Auto Shrink Enabled | string | Shows whether auto shrink is enabled. |
| Microsoft SQL Server Database | Config | Auto Update Statistics | string | Shows whether auto update statistics is enabled. |
| Microsoft SQL Server Database | Config | Read Only | string | Shows whether read only mode is enabled. |
| Microsoft SQL Server Database | Config | Single User Mode | string | Shows whether single user mode is enabled. |
| Microsoft SQL Server Database | Disk Access | Bytes Read | float | The Bytes Read of the Database. |
| Microsoft SQL Server Database | Disk Access | Bytes Written | float | The Bytes Written of the Database. |
| Microsoft SQL Server Database | Disk Access | IOPS | float | The IOPS of the Database. |
| Microsoft SQL Server Database | Disk Access | Read Delay | float | The Read Delay of the Database. |
| Microsoft SQL Server Database | Disk Access | Read Operations | float | The Read Operations of the Database. |
| Microsoft SQL Server Database | Disk Access | Total bytes | float | The Total bytes of the Database. |
| Microsoft SQL Server Database | Disk Access | Write Delay | float | The Write Delay of the Database. |
| Microsoft SQL Server Database | Disk Access | Write Operations | float | The Write Operations of the Database. |
| Microsoft SQL Server Database | General | Availability Group Name | string | Name of availability group. |
| Microsoft SQL Server Database | General | Creation Date | string | The database creation date. |
| Microsoft SQL Server Database | General | Recovery Model | string | The recovery model of the database. |
| Microsoft SQL Server Database | General | Resource Name | string | The Resource Name of the Database. |
| Microsoft SQL Server Database | General | Status | string | The current status of the database. |
| Microsoft SQL Server Database | General | Time Since Creation | float | Days since database was created. |
| Microsoft SQL Server Database | Growth | Smallest Fixed Size File Growth Increment | float | The Smallest Fixed Size File Growth Increment of the Database. |
| Microsoft SQL Server Database | Growth | Smallest Percent-based File Growth Increment | float | The Smallest Percent-based File Growth Increment of the Database. |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| Microsoft SQL Server Database | Index Performance | User Lookups | float | The User Lookups of the Database. |
| Microsoft SQL Server Database | Index Performance | User Scans | float | The User Scans of the Database. |
| Microsoft SQL Server Database | Index Performance | User Seeks | float | The User Seeks of the Database. |
| Microsoft SQL Server Database | Index Performance | User Updates | float | The User Updates of the Database. |
| Microsoft SQL Server Database | Memory | Buffer Pool Size | float | The Buffer Pool Size of the Database. |
| Microsoft SQL Server Database | Memory | Effective Server Total | float | Total effective memory available to the SQL server. |
| Microsoft SQL Server Database | Memory | Effective Used | float | The amount of effective server memory in MB used by the database. |
| Microsoft SQL Server Database | Memory | Effective Used Pages | float | The number of effective server memory pages used by the database. |
| Microsoft SQL Server Database | Performance | Data File Size | float | Size of the data file on a database. |
| Microsoft SQL Server Database | Performance | Log Cache Hit Ratio | float | Database log cache hit ratio. |
| Microsoft SQL Server Database | Performance | Log Cache Hit Ratio Base | float | Database base log chache hit ratio. |
| Microsoft SQL Server Database | Performance | Log Cache Reads | float | Database log cache reads over time. |
| Microsoft SQL Server Database | Performance | Log File Size | float | Size of log file on database. |
| Microsoft SQL Server Database | Performance | Log File Used | float | Percentage of log file used on database. |
| Microsoft SQL Server Database | Performance | Log File Used Size | float | Log file size used on database. |
| Microsoft SQL Server Database | Performance | Log Pool Cache Misses | float | Database log cache misses over time. |
| Microsoft SQL Server Database | Performance | Log Pool Disk Reads | float | Number of disk reads due to log cache misses for a database. |
| Microsoft SQL Server Database | Performance | Log Pool Requests | float | Database log cache requests over time. |
| Microsoft SQL Server Database | Performance | Tracked Transactions | float | Database tracked transactions over time. |
| Microsoft SQL Server Database | Performance | Transactions | float | Database transactions over time. |
| Microsoft SQL Server Database | Performance | Write Transactions | float | Database write transactions over time. |
| Microsoft SQL Server Database | Size | Database Backup Size | float | Database backup size |
| Microsoft SQL Server Database | Size | Database Log Size | float | Database log size. |
| Microsoft SQL Server Database | Size | Database Row Size | float | Database row size. |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| Microsoft SQL Server Database | Size | Total Used Disk Size | float | Total database file size on disk, this includes allocated but unused. |
| Microsoft SQL Server Database | Relationships | Always On Availability Group Parents | string | List of parent Always On Availability Group resources |
| Microsoft SQL Server Database | Relationships | File Children | string | List of child File resources |
| Microsoft SQL Server Database | Relationships | Instance Parents | string | List of parent Instance resources |
| Microsoft SQL Server Database | Object | Database Name | string | The Database Name of the Object. |
| Microsoft SQL Server Database | Object | Host | string | The Host of the Object. |
| Microsoft SQL Server Database | Object | Instance | string | The Instance of the Object. |
| Microsoft SQL Server Database | Object | Object Name | string | The Object Name of the Object. |
| Microsoft SQL Server Database | Object | Average Page Lock Wait Duration | float | Average page log wait time on database. |
| Microsoft SQL Server Database | Object | Average Row Lock Wait Duration | float | Average row lock wait time on database. |
| Microsoft SQL Server Database | Object | Page Locks | float | Number of page locks used by database. |
| Microsoft SQL Server Database | Object | Page Lock Waits | float | Number of page locks currently in waiting state on database. |
| Microsoft SQL Server Database | Object | Row Locks | float | Number of row locks on database. |
| Microsoft SQL Server Database | Object | Row Lock Waits | float | Number of row locks currently in waiting state on database. |
| Microsoft SQL Server Database | Object | Database Parents | string | List of parent Database resources |
| Microsoft SQL Server Instance | Buffer | Buffer Cache Hit Ratio | float | The Buffer Cache Hit Ratio of the Instance. |
| Microsoft SQL Server Instance | Buffer | Buffer Ideal Page Life Expectancy | float | The Buffer Ideal Page Life Expectancy of the Instance. |
| Microsoft SQL Server Instance | Buffer | Buffer Page Life Expectancy | integer | The Buffer Page Life Expectancy of the Instance. |
| Microsoft SQL Server Instance | Buffer | Buffer Pool Hit Ratio | float | The Buffer Pool Hit Ratio of the Instance. |
| Microsoft SQL Server Instance | Buffer | Buffer Pool Size | float | The Buffer Pool Size of the Instance. |
| Microsoft SQL Server Instance | Buffer | Checkpoint Pages | float | The Checkpoint Pages of the Instance. |
| Microsoft SQL Server Instance | Buffer | Database Pages | integer | The Database Pages of the Instance. |
| Microsoft SQL Server Instance | Buffer | Free List Stalls | float | The Free List Stalls of the Instance. |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| Microsoft SQL Server Instance | Buffer | Lazy Writes | float | The Lazy Writes of the Instance. |
| Microsoft SQL Server Instance | Buffer | Page Lookups | float | The Page Lookups of the Instance. |
| Microsoft SQL Server Instance | Buffer | Page Reads | float | The Page Reads of the Instance. |
| Microsoft SQL Server Instance | Buffer | Page Splits | float | The Page Splits of the Instance. |
| Microsoft SQL Server Instance | Buffer | Page Writes | float | The Page Writes of the Instance. |
| Microsoft SQL Server Instance | Buffer | Procedure Cached Pages | integer | The Procedure Cached Pages of the Instance. |
| Microsoft SQL Server Instance | Buffer | Read Aheads | float | The Read Aheads of the Instance. |
| Microsoft SQL Server Instance | Buffer | Target Pages | integer | The Target Pages of the Instance. |
| Microsoft SQL Server Instance | CPU | CPU Usage | float | The CPU Usage of the Instance. |
| Microsoft SQL Server Instance | CPU | Max CPU Percent | float | This is a hard limit of 100% used for capacity calculations. |
| Microsoft SQL Server Instance | Cluster State | Cluster Owner Status | string | The status of the owner cluster node. |
| Microsoft SQL Server Instance | Cluster State | Cluster Status | string | The worst status across all nodes in the cluster. |
| Microsoft SQL Server Instance | Cluster State | Number of Healthy Cluster Nodes | integer | The number of nodes in a cluster in the up state. |
| Microsoft SQL Server Instance | Cluster State | Number of Unhealthy Cluster Nodes | integer | The number of nodes in a cluster in a down, paused, or unknown state. |
| Microsoft SQL Server Instance | Cluster State | Is Current Owner (Deprecated) | string | |
| Microsoft SQL Server Instance | Cluster State | Status Text (Deprecated) | string | |
| Microsoft SQL Server Instance | Cluster State | Status Value (Deprecated) | integer | |
| Microsoft SQL Server Instance | Connection Capacity | Active Connections | integer | The Active Connections of the Instance. |
| Microsoft SQL Server Instance | Connection Capacity | Connections | integer | The Connections of the Instance. |
| Microsoft SQL Server Instance | Connection Capacity | Possible Connections | integer | The Possible Connections of the Instance. |
| Microsoft SQL Server Instance | Connections | Connections Reset | float | The Connections Reset of the Instance. |
| Microsoft SQL Server Instance | Connections | Logins | float | The Logins of the Instance. |
| Microsoft SQL Server Instance | Connections | Logouts | float | The Logouts of the Instance. |
| Microsoft SQL Server Instance | Connections | User Connections | integer | The User Connections of the Instance. |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| Microsoft SQL Server Instance | Database Performance | Data File Size | integer | The Data File Size of the Instance. |
| Microsoft SQL Server Instance | Database Performance | Log Cache Hit Ratio | float | The Log Cache Hit Ratio of the Instance. |
| Microsoft SQL Server Instance | Database Performance | Log Cache Hit Ratio Base | integer | The Log Cache Hit Ratio Base of the Instance. |
| Microsoft SQL Server Instance | Database Performance | Log Cache Misses | float | The Log Cache Misses of the Instance. |
| Microsoft SQL Server Instance | Database Performance | Log Cache Reads | float | The Log Cache Reads of the Instance. |
| Microsoft SQL Server Instance | Database Performance | Log Cache Requests | float | The Log Cache Requests of the Instance. |
| Microsoft SQL Server Instance | Database Performance | Log File Size | integer | The Log File Size of the Instance. |
| Microsoft SQL Server Instance | Database Performance | Log File Usage | integer | The Log File Usage of the Instance. |
| Microsoft SQL Server Instance | Database Performance | Log File Used Size | integer | The Log File Used Size of the Instance. |
| Microsoft SQL Server Instance | Database Performance | Log Pool Disk Reads | float | The Log Pool Disk Reads of the Instance. |
| Microsoft SQL Server Instance | Database Performance | Track Transactions | float | The Track Transactions of the Instance. |
| Microsoft SQL Server Instance | Database Performance | Transaction Summary | float | SQL Instance Transaction Summary |
| Microsoft SQL Server Instance | Database Performance | Transactions | float | The Transactions of the Instance. |
| Microsoft SQL Server Instance | Database Performance | Write Transactions | float | The Write Transactions of the Instance. |
| Microsoft SQL Server Instance | Disk Access | Bytes Read | float | Disk read rate of MS SQL Server. |
| Microsoft SQL Server Instance | Disk Access | Bytes Written | float | Disk write rate of MS SQL Server. |
| Microsoft SQL Server Instance | Disk Access | IOPS | float | Total amount of disk write and read operations. |
| Microsoft SQL Server Instance | Disk Access | Read Operations | float | This metric represents the amount of disk read operations over time. |
| Microsoft SQL Server Instance | Disk Access | Total bytes | float | Total throughput on disk. |
| Microsoft SQL Server Instance | Disk Access | Write Operations | float | This metrice represents the amount of disk write operations over time. |
| Microsoft SQL Server Instance | Disk Capacity | Total Disk Space | integer | The Total Disk Space of the Instance. |
| Microsoft SQL Server Instance | Disk Capacity | Used Disk Space | integer | The Used Disk Space of the Instance. |
| Microsoft SQL Server Instance | General | AWE Enabled | string | AWE Enabled |
| Microsoft SQL Server Instance | General | CPU Count | integer | CPU Count |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| Microsoft SQL Server Instance | General | Current Cluster Owner | string | The name of the node in the cluster that is the current owner. |
| Microsoft SQL Server Instance | General | MSSQL Architecture Type | string | MSSQL Architecture Type |
| Microsoft SQL Server Instance | General | Max Thread Count | integer | Max Thread Count |
| Microsoft SQL Server Instance | General | OS Architecture Type | string | OS Architecture Type |
| Microsoft SQL Server Instance | General | OS Memory | integer | OS Memory |
| Microsoft SQL Server Instance | General | Resource Name | string | The Resource Name of the Instance. |
| Microsoft SQL Server Instance | General | SQL Version | string | The SQL Version of the Instance. |
| Microsoft SQL Server Instance | General | Server Name | string | The Server Name of the Instance. |
| Microsoft SQL Server Instance | General | Service Name | string | The Service Name of the Instance. |
| Microsoft SQL Server Instance | General | Thread Size | integer | Thread Size |
| Microsoft SQL Server Instance | General | Type | string | The Type of the Instance. |
| Microsoft SQL Server Instance | General | Type | string | The Type of the Instance. |
| Microsoft SQL Server Instance | General | Windows Version | string | The Windows Version of the Instance. |
| Microsoft SQL Server Instance | History | Creation Date | string | The Creation Date of the Instance. |
| Microsoft SQL Server Instance | History | Days Since Last Start | integer | The Days Since Last Start of the Instance. |
| Microsoft SQL Server Instance | History | Most Recent Start Date | string | The Most Recent Start Date of the Instance. |
| Microsoft SQL Server Instance | History | Time Since Creation | integer | The Time Since Creation of the Instance. |
| Microsoft SQL Server Instance | Jobs | Failed Jobs | integer | The Failed Jobs of the Instance. |
| Microsoft SQL Server Instance | Jobs | Successful Jobs | integer | The Successful Jobs of the Instance. |
| Microsoft SQL Server Instance | Locks | Average Wait Time | integer | The Average Wait Time of the Instance. |
| Microsoft SQL Server Instance | Locks | Average Wait Time Base | integer | The Average Wait Time Base of the Instance. |
| Microsoft SQL Server Instance | Locks | Deadlocks | float | The Deadlocks of the Instance. |
| Microsoft SQL Server Instance | Locks | Instance Wait Time | float | The Instance Wait Time of the Instance. |
| Microsoft SQL Server Instance | Locks | Lock Requests | float | The Lock Requests of the Instance. |
| Microsoft SQL Server Instance | Locks | Lock Timeouts | float | The Lock Timeouts of the Instance. |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| Microsoft SQL Server Instance | Locks | Lock Wait Time | integer | The Lock Wait Time of the Instance. |
| Microsoft SQL Server Instance | Locks | Lock Waits | float | The Lock Waits of the Instance. |
| Microsoft SQL Server Instance | MAC Addresses | Agent Running | string | Indicates whether or not the agent is running. Agent must run to retrieve MAC Address. |
| Microsoft SQL Server Instance | MAC Addresses | MAC Addresses | string | Connected MAC Addresses |
| Microsoft SQL Server Instance | Memory | Available Physical Memory | float | The Available Physical Memory of the Instance. |
| Microsoft SQL Server Instance | Memory | Connection Memory | integer | The Connection Memory of the Instance. |
| Microsoft SQL Server Instance | Memory | Database Cache Memory | integer | The Database Cache Memory of the Instance. |
| Microsoft SQL Server Instance | Memory | Effective Total Memory | float | The total amount of memory available between SQL Server and unused memory on the host. |
| Microsoft SQL Server Instance | Memory | Free Memory | integer | The Free Memory of the Instance. |
| Microsoft SQL Server Instance | Memory | Granted Workspace Memory | integer | The Granted Workspace Memory of the Instance. |
| Microsoft SQL Server Instance | Memory | Lock Memory | integer | The Lock Memory of the Instance. |
| Microsoft SQL Server Instance | Memory | Log Pool Memory | integer | The Log Pool Memory of the Instance. |
| Microsoft SQL Server Instance | Memory | Max Server Memory | integer | The Max Server Memory |
| Microsoft SQL Server Instance | Memory | Maximum Workspace Memory | integer | The Maximum Workspace Memory of the Instance. |
| Microsoft SQL Server Instance | Memory | Memory Usage | float | The percentage of memory that is being utilized by SQL Server. |
| Microsoft SQL Server Instance | Memory | Optimizer Memory | integer | The Optimizer Memory of the Instance. |
| Microsoft SQL Server Instance | Memory | Physical Memory Max | float | The Physical Memory Max of the Instance. |
| Microsoft SQL Server Instance | Memory | Physical Memory Recommended | float | The Physical Memory Recommended of the Instance. |
| Microsoft SQL Server Instance | Memory | Physical Memory Used | float | The Physical Memory Used of the Instance. |
| Microsoft SQL Server Instance | Memory | Reserved Server Memory | integer | The Reserved Server Memory of the Instance. |
| Microsoft SQL Server Instance | Memory | SQL Cache Memory | integer | The SQL Cache Memory of the Instance. |
| Microsoft SQL Server Instance | Memory | Stolen Server Memory | float | The Stolen Server Memory of the Instance. |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| Microsoft SQL Server Instance | Memory | Total Physical Memory | float | The Total Physical Memory of the Instance. |
| Microsoft SQL Server Instance | Memory | Total Server Memory | float | The Total System Memory |
| Microsoft SQL Server Instance | Process States | Background Processes | integer | The Background Processes of the Instance. |
| Microsoft SQL Server Instance | Process States | Blocked Processes | integer | The Blocked Processes of the Instance. |
| Microsoft SQL Server Instance | Process States | Dormant Processes | integer | The Dormant Processes of the Instance. |
| Microsoft SQL Server Instance | Process States | Preconnect Processes | integer | The Preconnect Processes of the Instance. |
| Microsoft SQL Server Instance | Process States | Runnable Processes | integer | The Runnable Processes of the Instance. |
| Microsoft SQL Server Instance | Process States | Running Processes | integer | The Running Processes of the Instance. |
| Microsoft SQL Server Instance | Process States | Sleeping Processes | integer | The Sleeping Processes of the Instance. |
| Microsoft SQL Server Instance | Process States | Suspended Processes | integer | The Suspended Processes of the Instance. |
| Microsoft SQL Server Instance | Queries | Average Query CPU | integer | The Average Query CPU of the Instance. |
| Microsoft SQL Server Instance | Queries | Average Query Executions | integer | The Average Query Executions of the Instance. |
| Microsoft SQL Server Instance | Queries | Dead Locked Queries | integer | The Dead Locked Queries of the Instance. |
| Microsoft SQL Server Instance | Queries | High Query CPU | integer | The High Query CPU of the Instance. |
| Microsoft SQL Server Instance | Queries | High Query Executions | integer | The High Query Executions of the Instance. |
| Microsoft SQL Server Instance | Queries | Total Query CPU | integer | The Total Query CPU of the Instance. |
| Microsoft SQL Server Instance | Queries | Total Query Executions | integer | The Total Query Executions of the Instance. |
| Microsoft SQL Server Instance | Statements | Batch Compilation Ratio | float | The Batch Compilation Ratio of the Instance. |
| Microsoft SQL Server Instance | Statements | Batch Requests | float | The Batch Requests of the Instance. |
| Microsoft SQL Server Instance | Statements | Forced Parameterizations | float | The Forced Parameterizations of the Instance. |
| Microsoft SQL Server Instance | Statements | SQL Compilations | float | The SQL Compilations of the Instance. |
| Microsoft SQL Server Instance | Statements | SQL Recompilation Ratio | float | The SQL Recompilation Ratio of the Instance. |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| Microsoft SQL Server Instance | Statements | SQL Recompilations | float | The SQL Recompilations of the Instance. |
| Microsoft SQL Server Instance | Tasks | Runnable Tasks Count | integer | The Runnable Tasks Count of the Instance. |
| Microsoft SQL Server Instance | Threads | Threads | integer | The Threads of the Instance. |
| Microsoft SQL Server Instance | Relationships | Cluster Node Children | string | List of child Cluster Node resources |
| Microsoft SQL Server Instance | Relationships | Database Children | string | List of child Database resources |
| Microsoft SQL Server Instance | Relationships | Job Children | string | List of child Job resources |
| Microsoft SQL Server Instance | Relationships | Microsoft SQL Server Parents | string | List of parent Microsoft SQL Server resources |
| Microsoft SQL Server Instance | Relationships | Microsoft SQL Server Tag Parents | string | List of parent Microsoft SQL Server Tag resources |
| Microsoft SQL Server Instance | Relationships | Query Children | string | List of child Query resources |
| Microsoft SQL Server Instance | Relationships | Virtual Machine Parents | string | List of parent Virtual Machine resources |
| Microsoft SQL Server Instance | Relationships | Wait Type Children | string | List of child Wait Type resources |
| Microsoft SQL Server Instance | Disk | Drive ID | string | The Drive ID of the Disk. |
| Microsoft SQL Server Instance | Disk | Host | string | The Host of the Disk. |
| Microsoft SQL Server Instance | Disk | Instance | string | The Instance of the Disk. |
| Microsoft SQL Server Instance | Disk | Disk Size | integer | The Disk Size of the Disk. |
| Microsoft SQL Server Instance | Disk | Available Disk Space | integer | The Available Disk Space of the Disk. |
| Microsoft SQL Server Instance | Disk | Resource Name | string | The Resource Name of the Disk. |
| Microsoft SQL Server Instance | Disk | Instance Parents | string | List of parent Instance resources |
| Microsoft SQL Server Query | General | Average Execution Time | float | The Average Execution Time of the Query. |
| Microsoft SQL Server Query | General | Execution Count | float | The Execution Count of the Query. |
| Microsoft SQL Server Query | General | Execution Count Counter | float | Execution Count Counter |
| Microsoft SQL Server Query | General | Execution Delta | integer | Execution Delta |
| Microsoft SQL Server Query | General | Execution Time | float | Execution Time |
| Microsoft SQL Server Query | General | Last Execution | float | The Last Execution of the Query. |
| Microsoft SQL Server Query | General | Last Update | double | Last time query was last updated |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| Microsoft SQL Server Query | General | Query HTML | string | The HTML version of the SQL text of this query |
| Microsoft SQL Server Query | General | Query Name | string | The Query Name of the Query. |
| Microsoft SQL Server Query | General | Query Text | string | The Query Text of the Query. |
| Microsoft SQL Server Query | General | Total Elapsed Time | float | Total Elapsed Time |
| Microsoft SQL Server Query | General | Total Logical Reads | float | Total Logical Reads |
| Microsoft SQL Server Query | General | Total Logical Writes | float | Total Logical Writes |
| Microsoft SQL Server Query | General | Total Physical Reads | float | Total Physical Reads |
| Microsoft SQL Server Query | General | Total Worker Time | integer | Total Worker Time |
| Microsoft SQL Server Query | General | Unique Query Plans | integer | The Unique Query Plans of the Query. |
| Microsoft SQL Server Query | Relationships | Instance Parents | string | List of parent Instance resources |
| Microsoft SQL Server Wait Type | General | Last Update | double | Last time wait type was last updated |
| Microsoft SQL Server Wait Type | General | Wait Time | integer | Total wait time for this wait type in milliseconds. |
| Microsoft SQL Server Wait Type | General | Waiting Tasks Count | float | Number of waits on this wait type. This counter is incremented at the start of each wait. |
| Microsoft SQL Server Wait Type | Relationships | Instance Parents | string | List of parent Instance resources |
| Microsoft SQL Server Job | General | Description | string | Description |
| Microsoft SQL Server Job | General | Name | string | Name |
| Microsoft SQL Server Job | General | Next Scheduled Run | string | Next Scheduled Run |
| Microsoft SQL Server Job | General | Steps | integer | Number of steps in the job |
| Microsoft SQL Server Job | Relationships | Instance Parents | string | List of Parent Instance Resources |

## Metrics by Permission

The following table indicates the metrics collected for each required permission indicated in Assigning MS SQL User Privileges.

| Permission | Resource Group | Metric |
|---|---|---|
| **VIEW SERVER STATE** | Instance | Buffer Pool Size<br>Buffer Ideal Page Life Expectancy<br>Buffer Pool Hit Ratio<br>Total Server Memory<br>Max Server Memory<br>Total Physical Memory<br>Available Physical Memory<br>Physical Memory Used<br>Physical Memory Recommended<br>Physical Memory Max<br>Stolen Server Memory<br>SQL Cache Memory<br>Reserved Server Memory<br>Optimizer Memory<br>Maximum Workspace Memory<br>Log Pool Memory<br>Lock Memory<br>Granted Workspace Memory<br>Free Memory<br>Database Cache Memory<br>Connection Memory<br>Background Processes<br>Runnable Processes<br>Running Processes<br>Dormant Processes<br>Blocked Processes<br>Preconnect Processes<br>Sleeping Processes<br>Suspended Processes<br>Transactions<br>Batch Requests<br>SQL Compilations<br>SQL Recompilations<br>Forced Parameterizations<br>SQL Recompilation Ratio<br>Batch Compilation Ratio<br>Lazy Writes<br>Page Writes<br>Free List Stalls<br>Checkpoint Pages<br>Buffer Page Life Expectancy<br>Page Lookups<br>Page Reads<br>Database Pages<br>Read Aheads<br>Procedure Cached Pages<br>Target Pages<br>Buffer Cache Hit Ratio<br>Logins<br>Logouts<br>Connections Reset<br>User Connections<br>Creation Date<br>Most Recent Start Date |
| | | Time Since Creation<br>Days Since Last Start<br>Max Thread Count<br>Thread Size |

| Permission | Resource Group | Metric |
|---|---|---|
| | Query | SQL Handle<br>Statement Start Offset<br>Statement End Offset<br>Query Name<br>Query Text<br>Unique Query Plans<br>Execution Count<br>Execution Time<br>Last Execution<br>Average Execution Time<br>Total Physical Reads<br>Total Logical Reads<br>Total Logical Writes<br>Total Elapsed Time |
| | Disk | Drive ID<br>Resource Name<br>Disk Size<br>Available Disk Space |

| Permission | Resource Group | Metric |
|---|---|---|
| | Database | Database Name<br>Effective Used Pages<br>Effective Used<br>Effective Server Total<br>User Scans<br>User Seeks<br>User Lookups<br>User Updates<br>Buffer Pool Size<br>IOPS<br>Total bytes<br>Read Operations<br>Bytes Read<br>Write Operations<br>Bytes Written<br>Read Delay<br>Write Delay<br>Data File Size<br>Log File Size<br>Log File Used Size<br>Log File Used<br>Log Cache Reads<br>Log Cache Hit Ratio<br>Log Cache Hit Ratio Base<br>Log Pool Requests<br>Log Pool Cache Misses<br>Log Pool Disk Reads<br>Transactions<br>Write Transactions<br>Tracked Transactions<br>Active Connections<br>Active Transactions<br>Size Used<br>Disk Space Available<br>Log Size<br>Space Available<br>Space Used<br>Total Space<br>Usable Disk Space<br>Synchronization State<br>Synchronization Health<br>Suspended Reason<br>Availability Group Name |
| | Cluster Node | Node Name<br>Status Value<br>Status Text<br>Is Current Owner |

| Permission | Resource Group | Metric |
|---|---|---|
| | Availability Group | Replica ID<br>Group ID<br>Name<br>Automated Backup Preference<br>Replicas<br>Primary Replica<br>Primary Recovery Health<br>Secondary Recovery Health<br>Synchronization Health<br>Synchronization Health Property |
| | HADR Replica | Availability Group Name<br>Group ID<br>Name<br>Availability Mode<br>Failover Mode |
| | Wait Type | Wait Type<br>Wait Time<br>Waiting Tasks Count |
| | Program | Program Name<br>Open Sessions<br>CPU Time<br>Memory Usage<br>Max Elapsed Time<br>Reads<br>Writes<br>Logical Reads |
| **public** | Instance | AWE Enabled<br>Total Server Memory<br>Max Server Memory<br>Creation Date<br>Most Recent Start Date<br>Time Since Creation<br>Days Since Last Start<br>Max Thread Count<br>Thread Size<br>CPU Count<br>OS Memory |
| | Database | Database Name<br>Data Space<br>Index Space<br>Reserved Space<br>Reserved Space Unused |

| Permission | Resource Group | Metric |
|---|---|---|
| | File | Database Name<br>File ID<br>Physical Name<br>Size<br>Used Space<br>Autogrow Type<br>Autogrow Percent<br>Autogrow Fixed<br>Autogrow Amount<br>Allocated Free Space<br>Free Space<br>Percent Used<br>Type<br>State<br>Max Size<br>Name |
| **VIEW ANY DEFINITION** | Instance | Creation Date<br>Most Recent Start Date<br>Time Since Creation<br>Days Since Last Start<br>Max Thread Count<br>Thread Size<br>CPU Count<br>OS Memory<br>Used Disk Space<br>Total Disk Space |
| | Disk | Drive ID<br>Resource Name<br>Disk Size<br>Available Disk Space |
| | Database | Database Name<br>Smallest Percent-based File Growth Increment<br>Smallest Fixed Size File Growth Increment<br>Database Row Size<br>Database Log Size<br>Total Used Disk Size<br>Size Used<br>Disk Space Available<br>Log Size<br>Space Available<br>Space Used<br>Total Space<br>Usable Disk Space<br>Data Space<br>Index Space<br>Reserved Space<br>Reserved Space Unused<br>Availability Group Name |

| Permission | Resource Group | Metric |
|---|---|---|
| | Availability Group | Replica ID<br>Group ID<br>Name<br>Automated Backup Preference<br>Replicas<br>Primary Replica<br>Primary Recovery Health<br>Secondary Recovery Health<br>Synchronization Health<br>Synchronization Health Property |
| | HADR Replica | Availability Group Name<br>Group ID<br>Name<br>Availability Mode<br>Failover Mode |
| **GRANT SELECT ON [dbo].[sysjobhistory]** | Instance | Successful Jobs<br>Failed Jobs |
| | Job | Job ID<br>Description<br>Last Run Time<br>Last Message<br>Last Step<br>Last Run Status<br>Raw Last Run Status<br>Next Scheduled Run<br>Last Run Duration<br>Name |

| Permission | Resource Group | Metric |
|---|---|---|
| **VIEW ANY DATABASE** | Database | Database Name<br>Resource Name<br>Recovery Model<br>Status<br>Creation Date<br>Time Since Creation<br>Auto Close<br>Auto Create Statistics<br>Auto Shrink<br>Auto Update Statistics<br>Single User Mode<br>Read Only<br>Effective Used Pages<br>Effective Used<br>Effective Server Total<br>User Scans<br>User Seeks<br>User Lookups<br>User Updates<br>Auto Close Enabled<br>Auto Shrink Enabled<br>Active Connections<br>Active Transactions<br>Size Used<br>Disk Space Available<br>Log Size<br>Space Available<br>Space Used<br>Total Space<br>Usable Disk Space<br>Synchronization State<br>Synchronization Health<br>Suspended Reason |
| **GRANT SELECT ON [dbo].[sysjobactivity]** | Job | Job ID<br>Description<br>Last Run Time<br>Last Message<br>Last Step<br>Last Run Status<br>Raw Last Run Status<br>Next Scheduled Run<br>Last Run Duration<br>Name |
| **GRANT SELECT ON [dbo].[sysjobs_view]** | Job | Job ID<br>Description<br>Last Run Time<br>Last Message<br>Last Step<br>Last Run Status<br>Raw Last Run Status<br>Next Scheduled Run<br>Last Run Duration<br>Name |

| Permission | Resource Group | Metric |
|---|---|---|
| **GRANT SELECT ON [dbo].[sysjobsteps]** | Job Step | Job ID<br>Step ID<br>Step UID<br>Step Name<br>Subsystem<br>Command<br>Last Run Outcome<br>Last Run Duration |

## Management Pack for MongoDB

The **Management Pack for MongoDB** is an embedded adapter for VCF Operations, collecting performance and capacity data from your MongoDB environment and providing predictive analytics and real-time information about problems in your infrastructure—all within the VCF Operations user interface.

### Notable Features

The Management Pack for MongoDB provides the following notable features:

- 8 comprehensive monitoring dashboards:
  - MongoDB Overview
  - MongoDB Environment
  - MongoDB Health Investigation
  - MongoDB IOPS
  - MongoDB Mongos Instances
  - MongoDB Databases
  - MongoDB Replica Sets
  - MongoDB Mongods
- **60 high-level views—and 10 exportable reports**— of MongoDB capacity, health, KPIs, and resource overviews (i.e., clusters, mongos, mongods, databases, config servers, replica sets, and shards)
- **450+ collected metrics** for a MongoDB resources
- **Key performance metrics** – such as read/write operations counters, replication operations counters, connections asserts, network, indexes, and extents.
- **Detailed alerts and recommendations** based on key metrics in MongoDB
- **Capacity definitions** for mongos and mongod resource connections
- **Relationship mapping** between the MongoDB and VMware virtual layers

### System Requirements for Management Pack for MongoDB

Before installing and configuring the Management Pack, ensure your system meets the following requirements.

For compatibility between products, refer to:

VMware Cloud Foundation Operations Interoperability for MongoDB

-OR-

Third Party Solution Interoperability for MongoDB

| | MongoDB Requirements |
|---|---|
| Connection | The Management Pack collects data from the **MongoDB database by way ofthe Mongo Client**.<br>Your **MongoDB hostnameorIPv4 address** is required. A comma-separated list of multiple hosts (mongos/mongod instances) in an environment is supported.<br>**Note:**<br>If using a port other than the default of 27017, you can append the non-standard port number to the hostname. |
| Credentials | **MongoDB User Name/Password** for the admin database<br>**Note:**<br>The same User Name/Password must exist for all mongos/mongods in your environment.<br>**Note:**<br>Your MongoDB configuration may not require authentication. When Creating a Credential for the Management Pack, there is an “Authentication Not Required” option to accommodate this scenario. For those configurations that **do** require authentication, the Management Pack for MongoDB requires a least-privileged MongoDB user account of **read-only**. |

Adding an SSL Certificate to the VMware Cloud Foundation Operations Truststore

**Note:**

Different levels of security can be configured in the Management Pack for MongoDB. The least secure configuration is to leave the default selection of **False** for the **Validate Certificate Against Truststore** Advanced Setting. In this configuration, the Management Pack will accept all SSL certificates. For added security, you can enable SSL by following the instructions below and selecting **True** for the **Validate Certificate Against Truststore** Advanced Setting.

1. Obtain the SSL certificate for your MongoDB host server from your Internet browser. Export the certificate as an **X.509 Certificate (PEM)**.
2. Copy the certificate to your VCF Operations machine.
3. Use ‘SSH’ to log in to the VCF Operations machine as the root user, then run the following command:
   **Linux**:
   $VCOPS_BASE/jre/bin/keytool -import -alias <product_alias> -file /tmp/<certfile> -keystore "$VCOPS_DATA_VCOPS/user/conf/ssl/tcserver.truststore" -storepass `grep ssltruststorePassword /storage/vcops/user/conf/ssl/storePass.properties | sed s/ssltruststorePassword=//` -trustcacerts
   **Windows**:
   %VCOPS_BASE%\jre\bin\keytool -import -alias <product_alias> C:\path\to\certfile -keystore “%VCOPS_DATA_VCOPS%\user\conf\ssl\tcserver.truststore” -storepass <truststore_password> -trustcacerts

   **Note:**

   The <truststore_password> is generated by VCF Operations and is located in/storage/vcops/user/conf/ssl/storePass.properties. Copy the password from the ssltruststorePassword=field and paste it in the <truststore_password> placeholder.

   **Parameter Descriptions**:
   – <product_alias> is a unique name for each key that you add (per host)
   – <certfile> is the location where the cert file was saved
4. Run the reboot command to re-start the VCF Operations machine for the changes to take effect.

### Troubleshooting the Management Pack

Once you have finished Installing and Configuring the Management Pack, check the **Collection State** and **Collection Status** for the adapter instance on the **Solutions** page. If they show as **Not Collecting** or **No Data Receiving**, respectively, the following troubleshooting topics can help you diagnose and correct common problems with your adapter instance.

- Test Connection Errors
- Adapter and Collector Log Files

#### Test Connection Errors

When Creating an Adapter Instance, click **Test Connection** after entering all of your configuration settings to ensure VCF Operations can connect to the specified host.

The following errors are possible:

Authentication Failed

**Error Message**: “Authentication failed. Check your credentials and/or SSL certificate.”

**Action:** Ensure that your MongoDB user name and password is correct the SSL certificate is valid if ‘Validate Certificate Against Truststore’ Advanced Settings configuration field is set to True.

SSL Authentication Required

**Error Message**: “SSL Authentication Required.”

**Action:** Ensure that SSL is enabled (set to True) in the appropriate Advanced Settings configuration field.

PKIX Path Building Failed

**Error Message**: “PKIX path building failed. Unable to find valid certification path to requested target.”

**Action:** Ensure that the SSL certificate is in the correct truststore.

Change Authentication Type to ‘Default’

**Error Message**: Change Authentication Type to ‘Default’. Unsupported mechanism PLAIN.”

**Action:** Change Authentication Type to ‘Default’ when adding your credential.

MongoClient does not have any server addresses

**Error Message**: “MongoClient does not have any server addresses.”

**Action:** Ensure that connections can be made to the hostnames listed in the Host configuration field.

Not authorized on admin to execute command

**Error Message**: “Not authorized on admin to execute command <command> with username=”<username>” to on <host>/admin>.”

**Action:** Make sure the user being used for this host has permissions to run the specified command.

Unable to find databases with server address

**Error Message**: “Unable to find databases with server address list <hosts>.”

**Action:** Make sure the specified host has databases.

Could not create any MongoClients

**Error Message**: “Could not create any MongoClients connecting to mongods.”

**Action:** Make sure the mongods are up.

#### Adapter and Collector Log Files

You can review adapter errors in the adapter and collector log files, which can be viewed in the VCF Operations UI or in an external log viewer.

- **Adapter log files** are located in the $VCOPS_BASE/user/log/adapters/mongodb_adapter3/ folder.
- **Collector log files** are located in the $VCOPS_BASE/user/log/ folder.

To set the base log level for the collector in VCF Operations:

1. Navigate to **Administration > Logs**.
2. Expand the **PRIMARY** folder, then select the **COLLECTOR** folder.
3. Click the **Edit Properties**icon.
4. In the **Root logger level** drop-down menu, select the desired option:
   - The default logging level is **warning.**
   - To troubleshoot issues, set the logging level to **info.**
   - To view detailed messages, including micro steps, queries, and returned results, set the logging level to **debug.**

     **Note:**

     If you set the logging level to **debug,** log files can become large very quickly. Set the logging level to **debug** only for short periods of time.

For more information about viewing log files and modifying levels, refer to the VCF Operations online help.

### Using the Management Pack

The Management Pack for MongoDB includes the following out-of-the-box features, harnessing the power of the VCF Operations user interface:

- Inventory Tree (Traversal Spec)
- Dashboards
- Views
- Reports
- Analysis Badges and Capacity Definitions
- Alerts
- Metrics

#### Inventory Tree (Traversal Spec)

The Inventory Tree (Traversal Spec) feature in VCF Operations allows you to easily navigate your MongoDB environment. The hierarchical structure implicitly shows relationships among resources and enables quick drill-downs to root-cause issues.

To access your MongoDB Inventory Tree, select **Environment** from the top navigation bar, then select **MongoDB** from the left panel. click a resource from the Inventory Tree to populate details about the resource on the right.

#### Dashboards

Dashboards are the primary feature for monitoring and troubleshooting MongoDB from within VCF Operations.

The following dashboards are included in the Management Pack for MongoDB:

- MongoDB Overview provides at-a-glance heatmaps depicting the overall health of your MongoDB resources (mongos instances, mongods, shards, databases, etc.).
- MongoDB Health Investigation displays the current active alerts in your MongoDB environment. When an alert is selected, health information and key performance indicators (KPIs) related to the affected resource are shown, along with a Topology (relationships) view of the resource and health information for its parent and child resources.
- MongoDB Environment depicts your MongoDB environment from your top-level resources, down through the hierarchy. Selecting any resource from the Environment Selector will populate health, KPIs, and properties information for that resource.
- MongoDB IOPS allows you to select a node to view IOPS metrics for that mongos/mongod resource. You can also select a related virtual machine or datastore view IOPS metrics for those resources.
- MongoDB Mongos Instances allows you to select a database and Mongos instance to drill-down into opcounters, asserts, memory, connections, network, and page faults information for that Mongos instance.

- MongoDB Databases allows you to select a Mongos instance and database to view health, KPIs, topology (relationships), and top alerts for the database.
- MongoDB Replica Setsallows you to select a replica set to view opcounters, network, memory, topology (relationships), cache, and connections information for the selected resource.
- MongoDB Mongods allows you to select a mongod resource to view opcounters, memory, topology (relationships), cache, connections, and network information for the selected resource.

### Views

The Management Pack for MongoDB creates views that allow the user to view statistics of metrics for MongoDB resources. The views help give a broad picture of the entire system, as opposed to a more in depth view.

The following views are available in the Management Pack:

- **MongoDB Cluster Overview – Asserts**
- **MongoDB Cluster Overview – Cache**
- **MongoDB Cluster Overview – Current Connections**
- **MongoDB Cluster Overview – Global Lock**
- **MongoDB Cluster Overview – Memory**
- **MongoDB Cluster Overview – Network**
- **MongoDB Cluster Overview – Opcounters**
- **MongoDB Cluster Overview – Page Faults**
- **MongoDB Cluster Overview – Replication Opcounters**
- **MongoDB Config Server Overview – Asserts**
- **MongoDB Config Server Overview – Cache**
- **MongoDB Config Server Overview – Current Connections**
- **MongoDB Config Server Overview – Global Lock**
- **MongoDB Config Server Overview – Memory**
- **MongoDB Config Server Overview – Network**
- **MongoDB Config Server Overview – Opcounters**
- **MongoDB Config Server Overview – Page Faults**
- **MongoDB Config Server Overview – Replication Opcounters**
- **MongoDB Databases Overview – Statistics**
- **MongoDB Health**
- **MongoDB Mongod Capacity**
- **MongoDB Mongod Overview – Asserts**
- **MongoDB Mongod Overview – Cache**
- **MongoDB Mongod Overview – Current Connections**
- **MongoDB Mongod Overview – Documents**
- **MongoDB Mongod Overview – Global Lock**
- **MongoDB Mongod Overview – Memory**
- **MongoDB Mongod Overview – Network**
- **MongoDB Mongod Overview – Opcounters**
- **MongoDB Mongod Overview – Page Faults**
- **MongoDB Mongod Overview – Replication Opcounters**
- **MongoDB Mongod Properties**
- **MongoDB Mongos Capacity**
- **MongoDB Mongos Overview – Asserts**
- **MongoDB Mongos Overview – Current Connections**
- **MongoDB Mongos Overview – Memory**
- **MongoDB Mongos Overview – Network**
- **MongoDB Mongos Overview – Opcounters**
- **MongoDB Mongos Overview – Page Faults**

- **MongoDB Mongos Properties**
- **MongoDB Replica Set Overview – Asserts**
- **MongoDB Replica Set Overview – Cache**
- **MongoDB Replica Set Overview – Current Connections**
- **MongoDB Replica Set Overview – Global Lock**
- **MongoDB Replica Set Overview – Memory**
- **MongoDB Replica Set Overview – Network**
- **MongoDB Replica Set Overview – Opcounters**
- **MongoDB Replica Set Overview – Page Faults**
- **MongoDB Replica Set Overview – Replication Opcounters**
- **MongoDB Replica Set Properties**
- **MongoDB Shard Overview – Asserts**
- **MongoDB Shard Overview – Cache**
- **MongoDB Shard Overview – Current Connections**
- **MongoDB Shard Overview – Global Lock**
- **MongoDB Shard Overview – Memory**
- **MongoDB Shard Overview – Network**
- **MongoDB Shard Overview – Opcounters**
- **MongoDB Shard Overview – Page Faults**
- **MongoDB Shard Overview – Replication Opcounters**
- **MongoDB Shard Properties**

To access the Management Pack views:

1. Navigate to **Environment > All Objects > MongoDB**.
2. Double-click on the desired object (resource).
3. Select the **Details** tab, then **Views.**

The available views for that resource are listed and can be selected.

### Reports

The Management Pack for MongoDB contains out-of-the-box reports—which contain one or more Views—that you can export and easily share with key stakeholders in either .pdf or .csv formats.

The following reports are available in the Management Pack:

- **MongoDB Capacity**
- **MongoDB Cluster Overview**
- **MongoDB Config Server Overview**
- **MongoDB Databases Overview**
- **MongoDB Health**
- **MongoDB KPIs**
- **MongoDB Mongod Overview**
- **MongoDB Mongos Overview**
- **MongoDB Replica Set Overview**
- **MongoDB Shard Overview**

To access Management Pack reports:

1. Navigate to **Environment > All Objects > MongoDB**.
2. Double-click on the desired object (resource).
3. Select the **Reports** tab, then **Report Templates**.

The available reports for the selected resource are listed and can be selected.

To run a selected report, click the **Run Template** icon.

Analysis Badges and Capacity Definitions

Using the predictive analytics capabilities of VCF Operations Analysis Badges through capacity definitions, the Management Pack for MongoDB populates capacities as indicated in the table below.

**Capacity Definitions List**

| Container | Use in Workload |
|---|---|
| **Mongod** | |
| VMware Cloud Foundation Operations Calculated Connections Capacity | yes |
| **Mongos** | |
| VMware Cloud Foundation Operations Calculated Connections Capacity | yes |

Alerts

The Management Pack for MongoDB creates alerts (and in some cases provides recommended actions) based on various symptoms it detects in your MongoDB Environment. See the table below for the list of alerts available in the Management Pack.

**Alerts List**

| Name | Description | Recommendation |
|---|---|---|
| Replica Set has even number of voters | This alert indicates that an Immediate alert was raised in MongoDB. | A replica set should always have an odd number of members so there is never a tie when voting for the primary host. Either add another replicated host, or add an arbiter to the replica set. |
| Replica Set Has Less than 3 Data Bearing Nodes | This alert indicates that an Immediate alert was raised in MongoDB. | Three data-bearing nodes are required for replica-set wide data durability. |
| Mongod Shares a Host | This alert indicates that an Immediate alert was raised in MongoDB. | Running multiple mongod instances on a single host increases the likelihood of having inaccessible data by reducing horizontal scaling. |
| Mongod Is Recovering From Delay | This alert indicates that an Immediate alert was raised in MongoDB. | This instance is in RECOVERING mode. |
| Mongod Has a Delay Behind the Primary | This alert indicates that an Immediate alert was raised in MongoDB. | This node has very high replication delays. Delayed nodes can cause out-of-date data to be presented. |
| Mongod Has a Delay Behind the Primary | This alert indicates that an Immediate alert was raised in MongoDB. | This node has very high replication delays. Delayed nodes can cause out-of-date data to be presented. |
| Mongod Journaling Disabled | This alert indicates that an Immediate alert was raised in MongoDB. | To get accurate snapshots of running mongod processes, journaling must be enabled. This is critical if using the MMAPv1 storage engine. |
| Mongos Total Connections Reaching Maxmimum | This alert indicates that an Immediate alert was raised in MongoDB. | Current connections approaching max connections. |
| Mongos Ratio of Page-Faults to Operations Greater than or Equal to 1 | This alert indicates that an Immediate alert was raised in MongoDB. | There are a large number of page faults. Consider increasing memory provisioning. |
| Mongos is Reaching a Congested Level of Connections | This alert indicates that an Immediate alert was raised in MongoDB. | Number of connections has greatly increased compared to normal. Large numbers of connections use large amounts of memory, so be sure to provision adequate memory on the host. |

| Name | Description | Recommendation |
|---|---|---|
| Config Server Has no Configured Replica Set | This alert indicates that an Immediate alert was raised in MongoDB. | A production config server should always be configured as a replica set; when configured as a single host, if that host goes down, the entire cluster becomes inaccessible. |
| Shard Has No Configured Replica Set | This alert indicates that an Immediate alert was raised in MongoDB. | If a shard is not configured with a replica set, then if the host it is configured with goes down, none of the data on that host will be available. Configuring the shard as a replica set provides better fallback, allowing for a system with higher-availability. |

Metrics

The Management Pack for MongoDB collects the metrics listed in the table below, according to MongoDB resource kinds.

| Resource Kind | Resource Group | Resource Attribute | Attribute Description |
|---|---|---|---|
| MongoDB Mongos | Extra Info | Page Faults | Total number of page faults |
| MongoDB Mongos | Extra Info | Heap Used | Heap used by this resources database processes |
| MongoDB Mongos | Network | Network Traffic Out | Data sent from Mongos |
| MongoDB Mongos | Network | Network Traffic In | Data received by Mongos |
| MongoDB Mongos | Network | Number of Requests | Number of requests Mongos has received |
| MongoDB Mongos | Properties | PID | Process ID number on this resource |
| MongoDB Mongos | Properties | Advisory Host FQDNs | Advisory host fully qualified domain names of the machine hosting this resource |
| MongoDB Mongos | Properties | Process | Current MongoDB process on this resource |
| MongoDB Mongos | Properties | Version | Version of the current MongoDB process |
| MongoDB Mongos | Properties | Uptime | Time the current MongoDB process has been active |
| MongoDB Mongos | Properties | Host Name | Hostname of the machine hosting this resource |
| MongoDB Mongos | Connections | Total Connections Created | Count of all incoming connections created to this resource |
| MongoDB Mongos | Connections | Current Connections | Number of incoming connections to this resource |
| MongoDB Mongos | Connections | Available Connections | Number of unused incoming connections available to this resource |
| MongoDB Mongos | Asserts | Asserts: User | Number of user assertions that have occurred since the MongoDB process started |
| MongoDB Mongos | Asserts | Asserts: Rollovers | Number of rollovers since the MongoDB process started |

| Resource Kind | Resource Group | Resource Attribute | Attribute Description |
|---|---|---|---|
| MongoDB Mongos | Asserts | Asserts: Message | Number of message assertions raised since the MongoDB process started |
| MongoDB Mongos | Asserts | Asserts: Warning | Number of warnings raised since the MongoDB process started |
| MongoDB Mongos | Asserts | Asserts: Regular | Number of regular assertions raised since the MongoDB process started |
| MongoDB Mongos | Background Flushing | Background Flushing: Last | Time it took the last flush operation to complete |
| MongoDB Mongos | Background Flushing | Background Flushing: Average | Average time for each flush to disk |
| MongoDB Mongos | Background Flushing | Background Flushing: Total | Time the mongod processes have spent writing data to disk |
| MongoDB Mongos | Background Flushing | Background Flushing: Flushes | Number of times the databases on this resource has flushed all writes to disk |
| MongoDB Mongos | Opcounters | Command Operations | Database commands of Mongos |
| MongoDB Mongos | Opcounters | Insert Operations | Insert operations of Mongos |
| MongoDB Mongos | Opcounters | Update Operations | Update operations of Mongos |
| MongoDB Mongos | Opcounters | Get More Operations | Getmore operations of Mongos |
| MongoDB Mongos | Opcounters | Query Operations | Query operations of Mongos |
| MongoDB Mongos | Opcounters | Delete Operations | Delete operations of Mongos |
| MongoDB Mongos | Opcounters | Read Operations | Read operations of Mongos |
| MongoDB Mongos | Opcounters | Write Operations | Write operations of Mongos |
| MongoDB Mongos | Opcounters | Read/Write Operations | All read and write operations of Mongos |
| MongoDB Mongos | Memory | Extended Memory Supported | Indicates if the machine hosting this resource supports extended memory information |
| MongoDB Mongos | Memory | Resident Memory | Amount of RAM used by this resource's database processes |
| MongoDB Mongos | Memory | Memory Architecture | Architecture of the machine hosting this resource |
| MongoDB Mongos | Memory | Virtual Memory | Virtual memory being used by the mongod process |
| MongoDB Mongos | Memory | Mapped Memory with Journal | Amount of mapped memory with journal by this resource's databases |
| MongoDB Mongos | Memory | Mapped Memory | Amount of mapped memory by this resource's databases |
| MongoDB Mongos | Children | Database Children | Database resource names of Child resources. |
| MongoDB Mongos | Children | Config Server Children | Config Server resource names of Child resources. |
| MongoDB Mongos | Children | Shard Children | Shard resource names of Child resources. |
| MongoDB Mongos | Parents | Cluster Parents | Cluster resource names of Parent resources. |

| Resource Kind | Resource Group | Resource Attribute | Attribute Description |
|---|---|---|---|
| MongoDB Mongos | Alerts | Total Connections Reaching Maximum | The percent of connections being used. |
| MongoDB Mongos | Alerts | Ratio of Page Faults to Operations | The ratio of Page Faults to Operations. |
| MongoDB Mongos | Alerts | Ratio of Page Faults to Operations is Greater Than Or Equal to 1 | The ratio of Page Faults to Operations is greater than or equal to one. |
| MongoDB Cluster | Aggregated Mongod Metrics\| Opcounters | Delete Operations | Total delete operations of related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Opcounters | Update Operations | Total update operations of related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Opcounters | Get More Operations | Total getmore operations of related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Opcounters | Query Operations | Total queries operations of related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Opcounters | Command Operations | Total database commands of related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Opcounters | Insert Operations | Total insert operations of related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Opcounters | Read Operations | Total read operations of related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Opcounters | Write Operations | Total write operations of related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Opcounters | Read/Write Operations | All read and write operations of related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Replication Opcounters | Replication Get More Operations | Total replicated getmore operations of related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Replication Opcounters | Replication Delete Operations | Total replicated delete operations of related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Replication Opcounters | Replication Command Operations | Total replicated database commands of related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Replication Opcounters | Replication Insert Operations | Total replicated insert operations of related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Replication Opcounters | Replication Query Operations | Total replicated query operations of related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Replication Opcounters | Replication Update Operations | Total replicated update operations of related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Replication Opcounters | Replication Read Operations | Total replicated read operations of related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Replication Opcounters | Replication Write Operations | Total replicated write operations of related shards and config servers |

| Resource Kind | Resource Group | Resource Attribute | Attribute Description |
|---|---|---|---|
| MongoDB Cluster | Aggregated Mongod Metrics\| Replication Opcounters | Replication Read/Write Operations | Total replicated read and write operations of related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Network | Network Traffic Out | Data sent from related config servers and shards |
| MongoDB Cluster | Aggregated Mongod Metrics\| Network | Network Traffic In | Data received by related config servers and shards |
| MongoDB Cluster | Aggregated Mongod Metrics\| Network | Number of Requests | Number of requests related config servers and shards have received |
| MongoDB Cluster | Aggregated Mongod Metrics\| Cursor | Cursors Timed Out | Total number of cursors that have timed out for all related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Cursor | Cursors Open | Total number of cursors this environment maintains for related all shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Cache | Written Cache | Total data written from cache in related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Cache | Dirty Cache | Total dirty data in cache in related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Cache | Current Cache | Total data currently in cache in related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Cache | Read Cache | Total data read into cache in related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Extra Info | Page Faults | Total number of page faults from related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Asserts | Asserts: Message | Number of message assertions raised from related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Asserts | Asserts: Rollovers | Number of rollovers from related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Asserts | Asserts: Warning | Number of warnings raised from related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Asserts | Asserts: User | Number of user assertions from related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Asserts | Asserts: Regular | Number of regular assertions from related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Global Lock | Operations Waiting to Write | Total queued operations from related shards and config servers waiting for a write lock |
| MongoDB Cluster | Aggregated Mongod Metrics\| Global Lock | Total Operations Waiting | Total queued operations from related shards and config servers waiting for a lock |

| Resource Kind | Resource Group | Resource Attribute | Attribute Description |
|---|---|---|---|
| MongoDB Cluster | Aggregated Mongod Metrics\| Global Lock | Operations Waiting to Read | Total queued operations from related shards and config servers waiting for a read lock |
| MongoDB Cluster | Aggregated Mongod Metrics\| Connections | Current Connections | Current number of incoming connections to all related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Connections | Available Connections | Current number of unused incoming connections available to all related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Connections | Total Connections Created | Total incoming connections to all related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Tickets | Write Tickets Available | Total amount of write tickets available to the WiredTiger storage engine of all related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Tickets | Read Tickets Available | Total amount of read tickets available to the WiredTiger storage engine of all related shards and config servers |
| MongoDB Cluster | Aggregated Mongod Metrics\| Memory | Virtual Memory | Virtual memory being used by the related shards' and config servers' |
| MongoDB Cluster | Aggregated Mongod Metrics\| Memory | Resident Memory | Amount of RAM used by the related shards' and config servers' database processes |
| MongoDB Cluster | Aggregated Mongod Metrics\| Memory | Mapped Memory with Journal | Amount of mapped memory with journal by the related shards' and config servers' databases |
| MongoDB Cluster | Aggregated Mongod Metrics\| Memory | Mapped Memory | Amount of mapped memory by the related shards' and config servers' databases |
| MongoDB Cluster | Children | Mongos Children | Mongos resource names of Child resources. |
| MongoDB Config Server | Aggregated Mongod Metrics\| Opcounters | Delete Operations | Total delete operations of related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Opcounters | Update Operations | Total update operations of related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Opcounters | Get More Operations | Total getmore operations of related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Opcounters | Query Operations | Total queries operations of related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Opcounters | Command Operations | Total database commands of related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Opcounters | Insert Operations | Total insert operations of related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Opcounters | Read Operations | Total read operations of related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Opcounters | Write Operations | Total write operations of related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Opcounters | Read/Write Operations | All read and write operations of related Mongods |

| Resource Kind | Resource Group | Resource Attribute | Attribute Description |
|---|---|---|---|
| MongoDB Config Server | Aggregated Mongod Metrics\| Replication Opcounters | Replication Get More Operations | Total replicated getmore operations of related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Replication Opcounters | Replication Delete Operations | Total replicated delete operations of related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Replication Opcounters | Replication Command Operations | Total replicated database commands of related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Replication Opcounters | Replication Insert Operations | Total replicated insert operations of related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Replication Opcounters | Replication Query Operations | Total replicated query operations of related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Replication Opcounters | Replication Update Operations | Total replicated update operations of related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Replication Opcounters | Replication Read Operations | Total replicated read operations of related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Replication Opcounters | Replication Write Operations | Total replicated write operations of related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Replication Opcounters | Replication Read/Write Operations | Total replicated read and write operations of related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Network | Network Traffic Out | Data sent from related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Network | Network Traffic In | Data received by related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Network | Number of Requests | Number of requests related Mongod have received |
| MongoDB Config Server | Aggregated Mongod Metrics\| Cursor | Cursors Timed Out | Total number of cursors that have timed out for all related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Cursor | Cursors Open | Total number of cursors this environment maintains for all related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Cache | Written Cache | Total data written from cache in related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Cache | Dirty Cache | Total dirty data in cache in related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Cache | Current Cache | Total data currently in cache in related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Cache | Read Cache | Total data read into cache in related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Extra Info | Page Faults | Total number of page faults from related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Asserts | Asserts: Message | Number of message assertions raised from related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Asserts | Asserts: Rollovers | Number of rollovers from related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Asserts | Asserts: Warning | Number of warnings raised from related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Asserts | Asserts: User | Number of user assertions from related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Asserts | Asserts: Regular | Number of regular assertions from related Mongods |

| Resource Kind | Resource Group | Resource Attribute | Attribute Description |
|---|---|---|---|
| MongoDB Config Server | Aggregated Mongod Metrics\| Global Lock | Operations Waiting to Write | Total queued operations from related Mongods waiting for a write lock |
| MongoDB Config Server | Aggregated Mongod Metrics\| Global Lock | Total Operations Waiting | Total queued operations from related Mongods waiting for a lock |
| MongoDB Config Server | Aggregated Mongod Metrics\| Global Lock | Operations Waiting to Read | Total queued operations from related Mongods waiting for a read lock |
| MongoDB Config Server | Aggregated Mongod Metrics\| Connections | Current Connections | Current number of incoming connections to all related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Connections | Available Connections | Current number of unused incoming connections available to all related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Connections | Total Connections Created | Total incoming connections to all related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Tickets | Write Tickets Available | Total amount of write tickets available to the WiredTiger storage engine of all related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Tickets | Read Tickets Available | Total amount of read tickets available to the WiredTiger storage engine of all related Mongods |
| MongoDB Config Server | Aggregated Mongod Metrics\| Memory | Virtual Memory | Virtual memory being used by the related Mongods' |
| MongoDB Config Server | Aggregated Mongod Metrics\| Memory | Resident Memory | Amount of RAM used by the related Mongods' database processes |
| MongoDB Config Server | Aggregated Mongod Metrics\| Memory | Mapped Memory with Journal | Amount of mapped memory with journal by the related Mongods' databases |
| MongoDB Config Server | Aggregated Mongod Metrics\| Memory | Mapped Memory | Amount of mapped memory by the related Mongods' databases |
| MongoDB Config Server | Children | Replica Set Children | Replica Set resource names of Child resources. |
| MongoDB Config Server | Children | Mongod Children | Mongod resource names of Child resources. |
| MongoDB Config Server | Parents | Mongos Parents | Mongos resource names of Parent resources. |
| MongoDB Config Server | Parents | Database Parents | Database resource names of Parent resources. |
| MongoDB Config Server | Alerts | Config Server Has Child Replica Set | Config Server Has a Replica Set as a Child. |
| MongoDB Replica Set | Properties | Chaining Allowed | Indicated if chaining is supported on this resource |
| MongoDB Replica Set | Properties | Version | Version of MongoDB this resource has |
| MongoDB Replica Set | Properties | Protocol Version | Protocol version of this resource |
| MongoDB Replica Set | Properties | Primary Replication | Indicates this resource is a primary replication set |
| MongoDB Replica Set | Properties | State | State of this resource |

| Resource Kind | Resource Group | Resource Attribute | Attribute Description |
|---|---|---|---|
| MongoDB Replica Set | Properties | Replica Set Request Votes | Total request votes in a replica set |
| MongoDB Replica Set | Last Error | Last Error Default | Last error default of this resource |
| MongoDB Replica Set | Last Error | Last Error Modes | Last error modes of this resource |
| MongoDB Replica Set | Replication | Replication Rollback ID | Rollback ID of this resource |
| MongoDB Replica Set | Replication | Replication Hosts | List of hosts within this replication set |
| MongoDB Replica Set | Replication | Replication Arbiters | List of arbiters within this replication set |
| MongoDB Replica Set | Election | Election Timeout | Duration of this resource's election timeout |
| MongoDB Replica Set | Election | Election ID | Election ID of this resource |
| MongoDB Replica Set | Heartbeat | Heartbeat Timeout | Duration of this resource's heartbeat timeout |
| MongoDB Replica Set | Heartbeat | Heartbeat Interval | Frequency of this resource's heartbeats |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Opcounters | Delete Operations | Total delete operations of related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Opcounters | Update Operations | Total update operations of related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Opcounters | Get More Operations | Total getmore operations of related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Opcounters | Query Operations | Total queries operations of related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Opcounters | Command Operations | Total database commands of related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Opcounters | Insert Operations | Total insert operations of related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Opcounters | Read Operations | Total read operations of related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Opcounters | Write Operations | Total write operations of related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Opcounters | Read/Write Operations | All read and write operations of related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Replication Opcounters | Replication Get More Operations | Total replicated getmore operations of related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Replication Opcounters | Replication Delete Operations | Total replicated delete operations of related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Replication Opcounters | Replication Command Operations | Total replicated database commands of related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Replication Opcounters | Replication Insert Operations | Total replicated insert operations of related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Replication Opcounters | Replication Query Operations | Total replicated query operations of related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Replication Opcounters | Replication Update Operations | Total replicated update operations of related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Replication Opcounters | Replication Read Operations | Total replicated read operations of related Mongods |

| Resource Kind | Resource Group | Resource Attribute | Attribute Description |
|---|---|---|---|
| MongoDB Replica Set | Aggregated Mongod Metrics\| Replication Opcounters | Replication Write Operations | Total replicated write operations of related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Replication Opcounters | Replication Read/Write Operations | Total replicated read and write operations of related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Network | Network Traffic Out | Data sent from related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Network | Network Traffic In | Data received by related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Network | Number of Requests | Number of requests related Mongod have received |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Cursor | Cursors Timed Out | Total number of cursors that have timed out for all related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Cursor | Cursors Open | Total number of cursors this environment maintains for all related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Cache | Written Cache | Total data written from cache in related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Cache | Dirty Cache | Total dirty data in cache in related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Cache | Current Cache | Total data currently in cache in related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Cache | Read Cache | Total data read into cache in related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Extra Info | Page Faults | Total number of page faults from related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Asserts | Asserts: Message | Number of message assertions raised from related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Asserts | Asserts: Rollovers | Number of rollovers from related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Asserts | Asserts: Warning | Number of warnings raised from related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Asserts | Asserts: User | Number of user assertions from related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Asserts | Asserts: Regular | Number of regular assertions from related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Global Lock | Operations Waiting to Write | Total queued operations from related Mongods waiting for a write lock |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Global Lock | Total Operations Waiting | Total queued operations from related Mongods waiting for a lock |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Global Lock | Operations Waiting to Read | Total queued operations from related Mongods waiting for a read lock |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Connections | Current Connections | Current number of incoming connections to all related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Connections | Available Connections | Current number of unused incoming connections available to all related Mongods |

| Resource Kind | Resource Group | Resource Attribute | Attribute Description |
|---|---|---|---|
| MongoDB Replica Set | Aggregated Mongod Metrics\| Connections | Total Connections Created | Total incoming connections to all related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Tickets | Write Tickets Available | Total amount of write tickets available to the WiredTiger storage engine of all related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Tickets | Read Tickets Available | Total amount of read tickets available to the WiredTiger storage engine of all related Mongods |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Memory | Virtual Memory | Virtual memory being used by the related Mongods' |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Memory | Resident Memory | Amount of RAM used by the related Mongods' database processes |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Memory | Mapped Memory with Journal | Amount of mapped memory with journal by the related Mongods' databases |
| MongoDB Replica Set | Aggregated Mongod Metrics\| Memory | Mapped Memory | Amount of mapped memory by the related Mongods' databases |
| MongoDB Replica Set | Children | Mongod Children | Mongod resource names of Child resources. |
| MongoDB Replica Set | Parents | Config Server Parents | Config Server resource names of Parent resources. |
| MongoDB Replica Set | Parents | Database Parents | Database resource names of Parent resources. |
| MongoDB Replica Set | Parents | Shard Parents | Shard resource names of Parent resources. |
| MongoDB Replica Set | Alerts | Replica Set Even Votes | Replica Set even or odd total votes. |
| MongoDB Replica Set | Alerts | Replica Set Total Votes | Replica Set total votes. |
| MongoDB Replica Set | Alerts | Number of Data Bearing Nodes | Number of Data Bearing Mongods that are not Arbiters. |
| MongoDB Database | Statistics | Data Size | Data size of this resource |
| MongoDB Database | Statistics | Namespace File Size | Namespace file size for this resource |
| MongoDB Database | Statistics | Extents | Number of extents for this resource |
| MongoDB Database | Statistics | Index Size | Total index size for this resource |
| MongoDB Database | Statistics | Total Objects | Number of objects for this resource |
| MongoDB Database | Statistics | Total File Size | Total file size for this resource |
| MongoDB Database | Statistics | Total Indexes | Number of indexes for this resource |
| MongoDB Database | Statistics | Total Collections | Number of collections for this resource |
| MongoDB Database | Statistics | Average Object Size | Average object size of this resource |
| MongoDB Database | Statistics | Storage Size | Storage size of this resource |
| MongoDB Database | Properties | Data File Version | Data file version for this resource |

| Resource Kind | Resource Group | Resource Attribute | Attribute Description |
|---|---|---|---|
| MongoDB Database | Collection | Collections Size | Size of all records in this resource's collections |
| MongoDB Database | Collection | Collections Total Index Size | Total size of all in this resource's collections |
| MongoDB Database | Collection | Collections Total Indexes | Number of indexes on this resource's collections |
| MongoDB Database | Collection | Collections Count | Number of objects or documents in this resource's collections |
| MongoDB Database | Collection | Collections Max Size | Maximum size of this resource's capped collections |
| MongoDB Database | Collection | Collections Storage Size | Storage size of all records in this resource's collections |
| MongoDB Database | Collection | Collections Average Object Size | Average size of objects in this resource's collections |
| MongoDB Database | Collection | Collections Max | Maximum number of documents in this resource's collections |
| MongoDB Database | Freelist | Freelist Extents Size | Size of the extents on the freelist for this resource |
| MongoDB Database | Freelist | Freelist Extents | Number of extents in the freelist for this resource |
| MongoDB Database | Children | Shard Children | Shard resource names of Child resources. |
| MongoDB Database | Children | Mongod Children | Mongod resource names of Child resources. |
| MongoDB Database | Children | Config Server Children | Config Server resource names of Child resources. |
| MongoDB Database | Children | Replica Set Children | Replica Set resource names of Child resources. |
| MongoDB Database | Parents | Mongos Parents | Mongos resource names of Parent resources. |
| MongoDB Shard | Properties | Version | Shard version of this resource |
| MongoDB Shard | Alerts | Shard Has Replica Set Child | Shard Has a Replica Set Child |
| MongoDB Shard | Aggregated Mongod Metrics\| Opcounters | Delete Operations | Total delete operations of related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Opcounters | Update Operations | Total update operations of related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Opcounters | Get More Operations | Total getmore operations of related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Opcounters | Query Operations | Total queries operations of related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Opcounters | Command Operations | Total database commands of related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Opcounters | Insert Operations | Total insert operations of related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Opcounters | Read Operations | Total read operations of related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Opcounters | Write Operations | Total write operations of related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Opcounters | Read/Write Operations | All read and write operations of related Mongods |

| Resource Kind | Resource Group | Resource Attribute | Attribute Description |
|---|---|---|---|
| MongoDB Shard | Aggregated Mongod Metrics\| Replication Opcounters | Replication Get More Operations | Total replicated getmore operations of related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Replication Opcounters | Replication Delete Operations | Total replicated delete operations of related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Replication Opcounters | Replication Command Operations | Total replicated database commands of related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Replication Opcounters | Replication Insert Operations | Total replicated insert operations of related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Replication Opcounters | Replication Query Operations | Total replicated query operations of related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Replication Opcounters | Replication Update Operations | Total replicated update operations of related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Replication Opcounters | Replication Read Operations | Total replicated read operations of related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Replication Opcounters | Replication Write Operations | Total replicated write operations of related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Replication Opcounters | Replication Read/Write Operations | Total replicated read and write operations of related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Network | Network Traffic Out | Data sent from related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Network | Network Traffic In | Data received by related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Network | Number of Requests | Number of requests related Mongod have received |
| MongoDB Shard | Aggregated Mongod Metrics\| Cursor | Cursors Timed Out | Total number of cursors that have timed out for all related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Cursor | Cursors Open | Total number of cursors this environment maintains for all related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Cache | Written Cache | Total data written from cache in related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Cache | Dirty Cache | Total dirty data in cache in related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Cache | Current Cache | Total data currently in cache in related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Cache | Read Cache | Total data read into cache in related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Extra Info | Page Faults | Total number of page faults from related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Asserts | Asserts: Message | Number of message assertions raised from related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Asserts | Asserts: Rollovers | Number of rollovers from related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Asserts | Asserts: Warning | Number of warnings raised from related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Asserts | Asserts: User | Number of user assertions from related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\| Asserts | Asserts: Regular | Number of regular assertions from related Mongods |

| Resource Kind | Resource Group | Resource Attribute | Attribute Description |
|---|---|---|---|
| MongoDB Shard | Aggregated Mongod Metrics\|Global Lock | Operations Waiting to Write | Total queued operations from related Mongods waiting for a write lock |
| MongoDB Shard | Aggregated Mongod Metrics\|Global Lock | Total Operations Waiting | Total queued operations from related Mongods waiting for a lock |
| MongoDB Shard | Aggregated Mongod Metrics\|Global Lock | Operations Waiting to Read | Total queued operations from related Mongods waiting for a read lock |
| MongoDB Shard | Aggregated Mongod Metrics\|Connections | Current Connections | Current number of incoming connections to all related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\|Connections | Available Connections | Current number of unused incoming connections available to all related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\|Connections | Total Connections Created | Total incoming connections to all related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\|Tickets | Write Tickets Available | Total amount of write tickets available to the WiredTiger storage engine of all related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\|Tickets | Read Tickets Available | Total amount of read tickets available to the WiredTiger storage engine of all related Mongods |
| MongoDB Shard | Aggregated Mongod Metrics\|Memory | Virtual Memory | Virtual memory being used by the related Mongods' |
| MongoDB Shard | Aggregated Mongod Metrics\|Memory | Resident Memory | Amount of RAM used by the related Mongods' database processes |
| MongoDB Shard | Aggregated Mongod Metrics\|Memory | Mapped Memory with Journal | Amount of mapped memory with journal by the related Mongods' databases |
| MongoDB Shard | Aggregated Mongod Metrics\|Memory | Mapped Memory | Amount of mapped memory by the related Mongods' databases |
| MongoDB Shard | Parents | Database Parents | Database resource names of Parent resources. |
| MongoDB Shard | Parents | Mongos Parents | Mongos resource names of Parent resources. |
| MongoDB Shard | Children | Replica Set Children | Replica Set resource names of Child resources. |
| MongoDB Shard | Children | Mongod Children | Mongod resource names of Child resources. |
| MongoDB Mongod | Extra Info | Page Faults | Total number of page faults |
| MongoDB Mongod | Extra Info | Heap Used | Heap used by this resources database processes |
| MongoDB Mongod | Locks\|Oplog | Oplog Deadlocks Intent Exclusive Locks | Number of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Oplog | Oplog Acquire Waits Intent Exclusive Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |

| Resource Kind | Resource Group | Resource Attribute | Attribute Description |
|---|---|---|---|
| MongoDB Mongod | Locks\|Oplog | Oplog Cumulative Wait Time Exclusive Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Oplog | Oplog Intent Shared Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Oplog | Oplog Exclusive Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Oplog | Oplog Cumulative Wait Time Intent Shared Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Oplog | Oplog Shared Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Oplog | Oplog Acquire Waits Exclusive Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |
| MongoDB Mongod | Locks\|Oplog | Oplog Intent Exclusive Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Oplog | Oplog Acquire Waits Shared Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |
| MongoDB Mongod | Locks\|Oplog | Oplog Acquire Waits Intent Shared Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |
| MongoDB Mongod | Locks\|Oplog | Oplog Cumulative Wait Time Intent Exclusive Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Oplog | Oplog Deadlocks Exclusive Locks | Number of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Oplog | Oplog Cumulative Wait Time Shared Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Oplog | Oplog Deadlocks Intent Shared Locks | Number of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Oplog | Oplog Deadlocks Shared Locks | Number of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Mmapv1 Journal | MMAPv1 Journal Deadlocks Intent Exclusive Locks | Number of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Mmapv1 Journal | MMAPv1 Journal Cumulative Wait Time Shared Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Mmapv1 Journal | MMAPv1 Journal Acquire Waits Intent Exclusive Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |
| MongoDB Mongod | Locks\|Mmapv1 Journal | MMAPv1 Journal Cumulative Wait Time Exclusive Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Mmapv1 Journal | MMAPv1 Journal Cumulative Wait Time Intent Shared Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Mmapv1 Journal | MMAPv1 Journal Acquire Waits Intent Shared Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |
| MongoDB Mongod | Locks\|Mmapv1 Journal | MMAPv1 Journal Acquire Waits Exclusive Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |

| Resource Kind | Resource Group | Resource Attribute | Attribute Description |
|---|---|---|---|
| MongoDB Mongod | Locks\|Mmapv1 Journal | MMAPv1 Journal Intent Exclusive Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Mmapv1 Journal | MMAPv1 Journal Cumulative Wait Time Intent Exclusive Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Mmapv1 Journal | MMAPv1 Journal Deadlocks Intent Shared Locks | Number of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Mmapv1 Journal | MMAPv1 Journal Deadlocks Shared Locks | Number of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Mmapv1 Journal | MMAPv1 Journal Intent Shared Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Mmapv1 Journal | MMAPv1 Journal Acquire Waits Shared Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |
| MongoDB Mongod | Locks\|Mmapv1 Journal | MMAPv1 Journal Exclusive Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Mmapv1 Journal | MMAPv1 Journal Deadlocks Exclusive Locks | Number of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Mmapv1 Journal | MMAPv1 Journal Shared Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Metadata | Metadata Cumulative Wait Time Exclusive Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Metadata | Metadata Shared Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Metadata | Metadata Intent Shared Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Metadata | Metadata Deadlocks Intent Exclusive Locks | Number of times lock acquisitions with this mode encountered deadlocksNumber of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Metadata | Metadata Acquire Waits Intent Exclusive Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |
| MongoDB Mongod | Locks\|Metadata | Metadata Exclusive Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Metadata | Metadata Deadlocks Exclusive Locks | Number of times lock acquisitions with this mode encountered deadlocksNumber of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Metadata | Metadata Acquire Waits Intent Shared Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |
| MongoDB Mongod | Locks\|Metadata | Metadata Cumulative Wait Time Intent Exclusive Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Metadata | Metadata Intent Exclusive Locks | Number of times the lock was acquired in this mode |

| Resource Kind | Resource Group | Resource Attribute | Attribute Description |
|---|---|---|---|
| MongoDB Mongod | Locks\|Metadata | Metadata Deadlocks Shared Locks | Number of times lock acquisitions with this mode encountered deadlocksNumber of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Metadata | Metadata Acquire Waits Exclusive Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |
| MongoDB Mongod | Locks\|Metadata | Metadata Deadlocks Intent Shared Locks | Number of times lock acquisitions with this mode encountered deadlocksNumber of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Metadata | Metadata Cumulative Wait Time Intent Shared Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Metadata | Metadata Cumulative Wait Time Shared Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Metadata | Metadata Acquire Waits Shared Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |
| MongoDB Mongod | Locks\|Collection | Collection Acquire Waits Intent Shared Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |
| MongoDB Mongod | Locks\|Collection | Collection Cumulative Wait Time Exclusive Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Collection | Collection Deadlocks Shared Locks | Number of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Collection | Collection Intent Shared Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Collection | Collection Shared Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Collection | Collection Cumulative Wait Time Intent Shared Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Collection | Collection Intent Exclusive Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Collection | Collection Acquire Waits Shared Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |
| MongoDB Mongod | Locks\|Collection | Collection Exclusive Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Collection | Collection Acquire Waits Intent Exclusive Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |
| MongoDB Mongod | Locks\|Collection | Collection Deadlocks Intent Shared Locks | Number of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Collection | Collection Acquire Waits Exclusive Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |

| Resource Kind | Resource Group | Resource Attribute | Attribute Description |
|---|---|---|---|
| MongoDB Mongod | Locks\|Collection | Collection Deadlocks Intent Exclusive Locks | Number of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Collection | Collection Cumulative Wait Time Shared Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Collection | Collection Deadlocks Exclusive Locks | Number of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Collection | Collection Cumulative Wait Time Intent Exclusive Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Database | Database Acquire Waits Shared Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |
| MongoDB Mongod | Locks\|Database | Database Cumulative Wait Time Intent Shared Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Database | Database Exclusive Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Database | Database Acquire Waits Intent Exclusive Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |
| MongoDB Mongod | Locks\|Database | Database Intent Exclusive Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Database | Database Deadlocks Shared Locks | Number of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Database | Database Deadlocks Intent Exclusive Locks | Number of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Database | Database Cumulative Wait Time Shared Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Database | Database Cumulative Wait Time Exclusive Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Database | Database Deadlocks Exclusive Locks | Number of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Database | Database Acquire Waits Exclusive Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |
| MongoDB Mongod | Locks\|Database | Database Deadlocks Intent Shared Locks | Number of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Database | Database Acquire Waits Intent Shared Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |
| MongoDB Mongod | Locks\|Database | Database Cumulative Wait Time Intent Exclusive Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Database | Database Intent Shared Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Database | Database Shared Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Global | Global Deadlocks Intent Shared Locks | Number of times lock acquisitions with this mode encountered deadlocks |

| Resource Kind | Resource Group | Resource Attribute | Attribute Description |
|---|---|---|---|
| MongoDB Mongod | Locks\|Global | Global Deadlocks Intent Exclusive Locks | Number of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Global | Global Shared Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Global | Global Intent Exclusive Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Global | Global Intent Shared Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Global | Global Acquire Waits Intent Exclusive Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |
| MongoDB Mongod | Locks\|Global | Global Exclusive Locks | Number of times the lock was acquired in this mode |
| MongoDB Mongod | Locks\|Global | Global Cumulative Wait Time Intent Exclusive Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Global | Global Cumulative Wait Time Exclusive Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Global | Global Cumulative Wait Time Intent Shared Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Locks\|Global | Global Acquire Waits Shared Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |
| MongoDB Mongod | Locks\|Global | Global Acquire Waits Intent Shared Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |
| MongoDB Mongod | Locks\|Global | Global Deadlocks Exclusive Locks | Number of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Global | Global Deadlocks Shared Locks | Number of times lock acquisitions with this mode encountered deadlocks |
| MongoDB Mongod | Locks\|Global | Global Acquire Waits Exclusive Locks | Number of lock acquisitions with this mode waits because locks were held in a conflicting mode |
| MongoDB Mongod | Locks\|Global | Global Cumulative Wait Time Shared Locks | Cumulative wait time for lock acquisitions with this mode |
| MongoDB Mongod | Storage Engine | Storage Engine Name | Name of the current storage engine |
| MongoDB Mongod | Storage Engine | Storage Engine Persistent | Indicates if the storage engine persists data to disk |
| MongoDB Mongod | Storage Engine | Storage Engine Supports Committed Reads | Indicates if the storage engine supports "majority" read concern |
| MongoDB Mongod | Global Lock | Active Clients Performing Reads | Number of active client connections performing read operations |
| MongoDB Mongod | Global Lock | Active Clients Performing Writes | Number of active client connections performing write operations |
| MongoDB Mongod | Global Lock | Operations Waiting to Write | Number of operations currently queued and waiting for the write lock |

| Resource Kind | Resource Group | Resource Attribute | Attribute Description |
|---|---|---|---|
| MongoDB Mongod | Global Lock | Total Operations Waiting | Number of operations queued waiting for the lock |
| MongoDB Mongod | Global Lock | Global Lock Total Time | Time since database last started and created the "globalLock" |
| MongoDB Mongod | Global Lock | Active Clients | Number of active client connections to the database |
| MongoDB Mongod | Global Lock | Operations Waiting to Read | Number of operations currently queued and waiting for the read lock |
| MongoDB Mongod | Properties | Slave Delay | Lag time behind the primary of this resource's replica set |
| MongoDB Mongod | Properties | Advisory Host FQDNs | Advisory host fully qualified domain names of the machine hosting this resource |
| MongoDB Mongod | Properties | ID | Name of this resource's replica set |
| MongoDB Mongod | Properties | Priority | Signifies eligibility of this resource to become primary |
| MongoDB Mongod | Properties | Version | Version of the current MongoDB process |
| MongoDB Mongod | Properties | Tags | Document that describe the replica sets settings |
| MongoDB Mongod | Properties | Is Primary | Indicates if this resource is writable |
| MongoDB Mongod | Properties | Arbiter Only | Indicates if this resource is an arbiter |
| MongoDB Mongod | Properties | Build Indexes | Indicates if the mongod builds indexes on this resource |
| MongoDB Mongod | Properties | Votes | Number of votes this resource will have in a replica set election |
| MongoDB Mongod | Properties | PID | Process ID number on this resource |
| MongoDB Mongod | Properties | Hidden | Indicates if this resource is hidden by its replica set |
| MongoDB Mongod | Properties | Process | Current MongoDB process on this resource |
| MongoDB Mongod | Properties | Uptime | Time the current MongoDB process has been active |
| MongoDB Mongod | Properties | Host Name | Hostname of the machine hosting this resource |
| MongoDB Mongod | Properties | State | Current state of the mongod in the replica set |
| MongoDB Mongod | Properties | Optime | Optime of the mongod |
| MongoDB Mongod | Cache | Written Cache | Amount of data written from cache |
| MongoDB Mongod | Cache | Dirty Cache | Amount of dirty data in cache |
| MongoDB Mongod | Cache | Current Cache | Amount of data currently in cache |
| MongoDB Mongod | Cache | Read Cache | Amount of data read into cache |
| MongoDB Mongod | Network | Network Traffic Out | Data sent from Mongod |
| MongoDB Mongod | Network | Network Traffic In | Data received by Mongod |

| Resource Kind | Resource Group | Resource Attribute | Attribute Description |
|---|---|---|---|
| MongoDB Mongod | Network | Number of Requests | Number of requests Mongod has received |
| MongoDB Mongod | Background Flushing | Background Flushing: Last | Time it took the last flush operation to complete |
| MongoDB Mongod | Background Flushing | Background Flushing: Total | Time the mongod processes have spent writing data to disk |
| MongoDB Mongod | Background Flushing | Background Flushing: Flushes | Number of times the databases on this resource has flushed all writes to disk |
| MongoDB Mongod | Background Flushing | Background Flushing: Average | Average time for each flush to disk |
| MongoDB Mongod | Asserts | Asserts: Message | Number of message assertions raised |
| MongoDB Mongod | Asserts | Asserts: Rollovers | Number of rollovers |
| MongoDB Mongod | Asserts | Asserts: Warning | Number of warnings raised |
| MongoDB Mongod | Asserts | Asserts: User | Number of user assertions |
| MongoDB Mongod | Asserts | Asserts: Regular | Number of regular assertions |
| MongoDB Mongod | Operations | Queries with ID Field | Number of queries that contain the "_id" field |
| MongoDB Mongod | Operations | Queries using Sort | Number of queries that return sorted numbers that cannot perform the sort operation using an index |
| MongoDB Mongod | Operations | Queries with Write Conflicts | Number of queries that encountered write conflicts |
| MongoDB Mongod | Operations | Fastmod Operations | Number of update operations that don't cause documents to grow nor require updates to the index |
| MongoDB Mongod | Connections | Current Connections | Number of incoming connections to this resource |
| MongoDB Mongod | Connections | Available Connections | Number of unused incoming connections available to this resource |
| MongoDB Mongod | Connections | Total Connections Created | Count of all incoming connections created to this resource |
| MongoDB Mongod | Memory | Memory Architecture | Architecture of the machine hosting the Mongod |
| MongoDB Mongod | Memory | Virtual Memory | Virtual memory being used by the Mongod |
| MongoDB Mongod | Memory | Extended Memory Supported | Indicates if the machine hosting Mongod supports extended memory information |
| MongoDB Mongod | Memory | Resident Memory | Amount of RAM used by this Mongod's database processes |
| MongoDB Mongod | Memory | Mapped Memory with Journal | Amount of mapped memory with journal by this Mongod's databases |
| MongoDB Mongod | Memory | Mapped Memory | Amount of mapped memory by this Mongod's databases |

| Resource Kind | Resource Group | Resource Attribute | Attribute Description |
|---|---|---|---|
| MongoDB Mongod | Replication Opcounters | Replication Get More Operations | Replicated getmore operations of Mongod |
| MongoDB Mongod | Replication Opcounters | Replication Delete Operations | Replicated delete operations of Mongod |
| MongoDB Mongod | Replication Opcounters | Replication Command Operations | Replicated database commands of Mongod |
| MongoDB Mongod | Replication Opcounters | Replication Insert Operations | Replicated insert operations of Mongod |
| MongoDB Mongod | Replication Opcounters | Replication Query Operations | Replicated query operations of Mongod |
| MongoDB Mongod | Replication Opcounters | Replication Update Operations | Replicated update operations of Mongod |
| MongoDB Mongod | Replication Opcounters | Replication Read Operations | Replicated read operations of Mongod |
| MongoDB Mongod | Replication Opcounters | Replication Write Operations | Replicated write operations of Mongod |
| MongoDB Mongod | Replication Opcounters | Replication Reads/Write Operations | All replicated read and write operations of Mongod |
| MongoDB Mongod | Journaling | Journaling Enabled | Signifies if Journaling is enabled |
| MongoDB Mongod | Journaling | Data Journaled | Data written to journal during latest journal group commit interval |
| MongoDB Mongod | Journaling | Commits | Transactions written to journal during latest journal group commit interval |
| MongoDB Mongod | Journaling | Compression Ratio | Compression Ratio of the data written to journal |
| MongoDB Mongod | Journaling | Commits In Write Lock | Commits that occurred while a write lock was held |
| MongoDB Mongod | Journaling | Written to Data Files | Data written from journal during latest journal group commit interval |
| MongoDB Mongod | Journaling | Early Commits | Requested commits before the scheduled journal group commit interval |
| MongoDB Mongod | Opcounters | Delete Operations | Delete operations of Mongod |
| MongoDB Mongod | Opcounters | Update Operations | Update operations of Mongod |
| MongoDB Mongod | Opcounters | Get More Operations | Getmore operations of Mongod |
| MongoDB Mongod | Opcounters | Query Operations | Query operations of Mongod |
| MongoDB Mongod | Opcounters | Command Operations | Database commands of Mongod |
| MongoDB Mongod | Opcounters | Insert Operations | Insert operations of Mongod |
| MongoDB Mongod | Opcounters | Read Operations | Read operations of Mongod |
| MongoDB Mongod | Opcounters | Write Operations | Write operations of Mongod |
| MongoDB Mongod | Opcounters | Read/Write Operations | All read and write operations of Mongod |
| MongoDB Mongod | Documents | Documents Moved | Number of times documents have moved in the on-disk representation of the MongoDB data set |
| MongoDB Mongod | Documents | Documents Deleted | Number of documents deleted |

| Resource Kind | Resource Group | Resource Attribute | Attribute Description |
|---|---|---|---|
| MongoDB Mongod | Documents | Documents Updated | Number of documents updated |
| MongoDB Mongod | Documents | Documents Inserted | Number of documents inserted |
| MongoDB Mongod | Documents | Documents Returned | Number of documents returned by queries |
| MongoDB Mongod | Security | Expiration Date | Expiration date of the TLS/SSL certificate |
| MongoDB Mongod | Security | Subject Name | Subject name associated with TLS/SSL certificate |
| MongoDB Mongod | Security | Has Certificate Authority | Indicates if the TLS/SSL certificate is associated with a certificate authority |
| MongoDB Mongod | Cursor | Cursors Timed Out | Total number of cursors that have timed out |
| MongoDB Mongod | Cursor | Cursors Open | Total number of cursors this environment maintains |
| MongoDB Mongod | Tickets | Write Tickets Available | Amount of write tickets available to the WiredTiger storage engine |
| MongoDB Mongod | Tickets | Read Tickets Available | Amount of read tickets available to the WiredTiger storage engine |
| MongoDB Mongod | Statistics | Write Backs Queued | Indicates if there are operations from a mongos instance queued for retrying |
| MongoDB Mongod | Parents | Replica Set Parents | Replica Set resource names of Parent resources. |
| MongoDB Mongod | Parents | Database Parents | Database resource names of Parent resources. |
| MongoDB Mongod | Parents | Config Server Parents | Config Server resource names of Parent resources. |
| MongoDB Mongod | Parents | Shard Parents | Shard resource names of Parent resources. |
| MongoDB Mongod | Alerts | Shares Host | Mongod Shares a Host with another Mongod. |
| MongoDB Mongod | Alerts | Mongod is Recovering | Mongod is recovering from a delay. |
| MongoDB Mongod | Alerts | Mongod Delay Begind the Primary | Mongod delay behind the primary not accounting for the secondary delay. |

## Management Pack for MySQL

The **Management Pack for MySQL** is an embedded adapter for VCF Operations, collecting performance data from your MySQL database management system and providing predictive analysis and real-time information about problems in your infrastructure—all within the VCF Operations user interface.

### Notable Features

The Management Pack harnesses powerful VCF Operations features such as:

- **Comprehensive monitoring dashboards**
- **800+ collected metrics** for MySQL resources
- **Key performance metrics** such as wait times, disk I/O, indexes, transactions, and connections

- **External relationships** to the VMware virtual layer
- **Detailed alerts and recommendations** for events such as high wait times, unused indexes, and aborted connections
- **VMware Cloud Foundation Operations Calculated Connection Capacity** for MySQL instances

### System Requirements for Management Pack for MySQL

Before installing and configuring the Management Pack, ensure your system meets the following requirements.

For compatibility between products, please refer to:

VMware Cloud Foundation Operations Interoperability for MySQL

-OR-

Third Party Solution Interoperability for MySQL

| | MySQL Requirements |
|---|---|
| Credentials | **MySQL User Name/Password**<br>**Note:**<br>For least-privileged user (LPU) requirements, see: Setting User Privileges |
| Connection | **Hostname** (or IPv4/IPv6 address) **of MySQL instance**<br>**Note:**<br>If using SSL, the SSL certificate must be copied to your VMware Cloud Foundation Operations system prior to configuring an adapter instance.<br>**Default Port**: 3306<br>**Optional**: Database name(s) to monitor (If specific databases are not specified, all databases on the MySQL instance will be returned.) |

#### Setting User Privileges

For the Management Pack to access and return certain metrics from your MySQL database system, different privileges must be granted to the MySQL user as outlined below.

**Note:**

The **REPLICATION CLIENT** permission is required for proper operation of the Management Pack. Even if you are not using replication, please note that if the **REPLICATION CLIENT** permission is not used, process data will not be collected, and an error will be listed in the log files.

| Metric(s) | Required Privileges |
|---|---|
| **MySQL Table & Database I/O** (only applicable on MySQL 5.7) | EXECUTE |
| **Slave Replication** | REPLICATION CLIENT |

| Metric(s) | Required Privileges |
| --- | --- |
| **MYSQL Instance**:<br>• Instance Adaptive Hash Index<br>• Buffer<br>• Buffer Page I/O<br>• Change Buffer<br>• Compression<br>• DDL<br>• DML<br>• File System<br>• ICP<br>• Index<br>• Lock<br>• Metadata<br>• OS<br>• Purge<br>• Recovery<br>• Server<br>• Transaction<br>• Tablespace | PROCESS |

### Upgrading the Management Pack

When upgrading to a newer version of the Management Pack, we recommend the following tasks:

- Delete existing MySQL dashboards
- Delete existing views
- Delete existing reports

#### Delete existing MySQL dashboards

If you do not remove the dashboards from a previous version of the Management Pack before installing the new version, you will see duplicate and/or deprecated MySQL dashboards in the **Dashboard List** drop-down menu after upgrading.

**To remove existing Management Pack dashboards:**

1. Navigate to **Content > Dashboards**.
2. Multi-select all MySQL dashboards.
3. Click **Delete Dashboard**.
4. Click **Yes** when the confirmation dialog box appears to delete the selected dashboards.

#### Delete existing views

We also recommend removing existing views from the previous version of the Management Pack.

**To remove existing Management Pack views:**

1. Navigate to **Content > Views**.
2. Multi-select all MySQL views.
3. **Click Delete View**.
4. Click **Yes** when the confirmation dialog box appears to delete the selected views.

#### Delete existing reports

We also recommend removing existing reports from the previous version of the Management Pack.

**To remove existing Management Pack reports:**

1. Navigate to **Content > Reports**.

2. Multi-select all MySQL reports.
3. Click **Delete template**.
4. Click **Yes** when the confirmation dialog box appears to delete the selected reports.

**What to do next**

Installing and Configuring the Management Pack

### Troubleshooting the Management Pack

Once you have finished Installing and Configuring the Management Pack , check the **Collection State** and **Collection Status** for the adapter instance on the **Solutions** page. If they show as **Not Collecting** or **No Data Receiving**, respectively, the following troubleshooting topics can help you diagnose and correct common problems with your adapter instance.

- Test Connection Errors
- Adapter and Collector Log Files

**Test Connection Errors**

When Creating an Adapter Instance , click **Test Connection** after entering all of your configuration settings to ensure VCF Operations can connect to the specified host.

The following errors are possible:

**Invalid Host**

**Error Message**: Unable to establish a valid connection to the target system. Unable to connect to the server. Verify your Host settings.

**Description:** Verify your host URL or port number are appropriate.

**Request Timed Out**

**Error Message**: Unable to establish a valid connection to the target system. Unable to connect to the server. Request timed out.

**Description:** Verify your host URL or port number are appropriate.

**User Authorization**

**Error Message**: User Authentication error. Please check your user credentials or SSL settings.

**Description:** Verify the user credentials and SSL settings are appropriate.

**Invalid Port Number**

**Error Message**: "Port number specified contains invalid characters." – OR – "Invalid port number." Check your port number.

**Description:** Verify the port number is the correct database port for the server.

**Invalid Connection Settings**

**Error Message**: Unable to connect to the server. Verify your host settings.

**Description:** Verify the hostname/IP address for MySQL server is correct.

**Invalid Database**

**Error Message**: The database ‘<DatabaseName>’ not exist.

**Description:** Verify the correct database name was entered in the Database field.

**Invalid SSL Certificate**

**Error Message**: The given SSL certificate is invalid or does not exist.

**Description:** Verify the file path is correct and is the correct certificate for the MySQL server.

### SSL Connection Error

**Error Message**: SSL Mode currently set is not supported by the server. Verify your SSL settings.

**Description:** The **Use SSL** field is set to **True,** but the server does not support SSL. Set the field to **False.**

## Adapter and Collector Log Files

You can review adapter errors in the adapter and collector log files, which can be viewed in the VCF Operations UI or in an external log viewer.

- **Adapter log files** are located in the $ALIVE_BASE/user/log/adapters/MySQLAdapter/ folder.
- **Collector log files** are located in the $ALIVE_BASE/user/log/ folder.

To set the base log level for the collector in VMware Cloud Foundation Operations:

1. Navigate to **Administration > Logs**.
2. Expand the **Primary** folder, then select the **COLLECTOR** folder.
3. Click the **Edit Properties** icon.
4. In the **Root logger level** drop-down menu, select the desired option.

- The default logging level is **warning.**
- To troubleshoot issues, set the logging level to **info.**
- To view detailed messages, including micro steps, queries, and returned results, set the logging level to **debug.**

> **Note:**
>
> If you set the logging level to **debug,** log files can become large very quickly. Set the logging level to **debug** only for short periods of time.

For more information about viewing log files and modifying levels, refer to the VCF Operations online help.

### Using the Management Pack

The Management Pack for MySQL includes the following out-of-the-box features, harnessing the power of the VCF Operations user interface:

- Inventory Tree (Traversal Spec)
- Dashboards
- Views
- Reports
- Analysis Badges and Capacity Definitions
- Alerts
- Metrics

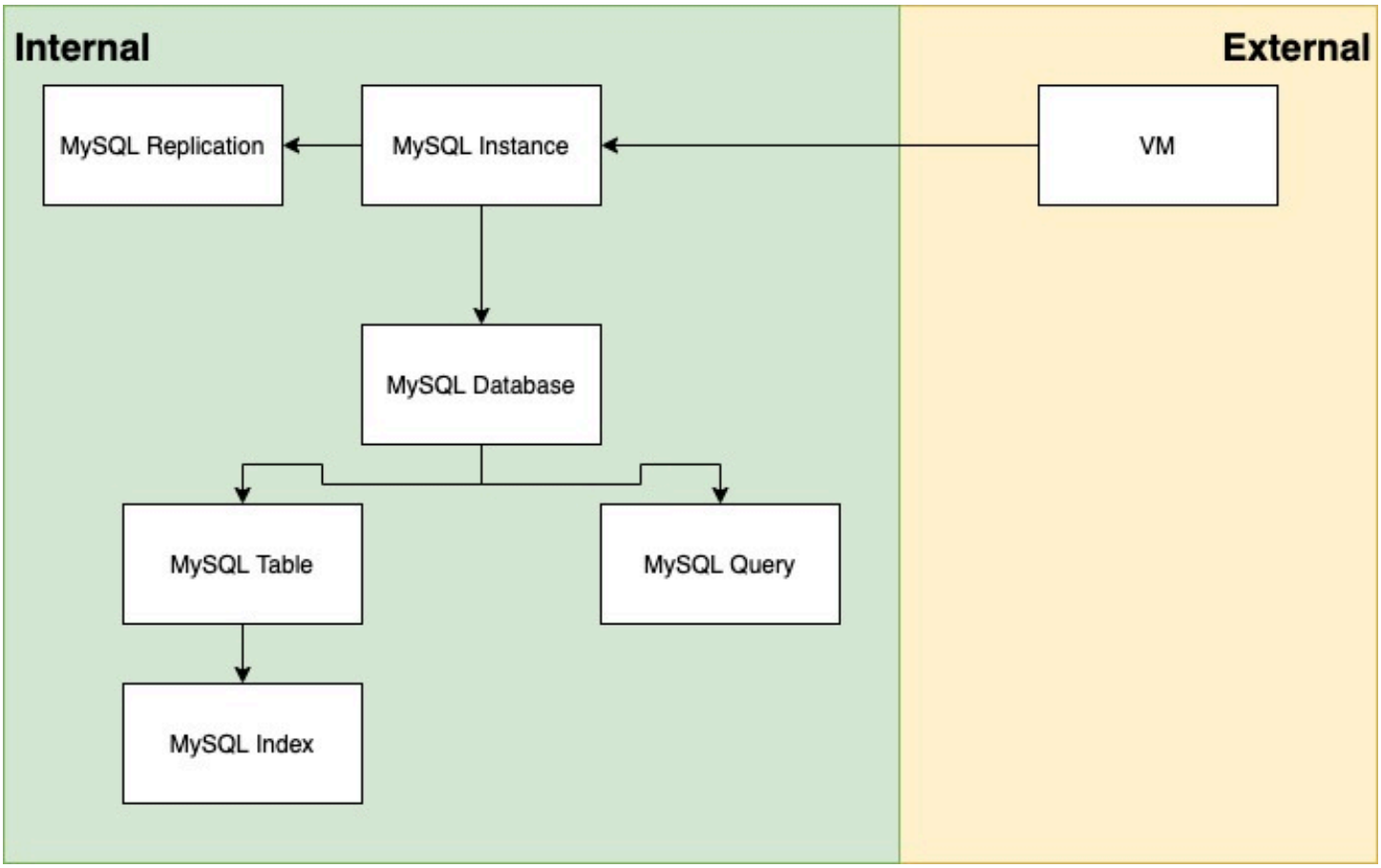
#### Inventory Tree (Traversal Spec), Resources and Relationships

The Inventory Tree (Traversal Spec) feature in VCF Operations allows you to easily navigate your MySQL environment. The hierarchical structure implicitly shows relationships among resources and enables quick drill-downs to root-cause issues.

To access your MySQL Inventory Tree, navigate to **Home > Environment > Inventory Trees**.

## Resources and Relationships

**Figure 484:**




### Dashboards

Dashboards are the primary feature for monitoring and troubleshooting MySQL data problems from within VCF Operations.

The following dashboards are included in the Management Pack for MySQL:

- MySQL Database Drill Down
- MySQL Infrastructure Drill Down

### Views

The Management Pack for MySQL creates views that allow the user to view statistics of metrics for MySQL resources. The views help give a broad picture of the entire system, as opposed to a more in depth view.

The following views are available in the Management Pack:

| View | Type | Description |
|---|---|---|
| **MySQL Adapter Active Alerts** | List | Shows active alerts for MySQL adapter resources and their descendants |
| **MySQL Adapter Database KPIs** | List | Shows all KPIs for MysQL adapter database |
| **MySQL Adapter Health** | List | Shows health and availability for MySQL adapter resources |
| **MySQL Adapter Instance KPIs** | List | Shows all KPIs for MySQL adapter instance |

| View | Type | Description |
|---|---|---|
| **MySQL Adapter Instance Properties** | List | Shows properties for MySQL adapter instance |
| **MySQL Adapter Query KPIs** | List | Shows all KPIs for MySQL adapter query |
| **MySQL Adapter Query Properties** | List | Shows properties for MySQL adapter query |
| **MySQL Adapter Replication Primary Properties** | List | Shows properties for MySQL adapter replication Primary |
| **MySQL Adapter Replication Slave Properties** | List | Shows properties for MySQL adapter replication slave |
| **MySQL Adapter Table KPIs** | List | Shows all KPIs for MySQL adapter table |
| **MySQL Adapter Tablespace Properties** | List | Shows properties for MySQL adapter tablespace |
| **MySQL Database Overview** | List | Shows an overview of MySQL databases |
| **MySQL Database Usage (Historic Data)** | Trend | Shows the trends in the number of rows inserted, updated, and deleted for a given database |
| **MySQL Instance Overview** | List | Shows an overview of MySQL instances |
| **MySQL Instance Usage (Historic Data)** | Trend | Shows the number of rows that have been inserted, updated, deleted, or selected for a given MySQL instance |
| **MySQL Primary** | List | Shows Primary properties |
| **MySQL Primary Stats** | List | Shows Primary stats |
| **MySQL Query** | List | Shows details for MySQL queries |
| **MySQL Query Text** | Text | Shows the query text for the selected MySQL query |
| **MySQL Slave** | List | Shows slave properties |
| **MySQL Slowest Queries** | List | Shows details for the slowest queries in a MySQL system |
| **MySQL Table Usage (Historic Data)** | Trend | Shows the number of rows inserted, updated, and deleted for a given MySQL table as a trend |
| **MySQL Throughput** | List | Shows I/O related information for MySQL systems |

To access the Management Pack views:

1. Navigate to **Environment > All Objects > MySQL**.
2. Double-click on the desired object (resource).
3. Select the **Details** tab, then **Views.**

The available views for that resource are listed and can be selected.

### Reports

The Management Pack for MySQL contains out-of-the-box reports that you can easily export and share with key stakeholders in either .pdf or .csv formats.

The following reports are available in the Management Pack:

| Report | Description |
|---|---|
| **MySQL Adapter Health and Availability** | Includes details on the overall health and availability of MySQL adapter resources |
| **MySQL Adapter KPIs** | Includes KPIs for MySQL adapter resources |

| Report | Description |
|---|---|
| **MySQL Adapter Properties** | Includes properties for MySQL adapter resources |
| **MySQL Databases Report** | Includes overview stats of MySQL databases |
| **MySQL Instances Report** | Includes overview stats of MySQL instances |
| **MySQL Replication Report** | Includes overview stats of MySQL replication resources |
| **MySQL Queries Report** | Includes overview stats of MySQL queries |
| **MySQL Throughput Report** | Includes I/O stats for a MySQL system |

To access Management Pack reports:

1. Navigate to **Environment > All Objects > MySQL**.
2. Double-click on the desired object (resource).
3. Select the **Reports** tab, then **Report Templates**.

The available report(s) for the selected resource are listed and can be selected.

To run a selected report, click the **Run Template**

### Analysis Badges and Capacity Definitions

Using the predictive analytics capabilities of the VCF Operations Analysis Badges through capacity definitions, the Management Pack populates Connection Capacity for MySQL instances.

**Capacity Definitions List**

| Container | Use in Workload |
|---|---|
| **MySQL Instance** | |
| VMware Cloud Foundation Operations Calculated Connection Capacity | yes |

### Alerts

The Management Pack for MySQL pulls MySQL database health indicators and displays them in VMware Cloud Foundation Operations as alerts. The critical levels of each alert are closely mapped from MySQL to VCF Operations. See the table below for the list of alerts available in the Management Pack.

| Name | Description | Recommendation |
|---|---|---|
| Buffer Pool Wait Free is Higher than Normal | The number of times waiting for a free buffer is higher than normal. | Make sure that the memory buffer is correctly configured for the number of writes being performed. |
| Number of Aborted Connections is Higher than Normal | The number of aborted connections are higher than normal. It is possible that clients are trying to connect and failing. | Ensure that clients have the right privileges/ credentials, and look into possible external attacks on the server. |
| Number of Spin Loop Rounds are Higher than Normal | rwlock spin loop rounds from exclusive latch request are higher than normal. It is possible that significant CPU resources are being wasted. | Consider reducing the spin delay. |
| Number of Unused Indexes are Higher than Normal | There are a large number of unused indexes. This can hurt the overall database performance. | Consider removing some of the unused indexes. |
| Total Wait Times of All Operations Higher than Normal | Total Wait Times of All Operations Higher than Normal | Verify table indexes are appropriate for the current operations. |

| Name | Description | Recommendation |
|---|---|---|
| Read Wait Times are Higher than Normal | Read Wait Times are Higher than Normal | Verify table indexes are appropriate for the current operations. |
| Write Wait Times are Higher than Normal | Write Wait Times are Higher than Normal | Verify table indexes are appropriate for the current operations. |
| Fetch Wait Times are Higher than Normal | Fetch Wait Times are Higher than Normal | Verify table indexes are appropriate for the current operations. |
| Insert Wait Times are Higher than Normal | Insert Wait Times are Higher than Normal | Verify table indexes are appropriate for the current operations. |
| Update Wait Times are Higher than Normal | Update Wait Times are Higher than Normal | Verify table indexes are appropriate for the current operations. |
| Delete Wait Times are Higher than Normal | Delete Wait Times are Higher than Normal | Verify table indexes are appropriate for the current operations. |

Metrics

The Management Pack for MySQL collects the metrics listed in the table below, according to MySQL resource kinds.

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Tag | Relationships | Replication Primary Children | string | List of children Replication Primary resources |
| MySQL Tag | Relationships | Database Children | string | List of children Database resources |
| MySQL Tag | Relationships | Index Children | string | List of children Index resources |
| MySQL Tag | Relationships | Instance Children | string | List of children Instance resources |
| MySQL Tag | Relationships | Query Children | string | List of children Query resources |
| MySQL Tag | Relationships | Children | string | List of childrenresources |
| MySQL Tag | Relationships | Table Children | string | List of children Table resources |
| MySQL Tag | Relationships | Tablespace Children | string | List of children Tablespace resources |
| MySQL Tag | Relationships | Replication Slave Children | string | List of children Replication Slave resources |
| MySQL Replication Primary | Details | Primary Host | string | The Primary host that the slave is connected to |
| MySQL Replication Primary | Details | Primary Port | integer | The port used to connect to the Primary |
| MySQL Replication Primary | Details | Primary UUID | string | The Universally Unique Identifier from the Primary |
| MySQL Replication Primary | Primary | Last I/O Error Message | string | The error message of the most recent error that caused the I/O thread to stop |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Replication Primary | Primary | Last I/O Error Number | integer | The error number of the most recent error that caused the I/O thread to stop. 0 means no error has been recorded |
| MySQL Replication Primary | Primary | Last I/O Error Timestamp | string | A timestamp in YYMMDD HH:MM:SS format that shows when the most recent I/O error took place |
| MySQL Replication Primary | Primary | Last SQL Error Message | string | The error message of the most recent error that caused the SQL thread to stop |
| MySQL Replication Primary | Primary | Last SQL Error Number | integer | The error number of the most recent error that caused the SQL thread to stop. 0 means no error has been recorded |
| MySQL Replication Primary | Primary | Last SQL Error Timestamp | string | A timestamp in YYMMDD HH:MM:SS format that shows when the last SQL error occurred |
| MySQL Replication Primary | Slave | Slave I/O Running | string | Whether the I/O thread is started and has connected successfully to the Primary |
| MySQL Replication Primary | Slave | Slave I/O State | string | A copy of the State field of the SHOW PROCESSLIST output for the slave I/O thread. This tells you what the thread is doing: trying to connect to the Primary, waiting for events from the Primary, reconnecting to the Primary, and so on |
| MySQL Replication Primary | Slave | Slave SQL Running | string | Whether the SQL thread is started |
| MySQL Replication Primary | Slave | Slave SQL Running State | string | The state of the SQL thread |
| MySQL Replication Primary | User Configuration | Auto Positioning | integer | Whether or not auto-positioning is in use |
| MySQL Replication Primary | User Configuration | Connect Retry Durations | integer | The number of seconds between connect retries |
| MySQL Replication Primary | User Configuration | Primary Retry Count | integer | The number of times the slave can attempt to reconnect to the Primary in the event of a lost connection |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Replication Primary | User Configuration | Primary SSL Allowed | string | Whether or not an SSL connection to the Primary is permitted. This value will report as 'Ignored' if an SSL connection is permitted but the slave server does not have SSL support enabled |
| MySQL Replication Primary | User Configuration | Primary SSL Verify Server Certificate | string | Whether or not SSL server certificates must be validated |
| MySQL Replication Primary | User Configuration | Primary User | string | The user name of the account used to connect to the Primary |
| MySQL Replication Primary | User Configuration | SQL Delay | integer | The number of seconds that the slave must lag the Primary |
| MySQL Replication Primary | Relationships | Instance Children | string | List of child Instance resources |
| MySQL Replication Primary | Relationships | MySQL Tag Parents | string | List of parent MySQL Tag resources |
| MySQL Database | Capacity | Database Data Size | integer | The size of the database's data |
| MySQL Database | Capacity | Database Index Size | integer | The size of the database's indexes |
| MySQL Database | Capacity | Database Total Size | integer | The total size of the database |
| MySQL Database | I/O | I/O Misc Latency | integer | The average latency of miscellaneous I/O |
| MySQL Database | I/O | I/O Misc Requests | float | The number of I/O miscellaneous requests |
| MySQL Database | I/O | I/O Read Latency | integer | The average latency of I/O reads |
| MySQL Database | I/O | I/O Read Requests | float | The number of I/O read requests |
| MySQL Database | I/O | I/O Reads | float | The number of I/O reads |
| MySQL Database | I/O | I/O Write Latency | integer | The average latency of I/O writes |
| MySQL Database | I/O | I/O Write Requests | float | The number of I/O write requests |
| MySQL Database | I/O | I/O Writes | float | The number of I/O writes |
| MySQL Database | Index | Unused Indexes | integer | The number of unused indexes |
| MySQL Database | Row Operations | Rows Deleted | float | The number of rows deleted |
| MySQL Database | Row Operations | Rows Fetched | float | The number of rows fetched |
| MySQL Database | Row Operations | Rows Inserted | float | The number of rows inserted |
| MySQL Database | Row Operations | Rows Read | float | The number of rows read |
| MySQL Database | Row Operations | Rows Read/Written | float | The total number of rows read and written |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Database | Row Operations | Rows Updated | float | The number of rows updated |
| MySQL Database | Row Operations | Rows Written | float | The number of rows written |
| MySQL Database | Threads | Binlog Dump Threads | integer | The number of Binlog Dump threads |
| MySQL Database | Threads | Change User Threads | integer | The number of Change User threads |
| MySQL Database | Threads | Close Statement Threads | integer | The number of Close Statement threads |
| MySQL Database | Threads | Connect Out Threads | integer | The number of Connect Out threads |
| MySQL Database | Threads | Connect Threads | integer | The number of Connect threads |
| MySQL Database | Threads | Connected Threads | integer | The number of Connected threads |
| MySQL Database | Threads | Create DB Threads | integer | The number of Create DB threads |
| MySQL Database | Threads | Daemon Threads | integer | The number of Daemon threads |
| MySQL Database | Threads | Debug Threads | integer | The number of Debug threads |
| MySQL Database | Threads | Delayed Insert Threads | integer | The number of Delayed Insert threads |
| MySQL Database | Threads | Drop DB Threads | integer | The number of Drop DB threads |
| MySQL Database | Threads | Error Threads | integer | The number of Error threads |
| MySQL Database | Threads | Execute Threads | integer | The number of Execute threads |
| MySQL Database | Threads | Fetch Threads | integer | The number of Fetch threads |
| MySQL Database | Threads | Field List Threads | integer | The number of Field List threads |
| MySQL Database | Threads | Init DB Threads | integer | The number of Init DB threads |
| MySQL Database | Threads | Kill Threads | integer | The number of Kill threads |
| MySQL Database | Threads | Long Data Threads | integer | The number of Long Data threads |
| MySQL Database | Threads | Misc Threads | integer | The number of Misc threads |
| MySQL Database | Threads | Ping Threads | integer | The number of Ping threads |
| MySQL Database | Threads | Prepare Threads | integer | The number of Prepare threads |
| MySQL Database | Threads | Processlist Threads | integer | The number of Processlist threads |
| MySQL Database | Threads | Query Threads | integer | The number of Query threads |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Database | Threads | Quit Threads | integer | The number of Quit threads |
| MySQL Database | Threads | Refresh Threads | integer | The number of Refresh threads |
| MySQL Database | Threads | Register Slave Threads | integer | The number of Register Slave threads |
| MySQL Database | Threads | Reset Statement Threads | integer | The number of Reset Statement threads |
| MySQL Database | Threads | Set Option Threads | integer | The number of Set Option threads |
| MySQL Database | Threads | Shutdown Threads | integer | The number of Shutdown threads |
| MySQL Database | Threads | Sleep Threads | integer | The number of Sleep threads |
| MySQL Database | Threads | Statistics Threads | integer | The number of Statistics threads |
| MySQL Database | Threads | Table Dump Threads | integer | The number of Table Dump threads |
| MySQL Database | Wait Time | Average Delete Wait Time | integer | The average Delete wait time |
| MySQL Database | Wait Time | Average Fetch Wait Time | integer | The average Fetch wait time |
| MySQL Database | Wait Time | Average Insert Wait Time | integer | The average Insert wait time |
| MySQL Database | Wait Time | Average Read Wait Time | integer | The average Read wait time |
| MySQL Database | Wait Time | Average Read/Write Wait Time | integer | The average R/W wait time |
| MySQL Database | Wait Time | Average Update Wait Time | integer | The average Update wait time |
| MySQL Database | Wait Time | Average Write Wait Time | integer | The average Write wait time |
| MySQL Database | Wait Time | Maximum Delete Wait Time | integer | The maximum Delete wait time |
| MySQL Database | Wait Time | Maximum Fetch Wait Time | integer | The maximum Fetch wait time |
| MySQL Database | Wait Time | Maximum Insert Wait Time | integer | The maximum Insert wait time |
| MySQL Database | Wait Time | Maximum Read Wait Time | integer | The maximum Read wait time |
| MySQL Database | Wait Time | Maximum Read/Write Wait Time | integer | The maximum R/W wait time |
| MySQL Database | Wait Time | Maximum Update Wait Time | integer | The maximum Update wait time |
| MySQL Database | Wait Time | Maximum Write Wait Time | integer | The maximum Write wait time |
| MySQL Database | Wait Time | Minimum Delete Wait Time | integer | The minimum Delete wait time |
| MySQL Database | Wait Time | Minimum Fetch Wait Time | integer | The minimum Fetch wait time |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Database | Wait Time | Minimum Insert Wait Time | integer | The minimum Insert wait time |
| MySQL Database | Wait Time | Minimum Read Wait Time | integer | The minimum Read wait time |
| MySQL Database | Wait Time | Minimum Read/Write Wait Time | integer | The minimum R/W wait time |
| MySQL Database | Wait Time | Minimum Update Wait Time | integer | The minimum Update wait time |
| MySQL Database | Wait Time | Minimum Write Wait Time | integer | The minimum Write wait time |
| MySQL Database | Wait Time | Total Delete Wait Time | float | The total Delete wait time |
| MySQL Database | Wait Time | Total Fetch Wait Time | float | The total Fetch wait time |
| MySQL Database | Wait Time | Total Insert Wait Time | float | The total Insert wait time |
| MySQL Database | Wait Time | Total Read Wait Time | float | The total Read wait time |
| MySQL Database | Wait Time | Total Read/Write Wait Time | float | The total R/W wait time |
| MySQL Database | Wait Time | Total Update Wait Time | float | The total Update wait time |
| MySQL Database | Wait Time | Total Write Wait Time | float | The total Write wait time |
| MySQL Database | Relationships | Instance Parents | string | List of parent Instance resources |
| MySQL Database | Relationships | MySQL Tag Parents | string | List of parent MySQL Tag resources |
| MySQL Database | Relationships | Query Children | string | List of child Query resources |
| MySQL Database | Relationships | Table Children | string | List of child Table resources |
| MySQL Database | | Hostname (Deprecated) | string | |
| MySQL Database | | Port (Deprecated) | integer | |
| MySQL Database | | Database Name (Deprecated) | string | |
| MySQL Database | | Database Total Size (Deprecated) | integer | |
| MySQL Database | | Database Data Size (Deprecated) | integer | |
| MySQL Database | | Database Index Size (Deprecated) | integer | |
| MySQL Database | | Unused Indexes (Deprecated) | integer | |
| MySQL Database | | Rows Read/Written (Deprecated) | integer | |
| MySQL Database | | R/W Total Wait Time (Deprecated) | integer | |
| MySQL Database | | R/W Min Wait Time (Deprecated) | integer | |
| MySQL Database | | R/W Average Wait Time (Deprecated) | integer | |
| MySQL Database | | R/W Max Wait Time (Deprecated) | integer | |
| MySQL Database | | Rows Read (Deprecated) | integer | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Database | | Read Total Wait Time (Deprecated) | integer | |
| MySQL Database | | Read Min Wait Time (Deprecated) | integer | |
| MySQL Database | | Read Average Wait Time (Deprecated) | integer | |
| MySQL Database | | Read Max Wait Time (Deprecated) | integer | |
| MySQL Database | | Rows Written (Deprecated) | integer | |
| MySQL Database | | Write Total Wait Time (Deprecated) | integer | |
| MySQL Database | | Write Min Wait Time (Deprecated) | integer | |
| MySQL Database | | Write Average Wait Time (Deprecated) | integer | |
| MySQL Database | | Write Max Wait Time (Deprecated) | integer | |
| MySQL Database | | Rows Fetched (Deprecated) | integer | |
| MySQL Database | | Fetch Total Wait Time (Deprecated) | integer | |
| MySQL Database | | Fetch Min Wait Time (Deprecated) | integer | |
| MySQL Database | | Fetch Average Wait Time (Deprecated) | integer | |
| MySQL Database | | Fetch Max Wait Time (Deprecated) | integer | |
| MySQL Database | | Rows Inserted (Deprecated) | integer | |
| MySQL Database | | Insert Total Wait Time (Deprecated) | integer | |
| MySQL Database | | Insert Min Wait Time (Deprecated) | integer | |
| MySQL Database | | Insert Average Wait Time (Deprecated) | integer | |
| MySQL Database | | Insert Max Wait Time (Deprecated) | integer | |
| MySQL Database | | Rows Updated (Deprecated) | integer | |
| MySQL Database | | Update Total Wait Time (Deprecated) | integer | |
| MySQL Database | | Update Min Wait Time (Deprecated) | integer | |
| MySQL Database | | Update Average Wait Time (Deprecated) | integer | |
| MySQL Database | | Update Max Wait Time (Deprecated) | integer | |
| MySQL Database | | Rows Deleted (Deprecated) | integer | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Database | | Delete Total Wait Time (Deprecated) | integer | |
| MySQL Database | | Delete Min Wait Time (Deprecated) | integer | |
| MySQL Database | | Delete Average Wait Time (Deprecated) | integer | |
| MySQL Database | | Delete Max Wait Time (Deprecated) | integer | |
| MySQL Database | | I/O Read Requests (Deprecated) | integer | |
| MySQL Database | | I/O Read (Deprecated) | integer | |
| MySQL Database | | I/O Read Latency (Deprecated) | integer | |
| MySQL Database | | I/O Write Requests (Deprecated) | integer | |
| MySQL Database | | I/O Write (Deprecated) | integer | |
| MySQL Database | | I/O Write Latency (Deprecated) | integer | |
| MySQL Database | | I/O Misc Requests (Deprecated) | integer | |
| MySQL Database | | I/O Misc Latency (Deprecated) | integer | |
| MySQL Index | Activity | Is Used | string | Whether or not the index is used |
| MySQL Index | Row Operations | Rows Deleted | float | The total number of rows deleted from the index |
| MySQL Index | Row Operations | Rows Inserted | float | The total number of rows inserted into the index |
| MySQL Index | Row Operations | Rows Selected | float | The total number of rows read using the index |
| MySQL Index | Row Operations | Rows Updated | float | The total number of rows updated in the index |
| MySQL Index | Wait Time | Delete Latency | float | The total wait time of timed deletes from the index |
| MySQL Index | Wait Time | Insert Latency | float | The total wait time of timed inserts into the index |
| MySQL Index | Wait Time | Select Latency | float | The total wait time of timed reads using the index |
| MySQL Index | Wait Time | Update Latency | float | The total wait time of timed updates in the index |
| MySQL Index | Relationships | MySQL Tag Parents | string | List of parent MySQL Tag resources |
| MySQL Index | Relationships | Table Parents | string | List of parent Table resources |
| MySQL Instance | Activity | ALTER TABLE's, CREATE INDEX's, DROP INDEX's In Progress | integer | Number of ALTER TABLE, CREATE INDEX, DROP INDEX in progress |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Instance | Activity | Indexes Being Created Online | integer | Number of indexes being created online |
| MySQL Instance | Activity | Indexes Waiting To Drop | integer | Number of indexes waiting to be dropped after failed index creation |
| MySQL Instance | Activity | Table Handles Closed | float | The number of table handles closed |
| MySQL Instance | Activity | Table Handles Opened | float | The number of table handles opened |
| MySQL Instance | Activity | Table Reference Count | float | The number of table references |
| MySQL Instance | Activity | Tables Waiting To Drop | integer | Number of tables in background drop table list |
| MySQL Instance | Adaptive Hash Index | Adaptive Hash B-tree Searches | float | The number of searches using B-tree on an index search |
| MySQL Instance | Adaptive Hash Index | Adaptive Hash Pages Added | float | The number of index pages on which the Adaptive Hash Index is built |
| MySQL Instance | Adaptive Hash Index | Adaptive Hash Pages Removed | float | The number of index pages whose corresponding Adaptive Hash Index entries were removed |
| MySQL Instance | Adaptive Hash Index | Adaptive Hash Rows Added | float | The number of Adaptive Hash Index rows added |
| MySQL Instance | Adaptive Hash Index | Adaptive Hash Rows Deleted With No Hash Entry | float | The number of rows deleted that did not have corresponding Adaptive Hash Index entries |
| MySQL Instance | Adaptive Hash Index | Adaptive Hash Rows Removed | float | The number of Adaptive Hash Index rows removed |
| MySQL Instance | Adaptive Hash Index | Adaptive Hash Rows Updated | float | The number of Adaptive Hash Index rows updated |
| MySQL Instance | Adaptive Hash Index | Adaptive Hash Searches | float | The number of successful searches using Adaptive Hash Index |
| MySQL Instance | Buffer | Buffer Data Reads | float | The amount of data read in bytes |
| MySQL Instance | Buffer | Buffer Data Written | float | The amount of data written in bytes |
| MySQL Instance | Buffer | Buffer Flush Adaptive | float | The number of adaptive batches |
| MySQL Instance | Buffer | Buffer Flush Adaptive Pages | float | The pages queued as an adaptive batch |
| MySQL Instance | Buffer | Buffer Flush Adaptive Total Pages | float | The total pages flushed as part of adaptive flushing |
| MySQL Instance | Buffer | Buffer Flush Average Page Rate | float | The average number of pages at which flushing is happening |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Instance | Buffer | Buffer Flush Background | float | The number of background batches |
| MySQL Instance | Buffer | Buffer Flush Background Pages | float | The number of pages queued as a background batch |
| MySQL Instance | Buffer | Buffer Flush Background Total Pages | float | The total pages flushed as part of background batches |
| MySQL Instance | Buffer | Buffer Flush Batch Num Scan | float | The number of times buffer flush list flush is called |
| MySQL Instance | Buffer | Buffer Flush Batch Pages | float | The number of pages queued as a flush batch |
| MySQL Instance | Buffer | Buffer Flush Batch Rescan | float | The number of times rescan of flush list forced |
| MySQL Instance | Buffer | Buffer Flush Batch Scanned | float | The total pages scanned as part of flush batch |
| MySQL Instance | Buffer | Buffer Flush Batch Scanned Per Call | float | The number of pages scanned per flush batch scan |
| MySQL Instance | Buffer | Buffer Flush Batch Total Pages | float | The total pages flushed as part of flush batch |
| MySQL Instance | Buffer | Buffer Flush Batches | float | The number of flush batches |
| MySQL Instance | Buffer | Buffer Flush LSN Average Rate | float | The average redo generation rate |
| MySQL Instance | Buffer | Buffer Flush Neighbor | float | The number of times neighbors flushing is invoked |
| MySQL Instance | Buffer | Buffer Flush Neighbor Pages | float | The number of pages queued as a neighbor batch |
| MySQL Instance | Buffer | Buffer Flush Neighbor Total Pages | float | The total neighbors flushed as part of neighbor flush |
| MySQL Instance | Buffer | Buffer Flush Number To Flush Requested | float | The number of pages requested for flushing |
| MySQL Instance | Buffer | Buffer Flush Percent For Dirty | float | The percent of I/O capacity used to avoid max dirty page limit. |
| MySQL Instance | Buffer | Buffer Flush Percent For LSN | float | The percent of I/O capacity used to avoid reusable redo space limit |
| MySQL Instance | Buffer | Buffer Flush Sync | float | The number of sync batches |
| MySQL Instance | Buffer | Buffer Flush Sync Pages | float | The pages queued as a sync batch |
| MySQL Instance | Buffer | Buffer Flush Sync Total Pages | float | The total pages flushed as part of sync batches |
| MySQL Instance | Buffer | Buffer Flush Sync Waits | float | The number of times a wait happens due to sync flushing |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Instance | Buffer | Buffer LRU Batch Evict Pages | float | The number of pages queued as a LRU evict batch |
| MySQL Instance | Buffer | Buffer LRU Batch Evict Total Pages | float | The total pages evicted as part of LRU evict batch |
| MySQL Instance | Buffer | Buffer LRU Batch Flush Pages | float | The number of pages queued as a LRU flush batch |
| MySQL Instance | Buffer | Buffer LRU Batch Flush Total Pages | float | The total pages flushed as part of LRU flush batch |
| MySQL Instance | Buffer | Buffer LRU Batch Number Scanned | float | The number of times LRU batch is called |
| MySQL Instance | Buffer | Buffer LRU Batch Pages | float | The number of pages queued as an LRU batch |
| MySQL Instance | Buffer | Buffer LRU Batch Scanned | float | The total pages scanned as part of LRU batch |
| MySQL Instance | Buffer | Buffer LRU Batch Scanned Per Call | float | The number of pages scanned per LRU batch call |
| MySQL Instance | Buffer | Buffer LRU Batch Total Pages | float | The total pages flushed as part of LRU batches |
| MySQL Instance | Buffer | Buffer LRU Batches | float | The number of LRU batches |
| MySQL Instance | Buffer | Buffer LRU Batches Evict | float | The number of LRU evict batches |
| MySQL Instance | Buffer | Buffer LRU Batches Flush | float | The number of LRU flush batches |
| MySQL Instance | Buffer | Buffer LRU Get Free Search | float | The number of searches performed for a clean page |
| MySQL Instance | Buffer | Buffer LRU Search Number Scan | float | The number of times LRU search is performed |
| MySQL Instance | Buffer | Buffer LRU Search Scanned | float | The total pages scanned as part of LRU search |
| MySQL Instance | Buffer | Buffer LRU Search Scanned Per Call | float | The number of pages scanned per single LRU search |
| MySQL Instance | Buffer | Buffer LRU Single Flush Failure Count | float | The number of attempts to flush a single page from LRU failed |
| MySQL Instance | Buffer | Buffer LRU Single Flush Number Scan | float | The number of times single page LRU flush is called |
| MySQL Instance | Buffer | Buffer LRU Single Flush Scanned | float | The total pages scanned as part of single page LRU flush |
| MySQL Instance | Buffer | Buffer LRU Single Flush Scanned Per Call | float | The number of pages scanned per single LRU flush |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Instance | Buffer | Buffer LRU Unzip Search Number Scan | float | The number of times LRU unzip search is performed |
| MySQL Instance | Buffer | Buffer LRU Unzip Search Scanned | float | The total pages scanned as part of LRU unzip search |
| MySQL Instance | Buffer | Buffer LRU Unzip Search Scanned Per Call | float | The number of pages scanned per single LRU unzip search |
| MySQL Instance | Buffer | Buffer Pages Created | float | The number of pages created |
| MySQL Instance | Buffer | Buffer Pages Read | float | The number of pages read |
| MySQL Instance | Buffer | Buffer Pages Written | float | The number of pages written |
| MySQL Instance | Buffer | Buffer Pool Bytes Data | integer | The number of buffer bytes containing data |
| MySQL Instance | Buffer | Buffer Pool Bytes Dirty | integer | The number of buffer bytes currently dirty |
| MySQL Instance | Buffer | Buffer Pool Pages Data | integer | The number of buffer pages containing data |
| MySQL Instance | Buffer | Buffer Pool Pages Dirty | integer | The number of buffer pages currently dirty |
| MySQL Instance | Buffer | Buffer Pool Pages Free | integer | The buffer pages currently free |
| MySQL Instance | Buffer | Buffer Pool Pages Miscellaneous | integer | The number of buffer pages for miscellaneous use such as row locks or the adaptive hash index |
| MySQL Instance | Buffer | Buffer Pool Pages Total | integer | The total buffer pool size in pages |
| MySQL Instance | Buffer | Buffer Pool Read Ahead | float | The number of pages read as read ahead |
| MySQL Instance | Buffer | Buffer Pool Read Ahead Evicted | float | The read-ahead pages evicted without being accessed |
| MySQL Instance | Buffer | Buffer Pool Read Requests | float | The number of logical read requests |
| MySQL Instance | Buffer | Buffer Pool Reads | float | The number of reads directly from disk |
| MySQL Instance | Buffer | Buffer Pool Size | integer | The server buffer pool size (all buffer pools) in bytes |
| MySQL Instance | Buffer | Buffer Pool Wait Free | float | The number of times waited for free buffer |
| MySQL Instance | Buffer | Buffer Pool Write Requests | float | The number of write requests |
| MySQL Instance | Buffer Page I/O | Buffer Page Read Extent Descriptor | float | The number of Extent Descriptor Pages read |
| MySQL Instance | Buffer Page I/O | Buffer Page Read File Space Header | float | The number of File Space Header Pages read |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Instance | Buffer Page I/O | Buffer Page Read First Compressed BLOB | float | The number of First Compressed BLOB Pages read |
| MySQL Instance | Buffer Page I/O | Buffer Page Read Index Inode | float | The number of Index Inode Pages read |
| MySQL Instance | Buffer Page I/O | Buffer Page Read Index Insert Buffer Leaf | float | The number of Insert Buffer Index Leaf Pages read |
| MySQL Instance | Buffer Page I/O | Buffer Page Read Index Insert Buffer Non Leaf | float | The number of Insert Buffer Index Non-Leaf Pages read |
| MySQL Instance | Buffer Page I/O | Buffer Page Read Index Leaf | float | The number of Index Leaf Pages read |
| MySQL Instance | Buffer Page I/O | Buffer Page Read Index Non Leaf | float | The number of Index Non-leaf Pages read |
| MySQL Instance | Buffer Page I/O | Buffer Page Read Insert Buffer Bitmap | float | The number of Insert Buffer Bitmap Pages read |
| MySQL Instance | Buffer Page I/O | Buffer Page Read Insert Buffer Free List | float | The number of Insert Buffer Free List Pages read |
| MySQL Instance | Buffer Page I/O | Buffer Page Read Other | float | The number of other/ unknown (old version of InnoDB) Pages read |
| MySQL Instance | Buffer Page I/O | Buffer Page Read Subsequent Compressed BLOB | float | The number of Subsequent Compressed BLOB Pages read |
| MySQL Instance | Buffer Page I/O | Buffer Page Read System Page | float | The number of System Pages read |
| MySQL Instance | Buffer Page I/O | Buffer Page Read Transaction System | float | The number of Transaction System Pages read |
| MySQL Instance | Buffer Page I/O | Buffer Page Read Uncompressed BLOB | float | The number of Uncompressed BLOB Pages read |
| MySQL Instance | Buffer Page I/O | Buffer Page Read Undo Log | float | The number of Undo Log Pages read |
| MySQL Instance | Buffer Page I/O | Buffer Page Written Extent Descriptor | float | The number of Extent Descriptor Pages written |
| MySQL Instance | Buffer Page I/O | Buffer Page Written File Space Header | float | The number of File Space Header Pages written |
| MySQL Instance | Buffer Page I/O | Buffer Page Written First Compressed BLOB | float | The number of First Compressed BLOB Pages written |
| MySQL Instance | Buffer Page I/O | Buffer Page Written Index Inode | float | The number of Index Inode Pages written |
| MySQL Instance | Buffer Page I/O | Buffer Page Written Index Insert Buffer Leaf | float | The number of Insert Buffer Index Leaf Pages written |
| MySQL Instance | Buffer Page I/O | Buffer Page Written Index Insert Buffer Non Leaf | float | The number of Insert Buffer Index Non-Leaf Pages written |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Instance | Buffer Page I/O | Buffer Page Written Index Leaf | float | The number of Index Leaf Pages written |
| MySQL Instance | Buffer Page I/O | Buffer Page Written Index Non Leaf | float | The number of Index Non-leaf Pages written |
| MySQL Instance | Buffer Page I/O | Buffer Page Written Insert Buffer Bitmap | float | The number of Insert Buffer Bitmap Pages written |
| MySQL Instance | Buffer Page I/O | Buffer Page Written Insert Buffer Free List | float | The number of Insert Buffer Free List Pages written |
| MySQL Instance | Buffer Page I/O | Buffer Page Written Other | float | The number of other/ unknown (old version InnoDB) Pages written |
| MySQL Instance | Buffer Page I/O | Buffer Page Written Subsequent Compressed BLOB | float | The number of Subsequent Compressed BLOB Pages written |
| MySQL Instance | Buffer Page I/O | Buffer Page Written System Page | float | The number of System Pages written |
| MySQL Instance | Buffer Page I/O | Buffer Page Written Transaction System | float | The number of Transaction System Pages written |
| MySQL Instance | Buffer Page I/O | Buffer Page Written Uncompressed BLOB | float | The number of Uncompressed BLOB Pages written |
| MySQL Instance | Buffer Page I/O | Buffer Page Written Undo Log | float | The number of Undo Log Pages written |
| MySQL Instance | Change Buffer | Change Buffer Deleted Merged Operations Discarded | float | The number of delete merged operations discarded |
| MySQL Instance | Change Buffer | Change Buffer Deleted Records Merged | float | The number of deleted records merged by change buffering |
| MySQL Instance | Change Buffer | Change Buffer Insert Merged Operations Discarded | float | The number of insert merged operations discarded |
| MySQL Instance | Change Buffer | Change Buffer Inserted Records Merged | float | The number of inserted records merged by change buffering |
| MySQL Instance | Change Buffer | Change Buffer Number Merges | float | The number of change buffer merges |
| MySQL Instance | Change Buffer | Change Buffer Purge Merged Records Discarded | float | The number of purge merged operations discarded |
| MySQL Instance | Change Buffer | Change Buffer Purge Records Merged | float | The number of purge records merged by change buffering |
| MySQL Instance | Change Buffer | Change Buffer Size | integer | The change buffer size in pages |
| MySQL Instance | Compression | Number of Padding Decrements Due To Compression | float | The number of times padding is decremented due to good compressibility |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Instance | Compression | Number of Padding Increments Due To Compression | float | The number of times padding is incremented to avoid compression failures |
| MySQL Instance | Compression | Number of Pages Compressed | float | The number of pages compressed |
| MySQL Instance | Compression | Number of Pages Decompressed | float | The number of pages decompressed |
| MySQL Instance | Connection | Aborted Connections | float | The number of failed attempts to connect to the MySQL server |
| MySQL Instance | Connection | Connections | float | The number of connection attempts (successful or not) to the MySQL server |
| MySQL Instance | Connection | Max Used Connections | integer | The maximum number of connections that have been in use simultaneously since the server started |
| MySQL Instance | Connection | SSL Client Connections | float | The number of SSL connection attempts to an SSL-enabled Primary |
| MySQL Instance | Connection | SSL Connection Renegotiates | float | The number of negotiates needed to establish the connection to an SSL-enabled Primary |
| MySQL Instance | Connection | SSL Finished Connections | float | The number of successful slave connections to an SSL-enabled Primary |
| MySQL Instance | Connection | Threads Connected | integer | The number of currently open connections |
| MySQL Instance | Details | Hostname | string | The Hostname |
| MySQL Instance | Details | Instance Name | string | The Instance Name |
| MySQL Instance | Details | Instance Total Disk Size | integer | The total size of all databases for this instance |
| MySQL Instance | Details | Port | integer | The Port |
| MySQL Instance | Details | Server ID | integer | The Identifier of the MySQL Instance |
| MySQL Instance | Details | Server UUID | string | The Universally Unique Identifier of the MySQL Instance |
| MySQL Instance | Details | Version | string | The Version |
| MySQL Instance | Index | Index Page Discards | float | The number of index pages discarded |
| MySQL Instance | Index | Index Page Merge Attempts | float | The number of index page merge attempts |
| MySQL Instance | Index | Index Page Merge Successes | float | The number of successful index page merges |
| MySQL Instance | Index | Index Page Reorganization Attempts | float | The number of index page reorganization attempts |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Instance | Index | Index Page Reorganization Successes | float | The number of successful index page reorganizations |
| MySQL Instance | Index | Index Page Splits | float | The number of index page splits |
| MySQL Instance | Locking | Deadlocks | float | The number of deadlocks |
| MySQL Instance | Locking | Lock Timeouts | float | The number of lock timeouts |
| MySQL Instance | Locking | Record Lock Requests | float | The number of record locks requested |
| MySQL Instance | Locking | Record Lock Waits | float | The number of record locks requested |
| MySQL Instance | Locking | Record Locks | integer | The current number of record locks on tables |
| MySQL Instance | Locking | Record Locks Created | float | The number of record locks created |
| MySQL Instance | Locking | Record Locks Removed | float | The number of record locks removed from the lock queue |
| MySQL Instance | Locking | Row Lock Current Waits | integer | Number of row locks currently being waited for |
| MySQL Instance | Locking | Row Lock Time | float | Time spent in acquiring row locks |
| MySQL Instance | Locking | Row Lock Time Average | float | The average time to acquire a row lock |
| MySQL Instance | Locking | Row Lock Time Max | float | The maximum time to acquire a row lock |
| MySQL Instance | Locking | Row Lock Waits | float | Number of times a row lock had to be waited for |
| MySQL Instance | Locking | Table Lock Waits | float | The number of record locks requested |
| MySQL Instance | Locking | Table Locks | float | The number of tables locks |
| MySQL Instance | Locking | Table Locks Created | float | The number of table locks created |
| MySQL Instance | Locking | Table Locks Removed | float | The number of table locks removed from the lock queue |
| MySQL Instance | Memory | Memory Pool Size | integer | Size of a memory pool InnoDB uses to store data dictionary and internal data structures in bytes |
| MySQL Instance | Memory | Total Allocated Memory | integer | The total allocated memory |
| MySQL Instance | Operating System | Data Reads | float | The number of reads initiated |
| MySQL Instance | Operating System | Data Writes | float | The number of writes initiated |
| MySQL Instance | Operating System | Files Currently Open | integer | The number of files currently open |
| MySQL Instance | Operating System | Log Bytes Written | float | The bytes of log written |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Instance | Operating System | Log Pending Writes | float | The number of pending log file writes |
| MySQL Instance | Operating System | Log Pending fsync Writes | float | The number of pending fsync writes |
| MySQL Instance | Operating System | Log fsync Writes | float | The number of fsync log writes |
| MySQL Instance | Operating System | Pending Reads | integer | The number of reads pending |
| MySQL Instance | Operating System | Pending Writes | integer | The number of writes pending |
| MySQL Instance | Operating System | fsync() Calls Count | float | The number of fsync() calls |
| MySQL Instance | Purge | Purge DML Delay Time | integer | The microseconds DML to be delayed due to purge lagging |
| MySQL Instance | Purge | Purge Delete Marked Records | float | The number of delete-marked rows purged |
| MySQL Instance | Purge | Purge Invoked | float | The number of times purge was invoked |
| MySQL Instance | Purge | Purge Resume Count | integer | The number of times purge was resumed |
| MySQL Instance | Purge | Purge Stop Count | integer | The number of times purge was stopped |
| MySQL Instance | Purge | Purge Undo Log Pages | float | The number of undo log pages handled by the purge |
| MySQL Instance | Purge | Purge Update Existing Or External Records | float | The number of purges on updates of existing records and updates on delete marked record with externally stored field |
| MySQL Instance | Recovery | Log Checkpoints | float | The number of checkpoints |
| MySQL Instance | Recovery | Log Current LSN | integer | The current log sequence number value |
| MySQL Instance | Recovery | Log LSN Checkpoint Age | float | The current log sequence number value minus log sequence number at last checkpoint |
| MySQL Instance | Recovery | Log LSN at Last Checkpoint | integer | The log sequence number at last checkpoint |
| MySQL Instance | Recovery | Log LSN of Last Flush | integer | The log sequence number of Last flush |
| MySQL Instance | Recovery | Log Number Log IO | float | The number of log I/Os |
| MySQL Instance | Recovery | Log Oldest LSN in Buffer Pool | integer | The oldest modified block log sequence number in the buffer pool |
| MySQL Instance | Recovery | Log Pending Checkpoint Writes | float | The pending checkpoints |
| MySQL Instance | Recovery | Log Pending Log Writes | float | The pending log writes |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Instance | Recovery | Log Waits | float | The number of log waits due to small log buffer |
| MySQL Instance | Recovery | Log Write Requests | float | The number of log write requests |
| MySQL Instance | Recovery | Log Writes | float | The number of log writes |
| MySQL Instance | Recovery | Max Modified LSN Age For ASYNC Preflush | integer | The maximum log sequence number difference; when exceeded, start asynchronous preflush |
| MySQL Instance | Recovery | Max Modified LSN Age For SYNC Preflush | integer | The maximum log sequence number difference; when exceeded, start synchronous preflush |
| MySQL Instance | Row Operations | Rows Deleted | float | The number of rows deleted |
| MySQL Instance | Row Operations | Rows Inserted | float | The number of rows inserted |
| MySQL Instance | Row Operations | Rows Read | float | The number of rows read |
| MySQL Instance | Row Operations | Rows Updated | float | The number of rows updated |
| MySQL Instance | Server | Background Drop Table Time | float | The time spent to process drop table list |
| MySQL Instance | Server | Change Buffer Merge Time | float | The time spent to process change buffer merge |
| MySQL Instance | Server | Checkpoint Time | float | The time spent by Primary thread to do checkpoint |
| MySQL Instance | Server | DICT LRU Processing Time | float | The time spent to process DICT LRU list |
| MySQL Instance | Server | Doublewrite Operation Pages Written | float | The number of pages that have been written for doublewrite operations (innodb_dblwr_pages_written) |
| MySQL Instance | Server | Doublewrite Writes | float | The number of doublewrite operations that have been performed (innodb_dblwr_writes) |
| MySQL Instance | Server | Log Flush Time | float | The time spent to flush log records |
| MySQL Instance | Server | Primary Active Loops | float | The number of times Primary thread performs its tasks when server is active |
| MySQL Instance | Server | Primary Idle Loops | float | The number of times Primary thread performs its tasks when server is idle |
| MySQL Instance | Server | Primary Purge Time | float | The time spent by Primary thread to purge records |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Instance | Server | Primary Thread Sleep Time | float | The amount of time Primary thread has slept |
| MySQL Instance | Server | Memory Validation Time | float | The time spent to do memory validation |
| MySQL Instance | Server | OS Waits From Exclusive Latch Request | float | The number of rwlock OS waits due to exclusive latch request |
| MySQL Instance | Server | OS Waits From Shared Latch Request | float | The number of rwlock OS waits due to shared latch request |
| MySQL Instance | Server | Server Activity Count | integer | The current server activity count |
| MySQL Instance | Server | rwlock Spin Loop Rounds From Exclusive Latch Request | float | The number of rwlock spin loop rounds due to exclusive latch request |
| MySQL Instance | Server | rwlock Spin Loop Rounds From Shared Latch Request | float | The number of rwlock spin loop rounds due to shared latch request |
| MySQL Instance | Server | rwlock Spin Waits From Exclusive Latch Request | float | The number of rwlock spin waits due to exclusive latch request |
| MySQL Instance | Server | rwlock Spin Waits From Shared Latch Request | float | The number of rwlock spin waits due to shared latch request |
| MySQL Instance | Transaction | Active Transactions | integer | The number of active transactions |
| MySQL Instance | Transaction | Active Transactions Rolled Back | float | The number of resurrected active transactions rolled back |
| MySQL Instance | Transaction | Current Rollback Size | integer | The current rollback segment size |
| MySQL Instance | Transaction | Insert And Update Commits | float | The number of transactions committed with inserts and updates |
| MySQL Instance | Transaction | Non-Locking Auto-Commit Read Only Commits | float | The number of non-locking auto-commit read-only transactions committed |
| MySQL Instance | Transaction | Read Only Transaction Commits | float | The number of read-only transactions committed |
| MySQL Instance | Transaction | Read Write Transaction Commits | float | The number of read-write transactions committed |
| MySQL Instance | Transaction | TRX_RSEG_HISTORY List Length | integer | The length of the TRX_RSEG_HISTORY list |
| MySQL Instance | Transaction | Transaction Rollbacks | float | The number of transactions rolled back |
| MySQL Instance | Transaction | Transactions Rolled Back To Savepoint | float | The number of transactions rolled back to savepoint |
| MySQL Instance | Transaction | Undo Slots Cached | float | The number of undo slots cached |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Instance | Transaction | Undo Slots Used | float | The number of undo slots used |
| MySQL Instance | Uncategorized | Index Push-Down Condition Check Attempts | float | The number of Index Push-Down Condition check attempts |
| MySQL Instance | Uncategorized | Index Push-Down Conditions Match | float | The number of Index Push-Down Conditions with a match |
| MySQL Instance | Uncategorized | Index Push-Down Conditions No Match | float | The number of Index Push-Down Conditions with no match |
| MySQL Instance | Uncategorized | Index Push-Down Conditions Out Of Range | float | The number of Index Push-Down Conditions out of range |
| MySQL Instance | User Configuration | Max Connections | integer | The Max Connections |
| MySQL Instance | User Configuration | Page Size | integer | The Page Size |
| MySQL Instance | Relationships | AIX Parents | string | List of parent AIX resources |
| MySQL Instance | Relationships | Database Children | string | List of child Database resources |
| MySQL Instance | Relationships | HPUX Parents | string | List of parent HPUX resources |
| MySQL Instance | Relationships | Linux Parents | string | List of parent Linux resources |
| MySQL Instance | Relationships | MySQL Tag Parents | string | List of parent MySQL Tag resources |
| MySQL Instance | Relationships | Query Children | string | List of child Query resources |
| MySQL Instance | Relationships | Replication Primary Parents | string | List of parent Replication Primary resources |
| MySQL Instance | Relationships | Replication Slave Children | string | List of child Replication Slave resources |
| MySQL Instance | Relationships | Solaris Parents | string | List of parent Solaris resources |
| MySQL Instance | Relationships | Tablespace Children | string | List of child Tablespace resources |
| MySQL Instance | Relationships | Virtual Machine Parents | string | List of parent Virtual Machine resources |
| MySQL Instance | Relationships | Windows Parents | string | List of parent Windows resources |
| MySQL Instance | | Hostname (Deprecated) | string | |
| MySQL Instance | | Port (Deprecated) | integer | |
| MySQL Instance | | Instance Name (Deprecated) | string | |
| MySQL Instance | | Total Allocated Memory (Deprecated) | integer | |
| MySQL Instance | | Version (Deprecated) | string | |
| MySQL Instance | | Max Connections (Deprecated) | integer | |
| MySQL Instance | | Instance Total Disk Size (Deprecated) | integer | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Instance | | System Memory (Deprecated) | integer | |
| MySQL Instance | | Adaptive Hash Pages Added (Deprecated) | integer | |
| MySQL Instance | | Adaptive Hash Pages Removed (Deprecated) | integer | |
| MySQL Instance | | Adaptive Hash Rows Added (Deprecated) | integer | |
| MySQL Instance | | Adaptive Hash Rows Deleted With No Hash Entry (Deprecated) | integer | |
| MySQL Instance | | Adaptive Hash Rows Removed (Deprecated) | integer | |
| MySQL Instance | | Adaptive Hash Rows Updated (Deprecated) | integer | |
| MySQL Instance | | Adaptive Hash Searches (Deprecated) | integer | |
| MySQL Instance | | Adaptive Hash B-tree Searches (Deprecated) | integer | |
| MySQL Instance | | Buffer Pages Read (Deprecated) | integer | |
| MySQL Instance | | Buffer Pool Read Ahead (Deprecated) | integer | |
| MySQL Instance | | Buffer Pool Bytes Dirty (Deprecated) | integer | |
| MySQL Instance | | Buffer Pool Reads (Deprecated) | integer | |
| MySQL Instance | | Buffer Pool Pages Misc (Deprecated) | integer | |
| MySQL Instance | | Buffer Pages Written (Deprecated) | integer | |
| MySQL Instance | | Buffer Pool Wait Free (Deprecated) | integer | |
| MySQL Instance | | Buffer Pool Pages Dirty (Deprecated) | integer | |
| MySQL Instance | | Buffer Data Written (Deprecated) | integer | |
| MySQL Instance | | Buffer Pool Pages Total (Deprecated) | integer | |
| MySQL Instance | | Buffer Pages Created (Deprecated) | integer | |
| MySQL Instance | | Buffer Pool Write Requests (Deprecated) | integer | |
| MySQL Instance | | Buffer Pool Bytes Data (Deprecated) | integer | |
| MySQL Instance | | Buffer Data Reads (Deprecated) | integer | |
| MySQL Instance | | Buffer Pool Read Ahead Evicted (Deprecated) | integer | |
| MySQL Instance | | Buffer Pool Pages Free (Deprecated) | integer | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Instance | | Buffer Pool Read Requests (Deprecated) | integer | |
| MySQL Instance | | Buffer Pool Pages Data (Deprecated) | integer | |
| MySQL Instance | | Buffer Flush Batch Scanned (Deprecated) | integer | |
| MySQL Instance | | Buffer Flush Batch Num Scan (Deprecated) | integer | |
| MySQL Instance | | Buffer Flush Batch Scanned Per Call (Deprecated) | integer | |
| MySQL Instance | | Buffer Flush Batch Rescan (Deprecated) | integer | |
| MySQL Instance | | Buffer Flush Batch Total Pages (Deprecated) | integer | |
| MySQL Instance | | Buffer Flush Batches (Deprecated) | integer | |
| MySQL Instance | | Buffer Flush Batch Pages (Deprecated) | integer | |
| MySQL Instance | | Buffer Flush Neighbor Total Pages (Deprecated) | integer | |
| MySQL Instance | | Buffer Flush Neighbor (Deprecated) | integer | |
| MySQL Instance | | Buffer Flush Neighbor Pages (Deprecated) | integer | |
| MySQL Instance | | Buffer Flush Num To Flush Requested (Deprecated) | integer | |
| MySQL Instance | | Buffer Flush Average Page Rate (Deprecated) | float | |
| MySQL Instance | | Buffer Flush LSN Average Rate (Deprecated) | float | |
| MySQL Instance | | Buffer Flush Percent For Dirty (Deprecated) | float | |
| MySQL Instance | | Buffer Flush Percent For LSN (Deprecated) | float | |
| MySQL Instance | | Buffer Flush Sync Waits (Deprecated) | integer | |
| MySQL Instance | | Buffer Flush Adaptive Total Pages (Deprecated) | integer | |
| MySQL Instance | | Buffer Flush Adaptive (Deprecated) | integer | |
| MySQL Instance | | Buffer Flush Adaptive Pages (Deprecated) | integer | |
| MySQL Instance | | Buffer Flush Sync Total Pages (Deprecated) | integer | |
| MySQL Instance | | Buffer Flush Background (Deprecated) | integer | |
| MySQL Instance | | Buffer Flush Background Pages (Deprecated) | integer | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Instance | | Buffer LRU Batch Scanned (Deprecated) | integer | |
| MySQL Instance | | Buffer LRU Batch Number Scan (Deprecated) | integer | |
| MySQL Instance | | Buffer LRU Batch Scanned Per Call (Deprecated) | integer | |
| MySQL Instance | | Buffer LRU Batch Total Pages (Deprecated) | integer | |
| MySQL Instance | | Buffer LRU Batch Evict Total Pages (Deprecated) | integer | |
| MySQL Instance | | Buffer LRU Batch Flush Total Pages (Deprecated) | integer | |
| MySQL Instance | | Buffer LRU Batches (Deprecated) | integer | |
| MySQL Instance | | Buffer LRU Batches Evict (Deprecated) | integer | |
| MySQL Instance | | Buffer LRU Batches Flush (Deprecated) | integer | |
| MySQL Instance | | Buffer LRU Batch Pages (Deprecated) | integer | |
| MySQL Instance | | Buffer LRU Batch Evict Pages (Deprecated) | integer | |
| MySQL Instance | | Buffer LRU Batch Flush Pages (Deprecated) | integer | |
| MySQL Instance | | Buffer LRU Single Flush Scanned (Deprecated) | integer | |
| MySQL Instance | | Buffer LRU Single Flush Number Scan (Deprecated) | integer | |
| MySQL Instance | | Buffer LRU Single Flush Scanned Per Call (Deprecated) | integer | |
| MySQL Instance | | Buffer LRU Get Free Search (Deprecated) | integer | |
| MySQL Instance | | Buffer LRU Search Scanned (Deprecated) | integer | |
| MySQL Instance | | Buffer LRU Search Number Scan (Deprecated) | integer | |
| MySQL Instance | | Buffer LRU Search Scanned Per Call (Deprecated) | integer | |
| MySQL Instance | | Buffer LRU Unzip Search Scanned (Deprecated) | integer | |
| MySQL Instance | | Buffer LRU Unzip Search Number Scan (Deprecated) | integer | |
| MySQL Instance | | Buffer LRU Unzip Search Scanned Per Call (Deprecated) | integer | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Instance | | Buffer Page Read Index Leaf (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Read Index Non Leaf (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Read Index Insert Buffer Leaf (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Read Index Insert Buffer Non Leaf (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Read Undo Log (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Read Index Inode (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Read Insert Buffer Free List (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Read Insert Buffer Bitmap (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Read System Page (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Read Transaction System (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Read File Space Header (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Read Extent Descriptor (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Read Uncompressed BLOB (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Read First Compressed BLOB (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Read Subsequent Compressed BLOB (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Read Other (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Written Index Leaf (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Written Index Non Leaf (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Written Index Insert Buffer Leaf (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Written Index Insert Buffer Non Leaf (Deprecated) | integer | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Instance | | Buffer Page Written Undo Log (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Written Index Inode (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Written Insert Buffer Free List (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Written Insert Buffer Bitmap (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Written System Page (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Written Transaction System (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Written File Space Header (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Written Extent Descriptor (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Written Uncompressed BLOB (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Written First Compressed BLOB (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Written Subsequent Compressed BLOB (Deprecated) | integer | |
| MySQL Instance | | Buffer Page Written Other (Deprecated) | integer | |
| MySQL Instance | | Change Buffer Deleted Records Merged (Deprecated) | integer | |
| MySQL Instance | | Change Buffer Deleted Merged Operations Discarded (Deprecated) | integer | |
| MySQL Instance | | Change Buffer Inserted Records Merged (Deprecated) | integer | |
| MySQL Instance | | Change Buffer Size (Deprecated) | integer | |
| MySQL Instance | | Change Buffer Insert Merged Operations Discarded (Deprecated) | integer | |
| MySQL Instance | | Change Buffer Number Merges (Deprecated) | integer | |
| MySQL Instance | | Change Buffer Purge Records Merged (Deprecated) | integer | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Instance | | Change Buffer Purge Merged Records Discarded (Deprecated) | integer | |
| MySQL Instance | | Number of Pages Compressed (Deprecated) | integer | |
| MySQL Instance | | Number of Pages Decompressed (Deprecated) | integer | |
| MySQL Instance | | Number of Padding Increments Due To Compression (Deprecated) | integer | |
| MySQL Instance | | Number of Padding Decrements Due To Compression (Deprecated) | integer | |
| MySQL Instance | | Indexes Waiting To Drop (Deprecated) | integer | |
| MySQL Instance | | Tables Waiting To Drop (Deprecated) | integer | |
| MySQL Instance | | Indexes Being Created Online (Deprecated) | integer | |
| MySQL Instance | | ALTER TABLE's, CREATE INDEX's, DROP INDEX's In Progress (Deprecated) | integer | |
| MySQL Instance | | Rows Inserted (Deprecated) | integer | |
| MySQL Instance | | Rows Read (Deprecated) | integer | |
| MySQL Instance | | Rows Updated (Deprecated) | integer | |
| MySQL Instance | | Rows Deleted (Deprecated) | integer | |
| MySQL Instance | | Files Currently Open (Deprecated) | integer | |
| MySQL Instance | | Index Push-Down Condition Check Attempts (Deprecated) | integer | |
| MySQL Instance | | Index Push-Down Conditions No Match (Deprecated) | integer | |
| MySQL Instance | | Index Push-Down Conditions Out Of Range (Deprecated) | integer | |
| MySQL Instance | | Index Push-Down Conditions Match (Deprecated) | integer | |
| MySQL Instance | | Index Page Splits (Deprecated) | integer | |
| MySQL Instance | | Index Page Merge Attempts (Deprecated) | integer | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Instance | | Index Page Merge Successes (Deprecated) | integer | |
| MySQL Instance | | Index Page Reorganization Attempts (Deprecated) | integer | |
| MySQL Instance | | Index Page Reorganization Successes (Deprecated) | integer | |
| MySQL Instance | | Index Page Discards (Deprecated) | integer | |
| MySQL Instance | | Deadlocks (Deprecated) | integer | |
| MySQL Instance | | Row Lock Waits (Deprecated) | integer | |
| MySQL Instance | | Row Lock Time Max (Deprecated) | integer | |
| MySQL Instance | | Row Lock Time (Deprecated) | integer | |
| MySQL Instance | | Lock Timeouts (Deprecated) | integer | |
| MySQL Instance | | Row Lock Time Average (Deprecated) | integer | |
| MySQL Instance | | Row Lock Current Waits (Deprecated) | integer | |
| MySQL Instance | | Record Lock Requests (Deprecated) | integer | |
| MySQL Instance | | Record Locks Created (Deprecated) | integer | |
| MySQL Instance | | Record Locks Removed (Deprecated) | integer | |
| MySQL Instance | | Record Locks (Deprecated) | integer | |
| MySQL Instance | | Table Locks Created (Deprecated) | integer | |
| MySQL Instance | | Table Locks Removed (Deprecated) | integer | |
| MySQL Instance | | Table Locks (Deprecated) | integer | |
| MySQL Instance | | Memory Pool Size (Deprecated) | integer | |
| MySQL Instance | | Table Handles Opened (Deprecated) | integer | |
| MySQL Instance | | Table Handles Closed (Deprecated) | integer | |
| MySQL Instance | | Table Reference Count (Deprecated) | integer | |
| MySQL Instance | | Log Bytes Written (Deprecated) | integer | |
| MySQL Instance | | Data Writes (Deprecated) | integer | |
| MySQL Instance | | Log Pending Writes (Deprecated) | integer | |
| MySQL Instance | | Data Reads (Deprecated) | integer | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Instance | | Log Pending fsync Writes (Deprecated) | integer | |
| MySQL Instance | | Log fsync Writes (Deprecated) | integer | |
| MySQL Instance | | fsync() Calls Count (Deprecated) | integer | |
| MySQL Instance | | Pending Reads (Deprecated) | integer | |
| MySQL Instance | | Pending Writes (Deprecated) | integer | |
| MySQL Instance | | Purge Delete Marked Records (Deprecated) | integer | |
| MySQL Instance | | Purge Update Existing Or External Records (Deprecated) | integer | |
| MySQL Instance | | Purge Invoked (Deprecated) | integer | |
| MySQL Instance | | Purge Undo Log Pages (Deprecated) | integer | |
| MySQL Instance | | Purge DML Delay Time (Deprecated) | integer | |
| MySQL Instance | | Purge Stop Count (Deprecated) | integer | |
| MySQL Instance | | Purge Resume Count (Deprecated) | integer | |
| MySQL Instance | | Log Write Requests (Deprecated) | integer | |
| MySQL Instance | | Log Waits (Deprecated) | integer | |
| MySQL Instance | | Log Writes (Deprecated) | integer | |
| MySQL Instance | | Log Checkpoints (Deprecated) | integer | |
| MySQL Instance | | Log LSN of Last Flush (Deprecated) | integer | |
| MySQL Instance | | Log LSN at Last Checkpoint (Deprecated) | integer | |
| MySQL Instance | | Log Current LSN (Deprecated) | integer | |
| MySQL Instance | | Log LSN Checkpoint Age (Deprecated) | integer | |
| MySQL Instance | | Log Oldest LSN in Buffer Pool (Deprecated) | integer | |
| MySQL Instance | | Max Modified LSN Age For ASYNC Preflush (Deprecated) | integer | |
| MySQL Instance | | Max Modified LSN Age For SYNC Preflush (Deprecated) | integer | |
| MySQL Instance | | Log Pending Log Writes (Deprecated) | integer | |
| MySQL Instance | | Log Pending Checkpoint Writes (Deprecated) | integer | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Instance | | Log Number Log IO (Deprecated) | integer | |
| MySQL Instance | | Buffer Pool Size (Deprecated) | integer | |
| MySQL Instance | | OS Waits From Shared Latch Request (Deprecated) | integer | |
| MySQL Instance | | rwlock Spin Waits From Shared Latch Request (Deprecated) | integer | |
| MySQL Instance | | rwlock Spin Loop Rounds From Exclusive Latch Request (Deprecated) | integer | |
| MySQL Instance | | Page Size (Deprecated) | integer | |
| MySQL Instance | | rwlock Spin Loop Rounds From Shared Latch Request (Deprecated) | integer | |
| MySQL Instance | | Doublewrite Operation Pages Written (Deprecated) | integer | |
| MySQL Instance | | OS Waits From Exclusive Latch Request (Deprecated) | integer | |
| MySQL Instance | | Server Activity Count (Deprecated) | integer | |
| MySQL Instance | | rwlock Spin Waits From Exclusive Latch Request (Deprecated) | integer | |
| MySQL Instance | | Doublewrite Writes (Deprecated) | integer | |
| MySQL Instance | | Primary Thread Sleep Time (Deprecated) | integer | |
| MySQL Instance | | Primary Active Loops (Deprecated) | integer | |
| MySQL Instance | | Primary Idle Loops (Deprecated) | integer | |
| MySQL Instance | | Background Drop Table Time (Deprecated) | integer | |
| MySQL Instance | | Change Buffer Merge Time (Deprecated) | integer | |
| MySQL Instance | | Log Flush Time (Deprecated) | integer | |
| MySQL Instance | | Memory Validation Time (Deprecated) | integer | |
| MySQL Instance | | Primary Purge Time (Deprecated) | integer | |
| MySQL Instance | | DICT LRU Processing Time (Deprecated) | integer | |
| MySQL Instance | | Checkpoint Time (Deprecated) | integer | |
| MySQL Instance | | TRX_RSEG_HISTORY List Length (Deprecated) | integer | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Instance | | Read Write Transaction Commits (Deprecated) | integer | |
| MySQL Instance | | Read Only Transaction Commits (Deprecated) | integer | |
| MySQL Instance | | Non-Locking Auto-Commit Read Only Commits (Deprecated) | integer | |
| MySQL Instance | | Insert And Update Commits (Deprecated) | integer | |
| MySQL Instance | | Transaction Rollbacks (Deprecated) | integer | |
| MySQL Instance | | Transactions Rolled Back To Savepoint (Deprecated) | integer | |
| MySQL Instance | | Active Transactions Rolled Back (Deprecated) | integer | |
| MySQL Instance | | Active Transactions (Deprecated) | integer | |
| MySQL Instance | | Undo Slots Used (Deprecated) | integer | |
| MySQL Instance | | Undo Slots Cached (Deprecated) | integer | |
| MySQL Instance | | Current Rollback Size (Pages) (Deprecated) | integer | |
| MySQL Instance | | Connections (Deprecated) | integer | |
| MySQL Instance | | Aborted Connections (Deprecated) | integer | |
| MySQL Instance | | Max Used Connections (Deprecated) | integer | |
| MySQL Instance | | Threads Connected (Deprecated) | integer | |
| MySQL Instance | | SSL Client Connections (Deprecated) | integer | |
| MySQL Instance | | SSL Connection Renegotiates (Deprecated) | integer | |
| MySQL Instance | | SSL Finished Connections (Deprecated) | integer | |
| MySQL Query | Activity | Average Errors | integer | The total number of errors produced by occurrences of the statement |
| MySQL Query | Activity | Average Sort Merge Passes | float | The average number of sort merge passes per occurrence of the statement |
| MySQL Query | Activity | Average Temporary Tables Created In-Memory | integer | The average number of internal in-memory temporary tables created per occurrence of the statement |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Query | Activity | Average Temporary Tables Created On-Disk | integer | The average number of internal on-disk temporary tables created per occurrence of the statement |
| MySQL Query | Activity | Average Warnings | integer | The average number of warnings produced by occurrences of the statement |
| MySQL Query | Activity | Error Count | float | The total number of errors produced by occurrences of the statement |
| MySQL Query | Activity | Execution Count | float | The total number of times the statement has executed |
| MySQL Query | Activity | First Seen | string | The first time at which the statement was seen |
| MySQL Query | Activity | Last Seen | string | The most recent time at which the statement was seen |
| MySQL Query | Activity | Rank | integer | The rank of this query, according to the way queries are ordered (usually by average latency) |
| MySQL Query | Activity | Sort Merge Passes | float | The total number of sort merge passes by occurrences of the statement |
| MySQL Query | Activity | Temporary Tables Created In-Memory | float | The total number of internal in-memory temporary tables created by occurrences of the statement |
| MySQL Query | Activity | Temporary Tables Created On-Disk | float | The total number of internal on-disk temporary tables created by occurrences of the statement |
| MySQL Query | Activity | Warning Count | float | The total number of warnings produced by occurrences of the statement |
| MySQL Query | Details | SQL HTML | string | The HTML version of the SQL text of this query |
| MySQL Query | Details | SQL Text | string | The normalized statement string |
| MySQL Query | Row Operations | Average Rows Affected | integer | The average number of rows affected per occurrence of the statement |
| MySQL Query | Row Operations | Average Rows Examined | integer | The average number of rows read from storage engines per occurrence of the statement |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Query | Row Operations | Average Rows Sent | integer | The average number of rows returned per occurrence of the statement |
| MySQL Query | Row Operations | Average Rows Sorted | integer | The average number of rows sorted per occurrence of the statement |
| MySQL Query | Row Operations | Rows Affected | float | The total number of rows affected by occurrences of the statement |
| MySQL Query | Row Operations | Rows Examined | float | The total number of rows read from storage engines by occurrences of the statement |
| MySQL Query | Row Operations | Rows Sent | float | The total number of rows returned by occurrences of the statement |
| MySQL Query | Row Operations | Rows Sorted | float | The total number of rows sorted by occurrences of the statement |
| MySQL Query | Wait Time | Average Latency | integer | The average wait time per timed occurrence of the statement |
| MySQL Query | Wait Time | Average Lock Latency | integer | The average time waiting for locks per timed occurrence of the statement |
| MySQL Query | Wait Time | Lock Latency | float | The total time waiting for locks by timed occurrences of the statement |
| MySQL Query | Wait Time | Max Latency | integer | The maximum single wait time of timed occurrences of the statement |
| MySQL Query | Wait Time | Total Latency | float | The total wait time of timed occurrences of the statement |
| MySQL Query | Relationships | Database Parents | string | List of parent Database resources |
| MySQL Query | Relationships | Instance Parents | string | List of parent Instance resources |
| MySQL Query | Relationships | MySQL Tag Parents | string | List of parent MySQL Tag resources |
| MySQL Query | | Hostname (Deprecated) | string | |
| MySQL Query | | Port (Deprecated) | integer | |
| MySQL Query | | Rank (Deprecated) | integer | |
| MySQL Query | | SQL Text (Deprecated) | string | |
| MySQL Query | | Database (Deprecated) | string | |
| MySQL Query | | Calls (Deprecated) | integer | |
| MySQL Query | | Average Rows Sent (Deprecated) | float | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Query | | Average Rows Examined (Deprecated) | float | |
| MySQL Query | | Average Time (Deprecated) | float | |
| MySQL Query | | Total Time (Deprecated) | integer | |
| MySQL Query | | Average Lock Time (Deprecated) | float | |
| MySQL Query | | Total Lock Time (Deprecated) | integer | |
| MySQL Table | Capacity | Table Data Size | integer | The size of the table's data |
| MySQL Table | Capacity | Table Index Size | integer | The size of the table's indexes |
| MySQL Table | Capacity | Table Total Size | integer | The total size of the table |
| MySQL Table | I/O | I/O Misc Latency | float | The total wait time of miscellaneous I/O requests for the table |
| MySQL Table | I/O | I/O Misc Requests | float | The total number of miscellaneous I/O requests for the table |
| MySQL Table | I/O | I/O Read | float | The total number of bytes read from the table |
| MySQL Table | I/O | I/O Read Latency | float | The total wait time of reads from the table |
| MySQL Table | I/O | I/O Read Requests | float | The total number of read requests for the table |
| MySQL Table | I/O | I/O Write | float | The total number of bytes written to the table |
| MySQL Table | I/O | I/O Write Latency | float | The total wait time of writes to the table |
| MySQL Table | I/O | I/O Write Requests | float | The total number of write requests for the table |
| MySQL Table | Row Operations | Delete Average Wait Time | integer | The average wait time of deletes |
| MySQL Table | Row Operations | Delete Max Wait Time | integer | The maximum wait time of deletes |
| MySQL Table | Row Operations | Delete Min Wait Time | integer | The minimum wait time of deletes |
| MySQL Table | Row Operations | Delete Total Wait Time | integer | The total wait time of deletes |
| MySQL Table | Row Operations | Fetch Average Wait Time | integer | The average wait time of fetches |
| MySQL Table | Row Operations | Fetch Max Wait Time | integer | The maximum wait time of fetches |
| MySQL Table | Row Operations | Fetch Min Wait Time | integer | The minimum wait time of fetches |
| MySQL Table | Row Operations | Fetch Total Wait Time | integer | The total wait time of fetches |
| MySQL Table | Row Operations | Full Rows Scanned | integer | The number of times the table has been fully scanned |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Table | Row Operations | Insert Average Wait Time | integer | The average wait time of inserts |
| MySQL Table | Row Operations | Insert Max Wait Time | integer | The maximum wait time of inserts |
| MySQL Table | Row Operations | Insert Min Wait Time | integer | The minimum wait time of inserts |
| MySQL Table | Row Operations | Insert Total Wait Time | integer | The total wait time of inserts |
| MySQL Table | Row Operations | R/W Average Wait Time | integer | The average wait time of reads and writes |
| MySQL Table | Row Operations | R/W Max Wait Time | integer | The maximum wait time of reads and writes |
| MySQL Table | Row Operations | R/W Min Wait Time | integer | The minimum wait time of reads and writes |
| MySQL Table | Row Operations | R/W Total Wait Time | integer | The total wait time of reads and writes |
| MySQL Table | Row Operations | Read Average Wait Time | integer | The average wait time of reads |
| MySQL Table | Row Operations | Read Max Wait Time | integer | The maximum wait time of reads |
| MySQL Table | Row Operations | Read Min Wait Time | integer | The minimum wait time of reads |
| MySQL Table | Row Operations | Read Total Wait Time | integer | The total wait time of reads |
| MySQL Table | Row Operations | Rows Deleted | float | The number of rows deleted |
| MySQL Table | Row Operations | Rows Fetched | float | The number of rows fetched |
| MySQL Table | Row Operations | Rows Inserted | float | The number of rows inserted |
| MySQL Table | Row Operations | Rows Read | float | The number of rows read |
| MySQL Table | Row Operations | Rows Read/Written | float | The total number of rows read and written |
| MySQL Table | Row Operations | Rows Updated | float | The number of rows updated |
| MySQL Table | Row Operations | Rows Written | float | The number of rows written |
| MySQL Table | Row Operations | Update Average Wait Time | integer | The average wait time of updates |
| MySQL Table | Row Operations | Update Max Wait Time | integer | The maximum wait time of updates |
| MySQL Table | Row Operations | Update Min Wait Time | integer | The minimum wait time of updates |
| MySQL Table | Row Operations | Update Total Wait Time | integer | The total wait time of updates |
| MySQL Table | Row Operations | Write Average Wait Time | integer | The average wait time of writes |
| MySQL Table | Row Operations | Write Max Wait Time | integer | The maximum wait time of writes |
| MySQL Table | Row Operations | Write Min Wait Time | integer | The minimum wait time of writes |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Table | Row Operations | Write Total Wait Time | integer | The total wait time of writes |
| MySQL Table | Relationships | Database Parents | string | List of parent Database resources |
| MySQL Table | Relationships | Index Children | string | List of child Index resources |
| MySQL Table | Relationships | MySQL Tag Parents | string | List of parent MySQL Tag resources |
| MySQL Table | | Hostname (Deprecated) | string | |
| MySQL Table | | Port (Deprecated) | integer | |
| MySQL Table | | Table Name (Deprecated) | string | |
| MySQL Table | | Database Name (Deprecated) | string | |
| MySQL Table | | Table Total Size (Deprecated) | integer | |
| MySQL Table | | Table Data Size (Deprecated) | integer | |
| MySQL Table | | Table Index Size (Deprecated) | integer | |
| MySQL Table | | Rows Read/Written (Deprecated) | integer | |
| MySQL Table | | R/W Total Wait Time (Deprecated) | integer | |
| MySQL Table | | R/W Min Wait Time (Deprecated) | integer | |
| MySQL Table | | R/W Average Wait Time (Deprecated) | integer | |
| MySQL Table | | R/W Max Wait Time (Deprecated) | integer | |
| MySQL Table | | Rows Read (Deprecated) | integer | |
| MySQL Table | | Read Total Wait Time (Deprecated) | integer | |
| MySQL Table | | Read Min Wait Time (Deprecated) | integer | |
| MySQL Table | | Read Average Wait Time (Deprecated) | integer | |
| MySQL Table | | Read Max Wait Time (Deprecated) | integer | |
| MySQL Table | | Rows Written (Deprecated) | integer | |
| MySQL Table | | Write Total Wait Time (Deprecated) | integer | |
| MySQL Table | | Write Min Wait Time (Deprecated) | integer | |
| MySQL Table | | Write Average Wait Time (Deprecated) | integer | |
| MySQL Table | | Write Max Wait Time (Deprecated) | integer | |
| MySQL Table | | Rows Fetched (Deprecated) | integer | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Table | | Fetch Total Wait Time (Deprecated) | integer | |
| MySQL Table | | Fetch Min Wait Time (Deprecated) | integer | |
| MySQL Table | | Fetch Average Wait Time (Deprecated) | integer | |
| MySQL Table | | Fetch Max Wait Time (Deprecated) | integer | |
| MySQL Table | | Rows Inserted (Deprecated) | integer | |
| MySQL Table | | Insert Total Wait Time (Deprecated) | integer | |
| MySQL Table | | Insert Min Wait Time (Deprecated) | integer | |
| MySQL Table | | Insert Average Wait Time (Deprecated) | integer | |
| MySQL Table | | Insert Max Wait Time (Deprecated) | integer | |
| MySQL Table | | Rows Updated (Deprecated) | integer | |
| MySQL Table | | Update Total Wait Time (Deprecated) | integer | |
| MySQL Table | | Update Min Wait Time (Deprecated) | integer | |
| MySQL Table | | Update Average Wait Time (Deprecated) | integer | |
| MySQL Table | | Update Max Wait Time (Deprecated) | integer | |
| MySQL Table | | Rows Deleted (Deprecated) | integer | |
| MySQL Table | | Delete Total Wait Time (Deprecated) | integer | |
| MySQL Table | | Delete Min Wait Time (Deprecated) | integer | |
| MySQL Table | | Delete Average Wait Time (Deprecated) | integer | |
| MySQL Table | | Delete Max Wait Time (Deprecated) | integer | |
| MySQL Table | | Full Rows Scanned (Deprecated) | integer | |
| MySQL Table | | I/O Read Requests (Deprecated) | integer | |
| MySQL Table | | I/O Read (Deprecated) | integer | |
| MySQL Table | | I/O Read Latency (Deprecated) | integer | |
| MySQL Table | | I/O Write Requests (Deprecated) | integer | |
| MySQL Table | | I/O Write (Deprecated) | integer | |
| MySQL Table | | I/O Write Latency (Deprecated) | integer | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Table | | I/O Misc Requests (Deprecated) | integer | |
| MySQL Table | | I/O Misc Latency (Deprecated) | integer | |
| MySQL Tablespace | Capacity | Current File Size | integer | The actual size of the file, which is the amount of space allocated on disk |
| MySQL Tablespace | Capacity | Maximum File Size | integer | The apparent size of the file, which represents the maximum size of the file, uncompressed |
| MySQL Tablespace | Details | Tablespace Name | string | The database and table name |
| MySQL Tablespace | User Configuration | File Format | string | The tablespace file format. The data in this field is interpreted from the tablespace flags information that resides in the .ibd file |
| MySQL Tablespace | User Configuration | Page Size | integer | The tablespace page size. The data in this field is interpreted from the tablespace flags information that resides in the .ibd file |
| MySQL Tablespace | User Configuration | Row Format | string | The tablespace row format (Compact or Redundant, Dynamic, or Compressed). The data in this field is interpreted from the tablespace flags information that resides in the .ibd file |
| MySQL Tablespace | User Configuration | Space Type | string | The type of tablespace. Possible values include General (for InnoDB general tablespaces created using CREATE TABLESPACE) and Single (for InnoDB file-per-table tablespaces) |
| MySQL Tablespace | User Configuration | Tablespace Flag Settings | integer | This value provides bit level information about tablespace format and storage characteristics |
| MySQL Tablespace | User Configuration | Zip Page Size | integer | The tablespace zip page size. The data in this field is interpreted from the tablespace flags information that resides in the .ibd file |
| MySQL Tablespace | Relationships | Instance Parents | string | List of parent Instance resources |
| MySQL Tablespace | Relationships | MySQL Tag Parents | string | List of parent MySQL Tag resources |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL Replication Slave | Details | Primary Server ID | string | The unique server ID of the Primary server that the slave server is replicating from |
| MySQL Replication Slave | Details | Slave UUID | string | The Universally Unique Identifier from the slave |
| MySQL Replication Slave | User Configuration | Primary Host | string | The host name of the slave server as specified on the slave with the --report-host option |
| MySQL Replication Slave | User Configuration | Primary Port | integer | The port on the Primary to which the slave server is listening, as specified on the slave with the --report-port option |
| MySQL Replication Slave | Relationships | Instance Parents | string | List of parent Instance resources |
| MySQL Replication Slave | Relationships | MySQL Tag Parents | string | List of parent MySQL Tag resources |
| MySQL MySQL Replication | Relationships | MySQL Tag Parents | string | List of parent MySQL Tag resources |
| MySQL MySQL Replication | | Slave Hostname (Deprecated) | string | |
| MySQL MySQL Replication | | Slave Port (Deprecated) | integer | |
| MySQL MySQL Replication | | Primary Host (Deprecated) | string | |
| MySQL MySQL Replication | | Primary User (Deprecated) | string | |
| MySQL MySQL Replication | | Primary Port (Deprecated) | integer | |
| MySQL MySQL Replication | | Connect Retry Time (Deprecated) | integer | |
| MySQL MySQL Replication | | Primary SSL Allowed (Deprecated) | string | |
| MySQL MySQL Replication | | Primary SSL Verify Server Cert (Deprecated) | string | |
| MySQL MySQL Replication | | Auto Position (Deprecated) | integer | |
| MySQL MySQL Replication | | Slave I/O State (Deprecated) | string | |
| MySQL MySQL Replication | | Slave I/O Running (Deprecated) | string | |
| MySQL MySQL Replication | | Slave SQL Running (Deprecated) | string | |
| MySQL MySQL Replication | | Last SQL Error Number (Deprecated) | integer | |
| MySQL MySQL Replication | | Last SQL Error (Deprecated) | string | |
| MySQL MySQL Replication | | Last I/O Error Number (Deprecated) | integer | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type | Attribute Description |
|---|---|---|---|---|
| MySQL MySQL Replication | | Last I/O Error (Deprecated) | string | |
| MySQL MySQL Replication | | SQL Delay (Deprecated) | string | |
| MySQL MySQL Replication | | Slave SQL Running State (Deprecated) | string | |
| MySQL MySQL Replication | | Primary Retry Count (Deprecated) | integer | |
| MySQL MySQL Replication | | Last I/O Error Time (Deprecated) | string | |
| MySQL MySQL Replication | | Last SQL Error Timestamp (Deprecated) | string | |

## Management Pack for Network Devices

The **Management Pack for Network Devices** is an embedded adapter for VMware Cloud Foundation Operations, collecting performance data from your Network systems and providing predictive analytics and real-time information about problems in your infrastructure—all within the VCF Operations user interface.

### Notable Features

The Management Pack harnesses powerful VCF Operations features such as:

- **Comprehensive monitoring dashboards**:
  - Networking Devices Overview
  - Networking Devices Investigation
  - Troubleshooting Networking Devices
  - Networking Devices Performance
  - Networking Devices Alerts
- **30+ number collected metrics** for Networking resources, including total throughput and out and in dropped packets
- **Detailed alerts and notifications**for several key events, including throughput total and in and out packet total
- **Key performance metrics** such as total throughput, and in and out dropped packets
- **External relationships** from the Host System to a Networking Device

### System Requirements Management Pack for Network Devices

Before installing and configuring the Management Pack, ensure your system meets the following requirements.

For compatibility between products, please refer to:

VMware Cloud Foundation Operations Interoperability for Network Devices

-OR-

Third Party Solution Interoperability for Network Devices

| | Network Devices Requirements |
|---|---|
| Connection | The default port used for this management pack is **161** |
| Credentials | The Management Pack requires one of the following credential types:<br>• **SNMP v2c read-only community string**<br>• **SNMP v3 User Name & Password** |

### Troubleshooting the Management Pack

Once you have finished Installing and Configuring the Management Pack , check the **Collection State** and **Collection Status** for the adapter instance on the **Solutions** page. If they show as **Not Collecting** or **No Data Receiving,** respectively, the following troubleshooting topics can help you diagnose and correct common problems with your adapter instance.

- Test Connection Errors
- Adapter and Collector Log Files

### Test Connection Errors

When Creating an Adapter Instance , click **Test Connection** after entering all of your configuration settings to ensure VCF Operations can connect to the specified host.

The following errors are possible:

**Unable to establish a valid connection to the target system**

**Error Message:** Unable to establish a valid connection to the target system. Could not connect to device(s): device1, device2.

**Description:** Check the following:

- The host and/or port are correct
- SNMP is enabled on the device
- The timeout allows sufficient time for data to be returned
- The community string or security parameters are correct
- A firewall isn't blocking UDP traffic on the specified port

**Unable to establish a valid connection to the target system**

**Error Message:** Unable to establish a valid connection to the target system. Could not resolve hostname for these device(s): device1

**Description:** Ensure that the hostname is correct.

### Adapter and Collector Log Files

You can review adapter errors in the adapter and collector log files, which can be viewed in the VCF Operations UI or in an external log viewer.

- **Adapter log files** are located in the $VCOPS_BASE/user/log/adapters/NetworkingDevicesAdapter/ folder.
- **Collector log files** are located in the $VCOPS_BASE/user/log/ folder.

To set the base log level for the collector in VCF Operations:

1. Navigate to **Administration > Logs**.
2. Expand the **PRIMARY** folder, then select the **COLLECTOR** folder.
3. Click the **Edit Properties**icon.
4. In the **Root logger level** drop-down menu, select the desired option.

- The default logging level is **warning.**
- To troubleshoot issues, set the logging level to **info.**
- To view detailed messages, including micro steps, queries, and returned results, set the logging level to debug.

  **CAUTION:**

  If you set the logging level to**debug**, log files can become large very quickly. Set the logging level to**debug**only for short periods of time.

For more information about viewing log files and modifying levels, refer to the VCF Operations online help.

### Using the Management Pack

The Management Pack for Network includes the following out-of-the-box features, harnessing the power of the VCF Operations user interface:

- Inventory Tree (Traversal Spec)
- Dashboards
- Views
- Reports
- Alerts
- Metrics


### Inventory Tree (Traversal Spec)

The Inventory Tree (Traversal Spec) feature in VCF Operations allows you to easily navigate your Network environment. The hierarchical structure implicitly shows relationships among resources and enables quick drill-downs to root-cause issues.

To access your Network Devices Inventory Tree, select **Environment** from the top navigation bar, then select **Networking Devices** from the left panel.

### Dashboards

Dashboards are the primary feature for monitoring and troubleshooting Network data problems from within VCF Operations.

The following dashboards are included in the Management Pack for Network Devices:

- **Networking Devices Overview** provides at-a-glance heatmaps depicting the overall health of your Networking devices.
- **Networking Devices Health Investigation** is the jumping-off point for examining the health of your networking environment. This dashboard displays at-a-glance health status for all resources in the Environment Overview widget. Selecting a resource from the widget will populate alerts and a health chart on the right.
- **Troubleshooting Network Devices** is the primary jumping-off point for troubleshooting specific network devices. Simply select a device from the Device widget to populate health and alert data about it.
- **Network Devices Performance**provides insight into the overall performance of your Networking Devices, including the top 10 devices by each of a variety of metrics.
- **Network Devices Alerts** shows all of the operational alerts specific to resources that have been discovered by the Adapter instances. Selecting an alert in the Alerts widget displays helpful information such as relationships and health data to allow you to more easily determine the origin and cause of the alert.

Additionally, the following dashboards are included for Cisco, Dell, HPE, and Lenovo:

- Device Summary
- Devices Traffic
- Networking Performance
- Networking Summary

### Views

The Management Pack for Network Devices creates views that allow the user to view statistics of metrics for various Network resources. The views help give a broad picture of the entire system, as opposed to a more in depth view.

The following views are available in the Management Pack:

| View | Type | Description |
|---|---|---|
| **Networking Devices Network Device Performance** | List | Displays performance of Network Devices |

To access the Management Pack views:

1. Navigate to **Environment > All Objects > Networking Devices**.
2. Double-click on the desired object (resource).
3. Select the **Details** tab, then **Views.**

The available views for that resource are listed and can be selected.

### Reports

The Management Pack for Network Devices contains out-of-the-box report(s) that you can export and easily share with key stakeholders in either .pdf or .csv formats.

The following reports are available in the Management Pack:

| Report | Description |
|---|---|
| **Networking Devices Network Device Performance** | Displays performance data for Network Devices |

To access Management Pack reports:

1. Navigate to **Environment > All Objects > Network Devices**.
2. Double-click on the desired object (resource).
3. Select the **Reports** tab, then **Report Templates.**

The available report(s) for the selected resource are listed and can be selected.

To run a selected report, click the **Run Template** icon.

### Alerts

The Management Pack pulls alerts and events from Network Devices and displays them in VCF Operations as alerts. See the table below for the list of alerts available in the Management Pack.

| Alert | Severity |
|---|---|
| Out Throughput Total | Immediate |
| Total Out Packets | Immediate |
| In Throughput Total | Immediate |
| Throughput Total | Immediate |
| Total In Packets | Immediate |

### Metrics

The Management Pack for Network Devices collects the metrics listed in the table below, according to Network Device resource kinds.

**CAUTION:**

The following list is generated with a standard MIB based on a large subset of devices; if any device is not using the standard MIB, some metrics may be unavailable.

| Vendors | Resource Type | Category | Metric | Description |
|---|---|---|---|---|
| All Vendors | entity | Details | Alias | The vendor-specific model name identifier string associated with this physical component. The preferred value is the customer-visible part number, which may be printed on the component itself. If the model name string associated with the physical component is unknown to the agent, then this object will contain a zero-length string. |

| | | | | |
|---|---|---|---|---|
| All Vendors | entity | Details | Asset ID | This object is a user-assigned asset tracking identifier for the physical entity as specified by a network manager, and provides non-volatile storage of this information. On the first instantiation of an physical entity, the value of entPhysicalAssetID associated with that entity is set to the zero-length string. Not every physical component will have a asset tracking identifier, or even need one. Physical entities for which the associated value of the entPhysicalIsFRU object is equal to 'false(2)' (e.g., the repeater ports within a repeater module), do not need their own unique asset tracking identifier. An agent does not have to provide write access for such entities, and may instead return a zero-length string. If write access is implemented for an instance of entPhysicalAssetID, and a value is written into the instance, the agent must retain the supplied value in the entPhysicalAssetID instance associated with the same physical entity for as long as that entity remains instantiated. This includes instantiations across all re- initializations/ reboots of the network management system, including those which result in a change of the physical entity's entPhysicalIndex value. If no asset tracking information is associated with the physical component, then this object will contain a zero-length string. |

| | | | | |
|---|---|---|---|---|
| All Vendors | entity | Details | Class | An indication of the general hardware type of the physical entity. An agent should set this object to the standard enumeration value which most accurately indicates the general class of the physical entity, or the primary class if there is more than one. If no appropriate standard registration identifier exists for this physical entity, then the value 'other(1)' is returned. If the value is unknown by this agent, then the value 'unknown(2)' is returned. |
| All Vendors | entity | Details | Contained In | The value of entPhysicalIndex for the physical entity which 'contains' this physical entity. A value of zero indicates this physical entity is not contained in any other physical entity. Note that the set of 'containment' relationships define a strict hierarchy; that is, recursion is not allowed. In the event a physical entity is contained by more than one physical entity (e.g., double-wide modules), this object should identify the containing entity with the lowest value of entPhysicalIndex. |
| All Vendors | entity | Details | Description | A textual description of physical entity. This object should contain a string which identifies the manufacturer's name for the physical entity, and should be set to a distinct value for each version or model of the physical entity. |

| | | | | |
|---|---|---|---|---|
| All Vendors | entity | Details | Firmware Revision | The vendor-specific firmware revision string for the physical entity. Note that if revision information is stored internally in a non-printable (e.g., binary) format, then the agent must convert such information to a printable format, in an implementation-specific manner. If no specific firmware programs are associated with the physical component, or this information is unknown to the agent, then this object will contain a zero-length string. |
| All Vendors | entity | Details | Hardware Revision | The vendor-specific hardware revision string for the physical entity. The preferred value is the hardware revision identifier actually printed on the component itself (if present). Note that if revision information is stored internally in a non-printable (e.g., binary) format, then the agent must convert such information to a printable format, in an implementation-specific manner. If no specific hardware revision string is associated with the physical component, or this information is unknown to the agent, then this object will contain a zero-length string. |
| All Vendors | entity | Details | Index | The index of the resource which uniquely identifies it on the SNMP tree. |

| | | | | |
|---|---|---|---|---|
| All Vendors | entity | Details | Is Field-Replaceable Unit | This object indicates whether or not this physical entity is considered a 'field replaceable unit' by the vendor. If this object contains the value 'true(1)' then this entPhysicalEntry identifies a field replaceable unit. For all entPhysicalEntries which represent components that are permanently contained within a field replaceable unit, the value 'false(2)' should be returned for this object. |
| All Vendors | entity | Details | Manufacturer Name | The name of the manufacturer of this physical component. The preferred value is the manufacturer name string actually printed on the component itself (if present). Note that comparisons between instances of the entPhysicalModelName, entPhysicalFirmwareRev, entPhysicalSoftwareRev, and the entPhysicalSerialNum objects, are only meaningful amongst entPhysicalEntries with the same value of entPhysicalMfgName. If the manufacturer name string associated with the physical component is unknown to the agent, then this object will contain a zero-length string. |
| All Vendors | entity | Details | Model Name | The vendor-specific model name identifier string associated with this physical component. The preferred value is the customer-visible part number, which may be printed on the component itself. If the model name string associated with the physical component is unknown to the agent, then this object will contain a zero-length string. |

| | | | | |
|---|---|---|---|---|
| All Vendors | entity | Details | Name | The textual name of the physical entity. The value of this object should be the name of the component as assigned by the local device and should be suitable for use in commands entered at the device's 'console'. This might be a text name, such as 'console' or a simple component number (e.g., port or module number), such as '1', depending on the physical component naming syntax of the device. If there is no local name, or this object is otherwise not applicable, then this object contains a zero-length string. Note that the value of entPhysicalName for two physical entities will be the same in the event that the console interface does not distinguish between them, e.g., slot-1 and the card in slot-1. |

| | | | | |
|---|---|---|---|---|
| All Vendors | entity | Details | Parent Relative Position | An indication of the relative position of this 'child' component among all its 'sibling' components. Sibling components are defined as entPhysicalEntries which share the same instance values of each of the entPhysicalContainedIn and entPhysicalClass objects. An NMS can use this object to identify the relative ordering for all sibling components of a particular parent (identified by the entPhysicalContainedIn instance in each sibling entry). This value should match any external labeling of the physical component if possible. For example, for a container (e.g., card slot) labeled as 'slot #3', entPhysicalParentRelPos should have the value '3'. Note that the entPhysicalEntry for the module plugged in slot 3 should have an entPhysicalParentRelPos value of '1'. If the physical position of this component does not match any external numbering or clearly visible ordering, then user documentation or other external reference material should be used to determine the parent-relative position. If this is not possible, then the the agent should assign a consistent (but possibly arbitrary) ordering to a given set of 'sibling' components, perhaps based on internal representation of the components. If the agent cannot determine the parent-relative position for some reason, or if the associated value of entPhysicalContainedIn is '0', then the value '-1' is returned. Otherwise a non-negative integer is returned, indicating the parent-relative position of this physical entity. Parent-relative ordering normally starts from '1' and continues to 'N', where 'N' represents the highest |

| | | | | |
|---|---|---|---|---|
| All Vendors | entity | Details | Serial Number | The vendor-specific serial number string for the physical entity. The preferred value is the serial number string actually printed on the component itself (if present). On the first instantiation of an physical entity, the value of entPhysicalSerialNum associated with that entity is set to the correct vendor-assigned serial number, if this information is available to the agent. If a serial number is unknown or non-existent, the entPhysicalSerialNum will be set to a zero-length string instead. Note that implementations which can correctly identify the serial numbers of all installed physical entities do not need to provide write access to the entPhysicalSerialNum object. Agents which cannot provide non-volatile storage for the entPhysicalSerialNum strings are not required to implement write access for this object. Not every physical component will have a serial number, or even need one. Physical entities for which the associated value of the entPhysicalIsFRU object is equal to 'false(2)' (e.g., the repeater ports within a repeater module), do not need their own unique serial number. An agent does not have to provide write access for such entities, and may return a zero-length string. If write access is implemented for an instance of entPhysicalSerialNum, and a value is written into the instance, the agent must retain the supplied value in the entPhysicalSerialNum instance associated with the same physical entity for as long as that entity remains instantiated. This includes instantiations across all re- initializations/ reboots of the network management system, including those which |

| | | | | |
|---|---|---|---|---|
| All Vendors | entity | Details | Software Revision | The vendor-specific software revision string for the physical entity. Note that if revision information is stored internally in a non-printable (e.g., binary) format, then the agent must convert such information to a printable format, in an implementation-specific manner. If no specific software programs are associated with the physical component, or this information is unknown to the agent, then this object will contain a zero-length string. |
| All Vendors | entity | Details | Vendor Type | An indication of the vendor-specific hardware type of the physical entity. Note that this is different from the definition of MIB-II's sysObjectID. An agent should set this object to a enterprise-specific registration identifier value indicating the specific equipment type in detail. The associated instance of entPhysicalClass is used to indicate the general type of hardware device. If no vendor-specific registration identifier exists for this physical entity, or the value is unknown by this agent, then the value { 0 0 } is returned. |

| | | | | |
|---|---|---|---|---|
| All Vendors | interface | Details | Alias | This object is an 'alias' name for the interface as specified by a network manager, and provides a non-volatile 'handle' for the interface. On the first instantiation of an interface, the value of ifAlias associated with that interface is the zero-length string. As and when a value is written into an instance of ifAlias through a network management set operation, then the agent must retain the supplied value in the ifAlias instance associated with the same interface for as long as that interface remains instantiated, including across all re-initializations/reboots of the network management system, including those which result in a change of the interface's ifIndex value. An example of the value which a network manager might store in this object for a WAN interface is the (Telco's) circuit number/ identifier of the interface. Some agents may support write-access only for interfaces having particular values of ifType. An agent which supports write access to this object is required to keep the value in non-volatile storage, but it may limit the length of new values depending on how much storage is already occupied by the current values for other interfaces. |
| All Vendors | interface | Details | Connector Present | This object has the value 'true(1)' if the interface sublayer has a physical connector and the value 'false(2)' otherwise. |

| | | | | |
|---|---|---|---|---|
| All Vendors | interface | Details | Description | A textual string containing information about the interface. This string should include the name of the manufacturer, the product name and the version of the interface hardware/software. |
| All Vendors | interface | Details | Index | A unique value, greater than zero, for each interface. It is recommended that values are assigned contiguously starting from 1. The value for each interface sub-layer must remain constant at least from one re-initialization of the entity's network management system to the next re- initialization. |
| All Vendors | interface | Details | Name | The textual name of the interface. The value of this object should be the name of the interface as assigned by the local device and should be suitable for use in commands entered at the device's 'console'. This might be a text name, such as 'le0' or a simple port number, such as '1', depending on the interface naming syntax of the device. If several entries in the ifTable together represent a single interface as named by the device, then each will have the same value of ifName. Note that for an agent which responds to SNMP queries concerning an interface on some other (proxied) device, then the value of ifName for such an interface is the proxied device's local name for it. If there is no local name, or this object is otherwise not applicable, then this object contains a zero-length string. |

| | | | | |
|---|---|---|---|---|
| All Vendors | interface | Details | Physical Address | The interface's address at its protocol sub-layer. For example, for an 802.x interface, this object normally contains a MAC address. The interface's media-specific MIB must define the bit and byte ordering and the format of the value of this object. For interfaces which do not have such an address (e.g., a serial line), this object should contain an octet string of zero length. |
| All Vendors | interface | Details | Promiscuous Mode | This object has a value of false(2) if this interface only accepts packets/frames that are addressed to this station. This object has a value of true(1) when the station accepts all packets/ frames transmitted on the media. The value true(1) is only legal on certain types of media. If legal, setting this object to a value of true(1) may require the interface to be reset before becoming effective. The value of ifPromiscuousMode does not affect the reception of broadcast and multicast packets/frames by the interface. |

| | | | | |
|---|---|---|---|---|
| All Vendors | interface | Details | Specific | A reference to MIB definitions specific to the particular media being used to realize the interface. It is recommended that this value point to an instance of a MIB object in the media-specific MIB, i.e., that this object have the semantics associated with the InstancePointer textual convention defined in RFC 2579. In fact, it is recommended that the media-specific MIB specify what value ifSpecific should/can take for values of ifType. If no MIB definitions specific to the particular media are available, the value should be set to the OBJECT IDENTIFIER { 0 0 }. |
| All Vendors | interface | Details | Type | The type of interface. Additional values for ifType are assigned by the Internet Assigned Numbers Authority (IANA), through updating the syntax of the IANAifType textual convention. |
| All Vendors | interface | Performance | Incoming Broadcast | The number of packets, delivered by this sub-layer to a higher (sub-)layer, which were addressed to a broadcast address at this sub-layer. Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime. |

| | | | | |
|---|---|---|---|---|
| All Vendors | interface | Performance | Incoming Discards | The number of inbound packets which were chosen to be discarded even though no errors had been detected to prevent their being deliverable to a higher-layer protocol. One possible reason for discarding such a packet could be to free up buffer space. Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime. |
| All Vendors | interface | Performance | Incoming Errors | For packet-oriented interfaces, the number of inbound packets that contained errors preventing them from being deliverable to a higher-layer protocol. For character- oriented or fixed-length interfaces, the number of inbound transmission units that contained errors preventing them from being deliverable to a higher-layer protocol. Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime. |
| All Vendors | interface | Performance | Incoming Multicast | The number of packets, delivered by this sub-layer to a higher (sub-)layer, which were addressed to a multicast address at this sub-layer. For a MAC layer protocol, this includes both Group and Functional addresses. Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime. |

| | | | | |
|---|---|---|---|---|
| All Vendors | interface | Performance | Incoming Non Unicast | The number of packets, delivered by this sub-layer to a higher (sub-)layer, which were addressed to a multicast or broadcast address at this sub-layer. Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime. This object is deprecated in favour of ifInMulticastPkts and ifInBroadcastPkts. |
| All Vendors | interface | Performance | Incoming Data | The total number of octets received on the interface, including framing characters. Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime. |
| All Vendors | interface | Performance | Incoming Unicast | The number of packets, delivered by this sub-layer to a higher (sub-)layer, which were not addressed to a multicast or broadcast address at this sub-layer. Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime. |

| | | | | |
|---|---|---|---|---|
| All Vendors | interface | Performance | Incoming Unknown Protocols | For packet-oriented interfaces, the number of packets received via the interface which were discarded because of an unknown or unsupported protocol. For character-oriented or fixed-length interfaces that support protocol multiplexing the number of transmission units received via the interface which were discarded because of an unknown or unsupported protocol. For any interface that does not support protocol multiplexing, this counter will always be 0. Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime. |
| All Vendors | interface | Performance | Outgoing Broadcast | The total number of packets that higher-level protocols requested be transmitted, and which were addressed to a broadcast address at this sub-layer, including those that were discarded or not sent. Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime. |
| All Vendors | interface | Performance | Outgoing Discards | The number of outbound packets which were chosen to be discarded even though no errors had been detected to prevent their being transmitted. One possible reason for discarding such a packet could be to free up buffer space. Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime. |

| | | | | |
|---|---|---|---|---|
| All Vendors | interface | Performance | Outgoing Errors | For packet-oriented interfaces, the number of outbound packets that could not be transmitted because of errors. For character-oriented or fixed-length interfaces, the number of outbound transmission units that could not be transmitted because of errors. Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime. |
| All Vendors | interface | Performance | Outgoing Multicast | The total number of packets that higher-level protocols requested be transmitted, and which were addressed to a multicast address at this sub-layer, including those that were discarded or not sent. For a MAC layer protocol, this includes both Group and Functional addresses. Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime. |
| All Vendors | interface | Performance | Outgoing Non Unicast | The total number of packets that higher-level protocols requested be transmitted, and which were addressed to a multicast or broadcast address at this sub-layer, including those that were discarded or not sent. Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime. This object is deprecated in favour of ifOutMulticastPkts and ifOutBroadcastPkts. |
| All Vendors | interface | Performance | Outgoing Queue Length | The length of the output packet queue. |

| | | | | |
|---|---|---|---|---|
| All Vendors | interface | Performance | Outgoing Data | The total number of octets transmitted out of the interface, including framing characters. Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime. |
| All Vendors | interface | Performance | Outgoing Unicast | The total number of packets that higher-level protocols requested be transmitted, and which were not addressed to a multicast or broadcast address at this sub-layer, including those that were discarded or not sent. Discontinuities in the value of this counter can occur at re-initialization of the management system, and at other times as indicated by the value of ifCounterDiscontinuityTime. |
| All Vendors | interface | Performance | Speed | An estimate of the interface's current bandwidth in units of 1,000,000 bits per second. If this object reports a value of 'n' then the speed of the interface is somewhere in the range of 'n-500,000' to 'n+499,999'. For interfaces which do not vary in bandwidth or for those where no accurate estimation can be made, this object should contain the nominal bandwidth. For a sub-layer which has no concept of bandwidth, this object should be zero. |
| All Vendors | interface | Status | Admin Status | Raw admin status value. |

| | | | | |
|---|---|---|---|---|
| All Vendors | interface | Status | Admin Status | The desired state of the interface. The testing(3) state indicates that no operational packets can be passed. When a managed system initializes, all interfaces start with ifAdminStatus in the down(2) state. As a result of either explicit management action or per configuration information retained by the managed system, ifAdminStatus is then changed to either the up(1) or testing(3) states (or remains in the down(2) state). |
| All Vendors | interface | Status | Last Change | The value of sysUpTime at the time the interface entered its current operational state. If the current state was entered prior to the last re-initialization of the local network management subsystem, then this object contains a zero value. |
| All Vendors | interface | Status | Operational Status | Raw operational status value. |
| All Vendors | interface | Status | Operational Status | The current operational state of the interface. The testing(3) state indicates that no operational packets can be passed. If ifAdminStatus is down(2) then ifOperStatus should be down(2). If ifAdminStatus is changed to up(1) then ifOperStatus should change to up(1) if the interface is ready to transmit and receive network traffic; it should change to dormant(5) if the interface is waiting for external actions (such as a serial line waiting for an incoming connection); it should remain in the down(2) state if and only if there is a fault that prevents it from going to the up(1) state; it should remain in the notPresent(6) state if the interface has missing (typically, hardware) components. |

| | | | | |
|---|---|---|---|---|
| All Vendors | interface | User Configuration | MTU | The size of the largest packet which can be sent/received on the interface, specified in octets. For interfaces that are used for transmitting network datagrams, this is the size of the largest network datagram that can be sent on the interface. |
| All Vendors | l3_interface | Network | IPv4 Broadcast Received Packets | The number of IP broadcast datagrams received. This metric applies to traffic over IPv4. |
| All Vendors | l3_interface | Network | IPv4 Broadcast Sent Packets | The number of IP broadcast datagrams sent. This metric applies to traffic over IPv4. |
| All Vendors | l3_interface | Network | IPv4 Forwards Received | The number of input datagrams for which this entity was not their final IP destination and for which this entity attempted to find a route to forward them to that final destination. This metric applies to traffic over IPv4. |
| All Vendors | l3_interface | Network | IPv4 Forwards Sent | The number of datagrams for which this entity was not their final IP destination and for which it was successful in finding a path to their final destination. This metric applies to traffic over IPv4. |
| All Vendors | l3_interface | Network | IPv4 Multicast Received Data | The total number of octets received in IP multicast datagram. This metric applies to traffic over IPv4. |
| All Vendors | l3_interface | Network | IPv4 Multicast Received Packets | The number of IP multicast datagrams received. This metric applies to traffic over IPv4. |
| All Vendors | l3_interface | Network | IPv4 Multicast Sent Data | The total number of octets transmitted in IP multicast datagram. This metric applies to traffic over IPv4. |
| All Vendors | l3_interface | Network | IPv4 Multicast Sent Packets | The number of IP multicast datagrams transmitted. This metric applies to traffic over IPv4. |

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|---|---|---|---|---|
| All Vendors | l3_interface | Network | IPv4 Packets Reassembled Correctly | The number of IP datagrams successfully reassembled. This metric applies to traffic over IPv4. |
| All Vendors | l3_interface | Network | IPv4 Packets Reassembly Failed | The number of failures detected by the IP re-assembly algorithm (for whatever reason:timed out, errors, etc.). This metric applies to traffic over IPv4. |
| All Vendors | l3_interface | Network | IPv4 Packets Requested for Reassembly | The number of IP fragments received that needed to be reassembled at this interface. This metric applies to traffic over IPv4. |
| All Vendors | l3_interface | Network | IPv4 Received Data | The total number of octets received in input IP datagrams, including those received in error. This metric applies to traffic over IPv4. |
| All Vendors | l3_interface | Network | IPv4 Received Packets | The total number of input IP datagrams received, including those received in error. This metric applies to traffic over IPv4. |
| All Vendors | l3_interface | Network | IPv4 Sent Data | The total number of octets in IP datagrams delivered to the lower layers for transmission. This metric applies to traffic over IPv4. |
| All Vendors | l3_interface | Network | IPv4 Sent Packets | The total number of IP datagrams that this entity supplied to the lower layers for transmission. This metric applies to traffic over IPv4. |
| All Vendors | l3_interface | Network | IPv6 Broadcast Received Packets | The number of IP broadcast datagrams received. This metric applies to traffic over IPv6. |
| All Vendors | l3_interface | Network | IPv6 Broadcast Sent Packets | The number of IP broadcast datagrams sent. This metric applies to traffic over IPv6. |

| | | | | |
|---|---|---|---|---|
| All Vendors | l3_interface | Network | IPv6 Forwards Received | The number of input datagrams for which this entity was not their final IP destination and for which this entity attempted to find a route to forward them to that final destination. This metric applies to traffic over IPv6. |
| All Vendors | l3_interface | Network | IPv6 Forwards Sent | The number of datagrams for which this entity was not their final IP destination and for which it was successful in finding a path to their final destination. This metric applies to traffic over IPv6. |
| All Vendors | l3_interface | Network | IPv6 Multicast Received Data | The total number of octets received in IP multicast datagram. This metric applies to traffic over IPv6. |
| All Vendors | l3_interface | Network | IPv6 Multicast Received Packets | The number of IP multicast datagrams received. This metric applies to traffic over IPv6. |
| All Vendors | l3_interface | Network | IPv6 Multicast Sent Data | The total number of octets transmitted in IP multicast datagram. This metric applies to traffic over IPv6. |
| All Vendors | l3_interface | Network | IPv6 Multicast Sent Packets | The number of IP multicast datagrams transmitted. This metric applies to traffic over IPv6. |
| All Vendors | l3_interface | Network | IPv6 Packets Reassembled Correctly | The number of IP datagrams successfully reassembled. This metric applies to traffic over IPv6. |
| All Vendors | l3_interface | Network | IPv6 Packets Reassembly Failed | The number of failures detected by the IP re-assembly algorithm (for whatever reason:timed out, errors, etc.). This metric applies to traffic over IPv6. |
| All Vendors | l3_interface | Network | IPv6 Packets Requested for Reassembly | The number of IP fragments received that needed to be reassembled at this interface. This metric applies to traffic over IPv6. |

| | | | | |
|---|---|---|---|---|
| All Vendors | l3_interface | Network | IPv6 Received Data | The total number of octets received in input IP datagrams, including those received in error. This metric applies to traffic over IPv6. |
| All Vendors | l3_interface | Network | IPv6 Received Packets | The total number of input IP datagrams received, including those received in error. This metric applies to traffic over IPv6. |
| All Vendors | l3_interface | Network | IPv6 Sent Data | The total number of octets in IP datagrams delivered to the lower layers for transmission. This metric applies to traffic over IPv6. |
| All Vendors | l3_interface | Network | IPv6 Sent Packets | The total number of IP datagrams that this entity supplied to the lower layers for transmission. This metric applies to traffic over IPv6. |
| All Vendors | l3_interface | Uncategorized | Index | The index of the resource which uniquely identifies it on the SNMP tree. |
| All Vendors | network_device | Details | Serial Number | The serial number as derived from the entity representing the module. |
| All Vendors | network_device | Details | System Contact | The textual identification of the contact person for this managed node, together with information on how to contact this person. If no contact information is known, the value is the zero-length string. |
| All Vendors | network_device | Details | System Description | A textual description of the entity. This value should include the full name and version identification of the system's hardware type, software operating-system, and networking software. |
| All Vendors | network_device | Details | System Location | The physical location of this node (e.g., 'telephone closet, 3rd floor'). If the location is unknown, the value is the zero-length string. |

| | | | | |
|---|---|---|---|---|
| All Vendors | network_device | Details | System Name | An administratively-assigned name for this managed node. By convention, this is the node's fully-qualified domain name. If the name is unknown, the value is the zero-length string. |
| All Vendors | network_device | Details | System Object ID | The vendor's authoritative identification of the network management subsystem contained in the entity. This value is allocated within the SMI enterprises subtree (1.3.6.1.4.1) and provides an easy and unambiguous means for determining 'what kind of box' is being managed. For example, if vendor 'Flintstones, Inc.' was assigned the subtree 1.3.6.1.4.1.424242, it could assign the identifier 1.3.6.1.4.1.424242.1.1 to its 'Fred Router'. |
| All Vendors | network_device | Details | System Uptime | The time since the network management porition of the system was last re-initialized. |
| All Vendors | network_device | Fan | Total Fan Count | The total number of Fan on this device. |
| All Vendors | network_device | Interface | Total Interface Count | The total number of interfaces on this device. |
| All Vendors | network_device | Module | Total Module Count | The total number of Module on this device. |
| All Vendors | network_device | Power Supply | Total Power Supply Count | The total number of Power Supply on this device. |
| All Vendors | network_device | Sensor | Total Sensor Count | The total number of Sensor on this device. |
| Cisco | hsrp_group | Details | Active Router | IP Address of the currently active router for this group. |
| Cisco | hsrp_group | Details | Group Address | The primary virtual IP address used by this group. |
| Cisco | hsrp_group | Details | Group Number | The number identifying this group. |
| Cisco | hsrp_group | Details | MAC Address | The MAC address assigned to this group. |
| Cisco | hsrp_group | Details | Standby Router | IP Address of the currently standby router for this group. |

| | | | | |
|---|---|---|---|---|
| Cisco | hsrp_group_member | Details | Active Router | IP Address of the currently active router for this group. |
| Cisco | hsrp_group_member | Details | Group Address | This is the primary virtual IP address used by this group. If this address is configured (i.e a non zero ip address), this value is used. Otherwise, the agent will attempt to discover the virtual address through a discovery process (which scans the hello messages). |
| Cisco | hsrp_group_member | Details | Group MAC Address | The MAC Address of the group on which this HSRP Group Member resides. |
| Cisco | hsrp_group_member | Details | Hello Time | If the Hellotime is not configured on a router, it can be learned from the Hello messages from active router, provided the Hello message is authenticated. If the Hellotime is not learned from a Hello message from the active router and it is not manually configured, a default value of 3 seconds is recommended. |
| Cisco | hsrp_group_member | Details | Hold Time | If the Holdtime is not configured on a router, it can be learned from the Hello message from the active router. Holdtime should be learned only if the Hello message is authenticated. If the Holdtime is not learned and it is not manually configured, a default value of 10 seconds is recommended. |
| Cisco | hsrp_group_member | Details | Preemption Delay | This delay is the time difference between a router power up and the time it can actually start preempting the currently active router. |

| | | | | |
|---|---|---|---|---|
| Cisco | hsrp_group_member | Details | Priority | The priority helps to select the active and the standby routers. The router with the highest priority is selected as the active router. In the priority range of 0 to 255, 0 is the lowest priority and 255 is the highest priority. |
| Cisco | hsrp_group_member | Details | Standby Router | IP Address of the currently standby router for this group. |
| Cisco | hsrp_group_member | Status | State | The current HSRP state of this group on this interface. |
| Cisco | hsrp_group_member | Uncategorized | Group Number | The group to which this HSRP Group Member belongs. |
| Cisco | hsrp_group_member | Uncategorized | Index | The index of the resource which uniquely identifies it on the SNMP tree. |
| Cisco | interface | Details | Connected WWNN | The world wide name of the Remote Node the logged-in Nx_Port belongs to. |
| Cisco | interface | Details | Connected WWPN | The world wide name of the logged-in Nx_Port. |
| Cisco | interface | Details | VSAN | Index of the VSAN to which this interface is statically assigned. |
| Cisco | interface | Network | Attached CDP IP Addresses | The (first) network-layer address of the device as reported in the Address TLV of the most recently received CDP message. For example, if the corresponding instance of cacheAddressType had the value 'ip(1)', then this object would be an IPv4-address. If the neighbor device is SNMP-manageable, it is supposed to generate its CDP messages such that this address is one at which it will receive SNMP messages. Use cdpCtAddressTable to extract the remaining addresses from the Address TLV received most recently. |
| Cisco | network_device | Memory | Memory Utilization | The average utilization of memory on the active supervisor. |

| | | | | |
|---|---|---|---|---|
| Cisco | network_device | Module | Module Critical | The total number of Module on this device that are in the 'critical' state. |
| Cisco | network_device | Module | Module Down | The total number of Module on this device that are in the 'down' state. |
| Cisco | network_device | Module | Module Normal | The total number of Module on this device that are in the 'normal' state. |
| Cisco | network_device | Module | Module Not Functioning | The total number of Module on this device that are in the 'notFunctioning' state. |
| Cisco | network_device | Module | Module Not Present | The total number of Module on this device that are in the 'notPresent' state. |
| Cisco | network_device | Module | Module Shutdown | The total number of Module on this device that are in the 'shutdown' state. |
| Cisco | network_device | Module | Module Unknown | The total number of Module on this device that are in the 'unknown' state. |
| Cisco | network_device | Module | Module Up | The total number of Module on this device that are in the 'up' state. |
| Cisco | network_device | Module | Module Warning | The total number of Module on this device that are in the 'warning' state. |
| Cisco | network_device | Network | Attached CDP IP Addresses | CDP IP Addresses attached to interfaces on this device. |
| Cisco | vsan | Details | In Order Delivery | The InorderDelivery guarantee flag for the local switch on this VSAN. |
| Cisco | vsan | Details | Interoperability Mode | The interoperability mode of the local switch on this VSAN. This object indicates if this device can inter-operate with devices from other vendors on this VSAN. |
| Cisco | vsan | Details | Interoperability Value | This object is used for configuring the interoperability value of the VSAN, to inter-operate with different vendor devices. |
| Cisco | vsan | Details | Load Balancing Type | The type of load balancing used on this VSAN. |

| | | | | |
|---|---|---|---|---|
| Cisco | vsan | Details | Media Type | The network media type on which this VSAN is defined. |
| Cisco | vsan | Details | MTU | The MTU of the VSAN. Normally, this object is 2112 for all VSANs for which object vsanMediaType has the value 'fibreChannel'. |
| Cisco | vsan | Details | Name | The name of the VSAN. |
| Cisco | vsan | Performance | Network Drop Latency | Network latency of this switch on this VSAN. This is the time interval after which the frames are dropped if they are not delivered in the order they were transmitted. |
| Cisco | vsan | Status | Admin State | Represents the configurable state of this VSAN. |
| Cisco | vsan | Status | Operational State | The operational state of the VSAN. Indicates whether traffic can pass thru this VSAN. |
| Cisco | vsan | Status | Row Status | The status of this conceptual row. |
| Cisco | vsan | Uncategorized | Index | The index of the resource which uniquely identifies it on the SNMP tree. |
| Cisco, Dell | network_device | CPU | CPU Utilization | The average utilization of CPU on the active supervisor. |
| Cisco, Dell, Radlan | entity | Status | State | The operational state of the entity. |
| Cisco,Dell | network_device | Fan | Fan Critical | The total number of Fan on this device that are in the 'critical' state. |
| Cisco,Dell | network_device | Fan | Fan Critical Ratio | The ratio of Fan on this device that are in the 'critical' state. |
| Cisco,Dell | network_device | Fan | Fan Down | The total number of Fan on this device that are in the 'down' state. |
| Cisco,Dell | network_device | Fan | Fan Down Ratio | The ratio of Fan on this device that are in the 'down' state. |
| Cisco,Dell | network_device | Fan | Fan Normal | The total number of Fan on this device that are in the 'normal' state. |
| Cisco,Dell | network_device | Fan | Fan Normal Ratio | The ratio of Fan on this device that are in the 'normal' state. |
| Cisco,Dell | network_device | Fan | Fan Not Functioning | The total number of Fan on this device that are in the 'notFunctioning' state. |

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|---|---|---|---|---|
| Cisco,Dell | network_device | Fan | Fan Not Functioning Ratio | The ratio of Fan on this device that are in the 'notFunctioning' state. |
| Cisco,Dell | network_device | Fan | Fan Not Present | The total number of Fan on this device that are in the 'notPresent' state. |
| Cisco,Dell | network_device | Fan | Fan Not Present Ratio | The ratio of Fan on this device that are in the 'notPresent' state. |
| Cisco,Dell | network_device | Fan | Fan Shutdown | The total number of Fan on this device that are in the 'shutdown' state. |
| Cisco,Dell | network_device | Fan | Fan Shutdown Ratio | The ratio of Fan on this device that are in the 'shutdown' state. |
| Cisco,Dell | network_device | Fan | Fan Unknown | The total number of Fan on this device that are in the 'unknown' state. |
| Cisco,Dell | network_device | Fan | Fan Unknown Ratio | The ratio of Fan on this device that are in the 'unknown' state. |
| Cisco,Dell | network_device | Fan | Fan Up | The total number of Fan on this device that are in the 'up' state. |
| Cisco,Dell | network_device | Fan | Fan Up Ratio | The ratio of Fan on this device that are in the 'up' state. |
| Cisco,Dell | network_device | Fan | Fan Warning | The total number of Fan on this device that are in the 'warning' state. |
| Cisco,Dell | network_device | Fan | Fan Warning Ratio | The ratio of Fan on this device that are in the 'warning' state. |
| Cisco,Radlan | network_device | Sensor | Sensor Non-Operational | The total number of Sensor on this device that are in the 'nonoperational' state. |
| Cisco,Radlan | network_device | Sensor | Sensor Non-Operational Ratio | The ratio of Sensor on this device that are in the 'nonoperational' state. |
| Cisco,Radlan | network_device | Sensor | Sensor Ok | The total number of Sensor on this device that are in the 'ok' state. |
| Cisco,Radlan | network_device | Sensor | Sensor Ok Ratio | The ratio of Sensor on this device that are in the 'ok' state. |
| Cisco,Radlan | network_device | Sensor | Sensor Unavailable | The total number of Sensor on this device that are in the 'unavailable' state. |
| Cisco,Radlan | network_device | Sensor | Sensor Unavailable Ratio | The ratio of Sensor on this device that are in the 'unavailable' state. |
| Dell | chassis | Details | Asset Tag | The asset tag of the product. |

| | | | | |
|---|---|---|---|---|
| Dell | chassis | Details | Index | The index of the resource which uniquely identifies it on the SNMP tree. |
| Dell | chassis | Details | Serial Number | The serial number of the product. |
| Dell | chassis | Details | Service Tag | The service tag of the product. |
| Dell | network_device | Details | Product Description | A short description of this product such as 'Ethernet Router Switch'. |
| Dell | network_device | Details | Product Version | The version of the product. |
| Dell | network_device | Memory | Memory Free | The total memory free for utilization. |
| Dell | network_device | Memory | Total Memory Available | The total memory available. |
| Dell | network_device | Power Supply | Power Supply Critical | The total number of Power Supply on this device that are in the 'critical' state. |
| Dell | network_device | Power Supply | Power Supply Critical Ratio | The ratio of Power Supply on this device that are in the 'critical' state. |
| Dell | network_device | Power Supply | Power Supply Normal | The total number of Power Supply on this device that are in the 'normal' state. |
| Dell | network_device | Power Supply | Power Supply Normal Ratio | The ratio of Power Supply on this device that are in the 'normal' state. |
| Dell | network_device | Power Supply | Power Supply Not Functioning | The total number of Power Supply on this device that are in the 'notFunctioning' state. |
| Dell | network_device | Power Supply | Power Supply Not Functioning Ratio | The ratio of Power Supply on this device that are in the 'notFunctioning' state. |
| Dell | network_device | Power Supply | Power Supply Not Present | The total number of Power Supply on this device that are in the 'notPresent' state. |
| Dell | network_device | Power Supply | Power Supply Not Present Ratio | The ratio of Power Supply on this device that are in the 'notPresent' state. |
| Dell | network_device | Power Supply | Power Supply Shutdown | The total number of Power Supply on this device that are in the 'shutdown' state. |
| Dell | network_device | Power Supply | Power Supply Shutdown Ratio | The ratio of Power Supply on this device that are in the 'shutdown' state. |

| | | | | |
|---|---|---|---|---|
| Dell | network_device | Power Supply | Power Supply Warning | The total number of Power Supply on this device that are in the 'warning' state. |
| Dell | network_device | Power Supply | Power Supply Warning Ratio | The ratio of Power Supply on this device that are in the 'warning' state. |
| Dell | network_device | Status | Global Status | Current status of the product. This is a rollup for the entire product. The status is intended to give initiative to a snmp monitor to get further data when this status is abnormal. |
| N/A | chassis | Details | Internal Vendor | The vendor(s) that our product is assigned based on the MiBs it can collect. |
| N/A | chassis | Details | System Name | An administratively-assigned name for this managed node. By convention, this is the node's fully-qualified domain name. If the name is unknown, the value is the zero-length string. |
| N/A | chassis | Details | System Object ID | The vendor's authoritative identification of the network management subsystem contained in the entity. This value is allocated within the SMI enterprises subtree (1.3.6.1.4.1) and provides an easy and unambiguous means for determining 'what kind of box' is being managed. For example, if vendor 'Flintstones, Inc.' was assigned the subtree 1.3.6.1.4.1.424242, it could assign the identifier 1.3.6.1.4.1.424242.1.1 to its 'Fred Router'. |
| N/A | entity | Details | Internal Vendor | The vendor(s) that our product is assigned based on the MiBs it can collect. |

| | | | | |
|---|---|---|---|---|
| N/A | entity | Details | System Name | An administratively-assigned name for this managed node. By convention, this is the node's fully-qualified domain name. If the name is unknown, the value is the zero-length string. |
| N/A | entity | Details | System Object ID | The vendor's authoritative identification of the network management subsystem contained in the entity. This value is allocated within the SMI enterprises subtree (1.3.6.1.4.1) and provides an easy and unambiguous means for determining 'what kind of box' is being managed. For example, if vendor 'Flintstones, Inc.' was assigned the subtree 1.3.6.1.4.1.424242, it could assign the identifier 1.3.6.1.4.1.424242.1.1 to its 'Fred Router'. |
| N/A | hsrp_group | Details | Internal Vendor | The vendor(s) that our product is assigned based on the MiBs it can collect. |
| N/A | hsrp_group_member | Details | Internal Vendor | The vendor(s) that our product is assigned based on the MiBs it can collect. |
| N/A | hsrp_group_member | Details | System Name | An administratively-assigned name for this managed node. By convention, this is the node's fully-qualified domain name. If the name is unknown, the value is the zero-length string. |

| | | | | |
|---|---|---|---|---|
| N/A | hsrp_group_member | Details | System Object ID | The vendor's authoritative identification of the network management subsystem contained in the entity. This value is allocated within the SMI enterprises subtree (1.3.6.1.4.1) and provides an easy and unambiguous means for determining 'what kind of box' is being managed. For example, if vendor 'Flintstones, Inc.' was assigned the subtree 1.3.6.1.4.1.424242, it could assign the identifier 1.3.6.1.4.1.424242.1.1 to its 'Fred Router'. |
| N/A | interface | Details | Internal Vendor | The vendor(s) that our product is assigned based on the MiBs it can collect. |
| N/A | interface | Details | System Name | An administratively-assigned name for this managed node. By convention, this is the node's fully-qualified domain name. If the name is unknown, the value is the zero-length string. |
| N/A | interface | Details | System Object ID | The vendor's authoritative identification of the network management subsystem contained in the entity. This value is allocated within the SMI enterprises subtree (1.3.6.1.4.1) and provides an easy and unambiguous means for determining 'what kind of box' is being managed. For example, if vendor 'Flintstones, Inc.' was assigned the subtree 1.3.6.1.4.1.424242, it could assign the identifier 1.3.6.1.4.1.424242.1.1 to its 'Fred Router'. |
| N/A | l3_interface | Details | Internal Vendor | The vendor(s) that our product is assigned based on the MiBs it can collect. |

| | | | | |
|---|---|---|---|---|
| N/A | l3_interface | Details | System Name | An administratively-assigned name for this managed node. By convention, this is the node's fully-qualified domain name. If the name is unknown, the value is the zero-length string. |
| N/A | l3_interface | Details | System Object ID | The vendor's authoritative identification of the network management subsystem contained in the entity. This value is allocated within the SMI enterprises subtree (1.3.6.1.4.1) and provides an easy and unambiguous means for determining 'what kind of box' is being managed. For example, if vendor 'Flintstones, Inc.' was assigned the subtree 1.3.6.1.4.1.424242, it could assign the identifier 1.3.6.1.4.1.424242.1.1 to its 'Fred Router'. |
| N/A | network_device | Details | Internal Vendor | The vendor(s) that our product is assigned based on the MiBs it can collect. |
| N/A | network_device | Network | Connected IP Addresses | IP Addresses of devices connected. |
| N/A | network_device | Network | IP Address | IP Address of this device. |
| N/A | vsan | Details | Internal Vendor | The vendor(s) that our product is assigned based on the MiBs it can collect. |
| N/A | vsan | Details | System Name | An administratively-assigned name for this managed node. By convention, this is the node's fully-qualified domain name. If the name is unknown, the value is the zero-length string. |

| | | | | |
|---|---|---|---|---|
| N/A | vsan | Details | System Object ID | The vendor's authoritative identification of the network management subsystem contained in the entity. This value is allocated within the SMI enterprises subtree (1.3.6.1.4.1) and provides an easy and unambiguous means for determining 'what kind of box' is being managed. For example, if vendor 'Flintstones, Inc.' was assigned the subtree 1.3.6.1.4.1.424242, it could assign the identifier 1.3.6.1.4.1.424242.1.1 to its 'Fred Router'. |
| Radlan | interface | Details | Admin Duplex Mode | This variable specifies whether this interface should operate in half duplex or full duplex mode. This specification will take effect only if swIfSpeedDuplexAutoNegotiation is disabled. A value of 'none' is returned if a value of the variable hasn't been set. |
| Radlan | interface | Details | Default VLAN Tag | This variable specifies the default VLAN tag which will be attached to outgoing frames if swIfTaggedMode for this interface is enabled. |
| Radlan | interface | Details | Duplex Mode | This variable indicates whether this interface operates in half duplex or full duplex mode. This variable can have the values hybrid or unknown only for a trunk. unknown - only if trunk operative status is not present. |

## Management Pack for Oracle Database

The **Management Pack for Oracle Database** is an embedded adapter for VMware Cloud Foundation Operations, collecting performance and capacity data from your Oracle Database environment and providing predictive analytics and real-time information about problems in your infrastructure—all within the VCF Operations user interface.

### Notable Features

The Management Pack for Oracle Database provides the following notable features:

- **Comprehensive monitoring dashboards**

- **Key performance metrics** such as I/O Throughput, Average I/O Time, Row Cache Hit Ratio, and Buffer Cache Hit Ratio
- **Out-of-the-box reports** for Oracle Database Health, Tablespace, Instance Session Capacity, and more
- **Relationship mapping** between Oracle Database and the virtual layer
- **Detailed alerts and recommendations** to help keep you on top of issues

**See also:**

System Requirements

### System Requirements for Management Pack for Oracle Database

Before installing and configuring the Management Pack, ensure your system meets the following requirements.

For compatibility between products, please refer to:

VMware Cloud Foundation Operations Interoperability for Oracle Database

-OR-

Third Party Solution Interoperability for Oracle Database

> **Restriction:**
>
> Oracle DB Management Pack does not support FIPS mode.

| | Oracle Database Requirements |
|---|---|
| Connection | **Oracle Database instance** hostname or IP address<br>**SID/Service Name** required to connect to the instance<br>**Default Ports**:<br>• **Non-SSL**: 1521 (TCP)<br>• **SSL:** 2484 (TCP) |
| Credentials | **Oracle Database User Name** and **Password**<br>**Note:**<br>For details on the minimum user privileges that must be granted for your monitoring user, see: Configuring a Least-Privileged User (LPU). |

Configuring a Least-Privileged User (LPU)

This topic contains the minimum Oracle Database user privileges required for creating a least-privileged user (LPU) for the management pack, as well as a script to help automate its creation.

#### Creating a Oracle Database Monitoring User

The following script can be used to help automate the creation of your least-privileged user. It includes the minimum Oracle Database user privileges required for the management pack.

To grant all necessary access to the LPU run the following queries.

> **Note:**
>
> For multitenant databases, a "common user" must be provisioned, which requires connection to the CDB root. The specified username must be prefixed with `C##`. The `CONTAINER_DATA` attribute must also be set after the user has been created, e.g.
>
> ```
> ALTER USER c##monitoring_user SET CONTAINER_DATA=ALL CONTAINER=CURRENT;
> ```

```
CREATE USER LPU IDENTIFIED BY LPU_PASSWORD;
GRANT CONNECT TO LPU;
GRANT CREATE SESSION TO LPU;
GRANT SELECT ON CDB_DATA_FILES TO LPU;
```

```
GRANT SELECT ON CDB_OUTSTANDING_ALERTS TO LPU;
GRANT SELECT ON CDB_TABLESPACES TO LPU;
GRANT SELECT ON CDB_TABLESPACE_USAGE_METRICS TO LPU;
GRANT SELECT ON DBA_DATA_FILES TO LPU;
GRANT SELECT ON DBA_FREE_SPACE TO LPU;
GRANT SELECT ON DBA_OUTSTANDING_ALERTS TO LPU;
GRANT SELECT ON DBA_TABLESPACES TO LPU;
GRANT SELECT ON DBA_TABLESPACE_USAGE_METRICS TO LPU;
GRANT SELECT ON DBA_TEMP_FILES TO LPU;
GRANT SELECT ON DBA_TEMP_FREE_SPACE TO LPU;
GRANT SELECT ON DBA_UNDO_EXTENTS TO LPU;
GRANT SELECT ON DUAL TO LPU;
GRANT SELECT ON GLOBAL_NAME TO LPU;
GRANT SELECT ON GV_$ARCHIVED_LOG TO LPU;
GRANT SELECT ON GV_$ARCHIVE_DEST TO LPU;
GRANT SELECT ON GV_$ASM_DISK_STAT TO LPU;
GRANT SELECT ON GV_$CON_SYSSTAT TO LPU ;
GRANT SELECT ON GV_$CON_SYSSTAT TO LPU;
GRANT SELECT ON GV_$FILESTAT TO LPU;
GRANT SELECT ON GV_$INSTANCE TO LPU;
GRANT SELECT ON GV_$PARAMETER TO LPU;
GRANT SELECT ON GV_$PDBS TO LPU;
GRANT SELECT ON GV_$RESOURCE_LIMIT TO LPU;
GRANT SELECT ON GV_$SERVICE_STATS TO LPU;
GRANT SELECT ON GV_$SESSION TO LPU;
GRANT SELECT ON GV_$SESSION_LONGOPS TO LPU;
GRANT SELECT ON GV_$SGAINFO TO LPU;
GRANT SELECT ON GV_$SQL TO LPU;
GRANT SELECT ON GV_$SYSMETRIC TO LPU;
GRANT SELECT ON GV_$SYSSTAT TO LPU;
GRANT SELECT ON GV_$SYSTEM_EVENT TO LPU;
GRANT SELECT ON GV_$SYSTEM_WAIT_CLASS TO LPU;
GRANT SELECT ON GV_$TEMPSTAT TO LPU;
GRANT SELECT ON V_$ARCHIVE_DEST TO LPU;
GRANT SELECT ON V_$CONTROLFILE TO LPU;
GRANT SELECT ON V_$DATABASE TO LPU;
GRANT SELECT ON V_$DATAFILE TO LPU;
GRANT SELECT ON V_$DB_CACHE_ADVICE TO LPU;
GRANT SELECT ON V_$FILESTAT TO LPU;
GRANT SELECT ON V_$INSTANCE TO LPU;
GRANT SELECT ON V_$LOG TO LPU;
GRANT SELECT ON V_$LOGFILE TO LPU;
GRANT SELECT ON V_$MEMORY_TARGET_ADVICE TO LPU;
GRANT SELECT ON V_$PARAMETER TO LPU;
GRANT SELECT ON V_$PGA_TARGET_ADVICE TO LPU;
GRANT SELECT ON V_$RECOVERY_AREA_USAGE TO LPU;
GRANT SELECT ON V_$RESOURCE_LIMIT TO LPU;
GRANT SELECT ON V_$RMAN_BACKUP_JOB_DETAILS TO LPU;
GRANT SELECT ON V_$SGA_TARGET_ADVICE TO LPU;
GRANT SELECT ON V_$SHARED_POOL_ADVICE TO LPU;
GRANT SELECT ON V_$TEMPFILE TO LPU;
GRANT SELECT ON V_$VERSION TO LPU;
```

### Troubleshooting the Management Pack

Once you have finished Installing and Configuring the Management Pack , check the **Collection State** and **Collection Status** for the adapter instance on the **Solutions** page. If they show as **Not Collecting** or **No Data Receiving**,

respectively, the following troubleshooting topics can help you diagnose and correct common problems with your adapter instance.

- Test Connection Errors
- Adapter and Collector Log Files

### Test Connection Errors

When Creating an Adapter Instance , click **Test Connection** after entering all of your configuration settings to ensure VMware Cloud Foundation Operations can connect to the specified host.

The following errors are possible:

User Authentication Error

**Error Message**: Unable to establish a valid connection to target system. User Authentication error.

**Description:** Ensure the correct Oracle Database User Name and Password were entered when Creating a Credential .

Host, port, or SSL settings are incorrect

**Error Message**: Host, port, or SSL settings are incorrect.

**Description:** Ensure the correct hostname, port number, and SSL settings for the Oracle Database instance were entered when adding your adapter instance. Ensure you are using the default port that corresponds to the applicable SSL settings (non-SSL = 1521; SSL = 2484).

SID/Service Name is invalid

**Error Message**: The given SID/Service Name is invalid.

**Description:** Check your Oracle Database database system and ensure your SID/Service name is the same as the one entered for your adapter instance.

Invalid character(s) found in SID/Service Name

**Error Message**: Invalid character(s) found in SID/Service Name.

**Description:** Make sure your SID/Service Name contains only alpha-numberic characters.

SSL RC4 Cipher Suites compromised

**Error Message**: SSL RC4 Cipher Suites are considered compromised (and disabled) with Java 1.8.0_60+.

**Description**: Configure the Oracle Database to not use SSL RC4 Cipher Suites. To configure valid SSL Cipher Suites for an Oracle database either use the netca utility installed with Oracle Database, or edit the sqlnet.ora file.

**Error Message**: ORA-00942 Table or view does not exist

**Description**: Database user may be missing privileges required for collecting data from this table. Please review the user privilages and compare them with the Least-Privileged User (LPU) recommendations.

### Adapter and Collector Log Files

You can review adapter errors in the adapter and collector log files, which can be viewed in the VMware Cloud Foundation Operations UI or in an external log viewer.

- **Adapter log files** are located in the $VCOPS_BASE/user/log/adapters/OracleDatabaseAdapter/ folder.
- **Collector log files** are located in the $VCOPS_BASE/user/log/ folder.

To set the base log level for the collector in VMware Cloud Foundation Operations:

1. Navigate to **Administration > Logs**.
2. Expand the **PRIMARY** folder, then select the **COLLECTOR** folder.
3. Click the **Edit Properties**icon.
4. In the **Root logger level** drop-down menu, select the desired option:
   – The default logging level is **warning.**

- To troubleshoot issues, set the logging level to **info.**
- To view detailed messages, including micro steps, queries, and returned results, set the logging level to **debug.**

  **Note:**

  If you set the logging level to **debug,** log files can become large very quickly. Set the logging level to **debug** only for short periods of time.

For more information about viewing log files and modifying levels, refer to the VMware Cloud Foundation Operations online help.

### Using the Management Pack

The Management Pack for Oracle Database includes the following out-of-the-box features, harnessing the power of the VCF Operations user interface:

- Resource Kinds and Relationships
- Inventory Tree (Traversal Spec)
- Dashboards
- Views
- Reports
- Analysis Badges and Capacity Definitions
- Alerts
- Metrics

#### Inventory Tree (Traversal Spec)

The Inventory Tree (Traversal Spec) feature in VCF Operations allows you to easily navigate your Oracle Database environment. The hierarchical structure implicitly shows relationships among resources and enables quick drill-downs to root-cause issues.

To access your Oracle Database Inventory Tree, select **Environment** from the top navigation bar, then select **Oracle Database** from the left panel. Click on a resource from the Inventory Tree to populate details about the resource on the right.

#### Resource Kinds and Relationships

The Management Pack for Oracle Database creates Resource Kinds and Relationships (associations) for critical resources in your Oracle Database environment.

**Resource Kinds**

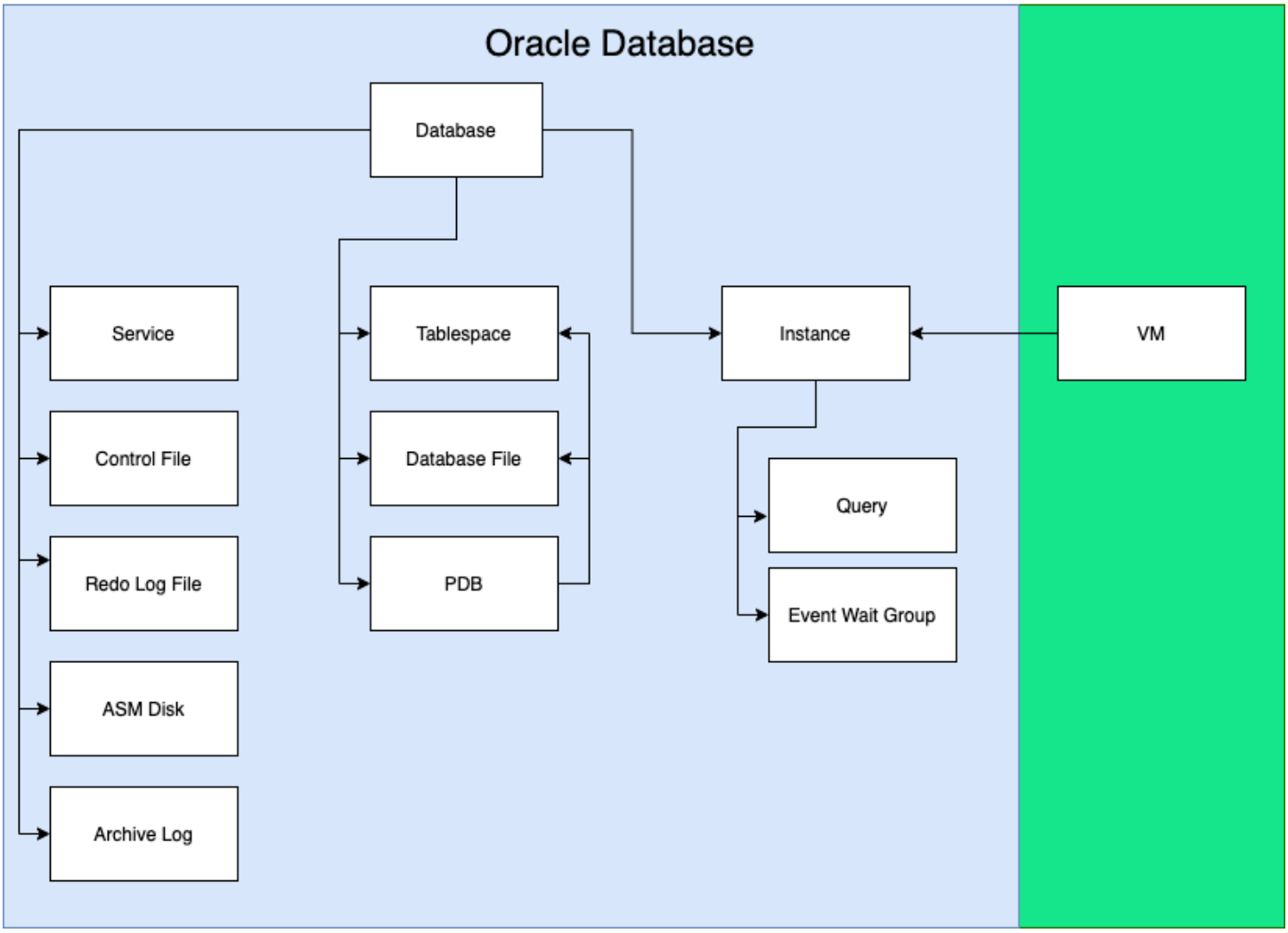
Performance, health, and availability data are collected for the following Oracle Database resource kinds:

1. Oracle Database Archive Log
2. Oracle Database ASM Disk
3. Oracle Database Control File
4. Oracle Database Database
5. Oracle Database Database File
6. Oracle Database Event Wait Group
7. Oracle Database Instance
8. Oracle Database PDB
9. Oracle Database Query
10. Oracle Database Redo Log File
11. Oracle Database Service
12. Oracle Database Tablespace

## Relationships

Relationships among discovered resources are also created by the Management Pack. The hierarchical structure of those relationships is depicted in the diagram below.

**Oracle Database Relationships Diagram**




## External Relationships

The Management Pack for Oracle Database also discovers the following external relationships:

- Oracle Database Instance > VMware Virtual Machine
- Oracle Database Instance > EP Ops

### Dashboards

Dashboards are the primary feature for monitoring and troubleshooting Oracle Database problems from within VCF Operations.

The following dashboards are included in the Management Pack for Oracle Database :

- **Oracle Database Database Drill Down**
- **Oracle Database Infrastructure Drill Down**

### Views

The Management Pack for Oracle Database creates views that allow the user to view statistics of metrics for Oracle Database resources. The views help give a broad picture of the entire system, as opposed to a more in depth view.

The following views are available in the Management Pack:

| View | Type | Description |
|---|---|---|
| **Oracle Database Configuration** | List | Shows Oracle Database Properties |
| **Oracle Database Overview** | List | Shows key metrics for Oracle Databases |
| **Oracle Health** | List | Shows health and availability for all Oracle objects |
| **Oracle Instance Overview** | List | Oracle Instance Overview |
| **Oracle Instance Waits** | List | Shows wait values for Oracle Instances |
| **Oracle Slow Queries** | List | Shows top slowest queries by Oracle Instance |
| **Oracle Tablespace** | List | Shows Tablespace metrics |
| **Oracle Instance Configuration** | List | Shows properties for Oracle Instances |

To access the Management Pack views:

1. Navigate to **Environment > All Objects > Oracle Database**.
2. Double-click on the desired object (resource).
3. Select the **Details** tab, then **Views.**

The available views for that resource are listed and can be selected.

### Reports

The Management Pack for Oracle Database contains out-of-the-box report(s)—which contain one or more Views—that you can export and easily share with key stakeholders in either .pdf or .csv formats.

The following reports are available in the Management Pack:

| Report | Description |
|---|---|
| **Oracle Database Overview** | Provides an overview of Oracle Database resources |
| **Oracle Health** | Provides health details about your Oracle Database resources |
| **Oracle Instance Overview** | Provides an overview specific to your Oracle Database instance |
| **Oracle KPIs** | Provides KPIs for your Oracle Database resources |
| **Oracle Tablespaces** | Provides details related to your Oracle Database Tablespaces |

To access Management Pack reports:

1. Navigate to **Environment > All Objects > Oracle Database**.
2. Double-click on the desired object (resource).
3. Select the **Reports** tab, then **Report Templates**.

The available report(s) for the selected resource are listed and can be selected.

To run a selected report, click the **Run Template** icon.

### Analysis Badges and Capacity Definitions

Using the predictive analytics capabilities of VCF Operations Analysis Badges through capacity definitions, the Management Pack for Oracle Database populates capacities as indicated in the table below.

### Capacity Definitions List

| Container | Use in Workload |
|---|---|
| **Oracle Database Instance** | |
| Calculated Session Capacity | yes |

Alerts

The VCF Operations Management Pack for Oracle Database pulls Oracle Database health indicators and displays them in VCF Operations as alerts. The critical levels of each alert are closely mapped from Oracle Database to VCF Operations. See the table below for the list of alerts available in the Management Pack.

### Alerts List

| Name | Description | Recommendation |
|---|---|---|
| ASM Disk Free Space is Low | ASM Disk Free Space is Low | ASM Disk Free Space Less Than 15%\|ASM Disk Free Space Less Than 10%\|ASM Disk Free Space Less Than 5% |
| ASM Status Not Normal | ASM Status Not Normal | Automatic Storage Management metadata indicates that the disk is known to be part of the Automatic Storage Management disk group, but no disk in the storage system was found with the indicated name\|Disk is present in the system, but is ignored by ASM |
| Archive Logging Disabled | Archive Logging Disabled | All databases should run in archive log mode. Enable archive logging |
| Average Datafile Read Duration Above Threshold | Average Datafile Read Duration Above Threshold | |
| Fast Recovery Area Not Configured | Fast Recovery Area Not Configured | Fast Recovery Area Should Be Configured |
| Database Flashback Disabled | Database Flashback Disabled | Database flashback is a useful facility for rapid recovery following a user error and should be enabled |
| Process Limit Above Threshold | Process Limit Above Threshold | Sessions will fail if the process limit is hit. Increase the limit for processes |
| Session Usage Above Threshold | Session Usage Above Threshold | Logins will begin to fail if the session limit is hit. Increase the limit for sessions |
| Datafile Not Available | Datafile Not Available | |
| High CPU Usage | High CPU Usage | |
| High Physical Read I/O Request | High Physical Read I/O Request | |
| High Volume Of User Calls | High Volume Of User Calls | |
| High Redo Logs Generated | High Redo Logs Generated | |
| High Average I/O Time | High Average I/O Time | |
| Long Read Time | Long Read Time | |
| Slow Write Time | Slow Write Time | |
| Database Not Active | Database Not Active | The database instance is not currently active. Connect to the instance and review the process status. |
| Low Buffer Cache Hit Ratio | Low Buffer Cache Hit Ratio | Review the buffer size configuration. |
| High Number of CR Blocks Created | High Number of CR Blocks Created | |
| Low Cursor Cache Hit Ratio | Low Cursor Cache Hit Ratio | |

| Name | Description | Recommendation |
|---|---|---|
| High Database CPU Time Usage | High Database CPU Time Usage | |
| High Database Wait Time | High Database Wait Time | |
| High Number of Enqueue Deadlocks | High Number of Enqueue Deadlocks | |
| Large Number of Executions | Large Number of Executions | |
| Number of Executions Per Second Exceeded Threshold | Number of Executions Per Second Exceeded Threshold | |
| High Volume of Executions Per Transaction | High Volume of Executions Per Transaction | |
| High I/O Throughput | High I/O Throughput | |
| Low Library Cache Hit Ratio | Low Library Cache Hit Ratio | |
| Slow Table Scans | Slow Table Scans | |
| Low PGA Cache Hit Percentage | Low PGA Cache Hit Percentage | |
| High Write Volume | High Write Volume | |
| Slow Response Time During Transactions | Slow Response Time During Transactions | |
| Low Cache Hit Ratio | Low Cache Hit Ratio | |
| Slow Service Response | Slow Service Response | |
| Large Number of Active Sessions | Large Number of Active Sessions | |
| Large Number of Applications | Large Number of Applications | |
| High Volume of Index Scans Per Transactions | High Volume of Index Scans Per Transactions | |
| Above Normal Query Time | Above Normal Query Time | |

Metrics

The Management Pack for Oracle Database collects the metrics listed in the table below, according to Oracle Database resource kinds.

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database Archive Log | Disk | Backup Count | Indicates the number of times this file has been backed up. Values range from 0-15. If the file has been backed up more than 15 times, the value remains 15 |
| Oracle Database Archive Log | Performance | Completion Time | Time when the archiving completed |
| Oracle Database Archive Log | General | Deleted | Indicates whether an RMAN DELETE command has physically deleted the archived log file from disk, as well as logically removing it from the control file of the target database and from the recovery catalog (YES) or not (NO) |
| Oracle Database Archive Log | General | Destination ID | Original destination from which the archive log was generated |
| Oracle Database Archive Log | General | Instance ID | |
| Oracle Database Archive Log | General | Name | Archived log file name |
| Oracle Database Archive Log | General | Record ID | Archived log record ID |
| Oracle Database Archive Log | Disk | Size | Size of the archive log |
| Oracle Database Archive Log | Status | Status | Status of the archived log |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database Archive Log | General | Thread Number | Redo Thread Number |
| Oracle Database ASM Disk | General | Compound Index | A 32-bit number consisting of a disk group number in the high-order 8 bits and a disk number in the low-order 24 bits |
| Oracle Database ASM Disk | General | Create Date | Date and time when the disk was added to the disk group |
| Oracle Database ASM Disk | Disk | Total Disk Capacity | Total capacity of the disk |
| Oracle Database ASM Disk | General | Disk Number | Number assigned to the disk withing its group |
| Oracle Database ASM Disk | General | Fail Group | Name of the failure group containing the disk |
| Oracle Database ASM Disk | Disk | Free Disk Space | Unused capacity of the disk |
| Oracle Database ASM Disk | Details | Group Number | Number of the disk group containing the disk |
| Oracle Database ASM Disk | Status | Header Status | Per-instance status of the disk as seen by discovery |
| Oracle Database ASM Disk | General | Instance ID | |
| Oracle Database ASM Disk | General | Discovery Label | Disk label portion of the name returned by discovery |
| Oracle Database ASM Disk | General | Library | Name of the library that discovered the disk |
| Oracle Database ASM Disk | Status | Mode Status | Global status about which kinds of I/O requests are allowed to the disk |
| Oracle Database ASM Disk | General | Mount Date | Date and time when the disk was was first mounted by the first instance |
| Oracle Database ASM Disk | Status | Mount Status | Per-instance status of the disk relative to group mounts |
| Oracle Database ASM Disk | General | Name | Name of the disk |
| Oracle Database ASM Disk | General | Path | Operating system pathname portion of the name returned by discovery |
| Oracle Database ASM Disk | General | Product | The name of the manufacturer and the name of the product |
| Oracle Database ASM Disk | Performance | Read | Total number of bytes read from the disk |
| Oracle Database ASM Disk | Performance | Read Errors | Total number of failed I/O read requests for the disk |
| Oracle Database ASM Disk | Performance | Read Time | Total I/O time for read requests for the disk |
| Oracle Database ASM Disk | Disk | Reads | Total number of I/O read requests for the disk |
| Oracle Database ASM Disk | General | Redundancy | External redundancy of the disk |
| Oracle Database ASM Disk | Status | State | Global state of the disk with respect to the disk group |
| Oracle Database ASM Disk | General | Universal Device ID | Universal Device ID portion of the name returned by discovery |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database ASM Disk | Performance | Bytes Written | Total number of bytes written to the disk |
| Oracle Database ASM Disk | Performance | Write Errors | Total number of failed I/O write requests for the disk |
| Oracle Database ASM Disk | Performance | Write Time | Total I/O time for write requests for the disk |
| Oracle Database ASM Disk | Disk | Writes | Total number of I/O write requests for the disk |
| Oracle Database ASM Disk | Disk | ASM Free Space | ASM free disk space percentage |
| Oracle Database Control File | General | Name | |
| Oracle Database Control File | General | Control File Size | |
| Oracle Database Database | Activity | Average Datafile Read Duration | |
| Oracle Database Database | General | Oracle Database Type | |
| Oracle Database Database | General | Cluster Database | |
| Oracle Database Database | General | Multiplexed Archive Log Files Count | |
| Oracle Database Database | General | Multiplexed Online Log Files Average Count | |
| Oracle Database Database | Activity | Connection Processes | |
| Oracle Database Database | General | Container ID | |
| Oracle Database Database | General | Controlfile Count | |
| Oracle Database Database | General | database_id | |
| Oracle Database Database | General | Database Type | Non-CDB, CDB, or PDB |
| Oracle Database Database | Undo | Expired Undo Extents Size | |
| Oracle Database Database | General | Fast Recovery Area | |
| Oracle Database Database | Activity | Fast Recovery Area Usage | |
| Oracle Database Database | General | Flashback Enabled | |
| Oracle Database Database | General | Global Name | |
| Oracle Database Database | General | Host List | |
| Oracle Database Database | General | Instances | |
| Oracle Database Database | Activity | Last RMAN Backup | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database Database | General | Archive Log Mode | |
| Oracle Database Database | Activity | Long Running Jobs | |
| Oracle Database Database | General | Oracle Managed Files | |
| Oracle Database Database | Activity | Processes Usage | |
| Oracle Database Database | Activity | Sessions Usage | |
| Oracle Database Database | Undo | Undo Tablespace Size | |
| Oracle Database Database | Undo | Unexpired Undo Extents Size | |
| Oracle Database Database File | General | Autoextensible | |
| Oracle Database Database File | Disk IO | Average I/O Time | |
| Oracle Database Database File | Disk IO | Average Read Time | |
| Oracle Database Database File | Disk IO | Average Write Time | |
| Oracle Database Database File | General | Container ID | |
| Oracle Database Database File | General | File Name | |
| Oracle Database Database File | Disk IO | Max Read Time | |
| Oracle Database Database File | General | Max Size | |
| Oracle Database Database File | Disk IO | Max Write Time | |
| Oracle Database Database File | Disk IO | Physical Blocks Read | |
| Oracle Database Database File | Disk IO | Physical Blocks Written | |
| Oracle Database Database File | Disk IO | Physical Reads | |
| Oracle Database Database File | Disk IO | Physical Writes | |
| Oracle Database Database File | Disk IO | Read Time | |
| Oracle Database Database File | Disk IO | Single Blocks Read | |
| Oracle Database Database File | General | Size | |
| Oracle Database Database File | Status | Status | |
| Oracle Database Database File | General | Tablespace Name | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database Database File | Disk IO | Write Time | |
| Oracle Database Event Wait Group | Activity | Foreground Time Waited | |
| Oracle Database Event Wait Group | Activity | Foreground Waits | |
| Oracle Database Event Wait Group | General | Instance ID | |
| Oracle Database Event Wait Group | Activity | Total Time Waited | |
| Oracle Database Event Wait Group | Activity | Total Waits | |
| Oracle Database Event Wait Group | General | Wait Class | |
| Oracle Database Instance | Performance | Active Parallel Sessions | |
| Oracle Database Instance | Performance | Active Serial Sessions | |
| Oracle Database Instance | General | Active State | |
| Oracle Database Instance | Session | Active User Sessions | |
| Oracle Database Instance | Session | Applications | |
| Oracle Database Instance | General | Archiver | |
| Oracle Database Instance | Performance | Average Active Sessions | |
| Oracle Database Instance | Disk IO | Average I/O Time | The average Input/Output time of the physical storage on the disk. |
| Oracle Database Instance | Disk IO | Average Read Time | Average read time to the physical storage on the disk. |
| Oracle Database Instance | Performance | Average Synchronous Single-Block Read Latency | |
| Oracle Database Instance | Disk IO | Average Write Time | Average write time to the physical storage on the disk. |
| Oracle Database Instance | Activity | Background Checkpoints | |
| Oracle Database Instance | Performance | Background Checkpoints | |
| Oracle Database Instance | Performance | Background CPU Usage | |
| Oracle Database Instance | Performance | Background Time | |
| Oracle Database Instance | Session | Background Types | |
| Oracle Database Instance | General | Blocked | |
| Oracle Database Instance | Session | Blocked by Unknown User Sessions | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database Instance | Session | Blocked User Sessions | |
| Oracle Database Instance | Activity | Branch Node Splits | |
| Oracle Database Instance | Performance | Branch Node Splits Per Transaction | |
| Oracle Database Instance | Performance | Branch Node Splits | |
| Oracle Database Instance | Performance | Buffer Cache Hit Ratio | |
| Oracle Database Instance | Shared Memory | Buffer Cache Size | |
| Oracle Database Instance | Session | Cached User Sessions | |
| Oracle Database Instance | Performance | Cell Physical I/O Interconnect | |
| Oracle Database Instance | Activity | Consistent Read Blocks Created | |
| Oracle Database Instance | Activity | Consistent Read Changes | |
| Oracle Database Instance | Performance | Consistent Read Changes Per Transaction | |
| Oracle Database Instance | Performance | Consistent Read Changes | |
| Oracle Database Instance | Activity | Consistent Read Gets | |
| Oracle Database Instance | Performance | Consistent Read Gets Per Transaction | |
| Oracle Database Instance | Performance | Consistent Read Gets | |
| Oracle Database Instance | Activity | Consistent Read Undo Records Applied | |
| Oracle Database Instance | Activity | CPU Usage | |
| Oracle Database Instance | Performance | CPU Usage Per Transaction | |
| Oracle Database Instance | Performance | CPU Usage | |
| Oracle Database Instance | Performance | Consistent Read Blocks Created | |
| Oracle Database Instance | Performance | Consistent Read Blocks Created Per Transaction | |
| Oracle Database Instance | Performance | Consistent Read Undo Records Applied | |
| Oracle Database Instance | Performance | Consistent Read Undo Records Applied Per Transaction | |
| Oracle Database Instance | Performance | Current Logons Count | |
| Oracle Database Instance | Performance | Current Open Cursors Count | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database Instance | Performance | Current OS Processes Load | |
| Oracle Database Instance | Performance | Cursor Cache Hit Ratio | |
| Oracle Database Instance | Activity | Data Definition Language Statements Parallelized | |
| Oracle Database Instance | Activity | Data Manipulation Language Statements Parallelized | |
| Oracle Database Instance | Shared Memory | Data Transfer Cache Size | |
| Oracle Database Instance | Activity | Database Block Changes | |
| Oracle Database Instance | Activity | Database Block Gets | |
| Oracle Database Instance | Performance | Database CPU Time Ratio | |
| Oracle Database Instance | General | Database Status | |
| Oracle Database Instance | Performance | Database Time | |
| Oracle Database Instance | Performance | Database Wait Time Ratio | |
| Oracle Database Instance | Activity | Database Writer Checkpoints | |
| Oracle Database Instance | Performance | Database Block Changes | |
| Oracle Database Instance | Performance | Database Block Changes Per Transaction | |
| Oracle Database Instance | Performance | Database Block Changes Per User Call | |
| Oracle Database Instance | Performance | Database Block Gets | |
| Oracle Database Instance | Performance | Database Block Gets Per Transaction | |
| Oracle Database Instance | Performance | Database Block Gets Per User Call | |
| Oracle Database Instance | Performance | Database Writer Checkpoints | |
| Oracle Database Instance | Performance | Data Definition Language Statements Parallelized | |
| Oracle Database Instance | Activity | Direct Read Full Index Scans | |
| Oracle Database Instance | Disk IO | Disk Sorts | |
| Oracle Database Instance | Performance | Disk Sort Per Transaction | |
| Oracle Database Instance | Performance | Disk Sort | |
| Oracle Database Instance | Performance | Data Manipulation Language Statements Parallelized | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database Instance | Activity | Enqueue Deadlocks | |
| Oracle Database Instance | Performance | Enqueue Deadlocks Per Transaction | |
| Oracle Database Instance | Performance | Enqueue Deadlocks | |
| Oracle Database Instance | General | Enqueue Locks Limit | |
| Oracle Database Instance | Activity | Enqueue Locks Utilization | |
| Oracle Database Instance | Activity | Enqueue Requests | |
| Oracle Database Instance | Performance | Enqueue Requests Per Transaction | |
| Oracle Database Instance | Performance | Enqueue Requests | |
| Oracle Database Instance | Activity | Enqueue Timeouts | |
| Oracle Database Instance | Performance | Enqueue Timeouts Per Transaction | |
| Oracle Database Instance | Performance | Enqueue Timeouts | |
| Oracle Database Instance | Activity | Enqueue Waits | |
| Oracle Database Instance | Performance | Enqueue Waits Per Transaction | |
| Oracle Database Instance | Performance | Enqueue Waits | |
| Oracle Database Instance | Performance | Execute Without Parse Ratio | |
| Oracle Database Instance | Activity | Executions | |
| Oracle Database Instance | Performance | Executions Per Transaction | |
| Oracle Database Instance | Performance | Executions Per User Call | |
| Oracle Database Instance | Performance | Executions | Represents the rate of SQL command executions over the sampling period. |
| Oracle Database Instance | Activity | Failed Parse Count | |
| Oracle Database Instance | Shared Memory | Fixed System Global Area Size | |
| Oracle Database Instance | Latch | Foreground Latch Time Waited | |
| Oracle Database Instance | Activity | Foreground Time Waited | |
| Oracle Database Instance | Shared Memory | Free System Global Area Memory Available | |
| Oracle Database Instance | Activity | Full Index Scans | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database Instance | Performance | Full Index Scans Per Transaction | |
| Oracle Database Instance | Performance | Full Index Scans | |
| Oracle Database Instance | Performance | Global Cache Consistent Read Block Received | |
| Oracle Database Instance | Performance | Global Cache Consistent Read Block Received Per Transaction | |
| Oracle Database Instance | Performance | Global Cache Current Block Received | |
| Oracle Database Instance | Performance | Global Cache Current Block Received Per Transaction | |
| Oracle Database Instance | Performance | Global Cache Average Consistent Read Get Time | |
| Oracle Database Instance | Performance | Global Cache Average Current Get Time | |
| Oracle Database Instance | Performance | Global Cache Blocks Corrupted | |
| Oracle Database Instance | Performance | Global Cache Blocks Lost | |
| Oracle Database Instance | Activity | Global Cache Consistent Read Blocks Received | |
| Oracle Database Instance | Activity | Global Cache Current Blocks Received | |
| Oracle Database Instance | Shared Memory | Granule Size | |
| Oracle Database Instance | Activity | Hard Parse Count | |
| Oracle Database Instance | Performance | Hard Parse Count Per Transaction | |
| Oracle Database Instance | Performance | Hard Parse Count | |
| Oracle Database Instance | Performance | Host CPU Usage | |
| Oracle Database Instance | Performance | Host CPU Utilization | |
| Oracle Database Instance | General | Host Name | |
| Oracle Database Instance | Session | Inactive User Sessions | |
| Oracle Database Instance | Performance | Total Index Scans | |
| Oracle Database Instance | Performance | Total Index Scans Per Transaction | |
| Oracle Database Instance | General | Instance ID | |
| Oracle Database Instance | General | Instance Name | |
| Oracle Database Instance | General | Instance Role | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database Instance | Performance | I/O Requests | |
| Oracle Database Instance | Performance | I/O Throughput | |
| Oracle Database Instance | Shared Memory | Java Pool Size | |
| Oracle Database Instance | Session | Killed User Sessions | |
| Oracle Database Instance | Shared Memory | Large Pool Size | |
| Oracle Database Instance | Activity | Leaf Node Splits | |
| Oracle Database Instance | Performance | Leaf Node Splits Per Transaction | |
| Oracle Database Instance | Performance | Leaf Node Splits | |
| Oracle Database Instance | Performance | Library Cache Hit Ratio | |
| Oracle Database Instance | Performance | Library Cache Miss Ratio | |
| Oracle Database Instance | Performance | Logical Reads | |
| Oracle Database Instance | Performance | Logical Reads Per Transaction | |
| Oracle Database Instance | Performance | Logical Reads Per User Call | |
| Oracle Database Instance | General | Login Mode | |
| Oracle Database Instance | Activity | Logons | |
| Oracle Database Instance | Performance | Logons Per Transaction | |
| Oracle Database Instance | Performance | Logons | |
| Oracle Database Instance | Activity | Long Table Scans | |
| Oracle Database Instance | Performance | Long Table Scans Per Transaction | |
| Oracle Database Instance | Performance | Long Table Scans | |
| Oracle Database Instance | Session | O/S Machines | |
| Oracle Database Instance | Disk IO | Max Read Time | |
| Oracle Database Instance | Disk IO | Max Write Time | |
| Oracle Database Instance | Shared Memory | Maximum System Global Area Size | |
| Oracle Database Instance | Performance | Memory Sorts Ratio | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database Instance | Activity | Network Data Received From Client | |
| Oracle Database Instance | Activity | Network Data Received From DBLink | |
| Oracle Database Instance | Activity | Network Data Sent to Client | |
| Oracle Database Instance | Activity | Network Data Sent to DBLink | |
| Oracle Database Instance | Performance | Network Traffic Volume | |
| Oracle Database Instance | Session | Non-Waiting User Sessions | |
| Oracle Database Instance | Activity | Open Cursors | |
| Oracle Database Instance | Performance | Open Cursors Per Transaction | |
| Oracle Database Instance | Performance | Open Cursors | |
| Oracle Database Instance | Session | O/S Client Users | |
| Oracle Database Instance | General | Parallel | |
| Oracle Database Instance | Activity | Parallel Execution downgraded 1 to 25 percent | |
| Oracle Database Instance | Activity | Parallel Execution downgraded 25 to 50 percent | |
| Oracle Database Instance | Activity | Parallel Execution downgraded 50 to 75 percent | |
| Oracle Database Instance | Activity | Parallel Execution downgraded 75 to 99 percent | |
| Oracle Database Instance | Activity | Parallel Execution downgraded to serial | |
| Oracle Database Instance | Activity | Parallel Execution Operations Not Downgraded | |
| Oracle Database Instance | Activity | Parallel Max Servers Utilization | |
| Oracle Database Instance | General | Parallel Max Servers Limit | |
| Oracle Database Instance | Performance | Total Parse Count | |
| Oracle Database Instance | Performance | Total Parse Count Per Transaction | |
| Oracle Database Instance | Performance | Parse Failure Count | |
| Oracle Database Instance | Performance | Parse Failure Count Per Transaction | |
| Oracle Database Instance | Performance | Program Global Area Cache Hit | |
| Oracle Database Instance | Disk IO | Physical Application Data Read | Total size of all disk reads by application activity (and not other instance activity) only. |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database Instance | Disk IO | Physical Application Data Written | Total size of all disk writes by application activity (and not other instance activity) only. |
| Oracle Database Instance | Disk IO | Physical Application Read I/O Requests | Number of read requests for application activity (mainly buffer cache and direct load operation) which read one or more database blocks per request. This is a subset of "Cumulative Physical Read IO Requests" statistic. |
| Oracle Database Instance | Activity | Physical Application Write I/O Requests | Number of write requests for application activity (mainly buffer cache and direct load operation) which write one or more database blocks per request. This is a subset of "Cumulative Physical Write IO Requests" statistic. |
| Oracle Database Instance | Disk IO | Physical Blocks Read | |
| Oracle Database Instance | Disk IO | Physical Blocks Written | |
| Oracle Database Instance | Disk IO | Physical Data Read | Total size of disk reads by all database instance activity including application reads, backup and recovery, and other utilities. The difference between this value and "Cumulative Physical Application Data Read" gives the total read size by non-application workload. |
| Oracle Database Instance | Disk IO | Physical Data Written | Total size of disk writes by all database instance activity including application writes, backup and recovery, and other utilities. The difference between this value and "Cumulative Physical Application Data Written" gives the total write size by non-application workload. |
| Oracle Database Instance | Disk IO | Physical Read I/O Requests | Number of read requests which read one or more database blocks for all instance activity including application, backup and recovery, and other utilities. |
| Oracle Database Instance | Performance | Physical Read I/O Requests | |
| Oracle Database Instance | Performance | Physical Read Rate | |
| Oracle Database Instance | Performance | Physical Read Total I/O Requests | |
| Oracle Database Instance | Disk IO | Physical Reads | |
| Oracle Database Instance | Activity | Physical Reads Direct | |
| Oracle Database Instance | Activity | Physical Reads Direct Lobs | |
| Oracle Database Instance | Performance | Physical Reads Direct Lobs Per Transaction | |
| Oracle Database Instance | Performance | Physical Reads Direct Lobs | |
| Oracle Database Instance | Performance | Physical Reads Direct Per Transaction | |
| Oracle Database Instance | Performance | Physical Reads Direct | |
| Oracle Database Instance | Performance | Physical Reads Per Transaction | |
| Oracle Database Instance | Performance | Total Physical Read Rate | |
| Oracle Database Instance | Disk IO | Physical Write I/O Requests | Number of write requests which write one or more database blocks for all instance activity including application, backup and recovery, and other utilities. |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database Instance | Performance | Physical Write I/O Requests | |
| Oracle Database Instance | Performance | Physical Write Rate | |
| Oracle Database Instance | Performance | Total Physical Write Rate | |
| Oracle Database Instance | Performance | Physical Write Total I/O Requests | |
| Oracle Database Instance | Disk IO | Physical Writes | |
| Oracle Database Instance | Disk IO | Physical Writes Direct | |
| Oracle Database Instance | Activity | Physical Writes Direct Lobs | |
| Oracle Database Instance | Performance | Physical Writes Direct Lobs Per Transaction | |
| Oracle Database Instance | Performance | Physical Writes Direct Lobs | |
| Oracle Database Instance | Performance | Physical Writes Direct Per Transaction | |
| Oracle Database Instance | Performance | Physical Writes Direct | |
| Oracle Database Instance | Performance | Physical Writes Per Transaction | |
| Oracle Database Instance | Performance | Parallel Statement Queuing Query Coordinator Sessions | |
| Oracle Database Instance | Performance | Parallel Statement Queuing Slave Sessions | |
| Oracle Database Instance | General | Processes Limit | |
| Oracle Database Instance | Performance | Process Limit | |
| Oracle Database Instance | Activity | Processes | |
| Oracle Database Instance | Performance | Parallel Execution Downgraded 1 to 25 Percent | |
| Oracle Database Instance | Performance | Parallel Execution Downgraded 25 to 50 Percent | |
| Oracle Database Instance | Performance | Parallel Execution Downgraded 50 to 75 Percent | |
| Oracle Database Instance | Performance | Parallel Execution Downgraded 75 to 99 Percent | |
| Oracle Database Instance | Performance | Parallel Execution downgraded to serial | |
| Oracle Database Instance | Performance | Parallel Execution Operations Not Downgraded | |
| Oracle Database Instance | Activity | Queries Parallelized | |
| Oracle Database Instance | Performance | Queries Parallelized | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database Instance | Disk IO | Read Time | |
| Oracle Database Instance | Activity | Recursive Calls | |
| Oracle Database Instance | Performance | Recursive Calls Per Transaction | |
| Oracle Database Instance | Performance | Recursive Calls | |
| Oracle Database Instance | Performance | Redo Allocation Hit Ratio | |
| Oracle Database Instance | Shared Memory | Redo Buffers Size | |
| Oracle Database Instance | Activity | Redo Generated | |
| Oracle Database Instance | Performance | Redo Generated Per Transaction | |
| Oracle Database Instance | Performance | Redo Generated | |
| Oracle Database Instance | Disk IO | Redo Writes | |
| Oracle Database Instance | Performance | Redo Writes Per Transaction | |
| Oracle Database Instance | Performance | Redo Writes | |
| Oracle Database Instance | Performance | Response Time Per Transaction | |
| Oracle Database Instance | Performance | Row Cache Hit Ratio | |
| Oracle Database Instance | Performance | Row Cache Miss Ratio | |
| Oracle Database Instance | Activity | RowID Ranges Full Index Scans | |
| Oracle Database Instance | Performance | Rows Per Sort | |
| Oracle Database Instance | Performance | Run Queue Processes | |
| Oracle Database Instance | General | Sessions Limit | |
| Oracle Database Instance | Performance | Session Limit | |
| Oracle Database Instance | Session | Sessions | |
| Oracle Database Instance | Shared Memory | Shared IO Pool Size | |
| Oracle Database Instance | Performance | Shared Pool Free | |
| Oracle Database Instance | Shared Memory | Shared Pool Size | |
| Oracle Database Instance | Activity | Short Table Scans | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database Instance | General | Shutdown Pending | |
| Oracle Database Instance | Disk IO | Single Blocks Read | |
| Oracle Database Instance | Session | Sniped User Sessions | |
| Oracle Database Instance | Performance | Soft Parse Ratio | |
| Oracle Database Instance | Performance | SQL Service Response Time | |
| Oracle Database Instance | Shared Memory | Startup Overhead in Shared Pool | |
| Oracle Database Instance | General | Startup Time | |
| Oracle Database Instance | General | Status | |
| Oracle Database Instance | Shared Memory | Streams Pool Size | |
| Oracle Database Instance | Performance | Streams Pool Usage | |
| Oracle Database Instance | Performance | Total Table Scans | |
| Oracle Database Instance | Performance | Total Table Scans Per Transaction | |
| Oracle Database Instance | Activity | Total Time Waited | |
| Oracle Database Instance | Latch | Total Foreground Latch Timeouts | |
| Oracle Database Instance | Latch | Total Foreground Latch Waits | |
| Oracle Database Instance | Activity | Total Foreground Timeouts | |
| Oracle Database Instance | Activity | Total Foreground Waits | |
| Oracle Database Instance | Latch | Total Latch Time Waited | |
| Oracle Database Instance | Latch | Total Latch Timeouts | |
| Oracle Database Instance | Latch | Total Latch Waits | |
| Oracle Database Instance | Activity | Total Parse Count | |
| Oracle Database Instance | Performance | Total Program Global Area Allocated | |
| Oracle Database Instance | Performance | Total Program Global Area Used by SQL Workareas | |
| Oracle Database Instance | Performance | Total Sorts Per User Call | |
| Oracle Database Instance | Performance | Total Table Scans Per User Call | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database Instance | Activity | Total Timeouts | |
| Oracle Database Instance | Activity | Total Waits | |
| Oracle Database Instance | Performance | Transactions Per Logon | |
| Oracle Database Instance | Session | Unblocked User Sessions | |
| Oracle Database Instance | Session | Unknown Wait Time User Sessions | |
| Oracle Database Instance | Activity | User Calls | |
| Oracle Database Instance | Performance | User Calls Per Transaction | |
| Oracle Database Instance | Performance | User Calls | |
| Oracle Database Instance | Performance | User Calls Ratio | |
| Oracle Database Instance | Activity | User Commits | |
| Oracle Database Instance | Performance | User Commits | |
| Oracle Database Instance | Performance | User Commits Ratio | |
| Oracle Database Instance | Performance | User Limit | |
| Oracle Database Instance | Activity | User Rollback Undo Records Applied | |
| Oracle Database Instance | Performance | User Rollback Undo Records Applied | |
| Oracle Database Instance | Performance | User Rollback Undo Records Applied Per Transaction | |
| Oracle Database Instance | Activity | User Rollbacks | |
| Oracle Database Instance | Performance | User Rollbacks | |
| Oracle Database Instance | Performance | User Rollbacks Ratio | |
| Oracle Database Instance | Performance | User Transactions | |
| Oracle Database Instance | Session | User Types | |
| Oracle Database Instance | General | Version | |
| Oracle Database Instance | Session | Waiting User Sessions | |
| Oracle Database Instance | Performance | Workload Capture and Replay Status | |
| Oracle Database Instance | Disk IO | Write Time | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database PDB | Activity | Physical Application Data Read | Total size of all disk reads by application activity (and not other instance activity) only. |
| Oracle Database PDB | Activity | Physical Writes Direct | |
| Oracle Database PDB | Session | Active User Sessions | |
| Oracle Database PDB | Session | Applications | |
| Oracle Database PDB | Disk IO | Average I/O Time | The average Input/Output time of the physical storage on the disk. |
| Oracle Database PDB | Disk IO | Average Read Time | Average read time to the physical storage on the disk. |
| Oracle Database PDB | Disk IO | Average Write Time | Average write time to the physical storage on the disk. |
| Oracle Database PDB | Activity | Background Checkpoints | |
| Oracle Database PDB | Session | Background Types | |
| Oracle Database PDB | Disk | Block Size | |
| Oracle Database PDB | Session | Blocked by Unknown User Sessions | |
| Oracle Database PDB | Session | Blocked User Sessions | |
| Oracle Database PDB | Uncategorized | Branch Node Splits | |
| Oracle Database PDB | Session | Cached User Sessions | |
| Oracle Database PDB | Activity | Consistent Read Blocks Created | |
| Oracle Database PDB | Activity | Consistent Read Changes | |
| Oracle Database PDB | Activity | Consistent Read Gets | |
| Oracle Database PDB | Activity | Consistent Read Undo Records Applied | |
| Oracle Database PDB | General | Container ID | |
| Oracle Database PDB | Activity | CPU Usage | |
| Oracle Database PDB | Activity | Data Definition Language Statements Parallelized | |
| Oracle Database PDB | Activity | Data Manipulation Language Statements Parallelized | |
| Oracle Database PDB | Activity | Database Block Changes | |
| Oracle Database PDB | Activity | Cumulative Database Block Gets | |
| Oracle Database PDB | General | Database ID | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database PDB | Activity | Database Writer Checkpoints | |
| Oracle Database PDB | Activity | Direct Read Full Index Scans | |
| Oracle Database PDB | Disk | Disk Sorts | |
| Oracle Database PDB | Activity | Enqueue Deadlocks | |
| Oracle Database PDB | Activity | Enqueue Requests | |
| Oracle Database PDB | Activity | Enqueue Timeouts | |
| Oracle Database PDB | Activity | Enqueue Waits | |
| Oracle Database PDB | Activity | Executions | |
| Oracle Database PDB | Activity | Failed Parses | |
| Oracle Database PDB | Activity | Full Index Scans | |
| Oracle Database PDB | Activity | Global Cache Consistent Read Blocks Received | |
| Oracle Database PDB | Activity | Global Cache Current Blocks Received | |
| Oracle Database PDB | General | GUID | |
| Oracle Database PDB | Activity | Hard Parse Count | |
| Oracle Database PDB | Session | Inactive User Sessions | |
| Oracle Database PDB | Session | Killed User Sessions | |
| Oracle Database PDB | Activity | Cumulative Leaf Node Splits | |
| Oracle Database PDB | Activity | Cumulative Logons | |
| Oracle Database PDB | Activity | Long Table Scans | |
| Oracle Database PDB | Session | O/S Machines | |
| Oracle Database PDB | Disk IO | Max Read Time | |
| Oracle Database PDB | Disk IO | Max Write Time | |
| Oracle Database PDB | Uncategorized | Name | |
| Oracle Database PDB | Activity | Network Data Received From Client | |
| Oracle Database PDB | Activity | Network Data Received From DBLink | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database PDB | Activity | Network Data Sent to Client | |
| Oracle Database PDB | Activity | Network Data Sent to DBLink | |
| Oracle Database PDB | Session | Non-Waiting User Sessions | |
| Oracle Database PDB | Activity | Open Cursors | |
| Oracle Database PDB | General | Open Mode | Open mode information. Possible values: MOUNTED, READ WRITE, READ ONLY, MIGRATE |
| Oracle Database PDB | Session | O/S Client Users | |
| Oracle Database PDB | Activity | Parallel Execution downgraded 1 to 25 percent | |
| Oracle Database PDB | Activity | Parallel Execution downgraded 25 to 50 percent | |
| Oracle Database PDB | Activity | Parallel Execution downgraded 50 to 75 percent | |
| Oracle Database PDB | Activity | Parallel Execution downgraded 75 to 99 percent | |
| Oracle Database PDB | Activity | Parallel Execution downgraded to serial | |
| Oracle Database PDB | Activity | Parallel Execution Operations Not Downgraded | |
| Oracle Database PDB | Activity | Physical Application Data Written | Total size of all disk writes by application activity (and not other instance activity) only. |
| Oracle Database PDB | Disk | Physical Application Read I/O Requests | Number of read requests for application activity (mainly buffer cache and direct load operation) which read one or more database blocks per request. This is a subset of "Cumulative Physical Read IO Requests" statistic. |
| Oracle Database PDB | Disk | Physical Application Write I/O Requests | Number of write requests for application activity (mainly buffer cache and direct load operation) which write one or more database blocks per request. This is a subset of "Cumulative Physical Write IO Requests" statistic. |
| Oracle Database PDB | Disk IO | Physical Blocks Read | |
| Oracle Database PDB | Disk IO | Physical Blocks Written | |
| Oracle Database PDB | Disk | Physical Data Read | Total size of disk reads by all database instance activity including application reads, backup and recovery, and other utilities. The difference between this value and "Cumulative Physical Application Data Read" gives the total read size by non-application workload. |
| Oracle Database PDB | Disk | Physical Data Written | Total size of disk writes by all database instance activity including application writes, backup and recovery, and other utilities. The difference between this value and "Cumulative Physical Application Data Written" gives the total write size by non-application workload. |
| Oracle Database PDB | Disk | Physical Read I/O Requests | Number of read requests which read one or more database blocks for all instance activity including application, backup and recovery, and other utilities. |
| Oracle Database PDB | Disk IO | Physical Reads | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database PDB | Activity | Physical Reads Direct | |
| Oracle Database PDB | Disk | Physical Reads Direct Lobs | |
| Oracle Database PDB | Disk | Physical Write I/O Requests | Number of write requests which write one or more database blocks for all instance activity including application, backup and recovery, and other utilities. |
| Oracle Database PDB | Disk IO | Physical Writes | |
| Oracle Database PDB | Disk | Physical Writes Direct Lobs | |
| Oracle Database PDB | Activity | Queries Parallelized | |
| Oracle Database PDB | Disk IO | Read Time | |
| Oracle Database PDB | Status | Recovery Status | Shows whether recovery is enabled or disabled for the PDB. Possible values: ENABLED, DISABLED |
| Oracle Database PDB | Activity | Recursive Calls | |
| Oracle Database PDB | Activity | Redo Generated | |
| Oracle Database PDB | Activity | Redo Writes | |
| Oracle Database PDB | General | Restricted | Indicates whether only users possessing RESTRICTED SESSION privilege can connect to the PDB. |
| Oracle Database PDB | Activity | Row ID Ranges Full Index Scans | |
| Oracle Database PDB | Session | Sessions Count | |
| Oracle Database PDB | Activity | Short Table Scans | |
| Oracle Database PDB | Disk IO | Single Blocks Read | |
| Oracle Database PDB | General | Snapshot Parent Container ID | This column shows the container ID of the primary PDB that this PDB is a snapshot clone of. This column shows a non-zero value only if the PDB is a snapshot clone. For all other cases, it shows a value of 0. |
| Oracle Database PDB | Session | Sniped User Sessions | |
| Oracle Database PDB | Activity | Total Parse Count | |
| Oracle Database PDB | Session | Unblocked User Sessions | |
| Oracle Database PDB | Session | Unknown Wait Time User Sessions | |
| Oracle Database PDB | Disk | Used Space | Disk space used by the container, including both data and temp files. |
| Oracle Database PDB | Activity | User Calls | |
| Oracle Database PDB | Activity | User Commits | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database PDB | Activity | User Rollback Undo Records Applied | |
| Oracle Database PDB | Activity | User Rollbacks | |
| Oracle Database PDB | Session | User Types | |
| Oracle Database PDB | Session | Waiting User Sessions | |
| Oracle Database PDB | Disk IO | Write Time | |
| Oracle Database Query | Activity | Application Wait Time | |
| Oracle Database Query | Activity | Average Time | |
| Oracle Database Query | Activity | Buffer Gets | |
| Oracle Database Query | Activity | Child Count | |
| Oracle Database Query | Activity | Concurrency Wait Time | |
| Oracle Database Query | Activity | CPU Time | |
| Oracle Database Query | Activity | Direct Writes | |
| Oracle Database Query | Activity | Disk Reads | |
| Oracle Database Query | Activity | Elapsed Time | |
| Oracle Database Query | Activity | Executions | |
| Oracle Database Query | General | Instance ID | |
| Oracle Database Query | Activity | I/O Interconnect Data | |
| Oracle Database Query | Activity | Last Active Time | |
| Oracle Database Query | Activity | Last Load Time | |
| Oracle Database Query | Activity | Name | |
| Oracle Database Query | Activity | Optimized Physical Read Requests | |
| Oracle Database Query | Activity | Parallel Server Executions | |
| Oracle Database Query | Activity | Parse Calls | |
| Oracle Database Query | Activity | Physical Read Data | |
| Oracle Database Query | Activity | Physical Read Requests | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database Query | Activity | Physical Write Data | |
| Oracle Database Query | Activity | Physical Write Requests | |
| Oracle Database Query | General | Rank | |
| Oracle Database Query | Activity | Rows Processed | |
| Oracle Database Query | Activity | Runtime Memory Used | |
| Oracle Database Query | General | Shareable Memory | |
| Oracle Database Query | General | SQL ID | |
| Oracle Database Query | Activity | SQL Text | |
| Oracle Database Query | Activity | Total Locks | |
| Oracle Database Query | Activity | Total Pins | |
| Oracle Database Query | Activity | User I/O Wait Time | |
| Oracle Database Query | Activity | Users Executing | |
| Oracle Database Redo Log File | Status | Status Extended | |
| Oracle Database Redo Log File | General | Group Number | |
| Oracle Database Redo Log File | General | Member Name | |
| Oracle Database Redo Log File | General | Sequence Number | |
| Oracle Database Redo Log File | General | Size | |
| Oracle Database Redo Log File | Status | Status | |
| Oracle Database Redo Log File | General | Thread Number | |
| Oracle Database Service | Performance | Application Wait Time | The total wait time for waits that belong to the Application wait class |
| Oracle Database Service | Performance | Cluster Wait Time | The total wait time for waits that belong to the Cluster wait class |
| Oracle Database Service | Memory | Multipass Work Area Executions | Total number of times the work area size was too small |
| Oracle Database Service | Performance | Concurrency Wait Time | The total wait time for waits that belong to the Concurrency wait class |
| Oracle Database Service | Disk | Database Block Changes | Total number of changes that were part of an update or delete operation that were made to all blocks in the SGA. |
| Oracle Database Service | Performance | Executions | Total number of calls (user and recursive) that executed SQL statements |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database Service | Disk | Global Cache Consistent Real Blocks Received | Total number of blocks received |
| Oracle Database Service | Disk | Global Cache Consistent Real Block Receive Time | The total time required for consistent read requests to complete |
| Oracle Database Service | Disk | Global Cache Current Block Receive Time | Total time required for consistent read requests to complete |
| Oracle Database Service | Disk | Global Cache Current Blocks Received | Number of current blocks received from the holding instance over the hardware interconnect. |
| Oracle Database Service | Details | Logons | Total number of logons since the instance started |
| Oracle Database Service | Memory | One Pass Work Area Executions | Total number of times the work area size was oversize |
| Oracle Database Service | Memory | Optimal Work Area Executions | Total number of times the work area size was optimal |
| Oracle Database Service | Disk | Physical Reads | Total number of data blocks read from disk |
| Oracle Database Service | Disk | Physical Writes | Total number of data blocks written to disk |
| Oracle Database Service | Redo | Redo Size | Total amount of redo generated in bytes |
| Oracle Database Service | General | Name | |
| Oracle Database Service | Disk | Session Logical Reads | The sum of "db block gets" plus "consistent gets". This includes logical reads of database blocks from either the buffer cache or process private memory. |
| Oracle Database Service | Performance | SQL Execute Elapsed Time | The amount of elapsed time SQL statements are executing |
| Oracle Database Service | Performance | User Calls | Number of user calls such as login, parse, fetch, or execute |
| Oracle Database Service | Redo | User Commits | Number of user commits |
| Oracle Database Service | Performance | User IO Wait Time | The total wait time for waits that belong to the User I/O wait class |
| Oracle Database Service | Redo | User Rollbacks | Number of times users manually issue the ROLLBACK statement or an error occurs during a user's transactions |
| Oracle Database Tablespace | General | Allocated Capacity Utilization | The percentage of the allocated capacity that is currently used |
| Oracle Database Tablespace | General | Allocated Size | Current size of the file |
| Oracle Database Tablespace | General | Capacity Utilization | The percentage of the maximum capacity that is currently used |
| Oracle Database Tablespace | General | Container ID | |
| Oracle Database Tablespace | General | Contents | |
| Oracle Database Tablespace | General | Free Allocated Space | Amount of currently allocated space that is unused |
| Oracle Database Tablespace | General | Free Space | Unused space left for user data. This will include as-yet unallocated space. |
| Oracle Database Tablespace | Uncategorized | Global Name | |

| Resource Kind | Resource Group | Resource Attribute | Attribute Type |
|---|---|---|---|
| Oracle Database Tablespace | General | Maximum Space | Maximum size to which the file can grow |
| Oracle Database Tablespace | Status | Status | |
| Oracle Database Tablespace | General | Tablespace Name | |
| Oracle Database Tablespace | General | Used Space | Amount of space that is currently in use |

## Management Pack for PostgreSQL

The **Management Pack for PostgreSQL** is an embedded adapter for VMware Cloud Foundation Operations, collecting performance data from your PostgreSQL systems and providing predictive analytics and real-time information about problems in your infrastructure—all within the VCF Operations user interface.

### Notable Features

The Management Pack harnesses powerful VCF Operations features such as:

- **Comprehensive monitoring dashboards**
- **Key performance metrics** such as Memory, CPU, Disk, Waits, Sessions, Queries, and Tablespace
- **Out-of-the-box reports** for Database, Instance, Slow Query, and Throughput
- **Relationship mapping** between PostgreSQL and the VMware virtual layer
- **Detailed alerts and recommendations**

### System Requirements for Management Pack for PostgreSQL

Before installing and configuring the Management Pack, ensure your system meets the following requirements.

For compatibility between products, refer to:

VMware Cloud Foundation Operations Interoperability for PostgreSQL

-OR-

Third Party Solution Interoperability for PostgreSQL

| | PostgreSQL Requirements |
|---|---|
| Connection | IP address or hostname of **PostgreSQL instance**<br>**Default Port**: 5432 (TCP)<br>**Database name**<br>**Note:**<br>By default, the Management Pack will connect to 'pg_ database' and retrieve available databases through it. If another database is specified when adding an adapter instance, it will connect to that database only. |
| Credentials | PostgreSQL **User Name/Password**<br>The Management Pack requires a least-privileged user (LPU) account of read-only*. For details on creating an LPU, see: Configuring Database Permissions and the Least-Privileged User.<br>**Note:**<br>*NOTE: Tablespace and query data are not returned for PostgreSQL read-only users. Assigning the monitoring user/role a "super user" attribute will correct this limitation. Alternatively, using our defined LPU script will allow you to bypass assigning the "super user" attribute. See: Configuring Database Permissions and the Least-Privileged User. |

**Note:**

Prior to configuring an adapter instance for the Management Pack, ensure the required PostgreSQL views, tables, and functions are added for resources being monitored. See: Configuring Database Permissions and the Least-Privileged User for more information.

Configuring Database Permissions and the Least-Privileged User

The following topic outlines the required views, tables, and functions needed to properly monitor each PostgreSQL database resource.

- PostgreSQL Required Views, Tables, and Functions
- Creating a Least-Privileged User (LPU)

## PostgreSQL Required Views, Tables, and Functions

The table below lists PostgreSQL database resource kinds and their required views, tables, or functions needed to monitor each resource correctly. Prior to configuring an adapter instance for the Management Pack, ensure the following PostgreSQL function/query functionality is added for your PostgreSQL resources.

**Note:**

To help automate the assignment of the following permissions, you can run the script provided in Creating a Least-Privileged User (LPU).

| Resource Type | View/Table/Function Name | Required? | Version |
|---|---|---|---|
| Instance | pg_stat_bgwriter | Yes | 9.0+ |
| Database | pg_stat_database | Yes | 9.0+ |
| Database | pg_database | Yes | 9.0+ |
| Database | pg_stat_database_conflicts | Yes | 9.1+ |
| Database | pg_tablespace | Yes | 9.0+ |

| Resource Type | View/Table/Function Name | Required? | Version |
|---|---|---|---|
| Tablespace* | pg_user* | Yes | 9.0+ |
| Table | pg_stat_user_tables | No | 9.0+ |
| Table | pg_statio_user_tables | No | 9.0+ |
| Table | pg_tables | No | 9.0+ |
| Trigger | pg_trigger | No | 9.1+ |
| Index | pg_stat_user_indexes | No | 9.0+ |
| Index | pg_statio_user_indexes | No | 9.0+ |
| Function | pg_stat_user_functions | No | 9.0+ |
| Sequence | pg_statio_all_sequences | Yes | 9.0+ |
| Session | pg_stat_activity | Yes | 9.2+ |
| Query* | pg_stat_statements* | Yes | 9.0+ |

**Note:**

*NOTE: For security, an administrative-level monitoring user (i.e., “super user”) is required to view the SQL text or queryid of queries executed by other users. An “insufficient privileges” error will be returned in the Query text field if a read-only user is used. Tablespace data also requires an administrative-level monitoring user. However, using our defined LPU script will allow you to bypass assigning the "super user" attribute. See: Creating a Least-Privileged User (LPU).

**Function Tracking**: To track functions, `pg_stat_user_functions` requires the `log_statement_stats` field to be modified (in postgresql.conf) to allow for any function tracking to happen.

**Reference:** https://www.postgresql.org/docs/current/static/runtime-config-statistics.html

**Query Tracking**: To track queries, these settings need to be added to the `shared_preload_libraries` settings:

```
shared_preload_libraries = 'pg_stat_statements'
pg_stat_statements.max = 10000
pg_stat_statements.track = all
```

You also need to create the extension in the database. The extension is database bound and must be created for each database, even though it pulls data from each:

```
CREATE EXTENSION IF NOT EXISTS pg_stat_statements;
```

You can check if it is installed on the database by calling:

```
SELECT * FROM pg_available_extensions WHERE name = 'pg_stat_statements' AND installed_version IS NOT NULL;
```

## Creating a Least-Privileged User (LPU)

To help automate the creation of a least-privileged user (LPU) for your PostgreSQL instance, run the following script on the database you plan to monitor. If you're monitoring multiple databases, run the script on the **Postgres** database.

```
CREATE SCHEMA bluemedora;
CREATE EXTENSION IF NOT EXISTS pg_stat_statements;
CREATE OR REPLACE FUNCTION bluemedora.pg_stat_statements() RETURNS SETOF pg_stat_statements AS
$$
SELECT * FROM public.pg_stat_statements;
$$ LANGUAGE sql VOLATILE SECURITY DEFINER;

CREATE OR REPLACE FUNCTION public.explain_this (
      l_query text,
      out explain json
    )
RETURNS SETOF json AS
```

```
$$
BEGIN
  RETURN QUERY EXECUTE 'explain (format json) ' || l_query;
END;
$$
LANGUAGE 'plpgsql'
VOLATILE
RETURNS NULL ON NULL INPUT
SECURITY DEFINER
COST 100 ROWS 1000;

CREATE USER lpu WITH PASSWORD 'tmppassword';
GRANT SELECT ON pg_database TO lpu;
GRANT SELECT ON pg_stat_bgwriter TO lpu;
GRANT SELECT ON pg_stat_database TO lpu;
GRANT SELECT ON pg_stat_user_indexes TO lpu;
GRANT SELECT ON pg_stat_user_tables TO lpu;
GRANT SELECT ON pg_statio_all_sequences TO lpu;
GRANT SELECT ON pg_statio_user_indexes TO lpu;
GRANT SELECT ON pg_statio_user_tables TO lpu;
GRANT SELECT ON pg_tables TO lpu;
GRANT SELECT ON pg_tablespace TO lpu;
GRANT SELECT ON pg_user TO lpu;
GRANT SELECT ON pg_stat_replication TO lpu;
GRANT SELECT ON pg_stat_database_conflicts TO lpu;
GRANT SELECT ON pg_trigger TO lpu;
GRANT SELECT ON pg_stat_activity TO lpu;
GRANT SELECT ON pg_stat_statements TO lpu;
GRANT USAGE ON SCHEMA bluemedora TO lpu;
GRANT EXECUTE ON FUNCTION public.explain_this(l_query text, out explain text) TO lpu;
```

**Note:**

PostgreSQL databases previous to 9.2 may need to edit the script above.

Once complete, you can ensure the user has the EXPLAIN privilege by running:

```
SELECT exists( SELECT 1 FROM information_schema.role_routine_grants WHERE routine_name='explain_this' AND
 privilege_type='EXECUTE' AND grantee='explainer');
```

This query should return a single row with a single column of "exists" with a value of "true".

**Note:**

If the user has permissions to read from your tables and views, they do not require this special EXPLAIN permission.

### Troubleshooting the Management Pack

Once you have finished Installing and Configuring the Management Pack, check the **Collection State** and **Collection Status** for the adapter instance on the **Solutions** page. If they show as **Not Collecting** or **No Data Receiving**, respectively, the following troubleshooting topics can help you diagnose and correct common problems with your adapter instance.

- Test Connection Errors
- Adapter and Collector Log Files

#### Test Connection Errors

When Creating an Adapter Instance, click **Test Connection** after entering all of your configuration settings to ensure VMware Cloud Foundation Operations can connect to the specified host.

The following errors are possible:

Wrong Credentials

**Error:** Unable to establish a valid connection to target system. Connection authorization failure occurred. Reason: User ID or Password invalid.

**Action:** Ensure the correct PostgreSQL and VCF Operations User Name and Password was entered when Creating a Credential.

Wrong Host

**Error:** Unable to establish a valid connection to target system. Required property is unknown host.

**Action:** Ensure the correct hostname or IP address of the PostgreSQL instance was entered when Creating an Adapter Instance.

Wrong Port

**Error:** Unable to establish a valid connection to target system. Connection authorization failure occurred. Error opening socket server <name> on port with message: Connection refused.

**Action:** Check your PostgreSQL database system and ensure the connection port number matches the port number entered (if other than default of 5432) when Creating an Adapter Instance

Wrong Database

**Error:** Connection authorization failure occurred. The application server rejected establishment of the connection. An attempt was made to access a database, <name>, which was either not found or does not support transactions.

**Action:** Ensure the database name provided matches the PostgreSQL system. Multiple databases should be separated by commas in the **Manage Solution** window.

SSL Not Enabled

**Error**: SSL connection is not enabled.

**Action**: Under Advanced Settings, set SSL mode to be Enabled (required).

Server Does Not Support SSL

**Error**: The server does not support SSL connections.

**Action**: Under Advanced Settings, set SSL mode setting to Disabled.

### Adapter and Collector Log Files

You can review adapter errors in the adapter and collector log files, which can be viewed in the VCF OperationsUI or in an external log viewer.

- **Adapter log files** are located in the $VCOPS_BASE/user/log/adapters/PostgreSQLAdapter/ folder.
- **Collector log files** are located in the $VCOPS_BASE/user/log/ folder.

To set the base log level for the collector in vROps:

1. Navigate to **Administration > Logs**.
2. Expand the **PRIMARY** folder, then select the **COLLECTOR** folder.
3. Click the **Edit Properties** icon.
4. In the **Root logger level** drop-down menu, select the desired option.

- The default logging level is **warning.**
- To troubleshoot issues, set the logging level to **info.**
- To view detailed messages, including micro steps, queries, and returned results, set the logging level to **debug.**

> **Note:**
>
> If you set the logging level to **debug,** log files can become large very quickly. Set the logging level to **debug** only for short periods of time.

For more information about viewing log files and modifying levels, refer to the VCF Operations online help.

### Using the Management Pack

The Management Pack for PostgreSQL includes the following out-of-the-box features, harnessing the power of the VCF Operations user interface:

- Inventory Tree (Traversal Spec)
- Dashboards
- Views
- Reports
- Alerts
- Metrics

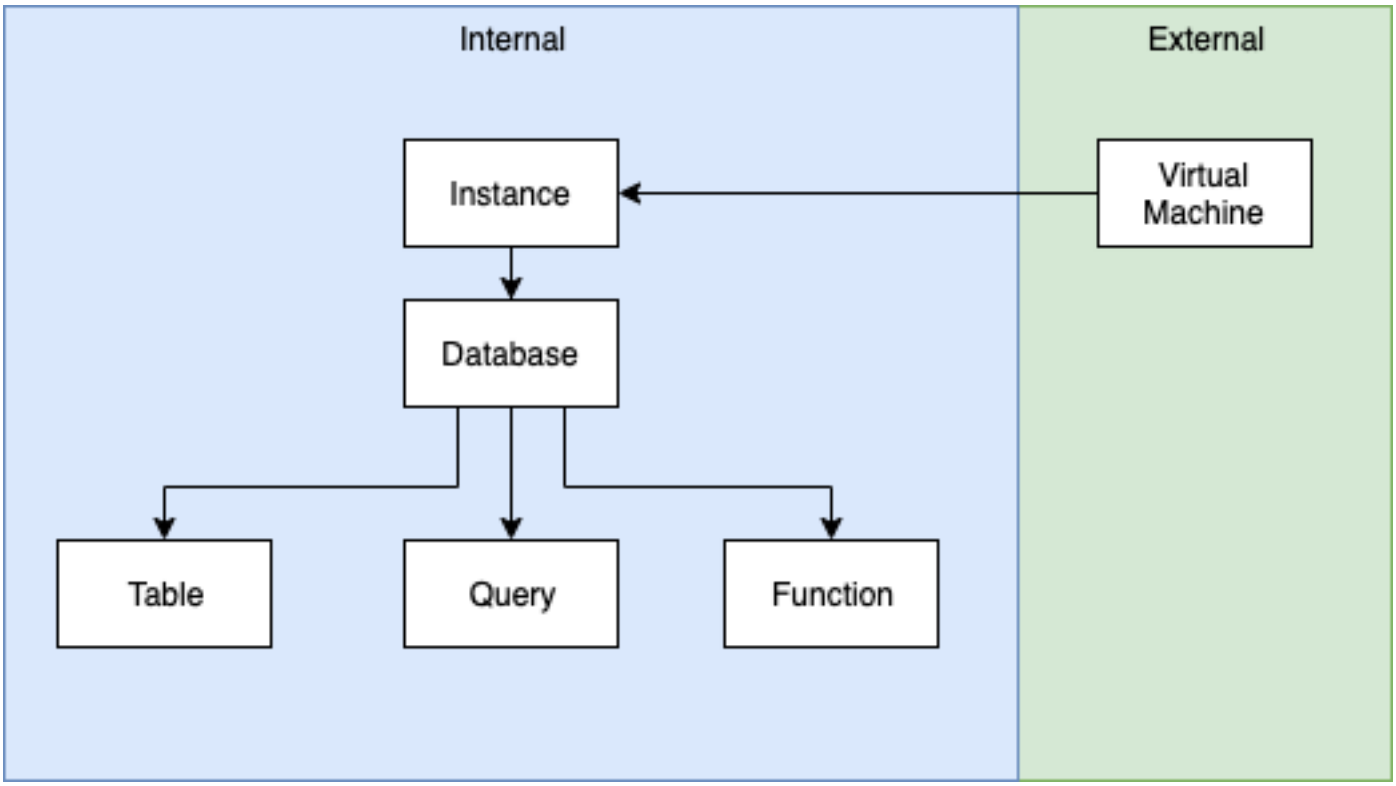
#### Inventory Tree (Traversal Spec), Resources and Relationships

The Inventory Tree (Traversal Spec) feature in VCF Operations allows you to easily navigate your PostgreSQL environment. The hierarchical structure implicitly shows relationships among resources and enables quick drill-downs to root-cause issues.

To access your PostgreSQL Inventory Tree, navigate to **Environment** > **Inventory Trees**.

**Resources and Relationships**

**Figure 485: PostgreSQL Relationships Diagram**




#### Dashboards

Dashboards are the primary feature for monitoring and troubleshooting PostgreSQL data problems from within VCF Operations.

The following dashboards are included in the Management Pack for PostgreSQL:

- **PostgreSQL Database Drill Down**

- **PostgreSQL Infrastructure Drill Down**

### Views

The Management Pack for PostgreSQL creates views that allow the user to view statistics of metrics for various PostgreSQL resources. The views help give a broad picture of the entire system, as opposed to a more in depth view.

The following views are available in the Management Pack:

| View | Type | Description |
|---|---|---|
| PostgreSQL Database Overview | List | Provides an overview of database performance metrics. |
| PostgreSQL Instance Overview | List | Provides an overview of instance performance metrics. |
| PostgreSQL Slowest Queries | List | Provides a list of slowest queries metrics. |
| PostgreSQL Throughput | List | Provides a list of throughput metrics. |

To access the Management Pack views:

1. Navigate to **Environment > All Objects > PostgreSQL**.
2. Double-click on the desired object (resource).
3. Select the **Details** tab, then **Views.**

The available views for that resource are listed and can be selected.

### Reports

The Management Pack for PostgreSQL contains four out-of-the-box reports that you can export and easily share with key stakeholders in either .pdf or .csv formats.

The following reports are available in the Management Pack:

| Report | Description |
|---|---|
| PostgreSQL Databases Report | Provides an overview of database performance metrics. |
| PostgreSQL Instance Report | Provides an overview of instance performance metrics. |
| PostgreSQL Slowest Queries Report | Provides a list of slowest queries metrics. |
| PostgreSQL Throughput Report | Provides a list of throughput metrics. |

To access Management Pack reports:

1. Navigate to **Environment > All Objects > PostgreSQL**.
2. Double-click on the desired object (resource).
3. Select the **Reports** tab, then **Report Templates**.

The available report for the selected resource is listed and can be selected.

To run a selected report, click the **Run Template** icon.

### Alerts

The Management Pack for PostgreSQL pulls PostgreSQL database health indicators and displays them in VCF Operations as alerts. See the table below for the list of alerts available in the Management Pack.

**Table 1668: Alerts List**

| Name | Description | Recommendation |
|---|---|---|
| Max Written Clean (Per Second) is Higher than Normal | The number of times the background writer has overwritten the maximum number of buffers (when a checkpoint is performed) is much higher than normal. | Consider increasing the value of “bgwriter_lru_maxpages” in postgresql.conf to increase performance. |
| Tablespace Size is Higher than Normal | The size of your Database is much higher than normal | Consider running a VACUUM to recover disk space. |
| Database Size is Higher than Normal | The size of your Database is much higher than normal | Consider running a VACUUM to recover disk space. |
| Table Size is Higher than Normal | The size of your Table is much higher than normal | Consider running a VACUUM to recover disk space. |
| Temp Files Created is Higher than Normal | The number of temporary files created is higher than normal. This can cause slower performance and I/O contention. | Consider increasing the “work_mem” in the postgresql.conf file. |
| Dead Rows is Higher than Normal | The dead rows for this table are higher than normal. | To clear the dead rows, consider running “VACUUM FULL” or “CLUSTER” during a maintenance window. |
| Queries Cancelled are Higher than Normal | The number of queries that are being canceled is higher than normal. | Consider enabling “hot_standby_feedback” in the postgresql.conf file to avoid canceled queries. Warning: This could create bloat in the primary database. |

Metrics

The Management Pack for PostgreSQL collects the metrics listed in the table below, according to PostgreSQL resource kinds.

| Resource Kind | Resource Group | Resource Attribute |
|---|---|---|
| PostgreSQL Instance | Tablespaces\|General | Tablespace Size |
| PostgreSQL Instance | Tablespaces\|Configuration | Tablespace Name |
| PostgreSQL Instance | Tablespaces\|Configuration | Tablespace Location |
| PostgreSQL Instance | Tablespaces\|Configuration | Tablespace Location |
| PostgreSQL Instance | Tablespaces\|Configuration | Tablespace Owner |
| PostgreSQL Instance | Tablespaces\|Configuration | Privileges |
| PostgreSQL Instance | Tablespaces\|Configuration | User Name |
| PostgreSQL Instance | Tablespaces\|Configuration | IP Address |
| PostgreSQL Instance | Tablespaces\|Configuration | Port |
| PostgreSQL Instance | Tablespaces\|Configuration | Instance Name |
| PostgreSQL Instance | Buffers | Buffers Allocated |
| PostgreSQL Instance | Buffers | Times Backend Executed Own fsync |
| PostgreSQL Instance | Buffers | Times Background Writer Stopped Due to too Many Buffers |
| PostgreSQL Instance | Buffers | Buffers Written by Backend |
| PostgreSQL Instance | Buffers | Buffers Written by Background Writer |
| PostgreSQL Instance | Buffers | Buffers Written During Checkpoint |
| PostgreSQL Instance | Configuration | Name |
| PostgreSQL Instance | Configuration | Server IP |
| PostgreSQL Instance | Configuration | Server Port |

| Resource Kind | Resource Group | Resource Attribute |
|---|---|---|
| PostgreSQL Instance | Configuration | Last Time Stats Reset |
| PostgreSQL Instance | Configuration | Version |
| PostgreSQL Instance | Checkpoints | Scheduled Checkpoints Performed |
| PostgreSQL Instance | Checkpoints | Time Synchronizing Checkpoint Files to Disk |
| PostgreSQL Instance | Checkpoints | Requested Checkpoints Performed |
| PostgreSQL Instance | Checkpoints | Time Writing Checkpoint Files to Disk |
| PostgreSQL Database | Sessions | Number of Active Connections |
| PostgreSQL Database | Sessions | Total Sessions |
| PostgreSQL Database | Sessions | Active Sessions |
| PostgreSQL Database | Sessions | Idle Sessions |
| PostgreSQL Database | Sessions | Idle Sessions in Transaction |
| PostgreSQL Database | Sessions | Idle Sessions in Transaction (Aborted) |
| PostgreSQL Database | Sessions | Sessions Executing Fastpath Function Call |
| PostgreSQL Database | Sessions | Sessions with Unknown State (Track Activities is Disabled) |
| PostgreSQL Database | General | Canceled Queries Due to Conflicts |
| PostgreSQL Database | General | Deadlock Count |
| PostgreSQL Database | General | Database Size |
| PostgreSQL Database | General | Last Timestamp Stats Reset |
| PostgreSQL Database | Activity | Rows Deleted |
| PostgreSQL Database | Activity | Rows Returned |
| PostgreSQL Database | Activity | Rows Updated |
| PostgreSQL Database | Activity | Rows Fetched |
| PostgreSQL Database | Activity | Rows Inserted |
| PostgreSQL Database | Configuration | Tablespace Name |
| PostgreSQL Database | Configuration | Database Name |
| PostgreSQL Database | Configuration | Database ID |
| PostgreSQL Database | Configuration | IP Address |
| PostgreSQL Database | Configuration | Port |
| PostgreSQL Database | Conflicts | Queries Canceled Due to Deadlocks |
| PostgreSQL Database | Conflicts | Queries Canceled Due to Old Snapshot |
| PostgreSQL Database | Conflicts | Queries Canceled Due to Lock Timeouts |
| PostgreSQL Database | Conflicts | Queries Canceled Due to Dropped Tablespaces |
| PostgreSQL Database | Conflicts | Queries Canceled Due to Pinned Buffers |
| PostgreSQL Database | Transactions | Transaction Rollbacks |
| PostgreSQL Database | Transactions | Transaction Commits |
| PostgreSQL Database | Disk IO | Temporary Bytes Written |
| PostgreSQL Database | Disk IO | Temporary Files Created |
| PostgreSQL Database | Disk IO | Buffer Hit Ratios |
| PostgreSQL Database | Disk IO | Block Reads/Buffer Misses |
| PostgreSQL Database | Disk IO | Time Spent Writing Data |
| PostgreSQL Database | Disk IO | Buffer Hits |

| Resource Kind | Resource Group | Resource Attribute |
|---|---|---|
| PostgreSQL Database | Disk IO | Time Spent Reading Data |
| PostgreSQL Database | Disk IO | Blocks Read by Sequences |
| PostgreSQL Database | Disk IO | Blocks Hit by Sequences |
| PostgreSQL Table | Triggers\|Configuration | Trigger Enabled Status |
| PostgreSQL Table | Triggers\|Configuration | Trigger ID |
| PostgreSQL Table | Triggers\|Configuration | Trigger Name |
| PostgreSQL Table | Triggers\|Configuration | Type of Trigger Fire |
| PostgreSQL Table | Triggers\|Configuration | Arguments Passed to Trigger |
| PostgreSQL Table | Triggers\|Configuration | Table ID |
| PostgreSQL Table | Triggers\|Configuration | Table Name |
| PostgreSQL Table | Triggers\|Configuration | Function Name |
| PostgreSQL Table | Triggers\|Configuration | Function ID |
| PostgreSQL Table | Triggers\|Configuration | IP Address |
| PostgreSQL Table | Triggers\|Configuration | Port |
| PostgreSQL Table | Indexes\|Activity | Number of Block Reads |
| PostgreSQL Table | Indexes\|Activity | Number of Table Rows Fetched by Scan |
| PostgreSQL Table | Indexes\|Activity | Number of Buffer Hits |
| PostgreSQL Table | Indexes\|Activity | Number of Index Entries Returned |
| PostgreSQL Table | Indexes\|Activity | Number of Scans |
| PostgreSQL Table | Indexes\|Configuration | Index ID |
| PostgreSQL Table | Indexes\|Configuration | Index Name |
| PostgreSQL Table | Indexes\|Configuration | Table Name |
| PostgreSQL Table | Indexes\|Configuration | Schema Name |
| PostgreSQL Table | Indexes\|Configuration | Table ID |
| PostgreSQL Table | Indexes\|Configuration | IP Address |
| PostgreSQL Table | Indexes\|Configuration | Port |
| PostgreSQL Table | General | Table Size |
| PostgreSQL Table | Activity | Number of Rows Updated |
| PostgreSQL Table | Activity | Number of Sequential Scans |
| PostgreSQL Table | Activity | Last Time Table Analyzed by Auto Daemon |
| PostgreSQL Table | Activity | Number of Dead Rows |
| PostgreSQL Table | Activity | Last Time Table Vacuumed by Auto Daemon |
| PostgreSQL Table | Activity | Number of Rows Deleted |
| PostgreSQL Table | Activity | Last Time Table Manually Analyzed |
| PostgreSQL Table | Activity | Number of Index Scans |
| PostgreSQL Table | Activity | Number of Rows Inserted |
| PostgreSQL Table | Activity | Number of Times Table Manually Vacuumed |
| PostgreSQL Table | Activity | Number of Times Table Vacuumed By Auto Daemon |
| PostgreSQL Table | Activity | Last Time Table Manually Vacuumed |
| PostgreSQL Table | Activity | Number of Rows Fetched by Index Scans |
| PostgreSQL Table | Activity | Number of Times Table Manually Analyzed |

| Resource Kind | Resource Group | Resource Attribute |
|---|---|---|
| PostgreSQL Table | Activity | Number of Rows Modified Since Last Analyzed |
| PostgreSQL Table | Activity | Number of Times Table Analyzed by Auto Daemon |
| PostgreSQL Table | Activity | Rows fetched by Sequential Scans |
| PostgreSQL Table | Activity | Number of Live Rows |
| PostgreSQL Table | Activity | Number of Rows Hot Updated (No separate index update requried) |
| PostgreSQL Table | Configuration | Table ID |
| PostgreSQL Table | Configuration | Tablespace Name |
| PostgreSQL Table | Configuration | Database Name |
| PostgreSQL Table | Configuration | Table Name |
| PostgreSQL Table | Configuration | Table Schema |
| PostgreSQL Table | Configuration | IP Address |
| PostgreSQL Table | Configuration | Port |
| PostgreSQL Table | Disk IO | Number of Block Reads from Indexes |
| PostgreSQL Table | Disk IO | Number of Block Reads |
| PostgreSQL Table | Disk IO | Number of Buffer Reads |
| PostgreSQL Table | Disk IO | Number of Block Reads from TOAST Table Indexes |
| PostgreSQL Table | Disk IO | Number of Buffer Hits from TOAST Table |
| PostgreSQL Table | Disk IO | Number of Buffer Hits from Indexes |
| PostgreSQL Table | Disk IO | Number of Buffer Hits from TOAST Table Indexes |
| PostgreSQL Table | Disk IO | Number of Block Reads from TOAST Table |
| PostgreSQL Function | Activity | Total Time Spent Calling This and All Called Functions |
| PostgreSQL Function | Activity | Average Total Call Time |
| PostgreSQL Function | Activity | Times Called |
| PostgreSQL Function | Activity | Time Spent on This Function Alone (not referenced functions) |
| PostgreSQL Function | Configuration | Schema Name |
| PostgreSQL Function | Configuration | Function Name |
| PostgreSQL Function | Configuration | Function ID |
| PostgreSQL Function | Configuration | IP Address |
| PostgreSQL Function | Configuration | Port |
| PostgreSQL Function | Configuration | Database Name |
| PostgreSQL Function | Configuration | Rank |
| PostgreSQL Query | Activity | Query Text |
| PostgreSQL Query | Activity | Total Time |
| PostgreSQL Query | Activity | Average Time Spent Per Call |
| PostgreSQL Query | Activity | Number of Calls |
| PostgreSQL Query | Activity | Rows Returned |
| PostgreSQL Query | Configuration | User Name |
| PostgreSQL Query | Configuration | Database Name |
| PostgreSQL Query | Configuration | IP Address |

| Resource Kind | Resource Group | Resource Attribute |
|---|---|---|
| PostgreSQL Query | Configuration | Port |
| PostgreSQL Query | Configuration | Rank |
| PostgreSQL Query | Disk IO | Local Block Cache Hits |
| PostgreSQL Query | Disk IO | Shared Block Reads |
| PostgreSQL Query | Disk IO | Shared Blocks Written |
| PostgreSQL Query | Disk IO | Total Time Spent Writing Blocks |
| PostgreSQL Query | Disk IO | Shared Blocks Dirty Read |
| PostgreSQL Query | Disk IO | Local Block Reads |
| PostgreSQL Query | Disk IO | Shared Block Cache Hits |
| PostgreSQL Query | Disk IO | Local Blocks Dirty Read |
| PostgreSQL Query | Disk IO | Total Time Spent Reading Blocks |
| PostgreSQL Query | Disk IO | Local Blocks Written |

## Management Pack for ServiceNow

The **Management Pack for ServiceNow** is an embedded adapter for VMware Cloud Foundation Operations, that contextualizes VCF Operations data by collecting ServiceNow CMDB configuration items (CIs) and mapping those VCF Operations objects based on a user-defined schema.

### Notable Features

The Management Pack for ServiceNow provides the following notable features:

- User-defined CMDB visibility in VCF Operations
- **Automatically generated dashboards** based on user-defined configuration

**See also:**

System Requirements

### System Requirements for Management Pack for ServiceNow

| | ServiceNow Requirements |
|---|---|
| Version(s) | All ServiceNow supported versions (Kingston+) |
| Connection | A ServiceNow host address<br>A configuration file<br>**Note:**<br>ServiceNow calls are made to the**Table API**and**CMDB Meta API**. |
| Credentials | A least privileged user with **itil** credentials is required (see: Creating a Least Privileged User |

Creating a Least Privileged User

A least privileged user account must have the following permissions:

All Configurations (required to validate the other ACLs)

- sys_user_has_role
- security_acl_detail
- sys_security_operation

Resource Grouping ACLs

- cmdb_metadata_hosting
- cmdb_metadata_reference
- cmdb_metadata_containment
- sys_dictionary
- sys_dictionary.*
- sys_glide_object
- svc_ci_assoc (only if you are using the association table in any of the group configurations)
- each table that is in the configuration json (`cmdb_ci_vmware_instance` for example)

Alerting ACLs:

- sys_choice
- sys_choice.*
- sys_dictionary
- sys_dictionary.*
- sys_glide_object
- each table that is in the configuration json if CI mapping is used (`cmdb_ci_vmware_instance` for example)
- One of the following depending on which option is specified in your configuration file:
  - incident (`read` and `write`)
  - em_alert (`read` and `write`)
  - em_event (`read` and `write`)

CMDB Sync ACLs

- sys_db_object
- cmdb_reconciliation_definition
- sys_choice
- sys_choice.*
- cmdb_rel_type
- each table in Synced Resources (`read` and `edit_ci_relations` are always required, `delete` is required if using a configuration which deletes CIs when they are removed from or are Not Existing in VCF Operations )

Role

- itil

> **Note:**
>
> This is a requirement for getting metadata about the tables, which we need in order to determine data types, allowable columns, etc.

### Adding an SSL Certificate to the VMware Cloud Foundation Operations Truststore

For added security, we recommend adding an SSL certificate to the VCF Operations truststore for your ServiceNow CMDB.

> **Note:**
>
> Different levels of security can be configured in the Management Pack for ServiceNow. The**least secure** configuration is to leave the default selection of**False**for the **ServiceNow SSL Configuration**Advanced Setting. In this configuration, the Management Pack will accept all SSL certificates.**For added security,**you can enable SSL by following the instructions below and selecting**True**for the**SSL Configuration**Advanced Setting when Creating an Adapter Instance.

1. Obtain the SSL certificate for your Unity host server from your Internet browser. Export the certificate as an **X.509 Certificate (PEM)**.
2. Copy the certificate to your VCF Operations machine.
3. Use ‘ssh’ to log in to the VCF Operations machine as the root user, then run the following command:

**Linux:**
$VCOPS_BASE/jre/bin/keytool -import -alias <product_alias> -file /tmp/<certfile> -keystore “$VCOPS_DATA_VCOPS/user/conf/ssl/tcserver.truststore” -storepass `grep ssltruststorePassword /storage/vcops/user/conf/ssl/storePass.properties | sed s/ssltruststorePassword=//` -trustcacerts
**Windows:**
%VCOPS_BASE%\jre\bin\keytool -import -alias <product_alias> C:\path\to\certfile -keystore “%VCOPS_DATA_VCOPS%\user\conf\ssl\tcserver.truststore” -storepass <truststore_password> -trustcacerts
**Parameter Descriptions:**
– -alias is a unique name for each key that you add (per host)
– certfile is the location where the cert file was saved

4. Run the reboot command to re-start the VCF Operations machine for the changes to take effect.

### Creating a Configuration File

By default, you must supply a configuration file in the work directory `$VCOPS_BASE/user/plugins/inbound/servicenow_adapter3/work`

The configuration file consists of an array of **resourceTypes**. Each element is explained below in detail.

```
{
  "cmdbSync": {},
  "resourceTypes": [
    {
      "vROps": {...},
      "serviceNow": {...},
      "identifierMapping": [],
      "groupTypes": [],
      "groupHierarchies": [[]]
    },
    ...
  ],
  "alertingConfigs": [
    {...},
    ...
  ]
}
```

### Examples of Configuration Files

Below are two examples of functional configuration files:

### Configuration File Example A

```
{
  "resourceTypes": [
    {
      "vROps": {
        "type": {
          "resourceKind": "VirtualMachine",
          "adapterKind": "VMWARE"
        },
        "id": {
          "type": "IDENTIFIER",
          "field": "VMEntityName"
        }
      },
      "serviceNow": {
        "type": {
          "cmdbCiClass": "cmdb_ci_vmware_instance"
```

```
        },
        "id": {
          "column": "name"
        }
      },
      "groupTypes": [
        {
          "serviceNowColumn": "u_application",
          "vROpsGroupType": "Application"
        },
        {
          "serviceNowColumn": "u_main_business_unit",
          "vROpsGroupType": "BusinessUnit",
          "booleanMapping":
            {
              "trueValue": "Main",
              "falseValue": "Other"
            }
        }
      ],
      "hierarchies": [
        [
          {
            "adapterKind": "Container",
            "resourceKind": "BusinessUnit"
          },
          {
            "adapterKind":"Container",
            "resourceKind": "Application"
          }
        ]
      ]
    }
  ]
}
```

## Configuration File Example B

```
{
  "resourceTypes": [
    {
      "vROps": {
        "type": {
          "resourceKind": "VirtualMachine",
          "adapterKind": "VMWARE"
        },
        "id": {
          "type": "PROPERTY",
          "field": "config|name"
        }
      },
      "serviceNow": {
        "type": {
          "cmdbCiClass": "cmdb_ci_vmware_instance"
        },
        "id": {
          "column": "name"
        }
```

```
      },
      "groupTypes": [
        {
          "serviceNowColumn": "u_application",
          "vROpsGroupType": "Application"
        },
        {
          "serviceNowColumn": "u_main_business_unit",
          "vROpsGroupType": "BusinessUnit",
          "booleanMapping":
          {
            "trueValue": "Main",
            "falseValue": "Other"
          }
        }
      ],
      "hierarchies": [
        [
          {
            "adapterKind": "Container",
            "resourceKind": "BusinessUnit"
          },
          {
            "adapterKind":"Container",
            "resourceKind": "Application"
          }
        ]
      ]
    }
  ]
}
```

### Configuration File Example C

```
{
 "resourceTypes": [
 {
 "vROps": {
 "type": {
 "resourceKind": "VirtualMachine",
 "adapterKind": "VMWARE"
 },
 "id": {
 "type": "IDENTIFIER",
 "field": "VMEntityName"
 }
 },
 "serviceNow": {
 "type": {
 "cmdbCiClass": "cmdb_ci_vmware_instance"
 },
 "id": {
 "column": "name"
 }
 },
 "groupTypes": [
 {
 "serviceNowColumn": "u_application",
```

```
  "vROpsGroupType": "Application"
  }
  ],
  "hierarchies": [ ]
  }
  ]
 }
```

### vROps

the `vROps` object in the resourceType defines the type of VCF Operations resource that will be assigned properties based on CMDB columns.

Example:

```
"vROps": { "type": {...}}
```

### Type

This object describes the resource and adapter kinds. Both are required.

```
"type": { "resourceKind": "VirtualMachine", "adapterKind": "VMWARE"}
```

### ServiceNow

The `serviceNow` object in the resourceType defines the type of ServiceNow CI that should be matched to the vROps resource.

```
"serviceNow": { "type": {...}}
```

### Type

This object describes the CI Class of the CI that will be matched to the defined vROps ResourceKind.

```
"type": { "cmdbCiClass": "cmdb_ci_vmware_instance"}
```

### Identifier Mapping

An array of mappings. Each mapping references one ServiceNow column which maps to a vROps Property, Identifier, Name, or vROps UUID. In most cases, using a unique identifier in combination with the vCenter UUID is preferred.

```
{ "identifierMapping": [ { "serviceNowId": { "column": "object_id" }, "vropsId":
{ "type": "IDENTIFIER", "field": "VMEntityObjectID" } }, { "serviceNowId": { "column":
"vcenter_uuid" }, "vropsId": { "type": "IDENTIFIER", "field": "VMEntityVCID" } } ]}
```

### ServiceNow ID

The column whose value will be matched to the id specified on the defined vROps ResourceKind.

```
"id": { "column": "vm_instance_uuid"}
```

### VROps ID

This field describes the identification mechanism used to match with a ServiceNow CI. There are 3 different ways this can be defined.

- Identifier: This method matches based on the value of the specified resource identifier on the resource. Set `type` to `IDENTIFIER` and `field` to the name of the identifier on the ResourceKind.

```
"id": { "type": "IDENTIFIER", "field": "VMEntityName"}
```

- PROPERTY: This method matches based on the latest value assigned to the specified property. Set `type` to `PROPERTY` and `field` to the name of the property on the ResourceKind.

```
"id": { "type": "PROPERTY", "field": "config|name"}
```

- Name: This method uses the name of the resource in vROps as the value matched by ServiceNow CI. Set `type` to `NAME` and do not specify a `field` in order to use this method.

```
"id": { "type": "NAME"}
```

- UUID: This method uses the unique ID generated by vROps as the value matched by ServiceNow CI. Set `type` to `UUID` and do not specify a `field` in order to use this method.

```
"id": { "type": "UUID"}
```

**Group Types**

The groupTypes array contains a list of mappings from serviceNow columns to vROps groups.

```
"groupTypes": [ {...}, {...}],
```

There are two ways to assign these mappings.

- Standard: This is used for string-based properties in which the literal string value of the `serviceNowColumn` will be used to create a group named the value specified by `vROpsGroupType`. All matched resources will be assigned to the group of type `vROpsGroupType` that matches the value of the column. In the example below, if a CI has `u_application` = `myApp` in the CMDB, the vROps resource will be assigned to a group of type `Application` named `myApp`.

```
{ "serviceNowColumn": "u_application", "vROpsGroupType": "Application"}
```

- Boolean: This is used for boolean properties only. The `serviceNowColumn` and `vROpsGroupType` work as explained above, but instead of assigning the literal value of the column, the adapter will translate it to the values assigned in the `booleanMapping`. In the example below, if a CI returns `u_main_business_unit` = `true`, the vROps resource will be assigned to a group of type `Business Unit` named `Main`.

```
{ "serviceNowColumn": "u_main_business_unit", "vROpsGroupType": "Business Unit",
"booleanMapping": { "trueValue": "Main", "falseValue": "Other" }}
```

Additionally, columns may be a reference to an object in another table. To do this, 'dot notation' is used.

```
{ "serviceNowColumn": "domain_manager.manager.name", "vROpsGroupType": "Domain Manager"}
```

Boolean mappings can be used (at the top level) as well. The value returned by the referenced table will use the boolean mapping in this case

```
{ "serviceNowColumn": "domain_manager.manager.active", "vROpsGroupType": "Domain Manager",
"booleanMapping": { "trueValue": "Active", "falseValue": "Inactive" }}
```

**Group Hierarchies**

Group hierarchies organize the groups. Objects at the top of the hierarchy are considered parents of objects lower on the hierarchy. The `resourceKind` is the `vROpsGroupType` from the `groupTypes` array.

```
"groupHierarchies": [ [ { "resourceKind": "Business Unit" }, { "resourceKind":
"Application" } ]]
```

It is possible to have multiple hierarchies that reference different `vROpsGroupType`s. It is also possible for them to share some of their group types.

```
"groupHierarchies": [ [ { "resourceKind": "Business Unit" }, { "resourceKind":
"Application" } ], [ { "resourceKind": "Business Unit" }, { "resourceKind": "Other
Group" } ]]
```

In the above scenario, `Application` and `Other Group` are both related to the `Business Unit`, but are not considered directly related to one another.

Constraints:

- Hierarchies must not be cyclic (`Business Unit` cannot both be a parent and a child of `Application`). This is for both individual resourceTypes and across all resourceTypes (`Business Unit` cannot be a parent of `Application` for a VM, but a child of `Application` for a Datastore). The hierarchy object can be assigned

```
"adapterKind":
        "Container"
```

  , but this can also be omitted since it is the only allowable adapterKind.
- All groups must be present in at least one hierarchy on the `resourceType` where it is defined.
- Similarly, groups in a hierarchy must be defined as a `vROpsGroupType` on the `resourceType` where it is defined.

**Alert Sync**

Alerting configurations for the alert-sync feature define what alerts will be synced between vROps and ServiceNow, and how they will be represented in ServiceNow. The three supported representations are INCIDENT (default), ALERT, and EVENT. Once the vROps alert is present in ServiceNow, the ServiceNow Management Pack performs the following additional features:

- If an alert is cancelled in vROps, the corresponding incident or alert in ServiceNow is closed.
- If an incident or alert is closed in ServiceNow, the corresponding alert in vROps is suspended. Alerts in vROps are generally triggered automatically by breaching thresholds. Because we cannot control this, if an incident or alert is closed in ServiceNow it should be due to the underlying issue being resolved. Thus, we suspend the Alert in vROps (for a configurable length of time) to allow the alert to resolve. If the threshold(s) is still breached when the alert suspend time has ended, the alert will re-trigger and reopen the ServiceNow incident or alert.
- If a synced vROps alert is on a vROps ResourceKind that is present in one of the `resourceTypes` sections above, the ServiceNow Management Pack will create a reference from the incident/alert/event `cmdb_ci` column to the CI representing the resource in ServiceNow. Note: A resourceType section can be created with empty `groupTypes` and `hierarchies` arrays for matching to a resource type, without creating any groupings and hierarchies.
- The ServiceNow Management Pack can be configured to watch a predefined set of incident/alert/event columns and display the current values in the corresponding vROps alert as a series of notes.

**Configuration**

If alert sync is not desired, `alertingConfigs` can be omitted entirely, or an empty list can be used:

```
"alertingConfigs": []
```

A typical incident config may look like this:

```
"alertingConfigs": [ { "callerId": "vROps Service User" "propagateAlertUpdates": true,
"retrieveIncidentUpdates": true, "incidentElementsToRetrieve": ["assigned_to.name",
"state", "cmdb_ci.sys_id"], "incidentReopenState": "In Progress", "incidentCloseState":
"Closed", "vropsSuspendMinutes": 30 }]
```

Each config contains a `callerId` which identifies which config the ServiceNow incident/alert/event comes from. The `callerId` does not need to be distinct between alerting configs, however there are some performance benefits if each `alertingConfig` has a unique `callerId`. Note that in the rare case where multiple vROps instances are monitoring the same vCenter and are pointing to the same ServiceNow instance, the `callerId`s should all be distinct between the vROps instances. Note that this list is ordered

```
"alertingConfigs": [ { "callerId": "VM Admin", ... }, { "callerId": "Storage
Admin", ... }]
```

In each alerting config, the following options are supported:

| Parameter | Type | Description |
|---|---|---|
| callerId | String | The `callerId` is the ServiceNow caller assigned in vROps ServiceNow plugin configuration. This is the only required field - defaults will be used for all other fields. All incidents in ServiceNow that have this caller id will be treated according to this configuration. Two incident configuration sections cannot have the same caller id. |
| serviceNowDestination | Enum | One of `INCIDENT`, `ALERT`, or `EVENT`. Specifies which type of Service now alerting record will be created for vROps alerts in this `alertingConfig`. |
| serviceNowOpenState | String | Specifies the ServiceNow state an alerting record will be in when it is opened. The default is `New`. The value must be present in the `sys_choice` table for the alerting record's state column. |
| serviceNowReopenState | String | Specifies the ServiceNow state an alerting record will be in when it is reopened. The default is `New` for Incidents and Events and `Reopen` for Alerts. The value must be present in the `sys_choice` table for the alerting record's state column. |
| serviceNowCloseState | String | Specifies the ServiceNow state an alerting record will be in when it is closed. The default is `Closed` for Incidents and Alerts and `Closing` for Events. The value must be present in the `sys_choice` table for the alerting record's state column. |
| category | String | Optional additional metadata for incidents/alerts/events that are opened in ServiceNow. |
| subcategory | String | Optional additional metadata for incidents/alerts/events that are opened in ServiceNow. |
| businessService | String | Optional additional metadata for incidents/alerts/events that are opened in ServiceNow. |
| contactType | String | Optional additional metadata for incidents/alerts/events that are opened in ServiceNow. |
| state | String | Optional additional metadata for incidents/alerts/events that are opened in ServiceNow. Defaults to the value of the corresponding vROps Alert, mapped using `ServiceNowOpenState`, `ServiceNowReopenState`, and `ServiceNowCloseState`. |
| resolutionCode | String | Optional additional metadata for incidents/alerts/events that are opened in ServiceNow. |
| resolutionNotes | String | Optional additional metadata for incidents/alerts/events that are opened in ServiceNow. |

| Parameter | Type | Description |
|---|---|---|
| onHoldReason | String | Optional additional metadata for incidents/alerts/events that are opened in ServiceNow. |
| impact | String | Optional additional metadata for incidents/alerts/events that are opened in ServiceNow. For Incident and Alert, defaults to the value of the vROps Alert's impact. |
| urgency | String | Optional additional metadata for incidents/alerts/events that are opened in ServiceNow. |
| filters | Object | Set of filters that determine which vROps Alerts will get processed in this `alertingConfig`. See detail below. |
| retrieveUpdates | Boolean | When set to true, vROps alerts will be suspended when an incident or alert is in a closed state. In addition, any elements specified in `incidentElementsToRetrieve` will be sent to the vROps alert as notes when they change. Defaults to `true`. |
| vropsSuspendMinutes | Int | Defines how long (in whole minutes) an alert should be suspended when the corresponding ServiceNow alerting record is marked as closed. |
| serviceNowElementsToRetrieve | Array | A list of strings corresponding to table columns that are surfaced in vROps alerts as notes. Columns with dot notation are allowed. The default list is `state`, `assigned_to.name`, `priority`, and `urgency` for Incidents; `state`, `assigned_to.name`, and `priority` for Alerts; and `state` and `resolution_state` for Events. |

## CMDB Sync

A cmdbSync configuration can be added to the configuration file in order to enable this adapter to populate the CMDB with resources. The configuration object can include the following, each of which will be described in greater detail.

```
"cmdbSync": {
   "cmdbSyncMethod": "IRE_API",
   "syncMode": "POPULATE_AND_DELETE_WHEN_NOT_EXISTING",
   "objectIdentifierSource": "MOID",
   "builtInTreeEnabled": true,
   "customTreesEnabled": false,
   "additionalColumns": {},
   "nameFilter": {}
}
```

Additionally, using CMDB sync requires the user to add a Data Source to the identification rule for each resource type, otherwise columns cannot be updated via IRE.

There are a few CI Class Identification Rules that must be addressed as well.

- cmdb_ci_vcenter_datacenter:
  - Add a rule that uses `object_id` and `vcenter_uuid` together to uniquely identify the datacenter
  - If you have ever used the Discovery plugin, it adds a dependent relationship to `cmdb_ci_cloud_service_account`. Remove this dependency and allow the Datacenter to be an Independent resource.

- cmdb_ci_vcenter_datastore
  - Add a dependent containment relationship to the Datacenter (`cmdb_ci_vcenter_datastoreContains::Contained` by `cmdb_ci_vcenter_datacenter`)
- cmdb_ci_esx_server
  - Add a rule that uses `object_id` and `vcenter_uuid` together to uniquely identify the datacenter
  - If Discovery is being used, you may also need to disable the `correlation_id` rule to avoid duplicate objects.
- cmdb_ci_vcenter
  - Add a rule that uses `instance_uuid` as the unique identifier

**Custom Trees**

By default, built-in trees are enabled and custom trees are disabled. If you need to change the identification of the built-in classes, or map these resources to different CIs, disable the built-in tree and create your own custom tree to represent vCenter resources.

> **Tip:**
>
> To add columns to a built-in tree, use `additionalColumns` rather than defining a custom tree. See Additional Columns below.

To use custom trees, you must enable custom trees in the CMDB Sync Configuration. You must define each custom tree in its own json file under 'work/custom_trees'. When you make the directory 'custom_trees', make sure your vROps user has permission to read from that directory.

Format:

```
{
    "treeName": "Sample Tree Name",
    "syncClasses": []
}
```

Tree Name

This is a unique identifier for the tree. It cannot match the name of any other custom trees, or the built-in tree if the built-in tree is enabled.

Sync Classes

Sync classes defined what should be synced to the CMDB. For a tree to be valid, the chosen sync classes must include all identifiers required by the CMDB. Test connection will inform you if there are any issues with any of the sync classes.

```
{
    "classIdentifier": "Sample Class ID",
    "vROps": {...},
    "serviceNow": {...},
    "identifierMapping: {...},
    "dataColumns": [],
    "relationships": [],
    "propertyFilter": {}
}
```

`Class Identifier` This is a unique identifier for the class. It cannot match the identifier of any other custom class, or the built-in class if the built-in tree is enabled.

`vROps Definition` This is the same object required for ResourceType vROps Definitions

`ServiceNow Definition` This is the same object required for ResourceType ServiceNow Definitions

`Identifier Definition` The identifier mapping is our way of linking our resources to ServiceNow's CIs. This is a combination of identifiers which uniquely identify the resource. For example, vROps resources should use the vCenter UUID in combination with the MOID (which is what is shown in the example in the link below).

These do not have to be identifiers on either the CMDB side or the vROps side, though they often will be.

This is the same object required for ResourceType Identifier Mapping

`Data Columns` Data columns represent the columns that we sync to the CMDB that are not used as part of the identification mapping. See vROps Types and Fields for available options for the vropsType and vropsField parameters.

```
{
  "cmdbColumn": "cpu_name",
  "vropsType": "PROPERTY",
  "vropsField": "cpu|cpuModel"
}
```

`Relationships` Defines all relationships to children. The `childClassIdentifier` is the unique Class Identifier assigned to the child class. The child class must be a part of this tree. The `relationshipType` is not required if the CI Class Definition defines only one possible relationship type between the two CI Classes, but is required if there are multiple options.

```
"relationships": [
    {
      "childClassIdentifier": "CustomDatastore",
      "relationshipType": "Contains::Contained By"
    }
]
```

`Property Filter` This is used to filter resources based on properties. An example use-case would be if you wanted to map VMWARE VirtualMachine resources to `cmdb_ci_vmware_instance` CIs only when the property `summary|config|isTemplate` is set to false. You could make a second Custom Class which maps to `cmdb_ci_vmware_template` when `summary|config|isTemplate` is set to true from the same vROps ResourceType.

If this is not needed for your class, omit the block.

This sample filters only resources that are not templates.

```
"propertyFilter": {
    "conditions": [
      {
        "vropsField": "summary|config|isTemplate",
        "value": "false",
        "filterOperation": "EQUALS",
        "default": false
      }
    ],
    "joiner": "AND or OR"
}
```

`Property Filter Value` Set this to the value to search for in the string returned from the vropsField. If your field is a boolean or an integer, set it to the string representation of that boolean or integer (such as "15" or "true")

`Property Filter Operation` This specifies how we determine whether the filter condition is met.

Available options:

| Option | Behavior |
|---|---|
| EQUALS | The condition is true if the property value returned by vROps exactly matches the provided value |
| DOES_NOT_EQUAL | The condition is true if the property value returned by vROps does not match the provided value |
| CONTAINS | The condition is true if the value returned from vROps contains the provided value |
| DOES_NOT_CONTAIN | The condition is true if the value returned from vROps does not contain the provided value |

`Property Filter Default`This specifies how we should evaluate the condition if we cannot retrieve the property in vROps. Since the most common reason this could occur is for vROps to discover a new resource but not complete a collection and thus assign properties yet, we recommend defaulting to false. Using false as default will cause the filter to exclude all resources that do not have this property assigned from the sync.

`Property Filter Joiner` This specifies whether we should include the resource if all of the conditions are met (AND) or if at least one of the conditions are met (OR)

Available options:

| Option | Behavior |
|---|---|
| AND | All property filters must resolve to true in order to include the resource |
| OR | At least one of the property filters must resolve to true in order to include the resource |

**Tip:**

To map a custom tree representing another technology that vROps can monitor, such as AWS, whilst simultaneously using the built-in tree, enable custom trees and leave the built-in tree also enabled.

Custom Tree Example

```
{
  "instructions": "Copy this file to work/custom_trees before you begin to modify it. Remove all
 'instructions' objects from this json before attempting to collect using the this custom tree.",
  "treeName": "Replace this with a unique string which describes what this tree represents (such as
 'vCenter')",
  "syncClasses": [
    {
      "classIdentifier": "Replace this with a unique string which describes what this class represents (such
 as 'Virtual Machine')",
      "vROps": {
        "type": {
          "resourceKind": "Replace this with the vROps Resource Kind (such as 'Datastore')",
          "adapterKind": "Replace this with the vROps Adapter Kind (such as 'VMWARE')"
        }
      },
      "serviceNow": {
        "type": {
          "cmdbCiClass": "Replace this with the CI class in the ServiceNow CMDB (such as
 'cmdb_ci_vcenter_datastore')"
        }
      },
      "instructions": "identifierMapping can include any number of identifiers which, when taken together,
 uniquely identify the resource. If you are syncing vCenter resources, a combination of VMEntityObjectID and
 VMEntityVCID is generally recommended. These are for this adapter to map back and forth between ServiceNow
 and vROps. The columns and vROps fields referenced do not have to be identifiers for their platform, though
 they often are.",
      "identifierMapping": [
        {
          "serviceNowId": {
            "column": "The CMDB column (such as 'object_id')"
          },
          "vropsId": {
            "instructions": "See documentation for a full description of possible types and how to specify
 fields for them.",
            "type": "IDENTIFIER",
            "field": "VMEntityObjectID"
          }
```

```
        }
      ],
      "instructions": "dataColumns includes all other fields that you which to sync to the CMDB that are not
  used for our identification.",
      "dataColumns": [
        {
          "instructions": "See documentation for a full description of possible types and how to specify
  fields for them.",
          "cmdbColumn": "sample_column",
          "vropsType": "METRIC",
          "vropsField": "sample|metric|key"
        }
      ],
      "instructions": "Use the relationships array to define children of the the class. You can only reference
  classes that are also members of this tree.",
      "relationships": [
        {
          "childClassIdentifier": "Reference To Another Class In This Tree",
          "relationshipType": "This is only required if there are multiple possible relationship types between
  two CI Classes - if it is not defined, the adapter will read the relationship type from the CMDB. The format
  for this string is 'Contains::Contained by'"
        }
      ],
      "propertyFilter": {
        "instructions": "This is optional. Remove the propertyFilter object if you don't need it for your
  resource. If you do need the filter, you can define any number of conditions and whether we should join them
  all together with AND or OR. Conditions can only be made from vROps properties. Metrics, Identifiers, etc
  will not work.",
        "conditions": [
          {
            "vropsField": "The property that we will used from the vROps resource to determine if the
  condition is met (formatted|like|this)",
            "value": "The expected value for the condition. Use a string representation even if this is an int
  or boolean property",
            "filterOperation": "Whether a resource's property should equal, not equal, contain, or not contain
  the value in order to be included for this class (acceptable values: EQUALS, DOES_NOT_EQUAL, CONTAINS,
  DOES_NOT_CONTAIN)",
            "default": "If the property is not present on the resource, the boolean default determines whether
  or not to include this resource. Our recommendation and the default for this field is false since it is safer
  to assume the property has not yet initialized for the resource and we should wait until it is before making
  the determination."
          }
        ],
        "joiner": "AND or OR"
      }
    },
    {
      "instructions": "Define a sync class for every class you want to define and add it to this array."
    }
  ]
}
```

## Additional Columns

To add properties to the CMDB that are not defined by the built-in tree definition, use the 'additionalColumns' parameter.

This is a mapping from class identifier (See table in Name Filter) to a list of additional columns. An additional column has the following fields:

- cmdbColumn: The key of the column that the vROps field will be send to on the CMDB CI Class
- vropsType: Where to get this data from in vROps. See the type table below for a list of options
- vropsField: Represents the field to sync. See the type table below for a list of options

This example syncs the vROps property 'cpu|cpuModel' to the CMDB column 'cpu_name' on all Host System resources:

```
"additionalColumns": {
  "builtInHostSystem": [
    {
      "cmdbColumn": "cpu_name",
      "vropsType": "PROPERTY",
      "vropsField": "cpu|cpuModel"
    }
  ]
}
```

## vROps Types and Fields

When defining a column, this describes available 'vropsType' enums and what should be provided as the 'vropsField' when that type is selected

| Type | Applicable Field |
|---|---|
| PROPERTY | The key of the vROps property to sync such as 'cpu\|cpuModel' |
| IDENTIFIER | The key of the vROps Identifier to sync sync as 'VMEntityVCID' |
| METRIC | The key of the vROps property to sync such as 'cpu\|demand\|stress' |
| NAME | This should be left empty. The name of the resource will be used. |
| UUID | This should be left empty. The UUID assigned by vROps for the resource will be used. This is unique to each vROps instances so if the same resource is monitored by multiple vROps instances, this is not an ideal metric to use |
| STATIC_STRING | The value you want assigned to the column. For instance, you could set 'Synced From vROps' to a description column and the column would be set to that value for all resources of the CI Class that we sync. |

## Data Source

In order to sync data, the following must be done on ServiceNow:

- Add data_source `VMwareTVS` to the

  `Choice List`

  (you can search `sys_choice.list` in the UI or query `sys_choice` in the REST API) with the `Element` (`element` in REST) attribute set to `discovery_source` and the `Value` (`value`) and `Label` (`label`) attributes set to `VMwareTVS`. The `Table` (`table`) must be set to `cmdb_ci`.
- Add a Reconciliation Rule for your new `data_source` (`VMwareTVS`) on each of the CI classes in >Synced Resources

If this is not done, IRE will not be able to update the resource, but the API still returns a successful response so we are not able to alert/fail collection due to this.

## Synced Resources

The following resources from the VMWARE adapter will be synced to the CMDB if CMDB Sync is enabled:

| vROps Type | CI Class |
|---|---|
| Datastore | cmdb_ci_vcenter_datastore |
| Datacenter | cmdb_ci_vcenter_datacenter |
| HostSystem | cmdb_ci_esx_server |

| vROps Type | CI Class |
|---|---|
| ClusterComputeResource | cmdb_ci_vcenter_cluster |
| VMwareAdapter Instance | cmdb_ci_vcenter |
| VirtualMachine | cmdb_ci_vmware_instance |

Additional information about what properties and relationships we sync can be found in CMDB Synced Metrics.

### CMDB Sync Method

Currently, the only supported sync method is the IRE API (IRE_API). This is the default value and specifying it is not required. IRE_API uses ServiceNow's Identification and Reconciliation Engine to resolve vROps resources to the appropriate ServiceNow resources. For this to work properly, the identification rules for the ci classes this adapter populates should not be altered from what ServiceNow includes by default. If rules are edited or removed, we cannot guarantee how the IRE will resolve resources that this adapter sends it.

CMDB Sync using the IRE is not compatible with ServiceNow's Discovery plugin.

Note: The TableAPI is also used for methods the IRE API cannot accomplish such as adding/deleting relationships that have changed which would affect identification and deleting resources if the sync mode is set to allow deletion.

### Sync Mode

This specifies how we should sync CIs to the CMDB.

There are 3 options for syncMode:

- POPULATE_ONLY: Populates the CMDB, but does not remove CIs. If a resource is Not Existing, it sets the operational status to '6' (retire).
- POPULATE_AND_DELETE_WHEN_NOT_EXISTING: Populates the CMDB and removes CIs that have been marked as `Not Existing` in vROps
- POPULATE_AND_DELETE_WHEN_REMOVED: Populates the CMDB and removes CIs that have been removed from vROps. If a resource is Not Existing, it sets the operational status to '6' (retire).

### Object Identifier Source

This is used to select which identifier should be used for objects which have both a UUID and a MOID. Currently, this only applies to Virtual Machine resources. If you are populating your CMDB for the first time, we recommend using MOID. If you are using this adapter to add data to a CMDB that has already been populated, check the format of the identifier in the `object_id` column for a `cmdb_ci_vmware_instance` object and choose the option below that matches it.

There are 2 options for Object Identifier Source:

- MOID (default): Also called MoRef ID, this is a required identifier in vROps and looks like `vm-111`. It includes a prefix and a number which is generated by a counter.
- UUID: This is an optional identifier in vROps and looks like a standard UUID. In most cases, this should be present, but there is a chance that it will not be. We have a few VMs in our test environment which are in bad states that don't have UUIDs, but they do have MOIDs.

For further info about identifiers in vCenter, see

- Identification Overview Part 1
- Identification Overview Part 2

### Name Filter

A map from ServiceNow CI Class to a regex filter. Every resource with a name that matches this regex will not be synced to the CMDB. This can be omitted if you intend to include all resources. It will look like this if you wish to ignore all VMs that end in the `dev`:

```
{ "cmdb_ci_vmware_instance": "^.*dev$"}
```

Note: VM templates are automatically filtered out so there is no need to add a name filter for VM templates.

### Configuration Samples

Some sample configuration files are provided for you. They are pre-installed with the Management Pack and located at `/usr/lib/vmware-vcops/user/plugins/inbound/servicenow_adapter3/conf/config_samples`. You can use these out-of-the-box, or you can modify the json as described in the (Configuration File)[#Configuration File] section. If you do modify any of the files, it is recommended that you move the modified file to the `work` directory. Otherwise, the modified file will be overwritten if you update the Management Pack.

If you wish to use an un-modified configuration file, you will need to specify the path starting with the directory inside `config_samples` in the Adapter Configuration.

For example, if you wish to use `moid_alerts.json`, you can specify `alert/moid_alerts.json`. If you copy it to the work directory so that you can modify it, you only need to specify the name of the file.

Since group configurations are custom to specific customer deployments, we do not include any out-of-the-box samples with group configurations

The following sample configurations are available:

#### Alerts No Mapping

Location in `config_samples`: `alert/alerts_no_mapping.json`

This is a basic configuration that will send all vROps alerts to ServiceNow as ServiceNow alerts. It does not apply any filters.

#### Alerts No Mapping

Location in `config_samples`: `alert/events_no_mapping.json`

This is a basic configuration that will send all vROps alerts to ServiceNow as ServiceNow events. It does not apply any filters.

#### Alerts No Mapping

Location in `config_samples`: `alert/incidents_no_mapping.json`

This is a basic configuration that will send all vROps alerts to ServiceNow as ServiceNow incidents. It does not apply any filters.

#### MOID Alerts

Location in `config_samples`: `alert/moid_alerts.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow alerts for these vROps alerts and associates them with their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds except VirtualMachine, which uses MOID and vCenter UUID.

#### MOID Events

Location in `config_samples`: `alert/moid_events.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow events for these vROps alerts and associates them with their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds except VirtualMachine, which uses MOID and vCenter UUID.

#### MOID Incidents

Location in `config_samples`: `alert/moid_incidents.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow incidents for these vROps alerts and associates them with

their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds except VirtualMachine, which uses MOID and vCenter UUID.

#### UUID Alerts

Location in `config_samples`: `alert/uuid_alerts.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow alerts for these vROps alerts and associates them with their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds.

#### UUID Events

Location in `config_samples`: `alert/uuid_events.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow events for these vROps alerts and associates them with their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds.

#### UUID Incidents

Location in `config_samples`: `alert/uuid_incidents.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow incidents for these vROps alerts and associates them with their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds.

#### CMDB Sync MOID Populate Only

Location in `config_samples`: `cmdb_sync/moid_populate_only.json`

This configuration syncs vROps resources to the CMDB. The MOID is used as the `object_id` for `cmdb_ci_vmware_instance` (Virtual Machine) CIs and the UUID is used as the `object_id` for all other CI Classes. This configuration populates the CMDB, but it does not delete resources from the CMDB.

#### CMDB Sync MOID Populate and Delete When Not Existing

Location in `config_samples`: `cmdb_sync/moid_populate_and_delete_when_not_existing.json`

This configuration syncs vROps resources to the CMDB. The MOID is used as the `object_id` for `cmdb_ci_vmware_instance` (Virtual Machine) CIs and the UUID is used as the `object_id` for all other CI Classes. This configuration populates the CMDB, and deletes resources from the CMDB when vROps marks them as `Not Existing`

#### CMDB Sync MOID Populate and Delete When Removed

Location in `config_samples`: `cmdb_sync/moid_populate_and_delete_when_removed.json`

This configuration syncs vROps resources to the CMDB. The MOID is used as the `object_id` for `cmdb_ci_vmware_instance` (Virtual Machine) CIs and the UUID is used as the `object_id` for all other CI Classes. This configuration populates the CMDB, and deletes resources from the CMDB when they are deleted from vROps.

#### CMDB Sync UUID Populate Only

Location in `config_samples`: `cmdb_sync/uuid_populate_only.json`

This configuration syncs vROps resources to the CMDB. The MOID is used as the UUID is used as the `object_id` for all CI Classes. It populates the CMDB, but it does not delete resources from the CMDB.

### CMDB Sync UUID Populate and Delete When Not Existing

Location in `config_samples`: `cmdb_sync/uuid_populate_and_delete_when_not_existing.json`

This configuration syncs vROps resources to the CMDB. The MOID is used as the UUID is used as the `object_id` for all CI Classes. It populates the CMDB, and deletes resources from the CMDB when vROps marks them as

```
Not
        Existing
```

### CMDB Sync UUID Populate and Delete When Removed

Location in `config_samples`: `cmdb_sync/uuid_populate_and_delete_when_removed.json`

This configuration syncs vROps resources to the CMDB. The MOID is used as the UUID is used as the `object_id` for all CI Classes. It populates the CMDB, and deletes resources from the CMDB when they are deleted from vROps.

### Combined: MOID Alerts and Populate Only

Location in `config_samples`: `combined/moid_alerts_and_populate_only.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow alerts for these vROps alerts and associates them with their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds.

Also, it syncs vROps resources to the CMDB. The MOID is used as the `object_id` for `cmdb_ci_vmware_instance` (Virtual Machine) CIs and the UUID is used as the `object_id` for all other CI Classes. This configuration populates the CMDB, and deletes resources from the CMDB when they are deleted from vROps.

### Combined: MOID Alerts and Populate Delete When Not Existing

Location in `config_samples`: `combined/moid_alerts_and_populate_when_not_existing.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow alerts for these vROps alerts and associates them with their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds.

Also, it syncs vROps resources to the CMDB. The MOID is used as the `object_id` for `cmdb_ci_vmware_instance` (Virtual Machine) CIs and the UUID is used as the `object_id` for all other CI Classes. This configuration populates the CMDB, and deletes resources from the CMDB when vROps marks them as `Not Existing`

### Combined: MOID Alerts and Populate When Removed

Location in `config_samples`: `combined/moid_alerts_and_populate_when_removed.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow alerts for these vROps alerts and associates them with their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds.

Also, it syncs vROps resources to the CMDB. The MOID is used as the `object_id` for `cmdb_ci_vmware_instance` (Virtual Machine) CIs and the UUID is used as the `object_id` for all other CI Classes. This configuration populates the CMDB, and deletes resources from the CMDB when they are deleted from vROps.

### Combined: MOID Events and Populate Only

Location in `config_samples`: `combined/moid_events_and_populate_only.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow events for these vROps alerts and associates them with their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds.

Also, it syncs vROps resources to the CMDB. The MOID is used as the `object_id` for `cmdb_ci_vmware_instance` (Virtual Machine) CIs and the UUID is used as the `object_id` for all other CI Classes. This configuration populates the CMDB, and deletes resources from the CMDB when they are deleted from vROps.

**Combined: MOID Events and Populate Delete When Not Existing**

Location in `config_samples`: `combined/moid_events_and_populate_when_not_existing.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow events for these vROps alerts and associates them with their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds.

Also, it syncs vROps resources to the CMDB. The MOID is used as the `object_id` for `cmdb_ci_vmware_instance` (Virtual Machine) CIs and the UUID is used as the `object_id` for all other CI Classes. This configuration populates the CMDB, and deletes resources from the CMDB when vROps marks them as `Not Existing`

**Combined: MOID Events and Populate When Removed**

Location in `config_samples`: `combined/moid_events_and_populate_when_removed.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow events for these vROps alerts and associates them with their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds.

Also, it syncs vROps resources to the CMDB. The MOID is used as the `object_id` for `cmdb_ci_vmware_instance` (Virtual Machine) CIs and the UUID is used as the `object_id` for all other CI Classes. This configuration populates the CMDB, and deletes resources from the CMDB when they are deleted from vROps.

**Combined: MOID Incidents and Populate Only**

Location in `config_samples`: `combined/moid_incidents_and_populate_only.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow incidents for these vROps alerts and associates them with their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds.

Also, it syncs vROps resources to the CMDB. The MOID is used as the `object_id` for `cmdb_ci_vmware_instance` (Virtual Machine) CIs and the UUID is used as the `object_id` for all other CI Classes. This configuration populates the CMDB, and deletes resources from the CMDB when they are deleted from vROps.

**Combined: MOID Incidents and Populate Delete When Not Existing**

Location in `config_samples`: `combined/moid_incidents_and_populate_when_not_existing.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow incidents for these vROps alerts and associates them with their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds.

Also, it syncs vROps resources to the CMDB. The MOID is used as the `object_id` for `cmdb_ci_vmware_instance` (Virtual Machine) CIs and the UUID is used as the `object_id` for all other CI Classes. This configuration populates the CMDB, and deletes resources from the CMDB when vROps marks them as `Not Existing`

**Combined: MOID Incidents and Populate When Removed**

Location in `config_samples`: `combined/moid_incidents_and_populate_when_removed.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow incidents for these vROps alerts and associates them with

their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds.

Also, it syncs vROps resources to the CMDB. The MOID is used as the `object_id` for `cmdb_ci_vmware_instance` (Virtual Machine) CIs and the UUID is used as the `object_id` for all other CI Classes. This configuration populates the CMDB, and deletes resources from the CMDB when they are deleted from vROps.

**Combined: UUID Alerts and Populate Only**

Location in `config_samples`: `combined/uuid_alerts_and_populate_only.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow alerts for these vROps alerts and associates them with their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds.

Also, it syncs vROps resources to the CMDB. The MOID is used as the UUID is used as the `object_id` for all CI Classes. It populates the CMDB, but it does not delete resources from the CMDB.

**Combined: UUID Alerts and Populate and Delete When Not Existing**

Location in `config_samples`: `combined/uuid_alerts_and_populate_when_not_existing.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow alerts for these vROps alerts and associates them with their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds.

Also, it syncs vROps resources to the CMDB. The MOID is used as the UUID is used as the `object_id` for all CI Classes. It populates the CMDB, and deletes resources from the CMDB when vROps marks them as

```
Not
        Existing
```

**Combined: UUID Alerts and Populate When Removed**

Location in `config_samples`: `combined/uuid_alerts_and_populate_when_removed.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow alerts for these vROps alerts and associates them with their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds.

Also, it syncs vROps resources to the CMDB. The MOID is used as the UUID is used as the `object_id` for all CI Classes. It populates the CMDB, and deletes resources from the CMDB when they are deleted from vROps.

**Combined: UUID Events and Populate Only**

Location in `config_samples`: `combined/uuid_events_and_populate_only.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow events for these vROps alerts and associates them with their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds.

Also, it syncs vROps resources to the CMDB. The MOID is used as the UUID is used as the `object_id` for all CI Classes. It populates the CMDB, but it does not delete resources from the CMDB.

**Combined: UUID Events and Populate and Delete When Not Existing**

Location in `config_samples`: `combined/uuid_events_and_populate_when_not_existing.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow events for these vROps alerts and associates them with

their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds.

Also, it syncs vROps resources to the CMDB. The MOID is used as the UUID is used as the `object_id` for all CI Classes. It populates the CMDB, and deletes resources from the CMDB when vROps marks them as

```
Not
        Existing
```

**Combined: UUID Events and Populate When Removed**

Location in `config_samples`: `combined/uuid_events_and_populate_when_removed.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow events for these vROps alerts and associates them with their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds.

Also, it syncs vROps resources to the CMDB. The MOID is used as the UUID is used as the `object_id` for all CI Classes. It populates the CMDB, and deletes resources from the CMDB when they are deleted from vROps.

**Combined: UUID Incidents and Populate Only**

Location in `config_samples`: `combined/uuid_incidents_and_populate_only.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow incidents for these vROps alerts and associates them with their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds.

Also, it syncs vROps resources to the CMDB. The MOID is used as the UUID is used as the `object_id` for all CI Classes. It populates the CMDB, but it does not delete resources from the CMDB.

**Combined: UUID Incidents and Populate and Delete When Not Existing**

Location in `config_samples`: `combined/uuid_incidents_and_populate_when_not_existing.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow incidents for these vROps alerts and associates them with their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds.

Also, it syncs vROps resources to the CMDB. The MOID is used as the UUID is used as the `object_id` for all CI Classes. It populates the CMDB, and deletes resources from the CMDB when vROps marks them as

```
Not
        Existing
```

**Combined: UUID Incidents and Populate When Removed**

Location in `config_samples`: `combined/uuid_incidents_and_populate_when_removed.json`

This configuration filters vROps alerts to the following ResourceKinds: VirtualMachine, ClusterComputeResource, Datacenter, Datastore, HostSystem. It creates ServiceNow incidents for these vROps alerts and associates them with their CI if it is present in the CMDB. The mapping uses a combination of UUID and vCenter UUID as identifiers for all ResourceKinds.

Also, it syncs vROps resources to the CMDB. The MOID is used as the UUID is used as the `object_id` for all CI Classes. It populates the CMDB, and deletes resources from the CMDB when they are deleted from vROps.

### Using the Management Pack

The Management Pack for ServiceNow includes the following out-of-the-box features, harnessing the power of the VCF Operations user interface:

- Dashboards
- Views
- Metrics and Statistics
- Alert Sync


## Dashboards

Dashboards are the primary feature for monitoring and troubleshooting ServiceNow problems from within VCF Operations.

The following dashboards are included in the Management Pack for ServiceNow:

**ServiceNow Statistics** provides a breakdown of all the relevant and important statistics from your monitored ServiceNow CMDB.

**ServiceNow Group Overview** provides an overview of different SNOW resources. Simply select a resource from the Environment Overview widget to populate the Alert List and KPI widgets with the relevant information for that resource.

**ServiceNow Details** provides status, relationships, health, alerts, and KPIs.

**ServiceNow Cost Analysis** provides monthly cost data for operating virtual machines, host systems, or datastores. ServiceNow Cost Analysis requires the user to set a currency. To do this, go to Administration/Management/Global Settings; find the Currency setting and set it to the appropriate value. Caution: Currency cannot be changed after it has been set.

> **Note:**
>
> Additional Details dashboards are generated based on your configuration file and will be installed automatically if you set the **Upload Dashboards** setting to true in the adapter configuration.

## Views

The Management Pack for ServiceNow creates views that allow the user to view statistics of metrics for ServiceNow resources. The views help give a broad picture of the entire system, as opposed to a more in depth view.

The following views are available in the Management Pack:

| View | Type | Description |
|---|---|---|
| Alerts that are currently active | List | Show alerts for the selected object and its descendants that are currently active |
| Environment Top Critical Alerts Summary | List | |
| Symptoms | List | This view provides a list of active symptoms for the selected object. It is useful for a dashboard that is focused on diagnosing problems that do not have an alert. |
| Datastore Cost Analysis | Summary | The total and average values of monthly cost metrics for Datastores related to the selected group. |
| Host System Cost Analysis | Summary | The total and average values of monthly cost metrics for Host Systems related to the selected group. |
| Virtual Machine Monthly Cost Analysis | Summary | The total and average values of monthly cost metrics for Virtual Machines related to the selected group. |

To access the Management Pack views:

1. Navigate to **Environment > All Objects > ServiceNow**.
2. Double-click on the desired object (resource).
3. Select the **Details** tab, then **Views.**

The available views for that resource are listed and can be selected.

CMDB Synced Metrics

**CMDB Sync Properties**

The following cmdb columns will be populated for the given CI Classes.

- ServiceNow Column: This is the column that will be populated in ServiceNow
- VCF Operations Type: This is indicates where the column value is taken from in VCF Operations.

The following options are available:

- IDENTIFIER: an identifier on the VCF Operations resource
- PROPERTY: a property on the VCF Operations resource
- METRIC: a metric on the VCF Operations resource
- STATIC_STRING: a static string that will always return the same value (that value is shown in the VCF Operations key column of the table)
- VCF Operations Key: The key for the attribute found in VCF Operations
- Conversion: A description of the conversion used on the VCF Operations attribute before sending it to ServiceNow
- Native Attribute: Returns true if this is an attribute that is present out-of-the-box on the ServiceNow CI Class. If this is false, it is added by the Discovery plugin. If you are not using the discovery plugin, but want to use the attribute, add the column to the CI Class definition cmdb_ci_vcenter_datastore (VMWARE::Datastore)

| ServiceNow Column | VCF Operations Type | VCF Operations Key | Conversion | Native Attribute |
|---|---|---|---|---|
| name | IDENTIFIER | VMEntityName | None | true |
| discovery_source | STATIC_STRING | VMwareTVS | None | true |
| morid | IDENTIFIER | VMEntityObjectID | None | true |
| accessible | PROPERTY | summary\|accessible | None | true |
| vcenter_ref | PROPERTY | summary\|parentVcenter | None | true |
| type | PROPERTY | summary\|type | None | true |
| capacity | PROPERTY | summary\|diskCapacity | None | true |
| freespace | METRIC | diskspace\|freespace | None | true |
| url | PROPERTY | summary\|path | None | true |
| object_id | IDENTIFIER | VMEntityObjectID | None | true |
| vcenter_uuid | IDENTIFIER | VMEntityVCID | None | true |

cmdb_ci_vmware_instance (VMWARE::VirtualMachine)

| ServiceNow Column | VCF Operations Type | VCF Operations Key | Conversion | Native Attribute |
|---|---|---|---|---|
| vm_instance_uuid | IDENTIFIER | VMEntityInstanceUUID | None | true |
| template | PROPERTY | summary\|config\|isTemplate | None | true |
| name | IDENTIFIER | VMEntityName | None | true |
| cpus | PROPERTY | config\|hardware\|numCpu | None | true |
| disks | PROPERTY | config\|numVMDKs | None | true |
| disks_size | PROPERTY | config\|hardware\|diskSpace | None | true |
| memory | PROPERTY | config\|hardware\|memoryKB | Convert KB to MB | true |
| nics | PROPERTY | summary\|config\|numEthernetCards | None | true |

| | | | | |
|---|---|---|---|---|
| state | PROPERTY | summary\|runtime\|powerState | None | true |
| discovery_source | STATIC_STRING | VMwareTVS | None | true |
| object_id | IDENTIFIER | VMEntityObjectID or VMEntityInstanceUUID | None | true |
| vcenter_uuid | IDENTIFIER | VMEntityVCID | None | true |

cmdb_ci_esx_server (VMWARE::HostSystem)

| ServiceNow Column | VCF Operations Type | VCF Operations Key | Conversion | Native Attribute |
|---|---|---|---|---|
| name | IDENTIFIER | VMEntityName | None | true |
| asset | PROPERTY | hardware\|vendorModel | None | true |
| connection_state | PROPERTY | runtime\|connectionState | None | false |
| connection_id | PROPERTY | summary\|hostuuid | None | false |
| cpu_core_count | PROPERTY | hardware\|cpuInfo\|numCpuCores | None | true |
| cpu_count | PROPERTY | hardware\|cpuInfo\|numCpuPackages | None | true |
| cpu_speed | PROPERTY | hardware\|cpuInfo\|hz | None | true |
| cpu_type | PROPERTY | cpu\|cpuModel | None | true |
| discovery_source | STATIC_STRING | VMwareTVS | None | true |
| disk_space | METRIC | diskspace\|total_capacity | None | true |
| hyper_threading | PROPERTY | config\|hyperThread\|available | None | false |
| ip_address | PROPERTY | net\|mgmt_address | None | true |
| manufacturer | PROPERTY | hardware\|vendor | None | true |
| model_id | PROPERTY | hardware\|vendorModel | None | true |
| os_version | PROPERTY | summary\|version | None | true |
| power_state | PROPERTY | runtime\|powerState | None | false |
| ram | PROPERTY | hardware\|memorySize | Convert KB to MB | true |
| serial_number | PROPERTY | hardware\|serialNumber | None | true |
| vcenter_ref | PROPERTY | summary\|parentVcenter | None | true |
| object_id | IDENTIFIER | VMEntityObjectID | None | true |
| vcenter_uuid | IDENTIFIER | VMEntityVCID | None | true |

cmdb_ci_vcenter_datacenter (VMWARE::Datacenter)

| ServiceNow Column | VCF Operations Type | VCF Operations Key | Conversion | Native Attribute |
|---|---|---|---|---|
| name | IDENTIFIER | VMEntityName | None | true |
| discovery_source | STATIC_STRING | VMwareTVS | None | true |
| morid | IDENTIFIER | VMEntityObjectID | None | true |
| vcenter_ref | PROPERTY | summary\|parentVcenter | None | true |
| object_id | IDENTIFIER | VMEntityObjectID | None | true |
| vcenter_uuid | IDENTIFIER | VMEntityVCID | None | true |

cmdb_ci_vcenter_cluster (VMWARE::ClusterComputeResource)

| ServiceNow Column | VCF Operations Type | VCF Operations Key | Conversion | Native Attribute |
|---|---|---|---|---|
| name | IDENTIFIER | VMEntityName | None | true |

| | | | | |
|---|---|---|---|---|
| discovery_source | STATIC_STRING | VMwareTVS | None | true |
| morid | IDENTIFIER | VMEntityObjectID | None | true |
| effectivehosts | METRIC | summary\|number_running_hosts | None | true |
| vcenter_ref | PROPERTY | summary\|parentVcenter | None | true |
| numhosts | METRIC | summary\|total_number_hosts | None | true |
| totalcpu | METRIC | cpu\|haTotalCapacity_average | None | true |
| totalmemory | METRIC | mem\|totalCapacity_average | None | true |
| numcpucores | METRIC | cpu\|corecount_provisioned | None | true |
| drs_behavior | PROPERTY | configuration\|drsConfig\|defaultVmBehavior | None | true |
| drs_enabled | PROPERTY | configuration\|drsconfig\|enabled | None | true |
| drs_vmotion_rate | PROPERTY | configuration\|drsConfig\|vmotionRate | None | true |
| object_id | IDENTIFIER | VMEntityObjectID | None | true |
| vcenter_uuid | IDENTIFIER | VMEntityVCID | None | true |

cmdb_ci_vcenter (VMWARE::VMwareAdapter Instance)

| ServiceNow Column | VCF Operations Type | VCF Operations Key | Conversion | Native Attribute |
|---|---|---|---|---|
| name | IDENTIFIER | VCURL | None | true |
| cmdb_ci_vcenter | STATIC_STRING | sys_class_name | None | true |
| fullName | PROPERTY | summary\|vcfullname | None | true |
| url | IDENTIFIER | VCURL | Builds SDK URL from hostname | true |
| discovery_source | STATIC_STRING | VMwareTVS | None | true |
| instance_uuid | IDENTIFIER | VMEntityVCID | None | true |

The following cmdb columns may be populated for the given Discovery CI Classes. Discovery classes can be either optional or required. Optional discovery classes are not required and if discovery has not been set up in the environment, they will not be included. If discovery is required, then then CI Class structure must include the required relation to the Cloud Service Account that discovery creates (even if you are not actively using discover).

Optional Discovery Class cmdb_ci_cloud_service_account (VMWARE::VMwareAdapter Instance)

| ServiceNow Column | VCF Operations Type | VCF Operations Key | Conversion |
|---|---|---|---|
| account_id | IDENTIFIER | account_id | None |
| object_id | IDENTIFIER | object_id | None |
| datacenter_url | IDENTIFIER | VCURL | Convert hostname or IP to URL |
| datacenter_type | STATIC_STRING | cmdb_ci_vcenter_datacenter | None |

CMDB Sync Relationships

The following relationships are made between CMDB CI sync classes. Relationships are displayed from the perspective of the parent object.

cmdb_ci_vcenter_datastore (VMWARE::Datastore)

| Child CI Class | CI Relationship Type |
|---|---|
| cmdb_ci_esx_server | Used by::Uses |
| cmdb_ci_vmware_instance | Provides storage for::Stored on |

cmdb_ci_vmware_instance (VMWARE::VirtualMachine)

| Child CI Class | CI Relationship Type |
|---|---|
| cmdb_ci_vcenter_datacenter | Registered on::Has registered |
| cmdb_ci_vcenter_datacenter | Hosted on::Hosts |

cmdb_ci_vcenter_datacenter (VMWARE::Datacenter)

| Child CI Class | CI Relationship Type |
|---|---|
| cmdb_ci_vcenter_cluster | Contains::Contained by |
| cmdb_ci_vcenter_datastore | Contains::Contained by |
| cmdb_ci_vmware_instance | Contains::Contained by |
| cmdb_ci_esx_server | Contains::Contained by |

cmdb_ci_vcenter_cluster (VMWARE::ClusterComputeResource)

| Child CI Class | CI Relationship Type |
|---|---|
| cmdb_ci_esx_server | Members::Member of |

cmdb_ci_vcenter (VMWARE::VMwareAdapter Instance)

| Child CI Class | CI Relationship Type |
|---|---|
| cmdb_ci_vcenter_datacenter | Contains::Contained by |
| cmdb_ci_esx_server | Runs on::Runs |

Discovery Resource cmdb_ci_cloud_service_account (VMWARE::VMwareAdapter Instance)

| Parent CI Class | CI Relationship Type |
|---|---|
| cmdb_ci_vcenter_datacenter | Hosted on::Hosts |

### Alert Sync

Incident configurations are used for the alert-sync feature and define what types of alerts will be synced between VCF Operations and ServiceNow. This feature only works on VCF Operations using the built in ServiceNow plugin. It depends on this plugin filter to send initial alerts from VCF Operations to ServiceNow. Once the VCF Operations alert is present in ServiceNow, the ServiceNow Management Pack performs the following additional features:

- If an alert is cancelled in VCF Operations, the corresponding incident in ServiceNow is closed.
- If an incident is closed in ServiceNow, the corresponding alert in VCF Operations is suspended. Alerts in VCF Operations are generally triggered automatically by breaching thresholds. Because we cannot control this, if an incident is closed in ServiceNow it should be due to the underlying issue being resolved. Thus, we suspend the Alert in VCF Operations (for a configurable length of time) to allow the alert to resolve. If the threshold(s) is still breached when the alert suspend time has ended, the alert will re-trigger and reopen the ServiceNow Incident.
- If a VCF Operations alert is triggered on a VCF Operations ResourceKind that is present in one of the `resourceTypes` sections above, the ServiceNow Management Pack will create a reference from the incident's `cmdb_ci` column to the CI representing the resource in ServiceNow the alert was triggered on. A resourceType section can be created with empty `groupTypes` and `hierarchies` arrays for matching to a resource type, without creating any groupings and hierarchies.

- The ServiceNow Management Pack can be configured to watch a predefined set of incident columns and display the current values in the corresponding VCF Operations alert as a series of notes.

  **Important:**

  VCF Operations alerts cannot be closed based on incidents, only suspended.

For more information, see Creating a Configuration File.

CMDB Synced Metrics

**CMDB Sync Properties**

The following cmdb columns will be populated for the given CI Classes.

- ServiceNow Column: This is the column that will be populated in ServiceNow
- VCF Operations Type: This is indicates where the column value is taken from in VCF Operations.

The following options are available:

- IDENTIFIER: an identifier on the VCF Operations resource
- PROPERTY: a property on the VCF Operations resource
- METRIC: a metric on the VCF Operations resource
- STATIC_STRING: a static string that will always return the same value (that value is shown in the VCF Operations key column of the table)
- VCF Operations Key: The key for the attribute found in VCF Operations
- Conversion: A description of the conversion used on the VCF Operations attribute before sending it to ServiceNow
- Native Attribute: Returns true if this is an attribute that is present out-of-the-box on the ServiceNow CI Class. If this is false, it is added by the Discovery plugin. If you are not using the discovery plugin, but want to use the attribute, add the column to the CI Class definition cmdb_ci_vcenter_datastore (VMWARE::Datastore)

| ServiceNow Column | VCF Operations Type | VCF Operations Key | Conversion | Native Attribute |
|---|---|---|---|---|
| name | IDENTIFIER | VMEntityName | None | true |
| discovery_source | STATIC_STRING | VMwareTVS | None | true |
| morid | IDENTIFIER | VMEntityObjectID | None | true |
| accessible | PROPERTY | summary\|accessible | None | true |
| vcenter_ref | PROPERTY | summary\|parentVcenter | None | true |
| type | PROPERTY | summary\|type | None | true |
| capacity | PROPERTY | summary\|diskCapacity | None | true |
| freespace | METRIC | diskspace\|freespace | None | true |
| url | PROPERTY | summary\|path | None | true |
| object_id | IDENTIFIER | VMEntityObjectID | None | true |
| vcenter_uuid | IDENTIFIER | VMEntityVCID | None | true |

cmdb_ci_vmware_instance (VMWARE::VirtualMachine)

| ServiceNow Column | VCF Operations Type | VCF Operations Key | Conversion | Native Attribute |
|---|---|---|---|---|
| vm_instance_uuid | IDENTIFIER | VMEntityInstanceUUID | None | true |
| template | PROPERTY | summary\|config\|isTemplate | None | true |
| name | IDENTIFIER | VMEntityName | None | true |
| cpus | PROPERTY | config\|hardware\|numCpu | None | true |
| disks | PROPERTY | config\|numVMDKs | None | true |
| disks_size | PROPERTY | config\|hardware\|diskSpace | None | true |

| | | | | |
|---|---|---|---|---|
| memory | PROPERTY | config\|hardware\|memory KB | Convert KB to MB | true |
| nics | PROPERTY | summary\|config\|numEthernetCards | None | true |
| state | PROPERTY | summary\|runtime\|powerState | None | true |
| discovery_source | STATIC_STRING | VMwareTVS | None | true |
| object_id | IDENTIFIER | VMEntityObjectID or VMEntityInstanceUUID | None | true |
| vcenter_uuid | IDENTIFIER | VMEntityVCID | None | true |

cmdb_ci_esx_server (VMWARE::HostSystem)

| ServiceNow Column | VCF Operations Type | VCF Operations Key | Conversion | Native Attribute |
|---|---|---|---|---|
| name | IDENTIFIER | VMEntityName | None | true |
| asset | PROPERTY | hardware\|vendorModel | None | true |
| connection_state | PROPERTY | runtime\|connectionState | None | false |
| connection_id | PROPERTY | summary\|hostuuid | None | false |
| cpu_core_count | PROPERTY | hardware\|cpuInfo\|numCpuCores | None | true |
| cpu_count | PROPERTY | hardware\|cpuInfo\|numCpuPackages | None | true |
| cpu_speed | PROPERTY | hardware\|cpuInfo\|hz | None | true |
| cpu_type | PROPERTY | cpu\|cpuModel | None | true |
| discovery_source | STATIC_STRING | VMwareTVS | None | true |
| disk_space | METRIC | diskspace\|total_capacity | None | true |
| hyper_threading | PROPERTY | config\|hyperThread\|available | None | false |
| ip_address | PROPERTY | net\|mgmt_address | None | true |
| manufacturer | PROPERTY | hardware\|vendor | None | true |
| model_id | PROPERTY | hardware\|vendorModel | None | true |
| os_version | PROPERTY | summary\|version | None | true |
| power_state | PROPERTY | runtime\|powerState | None | false |
| ram | PROPERTY | hardware\|memorySize | Convert KB to MB | true |
| serial_number | PROPERTY | hardware\|serialNumber | None | true |
| vcenter_ref | PROPERTY | summary\|parentVcenter | None | true |
| object_id | IDENTIFIER | VMEntityObjectID | None | true |
| vcenter_uuid | IDENTIFIER | VMEntityVCID | None | true |

cmdb_ci_vcenter_datacenter (VMWARE::Datacenter)

| ServiceNow Column | VCF Operations Type | VCF Operations Key | Conversion | Native Attribute |
|---|---|---|---|---|
| name | IDENTIFIER | VMEntityName | None | true |
| discovery_source | STATIC_STRING | VMwareTVS | None | true |
| morid | IDENTIFIER | VMEntityObjectID | None | true |
| vcenter_ref | PROPERTY | summary\|parentVcenter | None | true |
| object_id | IDENTIFIER | VMEntityObjectID | None | true |
| vcenter_uuid | IDENTIFIER | VMEntityVCID | None | true |

cmdb_ci_vcenter_cluster (VMWARE::ClusterComputeResource)

| ServiceNow Column | VCF Operations Type | VCF Operations Key | Conversion | Native Attribute |
|---|---|---|---|---|
| name | IDENTIFIER | VMEntityName | None | true |
| discovery_source | STATIC_STRING | VMwareTVS | None | true |
| morid | IDENTIFIER | VMEntityObjectID | None | true |
| effectivehosts | METRIC | summary\|number_running_hosts | None | true |
| vcenter_ref | PROPERTY | summary\|parentVcenter | None | true |
| numhosts | METRIC | summary\|total_number_hosts | None | true |
| totalcpu | METRIC | cpu\|haTotalCapacity_average | None | true |
| totalmemory | METRIC | mem\|totalCapacity_average | None | true |
| numcpucores | METRIC | cpu\|corecount_provisioned | None | true |
| drs_behavior | PROPERTY | configuration\|drsConfig\|defaultVmBehavior | None | true |
| drs_enabled | PROPERTY | configuration\|drsconfig\|enabled | None | true |
| drs_vmotion_rate | PROPERTY | configuration\|drsConfig\|vmotionRate | None | true |
| object_id | IDENTIFIER | VMEntityObjectID | None | true |
| vcenter_uuid | IDENTIFIER | VMEntityVCID | None | true |

cmdb_ci_vcenter (VMWARE::VMwareAdapter Instance)

| ServiceNow Column | VCF Operations Type | VCF Operations Key | Conversion | Native Attribute |
|---|---|---|---|---|
| name | IDENTIFIER | VCURL | None | true |
| cmdb_ci_vcenter | STATIC_STRING | sys_class_name | None | true |
| fullName | PROPERTY | summary\|vcfullname | None | true |
| url | IDENTIFIER | VCURL | Builds SDK URL from hostname | true |
| discovery_source | STATIC_STRING | VMwareTVS | None | true |
| instance_uuid | IDENTIFIER | VMEntityVCID | None | true |

The following cmdb columns may be populated for the given Discovery CI Classes. Discovery classes can be either optional or required. Optional discovery classes are not required and if discovery has not been set up in the environment, they will not be included. If discovery is required, then then CI Class structure must include the required relation to the Cloud Service Account that discovery creates (even if you are not actively using discover).

Optional Discovery Class cmdb_ci_cloud_service_account (VMWARE::VMwareAdapter Instance)

| ServiceNow Column | VCF Operations Type | VCF Operations Key | Conversion |
|---|---|---|---|
| account_id | IDENTIFIER | account_id | None |
| object_id | IDENTIFIER | object_id | None |
| datacenter_url | IDENTIFIER | VCURL | Convert hostname or IP to URL |
| datacenter_type | STATIC_STRING | cmdb_ci_vcenter_datacenter | None |

CMDB Sync Relationships

The following relationships are made between CMDB CI sync classes. Relationships are displayed from the perspective of the parent object.

cmdb_ci_vcenter_datastore (VMWARE::Datastore)

| Child CI Class | CI Relationship Type |
|---|---|
| cmdb_ci_esx_server | Used by::Uses |
| cmdb_ci_vmware_instance | Provides storage for::Stored on |

cmdb_ci_vmware_instance (VMWARE::VirtualMachine)

| Child CI Class | CI Relationship Type |
|---|---|
| cmdb_ci_vcenter_datacenter | Registered on::Has registered |
| cmdb_ci_vcenter_datacenter | Hosted on::Hosts |

cmdb_ci_vcenter_datacenter (VMWARE::Datacenter)

| Child CI Class | CI Relationship Type |
|---|---|
| cmdb_ci_vcenter_cluster | Contains::Contained by |
| cmdb_ci_vcenter_datastore | Contains::Contained by |
| cmdb_ci_vmware_instance | Contains::Contained by |
| cmdb_ci_esx_server | Contains::Contained by |

cmdb_ci_vcenter_cluster (VMWARE::ClusterComputeResource)

| Child CI Class | CI Relationship Type |
|---|---|
| cmdb_ci_esx_server | Members::Member of |

cmdb_ci_vcenter (VMWARE::VMwareAdapter Instance)

| Child CI Class | CI Relationship Type |
|---|---|
| cmdb_ci_vcenter_datacenter | Contains::Contained by |
| cmdb_ci_esx_server | Runs on::Runs |

Discovery Resource cmdb_ci_cloud_service_account (VMWARE::VMwareAdapter Instance)

| Parent CI Class | CI Relationship Type |
|---|---|
| cmdb_ci_vcenter_datacenter | Hosted on::Hosts |

## Management Pack for VMware Avi Load Balancer

This management pack provides a single pane view of all the Avi Load Balancer objects, their metrics, and properties. You can monitor and troubleshoot resource objects deployed across multiple Avi Load Balancer instances.

### Intended Audience

The information in this guide is intended for IT operations and security personnel who set up and support the VCF Operations infrastructure.

### Introduction

The VCF Operations Management Pack for Avi Load Balancer provides end-to-end visibility of the infrastructure and effective monitoring of the supported objects. While the Avi Load Balancer application uses next generation architecture, delivering application services beyond load balancing, such as application analytics, predictive autoscaling,

micro-segmentation, and self-service for app owners in both on-premises or cloud environments, the VCF Operations Management Pack for Avi Load Balancer allows you to intelligently monitor and troubleshoot Avi Load Balancer instances and its entities for smooth operations management.

Here are some notable features of this management pack:

- End-to-end visibility of load balanced resources across geographies, spread across both on-premises or cloud environments.
- Single pane view of the Avi Load Balancer objects such as, controllers, tenants, clouds, virtual services, pools, servers, service engines, service engine groups, and network, and their metrics, and properties.
- Continuous monitoring of the resource objects deployed in the Avi Load Balancer. This leads to effective and prompt troubleshooting.
- One-stop view of the availability and health of the supported objects from dashboards and alerts.
- Visualization of object relationships helps to troubleshoot issues by leveraging hierarchical insights using metrics, properties, and health score analytics.
- Unified alert reporting across multiple instances of the Avi Load Balancer.

**Object Types and Object Hierarchy**

The VCF Operations Management Pack for Avi Load Balancer collects metrics and properties for several object types.

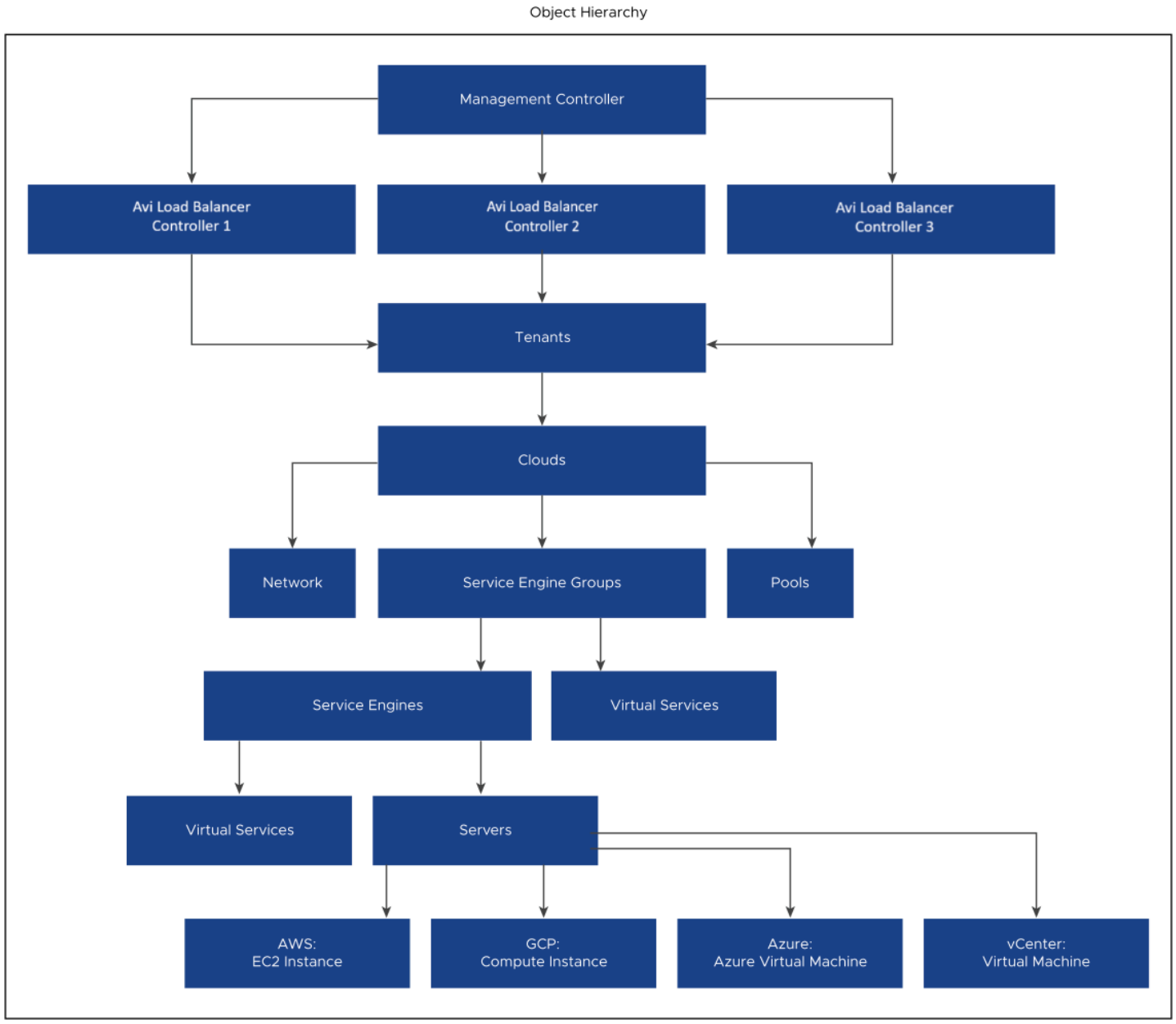
**Table 1669: List of Object Types**

| Object Types | Description |
|---|---|
| Controllers | Single point of management and control that serves as the brain of the entire system. For high availability, the controller is typically deployed as a three-node cluster. The controller implements the control plane. |
| Tenants | Isolated instances of the Avi Load Balancer that are associated with a user account that defines the resources that a user has access to. |
| Clouds | Containers for the environment that the Avi Load Balancer is installed in or operating within. |
| Virtual Services | Objects that advertise an IP address and ports to the external world and listen for client traffic. When the virtual service receives traffic, it can be configured for different reasons. |
| Pools | Objects that maintain the list of servers assigned to them and performs health monitoring, load balancing, persistence, and functions that involve Avi Load Balancer-to-server interaction. |
| Servers | The service engine machine. This can either be an EC2 instance/ Azure Virtual Machine/Compute Instance/vCenter Virtual Machine.<br>**Note:**<br>A prerequisite to view this object type is to add the cloud account, which is, vCenter/AWS/Azure/GCP, in VCF Operations where the service engine is deployed. This helps to monitor the actual service engine machine. |
| Service Engines | Machines that host the virtual services and require either direct or routable access to the client and server network. |
| Service Engine Groups | Objects that contain the definition of how the service engines should be sized, placed, and made highly available. |
| Network | Network resources used by service engines to communicate with the controller. This may be the same network as one of the data networks used for load balancing. |

For more details about object types, see the Avi Documentation.

## Object Hierarchy

The flow chart depicts the hierarchy of the supported objects types.




**Note:**
When a virtual service is initially created, it is associated with the service engine group. However, when it is hosted in the service engine, it is associated with all the service engines that are hosting the virtual service.

### Tenant Information

Each tenant that is supported in the instance of the Avi Load Balancer, is represented as an object in the object hierarchy. VCF Operations has the capability of representing the various tenants created in each of the Avi Load Balancer instances and represents them as part of the object hierarchy.

### Clouds

The VCF Operations Management Pack for Avi Load Balancer collects and monitors load balancer resources. This version of the management pack is certified for vCenter, NSX, AWS, Azure and GCP deployment options. The VCF Operations Management Pack for Avi Load Balancer collects metrics and properties for each of the public clouds. Although other deployment options supported in the Avi Load Balancer are not certified, that should not impact the

collection of inventory in terms of object metrics and associated alerts and dashboards. Deployments in native VMware environments such as NSX, vSphere, and vSAN are also supported.

**Getting Started**

Install and configure the VCF Operations Management Pack for Avi Load Balancer to collect metrics, properties, view dashboards, and troubleshoot alerts for the configured Avi Load Balancer instance.

### Install the VCF Operations Management Pack for Avi Load Balancer

The management pack consists of a PAK file that contains default dashboards for the objects that the management pack identifies. After installation, configure the management pack. For install information, see Installing and Configuring Integrations in VMware Cloud Foundation Operations.

### Configure the VCF Operations Management Pack for Avi Load Balancer

After you install the management pack, add and configure the Avi Load Balancer instance.

- As a non-admin user, to monitor the Avi Load Balancer objects in VCF Operations, the configured user must have at least read only access to the following objects to view the objects and their hierarchy.
  - Controller
  - Tenant
  - Cloud
  - Service Engine
  - Service Engine Group
  - Network
  - Pool
  - Virtual Service
  - Alert
  - Controller Site
  - System setting
  - Security admin role at admin tenant
- Ensure that the tenant access to the provider service engine is enabled for the tenants on the Avi Load Balancer instance: **Administration** > **Accounts** > **Tenants** > **Edit** > **Tenant Access to Provider Service Engine**.

1. From the left the menu, select **Data Sources** > **Integrations** and then select the **Accounts** tab from the left pane.
2. From the **Accounts** tab, click **Add Account**.
3. From the **Accounts Types** page, click **Avi Load Balancer**.
4. Enter a display name and description for the cloud account.
   a) Enter the **Name** for the management pack instance as you want it to appear in VCF Operations.
   b) Enter the **Description** for the management packs with any additional information that helps you manage your instances.
5. Enter the **Controller URL** which can be the IP address of an individual Avi Load Balancer controller, or the controller cluster IP, or the DNS name of the controller. Add the controller URL as *`https://<Controller_IP>`*.
6. Select the **Credential** you want to use to sign on to the environment from the drop-down menu. To add new credentials to access the management pack, click the plus sign and create a new credential.
   a) In the **Manage Credential** dialog box, enter the user name and password of the Avi Load Balancer controller instance.
   b) In the **Manage Credential** dialog box, to configure a proxy, enter the proxy details such as the IP, port, user name, and password.
   c) Click **OK**.
7. Click **VALIDATE CONNECTION**.
8. Enter **Collector/Group** to manage the management pack on VCF Operations. If you have only one account, select Default collector group.
9. Click **ADD**.

Verify that the adapter is configured and collecting data.

**Dashboards**
Use the dashboards to understand the availability and health of virtual services and to ensure consistency in the configuration of the virtual services.

Avi Load Balancer Availability & Health Dashboard
Use the **Avi Load Balancer Availability & Health** dashboard to understand the availability of a virtual service and associated service engines and the current availability status and health score as reported in the recent collection cycle.

**How to Use the Dashboard**

- Select an Avi Load Balancer instance from the **Avi LB Instances** widget.
- You see the availability and health of the virtual services for the selected instance in the **Virtual Service Availability Heatmap** and **Virtual Service Health Heatmap** widgets.
- Similarly, you see the availability and health of the service engines for the selected instance in the **Service Engine Availability Heatmap** and **Service Engine Health Heatmap** widgets.

By default, the heatmap widgets have three color codes. Green indicates that the health score of the virtual service or service engine is between 90–100. Yellow/orange means that the health score of the virtual service or service engine is between 70–89 and there are issues that should be identified and fixed. Red indicates that the health score of the virtual service or service engine is less than 70, has deteriorated, and needs immediate attention.

> **Note:**
> The **Virtual Services Health Overview** dashboard complements the **Avi Load Balancer Availability & Health** dashboard. For more information, see the Virtual Services Health Overview Dashboard.

Virtual Services Health Overview Dashboard
Use the **Virtual Services Health Overview** dashboard to get an overview of the health of the virtual services across all of your Avi Load Balancer instances.

**How To Use the Dashboard**

- Select an Avi Load Balancer instance from the **Avi LB Instances** widget.
- You see a list of corresponding virtual services for the selected Avi Load Balancer instance in the **Virtual Services** widget. Select a virtual service from the list. The **Virtual Services** widget also provides details of all the virtual services in the selected instance. It can be used to quickly check the status and view high level information and important metrics of the virtual service.
- Use the **Health Score Details** widget to understand details on specific aspects, such as performance, resource penalty, anomaly penalty, and security penalty that are the factors that determine the health score of a virtual service. This provides you with a starting point to troubleshoot the virtual service and eventually the application itself.
- Use the **Virtual Service End to End timing** widget to view the time taken for the application to respond at the server level, application level, and data transfer level. You can also view the average total time taken for the application to respond to client requests.
- Use the **Health Score Distribution Chart** widget to view the distribution of virtual services based on their current health score. For example, you might want to prioritise troubleshooting virtual services with the lowest health score.
- Use the **DNS Type Virtual Service Details** widget to view the DNS type virtual services and the related DNS queries, DNS query type, the DNS percentage errors, and other details.
- Use the **Virtual Service Metrics** widget to view the selected virtual service and related metrics.

Virtual Services Configuration Overview Dashboard

Use the **Virtual Services Configuration Overview** dashboard to ensure configuration consistency across all groups of virtual services.

**How to Use the Dashboard**

- Select an Avi Load Balancer instance from the **Avi LB Instances** widget.
- You see a list of corresponding virtual services for the selected Avi Load Balancer instance in the **Virtual Services** widget. Select a virtual service from the list.
- Use the **Profiles** widget to view the list of objects and other details.
- Use the **Service Ports** widget to view the ports that are open to service the client requests.
- Use the **Analytics** widget to view settings related to analytics across each virtual service.
- Use the **Advanced Settings** widget to view other configuration details for the virtual service selected.

Avi Load Balancer License Overview Dashboard

**Avi Load Balancer License Overview** dashboard provides an overview of licenses across different licensing models.

**How to Use the Dashboard**

- Select an Avi Load Balancer instance from the **Avi LB Instances** widget.
- In the **Current License Tier** widget, you can view the license tier option (Basic/Enterprise/Enterprise with Cloud Services/Essentials) selected in Avi LB instance:
  - Current Tier: Selected license type.
  - Message: License related information.
  - Service Core Usage (%): Total consumed service core capacity percentage.
  - Service Core Usage: Total service core consumed out of the total core capacity.
  - Reserved Service Licenses: Minimum service units that always remain reserved on the controller.
  - Max Service Licenses: Maximum service units limit for the controller.
  - Cloud Services Available Licenses: Available service units in the org.
- Similarly, you can see the capacity and consumption information for the other available license types in the **Available License Tiers** widget.

**Alerts**

Alerts occur in two ways, based on the health score and the alert forwarding mechanism from the Avi Load Balancer.

- **Based on the health score**

Alerts are triggered for virtual services and service engines based on the health score, using the same color code used in the Avi Load Balancer. The default Wait/Cancel cycle is set to 1. Alerts are grouped as following:

- Critical, when the total health score is less than 70.
- Warning, when the health score is between 70 and 89.

- **Alert forwarding from the Avi Load Balancer**

  The management pack has an alert forwarding mechanism that mirrors the alerts and sends them from the Avi Load Balancer to VCF Operations.

  Any new alert generated in the Avi Load Balancer instance, after the previous collection cycle, is forwarded to the corresponding object in VCF Operations as a "Notification Event". If the same alert does not recur in the next collection cycle, then it is assumed that the alert has been canceled. Alerts are grouped into Critical, Warning, and Info, corresponding to the alert levels from the Avi Load Balancer which are "ALERT_HIGH", "ALERT_MEDIUM", "ALERT_LOW.

  These alerts are currently not supported on the Management Controller and Controller objects because of an ambiguity while mapping alerts to these two objects.

**Metrics and Properties**

The management pack collects metrics and properties for the supported objects.

Metrics

VCF Operations connects to an instance of the Avi Load Balancer and collects metrics for the management controller, controller, tenant, cloud, virtual service, pool, service engine, service engine groups, and network object types.

Management Controller Metrics

Metrics are collected for the management controller object type.

Here is the list of management controller metrics:

- Controller Count
- Average CPU Usage (%)
- Average Disk Usage (%)
- Average Memory Usage (%)
- Number Of Metrics Collected
- Number Of Objects Collected
- Number Of Down Objects
- Number Of Tenants
- Number Of Clouds

Controller Metrics

Metrics are collected for the controller object type.

**Controller Metrics**

| Metric Name | Category | Description |
|---|---|---|
| CPU Usage | Controller Statistics | The host's view of the CPU usage as the amount of actively used virtual CPU, as a percentage of the total available CPU. |
| Disk Usage | Controller Statistics | Average disk usage. |
| Memory Usage | Controller Statistics | Memory usage as percentage of the total configured or available memory. |

Cloud Metrics

Metrics are collected for the vCenter, AWS, GCP, and Azure object types.

**vCenter, AWS, Azure, and GCP Metrics**

- Num of Virtual Services
- Num of Service Engines
- Num of Pools
- Number Of Down Objects

Pool Metrics

Metrics are collected for the pool object type.

**Pool Metrics**

| Metric Name | Category | Description |
|---|---|---|
| Open Connections | Pool | Average per server open connections in the pool. |
| Throughput | Pool | Average transmit and receive network bandwidth between client and virtual service. |
| Server RTT | Pool | Averaged RTT across all connections closed. |
| Request Errors | Pool | Rate of HTTP error responses sent per second. It does not include errors excluded in the analytics profile. |
| RPS | Pool | Average rate of HTTP requests received by a pool's server per second. |
| Request Errors % | Pool | Percent of HTTP 4xx and 5xx responses. |
| Total Score | Health Score | Total health score of the pool. |
| Security Threat Level | Health Score | Security penalty score of the pool. |
| Anomaly Score | Health Score | Anomaly penalty score of the pool. |
| Resource Score | Health Score | Resource penalty score of the pool. |
| Performance Score | Health Score | Performance score of the pool. |

Service Engine Metrics

Metrics are collected for the service engine object type.

**Service Engine Metrics**

| Metric Name | Category | Description |
|---|---|---|
| Connections | Service Engine | Average number of connections including the dropped connections and ones due to policy drops. This would be the same as the number of SYNS seen by service engines on any virtual service. |
| Dropped Connections % | Service Engine | Total number of connections dropped excluding the dropped connections and ones due to policy drops. This would be same as number of SYNS seen by the service engine on any virtual service. It include both connections that failed to establish. |
| Throughput | Service Engine | Average transmit and receive network bandwidth for an interface. |

| Metric Name | Category | Description |
|---|---|---|
| CPU Usage | Service Engine | The host's view of the CPU usage. The amount of actively used virtual CPU, as a percentage of total available CPU. |
| Memory Usage | Service Engine | Memory usage as percentage of total configured or available memory. |
| Disk1 Usage | Service Engine | Service engine disk 1 capacity usage. |
| RX Packets | Service Engine | Average rate of received packets per second. |
| RX Bytes | Service Engine | Average rate of received bytes on an interface. |
| RX packets Drop | Service Engine | Average number of received packets dropped per sec. It includes packets across all virtual services and non-virtual services. |
| TX Bytes | Service Engine | Average rate of transmitted bytes on an interface. |
| TX Packets | Service Engine | Average rate of transmit packets per second. |
| Interface Throughput | Service Engine | Average transmit and receive network bandwidth for an interface. |
| Connection Memory Usage | Service Engine | Connection memory usage as percentage of total allowed for connections. |
| Dynamic Memory Usage | Service Engine | Percentage of service engine's dynamic memory usage. |
| SSL Cache Usage | Service Engine | SSL session cache usage in percent. |
| Buffer Usage | Service Engine | Packet Buffer usage as percentage of total configured packet buffers. |
| Persistent Table Usage | Service Engine | Session persistent table entries usage percent. |
| Syn Cache Usage | Service Engine | Percentage of SYN cache usage. Higher usage indicates too many connection attempts and open at service engine. |
| Total Score | Health Score | Total health score of the service engine. |
| Anomaly Score | Health Score | Anomaly penalty of the service engine. |
| Performance Score | Health Score | Performance score of the service engine. |
| Resources Score | Health Score | Resource penalty of the service engine. |
| Security Threat Level | Health Score | Security penalty score of the service engine. |

### Tenant Metrics

Metrics are collected for the tenant object type.

Here is the list of tenant metrics:

- Num of Virtual Services
- Num of Service Engines
- Num of Pools
- Number Of Down Objects

Virtual Service Metrics
Metrics are collected for the virtual service object type.

**Virtual Service Metrics**

| Metric Name | Category | Description |
|---|---|---|
| Average Bandwidth | L4 statistics | Average transmit and receive network bandwidth between client and virtual service. |
| Average New Established Connections | L4 statistics | Averaged rate of new client connections per second. |
| Percent Connections DOS Attacks | L4 statistics | Fraction of L4 connections owing to DoS. |
| Average New SSL Handshakes | L7 statistics | Average new successful SSL sessions. |
| Max SSL Open Connections | L7 statistics | Maximum number of open SSL sessions. |
| RPS | Virtual Service Metrics | Average rate of client HTTP requests received by the virtual service per second. |
| CPS | Virtual Service Metrics | Rate of total connections per second. |
| Open Connections | Virtual Service Metrics | Max number of open connections. |
| Client RTT | Virtual Service Metrics | Average network round trip time between client and virtual service. |
| Server RTT | Virtual Service Metrics | Averaged RTT across all connections closed. |
| App response | Virtual Service Metrics | Average latency measured for pool server. |
| Data transfer time | Virtual Service Metrics | Average client data transfer time that represents latency of sending response to the client excluding the RTT time. Higher client data transfer time signifies lower bandwidth between client and the service engine. |
| End to end timing | Virtual Service Metrics | Average network round trip time between the client and virtual service. |
| Request packets | Virtual Service Metrics | Average rate of packets received per second. |
| Tx packets | Virtual Service Metrics | Average rate of packets transmitted per second. |
| Requests | Virtual Service Metrics | Average rate of client HTTP requests received by the virtual service per second. |
| Request Errors % | Virtual Service Metrics | Percent of 4xx and 5xx responses. |
| Rx Packets | Virtual Service Metrics | Average rate of packets received per second. |
| Tx Packets | Virtual Service Metrics | Average rate of packets transmitted per second. |
| DoS attacks | Virtual Service Metrics | Number DDOS attacks occurring. |
| Throughput | Virtual Service Metrics | Average transmit and receive network bandwidth between client and virtual service. |

| Metric Name | Category | Description |
|---|---|---|
| HTTP / HTTP2 Request | Virtual Service Metrics | Rate of HTTP responses sent per second and average number of client HTTP2 requests received by the Virtual Service per second. |
| DNS Query Types | Virtual Service Metrics | Rate of DNS Responses of type A per second, rate of DNS Requests of type AAAA per second, rate of DNS Requests of type NS per second, rate of DNS requests of type SRV per second, rate of DNS requests of type MX per second, and rate of DNS requests of type OTHER per second. |
| DNS Queries, % errors | Virtual Service Metrics | Rate of DNS responses per second, including errors and Percentage of DNS responses with errors. |
| TCP vs UDP | Virtual Service Metrics | Rate of TCP queries per second and rate of UDP queries per second. |
| Total Score | Health Score | Total health score of the virtual service. The final health score for the virtual service equals the performance score minus the various penalty scores. |
| Security Threat Level | Health Score | Security penalty score of the virtual service. The security penalty shows risks due to configuration settings that leave a site vulnerable, as well as vulnerability to attacks such as DDoS. |
| Anomaly Score | Health Score | Anomaly penalty score of the virtual service. Anomaly is defined as traffic that is outside the statistical norm for the time period. |
| Resource Score | Health Score | Resource penalty score of the virtual service. Any penalty assessed because of resource availability issues is assigned a score, which is then subtracted from the performance score. |
| Performance Score | Health Score | Performance score of the virtual service. Performance is comprised of end user's experience and server response. The performance score is a positive number from 0 (down) to 100 (ideal). |

### Properties

VCF Operations connects to an instance of the Avi Load Balancer and collects metrics for the controller, management controller, tenant, cloud, virtual service, pool, service engine, service engine groups, and network object types.

Some properties are listed in VCF Operations only if the corresponding properties are configured in the Avi Load Balancer.

### Management Controller Properties

Properties are collected for the management controller object type.

Here is the list of management controller properties:

- Controller Version

- Count of Regions
- Total Service Core Usage
- Total Service Core Usage (%)
  - Basic
    - Current Tier
    - Message
    - Service Core Usage
    - Service Core Usage (%)
  - Enterprise
    - Current Tier
    - Message
    - Service Core Usage
    - Service Core Usage (%)
  - Enterprise with Cloud Services
    - Current Tier
    - Message
    - Service Core Usage
    - Service Core Usage (%)
    - Cloud Services Available Licenses
    - Max Service Licenses
    - Reserved Service Licenses
  - Essentials
    - Current Tier
    - Message
    - Service Core Usage
    - Service Core Usage (%)

### Controller Properties

Properties are collected for the controller object type.

**Controller Properties**

| Property Name | Category | Description |
|---|---|---|
| Name | Summary | Name of the controller. |
| URL | Summary | URL of the controller. |
| Port | Summary | Port of the controller. |
| IP Address | Summary | IP address of the controller. |
| Public IP | Summary | Value of controller tag 'public_ip_or_name'. |
| UUID | Summary | UUID of the controller. |
| Version | Summary | Controller version. |
| Role | Summary | Controller role: whether leader or follower. |
| Status | Summary | Current status of the controller. |
| Current Tier | Basic | Selected Licence tier option. |
| Message | Basic | License related information. |
| Service Core Usage | Basic | No of consumed core out of the total core capacity. |
| Service Core Usage (%) | Basic | Total core consumption percentage. |
| Current Tier | Enterprise | Selected Licence tier option. |

| Property Name | Category | Description |
|---|---|---|
| Message | Enterprise | License related information. |
| Service Core Usage | Enterprise | No of consumed core out of the total core capacity. |
| Service Core Usage (%) | Enterprise | Total core consumption percentage. |
| Current Tier | Enterprise with Cloud Services | Selected Licence tier option. |
| Message | Enterprise with Cloud Services | License related information. |
| Service Core Usage | Enterprise with Cloud Services | No of consumed core out of the total core capacity. |
| Service Core Usage (%) | Enterprise with Cloud Services | Total core consumption percentage. |
| Cloud Services Available Licenses | Enterprise with Cloud Services | Available service units in the org. |
| Max Service Licenses | Enterprise with Cloud Services | Maximum service units limit for the controller. |
| Reserved Service Licenses | Enterprise with Cloud Services | Minimum service units that always remain reserved on the controller. |
| Current Tier | Essentials | Selected Licence tier option. |
| Message | Essentials | License related information. |
| Service Core Usage | Essentials | No of consumed core out of the total core capacity. |
| Service Core Usage (%) | Essentials | Total core consumption percentage. |

Cloud Properties

Cloud properties are collected for the vCenter, AWS, GCP, and Azure object types.

**vCenter Properties**

| Property Name | Category | Description |
|---|---|---|
| Name | Summary | Name of the object. |
| Cloud Type | Summary | Type of cloud. |
| Status | | Current status of the cloud. |
| vCenter Hostname Address | Summary | vCenter hostname or IP address . |
| UUID | Summary | UUID of the object to fetch. |
| DHCP Enabled | Summary | The IP address management scheme. |
| Management Network | Summary | Management network to use for the service engines. Reference to an object of type VIMgrNWRuntime. |
| Access Permission | Summary | Access mode to vCenterwhich is either Read, which allows the Avi Load Balancer to discover networks and servers, or Write, which also allows the Avi Load Balancer to create service engines and configure their network properties. |
| Tenant | Summary | Reference to an object of type Tenant. |
| License Type | Infrastructure | If no license type is specified then default license enforcement for the cloud type is chosen. |

| Property Name | Category | Description |
|---|---|---|
| IPAM Profile | IPAM/DNS | IPAM Profile for the cloud. Reference to an object of type IpamDnsProviderProfile |
| DNS Profile | IPAM/DNS | DNS Profile for the cloud. Reference to an object of type IpamDnsProviderProfile |
| State Based DNS Registration | IPAM/DNS | DNS records for VIPs are added/deleted based on the operational state of the VIPs. |
| Data Center | Data Center Properties | Datacenter for virtual infrastructure discovery. |
| IPv6 Auto Configuration | Data Center Properties | Enable IPv6 auto configuration. |
| Prefer Static Routes vs Directly Connected Network | Data Center Properties | Prefer static routes over interface routes during VirtualService placement. |
| Use Static Routes for Network Resolution of VIP | Data Center Properties | Use static routes for VIP side network resolution during VirtualService placement. |
| Template Service Engine Group | Service Engine | The service engine group to use as template. Reference to an object of type ServiceEngineGroup. |
| IP Subnet | Service Engine | Management subnet to use for service Engines |

## AWS Properties

| Property Name | Category | Description |
|---|---|---|
| Name | Summary | Name of the object type. |
| Cloud Type | Summary | Type of cloud. |
| Region | Summary | AWS region. |
| Route53 Integration | Summary | If enabled, create/update DNS entries in Amazon Route 53 zones. |
| VPC | Summary | VPC name. |
| VPC Id | Summary | VPC ID. |
| UUID | Summary | UUID of the object to fetch. |
| Availability Zone | AWS Zone Config | Availability zone. |
| Management Network Name | AWS Zone Config | Name or CIDR of the network in the Availability Zone that will be used as management network. |
| Management Network Id | AWS Zone Config | UUID of the network in the Availability Zone that will be used as management network. |
| License Type | Infrastructure | If no license type is specified then default license enforcement for the cloud type is chosen. |
| DNS registration for VS Names | DNS | DNS Profile for the cloud. It is a reference to an object of type IpamDnsProviderProfile. |
| State Based DNS Registration | DNS | DNS records for VIPs are added/deleted based on the operational state of the VIPs. |
| SQS for Autoscale Groups Monitoring | AWS VPC and Availability Zones | Use SNS/SQS based notifications for monitoring Auto Scaling Groups. |
| Template Service Engine Group | AWS VPC and Availability Zones | The service engine group to use as template. Reference to an object of type ServiceEngineGroup. |

| Property Name | Category | Description |
|---|---|---|
| Encryption for SE S3 Bucket | AWS VPC and Availability Zones | S3 encryption mode and the master key to be used for encrypting S3 buckets during SE AMI upload. Only SSE-KMS mode is supported. |
| Encryption for SE AMI/EBS volumes | AWS VPC and Availability Zones | EBS encryption mode and the master key to be used for encrypting SE AMI, Volumes, and Snapshots. |
| ASG Polling Interval | AWS VPC and Availability Zones | Time interval between periodic polling of all Auto Scaling Groups. |

## Azure Properties

| Property Name | Category | Description |
|---|---|---|
| Cloud Type | Summary | Type of cloud. |
| Name | Summary | Name of the object type. |
| Autoscale Polling Interval | Summary | CloudConnector polling interval in seconds for external autoscale groups, minimum 60 seconds. |
| Azure Subscription ID | Summary | Subscription ID for the Azure subscriptions. |
| UUID | Summary | UUID of the object to fetch. |
| License Type | Infrastructure | If no license type is specified then default license enforcement for the cloud type is chosen. |
| DNS registration | DNS | Whether Azure is the DNS provider. |
| DNS Provider | DNS | DNS Profile for the cloud. Reference to an object of type IpamDnsProviderProfile. |
| State Based DNS Registration | DNS | DNS records for VIPs are added/deleted based on the operational state of the VIPs. |
| Location | Location Or Network | Azure location where this cloud is located.. |
| Resource Group | Location Or Network | Azure resource group dedicated for the controller. The controller create all its resources in this resource group. |
| VNet | Location Or Network | Virtual network where virtual IPs belong. |
| Service Engine Network | Location Or Network | ID of the Azure subnet where the controller creates the service engines. |
| Use Dedicated Management Interface | Location Or Network | ID of the Azure subnet used as management network for the service engines. If set, service engines have a dedicated management NIC. Otherwise, they operate in inband mode. |
| Template Service Engine Group | Service Engine | The service engine group association for the service engine. If None, then the Default-Group, which is SEGroup, is associated with the service engine. Reference to an object of type ServiceEngineGroup. |
| Disk Encryption Set | Disk | Disks Encryption Set resource ID (des_id) to encrypt service engine images and managed disks using customer-managed-keys. |

## GCP Properties

| Property Name | Category | Description |
|---|---|---|
| Cloud Type | Summary | Type of cloud. |
| Name | Summary | Name of the object. |
| UUID | Summary | UUID of the object to fetch. |
| Service Account Email | GCP Account | Email of the GCP Service Account to be associated to the service engines. |
| GCS Bucket Name | GCP Account | Google Cloud Storage Bucket name where the service engine image is uploaded. This image will be deleted once the image is created in Google compute images. By default, a bucket will be created if this field is not specified. |
| SE Project Id | GCP Account | Google Cloud Platform Project ID where service engines will be spawned. |
| VPC Network Name | GCP Account | Google Cloud Platform VPC network configuration for the service engines. |
| GCS Project Id | GCP Account | Google Cloud Storage Project ID where the service engine image is uploaded. This image is deleted once the image is created in Google compute images. By default, the service engine Project ID is used. |
| Name | Zones | Google Cloud Platform region name where service engines will be spawned. |
| Regions | Zones | Google Cloud Platform zones where service engines will be distributed for HA. |

Network Properties

Properties are collected for the network object type.

## Network Properties

| Property Name | Category | Description |
|---|---|---|
| Name | Summary | Name of the object. |
| UUID | Summary | UUID of the object to fetch. |
| Discovered Subnets | Summary | A list of one or more IP networks that are auto-discovered via the virtualization orchestrator. |
| VRF Context | Summary | Reference to an object of type VrfContext. |
| DHCP Enabled | Summary | The IP address management scheme for this network. |
| IPv6 Auto Configuration | Summary | Enable IPv6 auto configuration. |

Pool Properties

Properties are collected for the pool object type.

## Pool Properties

| Property Name | Category | Description |
|---|---|---|
| Name | Summary | Name of the object. |
| UUID | Summary | UUID of the object to fetch. |
| Status | Summary | Enable or disable the pool. Disabling will terminate all open connections and pause health monitors. |
| Active Monitors | Summary | Verifies server health by applying one or more health monitors. Active monitors generate synthetic traffic from each service engine and mark a server up or down based on the response. The passive monitor listens only to client to server communication. It raises or lowers the ratio of traffic destined to a server based on successful responses. Reference to an object of type HealthMonitor. |
| Server Name | Summary | DNS resolvable name of the server. May be used in place of the IP address. |
| IP Address | Summary | IP address of the server. Required if there is no resolvable host name. |
| Port | Summary | Optionally specify the servers port number. This will override the pool's default server port attribute. |
| Row | Summary | Ratio of selecting eligible servers in the pool. Allowed values are 1-20. |
| Cloud | Summary | Reference to an object of type Cloud. |
| Tenant | Summary | Reference to an object of type Tenant. |
| Virtual Services Count | Summary | Count of virtual services using this pool. |
| Virtual Services | Summary | Name of the virtual services using this pool. |
| Servers (Up/Total) | Summary | Ratio of the total number of servers in the pool that are up, to the total number of servers configured in the pool. |
| VRF Context | Summary | Virtual Routing Context that the pool is bound to. Used to provide isolation of the set of networks the pool is attached to. Reference to an object of type VrfContext |
| Active Health Monitors | Summary | Verifies server health by applying one or more health monitors. Reference to an object of type HealthMonitor. |
| Server Name | Summary | DNS resolvable name of the server. May be used in place of the IP address. |
| Status | Summary | Enable, disable, or graceful disable determine if new or existing connections to the server are allowed. |
| IP Address | Summary | IP Address of the server. |
| Port | Summary | Port number of the server. Overrides the pool's default server port attribute. |
| Ratio | Summary | Ratio of selecting eligible servers in the pool. Allowed values are 1-20. |
| Default Server Port | Configuration | Traffic sent to servers will use this destination server port unless overridden by the server's specific port attribute. |

| Property Name | Category | Description |
|---|---|---|
| Graceful Disable Timeout | Configuration | Virtual service waits for the specified time before terminating the existing connections to the servers that are disabled. |
| Load Balance Algorithm | Configuration | The load balancing algorithm will pick a server within the pool's list of available servers. |
| Passive Health Monitor | Configuration | The Passive monitor will monitor client to server connections and requests and adjust traffic load to servers based on successful responses. |
| Min. Health Monitors to consider server 'up' | Configuration | Minimum number of health monitors in UP state to mark server UP. |
| AutoScale Policy | Configuration | Reference to Server Autoscale Policy. Reference to an object of type ServerAutoScalePolicy. |
| AutoScale Launch Config | Configuration | If configured, then the Avi Load Balancer will trigger an orchestration of pool server creation and deletion. Reference to an object of type AutoScaleLaunchConfig. |
| Persistence | Configuration | Persistence ensures that a user sticks to the same server for a desired duration of time. Reference to an object of type ApplicationPersistenceProfile. |
| Analytics Profile | Configuration | Specifies settings related to analytics. Reference to an object of type AnalyticsProfile. |
| Append Port | Configuration | The option to append port to hostname in the host header while sending a request to the server. By default, port is appended for non-default ports. |
| Lookup Server by Name | Configuration | Server lookup by name. |
| Rewrite Host Header to Server Name | Configuration | Rewrite incoming Host Header to server name of the server to which the request is proxied. |
| Enable real time metrics | Configuration | Enable real time metrics for server and pool metrics. |
| SSL | SSL to backend servers | When enabled, the Avi Load Balancer re-encrypts traffic to the backend servers. The specific SSL profile defines which ciphers and SSL versions will be supported. Reference to an object of type SSLProfile. |
| Server Disable Type | SSL to backend servers | Server graceful disable timeout behavior. |
| Request Queuing | Pool Full Settings | Enable request queue when pool is full. |
| Queue Length | Pool Full Settings | Minimum number of requests to be queued when pool is full. |
| Fail Action | Pool Full Settings | Enables a response to the client when the pool experiences a failure. By default TCP connection is closed. |
| Placement Settings/Server Network | Pool Full Settings | Manually select the networks and subnets used to provide reachability to the pool's servers. |
| Connection Idle Timeout | Connection Pool Settings | Connection idle timeout. |
| Connection Life Timeout | Connection Pool Settings | Connection life timeout. |

| Property Name | Category | Description |
|---|---|---|
| Connection Max Used Times | Connection Pool Settings | Maximum number of times a connection can be reused. |
| Max Cache Connections Per Server | Connection Pool Settings | Maximum number of connections a server can cache. |
| Disable Port Translation | Other Settings | Does not translate the client's destination port when sending the connection to the server. |
| Default Server Timeout | Other Settings | Server timeout value specifies the time within which a server connection needs to be established and a request-response exchange completes between the Avi Load Balancer and the server. |
| Connection Ramp | Other Settings | Duration for which new connections will be gradually ramped up to a server recently brought online. |
| Max Connections per Server | Other Settings | The maximum number of concurrent connections allowed to each server within the pool. |
| Routing Pool | Other Settings | Routing is enabled when this pool is selected to send traffic. |
| HTTP Server Reselect | Other Settings | Server reselect configuration for HTTP requests. |

Service Engine Properties

Properties collected for the service engine object type.

**Service Engine Properties**

| Property Name | Category | Description |
|---|---|---|
| Name | Summary | Name of the object. |
| UUID | Summary | UUID of the object to fetch. |
| Status | Summary | Boolean value to indicate if the service engine is up. |
| Cloud | Summary | Reference to an object of type Cloud. |
| Tenant | Summary | Reference to an object of type Tenant. |
| Virtual Services Count | Summary | Count of virtual services in the service engine. |
| Virtual Services | Summary | Virtual services present within the service engine. |
| Enable State | Summary | Set this field to disable the service engine. |
| SE Group | Summary | Reference to an object of type ServiceEngineGroup. |
| Management IP | Summary | Management IP address. |
| Availability Zone | Summary | The availability_zone of the service engine. |
| Host | Summary | Reference to an object of type VIMgrHostRuntime. |
| **Management Network** (Network used by the service engine to communicate with the controller) | | |
| Interface | Summary | Interface name of the management network. |

| Property Name | Category | Description |
|---|---|---|
| Network Name | Summary | Network name of the management network. |
| MAC Address | Summary | MAC address of the management network. |
| DHCP | Summary | DHCP of the management network. |
| IP Address[Mode on IP] | Summary | IP address of the management network. |
| VRF | Summary | Reference to an object of type VrfContext. |
| **Data Networks** (Network used for load balancing) | | |
| Interface | Summary | Interface name of the data network. |
| Network Name | Summary | Network name of the data network. |
| MAC Address | Summary | MAC address of the data network. |
| DHCP | Summary | DHCP of the data network. |
| IP Address | Summary | IP address of the data network. |
| VRF | Summary | Reference to an object of type VrfContext. |

Service Engine Group Properties

Properties are collected for the service engine group object type.

**Service Engine Group Properties**

| Property Name | Category | Description |
|---|---|---|
| Name | Summary | Name of the object. |
| UUID | Summary | UUID of the object to fetch. |
| Tenant | Summary | Reference to an object of type Tenant. |
| Cloud | Summary | Reference to an object of type Cloud. |
| Version | Summary | Version and build of the service engine group created. |
| Num of Service Engines | Summary | Number of service engines belonging to a particular service engine group. |
| Max Num of Service Engines | Summary | Maximum number of services engines in the group. |
| Num of Virtual Service | Summary | Total number of virtual services hosted in the service engine that belongs to a particular service engine group. |
| Max Num of Virtual Services per Service Engine | Summary | Maximum number of virtual services that can be placed on a single service engine. |
| RealTime Metrics | Summary | Enable or deactivate real time service engine metrics. |
| HA Mode | High Availability & Placement Settings | High Availability mode for all the virtual services using the service engine group. |
| SE Self-Election | High Availability & Placement Settings | Enables service engines to elect a primary amongst themselves in the absence of connectivity to a controller. |
| VS Placement across Ses | High Availability & Placement Settings | If placement mode is 'Auto',virtual services are automatically placed on service engines. |

| Property Name | Category | Description |
|---|---|---|
| Max Number of Service Engine | Service Engine Capacity and Limit Settings | Maximum number of services engines in the group. |
| Memory per Service Engine | Service Engine Capacity and Limit Settings | Amount of memory for each of the service engine virtual machines. |
| Memory Reserve | Service Engine Capacity and Limit Settings | Description of the property mem_reserve of the object type ServiceEngineGroup. |
| vCPU per Service Engine | Service Engine Capacity and Limit Settings | Number of vCPUs for each of the service engine virtual machines. |
| CPU Reserve | Service Engine Capacity and Limit Settings | Description of property cpu_reserve of the object type ServiceEngineGroup. |
| Disk per Service Engine | Service Engine Capacity and Limit Settings | Amount of disk space for each of the service engine virtual machines. |
| Host Geolocation Profile | Memory Allocation | Extra configuration memory to support large Geo DB configurations. |
| Memory for Caching | Memory Allocation | A percent value of total service engine memory reserved for application caching. |
| Available Memory for Connections and Buffers | Memory Allocation | Percentage of memory for the connection state. |
| Connections Memory Distribution | Memory Allocation | Percentage of memory for connection state. |
| Buffers Memory Distribution | Memory Allocation | Percentage of memory for buffers state. |
| License Type | License | If no license type is specified, then the default license enforcement for the cloud type is chosen. |
| Enable Per-app SE Mode | License | Per-app service engine mode is designed for deploying dedicated load balancers per app (virtual service). |
| SE Bandwidth Type | License | The bandwidth of the service engine for the bandwidth license. |
| Number of SE Data Paths | License | Configures the maximum number of se_dp processes that handles traffic. |
| Hyperthreading | License | Controls the distribution of service engine data path processes on CPUs which support hyper-threading. |
| Cluster Included | Host & Data Store Scope | List of clusters where service engines will be deployed. |
| Cluster Excluded | Host & Data Store Scope | List of clusters where service engines are not deployed. |
| Data Store Scope for Service Engine Virtual Machine | Host & Data Store Scope | Enum options - VCENTER_DATASTORE_ANY, VCENTER_DATASTORE_LOCAL, VCENTER_DATASTORE_SHARED |
| Buffer Service Engines | Advanced HA & Placement | Excess service engine capacity provisioned for HA failover. |
| Scale per Virtual Service (Min) | Advanced HA & Placement | Minimum number of active service engines for the virtual service. |
| Scale per Virtual Service (max) | Advanced HA & Placement | Maximum number of active service engines for the virtual service. |

| Property Name | Category | Description |
|---|---|---|
| CPU socket Affinity | Advanced HA & Placement | Allocate all the CPU cores for the service engine VMs on the same CPU socket. |
| Dedicated dispatcher CPU | Advanced HA & Placement | Dedicate the core that handles packet receive/transmit from the network to just the dispatching function. Not used for TCP/IP and SSL functions. |
| Override Management Network | Advanced HA & Placement | Management network to use for service engines. Reference to an object of type Network. |
| Significant Log Throttle | Log Collection and Streaming Settings | The setting limits the number of significant logs generated per second per core on this service engine. |
| UDF Log Throttle | Log Collection and Streaming Settings | The setting limits the number of UDF logs generated per second per core on this service engine. |
| Non-significant Log Throttle | Log Collection and Streaming Settings | The setting limits the number of non-significant logs generated per second per core on this service engine. |
| Number of Streaming Threads | Log Collection and Streaming Settings | Number of threads to use for log streaming. |
| Instance Flavor | AWS and Azure | Instance/Flavor name for the service engine instance. |
| Service Engine Name Prefix | AWS and Azure | Prefix to use for the virtual machine name of service engines. |
| Delete Unused Service Engines After | AWS and Azure | Duration to preserve unused service engine virtual machines before deleting them. If traffic to a virtual service were to spike up abruptly, this service engine would be available to be utilized again rather than creating a new service engine. |
| Avi Managed Security Group | AWS and Azure | By default, security groups are created and managed along with the custom security group provided by the user. Set this to True to disallow the creation and management of new security groups. Only custom security groups provided by user are used. |
| Management vNIC Custom Security Groups | AWS and Azure | Custom security groups to be associated with management vNics for service engine instances. |
| Data vNIC Custom Security Groups | AWS and Azure | Custom security groups to be associated with data vNICs for service engine instances. |
| Auto Rebalance | AWS and Azure | If set, virtual services are automatically migrated when load on a service engine is less than minimum or more than maximum thresholds. Only alerts are generated when the auto_rebalance is not set. |

Tenant Properties

Properties are collected for the tenant object type.

**Tenant Properties**

| Property Name | Category | Description |
|---|---|---|
| Name | Tenant | Name of the object. |
| Description | Tenant | User defined description for the object. |
| URL | Tenant | URL of the object. |
| UUID | Tenant | Unique object identifier of the object. |
| Tenant Access to Provider Service Engine | Tenant | Description of the property `tenant_access_to_provider_se` in the TenantConfiguration field. |
| Tenant VRF | Tenant | When 'Per Tenant IP Domain' is selected, each tenant gets its own routing domain that is not shared with any other tenant. When 'Share IP Domain across all tenants' is selected, all tenants share the same routing domain. |

Virtual Service Properties

Properties are collected for the virtual service object type.

**Virtual Service Properties**

| Property Name | Category | Description |
|---|---|---|
| Name | Summary | Name of the object. |
| UUID | Summary | UUID of the object to be fetched. |
| Status | Summary | Enable or disable the Virtual Service. |
| Traffic Enabled | Summary | Knob to enable the Virtual Service traffic on its assigned service engines. |
| VIP Address | Summary | Virtual IP address. |
| IPv4 Address | Summary | IPv4 address of the virtual IP. |
| IPv6 Address | Summary | IPv6 address of the virtual IP. |
| Application Domain Name | Summary | Fully qualified domain name. |
| VS VIP | Summary | Indicates the vip_id that needs placement retrial. |
| Cloud | Summary | Reference to an object of type Cloud. |
| Status | Summary | Boolean value suggesting whether the virtual service is up or not. |
| Pools | Summary | Pools to which the virtual service has to load balance to. |
| Virtual Hosting | Summary | Specifies the virtual service acting as virtual hosting (SNI) parent. Reference to an object of type VirtualService. |
| VRF Context | Summary | Virtual Routing Context that the virtual service is bound to. This is used to provide the isolation of the set of networks the application is attached to. Reference to an object of type VrfContext. |
| Service Port 1 | Service Port | The Virtual Service's port number. |

| Property Name | Category | Description |
|---|---|---|
| Service Port 2 | Service Port | The Virtual Service's port number. |
| Service Port N | Service Port | The Virtual Service's port number. |
| TCP/UDP Profile | Profiles | Determines network settings such as protocol, TCP or UDP, and related options for the protocol. Reference to an object of type NetworkProfile. |
| Application Profile | Profiles | Enables application layer specific features for the Virtual Service. Reference to an object of type ApplicationProfile. |
| Error Page Profile | Profiles | This profile is used to send the custom error page to the client generated by the proxy. Reference to an object of type ErrorPageProfile. |
| Name | Pool Type | An object that contains destination servers and related attributes such as load-balancing and persistence. Reference to an object of type Pool. |
| Pool Group | Pool Type | Reference to an object of type Pool. |
| Analytics Profile | Analytics | Specifies settings related to analytics. Reference to an object of type AnalyticsProfile. |
| Metrics Update Frequency | Analytics | Settings to set duration for real time updates. |
| Real Time Metrics | Analytics | Settings to turn on real time metrics. |
| Significant Log Throttle | Client Log Settings | This setting limits the number of significant logs generated per second for the virtual service on each service engine. |
| User defined filters Log Throttle | Client Log Settings | This setting limits the total number of UDF logs generated per second for the virtual service on each service engine. UDF logs are generated because of the configured client log filters or the rules with logging enabled. |
| Non Significant Logs | Client Log Settings | Captures all client logs including connections and requests. |
| Non-significant log throttle | Client Log Settings | Limits the number of non-significant logs generated per second for this virtual service on each service engine. Default is 10 logs per second. |
| Non-significant log duration | Client Log Settings | Amount of time the system captures all the logs, measured in minutes. |
| Name 1 | Client Log Filter | Description of `property client_log_filters` of object type AnalyticsPolicy. |
| Name 2 | Client Log Filter | Description of `property client_log_filters` of object type AnalyticsPolicy. |

| Property Name | Category | Description |
|---|---|---|
| Name 3 | Client Log Filter | Description of `property client_log_filters` of object type AnalyticsPolicy. |
| Weight | Quality of Service | The Quality of Service weight to assign to traffic transmitted from the virtual service. A higher weight will prioritize traffic versus other virtual services sharing the same service engines. |
| Fairness | Quality of Service | Determines how multiple virtual services sharing the same service engines will prioritize traffic over a congested network. |
| Auto Gateway | Other Settings | Response traffic to clients will be sent back to the source MAC address of the connection, rather than statically sent to a default gateway. |
| Use VIP as SNAT | Other Settings | Uses the virtual IP as the SNAT IP for health monitoring and sends traffic to the backend servers instead of the service engine interface IP. |
| Advertise VIP via BGP | Other Settings | Advertises a virtual service via BGP even if it is marked down by the health monitor. |
| Advertise SNAT via BGP | Other Settings | Enables Route Health Injection for Source NAT'ted floating IP Address using the BGP Config in the vrf context |
| SE Group | Other Settings | The service engine group to use for this virtual service. Reference to an object of type ServiceEngineGroup. |
| SNAT IP Address | Other Settings | Natted floating source IP Address(es) for upstream connection to servers. |
| Traffic Clone Profile | Other Settings | Server network or list of servers for cloning traffic. Reference to an object of type TrafficCloneProfile. |
| Remove Listening Port when VS down | Other Settings | Removes listening port if the virtual service is down. |
| Scale Out ECMP | Other Settings | Disable re-distribution of flows across service engines for a virtual service. Enable if the network itself performs flow hashing with ECMP in environments such as GCP. |
| Service Ports | ServicePort | Port number. |
| Protocol | ServicePort | Type of protocol. |

**Known Issues**

This section provides details of the known issues.

**Hierarchy mismatch in case of multi-tenancy**

The VCF Operations Management Pack for Avi Load Balancer collects all the resources of a tenant. However, if the same cloud is used across tenants, the relationship hierarchy lists all the child objects of the cloud collected across all tenants. This affects the count of virtual services displayed when the service engine group is shared across tenants.

**Workaround:** None

#### Certain objects are not collected

Upgrade of the Avi Load Balancer or the addition or removal of a tenant by the user, leads to collection cycles that do not work as expected.

**Workaround:** Reconfigure the VCF Operations Management Pack for Avi Load Balancer.

#### Some metrics/properties may return incorrect results

In some versions of the Avi Load Balancer, because of a bug in the API exposed by the Avi Load Balancer, the following metrics/properties may return incorrect results:

- VS VIP, Virtual Hosting, Application Domain Name
- App Response and Server RTT for Virtual Service
- Pool metrics in some cases

In some cases, intermittent API failures from the Avi Load Balancer occur, because of which metrics may not be collected for that particular collection cycle.

**Workaround:** None

#### The service engine object type may be associated with the wrong machine

In rare scenarios, when a service engine such as an EC2 Instance/virtual machine/Azure VM/Compute Instance in VCF Operations is created with the same name and within the same cloud, the service engine object type may be associated with the wrong machine.

**Workaround:** Use different name prefixes if multiple cloud accounts of the same type are used.

## Management Pack for SNMP

The Management Pack for SNMP guide provides information about installing and configuring the SNMP.

#### Introduction to SNMP

The Simple Network Management Protocol (SNMP) is a standard Internet protocol for managing devices on IP networks. The SNMP adapter is an embedded adapter for VCF Operations that collects metrics from devices that support SNMP. These devices typically include routers, switches, servers, workstations, and printers.

SNMP displays management data on SNMP managed devices as variables, which are organized in hierarchies. Management Information Base (MIB) files describe these hierarchies and contain object identifiers (OIDs) that identify each variable.

The SNMP adapter uses MIB files to identify the metrics that have to be collected from a particular device. When you install the SNMP adapter, the installation utility places a set of standard MIB files in the `conf/mib` folder. You can import additional MIB files after you install the SNMP adapter. The MIB files that are added will appear in the MIB list drop-down menu.

The SNMP adapter supports devices that are configured with SNMP v2C and V3. The SNMP adapter supports auto-discovery of devices when you add a range or pool of IP addresses while configuring the adapter. The SNMP adapter does not support the device with V3 Auth Only, the v3 device must have both Authentication and Privacy Protocols enabled.

The SNMP adapter ships a range of MIB files, and metrics are created for all the OIDs that have the data type that the Management Pack supports. By default, all metrics are created but none are enabled. You can enable metrics by using Policies to monitor network devices and its components.

#### Intended Audience

The information in this guide is intended for anyone who wants to install and configure the SNMP.

### Installing SNMP

Install the Management Pack for SNMP from the VCF Operations interface.

- Download the PAK file from VCF Solutions Catalog.
- Save the PAK file to a temporary folder on your local system.

1. Log in to the VCF Operations user interface with administrator privileges.
2. In the menu, select **Administration** and in the left pane select **Solutions** > **Repository**.
3. On the **Repository** tab, click **Add/Upgrade**.
4. Browse to locate the temporary folder and select the PAK file.
5. Click **Upload**. The upload might take several minutes.
6. Read and accept the EULA, and click **Next**.
   Installation details appear in the window during the process.
7. When the installation is complete, click **Finish**.

Configure an adapter instance for the management pack.

### Configuring SNMP

After you install the Management Pack, configure an instance for SNMP.

To monitor any network device, ensure that the SNMP adapter is enabled and is running.

1. In the menu, click **Administration** and then in the left pane, click **Other Accounts**.
2. Click **Add Account** and then click **SNMP Adapter**.
3. Configure the adapter instance.

| Option | Description |
|---|---|
| **Name** | Enter a name for the adapter instance. |
| **Description** | Enter a description for the adapter instance. |
| **Start IP** | Enter a start IP address.<br><br>**Note:**<br><br>The test connection fails if the Start IP does not belong to the defined Credential. The Start IP provided should match the defined credentials. |
| **End IP** | Enter an end IP address. The discovery process finds all the devices that have IP addresses within this range.<br><br>**Note:**<br><br>In a range of IPs, only those IPs that belong to the defined Credential are monitored. The IPs that you define should match with credentials that you have selected. The IPs that do not match the credentials are skipped. |
| **Port** | The default port for SNMP is 161. |
| **MIB File** | Select the MIB file specific to the component you want to monitor.<br><br>The SNMP adapter supports only specific data types with OIDs that it collects from the attached MIB file.You can attach any MIB file but the SNMP adapter can not create metrics for all OIDs even when the data for a component exists with that OID. The SNMP adapter supports the following datatypes for the MIBs.<br><br>• integer |

| Option | Description |
|---|---|
| | • integer32<br>• counter<br>• counter32<br>• gauge<br>• gauge32<br>• timeticks<br>• counter64<br><br>For details of MIB files that are supported and validated, see List of MIBs in SNMP Adapter.<br><br>For details on adding custom MIBs in VCF Operations for SNMP adapter, see Loading Custom MIBs to vRealize Operations Manager for SNMP. |
| **Credential** | Select the credential from the drop-down menu or click the plus icon to add new credentials. For details, see Adding a Credential.<br><br>**Note:**<br><br>With SNMP V3, the Authentication Protocols supported are MD5 and SHA and the Privacy Protocols supported are AES and DES. |
| **Collector / Group** | Select the collector on which you want to run the adapter instance. A collector gathers objects into its inventory for monitoring. The collector specified by default is selected for optimal data collecting.<br><br>**Note:**<br><br>Ensure that the network latency between the collector and the device is as per the VCF Operations guidelines. |

4. Click **Validate Connection**.

   The validation starts with the defined Start IP. The connection fails if the Start IP is invalid or if it does not match with the selected credential. For all the devices that the credentials match with, objects for each device will be created under **Inventory** > **SNMP Adapter Instance** .

Adding a Credential

You must add a credential for an SNMP adapter instance. The credential can be of SNMP v2C or V3.

You can add the credential before you create the adapter instance and select the correct credential while configuring the adapter instance. You can also add the credential while configuring the adapter instance.

1. In the menu, click **Administration** and then in the left pane, click **Other Accounts**.
2. Click **Add Account** and then, click **SNMP Adapter**.
3. To add new credentials, click the plus icon next to the **Credential** drop-down menu.
4. Enter the credential details.

| Option | Description |
|---|---|
| **Credential Kind** | Select **SNMPv2c** or **SNMPv3** from the drop-down menu. |
| **Credential Name** | Enter a name for the credential. |
| **Community** | Enter the community string for the v2c credential kind. |

   The following options are available only when you select **SNMPv3** as **Credential Kind**.

| Option | Description |
|---|---|
| **Username** | Enter the username for the credential. |
| **Authentication Protocol** | Select the authentication protocol from the list.<br>• MD5<br>• SHA<br>• SHA224<br>• SHA256<br>• SHA384<br>• SHA512 |
| **Authentication Password** | Enter the authentication password |
| **Privacy Protocol** | Select the privacy protocol from the list.<br>• AES<br>• AES192<br>• AES256<br>• DES<br>• TRIPLE_DES |
| **Privacy Password** | Enter a password for the credential. |

5. Click **Ok** to save the credential.

### Loading Custom MIBs to VCF Operations for SNMP

You can load your custom MIB if a device requires MIB files that are not shipped with SNMP Management Pack.

Copy the MIB files and all itsdependent MIB files under `conf/mibs`. The default installation location of SNMP is `$ALIVE_BASE/user/plugins/inbound/snmp_adapter3/conf/mibs` folder.

1. For a multinode environment, enable SSH to all VCF Operations nodes.
2. Provide executive permission to update the `Mibs.sh` file and run the MIB file update script to import the MIB files. The MIB update script is available in the `$ALIVE_BASE/user/plugins/inbound/snmp_adapter3/conf` folder.
3. Check the mib-import.log file for any errors. The script generates the mib-import.log file in the `conf` folder. Errors commonly occur when the dependent MIB files are not present in the `conf/mibs` folder.
4. Open the conf/describe.xml file in a text editor and increase the version number. For example, if the version is '7', then change it to '8'.
5. Repeat the procedure from Step 1 through Step 4 in all VCF Operations nodes.
6. Log in to VCF Operations as an administrator.
7. In the **Administration** page, click **Control Panel** and then, click **REDESCRIBE**.

   The custom MIB files are added to the SNMP Management Pack. You can select the custom MIB from the **MIB File** drop-down menu while configuring the adapter instance.

   **Note:**

   Only those MIB files which have levels with numeric data will appear in drop-down menu.

### List of MIBs in SNMP Adapter

The SNMP adapter uses MIB files to determine the metrics to collect from a particular device. When you install the SNMP adapter, the installation utility places a set of standard MIB files in the `conf/mib` folder. You can import additional MIB files after you install the SNMP adapter.

**List of MIBs Supported and Validated**

- RFC1158-MIB.mib
- RFC1213-MIB.mib
- RFC1229-MIB.mib
- RFC1231-MIB.mib
- RFC1243-MIB.mib
- RFC1253-MIB.mib
- RFC1269-MIB.mib
- RFC1271-MIB.mib
- RFC1285-MIB.mib
- RFC1315-MIB.mib
- RFC1316-MIB.mib
- RFC1381-MIB.mib
- RFC1382-MIB.mib
- RFC1398-MIB.mib
- RFC1406-MIB.mib
- RFC1407-MIB.mib
- RFC1414-MIB.mib
- RFC1493-MIB.mib
- CISCO-ENVMON-MIB.mib
- CISCO-COMMON-MGMT-MIB.mib
- CISCO-PROCESS-MIB.mib
- CISCO-MEMORY-POOL-MIB.mib
- CISCO-ENTITY-FRU-CONTROL-MIB.mib
- ENTITY-MIB.mib IF-MIB.mib

**List of MIBs Supported But Not Validated**

- VMWARE-TUNNEL-SERVER-MIB.mib
- UDP-MIB.mib SNMPv2-CONF.mib
- RFC1229-MIB.mib
- RFC-1155.mib
- dcs3rmt.mib
- CISCO-COMMON-MGMT-MIB.mib
- VMWARE-VROPS-AGENTCAP-MIB.mib
- VMWARE-VRNI-MIB.mib
- VMWARE-TC-MIB.mib
- VMWARE-SRM-EVENT-MIB.mib
- VMWARE-HEARTBEAT-MIB.mib
- VMWARE-ESX-AGENTCAP-MIB.mib
- SNMP-USER-BASED-SM-MIB.mib
- SNMP-COMMUNITY-MIB.mib
- RFC1382-MIB.mib
- qlaspCfg.mib
- LLDP-V2-MIB.mib
- IP-MIB.mib
- INET-ADDRESS-MIB.mib
- IEEE8021-TC-MIB.mib
- CISCO-VTP-MIB.mib
- CISCO-ENVMON-MIB.mib
- 10892.mib
- RFC1406-MIB.mib
- DELL-RAC-MIB_NPS_6.1.mib
- baspStat.mib
- VMWARE-VA-AGENTCAP-MIB.mib
- VMWARE-RESOURCES-MIB.mib
- VMWARE-NSX-MANAGER-MIB.mib

**Monitoring Network Devices Selectively**

With SNMP adapter, metrics are created using MIB as the base object. Metrics are created only for the OIDs that Management Pack supports. These metrics are then monitored for all the devices that are attached with MIB file while configuring the adapter instance.

While creating a symptom, the SNMP adapter does not allow you to choose a specific device, you can only choose a specific component at a specific index from the existing inventory. For example, if you have to monitor all three Cisco fan modules for 'shutdown' status, you can create three symptoms for each fan module at each index(1,2 and 3), but you cannot choose a specific device. The symptoms will be generated for all devices that has fan modules at index (1,2 and 3).

If there are three Cisco switches (10.107.119.1, 10.107.119.2 and 10.107.119.3) in the environment, then each device has three fan modules at index 1, 2 and 3. When you create an alert using the symptoms, these alerts will be generated for all three Cisco switches. The following scenario explains how to selectively monitor a specific device using VCF Operations 'Custom Groups' and 'Policies'.

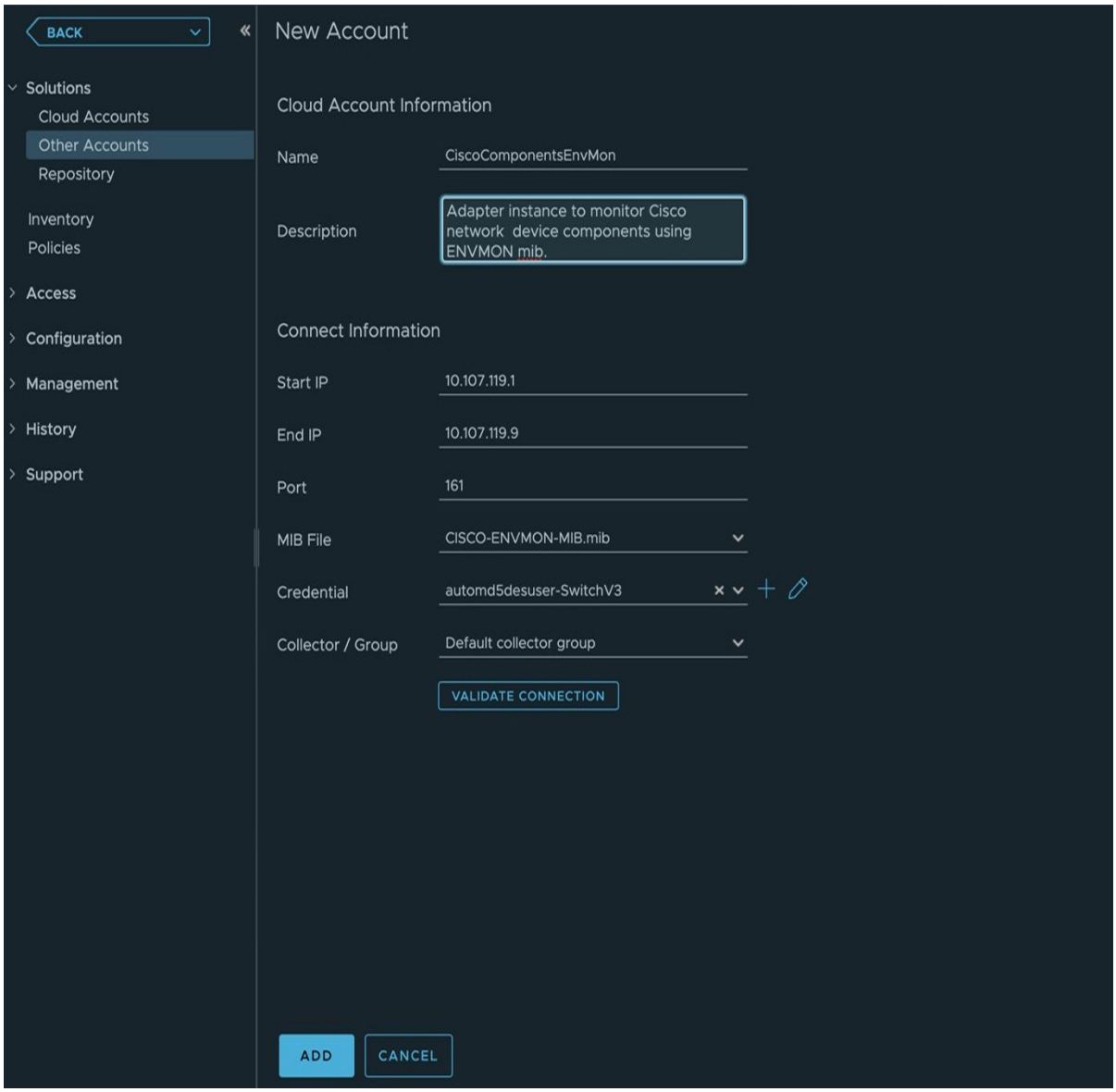
Example: Monitoring Fan State for a Cisco Switch

Let us consider the scenario of monitoring fan state for a Cisco Switch alone using VCF Operations 'Custom Groups' and 'Policies'.

In this example:

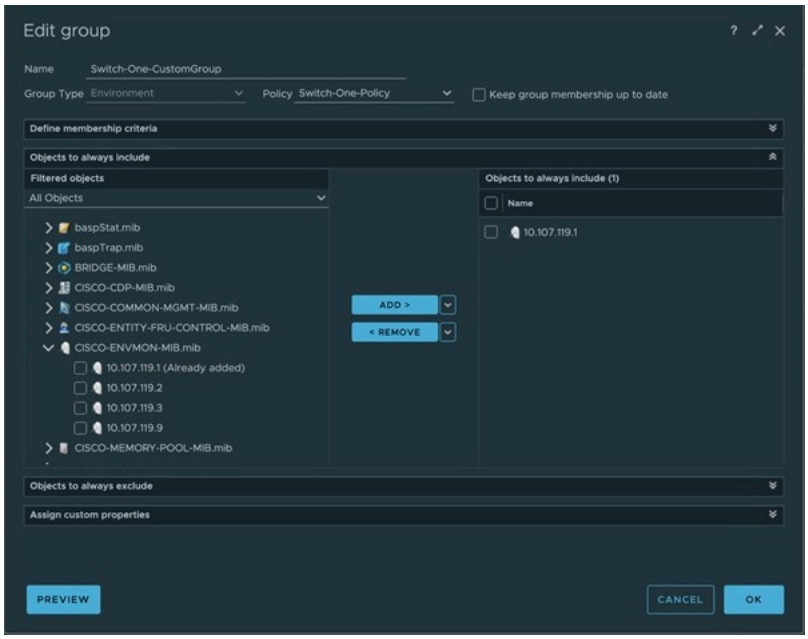
- CISCO-ENVMON-MIB gives the current state of the fan present at OID 1.3.6.1.4.1.9.9.13.1.4.1.3 (ciscoEnvMonFanState)
- For Cisco Switch with IP 10.107.119.1, there are three Fan components and you want to monitor Fan state for Cisco Switch 10.107.119.1 for any/all Fan modules
- The possible states of a Fan module are normal, warning, critical, shutdown, not present, and not functioning

1. Create an adapter instance for the range of devices that you want to monitor for Fan State. While creating the adapter instance, attach **CISCO-ENVMON-MIB** to adapter instance as shown in the following figure.

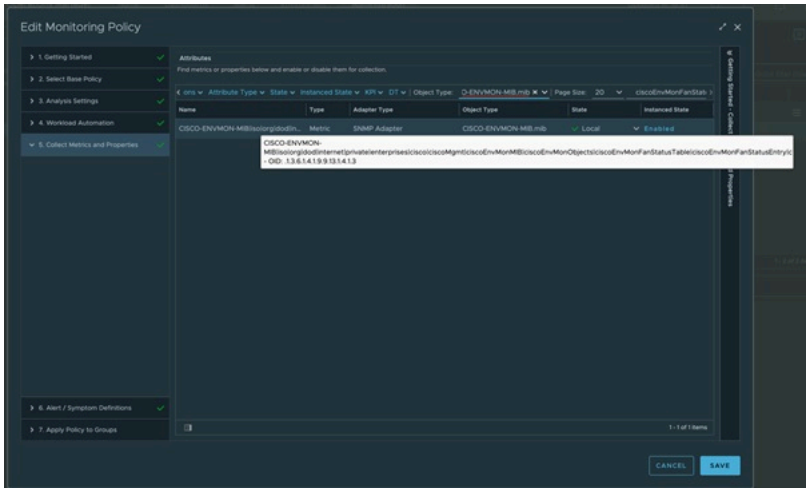



You have added a range of devices. For the credentials that match, the objects for each device are created under the adapter instance.

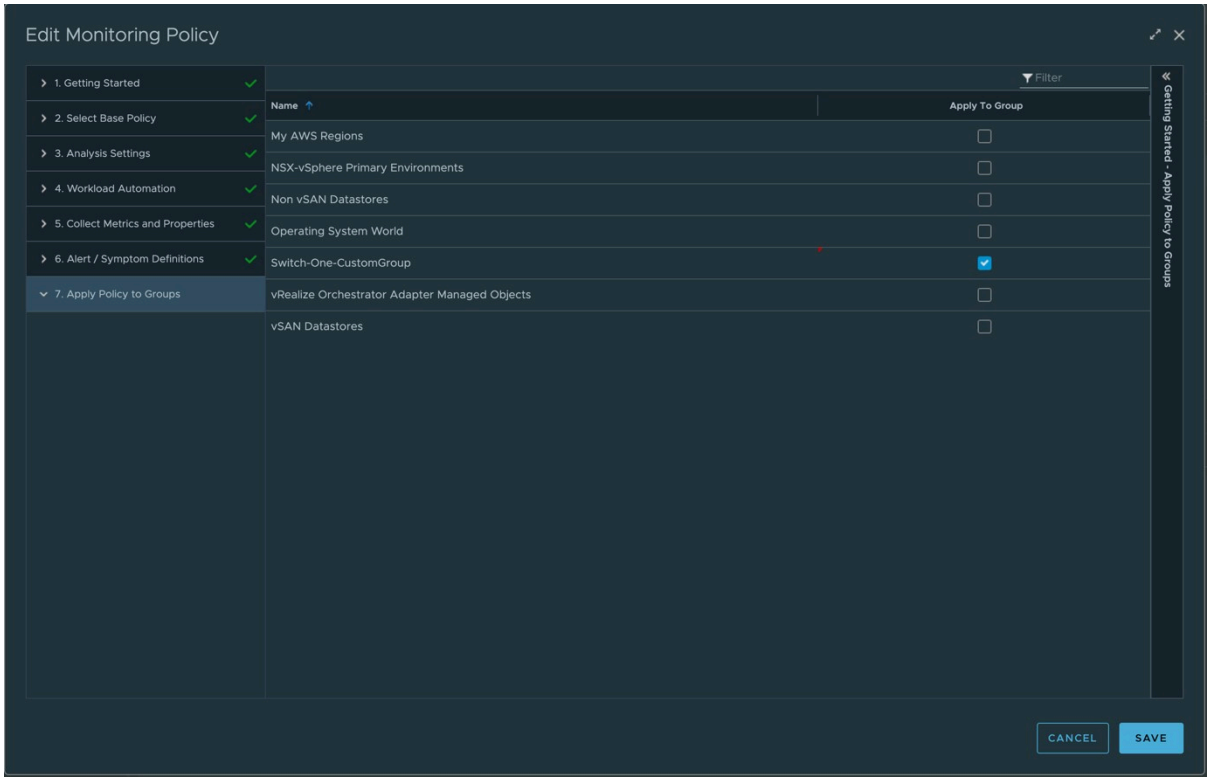
2. Create a new custom group in **Environment** > **Custom Groups** . While creating a custom group, select the policy and add only the device(s) that have to be monitored. In this example, add **10.107.119.1**.



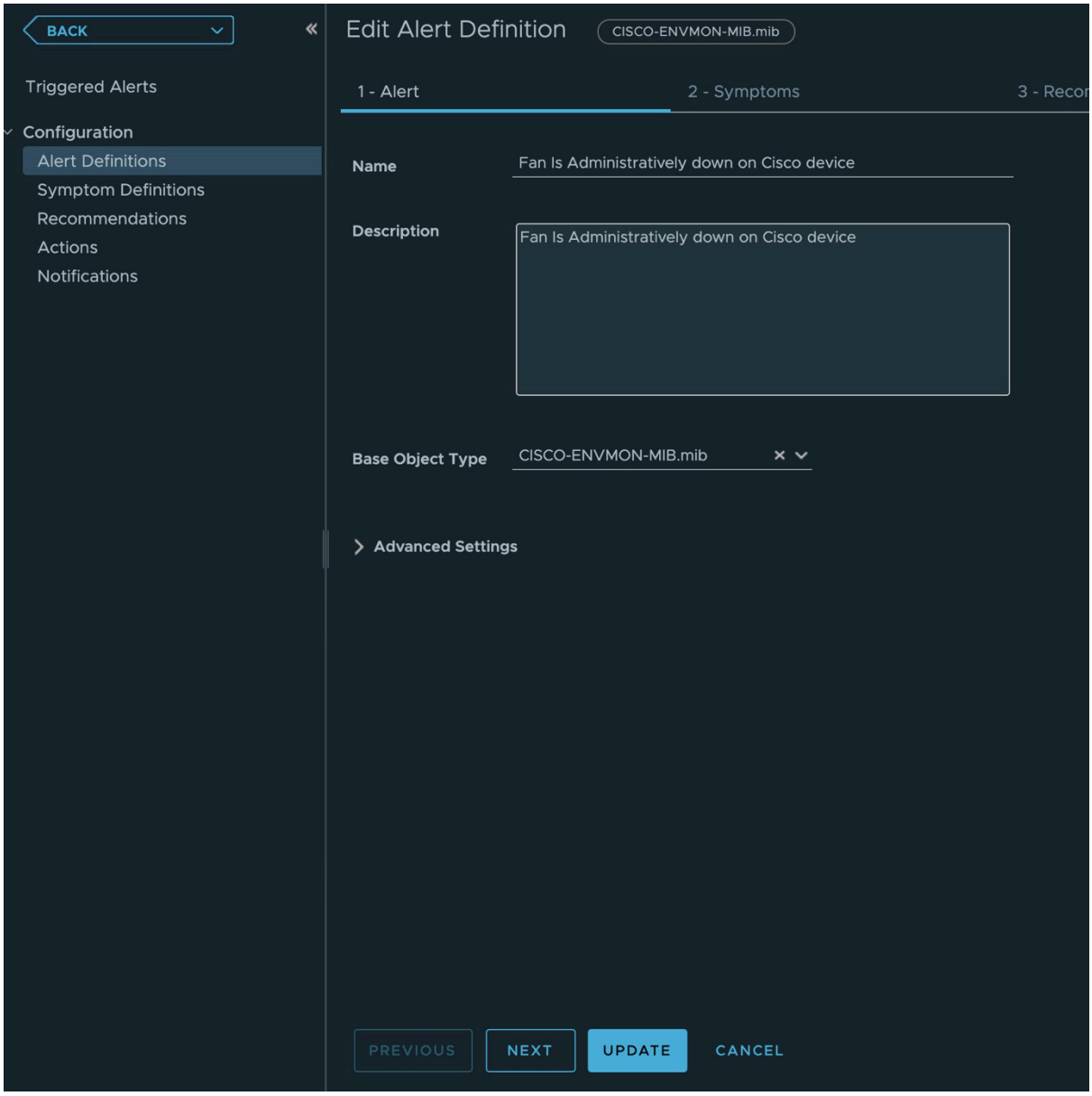

3. Go to **Administration** > **Policies** and click **Collect Metrics and Policies**. Search for `1.3.6.1.4.1.9.9.13.1.4.1.3` or `ciscoEnvMonFanState` and then set the state to `Local` for the OID as shown in the following figure.



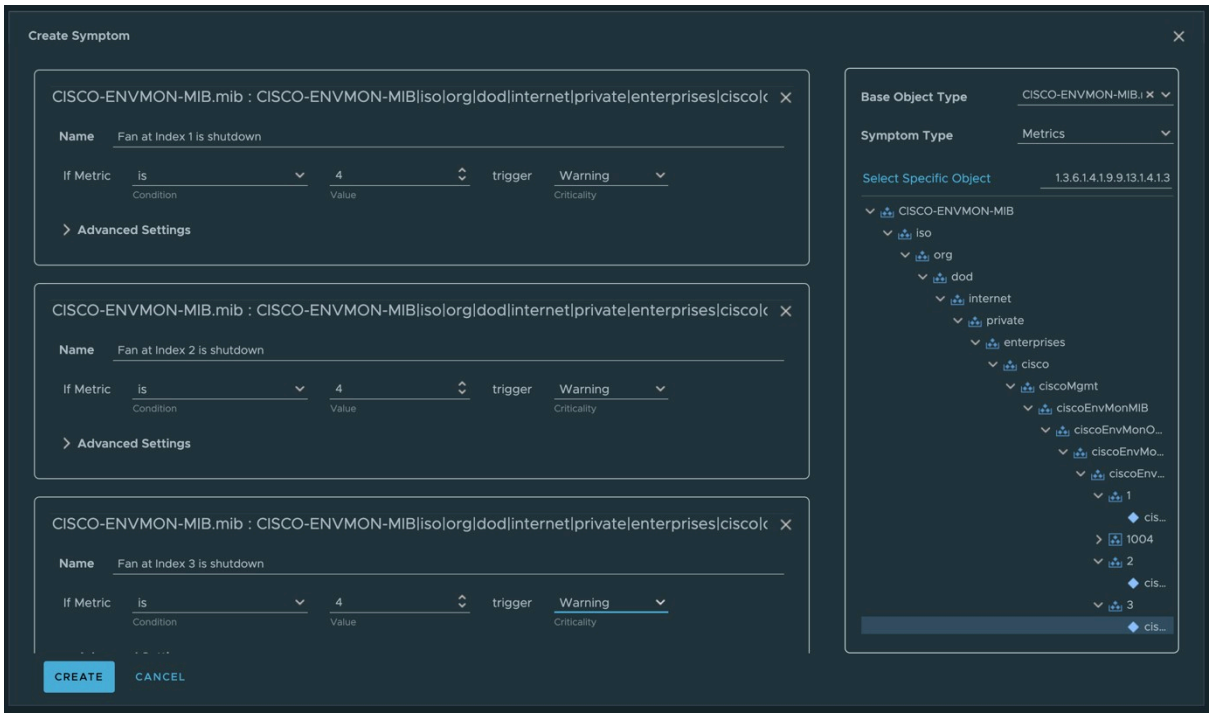
4. In the **Apply Policy to Groups** tab, select **Switch-One-CustomGroup**.



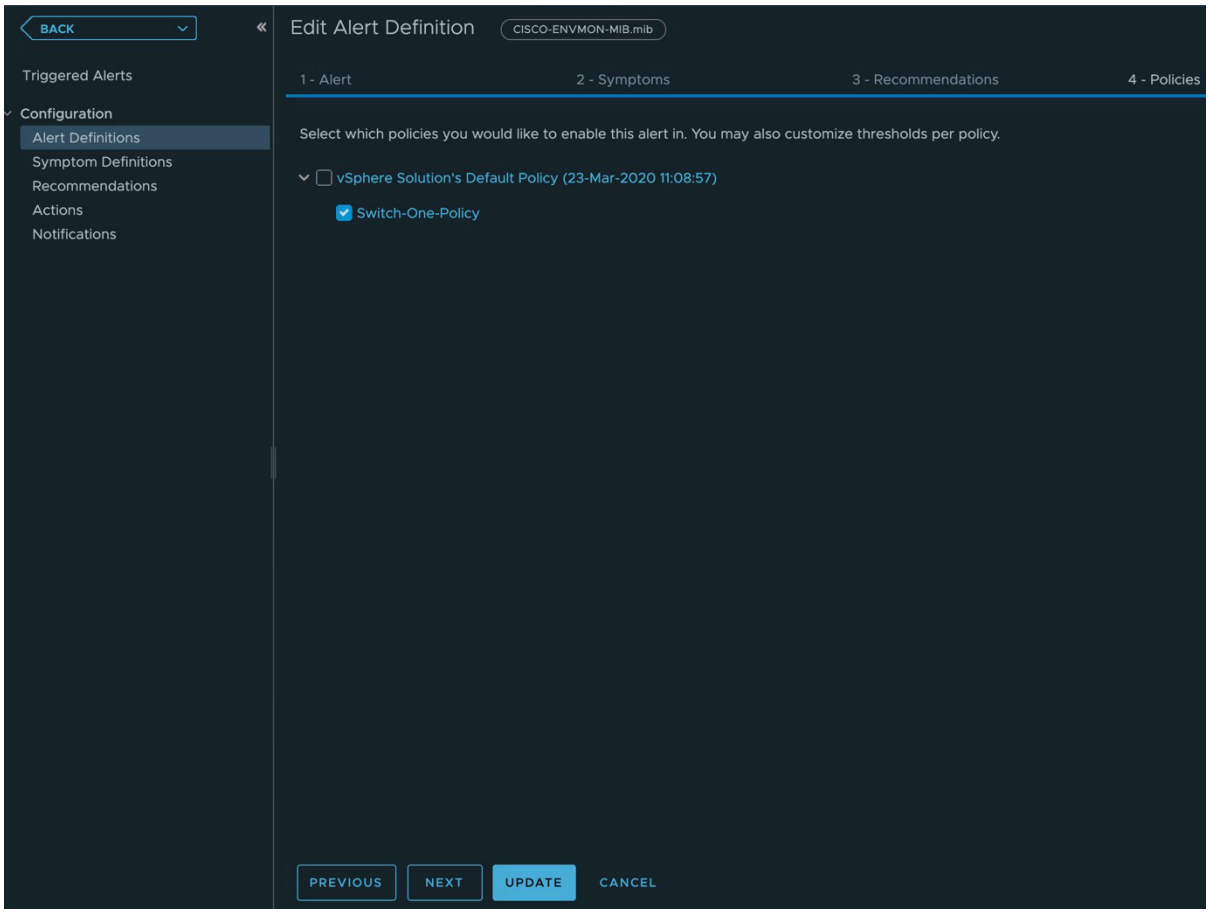
5. In the **Alerts** page, click **Configuration** > **Alert Definition**.
6. Create a new alert definition with the name 'Fan Is Administratively down on Cisco device' and select the **Base Object Type** as 'CISCO-ENVMON-MIB.mib'.



7. Create Symptoms for each fan module at index 1, 2 and 3 for status 'shutdown' at OID 1.3.6.1.4.1.9.9.13.1.4.1.3. Attach these symptoms to alert definition.

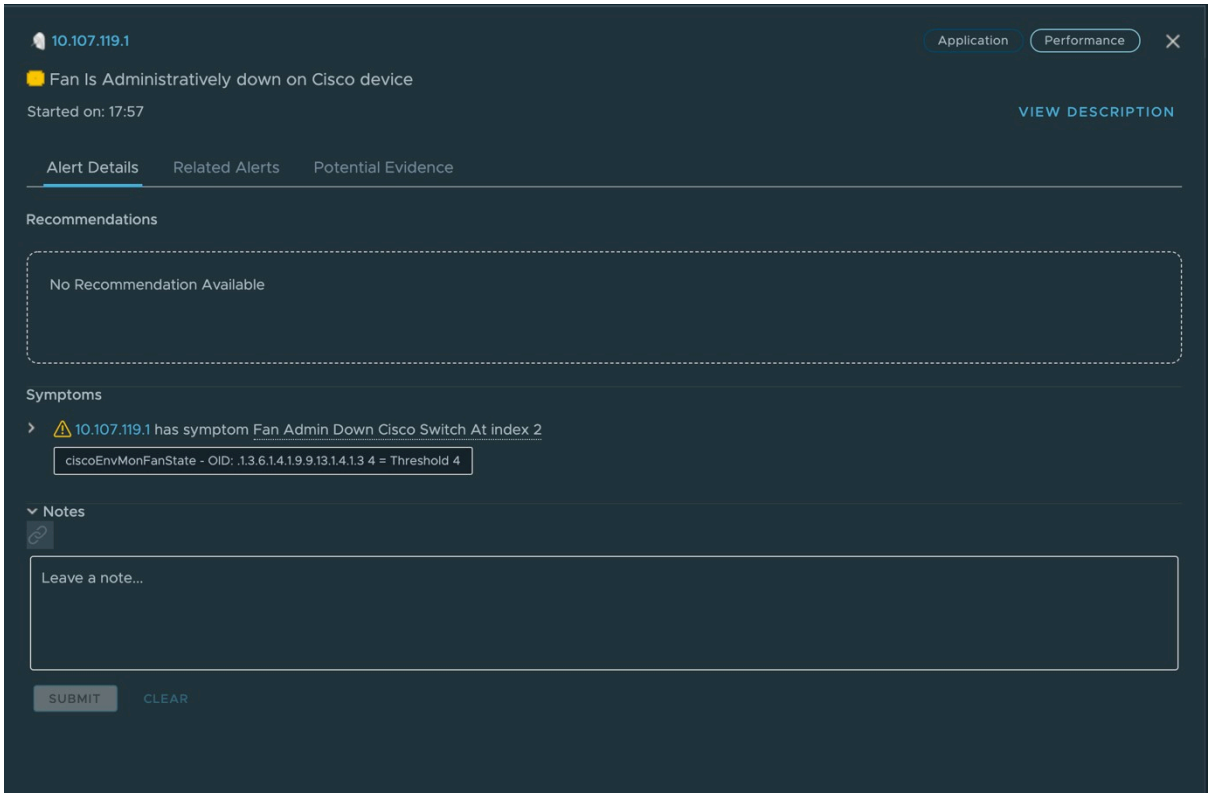



8. Restrict the alert to the policy 'Switch-One-Policy' so that this alert will be generated only for the Cisco Switch with IP 10.107.119.1.



9. Click **Update**.

   After a few collection cycles, alert is generated only for Cisco Switch with IP 10.107.119.1 as shown in the following figure.




**Metrics in SNMP Adapter**
The SNMP Adapter collects metrics for objects within its plug-in.

**Table 1670: SNMP Metrics**

| Resource | Metric Key | Metric Value |
|---|---|---|
| Predefined MIB Statistics (Interface Level) | Input Bandwidth Utilization | Double |
| Predefined MIB Statistics (Interface Level) | Output Bandwidth Utilization | Double |
| Predefined MIB Statistics (Interface Level) | Nominal Bandwidth | Double |

## Management Pack for Telco Cloud Automation Platform

Configuration Guide for this Management Pack is available at Management Pack for VMware Telco Cloud Automation Platform Guide.

## Management Pack for vSphere Replication

Configuration Guide for this Management Pack is available at Management Pack for vSphere Replication.

## Getting Started with Amazon Web Services

Amazon Web Services (AWS) is an embedded adapter with diagnostic dashboards for VCF Operations. The adapter collects metrics from AWS.

### Supported AWS Services

AWS supports the following services in VCF Operations.

| Service | Object | Description |
|---|---|---|
| Amazon MQ* | Amazon MQ Broker<br>Amazon MQ Node<br>Amazon MQ Topic<br>Amazon MQ Queue<br>Amazon MQ Virtual Host | Provides fully managed service for open source message brokers |
| Amazon Simple Queue Service (SQS) | SQS Queue | Provides fully managed message queues for microservices, distributed systems, and serverless applications |
| Amazon EC2 | EC2 Instance<br>EC2 Instance Type | Provides a secure and resizable compute capacity to support virtually any workload |
| Amazon EC2 Auto Scaling | EC2 Auto Scaling Group | Allows you to add or remove compute capacity to meet changes in demand |
| AWS Lambda | Lambda Function | Provides a serverless compute service that lets you run code without provisioning or managing servers, creating workload-aware cluster scaling logic, maintaining event integrations, or managing runtimes |
| AWS Elastic Beanstalk | Elastic Beanstalk Application<br>Elastic Beanstalk Environment | Provides an easy-to-use service for deploying and scaling web applications and services developed with Java, .NET, PHP, Node.js, Python, Ruby, Go, and Docker on familiar servers such as Apache, Nginx, Passenger, and IIS |
| Amazon Elastic Container Registry (ECR) | ECR Container Repository<br>ECR Container Image | Allows you to share and deploy container software, publicly or privately |
| Amazon Elastic Container Service (ECS) | ECS Cluster<br>ECS Service | Provides a highly secure, reliable, and scalable way to run containers |
| Amazon Elastic Kubernetes Service (EKS) | EKS Cluster<br>EKS Cluster ARN | Provides the flexibility to start, run, and scale Kubernetes applications in the AWS cloud or on-premises |
| Amazon Relational Database Service (RDS) | RDS DB Instance<br>RDS Instance Type | Allows you to set up, operate, and scale a relational database in the cloud |
| Amazon DynamoDB | DynamoDB<br>DynamoDB Accelerator Cluster<br>DynamoDB Accelerator Node | Provides fast and flexible NoSQL database service for any scale |
| Amazon ElastiCache | ElastiCache Cluster<br>ElastiCache Node | Provides fully managed in-memory data store, compatible with Redis or Memcached |
| Amazon WorkSpaces | Workspace | Provides a managed, secure Desktop-as-a-Service (DaaS) solution |
| AWS CloudFormation | CloudFormation Stack | Allows you to speed up cloud provisioning with infrastructure as code |
| Amazon Virtual Private Cloud (Amazon VPC) | VPC<br>VPC NAT Gateway<br>VPC VPN Connection | Allows you to build on a logically isolated virtual network in the AWS cloud |

| Service | Object | Description |
|---|---|---|
| Amazon CloudFront | CloudFront Distribution | Provides a fast, highly secure, and programmable content delivery network (CDN) |
| Amazon Route 53 | Route53 Hosted Zone | A highly available and scalable cloud Domain Name System (DNS) web service |
| AWS Direct Connect | AWS Direct Connect | A cloud service solution that makes it easy to establish a dedicated network connection from your premises to AWS |
| Elastic Load Balancing | Elastic Load Balancing | Allows you to distribute network traffic to improve the scalability of your applications |
| Amazon Simple Storage Service (S3) | S3 Bucket | Provides object storage built to store and retrieve any amount of data from anywhere |
| Amazon Fargate/ECS | ECS Cluster<br>ECS Service | Provides serverless compute for containers |
| Amazon Elastic Block Store (EBS) | EBS | An easy-to-use, high-performance, block-storage service designed for use with Amazon Elastic Compute Cloud (EC2) for both throughput and transaction-intensive workloads at any scale |
| Amazon Redshift | Redshift<br>Redshift Cluster<br>Redshift Node | Allows you to analyze all of your data with the fastest and most widely used cloud data warehouse |
| Amazon EMR | EMR Job Flow | Provides easily run and scale Apache Spark, Hive, Presto, and other big data frameworks |
| Amazon Elastic File System (EFS) | EFS | A simple, scalable, fully managed elastic NFS file system for use with AWS Cloud services and on-premises resources |
| Amazon Athena | Amazon Athena WorkGroup | An interactive query service that makes it easy to analyze data in Amazon S3 using standard SQL |
| Amazon CloudSearch | Amazon CloudSearch Domain | A managed service in the AWS Cloud that makes it simple and cost-effective to set up, manage, and scale a search solution for your website or application |
| Amazon OpenSearch | Amazon ElasticSearch Domain<br>Amazon ElasticSearch Node | Provides a managed service to easily deploy, operate, and scale OpenSearch clusters in the AWS Cloud |
| Amazon Elasticsearch Service* | Amazon ElasticSearch Domain<br>Amazon ElasticSearch Node | Provides fully managed, scalable, and secure Elasticsearch service |
| Amazon Kinesis Data Firehose | Amazon Kinesis Firehose | Allows you to prepare and load real-time data streams into data stores and analytics services |
| Amazon Kinesis Data Analytics | Amazon Kinesis Analytics | Provides actionable insights from streaming data with serverless Apache Flink |
| Amazon Kinesis Data Streams | Amazon Kinesis Stream | Allows you to easily collect, process, and analyze video and data streams in real-time |
| Amazon Kinesis Video Streams | Amazon Kinesis Video Stream | Provides a fully managed AWS service that can be used to stream live video from other devices to AWS Cloud or build applications for real-time video processing or batch-oriented video analytics. |
| Amazon Managed Streaming for Apache Kafka | Amazon MSK Cluster | Provides fully managed, highly available, and secure Apache Kafka service |

| Service | Object | Description |
|---|---|---|
| AWS Step Functions* | Amazon Step Functions State Machine | Allows you to assemble functions into business-critical applications |
| Amazon EventBridge* | Amazon EventBridge Rule | Provides serverless event bus that connects application data from your apps, SaaS, and AWS services |
| Amazon Simple Notification Service (SNS)* | Amazon SNS Topic | Provides fully managed pub/sub messaging, SMS, email, and mobile push notifications |
| Amazon Simple Workflow Service (SWF)* | Amazon SWF Workflow Type<br>Amazon SWF Activity Type<br>Amazon SWF Task List | Allows developers to build, run, and scale background jobs that have parallel or sequential steps |
| Amazon WorkMail* | Amazon WorkMail Organization | A secure, managed business email and calendar service with support for existing desktop and mobile email client applications |
| Amazon Connect* | Amazon Connect Instance | An easy-to-use omnichannel cloud contact center |
| Amazon Pinpoint* | Amazon Pinpoint Application | A multichannel marketing communication service |
| Amazon Neptune | Amazon Neptune DB Instance<br>Amazon Neptune DB Cluster<br>Amazon Neptune DB Engine | Provides fast, reliable graph database built for the cloud |
| Amazon Quantum Ledger Database (QLDB) | Amazon QLDB Ledger<br>Amazon QLDB Stream | Provides a fully managed ledger database that provides a transparent, immutable, and cryptographically verifiable transaction log |
| Amazon DocumentDB | Amazon DocDB DB Instance<br>Amazon DocDB DB Cluster<br>Amazon DocDB DB Engine | Provides a fast, scalable, and highly available MongoDB-compatible database service |
| Amazon Keyspaces* | Amazon keyspace | A scalable, highly available, and managed Apache Cassandra–compatible database service |
| Amazon Timestream | Amazon Timestream Database | Provides fast, scalable, serverless time series database |
| AWS CodeBuild* | Amazon CodeBuild Project | Allows you to build and test code with continuous scaling. Pay only for the build time you use |
| Amazon AppStream 2.0* | Amazon AppStream Fleet | A fully managed non-persistent application and desktop streaming service |
| Amazon GameLift* | Amazon GameLift Fleet<br>Amazon GameLift Queue | Allows you to enhance multiplayer experiences with dedicated cloud servers |
| AWS IoT* | Amazon IoT Protocol | Provides easily and securely connect devices to the cloud |
| AWS IoT Analytics* | Amazon IoTAnalytics Data Set | Provides analytics for IoT devices |
| Amazon Kendra* | Amazon Kendra Index<br>Amazon Kendra DataSource | Provides highly accurate intelligent search service powered by machine learning |
| Amazon Lex* | Amazon Lex Bot | A conversational AI for Chatbots |
| AWS AppSync* | AWS AppSync GraphQL API | Allows you to accelerate application development with scalable GraphQL APIs |
| Amazon Cognito* | Amazon Cognito User Pool | A simple and secure user sign-up, sign-in, and access control |

| Service | Object | Description |
|---|---|---|
| AWS WAF - Web Application Firewall* | AWS WAF WebACL | A web application firewall that helps protect your web applications or APIs against common web exploits that may affect availability, compromise security, or consume excessive resources |
| AWS Storage Gateway* | Amazon Storage Gateway | A hybrid cloud storage service that gives you on-premises access to virtually unlimited cloud storage |
| Amazon Glue | Amazon Glue | A simple, scalable, and serverless data integration |
| Amazon Aurora | Amazon Aurora DB Cluster<br>Amazon Aurora DB Instance | A MySQL and PostgreSQL-compatible relational database built for the cloud |
| Amazon SageMaker | Amazon SageMaker Endpoint<br>Amazon SageMaker Batch Transform Jobs<br>Amazon SageMaker Training Jobs<br>Amazon SageMaker Ground Truth Labeling Jobs<br>Amazon SageMaker Ground Truth Work Team | Helps data scientists and developers to prepare, build, train, and deploy high-quality machine learning (ML) models quickly by bringing together a broad set of capabilities purpose-built for ML |
| Amazon Personalize | Amazon Personalize DataSet Import Jobs<br>Amazon Personalize Solutions<br>Amazon Personalize Campaigns<br>Amazon Personalize Event Tracker | Allows you to create real-time personalized user experiences faster at scale |
| Amazon Elastic Inference | Amazon Elastic Inference | Allows you to attach low-cost GPU-powered acceleration to Amazon EC2 and Sagemaker instances or Amazon ECS tasks, to reduce the cost of running deep learning inference |
| Amazon API Gateway | Amazon API Gateway | Allows you to create, maintain, and secure APIs at any scale |
| Amazon Global Accelerator | Amazon Global Accelerator | Allows you to improve global application availability and performance using the AWS global network |
| Amazon FSx for Lustre | Amazon FSx for Lustre | Provides fast and scalable shared storage to power your compute workloads |
| Amazon FSx for Windows File Server | Amazon FSx for Windows File Server | Provides fully managed file storage built on Windows Server |
| AWS App Mesh (App Mesh Metrics for EKS & ECS) | **Note:**<br>These services appear under App Mesh Metrics for EKS & ECS. | A service mesh that provides application-level networking to make it easy for your services to communicate with each other across multiple types of compute infrastructure |
| Amazon S3 Glacier (Scale object of S3) | **Note:**<br>These services appear under Scale object of S3. | A secure, durable, and extremely low-cost Amazon S3 cloud storage classes for data archiving and long-term backup |
| Amazon Kinesis (Data Analytics, Data Firehose, and Data Stream) | **Note:**<br>These services appear under Data Analytics, Data Firehose, and Data Stream. | Allows you to easily collect, process, and analyze video and data streams in real-time |
| AWS Auto Scaling (EC2 Auto Scaling) | **Note:**<br>These services appear under EC2 Auto Scaling. | Allows you to add or remove compute capacity to meet changes in demand |

**Note:**
Services marked with * indicate that the objects have relationships only with regions.

**Note:**
All services are created with the following Service Descriptors:

- Account ID
- Region
- Service Type

For more information about Amazon Web Services, go to the Amazon Web Services site at https://aws.amazon.com/products.

### Charges for AWS Metrics

Amazon charges you for the metrics you collect. You can reduce costs by selecting only the metrics that are most helpful and filtering out those that are of less interest.

By default, the Management Pack for AWS requests data every five minutes. Every collection cycle makes one CloudWatch call per metric, for an object.

For information about metric costs, see Amazon CloudWatch Pricing.

Based on the costs associated with running the adapter, you can take advantage of some of the features that limit the amount of data you collect from AWS.

- Turn off auto discovery and use manual discovery.
- Go to **Advanced Settings** and select only those services that are critical to your system.
- Subscribe only to specific critical regions.
- Use acceptlist and blocklist filtering to select object import by name.
- Use acceptlist or blocklist to selectively import specific service instances using instance names. Using regex, you can filter services by providing the partial name of the instance.

### Configuring AWS

Configure the AWS in VCF Operations and optionally change its properties to customize the management pack's operation.

An Amazon Web Services account has multiple types of credentials associated with the account. Sign-in credentials are used to access the Amazon Web Services Web-based console, key pairs are used to access EC2 instances, and access keys are used in the REST API that Amazon Web Services exposes.

Because the AWS adapter is based on the REST API, you must use access keys when you set up the adapter. You generate access keys from the Amazon Web Services console. You can create credentials on a per user basis. Access keys are not a username-password pair, but a generated sequence of characters.

**Note:**
While it is not required, it is recommended that you create a guest type account, which has a read-only access to Amazon Web Services, and use the access keys associated with this account. When you create a guest group with default permissions, they do not include read access to the Elastic Map Reduce (EMR) service. You must use the IAM console to add the following permission:

```
elasticmapreduce:DescribeJobFlows
```

#### Generate Required Access Keys

To configure AWS, you must acquire an access key and secret key from the Amazon server. You can acquire these keys as an Amazon Web Services Admin user or as an Amazon Identity and Access Management (IAM) user. For the latest instructions,

- Ensure that you are using Amazon Web Services.
- Ensure that you have the valid permissions and roles in Amazon Web Services.

1. Log in to Amazon Web Services.
2. To generate access keys, see the online documentation on the https://docs.aws.amazon.com/ site.
   Complete the following tasks:
   - Generate access keys as an Amazon Web Services Administrator.
   - Generate access keys as Amazon Web Services Identity and Access Management User.

### Configuring IAM Permissions

When you set up IAM users and groups, you can stipulate which permissions the account has for API calls. The keys you use when you set up the adapter instance must have certain permissions activated.

For each supported AWS Service, the `ReadOnlyAccess` permission is enough to collect metrics. Use the permission to create a IAM Policy for all supported services and their related services.

To use resource groups tagging API operations, see Resource Groups Tagging API Reference and Services that support the Resource Groups Tagging API.

Log in to the AWS console and create a json similar to the following to get the list of privileges for the service:

```
{
    "Version": "2012-10-17",
    "Statement": [
       {
         "Action": [
          "autoscaling:Describe*",
          "cloudwatch:Describe*",
          "cloudwatch:Get*",
          "cloudwatch:List*",
          "logs:Get*",
          "logs:List*",
          "logs:Describe*",
          "logs:TestMetricFilter",
          "logs:FilterLogEvents",
          "sns:Get*",
          "sns:List*"
        ],
        "Effect": "Allow",
        "Resource": "*"
      }
   ]
}
```

**Table 1671: IAM Permissions**

| Service | Required | Permissions |
|---|---|---|
| Cloudwatch | Yes. | For the list of permissions, see Cloud Watch Read Only Access json. |
| EC2 | describeRegions is required. describeInstances and describeVolumes are only required if you subscribe to the EC2 service. | For more information, see EC2 Read Only Access json. |
| ELB (Elastic Load Balancing) | Required if subscribing to the ELB service. | For the list of permissions, see Elastic Load Balancing Read Only Access json. |

| Service | Required | Permissions |
|---|---|---|
| ELB V2 | Required for application load balancer service. | <pre>{<br>"Version": "2012-10-17",<br>"Statement": [<br>{<br>"Sid":<br>"VisualEditor0",<br>"Effect": "Allow",<br>"Action": [<br>"elasticloadbalancing:DescribeLoadBalancer<br>"elasticloadbalancing:DescribeListeners",<br>"elasticloadbalancing:DescribeTargetHealth<br>"elasticloadbalancing:DescribeTargetGroups<br>],<br>"Resource": "*"<br>}<br>]<br>}</pre> |
| EMR | Required if subscribing to the EMR service. | describe*<br><pre>{<br>"Effect": "Allow",<br>"Action": [<br>"elasticmapreduce:Describe*",<br>"elasticmapreduce:List*",<br>"elasticmapreduce:ViewEventsFromAllCluster<br>"s3:GetObject",<br>"s3:ListAllMyBuckets",<br>"s3:ListBucket",<br>"sdb:Select",<br>"cloudwatch:GetMetricStatistics"<br>],<br>"Resource": "*"<br>}</pre> |
| RDS | Required if subscribing to RDS service. | For the list of permissions, see RDS Read Only Access json. |
| ElasticCache | Required if subscribing to ElasticCache service. | For the list of permissions, see Elastic Cache Read Only Access json. |
| SQS | Required if subscribing to SQS service. | For the list of permissions, see SQS Read Only Access json. |
| Elastic Container Registry | | For the list of permissions, see Elastic Container Read Only Access json. |
| Elastic Container Service | | list* |
| Lambda | | For the list of permissions, see Lambda Read Only Access json and refer to the AWS Lambda policy. |
| DynamoDB | | For the list of permissions, see Dynamo DB Read Only Access json. |

| Service | Required | Permissions |
|---|---|---|
| DAX | | describe*<br>list* |
| Redshift | | For the list of permissions, see Redshift Read Only Access json. |
| Virtual Private Cloud | | For the list of permissions, see VPC Read Only Access json. |
| Cloud Front Distribution | | For the list of permissions, see Cloud Front Distribution Read Only Access json. |
| Direct Connect | | For the list of permissions, see Direct Connect Read Only Access json. |
| VPN Connection | | describe* |
| VPC NAT Gateway | | describe* |
| Elastic IP | | describe* |
| CloudformationStack | | For the list of permissions, see Cloud Formation Read Only Access json. |
| S3 | | For the list of permissions, see S3 Read Only Access json. |
| Workspaces | | describe* |
| Hosted Zone | | list* |
| Health Checks | | list* |
| Neptune DB | | For the list of permissions, see Neptune Read Only Access |
| Personalzie | | list*<br>describe* |
| Sagemaker | | For the list of permissions, see SageMaker Read Only |
| Fsx | | For the list of permissions, see FSx Read Only Access |
| Global Accelerator | | For the list of permissions, see Global Accelerator Read Only Access |
| APIGateway | | get* |
| Elastic Inference | | describe* |
| Glue | | get* |
| DocumentDB | | For the list of permissions, see Doc DB Read Only Access |
| QLDB | | For the list of permissions, see QLDB Read Only |
| Aurora DB | | For the list of permissions, see RDS Read Only Access |
| Storage Gateway | Required if subscribing to the storage gateway service. | listGateways<br>describeGatewayInformation<br>listFileShares<br>describeSMBFileShares<br>describeNFSFileShares<br>listVolumes |

Update Configuration Settings in the Properties File

The `amazonaws.properties` file provides configuration options. You can optionally change the following configuration properties if required.

**Table 1672: Amazon Web Services Property Settings**

| Property | Description |
|---|---|
| firstcollecthistoryhours | Determines how far in the past to collect data when the adapter starts. The default is 0, meaning no historical collection. |
| maxquerywindowminutes | The maximum query window for collections, in minutes. The default is 60. The adapter asks AWS for metrics for a maximum of this many minutes. |
| maxhoursback | The maximum number of hours back from the current time that the adapter attempts to collect. The default value is 336, or two weeks, because Cloudwatch keeps only two weeks worth of metrics. |
| includetransient | False by default. Set to true to allow the adapter to import known transient objects. Transient objects currently include any EMR job that is set to terminate on completion and all of the supporting cluster EC2 instances that belong to that job. |
| threadcount | Default is 4. Controls how many threads are active while making calls to cloudwatch to get metrics. This threadcount is per region. The total number of threads is this value times the number of regions. |
| collecttimeout | Controls how long the adapter waits for all metric collection calls to return from AWS during a collection cycle. The value is measured in seconds. The default value is 240 seconds, which is in line with the default 5 minute collection cycle. |

Installing and Configuring a VCF Operations Management Pack for AWS

You can add a AWS cloud account instance to your VCF Operations implementation. VCF Operations collects data from Amazon Web Services (AWS) and supports both commercial and AWS GovCloud accounts. It covers multiple services and regions. You can also add application metrics and other custom metrics from your AWS services and use them in dashboards, alerts and reports.

- You must install the Management Pack for AWS before you can configure it. Download the PAK file from the VCF Solutions Catalog to a temporary folder on your local system. For more information, see Installing and Configuring Integrations in VMware Cloud Foundation Operations.
- For key-based authentication, obtain the Access Key and Secret Key values. See Generate Required Access Keys. These values are not the same as your log in credentials for the Amazon Web Services site.
- Determine the services for which you collect metrics. See Supported AWS Services. The default value * includes all services in your subscription. If you do not want to subscribe to all services, you can specify region identifiers in the Regions text box.
- Determine the regions to which you subscribe. Amazon Web Services are divided into different regions. The default value * includes all regions in your subscription. If you do not want to subscribe to all regions, you can specify region identifiers in the Regions text box.

**Table 1673: Amazon Web Services Regions for AWS Standard Account**

| Region-Friendly Name | Region Identifier |
|---|---|
| US East (N. Virginia) | us-east-1 |
| US East (Ohio) | us-east-2 |
| US West (N. California) | us-west-1 |
| US West (Oregon) | us-west-2 |
| Asia Pacific (Hong Kong) | ap-east-1 |
| Asia Pacific (Tokyo) | ap-northeast-1 |
| Asia Pacific (Seoul) | ap-northeast-2 |

| Region-Friendly Name | Region Identifier |
|---|---|
| Asia Pacific (Osaka-Local) | ap-northeast-3 |
| Asia Pacific (Mumbai) | ap-south-1 |
| Asia Pacific (Singapore) | ap-southeast-1 |
| Asia Pacific (Sydney) | ap-southeast-2 |
| Asia Pacific (Jakarta) | ap-southeast-3 |
| Canada (Central) | ca-central-1 |
| China (Beijing) | cn-north-1 |
| China (Ningxia) | cn-northwest-1 |
| Europe (Frankfurt) | eu-central-1 |
| Europe (Zurich) | eu-central-2 |
| Europe (Ireland) | eu-west-1 |
| Europe (London) | eu-west-2 |
| Europe (Paris) | eu-west-3 |
| Europe (Stockholm) | eu-north-1 |
| Europe (Milan) | eu-south-1 |
| Europe (Spain) | eu-south-2 |
| South America (São Paulo) | sa-east-1 |
| Africa (Cape Town) | af-south-1 |
| Middle East (Bahrain) | me-south-1 |
| Middle East (UAE) | me-central-1 |

**Table 1674: Amazon Web Services Regions for AWS GovCloud (US) Account**

| Region-Friendly Name | Region Identifier |
|---|---|
| AWS GovCloud (US-East) | us-gov-east-1 |
| AWS GovCloud (US) | us-gov-west-1 |

- Determine any blocked list or allowed list filters. These filters use regular expressions to filter in or out specific objects by name. For example, an allowed list filter of `.*indows.*` allows only objects with a name including "indows". A blocked list filter of `.*indows.*` filters out all objects with that string in their name.
- To publish custom metrics to CloudWatch, see Publishing Custom Metrics.
- To collect additional metrics for EC2 through CloudWatch agent, set up the agent. For details, see Installing CloudWatch Agent.

1. In the left menu, click **Administration** > **Integrations**.
2. On the Accounts tab, click **ADD**.
3. On the Account Types page, click **AWS**.
4. Configure the instance settings.

| Option | Action |
|---|---|
| **Name** | Enter a name for the adapter instance. |
| **Description** | Enter a description. |
| **Account Type** | Select the AWS account type.<br>Select **AWS Standard Account** to set up a commercial AWS account. |

| Option | Action |
|---|---|
| | Select **AWS GovCloud (US) Account** to set up a AWS GovCloud (US) account to monitor GovCloud services and regions. |
| **Services/ GovCloud Services** | Select the services from which you want to capture metrics. If you want to collect metrics for specific services, then click the drop-down icon and select one or more services. For example, `Amazon CloudFormation, Amazon EC2`. If you do not select any of the services, the metrics for all the services get collected.<br><br>The services marked with an asterix* for example, `AWS AppSync*` are grouped together under AWS Other Services. These services display the relationship with the regions only. For more information on supported AWS services, see Supported AWS Services.<br><br>**Note:**<br>If you select **AWS GovCloud (US) Account** as the account type, you can only select the services applicable for AWS GovCloud (US). |
| **Regions/ GovCloud Regions** | Select the regions you want to subscribe to. If you want to subscribe to specific regions, then click the drop-down icon and select one or more regions. For example, `US East (N. Virginia),US East (Ohio)`. If you want to subscribe to all the regions, do not select any of the regions.<br><br>**Note:**<br>If you select **AWS GovCloud (US) Account** as the account type, you can only select the regions applicable for AWS GovCloud (US). The regions for AWS GovCloud (US) are AWS GovCloud (US-East) and AWS GovCloud (US). |
| **Credential** | Add the credentials used to access the AWS environment by clicking the plus sign and select the **Credential Kind**.<br><br>**Note:**<br>For each credential kind, you must enter a **Credential name**. This name is not the name of the adapter instance, but a friendly name to identify the credential with ease.<br><br>• **Key-based authentication**: Use the AWS key-based authentication an application (running in an AWS Account) using access key and secret key.<br><br>**Note:**<br>If you are configuring an AWS GovCloud (US) Account, then the access key and secret key values must be specific to AWS GovCloud.<br>Enter the **Credential name** and the **Access Key** and **Secret Key** values.<br>Optionally, enter any required local proxy information for your network.<br><br>**Note:**<br>This proxy information activates VCF Operations that is deployed on a local network to have a public network access for AWS to collect statistics for its accounts. |
| **Collector / Group** | Select the collector upon which you want to run the adapter instance. A collector gathers objects into its inventory for |

| Option | Action |
|---|---|
| | monitoring. The collector specified by default has been selected for optimal data collecting. |

5. Click **Test Connection** to validate the connection.
6. Click the arrow to the left of the **Advanced Settings** to configure advanced settings.

| Option | Action |
|---|---|
| **Collect Custom Metrics** | Set this option to true if you want to import all the custom metrics from your AWS account.<br>To publish custom metrics in VCF Operations, the metrics dimension names should match the following service mappings: |

| Service Name | Dimension Name |
|---|---|
| dax_cluster | ClusterId |
| dax_node | NodeId |
| dynamodb | TableName |
| efs | FileSystemId |
| eks | ClusterName |
| elasticbeanstalk_env | EnvironmentName |
| redshift_node | NodeID |
| redshift_cluster | ClusterIdentifier |
| s3_bucket | BucketName |
| vpc_nat_gateway | NatGatewayId |
| vpc_vpn | VpnId |
| workspace | WorkspaceId |
| ec2_auto_scale_group | AutoScalingGroupName |
| cloudfront_distribution | DistributionId |
| direct_connect | ConnectionId |
| ec2_instance | InstanceId |
| ec2_volume | VolumeId |
| transit_gateway | TransitGateway |
| ecs_cluster | ClusterName |
| ecs_service | ServiceName |
| elasticache_cachecluster | CacheClusterId |
| elasticache_cachenode | CacheNodeId |
| ec2_load_balancer | LoadBalancerName |
| application_load_balancer | LoadBalancer |
| network_load_balancer | LoadBalancer |
| emr_job_flow | JobFlowId |
| lambda_function | FunctionName |
| rds_dbinstance | DBInstanceIdentifier |
| hosted_zone | HostedZoneId |
| health_check | HealthCheckId |
| sqs_queue | QueueName |
| amazon_neptune_db_instance | DBInstanceIdentifier |
| amazon_neptune_db_cluster | DBClusterIdentifier |

| Option | Action |
|---|---|
| | <table><tr><th>Service Name</th><th>Dimension Name</th></tr><tr><td>amazon_personalize_data_import</td><td>DatasetimportjobArn</td></tr><tr><td>amazon_personalize_event_tracker</td><td>EventTrackerArn</td></tr><tr><td>amazon_personalize_solution</td><td>SolutionArn</td></tr><tr><td>amazon_personalize_campaign</td><td>CampaignArn</td></tr><tr><td>amazon_sagemaker_endpoint</td><td>EndpointName</td></tr><tr><td>amazon_sagemaker_batch_transform_job</td><td>Host</td></tr><tr><td>amazon_sagemaker_ground_truth_labeling_job</td><td>LabelingJobName</td></tr><tr><td>amazon_sagemaker_ground_truth_work_team</td><td>Workteam</td></tr><tr><td>amazon_global_accelerator</td><td>Accelerator</td></tr><tr><td>amazon_api_gw</td><td>ApiName</td></tr><tr><td>amazon_elastic_inference</td><td>ElasticInferenceAcceleratorId</td></tr><tr><td>amazon_glue_job</td><td>JobName</td></tr><tr><td>amazon_qldb_ledger</td><td>LedgerName</td></tr><tr><td>amazon_qldb_stream</td><td>StreamId</td></tr></table> |
| **Support Auto Discovery** | Set this option to true for automatic discovery of AWS services. If you set this value to false, when you create an adapter instance you must perform a manual discovery of services. |
| **Allowed List Regex** | Add regular expressions to allow only objects with names that fit the criteria you specify. |
| **Blocked List Regex** | Add regular expressions to filter out objects by name. |
| **Actions Activated** | Activate this option to run actions specific to the AWS virtual machine using VCF Operations. By default, this option is set to **true**.<br>For details on running actions, see Power On, Power Off, and Reboot Actions. |
| **Account ID** | The AWS account ID is auto populated. Each AWS account uses a unique account ID to run actions. The account ID is the same one used to log in Amazon Web Services site. |

7. Click **Save Settings**.

Make sure that VCF Operations is collecting data.

| Where to View the Information | Information to View |
|---|---|
| Collection Status and Collection State columns in the MP for AWS Solution Details pane on the **Accounts** tab on the **Integrations** page. | The collection status appears approximately 10 minutes after you have configured the adapter. |
| Environment Overview | The objects related to AWS are added to the inventory trees. |
| Dashboards | AWS dashboards are added to VCF Operations. |

Tagging Groups

AWS uses tagging groups. The tagging groups appear under the AWS Entity Status in the Inventory page.

**Table 1675: Tagging Groups**

| Group Name | Description |
|---|---|
| PoweredOn | Objects with this tag are in the running state. |
| PoweredOff | Objects with this tag are in the stopped state. |
| Transient | Objects with this tag are not expected to persist for long periods of time. |
| NotExisting | Objects with this tag do not exist in the Amazon Web Services system. You can use this tag to take advantage of the periodic purge feature of VCF Operations, that the `controller.properties` file on the Analytics server controls. |

**View AWS Objects**

You can use the inventory tree to browse and select objects. The inventory tree shows a hierarchical arrangement of the AWS objects by region.

1. In the left pane of VCF Operations, click the **Environment** icon.
   - Click **Accounts** to list the AWS adapter instances.
   - Click **Regions** to to list the AWS regions.
2. To view the child objects, expand the regions and then expand the regions per account.

   **Note:**
   All the account-specific objects related to a region are grouped under the region per account section.
3. To display information about the object, select an object in the inventory tree.

## AWS Dashboards

Dashboards provide the user interface you use to monitor and troubleshoot Amazon Web Services problems in VCF Operations.

You can access the dashboards by selecting **Dashboards**, and then selecting **AWS**.

**Table 1676: AWS Dashboards**

| Dashboard Name | Purpose |
|---|---|
| AWS Alerts | The Alerts dashboard reports system-generated performance information for Amazon Web Services. In VCF Operations 5.8 and later, the dashboard also displays alerts received from Amazon Web Services Cloudwatch. |
| AWS ASG Utilization | Use the Auto Scaling Group (ASG) dashboard to identify which ASG groups have a high utilization across the metrics CPU, Disk IO, Network Transmissions, Received/Sent, and Number of Instances in the ASG. Use that information to determine whether any action is needed to adjust the ASG parameters. For example, you might need to raise or lower the scaling threshold for the CPU metric.<br>ASG metrics are not collected by default. You must activate them when creating the group. This applies only to the metrics belonging directly to the auto scale group, for example GroupDesiredCapacity. It does not apply to the aggregate instance metrics for the ASG, for example Instance Aggregate CPU Utilization. |

| Dashboard Name | Purpose |
|---|---|
| AWS Disk Space | Use the Disk Space dashboard to monitor EBS volumes to see whether they are running out of disk space and take appropriate action to anticipate future storage needs. Amazon Web Services does not report disk space by default.<br>For more information on accessing additional metrics, including disk space, and corresponding pricing, go to the Amazon Web Services documentation page at http://docs.aws.amazon.com/AmazonCloudWatch/latest/DeveloperGuide/mon-scripts.html |
| AWS Instance Heatmap | Use the Instance Heatmap to monitor CPU/Disk/Network metric elements and identify instances that perform poorly. |
| AWS Instance Utilization | Use to identify which EC2 instances have high use across the metrics for CPU, Disk IO, Network Transmissions, Received/Sent, and Memory. Use that information to determine whether you can optimize the system by making adjustments to EC2 instances. |
| AWS Troubleshooting | This dashboard is most helpful when someone calls in with a problem and you know which device they are using. You can search for that type of device or the specific device, if you know the name.<br>When you select the device, the relationship tree displays the item, its parents, and children. You can observe the Health, Workload, Anomalies, and Faults to get an overview of how the system is functioning in those areas. You can use information in the Interesting Metrics widget to help identify the root cause of issues. The Health, Anomalies, and Events Mash-up widget allows you to compare changes in the system to see how they might affect one another. |
| AWS Volume Performance | Use the Volume Performance dashboard to identify Elastic Block Store (EBS) volumes that are experiencing high disk read time, high disk write time, a high volume of disk read operations, or a high volume of disk write operations. |
| AWS Availability | Use this dashboard to view the availability of each AWS service. |
| AWS Inventory | Use this dashboard to view the count of each AWS service instance in each region. |
| AWS Optimization | Use this dashboard to view if you are effectively using AWS services. |

**Table 1677: AWS - All Other Dashboards**

| Dashboard Name | Purpose |
|---|---|
| AWS Services<br>• CloudFormation Stacks<br>• Compute: EC2<br>• Compute: Elastic Containers<br>• Compute: Lambda Functions<br>• Database: Dynamo<br>• Database: Elasticache<br>• Database: RDS<br>• Database: Redshift<br>• Desktop: Workspaces<br>• Network: Load Balancers<br>• Network: VPS<br>• Simple Queue Services<br>• Storage | Select **AWS Services** and then select a dashboard to view a specific service-related information. |

**AWS Instance Utilization Dashboard**
Use the AWS Instance Utilization dashboard to identify which EC2 instances have a high usage across the metrics for CPU, Disk IO, Network Transmissions, Received/Sent, and Memory. Use that information to determine whether you can optimize the system by adjusting the EC2 instances.

For example, you might determine that you need to resize the EC2 instance to make it larger or smaller.

You most often use this dashboard to troubleshoot issues with the listed metrics based on a support request from a user.

You can also identify which EC2 instances have been running for the longest and shortest amount of time. Then, you can use that information to determine whether EC2 instances can be decommissioned, or discover instances that have been added and need to be tracked in inventory.

Memory metrics require that you implement an add-on for each EC2 instance. These add-ons cost extra, and are not included by default.

**AWS Auto Scaling Group Dashboard**
Use the AWS Auto Scaling Group (ASG) dashboard to identify which ASG groups have a high utilization across the metrics CPU, Disk IO, Network Transmissions, Received/Sent, and Number of Instances in the ASG. Use that information to determine whether any action is needed to adjust the ASG parameters. For example, you might need to raise or lower the scaling threshold for the CPU metric.

**AWS Troubleshooting Dashboard**
When a user calls in with a problem and you know the name of the device they are using, can search for that type of device or the specific device and use the AWS Troubleshooting dashboard to get an overview of the system functionality.

When you select the device, the relationship tree displays the item, its parents, and children. You can observe the Health, Workload, Anomalies, and Faults to get an overview of how the system is functioning in those areas.

Use information in the Interesting Metrics widget to help identify the root cause of issues. The Health, Anomalies, and Events Mash-up widget allows you to compare changes in the system to see how they might affect one another.

There is a suggested flow to using the widgets in this dashboard.

1. Start with only the AWS Object widget open, and find the item you want to inspect.
2. Select the item, then expand the AWS Relationship widget to view the item's status.
3. Select one or all the related objects, then view the Ordered Symptoms, Interesting Metrics, and Mash-up.
4. Optionally, drag widgets into a new configuration if it makes it easier for you to compare information that is meaningful to you.
5. Examine the list of ordered symptoms and determine which of these events, in the given order might cause the problem to occur.

**AWS Instance Heatmap Dashboard**
Use the AWS Instance Heatmap dashboard to monitor CPU/Disk/Network metric elements and identify instances that perform poorly.

You can use the Troubleshooting dashboard to find more detail, and research the root cause of issues. Then you can view the specific object instance to identify faulty processes and take a corrective action.

**AWS Volume Performance Dashboard**
Use the AWS Volume Performance dashboard to identify Elastic Block Store (EBS) volumes that are experiencing high disk read time, high disk write time, a high volume of disk read operations, or a high volume of disk write operations. When you identify the EC2 instance that generates the load, use the Troubleshooting dashboard to investigate further.

**AWS Disk Space Dashboard**
Use the AWS Disk Space dashboard to monitor EBS volumes to see whether they are running out of disk space and take appropriate action to anticipate future storage needs. Amazon Web Services does not report disk space by default.

For more information on accessing additional metrics, including disk space, and corresponding pricing, go to the Amazon Web Services documentation page at http://docs.aws.amazon.com/AmazonCloudWatch/latest/DeveloperGuide/mon-scripts.html.

**AWS Alerts Dashboard**
The AWS Alerts dashboard reports system-generated performance information for Amazon Web Services. In VCF Operations 6.6 and later, the dashboard also displays alerts received from Amazon Web Services Cloud watch.

**AWS Inventory Dashboard**
Using the **AWS Inventory** dashboard you can navigate through the AWS inventory and view service details for each AWS account.

The **AWS Inventory** dashboard is built to provide an overview of your AWS inventory. By default, the total number of accounts and services displayed is at the AWS World level. You can view the number and types of services and also the growth or decline of a service within a specific period for individual accounts. You can also drill down and view the service summary per region for a specific account. Using this information you can understand growth trends, and manage and utilize your inventory cost-effectively.

The Inventory dashboard helps you answer several key questions:

- How many accounts are being monitored?
- How many services are running?
- What services are being used?
- What is the growth trend of the services?

**How to Use the Dashboard**

The **AWS Inventory** dashboard displays the total number of accounts, the account names, available regions, and the total number of services running at the AWS World level.

- **Total Number of Accounts Being Monitored** widget: Displays the total number of AWS accounts being monitored at the AWS World level.
- **Accounts** widget: Select an account from the table to refresh the service count and view the services specific to that account.
- **Regions** widget: Select a region to filter the service count and view service information per region.
- **Services** widget: Services are arranged in the alphabetical order, double click a service to view the growth trend and details regarding the service.

**Compute EC2 Dashboard**
Using the **Compute EC2** dashboard you can cater to the network, memory, and CPU utilization of your EC2 instances at the AWS World level.

You can use the **Compute EC2** dashboard to monitor the health of all your EC2 instances in the AWS inventory and filter them based on accounts and regions. The dashboard displays the total number of EC2 instances and the number of active instances. You can also view the size of the instances and their CPU, network, and memory utilization and distribution at the AWS World level.

The **Compute EC2** dashboard displays the top 10 EC2 instances with the highest and the lowest CPU, network, and memory utilization and distribution at the AWS World level. You can filter this information and view it based on accounts and regions per account.

The **Compute EC2** dashboard helps you answer several key questions:

- How many EC2 instances are there?
- How many EC2 instances are running or shut down?
- What alerts are generated for the EC2 instances?
- What are the key metrics for specific VMs in an EC2 instance?

**How to Use the Dashboard?**

The **Compute EC2** dashboard displays the account names, available regions, EC2 instances by size, and the total number of EC2 instances at the AWS World level. It also displays the top ten EC2 instances with the highest and lowest CPU utilization and CPU distribution, the top ten EC2 instances with the highest and lowest network bandwidth and network distribution, and the top ten EC2 instances with the highest and lowest memory usage and distribution. To view memory-related data for any EC2 instances, the AWS CloudWatch agent must be installed on the EC2 instances.

You can also view the EC2 instances according to their power state and running time and view the alerts for the EC2 instances. You can also view the list of EC2 instances with their key metrics at the AWS world level.

- **Accounts** widget: Displays the total number of AWS accounts being monitored at the AWS World level. Select an account from the table to refresh the dashboard and view the EC2 instance information for the specific account.
- **Regions** widget: Select a region to filter the displayed information and view the data for EC2 instances per region.
- **EC2 Instances with Key Metrics** widget: Displays the list of EC2 instances with key metrics at the AWS World level. The list of EC2 instances gets updated according to the account and region you select. Select an EC2 instance to view its key metrics, CPU utilization, network bandwidth, and memory usage for a particular EC2 instance.

**AWS Network Services Overview Dashboard**

The **AWS Network Services Overview** dashboard is built to provide an overview of your AWS VPCs and key network services linked to it.

You can view the number of VPCs, Subnets, NAT Gateways, and VPN connections.

The **AWS Network Services Overview** dashboard helps you answer several key questions.

- What is the network service count for a specific account or region?
- How many VPCs are used for a specific account or region?
- What are the NAT gateways for a specific VPC?
- How many VPNs and Subnets are connected to a specific VPC?

**How to Use the Dashboard?**

In the **AWS Network Services Overview** dashboard, the numbers of VPCs, Subnets, NAT Gateways, and VPN connections are displayed at the AWS World level by default. You can filter them by account and by region. You can also drill down to a specific VPC and check the NAT Gateways and VPN Connections linked to the selected VPC.

- **Accounts** widget: Lists the AWS accounts. Select an account from the table to refresh the dashboard and view the VPC information for the specific account.
- **Regions** widget: Lists the AWS regions. Select a region to filter the displayed information and view the data for VPCs per region.
- **Network Service Count** widget: Displays the total number of VPCs, number of VPC NAT Gateways, and the number of VPN connections.
- **VPC List**: Displays the names and regions of the VPCs deployed. The first VPC in the list is selected by default. The list of VPCs gets updated according to the account and region you select. Select a VPC instance to view the associated VPC NAT gateways, VPN connection, and connected services.
- **VPC NAT Gateways**: Displays the VPC NAT Gateway information for the first VPC in the VPC list by default. The VPC NAT gateways list gets updated according to the VPC you select.
- **VPN Connection**: Displays the number of VPN connections for the first VPC in the VPC list by default. The VPN connections information gets updated according to the VPC you select.
- **Connected Services**: Displays the number of connected services for the first VPC in the VPC list by default. It displays the number of VPNs and the number of subnets connected to the VPC. The information gets updated according to the VPC you select.

**AWS Storage Services Overview Dashboard**

Using the **AWS Storage Services Overview** dashboard you view the total storage service count (EBS volumes and S3 buckets).

The **AWS Storage Services Overview** dashboard provides an overview of your EBS volumes and S3 buckets. You can view the number of EBS and S3 buckets used in your environment.

The **AWS Storage Services Overview** dashboard helps you answer several key questions.

- What is the usage of EBS volumes?
- Which are the top read and top written EBS volumes?
- Which S3 buckets have maximum usage?

**How to Use the Dashboard?**

The **AWS Storage Services Overview** dashboard is divided into two sections based on the two storage types, **EBS Volumes** and **S3 Buckets**. By default, the numbers are displayed at the AWS World level and you can filter the information by account and by region. You can check the usage of the EBS volumes and drill down to individual volumes. For S3 buckets, you can view the buckets with maximum usage and errors. You can also select an individual bucket to check the traffic and errors associated with it.

- **Accounts**: Lists the AWS accounts. Select an account from the table to refresh the storage information and view the EBS volumes and S3 buckets specific to that account.
- **Regions**: Lists the AWS regions. Select a region to filter the storage information and view the EBS volumes and S3 buckets specific to that region.
- **EBS Volumes**: The EBS volumes section displays the top read EBS volumes, top written EBS volumes, EBs volumes with maximum idle time, and the EBS volumes list based on the load that they have. The lists get updated according to the account or region you select.

  The **EBS Volumes List** displays the name, region, capacity, maximum queue, and maximum write time of the EBS Volumes storage. The first EBS volume in the list is selected by default. The list of EBS volumes gets updated according to the account and region you select. Select an EBS volume to view the associated read time and write time.
- **S3 Buckets**: The S3 buckets section displays the S3 buckets with maximum total latency, S3 buckets with maximum first byte latency, S3 buckets with most 4xx errors, S3 buckets with most 5xx errors, and the S3 bucket lists. The lists get updated according to the account or region you select.

  The **S3 Buckets List** displays the name, region, object, and size of the S3 buckets storage. The first S3 bucket in the list is selected by default. The list of S3 buckets gets updated according to the account and region you select. Select a S3 bucket to view the associated requests and the 4xx and 5xx errors related to it.

## Summary Tab for Management Pack for AWS

The summary tab of Management Pack for AWS provide subscription details, overall public cloud inventory of the subscription including key components, aggregated metrics, and so on.

**World Summary Tab**
The World Summary tab of Management Pack for AWS provides a summary and information about the overall performance of all the multi-regional objects that you monitor.

Where To View the World Summary Tab

From the left menu, click **Administration** > **Integrations**. Click the vertical ellipses against the AWS and then click **Go to Details**

**Table 1678: World Summary Options**

| Option | Description |
|---|---|
| World | This widget displays details such as object type, number of accounts, and the number of regions that have objects for an adapter instance. |
| Instances | Displays the regions where services are provided and gives an inventory of the top 10 services sorted by the number of accounts in descending order. Click any region on the map to view details of the selected region. |
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the labels of the alert. |
| Relationship | Displays the object relationship of the selected object. |

### Region Summary Tab

The Region Summary tab of Management Pack for AWS provides a summary and overall performance of the object where services are configured and allows you to troubleshoot performance issues, if any.

Where To View the Region Summary Tab

From the left menu, click **Administration** > **Integrations**. Click the vertical ellipses against the AWS Cloud Account, and then click **Go to Details**.

**Table 1679: Region Summary Options**

| Option | Description |
|---|---|
| Region | This widget displays details such as the name of the region, number of projects in the region, and region the availability zone belongs to. |
| Instances | Displays the total number of instances and gives an inventory of the top 10 services sorted by the number of accounts in descending order. Click any region on the map to view details of the selected region. |
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the labels of the alert. |
| Relationship | Displays the object relationship of the selected object. |

### Service Summary Tab

The Service Summary tab of Management Pack for AWS, provide the summary of the selected object and allows you to troubleshoot the performance issues.

Where To View the Service Summary Tab

From the left menu, click **Data Sources** > **Integrations**. Click the vertical ellipses against the AWS, and then click **Go to Details**.

**Table 1680: Service Summary Options**

| Option | Description |
|---|---|
| Instance | Displays the object name, type, subscription, region, availability zone, VPC, subnet, and autoscaling group of the adapter instance. |
| Active Alerts | This widget provides a visual indicator of the alert status for the following alert types.<br>• Health alerts that usually require immediate attention.<br>• Risk alerts indicating that you must look into any problems shortly.<br>• Efficiency alerts indicating that you can reclaim resources.<br>To see the alerts for the object, click the labels of the alert. |
| Metrics | Displays the metrics available for the adapter instance. |
| Tags/Labels | Displays the tags or labels available for the object. |
| Relationship | Displays the object relationship of the selected object. |
| The following widgets are available for AWS Management Pack. | |

| Option | Description |
|---|---|
| Time Remaining | This widget displays the number of days remaining till the projected resource utilization crosses the threshold for the usable capacity. |
| Capacity Remaining | This widget displays the unused capacity of your virtual environment to accommodate new virtual machines. |

# Management Pack Builder

The Management Pack Builder enables the creation of custom management packs for use in VCF Operations. This is a no-code solution for bringing in data from an external API and either creating new resources or extending VCF Operations and third party objects with new data, relationships, and events.

### Backwards Compatibility

In VCF Operations, there is no separate installation required. Previously, you needed to deploy a separate OVA virtual appliance, which involved extra setup steps. Now, it is pre-integrated into the VCF Operations platform, reducing the initial installation time and eliminating compatibility issues.

The Management Pack Builder retains backward compatibility. Existing designs can be imported directly, preserving previous work while leveraging new efficiencies and improved performance. This ensures a smooth transition for teams who previously relied on the standalone Management Pack Builder.

### Where you find the Management Pack Builder

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Developer Center** > **Management Pack Builder**.

### What Can You Do in the Management Pack Builder page

You can use the **Management Pack Builder** page to design a new management pack, import or export a design, edit an existing design, or delete an existing design. After you design a management pack, you can install it.

**Table 1681: Accounts Page Options**

| Option | Description |
|---|---|
| Create New | To create a new management pack, you must perform the following steps:<br>• Connect: Connect to a data source to collect data.<br>• Transform: Use the data from the connected source to define the VCF Operations objects.<br>• Verify: Review and validate the management pack design.<br>• Install: Install the management pack.<br>To create a new management pack, click **Create New**, enter the **Name** and **Description**, and select the **Type**. You cannot change the management pack type after you create it.<br>**HTTP API**: Use the HTTP API type to create a management pack that collects data from an HTTP API to create objects, relationships, and events in VCF Operations. To design a management pack using HTTP, see Designing a Management Pack using HTTP.<br>**Prometheus**: Use the Prometheus type to create a management pack that collects data from Prometheus which includes objects and relationships. To design a management pack using Prometheus, see Designing a Management Pack using Prometheus. |
| Import Design | If you do not have any designs, you can import a JSON design file in VCF Operations. For more information, see Importing or Exporting a Design. |

After you create or import a design, you can view, export, edit, or delete it in the **Management Pack Builder** page.

**Table 1682: Designs Grid Options**

| Option | Description |
|---|---|
| Design | Displays the name of the design. |
| Type | Displays the design type: HTTP API or Prometheus. |
| Design Status | Indicates the status of the design.<br>• Valid: The design contains all components required to create a management pack. It has not been tested using the "Verify" step.<br>• Invalid: The design has an issue that needs to be resolved before it can be tested and installed.<br>• Verified: The design contains all components required to create a management pack and the "Verify" step has run successfully.<br>• Incomplete: The design does not contain all the components needed to create a management pack. |
| Install Status | Indicates installation status of the design.<br>• Installed<br>• Uninstalled<br>• Draft<br>• Out of date |
| Last Edited | Indicates when the design was last modified. |
| Edit | Click a design to edit it. |
| Export | Export a design as a JSON file. For more information, see Importing or Exporting a Design. |
| Delete | Delete the design. |

## Before you Start your Design

Before starting, you should make sure you have access to the reference environment including hostname, user information, and API documentation. It is important to think about what you want to include in it. The main items to understand are objects, requests, and relationships.

### Things to Consider for designing a Management Pack using HTTP

- Objects: Objects are the core item that we will be adding or updating in VCF Operations . You should think of them as the nouns of your technology. An example is a network switch. Your nouns for a switch could be the switch, ports, CPUs, and fans. These would all become objects in your design. For every object, you will provide the metrics and properties by adding one or more requests. VCF Operations manages metrics by the objects and has limits based on number of objects and metrics. Please review the sizing guide for the appropriate sizing of VCF Operations when designing your objects.
  You can also extend existing objects in your VCF Operations environment such as VMs, Host Systems, or third-party resources. By extending those objects, you can add metrics or properties to the existing object allowing custom information to be used as if it was part of the original resource.

  **Note:**

  VCF Operations manages metrics by objects and imposes limits based on the number of objects.
- Requests: These are the API requests used to add metrics and properties. To understand what is available on these requests, reference the API documentation. In our switch example, the requests to create port objects could include an interface endpoint that lists all the ports and a stats endpoint that show the performance of the port.
- Relationships: Relationships associate objects to each other in a parent to child relationship. This creates a topology for your objects allowing VCF Operations to understand the connectivity between your objects. In our switch example, the switch would be the parent to the ports, CPUs, and fans. You can also add relationships to VCF Operations objects like hosts, VMs, or third-party resources like data arrays.
- Events: Events represent a message reported by your endpoint pertaining to a VCF Operations object with an associated severity. Events can be used to impact the health of an object using symptoms. These symptoms also can be used to create alerts.

**Things to Consider for designing a Management Pack using Prometheus**

- Objects: Objects are the core item that we will be adding or updating in VCF Operations. You should think of them as the nouns of your technology. In a Prometheus design, objects are associated with a specific Prometheus job. For every object type, you will select the Prometheus metrics you would like to include and the labels on those metrics which will be mapped to identifiers and properties on the object. VCF Operations manages metrics by the objects and has limits based on number of objects and metrics. Please review the sizing guide for the appropriate sizing of VCF Operations when designing your objects.

  You can also extend existing objects in your environment such as VMs, Host Systems, or third-party resources. By extending those objects, you can add metrics and properties to the existing object, allowing custom information to be used as if it was part of the original resource.

  **Note:**

  VCF Operations manages metrics by objects and imposes limits based on the number of objects.
- Relationships: Relationships associate objects to each other in a parent to child relationship. This creates a topology for your objects allowing VCF Operations to understand the connectivity between your objects. You can also add relationships to objects like hosts, VMs, or third-party resources.

### Creating A Regular Expression

A regular expression is a sequence of characters that define a search pattern, mainly for use in string matching. Management Pack Builder supports Java-style regular expressions. When using a **Regular Expression** in Management Pack Builder you can enter in the expression in the **Regex** field. When the test button is pressed the **Sample Value** will be run through the expression and the result will be displayed in the **Sample Output**.

Regular expressions can be used in the following contexts:

- When defining fields that should match between objects when creating a relationship.
- When defining fields that should match between an event and the object it could be assigned to when creating an event.

### Variables

Custom authentication allows the user to add custom variables to be used in the advanced section of the source creation. This is generally used with permanent session tokens, or any variable sent with every request.

## Designing a Management Pack using HTTP

VCF Operations offers a wizard that makes it easy for you to design a management pack based on an HTTP API.

Building a new HTTP API-based management pack consists of the following steps. Once you start your design, you can save your work and pick up from where you left off. You can edit an existing design at any step.

- Define a Data Source: In this initial step, you define the data source from which the management pack will collect information. This involves specifying the external API or system that the management pack will integrate with, setting up the necessary connection parameters, and ensuring that the data source is accessible for data retrieval.
- Set up API Requests: Here, you configure the specific API requests or queries that will be used to fetch data from the defined source. This includes detailing the endpoints, request methods, parameters, and any necessary authentication details to ensure successful data extraction.
- Define Objects: In this step, you define the objects that represent the data entities retrieved from the source. This involves mapping the data to meaningful resource types within VCF Operations, specifying their properties, and determining how they will be represented in the monitoring environment.
- Specify Relationships (Optional): This step focuses on establishing relationships between the defined objects. By specifying how these objects interrelate, you can create a structured representation of the data, which aids in understanding dependencies and interactions within the monitored environment.

- Define Events (Optional): Here, you create an event from an API by defining the message, severity, and assoicated objects. You can configure symptoms and alerts for significant events in VCF Operations after designing the management pack.
- Verify your Design: In this step, you validate the configurations made in the previous steps. This involves testing the data collection, ensuring that objects and relationships are correctly defined, and confirming that events trigger as expected. It’s a crucial step to ensure the management pack functions as intended before deployment.
- Install: This is the final step. After you have reviewed and verified the design, you can install the management pack.

## Define a Source

Start by identifying the source system or API from which you intend to pull data. This step includes specifying host name, connection settings, and authentication protocols.

**Sub-sections in Define Source**

- Data Source Connections
- Get Session (visible if you turn on session authentication)
- Global Settings
- Release Session (visible if you turn on session authentication)
- Test Connection

**Data Source Connections**

- Collector: The collector node or cloud proxy from which the Management Pack Builder adapter instance will make API requests during the design process.
- Host Name:The target you're pulling data from.
- Port:The port to use.
- TLS Configuration: Select from one of the following options:
  - Verify: Use TLS and verify the target's certificate
  - No Verify: Use TLS but don't verify the target's certificate
  - No TLS: Do not use TLS at all
- Base Path: Path to be added to all API calls defined in this design. You can leave it blank and use the full path for all requests if there is no shared prefix.
- Preview API Path: A preview of your full path.

**Authentication Details**

Configure the authentication details for your API requests.

Choose between basic authentication, custom authentication, and no authentication. The configuration here is used for every subsequent call, so you only need to configure authentication once. If you need to create a session token, before making connection requests, turn on **Use Session Authentication**.

The Management Pack Builder can use three different authentication source types:

- Basic: Specify a username and password. The header is automatically created.
  - Username: The user on reference system that is authorized to query the intended API endpoint.
  - Password: User’s password for authentication.
- Custom**:** Add authentication fields and headers.
  - Use custom authentication when you need to add custom variables to be used in the advanced section of the source creation.
  - This is generally used with permanent session tokens or any variable sent with every request if session authentication is not used.
  - It is generally used for login credentials that will be used in POST a post body to get the session token when using session authenication.
- No Authentication**:** Use for public and non-secured APIs.

You can next choose to turn on **Session Authentication**. Session authentication is a method in which a session is created on login and stores information from the response body and/or headers as defined by the user. Common fields include a token or a session ID. When running a collection, the session fields are retrieved once, then used on all subsequent requests. If a release session request is defined, the session is released at the end of the collection.

Once you are done, click **NEXT**.

**Get Session**

Define the **HTTP Method** and **API Path** when using session authentication.

- HTTP Method: Based on the API requirements this will be either a GET, POST, PUT, PATCH.
- API Path: This is the path to the session creation resource as defined by the API documentation.

Next, set up additional request settings, such as headers, query parameters, and body

- HTTP Headers: This is used to assign headers to the request that is used to retrieve the session. This only needs to be used if the API requires custom values in the header. If using basic credentials, the Authorization header is automatically added and thus does not need further configuration.
- Query Parameters: This section allows the user to add custom query prompts to the end of the URL request. This is optional and only necessary if the API documentation calls for it.
- Body: This allows you to create a body to be sent to the session creation request. Some APIs require the body to contain information like user and password. This is optional and only needs to be used if the API documentation indicates it.

Click **Test** to see the test response. VCF Operations displays the **Body**, **Headers** and **Log**.

Once you are done, click **NEXT**.

**Global Settings**

Define the headers which are used to pass data with every call in the request header. Session data should be put here so that every call includes the session ID or token information.

Once you are done, click **NEXT**.

**Release Session**

Defining a **Release Session Request** is recommended when using session authentication. This request will log out the session, reduce session leaks, and improve security.

- HTTP Method: Based on the API requirements this will be either a GET, POST, PUT, PATCH, DELETE.
- API Path: This is the path to the API endpoint you want to use to release the session.

Once you are done, click **NEXT**.

**Test Session**

In this step, you to define the **HTTP Method** being used (GET, POST, PUT, PATCH), the API path, and see a preview of the fully qualified path. Then, you click **TEST CONNECTION**.

- HTTP Method: Based on the API requirements this will be either a GET, POST, PUT, PATCH.
- API Path: This is the path to test that your defined authentication works for your API.

The test connection confirms that all the source data is set up correctly. If using basic authentication, this will send one request including the basic authentication header. If using the session, this will make up to three requests to create the session, make a call to the test endpoint, then make a call to delete the session.

The response section will show the response to the test connection endpoint. You can see the body and headers of the response as well as the logs generated by the request. You can download the test response as a JSON file.

Once you are done, click **SAVE SOURCE**.

## Add Requests

This is the second step of the wizard. You can add all the API calls used by the design to provide data to the objects and events. You will need to reference your API documentation to determine the endpoint and settings necessary for this section.

### Add API Request and API Pagination

In the **Add API Request** section, define the endpoint to be used for the request. Note: the source hostname, port, and base path will be prepended to this path for the request URL.

- HTTP Method: This is the method used for the request. It will be either a GET, PUT, PATCH, or POST.
- API Path: Endpoint that needs to be call to retrieve the data.
- Name: Name for this request used in future settings in the design.

Advanced (optional)

In this section you can add specific data to the requests body, headers, and query parameters, as defined by the API documentation.

Once you are done, click **SAVE REQUEST**.

You can also add paging to your API call in the **Add API Pagination** section. You must make the request prior to configuring paging. Management Pack Builder supports two main methods of pagination: offset and pages. Reference your API documentation to determine whether paging is necessary and, if so, which method to use.

- Paging Parameter: This is the parameter name to use to indicate which page we are requesting.
- Starting Offset: This is the number that represents the beginning of the API's results. Often, "Pages" start at 1 and "Offset" starts at 0. Confirm the starting number in the API documentation.
- Limit Parameter: This is the parameter name for the number of records per page.
- Limit Value: The number of records returned per page.
- Data List Returned from Paged Request: List of data being returned. When the count of this is less than the limit value, it means that the end of the data has been reached and no more paged requests will be made.

### Chain From Other API Calls

To add a chained request, return to the wizard, go to the **Requests** step, and select the parent request from requests list, then select **Add Chained**.

- In the **Add Chaining** section, select a list from the parent request's response which contains the attribute(s) to be used in the child request.
- System Attributes: For each attribute that you would like to use in the child request, select **Add Field**. Select the **Attribute**. If desired, update the **Parameter**. The **Parameter** will be used to define the usage key, so make sure that it is meaningful in the context of the design. If the parent request had a value for the selected attribute, it will populate by default. Regardless, you can manually set the value here. This is the value that will be substituted for the usage key later when testing the child request, so make sure that it is a valid value which will cause the child request to succeed.
- In the **Add API Request** section, define the child request like any other request, but now you can use the attribute(s) defined in the previous step. You can insert the usage key in the path, headers, query parameters, and/or body of the child request. To insert the attribute usage key, click the clipboard icon on the right side of the field, then select the attribute you want to insert. It will be inserted where the cursor is currently located in the field. When the request is run via **MAKE REQUEST**, all usage keys will be replaced with the sample values defined in the previous step.

Once you are done, click **SAVE REQUEST**.

## Add Objects

In the third step, you define the objects that will be part of your custom management pack. You can define new objects or extend existing objects from VCF Operations with new metrics and properties.

### Sub-Sections in Add New and Add Existing Objects

- Object Type
- Attributes from API
- Properties and Metrics
- Name and Identifiers (Section appears only when adding new objects.)
- Object Bindings (Section appears when there is an existing object or when the attribue selections require it)

### Object Type

- Add New Object Type: The object type is the user-facing label that will be displayed for this type of object in VCF Operations when this management pack is installed. Example object types are Switch, Port, etc. You can also choose the icon that will be used in VCF Operations. Once you are done, click **NEXT**.
- Add Existing Object: To extend an existing object, choose a resource type to extend from the list of resource types on your VCF Operations instance. You can filter by the resource type or page through the list of available options.

### Attributes from API

From the list of **Requests**, you select one or more requests to pull into this object. You will then select all the data entries you want to add to your object from the list. You can search the list of attributes for a specific attribute or scroll through the list.

Once you are done, click **NEXT**.

### Properties and Metrics

Here you will see the metric and property definitions you have attached to the object.

- Label: The name of the metric or property visible in VCF Operations.
- Datatype: Defines how the data is rendered. If you select decimal or one of the counter options and the data cannot be converted to a number, the data will be discarded.
  - String: Returns the value of the attribute as a string. This datatype is allowed only on properties, and not metrics.
  - Decimal: Returns the value of the attribute as a number.
  - Counter Delta: Calculates the change in the value between collections.
  - Counter Rate: Calculates the change in value between collections over time.
- Unit: If applicable, this is where you can define the unit for the data point. This will allow auto-scaling in dashboards and reports.
- Property: This drop-down list indicates whether the data is a property.
- KPI: This toggle indicates whether the data will be treated as a KPI. KPIs are key metrics that are used in dashboards and resource detail pages. They should be metrics that are important to the status, performance, and health of the object.
- Expanded row view:
  - Request: Shows the call and data path from which this metric is pulled.
  - List: List in the request response from which the data is pulled.
  - Attribute: Name of the attribute in the response being pulled.
  - Example: Shows the value of the first returned data point from the request.

  Once you are done, click **NEXT**.

### Name and Identifiers

Define the identity of the object in this step.

- Object Instance Name: This is the property from the list above that is used as the object's name in VCF Operations.
- Object Identifiers: Define the properties using the smallest set of properties that will accurately and uniquely identify each object.

Once you are done, click **SAVE OBJECT**. If you see Attribute Bindings, proceed to the next step.

#### Attribute Bindings

When combining data from multiple JSON or XML arrays into a single new object, it's necessary to establish a base array and select a corresponding attribute from each of the other arrays for mapping. You will only see this section if it is required.

When extending an existing object, all arrays must link to a property or identifier on the existing object in VCF Operations.

- API Attribute: The data from the selected JSON or XML list in the API response that you match to an existing property or identifier.
- Object Property: A matching property or identifier on the object. For a new object, it is selected from the base list. For an extended object, it is selected from the identifiers and properties of the existing object.

## Best Practices For Metrics and Properties

Follow these best practices when you create objects from new metrics and properties.

#### Best Practices

- Objects created from new metrics and properties should be represent stable objects, not transient objects.
- Metrics should represent measurable performance characteristics. Example: CPU Utilization %, Network Throughput, etc. Properties should represent relatively stable configuration values. Example: version, power status, etc.
- Properties should have a small number of possible values.
- Property values should not change frequently from collection to collection.
- Opt for defining metrics using reasonable unit sizes and prefix multipliers to avoid publishing excessively large values.
- The depth of the relationships of the objects should not be too deep (for example 100).
- Avoid defining artificial relationships that do not exist in the end-point.

## Define Object Relationships

Relationships are a parent to child connection that is used in VCF Operations to define the interaction between interconnected resources. Parent to child relationships are the only supported type of relationships with the Management Pack Builder.

#### Parent and Child object

Define the objects used to create the relationships. Parent objects are higher in the topology.

#### Matching Properties

Relationships in Management Pack Builder are created by matching all resources with matching properties. For example, a Switch has a **switch ID** property and the Port has a **parent switch ID** property. The parent object properties and the properties of any relevant child object(s) will always match, e.g. **switch ID** with **parent switch ID**. With multiple matching sets, all sets of properties must match for the relationship to be created. This can be either case-sensitive or insensitive and use basic or advanced matching.

Once you are done, click **SAVE RELATIONSHIP**.

## Add New Events

Map events from the target system into the VCF Operations event model. This includes defining event triggers and mapping severity levels, and linking them to VCF Operations objects.

#### Sub-sections in this step

- Add Events from API

- Severity
- Related Objects

**Add Events from API**

- Event Name: The user-friendly name of the event. This will be prefixed on the event message in VCF Operations.
- API Request: This is the request endpoint setup in the **Requests** section to use to create events.
- Event List: The location of events in the API response.
- Event Message: This is the event message that will show in VCF Operations. Often in APIs this would be the event description. You can create the message using one or more fields from the response and type any static information as well. You can also edit the value of the fields using regular expressions. See the section on regular expressions for further help on this.

  Once you are done, click **NEXT**.

**Severity**

Events in VCF Operations are ranked with a severity from Info to Immediate. Please reference your API documentation for a list of valid severities they support.

- Event Severity: Define API response fields and static text which form the event severity definition. You can also adjust the value of the fields using regular expression.
- Map Severities: Add mappings from the VCF event severity (Event Severity) to the corresponding value expected from the API (API Response Severity). For example, if the API returns "critical," you would map to the VCF "Critical" severity. Sometimes an API will have a severity that does not directly correspond to a VCF severity. In that case, the mapping is up to your discretion. For example, if any API returns an event severity of Severe, you must map that to the VCF severity "Critical," or "Immediate."
- Set all unmatched API response Severities to: Specify a default severity. If an event severity is returned from the API that does not match one of the mappings you created, the associated event will be set to the default severity specified here.

Once you are done, click **NEXT**.

**Related Objects**

Relate events to objects by matching. Select "All" if you want to related events to more than one object in the case that multiple related objects match. Select "First" if you only want the first related object to match.

The order defined in this section is the order in which the management pack will evaluate matches.

Add related object: For each related object, first select the object type (**Select object**) to relate to. By default, you can choose any object types included in the current design. If you select "All objects", you can also choose from existing object types in VCF Operations. Next, create a mapping from the object properties to the API response. You can use any combination of object properties, API response fields, static text, and regular expressions to make the mapping work. In the top-right, you can choose whether the matching logic should be case-sensitive. If you define multiple related objects and selected "Relate events to objects by matching: First", you can use the arrows on the left to change the order in which the mappings will be evaluated.

Once you are done, click **SAVE EVENT**.

## Verify your Design

Before final deployment, you can test your design to ensure that the management pack collects the correct data and accurately represents the target system within VCF Operations.

**Perform Collection**

Click **PERFORM COLLECTION** to verify your design.

VCF Operations runs the collection and displays the following information:

- Summary: Displays the **Collection Status**, **Collection Time**, **Objects Collected**, **Metrics Collected**, **Properties Collected**, **Relationships Found**, and **Events Collected**.
- Log: Displays the last 1000 lines. You can download the log to view all the lines. You can change the logging level to see more detailed logs.
- Objects: Displays the details for the objects.
- Events: Displays the events for the objects.

Once you are done, go to the final step, **INSTALL**.

### Install

Once testing is complete, click **REVIEW CONFIGURATION** or directly click **INSTALL**. VCF Operations installs the management pack, making it available for live monitoring.

## Designing a Management Pack using Prometheus

VCF Operations offers a wizard that makes it easy for you to design a management pack from Prometheus data.

Building a Prometheus-based management pack consists of the following steps. Once you start your design, you can save your work and pick up from where you left off. You can edit an existing design at any step.

- Define a Data Source: In this initial step, you define the Prometheus data source from which the management pack will collect information. You must specify the Prometheus instance that the management pack will integrate with, setting up the authentication, and ensuring that Prometheus is accessible for data retrieval. For more information, see Define a Source.
- Define Objects: In this step, you map metrics that are retrieved from Prometheus to meaningful objects and metrics within VCF Operations. For more information, see Add Objects.
- Specify Relationships: This step focuses on establishing relationships between the defined objects. By specifying how these objects interrelate, you can create a structured representation of the data, which aids in understanding dependencies and interactions within the monitored environment. For more information, see Define Object Relationships.
- Verify your Design: In this step, you validate the configurations that were made in the previous steps. It also involves testing the data collection and ensuring that objects and relationships are correctly defined. It’s a crucial step to ensure the management pack functions as intended before deployment. For more information, see Define Object Relationships.
- Install: As the final step, after you have reviewed and verified the design, you can install the management pack. For more information, see Install.

### Define a Source

Start by identifying the Prometheus instance from which you intend to pull data. This step includes specifying the IP address or FQDN, API port, TLS configuration, and authentication settings.

#### Data Source Connections

- Collector: The collector node or cloud proxy from which the Management Pack Builder adapter instance will make requests to Prometheus during the design process.
- Prometheus Host: the IP address, hostname, or FQDN for the Prometheus instance that will be used during the design process.
- Port: The port hosting the Prometheus APIs. The default is 9090.
- TLS Configuration: Select from one of the following options:
  - Verify: Use TLS and verify the target's certificate
  - No Verify: Use TLS but do not verify the target's certificate
  - No TLS: Do not use TLS at all
- Authentication Type: Select from one of the following options:

- Basic: Specify a username and password
- No Authentication: No credentials are required

Once the form is complete, click **Test Connection**. The test connection confirms that the Prometheus instance is accessible with the provided configuration information and the API is returning the expected data.

The **Test Response** section shows you the list of jobs available on the Prometheus instance. In Prometheus, each type of monitored technology typically corresponds to one job. For example, all Clickhouse-related metrics are defined with the “clickhouse” job. By default, Prometheus defines its own self-monitoring metrics under the “prometheus” job.

The **Test Response** section also shows a **Log** tab, where you can review the test connection logs. If you are seeing a failed test connection, review the logs to determine the root cause of the failure.

Once you are done, click **Save Source**.

## Add Objects

Next, select Prometheus metrics that you want to convert into VCF Operations metrics and objects. Prometheus metrics have labels which can be used to uniquely identify objects. Prometheus metrics can be filtered and processed during their conversion to VCF Operations metrics.

### Add New Object Type

1. Object Type:
   - Object Type: The type which will be assigned and visible in VCF Operations.
   - Prometheus Job: In Prometheus, each type of monitored technology typically corresponds to one job. Select which type of technology will be associated with this object type.
   - Object Icon: Select an icon to represent this object type. By default, all icons are shown. Select a category to filter the icons shown. After a management pack is installed, custom icons can optionally be uploaded in **Administration** > **Control Panel** > **Icons**.
2. Prometheus Data: Prometheus metrics related to the job selected previously are shown. Select the metrics that should comprise the new VCF Operations object type. Use the search bar to filter on metric name and label.

   **Note:**

   When defining object identifiers, you will only be able to select labels that are shared across all the selected metrics. All Prometheus metrics have the “instance” label by default.
3. Identifiers and Name: Select one or more labels to uniquely identify the objects in VCF Operations. You can optionally update the text that will be shown in the VCF Operations UI.

   After the identifying labels are defined, select one of them to be used as the object name.
4. Metrics Mapping: Each Prometheus metric selected previously is shown. A Prometheus metric can be mapped to one or more VCF Operations metrics on the new object type.
   - Metric Label: Optionally update the metric label to be shown in VCF Operations UI.
   - Function: The selection for function varies depending on the type of Prometheus metric:
     - Gauge metrics always use the **Current Total** function. This means the VCF Operations metric will report the sum of the latest values returned.
     - Counter and summary metrics can use a **Delta** or **Rate** function.
       - Delta: The VCF Operations metric will report the difference between the current value and previous value of the Prometheus metric.
       - Rate: The VCF Operations metric will report the difference between the current value and previous value of the Prometheus metric, divided by the time elapsed between collections.
   - Units: Specify the unit for this metric.
   - Filters: Optionally define one or more filters to include only a subset of the Prometheus metric’s time series in the VCF Operations metric calculation. For example, you may want to split an HTTP Response metric into multiple VCF Operations metrics by filtering on response codes (2xx -> Successes, 4xx -> Failures, etc.).
     - =: Include time series where the specified label matches the provided text exactly.

- !=: Include time series where the specified label does not match the provided text exactly.
- =~: Include time series where the specified label matches the provided regex. Supports the same RE2 regex syntax that Prometheus supports.
- !~: Include time series where the specified label does not match the provided regex. Supports the same RE2 regex syntax that Prometheus supports.

5. KPI: Indicates whether the data will be treated as a key performance indicator. KPIs are key metrics that are used in dashboards and resource detail pages. They should be metrics that are important to the status, performance, and health of the object.

**Extend Existing Object Type**

To extend an object type that is defined by another management pack already installed in VCF Operations with Prometheus data, select **Extend Existing**.

1. Object Type:
   – Select Existing Object Type: Select the type of object that will be extended with additional metrics. Use the search bar in the top-right to filter on **Object Type** and **Adapter**.
   – Prometheus Job: In Prometheus, each type of monitored technology typically corresponds to one job. Select which type of technology will be associated with this object type.
2. Prometheus Data: Prometheus metrics related to the job selected previously are shown. Select the metrics that should be added to the existing VCF Operations object type. Use the search bar to filter on metric name and label.

   **Note:**

   When defining object identifiers, you will only be able to select labels that are shared across all the selected metrics. All Prometheus metrics have the “instance” label by default.
3. Binding Labels: Select one or more binding labels from the selected Prometheus metrics to link to the existing objects in VCF Operations.
4. Object Binding: Use a combination of one or more binding labels and static text to define a matching pattern between the Prometheus metric labels and the existing object type’s properties and identifiers. You may optionally apply a regex to any label, property, or identifier to modify the text that will be used for matching.
5. Metrics Mapping: Each Prometheus metric selected previously is shown. A Prometheus metric can be mapped to one or more VCF Operations metrics on the new object type.
   – Metric Label: Optionally update the metric label to be shown in VCF Operations UI.
   – Function: The selection for function varies depending on the type of Prometheus metric:
     - Gauge metrics always use the **Current Total** function. This means the VCF Operations metric will report the sum of the latest values returned.
     - Counter and summary metrics can use a **Delta** or **Rate** function.
       - Delta: The VCF Operations metric will report the difference between the current value and previous value of the Prometheus metric.
       - Rate: The VCF Operations metric will report the difference between the current value and previous value of the Prometheus metric, divided by the time elapsed between collections.
   – Units: Specify the unit for this metric.
   – Filters: Optionally define one or more filters to include only a subset of the Prometheus metric’s time series in the VCF Operations metric calculation. For example, you may want to split an HTTP Response metric into multiple VCF Operations metrics by filtering on response codes (2xx -> Successes, 4xx -> Failures, etc.).
     - =: Include time series where the specified label matches the provided text exactly.
     - !=: Include time series where the specified label does not match the provided text exactly.
     - =~: Include time series where the specified label matches the provided regex. Supports the same RE2 regex syntax that Prometheus supports.
     - !~: Include time series where the specified label does not match the provided regex. Supports the same RE2 regex syntax that Prometheus supports.
6. KPI: Indicates whether the data will be treated as a key performance indicator. KPIs are key metrics that are used in dashboards and resource detail pages. They should be metrics that are important to the status, performance, and health of the object.

### What To Do Next

- Define Object Relationships: For more information, see Define Object Relationships.
- Verify your Design: For more information, see Verify your Design.
- Install: For more information, see Install.

## Importing or Exporting a Design

A JSON file is created when exporting a design. A JSON file can be imported into VCF Operations. The file is verified as a valid design file and the author and description is shown. Only valid designs can be exported.

### Importing a Design

To import a design, click the **Import** button in the Management Pack Builder page. Browse to the location of the JSON file in your system. Only one design can be imported at a time.

VCF Operations verifies the validity and displays the name and description of the design. Ensure that another design of the same name does not exist. The design that you are importing must have a unique name.

After you import the design, it is displayed in the Management Pack Builder page. Verify the information in the **Design Status** column. You must provide the reference environment to test the design against. Click the imported design, and in the **Source** step of the wizard, specify the required fields.

### Exporting a Design

Before you export a design, ensure that it does not contain private information, such as IP addresses, credentials, or IDs, outside of end-user configurable fields. Before sharing or storing your exported design in an unsecured location, review the export file to ensure it does not include any private information.

A zip file containing the JSON file is exported. You must extract the JSON file from the zip file before you can import it.

## Troubleshooting a Design

1. Request Log: Every time you make a test request or perform a test collection in Management Pack Builder, there are tabs which show the body, headers, and logs. In the case of an error response code, review the logs and headers for information regarding the cause of the error.
2. Third-party API documentation: ensure that the request you defined complies with the requirements described in the third-party API documentation.
3. Management Pack Builder Adapter logs: located under `$VCOPS_BASE/user/log/adapters/MPBAdapter` on the node or cloud proxy selected under **Source**. Review these logs for any additional errors or warnings.
4. If you suspect an error with the Management Pack Builder tool itself, you can open a support case, or review the server logs at `/usr/lib/vmware-vcops/user/log/integration-designer-controller.log`. You can also review the Suite API logs at `/usr/lib/vmware-vcops/user/log/suite-api for additional errors`.

   **Note:**

   VMware Support is limited to the functionality and performance of the Management Pack Builder application. Support does not include troubleshooting or designing customer-specific management packs or custom API integrations. For questions about specific management pack designs, please post in the Management Pack Builder Community Page.